FOUR-DIMENSIONAL SU(2) YANG–MILLS THEORY A PROOF OUTLINE FOR $\mathrm{tr}(F^2)$

Abstract. Under the stated uniform renormalization and short-distance hypotheses, we outline the continuum construction of the renormalized $\mathrm{tr}(F^2)$ observable in four-dimensional Wilson SU(2) theory at sufficiently small reference coupling. The proof follows the source-dependent multiscale integration through uniform correlation estimates, volume-first limits and canonical decomposition to scalar Osterwalder–Schrader reconstruction.

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1. Introduction

Theorem 0. Assume the uniform renormalization and short-distance estimates stated below. For sufficiently small reference coupling, the correlations of $\mathrm{tr}(F^2)$ in four-dimensional Wilson SU(2) theory have nontrivial volume-first subsequential continuum limits. In every canonical component, the Schwinger functions satisfy the Osterwalder–Schrader axioms and reconstruct a local Hermitian scalar Wightman theory with positive energy and a unique vacuum. ■□

The precise statements are Thms. A to C.

Here $\mathrm{tr}(F^2)$ is the scalar curvature-squared observable with Wilson normalization. At each lattice cutoff we subtract the expectation of its smeared version $\mathcal{O}(f)$. A volume-first limit takes infinite volume at each fixed spacing before sending the spacing to zero along the weak-coupling sequence defined in § 2.3. The Euclidean functionals are first constructed on tests with separated supports and then extended to Schwartz tests flat on all coincidence diagonals. The reconstructed field generates the scalar sector associated with this observable.

The main task is uniform control of the source-dependent multiscale integration. The background field, its constrained inverse, the polymer sum and the large-field history sum must remain compatible on one analytic domain. In particular, a derivative of the normalizer and a change of exterior configuration must satisfy the same estimates used to bound local correlations. The proof develops these estimates in §§3 and 4 before passing to limits.

§5 controls the number of insertions and extends the limits to diagonal-flat tests. §6 gives the short-distance law and covariance. §7 treats spectral representations, canonical components and scalar reconstruction. Appendix A contains the auxiliary lattice estimates.

2. Lattice fields and the main results

This section fixes the Wilson measure, the normalization of $\mathrm{tr}(F^2)$, the admissible limiting sequences and the test spaces. The two main continuum statements follow the setting. The final subsection states the analytic inputs used in the proof, with their domains and uniformity requirements.

2.1. **Finite-cutoff identities and compactness.** This section establishes the finite-dimensional identities and compactness arguments used throughout the construction. None of its statements assumes a continuum Yang–Mills measure. Uniformity in the cutoff requires constants independent of the number of plaquettes, even though each integral is finite.

2.1.1. *The configuration space and its probability measure.* Let $G = \mathrm{SU}(2)$, the compact group of complex 2×2 matrices U with $U^*U = 1$ and $\det U = 1$. Its normalized Haar probability measure is denoted by dg. Fix a finite oriented graph obtained from a cubic lattice on a four-dimensional torus. Choose one orientation of each geometric bond; write B_+ for this set and impose $U_{\bar{b}} = U_b^{-1}$ for the reverse orientation. The configuration space is the compact space $\mathcal{V} = G^{B_+}$, with product probability measure dU.

A path is an ordered list of consecutive oriented bonds. Its transport is the ordered product of their matrices. For each geometrical elementary square choose one boundary orientation and one starting vertex, and denote its transport by U_p. Define

$$a_p(U) = 1 - \tfrac{1}{2}\,\mathrm{Re}\,\mathrm{tr}\,U_p,$$
$$A(U) = \sum_{p \in P} a_p(U),$$
$$N_P = |P|.$$

Changing the starting vertex conjugates U_p, and reversing orientation inverts it. Both changes preserve a_p.

Proposition 2.1 (Finite-cutoff existence and elementary bounds)**.** For every real $\beta \geq 0$,

$$e^{-2\beta N_P} \leq Z(\beta) := \int e^{-\beta A(U)} dU \leq 1.$$

Consequently $Z(\beta)^{-1} e^{-\beta A(U)} dU$ is a well-defined probability measure. Every bounded Borel observable has a finite expectation. ■

Proof. The eigenvalues of an element of G are e^{it}, e^{-it} for a real t. Thus $\frac{1}{2}\,\mathrm{Re}\,\mathrm{tr}\,U_p = \cos t$ lies in $[-1, 1]$, so $0 \leq a_p \leq 2$ and $0 \leq A \leq 2N_P$. Exponentiate these inequalities and integrate against a probability measure. The lower bound is strictly positive and supplies the normalization. □

The proof uses only compactness and positivity. Its constants degenerate when $N_P \to \infty$. In particular it supplies neither tightness of renormalized local fields nor a useful estimate for a continuum limit.

A gauge transformation is a family (h_x) of group elements indexed by vertices. It acts by $U_b \mapsto h_{b_-} U_b h_{b_+}^{-1}$. Successive factors cancel along a path, so a closed transport based at x is conjugated by h_x. Haar measure on each bond is unchanged by its independent left and right multiplications. This proves gauge invariance of the action and of the probability measure. A permutation of vertices and bonds coming from a lattice isometry also preserves the product measure and permutes the plaquettes. Therefore the finite torus measure has every isometry of that torus. This conclusion concerns these finite torus measures; it is not a proof that every infinite-volume periodic Gibbs state has every reflection.

2.1.2. *The continuum normalization is a local expansion.* Represent the Lie algebra by Hermitian traceless matrices and set $|F|^2 = \frac{1}{2}\operatorname{tr}(F^2)$. For a fixed such matrix,

$$1 - \tfrac{1}{2}\operatorname{Re}\operatorname{tr} e^{itF} = \tfrac{t^2}{4}\operatorname{tr}(F^2) + O(t^4|F|^4) = \tfrac{1}{2}t^2|F|^2 + O(t^4|F|^4).$$

The expansion follows by the matrix exponential series: the linear trace vanishes and the odd powers have zero real trace. If a smooth connection has elementary holonomy $U_p = \exp(ia^2 F_{\mu\nu}(x_p) + O(a^3))$, with all required derivatives uniformly bounded, it follows that

$$a_p = \tfrac{1}{2}a^4|F_{\mu\nu}(x_p)|^2 + O(a^5).$$

Hence, on a fixed compact region, summing $f(x_p)a_p$ over plaquettes gives the corresponding Riemann integral of $\frac{1}{2}f\sum_{\mu<\nu}|F_{\mu\nu}|^2$ as $a \downarrow 0$.

This is a deterministic smooth-field calculation. Typical configurations under a quantum measure have not been shown to satisfy its hypotheses. The calculation fixes the intended normalization of the observable; it does not prove convergence of that observable under the lattice measures.

2.1.3. *Sources, cumulants, and centering.* For bounded functions f_i on the plaquette barycentres put $O_i = \sum_p f_i(x_p)a_p$ and

$$Z(z) = \int \exp\{-\beta A(U) + \sum_{i=1}^{n} z_i O_i(U)\}\, dU,$$
$$z \in \mathbb{C}^n.$$

The estimate $|O_i| \le 2\sum_p |f_i(x_p)|$ gives an integrable uniform majorant on every compact source set. Termwise differentiation of the exponential series is therefore valid, and Z is entire. Since $Z(0) > 0$, continuity gives a sufficiently small polydisc about zero where Z has no zeros. On that polydisc there is a unique holomorphic logarithm W with $W(0) = \log Z(0)$.

The joint cumulant is

$$\kappa(O_1, \dots, O_n) = \partial_{z_1} \cdots \partial_{z_n} W(z)\big|_{z=0}.$$

Its expression in moments is

$$\kappa(O_1, \dots, O_n) = \sum_{\pi \in \mathrm{Part}(n)} (|\pi| - 1)!(-1)^{|\pi|-1} \prod_{J \in \pi} \mathbb{E} \prod_{i \in J} O_i.$$

Indeed, write $Z(z)/Z(0) = 1 + u(z)$ and use $\log(1+u) = \sum_{m \ge 1}(-1)^{m-1}u^m/m$ near zero. The coefficient of $z_1 \cdots z_n$ in u^m sums over ordered partitions into m nonempty blocks. Dividing by m turns their $m!$ orderings into $(m-1)!$. Conversely, expanding $e^{W(z)-W(0)}$ proves the moment–cumulant relation

$$\mathbb{E}\prod_i O_i = \sum_{\pi \in \mathrm{Part}(n)} \prod_{J \in \pi} \kappa((O_i)_{i \in J}).$$

Replacing O_i by $O_i - c_i$ changes the logarithm only by $-\sum_i z_i c_i$. Thus every cumulant of order at least two is unchanged. If $c_i = \mathbb{E}O_i$, every singleton cumulant vanishes. The centered moment functional is consequently

$$\mathbb{E}\prod_i (O_i - \mathbb{E}O_i) = \sum_{\pi \in \mathrm{Part}_{\ge 2}(n)} \prod_{J \in \pi} \kappa((O_i)_{i \in J}).$$

There are finitely many partitions at fixed n. Convergence of every separated cumulant of order at least two therefore implies convergence of the centered moments on the same tuples. Positivity must be applied to these moments, not to the cumulants separately.

The source is exactly a local coupling shift:

$$-\beta A + \sum_i z_i O_i = -\sum_p (\beta - w(x_p)) a_p,$$

$$w = \sum_i z_i f_i.$$

For a constant source c, $Z_\beta(c + w) = Z_{\beta - c}(w)$. This algebraic identity holds before any renormalization procedure is chosen.

2.1.4. *What a mass-preserving transformation preserves.* Let T be a fixed linear map on finite measures on a measurable space, with $(T\nu)(1) = \nu(1)$ for all measures in its domain. If $\nu_z = e^{\sum_i z_i O_i}\nu$ belongs to that domain, then

$$(T\nu_z)(1) = \nu_z(1).$$

This proves exact equality of sourced partition functions, and hence of their derivatives when the analytic dependence can be justified. It does not identify $T\nu$ as the distribution of a predetermined block variable. For that stronger conclusion one needs a measurable map M and the identity $(T\nu)(F) = \nu(F \circ M)$ for every bounded measurable F.

For example, on the two-point set $\{0, 1\}$ let $T\nu = \nu(\{0, 1\})\delta_0$. This positive linear map preserves total mass but changes the expectation of the coordinate from $\nu(\{1\})$ to zero when ν is a probability measure. Thus mass preservation alone contains no statement about observable expectations. Sources must be transported through the exact integral, with the choices of the transformation fixed throughout the source family.

2.1.5. *A fixed-spacing infinite-volume limit.* Consider the countable cubic lattice at a fixed positive spacing. Its configuration space is $\mathcal{V}_\infty = G^{B_+}$ with the product topology. This is a compact metrizable space: enumerate the bonds and use, for example, the sum of 2^{-j} times bounded compatible distances on their group coordinates. The space of Borel probability measures is sequentially compact in the weak topology.

For a finite set of bonds D, define the conditional specification

$$(\gamma_D F)(U) = \frac{\int F(V_D, U_{D^c}) \exp[-\beta A_D(V_D, U_{D^c})]\, dV_D}{\int \exp[-\beta A_D(V_D, U_{D^c})]\, dV_D},$$

$$A_D = \sum_{p : p \cap D \neq \varnothing} a_p.$$

The sum is finite and the denominator is bounded below by $e^{-2\beta N(D)}$, where $N(D)$ is the number of plaquettes meeting D. For continuous cylinder F, numerator and denominator depend continuously on only finitely many boundary bonds; dominated convergence proves that $\gamma_D F$ is another continuous cylinder function.

Proposition 2.2 (DLR existence at fixed lattice parameters)**.** Take torus Wilson measures at fixed spacing and fixed β, and extend their bond configurations periodically to the infinite lattice. Every sequence of torus side lengths tending to infinity has a weakly convergent subsequence. Each limit ν satisfies $\nu(\gamma_D F) = \nu(F)$ for every finite D and bounded Borel cylinder function F. It is invariant under all lattice translations and all lattice isometries preserved by the finite torus measures. ■

Proof. Compactness gives the subsequence. Fix D and a continuous cylinder F. For sufficiently large tori the finite set consisting of D, all adjacent plaquettes and the support of F embeds without identifications. The finite-volume conditional expectation identity then reads $\mu_\ell(\gamma_D F) = \mu_\ell(F)$ after periodic extension. Both integrands are continuous cylinder functions, so the identity passes to the weak limit. Continuous functions on the finite-dimensional compact coordinate spaces determine measures. The identity therefore extends first to bounded Borel cylinder functions, and, if desired, to all bounded Borel functions by the monotone class theorem. Invariance passes to the limit by the same continuous-cylinder argument, since a fixed lattice isometry acts continuously on the product space. □

This proposition does not change the spacing or the coupling. It provides no bounds uniform as the spacing tends to zero and gives no uniqueness or extremality of the limiting state. A theorem about all periodic Gibbs states is stronger than a theorem about these torus limit points.

2.1.6. *Compactness of separated correlation forms.* Let $\mathbb{E}$ be $\mathbb{R}^4$ or a fixed torus. A separated tuple consists of finitely many compactly supported C^1 functions with pairwise disjoint supports. Their compactness makes the minimum distance between nonempty supports positive. Fix the norm $\|f\|_1 = \max(\|f\|_\infty, \|\nabla f\|_\infty)$.

Proposition 2.3 (A diagonal extraction principle)**.** Suppose $T_{n,j}$ are symmetric multilinear forms on separated tuples and for every fixed n, $R < \infty$ and $d > 0$ there is a finite $C(n, R, d)$, independent of all sufficiently large j, such that

$$|T_{n,j}(f_1, \dots, f_n)| \le C(n, R, d) \prod_i \|f_i\|_1$$

whenever the supports lie in $B(0, R)$ and are separated by at least d. Then a subsequence converges at every separated tuple, simultaneously for all n. The limits obey the same bounds. ■

Proof. Choose all finite families of finite unions of rational closed boxes with positive mutual distance. The C^1 functions supported in each union form a separable normed space, being a subspace of a finite product of spaces of continuous functions on a compact set. Choose countable dense subsets. There are countably many finite tuples drawn from these subsets. Each sequence of values is eventually bounded by the assumed estimate. Bolzano–Weierstrass and a successive diagonal extraction give a common subsequence on which all these values converge.

Given any separated tuple, enclose its supports in pairwise separated finite unions of rational boxes, still inside a fixed bounded region. Approximate each function in C^1 by elements of the chosen dense sets. The telescoping multilinear identity

$$T(f_1, \dots, f_n) - T(g_1, \dots, g_n) = \sum_{i=1}^{n} T(g_1, \dots, g_{i-1}, f_i - g_i, f_{i+1}, \dots, f_n)$$

and the uniform bound make the approximation error uniformly small along the extracted subsequence. This proves the Cauchy property for the original tuple. Multilinearity, symmetry and the bound pass to the limit on each product of spaces with separated supports. □

The difficult Yang–Mills input is the hypothesis supplying $C(n, R, d)$. Finite-cutoff boundedness with a constant involving the number of plaquettes does not meet it. An exponentially large dependence on the physical torus volume also does not supply a bound uniform on $\mathbb{R}^4$. These are different compactness problems.

2.1.7. *Reflection positivity and closed limits.* Let θ be a geometric reflection, and let $\overline{X}$ denote componentwise complex conjugation of a test tuple. Real reflection-invariant centering preserves reflection positivity: replacing each $O_j(f)$ by $O_j(f) - m_j(f)$ maps every positive-half polynomial to another such polynomial and commutes with reflection when $m_j(\theta f) = m_j(f)$. Suppose for each cutoff a centered moment functional σ_j satisfies

$$\sum_{r,s=1}^{m} \overline{c_r} c_s \, \sigma_j(\theta \overline{X}_r \cdot X_s) \geq 0$$

for every finite collection of tuples X_r in the positive half-space. Within each X_r the supports are separated; the reflected negative tuple and the positive tuple are automatically separated from one another. Overlap between supports from X_r and X_s on the same side causes no problem, because these supports do not occur together in the same concatenated pair $\theta X_r \cdot X_s$.

If all the entries converge, the finite sum converges and remains nonnegative. Thus the limiting moment functional is reflection positive on this vector space of separated monomials. This elementary closedness statement is the precise sense in which positivity passes to limits. Passing from that domain to a Schwartz test space requires a density result and a continuous bound for the limiting distributions; it cannot be justified by closedness alone.

The algebraic form $\langle F, G\rangle = \sigma(\theta\overline{F} \cdot G)$ is positive semidefinite. Positivity of $\langle F + zG, F + zG\rangle$ for every complex z gives $|\langle F, G\rangle|^2 \leq \langle F, F\rangle\langle G, G\rangle$. Consequently a null vector is orthogonal to every vector. The quotient by the null space is an inner-product space, and its completion is a Hilbert space. Defining translations on this quotient additionally requires that they preserve the null space and satisfy the appropriate bounds. Strong continuity and a nonnegative generator are later conclusions with their own hypotheses.

2.1.8. *Two limits of what scalar correlations prove.* Let X be a real standard Gaussian variable and put $Y = X^2 - 1$. For $t < 1/2$ near zero,

$$\log \mathbb{E} e^{tY} = -t - \tfrac{1}{2}\log(1 - 2t) = \sum_{n \geq 2} \frac{2^{n-1}}{n} t^n.$$

It follows that $\kappa_n(Y) = 2^{n-1}(n-1)!$ for every $n \geq 2$. In particular a nonzero connected four-point function of a quadratic observable is compatible with a Gaussian underlying theory. For distinct centered jointly Gaussian variables the same conclusion is expressed by the connected cycle contractions of their Wick squares.

Accordingly a non-Gaussian $\mathrm{tr}(F^2)$ field is an important property of that field, but is not by itself a characterization of an interacting Yang–Mills gauge theory. Reconstruction from the $\mathrm{tr}(F^2)$ correlations produces the cyclic sector generated by that observable. Identifying a richer gauge-invariant algebra requires additional observables and compatibility arguments.

A polynomial cluster estimate likewise gives no positive mass gap. In a Hilbert space with a nonnegative self-adjoint H, a connected time correlation has the spectral form $\langle v, e^{-tH} v\rangle = \int_0^\infty e^{-tE}\, d\mu_v(E)$ for a vector orthogonal to the vacuum. If $d\mu_v(E) = Ee^{-E}\, dE$, the integral is $(1 + t)^{-2}$. It decays, yet its spectral measure meets every interval $(0, \varepsilon)$. The support condition required for a gap would instead exclude all such intervals below one fixed positive threshold.

These examples delimit the conclusions of the conditional construction. They do not contradict the proposed ultraviolet estimates, and they do not replace the work of proving those estimates.

2.2. **Renormalisation objects and norms.** The renormalisation step must represent the same integral after one scale is integrated, with local interactions whose bounds remain uniform in the number of scales. We describe those interactions by their local functions and measures, their field domains, and their norms. This gives a precise meaning to preservation of the class under a step.

Constructing backgrounds and keeping every substituted field inside its stated domain require analytic estimates beyond these definitions.

2.2.1. *Lattices, currents, and auxiliary geometric data.* The normalization in this section is that of $G = \mathrm{SU}(2)$. Its Lie algebra is represented by traceless Hermitian matrices, with $\langle A, B\rangle = \frac{1}{2}\operatorname{tr}(AB)$ and complexification $\mathfrak{g}^{\mathbb{C}}$. Fix an odd integer $L > 1$. The level-j lattice, denoted $\mathbb{T}^{(j)}$, consists of the sites retained after j blockings of a periodic bare lattice. Its bare spacing, expressed in level-j units, is $\eta_j = L^{-j}$. A bond configuration satisfies $U(\bar{b}) = U(b)^{-1}$. Products along oriented paths are ordered in path order. Positive coordinate bonds suffice to specify a configuration. All suprema over bonds below use these bonds. In the left multiplication chart, reversal is $J(\bar{b}) = -\operatorname{Ad}(U_b^{-1})J(b)$. The complex configuration group is $\mathrm{SL}(2,\mathbb{C})$.

For a positively oriented plaquette p, based at its chosen corner, write U_p for its holonomy and set

$$a_p(U) = 1 - \tfrac{1}{2}\operatorname{tr} U_p,$$
$$A(U) = \sum_p a_p(U),$$
$$A(c,U) = \sum_p c(x(p))a_p(U).$$

Here $x(p)$ is the plaquette midpoint. On real configurations this is the real Wilson action; the same formula is its holomorphic extension. For definiteness the current on a lattice of spacing η is specified by its variational normalization:

$$\frac{d}{dt}A(e^{i\eta tB}U)\Big|_{t=0} = \eta^4 \sum_b \langle J_\eta(U)(b), B(b)\rangle. \tag{2.i}$$

The sum is over positive bonds and B is arbitrary. This identity defines every component of J_η, also for complex U by analytic continuation. It fixes the transports and factors implicit in the notation $D_U^* F_U$. One must use the plaquette divergence associated with eq. 2.i in that notation; the adjoint of the site gradient alone has a different domain. At $U = \mathbf{1}$, differentiation in the potential coordinate $U(t) = e^{i\eta tB}$ gives $J'_\eta(\mathbf{1})B = \mathrm{d}_\eta^* \mathrm{d}_\eta B$, where d_η here is the oriented plaquette difference on one-forms divided by η. If the numerical mesh changes from η to $\eta' = L^m\eta$ while the group configuration stays fixed, potentials are multiplied by L^{-m} and currents by L^{-3m}. This follows directly from eq. 2.i and the chart $e^{i\eta A}$, and specifies the direction of the unit change. The current variable of a complex pair, introduced below, is independent of this particular function of U.

Fix powers of L called $\mathfrak{M}, M_1', M$, a gap $\rho = R_1\mathfrak{M}$, and constants $C_{\mathrm{box}}, C_{\mathrm{pot}}$. They are part of the auxiliary data, selected before domains or norms are formed. The first two sizes concern backgrounds and walks; M is the localization cube size. They are not interchangeable. The partition π_j consists of the centered, half-open M-cubes in level-j units. Boundary conventions are fixed once on every lattice. For a cubical region Y, $Y^{\sim n}$ means addition of n layers in the partition currently named; $Y^{\sim -n}$ means removal of those layers. A hull is the smallest union of cubes of the named partition containing the set. These operations include their periodic versions.

The two erosion assignments are fixed as follows. Take $L \geq 13$ odd, $R_1 \geq L$, $M_1' \geq R_1\mathfrak{M}$ and $M \geq L^4 M_1'$, in addition to the background floors. For $Z \in \mathcal{D}_{p-1}$ put $\Omega_0^{(1)} = \Omega_0^{(2)} = Z$. In bare

units their successive members are

$$\begin{aligned}\Omega^{(1)}_{q+1} &= \bigcup_{\substack{C \text{ of side } \mathfrak{M}L^{q+1}\\ C\subset\Omega^{(1)}_q,\ \mathrm{dist}_\infty(C,(\Omega^{(1)}_q)^c)>R_1\mathfrak{M}L^q}} C,\\ \Omega^{(2)}_{q+1} &= \bigcup_{\substack{C \text{ of side } M_1'L^{q+1}\\ C\subset\Omega^{(2)}_q,\ \mathrm{dist}_\infty(C,(\Omega^{(2)}_q)^c)\geq M_1'L^{q+1}}} C, \qquad 0\leq q<p.\end{aligned} \tag{2.ii}$$

Each union contains every cube of the named aligned partition satisfying its test. A tower of depth $n \leq p$ is the truncation of this construction at n. These definitions commute with the allowed translations and common lifts. Empty members remain empty; for a full torus the complement distance is infinite and the members remain full. The maximal-cube rule is essential: it gives the upper bound on total erosion proved in Lem. 3.126.

Here is the determining set of a nested sequence. Work temporarily in level-k units, so the spacing of level j is $\xi_j = L^{j-k}$. Let $\Omega_0 \supset \cdots \supset \Omega_k$ and $\Omega_{k+1} = \varnothing$. The set Γ_j consists of bonds of $\mathbb{T}^{(j)}$ whose endpoints lie in Ω_0, at least one endpoint lies in Ω_j, and neither lies in Ω_{j+1}. Thus inward bonds crossing into the next layer are excluded. Put

$$\mathbf{B} = \bigcup_j \Gamma_j,$$

$$M_{\mathbf{B}}(U) = (\mathcal{M}^j U|_{\Gamma_j})_j. \tag{2.iii}$$

The endpoints of these bonds are the nodes. They are the tops of the blocks, or pillars, partitioning the determining region. A residual gauge function equals $\mathbf{1}$ at all nodes. In particular, imposing additional inward-bond constraints changes this problem.

The averaging map $\mathcal{M}$ used in eq. 2.iii is the symmetric contour average. For group elements V_i of small diameter their mean is the unique nearby V satisfying $\sum_i \log(V_i V^{-1}) = 0$. The contour transport $U(y,x)$ is the mean over the shortest paths from y to x in its centered block that traverse the coordinate directions in one of their orders. Set $U(x,y) = U(y,x)^{-1}$. For a coarse bond c, $x' = x + (c_+ - c_-)$, and straight transports $U[c]$ and $U([x,x'])$,

$$\mathcal{M}(U)(c) = \exp\left(L^{-4}\sum_{x\in B(c_-)} \log\{U(c_-,x)U([x,x'])U(x',c_+)U[c]^{-1}\}\right) U[c]. \tag{2.iv}$$

All logarithms in regular neighborhoods are their near-identity branches. Away from those neighborhoods a measurable extension of the contour means and logarithms is auxiliary input. Exact identities using that extension require its gauge covariance; small-field analyticity concerns only the displayed branches. Existence of a globally symmetric extension with every desired property is not asserted merely by writing eq. 2.iv.

For every determining sequence and real regular datum $\mathbf{V}$, the background input is a regular critical orbit for $\sum_{p\subset\Omega_0} a_p$ on $\{U : M_{\mathbf{B}}(U) = \mathbf{V}\}$, together with a chosen analytic continuation $\Phi_{\mathbf{V}}(B)$ at datum $e^{iB}\mathbf{V}$. The representatives, gauge, analytic domain and transport under local charts are included in this input. Uniqueness, existence and uniform estimates for these maps are separate background theorems. The word "background" does not substitute for them.

2.2.2. *Hereditary domains and their shifts.* A triple of level j on X is defined on the bare mesh η_j when lengths are expressed in level-j units. It is $P = (U, A', \mathbf{J})$, with U real and $A', \mathbf{J}$ complex Lie-algebra-valued bond forms. Its pair is $\mathbf{p}(P) = (\mathbf{U}, \mathbf{J})$, where $\mathbf{U} = e^{i\eta_j A'}U$. A triple is a splitting of its pair. Fix positive radii a_0, a_1, a_2. The simple conditions are

$$\begin{aligned} &|\partial U - \mathbf{1}|, |\partial \mathbf{U} - \mathbf{1}| < a_0 \eta_j^2, \qquad |A'|, |\nabla_U A'| < a_1, \qquad |\mathbf{J}| < a_2; \\ &|A|, |\nabla A| < C_{\text{pot}} M a_0 \quad \text{in the real gauges described below.} \end{aligned} \tag{2.v}$$

For each cubical box contained in X with side at most $C_{\text{box}} LM$, such a real gauge must exist, with $U = e^{i\eta_j A}$. The norms are maximum matrix norms in the named local units. For a one-form component A_μ based at its initial site, fix the forward convention

$$(\nabla_\nu^U A)_\mu(x) = \eta_j^{-1}\{\mathrm{Ad}(U_\nu(x))A_\mu(x + \eta_j e_\nu) - A_\mu(x)\}.$$

The maximum of these component norms defines $|\nabla_U A|$. In the usual two-radius notation $a_2 = a_0$.

For every subdomain $Z \subset X$ at every level $p \le j$, for both erosion assignments, and every truncation depth $1 \le n \le p$, form its determining set $\mathbf{B}'$. Define its tower configuration by

$$U_n(Z; P) = \Phi_{M_{\mathbf{B}'}(U)} \left(\frac{1}{i} \log\{M_{\mathbf{B}'}(\mathbf{U}) M_{\mathbf{B}'}(U)^{-1}\} \right). \tag{2.vi}$$

This expression is required to exist in the chosen continuation domain. On Z with two layers of the M-cubes of level $p-1$ removed, require

$$\begin{aligned} |\partial U_n(Z; P) - \mathbf{1}| &< a_0 L^{-(j-p)} \eta_p^2, \\ |J(U_n(Z; P))| &< a_0 L^{-(j-p)} L^{2(n-p)}. \end{aligned} \tag{2.vii}$$

The current in the second line is formed in level-n units. Impose the same conditions with $\mathbf{U}$ replaced by U. These are quantifiers over every allowed Z, p, n, not just the domain on which an activity is eventually evaluated. The hereditary core $\mathcal{U}_j^\circ(X; a_0, a_1, a_2)$ is the set of triples satisfying eq. 2.v–eq. 2.vii. Its pair image is the analytic pair domain; holomorphy means holomorphy in $(A', \mathbf{J})$ on every open slice with U fixed. The inequalities are strict. Requiring the tower maps to exist in their open continuation domains is part of membership.

An activity on this core must have the tie: its values at any two splittings of the same complex pair agree. Gauge invariance means invariance under simultaneous transformation of both entries of the pair; the current at a bond is based at its initial site and transforms by conjugation there. The core itself is not saturated by complex gauge transformations. Extension of a function to the union of complex orbits requires equality on all intersections and holomorphic descent; it is an additional statement when used.

Restriction ρ_X discards bonds and variables outside X without extending them across the boundary. If an evaluation changes its numerical mesh from η_j to $\eta' = L^m \eta_j$, the coordinate change is explicitly

$$\begin{aligned} (U, A', \mathbf{J}) &\longmapsto (U, L^{-m} A', L^{-3m} \mathbf{J}), \\ m &\ge 0. \end{aligned} \tag{2.viii}$$

For a shift $(\mathbf{H}, \mathrm{j})$ on the same lattice define

$$\begin{aligned} P \dotplus (\mathbf{H}, \mathrm{j}) &= (U, A' \boxplus \mathbf{H}, \mathbf{J} + \mathrm{j}), \\ e^{i\eta_j (A' \boxplus \mathbf{H})} &= e^{i\eta_j \mathbf{H}} e^{i\eta_j A'}. \end{aligned} \tag{2.ix}$$

The logarithm defining $\boxplus$ is the small branch; inclusion in its domain is required. The independent current is shifted additively. Replacing it by $J(e^{i\eta_j \mathbf{H}}\mathbf{U})$ at every complex argument would change the construction.

The decoupled response on a cubical region Y is the pair of maps $\mathfrak{h}_Y(s;P,B') = (\mathbf{H},\mathsf{j})$, holomorphic in complex cube parameters $|s(\Delta)| < e^{\bar\kappa}$, the complex triple variables, and the datum B'. It vanishes at $B' = 0$. Its first component is the solution of the actual background equations with the propagators decoupled by these parameters:

$$\begin{aligned} \mathbf{H} &= \mathbf{a} - \mathsf{H}(s)\mathsf{D}, \qquad & \mathsf{D} &= C_k(\mathbf{a} - \mathsf{H}(s)\mathsf{D}), \\ \mathbf{a} &= \mathsf{H}_1(s)B' + \mathbf{A}_1, \qquad & \mathbf{A}_1 &+ \mathfrak{G}(s)\mathsf{W}_s(\mathbf{a}) = 0. \end{aligned} \tag{2.x}$$

Here C_k is the nonlinear part of the straightened constraint map; H, H_1 are the constrained linear minimizers; $\mathfrak{G}$ is the inverse on the simultaneous Landau and zero-average slice; and W_s is the nonlinear gradient remainder. The actual raw-coordinate force and complete carrier prescription are given in Prop. 3.171 and Lem. 3.172 and eq. 3.mccxv. Their first-order regular-pair domain, independent current radius, and parameter discs are part of that construction. The reduced current is the multiplier expression eq. 3.mccxviii, with the uniform bound eq. 3.mccxix. At one change of units its value is $L^{-3}\mathbf{J} + \mathsf{j}$.

The response has realness, gauge covariance, compatibility under restriction to wall-components, and transport by common lifts. A wall-component is a component of cubes sharing three-dimensional faces. When s vanishes on separating cubes, the equations split over the remaining components, and the response on a component with no datum is zero. In local first-order norms its size is at most $Ce^{C\bar\kappa}\sup|B'|$; a response viewed $k-j$ levels below has the additional potential factor $L^{-(k-j)}$. At the whole-sequence all-one point over a physical critical real reference, its shifted pair is the corresponding background and its own current, with the recorded node gauge factor, by Prop. 3.176. Existence, uniqueness and the stated properties are proved in Props. 3.174 and 3.175.

For a term F on X let X_0 be its hull in the cubes of the decoupling scale. The interpolation is

$$\Phi_Y[F](s,\nu) = F\big((\rho_X P) \dot{+} \nu\mathfrak{h}_Y(s;P,B')\big). \tag{2.xi}$$

Every occurrence of eq. 2.xi requires a domain-inclusion hypothesis: for all the indicated s and $|\nu| < \varrho_F$, its triple lies in the core of F. One sufficient quantitative hypothesis, with a fixed constant $c_F > 0$, is inclusion throughout

$$|\nu| < c_F L^{k-j} a_{0,j}\delta(\theta_k)^{-1}e^{-C_F\bar\kappa} =: \varrho_F > 1. \tag{2.xii}$$

In inner and removed substitutions separate inclusion bounds and their own radii are required. eq. 2.xii is not a consequence of the formal shift law in eq. 2.ix.

For disjoint finite cube sets $\mathcal{K},\mathcal{N}$ define

$$F[\mathcal{K}\mid\mathcal{N}] = \sum_{\mathcal{K}'\subset\mathcal{K}} (-1)^{|\mathcal{K}|-|\mathcal{K}'|}F(\mathbf{1}_{\mathcal{K}'\cup\mathcal{N}}). \tag{2.xiii}$$

This equals the integral of $\prod_{\Delta\in\mathcal{K}}\partial_{s(\Delta)}F(s+\mathbf{1}_{\mathcal{N}})$ over the unit cube, by iterating the one-variable fundamental theorem of calculus. Summing eq. 2.xiii over subsets gives its exact resummation identity. For the rooting assertion, use the wall-components of $\mathcal{K}\cup\mathcal{N}\cup X_0$. If such a component contains a cube of the differenced set $\mathcal{K}$ and misses the base X_0, then locality and splitting make the function independent of that component's parameters, so the alternating bracket is zero. An additional component made only of fixed $\mathcal{N}$ cubes need not make a bracket zero. For $\Phi_Y[F](s,\nu) - \Phi_Y[F](s,0)$ the expression vanishes if the component of the base has no datum for the shifted

response, since that response is then zero on the base. These conclusions require the stated splitting and support properties of the actual response. If $|\Phi_Y[F]| \le M_F$ on the stated polydisc, Schwarz's lemma gives $|\Phi_Y[F](s,1) - \Phi_Y[F](s,0)| \le 2M_F/\varrho_F$. Cauchy's formula on the s discs then bounds the bracket of this excess by $2M_F\varrho_F^{-1}(e^{\bar\kappa}-1)^{-|\mathcal{K}|}$. For $\bar\kappa \ge 1$ this is at most $2M_F\varrho_F^{-1}e^{-(\bar\kappa-1)|\mathcal{K}|}$. When $\mathcal{K}$ is nonempty, this also bounds the bracket of $\Phi_Y[F](s,1)$ itself: its value at $\nu = 0$ is the s-independent constant $F(\rho_X P)$, which the bracket annihilates. For an empty differenced set, an un-subtracted constant does not acquire ϱ_F^{-1}. Thus the algebraic bracket bound is proved once the response and inclusion hypotheses have been supplied.

2.2.3. *Pointed families, marginal coefficients, and list norms.* Let $\mathcal{D}_j$ be the connected unions of cubes of π_j. Set $d_j(X) = M^{-1}$ times the length of a shortest tree in X meeting each cube; it is zero for one cube. A pointed family is the collection $E(X, \mathbf{p}, z)$ for $X \in \mathcal{D}_j$ and unit sites $z \in X \cap \mathbb{T}^{(j)}$, with each term analytic on its core, gauge invariant and tied. Its norm is

$$\|E\|_{j,\kappa} = \sup_{\text{admissible tori and their companion}} \sup_{X, z\in X} e^{\kappa d_j(X)} \sup_{P\in\mathcal{U}_j^\circ(X)} |E(X, \mathbf{p}(P), z)|. \tag{2.xiv}$$

The supremum in z is not a sum over base points. For an unpointed family omit z from this formula. Sums over domains require the separate lattice-animal summability bound at the chosen decay rate.

A companion is a single family $\widehat{E}$ on $\mathbb{Z}^4$, agreeing with every torus term on every common lift of a non-wrapping domain with the necessary collar. For a fixed level, fixed frame prefix and fixed initial weight, the same companion serves every cutoff on which those input data coincide. Its existence is compatibility between tori, not a property that can be tested on just one torus. Individual terms are required to be covariant under the translations preserving π_j. The pointed total has the stronger property

$$\mathcal{P}_z^{\mathsf{T}}(W) = \sum_{X\ni z} E(X, \mathbf{p}_W, z),$$
$$\mathcal{P}_{rz}^{\mathsf{T}}(rW) = \mathcal{P}_z^{\mathsf{T}}(W) \tag{2.xv}$$

for every lattice isometry r, including unit translations. Here

$$\mathbf{p}_W = (U_j(W), J(U_j(W))).$$

The covariance class consists of the W for which these backgrounds exist, every restriction lies in the required core, and

$$L^{2j} \max_p |\partial U_j(W)(p) - \mathbf{1}| < c \min\{\varepsilon(\theta_{j-1}), \varepsilon(\theta_j)\}. \tag{2.xvi}$$

Both W and rW must be in this class. Invariance of the sum over z alone is weaker than eq. 2.xv.

The site tent is $h_z(x) = \prod_{\mu=1}^4 \max\{1 - |x_\mu - z_\mu|, 0\}$. Evaluate it at bond and plaquette midpoints. Then $\sum_z h_z(x) = 1$ and $h_{rz}(rx) = h_z(x)$. A bond-rooted term is pointed by $E(X, \cdot, z) = \sum_b h_z(x_b) E(X, b; \cdot)$; its collar must contain the supports of these tents. The pointed Wilson family has terms $A(h_z, \mathbf{U})$ on the cube of z enlarged by one layer, and zero elsewhere. This convention distributes a coupling counterterm by the same base points as the regular family.

To define the coefficient, use the chart $\mathsf{e}_X(A, J) = (e^{i\eta_j A}, J)$ at $(\mathbf{1}, 0)$. Directions are pairs of complex bond forms, with norm $\max\{\sup|A|, \sup|\nabla A|, \sup|J|\}$. Let H_∞ be the derivative of the infinite-lattice background map at its trivial datum, in its fixed gauge, and put

$$\mathfrak{h}_\infty B = (\mathsf{H}_\infty B, \mathrm{d}^*\mathrm{d}\mathsf{H}_\infty B).$$

This linear response is an input from the background theory; it must include its domain and decay estimates. Choose an orthonormal color basis t_1, t_2, t_3, and let δ_c^a be t_a on the unit bond c and zero elsewhere. With Hessians taken in the displayed chart,

$$\begin{aligned} \Pi^E(c,c';z) &= \tfrac{1}{3}\sum_{a=1}^{3}\sum_{\widehat{X}\ni z} \mathrm{Hess}\,\widehat{E}(\widehat{X},\cdot,z)[\mathfrak{h}_\infty\delta_c^a, \mathfrak{h}_\infty\delta_{c'}^a], \\ \Pi^E(c,c') &= \sum_z \Pi^E(c,c';z) =: \Pi^E_{\mu\nu}(c'_- - c_-), \\ b[\widehat{E}] &= \sum_{u\in\mathbb{Z}^4} \Pi^E_{\mu\nu}(u)u_\mu u_\nu, \qquad \mu\neq\nu. \end{aligned} \tag{2.xvii}$$

The coefficient is defined only when all these sums converge absolutely, the middle expression depends just on directions and displacement, and the last value is independent of the distinct directions chosen. Gauge independence, the Ward identities and the reduction of the second-moment tensor to this scalar require proof. They are not incorporated by declaring an arbitrary Hessian to have a coefficient. In particular, a claim that the Wilson tent family has coefficient one requires the linear-response identities with the normalization in eq. 2.i.

A member is a triple $(E, \widehat{E}, e)$ satisfying eq. 2.xiv–eq. 2.xvii, with $e = b[\widehat{E}]$. Activity bounds and coefficient bounds use their declared physical radii, including the inverse-square readout factor of § 4.23. A list at step $k+1$ contains such members for $j \le k$ and unpointed slot families $H^{(j)}$ with no coupling counterterm. Its action is

$$\begin{aligned} S[\mathfrak{f}](U) = &\sum_{j\le k}\left[\sum_z\sum_{X\ni z}\{E^{(j)}(X,U,z) - E^{(j)}(X,\mathbf{1},z)\} - e_j A(U)\right] \\ &+ \sum_{j,X}\{H^{(j)}(X,U) - H^{(j)}(X,\mathbf{1})\}. \end{aligned} \tag{2.xviii}$$

Every evaluation uses the background and units appropriate to its level. Slot families have companions, term covariance under the localization translations, and invariant totals on the stated background class, but no requirement of a pointed coefficient. For $0 < \alpha_{\mathrm{H}} < 1$ their list size is

$$\begin{aligned} N_{\mathfrak{a}}(\mathfrak{f}) &= \sum_{j\le k}(k+2-j)^{3/2} L^{-\alpha_{\mathrm{H}}(k-j)}\|E^{(j)}\|_{j,\kappa} + \sum_j \varrho_j^2 \|H^{(j)}\|_{j,\kappa}, \\ \varrho_j &= \max_\iota \frac{(1+\beta_{\mathrm{s}})a_{\iota,k+1}}{a_{\iota,j}}. \end{aligned} \tag{2.xix}$$

All three radii, including the independent-current radius, enter this maximum. Admissible radius systems require $CMa_{0,j} < a_{1,j}$, $a_{1,j}^2 \le Ca_{0,j}$, $a_{\iota,j} \le (1+\beta_{\mathrm{s}})a_{\iota,k+1} \le C(k+2-j)^{1/2}a_{\iota,j}$, and a fixed small upper bound on the absolute radii. They also require $\mathsf{s}_{\mathfrak{a}} := \delta(\theta_k)/a_{0,k+1} \le c$. The constants and the admissible systems are specified before taking eq. 2.xix. For the running system use $a_{\iota,j} = \check{\alpha}_{\iota,j} := \min\{\alpha_{\iota,j-1}, \alpha_{\iota,j}\}$ at $j \le k$ and $a_{\iota,k+1} = \alpha_{\iota,k}$. For the absolute system use fixed α_ι at $j \le k$ and $a_{\iota,k+1} = \alpha_\iota/(1+\beta_{\mathrm{s}})$.

2.2.4. *Frames, histories, structures, and stored variables.* A frame is a real sequence $\theta_0, \dots, \theta_k \ge \gamma^{-2}$, with $0 < \gamma < e^{-1/2}$, such that, for $i \le j$,

$$\underline{\lambda}(j-i) - c_5 \le \theta_i - \theta_j \le \bar{B}(j-i),$$
$$\check{R}_j = R(\min_{i\le j} \theta_i),$$
$$\check{R}_j \le L\check{R}_{j+1}, \tag{2.xx}$$

where $R(y)$ is the smallest power of L not below $(\log y)^r$. Set $\bar{g}_j = \theta_j^{-1/2}$, $p_0(g) = A_0(\log g^{-2})^{\mathfrak{p}_0}$, $p_1(g) = (\log g^{-2})^{\mathfrak{p}_1}$, $\varepsilon_j = \bar{g}_j p_0(\bar{g}_j)$, $\delta_j = (A_1/A_0)\varepsilon_j$ and $\alpha_{\iota,j} = C_\iota \theta_j^{-1/2}(\log\theta_j)^q$. Here $\mathfrak{p}_0 > \mathfrak{p}_1$, $A_1 < A_0$, and all constants are fixed. The waiting depth is $\mathsf{N}_j = \check{R}_j$. The barred couplings belong only to the auxiliary frame. The actual running couplings $x_j^{-1/2}$ are separate. The weights x_j are separate entries, with $x_0 = \beta$ and $x_j = x_{j-1} - b_j$. A tube requires

$$|\theta_j - x_j| \le \mathsf{t}_j,$$
$$\mathsf{t}_j \le c\theta_{j+1}(\log\theta_{j+1})^{-2q_{\mathrm{t}}},$$
$$q_{\mathrm{t}} \ge q,$$
$$2q_{\mathrm{t}} > q + \mathfrak{p}_0. \tag{2.xxi}$$

Its radii are non-increasing. At a terminal frame with no next member the radius must instead be supplied explicitly; for a one-member frame require $\mathsf{t}_0 \le 2c\theta_0(\log\theta_0)^{-2q_{\mathrm{t}}}$. All tests, regions and kernels are formed from θ, not from the unknown entries x_j.

A history is a finite record of every outcome of the tests of the steps and of the operations inside large-field components, including coordinate frames in the signed-permutation group W_4. The formation data assign to each record its regions, stored bonds and integration variables. In particular they give total tests $\chi, 1-\chi$ as Borel functions on the actual variable space, with each value in $\{0,1\}$. A plaquette test is $\mathbf{1}\{|\partial U_{j,\square} - \mathbf{1}| < \varepsilon_j\eta_j^2\}$; a fluctuation test is $\mathbf{1}\{|\mathcal{A}_j| < \delta_j\}$. The local background map, the neighborhoods of each test, and the collar and rim tests with their sharper thresholds are part of the formation data. Listing the names of tests does not define them.

The label is $\mathcal{S} = (\Omega_j, \Lambda_j, S_j)_{j\le k}$ with $T_\ell \supset \Omega_1 \supset \Lambda_1 \supset \cdots \supset \Lambda_k$; $S_j \subset \Omega_j \setminus \Lambda_j$ records strips and failed fluctuation regions. Write $Z_j = \Lambda_j^c$. The step creates unions of $M\check{R}_j$-cubes, with gaps at least $cM\check{R}_j$ in local units. Nests inside an operation may have gaps of only one M-cube of the later level. The formation data must state the actual enlargements and test neighborhoods; bounds described by a fixed number of layers leave that number as a required datum.

An operation at level t acts on the prescribed class of components which are small, quiet for N_t steps, and satisfy the waiting and quarantine conditions. It first integrates stored fields in N_t stages, then tests extension of the exterior field, integrates the inner box in an axial gauge with frame in W_4, restores block variables through normalized reference laws, localizes the exponent, and performs its last test. A healed component leaves the label; an island passes the earlier tests and fails the last; a drop-out fails after $n \le \mathsf{N}_t$ stages and keeps those stages. These alternatives and every operation kernel must be given as measurable maps and positive measures. Their exactness is a separate equality of represented measures.

An operation can use its waiting window only when $t_\flat = t - \mathsf{N}_t \ge 0$. Before then it is not eligible; no negative-level history is introduced. A drop-out C records its former regions $\Omega_l^\circ \cap C$ and its stage boxes $Z_l''(C)$. For each integer $0 \le \nu(C) \le n(C)$ define

$$\Omega_l^{\boldsymbol{\nu}} \cap C = \begin{cases} C \setminus Z_l''(C), & t_\flat < l \le t_\flat + \nu(C), \\ \Omega_l^\circ \cap C, & t_\flat + \nu(C) < l \le t_\flat + n(C). \end{cases} \tag{2.xxii}$$

Outside the affected levels retain the label regions. The choice $\boldsymbol{\nu} = n$ is the determining sequence; it governs backgrounds. The choice $\boldsymbol{\nu} = 0$ is the action sequence; it governs the sums and the coupling profiles. Standing marks record the analogous Λ_l, S_l data. Their waiting windows are required to be disjoint at a point, and old marks disappear when swallowed by a healed component or an island. These geometric facts require verification for the actual operation.

A structure $\sigma = (\mathcal{S}, \boldsymbol{\nu}; t, \phi)$ also records a phase. Its phase weight is 0 at output, 1 at the step input, and $1 - n/(2\mathsf{N}_t)$ after n stages. A structure precedes another if it counts no more completed stages and has lower level, or the same level with larger phase weight. At an index $0 \le m \le t$ the radius coefficient is

$$c_\sigma(x \mid m) = 1 - \beta_{\mathrm{s}}(1 - 2^{-(t-m)}) + \varpi_\phi \mathsf{g}_t(m) - \beta_{\mathrm{R}} \sum_{l \in \mathrm{Re}_\sigma(x)} 2^{-|m-l|},$$
$$\mathsf{g}_t(m) = \begin{cases} \theta_{\mathrm{s}}\beta_{\mathrm{s}} 2^{-(t-m)}, & m < t, \\ \beta_{\mathrm{s}}, & m = t. \end{cases} \tag{2.xxiii}$$

Here $0 < \theta_{\mathrm{s}} < 1$ and $\beta_{\mathrm{R}} = c_{\mathrm{R}}\beta_{\mathrm{s}}$ are fixed fractions; $\mathrm{Re}_\sigma(x)$ consists exactly of the levels of the first $\nu(C)$ stages of each drop-out containing x. The layer-radius formation data give the original regularity layer and the layer after each stage. Their nonnegative difference is $e_\sigma(x)$. The lift is $L_\flat^{2e_\sigma(x)}$ for a fixed $L_\flat \ge 1$. The layer-m radii are multiplied by this lift and, except for the real-potential gauge bounds, by eq. 2.xxiii. Unfinished structures here have $t \ge 1$: at layer zero use level-one radii; at the unfinished top use level-$t-1$ radii. The initial $t = 0$ base structure is separate input with its own specified radii. The data must ensure $c \le c_\sigma \le 1 + \beta_{\mathrm{s}}$.

The space $\widetilde{U}^c_\sigma(X)$ imposes the simple, tower and small-field-core clauses with those layer radii, with one splitting, on every subdomain and every preceding structure. The layer assignments, erosion rules, continuation maps and recorded remade levels are thus indispensable parts of its definition. The claims that restriction or a stage substitution maps one such space into another are analytic inclusion hypotheses. They require the specified radii and constants; a formal partial order by itself does not prove them.

2.2.4.1. Pointwise and local gauge clauses. For a structure with first member Ω_0, let $\mathsf{i}(Y)$ consist of the bare bonds with both endpoints in $Y \cap \Omega_0$. The pointwise part of the structured space imposes the real and complex curvature bounds on every plaquette formed by these bonds. The complex potential, its covariant differences on their retained stencils and the independent current have their layer bounds on the same carrier. In particular the pointwise bounds extend through $\Omega_0 \setminus \Omega_1$, with the layer-zero radii above.

In addition to the running gauge bounds, for each address (σ'', i, Z) of a preceding or current structure with $Z \subset Y$, the structured space requires good real gauges on a fixed family of cubes in two consecutive layers of that address. Each cube lies in $Z \cap \Omega_0$ for the corresponding first member. In using the strip response choose this fixed family to include all its gauge cubes of side at most $12\mathfrak{M}\xi_n$ at index n, including cubes contained in layer $n+1$. The gauge radius $a_\flat$ is below the response ceiling. The width and radius are fixed before the coupling ceiling, which is chosen for that family. In original bare units the good gauge is $U^v = e^{i\mathsf{v}}$ with $L^n|\mathsf{v}|, L^{2n}|\delta\mathsf{v}| \le a_\flat$. The same condition in level units uses $\xi_n|A|$ and $\xi_n^2|\nabla^\xi A|$, with $\mathsf{v} = \xi A$.

Use an address whose domain covers the full carrier of the response, then restrict its gauges and stencils to the active union of parameter cubes. When applying a substitution or taking a restriction, we must verify that the resulting triple has these clauses on that address. The layered bounds on $Y \cap \Omega_1$ alone give only that part of this verification. In particular an independent current at a layer-zero bond is unconstrained by those bounds, even if the configuration and the tower backgrounds are flat.

Stored fluctuation variables are real bond forms $\mathcal{A}_l^{\mathrm{s}}(b)$ belonging to the kernels outside the exponent. They transform by $\operatorname{Ad} u(b_-)$. Near bonds are the bonds in collars, rims and the nonanalytic cutoff zone; all other stored bonds are far. Fix $\lambda_l^{\mathrm{st}} = (\log \theta_l)^{6r+1} > 1$. A stored-field function is Borel in the near variables and is the value at $\mu_b = 1$ of a function jointly analytic in the pair, the far variables, and switches $|\mu_b| < \lambda_l^{\mathrm{st}}$. It is independent of a near variable when its switch is zero, and independent of that switch when the variable is zero. For far variables the analytic ball has radius λ_l^{st} times their specified real range. That range is part of the kernel data. The pair tie is required at every switch value.

2.2.5. *Presentations, standing operations, and their bounds.* A real presentation consists of the frame and formation data above, weights x_j, regular members $(E^{(j),\mathrm{u}}, E^{(j),\mathrm{m}}, \widehat{E}^{(j)}, b_j^{\mathrm{u}}, b_j^{\mathrm{m}})$, remainder families $(R^{(j)}, \widehat{R}^{(j)})$, boundary functions $B_{\mathcal{S}}(j, X, o)$, and the standing-operation exponents below. Here $E = E^{\mathrm{u}} + E^{\mathrm{m}}$, $b_j = b_j^{\mathrm{u}} + b_j^{\mathrm{m}} = x_{j-1} - x_j = b[\widehat{E}^{(j)}]$. Boundary origins o range over five types: the step, the last layer of a regular family, a step remainder beside an island, a stage, and exponentiation. A slot is the tuple (j, X, o) together with the structure-domain on which the function was formed. It retains that domain at later steps.

For a finite exempt family H of nonempty regions, define $d_{j,\mathsf{H}}^{\natural}(X)$ by minimizing M^{-1} times tree length also over subfamilies $\mathsf{F} \subset \mathsf{H}$ contained in X: the tree meets each cube outside $\bigcup \mathsf{F}$ and each member of F, but need not fill those members. For a label use the level-j hulls of its standing islands as H. Denote the result by $d_j^{\natural}(\mathcal{S}; X)$. It is at most $d_j(X)$ and asserts no decay across an exempt island. The real class requires

$$\begin{aligned} \|E^{(j)}\|_{j,\kappa} &\le E_0, \\ \|R^{(j)}\|_{j,\kappa} &\le \bar{g}_j^{\kappa_0}, \\ |B_{\mathcal{S}}(j, X, o)| &\le \mathsf{B} e^{-\kappa d_j^{\natural}(\mathcal{S}; X)}. \end{aligned} \tag{2.xxiv}$$

Boundary functions have the stored-field property. All entries are real on real arguments and invariant under the fixed simultaneous action on pairs and stored data. For regular families in eq. 2.xxiv is evaluated at $\check{\alpha}_{\iota,j}$; for the unmarked family also require a member of size at most E_0 on fixed absolute radii. The whole-lattice step is additional formed data: its positive tame kernel is $\mathbf{T}_{j-1}^{\mathrm{sf}}[\theta_{j-1}]$, with its non-pivot bond coordinates and sharp fluctuation box. For a list action S its step functional is

$$\begin{aligned} \mathcal{E}^{(j)}[\theta; x; S](W) = {} & \log \left(\mathbf{T}_{j-1}^{\mathrm{sf}}[\theta] e^{-xA(U_{j-1}) + S(U_{j-1})} \right)(W) \\ & + xA(U_j(W)) - S(U_j(W)). \end{aligned}$$

The kernel, the fibre coordinates, the normalization and the near-real logarithm branch must be given for this definition. The localization rule producing a pointed family from this functional is supplied once, for every torus and list; it is not determined by its total. The identification clause requires $E^{(j)}$ to be exactly this localized output from the earlier entries. Its unmarked version is the same output using only their unmarked entries.

The formed data also contain pointed Gaussian families $\mathbf{N}_j^{\circ}$ and $\mathbf{G}_{j,\tau}$ on those absolute cores, the first localizing the Gaussian logarithmic normalizer and its trivial-field local numbers, and the second localizing the change of its quadratic form when $x/\theta = \tau$. Their finite-dimensional covariance matrices, localization maps and logarithm branches are typed input data, so the following conditions have definite meanings. Set $b_j^0 = b[\mathbf{N}_j^{\circ}]$ and $\tau_{j-1} = x_{j-1}/\theta_{j-1}$. Require, for fixed constants $C_{\mathrm{m}}, C_{\mathrm{u}}, C_b$,

$$\|E^{(j),\mathrm{u}} - \mathbf{N}_j^\circ - \mathbf{G}_{j,\tau_{j-1}}\|_{j,\kappa} \le C_{\mathrm{u}} \bar{g}_{j-1} (\log \theta_{j-1})^{\mathfrak{p}_0},$$
$$\|E^{(j),\mathrm{m}}\|_{j,\kappa} \le C_{\mathrm{m}} \bar{g}_{j-1}^{\kappa_0 - C_{\mathrm{m}}},$$
$$|b_j^{\mathrm{m}}| \le C_b \bar{g}_{j-1}^{\kappa_0 - 4}, \qquad |b_j^{\mathrm{u}} - b_j^0| \le C_b \bar{g}_{j-1} (\log \theta_{j-1})^{\mathfrak{p}_0}.$$

No universality formula or asymptotic-freedom estimate for b_j^0 is implied by calling these families Gaussian. Their covariance construction and coefficient estimates are separate analytic input.

A standing operation at an island (t, Y) has a countable or finite local-history index ω, a Borel subset S_ω of a finite product of group variables, real bond variables and finite frame sets, and a positive Borel measure μ_ω on that set. These sets and measures are formed independently of the entries. Its exponents are Borel in $\xi \in S_\omega$, holomorphic in the pair, real at real arguments, and have the form

$$\mathbf{T}'_t(Y) F = \frac{1}{|W_4|} \sum_\omega \int_{S_\omega} F(\xi) e^{\Phi_\omega(\xi;\mathbf{p})} \mu_\omega(d\xi),$$
$$\Phi_\omega = \Phi_\omega^\circ(\xi) + \sum_f \Phi_{\omega,f}(\xi; \mathbf{p}|_{R_f}). \tag{2.xxv}$$

Each carrier $R_f \subset Y^\sim$ supports both dependencies of its factor. The finite factor index set, the measurable coordinate projections onto stored variables in each carrier, and any nested island variables are part of the data. No variable is integrated twice: swallowed-operation variables retain their earlier labels inside the product assigned to the swallowing component.

If (s, D) ranges over the forest of nested stored components, set $v(\omega) = \sum_{(s,D)} \log \theta_s \, \#_s(D)$, where $\#_s$ counts sites in their own level. For a fixed creation-time complex domain $\mathcal{K}_{t,Y}$ require

$$\frac{1}{|W_4|} \sum_\omega e^{c_{\mathrm{r}} v(\omega)} \int_{S_\omega}^* \sup_{\mathbf{p} \in \mathcal{K}_{t,Y}} |e^{\Phi_\omega(\xi;\mathbf{p})}| \mu_\omega(d\xi) \le C e^{-2\mathfrak{p}_0(\bar{g}_t)/(1+\beta_0)},$$
$$\sum_{f : R_f = R} \operatorname{osc} \Phi_{\omega,f} \le \mathsf{B}_\Phi (\log \theta_t)^{2\mathfrak{p}_0} e^{-\kappa d_t^\natural(\mathcal{S};R)}. \tag{2.xxvi}$$

The first integral is an upper integral, allowing a nonmeasurable supremum. The oscillation is the supremum of $|\Phi_{\omega,f}(\xi; p) - \Phi_{\omega,f}(\xi; p')|$ over ξ and both pairs in that fixed domain. The domain is supplied from the data at creation and must contain every later evaluation; this last inclusion is an independent requirement. $c_{\mathrm{r}} \ge 0$ is fixed before the coupling ceiling, and $\beta_0 > 0$ satisfies $\mathfrak{p}_0(\bar{g}_m) \ge (1+\beta_0)^{-1} \mathfrak{p}_0(\bar{g}_n)$ for $m < n$. The supremum remains inside the integral and sum. There is no asserted small factor in the size of Y. A drop-out has no new free entry of type eq. 2.xxv: it keeps its partially formed operation, and its carrying Wilson factor is included in its separate majorant. Such a majorant must be supplied to use a drop-out estimate.

The effective exponent additionally requires finite index maps and vacuum numbers, which are formed data. At label $\mathcal{S}$ the map $\mathcal{I}_j^E(\mathcal{S})$ is a subset of $\{(z, X) : z \in \Lambda_j \cap \mathbb{T}^{(j)}, \ X \in \mathcal{D}_j, z \in X\}$; it gives the actual enlarged-region rule for the regular terms. The enlargement width and treatment of boundary-crossing domains must be included in that map. The remainder index map is $\mathcal{I}_j^R(\mathcal{S}) = \{X \in \mathcal{D}_j : X \subset \Lambda_j^{\sim -1}\}$. The boundary index map selects the slots meeting the one-layer enlargement of Z_j. These maps use the action sequence in eq. 2.xxii. The profiles use the small-field regions of the action sequence. Fix a nonnegative smooth function ψ on $\mathbb{R}^4$, invariant under signed coordinate permutations, supported in $\{|u|_\infty \le 1/4\}$, and normalized by $\int \psi = 1$. For example,

normalize the product of the four functions $b(u_\nu)$, where

$$b(v) = \begin{cases} \exp[-1/(1-16v^2)], & |v| < 1/4, \\ 0, & |v| \geq 1/4. \end{cases}$$

At level j, put $\mathsf{s}_j = M\check{R}_j$ when Λ_j is a union of those cubes, and $\mathsf{s}_j = M$ otherwise. The latter region is a union of M-cubes. Measure distances and translations in level-j units and set $D_j(x) = \mathrm{dist}_\infty(x, \Lambda_j^c)$, with $D_j = +\infty$ when $\Lambda_j = T_\ell$. Define on the torus

$$\varphi_j(x) = \int_{\mathbb{R}^4} \psi(u) \mathbf{1}\{D_j(x - \mathsf{s}_j u) \geq \mathsf{s}_j/2\}\, du, \tag{2.xxvii}$$

and evaluate it at plaquette midpoints in the action. The geometry and this rule are held fixed as a source varies.

The distance function is one-Lipschitz when the complement is nonempty. Thus $0 \leq \varphi_j \leq 1$, it vanishes where $D_j \leq \mathsf{s}_j/4$, and it equals one where $D_j \geq 3\mathsf{s}_j/4$. Indeed, wherever $\psi(u) \neq 0$ one has $|u|_\infty < 1/4$, so the argument of D_j moves its value by strictly less than $\mathsf{s}_j/4$. In particular $\varphi_j = 1$ one layer inside Λ_j, with that layer taken in the s_j-cube partition. Its support is contained in $\{D_j \geq \mathsf{s}_j/4\}$, which is a compact subset of the interior of Λ_j. For the whole torus the formula gives $\varphi_j = 1$. On the periodic lift, changing variables in the convolution and differentiating its smooth kernel gives, for every multi-index,

$$\|\partial^\alpha \varphi_j\|_\infty \leq \mathsf{s}_j^{-|\alpha|} \|\partial^\alpha \psi\|_{L^1(\mathbb{R}^4)}.$$

This also proves smoothness, since the indicator is bounded and each kernel derivative has compact support. Cross-level ordering, where used below, is an additional condition. It follows, for example, if $\Lambda_{j+1} \subset \{D_j \geq 3\mathsf{s}_j/4\}$: then $\varphi_j = 1$ throughout the support of φ_{j+1}. With $x_k(p) = \beta - \sum_{j \leq k} b_j \varphi_j(p)$, the exponent is

$$\begin{aligned} A_{k,\mathcal{S}}(U) = & -A(x_k(\cdot), U) + \sum_{j,(z,X) \in \mathcal{I}_j^E(\mathcal{S})} \{E^{(j)}(X, U, z) - E^{(j)}(X, \mathbf{1}, z)\} \\ & - \sum_{j \leq k} b_j A(\varphi_j, U) + \sum_{j, X \in \mathcal{I}_j^R(\mathcal{S})} \{R^{(j)}(X, U) - R^{(j)}(X, \mathbf{1})\} \\ & + \sum_{(j,X,o) \text{ in boundary index map}} B_{\mathcal{S}}(j, X, o; U, \text{stored data}) - \mathcal{E}_{k,\mathcal{S}}. \end{aligned}$$

The real scalar $\mathcal{E}_{k,\mathcal{S}}$ is supplied with its vacuum bookkeeping rule, including the stage numbers retained by drop-outs. The pointwise identity $x_k(p) + \sum_{j \leq k} b_j \varphi_j(p) = \beta$ is exact. It does not authorize changing the index maps or the vacuum rule. These definitions specify the objects; their construction and uniform estimates remain hypotheses where no proof is given.

Erase every healed component together with its ancestral cluster from a history h; the result is its standing part $h^\flat$. Erasure being an admissible history with the same label, and the hard-core decomposition used to sum healed clusters, are structural requirements on the formation data. They are not assumed merely from the word "erase." The never-large standing part has $\Omega_j = \Lambda_j = T_\ell$. The presentation gives finite positive measures $\widehat{\tau}_{h^\flat}$ on V_k, stored fields and coordinate frames, with marginals $\tau_{h^\flat}$ on V_k. It also retains the individual-history measures $\widehat{\tau}_h$ produced by the exact kernels. Their identities are

$$\begin{aligned} \rho_k(V_k) dV_k &= \sum_{h^\flat} \tau_{h^\flat} = \sum_h \tau_h, \\ \sum_h \tau_h(\mathcal{V}_k) &= Z. \end{aligned} \tag{2.xxviii}$$

The first equality uses the summed healed operations; the exponent of an individual h need not have the effective-action form.

The reading at X records the standing regions, outcomes, marks, stored test bonds and frames within the hull of X enlarged by one level-l cell at every scale l. A chart is a translation by top-level cells, a common lift between tori, or their composite. Coherence requires a chart matching readings to carry the terms, carriers, Borel sets and measures to their counterparts. Factorwise,

$$\Phi_{\iota\omega,\iota f}(\iota\xi;\iota p)=\Phi_{\omega,f}(\xi;p). \tag{2.xxix}$$

Unit translations not preserving the partitions are covered by the pointed-total requirement eq. 2.xv, not by this chart definition. The simultaneous lattice symmetries preserving level-j cells must also transport the measures with their variables.

For a datum at a just below a pillar top y, let $\tau[U](a)=(\mathcal{M}^{m-1}U)(y,a)$ be its contour transport. An adapted background satisfies $\tau[U](a)=\mathbf{1}$ at every stored datum site. To define $\widehat{\tau}$ require that all backgrounds exist on its support, every argument belongs to its entry's domain, and the exponential has one value at all adapted representatives. The last clause is independent of simultaneous gauge invariance: an orbit transformation need not fix every datum site. These are part of class membership, not formal consequences of the norm bounds.

Finally let $\mathcal{Z}_k(h)$ be the live region after healing. For a live component X, define its lineage downwards by $\mathsf{Z}_k(X)=X$ and by taking at level j all live components contained in $\mathsf{Z}_{j+1}(X)$. Write $|C|_j$ for its number of level-j cells and $d'_j(W)$ for tree length in those cells. The created sets in a lineage are the components W of the step's creation set there; operation marks do not enter its deposit. With one fixed deposit constant $c>0$, set

$$\begin{aligned}
\mathrm{Vol}_k(X;h)&=\sum_{(j,C)\text{ in its lineage}} 2M^4\check{R}_j^5|C|_j,\\
\mathrm{Dep}_k(X;h)&=\sum_{(j,W)\text{ created in its lineage}} cp_0(\bar{g}_j)^2(d'_j(W)+1),\\
\Theta^{(k)}(h)&=\sum_{X\text{ live at }k}\{w\,\mathrm{Dep}_k(X;h)+c_a\,\mathrm{Vol}_k(X;h)\},\quad 0\le w<1,\quad c_a\ge 0.
\end{aligned} \tag{2.xxx}$$

The end mark is $\Theta^{(K)}$. Label marks and this numerical weight are different objects. Prepaid stability would assert $\sum_h e^{-E}\tau_h(\mathcal{V}_k)e^{\Theta^{(k)}(h)}\le e^{E_+\ell^4}$, with the stipulated normalization and dials; it requires a proof from creation deposits and the operation majorants. Nothing in eq. 2.xxx establishes that assertion.

Thus the class $\mathfrak{C}_k(\theta)$ is the set of these complete typed presentations satisfying eq. 2.xiv–eq. 2.xxx, all their stated structural and positivity clauses, and the specified analytic domains. A step-closure theorem must provide the next presentation, prove those clauses with the same constants and margins, and prove equality of the represented measures under its kernels. The background, response, substitution, island-mass, drop-out-majorant and closure hypotheses remain the substantive unresolved inputs where independent proofs have not been supplied.

2.3. **The weak-coupling sequence and continuum statements.**

2.3.1. *The object of the construction and the source of its difficulty.* We state the continuum and phase theorems. The coupling flow fixes the physical scale, and source derivatives define the connected correlations.

Objects. We take $G=\mathrm{SU}(2)$ and $N=2$. Put $T_\ell=(\mathbb{R}/\ell\mathbb{Z})^4$ and $\Lambda_{K,\ell}=a_K\mathbb{Z}^4/\ell\mathbb{Z}^4$, where $a_K=L^{-K}$. The compatible sides are $\mathfrak{L}=\{2L^m : m\in\mathbb{Z}_{\ge 0}\}$. At reference coupling g we use $\mathfrak{L}(g)=\{\ell\in\mathfrak{L}:\ell\ge\ell_\star(g)\}$; the floor in eq. 5.lii makes room for the localization cubes at every level. On such a torus we consider the following objects: the lattice $\Lambda_{K,\ell}$, whose spacing is $a_K=L^{-K}$;

the configurations U on it; the quantity $a_p(U) = 1 - \frac{1}{2}\operatorname{Re}\operatorname{tr} U_p$; Wilson's measure; the smeared curvature density; and the connected correlation functions of that density:

$$\mu_{K,\ell,\beta}(dU) = Z_{K,\ell,\beta}^{-1}\, e^{-\beta \sum_p a_p(U)}\, dU,$$

$$\mathcal{O}(f) = \sum_p f(x(p))\, a_p(U),$$

$$S^T_{n;K,\ell,\beta}(f_1, \dots, f_n) = \langle \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle_{K,\ell,\beta}.$$

In view of $a_p \approx \frac{1}{2} a_K^4 |F_U(p)|^2$, the sum $\mathcal{O}(f)$, which carries no prefactor involving the spacing, is a Riemann-sum approximation of $\frac{1}{2}\int f \sum_{\mu<\nu} |F_{\mu\nu}|^2$. It is therefore the $\operatorname{tr}(F^2)$ observable. The test functions are restricted to separated tuples $(f_i) \in \mathfrak{S}_n(\mathbb{E})$: tuples of real C_c^1 functions with pairwise disjoint supports. We use $\|f\|_\sharp = \max\{\|f\|_\infty, c_M \operatorname{Lip} f\}$, where $c_M > 0$ is the fixed multiple of the localization size M used in § 5.20. The reason is given under "The frame" in Thm. A.

The cutoff is tied to the coupling. The renormalisation group of §§ 3.1 and 4.17 assigns to $\beta = g_0^{-2}$ a sequence of running inverse couplings $x_0 = \beta$, $x_k = x_{k-1} - b_k$, whose increments satisfy $b_k \le B_\sharp$. These increments are independent of the torus § 5.2.2, and also of the cutoff $K' \ge k$ used to carry out the steps § 5.2.4. We fix a small reference coupling g. As the unit of length we take the scale at which the flow reaches, for the first time, the reference window $g^{-2} \le x \le g^{-2} + B_\sharp$ belonging to g:

$$K(\beta, g) = \min\{k : x_k(\beta) \le g^{-2} + B_\sharp\},$$

$$\mathcal{W}_K(g) = \{\beta \ge g^{-2} + B_\sharp : K(\beta, g) = K\};$$

a pair (K, β) such that $\beta \in \mathcal{W}_K(g)$ is called a first-passage datum.

The remaining pairs with $\beta \ge g^{-2} + B_\sharp$ are accounted for as follows. Suppose that $K \ge K' = K(\beta, g)$, put $b = K - K'$, and let $f^{[b]}(x) = f(L^{-b}x)$ on $T_{L^b\ell}$. Then

$$\langle \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle_{K,\ell,\beta} = \langle \mathcal{O}(f_1^{[b]}); \dots; \mathcal{O}(f_n^{[b]})\rangle_{K',\, L^b\ell,\, \beta}$$

holds exactly, for all bounded f_i. To see this, note that the dilation $x \mapsto L^b x$ maps $\Lambda_{K,\ell}$ onto $\Lambda_{K',L^b\ell}$, plaquettes to plaquettes and barycentres to barycentres. Wilson's weight depends only on the plaquette variables and on Haar measures, and no power of the spacing enters $\mathcal{O}$. The same identity holds between a periodic Gibbs state at β on $\mathbb{L}_K$ and its image on $\mathbb{L}_{K'}$ § 5.2.4. Consequently a pair with $K > K(\beta, g)$ amounts to the first-passage datum of β on a dilated torus, on which $\|f^{[b]}\|_\sharp \le \|f\|_\sharp$ and every distance is multiplied by L^b. The restriction to first passage therefore loses none of these lattice measures; its only effect is to fix the unit of length. (When $K < K(\beta, g)$ the identity is used in the opposite direction, on a contracted torus. This requires a lower bound on ℓ, and the norms grow by at most the factor $L^{K(\beta,g)-K}$ § 5.2.4.)

The final value $x_K(\beta)$ lies in the window but is not fixed at g^{-2}. This is why (a)(ii) passes to subsequences, and it is the purpose of clause (d). On each torus, provided the conditions of (d)(1) are satisfied, that clause dispenses with the subsequences as soon as the renormalised coupling, defined below, converges along the sequence. On $\mathbb{R}^4$, in (a)(iii), a subsequence is taken in every case, including the case in which the renormalised coupling converges (as stated in Thm. A(a)(iii)). The renormalised inverse coupling $x_K^\natural(\beta)$ is the final value of a particular run of K steps starting at β. In that run, the auxiliary choices of level k that depend on the coupling are made with $\theta_k = g^{-2} + c_0(K - k)$ instead of x_k, where $c_0 = \frac{11N^2}{12\pi^2}\log L$. Here and in all steps, log denotes the natural logarithm. A second kind of run is needed for the following reason. Continuity and holomorphy in the bare coupling are established by setting the constructions at two bare couplings

side by side and comparing them entry by entry, and by letting a single fixed linear operation act on a whole family of bare densities. Each of these two devices needs the choices to coincide at the two couplings. In a run made with its own couplings, however, the choices vary with β. Some of them, namely the cube sizes $\check{R}_k$, change through integer values § 5.2.3. Freezing the choices leaves $Z_{K,\ell,\beta}$ and S_n^T unchanged, since these are quantities of the measure. At first-passage data, and under the conditions of (d)(1), the shift that freezing produces in the couplings is at most g times a power of $\log g^{-2}$ § 5.2.3, § 5.2 and appendix A.8. By a continuum limit at g we mean a family $S = (S_n)_{n\geq 2}$ with

$$S_n(X) = \lim_{j\to\infty} S^T_{n;K_j,\ell_j,\beta_j}(X),$$
$$\beta_j \in \mathcal{W}_{K_j}(g),$$
$$K_j \to \infty,$$

taken either on T_ℓ (with all $\ell_j = \ell$) or on $\mathbb{R}^4$ (with $\ell_j \to \infty$). Limits on a torus along the tuned couplings of (a)(i) are also counted as continuum limits. A volume-first limit at g is a limit $\lim_j S_n^T[\nu_j]$, taken along first-passage data (K_j, β_j) with $K_j \to \infty$. In it each $\nu_j \in \mathcal{G}^{\mathrm{tor}}_{K_j,\beta_j}$ is a limit point of the periodically extended torus measures as $\ell \to \infty$, with spacing and coupling held fixed. Thus the volume is removed first and the spacing only afterwards.

The two theorems in brief. The exponents are fixed first. Then L is taken large, then the remaining constants are chosen in their order, and finally g is taken small. Thm. A states the following. The two assertions below hold uniformly in the first-passage datum, and the left-hand one holds also uniformly in the torus as soon as $\ell \geq Cn(R+1)$:

$$|S^T_{n;K,\ell,\beta}(f_1,\dots,f_n)| \leq C_n^\infty(\mathsf{d};R)\prod_i \|f_i\|_\sharp,$$
$$S_2(f,h) = \frac{3(N^2-1)}{\pi^4}\Big(\frac{12\pi^2}{11N^2}\Big)^2 \frac{\mathcal{K}_{\mathrm{free}}(f,h)}{(\log(1/\mathsf{d}))^2}\,(1+\eta) > 0.$$

In the left-hand bound, $n \geq 2$, and the supports lie in balls of radius R and are pairwise at distance $\geq \mathsf{d}$. In the right-hand formula, S is a limit along first-passage data, $f, h \geq 0$ are bumps whose size and separation are of the order of a small d, and $\mathcal{K}_{\mathrm{free}}(f,h) = \iint f(x)h(y)|x-y|^{-8}\,dx\,dy$. The factor $1+\eta$ is bounded above and below by positive constants, and on each torus $\eta \to 0$ as $\mathsf{d} \to 0$; on $\mathbb{R}^4$ the clause consists of the bounds by the two constants and nothing more. The five clauses of this theorem are the following:

- limits exist on the tori and on $\mathbb{R}^4$ (a);
- they inherit translation and hypercubic invariance, reflection positivity, symmetry, regularity and growth (b);
- they are non-trivial and have, at short distance, the normalisation of asymptotic freedom (c);
- on a torus they depend only on the renormalised coupling (d);
- on $\mathbb{R}^4$ they are rotation invariant if and only if a flux through far shells vanishes (e).

Thm. B states that a volume-first limit S is O(4)-invariant. Its centred functional $\sigma_S = \sum_{\pi\in\mathrm{Part}_{\geq 2}} \prod_{J\in\pi} S_{|J|}$ is precisely the system of moments obtained when the one-point function is set to zero, which is to say the moments of the centred density. This functional admits a canonical decomposition, which consists of a probability measure λ, carried by a compact set $\mathsf{X} = \operatorname{supp}\lambda$, together with a continuous injective map $\xi \mapsto \sigma_\xi$ such that

$$\int_{\mathsf{X}} \prod_{i=1}^{q} \sigma_\xi(X^i)\,\lambda(d\xi) = \lim_{|v_q|\to\infty} \cdots \lim_{|v_2|\to\infty} \sigma_S\big(X^1 \cdot \tau_{v_2} X^2 \cdots \tau_{v_q} X^q\big)$$

$$(q \geq 1),$$

so that

$$\sigma_S = \int_{\mathsf{X}} \sigma_\xi\,\lambda(d\xi).$$

Each ergodic component σ_ξ satisfies the full set of axioms of Osterwalder and Schrader.

2.3.1.1. The theorem, stated clause by clause. There exist a shape constant $\vartheta_0 \in (0,1]$ and, for each source radius $\varrho > 0$ (§ 5.20) and each shape $\vartheta \in (0,1]$, thresholds $0 < \gamma_{U'} \leq \gamma_U \leq \gamma_J$ and $0 < \gamma_W(\vartheta) \leq \gamma_J$ such that Thm. B and the clauses below hold. Throughout, $0 < g \leq \gamma_J$, $\ell \in \mathfrak{L}(g)$, and (K, β) is a first-passage datum. Apart from dependences that are displayed, the constants do not vary with K, β, ℓ or with positions. The ceilings $\gamma_{U'}, \gamma_U, \gamma_J$ depend only on the constants fixed before them, among which is ϱ, and $\gamma_W(\vartheta)$ depends only on these and on ϑ. Each constant, threshold and function introduced by a clause of the two theorems may in addition depend on g and on the quantities displayed with it.

The frame is adapted to the tools. When the side is $\ell = 2L^m$ with $m \in \mathbb{N}$, each axis carries an even number $2L^{m+K}$ of sites. Each mirror pair $\{y_\mu = c\} \cup \{y_\mu = c + \ell/2\}$ therefore cuts the torus into two halves, and the chessboard cells come in pairs §§ 5.20 and 5.22. Because L is odd, the mirrors that bisect bonds at successive cutoffs are nested: $a_K(\mathbb{Z} + \frac{1}{2}) \subset a_{K+1}(\mathbb{Z} + \frac{1}{2})$. Because barycentres are used, the identity $\mathcal{O}(f) \circ \mathsf{r}^* = \mathcal{O}(\mathsf{r} f)$ is exact for each lattice isometry r, where r^* denotes its action on configurations. If the support of f has distance greater than Ca_K from the mirror, every contributing plaquette and all its bonds stay on that same side. A barycentre alone does not guarantee this for a support touching the mirror; the continuum reflection-positivity passage uses fixed positive support gaps. In the Gaussian curvature model of Prop. 3.21, let $V_{\mathrm{lat}} = (\ell/a_K)^4$ be the number of sites. Its exact two-cochain space has dimension $3(V_{\mathrm{lat}} - 1)$ per colour: the kernel of the one-cochain coboundary consists of the $V_{\mathrm{lat}} - 1$ gradients and four harmonic one-cochains. Translation and hypercubic symmetry make the diagonal of its orthogonal projector constant on the $6V_{\mathrm{lat}}$ plaquettes. Thus the quadratic observable has expectation

$$\langle \mathcal{O}_{\mathrm{quad}}(f) \rangle_{\mathrm{G}} = \frac{N^2 - 1}{4\beta} \left(1 - \frac{1}{V_{\mathrm{lat}}}\right) \sum_p f(x(p)).$$

For fixed ℓ and smooth f, its leading Riemann-sum term is $\frac{3(N^2-1)}{2\beta} a_K^{-4} \int f$. Along the weak-coupling sequence $\beta \asymp K$, this term diverges when $\int f \neq 0$. This is a calculation in the Gaussian model; the interacting continuum construction uses the exact lattice expectation for centering. Moreover $|x - y|^{-8}$ is not locally integrable. For these reasons only $n \geq 2$ and only separated tuples are considered, and positivity is stated for the centred σ_S. The axioms of Osterwalder and Schrader require nothing more, because their test functions vanish to infinite order wherever points coincide. By a mirror Σ of $\mathbb{R}^4$ we mean a hyperplane $\{u \cdot x = c\}$ with either $u = e_\mu$, the unit vector of the μ-th coordinate (coordinate), or $u = (e_\mu \pm e_\nu)/\sqrt{2}$ for some $1 \leq \mu < \nu \leq 4$ (diagonal). Its reflection is θ_Σ and its open side is $\mathbb{E}_+(\Sigma)$. A torus has only the coordinate pairs as mirrors, and there the open side is a slab.

Theorem A (Continuum correlations)**.** Assume the uniform background, interaction, history, flow and ordinary-source estimates of § 2.4, with their stated domains and normalization. For the short-distance clauses assume the interacting comparison and running-normalization estimates there; for the matched-cutoff and Ward clauses assume their additional estimates. Then the following conclusions hold. **(a) Existence.** Each run started from $\beta \geq g^{-2} + B_\sharp$ reaches first passage. For

such a run, $c(\beta - g^{-2}) - C \le K(\beta, g) \le C(\beta - g^{-2}) + C$, and since $b_k \le B_\sharp$ also $g^{-2} < x_K(\beta) \le g^{-2} + B_\sharp$. Hence $K(\beta, g) \to \infty$ as $\beta \to \infty$, and so $\mathcal{W}_K(g) \neq \varnothing$ for infinitely many K. The values $x_k(\beta)$ are the same whatever torus the steps are carried out on, and $\beta/\log(1/a_K) \to \frac{11N^2}{12\pi^2}$ uniformly on $\mathcal{W}_K(g)$. (i) Let $g \le \gamma_U$. Then for each K there is a tuned $\beta_K^\natural$, that is, one with $x_K^\natural(\beta_K^\natural) = g^{-2}$, and it satisfies $\frac{1}{2}c_0 K - c_5 \le \beta_K^\natural - g^{-2} \le \frac{3}{2}c_0 K + c_5$. Moreover $S^T_{n;K,\ell,\beta_K^\natural} \to S_n^{\natural,\ell}$ as $K \to \infty$ along the whole sequence. This convergence holds on each torus of the family and on tuples of Lipschitz functions whose supports are pairwise disjoint. (ii) On a fixed torus, from every sequence of first-passage data with $K_j \to \infty$ one can extract a subsequence that converges on $\mathfrak{S}(T_\ell)$. (iii) The same holds on $\mathfrak{S}(\mathbb{R}^4)$ when in addition $\ell_j \to \infty$. The limits obtained in this way along tori, with defining sequence (K_j, ℓ_j, β_j), are what the phrase "a limit of (a)(iii)" means. The same also holds for $S_n^T[\nu_j]$ with $\nu_j \in \mathcal{G}^{\mathrm{tor}}_{K_j,\beta_j}$, so volume-first limits exist.

(b) Symmetry, positivity and growth. Each continuum limit has the following properties. (i) It is invariant under W_4 and under the translations of $\mathbb{E}$. (ii) It is reflection positive at each coordinate mirror: $\sum_{i,j} \bar{c}_i c_j \, \sigma_S(\theta_\Sigma X_i \cdot X_j) \ge 0$ whenever $X_i \in \mathfrak{S}(\mathbb{E}_+(\Sigma))$. (iii) It is symmetric. (iv) It is regular: there is a unique distribution of locally finite order on $\mathbb{E}^n_{\neq}$ of which S_n is the restriction. On C^1 functions, and with the same constants, it satisfies (c)(1) on T_ℓ, and it satisfies both bounds (c)(2) when it is a limit of (a)(ii)–(iii). (v) On $\mathbb{R}^4$ it satisfies the growth condition

$$|\mathsf{S}_n^{\sigma_S}(\varphi)| \le c_{\mathrm{E}} C_{\mathrm{E}}^n (n!)^{\varkappa} |\varphi|_{k_{\mathrm{E}} n}$$

on $\mathcal{S}_{\neq}(\mathbb{R}^{4n})$. Take the form $\langle F, G\rangle = \sigma_S(\theta \bar{F} \cdot G)$ on $\mathbb{C}\mathfrak{S}(\{x_4 > 0\})$. The separable space $\mathcal{H}(S)$ then carries three structures:

- $\mathcal{T}_t[F] = [\tau_{te_4} F]$, $\mathcal{T}_t = e^{-tH}$, which form a strongly continuous semigroup of self-adjoint contractions, where $H \ge 0$ and $H\Omega = 0$;
- a strongly continuous unitary representation $e^{iv\cdot P}$ of $\mathbb{R}^3$ that leaves Ω fixed, the P_j commuting strongly with H;
- a unitary representation of W_3 that leaves Ω fixed and commutes with H.

Already for $g \le \gamma_J$, each volume-first limit is a limit of (a)(iii) and is, in addition, reflection positive at each diagonal mirror.

(c) Correlation bounds and nontriviality. Let $n \ge 2$, let the f_i be Lipschitz with supports pairwise at distance $\ge \mathsf{d} > 0$, and in (2) let the supports also lie in balls of radius R. Then

$$(1)\ |S_n^T| \le C_n(\mathsf{d})\, e^{n c_V \ell^4} \prod_i \|f_i\|_\sharp;$$

$$(2)\ |S_n^T| \le C_n^\infty(\mathsf{d}; R) \prod_i \|f_i\|_\sharp \ (\ell \ge Cn(R+1)),$$

$$|S_n^T| \le A(\mathsf{d}; R) B(R)^n (n!)^{\varkappa_\star} \prod_i \|f_i\|_\sharp \ (\ell \ge \ell_G);$$

Here $C_n^\infty(\mathsf{d}; R)$ does not increase as d grows. The bounds (2) also hold, with no condition on a volume, for $S_n^T[\nu]$, where ν is an arbitrary periodic Gibbs state on $\mathbb{L}_K$ of Wilson's weight at β, and where (K, β) is a first-passage datum.

(3) The class $\mathfrak{T}_{\mathbb{E}}(\mathsf{d}; \vartheta)$ consists of the pairs of Lipschitz functions $f, h \ge 0$, neither of which is identically zero, that satisfy the following conditions. Each is supported in a ball of radius d, and $\mathsf{d} \le \mathrm{dist}(\mathrm{supp}\, f, \mathrm{supp}\, h) \le 6\mathsf{d}$. In addition, $\int f \ge \vartheta \|f\|_\infty \mathsf{d}^4$ and $\mathrm{Lip}\, f \le \|f\|_\infty/(\vartheta \mathsf{d})$, and the same two conditions hold for h. Assume $g \le \gamma_W(\vartheta)$, $\mathsf{d} \le \mathsf{d}_0$ and $(f, h) \in \mathfrak{T}_{T_\ell}(\mathsf{d}; \vartheta)$. Assume also $K - s(\mathsf{d}) \ge n_0$, so that at least n_0 steps lie between the scale d and the cutoff. Then

$$\langle \mathcal{O}(f); \mathcal{O}(h) \rangle_{K,\ell,\beta} = \zeta_K(\mathsf{d})^2 \, \tfrac{3(N^2-1)}{\pi^4} \, g(\mathsf{d})^4 \, \mathcal{K}_{\mathrm{free}}(f,h) \, (1+\varepsilon),$$
$$|\varepsilon| \le \min\{\tfrac{1}{2}, \bar{\varepsilon}_\ell(\mathsf{d})\}.$$

In this formula $g(\mathsf{d}) = g_{K-s(\mathsf{d})}(\beta)$, where $L^{-s(\mathsf{d})} \approx \mathsf{d}$ up to logarithmic factors. Also $\underline{x}_s \le g(\mathsf{d})^{-2} \le \bar{x}_s$, and both bounds are affine in $s = s(\mathsf{d})$. The factor $\zeta_K(\mathsf{d}) = \zeta'_{K-s(\mathsf{d})}$ is the running normalisation, that is, it measures how $x_{K-s(\mathsf{d})}$ responds when β is shifted by a constant §5.20.1.2. One has $|\zeta_K(\mathsf{d}) - 1| \le \bar{\eta}(\mathsf{d})$ and $|\frac{11N^2}{12\pi^2} g(\mathsf{d})^2 \log \frac{1}{\mathsf{d}} - 1| \le \bar{\eta}(\mathsf{d})$. Here $\mathsf{d}_0 \in (0,1)$, and the two functions $\bar{\eta} : (0, \mathsf{d}_0] \to [0, \frac{1}{3}]$ and $\bar{\varepsilon}_\ell : (0, \mathsf{d}_0] \to [0, \frac{1}{2}]$ are non-decreasing; moreover $\bar{\eta}, \bar{\varepsilon}_\ell \to 0$ as $\mathsf{d} \to 0$. Hence, on each torus, $\varepsilon \to 0$ uniformly with respect to the cutoff, to the bare coupling as it ranges over its window, to the pair, and to where the pair is placed. For periodic Gibbs states the identity remains true, but only with $|\varepsilon| \le \frac{1}{2}$.

(4) Consequently the limits of (a)(ii)–(iii) obey the law stated at the opening of the text, for those pairs in $\mathfrak{T}_{\mathbb{E}}(\mathsf{d}; \vartheta)$ in which f, h are of class C^1; the tuned $S^{\natural,\ell}$ are not included. The error satisfies $(1-\bar{\varepsilon})(1-\bar{\eta})^4 \le 1 + \eta \le (1+\bar{\varepsilon})(1+\bar{\eta})^4$, with $\bar{\varepsilon} = \bar{\varepsilon}_\ell(\mathsf{d})$ on T_ℓ and $\bar{\varepsilon} = \frac{1}{2}$ on $\mathbb{R}^4$, where in addition $\dim \mathcal{H}(S) \ge 2$.

For clause (5), assume in addition the comparison eq. 6.ccxxxiii for the actual normalized history/block-field law, with $x = g(\mathsf{d})^{-2}$ and $\zeta = \zeta_K(\mathsf{d})$. On the whole prescribed four-test class, its remainder must satisfy

$$|\mathcal{R}_4(\underline{f})| \le \epsilon_{4,\ell}(\mathsf{d}) \, g(\mathsf{d})^8 \prod_i \|f_i\|_\infty, \qquad \epsilon_{4,\ell}(\mathsf{d}) \longrightarrow 0 \quad (\mathsf{d} \downarrow 0),$$

for each fixed admissible torus, uniformly in the permitted cutoff, bare coupling and support positions. The comparison retains the actual constraint, gauge-normalization and closed-mode identifications accompanying eq. 6.ccxxix, the coarse-complement estimate eq. 6.ccxxx, and the weighted lineage-age estimate of Prop. 5.71 at a rate strictly greater than 16, or a stronger direct estimate for these four insertions. All the nonquadratic, source, boundary and excluded-history terms listed after eq. 6.ccxxxiii belong in $\mathcal{R}_4$. This additional input has the torus scope of clause (5).

(5) The class $\mathfrak{T}^{(4)}_{\mathbb{E}}(\mathsf{d}; \vartheta)$ consists of the quadruples whose pairs all belong to $\mathfrak{T}_{\mathbb{E}}(\mathsf{d}; \vartheta)$ and for which $\operatorname{supp} f_i \subset B(c_i, \mathsf{d}/16)$, the four centres c_i lying on one line. On the tori there exists $\vartheta_\square \in (0,1]$ with the following property. Let $\vartheta \le \vartheta_\square$ and $g \le \gamma_W(\vartheta)$. Then there are constants $0 < c_\square \le C_\square$, which may depend on g and ϑ, such that the bound below holds whenever $K - s(\mathsf{d}) \ge n_4$, where $n_4 \ge n_0$ is an integer, d is smaller than a threshold, which may depend on ℓ, and the quadruple belongs to $\mathfrak{T}^{(4)}_{T_\ell}(\mathsf{d}; \vartheta)$:

$$c_\square (N^2-1) \, g(\mathsf{d})^8 \prod_i \|f_i\|_\infty \le S^T_{4;K,\ell,\beta}(f_1, \dots, f_4) \le C_\square (N^2-1) \, g(\mathsf{d})^8 \prod_i \|f_i\|_\infty :$$

on these quadruples the connected four-point function is therefore strictly positive. On rescaled quadruples it decreases as the fourth power of the reciprocal of the logarithm of the scale.

(d) Fixed-torus uniqueness. Assume $g \le \gamma_U$. Take $(\beta^\natural_K)$ and $S^{\natural,\ell}$ as in (a)(i). (1) Let ϱ be large and $g \le \gamma_{U'}$. Then the first-passage windows lie close to the tuned couplings: $\mathcal{W}_K(g) \subset [\beta^\natural_K - \frac{1}{4}\varrho, \ \beta^\natural_K + \frac{1}{4}\varrho]$ for each K. (2) For $|\beta - \beta^\natural_K| \le \frac{1}{4}\varrho$ the reference runs of length K are well defined, and $\beta \mapsto x^\natural_K(\beta)$ is bi-Lipschitz uniformly in K. As $K \to \infty$, the functions $t \mapsto x^\natural_K(\beta^\natural_K - t)$ converge uniformly on $|t| \le \frac{1}{4}\varrho$. The limit function has a compact interval I as its range, and this interval contains $[g^{-2} - c'\varrho, g^{-2} + c'\varrho]$. (3) For each torus of the family, one has a family $(S^{(\ell,x)})_{x \in I}$. Each of its members is a family of connected functions, defined on tuples of Lipschitz functions whose supports are pairwise disjoint. It satisfies $S^{(\ell,g^{-2})} = S^{\natural,\ell}$, and $x \mapsto S^{(\ell,x)}_n$ is continuous on

I. Moreover, whenever $K_j \to \infty$, $|\beta_j - \beta^\natural_{K_j}| \le \frac{1}{4}\varrho$ and $x^\natural_{K_j}(\beta_j) \to x$, it follows that $x \in I$ and $S^T_{n;K_j,\ell,\beta_j} \to S^{(\ell,x)}_n$, with no passage to a subsequence.

Consequently, take two sequences of bare data close to the tuned couplings whose renormalised couplings converge to the same value. They have the same limit on T_ℓ. Under the conditions of (1) this applies to every sequence of first-passage data, and every limit of (a)(ii) is equal to some $S^{(\ell,x)}$. On each torus, the renormalised coupling by itself determines the limit theory.

(e) Rotations. Let $\omega \in \mathfrak{so}(4)$, and let χ be a cutoff whose support lies in $(-\ell/2 + 2a_K, \ell/2 - 2a_K)^4$. Consider the rotational Ward identity satisfied by $\mu_{K,\ell,\beta}$ with respect to the deformation field $\chi(y)\,\omega y$. The shell flux is the boundary term of this identity:

$$\mathfrak{B}_{\omega,\chi} = \beta \sum_p a_p\,(\nabla^{\mathrm{s}}_\omega \chi)(x(p)) - \beta \sum_p \mathfrak{h}_{\omega,\chi}(p) - \tfrac{3}{2} \sum_p a_p \sum_{\rho \in \{\mu,\nu\}} (\omega x(p))_\rho\,(\nabla^{\mathrm{c}}_\rho \chi)(x(p)),$$

where $\mu < \nu$ are the two directions spanning p and $\nabla^{\mathrm{s}}_\omega, \nabla^{\mathrm{c}}_\rho$ are lattice differences, and where

$$\mathfrak{h}_{\omega,\chi}(p) = a_K^{-1} \sum_{b \subset \partial p} \sum_\rho [\chi(x_b) - \chi(x(p))]\,(\omega x_b)_\rho\,\mathfrak{d}_{b,p,\rho}$$

, the $\mathfrak{d}_{b,p,\rho}$ being bounded, gauge-invariant local densities. Each term contains a difference of χ. For $\mathfrak{B}_{\omega,\mathcal{R}} := \mathfrak{B}_{\omega,\chi_\mathcal{R}}$, only terms whose plaquette anchors lie in $\{\mathcal{R} - a \le |x(p)| \le 2\mathcal{R} + a\}$ can contribute, where $a = a_K$. The local densities also read neighboring plaquettes. Their full field dependence lies in the enlarged shell $\{\mathcal{R} - 3a \le |x| \le 2\mathcal{R} + 3a\}$, as in eq. 6.cccxciii. These shells surround the transition region $\mathcal{R} \le |y| \le 2\mathcal{R}$, where $\chi_\mathcal{R}$ falls from 1 to 0. A limit of (a)(iii) is O(4)-invariant if and only if the following holds for one $\omega \neq 0$ (equivalently, for all ω), for all $n \ge 2$ and for all separated tuples with smooth entries:

$$\liminf_{\mathcal{R}\to\infty} \limsup_{j\to\infty} \left|\langle \mathfrak{B}_{\omega,\mathcal{R}}; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle_{K_j,\ell_j,\beta_j}\right| = 0.$$

In that case $\mathcal{H}(S)$ carries a strongly continuous unitary representation of O(3). This representation leaves Ω fixed, commutes with H, and extends the representation of W_3. This equivalence singles out the one infrared input, and for volume-first limits Thm. B settles it. ■□

What the equivalence uses. When $\chi = 1$ near the supports, the identity takes the following form (the exact lattice Ward hypothesis in eq. A.lxxxi):

$$\sum_{i=1}^n \langle \mathcal{O}(f_1); \dots; \mathcal{O}(\nabla^{\mathrm{s}}_\omega f_i); \dots; \mathcal{O}(f_n)\rangle_{K,\ell,\beta} = \mathfrak{U}_{\omega,\chi}(f_1, \dots, f_n)$$
$$+ \langle \mathfrak{B}_{\omega,\chi}; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle_{K,\ell,\beta}\,:$$

besides the flux, it contains a bulk term $\mathfrak{U}_{\omega,\chi}$. This term is the correlation between the $\mathcal{O}(f_i)$ and lattice artefacts of dimension six, distributed over $\operatorname{supp}\chi$. Once $\mathfrak{U}_{\omega,\chi_\mathcal{R}} \to 0$ along every defining sequence, the rotation defect of S, namely the derivative $\frac{d}{dt} S_n(e^{t\omega} f_1, \dots, e^{t\omega} f_n)|_{t=0}$, equals minus the limit of the correlation between the flux and the $\mathcal{O}(f_i)$, and clause (e) follows.

This vanishing requires the typed first-order insertion estimates, the joint-insertion moment bound, and the volume-transfer estimate in the Ward hypotheses. The first-order estimates are needed on fixed tori and slowly growing volumes; the joint-insertion bound and volume transfer are needed for arbitrary sequences with $\ell_j \to \infty$. On this route the anisotropy estimate also requires the exponent stated next.

One marginal operator remains in $\mathfrak{U}$: a traceless diagonal anisotropy of Wilson's weight. Its coefficient is of the size $g_0^{-1} \approx K^{1/2}$, and it disappears only through its own decay K^{-c_1} along the flow. What is needed is $c_1 > \frac{1}{2}$.

This number is expressed through one-loop data. Put the weight $1 + \mathsf{e}_{\pi(p)}$ on each a_p, where $\pi(p)$ is the coordinate plane of p. Differentiating the one-loop coefficient of step l plane by plane at $\mathsf{e} = 0$ gives a derivative that commutes with W_4. Let $\mathsf{m}(l)$ denote the eigenvalue of this derivative on the traceless diagonal weights $\mathsf{e}_{\{\mu,\nu\}} = \mathsf{e}_\mu + \mathsf{e}_\nu$, $\sum_\mu \mathsf{e}_\mu = 0$. Then $c_1 \in (\frac{1}{2}, 1]$ is an exponent such that $\sum_{i<l\le j} \mathsf{m}(l) \ge c_1 c_0 (j-i) - C$. The one-loop law in this channel, $|\sum_{l\le k} \mathsf{m}(l) - c_0 k| \le C$, gives $c_1 = 1$.

The law comes from a telescoping sum. By the law of total covariance, the sum of the one-loop responses of the steps $l \le k$ is a single trace on the bare lattice. The logarithmic part of this trace is $\int_1^{L^{2k}} d\sigma/\sigma = 2k \log L$, in proper time σ, multiplied by the heat coefficient on a constant abelian background. It equals kc_0, with the same c_0 as the coupling itself. In the continuum, an anisotropy of the weight amounts, to first order, to a change of coordinates, and it does not run (see the Ward comparison hypotheses). Rotation invariance in Thm. B (1) follows from Prop. 6.26, whose proof uses no Ward identity and does not depend on c_1.

Theorem B (Canonical components)**.** Assume the background, interaction, history and flow inputs of § 2.4, the ordinary-source hypothesis D1 in its stated rooted-domain geometry, and the interacting two-point and running-normalization inputs F2–F3 there. The latter inputs are required uniformly over the periodic Gibbs states used to realize the limit and its components, on the full test class specified in Prop. 6.25, including $\vartheta_* = 1/12000$. Let $g \le \min\{\gamma_W(\vartheta), \gamma_W(\vartheta_0)\}$, and let S be a volume-first limit with defining sequence (K_j, β_j, ν_j). (1) Every ν_j is a periodic Gibbs state and is invariant under all translations of the lattice; moreover S is a limit of (a)(iii). In addition to (b) and (c)(4), it satisfies $H \ge |P_j|$ for $j = 1, 2, 3$, is O(4)-invariant, and is reflection positive at the diagonal mirrors. (2) Its canonical decomposition exists and is unique up to isomorphism. Here an isomorphism is a homeomorphism of the spaces X that carries λ and the σ_ξ along with it. Let P_0 be the projection onto $\ker H$. The map $P_0[F] \mapsto (\xi \mapsto \sigma_\xi(F))$ extends to a unitary map $\ker H \to L^2(\mathsf{X}, \lambda)$ with $\Omega \mapsto 1$, by the zero-energy representation theorem of § 7.8. If X has exactly $m < \infty$ points, then $\dim \ker H = m$; if X is infinite, then $\ker H$ has countably infinite Hilbert dimension. This does not identify the set-theoretic cardinality of an infinite compact phase space with its Hilbert dimension. The components are permuted, and λ is preserved, by the isometries that leave S invariant (translations, W_4, rotations). (3) Each σ_ξ has the following properties. (i) It has the same invariances and positivity properties, and it is symmetric and regular. (ii) Its one-point function is $S_1^{\sigma_\xi}(f) = \mathfrak{c}_\xi \int f$, where $\mathfrak{c}_\xi$ is real and continuous in ξ and $\int \mathfrak{c}_\xi \, \lambda(d\xi) = 0$. With the same constants, it satisfies the growth condition and the bounds (c)(2); it also satisfies the law (c)(4) in its form on $\mathbb{R}^4$. (iii) On $\mathcal{H}(\sigma_\xi)$ it has the operators of (b)(v) and a unique vacuum, $\ker H_{\sigma_\xi} = \mathbb{C}\Omega_{\sigma_\xi}$. Its joint spectrum lies in the closed forward cone $\{E \ge |q|\}$. It also has a strongly continuous unitary representation of O(3) that leaves Ω_{σ_ξ} fixed and commutes with H_{σ_ξ}. (iv) It has the cluster property:

$$|\sigma_\xi(X \cdot \tau_v Y) - \sigma_\xi(X)\,\sigma_\xi(Y)| \le C(X,Y)\,|v|^{-2}$$
$$(|v| \ge r(X,Y)),$$

where $C(X,Y)$ and $r(X,Y)$ are the same for all ξ and for all volume-first limits at g. (4) Consequently every Osterwalder–Schrader axiom holds for σ_ξ: growth, invariance under $\mathbb{R}^4 \rtimes \mathrm{O}(4)$, reflection positivity, symmetry and clustering. Its Schwinger functions coincide with the Schwinger functions of a Wightman theory. This theory has a single Hermitian scalar field on a separable Hilbert space and a unique vacuum. It is covariant under the proper orthochronous Poincaré group, its spectrum lies in the closed forward cone, and it is local. ■□

2.4. **Technical lemmas and proof dependencies.**

2.4.1. *Lemmas and dependencies of the proof outline.* This catalogue gives the technical statements and dependencies of the author's proof outline. It fixes the full domains, quantifiers and uniform constants of each lemma invoked in the argument. The accompanying sections expand some proofs; other technical statements are quoted here without their detailed proofs. A reference to one of these statements always retains its complete hypotheses and only the conclusions stated below. The catalogue identifies where the principal multiscale estimates enter. Those estimates require proofs beyond the finite-dimensional and reference calculations given here.

Different routes use different groups of hypotheses. On an existing sequence of arbitrarily fine admissible data, the scalar route uses D1 with its actual rooted geometry, G1, and the ultraviolet inputs F2–F3. The bridge sections derive compactness, D2, H1, compact Hilbert evolution and G2 from those data. The joint induction outlined in A–C and the flow section is responsible for constructing these data with the stated uniformity. Weighted and matched-run comparisons and typed Ward estimates remain requirements for the stronger statements that invoke them; the scalar route requires only its ordinary D1 and ultraviolet premises, together with the consequences derived from them.

Here are conventions shared by the packages. For numerical coefficients written in terms of N, the group is $\mathrm{SU}(N)$ and here $N = 2$. Fix an odd integer $L > 1$, a compact simple matrix group G, and dimension four. The principal example is $G = \mathrm{SU}(2)$. The lattice spacing is $a_K = L^{-K}$, and a finite torus has physical side ℓ. There is one group variable U_b per positively oriented bond, $U_{b^{-1}} = U_b^{-1}$, and normalized product Haar measure dU. Write

$$a_p(U) = 1 - \tfrac{1}{2}\,\mathrm{Re}\,\mathrm{tr}\,U_p,$$
$$A(w,U) = \sum_p w(x(p))a_p(U),$$
$$Z_{K,\ell}(w) = \int e^{-\beta A(1,U)+A(w,U)}dU.$$

The trace convention in this display is for SU(2); another group requires its own normalization. Let $\mathcal{O}_K(f) = A(f,U)$. The connected functions are the mixed derivatives at zero of $\log Z_{K,\ell}(\sum_i z_i f_i)$. Test functions used together have pairwise disjoint compact supports, separation at least $d > 0$, and individual supports in balls of radius R. Set $\|f\|_\sharp = \max(\|f\|_\infty, c_M\|\nabla f\|_\infty)$, with c_M fixed. All uniform constants below must be independent of K, the torus, the history, the positions of the tests, and the relevant Gibbs state. Dependence on g, n, d, R must be displayed. A lower bound on admissible ℓ may depend on g and on constants fixed earlier. Replacing this uniformity by a bound at one fixed cutoff does not satisfy an input.

A. Background fields and propagators: four inputs. **Technical statement A1: the constrained critical orbit.** For every allowed nested family of small-field regions and every regular coarse datum V, there must exist a regular critical orbit $[U(V)]$ for the Wilson action with the prescribed block averages. It is unique among regular critical orbits in a specified larger neighborhood. A chosen gauge gives a representative which is Borel in all real data and holomorphic in a complex neighborhood of each regular real datum. Both neighborhoods have stated radii on the local scale; their radii and constants are uniform in the number of scales. Existence of an unrestricted global minimizer is insufficient: the needed assertion concerns the constrained regular orbit and its gauge representative. Compatibility with restriction to subregions, with seam constraints, and with changing the torus must be part of the statement. This is the background-orbit input behind the local assertions numbered Prop. 3.25, Prop. 3.27 and the background theorems. For the pointed window deduction, the actual restrictions and substitutions must remain in their prescribed pair cores. These cores contain the usable Taylor chart ball and obey the curvature and own tent-action

bounds on the whole admitted core, including the finitely many young ages. § 3.3 proves a local chart for the actual one-step average near a pure-gauge double block, including the pivot derivative, exact straightening and measure factors. Its explicit chart radius and nonwrapping premise do not imply existence, uniformity or all-substitution room for the many-level critical orbit.

2.4.1.1. The regular proper background. §§ 3.7 and 3.8 realize the first physical blocking step at fixed L, with arbitrary admitted real flat holonomies. For the separated determining sets eq. 3.cc-cvi, which retain interior and outward rows and omit inward rows, Prop. 3.124 proves existence and uniqueness among all regular critical orbits. Its proof uses the actual proper strong inverse, forced trial correction, datum descent, and a reset of curvature and current after every ascent. Lem. 3.122 places arbitrary regular competitors in one actual Landau chart; Lem. 3.121 obtains the required strong norm from their actual currents. Cor. 3.125 then supplies a fixed physical holomorphic continuation ball at every such real datum, uniformly in depth and volume. The exact comb construction in § 3.11 selects its Borel real representative. The larger history input A1 also specifies its actual restriction, substitution and core requirements; those clauses retain their separate hypotheses and proofs.

Technical statement A2: coercivity on a fluctuation fiber. Let Q be the derivative of the chosen block average, and let $\mathcal{E}$ be the gauge-fixed Hessian on the fluctuation variables at a regular background. The chosen coordinates must remove all residual gauge directions. On each conditioning set and with its specified boundary conditions require

$$\mathrm{Re}\langle B, \mathcal{E}B\rangle \geq c\|B\|_2^2,$$
$$QB = 0,$$

with the norm in the units of that fluctuation scale. The same estimate, with $c/2$, must hold throughout the claimed complex neighborhoods. Zero modes, torus holonomies, and node constraints must either be absent from this space or treated separately. A mass for this constrained operator is not a mass gap for the physical continuum Hamiltonian. This is the quantitative content required by the conditioning and covariance assertions numbered 1c.6 and related claims. For the actual unweighted classical flat reference, §§ 3.4 and 3.5 prove $c(L)I \leq \mathsf{A}_k^{\mathrm{flat}} \leq CL^8 I$, uniformly in the number of previous classical blocking steps and in compatible tori, including real flat holonomies. They use the actual plain constraint and the soft internal-contour gauge. The curl-energy comparison is $\|\mathrm{d}a\|^2 \leq E_k(a) \leq 256\|\mathrm{d}a\|^2$. These are the flat classical estimates used in the nonflat comparisons below; their statements retain that reference-case scope.

2.4.1.2. The physical first step and a specified current extension. The nonflat first-step precision theorem supplies positivity for the actual Wilson action with its soft gauge term on the curved constraint fibre, at an admitted real block-axial trial background. Its radius and current ceilings are retained. § 3.10 then defines one full relative auxiliary-current extension, identifies its physical action at a critical physical pair, and proves the complete first-step half-disc estimate with the gauge part included. This chosen definition must be used consistently when invoking that result. Neither zero coarse readout nor a Ward identity identifies a separately unspecified completion. For the canonical proper precision eq. 5.cclxxviii, Prop. 5.45 proves uniform real coercivity and holomorphic invertibility on a common weighted C^1 complex background tube and independent-current ball. The estimate includes the outer second-jet correction and relative gauge, is uniform in the proper inner hierarchy and conditioning set, and also covers the stated strict tree-zero subspace with the gauge term omitted. The local inverse, logarithm and normalized Gaussian families are constructed in § 5.13. Invoking these results requires the actual precision, readouts and conditioning coordinates to be those specified there, with their stated regular-pair ceilings. Signed profiles, counterterm-modified precisions and interacting Hessians require their own comparisons on the joint history cores.

2.4.1.3. Layered site and projected-form reductions. § 3.13 proves the site intertwining, site inverse/Green/Gram and projected physical form/readout reductions within its separated geometry and regular-pair class, with field-row/Feynman and depth-uniform density-coefficient comparisons. Verifying that every actual background and history substitution lies in this class remains necessary. The fine Landau coercivity and its complex moving-slice continuation are proved in Props. 3.70 and 3.77; the specified current completion is identified in Prop. 3.90. The named propagator and decoupled families are supplied by Props. 3.112 and 3.171 and Lem. 3.172. Each result retains its own geometry, complete-read and regular-pair hypotheses. The six-propagator normalizations are recorded in § 3.12.

Technical statement A3: local inverse and response bounds. The inverses and response operators actually used, including conditional covariances and derivatives of $U(V)$, must obey their declared pointwise and source-norm bounds with exponential decay in scale units. The shorthand $Ce^{-c\,\mathrm{dist}(x,y)}$ applies after the local source and output normalizations have been specified. In § 3.12 the kernels use counting measure: the fine Green source estimates retain the factors $[\xi_y^2, \xi_y, \xi_y, 1]$, the inverse site Gram entries retain ξ_y^{-4}, and the inverse coarse field-row Schur entries retain ξ_c^{-2}. The stated short-distance, first-difference and Hölder factors must also be kept. For a fine-to-fine Green map, its density against physical fine volume is η^{-4} times its counting kernel; a counting-entry bound is not the same density bound. Coarse row operators instead use their own row weights and rescalings, as in eq. 3.miv. Weighted cell bounds in norms such as eq. 3.mclxxxiii are operator estimates, not pointwise density estimates. All required bounds must hold throughout the full complex datum polydisc, not only at the identity or at real data. For the pointed window this is the product bond-log domain eq. 4.xxv, with the scale-weighted potential, gradient and current response in eq. 4.xxvi–eq. 4.xxvii. For a region with several local scales, use a precisely defined weighted path distance and the corresponding powers of the local spacings. Holomorphic derivatives in each decoupling parameter must satisfy Cauchy bounds on a fixed enlarged complex disc. A derivative joining distant sets must retain the decay needed by the polymer norm. The mesh-uniform scalar estimate and its Dirichlet compressions are proved in the block-resolvent section. On the admitted separated regular-pair domains, Props. 3.171, 3.174 and 3.175 and Lem. 3.172 prove the non-Abelian decoupled response, its screened field and current bounds, holomorphy, and wall decomposition. The all-one physical identification and coupled response are proved in Props. 3.176 and 3.177; Cor. 3.179 identifies the prescribed response maps. This supplies the assertion numbered 1c.10 on those domains. Each application must still verify that its actual history data, complete determining reads and substitution lie in the stated domains with the required radius margins. § 3.6 proves exponential kernels for the actual unweighted classical flat precision and its constrained covariance, uniformly in depth and compatible volume. It derives the exact alias symbol, cancels its apparent gauge denominator on the full complex zero cone, proves a uniform strip, and then inverts using the flat gap. Periodization and unitary flat holonomies preserve the bounds. This establishes the stated flat reference kernels, not the nonlinear response or decoupling bounds required on the full interacting domains. The finite-box affine comparison in eq. 4.xxxvii prescribes the affine test field's exact averages on the same determining set, including the seam rows. The weighted first response controls the effect of discarding its remote rows.

2.4.1.4. Primitive inversion versus nonlinear response. § 3.14 gives the exact coupled field–multiplier inverse, with the true datum differential, weighted transposes and constrained Hessian. It proves the response and complex-ball deductions from stated two-sided primitive inverse bounds. Boundary forcing is kept in its own norm. § 3.15 proves mesh-uniform local scalar estimates and their flat-unitary covariant version. Applying them to a nonlinear response requires the actual connection, forcing and covariant Laplacian point bounds recorded there. The nonlinear application uses those primitive equations and bounds for the actual response on every allowed core. Once they hold, the Hölder conclusion follows from the displayed argument; no further Hölder lemma is invoked.

Technical statement A4: averaging schemes and locality. Any transfer from one averaging scheme to another must provide a uniform intertwining map, including its coarse gauge factor. On fluctuation fibers the transferred Hessians, inverses and constraints must obey A1–A3 with the same allowed losses, for every number of blocking steps. Restrictions to a region must depend only on an explicitly bounded enlargement of that region. Equality of two linearized averages only at the identity is insufficient. A proof must bound the accumulation of their differences at nonflat backgrounds and must state the gauge in which the comparison is made.

2.4.1.5. Exact restriction with its boundary price. The finite factorization in § 3.11 shows how a new constrained variation fixes the old data and why restriction preserves criticality. Cutting the plaquette action can change its boundary current. The changed bonds are in the new bare boundary layer, where the local scale is η; the at-most-six-plaquette estimate pays a fixed current constant. That constant and the datum/seam losses are included in the actual uniqueness-input ceiling. Under this admission and regular critical uniqueness, the restriction identity is a deduction. More general changes of averaging scheme still need the stated physical and complex-domain intertwining estimates; the restriction proof alone does not supply such a comparison.

B. Cluster expansion and power counting: four requirements. **Requirement B1: a conditional polymer majorant.** Let polymers be finite connected sets of scale cubes, with incompatibility meaning intersecting supports. A specified positive size $a(X)$ and activities $z(X)$ must satisfy, uniformly in all external data,

$$\sum_{Y:Y\not\sim X} |z(Y)|e^{a(Y)} \le a(X).$$

The same inequality, with strict room, must hold for complex sources and decoupling parameters on the domains used for differentiation. Require the absolutely convergent connected-cluster formula for the logarithm, including termwise differentiation and a uniform majorant for each derivative used later. The finite-volume identity must be exact before taking limits. This is the content needed from the convergence statement numbered 1d.4.1; a formal graph expansion is not sufficient. The unmarked polymer part of § 4.5 derives the local unmarked activity and union-filed logarithm bounds from its actual enlarged Gaussian matrix/core hypotheses, sharp-cutoff structure and local potential budgets. The Mayer-parameter discs are established at the Gaussian integral, before zero-freeness is inferred. Gaussian volume costs, failed bonds, actual neighbor-layer closures and the spare potential slot are all paid for in the rate allocation. This also supplies eq. 4.xciii under those primitive hypotheses. Their verification for the actual complex interacting matrices and all allowed cores remains necessary; convergence of an unrelated model gas would not supply it.

Requirement B2: conditional pointed marginal extraction. §§ 4.2 and 4.3 prove the following implication for an actual regular member with no stored color spectators. Its pointed total and differentiated sums must converge normally; it is locally gauge invariant, and the individual member's pointed totals are covariant under unit translations and the full hypercubic group. Use the actual free response, with gauge covariance and affine constant-curvature reproduction modulo gauge, with zero linear current. The usable pair core, localized response and entropy gap must give the absolute Hessian moments in eq. 4.v. Then eq. 4.xix defines its coefficient and eq. 4.xxiv proves cancellation by that same member's own midpoint tent action. Different members may be chosen at different sites; their coefficients cannot be interchanged.

On physical radius a, the readout retains $|b(E)| + \|T[E]\| \le Ca^{-2}\|E\|$. § 4.23 supplies its stated near-zero inclusion and normalization. For the nonlinear remainder also require A1–A3 throughout the complex datum domain, the actual core inclusions, spatial Hölder regularity, compatible boundary equations and the whole-core bound on the own tent action. Put $\sigma = L^{-(n-j)}$,

$\rho = \max(1, a_{\mathrm{out}}/a)$ and $0 < \alpha_H \le 1$. Under the window domain conditions the near-domain sum at one base point, after its vacuum and own counterterm subtraction, has bound

$$C\|E\| \left[\rho^2\sigma^{4+\alpha_H} + \rho^3\sigma^6 + (1+\rho^2\sigma^4)e^{-c/\sigma}\right].$$

The final exponential term includes remote domains without a nonwrapping chart. No factor proportional to the actual curvature is asserted for that tail. Under the ordinary age bound $\rho^2 \le C(1+n-j)$, the finite young-age core fallback and the $O(\sigma^{-4})$ site count give the summable weight $C\|E\|(1+n-j)^{3/2}L^{-\alpha_H(n-j)}$.

These are conditional deductions, not verification that all actual RG formed members have the required hypotheses. They do not establish arbitrary torus-sensitive or field-sensitive tails, source derivatives of the formation map, or the fluctuation disintegration laws in F2. Stronger assertions in 1e.1.13, 1e.1.15 and 1e.1.16 require separate estimates.

Technical statement B3: the one-loop law with a summable lattice correction and controlled nonlinear errors. For the chosen normalization let $c_0 = 11N^2 \log L/(12\pi^2)$. The Gaussian contribution b_j^0 must be independent of the bare coupling and obey $|b_j^0 - c_0| \le CL^{-\sigma j}$, or a stated summable substitute. The full increment must have a decomposition

$$b_j = b_j^0 + f_j + q_j,$$
$$|f_j| \le C x_{j-1}^{-1/2} (\log x_{j-1})^p,$$
$$|q_j| \le C x_{j-1}^{-(\kappa_0-4)/2},$$
$$\kappa_0 > 4.$$

If a later deduction needs a stronger error, that stronger estimate must be proved separately. The coefficient c_0 requires an actual lattice trace calculation with the stated zero-mode convention. A continuum perturbative coefficient quoted without this comparison does not prove this input.

Technical statement B4: the ordinary step and its finer derivatives. The ordinary source box allows a separate Euclidean regular member at each base point, its own scalar counterterm, a variable Wilson weight, and scale-small noncovariant defects. The step must be holomorphic on this product box with a majorant independent of the number of labels. It is not necessary that the whole spatial array be translation invariant. The ordinary regular-output estimate does not by itself prove every marked or large-field derivative.

For the stronger comparisons actually invoked, require the potential derivative bound in eq. 4.dxciii and the one-zone complex-mass bound in eq. 4.dxciv, on the domains stated in that section, with constants independent of the levels and source positions. The regular irrelevance weight, squared-radius weights for defects, and large-field survival weights have different roles. In particular the survival factor belongs in the bound on a zone's output; it does not enlarge its input radius by its reciprocal. Joint marked derivatives and typed tensor counter-insertions must satisfy their corresponding estimates. For the stronger routes these finer conditions are stated technical lemmas; entire dependence of the bare partition function alone does not imply them. They are required for the retained stronger comparison and Ward routes. The ordinary-source insertion and angular proofs below do not obtain their operator estimates by assuming these typed Ward derivatives. § 4.20 proves ordinary old-level summations, scalar action-load and reserve implications. The source-insertion and renewed-pointing parts of § 4.5 derive the local first response under their local Gaussian and bond-split hypotheses; its unmarked polymer part derives the needed majorant from the stronger primitive matrix/core and potential package. Its window-formation part derives the ordinary formation bound using one near window member at

each old base point before its bracket is applied. The linear bootstrap retains the source norm and gradient gain under these inputs, the constant-source slopes and the old-remainder bounds. Actual nonlinear background/core inclusions and history formation remain to be verified; these implications do not supply them.

C. Large-field histories and closed induction: four inputs. **Technical statement C1: exact measurable operations.** Every move must be a specified Borel kernel, change of variables, or partition of unity on an explicitly defined space. The tests are total, including at configurations where logarithms or minimizers fail to exist. Every stored variable is integrated exactly once or retained exactly once. For every bounded gauge-invariant insertion F, the sourced construction must preserve the total integral of $Fe^{-\beta A}$. An operation preserving only the integral without F cannot be used to transport that observable. For real inputs every summand used as a probability weight must be nonnegative.

Technical statement C2: geometric margins and determination. The allowed histories must have a constructive definition. At level j, tests, local backgrounds and retained terms depend on at most a stated enlargement of the determining regions. Every enlargement lies inside the margin supplied by the next region. Healing, restriction, joining and continuation must preserve these properties, including near holes, seams, and waiting windows. On an admissible torus each cell, each box on which the operation acts, and its prescribed bounded neighborhood admit a common lift. Use the companion identities on non-wrapping domains in matching step units. Other carriers enter the torus estimates with their winding retained. The number of histories with a specified size must have a proved entropy bound, with the same size function used in the rarity estimate.

Technical statement C3: rarity with entropy and insertions paid for. For each created large-field component X at level j, require a mass bound $\mathfrak{M}_j(X) \le \exp[-cp_0(g_j) - \kappa|X|_j]$ with the stated supremum over external complex data inside its internal integral when component majorants are multiplied. Here $p_0(g) = (\log g^{-2})^p$, and κ exceeds the entropy rate of C2. The bound must include the Wilson factor actually available to pay for that component, all stored Jacobians and gauge factors, and the designated insertion weights. A failure of existence of a background, as opposed to a quantitative large-field inequality, must also be covered. Drop-outs without their own Wilson factor need a separate bound after attachment to their carrying term.

Technical statement C4: invariance of the inductive class. There must be one nonempty class $\mathfrak{C}_j$ of complete presentations, with the same constants at every level, containing the bare measure. Under the increment and tube conditions, the exact step maps $\mathfrak{C}_j$ into $\mathfrak{C}_{j+1}$. Its conclusions include the analytic polymer bounds, C1–C3, boundary bounds, covariance, and the required stability floor. All constants must admit one simultaneous order of choice, without a parameter needing to be both smaller and larger than quantities chosen later. The step assertion 2.11 requires all these properties; each must be proved.

D. Sources and local correlation bounds: four requirements. **Technical statement D1: the ordinary localized source bound and stability.** Require the precise torus hypotheses in eq. 5.cdxxxv–eq. 5.cdxxxvi of § 5.22: holomorphic point counterterms and localized source energies, a strict entropy gap, source dependence only on the specified regions, and independence of the representation of a source function by a list of tests. On the ordinary source ball, require the normalized upper stability bound and unsourced partition-function floor there, with constants uniform in cutoff and allowed torus. The source class must include the global mirror-periodized Lipschitz profiles used in the chessboard construction. Fixed-cutoff bounds for point counterterms may depend on the cutoff but must be uniform in the torus.

For the graded deduction, this input also specifies the actual domain indexing of § 5.23. Take $M = L^{m_0}$ with fixed integer $m_0 \ge 0$, and the cofinal compatible tori $\ell = 2L^m$, $m \ge m_0$. At scale $a_s = L^{-s}$ the localization cubes have side Ma_s; their counts in every direction are even integers. Every fixed power-of-L chessboard tiles all sufficiently large members of this family. All actual

scales $0 \le s \le K$ are retained. A domain X is indexed by its union of these cubes, with $m(X)$ the minimum number of edges of a finite connected cubical edge tree meeting every cube. Its source dependence is contained in a fixed number of neighboring cube layers. The source-energy bound uses $e^{-\kappa m(X)}$, with κ above the rooted entropy rate for this actual indexing. If extra domain types are present, their multiplicity must be included in that stated entropy gap.

The closed-walk count in eq. 5.cdxlvi, the ball-root count in eq. 5.cdxlvii, and the global weighted count in eq. 5.cdxlviii are then proved geometric consequences. In particular, the sharp local count is $C(1+u/a_s)^4 e^{-ct}$, and the global count is $C\ell^4 a_s^{-4}$. The weaker abstract estimate in eq. 5.cdxxxvi alone does not imply these sharper bounds and is insufficient for the graded conclusion below. Checking that the localized RG energies use this specified geometry is part of D1, not supplied by writing an abstract domain sum.

These are raw partition-function bounds. A zero-free logarithm on the whole global source ball is not assumed. The fixed-volume normalized zero-free lemma gives a suitable neighborhood when needed; the all-state transfer theorem instead constructs common holomorphic cell normalizers and obtains the derivatives at zero directly. The elementary transfer does not prove the ordinary RG source estimates in this input. § 5.21 proves explicit sufficient criteria for the upper bound and floor from the normalized positive label/history representation: total-variation bounds with a weighted label sum, and a regular all-small-field label with its own action bound. Those criteria, including the reserve and allocation of Wilson factors, remain to be verified for the construction. A pointwise polymer logarithm cannot replace that history representation without an exact common integral formula.

2.4.1.6. The actual history norm and the remaining local checks. § 4.18 proves the weighted genealogy count, reserve arithmetic, gauge-normalizer half-disc bound, strong-mass induction and loaded label sum from their explicit finite-operation hypotheses. Its norm integrates the supremum over all admitted reads; a bound at each fixed read is insufficient. To apply the result one must identify the literal stored-coordinate operation and its fixed reference measure, admit every later read to the same core, prove the complete quadratic domination and localized energy envelope, and construct owned witnesses for the failed tests. The original inner Wilson weight is retained. The source extension has two proved routes: an actual pointwise excess bound, or an ordinary kernel holomorphic and bounded in its Banach norm on a complex source ball over a fixed reference measure, together with a real estimate with a sufficiently large prepaid mass-weight coefficient on a full real source ball and one common actual field/read core for both estimates. Both routes retain their specified formation and domain premises. The lower partition floor also needs its admitted positive regular label and common normalization; an upper history-mass estimate does not prove that floor.

Requirement D2: graded all-state bounds derived from D1 and G1. Prop. 5.76 uses D1 in its actual rooted setting above and the proved chessboard domination from G1. It yields one fixed finite integer k and $B(R) = C(R+1)^4$ such that, for periodic Gibbs states ρ, ν at the same datum,

$$\left| \rho \prod_{i=1}^{n} [\mathcal{O}_K(f_i) - \nu(\mathcal{O}_K(f_i))] \right| \le B(R)^n (n!)^k \max\{1, d^{-5n}\} \prod_i \|f_i\|_\sharp .$$

For $n \ge 2$ the cumulants under ρ satisfy the same estimate with $[2B(R)]^n$ in place of $B(R)^n$. Each support lies in its own radius-R ball, placed anywhere, and different supports have distance at least d. The first-order case uses the one-point difference bound. All constants are uniform in cutoff, states and support positions.

The proof gives a polynomial cell-normalizer loss $Q_i \le Cn^p$ and $k = \lceil p \rceil + 5$. It uses a genuine product source polydisc from separate chessboard factors and common univariate normalizers. The separation singularity is explicitly d^{-5n}, not a single d^{-5} prefactor independent of n. D2 is therefore a conditional deduction of the ordinary analytic input D1 with its specified geometry and G1; it is

not an additional graded-source hypothesis. D1's analytic estimates and actual RG realization are not proved by this deduction.

Technical statement D3: age-resolved uniform rarity. The probabilities or positive weights of live regions persisting from a given level to a later level must have a summable bound uniform in volume and in the location of a test. Witness selection, replication by reflections, and pattern counting must preserve the insertion and source weights. State the resulting tail function $r(m)$ with $\sum_m r(m) < \infty$, together with every additional moment in m used later. A rarity estimate for one newly created component is insufficient for this persistence assertion. This input represents the age and survival arguments wherever actual history persistence or weighted comparisons are used. The ordinary graded all-state estimate D2 is obtained by its GS route above and does not require this separate history-persistence hypothesis.

For deductions from a fixed-bump Riemann sum, the cutoff is required to be sufficiently fine.

Requirement D4: one-point transfer derived from D1. Prop. 5.75 proves, under D1, that periodic Gibbs states ν, ρ at the same datum satisfy

$$|\nu(\mathcal{O}_K(f)) - \rho(\mathcal{O}_K(f))| \le C_1(R)\|f\|_\sharp.$$

If both states are invariant under all plaquette isometries, the bound implies $|\nu(a_p) - \rho(a_p)| \le Ca_K^4$ by testing a fixed nonnegative bump of nonzero integral. The uniform bound on differences follows from a common holomorphic normalizer; neither individual uncentered mean is required to remain bounded. This is essential when a pure component is centered with respect to a mixture.
E. Flow, tuning and comparison of cutoffs: four inputs. **Technical statement E1: a consistent flow.** The recursion $x_j = \beta - \sum_{i\le j} b_i$ must be defined up to first passage, with $b_i \le B_\sharp$ and $\lambda(j-i) - C \le x_i - x_j \le B_\sharp(j-i)$, $\lambda > 0$. Its increments must be independent of the admissible torus and of the unused finer cutoff. These are hypotheses for the elementary first-passage argument; they cannot be deduced from a recursion whose next step is itself undefined outside the invariant tube specified by the flow lemmas. Together with B3 they imply the leading logarithmic relation between β and the cutoff.

Technical statement E2: a frozen calibration and tuning map. For auxiliary choices fixed independently of β, the terminal map $\beta \mapsto x_K^\natural(\beta)$ must be holomorphic on a stated complex neighborhood of a real interval. On that interval require $0 < c \le \partial_\beta x_K^\natural \le C$, and endpoint values bracketing the prescribed renormalized inverse coupling. All intermediate values must remain in the allowed tubes. This supplies existence and local uniqueness of a tuned bare coupling by elementary analysis. The holomorphy assertion is not valid merely because the original finite-lattice partition function is entire. The flow section proves the slope implication both from the stronger independent coupling polydiscs and from the weaker marked bare-disc route. In the latter, the marked forcing is of size $C\epsilon_i'/\varrho$, not $C\epsilon_i'/x_{i-1}$, and is retained in the discrete Gronwall estimate. Its summability needs the stated exponent restriction. Canonical tuning uses the admitted real interval; broader presentation gluing additionally needs an admitted real overlap. Overlapping geometric complex discs alone do not establish that admission.

Technical statement E3: a summable matched-cutoff comparison. At matched terminal coupling, require for each fixed torus and separated tuple

$$|S^T_{n;K+1}(X) - S^T_{n;K}(X)| \le C(n,d,g,\ell)\epsilon_K \prod_i \|f_i\|_\sharp,$$

$$\sum_K \epsilon_K < \infty.$$

The floor in K may depend on the fixed torus. To compare two nearby terminal couplings, add a uniform Lipschitz term proportional to their difference. Neither assertion implies infinite-volume uniqueness. This is the actual analytic input behind the matched-run claims numbered 8a and the tuned Cauchy sequence; a bounded difference at each cutoff does not suffice.

Technical statement E4: compatibility of calibrations. When moving between a self-calibrated first-passage run and a frozen run, the maps must represent the same bare integral and satisfy a stated uniform bound on their coupling discrepancy. That discrepancy must be small enough for every comparison domain and source radius used above. Exact dilation identifies the same finite lattice in two units, but supplies no estimate on this calibration discrepancy. Uniqueness statements must state which terminal coupling and which calibration they use.

F. Short-distance normalization and nontriviality: four requirements. **Requirement F1: the proved free-kernel comparison.** The exact continuum coefficient is proved in appendix A.3. For the remaining lattice comparison, let $a = \ell/n$, with integer n, and use plain coordinate inner products on site, bond and positively oriented plaquette cochains of the periodic lattice. Let d_1 be its unscaled circulation matrix and P_{ex} the orthogonal projection onto $\operatorname{ran}\mathrm{d}_1$. A standard Gaussian on that range has covariance P_{ex}. Its physical curvature kernel is the 6×6 matrix of components

$$K_{a,\ell;P,Q}(p,q) = a^{-4} P_{\mathrm{ex}}(p_P, q_Q),$$

with each component sampled at its own plaquette barycentre. Appendix A.4 proves the following comparison with the continuum matrix K, with $\gamma = 1/2$:

$$\begin{gathered} |K_{a,\ell;P,Q}(p,q) - K_{P,Q}(x(p) - x(q))| \le C r^{-4}\big[(a/r)^{\gamma} + (r/\ell)^4\big], \\ a \le r/10, \\ r \le \ell/10, \\ r = |x(p) - x(q)|, \end{gathered}$$

for some fixed $\gamma > 0$, uniformly in a, ℓ, plaquette positions and orientations. Periodic differences are represented by the short displacement in this range. The same section proves separated smearing convergence and the resulting quadratic-density coefficient. This closes the free exact-cochain Gaussian requirement; the interacting extraction F2 remains an independent input.

The projection convention in this input is distinct from physical weighted block means. When passing to unit physical blocks use the adjoints and physical difference d_η established in the block-normalization section. Under plain fine and coarse inner products an arithmetic block mean does not have an isometric adjoint; no projection identity may silently interchange these conventions. Neither normalization proves the interacting extraction F2.

Technical statement F2: extraction of the leading two-point term. For the prescribed short-distance test pairs require the finite-cutoff identity

$$S^T_{2;K,\ell}(f,h) = C_G \zeta_K(d)^2 g_K(d)^4 \mathcal{K}(f,h)(1 + r_{K,\ell}(f,h)),$$

where $C_G > 0$, the real-data error satisfies $|r| \le 1/2$, and all lower cutoff requirements are explicit. For the angular deduction this input must cover the full class $\mathfrak{T}(d;\vartheta_*)$ in eq. 6.cccxxvi–eq. 6.cccxxx of § 6.12, with $\vartheta_* = 1/12000$, or a broader class. These are nonnegative C^1 tests in radius-d balls, with support distance between d and $6d$, the stated lower integral and upper Lipschitz bounds, and arbitrary positions. A law proved only for a narrower shape class is insufficient without a further inclusion argument.

The same bound is required for $S_2^T[\rho]$ in every periodic Gibbs state used to approximate the limit or any component. At fixed d its eventual cutoff threshold and constants must be uniform in those states and support positions. This quantifier is part of the stated F2 lemma, not a consequence of a torus-only law. On each fixed torus any asserted $r \to 0$ needs a separate uniform estimate; it must not be silently promoted to a statement uniform in volume. This is the substantive Gaussian-extraction input behind the short-distance law, including all interaction and large-field remainders. The finite source insertion and renewed-pointing proofs retain the $\alpha^2 s$ potential norm under the actual unmarked-majorant, leg-path and complex-domain hypotheses. The window formation and linear bootstrap propagate it under their explicit ordinary inputs; the resulting near-field estimate requires the complete field to have size δ, not merely a δ-sized increment on an arbitrary α-background. Remote and holonomy tails retain their separate bounds. These deductions do not verify the background and history inputs for the interacting construction.

The actual clean-history normal form must still control every old activity's localized Hessian, the conditional exterior force and pointwise deletion with the same exterior field and correct flag mass. That route also requires the nonlinear averaging-fibre comparison; gauge fixing the actual run provides a route that avoids it. The corresponding normal-form estimates are invoked in F2. An arbitrary resolved coupling supplies neither law. § 6.2 proves the covariance arithmetic and common-normalizer transfer once their inputs hold. The old-activity part of § 4.5 derives the localized Hessian from the actual all-substitution window room, window-first value bounds and holomorphy on a Gaussian-coordinate ball of radius $c\alpha/g$. The resulting scale-small bound has factor $(g/\alpha)^2$. Establishing that room, and the conditional exterior-force and pointwise clean-class laws, remains necessary. The local one-step chart or a value bound without a complex domain does not supply those conditions.

The transfer of a Gaussian reference law has two different scopes. D1's common cell normalizer compares any two periodic infinite-volume states; an infinite reference obtained with a Gaussian estimate uniform along arbitrarily large tori therefore needs no finite-volume vacuum comparison. A law proved only on one prescribed finite reference torus instead requires a comparison with those states. Compatible local counterterms, the floor and vacuum-density comparison give one sufficient route. The local companion criterion in eq. 6.vii–eq. 6.ix proves that vacuum comparison from explicit nonwrapping and normalization data; it is not required in the infinite-reference route. Per-torus estimates whose distance thresholds decrease with the volume do not establish the latter reference law.

2.4.1.7. The conditional law cannot be replaced by a formal Gaussian. § 6.3 gives the actual coarea/conditional-density bookkeeping and the exact Stein identities, including cutoff boundary fluxes. It retains the exterior force, the conditional mean, the third and fourth cumulants, and the change of conditional means when clean and bad classes are mixed. A small Hessian for old activities bounds none of these other quantities by itself. The remaining comparison must identify one actual positive conditional measure and its normalizer, and bound these terms uniformly on the specified exterior fields before applying the covariance arithmetic. The localization of both the observable and its deletion must be the one in that identity. The test transfer additionally uses a uniform scaled C^1 shape bound and the stated lattice-to-support ratio, so a positive continuum mass cannot disappear at every sampling point. These are explicit remaining law and boundary estimates for the F2 route.

2.4.1.8. The Gaussian cutoff reference. § 6.6 proves uniform normalized boundary-flux and quadratic Wick-error estimates for centered Gaussian laws with the stated convex physical cutoffs, including dependent blocks. The actual flat first-step elimination is proved at $x = \theta$, with curvature covariance $P_{\mathrm{d}_1 \ker P_L}$. Comparison with the full P_{ex} reference also requires its coarse complement. The interacting conditional law, complete Hessian, exterior force, normalized force/boundary cumulants and common-reference means remain model inputs.

Technical statement F3: the running normalization at the witness scale. The witness depth must be defined without reference to a desired conclusion, and its lattice separation must lie within the permitted comparison range. Require

$$g_K(d)^{-2} = c_* \log(1/d)(1 + o(1)),$$
$$\zeta_K(d) = 1 + o(1),$$
$$c_* > 0,$$

with the precise uniformity needed by F2. A derivative of a running coupling with respect to a constant source must be shown to equal this normalization; it is not an independent choice of field renormalization. F2–F3 give a nonzero two-point limit by a closed lower bound. Appendix A.8 proves summability of the normalization slopes from the actual increment estimates on coupling-sized independent complex polydiscs and control of the large-field remainder loss. The flow section preserves the alternative with an actual marked remainder on a fixed bare-coupling disc and its separate summable forcing. A sharp $O(g \log^p g^{-2})$ slope rate on this weaker route needs $p_R \geq 3/2$; summability alone needs $p_R > 1$. Analyticity on an ordinary source disc alone does not provide the required marked size and history bounds.

Here is their sufficient ultraviolet consequence, with its size estimate included. Use

$$\mathcal{K}(f, h) = \iint f(x)h(y)|x - y|^{-8}\, dx\, dy.$$

If the nonnegative supports lie in radius-d balls at distance at least d, and v_4 is the volume of the unit ball in $\mathbb{R}^4$, then

$$0 \leq \mathcal{K}(f, h) \leq d^{-8}\|f\|_{L^1}\|h\|_{L^1} \leq v_4^2\|f\|_\infty\|h\|_\infty.$$

The uniform F3 bounds imply $|\zeta_K(d)| \leq Z$ and $g_K(d)^4 \leq C[\log(1/d)]^{-2}$ for small d and all permitted sufficiently fine cutoffs. Thus F2 gives

$$0 \leq S_2^T[\rho](f, h) \leq C'[\log(1/d)]^{-2}\|f\|_\infty\|h\|_\infty.$$

The lower inequality uses $1 + r \geq 1/2$; the upper inequality uses $1 + r \leq 3/2$. At each fixed pair these closed inequalities pass to every limit and to the all-state phase realizations. Taking $u(d) = C'[\log(1/d)]^{-2}$ proves precisely eq. 6.cccxxvii, with $u(d) \to 0$. This deduction does not verify the state-uniform extraction or running normalization required in F2–F3.

2.4.1.9. Flow bookkeeping and its actual coefficients. § 5.2 proves the arithmetic first-passage, forced-slope and tuning implications under the stated coefficient, source-disc and history-error bounds. The reference frame, initial physical weight and vacuum normalization are kept fixed in each comparison. Tensor reduction uses the period average where only periodic covariance has been established; a pointwise own counterterm requires its full pointed covariance. What remains for the interacting flow is to identify and estimate the literal formed coefficients and their admitted overlap domains. An alternative auxiliary pointing or current convention cannot supply those coefficient identities without its own comparison.

Technical statement F4: interpretation of the observable sector. The conclusions of F1–F3 concern the scalar $\mathrm{tr}(F^2)$ sector. To identify a reconstructed theory as the full Yang–Mills theory, specify a sufficient gauge-invariant observable family, its consistency relations, renormalization, and its relation to this scalar sector. Supply the needed continuum gauge identities or equivalent identification. A nonzero scalar Wightman field alone does not provide this identification. If no

such input is verified, the conclusion must be stated as existence of the conditional scalar observable theory, with full gauge-theory identification left open.

2.4.1.10. A complete cutoff family and a precise extension problem. §§ 7.10 and 7.11 construct the full cutoff invariant algebra. Finite Wilson trace words separate every finite SU(2) gauge orbit, including degenerate ranks, and their polynomials are uniformly dense in the invariant continuous functions. § 7.12 exhibits configurations with arbitrarily small plaquette curvature and identical action data but distinct Wilson traces. Its example states are not asserted to be Wilson Gibbs or continuum states. The conclusion is the precise information distinction between the two algebras. § 7.13 proves the mixed positivity, relation, consistency, support, graph-domain and representation consequences of specified joint readouts. The remaining F4 task is to construct a sufficient family of such readouts from this same lattice theory, prove its compatible joint continuum estimates and reconstruction, and justify sufficiency for the intended gauge theory. Cutoff density alone is not a uniform continuum approximation theorem.

G. Symmetries, rotations and phases: four requirements. **Requirement G1: proved finite-lattice symmetries and positivity.** appendix A.5 proves link-reflection invariance and the common period $2\mathbb{Z}^4$ for every periodic SU(2) Wilson DLR state at fixed $\beta > 0$. Appendix A.6 then proves invariance and reflection positivity at every coordinate site, coordinate link, and diagonal lattice mirror, on the full bounded Borel half-space bond algebra. No restriction to torus-selected states is made. The compositions give all unit translations and signed permutations. At $\beta = 0$ the unique product Haar state has the same properties. Adapted split tori are used for diagonal mirrors; an invalid diagonal split of an ordinary cubic torus is not assumed.

Requirement G2: scalar rotations derived from ordinary sources and the ultraviolet law. Consider limits through periodic Gibbs states, including the canonical components with their all-state realizations. G1 and the compactness inheritance argument provide their coordinate and diagonal reflection positivity; § 6.16 provides the exact coordinate cube cone. A component with mean $c \int f$ is first centered at its own mean. Its self-centered cutoff moments converge to this centered functional by the finite subset expansion and the convergent one-point differences; the same expansion restores the invariant mean after the proof. Under D1 in its actual rooted geometry, Prop. 6.23 proves a Hilbert insertion bound uniform in the number of surrounding fields. This operator estimate does not follow from D2 alone.

Prop. 6.24 converts that bound to an analytic L^1 plate density, with the explicit loss of five powers of the time margin. § 6.11 constructs the holomorphic product kernels and vectors, glues their frames, and proves the finite angular band and continued Gram positivity. Prop. 6.25 uses eq. 6.cccxxxv to remove every nonconstant two-point harmonic, including the critical frequency. Its diagonal measure and cone support are constructed in the proof. The shape bound eq. 6.cccxxvii gives the required limit. Alternatively § 6.12.6 deduces it from the ordinary source estimates at the running stops, with their stated logarithmic dependence on the inverse coupling. Prop. 6.26 proves polygon divisibility, integer convexity, removal of exceptional distributional support by common cuts, and full O(4) covariance. QF transfers this covariance to the diagonal-flat Schwartz domain. For the canonical components, the symmetry of the decomposition also transfers covariance once it has been established for the mixture. The separate component two-point law uses the state-uniform estimates on their realizing states.

This proves G2 conditionally for this state class, using either supplier of eq. 6.cccxxxv. The ordinary-source route uses the quantitative stopped estimates. Typed Ward counter-insertions and distant-shell flux belong to the separate route below. The density comparison also applies the route, with the same hypotheses, to limits of first-passage sequences above the cutoff-dependent side threshold. For a general growing cubic-torus limit one must establish the mirror and cone interface separately.

The stronger Ward route requires the following: an exact lattice identity must pass to $\delta_\omega S_n(X) = -\lim_{R\to\infty}\lim_j \langle \mathfrak{B}_{\omega,R}; X\rangle_j$, with every bulk and discretization error controlled and the claimed boundary limit existing and vanishing. The order of those limits and the typed estimates needed for them remain assumptions wherever that route is invoked. The angular proof does not establish those separate comparison estimates.

Requirement G3: the proved finite-spacing plateau law. For a spacing $a > 0$, let $\Omega_a = \mathrm{SU}(2)^{\mathcal{B}_a}$, where $\mathcal{B}_a$ is the countable set of positively oriented bonds of $a\mathbb{Z}^4$. Give this space its product topology. Let $\mathcal{G}_{a,\beta}$ be the translation-invariant probability measures satisfying the Wilson DLR equations at inverse coupling β. The finite-bond conditional kernel integrates the bonds in W against the normalized density $\exp[-\beta\sum_{p:p\cap W\neq\varnothing} a_p]$ and holds the other bonds fixed. These kernels take continuous local functions to continuous local functions. Thus $\mathcal{G}_{a,\beta}$ is a compact convex set. Its reflection invariance and reflection positivity are proved under G1.

Let $\mathcal{I}$ be the completed σ-field of events invariant, modulo the measure, under all translations in $a\mathbb{Z}^4$. For bounded local functions A, B define

$$\mathrm{Pl}_\nu(A,B) = \nu(\overline{E_\nu[A \mid \mathcal{I}]}E_\nu[B \mid \mathcal{I}]).$$

Prop. A.12 proves the following law, with one numerical constant C independent of $a, \beta, \nu, r, n, A, B$. If $r \geq 1$ is an integer, A, B depend only on bonds with both endpoints in $a[-r,r]^4$, and $y \in a\mathbb{Z}^4$ has $n = |y|_\infty/a - 2r \geq 1$, then

$$|\nu(\overline{A}\,\tau_y B) - \mathrm{Pl}_\nu(A,B)| \leq C\left[(r/n)^2 + (r/n)^3\right]\mathcal{N}_{\nu,r}(A)^{1/2}\mathcal{N}_{\nu,r}(B)^{1/2},$$

where

$$\mathcal{N}_{\nu,r}(A) = \max_{\mu=1,\dots,4;\,\gamma\in\Gamma_{16}} \nu(\overline{\theta_\mu\gamma A}\tau_{2rae_\mu}\gamma A) \leq \|A\|_\infty^2.$$

Here θ_μ reflects x_μ to $-x_\mu$, and Γ_{16} is the group of coordinate sign changes. Nonnegativity of these reflection sizes follows from G1. The law holds for every $\nu \in \mathcal{G}_{a,\beta}$, including mixtures. It is a statement before any continuum component is constructed.

Prop. 7.7 constructs the required conditional probabilities on this compact metric space, including Borel versions and simultaneous probability-valued uniqueness. Applied to $\mathcal{I}$, this theorem, the DLR equations, and the law above yield a unique decomposition into ergodic DLR states. The argument and the required closedness of the pure state set are proved in the phase-interface section. Compactness of that pure set and existence of continuum components are conclusions, not assumptions.

Requirement G4: all-state moments derived from ordinary sources. Choose $a_j = L^{-K_j} \downarrow 0$, couplings β_j, and centering states $\nu_j \in \mathcal{G}_{a_j,\beta_j}$. Write $\mathcal{O}_j(f) = \sum_p f(x(p))a_p$. Under the precise ordinary source hypothesis D1, Prop. 5.75 proves for every $\rho \in \mathcal{G}_{a_j,\beta_j}$ that

$$|\rho(\mathcal{O}_j(f)) - \nu_j(\mathcal{O}_j(f))| \leq C_1(R)\|f\|_\sharp,$$

$$|\kappa_n^\rho(\mathcal{O}_j(f_1),\dots,\mathcal{O}_j(f_n))| \leq C_n(d;R)\prod_i \|f_i\|_\sharp$$

$$(n \geq 2).$$

The tests are real C^1 functions; each support is contained in a ball of radius R, and different supports have distance at least $d > 0$ in the second bound. Here $\|f\|_\sharp = \max(\|f\|_\infty, c_M\|\nabla f\|_\infty)$. The constants are independent of j, ρ and support positions. They also apply to the separated reflected copies used in G3. For the chosen sequence, assume that the moments centered at ν_j

converge on every separated tuple; alternatively obtain such a sequence from the preceding compactness theorem. The per-order transfer alone would not establish factorial growth. In the actual rooted setting of D1, Prop. 5.76 also provides D2, and Prop. 5.78 gives H1. The bounds displayed here and G3 suffice for the canonical decomposition constructed below. They do not assume that decomposition or clustering of its future components.

H. Test spaces and reconstruction: four requirements. **Requirement H1: quantitative diagonal-flat extension derived from D2.** Let $\mathcal{S}_{\neq}(\mathbb{R}^{4n})$ be the Schwartz functions flat, to all orders, whenever two arguments coincide. Under the D2 moment bound, passed to a separated-tuple limit, Prop. 5.78 constructs its unique continuous extension to this space and proves

$$|\mathsf{S}_n(\varphi)| \le C_{\mathrm{E}}^n (n!)^{\kappa_{\mathrm{E}}} \sup_{|\alpha| \le k_{\mathrm{E}} n, x} (1+|x|^2)^{k_{\mathrm{E}} n/2} |\partial^\alpha \varphi(x)|,$$

$$k_{\mathrm{E}} = 20,$$

$$\kappa_{\mathrm{E}} = k + 32.$$

The constants are independent of n. The same uniform statement applies to component limits obtained from the all-state bounds. Only the position-uniform radius-one moment bound is used; a unit partition already gives an effective $B(R) \le CB(1)(1+R)^4$. The explicit dyadic/product-box proof includes tensor-relation independence, all diagonal cutoffs, infinity cutoffs, and the positive-time boundary. Its constrained tensor density transfers reflection positivity to finite positive-time Schwartz sequences. Thus H1 follows from D1 in its GS geometry and from G1.

A distribution only on the complement of the diagonals with unspecified constants would not establish this result. Diagonal-flat functions are not dense in the full Schwartz space; an extension across diagonals or contact-term identities still need a separate renormalization statement.

Requirement H2: positivity and continuous Euclidean evolution. § 6.16 proves the compact-tuple Hilbert construction from invariant separated moments, full reflection positivity and position-independent per-ball bounds. It proves strong continuity, time contraction and the cube spectral cone for the stated lattice approximation. For the full positive-time test sequences in the precise reconstruction theorem, the required finite quadratic inequalities are $\sum_{p,q} \mathsf{S}_{p+q}(\Theta\overline{\varphi}_p \otimes \varphi_q) \ge 0$. All tensors in this expression must belong to the declared domain. Under H1 and the constrained tensor-density lemma, those full sequence inequalities and the same Hilbert evolution follow from the compact-tuple construction. The proof bounds all translated matrix elements before iterating Cauchy–Schwarz. Its position-independent per-ball hypotheses are supplied by D1 for the ordinary phase moment bounds. The compact argument does not by itself give quantitative Schwartz growth; that additional conclusion follows from the GS/QF estimates in H1.

Requirement H3: the checked reconstruction theorem import. § 7.3 states the corrected Osterwalder–Schrader theorem in its diagonal-flat domain and verifies its quantitative hypotheses. The norm comparison tracks the dimension-dependent constants as the field number grows, so both derivative order and factorial coefficient growth have the required form. H1–H2 supply their stated flat test continuity and positive ordered form. G2 supplies full Euclidean covariance and permutation symmetry on the declared ordinary-source and ultraviolet route. Clustering is used for the unique-vacuum component; the cluster-free version retains the stated possibly nonunique vacuum.

The energy-band and test-space sections derive the actual fields, positive unrestricted tempered moments and joint boundary values from the damped insertions, and derive Lorentz covariance from G2. Their identification with the imported realization is proved before locality is transferred. The resulting field is on $\mathcal{K}_{\mathrm{ord}}$. Equality with the whole Euclidean Hilbert space is not required.

The representation and phase sections give the remaining domain and scalar direct-integral identifications.

The corrected theorem applies by the hypothesis checks above. We cite its continuation and locality conclusions; the model estimates A–F require their own proofs. Contact terms invisible to diagonal-flat Euclidean data are not assigned arbitrary values. Actual D1, F2–F3 and the existence of the admissible data supplying these hypotheses remain the model requirements.

2.4.1.11. What is reconstructed directly. The energy-band construction gives the actual fields and their common invariant energy domain. The test-space estimates supply full non-tensor Schwartz extensions, joint boundary limits and the needed graph approximations. The representation section identifies their canonical closures and generators. § 7.9 constructs the measurable direct integral of the ordered component sectors and proves that its mixed cyclic range is onto that integral. The remaining named reconstruction import supplies locality and its stated stronger output order after the verified hypothesis matching. Its proof is an identified functional-analysis import, not an additional unknown model estimate.

Technical statement H4: scope of the final conditional assertion. The conditional existence statement may use only verified deductions on its declared route. For periodic-state limits the ordinary D1 hypothesis in its actual geometry, the ultraviolet F2–F3 hypotheses, and the proved bridges give the scalar conditions on a sequence of admissible data; H3 is supplied by the checked theorem import just stated. Actual verification of D1 and of the interacting short-distance law is not supplied here. The additional weighted, matched-cutoff and Ward conclusions require their stated hypotheses. Full gauge-theory identification additionally needs F4. A mass-gap conclusion would require a new estimate, for example a bound yielding spectrum outside zero contained in $[m, \infty)$ for some $m > 0$. None of the packages above supplies that estimate. A finite lattice measure, a subsequential separated-point limit, an axiomatic scalar theory, and its Yang–Mills identification are successive claims, each requiring its own proof.

2.4.1.12. The exact spectral question left open. The scalar direct-integral theorem identifies the mixed zero-energy space with $L^2(X, \lambda)$. It also proves that the gap above that entire space is the essential infimum of the component gaps. Positivity of a component gap at each phase does not imply a positive uniform lower bound. An equivalent sufficient target is a common positive exponential decay rate for the positive heat-semigroup matrix elements on a dense core, after subtracting the full zero-energy projection. No such rate is proved here. Enlarging the observable family requires the corresponding joint spectral and vacuum assertions on the larger Hilbert space, rather than only on its scalar subspace.

2.5. **Domains and dependencies of the analytic construction.** The inverse, background and integration results use different analytic domains. We specify their order here, including the two changes of domain which enter the regular trajectory. This also fixes the meaning of the physical identifications used below. All constants are uniform in depth and compatible volume after the displayed choices have been made. They need not be uniform as an exponent approaches an endpoint or as the blocking integer is changed.

2.5.1. *The reference for the inverse.* An admitted proper geometry specifies the field and site rows, their assigned scales, bare components and complete determining collars. It is given before an inverse is constructed. Choose a real current-regular reference U_* in that geometry. A nearby pair is

$$U = e^{i\eta E} U_*, \qquad \|E\|_{\mathcal{A}_\alpha} < r_*, \qquad \|J\|_{\infty,3} < \gamma_*. \tag{2.xxxi}$$

Here r_*, γ_* are the smaller strong-family radii chosen after $\alpha, p_{\mathrm{ell}}, \kappa_+$ and the final separation and reblocking parameters. The admitted geometry includes these separation choices. The energy current ceiling alone does not give these strong families. The frames and norms are fixed at U_*. The present field U, its restoration and its actual constraint are allowed to vary. The current J is an independent parameter. It is not defined by differentiating the action of U.

The order of the operators is consequential. Positive local Dirichlet columns first give the field and site primitives. Positive coarse inversion then gives the projected zero-current inverse. The first current correction is performed before its row Schur inverse and first lift are formed. That lift supplies the multiplier readout used in the second current correction. In the notation of Prop. 3.112, this order is

$$G_0, \quad G, \quad H, \quad \omega_* = H^\sharp J, \quad G_1, \quad H_1. \tag{2.xxxii}$$

The second lift is not substituted for H in the definition of ω_*. The physical transposes include the row lengths and fine counting densities before any supremum norm is estimated.

The fourth-order construction in Prop. 3.160 provides a new coefficient family for the site restoration. It supplies two Hölder derivatives of the gauge parameter. The strong current construction in Prop. 3.163 then provides the full field–multiplier saddle in the required finite-p force classes. These are estimates on coefficients and their complete carriers. An endpoint decay bound for the completed operator would not suffice for this step.

There is a second distinction in the norm. For joint Taylor expansions of local base data the required positive majorant is

$$\sup_y \sum_X e^{\kappa_+ \sigma_c(X)} \sup_{P_X \in \mathfrak{D}_X^+} \|K_y[X](P_X)\|. \tag{2.xxxiii}$$

The supremum over each local coefficient domain precedes the carrier sum. A bound with one global supremum outside that sum does not imply this bound. Lem. 3.164 proves the stronger order of operations. In a coarse short-power stage it uses the uniform entry bound and the full endpoint count, giving $C(1+B_m)^4(q')^m$. The exponential contraction pays that polynomial after the fixed construction parameters are enlarged.

2.5.2. *The nonlinear germ and its physical restriction.* The auxiliary scalar action is specified explicitly in eq. 3.mcxlviii. Its current convention is part of its definition. The nonlinear equation is solved on the domain eq. 2.xxxi with a smaller strong radius. At zero reduced datum its auxiliary potential is zero. Put $C(B) = \Pi_U(A(b_U(B)))$ for the completed field before the final literal node gauge. The derivatives are

$$D_b A(0) = H_1, \qquad D_b \Pi_U(A(b))\big|_{b=0} = SH_1, \qquad D_B C(0) = SH_1 D b_U(0). \tag{2.xxxiv}$$

The last expression includes the actual literal-to-reduced datum map. A further literal node gauge contributes its specified fine gauge vector to the literal field derivative.

The remainders are formed before they are assigned to carriers. In particular, $QS = Q$ is used to write the constraint remainder as $Q\Xi + N_q(\Pi)$, so each of its terms has quadratic field degree. The current and multiplier remainders are rewritten in their actual point-force classes before restoration adjoints are applied. Thus the tree series has no linear term that must cancel between different carriers. Props. 3.167 and 3.169 prove convergence over every datum degree and every local base degree. The base variables in that statement are supported strong Banach slots, with bounded-overlap reconstruction. They are not independent discs for every raw fine link.

The physical restriction is an exact scalar identity. At

$$J = j_U = Q^\sharp \omega_{\mathrm{phys}}, \tag{2.xxxv}$$

the selected auxiliary action equals the physical action on the finite gauge completion, with its constraint multiplier and explicit penalties. The gauge tests remove the gauge penalty. The artificial row mass is constant on the prescribed constraint and is removed from the physical multiplier by the shift $\nu_{\mathrm{phys}} = \nu - a_f b$. Uniqueness then identifies the solution with the physical constrained orbit. This is the content of Prop. 3.150. The off-current construction remains the selected convention just defined; its existence does not identify any other prescribed auxiliary action.

2.5.3. *Producing the homogeneous reference.* The reference in eq. 2.xxxi is an input for a proper geometry. In the homogeneous case it can instead be constructed from coarse curvature alone. For a real coarse field V with sufficiently small plaquettes, Prop. 3.193 constructs a smooth quaternionic bundle connection whose original coarse-edge transports are exactly those of V. Its fine samples have physical curvature and Wilson current $O(\kappa)$. Their actual hierarchical means differ from V by $O(\kappa)$, uniformly in the number of means.

The current estimate pairs opposite plaquettes before division by the third power of the fine spacing. The mean estimate uses the full plain-mean row majorant. These two operations prevent,

respectively, a loss of one inverse fine spacing and a factor growing exponentially with the number of blocking steps. The forced trial equation retains the trial current as a source and corrects both defects. Prop. 3.194 therefore produces the physical homogeneous orbit without assuming a global gauge close to a flat connection.

The bundle construction also supplies the comparison of nearby real references. Its Kato map aligns the connections before fine edges are sampled. A change of a coarse datum then gives a relative fine potential of size $O(\|B\|)$, rather than a frame jump divided by the fine spacing. The resulting strong comparison, with the larger uniqueness margin, identifies complex germs on overlaps. There is no need to assume that an arbitrary complex overlap contains a real point.

For constant diagonal plaquette weights, Prop. 3.199 uses the same smooth trial. Its principal perturbation is estimated from $W^{2,p}$ to L^p. The full joint inverse is first proved in that stronger field norm. It is not inferred from a $C^{1,\alpha}$ output estimate. The weighted current of the corrected physical orbit is small because it is the actual row transpose of the bounded physical multiplier. No point bound for its unweighted current is inserted into this argument.

These statements produce homogeneous references, including large holonomies and nontrivial bundle topology. The definition and construction of mixed-scale proper histories still use their stated geometry and admission hypotheses.

2.5.4. *The physical Gaussian and the integral domain.* At a physical critical pair the mass-added second inverse has a positive energy gap. Its effective coarse action Hessian is

$$D^2S = (QG_1Q^\sharp)^{-1} - a_f. \tag{2.xxxvi}$$

The artificial row mass is subtracted. The actual outer precision also contains the first action gradient paired with the second derivative of the outer chart:

$$K_\tau = \tau\{\mathsf{C}_V^\sharp(D^2S)\mathsf{C}_V + \langle\omega, \Phi''(0)\rangle\} + K_G. \tag{2.xxxvii}$$

Here $\mathsf{C}_V = D\Phi_V(0)$ is the linear outer chart map. The lower bound for this matrix is proved by flux transfer and the geometric curl–gauge estimate. It does not follow merely from positivity of the mass-added coarse Schur matrix. Props. 3.201 and 5.41 prove the real physical gaps, and Prop. 5.42 constructs the local Gaussian families with their sector and normalization bounds on its declared unweighted canonical domain near a physical pair with small unweighted current. The weighted physical continuation has the sector proved in Prop. 3.202. That sector statement does not by itself supply independent weighted Gaussian cube families.

For the regular trajectory the old interaction is defined on a fixed raw field ball of radius r. It must be evaluated there throughout the integration. Let $R > r$ be the larger output radius. The literal intermediate field and the outer graph satisfy the two separate bounds

$$\|\log \mathsf{p}_j(W)\|_\infty \le C_pR/L < r/64, \qquad \|\log Z_V(\gamma B) - \log V\|_\infty \le C_{\mathrm{dom}}(L)|\gamma|b < r/64. \tag{2.xxxviii}$$

Thus an output argument on the larger ball is sent well inside the old ball before the fluctuation is added. The complete old functions are evaluated there. Gauge identities on admitted overlaps are used after that inclusion has been proved.

Here is a useful explicit order for these choices. First fix $0 < \alpha < 1$ and $p_{\mathrm{ell}} > 4/(1-\alpha)$ for the strong estimates. Fix the desired transport shrink t, final spatial and period rates, and a larger intrinsic rate. Choose the odd blocking integer L from eq. 4.cccxc; the literal amplitude constant in that choice is uniform in L. Choose the Gaussian and carrier localization parameters for those fixed rates. Choose R within the resulting common analytic radii and put $r = tR$. The constants $C_{\mathrm{dom}}(L)$ and the positive Gaussian margins are now fixed. Finally enlarge the coupling floor until eq. 2.xxxviii holds at the physical $\gamma = \theta^{-1/2}$ with $b = A_1(\log\theta)^{p_{\mathrm{cut}}}$, where $p_{\mathrm{cut}} > 1/2$. The last choice is possible since $\theta^{-1/2}(\log\theta)^{p_{\mathrm{cut}}}$ tends to zero. It changes no previous geometric parameter.

The finite exponent used in the elliptic estimates is a different parameter, fixed before the strong inverse constants and the blocking choice. The logarithmic exponent p_{cut} belongs to the real

fluctuation cutoff. Keeping these roles distinct prevents a false endpoint conclusion for the elliptic estimates.

2.5.5. *Formation, tuning and differentiated cutoffs.* For a fixed profile ceiling the formation bounds give

$$\|\mathbf{F}_j\| + \|D_h\mathbf{F}_j\| \le C_H\epsilon(\theta_j), \qquad \|\partial_x\mathbf{F}_j\| \le C_H\epsilon(\theta_j)/\theta_j, \qquad \epsilon(t) = (\log t)^m/t. \tag{2.xxxix}$$

The reference θ_j, its cutoff and the Gaussian coordinate scaling are frozen. Differentiation in $x = \theta_j\tau$ means $\theta_j^{-1}\partial_\tau$. The common-radius transport has an exact semigroup and exponential age decay. Its adapted norm gives a one-step contraction. These are the inputs for the opposite-end conditions $h_0 = 0$, $x_N = X$.

The scale sums needed for tuning have different sizes:

$$\sum_{j<N} \epsilon(a_j) \le C\{1 + (\log a_0)^{m+1}\}, \qquad \sum_{j<N} \frac{\epsilon(a_j)}{a_j} \le C(\log X)^m/X, \qquad a_j \asymp X + N - j. \tag{2.xl}$$

The first sum need not be small. It is a sublinear correction to the bare coupling. The second sum controls the contraction and can be made small independently of N. Prop. 5.51 uses these two sums separately. Props. 5.52 and 5.53 allow a bounded irrelevant initial profile and estimate its effect after retuning to the same endpoint.

Changing a plaquette weight is a different derivative. The moving weighted background changes the centre of the free colour balls. For its first weight derivative, Prop. 4.54 compares it with the fixed isotropic kernel on the same Haar-disintegrated fibre. The difference is a normalized rooted sphere functional with bound

$$C_H\theta^{1/2}(1+b)^K e^{-cb^2}. \tag{2.xli}$$

This is smaller than every fixed inverse power of θ. Its six curvature readouts vanish. It must nevertheless be included in the profile forcing before the diagonal stress recurrence is estimated. Prop. 5.54 uses this actual fixed-kernel first jet. It makes no claim about a uniform fixed-centre complex weight disc.

2.5.6. *The interface with the continuum statements.* The analytic constructions above give the specified inverses, physical homogeneous backgrounds, regular integration maps and tuned regular trajectories. Their identities retain their reference domains and the complete functions to which pointings were assigned. They provide concrete inputs to the multiscale source argument.

The main continuum statements also require the stated estimates for complementary histories, exterior changes, source normalizers and the short-distance comparison. The regular trajectory does not replace those operations by a bound for a single positive quadratic form. Nor does a classical or Gaussian descent identity identify quantum partition functions on different volumes. The hypotheses in Thms. A to C retain those specified model estimates. Each use below refers to the corresponding quantitative statement, with its original source and geometry class.

3. Background fields and constrained inverses

The renormalization step integrates fluctuations around a constrained background field. We construct the averaging maps and their coordinate charts, compare the flat and background-dependent quadratic forms, and state the uniform inverse estimates on the required current spaces. The nonlinear response estimates then control differentiation of the background and of its propagator. These are the inputs to the local integration in §4.

3.1. **Block averages and gauge fixing.**

3.1.1. *Block averages, gauge fixing, and the renormalisation group of this construction.* A block average must turn a fine gauge field into a coarse one while preserving gauge covariance and the coarse-lattice symmetries. The associated change of variables must preserve the integral and provide local fluctuation coordinates. We first specify the averaging map and its gauge choices, which are the geometric ingredients of the renormalisation step.

Setting and objects. *The numbers.* Each statement of this part bears a number belonging to one of three groups. The group 1a.1 concerns the block average of a gauge field; the group 1a.2 concerns the renormalisation transformation, namely its exact identities, its invariant form and the anatomy of a single step; the group 1a.3 concerns gauge fixing and regular configurations. A label "Theorem, Proposition, Lemma 1a.n.m" denotes statement m in group 1a.n, while "1a.n" standing alone denotes the group itself. *Requirements on an average.* We work in level-k units, with L odd and centred blocks $B(y) = \{x : |x - y|_\infty \le \frac{1}{2}(L-1)\}$. We put $\varpi(W) = \sup_p |\partial W(p) - \mathbf{1}|$, and ϖ_c denotes the same supremum taken over the double block $B(c_-) \cup B(c_+)$. For kernels we use the norms $\|K\|_{\mathrm{row}} = \sup_c \sum_b |K(c,b)|$ and $\|K\|_{\mathrm{col}} = \sup_b \sum_c |K(c,b)|$. Elements of the group cannot be added, and transports joining different pairs of points cannot even be compared. A block average $\mathcal{M} : \mathcal{V}_k \to \mathcal{V}_{k+1}$ is required to have the following properties: gauge covariance; locality; linearity at first order; analyticity near flat fields, on G^c as well; symmetry under the isometries of the coarse lattice; and Borel measurability on the whole of $\mathcal{V}_k$, because $\mathbf{T}$ integrates over all configurations. For the mean of group elements Bałaban gives axioms (109 (1987), (0.5)–(0.9)). Such a mean is any analytic G^c-valued function M on families of small diameter that satisfies

$$M(\{\mathbf{U}_j^{-1}\}) = M(\{\mathbf{U}_j\})^{-1},$$
$$M(\{u\mathbf{U}_j v\}) = u\, M(\{\mathbf{U}_j\})\, v,$$
$$M(\pi\{\mathbf{U}_j\}) = M(\{\mathbf{U}_j\}),$$

$$\tfrac{1}{i}\log M(\{e^{iA_j}\}) = \tfrac{1}{n}\sum_j A_j + (\text{higher orders}),$$
$$\mathbf{U}_j \in G \text{ for all } j \;\Rightarrow\; M(\{\mathbf{U}_j\}) \in G.$$

According to Bałaban, the results of his 1987 paper and of the earlier papers hold for every mean of this kind. For the map $\mathcal{M}$, the second of the two symmetric block averages that he introduces, he states that every result of the paper on averaging operations (Comm. Math. Phys. 98) holds, with proofs that are for the most part unchanged. This statement concerns that single paper; the broader sentence, which refers to "this, and previous papers", is the one about means. Of the two symmetric averages together he states that they are equally good, and that many other averages can also be used. He also states that the block average admits an axiomatisation as well. Already in 1985 he states that the analysis of the paper on averaging operations, together with its propositions, extends easily to other averages that satisfy the two requirements set out in its introduction, namely gauge covariance and agreement with the linear block average at first order. No proof accompanies any of these statements. For the theorems of 1984–85 on averaging, gauge fixing, propagators and backgrounds, the proofs are carried out for an average that is not symmetric. Prop. 3.12 states which properties of an average the proofs use, and it compares the linear parts (1a.1).

The mean, the contours, the map. The *mean* $\mathrm{av}(V_\bullet)$ (Federbush) is the solution, close to the family, of $\sum_i \log(V_i V^{-1}) = 0$. By $\mathcal{G}(y,x)$ we denote the family of shortest paths from y to $x \in B(y)$ that complete the displacement in one direction before passing to the next. There is one such path for each ordering of the directions, so the family has at most 4! members and does not contain every shortest path. The *comb* $\Gamma_{y,x}$ is the member with the order $4,3,2,1$. The averaged contour

variable is $U(y,x) := \mathrm{av}(U(\Gamma))_{\Gamma\in\mathcal{G}(y,x)}$, and we set $U(x,y) = U(y,x)^{-1}$ and $U(y,y) = \mathbf{1}$. With $x' = x + (c_+ - c_-)$, one defines

$$W_x = U(c_-,x)\,U([x,x'])\,U(x',c_+)\,U[c]^{-1},$$
$$\mathcal{M}(U)(c) = \exp\Big(L^{-4}\sum_{x\in B(c_-)}\log W_x\Big)U[c]$$
$$\text{on } \mathrm{Dom}(c) = \{\varpi_c < C^{-1}L^{-2}\}:$$

the W_x are closed transports based at c_- and lying near $\mathbf{1}$, so their logarithms can be added. Where $U(y,x) = \mathbf{1}$, only straight segments are left, so the form is adapted to the block axial gauge. Off $\mathrm{Dom}(c)$ the map is given by two cruder rules, both covariant. When none of the elements $Y_{x,\Gamma,\Gamma'} = U(\Gamma)U([x,x'])U(\Gamma')^{-1}U[c]^{-1}$ equals $-\mathbf{1}$, one replaces $\log W_x$ by the arithmetic mean, over (Γ,Γ'), of the intrinsic logarithms of the $Y_{x,\Gamma,\Gamma'}$; this is Bałaban's first symmetrised average. In the remaining case one sets $\mathcal{M}(U)(c) = U[c]$. This third rule is required because Bałaban defines his average on all configurations by means of the spectral logarithm, which satisfies $\log(-\mathbf{1}) = i\pi\mathbf{1} \notin i\mathfrak{g}$ and so leads out of SU(2) (1a.1). By $\mathcal{M}^k$ we mean the iterate only, and not a mean taken in a single stroke over B^k. In Bałaban's paper of 1987 the lattices consist of centres of cubes and have coordinates in $a_j(\mathbb{Z}+\frac12)$. The single translation $x \mapsto x+\frac12 L^m\sum_\mu e_\mu$ of T_ℓ, $\ell = 2L^m$, maps them onto the $\Lambda_{j,\ell}$ at every level simultaneously, since $\frac12 L^m = \frac12 L^{m+j}a_j$ and L^{m+j} is odd. This is the second place where the oddness of L is used, the first being the centring of the blocks (1a.1). Bałaban also remarks that all the results were obtained first for the two-dimensional chiral models, in which the fields sit on the sites and the action is $\sum_b[1-\frac12\,\mathrm{Re}\,\mathrm{tr}\,U(b_-)U(b_+)^{-1}]$. In those models block averages are means of group elements, and neither contours nor a gauge group enter.

Linear averages of bond functions a, which enter in the form $e^{i\mathsf{a}}W$ without any length factor, are defined as follows, where $\bar{\mathsf{a}}(y,x)$ denotes the arithmetic mean of $\mathsf{a}(\Gamma)$ over $\mathcal{G}(y,x)$:

$$(\mathsf{Q}^{\mathrm{s}}\mathsf{a})(c) = L^{-4}\sum_{x\in B(c_-)}\mathsf{a}([x,x']),$$
$$\sigma(y) = L^{-4}\sum_{x\in B(y)}\bar{\mathsf{a}}(y,x),$$
$$(\mathsf{Q}^{\mathrm{pl}}\mathsf{a})(c) = \sigma(c_-) + (\mathsf{Q}^{\mathrm{s}}\mathsf{a})(c) - \sigma(c_+),$$
$$(q\lambda)(y) = L^{-4}\sum_{x\in B(y)}\lambda(x);$$

here Q^{pl} is the derivative of $\mathcal{M}$ at $\mathbf{1}$. The operator $\mathsf{S}(W_0)$ is Q^{s} formed with parallel transport to c_-, namely

$$(\mathsf{S}(W_0)\mathsf{a})(c) = L^{-4}\sum_{x\in B(c_-)}\sum_{b\in[x,x']}\mathrm{Ad}\,(W_0(c_-,x)W_0([x,b_-]))\mathsf{a}(b)$$

.

The straightened average of $\mathbf{W} = e^{i\mathsf{a}}W_0$ relative to W_0 and its *block factor* u are given by

$$\mathsf{u}(y) = \exp\Big(L^{-4}\sum_{x\in B(y)}\log\,[\mathbf{W}(y,x)W_0(y,x)^{-1}]\Big),$$
$$e^{i\mathcal{Q}^1(W_0,\mathsf{a})(c)}\mathcal{M}(W_0)(c) = \mathsf{u}(c_-)^{-1}\mathcal{M}(\mathbf{W})(c)\,\mathsf{u}(c_+).$$

The right-hand side equals $\mathcal{M}(\mathbf{W}^v)$, where v is the gauge transformation with $\mathbf{W}^v(y,x) = W_0(y,x)$ (the block axial gauge relative to W_0) and with trivial average $\bar{R}_{W_0} v = \mathbf{1}$; here $\bar{R}_W$ is the averaging of gauge transformations that is defined below under *Gauge transformations*. Together these two conditions imply $v = \mathsf{u}^{-1}$ at the coarse sites. Bałaban gives two reasons for the straightening. First, the variational problems of § 3.11 involve composing the average with the passage from the axial gauge to the Landau gauge. Second, only the straight linear average admits the explicit representations of the propagators § 3.12.1; this average is compatible with composition, has an explicit Fourier symbol and does not change under $q\lambda = 0$. The change of gauge has to preserve this form, and the non-linear counterpart of $q\lambda = 0$ is $\bar{R}_{W_0} v = \mathbf{1}$. A third reason appears in the proof of Prop. 3.6.

Order k; the maps denoted by Q. Along the sequence $W_{0,n} = \mathcal{M}^n U_0$ we set $\mathsf{a}_0 = \eta_k A$, $\mathsf{a}_{n+1} = \mathcal{Q}^1(W_{0,n}, \mathsf{a}_n)$, $v_0 = \mathbf{1}$, $v_{n+1} = v_n\, \mathsf{u}(W_{0,n}, \mathsf{a}_n)$, and define

$$\mathcal{Q}_k^\sharp(U_0; A) := \mathsf{a}_k = Q_k^\sharp(U_0) A + C_k(U_0, A),$$

$$\mathcal{Q}_k(U_0; A) = \tfrac{1}{i} \log \left[\mathcal{M}^k(e^{i\eta_k A} U_0) \mathcal{M}^k(U_0)^{-1}\right] = Q_k(U_0) A + C_k^{\mathrm{pl}}(U_0, A).$$

A sharp indicates the straightened objects, which are the ones that enter the propagators; Bałaban's $Q_k(U_0, \eta A)$ is the same object built with combs. Without the sharp, $\mathcal{Q}_k$ and Q_k refer to the plain average, whose remainder is denoted C_k^{pl}, while C_k denotes the remainder of the straightened average. The product $v_k = v_k[A]$ of the k block factors is the *node factor*, and $\dot{v}_k$ denotes the linear part of $\frac{1}{i} \log v_k$. The letter Q is used for maps of four kinds (1a.1). (1) The non-linear constraint map $\mathcal{Q}_k$ of the plain iterate $\mathcal{M}^k$, and next to it its straightened form $\mathcal{Q}_k^\sharp$. (2) $Q_k(U) = \partial_A \mathcal{Q}_k(U; A)|_{A=0}$, together with $Q = Q_1(\mathbf{1})$ and $Q_V = Q_1(V)$; this is the operator in the delta function of a step. (3) $Q_k^\sharp(U)$, the operator that enters the propagators and the Landau gauge. Its leading term is the *straight* average with parallel transport, $Q_k^{\mathrm{st}}(U) = \eta_k \mathsf{S}(W_{0,k-1}) \cdots \mathsf{S}(W_{0,0})$. The two differ by $O(\alpha_0)$ in row norm, and they are equal at $U = \mathbf{1}$. There, after the translation of Prop. 3.12, each of them is Bałaban's Q_k of 1984. (4) $Q'_k(U)$, which averages site functions. In a background, the linear average in Bałaban's papers (99 (1985) 389, (3.15)) is $Q_j^{\mathrm{B}}(U) := Q(\bar{U}^{j-1}) \cdots Q(\bar{U}) Q(U)$. It is the linear part of his gauge-corrected average built with combs, and each of its factors equals the straight covariant average up to $O(L^2 \alpha_0)$. Props. 3.6 and 3.12 provide two comparisons, that of (2) with (3) and that of Q_k^{B} with $Q_k^\sharp$, and § 3.12.1 relies on both.

Gauge transformations are averaged by means of the same contours,

$$(\bar{R}_W v)(y) = v(y) \exp\Big(L^{-4} \sum_{x \in B(y)} \log\left[v(y)^{-1} \operatorname{Ad}(W(y,x)) v(x)\right]\Big),$$

and the iterates of this map give $\bar{R}_U^{(j)}$. The linear part, $Q'_j(U)\lambda$, is the mean over $B^j(y)$ of λ after transport to y.

Isometries on 1-forms. An isometry r acts by $(\mathsf{r}U)(b) = U(\mathsf{r}^{-1} b)$. Bond functions are specified on the positively oriented bonds, whereas a reflection reverses orientation. Since $(e^{i\mathsf{a}} W_0)(\bar{b}) = \exp\big(-i \operatorname{Ad}(W_0(b)^{-1}) \mathsf{a}(b)\big) W_0(\bar{b})$, the action relative to W_0 must include a twist:

$$(\mathsf{r}_{W_0} \mathsf{a})(b') = \mathsf{a}(b) \ \text{ if } \mathsf{r}^{-1} b' = b,$$

$$(\mathsf{r}_{W_0} \mathsf{a})(b') = -\operatorname{Ad}\left(W_0(b)^{-1}\right) \mathsf{a}(b) \ \text{ if } \mathsf{r}^{-1} b' = \bar{b};$$

on pairs the action is $\mathsf{r}(\mathbf{U}, \mathbf{J}) = (\mathsf{r}\mathbf{U}, \mathsf{r}_{\mathbf{U}} \mathbf{J})$ (1a.1).

Lift, regions, determining sets. The *lift* is defined by $(Q_k^{*} V)(b) = V(c)$ when $b_\pm \in B^k(c_\pm)$; on every bond contained in a single block of order k its value is $\mathbf{1}$. For a bond function B on $\mathbb{T}^{(k)}$,

its linear lift is the 1-form that takes the value $\eta_k^{-1}B(c)$ on these crossing bonds and the value 0 on bonds lying within a block. The value $\mathcal{M}^q(W)(c)$, and also which rule applies, depend on W only in $B^q(c_-)\cup B^q(c_+)$. Averages are therefore defined for a field given on part of a lattice; they remain covariant and obey the semigroup law $\mathcal{M}^{q+1}W(c')=\mathcal{M}(\mathcal{M}^qW)(c')$. Understood in this way, $M_{\mathbf{B}}(U):=(\mathcal{M}^jU|_{\Gamma_j})_j$ is defined on configurations given on a region, and the problems that define the localised backgrounds $U_{k,\square}$ are set up near a cube without reference to the torus (1a.1).

The transformation. Here δ denotes the Dirac distribution at $\mathbf{1}\in G$ with respect to Haar measure. It is a class function, and it is the limit $\mathfrak{a}\to\infty$ of the normalised weights $\propto e^{-\mathfrak{a}(1-\frac{1}{2}\operatorname{Re}\operatorname{tr}W)}$. The transformation sends a complex Borel measure ν to its image $\mathbf{T}\nu=\mathcal{M}_*\nu$, which is written with the kernel $\delta(\mathcal{M}(U)V^{-1})$. Thus a coarse configuration receives the total weight of the fine configurations whose average it is. Wilson and Kadanoff introduced the block-spin transformations $(\mathbf{T}_t\rho)(V)=\int dU\,t(V,U)\rho(U)$, with $t\ge0$, $\int dV\,t(V,U)=1$ and $t(V^{u|\mathrm{coarse}},U^u)=t(V,U)$; the transformation here is the member of this family with a hard constraint. Decompositions of unity and the factor $\mathfrak{G}_k$ are inserted into the integral. They are built with the auxiliary choices $\varepsilon_k,\delta_k,\alpha_{i,k},\check{R}_k$ and with the weight of $\mathfrak{G}_k$. All of these depend on a single number $\mathtt{t}$, which is x_k in the run and θ_k in a reference run. Only $\check{R}_k$ is formed with the minimum of these numbers taken over all levels $i\le k$, which makes it non-increasing in k. A history h records outcomes, and these yield the term τ_h, or the term $\tau_h^{\mathfrak{w}}$ when ρ_0 is multiplied by a bounded complex weight $\mathfrak{w}(U)$. In addition to χ_k and $\chi^{(k)}$ there is the approximate fluctuation test $\chi'_k(\square)=\mathbf{1}\{|V_k(V_{k,\square}^{\mathrm{bg}})^{-1}-\mathbf{1}|<2\delta_k \text{ near } \square\}$, which is built with the local averaged background $V_{k,\square}^{\mathrm{bg}}=\mathcal{M}^kU_{k+1,\square}$. Braces denote a difference between backgrounds: $\{F\}:=F(U_k(V_k))-F(U_{k+1})$.

The block algebra. The block field of depth s, $V^{(s)}:=\mathcal{M}^{K-s}U$, is defined on $\Lambda_{s,\ell}$, a lattice that does not depend on the cutoff $K\ge s$. The *block algebra* $\mathfrak{A}$ is the set of gauge-invariant functions F, bounded and continuous, that depend on only finitely many block variables $V^{(s)}(b)$. The composite $F\circ\mathcal{M}^{K-s}$ is defined for every $K\ge s$, so these observables exist at all cutoffs simultaneously (1a.2). The block algebra is a notion separate from Thms. A and B, whose observables are the smeared curvature evaluated at the bare scale; neither theorem refers to $\mathfrak{A}$.

Two operations (1a.2). Let $\Delta=\mathsf{C}^{\mathsf{T}}\Delta^{(k)}\mathsf{C}$, let $z^{(k)}$ be a product of Jacobians, and let $\chi(S_{k+1})$ denote the tests on $\Omega_{k+1}\setminus\Lambda_{k+1}$. To obtain $\zeta(\Omega_{k+1}^c)$, form for each outcome that determines this same Ω_{k+1} the product of the tests and gauge-fixing factors localised in $\Omega_k\setminus\Omega_{k+1}$, and add these products over all such outcomes. The one-step operation collects everything that step $k+1$ does not carry out:

$$\mathbf{T}^{(k)}(Z_{k+1})F=\int dV_k|_{\Omega_{k+1}^c}\,\delta(\mathcal{M}(V_k)V_{k+1}^{-1})\,\zeta(\Omega_{k+1}^c)\;z^{(k)}\int d\mathcal{A}_k^{\mathrm{s}}|_{\Omega_{k+1}\setminus\Lambda_{k+1}}\chi(S_{k+1})\,e^{-\frac{1}{2}\langle\mathcal{A}_k^{\mathrm{s}},[\Delta-\Delta C^{(k)}(\Lambda_{k+1})\Delta]\mathcal{A}_k^{\mathrm{s}}\rangle}F,$$

where, more precisely, the Gaussian exponent of the operation is a part of this quadratic form, namely a part that couples no two components of $\Omega_{k+1}\setminus\Lambda_{k+1}$; the remainder of the form is assigned to the boundary terms $B^{(k+1)}$ §4.17.1.3. As long as $\mathbf{R}$ does not intervene, $\mathbf{T}'_{k+1}=\mathbf{T}^{(k)}(Z_{k+1})\mathbf{T}'_k$. For $x\in\mathbb{C}$ and a function S of the bare fields, we define the pure small-field integral and then the regular part of step $k+1$ by

$$(\mathbf{T}_k^{\mathrm{sf}}[\mathtt{t}]F)(W)=\int dV\,\delta(\mathcal{M}(V)W^{-1})\,\mathfrak{G}_k(V)\,\chi^{(k)}(T_\ell)\,F(V),$$

$$\mathfrak{E}_{k+1}[\mathtt{t};x;S](W)=\log\big(\mathbf{T}_k^{\mathrm{sf}}[\mathtt{t}]\,e^{-xA(U_k(\cdot))+S(U_k(\cdot))}\big)(W)+xA(U_{k+1}(W))-S(U_{k+1}(W)).$$

Regular configurations. We consider sequences $\Omega_0\supset\cdots\supset\Omega_k$, with $\Omega_{k+1}=\emptyset$, in which each Ω_j is a union of M_1-cubes of level j and $\operatorname{dist}(\Omega_j^c,\Omega_{j+1})\ge CM_1L^j\eta_k$. The small-field case is the constant sequence $\Omega_j=T_\ell$. Layer j is the set $\Omega_j\setminus\Omega_{j+1}$. It has the local scale $\xi_j=L^j\eta_k$ and the

tops $\mathbb{Y}_j = \Lambda_j(\mathbf{B})$, which are its sites in $\mathbb{T}^{(j)}$. Its pillars are the $B^j(y)$, $y \in \mathbb{Y}_j$, and over all $j \le k$ the pillars partition the bare sites of Ω_0. We define

$$\mathfrak{U}_k(\Omega_\bullet, \alpha) = \{U : |\partial U - \mathbf{1}| < \alpha\, \eta_k^2 \xi_j^{-2},\ |J(U)| < \alpha\, \xi_j^{-3} \text{ on layer } j\},$$
$$|f|_{(-s)} = \sup_j \sup_{\text{layer } j} \xi_j^s |f|,$$

together with

$$\|A\|_1 = \max\{|A|_{(-1)}, |\nabla_{U_0} A|_{(-2)}\}$$

and

$$\|\lambda\|_0 = \max\{\sup|\lambda|, |D_{U_0}\lambda|_{(-1)}\}$$

. Both conditions are invariant under gauge transformations. Combined with the Bianchi identity, the second condition is an elliptic system for the field strength F_U, and all bounds on derivatives come from this system; minimisers satisfy the second condition. The complex neighbourhood of radius α_1 is the set of $\mathbf{U} = e^{i\eta_k A'} U$ with U real and regular and $\|A'\|_1 < \alpha_1$. The norm $\|A'\|_1$ controls only first differences, whereas the current of $\mathbf{U}$ is a second difference. Thus the current enters the domains $U_j^c(X, \alpha_0, \alpha_1)$ as an independent variable $\mathbf{J}$ with $|\mathbf{J}| < \alpha_0$. The real centre U is required to have small curvature and small local potentials, but not a small current. The backgrounds of the averages of $\mathbf{U}$ are required to satisfy what the condition on the current provided, and U_j^c is the set of all such pairs. This set is not saturated with respect to the G^c-valued u, which act by

$$(\mathbf{U}, \mathbf{J}) \to (\mathbf{U}^u, \mathrm{Ad}(u)\mathbf{J})$$

. The extent to which a term is invariant under these transformations is given in Prop. 3.28 and in § 4.5.

Gauges. Because G is compact, the gauge is not fixed in order to obtain finiteness. The reason for fixing it is that the integrand of $\mathbf{T}$ does not change under the block group $\mathcal{G}_{\mathrm{bl}} = \{u : u = \mathbf{1} \text{ on } \mathbb{T}^{(k+1)}\}$. Consequently, in each block there are $3(L^4 - 1)$ directions $D_V\lambda$ along which the Hessian at the minimum vanishes, and there is then no Gaussian measure, no covariance and no decay. Similarly, the residual group $\mathcal{G}^{\mathrm{res}}$ leaves the constraints $\mathcal{M}^j U = V_j$ on Γ_j unchanged. This group consists of the bare u that equal $\mathbf{1}$ at the *nodes*, which are the end points of the bonds of $\mathbf{B}[\Omega_\bullet]$. Hence a background is an orbit and not a point. No gauge has all of the following properties at once: locality (on a torus, no global gauge turns small curvature into a small potential), compatibility with the averages, covariance under the remaining gauge group, symmetry under the hypercubic group W_4, and regularity uniform in k. Axial gauges are algebraic and strictly local and have Jacobian one; they appear in the integrals and serve to select representatives. *Box axial:* the condition $V = \mathbf{1}$ on the tree of combs that issues from the base point y_P of a box. *Block axial:* the condition $V(y, x) = \mathbf{1}$ on the averaged contour variables. It is imposed softly on a set P'' of blocks, at those pairs (y, x) for which the mean $V(y, x)$ is defined (no factor is inserted at the other pairs), by means of

$$\mathfrak{G}_k(V; P'') = \prod_{y \in P''} \prod_{x \in B(y) \setminus y} z_{\mathfrak{G}}^{-1}\, \mathbf{1}\{|V(y,x) - \mathbf{1}| < \varepsilon_k\}\, e^{-x_k(1 - \frac{1}{2}\operatorname{Re}\operatorname{tr} V(y,x))},$$
$$\mathrm{GF}_k(V) = \sum_{y,x} [1 - \tfrac{1}{2}\operatorname{Re}\operatorname{tr} V(y,x)] \ge 0.$$

Tower axial relative to U_0, written $\mathrm{Ax}(\Omega_\bullet, U_0)$: in every pillar and at every level $n < j$, the contour variables of $\mathcal{M}^n U$ coincide with those of $\mathcal{M}^n U_0$. When the tops are the only nodes, solving this condition from the top down selects one representative in each orbit of $\mathcal{G}^{\mathrm{res}}$. An example, for $U_0 = \mathbf{1}$, is the representative of the orbit $U_k(\mathbf{V})$ of the regular critical configurations of § 3.11. Suppose that an inward bond of layer $j-1$, that is, a bond of $\mathbb{T}^{(j-1)}$ joining a point of layer $j-1$ to a point of Ω_j, is retained as an additional constraint (see § 3.11, *Inward bonds*; by the Notation, §4, the determining set contains no such bond). Its end point then lies inside a pillar of layer j. That end point is a node but not a top, and it is called a *junction node*; in § 3.11 it is the seam node. The transport from the top to a junction node is invariant under $\mathcal{G}^{\mathrm{res}}$, so the condition cannot be achieved there. Leaving out these conditions gives the gauge $\mathrm{Ax}^\circ(\Omega_\bullet, U_0)$, which meets in exactly one point every orbit where the contour variables are defined. Representatives are covariant. Given a gauge transformation v on the nodes, its lift is the bare transformation $\tilde{v}$ whose value at a bare site is the value of v at the first node reached when one ascends from that site through the centres of its blocks. The map $W \mapsto W^{\tilde{v}}$ then sends the constraint set of $\mathbf{V}$ onto the constraint set of $\mathbf{V}^v$; it also sends orbits to orbits and maps $\mathrm{Ax}^\circ(\Omega_\bullet, \mathbf{1})$ onto itself. Hence, for the representative taken at the reference $U_0 = \mathbf{1}$, $U_k(\mathbf{V}^v) = U_k(\mathbf{V})^{\tilde{v}}$. For nests of boxes whose lattices are of different levels §4.17.1.3, the tree consists of shell trees, connected by one entrance bond in each box. Trees depend on the choice of a frame in W_4, and averaging over the $2^4\,4!$ frames restores the symmetry (1a.3).

The estimates are carried out in the elliptic Landau gauge, and this gauge comes with a determinant. We put $Q^\sharp = (Q_j^\sharp|_{\Gamma_j})_j$, with each component normalised to total weight $1 + O(\alpha_0)$, so that $\mathcal{Q}_j^\sharp(U_0; A) = \xi_j Q_j^\sharp A + C_j(A)$. Further, $Q' = (Q'_j|_{\mathbb{Y}_j})_j$, and its null space is denoted $N(Q')$. On layer j we set $\xi = \xi_j$ and $a = \xi_j^{-2}$. We also put $G' = (\Delta_{U_0} + Q'^* a Q')^{-1}$, and R is the orthogonal projection onto $\Delta_{U_0} N(Q')$. Relative to U_0, a *Landau pair* (A, u) of U is a pair satisfying

$$U^{u^{-1}} = e^{i\eta_k A} U_0,$$

$$R\, D^*_{U_0} A = 0,$$

$$\bar{R}^{(j)}_{U_0} u = \mathbf{1} \ \text{ on } \mathbb{Y}_j,$$

in which conditions on averages take the place of the condition $u = \mathbf{1}$ at the nodes, since that condition cannot be handled analytically. This has a cost. Transformations u of this kind do not form a group: $\bar{R}u_1 = \bar{R}u_2 = \mathbf{1}$ does not imply $\bar{R}(u_1 u_2) = \mathbf{1}$. The word "orbit" therefore has no meaning, and the set that such transformations generate depends on its base point. For this reason the operations come in a fixed order. First, the tower axial gauge Ax° selects a base point in each orbit of $\mathcal{G}^{\mathrm{res}}$. Next, the set is generated from that base point. Finally, the set is intersected with the slice. The average that belongs to this gauge is the *reduced average*

$$\mathcal{M}_{\mathrm{red}}(W; U_0) = \mathcal{M}(W)^{\mathsf{n}_W^{-1}} \mathcal{M}(U_0)^{-1} = e^{i\mathcal{Q}^1(U_0, \mathsf{a})}$$

; here n_W stands for the block factor $\mathsf{u}(U_0, \mathsf{a})$ taken at $W = e^{i\mathsf{a}} U_0$. The reduced average is the straightened average, under the name it bears in the theory of gauges. The contour variable is required to be the same in all four objects $\mathcal{M}$, n_W, $\bar{R}$ and Ax: it must be the mean in each of them, or the comb in each of them.

Statements.

Lemma 3.1 (the mean)**.** Consider families $V_\bullet = (V_1, \dots, V_n)$ in G^c of small diameter $\operatorname{diam} V_\bullet := \max_{i,i'} |V_i V_{i'}^{-1} - \mathbf{1}|$, where the radii do not depend on n. For such families the mean near the family is unique. It is holomorphic, with bounds independent of n (by the contraction principle), and it is symmetric. On families in G it takes values in G, and there it is the only mean lying in G. Here

a mean of $V_\bullet$ is an element V with $\max_i |V_i V^{-1} - \mathbf{1}| < 1$ and $\sum_i \log(V_i V^{-1}) = 0$; the logarithm is the series logarithm, which has a meaning because of the first condition. Next, $\mathrm{av}(TV_\bullet T') = T\,\mathrm{av}(V_\bullet)T'$ holds for $T, T' \in G$. It also holds for $T \in G^c$ as long as $\mathrm{cn}(T)\,\mathrm{diam}\, V_\bullet$ remains below the radius, where $\mathrm{cn}(T) := |T|\,|T^{-1}|$ is the condition number. The identity $\mathrm{av}(V_\bullet^{-1}) = \mathrm{av}(V_\bullet)^{-1}$ holds in G. It holds in G^c when $\mathrm{diam}\, V_\bullet$ is below half the radius and $\mathrm{cn}(V_i) \le 2$ for all i. Finally, $\frac{1}{i}\log \mathrm{av}(e^{iA_\bullet}) = \frac{1}{n}\sum_j A_j + O(\max_j |A_j|^2)$. ■

Lemma 3.2 (small curvature: nearby transports, comb gauge)**.** Consider a box that contains at most s sites along each direction and has plaquette size ϖ. Suppose the m steps of a path are rearranged so that the rearranged paths stay inside the box. Then the transport along the path changes by at most $(1+\varpi)^{m(m-1)/2} - 1$. Moreover, with $h(x) = W(\Gamma_{y,x})$,

$$|W^h(b) - \mathbf{1}| \le |b_- - y|_1\, \varpi,$$
$$W^h = e^{i\mathsf{a}^0},$$
$$|\mathsf{a}^0| \le Cs\varpi.$$

Consequently the contour families on $\mathrm{Dom}(c)$ are of diameter $\le CL^2\varpi_c$, and one has $\mathcal{M}(W)(c) = e^{iY} W[c]$ where $|Y| \le CL^2\varpi_c$. ■

Proposition 3.3 (the averaging map)**.** Let U, u be arbitrary, and let r be any isometry that preserves $\mathbb{T}^{(k+1)}$ (translations, W_4, coordinate and diagonal reflections). Then

$$\mathcal{M}(U^u) = \mathcal{M}(U)^{u|_{\mathrm{coarse}}},$$
$$\mathcal{M}(\mathsf{r}U) = \mathsf{r}\,\mathcal{M}(U);$$

and $\mathcal{M}^n(U)(c)$ depends on U only in $B^n(c_-) \cup B^n(c_+)$. The averaging map has four further properties.

- It is Borel. Each of the three rules applies on a Borel set, and $\mathcal{M}$ is continuous on each of these sets but not across their boundaries. Outside $\mathrm{Dom}(c)$ no further regularity is required.
- It is holomorphic at $e^{i\mathsf{a}}W_0$ for $\varpi_c(W_0), |\mathsf{a}| < c_L$.
- It is onto: $\mathcal{M}^k(Q_k^{s*}V) = V$.
- On small fields of one colour it is linear: $\mathcal{M}(e^{ifT}) = e^{i(\mathsf{Q}^{\mathrm{pl}}f)T}$ whenever $T \in \mathfrak{g}$, $|T| = 1$, and f is real with $\sup|f| < c_L$.

For a G^c-valued g, covariance is obtained at the expense of room in the domain, and the amount of room lost is $\mathrm{cn}(g)$. Precisely, consider the graded domains $\mathrm{Dom}^c_\theta(c)$, $0 < \theta \le 1$, on which the contour families and the W_x lie within θ times the radii. If $\mathrm{cn}(g(c_\pm)) \le \varkappa$ and $\varkappa\theta \le 1$, then $\mathbf{W}^g \in \mathrm{Dom}^c_{\varkappa\theta}(c)$ and

$$\mathcal{M}(\mathbf{W}^g)(c) = g(c_-)\,\mathcal{M}(\mathbf{W})(c)\,g(c_+)^{-1}.$$ ■

Proposition 3.4 (one averaging multiplies curvature by L^2)**.**

$$|\partial\mathcal{M}(W)(p') - \mathbf{1}| \le L^2\varpi + C(L^2\varpi)^2;$$
$$|\partial U - \mathbf{1}| \le \alpha_0\eta_k^2,\ \alpha_0 \le c$$
$$\Rightarrow$$
$$\varpi(\mathcal{M}^n U) \le 2\alpha_0(L^n\eta_k)^2$$
$$(n \le k),$$

and $\mathcal{M}^n U(c)$ differs from the straight bare transport by at most $C\alpha_0(L^n\eta_k)^2$. Consequently the first rule is applicable on every level, and on $\{|\partial U - \mathbf{1}| < c\,\eta_k^2\}$ the map $\mathcal{M}^k$ is real-analytic. ∎

Proposition 3.5 (one step relative to a reference)**.** Let $\varpi = \varpi(W_0) \le c_L$, and consider complex $|\mathsf{a}| < c_L$. Then $\mathcal{Q}^1$ and $\frac{1}{i}\log \mathsf{u}$ depend holomorphically on a, are covariant and do not exceed $C|\mathsf{a}|$ in size; moreover

$$\mathcal{Q}^1(W_0, \mathsf{a}) = \mathsf{Q}(W_0)\mathsf{a} + C^1(W_0, \mathsf{a}),$$
$$\mathsf{Q}(\mathbf{1}) = \mathsf{Q}^{\mathrm{s}},$$
$$\|\mathsf{Q}(W_0) - \mathsf{S}(W_0)\|_{\mathrm{row,\,col}} \le C\varpi,$$

$$\|\mathsf{S}(W_0)\|_{\mathrm{row}} \le L,$$
$$\|\mathsf{S}(W_0)\|_{\mathrm{col}} \le L^{-3},$$
$$|C^1| \le C|\mathsf{a}|^2.$$ ∎

Proposition 3.6 (k steps, uniformly in k)**.** Assume $|\partial U_0 - \mathbf{1}| \le \alpha_0\eta_k^2$ with $\alpha_0 \le c$, and let A be complex with $\sup|A| < c$. Here c, C depend only on L. The following hold holomorphically in A and in the complex shifts $\mathbf{U} = e^{i\eta_k A'}U_0$ of U_0 with $\sup|A'| < \alpha_1 \le c$. The bounds are uniform in A'; in the bounds on the row norm and on the functional derivative, $\alpha_0 + \alpha_1$ then replaces α_0.

$$\mathcal{M}^n(e^{i\eta_k A}U_0) = \big(e^{i\mathsf{a}_n}\mathcal{M}^n U_0\big)^{v_n},$$
$$|\mathsf{a}_n| \le 2L^n\eta_k \sup|A|,$$
$$|v_n - \mathbf{1}| \le CL^n\eta_k \sup|A|,$$

$$Q_k^\sharp(U_0) = \eta_k\,\mathsf{Q}(W_{0,k-1})\cdots\mathsf{Q}(W_{0,0}),$$
$$\|Q_k^\sharp(U_0)\|_{\mathrm{row}} \le 1 + C\alpha_0,$$
$$|C_k| \le C\sup|A|^2,$$

$$Q_k(U_0)A = Q_k^\sharp(U_0)A - D_{\mathcal{M}^k U_0}\dot{v}_k(A),$$
$$|Q_k A| \le C\sup|A|,$$
$$|C_k^{\mathrm{pl}}| \le C\sup|A|^2 :$$

thus a coarse gradient separates the two linear parts, and this gradient does not become small as the curvature does. More generally, let h be G^c-valued with $\mathrm{cn}(h)\,\mathrm{cn}(v_n) \le 2$. Then $(e^{i\eta_k A}U_0)^h$ has well-defined averages, and $\mathcal{M}^n\big((e^{i\eta_k A}U_0)^h\big) = \big(e^{i\mathsf{a}_n}\mathcal{M}^n U_0\big)^{hv_n}$. This ball is much larger than the regular class, because the plaquettes of $e^{i\eta_k A}U_0$ are $O(\eta_k)$ rather than $O(\eta_k^2)$. Accordingly, the validity of the first rule on every level and the analyticity are consequences of the factorisation, not of Prop. 3.4. The remainder decomposes as $C_k = C_k^{(2)} + C_k^{(3)} + \cdots$, where

$$C_k^{(2)}(U_0, A) = \sum_{n<k} \mathsf{Q}(W_{0,k-1})\cdots\mathsf{Q}(W_{0,n+1})\,C^{1,(2)}(W_{0,n}, \mathsf{a}_n^{\mathrm{lin}}),$$

$$|n\text{-th term}| \le CL^{n-k} \sup |A|^2 :$$

the non-linearity consists of commutators and is concentrated at the coarsest of the k scales. The derivative $K_A(c,b) := \partial \mathcal{Q}_k^\sharp(c)/\partial A(b)$ satisfies $\|K_A\|_{\mathrm{col}} \le 2\eta_k^4$ and the Lipschitz bound $\|K_A - K_{A'}\|_{\mathrm{col}} \le C\eta_k^4 \sup|A - A'|$. Consequently $1 + C\alpha_0 + C|A|$ is a bound for the functional derivative $\eta_k^{-4} K_A$, the adjoint K_A^* is a bounded operator for the sup norm, and K_A acts boundedly from L^2 to ℓ^2. The Lagrange equations of § 3.11 involve K_A^*. Complex regular fields $\mathbf{U} = e^{i\eta_k A'} U_0$ with $\sup|A'| < \alpha_1 \le c$ have averages that are regular, each on its own scale:

$$|\partial \mathbf{U} - \mathbf{1}| \le \alpha_0' \eta_k^2, \quad \alpha_0' \le c$$

$$\implies \quad |\partial \mathcal{M}^n \mathbf{U} - \mathbf{1}| \le 2\alpha_0' (L^n \eta_k)^2.$$

Here the complex curvature bound is proved directly. Fix a level-n plaquette p_n and let Y be the union of the four complete backward blocks of its corners, with all bonds and plaquettes whose vertices lie there. If this read wraps, lift its literal averaging formulas to a periodic cover on which the read embeds. Put $\ell = L^n$, $t = \ell\eta_k$ and $r = \alpha_0' t^2$. The case $n = 0$ is the hypothesis. For $n \ge 1$, use the coordinate comb in this box. Its paths have at most $C\ell$ bare links. The real factors are unitary, and each complex increment and its inverse have norm at most $e^{\eta_k \alpha_1}$. The resulting gauge g therefore satisfies

$$\sup_x \mathrm{cn}(g(x)) \le K := e^{C\alpha_1 t}.$$

The strip between neighboring combs contains at most $C\ell$ plaquettes. Their conjugating paths have the same bound on their condition numbers. The product and logarithm estimates give

$$\mathbf{U}^g = e^{ia}, \qquad \|a\|_\infty \le C\ell\alpha_0' \eta_k^2 = Cr/\ell.$$

All constants here depend only on L; the fixed ceiling is decreased so that the products stay in their logarithm charts.

Extend a by zero outside Y only to use the flat supremum-ball Taylor estimate; its supremum does not increase, and the four output rows do not change. No curvature bound across this new boundary is asserted. Let $P_n(a) = i^{-1} \log \mathcal{M}^n(e^{ia})$ and let L_n be its linear part at zero. The flat recursion eq. 3.mccliii and its node factors give the uniform holomorphic ball of Lem. 5.26. Cauchy on a fixed larger ball therefore gives

$$\|P_n(a)\|_\infty \le Cr, \qquad \|P_n(a) - L_n a\|_\infty \le Cr^2.$$

Indeed its normalized input is ℓa, of size at most Cr, on a fixed larger ball. Let d_0 and d_n denote oriented sums around fine and level-n plaquettes, respectively. In the one-step linear formula the two contour terms at each corner cancel. Its curl is the average of the sums over the L^2 fine plaquettes filling the corresponding square. Iterating this exact identity gives a plaquette average with nonnegative weights and total weight ℓ^2. For the selected plaquette every fine plaquette in this identity lies in Y, so

$$|(\mathrm{d}_n L_n a)(p_n)| \le \ell^2 \sup_{p \subset Y} |(\mathrm{d}_0 a)(p)|.$$

On precisely this stock the finite plaquette product gives $\sup_{p\subset Y} |(\mathrm{d}_0 a)(p)| \le K\alpha_0' \eta_k^2 + C\|a\|_\infty^2$. Apply the same product estimate to the four coarse logarithms. Complex covariance then conjugates the result back by the gauge at the base point. This covariance is valid throughout the retained chart. The flat raw-angle recursion bounds the intermediate plain logarithms by CrL^{s-n}, so every one-step contour family has a fixed reserve. Conjugation by g or g^{-1} spends at most K of that reserve, and K is as close to one as required by decreasing c. This admission uses the flat recursion, independently of the complex curvature estimate being proved. Consequently

$$|\partial \mathcal{M}^n \mathbf{U} - 1| \le K^2 r + Cr^2 \le r(1 + C\alpha_1 t + Cr) \le 2r.$$

The choice is uniform because $0 < t \le 1$. The formulas commute with pullback, so the same bound descends from the cover. From $|\nabla_{U_0} A'| < \alpha_1$ one obtains only $\alpha_0' = \alpha_0 + C\alpha_1$. Thus the domains U_j^c require, as a separate condition, that the complex plaquettes be small; the estimate just proved preserves their prescribed curvature scale.

All of this holds on determining sets, layer by layer. At the bare bonds of layer j the local scale is $\xi(b) = \xi_j = L^j \eta_k$; it is the local scale of layer j in the sense of "Regular configurations". On determining sets the ceilings on α_0 and on $\sup_b \xi(b)|A(b)|$ are lowered by the factors L^{-2} and L^{-1}. At a bond of Γ_j that lies astride a seam, one factor is lost. Writing $K_A(c,b) = \partial \mathsf{a}_j(c)/\partial A(b)$ for $c \in \Gamma_j$, one has

$$|\partial \mathcal{M}^j U_0 - \mathbf{1}| \le 2L^2 \alpha_0,$$

$$|\mathsf{a}_j| \le 2L \sup_b \xi(b)|A(b)|,$$

$$\sup_c \sum_b \frac{|K_0(c,b)|}{\xi(b)} \le 2L,$$

$$\sum_j \sum_{c \in \Gamma_j} |K_A(c,b)| \le \frac{C\eta_k^4}{\xi(b)^3},$$

and this factor is lost once, without compounding; it cannot be removed (Prop. 3.13). The last bound is a sum over the levels at which a given bare bond enters. ∎

Proposition 3.7 (linear averages; gauge transformations)**.** The operator

$$Q_k^{\mathrm{st}} := Q_k^{\mathrm{st}}(\mathbf{1}) = Q_k^{\sharp}(\mathbf{1}) = \eta_k (\mathsf{Q}^{\mathrm{s}})^k$$

is the straight average taken over the blocks of order k, and

$$(Q_k^{\mathrm{st}} A)(c) = \eta_k^5 \sum_{x \in B^k(c_-)} A([x, x']),$$

$$u_k(p) = \prod_\mu \frac{\eta_k (e^{ip_\mu} - 1)}{e^{i\eta_k p_\mu} - 1},$$

$$c \le Q_k^{\mathrm{st}} Q_k^{\mathrm{st}*} \le 1,$$

where u_k coincides with the Fourier symbol of Q_k' up to a phase factor. The normalisation is as follows. Constants are mapped to the same constants, and $\|Q_k^{\mathrm{st}}\|_{\mathrm{row}} = 1$. Composing Q_k^{st} with the linear lift gives the identity, and $Q_k' Q_k'^* = 1$. The adjoint does not coincide with the lift. The operator $Q_k^{\mathrm{st}} Q_k^{\mathrm{st}*}$ is not a multiple of the identity: it is a convolution that couples each coarse bond to its two neighbours on the same line. This is why the display gives only a two-sided bound. In a background one has $|Q_k^{\sharp}(U_0)A| \le Q_k^{\mathrm{st}}|A| + C\alpha_0 Q_k''|A|$, where Q_k'' takes the mean of $|A|$ over every bond in the double block of order k. With gradients $(\mathrm{d}\lambda)(b) = \lambda(b_+) - \lambda(b_-)$, the two averages intertwine gradients in different ways:

$$\mathsf{Q}^{\mathrm{pl}} \mathrm{d}\lambda = \mathrm{d}(\lambda|_{\mathrm{coarse}}),$$

$$\mathsf{Q}^{\mathrm{s}} \mathrm{d}\lambda = \mathrm{d}(q\lambda),$$

$$Q_k^{\sharp}(U) D_U \lambda = D_{\mathcal{M}^k U} Q_k'(U) \lambda,$$

the last of these relations being exact and valid for the straightened average only. In a background the plain average satisfies, likewise exactly, $Q_k(U)D_U\lambda = D_{\mathcal{M}^k U}(\lambda|_{\mathbb{T}^{(k)}})$, where the restriction stands in place of Q'_k. For the main term $Q_k^{\mathrm{st}}(U)$, by contrast, the relation involving Q'_k is valid only up to $O(\alpha_0)$.

It follows that there are two residual invariances. The constraint $\mathcal{M}^k U = V$ is preserved by every transformation with $u = \mathbf{1}$ on $\mathbb{T}^{(k)}$. The straightened constraint is preserved by the w with $r_k = \mathbf{1}$, where r_k is defined below. In linearised form this condition is $Q'_k\lambda = 0$. When $A = 0$, and likewise at a field tower axial relative to U_0, these transformations are precisely the w whose averages are trivial. Of the two sets, only the first forms a group. The set generated by the second depends on its base point, and the tower axial gauge selects that base point in each orbit of the first. The two invariances correspond to each other through $v_k[A]$.

Now let A be real with $\sup|A| \le c$. Put $w = e^{i\lambda}$ and $e^{i\eta_k A^w}U_0 := (e^{i\eta_k A}U_0)^w$, and for a site function assume $\|\lambda\|_1 := \max\{\sup|\lambda|, \sup|D_{U_0}\lambda|\} \le c$. This norm is taken in level-k units, without local scales, and it differs from the norm $\|\cdot\|_1$ of 1-forms. It equals $\|\lambda\|_0$ for the constant sequence $\Omega_j = T_\ell$, whose only layer is k, with $\xi_k = 1$. The straightened averages change by r_n, where $r_0 = w$ and $r_{n+1} = \mathsf{u}(W_{0,n}, \mathsf{a}_n(A^w))^{-1}\, r_n\, \mathsf{u}(W_{0,n}, \mathsf{a}_n(A))$:

$$e^{i\mathsf{a}_n(A^w)}W_{0,n} = \big(e^{i\mathsf{a}_n(A)}W_{0,n}\big)^{r_n},$$
$$w|_{\mathbb{T}^{(n)}}\, v_n[A] = v_n[A^w]\, r_n,$$
$$r_n|_{A=0} = \bar{R}^{(n)}_{U_0} w,$$
$$\tfrac{1}{i}\log r_k = Q'_k(U_0)\lambda + O\big((\sup|A| + \|\lambda\|_1)\|\lambda\|_1\big).$$

We say that u is of class α when each $\bar{R}^{(j)}u$ is within α of $\mathbf{1}$ and its variation inside any block is less than $\alpha L^{j+1}\eta_k$. If u_1, u_2 are of class $\alpha \le c$, the averages are multiplicative up to second order,

$$\bar{R}^{(j)}(u_1u_2) = \bar{R}^{(j)}u_1\, \bar{R}^{(j)}u_2\, e^{i\mathsf{z}_j},$$
$$|\mathsf{z}_j| \le C(\alpha L^j\eta_k)^2,$$

and u_1u_2 is of class $2\alpha + C\alpha^2$. The averages of products depend holomorphically on complex λ. Precisely, let $u' = e^{i\lambda}$ with $\|\lambda\|_1 \le \alpha' \le c$, and let u_1 be of class α. Then the quotients $\tilde{u}^{(j)} := \bar{R}^{(j)}(u'u_1)(\bar{R}^{(j)}u_1)^{-1}$ have four properties: they are holomorphic in λ; they lie within $C\alpha'$ of $\mathbf{1}$; their covariant differences are $\le C\alpha' L^j\eta_k$; and $\frac{1}{i}\log\tilde{u}^{(k)} = Q'_k(U_0)\lambda + O((\alpha+\alpha')\alpha')$. 1a.3 uses this to show that the Landau gauge depends analytically on the fields. ■

Lemma 3.8 (rigidity of one step)**.** Suppose that W, W_0 have plaquette size $\le \varpi$, that their contour variables agree up to a defect δ, and that $|\mathcal{M}(W)\mathcal{M}(W_0)^{-1} - \mathbf{1}| \le \alpha_1$. Then

$$|W(b)W_0(b)^{-1} - \mathbf{1}| \le \alpha_1 + C\varpi + 2\delta$$

at every bond of the double block. Put differently, modulo the curvature, each bond variable is a function of the contour variables together with the average:

$$|W(b) - W(y, b_-)^{-1}W(y, b_+)| \le C\varpi,$$
$$|W(b') - W(c_-, b'_-)^{-1}\,\mathcal{M}(W)(c)\,W(c_+, b'_+)| \le C\varpi,$$

for b inside $B(y)$ with $y = c_\pm$, and for b' in the corridor of c, that is, among the bonds parallel to c joining $B(c_-)$ to $B(c_+)$. The defect has two sources: soft gauge fixing gives $|V(y,x) - \mathbf{1}| < \varepsilon_k$ instead of equality, and seams contribute the defect of Lem. 3.29. ■

Proposition 3.9 (the fibre is curved; the current enters)**.** Let $|\partial U - \mathbf{1}| \le \alpha_0 \eta_k^2$ with $\alpha_0 \le c$ (the range of Prop. 3.6). Let moreover $U \in \mathfrak{U}_k(\Omega_\bullet, \alpha_0)$ at the constant sequence $\Omega_j = T_\ell$; this membership brings in the further condition $|J(U)| < \alpha_0$. Suppose that M_1, $M\alpha_0$ and the number of sites of $\mathbb{T}^{(k)}$ satisfy the hypotheses of Lem. 3.24 [stated technical lemma; Group A], which provides G and H_k. Finally, let A be small. On $Q_k^\sharp(U)A + C_k(U,A) = 0$, in the Landau gauge, one has $A = A_1 - \mathsf{H}_k C_k^{(2)}(A_1) + \dots$ with $Q_k^\sharp(U)A_1 = 0$, and

$$A(e^{i\eta_k A}U) = A(U) + \langle J(U), A_1\rangle + \tfrac{1}{2}\langle A_1, \Delta_U^{\mathrm{W}} A_1\rangle - \langle \mathsf{H}_k C_k^{(2)}(A_1), J(U)\rangle + O(|A_1|^3).$$

In this expansion H_k denotes the right inverse $\mathsf{H} = GQ^*(QGQ^*)^{-1}$ of $Q = Q_k^\sharp(U)$, built from the G of § 3.12.1; with this choice the relation $RD^*GQ^* = 0$ is exact. For the derivative of $V \mapsto U_k(V)$ the corresponding objects are G_1, H_1. The term $O(|A_1|^3)$ collects all terms of order ≥ 3 in A_1. Indeed, the constraint is solved up to second order, and the solution is substituted into the Taylor expansion of the action. ∎

Proposition 3.10 (isometries and linearised objects)**.** Suppose that r preserves every lattice involved. Then $\mathsf{r}(e^{i\mathsf{a}}W_0) = e^{i\mathsf{r}W_0\,\mathsf{a}}\,\mathsf{r}W_0$, and

$$\mathcal{Q}^1(\mathsf{r}W_0, \mathsf{r}_{W_0}\mathsf{a}) = \mathsf{r}_{\mathcal{M}(W_0)}\mathcal{Q}^1(W_0,\mathsf{a}),$$
$$Q_k^\sharp(\mathsf{r}U)\,\mathsf{r}_U = \mathsf{r}_{\mathcal{M}^k U}\,Q_k^\sharp(U),$$
$$J(\mathsf{r}U) = \mathsf{r}_U J(U),$$
$$\Delta^{\mathrm{W}}_{\mathsf{r}U}\mathsf{r}_U = \mathsf{r}_U\Delta^{\mathrm{W}}_U,$$

and the corresponding relations hold for Q_k, Q_k', $\bar{R}$, u. For real W_0, the map r_{W_0} is an isometry both for the sup norm and for $\langle\cdot,\cdot\rangle$. These maps compose according to $(\mathsf{r}\mathsf{r}')_{W_0} = \mathsf{r}_{\mathsf{r}'W_0}\mathsf{r}'_{W_0}$. If the twisted action is replaced by the plain one (the action at $W_0 = \mathbf{1}$), these relations in general fail for a reflection reversing bonds. The plain and the twisted action are equal where $\mathrm{Ad}(W_0(b))$ acts as the identity on the reversed bonds b. Consequently, propagators, H_k, $\Delta^{(k)}$ and $C^{(k)}$ are covariant under all of W_4 only in the twisted sense, and § 3.12.1, § 4.5 and § 4.2 use the symmetry in this form. ∎

Proposition 3.11 (the fundamental equality; the transformation to the tower axial gauge)**.** Let U_0, A be as in Prop. 3.6. There exists exactly one u, defined on the bare sites, for which $(e^{i\eta_k A}U_0)^u$ is tower axial relative to U_0 and $\bar{R}_{U_0}^{(k)}u = \mathbf{1}$. Its averages are given by $\bar{R}_{U_0}^{(n)}u = u\,v_n$, so that $u = v_k^{-1}$ on $\mathbb{T}^{(k)}$, and

$$\mathcal{M}^k\big((e^{i\eta_k A}U_0)^u\big) = e^{i\mathcal{Q}_k^\sharp(U_0;A)}\,\mathcal{M}^k U_0\,:$$

once the gauge has been changed in this way, averaging of order k amounts to composing the straightened one-step averages k times. The transformation can be written explicitly. Let $x = x_0, \dots, x_k = y$ be the tower of block centres. Let $\mathsf{w}_j^{\mathrm{rel}}(z,x) = (e^{i\mathsf{a}_j}W_{0,j})(z,x)W_{0,j}(z,x)^{-1}$ be the relative contour variables of the straightened averages, let u_j be the block factors, and let T_{U_0} be the transport down the tower. Then

$$u(x) = \prod_{j=k-1}^{0} \mathrm{Ad}\left(\mathsf{T}_{U_0}(x_{j+1},x)\right)^{-1}\Big[\mathsf{u}_j(x_{j+1})^{-1}\,\mathsf{w}_j^{\mathrm{rel}}(x_{j+1},x_j)\Big],$$

$$|u - \mathbf{1}| \le C\sup|A|.$$

Consequently u depends locally and holomorphically on A and is of class $C\sup|A|$. At the bare scale, however, it fails to be regular: it jumps by $O(L^{j}\eta_{k}\sup|A|)$ between neighbouring bare sites belonging to different blocks of order j. ∎

Proposition 3.12 (two schemes; the linear parts compared; what carries over)**.** Bałaban's theorems of 1984–85 are proved for blocks that have their site at a corner and for single combs; this is scheme c. The map $\mathcal{M}$ uses centred blocks and means; this is scheme m. Since L is odd, one translation of his entire setting, by $\mathsf{e}_{j}=\frac{1}{2}(L^{j}-1)a_{K}(1,1,1,1)$ at level j, carries his blocks onto the centred blocks. Of the contour variable, the proofs of Lem. 3.2–Lem. 3.8 use only the following properties:

(i) there is one contour variable for each pair (y,x), and it is covariant;
(ii) its paths stay in the block and make fewer than L steps in each direction;
(iii) coarse translations carry the contour variables into one another;
(iv) it is G-valued on G;
(v) it is holomorphic near $\mathbf{1}$, with a mean of the $\mathsf{a}(\Gamma)$ as its linear part;
(vi) where the bond variables commute with one another, it is the exponential of that mean.

This list is one way of carrying out the axiomatisation that Bałaban mentions. The results of Lem. 3.2–Lem. 3.8 therefore hold in both schemes, with two exceptions. The first exception is covariance under isometries other than coarse translations. It fails for combs, since under an element of W_{4} a comb goes over into a comb of a different order, and a lowest corner into a different corner; this failure is precisely why the mean is used. The second exception is the three rules, which are formulated for $\mathcal{M}$ only. When the background is trivial, the two schemes have the same straightened linear part, $\mathsf{Q}_{\star}(\mathbf{1})=\mathsf{Q}^{\mathrm{s}}$, but different plain linear parts. In a background, let t denote a transport of U_{0} joining the two base sites of a block. Then

$$Q_{k}^{\sharp}(U_{0})=\mathrm{Ad}(t)\big(Q_{k}^{\mathrm{B}}(U_{0})+\mathsf{D}_{k}\big),$$
$$\|\mathsf{D}_{k}\|_{\mathrm{row}}\le C\alpha_{0},$$
$$\|\mathsf{D}_{k}\|_{\mathrm{col}}\le C\alpha_{0}\eta_{k}^{4},$$

and the same holds for Q_{k}'. Moreover, $\mathsf{D}_{k}(c,b)=0$ unless b lies under c. In each scheme, the linear part and the straight average of that scheme differ by a kernel of row norm $C\alpha_{0}$. In this leading part, a bare bond b parallel to c receives as its weight η_{k}^{5} multiplied by the count of segments $[x,x']$, $x\in B^{k}(c_{-})$, passing through b (a tent along the direction of c). Every such segment enters with the transport of U_{0} taken down a chain of contours and segments. The exact identity $Q_{k}=Q_{k}^{\sharp}-D\dot{v}_{k}$ is supplied in the present text. The identity involving D_{k} takes the place of the assertion of 1987 (109, p. 254) that the new average satisfies all the results of the paper on averaging operations. For real U_{0} the factor $\mathrm{Ad}(t)$ is orthogonal, so it cancels from $Q_{k}^{\sharp}(U_{0})^{*}\,a\,Q_{k}^{\sharp}(U_{0})$. Consequently, in the two schemes the forms defining G', R, G_{k}, H_{k} differ from each other by local forms whose relative size is $C\alpha_{0}$. Through this comparison, the operators built from $Q_{k}^{\mathrm{B}}(U_{0})$ are used in § 3.11. The operators are built afresh from $Q_{k}^{\sharp}(U_{0})$, and their bounds are derived from the case $U_{0}=\mathbf{1}$, in which, after the blocks have been aligned, the two linear parts are equal § 3.12.1. ∎

Proposition 3.13 (sharpness: fields of one colour)**.** Let $T\in\mathfrak{g}$ with $|T|=1$. For $U=e^{ifT}$, with f real and small (Prop. 3.3), everything is linear and explicitly computable. A constant curvature φ gives $\partial\mathcal{M}U(p')=e^{iL^{2}\varphi T}$, so the factor L^{2} of Prop. 3.4 is attained.

Now consider a seam. Let Ω_{1} be, locally, a slab $\{x_{1}\ge\mathrm{const}\}$. Let $z\in\mathbb{T}^{(1)}$ be the last coarse site outside it, and let p' be the coarse plaquette with corners $z,z+Le_{1},z+Le_{1}+Le_{2},z+Le_{2}$. Set $f(\langle x,x+e_{2}\rangle)=\mathsf{f}^{\Sigma}(x_{1})$ and $f=0$ on the other bonds, where $\mathsf{f}^{\Sigma}(t):=\sum_{t'<t}\mathsf{f}(t')$. Here $\mathsf{f}=\varphi$ on the $L-1$ bonds in direction 1 that lie in the block of z; on the $L-1$ bonds preceding them

the value is $-\varphi$, and everywhere else it is 0. The curvature has size at most φ and vanishes in Ω_1, while $\partial\mathcal{M}U(p') = e^{i\Theta T}$ with $\Theta = \frac{1}{2}L(L-1)\varphi$. Thus no bound $C\alpha_0$ with C independent of L can take the place of the bound $2L^2\alpha_0$ of Prop. 3.6.

At $U_0 = \mathbf{1}$, take a bond $c \in \Gamma_j$ astride a seam with outer end c_-. Such a bond has one end, its outer end, outside Ω_j; in § 3.11 it is called an outward bond. The weighted row sum equals $\sum_b \frac{|K_0(c,b)|}{\xi(b)} = \frac{1}{2}L(1-L^{-j}) + \frac{1}{2}(1+L^{-j})$, and its limit as $j \to \infty$ is $\frac{1}{2}(L+1)$. Indeed, along a segment $[x, x']$, a bare bond contained in $B^j(c_-)$ has $\xi(b) = L^{j-1}\eta_k$, while the remaining ones extend into $B^j(c_+)$, where the scale equals $L^j\eta_k$.

For a bond of Γ_j astride a seam, an additional block factor arises at its outer end. Moreover, the datum entering the straightened constraint (the reduced datum of § 3.11) is not the same as the plain datum. The factor is genuinely present. At $U_0 = \mathbf{1}$, take $\eta_k A = tT$ on the single bare bond $\langle z, z + e_\nu\rangle$, where z is the outer end of a bond of Γ_1, and $A = 0$ elsewhere. Then $\mathsf{u}(z) = \exp\left(it\frac{L^4-1}{8L^4}T\right) \neq \mathbf{1}$. Finally, $\|\mathsf{Q}^{\mathrm{pl}}\|_{\mathrm{row}} > 2L$. ■

Proposition 3.14 (the exact identities)**.** The transformation $\mathbf{T}$ preserves mass, positivity ($|\mathbf{T}\nu| \leq \mathbf{T}|\nu|$) and gauge invariance. It commutes with the isometries that map $\mathbb{T}^{(k+1)}$ onto itself, and

$$\mathbf{T}^n\nu = (\mathcal{M}^n)_*\nu,$$

$$\langle F(\mathcal{M}^k U)\rangle_{K,\ell,\beta} = \int F\, d(\mathbf{T}^k\rho_0) \Big/ \int d(\mathbf{T}^k\rho_0).$$

Consequently, whenever $F \in \mathfrak{A}$ has largest depth s, each of the numbers $\langle F \circ \mathcal{M}^{K-s}\rangle_{K,\ell,\beta}$, $K \geq s$, is the integral of one fixed function with respect to the normalised measures $\mathbf{T}^{K-s}\rho_0$, all on the same compact space $\mathcal{V}(\Lambda_{s,\ell})$. ■

Proof. The Borel property of the actual average follows from its three defining rules. Fix a coarse bond c, and put

$$\mathcal{N}_c = \bigcup_{x,\Gamma,\Gamma'} \{U : Y_{x,\Gamma,\Gamma'}(U) = -\mathbf{1}\},$$

with the finite index sets used in the definition of the average. Each holonomy is continuous, so $\mathcal{N}_c$ is closed. The small-curvature construction makes the first rule valid on the open set $\mathrm{Dom}(c)$. The other two rules apply respectively on

$$\mathrm{Dom}(c)^c \setminus \mathcal{N}_c, \qquad \mathrm{Dom}(c)^c \cap \mathcal{N}_c.$$

The first rule is analytic; the second is continuous on its domain because the intrinsic logarithm is continuous on $G \setminus \{-\mathbf{1}\}$; the last is the continuous straight-path transport $U[c]$. Thus the bond average is Borel on this finite partition. The configuration space has finitely many independent positive-bond coordinates, so $\mathcal{M}$ is a Borel map on the whole configuration space.

In particular $K(U, \cdot) = \delta_{\mathcal{M}(U)}$ is a probability kernel: $K(U, B) = \mathbf{1}_B(\mathcal{M}(U))$ is Borel for each Borel set B. Its defining identity is

$$\int F(V)\, K(U, \mathrm{d}V) = F(\mathcal{M}(U))$$

for every bounded Borel F. Integration against the input measure gives the asserted pushforward identity. Kernel normalization proves mass preservation, and positivity gives $|\mathbf{T}\nu| \leq \mathbf{T}|\nu|$. Iterating the same identity gives $\mathbf{T}^n\nu = (\mathcal{M}^n)_*\nu$. Gauge covariance gives gauge invariance of the pushforward, since every coarse gauge transformation extends to a fine one. The same argument with lattice isometries gives their commutation with $\mathbf{T}$. Apply the iterated identity to $\nu = \rho_0\, \mathrm{d}U$ and divide by its positive total mass to obtain the expectation formula. □

Proposition 3.15 (soft and hard constraints)**.** Each kernel t of the kind described above preserves mass, positivity and gauge invariance. In the case of linear site fields and of block means Q, the Gaussian kernels $t_{\mathfrak{a}} \propto e^{-\frac{\mathfrak{a}}{2}|\Psi - Q\Phi|^2}$, taken after the canonical rescaling by $L^{(d-2)/2}$, form a family that is closed under composition. In these kernels Φ, Ψ are site functions in dimension d, Q is the rescaled block mean, and Q_k is the mean over B^k, rescaled in the same way; the composition law reads

$$t_{\mathfrak{a}} * t_{\mathfrak{a}_k} = t_{\mathfrak{a}_{k+1}},$$
$$\mathfrak{a}_{k+1}^{-1} = \mathfrak{a}^{-1} + L^{-2}\mathfrak{a}_k^{-1},$$
$$\mathfrak{a}_k \to \mathfrak{a}(1 - L^{-2}),$$

and these kernels give rise to the propagators $(\Delta + \mathfrak{a}_k Q_k^* Q_k)^{-1}$, where $\Delta = D^*D$ acts on site functions. For the delta kernel the fibres are nested, and over regular data, up to gauge,

$$\{\mathcal{M}^{k+1}U = W\} = \bigsqcup_{\mathcal{M}(V)=W} \{\mathcal{M}^k U = V\}$$
$$\Rightarrow$$
$$U_k(V_k^{\mathrm{bg}}) = U_{k+1}(W),$$
$$V_k^{\mathrm{bg}} = \mathcal{M}^k U_{k+1}(W),$$
$$U_j(\mathcal{M}^j U_k(V)) = U_k(V) \ (j \le k) :$$

every background therefore coincides with the background of its own averages, and V_k^{bg}, with $\mathcal{M}^k U$ taken in the block axial gauge, is in that gauge the critical point of $V \mapsto A(U_k(V))$ on $\{\mathcal{M}(V) = W\}$. Suppose that $\Delta \ge 0$ is positive on $\ker Q$ and that Q is surjective. Then, under the constraint $QA = B$, the form $\frac{1}{2}\langle A, \Delta A\rangle$ attains its minimum at $\mathsf{H}B$, where $\mathsf{H} = \mathsf{G}Q^*(Q\mathsf{G}Q^*)^{-1}$ and $\mathsf{G} = (\Delta + aQ^*Q)^{-1}$. This holds for every $a > 0$. *Remark.* Let W be real and lie in a neighbourhood of $\mathbf{1}$; this neighbourhood may depend on k and on the torus. Then V_k^{bg} is, in that gauge, the unique point of minimum of $V \mapsto A(U_k(V))$ on $\{\mathcal{M}(V) = W\}$. ∎

Proposition 3.16 (the steps with their insertions are exact)**.** Each operation of **T** and of **R** consists of finitely many positive kernels, one for each outcome of the operation, whose sum is a probability kernel. These kernels depend on the auxiliary choices. Once the choices are fixed, the kernels depend neither on β nor on the density. When the run makes its own choices, the operations themselves vary with β. For a bounded, gauge-invariant, complex-valued $\mathfrak{w}$,

$$\tau_h^{\mathfrak{w}}(dV) = \int dU\, \mathfrak{w}(U)\rho_0(U)\, \mathsf{K}_h(U, dV),$$
$$\sum_h \mathsf{K}_h(U, \mathcal{V}_k) = 1.$$

Consequently $\tau_h \ge 0$ and $|\tau_h^{\mathfrak{w}}| \le \sup|\mathfrak{w}|\,\tau_h$. Moreover $\int \rho_k = Z_{K,\ell,\beta}$ and $\int \rho_k^w = Z(w)$ at every level, on a torus that is large compared with the cubes of side $M\check{R}_j$ of every level. This holds irrespective of the gauges and representatives chosen. It also holds for every real sequence of numbers t among those admitted in §4.17.1.45. These are the sequences whose terms are all $\ge \gamma^{-2}$, whose differences obey two-sided bounds of the same type as the flow inequalities of Prop. 3.18, and for which $\check{R}_j \le L\check{R}_{j+1}$. The sum over outcomes of the operations of **T** is $\mathcal{M}_*$. By contrast,

$\mathbf{R}$ preserves only those integrals of F that do not depend on the components on which it acts; in general, therefore, $\rho_k \neq \mathbf{T}^k\rho_0$. *Exact insertion:* for every gauge-invariant F,

$$\langle F(\mathcal{M}^k U)\rangle_{K,\ell,\beta} = \sum_h \int F^{(h)} d\tau_h \Big/ \sum_h \int d\tau_h,$$

$$|F^{(h)}| \leq \sup |F|,$$

where $F^{(h)}$ denotes the conditional expectation, under the joint law constructed from the kernels, of $F(\mathcal{M}^k U)$ given (h, V_k). If F depends only on fields away from those components, then $F^{(h)} = F$. *Covariance:* $\mathsf{r}_*\tau_h = \tau_{\mathsf{r}h}$ for every isometry that preserves all the partitions involved. An individual $E^{(j)}(X,\cdot)$ is covariant under the translations that preserve π_j. An individual $R^{(j)}(X,\cdot)$ is covariant under the translations that preserve the $M\check{R}_j$-cubes. Covariance under all isometries r of $\mathbb{T}^{(j)}$ holds only for the unrestricted pointed totals $\sum_{X\ni z} E^{(j)}(X, U_j, z)$. These totals are taken at $U_j = U_j(V)$, where V and $\mathsf{r}V$ both belong to a class of real block fields of level j lying near the regular ones § 4.5. Wherever the covariance of pointed totals is invoked below, it has this meaning. ■

Lemma 3.17 (units; the delta function resolved; regrouping)**.** When the plaquette variables are close to $\mathbf{1}$, $\xi^{d-4}a_p = \frac{1}{2}\xi^d\langle F_U, F_U\rangle(1 + O(\xi^4|F_U|^2))$. Hence, for $d = 4$, Wilson's action is a Riemann sum for $\frac{1}{2}\int|F|^2$ whatever the unit of length. A change of unit leaves A, ρ_k and x_k unchanged. On small fields, $\mathcal{M}(e^{i\mathcal{A}}V_k^{\mathrm{bg}})V_{k+1}^{-1} = \exp i\mathcal{Q}_1(L\mathcal{A})$ and $\mathcal{Q}_1(L\mathcal{A}) = LQ\mathcal{A} + C^{\mathrm{pl},(2)}(\mathcal{A}) + \dots$. Here $\mathcal{A}$ carries no factor of length. Up to the scaling of its argument by L, the map is $\mathcal{Q}_1(\cdot) = \mathcal{Q}_1(V_k^{\mathrm{bg}};\cdot)$, and this scaling accounts for the factor L in front of its linear part. The average here is the plain $Q = Q_{V_k^{\mathrm{bg}}}$, and $C^{\mathrm{pl},(2)}$ is the second-order part of the plain remainder $C_1^{\mathrm{pl}}(V_k^{\mathrm{bg}}, L\mathcal{A})$. The symbol $C_k^{(2)}$ is reserved for the straightened average.

An analytic substitution $\mathcal{A} = \psi^{\mathrm{lin}}(\mathcal{A}') = \mathcal{A}' - \mathsf{h}\tilde{D}(\mathcal{A}')$ changes only the variables on the pivot bonds. It turns the delta function into the linear one, $\delta(LQ\mathcal{A}')$, which is solved for a single pivot bond $b_0(c)$ for each coarse bond. The substitution comes from the implicit function theorem. In it, h inverts LQ on the pivot bonds, $LQ\mathsf{h} = 1$. The linearising correction $\tilde{D}$ is of second order and is written $\widetilde{\mathsf{D}}$ in § 4.5. It is the unique small solution of $\mathcal{Q}_1\big(L(\mathcal{A}' - \mathsf{h}\tilde{D}(\mathcal{A}'))\big) = LQ\mathcal{A}'$, obtained as the fixed point of a contraction. For F supported in this regular chart, and with the free integral restricted to its inverse image, the local coarea identity is

$$\int dV_k\, \delta(\mathcal{M}(V_k)V_{k+1}^{-1})F(V_k) = z^{(k)} \int \prod_{b\neq b_0(c)} d\mathcal{A}'(b)\; e^{\mathrm{Tr}\log(1-\mathsf{h}\,\partial\tilde{D}) + \sum_b \log\sigma_{\mathrm{H}}(\mathcal{A}(b))}\, F\big(e^{i\psi^{\mathrm{lin}}(\mathsf{C}\mathcal{A}')}V_k^{\mathrm{bg}}\big),$$

where σ_{H} is the density of Haar measure in exponential coordinates. Once this density is divided by its value $\sigma_{\mathrm{H}}(0)$ at $\mathbf{1}$, the Jacobians are $1 + O(\delta_k)$ for each bond. They are local and analytic, and they are included in $\mathbb{P}^{(k)}$. The factor $\sigma_{\mathrm{H}}(0)$, taken once for each non-pivot bond, is absorbed into the constant of the step. For every number b,

$$-x_k A + \sum_X E^{(k+1)}(X,\cdot) = -(x_k - b)A + \Big(\sum_X [E^{(k+1)}(X,\cdot) - E^{(k+1)}(X,\mathbf{1})] - bA\Big) + \sum_X E^{(k+1)}(X,\mathbf{1}),$$

and, summed over the levels, $-A(x_k(\cdot), U) - \sum_{j\leq k} b_j A(\varphi_j, U) = -\beta A(U)$. The constant of a step depends on the regions of that step. Let $|\Omega^{(k)*}|$ be the number of non-pivot bonds of $\mathbb{T}^{(k)}$ in Ω, and $\#_{k+1}(\Omega)$ the corresponding number of sites of $\mathbb{T}^{(k+1)}$. Let $z_{\mathfrak{G}} = \mathfrak{z}(x_k, \varepsilon_k)$ be the normaliser of $\mathfrak{G}_k$ for each block site (1a.3), and $\nu_{\mathfrak{g}} = \log\int_{\mathfrak{g}} e^{-|Y|^2/2}dY$. Then

$$\mathsf{E}^{(k)}(\Omega,\Lambda) = \big(\log\sigma_{\mathrm{H}}(0) - \tfrac{1}{2}(N^2-1)\log x_k\big)|\Omega^{(k)*}| - (L^4-1)\log z_{\mathfrak{G}}\,\#_{k+1}(\Omega) + \nu_{\mathfrak{g}}|\Lambda^{(k)*}| + \sum_{X\subset\Lambda}\big[E^{(k+1)} + R^{(k+1)}\big](X,\mathbf{1}),$$

together with $-\mathcal{E}_k = \sum_{j<k}\mathsf{E}^{(j)}(\Omega_{j+1},\Lambda_{j+1})$ and $E = \sum_{j<K}\mathsf{E}^{(j)}(T_\ell,T_\ell)$. Here $\mathcal{E}_k$ depends on the label, and at the all-small-field label the vacuum constant E equals $-\mathcal{E}_K$. ■

3.1.1.1. Primitive hypotheses for the joint step. Write $\mathcal{H}_{\mathrm{step}}$ for the following inputs on every actual determining region, stored-field fibre and allowed parameter domain. Retain the background and substitution admission in A1–A4 of § 2.4, and the actual matrix/core, potential and window estimates used in § 4.5. Their operations and local families must obey the declared covariance and common-lift compatibility rules.

For the history operations retain the common-core integrable envelope, kept Gaussian integral and owned failure-energy certificates in eq. 4.cdlxviii, eq. 4.dii and eq. 4.div of § 4.18. Use the stated label representation, common vacuum normalization and finite read ownership. Drop-out evaluations must preserve their failure certificates under the supremum inside the integral. In the sourced application also retain eq. 4.dxiii on the same fixed representation, or the ordinary complex/prepaid real extension hypotheses stated there that replace it.

The formation, geometric counting and integration arguments give their conditional successor bounds under these inputs. Any primitive estimate not yet verified for all actual histories remains a technical hypothesis of the closure assertions below. Class invariance C4 is their target, not an assumption used to prove them. Exact transport and mass preservation follow separately from the normalized measurable moves in C1. A homogeneous regular-sector construction alone does not verify the all-history premises in $\mathcal{H}_{\mathrm{step}}$.

Proposition 3.18 (the invariant form)**.** Assume $\mathcal{H}_{\mathrm{step}}$. Let $\mathfrak{F}_k$ denote the complete presentations in $\mathfrak{C}_k(\theta)$, written through their represented measures of the form below. Membership includes the specified domains, stored variables, measures, marks and reading rules. The step acts on this presentation; $\mathbf{RT}\mathfrak{F}_k$ below denotes its successor presentations. In it the $E^{(j)}$ are pointed families, the pointed totals of which are covariant with respect to every isometry of $\mathbb{T}^{(j)}$, and b_j is their coefficient § 4.2. A step does not preserve the form and the bounds alone §4.17.1.45. The measures of the class, with their bounds, are

$$\sum_{\mathcal{S}} \mathbf{T}'_k(\mathcal{S})\big[\chi_{k,\mathcal{S}}\, e^{A_{k,\mathcal{S}}(U_k(\mathbf{V}))}\big],$$

$$\|E^{(j)}\|_{j,\kappa} \le E_0,$$

$$\|R^{(j)}\|_{j,\kappa} \le g_j^{\kappa_0},$$

$$|B^{(j)}(X,\cdot)| \le \mathsf{B}\, e^{-\kappa d_j^\natural(\mathcal{S};X)},$$

$$\mathsf{M}(i,X) \le e^{-(2-c')\,p_0(g_i)},$$

where B is a constant chosen before γ (§ 4.17, §4.17.1.45), and where the last inequality, the *mass bound*, concerns those components X that $\mathbf{R}$ has acted on at some level $i \le k$. Such a component was healed, or left standing as an island, or else it dropped out at a failed test, either of a stage or of the main step of $\mathbf{R}$. No assumption is made on a drop-out: its bound follows from the bound for the islands of the lower levels, the Wilson weight of the carrying term over X being counted together with its operation. Let the constants be chosen in order, and let the torus be large compared with the $M\check{R}_j$-cubes at each level. If, in addition, $g_j \le \gamma$ and

$$\lambda_\sharp(j-i) - c_5 \le x_i - x_j \le B_\sharp(j-i)$$

$$(i \le j \le k+1),$$

hold, then $\rho_0 \in \mathfrak{F}_0$ and $\mathbf{RT}\mathfrak{F}_k \subset \mathfrak{F}_{k+1}$ *with the same constants.* No family of a level $j \le k$ is altered; only the background at which they are evaluated changes, $U_k \to U_{k+1}$. The new families come out of $\mathbf{T}$ with a surplus, and $\mathbf{R}$ uses it up: radii $(1+\beta_{\mathrm{s}})\alpha_{i,k} \ge \alpha_{i,k+1}$ and rate $(1+4\beta_{\mathrm{s}})\kappa$, where β_{s} is a fixed fraction. For given $(x_0,\dots,x_j)$, the activities $E^{(j)}(X,\cdot)$, $R^{(j)}(X,\cdot)$ do not depend on the label. When X does not wrap around the torus, they depend neither on the torus nor on the cutoff. Further, $x_{k+1} = x_k - b_{k+1}(x_0,\dots,x_k)$ with $|b_{k+1}| \le B_\sharp$. When the label has no large fields, $|\mathbf{E}_k(U_k)| + |\mathbf{R}_k(U_k)| \le C|\mathbb{T}^{(k)}|$. For a general label, every layer is measured in its own scale, and for each $j \le k$ one adds the sites of level j that lie in a rim around the large-field region Z_j (§ 4.2, § 5.2.1). Finally, stability holds if the flow inequalities hold for all $i \le j \le K$:

$$\chi_k\, e^{-x_k A(U_k) - E_- |\mathbb{T}^{(k)}|} \le \bar\rho_k,$$

$$\int \bar\rho_k \le e^{E_+ |\mathbb{T}^{(k)}|},$$

$$E_+ \le \tfrac{3}{2}(N^2-1)\log x_k + O(1).$$

The large-field factor (part §4.17.1.40). The operation $\mathbf{T}_k(Z)$ contains delta functions, tests and conditional Gaussian weights. Apart from the core factors of the islands in Z, it contains no Wilson weight. Hence, for a component produced by failed plaquette tests, $\mathbf{T}_k(Z)1$ carries no small factor. If no $\mathbf{R}$ has acted on Z, then under the tests of the term it obeys only the bound $\le \exp[C\sum_{t\le k} \#_t(Z_t \cap Z)]$. The small factor lies in three places. They depend on pairwise disjoint sets of plaquettes, factors and bonds, so no part of the small factor is used twice. The first is the Wilson weight: near a failed plaquette test of step j, $x_j A \ge c\, p_0(g_j)^2$. The second is the kept gauge-fixing exponents; together with the Wilson sum, they pay for a failed approximate fluctuation test. The third is the Gaussian forms of the fluctuation fields, which pay for a failed fluctuation test.

(1) Fix a history, and a component X of Z_k that has not been acted on by any $\mathbf{R}$. Let $(j_i, \mathfrak{c}_i)$ be the creation cells of the history in X, namely those cells of the creation set of step j_i, which are $M\check{R}_{j_i}$-cubes, that meet a failed test §4.17.1.40. Then, for all values of the exterior variables, and with the norm taken as the total variation of that part which the term of the history carries,

$$\big\|\mathbf{T}_k(X)\big[\chi_{k,\mathcal{S}}\, e^{-A(x_k(\cdot),U_k)}\big]\big\| \le \prod_i e^{-c\,p_0(g_{j_i})^2}\, \exp\Big[C\sum_{t\le k} \#_t(Z_t\cap X)\Big]:$$

there is one factor for each creation, taken at the coupling of its own step, and one constant for each site.

(2) Bounds of this kind do not multiply over the components, because the background couples the components to one another.

(3) What does multiply, piece by piece, is the pointwise majorant on which (1) rests. A piece (t, D) consists of a component D of the large-field region formed at step t, whose enlargement $D^\sim$ shares no cube with any component that $\mathbf{R}$ has acted on at a level in $[t,k]$, together with the stored variables $\xi_{t,D}$ belonging to it and the weight $\mathsf{w}_{t,D}$ of those variables. Throughout the support of the history,

$$A(x_k(\cdot),U_k) \ge A_{\mathrm{cr}},$$

$$e^{-A_{\mathrm{cr}}} \prod_{(t,D)} \mathsf{w}_{t,D} \le \prod_{\text{creations}} e^{-c\,p_0(g_t)^2} \prod_{(t,D)} \bar{\mathsf{w}}_{t,D}(\xi_{t,D}),$$

$$\int \bar{\mathsf{w}}_{t,D} \le e^{C\#_t(D)},$$

where A_{cr} denotes the Wilson sums near the creations in these pieces. Only those creations are counted on the right-hand side. Each $\bar{\mathsf{w}}_{t,D}$ is a function of the variables of its own piece only.

(4) After integration, take a whole label together with the list of those outermost components (i, X) that $\mathbf{R}$ has acted on and that still stand in the label. For some history having this label and this list, the bound carries one factor for each of its creation cells lying outside the components that $\mathbf{R}$ has acted on and that still stand in the label (islands and drop-outs; the creations inside such a component are spent in its mass), and one mass for each outermost one among these components:

$$\int \mathbf{T}'_k(\mathcal{S})\big[\chi_{k,\mathcal{S}}\, e^{-A(x_k(\cdot),U_k)}\big]dV_k \le \exp\Big[C\sum_j (1+\log x_j)\#_j(Z_j)\Big] \prod_{\text{creation cells } (t,\mathfrak{c})} e^{-c\,p_0(g_t)} \prod_{(i,X)} \mathsf{M}(i,X),$$

$$\mathsf{M}(i,X) = \frac{1}{|W_4|}\sum_{\omega} e^{c\,\mathsf{v}(\omega)} \int \sup_{\mathbf{p}} \big|e^{\Phi_\omega(\xi;\mathbf{p})}\big|\,\mu_\omega(d\xi),$$

where ω ranges over the local histories of X, each equipped with a hypercubic frame in W_4 of the gauge tree, and ξ denotes the stored variables. The variable $\mathbf{p} = (\mathbf{U}, \mathbf{J})$ ranges over the complex pairs on $X^\sim$ at which a term evaluates its factor e^{Φ_ω}. The exponent has the form $\Phi_\omega = \Phi^\circ_\omega(\xi) + \sum_f \Phi_{\omega,f}(\xi; \mathbf{p}|_{R_f})$ with $R_f \subset X^\sim$, and $\mathsf{v}(\omega)$ is a space-time volume. The supremum is taken inside the integral and the sum. This is the form of the mass for a healed component or an island. In the case of a drop-out, the Wilson weight of the carrying term over X appears inside the integral, and the supremum extends over the evaluations at ξ. These are the real configurations that are compatible with the stored variables and regular layer by layer, and at which every test listed by ω takes its recorded value, with the exception of the test whose failure made X a drop-out. They form a set fibred over ξ §4.17.1.40. The induction carries the mass bound at the coupling of level i of the action. In this bound the c of $2 - c$ is small, whatever the c in the weight $e^{c\,\mathsf{v}(\omega)}$. The mass evaluated at a later level does not exceed the mass at level i.

(5) The integral operation $\mathbf{T}'_k(X)$ on a component on which $\mathbf{R}$ acts contains, except in the case of a drop-out, a Wilson factor of its own, namely that of the core. When the component is healed or remains as an island, the total mass that is bounded is that of this operation, not that of $\mathbf{T}_k(Z)$:

$$\mathbf{T}'_k(X)\,1 \le \sup \exp\Big\{\sum_{j\le k} CM^4 \check{R}_j^5\, d'_j(Z_j)\Big\} \prod_{j\le k}\prod_i \exp\Big(-c\,p_0(g_j)^2\big(d'_j(Z_j^{(i)})+1\big) - 2p_0(g_j)\Big),$$

where the supremum extends over every admissible choice of the domains Z_j, $Z_j^{(i)}$ (Bałaban, Comm. Math. Phys. 122 (1989) 355, (1.79)), and the sums over the admissible sequences are counted among the costs. The inner product is taken over the regions $Z_j^{(i)}$ created at step j. For a component Z of Z_j which has been neither healed nor marked, the factor is written $\exp(-\kappa_j(Z) - 2p_0(g_{j(Z)}))$; here $j(Z)$ is the level at which the first creation in Z occurs. The *credit* satisfies $\kappa_j(Z) \ge \sum_{n=j+1}^{j+\mathsf{k}} CM^4 \check{R}_n^5\, d'_n(S^{n-j}Z)$, where the sum runs over the k further steps at which nothing is created. Here S denotes the hull in the next partition with a margin added. The number of $M\check{R}_n$-cubes that meet the region takes the place of the linear size d'_n, because a single cube has $d' = 0$ and still carries a cost. The credit covers the volume costs of every step until $\mathbf{R}$ acts on Z. Once no further step is owed, $\mathbf{T}'_k(X)1 \le e^{-(2-c)\,p_0(g_k)}$. Moreover, in the weighted sum over the labels, every component C of Z_k keeps a factor $\exp[-c\,p_0(g_k) - \kappa_1|C|_k]$, where $|C|_k$ is its number of cells. ∎

Proposition 3.19 (the small-field step as a map on actions)**.** Suppose that the tests and $\mathfrak{G}_k$ are formed at t, where t is large, and let $x \in \mathbb{C}$ with $|x - \mathsf{t}| \le c\,\mathsf{t}$. Suppose that the analyticity radii (α_0, α_1) of the levels $j \le k+1$ are small, with $CM\alpha_0 < \alpha_1$ and $\alpha_1^2 \le C\alpha_0$ at each level. Suppose also that they are comparable: the quotient of the radii of level $k+1$ by those of any level $j \le k$

lies between $(1+\beta_{\mathrm{s}})^{-1}$ and $(1+\beta_{\mathrm{s}})(k+2-j)^{1/2}$. These are the admissible systems of § 4.5. Let S be local, analytic and *renormalised* in the same way as $\mathbf{E}_k+\mathbf{R}_k$:

$$S=\sum_{j\le k}\Big[\sum_{z}\sum_{X\ni z}\big(G^{(j)}(X,\cdot,z)-G^{(j)}(X,\mathbf{1},z)\big)-e_jA\Big]+\sum_{j\in\mathcal{J}}\sum_{X}\big(H^{(j)}(X,\cdot)-H^{(j)}(X,\mathbf{1})\big),$$

$$N(S)=\sum_{j\le k}\omega_{k+1-j}\|G^{(j)}\|_{j,\kappa}+\sum_{j\in\mathcal{J}}\mathsf{q}_j^2\|H^{(j)}\|_{j,\kappa},$$

$$\omega_n\le Cn^CL^{-\alpha_{\mathrm{H}}n},$$

where the pointed totals of $G^{(j)}$ are covariant under all isometries of $\mathbb{T}^{(j)}$, and e_j is the coefficient § 4.2 of $G^{(j)}$. The totals of $H^{(j)}$ are invariant under the isometries of $\mathbb{T}^{(k+1)}$. Finally, q_j is the ratio of the analyticity radii of level $k+1$ to those of level j. Geometric weights occur only in the first sum. Both symmetries are part of the definition of "renormalised". The covariance of the pointed totals is what makes the bracket left after the subtraction of e_jA irrelevant § 4.2. Consider W in a class of real block fields, namely those at which every background formed in the step exists and yields pairs in the domains of the terms. Then, on the enlarged radii and uniformly in k,

$$\mathfrak{E}_{k+1}[\mathsf{t};x;S]=\log Z^{(k)}+\sum_{X\in\mathcal{D}_{k+1}}\mathsf{F}(X,U_{k+1})+\mathrm{const}$$

, where

$$|\mathsf{F}(X,\cdot)|\le C\big(|1-x/\mathsf{t}|+\delta(\mathsf{t})+\mathsf{s}\,N(S)+e^{-c\,\mathsf{t}\delta(\mathsf{t})^2}\big)\,e^{-(1+4\beta_{\mathrm{s}})\kappa d_{k+1}(X)},$$

with holomorphic dependence on x and on the parameters of S. The map $S\mapsto\mathsf{F}$ satisfies a Lipschitz bound with constant $C\mathsf{s}$; here $\mathsf{s}\ll1$ denotes the quotient of the fluctuation threshold by the analyticity radius. Everything above holds under the condition $\mathsf{s}\,(1+N(S))\le c$. Along the run, $G^{(j)}=E^{(j)}$ and $e_j=b_j$, while the $H^{(j)}$ are the aged $R^{(j)}$ of Prop. 3.20. ■

Proposition 3.20 (anatomy of step $k+1$)**.** The insertions are made in the following order. First, the plaquettes of the local backgrounds of V_{k+1} are tested. Next, on the blocks P'' that lie away from failures and from Z_k, where the $V(y,x)$ are defined, the factor $\mathfrak{G}_k(\cdot;P'')$ is inserted. Then the approximate fluctuation $V_k(V_{k,\square}^{\mathrm{bg}})^{-1}$ is tested, and this determines Ω_{k+1}. On that region V_k^{bg} is defined and $V_k=e^{i\mathcal{A}_k}V_k^{\mathrm{bg}}$ is set. Finally, $\mathcal{A}_k$ is tested, and this determines Λ_{k+1}. The insertions and tests just described, together with the regions that they determine, are those of Bałaban's form, whereas the sharper tests on collar and rim, together with their outcomes, are given in §4.17.1.1. As a formula,

$$(\mathbf{T}\rho_k)(V_{k+1})=\sum_{P,Q,R}\int dV_k\,\delta(\mathcal{M}(V_k)V_{k+1}^{-1})\;\chi_{k+1}\chi_{k+1}^{c}(P)\;\mathfrak{G}_k(V_k)\;\chi_k'\chi_k'^{c}(Q)\;\chi^{(k)}\chi^{(k)c}(R)\;\rho_k(V_k).$$

After Lem. 3.17 has been applied and the field has been scaled, $\mathcal{A}_k=g_kB$, a Taylor expansion about the critical point yields

$$x_k\big[A(U_k(V_k))+\mathrm{GF}_k(V_k)\big]=x_kA(U_{k+1})+\tfrac12\langle B,\Delta^{(k)}B\rangle+\sum_b O(g_k|B|^3).$$

Completing the square in $B|_{\Lambda_{k+1}}$ produces $C^{(k)}(\Lambda_{k+1})$, together with the conditional Gaussian that is kept in $\mathbf{T}^{(k)}(Z_{k+1})$:

$$(\mathbf{T}\rho_k)(V_{k+1}) = \sum_{\mathcal{S}'} \chi_{k+1}(\Omega_{k+1})\, \mathbf{T}^{(k)}(Z_{k+1})\mathbf{T}'_k(\mathcal{S})\Big[e^{A_{k,\mathcal{S}}(U_{k+1})+E_0^{(k)}(\Omega_{k+1})}\,\mathsf{I}\Big],$$

$$\mathsf{I} = Z^{(k)}(\Lambda_{k+1})\int d\mu_{C^{(k)}(\Lambda_{k+1})}(B)\,\chi^{(k)}(\Lambda_{k+1})\,e^{\mathbb{P}^{(k)}+\{A_{k,\mathcal{S}}\}'+\mathsf{W}^{(k)}},$$

where $E_0^{(k)}(\Omega)$ consists of the first two terms of $\mathsf{E}^{(k)}(\Omega,\Lambda)$. The measure $\mu_{C^{(k)}(\Lambda_{k+1})}$ is the Gaussian measure conditioned on the exterior fluctuation field, and its mean is linear in that field. The prime indicates that the quadratic part of $-x_k A$ has been absorbed into the measure. The term $\mathsf{W}^{(k)}$ is left over from a change of variables on the strip, that is, on the part of S_{k+1} lying along $\partial\Lambda_{k+1}$. Before any expansion, the logarithm is divided into three parts by interpolation:

$$\log\mathsf{I} = \int_0^1 dt_1\,\langle\{\mathbf{B}_k - A(x_k(\cdot)-x_k,\cdot)\}+\mathsf{W}^{(k)}\rangle_{t_1,1} + \int_0^1 dt_2\,\langle\{\mathbf{R}''_k\}\rangle_{0,t_2} + \log\mathsf{I}\big|_{t_1=t_2=0},$$

where $\mathbf{R}''_k$ contains the young $R^{(j)}$, namely those whose cubes are larger than a block. The logarithm is expanded § 4.5 and regrouped (Lem. 3.17, with b_{k+1}, φ_{k+1} and $\mathsf{E}^{(k)}(\Omega_{k+1},\Lambda_{k+1})$). This gives the form of level $k+1$:

$$\begin{aligned}(\mathbf{T}\rho_k)(V_{k+1}) &= \sum_{\mathcal{S}'}\chi_{k+1}\,\mathbf{T}'_{k+1}(\mathcal{S}')\exp\Big[A_{k,\mathcal{S}}(U_{k+1}) + \sum_{X\subset\Lambda_{k+1}}[E^{(k+1)}+R^{(k+1)}](X,U_{k+1})\\ &\quad + \sum_{X\cap Z^{\sim}_{k+1}\neq\emptyset} B^{(k+1)}(X;U_{k+1},\underline{\mathrm{f}}) + E_0^{(k)}(\Omega_{k+1}) + \nu_{\mathfrak{g}}|\Lambda^{(k)*}_{k+1}|\Big]\\ &= \sum_{\mathcal{S}'}\chi_{k+1}\,\mathbf{T}'_{k+1}(\mathcal{S}')\exp A_{k+1,\mathcal{S}'}(U_{k+1});\end{aligned}$$

after this, $\mathbf{R}$ acts. The terms with $X\subset\Lambda_{k+1}$ coincide with the terms of the pure small-field integral (among the $R^{(k+1)}$, here and in $\mathsf{E}^{(k)}$, a label keeps those whose X lies one cell inside Λ_{k+1}, and the others are counted among the $B^{(k+1)}$); this coincidence is the locality stated in Prop. 3.18. ■

Proposition 3.21 (a model case: the free field; the renormalisation group as nested orthogonal projections)**.** Let the *free curvature* F be the standard Gaussian vector in the Euclidean space $\Omega^2_{\mathrm{ex}} = \operatorname{ran}\mathrm{d}$ of exact 2-cochains on the bare lattice. Here d is the coboundary taking real 1-cochains to real 2-cochains on the bare lattice, and $\mathrm{d}_{(j)}$ is the coboundary of $\mathbb{T}^{(j)}$. This vector stands for one colour of the gauge field, taken in the quadratic approximation and up to closed forms. The plain linear average and the straight linear average induce the same average of curvatures, called the *plaquette average*:

$$\mathrm{d}_{(j)}(\mathsf{Q}^{\mathrm{pl}})^j = \mathrm{d}_{(j)}(\mathsf{Q}^{\mathrm{s}})^j = \mathsf{A}^{(j)}\mathrm{d},$$

$$(\mathsf{A}^{(j)}F)(P) = L^{-4j}\sum_{z\in B^j(y)}\ \sum_{p\in\mathrm{Sq}^{(j)}(z;P)} F(p),$$

where $\mathrm{Sq}^{(j)}(z;P)$ is the set of L^{2j} bare plaquettes which together cover the plaquette P of $\mathbb{T}^{(j)}$ with corner y, once it is translated to z. Put $\mathsf{W}_j := \Omega^2_{\mathrm{ex}} \ominus (\Omega^2_{\mathrm{ex}}\cap\ker\mathsf{A}^{(j)})$. These are subspaces of Ω^2_{ex} whose level is written as a subscript, and they are distinct from the term $\mathsf{W}^{(k)}$ of Prop. 3.20, whose level is written as a superscript. Let P_{W} denote the orthogonal projections. Then the spaces are nested, $\mathsf{W}_0\supset\mathsf{W}_1\supset\cdots\supset\mathsf{W}_K$. Given the block field of level k, the conditional expectation of

F is $P_{\mathsf{W}_k}F$. This is the curvature of the linear background, namely the exact cochain of smallest norm that has the prescribed plaquette averages. The covariances of the background and of the fluctuation are two complementary orthogonal projections, $\mathbb{E}\big[P_{\mathsf{W}_k}F(p)\,P_{\mathsf{W}_k}F(q)\big] = P_{\mathsf{W}_k}(p,q)$ and $P_{\mathrm{ex}} = P_{\mathsf{W}_k} + P_{\mathsf{W}_k^{\perp}}$, where P_{ex} is the projection onto Ω^2_{ex}. Finally,

$$F = P_{\mathsf{W}_K}F + \sum_{j=1}^{K} \mathcal{C}_{(j)}F,$$

$$\mathcal{C}_{(j)} = P_{\mathsf{W}_{j-1}} - P_{\mathsf{W}_j} = \mathcal{H}_{j-1}C_j^{\mathrm{tr}}\mathcal{H}_{j-1}^{*},$$

$$|\mathcal{C}_{(j)}(p,q)| \le C\eta_{j-1}^{4} e^{-\delta|x(p)-x(q)|/a_{K-j+1}},$$

where the summands are independent and the covariance of each summand is its own projection. Here, for a block field B of level $j-1$, $\mathcal{H}_{j-1}B$ denotes the curvature of its linear background, namely the exact 2-cochain on the bare lattice of smallest norm satisfying $\mathsf{A}^{(j-1)}\mathcal{H}_{j-1}B = \mathrm{d}_{(j-1)}B$; and C_j^{tr} is the fluctuation covariance, in a tree gauge, of step j. Thus step j integrates out precisely $\mathcal{C}_{(j)}F$, and the amplitude and range of each slice are those of its own scale. In this model the semigroup law is the nesting of the spaces, and the delta constraint is conditioning. The nesting of backgrounds takes the form $P_{\mathsf{W}_j}P_{\mathsf{W}_k} = P_{\mathsf{W}_k}$ $(j \le k)$. Three features of the full theory are absent from the model: the fibre is flat, no coupling runs, and no field is large. The current of a linear background is not zero: $\mathrm{d}^{*}P_{\mathsf{W}_k}F = ((\mathsf{Q}^{\mathrm{s}})^{k})^{*}\mathrm{d}^{*}_{(k)}\phi$, which is the image of a Lagrange multiplier. What is missing is the term $\langle \mathsf{H}_k C_k^{(2)}, J\rangle$ of Prop. 3.9, because $C_k^{(2)} = 0$ on a flat fibre. In the non-linear theory the current and the multiplier do vanish, at $U = \mathbf{1}$. For this reason the linearisation at that point of the minimiser of A subject to $\mathcal{M}^{k}U = V$ is the minimiser of $\frac{1}{2}\|\mathsf{da}\|^2$ subject to the linearised constraint, and the second-order part of the averages makes no contribution. Hence the model is the linearisation at $\mathbf{1}$. It serves as the template for §6.1.1.1, §6.1.1.2. ∎

Lemma 3.22 (axial gauges)**.** Consider a box of side D on which $|\partial V - \mathbf{1}| \le \epsilon$. Among the representatives $V^{\psi'}$ with $\psi'(y_P) = \mathbf{1}$, the configuration $V^{\mathrm{ax}} = V^{\psi}$, $\psi(w) = V(\Gamma_{y_P,w})$, is the only one that equals $\mathbf{1}$ on the tree. The non-abelian Stokes theorem expresses $V^{\mathrm{ax}}(b)$ as a product of conjugated plaquette variables, at most $|b_- - y_P|_1$ in number:

$$V^{\mathrm{ax}}(b) = \prod_{i=1}^{m} \psi(w_i)\,\partial V(p_i)^{\pm 1}\psi(w_i)^{-1},$$

$$|V^{\mathrm{ax}}(b) - \mathbf{1}| \le |b_- - y_P|_1\,\epsilon \le 4D\epsilon;$$

Suppose that $D\epsilon \le c$ and that $F = \frac{1}{i}\log \partial V$ is Hölder along combs; it is distinct from the field strength F_V introduced in the Notation, §2, being the logarithm of each plaquette variable. Then $\frac{1}{i}\log V_{\mu}^{\mathrm{ax}}(x) = \sum_{\rho<\mu}(x_\rho - y_\rho)F_{\rho\mu}(y_P) + O(|x-y_P|^2)\times(\text{Hölder constant} + \epsilon^2)$, which is the expansion used in § 4.2 and §4.17.1.3. On a separated sequence suppose that $U_0, U \in \mathfrak{U}_k(\Omega_\bullet, \alpha_0)$, with $\alpha_0 > 0$ below the fixed geometric ceiling, and that their raw averaged data differ by at most $\alpha_1 \ge 0$ on every interior and outward determining row. Let $\hat{g}$ be the real gauge, equal to one at every layer top, whose descending construction aligns the averaged contour variables of $U^{\hat{g}}$ with those of U_0. The construction following eq. 3.cccxx gives $|\mathcal{M}^{j}(U^{\hat{g}})(b)\mathcal{M}^{j}U_0(b)^{-1} - \mathbf{1}| \le C_L(\alpha_0+\alpha_1)$ for $0 \le j \le k$ and every carried level-j bond b inner to Ω_0 with an endpoint in Ω_j, uniformly in k. This is a raw group-element bound, before division by the local spacing. On a single level (compare the rigidity Lem. 3.8), $\mathcal{A}_k$ is defined, and is $O(\varepsilon_k)$, only after the gauge has been fixed. ∎

Proposition 3.23 (Faddeev–Popov identities with Jacobian one)**.** Let a compact group H act, and let $\mathsf{k} \ge 0$ satisfy $\int_H \mathsf{k}(h\cdot x)\,dh = 1$. Then for every finite complex measure ν and every bounded invariant F,

$$\int \nu(dx) \int_H dh\, \mathsf{k}(h \cdot x)\, F(h \cdot x) = \int F\, d\nu;$$

The measure obtained in this way is constructed as follows: one chooses h with law $\mathsf{k}(h \cdot x)dh$ and replaces x by $h \cdot x$. This measure equals $\mathsf{k}\,\bar{\nu}$, where $\bar{\nu}$ is the gauge average of ν. When ν is invariant, it is the product $\mathsf{k}\,\nu$. If $u = \mathbf{1}$ on the coarse sites, then $V^u(y,x) = V(y,x)u(x)^{-1}$. Where $V(y,x)$ is undefined, no factor is inserted at the pair (y,x), and with this convention

$$\int \prod_x du(x)\, \mathfrak{G}_k(V^u; P'') = 1,$$

$$(\mathbf{T}\rho_k)(V_{k+1}) = \int dV\, \delta(\mathcal{M}(V)V_{k+1}^{-1})\, \mathfrak{G}_k(V; P'')\, \rho_k(V).$$

In addition, $\mathfrak{G}_k(V^v) = \mathfrak{G}_k(V)$ whenever v is constant on blocks. The same holds for delta functions on spanning trees. Let T be a tree whose root lies outside the sites of the group, and let V_0 be any configuration on T. Then a unique u satisfies $(u \cdot V)|_T = V_0$, and the measure dV splits into factors along the tree:

$$\int f\, dV = \int du \int \prod_{b \notin T} dS(b)\, f(u \cdot (S \sqcup V_0)),$$

so

$$\int dV\, \delta_T(VV_0^{-1})\, \rho F = \int dV\, \rho F$$

for invariant ρ, F. Finally, the delta function can be removed: the gauge average of $\Psi\, \delta_T(VV_0^{-1})dV$ is $(\Psi \circ \pi_T)\, dV$, where π_T is the projection onto the slice along the orbits. ■

Lemma 3.24 ([stated technical lemma; Group A)**.** (the Landau gauge)] Let U be tower axial and let $W = U^{u^{-1}}$. Then $\mathsf{n}_W = u^{-1}\bar{R}_{U_0}u$, so a u with trivial averages leaves the straightened averages exactly unchanged. In linearised form this reads $Q_j^\sharp(U_0)D\lambda = D_{\mathcal{M}^j U_0} Q_j'(U_0)\lambda$. Hence $Q^\sharp A$ does not change under $A \to A - D\lambda$, $\lambda \in N(Q')$. Up to terms that contain $J(U_0)$, gradients are null directions of Δ^{W}. Indeed $D^*J(U_0) = 0$, and, writing λ_+ for $\lambda(b_+)$ transported to b_-,

$$\langle A - D\lambda, \Delta^{\mathrm{W}}(A - D\lambda)\rangle = \langle A, \Delta^{\mathrm{W}}A\rangle - \langle i[\lambda, A] - i[A, \lambda_+] - i[\lambda, D\lambda],\, J(U_0)\rangle.$$

The projection is given by

$$R = 1 - G'Q'^*(Q'G'^2Q'^*)^{-1}Q'G',$$

$$\operatorname{ran} R = \{f : Q'G'f = 0\},$$

$$\inf_{\lambda \in N(Q')} \|D^*A - \Delta\lambda\|^2 = \|(1-R)D^*A\|^2,$$

and $R\Delta\lambda = \Delta\lambda = G'^{-1}\lambda$ for $\lambda \in N(Q')$. The map $\lambda \mapsto (R\Delta\lambda, Q'\lambda)$ is a bijection onto $\operatorname{ran} R \times \{\text{functions on tops}\}$. Its inverse is $(f,\nu) \mapsto G'f + H'\nu$, where ν is a function on the tops and H' is a right inverse of Q', constructed from G', with $R\Delta H' = 0$. Let $U \in \mathfrak{U}_k(\Omega_\bullet, \alpha_0)$ and let M_1 be large. Assume that $M\alpha_0$ (and therefore $M_1\alpha_0$) lies below a number that depends only on L, and that $\mathbb{T}^{(k)}$ has at least CM_1 sites in each coordinate direction. Then the form

$$\langle A, \Delta_U^{\mathrm{W}}A\rangle + \|RD_U^*A\|^2 + \langle Q^\sharp A, aQ^\sharp A\rangle$$

is positive definite. Its inverse $\mathsf{G}_k(U)$ maps $|\cdot|_{(-3)}$ into $\|\cdot\|_1$, with a norm that depends only on L. On the set $\{RD^*A = 0,\ \xi Q^\sharp A = B\}$ the form differs from the Hessian by a constant. Consequently $\mathsf{H}_k B = \mathsf{G}Q^{\sharp *}(Q^\sharp \mathsf{G} Q^{\sharp *})^{-1}\xi^{-1}B$ minimises the Hessian itself under the constraint, and $RD^*\mathsf{G}Q^{\sharp *} = 0$. Here G stands for Bałaban's G, and not for the inverse of the displayed form, which is his G_0. The formula for $\mathsf{H}_k B$ and the identity just stated hold exactly only when the Hessian in the form is replaced by its invariant extension Δ_π. This extension is defined by $\langle A, \Delta_\pi A\rangle = \langle A - D\lambda, \Delta^{\mathrm{W}}(A - D\lambda)\rangle$, where $\lambda \in N(Q')$ is the one with $RD^*(A - D\lambda) = 0$. The replacement is a perturbation of size $CM\alpha_0$, which is small together with the current because α_0 is chosen after M. The bounds persist under this perturbation, and it vanishes where $J = 0$. In the symbols of the Notation, §5, this operator equals $\mathsf{H}\,\xi^{-1}$. There H denotes the linear minimiser constructed from G, and it satisfies $Q^\sharp \mathsf{H} = 1$. Where $J = 0$ and $\xi = 1$, the operator coincides with the H_k of the Notation. It carries the normalisation for the constraint $\xi Q^\sharp A = B$. With his Q^{B}, Bałaban uses both normalisations: the minimiser under $QA = B$ and the minimiser under $L^j \eta\, Q_j A = B$. On layer j the two differ by the factor $\xi^{-1} = (L^j \eta_k)^{-1}$. The normalisation with ξ is the one taken in his paper on the variational problem. ■

Proposition 3.25 (from the axial to the Landau gauge)**.** Work in the scheme c, in which Bałaban's average of 1985 replaces $\mathcal{M}$ in every object, and fix the block and gap numbers $\mathfrak{M}, R_1$. The constants $c_1, B_1 > 0$ may depend on $L, \mathfrak{M}, R_1$, but are independent of depth and compatible volume. Assume the hypotheses of Lem. 3.22 [stated technical lemma; Group A], with the full gauge $\mathrm{Ax}(\Omega_\bullet, U_0)$ and not Ax°, and assume $\alpha_0 + \alpha_1 \le c_1$. Then U has exactly one Landau pair that satisfies

$$|A|_{(-1)},\ |\nabla_{U_0} A|_{(-2)},\ |D^*DA|_{(-3)},\ |\Delta_{U_0} A|_{(-3)} \le B_1(\alpha_0 + \alpha_1),$$
$$\sup |u - \mathbf{1}| \le C B_1(\alpha_0 + \alpha_1),$$

that satisfies Hölder bounds on ∇A (these bounds give rise to the α_{H} below), and that has the straightened averages of U,

$$\mathcal{Q}_j^\sharp(U_0; A) = B \ \text{ on the bonds at } \mathbb{Y}_j,$$
$$B = \tfrac{1}{i}\log\left[\mathcal{M}^j U (\mathcal{M}^j U_0)^{-1}\right],$$
$$|B| \le C\alpha_1$$

(with an explicit correction entering on bonds that join two layers). For $\mathcal{M}$ itself, under the same hypotheses with $\alpha_0 + \alpha_1 \le c$, there exists a pair (A, u) in the slice of § 3.11 that has these averages and satisfies $\|A\|_1,\ |D^*DA|_{(-3)} \le C(\alpha_0 + \alpha_1)$. That slice has the same form as the Landau gauge but a different projection, whose site averages run along combs from the block corners. ■

Proposition 3.26 (a priori bounds on the slice)**.** Assume that $U_0, e^{i\eta_k A}U_0 \in \mathfrak{U}_k(\Omega_\bullet, \alpha_0)$, that $RD^*A = 0$, and that the averages are as above. Assume further that the only bound given is $|A|_{(-1)} < \alpha_2$, with $\alpha_0, \alpha_1, \alpha_2 \le c$ (c chosen for the fixed block and gap geometry of Prop. 3.25) and $C\alpha_2^2 \le \alpha_0 + \alpha_1$. Then, in the scheme c, all the bounds of the theorem on A hold, with $B_1 = CL\times$ the norms of $\mathsf{G}_k(U_0)$, $\mathsf{H}_k(U_0)$. For $\mathcal{M}$, the bounds on $\|A\|_1$ and $|D^*DA|_{(-3)}$ hold, in the slice of § 3.11. ■

Proposition 3.27 (regular means small potential, locally)**.** Let $U \in \mathfrak{U}_k(\Omega_\bullet, \alpha_0)$ and impose the fixed background geometry floors. Take M to be a positive integer multiple of $\mathfrak{r}_{\mathrm{p}}$ and take a cube of side $M\xi_j$ aligned on the $\mathfrak{r}_{\mathrm{p}}\xi_j$ grid whose concentric enlargement of side $(M + 4\rho)\xi_j$ lies in Ω_j and is shorter than half the torus. The curvature and full-current bounds are required on every complete stencil meeting this enlargement, as in eq. 3.cdxxxiii. Assume $CM\alpha_0 \le c_1$. Then on the original cube, for some u

$$U^{u^{-1}} = e^{i\eta_k A},$$

$$\xi_j|A|,\ \xi_j^2|\nabla A|,\ \xi_j^3|D_{\mathbf{1}}^* D_{\mathbf{1}} A|,\ \xi_j^3|\Delta A| \le CM\alpha_0,$$

For the additional normalization in scheme c, choose the gauge by Prop. 3.25. First put the top box datum in its box axial gauge, then impose the lower full tower gauge while fixing the top nodes, and finally take the normalized Landau pair relative to $\mathbf{1}$. The lower tower changes preserve the top datum. The Landau change has trivial nonlinear site averages, so the reduced-average identity of Lem. 3.24 preserves its straightened value. On the homogeneous top rows this gives exactly $\mathcal{Q}_j^\sharp(\mathbf{1}; A) = \frac{1}{i}\log(\mathcal{M}^j U)^{\mathrm{ax}}$. This identity uses the stated axial-to-Landau theorem and its choice of gauge. Conversely, let $U_0 \in \mathfrak{U}_k(\Omega_\bullet, \alpha_0)$ and $\|A\|_1, |D^* DA|_{(-3)} < \alpha_2$, with α_0, α_2 small. Then $e^{i\eta_k A} U_0 \in \mathfrak{U}_k(\Omega_\bullet, \alpha_0 + 3\alpha_2)$. Now take a complex $\mathbf{U} = e^{i\eta_k A'} U$ with $\xi_j|A'| < \alpha_1$ whose own plaquettes are small, $|\partial \mathbf{U} - \mathbf{1}| < \alpha_0 \eta_k^2 \xi_j^{-2}$ on the cube. (From $\|A'\|_1 < \alpha_1$ this follows only with $\alpha_0 + C\alpha_1$ in place of α_0, Prop. 3.6.) Assume also $CM(\alpha_0 + \alpha_1) \le c$. Then the averages are V^v. Here V is what the averages of $\mathbf{U}$ become in the comb gauge of the cube, and $v = v' v_{\mathrm{r}}$ is a G^c-valued function on the nodes. The factor v_{r} is G-valued and does not depend on A', the factor v' is holomorphic in A', and

$$|V - \mathbf{1}| \le CM\alpha_0,$$

$$\mathrm{cn}(v) = |v|\,|v^{-1}| \le e^{CM\alpha_1},$$

$$|v' - \mathbf{1}| \le CM\alpha_1.$$

The third bound of Bałaban, $|v - \mathbf{1}| \le CM\alpha_1$, holds for v'. For v it holds only at pairs normalised by $U(\Gamma_{y_\square, x}) = \mathbf{1}$. Indeed, for $U = \mathbf{1}^g$, $A' = 0$ one has $v = g(\cdot) g(y_\square)^{-1}$, which is arbitrary, whereas $\mathrm{cn}(v) = 1$. ■

Proposition 3.28 (transversality, locality, node factor; the choices disappear)**.** The slice is transversal. Let $\tilde{A}$ be real with $\|\tilde{A}\|_1 \le c$, and let a G-valued y near $\mathbf{1}$ be prescribed on the tops. Then some $w = e^{i\lambda}$ brings $\tilde{A}$ into the slice with $r_j = \mathsf{y}$ on $\mathbb{Y}_j$, and $\|\lambda\|_0 \le C(|D^* \tilde{A}|_{(-2)} + \sup|\mathsf{y} - \mathbf{1}|)$. Here r_j is defined as in Prop. 3.7 and formed at $\tilde{A}$; at $\tilde{A} = 0$ it is $\bar{R}_{U_0}^{(j)} w$. The pair (A, u) depends locally on the data $(J(U), B)$, with tails $Ce^{-\delta d(y,y')}$ and $\delta < \delta_0$. Distances are measured in local scales, $J(U)$ in $|\cdot|_{(-3)}$, B in the sup norm and A in $\|\cdot\|_1$. Along differentiable families of U, this locality follows from the relation

$$\delta A = \mathsf{G}_k(U_0)\big[\,\mathrm{Ad}(u^{-1})\delta J(U) + \mathcal{L}\delta A\big] + \mathsf{G}_k(U_0) Q^{\sharp *} \xi^{-3}\big[\delta B - \partial_A C(A)\delta A\big].$$

For backgrounds over a real datum $\mathbf{V}$, assume that the datum is regular in the sense of § 3.11: its test loops lie within a small ε of $\mathbf{1}$, and, where inward bonds are kept, so do the two-scale loops at the seams. For the retained-inward variant also assume the original and peeled admissions of Lem. 3.36. Under these hypotheses the pair depends holomorphically on the datum within a ball of radius c, the representative at $e^{iB}\mathbf{V}$ being $(e^{i\eta_k A(B)} U_k(\mathbf{V}))^{h(B)}$. None of these statements depends on k. The plain averages of $e^{i\eta_k A} U_0$ are the datum transformed by the *node factor*: $\mathcal{M}^j(e^{i\eta_k A} U_0)(c) = \mathsf{n}(c_-) V_j(c) \mathsf{n}(c_+)^{-1}$, where $\mathsf{n} = u^{-1}|_{\mathrm{nodes}}$. At the tops the node factor is the iterated n_W, namely $v_j[A]$. At a junction node it contains one further factor, the quotient of the datum by the reduced average on one bond ending at that node. Every ingredient of a step is gauge covariant, and the map $(A, u) \to (\mathrm{Ad}(v)A, vuv^{-1})$ takes the set of Landau pairs of U with respect to U_0 onto the set of Landau pairs of U^v with respect to U_0^v. Consequently $E^{(k+1)}(X, \cdot)$ is invariant under

$$(\mathbf{U},\mathbf{J}) \to (\mathbf{U}^u, \mathrm{Ad}(u)\mathbf{J})$$

for G-valued u. It is therefore also invariant for the G^c-valued u that are connected to these within the domain, and it can be evaluated in any gauge. A function F constructed on a slice is invariant under the subgroup that preserves the slice, and it has a unique invariant extension, $F^\natural(\mathfrak{z}) = F(\mathsf{c}[\mathfrak{z}]\cdot\mathfrak{z})$. Here $\mathsf{c}[\mathfrak{z}]$ is the transformation that takes $\mathfrak{z} = (U, A', \mathbf{J})$ to the slice; this symbol, which carries an argument, is distinct from the scheme c of Prop. 3.12. The extension retains locality, the bounds, and holomorphy in $(A', \mathbf{J})$ at fixed real U. ■

Lemma 3.29 (the block axial gauge outside the nodes)**.** Near a seam, the block of a coarse site contains nodes, and every element of $\mathcal{G}^{\mathrm{res}}$ is pinned at these nodes. In that case the gauge condition is $(\mathcal{M}^i U)(y,x) = \mathbf{1}$ for $y \in P''$ and for those $x \in B(y)\setminus y$ other than nodes. For a pair of nodes, the value $(\mathcal{M}^i U)(y,x)$ is invariant on the orbit, and the datum does not contain it. If P'' consists of nodes, the gauge is met once by every orbit of $\mathcal{G}^{\mathrm{res}}$, meaning that the averages $\mathcal{M}^i U$ on the bonds within $B(P'')$ and the nodes depend only on the orbit. Hence $V_k^{\mathrm{bg}} = \mathcal{M}^k U$, with U in the orbit $U_{k+1}(\mathbf{V})$ and in the gauge, is a function of the datum; this is the case $i = k$, with $k+1$ in place of k. In this situation

$$\mathrm{GF}_i(\mathcal{M}^i U; P'') = \sum_{y\in P''} \sum_{x\in B(y)\setminus y,\, x \text{ a node}} \big[1 - \tfrac{1}{2}\,\mathrm{Re}\,\mathrm{tr}(\mathcal{M}^i U)(y,x)\big],$$

which is constant on the orbit. Near a seam, this is what is left of the identity $\mathrm{GF}_k(V_k^{\mathrm{bg}}) = 0$. ■

Proposition 3.30 (a generalisation of Prop. 3.25, in the scheme c)**.** Fix $\gamma' > 0$. Where the constant B_1 of the theorem enters, γ' is not large; $\gamma' = 1$ is an example. Assume the hypotheses of the theorem, with c_1 depending on γ' as well; for arbitrary γ', the constant B_1 must also depend on γ'. Let $f \in \mathrm{ran}\, R$ with $|f|_{(-2)} < \gamma'(\alpha_0 + \alpha_1)$. Then the contraction in $\|\lambda\|_0$ from the proof of the theorem produces, for λ in a ball of fixed radius, a unique u whose averages are trivial and for which $R\,D^*_{U_0} A = f$. The consequence below uses only this conclusion. Bałaban's Theorem 8 [stated technical lemma; Group E] also asserts, without proof, all the bounds of the theorem, with the same B_1. The second-order and Hölder bounds require, in addition, a bound on Df in $|\cdot|_{(-3)}$. As a consequence, the set $\{A : \mathcal{Q}^\sharp(U_0; A) = B,\ RD^*A = f \text{ for some small } f\}$ consists of pieces of gauge orbits, and each piece meets the slice $f = 0$ in exactly one point. On each piece the action takes a single value. Hence a point of the slice that is critical there is critical on the thickened set as well, so that the Lagrange equations no longer involve the gauge condition. Bałaban makes use of this a single time, to obtain the second-order bounds on the minimiser. In § 3.12.1 the corresponding step, which passes from the slice to a multiplier attached to the averages only, is carried out with gauge directions the images of which under RD^* fill all of $\mathrm{ran}\, R$. The domains U_j^c have real centres that are regular, though not nearly critical, and at these centres the construction demands small local potentials outright. ■

Proposition 3.31 (a seam in dimension one)**.** Take all definitions on $\mathbb{Z} \supset L\mathbb{Z}$, and perform one step on the weight 1 on two blocks, the second of which is centred at y; the third block, $B(y')$, remains fine. The seam bond s is kept in the datum, p is the transport from s to y', and $f_{\mathfrak{G}}$ is the one-site factor of $\mathfrak{G}_k$. Let $V^+(c)$ be the coarse variable on $c = \langle y, y'\rangle$. Then the integral of the step can be computed exactly:

$$\Phi = f_{\mathfrak{G}}\big(V^+(c)\,\mathsf{p}^{-1} V(s)^{-1}\big).$$

Under the node-wise gauge group of the datum, this integral fails to be invariant. Let a transformation be g at the centre y and $\mathbf{1}$ at all remaining end points of bonds of the datum, the end of

s among them. It changes the integral into $f_{\mathfrak{G}}(gV^{+}(c)\mathsf{p}^{-1}V(s)^{-1})$. The integral is invariant under the action in which the end of s moves with the centre of its block, and with respect to this action the tree insertion is a valid identity. Under the node-wise action, Φ has gauge average equal to the constant 1. The procedure of Prop. 3.23 therefore produces the bare slice measure, not Φ times that measure, and in this case the insertion is not an identity. If s is integrated together with its block, then $\int dV(s)\Phi = 1$, and nothing remains to vary. ∎

Proofs. *Averaging.* Every covariance property follows from the identity $\log(TXT^{-1}) = T(\log X)T^{-1}$. Each expression under a logarithm is conjugated at a single site (c_- for $\log W_x$). The norms are gauge invariant, and so therefore is the choice of rule. For Lem. 3.1, in the chart $V = e^{Y}V_1$ the means are exactly the fixed points of the map $Y \mapsto Y + \frac{1}{n}\sum_i \log(V_i V_1^{-1} e^{-Y})$.

Denote this fixed-point map by Φ. For explicit radii, put $d = \operatorname{diam} V_\bullet \le 1/40$ and $T_i = V_i V_1^{-1}$. On $|Y| \le 1/8$ one has $|T_i e^{-Y} - 1| \le (1+d)e^{1/8} - 1 < 1/6$. The exponential series and the logarithmic resolvent identity give

$$\operatorname{Lip}(\Phi) \le d + (1+d)(e^{1/8}-1) + \tfrac{1}{5}(1+d)e^{1/8} < \tfrac{2}{5}, \qquad |\Phi(0)| \le \frac{d}{1-d}.$$

Thus Φ maps this closed ball into itself and its fixed point satisfies $|Y_*| \le (5/3)d/(1-d) \le 2d$. Its iterates from zero converge uniformly, which proves holomorphy in the family on the open diameter domain. For determinant-one inputs the iterates are traceless: every logarithmic argument has determinant one and lies within $1/6$ of one, so the trace of its series logarithm is zero. For unitary inputs they are also anti-Hermitian. The constructed mean therefore belongs to G^c, and to G on real inputs.

To prove uniqueness among all admissible real means, let $V' \in G$ be any mean and put $Z_i = V_i V'^{-1}$. Then $|Z_i - 1| < 1$, $\sum_i \log Z_i = 0$, and $|Z_i - Z_1| \le d$. For unitary A, B with $|A-1|, |B-1| < 1$, the identity

$$\log A - \log B = \int_0^1 (1 + t(A-1))^{-1}(A-B)(1+t(B-1))^{-1}\,dt$$

implies $|\log A - \log B| \le 2|A-B|$. Indeed each resolvent has norm at most $(1-t/2)^{-1}$, since the real parts of the eigenvalues of A and B exceed $1/2$, and $\int_0^1 (1-t/2)^{-2}\,dt = 2$. Consequently

$$|\log Z_1| = \left| \frac{1}{n}\sum_i (\log Z_1 - \log Z_i) \right| \le 2d < \tfrac{1}{8}.$$

With $Y = -\log Z_1$ we have $V' = e^{Y}V_1$; its mean equation says precisely $\Phi(Y) = Y$. Hence $Y = Y_*$ and $V' = \operatorname{av}(V_\bullet)$. All constants above are independent of the number of family members.

Under a conjugation that is not unitary, diameters change by at most $\operatorname{cn}(T)$. The isometry r permutes the centred blocks and the families $\mathcal{G}$, but it carries a comb to a comb in a different order. This is why Bałaban's first average, built on cornered blocks and combs, needed to be symmetrised. Lem. 3.2 (lattice homotopy): each interchange of two steps in a path costs one plaquette. The contours sweep out $O(L^2)$ plaquettes, and this determines $\operatorname{Dom}(c)$. In the comb gauge the bound increases with the size of the box, so small curvature yields a small potential only on boxes of bounded size. In the comb gauge one has $\mathcal{M}(e^{i\mathsf{a}}) = \exp(i\mathsf{Q}^{\mathrm{pl}}\mathsf{a} + O(|\mathsf{a}|^2))$. Since the contour terms σ cancel at the corners of p', the lattice Stokes formula yields

$$\sum_{c \in \partial p'} \pm(\mathsf{Q}^{\mathrm{pl}}\mathsf{a})(c) = L^{-4} \sum_{x \in B(y)} \sum_{p \subset S_x} \mathsf{a}(\partial p),$$

$S_x = p'$, with corner y, translated to x, L^2 plaquettes.

Consequently $\hat{\varpi}_n = \varpi(\mathcal{M}^n U)/\alpha_0(L^n\eta_k)^2$ satisfies $\hat{\varpi}_{n+1} \le \hat{\varpi}_n(1 + C\alpha_0 L^{-2(k-n)})$. Measured in its own scale, the curvature of level n is summable, and this outweighs the accumulation of k errors.

A single step (Prop. 3.5) is proved by universality and Cauchy bounds, whereas Bałaban computes $\mathsf{Q}(W_0)$ in closed form from the Campbell–Hausdorff formula. In the comb gauge everything reduces to finitely many holomorphic functions that depend on L alone, and hence $\mathcal{Q}^1 = \mathsf{Q}(W_0)\mathsf{a} + O(|\mathsf{a}|^2)$. At $W_0 = \mathbf{1}$ one has $\frac{1}{i}\log \mathsf{u} = \sigma + O(|\mathsf{a}|^2)$. Thus the block factors take away precisely the contour terms of Q^{pl}, and $\mathsf{Q}(\mathbf{1}) = \mathsf{Q}^{\mathrm{s}}$. The kernels of $\mathsf{Q}(W_0)$ and $\mathsf{S}(W_0)$ are holomorphic and coincide at $\mathbf{1}$, so they differ by $O(\varpi)$. The norms of S reduce to counting: there are $L^4 \cdot L$ pairs (x, b) for each coarse bond, each fine bond is contained in L segments, and every pair carries the weight L^{-4}. For k steps, the factorisation follows from covariance by induction. At a complex 1-form A, covariance under the G^c-valued v_n uses up part of the room in the domain, and it is applied with $\mathrm{cn}(v_n) = |v_n|\,|v_n^{-1}| \le 2$. Setting $t_n = \sup |\mathsf{a}_n|/L^n\eta_k$ and $\varpi_n = \varpi(W_{0,n})$, one has

$$t_{n+1} \le t_n(1 + C\varpi_n + CL^n\eta_k t_n),$$
$$\sum_n \varpi_n \le C\alpha_0,$$
$$\sum_{n<k} L^n\eta_k \le 2L^{-1}.$$

The recursion closes because $\|\mathsf{S}(W_{0,n})\|_{\mathrm{row}}$ does not exceed L, the ratio of the spacings, with no loss. This is the third gain of straightening. Since $\|\mathsf{Q}^{\mathrm{pl}}\|_{\mathrm{row}} \ge L + \frac{3(L^2-1)}{2L} > 2L$, a product of one-step bounds for the plain average grows geometrically in k. The derivative is bounded through column sums and Schur's test. When the plaquettes are complex, the real comb gauge does not work: at every level its error is as large as the bound itself, which gives only $1 + Ck\alpha_1$. The G^c-valued comb gauge of $\mathcal{M}^n\mathbf{U}$, whose condition number is $1 + C\alpha_1 L^n\eta_k$, gives $\varpi^c_{n+1} \le L^2\varpi^c_n(1 + C\alpha_1 L^n\eta_k)(1 + CL^2\varpi^c_n)$. The price of the non-unitary conjugation is therefore one factor per level, and these factors are summable. The determining sets are obtained by folding with reflections. The double block of order j beneath a bond of Γ_j lies in Ω_{j-1}. There the fields are, in the worst case, regular only in the next finer scale, and this costs one factor, a single time.

Intertwining: at a pure gauge $e^{i\mathsf{a}^w}W_0 = W_0^w$ one has $\mathsf{u}(W_0, \mathsf{a}^w) = w\,(\bar{R}_{W_0} w)^{-1}$. It is for the sake of this identity that $\bar{R}$ is defined as it is. Prop. 3.7 rests on a coupled recursion for the size of r_n and for its covariant difference, which the same geometric sums close. For products one has $|\mathsf{z}_{j+1}| \le |\mathsf{z}_j| + C(|\mathsf{z}_j| + \alpha L^{j+1}\eta_k)^2$, because the defect added at each step is quadratic in the variation within a block. This is the origin of the second condition in the definition of a class. Lem. 3.8: up to an error $C\varpi$, a bond inside a block is the quotient of two contour transports, and the average then determines the crossing bonds. The comparison is made factor by factor. Prop. 3.9: the constraint is solved to second order by means of the right inverse H_k. At a critical point under the constraint, $J(U) = Q_k^\sharp(U)^*\omega$ is in general not zero. It is the image of a Lagrange multiplier ω, whose size is that of the curvature of V. The extra term does not occur in abelian theories, and it gives rise to G_1, H_1 in § 3.12.1. Prop. 3.10 follows from the first identity together with $\mathcal{M}(\mathsf{r}U) = \mathsf{r}\mathcal{M}(U)$ and $(\mathsf{r}U)(\mathsf{r}y, \mathsf{r}x) = U(y, x)$. Prop. 3.11 is purely algebraic. Writing $\hat{u}_n = u\,v_n$, the axial condition at level n takes the form

$$\hat{u}_n(x) = \mathrm{Ad}(W_{0,n}(z, x))^{-1}[\hat{u}_{n+1}(z)\mathsf{u}_n(z)^{-1}\mathsf{w}_n^{\mathrm{rel}}(z, x)]$$

. This fixes u level by level, starting at the top and going down, and it yields $\bar{R}_{W_{0,n}}\hat{u}_n = \hat{u}_{n+1}$. Prop. 3.12: two chains of transports with common end points differ by the curvature that they enclose. At level n this is $O(\alpha_0 L^{-2(k-n)})$, so the differences form a geometric series. The transport t is unavoidable, because in each block the two schemes see a gauge transformation at two different sites. Prop. 3.13 follows from linearity and the Stokes formula, which give $\Theta = L^{-4}\sum_{x\in B(z)} L[\mathsf{f}^\Sigma(x_1 + L) - \mathsf{f}^\Sigma(x_1)]$, together with a count.

The transformation. Prop. 3.14 combines the definition of an image measure with a property of $\mathcal{M}$. The renormalisation transformations obey the semigroup law $\mathbf{T}^n\nu = (\mathcal{M}^n)_*\nu$, which relates different spaces. In physical units the symmetry group becomes smaller as k grows; in the units of the level it is the symmetry group of the unit lattice. Finite range does not survive, since $\mathbf{T}\rho_0$ couples all bonds. Reflection positivity is required of the bare measure only §§ 5.20 and 5.22, appendix A.6, § 5.24 and appendix A.6. Prop. 3.15: the variances are additive (taking a block mean divides by L^d, the rescaling multiplies by L^{d-2}, and the new kernel contributes $\mathfrak{a}^{-1}$). Nested backgrounds: backgrounds exist only over regular data, so minimising in two stages is not enough. The argument is instead as follows. A point that is critical on the larger fibre is critical on the smaller one, regularity improves in finer units, and uniqueness § 3.11 completes the proof. Because the argument involves critical points alone, it extends to complex data. The minimiser $\mathsf{H}B$ is the one given by the method of Lagrange multipliers. The delta kernel secures four things:

- closure under composition. Since $\mathcal{M}$ is non-linear, the composite of two soft kernels on G is not a function of W and $\mathcal{M}^2U$ alone.
- a single large term. The critical point does not depend on the coupling, the leading term is exactly $x_kA(U_{k+1})$, and terms produced at U_j can be evaluated at U_k at every later level.
- exact averages. Since $V_k = \mathcal{M}^kU$, a test on the block fields is a test on the bare field.
- no analytic loss. The mass $aQ^{\sharp *}Q^{\sharp}$ remains in G_k as a device, with $a > 0$ arbitrary.

The cost is that the fibre is curved. For the gauge fixing the opposite choice is made: it enters through a soft weight.

Prop. 3.16 (stochastic kernels, balance of masses). The mass, summed over the outcomes, is preserved by operations of three kinds. *Tests:* at any given state exactly one outcome takes the value 1. A test is counted as failed whenever the object it examines (a mean, a logarithm, a minimal orbit, a minimiser) is undefined, so every test is defined on all configurations. The domains and values of the real tests for the block average, regular critical orbit and unique attained minimum are Borel by the finite formulas for those objects in the measurability argument below. *Draws:* the kernel $\kappa_{\mathfrak{G}}(V,\cdot) = \int du\, \mathfrak{G}_k(V^u)\delta_{V^u}$. *Maps:* adjoining $\mathcal{M}(V_k)$, and forgetting a variable. No smallness is involved in any of this. The operation $\mathbf{R}$ draws the variables of a component afresh using the reference kernel specified below. Because of the partitions, single terms have only restricted symmetry. Only the totals at a base point are covariant under the full group W_4, which § 4.2 requires. Base points are required because the sum $\sum_{X\subset\Lambda_j}$ is not invariant and the cancellation against $b_jA(\varphi_j,\cdot)$ must be made at each point separately. When z lies one layer of $M\check{R}_j$-cubes inside Λ_j, the restricted pointed total differs from the unrestricted one by $O(e^{-\frac{1}{2}\kappa\check{R}_j})$.

Lem. 3.17: a compact group admits no dilation. Hence x_k *is* the normalisation of the kinetic term. Gauge invariance links the quadratic, cubic and quartic parts of the marginal term, so the renormalisations of the field strength and of the coupling are given by a single number. The substitution $\mathcal{A}_k = g_k\mathcal{A}_k^{\mathrm{s}}$ takes the place of the rescaling. Consequently there is no scaling transformation, no fixed point and no eigen-direction. The word "irrelevant" can then mean only a bound $O(\eta_{k-j}^{4+\alpha_{\mathrm{H}}})$ at regular backgrounds, where $\alpha_{\mathrm{H}} < 1$ is a Hölder exponent § 4.2. Level k is a problem on a unit lattice that carries a fine lattice of spacing $\eta_k \to 0$. Every estimate in § 3.11, § 3.12.1 must therefore hold uniformly in η_k. Lem. 3.17: the density itself does not change. Its exponent is always $-\beta A(U_k)$ together with the terms in their unrenormalised form. The point is which grouping of the terms admits uniform bounds:

- without grouping, the bound $\|E^{(j)}\| \le E_0$ yields no more than $CE_0\eta_k^{-4}|\mathbb{T}^{(k)}|$;
- the number $b = b_{k+1}$ by itself leaves a geometric sum over j;
- if the marginal terms are left among the corrections, they add up to c_0kA to leading order;
- once absorbed into x_k, they fix the next thresholds and g_{k+1}, the width of the following Gaussian.

The profiles φ_j keep track of the place at which step j was performed. In the constant, the contribution $-\frac{1}{2}(N^2-1)\log x_k$ for each free bond is the Jacobian of $\mathcal{A}_k = g_k B$. Laplace's method gives $z_{\mathfrak{G}} \approx C x_k^{-(N^2-1)/2}$. With four bonds per site, less one gauge direction, this gives $\mathsf{E}^{(j)}(T_\ell, T_\ell) = -(\frac{3}{2}(N^2-1)\log x_j + O(1))(|\mathbb{T}^{(j)}| - |\mathbb{T}^{(j+1)}|)$. The term $\log x_k$ in E_+ arises from the constants $\mathsf{E}^{(j)}(T_\ell, T_\ell)$, $k \le j < K$, which belong to the steps $k+1, \dots, K$ and remain owed in $\bar\rho_k = e^{-E}\rho_k$ (when $k = K$ no constant remains owed, and $E_+ = O(1)$). Over a large-field region, the constants differ by a volume factor $e^{C \log x_j \#_j(Z_{j+1})}$, and the small factors of §4.17.1.40 must outweigh it.

Prop. 3.18 (the division of the work among the parts). Each part supplies the following:

- § 3.11: the backgrounds;
- § 3.12.1: the propagators;
- § 4.5: the new $E^{(k+1)}$, by a polymer expansion under a criterion of Kotecký–Preiss type;
- § 4.2: the coupling and the bounds on the sums;
- §4.17.1.1: tests, regions and histories;
- §4.17.1.3: $\mathbf{R}$;
- §4.17.1.40: the large-field factor and stability;
- § 4.17: the terms that $\mathbf{R}$ leaves behind;
- §4.17.1.45: the order of the constants and the inductive form;
- § 5.2.1: the flow;
- § 5.2.2: independence from the label, the torus and the cutoff.

The last item rests on locality. A term localised in X is assembled from walks, potentials and tests within X. It therefore does not depend on the label, and, when X does not wrap around the torus, it depends on neither the torus nor the cutoff. The small-field half requires $g_j \le \gamma$. It also requires $-C \le x_i - x_j \le C(j-i)$ for $i \le j \le k$, because the analyticity radii of level j run with g_j and the radii of two levels must be comparable. For radii that do not run, no relation between the couplings is required; Bałaban's remark that asymptotic freedom is not assumed refers to this case. For the complete model, what is used is the lower half of asymptotic freedom. At the levels of creation $j \le k$ the small factors were deposited, and since $x_j \ge x_k - C$ one has $p_0(g_j) \ge c\, p_0(g_k)$. The terms $R^{(j)}$ carry the vacuum subtraction but no counterterm for the coupling, and they are marginal. On the support of the tests, the following bound holds for every $X \in \mathcal{D}_j$ that contains a site z of Ω_k and remains within cML^{k-j} of z in level-j units (a longer X satisfies $d_j(X) \ge cL^{k-j} - C$ and has only the bound $2g_j^{\kappa_0} e^{-\kappa d_j(X)}$; around a site of $\Omega_n \setminus \Omega_{n+1}$, $j \le n < k$, the right-hand side holds with n in place of k):

$$|R^{(j)}(X, U_k) - R^{(j)}(X, \mathbf{1})| \le C(d_j(X)+3)^2 (x_j/x_k)^2 \eta_{k-j}^4\, g_j^{\kappa_0} e^{-\kappa d_j(X)},$$
$$\sum_{j \le k} (x_j/x_k)^2 g_j^{\kappa_0} \le C g_k^{\kappa_0 - 2} :$$

these terms do not decrease as $k-j$ grows. Only the linear growth of x_j makes them summable, and it does so for $\kappa_0 > 6$. If one assumes no more than $x_j \ge x_k - C$, the sum can grow to order $k g_k^{\kappa_0}$. It is this summability that makes a counterterm for them unnecessary. It is also why aged R-terms change b_j by no more than $C\gamma^{\kappa_0 - C}$, so that $\lambda_\sharp > 0$ is preserved § 5.2.1. The upper half ensures that the thresholds of adjacent levels are comparable: $\varepsilon_{k+1} \le C\varepsilon_k$.

The large-field factor: the reasons for these forms. The fields at which a plaquette test fails do not form a small set for the measures contained in $\mathbf{T}_k(Z)$. They do form a small set for the Wilson weight, since $e^{-x_j A} \le e^{-c\, x_j \varepsilon_j^2} = e^{-c\, p_0(g_j)^2}$. Therefore, if no $\mathbf{R}$ has acted on a component X, the small factor that a failed plaquette test carries there is obtained from $\mathbf{T}_k(X)$ applied to $\chi\, e^{-A(x_k(\cdot), U_k)}$, and never from $\mathbf{T}_k(X)1$. This weight is evaluated at the background, which depends on the variables of all the components simultaneously. For this reason, bounds that are uniform in

the exterior variables do not combine multiplicatively. The phenomenon already occurs in measure theory. Take Lebesgue measure on $[0,\mathsf{l}]$ in each of two variables, $0<\epsilon\ll\mathsf{l}$, and $F=\mathbf{1}\{|\xi_1-\xi_2|<\epsilon\}$. Then

$$\sup_{\xi_2}\int F\,d\xi_1=2\epsilon,$$

$$\iint F\,d\xi_1 d\xi_2=2\epsilon\,\mathsf{l}-\epsilon^2\gg(2\epsilon)^2.$$

The remedy is a pointwise bound by a product of functions, each depending only on its own group of variables, the groups being disjoint, combined with Tonelli's theorem. The small factors come out as numbers from the Wilson sums near the creations, joined with a fixed fraction both of the Gaussian forms and of the kept gauge-fixing exponents. What remains of the weights is bounded by functions that no longer depend on the background. The supremum in M is placed under the integral for the same reason. The factor of X then becomes a function of ξ only, and the masses multiply. When a component has dropped out, the supremum is taken over the evaluations at ξ. These are the real configurations that are compatible with the stored variables and regular layer by layer, and at which every test listed by ω takes its recorded value, with one exception: the test whose failure turned X into a drop-out. They form a set fibred over ξ. A product set contains a flat $\mathbf{U}$, at which $e^{-A}=1$, so taking the supremum over it eliminates the Wilson weight. For a live component, a bound $e^{-c\,p_0(g_k)}$ does not reproduce itself, since a step without a creation costs $e^{+CM^4\check R_k^5(d'_k(Z)+1)}$. What is carried along is the credit. As long as nothing is created, linear sizes counted in cells decrease geometrically. At a drop of $\check R$ the two partitions coincide and nothing shrinks, and no two consecutive steps are both drops. The waiting lasts at most $\check R_j$ steps, unless two components join or a mark is set. In that case the waiting begins anew. In the count of §4.17.1.40, every joining is charged to a separate created cell belonging to it alone, and a mark sets one factor $e^{-p_0(g_t)}$ of its mass against the same $\check R_t^6$. Hence the costs still owed add up to at most $CM^4L^5\check R_j^6(d'_j(Z)+1)$. The $p_0(g_j)^2$ per cube deposited at the creation covers $\check R_j^6$ per cube, and the powers of $\log g^{-2}$ are chosen in this order. After $\mathbf{R}$ has acted, the mass bound is what is carried forward.

Three losses are made good:

- The expansions use up part of κ and of the radii. Cubes that are L times larger give back a factor of order L in the rate, and the curvature decreases by L^{-2} at each level. $\mathbf{T}$ reserves a surplus of this gain for $\mathbf{R}$, whose re-expansions consume that surplus and restore exactly κ, α_i.
- The marginal term is absorbed into x_k.
- $e^{-p_0(g_j)}$ is smaller than any power of g_j, but it is not a power of the spacing when $d=4$, so it cannot pay for all later steps. This is the reason for $\mathbf{R}$.

In the presence of large fields, $\log\rho_k$ is not a sum of local terms, and this is why the construction uses densities and sums over labels. Analyticity serves as a device. A single supremum bound on a complex domain stands in for bounds on all derivatives, and it is preserved under substitution, integration and taking logarithms.

Prop. 3.19 is § 4.5 carried out with parameters. The size of $\mathbb{P}^{(k)}$ is $g|B|^3$, and under the test $g|B|<\delta(\mathsf{t})$ this is $\le\delta(\mathsf{t})|B|^2$. The bracket of level j in $\{S\}$ is $O(\mathsf{q}_j^3\,\eta_{k-j}^{4+\alpha_{\mathrm H}})$. Here, as in Prop. 3.19, q_j is the quotient of the analyticity radii at level $k+1$ by those at level j, and it is $\le C(k+2-j)^{1/2}$ thanks to the comparability assumed there. In § 4.5 the quotient is taken with the output radii, which are larger than those of level $k+1$ by a fixed factor, and in that part it goes by the name of the ratio. What the remainders contribute to the increment b_j is a different quantity, even though § 4.2, §4.17.1.45, § 5.2.1 and § 5.2.3 write it with the same letter. Each Cauchy estimate taken

at a later background costs one power of q_j, up to the third, and on radii that do not run there is no such loss. Applied to a shift of size δ, Cauchy's estimate produces s, and a cube of level k contains η_{k-j}^{-4} cubes of π_j. What is left at the age $n = k+1-j$ is $\mathsf{q}_j^3\,\eta_{k-j}^{\alpha_{\mathrm{H}}} \le C(n+1)^{3/2}L^{-\alpha_{\mathrm{H}}(n-1)}$, and these are the weights ω_n. The $H^{(j)}$ are marginal, and their sum is controlled only through their norms. If the raw local terms are taken without the counterterms b_jA, then $\{S\}$ contains $\sum_{j\le k} b_j\{A\} \approx c_0 k\{A\}$. In that case no bound holds uniformly in k, and the statement does not hold. Keeping t (which governs the tests, the gauge-fixing weight and the scaling) separate from x, the coefficient of A, comes at the price of a quadratic term proportional to $1 - x/\mathsf{t}$. In return it allows frozen choices § 5.2.3 and complex weights §5.20.1.1, appendices A.8 and A.9. Once the choices are fixed, the steps are fixed operations that are linear in the measure, and β enters only through the weight of the action.

Prop. 3.20: $\mathrm{GF}_k(V_k^{\mathrm{bg}}) = 0$. The linear term $x_k\langle \mathsf{H}_k\mathcal{A}', J(U_{k+1})\rangle$ vanishes on $Q_{V_k^{\mathrm{bg}}}\mathcal{A}' = 0$, since the current is the image of a Lagrange multiplier. Because the fibre is curved, the term $\langle \mathsf{H}_k \mathsf{h} C^{\mathrm{pl},(2)}(B), J\rangle$ remains, and it is included in $\Delta^{(k)}$. Hence $x_k A(U_{k+1})$ is the only place where a negative power of g_k occurs. The choices are made in the following order. b_{k+1} is obtained on the infinite lattice from the choices of level k, so x_{k+1}, ε_{k+1}, $\check{R}_{k+1}$ are fixed in advance of testing V_{k+1}. *The new tests imply the old:* on Ω_{k+1}, wherever the new functions are supported, the old ones take the value 1. By the nesting, $U_{k,\square} = (e^{i\eta_k \mathbf{H}} U_{k+1,\square'})^u$. Here $\mathbf{H}$ denotes the response of § 3.11 when the datum is shifted, the shift being $O(\delta_k)$ everywhere but in a boundary layer at a distance of order $M_2\check{R}_k$. Therefore

$$|\mathbf{H}|, |\nabla \mathbf{H}| \le C\big(\delta_k + e^{-\check{R}_k}\varepsilon_k\big) \le c\,\varepsilon_k,$$
$$|\partial U_{k,\square} - \mathbf{1}| \le 2\varepsilon_{k+1}\eta_{k+1}^2 + c\,\varepsilon_k\eta_k^2 < \varepsilon_k\eta_k^2 :$$

so χ_k passes into χ_{k+1}. This holds because the new threshold is smaller by the factor $L^{-2}\varepsilon_{k+1}/\varepsilon_k \le CL^{-2}$ (by the upper half of the flow inequalities, thresholds of neighbouring levels remain comparable), and because $\delta_k = (A_1/A_0)\varepsilon_k \ll \varepsilon_k$ (both carry the same power of the logarithm, and their ratio is small and fixed). Likewise every indicator in $\mathfrak{G}_k$ is equal to 1, and χ_k' follows from $\chi^{(k)}$. Since the local backgrounds are so close to the global ones, the tests can be carried out on local objects, which factorise, while the expansions are made around global ones. *The split:* an aged $R^{(j)}$ has each of its cubes inside a single block, its total transforms covariantly under the isometries of $\mathbb{T}^{(k+1)}$, and it is counted together with $\mathbf{E}_k$. For a young one this is not so, and a young one may not be admitted to the family from which b_{k+1} is computed. The t_1-integral, which comprises boundary terms, the part of the coupling that depends on position, and the strip term $\mathsf{W}^{(k)}$, contributes only to $B^{(k+1)}$. The t_2-integral takes the expectation of the young terms in a measure which itself contains them. Consequently at least one factor $g_j^{\kappa_0}$ appears in every term of its expansion, and this integral contributes to both $R^{(k+1)}$ and $B^{(k+1)}$. The third term contains only regular terms. A further interpolation makes it independent of the exterior fluctuation field, and the resulting contribution goes to $B^{(k+1)}$. Its cluster expansion then produces, for $X \subset \Lambda_{k+1}$, the pure small-field integral's terms, $E^{(k+1)}$, and in addition boundary terms. b_{k+1} is computed from this symmetric remainder. Thus a contribution to $R^{(k+1)}$ comes from $\mathbf{T}$ already, ahead of the one that $\mathbf{R}$ adds. Prop. 3.21: iterating the Stokes formula of Prop. 3.4 gives the first display. When a standard Gaussian vector is conditioned on the value of a linear map, its conditional expectation projects orthogonally onto the complement of the kernel of that map, and its components in mutually orthogonal subspaces are independent. The slice is a Schur complement, and $\mathcal{H}_{j-1}$ transports it. The only property of it that has to be analysed is its decay, and this is the analysis of § 3.12.1 at $\mathbf{1}$.

Gauges. Lem. 3.22 is Stokes' formula on a tree. Because of the factor $|b_- - y_P|$, boxes have side $O(M)$ and $M\varepsilon_k$ has to be small; that is, the coupling is taken small after M. Lem. 3.22 is a descent through the levels, with increments $C\alpha_0 \sum_i L^{-2i}$. Prop. 3.23 follows from Fubini's theorem and the invariance of Haar measure. No determinant arises, because the gauge function is a group variable translated by $u(x)$. The mean is used because single combs break W_4. The constraint is taken soft because fixing the mean by a delta function imposes a non-linear restriction which couples all the bonds of the paths to one another and has no explicit solution. The weight is x_k, and with this weight

$$x_k \,\mathrm{GF}_k(e^{ig_k B} V_k^{\mathrm{bg}}) = \tfrac{1}{2} \sum_{y,x} \left[|B(y,x)|^2 + O(g_k |B|^3) \right]$$

is of order one. Here $g_k B(y,x)$ denotes the linear part in B of $\frac{1}{i}\log$ applied to the mean $(e^{ig_k B} V_k^{\mathrm{bg}})(y,x)$, and when $V_k^{\mathrm{bg}} = \mathbf{1}$, $B(y,x)$ coincides with the mean of $B(\Gamma)$ taken over $\mathcal{G}(y,x)$. This quantity is strictly positive on each direction $D\lambda$ with $\lambda|_{\mathrm{coarse}} = 0$ (on these $|B(y,x)| = |\lambda(x)|$), and these directions exhaust the null directions of the Hessian. The weight itself is auxiliary: the first identity of Prop. 3.23 yields an identity for every choice of weight, so the weight may be frozen § 5.2.3 and need not depend on the source §5.20.1.1. *Procedure and product:* the insertion, taken as a draw, is exact on any measure, and on invariant measures it agrees with the written product. A term with stored fields is invariant. Indeed, the distance from P'' to Z_k is $\geq cM\check{R}_k$. Hence every transformation over P'' can be lifted to a bare transformation which is constant on pillars and equal to $\mathbf{1}$ near Z_k, the region where all stored quantities enter. In the case of the trees of §4.17.1.3, the stored fields are acted on by the group via the tops of their pillars. The exact model Prop. 3.31 shows that this verification is not idle: the group that acts is decided by whether or not the seam bonds lie in the determining set. The removal identity shows that dropping δ_T from the density gives an equivalent density, not an equal one: the two coincide when integrated against invariant functions.

Lem. 3.24: in the pillars, U and U_0 have the same contour variables. Hence $W(y,x)U_0(y,x)^{-1} = u(y)^{-1}\,\mathrm{Ad}(U_0(y,x))u(x)$, and the block factor n_W becomes the definition of $u^{-1}\bar{R}_{U_0}u$. This is Bałaban's deeper reason for imposing the axial gauge first and for keeping a single contour variable throughout (with two of them, circulations of A survive in the linear part). *Origin of R:* the term sought is Feynman's $\|\partial^* A\|^2$, but only the $\lambda \in N(Q')$ leave $\delta(B - Q^\sharp A)$ unchanged. Put $N = \ker Q'$, equipped with the induced physical inner product and its Lebesgue measure $d\lambda_N$. In the two gauge integrals below we integrate with this measure on N. The Faddeev–Popov quotient with gauge parameter α_{g} is

$$e^{-\frac{1}{2\alpha_{\mathrm{g}}}\|\partial^* A\|^2} \Big/ \int_N d\lambda_N \, e^{-\frac{1}{2\alpha_{\mathrm{g}}}\|\partial^* A - \Delta\lambda\|^2} = \mathrm{const} \cdot e^{-\frac{1}{2\alpha_{\mathrm{g}}}\|R\,\partial^* A\|^2}.$$

In a real background, let $E_R = \mathrm{ran}\, R$ carry its induced physical inner product, and let δ_{E_R} be its unit delta distribution. The restriction $T = \Delta_{U_0}|_N : N \longrightarrow E_R$ is a bijection; its absolute determinant is $J_T = \det(T^*T)^{1/2}$, with the determinant taken on N. Then

$$e^{-\frac{1}{2}\langle A, \Delta_\pi A\rangle} = J_T \int_N d\lambda_N \, \delta_{E_R}(RD^*A - T\lambda)\, e^{-\frac{1}{2}\langle A - D\lambda, \Delta^{\mathrm{W}}(A - D\lambda)\rangle}.$$

The change of variables $y = T\lambda$ evaluates the integral at $\lambda_* = T^{-1}RD^*A$, with factor J_T^{-1}, and the definition of Δ_π gives the identity. The determinant here is the Landau gauge's Faddeev–Popov determinant, and axial gauges have no such determinant. In normalised expressions such as $\mathsf{H}_k B$ it cancels; in Gaussian normalisers it does not. In the one-loop coefficient § 4.2, § 5.2.1 it enters together with the determinant of the gauge-fixed Hessian. Positivity (Lem. 3.24): consider $U = \mathbf{1}$

(the remaining cases are perturbations of it). A gradient $\partial\lambda$ ($\partial = D_{\mathbf{1}}$) is detected by $Q^\sharp$ if $\partial Q'\lambda \neq 0$, and by $R\,\partial^*$ otherwise, since in that case $\lambda \in N(Q')$ once a constant is subtracted. The full DD^* is positive as well, but admissible transformations cannot reach $D^*A = 0$, and with the full operator $\mathsf{G}Q^{\sharp *}(Q^\sharp \mathsf{G} Q^{\sharp *})^{-1}$ is no longer the minimiser of the Hessian. $RD^*\mathsf{G}Q^{\sharp *} = 0$ requires the invariant extension Δ_π, which is a perturbation of size $CM\alpha_0$. (Bałaban's G_0 is defined by the form of the statement, and his G goes with the identity. Adding the term of Prop. 3.9 gives G_1. The three satisfy the same bounds, with the exception of the one with the covariant Laplacian, and they coincide where $J = 0$.)

Prop. 3.26 (elliptic regularity through $\mathsf{G}_k(U_0)$): both currents, each smaller than α_0, give $D^*DA = \mathcal{J}$ with $|\mathcal{J}|_{(-3)} \leq 2\alpha_0 + C\alpha_2|\nabla A|_{(-2)} + \dots$. The averages give $\xi Q^\sharp A = B - C(A) =: \mathsf{b}$ with $|C(A)| \leq C\alpha_2^2$ (Prop. 3.6 leaves a remainder that is of second order), and this accounts for the α_1 in the result and for the condition on α_2^2. Put $\Delta' = \Delta^{\mathrm{W}}_{U_0} - D^*D$; the plaquette Hessian comparison gives $|\Delta' A|_{(-3)} \leq C_L\alpha_0|A|_{(-1)}$. Use the fixed auxiliary row weights $a = \xi^{-2}$ in G_k for this estimate. On the slice,

$$A = \mathsf{G}_k(U_0)(\mathcal{J} + \Delta' A) + \mathsf{G}_k(U_0)Q^{\sharp *} a\xi^{-1}\mathsf{b},$$
$$\|A\|_1 \leq B_0\big(|\mathcal{J}|_{(-3)} + C_L\alpha_0|A|_{(-1)} + \sup|\mathsf{b}|\big),$$

where $a\xi^{-1} = \xi^{-3}$ for the chosen row weights. Since $|A|_{(-1)} \leq \|A\|_1$, absorption follows once $B_0C_L(\alpha_0 + \alpha_2) \leq \frac{1}{2}$. Prop. 3.25 (Bałaban's argument, written out for the scheme c) is proved by induction on k. On Ω_k the pair (A, u_1) of depth $k-1$ is a factor L worse; here u_1 denotes the explicit product from Prop. 3.11. One seeks $u = u_1 e^{i\lambda}$. By Campbell–Hausdorff, once the non-linear average has been straightened to $Q'\lambda = 0$,

$$\lambda = -G'RD^*A + G'R\,\mathfrak{N}_4(\lambda, D\lambda, A, D^*A),$$
$$|R\mathfrak{N}_4|_{(-2)} \leq CB_1(\alpha_0 + \alpha_1)\|\lambda\|_0,$$

and this is a contraction on complex λ in $\|\lambda\|_0$. (The bound is valid on the ball of radius $CB_1(\alpha_0 + \alpha_1)$, which contains the solution; within a ball of fixed radius the solution is unique.) Curvature and mismatch thus have to beat fixed norms for the chosen block and gap geometry. These norms do not grow with the depth. The new constant is of order LB_1, and Prop. 3.26 resets it to B_1; without this reset its growth in k is geometric. Uniqueness holds among small A, so no Gribov ambiguity arises. In § 3.11 the passage to the slice is redone for $\mathcal{M}$. There Brouwer's theorem replaces the contraction. It provides a pair, though not its uniqueness, and with it the bounds on $\|A\|_1$ and $|D^*DA|_{(-3)}$. For a sequence with layers, the induction is replaced by a single preliminary gauge followed by an ascent through the truncations $(\Omega_0, \dots, \Omega_j)$. Prop. 3.27: the cube is surrounded by a tower that descends to the bare scale. After passing to tower and box axial gauges, one applies the theorem to $(\mathbf{1}, U)$ within the scheme c. Only propagators at $\mathbf{1}$ enter, and this breaks the circularity with § 3.12.1. At complex fields we have $v = w^{-1}$ and $w(x) = \mathbf{U}(\Gamma_{y_\square, x}) = \Pi\, v_{\mathrm{r}}^{-1}$. Here Π is a product of factors $\exp(i\eta_k\, tA'(b)t^{-1})$, where each t is a unitary transport of U, so $\mathrm{cn}(v) = \mathrm{cn}(\Pi)$. The potential of the background inside a large-field box is obtained in §4.17.1.3 by the same sequence of steps: the box axial gauge, used for the average, beneath it the tower axial gauge, and then Landau. Every change of its gauge is compensated on the stored variables. Local gauges $\hat{k}_y$ on overlapping boxes are glued together by $k(x) = \mathcal{N}\big(\sum_y \chi_y(x)\hat{k}_y(x)\big)$, $\mathcal{N}(X) = X/\sqrt{\det X}$. In SU(2), a real combination of group elements close to one another is a positive number times a group element, and the map $\mathcal{N}$ is holomorphic.

Prop. 3.28: with weighted norms, or on complex balls, the induction incurs a loss at every level, and there is no reset. Instead one differentiates the identity of Prop. 3.26, with G_k bounded in the

sup norms that carry the weight $e^{\delta d(\cdot,y')}$. This is a single linear estimate and needs no induction. That the averages are trivial does not force u to be trivial at the nodes, and this gives rise to the node factor. Invariant functions do not detect it. At sites that carry stored fields but have ceased to be tops, n^{-1} is continued into each pillar by means of a smooth bump, and in this way the stored data are not acted on. Prop. 3.28: the objects are words in transports and logarithms, each typed at a site, and fixed points are covariant by uniqueness. Prop. 3.25 gives the uniqueness of its Landau pairs only in the scheme c; the set of these pairs is nevertheless covariant, since every one of the three defining conditions of a pair is covariant. The case of complex u follows by the identity theorem, because the G-valued u form a totally real submanifold of full dimension. Lem. 3.29 is a matter of covariance only: $(\mathcal{M}^i U^w)(y,x) = w(y)(\mathcal{M}^i U)(y,x)w(x)^{-1}$ with $w = \mathbf{1}$ at the nodes. Prop. 3.30: the equation becomes $\lambda = G'(f - RD^*A) + G'R\,\mathfrak{N}_4$, whose inhomogeneity is of the size of the other terms. Bałaban states that the remaining proofs carry over with only minor changes. However, $\Delta_{U_0}A - D^*DA$ equals $Df + D(1-R)D^*A$ up to curvature terms, so the bounds of the theorem force $|Df|_{(-3)} \le C(\alpha_0 + \alpha_1)$. On the other hand, $|Df|$ is of order $\eta_k^{-1}|f|$ for $f = \Delta\lambda$ with $\lambda \in N(Q')$ oscillating at the bare spacing. Prop. 3.31: the contour variables of a block are coordinates that preserve Haar measure. All the integrals except one give $\int f_{\mathfrak{G}} = 1$, and the remaining one, $\int dt\, f_{\mathfrak{G}}(t)\, \delta(tV(s)\mathsf{p}V^+(c)^{-1})$, is the display. In four dimensions the same occurs at each coarse site whose block contains an end of a seam bond.

Routine and delicate. Routine are the algebra of means, of logarithms and of covariance, the exact identities, Stokes' formula and the Haar identities. The delicate points are the following:

- uniformity in k, which is reached solely via the straightened average, and only because curvature and potentials are measured on the scale of each level;
- complex arguments, where each use of covariance costs room in the domain;
- the two averages, together with the invariances of the constraints they impose, which have to be kept apart;
- normalisation by averages instead of pointwise normalisation, the group property that is lost, and the node factor that is left over;
- the junctions of layers;
- the invariance of whatever an insertion encounters after fields have been stored.

For the construction as a whole, the difficulty lies in the words "with the same constants".

3.2. **Nonlinear averaging and the Wilson energy.** The four-dimensional normalization of the contour average gives an almost-contraction of the Wilson energy. Its leading constant is one. This matters when the average is repeated: a fixed larger constant at each step would grow with the number of levels. We prove the estimate for the symmetric contour map in eq. 2.iv, using its near-identity means as in [B85a].

Work first on a periodic lattice of spacing one. Its side lengths are divisible by a fixed odd integer $L \ge 3$. The coarse sites form $L\mathbb{Z}^4$, and $B(y) = y + \mathcal{B}$ is the translated centred block of L^4 fine sites belonging to y. A translated choice of the block origins gives the same proof. Each independently oriented fine plaquette is counted once. For a real SU(2) configuration W, write

$$F_p = -i \log \partial W(p), \qquad \varepsilon(W) = \max_p |F_p|, \qquad A(W) = \sum_p \Big(1 - \tfrac{1}{2}\operatorname{Tr} \partial W(p)\Big). \tag{3.i}$$

The logarithms are the small branches. Lie algebra norms use $\langle X, Y\rangle = \frac{1}{2}\operatorname{Tr}(XY)$, so $1 - \frac{1}{2}\operatorname{Tr} e^{iF} = 1 - \cos|F|$. All constants below can depend on L, and none depends on the fine spacing, number of blocks or number of averaging steps.

Proposition 3.32 (One averaging step)**.** There are $\varepsilon_L > 0$ and $C_L < \infty$ such that $\varepsilon(W) \le \varepsilon_L$ admits every local contour mean and logarithm defining $\mathcal{M}(W)$, and

$$\varepsilon(\mathcal{M}(W)) \le L^2 \varepsilon(W) + C_L \varepsilon(W)^2, \tag{3.ii}$$

$$A(\mathcal{M}(W))^{1/2} \le (1 + C_L\varepsilon(W))A(W)^{1/2}. \tag{3.iii}$$

These estimates hold for real fields. No critical-point equation or inverse Hessian estimate is required. ■

3.2.1. *The linear flux weights.* Let a be a Lie algebra one-cochain. If $t_a(y,x)$ denotes the mean of its line integrals over the prescribed shortest contour paths, differentiation of the actual contour formula at the identity gives, for $c = (y, y + Le_\mu)$,

$$\begin{aligned}(Qa)_\mu(y) &= L^{-4} \sum_{x\in B(y)} \sum_{i=0}^{L-1} a_\mu(x + ie_\mu) + h_a(y) - h_a(y + Le_\mu),\\ h_a(y) &= L^{-4} \sum_{x\in B(y)} t_a(y,x).\end{aligned} \tag{3.iv}$$

Indeed the logarithm of a product has, at the identity, the sum of the logarithmic increments. The derivative of a symmetric mean of nearby transports is their arithmetic mean. The central straight transport in the contour formula cancels its copy inside the averaged logarithm. The two endpoint transports give the displayed coarse gradient of h_a.

The curl of this gradient vanishes. The remaining straight transports bound an L-by-L plaquette, so exact discrete Stokes gives

$$(\mathrm{d}_c Qa)_{\mu\nu}(y) = L^{-4} \sum_{x\in B(y)} \sum_{i,j=0}^{L-1} (\mathrm{d}a)_{\mu\nu}(x + ie_\mu + je_\nu) = \sum_p T_{C,p}(\mathrm{d}a)_p. \tag{3.v}$$

Here $C = (y, \mu, \nu)$ is a coarse plaquette, and $T_{C,p}$ is zero for other fine orientations. Every coefficient is nonnegative. Its row and column sums are exactly

$$\sum_p T_{C,p} = L^2, \qquad \sum_C T_{C,p} = L^{-2}. \tag{3.vi}$$

The row sum counts L^4 starting sites and L^2 plaquettes per surface, each with weight L^{-4}. For the column sum fix a fine plaquette and its orientation. In each of its two coordinate directions, each of the L choices of surface offset determines one block residue and one coarse origin. In the transverse directions the block residue determines the coarse origin uniquely. There are therefore L^2 representations, with total weight L^2L^{-4}. This count also holds modulo the periodic side lengths, with multiplicity.

More generally let $O_{C,p}$ be arbitrary orthogonal color matrices. Cauchy–Schwarz and eq. 3.vi imply

$$\sum_C \left| \sum_p T_{C,p} O_{C,p} f_p \right|^2 \le L^2 \sum_{C,p} T_{C,p}|f_p|^2 = \sum_p |f_p|^2. \tag{3.vii}$$

Thus independent local gauge frames do not change the leading constant. In dimension d the same argument gives norm $L^{2-d/2}$; it equals one in dimension four.

3.2.2. *Local gauges and the nonlinear remainder.* For each coarse plaquette C, choose a rectangular fine box D_C containing all paths in its four contour averages, with a fixed additional layer. Its side lengths are at most cL. The family of these boxes has bounded overlap. If a box crosses a periodic seam, perform this construction on a lift to the covering lattice. Path identities and the following overlap counts include the resulting periodic multiplicities.

Choose a coordinate comb tree in D_C, rooted at the initial vertex of C, and gauge its edge transports to one. Write the resulting local field as $W_b^{u_C} = e^{ia_b}$. There are constants depending

only on L such that

$$\max_{b\subset D_C}|a_b|\le C_L\varepsilon(W),\qquad \left(\sum_{b\subset D_C}|a_b|^2\right)^{1/2}\le C_L\left(\sum_{p\subset D_C}|F_p|^2\right)^{1/2}. \tag{3.viii}$$

Enlarge D_C by another fixed layer if needed for the plaquettes in the right side.

Here are finite proofs of these bounds. A remaining edge, together with its two comb paths, forms a loop. Move the coordinate steps of one path past those of the other, cancelling immediate backtracks. Each interchange inserts one elementary plaquette, conjugated by a prefix transport. There are at most C_L such interchanges. This gives an exact product of conjugated plaquette holonomies for $W_b^{u_C}$. All factors and conjugating transports are unitary, and $\|1-\prod_i V_i\|_{\mathrm{op}}\le\sum_i\|1-V_i\|_{\mathrm{op}}$. The small logarithm has a uniformly bounded derivative near one. Consequently $|a_b|\le C\sum_p n_{b,p}|F_p|$, where the nonnegative integers $n_{b,p}$ belong to a finite matrix of bounded row and column sums depending only on L. Its Schur bound gives the second inequality in eq. 3.viii; its row bound gives the first.

These same local charts admit the actual means. For finitely many nearby transports V_i, the defining equation $\sum_i\log(V_iV^{-1})=0$ has derivative minus the number of transports times the identity in the logarithmic coordinate of V, at the identity. The analytic implicit function theorem therefore gives a unique small mean on a fixed ball. There are only finitely many local patterns for fixed L. Decrease ε_L so that eq. 3.viii places all their transports in these balls. Gauge covariance of the mean follows from uniqueness and conjugation of the small logarithm. Means constructed in different local gauges therefore give the same original coarse transport.

Let $F_p^{(C)}=\mathrm{Ad}(u_C(p_-))F_p$. Taylor's formula for the fine plaquette logarithm on this local chart gives

$$F_p^{(C)}=(\mathrm{d}a)_p+r_p(a),\qquad \left(\sum_{p\subset D_C}|r_p(a)|^2\right)^{1/2}\le C_L\varepsilon(W)\left(\sum_{p\subset D_C}|F_p|^2\right)^{1/2}. \tag{3.ix}$$

For clarity, the Taylor remainder of one plaquette is bounded by $C\max_b|a_b|\sum_{b\subset\partial p}|a_b|$ on this fixed matrix ball. Sum its square, use the bounded incidence of edges and plaquettes, and apply eq. 3.viii. This proves the displayed estimate, including its dependence on the curvature rather than merely on the number of edges.

The coarse plaquette logarithm is an analytic function of the same finite list a, vanishes at zero, and has derivative $(\mathrm{d}_cQa)_C$ by eq. 3.iv. Its Taylor remainder has the bound

$$|r_C^{\mathrm{av}}(a)|\le C_L\max_b|a_b|\left(\sum_{b\subset D_C}|a_b|^2\right)^{1/2}.$$

Indeed the second derivative is bounded on a smaller fixed analytic ball, and the equivalence of the finite-dimensional norms costs only a constant depending on L. Substitute eq. 3.ix in eq. 3.v. After conjugating the coarse logarithm back, this gives

$$F_C^{\mathrm{av}}=\sum_p T_{C,p}O_{C,p}F_p+R_C,\qquad |R_C|\le C_L\varepsilon(W)\left(\sum_{p\subset D_C}|F_p|^2\right)^{1/2}, \tag{3.x}$$

Here the sum keeps separate the fine plaquette occurrences in the chosen lift of D_C. Each $O_{C,p}$ is an adjoint rotation and hence orthogonal. If several lifted occurrences project to the same physical plaquette, their rotations need not agree. In the global column sum group their scalar weights, retaining the separate rotations in the flux sum. The resulting physical column weight is still L^{-2}, and every row weight is still L^2. Thus the coefficients are exactly the weights of eq. 3.v, with their periodic multiplicities; no nonlinear coefficient replaces either leading sum.

Proof of Prop. 3.32. The row bound in eq. 3.vi and the finite size of D_C give eq. 3.ii from eq. 3.x. Bounded overlap and eq. 3.vii give

$$\left(\sum_C |F_C^{\mathrm{av}}|^2\right)^{1/2} \le (1 + C_L\varepsilon(W)) \left(\sum_p |F_p|^2\right)^{1/2}. \tag{3.xi}$$

The scalar Taylor estimate

$$\tfrac{1}{2}t^2(1 - Ct^2) \le 1 - \cos t \le \tfrac{1}{2}t^2$$

holds for small real t. Apply the upper estimate to the coarse field and the lower estimate to the fine field. Absorbing $(1 - C\varepsilon(W)^2)^{-1/2}$ in $1 + C_L\varepsilon(W)$ proves eq. 3.iii. □

3.2.3. *Iteration without a loss in the number of levels.*

Corollary 3.33 (An all-depth energy bound)**.** There is $\alpha_L > 0$ such that, for every integer $k \ge 0$ and every lattice admitting k block averages, the condition

$$\varepsilon(U) \le \alpha L^{-2k}, \qquad 0 < \alpha \le \alpha_L, \tag{3.xii}$$

admits all the near-identity branches used in those averages and implies

$$\varepsilon(\mathcal{M}^j U) \le 2\alpha L^{-2(k-j)} \quad (0 \le j \le k), \qquad A(\mathcal{M}^k U) \le 2A(U). \tag{3.xiii}$$

■□

Proof. Write $\varepsilon_j = \varepsilon(\mathcal{M}^j U)$. The first recurrence of Prop. 3.32 gives

$$\varepsilon_{j+1} \le L^2\varepsilon_j(1 + C_L\varepsilon_j/L^2).$$

Suppose the curvature bound in eq. 3.xiii holds up to a given level. Up to that level,

$$\sum_{j<k} \varepsilon_j \le 2\alpha \sum_{s\ge 1} L^{-2s} = \frac{2\alpha}{L^2 - 1}.$$

Iterating the recurrence therefore bounds $\varepsilon_j/(\alpha L^{-2(k-j)})$ by $\exp[2C_L\alpha/(L^2(L^2-1))]$. Choose α_L so that this is strictly less than two and $2\alpha_L < \varepsilon_L$. The first possible failure of the curvature bound is then impossible, and every mean used at the next level is admitted.

Iterate eq. 3.iii, retaining its leading constant one. It gives

$$A(\mathcal{M}^k U)^{1/2} \le \exp\left(C_L \sum_{j<k} \varepsilon_j\right) A(U)^{1/2} \le \exp\left(\frac{2C_L\alpha}{L^2 - 1}\right) A(U)^{1/2}.$$

Decrease α_L so that the last exponential is at most $\sqrt{2}$. Squaring proves the energy assertion. □

For example, if $\nu(p) \ge \nu_* > 0$ and one coarse plaquette obeys $\|1 - \partial\mathcal{M}^k U(C)\|_{\mathrm{op}} \ge a$, then

$$A(\nu, U) \ge \nu_* A(U) \ge \tfrac{1}{4}\nu_* a^2. \tag{3.xiv}$$

The last inequality uses $1 - \frac{1}{2}\operatorname{Tr} V = \frac{1}{2}\|1 - V\|_{\mathrm{op}}^2$ for $V \in \mathrm{SU}(2)$, and eq. 3.xiii. This is a global one-event certificate under eq. 3.xii. Several history events require disjoint ownership or a bounded-overlap allocation of the action. A field containing an unregularized old core does not satisfy the hypotheses of this corollary.

3.3. **The block average and its pivot chart.** The finite calculation here concerns the actual centered one-step average. Its definition uses an implicit group mean for each internal contour transport, followed by the exponential of an arithmetic mean of closed logarithms for the block bond. We use both operations. The calculation proves the local pivot and quadratic-remainder estimates; it does not prove a many-level background theorem or an exterior-force estimate.

The exact finite map. Work on a unit-spacing lattice in dimension four, with odd $L \geq 3$. The block at a coarse site y is $B(y) = \{x : |x-y|_\infty \leq (L-1)/2\}$. For a positive coarse bond $c = (y, y+Le_\mu)$, put $y' = y + Le_\mu$ and $D_c = B(y) \cup B(y')$. On a torus we assume that this double block and its one-bond neighborhood have an embedded rectangular lift preserving their unit bond graph, with no periodic edge joining opposite faces of the patch. A sufficient condition for a cubic torus is that its period exceeds $4L+4$. All bonds used below have both endpoints in D_c, with the rectangular bond graph of this lift.

For $x \in B(y)$, let $\mathcal{G}(y,x)$ be the shortest paths which complete one coordinate displacement before starting the next. There are at most 24 such paths, each of length at most $2L$. Repeated paths, if retained, are counted with their prescribed equal weights. The transport along an oriented path is the ordered product of its bond matrices; reversed bonds use the inverse matrix. For nearby matrices $T_1, \dots, T_n$, define their contour mean as the solution near them of

$$\frac{1}{n}\sum_{j=1}^{n} \log(T_j T^{-1}) = 0. \tag{3.xv}$$

All logarithms below are the local analytic logarithm near the identity. Set

$$U(y,x) = \operatorname{av}\big(U(\Gamma) : \Gamma \in \mathcal{G}(y,x)\big), \qquad U(x,y) = U(y,x)^{-1}. \tag{3.xvi}$$

With $x' = x + Le_\mu$, the actual block map is

$$\begin{aligned} Z_x(U) &= U(y,x)U([x,x'])U(x',y')U[c]^{-1}, \\ \mathcal{M}(U)(c) &= \exp\left[L^{-4}\sum_{x\in B(y)} \log Z_x(U)\right] U[c]. \end{aligned} \tag{3.xvii}$$

Here $[x,x']$ and $[c]$ are straight segments of length L. In particular, the outer operation in eq. 3.xvii is not the implicit mean eq. 3.xv. The cruder Borel definitions of the block map away from this analytic neighborhood do not enter the calculation.

For a background V, let $W = \mathcal{M}(V)$. In physical bond logarithms, with no scale factor in the exponential, put

$$q_{V,c}(a) = \frac{1}{i}\log\left[\mathcal{M}(e^{ia}V)(c)\mathcal{M}(V)(c)^{-1}\right]. \tag{3.xviii}$$

The matrices $a(b)$ are complex traceless 2 by 2 matrices, and their norm is the supremum of the operator norms over positive bonds. On real variables they are Hermitian. All constants denoted C, c below are universal; their values may be enlarged or decreased at successive occurrences.

The contour mean, with quantitative bounds. The elementary matrix inequalities

$$\|e^X - I - X\| \leq \tfrac{1}{2}e^{\|X\|}\|X\|^2, \qquad \|\log(I+H) - H\| \leq \frac{\|H\|^2}{2(1-\|H\|)} \tag{3.xix}$$

hold for $\|H\| < 1$, by the power series and submultiplicativity. They show, on a fixed small ball, that

$$\begin{aligned} \log(e^X e^{-Z}) &= X - Z + E(X,Z), \\ \|E(X,Z)\| &\leq C(\|X\| + \|Z\|)^2, \\ \|D_Z E(X,Z)\| &\leq C(\|X\| + \|Z\|). \end{aligned} \tag{3.xx}$$

For the derivative estimate one can differentiate the convergent logarithm series term by term and use $De^Z[H] = \int_0^1 e^{(1-t)Z} H e^{tZ}\, dt$. The derivative of each remainder in eq. 3.xix is bounded by a constant times the radius. Thus the constants in eq. 3.xx depend on no number of contour paths.

Write $T_j = e^{X_j}$, with $\max_j \|X_j\| \leq t$, and seek $T = e^Z$. eq. 3.xv is equivalent to

$$Z = \overline{X} + \frac{1}{n}\sum_j E(X_j, Z), \qquad \overline{X} = \frac{1}{n}\sum_j X_j. \tag{3.xxi}$$

For a fixed sufficiently small t, the right side maps $\|Z\| \le 2t$ into itself and has Lipschitz constant at most $1/2$. Iteration from $\overline{X}$ proves existence and uniqueness there. It also proves $Z = \overline{X} + O(t^2)$, uniformly in n. The iterates are holomorphic in all X_j, and convergence is locally uniform on a slightly smaller fixed ball. Their limit is therefore holomorphic. Trace zero is preserved in eq. 3.xxi. For anti-Hermitian X_j, anti-Hermiticity is preserved as well, so the mean lies in SU(2) on real group inputs.

Uniqueness proves permutation symmetry and left-right covariance. Indeed under $T_j \mapsto uT_jv$, with $T \mapsto uTv$, every logarithm in eq. 3.xv is conjugated by u, whenever the transformed families remain in their analytic domains. The inverse property follows from $\log(T_j^{-1}T) = -\mathrm{Ad}(T^{-1})\log(T_jT^{-1})$. Thus the locally constructed mean is the mean in eq. 3.xvi, including families centered at a unitary matrix other than I.

Local gauges do not constrain coarse holonomies. For a real background V_0 on D_c, put $\varpi_c = \sup_{p\subset D_c}\|\partial V_0(p) - I\|$. There is a unitary gauge s_c on this rectangle such that

$$\sup_{b\subset D_c}\|V_0^{s_c}(b) - I\| \le CL\varpi_c. \tag{3.xxii}$$

Here $V^s(b) = s(b_-)V(b)s(b_+)^{-1}$. To prove eq. 3.xxii, fix a coordinate-order comb from y to every site of the rectangle and set $s_c(x) = V_0(\Gamma_{y,x})$. For any bond b, the product $s_c(b_-)V_0(b)s_c(b_+)^{-1}$ is its comb loop. Slide the additional bond past the later coordinate segments of the comb. Each slide exchanges two adjacent perpendicular steps and inserts one plaquette, conjugated by a partial transport. There are at most CL slides, since every coordinate interval in the rectangle has length at most $2L$. All these conjugations are unitary. The norm of a product of such plaquette factors minus I is at most the sum of their deviations, proving eq. 3.xxii.

The same gauge, fixed from the real background, works for $V = e^{iE}V_0$: its distance from I is at most $C(L\varpi_c + \|E\|_\infty)$, when these quantities are small. More generally we can start directly with a unitary local gauge in which

$$\sup_{b\subset D_c}\|V^{s_c}(b) - I\| \le \epsilon. \tag{3.xxiii}$$

Different double blocks may use different gauges. We never require one gauge close to I on an entire observation window.

The group mean and eq. 3.xvii are gauge covariant. The internal transport transforms with its two endpoint factors; each Z_x transforms by conjugation at y. Consequently

$$q_{V^{s_c},c}(a^{s_c}) = \mathrm{Ad}(s_c(y))q_{V,c}(a), \qquad a^{s_c}(b) = \mathrm{Ad}(s_c(b_-))a(b). \tag{3.xxiv}$$

All these adjoint actions have norm one. A coarse bond value $\mathcal{M}(V)(c)$ in the original gauge may be anywhere in SU(2). It is the ratio in eq. 3.xviii, rather than that coarse bond itself, whose logarithm is taken near I. Nontrivial coarse holonomies therefore cause no loss in these local estimates.

Uniform Taylor estimates for the actual average. We first work in the gauge eq. 3.xxiii, so $V = e^{iv}$ with $\|v\|_\infty \le C\epsilon$. If $\|v\|_\infty, \|a\|_\infty < c_0/L$, every path product entering eq. 3.xvi stays in a fixed small neighborhood of I. For example, a product of N factors, each within t of I, is within $e^{CNt} - 1$ of I. Its reversed factors satisfy the same bound. Here the internal paths have $N \le 2L$. The contour-mean estimate just proved applies, and the closed products in eq. 3.xvii, of total path length at most $6L$, are also in a fixed logarithm neighborhood. The outer logarithm average is normalized by L^{-4}; its size is bounded by its largest summand, without a factor L^4.

It follows that

$$F_c(v,a) = \frac{1}{i}\log\left[\mathcal{M}(e^{ia}e^{iv})(c)\mathcal{M}(e^{iv})(c)^{-1}\right] \tag{3.xxv}$$

is holomorphic and bounded by a universal constant on the joint supremum balls of radius c_0/L. It vanishes at $a = 0$. This establishes the domain before any derivative estimate. On balls a fixed

fraction smaller, two scalar Cauchy formulas give

$$\|D_a F_c\|_{\infty\to\mathrm{op}} \le CL, \qquad \|D_a^2 F_c\| + \|D_v D_a F_c\| \le CL^2. \tag{3.xxvi}$$

For example, for unit supremum directions u, w, apply Cauchy to $F_c(v, a + zu + tw)$, with both scalar circles of radius c/L. This proves the mixed bilinear bound without dividing the radius by the number of bond coordinates.

Let $K_V = Dq_V(0)$ and $R_V(a) = q_V(a) - K_V a$. Taylor's formula with integral remainder gives

$$\begin{aligned} \|K_V\|_{\infty\to\infty} &\le CL, \\ \|R_V(a)\|_\infty &\le CL^2 \|a\|_\infty^2, \\ \|DR_V(a)\|_{\infty\to\infty} &\le CL^2 \|a\|_\infty. \end{aligned} \tag{3.xxvii}$$

for $\|a\|_\infty < c/L$ and $\epsilon < c/L$. The estimates transfer to the original frame by eq. 3.xxiv. They are uniform over all coarse bonds, hence over the volume. In addition, varying the background in eq. 3.xxvi gives

$$\|Dq_{e^{iv}}(0) - Dq_I(0)\|_{\infty\to\infty} \le CL^2 \|v\|_\infty. \tag{3.xxviii}$$

These are estimates for the explicit nonlinear average in eq. 3.xvii, not for a model linear average.

The pivot coefficient and finite elimination. Write $h = (L-1)/2$. The pivot of $c = (y, y + Le_\mu)$ is $b_0(c) = (y + he_\mu, y + (h+1)e_\mu)$. It crosses from $B(y)$ to $B(y')$. No internal contour path contains it. Among all coarse pivots, only $b_0(c)$ occurs in the support of $q_{V,c}$. Indeed internal contours stay inside one block; each straight segment in eq. 3.xvii crosses only the face between these two blocks and is parallel to e_μ. The only coarse pivot on that face is $b_0(c)$. This support assertion, rather than a claim about all bonds joining two vertices of a periodic double block, is what is used below.

At the identity background define $\overline{a}(y, x)$ to be the arithmetic mean of the oriented path sums of a over $\mathcal{G}(y, x)$, and $\sigma_a(y) = L^{-4} \sum_{x\in B(y)} \overline{a}(y, x)$. The first derivative of the contour mean is this arithmetic mean. Differentiating eq. 3.xvii, the negative derivative of $U[c]$ in each closed logarithm cancels its final positive derivative. Thus

$$(K_I a)(c) = \sigma_a(y) + L^{-4} \sum_{x\in B(y)} a([x, x']) - \sigma_a(y'). \tag{3.xxix}$$

The two σ_a terms contain no pivot. Exactly L segments in the remaining sum contain $b_0(c)$: their transverse coordinates equal those of y, and their longitudinal starting coordinate can be any of the L coordinates in the block. All occurrences have positive orientation. Therefore

$$J_c(I) := D_{a(b_0(c))} q_{I,c}(0) = L^{-3} I_3. \tag{3.xxx}$$

For a pure gauge $V^s = I$, the coefficient in its original frame is

$$J_c(V) = L^{-3} \operatorname{Ad}(s(y)^{-1} s(b_0(c)_-)). \tag{3.xxxi}$$

The unitary matrix in this formula is the prefix transport from y to the start of the pivot. Replacing it by I in the original gauge would be incorrect.

From eq. 3.xxviii, after the unitary local change of frame, we have $\|J_c(V^{s_c}) - L^{-3} I_3\| \le CL^2\epsilon$. Consequently, under the conservative condition

$$\epsilon \le cL^{-5}, \qquad \text{in particular if} \quad \|E\|_\infty + L\varpi_c \le cL^{-5}, \tag{3.xxxii}$$

every J_c is invertible and $\|J_c^{-1}\| \le 2L^3$. There is no off-diagonal pivot coefficient. Set

$$(h_V \nu)(b_0(c)) = J_c^{-1} \nu(c), \qquad (h_V \nu)(b) = 0 \ (b \text{ nonpivot}), \qquad \mathsf{C}_V = \iota - h_V K_V \iota, \tag{3.xxxiii}$$

where ι extends the nonpivot coordinates by zero. Then

$$K_V h_V = 1, \quad K_V \mathsf{C}_V = 0, \quad (\mathsf{C}_V u)|_{\mathrm{free}} = u, \qquad \|h_V\| \le 2L^3, \quad \|\mathsf{C}_V\| \le CL^4. \tag{3.xxxiv}$$

These are supremum operator norms. The same powers bound column norms. Indeed a fine bond belongs to at most a fixed number of double blocks, and eq. 3.xxvi bounds the coefficient in each of

their rows. The maps are local: a pivot row uses only its own double block. They are holomorphic in the background, by the Neumann series for J_c^{-1}.

For comparison with a convention which inserts a/L in the one-step exponential, its linear average is $Q_V = K_V/L$, and its pivot coefficient is

$$\mathsf{m}_c = J_c/L, \qquad \mathsf{m}_c(I) = L^{-4} I_3, \qquad \det J_c = L^3 \det \mathsf{m}_c. \tag{3.xxxv}$$

This accounts for the factor L^{-3} in a real elimination Jacobian; it is not a change in the number of color coordinates.

The nonlinear graph, constructed rather than assumed. For a full physical bond variable t, solve

$$d = R_V(t - h_V d), \qquad \Psi_V(t) = t - h_V d(t). \tag{3.xxxvi}$$

On $\|t\|_\infty < r_1 = cL^{-5}$, eq. 3.xxvii maps the ball $\|d\|_\infty \le CL^2\|t\|_\infty^2$ into itself, after choosing its fixed prefactor large enough. Its Lipschitz constant in d is at most $CL^5 r_1 < 1/2$. Iteration proves the unique solution, its holomorphic dependence, and

$$\begin{aligned} q_V(\Psi_V(t)) &= K_V t, & \|d(t)\| &\le CL^2\|t\|^2, \\ \|D_t d(t)\| &\le CL^2\|t\|, & \|D\Psi_V(t) - 1\| &\le CL^5\|t\|. \end{aligned} \tag{3.xxxvii}$$

The derivative estimate follows by differentiating the contraction and inverting $1 + DR_V h_V$ by its convergent Neumann series. Only the pivot values are changed. There is even an explicit inverse, on a smaller neighborhood: $\Psi_V^{-1}(a) = a + h_V R_V(a)$. Substitution in eq. 3.xxxvi verifies this formula by uniqueness.

Each equation for $d(c)$ uses only its own pivot value and the nonpivot data in D_c. The equations for different coarse bonds are independent once the nonpivot data are fixed. Thus the graph is exactly finite range, not merely exponentially local. In particular

$$a(u) = \Psi_V(\mathsf{C}_V u), \quad q_V(a(u)) = 0, \quad a(u)|_{\mathrm{free}} = u, \qquad \|u\|_\infty < cL^{-9} \tag{3.xxxviii}$$

is the actual small fibre chart. It obeys $\|a(u)\|_\infty + \|D_u a(u)\|\|u\|_\infty \le CL^4\|u\|_\infty$. Conversely, any sufficiently small solution of $q_V(a) = 0$ has this form: $t = a + h_V R_V(a)$ satisfies $K_V t = 0$, hence $t = \mathsf{C}_V(a|_{\mathrm{free}})$, and eq. 3.xxxvii gives uniqueness.

For the canonical Gaussian coordinate $u = gy$, the physical chart therefore contains $\|y\|_\infty < cL^{-9}/g$. At weak coupling it contains any radius $c_*\alpha/g$ with fixed c_* and sufficiently small α. This establishes the one-step primitive radius; inclusion in all the older background domains remains a different estimate.

Real coarea normalization and complex continuation. We record the normalization only on the real positive chart first. Let $\mathcal{E}, \mathcal{C}, \mathcal{F}$ be the fine positive bonds, coarse positive bonds and nonpivots, respectively. Thus $|\mathcal{F}| = |\mathcal{E}| - |\mathcal{C}|$. In real Lie algebra coordinates Haar measure is $d(e^{ia}) = \sigma(a) d^3a$, where $\sigma(a) = \sigma_0(\sin|a|/|a|)^2$ and $\sigma_0 > 0$. Group delta in the same chart satisfies $\delta_G(e^{iq}) = \sigma_0^{-1}\delta_{\mathbb{R}^3}(q)$.

In the linear variables $t = \mathsf{C}_V u + h_V v$, free coordinates are u and $K_V t = v$. Its determinant is $\prod_c \det J_c^{-1}$. The full straightening $a = \Psi_V(t)$ has derivative close to the identity by eq. 3.xxxvii. At real backgrounds, the determinants $\det J_c$ are positive: in the local unitary frame J_c is within half of the positive scalar matrix $L^{-3} I_3$, and the real adjoint frame transformations have determinant one. Also $\det D\Psi_V > 0$, by continuity from its value one at zero and the uniform derivative bound. The ordinary real change of variables therefore gives, on the fibre,

$$\begin{aligned} &\prod_{b\in\mathcal{E}} dU_b \prod_{c\in\mathcal{C}} \delta_G(\mathcal{M}(U)(c) W(c)^{-1}) \\ &\quad = \sigma_0^{|\mathcal{F}|} \prod_c (\det J_c)^{-1} \det D\Psi_V(\mathsf{C}_V u) \prod_{b\in\mathcal{E}} \frac{\sigma(a_b(u))}{\sigma_0} \prod_{b\in\mathcal{F}} d^3u_b. \end{aligned} \tag{3.xxxix}$$

This is an identity of local fibre measures, with the group delta interpreted by this change of coordinates. The straightening derivative is block triangular: its free block is the identity, and its pivot diagonal has one three by three block per coarse bond. Hence its log determinant is a sum of local trace logarithms. Substituting $u = gy$ in eq. 3.xxxix supplies $(\sigma_0 g^3)^{|\mathcal{F}|}$, while $(\det J_c)^{-1} = L^{-3}(\det \mathsf{m}_c)^{-1}$.

For complex backgrounds the coefficient functions in eq. 3.xxxix have their coherent holomorphic continuations on the chart just constructed. In particular $(\det J_c)^{-1}$ and the branch of $\log\det J_c$ continuing its real value are used, not an absolute value of a complex determinant. The Haar density is continued by the entire power series $\sigma_0[\sin\sqrt{z}/\sqrt{z}]^2$, $z = \frac{1}{2}\operatorname{tr} a^2$, rather than by a complex operator norm. No complex measure or complex delta function is being asserted by this continuation.

Dependence on the remaining scales. The average, its raw Taylor constant CL^2, and the local pivot construction contain no M, observation radius or number of free variables. If the supplied local near-pure-gauge bound is only $\|V^s - I\| \le C_b M a$, eq. 3.xxxii requires $a \le c/(C_b M L^5)$. Alternatively the direct real condition $\varpi_c \le cL^{-6}$, with a small complex perturbation, suffices. The input-to-chart constants are conservative fixed powers of L; no claim of optimal powers is needed.

If only a complex gauge with condition numbers at most $K \ge 1$ is available, conjugation charges one factor K per input direction and one for the output. Thus safe bounds are CK^3L^2 for the quadratic residual, $2K^2L^3$ for h_V, and CK^4L^4 for C_V. The conservative physical nonpivot radius becomes $cK^{-9}L^{-9}$. A condition number growing with a larger window must therefore be charged; it cannot be discarded by calling the one-step chart local.

The calculation discharges the one-step pivot and Taylor primitives in these actual physical coordinates. It does not bound the many-level response or a further coordinate transformation on a large run window. In particular, a polynomial loss in that window radius in the background/core inclusion must still be retained in a later Cauchy radius. The free precision gap, the all-window core and tower estimates, the conditional exterior forces and the clean-class law remain separate assertions.

3.4. **A gap for the classical flat precision.** The gauge form in this section is the actual soft contour gauge. It is not a divergence projector. At the flat field its quadratic form is the sum of the squares of the arithmetic internal contour path sums. The constraint is the plain derivative of the nonlinear block average, rather than the straight average alone. With these operators the classical flat precision has a gap uniform in the torus volume and in the number of finer averaging levels.

The statement concerns the unweighted Wilson action at a flat background, with its prescribed positive soft gauge coefficient. Signed counterterms, nonconstant layer weights, nonzero-current constraint corrections and complex decoupling matrices are not included in the conclusion.

The actual quadratic forms and their normalization. Use raw bond logarithms on a unit-spacing four-dimensional lattice. For a scalar site function f and a positively oriented bond field a, set

$$\begin{aligned}(\mathrm{d}f)_\mu(x) &= f(x+e_\mu) - f(x),\\ (\mathrm{d}a)_{\mu\nu}(x) &= a_\mu(x) + a_\nu(x+e_\mu) - a_\mu(x+e_\nu) - a_\nu(x).\end{aligned} \tag{3.xl}$$

The same convention, with step n, is written d_n on a lattice of spacing n. Norms are counting ℓ^2 norms of sites, positive bonds or plaquettes, with each unordered direction pair counted once. One color suffices for the proof; the flat SU(2) forms are three identical copies.

The Wilson convention $A(U) = \sum_p[1 - \frac{1}{2}\operatorname{Re}\operatorname{tr} U_p]$ gives $A(e^{ita}) = \frac{1}{2}t^2\|\mathrm{d}a\|^2 + O(t^3)$. For the actual soft gauge,

$$\mathrm{GF}(V) = \sum_y \sum_{x\in B(y)\setminus\{y\}} [1 - \tfrac{1}{2}\operatorname{Re}\operatorname{tr} V(y,x)]. \tag{3.xli}$$

Here $V(y,x)$ is the implicit mean of the transports along the coordinate-order internal paths. Its first derivative at the identity is the arithmetic mean of the corresponding oriented path sums. Denote this linear functional by $\gamma_{y,x}(a)$, and put $(Ga)_{y,x} = \gamma_{y,x}(a)$. Consequently

$$\mathrm{GF}(e^{ita}) = \tfrac{1}{2}t^2\|Ga\|^2 + O(t^3). \tag{3.xlii}$$

Internal paths contain no crossing pivot. Thus the gauge form on nonpivot variables is local to each block and has no pivot coordinate. For a gradient, $\gamma_{y,x}(\mathrm{d}f) = f(x) - f(y)$.

The physical step uses the scale $g = \theta^{-1/2}$ and gauge weight θ. For the real unweighted step the action coefficient is also θ. Hence $\theta g^2 = 1$, and the two quadratic forms above have coefficient one in the canonical Gaussian variables. Normalizing constants and the Haar Jacobian belong to other parts of the step; they do not change this definition of its free precision.

Plain and straight averaging. For odd n, let $B_n(y)$ be the centered block of n^4 sites. The raw straight bond average is

$$(S_n a)_\mu(y) = n^{-4} \sum_{x \in B_n(y)} \sum_{t=0}^{n-1} a_\mu(x + te_\mu). \tag{3.xliii}$$

At one step define $\sigma(a)(y) = L^{-4}\sum_{x\in B_L(y)} \gamma_{y,x}(a)$. Direct differentiation of the actual outer exponential of closed logarithms gives the plain derivative

$$P_L a = S_L a - \mathrm{d}_L \sigma(a). \tag{3.xliv}$$

The derivative of its reference straight segment cancels its negative occurrence in every closed logarithm. This explains why no extra reference-segment term is present in eq. 3.xliv.

Let $q_L f$ be the arithmetic site average over $B_L(y)$, and $r_L f = f|_{\text{block centers}}$. Telescoping along paths proves

$$S_L \mathrm{d}f = \mathrm{d}_L q_L f, \qquad \sigma(\mathrm{d}f) = q_L f - r_L f, \qquad P_L \mathrm{d}f = \mathrm{d}_L r_L f. \tag{3.xlv}$$

In particular it is restriction at the centers, not a site average, that intertwines with the plain constraint.

Write $P^{[k]}$ for the derivative at the identity of the plain k-fold nonlinear average, with raw logarithms at both ends. For $n = L^k$,

$$P^{[k]} a = S_n a + \mathrm{d}_n \psi_k(a) \tag{3.xlvi}$$

for a linear site functional ψ_k. Here is a direct verification. Composition of the straight averages equals S_{nL}: in each coordinate the centered digits $nr + u$, with $r \in \{-(L-1)/2, \dots, (L-1)/2\}$ and $u \in \{-(n-1)/2, \dots, (n-1)/2\}$, tile the larger centered block, and the segment indices $nt + j$, $0 \le t < L$, $0 \le j < n$, tile the longer straight segment. All weights multiply to $(nL)^{-4}$. Applying the next plain step to the gradient in eq. 3.xlvi gives another coarse gradient by eq. 3.xlv. Its application to $S_n a$ differs from the next straight average by a gradient, by eq. 3.xliv. This proves eq. 3.xlvi inductively, starting with eq. 3.xliv.

Curvature averaging is a contraction in dimension four. Expanding the boundary of an n by n square gives

$$(\mathrm{d}_n S_n a)_{\mu\nu}(y) = n^{-4} \sum_{x \in B_n(y)} \sum_{r,s=0}^{n-1} (\mathrm{d}a)_{\mu\nu}(x + re_\mu + se_\nu). \tag{3.xlvii}$$

Every interior bond cancels in this expansion; its four remaining sides are exactly the four straight averages in $\mathrm{d}_n S_n$. Call the positive averaging operator on plaquettes on the right T_n. Each row has sum n^2. Each fine plaquette has total column weight n^{-2}: for each pair (r,s) its starting point determines exactly one block $B_n(y)$, since those blocks partition the fine sites. There are n^2 such pairs, each with weight n^{-4}.

For nonnegative weights t_{Pp}, weighted Cauchy gives

$$\sum_P \left|\sum_p t_{Pp}F_p\right|^2 \le \sum_P \left(\sum_p t_{Pp}\right)\left(\sum_p t_{Pp}|F_p|^2\right) \le n^2 n^{-2}\sum_p |F_p|^2. \tag{3.xlviii}$$

Thus $\|T_n\|_{2\to 2} \le 1$. Counting with multiplicities makes the same proof valid on a periodic lattice. The product of row and column sums in dimension d would be n^{4-d}; its cancellation here uses dimension four. Taking curls of eq. 3.xlvi therefore proves the exact uniform bound

$$\|\mathrm{d}_n P^{[k]} w\| \le \|\mathrm{d}w\|. \tag{3.xlix}$$

No bound on a background propagator was used.

The classical constrained pullback. For current bond data a, on the level-k lattice, define the classical flat quadratic value

$$E_k(a) = \min\{\|\mathrm{d}w\|^2 : P^{[k]}w = a\}. \tag{3.l}$$

The affine set is nonempty. For instance place the value of a coarse bond on every fine bond crossing its block face and put zero on fine bonds internal to a block of side $n = L^k$. The nonlinear group lift with internal bonds I has exactly the prescribed average: all parallel segments cross the same face value and all closed logarithms vanish. Applying this observation successively through the centered subdivisions gives the same assertion at k levels. Its derivative is a right inverse for $P^{[k]}$.

Existence of the minimum in eq. 3.l needs no uniform inverse theorem. The image under d of the finite-dimensional affine constraint set is a closed affine subspace; its point nearest zero exists. Equivalently, quotient $\ker P^{[k]} \cap \ker \mathrm{d}$, restrict to its orthogonal complement, and solve the positive finite-dimensional normal equations there. This also shows that the minimum is a quadratic form in a. Its construction at this stage has no claimed uniform inverse bound. eq. 3.xlix, applied to every admissible lift, gives

$$E_k(a) \ge \|\mathrm{d}_n a\|^2. \tag{3.li}$$

We henceforth express the current lattice in units of its own spacing, so the curl on the right is written $\mathrm{d}a$.

This is the classical action part of the actual flat precision. Indeed the first variation of the bare action at the identity is zero. A differentiable exact background lift has a tangent w with $P^{[k]}w = a$; its action Hessian is $\|\mathrm{d}w\|^2$, because its second derivative is multiplied by that zero first variation. Linearized constrained stationarity makes this tangent minimize the quadratic form in eq. 3.l, modulo its gauge null space. The same observation eliminates the second-order one-step constraint correction at a flat point. In particular, no nonzero-current correction has been dropped in establishing eq. 3.l at the flat point. The finite Gaussian form in eq. 3.l is defined without assuming a nonlinear background theorem; when that background is used, this argument identifies its flat quadratic pullback. Even an arbitrary exact differentiable lift would suffice for the lower bound in eq. 3.li.

Let C be the actual flat pivot inclusion, with identity on nonpivot coordinates. The physical flat free precision is characterized by

$$\langle u, A_k^{\mathrm{flat}} u\rangle = E_k(Cu) + \|GCu\|^2 \ge \|\mathrm{d}Cu\|^2 + \|GCu\|^2. \tag{3.lii}$$

This formula is for the actual contour gauge and the actual plain fibre. It is not obtained by substituting a Landau projection for G.

A fixed-cell Fourier calculation. Group the current lattice into blocks of side L. In one cell let $R = \{-(L-1)/2, \dots, (L-1)/2\}^4$. A bond is indexed by its starting site $r \in R$ and direction. There are $4L^4$ bond coordinates and four pivot coordinates. The free space consequently has dimension $4L^4 - 4$, independently of the volume. For a coarse Bloch parameter $p \in (\mathbb{R}/2\pi\mathbb{Z})^4$, extend a cell field by

$$a(Lm + r, \mu) = e^{ip\cdot m} a(r, \mu). \tag{3.liii}$$

The curl matrix d_p has entries which are constants or $e^{\pm ip_\mu}$. The internal-contour matrix G uses only edges within its cell, hence has no p dependence. The plain constraint matrix $P(p)$, obtained from eq. 3.xliv, is also a finite matrix with such phase entries.

The middle pivot of direction μ is the crossing bond from he_μ to $(h+1)e_\mu$, $h=(L-1)/2$. At the identity its coefficient in the raw constraint is L^{-3}, since exactly L of the L^4 parallel segments contain it. There is no other pivot in that constraint. It is the own pivot in the cell of the coarse bond's start, so this coefficient has no Bloch phase. Thus the pivot block of $P(p)$ is $L^{-3}I_4$ for every p. If h_0 inserts L^3 on those pivots and ι inserts free coordinates with zero pivots, then

$$C(p)=\iota-h_0P(p)\iota,\quad P(p)C(p)=0,\quad C(p)|_{\mathrm{free}}=1,\quad \operatorname{Ran}C(p)=\ker P(p). \tag{3.liv}$$

The last equality follows by solving each pivot equation. It also proves the rank $4L^4-4$ for all p. All entries of $C(p)$ are continuous Laurent polynomials in the Bloch phases. This is the same finite-range pivot map as in the physical chart, evaluated at the flat field.

Define the Hermitian finite matrix

$$B(p)=C(p)^*(\mathrm{d}_p^*\mathrm{d}_p+G^*G)C(p). \tag{3.lv}$$

We prove that it has no zero vector for every p, including $p=0$. Suppose $a=C(p)u$ and $\mathrm{d}_pa=0$, $Ga=0$. Extend it by eq. 3.liii. Within a block, two coordinate-order paths to the same site differ by plaquette swaps. Zero curl makes their sums equal. The same swaps express a comb loop with one additional internal bond as a sum of internal plaquette curls. Consequently zero G forces every internal path sum, and then every internal bond, to vanish.

Across a face perpendicular to e_μ, a plaquette with one transverse step compares two neighboring crossing values. Its other two bonds are internal to the two blocks, hence zero. The crossing values are therefore equal. The transverse face grid is connected, so all crossing bonds on that face have a common value $\beta_{y,\mu}$. Every straight segment from $x\in B(y)$ to $x+Le_\mu$ then has sum $\beta_{y,\mu}$, and all internal contour sums are zero. The actual constraint gives $(P_La)(y,\mu)=\beta_{y,\mu}$. It vanishes by eq. 3.liv, killing every crossing bond as well. Thus $a=0$, and $u=0$ by its free restriction. The argument is valid for every real Bloch parameter, with complex fiber vectors.

One can also describe the possible closed fields explicitly. On the cell, the L^4 functions $L^{-2}\exp(i\xi_\ell\cdot r)$, $\xi_\ell=(p+2\pi\ell)/L$, form an orthonormal basis, where each component of ℓ runs from 0 to $L-1$. Orthogonality is the finite geometric-series identity. Fine differences multiply a mode by $t_\mu=e^{i\xi_{\ell,\mu}}-1$. The closedness equations are $t_\mu a_\nu-t_\nu a_\mu=0$. If one $t_\mu\neq0$, they say exactly $a=t\phi$, a gradient. The only remaining case is $p=0,\ell=0$, giving a constant harmonic vector. The local argument above kills that case as well; no zero-frequency mode has been silently omitted.

The entries of $B(p)$ are continuous on the compact Bloch torus. Its least Rayleigh quotient therefore has a positive uniform minimum $c(L)>0$. Explicitly, if the infimum were zero, a sequence of parameters and unit free vectors would have a convergent subsequence. Their limit would be a unit zero vector for the limiting $B(p)$, contradicting the previous paragraph. Thus

$$B(p)\geq c(L)1\qquad\text{for every } p. \tag{3.lvi}$$

On a torus with N_μ coarse blocks in each direction, the allowed parameters are $p_\mu=2\pi j_\mu/N_\mu$. The normalized discrete Fourier transform in the block index is unitary, again by the same geometric-series identity. It converts the comparison form on the right of eq. 3.lii into the sum of eq. 3.lv over these parameters. Parseval and eq. 3.lvi therefore prove

$$A_k^{\mathrm{flat}}\geq c(L)1 \tag{3.lvii}$$

on the actual free nonpivot coordinates, uniformly in the torus volume and in k. Local formulas can first be used on the admissible large tori; no small-torus identification of different pivots is needed. The three color copies have the same constant.

Flat holonomies and the limit of the conclusion. The same conclusion holds for a real flat background in its prescribed block axial representative. Flatness makes internal transports path independent. Since their means in this representative are I, every internal bond is I. Flat plaquettes make the crossing value constant on each block face. The background is therefore the lift of a flat coarse connection. A gauge constant on each block flattens it on the universal cover and preserves the soft contour gauge norm. Only twisted periodicity remains.

The four torus holonomies commute: their commutator is the product of the plaquettes of a fundamental rectangle, all equal to I. Commuting unitary matrices have a common orthonormal eigenbasis; choose an eigenspace of any noncentral holonomy and use its invariance under the others, or any basis if all are central. Their adjoint actions are consequently commuting unitaries on the complexified color space. Decomposing into their common eigenspaces gives characters $e^{i\chi_\mu}$. The allowed block momenta in that character are $(2\pi j_\mu + \chi_\mu)/N_\mu$, up to the sign convention for the boundary twist. The local matrices and the curvature averaging proof are unchanged: the boundary phases have modulus one, so the absolute row and column counts in eq. 3.xlviii still apply. Since eq. 3.lvi holds for every real Bloch parameter, all these shifted sets satisfy the same gap. No smallness of global holonomies is required.

The result proves the classical flat gap for the actual contour gauge and plain constraint. It is a lower bound, and does not prove a uniform upper bound or a decay estimate for a dressed precision. Nonzero-current terms from curved constraints, signed layer profiles, counterterm arrays, non-Abelian perturbations and enlarged complex decoupling parameters still need their own quantitative comparison with this positive flat form. No full background or interacting Gaussian-law assertion follows from eq. 3.lvii alone.

3.5. **An upper bound for the classical flat action.** This section concerns the same unweighted classical Gaussian value as the flat-gap calculation. Let $n = L^k$, with L odd, and let $P^{[k]}$ be the raw derivative of the actual plain iterated average. For a coarse bond field a, put

$$E_k(a) = \min\{\|\mathrm{d}w\|^2 : P^{[k]}w = a\}. \tag{3.lviii}$$

We construct an exact lift with energy at most $256\|\mathrm{d}_n a\|^2$, uniformly in k and in the volume. No bound on a non-Abelian background response is used.

Discrete interpolation on the actual centered grids. Fine sites are integers and coarse sites are nm, $m \in \mathbb{Z}^4$. Write

$$\varphi_n(t) = \max\{1 - |t|/n, 0\}, \qquad \beta_n(t) = \mathbf{1}_{\{0 \le t < n\}}, \quad t \in \mathbb{Z}. \tag{3.lix}$$

Periodize these functions on a finite torus. The identities $\sum_m \varphi_n(x - nm) = 1$, $\sum_m \beta_n(x - nm) = 1$, and $\sum_{x\in\mathbb{Z}} \varphi_n(x) = n$ follow by summing the two linear pieces. Also

$$\varphi_n(t+1) - \varphi_n(t) = n^{-1}\big[\beta_n(t+n) - \beta_n(t)\big]. \tag{3.lx}$$

Define interpolation of sites, bonds and plaquettes by

$$\begin{aligned}
(I_0 f)(x) &= \sum_m f(m) \prod_\lambda \varphi_n(x_\lambda - nm_\lambda),\\
(I_1 a)_\mu(x) &= n^{-1} \sum_m a_\mu(m)\beta_n(x_\mu - nm_\mu) \prod_{\lambda\neq\mu} \varphi_n(x_\lambda - nm_\lambda),\\
(I_2 F)_{\mu\nu}(x) &= n^{-2} \sum_m F_{\mu\nu}(m)\beta_n(x_\mu - nm_\mu)\beta_n(x_\nu - nm_\nu)\\
&\qquad \times \prod_{\lambda\notin\{\mu,\nu\}} \varphi_n(x_\lambda - nm_\lambda).
\end{aligned} \tag{3.lxi}$$

The coarse arrays in these formulas are indexed by their starting node m. Their values are raw bond or plaquette values; no length factor has been included in the coarse difference d_n. eq. 3.lx,

followed by reindexing m, proves

$$\mathrm{d}I_0 = I_1\mathrm{d}_n, \qquad \mathrm{d}I_1 = I_2\mathrm{d}_n. \tag{3.lxii}$$

For example, differentiating a transverse nodal hat in I_1 replaces it by the difference of two interval indicators divided by n; their coefficients are precisely the two terms of the corresponding coarse plaquette boundary. This proves the second identity with the same orientation as $(\mathrm{d}a)_{\mu\nu} = a_\mu + \nabla_\mu a_\nu - a_\mu(\cdot + e_\nu)$.

For a fixed plaquette direction, I_2 has nonnegative weights. Their row sum is n^{-2}, by the two partition identities. Their column sum is n^2: each interval contributes n fine positions, each of the remaining two nodal hats has sum n, and the prefactor is n^{-2}. Weighted Cauchy therefore gives

$$\|I_2F\|^2 \le n^{-2}n^2\|F\|^2 = \|F\|^2. \tag{3.lxiii}$$

These counting norms are those of the four-dimensional Wilson quadratic energy. As for curvature averaging, dimension four is used in the cancellation of the powers of n.

The exact averaging symbol. With centered blocks, the raw straight average is

$$(S_nw)_\mu(y) = n^{-4}\sum_{x\in B_n(y)}\sum_{j=0}^{n-1} w_\mu(x + je_\mu).$$

We compute S_nI_1 exactly to obtain a right inverse. Put $h = (n-1)/2$. In a transverse direction the coefficient of the neighboring coarse index m is $n^{-1}\sum_{x=-h}^{h}\varphi_n(x - nm)$. In the component direction it is $n^{-2}\#\{(x,j) : -h \le x \le h,\ 0 \le j < n,\ nm \le x + j < n(m+1)\}$. Both expressions are zero unless $m = 0, \pm 1$, and both have the same three values:

$$a_0 = 1 - \frac{h(h+1)}{n^2} = \frac{3}{4} + \frac{1}{4n^2}, \qquad a_1 = \frac{h(h+1)}{2n^2} = \frac{1-n^{-2}}{8} \quad (m = \pm 1). \tag{3.lxiv}$$

For the longitudinal count, the negative and positive overshoots each number $1+\cdots+h = h(h+1)/2$; the remaining $n^2 - h(h+1)$ pairs give the central coefficient. For the transverse hat, its positive-side and negative-side linear sums give the same two triangular numbers.

Let T_λ translate a coarse array by one node in direction λ. The tensor product in eq. 3.lxi and the identical one-dimensional coefficients just computed show

$$S_nI_1 = \mathcal{A}_n, \qquad \mathcal{A}_n = \prod_{\lambda=1}^{4}[a_0I + a_1(T_\lambda + T_\lambda^{-1})]. \tag{3.lxv}$$

Crucially, this is the same scalar convolution on all four bond components. Each one-dimensional symbol is

$$a_n(t) = \frac{3}{4} + \frac{1}{4n^2} + \frac{1-n^{-2}}{4}\cos t, \qquad \frac{1}{2} \le a_n(t) \le 1. \tag{3.lxvi}$$

Fourier transform on the coarse torus, or the quadratic-form bound $a_0 - 2a_1 \ge 1/2$ for a unitary shift, proves

$$\|\mathcal{A}_n^{-1}\|_{2\to 2} \le 16. \tag{3.lxvii}$$

All shifts commute. Their scalar convolution commutes with every coarse difference, on sites, bonds and plaquettes. Its inverse has the same property.

An exact lift for the plain constraint. For prescribed coarse a, take $w_0 = I_1\mathcal{A}_n^{-1}a$. eq. 3.lxii, eq. 3.lxiii, eq. 3.lxv and eq. 3.lxvii imply

$$S_nw_0 = a, \qquad \|\mathrm{d}w_0\| \le \|\mathrm{d}_n\mathcal{A}_n^{-1}a\| \le 16\|\mathrm{d}_na\|. \tag{3.lxviii}$$

This is an exact straight lift. The actual constraint is plain. For completeness its two relevant identities are

$$P^{[k]}w = S_nw + \mathrm{d}_n\psi_k(w), \qquad P^{[k]}\mathrm{d}f = \mathrm{d}_n(r_nf), \tag{3.lxix}$$

where r_n restricts a fine site function to the centers $n\mathbb{Z}^4$. At one step these follow by differentiating the actual closed-logarithm formula: $P_L = S_L - \mathrm{d}_L\sigma$, and on gradients an internal contour is $f(x) - f(y)$, so $P_L\mathrm{d}f = \mathrm{d}_L r_L f$. Iterating proves the second identity. Composition of straight averages is S_n; applying a plain average to a gradient again gives a gradient, proving the first by induction.

Nodal interpolation satisfies $r_n I_0 f = f$. Set

$$f = -\psi_k(w_0), \qquad w = w_0 + \mathrm{d}I_0 f. \tag{3.lxx}$$

Then eq. 3.lxviii–eq. 3.lxix give $P^{[k]}w = a$ exactly, while $\mathrm{d}w = \mathrm{d}w_0$. The gradient correction has no classical curvature cost. The energy estimate itself needs no amplitude bound; such a bound is proved below. We have proved

$$E_k(a) \le 256\|\mathrm{d}_n a\|^2. \tag{3.lxxi}$$

The complementary lower bound follows from the exact curvature average: $\mathrm{d}_n P^{[k]}w = T_n\mathrm{d}w$, where T_n has row sum n^2 and total column weight n^{-2}. The same weighted Cauchy argument gives $\|T_n\| \le 1$. Thus

$$\boxed{\|\mathrm{d}_n a\|^2 \le E_k(a) \le 256\|\mathrm{d}_n a\|^2}. \tag{3.lxxii}$$

All statements include $n = 1$, where the interpolations and $\mathcal{A}_1$ are identities. No zero-frequency inverse is taken: $\mathcal{A}_n$ itself is strictly positive even on constants.

Consequences and scope. For the actual flat free nonpivot precision, $\langle u, A_k^{\mathrm{flat}}u\rangle = E_k(Cu) + \|GCu\|^2$. The local pivot basis has counting ℓ^2 norm at most CL^4, from its row and column bounds. Also $\|\mathrm{d}\|^2 \le 24$ on one-forms in four dimensions. The internal contour matrix has row sum at most CL and column sum at most CL^4, hence $\|G\|^2 \le CL^5$. It contains no pivot, so $GC = G\iota$, where ι inserts free coordinates. eq. 3.lxxi and these bounds give the conservative uniform upper bound

$$A_k^{\mathrm{flat}} \le CL^8 I. \tag{3.lxxiii}$$

Together with the separate flat-gap theorem this bounds the same classical precision above and below independently of the number of finer levels and the volume.

Supremum norms of the constructed linear lift. The same scalar inverse is bounded in the supremum norm. For one direction the off-diagonal part of $a_0 I + a_1(T + T^{-1})$, divided by a_0, has supremum operator norm at most $2a_1/a_0 < 1$. Its inverse Neumann series therefore has norm at most $(a_0 - 2a_1)^{-1} \le 2$. Multiplying the four commuting direction inverses gives

$$\|\mathcal{A}_n^{-1}\|_{\infty\to\infty} \le 16, \qquad \|w_0\|_\infty \le \frac{16}{n}\|a\|_\infty. \tag{3.lxxiv}$$

The second inequality uses the row sum $1/n$ of I_1.

There is an explicit choice of the scalar in eq. 3.lxix. Let $\psi_0 = 0$. When the current spacing is L^j, apply $P_L = S_L - \mathrm{d}_L\sigma_L$ to $P^{[j]}w_0 = S_{L^j}w_0 + \mathrm{d}_{L^j}\psi_j$. The next plain step sends the last gradient to the gradient of the center restriction. Thus

$$\psi_{j+1} = r_L\psi_j - \sigma_L(S_{L^j}w_0). \tag{3.lxxv}$$

All fields in this recurrence are raw bond or site values on their current lattices. There is no missing spacing factor. Internal paths have at most $2(L-1)$ steps, so $\|\sigma_L\|_{\infty\to\infty} \le 2(L-1)$. Also $\|S_{L^j}\|_{\infty\to\infty} = L^j$, whereas restriction has norm one. The geometric sum yields $\|\psi_k\|_\infty \le 2(L-1)\sum_{j<k} L^j\|w_0\|_\infty = 2(n-1)\|w_0\|_\infty \le 32\|a\|_\infty$.

Since $\mathrm{d}I_0 f = I_1\mathrm{d}_n f$, the correcting gradient in eq. 3.lxx has norm at most $2\|f\|_\infty/n$. Consequently the particular exact lift constructed above obeys

$$\|\mathrm{d}I_0 f\|_\infty \le \frac{64}{n}\|a\|_\infty, \qquad \|w\|_\infty \le \frac{80}{n}\|a\|_\infty. \tag{3.lxxvi}$$

The scalar convolution and its inverse commute with curl. The row sum n^{-2} of I_2 and eq. 3.lxxiv also give

$$\|\mathrm{d}w\|_\infty = \|\mathrm{d}w_0\|_\infty \le \frac{16}{n^2}\|\mathrm{d}_n a\|_\infty. \tag{3.lxxvii}$$

These are norms of one explicitly defined linear right inverse. They are not estimates for a minimizing response.

The interpolation identities also hold with commuting unitary boundary twists. Work on the universal cover and periodize with the corresponding character; absolute row and column sums are unchanged. Twisted shifts remain unitary and commuting, so eq. 3.lxvi–eq. 3.lxvii retain their lower bound for every shifted Bloch parameter. Thus the same energy comparison applies to real flat holonomies in the block axial gauge.

The construction gives energy and supremum estimates for a particular linear lift. It does not bound a minimizing or nonlinear background map, or its complex core and tower substitutions. It does not prove exponential decay of the effective Hessian. Nor does it control signed layer profiles, nonzero-current corrections, counterterm perturbations or complex decoupling arrays. Those estimates must compare their actual operators with this unweighted classical flat form.

3.6. **Exponential kernels of the classical flat precision.** We compute the classical flat quadratic form of the actual plain iterated average, and then the covariance on its next-step nonpivot fiber with the actual soft contour gauge. A spectral gap by itself does not imply kernel decay. The proof below supplies a uniform complex strip for the exact symbols before using the flat gap. All fields are in raw bond coordinates, indexed by their starting sites. No midpoint phase is suppressed.

Let $n = L^k$, with L odd. The current lattice is in units of its own spacing; its finer lattice has n subdivisions per direction. We first use one scalar color. At the identity the three SU(2) colors give identical copies. The normalization of the action is $A(e^{ita}) = \frac{1}{2}t^2\|\mathrm{d}a\|^2 + O(t^3)$. After the canonical scaling $g = \theta^{-1/2}$, the real unweighted step has $\theta g^2 = 1$. Thus the matrices calculated here are precisely its classical quadratic precision matrices.

The two constrained minima are equal. Write $P^{[k]}$ for the raw derivative of the actual plain k-fold average. The straight average is

$$(S_n w)_\mu(ny) = n^{-4} \sum_{r\in B_n(0)} \sum_{j=0}^{n-1} w_\mu(ny + r + je_\mu). \tag{3.lxxviii}$$

Here $B_n(0) = \{-(n-1)/2, \dots, (n-1)/2\}^4$. The actual one-step map is that of § 3.3. Let $\mathcal{G}_L(y,x)$ be its coordinate-order shortest-path multiset from y to $x \in B_L(y)$, with its prescribed multiplicities and equal weights. In positive bond coordinates set

$$\begin{aligned} \gamma_{y,x}(a) &= \frac{1}{\#\mathcal{G}_L(y,x)} \sum_{\Gamma\in\mathcal{G}_L(y,x)} \sum_{b\in\Gamma} \varepsilon_\Gamma(b)a(b), \\ (Ga)_{y,x} &= \gamma_{y,x}(a) \quad (x \ne y), \qquad \sigma(a)(y) = L^{-4} \sum_{x\in B_L(y)} \gamma_{y,x}(a). \end{aligned} \tag{3.lxxix}$$

Here the signs record the path orientation and $\gamma_{y,y} = 0$. Differentiation gives $P_L = S_L - \mathrm{d}_L\sigma$. On gradients the plain map intertwines with restriction at the centers, whereas the straight map intertwines with the site average. Iteration gives

$$P^{[k]}w = S_n w + \mathrm{d}_n\psi_k(w), \qquad P^{[k]}\mathrm{d}f = \mathrm{d}_n r_n f, \qquad S_n\mathrm{d}f = \mathrm{d}_n q_n f. \tag{3.lxxx}$$

Both site maps are onto: for r_n assign the values at the centers, and for q_n use a function constant on each centered block. These are statements about finite linear cochains; no regularity bound for those corrections is required.

Suppose $P^{[k]}w = a$. Then $S_n w = a - \mathrm{d}_n \psi_k(w)$. Choose $q_n f = \psi_k(w)$. The field $w + \mathrm{d}f$ has straight average a and the same curl. Conversely, for $S_n w = a$, choose $r_n f = -\psi_k(w)$ to correct the plain average. Consequently the actual classical quadratic value is

$$E_n(a) := \min_{P^{[k]}w=a} \|\mathrm{d}w\|^2 = \min_{S_n w=a} \|\mathrm{d}w\|^2. \tag{3.lxxxi}$$

Nonemptiness also follows from the filter calculation below: its central alias is nonzero in every component at $p \neq 0$, and a constant fine field lifts a constant coarse field. The two gradient corrections transfer feasibility between the constraints. Finite dimensional projection after quotienting the curl-null directions then gives both minima. The equality does not change the actual soft gauge, which is added later on the current one-step fiber.

Fourier normalization and the auxiliary covariance. Take a coarse torus with s sites per direction and a fine torus with $n\mathsf{s}$. Use the unitary transforms

$$\widehat{w}(\xi) = (n\mathsf{s})^{-2} \sum_x e^{-i\xi\cdot x} w(x), \qquad \widehat{a}(p) = \mathsf{s}^{-2} \sum_y e^{-ip\cdot y} a(ny). \tag{3.lxxxii}$$

Fine aliases of a coarse momentum p are $\xi_m = (p+2\pi m)/n$, with $m \in I_n^4$, $I_n = \{-(n-1)/2, \dots, (n-1)/2\}$. Put

$$b_n(z) = \frac{\sin(nz/2)}{n\sin(z/2)}, \qquad B_n(\xi) = \prod_\nu b_n(\xi_\nu), \qquad h = (n-1)/2. \tag{3.lxxxiii}$$

The centered block sum is B_n, and the straight segment sum is $ne^{ih\xi_\mu} b_n(\xi_\mu)$. Thus the exact filter of component μ in eq. 3.lxxviii is

$$s_{\mu,n}(\xi) = ne^{ih\xi_\mu} B_n(\xi) b_n(\xi_\mu), \qquad \widehat{a}_\mu(p) = n^{-2} \sum_{m\in I_n^4} s_{\mu,n}(\xi_m)\widehat{w}_\mu(\xi_m). \tag{3.lxxxiv}$$

The factor n^{-2} is the square root of the ratio of the coarse and fine numbers of sites. In particular its square, not an unnormalized alias sum, enters a covariance.

The auxiliary fine Feynman form is $\|\mathrm{d}w\|^2 + \|\mathrm{d}^* w\|^2$. Its symbol is component-diagonal, with $\lambda(\xi) = 4\sum_\nu \sin^2(\xi_\nu/2)$. For $p \neq 0$, every alias has positive λ. The coarse covariance of this auxiliary form is diagonal:

$$D_{\mu,n}(p) = n^{-4} \sum_m \frac{|s_{\mu,n}(\xi_m)|^2}{\lambda(\xi_m)}. \tag{3.lxxxv}$$

This formula can equally be obtained by minimizing the positive fine Feynman form under its linear constraint; no probability measure on the fine constant mode is needed.

For complex continuation define

$$\begin{aligned} q_{\nu,m}(p) &= \frac{\sin((p_\nu + 2\pi m_\nu)/2)}{n\sin((p_\nu + 2\pi m_\nu)/(2n))}, \\ f_{\mu,m}(p) &= \left(\prod_\nu q_{\nu,m}(p)^2\right) q_{\mu,m}(p)^2, \\ s_{n,m}(p) &= 4n^2 \sum_\nu \sin^2((p_\nu + 2\pi m_\nu)/(2n)). \end{aligned} \tag{3.lxxxvi}$$

Removable ratios are filled in at zero. On real momenta the phase $e^{ih\xi_\mu}$ in eq. 3.lxxxiv cancels with its conjugate. Its analytic version is the product of the filters at ξ and $-\xi$, giving exactly

$$D_{\mu,n}(p) = \sum_{m\in I_n^4} \frac{f_{\mu,m}(p)}{s_{n,m}(p)}. \tag{3.lxxxvii}$$

Indeed $s_{\mu,n}(\xi)s_{\mu,n}(-\xi) = n^2 B_n(\xi)^2 b_n(\xi_\mu)^2$, and $s_{n,m} = n^2\lambda(\xi_m)$. This checks every power of n. Squared absolute values in eq. 3.lxxxv are not used as holomorphic functions.

Recovering the gauge-invariant energy. For real p, let $t^+_\mu(p) = e^{ip_\mu} - 1$ and $t^-_\mu(p) = e^{-ip_\mu} - 1$. The latter is the conjugate of the former only on real momenta. In analytic formulas below $(t^-)^{\mathsf{T}}$ is a holomorphic row, not a conjugation of a complex variable.

At nonzero fine frequency, the elementary Hodge identity is $\|\mathrm{d}\widehat{w}\|^2 + \|\mathrm{d}^*\widehat{w}\|^2 = \lambda\|\widehat{w}\|^2$. Subtracting a fine gradient makes a field divergence free without changing its curl. Its straight average changes by a coarse gradient, by eq. 3.lxxx. Conversely any coarse gradient can be corrected by a fine gradient, since q_n is onto. It follows that the curl minimum in eq. 3.lxxxi equals the minimum of the Feynman quadratic value over coarse gauge changes:

$$E_n(p;a) = \min_{z\in\mathbb{C}} (a + t^+ z)^* D_n(p)^{-1} (a + t^+ z). \tag{3.lxxxviii}$$

For clarity, the two inequalities use different simple steps. A divergence-free representative of a curl-minimizing field has Feynman energy equal to its curl energy, with coarse value $a + t^+ z$. In the other direction a Feynman minimizer at that coarse value can be gradient-corrected back to a, and its curl energy is no greater than its Feynman energy.

Solving the scalar minimum gives the precision symbol

$$\mathcal{E}_n = D_n^{-1} - \frac{D_n^{-1} t^+ (t^-)^{\mathsf{T}} D_n^{-1}}{(t^-)^{\mathsf{T}} D_n^{-1} t^+}. \tag{3.lxxxix}$$

Here $D_n = \mathrm{diag}(D_{\mu,n})$. At the real zero momentum the classical value is zero on every constant coarse one-form: its constant fine lift is a/n. We accordingly set $\mathcal{E}_n(0) = 0$. This zero mode is retained, not removed by a mass term. We next prove that this definition is the value of a uniformly holomorphic symbol.

Uniform noncentral alias bounds. Fix a small $\rho > 0$ and allow $|\operatorname{Re} p_\nu| < \pi + \rho$, $|\operatorname{Im} p_\nu| < \rho$. For $m_\nu \neq 0$,

$$|n \sin((p_\nu + 2\pi m_\nu)/(2n))| \geq c(1 + |m_\nu|). \tag{3.xc}$$

To see this, the real part of the sine argument stays in a fixed subinterval of $(-\pi, \pi)$, and its absolute value is at least $c|m_\nu|/n$. On that interval $|\sin x| \geq c|x|$, and $|\sin(x + iy)|^2 = \sin^2 x + \sinh^2 y$. The numerator in eq. 3.lxxxvi is bounded. For $m_\nu = 0$ the ratio is a quotient of two sinc functions on a fixed compact zero-free neighborhood. Hence, uniformly in n,

$$|q_{\nu,m}(p)| \leq \frac{C}{1 + |m_\nu|}. \tag{3.xci}$$

For $m \neq 0$ the denominator in eq. 3.lxxxvi has a uniform positive real part. In fact $\operatorname{Re} \sin^2(x + iy) \geq \sin^2 x - \sinh^2 y$. After multiplication by $4n^2$, the imaginary displacement costs at most $C\rho^2$, whereas the real sum is at least $c|m|^2$. Decrease ρ to obtain

$$\operatorname{Re} s_{n,m}(p) \geq c(1 + |m|^2) \qquad (m \neq 0). \tag{3.xcii}$$

The constants do not grow as the alias set grows.

Write $s_n = s_{n,0}$, $q_\nu = q_{\nu,0}$, $Q_n = \prod_\nu q_\nu^2$, and $f_\mu = Q_n q_\mu^2$. Then

$$D_{\mu,n} = \frac{f_\mu}{s_n} + R_{\mu,n}, \qquad R_{\mu,n} = \sum_{m\neq 0} \frac{f_{\mu,m}}{s_{n,m}}. \tag{3.xciii}$$

Each summand of R is bounded by the absolutely summable majorant

$$\frac{C}{1 + |m|^2} \prod_{\nu=1}^{4} (1 + |m_\nu|)^{-2}. \tag{3.xciv}$$

The extra factor $(1 + |m_\mu|)^{-2}$ may be discarded. Thus R is holomorphic and uniformly bounded on the stated box, independently of n. Cauchy on a smaller box bounds all its fixed-order derivatives

uniformly as well. For $n = 1$ the sum is empty. Every noncentral summand vanishes at $p = 0$, so $R_{\mu,n}(0) = 0$; only boundedness is needed below.

Cancellation of the apparent gauge denominator. Near $p = 0$, uniformly in n, $q_\nu = 1 + O(|p|^2)$, $s_n = O(|p|^2)$, and $f_\mu = 1 + O(|p|^2)$. Define

$$h_\mu = (f_\mu + s_n R_{\mu,n})^{-1}, \qquad H_n = \operatorname{diag}(h_\mu). \tag{3.xcv}$$

On a sufficiently small fixed complex polydisc these are holomorphic and uniformly bounded. The exact filter identity

$$\sum_\mu \frac{t_\mu^+ t_\mu^-}{f_\mu} = \frac{4n^2 \sum_\mu \sin^2(p_\mu/(2n))}{Q_n} = \frac{s_n}{Q_n} \tag{3.xcvi}$$

is essential. It follows directly by canceling the numerator $\sin^2(p_\mu/2)$ against q_μ^2, including removable values. It is not an approximation to a gauge denominator.

Since $h_\mu = f_\mu^{-1} - s_n R_{\mu,n} h_\mu / f_\mu$, eq. 3.xcvi gives

$$\begin{aligned} F_n &:= \sum_\mu t_\mu^+ t_\mu^- h_\mu = s_n J_n, \\ J_n &= Q_n^{-1} - \sum_\mu \frac{t_\mu^+ t_\mu^- R_{\mu,n} h_\mu}{f_\mu}. \end{aligned} \tag{3.xcvii}$$

Here $J_n(0) = 1$ and $|J_n - 1| \le C|p|^2$ uniformly. It is therefore nonzero on a smaller fixed polydisc. On its meromorphic domain $D_n^{-1} = s_n H_n$, and the denominator in eq. 3.lxxxix equals $s_n F_n = s_n^2 J_n$. The actual precision consequently has the holomorphic form

$$\mathcal{E}_n(p) = s_n H_n - \frac{H_n t^+ (t^-)^{\mathsf{T}} H_n}{J_n}. \tag{3.xcviii}$$

This removes the whole complex zero cone of s_n, not merely the value at the real origin. The expression is uniformly bounded there, is $O(|p|^2)$, and gives $\mathcal{E}_n(0) = 0$. It also obeys the exact Ward identity $\mathcal{E}_n t^+ = 0$, by eq. 3.xcvii. For $n = 1$, it reduces to $\lambda(p) I - t^+ (t^-)^{\mathsf{T}}$, the ordinary curl quadratic form. This checks the normalization and phases.

A uniform strip and its Fourier kernels. We establish a complex domain away from zero explicitly. On the real Brillouin cube the central q_ν are positive and bounded above and below uniformly in n. Also $c|p|^2 \le s_n(p) \le C|p|^2$. For real $|p| \ge r > 0$, the central term in eq. 3.xciii and the nonnegative alias terms give

$$c_D \le D_{\mu,n}(p) \le C_r, \qquad (t^-)^{\mathsf{T}} D_n^{-1} t^+ \ge c_r > 0. \tag{3.xcix}$$

The second inequality uses $\sum_\mu |t_\mu^+|^2 \ge c|p|^2$ and the upper bound on D.

On a sufficiently thin complex neighborhood of this real compact set, $\operatorname{Re} s_n \ge cr^2 - C|\operatorname{Im} p|^2$. Thus the central term is holomorphic and bounded there; eq. 3.xc–eq. 3.xciv do the same for the aliases. Cauchy bounds the first derivatives uniformly in n. Shrinking the neighborhood by a fixed amount depending on r, eq. 3.xcix therefore keeps every D_μ and its scalar gauge denominator nonzero. The formula in eq. 3.lxxxix is holomorphic and uniformly bounded on that neighborhood. This is a quantitative perturbation of uniformly separated real denominators.

Choose r inside the origin polydisc of eq. 3.xcviii, then reduce the common width once. The two formulas agree wherever their denominators are nonzero, by their exact algebraic derivation, and hence on their overlaps by continuity. Together they give constants $\eta > 0, M_E < \infty$, independent of n, such that

$$\mathcal{E}_n \text{ is holomorphic and } \sup_{|\operatorname{Im} p_\nu| < \eta} \|\mathcal{E}_n(p)\| \le M_E. \tag{3.c}$$

The symbol is 2π-periodic. In eq. 3.lxxxvii, changing a coarse momentum by 2π reindexes the aliases; an endpoint alias changes its fine frequency by 2π, under which the actual filter and denominator are periodic. The same is true of eq. 3.lxxxix, and its analytic continuation has that periodicity.

Define its infinite-lattice kernel by

$$\mathcal{K}_n(z) = (2\pi)^{-4} \int_{[-\pi,\pi]^4} e^{ip\cdot z} \mathcal{E}_n(p)\, dp. \tag{3.ci}$$

For $0 < \eta' < \eta$, shift the ν-th contour to $\operatorname{Im} p_\nu = \eta' \operatorname{sgn} z_\nu$. Periodicity cancels the vertical faces. The exponential then gives

$$\|\mathcal{K}_n(z)\| \le M_E e^{-\eta'|z|_1}. \tag{3.cii}$$

All diagonal terms are retained in this Fourier definition. The kernel and its weighted absolute row sums are bounded uniformly in k. Its zero-frequency value and Ward identity are those already proved; no contact term is added afterward.

The actual next-step precision and its inverse. Now group the current lattice into the fixed blocks of side L used by the next RG step. For block Bloch momentum P, let $r \in B_L(0)$, $j \in I_L^4$, and $\zeta_j = (P + 2\pi j)/L$. The finite cell transform and its analytic inverse are

$$U(P)_{rj} = L^{-2} e^{i\zeta_j \cdot r}, \qquad V(P)_{jr} = L^{-2} e^{-i\zeta_j \cdot r}. \tag{3.ciii}$$

Finite geometric-series orthogonality proves $VU = UV = 1$. For real P, $V = U^*$. For complex P the displayed inverse, not literal conjugation, is used. Their norms are bounded on a fixed strip, since their extra diagonal factors are $e^{\pm iP\cdot r/L}$ and $|r_\nu|/L < 1/2$. The folded classical precision is

$$\mathcal{E}_n^{\mathrm{cell}}(P) = U(P)\, \mathrm{diag}_j(\mathcal{E}_n(\zeta_j)) V(P), \tag{3.civ}$$

with the identity on component indices understood. It is periodic and holomorphic on a fixed block strip, uniformly in n. Periodicity again follows by the alias permutation.

Let $C(P)$ be the actual flat middle-pivot basis for the plain one-step constraint. Its free restriction is the identity and its pivot block solves the coefficient $L^{-3} I_4$; hence its entries are fixed Laurent polynomials in the block phases. Let G be the internal contour matrix defined in eq. 3.lxxix, with rows (y, x), $x \ne y$. Define $C^\sharp(P) = C(-P)^{\mathsf{T}}$; on real P this is $C(P)^*$. The exact classical free nonpivot precision is

$$\mathsf{A}_n(P) = C^\sharp(P)[\mathcal{E}_n^{\mathrm{cell}}(P) + G^{\mathsf{T}} G] C(P). \tag{3.cv}$$

This is the action pullback plus the actual soft gauge. The Feynman form was used only to compute the gauge-invariant minimum in eq. 3.lxxxi; it has not replaced the soft gauge in eq. 3.cv.

For clarity the uniform real gap used next can be obtained independently of this symbol calculation. Expanding the curl of eq. 3.lxxviii gives an average of n by n fine plaquette sheets. Its nonnegative row weights sum to n^2 and its total column weights to n^{-2}. Weighted Cauchy gives $\|\mathrm{d}_n S_n w\| \le \|\mathrm{d} w\|$, hence $E_n(a) \ge \|\mathrm{d} a\|^2$ in current units. Consequently eq. 3.cv dominates the fixed-cell form $C(P)^*[d_P^* d_P + G^{\mathsf{T}} G] C(P)$. This form has no zero vector: zero curl makes internal path sums independent of the path, so zero G kills every internal bond. A plaquette crossing a block face then makes its crossing values equal tangentially. The plain constraint kills that common face value. The free restriction of C is the identity, so the original free vector is zero. The matrices in this comparison are continuous on the compact block Bloch torus, with fixed dimension and fixed rank. Their least Rayleigh quotient on the unit sphere therefore has a positive minimum $c(L)$. This proves $\mathsf{A}_n(P) \ge c(L) 1$ uniformly in n, including $P = 0$. eq. 3.c, the finite transform and the fixed polynomial matrices in eq. 3.cv give a uniform bound for A_n on a complex strip. Cauchy on a smaller strip bounds its first derivatives by a constant $M_A(L)$. Choose its width so that

$$\|\mathsf{A}_n(P + iY) - \mathsf{A}_n(P)\| \le 4 M_A(L) |Y|_\infty < c(L)/2. \tag{3.cvi}$$

The Neumann series around the real positive matrix then proves

$$\mathsf{A}_n^{-1} \text{ is holomorphic}, \qquad \|\mathsf{A}_n(P + iY)^{-1}\| \le 2/c(L) \tag{3.cvii}$$

on one strip independent of n. This step uses both the proved analytic bounds and the gap.

Contour shifting as in eq. 3.ci–eq. 3.cii now bounds the Fourier coefficients of the block precision and its inverse by $C(L)e^{-c(L)|m|_1}$. The cell has fixed size, so converting its indices back to current nonpivot bonds only changes the constants. We obtain

$$|\mathsf{A}_k^{\mathrm{flat}}(b,b')| + |\mathsf{C}_k^{\mathrm{flat}}(b,b')| \le C_L e^{-\mu_L d(b,b')}, \qquad \mathsf{C}_k^{\mathrm{flat}} = (\mathsf{A}_k^{\mathrm{flat}})^{-1}. \tag{3.cviii}$$

Here $d(b,b')$ is bond distance in current lattice units; using starting-site distance or midpoint distance only changes the fixed constants. The symbols $\mathsf{C}_k^{\mathrm{flat}}$ in this conclusion are covariances; the undecorated pivot map $C(P)$ is a different rectangular matrix. Exponential weighted row and column sums follow by summing the fixed-dimensional lattice shells.

Finite tori, holonomies, and scope. For a periodic lattice the exact quadratic minimization is diagonal on its allowed Fourier momenta. Its symbols are the samples calculated above, with no volume correction. If $K_n(m)$ is any infinite block kernel just obtained and the block period is s_b, discrete Fourier orthogonality and absolute convergence give

$$K_{n,\mathsf{s}_\mathrm{b}}(m) = \sum_{v\in\mathbb{Z}^4} K_n(m+\mathsf{s}_\mathrm{b}v). \tag{3.cix}$$

The image sum preserves an exponential bound in periodic distance, uniformly over the admissible periods. It identifies the infinite-lattice kernel determined by the same symbol, with exponentially small image errors on fixed nonwrapping neighborhoods. It does not assert exact equality of a full torus kernel and an infinite kernel on a lift. The same argument applies to the inverse symbol, since its sampled matrix is the inverse of each sampled positive precision.

For real flat holonomies in the prescribed block axial representative, a unitary gauge constant on each block flattens the connection on the universal cover and preserves the contour gauge norm. The commuting adjoint holonomies split into unitary characters. Each character shifts the allowed block momenta by a real amount; the infinite symbols are unchanged. Equivalently the image sum in eq. 3.cix receives unit-modulus factors. All absolute kernel bounds remain valid, without requiring small global holonomies.

The conclusion is the exponential precision and covariance bound for the actual unweighted classical flat quadratic problem. The auxiliary D_n itself is massless and is not asserted to have an exponential kernel. The gap and decay in eq. 3.cviii concern the next-step constrained nonpivot variables with the soft gauge, not a physical Yang–Mills mass gap. No signed profile, counterterm, nonzero-current, non-Abelian or enlarged complex-decoupling comparison has been proved by this calculation. Those require their actual operators and uniform perturbation estimates.

3.7. **First-step precision at small curvature.** This section concerns one unweighted step on a unit-spacing lattice. It starts with a given real field and the actual nonlinear block average. It proves a gap and exponential covariance bounds for the physical curved-fibre Hessian at a given constrained-critical background. Small plaquettes suffice; a small global potential or small torus holonomies are not required. The argument does not construct the background, iterate the conclusion through lower levels, or control signed profiles and complex decoupling parameters.

A fine current and a coarse multiplier have different types. We keep them distinct throughout. An auxiliary extension with an independently prescribed fine current will be useful in the proof. Its agreement with any separately named off-shell completion is an additional identification, stated precisely at the end.

The actual map and the two currents. Fix odd $L \ge 3$. Positive fine bonds carry Hermitian traceless matrices with colour pairing $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr}(XY)$. Complexification uses this bilinear pairing and its transpose. The ℓ^2 norm on real fields uses the associated Euclidean colour norm. Operator estimates on complex fields use its Hermitian extension. No bare-spacing factor occurs in the exponential.

Use the centered blocks, contour-path multiset and middle pivots of § 3.3. Denote the actual contour mean from y to $x \in B(y)$ by $T_{y,x}(U)$. Thus it solves $\sum_\Gamma \log(U(\Gamma)T_{y,x}^{-1}) = 0$, with the prescribed equal weights. The block bond $c = (y, y')$, $y' = y + Le_\mu$, is

$$\begin{aligned} Z_x(U) &= T_{y,x}(U)U([x, x + Le_\mu])T_{y',x+Le_\mu}(U)^{-1}U[c]^{-1}, \\ \mathcal{M}(U)(c) &= \exp\left\{L^{-4}\sum_{x\in B(y)} \log Z_x(U)\right\} U[c]. \end{aligned} \tag{3.cx}$$

The internal implicit mean and the outer explicit exponential are different operations. The analytic branches are the ones fixed by the local pivot construction. On a torus every relevant double block and its bond neighbourhood must have the embedded rectangular lift used there. Periods divisible by L and exceeding $4L + 4$ are sufficient. Work on the periodic cover whenever a larger local patch is needed.

The unweighted action and actual soft gauge are

$$A(U) = \sum_p \left[1 - \tfrac{1}{2}\,\mathrm{tr}\, U(\partial p)\right], \qquad \mathrm{GF}(U) = \sum_y \sum_{x\in B(y)\setminus\{y\}} \left[1 - \tfrac{1}{2}\,\mathrm{tr}\, T_{y,x}(U)\right]. \tag{3.cxi}$$

These expressions equal their usual real-trace versions on SU(2). Their complex continuations use the trace in eq. 3.cxi, without taking a real part. All internal contour paths avoid every middle crossing pivot. Hence GF depends on no pivot variable. On the infinite lattice these sums specify local interactions: derivatives are taken of their finite differences under compactly supported variations. Only finitely many local terms then change. The coefficient bounds below extend the resulting matrices to ℓ^2; no finite value for the whole background action is assumed.

Let $\mathfrak{X}$ be the positive nonpivot bonds and ι their inclusion into all fine bonds. For a background U, define

$$\begin{aligned} q_U(a) &= \frac{1}{i}\log\left[\mathcal{M}(e^{ia}U)\mathcal{M}(U)^{-1}\right], \\ P_U = Dq_U(0), &\qquad P_U h_U = 1, \qquad C_U = \iota - h_U P_U \iota. \end{aligned} \tag{3.cxii}$$

Here h_U is supported only on the pivot of each coarse bond. The actual fibre chart Φ_U has free restriction u, satisfies $q_U(\Phi_U(u)) = 0$, and obeys

$$\Phi_U(0) = 0, \quad D\Phi_U(0) = C_U, \quad D^2\Phi_U(0)[u, v] = -h_U D^2 q_U(0)[C_U u, C_U v]. \tag{3.cxiii}$$

Indeed differentiating the constraint gives its displayed linear and quadratic identities. The second derivative of the chart is supported on pivots, so applying $h_U P_U$ to it is the identity.

Write $J_A(U)$ for the Wilson gradient in the left exponential coordinates. For an arbitrary fine current J, set $\mu_U(J) = h_U^{\mathsf{T}} J$; this is a coarse row covector. Define the auxiliary matrix by its bilinear form

$$\begin{aligned} \mathscr{K}_U[J](u, v) = {} & D^2 A_U[C_U u, C_U v] + D^2\mathrm{GF}_U[\iota u, \iota v] \\ & - \langle \mu_U(J), D^2 q_U(0)[C_U u, C_U v]\rangle. \end{aligned} \tag{3.cxiv}$$

The equality $D^2\mathrm{GF}[C_U u, C_U v] = D^2\mathrm{GF}[\iota u, \iota v]$ holds because GF has no pivot coordinate. The same fact gives $h_U^{\mathsf{T}} J_{\mathrm{GF}}(U) = 0$ identically, even if the contour means are not the identity.

The chain rule and eq. 3.cxiii now give the exact physical identity

$$\begin{aligned} K_{\mathrm{phys}}(U) &:= D_u^2\{A(e^{i\Phi_U(u)}U) + \mathrm{GF}(e^{i\Phi_U(u)}U)\}_{u=0}, \\ K_{\mathrm{phys}}(U) &= \mathscr{K}_U[J_A(U)], \\ \mathscr{K}_U[J] - K_{\mathrm{phys}}(U) &= \langle J - J_A(U), D^2\Phi_U(0)\rangle. \end{aligned} \tag{3.cxv}$$

No criticality is needed for this coordinate identity. Its Gaussian interpretation uses a given constrained-critical background in the block-axial representative. All internal contour means then

equal one. The GF gradient vanishes and its Hessian is the sum of the squares of the derivatives of the actual contour means. Constrained criticality also gives

$$J_A(U) = P_U^{\mathsf{T}} \mu_U(J_A(U)). \tag{3.cxvi}$$

To see this, the gradient annihilates $\ker P_U$, and P_U is onto by its pivot right inverse. Thus it is $P_U^{\mathsf{T}}\mu$; applying h_U^{T} identifies μ. An independent J is not asserted to satisfy eq. 3.cxvi.

At this first step there is no inner background minimization: the zero-fold averaging and site maps are identities. In a formulation using an intermediate Landau projector, that projector is zero, and the zero-fold constraint has no quadratic part. These facts do not by themselves identify or annihilate a separately defined invariant-Hessian correction such as $\Delta'_\pi[J]$, even after projection. The physical identity above follows directly from the actual chart chain rule, independently of that normal-form check. In canonical Gaussian coordinates $a = gu$, an action coefficient x with $xg^2 = 1$ gives precisely this same unweighted Hessian.

A joint analytic domain and explicit support counts. We record the finite bounds used below. Constants c, C in this section are universal, with their value allowed to change. There are radii

$$r_b = c_b L^{-5}, \qquad r_u = c_u L^{-9} \tag{3.cxvii}$$

with universal positive c_b, c_u, the latter fixed after the former, such that, for $U = e^{iX}$, the actual chart is jointly holomorphic on $\|X\|_\infty < 8r_b$, $\|u\|_\infty < 8r_u$, and

$$\begin{aligned} \|h_U\|_{\infty\to\infty},\ \|h_U\|_{1\to 1} &\le CL^3, \\ \|C_U\|_{\infty\to\infty},\ \|C_U\|_{1\to 1} &\le CL^4, \\ \|\Phi_U(u)\|_\infty &\le CL^4\|u\|_\infty. \end{aligned} \tag{3.cxviii}$$

The notation for ℓ^1 and ℓ^∞ here includes the fixed three-dimensional colour space; changing among its coordinate norms changes only C.

Here are the domain and locality details. Products along a contour of length $O(L)$, the implicit internal mean, and the logarithms in eq. 3.cx are jointly holomorphic for $\|X\|, \|a\| < c/L$. The product and logarithm estimates of the pivot construction give $\|P_U\|_{\infty\to\infty} \le CL$ and a quadratic constraint remainder at most $CL^2\|a\|^2$. A background derivative of P_U, obtained on this larger joint domain by Cauchy's formula, is at most CL^2. Consequently the pivot coefficient differs from $L^{-3}1$ by at most $CL^2\|X\|$. For the first radius in eq. 3.cxvii, its inverse has norm at most CL^3. The pivot matrix is block diagonal over coarse bonds: no other pivot occurs in the support of a given constraint row. This proves the bound for h_U.

A constraint row uses at most CL^4 fine bonds. A fine bond lies in at most eight relevant double blocks. These facts and the same directional Cauchy estimate give the column bound CL for P_U, as well as its row bound. The formula in eq. 3.cxii proves both bounds for C_U. Solving the nonlinear remainder with the pivot inverse has contraction constant at most $CL^3L^2(L^4\|u\|) = CL^9\|u\|$. Taking the second radius in eq. 3.cxvii small secures a contraction on the whole indicated joint domain. Its iterates are holomorphic in both parameters and converge uniformly there on smaller closed polydiscs. Passing their Cauchy integrals to the limit proves joint holomorphy.

For locality, the pivot value $\Phi_U(u)(b_0(c))$ depends only on the background and free variables in its own double block. The component contraction for this pivot does not involve another pivot. All nonpivot values are exactly u. In particular:

(i) A given free coordinate affects its own full bond and at most eight pivot components. Each of these bonds belongs to six plaquettes. Thus at most 54 Wilson plaquette terms can depend on that free coordinate through the chart.

(ii) GF has no pivot. A free coordinate can occur only in the $L^4 - 1$ internal means of its own block. Counting contour-path multiplicities does not increase the number of these terms.

(iii) Each such composed term depends on at most CL^4 scalar coordinates. For a fixed free bond, the union of all possible other free coordinates coupled to it still lies in a fixed number of neighbouring blocks and has size CL^4. This union bound does not multiply the two preceding counts together.

All matrix coefficients are therefore finite range. A double block has ℓ^1 diameter at most $5L$. Two pivot bonds entering one plaquette have starting sites at distance at most two. Thus the union of their chart supports has diameter at most $10L+2 \le 12L$. The safe range $12L$, measured by starting sites of fine bonds, also accommodates the internal contour terms. The same bound applies to the background variables on which a coefficient depends.

Each composed local action or gauge term is bounded by a universal constant on the joint domain above. Indeed all the products have $O(L)$ factors, with X and Φ of size $O(L^{-5})$, and the contour mean remains in its fixed analytic neighbourhood. Cauchy's formula in two free scalar directions bounds its Hessian coefficient by $Cr_u^{-2} \le CL^{18}$. The CL^4 possible terms give CL^{22} per matrix entry, and the CL^4 possible coupled coordinates give the Schur bound CL^{26}. One more background-direction Cauchy estimate costs $r_b^{-1} \le CL^5$. Here and below the Schur norm is the maximum of the absolute row and column sums of colour block norms. We have proved

$$\begin{aligned} \|K_{\mathrm{phys}}(e^{iX})\|_{\mathrm{Sch}} &\le CL^{26}, \\ \|K_{\mathrm{phys}}(e^{iX_1}) - K_{\mathrm{phys}}(e^{iX_0})\|_{\mathrm{Sch}} &\le CL^{31}\|X_1 - X_0\|_\infty. \end{aligned} \tag{3.cxix}$$

on a fixed smaller background domain. Only local terms were estimated; no extensive action bound independent of the volume was used.

For the current term, Cauchy's formula and eq. 3.cxviii give $|\partial_{u_b}\partial_{u_{b'}}\Phi(b_0(c))| \le CL^4/r_u \le CL^{13}$. A fixed free coordinate occurs in at most eight such pivot components and couples to CL^4 others. Hence

$$\|\langle J, D^2\Phi_U(0)\rangle\|_{\mathrm{Sch}} \le CL^{17}\|J\|_\infty. \tag{3.cxx}$$

The corresponding background derivative costs another CL^5. The Wilson gradient near the identity obeys $\|J_A(e^{iX})\|_\infty \le C\|X\|_\infty$, by differentiating the six adjacent plaquette products. Combining eq. 3.cxv, eq. 3.cxix and eq. 3.cxx proves, for $\|J_i\| \le 1$,

$$\begin{aligned} \|\mathscr{K}_{e^{iX}}[J] - K_0\|_{\mathrm{Sch}} &\le CL^{32}(\|X\|_\infty + \|J\|_\infty), \\ \|\mathscr{K}_{e^{iX_1}}[J_1] - \mathscr{K}_{e^{iX_0}}[J_0]\|_{\mathrm{Sch}} &\le CL^{32}(\|X_1 - X_0\|_\infty + \|J_1 - J_0\|_\infty). \end{aligned} \tag{3.cxxi}$$

Here $K_0 = \mathscr{K}_1[0]$ is the actual first flat precision. These are derived coefficient estimates, not assumptions about a matrix perturbation. All counts and domains are independent of the torus volume.

Local frames which preserve the actual soft gauge. Suppose now that U is real and block axial, and $\sup_p \|U(\partial p) - 1\| \le \epsilon$. The relevant symmetry of GF is a unitary gauge transformation constant on each centered block. Its values may vary from block to block. If v is such a transformation and $(V_v a)(b) = \mathrm{Ad}(v(b_-))a(b)$, then

$$\Phi_{U^v}(V_v u) = V_v\Phi_U(u), \qquad \mathscr{K}_{U^v}[V_v J] = V_v\mathscr{K}_U[J]V_v^{-1} \tag{3.cxxii}$$

on free coordinates. In this formula V_v is restricted to the appropriate full or free space. To verify it, internal means transform by conjugation with the block value, so GF is unchanged. The relative coarse logarithm in eq. 3.cxii transforms by that same adjoint action at the coarse bond's starting site. The pivot's starting site belongs to that block. The pivot right inverse and free-identity basis therefore intertwine these actions. Uniqueness of the actual chart proves its first identity, and differentiating the invariant local functions proves the second. The transformations are unitary in the colour norm. No covariance of GF under an arbitrary site gauge is used.

Consider a rectangular patch of diameter $O(LR)$, with a fixed additional margin of complete blocks. A unitary comb gauge g on that patch makes

$$\|U^g(b) - 1\| \le CLR\epsilon. \tag{3.cxxiii}$$

For completeness, compare the two comb paths to the endpoints of a bond. Moving that bond successively past the later coordinate segments expresses the resulting closed transport as a product of at most CLR conjugated elementary plaquettes. Unitarity and the telescoping identity for a product of unitary matrices bound its distance from the identity by the sum of their distances. This proves eq. 3.cxxiii, with no bound on $g - 1$.

Although g need not be block constant, block axiality supplies the missing adjustment. The contour means transform exactly as

$$T_{y,x}(U^g) = g(y)g(x)^{-1}. \tag{3.cxxiv}$$

Every path in such a mean has length $O(L)$. By eq. 3.cxxiii its transport, and therefore its implicit mean, is within $CL^2R\epsilon$ of the identity. This last statement follows directly from the contraction defining the mean: inputs within t of the identity have mean within Ct. Let $v(x) = g(y)$ on $B(y)$. Then $v(x)g(x)^{-1} = T_{y,x}(U^g)$, so eq. 3.cxxiii–eq. 3.cxxiv imply

$$\|U^v(b) - 1\| \le CL^2R\epsilon, \qquad v \text{ is block constant.} \tag{3.cxxv}$$

We can take its logarithm with the same bound. Thus the local frame preserves the actual soft gauge and gives a small potential on the patch. Arbitrary coarse holonomies have been retained outside it.

If a real field is initially given without the axial condition, small plaquettes make the contour-path transports mutually close: the same plaquette sliding uses at most CL^2 plaquettes. The actual implicit means therefore exist. Taking the site gauge $u(x) = T_{y,x}(U)$, $u(y) = 1$, makes all internal means equal to one and leaves the coarse datum unchanged. This supplies the representative used here. It does not assert existence of a constrained-critical orbit.

The gap from a covariant partition of unity. Let $c_0 = c_0(L) > 0$ be the proved flat first-step gap $K_0 \ge c_0$, as in eq. 3.lvii. Inequalities for real symmetric matrices in this section use their Hermitian quadratic forms on the complexified ℓ^2 space. Increase a universal constant once so that $K_L = CL^{40}$ dominates the coefficient, support and partition constants just obtained. Choose

$$\begin{aligned} R \ge 40, \qquad & K_L/R^2 \le c_0/8, \\ CL^2R\epsilon \le r_b/8, \qquad & K_L(R\epsilon + \gamma) \le c_0/8. \end{aligned} \tag{3.cxxvi}$$

Here γ bounds the real fine current J, and it is also at most one. These are explicit smallness requirements in terms of L, the finite flat gap and the already proved chart radii.

Construct a scalar squared partition on the coarse lattice. Let $h(t) = \max(1 - |t|, 0)$, put $h_z(x) = \prod_\nu h(x_\nu/R - z_\nu)$, and divide by $(\sum_z h_z(x)^2)^{1/2}$. At least one factor product is $\ge 2^{-4}$, and at most 2^4 are nonzero. The resulting χ_z satisfy $\sum_z \chi_z^2 = 1$, have support of radius R, and obey

$$\sum_z |\chi_z(x) - \chi_z(x')|^2 \le C|x - x'|_1^2/R^2 \quad \text{if } |x - x'|_1 \le 20. \tag{3.cxxvii}$$

The estimate follows by differentiating the normalized hats along the segment; only a bounded number of them occur there. Evaluate them at the block containing a free bond's starting site.

For a compactly supported free field f, the exact matrix identity and its bound are

$$\begin{aligned} \langle f, \mathscr{K}_U[J]f\rangle &= \sum_z \langle \chi_z f, \mathscr{K}_U[J]\chi_z f\rangle + \mathcal{E}, \\ |\mathcal{E}| &\le (K_L/R^2)\|f\|_2^2. \end{aligned} \tag{3.cxxviii}$$

Indeed $1 - \sum_z \chi_z(b)\chi_z(b') = \frac{1}{2}\sum_z(\chi_z(b) - \chi_z(b'))^2$. The matrix has range $12L$, so the block indices differ by at most a fixed number. Apply eq. 3.cxxvii, the row and column bounds, and

$2|f_b||f_{b'}| \le |f_b|^2 + |f_{b'}|^2$. The localization is on free variables. Its full tangent field is $C_U(\chi_z f)$; it is not replaced by $\chi_z C_U f$.

Use eq. 3.cxxv on a larger patch around each support. The margin contains every background variable of its matrix coefficients. eq. 3.cxxii carries the localized form into that block-constant frame. Extend the small background logarithms by zero outside the larger patch, and the transformed current by zero. The coefficients in the localized form are unchanged, by their strict locality. The near-identity bound in eq. 3.cxxi and the flat gap therefore give

$$\langle \chi_z f, \mathscr{K}_U[J]\chi_z f\rangle \ge \{c_0 - K_L(R\epsilon+\gamma)\}\|\chi_z f\|_2^2. \tag{3.cxxix}$$

The harmless L^2 in eq. 3.cxxv has already been absorbed in $K_L = CL^{40}$. On a double block the same local-frame argument also proves the uniform coefficient bounds and existence of its pivot inverse before eq. 3.cxxviii is used; this involves only the smaller $R = O(1)$ patch and has no dependence on the desired global gap. Thus no matrix bound needed in the IMS error has been assumed. Summing eq. 3.cxxix and using eq. 3.cxxvi–eq. 3.cxxviii proves

$$\mathscr{K}_U[J] \ge \tfrac{1}{2}c_0\, 1. \tag{3.cxxx}$$

This proof takes place on the periodic cover. It also gives the torus inequality without requiring one local frame to descend to the torus. Lift a fixed periodic vector to the cover and restrict it to a box of N^4 period cells. Its squared norm is N^4 times the period norm. Finite range makes its quadratic form equal N^4 times the torus form, up to a boundary error of order N^3, for that fixed vector and period. Divide the cover inequality by N^4 and let N tend to infinity. The constant in eq. 3.cxxx is independent of the period. This also explains why no assumption on noncontractible holonomies entered the proof.

For the physical current, each first variation of one plaquette vanishes when its plaquette transport is the identity. Differentiating its four factors and using unitarity bounds its contribution by $C\|U(\partial p) - 1\|$. There are six adjacent plaquettes, hence

$$\|J_A(U)\|_\infty \le C\epsilon. \tag{3.cxxxi}$$

Consequently a sufficiently small ϵ, depending only on L, satisfies eq. 3.cxxvi with $J = J_A(U)$. eq. 3.cxv and eq. 3.cxxx prove the actual physical first-step gap at every given real block-axial constrained-critical field of this curvature size.

Complex neighbourhood and exponential kernels. The same local frames yield a quantitative complex statement. Fix a real pair satisfying eq. 3.cxxvi, with room in those inequalities, and put $\mathbf{U} = e^{iE}U$, $\mathbf{J} = J + Z$, for complex fine fields. The local unitary gauges preserve $\|E\|_\infty$ and $\|Z\|_\infty$. The logarithm in a small local frame has bounded derivative, so the joint-domain estimate in eq. 3.cxxi, now applied to the background difference, proves

$$\|\mathscr{K}_{\mathbf{U}}[\mathbf{J}] - \mathscr{K}_U[J]\|_{\mathrm{Sch}} \le K_L(\|E\|_\infty + \|Z\|_\infty) \tag{3.cxxxii}$$

provided $\|E\|_\infty \le r_b/16$ and the total current remains below one. In particular require

$$K_L(\|E\|_\infty + \|Z\|_\infty) \le c_0/8. \tag{3.cxxxiii}$$

Schur's inequality bounds the ℓ^2 perturbation by the same quantity. The inverse follows by a Neumann series about the real matrix and has norm at most $4/c_0$. The coefficients are holomorphic local functions, and their uniform Cauchy bounds make that Neumann inverse holomorphic in every admitted finite complex parameter family.

For physical analytic continuation take $\mathbf{J} = J_A(\mathbf{U})$. The adjacent-plaquette formula also gives $\|J_A(e^{iE}U) - J_A(U)\|_\infty \le C\|E\|_\infty$ on the stated unitary-centred complex neighbourhood. Thus a smaller explicit bound $\|E\| \le c_0/(CK_L)$, together with the chart bound, implies eq. 3.cxxxiii. If such a field is a given block-axial complex critical background, its physical Gaussian is the analytic matrix in eq. 3.cxv. Existence of that background is still a separate question.

Write $r_L = 12L$, increase K_L to bound the complex row and column norms by $2K_L$, and choose

$$\tau_L = r_L^{-1} \log\left(1 + c_0/(16K_L)\right) > 0. \tag{3.cxxxiv}$$

For a bounded real function φ which is one-Lipschitz in starting-site distance, multiplication by $e^{\tau_L \varphi}$ changes the matrix by an operator of norm at most

$$2K_L(e^{\tau_L r_L} - 1) \le c_0/8. \tag{3.cxxxv}$$

This follows entry by entry, from finite range, and then from the row and column bounds. The inverse of the conjugated matrix has norm at most $8/c_0$. Take $\varphi(b) = \min\{d(b, b'), N\}$, apply this estimate between coordinate vectors and then let N increase. On the infinite lattice the weights used before this limit are bounded; no unbounded conjugation identity is assumed. We obtain

$$\|\mathscr{K}_{\mathbf{U}}[\mathbf{J}]^{-1}(b, b')\| \le (8/c_0)e^{-\tau_L d(b,b')}. \tag{3.cxxxvi}$$

The same proof on a torus uses its periodic distance. Fixed-matrix Dirichlet compressions retain the real lower bound, coefficient estimates and conjugation bound, so eq. 3.cxxxvi also holds for them. This does not identify a matrix rebuilt from a different local background or a different sequence of regions with that compression. Finite range gives an exponential bound for the precision itself. Summing the kernel bound with the smaller weight $e^{\tau_L d/2}$ gives absolute weighted row and column bounds, since a four-dimensional lattice has at most $C(m+1)^3$ bonds at distance in $[m, m+1)$. All these constants depend only on L and the fixed flat gap, not on volume or global holonomies.

The remaining identification and the exact scope. The physical conclusion is fully determined by the actual chart and the actual Wilson and contour-gauge functionals. An extension to independent-current cores must specify its complete projected current and gauge form. If its isolated one-step constraint subtraction is called $E_U^{(2)}[J]$, the necessary constraint-readout condition for agreement with eq. 3.cxiv is

$$\langle C_U u, E_U^{(2)}[J] C_U v\rangle = \langle h_U^{\mathsf{T}} J, D^2 q_U(0)[C_U u, C_U v]\rangle. \tag{3.cxxxvii}$$

Equivalently, if $q_U^{(2)}$ denotes the homogeneous quadratic part, its diagonal readout is $2\langle h_U^{\mathsf{T}} J, q_U^{(2)}(C_U u)\rangle$. This fixes the factor of two and the pivot-right-inverse composition. The first-step calculation below verifies eq. 3.cxxxvii for the named form and checks its remaining terms on the axial locus. The condition alone does not identify a general all-core completion. Its projected gauge block and every remaining current or gauge-restoration contribution, including a term denoted $\Delta'_\pi[J]$, must also match. The Hessian of the absolute contour gauge used here agrees with the relative-contour gauge Hessian on the axial locus, but need not agree away from it. Agreement at physical critical pairs alone proves none of these off-core identities. We have defined and bounded eq. 3.cxiv explicitly, without assuming the identification being tested.

Thus the actual on-shell first unweighted Gaussian, and its admitted physical analytic continuation, have the proved gap and kernels. The auxiliary independent-current extension has the same bounds under its explicit current ceiling. No assertion about all lower levels, signed layer coefficients, nonzero-current many-level Schur complements, enlarged complex interpolation domains, or the conditional exterior-force law follows from this one-step result.

3.8. **The first-step background and its regular minimum.** We construct a background for one raw, unweighted step. The input is a real coarse connection with sufficiently small plaquettes. Its potential and its noncontractible holonomies need not be small. The output is an actual block-axial fine connection, with the prescribed nonlinear average, which is constrained-critical for the Wilson action and minimizes it within the explicit regular class proved below. The construction gives exponential first responses, product-topology continuity, and compatible local complex continuations. It uses the actual contour soft gauge as an auxiliary function in the construction.

The constants below depend on the fixed odd integer $L \ge 3$, but not on volume. We use the actual average, middle pivots and physical left bond logarithms of § 3.7. On a periodic lattice

we retain its embedded-double-block condition and use the periodic cover for larger patches. The infinite-lattice action is interpreted through finite local variations. No finite total action is assumed.

An exact initial field. For a coarse connection V, define its face lift $U_0(V)$ as follows. A fine bond internal to a centered block has value I. Every positive fine bond crossing from $B(y)$ to $B(y+Le_\mu)$ has value $V(y,\mu)$. The value on a reversed bond is the inverse. Every internal contour transport, and therefore every actual implicit contour mean, is I. Every straight length-L segment used in the average crosses its designated coarse face once and has transport $V(y,\mu)$, as does its reference segment. Consequently

$$T_{y,x}(U_0(V)) = I, \qquad Z_x(U_0(V)) = I, \qquad \mathcal{M}_L U_0(V) = V. \tag{3.cxxxviii}$$

These are exact nonlinear identities. A fine plaquette whose two directions are not both crossing directions is I. If both directions cross, its four factors are precisely the four factors of the corresponding coarse plaquette. In particular, for real V,

$$\sup_p \|U_0(V)(\partial p) - I\| \le \sup_P \|V(\partial P) - I\| =: \epsilon. \tag{3.cxxxix}$$

There is no holonomy restriction in this construction.

Write $\Phi_V := \Phi_{U_0(V)}$ for the actual fibre chart, with identity free restriction. Its free bond set $\mathfrak{X}$ depends only on the fixed block partition. Define

$$U_V(u) = e^{i\Phi_V(u)} U_0(V), \qquad F_V(u) = D_u\{A(U_V(u)) + \mathrm{GF}(U_V(u))\}. \tag{3.cxl}$$

We identify a row derivative with a field using the fixed bilinear colour pairing. On the infinite lattice, a component of F_V is the derivative under its compactly supported free coordinate variation. Only finitely many local terms enter that component. The estimates below give a bounded analytic map on the stated ℓ^∞ ball; eq. 3.cxl does not assign a value to an infinite sum of action terms.

Let $c_0 = c_0(L) > 0$ be the actual flat gap. Fix a physical curvature ceiling $\alpha_L > 0$ so that every real block-axial field of curvature at most α_L, together with its Wilson current, satisfies the smallness conditions of the preceding section with room. At the face lift, the actual GF gradient vanishes. Thus

$$\begin{aligned} F_V(0) &= C^{\mathsf{T}}_{U_0(V)} J_A(U_0(V)), & \|F_V(0)\|_\infty &\le A_L\epsilon, \\ K_V &:= DF_V(0) = K_{\mathrm{phys}}(U_0(V)), & K_V &\ge \tfrac{1}{2}c_0 I, \qquad A_L = C_* L^4. \end{aligned} \tag{3.cxli}$$

Here and below C_* is a sufficiently large fixed universal constant. The gap is legitimate before criticality is established: the exact chain rule eq. 3.cxv and the proof of eq. 3.cxxx require a real block-axial small-curvature field and its small physical current. Criticality was required there only to interpret the resulting matrix as a background Gaussian. No such interpretation is used to obtain eq. 3.cxli.

The bounded-field inverse and the nonlinear estimates. The matrix K_V is real symmetric, has range $12L$, and has uniform absolute row and column bounds. Its ℓ^2 inverse has the actual exponential kernels proved in eq. 3.cxxxvi. If $\tau_L > 0$ is the rate there, put

$$S_4(\tau_L) = \sum_{m\ge 0} C_4(m+1)^3 e^{-\tau_L m}, \qquad B_L = 1 + \frac{8}{c_0} S_4(\tau_L). \tag{3.cxlii}$$

The universal C_4 bounds the number of positive fine bonds in a distance shell; it is valid also for periodic distance. We may enlarge it so $B_L \ge 8/c_0$. Absolute kernel rows define an operator H_V on bounded free fields, with

$$\|H_V\|_{\infty\to\infty} \le B_L, \qquad K_V H_V = H_V K_V = I \quad \text{on } \ell^\infty. \tag{3.cxliii}$$

For clarity, both identities are proved here. Each matrix entry of $K_V H_V$ or $H_V K_V$ equals the identity entry by the ℓ^2 inverse identities. Applying the former to a bounded field involves finitely

many K_V entries in each row and absolute H_V rows, so the sums interchange. For the latter, the absolute double sum is bounded by

$$\sum_j \|H_V(b,j)\| \sum_{b'} \|K_V(j,b')\| \, \|f\|_\infty \le B_L \|K_V\|_{\mathrm{Sch}} \|f\|_\infty .$$

It therefore interchanges as well. This proves eq. 3.cxliii on ℓ^∞, rather than merely extending an ℓ^2 notation to that space.

We need nonlinear estimates for the actual local functions. The joint background and free radii $r_b = c_b L^{-5}$, $r_u = c_u L^{-9}$ of eq. 3.cxvii are retained. In each local real block-constant frame the face lift has small potential. The actual joint chart therefore exists on a fixed multiple of these radii, with uniform local bounds. Its component dependence is the same as in the preceding section. A fixed free coordinate enters at most 54 composed Wilson terms and $L^4 - 1$ actual GF terms, and its coupled free coordinates lie in a fixed number of blocks. Two scalar free Cauchy formulas cost r_u^{-2}, the term count costs CL^4, and the coupled-coordinate count costs CL^4. Hence, on a fixed smaller ball,

$$\begin{gathered} \|DF_V(u)\|_{\mathrm{Sch}} \le M_L, \qquad \|DF_V(u) - K_V\|_{\mathrm{Sch}} \le N_L \|u\|_\infty, \\ M_L = C_* L^{26}, \qquad N_L = C_* L^{35}. \end{gathered} \tag{3.cxliv}$$

To obtain the second estimate, apply one more scalar Cauchy formula in an arbitrary unit supremum free direction. It costs $r_u^{-1} = CL^9$. Integrate that derivative along the free segment from zero to u. No count of all coordinates in the direction is inserted. All these Hessians have range $12L$, and their coefficients depend on background variables within a fixed neighbourhood of the same size. These are derived bounds for the composed actual action and actual contour GF; a bound on their desired matrix perturbation is not an input.

We will also vary the coarse datum. Fix a real V as above, and write

$$\begin{gathered} V_B(c) = e^{iB(c)} V(c), \qquad U_0(V_B) = e^{iE_0(B)} U_0(V), \\ E_0(B)(b) = \begin{cases} B(c), & b \text{ crosses face } c, \\ 0, & b \text{ is internal.} \end{cases} \end{gathered} \tag{3.cxlv}$$

In particular $\|E_0(B)\|_\infty \le \|B\|_\infty$. The real local frames are fixed while B varies. Write $F(B,u) = F_{V_B}(u)$. On $\|B\|_\infty < r_b / C_*$ and the retained free ball, the actual chart and all component functions are jointly holomorphic, and

$$\begin{aligned} &\|F(B,0) - F(0,0)\|_\infty \le Q_L \|B\|_\infty, && Q_L = C_* L^{18}, \\ &\|D_u F(B,u) - K_V\|_{\mathrm{Sch}} \le D_L \|B\|_\infty + N_L \|u\|_\infty, && D_L = C_* L^{31}. \end{aligned} \tag{3.cxlvi}$$

For the first estimate the local scalar Cauchy costs are L^9 for one free derivative and L^5 for one background direction, followed by the L^4 term count. For the second, apply the background Cauchy loss L^5 to the L^{26} Schur estimate and combine with eq. 3.cxliv. The face replication in eq. 3.cxlv has supremum norm one, so no number of crossing bonds is charged to the directional background Cauchy radius. Local real unitary frames preserve all input norms. These estimates are consequently uniform even when the coarse holonomies are large.

Explicit radii and the real fixed point. Enlarge $S_L = C_* L^4$ so that on the retained chart

$$\|\Phi_V(u)\|_\infty \le S_L \|u\|_\infty, \qquad \sup_p \|U_V(u)(\partial p) - I\| \le \epsilon + S_L \|u\|_\infty \tag{3.cxlvii}$$

for real u. The second estimate follows by telescoping four unitary bond factors and using the first. Set

$$\begin{aligned} r_N &= \min\left\{r_u/16,\ (16B_LN_L)^{-1},\ \alpha_L/(16S_L)\right\},\\ \eta_{\mathrm{reg}} &= \min\left\{\alpha_L/4,\ c_*L^{-5},\ r_N/(16C_{\mathrm{cmp}}L^3)\right\},\\ P_L &= 1+2B_LS_LA_L, \qquad \epsilon_* = \min\left\{\alpha_L/16,\ r_N/(16B_LA_L), \eta_{\mathrm{reg}}/(16P_L)\right\}. \end{aligned} \tag{3.cxlviii}$$

Here C_{cmp} is the universal comparison constant proved below, and c_* is small enough for its local logarithm branches. This is a sequential choice: r_N is fixed first, then η_{reg}, then ϵ_*. All are positive and depend only on L.

Suppose $\epsilon \le \epsilon_*$. On the closed bounded-field ball $\|u\|_\infty \le r_N$, let

$$\mathcal{T}_V(u) = u - H_V F_V(u). \tag{3.cxlix}$$

By eq. 3.cxliii–eq. 3.cxliv,

$$\|D\mathcal{T}_V(u)\|_{\infty\to\infty} \le B_LN_Lr_N \le 1/16, \qquad \|\mathcal{T}_V(0)\|_\infty \le B_LA_L\epsilon \le r_N/16. \tag{3.cl}$$

Thus $\mathcal{T}_V$ maps the ball strictly into itself and is a contraction. Its iterates from zero are real and converge to a unique root $u(V)$ in that ball. The two-sided identity in eq. 3.cxliii makes the fixed-point equation equivalent to $F_V(u(V)) = 0$. Moreover

$$\|u(V)\|_\infty \le 2B_LA_L\epsilon, \qquad \|\Phi_V(u(V))\|_\infty \le 2S_LB_LA_L\epsilon. \tag{3.cli}$$

The resulting real field has curvature at most $P_L\epsilon \le \eta_{\mathrm{reg}}/16$. In particular it remains inside the physical NFP curvature domain. No critical point was assumed to start this iteration.

Why the root is a Wilson critical background. We first check the tangent space used by the stationarity equation. On the retained joint domain, the pivot derivative of the actual constraint stays invertible. The chart has identity free restriction. Differentiating its constraint and solving the pivot part therefore shows that its derivative spans the entire tangent kernel of that constraint, at every point in the ball. The component derivative is strictly local. In particular a compactly supported full tangent variation is represented by its compactly supported free restriction; no tangent directions are omitted in eq. 3.cxl.

Let $x \in B(y)\setminus\{y\}$. Apply a site gauge supported at x, with $s_t(x) = e^{itX}$, and identity at every other site. It fixes all coarse centers, so the actual nonlinear average is unchanged. It is an admitted fibre variation for sufficiently small t. The Wilson derivative along it is zero by gauge invariance. Endpoint covariance of the actual contour mean gives

$$T_{y,x}(U^{s_t}) = T_{y,x}(U)e^{-itX}. \tag{3.clii}$$

Every other internal mean is unchanged: any occurrence of x inside a contour cancels between its two adjacent factors. Consequently stationarity of eq. 3.cxl at the root implies

$$0 = \frac{i}{2}\,\mathrm{tr}\left(T_{y,x}(U)X\right) \quad \text{for every traceless Hermitian } X. \tag{3.cliii}$$

The trace pairing then says that $T_{y,x}(U)$ is scalar. Its determinant is one, so it is I or $-I$. The chart keeps all internal bonds, their $O(L)$-length contour transports and their means in the retained neighbourhood of I; the choice $r_N \le r_u/16$ secures this uniformly. The minus sign is excluded. Thus

$$T_{y,x}(U(V)) = I \quad \text{for every internal pair}, \qquad D\mathrm{GF}(U(V)) = 0. \tag{3.cliv}$$

The second identity holds on the full fine-bond space: the trace derivative of an SL(2) curve through I is zero. The root equation is therefore Wilson stationarity along the entire actual fibre. We have constructed

$$\boxed{\mathcal{M}_LU(V) = V, \qquad U(V) \text{ is block axial}, \qquad J_A(U(V)) \perp \ker P_{U(V)}.} \tag{3.clv}$$

All arguments on the infinite lattice used compactly supported free or site variations; they require no value for the background action. Here the perpendicular notation means annihilation of compactly supported tangent fields. The bounded pointwise multiplier identity, which implies no ℓ^2 membership, is proved in eq. 3.clxxi–eq. 3.clxxii. At this actual critical background the physical gap and covariance bounds of the preceding section now have their stated Gaussian interpretation.

Uniqueness and the comparison with the face lift. An axial real Wilson critical field represented by $U_V(u)$ in the closed r_N-ball has zero GF gradient and hence satisfies $F_V(u) = 0$. The contraction proves that it equals $U(V)$. The comparison in eq. 3.cxlviii identifies a geometric uniqueness class without assuming chart membership.

Let W be a real block-axial field with fine plaquette deviation at most η. On one block a unitary comb gauge g, anchored at its center, makes its internal bonds $CL\eta$-close to I. Every transformed internal path has length $O(L)$, so its actual mean is $CL^2\eta$-close to I. But axiality and covariance give that mean as $g(y)g(x)^{-1} = g(x)^{-1}$. Thus the original internal bonds obey

$$\|W(b) - I\| \le CL^2\eta \quad (b \text{ internal}). \tag{3.clvi}$$

For a coarse face c, write $S_x = W([x, x + Le_\mu])$. Coordinate-path swaps express each contour/reference loop in the double block as a product of at most CL^2 conjugated elementary plaquettes. Unitarity makes its deviation at most $CL^2\eta$. Two internal coordinate-order paths differ by the same number of swaps, so their transports differ by that amount. The contraction for the implicit mean, recentered at any one unitary input, shows that every such path transport differs from its actual mean by $CL^2\eta$. Since those means are I, replacing the two contour transports in the closed loop by their means gives

$$\|S_x W[c]^{-1} - I\| \le CL^2\eta. \tag{3.clvii}$$

The actual outer exponential of the arithmetic logarithm mean has the same bound relative to $W[c]$. Hence

$$\|S_x \mathcal{M}_L(W)(c)^{-1} - I\| \le CL^2\eta. \tag{3.clviii}$$

Each S_x contains one crossing bond and at most $L - 1$ internal bonds. Stripping those unitary internal factors with eq. 3.clvi proves the safe bound

$$\sup_b \|W(b) - U_0(\mathcal{M}_L W)(b)\| \le C_{\mathrm{cmp}} L^3 \eta. \tag{3.clix}$$

Enlarge C_{cmp} to give the same estimate for the physical left logarithm of the displayed ratio on its near-identity branch. No global small gauge or small coarse potential was used.

If $\mathcal{M}_L W = V$ and $\eta \le \eta_{\mathrm{reg}}$, that logarithm is within the ordinary local graph uniqueness radius and its free restriction is within $r_N/16$. The actual fibre chart therefore represents W; if W is constrained-critical it equals $U(V)$. For a real field not initially axial, small plaquettes first make its internal contour families mutually close. Their actual means exist, even if centered at a large unitary transport. The site gauge $s(x) = T_{y,x}(W)$, $s(y) = I$, sets every mean to I and preserves the coarse datum. Curvature and constrained criticality are preserved. Applying eq. 3.clix to this representative proves uniqueness of the center-fixing gauge orbit among real regular fields of curvature at most η_{reg}. The constructed field belongs to this same class by eq. 3.cli and the final smallness choice in eq. 3.cxlviii. Two axial representatives in a center-fixing orbit are equal: covariance of their means forces the site gauge itself to be identity.

Complex continuation and first response kernels. Keep the real V fixed. Enlarge a universal constant in $P'_L = C_* + 2B_L S_L Q_L$, and set

$$b_N = \min\left\{ r_b/(16C_*),\ (16B_L D_L)^{-1},\ r_N/(16B_L Q_L), \eta_{\mathrm{reg}}/(16P'_L) \right\}. \tag{3.clx}$$

For $\|B\|_\infty < b_N$, use the same fixed real inverse H_V in

$$\mathcal{T}_B(u) = u - H_V F(B,u).$$

eq. 3.cxlvi–eq. 3.cl give

$$\|D\mathcal{T}_B(u)\|_{\infty\to\infty} \le 1/8, \qquad \|\mathcal{T}_B(0)\|_\infty \le r_N/8. \tag{3.clxi}$$

The contraction is uniform on the complex ball and maps it strictly inside. Its iterates are holomorphic and converge uniformly on smaller closed parameter domains. Thus $u(B)$, and $U(V_B) = e^{i\Phi_{V_B}(u(B))}U_0(V_B)$, are holomorphic in every admitted finite complex parameter family. Reality is preserved when B is real. The root satisfies

$$\|u(B)\|_\infty \le 2B_L(A_L\epsilon + Q_L\|B\|_\infty). \tag{3.clxii}$$

The site variation argument in eq. 3.clii–eq. 3.cliv applies to a complex root as well. The equation pairs with every complexified traceless direction; a scalar mean in $\mathrm{SL}(2,\mathbb{C})$ is $\pm I$, and the retained branch selects I. Hence this is an actual complex block-axial Wilson critical background, rather than a critical point only of $A + \mathrm{GF}$. No independent fine current is introduced.

For completeness we derive response locality from actual finite matrices. At the root put $\mathcal{H}_B = D_u F(B,u(B))$. The preceding estimates give

$$\|\mathcal{H}_B - K_V\|_{\mathrm{Sch}} \le (8B_L)^{-1} \le c_0/64, \qquad \|\mathcal{H}_B\|_{\mathrm{Sch}} \le M_L. \tag{3.clxiii}$$

They also retain range $r_0 = 12L$. Choose

$$\sigma_L = r_0^{-1}\log\left(1 + c_0/(64M_L)\right) > 0. \tag{3.clxiv}$$

Multiplication by $e^{\sigma_L\varphi}$, for a bounded real one-Lipschitz φ, changes $\mathcal{H}_B$ by at most $M_L(e^{\sigma_L r_0}-1) = c_0/64$ in Schur norm. A Neumann argument about $K_V \ge c_0/2$, first for the complex perturbation and then for this conjugation, gives an inverse norm at most $8/c_0$. Bounded weights $\varphi(b) = \min\{d(b,b'),N\}$, followed by $N\to\infty$, give the actual kernel bound

$$\|\mathcal{H}_B^{-1}(b,b')\| \le (8/c_0)e^{-\sigma_L d(b,b')}. \tag{3.clxv}$$

Absolute rows again give both inverse identities on bounded fields. At a real root this matrix is a positive coordinate Hessian. At any root the physical free coordinates centered at that root differ from the original free coordinates by pointwise invertible three-by-three exponential-coordinate Jacobians. Their norms and inverse norms are uniformly bounded on the retained ball. Since the first derivative vanishes, the Hessians are related by ordinary congruence with these pointwise matrices. Thus the physical centered Hessian also has the corresponding inverse kernel bounds.

Define the source derivative by first differentiating the first argument of $F(B,u)$ at fixed u, and only then substituting the root: $\mathcal{R}_B = (\partial_1 F)(B,u(B))$. Thus the total derivative of the root equation is $0 = \mathcal{R}_B + \mathcal{H}_B D_B u$. This partial source derivative is strictly local. A free row sees only a fixed number of coarse faces in its dependence neighbourhood. The Cauchy count giving Q_L bounds each coefficient by CL^{18}, while a coarse face can reach at most CL^4 free coordinates. Therefore

$$\|\mathcal{R}_B\|_{\mathrm{Sch}} \le C_* L^{22}, \qquad D_B u = -\mathcal{H}_B^{-1}\mathcal{R}_B. \tag{3.clxvi}$$

The support distance between a free bond and a contributing coarse face, represented by its pivot, is at most a fixed multiple of L. Combining this fact with eq. 3.clxv gives

$$\|D_{B(c)}u(b)\| \le C_L e^{-\sigma_L d(b,b_0(c))}. \tag{3.clxvii}$$

The derivative $D_u\Phi$ has the same strict locality and absolute row and column bounds CL^4 on the shrunken domain, by one free Cauchy formula. Its direct background derivative is local and bounded by $CL^9\|u\|_\infty$, using a background Cauchy formula and $\Phi_B(0) = 0$. The exponential map

and the direct face lift have bounded local derivatives. Consequently the actual fine-field derivative in left trivialization satisfies

$$\left\| \frac{1}{i}(D_{B(c)}U(V_B)(b))U(V_B)(b)^{-1} \right\| \le C_L e^{-\sigma_L d(b,b_0(c))}. \tag{3.clxviii}$$

The constants in these response bounds are independent of volume and holonomies. The response formulas concern this first-step solution; they are not many-level background derivative estimates.

Interior locality, periodicity and exact scope. There is a useful finite comparison without interpolating between two distant coarse connections. Let V, V' be admitted real coarse fields which agree on a coarse patch, and let u, u' be their constructed free solutions. Both arrays have the same free coordinate labels and lie in the uniform r_N-ball. The function $F_V(u')$ is defined even where the two face lifts differ. Strict locality gives $F_V(u') = F_{V'}(u') = 0$ on every row whose entire coarse dependence neighbourhood lies in the common patch. Call this set of rows I. On the convex free segment define

$$\overline{\mathcal{H}} = \int_0^1 DF_V(u' + t(u - u'))\, dt. \quad \text{Then} \quad \overline{\mathcal{H}}(u - u') = -F_V(u'). \tag{3.clxix}$$

Every integrand is a real symmetric range-$12L$ matrix with the same Schur bound and the same positive gap: it differs from K_V by at most $N_L r_N \le c_0/128$. Its average has those properties as well, so the bounded-conjugation argument gives exponential inverse kernels. The residual in eq. 3.clxix is supported in I^c and has norm at most $2M_L r_N$, by the derivative bound along the same segment. Summing the inverse with the smaller exponential weight gives

$$|u(b) - u'(b)| \le C_L r_N e^{-\sigma_L d(b,I^c)/2}. \tag{3.clxx}$$

The local chart derivative gives the analogous bound for fine fields whose background dependence neighbourhoods are in the common patch; the fixed additional range is absorbed in C_L. No path from V to V' through small-curvature coarse fields has been assumed. This is a deterministic locality statement for these canonical first-step axial representatives.

The face lift, chart and actual GF intertwine every real unitary gauge transformation constant on blocks. The unique fixed point therefore has the corresponding coarse gauge covariance. The same uniqueness proves compatibility with translations preserving the block partition. If the coarse input is periodic, translating the cover root by a period gives another root in the same ball; it is therefore periodic. Conversely a finite-torus root lifts to such a cover root, so the two constructions agree. This uses locality and uniqueness of the bounded-field equation, rather than silently periodizing an ℓ^2 inverse. All local frames and kernel estimates are valid on the cover, and their constants do not depend on the noncontractible holonomies.

Bounded currents and pointwise constrained stationarity. Let $U = U(V)$, and use the actual chart centered at U for this paragraph. Write $P = P_U$, $h = h_U$, $C = C_U = \iota - hP\iota$, and $J = J_A(U)$. Every component of J is bounded by the physical plaquette estimate. The finite-range maps P, h, C, and their transposes, act on bounded fields by finite component sums. In particular the coarse covector $\mu = h^{\mathsf{T}} J$ is pointwise well-defined and obeys

$$\mu(c) = J_c(U)^{-\mathsf{T}} J(b_0(c)), \qquad \|\mu\|_\infty \le CL^3 \|J\|_\infty, \quad J_c(U) = D_{a(b_0(c))} q_{U,c}(0). \tag{3.clxxi}$$

The symbol $J_c(U)$ here is the three-by-three pivot coefficient, whereas $J(b)$ is a fine Wilson current.

For each compactly supported free field v, Cv is a compactly supported tangent field. Wilson stationarity gives $\langle J, Cv \rangle = 0$; testing one free coordinate at a time therefore gives $C^{\mathsf{T}} J = 0$ pointwise. Set $R = J - P^{\mathsf{T}} h^{\mathsf{T}} J$. Its free restriction is exactly $C^{\mathsf{T}} J$, hence zero. On a pivot $b_0(c)$, no other constraint row contains that pivot, and eq. 3.clxxi gives

$$R(b_0(c)) = J(b_0(c)) - J_c(U)^{\mathsf{T}} J_c(U)^{-\mathsf{T}} J(b_0(c)) = 0.$$

Both restrictions vanish. Thus the exact bounded-current statement is

$$\boxed{J_A(U) = P_U^{\mathsf{T}} h_U^{\mathsf{T}} J_A(U) \quad \text{pointwise on fine bonds.}} \tag{3.clxxii}$$

Conversely, this pointwise identity annihilates every compactly supported $a \in \ker P_U$: both a and $P_U a$ have finite support, so transposition is a finite-sum identity and $\langle J_A(U), a\rangle = \langle h_U^{\mathsf{T}} J_A(U), P_U a\rangle = 0$. The multiplier is unique since $Ph = I$. These equivalences are statements about bounded currents and compact variations; their proof does not place $J_A(U)$ or μ in ℓ^2.

Local interactions and bounded-field differentiability. We specify the derivatives used in the iteration, source response and comparison arguments. For a finite-support increment of free variables, each changed pivot is in a fixed neighbouring double block. Only finitely many Wilson and GF terms change, so the relative value of $A + \mathrm{GF}$, its first variation, and its second variation are ordinary finite sums. The b-component of $F(B, u)$ is the derivative under a variation of the single free coordinate b. It is a function of finitely many local free and coarse coordinates. This defines the same component even when u and the background do not have finite support. The row/column estimates previously proved extend its derivative matrices to bounded fields.

Here is a uniform remainder check that proves actual Fréchet differentiability on ℓ^∞. Put

$$\rho_b = c_{\mathrm{loc}} r_b, \qquad \rho_u = r_u/4, \qquad G_L = C_* L^{13}. \tag{3.clxxiii}$$

Here $c_{\mathrm{loc}} > 0$ is a universal constant fixed by the matrix-logarithm and local-frame domain estimates before C_* is chosen. Take C_* large enough that $b_N \le \rho_b/4$ and to cover all derivative constants below. Each component of F is holomorphic and bounded by G_L on the joint local domain of radii $2\rho_b, 2\rho_u$. Indeed one free Cauchy derivative of a bounded local interaction costs CL^9, and at most CL^4 terms enter that component. The estimate is independent of the number of coordinates elsewhere on the lattice. The radii in eq. 3.cxlviii and eq. 3.clx, with its reduced first entry, lie strictly inside half these local radii.

At such an interior point define $\nu = \|\Delta B\|_\infty/\rho_b + \|\Delta u\|_\infty/\rho_u \le 1/4$. For each row apply scalar Cauchy to $t \mapsto F_b(B + t\Delta B, u + t\Delta u)$ on a circle of radius $(2\nu)^{-1}$; the case $\nu = 0$ is immediate. The circle stays in the stated local domain. Summing the scalar Taylor remainder of order at least two, uniformly in the row, gives

$$\begin{aligned}\|F(B + \Delta B, u + \Delta u) - F(B, u) - (\partial_1 F)(B, u)\Delta B - (\partial_2 F)(B, u)\Delta u\|_\infty \\ \le C G_L \nu^2.\end{aligned} \tag{3.clxxiv}$$

The first Taylor coefficient is a linear map of the increments because each component is an ordinary finite-dimensional derivative. Its uniform Cauchy bound makes that map bounded on ℓ^∞. eq. 3.clxxiv is the required uniform Fréchet remainder. Applying the same argument to the derivatives proves their continuity and higher local analytic remainders. It also applies to the chart components, using their uniform bound $S_L\|u\|_\infty$. Consequently the iteration is a genuine bounded-field equation. Its holomorphic iterates converge uniformly; their component Cauchy integrals and uniform remainders pass to the limit. The resulting bounded-field maps are locally holomorphic, as are the finite parameter families already stated. No extensive action estimate or formal derivative of an infinite action value is needed.

The regular minimum and its quantitative strictness. For every real u in the whole closed r_N-ball, eq. 3.cxliv and the radius choice give the Hermitian form bound

$$DF_V(u) \ge (c_0/2 - N_L r_N)\, I \ge c_0/4\, I. \tag{3.clxxv}$$

In fact $N_L r_N \le (16 B_L)^{-1} \le c_0/128$. On a finite torus this proves uniform strict convexity of the actual $A + \mathrm{GF}$ in free coordinates. On the infinite lattice the same quadratic inequality applies to compact variations.

On a finite torus let W be any real fibre field of curvature at most η_{reg}, with $\mathcal{M}_L W = V$. Its canonical gauge fixing, equal to identity at the coarse centers, gives an axial representative W_{ax}. By eq. 3.clix it has actual coordinates u_W in the same ball. Let $u_* = u(V)$ and $d = u_W - u_*$. The full free segment $u_* + sd$, $0 \le s \le 1$, remains in the convex ball. The segment is used for $A + \mathrm{GF}$; axiality is needed at its two endpoints. Since $F_V(u_*) = 0$, Taylor's integral identity and eq. 3.clxxv give

$$\begin{aligned}&[A + \mathrm{GF}](U_V(u_W)) - [A + \mathrm{GF}](U_V(u_*))\\ &\quad = \int_0^1 (1-s)\langle d, DF_V(u_* + sd)d\rangle\, ds \ge (c_0/8)\|d\|_2^2.\end{aligned} \tag{3.clxxvi}$$

Both endpoints are axial, so their actual GF values vanish. Wilson gauge invariance gives $A(W) = A(W_{\mathrm{ax}})$. Therefore

$$\boxed{A(W) - A(U(V)) \ge (c_0/8)\|u_W - u(V)\|_2^2.} \tag{3.clxxvii}$$

Equality forces the canonical axial representatives to coincide, so equality occurs precisely in the constructed center-fixing gauge orbit. The constructed field belongs to this regular class by eq. 3.cxlviii–eq. 3.cli. Hence the construction gives its minimizing orbit among the stated regular fibre fields; no minimum over fields outside that class is claimed.

On the infinite lattice take instead a compact perturbation W of $U(V)$, still real, regular and in the same fibre. Only finitely many blocks can have changed internal means. Its canonical axializing site gauge is therefore identity outside finitely many blocks, because $U(V)$ is already axial. Thus $W_{\mathrm{ax}} - U(V)$, and hence $u_W - u_*$, have finite support. Chart locality makes the entire segment in eq. 3.clxxvi a variation in a fixed finite fine neighbourhood. All relative action values and derivatives in that equation are finite interaction differences. The same Taylor argument proves eq. 3.clxxvii with exactly that meaning of the left-hand side. It does not compare infinite extensive action values.

Dependence on coarse data in the product topology. For a free row b, let D_b be its finite set of coarse face variables on which $F_V(u)_b$ depends. This set is contained in a fixed number of neighbouring coarse blocks, for every admitted u. For a finite coarse patch $\mathcal{P}$, put

$$I_{\mathcal{P}} = \{b : \mathsf{D}_b \subset \mathcal{P}\}, \qquad \delta_{\mathcal{P}} = \sup_{c\in\mathcal{P}} \|V(c) - V'(c)\|, \qquad \delta_L = \min\{1/8, r_b/(8C_*)\}. \tag{3.clxxviii}$$

Suppose V, V' are both admitted real data and $\delta_{\mathcal{P}} \le \delta_L$. Write $u = u(V)$, $u' = u(V')$, and evaluate the latter array in F_V. It is in the common free ball even though the data can differ arbitrarily outside $\mathcal{P}$.

We justify a local source estimate without a patch-volume loss. For $c \in \mathcal{P}$ take the principal left logarithm $B(c) = i^{-1}\log(V'(c)V(c)^{-1})$; it has norm at most $C\delta_{\mathcal{P}}$. Extend it by zero outside the patch. On a row in $I_{\mathcal{P}}$, strict locality identifies the endpoint data with V'. Uniqueness of each actual local pivot graph identifies that endpoint with the chart used for V', independently of data outside its dependence set. Fixed local real frames for V keep the background interpolation $e^{itB}V$ inside the actual joint analytic chart. This interpolation need not stay in the admitted small-curvature class. One background-direction and one free-direction Cauchy formula, with u' held fixed, give

$$\sup_{b\in I_{\mathcal{P}}} |F_V(u')_b - F_{V'}(u')_b| \le Q'_L\delta_{\mathcal{P}}, \qquad Q'_L = C_* L^{18}. \tag{3.clxxix}$$

The count is L^5 for the background radius, L^9 for the free radius and L^4 local terms. The background direction has supremum norm at most $C\delta_{\mathcal{P}}$, so its number of nonzero faces is absent from this estimate.

The same real averaged Hessian as in eq. 3.clxix, with the data V fixed, satisfies

$$\overline{\mathcal{H}}(u - u') = -F_V(u'), \qquad |F_V(u')_b| \le \begin{cases} Q'_L\delta_{\mathcal{P}}, & b \in I_{\mathcal{P}},\\ 2M_L r_N, & b \notin I_{\mathcal{P}}.\end{cases} \tag{3.clxxx}$$

The interior estimate uses $F_{V'}(u') = 0$; the exterior estimate uses $F_V(u) = 0$ and the uniform derivative bound along the free segment. No comparison of the distant coarse data is used there. Define the finite weighted kernel-row constant

$$\mathcal{B}_L = \frac{8}{c_0} \sum_{m \geq 0} C_4 (m+1)^3 e^{-\sigma_L m/2}, \qquad C_{\mathrm{cont},L} = \mathcal{B}_L \max\{Q'_L, 2M_L\}. \tag{3.clxxxi}$$

The conjugation proof applies to $\overline{\mathcal{H}}$ by eq. 3.clxxv. Its exponentially weighted absolute rows, multiplied by the two parts of eq. 3.clxxx, yield

$$|u(b) - u'(b)| \leq C_{\mathrm{cont},L} \delta_{\mathcal{P}} + C_{\mathrm{cont},L} r_N e^{-\sigma_L d(b, I^c_{\mathcal{P}})/2}. \tag{3.clxxxii}$$

If the complement is empty, its distance is infinity and the second term is zero. The estimate contains neither the patch cardinality nor the torus volume. Local chart derivatives give the analogous fine-field estimate, with a fixed additional neighbourhood absorbed in the constant and in the distance margin.

For clarity the admitted infinite-volume real data set is

$$\mathcal{A}_L = \{V \in \mathrm{SU}(2)^{\mathcal{E}_{\mathrm{coarse}}} : \|V(\partial P) - I\| \leq \epsilon_* \text{ for every coarse plaquette } P\}. \tag{3.clxxxiii}$$

The bond set is countable. Each displayed condition is continuous in finitely many product coordinates, so $\mathcal{A}_L$ is closed in the compact metrizable product configuration space. Given a fixed free bond and an error tolerance, first choose a finite patch with distance to its exterior large enough to make the second term of eq. 3.clxxxii small. Then constrain its finitely many face variables so $\delta_{\mathcal{P}}$ makes the first term small. This proves coordinatewise continuity of $V \mapsto u(V)$ on $\mathcal{A}_L$ in its relative product topology. Every fine output component depends continuously on finitely many data and chart coordinates, so $V \mapsto U(V)$ is continuous in the fine product topology. It is consequently Borel. The Borel structures here are those of the two countable product configuration spaces.

Compatibility of the complex germs. For each admitted real V, use the complex neighbourhood with principal coarse coordinates $W = e^{iB}V$, $\|B\|_\infty < b_N$. The fixed-real-base construction gives a root $u_V(W)$; denote its associated fine field by $\mathcal{U}_V(W)$. Its improved bound follows directly from eq. 3.clxi:

$$\|u_V(W)\|_\infty \leq \frac{\|\mathcal{T}_B(0)\|_\infty}{1 - 1/8} \leq r_N/7 < r_N/4. \tag{3.clxxxiv}$$

Consider a common complex datum $W = e^{iB}V = e^{iB'}V'$ in two such neighbourhoods. The face lift $U_0(W)$ is the same exact bond configuration in both presentations. Its internal bonds and their means are I, and its exact average is W. The free coordinates are the principal small left bond logarithms relative to this same lift.

We check that the actual chart is shared. At zero, all closed products in its average are I. For the retained small full chart variables, the internal path transports, the closed products, and the relative coarse products stay in their same near-identity logarithm neighbourhoods. The face values of W and their inverses are uniformly bounded, because W is in a retained complex neighbourhood of a real unitary datum. Internal means use the unique near-identity implicit mean. The closed and relative logarithms use the unique principal branch there. Unitary local frames conjugate these branches; they do not select new ones. It follows that both presentations have the same actual function

$$\begin{aligned} q_W(a) &= i^{-1} \log[\mathcal{M}_L(e^{ia} U_0(W)) W^{-1}], \\ P_W = Dq_W(0), &\qquad R_W(a) = q_W(a) - P_W a. \end{aligned} \tag{3.clxxxv}$$

Here h_W is the inverse of the same diagonal pivot coefficients, inserted at the same pivots. For a fixed free u, either chart satisfies the identical local equation

$$a = C_W u - h_W R_W(a), \qquad a|_{\mathrm{free}} = u. \tag{3.clxxxvi}$$

The existing joint pivot contraction has Lipschitz constant less than one on its retained full-variable neighbourhood. Both chart outputs are in that neighbourhood, since $\|a\|_\infty \le S_L r_N$ and $r_N \le r_u/16$. Its uniqueness therefore proves equality of the two graphs for $\|u\|_\infty \le r_N$, component by component. The resulting equation $F_W(u) = 0$ is identical in the two constructions.

Both roots in eq. 3.clxxxiv are in the entire r_N-ball on which either fixed-real-base map is a contraction. Either inverse H_V or $H_{V'}$ is two-sided on bounded fields, so its fixed points in that ball are exactly the roots of this common equation. Uniqueness at the common complex datum gives

$$u_V(W) = u_{V'}(W), \qquad \mathcal{U}_V(W) = \mathcal{U}_{V'}(W). \tag{3.clxxxvii}$$

The uniform local Fréchet estimates and holomorphic contractions then glue these continuations on overlaps. This uses the common complex datum and the actual pivot-graph uniqueness; it does not assume that the overlap contains an admitted real point or extend the argument to unrelated average maps or logarithm branches.

We have proved existence, a quantitative regular minimizing orbit, bounded pointwise current stationarity, product-topology continuity, compatible complex continuation and exponential first responses for an actual first-step background. The named comparison below checks independent currents on this first-step axial locus. Nothing here constructs a many-level background, treats signed layer coefficients or complex decoupling arrays, or proves the conditional exterior-force, history-class or interacting Gaussian-law assertions.

3.9. **Domains of local substitutions.** The ordinary step uses three coordinate systems at once. A bond logarithm is a group coordinate, a potential is a group coordinate divided by a length, and a Gaussian coordinate is a physical logarithm divided by a coupling. Its analytic domains also have two distinct parts: bounds on a field and bounds on every background obtained from its restricted averages. We fix these conventions and give quantitative admission tests. The tests apply to the nonlinear local maps themselves.

The finite averaging chart, the classical flat comparison, the first-step physical Hessian and the first-step background construction are used in their stated scopes. The last construction includes its regular minimum, product continuity and complex gluing conclusions. The many-level background and response estimates are not conclusions of this section. They occur below as explicitly named domain estimates. Once those estimates hold, all the inclusions, changes of units and Cauchy estimates used here are proved with constants independent of depth and compatible volume.

3.9.1. *Lengths, covectors and quadratic coefficients.* Work in the units of a fixed level. The bare spacing is η, with $0 < \eta \le 1$; a bond configuration is written

$$\mathbf{U}(b) = e^{ia(b)}U(b) = e^{i\eta A'(b)}U(b), \qquad a = \eta A'. \tag{3.clxxxviii}$$

Here U is real. The independent analytic current will be denoted by $\mathbf{J}$. The Wilson gradient is denoted by $J_A(U)$ when its normalization has been specified. These are different variables. The color pairing is $(X, Y) = \frac{1}{2}\operatorname{tr}(XY)$. It is complex bilinear. Operator norms on complex coordinate spaces use the corresponding Hermitian norm, and therefore use adjoints rather than bilinear transposes in positivity arguments.

Suppose that the current density is normalized by

$$DA_W(U)[\eta A] = \eta^4 \sum_b (J_A(U)(b), A(b)). \tag{3.clxxxix}$$

The covector for the raw logarithm a, with counting pairings, is $\jmath_A(U) = \eta^3 J_A(U)$. Indeed substitution of $A = a/\eta$ in eq. 3.clxxxix gives $DA_W(U)[a] = \sum_b (\eta^3 J_A(U)(b), a(b))$. At unit spacing, these two current conventions coincide. The formula in eq. 3.clxxxix also explains the dimension three weight on a current: the action has dimension zero, while a potential has dimension one in four dimensions.

More generally let a linear response H map datum coordinates to field coordinates with positive diagonal pairing matrices W_r, W_f. Its bilinear transpose and its transpose after length normalization are

$$H^{\mathsf{T}} = W_r^{-1} H^{\mathrm{t}} W_f, \qquad P = \mathsf{m}Q, \qquad P^{\mathsf{T}} = Q^{\mathsf{T}}\mathsf{m}^{\mathsf{T}}. \tag{3.cxc}$$

The matrix m multiplies a row by its length ξ_c. This formula is valid for arbitrary specified row weights; no row-volume factor is inferred from the letter Q. To verify it, write $u^{\mathrm{t}} W_f H v = (H^{\mathrm{t}} W_f u)^{\mathrm{t}} v$ and compare with the row pairing. Thus every transpose in a field-current identity must refer to its two declared pairings. Counting row pairings and volume field pairings can be used together, provided eq. 3.cxc is used.

Let h be a right inverse for the raw constraint derivative P. The right inverse for the per-length derivative Q is $h_Q = h\mathsf{m}$. For the corresponding raw and per-length multipliers one obtains

$$Ph = 1, \quad Qh_Q = 1, \qquad \mu = h^{\mathsf{T}} J, \quad \mu_Q = h_Q^{\mathsf{T}} J = \mathsf{m}^{\mathsf{T}} \mu. \tag{3.cxci}$$

If $q_Q = \mathsf{m}^{-1} q$, then $\langle \mu_Q, D^2 q_Q \rangle = \langle \mu, D^2 q \rangle$. Consequently changing a constraint from raw to per-length coordinates changes its multiplier at the same time. It does not change the physical curvature pairing.

For a change of length unit, let the new unit be L^{-n} times the old one. Numerical lengths multiply by L^n. The same configuration and current are represented by

$$\eta_{\mathrm{new}} = L^n \eta_{\mathrm{old}}, \quad A_{\mathrm{new}} = L^{-n} A_{\mathrm{old}}, \quad J_{\mathrm{new}} = L^{-3n} J_{\mathrm{old}}, \quad \nabla_{\mathrm{new}} A_{\mathrm{new}} = L^{-2n} \nabla_{\mathrm{old}} A_{\mathrm{old}}. \tag{3.cxcii}$$

The first identity and $a = \eta A$ prove the second. A difference quotient contains one inverse numerical length, which proves the fourth. The third follows either from eq. 3.clxxxix, or from the dimension three of its covector. In particular $\eta^4 \langle J, A \rangle$ is unchanged. If a bond lies in a layer of local scale ξ, the quantities

$$\xi |A|, \qquad \xi^2 |\nabla_U A|, \qquad \xi^3 |J|, \qquad (\xi/\eta)^2 |\partial U - 1| \tag{3.cxciii}$$

are unchanged as well. These are the quantities that enter a layered regularity test. Applying the same radius to their unweighted numerical coordinates would describe a different domain.

At a step whose fixed reference inverse coupling is t, put $g = \mathsf{t}^{-1/2}$. If y is the Gaussian free coordinate and Φ_U the actual physical fibre chart, the configuration is

$$U_y = e^{i\Phi_U(gy)} U, \qquad |y| < \mathfrak{b}, \qquad \delta = g\mathfrak{b}. \tag{3.cxciv}$$

Thus a physical free-coordinate radius r becomes r/g, before allowances for the norm of the chart and the background response. The chart is applied to gy, not to y.

For a holomorphic scalar functional $F(B)$ of a physical datum, let $F_s(B_s) = F(gB_s)$. The chain rule gives

$$D^2 F_s(0) = g^2 D^2 F(0), \qquad |b(F_s)| \le CM(g/r)^2 \iff |b(F)| \le CMr^{-2}. \tag{3.cxcv}$$

Here the last equivalence means the corresponding bounds under the same linear coefficient readout. It is not an assertion that a general coefficient is saturated by its Cauchy bound. The free-response and spatial-moment sums in that readout have their own fixed constants. In particular a radius proportional to $g(\log g^{-2})^q$ is a growing logarithmic radius in the Gaussian variable. Its physical extraction factor remains $g^{-2}(\log g^{-2})^{-2q}$.

3.9.2. *The actual average and its three linearizations.* The internal transport $T_{y,x}(U)$, $x \in B(y)$, is the mean of the transports along coordinate-order shortest paths from y to x. There is one path for each ordering of the four directions, with multiplicity; there are at most 4! such paths. It is the solution near that family of $\sum_\Gamma \log(U(\Gamma) T_{y,x}^{-1}) = 0$. It is not an arithmetic average of matrices.

With $x' = x + c_+ - c_-$, set

$$W_x(U) = T_{c_-,x}(U)U([x,x'])T_{c_+,x'}(U)^{-1}U[c]^{-1},$$
$$\mathcal{M}(U)(c) = \exp\left(L^{-4}\sum_{x\in B(c_-)}\log W_x(U)\right)U[c]. \tag{3.cxcvi}$$

This formula uses the regular local logarithm. The total measurable average also has its prescribed fallback rules. Those rules make real measure identities meaningful on every configuration. Cauchy estimates below use only the analytic extension of eq. 3.cxcvi, with all physical substitution values remaining in its regular branch.

At the identity let $\overline{a}(y,x)$ be the arithmetic mean of the signed path sums. Define the raw straight and plain derivatives by

$$\begin{aligned}(S_La)(c) &= L^{-4}\sum_{x\in B(c_-)} a([x,x']),\\ (\Sigma_La)(y) &= L^{-4}\sum_{x\in B(y)}\overline{a}(y,x),\\ P_La &= S_La - \mathrm{d}_c\Sigma_La, \qquad (q_L\lambda)(y) = L^{-4}\sum_{x\in B(y)}\lambda(x),\end{aligned} \tag{3.cxcvii}$$

where $(\mathrm{d}_cf)(c) = f(c_+) - f(c_-)$. The sign in eq. 3.cxcvii follows by differentiating the four transports in eq. 3.cxcvi. For a gradient $a = \mathrm{d}\lambda$, every path sum telescopes. Consequently

$$\Sigma_L\mathrm{d}\lambda = q_L\lambda - \lambda|_{\text{centers}}, \qquad S_L\mathrm{d}\lambda = \mathrm{d}_cq_L\lambda, \qquad P_L\mathrm{d}\lambda = \mathrm{d}_c(\lambda|_{\text{centers}}). \tag{3.cxcviii}$$

This proves both gradient identities directly. The scalar site average and restriction to centers have different kernels.

The straightening relative to a real background U_0 removes a block factor. In raw coordinates its one-step definition is

$$\begin{aligned}u_y(a) &= \exp\left(L^{-4}\sum_{x\in B(y)}\log[T_{y,x}(e^{ia}U_0)T_{y,x}(U_0)^{-1}]\right),\\ e^{i\mathcal{Q}^1(U_0,a)(c)}\mathcal{M}(U_0)(c) &= u_{c_-}(a)^{-1}\mathcal{M}(e^{ia}U_0)(c)u_{c_+}(a).\end{aligned} \tag{3.cxcix}$$

For several steps this definition is iterated, including the accumulated node factors. Differentiating at a flat background removes precisely the $\mathrm{d}_c\Sigma_L$ term and gives the straight derivative. For a general background, the resulting derivative is the straightened derivative $Q^\sharp$. Its leading transported straight derivative is a third operator, Q^{st}. The plain derivative Q^{pl}, $Q^\sharp$, and Q^{st} must therefore have separate names.

In physical potential coordinates, the derivative also contains the input length factor. After k straight steps at the identity it is $\eta_kS_L^k$, where $\eta_k = L^{-k}$. Its row sum is one, because the raw straight row sum is L at every step. With covariant difference quotients and the appropriate coarse units, differentiation of gauge covariance gives

$$Q_k^{\mathrm{pl}}(U)D_U\lambda = D_{\mathcal{M}^kU}(\lambda|_{\text{centers}}), \qquad Q_k^\sharp(U)D_U\lambda = D_{\mathcal{M}^kU}Q_k'(U)\lambda. \tag{3.cc}$$

The first formula follows from the exact bi-equivariance of $\mathcal{M}$, including its node endpoints. The second follows from the same calculation after eq. 3.cxcix; its site factor is the derivative of the averaged gauge transformation. A transported straight main term does not satisfy the second identity exactly unless the omitted straightening correction is zero.

There is a quantitative reason to keep these distinctions even at a flat field. For the raw straight map, every fixed fine bond occurs in exactly L of the intervals $[x, x+Le_\mu]$, counted over all block starts. Its unsigned column sum is L^{-3}. Its row sum is L. In contrast choose the first internal bond

from a block center in direction μ. Among the path orderings, at least a fraction $1/4$ start in that direction. There are $(L-1)L^3/2$ endpoints with positive μ-coordinate. Therefore its coefficient in $\Sigma_L a(y)$ is at least

$$L^{-4}\frac{1}{4}\frac{L-1}{2}L^3 = \frac{L-1}{8L} \geq \frac{1}{12} \qquad (L \geq 3). \tag{3.cci}$$

Take an outgoing coarse bond in another direction. Its straight part does not see this fine component, and its plain derivative contains this Σ_L coefficient. Hence the plain column bound is not the straight column bound. Zero displacement in some path directions causes no problem, since the coordinate orders form a multiset. This calculation uses a nonwrapping local lift. Periodic tori large enough for the same double block have the identical calculation.

On curvature quotients the difference in eq. 3.cxcvii is a gradient, so the classical energy comparison can transfer after an exact gradient correction. That quotient statement does not identify the full gauge, current or node-factor operators. In particular a column estimate for one of those operators is an estimate in its declared coordinates, not an estimate for an unnamed map denoted by Q.

3.9.3. *The fibre chart, current subtraction and contour gauges.* On the regular logarithm branch, define

$$q_U(a) = i^{-1}\log[\mathcal{M}(e^{ia}U)\mathcal{M}(U)^{-1}].$$

Put $P_U = Dq_U(0)$. The pivot construction gives a right inverse h_U, supported on the crossing pivots. If ι inserts the free coordinates and leaves pivots zero, set

$$C_U = \iota - h_U P_U \iota, \qquad q_U(\Phi_U(u)) = 0, \qquad \Phi_U(0) = 0, \quad D\Phi_U(0) = C_U. \tag{3.ccii}$$

The chart has identity restriction to the free coordinates; its nonlinear correction is on the pivots. Two differentiations of the constraint give

$$D^2\Phi_U(0)[u,v] = -h_U D^2 q_U(0)[C_U u, C_U v]. \tag{3.cciii}$$

Indeed its image is on the pivots, where P_U has inverse h_U. This also proves the sign without appealing to a convention for a "quadratic part." If $q_U^{(2)}$ is the homogeneous quadratic polynomial, then $D^2 q_U(0)[a,a] = 2q_U^{(2)}(a)$.

Let $\mathscr{A}_U$ denote the raw Wilson Hessian, and let J be a raw fine covector in this paragraph. The explicitly defined auxiliary form and its physical value are

$$\begin{aligned} K_U^{\mathrm{aux}}[J](u,v) &= \langle C_U u, \mathscr{A}_U C_U v\rangle + G_U^{\mathrm{abs}}(u,v) - \langle h_U^{\mathsf{T}} J, D^2 q_U[C_U u, C_U v]\rangle, \\ K_U^{\mathrm{phys}} &= K_U^{\mathrm{aux}}[\jmath_A(U)]. \end{aligned} \tag{3.cciv}$$

Here G_U^{abs} is the Hessian of the absolute contour gauge composed with the chart; pivot avoidance also identifies it with $C_U^{\mathsf{T}}(\mathrm{Hess}\,\mathrm{GF})C_U$. The equality is the ordinary chain rule, since the pivots occur in no internal contour path and hence $h_U^{\mathsf{T}}\jmath_{\mathrm{GF}} = 0$. It does not require the starting field to be a critical point. Criticality is used for the stationary Gaussian interpretation. The proved small-curvature physical gap also applies to the noncritical starting face lift used to construct the first-step background.

For clarity the two contour functionals are

$$\begin{aligned} \mathrm{GF}(V) &= \sum_{y,x\neq y}\left(1 - \tfrac{1}{2}\,\mathrm{tr}\,T_{y,x}(V)\right), \\ \mathrm{GF}_{\mathrm{rel},U}(a) &= \sum_{y,x\neq y}\left(1 - \tfrac{1}{2}\,\mathrm{tr}[T_{y,x}(e^{ia}U)T_{y,x}(U)^{-1}]\right). \end{aligned} \tag{3.ccv}$$

The second has zero first derivative at every U. If $G_U a$ is the first derivative of the relative contour logarithm, its Hessian is $G_U^{\mathsf{T}} G_U$, with the declared color pairings. This follows by expanding $1 - \frac{1}{2}\,\mathrm{tr}\,e^{iX} = \frac{1}{2}|X|^2 + O(|X|^4)$. The first functional has that Hessian only on the axial locus

$T_{y,x}(U) = 1$. For example the second derivative of $1 - \frac{1}{2}\operatorname{tr}(e^{it\sigma_3}e^{i\vartheta\sigma_3})$ is $\cos\vartheta$, whereas the relative expression has second derivative one. Absolute gauge is invariant under block-constant gauges. The relative function is jointly invariant under arbitrary site gauges: the product of its two contour factors transforms by conjugation at y. These are different covariance statements.

The gauge-restoration term in a named propagator requires an additional check. Let Q' be the assigned average of site functions, $\Delta_{\text{site}} = D^{\mathsf{T}}D$, and $G' = (\Delta_{\text{site}} + Q'^{\mathsf{T}}aQ')^{-1}$, with the specified positive site weight a. The real gauge projection R projects onto $\Delta_{\text{site}}\ker Q'$; on the admitted complex domain the same operators use their bilinear analytic formulas. Their existence on that domain is part of the primitive operator requirement. Put $S = 1 - DG'RD^{\mathsf{T}}$. On physical currents its projected Hessian is expressed by

$$\Delta_\pi = S^{\mathsf{T}}\Delta^{\mathrm{W}}S, \qquad \Delta'_\pi = \Delta^{\mathrm{W}} - \Delta_\pi. \tag{3.ccvi}$$

Thus $R = 0$ makes this physical correction zero. For an independent-current completion $\Delta'_\pi[J]$, it is necessary to inspect its actual formula. If it is obtained by expanding eq. 3.ccvi and extending its current factors without removing the factors of R, then it is zero when $R = 0$ for every J: every term of $\Delta^{\mathrm{W}} - (1-T)^{\mathsf{T}}\Delta^{\mathrm{W}}(1-T)$, $T = DG'RD^{\mathsf{T}}$, contains a T. This is a sufficient algebraic criterion, not a conclusion from agreement on the physical graph alone.

Normalize every quadratic matrix by the exponent $-\frac{1}{2}\langle u, Ku\rangle$. At the first step, where the zero-fold average is the identity, the difference between a named completion and eq. 3.cciv has the complete form

$$\begin{aligned}\mathcal{D}_U[J] &= C_U^{\mathsf{T}}(\mathscr{G}_U^{\text{named}} - \mathscr{G}_U^{\text{abs}} - \Delta'_\pi[J])C_U \\ &\quad - C_U^{\mathsf{T}}E_U^{(2)}[J]C_U + \mathcal{E}_U[J], \\ \mathcal{E}_U[J](u,v) &= \langle h_U^{\mathsf{T}}J, D^2q_U[C_Uu, C_Uv]\rangle.\end{aligned} \tag{3.ccvii}$$

The notation for the current readout inside $E^{(2)}$ must first be converted by eq. 3.cxc–eq. 3.cxci. The formula in eq. 3.ccvii is obtained by subtracting the two quadratic forms term by term. Matching just its last line is the constraint-readout condition; it is not the complete comparison. In particular a relative gauge prescription at a seam-free stage must be reconciled with the named matrix used at that stage.

3.9.3.1. Domain estimate I: the named completion. For the actual regular pair on each required region, compute all terms of eq. 3.ccvii with one declared pairing and Hessian normalization. Prove either $\mathcal{D}_U[J] = 0$, or an explicit local coefficient estimate small enough for the available gap. The quantifiers include independent complex J on its own radius, the chosen gauge representatives and all admitted local restrictions. The physical identity in eq. 3.cciv and the first-step physical gap already hold independently of this estimate. The calculation at zero inner depth can be done directly.

3.9.3.2. The named form at the first step. Take the unweighted bare torus with no lower layers, and use bare units. In the lower sequence all bond and site rows are the zero-fold rows $Q = 1$ and $Q' = 1$. The Landau projection R is zero. In the definition of the gauge-restoration term, the gauge read is $\mathsf{V}A = G'RD^{\mathsf{T}}A$. Its vanishing gives $\Delta'_\pi[J] = 0$ directly from the defining quadratic form, for independent J as well as for the physical current. The lower straightened constraint $\mathcal{Q}_0(U;A) = A$ is linear. Its quadratic correction $\Delta^{(2)}$ is therefore zero. With the bare row weight one, the ordered inverse formulas give

$$\mathsf{G}_1^{-1} = \mathscr{A}_U + 1, \qquad \mathsf{H}_1 = 1, \qquad (Q\mathsf{G}_1Q^{\mathsf{T}})^{-1} - 1 = \mathscr{A}_U.$$

These identities are on the domain of the named inverses.

The outer row is still the genuine one-step average. The source's per-length derivative is $L^{-1}Dq_U(0)$, so at a pivot its coefficient is L^{-1} times that of the raw derivative. The pivot operator with coefficient $L^{-1}\kappa_U(c)^{-1}$ is exactly h_U in raw constraint coordinates. The polarized

quadratic coefficient of the raw logarithm is $\frac{1}{2}D^2q_U(0)$. In the bare pairing the lower current read $\mathsf{H}_1^{\mathsf{T}}J$ is J itself, restricted to the bonds of the next step. The named correction pairs twice that polarized coefficient, after h_U, with this current. Consequently

$$\langle C_U u, \mathsf{E}^{(2)}[\mathsf{H}_1^{\mathsf{T}}J]C_U v\rangle = \langle h_U^{\mathsf{T}}J, D^2q_U(0)[C_U u, C_U v]\rangle.$$

The named gauge term is the relative contour Hessian in eq. 3.ccv. On the axial locus the base contour values equal one, so the relative and absolute functionals agree there as functions of the perturbation. With the common free chart, all terms of the named form therefore agree with eq. 3.cciv on this first-step locus for independent admitted currents. Away from that locus the calculation gives the named relative completion, with the absolute-to-relative difference explicitly displayed in eq. 3.ccvii.

3.9.4. *Triples, simple bounds and hereditary cores.* Write $\eta_j = L^{-j}$ in level-j units. A core is initially a domain of triples $(U, A', \mathbf{J})$, with U real and $\mathbf{U} = e^{i\eta_j A'}U$. For fixed U, analyticity means holomorphy in $(A', \mathbf{J})$. On a single layer its simple conditions at radii (b_0, b_1) are

$$\begin{gathered}|\partial U - 1|,\ |\partial \mathbf{U} - 1| < b_0\eta_j^2, \qquad |A'|,\ |\nabla_U A'| < b_1, \qquad |\mathbf{J}| < b_0,\\ U = e^{i\eta_j A} \text{ in real gauges on the specified } O(LM) \text{ cubes}, \qquad |A|, |\nabla A| < C_g M b_0.\end{gathered} \tag{3.ccviii}$$

A convention with LM instead of M in the last constant is absorbed in $C_g(L)$, not in a level-dependent radius. On a layered domain the inequalities are written in the dimensionless quantities eq. 3.cxciii, with the assigned radii of that layer.

For every subdomain Z of level $p \le j$, form each of the two assigned eroded towers. The first uses its prescribed cube size and outer-unit gap, the second its prescribed minimal determining size and inner-unit gap. If $\mathbf{B}'$ is the determining set of a truncated tower, its configuration is the background continued from the real datum, evaluated at

$$B_Z = \frac{1}{i}\log\left[M_{\mathbf{B}'}(e^{i\eta_j A'}U)M_{\mathbf{B}'}(U)^{-1}\right]. \tag{3.ccix}$$

This definition keeps the real base U. Existence of a continuation from one base does not by itself identify it with a continuation from another base representing the same complex pair.

Let $Z^{\ominus 2}$ mean removal of the prescribed two cube layers. The hereditary clauses, for every truncation $n \le p$ of both rules, are existence and the bounds

$$\begin{aligned}&|\partial U_n(Z;\mathbf{U}) - 1| < b_0 L^{-(j-p)}\eta_p^2,\\ &|J_A(U_n(Z;\mathbf{U}))| < b_0 L^{-(j-p)}L^{2(n-p)} \qquad \text{on } Z^{\ominus 2}.\end{aligned} \tag{3.ccx}$$

The current in the second line is formed in level-n units. The same clauses are imposed on the real factor U. This current is not the independent argument $\mathbf{J}$ of an activity. The two kinds of current enter different tests.

We denote the resulting hereditary core by $\mathcal{C}_j(X; b_0, b_1)$. A local function descends from triples to pairs only when its actual definition gives the same value on two admitted splittings of that pair. In collar expressions this requirement may belong to a specified difference of terms rather than each term. Restriction to a smaller set is permitted only with its required collars and determining neighborhoods. In particular it is not an operation that discards some of the quantifiers in eq. 3.ccx.

The linear regular-pair class has a related but different convention. On a layer of scale ξ it asks for curvature $\alpha_0\eta^2\xi^{-2}$, imaginary potential and first difference $\alpha_1\xi^{-1}, \alpha_1\xi^{-2}$, independent current $\gamma_0\xi^{-3}$, and real gauges on $O(M_1\xi)$ cubes with $\xi|A|, \xi^2|\nabla A| \le \alpha_0$. If those cubes fit into the core's gauge cubes, the simple part of eq. 3.ccviii enters this linear class whenever

$$b_0 \le \min\{\alpha_0, \gamma_0\}, \qquad b_1 \le \alpha_1, \qquad C_g M b_0 \le \alpha_0. \tag{3.ccxi}$$

These inequalities follow by substitution in the four tests. The last one is essential: the linear class's gauge bound is α_0 on the entire cube, not $M\alpha_0$. The required containment of gauge cubes, and

their nonwrapping torus lift, are geometric conditions that must be checked before this inclusion is used.

3.9.5. *Quantitative room inside a core.* Here is a precise meaning of spending a fraction of a core. Divide each left side of eq. 3.ccviii and eq. 3.ccx by its stated radius and scale factor. Include the corresponding real-gauge bounds and the datum-radius tests for each atomic continuation used in eq. 3.ccix. Call the resulting nonnegative tests N_β. A central core of fraction ϑ means $N_\beta \le \vartheta < 1$ for every test, on the same declared atomic domains. At fixed finite world there are finitely many tests, although their number need not have a useful bound independent of depth.

Margin lemma. Suppose a proposed substitution p_z has all its atomic maps defined on a joint parameter domain and

$$N_\beta(p_0) \le \vartheta, \qquad |N_\beta(p_z) - N_\beta(p_0)| \le \epsilon \quad \text{for every } \beta, \qquad \vartheta + \epsilon < 1. \tag{3.ccxii}$$

Then the substitution belongs to the core throughout that domain. This is immediate for each test, and the same bound holds after taking their supremum. No factor counting the tests occurs. The hard part of an application is proving the uniform test difference, including every normalized tower test. Simple-condition bounds alone are not that proof.

Near the identity a useful inclusion follows from a precise tower estimate. Put $\|v\|_1 = \max(\|A'\|_\infty, \|\nabla A'\|_\infty, \|\mathbf{J}\|_\infty)$. The plaquette expansion gives

$$|\partial e^{i\eta A'} - 1| \le C\eta^2(\|\nabla A'\|_\infty + \|A'\|_\infty^2). \tag{3.ccxiii}$$

To prove it, the terms linear in the four exponentials add to the discrete curl times $i\eta^2$. The remaining series has at least two factors $\eta A'$ and is bounded by $C\eta^2\|A'\|_\infty^2$ on a fixed small chart. This proof is uniform as $\eta \downarrow 0$. A supremum bound on A' without its difference quotient would instead allow a linear term of order $\eta\|A'\|_\infty$.

3.9.5.1. Domain estimate II: uniform atomic towers. Cor. 3.130 proves that simple bounds at sufficiently small radii (e_0, e_1) give every continued tower in eq. 3.ccix, for both erosion rules and every truncation. The tests in eq. 3.ccx, divided by their displayed length and scale factors, are at most $K_T(e_0 + e_1^2)$. The constant is independent of j, p, n, Z, compatible volume and the real base. The common complex ball and the physical-current estimate are proved in Lem. 3.127 and Prop. 3.129.

Consequently there is $c > 0$ such that

$$\|v\|_1 < c\min\{b_0, b_1, 1\} \quad \Longrightarrow \quad (1, A', \mathbf{J}) \in \mathcal{C}_j(X; b_0, b_1). \tag{3.ccxiv}$$

For a proof choose $e_0 = \min\{b_0/(4K_T), a_T/4\}$ and $e_1 = \min\{b_1/2, \sqrt{b_0/(4K_T)}, a_T/4\}$. Then $e_0 + e_1 < a_T$, as required by Cor. 3.130. The right side of eq. 3.ccxiii divided by η^2, as well as $|\mathbf{J}|$, is less than e_0 when c is small. The potential tests are less than e_1. The real field is the identity, so its gauge tests vanish. Finally $K_T(e_0 + e_1^2) \le b_0/2$, which proves every tower clause. This argument also proves a fixed central-core margin by reducing c.

If a continued background pair $p(B)$, on the physical datum ball $\|B\|_\infty < r_B$, satisfies $\|p(B) - (1,0)\|_1 \le C_B\|B\|_\infty$, eq. 3.ccxiv proves admission on

$$\|B\|_\infty < \min\{r_B/2,\ c\min(b_0, b_1, 1)/(2C_B)\}. \tag{3.ccxv}$$

The radius is in the physical datum. In a Gaussian datum it is divided by g. Thus the background estimate and the tower estimate supply a near-zero ball directly; a further radius loss is not inferred just because the physical radius runs with the coupling.

For two local holomorphic representatives, equality on a real open subset of a common connected datum domain proves equality throughout that domain. One proof expands at a real interior point: every real partial derivative of their difference vanishes, hence every complex Taylor coefficient vanishes. Analytic continuation along overlapping balls in the connected domain finishes the proof. This applies to the first-step background's admitted common complex charts. It applies to functions of independent (A', J) only when the real agreement is open in both variables. Equality just at

$J = J_A(U)$ is not such an agreement; the holomorphic function $J - J_A(U)$ already demonstrates the distinction.

3.9.6. *Absolute and running systems of radii.* Set $f_q(t) = t^{-1/2}(\log t)^q$. The two running field radii have the form $\alpha_{i,j} = C_i f_q(\theta_j)$, $i = 0, 1$, while $\delta_j = C_\delta \theta_j^{-1/2}(\log \theta_j)^{p_0}$, with $q > p_0$. The constants C_i are fixed after the geometric parameters. They are not changed when a complex source is introduced. For a step of depth k, a system $\mathfrak{a}$ is admissible when

$$\begin{gathered} C_s M a_{0,j} < a_{1,j}, \qquad a_{1,j}^2 \le C_s a_{0,j}, \\ a_{i,j} \le \bar a_i := (1 + c_o) a_{i,k+1} \le C_o \sqrt{k + 2 - j}\, a_{i,j}, \\ s_{\mathfrak{a}} := \delta_k / a_{0,k+1} \le s_*, \qquad \bar a_0 + \bar a_1 \le a_*. \end{gathered} \tag{3.ccxvi}$$

The first two conditions govern the shape of the core; the next line compares every input radius with the output radius. The last line gives fluctuation room and an absolute ceiling. An output radius larger than an input radius is permitted; its ratio must be recorded in estimates.

The absolute system has entries $a_{i,j} = \alpha_i$ for $j \le k$ and $a_{i,k+1} = \alpha_i/(1 + c_o)$. Its output radii are $\bar a_i = \alpha_i$. The running system has

$$a_{i,j} = \min\{\alpha_{i,j-1}, \alpha_{i,j}\} \quad (1 \le j \le k), \qquad a_{i,k+1} = \alpha_{i,k}, \qquad \varrho_j = \max_i \frac{\bar a_i}{a_{i,j}}. \tag{3.ccxvii}$$

If a level-zero entry is needed, set $a_{i,0} = \alpha_{i,0}$; there is no negative-index radius. For $k = 0$ the old-member list is empty and its output entry is $a_{i,1} = \alpha_{i,0}$. The adjacent minimum admits the same old member at the two adjacent steps. It is not replaced by a radius chosen after that member is read.

Radii lemma. Suppose the real frame satisfies

$$\theta_j \ge T, \qquad -c_5 \le \theta_i - \theta_j \le B(j - i) \quad (i \le j). \tag{3.ccxviii}$$

Choose $C_1 > C_s M C_0$. For sufficiently large T, the absolute system with sufficiently small fixed α_i, and the running system below it, satisfy eq. 3.ccxvi. Moreover

$$\varrho_j \le C\sqrt{k + 2 - j}, \qquad s_{\mathfrak{a}} \le C(\log \theta_k)^{p_0 - q}. \tag{3.ccxix}$$

For the proof, $(\log f_q)'(t) = (-\frac{1}{2} + q/\log t)/t$, so f_q is decreasing above e^{2q}. Its logarithmic derivative has absolute value at most $1/t$ there. Thus bounded displacements of t change $f_q(t)$ by $1 + O(T^{-1})$, uniformly for $t \ge T$. The lower inequality in eq. 3.ccxviii then proves $a_{i,j} \le (1 + c_o)\alpha_{i,k}$, after increasing T. The adjacent minimum is at least $C_i f_q(\theta_j + B)$. For $u \ge \theta_k$, $f_q(\theta_k)/f_q(u) \le \sqrt{u/\theta_k}$; for $\theta_k - c_5 \le u < \theta_k$ its ratio is uniformly bounded. Since $\theta_j + B \le \theta_k + B(k + 1 - j)$, this proves the first bound in eq. 3.ccxix.

All entries of the running system are a common scalar times C_i. Its first shape condition follows from the choice of C_1/C_0. Its second follows from $a_{1,j}^2/a_{0,j} = (C_1^2/C_0)\min f_q(\theta) \to 0$ uniformly as $T \to \infty$. The remaining ceiling follows the same way, and division of δ_k by $\alpha_{0,k}$ proves the second bound in eq. 3.ccxix. For the absolute system choose the fixed α_i with the same shape, then increase T until the running system lies below it and $\delta_k/a_{0,k+1}$ is small. Every choice is independent of k.

There are two useful radius ratios for an old member. Its general unmarked bound may use $\max\{1, \delta_k/\alpha_j\}$. A homogeneous near-field Taylor bound uses the actual ratio $\rho_j = \delta_k/\alpha_j$, which can be less than one. With $a = k - j$, $\sigma_a = L^{-a}$, the frame bounds imply

$$\frac{C_{\mathrm{ax}} \delta_k \sigma_a}{r_{B,j}} \le C s_{\mathfrak{a}} \sqrt{a + 2}\, L^{-a}, \qquad \text{provided } \alpha_j / r_{B,j} \le C. \tag{3.ccxx}$$

The supremum of $\sqrt{a + 2}L^{-a}$ is finite: it is bounded by $\sum_{a \ge 0}(a + 2)L^{-a}$. Consequently a single small choice of s_* puts the central axial rows in a half datum disc at every age. The remote rows have no such age gain. Their admission uses the fixed absolute whole-datum ceiling eq. 4.xxx for

the actual window maps. For a different datum chart this inclusion must be verified separately; the displayed radius ratio does not imply it.

The corresponding quadratic extraction weights obey the exact identity

$$w_{n,j} = (\alpha_n/\alpha_j)^2, \qquad \alpha_n^{-2} w_{n,j} = \alpha_j^{-2}. \tag{3.ccxxi}$$

This identifies where the physical readout is paid. For example, if $r_j \le C\theta_j^{-p}$, $p > 2$, and the stronger frame bound $\theta_j \ge \theta_n + \lambda(n-j) - c_5$ holds, then

$$\alpha_n^{-2} \sum_{j \le n} w_{n,j} r_j \le C\theta_n^{2-p} (\log \theta_n)^{-2q}. \tag{3.ccxxii}$$

Indeed the left side is bounded by $C \sum_{a \ge 0} (\theta_n + ca)^{1-p} (\log(\theta_n + ca))^{-2q}$, with bounded changes of the lower floor absorbed in C. Pull out $(\log \theta_n)^{-2q}$ and compare the decreasing power with its integral. The integral is a constant times θ_n^{2-p}. Thus summing ages and extracting a quadratic coefficient are two specified operations, not two additional copies of the same radius loss.

3.9.7. *Admission on a joint parameter domain.* Several distinct parameters occur in a local expression. A fine decoupling parameter s_Δ varies on $|s_\Delta| < R$, $R = e^{\bar\kappa} > 1$. A multiplier ν amplifies an actual response. A reference ratio $\tau = x/\mathtt{t}$ lies in a fixed disc $|\tau - 1| < c_\tau$. Physical sources vary the exponent members on their ordinary source domain. An auxiliary potential parameter can have a larger domain. The real frame θ, numerical cutoffs, Gaussian scaling, reference operations and geometry are frozen before any of these source derivatives are taken.

One cannot choose these domains independently by checking one real parameter at a time. A useful exact calculus is the following. Consider a finite ordered collection of atomic holomorphic maps F_v, each already defined on the ball $\|d - d_v^0\| < r_v$. The input d_v of a later map is a specified holomorphic expression in the external parameters and earlier outputs. Suppose normalized displacements satisfy majorants

$$u_v \le e_v + \sum_{w<v} A_{vw} u_w, \qquad A_{vw} \ge 0, \qquad u_v = \|d_v - d_v^0\|/r_v. \tag{3.ccxxiii}$$

These inequalities must be verified on the proposed joint domain, using primitive derivative estimates inside the atomic balls. If the recursively computed right sides are at most $\vartheta < 1$, then every composition is holomorphic there. This follows by induction: the previous compositions exist, their values put the next input in its already assigned ball, and composition of holomorphic maps is holomorphic. There is no step of continuing an unknown map past its domain.

Writing A for the strictly lower triangular matrix, the exact finite bound is

$$u \le (1 + A + \cdots + A^{N-1}) e. \tag{3.ccxxiv}$$

Each matrix product counts ordered paths through the dependency graph. If its row norm is at most $\zeta < 1$, then $\sup u_v \le \sup e_v/(1-\zeta)$, independently of the number of maps. Alternatively actual scale weights can bound the path sum even when the unweighted row norm is not less than one. This bound must be proved. For example the scalar recursion $d_v = d_{v-1} + z$ has input Nz at its last node; an individual bound $|z| < r$ supplies no depth-independent admission disc.

For a localized expression at old level $j \le k$, the actual substitution is of the form

$$p_{j,Y}(s, \nu; B) = \rho_X p \mathbin{\dot+} \nu\, \mathfrak{h}_Y(s; B), \qquad |s_\Delta| < e^{\bar\kappa}. \tag{3.ccxxv}$$

The operation $\dot+$ changes both the potential and current by the specified nonlinear local response; ρ_X is the restriction appropriate to that term. It is not addition only to its potential component. For an inner window the background is first rewritten on the assigned box. Its datum is the actual nonlinear average

$$M_{\mathbf{B}_j}\left(e^{i\eta_k(t + t_\square \zeta_\square)\mathbf{H}_k(B')} U_+\right), \qquad 0 \le t \le 1, \tag{3.ccxxvi}$$

with its declared continuation in $t_{\square}$. The background of this datum is inserted before the bracket interpolation. This is where the pointed marginal cancellation acts. Its datum is not obtained by replacing $M_{\mathbf{B}_j}$ by its linearization.

3.9.7.1. Domain estimate III: the full local substitution. For each inner, outer and removed origin, each allowed localization domain Y, all its restricted towers and the entire joint polydisc in eq. 3.ccxxv, prove admission to the core of the term being evaluated. In the normalized tests of eq. 3.ccxii, a sufficient quantitative bound is

$$\sup_{\beta} |N_{\beta}(p_{j,Y}(s,\nu;B)) - N_{\beta}(\rho_X p)| \le C_R e^{C_R\bar{\kappa}} |\nu|\, \delta_k L^{-(k-j)}/a_{0,j}. \tag{3.ccxxvii}$$

The domain includes every admitted external complex pair, all real free fields on their prescribed window box and every auxiliary parameter of the expression. Atomic datum distances must satisfy their own radii as in eq. 3.ccxxiii. If the actual bound has a further geometric loss, that loss must be included in the right side before choosing the parameters.

The numerical consequence of eq. 3.ccxxvii is complete. On a central core of fraction ϑ, choose

$$\rho_{j,k} = c_R e^{-C_R\bar{\kappa}} \frac{a_{0,j}}{\delta_k L^{-(k-j)}}. \tag{3.ccxxviii}$$

For $|\nu| < \rho_{j,k}$, a sufficiently small c_R spends less than half the initial margin. eq. 3.ccxvii–eq. 3.ccxix give

$$\rho_{j,k}^{-1} \le C e^{C_R\bar{\kappa}} s_{\mathfrak{a}} \sqrt{k+2-j}\, L^{-(k-j)}. \tag{3.ccxxix}$$

Hence one small, depth-independent choice of s_* makes every $\rho_{j,k} > 2$, so $\nu = 1$ is an interior point. The gain $\rho_{j,k}^{-1}$ is obtained by a Schwarz estimate after admission; it is not a device to create admission. The scale factor in eq. 3.ccxxviii is what prevents a bound growing with the number of old levels.

3.9.8. *Cauchy estimates and the count of parameters.* Let F be holomorphic and bounded by M_F on a supremum ball of radius r. For two arbitrary directions u, v, apply Cauchy to $F(zu/\|u\|_\infty + wv/\|v\|_\infty)$ on $|z|, |w| < r/3$. It proves

$$|D^2F(0)[u,v]| \le 9M_F r^{-2} \|u\|_\infty \|v\|_\infty. \tag{3.ccxxx}$$

The same proof on a smaller centered ball works uniformly away from the origin. It has no dimension factor. If instead only entrywise Hessian bounds $|\partial_i\partial_j F| \le M_2$ are known for m coordinates, the elementary bound on the Euclidean operator norm is mM_2. A better bound needs information about the full scalar supremum or the sum of the local entries. In a sum of local energies, the global scalar supremum usually grows with volume; eq. 3.ccxxx does not turn that extensive bound into a volume-independent local one.

For independent decoupling coordinates the calculation is different. If $F(s,\nu)$ is bounded by M_F on $|s_\Delta| < R$, $|\nu| < \rho$ with $\rho > 1$, and $F(s,0)$ is subtracted, Schwarz followed by coordinate Cauchy gives

$$\left| \int_{[0,1]^K} \partial_K [F(s+1_N,1) - F(s+1_N,0)]\, ds \right| \le \frac{2M_F}{\rho} (R-1)^{-|K|}. \tag{3.ccxxxi}$$

Here K, N are disjoint sets of coordinates and ∂_K differentiates each coordinate in K once. A circle of radius arbitrarily close to $R-1$ fits at every real center in $[0,1]$. Successive one-variable Cauchy formulae prove eq. 3.ccxxxi, and the limit in the radii is harmless. There is no $|K|!$, because no coordinate is differentiated repeatedly. By contrast, m derivatives in one coordinate produce $m! r^{-m}$. With $R = e^{\bar{\kappa}}$ and $\bar{\kappa} \ge 1$,

$$(R-1)^{-|K|} \le e^{-(\bar{\kappa}-1)|K|}. \tag{3.ccxxxii}$$

This follows from $1 - e^{-\bar{\kappa}} \ge e^{-1}$.

If local domains of tree length between m and $m+1$, rooted at a fixed cube, number at most $C_D e^{\kappa_D m}$, then every fixed polynomial multiplicity is summable with a strict rate surplus:

$$\sum_{m\ge 0}(m+1)^A e^{-(\kappa-\kappa_D)m}<\infty \qquad (\kappa>\kappa_D). \tag{3.ccxxxiii}$$

For completeness, put $h=(\kappa-\kappa_D)/2$. The continuous function $(t+1)^A e^{-ht}$ is bounded on $[0,\infty)$, so the series is bounded by a constant times $\sum e^{-hm}$. Using eq. 3.ccxxxii in a rooted bracket expansion requires the geometric comparison between its number of parameter cubes and its output tree length. It also requires the actual rooting identity: a component disjoint from the term's base contributes zero. Analyticity by itself does not give this identity.

For a family with potential norm ε, one spare finite potential slot of size w_* admits an auxiliary parameter on $|z|<w_*/\varepsilon$. If the uninserted local logarithm has norm at most C_* on that slot, Cauchy proves

$$\|DE[V]v\|_{\mathrm{out}}\le (2C_*/w_*)\|v\|_{\mathrm{pot}}. \tag{3.ccxxxiv}$$

There is no dependence on the number of field coordinates or the total volume in this estimate when the local majorant has no such dependence. For two directions of norms $\varepsilon_1,\varepsilon_2$, use $|z_i|<w_*/(4\varepsilon_i)$. The sum of the two potentials fits in half the slot, and

$$\|D^2E[V](v_1,v_2)\|_{\mathrm{out}}\le 16C_*w_*^{-2}\varepsilon_1\varepsilon_2. \tag{3.ccxxxv}$$

For n unrelated simultaneous directions the analogous sum-budget disc has radius proportional to $w_*/(n\varepsilon_i)$ in each coordinate. Reusing the two-direction radii for arbitrary n would exceed the slot. This count concerns potential allowance, not the distinct-cube calculation in eq. 3.ccxxxi.

The auxiliary slot need not be an enlarged physical source domain for histories. Suppose the actual source curve has $V(w)=V(0)+wv+O(w^2)$ in potential norm on its ordinary fixed disc. The norm Taylor expansion of the holomorphic local map gives $\partial_w E(V(w))|_0=DE[V(0)]v$. Its quadratic remainder is bounded by $C_*\|V(w)-V(0)\|_{\mathrm{pot}}^2/w_*^2$, by one-variable Cauchy on the line through that difference. Thus the large auxiliary disc computes the actual first jet, even if the nonlinear physical curve is defined only on its smaller source disc.

3.9.9. *Support, Schur norms, and finite coordinate counts.* For a matrix of color blocks, normalize the input and output coordinates by their declared positive weights. Write $B=D_oAD_i^{-1}$ for the normalized matrix and put

$$R_B=\sup_i\sum_j\|B_{ij}\|, \qquad C_B=\sup_j\sum_i\|B_{ij}\|. \tag{3.ccxxxvi}$$

Then $\|B\|_{2\to2}\le\sqrt{R_BC_B}$. To prove this, set $b_{ij}=\|B_{ij}\|$. For finitely supported vectors, $\|(Bx)_i\|^2\le(\sum_j b_{ij})(\sum_j b_{ij}\|x_j\|^2)$. Sum over i, use the column bound, and then complete. This also proves the assertion on a countable lattice. A row estimate alone is not the required Schur estimate. For example the matrix with a single nonzero column containing m ones has row sum one and norm $\sqrt{m}$.

Suppose one scalar local energy has modulus at most M_0 on a joint coordinate domain with free radius h_u and background radius h_b. Suppose at most T such energies contribute to any fixed row and at most N columns couple to that row, with the same bounds for columns. Cauchy, first in two free directions and then in a background direction, proves

$$\|K\|_{\mathrm{Schur}}\le CNTM_0h_u^{-2}, \qquad \|K(U')-K(U)\|_{\mathrm{Schur}}\le CNTM_0h_u^{-2}h_b^{-1}\|U'-U\|. \tag{3.ccxxxvii}$$

The backgrounds lie in a smaller common analytic chart, and the segment between their coordinate values stays there. The first bound counts the terms in an entry, then entries in a row. The second integrates the third derivative along that segment. This proof specifies where coordinate counts enter; they are not powers of the number of parameters in eq. 3.ccxxxi.

For the actual first-step chart, the verified finite geometry gives

$$h_u = cL^{-9}, \quad h_b = c'L^{-5}, \quad T, N \le CL^4. \tag{3.ccxxxviii}$$

A free coordinate changes its own bond and a fixed number of nearby crossing pivots. Wilson terms have fixed plaquette incidence, and a gauge block contains at most $L^4 - 1$ internal contour rows. Thus the number of scalar terms affecting one row and the number of coupled free coordinates are each CL^4, not their product before taking the row sum. With $M_0 \le C$, eq. 3.ccxxxvii gives CL^{26} and $CL^{31}\|U' - U\|$. Also $\|\Phi_U(u)\|_\infty \le CL^4\|u\|_\infty$ on its free ball implies, by two-direction Cauchy, $\|D^2\Phi_U(0)[e_i, e_j]\| \le CL^{13}$. Only a fixed number of pivot outputs occur for a given row. After the CL^4 column count, the independent-current part is bounded by $CL^{17}\|J\|_\infty$. These are local first-step estimates. They provide no hidden bound on a many-level window chart with a number of coordinates growing with its side.

The actual constraint basis must also be used when localizing fields. For a free-coordinate cutoff χ, the tangent field is $C_U(\chi u)$. It need not equal $\chi C_U u$, since pivot values depend on neighboring free coordinates. A quadratic partition argument on free variables therefore produces the commutator of the actual compressed free-coordinate matrix. It is not obtained by clipping the fine constraint rows. If $H_0 \ge c_0$ and a derived local coefficient estimate gives $\|H - H_0\| \le \varepsilon < c_0$, then

$$\mathrm{Re}(u, Hu)_{\mathrm{H}} \ge (c_0 - \varepsilon)\|u\|^2, \qquad \|H^{-1}\| \le (c_0 - \varepsilon)^{-1}. \tag{3.ccxxxix}$$

The inverse bound follows from the Neumann series with H_0^{-1}. A compression of this same matrix inherits the form inequality. An operator rebuilt with different block means or boundary constraints requires its own coefficient comparison; it is not that compression.

3.9.9.1. Domain estimate IV: coordinate counts across scales. On the actual run window of side parameter r, determine its free coordinate set, its constraint inverse and its joint background domain. Prove explicitly the bounds replacing h_u, h_b, T, N in eq. 3.ccxxxvii, including their dependence on r, L, M, the region and the local layer. Polynomial losses in r are allowed. The normalization matrices D_i, D_o, the compatible boundary rows and the nonwrapping lift are part of this estimate. The finite one-step constants in eq. 3.ccxxxviii are not evidence that these window constants are independent of r.

3.9.10. *Gauge changes, pivots and the real coarea factor.* Real unitary changes of local gauge preserve all the norm tests that are written covariantly. A complex change has a cost. If $K = \sup_x \|g(x)\|\|g(x)^{-1}\|$, then $\|\operatorname{Ad}(g(x))X\| \le K\|X\|$. For the actual average, $W_x(U^g) = g(c_-)W_x(U)g(c_-)^{-1}$. Its logarithm has the same covariance as long as both points belong to the regular analytic branch. The internal mean has the analogous bi-equivariance, since its defining logarithmic equation is conjugated at its initial endpoint. Therefore

$$\max_x \|W_x(U) - 1\| < r_{\log}/(2K) \quad \Longrightarrow \quad \max_x \|W_x(U^g) - 1\| < r_{\log}/2. \tag{3.ccxl}$$

This requires a smaller analytic domain; it does not imply that a complex gauge orbit saturates the original core. It also avoids multiplying a condition number separately at every bond of a closed path: covariance reduces that product to one conjugation. A real flat holonomy need not be close to one for this calculation, since the closed comparison transports are the arguments of the logarithms.

At a flat block-axial reference, the crossing pivot appears once in each of L longitudinal segments and in no internal contour path. Its raw constraint derivative is $L^{-3}\operatorname{Ad}(V_c)$, with the actual prefix transport V_c. No other pivot occurs in that constraint's support, under the nonwrapping double-block geometry. Thus

$$J_c := \partial_{a_{p(c)}} q_U(c) = L^{-3}\operatorname{Ad}(V_c), \qquad \det_{\mathbb{R}^3} J_c = L^{-9} \quad \text{at a real flat reference.} \tag{3.ccxli}$$

The determinant of a real adjoint transport is one: conjugation preserves the color metric and is connected to the identity in SU(2). For the per-length pivot matrix $\mathsf{m}_c = J_c/L$, $\det J_c = L^3 \det \mathsf{m}_c$. These two determinants cannot be interchanged in a coarea factor.

Let $\sigma_H(a)d^3a$ be Haar measure in its regular exponential chart. In the same chart $\delta_G(e^{iq}) = \delta_{\mathbb{R}^3}(q)/\sigma_H(0)$. Integrating one real pivot from a constrained integral therefore leaves

$$\frac{\sigma_H(a_{p(c)})}{\sigma_H(0)\,|\det \partial_{a_{p(c)}} q_U(c)|}. \tag{3.ccxlii}$$

At the flat reference this is L^9. The remaining free bonds still carry their Haar densities, and $a_b = g y_b$ contributes g^{3N_f} for N_f free bonds. The positivity and absolute determinant in eq. 3.ccxlii belong to real coarea. On a connected small real chart its oriented determinant stays positive. The analytic factor used later is its holomorphic determinant inverse, times the continued Haar density, on the specified complex chart where the determinant is nonzero. One does not analytically continue an absolute value.

3.9.11. *Finite potentials, cutoffs and renewed pointing.* The local potential norm used by the ordinary finite integral is

$$\|\mathbf{V}\|_\lambda = \sup_{Y,P,B} \frac{e^{\lambda d(Y)}|V(Y;P,B)|}{Q_Y(B)}, \qquad Q_Y(B) = 1 + M^{-4}\sum_{b\subset Y}|B_b|^2. \tag{3.ccxliii}$$

The supremum uses its specified complex local cores and field box. The coordinate B in this paragraph is the Gaussian variable of eq. 3.cxciv. It is neither the physical datum of eq. 3.cxcv nor the pivot basis C_U. An origin tag distinguishes several potential members with the same geometric carrier. The tag changes neither $d(Y)$ nor the geometric support, but its norm is charged in the potential allowance.

Use nested boxes $\mathcal{B}_0 \subset \mathcal{B}_1$ with bounds $\mathfrak{b}$ and $(1+\epsilon_b)\mathfrak{b}$, $\mathfrak{b} \geq 1$. Potentials to be differentiated are holomorphic on the larger box; the finite slot majorant accepts the resulting potentials on the smaller box. The real cutoff is kept on that smaller box. For $t \in [0,1]$, a coordinate Cauchy circle at tB of radius

$$r_{b,a} = c_b(1 + |B_b^a|), \qquad c_b < \epsilon_b/4, \tag{3.ccxliv}$$

fits in the larger box after reducing the fixed constant if needed. The Euclidean color norm differs from the coordinate supremum by a fixed factor. On that circle Q_Y is at most $C_b Q_Y(B)$, because the changed squared coordinate is at most $C(|B_b^a|^2 + 1)$. Consequently

$$|B_b^a \partial_{b,a} v(Y;tB)| \leq C_b \|\mathbf{v}\|_\lambda e^{-\lambda d(Y)} Q_Y(B). \tag{3.ccxlv}$$

This estimate is uniform in $\mathfrak{b}$. A fixed circle of radius proportional to $\mathfrak{b}$, estimated with the quadratic weight at its largest point, would give an unnecessary $\mathfrak{b}^2$ when the original coordinate is small. The formula in eq. 3.ccxliv retains the quadratic norm of the actual field.

An older beta-source member follows the path $v(tB)$. The intrinsic action and gauge-fixing remainders, which vanish through order two, instead have the normalization $t^{-2}V(tB)$. This distinction also has a direct analytic check. If $V(tB) = \sum_{m\geq 3} t^m V_m(B)$ is bounded on a fixed larger t-disc $|t| < 1 + \epsilon_t$, then

$$\int_0^1 \left| \frac{d}{dt}\{t^{-2}V(tB)\} \right| dt \leq \sum_{m\geq 3} |V_m(B)| \leq C_{\epsilon_t} \sup_{|z|<1+\epsilon_t} |V(zB)|. \tag{3.ccxlvi}$$

The first inequality integrates $(m-2)t^{m-3}$, whose integral is one. Cauchy bounds the final series geometrically. Applying t^{-2} to a general older source with a linear or quadratic part would not give this integrability. The actual bond-indexed split of the three interpolation legs must still be used: a bound on their total does not bound the integral of the absolute values of an arbitrarily chosen split.

3.9.11.1. Domain estimate V: the actual root slot. For each of the three assigned interpolation legs, prove the integrated potential norm of its actual bond-root density, on $\mathcal{B}_0$, uniformly in the external core, carrier and interpolation time. The larger box $\mathcal{B}_1$ must provide the coordinate circles in eq. 3.ccxliv. Intrinsic legs use their actual third-order cancellation; older source legs use eq. 3.ccxlv. This is the precise root input to renewed pointing, not a replacement of that pointing by an Euler split.

There is a simple exact interpretation of the connected term. On a complete physical pair let the fixed reference integral be

$$\Xi(z) = \int \chi(B) e^{V(B)+zv(B)}\, d\mu(B). \tag{3.ccxlvii}$$

The reference Gaussian and cutoffs have no z-derivative. Wherever the local expansion supplies its nonzero denominator,

$$\frac{d}{dz}\langle a_z\rangle_z\Big|_0 = \langle \dot a_0\rangle_0 + \langle a_0 v\rangle_0 - \langle a_0\rangle_0\langle v\rangle_0. \tag{3.ccxlviii}$$

This follows by differentiating numerator and denominator, with the same normalizer. The double-direction estimate in eq. 3.ccxxxv, union filing and the rooted cluster bound estimate this root-plus-source term on the actual local cores. On a general independent-current core the localized logarithmic terms are still defined and differentiated by their local finite formulas. They need not be marginals of one global Ξ; eq. 3.ccxlviii is the physical sum interpretation, not an off-graph identity inferred from holomorphy.

The number of scalar Gaussian coordinates in a local set is

$$N(Z) = 3\#\{\text{free bonds in } Z\}$$

and is of order $(LM)^4 N_c(Z)$ when N_c counts coarse cubes. Thus a Gaussian majorant $e^{CuN(Z)}$ equals a bound of the form $e^{C'(LM)^4 uN_c(Z)}$. The dimension factor is counted once. Neither holomorphy of an integrand nor smallness of u alone gives domination: the real reference precision, the fixed integration sets and the available Gaussian quadratic reserve must be specified.

3.9.12. *The real weight, the reference Gaussian and domination.* The reference inverse coupling θ and the real action weight x have different roles. Put $\tau = x/\theta$. In Gaussian coordinates the actual quadratic form consists of τ times the physical action Hessian and the gauge Hessian at the reference weight. Write these two free-coordinate matrices as K_A, K_G. Then

$$K_\tau = \tau K_A + K_G, \qquad K = K_A + K_G, \qquad e^{-\frac12 B^{\mathsf T} K_\tau B} = e^{-\frac\tau2 B^{\mathsf T} K B} e^{\frac{\tau-1}{2} B^{\mathsf T} K_G B}. \tag{3.ccxlix}$$

Thus using the normalized Gaussian with covariance $\tau^{-1}K^{-1}$ is a presentation choice with a compensating free potential. It does not multiply the actual gauge weight by τ. Its normalizing determinant contributes the factor $\tau^{-N/2}$ for N real scalar coordinates. The logarithm is chosen continuously from $\tau = 1$. Physical sources are not inserted in x in this presentation; their Wilson terms occur as potential members. Consequently neither this determinant nor g has a physical-source derivative.

Here is the finite domination calculation with all its premises. Let A_0 be a real positive precision on N coordinates, $A_0 \ge m_0$. Suppose the actual complex symmetric matrix satisfies

$$\|A_0^{-1/2}(A - A_0)A_0^{-1/2}\| \le \epsilon, \qquad \rho \ge 0, \qquad \epsilon + 2\rho/m_0 < \tfrac12. \tag{3.ccl}$$

If a measurable integrand has modulus at most $F_0 e^{\rho|B|^2}$, then the coherently normalized complex Gaussian gives

$$\left|\int F(B)\, d\mu_A(B)\right| \le F_0 \left(\frac{1+\epsilon}{1-\epsilon-2\rho/m_0}\right)^{N/2} \le F_0 e^{C(\epsilon+\rho/m_0)N}. \tag{3.ccli}$$

Indeed the determinant ratio $|\det A|/\det A_0$ is at most $(1+\epsilon)^N$, while $\operatorname{Re} A - 2\rho I \ge (1-\epsilon-2\rho/m_0)A_0$. Integrating that real Gaussian proves the first bound. The logarithms of numerator

and denominator prove the second. The same computation works with an integrand restricted to a fixed Borel set. On smaller parameter polydiscs its right side is an integrable bound independent of the parameters. If the densities and integrands are holomorphic there, dominated Cauchy formulae or Morera's theorem prove holomorphy of the integral.

The hypothesis on parameter-independent sets is substantive. A moving sharp cutoff is not covered by the preceding proof. It must either be frozen, as it is for the ordinary source construction, or be transformed by an explicitly controlled change of variables, including its boundary and Jacobian. Likewise holomorphy of a normalized integral uses the coherent determinant branch and its nonvanishing matrix domain.

There is an explicit coefficient route to eq. 3.ccl. If the actual entries of $A - A_0$ satisfy row and column sums at most ϵm_0, eq. 3.ccxxxvi gives eq. 3.ccl. For a ratio perturbation in eq. 3.ccxlix, the additional relative norm is at most

$$|\tau - 1|\,\|K^{-1/2} K_A K^{-1/2}\|. \tag{3.cclii}$$

At a flat reference $0 \le K_A \le K$, so this factor is at most $|\tau - 1|$. Away from that reference its bound follows from the actual local action coefficients and the proved gap, not from the name of the coupling ratio. Here K is the real reference precision; an additional complex-pair perturbation is tested by eq. 3.ccl. Thus the allowed τ-disc is chosen after these constants and the unused quadratic potential reserve.

3.9.12.1. Domain estimate VI: primitive Gaussian and potential room. For every local conditioned matrix used by the assigned decoupling, derive the normalized entry sums needed for eq. 3.ccl, on its enlarged parameter discs and its external core. Determine the scalar coordinate count, the reference precision and the total positive quadratic growth of the admitted potentials. Verify that the sum leaves the strict inequality in eq. 3.ccl, with the small per-cube cost required by the ordinary polymer criterion. This is one matrix/coefficient and potential-budget estimate. Once it holds, eq. 3.ccli, the spare-slot derivatives and domination of parameter integrals are conclusions. It is not necessary to identify all these local matrices with marginals of one Gaussian at a general independent-current pair.

3.9.13. *The site-frozen complex source box.* The ordinary complex data allow each site to carry its own Euclidean member. Such a member consists of a pointed local family, its infinite companion and its own scalar coefficient, with covariance of the pointed total under the required site symmetries. Members chosen at different sites need not be related. For a label $\mathcal{S}$, let $h_z^{(j)}$ be its fixed nonnegative tents, with $\sum_z h_z^{(j)} = 1$, and let $0 \le \varphi_j \le 1$ be the nested profiles. Write b_j for the real unsourced increments. Define

$$\begin{aligned} c_{\mathcal{S}}(p) &= u(p) + \sum_{j \le k} \varphi_j(p) \sum_z h_z^{(j)}(p)[e_j(z) - b_j], \\ c_{j,\mathcal{S}}(p) &= \varphi_j(p) \sum_z h_z^{(j)}(p) e_j(z), \\ x_k(p) &= \beta - \sum_{j \le k} b_j \varphi_j(p). \end{aligned} \tag{3.ccliii}$$

Each site uses the coefficient $e_j(z)$ of its own full Euclidean member, even though the exponent reads just that member's terms pointed at z. A defect does not inherit this symmetry by being placed at that site.

The exact cancellation of the scalar action weights is

$$-A(x_k - c_{\mathcal{S}}, U) - \sum_{j \le k} A(c_{j,\mathcal{S}}, U) = -A(\beta - u, U). \tag{3.ccliv}$$

Expand eq. 3.ccliii, use $\sum_z h_z^{(j)} = 1$, and use linearity of the Wilson action in its weight. This proves the identity label by label. It does not depend on the number of sites. It also shows why replacing an own counterterm by a coefficient from another member changes the presentation.

The scalar part of the box is measured by

$$\mathsf{u} = \|u\|_\infty + \sum_{j\le k} \sup_z |e_j(z) - b_j|. \qquad \|c_{\mathcal{S}}\|_\infty \le \mathsf{u}. \tag{3.cclv}$$

The inequality uses only the tents and $0 \le \varphi_j \le 1$. For ordered profiles, $x_k(p)$ is a convex combination of $\beta, x_1, \dots, x_k$: its coefficients are $1-\varphi_1, \varphi_1-\varphi_2, \dots, \varphi_k$. Thus on real field integrals $\mathrm{Re}(x_k(p) - c_{\mathcal{S}}(p)) \ge \min_i x_i - \mathsf{u}$. A fixed scalar box $\mathsf{u} \le \bar{c}$ preserves a positive Wilson coefficient when the real floor exceeds $\bar{c}$. This is a weight calculation; it is not a positivity assertion for arbitrary complex local activities.

Let $d_j(X)$ be the localization tree length and $d_j^\natural(X)$ the hull length with the specified standing islands exempted. The remaining coordinates of the ordinary box have sizes

$$\begin{aligned} \mathsf{g}_j &= \sup_{z_0,X,z,P} e^{\kappa d_j(X)} |G_{z_0}^{(j)}(X,P,z)|, \quad \mathsf{a}_j = \sup_{X,z,P} e^{\kappa d_j(X)} |D^{(j)}(X,P,z)|, \\ \mathsf{r}_j &= \sup_{X,P} e^{\kappa d_j(X)} |H^{(j)}(X,P)|, \qquad \mathsf{h} = \sup \frac{|B_{\mathcal{S}}^{(j)}(X)|}{\bar{B}_{\mathcal{S}}^{(j)}(X)}, \qquad \mathsf{f} = \sup_\omega \frac{|\Psi_\omega|}{n(\omega)}. \end{aligned} \tag{3.cclvi}$$

The external P ranges over the prescribed hereditary core. The boundary denominator is its actual unsourced majorant, proportional to $e^{-\kappa d_j^\natural}$. If $\Psi_\omega = \Psi_\omega^\circ + \sum_f \psi_f(P|_{R_f})$, a separate oscillation size is

$$\mathsf{f}_\Phi = \sup_{\omega,R} \frac{\sum_{f:R_f=R} \mathrm{osc}_P\, \psi_f}{(\log\theta_i)^{2p_0} e^{-\kappa d_i^\natural(R)}}. \tag{3.cclvii}$$

Amplitude and oscillation are different coordinates: a constant exponent has zero oscillation and may have arbitrarily large amplitude.

At input or output index $m \in \{k, k+1\}$, the actual box bounds are

$$\begin{gathered} \mathsf{u} \le \bar{c}, \quad \mathsf{g}_j \le E_0 + 1, \quad \mathsf{a}_j \le L^{-(m-j)}, \quad \mathsf{r}_j \le C\theta_j^{-\kappa_0/2}, \\ \mathsf{h} \le C, \qquad \mathsf{f} \le \bar{f}, \qquad \mathsf{f}_\Phi \le C c_\times(\bar{c}). \end{gathered} \tag{3.cclviii}$$

The label spaces, tests, real backgrounds, reference operations, measures and all numerical Gaussian constants are the real ones. The complex variables are exponent members and their own coefficients. This is why local functions can be differentiated without differentiating a sharp test or an adaptively chosen threshold.

For a physical source w, the site's member is the one for the constant source $\zeta = w(p_z)$. The defect is $E^{(j),w}(X,P,z) - E^{(j),w(p_z)}(X,P,z)$, not an independent Euclidean member. A localization that sees only w within a domain of diameter D gives the elementary difference bound $\sup|w - w(p_z)| \le D\ \mathrm{Lip}\, w$. Cauchy in this localized source direction then gives the source-gradient gain when its actual coefficient norm is known. The spatial inequality supplies the gain in $\mathrm{Lip}\, w$; it does not supply a missing α_j^2 in the coefficient norm.

The fixed scalar ceiling in eq. 3.cclviii also explains the two source discs. The potential uA has size $O(\alpha_j^2\|u\|_\infty)$ on a small-field chart, and therefore fits an enlarged auxiliary finite slot. The same $\|u\|_\infty$ occurs without this factor in eq. 3.cclv. It can also occur repeatedly in a history exponent. Hence finite-slot admission cannot enlarge the full source box by α_j^{-2}. The first-jet calculation in eq. 3.ccxxxiv remains valid.

There is an exact elementary reserve calculation for history amplitudes. Write $n(\omega) = \sum_j n_j(\omega)$, including each swallowed history exactly once, and $v(\omega) = \sum_j \log\theta_j\, n_j(\omega)$. If $\log\theta_j \ge 2\bar{f}$, then

$$|\exp\Psi_\omega| \le e^{\bar{f} n(\omega)} \le e^{v(\omega)/2}. \tag{3.cclix}$$

It can spend half of a previously established $e^{-v(\omega)}$ reserve, while keeping the remaining half. The retained inner Wilson factor and component rarity must still be present in the original mass estimate. The formula in eq. 3.cclix does not manufacture that estimate.

3.9.13.1. Domain estimate VII: source incidence and excess exponents. For one actual step or one large-field component, list every use of a given complex member, its own coefficient and its scalar anchor. Prove the aggregate site bound and the bounds in eq. 3.cclvi–eq. 3.cclviii after formation, using a single unused action reserve. For a component mass the supremum over its admitted external pair is placed inside its stored-variable integral. For drop-outs it is over the actual fibre of stored variables and recorded tests, not a product set containing an unrelated flat field. These specified incidence and fibre estimates are the remaining source-domain calculation; independent one-factor disc checks do not give the joint box.

3.9.14. *Large run windows and old activities.* The actual free-coordinate window may have $m_I \le C(r+D)^4$ scalar coordinates. A useful larger analytic chart is constructed in physical coordinates and then divided by g_k; it is not obtained by continuing the cutoff indicator. Suppose the actual physical pivot map on this window and every atomic background composition have radius at least

$$h_W \ge c\,\alpha_k r^{-A_c}, \qquad R_y = h_W/g_k, \tag{3.cclx}$$

and their normalized displacements obey the admitted graph bounds in eq. 3.ccxxiii–eq. 3.ccxxiv. This defines a precise chart target, including the polynomial loss A_c. It is not a uniform one-block constant. For old level i, the output displacement appearing in that graph has the scale ratio $L^{-(k-i)}\alpha_k/\alpha_i$; by eq. 3.ccxix its supremum is bounded. Amplified window directions use the same estimate with their actual multiplier, not a separate analytic continuation argument for each real direction.

If the cutoff is $\mathfrak{b}_k \asymp (\log\theta_k)^{p_0}$ and $r \le C(\log\theta_k)^{p_r}$, then the cutoff and two fixed-fraction coordinate circles fit in eq. 3.cclx whenever

$$q > p_0 + A_c p_r, \qquad \theta_k \ge T_*. \tag{3.cclxi}$$

Indeed $R_y \ge c(\log\theta_k)^{q-A_c p_r}$, so $\mathfrak{b}_k/R_y \to 0$. A finite increase of T_* gives any prescribed strict fractional room. If the complete old exponent has scalar supremum at most Cr^{A_0} on this larger chart, its directional Hessian is bounded by eq. 3.ccxxx; an entrywise and row-count proof can instead yield a bound of the form

$$\|\operatorname{Hess}_{y_I} P_{\mathrm{old}}\| \le C r^{A_{\mathrm{tot}}} (\log\theta_k)^{-2q}. \tag{3.cclxii}$$

Here A_{tot} must include the two losses $2A_c$, the actual amplitude and coordinate-count losses, and any response losses. For $r \le C(\log\theta_k)^{p_r}$, this tends to zero if $2q > A_{\mathrm{tot}} p_r$. If the free precision has a fixed gap m_*, increasing T_* makes the bound less than $m_*/4$.

3.9.14.1. Domain estimate VIII: the actual window chart and exponent. Prove eq. 3.cclx for the actual constrained window, exterior coordinates and gauge quotient; prove the graph bounds for all local substitutions; and derive its complete old-exponent amplitude or localized entry bounds before using eq. 3.cclxii. The count m_I, the old-age summation and the normalization of each old activity are part of this estimate. The localized old-activity theorem supplies the Hessian implication from these primitive data and the pointed remainder. It does not identify a conditional exterior law.

For the near, nonwrapping part there is a sharper estimate. Let a regular source derivative born at level j have norm $A_j \le C\alpha_j^2 s_j$, and put $\rho_j = \delta_n/\alpha_j$, $a = n-j$. The actual pointed window theorem, after its fine-site count, gives

$$C A_j \left(\rho_j^2 L^{-\alpha_H a} + \rho_j^3 L^{-2a}\right). \tag{3.cclxiii}$$

The needed hypothesis is a bound $C\delta_n$ on the complete background-plus-fluctuation field used in the near Taylor estimate. A bound only on its increment does not give it. Under that hypothesis, eq. 3.ccxx admits the central half-disc at every age, even when $\rho_j < 1$; the remote datum uses the

separate absolute ceiling eq. 4.xxx. On the complete homogeneous window class, eq. 4.cxlv and the full-profile calculation preceding eq. 4.clxxx supply the proportional $C\delta_n$ remote-datum bound as well. For a class without that bound its response is retained as a separate tail. The frame gives $A_j/\alpha_j^2 \le Cs_n$ and $\rho_j \le Cs_n\sqrt{a+2}$. Therefore the sum of eq. 3.cclxiii is at most $C\delta_n^2 s_n$, by geometric summation. A defect with an additional factor $h_j \operatorname{Lip} w$ sums the same way since $h_j = h_n L^{-a}$. A field-independent remote or winding term has no such homogeneous radius factor and is assigned its own decaying class.

The exterior-force estimate is a separate bounded target. In the assigned conditional coordinates I, O, it must control

$$Q^{\mathrm{sc}}y, \quad Q^e_{IO}y_O(e), \quad (Q^e_{II} - Q^{e_0}_{II})y, \quad \nabla_I(P_e - P_{e_0}), \qquad Q^{\mathrm{sc}} = Q_{IO}Q_{OO}^{-1}Q_{OI}. \tag{3.cclxiv}$$

For example the third term needs the actual bound $|((Q^e_{II} - Q^{e_0}_{II})y)_b| \le C\mathfrak{b}\, e^{-c\,d(b,O)}$, uniformly over the admitted exterior data and the real window; the fourth needs the corresponding exponential derivative bound without an uncontrolled exterior factor. They require control of the actual exterior field and conditional exponent. Neither eq. 3.cclxii nor decay of one covariance proves them.

3.9.15. *Carriers, world floors and the source norm.* A world is a torus with physical side ℓ and cutoff K. Its numerical side at level j is $\mathfrak{s}_j = \ell L^{K-j}$. A comparison of local formulas requires one injective rectangular lift containing the whole bond/path neighborhood being compared. A separate lift for each endpoint is not enough. In particular the pivot determinant calculation requires the support of the relevant constraint to contain no other crossing pivot. That support, including its contour paths, is what is lifted.

For the actual finite average the determining neighborhood grows by a fixed number of blocks at each scale. If R_j is its radius in bare numerical units after j steps, the elementary support recursion is

$$R_{j+1} \le R_j + C_L L^{j+1}\eta, \qquad L^{-k}\eta^{-1}R_k \le C_L \sum_{a=0}^{k-1} L^{-a} \le \frac{C_L}{1 - L^{-1}}. \tag{3.cclxv}$$

Every internal path lies in its assigned block and every straight segment in the assigned double block, which proves the first inequality. Composing supports proves the recursion; the displayed geometric sum proves the depth-independent radius in final units. The differentiated map has the same determining neighborhood. Taking a local plaquette or current adds its explicit finite range.

This proof concerns the average, not the full nonlinear minimizing background. The latter has inverse tails. A localized walk term has its own finite surroundings; the complete inverse is an absolutely convergent sum of such terms. Carrier identities for a localized substitution use its decoupling and rooting rule. They cannot be deduced by applying eq. 3.cclxv to an unlocalized inverse. For instance a scale-dependent collar of width e_j contributes $\sum_{j\le k} e_j L^{j-k}$ in final units. If e_j is bounded, this is a geometric sum. The weaker relation $e_j \le Le_{j+1}$ alone does not give a depth-independent sum: $e_j = L^{k-j}$ makes every summand one. Thus changes of localization size must be checked in the actual carrier construction.

The nonwrapping conditions have three separate uses. The double blocks of an average and the cubes of local gauges must be shorter than half the torus. A gauge cube enlarged by its comparison collar must still have that property. Finally each tower is formed only to a depth at which its prescribed inner cube size fits in half the torus. If the largest required numerical size in the relevant level is D_j, the test is

$$2D_j < \ell L^{K-j}. \tag{3.cclxvi}$$

This condition is unchanged by moving the lifted patch around the torus. It asserts no smallness of holonomy around a noncontractible cycle. The first-step curved physical gap handles arbitrary real holonomies by its local gauge and partition argument; it is not obtained by choosing one globally small potential.

3.9.15.1. Domain estimate IX: compatible carriers and floors. For each average, tower, conditioned inverse term and inner, outer or removed substitution, give its actual determining neighborhood and collars. Prove that its required lifts satisfy eq. 3.cclxvi, and that each restriction or transport carries all its defining rows and parameter cubes. For full inverses, give the separately summed boundary/image tail. This finite-list geometric estimate must accompany the analytic estimate III; replacing a region operator by a clipped average does not preserve its determining neighborhood.

Let $h_j = L^{j-K}$ be the physical spacing of level j, and let f be a Lipschitz test on the physical torus. Use

$$\|f\|_{\sharp,j} = \max\{\|f\|_\infty, c_M h_j \operatorname{Lip} f\}, \qquad c_M \geq C_{\mathrm{geo}} M. \tag{3.cclxvii}$$

The constant c_M contains the fixed block-size factor. If a domain of tree length $d_j(X)$ sees plaquettes within physical distance $C_{\mathrm{geo}} M h_j(d_j(X)+1)$ of its anchor, then

$$|f(p) - f(p_z)| \leq C_{\mathrm{geo}} M h_j (d_j(X)+1) \operatorname{Lip} f \leq (d_j(X)+1)\|f\|_{\sharp,j}. \tag{3.cclxviii}$$

This follows from the Lipschitz definition and the diameter of a connected tree of cubes. The polynomial $d_j(X)+1$ is absorbed by the strict entropy surplus in eq. 3.ccxxxiii. Moreover $\|f\|_{\sharp,j} \leq \|f\|_{\sharp,n}$ for $j \leq n$, and $h_j/h_n = L^{-(n-j)}$. These are exactly the scale factors of a source-gradient defect in a later window. No second block factor is removed from eq. 3.cclxviii; M/c_M, rather than M, is the fixed ratio being bounded.

3.9.16. *Tubes of frozen couplings and real admission.* The frozen reference choices are formed from a real frame θ. Let the real recursion have weights $x_j^\natural = \beta - \sum_{i\leq j} b_i$, deviations $z_j = \theta_j - x_j^\natural$, and nonincreasing tube radii t_j. The actual tube requires $t_j(\log \theta_{j+1})^{C_t} \leq c_t \theta_{j+1}$. Room $2r$ means $|z_j| + 2r \leq t_j$ at every intermediate level. It is a simultaneous condition, not just a condition at the endpoint.

Suppose the constant-source coefficients are holomorphic on $|\zeta| < 2r$, and their difference quotients have bounds d_i with $\sum_i d_i \leq B_0$. Their exact running shift is

$$\zeta_j(\zeta) = \zeta + \sum_{i\leq j} [b_i(\zeta) - b_i(0)], \qquad |\zeta_j(\zeta)| \leq (1+B_0)|\zeta|. \tag{3.cclxix}$$

Thus $|\zeta| < r/(1+B_0)$ leaves room r inside every tube. The estimate is just the triangle inequality for the displayed recursion and is independent of the number of levels. The substantive input is the summable quotient bound on its actual disc, including the marked bare-disc forcing if that is how the bound is obtained.

A constant source may be read either in the fixed source presentation of eq. 3.ccliii, or as an alternative frozen-weight presentation. In the former, it remains absent from the Gaussian structures; in the latter, the compensating presentation identity in eq. 3.ccxlix must be used with the new weights. Equality of the finite integrals includes all these normalizing and interaction factors. It is not justified by changing τ in only the Gaussian determinant.

Real admission also matters when comparing two holomorphic presentations. On a connected complex overlap that contains a real open set where both are admitted and represent the same bare integral, the real identity extends holomorphically. The proof is the Taylor argument following eq. 3.ccxv. Geometric overlap of two complex discs without a common admitted real set gives no such conclusion for the actual presentations. Endpoint bracketing and slope bounds on a canonical admitted interval can therefore be used for tuning on that interval; broader calibration gluing requires its own real-admission estimate.

3.9.16.1. Domain estimate X: the actual coupling tube and overlap. Prove the coefficient quotient bounds used in eq. 3.cclxix on the declared frozen discs, preservation of the intermediate tube with its prescribed slack, and the actual real admission of any overlap on which gluing is claimed. The objects are the unchanged frozen choices and the actual constant-source coefficients. This estimate

is separate from the many-level field core: a bound on a physical field radius does not give a bound on an analytic coupling disc.

3.9.17. *A simultaneous choice of the remaining small parameters.* The elementary choice problems can now be separated from their model estimates. First determine the polynomial degrees and their dependencies. The logarithmic exponents are chosen before geometric parameters when the actual construction requires that order. The fixed block size, its finite-map constants, the propagator rates, tower and collar sizes, remaining analytic radii and localization sizes are then chosen in their declared order; the weak-coupling floor is last. For the numerical calculation below these geometric data and a strict entropy surplus have already been fixed. Choose the enlarged decoupling radius $e^{\bar\kappa}$ so that the remaining rate in eq. 3.ccxxxii, after geometric and entropy charges, is still positive. Choose the shape constants C_0, C_1 and the absolute ceiling to satisfy eq. 3.ccxi and eq. 3.ccxvi. The quantities $L, M, \bar\kappa$ are then fixed; a constant such as $e^{C_R\bar\kappa}$ is not allowed to depend on the depth.

The analytic logarithmic exponent q has to satisfy every required finite lower bound, including

$$q > p_0, \qquad q > p_0 + A_c p_r, \qquad 2q > A_{\mathrm{tot}} p_r. \tag{3.cclxx}$$

Any additional power of the window radius or history reserve used by the actual estimates must be added before that exponent is chosen. The elementary feasibility conclusion assumes that the displayed polynomial degrees do not themselves increase with the new choice of q. When q must be chosen before later geometry, these degree bounds must also be uniform over the later admitted choices, or a simultaneous feasibility list must be solved in the required order. Fixed-geometry logarithmic smallness alone does not prove that earlier choice. If the degrees do depend on q, the resulting simultaneous inequalities must first be solved with that dependence; increasing q alone is no proof of feasibility. For finitely many conditions $C_\nu(\log T)^{-b_\nu} < \epsilon_\nu$, with $b_\nu > 0$, a single sufficiently large T satisfies them all: take $\log T > \max_\nu (C_\nu/\epsilon_\nu)^{1/b_\nu}$. The conditions $CMf_q(T) < c$ hold after a further increase, since every power of $\log T$ is $o(T^{1/2})$. This proves simultaneous smallness of the radius ceilings, $s_{\mathfrak{a}}$, the factors in eq. 3.ccxxix, the Gaussian ratio perturbation and every explicitly listed logarithmic loss. It is a finite feasibility calculation after the exponents of the actual estimates have been obtained.

Finally fix the source scalar ceiling and excess-exponent ceiling, and increase the real floor so that $\log T \geq 2\bar f$ and the Wilson weights in eq. 3.cclv remain positive on the required real integrals. The spare potential allowance is chosen inside the already established Gaussian/polymer reserve; the nested field boxes spend their fixed fractional slack once. None of these choices alters an earlier real cutoff or reference operation under a source derivative.

At this point the algebraic domain reductions are closed. Domain estimates I–X specify what remains to be proved for the actual many-level operators, towers, substitutions, roots, Gaussian matrices, source arrays and run windows. They have explicit objects, radii and uniformities. The finite chart and physical background results realize only their stated first-step chart, background, gap and radius instances. Independent-current normal-form matching, actual integrated roots, complete local Gaussian budgets and source/history estimates remain separate wherever they have not been supplied, even at a first step. At further levels the action is the actual composed action $A(U_k(V))$, with its constraint and current terms. Reusing the numerical first-step constants without the corresponding many-level estimate would not be an induction.

3.10. **A relative-gauge current extension at the first step.** An independent-current operator is not determined by its values on the physical current graph. We therefore give a concrete alternative definition at one step and prove what it supplies. Its complete nonlinear action, quadratic form, gauge covariance, real gap and complex inverse are specified. At a constrained-critical axial background it has the actual physical action as its value, throughout the admitted fibre chart. Agreement with any other current completion requires a separate identification. Using it at later scales requires the further consistency and closure checks stated below.

The raw fibre and the relative contour functional. Use fine lattice spacing one in this section. Currents are raw covectors for the bond logarithms with the counting pairing. The conversion to a potential or a different length unit is the domain section's conversion, including its dual cell weights. For the actual one-step group average put

$$q_U(a) = \frac{1}{i}\log[\mathcal{M}(e^{ia}U)\mathcal{M}(U)^{-1}], \qquad P_U = Dq_U(0). \tag{3.cclxxi}$$

All logarithms and contour means use their admitted regular branches. The geometry is the embedded double-block geometry of the pivot chart, on the infinite lattice or a compatible torus. In particular no second crossing pivot enters a row.

Let h_U be the pivot right inverse, let ι insert free coordinates with zero pivots, and set

$$C_U = \iota - h_U P_U \iota, \qquad q_U(\Phi_U(u)) = 0, \qquad \Phi_U(0) = 0, \qquad D\Phi_U(0) = C_U. \tag{3.cclxxii}$$

The free restriction of $\Phi_U(u)$ is exactly u. The quantitative chart gives a joint background and free-variable neighbourhood, with free radius $r_u = cL^{-9}$ and background radius $r_b = c'L^{-5}$ after its specified fixed shrinkages. Two differentiations give

$$D^2\Phi_U(0)[u,v] = -h_U D^2 q_U(0)[C_U u, C_U v]. \tag{3.cclxxiii}$$

The second derivative has only pivot components, so the right inverse on those components proves the identity and its sign.

Write $f_U(a) = A_W(e^{ia}U)$ and let $j_U = Df_U(0)$ be its raw gradient. On a finite torus these are ordinary finite-dimensional functions. On the infinite lattice every scalar action difference below is first evaluated for finitely supported free variations. The locally finite decomposition and its derivatives then define the kernels. No extensive scalar action is assigned a finite supremum on all bounded fields.

For every internal contour row define its relative logarithm by

$$\ell_{U;y,x}(a) = \frac{1}{i}\log[T_{y,x}(e^{ia}U)T_{y,x}(U)^{-1}], \qquad G_U a = (D\ell_{U;y,x}(0)a)_{y,x\neq y}. \tag{3.cclxxiv}$$

The relative gauge function is

$$\mathcal{G}_U(a) = \sum_{y,x\neq y}\left(1 - \tfrac{1}{2}\operatorname{tr}[T_{y,x}(e^{ia}U)T_{y,x}(U)^{-1}]\right). \tag{3.cclxxv}$$

It has value and first derivative zero at $a = 0$. Since $1 - \frac{1}{2}\operatorname{tr} e^{iX} = \frac{1}{2}|X|^2 + O(|X|^4)$, its Hessian is $G_U^{\mathsf{T}}G_U$. Here G_U acts on full bond variations; on free coordinates the contour differential is $G_U C_U$. This distinction fixes the placement of both copies of C_U.

A stationary extension for every independent current. For a declared independent raw current J define

$$\begin{aligned} S_U^A[J](u) &= f_U(\Phi_U(u)) - f_U(0) - \langle j_U, \Phi_U(u)\rangle \\ &\quad + \langle J, \Phi_U(u) - C_U u\rangle, \\ S_U^{\mathrm{rel}}[J](u) &= S_U^A[J](u) + \mathcal{G}_U(\Phi_U(u)). \end{aligned} \tag{3.cclxxvi}$$

The pairings are complex bilinear. These formulas are definitions on the actual pivot chart, rather than an extension by agreement on a lower-dimensional graph.

Every term in the first derivative cancels: $Df_U(0)C_U = j_U C_U$, the next term subtracts it, and $D(\Phi_U - C_U u)(0) = 0$. The relative gauge also has zero first derivative. Thus

$$S_U^{\mathrm{rel}}[J](0) = 0, \qquad DS_U^{\mathrm{rel}}[J](0) = 0 \quad \text{for every admitted } (U,J). \tag{3.cclxxvii}$$

The cancellation uses the actual j_U in the subtraction. Replacing it by J there would give a different linear term.

Put $\mathscr{A}_U = D^2 f_U(0)$. The chain rule and eq. 3.cclxxiii give

$$\begin{aligned} K_U^A[J](u,v) &= \langle C_U u, \mathscr{A}_U C_U v\rangle - \langle h_U^{\mathsf{T}} J, D^2 q_U(0)[C_U u, C_U v]\rangle, \\ K_U^{\mathrm{rel}}[J] &= K_U^A[J] + (G_U C_U)^{\mathsf{T}}(G_U C_U). \end{aligned} \tag{3.cclxxviii}$$

Indeed the $j_U D^2\Phi_U$ term from the first composition is cancelled by $-\langle j_U, \Phi_U\rangle$. The J term supplies $\langle J, D^2\Phi_U\rangle$. There is no constraint-curvature term from the relative gauge because its full-bond first derivative is zero. All matrices are normalized as Hessians, with quadratic exponent $-\frac{1}{2}\langle u, Ku\rangle$.

For a fixed reference inverse coupling $\theta = g^{-2}$ and physical action weight x put $\tau = x/\theta$. The gauge part keeps its reference weight. Consequently the corresponding free precision is

$$K_{U,\tau}^{\mathrm{rel}}[J] = \tau K_U^A[J] + (G_U C_U)^{\mathsf{T}}(G_U C_U). \tag{3.cclxxix}$$

The factor τ has not multiplied the gauge block. This is the actual distinction needed in a complex-weight comparison at a frozen frame.

Exact physical agreement and the noncritical linear term. If U is constrained-critical, C_U parametrizes its fibre tangent and therefore

$$C_U^{\mathsf{T}} j_U = 0. \tag{3.cclxxx}$$

At the physical current $J = j_U$, eq. 3.cclxxvi then gives, for every free u in the chart,

$$S_U^A[j_U](u) = A_W(e^{i\Phi_U(u)}U) - A_W(U). \tag{3.cclxxxi}$$

This is an equality of the whole action difference. It is stronger than equality of Hessians.

If U is also contour-axial, $T_{y,x}(U) = 1$. Hence $\mathcal{G}_U(a) = \mathrm{GF}(e^{ia}U)$ identically on the chart, and

$$\begin{aligned} g^{-2}\{\tau S_U^A[j_U](gy) + \mathcal{G}_U(\Phi_U(gy))\} = x\{A_W(e^{i\Phi_U(gy)}U) - A_W(U)\} \\ + \theta\,\mathrm{GF}(e^{i\Phi_U(gy)}U). \end{aligned} \tag{3.cclxxxii}$$

Thus the extension produces the actual weighted constrained action and the reference gauge weight at this physical background.

For a noncritical U the exact identity instead has

$$S_U^A[j_U](u) = A_W(e^{i\Phi_U(u)}U) - A_W(U) - \langle C_U^{\mathsf{T}} j_U, u\rangle. \tag{3.cclxxxiii}$$

The missing physical linear force must be restored. It has zero Hessian, so at $J = j_U$ the quadratic comparison is still the physical chain-rule comparison. But its contribution after Gaussian rescaling is of order g^{-1} unless that tangent force vanishes. No coupling-independent holomorphic potential claim for the whole noncritical physical action follows from eq. 3.cclxxvii.

Gauge covariance from the actual graph. For a site gauge s the left logarithm transforms as $a^s(b) = \mathrm{Ad}(s(b_-))a(b)$. Denote this full-bond linear map by $\mathcal{U}_s$ and its free restriction by $\mathcal{U}_s^f$. The actual average is bi-equivariant. On common regular branches this gives

$$q_{U^s}(\mathcal{U}_s a) = \mathcal{U}_s^{\mathrm{coarse}} q_U(a). \tag{3.cclxxxiv}$$

Free restriction and pivot insertion are stable under this bondwise transformation. The vector $\mathcal{U}_s\Phi_U(u)$ solves the transformed constraint and has free restriction $\mathcal{U}_s^f u$. Uniqueness of the pivot graph proves

$$\Phi_{U^s}(\mathcal{U}_s^f u) = \mathcal{U}_s\Phi_U(u), \qquad C_{U^s}\mathcal{U}_s^f = \mathcal{U}_s C_U. \tag{3.cclxxxv}$$

For complex gauges this assertion is made only on the common admitted graph branch. It does not say that a complex orbit saturates a core.

Wilson gauge invariance gives $j_{U^s} = \mathcal{U}_s j_U$ under the invariant bilinear colour pairing. The product in eq. 3.cclxxv is conjugated at its contour's initial node. Combining these facts proves

$$S_{U^s}^{\mathrm{rel}}[\mathcal{U}_s J](\mathcal{U}_s^f u) = S_U^{\mathrm{rel}}[J](u). \tag{3.cclxxxvi}$$

The same holds separately for the action and gauge parts, and hence for every fixed τ in eq. 3.cclxxix. In an orthonormal colour basis $\mathcal{U}_s^f$ is bilinearly orthogonal even for admitted complex gauges. For real unitary gauges it is also unitary in the Hermitian Hilbert norm. Therefore

$$K^{\mathrm{rel}}_{U^s,\tau}[\mathcal{U}_s J] = \mathcal{U}_s^f K^{\mathrm{rel}}_{U,\tau}[J](\mathcal{U}_s^f)^{-1}. \tag{3.cclxxxvii}$$

This proves matrix covariance directly. It does not use an integral change by a complex gauge through a sharp real cutoff.

Axialization and the independent-current real gap. For a real regular field, define within each block $s(y) = 1$ at its centre and $s(x) = T_{y,x}(U)$. The internal contour means are unitary. Their bi-equivariance gives

$$T_{y,x}(U^s) = 1, \qquad \mathcal{M}(U^s) = \mathcal{M}(U). \tag{3.cclxxxviii}$$

The second equality uses $s = 1$ at the coarse centres. The regular logarithm branches remain the same: after this transformation all path transports are near identity by the small-plaquette comparison, and uniqueness of the internal mean identifies the transformed mean. This is the finite averaging chart's real gauge-covariance construction.

Plaquette norms and the norm of the transformed current are unchanged. The gauge is periodic on a compatible torus and is defined on the full cover as well. No noncontractible holonomy is made small.

Let $c_0(L)$ be the proved flat gap. Choose R, ϵ, γ as in the first-step gap theorem, in particular

$$R \geq 40, \quad K_L/R^2 \leq c_0/8, \quad CL^2 R\epsilon \leq r_b/8, \quad K_L(R\epsilon+\gamma) \leq c_0/8, \quad K_L = CL^{40}, \quad \gamma \leq 1. \tag{3.cclxxxix}$$

Suppose all fine plaquettes have size at most ϵ and the real raw current has norm at most γ. At the axialized field the relative and absolute contour Hessians coincide. The formula in eq. 3.cclxxviii is then exactly the explicit axial auxiliary form of the first-step gap theorem. That theorem gives $K^{\mathrm{rel}}_{U^s}[\mathcal{U}_s J] \geq c_0/2$. Unitary conjugation in eq. 3.cclxxxvii proves

$$K^{\mathrm{rel}}_U[J] \geq c_0/2 \tag{3.ccxc}$$

for every such real regular U, without assuming that U was axial or critical. The inequality uses the Hermitian quadratic form of the real symmetric matrix on its complexification. The current is the declared raw current of this one-step problem.

This transfer is possible because the relative gauge is jointly site-gauge covariant. Applying it to an arbitrary absolute-gauge operator off the axial locus would not be valid.

The full half-disc with the gauge weight retained. There is a stronger weight statement than a small operator-norm neighbourhood of $\tau = 1$. Choose the universal constant in K_L in eq. 3.cclxxxix to control the action and gauge blocks separately as well as their sum. The same free-coordinate partition proof gives

$$K^A_U[J] \geq -K_L\{R^{-2} + R\epsilon + \gamma\} \geq -c_0/4. \tag{3.ccxci}$$

Here the flat action block is nonnegative: it is $C_0^{\mathsf{T}} \mathscr{A}_{\mathbf{1}} C_0$, the compression of the free Wilson curl form. In each localized patch the action block differs from it by at most $K_L(R\epsilon+\gamma)$, by the same actual chart coefficient comparison used for the complete form. Localizing the action block on free variables costs at most K_L/R^2. Summing the squared partition therefore proves eq. 3.ccxci. No positive gap for the action block alone has been assumed. Real gauge covariance transfers this statement from the axial representative just as before.

Write $K = K^A_U[J] + K^G_U$, where $K^G_U = (G_U C_U)^{\mathsf{T}}(G_U C_U) \geq 0$. eq. 3.ccxc and eq. 3.ccxci imply

$$B = K^{-1/2} K^A_U[J] K^{-1/2}, \qquad -\tfrac{1}{2} I \leq B \leq I. \tag{3.ccxcii}$$

The upper bound follows from $K_U^G \geq 0$. The lower bound uses $K^{-1} \leq 2/c_0$. In particular B is a bounded self-adjoint operator with norm at most one, and

$$K_{U,\tau}^{\mathrm{rel}}[J] = K^{1/2}[I + (\tau - 1)B]K^{1/2}. \tag{3.ccxciii}$$

For $|\tau - 1| \leq 1/2$ this proves $\mathrm{Re}(v, K_{U,\tau}^{\mathrm{rel}}[J]v)_{\mathrm{H}} \geq \frac{1}{2}(v, Kv)_{\mathrm{H}} \geq (c_0/4)\|v\|_2^2$ and inverse norm at most $4/c_0$. The inverse is holomorphic on $|\tau - 1| < 1$ by the Neumann series for its middle factor. The gauge block has stayed at its original weight.

In finite dimension let $\beta_a \in [-1/2, 1]$ be the eigenvalues of B. Then $\det K_\tau = \det K \prod_a [1 + (\tau - 1)\beta_a]$. Every factor has positive real part on that disc. Its logarithm continued from one defines the coherent Gaussian determinant branch. This is a matrix normalizer statement, not a zero-free theorem for an interacting partition function with cutoffs.

One also obtains a useful absolute bound. For N real coordinates, $0 \leq \rho < 1/4$, and $|F(y)| \leq F_0 e^{\rho(y,Ky)}$,

$$\left| \int F \, d\mu_{K_\tau} \right| \leq F_0 \left(\frac{3/2}{1/2 - 2\rho} \right)^{N/2}, \qquad |\tau - 1| \leq 1/2. \tag{3.ccxciv}$$

The determinant modulus is at most $(3/2)^N \det K$, and $\mathrm{Re}\, K_\tau - 2\rho K \geq (1/2 - 2\rho)K$. Integrating this real Gaussian proves the inequality. Its cost is exponential in the actual number of coordinates. A component mass estimate must pay that cost from its stated reserve; it has not become a volume-independent scalar bound.

A complex neighbourhood and exponentially local inverse. The matrix has the fixed finite range of the actual one-step chart. Relative contour factors add no new geometric range: both factors use the same internal paths. On the joint chart their analytic values and inverses are bounded in a local unitary frame. The local Cauchy estimates and coordinate counts give

$$\|K_{e^{iE}U,\tau}^{\mathrm{rel}}[J + Z] - K_U^{\mathrm{rel}}[J]\|_{\mathrm{Sch}} \leq \widehat{K}_L(\|E\|_\infty + \|Z\|_\infty + |\tau - 1|), \qquad \widehat{K}_L = C' L^{40}. \tag{3.ccxcv}$$

Here use $\|E\|_\infty \leq r_b/16$, retained chart room and total current at most one. To justify the same safe power, each scalar relative contour term is bounded on the same free radius cL^{-9} and background radius $c'L^{-5}$. At most CL^4 terms affect one row and at most CL^4 columns couple to it. Its free Hessian costs CL^{26} and one background difference CL^{31}. The current curvature term has the previously counted CL^{17} current cost. The action block in the τ derivative has a bounded row sum of this same safe order. Increasing C' gives eq. 3.ccxcv, independently of volume.

If the right side is at most $c_0/8$, the Neumann series about eq. 3.ccxc gives inverse norm at most $4/c_0$. Finite range and its uniform row and column sums also give a local exponential inverse. For clarity let d be the fine lattice distance between free coordinates and let ψ be any bounded one-Lipschitz function. For small $m > 0$,

$$\|e^{m\psi} K e^{-m\psi} - K\|_{\mathrm{Sch}} \leq C_L(e^{mr_0} - 1), \qquad r_0 \leq 12L. \tag{3.ccxcvi}$$

This follows entry by entry from the range bound. Choose $m = m(L) > 0$ so this perturbation is at most $c_0/8$. The conjugated complex matrix is still invertible by comparison with the same real matrix. Apply its inverse bound with $\psi(b) = \min\{d(b, b'), N\}$ and let N increase. One obtains

$$\|(K^{-1})_{bb'}\| \leq (4/c_0)e^{-md(b,b')}. \tag{3.ccxcvii}$$

The distance is periodic distance on a torus. Counting its shells in the four-dimensional cover shows that row and column sums of this kernel are bounded uniformly in compatible volume. Consequently it is a two-sided inverse on bounded fields as well. The left identity uses the finite range of K; the right identity follows by absolute summation using that row bound. This also proves uniqueness of a bounded solution.

The half-disc has a uniform complex-field version as well. Bound the action-block and gauge-block differences separately by the same local Cauchy estimates. Since $|\tau| \leq 3/2$, their sum has norm at

most $C_L(\|E\|_\infty + \|Z\|_\infty)$, uniformly on $|\tau - 1| \le 1/2$. Make this norm at most $c_0/16$. Comparison with the inverse in eq. 3.ccxciii gives the inverse for the complex pair. For the weighted conjugation choose $m(L) > 0$ so its additional row/column cost is at most $c_0/16$. The total perturbation of the same real-pair half-disc operator is then at most $c_0/8$. Its inverse norm and kernel satisfy

$$\|K^{\mathrm{rel}}_{\mathbf{U},\tau}[\mathbf{J}]^{-1}\| \le 8/c_0, \qquad \|(K^{\mathrm{rel}}_{\mathbf{U},\tau}[\mathbf{J}]^{-1})_{bb'}\| \le (8/c_0)e^{-m(L)d(b,b')}. \tag{3.ccxcviii}$$

This argument is uniform in τ on the full closed half-disc. The smaller near-one comparison in eq. 3.ccxcv is a sufficient alternative; it is not needed to restrict τ in this stronger version.

Local intrinsic remainders and an admitted coupling disc. There is a locally finite decomposition of eq. 3.cclxxvi in which each scalar term has value and derivative zero at zero. For every Wilson plaquette take its action difference minus its own raw gradient paired with Φ_U. For every bond take $\langle J_b, (\Phi_U - C_U u)_b\rangle$. Add each individual relative contour term. The plaquette gradient terms sum to j_U, so this is exactly eq. 3.cclxxvi.

Each term depends on a fixed neighbourhood of its plaquette, bond or contour. On the common free ball of a fixed fraction of r_u its holomorphic supremum is bounded by a constant M_L. The unitary local frames and the actual chart give this bound uniformly in the real regular base, with the retained complex perturbation and current ceilings. Subtract half its local Hessian and denote the result by $R_\alpha(u)$. It vanishes through degree two. One-variable Cauchy on the line through u gives

$$|R_\alpha(u)| \le 2M_L(\|u\|_\infty/r)^3, \qquad \|u\|_\infty \le r/2, \tag{3.ccxcix}$$

for a fixed smaller radius $r = r(L) > 0$. Indeed the Taylor coefficients of order n are bounded by $M_L r^{-n}$ and the sum from $n = 3$ is geometric.

Let the fixed Gaussian field box have $\|y\|_\infty \le b$, $b \ge 1$. For an auxiliary coupling q with $|q|b \le r/2$, the singularity at zero in the following expression is removable:

$$q^{-2}R_\alpha(qy), \qquad |q^{-2}R_\alpha(qy)| \le 2M_L|q|b^3/r^3. \tag{3.ccc}$$

The gauge and action parts may have the separate fixed coefficients $1, \tau$ of eq. 3.cclxxix. All constants are uniform for τ on its retained disc.

The actual Haar and pivot-coarea factors also give local analytic potentials. Factor their value at $u = 0$ and take the holomorphic logarithm of the ratio on a smaller common chart. Each factor and its inverse are bounded there by the quantitative pivot inverse and Haar chart. Cauchy bounds its logarithmic derivative on a half ball; integration along the radial segment proves $|\log(J_\alpha(u)/J_\alpha(0))| \le C_L\|u\|_\infty/r$. This does not analytically continue an absolute value: the real coarea orientation is positive, and the complex factor is its coherent determinant inverse.

Group these fixed-range terms into their actual local carriers at a fixed block size M. Only $C_{L,M}$ terms enter a carrier or a rooted carrier sum. The potential weight $Q_Y(y) \ge 1$ and the bounded tree length of these carriers then give

$$\|V_{q,\tau}\|_\lambda \le C_{L,M,\lambda}|q|b^3/r^3. \tag{3.ccci}$$

This is a potential-norm estimate, independent of the number of carriers. For any fixed allowed potential budget w_* one can therefore take an admitted disc $|q| < c_{L,M,\lambda,w_*}b^{-3}$, also shrunk to ensure $|q|b < r/2$. This is an actual lower bound on an admissible radius for these first-step intrinsic and measure potentials. It is not just an upper restriction on a radius that has not been constructed.

The finite integral and the factors that remain outside. Here I denotes the constrained integral with this specified soft gauge weight and regular field box. Its identification with a regular part of the original Wilson measure uses the actual gauge-fixing identity and its prefactor; neither is created by the pivot change of variables.

At a physical critical axial pair, let $\mathcal{J}_U(u)$ be the whole real Haar/coarea density in the free chart and let N be the number of real free coordinates. On its fixed Gaussian box the exact integral is

$$I = e^{-xA_W(U)} g^N \mathcal{J}_U(0)(2\pi)^{N/2}(\det K^{\mathrm{rel}}_{U,\tau}[j_U])^{-1/2} \cdot \int \chi(y) e^{V_{g,\tau}(y)} d\mu_{K^{\mathrm{rel}}_{U,\tau}[j_U]}(y). \tag{3.cccii}$$

This follows from eq. 3.cclxxxii, the exact chart coarea factor and subtraction of its quadratic exponent. The nonconstant potential consists of the negative intrinsic remainders divided by g^2 and the logarithm of $\mathcal{J}_U(gy)/\mathcal{J}_U(0)$. The Gaussian determinant uses its coherent branch from the real positive matrix.

The prefactor is independent of the integrated field y, but generally depends on the coarse datum U and on the specified current and weight. It is not automatically a numerical vacuum constant. Its field-dependent part still needs the assigned localization; its value at the reference pair is counted once in the scalar normalization. The powers of g have been factored explicitly before introducing the auxiliary analytic coupling q in eq. 3.ccc.

The fixed box may be chosen as a product of colour-norm balls, so real site gauges preserve it. Complex gauge covariance of matrices, already proved in eq. 3.cclxxxvii, is not by itself a contour-deformation theorem for a sharp real integral. Such an integral's local holomorphic gauge identity follows on an admitted connected continuation from its real gauge identity; arbitrary full-core orbit descent requires the additional domain comparison.

We have supplied an explicit consistent first-step extension, the physical action identity, its real gap, complex inverse, and the intrinsic potential radius. The enlarged decoupling matrices, integrated bond-root bounds, all-substitution core admission, localization of the prefactor, and the many-level effective action and history operations remain separate. If this extension is used beyond the first step, each of them must use the same definition and prove its physical sum identity. It is not legitimate to import estimates for a differently defined current completion without the complete comparison. Conversely, the named-completion comparison need not be imposed in addition when the entire alternate construction has actually been rebuilt from eq. 3.cclxxvi.

3.11. **Constrained backgrounds at several scales.** A background is a regular critical orbit of the fine Wilson action with its assigned block averages, not a new independent coarse action. Its construction has three separate ingredients. The determining rows and their residual gauge group are fixed geometrically. The critical equations are then solved on a slice by a coupled field and multiplier argument. Finally a local regularity improvement identifies the solution among all regular orbits and gives the bounds needed for restriction and analytic continuation.

This section develops the geometric and nonlinear construction using the actual finite averaging map. Its proper-region inverse and local coefficient estimates are proved subsequently; their application to all sufficiently small regular mixed data is completed in Prop. 3.124. The construction includes arbitrary regular competitors, the action-boundary current, and the outward node factors. The physical constrained Hessian agrees with an independently defined current operator only when their complete projected forms agree.

The six background statements concern the regular critical orbit, local regularity, complex continuation, response, hereditary complex domains, and restriction with boundary comparison. A proved reduction to an explicitly stated primitive estimate is distinguished from a proof of that estimate.

3.11.1. *Determining rows and the geometry of a seam.* Fix an odd integer $L \geq 3$ and a depth k. Work in level-k units, so the bare spacing and intermediate spacings are

$$\eta = L^{-k}, \qquad \xi_j = L^{j-k}, \qquad 0 \leq j \leq k. \tag{3.ccciii}$$

The regions $\Omega_0 \supset \cdots \supset \Omega_k$ are unions of the assigned aligned cubes of side $\mathfrak{M}\xi_j$, where $\mathfrak{M}$ is an odd power of L. Set $\Omega_{k+1} = \varnothing$. Half-open cells specify boundary membership. The separation condition is

$$\operatorname{dist}_\infty(\Omega_j^c, \Omega_{j+1}) > \rho\xi_j, \qquad \rho \geq C_{\text{geom}} L. \tag{3.ccciv}$$

The number C_{geom} is chosen large enough for the fixed one-step path neighborhoods and collars used below. This condition does not require a region to be connected or simply connected.

Because all cubes are aligned and their sizes are powers of L, each Ω_j is a union of the blocks $B^j(y)$ of side ξ_j, and each Ω_{j+1} is also a union of those blocks. A top is a site $y \in \mathbb{T}^{(j)} \cap (\Omega_j \setminus \Omega_{j+1})$; its pillar is $B^j(y)$. These pillars partition Ω_0. To prove this, take a bare cell and the largest j for which it lies in Ω_j. Its level-j block is contained in Ω_j, while Ω_{j+1}, being a union of level-j blocks, contains none of that block. Its center is therefore a top of level j. Two such blocks at one level are disjoint. Blocks at different levels are nested if they meet, which is incompatible with the corresponding disjoint layers. This proves existence and uniqueness of its pillar.

For a bare bond or plaquette e, let $\mathsf{L}(e)$ be the largest top scale attached to its corners. The numerical action and regularity tests use

$$A_{\Omega_0}(U) = \sum_{p \subset \Omega_0} \left(1 - \tfrac{1}{2} \operatorname{tr} \partial U(p)\right), \qquad \eta^{-2}\mathsf{L}(p)^2 |\partial U(p) - 1|, \quad \mathsf{L}(b)^3 |J_A(U)(b)|. \tag{3.cccv}$$

Only inner plaquettes contribute to this action. Its boundary current therefore need not contain all six plaquettes of a bond. The current is its actual derivative with the length and pairing normalization of § 3.9.

The determining bonds of level j are

$$\Gamma_j = \left\{ c \subset \Omega_0 : \begin{array}{l} c \text{ is a level-}j \text{ bond, } c \text{ touches } \Omega_j, \\ c \text{ has no endpoint in } \Omega_{j+1} \end{array} \right\},$$
$$\mathbf{B} = \bigcup_{j=0}^{k} \Gamma_j. \tag{3.cccvi}$$

No inward row is kept. A row whose two endpoints lie in Ω_j is interior. An outward row has one endpoint outside Ω_j. Every endpoint is a top, possibly of the preceding level.

Here is the precise local geometry behind the last assertion. Let $c = (y, x) \in \Gamma_j$, $j \geq 1$, with $y \in \Omega_j$ and $x \notin \Omega_j$. Their distance is $\xi_j = L\xi_{j-1}$. By eq. 3.ccciv, neither endpoint nor its fixed double-block neighborhood can leave Ω_{j-1}. The endpoint x consequently belongs to $\Omega_{j-1} \setminus \Omega_j$, and it is a site of $\mathbb{T}^{(j-1)}$. It is a top of level $j-1$, even when it also happens to be a site of a coarser lattice. Its centered level-j block is entirely outside Ω_j, because that region is a union of level-j blocks. The separation puts this block entirely in Ω_{j-1}. Hence every level-$(j-1)$ bond internal to that block is a prescribed interior row of Γ_{j-1}.

The same argument shows why a row's double block reads no third layer. Each endpoint block misses Ω_{j+1}, by the no-inward condition and block alignment, and the neighborhood is inside Ω_{j-1} by separation. A one-step seam calculation therefore contains the levels $j-1, j$, whose length ratio is the fixed number L. It does not contain a product of ratios over all earlier levels. Larger constructions must explicitly preserve these determining neighborhoods; the assertion is not true for an arbitrary enlarged box at a seam.

The datum and residual group are

$$M_{\mathbf{B}}(U) = (\mathcal{M}^j U|_{\Gamma_j})_{j=0}^k, \qquad \mathcal{F}(\mathbf{V}) = \{U : M_{\mathbf{B}}(U) = \mathbf{V}\},$$
$$\mathcal{G}_{\text{res}} = \{g : g(y) = 1 \text{ at every node}\}. \tag{3.cccvii}$$

Exact gauge covariance of the iterated average proves that this group preserves the fibre. Its definition involves all nodes, not only the centers of the last lattice.

3.11.1.1. Test loops and the missing inward rows. At an ordinary coarse plaquette the datum loop uses V_j. Where the loop crosses into the preceding layer it uses the actual one-step averages of V_{j-1} on those edges. These are the mixed test loops. Their values transform by conjugation at their starting node; their distance from one is invariant under real unitary gauges. This is the local compatibility information that a regular datum supplies. It is not a bound on all individual $V_j(c) - 1$.

If inward rows are retained, their endpoint in Ω_{j+1} becomes an additional seam node. There is then a genuine distinction between fixing all nodes and imposing a full block-axial gauge in the inner block. The following example is exact. Take an inner top y and an additional seam node $x \neq y$ in its block. Choose a site gauge with $g(y) = 1$ and $g(x) \neq 1$, as close to one as desired, and set $U = 1^g$. All plaquettes and the action current are zero. All closed datum tests are also one. But

$$T_{y,x}(\mathcal{M}^j U) = g(y)g(x)^{-1} \neq 1. \tag{3.cccviii}$$

Every residual transformation fixing both nodes leaves this transport unchanged. Thus this pure-gauge orbit cannot be made fully block axial there. The example does not obstruct an axial gauge *outside* the nodes; that gauge is proved next. It explains why a step that requires a full inner block-axial representative omits the inward rows, or else states its exceptional-node gauge explicitly.

With two kept inward edges meeting the same seam node, compatibility also records their relative frames. In an adapted real datum frame one can write these as products of a coarse edge, the internal fine contour transport and the kept inward edge. The product of one such route with the inverse of another is a closed two-scale loop. It is a datum test independent of the fine representative. Omitting these tests while retaining the edges changes the regular-datum class; elementary plaquettes in each separate layer do not record their relative frame.

3.11.2. *An exact representative outside the nodes.* The residual orbit has a canonical comb representative independently of any elliptic equation. A comb is the fixed coordinate-order path inside a block. The only properties needed here are locality, bi-equivariance and the identity transport at a zero-length path. The construction works as well with an admitted mean transport having those properties.

For a nonnode x, let $n(x)$ be its largest lattice index: $x \in \mathbb{T}^{(n)} \setminus \mathbb{T}^{(n+1)}$. Its parent $z(x)$ is the center of its level-$(n+1)$ block. Its ascent terminates at its pillar top. We seek a residual gauge such that, for every such site,

$$(\mathcal{M}^{n(x)}U)(\Gamma_{z(x),x}) = 1. \tag{3.cccix}$$

Process the indices in descending order. At index n, leave the gauge one at all nodes and at $\mathbb{T}^{(n+1)}$. At a nonnode of exact index n, assign it the current transport $(\mathcal{M}^n U)(\Gamma_{z(x),x})$. Extend this gauge by one at the sites not being processed. Bi-equivariance makes that transport one. Gauge covariance of $\mathcal{M}^m$, $m > n$, sees only values at $\mathbb{T}^{(m)}$, which have not changed. Thus previously imposed conditions remain true. After finitely many stages every condition in eq. 3.cccix holds, and every prescribed node value has stayed one.

Uniqueness is equally direct. Suppose two representatives in one residual orbit satisfy eq. 3.cccix, and let h take one to the other. It is one at the tops. Descending an ascent, if $h(z) = 1$, covariance of a transport already equal to one gives

$$1 = h(z)\,1\,h(x)^{-1}, \qquad h(x) = 1. \tag{3.cccx}$$

Every nonnode is reached, so $h = 1$. The representative is therefore a function of the orbit. On real configurations the construction is Borel: the total average is Borel, the comb transports are finite products, and all inversions and multiplications are continuous group operations. On an admitted regular complex chart it is holomorphic, since there the assigned averages are holomorphic.

This argument gives no depth-independent bound for the derivative of the gauge. A real pure gauge may have arbitrary values at fine sites while fixing the tops. Norm bounds on a slice require

the separate estimates below. Likewise a complex representative is used only when its values remain in the admitted analytic branches and its condition numbers have the required bounds. The Borel orbit construction alone does not establish that admission.

For a sequence of depth $k+1$ with no inward rows, an inner level-$(k+1)$ block has no node except its center. Indeed every lower-level site of the block lies in Ω_{k+1}, so cannot be a top of a preceding layer, and alignment supplies no other coarse center in that block. A residual gauge can consequently set all its level-k internal contour variables to one while fixing the datum. This is the exact node-counting fact used in the definition of the next step's physical background.

3.11.3. *The reduced datum has an explicit local inverse.* Fix a real reference field U_0 whose used averages lie on the regular branches of the actual finite mean. For an interior row the reduced datum is $b(B)(c) = B(c)$. On an outward row oriented toward its outer endpoint z, it is

$$e^{ib(B)(c)} = e^{iB(c)} Z_c(B_{\mathrm{int}}), \qquad Z_c(B_{\mathrm{int}}) = \mathrm{Ad}(V_0(c))\, v_z(B_{\mathrm{int}}). \tag{3.cccxi}$$

The factor v_z is the actual one-step block factor made from the interior data in the finer block centered at z. The geometry following eq. 3.cccvi proves that these are prescribed interior rows, not another set of unknown outward rows. For the reverse orientation use the inverse group identity.

There is therefore a direct inverse. Recover $B_{\mathrm{int}} = b_{\mathrm{int}}$, then set

$$B(c) = \frac{1}{i} \log \left(e^{ib(c)} Z_c(b_{\mathrm{int}})^{-1} \right) \quad \text{on outward rows.} \tag{3.cccxii}$$

All logarithms are on the common small datum chart. Conjugation by the real $V_0(c)$ has norm one; it imposes no smallness of that coarse edge or of a global holonomy. The finite mean construction gives a radius $r_0(L) > 0$ on its admitted local reference class. On a fixed smaller ball, its block factors and their first two derivatives are bounded by constants depending on that finite construction. Thus eq. 3.cccxi–eq. 3.cccxii are admitted on balls $\|B\|_\infty < r_0/C_L$, respectively $\|b\|_\infty < r_0/C_L'$, and satisfy

$$\|b(B)\|_\infty \le C_L \|B\|_\infty, \qquad \|b^{-1}(b)\|_\infty \le C_L' \|b\|_\infty. \tag{3.cccxiii}$$

The uniform statement is over local reference patches on which that same finite chart and its radius are admitted. Establishing those patches for a multilevel background is part of its regular-domain estimate; it is not inferred just from the existence of U_0.

It is useful to see exactly why the inverse does not contain a long product of triangular matrices. Define the right logarithmic differential $\mathcal{E}_X(H) = (1/i)(d/dt) e^{i(X+tH)}|_{t=0} e^{-iX}$. Differentiating eq. 3.cccxi gives the outward diagonal $D_c = \mathcal{E}_{b(c)}^{-1} \mathcal{E}_{B(c)}$, not the identity in general. Let D be this block diagonal matrix, with identity on interior rows, and let T contain all derivatives of block factors. Its only nonzero blocks have interior columns and outward rows. Consequently

$$Db = D + T, \qquad TD^{-1}T = 0, \qquad (Db)^{-1} = D^{-1} - D^{-1}TD^{-1}. \tag{3.cccxiv}$$

Multiplying the last two matrices verifies both inverse identities. This nilpotence is a property of the actual reduced datum. An arbitrary lower triangular coupling between successive levels would not have it.

For the row count, one outward factor reads the $O(L^4)$ interior data of its single finer block. For the column count, an interior bond belongs to one assigned finer block, and that block can be the outer endpoint block of at most eight coarse incident rows. Thus both row and column counts are bounded at fixed L, independently of the number of layers. After inserting the actual cell weights and amplitudes, an interaction changes scale only from $j-1$ to j; its weight ratios are fixed powers of L. eq. 3.cccxiii–eq. 3.cccxiv therefore also give two-sided weighted Schur bounds in those declared norms. Higher derivatives follow from the same finite group formula and its fixed datum disc.

This local datum calculation is different from constructing a right inverse of all the field constraint rows. The latter must also lift fields and control their energy and gauge slice at a seam. That

primitive linear task is retained in the precision section. The exact derivative Db, including its nonidentity outward diagonal, is the datum forcing in the response equations.

3.11.4. *Real reconstruction on one block.* The next estimates are finite group calculations. They concern real unitary fields on a rectangular lift of one or two blocks. They do not require those fields to be close to one on every bond. Let $\varpi(U) = \sup_p |\partial U(p) - 1|$ on that neighborhood.

A coordinate-order path can be changed to another such path by interchanging adjacent steps in different directions and canceling backtracks. Each interchange inserts one plaquette, conjugated by the transport of its preceding path. On a box of side $O(L)$, the number of interchanges is bounded in terms of L, and more specifically at most a dimensional constant times L^2 for the fixed path families used here. Thus a contractible comparison loop γ in this family is a product of transported plaquette factors and

$$|U(\gamma) - 1| \le C_L \varpi(U). \tag{3.cccxv}$$

Indeed for unitary matrices $|AB - 1| \le |A - 1| + |B - 1|$, and conjugation by a unitary transport preserves the norm. Inducting over the inserted factors proves eq. 3.cccxv. The constant is not multiplied by the size of an individual open transport. This is why arbitrary real holonomies cause no loss in the local argument.

Fix the comb from the block center. The relative transports of the coordinate-order path family differ from that comb by eq. 3.cccxv. The local mean lemma, on its fixed regular radius, then implies

$$|T_{y,x}(U) - U(\Gamma_{y,x})| \le C_L \varpi(U). \tag{3.cccxvi}$$

To see that the mean lemma applies without a small open transport, multiply every member on the right by the inverse of the comb transport. The resulting family is near one. The implicit logarithmic mean has a uniformly bounded derivative on a smaller fixed ball, by its finite-dimensional inverse construction. Its value therefore differs from one by $C_L \varpi(U)$. Right equivariance restores the original transport and proves eq. 3.cccxvi.

For a fine bond b, let y_-, y_+ be the centers of its endpoint blocks and use the same center when both endpoints lie in one block. Define its transported value by

$$\widetilde{U}(b) = T_{y_-,b_-}(U)\, U(b)\, T_{y_+,b_+}(U)^{-1}. \tag{3.cccxvii}$$

If the centers coincide, this is a comparison loop, so it is within $C_L \varpi(U)$ of one. Otherwise the centers are joined by a coarse bond c. Replace the means by combs using eq. 3.cccxvi. The resulting open route and the central straight route from y_- to y_+ differ by a loop of the type in eq. 3.cccxv. The closed comparison transports in the definition of $\mathcal{M}(U)(c)$ satisfy the same bound. The finite exponential and logarithm estimate consequently gives

$$|\widetilde{U}(b) - \mathcal{M}(U)(c)| \le C_L \varpi(U) \quad (y_- \ne y_+). \tag{3.cccxviii}$$

All paths used are in the fixed double block. The regularity threshold makes the closed logarithms well defined; the final coarse bond itself may be far from one.

For two such fields U, U_0, put $d_T(x) = |T_{y,x}(U) - T_{y,x}(U_0)|$, using the endpoint's block center. Combining eq. 3.cccxvii–eq. 3.cccxviii and using unitarity gives

$$\begin{aligned} |U(b) - U_0(b)| \le{}& |\mathcal{M}(U)(c) - \mathcal{M}(U_0)(c)| + d_T(b_-) + d_T(b_+) \\ &+ C_L\{\varpi(U) + \varpi(U_0)\}. \end{aligned} \tag{3.cccxix}$$

For a bond internal to one block, omit the coarse-bond difference. The coefficient of the coarse difference is exactly one, since it is multiplied only by unitary transports. In a comparison where the contour means agree the two d_T terms vanish. If only the combs agree, eq. 3.cccxvi absorbs them into the curvature error.

An iteration with this coefficient is qualitatively different from an iteration with an unspecified constant in front of the coarse error. If a bond is in a pillar of height j, and the regularity at intermediate index $n < j$ is $\varpi_n \le \varepsilon L^{2(n-j)}$, eq. 3.cccxix yields, along an aligned descent,

$$e_n \le e_{n+1} + C_L \varepsilon L^{2(n-j)}, \qquad e_n \le e_j + \frac{C_L \varepsilon}{L^2 - 1}. \tag{3.cccxx}$$

The second inequality is obtained by summing the first. An uncontrolled endpoint contour defect would have to be added at each step; it cannot be omitted. For the preliminary gauge on a separated real sequence, let $W_n = \mathcal{M}^n U$ and $W_{0,n} = \mathcal{M}^n U_0$. Begin with $g_k = 1$ and descend. Set $g_n = 1$ on the tops Λ_n of that layer. On the block of an interior next-level center y, set

$$g_n(x) = W_{0,n}(y, x)^{-1} g_{n+1}(y) W_n(y, x).$$

These blocks and the layer tops partition the level-n sites of Ω_n. Since the contour from y to itself is one, the formula extends g_{n+1} and gives a single gauge on the bare sites. Covariance of the mean then makes the two contour values equal in every such block. The interior and outward determining bonds have both endpoints at tops, so their original datum mismatch is unchanged.

At an inward bond orient it towards the seam node. Its corridor belongs to the outward bond one level above, whose outer endpoint block lies wholly in Λ_n. The inward end uses the aligned contour. At the outer end each contour reads at most $2(L-1)$ interior bonds of the recorded block. With a mismatch at most α_1 on the proper determining rows, telescoping along the path and replacing the mean by the path costs at most $2(L-1)\alpha_1 + C_L\alpha_0$. Here α_0 bounds the regularity radius and the double block lies in Ω_n. Including the outward bond one level above gives the inward bound $(2L-1)\alpha_1 + C_L\alpha_0$. This applies whether the inward row is retained or omitted. For a bond with both ends in a deeper layer, the endpoint blocks follow the same largest common layer as the bond, by alignment. Its double-block curvature at level n is bounded by $C\alpha_0 L^{2(n-j)}$ when j is that largest layer. Applying eq. 3.cccxx from the proper or inward bound at j then gives a uniform bound $C_L(\alpha_0 + \alpha_1)$ at every carried link.

The estimate in eq. 3.cccxx is a bound on raw relative group elements. It does not divide those differences by the fine spacing and does not prove a bound on a first difference quotient. The elliptic slice estimate used later is precisely what improves this preliminary closeness to the natural physical norm.

3.11.4.1. The sharp linear curvature factor. A rough application of eq. 3.cccxv to each coarse edge gives $C_L\varpi$, which is not enough for a uniform iteration in physical units. The leading coefficient can be computed exactly. On the same finite rectangular patch take a comb tree gauge. Each remaining bond is a comparison loop with the tree, hence is within $C_L\varpi$ of one. For small ϖ write it as e^{ia_b} with $\|a\|_\infty \le C_L\varpi$. Taylor expansion of the finite products gives

$$|\mathrm{d}a(p)| \le \varpi + C_L\varpi^2, \qquad \log \partial\mathcal{M}(e^{ia})(p')/i = \mathrm{d}_c(P_L a)(p') + O(C_L\varpi^2). \tag{3.cccxxi}$$

The first linear term is the discrete curl; every other product term has at least two a's. The same is true of the finite implicit mean and logarithm on their fixed chart. These statements are estimates for a fixed number of coordinates depending on L, so the quadratic constants are finite and independent of depth.

The flat identity $P_L = S_L - \mathrm{d}_c\Sigma_L$ implies $\mathrm{d}_c P_L = \mathrm{d}_c S_L$. For a $\mu\nu$ coarse plaquette, direct telescoping of the straight sums gives the exact flux formula

$$(\mathrm{d}_c S_L a)_{\mu\nu}(y) = L^{-4} \sum_{x \in B(y)} \sum_{r,s=0}^{L-1} (\mathrm{d}a)_{\mu\nu}(x + re_\mu + se_\nu). \tag{3.cccxxii}$$

For example the difference of the two ν-edge sums separated by Le_μ telescopes in the μ direction; the other two edge sums telescope in ν with the opposite sign. Their combination is the displayed sheet curl. There are L^4L^2 terms with weight L^{-4}, so its supremum norm is at most $L^2\|\mathrm{d}a\|_\infty$.

Finally the exponential changes a small coarse logarithmic curvature into its group value with a quadratic error. Together with eq. 3.cccxxi this proves the real one-step estimate

$$\varpi(\mathcal{M}U) \le L^2\varpi(U) + C_L\varpi(U)^2. \tag{3.cccxxiii}$$

The tree gauge was local and unitary, so the conclusion holds in the original frame.

For a homogeneous regular patch let $\varpi(U) \le \varepsilon L^{-2k}$. Suppose all determining neighborhoods stay in that patch. Put $v_n = L^{2(k-n)}\varpi(\mathcal{M}^n U)$. eq. 3.cccxxiii gives

$$v_{n+1} \le v_n + C_L L^{-2(k-n)-2} v_n^2. \tag{3.cccxxiv}$$

The coefficients in this nonlinear recursion have a sum bounded independently of k. Choose ε small enough that four times this sum times ε is less than one. Inductively, if $v_q \le 2\varepsilon$ up to n, summing the recursion gives $v_{n+1} < 2\varepsilon$. Thus

$$|\partial\mathcal{M}^n U - 1| \le 2\varepsilon L^{2(n-k)} \quad (0 \le n \le k). \tag{3.cccxxv}$$

The same induction establishes admission to every one-step regular mean branch, by reducing ε below its fixed threshold. On a proper region it is used only at points whose full determining neighborhood remains in the region.

Across a seam the comparison uses the finer of the two adjacent length scales. A curvature measured in the coarser scale can therefore lose a fixed factor L^2. The geometry of eq. 3.cccvi permits that local loss and no chain of unexamined intermediate layers in one row. Proving an entire tower or joined-region statement still requires its determining neighborhoods to have the advertised alignment and collars. The formula in eq. 3.cccxxv is the averaging estimate; it does not solve a constrained variational problem.

3.11.5. *A domain specified before the unknown background.* An inverse estimate cannot be used to define its own range of application. We therefore first specify a class of possible reference fields without referring to the unknown solution. On each fixed determining patch use a real unitary local gauge and ask that its elementary plaquettes satisfy the bound in eq. 3.cccv, with a small specified constant ϵ_0. The bound is a test of the given field. Its coarse tests are obtained from the actual averages. For real homogeneous subpatches eq. 3.cccxxv proves admission to their regular branches. At a seam the preceding two-scale neighborhood is tested, and its finite averaging factors are tested in the same regular chart. The latter tests are explicit finite group inequalities, not assertions about the critical field.

Let $\mathcal{R}_{\mathrm{lin}}(\epsilon_0, \gamma_0)$ consist of such real fields and independent currents with

$$\sup_{p\subset\Omega_0} \eta^{-2}\mathsf{L}(p)^2|\partial U(p) - 1| < \epsilon_0, \qquad \sup_b \mathsf{L}(b)^3|J(b)| < \gamma_0. \tag{3.cccxxvi}$$

Include the actual finite-chart and row-rank inequalities needed by the chosen slice, and name their radii. These conditions can be checked on (U, J) before stationarity is imposed. For a physical trial field the current used in a physical curved-fibre Hessian is $J_A(U)$. A separately prescribed J in eq. 3.cccxxvi is not set equal to it. The two domains may coincide on a physical submanifold while their off-submanifold completions differ.

Complex fields are introduced relative to one such real reference by $U' = e^{i\eta A'}U$. Their domain records, separately, bounds on the physical logarithm and on every actual finite averaging or gauge formula used:

$$\|\mathsf{L}A'\|_\infty < r_A, \qquad \|\mathsf{L}^2 D_U A'\|_\infty < r_D, \qquad \|\mathsf{L}^3 J'\|_\infty < r_J, \qquad \mathcal{Q}_\beta(A', J') < r_\beta. \tag{3.cccxxvii}$$

Here $\mathcal{Q}_\beta$ ranges over a declared finite list of local readouts and atomic branch tests, including the outward datum factors, slice maps, current readouts and inverses actually used. The tests are normalized by length and scale, not by their target radii. The separate inequalities and their margins are essential: a point satisfying the curvature inequality need not satisfy a complex logarithm or multiplier inequality.

The linearized operator has to be defined on this primitive domain. If its definition uses a nonlinear background at a smaller scale, that background must already have been constructed in the induction, with its own admitted domain and margins. It cannot use the background presently being solved for. The induction is a directed finite list: an action or readout at level j reads only completed levels below j, and its new constraint rows are the specified level-j rows. This is a condition on the actual construction, not on notation.

There are two useful consequences of this arrangement. A primitive matrix theorem on eq. 3.cccxxvi–eq. 3.cccxxvii can be applied at a noncritical trial field. Also the complex inverse can be continued on an entire admitted ball by the two-sided Schur argument, rather than being asserted only along the eventual solution. Both consequences are needed in a first-exit proof of admission.
3.11.5.1. A regular section and what it must prove. A regular datum is defined by its elementary and two-scale test loops. Before the variational construction one needs a configuration realizing those data sufficiently accurately, or exactly, in the primitive domain. There are four concrete tasks in such a construction.

First, on the finite coarse patches, choose local unitary frames from the datum transports. Their transition matrices must obey the actual node cocycle. Products around contractible test loops are small, while noncontractible holonomies are retained as unitary transition matrices. Replacing them by the identity would change the fibre.

Second, extend the local coarse connection through a block with smooth fields on its free coordinates. The required estimates are on the physical potential, its first differences and its current or covariant Laplacian, at that block's scale. A raw face-supported lift need not have these estimates. For instance a jump in its plaquette field on a coarse face can produce a fine current of order η^{-1} times that jump.

Third, evaluate the *actual* finite mean of this trial field. The discrepancy from the datum is a named local function $\mathfrak{e}(B)$, not the discrepancy of a linear surrogate. The finite chart gives its derivative and quadratic remainder. The linear row right inverse must map this discrepancy into the same potential and current norms.

Fourth, the local corrections must agree on overlaps and leave previously frozen rows unchanged. At a seam this includes the outward factor in eq. 3.cccxi and its interior dependence, not merely the interior average.

The nonlinear correction in the third task is elementary once its linear and finite coefficient estimates have been proved. In declared field and datum norms let

$$\begin{aligned} q(A) &= Q_0 A + N_q(A), \qquad Q_0 E = 1, \qquad \|Eb\|_X \le C_E \|b\|_Y, \\ \|N_q(A) - N_q(A')\|_Y &\le C_q(\|A\|_X + \|A'\|_X)\|A - A'\|_X. \end{aligned} \tag{3.cccxxviii}$$

All maps are the actual constraint and its actual derivative. Writing $A = Eu$ reduces $q(A) = b$ to

$$u = b - N_q(Eu). \tag{3.cccxxix}$$

On $\|u\|_Y \le 2\|b\|_Y$ the right side is a self-map and has Lipschitz constant at most $4C_qC_E^2\|b\|_Y$. If that number is less than one half, successive substitution converges geometrically, uniformly in the finite lattice. The resulting correction is at most $2C_E\|b\|_Y$ in X. For complex data, uniform convergence on a smaller open ball makes it holomorphic: each iterate is holomorphic, and applying continuous linear functionals and Cauchy's integral formula proves holomorphy of the limit.

This proves the constraint correction from eq. 3.cccxxviii. It does not prove the scale-uniform right-inverse estimate, the smoothed trial estimate, or the overlap cocycle just listed. Those are bounded geometric and operator tasks for the actual map. The target is that their constants are independent of depth, volume, region shape and real holonomy, for the fixed allowed local geometry. A proof for only one block realizes the corresponding first-step instance.

3.11.6. *The actual critical equations.* Use fixed reference coordinates $U(A) = e^{i\eta A}U_0$, with the prescribed slice imposed. Write the constraint in the coordinates just specified as $q(A) = b(B)$, and

set $F(A) = A_{\Omega_0}(U(A))$. Here q is the chosen actual plain constraint, or the straightened constraint with the accompanying node factors and reduced datum identified below. These choices have different derivatives; every use of a linear inverse specifies which is used. All transposes below are bilinear cell-weighted transposes. For real positivity the pairing is Hermitian instead. In particular a change from raw bond logarithms to physical potentials also changes the dual current weights.

Let $J_A(U)$ be the physical left-variation current:

$$\frac{d}{dt} A_{\Omega_0}(e^{i\eta t h} U)\Big|_{t=0} = \eta^4 \sum_b (J_A(U)(b), h(b)). \tag{3.cccxxx}$$

For the reference-coordinate derivative this gives

$$DF(A)[h] = \eta^4 \sum_b (J_A(U(A))(b), \mathcal{E}_{\eta A(b)} h(b)). \tag{3.cccxxxi}$$

Indeed $(d/dt)e^{i\eta(A+th)}|_0 = i\eta \mathcal{E}_{\eta A}(h) e^{i\eta A}$. Thus the reference covector at nonzero A includes a right-logarithmic differential. Dropping this factor from a second derivative changes its current terms.

The coupled critical equations are

$$DF(A) - Dq(A)^\sharp \omega = 0, \qquad q(A) = b(B). \tag{3.cccxxxii}$$

Here ω is a coarse covector in the normalization dual to q. For raw constraints $q = mQ$, one has $\omega_Q = m^\sharp \omega$. The scalar $\langle \omega, D^2 q \rangle = \langle \omega_Q, D^2 Q \rangle$ is unchanged.

These equations are equivalent to constrained criticality when Dq is onto on the chosen slice. One direction follows by testing eq. 3.cccxxxii on $\ker Dq$. Conversely choose any right inverse E_A and suppose $DF(A)$ annihilates that kernel. Since $h - E_A Dq\, h \in \ker Dq$,

$$DF(A)[h] = DF(A)[E_A Dq\, h] = \langle E_A^\sharp DF(A), Dq\, h \rangle. \tag{3.cccxxxiii}$$

This proves the assertion and the uniqueness of the multiplier: if $Dq^\sharp \omega = 0$, testing on $E_A v$ gives $\langle \omega, v \rangle = 0$ for every v. Surjectivity and its quantitative bound are therefore genuine parts of the background theorem.

The Lagrangian Hessian is

$$K(A, \omega) = D^2 F(A) - \langle \omega, D^2 q(A) \rangle. \tag{3.cccxxxiv}$$

It is its restriction to tangent fields, not $D^2 F$ alone, that controls the critical fibre. This follows either by differentiating the multiplier equation or directly from a curve $q(A(t)) = b$: $Dq\, \ddot{A} = -D^2 q[\dot{A}, \dot{A}]$ and hence

$$\frac{d^2}{dt^2} F(A(t))\big|_0 = \langle \dot{A}, K(A, \omega) \dot{A} \rangle \quad \text{at a critical point.} \tag{3.cccxxxv}$$

At a noncritical point the same fibre coordinate has an additional term involving its acceleration and DF. For the exact pivot chart Φ, differentiating $q(\Phi(u)) = q(0)$ gives $D^2\Phi = -h\, D^2 q[D\Phi, D\Phi]$. Consequently the Hessian of $F \circ \Phi$ is the projected action Hessian minus the contraction with $h^\sharp DF$, with its correct physical current and chart differentials. This identity follows from the finite chart differentiation. The first-step positivity actually proved by the finite precision results is for $D^2[(F + \mathrm{GF}_{\mathrm{actual}}) \circ \Phi]$ at an admitted real block-axial trial background, under the stated first-step radius and current ceilings. The actual soft gauge term and its identification are retained; at the physical current this is the corresponding action-plus-gauge fibre Hessian. A positive action-only Hessian on a separately chosen slice would require its own slice estimate and is not imported from that result. Criticality is needed to identify the multiplier interpretation in eq. 3.cccxxxiv, not for the stated action-plus-gauge trial-field estimate.

3.11.6.1. Absolute and relative soft gauges. The internal transport is a mean of coordinate-order paths. The absolute gauge term is

$$\mathrm{GF}_{\mathrm{abs}}(U) = \sum_{y,x} \left(1 - \tfrac{1}{2} \operatorname{tr} T_{y,x}(U)\right). \tag{3.cccxxxvi}$$

Its full Hessian is the one in a physical first-step calculation on an axial field. If a construction instead uses the relative gauge with reference U,

$$\mathrm{GF}_{\mathrm{rel},U}(a) = \sum_{y,x} \left(1 - \tfrac{1}{2} \operatorname{tr}[T_{y,x}(e^{ia}U) T_{y,x}(U)^{-1}]\right), \tag{3.cccxxxvii}$$

its gradient at $a = 0$ vanishes for every background. If $G_U a$ is the right-logarithmic first derivative of its transport, its quadratic form is the corresponding positive sum of squares, $G_U^\sharp G_U$, with the actual trace normalization. The second transport derivative has zero contribution because it is paired with the derivative of the scalar function at the identity.

The absolute Hessian has the same form only where $T_{y,x}(U) = 1$. For example along $T(t) = e^{i(\theta+t)\sigma_3}$ the absolute second derivative of $1-\cos(\theta+t)$ is $\cos\theta$, whereas the relative second derivative is one. They differ at arbitrarily small nonzero θ. Both absolute and relative terms are actual finite formulas; one cannot select between them by positivity alone.

The relative expression is covariant under a joint site gauge of its reference and variable field: both transports transform bi-equivariantly, and their product transforms by conjugation at the center. The absolute expression has the corresponding invariance only when the endpoint and center gauge values agree, or when the prescribed gauge convention otherwise makes its transport a closed quantity. At a seam, removing bonds from a determining set can make the relative convention the appropriate one. A retained boundary datum may instead require the absolute convention. The local row definition must decide which formula is present.

Thus an independent-current precision theorem must compare its entire projected form with the physical or prescribed auxiliary one. Write $\mathcal{J} = DF$ for the physical comparison, and let $\mathcal{J}$ be the declared independent fine covector, after the same dual-coordinate conversion, for an auxiliary comparison:

$$C^\sharp[\Delta_{\mathrm{named}}(U,J) - D^2F(U) + \langle h^\sharp \mathcal{J}, D^2 q\rangle - K_{\mathrm{gauge,actual}}(U,J)]C = 0, \tag{3.cccxxxviii}$$

after every term has been expressed in the same raw or physical coordinates. Equivalently the difference may be estimated with an explicit error and charged to the gap. The displayed subtraction is schematic only in its *name*: each named term must have its literal local definition, including every projected extra-current term. The comparison is an equality of fully typed bilinear forms. Constraint-curvature matching alone checks only one summand. Zero-fold averaging maps do not imply that a separate projected correction $\Delta'_\pi[J]$ vanishes. The precision section retains this complete matching task.

3.11.7. *The action induced on the coarse datum.* Suppose the critical solution of eq. 3.cccxxxii has been constructed on an open real datum ball and admits differentiation there. The induced action is $S(B) = F(A(B))$. Let $Q = Dq(A)$, and let T denote the energetic tangent lift: $QT = 1$, while KTv annihilates $\ker Q$. The response equations give

$$\dot{A} = T\, Db\, \dot{B}, \qquad \dot{\omega} = \mathcal{M}\, Db\, \dot{B}, \qquad KT = Q^\sharp \mathcal{M}. \tag{3.cccxxxix}$$

The letter $\mathcal{M}$ here denotes the multiplier response, not the group averaging map. It is determined by these equations and is symmetric in the real bilinear pairing. Indeed $\langle v, \mathcal{M} w\rangle = \langle Tv, KTw\rangle$ by $QT = 1$.

Differentiating S and using eq. 3.cccxxxii proves

$$\begin{aligned} DS(B)[v] &= \langle \omega, Db\, v\rangle, \\ D^2S(B)[v,w] &= \langle Db\, v, \mathcal{M}\, Db\, w\rangle + \langle \omega, D^2b[v,w]\rangle. \end{aligned} \tag{3.cccxl}$$

In particular an outward seam has the second term even when its interior datum coordinates are linear. An argument using only $\mathcal{M}$ would lose this actual datum-curvature term. There is no such term in a chart where the datum itself is the constraint coordinate; converting that chart back to B restores it.

There is also an exact formula from an admitted soft completion. Let a be the coarse gauge or penalty matrix in that completion and assume

$$G = (K + Q^\sharp a Q)^{-1}, \qquad W = QGQ^\sharp \quad \text{are defined, with } W \text{ invertible.} \tag{3.cccxli}$$

Set $H = GQ^\sharp W^{-1}$. Then $QH = 1$, and

$$KH = Q^\sharp(W^{-1} - a), \qquad H = T, \qquad \mathcal{M} = W^{-1} - a. \tag{3.cccxlii}$$

The first identity follows by multiplying $(K + Q^\sharp a Q)G = 1$ by $Q^\sharp W^{-1}$. It proves the energetic property of H. Uniqueness follows from invertibility of the constrained tangent operator: the difference of two energetic lifts lies in $\ker Q$ and is orthogonal under K to that kernel. This proves eq. 3.cccxlii.

These are algebraic identities on the admitted domain. They supply no bound for W^{-1}, no positivity for a completion not actually present, and no identification of an independent-current operator. Once those primitive facts have been proved, however, the coarse readout and its correct subtraction are forced by eq. 3.cccxl–eq. 3.cccxlii. On the next scale the action is this S, that is the original fine Wilson action evaluated at its constrained critical background. It must not be replaced by the bare Wilson formula on the coarse variables.

3.11.8. *Straightening and the named current completion.* Plain, straight and straightened averages must remain distinct. Write $q_n^{\mathrm{st}}(A)$ for the nonlinear straightened raw constraint at level n. Its defining node factors satisfy

$$\mathcal{M}^n(e^{i\eta A}U_0) = \big(e^{iq_n^{\mathrm{st}}(A)}\mathcal{M}^n U_0\big)^{v_n[A]}. \tag{3.cccxliii}$$

For each top of level n, prescribe $u[A](y)v_n[A](y) = 1$. Interior rows then have physical datum $e^{iq_n^{\mathrm{st}}(A)}V_0$. At an outward endpoint of level $n-1$ the value prescribed for $u[A]$ uses v_{n-1}, not v_n. With the orientation of eq. 3.cccxi, covariance gives

$$e^{iq_n^{\mathrm{st}}(A)(c)} = e^{iB(c)}\,\mathrm{Ad}(V_0(c))[v_{n-1}[A](z)^{-1}v_n[A](z)]. \tag{3.cccxliv}$$

On the fully prescribed outer block the bracket is exactly the one-step factor $v_z(B_{\mathrm{int}})$ in eq. 3.cccxi. This follows by evaluating its defining one-step averaging identity on the already prescribed level-$(n-1)$ interior data. Therefore the straightened equation is $q^{\mathrm{st}}(A) = b(B)$ with the reduced datum, including its outward factors. eq. 3.cccxliv also checks orientation and multiplication order.

After an admitted Landau slice has been constructed, the physical representative is

$$\Phi(B) = (e^{i\eta\mathcal{H}(B)}U_0)^{u_B}, \qquad RD^\sharp\mathcal{H}(B) = 0, \qquad u_B v_n[\mathcal{H}(B)] = 1 \text{ at level-}n \text{ tops.} \tag{3.cccxlv}$$

An independent comb slice gives a different potential and a different gauge factor. It has the same physical orbit and the same reduced datum, but need not coincide with the pair in eq. 3.cccxlv. Equality requires that its potential already obey the Landau condition. A bound on $u_B - 1$ only at the tops cannot be extended to all bare sites without the separate comb gauge estimate.

Gauge invariance of the fine action makes $F(\mathcal{H}(B))$ the action of $\Phi(B)$. Constrained criticality is therefore described by eq. 3.cccxxxii on this slice, provided that the slice and node construction parametrize the local physical fibre modulo the residual group. The primitive gauge estimate must

prove that parametrization, its uniqueness and its admitted ball. The node identities above prove the datum correspondence; they alone do not give the quantitative gauge estimate.

For reference the actual named propagator family uses the linear part Q per unit row length of this straightened constraint. Let $D = D_U$ be the covariant site difference, let Q' be the corresponding site average to the tops, and let a have the actual row or site weight. On its admitted domain use the zero-extension site Laplacian Δ_{s}, and set

$$\begin{aligned} G' &= (\Delta_{\mathrm{s}} + Q'^{\sharp} a Q')^{-1}, \\ R &= 1 - G'Q'^{\sharp}(Q'G'^{2}Q'^{\sharp})^{-1}Q'G', \\ \mathsf{S} &= 1 - DG'RD^{\sharp}, \qquad \mathsf{G}_0^{-1} = \Delta_U^{\mathrm{W}} + DRD^{\sharp} + Q^{\sharp} a Q. \end{aligned} \tag{3.cccxlvi}$$

At a real field R is the declared orthogonal site projector; on complex fields the formula uses bilinear transposes and analytic continuation on its admitted domain. Its inverses and its relation to the chosen slice are estimates to be established, not consequences of displaying the formula.

Let $\Delta'_{\pi}[J]$ be the specified linear-current correction and let $\Delta^{(2)}[\mathsf{j}]$ be the constraint quadratic form, normalized by $\langle A, \Delta^{(2)}[\mathsf{j}]A\rangle = 2\langle C^{(2)}(A), \mathsf{j}\rangle$. The subsequent definitions are ordered:

$$\begin{aligned} \mathsf{G}^{-1} &= \mathsf{G}_0^{-1} - \Delta'_{\pi}[J], \\ \mathsf{H} &= \mathsf{G}Q^{\sharp}(Q\mathsf{G}Q^{\sharp})^{-1}, \\ \mathsf{G}_1^{-1} &= \mathsf{G}^{-1} - \mathsf{S}^{\sharp}\Delta^{(2)}[\mathsf{H}^{\sharp}J]\mathsf{S}, \\ \mathsf{H}_1 &= \mathsf{G}_1 Q^{\sharp}(Q\mathsf{G}_1 Q^{\sharp})^{-1}. \end{aligned} \tag{3.cccxlvii}$$

In particular the H inside the definition of G_1 is formed first. Replacing it by H_1 there gives a different implicit operator. Likewise Q in these formulas is not the plain derivative of $\mathcal{M}^n$, and the gauge term is not replaced by the contour soft gauge of the next step.

The next-step coarse form has the readout

$$\begin{aligned} \Delta_k &= (Q\mathsf{G}_1 Q^{\sharp})^{-1} - a - \mathsf{E}^{(2)}[\mathsf{H}_1^{\sharp}J], \\ \mathsf{A} &= \mathsf{C}_V^{\sharp}(\mathsf{G}_V^{(2)} + \Delta_k)\mathsf{C}_V, \qquad V = \mathcal{M}^k U. \end{aligned} \tag{3.cccxlviii}$$

Here C_V eliminates the actual plain-average pivots, $\mathsf{E}^{(2)}$ is that one-step plain constraint's quadratic form, and $\mathsf{G}_V^{(2)}$ is the prescribed soft contour gauge there. The last current contraction is the datum-curvature subtraction obtained when the coarse datum is itself restricted to the next-step fibre. It is not the earlier straightened $\Delta^{(2)}$.

On a physical critical pair, identifying G_1^{-1} with the actual Lagrangian Hessian plus its actual gauge and row terms makes eq. 3.cccxl–eq. 3.cccxlii the derivation of this coarse readout. If an alternative expression subtracts an additional $\mathsf{H}_1^{\sharp}DRD^{\sharp}\mathsf{H}_1$, it agrees only after that term has been shown to vanish, for example by the proved physical slice relation $RD^{\sharp}\mathsf{H}_1 = 0$. There is no corresponding automatic identity for an independent J.

Thus the precision problem contains two different tasks: prove bounds for the literal ordered family in eq. 3.cccxlvi–eq. 3.cccxlviii on its primitive independent-current domain, and identify its physical readout with the Hessian of the induced action. The latter uses the full projected-current and gauge comparison, not merely the constraint quadratic term. A proof of the first-step Wilson curved-fibre gap does not by itself perform either multilevel task.

3.11.9. *A quantitative coupled construction.* Here the primitive inverse and coefficient estimates are stated with their actual objects. This makes the existence argument a finite, checkable nonlinear reduction rather than an invocation of an unspecified implicit-function theorem.

Fix a real trial field U_0, not necessarily critical, in eq. 3.cccxxvi, with admitted constraint and slice charts. Let $Q_0 = Dq(0)$. Choose a coarse trial covector ω_0 and put

$$\begin{aligned} f_0 &= -DF(0) + Q_0^\sharp \omega_0, \qquad K_0 = D^2F(0) - \langle \omega_0, D^2q(0)\rangle, \\ \mathcal{S}_0 &= \begin{pmatrix} K_0 & -Q_0^\sharp \\ Q_0 & 0 \end{pmatrix}. \end{aligned} \tag{3.cccxlix}$$

The field is restricted to the fixed slice, and the force covectors are restricted to its dual. If the gauge kernel has not been quotiented, its gauge equation and its actual site inverse have to be included in this system. There is no inverse on an unremoved gauge kernel.

Let X, Z, Y, X^* be the declared field, multiplier, datum and force spaces with their physical weights. The primitive inverse estimate to be proved is a two-sided kernel estimate for $\mathcal{S}_0^{-1} : X^* \oplus Y \to X \oplus Z$. Its consequence used in this section is

$$\|\mathcal{S}_0^{-1}(f,b)\|_{X\oplus Z} \le C_0(\|f\|_{X^*} + \|b\|_Y), \tag{3.cccl}$$

with constants uniform over volume, depth, admissible regions, local scales and real holonomies. The precision section separates its gauge, tangent, row and seam estimates. It must also prove that the complete matrix in eq. 3.cccxlix, rather than a convenient surrogate, has this bound on the primitive trial class.

Write A for the field and $\nu = \omega - \omega_0$. Taylor's formula gives the exact equations

$$\begin{aligned} \mathcal{S}_0(A,\nu) &= \big(f_0 - R_F(A) + R_{q,\omega_0}(A) + (Dq(A) - Q_0)^\sharp \nu, \quad b - N_q(A)\big), \\ R_F(A) &= DF(A) - DF(0) - D^2F(0)A, \\ R_{q,\omega_0}(A) &= [Dq(A) - Q_0 - D^2q(0)A]^\sharp \omega_0, \\ N_q(A) &= q(A) - q(0) - Q_0 A. \end{aligned} \tag{3.ccli}$$

The right datum b here is measured from $q(0)$. The term linear in A and in ω_0 was included in K_0. Retaining it again on the right would double count the constraint curvature.

For clarity, the finite coefficient task is to verify, on the entire declared complex ball, the inequalities

$$\begin{aligned} \|R_F(A) - R_F(A')\|_{X^*} &\le c_F(r)\|A - A'\|_X, \\ \|R_{q,\omega_0}(A) - R_{q,\omega_0}(A')\|_{X^*} &\le c_{q\omega}(r)\|A - A'\|_X, \\ \|N_q(A) - N_q(A')\|_Y &\le c_q(r)\|A - A'\|_X, \\ \|(Dq(A) - Q_0)^\sharp \nu - (Dq(A') - Q_0)^\sharp \nu'\|_{X^*} &\le c_{\mathrm{mix}}(r)(\|A - A'\|_X + \|\nu - \nu'\|_Z). \end{aligned} \tag{3.ccclii}$$

Here $\|A\|_X, \|A'\|_X, \|\nu\|_Z, \|\nu'\|_Z \le r$. At a fixed finite lattice, Taylor integrals prove the inequalities with finite constants. For example

$$R_F(A) = \int_0^1 (1-t)\, D^3F(tA)[A,A]\, dt. \tag{3.ccliii}$$

The notation contracts two slots, leaving a covector. The analogous integral for $Dq - Q_0$ has one slot contracted, and that for R_{q,ω_0} has two. A uniform estimate must count the actual local coordinates in these tensors and use the force and field weights before summing. A scalar supremum bound times the total number of fine variables is not uniform in this problem.

Set $c(r) = c_F(r) + c_{q\omega}(r) + c_q(r) + c_{\mathrm{mix}}(r)$. Assume the verified inequalities give

$$C_0 c(r) \le \tfrac{1}{2}, \qquad C_0(\|f_0\|_{X^*} + \|b\|_Y) \le r/2. \tag{3.ccliv}$$

Applying $\mathcal{S}_0^{-1}$ to eq. 3.ccli defines a contraction of the closed radius-r ball into itself. Starting at zero, its successive iterates satisfy $\|x_{n+1} - x_n\| \le 2^{-n}\|x_1\|$; summing this geometric series proves convergence in $X \oplus Z$. Continuity of the map proves the equation. The same estimate on the difference of two fixed points proves uniqueness in that ball.

For two forcings whose actual maps share this ball, subtracting their fixed-point equations gives

$$\|A-\widetilde{A}\|_X+\|\nu-\widetilde{\nu}\|_Z\le 2C_0(\|f_0-\widetilde{f}_0\|_{X^*}+\|b-\widetilde{b}\|_Y) \tag{3.ccclv}$$

when their nonlinear coefficient maps agree. If those maps also change, add their evaluated difference on the common ball to the right-hand side. Its smallness must be proved from the actual coefficient comparison, not from a difference of the desired solutions.

The argument permits a noncritical trial field through the explicit f_0. It does not establish that an arbitrary regular critical field lies in this ball. That additional uniqueness statement needs a preliminary slice-closeness and local regularity argument. The raw group closeness eq. 3.cccxx is one ingredient, but does not by itself give X-closeness.

3.11.9.1. Whole-ball holomorphy and reality. Assume the finite products, means, logs, slice readouts and operator inverse are admitted on the whole complex ball used in eq. 3.cccli, with a fixed margin. If the datum parameter varies over an open disc for which eq. 3.ccccliv holds uniformly, all iterates of the fixed-point map are holomorphic and converge uniformly on smaller closed discs. Cauchy's formula on each finite collection of parameter directions then passes to the limit. The solution is holomorphic and has the same bound r. This proof does not require a volume-dependent radius.

For real reference data, conjugating the equations by $A\mapsto A^*$, $\omega\mapsto\omega^*$ and $g\mapsto(g^*)^{-1}$ preserves their finite formulas. The conjugate solution solves the equation at the conjugate datum and remains in the same ball. Uniqueness proves

$$A(B^{\mathfrak{c}})=A(B)^{\mathfrak{c}},\qquad \omega(B^{\mathfrak{c}})=\omega(B)^{\mathfrak{c}},\qquad \Phi(B^{\mathfrak{c}})=\Phi(B)^{\mathfrak{c}}. \tag{3.ccclvi}$$

At real B this is the required real solution. Gauge covariance is proved in the same way, provided the gauge maps both primitive domains and their slice normalizations to one another. The uniqueness theorem then identifies the transformed solution. For a complex gauge this proviso includes its condition number and every branch inequality; algebraic covariance alone does not make the complex core gauge saturated.

3.11.10. *Local gauges can be glued without a large holonomy assumption.* We next prove the gluing calculation used by a local regularity argument. Its hypotheses are local potential and first-difference estimates on patches of fixed size. Its conclusion does not include a componentwise Laplacian estimate; that estimate has a separate elliptic role.

Let a rectangular cube have side $m\xi$, with $\eta\le\xi$, and enlarge it by a fixed collar. Cover it by overlapping patches W_a of side $C\rho\xi$, whose centers lie on a rectangular grid of spacing comparable to $\rho\xi$. Here ρ is fixed. Assume bounded overlap and a partition of unity with

$$\sum_a\chi_a=1,\qquad \chi_a\ge 0,\qquad |\nabla^r\chi_a|\le C_r\xi^{-r}\quad (r=1,2). \tag{3.ccclvii}$$

Such a partition is obtained by sampling fixed smooth tensor bumps and dividing by their positive sum. The finite-difference product formula, or the mean value theorem applied to the smooth bumps, proves the bounds uniformly in η.

Suppose the real field has $|\partial U-1|\le\epsilon\eta^2\xi^{-2}$ and on each W_a there is a real unitary gauge k_a in which

$$U^{k_a}=e^{i\eta A_a},\qquad \xi|A_a|+\xi^2|\nabla A_a|\le C\epsilon. \tag{3.ccclviii}$$

On the real cubes used below, the elliptic section construction supplies eq. 3.ccclviii from curvature and full-current bounds. The actual current transforms by conjugation and can be carried along in the patch comparison.

We first align the constant frames. Choose a corner of the large rectangular lift as root. For each center y_a, choose the coordinate-order path from the root to y_a, and multiply k_a by the constant

$$c_a=U(\Gamma_{\mathrm{root},y_a})\,k_a(y_a)^{-1},\qquad \widehat{k}_a=c_ak_a. \tag{3.ccclix}$$

This does not change either norm in eq. 3.ccclviii, since it conjugates A_a by a constant unitary matrix. If $x\in W_a$, the transport of the locally gauged field from y_a to x along a path of length

$O(\rho\xi)$ differs from one by $C\epsilon$. The unitary product bound proves this by summing $\eta|A_a|$ on that path. Therefore $\widehat{k}_a(x)$ differs by $C\epsilon$ from the original transport along the root path followed by that local path.

For overlapping W_a, W_b, those two rooted paths differ by a strip, not an arbitrary surface spanning the whole large cube. To verify this, change their center coordinates one at a time. The difference between the centers is $O(\rho\xi)$ in each coordinate. Commuting this short displacement through the four ordered root segments sweeps at most $Cm\rho\xi^2$ area. The extra local paths sweep only $C\rho^2\xi^2$. The lattice interchange proof of eq. 3.cccxv, now counting fine plaquettes in these strips, gives

$$|\widehat{k}_a(x)\widehat{k}_b(x)^{-1} - 1| \le C_\rho m\epsilon \quad (x \in W_a \cap W_b). \tag{3.ccclx}$$

The estimate is valid on the rectangular lift that contains those paths. A periodic cube wider than half the torus is excluded from this local assertion. No global Wilson line was replaced by one; the possibly large root transports cancel in a comparison loop and conjugate its plaquettes unitarily.

The transition also has controlled derivatives. Put $r_{ab} = \widehat{k}_a\widehat{k}_b^{-1}$. For an oriented fine bond, covariance of U says

$$r_{ab}(x + \eta e_\nu) = e^{-i\eta\widehat{A}_{a,\nu}(x)} r_{ab}(x) e^{i\eta\widehat{A}_{b,\nu}(x)}. \tag{3.ccclxi}$$

Subtract $r_{ab}(x)$ and divide by η. Unitarity and eq. 3.ccclviii give $|\nabla r_{ab}| \le C\epsilon/\xi$. Taking another difference of this exact formula uses only $\nabla A_a, \nabla A_b, \nabla r_{ab}$. The derivative bound for the exponential on a fixed small raw ball gives

$$|\nabla r_{ab}| \le C\epsilon\xi^{-1}, \qquad |\nabla^2 r_{ab}| \le C(\epsilon + \epsilon^2)\xi^{-2}. \tag{3.ccclxii}$$

This proves the two derivative bounds without differentiating the long root paths.

Form $X(x) = \sum_a \chi_a(x)\widehat{k}_a(x)$. A real linear combination of SU(2) matrices is a quaternionic matrix. Its determinant is the squared Euclidean norm of its four real quaternion coordinates, so, when it is nonzero,

$$\mathcal{N}(X) = X/\sqrt{\det X} \in \mathrm{SU}(2). \tag{3.ccclxiii}$$

At a point select any active anchor a_0, and write $X = Y\widehat{k}_{a_0}$, where $Y = \sum_a \chi_a r_{aa_0}$. eq. 3.ccclx gives $\|Y - 1\| \le C_\rho m\epsilon$. If this is less than a sufficiently small fixed number, then $\det Y$ stays in a neighborhood of one not meeting zero. Thus $\mathcal{N}(X)$ is defined, independently of the selected anchor, and

$$k = \mathcal{N}(X) = \mathcal{N}(Y)\widehat{k}_{a_0}, \qquad |\mathcal{N}(Y) - 1| + \xi|\nabla\mathcal{N}(Y)| + \xi^2|\nabla^2\mathcal{N}(Y)| \le C_\rho m\epsilon. \tag{3.ccclxiv}$$

The derivative estimates follow from eq. 3.ccclvii, eq. 3.ccclx–eq. 3.ccclxii and the uniformly bounded first two derivatives of $Y \mapsto Y/\sqrt{\det Y}$ near one. When taking differences one keeps the anchor on a neighborhood containing those differences; the bounded enlarged overlap makes this possible. The resulting k itself does not depend on that choice.

In the anchor frame, U^k is

$$\mathcal{N}(Y)(x)\, e^{i\eta\widehat{A}_{a_0}(b)} \mathcal{N}(Y)(x + \eta e_\nu)^{-1}. \tag{3.ccclxv}$$

The relative endpoint product is $1 + O(\eta m\epsilon/\xi)$. The admitted small logarithm therefore defines A. Taking one difference, using eq. 3.ccclxiv, gives

$$U^k = e^{i\eta A}, \qquad \xi|A| + \xi^2|\nabla A| \le C_\rho m\epsilon. \tag{3.ccclxvi}$$

These bounds can be verified directly by differentiating the logarithm on its small fixed raw ball: the endpoint increment divided by η is $O(m\epsilon/\xi)$; its first difference is $O(m\epsilon/\xi^2)$. All remainders have an extra factor $\eta m\epsilon/\xi \le m\epsilon$, which is small. If $\xi^3|J_A(U)| \le \epsilon$, the same bound holds after k, since the current is conjugated at its fine endpoint. In particular no factor m is needed for that covariant current bound.

The normalization has an admitted holomorphic version. If the local gauges depend holomorphically on parameters and their anchored Y's remain in the same small complex neighborhood of one,

the branch of $\sqrt{\det Y}$ is the principal branch there. The formula is holomorphic and the relative normalizations agree on overlaps by the exact identity $\det \widehat{k}_a = 1$. For real gauges it is the positive square root. No absolute value of a complex determinant occurs.

The overlap argument gives the potential, first difference and covariant current bounds. For the componentwise Laplacian of this particular patched potential one would need a further estimate. The real cubes in the layered profile instead use the single gauge constructed below. Given the componentwise Laplacian bound, the discrete interior estimates give

$$[\nabla A]_{\alpha;\square} \le C_{\alpha,\rho} m\epsilon\, \xi^{-2-\alpha}, \qquad 0 < \alpha < 1. \tag{3.ccclxvii}$$

Here the input is the bounded potential and componentwise Laplacian on the larger cube, with their displayed physical scales. The constant may diverge as $\alpha \uparrow 1$. Thus this Hölder conclusion follows from those two bounds.

3.11.10.1. A single real gauge with second-order bounds. Work first in units of the cube scale, with bare spacing $\eta \le 1/13$. On a lifted box of side $s \ge 5$, assume $|\partial U - 1| \le \alpha\eta^2$ on its plaquettes and $|D_U^{\mathsf{T}} \partial U| \le \alpha\eta^2$ at its full bonds. Put $e = s\alpha$ and take it below a fixed small constant. The comb section ϕ_0 is unitary. Comparing the two comb paths at the endpoints of a bond inserts at most $3s/\eta$ plaquettes, so $\|D_0\phi_0\|_\infty \le 3e$. Here D_0 is the covariant difference of a section, formed only on bonds of the box. Smooth ϕ_0 by solving

$$(D_0^* D_0 + 1)\phi = \phi_0, \qquad \psi = D_0\phi.$$

The operator is positive and invertible in the real quaternionic space. At a site of maximum section norm the inner product of $D_0^* D_0 \phi$ with ϕ is nonnegative, by unitarity of each transport. It follows that $\|\phi\|_\infty \le 1$.

Differentiate the equation. Moving a transported difference past a perpendicular one inserts the corresponding plaquette holonomy. In the equation for ψ_μ the difference of the opposite plaquette contributions gives the covariant plaquette divergence. The remaining commutators have one curvature factor and one first difference of ϕ. At a missing perpendicular neighbor there is also an unpaired plaquette contribution of size $5\alpha/\eta$. More precisely, with $\Sigma = \|\psi\|_\infty$, the modulus is a subsolution of

$$(\mathsf{L}_{0,\mu} + 1)|\psi_\mu|(x) \le \|D_0\phi_0\|_\infty + 2\alpha + 12\alpha\Sigma + 5\alpha\eta^{-1} n_\mu(x),$$

where n_μ counts the missing perpendicular neighbors. The scalar bond Laplacian $\mathsf{L}_{0,\mu}$ uses only the available perpendicular neighbors and zero extension along direction μ. This inequality follows by taking the inner product with the unit direction of ψ_μ, using unitarity on the transported terms. At a zero the scalar left-hand side is nonpositive. To pay the boundary term, for each perpendicular direction take the one-dimensional function on N sites

$$\beta(l) = \frac{\cosh(\kappa\eta(l - (N-1)/2))}{\sinh(\kappa\eta N/2)}, \qquad 2\sinh(\kappa\eta/2) = \eta.$$

It is bounded by two, solves the massive homogeneous difference equation in the interior and gives η^{-1} at each endpoint with the absent neighbor omitted. The sum of the perpendicular barriers is a supersolution for $\eta^{-1} n_\mu$, because the zero extension in direction μ adds a nonnegative term. Scalar comparison yields $\Sigma \le \|D_0\phi_0\|_\infty + 32\alpha + 12\alpha\Sigma$. In particular $\Sigma \le 19e$ for the admitted smallness.

In the comb frame, $\phi_0^{-1}(\phi - \phi_0)$ is the flat divergence of $\phi_0^{-1}\psi$, up to an error of size $4\|D_0\phi_0\|_\infty\Sigma$. Average on a subbox of unit side up to one mesh spacing. The divergence telescopes in each coordinate, so its mean is at most 8Σ. Across one bond the section difference changes by at most $\eta(\Sigma + \|D_0\phi_0\|_\infty)$. Comparing the mean with the value at an included site gives

$$\|\phi - \phi_0\|_\infty \le 12\Sigma + 4\|D_0\phi_0\|_\infty + 4\|D_0\phi_0\|_\infty\Sigma.$$

Thus $\rho_\phi = |\phi| \ge 1/2$ and $u = \phi/\rho_\phi$ is a real gauge. On the bond $b = (x, x + \eta e_\mu)$ put $\omega_\mu(x) = \rho_\phi(x)^{-1} u(x)^{-1} \psi_\mu(x)$. The transformed link is $(1 + \eta\omega_\mu)/|1 + \eta\omega_\mu|$. In the interior the product rule

in the differentiated section equation gives

$$|\Delta_\eta \omega_\mu(x)| \le 102e + 304e \max_{x' \sim x} \eta^{-1} |\omega_\mu(x') - \omega_\mu(x)|.$$

Indeed the transported Laplacian of ψ is bounded by $\Sigma + \|D_0\phi_0\|_\infty + 2\alpha + 12\alpha\Sigma$ there. The derivative of the normalizing section contributes one term bounded by $|\phi - \phi_0||\omega|$ and an edge sum bounded by $8\Sigma \max_{x' \sim x} |\omega(x') - \omega(x)|/\eta$ before division by ρ_ϕ. To absorb the first-order term, maximize the size of a first difference multiplied by the smaller of one and the distance to the boundary. When this distance is at most 64η, use the sup bound for ω. Otherwise use the interior gradient estimate in a cube of radius one quarter of the weighted distance. The distances there and on the neighboring stencil are at least half the maximizing distance. If H is the maximum, the gradient estimate gives $H \le Ce + CeH$. Decrease the ceiling for e to absorb the last term. At distance at least one this gives $|\nabla\omega| \le Ce$, and the equation bounds $|\Delta_\eta\omega|$ on the next interior stencil by Ce.

Take the small logarithm of the normalized link, divided by $i\eta$, to obtain A. Its first difference is controlled by $\nabla\omega$, and the Laplacian has one additional chain-rule term bounded by $C\eta|\nabla\omega|^2$. The flat curl divergence is obtained separately. Expand the four factors of a plaquette to first order and take the transported differences of the three opposite plaquette pairs of a full bond. The transported linear parts give $\partial^\sharp \partial A$ up to products of A and its first differences. In the nonlinear parts the difference of opposite remainders is bounded by $CA_0A_1\eta^3$, and transport of a quadratic remainder costs $CA_0^3\eta^3$. Here A_0, A_1 are stencil suprema of $|A|, |\nabla A|$. The resulting bound is

$$|\partial^\sharp \partial A|(b) \le \eta^{-2} |(D_U^{\mathsf{T}} \partial U)(b)| + C(A_0A_1 + A_0^3).$$

Both second-order bounds are therefore Ce on the prescribed interior. This supplies the real gauge in eq. 3.ccclviii with the full current hypothesis.

3.11.10.2. The cubes used in the layered profile. Let the target cube have side $2m'\xi_j$, with m' a multiple of the separation radius ρ. Its concentric enlargement of side $(2m' + 4\rho)\xi_j$ lies in Ω_j and is shorter than half the torus. Suppose the curvature and full-current radii on elements meeting that enlargement are at most ϵ. In level-j units the radius bound gives $|\partial U - 1| \le \epsilon\eta^2$ and $|J(U)| \le \epsilon$. Writing $\partial U(p) = c(p)1 + i\eta^2 F(p)$, the scalar coefficient satisfies $1 - c(p) = |\partial U(p) - 1|^2/2$. In the transported divergence the imaginary coefficient is $i\eta^2 J(U)$. The scalar difference of each opposite pair is bounded by $\epsilon^2\eta^4/2$, so at a full bond

$$|D_U^{\mathsf{T}} \partial U| \le (\epsilon + \tfrac{3}{2}\epsilon^2)\eta^2.$$

For $j \ge 1$, use a cube of side $2m' + 2\mathfrak{r}_{\mathrm{p}}$ aligned to the $\mathfrak{r}_{\mathrm{p}}$ grid. Its left face in each coordinate is the last grid face at least four units to the left of the target. The cube then contains the full four-unit neighborhood of the target. Its further concentric enlargement by $2L\mathfrak{r}_{\mathrm{p}}$ on each side and the full plaquette stencils fit in the original 2ρ margin. These inclusions use the separation floor and $\mathfrak{r}_{\mathrm{p}} \ge 13$. Apply the real cube gauge with $\alpha = 2\epsilon$. The side of the aligned cube is at most $3m'$, so $m'\epsilon \le c$ gives all four bounds of size $Cm'\epsilon$ on that cube. At level zero the mesh is one in these units. The comb gauge already bounds the potential by $Cm'\epsilon$, and each difference in the displayed operators is a bounded stencil of these values.

For the gradient Hölder bound compare sites within distance one quarter in the scale units. Multiply the relevant component A_μ by a cutoff equal to one within distance one and supported within distance two of the first site. The cutoff and its neighbor stencils lie in the aligned cube. Write the result as $f = G'h$ for the constant-layer site inverse $G' = (\Delta + \varkappa_j Q'^* Q')^{-1}$, with Δ the positive flat Laplacian. The product rule and the four bounds give $\|h\|_\infty \le Cm'\epsilon$. The support of h meets at most 8^4 unit blocks. Apply the flat gradient Hölder bound to each supported part, using the same local gradient at both sites since the cutoff is one there and on their forward stencils. At larger distances the sup bound on the gradient suffices. Rescaling gives eq. 3.ccclxvii with $m = 2m'$. On a critical field the radius ϵ can be taken from eq. 3.ccclxxix on the entire enlargement.

3.11.11. *The site projector and the slice estimate.* Some of the gauge-slice algebra can be completed before the quantitative kernel bounds. Put $\Delta = \Delta_{\mathrm{s}}$, and suppose $G' = (\Delta + Q'^{\sharp} a Q')^{-1}$ exists, Q' is onto, and the site Gram matrix $B' = Q' G'^2 Q'^{\sharp}$ is invertible. At real data G' is symmetric. The formula in eq. 3.cccxlvi then makes R the orthogonal projection onto

$$\ker(Q'G') = \Delta \ker Q'. \tag{3.ccclxviii}$$

Indeed $G'Q'^{\sharp}(B')^{-1}Q'G'$ is the orthogonal projection onto $\mathrm{ran}(G'Q'^{\sharp})$. If $Q'\lambda = 0$, then $G'\Delta\lambda = \lambda$, so $Q'G'\Delta\lambda = 0$. Conversely Δ is injective on $\ker Q'$: if it annihilates such a λ, the invertible $\Delta + Q'^{\sharp} a Q'$ also annihilates it. The two subspaces in eq. 3.ccclxviii have the same dimension. This proves equality. On an admitted complex domain the identities remain bilinear matrix identities.

In particular

$$Q'G'R = 0, \qquad \Delta G'R = R, \qquad R\Delta G'R = R. \tag{3.ccclxix}$$

The first follows from the projector formula, and the second from $\Delta G' = 1 - Q'^{\sharp} a Q' G'$. For the actual regional sequence the site and inner-bond Laplacians satisfy $\Delta - D^{\sharp} D = \eta^{-2} n$, where $n(x)$ counts missing bare neighbors at the outer face. The separation puts these sites in the bare layer, with all their inner neighbors also in that layer. If f is supported on these sites, $G'^{-1} f$ is supported on bare tops. The site row there is evaluation, so $G'^{-1} f$ lies in $\mathrm{ran}\, Q'^{\sharp}$. Hence $Rf = 0$ and $RD^{\sharp} D = R\Delta$. Also $Q'\lambda = 0$ forces λ to vanish on the bare tops. These facts give

$$\mathsf{S}^2 = \mathsf{S}, \qquad RD^{\sharp}\mathsf{S} = 0, \qquad \mathsf{S}D\lambda = 0 \quad (Q'\lambda = 0). \tag{3.ccclxx}$$

For example the quadratic term in the square of $1 - DG'RD^{\sharp}$ reduces using $R\Delta G'R = R$. The last identity uses $\Delta\lambda \in \mathrm{ran} R$ and $G'\Delta\lambda = \lambda$. Thus S projects onto the fixed Landau slice along the infinitesimal residual gauges.

If the actual straightened rows and site readout satisfy $QD = \bar{D}Q'$, then also

$$Q\mathsf{S} = Q. \tag{3.ccclxxi}$$

This is immediate from $Q'G'R = 0$. The intertwining has to be checked with the actual outer-endpoint readout of every outward row. Scale-tagged endpoint values cannot be silently identified when they are averages at different levels. For plain rows the corresponding site readout is different even at a flat field. eq. 3.ccclxxi is therefore a consequence of the specified intertwining, not a reason to assume it.

There is a complete linear inverse for the site equation. Let E_0 be any admitted right inverse of Q'. For $f \in \mathrm{ran} R$ the unique solution of $R\Delta\lambda = f,\ Q'\lambda = b$ is

$$\lambda = G'f + (1 - G'R\Delta)E_0 b. \tag{3.ccclxxii}$$

The first constraint follows from $Q'G'R = 0$. For the other, eq. 3.ccclxix cancels the two terms $R\Delta E_0 b$. A homogeneous solution has $\lambda \in \ker Q'$ and $\Delta\lambda \in \mathrm{ran} R$, hence $\Delta\lambda = 0$, and is zero by the injectivity already proved. The formula is independent of the chosen E_0, since uniqueness identifies all choices.

For a real small field, the actual nonlinear gauge equations are the Landau condition on the logarithm of $U^g U_0^{-1}$ and the pillar-top normalization $g(y) = v_n[A](y)$ at the top of a pillar of level n. The retained seam nodes are handled by the outward factors in eq. 3.cccxliv. Their derivative consists of the site pair in eq. 3.ccclxxii, with the signs determined by the chosen infinitesimal gauge convention. The infinitesimal gauge is defined by $(d/dt)U^{e^{it\lambda}}|_0$, so the same sign is used in D, in the node derivative and in $\bar{D}$. The derivative of the node equation is the declared $Q'\lambda$, as checked from the straightening identity in eq. 3.cccxliii.

3.11.11.1. The real site correction. Use the weighted site inverse on the separated real sequence. Write $|f|_{(-2)}$ for the site sup norm with weight ξ^2. For $f \in \mathrm{ran}\, R$ and node data ν, let $\lambda(f,\nu)$ be the solution of $R\Delta\lambda = f$, $Q'\lambda = \nu$ given by eq. 3.ccclxxii. In the regional norms this solution and R satisfy

$$\|\lambda(f,\nu)\|_{Y_{\mathrm{site}}} \le C(|f|_{(-2)} + \|\nu\|_{\infty}), \qquad |Rf|_{(-2)} \le C|f|_{(-2)}.$$

The site norm includes the weighted first difference and the zero-extension Laplacian. It controls the corresponding norm $Y^{\circ}_{\mathrm{site}}$ with the inner-bond Laplacian $D^{\sharp}D$. Indeed their difference is $\eta^{-2}n$ on the bare layer, with $n \le 8$, so the comparison constant is at most nine.

Let A_0 be a preliminary real chart potential, and write A_0^w for the logarithm after applying $w = e^{i\lambda}$, with the reference U_0 kept fixed. For a top of level j the node factor of this change is $r_j = v_j[A_0^w]^{-1} w v_j[A_0]$. Let T be the node transport rotation in the straightened linearization. The nonlinear divergence and node equations use

$$D^{\sharp}A_0^w = D^{\sharp}A_0 - D^{\sharp}D\lambda + \mathcal{N}(A_0,\lambda), \qquad \mathsf{T}^{-1}\tfrac{1}{i}\log r_j = Q'\lambda + q(A_0,\lambda).$$

For the first remainder the logarithmic gauge change gives

$$|\mathcal{N}(A_0,\lambda)|_{(-2)} \le C(\|A_0\|_X + \|\lambda\|_{Y^{\circ}_{\mathrm{site}}})\|\lambda\|_{Y^{\circ}_{\mathrm{site}}}.$$

It also gives $|A_0^w - A_0|_{(-1)} \le C\|\lambda\|_{Y_{\mathrm{site}}}$. In the second remainder the straightened node derivative differs from Q' by the curvature of the reference and the variations of the averaging factors. If α_0 is the reference radius, the resulting bound is

$$\|q(A_0,\lambda)\|_{\infty} \le C(\alpha_0 + |A_0|_{(-1)} + \|\lambda\|_{Y_{\mathrm{site}}})\|\lambda\|_{Y_{\mathrm{site}}}.$$

The factor at a bare top is exactly w, so the node remainder there vanishes. These bounds use the logarithms on the full admitted ball and the literal recursive node factors.

Prescribe $r_j = \mathsf{r}$ on the corresponding tops and put $\nu_* = \mathsf{T}^{-1}(\log \mathsf{r})/i$. Let $m_0 = |D^{\sharp}A_0|_{(-2)} + \|\mathsf{r} - 1\|_{\infty}$. On the closed real ball of radius $C_0 m_0$ in the pair (f,ν), use the map

$$(f,\nu) \longmapsto \big(R[D^{\sharp}A_0 + \mathcal{N}(A_0,\lambda(f,\nu))], \nu_* - q(A_0,\lambda(f,\nu))\big).$$

Choose C_0 first to pay for the two linear forcing terms. Then decrease the ceiling for $\alpha_0, \|A_0\|_X$ and $\|\mathsf{r}-1\|_{\infty}$ so the remainders are at most half the ball radius. The continuous map sends the ball into itself, so the finite-dimensional fixed point gives the real correction. Use $RD^{\sharp}D = R\Delta$ to get $RD^{\sharp}A_0^w = 0$, and the second equation gives the prescribed top factors. The two estimates are $\|\lambda\|_{Y_{\mathrm{site}}} \le Cm_0$ and $|A_0^w - A_0|_{(-1)} \le Cm_0$. When $m_0 = 0$ take $\lambda = 0$. The construction specifies top factors. The retained seam factors follow from the row equations.

The a priori improvement in that slice is a different statement. It concerns *any* two regular configurations $U_0, e^{i\eta A}U_0$ already in the slice and with $\|\mathsf{L}A\|_{\infty}$ below a fixed admitted ceiling. Its target is

$$\|A\|_X + \|\mathsf{L}^3 D^{\sharp}DA\|_{\infty} \le B_1(\epsilon_0 + \|q^{\mathrm{st}}(A)\|_{\infty}), \tag{3.ccclxxiii}$$

where B_1 is independent of the depth and does not depend on the larger preliminary constant used to put A under the ceiling. No criticality of these two configurations is assumed. The currents of both configurations control the covariant second-order forcing. For the nonlinear group formula the forcing must be computed with the reference logarithmic differential eq. 3.cccxxxi.

Put $a_2 = \|\mathsf{L}A\|_{\infty}$, and let α_0 be the common regularity bound of the two configurations, in the normalization of the slice. Write β_0 for the maximum of the logarithmic row coordinates $|a_0(A)|$ on interior level-zero rows, and β_1 for the corresponding maximum of $|a_j(A)|$ on all interior and outward rows. Write $|f|_{(-3)} = \|\mathsf{L}^3 f\|_{\infty}$ for the bond norm. On a full bond the Wilson currents at the two configurations and the Taylor comparison of the logarithmic action bound the linear second difference by $3\alpha_0 + C_L a_2\|A\|_X$ in this norm. At an inner bond that is not full, every bond of its inner plaquettes is a level-zero interior bond. There $a_0(A)(c) = \eta A(c)$, so the stencil gives the separate bound $24\beta_0$. Thus, with $J' = D^{\sharp}DA$,

$$|J'|_{(-3)} \le \max\{3\alpha_0 + C_L a_2\|A\|_X, 24\beta_0\}.$$

In the real slice $RD^{\sharp}A = 0$. Straightening the proper rows gives $QA = \widehat{a} - C^{+}(A)$, where $\widehat{a}$ is the conjugated row coordinate, with the same norm, and $\|C^{+}(A)\|_{\infty} \le C_L(\alpha_0 + a_2)a_2$. For data on these rows, extended by zero to all other rows, $|Q^{*}\omega|_{(-3)} \le 2(1 + L^{-3})\|\omega\|_{\infty}$. Indeed the finer

row that could read a bond in the next layer would be inward, so it is absent. The same-layer and next-coarser row counts give the two displayed weights.

Use the actual inverse G_0 of the regional Wilson Hessian completed by $DRD^\sharp + Q^*Q$. If Δ' is the Wilson Hessian minus $D^\sharp D$, the slice equation is

$$G_0^{-1}A = J' + \Delta' A + Q^*(\widehat{a} - C^+(A)), \qquad |\Delta' A|_{(-3)} \le C_L \alpha_0 a_2.$$

The weighted regional inverse bound and $a_2 \le \|A\|_X$ therefore give

$$\|A\|_X \le C_1(\alpha_0 + \beta_0 + \beta_1) + C_2(\alpha_0 + a_2)\|A\|_X. \tag{3.ccclxxiv}$$

For an interior row its logarithmic coordinate is the reduced datum q^{st}. At an outward row the additional seam factor is the block average at the outer end. It reads only interior rows one level below, so the group angle inequality gives $\beta_1 \le (1+\kappa_6)\|q^{\mathrm{st}}(A)\|_\infty$. In particular no bound on inward or edge rows is needed. With α_0, a_2 below a fixed ceiling for which the last coefficient is at most one half, absorption followed by the bound on J' proves eq. 3.ccclxxiii.

To reach that slice from a real regular pair, first align the mean contours level by level with the gauge equal to one on the final pillar tops. The contour comparison gives a relative link bound $C_{\mathrm{prod}}(\alpha_0 + \alpha_1)$ on the carried links at each level, where α_1 bounds the mismatch on the retained datum. Use the truncated sequences up to each level in turn. At level zero the link logarithm in this preliminary gauge has the same bound in the field norm, since the mesh is one in its units.

At ascent to level j replace the previous logarithm by $A_0 = LA^{(j-1)}$ in the new units. The products $\eta_j A_0$ and $\eta_{j-1}A^{(j-1)}$ are equal, so the old node factors are unchanged. The field norm costs at most L^2. At the old tops the correction target r is one. At the newly promoted tops the old representative uses $v_{j-1}[A_0]$ in place of $v_j[A_0]$. The correction target is the last one-step node factor $v_{j-1}[A_0]^{-1}v_j[A_0]$, with defect bounded by $C_L|A_0|_{(-1)}$ by the full block estimate. With this target the correction cancels the initial node mismatch. Its potential stays under the fixed ceiling when $C_L L^2$ times the previous field bound fits that ceiling.

The reduced proper rows now compare the two averaged data directly. The edge row is a quotient of two links, so the mismatch there costs at most twice the link bound. At intermediate levels use the preliminary bound $C_{\mathrm{prod}}(\alpha_0 + \alpha_1)$. At the final level use α_1 on the actual retained rows: the preliminary gauge fixes interior and outward endpoints and cancels at the common seam node of an edge row. Then eq. 3.ccclxxiii resets the norm at every ascent to a fixed multiple of $\alpha_0 + \alpha_1$. At the final seam the remaining node factor is fixed by its inward-link equation, with the retained endpoint and the appropriate orientation. In particular, when the retained data are the same and the coupled system identifies $A = 0$, the normalized gauge fixes the tops and that link equation fixes every retained seam node. This yields the residual orbit on the full retained datum.

3.11.12. *The local improvement and its weighted maximum argument.* Define the distance $d_{\mathbf{B}}$ by assigning a path the length $\int ds/\xi(s)$, where $\xi(s)$ is the local pillar scale, and taking the infimum over paths. One may equivalently use the corresponding weighted lattice path distance, with a fixed comparison constant. At a seam the adjacent scales differ by L; the comparison is made using the same convention in every kernel estimate. Test plaquettes are placed at a fixed corner or center, at bounded distance in this metric.

Let $r(x)$ be the maximum local dimensionless curvature and current radius of a regular critical field, as in eq. 3.cccv, and let $v(\tau) = |\partial \mathbf{V}(\tau) - 1|$ be its assigned datum tests. For a radius R in this metric write $r_R(x)$ for the supremum over the R-ball and $v_R(x)$ for the supremum of tests in that ball. The actual local improvement estimate to be established is

$$r(x) \le C v_R(x) + \{C\sigma^2 e^{-c\delta_*\sigma} + C\sigma^4 r_R(x)\}\, r_R(x), \qquad R \le C_R L\sigma. \tag{3.ccclxxv}$$

The constants $C, c, C_R, \delta_* > 0$ must be independent of σ, depth, volume and region shape. The patch must contain the actual determining neighborhood of its cutoff fields and rows. This is a critical-field estimate. It is not an inequality for every field with small plaquettes.

The local proof is given under the recovery of the regularity radius. It uses an auxiliary separated sequence whose positive-level members are inside the trivialized cube. Extend the local potential by zero on the auxiliary torus. Constrained variations of this extension patch to constrained variations of the original field, which is how the critical equation is obtained in the auxiliary flat slice. The current created by the extension is paid in the bare-scale norm outside the positive-level members.

The datum gauge costs one further factor of σ. The datum and preliminary slice bounds are both at most $C\sigma^2 r_R(x)$. Split the flat kernel at the inner cube, then make the same split in the multiplier equation. The central datum rows grow only linearly with distance from the observation, while the outer rows and outer differentiated fields are at distance at least $c\sigma$. The field and row remainders have two small factors in the preliminary slice. Thus the contribution of the central datum, the near remainders and the far reads to the plaquette and current estimates is bounded, respectively, by

$$Cv_R(x), \qquad C\sigma^4 r_R(x)^2, \qquad C\sigma^2 e^{-c\delta_* \sigma} r_R(x). \tag{3.ccclxxvi}$$

This calculation is on the real separated geometry with the auxiliary flat-slice and local gauge estimates. Here the distance to a tested plaquette is the minimum over its corners. When an inward row is kept one also counts the pinch tests, with distance measured to their seam node. On a determining set in force all inward rows have been removed, so only tested plaquettes remain.

We now give the entire maximum argument from eq. 3.ccclxxv. Choose σ so large that $C\sigma^2 e^{-c\delta_*\sigma} \le 1/8$. Then choose the regularity ceiling r_* so small that $C\sigma^4 r_* \le 1/8$. For $r \le r_*$, eq. 3.ccclxxv becomes

$$r(x) \le Cv_R(x) + \theta r_R(x), \qquad \theta \le \tfrac{1}{4}. \tag{3.ccclxxvii}$$

Choose $\delta > 0$ with $\theta e^{\delta R} \le 1/2$, and define

$$\widetilde{v}_\delta(x) = \sup_\tau e^{-\delta d_{\mathbf{B}}(x,\tau)} v(\tau). \tag{3.ccclxxviii}$$

The triangle inequality gives $\widetilde{v}_\delta(y) \le e^{\delta d_{\mathbf{B}}(x,y)} \widetilde{v}_\delta(x)$. In particular $v_R(x) \le e^{\delta R} \widetilde{v}_\delta(x)$.

For $s > 0$ let $M_s = \sup_x r(x)/(\widetilde{v}_\delta(x) + s)$. It is finite since $r \le r_*$. eq. 3.ccclxxvii gives

$$M_s \le Ce^{\delta R} + \theta e^{\delta R} M_s, \qquad r(x) \le 2Ce^{\delta R} \widetilde{v}_\delta(x). \tag{3.ccclxxix}$$

The second inequality follows by absorption and $s \downarrow 0$. No maximum has to be attained, so the proof works on infinite volume as well when the input radius is bounded. If the datum tests in layer q are at most $\epsilon^{(q)}$, then

$$r(x) \le C' \sup_q \epsilon^{(q)} e^{-\delta d_{\mathbf{B}}(x,\text{layer } q)}. \tag{3.ccclxxx}$$

This proves the profile assertion from the local improvement, with its exact choice order.

The unweighted estimate eq. 3.ccclxxvi closes the radius in the admitted real induction. If a provisional construction is regular below the input ceiling, its output radius is $C'\|v\|_\infty$, independently of the larger provisional constant. The conditions for applying that improvement are checked before it is used. One cannot use the smaller output radius to justify a construction already outside its input ceiling.

If $v = 0$, eq. 3.ccclxxix proves that curvature and current vanish. This alone does not identify global holonomies. If, in addition, the entire retained datum is trivial, $M_{\mathbf{B}}(U) = 1$, the separated determining rows eq. 3.cccvi imply $U = 1^g$ for $g \in \mathcal{G}_{\mathrm{res}}$, including when Ω_0 has holes or several components.

Here is the geometric argument. Extend U by one outside the inner bonds of Ω_0. A plaquette meeting its complement has no corner in Ω_1, by separation; each of its bonds is therefore either an original trivial level-zero row or an added bond with value one. The extension is flat and has trivial datum for the sequence with first member replaced by the torus. If a periodic coordinate is too short for a required pillar or double block to embed, make this local construction on a compatible periodic cover. The rows and their complete averaging reads lift with the same values. The pillar

transport rule below commutes with the deck translations, so the resulting gauge descends. On each pillar with top y, define $t(x)$ as the transport from y to x inside that pillar. Flatness makes this transport independent of the path. Also every retained average equals its straight path transport: on its complete double block a flat field is a pure gauge, and exact covariance of the actual average reduces this assertion to $\mathcal{M}^q 1 = 1$.

For a bond $b = (x, x')$ inside one pillar, flatness gives $t(x)U(b)t(x')^{-1} = 1$. For adjacent pillars of the same order, close this path with their retained interior row inside its double block; that row has transport one. Adjacent pillars of different orders have orders $q, q+1$. The seam geometry gives a retained outward row from the upper top to a site w whose level-$(q+1)$ block contains the lower top. Join w to the lower top by level-q interior rows in that block. These rows and the outward row all have transport one, and they close the path through b inside the outward row's complete double block. Thus $t(x)U(b)t(x')^{-1} = 1$ in this last case as well; no omitted inward row is used. Every node is a pillar top, so $t = 1$ at the nodes. Restricting to Ω_0 and setting $g = t^{-1}$ yields $U(b) = t(x)^{-1}t(x') = 1^g(b)$ with $g \in \mathcal{G}_{\mathrm{res}}$. The argument uses the separated determining geometry, not a topological condition imposed separately on the region.

3.11.12.1. Second-order and Hölder readouts. For a local field h, exponential decay of the inverse gives a bound on h in its declared field norm. It does not automatically give a bound on a second derivative or on the ordinary Laplacian. The actual field equation must be used. In a real local gauge the covariant Laplacian and the componentwise Laplacian differ by terms containing A, a first difference of A, and at most one difference of h. On an admitted potential ball these satisfy the local numerical comparison

$$|(L_U + \Delta_\eta)h| \le C(|A|\,|\nabla h| + (|\nabla A| + |A|^2)|h|), \qquad L_U = \sum_\nu \nabla^\sharp_{U,\nu}\nabla_{U,\nu}. \tag{3.ccclxxxi}$$

Here Δ_η is the nonpositive scalar heat generator. The signs in eq. 3.ccclxxxi are therefore a plus on its left. The estimate follows by expanding each transported difference and its transpose and using $|e^{i\eta A} - 1| \le C\eta|A|$. For the second-order coefficient difference, subtract the two endpoint potentials and use $|\nabla A|$; the remaining product has two A's. Both contributions have the displayed dimensions.

Given bounds on $h, \nabla_U h, L_U h$ and on the actual gauge coefficients, eq. 3.ccclxxxi supplies a componentwise Laplacian bound. The discrete interior theorem then gives the first-difference Hölder bound. It also supplies the coarse-mesh fallback when the number of sites in the interior ball is bounded. This removes a separate Hölder estimate from the primitive list, but not the pointwise gradient bound, the covariant Laplacian bound or the local gauge construction.

For a differentiated background at a critical pair, the slice can give the separate second-order member $D^\sharp Dh$. At a noncritical independent-current pair the actual linear equation instead controls the combinations

$$(D^\sharp D + DRD^\sharp)h, \qquad D(1-R)D^\sharp h. \tag{3.ccclxxxii}$$

Their sum has to be compared to the covariant componentwise Laplacian with the actual curvature and current terms. The critical slice identity cannot be inserted at an independent-current pair. Near the outer boundary all these claims are restricted to bonds with the required neighbors; the missing-plaquette terms of the action are separate boundary forcing.

3.11.13. *The quadratic real-to-complex reduction.* The square of the complex radius is forced by the actual plaquette product. Consider a real unitary U, a complex physical potential $A = A_{\mathrm{r}} + iA_{\mathrm{i}}$, and $U' = e^{i\eta A}U$. On a patch of scale ξ, assume

$$\xi|A| \le e_1, \qquad |\partial U - 1| + |\partial U' - 1| \le e_0\eta^2\xi^{-2}, \qquad e_0, e_1 \le c. \tag{3.ccclxxxiii}$$

Transport the four oriented increments to the starting corner of a plaquette, using the real base field. With raw increments $a_b = \eta A_b$, reversing an edge uses $a_{\bar b} = -\operatorname{Ad}(U_b^{-1})a_b$. The exact product

identity is

$$\partial U'\,\partial U^{-1} = \prod_{r=1}^{4} e^{iX_r}. \tag{3.ccclxxxiv}$$

Each X_r is an oriented a_b conjugated by the preceding real base transport. Consequently $|X_r| \le \eta e_1/\xi$. The finite product expansion gives

$$\left|\prod_{r=1}^{4} e^{iX_r} - 1 - i\sum_{r=1}^{4} X_r\right| \le C\eta^2 e_1^2 \xi^{-2}. \tag{3.ccclxxxv}$$

One can prove this without a series rearrangement: the second derivative of the product with all X_r multiplied by one scalar is bounded by $C(\sum |X_r|)^2$ on the unit disc, and the integral Taylor remainder has the same bound.

The sum of transported increments is $\eta^2 \mathrm{d}_U A$, up to the terms changing the closing base transport to the identity. Those terms are bounded by $C\eta|A|\,|\partial U - 1|$. Dividing eq. 3.ccclxxxiv–eq. 3.ccclxxxv by η^2 and using eq. 3.ccclxxxiii proves

$$\xi^2 |\mathrm{d}_U A| \le C(e_0 + e_1^2). \tag{3.ccclxxxvi}$$

Indeed the closing error is at most $Ce_0e_1\eta\xi^{-3}$, bounded by $Ce_0\xi^{-2}$ since $\eta \le \xi$. The real base transports make d_U real linear on the Hermitian Lie algebra. The same estimate therefore holds for A_{r} and A_{i} separately. Applying the product expansion again shows that the real unitary configurations $e^{i\eta A_{\mathrm{r}}}U$ and $e^{i\eta A_{\mathrm{i}}}U$ have curvature at most $C(e_0 + e_1^2)\eta^2\xi^{-2}$. This is the real comparison used by the continuation argument.

Here is the precise analytic step. Let $\mathcal{F}(z_1, z_2)$ be any one of the fixed holomorphic background readouts obtained from $e^{i\eta(z_1 A_{\mathrm{r}} + z_2 A_{\mathrm{i}})}U$, with all its determining maps and gauges admitted. Suppose

$$\|\mathcal{F}(z_1, z_2)\|_{\mathrm{sc}} \le M \quad \text{for } |z_1|, |z_2| < r_0/(2e_1), \qquad e_1 \le r_0/16. \tag{3.ccclxxxvii}$$

The subscript sc means that the output is normalized by its physical scale. It does not divide by the desired smaller target radius. A two-variable Cauchy estimate gives

$$\|\mathcal{F}(1,i) - \mathcal{F}(1,0) - i\mathcal{F}(0,1) + i\mathcal{F}(0,0)\|_{\mathrm{sc}} \le CMe_1^2/r_0^2. \tag{3.ccclxxxviii}$$

For completeness, expand on a polydisc of radius $R = r_0/(4e_1) \ge 4$. Each coefficient of total degree n is at most MR^{-n}. The displayed linear combination annihilates degrees zero and one. There are $n+1$ monomials of degree n; the sum over $n \ge 2$ is bounded by CMR^{-2}. This proves eq. 3.ccclxxxviii with no dependence on the number of lattice variables.

The three real configurations on the right of eq. 3.ccclxxxviii are treated by the real background estimate, using eq. 3.ccclxxxvi. If the actual crude analytic estimate has $M/r_0^2 \le C_{\mathrm{geom}}$, this gives

$$\|\mathcal{F}(1,i)\|_{\mathrm{sc}} \le C_{\mathrm{geom}}(e_0 + e_1^2). \tag{3.ccclxxxix}$$

Thus the hereditary complex estimate is reduced to a fixed-radius crude analytic bound and the real estimate. It is not an independent assumption of the desired improved radius. If r_0 depends on a running field normalization, its actual factor in M/r_0^2 must be retained. A bound on a shrinking disc without this ratio does not prove eq. 3.ccclxxxix. The physical and Gaussian-coordinate conversion in the domain estimates determines which radius enters that ratio.

3.11.14. *Restriction is an exact constrained-variation argument.* The iterated average has a stronger locality than an arbitrary composition of finite-range maps. A level-j row reads the two level-j endpoint blocks and their joining bonds. To prove this, proceed by induction. At one step this is its definition. At the next step, a block is the disjoint union of its L^4 subblocks. An internal contour reads only subblocks inside that block. The straight segment between corresponding points of two adjacent blocks reads subblocks in their union and their joining bonds. The lower-step locality closes the induction. The fixed local lift includes all these paths.

For a determining row these two blocks miss the next region, and they lie in the preceding region, by the seam geometry. Their diameter in $d_{\mathbf{B}}$ is bounded by a constant depending only on the fixed local geometry. The number of bare variables can be large; the bound is on the physical determining neighborhood, not on that number.

Consider two aligned separated sequences with $\Omega'_q \subset \Omega_q$, and let $\mathbf{B}'$ be the new determining set. The smaller regions give a refinement of the local top scales. An old row's value can be evaluated from the new determining values and the fixed field outside Ω'_0. Here is a terminating algorithm that proves the assertion.

Start with the averaging expression for an old row of level j. No endpoint block lies in Ω'_{j+1}, since it misses Ω_{j+1}. If the row touches Ω'_j, it is a new determining row and its value is prescribed. Its full neighborhood is inside Ω'_0 by separation when $j \geq 1$. If neither endpoint lies in Ω'_j, both endpoint blocks are entirely outside that region, by alignment. Expand its one-step formula into level-$(j-1)$ row values. Their endpoint blocks have no point in Ω'_j. A lower row touching Ω'_{j-1} is prescribed; otherwise expand it again. The index strictly decreases. At level zero the bond is either a prescribed new row or a bond of the fixed exterior.

Rows whose determining neighborhood meets the boundary of Ω'_0 cause no exception. If a level-r endpoint were in Ω'_r, the sum of the gaps below that level and the chosen $\rho \gg L$ would put its two-block neighborhood inside Ω'_0. Thus such a row is expanded until its bare exterior bonds are fixed. This proves the finite factorization

$$M_{\mathbf{B}}(U) = \mathcal{F}_{\mathbf{B},\mathbf{B}'}(M_{\mathbf{B}'}(U|_{\Omega'_0}), U|_{\text{exterior}}) \tag{3.cccxc}$$

for those rows affected by an interior variation. The function $\mathcal{F}$ is built from the actual products, means and averages just evaluated. No linearization is involved.

Let U_t be a differentiable local variation on the new fibre, and extend it by the unchanged field outside Ω'_0. The level-zero new datum fixes every boundary-layer bond: its free bonds can touch only Ω'_1, which is separated from $(\Omega'_0)^c$ by more than two bare spacings. Consequently every old plaquette changed by the variation is an inner plaquette of Ω'_0. The remaining old action terms are constant, and eq. 3.cccxc fixes every old constraint. Therefore

$$\frac{d}{dt} A_{\Omega'_0}(U_t)\Big|_0 = \frac{d}{dt} A_{\Omega_0}(\widetilde{U}_t)\Big|_0 = 0 \tag{3.cccxci}$$

whenever U is critical for the old problem. This also explains why an arbitrary cut with freely varying boundary bonds would not prove the same statement.

The new local scales are no larger than the old scales at their elements, so the normalized fine curvature bound improves. The current needs a separate boundary check, because the new action omits some old plaquettes. Write j_{old} and j_{new} for the two left-variation currents, in the physical normalization used above. If a bond has the same incident retained plaquettes in both actions, these currents agree and its new current weight improves.

Every remaining changed-current bond is within a fixed bare plaquette distance of the new boundary. The gap between Ω'_1 and $(\Omega'_0)^c$ therefore puts it in the bare boundary layer, where its new local scale is η. The left-variation formula has at most six retained plaquette contributions. Unitarity and the fixed trace normalization give

$$\eta^3 |j_{\text{new}}(b)| \leq C_\partial \max_{\substack{p \subset \Omega'_0 \\ p \ni b}} |U_p - I| \leq C_\partial \epsilon_{\text{old}}. \tag{3.cccxcii}$$

The maximum is zero when there is no retained incident plaquette. Here ϵ_{old} is the old regularity parameter; the normalized fine curvature estimate bounds each displayed bare plaquette by that parameter. This proves current regularity with a fixed boundary constant. It does not require monotonicity of the current under restriction of the action. The new datum is regular by the actual averaging estimate. On its two-block neighborhoods the smallest old scale is at least its new scale

divided by L, giving the fixed seam loss $C_L L^2$. Mixed tests are the same physical averages by the finite factorization. Let C_{res} dominate 1, C_∂, and the stated datum and seam constant $C_L L^2$. The restriction identity is asserted when $C_{\mathrm{res}}\epsilon_{\mathrm{old}}$ lies below all the corresponding regularity input ceilings of the real uniqueness theorem. Under this actual admission condition,

$$U_{\mathbf{B}'}\Big(M_{\mathbf{B}'}(U_{\mathbf{B}}(\mathbf{V})|_{\Omega_0'})\Big) = U_{\mathbf{B}}(\mathbf{V})|_{\Omega_0'} \pmod{\mathcal{G}_{\mathrm{res}}(\mathbf{B}')}. \tag{3.cccxciii}$$

This conclusion uses regular critical uniqueness, not global minimization. It is a complete reduction of the restriction identity to that theorem.

The same argument proves the background nesting used for the next step. In a depth-$(k+1)$ problem the single inner node of each last block permits an axial mean representative there. Its k-fold average on the bonds touching Ω_{k+1} is independent of the representative chosen: the gauge relating two such representatives is one at the block centers and, by their axial means, at every level-k site in the block. Covariance fixes those coarse bonds. Freeze that averaged field as the new last datum and apply eq. 3.cccxciii to the truncation. Thus the shorter background and the longer one are the same orbit on their common region. No new effective action is introduced by this identity.

3.11.15. *Collars and joined sequences.* Take $\mu \geq \max\{\rho, \mathfrak{M}, L^2\}$ to be a power of L, with $\rho \geq L^2$. A part A_a is a union of cubes of side $s_a = \mu\xi_{j_a}$ at level j_a. Write $A_a^{(-r)}$ for the result of removing r layers of these cubes, and use cube-wise meets throughout. For distances use the closed representatives of the cubes. Suppose that the four-layer region $F_a = A_a \setminus A_a^{(-4)}$ lies in the ambient layer j_a and that for $a \neq b$,

$$\mathrm{dist}_\infty(A_a^c, A_b^c) \geq 10\max(s_a, s_b).$$

Put $W = \bigsqcap_a A_a$, $W^{(-r)} = \bigsqcap_a A_a^{(-r)}$, and $Q_a = A_a \setminus A_a^{(-3)}$. If $x \in F_a$, its distance from A_a^c is at most $4s_a$, so its distance from A_b^c is at least $6s_b$. The entire s_b-cube through x is then in $A_b^{(-4)}$. In particular $Q_a \subset F_a \subset W$, and the wide collars Q_a are pairwise disjoint.

The μ-tower of A_a has $A_{a,0} = A_a$. For $0 \leq q < j_a$, take

$$A_{a,q+1} = \bigcup_{\substack{C \text{ of side } \mu\xi_{q+1},\ C\subset A_{a,q} \\ \mathrm{dist}_\infty(C, A_{a,q}^c)\geq \mu\xi_{q+1}}} C. \tag{3.cccxciv}$$

Here the indicated aligned partitions exist whenever the level-j_a partition of side s_a exists. The construction gives

$$\mathrm{dist}_\infty(A_{a,q}^c, A_{a,q+1}) \geq \mu\xi_{q+1} > \rho\xi_q. \tag{3.cccxcv}$$

At depth q, every cube of side $\mu\xi_q$ at distance at least $2\mu\xi_q$ from A_a^c lies in the tower. Indeed, let C be such a cube at depth $q+1$. Every depth-q cube meeting its open $\mu\xi_{q+1}$-neighborhood has distance at least $\mu\xi_{q+1} - \mu\xi_q \geq 2\mu\xi_q$ from A_a^c. By induction these cubes lie in $A_{a,q}$, which proves that C passes the next erosion. Thus $A_a^{(-2)}$ lies in every member of the tower of A_a up to depth j_a.

Define

$$\Theta_q^a = \begin{cases} A_{a,q}, & q \leq j_a, \\ A_a^{(-2)}, & q > j_a, \end{cases} \qquad \Omega_q^W = \Omega_q \sqcap \bigsqcap_a \Theta_q^a.$$

The sequence decreases and has first member $\Omega_0 \sqcap W$. For $q \leq j_a$ the mask is a union of ambient level-q cubes by grid refinement. For $q > j_a$, the ambient region misses F_a. Intersecting it with $A_a^{(-2)}$ therefore selects exactly those ambient cubes on the interior side of the collar. This proves alignment.

To check separation at q, first use the ambient gap for the complement of Ω_q. For a mask with $q < j_a$ use eq. 3.cccxcv. For $q \geq j_a$, the points of Ω_{q+1}^W lie in $A_a^{(-4)}$, while the complement of Θ_q^a lies at distance at most $2s_a$ from A_a^c. At $q = j_a$ this gives a gap of at least $2s_a$. At $q > j_a$ a shortest

segment from Ω_{q+1}^W to the complement of Θ_q^a crosses the surface at distance $3s_a$ from A_a^c. That surface lies in $F_a \subset \Omega_q^c$. The ambient gap then bounds the segment's length from below. The same argument applies part by part, so the joined sequence has the prescribed separation. A smaller first member can be used when Ω_1^W is a union of $\mu\xi_1$-cubes and is at distance at least $\mu\xi_1$ from $(\Omega_0 \sqcap W)^c$. In that case enlarge Ω_1^W by one layer of those cubes to obtain the collared first member. It stays inside the full first member and keeps a gap of at least $\mu\xi_1$ from its own complement to Ω_1^W.

On $W^{(-2)}$ the positive-index masks leave the ambient regions unchanged. In particular, on bonds of the joined first member with both ends in $W^{(-2)}$, the determining rows are the ambient rows. A new row of level q can arise only when an end that was in Ω_{q+1} is cut out of the new region at index $q+1$. That cut must occur in a tower with $q < j_a$, since the level-j_a boundary collar misses Ω_{j_a+1}. The removed end is at distance at most $2\mu\xi_{q+1} \le 2s_a$ from A_a^c, by the interior inclusion just proved. It lies in $A_a \setminus A_a^{(-2)}$. Its other end is within one level-q spacing and lies in Q_a. Both ends are therefore in the unique Q_a that contains the row. These are the collar rows.

The localized datum agrees with the original datum on old rows. On a new collar row it is the exact face lift of the ambient level-j_a datum:

$$V^{\mathrm{lift}}(b) = \begin{cases} 1, & b \text{ stays in one ambient } j_a\text{-block}, \\ V_{j_a}(c), & b \text{ crosses from the block of } c_- \text{ to that of } c_+. \end{cases} \tag{3.cccxcvi}$$

The orientation of a reverse bond uses the group inverse. On each successive grid the central straight route crosses a fixed coarse face once. An internal contour is one. The one-step average of this lift is therefore exactly the preceding lifted datum, by its defining exponential average. This proves constraint compatibility. The lift can concentrate curvature on its faces; it is not asserted to be a regular fine configuration. Its role is to prescribe the collar data for the localized background.

For a tested plaquette p of level q, let $\mathcal{R}(p)$ be the union of the spacing-ξ_q cubes of its corners. Its diameter is at most $2\xi_q$ in the sup distance. It is contained in W, by separation at positive levels and by the inner-bond condition at level zero. Every joined test is of one of two kinds. Either $\mathcal{R}(p) \subset W^{(-2)}$, or $\mathcal{R}(p) \subset Q_a$ for a part with $q \le j_a$. To check this, fix a. If $q \le j_a$ and some point of $\mathcal{R}(p)$ has distance at most $2s_a$ from A_a^c, the whole test lies at distance at most $2s_a + 2\xi_q < 3s_a$, hence in Q_a. If no such point exists, the test lies in $A_a^{(-2)}$. If $q > j_a$, a corner in Ω_q^W has distance at least $4s_a$ from A_a^c. When $q = j_a + 1$, the test diameter keeps the whole test at distance greater than $2s_a$. When $q > j_a + 1$, a segment to a point within distance $2s_a$ would meet $F_a \subset \Omega_{q-1}^c$ within $2\xi_q$ of the corner. The ambient gap excludes this since $\rho > 2L$.

In the interior case all determining rows used by the test are ambient rows. The holonomy, including its one-step averages on mixed rows, is the original holonomy. In the collar case the joined datum on the test is the lift from level j_a. A common row in that collar is necessarily of level j_a, so its prescribed value agrees with the lift as well. The exact one-step averaging identity for eq. 3.cccxcvi reduces a mixed test to a test of this same lift. Its holonomy is one unless the plaquette crosses two ambient faces, in which case it is the holonomy of the corresponding level-j_a plaquette of Q_a. All its ambient defining rows lie in W. Thus the joined datum is determined by the ambient rows with both ends in W. It inherits their test bound ε_1 with no increase, provided the bound is imposed on the ambient tests whose corner cubes are contained in W. The real background theorem applies to this regular datum on the joined sequence when its geometric floors and regularity radii hold. The reference field is the normalized representative of the resulting regular critical orbit.

3.11.16. *Response, collars and a common admitted path.* The source direction in a datum derivative includes the reduced-datum map. At a critical solution, differentiating eq. 3.cccxxxii gives

$$\begin{pmatrix} K & -Q^\sharp \\ Q & 0 \end{pmatrix} \begin{pmatrix} \dot{A} \\ \dot{\omega} \end{pmatrix} = \begin{pmatrix} 0 \\ Db\,\dot{B} \end{pmatrix}. \tag{3.cccxcvii}$$

No derivative of F with respect to B occurs in fixed reference coordinates. If the reference or an independent current also changes, its actual derivative is a force in the first row and a datum force in the second. The fixed-reference response calculation supplies those terms.

Assume the actual coupled inverse has weighted pointwise kernels for both columns and all declared readouts. In normalized coordinates write its field–datum entries as T_{yd}. For a point source at row c, eq. 3.cccxiv gives

$$\dot{A}(y) = \sum_d T_{yd}(Db)_{dc}e, \qquad \|\dot{A}(y)\|_{X(y)} \le Ce^{-md_{\mathbf{B}}(y,y_c)}|e|. \tag{3.cccxcviii}$$

The sum contains only $d = c$ and outward rows whose one-step outer block contains the interior row c. Its number and distance from c are bounded at fixed geometry; its length weights change by only one factor of L. The triangle inequality proves the exponential estimate from the inverse's entry bound. The same calculation applies to the multiplier and to each proved differentiated field readout.

For an extended source, the required bound is a row *sum*, not just a bound on each entry. If the normalized matrix satisfies $\sup_y \sum_c e^{md(y,y_c)}\|T_{yc}\| \le C$, then a source supported on $\mathcal{C}$ obeys

$$\|\dot{A}(y)\|_{X(y)} \le Ce^{-md_{\mathbf{B}}(y,\mathcal{C})}\|\dot{B}\|_\infty. \tag{3.cccxcix}$$

If kernels are written with cell measures, those measures are included in this Schur row sum. Mixing a measure-weighted kernel convention with unweighted point entries would give the wrong source normalization.

The current derivative must be read from the actual local Wilson current at $e^{i\eta A}U_0$, including its logarithmic differential. Its stencil contains second-order field terms. The same is true of the covariant Laplacian member. Once these readouts and the pointwise gradient have the weighted bounds, the interior theorem gives their gradient Hölder member with any fixed exponent below one. The node derivative follows from the actual differentiated node factor. Its weighted bound is another readout of the primitive operator package. The mere algebraic existence of $v_n[A]$ does not bound the product of its derivatives over many levels.

Suppose two real data sets on a joined sequence differ only on its collar rows and have been put into a common node frame. Write them there as $V_1 = e^{iB}V_0$, where $\|B\|_\infty \le C_\mu\epsilon$. The frame comparison is made by the exact finite contour reconstruction eq. 3.cccxvii–eq. 3.cccxx. At every collar it involves one ambient level and its finer tower; it must respect the recorded old edges. This is the datum alignment step. The smallness condition on ϵ depends on μ.

For this comparison, the output norm $X(y)$ uses the local scale of the joined sequence, whereas $d_{\mathbf{B}}$ is the ambient metric. Joined scales are no larger than ambient scales, so the native joined path distance dominates $d_{\mathbf{B}}$. The native weighted Schur bound therefore implies the same bound with this ambient exponential weight and unchanged joined source and output normalizations.

Take the actual path $V_t = e^{itB}V_0$. Assume it stays in the common admitted datum ball. Integrating eq. 3.cccxcix gives a field comparison whose forcing is supported only on the collar. In the ambient metric each collar point is within $C\mu$ of W^c, because the supplied collar lies in one ambient layer. Hence

$$\|f(y)\|_{X(y)} \le C_\mu\epsilon e^{-md_{\mathbf{B}}(y,W^c)}, \qquad C_\mu \le C_\mu^{(0)}e^{Cm\mu}. \tag{3.cd}$$

Indeed $d_{\mathbf{B}}(y,\mathcal{C}) \ge d_{\mathbf{B}}(y,W^c) - C\mu$; eq. 3.cccxcix and the fundamental theorem of calculus prove the assertion. The factor $C_\mu^{(0)}$ is the explicit finite alignment and path constant. It is not claimed uniform as μ tends to infinity. Where the joined and ambient scales agree, this is also an ambient output-norm bound. At a finer tower element, stronger ambient readout weights require the separate scale factors and lag or small-source estimate in eq. 3.cdxix–eq. 3.cdxxi; they are not absorbed into C_μ.

The physical comparison has the form $U|_W = (e^{i\eta f}U_W^\circ)^w$. At tops away from the collar the node identity prescribes $w = v_q[f]^{-1}$. At other points it has no specified normalization. In particular

eq. 3.cd is a bound on the comparison potential and its declared readouts, not a bound on that gauge at every fine site.

3.11.16.1. Comparison with the real joined background. Let V be a real datum with all test holonomies within ε_1 of one. Fix the collar width μ. Take the no-inward determining sets, the separation and weighted-response floors of the real background theorem, and at least 6ρ sites in each direction at the last level, with ρ taken above those floors. Decrease ε_1 so that the radii used below are admitted. Let U be a critical configuration in the regular orbit of V, and let U°_W be the normalized representative of the regular orbit of V^W. The orbit radius ε_0 obeys $B_3\varepsilon_1 \le \varepsilon_0 \le a_0$. Regularity improvement puts U at radius $\epsilon = B_3\varepsilon_1$. By eq. 3.cccxciii, $U|_{\Omega_0^W}$ is critical at its induced datum $M_{\mathbf{B}^W}(U|_{\Omega_0^W})$, which agrees with V^W on common rows. We compare the induced datum with V^W in the same node frame.

In Q_a with $j = j_a \ge 1$, each level-q average $W_q = \mathcal{M}^q U$, $0 \le q \le j$, obeys

$$\max_{p \text{ inner in } Q_a} |\partial W_q(p) - 1| \le 2\epsilon L^{2(q-j)}.$$

This is the averaging curvature bound in the ambient layer j. At $q = j$ the inner bonds of Q_a are determining rows, so $W_j = V_j$ there. With C_{lift} the constant of the finite alignment estimate eq. 3.cccxvii–eq. 3.cccxx there is a real gauge on Q_a, equal to one on level-j sites, whose level-q averages for $q < j$ differ from the lift of W_j by at most

$$2C_{\mathrm{lift}}\epsilon/(L^2-1).$$

The denominator comes from summing the smaller curvature bounds over $q < j$. The gauges on the wide collars combine with the identity elsewhere to give a gauge g on the joined first member. It leaves all common rows unchanged. Indeed, if a common row has an end in Q_a, its layer end is either in F_a, in which case the row has level j_a, or outside F_a. In the latter case the row would have length at least s_a, hence level at least $j_a + 2$ since $\mu \ge L^2$. Separation would put its end in Q_a inside Ω_{j_a+1}, which is impossible. Thus on every common row meeting a gauge support, the gauge equals one at the relevant endpoints. There are no new rows of a part of level zero.

Consequently the induced datum of $(U|_{\Omega_0^W})^g$ is $e^{iB}V^W$, with $B = 0$ away from the collar rows. For the discrepancy on a collar row use the intrinsic logarithm. Once $2C_{\mathrm{lift}}\epsilon/(L^2-1) < 1$, the logarithm is bounded by $\pi/3$ times the group discrepancy. In particular

$$\|B\|_\infty \le \frac{2\pi C_{\mathrm{lift}} B_3}{3(L^2-1)}\varepsilon_1. \tag{3.cdi}$$

Take the right-hand side below the radius r_{B} of the joined response polydisc. The line tB now lies in that polydisc. The real endpoints represent the regular critical orbits at V^W and $e^{iB}V^W$. This uses gauge covariance of restriction and uniqueness at the enlarged regularity radius of the induced datum. Let $\mathcal{H}_W$ denote the response in its normalized slice. The comparison field is $f = \mathcal{H}_W(B)$, and restoring the gauges gives

$$U|_{\Omega_0^W} = (e^{i\eta f}U^{\circ}_W)^w.$$

Outside the wide collars $g = 1$. The residual gauge fixes the nodes, so at the level-q tops there $w = v_q[U^{\circ}_W; f]^{-1}$. If all parts have level zero, $B = f = 0$ and the comparison is by a residual gauge.

Write $\mathsf{N}_x(f)$ for the maximum of the potential, first covariant difference and covariant $D^{\mathsf{T}}D$ readouts at a joined bond x, multiplied by the first, second and third powers of the joined top spacing, respectively. The point response at a collar row c with top y_c is bounded in this norm at the top y of x by $B_5 e^{-\delta_0 d_{\mathbf{B}^W}(y,y_c)}$ times the size of the row increment. Use the part of the rate $\delta_0/2$ that gives a uniform sum of row weights, as in eq. 3.cccxcix. For a collar point, the distance to W^c in the ambient metric is at most 12μ. Indeed the path to the corresponding A^c_a stays in the single layer j_a until it reaches the boundary and has bare length at most $12\mu L^{j_a}$. For a path from y to a collar row, stop the path when it first reaches the boundary of $W^{(-3)}$. Up to that point the

ambient and joined layer weights agree. The stopping point is in a wide collar, and the preceding bound and the triangle inequality therefore give

$$\inf_c d_{\mathbf{B}^W}(y, y_c) \;\geq\; d_{\mathbf{B}}(y, W^c) - 12\mu.$$

The inequality also holds if y already lies in a wide collar. Integrating the derivative of $\mathcal{H}_W(tB)$ over $0 \leq t \leq 1$, using eq. 3.cdi, gives

$$\mathsf{N}_x(f) \leq C_\mu \varepsilon_1 e^{-\delta_0 d_{\mathbf{B}}(y, W^c)/2}. \tag{3.cdii}$$

Here the row sum and the alignment constant are independent of the depth and volume, and the conversion of distances costs at most $e^{6\mu\delta_0}$.

In this gauge the difference of plaquette variables is bounded by the first covariant differences and the quadratic remainder in f. The current difference also uses $D^{\mathsf{T}}Df$. On these fixed stencils the assigned scales differ by at most one level, and changing the observation point within the stencil costs a fixed factor in the exponential. Hence the plaquette difference at p and the bare-unit current difference at a full bond b have, in addition to the right-hand side of eq. 3.cdii measured at the corner or endpoint, the factors $L^{-2\lambda^W(p)}$ and $L^{-3\lambda^W(b)}$, respectively. Here λ^W is the assigned level in the joined sequence. This last power uses $J^{\mathrm{bare}} = L^{-3i}J^{(i)}$ when the current is calculated in level-i units.

Two separated sequences which agree on W, with real data agreeing on all determining bonds with both ends in W, have the same joined datum and normalized joined background. The distance from an interior point to W^c is the same for the two sequences, since the initial part of any path up to its first exit has the same layer weights. Apply the estimate to each background against this common joined background. For plaquettes and full-bond current stencils in $W^{(-2)} \sqcap \Omega_0^W$ the assigned levels there are the ambient ones. The triangle inequality gives the comparison of the two readouts, with twice the constant, in the gauges just specified.

The real comparison supplies the endpoint estimate on these joined domains. To use it for complex retained forms, we must first admit the full and restricted histories on one domain of independent complex potentials and currents, with the actual collar substitutions and their remaining radius margins. The real datum path above varies neither an independent current nor the history's stored variables. The fixed-tower complex response also keeps its determining sequence fixed. A comparison of the complete forms on that common domain, including the error from discarded chains, is needed before transferring a positive lower bound to a retained form.

3.11.16.2. The real base of a layered domain. Let $e(x)$ be a regularity profile for a real critical configuration, so it bounds the scaled curvature and the current at full bonds based at x. Use a separated sequence with gaps greater than $8\rho L^q$ between levels q and $q+1$, and with at least 6ρ top sites in each torus direction. On a nonempty aligned localization X' of M_1L^i-cubes, intersect the sequence with the M_1-tower of X'. For each depth $p \leq i$, restriction of the critical configuration is critical at the induced datum by the restriction argument for eq. 3.cccxciii. The profile bound persists for the plaquettes and full bonds of the localized first member. Their assigned level can only decrease, and the current at those full bonds has the same incident plaquettes. A current of the restricted action at a changed stencil instead uses eq. 3.cccxcii and its bare-layer bound. If e' is the profile supremum on the localized first member, the tested plaquettes of the induced datum have defects at most $C_L e'$, by averaging on their full backward cells. Use the same datum ceiling for the tested seam holonomies if present. Take e' below the test and orbit ceilings, including $L^2 e'$ below the curvature radius of one-step averaging and $C_L e'$ below the datum radius of the tower. The same restriction then lies in the regular orbit of the induced datum. This identifies the normalized tower configuration with the restriction up to its residual gauge.

For a point of original level $\lambda(x)$, the layer of the localization is $r = \min\{\lambda(x), i\}$. A sufficient profile inequality for layer radius b_r is

$$e(x)L^{-2(\lambda(x)-i)_+} \leq b_r. \tag{3.cdiii}$$

In level-i units let $\xi = L^{-i}$ and $\xi_r = L^{r-i}$. The curvature bound on this layer is $b_r\xi^2\xi_r^{-2}$ at plaquettes in the field's domain. In converting the full-bond current from bare units, the factor L^{3i} multiplies its bound $e(x)L^{-3\lambda(x)}$. The extra suppression at $\lambda(x) > i$ has exponent three, so the same condition gives $b_r\xi_r^{-3}$. At a smaller observation level i'', the radii on layers $r < i''$ can be kept, and the new top radius can be chosen to dominate

$$\max_{i''\le r\le i} b_r L^{-2(r-i'')}.$$

This is the profile comparison for all nested subdomains whose tower grids are aligned.

The gauge clauses require a separate local estimate. For a cube of side $C_{\mathrm{g}}ML^r$ in bare units, take the profile supremum $e_\square$ on its $8\rho L^{r-1}$ neighborhood. In the nonwrapping geometry the local transport and critical background estimate supplies a real gauge with both $\xi_r|A|$ and $\xi_r^2|\nabla A|$ bounded by $C_L C_{\mathrm{g}} M e_\square$. In using it require that the profile is below the one-step curvature ceiling and that $2\mathsf{k}_L C_{\mathrm{g}} M L e_\square \le \rho a_1$, where k_L is the transport constant and a_1 the regular datum ceiling. Require also that this gauge bound fits the gauge radius of the layer. These checks, together with eq. 3.cdiii, give the real initial triple $(U, 0, J(U))$ in the layered domain with the plaquette and full-bond tests just described. If the domain also tests the restricted current at changed stencils, its bare-layer radius must allow the boundary bound above. The potential slot is zero here. Admission of the varied potential and independent current along the complex path requires the margins in the following argument.

3.11.16.3. Admission is checked for every object. The assumption on the common path can be reduced to a finite first-exit calculation. List all the actual core quantities $\mathcal{R}_\beta(t)$, their strict bounds r_β, and initial margins $s_\beta = r_\beta - \mathcal{R}_\beta(0) > 0$. Include small logarithm branches, mean Jacobians, current ceilings, gauge condition numbers, site Gram inverses, field inverses and tower clauses. If an inverse is an atomic object, its norm or its distance from singularity is one of these tests.

While the path is admitted, suppose its actual response and coefficient estimates give

$$|\mathcal{R}_\beta(t) - \mathcal{R}_\beta(0)| \le K_\beta t\,\|B\|, \qquad K_\beta\|B\| \le s_\beta/2 \quad \text{for every } \beta. \tag{3.cdiv}$$

At a proposed first exit every inequality still has half its margin. On a finite lattice the bounded coordinates have a limiting point. The branch and inverse margins extend every atomic formula there, and the uniform coupled inverse and coefficient bound extend the solution by its local contraction. This contradicts a first exit before $t = 1$. The constants must be uniform in the lattice, but compactness here is only the finite-dimensional continuation argument.

Checking only curvature in eq. 3.cdiv would not suffice. Nor does a response bound at already admitted points establish that every substitution is admitted. The actual inner substitution also includes the fluctuation, its amplified window datum, the coarse background, the cutoff, and the independent-current readout. Each has its own test in this finite list. The domain estimates give the physical/Gaussian radius conversion and the slow-radius inequalities used to verify those numerical margins.

3.11.17. *Gluing real constructions and their analytic maps.* There are two different gluing arguments. For real data, uniqueness among all regular critical orbits identifies constructions made in different neighborhoods. Applying the unique comb normalization outside the nodes identifies the representatives themselves. For complex data one uses either a common contraction neighborhood or analytic uniqueness on the relevant connected overlap.

The analytic uniqueness fact can be proved directly. If two holomorphic maps on a connected complex datum ball agree on a real open ball, their difference vanishes on a complex polydisc. Fix all coordinates but one to be real and apply the one-variable identity theorem. Then allow that coordinate to be complex and repeat with the next. This proves vanishing on the polydisc.

Continuation along a finite chain of intersecting polydiscs in the connected domain proves equality everywhere.

For maps with different real centers this argument applies to each connected component of their complex overlap that contains the required real open overlap. It does not identify an unrelated overlap component without such an argument. An alternative is to put both solutions into a single admitted reference chart and the single uniqueness ball of eq. 3.cccliv. That gives equality pointwise on the common chart, without an overlap connectedness claim. The chart transition includes its node factors and gauge, not just a difference of coarse logarithms.

Let $\mathcal{D}$ be the real admitted datum domain covered by these local constructions. It is open in the finite product of real group coordinates. Every local fixed point is continuous in its datum by the uniform contraction estimate. The comb representative is continuous on the admitted mean branches. On overlaps the representatives agree, so these give a continuous, hence Borel, real background map on $\mathcal{D}$. Equivalently choose a countable chart cover from the countable basis of this finite-dimensional manifold and use the first chart containing each point; overlap agreement makes this a Borel map.

If an integral uses this map only under a cutoff supported in $\mathcal{D}$, one can assign the identity outside $\mathcal{D}$ to obtain a Borel extension. This is only a definition outside the regular problem. It makes no claim about its fibre or criticality. Holomorphic integrals use the actual common complex domain and its parameter-independent reference measure; the Borel extension is not an analytic substitute for that domain.

3.11.17.1. Measurability of the real background tests. Fix the finite lattice, determining regions, and numerical parameters. In real matrix coordinates, G is algebraic. The intrinsic logarithm has the graph $X \in \mathfrak{g}$, $|X| < \pi$, $e^{iX} = H$; the exponential on this bounded set is restricted analytic. The finite equations defining a mean, together with the requirement that their solution be unique, therefore define its graph and its domain in $\mathbb{R}_{\mathrm{an}}$. The actual three branches of $\mathcal{M}$, its finite iterates, the determining-data map, and the prescribed axial gauge conditions are definable as well. We use closure under real quantification and cell decomposition in this structure [Edm00].

It remains to check the criticality condition, which quantifies over all differentiable curves in a fibre. Write $C_d = \{U : M_{\mathbf{B}}(U) = d\}$ and regard it as a definable subset of a finite-dimensional real matrix space. At $U \in C_d$ define the contingent cone by the finite formula

$$\begin{aligned} v \in T_d(U) \quad \Longleftrightarrow \quad & \forall \epsilon > 0 \; \forall \delta > 0 \; \exists t, z : \\ & 0 < t < \delta, \quad |z - v| < \epsilon, \quad U + tz \in C_d. \end{aligned}$$

The velocities of two-sided differentiable curves in C_d through U are exactly $T_d(U) \cap (-T_d(U))$. One inclusion follows from the difference quotients. For the converse, if $v \in T_d(U)$, then $(0, v)$ belongs to the closure of the definable set $\{(t, z) : t > 0,\ U + tz \in C_d\}$. Definable curve selection and one-variable monotonicity [Kur98, Lemma 2 and Proposition 1] give, after reparametrizing by the positive coordinate t, a differentiable curve $z(t)$ tending to v. Thus $U + tz(t)$ lies in C_d and has right derivative v at 0. If also $-v \in T_d(U)$, apply the same argument to $-v$ and reverse time on that branch. The two branches joined at U are differentiable on an open interval and have derivative v at 0.

The Wilson action has a polynomial extension in real matrix coordinates, using the conjugate transpose for inverse link variables. Hence the source criticality condition is exactly

$$\forall v \in T_d(U) \cap (-T_d(U)), \qquad DA_{\Omega_0}(U)v = 0.$$

It is therefore definable even at a singular fibre. Adding the prescribed regularity inequalities defines the relation $\mathscr{C}(d, U)$ of all regular critical configurations. The minimal-orbit domain is given by

$$\begin{aligned} & \exists U \; \mathscr{C}(d, U), \\ & \forall U_1, U_2 \quad \mathscr{C}(d, U_1) \wedge \mathscr{C}(d, U_2) \Longrightarrow \exists g \in \mathcal{G}^{\mathrm{res}}(\mathbf{B}) \quad U_2 = U_1^g. \end{aligned}$$

The residual group fixes every prescribed node, including seam nodes; it is a finite-dimensional compact algebraic group. Gauge invariance of $\mathscr{C}$ makes this formula precisely the condition that the nonempty set of regular critical configurations be one orbit.

The gauge-invariant action and plaquette readouts have definable graphs obtained by quantifying over this orbit. The prescribed representative in Ax° has a definable graph on its exact domain of existence and uniqueness. For a block axial readout on Y, retain the conditions that Y consists of nodes, that $B(Y) \setminus Y$ contains no nodes, and that the required means exist. Gauge covariance supplies a representative in this gauge; two such representatives differ by a residual gauge transformation equal to the identity on $B(Y)$. Their averaged fields therefore agree on the interior bonds used by the test. Existential quantification over these representatives defines that same single-valued readout, without choosing a different representative.

Finally, in the averaged-background problem let $\mathsf{W}_d^{\mathrm{reg}}$ be the prescribed constrained regular set, and write $a_d(w) = A_K(U_k(w))$. The preceding orbit formula and gauge invariance make both this set and this action readout definable. The graph of its unique attained minimum is exactly

$$\begin{aligned} w \in \mathsf{W}_d^{\mathrm{reg}}, \qquad & \forall z \in \mathsf{W}_d^{\mathrm{reg}}, \quad a_d(w) \le a_d(z), \\ & \forall z \in \mathsf{W}_d^{\mathrm{reg}}, \quad a_d(z) = a_d(w) \Longrightarrow z = w. \end{aligned}$$

Its projection is the exact existence set used in the definition. Cell decomposition makes these domains and their single-valued maps Borel. Consequently the real background tests, with their prescribed failure values when an object is undefined, are Borel. Existence of the required orbit and attainment of the minimum on the intended supports use the corresponding real estimates.

3.11.18. *Two finite mechanisms that prevent growth with depth.*

3.11.18.1. Eliminating the multiplier. It is unnecessary to assume an a priori small multiplier in a uniqueness statement once its actual row lift is bounded. Fix $Q_0E_0 = 1$, and write $Dq(A) = Q_0 + C'(A)$. At a critical field, eq. 3.cccxxxiii gives

$$[1 + (C'(A)E_0)^\sharp]\omega = E_0^\sharp DF(A). \tag{3.cdv}$$

If $\|(C'(A)E_0)^\sharp\|_{Z\to Z} < 1/2$, the bracket has its convergent Neumann inverse, and the multiplier is the explicit function

$$\omega(A) = [1 + (C'(A)E_0)^\sharp]^{-1} E_0^\sharp DF(A). \tag{3.cdvi}$$

The required transpose bound is a consequence of the column as well as row estimate of the actual lift, with the declared force and multiplier amplitudes. It does not follow from an unweighted supremum bound on E_0.

Suppose on a field ball $\|DF(A)\|_{X^*} \le \gamma$, $\|DF(A) - DF(A')\|_{X^*} \le C_F\|A - A'\|_X$, $\|E_0^\sharp\| \le C_E$, and the bracket's perturbation has Lipschitz constant C_N. The inverse difference identity gives

$$\begin{aligned} \|\omega(A)\|_Z &\le 2C_E\gamma, \\ \|\omega(A) - \omega(A')\|_Z &\le (2C_EC_F + 4C_E\gamma C_N)\|A - A'\|_X. \end{aligned} \tag{3.cdvii}$$

The second constant need not be small. However a field fixed-point equation reads it through the small factor $C'(A)^\sharp$. If $\|C'(A)^\sharp\| \le cr$ and its Lipschitz constant is c, then

$$\|C'(A)^\sharp\omega(A) - C'(A')^\sharp\omega(A')\|_{X^*} \le \{cr\,\mathrm{Lip}(\omega) + 2cC_E\gamma\}\|A - A'\|_X. \tag{3.cdviii}$$

The coefficient is small with $r + \gamma$. This is the field-only form of the coupled contraction. It shows why a non-small derivative of the multiplier is not an obstruction.

Alternatively suppose the two-component iteration has a Lipschitz matrix bounded by $\left(\begin{smallmatrix} \epsilon & \epsilon \\ C & \epsilon \end{smallmatrix}\right)$. In the norm $\max(\|A\|_X, \Theta^{-1}\|\omega\|_Z)$ its row sums are at most $\epsilon + \epsilon\Theta$ and $C/\Theta + \epsilon$. Choose a fixed $\Theta \ge 4C$ first, then $\epsilon(1 + \Theta) \le 1/4$. Both row sums are less than one half. The admitted multiplier radius must be at least Θ times the field radius used by this ball. These choices are made with the actual physical norms, not by suppressing the multiplier dimension in notation.

3.11.18.2. Separation certificates for the region operations. For sets, write the gap condition as $N_{\rho\xi_q}(\Omega_{q+1}) \subset \Omega_q$, where N_r is the metric neighborhood. Unions preserve it, since $N_r(\bigcup A) = \bigcup N_r(A)$. Intersections also preserve it, since $N_r(\bigcap A) \subset \bigcap N_r(A)$. Thus levelwise cube-wise unions and meets of separated aligned sequences are separated and aligned.

Enlargement and rounding spend a computable part of the gap. If the next region is enlarged by at most $a\xi_{q+1}$, its distance to the old complement decreases by at most $aL\xi_q$. Rounding to aligned $\mathfrak{M}\xi_{q+1}$ cubes adds at most $C\mathfrak{M}L\xi_q$. Hence

$$\mathrm{gap}_{\mathrm{new}}(q) \geq \{\rho - L(a + C\mathfrak{M})\}\xi_q. \tag{3.cdix}$$

An enlargement of the current region can only improve this lower bound. If the total finite list of enlargements spends at most half the initial coefficient, all resulting gaps remain at least $\rho\xi_q/2$. The list must be counted for the actual construction; repeating an unrecorded erosion arbitrarily many times is not covered by this claim.

This gives a mathematical certificate for each sequence used in the induction: its levels are unions and meets of the declared cube families, its finite enlargements have the total cost eq. 3.cdix, and the only change of local scale in a collar is the single-layer change proved in eq. 3.cccxciv–eq. 3.cccxcv. The original label sequence, the extension by the next-step domain, healed sequences, prepared sequences and stage sequences, towers, joined strip and glued regions, and cube-wise localizations all require such a certificate from their actual definitions. The history construction supplies the first cube families; the elementary closure facts above supply the geometric conclusion once that finite certificate is verified.

3.11.18.3. The two-radius datum descent. A section may lose a constant $C_\downarrow \geq 1$ on tests near its seam and be exact away from that seam. It would be incorrect to iterate the global bound $\epsilon \mapsto C_\downarrow\epsilon$. The relevant object is the finite dependency graph of tests.

At every descent stage assign an input vertex to each actual test read by the section and an output vertex to each resulting test. A good edge carries coefficient one. A bad edge carries at most $C_\downarrow$ and has its input in the seam determining collar. The finite section estimate has the form

$$t_{\mathrm{out}}(\tau) \leq \max_{\tau' \to \tau} c(\tau', \tau) t_{\mathrm{in}}(\tau'), \qquad c \in \{1, C_\downarrow\}. \tag{3.cdx}$$

A sum over finitely many local tests is absorbed in $C_\downarrow$ at the bad edge, not added at each later exact transfer.

Assume that a seam collar's full test-determining neighborhood is disjoint from the output footprints of the previously spoiled collars. This is the specific collar condition to check using eq. 3.cdix; disjoint geometric faces without their determining neighborhoods would not suffice. Then a directed dependency path meets at most one bad edge. Iterating eq. 3.cdx gives

$$t_{\mathrm{final}}(\tau) \leq \max_{\pi \text{ ending at } \tau} \prod_{e \in \pi} c(e) \, \max t_{\mathrm{initial}} \leq C_\downarrow\epsilon. \tag{3.cdxi}$$

This proves the uniform descent bound. At the input to each fresh collar the radius is still ϵ. Away from those collars the carried radius may be $C_\downarrow\epsilon$, but the next exact transfer does not multiply it.

The estimate is a complete finite graph argument. Its model inputs are the actual one-step section estimate and the collar-neighborhood certificate. A pointwise regularity estimate for a section with no record of where it loses its constant does not imply this two-radius conclusion. The same distinction is needed when sections are folded onto boxes: every reflected or translated determining neighborhood must remain in its recorded collar.

3.11.19. *Criticality through two actual averaging levels.* The induction changes the number of constraints. Its chain rule must use the induced fine action, not a fresh coarse Wilson action. Let m be the actual datum map for a truncation, let r be its next-step average, including the actual outward reduced datum, and suppose the weaker constraint is $r \circ m$. Let $\Psi(V)$ be a differentiable section with $m(\Psi(V)) = V$, critical for the fine action F on each m-fibre. Define $S(V) = F(\Psi(V))$.

Every fine tangent vector h at $\Psi(V)$ has the exact decomposition

$$h = D\Psi(V)\, Dm(h) + h_{\mathrm{vert}}, \qquad Dm(h_{\mathrm{vert}}) = 0. \tag{3.cdxii}$$

This follows by differentiating $m \circ \Psi = 1$. The derivative of F annihilates h_{vert}. Consequently

$$DF[h] = DS[Dm(h)]. \tag{3.cdxiii}$$

If $Dr\, Dm(h) = 0$ and V is critical for S on its r-fibre, this is zero. Conversely, if $\Psi(V)$ is critical on the $(r \circ m)$-fibre, test with $h = D\Psi\, v$, $Dr\, v = 0$, to see that V is critical for S. Thus

$$\Psi(V) \text{ is critical on } (r \circ m)^{-1}(r(V)) \iff V \text{ is critical for } S \text{ on } r^{-1}(r(V)). \tag{3.cdxiv}$$

In orbit coordinates, choose the admitted representatives and carry the node gauges through both maps. The residual gauge directions contribute zero by invariance. The same proof then applies to the quotient tangent vectors. Its geometric premise, $r \circ m$ equal to the desired weak constraint, is the finite averaging identity and the outward factor in eq. 3.cccxliv; it is not a formal convention about the names of the two data sets.

This supplies a precise noncircular order for constructing depth $k+1$. Assume the real backgrounds through depth k have already been constructed on their admitted domains, together with the needed actual readouts. Complete the last coarse datum to a datum for the depth-k truncation by the one-step section, leaving all earlier rows unchanged. At an outward last row its pivot crosses from an inner block to the outer block. That pivot is not an earlier kept inward row, precisely because those rows were omitted. The finite pivot chart therefore supplies a local constraint correction. Its regular-test and collar estimates are the section task in eq. 3.cccxxviii and eq. 3.cdx, not an assumption of a critical field.

The available background at the completed datum is a trial field for the weaker problem. It is already critical in its vertical directions, but need not be critical for the weak constraint. Its remaining force is the derivative of the induced action S_k on the last coarse datum. Its Hessian is the actual readout eq. 3.cccxl–eq. 3.cccxlviii. The primitive inverse and finite coefficient estimates for that readout are established on the real primitive class before solving the weaker problem. They may use the backgrounds of depths at most k; they must not use the unknown depth-$(k+1)$ critical field.

The lower background is evaluated on the admitted analytic datum neighborhood already constructed at depth k. Exploring this coarse variation does not restart the real datum descent at every point. The marked two-radius class is needed to construct its real center; the actual whole-disc continuation and its fixed margins control the subsequent coarse variation.

Apply the one-step coupled construction to S_k and its actual plain constraint chart, with its prescribed soft gauge. The soft gauge has to be removed or identified by the actual gauge equation on the physical solution; its mere positivity does not make a critical point of the soft action critical for S_k. Once that identity is proved, eq. 3.cdxiv makes the lifted fine field critical for the desired depth-$(k+1)$ problem.

Suppose its provisional regular radius is $C_{\mathrm{up}}C_{\downarrow}\epsilon$. All constants here are the fixed section, inverse and coefficient constants, independent of k. Require

$$C_{\mathrm{up}}C_{\downarrow}\epsilon < r_*, \qquad C_{\mathrm{prof}}\epsilon < r_{\mathrm{induct}}, \tag{3.cdxv}$$

where r_* is the input ceiling of eq. 3.ccclxxv, and r_{induct} is the radius promised to the next induction step. The profile estimate then resets the output to $C_{\mathrm{prof}}\epsilon$. The two-radius datum certificate prevents $C_{\downarrow}$ from compounding during the completion. This proves closure of the radius induction from its listed estimates.

At depth zero all the fine bonds are datum: there are no free fibre directions. The finite plaquette derivative gives $\xi_0^3|J_A| \le C \sup|\partial V - 1|$, because a bond meets at most six inner plaquettes and $\xi_0 = \eta$. This is the base regularity assertion. The first nontrivial constraint chart and physical

background step are the finite results already proved. They realize those instances of the induction, not the uniform multilevel estimates that its later steps require.

3.11.19.1. Uniqueness among regular orbits. Take two regular critical fields for the same datum. Put the first in the reference role. The preliminary contour alignment and eq. 3.cccxx give a raw closeness below the fixed gauge ceiling after choosing the common regularity radius small enough. The actual site correction puts the second field in the reference slice. Its a priori estimate in eq. 3.ccclxxiii gives $\|A\|_X \le B_1\epsilon$, independent of the preliminary constants. Both configurations then lie in the single uniqueness ball of the coupled construction.

There is no independent multiplier hypothesis: eq. 3.cdvi–eq. 3.cdvii bound the multiplier from the actual current and row lift. Thus the two solutions of eq. 3.cccxxxii, with the same reduced datum, are equal in that slice. Their relating gauge is one at the nodes by the node normalization. They belong to one residual orbit. On the admitted real regional sequence the site correction and eq. 3.ccclxxiii give the field in the ball used here. The uniqueness assertion also uses the admitted coupled system for the full set of retained rows.

3.11.20. *Why the physical soft gauge disappears from stationarity.* There is an exact local argument for the gauge step used in the physical construction. Let $S(V)$ be a gauge-invariant coarse action and fix its plain one-step averages. Assume each interior block has no prescribed node except its center. For each noncenter x let $T_{y,x}(V)$ be the admitted mean transport from its center. Let $T^0_{y,x}$ be a fixed reference transport, and put

$$G_{\mathrm{soft}}(V) = \sum_{y,x\neq y} \kappa_{y,x}\left(1 - \tfrac{1}{2}\,\mathrm{tr}[T_{y,x}(V)(T^0_{y,x})^{-1}]\right), \qquad \kappa_{y,x} > 0. \tag{3.cdxvi}$$

The coefficients are fixed on the variation under consideration. The absolute gauge is the case $T^0_{y,x} = 1$. All transports and reference ratios lie on the regular branch near the identity.

A gauge h equal to one at the block centers preserves the plain average by covariance. Vary only $h(x) = e^{itH}$ at one noncenter site. Every other mean transport is unchanged, and $T_{y,x}$ changes to $T_{y,x}e^{-itH}$. Gauge invariance gives no derivative of S. If $S + G_{\mathrm{soft}}$ is critical on this fibre,

$$\mathrm{tr}[(T^0_{y,x})^{-1}T_{y,x}H] = 0 \quad \text{for all traceless } H. \tag{3.cdxvii}$$

The orthogonal complement of the traceless 2 by 2 matrices under the trace pairing is the scalar matrices. The determinant is one, so $(T^0)^{-1}T$ equals 1 or -1. The admitted branch excludes -1. Thus

$$T_{y,x}(V) = T^0_{y,x} \quad \text{at every such site.} \tag{3.cdxviii}$$

At these values the differential of G_{soft} is zero for every field variation, not only a gauge variation: the derivative of the trace at the identity annihilates the traceless tangent. Consequently the critical point is critical for S on the fibre.

Conversely a physical critical point in this prescribed transport gauge is critical for $S + G_{\mathrm{soft}}$. The proof holds for real unitary fields and on an admitted complex branch using the bilinear holomorphic derivative. The positive real coefficient may be replaced in the complex statement by an admitted nonzero coefficient.

This argument explains all its scope requirements. A retained seam node cannot be varied, so eq. 3.cdxvii need not hold there. The no-inward geometry removes that obstruction. A sharp regular-box cutoff must be constant on the stationary neighborhood; otherwise its derivative is an extra term. An independent-current auxiliary action containing linear source subtractions is not the gauge-invariant physical S in this argument. Its physical matching must first identify the action and gauge terms. Finally the choice of relative or absolute gauge still matters off the axial locus, as eq. 3.cccxxxvi–eq. 3.cccxxxvii show.

3.11.21. *Ambient towers and their scale factors.* The tower of a localized field may sit inside a deeper separated sequence. At an element write l for the ambient scale and $\xi' \le l$ for the tower scale. The natural tower curvature and current readouts are multiplied by the same squared scale ratio:

$$\left(\frac{l}{\xi'}\right)^2 (\eta^{-2}\xi'^2|\partial U - 1|, \xi'^3|J|) = (\eta^{-2}l^2|\partial U - 1|, l^2\xi'|J|). \tag{3.cdxix}$$

The current factor is $l^2\xi'$, not l^3. This is the two-power improvement asserted by the tower estimate, not the three-power dimensional weight of a fully ambient current. Under a unit change by t, η, l, ξ' multiply by t, A by t^{-1}, and J by t^{-3}. Both entries of eq. 3.cdxix are invariant. This calculation fixes the weights in a joined sequence independently of an index convention.

The ambient estimate is proved in Prop. 3.138 for every fixed pair of contained separated sequences. At a settled element write $s(x) = l(x)/\xi'(x)$. The near input is cut off on a ball of radius $l(x)$; its weighted tower norm gains $s(x)^{-1}$, so the quadratic Cauchy remainder pays the factor $s(x)^2$. For a far input bond b, the actual geometry gives

$$d'(x, x_b) \ge d^\flat(x, x_b), \qquad s(x) \le 1 + d'(x, x_b), \qquad \frac{s(x)}{s(x_b)} \le L^{2+2d'(x,x_b)/\rho}. \tag{3.cdxx}$$

The complete row read has bounded diameter in d', and its raw logarithmic derivative contributes $s(x_b)^{-1}$. Thus, with $D = d'(x, x_b)$ and the fixed rate reserve,

$$s(x)^2 e^{-md'(x,y_c)} s(x_b)^{-1} \le C(1 + D)e^{-(m-2\log L/\rho)D} \le Ce^{-2\delta D}. \tag{3.cdxxi}$$

This is the far-input estimate; a positive distance gain is not needed for the near input. The physical current follows from the differentiated constrained Euler equation, with all six plaquettes and the actual row-transpose densities retained. The output weights are precisely eq. 3.cdxix. The real reference, determining tower and ambient sequence remain fixed in the comparison. Constants are uniform over those geometries. The smaller rate in Prop. 3.138 includes the scale-ratio and carrier losses. Fixed higher derivatives follow on smaller analytic domains with their own Cauchy constants.

3.11.21.1. Outer and inner substitutions. For an outer substitution $\mathbf{U} \mapsto e^{i\eta\mathbf{A}}\mathbf{U}$, the finite product formula gives, in normalized first-order norms,

$$e_{0,\mathrm{new}} \le C(e_{0,\mathrm{old}} + \|\mathbf{A}\|_1 + \|\mathbf{A}\|_1^2), \qquad e_{1,\mathrm{new}} \le e_{1,\mathrm{old}} + C\|\mathbf{A}\|_1. \tag{3.cdxxii}$$

Here $\|\mathbf{A}\|_1$ uses the local length and derivative factors, and e_0 includes all actual tower curvature readouts. For a tower entry the same inequality uses its applicable admitted readout estimate: the settled estimate at a settled element and the boundary estimate otherwise. It does not follow merely from the elementary plaquette formula. Current and atomic branch tests have their own finite Lipschitz estimates. If the old tests use half the target radii, the inequalities $\alpha_0 \le \alpha_1$, $\alpha_1^2 \le C\alpha_0$, and $\|\mathbf{A}\|_1 \le c\alpha_0$ leave the needed margin once c is chosen from these actual constants.

For the inner substitution, the datum is the actual window expression $\varsigma B + B'$, $0 \le \varsigma \le 1$. The first term comes from the averages of the sum of the coarse background and the fluctuation in the specified order of multiplication. The second is its boundary correction. It is not an independent scalar source. The entire nonlinear datum must first be placed in its absolute physical disc. The free response and its amplification are then applied, with the current computed from the same composed field.

The numerical age reduction can be stated without suppressing its source of smallness. Write $\delta_n = \alpha_n s_n$ for the final physical fluctuation amplitude, $\rho_j = \delta_n/\alpha_j$, $a = n - j$, and now $\sigma = L^{-a}$. The ordinary slow-radius estimate is $\alpha_n/\alpha_j \le C\sqrt{1 + a}$. The independent-coordinate datum radius satisfies $\alpha_j/r_{B,j} \le C$ on the actual near window class. If the actual fluctuation amplitude and

window response obey

$$\frac{\delta_n}{r_{B,j}} \le C\rho_j, \qquad \frac{C_{\mathrm{ax}}\delta_n\sigma}{r_{B,j}} \le Cs_n\sqrt{1+a}\,L^{-a}, \qquad a = n-j \ge 0, \tag{3.cdxxiii}$$

then the second quantity is uniformly small at all ages by choosing the common small fluctuation constant first. Indeed $\sup_{a\ge 0}\sqrt{1+a}L^{-a} < \infty$. Young ages cause no exception once that constant is chosen with the actual response and radius factors.

This verifies the central datum-disc margin. The full datum also needs the separate absolute ceiling eq. 4.xxx. To infer that the complete inner substitution belongs to every running core, the actual local window map must put its background-plus- fluctuation pair in the primitive class, including the entire outgoing physical amplitude, its gradient, the independent current, the tower and the atomic branch tests. An increment bound alone does not control that outgoing amplitude. The first-exit list eq. 3.cdiv and the domain estimates isolate those remaining finite inequalities.

3.11.22. *A real one-step section at a seam.* The finite section used in the descent can be made explicit. Work for this paragraph on the level-k lattice, so a new block contains L^4 sites. The old level-k data are prescribed outside the next region and on its exterior bonds; the newly prescribed data are on the level-$(k+1)$ rows touching the next region. Assume all ordinary and mixed plaquette tests are at most ϵ, and use an embedded two-block neighborhood for every new pivot.

For a block outside the next region, all its internal level-k bonds are old prescribed data. Compute their real mean transports and use the gauge $h(x) = T_{y,x}$, $h(y) = 1$, inside that block. The average covariance fixes all level-$(k+1)$ endpoint values. The internal transport means become one, exactly. eq. 3.cccxvi–eq. 3.cccxviii then put every internal old bond within $C_L\epsilon$ of one. This intermediate gauge is permitted to change the old datum; it will be undone at the end. It is not a residual gauge of the original fixed datum.

Complete the unassigned level-k field in this frame as follows. Inside each new block put its internal bonds equal to one. On every fine bond crossing a new coarse face put the given coarse value $V(c)$. Leave all old prescribed bonds unchanged. The no-inward convention ensures that these instructions do not prescribe an old fixed bond twice: every crossing bond with an endpoint in the new region was omitted from the old determining set.

The curvature of this trial completion is at most $C_L\epsilon$. Here is the full local check. A fine plaquette crossing no coarse face is either an old plaquette or has value one. A plaquette crossing one face has equal coarse transports on its two parallel crossing bonds; they cancel, up to internal old transports already within $C_L\epsilon$ of one. A plaquette crossing two faces corresponds to one coarse corner. If all of its coarse edges are new, its product is the new coarse plaquette. If some are old exterior edges, eq. 3.cccxviii replaces each old crossing bond by its actual coarse mean with error $C_L\epsilon$. The resulting product is exactly the assigned mixed test. These alternatives exhaust the 0, 1 and 2 coordinate faces that a fine plaquette can cross. All prefactors are unitary.

For a new coarse row, its inner internal transports are one and its outer internal transports are one or within $C_L\epsilon$ of one. Each straight route crosses its coarse face once. The defining closed logarithmic factors are therefore $O(C_L\epsilon)$. Its actual average differs from the desired $V(c)$ by that amount. There is no smallness assumption on $V(c) - 1$.

Correct this discrepancy on the assigned crossing pivot. The finite pivot chart gives a uniformly invertible real derivative and a fixed admitted radius depending on L. For ϵ below its fixed threshold it gives a pivot logarithm t_c with

$$\mathcal{M}(V^{\mathrm{trial}}[t_c])(c) = V(c), \qquad |t_c| \le C_L\epsilon. \tag{3.cdxxiv}$$

The internal transport paths contain no crossing pivot, and a straight segment of another coarse row contains no such pivot: the segment either has a different orientation or crosses a different coarse face. Thus its average does not depend on t_c. All these corrections can be performed independently. No norm cost is proportional to the number of new rows.

A fine plaquette has at most four corrected edges. Its unitary product changes by at most $4\sup_c |t_c|$ times a fixed small-chart constant. The completed datum therefore satisfies

$$\sup|\partial V^{\downarrow} - 1| \le C_{\downarrow}\epsilon, \qquad \mathcal{M}(V^{\downarrow})|_{\Gamma_{k+1}} = V_{k+1}, \qquad V^{\downarrow}|_{\Gamma_k} = V_k. \tag{3.cdxxv}$$

Undoing the intermediate unitary gauge restores the old prescribed values and preserves these test bounds. Mixed tests for the completed truncation are checked by the same corner argument. The one-step coarse constraint is exact.

This is a real finite section with a fixed constant $C_{\downarrow}(L)$. Its hypotheses include the actual ordinary and mixed tests, the regular finite mean branch and the pivot support premise. Its output is a coarse datum, not a regular fine critical field. In a refinement where no seam is present, the face lift is already exact before the pivot corrections. To use a coefficient one in the two-radius graph eq. 3.cdx, one records only the test coordinates on which that exact transfer preserves the original test; the remaining corner and pivot coordinates belong to the spoiled collar footprint. Their future determining neighborhoods must satisfy the disjointness certificate of that graph.

The finite construction does not prove a smoothed trial bound for the bare current at arbitrary depth. It supplies the one-step datum completion used before the lower background is evaluated. The regular fine trial then comes from that already constructed lower background, as in eq. 3.cdxii–eq. 3.cdxv. This avoids mistaking the face-supported datum for the desired continuum scale potential.

3.11.23. *Descent of the datum and ascent of the critical field.* The induction on the number of levels need not preserve an exact one-step average during the descent of the datum. It must preserve the older rows exactly and leave a uniformly small error on the new rows. The coupled critical equation removes this error during the ascent. The following argument keeps the error from acquiring a factor at each level.

We use the no-inward determining sets of eq. 3.cccvi. In addition to the geometry already specified, fix powers $\mathfrak{M}, R_1, \rho$ of L with

$$L \ge 13, \qquad \mathfrak{M} \ge \mathfrak{m}^{\mathrm{s}}, \qquad R_1 \ge L, \qquad R_1\mathfrak{M} \ge L^2\rho, \qquad \rho \ge \mathfrak{r}^{\mathrm{s}}.$$

Every level-k coordinate circle has at least 6ρ sites. The gap in eq. 3.ccciv is greater than $R_1\mathfrak{M}\xi_j$. The fixed floors $\mathfrak{m}^{\mathrm{s}}, \mathfrak{r}^{\mathrm{s}}$ depend on L. Empty regions and regions equal to the whole torus are allowed. These are quantitative restrictions on the geometry; separation with an unspecified constant is not a substitute for them.

Let $\mathbf{B}^{(n)}$ be the determining set of the truncated sequence $(\Omega_0, \dots, \Omega_n)$, with all its inward rows removed. Its last member contains the level-n bonds inner to Ω_0 that touch Ω_n. Its earlier members agree with the corresponding retained members of $\mathbf{B}$.

3.11.23.1. The two radii for a datum. Fix the constant $C_{\downarrow} \ge 1$ of the finite seam construction. A datum belongs to $\mathcal{D}(\varepsilon)$ if all its ordinary and mixed test holonomies are defined and within $C_{\downarrow}\varepsilon$ of one, and the tested plaquettes of levels $j-1, j$ within $6\xi_j$ of $\partial\Omega_j$ are within ε of one. All distances in this paragraph use level-k units. For a determining set retaining inward rows, one must also impose the ε bound on each pinch holonomy. Such holonomies compare the two retained transports ending at the same seam node; they are additional compatibility tests. They are absent for the determining sets used here.

For $C_{\downarrow}\varepsilon \le \varepsilon_{\downarrow}$ the finite seam construction gives data $V^{(n)}$ on $\mathbf{B}^{(n)}$, with $V^{(k)} = V$, such that

$$V_j^{(n)} = V_j \qquad \text{on the original rows of level } j \le n, \tag{3.cdxxvi}$$

$$|\mathcal{M}(V_{n-1}^{(n-1)})(c)(V_n^{(n)}(c))^{-1} - 1| \le C_{\downarrow}\varepsilon \qquad (c \in \Gamma_n^{(n)}). \tag{3.cdxxvii}$$

The descended datum satisfies $V^{(n)} \in \mathcal{D}(\varepsilon)$.

Here is the reason for using two radii. In a last-level cell the unassigned fine bonds are filled by the face lift: internal bonds are one, and bonds crossing a coarse face carry its coarse value, in the local

transport frame. A fine plaquette deep inside the new region has value one or the corresponding coarse plaquette. It therefore inherits the larger bound without another factor. Near the seam the comparison with the old prescribed bonds costs $C_{\downarrow}$. All tests used in that comparison lie within three coarse spacings of the row or plaquette under consideration. They are within six spacings of the current boundary and hence obey the smaller bound. Separation keeps these tests outside the region altered in the preceding descent step. Older tests remain unchanged. This proves the assertions by downward induction; the bound is $C_{\downarrow}\varepsilon$, not $C_{\downarrow}^{k-n}\varepsilon$.

This descent only asserts eq. 3.cdxxvii at a general seam row. Its average is exact when both row endpoints and their sup-norm neighbor shells of one level-n spacing lie in the new region. The independent pivot corrections of eq. 3.cdxxiv produce a different finite section under their stated support premise. The present induction uses neither equality of these sections nor exact seam averages during descent.

3.11.23.2. The one-level analytic input. Write $\mathfrak{U}'_n(r)$ for the regularity class with curvature and full-bond current at radius r. In bare units ($\eta = 1$), if $j_n(e)$ is the largest $j \le n$ for which e has a corner in Ω_j, its tests are

$$|\partial U(p) - 1| < rL^{-2j_n(p)}, \qquad |\eta D_U^{\mathsf{T}} \partial U(b)| < rL^{-3j_n(b)}.$$

The second expression is the covariant plaquette divergence on full bonds; the truncated boundary action has its separate boundary current. This convention is needed in applying the local regularity estimate.

The analytic step needed in this induction has the following precise form. Let $\varepsilon' > 0$ be small enough that $L^3 B_3^{\mathrm{s}} \varepsilon' \le c_{10}$ and $K_9 L^3 B_3^{\mathrm{s}} \varepsilon' \le a_5(\rho)$, where K_9 is the provisional regularity constant below. Suppose the new datum has all test holonomies defined and strictly within ε' of one, and a regular field $U_0 \in \mathfrak{U}'_{n-1}(B_3^{\mathrm{s}}\varepsilon')$ has the prescribed averages on all older rows and differs from the desired level-n average by at most ε'. The coupled equation, followed by regularity improvement, must produce a critical field for the new datum in $\mathfrak{U}'_n(B_3^{\mathrm{s}}\varepsilon')$. The same constants and smallness thresholds must apply at every truncation. This is an assertion about the actual row equations, their node factors and the Wilson current.

The quantitative use of the coupled equation is as follows. In level-n units put $\alpha_0 = L^3 B_3^{\mathrm{s}} \varepsilon'$. The old field is in the new regularity class at radius α_0. The datum displacement vanishes at every older row, and at the new rows it is

$$B_n(c) = \frac{1}{i} \log\{V_n(c)(\mathcal{M}^n U_0(c))^{-1}\}, \qquad \sup |B_n| \le \frac{\pi}{2}\varepsilon' \le \alpha_0.$$

The boundary derivative of the inner Wilson action is bounded in the force norm by $6\alpha_0$: at a full bond this is the current estimate, and at a boundary bond it follows by summing its at most six incident inner plaquettes. Such a bond is on the bare layer, so the three required powers of the local spacing are exactly the bare powers.

Work on the original bare region of the new truncation. Write $q(A)$ for the straightened rows and $b^\natural$ for their prescribed values. The outward factor in eq. 3.cccxliv is determined by the already prescribed interior rows of the next finer block. Those displacements are zero in this ascent, so the factor is one and $\|b^\natural\|_\infty \le \alpha_0$. Write $K_A = q'(A)$ and F for the inner Wilson action in the logarithmic chart based at U_0. The equations are

$$F'(A) + DRD^\sharp A = K_A^\sharp \omega, \qquad q(A) = b^\natural.$$

In the isometric row frames of the regional inverse, write $T^{-1}q(A) = Q^\natural A + C(A)$ and $\widehat{b} = T^{-1}b^\natural$, using dimensionless row coordinates. Then

$$F'(A) + DRD^\sharp A = F'(0) + \Delta_R A + \mathcal{V}(A), \qquad \Delta_R = D^\sharp D + DRD^\sharp + \Delta'.$$

Here D in the curl term is covariant at U_0, and Δ' is the curvature correction to the covariant curl Hessian of the inner Wilson action. Let H_0 be the energetic row lift for the regional inverse

and $\widetilde{G}_0$ its constrained inverse. Thus $Q^\natural H_0 = 1$, $Q^\natural \widetilde{G}_0 = 0$, and $\ker \widetilde{G}_0 = \operatorname{ran} Q^{\natural\sharp}$. With $\mathcal{G}(A) = F'(A) + DRD^\sharp A$, the field equation is

$$w(A) = (1 + H_0^\sharp C'(A)^\sharp)^{-1} H_0^\sharp \mathcal{G}(A),$$
$$\Psi(A) = \widetilde{G}_0[-F'(0) - \mathcal{V}(A) + C'(A)^\sharp w(A)] + H_0[\widehat{b} - C(A)].$$

Applying $Q^\natural$ to $A = \Psi(A)$ gives the exact row equation. The identity $\widetilde{G}_0 \Delta_R = 1 - H_0 Q^\natural$ puts the residual of the linear field equation in $\ker \widetilde{G}_0$. Applying $H_0^\sharp$ identifies its multiplier as $w(A)$, giving $\omega = Tw(A)$.

The multiplier estimate uses only first differences of A, by the identity

$$H_0^\sharp \mathcal{G}(A) = H_0^\sharp[F'(0) + \Delta' A + \mathcal{V}(A)] + (DH_0)^\sharp DA + (D^\sharp H_0)^\sharp RD^\sharp A.$$

The weighted column estimates for the lifts bound this by $K_8(\mathsf{j} + \|A\|_X + |\mathcal{V}(A)|_{(-3)})$, where $\mathsf{j} = |F'(0)|_{(-3)}$. In the difference of the expressions at two fields, $F'(0)$ cancels. Let K_1, K_7 be the bounds of $H_0, \widetilde{G}_0$ in the row-to-field and force-to-field norms, and set $K_8 = K_1(19 + 12K_1^3)$, $K_5 = 2(K_7 + K_1)$ and $\Theta = 4K_8$. The bound on the Neumann perturbation is $4K_1 L^3 \epsilon^* \le 1/8$, where $\|C'(A)^\sharp\|_{\infty\to(-3)} \le 4L^3\epsilon^*$. Use $\operatorname{Lip}(\mathcal{V}) \le 4C_{4,\mathrm{svar}} L^4 r$, $\operatorname{Lip}(C'^\sharp) \le 2L^4 C_3^\natural$ and $\operatorname{Lip}(C) \le L\epsilon$. Here ϵ, ϵ^* are the row and transpose error bounds at $(\alpha_0, r_\circ)$, both $O(\alpha_0 + r_\circ)$. The inverse difference identity gives $\|w(A)\|_\infty \le (17/7)K_8 r$ and $\operatorname{Lip}(w) \le 3K_8$ for $\mathsf{j} \le r \le r_\circ$. In the difference of $C'(A)^\sharp w(A)$ use the small bound on $C'(A)^\sharp$ with the difference of w, and the small bound on w with the difference of C'. The resulting contraction constant is at most

$$K_7[4C_{4,\mathrm{svar}} L^4 r_\circ + 6K_8 L^4 C_3^\natural r_\circ + 12K_8 L^3 \epsilon^*] + K_1 L\epsilon.$$

Choose the fixed chart thresholds as in the regional contraction estimates, so this is at most $1/2$. They also ensure that $\alpha_0 + 3L^2 r_\circ < L^{-2} c_{2,\mathrm{sobj}}$, the curvature ceiling for the physical stationarity argument. At zero the inverse bounds give $\|\Psi(0)\|_X \le (K_5/2)(\mathsf{j} + \|b^\natural\|_\infty)$. The threshold c_{10} keeps $r = K_5(\mathsf{j} + \|b^\natural\|_\infty) \le 7K_5\alpha_0$ within the fixed contraction radius $r_\circ$. Both closed balls are mapped into themselves, so their unique fixed points coincide.

For a real infinitesimal gauge λ, write $\delta_\lambda A$ for its bond tangent and $\varrho_A \lambda$ for the derivative of the normalized pillar-top values. After the site row conjugation the linear part of the site pair is $(-R\Delta_{U_0}\lambda, Q'\lambda)$. Differentiating the real gauge change and the node factors gives errors bounded by $C(\alpha_0 + \|A\|_X)\|\lambda\|_Y$ in the weighted site force and top-row norms. Compose with the covariant site inverse eq. 3.ccclxxii with the first sign reversed. Its uniform site bound gives identity plus an error of norm less than $1/2$ on the fixed chart. This proves that $\lambda \mapsto (RD^\sharp \delta_\lambda A, \varrho_A \lambda)$ is onto. When $\varrho_A \lambda = 0$, the derivative $K_A^\flat$ of the normalized reduced rows vanishes on $\delta_\lambda A$ by covariance. The node relation changes each outward row by a factor depending only on the already retained interior rows of the next finer block, giving a triangular change of row coordinates with invertible diagonal. Hence the kernel of $K_A^\flat$ is $\ker K_A$. Gauge invariance also gives $\langle F'(A), \delta_\lambda A\rangle = 0$. Pairing the field equation with these tangents gives $\langle RD^\sharp A, RD^\sharp \delta_\lambda A\rangle = 0$. With $\varrho_A\lambda = 0$ the first entry of the site pair can be any vector in $\operatorname{ran} R$. Taking it to be $RD^\sharp A$ gives $RD^\sharp A = 0$. Hence $F'(A) = K_A^\sharp \omega$. Differentiating the physical rows adds the infinitesimal top-site rotations to the normalized derivative. The derivative of the normalized top map on gauge tangents is onto by the site pair above. Consequently a physical constraint tangent can be made a normalized reduced constraint tangent by adding a gauge tangent. Both parts are annihilated by $F'(A)$, giving stationarity on the physical constraint fibre. In the solved field equation the second difference is now $K_A^\sharp \omega - F'(0) - \Delta' A - \mathcal{V}(A)$. The regional row and force estimates bound this by $4(1 + L^3)\Theta r$ in the weighted force norm. This gives

$$\|A\|_{X,\mathbf{B}} \le 7K_5\alpha_0, \qquad |D^\mathsf{T} DA|_{(-3)} \le 28(1 + L^3)\Theta K_5 \alpha_0.$$

Restore the node values by taking the gauge equal to $v_j[A]^{-1}$ at tops of level j. On interior rows eq. 3.cccxliii gives the prescribed averages. On an outward row the gauge at its outer end is $v_{j-1}[A]^{-1}$. The one-step node factor in eq. 3.cccxliv cancels against the same factor of the straightened datum, giving the exact physical datum for both orientations. Gauge covariance carries constrained stationarity to the restored field.

Set $\alpha_2 = 7L^2 \max\{1, 4(1+L^3)\Theta\}K_5\alpha_0$. The threshold $c_{10} \le c_8, c_9$ also ensures $\alpha_2 \le \rho_{1,\mathrm{svar}}$, $C_{4,\mathrm{svar}}\alpha_2 \le 1$, $4\alpha_2 + C_{4,\mathrm{svar}}(\alpha_0+\alpha_2) \le 1/2$, and $2\alpha_0 + 3\alpha_2 \le 1/3$. The curvature and current expansions with the displayed first and second difference bounds give provisional regularity radius $2(2\alpha_0+3\alpha_2) = K_9\alpha_0$. The same bounds hold after node restoration by covariance. By the datum threshold $K_9\alpha_0 \le a_5(\rho)$. To reduce this provisional radius use the improvement in the class with full matrix divergence. On a critical field with input radius $e_0 \le a_5(\rho)$ and datum bound $\varepsilon' \le e_0$, its output radius is

$$\max\{6\varepsilon',\, 3K^\sharp\varepsilon' + 3\vartheta e_0\}, \qquad \vartheta = C_b\rho^2 e^{-\delta_*\rho/24} + C_c\rho^4 e_0.$$

The estimate at positive index uses the localized critical equation. At index zero the plaquette is prescribed, and the full-bond divergence is bounded by the six incident plaquettes of index at most one. Choose the separation floor and $a_5(\rho)$ so the two terms of ϑ are at most $1/24$ each. Since $B_{\mathrm{I}} \ge \max\{6, L^2, 12K^\sharp\}$, the output radius is at most $\max\{B_{\mathrm{I}}\varepsilon', e_0/2\}$. Iterating until the first term dominates gives radius $B_{\mathrm{I}}\varepsilon' \le B_3^{\mathrm{s}}\varepsilon'$. This last return, including its current estimate, is essential; the factor L^3 cannot be iterated as a loss of regularity.

Proposition 3.34 (The finite-depth induction)**.** Assume the one-level analytic input just stated, uniformly in the admitted geometry, and the regularity improvement and regular-orbit comparison estimates used in that input. There are radii $a_0^{\mathrm{s}} > 0$ and $a_1^{\mathrm{s}}(\rho) > 0$, independent of the depth and torus, with $B_3^{\mathrm{s}} a_1^{\mathrm{s}}(\rho) \le a_0^{\mathrm{s}}$, such that a real datum whose test holonomies are all within $0 < \varepsilon_1 \le a_1^{\mathrm{s}}(\rho)$ one has a critical configuration in $\mathfrak{U}'_k(B_3^{\mathrm{s}}\varepsilon_1)$. Any two critical configurations for that datum in $\mathfrak{U}'_k(\varepsilon_0)$, with $\varepsilon_0 \le a_0^{\mathrm{s}}$, belong to the same residual gauge orbit. ■

Proof. Take $\varepsilon' > C_\downarrow\varepsilon_1$ in the uniform smallness range, with $\varepsilon' \le 2C_\downarrow\varepsilon_1$, and first descend the datum as in eq. 3.cdxxvi– eq. 3.cdxxvii. At depth zero the datum itself specifies every inner bare bond. Its curvature is bounded by ε'; its covariant plaquette divergence is bounded by $6\varepsilon'$. Since $B_3^{\mathrm{s}} \ge 6$, this starts the upward induction.

Suppose $U^{(n-1)}$ has been constructed. On every double block read by a level-n row, all level-$(n-1)$ bonds belong to the last member of the preceding truncation. Exact locality of the one-step average therefore gives

$$\mathcal{M}^n U^{(n-1)}(c) = \mathcal{M}(V_{n-1}^{(n-1)})(c).$$

Together with eq. 3.cdxxvii, this verifies the new datum error required by the analytic step. That step produces $U^{(n)} \in \mathfrak{U}'_n(B_3^{\mathrm{s}}\varepsilon')$. The older constraints are exact by eq. 3.cdxxvi. The same bound consequently holds at every depth. At depth k the radius is at most $B_3^{\mathrm{s}}\varepsilon'$. Choose $a_1^{\mathrm{s}}(\rho)$ small enough that $2B_3^{\mathrm{s}}C_\downarrow a_1^{\mathrm{s}}(\rho)$ is below the initial radius allowed by the improvement estimate on the full sequence. Applying that estimate at the original datum radius ε_1 gives $B_3^{\mathrm{s}}\varepsilon_1$.

For uniqueness, take one regular critical field as reference. The regular-orbit comparison places any second one in its Landau chart, with zero reduced datum and field norm at most $\mathsf{B}_1\varepsilon_0$. Choose $a_0^{\mathrm{s}} \le r_\circ/\mathsf{B}_1$ and $6K_5 a_0^{\mathrm{s}} \le r_\circ$, in addition to the curvature ceilings of that chart. Both the zero field and the second field then solve the same coupled equation in its uniqueness ball. The field component of the second solution is zero. Its gauge fixes the tops, and the common prescribed datum fixes the retained seam nodes; hence it is a residual gauge. This proves orbit uniqueness. □

For the actual determining sets eq. 3.cccvi, Prop. 3.124 verifies these analytic premises using Props. 3.88 and 3.120 and the comparison and reset estimates Lems. 3.121 to 3.123. The proof retains the complete outward blocks and the inner-action boundary force. Its uniqueness conclusion

concerns all regular critical configurations in the stated radius. It does not identify an unrestricted global minimum or assert an infinite-volume or continuum limit.

3.11.23.3. Recovering the regularity radius from the datum. In the separated geometry choose a power σ of L satisfying $\sigma \geq L^2\rho_{\mathrm{c}}$, with $\mathfrak{M} \geq \mathfrak{M}_{\mathrm{c}}$ and $\rho \geq L^2\sigma$, and with at least 6ρ sites in each direction at the top level. Let U be critical and already regular at a radius a_{c}. The fixed floors $\mathfrak{M}_{\mathrm{c}}, \rho_{\mathrm{c}}$ come from the local flat inverse construction. At a bare site x let $r_U(x)$ be the maximum of $\eta^{-2}\mathsf{L}(p)^2|\partial U(p) - 1|$ and $\mathsf{L}(b)^3|J(U)(b)|$ over the plaquettes and full bonds based at x. Here L is the local layer spacing and the current in this test is formed at full bonds. For $R = 20L\sigma$ in the weighted distance, let $r_R(x)$ be the supremum of r_U over the ball and $v_R(x)$ the largest defect in the datum tests meeting it, including the pinch tests if inward rows are kept. The local estimate at a regular critical field has the form

$$r_U(x) \leq K_{\mathrm{loc}} v_R(x) + \tfrac{1}{2} r_R(x).$$

Here are the local estimates behind this inequality. If x is in the bare layer away from the next layer, the bonds in the relevant plaquettes are datum bonds. The current there is bounded by the sum of at most six such plaquette defects. Otherwise choose level k to be the layer of x, or the next layer when x is within 5σ in the units of its own layer of that next member. In the units of level k, the enlarged cube reads only layers $k-1, k$. Write Q_s for the s-neighborhood of the partition cube of side σ containing x. For the auxiliary sequence take the full torus at level zero and Q_{s_i} for $1 \leq i < k$, where

$$s_k = \sigma - \rho_{\mathrm{c}}, \qquad s_i = s_k + \sum_{i \leq l < k} (\rho_{\mathrm{c}} + \mathfrak{M}_{\mathrm{c}}) L^{l-k}.$$

The member at level k is $Q_{s_k} \cap \Omega_k$. Keep an inward row of this auxiliary sequence only when the original determining set keeps it. Its positive-level members are subordinate to the original ones, and the largest is inside $Q_{\sigma-2}$. If $v_R(x)$ is above the fixed ceiling of the datum transport gauge, the local inequality follows from $r_U(x) \leq a_{\mathrm{c}}$ by increasing K_{loc}. Use that gauge below its ceiling.

For the real local gauge on Q_σ, take the enlargement of side $3\sigma + 4L\mathfrak{r}_{\mathrm{p}}$, where $\mathfrak{r}_{\mathrm{p}}$ is the fixed cube floor. It lies in $Q_{3\sigma}$ and lifts to a single cube. The two local scales give curvature at most $L^2 r_R(x)\eta^2$, and the current comparison gives covariant divergence at most $2L^3 r_R(x)\eta^2$. The local regular-gauge estimate with these two inputs bounds the potential, its first difference and its flat second-order reads on Q_σ by $C\sigma r_R(x)$ in level-k units. Its smallness condition $C\sigma r_R(x) \leq c$ is included in the radius choice. Extend this potential by zero on the other bonds of the auxiliary torus. A constrained variation there is fixed on the bare determining bonds outside the largest positive-level member. Patch it to the original field across Q_σ. The patched curve fixes each old row. A row whose two endpoints are below its reading level in the auxiliary sequence is fixed by induction in the averaging equation. A row with an endpoint of exactly its reading level is an auxiliary constraint, and its two-block read is inside Q_σ. The old inward rows meeting the varied set, when present, are inherited auxiliary constraints at the interface of levels $k-1, k$. Every varied plaquette lies within Q_σ, so the two actions have the same derivative along the curves. This proves criticality for the extension at its own auxiliary datum. The cutoff region retains the actual current of the extension. Its local length there is the bare spacing, so the zero extension costs $C\sigma r_R(x)$ in the weighted current norm. Passing to the flat Landau slice with the prescribed node factors gives a datum bound $b_{\mathrm{f}} \leq C\sigma^2 r_R(x)$ and the flat response bound $r_{\mathrm{f}} \leq C\sigma^2 r_R(x)$.

In the inner cube the datum in that slice is smaller. The transport gauge of the datum writes each nearby row in terms of the tested plaquettes and, at a retained seam, the tested holonomies. A row based at y has size at most $C(1 + |y - x|)v_R(x)$ in the chosen scale. At the outer endpoint of a coarse row the extra straightening factor is one, since the prescribed datum gauge makes the corresponding lower contour one. At an edge row the two factors at the common seam node cancel, leaving the tested seam transport.

The flat response uses the full flat linear package of Lem. 3.36 when inward edge rows are retained. It bounds the weighted potential and its first difference at $|y-x| \le \sigma/8$ by

$$C(1+|y-x|)v_R(x) + Ce^{-c\sigma}b_{\mathrm{f}} + Cr_{\mathrm{f}}^2.$$

This is the near and far split of the datum sum. The linear distance factor is summable against the kernel, and the far rows are at distance at least $\sigma/8-2$. In the multiplier equation split the differentiated potential sum at the same inner cube. For rows within $4L+2$ of x this gives

$$|\omega(r)| \le Cv_R(x) + Ce^{-c\sigma}(b_{\mathrm{f}} + r_{\mathrm{f}}) + Cr_{\mathrm{f}}^2.$$

The actual row differential in $F'(A) = K_A^{\natural *}\omega$ reads only these nearby multipliers at the full bond. The plaquette estimate uses the potential bound on the neighboring tops. The two readouts agree with those of the original field up to gauge, and the old local length is at most L times the auxiliary length at these reads. The resulting coefficient of $r_R(x)$ is at most $C\sigma^2 e^{-c\sigma} + C\sigma^4 a_{\mathrm{c}}$. Choose σ first and then a_{c} to make this at most $1/2$, with the ceilings of the local gauge and flat inverse also imposed at that choice.

The estimate implies regularity improvement with no accumulation in the number of layers. Let

$$P(x) = \sup_{\tau} e^{-\delta d_{\mathbf{B}}(x,\tau)}|\partial V(\tau) - \mathbf{1}|, \qquad e^{\delta R} = \sqrt{2},$$

where τ ranges over the datum tests. The triangle inequality gives $v_R(x) \le \sqrt{2}P(x)$ and $P(y) \le \sqrt{2}P(x)$ for y in the ball. For $a > 0$, maximize $r_U(x)/(P(x)+a)$ over the bare sites. If S is the maximum, the local inequality at a maximizing site gives $S \le \sqrt{2}K_{\mathrm{loc}} + S/\sqrt{2}$. Letting $a \downarrow 0$ yields $r_U(x) \le 5K_{\mathrm{loc}}P(x)$. Moving from the base point of a bond or plaquette to a corner costs at most two bare steps in the weighted distance, so the same estimate holds there with an enlarged fixed constant. In particular, when every datum test is bounded by ε', the regularity radius is at most $C\varepsilon'$. The flat chart admission in this calculation is verified in Lem. 3.123; the provisional one-level output lies in its permitted radius by eq. 3.mlxxxix.

3.11.24. *Norms and the remaining finite estimates.* The natural weights in the nonlinear estimates can be stated explicitly. One field norm and force norm are

$$\|A\|_{X_1} = \max\{\|\mathsf{L}A\|_\infty, \|\mathsf{L}^2\nabla_{U_0}A\|_\infty\}, \qquad \|f\|_{\mathcal{F}} = \|\mathsf{L}^3 f\|_\infty. \tag{3.cdxxviii}$$

The force is the physical density in the fine pairing $\eta^4\sum_b(f_b, A_b)$. The datum b is a raw row logarithm, with norm $\|b\|_Y = \sup_c|b_c|$. Use the counting pairing $\sum_c(\omega_c, b_c)$ for its multiplier and $\|\omega\|_Z = \sup_c|\omega_c|$. For differentiated conclusions append the separate natural second-order and current readouts; do not replace them by the supremum of every second difference.

This convention explains a normalization in the named propagator formulas. The per-length row field is $B_Q(c) = \xi_c^{-1}b_c$ with cell pairing $\sum_c \xi_c^4(\omega_Q(c), B_Q(c))$. Equality with the raw pairing requires

$$\omega_Q(c) = \xi_c^{-3}\omega_c, \qquad \|\omega\|_Z = \|\xi_c^3\omega_Q\|_\infty. \tag{3.cdxxix}$$

It is the bilinear weighted transpose of $b = \mathsf{m}B_Q$. Similarly the per-length penalty $\sum_c \xi_c^4 a_0 \xi_c^{-2}|B_Q(c)|^2$ equals $\sum_c a_0|b_c|^2$. Thus its raw mass is a_0, while its per-length mass is $a_0\xi_c^{-2}$. The generic identity in eq. 3.cccxlii uses one consistent pairing; eq. 3.cccxlvi–eq. 3.cccxlviii use the per-length convention. Converting their multipliers and quadratic readouts uses eq. 3.cdxxix.

The requirements for a general background construction can be organized as the local inequalities below. For the proper determining sets in force, the later construction supplies the estimates needed for the physical problem on its stated domains. The strong row saddle is proved in Prop. 3.88; the actual Wilson and constraint remainders are bounded in eq. 3.mxxxv and eq. 3.mxxxviii. The site inverse and finite gauge construction are given by eq. 3.mxliv–eq. 3.mlviii. These estimates use the physical amplitudes eq. 3.mxxix, retain every outward factor, and apply about real references with the required actual curvature and current bounds. Their uniform holomorphic continuation and

weighted response are proved in Prop. 3.117 and eq. 3.mlxvii. Props. 3.120 and 3.124 then supply the regular-trial correction and the real background induction on those determining sets.

The canonical independent-current completion is fixed separately in Prop. 3.90, with its strong bounds in Prop. 3.92. These results apply to that specified operator family. For any other primitive package, the list and conditional assembly below retain its own domain, norms and complete projected comparison eq. 3.cccxxxviii; no identification follows merely from using the same operator names.

The site estimate. For the actual Q', including its outward endpoint convention, the real site correction uses the weighted inverse on the admitted separated sequence. On the primitive complex ball one needs the corresponding weighted kernel bounds for eq. 3.ccclxxii and its first differences. Prove the nonlinear gauge and node remainders there with fixed admission margins. This is the quantitative version of the completely proved projector algebra in eq. 3.ccclxviii–eq. 3.ccclxxii.

The constrained inverse. For the actual Lagrangian and all actual row and gauge terms, prove

$$\|\mathcal{S}_0^{-1}\|_{\mathcal{F}\oplus Y\longrightarrow X_1\oplus Z}\le C,\qquad \sup_y\sum_c e^{md_{\mathbf{B}}(y,y_c)}\|(\mathcal{S}_0^{-1})_{yc}\|_{\mathrm{norm}}\le C, \tag{3.cdxxx}$$

and the corresponding column and readout estimates. The normalized entry includes the physical cell weights of its source species. This is required first on the primitive real class, then by a whole-ball complex perturbation. The complete projected current and gauge comparison in eq. 3.cccxxxviii, with every term of eq. 3.cccxlvi–eq. 3.cccxlviii, is a separate identification task.

The local coefficient estimate. For the actual fine Wilson action, actual straightened constraint and actual datum factors, prove eq. 3.ccclii in these norms, with

$$c_F(r)+c_q(r)+c_{q\omega}(r)+c_{\mathrm{mix}}(r)\le C(r+\gamma_0+\epsilon_0) \tag{3.cdxxxi}$$

in the chosen linearization, or the corresponding small triangular matrix just treated. The finite coordinate remainders must be summed using their actual supports. For example bare second differences created by a group product have to be paired with their actual powers of η, or summed by parts before a kernel is applied. Counting all fine variables of a pillar as independent losses is not a proof of eq. 3.cdxxxi.

The slice improvement. The current comparison and eq. 3.ccclxxiv give the reset in the real slice under its fixed potential ceiling. The descent and ascent also require the preliminary contour alignment at seams with its single defect cost. The exact rigidity and summability eq. 3.cccxix–eq. 3.cccxx already supply the group-valued part.

The critical patch estimate. The auxiliary sequence and the flat row readouts give eq. 3.ccclxxvi and the pointwise profile on the admitted real geometry. The real cube gauge supplies the potential and second-order bounds used on that geometry. The maximum argument transfers the local inequality to the profile.

The continuation and tower estimate. Lem. 3.127 and Prop. 3.129 construct the common analytic ball and bound the ratio M/r_0^2 in eq. 3.ccclxxxvii, including the actual representative gauge and physical current. Cor. 3.130 supplies both erosion rules and every truncation. The outer and inner substitutions, with their full parameter domains and nested tower tests, are proved in Props. 3.132 and 3.135. The geometric and radius comparisons below use these estimates with their stated localization collars.

These are finite lists of specific actual inequalities and identities with their uniformities. Their proofs can share the same matrix estimates, and some can be proved in a different order. They are not all extra assumptions on every possible route. For example a direct smooth fine section can be replaced by the real coarse datum section eq. 3.cdxxv and the lower background induction. The latter avoids requiring an independent fine trial construction.

3.11.25. *The six conditional background conclusions.* Here is a precise way to assemble the reductions without promoting an estimate that has not been proved. Suppose the preceding finite

estimate list is established on the declared primitive domains, with its actual normal-form matches and fixed margins. Suppose also that the history regions satisfy their finite separation certificates, and that the one-step section uses the proved pivot support geometry. All constants below may depend on $L, \rho, \mathfrak{M}$ and the stated local geometry, but not on depth, volume, region shape, datum position or real holonomy. Then the following six conclusions hold.

3.11.25.1. Regular orbit. Every datum in the specified small regular class has one regular critical residual orbit, with

$$\sup_p \eta^{-2}\mathsf{L}(p)^2|\partial U(p) - 1| + \sup_b \mathsf{L}(b)^3|J_A(U)(b)| \le C \sup_\tau |\partial \mathbf{V}(\tau) - 1|. \tag{3.cdxxxii}$$

It has the unique comb representative outside all nodes. Existence is the datum descent, the actual induced action step and eq. 3.cdxiv, with the uniform radius reset. The induction is on the marked two-radius data class of eq. 3.cdx, so each recursive completion encounters at most one loss on a test path. The single global small base radius is chosen after that fixed $C_\downarrow$. Uniqueness is the slice argument following eq. 3.cdxv. The representative is the exact construction in eq. 3.cccix–eq. 3.cccx. At trivial complete datum the orbit is flat and is an absolute minimum of the nonnegative Wilson action. At other data the conclusion is regular criticality.

3.11.25.2. Local regularity. The field has the profile eq. 3.ccclxxix–eq. 3.ccclxxx. On a cube of side $m\xi_j$ with its prescribed enlargement inside Ω_j and shorter than half the torus, $m\epsilon \le c$ gives a real local gauge with

$$\xi_j|A| + \xi_j^2|\nabla A| + \xi_j^3|\Delta A| + \xi_j^3|\partial^\sharp \partial A| \le Cm\epsilon. \tag{3.cdxxxiii}$$

Its first differences have the Hölder bound in eq. 3.ccclxvii. Here $\Delta = -\Delta_\eta$ is the positive componentwise Laplacian; the heat generator Δ_η in eq. 3.ccclxxxi has the opposite sign. The full-current radius is required on the enlarged cube. The single real gauge construction supplies the ordinary Laplacian in eq. 3.cdxxxiii, and the covariant divergence comparison supplies its flat curl term. The constant-layer site estimate then gives the Hölder bound.

3.11.25.3. Complex continuation. For each admitted real datum center there is one holomorphic comb background on its specified physical datum ball, agreeing with the real representative. It has a Landau pair in eq. 3.cccxlv with its actual node normalization, and a possibly different comb pair. The real conjugation and admitted gauge covariance follow from uniqueness. The coupled construction, site correction and real/complex gluing prove these assertions. A global all-site bound on the comb gauge is included only when its actual readout estimate is supplied; a top bound for the Landau gauge is not silently enlarged.

3.11.25.4. Response. On the entire admitted datum ball, a unit source e at row c has the bounds

$$\begin{aligned}\xi_y|h| + \xi_y^2|\nabla h| + \xi_y^3|D^\sharp Dh| + \xi_y^3|\Delta_{U_0} h| + \xi_y^3|\delta J_A| \\ \le Ce^{-md_{\mathbf{B}}(y,y_c)}|e|.\end{aligned} \tag{3.cdxxxiv}$$

The multiplier and top node readouts have their corresponding normalized bounds. The Hölder member follows from the actual pointwise gradient and Laplacian estimates. At a noncritical independent-current pair the second-order members are the combinations eq. 3.ccclxxxii; the critical slice identity is not asserted there. eq. 3.cccxcvii, the actual kernel estimates and the local readout formulas prove these claims.

3.11.25.5. Hereditary complex domains. For sufficiently small radii (e_0, e_1), Cor. 3.130 gives tower curvature and current bounded by $C(e_0 + e_1^2)$ in their declared scales. The common complex ball and its uniform crude readout bound are proved in Lem. 3.127. At settled elements of an ambient sequence the same readouts use eq. 3.cdxix and have the ambient exponentially weighted Lipschitz bound. Outer and inner substitutions preserve the specified cores under their all-clause numerical margins. These follow from eq. 3.ccclxxxvi–eq. 3.ccclxxxix, Prop. 3.138, and eq. 3.cdiv, with the true radius factors retained.

3.11.25.6. Construction, restriction and comparison. The certified label, healed, prepared, stage, tower and joined sequences are separated. Restriction gives eq. 3.cccxciii, and the next-step background nests with its truncation. For a collared region the joined problem and exact lifted datum define its localized background. The common admitted path then gives eq. 3.cd, with its potential, gradient and second-order readouts. The comparison gauge is prescribed only at the stated tops. Separation closure, the adaptive factorization, constrained criticality, regular uniqueness and the weighted response prove this conclusion.

The numerical constants in this theorem must have a solved order of choice. One admissible order fixes the local geometry and its uniform coefficient degrees, the inverse and response constants, then the cutoff scale σ, and finally the small regularity and analytic radii. If an exponent or reserve parameter must be chosen before later geometry, the needed degree bound must be uniform in that geometry, or the whole finite feasibility list must be solved jointly. Logarithmic smallness at already fixed geometry does not prove a different choice order.

The first-step chart, classical flat inverse and kernels, and physical first-step backgrounds provide proved instances of these reductions. The site, layered, full-current and window estimates listed above remain their precise multilevel application tasks. No claim about an all-current completion, an interacting Gaussian law or a history formation bound follows from the conditional background theorem without those separate matches and estimates.

3.12. **Constrained propagators and local decay.**

3.12.1. *Propagators in a background field: exponential decay produced by the averages.* Every object produced by a renormalisation step is assembled from inverses of "Hessian of the action + gauge term + mass made of block averages". These inverses are taken around a background on a decreasing sequence of regions, in which each layer has a scale of its own. The propagator hypotheses are: they exist; they depend analytically on the background; in each layer they behave like massive Green's functions measured in the unit of that layer; they decay exponentially; and they are sums of terms that are local both in the field and in the regions. A single requirement governs the whole construction. Since the regions are supplied by the large-field analysis, none of the constants may vary with their shape or their number, nor with the level, the cutoff or the torus. Two numbers are exceptions, and neither of them enters any localised bound, rate, weight or sum of weights. (Bałaban's papers are referred to by the volume of Commun. Math. Phys. in bold. Where two papers must be told apart, the first page is added: **99**, 75 and **99**, 389; **102**, 277; **122**, 175. The volume and page references identify the published papers. Clauses 0–5 appear below; the following list defines the references to later requirements. The mechanisms in (iii) carry the letters (a)–(k).)

The numbered requirements concern six propagators, their covariances and walks, and parameter decoupling. The vector and current conclusions require the specified operator and domain estimates. § 3.13 proves its stated site, row, averaging-coefficient and physical-Hessian reductions; those reductions establish only the requirements stated there.

3.12.1.1. References to the propagator hypotheses. References to statements 6–8 mean the requirements below. The labels 1c.1–1c.10 in the strategy name stages of the conditional argument, as follows.

Conditioned covariance (statement 6).: This means the actual fluctuation precision A, its conditioning $\chi_Y \mathsf{A} \chi_Y$, and the inverse $C^{(k)}(Y)$ defined below, with the coercivity, complex-domain and boundary requirements of Input A2 and the kernel/locality requirements of Input A3 in the dependency catalogue. Their walk requirements are included in statement 5 here. Physical identification uses the complete current and gauge comparison below and the three Hessian readouts of § 3.13, not a Ward identity alone. The canonical proper covariance, including its outer constraint curvature and relative contour form, is constructed in Prop. 5.45 and § 5.13.

Walks as parameter-dependent operator data (statement 7).: For a fixed cube partition P, each supplied walk term has its specified finite parameter carrier $\mathsf{P}(\omega)$. Write $s^\omega = \prod_{\Delta\in\mathsf{P}(\omega)} s(\Delta)$ and $\mathsf{O}(s) = \sum_\omega s^\omega \mathsf{O}_\omega$. The required data include the total-incidence bound in mechanism (i), absolute locally uniform convergence on $|s(\Delta)| < e^{\bar\kappa}$, the stated weighted bounds, reality and covariance, compatibility on common lifts, and wall-component splitting when separating parameters vanish. The exceptional convergence-only bound mentioned below is part of these supplied data. These are the operator-data hypotheses following eq. 2.x in § 2.2; the substitutions, their domain condition and their alternating brackets are eq. 2.xi–eq. 2.xiii there. None of this asserts $QH(s) = 1$.

Decoupled response (statement 8 and system (I)).: The operative system (I) is eq. 2.x in § 2.2. There B' and D are raw rows; each map written $\mathsf{H}_{(1)}(s)$ means the raw-datum version $\mathsf{H}_{(1)}(s)\mathsf{m}^{-1}$ of the per-length map used here, and C_k means the raw nonlinear part $\mathcal{C}$ defined below. Use the following hypotheses there on the reduced current, holomorphy, bounds, splitting, transport and actual substitution domains. Their existence and uniqueness remain response inputs of A3. At $s = 1$, identification with the physical branch uses the fixed-reference stationarity and constraint equations in eq. 3.dcviii, the actual Lagrangian Hessian eq. 3.dcxi, and the coupled inverse in eq. 3.dcxii–eq. 3.dcxvi in § 3.14.

The coupled alternative (system in eq. 3.cdxliv).: This is the explicitly defined operator-data map below. Its coefficient bounds, radius and current requirements are additional conditional data; it is not identified with eq. 3.dcviii at arbitrary s or at an independent current. Equality of the two response constructions is used only on their common domain at $s = 1$ over the stated critical physical pair, after full operator matching and the required uniqueness have been established.

(i) The setting and the objects.

Geometry. The integer L is odd and large. The cube size of the walk expansions, M_1, and the gap number, R_1, are both powers of L, with $R_1 \geq L$. We work in level-k units, in which $\mathbb{T}^{(j)}$ has spacing $\xi_j := L^{j-k}$; moreover $\eta = \eta_k = \xi_0$. A sequence $\Omega_0 \supset \cdots \supset \Omega_k$ is called *separated* when every Ω_j is a union of M_1-cubes at level j and

$$\mathrm{dist}_\infty(\Omega_j^c, \Omega_{j+1}) > R_1 M_1 \xi_j.$$

No further condition is placed on the regions. In particular Ω_0 is arbitrary, and Dirichlet conditions are imposed outside it. *Layers* are the sets $\mathcal{L}_j = \overline{\Omega_j \setminus \Omega_{j+1}}$. The *tops* y are the sites of $\mathbb{T}^{(j)}$ lying in $\mathcal{L}_j$. Such a top has local scale $\xi_y = \xi_j$ and block $\Delta(y)$, which is the cube of side ξ_y with centre y. The *rows* c are the bonds of $\mathbb{T}^{(j)}$ that have one end point in $\mathcal{L}_j$ and the other end point either in $\mathcal{L}_j$ (the *interior* rows) or, when $j \geq 1$, outside Ω_j (the *outward* rows); for them $\xi_c = \xi_j$. The *inward* bonds, those that enter Ω_{j+1}, carry no row. The distance

$$d_{\mathbf{B}}(y, y') = \inf_{\Gamma \ni y, y'} |\Gamma|_{\mathbf{B}},$$

$$|\Gamma|_{\mathbf{B}} = \sum_{j\geq 1} \xi_j^{-1} \,\mathrm{length}(\Gamma \cap \mathcal{L}_j) + \xi_0^{-1} \,\mathrm{length}(\Gamma \setminus \Omega_1)$$

(Γ connected)

measures each layer in the unit of that layer. Outside Ω_1, and hence also outside Ω_0, length is counted in the bare unit. Therefore $d_{\mathbf{B}}$ is independent of Ω_0, and a path that leaves Ω_0 finds no cheaper unit of length out there. This distance differs from the distance (2.46) of **96**, whose contours are made up, in each layer, of bonds of the lattice of that layer. A larger class of paths is allowed here, and so $d_{\mathbf{B}}$ is the smaller of the two.

Regular pairs $(\mathbf{U}, \mathbf{J})$ consist of $\mathbf{U} = e^{i\eta A'}U$, with U real, together with an **independent** $\mathfrak{g}^c$-valued 1-form $\mathbf{J}$ that represents the current. Such a pair is required to satisfy, on $\mathcal{L}_j$,

$$|\partial U - \mathbf{1}|, |\partial \mathbf{U} - \mathbf{1}| < \alpha_0 \eta^2 \xi_j^{-2},$$
$$\xi_j |A'|,\ \xi_j^2 |\nabla_U A'| < \alpha_1,$$
$$|\mathbf{J}| < \gamma_0 \xi_j^{-3},$$

and, on every cube of side $O(M_1 \xi_j)$ contained in $\mathcal{L}_j \cup \mathcal{L}_{j+1}$, to admit a real gauge $U = e^{i\eta A}$ with $\xi_j |A|, \xi_j^2 |\nabla A| \le \alpha_0$. In this condition the bound is α_0 on the entire cube, and not $O(M_1)\alpha_0$. Pairs in which the current is a separate variable, with a radius γ_0 of its own, are due to Bałaban (**109**, p. 262, (1.10)–(1.14)). In the present setting they are formed on a sequence, with the local scales. Throughout, $J(\mathbf{U})$ is the first variation of the action of the *inner* plaquettes, namely those inside Ω_0. At every bond whose six plaquettes are all inner, it coincides with the current $D^*_{\mathbf{U}} F_{\mathbf{U}}$; when $\Omega_0 = T_\ell$, every bond is of this kind. *On shell*: the pair satisfies $\mathbf{J} = J(\mathbf{U})$ at the bonds that touch Ω_1. *Real*: $A' = 0$, and $\mathbf{J}$ is real. *Local class on* W: the same conditions, required inside W.

Operators. Relative to $\mathbf{U}$, the *straightened* j-fold average of $e^{i\eta A}\mathbf{U}$ takes, at a row c of level j, the value $\exp i\mathcal{Q}(A)(c)$. Here $\mathcal{Q} = \mathsf{m}\,(Q + C^{(2)} + \dots)$, and m is multiplication by ξ_c. Thus Q, which maps 1-forms to rows, is the linear part per unit length, and $C^{(2)}$ is the quadratic part per unit length. When the level has to be displayed, the straightened average is written $\mathcal{Q}_j^{\mathrm{st}}(\mathbf{U}; A)$. In the units of level j, its linear part is the operator $Q_j^\sharp(\mathbf{U})$ of the Notation, §3, which is built from the means over the contour families of $\mathcal{M}$. The index st stands for "straightened"; the *straight* linear average $Q_j^{\mathrm{st}}(\mathbf{U})$ of the Notation is a different operator. Consequently, in this part the unadorned letters $\mathcal{Q}$ and Q differ from those of §3 of the Notation: they denote neither the logarithm $\mathcal{Q}_j(\mathbf{U}; A)$ of $\mathcal{M}^j(e^{i\eta A}\mathbf{U})\,\mathcal{M}^j(\mathbf{U})^{-1}$ nor its linear part $Q_j(\mathbf{U})$. The straightened and the plain averages are related by *site factors* v_n; these are the node factors $v_n = v_n[A]$ of the Notation, §3, and in 1c.10 they are also called node factors. These are group-valued functions on the sites of $\mathbb{T}^{(n)}$, and they accumulate from one step to the next (statement 3: $\mathcal{M}^n(e^{i\eta A}\mathbf{U}) = \big(e^{i\mathcal{Q}_n^{\mathrm{st}}(A)}\mathcal{M}^n\mathbf{U}\big)^{v_n}$). The linear parts of the straightened and of the plain average already differ at $\mathbf{1}$, by a gauge term on the coarse lattice. For the plain averages, two facts of statement 3 fail: the identity $QD = \bar{D}Q'$, and the bound $1 + C(\dots)$ on the derivative. In that identity, at a row of level j, $\bar{D}$ is the covariant difference on $\mathbb{T}^{(j)}$ formed with $\mathcal{M}^j\mathbf{U}$, and Q' takes, at each of the two end points of the row, the j-fold average of a site function; this applies also to the outer end point of an outward row, which is a top of level $j-1$. The operator Q' maps site functions to tops. It is the linearisation of the average of a gauge transformation, taken with the same contour variables. At a row of level j, the notation $\bar{D}Q'$ uses the j-fold site average at both endpoints. At the outer endpoint of an outward row, which is a top of level $j-1$, this value is the additional transported L^4-block average of its own-level top values. Thus $\bar{D}$ means the extended endpoint operator of § 3.13, not an ordinary difference of the two own-top components. The symbol ^{T} denotes the transpose with respect to bilinear pairings, so that everything remains holomorphic. The operator a acts on rows as multiplication by $a_0\xi_c^{-2}$, where a_0 is a fixed number (equal to 1 in the sub-parts). On tops it acts as multiplication by a comparable weight $\approx \xi_y^{-2}$. We write $D = D_{\mathbf{U}}$. Finally, $\Delta^{\mathrm{W}}_{\mathbf{U}}$ is the Hessian of the action in Ω_0. In statements 3–8 it is the Hessian of the action of the inner plaquettes, which is an exactly invariant function. In statements 1, 2 it is obtained by taking the Hessian on the entire lattice and compressing it to the bonds of Ω_0.

$$G' = (\Delta_{\mathbf{U}} + Q'^{\mathsf{T}} a Q')^{-1},$$
$$R = 1 - G'Q'^{\mathsf{T}}(Q'G'^2Q'^{\mathsf{T}})^{-1}Q'G',$$

$$\mathsf{S} = 1 - DG'RD^{\mathsf{T}},$$

$$\mathsf{G}_0 = (\Delta_{\mathbf{U}}^{\mathrm{W}} + DRD^{\mathsf{T}} + Q^{\mathsf{T}}aQ)^{-1},$$
$$\mathsf{G} = (\mathsf{G}_0^{-1} - \Delta'_\pi[\mathbf{J}])^{-1},$$
$$\mathsf{G}_1 = (\mathsf{G}^{-1} - \mathsf{S}^{\mathsf{T}}\Delta^{(2)}[\mathsf{H}^{\mathsf{T}}\mathbf{J}]\mathsf{S})^{-1},$$
$$\mathsf{H}_{(1)} = \mathsf{G}_{(1)}Q^{\mathsf{T}}(Q\mathsf{G}_{(1)}Q^{\mathsf{T}})^{-1}$$

(where a bracketed index occurs, the formula holds both with that index and without it). The operators are defined in this order: G, then H from G, then G_1, then H_1 from G_1. The symbol $\Delta'_\pi[\mathbf{J}]$ denotes a specified current-linear completion whose required physical Ward identity is stated below. Its complete form is part of the operator data: linearity and that Ward identity alone do not determine its horizontal or independent-current terms. The distinct mean-constraint correction $\Delta^{(2)}$ is retained separately. Moreover, $(A, \Delta^{(2)}[\mathsf{j}]A) = 2(C^{(2)}(A), \mathsf{j})$. The *six propagators* are

$$\mathsf{G}, \mathsf{G}_1, \mathsf{H}, \mathsf{H}_1, (Q\mathsf{G}Q^{\mathsf{T}})^{-1}, (Q\mathsf{G}_1Q^{\mathsf{T}})^{-1}$$

. In addition we set $\mathfrak{G} = \mathsf{G}_1 - \mathsf{G}_1DRD^{\mathsf{T}}\mathsf{G}_1 - \mathsf{H}_1Q\mathsf{G}_1$. For step $k+1$ we work on the bonds of $\mathbb{T}^{(k)}$ that touch Ω_{k+1}, with $V = \mathcal{M}^k\mathbf{U}$. Let C_V be the elimination of the pivot bonds, and let $\mathsf{G}_V^{(2)}$ be the quadratic part of the soft axial gauge fixing. Let $\mathsf{E}^{(2)}$ be the quadratic current form of the one-step constraint, which is imposed through the plain $\mathcal{M}$ at V and whose linear part is the Q_V of the Notation, §3. Its notation includes the actual pivot multiplier. More precisely, write $\mathfrak{q}_V$ for the raw constraint logarithm and h_V for the pivot right inverse of $D\mathfrak{q}_V(0)$. In raw counting coordinates, its bilinear form contracts $D^2\mathfrak{q}_V(0)[u,v]$ with $h_V^{\mathsf{T}}\widehat{j}$, where $\widehat{j}$ is the raw coarse-bond current. The unhatted form uses the corresponding raw/per-length dual conversion. This is the curved-fibre readout derived in § 3.13; the coarse-bond current is not directly identified with a new-row multiplier. We set

$$\Delta_k = (Q\mathsf{G}_1Q^{\mathsf{T}})^{-1} - a - \mathsf{E}^{(2)}[\mathsf{H}_1^{\mathsf{T}}\mathbf{J}],$$
$$\mathsf{A} = \mathsf{C}_V^{\mathsf{T}}(\mathsf{G}_V^{(2)} + \Delta_k)\mathsf{C}_V,$$
$$C^{(k)}(Y) = (\chi_Y\mathsf{A}\chi_Y)^{-1}.$$

At the physical critical pair $(U_{k+1}, J(U_{k+1}))$ of the step, the required identification is $\mathsf{A} = \mathsf{C}^{\mathsf{T}}\Delta^{(k)}\mathsf{C}$. This identity must hold for the complete projected forms, including the pivot graph, all current and mean-constraint corrections, and the specified soft-gauge term. The absolute or relative contour-gauge convention must be checked in that comparison; a Ward identity alone does not establish the identification. Its action-value component is the quadratic term of the action of the background, regarded as a function of its datum. This is Bałaban's expansion **102**, 277, (174), together with **109**, (2.4)–(2.8), (2.11). There the derivation is carried out on the whole lattice, using Bałaban's own rows and his averages. For $\mathcal{M}$ and the straightened rows, on the sequence of a label, the expansion is obtained from the system with a multiplier of § 3.11. That multiplier coincides with the gradient of the action of the background with respect to its datum, as one sees by differentiating the constraint and pairing the result with the Euler–Lagrange equation. A single differentiation of the system yields the derivative of the multiplier. Here ω is the physical row multiplier in $J(\mathbf{U}) = Q^{\mathsf{T}}\omega$, with the declared cell-weighted transpose; it is unique on the independent datum space when Q is onto. For the critical pair, the required identified form of G_1^{-1} is the projected Lagrangian Hessian $\mathsf{S}^{\mathsf{T}}K_{\mathrm{L}}\mathsf{S}$, together with the gauge term and the row mass,

where $K_{\mathrm{L}} = \Delta^{\mathrm{W}} - \Delta^{(2)}[\omega]$. On the declared Landau slice the projections act as the identity. For the canonical completion this identity follows from Prop. 3.90; its proper outer-fibre identification, including the pivot second derivative, is proved in § 5.13. Hence this derivative is obtained by applying $(Q\mathsf{G}_1Q^{\mathsf{T}})^{-1} - a$ to the datum. When the datum moves along a curve contained in the fibre of the one-step average, it picks up a second-order term, and this term is determined by the quadratic part of the one-step constraint. Paired with the gradient, this term yields $-\mathsf{E}^{(2)}[\mathsf{H}_1^{\mathsf{T}}\mathbf{J}]$. Bałaban also gives a second expression for the first two terms of Δ_k, namely $\mathsf{H}_1^{\mathsf{T}}[\mathsf{G}_1^{-1} - DRD^{\mathsf{T}} - Q^{\mathsf{T}}aQ]\mathsf{H}_1$. It differs from them by $-\mathsf{H}_1^{\mathsf{T}}DRD^{\mathsf{T}}\mathsf{H}_1$, a term that vanishes on shell. The bounds and the positivity in statement 6 concern the form displayed above.

3.12.1.2. Positivity of the compressed fluctuation form. The direct argument in Lems. 5.43 and 5.44 and Prop. 5.45 proves the proper lower bound, complex continuation and local inverse families for the precise relative-contour form below. The following local-system comparison describes an alternative positivity mechanism. Consider an admitted real layered pair in statement 6 with the complex potential slot equal to zero and with real current. In this argument the local pieces have width $\mathfrak{M}$ on the fluctuation scale, with $\mathfrak{M} \geq M^{\mathrm{pcs}}$, $R_1 \geq L$, and $R_1\mathfrak{M}$ at least the floor of the local propagator construction. The torus side in level-k units is at least $2L\mathfrak{M}$. Take Y to be the free-bond set for a subfamily of the retained top blocks in the conditioning geometry. The form $T_Y = \chi_Y \mathsf{A} \chi_Y$ acts on the free bonds in Y, after pivot elimination. To compare it with a positive form, use a partition $\sum_\kappa h_\kappa^2 = 1$ on those bonds, with each cutoff supported inside a larger piece on which the local system is defined. In each piece, continue the field in a gauge with a small potential and keep the three current slots, including the two multiplier readouts $\mathsf{H}^{\mathsf{T}}\mathbf{J}$ and $\mathsf{H}_1^{\mathsf{T}}\mathbf{J}$. These readouts obey the smallness bounds in the units of their rows by the weighted operator estimates. Perturbation of the flat system gives the inverse C_κ in the fluctuation slot with $cI \leq C_\kappa \leq pI$, where $c, p > 0$ are independent of the piece. The comparison is made for the completed local system, including the pivot graph and gauge block.

Write $P = \sum_\kappa h_\kappa C_\kappa h_\kappa$, with restriction and extension by zero in each summand. Then $cI \leq P \leq pI$. To compute PT_Y, take a free field B, set $f = T_Y B$, and complete B to the solution Φ of the coupled equations with right-hand side f in the fluctuation slot and zero in the other slots. The uniform auxiliary inverse and pivot bounds give $\|\Phi\|_\infty \leq C\|B\|_\infty$ in local units. The local and global systems agree on inputs supported in the intermediate piece, with output in the larger piece. Choose $\widetilde{h}_\kappa$ supported in that intermediate piece and equal to one on the cutoff support and its interaction neighborhood. The local system applied to $h_\kappa\Phi$ has right-hand side $h_\kappa f$ in the fluctuation slot plus the cutoff commutator applied to $\widetilde{h}_\kappa\Phi$. Solving the local equations and summing their free-field slots yields $PT_Y = I - E$. The commutator costs $C/\mathfrak{M}$ in the scale-weighted source norm, and bounded overlap of the cutoffs gives $\|E\|_{\infty\to\infty} \leq C/\mathfrak{M} \leq 1/8$.

This last bound controls the eigenvalues of PT_Y. Since PT_Y is similar to the real symmetric matrix $P^{1/2}T_YP^{1/2}$, that matrix is at least $7I/8$. Thus $T_Y \geq 7I/(8p)$, and the inverse on the free bonds has norm at most $8p/7$. On the axial slice the same calculation uses the free fields vanishing on the box trees. The reconstructed fields are supported on the free bonds together with their pivot bonds. The bounded norm of the pivot lift converts the free-field lower bound into the lower bound in the norm of the reconstructed field.

The walk terms for the covariance of the compressed form are written against counting measure on its free bonds. Their local comparison uses the traces of the regions and of Y on the carrier, and the regular pair on the enlarged pieces used in the two local systems. For the history induction the forms to be integrated must be identified with this compressed form, or compared with it with an error smaller than its lower bound. In particular, selecting part of a walk sum does not preserve its positivity by this argument. To extend the lower bound to the complex integration cores one must bound the change in the complete form on the specified cores.

Walks. A *cell* is a union of $O(1)$ cubes of side $O(M_1\xi_j)$ lying in two consecutive layers. In a *walk* $\omega = (X_0, \dots, X_m)$, any two consecutive cells overlap. In a *tree walk*, each X_i overlaps an earlier

cell, called its parent, and the cells are enumerated depth first. With arbitrary parent maps, one sequence of length m is counted as many as $m!$ times. The *hull* $[\![\omega]\!]$ enlarges the walk by $O(M_1)$ local units and is not cut off at Ω_0. The *carrier* is $[\omega] = [\![\omega]\!] \cap \Omega_0$. Finally, $d_{\mathbf{B}}(\omega; y, y')$ is the smallest weighted length of a tree that passes through y, y' and meets the cells $X_1, \dots, X_m$. The tree is not required to meet X_0.

Several letters have a different meaning in the sub-parts 1c.1, 1c.6 and 1c.10, as follows. The cube size of the walks is denoted $\mathfrak{M}$ in 1c.1. In 1c.10 the cubes $\square$ that carry parameters (statement 7), whose side is ℓ_{P}, are denoted Δ. Also there, Γ_j denotes the set of rows of level j. In 1c.10, Q stands for $\mathsf{m}Q$, and slanted H, H_1 stand for $\mathsf{Hm}^{-1}, \mathsf{H}_1\mathsf{m}^{-1}$. Within 1c.6, which treats averaging operators at complex configurations together with the gauge identities on which the propagators rest, the operator $Q_k^{\mathrm{st}}(\mathbf{U}) = \eta\, \mathbf{Q}(W_{k-1}) \cdots \mathbf{Q}(W_0)$, where $\mathbf{Q}(W)$ denotes the linear part of a single step, stands everywhere for the linear part of the straightened average, namely the $Q_k^{\sharp}(\mathbf{U})$ of §3 of the Notation; it is not the straight average defined in that section of the Notation.

In comparison with the Notation:

- R_1 denotes a number (as in **116**, p. 3) and not a region R_t.
- A cell means a cell of a walk, as opposed to a level-k cell of §9.
- The hull of a walk differs from the hull in a coarser partition.
- $\mathfrak{G}$ denotes an operator, not the gauge-fixing factor $\mathfrak{G}_k$.
- A is the fluctuation operator, which is written K in §5 of the Notation.
- $\Delta(y)$ is the block $B^j(y)$, regarded as a cube.
- $s(\square)$ is a parameter, not a depth.
- $\bar\kappa$ is the exponent in the radius of the parameters, the same $\bar\kappa$ as in the order of the constants fixed in the Notation. **116** and 1c.10 write this exponent as κ_1; that κ_1 is not the κ_1 of the Notation, which is the rate at which the large-field factor decays.
- $\gamma, \gamma^{\mathrm{pos}}$ are lower bounds, not ceilings on couplings.
- γ_0 is the radius of the current, which is also written α_0 in §11.
- λ denotes a site function (an infinitesimal gauge transformation); in statement 8 and in mechanisms (j), (k), however, it denotes a multiplier, that is, a function on rows.
- X denotes a localisation domain in statement 0, X_i is a cell, X in (I) is a function on rows, and X_χ is a gauge direction.
- A' is the complex factor of the triple $(U, A', \mathbf{J})$; the one exception is 1c.10, where, in the system of clause 1 of the Theorem (the decoupled response at complex parameters), it stands for the unknown $H_1(s)B' + \mathbf{A}$.
- S denotes, in statement 6, a set of coarse cubes and, in eq. 3.cdxliv, an operator; S denotes the operator $1 - DG'RD^{\mathsf{T}}$.

In 1c.10 the following further conventions hold:

- θ_0 denotes the decay rate with respect to $d_{\mathbf{B}}$; it is not one of the members of the sequence θ of §6.
- $\mathsf{r} = v[\mathbf{H}]$ is a function on the nodes with values in G, and not an isometry.
- β and $\beta_\natural$ denote a small real datum, not the bare inverse coupling.
- $\omega(\cdot)$ is the map that sends a field to its multiplier; it is not a walk.
- $\mathsf{E}_1 = G_1 DRD^{\mathsf{T}} G_1$ denotes an operator rather than an event.
- $P_1(s)$ is the operator built from the walk data of $D(1-R)D^{\mathsf{T}}$, and neither a region nor a momentum.

(ii) The principal statements. All constants depend only on L (and on Hölder exponents). There are two exceptions, and neither enters any localised bound, rate, weight or sum of weights. The first is the bound of statement 7, which serves only for convergence. The second is the entrywise bound of mechanism (i), which makes the family normal. The numbers γ, γ^{pos}, δ_0, δ_*, δ' are positive constants of this kind.

In statement 2 the side of the torus, measured in top units, is at least a small fixed multiple of R_1M_1; in the sentence of that statement which concerns a single lattice, it is at least C. From statement 4 onwards the side is at least CM_1. In 5, the single assignment of terms valid for all tori needs room for three cubes of side $O(M_1)$ in each direction. Statements 7, 8 need room for three cubes of side ℓ_{P}. These are lower bounds on the volume, and no constant depends on the volume except the bound of statement 7 just mentioned.

In what follows:

- $x \in \Delta(y)$, and the sources f lie near $\Delta(y')$;
- $\mathsf{e} = e^{-\delta_0 d_{\mathbf{B}}(y,y')}$, up to a fixed fraction of the rate;
- $\|\cdot\|_\varepsilon$ denotes a Hölder seminorm;
- kernels are taken with respect to the counting measure;
- on 1-forms, the covariant Laplacian is $\Delta_{\mathbf{U}} = \sum_\nu \nabla_\nu^{\mathsf{T}} \nabla_\nu$, where ∇_ν are the covariant differences of the components;
- $\Delta = \Delta_{\mathbf{1}}$ is the full vector Laplacian $\partial^*\partial + \partial\partial^*$, with $\partial = D_{\mathbf{1}}$. It acts componentwise and is not the $\Delta = \Delta_U^{\mathrm{W}}$ of the Notation, §5.

0. The domains of analyticity are contained in the classes. Consider a pair in the domain of an activity on $X \in \mathcal{D}_j$. Such a pair satisfies $|\partial U - \mathbf{1}|, |\partial \mathbf{U} - \mathbf{1}| < \alpha_0\eta^2$, $|A'|, |\nabla_U A'| < \alpha_1$ and $|\mathbf{J}| < \alpha_0$, and on cubes of side $O(LM)$ it admits real gauges with $|A|, |\nabla A| < O(LM)\alpha_0$. It is a regular pair for the tower of regions obtained from X by erosion (1b.8), and the radii with which it is regular are

$$(O(LM)\,\alpha_0,\, \alpha_1,\, \alpha_0).$$

Let a label be produced by steps in which each region Ω_n, $n \geq 1$, is a union of LM-cubes of level n. By a *layered domain* we mean a domain defined by the layered bounds of §4.17.1.1, *Structures*, whose radii do not exceed the running radii of the level of each layer. Only the following of these bounds enter. On the layer $\mathcal{L}_n$ of the label, $n \geq 1$, one has $|\partial U - \mathbf{1}|, |\partial \mathbf{U} - \mathbf{1}| < \alpha_{0,n}\eta^2\xi_n^{-2}$, $|\mathbf{J}| < \alpha_{0,n}\xi_n^{-3}$ and $\xi_n|A'|, \xi_n^2|\nabla_U A'| < \alpha_{1,n}$. The cubes of level n of the domain are the closed cubes of side $CM\xi_n$ that are contained in $\mathcal{L}_n \cup \mathcal{L}_{n+1}$ and meet Ω_{n+1}^c; on the part in X of each of them there is a real gauge with $\xi_n|A|, \xi_n^2|\nabla A| < CM\alpha_{0,n}$. Then on $X \cap \Omega_1$ a pair of a layered domain belongs to the local class of the *label sequence*, which is the sequence formed by the Ω_n of the label. The radii are $O(M)\max_n \alpha_{0,n}$ and $\max_n \alpha_{1,n}$, together with $\max_n \alpha_{0,n}$ for the current. In that class the gauge clause is derived for those cubes of level-n size which meet $\mathcal{L}_n$. For $n = 0$, on the cubes inside $\mathcal{L}_1$, it is derived from the gauge of a cube of level 1, which can be used at the index 0 since $\xi_0 < \xi_1$. The same remark applies to a cube of level-n size inside $\mathcal{L}_{n+1}$, because some cube of level $n+1$ of the domain contains it. This smaller cube is contained in Ω_{n+1}, intersects Ω_{n+2}^c, and its side is below $M\xi_{n+1}$. The derivation of the clause on the cubes meeting $\mathcal{L}_n$ uses no other property of a cube; hence repeating the same steps with the grids of level $n+1$ produces the larger cube. The factors $O(LM)$, $O(M)$ are the cost of the gauge clause, since on a cube of side $O(M)$ a gauge has $|A| \leq O(M)\alpha_0$. The field strengths are fixed after M.

1. The flat theorem. Let $\mathbf{U} = \mathbf{1}$, $\mathbf{J} = 0$, and let Q, Q' be the straight averages taken over the centred blocks. Then on every separated sequence

$$|\mathsf{G}_0 f|, |\nabla \mathsf{G}_0 f|, |\mathsf{G}_0 \nabla^* f|, |\Delta \mathsf{G}_0 f|\,(x) \leq C[\xi_y^2, \xi_y, \xi_y, 1]\,\mathsf{e}\,|f|,$$
$$|\nabla \mathsf{G}_0 \nabla^* f(x)| \leq C_\varepsilon\,\mathsf{e}\,(\xi_{y'}^\varepsilon \|f\|_\varepsilon + |f|),$$

together with matching bounds in Hölder and L^2 norms. The same bounds hold for G', and

$$|(Q'G'^2Q'^*)^{-1}(y,y')| \leq C\xi_y^{-4}\mathsf{e},$$

$$|(Q\mathsf{G}_0Q^*)^{-1}(c,c')| \le C\xi_c^{-2}\mathsf{e},$$
$$|\mathsf{H}(b,c)|, |\nabla\mathsf{H}(b,c)| \le C[1,\xi_y^{-1}]\mathsf{e},$$

$$Q\mathsf{G}_0Q^* \ge \gamma\,\xi_c^2,$$
$$\langle B, [(Q\mathsf{G}_0Q^*)^{-1} - a]B\rangle \;\ge\; c\,L^{-5}\|B\|^2$$

for every B with three properties: it lives on the bonds of $\mathbb{T}^{(k)}$ that touch a union of L-blocks lying well inside Ω_k; its straight one-step average vanishes; and it vanishes on the axial trees of the blocks. Each block has one such tree. It consists of the contours joining the site $y \in \mathbb{T}^{(k+1)}$ of the block to each of its points x, along which the coordinates are changed successively, always in the same fixed order. The gauge so defined is the strict axial gauge, which is a hard gauge fixing. It is not the block axial gauge (Notation, §3), where the means $V(y,x)$ over contour families are all equal to $\mathbf{1}$. For the plain one-step constraint, in that gauge or on trees, the lower bound is proved again in 1c.8. If the gauge condition is dropped, the form has null directions.

2. Real backgrounds with Bałaban's average. Let Q, Q' be formed with one contour per site and moved to centred blocks by a translation. Let the Hessian be the compressed one. Let U be real, with local gauges and with small plaquettes; for G_0 this includes the plaquettes that stick out of Ω_0. Then $\Delta_U + Q'^*aQ'$, $Q'G'^2Q'^*$ and G_0^{-1} are positive. The operators $G', \nabla G', \mathsf{G}_0, \nabla\mathsf{G}_0, (Q'G'^2Q'^*)^{-1}$ satisfy the bounds of 1 for every Ω_0, including a collar of Ω_1 and a concentric cube. On a single lattice, and for U with $|\partial U - \mathbf{1}|, |D_U^*\partial U| < \alpha\eta^2$, where α is small, also $Q\mathsf{G}_0Q^* \ge \gamma > 0$, and $\mathsf{G}_0Q^*(Q\mathsf{G}_0Q^*)^{-1}$ is a right inverse of the linearised constraint; this right inverse and its gradient decay like $e^{-\delta_*|y-y'|}$. The rows of $\mathcal{M}$ agree with these rows up to a transport and an error $O(\alpha_0)$ in the row and column norms, uniformly in the level. Statements 3–8 concern different operators, built from the rows of $\mathcal{M}$ and the Hessian of the inner plaquettes. At $\mathbf{1}$ the two kinds of rows coincide.

3. The averages and the gauge identities. The operators Q, Q' are local and holomorphic in $\mathbf{U}$. Their row sums are $\le 1 + C(\alpha_0 + \alpha_1)$ and their entries are $\le C(\eta/\xi_c)^4$, independently of the number of levels. At each row the non-linear average is holomorphic in (A', A) on a polydisc $\sup|A'|, \sup|A| < r_L$, where r_L is determined by L and measured in units of the row's level. Fix a bond of $\mathbb{T}^{(k)}$, and write $\delta/\delta A_b = \eta^{-4}\partial/\partial A(b)$. Let $\mathcal{Q}_n^{\mathrm{st}}$ be the n-fold straightened average, and let v_n be the accumulated site factors belonging to it. Then on this polydisc

$$\Big|\frac{\delta\mathcal{Q}(A)(c)}{\delta A_b}\Big| \le 1 + C(\alpha_0 + \sup|A'| + \sup|A|),$$
$$\Big|\frac{\delta(\mathcal{Q} - \mathsf{m}Q)(A)(c)}{\delta A_b}\Big| \le C\sup|A|,$$
$$\mathcal{M}^n(e^{i\eta A}\mathbf{U}) = \big(e^{i\mathcal{Q}_n^{\mathrm{st}}(A)}\mathcal{M}^n\mathbf{U}\big)^{v_n}.$$

Let $\bar{D}$ be the covariant difference of $\mathcal{M}^j\mathbf{U}$ on $\mathbb{T}^{(j)}$. At a row of level j, let $Q'\lambda$ denote the j-fold average taken at both ends of the row. For an outward row, the outer end is a top of level $j-1$, not a top of level j. Then

$$QD\lambda = \bar{D}Q'\lambda,$$
$$\Delta_{\mathbf{U}}^{\mathrm{W}} - \Delta'_\pi[J(\mathbf{U})] = \mathsf{S}^{\mathsf{T}}\Delta_{\mathbf{U}}^{\mathrm{W}}\mathsf{S},$$
$$\mathsf{G}_{(1)}DR = DG'R + \mathsf{G}_{(1)}\Delta'_\pi[\mathbf{J} - J(\mathbf{U})]DG'R.$$

On shell, the defect is zero, $RD^{\mathsf{T}}\mathsf{H}_{(1)} = 0$, and $\mathsf{H}B$ is a critical point of $\frac{1}{2}(A, \Delta_{\mathbf{U}}^{\mathrm{W}} A)$ on $\{QA = B,\ RD^{\mathsf{T}}A = 0\}$, with critical value $\frac{1}{2}(B, [(Q\mathsf{G}Q^{\mathsf{T}})^{-1} - a]B)$.

4. The six propagators at regular pairs. The ceiling on the radii $\alpha_0, \alpha_1, \gamma_0$ is fixed before M_1 and is **independent of** M_1. At every regular pair all the inverses exist, they are holomorphic in $(\mathbf{U}, \mathbf{J})$, and they are real at real pairs. At real pairs $\mathsf{G}_0, \mathsf{G}, \mathsf{G}_1 > 0$, both on shell and off shell. The six propagators and $(Q'G'^2Q'^{\mathsf{T}})^{-1}$ satisfy the upper bounds of 1 with covariant differences, and G and G_1 satisfy those of G_0. The exception is the bound on $\Delta_{\mathbf{U}}\mathsf{G}_{(1)}$, and the Hölder quotients are taken only over $|x - x'| \le \xi_y$. For G' at complex pairs the bounds reduce to the supremum bounds on $G'f$ and $DG'f$. Moreover $|D(1-R)D^{\mathsf{T}}(b, b')| \le C(\eta/\xi_{y'})^4 \xi_y^{-2} \mathsf{e}$.

5. Walk expansions. Let O be any of the following operators:

- one of the six propagators;
- $1 - R$ or one of its derivatives;
- $\mathfrak{G}$;
- Bałaban's $H_0, \tilde{G}$, namely $\mathsf{H}\mathsf{m}^{-1}$ and $\mathsf{G} - \mathsf{H}Q\mathsf{G}$ at $\mathbf{J} = 0$;
- $C^{(k)}(Y)$ or $\chi_Y \mathsf{A} \chi_Y$, these two taken at the pairs that are regular for the longer sequence $(\Omega_j)_{j \le k+1}$, where M_1 is larger and the ceiling on the radii is smaller than in 4.

For $\tilde{G}$ the indices are the walks together with the pairs of walks in which the second walk starts at a cell that a row joins to some cell of the first walk, its cells being placed after those of the first. In the display, $|\omega|$ is then the total length of the two walks, and the step that joins them contributes no small factor. Each such operator is then a sum $\sum_\omega \mathsf{O}_\omega$, where, in the norms in which O is bounded,

$$\|\mathsf{O}_\omega\| \le (\text{bound of } \mathsf{O}) \cdot (C M_1^{-1/2})^{|\omega|} M_1^{-|\omega|/2},$$

$$d_{\mathbf{B}}(y, y') \text{ replaced by } d_{\mathbf{B}}(\omega; y, y').$$

The first factor makes the sum over the walks converge. The second factor is left over, and it absorbs the factor $|\omega| + 1$ of mechanism (i). The bounds of O are as follows:

- For $\mathfrak{G}$ and $\tilde{G}$ there are exactly four: the bounds on $|\mathsf{O}f|$, $|\nabla \mathsf{O}f|$, $|\mathsf{O}\nabla^{\mathsf{T}} f|$ and on the Hölder seminorm of $\nabla \mathsf{O}f$ that 4 provides for G, G_1. For $\tilde{G}$ they hold with a smaller fraction of δ_0.
- For $1 - R$ and for its derivatives $D(1-R)$, $(1-R)D^{\mathsf{T}}$, $D(1-R)D^{\mathsf{T}}$, the relevant bounds are the kernel bounds $C(\eta/\xi_{y'})^4 [1, \xi_y^{-1}, \xi_y^{-1}, \xi_y^{-2}]\, \mathsf{e}$. The last of these is displayed in 4.
- For $\chi_Y \mathsf{A} \chi_Y$, the bound is the kernel bound that 6 states for $C^{(k)}(Y)$.

Each O_ω has the following properties:

- it is holomorphic on the local class on $[\omega]$;
- its dependence on the regions is through their traces on $[\![\omega]\!]$ alone;
- it is gauge covariant and real at real pairs;
- it is **the same function** for any two sequences, tori or cutoffs that a *shift*, namely a translation in $M_1 \xi_k \mathbb{Z}^4$, relates on the hull of (i), *Walks*, which is the set just named; in the Notation, §5, this hull is called the surroundings of ω. For $C^{(k)}(Y)$, $\chi_Y \mathsf{A} \chi_Y$ the translation is in $M_1 \xi_{k+1} \mathbb{Z}^4$ and must also carry Ω_{k+1} and Y.

The operators are covariant under every isometry that preserves $\mathbb{T}^{(k)}$ and the regions, using the induced action on each of their input and output spaces. At a reversed bare bond b this action sends $A(b)$ to $-\operatorname{Ad}(\mathbf{U}(b)^{-1})A(b)$. At a reversed row or coarse bond c of level n, it instead sends $B(c)$ to $-\operatorname{Ad}(V_n(c)^{-1})B(c)$, where $V_n(c) = (\mathcal{M}^n \mathbf{U})(c)$ is the actual level-n average; in particular, the covariance and fluctuation operators on level k use V_k. Site slots are transported without an orientation factor. These formulas act on the independent positively oriented tangent coordinates. The skew extension $\nu(\bar{c}) = -\nu(c)$ within a fixed choice of orientation remains a separate convention. Thus an operator from one slot to another transforms as $\mathsf{r}_{\mathrm{out}} \mathsf{O} \mathsf{r}_{\mathrm{in}}^{-1}$. For a term evaluated at a pair

given only on its carrier, every nonzero kernel entry has the complete averaging reads needed to form both orientation factors. The transformed kernel is zero at zero entries; filling undefined averages outside those reads by arbitrary elements of G^c does not change it. The maps preserve the complex bilinear pairings: $\mathrm{Ad}(g)^{\mathsf{T}} = \mathrm{Ad}(g^{-1})$, and the isometry preserves each row's level and hence its cell weight. Accordingly the same transformation law applies to the declared transposes, with either physical or counting row measure, without an additional density factor. At complex pairs this is a bilinear identity, not a claim that the action preserves the norm. For $C^{(k)}(Y)$, $\chi_Y \mathsf{A} \chi_Y$ the isometry must also preserve $\mathbb{T}^{(k+1)}$ and Ω_{k+1}, and Y is carried along. Pivot bonds go to pivot bonds, since each pivot bond is the middle bond of its path. For single terms the covariance holds only under those isometries which in addition preserve the M_1-cubes of the top level. For $C^{(k)}(Y)$, $\chi_Y \mathsf{A} \chi_Y$ that level is $k+1$, and in the strict axial gauge the isometry must also preserve the axial trees. This covariance holds at the pairs whose image again lies in the class.

(iii) The strategy.

The chain. The parts have the following content.

- 1c.1 (the operators and the statements about them) fixes the operators, the classes and the form of the statements. It counts the walks and derives statement 0. It also shows that the identification of A at the step reduces to Bałaban's expansion of the background action.
- 1c.2 (the flat case, I) states the flat theorem and compares it clause by clause with the published statements. It also contains the exact algebra at $\mathbf{1}$: existence, identities with boundary terms, and the independence of R and H from Ω_0.
- In 1c.3 (the second flat case) the one-scale operator is treated uniformly in the spacing; then come G', $(Q'G'^2Q'^*)^{-1}$ and R.
- In 1c.4 (the flat case, III) the local two-scale operators are treated, followed by G_0, $(Q\mathsf{G}_0Q^*)^{-1}$ and H. With this, statement 1 is complete.
- 1c.5 (Bałaban's theorems in a background) yields statement 2. On the whole lattice it comes from the published theorems; on sequences it comes from the flat case, by a shortened form of mechanism (f).
- 1c.6 (the averaging operators at complex pairs; the algebraic identities) yields statement 3.
- 1c.7 (the six propagators) uses mechanisms (f), (g) to derive, from the flat bounds and 1c.6, statement 4 and the expansions of 5.
- In 1c.8 (the covariance; positivity) these two mechanisms are applied to a larger system. This yields statement 6, except for the expansion of A over sets of coarse cubes.
- In 1c.9 (walk expansions as data) an expansion is regarded as a datum enjoying the short list of properties of 7. Statement 7 is deduced from these properties together with the geometry. Moreover, 1c.9 *makes* terms that are real, covariant and identical on all tori with room; these properties do not follow from the list.
- 1c.10 (the decoupled response) takes the operators as data and solves the systems (I), eq. 3.cdxliv. Next it inserts the local maps and operators of 1c.6–1c.9; for $\mathfrak{G}$ and $D(1-R)D^{\mathsf{T}}$ the inserted data are those named at the end of 7. Last, it treats the case $s \equiv 1$.

The order of § 3.11 and § 3.12.1, which involves no circle. None of statements 1–7 mentions a background of § 3.11, since the classes are defined by inequalities. The results are obtained in the following order:

- the geometry and the real calculus of $\mathcal{M}$ (1b.1–1b.2);
- the flat case;
- the lemmas of 1b.3 that are geometric or algebraic (folding, cells and rows, the algebra of Q', Q, R, sums over tops, weights for a general metric), and none of them uses any bound on a propagator;
- 1c.5;

- the rest of 1b.3; then existence, regularity (which provides the backgrounds of § 3.11 with their local gauges) and continuation (1b.4–1b.6);
- 1c.6–1c.9. Their inputs are the flat case, the alignment of the blocks in 1c.5, and 1b.1–1b.2, together with, besides these, geometric facts in which neither a propagator nor a background of § 3.11 enters (meets of sequences; the covariance of $\mathcal{M}$ with respect to the isometries of the lattice);
- the decay of the response, and the membership in the classes of the backgrounds of § 3.11 (1b.7–1b.9);
- finally, the identification at $s \equiv 1$.

Mechanisms.

(a) **The metric matches the mass.** The form satisfies $\langle \lambda, (\Delta + Q'^{*} a Q')\lambda\rangle \geq \gamma \sum_j \xi_j^{-2} \|\lambda\|^2_{\mathcal{L}_j}$, and on each layer $d_{\mathbf{B}}$ has slope ξ_j^{-1}. A single conjugation by $e^{\varsigma\varphi}$, with φ Lipschitz with respect to $d_{\mathbf{B}}$ (Combes–Thomas), therefore serves for all layers at once. It costs $O(\varsigma L^2)$, and this cost stays below γ under a ceiling on the rate, $\varsigma \leq c\gamma L^{-2}$, where L is chosen first. Every layer that has to be traversed in full costs the gap, that is, the separation of the boundaries of consecutive regions, which exceeds $R_1 M_1$ when measured in the unit of that layer; thus $d_{\mathbf{B}}(y, y') \geq R_1 M_1 (|j - j'| - 1)_+$. Let a rate $\tau > 0$ and a power $q \geq 0$ be given, and choose the product $R_1 M_1$ after them so that $\tau R_1 M_1 \geq C(1+q)\log L$. Then

$$\sum_{y'} e^{-\tau d_{\mathbf{B}}(y,y')} \leq C(\tau, L),$$

$$(\xi_{y'}/\xi_y)^q \leq L^q e^{\tau d_{\mathbf{B}}(y,y')} :$$

the factor $e^{-c\tau R_1 M_1}$ for each level outweighs the growth L^4 in the number of sites. Conjugation and walk expansions do different work. Conjugation yields decay of the inverse as a whole, and it requires no cube size to be large; its rate is a fraction of the spectral gap, which is the lower bound γ of the form. A walk expansion gives, in addition, a sum of terms, each of which depends on the background and on the regions only near its own walk. This property is what statements 5–8, the comparison of tori and § 4.5 make use of. Both methods are used. Conjugation serves for the scalar, one-scale and local operators and for the inverses on tops and rows. Walks serve for the assembly, namely of G_0 in (d) and of the propagators in a background in (f). In **95**, p. 36, Bałaban describes conjugation as "probably the simplest proof" of the decay; he builds a walk representation instead, since "such representations will be our basic tool in investigation of more complicated operators".

(b) **No Dirichlet condition on a lattice finer than the blocks.** Because of the gap that separates Ω_0^c from Ω_1, $\partial\Omega_0$ lies in $\mathcal{L}_0$. On that layer $Q = Q' = 1$ and the gauge term is zero, and the local operator $G_\square$, which inverts the form on a cube $\square$ (see (d)), obeys $a_0 \leq \eta^2 G_\square^{-1} \leq a_0 + 16$. Finite range together with a mass comparable with the norm gives decay for an arbitrary set of bonds. The following examples show that this placement of the boundary matters. Let G be a one-scale operator (spacing η, unit blocks) whose order is ≥ 1, with Dirichlet conditions imposed outside a union of *whole* blocks, so that the lattice is finer than the blocks and no block is cut. Two examples, a flat face and a re-entrant edge, rule out constants uniform in the order, by powers of η^{-1}. This applies to the L^2 bounds on second differences, to the bounds on ΔG, to those on ∇G and to its Hölder norm. It applies equally to the pointwise bound and to the Hölder bound involving $\nabla G \nabla^*$, taken in the form of statement 1, the powers being those that occur for ∇G and for its Hölder norm. Indeed, on the region every bounded f supported in a block can be written as $\nabla^* F$ with $F = \nabla \Delta^{-1} f$, where Δ^{-1} denotes the free inverse on $\eta\mathbb{Z}^4$ (on a torus, the inverse on functions of mean zero, after f has been modified on a block lying outside the region); moreover $|F|, \|F\|_\varepsilon \leq C_\varepsilon \sup|f|$ uniformly in η, and the sum over the blocks is controlled by the exponential

factor in the bound. For the L^2 bound involving $\nabla G \nabla^*$, by contrast, the two examples rule out no uniform constant. On a separated sequence no operator of this kind arises.

(c) **One scale, two scales.** Uniformity in η comes from two sources. The first is the representation $(\Delta+1)^{-1} = \int_0^\infty e^{-t} e^{-t\Delta} dt$, combined with heat-kernel bounds that hold both below and above $t = \eta^2$. The second is a factorisation of $\partial(1-R)\partial^*$ through the unit lattice. At a seam one makes an exact change of gauge $A = A_2 + H_j B - \partial H_j' \vartheta$. Here $A_2 \in \ker Q_j$, and B is a 1-form on the coarse lattice whose values on the axial trees of the L-blocks of the coarse region are zero. Further, ϑ is a site function on the coarse lattice, supported in the coarse region, whose mean over every one of those blocks is zero; for these two classes the representation is unique. Finally, H_j, H_j' are the one-scale minimisers, with H_j' the minimiser of $\|\Delta\lambda\|^2$ rather than of $\|\partial\lambda\|^2$. This change of gauge splits the form orthogonally into three parts: a form in A_2, a form in B and a form in $\vartheta - \vartheta_1$, where ϑ_1 is linear in A_2 and B. If the coarse region is all of the torus, the two-scale representation of the inverse of the form so obtained becomes a recursion $G_j \to G_{j+1}$, where G_j denotes the one-scale propagator of order j. It is applied for a single step and is never iterated. In the scalar theory, recursions of this kind add up to a series over the scales, and they served as the method of proof of regularity (**89**, (2.34)). Here G_j occurs multiplied by other operators. Iteration then leads to long products of operators whose gauge conditions are not the same at different scales (a remark of Bałaban, **96**, pp. 246–247). For gauge propagators, accordingly, no series over the scales is employed; instead the one-scale G_j has to be known simultaneously for all j (1c.3). Three spectral gaps on the unit lattice take the place of a small parameter. The sharp one is stated for B on a unit lattice that vanishes on the axial trees. In it, Q_1 and ∂_1 are the straight one-step average and the exterior derivative; the $Q_1(\mathbf{1})$ of the Notation, §3, also contains site terms. The sharp gap reads

$$L^2 \sum_c |(Q_1 B)(c)|^2 + \sum_p |(\partial_1 B)(p)|^2 \geq c L^{-5} \|B\|^2.$$

(d) **Many scales at 1.** No expansion of G' is required. For $u = G'(f + \nabla^* F)$ the conjugation of (a) yields

$$\|\xi^{-1} e^{\varsigma\varphi} u\| + \|e^{\varsigma\varphi} \nabla u\| \leq C\gamma^{-1} (\|\xi e^{\varsigma\varphi} f\| + \|e^{\varsigma\varphi} F\|)$$

, where ξ denotes multiplication by the local scale. From this one passes to local L^2 norms, and then to interior estimates at the scale ξ_y, with the mass term treated as a source. The walk expansions are obtained afterwards, as data. For G_0 we take $\sum_\square h_\square^2 = 1$, with $\zeta_\square = 1$ near $\operatorname{supp} h_\square$, and we let $G_\square$ be the inverse, on the periodised cube $\square$, of the form of (c), taken with one scale or with two. For the cubes of level 0 that extend beyond Ω_0, $G_\square$ is instead the operator of (b). For a cube of level ≥ 1 the periodic two-scale model is retained, irrespective of where Ω_0^c and the finer layers lie; this is legitimate since across the two seams to $\mathcal{L}_j \cup \mathcal{L}_{j+1}$ the function $h_\square$ is cut off. With $P = 1 - R$ and $\mathcal{P} = \partial P \partial^*$ one has

$$\mathsf{G}_0^{-1} \sum_\square h_\square G_\square h_\square = 1 - \mathcal{R},$$

$$\mathcal{P} h - h \mathcal{P}_\square = \zeta [\mathcal{P}_\square, h] + \zeta(\mathcal{P} - \mathcal{P}_\square) h + (1-\zeta)\mathcal{P} h :$$

of all these operators, only the non-local $\mathcal{P}$ cannot be transplanted. Each commutator costs M_1^{-1}, because $|\nabla h| \leq C(M_1 \xi_j)^{-1}$ stands against $|\nabla G_\square| \leq C\xi_j$. The change of domain costs e^{-cM_1}; this follows from resolvent identities on the two domains, recombined by means of $1 - P = R$. The tail is likewise exponentially small, and it is turned into a step $\square \to \square'$ by means of $\sum h_{\square'}^2 = 1$. The factor M_1^{-1} for each step outweighs the number $C(\tau, L)$ of neighbours.

Lower bounds independent of the number of levels underlie the inverses on tops and rows. The bound $Q'G'^2Q'^* \geq \gamma\xi_y^4$ follows by duality with test functions ψ_y on the blocks, with $Q'\psi_y = \delta_y$, whose second differences are $\leq C\xi_y^{-2}$, across the faces as well. The bound $Q\mathsf{G}_0Q^* \geq \gamma\xi_c^2$, with γ independent of L, follows from a right inverse E for all rows of all levels at once. This right inverse has scale-invariant energy, $\|\nabla\mathsf{E}B\|^2 \leq C\|\mathsf{m}^{-1}B\|^2$. The construction below gives the finite energy lift for the cornered-comb rows. Applying its estimate to the actual averaged rows requires the row comparison and the indicated pairing change. In 2, where there is a single lattice, the test forms are placed on the face separating the two blocks of a bond; there the share $\theta_0 > \frac{1}{2}$ of its own chains outweighs $1 - \theta_0$. Neither matrix is bounded below by a number independent of the scales ξ. They are therefore rescaled, to $\Sigma = \xi^{-2}Q'G'^2Q'^*\xi^{-2}$ or $\xi^{-1}Q\mathsf{G}_0Q^*\xi^{-1}$. A matrix $\Sigma \geq \gamma$ with $|\Sigma| \leq Ce^{-\tau d_{\mathbf{B}}}$ then satisfies $|\Sigma^{-1}| \leq 2\gamma^{-1}e^{-\tau' d_{\mathbf{B}}}$, where τ' is small compared with the spectral gap γ (Combes–Thomas, Schur). Thus both inverses decay without any expansion, and H decays column by column. A sequence is compared with the same sequence cut down around a cube. The cut-down sequence is again separated and coincides with the given one on the doubled cube, and its $Q\mathsf{G}_0Q^*$ provides the local operators that enter the walk expansion of $(Q\mathsf{G}_0Q^*)^{-1}$. The comparison uses Green's identity with a cut-off, by way of $(\partial^*\partial + Q^*aQ)\,\mathsf{G}_0Q^* = Q^*$, and the gauge term vanishes on the columns.

3.12.1.3. An energy-bounded lift of all determining rows. The row estimate used to invert the coarse Schur operator has a finite construction. It is useful to retain the distinction between the cornered-comb rows used in this construction and the differential of the averaged nonlinear constraint. Write $Q_{\mathrm{c}}^+ = (Q_j^{\mathrm{c}}|_{\widehat{\Gamma}_j})_j$ for the former, with all inward rows retained in $\widehat{\Gamma}_j$. Its output is a raw row logarithm, with the counting pairing $\sum_c\langle\omega(c), b(c)\rangle$. In particular, $Q_0^{\mathrm{c}}A = \eta A$. Fine fields have the pairing $\eta^4\sum_b\langle A(b), A'(b)\rangle$. The rows here are the linear parts of the iterated cornered-comb average, with the endpoint factors kept in its nonlinear factorization. They are not the response operator of the critical background.

Assume $L \geq 13$, the separated aligned geometry of eq. 3.ccciii–eq. 3.ccciv, with gap at least $L^2\xi_j$, and a real background satisfying

$$|\partial U(p) - 1| \leq \alpha\eta^2\mathsf{L}(p)^{-2}.$$

The required smallness threshold for α depends on L, but not on the depth. In the construction below all norms of row matrices are the counting ℓ^2 norms.

For a positive row c of level $j \geq 1$, choose its endpoint y_c in the level-j layer. Put $N = L^j$, $n = (N-1)/2$, and $w = \lfloor N/8\rfloor$. In integer coordinates P_ν measured from the lowest corner of $B^j(y_c)$, set

$$\begin{aligned} h(P) &= (w - |P - n|)_+, \\ \varphi_\pm(P) &= (w - |P - (n \pm w)|)_+, \\ g^+(P) &= (n+w)\varphi_+(P) - (n-w)\varphi_-(P), \\ g^-(P) &= (n+w+1)\varphi_-(P) - (n-w+1)\varphi_+(P). \end{aligned}$$

The sign is positive when $y_c = c_-$, and negative otherwise. For a bare bond b in the direction μ of c, define

$$\psi_c(b) = \frac{N^3}{2\eta w^9} g^{\pm}(P_\mu(b_-)) \prod_{\nu\neq\mu} h(P_\nu(b_-)),$$

and set it equal to zero on the other bonds. Transport a row vector to b_- along the cornered comb in this cell; for a negative row first transport it along the straight segment to that corner. The resulting test form is denoted by $Z_c e$. At level zero put $Z_c e = \eta^{-1}e$ on c, and zero elsewhere. Set $Z\omega = \sum_c Z_c\omega(c)$ and $\Theta = Q_{\mathrm{c}}^+ Z$.

The profiles are supported a positive fraction of the cell width away from its faces. Extending them by zero introduces no jump across a cell face. The identities

$$\sum_P h(P) = w^2,$$
$$\sum_P g^+(P)(P+1) = \sum_P g^-(P)(N-1-P) = 2Nw^3,$$
$$\sum_P g^+(P)(N-1-P) = \sum_P g^-(P)(P+1) = 0$$

follow by summing each triangular profile about its center. These are the two moments needed for the positive and negative rows at one site. The chain multiplicities in the cornered comb are $P+1$ and $N-1-P$, respectively. Thus at a flat field the test forms are biorthogonal to the rows at the same level.

For a small-curvature field, comparing each chain transport with the cornered comb changes this calculation by $O_L(\alpha)$. There are at most sixteen nonzero entries in each row and eight in each column of the same-level error. Decreasing the curvature threshold gives

$$\Theta = D_0 + N_0, \qquad \|D_0 - 1\| \le \frac{1}{48}, \qquad \|N_0\| \le \frac{288}{L} < 23, \qquad N_0^2 = 0. \tag{3.cdxxxv}$$

Here D_0 preserves the level and N_0 raises it by one. To check the last two assertions, a level-j test form can be read at a higher level only by an outward row of level $j+1$. Its entry has size at most $36L^{-3}$; at most $8L^4$ entries occur in one row and at most eight in one column. The Schur test gives $288/L$. Two consecutive upward transitions would place Ω_{j+2} within $(1+(L-1)/2+L)\xi_{j+1} < L^2\xi_{j+1}$ of Ω_{j+1}^c, contrary to separation. Downward transitions vanish because the supports avoid the outer sheet of each cell. This proves the asserted form and nilpotence.

Proposition 3.35 (A lift in the energy norm)**.** For sufficiently small α, there is a right inverse $Z^\sharp$ of Q_{c}^+ satisfying

$$Q_{\mathrm{c}}^+ Z^\sharp = 1, \qquad \mathcal{N}(Z^\sharp\omega) \le 48^2 c_Z \|\omega\|_2^2, \qquad c_Z = 21 \cdot 18^{12}.$$

Here

$$\mathcal{N}(A) = \|DA\|_{2,\sim}^2 + \|D^*A\|_2^2 + \|\mathsf{L}^{-1}A\|_2^2 + \|Q_{\mathrm{c}}^+ A\|_2^2.$$

The first term is the usual inner-plaquette energy together with $\eta^2 \sum_p (\sum_{b\subset\partial p} |A(b)|)^2$ over plaquettes which meet an inner bond and are not inner plaquettes. All constants are uniform in the depth and admitted torus. ∎

Proof. Since $N_0^2 = 0$, the exact factorization is

$$\Theta = (1+N_0)\{1 + (1-N_0)(D_0-1)\}.$$

The perturbation in braces has norm at most $24/48 = 1/2$. Its Neumann inverse therefore has norm at most two, and $\|\Theta^{-1}\| \le 48$. Notice that the inverse is

$$\Theta^{-1} = \{1 + (1-N_0)(D_0-1)\}^{-1}(1-N_0).$$

The same-level error must be included at a curved field.

For completeness, the energy estimate is local. The profile bounds and the curvature comparison give

$$|Z_c e| \le \tfrac{1}{2} 18^5 \xi_j^{-1} |e|, \qquad |\nabla_U Z_c e| \le 18^6 \xi_j^{-2} |e|.$$

A cell contains N^4 sites and has physical volume $\eta^4 N^4 = \xi_j^4$. At most two rows of each direction are assigned to one cell. The curl, divergence and weighted field terms consequently contribute at most $20 \cdot 18^{12} + 2(18^5/2)^2$ times the squared row norm. The level-zero terms obey the smaller bounds 24, 16 and 1. Disjoint cell cores prevent sums over different levels in this calculation.

Finally $\|Q_{\mathrm{c}}^{+}Z\omega\|_2^2 = \|\Theta\omega\|_2^2 \le 625\|\omega\|_2^2$. Adding the estimates gives $\mathcal{N}(Z\omega) \le c_Z\|\omega\|_2^2$. Take $Z^\sharp = Z\Theta^{-1}$. □

This lift also gives the lower bound for the coarse Gram operator. Suppose the real positive operator G_0 is defined by the Wilson Hessian, its specified gauge term and the retained row penalty, and suppose its form obeys

$$\langle A, G_0^{-1}A\rangle \le C_E\mathcal{N}(A).$$

For the inner Wilson Hessian one can take $C_E = 1 + 19C_{4,\mathrm{svar}}$, where $C_{4,\mathrm{svar}}$ is the constant in its local curvature-remainder estimate. With $\gamma_Q = (48^2 C_E c_Z)^{-1}$, Cauchy–Schwarz gives

$$\begin{aligned}\|\omega\|_2^2 &= \langle G_0^{-1}Z^\sharp\omega, G_0(Q_{\mathrm{c}}^{+})^*\omega\rangle \\ &\le \gamma_Q^{-1/2}\|\omega\|_2\langle\omega, Q_{\mathrm{c}}^{+}G_0(Q_{\mathrm{c}}^{+})^*\omega\rangle^{1/2}.\end{aligned}$$

Thus $Q_{\mathrm{c}}^{+}G_0(Q_{\mathrm{c}}^{+})^* \ge \gamma_Q$. This conclusion uses positivity of the specified fine operator; the right inverse by itself does not prove that positivity.

Removing inward rows is restriction in the counting row space. The lift and the Gram estimate restrict with the same constants. Their use for the averaged constraint additionally requires the comparison of its rows and endpoint transports with the cornered-comb rows. Their use for a current-corrected inverse requires the corresponding form comparison. Neither comparison is an identification of this lift with the derivative of the nonlinear background map.

For the scale convention in this section, the row pairing is weighted by m^4, where $\mathsf{m}(c) = \xi_c$. If $\widehat{Q}$ is the raw operator and $Q = \mathsf{m}^{-1}\widehat{Q}$, then $Q^* = \widehat{Q}^{\mathsf{T}}\mathsf{m}^3$, with $\widehat{Q}^{\mathsf{T}}$ taken in the counting row pairing. Thus multiplication by m^2, which identifies the weighted row Hilbert space with the counting row space, conjugates QG_0Q^* to $\mathsf{m}(\widehat{Q}G_0\widehat{Q}^{\mathsf{T}})\mathsf{m}$. This is the source of the two factors of the local spacing in the target Gram lower bound. The constants from the energy comparison must also have the uniformity required for the intended application.

3.12.1.4. From comb rows to the averaging constraint. The row lift above applies first to a specified linear family. We describe the change to the straightened mean constraint, including its error at the reference field. Throughout this comparison the reference configuration U_0 is real, the regions satisfy the separated-layer hypotheses, and all local boxes used by an average are admitted boxes. Constants depend on L, and the smallness and separation thresholds are fixed independently of the number of layers.

Write Q_{c}^{+} for the raw cornered-comb rows on all retained positively oriented bonds and Q_{c} for their restriction to the interior and outward bonds. The pairing of raw rows is counting measure. The fine-field pairing has weight η^4, and the site pairing on tops has weight ξ_y^4. In these conventions the comparison at level j reads

$$Q_{\mathrm{m},j} = T_j(Q_{\mathrm{c},j} + E_j), \qquad Q'_{\mathrm{m},j} = T'_j(Q'_{\mathrm{c},j} + E'_j). \tag{3.cdxxxvi}$$

Here $T_j\omega(c) = \mathrm{Ad}(t_j(c_-))\omega(c)$ and $T'_j\mu(y) = \mathrm{Ad}(t_j(y))\mu(y)$, with transports of U_0 in the appropriate local boxes. They preserve the indicated real pairings. The field-row error vanishes outside the double block under c, and

$$\sum_b |E_j(c,b)| \le \kappa_F\alpha_0\xi_j, \qquad \sum_{c\in\Gamma_j^{+}} |E_j(c,b)| \le \kappa_F\alpha_0\eta^4\xi_j^{-3}, \tag{3.cdxxxvii}$$

$$|E'_j\lambda(y)| \le \kappa_F\alpha_0 \max_{B^j(y)}|\lambda|.$$

At level zero these errors vanish. At higher levels, the curvature bound on the double block is $L^2\alpha_0 L^{-2j}$. The local comparison of the two averages gives the first bound and the corresponding bound for each kernel entry. A fine bond belongs to at most eight such rows at a fixed level, which gives the second bound. In level-j units the field is $\xi_j A$, so the row and column bounds become $\kappa_F\alpha_0$ and $\kappa_F\alpha_0(\eta/\xi_j)^4$, respectively.

The positive fine operator used for the comb Gram estimate is

$$G_{\mathrm{c}}^{-1} = \Delta_{U_0}^{\mathrm{W}} + B + DR_{\mathrm{c}}D^* + Q_{\mathrm{c}}^*Q_{\mathrm{c}}, \qquad R_{\mathrm{c}} = \mathrm{proj}_{\Delta_{U_0}\ker Q_{\mathrm{c}}'}. \tag{3.cdxxxviii}$$

The boundary operator B is positive and symmetric, its kernel joins only bonds of a common non-inner plaquette, and its row sums are at most $12\eta^{-2}$. The projection in eq. 3.cdxxxviii is taken in the fine site pairing. The scale-weighted comb propagator estimates give

$$M_{\mathrm{c}} = Q_{\mathrm{c}}G_{\mathrm{c}}(Q_{\mathrm{c}})^* \geq \gamma, \qquad H_{\mathrm{c}} = G_{\mathrm{c}}(Q_{\mathrm{c}})^*M_{\mathrm{c}}^{-1}, \qquad Q_{\mathrm{c}}H_{\mathrm{c}} = 1, \tag{3.cdxxxix}$$

and the scale-weighted bounds

$$\|H_{\mathrm{c}}\beta\|_X \leq K_1\|\beta\|_\infty, \qquad \|H_{\mathrm{c}}^*f\|_\infty \leq K_1|f|_{(-3)}.$$

These are estimates for the operator in eq. 3.cdxxxviii. In the nonlinear estimates below, $|f|_{(-s)} = \sup \mathsf{L}^s|f|$ uses the scale assigned to the fine bond or plaquette, and $\|A\|_X = \max\{|A|_{(-1)}, |\nabla A|_{(-2)}\}$. The version using the scale of the top of a cell is equivalent with factors at most L^s. Enlarge K_1 by these fixed factors when using the linear estimates in these norms.

Let $\mathcal{Q}^\natural(A)$ be the logarithmic straightened mean on the interior and outward rows, with $K_A = (\mathcal{Q}^\natural)'(A)$. With the corresponding fixed transport T, one has

$$T^{-1}\mathcal{Q}^\natural(A) = Q_{\mathrm{c}}A + C(A), \qquad T^{-1}K_A = Q_{\mathrm{c}} + C'(A), \qquad C'(0) = E. \tag{3.cdxl}$$

In particular, the remainder contains a linear term at a curved reference field.

Here are the quantitative estimates in these row coordinates. Let c_4, C_2 be the local logarithm and mean kernel constants, let $\kappa_{F,\mathrm{slin}}$ be the constant in the corresponding local comparison, and put

$$C_3^\natural = 104/c_4, \qquad \epsilon = \kappa_{F,\mathrm{slin}}L^2\alpha_0 + 4(C_2+1)Lr, \qquad \epsilon^* = \kappa_{F,\mathrm{slin}}L^2\alpha_0 + C_3^\natural Lr.$$

For $|A|_{(-1)}, |A_i|_{(-1)} \leq r \leq c_4/(4L)$,

$$\|C(A)\|_\infty \leq L\epsilon|A|_{(-1)}, \qquad \|C'(A)a\|_\infty \leq L\epsilon|a|_{(-1)}, \tag{3.cdxli}$$

$$|C'(A)^*\omega|_{(-3)} \leq 4L^3\epsilon^*\|\omega\|_\infty, \qquad |(C'(A_1)^* - C'(A_2)^*)\omega|_{(-3)} \leq 2L^4C_3^\natural|A_1 - A_2|_{(-1)}\|\omega\|_\infty.$$

The first line also gives the Lipschitz bound for C. For the transpose estimates, a fine bond of scale ξ_n occurs only in rows of levels $n-1, n, n+1$; their weights sum to at most $(L^3+2)\xi_n^{-3}$. The derivatives of the local mean kernel bound the column sum. Applying Cauchy's estimate in the local field ball to that column norm gives the last bound in eq. 3.cdxli. In particular these estimates do not introduce a factor equal to the depth.

The right inverse now transfers on the small chart. If $K_1L\epsilon \leq 1/2$, then

$$T^{-1}K_AH_{\mathrm{c}} = 1 + C'(A)H_{\mathrm{c}}, \qquad \|C'(A)H_{\mathrm{c}}\|_{\infty\to\infty} \leq 1/2.$$

Its Neumann inverse therefore gives

$$H_A = H_{\mathrm{c}}(1 + C'(A)H_{\mathrm{c}})^{-1}T^{-1}, \qquad K_AH_A = 1, \qquad \|H_A\beta\|_X \leq 2K_1\|\beta\|_\infty. \tag{3.cdxlii}$$

The dual calculation controls the multiplier. If $q = 4K_1L^3\epsilon^* \leq 1/8$, then

$$S_A = 1 + H_{\mathrm{c}}^*C'(A)^*, \qquad \|S_A^{-1}\|_{\infty\to\infty} \leq 8/7, \qquad H_{\mathrm{c}}^*(T^{-1}K_A)^* = S_A.$$

Consequently, whenever $F'(A) + DR_{\mathrm{c}}D^*A = K_A^*\omega$, its multiplier is uniquely determined by

$$T^{-1}\omega = S_A^{-1}H_{\mathrm{c}}^*(F'(A) + DR_{\mathrm{c}}D^*A). \tag{3.cdxliii}$$

This argument does not assume a bound for the multiplier before solving for it. Here $F(A)$ is the inner Wilson action evaluated at $e^{i\eta A}U_0$.

There is a further coordinate change when the datum is written with its endpoint gauges. The seam coordinate change Ξ gives $K_A^\flat = D\Xi\, K_A$. For the actual reduced datum, write $n(A) = x(A)^{v[A]}$. Its derivative is

$$dn[A]a = [dx[A]a]^{v[A]} + \mathrm{rot}_{n(A)}(\mu), \qquad \mu = i^{-1}(dv[A]a)v[A]^{-1}.$$

Gauge covariance gives $dn[A]\delta_\lambda A = \mathrm{rot}_{n(A)}(\lambda|_{\Lambda_*})$, where $\delta_\lambda A$ is the infinitesimal bare gauge variation and rot denotes the infinitesimal action on the datum. To attain a prescribed tangent datum, first choose a using the surjectivity of the straightened rows and then choose $\lambda = \mu$ on the tops. The vector $a - \delta_\lambda A$ attains the datum. This proves surjectivity of the finite-dimensional datum map. A uniform bound for its right inverse additionally uses the estimates for this gauge extension.

The first line of eq. 3.cdxli holds for complex increments around the fixed real reference field. The transpose bounds hold with an additional factor two, using the complex bilinear transpose. Extension of a real linear estimate to complex vectors is distinct from allowing the reference field and the independent current to vary on the full complex domain of the propagator statements.

Thus the local averaging constraint has a bounded right inverse on its specified chart. To obtain the target Gram estimate for the corrected rows and current-completed fine operator, one must still match the row coordinates and pairings, establish the energy bound for the resulting lift, and compare that fine quadratic form with eq. 3.cdxxxviii. The construction in eq. 3.dcv addresses the actual row energy; the row comparison above supplies its source interpretation when the averaging conventions agree. Neither surjectivity in the supremum norm nor eq. 3.cdxxxix alone performs the remaining quadratic-form comparison.

(e) **Gauge algebra without Gaussian integrals.** On a pure gauge, take the derivative of the covariance of the straightened average; the site factors then convert point values into block means: $QD = \bar{D}Q'$. Consequently the rows annihilate $D\ker Q'$. Combine this with the invariance of the action to second order, which is exact for the inner plaquettes; this accounts for that Hessian. The result is $\mathsf{G}^{-1}D\lambda = D\Delta_{\mathbf{U}}\lambda - \Delta'_\pi[\mathbf{J} - J(\mathbf{U})]D\lambda$ on $\ker Q'$. The propagators are taken at pairs whose current is an independent variable carrying a bound of its own, as in the spaces of **109**, p. 262. The reason is that bounds on the plaquettes, on the potential and on the first differences do not control $J(U)$ uniformly in the level: a staggered field satisfies all of them with $|J(U)| \geq \alpha\eta^{-1}$. In **99**, 389, by contrast, $J = J(U)$, and a condition on second differences is imposed in addition ((3.36), (3.38)). The price of a free current is the defect in 3. Uniformity in the level holds for the following reason. Row norms and column norms are both submultiplicative, and for a single step they lie near L and L^{-3}. The relative errors $O(\alpha_0(L^n\eta)^2 + \alpha_1 L^n\eta)$ of the steps form geometric series that are dominated by the top level.

3.12.1.5. Matching the local factors of the background propagator. For the native propagator construction use the same candidate cubes, trace-local bump rule and rules for modifying the cover in the two nests. Let $G_0 = \sum_\square h_\square G_\square h_\square$ be the native vector parametrix. Divide each candidate bump by the square root of the sum of all retained and inserted bump squares at that point, so that $\sum_\square h_\square^2 = 1$. Use local inverses whose levels agree with the global levels on the tested stencils. For a level-j cube this agreement holds when its twice-enlarged cube lies in $\Omega_j \setminus \Omega_{j+2}$. Separation supplies the inner margin. At the outer face one has to check the same levels or use the corresponding boundary model. Write D for the covariant site derivative and P for the projection occurring in the gauge term. Write $\Delta_a G_0 = I - R$. The four summands of R at a cube are

$$K(h_\square)G_\square h_\square, \qquad (1-\zeta_\square)DPD^* h_\square G_\square h_\square,$$
$$\zeta_\square(DPD^* - DP_\square D^*)h_\square G_\square h_\square, \qquad \zeta_\square[DP_\square D^*, h_\square]G_\square h_\square.$$

Here $K(h_\square)$ is the local differential and averaging commutator. The last term uses the local projection and the first difference of the bump. In the two occurrences of the global projection substitute

$$P = G'Q'^*(Q'G'^2Q'^*)^{-1}Q'G'.$$

For G', the scalar parametrix with the same square partition gives the series of local inverse and commutator factors. For the site Gram inverse, glue its local inverses. The resulting error consists of a tail outside the cut, a difference of scalar propagator squares and a local commutator. Insert the scalar expansion with the cover modified at the cut. The difference of squares is expanded as $(A-B)A + B(A-B)$, with the corresponding local scalar propagators A, B. In the vector error

with $\zeta_\square$, the term containing no small commutator or separated supports cancels against the local projection term. To get this cancellation, replace the inserted bumps by $\zeta_\square$, compare the modified local inverses with the original ones, and commute the cutoffs to the right. The errors have a cutoff difference, a localized inverse difference or a commutator. The term left after these operations uses $\zeta_\square h_\square = h_\square$. Consequently $G_0 \sum_{m\ge 0} R^m$, with these substitutions, has only local inverse, projection and commutator factors [B85c, (3.87)–(3.108)].

For comparison on a common region, fix one list of possible words before imposing either nest. Take as the nominal footprint the union of the constituent cubes with five layers of padding at their own scales. Admission of the word also reads the geometric decisions that modify the cover. In top units the index rule reads the region data less than $2M$ beyond the footprint, since it tests the neighboring cube in deciding which cover to use. Enlargement by $2M$ is also sufficient for the coefficients, including the bumps and the full field reads. If the outer averaging and derivative factors read entirely inside the footprint, the field reads require only the original footprint. Use this enlarged region for comparison. Equality of the two nests and their region data there gives the same cover decisions, local sequences and cutoffs. The local operators are therefore the same functions of the common background reads. At the same background on those reads their values are equal. Thus all the words whose enlarged reads lie in the common region cancel, term by term. Under the native regularity and separation assumptions the absolute walk majorant permits this cancellation inside the sum. It estimates the remaining terms by the first escape from the common region, with the geometric padding subtracted before measuring the escape. An operator formed with an additional region restriction uses its required margin to that region's boundary. For the history comparison, one neighboring cell supplies this geometric padding when $\check{R}_s \ge 2$. The trace data must record the full nest inside each component meeting the filed domain. The clean neighboring cells have trivial trace when they belong to the regular region of every nest in force. Where the padding reaches a different component, the comparison uses its nest and region trace on the cells met by the padding. For an application to a switched or compressed inverse one must also match its actual masks and the data read by its additional vertices.

(f) **Unfolding and parametrix.** With auxiliary unknowns

$$\Psi = (A; v, w, \sigma; p, q, r; \nu)$$

, the operators R, Δ'_π, $\mathsf{S}^{\mathsf{T}}\Delta^{(2)}\mathsf{S}$ and the constraint are combined into a single system $\mathbb{K}$ of finite range. The inverse of this system has the propagators as blocks. Let $\varkappa$ be an M_1-cube; these cubes have side M_1 in the local unit and tile Ω_0, whereas the $\square$ of (b), (d) are the cubes of the flat expansions. On a regular interior piece, in the gauge of the class, $\mathbb{K}_\varkappa$ equals the flat system plus $O(\alpha_0 + \alpha_1 + \gamma_0)$, and this perturbation need only be small compared with flat constants. A piece meeting the true bare boundary instead uses the direct massive inverses of the actual compressed Wilson Hessian and the covariant site Laplacian. In bare units the real operators $1 + \eta^2\Delta_Y(U)$ and $1 + \eta^2\Delta_U^\circ$ have uniform positive lower bounds, respectively $1/2$ and 1, and uniformly bounded finite-range kernels. A small complex perturbation preserves their inverses and exponential decay; local elimination then supplies the remaining slots. This boundary construction uses the actual incident inner plaquettes and requires no gauge across the boundary. Let $\theta_\varkappa$ be a partition of unity, let the cut-offs $\tilde{h}_\varkappa = 1$ near $\operatorname{supp}\theta_\varkappa$ be such that $\mathbb{K} = \mathbb{K}_\varkappa$ on $\operatorname{supp}\tilde{h}_\varkappa$, and let δ denote the rate of the local inverses. Then

$$\mathbb{K}\sum_\varkappa \tilde{h}_\varkappa \mathbb{K}_\varkappa^{-1}\theta_\varkappa = 1 - \sum_\varkappa \mathbb{R}_\varkappa,$$

$$\mathbb{R}_\varkappa = -[\mathbb{K}_\varkappa, \tilde{h}_\varkappa]\mathbb{K}_\varkappa^{-1}\theta_\varkappa = O\big(M_1^{-1}e^{-c\delta M_1/L}\big),$$

where the power comes from the commutator, and the exponential from how far $\operatorname{supp}\theta_\varkappa$ lies from $\operatorname{supp}\nabla\tilde h_\varkappa$. This bound outweighs the $O(1)$ neighbours. Three features keep the perturbation and the commutators from losing a derivative: scaling weights, one component of the source that admits a divergence, and the absence of any derivative on p.

(g) **Trees.** The system $\mathbb{K}$ depends affinely on the non-local coefficient $\mathsf{H}^{\mathsf{T}}\mathbf{J}$. The part of walk length zero stays in the local operators. The remainder, of size $O(\gamma_0 M_1^{-1}e^{-cM_1/L})$, forms a vertex that carries a walk of H. Thus vertices are small because of M_1 and not because of the field, and each term is indexed by a trunk with branches. When the cells are listed depth first, the number of such trees grows geometrically: for each first cell, there are C^m tree walks of length m starting from it. Both this count and the size of M_1 needed for the sum over trees to converge coincide with the ones in 1c.9. In 1c.7 and 1c.8 a branch is listed as soon as its cut is reached, and the trunk is resumed afterwards. For the data of 1c.9 the listing first continues at every cell along the trunk (within a branch, along the branch). So the whole trunk comes first, and then the branches come in the reverse order of their cuts, beginning with the last cut and ending with the first. Each of the two orders is a depth-first order. The order of listing changes neither the cells nor their parents, and so it changes neither the hull, nor the tree length, nor the bounds.

(h) **Multiplier and positivity.** Adding the equation "QA = fluctuation field" brings in $(Q\mathsf{G}_1Q^{\mathsf{T}})^{-1}$ as a Lagrange multiplier. Consequently $C^{(k)}(Y)$ occurs as a block in the inverse of a system of finite range. In the flat case, positivity follows from $\sum_{p'}|\bar\partial QA(p')|^2 \le \eta^4\sum_p|\partial A(p)|^2$, with $\bar\partial$ taken on $\mathbb{T}^{(k)}$, together with a definite form on double blocks. In a background, a homotopy to $\mathbf{1}$ preserves $C_\varkappa > 0$ for the local inverses. The eigenvalues z of $\chi_Y\mathsf{A}\chi_Y\sum h_\varkappa C_\varkappa h_\varkappa$ satisfy $|1-z|\le CM_1^{-1}$.

(i) **Length against parameters.** A tree of length t comes close to at most $2^4\max\{1, Ct/\ell_{\mathsf{P}}\}$ cubes $\square$ of the partition of side ℓ_{P} in statement 7; these are not the cubes $\square$ of (b), (d). Hence $\bar\kappa\#\mathsf{P}(\omega) \le 2^4\bar\kappa + (C\bar\kappa/\ell_{\mathsf{P}})|\Gamma|_{\mathbf{B}}$ for every tree Γ admitted in $d_{\mathbf{B}}(\omega; y, y')$, provided that X_0, which Γ need not meet, lies within $O(1)$ top units of y (no other term enters a functional near y, since the left argument of the kernel of O_ω touches X_0). Consequently, for $|s(\square)| \le e^{\bar\kappa}$ and $\delta \ge C\bar\kappa/\ell_{\mathsf{P}}$,

$$|s^\omega|\, e^{-\delta d_{\mathbf{B}}(\omega;y,y')} \le e^{2^4\bar\kappa}\, e^{-(\delta - C\bar\kappa/\ell_{\mathsf{P}})d_{\mathbf{B}}(y,y')}.$$

Each cube beyond the first 2^4 adds to the tree a length $\ge c\ell_{\mathsf{P}}$. Since ℓ_{P} is chosen last, $e^{-c\delta\ell_{\mathsf{P}}}$ outweighs $e^{\bar\kappa}$. The factor $e^{2^4\bar\kappa}$ is left over and cannot be avoided, as a cell at a corner of the grid shows. The displayed bound uses only the total-incidence count: $|s^\omega| \le e^{\bar\kappa\#\mathsf{P}(\omega)}$, followed by $d_{\mathbf{B}}(\omega; y, y') \ge d_{\mathbf{B}}(y, y')$. Its parameter loss is the stated reduction of the spatial decay rate. The estimate is Bałaban's (**116**, p. 5, after (1.11)); in his setting as well, the exponential by itself does the paying, and $e^{16\bar\kappa}$ remains. He keeps the factor $M_1^{-|\omega|/2}$ in reserve together with the exponential (p. 3, after (1.7)). This factor is not used up on the parameters, and here it is still at our disposal for the factor $|\omega|+1$ that occurs below. Since the uncut hull, which is his localisation domain, is connected, $\mathsf{P}(\omega)$ is connected through walls; this yields the splitting. The carrier is needed because the fields are given only on Ω_0, and it need not be connected when $\Omega_0 \ne T_\ell$.

The properties of single terms are produced by construction; they are not deduced. The sum is itself covariant and real. Therefore the Haar mean of the terms, made symmetric under the conjugation fixing the real pairs, is once more an expansion of the sum. By convexity it satisfies the same bounds, provided that the family of bounds is invariant under the real gauge transformations and under this conjugation; the family of localised bounds has this invariance. A single assignment for all tori is obtained in three steps: a mean over translations; a push-forward from covering tori; and a passage to the limit by Montel's theorem, along a subsequence fixed once, at the given top level and units. In the push-forward, a walk wrapping around the small torus gains at most $|\omega|+1$ blocks; the weight therefore loses a factor $|\omega|+1$, and nothing else is lost. The family is made

normal by an entrywise bound, which is one and the same number for all tori with room. This number is allowed to depend on the level, since the entries of $(Q\mathsf{G}Q^{\mathsf{T}})^{-1}$ are $O(\eta^{-2})$, and none of the norms, rates, weights or sums of weights involves it. If a family of terms is already carried by all shifts and compatible with all coverings, it is not changed.

3.12.1.6. The coupled map eq. 3.cdxliv. The datum in this map is the raw row logarithm B'. The fine pairing retains its physical cell weights, while the raw row pairing is counting measure. At $s=1$, define $\hat{Q}=\mathsf{m}Q$, $H=\mathsf{H}\mathsf{m}^{-1}$, $S=(\hat{Q}\mathsf{G}\hat{Q}^{\mathsf{T}})^{-1}$, $H_0=\mathsf{G}_0\hat{Q}^{\mathsf{T}}(\hat{Q}\mathsf{G}_0\hat{Q}^{\mathsf{T}})^{-1}$, and $\tilde{G}=\mathsf{G}_0-H_0\hat{Q}\mathsf{G}_0$. Here S acts on raw rows and is distinct from the fine projector S. The multiplication operator a is fixed by $\hat{Q}^{\mathsf{T}}\mathsf{a}\hat{Q}=Q^{\mathsf{T}}aQ$; for the row volumes ξ_c^4 and $a(c)=a_0\xi_c^{-2}$, it is $a_0 I$. All transposes in these formulas use the indicated two pairings.

Let $F(T)=A_{\Omega_0}(e^{i\eta T}\mathbf{U})$, using its regular holomorphic continuation, and define $\mathcal{C}(T)=\mathcal{Q}(T)-\hat{Q}T$ and $\mathcal{N}(T)=F'(T)-F'(0)-\Delta_{\mathbf{U}}^{\mathrm{W}}T$. Both remainders vanish to second order at zero on their actual admitted balls. The independent $\mathbf{J}$ is not set equal to $F'(0)$. The transpose of $\mathcal{C}'_T$ maps a raw row multiplier λ to a fine force. For $s\neq 1$, the four maps $H_0(s),\tilde{G}(s),H(s),S(s)$ are their specified walk-data sums, not inverses recomputed using identities that may fail after decoupling.

The map on a fine potential T and raw row multiplier λ is

$$\begin{aligned} T &= H_0(s)[B'-\mathcal{C}(T)] \\ &\quad -\tilde{G}(s)[\mathcal{N}(T)-{\mathcal{C}'_T}^{\mathsf{T}}\lambda], \\ \lambda &= H(s)^{\mathsf{T}}\mathbf{J}+[S(s)-\mathsf{a}][B'-\mathcal{C}(T)] \\ &\quad +H(s)^{\mathsf{T}}[\mathcal{N}(T)-{\mathcal{C}'_T}^{\mathsf{T}}\lambda]. \end{aligned} \tag{3.cdxliv}$$

Thus $H_0,\tilde{G}$ are the zero-current slots, whereas H,S use the current-dependent propagator. The centred multiplier variable is $\lambda-H(s)^{\mathsf{T}}\mathbf{J}$. With these raw-row pairings, the physical multiplier in the identified case is $\mathsf{m}^{-3}\lambda$. Prop. 3.177 proves convergence and uniqueness on the full first-order regular-pair domain and enlarged parameter discs, with the additional independent-current smallness required by this coupled system. Its decorated maps are the actual carrier sums of Lem. 3.172.

(j) **Operators as data.** The principle goes back to Bałaban, who observes that in the equations for the response "we can replace the propagators by arbitrary operators having the same regularity properties and satisfying the same bounds" (**116**, p. 3, (1.5)). In its published form, however, the second equation relies on the identities $RD^{\mathsf{T}}\mathsf{H}=0$ and $\mathsf{G}^{-1}\mathsf{H}=Q^{\mathsf{T}}(Q\mathsf{G}Q^{\mathsf{T}})^{-1}$, and therefore on their consequences $Q\mathsf{H}=1$ and $\Delta_\pi\mathsf{H}=Q^{\mathsf{T}}[(Q\mathsf{G}Q^{\mathsf{T}})^{-1}-a]$ ($\Delta_\pi=\Delta_{\mathbf{U}}^{\mathrm{W}}-\Delta'_\pi$). None of these is available for the sums $\sum_\omega s^\omega\mathsf{O}_\omega$: on the cubes around a row where s vanishes, every term is absent, so that $Q\mathsf{H}(s)$ equals 0 there and not 1. The first identity, moreover, holds only for the pairs on shell, even at $s\equiv 1$. Consequently no identity enters the argument. The identities are applied earlier, where they hold. With their help the gradient of the non-quadratic part of the action is put into a shape from which its second-order term $\Delta_\pi\mathsf{H}$ is absent, and this expression *defines* W_s for every s and every pair (it is a gradient only at $s\equiv 1$ and on shell). From then on only bounds are used, in the norms

$$\|A\|_\varpi=\sup_y \varpi(y)\max\{\xi_y|A|,\,\xi_y^2|\nabla_{\mathbf{U}}A|\},$$

$$|f|_\varpi^\sharp=\sup_y \varpi(y)\,\xi_y^3|f|,$$

$$\varpi(y)\le e^{\theta d_{\mathbf{B}}(y,y')}\varpi(y'):$$

the first measures a potential together with its first derivative, the second a force. The operator $\mathfrak{G}(s)$ takes the second norm back to the first, with norm $\hat{\mathsf{b}} \sim e^{2^4\bar{\kappa}}$, which is the size of every one of the operators; this closes the fixed-point argument.

3.12.1.7. The reduced force on the complex ball. Use raw row coordinates in this calculation. Write $\widehat{Q} = \mathsf{m}Q$ and let H_s be the walk sum for the raw lift $\mathsf{H}\mathsf{m}^{-1}$. Let S_s be the row inverse sum in these coordinates. At $s = 1$ it equals $(\widehat{Q}\mathsf{G}\widehat{Q}^{\mathsf{T}})^{-1}$, where the transpose uses the raw row pairing. If λ is a raw multiplier, the multiplier in the row-volume pairing is $\mathsf{m}^{-3}\lambda$. Indeed the row amplitude is multiplied by m and the row volume is m^4, so the corresponding per-length transpose is $\widehat{Q}^{\mathsf{T}}\mathsf{m}^3$. Put $P_s = S_s - \mathsf{a}$ with the row mass a used in eq. 3.cdxliv.

Let $\mathcal{C}_2$ be the quadratic part of the raw remainder $\mathcal{C}$. Given an auxiliary potential T, solve

$$X(T) = \mathcal{C}(T - H_sX(T)), \qquad Y(T) = T - H_sX(T), \qquad E_T = (\mathrm{d}X(T))^{\mathsf{T}}.$$

The bounded local row map and the quadratic estimate on $\mathcal{C}$ make this a contraction in X on a ball of radius Cr^2 when $\|T\| \le r \le c/\hat{\mathsf{b}}$. Indeed the Lipschitz constant is $C\hat{\mathsf{b}}r$, and $\|Y(T)\| \le 2r$ after decreasing c. Differentiating the equation for X gives

$$E_T = {\mathcal{C}'_{Y(T)}}^{\mathsf{T}}(1 + H_s^{\mathsf{T}}{\mathcal{C}'_{Y(T)}}^{\mathsf{T}})^{-1}.$$

In particular X starts with $\mathcal{C}_2(T)$. In the following formulas $\mathcal{V}$ is the quadratic remainder of the action gradient, restricted to the force bonds of the local calculation as in $\mathcal{N}$ above. Write $\phi = H_s^{\mathsf{T}}\mathbf{J}$. The formula used with the walk data is

$$\begin{aligned}\mathsf{W}_s(T) = \; & -(E_T - (\mathrm{d}\mathcal{C}_2(T))^{\mathsf{T}})\phi - \widehat{Q}^{\mathsf{T}}P_sX(T) \\ & - E_TP_s\widehat{Q}T + E_TP_sX(T) + \mathcal{V}(Y(T)) - E_TH_s^{\mathsf{T}}\mathcal{V}(Y(T)).\end{aligned} \tag{3.cdxlv}$$

For the reduction take the full carrier at $s = 1$, with the on-shell identity $RD^{\mathsf{T}}H_{s=1} = 0$. Use $\Delta_\pi H_{s=1} = \widehat{Q}^{\mathsf{T}}P_{s=1}$, where $H_{s=1}$ is the raw lift from G. Thus $H_{s=1}^{\mathsf{T}}\Delta_\pi H_{s=1} = P_{s=1}$. The terms in the differentiated reduced action come from

$$-(\phi, X - \mathcal{C}_2(T)) - (\widehat{Q}T, P_{s=1}X) + \tfrac{1}{2}(X, P_{s=1}X) + V_0(Y), \qquad V_0' = \mathcal{V}$$

on the full physical domain. Symmetry of $P_{s=1}$ gives the fourth term in eq. 3.cdxlv, and the chain rule for Y gives its last two terms. For general s eq. 3.cdxlv is evaluated directly using the designated operator sums.

For the estimate, the two small arguments T_i have $\|T_i\| \le r$. Set $d = \|T_1 - T_2\|_\varpi$. In the raw row norm the fixed point and its derivative satisfy

$$|X_i| \le Cr^2, \quad |X_1 - X_2|_\varpi \le Crd, \quad |E_{T_i}\psi|_\varpi^\sharp \le Cr|\psi|_\varpi, \quad |(E_{T_1} - E_{T_2})\psi|_\varpi^\sharp \le Cd|\psi|.$$

Here $|\cdot|$ on rows is the unweighted supremum and $|\cdot|_\varpi$ uses the weight at the assigned top. The first two bounds follow from the row contraction. The two derivative bounds use the local estimate for the transpose of the average differential and the resolvent identity for $1 + H_s^{\mathsf{T}}{\mathcal{C}'_Y}^{\mathsf{T}}$. Each difference in the resolvent identity contains one difference of $\mathcal{C}'_Y$, bounded by Cd with its weight on that factor. Both resolvents are bounded on the same ball since $C\hat{\mathsf{b}}r$ is small.

The second and third terms of eq. 3.cdxlv have differences bounded by $C\hat{\mathsf{b}}rd$. In the fourth the additional small factor r comes from E_T acting on the quadratic X, or from X when the difference falls on E_T. The fifth term is bounded by Crd by the local action remainder estimate and $\|Y(T_1) - Y(T_2)\|_\varpi \le 2d$. For the sixth, the undifferenced action remainder is $O(r^2)$, its difference is $O(rd)$, and H_s^{T} costs $C\hat{\mathsf{b}}$. These contributions are therefore bounded by $C\hat{\mathsf{b}}r^2d$.

For the first term let $r_0 = c/\hat{\mathsf{b}}$ be the radius of the full holomorphic ball and assume $0 < r < r_0$. Put $N(T) = E_T - (\mathrm{d}\mathcal{C}_2(T))^{\mathsf{T}}$. The function $g(z) = (N(zT_1) - N(zT_2))\phi$ is holomorphic on $|z| < r_0/r$, with $g(0) = g'(0) = 0$, because $\mathcal{C}_2$ is the quadratic term of X. The derivative difference bound gives

$|g(z)|_{\varpi}^{\sharp} \le C|z|d\,|\phi|$. Apply Schwarz's lemma to $g(z)/z$, after pairing with a dual functional of norm one, and take the supremum over the dual unit ball. This gives

$$|g(1)|_{\varpi}^{\sharp} \le C(r/r_0)d\,|\phi| \le C\hat{\mathsf{b}}^{\,2}\gamma_0 rd.$$

The resulting bound is $C\hat{\mathsf{b}}(1+\gamma_0\hat{\mathsf{b}})rd$. The coefficient and operator bounds here are on the local class used for the walk sums, with the force bonds, row supports, and the neighboring plaquettes inside its enlarged carrier. In applying this calculation to a history, membership of the actual restricted fields in that common class must still be checked.

The Lipschitz constant of the fixed-point map is $\hat{\mathsf{b}}\cdot\hat{\mathsf{b}}^2\cdot\hat{\mathsf{b}}\sup|B'|$, with contributions from $\mathfrak{G}$, W_s and H_1, and this is why the radius is $\hat{\mathsf{b}}^{-4}$. Decay amounts to choosing the weight $\varpi = e^{\theta d_{\mathbf{B}}(\cdot,y')}$. When $s \not\equiv 1$, bounds of first order on the operators do not yield a bound on $\Delta_{\mathbf{U}}\mathbf{H}$ independent of the level; an oscillating kernel belonging to the class of data is an example. The reduced current takes its place. It is built from the multiplier, and in it the second-order part of the gradient occurs only in paired form, $(D^{\mathsf{T}}DA, \mathsf{H}e_c) = (DA, D\mathsf{H}e_c)$, where e_c is an element of $\mathfrak{g}^c$ placed at the row c. As a result, only first-order norms occur.

3.12.1.8. Restriction of the response. A surviving walk has its full parameter support in one wall component of $\{s \neq 0\}$. It gives the same summand when all parameters outside that component are set to zero. For the nonlinear part, use the two nested collars in the definition of the local class. The potential and datum slots of the walk operators are supported inside the first collar boundary, at distance greater than the reach of the local average and force from the other components. Thus an iterate supported there has its averages, action remainder and transposed differential computed from data in the enlarged collar. The next iterate is again supported in the same slots. Starting at zero, the iteration for the union is the sum of the component iterations. Uniform contraction proves the same identity for the limits and allows comparison of settings by checking the walk terms and the local coefficients on these collars. A match of the decay rates alone would not identify the two responses.

For identification with the actual background, use all parameters equal to one at a real critical reference with its actual current. The reduced datum is obtained from the physical datum by the seam and node factors of the straightened average. Let C_{b} bound that map, r_{b} be its chart radius, r_B the radius of the holomorphic background chart, and ε_{D} the response datum radius at $s = 1$. Take the physical datum in the ball of radius $\min\{r_B, r_{\mathsf{b}}, \varepsilon_{\mathrm{D}}/C_{\mathsf{b}}\}$. The reference pair must belong to the common regular class at the admitted response radii. The holomorphic background in the normalized node gauge must lie in the response uniqueness ball and solve the constraint and slice equations there. The multiplier argument below and uniqueness identify it with the response to the reduced datum. On the constant sequence the two datum coordinates agree. The identification on a history requires the same determining rows and the same current correction in both constructions.

3.12.1.9. The coupled response in raw row coordinates. For eq. 3.cdxliv the walk sums in the four slots are $H_0(s), \tilde{G}(s), H(s)^{\mathsf{T}}$, and $S(s) - \mathsf{a}$. The mass subtraction is a local diagonal term and does not carry a decoupling parameter. In the estimate of these slots, the potentials and the row outputs of a datum at y' carry the factor $\xi_y/\xi_{y'}$. The potential from a force supported over y' carries $(\xi_y/\xi_{y'})^3$. The two bounds have the same exponential decay in $d_{\mathbf{B}}(y, y')$ and hold for partial sums. To see how the normalization enters, the datum at y' is a raw row logarithm, whereas the output potential and its first difference are measured in the units at y. Transpose the H-kernel with its fine pairing and the counting row pairing before estimating the multiplier slot. To absorb a ratio of power $p \le 3$, use the distance lower bound $\rho(|j - j'| - 1)_+$ for tops on levels j, j', with $\rho = R_1\mathfrak{M}$. Then the ratio costs at most $L^p e^{p(\log L)d_{\mathbf{B}}/\rho}$. For the kernel rate $3\dot{\delta}_0/16$, the separation floor $\dot{\delta}_0\rho \ge 512\log L$ leaves decay at rate $11\dot{\delta}_0/64$. Summing with the admissible weights at smaller rates gives the common operator bound $\hat{\mathsf{b}}$ of order $e^{16\bar{\kappa}}$. The carrier construction uses parameter cubes of side at least $88\,\mathsf{r}_{\mathrm{w}}\mathfrak{M}$, where r_{w} bounds the number of cubes in a walk cell, with at least three parameter cubes in each direction. The side is also chosen so that its product with $\dot{\delta}_0$ is at

least $4352\bar{\kappa}$. These are floors on the regular pair and walk construction, which is used throughout the contraction.

Put $\lambda_0 = H(s)^{\mathsf{T}}\mathbf{J}$ and $\nu = \lambda - \lambda_0$. For $|B'|_\infty \le r$ consider the ball of radius $R = 2\hat{\mathsf{b}}r$ in $\max\{\|T\|, |\nu|_\infty\}$. On this ball let Δ be the maximum of the weighted differences of T and ν at two points. The differences of the two quadratic remainders $\mathcal{C}, \mathcal{N}$ are bounded by $CR\Delta$. For the remaining term take the difference of ${\mathcal{C}'_T}^{\mathsf{T}}(\lambda_0 + \nu)$ by putting the weight on the changed factor. A difference in T costs $C(|\lambda_0|_\infty + R)\Delta$, and a difference in ν costs $CR\Delta$. Since $|\lambda_0|_\infty \le \hat{\mathsf{b}}\gamma_0$, the Lipschitz constant of the map is bounded by

$$C\hat{\mathsf{b}}(R + \hat{\mathsf{b}}\gamma_0).$$

Choose both r and γ_0 at most a sufficiently small constant times $\hat{\mathsf{b}}^{-2}$. The map has Lipschitz constant at most $1/2$, and its value at zero has norm at most $\hat{\mathsf{b}}r$. It therefore preserves the ball. Iteration gives the holomorphic solution, with the same contraction estimates in the weighted norms. Subtracting the equations at two data then bounds their solution difference by $2\hat{\mathsf{b}}|B'_1 - B'_2|_\varpi$. The choice of weight centred at the changed row gives the decay.

These estimates concern the reduced system on the interior and outward rows. The identification at parameters one is proved below at a real critical reference with its own current.

(k) **From the slice to a multiplier.** Let $s \equiv 1$, and fix a critical real U_0 such that $(U_0, J(U_0))$ is a regular pair whose radii lie below the ceiling. Then (I) is equivalent to the constraint, the Landau condition and $\mathsf{z} := F'(\mathbf{H}) - K_{\mathbf{H}}^{\mathsf{T}}\lambda \in DR(\text{site functions})$, where λ is the multiplier formed from $\mathbf{H}$. In other words, (I) amounts to stationarity on the gauge slice alone. If B' is real and close to 0, then $\mathbf{H}(B')$ is real and lies in a neighbourhood of 0 whose size is not uniform. On that neighbourhood the invariance of the action and the factorisation in 3 provide gauge directions X_χ with $(F'(\mathbf{H}), X_\chi) = 0$ and $K_{\mathbf{H}}X_\chi = 0$, and the $RD^{\mathsf{T}}X_\chi$ fill out the range of R. Therefore $\mathsf{z} = 0$ on this neighbourhood, and by the identity theorem also on the whole polydisc, so the neighbourhood enters no constant. The factorisation of the averages holds only up to site factors and, at rows that leave their layer, up to seam factors; this is the reason for the row datum.

3.12.1.10. The reduced response over a real critical reference. For the remainder of this argument put $s = \mathbf{1}$ and take the real regular pair (U_0, J) with $J = J_{\Omega_0}(U_0) = \hat{Q}^{\mathsf{T}}\lambda_0$. Use the closed row set: the outer end of an outward row has its full block of interior rows at the preceding level. The transposes use the fine weighted pairing and the counting row pairing. Put $\mathcal{Q}(T) = \hat{Q}T + \mathcal{C}(T)$ and $K_T = \hat{Q} + \mathcal{C}'_T$. If $F(T)$ is the action of the inner plaquettes at $e^{i\xi T}U_0$, then $F'(T) = J + \Delta T + \mathcal{V}(T)$, with the same Hessian and current convention as in the propagators. Here $\mathcal{V}$ is the nonlinear force remainder.

Define $\mathsf{z} = F'(T) - K_T^{\mathsf{T}}\lambda$ and $\phi = RD^{\mathsf{T}}T$. Write r_1, r_2 for the left sides minus the right sides of the two reduced equations, evaluated at $B' = \mathcal{Q}(T)$. Their exact expressions are

$$r_1 = \widetilde{G}(\mathsf{z} + D\phi), \qquad r_2 = \mathsf{L}G'\phi - H^{\mathsf{T}}(\mathsf{z} + D\phi). \tag{3.cdxlvi}$$

Indeed $\widetilde{G}\hat{Q}^{\mathsf{T}} = 0$ and $T = H_0\hat{Q}T + \widetilde{G}G_0^{-1}T$, where $G_0^{-1}T = \Delta T + D\phi + \hat{Q}^{\mathsf{T}}\mathsf{a}\hat{Q}T$. This gives r_1. For r_2 use $\hat{Q}H = 1$ and

$$H^{\mathsf{T}}\Delta T = (\hat{S} - \mathsf{a})\hat{Q}T + H^{\mathsf{T}}\Delta'_\pi T - H^{\mathsf{T}}D\phi.$$

To identify the middle term, set $\Upsilon = G'RD^{\mathsf{T}}$. The quadratic form of Δ'_π has terms bilinear in $(A, \Upsilon A)$ or quadratic in ΥA. In the polarization at $(H\beta, T)$ only the term with $(H\beta, \Upsilon T)$ remains, because $RD^{\mathsf{T}}H = 0$ at the real current. If $\mathfrak{b}(v, A)$ denotes this bilinear part with the arguments in that order, define $\langle \beta, \mathsf{L}v\rangle = \mathfrak{b}(v, H\beta)$. This gives $H^{\mathsf{T}}\Delta'_\pi T = \mathsf{L}G'\phi$. Also $H^{\mathsf{T}}J = \lambda_0$.

For the reduced solution the first equation gives $\hat{Q}T = B' - \mathcal{C}(T)$. Since the kernel of $\widetilde{G}$ is $\operatorname{ran}\hat{Q}^{\mathsf{T}}$, the two vanishing defects in eq. 3.cdxlvi give

$$F'(T) + D\phi = K_T^{\mathsf{T}}\lambda + \hat{Q}^{\mathsf{T}}\mathsf{L}G'\phi. \tag{3.cdxlvii}$$

For real B' near zero, let $X_\chi(T)$ be the variation of T by the real gauge $e^{i\tau\chi}$. The accompanying node variation is $\varrho(T;\chi)$, obtained by undoing the averaging factors of the two fields. The straightened constraint varies by $\ell^\natural_{B'}[\varrho(T;\chi)]$, the infinitesimal row action with its outward endpoint transport. At $T=0$ the maps X_χ and ϱ are $-D\chi$ and $Q'\chi$. The map

$$\chi \longmapsto (\varrho(T;\chi), RD^{\mathsf T} X_\chi(T))$$

is invertible near zero. At zero this follows from the onto site average and the inverse of $RD^{\mathsf T}D$ on $\ker Q'$. The finite-dimensional inverse persists on an open real neighborhood. Choose χ with first component zero and second component ϕ. Then $K_T X_\chi = 0$ and $(F'(T), X_\chi)_\xi = 0$ by gauge invariance. Pairing eq. 3.cdxlvii with X_χ gives

$$(\phi,\phi)_\xi = -\langle \mathsf{L} G'\phi, \mathcal{C}'_T X_\chi\rangle.$$

At fixed finite volume the right side is bounded by $C\|T\|(\phi,\phi)_\xi$, since $\mathcal{C}'_0 = 0$ and $\|X_\chi\| \le C\|\phi\|$. Thus $\phi = 0$ near the real origin. Holomorphy extends this identity to the entire reduced datum ball, so $F'(T) = K_T^{\mathsf T}\lambda$ there. This argument uses no uniform size for the real neighborhood. In the reverse direction a solution of the three equations has both defects zero. A sufficiently small T puts its multiplier in the same uniqueness ball: the second equation bounds $|\lambda-\lambda_0|$ by $C\hat{\mathsf{b}}\|T\|(1+|\lambda_0|+|\lambda-\lambda_0|)$, where the last term is absorbed. This gives the comparison on a common smaller uniform radius.

To return to physical data, the factorization of $\mathcal{Q}$ gives the plain datum $(e^{iB'^\natural}V_0)^{v[T]}$, where $V_0 = M_{\mathbf{B}}(U_0)$. At an outward row the right endpoint factor is $\theta_z(B')^{-1}$ if the outer endpoint is $z = c_+$, and the left factor is $\theta_z(B')$ if $z = c_-$. This is the mean formed from the interior rows in the block at z. Use the inverse conversion of the physical logarithm to B' on the common smaller chart. To check criticality on the physical fibre, pair λ with $\ell^\natural_{B'}[\sigma]$. At small real B' the map of gauge variations above supplies every node value σ. The pairing is zero since it equals $(F'(T), X_\chi)_\xi$. Holomorphy extends its vanishing to the datum ball. On a differentiable curve of real fields with fixed physical datum, the derivative of the reduced datum is a node variation of this form, obtained by differentiating the factors $v[T]$ with the physical datum held fixed. The derivative of the action on the curve therefore vanishes. After undoing $v[T]$ at the nodes this gives a critical field at the specified physical datum. Where the response lies in the regularity class, uniqueness of the real orbit identifies it with the continued regular background in the normalized slice. For the decoupled response at an independent current we use the reduced system and its analytic estimates. The identification with a physical critical orbit uses the real critical reference and the closed rows as above.

Constants are chosen in the following order, each large or small enough relative to those chosen before it. First comes L, and with it, as a single group, the numbers determined by L alone: the flat constants and the flat rate, the lower thresholds that the flat case imposes on M_1 and on R_1M_1, the radius of the local perturbation theory, the rates δ_0, δ_*, δ', and the ceiling on $\alpha_0, \alpha_1, \gamma_0$. Next come M_1 and R_1M_1; the ceiling does not depend on them, and they do not depend on the ceiling. Then comes $\bar\kappa$. It does not depend on M_1, R_1, and they do not depend on it, so it may just as well be chosen before them. After that come $\ell_{\mathsf P}$, the radius in B', and the second ceiling on γ_0, which concerns eq. 3.cdxliv. The final choices are M and the field strengths of the construction, taken so small that $O(LM)\alpha_0$ is below the ceiling. Two numbers are exceptions: among the bounds of statement 7, the one bound that is used for convergence alone; and, in mechanism (i), the entrywise bound by which the family is normal. Neither of them appears in a localised bound, a rate, a weight or a sum of weights. With the exception of these two, every constant, rate and ceiling, and the radius in B', is independent of k, K, ℓ, of the sequence (in particular of Ω_0), of Y and of the pair, and also of M_1, R_1 once these are large enough. The sizes $\ell_{\mathsf P}$ and M, and the side of the coarse cubes of 6, are each at least a multiple of M_1. The field strengths are small compared with

$(LM)^{-1}$. Where statement 8 makes the identification with the response of § 3.11, the datum of the background is small compared with M_1^{-1}.

3.12.2. *The additional reduced rows at a retained inward seam.* This paragraph supplies the linear inverse for the optional convention that retains inward bonds. The determining sets in eq. 3.cccvi still use only interior and outward rows. At each additional seam node choose one retained inward bond as its tying bond, oriented towards that node. The tying equation fixes its node factor. The remaining independent equations are the quotients of the other retained inward bonds by this tying bond. Call these the edge rows. If only one inward bond is retained at a node, it adds a node factor but no edge row.

Write K_{p} for the differential of the proper reduced rows and K_{e} for that of these edge rows, all in raw dimensionless logarithmic coordinates. Both row spaces have counting measure. In this paragraph K^* denotes the transpose for that counting measure and the physical fine pairing $\eta^4 \sum_b \langle A(b), A'(b)\rangle$. In particular it is not the physical top transpose with its row-length weight omitted. Let $a_q(A)$ be the iterated reduced logarithm at level q in the reference chart, and extend it to reversed bonds by $\widehat{a}_q(\bar{b}) = -\operatorname{Ad}(W_q(b)^{-1})a_q(b)$, where $W_q = \mathcal{M}^q U_0$. For an edge $e = (b_i(x), b_1(x))$, put $V_0(e) = W_q(b_i(x))W_q(b_1(x))^{-1}$. Differentiation of its quotient gives the exact formula

$$K_{\mathrm{e}} A(e) = \widehat{\dot{a}_q[A]}(b_i(x)) - \operatorname{Ad}(V_0(e))\widehat{\dot{a}_q[A]}(b_1(x)). \tag{3.cdxlviii}$$

Put $\mathfrak{M} = M_1$. The auxiliary peeled sequence has $\Omega'_0 = \Omega_0$; for $q \geq 1$, Ω'_q is the union of the aligned $\mathfrak{M}\xi_q$-cubes in Ω_q whose sup distance from Ω_q^c is at least $\mathfrak{M}\xi_q$. Its determining set contains only proper rows. A full torus is unchanged, and empty members are allowed. Use the same large separation floors for both sequences. The elementary consequences are

$$\Omega_{q+1} \subset \Omega'_q \subset \Omega_q, \qquad \xi'(z) \leq \xi(z) \leq L\xi'(z).$$

Indeed a discarded cube touches the old boundary and has width $\mathfrak{M}\xi_q$. Separation puts the next member beyond this discarded strip; erosion of that next member adds $\mathfrak{M}\xi_{q+1}$ to its distance from the old boundary, so the original separation is preserved. A site loses at most one level. Each peeled pillar is contained in its old pillar.

Lemma 3.36 (Uniform edge Schur inverse)**.** Let U_0 be real. Assume that $(U_0, J(U_0))$ is a real regular pair on the original sequence and that $(U_0, 0)$ is a real regular pair on the peeled sequence, in the preceding definition with $A' = 0$ and $\alpha_0 = \alpha_1 = \gamma_0 = \alpha$. Require the local-gauge bounds on every complete box used in the proper-row inverse estimates, with a common sufficiently small radius α. These are two primitive admissions: the original pair uses the actual physical current, and the peeled pair uses independent current zero. The original physical-current premise ensures positivity of the zero-independent-current completed Wilson form. Both admissions hold at the flat reference. The separation floor may be increased by a fixed number depending only on the inverse constants and L.

Let $G_{\mathrm{p}} = T^{-1}$ be the original positive completed Wilson inverse, in raw row normalization, and set

$$\begin{aligned} T &= \Delta_{U_0}^{\mathrm{W}} + DRD^\sharp + K_{\mathrm{p}}^* K_{\mathrm{p}}, \\ H_{\mathrm{p}} &= G_{\mathrm{p}} K_{\mathrm{p}}^* (K_{\mathrm{p}} G_{\mathrm{p}} K_{\mathrm{p}}^*)^{-1}, \qquad C_{\mathrm{p}} = G_{\mathrm{p}} - H_{\mathrm{p}} K_{\mathrm{p}} G_{\mathrm{p}}, \\ M_{\mathrm{e}} &= K_{\mathrm{e}} C_{\mathrm{p}} K_{\mathrm{e}}^*. \end{aligned}$$

There are $c, C, s > 0$, independent of depth, mesh, volume and the subset of inward bonds retained, such that

$$cI \leq M_{\mathrm{e}} \leq CI, \qquad |M_{\mathrm{e}}^{-1}(e, e')| \leq Ce^{-sd(y_e, y_{e'})}.$$

Here y_e is the outer top of the nontying bond of e, and d is the original weighted distance. If there are no edge rows, the edge space is zero and the following corrections are zero. Otherwise define

$$\begin{aligned} H_{\mathrm{all}}(b_{\mathrm{p}}, b_{\mathrm{e}}) &= H_{\mathrm{p}} b_{\mathrm{p}} + C_{\mathrm{p}} K_{\mathrm{e}}^{*} M_{\mathrm{e}}^{-1} (b_{\mathrm{e}} - K_{\mathrm{e}} H_{\mathrm{p}} b_{\mathrm{p}}), \\ C_{\mathrm{all}} &= C_{\mathrm{p}} - C_{\mathrm{p}} K_{\mathrm{e}}^{*} M_{\mathrm{e}}^{-1} K_{\mathrm{e}} C_{\mathrm{p}}. \end{aligned} \tag{3.cdxlix}$$

With $K = (K_{\mathrm{p}}, K_{\mathrm{e}})$ and $\mathcal{X} = \Delta_{U_0}^{\mathrm{W}} + DRD^{\sharp}$, these maps satisfy

$$K H_{\mathrm{all}} = 1, \quad K C_{\mathrm{all}} = 0, \quad C_{\mathrm{all}} \mathcal{X} = 1 - H_{\mathrm{all}} K, \quad \ker C_{\mathrm{all}} = \operatorname{ran} K^{*}.$$

They have the same raw-datum-to-field and force-to-field bounds as the proper package, with a smaller positive exponential rate. More precisely, use the cell field norm $\|A\|_y = \max\{\xi_y |A|_y, \xi_y^2 |\nabla_{U_0} A|_y\}$ and force norm $|f|_y^{\sharp} = \xi_y^3 |f|_y$. Here the maxima are over bonds or difference sites assigned to the pillar of y, and $|b|_y$ is the maximum of the Euclidean raw row norms at that top, taken as zero if there is no row. Complete fixed enlargements in local estimates have bounded overlap. Then

$$\begin{aligned} \|H_{\mathrm{all}} b\|_y &\le C \sum_z e^{-s d(y,z)} |b|_z, \\ \|C_{\mathrm{all}} f\|_y &\le C \sum_z e^{-s d(y,z)} |f|_z^{\sharp}. \end{aligned}$$

The localized gradient Hölder bounds hold for each fixed $0 < \beta < 1$, with constants additionally depending on β. This is a linear package for the completed Wilson Hessian; it is not an identification with a current-corrected background response. ∎

Proof. Let H_{p}' be the proper raw-row lift on the peeled sequence. Use the Euclidean Lie algebra norm induced by the real trace pairing and put $\|A\|_{\mathcal{E}}^2 = \sum_y \|A\|_y^2$. The proper exponential estimates and the bounded weighted metric volume imply

$$\|H_{\mathrm{p}} b\|_{\mathcal{E}} \le C \|b\|_2, \quad \|H_{\mathrm{p}}' b'\|_{\mathcal{E}'} \le C \|b'\|_2, \quad \|A\|_{\mathcal{E}} \le L^2 \|A\|_{\mathcal{E}'}.$$

The last inequality follows by replacing the maximum on an old pillar by the sum of squares on its disjoint peeled subpillars. All constants in this proof are fixed before the depth is chosen.

We construct a test for each edge row. Write $b_i(x) = (y_i, x)$, with $x = y_i + \sigma \xi_q e_\mu$, and let $B(X)$ be the level-$(q+1)$ block containing x. The bonds

$$f_0 = (y_i, x), \quad f_1 = (x, x + \sigma \xi_q e_\mu), \quad f_2 = (x + \sigma \xi_q e_\mu, x + 2\sigma \xi_q e_\mu)$$

are proper interior rows of the peeled sequence. In fact this entire block is in the strip removed from Ω_{q+1}, while it and y_i lie in Ω_q'. On their common line transport a vector v from y_i to the initial point of each bond, and assign the oriented values $(1, -2, 1)v$ in these transported frames. On a reversed bond use the covariant minus sign. Denote this pattern by $J_e v$. Different edge rows have disjoint pattern supports: inward bonds at a common seam node have distinct directions, and parallel inward bonds on the same block line are separated by at least $L - 1$ bonds. Since $L \ge 13$, the three-bond patterns cannot meet. None uses a tying bond. Thus

$$\|Jb\|_2^2 = 6 \|b\|_2^2, \qquad Jb = \sum_e J_e b(e).$$

The cancellation at the next scale is essential. Along a line, the L-step paths starting in the first endpoint block meet the pattern in the sets $\{0\}, \{0,1\}, \{0,1,2\}$. Their sums are $1, -1, 0$, the first two occurring once. For the next coarse bond the nonzero sums are $-1, 1$, again once each. At a flat reference each pair cancels exactly in the L^{-4}-normalized straight-path average. At a curved reference the two transports differ by a loop in the same double block. The loop comparison bounds that difference by $C_L \alpha$. Let $\mathbf{Q}_q = D_a F_{W_q}(0)$, where F_W is the straightened one-step logarithm of eq. 3.dxc. It is the derivative in the chain rule for the reduced logarithms a_{q+1}. At a flat reference, differentiating the two endpoint factors in that formula cancels the internal contour derivatives

of the plain mean. The remaining map is the covariant straight-path average. For curvature $O(\alpha)$, differentiate the same endpoint factors, local logarithms and defining mean equation on the complete double block. In its real comb gauge their coefficients and the inverse derivative of the mean equation differ from their flat values by $O_L(\alpha)$, using the logarithm series on its fixed smaller ball. Unitary covariance restores the original frames. Thus $\mathbf{Q}_q$ differs from the covariant straight-path map by $C_L\alpha$ in its fixed finite row and column sums. The one-step row has at most $8L^4$ input bonds, and each input belongs to at most eight double blocks. These are one-step counts, with no dependence on q.

Set $A_b = H'_{\mathrm{p}} Jb$. The iterated reduced logarithms for the two sequences agree on every common complete read, by induction in the averaging equation. Hence

$$K_{\mathrm{e}} A_b = b, \qquad \|K_{\mathrm{p}} A_b\|_2 \le C_L\alpha \|b\|_2, \qquad \|A_b\|_{\mathcal{E}} \le C\|b\|_2.$$

For the first identity, eq. 3.cdxlviii reads the assigned value on b_i and zero on the tying bond. An old proper row shared by the peeled sequence reads zero. Every other old proper row has its full predecessor double block among the peeled interior rows, so the preceding cancellation and the chain rule give the second inequality. Bounded double-block overlap gives both Schur sums needed for that ℓ^2 estimate.

Correct into the proper kernel by $A = A_b - H_{\mathrm{p}} K_{\mathrm{p}} A_b$. The actual local row bounds give

$$K_{\mathrm{p}} A = 0, \qquad \|A\|_{\mathcal{E}} \le C_A\|b\|_2, \qquad \|K_{\mathrm{e}} A - b\|_2 \le C_{\mathrm{e}}\alpha\|b\|_2.$$

Choose $C_{\mathrm{e}}\alpha \le 1/2$. On this proper kernel,

$$(A, TA)_\eta \le C_T \|A\|_{\mathcal{E}}^2.$$

To check this bound, the row penalty vanishes, R is an orthogonal projection, the curl and divergence use finitely many neighboring differences, and the Wilson-minus-curl remainder is bounded by $C_L\alpha\xi_y^{-2}|A|$. The physical number of bonds in a pillar is $O(\xi_y^4)$; this cancels the four powers in the squared first-difference bound. Neighboring layers differ by at most L.

Since T is positive, its constrained inverse satisfies

$$(f, C_{\mathrm{p}} f)_\eta = \sup_{K_{\mathrm{p}} u = 0} \{2(f,u)_\eta - (u, Tu)_\eta\}.$$

Put $f = K_{\mathrm{e}}^* b$, test with $u = tA$, and use $\langle b, K_{\mathrm{e}} A\rangle \ge \|b\|_2^2/2$. Taking $t = (2C_T C_A^2)^{-1}$ proves

$$\langle b, M_{\mathrm{e}} b\rangle \ge (4C_T C_A^2)^{-1} \|b\|_2^2.$$

The upper bound and an exponentially decreasing raw kernel for M_{e} follow by composing the proper constrained inverse with the two actual local edge maps. Their counting transposes have the physical fine density; no mesh-counting factor is introduced.

For completeness, fix an edge e_0 and conjugate this raw matrix by multiplication by $\exp(s \min\{d(y_e, y_{e_0}), N\})$. The distance is one-Lipschitz in itself. Let $a > 0$ be the kernel rate already obtained. For $0 < s \le a/4$, the exponential kernel bound and the metric sum at the fixed rate $a/2$ bound the row and column sums of the conjugation error by Cs, uniformly in N. The constant C is fixed once the separation exceeds the floor for this rate. Choose $s \le \min\{a/4, c/(2C)\}$. The conjugated inverse has norm at most $2/c$. Apply it to a unit coordinate at e_0 and let N increase. This proves the displayed decay of M_{e}^{-1}. Now increase the separation floor also to bound the metric exponential sums at this smaller rate, using the same constants already fixed above. Only a fixed number of edge species occur at a top.

Direct multiplication in eq. 3.cdxlix proves the first three identities. Positivity on $\ker K$ identifies C_{all} as its constrained inverse, giving $\ker C_{\mathrm{all}} = \operatorname{ran} K^*$. Composition of the exponential kernels proves the two cell bounds after one further fixed decrease of the rate. For the localized gradient Hölder estimate, write each correction as $C_{\mathrm{p}} f$ and apply the proper estimate to this force; its bound follows from the same convolution. This uses no derivative of a row datum. All maps are real at

real coefficients. On a common smaller admitted complex tube their bilinear matrix formulas are holomorphic; the uniform weighted perturbation estimates for the proper inverse and local rows preserve the edge gap by Neumann series. At the flat reference the loop error is zero and both primitive admissions are automatic. The resulting flat package therefore includes any retained inward edge rows in the local regularity argument. □

3.12.2.1. The derivative at a retained-inward critical background. Suppose additionally that U_0 is a real critical reference for these reduced rows. Write $q(A)$ for their raw nonlinear logarithm, so $q'(0) = K$, and write F for the inner Wilson action in this chart. In this paragraph put $H = H_{\mathrm{all}}$, $C = C_{\mathrm{all}}$ and $Z = DRD^\sharp$. The nonlinear system to be solved is

$$F'(A) + ZA = q'(A)^*\omega, \qquad q(A) = b.$$

The gauge term is retained until the physical-fibre argument. Since $KH = 1$, the multiplier at the reference is unique and

$$F'(0) = K^*\omega_0, \qquad \omega_0 = H^*F'(0), \qquad |\omega_0|_\infty \le C_0\alpha.$$

Here and below the constants are uniform over the admitted data. The last bound uses the physical current admission and the absolute column estimate: in a cell the pairing has scale $\xi_y^{-1}\xi_y^{-3}\xi_y^4 = 1$.

Define B by

$$(v, Bu)_\eta = \langle \omega_0, q''(0)[u, v]\rangle.$$

The local two-leg bounds for the actual logarithms, with both leg sums and the fine density, give $|Bu|_{(-3)} \le C_1|\omega_0|_\infty\|u\|_1$. Differentiating both factors of an edge quotient gives the same bound; only a fixed number of inward bonds share a tying bond. The differential of the system at $(0, \omega_0)$ is

$$\mathcal{L}(u, \nu) = ((\mathcal{X} - B)u - K^*\nu, Ku).$$

For a force f and raw datum b, its inverse has field part

$$u = (1 - CB)^{-1}(Cf + Hb), \qquad \nu = H^*((\mathcal{X} - B)u - f).$$

Applying C proves the first formula. Conversely it gives $Ku = b$, and the first-row residual belongs to $\ker C = \mathrm{ran}\, K^*$; applying H^* then gives the unique multiplier in the second formula. Decrease α so that $\|CB\|_{1\to1} \le 1/2$, also in a fixed smaller exponentially weighted norm.

The weak multiplier estimate uses the unchanged site maps of the proper package. The proper exponential bounds for $P = 1 - R$ on site functions and for $DPD^\sharp$ remain valid. Write d for the covariant bond-to-plaquette curl, reserving D for the site-to-bond derivative. The divergence maps a field controlled in $\|\cdot\|_1$ to a site function with its two powers of local length. In $\langle e, H^*\mathrm{d}^\sharp \mathrm{d}u\rangle = (\mathrm{d}He, \mathrm{d}u)_\eta$ move one difference to each field; for the gauge term use $\langle e, H^*DRD^\sharp u\rangle = (D^\sharp He, RD^\sharp u)_\eta$ and $R = 1 - P$. The cell counts cancel the four inverse powers of length. The Wilson-minus-curl term uses its local force bound. Together with the column bound for H^*f, this proves

$$\max\{\|u\|_1, |\nu|_\infty\} \le C_I \max\{|f|_{(-3)}, |b|_\infty\}.$$

The same argument with the weight on the changed factor proves the weighted version, with a fixed smaller rate.

Here is an explicit contraction using this inverse. Write $\omega = \omega_0 + \nu$, $X = (A, \nu)$, $\|X\| = \max\{\|A\|_1, |\nu|_\infty\}$, and put

$$N_1(X) = F'(A) + ZA - q'(A)^*(\omega_0 + \nu) - [(\mathcal{X} - B)A - K^*\nu],$$
$$N_2(X) = q(A) - KA.$$

Both remainders and their first derivatives vanish at zero. The local Taylor and transpose bounds give, on $\|X_i\| \le r$ in the common chart,

$$\max\{|N_1(X_2) - N_1(X_1)|_{(-3)}, |N_2(X_2) - N_2(X_1)|_\infty\} \le C_N r\|X_2 - X_1\|.$$

The mixed term is $-(q'(A)-K)^*\nu$; its difference has one small factor and one difference. The part paired with ω_0 uses the second difference of q'. Choose r below the common row and action chart radii and so that $C_I C_N r \le 1/4$. For $|b|_\infty < r/(2C_I)$, the map

$$X \longmapsto \mathcal{L}^{-1}(-N_1(X), b - N_2(X))$$

maps the closed r-ball into itself and contracts by at most $1/4$. Iteration gives its unique solution there, holomorphic on the open datum ball and real for real data. Weighted differences have the same bound after absorption. All constants and radii were fixed before the depth and the retained subset were chosen.

Physical identification uses the same original and peeled admissions, the actual node and seam datum conversion, and the common smaller chart described above. The real gauge-variation argument removes ZA near the origin and identifies criticality on the physical fibre; its analytic identities extend over the connected datum ball. The compared physical branch must lie in this response uniqueness ball and in the admitted regularity class. Uniqueness of its real critical orbit then gives the normalized branch identification. Thus the optional continuation uses the corrected derivative $\mathcal{L}$ and the stated admissions throughout.

3.13. **Layered precision and current terms.** The precision of a constrained fluctuation is obtained only after three different operations. One forms an operator on the fine field, reads its induced action on the coarse datum, and then restricts that coarse action to the next actual fibre. Gauge and current terms enter at all three places. A positive matrix at one of them does not identify the others.

We retain this order. The fine site and field operators, their row lift and their constraint-curvature terms are typed separately. Their complete physical match is an identity of projected forms, with the actual soft gauge. The alternative explicitly defined relative-current first-step extension gives another coherent starting point. It supplies its stated first-step estimates; using that definition at later levels requires its own consistency and closure.

The reductions below use the domain and background sections. In particular, all transposes are bilinear weighted transposes on complex domains, while real lower bounds use Hermitian quadratic forms. The primitive domains are specified before the unknown critical field. No arbitrary independent current is replaced by the physical action current.

3.13.1. *Cell norms retain the ultraviolet scale.* Work on a finite compatible torus, with a separated aligned sequence and the spacings of § 3.11. Let $\Delta(y)$ be the pillar of a top y, of side ξ_y. Outside the action region one can add bare cells and extend the local scale to η; a compressed operator is zero on that added space. The point scale $\xi(x)$ is the scale of its pillar. Use the exact path metric

$$d(x,z) = \inf_{\gamma: x \to z} \int_\gamma \frac{ds_\infty}{\xi(s)}. \tag{3.cdl}$$

The half-open convention only assigns a measure-zero cell boundary; paths and their infima can be approximated on either side. The corresponding lattice metric is comparable after the fixed collar constants have been inserted.

For each field species s, declare a positive amplitude $a_s(y)$ and use $\|f\|_{s,y} = a_s(y)^{-1}\|f\|_{\infty,\Delta(y)}$. For example a physical potential has amplitude ξ_y^{-1}, a physical force density ξ_y^{-3}, and a raw datum logarithm amplitude one. The site gauge parameter also has amplitude one. A raw multiplier uses counting measure and amplitude one; its per-length density has amplitude ξ_y^{-3}. The conversion is the weighted dual conversion of the domain section.

If P_y is restriction to a cell, define the normalized cell entry of an operator $K : t \to s$ by

$$k_{st}(y,z) = \|P_y K P_z\|_{\ell^\infty(t,z)\to\ell^\infty(s,y)}. \tag{3.cdli}$$

All physical cell measures in the action of K are included before taking this norm. It is a block operator norm, not a pointwise bound on the continuum kernel. A Green kernel with its fine

Coulomb singularity can have a bounded cell norm. Replacing it by a pointwise kernel bound independent of η would be a stronger and generally false assertion.

For compatible species, the cell partition and triangle inequality give

$$(k\ell)_{su}(y,z) \le \sum_w k_{st}(y,w)\ell_{tu}(w,z). \tag{3.cdlii}$$

This follows first for finite lattices by inserting $\sum_w P_w = 1$. In an infinite-volume application the same formula holds when the row sums below give absolute convergence. Thus a two-sided exponential cell norm is an algebra norm, without discarding the fine singularity inside a block.

3.13.2. *The growth of a ball in the layer metric.* The layer metric is not assumed to have polynomial volume growth. A small-cell region can contain many more pillars than a nearby large-cell region. Separation controls that growth quantitatively.

Suppose x has layer index j and z has index i. Any path between them crosses each complete intervening layer. For a layer r, a crossing from Ω_{r+1} to Ω_r^c has an arc whose interior lies in $\Omega_r \setminus \Omega_{r+1}$. The separation makes its physical length greater than $\rho\xi_r$. On that arc the metric density is ξ_r^{-1}, so its length in eq. 3.cdl is at least ρ. Arcs belonging to distinct layers have disjoint interiors. Consequently

$$d(x,z) \ge \rho(|i-j|-1)_+. \tag{3.cdliii}$$

One layer is lost because an endpoint can already lie next to its neighboring layer. The same proof applies to an intermediate point of a path, so a path of metric length at most $R+1$ starting in layer j never uses a scale larger than

$$\xi_{\max} \le L\xi_j L^{(R+1)/\rho}. \tag{3.cdliv}$$

If the top index would exceed the depth, this is simply an overestimate.

Its physical length is at most $(R+1)\xi_{\max}$. Every pillar center in the metric R-ball therefore has a lift in the physical sup-norm ball of radius

$$H_R = L(R+1)\xi_j L^{(R+1)/\rho}. \tag{3.cdlv}$$

For each fixed level i the centers lie on its grid of spacing ξ_i. The number of grid points in that physical ball is at most $C(1+H_R/\xi_i)^4$. eq. 3.cdliii restricts the possible indices to $|i-j| \le 1+(R+1)/\rho$. Using the smallest such ξ_i in the grid count and then summing the possible levels gives

$$\#\{z : d(y,z) \le R\} \le C_L(1+R)^5 \exp\left(\frac{8\log L}{\rho}R\right). \tag{3.cdlvi}$$

Indeed $H_R/\xi_i \le C_L(R+1)L^{2(R+1)/\rho}$; its fourth power gives the exponential in eq. 3.cdlvi, and the number of levels costs one more polynomial power. Since $\rho \ge 1$, the harmless factors $L^{8/\rho}$ and the bound on the number of levels can be included in C_L uniformly in ρ.

The argument works on a torus without a nonwrapping-ball assumption. For every reached center choose the lift reached by its path. Distinct periodic centers give distinct lifted grid points. Counting all grid points in the cover ball is therefore an upper bound, even if that ball crosses many fundamental domains. Restricting the allowed regions only decreases the number counted.

An entry bound of the form

$$k_{st}(y,z) \le C\left(\frac{\xi_y}{\xi_z}\right)^p e^{-md(y,z)} \tag{3.cdlvii}$$

now implies a weighted row and column sum, provided the rate pays for scale growth. By eq. 3.cdliii the scale ratio on an annulus of metric radius R is at most $L^{|p|}e^{|p|(\log L)(R+1)/\rho}$. Break the space into unit annuli and use eq. 3.cdlvi. Thus

$$\sup_y \sum_z e^{\delta d(y,z)}k_{st}(y,z) + \sup_z \sum_y e^{\delta d(y,z)}k_{st}(y,z) \le C' \tag{3.cdlviii}$$

whenever

$$m-\delta>\frac{(8+|p|)\log L}{\rho}. \tag{3.cdlix}$$

For the column sum use the same ball argument centered at z; the absolute scale exponent is unchanged. The convergent scalar sum is bounded by $C\sum_{n\ge0}(n+2)^5e^{-[m-\delta-(8+|p|)\log L/\rho]n}$. This proves the claim with its precise uniformity.

If the amplitude normalization has already removed the scale ratio, take $p=0$. For several species use the largest of their finitely many exponents. An exponential bound with an arbitrarily small m does not automatically imply eq. 3.cdlviii. The geometry and rate must satisfy eq. 3.cdlix, or a stronger actual row estimate must be proved directly. This is a numerical condition in the order of choice of separation and kernel rates.

3.13.3. *The exponential-chart Ward identity.* We derive a complete current-linear normal form. This also states exactly what a separately specified current completion would have to match. For this calculation use raw bond logarithms and counting pairings. Equivalently use physical potentials and their dual fine cell weights, with the conversions stated below.

Let $F_U(a)=A_{\Omega_0}(e^{ia}U)$, with all plaquettes of this action inner to its gauge-invariant active graph, and put $j_U=DF_U(0)$, $\mathscr{D}_U=D^2F_U(0)$. For a site function λ, define the raw infinitesimal gauge generator by

$$(D_U\lambda)(b)=\lambda(b_-)-\mathrm{Ad}(U_b)\lambda(b_+). \tag{3.cdlx}$$

This equation fixes the sign of D_U in this section. A convention with the opposite covariant difference changes both occurrences consistently. The site Laplacian $D_U^\sharp D_U$ is unchanged.

At a nonzero reference logarithm a the gauge generator in that same logarithmic coordinate is

$$\mathfrak{d}_U(a)\lambda=\mathcal{E}_a^{-1}\{\lambda(b_-)-\mathrm{Ad}(e^{ia_b}U_b)\lambda(b_+)\},\qquad \mathcal{E}_ah=\int_0^1\mathrm{Ad}(e^{ita})h\,dt. \tag{3.cdlxi}$$

The equality follows by differentiating $e^{it\lambda_-}e^{ia_b}U_be^{-it\lambda_+}$ and multiplying by its inverse on the right. The factor $1/i$ in the right-logarithmic derivative is included in $\mathcal{E}_a$. The integral formula follows by differentiating the exponential series, or by integrating its constant-coefficient differential equation. It is holomorphic on the admitted logarithm chart.

The derivative at zero is

$$D\mathfrak{d}_U(0)[v]\lambda=-\frac{i}{2}[v_b,\lambda(b_-)+\mathrm{Ad}(U_b)\lambda(b_+)]. \tag{3.cdlxii}$$

Indeed $D\mathcal{E}_0[v]h=(i/2)[v,h]$, while differentiating the endpoint transport contributes $-i[v,\mathrm{Ad}(U_b)\lambda_+]$. Subtracting the derivative of $\mathcal{E}_a$ acting on $\lambda_--\mathrm{Ad}(U_b)\lambda_+$ gives exactly eq. 3.cdlxii, with a plus between the two endpoint site values inside its commutator.

Gauge invariance implies $DF_U(a)[\mathfrak{d}_U(a)\lambda]=0$. Differentiating once at zero proves

$$D_U^\sharp j_U=0,\qquad \mathscr{D}_UD_U=\mathcal{W}_U[j_U], \tag{3.cdlxiii}$$

where the explicit linear-current operator is

$$(\mathcal{W}_U[J]\lambda)(b)=\frac{i}{2}[\lambda(b_-)+\mathrm{Ad}(U_b)\lambda(b_+),J_b]. \tag{3.cdlxiv}$$

To verify its order and sign, cyclicity gives $\langle J,(i/2)[v,S]\rangle=\langle v,(i/2)[S,J]\rangle$. Together with eq. 3.cdlxii this is the differentiated invariance identity. The formula uses the actual action current. No constrained criticality has been assumed.

The operator $\mathcal{W}_U[J]$ is defined for an independent J by eq. 3.cdlxiv. It is local and linear in that current. It does not differentiate J. For an arbitrary independent current, $D_U^\sharp J$ need not vanish, and $D_U^\sharp\mathcal{W}_U[J]$ need not be symmetric. Those physical identities cannot be inserted in an off-current definition.

If the spacing is η, the physical generator is $D_U^{\mathrm{phys}} = \eta^{-1} D_U$. The raw and physical currents satisfy $j_{\mathrm{raw}} = \eta^3 J_{\mathrm{phys}}$, and the physical Hessian operator is $\eta^{-2}\mathscr{D}_{\mathrm{raw}}$ with respect to the η^4 pairing. Thus

$$\mathscr{D}_{\mathrm{phys}} D_U^{\mathrm{phys}} = \mathcal{W}_U[J_{\mathrm{phys}}]. \tag{3.cdlxv}$$

There is no extra inverse power of η in $\mathcal{W}_U[J_{\mathrm{phys}}]$. Both sides have the dimension of a force density applied to a dimensionless site gauge.

For the operator formulas from this point onward restore the physical cell pairings and write D_U for D_U^{phys}. Write j_U for the physical current density and $\mathscr{D}_U$ for the physical Hessian operator. Their conversion is exactly the one in eq. 3.cdlxv. The formula in eq. 3.cdlxiv is unchanged. In bare units $\eta = 1$ all these conventions coincide.

3.13.4. *An explicit symmetric current correction.* Use the site operators of the background section and set

$$\Lambda = G' R D_U^\sharp, \qquad \mathsf{S} = 1 - D_U \Lambda. \tag{3.cdlxvi}$$

On their admitted domain the proved projector identities give $\mathsf{S} D_U G' R = 0$. Define the following specific symmetric current-linear form:

$$\begin{aligned} \mathcal{B}_U[J] = {} & \mathcal{W}_U[J]\Lambda + \Lambda^\sharp \mathcal{W}_U[J]^\sharp \\ & - \tfrac{1}{2}\Lambda^\sharp \{ D_U^\sharp \mathcal{W}_U[J] + \mathcal{W}_U[J]^\sharp D_U \} \Lambda. \end{aligned} \tag{3.cdlxvii}$$

This is a definition on the whole independent-current domain. Its symmetry is visible term by term. For the physical current, eq. 3.cdlxiii and symmetry of $\mathscr{D}_U$ imply $D_U^\sharp \mathcal{W}_U[j_U] = D_U^\sharp \mathscr{D}_U D_U$, which is symmetric. Expanding $(1 - D_U\Lambda)^\sharp \mathscr{D}_U (1 - D_U\Lambda)$ therefore proves

$$\mathscr{D}_U - \mathcal{B}_U[j_U] = \mathsf{S}^\sharp \mathscr{D}_U \mathsf{S}. \tag{3.cdlxviii}$$

This is an exact identity at every admitted real or complex field with its physical current. It requires no inner minimization.

The apparent derivative of J in the middle term of eq. 3.cdlxvii is never needed for a norm estimate. Its bilinear form can be written

$$\begin{aligned} \langle u, \mathcal{B}_U[J] v \rangle = {} & \langle u, \mathcal{W}_U[J]\Lambda v \rangle + \langle \mathcal{W}_U[J]\Lambda u, v \rangle \\ & - \tfrac{1}{2}\langle D_U \Lambda u, \mathcal{W}_U[J]\Lambda v \rangle - \tfrac{1}{2}\langle \mathcal{W}_U[J]\Lambda u, D_U \Lambda v \rangle. \end{aligned} \tag{3.cdlxix}$$

Derivatives are kept on Λ. Thus bounded current, a site inverse bound and its differentiated field readouts are the actual ingredients in a current perturbation estimate. Replacing this form by a pointwise bound on $D_U^\sharp \mathcal{W}_U[J]$ would introduce an unassumed derivative of the current.

3.13.4.1. Comparison with the named gauge-restoration form. Use the same active graph, background, site inverse and projection in the two definitions. The named form uses the covariant difference $D_{\mathrm{n}} = -D_U$. This leaves the site operators unchanged and gives its gauge read $G' R D_{\mathrm{n}}^\sharp A = -\Lambda A$. Write $w = \Lambda A$ and $w_+ = \mathrm{Ad}(U_b) w(b_+)$, $w_- = w(b_-)$ on a bond. With the spacing ξ, $D_U w = \xi^{-1}(w_- - w_+)$. Substitution of the gauge read $-w$ in the named quadratic form gives the bond expression

$$i[A, w_- + w_+] - i\xi^{-1}[w_-, w_+],$$

paired with J in the physical cell pairing.

The diagonal of eq. 3.cdlxvii is $2\langle A, \mathcal{W}_U[J] w \rangle - \langle D_U w, \mathcal{W}_U[J] w \rangle$. Cyclicity in the first pairing gives $i[A, w_- + w_+]$. In the second, use $[w_- - w_+, w_- + w_+] = 2[w_-, w_+]$. Thus its bond expression is the same. Polarization proves $\mathcal{B}_U[J] = \Delta'_\pi[U, J]$ for every independent current on the common domain. In particular the equality uses the named quadratic form itself. The physical Ward identity is used separately when $J = j_U$.

The site output w lies in $\ker Q'$. At each site of the bare layer outside Ω_1 the site row is the identity, so w vanishes there. The displayed commutators therefore use the current only on the

inner bonds with an endpoint in Ω_1. This is the read set of the gauge-restoration term, before adding the separate constraint-curvature subtraction.

For the explicit choice eq. 3.cdlxvii, $R = 0$ implies $\Lambda = 0$ and $\mathcal{B}_U[J] = 0$ for every J. For an unspecified completion this conclusion does not follow merely from zero-fold averaging maps. The physical identity in eq. 3.cdlxviii and the choice of an off-current extension are different statements.

In particular $\mathcal{B}_U[j_U]$ is a gauge-Ward correction, not the full constrained-action curvature correction even on the physical graph. For horizontal vectors u, v with $\Lambda u = \Lambda v = 0$, its bilinear form is zero. The Lagrangian Hessian still contains $-\langle \omega, D^2 q[u, v]\rangle$. The distinct subtraction in eq. 3.cdlxxvi is precisely where that mean-constraint term belongs after its actual multiplier has been identified. No gap for a constrained fluctuation is transferred from the Ward identity alone.

3.13.5. *The off-current gauge defect is computable.* The preceding explicit correction has useful identities even before inverse estimates. Use the actual row intertwining $QD_U = \bar{D}Q'$, on the specified site and row spaces. This is an input identity to be checked for the actual straightened rows, including all outward endpoint values. Set

$$\widehat{G}[J]^{-1} = \mathscr{D}_U - \mathcal{B}_U[J] + D_U R D_U^\sharp + Q^\sharp a Q, \qquad \delta J = J - j_U. \tag{3.cdlxx}$$

Every assertion about an inverse is restricted to its admitted invertibility domain. The identities $\mathsf{S} D_U G' R = 0$, $Q' G' R = 0$ and $\Delta G' R = R$, with $\Delta = D_U^\sharp D_U$, give

$$\widehat{G}[J]^{-1} D_U G' R = D_U R - \mathcal{B}_U[\delta J] D_U G' R. \tag{3.cdlxxi}$$

For the action part use eq. 3.cdlxviii, whose right side annihilates $D_U G' R$. The gauge part is $D_U R \Delta G' R = D_U R$. The row mass is zero because $Q D_U G' R = \bar{D} Q' G' R = 0$. Linearity of $\mathcal{B}$ supplies the remaining current difference. Multiplying by the inverse proves

$$\widehat{G}[J] D_U R = D_U G' R + \widehat{G}[J] \mathcal{B}_U[\delta J] D_U G' R. \tag{3.cdlxxii}$$

Thus the failure of the physical gauge identity is exactly linear in the independent-current defect before inversion, and has the displayed resolvent dependence after inversion.

If $Q\widehat{G}[J]Q^\sharp$ is invertible, define

$$\widehat{H}[J] = \widehat{G}[J] Q^\sharp (Q \widehat{G}[J] Q^\sharp)^{-1}. \tag{3.cdlxxiii}$$

Transposing eq. 3.cdlxxii and using $R G' Q'^\sharp = 0$ proves

$$R D_U^\sharp \widehat{H}[J] = R G' D_U^\sharp \mathcal{B}_U[\delta J] \widehat{H}[J]. \tag{3.cdlxxiv}$$

The first term after transposition vanishes because $D_U^\sharp Q^\sharp = Q'^\sharp \bar{D}^\sharp$. At $J = j_U$ the right side is zero. An arbitrary independent current does not obey that slice condition. Norm estimates for its defect use the actual site and field kernels and the form in eq. 3.cdlxix.

There is an equally explicit second variant. Let $\mathcal{C}_U^{(2)}$ be the homogeneous quadratic part per unit length of the actual nonlinear straightened average, and normalize its current form by

$$\langle v, \Delta^{(2)}[\omega] v\rangle = 2\langle \omega, \mathcal{C}_U^{(2)}(v)\rangle. \tag{3.cdlxxv}$$

Define it bilinearly by polarization, and set

$$\widehat{G}_1[J]^{-1} = \widehat{G}[J]^{-1} - \mathsf{S}^\sharp \Delta^{(2)}[\widehat{H}[J]^\sharp J] \mathsf{S}. \tag{3.cdlxxvi}$$

The added form annihilates $D_U G' R$ on its right, by $\mathsf{S} D_U G' R = 0$. Therefore eq. 3.cdlxxi–eq. 3.cdlxxii hold with $\widehat{G}_1$ in place of $\widehat{G}$. Its row lift has the corresponding slice-defect identity. The $\widehat{H}$ inside eq. 3.cdlxxvi is formed from the first variant; replacing it by the new row lift would change the definition.

These identities do not prove invertibility or bounds. They do prove the correct physical cancellations once those estimates hold. They therefore apply to the named current completion on the

common primitive domain. If a different extension is chosen, its own defect must be calculated from its definition.

3.13.6. *An inner action and a compression have different Ward terms.* For an inner-plaquette action the gauge calculation in eq. 3.cdlxiii is exact on its active graph. A different operation first takes the Hessian of a larger action and then compresses it. Let χ be restriction of fine fields to a set of bonds, and let χ_0 restrict sites. Writing D_{loc} for a proposed local gauge generator, the defect of commuting compression and gauge variation is

$$E_D = \chi D_U \chi_0^\sharp - D_{\mathrm{loc}}. \tag{3.cdlxxvii}$$

Even if $E_D = 0$, the full Hessian can couple through removed bonds. The compressed Ward identity reads

$$\begin{aligned}(\chi \mathscr{D}_U \chi^\sharp) D_{\mathrm{loc}} = \chi \mathcal{W}_U[j_U]\chi_0^\sharp \\ - \chi \mathscr{D}_U (1 - \chi^\sharp\chi) D_U \chi_0^\sharp - \chi \mathscr{D}_U \chi^\sharp E_D.\end{aligned} \tag{3.cdlxxviii}$$

This follows by substituting eq. 3.cdlxxvii, inserting $1 = \chi^\sharp\chi + (1 - \chi^\sharp\chi)$, and applying eq. 3.cdlxiii. The extra terms are supported near the boundary when the full operator is local; with an exponentially local operator they have the corresponding boundary tails. They are not part of the physical interior current unless the action has actually been changed to the inner action.

There is a parallel distinction for lower bounds. Compressing a positive form to free coordinates by zero extension preserves its gap:

$$A \geq cI \quad \Longrightarrow \quad \chi A \chi^\sharp \geq cI. \tag{3.cdlxxix}$$

Rebuilding its Laplacian, row averages or block means on the clipped region is another operator. For example take a unit physical block with fine spacing η, its discrete Laplacian and its full-block constant projection. Their sum has a fixed physical gap after a fixed normalization. Keep one fine site, drop all crossing Laplacian edges as in a Neumann cut, and retain only the compressed full-block mean. The remaining eigenvalue is the fine-volume fraction η^4, which tends to zero. In the true Dirichlet compression the crossing-edge energy remains; this example does not contradict eq. 3.cdlxxix.

Thus every local inverse theorem specifies the order: form the actual operator with its declared rows and gauge, then compress to its free region. A local construction with modified rows instead supplies its complete boundary comparison and its own gap proof. A collar can make those differences small at the rows that are used; it does not erase them as identities.

3.13.7. *The site readout at an outward row.* The site average used in the Landau projector has one value at each top, taken at that top's assigned level. The endpoint readout of a row is a different list when the row is outward. We now give their exact relation.

Let $\mathsf{T}_j(y,x)$ be the product of the admitted internal mean transports from a level-j center y down its pillar to a bare site x. For a top of level j define

$$(Q'_{\mathrm{pill}}\lambda)(y) = L^{-4j} \sum_{x \in B^j(y)} \mathrm{Ad}(\mathsf{T}_j(y,x))\lambda(x). \tag{3.cdlxxx}$$

These pillars are disjoint. There is therefore an explicit right inverse on node fields:

$$(E_{\mathrm{pill}}\psi)(x) = \mathrm{Ad}(\mathsf{T}_j(y,x))^{-1}\psi(y) \quad (x \in B^j(y)), \qquad Q'_{\mathrm{pill}}E_{\mathrm{pill}} = 1. \tag{3.cdlxxxi}$$

This proves rank without an elliptic estimate. For real transports the right inverse has supremum norm one. With fine site measure η^4 and top measure ξ_y^4, it is also an isometry in L^2. Indeed a pillar contains L^{4j} bare sites, so $\eta^4 L^{4j} = \xi_y^4$. The invariant bilinear color pairing gives

$$Q'_{\mathrm{pill}} = E_{\mathrm{pill}}^\sharp, \qquad Q'_{\mathrm{pill}}(Q'_{\mathrm{pill}})^\sharp = 1. \tag{3.cdlxxxii}$$

The algebraic identities hold for admitted complex transports as well. Their Hermitian norm bounds then include the actual condition numbers of those transports. A counting kernel must include the ratio ξ_y^4/η^4 in the transpose; an ordinary unweighted array transpose would not be eq. 3.cdlxxxii.

For any level-j center z, write $Q'_j\lambda(z)$ for its full j-fold transported site average, whether or not its assigned top level is j. If z is the outer endpoint of a determining level-j row, its assigned level is $j-1$. The entire block $B^j(z)$ lies in that outer layer. Decompose it into its L^4 level-$(j-1)$ pillars. The product definition of the transport gives

$$Q'_j\lambda(z) = L^{-4} \sum_{w\in B(z)\cap\mathbb{T}^{(j-1)}} \mathrm{Ad}(T_{z,w}(\mathcal{M}^{j-1}U))(Q'_{\mathrm{pill}}\lambda)(w). \tag{3.cdlxxxiii}$$

To verify the coefficient, each smaller average has weight $L^{-4(j-1)}$; multiplying it by L^{-4} gives L^{-4j}. The two transport factors multiply in their path order. This proves eq. 3.cdlxxxiii exactly, without an asymptotic or small-field approximation.

Define $\mathcal{E}_{j,z}\psi$ to be $\psi(z)$ at an assigned level-j top, and to be the finite average on the right of eq. 3.cdlxxxiii at an outer endpoint. Then define the extended coarse difference by

$$(\bar{D}_{\mathrm{ext}}\psi)(c) = \xi_c^{-1}\{\mathcal{E}_{j,c_-}\psi - \mathrm{Ad}((\mathcal{M}^jU)(c))\mathcal{E}_{j,c_+}\psi\}, \qquad c\in\Gamma_j. \tag{3.cdlxxxiv}$$

For an interior row it is the ordinary covariant difference of its two top values. For an outward row it first averages the outer finer tops. There is no need to duplicate that site as a new independent top in the site Gram matrix.

Here is a direct verification of the field intertwining. Under an infinitesimal gauge $e^{it\lambda}$ the derivative of the level-j node factor is

$$\frac{1}{i}\frac{d}{dt}v_j[tD_U^{\mathrm{phys}}\lambda](y)\Big|_{t=0} = \lambda(y) - Q'_j\lambda(y). \tag{3.cdlxxxv}$$

At one step the change in each relative contour has derivative $\lambda(y) - \mathrm{Ad}(T_{y,x})\lambda(x)$. The implicit mean at the identity has differential the arithmetic mean of its inputs, so this gives the one-step formula. Iterating and using the product rule for the accumulated node factors gives eq. 3.cdlxxxv, with the hierarchical site average in eq. 3.cdlxxx.

Differentiate the actual straightening identity $\mathcal{M}^j(e^{i\eta A}U) = v_j[A]\,e^{iq_j^{\mathrm{st}}(A)}\mathcal{M}^jU\,v_j[A]^{-1}$ on a row. The plain coarse variation is $\lambda(c_-) - \mathrm{Ad}(\mathcal{M}^jU(c))\lambda(c_+)$. Subtract the two node derivatives eq. 3.cdlxxxv, with their endpoint transports. The remaining variation is

$$Dq_j^{\mathrm{st}}(0)\,D_U^{\mathrm{phys}}\lambda = Q'_j\lambda(c_-) - \mathrm{Ad}(\mathcal{M}^jU(c))Q'_j\lambda(c_+). \tag{3.cdlxxxvi}$$

Divide by ξ_c to obtain the per-length row Q. Using eq. 3.cdlxxxiii proves the fully typed identity

$$QD_U^{\mathrm{phys}} = \bar{D}_{\mathrm{ext}}Q'_{\mathrm{pill}}. \tag{3.cdlxxxvii}$$

This supplies the intertwining needed in eq. 3.cdlxx–eq. 3.cdlxxiv. There $\bar{D}$ means this extended row difference and Q' means the own-pillar site map. Using a single unextended value of the outer top would give the wrong formula.

Each endpoint extension reads at most L^4 tops in one adjacent layer. Its real row sum is at most one, since the transports are unitary and the weights add to one. A finer top occurs in one such outer block and at most eight incident coarse rows. Thus its two-sided normalized cell counts are bounded at fixed L, without a factor depending on the depth. For complex pairs the same statement uses the actual bounded transport condition numbers. The latter are atomic domain estimates; the algebraic intertwining does not prove them.

3.13.8. *Finite real existence of the site operators.* There is also a real finite positivity statement independent of a quantitative inverse bound. Assume the actual mean transports are admitted and unitary, and that every active fine site lies in a connected pillar with its top. Let a be strictly positive on the top space. Then

$$L_{\mathrm{site}} = D_U^\sharp D_U + (Q'_{\mathrm{pill}})^\sharp a Q'_{\mathrm{pill}} \quad \text{is strictly positive.} \tag{3.cdlxxxviii}$$

To prove it, a vector of zero Hermitian energy must obey $D_U\lambda = 0$ and $Q'_{\mathrm{pill}}\lambda = 0$. Split into Hermitian real and imaginary parts; it suffices to treat real λ. The first identity means that $e^{it\lambda(x)}$ leaves every active bond unchanged:

$$e^{it\lambda(b_-)} U_b e^{-it\lambda(b_+)} = U_b. \tag{3.cdlxxxix}$$

Gauge covariance of each actual mean transport therefore gives $\mathrm{Ad}(\mathsf{T}_j(y,x))\lambda(x) = \lambda(y)$. Average over the pillar: $Q'_{\mathrm{pill}}\lambda(y) = \lambda(y)$. Its zero value makes $\lambda(y) = 0$. Parallel transport along the active graph from the top to every site makes $\lambda = 0$. This proves eq. 3.cdlxxxviii.

It follows that $G' = L_{\mathrm{site}}^{-1}$ exists on every finite real problem of this type. The site Gram matrix

$$Q'_{\mathrm{pill}} G'^2 (Q'_{\mathrm{pill}})^\sharp \tag{3.cdxc}$$

is strictly positive as well. For a nonzero node vector ψ, surjectivity makes $(Q'_{\mathrm{pill}})^\sharp\psi \neq 0$; invertibility of G' then makes its squared norm positive. Thus the formulas for R and S are genuine finite real matrices before their uniform bounds are established.

No volume-uniform or scale-weighted lower bound has been deduced from this nullspace argument. That bound needs the quantitative site Poincaré and local-gauge estimates. Nor does eq. 3.cdlxxxviii prove positivity of the fine Wilson Hessian or of its independent-current completion. The distinctions between finite invertibility, uniform coercivity and exponentially local readouts remain necessary.

3.13.9. *Transport control on one pillar.* The site gap has a quantitative proof from the actual averages. Let a pillar have side $\xi = n\eta$, where $n = L^j$, and assume

$$|\partial U(p) - 1| \le \epsilon\, \eta^2 \xi^{-2} \tag{3.cdxci}$$

on its determining neighborhood. The fixed path neighborhood has an embedded rectangular lift. If an extra fine collar is needed, the regularity bounds in the adjacent layer give the same inequality with a fixed L-dependent factor. We include that factor in the constants below.

Take the fine coordinate comb from the pillar center and put $g(x) = U(\Gamma_{y,x})$. For adjacent $x, x + \eta e_\mu$, the loop made by the two root combs and that bond is filled by strips of width one fine cell. Commuting the extra μ-step through the later coordinate segments uses at most Cn plaquettes, not Cn^2. The unitary Stokes product inequality therefore gives

$$|U^g(b) - 1| \le C\epsilon/n \quad \text{on internal fine bonds.} \tag{3.cdxcii}$$

Here $g(y) = 1$. Every curvature norm is unchanged. The gauge need not be close to one, and its derivatives are not estimated in this step.

Write $\xi_r = L^r\eta$, and let a_r be the supremum of $|\mathcal{M}^r(U^g)(c) - 1|$ on the level-r internal edges used in the hierarchical transport. Let ϖ_r be the corresponding coarse plaquette bound. The determining neighborhoods remain in the rectangular pillar neighborhood: each parent block is a union of its child blocks, and each contour stays in its assigned block. The real curvature iteration proved in the background section gives

$$\varpi_r \le C_L \epsilon (\xi_r/\xi)^2. \tag{3.cdxciii}$$

The same finite comparison shows that one actual averaged edge differs from its central straight transport by at most $C_L\varpi_r$. Indeed every closed logarithm in its definition is a comparison loop

with that central transport. The unitary strip estimate bounds those logarithms, and then their exponential average. The central route has L edges, so

$$a_{r+1} \le L a_r + C_L \varpi_r. \tag{3.cdxciv}$$

The coefficient of a_r is exactly L, by the unitary product inequality.

Iterating eq. 3.cdxciv, using $a_0 \le C\epsilon\eta/\xi$, gives

$$\begin{aligned} a_r &\le L^r a_0 + C_L \epsilon \sum_{s<r} L^{r-1-s} (\xi_s/\xi)^2 \\ &\le C_L \epsilon\, \xi_r/\xi. \end{aligned} \tag{3.cdxcv}$$

For the sum, insert $\xi_s = \eta L^s$; its geometric factor is bounded by a constant times $(\xi_r/\xi)^2$, which is no larger than ξ_r/ξ. The constants do not multiply with the number of levels.

The internal mean on level r is a finite smooth mean of paths of $O(L)$ edges. On its regular near-identity chart it differs from one by at most $C_L a_r$. The hierarchical transport has one such factor at each level below j. All are unitary, hence

$$|\mathsf{T}_j(U^g; y, x) - 1| \le C_L \sum_{r<j} a_r \le C_L \epsilon \quad (x \in B^j(y)). \tag{3.cdxcvi}$$

The last sum uses $\sum_{r<j} \xi_r/\xi \le 1/(L-1)$. The fixed smallness of ϵ also admits every mean used in this argument. Thus eq. 3.cdxcvi is a uniform-depth property of the actual iterated transports, not a hypothesis of such a property.

3.13.10. *A real weighted site gap.* We first prove the flat block inequality with its scale. Let $\bar\lambda$ be the arithmetic mean on the n^4 sites of a block. The exact variance identity is

$$\sum_x |\lambda(x) - \bar\lambda|^2 = \frac{1}{2n^4} \sum_{x,z} |\lambda(x) - \lambda(z)|^2. \tag{3.cdxcvii}$$

Connect each pair by the coordinate-order path. Its length is at most $4n$. A fixed oriented edge lies in at most n^5 of these paths: its two longitudinal endpoint coordinates give at most n^2 choices, and the three other coordinates give at most n^3. Cauchy on the path and then eq. 3.cdxcvii give a constant times n^2 times the sum of squared edge differences. Multiplying by $\eta^4\xi^{-2}$ proves

$$\xi^{-2}\|\lambda\|_{2,B}^2 \le C\{\|\partial_{\mathrm{int}}\lambda\|_{2,B}^2 + \xi^2|\bar\lambda|^2\}. \tag{3.cdxcviii}$$

Here the fine norm uses η^4, and ∂ divides a bond difference by η. The mean part is $\eta^4 n^4 \xi^{-2}|\bar\lambda|^2 = \xi^2|\bar\lambda|^2$. For $n = 1$ the variance vanishes and the same inequality holds.

Apply the real comb gauge of eq. 3.cdxcii in each pillar. Only internal bonds are used in the block inequality, so the gauges of neighboring pillars may be different. They define a unitary site gauge globally, and all covariant norms are unchanged. In this frame

$$\|(\partial_{\mathrm{int}} - D_{U,\mathrm{int}})\lambda\|_{2,B} \le C_L \epsilon\, \xi^{-1}\|\lambda\|_{2,B}. \tag{3.cdxcix}$$

The estimate follows from $|\operatorname{Ad}(U_b) - 1| \le C_L\epsilon\eta/\xi$ and the at-most-eight incidences at a site.

Let Q_0' be arithmetic block average. eq. 3.cdxcvi, Cauchy for the finite average and $\xi^4 = n^4\eta^4$ give

$$\xi^4|(Q'_{\mathrm{pill}} - Q_0')\lambda|^2 \le C_L\epsilon^2\|\lambda\|_{2,B}^2. \tag{3.d}$$

Use eq. 3.cdxcix–eq. 3.d on the right of eq. 3.cdxcviii. The result is the same two terms with D_U and Q'_{pill}, plus $C_L\epsilon^2\xi^{-2}\|\lambda\|_{2,B}^2$. Choose the fixed curvature ceiling so that this coefficient is at most one half of the left coefficient. Absorption proves

$$\xi^{-2}\|\lambda\|_{2,B}^2 \le C_L\{\|D_{U,\mathrm{int}}\lambda\|_{2,B}^2 + \xi^2|Q'_{\mathrm{pill}}\lambda|^2\}. \tag{3.di}$$

Sum over the disjoint pillars. Every internal fine bond is counted at most once; all other gradient terms in the full energy are nonnegative. Suppose $a(y) = c_y \xi_y^{-2}$ with $c_y \geq a_* > 0$. Then

$$\begin{aligned} \sum_x \eta^4 \xi(x)^{-2} |\lambda(x)|^2 &\leq C_{L,a_*} \{ \|D_U \lambda\|_2^2 + \langle Q'_{\mathrm{pill}} \lambda, a Q'_{\mathrm{pill}} \lambda \rangle_{\mathrm{tops}} \} \\ &= C_{L,a_*} \langle \lambda, L_{\mathrm{site}} \lambda \rangle. \end{aligned} \tag{3.dii}$$

This is the actual site weighted gap, uniform in depth, volume, region shape and real holonomy. It uses the small real curvature and actual transport construction; no physical current or criticality is used. Crossing-edge energy is kept, not replaced by a clipped block mean.

For a complex site vector split into Hermitian real and imaginary parts. Since the real unitary connection is a real matrix in an orthonormal Lie-algebra basis, the same inequality holds for its Hermitian energy. If the top scale is at most one, it in particular gives an ordinary finite L^2 gap independent of the bare mesh. In other physical units its smallest unweighted gap is scaled by the inverse square of the largest local scale.

3.13.11. *The complex site domain uses the actual iterated means.* The preceding transport argument has a complex version with a fixed physical radius, independent of the number of fine sites. We give it to avoid assuming a uniform radius for the iterated site average.

Consider one step on a fixed two-block neighborhood. Suppose all its edge matrices satisfy $|W_b - 1| \leq a$, and let $\varpi = \sup |\partial W - 1|$. There are constants r_L, C_L for which the actual means and logarithms are admitted if $a, \varpi < r_L$, and

$$\begin{aligned} a_{\mathrm{next}} &\leq La + C_L a^2 + C_L \varpi, \\ \varpi_{\mathrm{next}} &\leq L^2 (1 + C_L a) \varpi + C_L \varpi^2. \end{aligned} \tag{3.diii}$$

For the first line, a central straight product differs from one by at most $La + C_L a^2$. Plaquette sliding bounds each closed comparison logarithm by $C_L \varpi$: all prefix transports have condition number at most $1 + C_L a$ on this finite neighborhood. The defining exponential average therefore changes the central product by at most that amount.

For the second line, take the local complex comb gauge. Its values and inverses are $1 + O(C_L a)$. It sets the tree bonds to one; each remaining bond is then $1 + O(C_L \varpi)$ by the finite plaquette sliding identity. In that gauge the Taylor/flux argument for the actual one-step map gives the coefficient L^2 and a quadratic error $C_L \varpi^2$. Conjugating the fine and coarse plaquettes through the small complex gauge changes their norms by $1 + C_L a$. This proves eq. 3.diii. Complex covariance here follows directly from the conjugation identity for the small logarithm, or from uniqueness of the admitted finite mean. No absolute-value formula is continued holomorphically.

These constants can be bounded before an iteration is run. The finite mean construction has a joint raw-edge holomorphic ball of radius c/L, with universal supremum bounds; two scalar Cauchy circles bound its second derivative by CL^2. A comb on the one-step patch sweeps at most CL^2 plaquettes, so its non-tree logarithms have size $CL^2 \varpi$. The quadratic Taylor error in the coarse plaquette is therefore at most $CL^2 (L^2 \varpi)^2 = CL^6 \varpi^2$. The fine logarithmic curl error times its L^2 flux factor has the same upper order. Prefix conjugations cost $1 + CLa$. For example $r_L = cL^{-3}$ and $C_L = CL^8$, with fixed universal shrinkages and enlargements, suffice in eq. 3.diii. This count uses directional Cauchy on the proved full supremum ball, not one coordinate estimate summed over all field coordinates.

On a pillar let $t_r = \xi_r / \xi$, and normalize $A_r = a_r / t_r$, $V_r = \varpi_r / t_r^2$. eq. 3.diii implies, with an increased fixed constant,

$$Z_{r+1} \leq Z_r + C_L t_r (Z_r + Z_r^2), \qquad Z_r = A_r + V_r, \qquad \sum_{r<j} t_r \leq \frac{1}{L-1}. \tag{3.div}$$

For $Z_r \le 1$, the factor is at most $1 + 2C_L t_r$. Its product is at most $\exp(2C_L/(L-1))$. Choose Z_0 below one half the minimum one-step chart radius times the inverse of this number. Induction then gives $Z_r \le C_L Z_0$ and keeps every step in its chart. The constants depend on L, not j. With the preceding finite choices a sufficient common small input radius has the explicit form $cL^{-3}\exp(-CL^8)$. The precise numerical constants are not needed below, but this gives a positive lower radius rather than an asserted upper cap. The product of the internal means and its inverse have norm at most $\exp(C_L \sum_r a_r)$. Hence

$$|\mathsf{T}_j(W;y,x) - 1| \le C_L Z_0, \qquad |\mathsf{T}_j(W;y,x)| + |\mathsf{T}_j(W;y,x)^{-1}| \le C_L. \tag{3.dv}$$

This proves the existence and boundedness of the whole finite hierarchical formula on that domain.

Now start with a real field satisfying eq. 3.cdxci, put it in the real comb gauge eq. 3.cdxcii, and let $W = e^{i\eta E}U$. Assume the actual physical first-order input satisfies

$$h = \max\{\|\xi E\|_\infty, \|\xi^2 \nabla_U E\|_\infty\} \le h_L. \tag{3.dvi}$$

In the comb frame the ordinary first difference of E differs from its covariant one by $O(\epsilon\xi^{-1}|E|)$. The elementary plaquette product then gives $a_0 \le C(\epsilon + h)\eta/\xi$ and $\varpi_0 \le C(\epsilon + h + h^2)(\eta/\xi)^2$. Thus the starting Z_0 in eq. 3.div is at most $C(\epsilon + h)$.

Choose a fixed $h_L > 0$ so that the same argument holds with E replaced by sE for $|s| < h_L/h$, and assume $h \le h_L/4$. Reduce the real curvature ceiling at the same time so that $\epsilon \le h_L$ and the resulting starting Z_0 satisfies eq. 3.div. The transport is holomorphic in s on this disc, with a uniform bound. Cauchy's formula on a smaller concentric disc, or the geometric sum of its Taylor coefficients, yields

$$|\mathsf{T}_j(e^{i\eta E}U;y,x) - \mathsf{T}_j(U;y,x)| \le C_L h. \tag{3.dvii}$$

The bound in eq. 3.dv is uniform in j and in the number of coordinates. This is a single directional Cauchy argument on an actual admitted physical disc. It does not assert an entrywise absolute Hessian sum for an arbitrary many-variable functional.

For the complex site average write Q'_E and for the real reference write Q'_U. The transported finite-average calculation used in eq. 3.d gives

$$\|a^{1/2}(Q'_E - Q'_U)\lambda\|_{2,\mathrm{tops}} \le C_L h\, \|\xi^{-1}\lambda\|_2. \tag{3.dviii}$$

Here the real row weights satisfy $a(y) \le a^* \xi_y^{-2}$. The proof squares the average difference in each disjoint pillar, uses eq. 3.dvii, and sums its fine volume. There is no factor equal to the number of fine sites. The same argument applies to the outer endpoint extension eq. 3.cdlxxxiii: its neighborhood uses one adjacent scale, so the input constants change by fixed powers of L.

3.13.12. *Relative forms give the complex site inverse.* It remains to compare the actual site operators, not their unscaled matrix coefficients. Let L_U be the real site operator and let $L_E = D^\sharp_{e^{i\eta E}U} D_{e^{i\eta E}U} + Q'^\sharp_E a Q'_E$. The transposes in L_E are bilinear analytic transposes. At a real reference they are the usual Hermitian adjoints.

For a fine bond the difference of the two site derivatives is bounded by $C|E_b||\lambda(b_+)|$. Adjacent pillar scales differ by at most the fixed factor L; the region gaps exclude a bare bond jumping over another layer. The finite incidence count therefore gives

$$\|(D_{e^{i\eta E}U} - D_U)\lambda\|_2 \le C_L h \|\xi^{-1}\lambda\|_2. \tag{3.dix}$$

Keeping derivatives on the two arguments, expand the kinetic difference. Use eq. 3.dix, then eq. 3.dviii for the mass difference, and finally the real weighted site gap in eq. 3.dii. For arbitrary complex vectors v, w, this gives

$$|\langle v, (L_E - L_U)w\rangle_{\mathrm{H}}| \le C_L h\, \langle v, L_U v\rangle_{\mathrm{H}}^{1/2} \langle w, L_U w\rangle_{\mathrm{H}}^{1/2}. \tag{3.dx}$$

For example the quadratic mass term is $(Q'_E - Q'_U)^\sharp a (Q'_E - Q'_U)$; the two cross terms contain only one difference. On replacing the left argument by its complex conjugate, the same absolute bounds

apply to the analytic bilinear transpose. No Hermitian conjugation of the complex coefficients is inserted into their definition.

Let $B_E = L_U^{-1/2}(L_E - L_U)L_U^{-1/2}$. eq. 3.dx proves $\|B_E\|_{2\to 2} \le C_L h$. For $C_L h \le 1/2$ one has

$$G'_E = L_U^{-1/2}(1 + B_E)^{-1} L_U^{-1/2}, \qquad \mathrm{Re}\langle \lambda, L_E \lambda\rangle_{\mathrm{H}} \ge \tfrac{1}{2}\langle \lambda, L_U \lambda\rangle_{\mathrm{H}}. \tag{3.dxi}$$

The inverse is holomorphic on the whole admitted physical ball. It has the same weighted coercivity, up to a fixed factor, uniformly in depth and volume. This is a relative-form argument. The unscaled operator norm of a coefficient perturbation need not be bounded as $\eta \downarrow 0$.

The proof establishes the site inverse part of the primitive package. It has not yet established the site Gram inverse, the fine vector Green function, or the layered current precision. Those require their separate row, readout and gap arguments.

3.13.13. *Exponential decay from the weighted site form.* Let V denote multiplication by $\xi(x)^{-2}$ on sites. The real gap in eq. 3.dii is $L_U \ge cV$. We prove a weighted inverse bound before converting it to pointwise regularity. This avoids estimating a bare kinetic operator by its unscaled norm.

Let ψ be bounded and one-Lipschitz for eq. 3.cdl, and let $M_s = e^{s\psi}$ on sites. Use the value $\psi(b_-)$ on an edge and $\psi(y)$ on a top. Put

$$D_s = M_s^{\mathrm{edge}} D_U M_s^{-1}, \qquad Q'_s = M_s^{\mathrm{top}} Q'_U M_s^{-1}. \tag{3.dxii}$$

On an edge the two endpoint values of ψ differ by at most $C_L\eta/\xi_b$. Inside a pillar their difference from the top value is at most a fixed constant. The difference from the unweighted row therefore satisfies

$$\|(D_s - D_U)\lambda\|_2 + \|a^{1/2}(Q'_s - Q'_U)\lambda\|_{2,\mathrm{tops}} \le C_L|s|\,\|V^{1/2}\lambda\|_2 \tag{3.dxiii}$$

for $|s| \le s_L$. The edge estimate follows from $|e^{st} - 1| \le C|s||t|$ on the stated bounded interval. For the site row use the same estimate in its finite transported average, then Cauchy with the actual pillar volume. Its constant is independent of $\|\psi\|_\infty$.

The conjugated form has the exact factorization

$$M_s L_U M_s^{-1} = D_{-s}^{\sharp} D_s + (Q'_{-s})^{\sharp} a Q'_s. \tag{3.dxiv}$$

Expanding it and using eq. 3.dxiii, eq. 3.dii proves a relative form error at most $C_L|s|$. With a complex field perturbation in the ball of eq. 3.dxi, the same calculation gives error $C_L(h + |s|)$ relative to the real L_U. Choose both small enough for that number to be at most one half. All these forms are invertible.

For any such inverse $\mathcal{G}_s$, testing the equation $L_s\lambda = V^{1/2}f$ against λ gives

$$\|V^{1/2}\mathcal{G}_s V^{1/2}\|_{2\to 2} \le 2/c. \tag{3.dxv}$$

Indeed its real part is at least $(c/2)\|V^{1/2}\lambda\|_2^2$, whereas the right pairing is at most $\|V^{1/2}\lambda\|_2\|f\|_2$. This proof uses neither a normal matrix nor a pointwise positive complex kernel.

Choose $\psi(x) = \min\{d(x, \Delta(z)), N\}$. It is one-Lipschitz. The diameter of a pillar in eq. 3.cdl is bounded, so on $\Delta(y)$ its value is at least $d(y, z) - C$ after N is large enough. eq. 3.dxv then yields

$$\|P_y G'_E P_z\|_{2\to 2} \le C_L \xi_y \xi_z e^{-m d(y,z)} \tag{3.dxvi}$$

for a fixed $m = m(L) > 0$. On a finite torus take the periodic distance. On the cover first use bounded N and then increase it. The constants are independent of this truncation.

In particular the normalized operator $V^{1/2}G'_E V^{1/2}$ has exponentially decaying cell L^2 entries. Its two-sided cell Schur norm follows from eq. 3.cdlvi–eq. 3.cdlix when the separation is chosen with

$$\frac{8\log L}{\rho} < m/2. \tag{3.dxvii}$$

The rate m above depends only on the fixed one-step constants and the site gap; it does not depend on ρ. Thus this is a genuine order of choice: increase the separation after those constants, or verify

that the already chosen separation satisfies this finite inequality. It is not an appeal to unspecified polynomial volume growth.

3.13.14. *A scalar subsolution estimate at the fine mesh.* We use a local estimate whose proof also explains why a fine Coulomb singularity causes no loss in eq. 3.dxvi. On the infinite scalar fine lattice in four dimensions, let $\mathcal{L} = -\Delta_\eta$. The positive heat kernel gives a positive Green kernel

$$G_0(x) = \int_0^\infty p_t^\eta(x)\,dt, \qquad |G_0(x)| \le C(|x|+\eta)^{-2}, \quad |\nabla G_0(x)| \le C(|x|+\eta)^{-3}. \tag{3.dxviii}$$

These inequalities follow by integrating the mesh-uniform heat kernel derivative bounds. For example split at $t = |x|^2$, use a polynomial off-diagonal tail before the split, and Ct^{-2}, respectively $Ct^{-5/2}$, afterwards. At $|x| \le \eta$ the bounds are the corresponding lattice scaling bounds. Also $\mathcal{L}G_0$ is the normalized lattice delta, by integrating $-\partial_t p_t^\eta$.

Suppose $w \ge 0$ and $\mathcal{L}w \le F_0$ on a ball of radius $2r$, $r \ge 8\eta$. Choose a nonnegative cutoff χ, equal to one on the r-ball, supported in the $2r$-ball, with its first two differences bounded by Cr^{-1}, Cr^{-2}. At the center x, summation by parts gives

$$w(x) = \langle \chi G_0(x-\cdot), \mathcal{L}w\rangle - \langle [\mathcal{L},\chi]G_0(x-\cdot), w\rangle. \tag{3.dxix}$$

All sums use η^4. This identity follows by moving $\mathcal{L}$ off w and using $\mathcal{L}G_0 = \delta_x$. There is no derivative of w in its last term. Since the first test is nonnegative, its contribution is at most CF_0r^2. Indeed the lattice sum of G_0 on that ball is at most Cr^2, by dyadic shells.

The commutator is supported on the cutoff annulus. Writing it as $(\mathcal{L}\chi)G_0$ plus products of first differences shows that its absolute value is at most Cr^{-4} there, by eq. 3.dxviii. Cauchy and the annulus volume therefore prove

$$w(x) \le C\{r^{-2}\|w\|_{2,B_{2r}} + r^2F_0\}. \tag{3.dxx}$$

If r/η is bounded, the point-value estimate $|w(x)| \le \eta^{-2}\|w\|_2$ gives the same inequality with a changed constant. Thus the local estimate is uniform at coarse meshes as well.

For a real unitary connection, let λ be Lie-algebra valued and put $w = |\lambda|$. For $\lambda(x) \ne 0$, pair each term $\lambda(x) - \mathrm{Ad}(U_{x,x'})\lambda(x')$ with $\lambda(x)/|\lambda(x)|$. Unitarity gives

$$-\Delta_\eta|\lambda|(x) \le |D_U^\sharp D_U\lambda(x)|. \tag{3.dxxi}$$

At a zero value of λ the left side is nonpositive, so the same inequality holds. This proves the covariant scalar subsolution inequality without differentiating a gauge. It is used only on a patch with the full covariant Laplacian stencil. Bare boundary cells will use the point-value fallback instead.

3.13.15. *Pointwise site Green bounds.* First take a real U and $\lambda = G_U'f$. At a site in pillar $\Delta(y)$, the mass part of its equation obeys

$$|(Q_U'^\sharp a Q_U'\lambda)(x)| \le C_L\xi_y^{-4}\|\lambda\|_{2,\Delta(y)}. \tag{3.dxxii}$$

In fact it is the transported vector $a(y)Q_U'\lambda(y)$, whose norm is at most $a^*\xi_y^{-2}|Q_U'\lambda(y)|$. Cauchy in the average gives the additional ξ_y^{-2}.

Take a local ball of radius comparable to ξ_y. It meets a bounded L-dependent number of pillars, all with scales in $[\xi_y/L, L\xi_y]$. The gaps exclude a second intervening layer in that ball once $\rho > C_LL$. Call the corresponding whole-pillar neighborhood $\mathcal{N}(y)$. Its cells are at bounded d-distance from y. Apply eq. 3.dxx–eq. 3.dxxii on a lift of this ball. This proves

$$\|\lambda\|_{\infty,\Delta(y)} \le C_L\xi_y^{-2}\sum_{w\in\mathcal{N}(y)}\|\lambda\|_{2,\Delta(w)} + C_L\xi_y^2\|f\|_{\infty,\mathcal{N}(y)}. \tag{3.dxxiii}$$

Near a boundary of the action region, every pillar lacking this full patch is bare: the nonbare regions have the prescribed gap from that boundary. The single-cell fallback in eq. 3.dxx proves

eq. 3.dxxiii there directly. On a torus the local ball is lifted to the periodic cover. Its fixed radius can meet only a fixed number of copies of a cell, since every period contains at least two top blocks. This changes only the constant.

If f is supported in $\Delta(z)$, use eq. 3.dxvi and $\|f\|_{2,\Delta(z)} \le \xi_z^2 \|f\|_\infty$ on the right side. The far contribution is $C\xi_z^3 \xi_y^{-1} e^{-md(y,z)} \|f\|_\infty$. The local forcing term has $d(y,z) \le C_L$. eq. 3.cdliii absorbs the scale ratio:

$$\|G'_U f\|_{\infty,\Delta(y)} \le C_L \xi_y^2 e^{-m_1 d(y,z)} \|f\|_\infty, \qquad 0 < m_1 < m - \frac{3\log L}{\rho}. \tag{3.dxxiv}$$

Choosing the separation large enough leaves a positive fixed m_1. This is a pointwise estimate for a bounded cell source. It is not a uniform bound on the Green kernel evaluated at two individual fine sites.

The actual equation also gives

$$\|D_U^\sharp D_U G'_U f\|_{\infty,\Delta(y)} \le C_L e^{-m_1 d(y,z)} \|f\|_\infty. \tag{3.dxxv}$$

Use $D_U^\sharp D_U \lambda = f - Q_U'^\sharp a Q'_U \lambda$, and bound its mass term by the supremum on its same pillar. The cell diameter in the layer metric is bounded.

3.13.16. *Site gradients and their complex continuation.* For differentiated pointwise bounds use the actual local gauge clause of the regular-pair class. On the required patch around a cell it supplies a unitary gauge with

$$U = e^{i\eta A}, \qquad \xi|A| + \xi^2 |\nabla A| \le a_{\mathrm{loc}}, \qquad a_{\mathrm{loc}} \le a_L. \tag{3.dxxvi}$$

The constant a_L is fixed small. Near one interface the patch may be described in the finer adjacent scale; converting to ξ changes these bounds by fixed powers of L. The region gaps and the fixed patch collar are chosen to include no third layer. This gauge clause is an input defining the regular-pair class. Deriving it for an actual background from weaker tests is the separate background estimate.

In this local gauge the site equation is a scalar componentwise Laplace equation with first- and zeroth-order errors. Expanding the transported differences gives

$$|\Delta_\eta \lambda| \le |f - Q'^\sharp a Q' \lambda| + C_L a_{\mathrm{loc}} \{\xi^{-1} |\nabla \lambda| + \xi^{-2} |\lambda|\}. \tag{3.dxxvii}$$

Each difference of a link exponential costs $\eta|\nabla A|$; each undifferentiated link error costs $\eta|A|$. After division by the two powers of η in the Laplacian the terms have exactly the displayed dimensions.

Here is the interior absorption that supplies the gradient. On a slightly larger ball of radius $C\xi$, put $M = \|\lambda\|_\infty$ and $F = \|f - Q'^\sharp a Q' \lambda\|_\infty$. Let d_b be distance of a fine edge to the ball boundary and take the finite maximum of $d_b |\nabla \lambda(b)|$. If d_b is only a fixed multiple of η, the point-value bound on a difference controls it by CM. Otherwise apply the scalar interior gradient estimate to a ball of radius $d_b/4$ around an edge where this maximum is approached. Distances in that smaller ball are comparable to d_b. Its gradient supremum is therefore at most the weighted maximum divided by cd_b. Using eq. 3.dxxvii in the scalar estimate gives

$$\sup_b d_b |\nabla \lambda(b)| \le C_L (M + \xi^2 F) + C_L a_{\mathrm{loc}} \sup_b d_b |\nabla \lambda(b)|. \tag{3.dxxviii}$$

Choose $C_L a_{\mathrm{loc}} \le 1/2$. Absorption, eq. 3.dxxiv–eq. 3.dxxv and the scalar interior Hölder estimate then give, for a bounded source in $\Delta(z)$,

$$\begin{aligned} \|\nabla_U G'_U f\|_{\infty,\Delta(y)} &\le C_L \xi_y e^{-\delta d(y,z)} \|f\|_\infty, \\ [\nabla_U G'_U f]_{\alpha,\Delta(y)} &\le C_{L,\alpha} \xi_y^{1-\alpha} e^{-\delta d(y,z)} \|f\|_\infty, \qquad 0 < \alpha < 1. \end{aligned} \tag{3.dxxix}$$

The Hölder comparison is made in the assigned local frame, or with its declared transport to a common base. The bounded potential and pointwise gradient give the transported version. The

constant is not asserted bounded as $\alpha \uparrow 1$. Cells with a bounded number of fine sites use the direct difference bound; near the action boundary they are bare, as already noted.

These pointwise estimates continue to the actual complex site operator by a stronger norm argument than a pointwise matrix coefficient estimate. For a fixed source cell z, let

$$\begin{aligned} \|\lambda\|_{\mathcal{Y}_z} &= \sup_y e^{\delta d(y,z)} \max\{\xi_y^{-2}\|\lambda\|_{\infty,\Delta(y)}, \xi_y^{-1}\|\nabla_U\lambda\|_{\infty,\Delta(y)}\}, \\ \|f\|_{\mathcal{F}_z} &= \sup_y e^{\delta d(y,z)}\|f\|_{\infty,\Delta(y)}. \end{aligned} \tag{3.dxxx}$$

Use a rate δ strictly below the cell Green rate, with its spare rate larger than the ball-growth rate in eq. 3.cdlvi. Summing the bounded-source cell estimates then proves $\|G'_U\|_{\mathcal{F}_z\to\mathcal{Y}_z} \le C_L$, uniformly in z.

For the actual left perturbation $U_E = e^{i\eta E}U$, expand the two covariant site differences and the two site averages. The kinetic difference has a coefficient bounded by $|E|$ on a first difference of λ, and a coefficient bounded by $|\nabla_U E| + |E|^2$ on λ. The mass difference uses eq. 3.dvii, has size $C_L h\xi^{-2}$, and reads only the same pillar. Hence

$$\|(L_E - L_U)\lambda\|_{\mathcal{F}_z} \le C_L h\|\lambda\|_{\mathcal{Y}_z}. \tag{3.dxxxi}$$

No second difference of λ occurs: the difference of two connections in a site Laplacian is lower order. The first-order input on E is used here; its supremum alone would not bound the zeroth-order coefficient. The metric diameter of a pillar is bounded, so the profile weight passes through the mass average with a fixed factor.

For h below the resulting fixed threshold, the series

$$G'_E = \sum_{n\ge 0}[-G'_U(L_E - L_U)]^n G'_U \tag{3.dxxxii}$$

converges from $\mathcal{F}_z$ to $\mathcal{Y}_z$. At finite volume it coincides with the inverse already constructed in eq. 3.dxi, by uniqueness. It therefore proves the complex versions of the pointwise potential and gradient estimates. The equation supplies the complex covariant Laplacian bound. In the same real local frame, $e^{i\eta E}e^{i\eta A}$ has a physical logarithm with the first-order bounds $C(a_{\mathrm{loc}} + h)$. The scalar interior estimate then gives the complex Hölder member as well.

The argument establishes the site estimates for bounded cell sources. A mixed gradient with a divergence source has a Calderón–Zygmund singularity and requires its specified cancellation or Hölder input. No uniform bound for that mixed operator on all bounded raw sources has been inferred.

3.13.17. *The site Gram inverse has its own lower-bound proof.* Invertibility of G' does not by itself give a uniform inverse for $Q'G'^2Q'^\sharp$. We prove that estimate using site bump functions. Let Ξ multiply a top value by ξ_y.

Choose a fixed smooth scalar bump χ_y, supported in the middle portion of its pillar and equal to one on a smaller middle portion. Its first two differences are bounded by $C\xi_y^{-1}$ and $C\xi_y^{-2}$. The arithmetic average $\beta_y = Q'_0\chi_y$ has a positive lower bound independent of the fine mesh: count the grid points in that smaller cube. The finitely coarse cases use the center point.

Use the real local gauge eq. 3.dxxvi on a neighborhood of the pillar and its one-fine-cell collar. In this gauge the initial open bond bound is $Ca_{\mathrm{loc}}\eta/\xi_y$. The recurrence eq. 3.cdxciv, with the same real curvature bound, therefore puts the hierarchical transports within $C_L(a_{\mathrm{loc}} + \epsilon)$ of one. Consequently the three by three matrix $m_y\psi = Q'_U(\chi_y\psi)$ equals $\beta_y I + O(C_L(a_{\mathrm{loc}} + \epsilon))$. For the fixed small class ceiling it is invertible with uniform bound.

Define a site lift on that pillar by $\chi_y m_y^{-1}\psi(y)$ in the local gauge and undo the gauge. At the node use its transformed frame as well. The resulting operator E_{sm} satisfies

$$Q'_U E_{\mathrm{sm}} = 1, \qquad \|E_{\mathrm{sm}}\psi\|_2 \le C_L\|\psi\|_{2,\mathrm{tops}}, \qquad \|L_U E_{\mathrm{sm}}\psi\|_2 \le C_L\|\Xi^{-2}\psi\|_{2,\mathrm{tops}}. \tag{3.dxxxiii}$$

The first equality holds pillar by pillar, since the site averages have disjoint supports. For the last inequality, the covariant Laplacian of one bump is bounded by $C_L\xi_y^{-2}|\psi(y)|$. Expand it in the local gauge: the second cutoff differences, the connection times first differences, and the differentiated connection all have that bound by eq. 3.dxxvi. Its mass term is the transported vector $a(y)\psi(y)$ on the same pillar. The volume is ξ_y^4, giving the last L^2 scale in eq. 3.dxxxiii. One fine collar of different bumps has bounded overlap, since each fine site has a fixed number of neighbors. For a bare pillar, take the single site bump; its Laplacian is bounded by $C\eta^{-2}|\psi|$, which is exactly the same scale. Thus no missing local gauge at the outer bare boundary is used.

For a real top vector ψ, use eq. 3.dxxxiii with $\Xi^4\psi$ and the equation $L_U G'_U = 1$:

$$\begin{aligned}\|\Xi^2\psi\|_{2,\mathrm{tops}}^2 &= \langle (Q'_U)^\sharp\psi, E_{\mathrm{sm}}\Xi^4\psi\rangle \\ &= \langle G'_U(Q'_U)^\sharp\psi, L_U E_{\mathrm{sm}}\Xi^4\psi\rangle \\ &\le C_L\|G'_U(Q'_U)^\sharp\psi\|_2\|\Xi^2\psi\|_{2,\mathrm{tops}}.\end{aligned} \tag{3.dxxxiv}$$

Divide by the last node norm, with the zero case immediate. This proves the operator lower bound

$$Q'_U G'^2_U (Q'_U)^\sharp \ge c_L\Xi^4. \tag{3.dxxxv}$$

The Ξ^4 test in eq. 3.dxxxiv is essential. Testing only with $E_{\mathrm{sm}}\psi$ would give a weaker ratio of two norms and would not prove the stated weighted lower bound at different local scales.

For the upper bound and locality, set

$$\mathcal{F}_U = G'_U(Q'_U)^\sharp\Xi^{-2}, \qquad M_U = \Xi^{-2}Q'_U G'^2_U(Q'_U)^\sharp\Xi^{-2} = \mathcal{F}_U^\sharp\mathcal{F}_U. \tag{3.dxxxvi}$$

The site injection $(Q'_U)^\sharp$ is an isometry between node L^2 and its transported constant pillar fields. eq. 3.dxvi therefore bounds the normalized cell entry of $\mathcal{F}_U$ by

$$C_L(\xi_y/\xi_z)e^{-md(y,z)}. \tag{3.dxxxvii}$$

The same estimate holds for the complex $\mathcal{F}_E$, with the bounded transport condition numbers already proved. The cell summability calculation with $p = 1$ gives a uniform exponential two-sided Schur norm of these operators, after increasing the fixed separation if needed. Their products therefore have the same type of norm. In particular $c_L I \le M_U \le C_L I$ at a real pair.

The complex perturbation is uniform as well. The finite matrix $\mathcal{F}_E$ is holomorphic on the actual physical first-order ball constructed above. Its operator norm is uniformly bounded there by eq. 3.dxxxvii. Apply one scalar Cauchy circle to the line through E. This gives $\|\mathcal{F}_E - \mathcal{F}_U\| \le C_L h$, independent of the number of coordinates. Hence

$$\|M_E - M_U\|_{2\to2} \le C_L h, \qquad \|M_E^{-1}\|_{2\to2} \le 2/c_L \tag{3.dxxxviii}$$

on a fixed smaller ball. Here $M_E = \mathcal{F}_E^\sharp\mathcal{F}_E$ uses the analytic bilinear transpose. Its norm difference is bounded by $(\|\mathcal{F}_E\| + \|\mathcal{F}_U\|)\|\mathcal{F}_E - \mathcal{F}_U\|$. The transpose has the same Hilbert operator norm in the real weighted orthonormal bases. No Hermitian positivity of the complex Gram matrix is assumed.

An exponentially weighted Schur norm also controls its conjugation. For a bounded node function ψ that is one-Lipschitz in d, its cell entries change by the factor $e^{s(\psi(y)-\psi(z))} - 1$. The inequality

$$|e^{st} - 1| \le |s|\,|t|e^{|s||t|} \tag{3.dxxxix}$$

and a spare fraction of the exponential rate bound this perturbation by $C_L|s|$ in operator norm. Indeed $\sup_{r\ge0} re^{-\delta r/2} < \infty$. Choose s small and use eq. 3.dxxxviii. The inverse of the conjugated M_E is uniformly bounded. The distance-weight argument used for eq. 3.dxvi gives exponential entries of M_E^{-1}.

In the physical top pairing the weighted orthonormal basis at y is ξ_y^{-2} times the color basis. If B_{yz} are the entries of M_E^{-1} in those orthonormal coordinates, then $\|B_{yz}\| \le C_L e^{-m_2 d(y,z)}$. Undoing

both Ξ^{-2} factors and the orthonormal bases gives the counting kernel

$$\left|(Q'_E G'^2_E (Q'_E)^\sharp)^{-1}(y,z)\right| \le C_L \xi_y^{-4} e^{-m_2 d(y,z)}. \tag{3.dxl}$$

Explicitly the unnormalized entries of M_E^{-1} are $(\xi_z^2/\xi_y^2)B_{yz}$. Multiplying on the left and right by ξ_y^{-2}, ξ_z^{-2} leaves $\xi_y^{-4}B_{yz}$. This explains the asymmetric fourth power in a counting-kernel formula, despite symmetry in the physical weighted pairing.

Finally the actual site projector has the representation

$$R_E = 1 - \mathcal{F}_E M_E^{-1} \mathcal{F}_E^\sharp. \tag{3.dxli}$$

It agrees with its original Gram formula by eq. 3.dxxxvi. The smoothing part has an exponential cell L^2 Schur norm, and the matrix is holomorphic on the whole smaller physical ball. At a real field it is the orthogonal projector already identified. This completes the site Gram and projector part of the package. Fine vector coercivity, mixed derivative source norms and the current constraint terms are not consequences of eq. 3.dxli.

3.13.18. *The physical fine-to-coarse readout is a Hessian identity.* We next identify the three quadratic forms in the physical calculation. This is a finite-dimensional identity on any admitted chart; it precedes a uniform inverse estimate. Let U be the reference critical field, j_U its action current, Q the derivative of the actual straightened constraint, and ω its multiplier. Take a fixed reference slice, so that its linear condition is $RD_U^\sharp a = 0$.

Criticality on that slice already gives the full stationarity equation

$$j_U = Q^\sharp \omega. \tag{3.dxlii}$$

To see this, first note that $Q\mathsf{S} = Q$, $\mathsf{S}a$ is in the slice, and $a - \mathsf{S}a = D_U G' R D_U^\sharp a$. For $a \in \ker Q$, criticality annihilates $\mathsf{S}a$; the Ward identity annihilates its remaining gauge part. Thus j_U annihilates $\ker Q$. The finite-dimensional annihilator identity gives eq. 3.dxlii. When the row space is the actual independent datum space, Q is onto and ω is unique. Equivalently it is unique whenever the row Gram inverse used in eq. 3.cdlxxiii exists.

Because $Q\widehat{H} = 1$, eq. 3.dxlii implies $\widehat{H}^\sharp j_U = \omega$. Define the actual Lagrangian Hessian

$$K_{\mathrm{L}} = \mathscr{D}_U - \Delta^{(2)}[\omega]. \tag{3.dxliii}$$

The quadratic constraint term here is the second derivative of the same straightened constraint as Q, with the same per-length pairing. It cannot be replaced by the quadratic part of a different row map.

For the explicitly chosen Ward correction eq. 3.cdlxvii, the physical second variant is therefore

$$\widehat{G}_1[j_U]^{-1} = \mathsf{S}^\sharp K_{\mathrm{L}} \mathsf{S} + D_U R D_U^\sharp + Q^\sharp a Q. \tag{3.dxliv}$$

Let $W_1 = Q\widehat{G}_1[j_U]Q^\sharp$ and $H_1 = \widehat{G}_1[j_U]Q^\sharp W_1^{-1}$. The physical version of eq. 3.cdlxxiv gives $RD_U^\sharp H_1 = 0$. Consequently $\mathsf{S}H_1 = H_1$, and direct multiplication of eq. 3.dxliv gives

$$H_1^\sharp K_{\mathrm{L}} H_1 = W_1^{-1} - a. \tag{3.dxlv}$$

For example, $H_1^\sharp \widehat{G}_1^{-1} H_1 = W_1^{-1}$, the gauge contribution is zero, and the row mass contributes a, since $QH_1 = 1$. This proves all signs in eq. 3.dxlv. It does not prove the existence of either inverse.

Let the actual local critical branch be parametrized by a coarse variation B, with reduced datum $b(B)$. Its derivative h_B at zero is taken in the fixed reference slice. Differentiated stationarity gives the constrained linear equations for K_{L}, with row value $Db(0)\dot{B}$. They imply

$$h_B = H_1 Db(0), \qquad D^2\mathcal{A}_{\mathrm{eff}}(0) = Db(0)^\sharp (W_1^{-1} - a) Db(0) + \langle \omega, D^2 b(0) \rangle. \tag{3.dxlvi}$$

Here $\mathcal{A}_{\mathrm{eff}}$ is the action evaluated on that branch. For the first assertion, the differentiated equation tested against slice vectors in $\ker Q$ says $\mathsf{S}^\sharp K_{\mathrm{L}} h_B$ lies in the range of $Q^\sharp$. Adding $Q^\sharp a Q h_B$ and using eq. 3.dxliv then solves the row problem by its defining inverse. For the second, differentiate the

value function twice. The second derivative of the branch contracts with $j_U = Q^\sharp\omega$; differentiating its constraint replaces that contraction by $\omega D^2 b - \omega D^2 q[h_B, h_B]$. The latter term is exactly the subtraction in K_{L}.

3.13.19. *Which coarse coordinates remove the datum curvature.* The statement $b(B) = B$ has a specific coordinate meaning. In the present chart the actual coarse group field is

$$V(B)(c) = e^{i\xi_c B(c)} V_0(c). \tag{3.dxlvii}$$

The fine straightened row logarithm is relative to the same reference V_0. On an interior row the exact node identity then gives $b(B)(c) = B(c)$. It holds for nontrivial $V_0(c)$ as well. No assertion about the absolute logarithm of $V(B)(c)$ is involved.

An outward row instead has its prescribed factor at the finer outer endpoint. In raw notation its equation is

$$e^{i\xi_c b(B)(c)} = e^{i\xi_c B(c)}\,\mathrm{Ad}_{V_0(c)}\, v_z(B_{\mathrm{int}}). \tag{3.dxlviii}$$

It can have both linear and quadratic datum corrections. If only the highest active level is varied, the lower-level data B_{int} in this factor are fixed to zero. Then $v_z(0) = 1$ and the factor is constant. More generally the next-step support mask must be checked against every outward carrier. Deep interior rows give an immediate instance; a mask by itself does not change a logarithmic coordinate.

For contrast, if an absolute coordinate $W_c = \log(V_0(c))/i$ is used and the variable is still a relative increment D, its new absolute logarithm is $\log(e^{iD}e^{iW_c})/i$. Its second derivative in D generally does not vanish. eq. 3.dxlvi then retains its datum-curvature term. This distinction is independent of the support mask.

Let ι inject the actual active next-step coarse bond variables into the row space. When the literal map in eq. 3.dxlvii–eq. 3.dxlviii satisfies

$$Db(0)\iota = \iota, \qquad D^2 b(0)[\iota\,\cdot, \iota\,\cdot] = 0, \tag{3.dxlix}$$

its coarse action precision on those variables is $\iota^\sharp(W_1^{-1} - a)\iota$. Its coarse current is $\iota^\sharp H_1^\sharp j_U$. The restriction $\iota^\sharp$ and the physical dual weights are part of the readout, even when notation suppresses them.

3.13.20. *The next curved fibre inserts a different multiplier.* At the next lattice, write raw bond increments as D, so that the group field is $e^{iD}V$. The actual plain constraint logarithm is

$$\mathfrak{q}_V(D) = \log\Big(M(e^{iD}V)M(V)^{-1}\Big)/i. \tag{3.dl}$$

Its local pivot graph has derivative C_V, and the pivot injection h_V satisfies $D\mathfrak{q}_V(0)h_V = 1$. Twice differentiating the graph equation gives

$$D^2\Phi_V(0)[u,v] = -h_V D^2\mathfrak{q}_V(0)[C_V u, C_V v]. \tag{3.dli}$$

This is the actual pivot graph, not a free extension chosen after the Hessian.

Let $\widehat{K}$ and $\widehat{J}$ be the coarse effective action Hessian and gradient in these raw counting coordinates. The chain rule on that graph is

$$\begin{aligned} D^2(\mathcal{A}_{\mathrm{eff}} \circ \Phi_V)(0)[u,v] = {} & \langle C_V u, \widehat{K} C_V v\rangle \\ & - \langle h_V^\sharp \widehat{J}, D^2\mathfrak{q}_V(0)[C_V u, C_V v]\rangle. \end{aligned} \tag{3.dlii}$$

Thus the next constraint multiplier is $h_V^\sharp\widehat{J}$. It is not $\widehat{J}$ on the new row space unless the notation already includes this $h_V^\sharp$. Define $E_V^{(2)}[\widehat{J}]$ by this last contraction. With the actual soft gauge Hessian $\widehat{K}_{\mathrm{GF}}$, the next free precision is

$$C_V^\sharp(\widehat{K} - E_V^{(2)}[\widehat{J}] + \widehat{K}_{\mathrm{GF}})C_V. \tag{3.dliii}$$

A gauge term relative to V and the absolute contour gauge have equal values on an axial physical reference. Off that locus their Hessians can differ, as shown in the domain section. eq. 3.dliii uses the actual gauge definition of the model being considered.

For a uniform next-lattice spacing ξ, the raw and physical conversions are

$$\widehat{K} = \xi^2 K_{\mathrm{phys,count}}, \qquad \widehat{J} = \xi^3 J_{\mathrm{phys}}. \tag{3.dliv}$$

Indeed $\xi^4 \sum B(KB) = \xi^2 \sum D(KD)$ when $D = \xi B$, and $\xi^4 \sum J_{\mathrm{phys}} B = \sum \xi^3 J_{\mathrm{phys}} D$. At nonuniform row lengths the corresponding diagonal length matrices must be kept on both sides; their commutation is not assumed. The coupling coefficients in the original action remain inside K and J.

Combining eq. 3.dxlv, eq. 3.dxlix and eq. 3.dlii gives the physical diagram: fine Lagrangian Hessian, coarse value Hessian, then next-fibre Hessian. Each arrow has its own row map, its own dual current, and its own gauge term. The physical identity proves none of those three objects interchangeable.

For completeness the chosen off-current extension has an exact additional readout. Put $h = H_1[J]$, $r = RD_U^\sharp h$, $\delta J = J - j_U$, and $K[J] = \mathscr{D}_U - \Delta^{(2)}[\widehat{H}[J]^\sharp J]$. Using eq. 3.cdlxviii in eq. 3.cdlxxvi gives

$$(Q\widehat{G}_1[J]Q^\sharp)^{-1} - a = h^\sharp \mathsf{S}^\sharp K[J] \mathsf{S} h - h^\sharp \mathcal{B}_U[\delta J] h + r^\sharp r. \tag{3.dlv}$$

All pairings are bilinear on the complex domain. On a real pair the last term is nonnegative; it vanishes on the physical graph. The middle term is a current defect and does not disappear from a Ward property alone. For a separately named normal form, the complete projected difference between eq. 3.dliii and that form must be identified, including its horizontal multiplier and gauge terms.

3.13.21. *A flat Feynman comparison on all the layers.* There is a useful vector estimate which does not involve the Landau projection or an independent current. It uses the actual flat straightened averages. We prove it to specify exactly what is available before those two additional operations.

First take the full periodic fine lattice as the ambient graph. Let $a_\mu(x)$ be a fine vector one-form, with physical L^2 measure η^4. For a positive level-j row of length $\xi_j = n\eta$, $n = L^j$, its flat per-length average is

$$(Q_j^0 a)_\mu(y) = n^{-5} \sum_{u \in \mathcal{B}_n} \sum_{t=0}^{n-1} a_\mu(y + \eta u + \eta t e_\mu). \tag{3.dlvi}$$

Here $\mathcal{B}_n$ is the centered set of n^4 fine block sites. The formula is the flat straightened map, not the plain path map. At $j = 0$ it is the identity. The coefficient of each fine bond is nonnegative, the coefficients sum to one, and each is at most n^{-4}. The last fact follows because at most n pairs (u_μ, t) give a fixed longitudinal position, while the three other coordinates are fixed.

Its support lies in the union of the two endpoint blocks. Consequently

$$|Q_j^0 a(c)| \le C \xi_j^{-2} \|a_\mu\|_{2,\mathcal{D}_c}. \tag{3.dlvii}$$

For example Cauchy gives $\sum_b w_b^2 \le n^{-4}$ from $\sum_b w_b = 1$ and $w_b \le n^{-4}$; converting counting L^2 to physical L^2 contributes η^{-2}. Thus the resulting coefficient is $\eta^{-2} n^{-2} = \xi_j^{-2}$. This bound is specific to this flat row formula.

For a fine bond b, put ξ_b equal to the larger of its endpoint pillar scales and define

$$\|a\|_{\mathrm{mass}}^2 = \sum_b \eta^4 \xi_b^{-2} |a(b)|^2, \qquad \mathcal{E}_{\mathrm{F}}^0(a) = \|\partial a\|_2^2 + \sum_{c \in \Gamma} c_c \xi_c^2 |Q^0 a(c)|^2, \quad 0 < c_* \le c_c \le c^*. \tag{3.dlviii}$$

The component gradient contains all fine coordinate differences on the ambient torus. The second term is exactly the physical row pairing with $a(c) = c_c \xi_c^{-2}$. In particular it has neither an extra fine volume factor nor an unweighted counting transpose.

For each nonbare top y and each direction μ, one of the adjacent level-j rows has y as an endpoint and is a determining row. Indeed the excluded higher region is a union of aligned blocks of side at least $L\xi_j$. Along that coordinate line, the excluded block containing an exterior point has at least $L \geq 3$ level-j sites. Both neighboring sites cannot belong to the higher region. Choose one that does not. If the other endpoint is outside Ω_j, this is precisely an outward row, which is retained. At the highest level there is no higher region to exclude.

Let $m_{y,\mu}$ be the arithmetic mean of a_μ on the pillar enlarged by one fine cell. A rectangular carrier containing that enlarged pillar and the chosen row's two endpoint blocks has side length at most $C\xi_j$. For $j \geq 1$ it lies inside the ambient fine graph by the nesting gap. The ordinary box Poincare inequality, proved by the coordinate-path count used for eq. 3.cdxcvi, gives

$$|Q_j^0 a(c(y,\mu)) - m_{y,\mu}| \leq C\xi_j^{-1} \|\partial a_\mu\|_{2,\mathcal{D}_{y,\mu}}. \tag{3.dlix}$$

To verify this without a pointwise trace estimate, subtract the carrier's arithmetic mean. Both functionals annihilate that constant in their difference and have L^2 norm at most $C\xi_j^{-2}$. Poincare then supplies one factor ξ_j.

A second use of Poincare on the enlarged pillar gives

$$\xi_j^{-2} \|a\|_{2,\mathcal{B}_y^+}^2 \leq C \sum_\mu \|\partial a_\mu\|_{2,\mathcal{D}_{y,\mu}}^2 + C\xi_j^2 \sum_\mu |Q_j^0 a(c(y,\mu))|^2. \tag{3.dlx}$$

Every fine bond with a nonbare endpoint can be assigned to a pillar of its larger endpoint scale and is included in that pillar's enlargement. If both endpoints are bare, its level-zero row is the identity and controls it directly.

The overlap in the gradient sum is uniform. For a fixed level, all carrier centers are on its grid and their diameters are $C\xi_j$, so a fine point belongs to a fixed number of them. A carrier of an outward level-j row is inside Ω_{j-1}, and its endpoint blocks are outside Ω_{j+1}. Its one-fine-cell collar can enter only an adjacent level. Thus a fixed fine point is counted at only a fixed number of levels, not at every level of the sequence. Each chosen row has at most its two endpoints as selecting tops. Summing eq. 3.dlx therefore proves

$$\|a\|_{\mathrm{mass}}^2 \leq C_L \mathcal{E}_{\mathrm{F}}^0(a). \tag{3.dlxi}$$

The constants are independent of depth, fine mesh, volume and the shapes of the separated regions. Periodic carriers which wrap are lifted to their rectangles; the compatible torus has at least two top blocks, so the multiplicity of that lift is bounded.

There is also the upper row estimate

$$\sum_{c \in \Gamma} \xi_c^2 |Q^0 a(c)|^2 \leq C_L \|a\|_{\mathrm{mass}}^2. \tag{3.dlxii}$$

Use eq. 3.dlvii on each row carrier. All fine endpoint scales there are comparable to ξ_c, within the fixed adjacent scale factor. The preceding overlap count then applies. eq. 3.dlxi and eq. 3.dlxii are estimates for the flat component Feynman form.

For a proper active region, extend the sequence by bare cells on the ambient torus and compress this full positive form by zero extension of its active bond coordinates. Its lower bound survives exactly: apply eq. 3.dlxi to the extended vector. This is the Dirichlet compression of the component Laplacian. An inner-plaquette Maxwell Hessian has a different boundary form. Nor has the nonlocal subtraction $D_U(1-R)D_U^\sharp$ been included. Those comparisons are separate in the vector precision problem.

3.13.22. *What small curvature does to the actual Wilson Hessian.* The Wilson part admits a relative form estimate without assuming a matrix norm bound. We derive it directly from the plaquette product. Use $\langle X, Y \rangle_{\mathrm{col}} = \frac{1}{2}\,\mathrm{Tr}(XY)$ on Hermitian traceless matrices. For a plaquette with oriented bonds $b_1, \ldots, b_4$, put $P = U(b_1)\cdots U(b_4)$ and $T_{r-1} = U(b_1)\cdots U(b_{r-1})$. A left variation $e^{i\eta v(b)}U(b)$

inserts $X_r = \eta \operatorname{Ad}(T_{r-1})v(b_r)$. On reversed bonds the variation is $v(\bar{b}) = -\operatorname{Ad}(U(b)^{-1})v(b)$, as follows by inverting the group field.

The varied plaquette is $\prod_{r=1}^{4} e^{iX_r} P$. At $P = 1$, the trace of the mixed second derivative is the trace of the product of the two sums of insertions. Cyclicity of the trace cancels their commutators. For general real P, subtract that calculation and use the factor $P - 1$ in every remaining term. Thus the plaquette action $1 - \frac{1}{2}\operatorname{Tr} P$ satisfies

$$\begin{aligned} D^2\mathcal{A}_p(U)[v,w] &= \left\langle \sum_r X_r, \sum_r Y_r \right\rangle_{\mathrm{col}} + \mathcal{R}_p(v,w), \\ |\mathcal{R}_p(v,w)| &\le C\|P-1\| \left(\sum_r |X_r|^2\right)^{1/2} \left(\sum_r |Y_r|^2\right)^{1/2}. \end{aligned} \tag{3.dlxiii}$$

There are only sixteen mixed insertion terms. Unitarity of the prefixes makes $\sum |X_r|^2 = \eta^2 \sum |v(b_r)|^2$. The same polynomial trace expression, with no real-part operation, is its holomorphic continuation.

Define the plaquette linearized curl by

$$(\mathfrak{d}_U v)(p) = \eta^{-1} \sum_{r=1}^{4} \operatorname{Ad}(T_{r-1})v(b_r). \tag{3.dlxiv}$$

Then $\sum_r X_r = \eta^2 \mathfrak{d}_U v(p)$. If $\|P-1\| \le \epsilon\eta^2\xi_p^{-2}$, with ξ_p the largest corner scale, summing eq. 3.dlxiii over the actual plaquettes gives

$$|\langle v, \mathscr{D}_U w\rangle - \langle \mathfrak{d}_U v, \mathfrak{d}_U w\rangle_2| \le C_L\epsilon \|v\|_{\mathrm{mass}} \|w\|_{\mathrm{mass}}. \tag{3.dlxv}$$

Every bond is in at most six unoriented plaquettes. Each error has the factor $\eta^2\epsilon\xi_p^{-2}$ from $P-1$ and another η^2 from the two insertions. Their product is the physical η^4 measure times the natural inverse scale squared. There is no loss $\epsilon\eta^{-2}$ in this form estimate.

This curl is close to the ordinary forward covariant curl $\mathrm{d}_U v = \nabla_{U,\mu} v_\nu - \nabla_{U,\nu} v_\mu$. For example on a $\mu\nu$ plaquette the difference of their numerator sums is

$$(1 - \operatorname{Ad} P)\big(\operatorname{Ad} U_\nu(x) v_\mu(x + \eta e_\nu) + v_\nu(x)\big). \tag{3.dlxvi}$$

Direct multiplication of the four prefixes proves this identity. It gives $\|\mathfrak{d}_U v - \mathrm{d}_U v\|_2 \le C_L\epsilon\|v\|_{\mathrm{mass}}$, since $\eta \le \xi_p$. In particular

$$\langle v, \mathscr{D}_U v\rangle \ge (1 - C_L\epsilon)\|\mathrm{d}_U v\|_2^2 - C_L\epsilon\|v\|_{\mathrm{mass}}^2 \tag{3.dlxvii}$$

for a real field, after decreasing the fixed curvature ceiling. Positive bounded plaquette weights give the same statement with their lower and upper bounds. Signed layer profiles or additional current subtractions must be estimated separately.

On the full periodic graph, the covariant Hodge calculation is also explicit:

$$\left| \|\mathrm{d}_U v\|_2^2 + \|D_U^\sharp v\|_2^2 - \sum_\mu \|\nabla_{U,\mu} v\|_2^2 \right| \le C_L\epsilon\|v\|_{\mathrm{mass}}^2. \tag{3.dlxviii}$$

Expand both squares. The diagonal terms agree because each covariant shift is unitary. For different directions the uncancelled terms are commutators of the two shifts, or of a shift and an inverse shift. Their coefficients are conjugates of $\operatorname{Ad} P - 1$. The two difference denominators contribute η^{-2}; the plaquette bound leaves $C\epsilon\xi^{-2}$. There are a fixed number of neighboring terms at each bond. Adjacent scales are comparable, so Cauchy proves eq. 3.dlxviii. On a proper inner graph the summation by parts has additional boundary terms.

The full Feynman gauge uses $D_U D_U^\sharp$. The actual fine Landau form instead uses $D_U R D_U^\sharp$. Its difference is the nonnegative real smoothing form

$$D_U(1 - R)D_U^\sharp = (D_U\mathcal{F}_U)M_U^{-1}(D_U\mathcal{F}_U)^\sharp. \tag{3.dlxix}$$

A lower bound for the Feynman form does not survive this subtraction by monotonicity. This is one reason the vector coercivity step remains distinct from the site gap and eq. 3.dlxi.

For the admitted complex perturbation E, the actual plaquette Taylor formula gives $\|P(U_E) - P(U)\| \le C_L h\eta^2 \xi_p^{-2}$. Indeed the first order boundary sum of E is its covariant curl; the remaining terms contain either two increments or one increment times a real connection on a local gauge patch. The first-order core bounds control each by that scale. Also $\|\mathrm{d}_{U_E} v - \mathrm{d}_U v\|_2 \le C_L h \|v\|_{\mathrm{mass}}$. Subtracting eq. 3.dlxiii therefore proves

$$|\langle v, (\mathscr{D}_{U_E} - \mathscr{D}_U) w\rangle| \le C_L h (\|\mathrm{d}_U v\|_2 + \|v\|_{\mathrm{mass}})(\|\mathrm{d}_U w\|_2 + \|w\|_{\mathrm{mass}}). \tag{3.dlxx}$$

This uses a physical first-order perturbation radius, not an unscaled matrix norm for the fine precision. It is a genuine part of the complex comparison. The corresponding actual field-row, projected-current and constraint form comparisons remain to be joined to it.

3.13.23. *A mixed derivative needs cancellation.* One cannot replace those comparisons by a universal supremum bound for every order-zero projection. Already at a flat field the operator $\partial_1\partial_2(-\Delta+1)^{-1}$ is not bounded on L^∞. Its kernel near zero is $c\, x_1 x_2 |x|^{-6}$ plus a locally integrable term in four dimensions. Let f_η equal the sign of this leading kernel on the annulus $C\eta < |x| < 1/4$, with a bounded smooth cutoff at its edges. Then

$$\|f_\eta\|_\infty \le 1, \qquad \left|(\partial_1\partial_2(-\Delta+1)^{-1} f_\eta)(0)\right| \ge c \log(\eta^{-1}) - C. \tag{3.dlxxi}$$

The radial integral is $\int_{C\eta}^{1/4} dr/r$; the angular integral of $|x_1 x_2|/|x|^2$ is positive. The integrable remainder is uniformly bounded. Discrete kernels have the same annular estimate away from a fixed number of fine cells, or the continuum example follows as a limit of any proposed uniform lattice supremum estimate.

The required replacement has a precise cancellation form. If a kernel $K(x,z)$ has size $C|x-z|^{-4}$ on a cell of scale ξ, and a uniformly bounded constant-source response, write its action as

$$\int K(x,z)(f(z) - f(x))\, dz + f(x) \int K(x,z)\, dz. \tag{3.dlxxii}$$

For a transported Hölder test, the first term is bounded by

$$C[f]_\alpha \int_0^{C\xi} r^{\alpha-1}\, dr \le (C_\alpha \xi^\alpha)[f]_\alpha. \tag{3.dlxxiii}$$

The same shell calculation applies to fine sums, with the first shell at η; the bound remains uniform as its lower endpoint decreases. The second term is the stated constant-source response. Thus an actual mixed-kernel estimate plus its cancellation clause gives the required Hölder-to-supremum bound. Size alone does not.

For the fine gauge projection $\mathsf{S} = 1 - D_U G' R D_U^\sharp$, this distinction is material. Its differentiated site inverse has order zero at the fine diagonal. Energy estimates use its L^2 form bounds; point estimates must either keep it composed with a smoothing field Green operator or use actual Hölder data and eq. 3.dlxxii. No unqualified supremum Schur norm of S on arbitrary bounded fine fields is being asserted.

3.13.24. *Compact dual bumps for the actual flat field rows.* We can also prove a uniform right inverse for the flat layered field rows. This is a different statement from the site lift eq. 3.dxxxiii. The proof uses two longitudinal moments and a strictly triangular correction between adjacent levels.

Fix a top y of level $j \ge 1$, put $n = L^j = 2m+1$, and work on its fine block $\eta\mathcal{B}_n + y$. Choose a field with just its μ-component nonzero there:

$$a_\mu(y + \eta u) = \psi\, \phi(u_\mu) \prod_{\nu \ne \mu} \chi(u_\nu). \tag{3.dlxxiv}$$

It is zero outside the block and on bonds whose other endpoint is outside. For sufficiently large n, take $\chi(u)$ from a fixed even smooth profile of u/n, supported in $(-1/4, 1/4)$ and positive on a smaller interval. Write

$$\beta = n^{-1}\sum_u \chi(u), \qquad \sigma = n^{-1}\sum_u (u/n)^2\chi(u). \tag{3.dlxxv}$$

Both have positive lower and finite upper bounds independent of n.

For a fine longitudinal position u in this block, the right coarse row in eq. 3.dlvi counts that bond $u+m+1$ times. The left row counts it $m-u$ times. No other coarse row sees it. After the transverse sums, their two response functionals are

$$\ell_+(\phi) = \beta^3 n^{-2}\sum_u (u+m+1)\phi(u), \qquad \ell_-(\phi) = \beta^3 n^{-2}\sum_u (m-u)\phi(u). \tag{3.dlxxvi}$$

Define

$$\begin{aligned} \phi_+(u) &= \beta^{-4}\chi(u) + \frac{m}{n}\frac{\beta^{-3}}{\sigma}\frac{u}{n}\chi(u), \\ \phi_-(u) &= \beta^{-4}\chi(u) - \frac{m+1}{n}\frac{\beta^{-3}}{\sigma}\frac{u}{n}\chi(u). \end{aligned} \tag{3.dlxxvii}$$

Evenness gives $\sum u\chi(u) = 0$. Substitution into eq. 3.dlxxvi now gives

$$(\ell_+(\phi_+), \ell_-(\phi_+)) = (1, 0), \qquad (\ell_+(\phi_-), \ell_-(\phi_-)) = (0, 1). \tag{3.dlxxviii}$$

For example the sum of the two functionals is $\beta^3 n^{-1}\sum\phi$. For either profile this is one; the odd term in ϕ_+ sets the left functional to zero. The right one is then one. The calculation for ϕ_- is the same with the other moment. The signed profiles are uniformly bounded and their first differences are $O(n^{-1})$.

The finitely many odd values below this smooth-profile range need no limiting argument. Put the three transverse profiles at their center point and use longitudinal bonds with positions $-1, 0$. Both endpoints of these bonds are in the block for $n \geq 3$. Their two response columns form

$$n^{-5}\begin{pmatrix} m & m+1 \\ m+1 & m \end{pmatrix}. \tag{3.dlxxix}$$

Its determinant is $-n^{-9}$. Its inverse therefore gives the required two exact dual profiles, with a bound depending only on the fixed cutoff between this finite set and the smooth range. That is a universal constant, not a depth dependence.

Assign a determining row c to one of its level-j top endpoints y. An outward row has its interior endpoint as that top. Use ϕ_+ or ϕ_- according to the chosen endpoint. Its seed $E_c^0\psi$ is supported on fine bonds strictly inside that pillar and satisfies, on the full regular level-j grid,

$$Q_j^0 E_c^0\psi = \delta_c\psi. \tag{3.dlxxx}$$

The transverse support lies in a single coarse transverse block, and eq. 3.dlxxviii removes the only other longitudinal row. For a bare row take its single fine bond value. It has both endpoints in the bare layer.

Let $E^0\psi$ be the sum of these seeds at every level. There are at most eight assigned seeds at a top. Their supports belong to distinct pillars, and the collars in their gradient norm have bounded overlap. The amplitude and first-difference bounds just proved give

$$\|E^0\psi\|_{\text{mass}}^2 + \|\partial E^0\psi\|_2^2 \leq C\sum_{c\in\Gamma}\xi_c^2|\psi(c)|^2. \tag{3.dlxxxi}$$

The factor on the right is forced by the dimensions: the supporting volume is ξ_c^4, and a first difference or mass factor is ξ_c^{-1}. For a bare seed the same calculation is $\eta^4\eta^{-2}|\psi|^2$.

3.13.25. *A contraction estimate between adjacent levels.* A seed of level j has zero value under every lower determining row. Indeed its fine support is in Ω_j, whereas a lower row's entire carrier is outside the next region of that lower level. For $i < j$, that next region contains Ω_j. It also has zero value under every row of level $i \geq j+2$: such a carrier is inside $\Omega_{i-1} \subset \Omega_{j+1}$, whereas the seed pillar is outside Ω_{j+1}. The same argument uses the two bare endpoints for $j = 0$. Thus only the next level can receive an unwanted value.

The full flat averages compose exactly:

$$Q_{j+1}^{0,\mathrm{full}} = Q_L^0 Q_j^{0,\mathrm{full}}. \tag{3.dlxxxii}$$

In the defining sums, the transverse digits are $nr + u$, where r is in the centered L-range and u in the centered n-range. They tile the centered Ln-range. The longitudinal digits are $nt + s$, with $0 \leq t < L$, $0 \leq s < n$. They tile $0, \dots, Ln-1$. The product normalization is $L^{-5}n^{-5}$. This proves eq. 3.dlxxxii, including the per-length normalization.

Let the row Hilbert norm be $\|\psi\|_{\mathrm{row}}^2 = \sum_c \xi_c^2 |\psi(c)|^2$. A regular L-step average has row coefficient sum one and column coefficient sum L^{-4}. The column identity can be checked directly: the two longitudinal counts add to L, while the three transverse coordinates select one block. The ordinary counting ℓ^2 operator norm is therefore at most L^{-2} by Schur. The row norm at the next level has one extra factor L in its square root. Hence the map between these row norms has norm at most L^{-1}.

Writing T for the next-level leakage, eq. 3.dlxxx–eq. 3.dlxxxii give

$$Q^0 E^0 = 1 + T, \qquad T_{ij} = 0 \ (i \neq j+1), \qquad \|T\|_{\mathrm{row}\to\mathrm{row}} \leq L^{-1}. \tag{3.dlxxxiii}$$

Restriction to actual determining rows and injection from their subset of the regular grid cannot increase the norm. Different level blocks of T have different input and output levels, so the whole direct-sum norm is the supremum of those block norms. No sum over the number of levels is introduced.

We have proved the explicit flat row lift

$$E_{\mathrm{row}}^0 = E^0(1+T)^{-1}, \qquad Q^0 E_{\mathrm{row}}^0 = 1, \qquad \|E_{\mathrm{row}}^0 \psi\|_{\mathrm{mass}}^2 + \|\partial E_{\mathrm{row}}^0 \psi\|_2^2 \leq \frac{C}{(1-L^{-1})^2} \|\psi\|_{\mathrm{row}}^2. \tag{3.dlxxxiv}$$

At finite depth the inverse is the finite triangular series; its norm is bounded by the displayed geometric series. The same series converges in the row Hilbert space for an infinite compatible sequence. The assertion is a right inverse with an energy bound, not an estimate of a nonlinear background map.

At a real flat connection the same construction uses a flat unitary frame on each pillar. The response at the other coarse endpoint differs by its fixed unitary transport. The two scalar moments still annihilate the unwanted row; the desired row is normalized in its own color frame. Schur uses the same positive scalar coefficient majorants, so its constants are unchanged. All profile supports lie inside their pillars. This handles periodic flat holonomies without choosing a single globally small logarithm. Under this gauge transfer the gradient in the energy bound is ∇_U, or equivalently the ordinary component gradient in its flat parallel frame. An ordinary gradient in an arbitrary original gauge is not bounded by that statement. No analogous bare L^2 field-row estimate at a general nonflat field has been assumed.

3.13.26. *What a field-row Gram estimate actually needs.* The last proof gives the flat instance of a general useful duality reduction. Let K be a real symmetric positive fine precision and $G = K^{-1}$. Let Ξ multiply each field row by its length. Suppose the actual row map has an extension E_{row} such that

$$Q E_{\mathrm{row}} = 1, \qquad \langle E_{\mathrm{row}}\psi, K E_{\mathrm{row}}\psi \rangle \leq C \|\Xi^{-1}\psi\|_{2,\mathrm{rows}}^2. \tag{3.dlxxxv}$$

The node pairing here has volume ξ_c^4; its last norm is just $\sum_c \xi_c^2|\psi(c)|^2$. Testing with $E_{\mathrm{row}}\Xi^2\psi$ gives

$$\|\Xi\psi\|_{2,\mathrm{rows}}^2 = \langle Q^\sharp\psi, E_{\mathrm{row}}\Xi^2\psi\rangle \le C^{1/2}\langle\psi, QGQ^\sharp\psi\rangle^{1/2}\|\Xi\psi\|_{2,\mathrm{rows}}. \tag{3.dlxxxvi}$$

Thus $QGQ^\sharp \ge C^{-1}\Xi^2$. No pointwise delta-function test in the fine field was used.

An upper bound needs the separate actual trace estimate

$$\|\Xi^{-1}Qa\|_{2,\mathrm{rows}}^2 \le C\langle a, Ka\rangle. \tag{3.dlxxxvii}$$

Indeed apply it to $a = G^{1/2}f$. It proves $\|\Xi^{-1}QG^{1/2}\| \le C^{1/2}$ and therefore $QGQ^\sharp \le C\Xi^2$. For the flat Feynman form, eq. 3.dlxii, eq. 3.dlxi and eq. 3.dlxxxiv prove both premises. For the flat Landau form, eq. 3.dlxxxiv proves the upper energy cost in eq. 3.dlxxxv, since $0 \le R \le 1$; the needed fine coercivity is still a distinct step.

Normalize the row covariance as $N = \Xi^{-1}QGQ^\sharp\Xi^{-1}$. If its exponential cell Schur bound has also been proved, the conjugation argument in eq. 3.dxxxix applies verbatim to N. It gives exponential entries of N^{-1}. In physical weighted orthonormal row coordinates, undoing normalization gives the counting kernel estimate

$$|(QGQ^\sharp)^{-1}(c,d)| \le C\,\xi_c^{-3}\xi_d\, e^{-m_3 d(c,d)}. \tag{3.dlxxxviii}$$

To check the powers, an orthonormal entry of N^{-1} becomes (ξ_d^2/ξ_c^2) times that entry on unnormalized values. The two exterior Ξ^{-1} factors leave $\xi_c^{-3}\xi_d$. This differs from the ξ_y^{-4} in eq. 3.dxl: the site Gram contains G'^2, whereas this row covariance contains G.

Finally a nonflat row extension is reduced to a definite error estimate. Starting from a specified flat or locally flat extension E_0, set $D_Q = QE_0 - 1$. If the actual primitive comparison proves

$$\|D_Q\|_{\mathrm{row}\to\mathrm{row}} \le \vartheta < 1, \tag{3.dlxxxix}$$

then $E_{\mathrm{row}} = E_0(1 + D_Q)^{-1}$ is an actual right inverse, with its energy cost multiplied by $(1-\vartheta)^{-2}$, provided that cost for E_0 has been proved in the actual precision. The remaining task is eq. 3.dlxxxix for the full actual row map on the admitted domain, together with eq. 3.dlxxxvii; it is not an assumed bound for the desired Green kernel.

3.13.27. *The actual straightened row is a finite analytic recursion.* We now obtain the needed row coefficient bounds from the actual formula, rather than from a proposed inverse. Use raw one-step link logarithms on a lattice whose spacing has been set to one. Let $T_{yx}(W)$ be the internal mean transport of the block chart, with its specified coordinate-order path multiset. For a reference W and a raw increment a, put

$$\begin{aligned} u_W(a)(y) &= \exp\left[L^{-4}\sum_{x\in B_L(y)} \log\Big(T_{yx}(e^{ia}W)T_{yx}(W)^{-1}\Big)\right], \\ F_W(a)(c) &= \frac{1}{i}\log\Big[u_W(a)(c_-)^{-1}M(e^{ia}W)(c)u_W(a)(c_+)M(W)(c)^{-1}\Big]. \end{aligned} \tag{3.dxc}$$

All logarithms are the regular branch. The first line is an exponential of averaged relative logarithms; it is not another unspecified mean. The internal transports themselves use the actual Federbush mean defined in the block chart. This distinction fixes both levels of averaging.

For $\|\log W/i\|_\infty$ and $\|a\|_\infty$ below c/L, these are holomorphic finite functions. Every path has at most CL bonds. Products of their exponentials remain in a fixed logarithmic neighborhood; the mean contraction proved in the block chart is uniform in the number of paths. After a fixed decrease of c, the expression inside the second logarithm remains in that same neighborhood. Thus $\|F_W(a)\|_\infty \le C$ on this joint raw ball, with universal C.

Differentiating at $W = 1, a = 0$ cancels the two internal path sums from the plain average against the two factors u. The remaining sum is exactly the straight segment average S_L, with raw row sum L and raw column sum L^{-3}. It is essential to use both factors in eq. 3.dxc; the plain average has a different derivative.

Let $A_W(a) = D_a F_W(a)$. Two scalar Cauchy circles, in arbitrary sup-norm directions of the reference logarithm and of a, give

$$\|A_W(a)\|_{\mathrm{row}} \le L(1 + C_L\delta), \quad \|A_W(a)\|_{\mathrm{col}} \le L^{-3}(1 + C_L\delta),$$
$$\sup_b \sum_{c,b'} \|D_b D_{b'} F_W(a)(c)\| \le C_L, \qquad \delta = \|\log W/i\|_\infty + \|a\|_\infty. \tag{3.dxci}$$

Here and below the norm of a color entry is its finite matrix norm. We explain why the derivative estimate is independent of a field-coordinate count. The two circles bound a second directional derivative by CL^2. For a fixed output color row, its operator norm from bounded input colors controls the sum of its entry norms, up to the fixed color dimension: sum the dual norms of its three scalar rows. A fixed input bond affects only the coarse rows at the endpoints of the one or two blocks containing it. Their number is fixed. Apply the same argument with one input bond fixed to obtain the last line. Dividing the row and column errors by their flat values costs only powers of L. For definiteness, all one-step constants in eq. 3.dxci may be enlarged to CL^{12}. This deliberately loose polynomial bound is before any number of levels is introduced.

The multilevel recursion is literal. Write $W_r = M^r(W_0)$, $a_0 = \eta A$, and

$$a_{r+1} = F_{W_r}(a_r), \qquad v_{r+1} = v_r|_{\mathrm{next}}\, u_{W_r}(a_r). \tag{3.dxcii}$$

Equivariance of the plain mean gives, by induction,

$$M^r(e^{i\eta A} W_0) = (e^{ia_r} W_r)^{v_r}. \tag{3.dxciii}$$

Indeed apply M to the previous factorization, restrict its gauge to the next nodes, and then use eq. 3.dxc. The gauge factors multiply in the order in eq. 3.dxcii. They do not alter the recursion for a_r.

3.13.28. *Summability of the raw radii gives a depth-uniform row bound.* For the row estimates from this point on, ϵ is a common ceiling for the real curvature and the local real-gauge bounds. Thus the earlier a_{loc} and the curvature radius are both at most ϵ; when they are supplied separately, take their maximum here. Existence of those local gauges remains the primitive regular-pair hypothesis. No bound for a_{loc} is inferred from curvature alone.

Fix a row of length $\xi = n\eta$, $n = L^j$. On its determining carrier, the regular-pair gauge clause gives a real local potential and its first difference at scale ξ, with at most the fixed adjacent-layer factor at a seam. The open-bond and curvature recurrences eq. 3.cdxciv, eq. 3.diii therefore give

$$\|\log W_r/i\|_\infty \le C_L(\epsilon + h)t_r, \qquad t_r = L^r/n, \qquad r < j. \tag{3.dxciv}$$

These estimates are used only on the dependence cone of that row. They also follow by extending its local small potential with a cutoff on a fixed larger cube and then using locality. The cutoff keeps its first difference at the same scale. The torus room required for nonflat local classes permits this cube; the flat periodic formulas need no such extension. No gauge bound growing with M_1 is used.

Allow an arbitrary fine increment with $\xi\|A\|_\infty \le r$. It need not have small fine derivatives. On the recursion ball, eq. 3.dxci and $F_W(0) = 0$ give

$$\|a_{r_0}\|_\infty \le 2r\, t_{r_0} \quad (0 \le r_0 \le j) \tag{3.dxcv}$$

when $\epsilon + h + r$ is below a fixed threshold depending on L. To avoid confusing the index with the radius, write $p_s = \|a_s\|_\infty/t_s$. As long as $p_s \le 2r$, Taylor's formula and eq. 3.dxci imply

$$p_{s+1} \le p_s[1 + C_L t_s(\epsilon + h + p_s)]. \tag{3.dxcvi}$$

The sum of t_s is below $(L-1)^{-1}$. Choose the fixed ceiling so that $C_L(\epsilon + h + 2r)/(L-1) < \log 2$. Multiplying the factors proves eq. 3.dxcv by first exit. The raw radii also stay strictly inside the c/L one-step balls. Finally $\|u_{W_s}(a_s) - 1\| \le C_L\|a_s\|_\infty$; their condition numbers have product at

most $\exp(C_L \sum_s \|a_s\|_\infty)$. A further fixed decrease makes this at most two, supplying the room for the complex covariance in eq. 3.dxciii.

Put $A_s = D_a F_{W_s}(a_s)$ and $P_{b,a} = A_{b-1} \cdots A_a$. The same summable radii in eq. 3.dxci give

$$\|P_{b,a}\|_{\mathrm{row}} \le C L^{b-a}, \qquad \|P_{b,a}\|_{\mathrm{col}} \le C L^{-3(b-a)}. \tag{3.dxcvii}$$

The constant is bounded by the product of $1 + C_L(\|\log W_s/i\|_\infty + \|a_s\|_\infty)$. It is independent of a, b, j. Multiplying only uniform bounds larger than one at each level would miss this point.

The per-length row logarithm is $q_j(A) = a_j/\xi$. Its derivative with respect to the physical fine coordinates is $(\eta/\xi)P_{j,0} = n^{-1}P_{j,0}$. Consequently

$$\|Dq_j(A)\|_{\mathrm{row}} \le C, \qquad \|Dq_j(A)\|_{\mathrm{col}} \le Cn^{-4}, \qquad |D_b q_j(A)(c)| \le C(\eta/\xi)^4. \tag{3.dxcviii}$$

These are counting-coordinate derivatives. A functional derivative uses the additional η^{-4}. Telescoping the products against their flat factors also gives row error $C_L(\epsilon + h + r)$ and column error $C_L(\epsilon + h + r)n^{-4}$, in the common local unitary frames.

An actual row carrier has only its two endpoint blocks. A fixed fine bond in a layer can occur in carriers of only adjacent levels: the outer block is inside the previous region and both endpoint blocks are outside the next. At each of those levels the number of rows is fixed. Thus eq. 3.dxcviii implies for the full layered field map

$$\|\Xi^{-1}Qa\|_{2,\mathrm{rows}} \le C_L \|a\|_{\mathrm{mass}}. \tag{3.dxcix}$$

For a single row, its coefficient sum is bounded and each entry is at most $C(\eta/\xi_c)^4$. Cauchy therefore gives the analogue of eq. 3.dlvii; the carrier overlap argument of eq. 3.dlxii then proves eq. 3.dxcix. This estimate has been obtained from the actual finite recursion, not by treating an infinity norm bound alone as an L^2 bound.

3.13.29. *The two-legged constraint coefficient has a uniform moment.* The same computation gives the coefficient estimate needed for the quadratic current term. Let $n_s = L^s$ and $m_s = L^{j-s-1}$. Twice differentiating the recursion gives

$$D^2_{a_0} a_j = \sum_{s<j} P_{j,s+1} D^2 F_{W_s}(a_s)[P_{s,0}\,\cdot, P_{s,0}\,\cdot]. \tag{3.dc}$$

Fix the first fine bond b and sum absolutely over the second. For the term indexed by s, the largest entry of the outer product is at most Cm_s^{-3}, by its column sum. The sum over the first inner leg is at most Cn_s^{-3}. The second inner row sum is at most Cn_s. The remaining sum over the one-step Hessian, with its first input fixed, is bounded by eq. 3.dxci. Thus

$$\sum_{b'} \|D_{a_0(b)} D_{a_0(b')} a_j(c)\| \le C_L \sum_{s<j} m_s^{-3} n_s^{-2}. \tag{3.dci}$$

This step uses a column sum on the fixed leg and a row sum on the summed leg. Using two row sums would lose the fine density.

Physical input derivatives contribute η^2; the output per-length normalization contributes ξ^{-1}. Since $m_s^{-3}n_s^{-2} = L^3 n_s/n^3$ and $\sum_{s<j} n_s \le n/(L-1)$, we obtain

$$\sup_b \sum_{b'} \|D_b D_{b'} q_j(A)(c)\| \le C_L\, \xi\, (\eta/\xi)^4. \tag{3.dcii}$$

There is no number-of-levels factor. The geometric sum is dominated by its final scales. Symmetry of the second derivative gives the corresponding other-leg sum.

Let $\omega(c)$ be an arbitrary row current with $|\omega(c)| \le \gamma_\omega \xi_c^{-3}$. Pair the Hessian at zero with the physical row volume. eq. 3.dcii, Schur and the same bounded carrier overlap give

$$\left| \sum_c \xi_c^4 \langle \omega(c), D^2 q_c(0)[u, v] \rangle_{\mathrm{col}} \right| \le C_L \gamma_\omega \|u\|_{\mathrm{mass}} \|v\|_{\mathrm{mass}}. \tag{3.dciii}$$

For clarity, the row prefactor $\xi_c^4|\omega(c)|$ is bounded by $\gamma_\omega\xi_c$. Multiplying it by the leg moment $C_L\xi_c(\eta/\xi_c)^4$ leaves $C_L\gamma_\omega\eta^4\xi_c^{-2}$ on each fine leg. Adjacent scales replace ξ_c by the fine bond scale at only a fixed cost.

The holomorphic ball was used to extract these coefficients. The quadratic form in eq. 3.dciii then extends to every finite fine vector by its coefficient formula. It does not require a Gaussian sample to lie in that holomorphic ball. For the second variant one must still bound the actual induced current $\widehat{H}[J]^\sharp J$ in this row norm, and bound S in the weighted fine energy spaces. Those are separate operator readouts.

3.13.30. *The nonflat field-row lift follows from the row comparison.* The coefficient estimates allow one to complete eq. 3.dlxxxix at a regular pair. Keep the real local gauges of its center. Construct the profiles eq. 3.dlxxiv in each pillar in such a gauge, and undo the gauge. For a seed assigned to the far endpoint of a row, normalize its color by the fixed unitary transport of that row. Denote this operator again by E^0. It is fixed at the real center while the complex increment varies.

Two local real gauges used on the same pillar differ there by a nearly constant unitary matrix. To prove this, write their connection equation on each fine bond and telescope along a coordinate path of length $C\xi_j$. Both gauge potentials have size at most $C_L\epsilon/\xi_j$. The transition therefore differs from its root value by at most $C_L\epsilon$. The first-order class on a fixed larger cube supplies both gauges. Only a path of size $C_L\xi_j$ is used, not the full $M_1\xi_j$ cube.

It follows from eq. 3.dxcviii and this transition bound that the same-level seed response is the identity with an error of row and column norm $C_L(\epsilon+h)$ in the row energy norm. Only a fixed number of same-level rows meet a seed pillar. The next-level response has a leading part with the positive scalar majorant of Q_L^0, multiplied by fixed unitary color matrices, and an error. There are at most CL^4 lower-level seeds in one such carrier; a seed affects at most a fixed number of next-level rows. An error entry of size $C_L(\epsilon+h)$ therefore has row sum $C_LL^5(\epsilon+h)$ and column sum $C_LL(\epsilon+h)$ after the row-energy normalization. Schur bounds this error by $C_LL^3(\epsilon+h)$. Absorb this fixed power into C_L.

All other level blocks are zero by the carrier argument preceding eq. 3.dlxxxii. The leading next-level part still has norm at most L^{-1}; Schur uses only the scalar majorants, not a compatible global choice of flat color frames. We have proved

$$\|Q_{\mathbf{U}}E^0-1\|_{\mathrm{row}\to\mathrm{row}}\le L^{-1}+C_L(\epsilon+h)=:\vartheta<1. \tag{3.dciv}$$

The ceiling needed here depends on L and not on depth, volume or M_1. The actual right inverse is

$$E_{\mathrm{row}}=E^0(Q_{\mathbf{U}}E^0)^{-1}. \tag{3.dcv}$$

Both its squared mass norm and its covariant gradient energy are bounded by $C_L(1-\vartheta)^{-2}\|\psi\|_{\mathrm{row}}^2$. For the gradient of a seed, expand in its local gauge: the profile derivative is of size $C\xi_j^{-1}|\psi|$ and the connection term has the same bound. Supports and their fine collars have bounded overlap. The small complex perturbation adds at most C_Lh times the mass norm. This proves the stated energy bound, including bare seeds.

There is a corresponding actual Feynman Poincare inequality. Repeat eq. 3.dlix in a common local real gauge for a pillar and its chosen row carriers. The row comparison after eq. 3.dxcviii has both row and column error bounds. Schur therefore bounds its local L^2 functional norm by $C_L\epsilon\xi_j^{-2}$. Ordinary gradients and covariant gradients differ there by $C_L\epsilon\xi_j^{-1}$ times the field. Consequently eq. 3.dlx, summed over the carriers, gives

$$\|a\|_{\mathrm{mass}}^2\le C_L(\|\nabla_U a\|_2^2+\|a^{1/2}Q_Ua\|_{2,\mathrm{rows}}^2)+C_L\epsilon^2\|a\|_{\mathrm{mass}}^2. \tag{3.dcvi}$$

The repeated letter a in $a^{1/2}$ is the fixed positive row weight, whereas the arguments are fine fields. Equivalently that term is $\sum_c c_c\xi_c^2|Q_Ua(c)|^2$. Absorb the last term by the fixed weak-field ceiling. Only carrier differences in the interior are used; the bare boundary fields are controlled by their

identity rows. Thus the same argument applies to an active component-gradient form containing all those interior differences.

The full complex Feynman form is stable on a fixed physical ball. The actual row comparison gives

$$\|a^{1/2}(Q_{\mathbf{U}} - Q_U)v\|_2 + \|(\nabla_{\mathbf{U}} - \nabla_U)v\|_2 \le C_L h\|v\|_{\mathrm{mass}}. \tag{3.dcvii}$$

For the first term use the row and column differences, the physical row volumes and the bounded carrier overlap. For the second use $\mathrm{Ad}(e^{i\eta E}) - 1 = O(\eta E)$. Expanding the two bilinear squares and using eq. 3.dcvi gives a relative form perturbation $C_L h$. Its inverse exists by the normalized Neumann series. This proves a component Feynman inverse statement. The actual Landau subtraction eq. 3.dlxix is still absent.

3.14. **Response equations and weighted inverse estimates.** The response of a constrained background is a solution of a coupled field–multiplier system. Bounds on the inverse of one positive quadratic form give only part of that response. Constraints, gauges, the current readout, the local length factors, and admissibility of the substituted data must all be included. This section derives the exact reductions needed for these tasks. Its linear algebra and analytic estimates are proved. On an admitted strong reference domain, Props. 3.112, 3.163 and 3.169 construct the primitive coefficients and the selected nonlinear current germ. The construction of a reference from homogeneous curvature-small data is Prop. 3.194. Applications to complementary histories retain their specified geometry, source and admission hypotheses.

All calculations are first made in a finite compatible geometry. The constants in an infinite-volume or arbitrary-depth application must be uniform in that geometry before a limit is taken. In particular, a finite-dimensional implicit-function theorem by itself supplies no uniform radius.

The actual equations and their differentiated right-hand sides. Use the real reference field U_0 and write the varied fine field as $e^{i\eta A}U_0$. For the multilevel determining set $\mathbf{B}$, the straightened average is denoted by $\mathcal{Q}^{\mathrm{st}}(A)$ and its derivative by K_A. This is the nonlinear straightening of the group average, rather than the straight linear average. With the reduced datum $\mathsf{b}(B)$, the physical critical pair satisfies

$$\begin{aligned} F'(A) &= K_A^{\mathsf{T}}\omega, & \mathcal{Q}^{\mathrm{st}}(A) &= \mathsf{b}(B), \\ RD^{\mathsf{T}}A &= 0, & F(A) &= A_{\Omega_0}(e^{i\eta A}U_0). \end{aligned} \tag{3.dcviii}$$

Here R and D are formed at the reference field. The first equation is the full physical stationarity identity when that identity has been proved; its restriction to the slice is sufficient for the algebra below. When the reference is changed with the datum, differentiating the slice also contributes $(\dot{R}D^{\mathsf{T}} + R\dot{D}^{\mathsf{T}})A$. We keep the reference fixed in eq. 3.dcviii. A calculation that moves the reference and then omits this contribution does not differentiate eq. 3.dcviii.

The reduced datum has a local triangular correction at an outward bond. For an outer endpoint $z = c_+$ it is defined by

$$e^{i\mathsf{b}(B)(c)} = e^{iB(c)}\,\mathrm{Ad}(\mathbf{V}_0(c))\mathsf{v}_z(B). \tag{3.dcix}$$

At an interior bond it equals $B(c)$. The block factor v_z depends on the interior bonds of its block. Consequently the derivative of b is triangular in the same ordering. On a smaller regular logarithm chart, differentiation of the exponential and logarithm gives

$$\dot{b} = D\mathsf{b}(B)\dot{B}, \qquad \dot{b}(c) = L_c(B)\dot{B}(c) + \sum_{c'\ \mathrm{interior}} T_{cc'}(B)\dot{B}(c') \tag{3.dcx}$$

on the outward rows. Define the right logarithmic differential $\mathcal{E}_X(V) = (1/i)[\frac{d}{dt}\exp(i(X + tV))]_{t=0}\exp(-iX)$. Differentiating eq. 3.dcix after right multiplication by $\exp(-i\mathsf{b})$ gives $L_c = \mathcal{E}^{-1}_{\mathsf{b}(B)(c)}\mathcal{E}_{B(c)}$. The same differential and the derivative of the block factor give T. Thus the outward diagonal is invertible near the regular chart; it need not be the identity. The

bounds require a fixed distance to the logarithm cut and the actual local block-factor estimate. Triangularity establishes invertibility once those diagonal logarithmic derivatives are invertible; it does not bound an unrestricted product of triangular operators over an arbitrary number of levels.

Let $\mathcal{V}$ be the fixed linear slice, after the residual node gauge has been fixed. Its fine pairing and the datum pairing are the actual cell-weighted pairings. Write $q = K_A|_{\mathcal{V}}$ and let $q^\sharp$ be its transpose for these pairings. The constrained second variation is

$$\mathcal{K} = F''(A) - \sum_a \mu_a \omega_a (\mathcal{Q}_a^{\mathrm{st}})''(A) \quad \text{on } \mathcal{V}. \tag{3.dcxi}$$

The weights μ_a in this coordinate expression are the datum cell weights. Equivalently the second term is the Hessian of the scalar pairing $\langle \omega, \mathcal{Q}^{\mathrm{st}}(A)\rangle$. This latter description fixes the convention if a different coordinate basis is used. The curvature term in eq. 3.dcxi cannot be replaced by the Hessian of F alone. It is the usual second fundamental form of the constraint fibre.

Differentiating eq. 3.dcviii at a fixed reference gives the exact system

$$\mathcal{K}\dot{A} - q^\sharp \dot{\omega} = 0, \qquad q\dot{A} = D\mathsf{b}(B)\dot{B}. \tag{3.dcxii}$$

More generally, for explicit parameters p in the action, gauge or constraints, put $\mathcal{C}(A,p) = 0$ and $\mathcal{E}(A,\omega,p) = F_p'(A) - \mathcal{C}_A^\sharp \omega$. The right-hand sides are then

$$f = -D_p\mathcal{E}\,\dot{p}, \qquad b = -D_p\mathcal{C}\,\dot{p}, \qquad \mathcal{K}\dot{A} - q^\sharp\dot{\omega} = f, \quad q\dot{A} = b. \tag{3.dcxiii}$$

Every differentiation in D_p is made before substituting the solution $A(p), \omega(p)$. This distinction prevents an implicit extra response term on the right-hand side.

An independent-current extension is an additional specified operator. The formula in eq. 3.dcxi identifies the physical constrained Hessian. It does not identify every separately defined $\Delta_\pi'[J]$ with that Hessian. Likewise, an absolute contour gauge functional and its relative version have the same quadratic gauge block at an axial field, but generally have different blocks away from that field. The complete gauge and current normal form must be matched before an independently named propagator can be inserted into eq. 3.dcxii.

A complete block inverse, including the multiplier. The following result is finite-dimensional and exact. It also applies to bounded operators when the displayed inverses and compositions exist on the stated spaces. Every transpose in the complex calculation is bilinear, with no conjugation.

Proposition 3.37 (Response from a primitive constrained inverse)**.** Suppose $q : \mathcal{V} \to \mathcal{D}$ is onto, $E : \mathcal{D} \to \mathcal{V}$ satisfies $qE = I$, and $P : \mathcal{X} \to \mathcal{V}$ is an isomorphism onto $\ker q$. Assume

$$H = P^\sharp \mathcal{K} P \quad \text{is invertible}, \qquad L = PH^{-1}P^\sharp, \qquad T = E - L\mathcal{K}E. \tag{3.dcxiv}$$

Then the unique solution of eq. 3.dcxiii is

$$\begin{aligned} \dot{A} &= Lf + Tb, \\ \dot{\omega} &= E^\sharp(\mathcal{K}L - I)f + E^\sharp \mathcal{K} T b. \end{aligned} \tag{3.dcxv}$$

If $\mathcal{K}^\sharp = \mathcal{K}$, this simplifies to

$$\dot{\omega} = -T^\sharp f + Mb, \qquad M = E^\sharp \mathcal{K} E - E^\sharp \mathcal{K} L \mathcal{K} E, \qquad M^\sharp = M. \tag{3.dcxvi}$$

■

Proof. Write $\dot{A} = Eb + Pv$. Applying $P^\sharp$ to the first equation of eq. 3.dcxiii, and using $qP = 0$, gives $Hv = P^\sharp(f - \mathcal{K}Eb)$. This determines v and gives the first formula of eq. 3.dcxv. The residual $\mathcal{K}\dot{A} - f$ annihilates $\ker q$. It therefore belongs to the range of $q^\sharp$. Since $E^\sharp q^\sharp = I$, its unique multiplier is the second formula of eq. 3.dcxv. Conversely these formulas satisfy both equations, so there is no lost multiplier equation. For a homogeneous solution, the same argument gives $v = 0$, then $\dot{A} = 0$ and $\dot{\omega} = 0$.

For the symmetric assertion, $H^\sharp = H$ and $L^\sharp = L$. Thus $T^\sharp = E^\sharp - E^\sharp \mathcal{K} L$, which gives eq. 3.dcxvi. Finally $E^\sharp \mathcal{K} T$ equals the displayed expression for M; both its terms are symmetric. Positivity of the full $\mathcal{K}$ was not used. Invertibility was needed on the actual constrained slice. □

The inverse L acts on fine forces; T lifts the constraint datum after the energetic correction; M is the multiplier response. Bounding L alone leaves the products $L\mathcal{K}E$ and $E^\sharp \mathcal{K} T$ to be controlled. The size and locality of E also matter. A right inverse with a norm growing in the number of layers would not prove a uniform response theorem.

At one step the actual raw pivot derivative has size L^{-3}, whereas the derivative after division of the coarse logarithm by L has size L^{-4}. Accordingly a change of the constraint normalization changes both E and the dual multiplier. For example, replacing q by $\widetilde{q} = q/L$ while keeping the same pairings replaces ω by $\widetilde{\omega} = L\omega$ in the scalar Lagrangian. Using the new right inverse with the old multiplier would change eq. 3.dcxi, not merely its notation.

Cell weights and a kernel algebra. We describe the norm in which the preceding formulas can be estimated without a bare-mesh counting loss. Let each component i have a positive cell weight μ_i and an amplitude $s_i > 0$. A vector is measured by $\|v\|_s = \sup_i |v_i|/s_i$. A kernel is written as

$$(Av)_i = \sum_j \mu_j A_{ij} v_j, \qquad \widehat{A}_{ij} = s_i^{-1} A_{ij} s_j. \tag{3.dcxvii}$$

For different input and output species their respective amplitudes and cell weights are used. The identity kernel is δ_{ij}/μ_j. Thus even the identity is not represented by δ_{ij} when it is used with a weighted summation convention.

For a metric d common to the component locations define

$$\|A\|_m = \max\Big\{ \sup_i \sum_j \mu_j |\widehat{A}_{ij}| e^{md(i,j)}, \\ \sup_j \sum_i \mu_i |\widehat{A}_{ij}| e^{md(i,j)} \Big\}. \tag{3.dcxviii}$$

Matrix norms in a colour or local finite-dimensional fibre are operator norms. The two sums use the correct weights for their two species. The triangle inequality for d and absolute convergence give

$$\|AB\|_m \le \|A\|_m \|B\|_m. \tag{3.dcxix}$$

Indeed, insert the intermediate sum with weight μ_k, use $d(i,j) \le d(i,k) + d(k,j)$ and cancel the intermediate amplitudes. For the row bound take the supremum of the inner B row sum; for the column bound use the corresponding A column sum. These are distinct calculations and neither counts the number of fine bonds.

The first row sum bounds A on the normalized supremum norm. The two sums together bound it on the normalized, cell-weighted ℓ^2 space by the Schur estimate. To see the latter directly, apply weighted Cauchy–Schwarz to each row with weights $\mu_j |\widehat{A}_{ij}|$, then sum in i and use the column bound. The result is $\|Av\|_{2,s} \le \|A\|_0 \|v\|_{2,s}$.

For the natural response components above a top of scale ξ_y, the amplitudes are

$$s_H(y) = \xi_y^{-1}, \qquad s_{\nabla H}(y) = \xi_y^{-2}, \qquad s_J(y) = \xi_y^{-3}. \tag{3.dcxx}$$

Equivalently the output is $(\xi_y H, \xi_y^2 \nabla H, \xi_y^3 J)$. The cell weights are determined by the fine, coarse and pillar pairings. Each constraint length factor belongs in the matrix before eq. 3.dcxvii is formed. The domain section gives the resulting transposes. These transpose conventions are needed when applying eq. 3.dcxix to a product that changes species at a seam.

For a raw logarithm $a = gy$, a physical radius α becomes α/g in the canonical Gaussian variable y. A derivative with respect to y contributes g times the derivative with respect to a. Thus the second

derivative of a scalar activity in Gaussian coordinates is g^2 times its raw-logarithm Hessian. This conversion occurs in addition to the length and cell-weight conversions in eq. 3.dcxvii–eq. 3.dcxx.

Corollary 3.38 (Uniform response after the primitive bounds)**.** Suppose the actual normalized kernels in eq. 3.dcxiv satisfy

$$\|L\|_m \le a, \quad \|E\|_m \le e, \quad \|\mathcal{K}E\|_m \le k, \quad \|E^\sharp\mathcal{K}\|_m \le k'. \tag{3.dcxxi}$$

Then

$$\|T\|_m \le e + ak, \qquad \|M\|_m \le k'(e + ak). \tag{3.dcxxii}$$

In the nonsymmetric case the force-to-multiplier block has norm at most $k'a + \|E^\sharp\|_m$. In the symmetric case it is $-T^\sharp$. Its norm equals the norm of T with reversed species if the dual amplitudes are chosen reciprocally for the same cell-weighted pairings. For other force and current amplitudes its weighted transpose bound must be checked separately. ■□

The last qualification is necessary. A raw transpose and a transpose after conjugation by amplitude weights are different matrices. If $\widehat{T}_{ij} = s_i^{-1}T_{ij}s_j$, its transposed normalized kernel is obtained by giving the dual components amplitudes s_i^{-1} and s_j^{-1}. A natural force weight of a different physical dimension does not satisfy this rule automatically. Its conversion belongs in the kernel before a Schur norm is claimed.

The bounds in eq. 3.dcxxi are the primitive targets. They must hold for the actual enlarged saddle system on the retained complex domain and at every allowed seam. The corollary does not assert them. In particular, an unweighted flat inverse does not supply k, k' on a current-dependent layered domain. It does establish an exact-model anchor from which a perturbative comparison can start.

Invertibility on the whole complex ball. Let $\mathcal{S}$ denote the full field–constraint operator in eq. 3.dcxiii, with all gauge rows included. Fix a reference operator $\mathcal{S}_0$ on the same spaces. If the primitive estimates give

$$\|\mathcal{S}_0^{-1}\|_m \le C_0, \qquad \|\mathcal{S}(B) - \mathcal{S}_0\|_m \le \varepsilon, \qquad C_0\varepsilon < 1, \tag{3.dcxxiii}$$

then the Neumann series converges in the kernel algebra and proves

$$\|\mathcal{S}(B)^{-1}\|_m \le \frac{C_0}{1 - C_0\varepsilon}. \tag{3.dcxxiv}$$

It is essential that the second bound in eq. 3.dcxxiii holds on the whole retained complex ball. A bound only along the real solutions does not give the analytic disc needed later for Cauchy estimates.

If $\mathcal{S}_0$ is unbounded in the chosen norm, the correct statement uses the relative perturbation $\mathcal{S}_0^{-1}(\mathcal{S} - \mathcal{S}_0)$ between its actual graph spaces. The same series applies there. In particular an elliptic second-order operator cannot be treated as a bounded supremum-norm matrix with a mesh-independent norm merely because its inverse is bounded.

If the coefficients of $\mathcal{S}(B)$ are holomorphic in the indicated operator norm, holomorphy follows term by term on a strictly smaller disc, or by the resolvent identity

$$\mathcal{S}(B')^{-1} - \mathcal{S}(B)^{-1} = -\mathcal{S}(B')^{-1}[\mathcal{S}(B') - \mathcal{S}(B)]\mathcal{S}(B)^{-1}. \tag{3.dcxxv}$$

The identity also proves continuity and the derivative formula wherever the coefficient derivative exists in the indicated operator norm. No complex positivity assertion is used. Real coercivity may construct $\mathcal{S}_0^{-1}$; the complex inverse then requires a genuine norm comparison or another explicitly convergent construction.

For a complete identification of the proposed response, eq. 3.dcxxiii has to be checked with the nonlinear K_A, the full constraint-curvature block, the prescribed relative or absolute gauge block, and the seam rows. Changing any of these while keeping the same notation for $\mathcal{S}$ invalidates the comparison.

From local forcing to first response. Suppose a change in one datum coordinate $B(c)$ gives a forcing column F_c in the enlarged system. The column consists of both the field force and the constraint force in eq. 3.dcxiii. If it is supported within distance r_0 of the top y_c and has normalized supremum norm at most b_0, then eq. 3.dcxxiv gives

$$|D_{B(c)}Z_y[e]|/s_y \le \frac{C_0 b_0 e^{mr_0}}{1-C_0\varepsilon} e^{-md(y,y_c)}|e|. \tag{3.dcxxvi}$$

Here Z contains the output species actually controlled by the inverse, with amplitude s_y from eq. 3.dcxvii. Equivalently apply the bound to the normalized readout Z_y/s_y; subsequent Cauchy estimates use normalized readouts. The proof is the row estimate: for each contributing component j, $d(y,y_c) \le d(y,j)+r_0$, so multiply by $e^{md(y,y_c)}$ and sum the kernel against F_c. A triangular datum correction from eq. 3.dcx changes b_0, r_0 by its actual block bounds. It must be included in F_c.

For an exponentially local forcing column, a bound $|F_c(j)|/s_j \le b_0 e^{-ad(j,y_c)}$ gives eq. 3.dcxxvi with any $0 < m' < \min(m,a)$ and m' in place of m. Indeed the triangle inequality leaves a factor $e^{-(a-m')d(j,y_c)} \le 1$, and the remaining sum is a weighted row sum at rate m'. No uncontrolled number of sites appears.

This proves response decay from three smaller tasks: construct the actual primitive inverse, verify its weighted row norm, and compute the actual datum forcing. It does not assume response decay as a premise. If the output inverse controls only H and the multiplier, the estimate has not yet been proved for ∇H or the current. Those outputs require their own readout maps.

For example the physical current is a specified nonlinear function $J(U)$. In a fixed logarithmic chart its derivative is

$$\dot J = D_A[J(e^{i\eta A}U_0)]\dot A. \tag{3.dcxxvii}$$

An implementation may replace this expression by the multiplier and lower-order terms through the differentiated field equation. Such a replacement must be derived from eq. 3.dcviii and the definition of J. The equality is not implied by the fact that both quantities have dimension three. On an independent-current core the variable $\mathbf{J}$ and the computed current $J(\mathbf{U})$ are distinct until the physical pair is imposed.

Why a kernel bound does not prove a difference bound. Uniform locality and a gap do not control variation at the bare scale. On a one-dimensional mesh, the positive diagonal inverse $R_{ii} = 1 + \frac{1}{2}(-1)^i$ has norm at most $3/2$ and zero range. Applied to a constant force it produces a difference of size η^{-1}. Its inverse is positive and uniformly bounded as well. This example shows exactly why a separate regularity estimate is needed.

There is a useful exact identity for that estimate. On a translation patch let T shift by one mesh step, $\nabla = (T-I)/\eta$ and $\mathcal{K}_T = T\mathcal{K}T^{-1}$. If $\mathcal{K}v = f$, then

$$\mathcal{K}_T\nabla v = \nabla f - \frac{\mathcal{K}_T - \mathcal{K}}{\eta} v. \tag{3.dcxxviii}$$

To prove it, write $T(\mathcal{K}v) = \mathcal{K}_T T v$, subtract $\mathcal{K}v$ and divide by η. At a boundary this calculation must be made on the common interior patch; it does not shift an inadmissible boundary row into the equation.

For a scalar divergence-form example $\mathcal{K} = D^\sharp a D + V$, with constant mesh difference D, the coefficient commutator in eq. 3.dcxxviii is

$$\frac{\mathcal{K}_T - \mathcal{K}}{\eta} = D^\sharp(\nabla a)D + \nabla V. \tag{3.dcxxix}$$

If the energy norm is $\|v\|_{\mathcal{E}}^2 = \|Dv\|_2^2 + \|v\|_2^2$, the bilinear pairing and Cauchy–Schwarz imply

$$\big\|[(\mathcal{K}_T - \mathcal{K})/\eta]v\big\|_{\mathcal{E}^*} \le (\|\nabla a\|_\infty + \|\nabla V\|_\infty)\|v\|_{\mathcal{E}}. \tag{3.dcxxx}$$

Thus a bounded inverse from $\mathcal{E}^*$ to $\mathcal{E}$, together with coefficient regularity, bounds ∇v in energy norm. This does not assert a supremum bound; passing to a supremum or Hölder bound requires the corresponding local elliptic estimate.

In the gauge system, shifting the covariant differences adds transport commutators, the average rows add contour terms, and a seam adds its own boundary comparison. The required finite list is: the translated interior equation, the gauge-row commutator, the constraint-row commutator, and each seam or exterior boundary row. Each term must be bounded between its actual force and output norms. The uniform primitive regularity estimate is precisely this list of bounds and the associated local elliptic estimate. It is not supplied by exponential decay of H^{-1} alone.

Joint Cauchy estimates with joining distances. Once eq. 3.dcxxvi has been established uniformly on a complex product ball, several useful higher response bounds require no further model estimate. We give the argument with all radius losses visible.

Let $\mathcal{B}_r$ be a ball in the supremum of the dimensionless bond logarithms, with a fixed norm on the three colour coordinates. Suppose a holomorphic map Z_y on $\mathcal{B}_r$ satisfies

$$|\partial_c Z_y(B)[e]| \le C e^{-m d(y,y_c)} |e|, \qquad B \in \mathcal{B}_r. \tag{3.dcxxxi}$$

Fix $0 < \theta < 1$. There is a colour constant c_* such that the coordinate polydisc of radius $c_*(1-\theta)r$ about every $B \in \mathcal{B}_{\theta r}$ lies in $\mathcal{B}_r$. The number of bonds does not enter this assertion, since the norm is a supremum over bonds; simultaneous variation of three colour components changes only c_*. Cauchy's formula applied to the first derivative gives

$$|\partial_{c_1} \cdots \partial_{c_q} Z_y(B)| \le C(q-1)! [c_*(1-\theta)r]^{-(q-1)} e^{-m \max_j d(y,y_{c_j})}. \tag{3.dcxxxii}$$

Repeated coordinates are allowed. The relevant multi-index factorial is at most $(q-1)!$. Holomorphic derivatives commute, so each one of the q coordinates can be selected as the first derivative in eq. 3.dcxxxi; taking the minimum of the resulting exponential bounds proves the maximum in eq. 3.dcxxxii.

Let $\mathcal{T}(y, c_1, \dots, c_q)$ be the minimum spanning-tree length in the underlying metric. The star through y is a competing tree, and therefore

$$\mathcal{T} \le \sum_{j=1}^{q} d(y, y_{c_j}) \le q \max_j d(y, y_{c_j}). \tag{3.dcxxxiii}$$

Consequently eq. 3.dcxxxii gives a joining-tree exponential with rate m/q. For the second derivative it also gives the more explicit bound

$$|\partial_c \partial_{c'} Z_y(B)| \le \frac{C}{(1-\theta)r} e^{-\frac{m}{2}[d(y,y_c)+d(y,y_{c'})]}. \tag{3.dcxxxiv}$$

This is the estimate used in the pointed Taylor remainder. It costs one datum radius. The formula in eq. 3.dcxxxii is not a uniform-in-q tree bound at a fixed rate: its rate deteriorates with q. If an arbitrary-order polymer estimate needs a fixed rate, one must use the local nonlinear equations and their tree expansion, with the resulting factorial and entropy costs. A fixed-order application may use eq. 3.dcxxxii directly.

The same Cauchy argument applies to an output Hölder seminorm if the first response is already bounded in that seminorm on the larger ball. Cauchy's formula can be applied to every normalized difference $[Z_y - Z_{y'}]/|y-y'|^\gamma$ and then a supremum taken. It cannot create spatial Hölder regularity from a pointwise bound.

Finite boxes and compatible boundary forcing. Let $\mathcal{S}$ be the actual linearized saddle operator and $\mathcal{S}_\Lambda$ its prescribed finite-box version. To compare solutions, their equations must be put on common variables and rows in an interior collar. Write the resulting exact difference equation as

$$\mathcal{S}_\Lambda \, \delta X = f_\Lambda. \tag{3.dcxxxv}$$

The force includes the difference of boundary rows, the datum restriction, the gauge-restoration term and any cutoff commutator. The identity in eq. 3.dcxxxv is checked before its right-hand side is bounded. Replacing an actual box boundary by a convenient Dirichlet boundary changes this force and requires a new comparison.

Suppose the inverse has weighted row bound C at rate m and the force is supported in a collar $\mathcal{C}$. If $\|f_\Lambda\|_s \le AR^p$, then for a component at y

$$|\delta X_y|/s_y \le CAR^p e^{-m\,\mathrm{dist}(y,\mathcal{C})}. \tag{3.dcxxxvi}$$

This follows by extracting the exponential distance from the weighted row sum. In particular no factor equal to the number of bare sites in the collar is necessary. Such a factor would generally destroy uniformity as the mesh tends to zero.

The same conclusion, at any rate $m' < \min(m, a)$, holds if

$$|f_\Lambda(j)|/s_j \le AR^p e^{-a\,\mathrm{dist}(j,\mathcal{C})}. \tag{3.dcxxxvii}$$

For each j, the distance from y to the collar is at most $d(y, j)+\mathrm{dist}(j, \mathcal{C})$. Multiplying the summand by $e^{m'\,\mathrm{dist}(y,\mathcal{C})}$ leaves a weighted row sum and a remaining exponential at most one. This proves the claim.

A cutoff must be made in physical coordinates. For a smooth function $\chi(x/R)$ on a mesh η, its first difference divided by η is $O(R^{-1})$ and its second difference divided by η^2 is $O(R^{-2})$, by the integral form of the mean-value formula. The unnormalized difference itself has an extra factor η. These are different quantities. A cutoff of an affine potential of size $O(R)$ can therefore have bounded first derivatives and polynomially bounded forcing, rather than a bare-mesh singularity.

For a nonlinear equation a cutoff of a solution is not a solution even where the cutoff is a constant strictly between zero and one. All such points belong to the transition collar. On that collar the force is computed from the actual nonlinear formula or its integral Taylor formula. The admissibility of the cutoff field, its averages, and its tower backgrounds is a separate domain check. A bounded formal commutator cannot replace it.

For two nonlinear backgrounds, subtracting their equations gives an averaged Jacobian along a common admissible coordinate segment. The inverse estimate for that averaged operator must be justified, for example by a uniform comparison of every Jacobian with the same reference as in eq. 3.dcxxiii. An arbitrary average of invertible complex matrices need not be invertible. The first-step background section uses just such a common comparison; a multilevel application needs it on the actual multilevel ball.

The full substitution and the test for staying inside a core. An old-origin source is evaluated on data of the form

$$V_j(\nu) = \mathcal{M}^j\big(e^{i\eta(t+\nu\zeta)H}U\big)|_{\Gamma_j}. \tag{3.dcxxxviii}$$

The parameters on the fine bonds in ζ may vary independently. The inner analytic argument uses $|\nu| < c/s$, not merely the physical segment between zero and one. There are also outer and removed substitutions. Each must satisfy the same list of field, independent current, node, gauge and tower conditions that defines its assigned core. A proof only of the curvature clause is incomplete.

Here is a precise useful criterion. Express the core as inequalities $\|\mathcal{R}_a(X)\| < r_a$ for its actual clauses, including all hereditary tower clauses. Let X_0 be the complete reference pair and assume margins $\|\mathcal{R}_a(X_0)\| \le (1-\epsilon_a)r_a$. If a common analytic curve satisfies

$$\sup_{\nu \text{ on the retained disc}} \|\mathcal{R}_a(X_\nu) - \mathcal{R}_a(X_0)\| < \epsilon_a r_a \quad \text{for every } a, \tag{3.dcxxxix}$$

then it stays in the core. The proof is the triangle inequality for every defining clause. Its substantive content in an application is the uniform bound for the infinite hereditary list, not that triangle inequality.

If the actual response and average derivatives imply

$$\frac{\|\mathcal{R}_a(X_\nu)-\mathcal{R}_a(X_0)\|}{r_a}\le C_a|\nu|\,\delta_k\sigma_j/\alpha_j, \tag{3.dcxl}$$

then the inner disc is admitted whenever $c\sup_a C_a\delta_k\sigma_j/(s\alpha_j)<\inf_a\epsilon_a$. This calculation shows the exact estimate required; it does not establish eq. 3.dcxl for the actual maps. The ratio σ_j is the old-to-current physical length ratio and must be inserted with the units of the relevant average. It cannot be guessed by identifying raw logarithms with potentials.

One part of this check is explicit on the fixed real-base tower class. At a positive bare bond b let $l(b)$ be its layer spacing in the ambient sequence. Let n_b be the maximum of the potential and fixed real-base covariant first-difference seminorms scaled by $l(b)$ and $l(b)^2$, respectively. For a shift $\mathbf{A}$ of $\mathbf{U}=e^{i\eta A'}U$, set

$$u(b)=\max\{l(b)|\mathbf{A}(b)|,\,l(b)^2|\nabla_{\mathbf{U}}\mathbf{A}|(b)\},\qquad A'_{\mathrm{new}}(b)=\frac{1}{i\eta}\log\bigl(e^{i\eta\mathbf{A}(b)}e^{i\eta A'(b)}\bigr).$$

Here the first differences are those inside the domain of the tower. Write $u^+(b)$ for the maximum of u at b and its available translates by one positive bare step. The chart estimate gives $n_b(A'_{\mathrm{new}}-A')\le 2u^+(b)$ when $u(b)\le(256L^2)^{-1}$ throughout the chart and the base potential belongs to the admitted tower class. The neighboring entries occur because a first difference of the product of the two exponentials uses both endpoints. Thus if $n_b(A')+2u^+(b)<a_{\mathrm{T}}$, the new potential lies in the same fixed-base class with radius a_{T}. The tower response estimate then gives the increment at an element whose tower index agrees with the ambient index capped at the tower height, in the form $2e^{\delta}C\sup_b e^{-\delta d^{\flat}(x,x_b)}u(b)$ at its base point x, using the weighted distance $d^{\flat}$ of the ambient sequence. The factor e^{δ} absorbs the one bare step in u^+. For the inner contour put $\mathbf{A}=\nu\zeta H$ and check these bounds on the full retained disc, using the actual fine-bond differences of ζH. This verifies the fixed real-base potential clause when there is the stated margin. The induced tower current is part of that response. The clauses for independently varied currents and outer and removed substitutions still require their own bounds.

There are two logically different uses of a displacement estimate. A bound for $X_\nu-X_0$ on a disc already known to be admitted proves a Taylor estimate. To prove admission, one needs the bound while continuing up to a possible first exit. On a finite geometry this is done on the connected part of the analytic solution branch before exit. If eq. 3.dcxxxix is strict up to its closure and the primitive inverse has a uniform continuation radius there, no first exit occurs. The uniform continuation radius and the strict margin are both needed. Without them the first-exit argument is circular.

At young ages the factor σ_j may be too large to make eq. 3.dcxl small. The permitted fallback is a direct estimate at the finitely many such ages, using their actual local disc and radius. It must bound the whole reference value as well as the increment. An increment that vanishes at $\nu=0$ does not prove that X_0 itself satisfies its core inequalities.

For removed substitutions the current and the background may change together. The path used in the comparison must then remain in the joint core. Separate coordinate Cauchy estimates at the two endpoints do not imply that the segment connecting them is admitted, especially when a hereditary condition is not convex. One can instead use the actual common analytic chart and the strict clause margins in eq. 3.dcxxxix.

The remaining quantitative tasks. The preceding arguments reduce the multilevel response problem to the following specific estimates on the primitive domain. First, the actual constraint rows and gauge slice must admit E,P with uniform weighted bounds, including every outward seam row. Second, the constrained inverse and its energetic compositions must satisfy eq. 3.dcxxi–eq. 3.dcxxiv on the whole retained complex ball. Third, the datum forcing and current readout must satisfy their local weighted bounds. Fourth, the translated equations and boundary rows must give the required

first-difference and Hölder estimates. Fifth, all inner, outer and removed substitutions must satisfy eq. 3.dcxxxix, with a supremum over the actual tower list and independent fine parameters.

These tasks have different mechanisms and should be checked separately. Proving the first-step versions does not prove their constants uniform in depth. Once these estimates hold, the field and multiplier response follows from the block inverse, joining distance estimates follow from Cauchy on the admitted ball, and window-background comparison follows from the exact boundary-force equation. These last implications are the proved part of the section and may be used without adding a new response assumption.

3.15. **A mesh-uniform interior Hölder estimate.** The response system needs spatial regularity as well as decay. We prove the particular scalar estimate that converts a bounded potential and a bounded componentwise Laplacian into a Hölder bound for its first differences. The estimate is uniform in the mesh. It also applies in a parallel frame for an unweighted flat unitary connection, including arbitrary flat holonomies on a compatible torus. The nonlinear gauge application is then reduced to explicit coefficient estimates. The construction of those local gauges and the layered equations supplying the Laplacian bound remain model-dependent tasks.

Fix $0 < \alpha < 1$. On $\eta\mathbb{Z}^4$ define

$$D_i^+ u(x) = \frac{u(x+\eta e_i) - u(x)}{\eta}, \quad D_i^- u(x) = \frac{u(x) - u(x-\eta e_i)}{\eta}, \quad \Delta_\eta = \sum_{i=1}^4 D_i^- D_i^+. \tag{3.dcxli}$$

The Laplacian has nonpositive Fourier symbol. The counting measure in all convolution kernels is η^4 times counting measure. We use the Euclidean distance, allowing dimension-dependent constants when a coordinate path is used. A derivative of order r means an arbitrary product of r of the forward or backward differences in eq. 3.dcxli.

Heat kernels with enough derivative and tail control. Start on the unit lattice. For $T \geq 0$ set

$$q_T(n) = \frac{1}{(2\pi)^4} \int_{[-\pi,\pi]^4} e^{-T\omega(\theta)} e^{in\cdot\theta}\, d\theta, \qquad \omega(\theta) = 4\sum_i \sin^2(\theta_i/2). \tag{3.dcxlii}$$

This is a nonnegative probability kernel. Indeed, if A is the sum of the eight nearest-neighbour translations, then $\Delta_1 = \mathsf{A} - 8I$ and

$$e^{T\Delta_1} = e^{-8T} \sum_{k=0}^\infty \frac{T^k \mathsf{A}^k}{k!}. \tag{3.dcxliii}$$

Every coefficient is nonnegative, and applying the operator to the constant function gives one. Fourier transformation gives eq. 3.dcxlii, so the two constructions agree. The physical heat-kernel density is

$$p_t^\eta(\eta n) = \eta^{-4} q_{t/\eta^2}(n). \tag{3.dcxliv}$$

It follows directly that convolution with p_t^η is a contraction on bounded functions.

The following polynomial tail estimates suffice.

Lemma 3.39 (Uniform differentiated heat-kernel tails)**.** For integers $r, N \geq 0$ there is $C_{r,N}$ independent of η, t, a such that

$$\begin{aligned} \sum_{|x|\geq a} \eta^4 |D^r p_t^\eta(x)| &\leq C_{r,N} t^{-r/2} (1 + a/\sqrt{t})^{-N}, && t \geq \eta^2, \\ \sum_{|x|\geq a} \eta^4 |D^r p_t^\eta(x)| &\leq C_{r,N} \eta^{-r} (1 + a/\eta)^{-N}, && 0 \leq t \leq \eta^2. \end{aligned} \tag{3.dcxlv}$$

■

Proof. A product of r unit differences has Fourier multiplier $m_r(\theta)$, a product of factors $e^{\pm i\theta_i}-1$. For every multi-index b,

$$|\partial^b m_r(\theta)| \le C_{r,b}|\theta|^{(r-|b|)_+}. \tag{3.dcxlvi}$$

The bound follows by assigning the derivatives to factors. A factor that has not been differentiated vanishes to first order at zero; all its positive-order derivatives are bounded. If more than r derivatives occur the asserted power is zero.

For $\theta \in [-\pi,\pi]^4$, $\omega(\theta) \ge c|\theta|^2$. Its first derivatives are $O(|\theta|)$ and its higher derivatives are bounded. Differentiating $e^{-T\omega}$ for $T \ge 1$ gives

$$|\partial^b e^{-T\omega(\theta)}| \le C_b T^{|b|/2}(1+|\sqrt{T}\theta|)^{M_b} e^{-c_b T|\theta|^2}. \tag{3.dcxlvii}$$

For completeness, each term of the repeated product rule consists of $e^{-T\omega}$ times factors $T\partial^{b_j}\omega$, with $b_j \ne 0$ and $\sum b_j = b$. A first derivative contributes $\sqrt{T}\,O(|\sqrt{T}\theta|)$. A factor of order at least two is at most a constant times $T^{|b_j|/2}$ because $T \ge 1$. Multiplying proves eq. 3.dcxlvii; the finite polynomial can be absorbed by a smaller Gaussian exponent when integrating.

Leibniz' rule, eq. 3.dcxlvi, and eq. 3.dcxlvii now imply

$$\int_{[-\pi,\pi]^4} |\partial^b(m_r e^{-T\omega})|^2\,d\theta \le C_{r,b}T^{|b|-r-2}, \qquad T \ge 1. \tag{3.dcxlviii}$$

When at most r derivatives hit the multiplier, scale $\theta = T^{-1/2}v$ in that term. When more than r hit it, its bound is smaller than the displayed power since $T \ge 1$. The four-dimensional volume factor is T^{-2}. This explains every power in eq. 3.dcxlviii.

Let δ^r denote the corresponding unit differences. Parseval's identity and differentiation of the periodic Fourier multiplier give, for every integer $M \ge 0$,

$$\sum_n (1+|n|^2/T)^M |\delta^r q_T(n)|^2 \le C_{r,M}T^{-r-2}, \qquad T \ge 1. \tag{3.dcxlix}$$

Expand the weight into coordinate monomials. The term $|n^b|^2 T^{-|b|}$ is bounded by eq. 3.dcxlviii times $T^{-|b|}$. All multiplier functions are smooth and periodic, so there are no boundary terms at $\pm\pi$.

Choose $M > N+2$, increasing it by one if necessary. Comparing the lattice sum with the radial integral in four dimensions gives

$$\sum_{|n|\ge A} (1+|n|^2/T)^{-M} \le C_{M,N}T^2(1+A/\sqrt{T})^{-2N}, \qquad T \ge 1. \tag{3.dcl}$$

One way to check the comparison is to cover the lattice by unit cubes. For $T \ge 1$ the values of $1+|x|/\sqrt{T}$ at any two points of a cube are comparable by a fixed dimensional factor. The integral of $(1+|x|^2/T)^{-M}$ outside a ball of radius comparable with A is T^2 times the corresponding integral in scaled coordinates. That integral decays at least as $(1+A/\sqrt{T})^{-2N}$ for this choice of M. Small values of A are absorbed by the full integral bound.

Cauchy–Schwarz with eq. 3.dcxlix and eq. 3.dcl proves $\sum_{|n|\ge A}|\delta^r q_T(n)| \le CT^{-r/2}(1+A/\sqrt{T})^{-N}$. For $0 \le T \le 1$, all derivatives of $m_r e^{-T\omega}$ of any fixed order are uniformly bounded. The same Parseval argument, with weight $(1+|n|^2)^M$, gives instead $C(1+A)^{-N}$. Finally eq. 3.dcxliv contributes η^{-r} to the weighted ℓ^1 norm. Substitution of $T = t/\eta^2$ proves eq. 3.dcxlv. □

These constants depend on the fixed derivative order and the desired tail power. They do not depend on the number of lattice sites. The small-time part of eq. 3.dcxlv is important: replacing it by a continuum heat-kernel formula all the way to $t = 0$ would not justify the uniform estimates for the discrete second and third differences.

A massive resolvent and its first-difference modulus. For $\mu > 0$ define

$$G_\mu^\eta(x) = \int_0^\infty e^{-\mu^2 t} p_t^\eta(x)\, dt, \qquad R_\mu f = G_\mu^\eta * f. \tag{3.dcli}$$

Its kernel is nonnegative and has total mass μ^{-2}. For fixed η, Δ_η is a bounded operator on bounded functions. Integrating the derivative of $e^{-\mu^2 t} e^{t\Delta_\eta}$ in operator norm shows

$$(-\Delta_\eta + \mu^2) R_\mu f = f, \qquad \|R_\mu f\|_\infty \le \mu^{-2} \|f\|_\infty. \tag{3.dclii}$$

Conversely, any bounded solution of this equation equals $R_\mu f$. Indeed integrate the same identity applied to the bounded solution; the terminal term tends to zero because of $e^{-\mu^2 t}$ and the heat contraction. This also proves uniqueness without a maximum attainment argument on an infinite lattice.

Assume $\mu\eta \le 1$. Integrating eq. 3.dcxlv for $r = 1$, first on $[0, \eta^2]$ and then on $[\eta^2, \infty)$, gives

$$\sum_x \eta^4 |DG_\mu^\eta(x)| \le C\mu^{-1}. \tag{3.dcliii}$$

The first interval contributes $C\eta$ and the second is bounded by $C \int_0^\infty e^{-\mu^2 t} t^{-1/2}\, dt = C'\mu^{-1}$.

For a lattice displacement h, telescoping along a coordinate path and using eq. 3.dcxlv with $r = 1, 2$ gives, for $t \ge \eta^2$,

$$\|\tau_h D p_t^\eta - D p_t^\eta\|_1 \le C \min(t^{-1/2}, |h| t^{-1}). \tag{3.dcliv}$$

The second bound follows by summing the second difference over the path of physical length at most $2|h|$ in four dimensions; the first is twice the first-difference norm. For $t \le \eta^2$ use $C\eta^{-1}$ instead. Every nonzero lattice displacement has $|h| \ge \eta$.

Suppose first that $\eta \le |h| \le \mu^{-1}$. Integrate eq. 3.dcliv on $[\eta^2, |h|^2]$, on $[|h|^2, \mu^{-2}]$ and on the remaining tail. Together with the small-time interval, this gives

$$\|\tau_h DG_\mu^\eta - DG_\mu^\eta\|_1 \le C|h|(1 + \log(1/(\mu|h|))). \tag{3.dclv}$$

On the middle interval the integral is $|h| \int_{|h|^2}^{\mu^{-2}} t^{-1} dt$. The first interval contributes $O(|h|)$, and the exponentially damped last interval does as well. If $|h| > \mu^{-1}$ use twice eq. 3.dcliii. Since for $0 < x \le 1$

$$x^{1-\alpha}(1 + \log(1/x)) \le 1 + \frac{1}{e(1-\alpha)}, \tag{3.dclvi}$$

both ranges imply

$$\|\tau_h DG_\mu^\eta - DG_\mu^\eta\|_1 \le C_\alpha \mu^{\alpha-1} |h|^\alpha. \tag{3.dclvii}$$

The value at $h = 0$ is zero. It follows for bounded f that

$$\|DR_\mu f\|_\infty \le C\mu^{-1} \|f\|_\infty, \qquad [DR_\mu f]_\alpha \le C_\alpha \mu^{\alpha-1} \|f\|_\infty. \tag{3.dclviii}$$

The seminorm compares lattice points, using their physical distance. The logarithmic modulus eq. 3.dclv explains why these estimates do not give a mesh-uniform Lipschitz bound for $DR_\mu f$ from bounded f alone. The constants are permitted to diverge as α tends to one.

Off-diagonal resolvent derivatives. Let $R \ge \eta$ and $\mu = R^{-1}$. For every fixed $c > 0$ and $r = 0, 1, 2, 3$,

$$\sum_{|x| \ge cR} \eta^4 |D^r G_{1/R}^\eta(x)| \le C_{r,c} R^{2-r}. \tag{3.dclix}$$

Choose $N > \max(0, r-2)$. On the time interval $[\eta^2, R^2]$, the large-time part of eq. 3.dcxlv is bounded by $Ct^{-r/2}(\sqrt{t}/R)^N$. Its integral is $O(R^{2-r})$. On $[R^2, \infty)$ the exponential e^{-t/R^2} gives the same bound by scaling. The interval $[0, \eta^2]$ contributes at most $C\eta^{2-r}(\eta/R)^N$, which is also $O(R^{2-r})$. This proves eq. 3.dclix, including the second and third derivatives for which the uncut global norms would have a worse small-scale behaviour.

There is no assertion here that the second resolvent derivative has a uniformly bounded global absolute sum. Its off-diagonal bound is what makes the interior cutoff argument work.

An interior estimate with no unknown cutoff gradient. Let Q_R be the cube of side $2R$ centred at a lattice point. The following formulation uses a fixed ratio of cubes; changing that ratio changes only the constants.

Proposition 3.40 (Interior first-difference Hölder bound)**.** Let $R \geq 64\eta$. Suppose u is defined on Q_{2R} and $\Delta_\eta u = f$ at every point needed by the equation there. Put $M = \|u\|_{\infty,Q_{2R}}$ and $F = \|f\|_{\infty,Q_{2R}}$. On $Q_{R/2}$, with all first differences and comparisons taken inside its one-mesh enlargement,

$$\begin{aligned} \|Du\|_\infty &\leq C(R^{-1}M + RF), \\ [Du]_\alpha &\leq C_\alpha(R^{-1-\alpha}M + R^{1-\alpha}F). \end{aligned} \tag{3.dclx}$$

The constants are independent of R, η, u. ■

Proof. Choose the restriction of a smooth cutoff $\chi(x/R)$, equal to one on Q_R and supported in $Q_{3R/2}$. It satisfies $|D\chi| \leq C/R$ and $|\Delta_\eta\chi| \leq C/R^2$. Its difference supports lie in a collar at distance at least cR from the evaluation cube and its one-mesh enlargement. The condition $R \geq 64\eta$ makes this separation uniform after every difference used below. Extend u arbitrarily outside Q_{2R}; the value of $v = \chi u$ and all forces below are independent of that extension.

With $\mu = 1/R$ the exact product identity is

$$(-\Delta_\eta + \mu^2)v = a - [\Delta_\eta, \chi]u, \qquad a = \chi(-f + \mu^2 u). \tag{3.dclxi}$$

Both sides are extended by their actual compactly supported formulas. The equation for u is used only where χ or its differences are nonzero, inside the given domain. By eq. 3.dclii, $v = R_\mu a - R_\mu[\Delta_\eta, \chi]u$. The first term has force norm at most $F + R^{-2}M$, so eq. 3.dclviii gives exactly its contributions to eq. 3.dclx.

To treat the commutator one must avoid estimating the unknown Du in the cutoff collar. Move the commutator onto the kernel. For scalar multiplication by χ and the constant cell-weighted pairing,

$$[\Delta_\eta, \chi]^{\mathsf{T}} = [\chi, \Delta_\eta] = -[\Delta_\eta, \chi]. \tag{3.dclxii}$$

For $j = 1, 2$, apply this identity to $K_j(y) = D_x^j G_\mu^\eta(x - y)$ in the convolution. The discrete product rule shows that its absolute value is bounded by

$$CM\left(R^{-2}\|1_{|\cdot|\geq cR}D^j G_\mu^\eta\|_1 + R^{-1}\|1_{|\cdot|\geq cR}D^{j+1}G_\mu^\eta\|_1\right). \tag{3.dclxiii}$$

To check the rule explicitly, $\Delta(\chi K) - \chi\Delta K - K\Delta\chi$ is the sum over i of $(D_i^+\chi)(D_i^+K) + (D_i^-\chi)(D_i^-K)$. All factors containing a difference of χ are supported in the separated collar. Changes of the evaluation point by at most the fixed number of mesh steps in a derivative merely change c. The kernel is absolutely summable on that collar by eq. 3.dclix, which also justifies the summation by parts.

For $j = 1$, eq. 3.dclix bounds in eq. 3.dclxiii by CM/R. For $j = 2$ it bounds it by CM/R^2. Let $w = R_\mu[\Delta_\eta, \chi]u$. A coordinate path between two points of $Q_{R/2}$ stays in that cube; telescoping the first differences of w gives $|Dw(x) - Dw(x')| \leq CMR^{-2}|x - x'|$. Its supremum bound is CM/R. Since $|x - x'| \leq CR$ in the cube, this gives $[Dw]_\alpha \leq C_\alpha MR^{-1-\alpha}$. Adding the contribution of $R_\mu a$ proves both estimates. On the evaluation cube $v = u$, including all the differences used, so the estimates apply to u itself. □

The proof has not used a previously known bound on Du in the outer collar. Estimating the original cutoff commutator directly by $R^{-1}\|Du\|$ would have introduced that extra premise. Moving the commutator onto the separated resolvent kernel is the step that removes it.

The result holds componentwise for vectors in any fixed finite-dimensional normed space. Equivalent coordinate norms contribute a constant depending on that fixed dimension. No number of lattice sites appears.

Tori and flat unitary holonomies. On a torus with periods ℓ_i, the scalar heat kernel and massive resolvent are periodizations of their infinite-lattice kernels. Their absolute derivative sums are bounded by the corresponding infinite sums. For a set at periodic distance at least cR from the origin, every lift has Euclidean distance at least cR. Thus eq. 3.dclix holds for periodic distance as well. If the cubes and their collars in the interior theorem embed without wrapping, the same cutoff proof gives eq. 3.dclx with the same constants. For example $\min_i \ell_i \geq 8R$ suffices for the stated cubes.

A flat unitary connection can be made trivial on the lifted lattice. The deck transformations act by its commuting unitary holonomies. The kernel on the torus is the image sum with the corresponding unitary factors. On a common parallel frame in the embedded cube, differences act on the lifted kernel. Taking the norm and using unitarity bounds the image sum by the scalar sum already estimated. Hence all the preceding local bounds remain uniform in the holonomies. Their distance from the identity is irrelevant.

The argument requires flatness. Curved parallel transports have additional commutators. Nor does it give the same estimate for an arbitrary nonlocal or layered positive operator: the proof used the componentwise constant-coefficient Laplacian in eq. 3.dcxli.

The local gauge application. Here is how eq. 3.dclx enters the background and response estimates. Suppose a local real gauge has supplied a potential on a cube enlarged by a fixed multiple of its local scale ξ, with

$$\|A\|_\infty \leq C_0 m\varepsilon\xi^{-1}, \qquad \|\Delta_\eta A\|_\infty \leq C_1 m\varepsilon\xi^{-3}. \tag{3.dclxiv}$$

Apply the theorem on cubes of radius a fixed fraction of ξ, covering the interior. It proves

$$[DA]_\alpha \leq C_\alpha m\varepsilon\xi^{-2-\alpha}, \qquad \|DA\|_\infty \leq Cm\varepsilon\xi^{-2}. \tag{3.dclxv}$$

For pairs farther apart than a fixed fraction of ξ, the supremum bound gives the same Hölder estimate. Thus local cubes give the full indicated interior seminorm. On the enlarged real cubes of the layered profile, the single-gauge construction supplies both premises of eq. 3.dclxiv from curvature and full-current bounds.

There is a finite coarse-mesh case in this application. If the chosen fraction of ξ is less than 64η, the ratio ξ/η is bounded by a fixed constant. Directly, $|DA| \leq 2\|A\|_\infty/\eta$ and $[DA]_\alpha \leq 4\|A\|_\infty\eta^{-1-\alpha}$. They give eq. 3.dclxv with the same powers of ξ and a larger fixed constant. The same fallback applies to the response application below. Thus no unmentioned lower bound on ξ/η is needed.

For a response, the available equation often controls the positive covariant componentwise Laplacian $\mathcal{L}_{U_0} = \sum_i \nabla_{U_0,i}^{\mathsf{T}}\nabla_{U_0,i}$. Thus $\mathcal{L}_{\mathbf{1}} = -\Delta_\eta$ with the heat-generator sign in eq. 3.dcxli. Its conversion can also be made quantitative. In a local gauge write $U_0(b) = e^{i\eta A_0(b)}$ and let $T_b = \mathrm{Ad}(U_0(b))$ in the chosen unitary colour representation. If $\eta|A_0|$ stays in a fixed small exponential chart, then

$$\|(T_b - I)/\eta\| \leq C|A_0(b)|. \tag{3.dclxvi}$$

Differencing T_b in a transverse direction uses the integral derivative of the exponential. It gives $|DT_b| \leq C\eta|DA_0|$, with the adjacent bond values interpreted in the same local frame. The inverse transport satisfies the same bound. Expanding the forward and backward covariant differences therefore gives

$$|(\mathcal{L}_{U_0} + \Delta_\eta)h| \leq C\big(|A_0|\,|Dh| + (|DA_0| + |A_0|^2)|h|\big). \tag{3.dclxvii}$$

One can verify this by first writing each covariant first difference as the ordinary difference plus $(T_b - I)/\eta$ times a neighbouring value. In the second difference the apparent η^{-1} terms combine into a first difference of h, a difference of the coefficient $(T_b - I)/\eta$, and a product of two such

coefficients. These are exactly the three terms on the right of eq. 3.dclxvii. There is no estimate of an individual uncombined $\eta^{-1}A_0h$ term.

Assume on a local-scale cube that

$$\xi|A_0| + \xi^2|DA_0| \le a_0,$$
$$\xi|h| + \xi^2|\nabla_{U_0}h| + \xi^3|\mathcal{L}_{U_0}h| \le B. \tag{3.dclxviii}$$

The formula in eq. 3.dclxvi first gives $\xi^2|Dh| \le C(1+a_0)B$. Then eq. 3.dclxvii implies

$$\xi^3|\Delta_\eta h| \le C(1+a_0+a_0^2)B. \tag{3.dclxix}$$

Applying eq. 3.dclx at scale ξ yields $[Dh]_\alpha \le C_{\alpha,a_0}B\xi^{-2-\alpha}$. The difference between Dh and $\nabla_{U_0}h$ is a bounded coefficient times a neighbouring h. Its Hölder seminorm is controlled by the same local bounds, using the Lipschitz coefficient bound from DA_0 and the first-difference bound for h. Thus

$$[\nabla_{U_0}h]_\alpha \le C_{\alpha,a_0}B\xi^{-2-\alpha}. \tag{3.dclxx}$$

This initially compares gradient components in the chosen local gauge. If the assigned Hölder seminorm transports one component to the other along paths of length at most $C|x-x'|$, unitary product telescoping and eq. 3.dclxviii give $\|\operatorname{Ad}U_0(\Gamma)-I\| \le Ca_0|x-x'|/\xi$. Multiplying by $|\nabla_{U_0}h| \le B/\xi^2$ adds at most $Ca_0B\xi^{-2-\alpha}|x-x'|^\alpha$ on the scale-ξ cube. Hence eq. 3.dclxx also holds in that transported seminorm. Its premise still includes the pointwise covariant-gradient bound in eq. 3.dclxviii; the argument removes an additional Hölder premise, rather than that gradient bound.

If the right-hand side of eq. 3.dclxviii has the response factor $B = Ce^{-md(y,y_c)}|e|$, that factor persists on a fixed-size local-scale cube up to the fixed exponential comparison of distances within the cube. This gives the corresponding Hölder member of the response estimate. The primitive model input is now explicit: the actual local gauge and the three bounds in eq. 3.dclxviii, with the proper current and boundary readouts. No separate Hölder estimate is needed once those inputs have been proved. At a seam the cube, frame and difference equation must be compatible on both sides; the scalar theorem cannot supply that compatibility.

3.16. **Local gauges from curvature and the Wilson current.** The background estimates require a local potential and its first differences in physical units. A bound on plaquettes gives a small potential in a comb gauge, but does not directly bound its first differences. We construct a Coulomb gauge in a finite L^p space and then use the actual Wilson current to obtain the additional derivative. The distinction between these two steps is essential: no boundedness of an order-zero singular integral on L^∞ is used.

The problem of choosing regular local representatives is part of Bałaban's gauge-fixing construction [B85b]. The proof here specifies a different local gauge and normalizes the current directly by differentiating the Wilson action. The representative need not preserve block averages or a previously imposed contour gauge. Those additional requirements are discussed at the end.

3.16.1. *The statement and its physical normalization.* Let $\eta > 0$ be a lattice spacing. For $R > 0$ with R/η an integer, write

$$Q_R = \{x \in \eta\mathbb{Z}^4 : |x|_\infty \le R\}.$$

An assertion on bonds or plaquettes of a cube concerns those having all their vertices in the cube. A connection assigns $U_\mu(x) \in \mathrm{SU}(2)$ to the positive bond from x to $x+\eta e_\mu$; the reverse bond has the inverse value. Gauge transformations act by

$$U_\mu^g(x) = g(x)U_\mu(x)g(x+\eta e_\mu)^{-1}. \tag{3.dclxxi}$$

Lie-algebra elements are traceless Hermitian matrices. We use $\langle X, Y\rangle = \frac{1}{2}\mathrm{tr}(XY)$ and its Euclidean norm. All estimates below remain valid for the equivalent operator norm, with a changed absolute constant.

For distinct μ,ν, put

$$\begin{aligned} P_{\mu\nu}(x) &= U_\mu(x)U_\nu(x+\eta e_\mu) \\ &\qquad \times U_\mu(x+\eta e_\nu)^{-1}U_\nu(x)^{-1}, \\ F_{\mu\nu}(x) &= \frac{1}{i\eta^2}\log P_{\mu\nu}(x), \\ W_{\mu\nu}(x) &= \frac{P_{\mu\nu}(x)-P_{\mu\nu}(x)^*}{2i\eta^2}. \end{aligned} \tag{3.dclxxii}$$

The logarithm is the branch near the identity. On this branch $F_{\nu\mu}=-F_{\mu\nu}$, and $W_{\mu\nu}$ is the matrix sine $\eta^{-2}\sin(\eta^2 F_{\mu\nu})$.

The Wilson current is the physical gradient of $\sum_p(1-\frac12\operatorname{tr}U_p)$. At every bond whose full plaquette stencil is present, it is normalized by

$$\frac{d}{dt}A(e^{i\eta t a}U)\Big|_{t=0} = \eta^4\sum_{x,\mu}\langle J_\mu(x),a_\mu(x)\rangle. \tag{3.dclxxiii}$$

This is a local definition: compactly supported variations change only finitely many plaquettes. No finite total action on an infinite lattice is assumed.

For a matrix-valued function let

$$D^+_\mu f(x)=\frac{f(x+\eta e_\mu)-f(x)}{\eta},\quad D^-_\mu f(x)=\frac{f(x)-f(x-\eta e_\mu)}{\eta},\quad L_\eta=-\sum_\mu D^-_\mu D^+_\mu. \tag{3.dclxxiv}$$

The Coulomb condition on a potential is $\operatorname{div}_\eta A:=\sum_\mu D^-_\mu A_\mu=0$. The notation DA includes every forward first difference of every component. For a function on a cube, $[f]_{\alpha,Q}$ is the ordinary discrete Hölder seminorm, with physical Euclidean distance.

Proposition 3.41 (A local gauge controlled by the physical current)**.** There are absolute constants $\varepsilon_*>0$ and $C<\infty$ with the following property. Let $0<\eta\le r$, $r/\eta\in\mathbb{N}$, and let a real connection be given on Q_{8r}. Assume that all its plaquettes lie in the logarithm branch of eq. 3.dclxxii, and that

$$\|F\|_{\infty,Q_{8r}}\le\kappa,\qquad r^2\kappa\le\varepsilon_*. \tag{3.dclxxv}$$

Suppose the actual current in eq. 3.dclxxiii satisfies $\|J\|_{\infty,Q_{7r}}\le j$. Then there is a real site gauge g such that $U^g=e^{i\eta A}$ on Q_{6r}, in the logarithm branch near the identity, and $\operatorname{div}_\eta A=0$ wherever its stencil lies in Q_{6r}. On Q_r,

$$\begin{aligned} \|A\|_\infty &\le Cr\kappa, \\ \|DA\|_\infty &\le C(\kappa+rj), \\ \|L_\eta A\|_\infty &\le C(j+r\kappa^2). \end{aligned} \tag{3.dclxxvi}$$

For each fixed $0<\alpha<1$,

$$[DA]_{\alpha,Q_r}\le C_\alpha r^{-\alpha}(\kappa+rj). \tag{3.dclxxvii}$$

The constants are independent of η, r, the connection and its holonomies outside the cube. No smallness condition on j is needed. The gauge and the bound on A use only the curvature assumption; the current enters the derivative estimates. ∎

The condition can equivalently be imposed using $\eta^{-2}\|P-1\|_\infty$: on a fixed small logarithm branch it is comparable with $\|F\|_\infty$. In particular the hypothesis has the physical curvature scaling of the background domains, rather than a bound on an unscaled bond potential.

We prove the proposition after the analytic estimates below. Under $x=r\widetilde{x}$, the scaled quantities are

$$\widetilde{\eta}=\eta/r,\quad \widetilde{A}=rA,\quad \widetilde{F}=r^2F,\quad \widetilde{J}=r^3J. \tag{3.dclxxviii}$$

Thus it suffices to prove the statement at $r=1$.

3.16.2. *Uniform finite-L^p estimates on a lattice torus.* The auxiliary torus in this proof has physical side 32. Its spacing is $h = 1/N$, $N \in \mathbb{N}$, and integration means h^4 times counting measure. All constants in this subsection may depend on p and on that fixed side, but not on N. Let Π be the arithmetic mean and let $G_h = L_h^{-1}(1 - \Pi)$.

Lemma 3.42 (Uniform Green and Riesz estimates)**.** For $1 < p < \infty$ and any choices of forward or backward differences,

$$\|D_i D_j G_h f\|_p \le C_p \|f\|_p. \tag{3.dclxxix}$$

For $p > 4$,

$$\|DG_h f\|_\infty \le C_p \|f\|_p, \qquad \|u - \Pi u\|_\infty \le C_p \|Du\|_p. \tag{3.dclxxx}$$

For a one-form a,

$$\|Da\|_p \le C_p(\|\mathrm{d}a\|_p + \|\operatorname{div}_h a\|_p), \quad (\mathrm{d}a)_{\mu\nu} = D^+_\mu a_\nu - D^+_\nu a_\mu. \tag{3.dclxxxi}$$

These estimates hold componentwise for the fixed colour space. ∎

Proof. We give the kernel argument, including the mesh uniformity at the diagonal. The Fourier symbol of L_h is

$$\lambda_h(k) = 4h^{-2} \sum_{\mu=1}^{4} \sin^2(hk_\mu/2), \qquad k \in (2\pi/32)\mathbb{Z}^4 \cap [-\pi/h, \pi/h)^4. \tag{3.dclxxxii}$$

It is comparable with $|k|^2$ on the Brillouin cube. Fourier orthogonality proves the L^2 estimate in eq. 3.dclxxix; each first-difference symbol has modulus at most $\lambda_h^{1/2}$.

Let p_t^h be the physical heat kernel. Direct Fourier integration gives, for a product of s differences,

$$\|D^s p_t^h\|_\infty \le C_s(t + h^2)^{-(4+s)/2}, \qquad 0 < t \le 1. \tag{3.dclxxxiii}$$

For $t \le h^2$, use the volume of the Brillouin cube and $|D| \le 2/h$. For $t \ge h^2$, put $q = k\sqrt{t}$ and use $\lambda_h \ge c|k|^2$. The corresponding physical L^1 tails are

$$\sum_{|x| \ge a} h^4 |D^s p_t^h(x)| \le C_{s,M}(t + h^2)^{-s/2}(1 + a/\sqrt{t + h^2})^{-M}. \tag{3.dclxxxiv}$$

Here and below the distance is periodic. One may obtain this estimate from the differentiated heat-kernel tail calculation in § 3.15. For clarity, its Fourier argument uses integration by parts in a coordinate with $|x_\mu| \ge |x|/2$ when $t \ge h^2$. Each differentiation of the Gaussian multiplier is bounded by a polynomial in $k\sqrt{t}$ times $e^{-ct|k|^2}$, and supplies one factor $\sqrt{t}$. At $t \le h^2$, the random-walk series $e^{t\Delta_h} = e^{-8t/h^2} \sum_{n \ge 0} (t/h^2)^n \mathcal{A}^n/n!$ has uniformly bounded moments of every order; a product of s differences costs $C_s h^{-s}$. Periodizing the infinite-lattice estimate proves the displayed torus version.

The semigroup law turns these two bounds into a pointwise bound. In $D^s p_t = (D^s p_{t/2}) * p_{t/2}$, either the first or the second variable has distance at least $|x|/2$. Use the L^1 tail for that factor and the L^∞ bound for the other. Consequently

$$|D^s p_t^h(x)| \le C_{s,M}(t + h^2)^{-(4+s)/2}(1 + |x|/\sqrt{t + h^2})^{-M}. \tag{3.dclxxxv}$$

For $t \ge 1$, subtract the constant mode. The nonzero frequencies have a fixed positive lower spectral bound, and the same Fourier estimate gives $C_s e^{-ct}$ for every fixed derivative order.

Integrate the heat kernel in time. With $\rho(x)$ the periodic distance from zero, the kernels with physical summation convention satisfy

$$\begin{aligned} |DG_h(x)| &\le C(\rho(x) + h)^{-3} + C, \\ |D_i D_j G_h(x)| &\le C(\rho(x) + h)^{-4} + C, \\ |D_\ell D_i D_j G_h(x)| &\le C(\rho(x) + h)^{-5} + C. \end{aligned} \tag{3.dclxxxvi}$$

For example split the integral for a second derivative at $t = (\rho + h)^2$, use $M > 6$ below that point and the power t^{-3} above it. The interval $0 < t < h^2$ contributes at most Ch^{-4} at the diagonal. The first and third derivative estimates follow with powers $t^{-5/2}$ and $t^{-7/2}$. Thus a diagonal matrix entry is large, but its physical mass h^4 is retained.

We now prove the L^p consequence for $T = D_i D_j G_h$. Its physical kernel has sum zero, its L^2 norm is at most one, and eq. 3.dclxxxvi implies

$$\sum_{\rho(x-y)>4|y-z|} h^4 |K_h(x-y) - K_h(x-z)| \le C \quad (y \neq z). \tag{3.dclxxxvii}$$

Indeed join y to z by a shortest coordinate path. The third difference estimate, summed along that path, bounds the difference by $C|y-z|\rho(x-y)^{-5}$ away from the displayed ball. Summing annuli in four dimensions gives a uniform constant. The bounded large-distance term is harmless on the fixed torus.

Use the Calderón–Zygmund decomposition [CZ52], which we spell out for these finite lattices. Split the coordinate intervals of the torus repeatedly into two integer intervals of lengths differing by at most one. The resulting rectangles have uniformly bounded aspect ratio, and each parent has volume at most 3^4 times that of any child. At the final generation they are single lattice sites. For a level λ above the mean of $|f|$, take the maximal rectangles Q with average greater than λ. They are disjoint, and

$$\sum_Q |Q| \le \lambda^{-1} \|f\|_1, \qquad f = g + \sum_Q b_Q, \quad |g| \le 3^4 \lambda, \quad \|g\|_2^2 \le C\lambda \|f\|_1. \tag{3.dclxxxviii}$$

Here $b_Q = (f - f_Q)1_Q$, so it has mean zero and $\sum_Q \|b_Q\|_1 \le 2\|f\|_1$. The enlarged rectangles have total measure at most $C\lambda^{-1}\|f\|_1$. Outside them, eq. 3.dclxxxvii, with the cancellation of each b_Q, gives $\|\sum_Q T b_Q\|_{L^1(\text{outside})} \le C\|f\|_1$. The L^2 estimate controls Tg. Chebyshev's inequality proves

$$|\{|Tf| > \lambda\}| \le C\lambda^{-1}\|f\|_1. \tag{3.dclxxxix}$$

If λ is below the mean, the same bound is immediate from the total volume of the torus. This includes the last atomic scale; no continuum limit has been used.

For completeness the interpolation is uniform as well. Split f at height proportional to t when estimating $|Tf| > t$, use the weak L^1 bound on the large part and the L^2 bound on the small part, and integrate pt^{p-1} in t. For $1 < p < 2$, the two integrals are bounded by constants times

$$\int |f(x)| \int_0^{|f(x)|} t^{p-2}\, dt\, dx, \qquad \int |f(x)|^2 \int_{|f(x)|}^{\infty} t^{p-3}\, dt\, dx.$$

They equal $\|f\|_p^p/(p-1)$ and $\|f\|_p^p/(2-p)$. Duality proves $2 < p < \infty$, since the adjoint has the same kernel estimates. This proves eq. 3.dclxxix for all forward/backward choices.

For $p > 4$, $p' = p/(p-1) < 4/3$. The first kernel in eq. 3.dclxxxvi has uniformly bounded physical $L^{p'}$ norm, because $\sum_x h^4(\rho(x)+h)^{-3p'} \le C_p$. Hölder's inequality therefore proves the first assertion in eq. 3.dclxxx. The identity $u - \Pi u = -\sum_i D_i^- G_h D_i^+ u$ proves the second.

Finally ordinary differences commute and

$$L_h a_\mu = \sum_\nu D_\nu^- (\mathrm{d}a)_{\mu\nu} - D_\mu^+ \operatorname{div}_h a. \tag{3.dcxc}$$

Apply $D_i^+ G_h$ and eq. 3.dclxxix. Constant one-forms have zero derivative, so their mean causes no additional term. This proves eq. 3.dclxxxi. □

The use of $p > 4$ in eq. 3.dclxxx is strict. The order-zero operator in eq. 3.dclxxix is never applied with $p = \infty$. Its kernel has the borderline singularity $|x|^{-4}$, whereas the kernel used to bound a first derivative from a Laplacian has the integrable power $|x|^{-3}$ in the required dual space.

3.16.3. *A small-potential Coulomb gauge in a finite L^p space.*

Lemma 3.43 (A uniform nonlinear gauge calculation)**.** There are absolute $a_* > 0$ and $C < \infty$ such that the following holds for Hermitian traceless matrices λ, A, w whenever $|\lambda| + h|A| + h|w| < a_*$. Set

$$\mathcal{B}_h(\lambda, A, w) = \frac{1}{ih} \log\left(e^{i\lambda} e^{ihA} e^{-i(\lambda + hw)}\right) = A - w + N_h(\lambda, A, w). \tag{3.dcxci}$$

Then

$$|N_h| \le C\big(|\lambda| + h(|A| + |w|)\big)(|A| + |w|). \tag{3.dcxcii}$$

On a fixed smaller domain its first two derivatives satisfy the corresponding product bounds obtained by differentiating the right side. All constants are uniform in $0 < h \le 1$. ■

Proof. Write $X = hA$, $Y = hw$ and consider $\mathcal{L}(\lambda, X, Y) = i^{-1} \log(e^{i\lambda} e^{iX} e^{-i(\lambda + Y)})$. This is analytic on a fixed matrix polydisc. It vanishes for $X = Y = 0$, for every admitted λ, and its differential in (X, Y) at $(0, 0, 0)$ is $X - Y$. Taylor's integral formula in (X, Y) therefore gives

$$|\mathcal{L} - (X - Y)| \le C|\lambda|(|X| + |Y|) + C(|X| + |Y|)^2.$$

Divide by h. The same integral formula and the Cauchy estimate on a fixed larger polydisc prove the derivative bounds. The argument uses finite matrix analyticity, so its constants do not depend on the number of lattice sites. □

Proposition 3.44 (Coulomb gauge from a small global potential)**.** Fix $p > 4$. On the physical side-32 torus there are constants $b_p > 0$ and $C_p < \infty$, uniform in $h = 1/N$, with this property. If $V_\mu = e^{iha_\mu}$ is real and $\|a\|_\infty \le b_p$, there is a mean-zero real site function λ such that $g = e^{i\lambda}$ and

$$V^g = e^{ihB}, \quad \operatorname{div}_h B = 0, \quad \|D\lambda\|_p + \|\lambda\|_\infty + \|B\|_p \le C_p \|a\|_\infty. \tag{3.dcxciii}$$

The solution is unique in the ball specified in the proof. This is a uniqueness assertion for this small gauge parameter, not for all gauge representatives of the connection. ■

Proof. At the bond (x, μ) use eq. 3.dcxci with $w = D^+_\mu \lambda(x)$ and $A = a_\mu(x)$. Thus

$$B = a - D\lambda + N(\lambda), \qquad \lambda = -G_h \operatorname{div}_h(a + N(\lambda)) \tag{3.dcxciv}$$

is exactly the Coulomb equation. The second map has mean zero. Use the Banach norm $\|D\lambda\|_p$ on mean-zero site functions. The Sobolev estimate in eq. 3.dclxxx controls $\|\lambda\|_\infty$ by this norm. The elementary atomic bound is

$$h\|D\lambda\|_\infty \le h^{1-4/p} \|D\lambda\|_p \le \|D\lambda\|_p. \tag{3.dcxcv}$$

It controls the group increment without asserting a uniform bound on the unscaled L^∞ derivative of λ.

On the ball $\|D\lambda\|_p \le R$ with $\|a\|_\infty \le b$ and $b + R$ small, Lem. 3.43 gives

$$\begin{aligned} \|N(\lambda)\|_p &\le C_p(b + R)(b + R), \\ \|N(\lambda) - N(\widetilde{\lambda})\|_p &\le C_p(b + R)\|D\lambda - D\widetilde{\lambda}\|_p. \end{aligned} \tag{3.dcxcvi}$$

To check the quadratic increment term, use $h\|w^2\|_p \le h\|w\|_\infty \|w\|_p$ and eq. 3.dcxcv. For a difference of two such terms use $w^2 - \widetilde{w}^2$ as two ordered matrix products. The terms containing λ use its uniform Sobolev bound. These arguments also prove the displayed Lipschitz estimate when the matrices do not commute.

By eq. 3.dclxxix, the map on the right of eq. 3.dcxciv is bounded in this norm by $C_p\|a + N\|_p$. Let K_p dominate this constant, the fixed torus volume factor and the constants in eq. 3.dcxcvi. Take $R = 2K_p b$, and then take b_p so small that the domain of Lem. 3.43 contains the whole ball and

$$K_p C_p(b_p + 2K_p b_p) \le \tfrac{1}{4}, \qquad K_p C_p(b_p + 2K_p b_p)^2 \le K_p b_p/2. \tag{3.dcxcvii}$$

The map is a contraction and maps the closed radius-R ball into itself. Its real iterates converge to a real solution. The logarithms remain on the common branch by the same inequalities. Taking the divergence of the first identity in eq. 3.dcxciv proves the Coulomb equation exactly. The estimates for λ and B now follow from the ball bound, Sobolev and eq. 3.dcxcvi. □

The parameter ball is uniform in the mesh. Its proof used a fixed finite p throughout. The smallness of $hD\lambda$ in eq. 3.dcxcv is enough to keep the link logarithm on its branch; an L^∞ estimate for the order-zero gauge projection was not needed.

3.16.4. *Comb gauges and an auxiliary periodic extension.*

Lemma 3.45 (A comb potential and its curvature-preserving extension)**.** Suppose U is real on Q_8, $h = 1/N \le 1$, and $\|F\|_{\infty,Q_8} \le \kappa$ with κ sufficiently small. There is a real gauge g_0 on Q_8 for which $U^{g_0} = e^{iha}$ on Q_7 and $\|a\|_\infty \le C\kappa$. There is also a side-32 periodic potential a^{ext} agreeing with a on Q_6, satisfying

$$\|a^{\mathrm{ext}}\|_\infty \le C\kappa, \qquad \|F(e^{iha^{\mathrm{ext}}})\|_\infty \le C\kappa. \tag{3.dcxcviii}$$

The extension does not assert a current bound outside Q_6. ■

Proof. Use the fixed coordinate-order comb from one corner of Q_8 to each site. Choose g_0 so its tree transports become one. The transformed value of a bond is the closed transport made from the two combs and that bond. Moving the bond past the later coordinate segments expresses this loop as a product of at most C/h conjugated plaquettes. The estimate $\|AB-1\| \le \|A-1\| + \|B-1\|$ for unitary matrices therefore gives

$$\|U^{g_0}(b) - 1\| \le Ch\kappa. \tag{3.dcxcix}$$

Take the logarithm and divide by h. No bound on $g_0 - 1$ is used.

We record a local analytic identity that is also used later. If $V_\mu = e^{ihB_\mu}$ and all $h|B_\mu|$ are in a fixed small ball, then

$$F_{\mu\nu}(V) = (\mathrm{d}B)_{\mu\nu} + E_{\mu\nu}(B), \qquad |E_{\mu\nu}(B)(x)| \le C \max_{b\in\partial p} |B(b)|^2. \tag{3.dcc}$$

Indeed apply the matrix logarithm to the four exponential factors of the plaquette. Its first-order term is $h[B_\mu(x) + B_\nu(x+he_\mu) - B_\mu(x+he_\nu) - B_\nu(x)] = h^2(\mathrm{d}B)_{\mu\nu}$. All remaining terms have degree at least two in these four variables, and division by h^2 gives the stated uniform bound. Its first derivatives with respect to the four B variables are bounded by $C\max|B|$ on a smaller chart.

It follows that $\|\mathrm{d}a\|_\infty \le C\kappa$ on Q_7. Choose a fixed smooth scalar cutoff χ, equal to one on Q_6 and zero outside the interior of Q_7, and put $a^{\mathrm{ext}}(b) = \chi(b_-)a(b)$ on the available bonds and zero elsewhere. Embed its support in the side-32 torus. The discrete product rule gives $\|da^{\mathrm{ext}}\|_\infty \le C\kappa$. Apply eq. 3.dcc once more to the extended potential. This proves eq. 3.dcxcviii. At the finest exceptional collar positions the same formula uses only bonds in Q_8, since $h \le 1$. Thus the construction has the claimed domain even when $N = 1$. □

The extension is an analytic tool on a fixed auxiliary torus. On the inner cube it agrees with a gauge transform of the given field. It imposes no condition on holonomies of any larger ambient torus. The current of the extension is deliberately not used near its cutoff.

3.16.5. *The exact current identity and its elliptic consequence.*

Lemma 3.46 (The Wilson current as a covariant difference)**.** With the normalization in eq. 3.dclxxiii,

$$J_\mu(x) = \frac{1}{h}\sum_{\nu\neq\mu}\left[W_{\mu\nu}(x) - \mathrm{Ad}(U_\nu(x-he_\nu)^{-1})W_{\mu\nu}(x-he_\nu)\right]. \tag{3.dcci}$$

This identity is exact for real SU(2) fields on the logarithm branch. It is not a small-field replacement of the action. If $U = e^{ihB}$, $\operatorname{div}_h B = 0$ and $M = \|F\|_\infty$, then, wherever the full local stencil is present,

$$\begin{aligned} L_h B_\mu &= J_\mu - \sum_{\nu\neq\mu} D_\nu^- E_{\mu\nu}(B) - \sum_{\nu\neq\mu} D_\nu^- S_h(F_{\mu\nu}) - \mathcal{R}_\mu, \\ S_h(X) &= h^{-2}\sin(h^2 X) - X, \\ |\mathcal{R}(x)| &\le C\|B\|_{\infty,\mathrm{stencil}} M, \qquad |\operatorname{div} S_h(F)(x)| \le C h^3 M^3. \end{aligned} \tag{3.dccii}$$

In the same local norms,

$$|DE(B)| \le C\|B\|_\infty \max|DB|. \tag{3.dcciii}$$

■

Proof. Differentiate a plaquette containing the positive bond (x,μ). For the plaquette based at x whose first edge is that bond, the derivative is $h\langle a_\mu(x), h^2 W_{\mu\nu}(x)\rangle$. For the plaquette based at $x - he_\nu$, the bond occurs reversed. Move the insertion to the initial point by cyclicity of the trace. Its contribution is

$$-h^3\left\langle a_\mu(x), \operatorname{Ad}(U_\nu(x-he_\nu)^{-1}) W_{\mu\nu}(x-he_\nu)\right\rangle.$$

The apparent additional conjugation by the plaquette itself drops out: $W(P) = (P - P^*)/(2ih^2)$ commutes with P. The same calculation with the inverse orientation handles $\nu < \mu$, using $W_{\nu\mu} = -W_{\mu\nu}$. Divide the total derivative by the physical factor h^4. This proves eq. 3.dcci, with its sign and all spacing powers fixed.

Replace its covariant difference by the ordinary backward difference. The error is bounded by $C\|B\|_\infty M$, since $\|1 - \operatorname{Ad}(e^{-ihB})\| \le Ch|B|$ and $|W(F)| \le C|F|$ on the branch. Next insert $F = \mathrm{d}B + E(B)$ and $W(F) = F + S_h(F)$, and use eq. 3.dcxc with zero divergence.

The sine series gives $|S_h(X)| \le Ch^4|X|^3$. An ordinary first difference of any bounded lattice function is at most twice its supremum divided by h. Therefore

$$|D^- S_h(F)| \le Ch^3 M^3. \tag{3.dcciv}$$

No derivative bound for F is hidden in this estimate. Finally $E(B)(x)$ is the same analytic function of the four neighbouring B values at every x. Subtract its values at two neighbouring plaquettes and use the integral derivative of that function, bounded as in eq. 3.dcc. This proves eq. 3.dcciii. □

The factor h^3 in eq. 3.dcciv is useful. A bound on the physical curvature is much stronger than a bound on an unscaled plaquette logarithm when the mesh is small. This factor permits the Wilson current to replace the divergence of logarithmic curvature without requiring a modulus of continuity for that curvature.

3.16.6. *Proof of the local gauge bounds.*

Proof of Prop. 3.41. Scale to $r = 1$ as in eq. 3.dclxxviii; write $h = \eta/r$. Apply Lem. 3.45, then Prop. 3.44 with $p = 16$ to the auxiliary connection. Write its transformed potential as B. Choose ε_* small enough that $C\kappa$ in the extension is below b_{16} and all logarithms stay in the fixed branches. On the torus,

$$\operatorname{div}_h B = 0, \qquad \|B\|_{16} \le C\kappa, \qquad \|F(B)\|_\infty \le C\kappa. \tag{3.dccv}$$

The last bound follows from gauge covariance of the actual plaquettes, not from a separate estimate of differentiated gauge parameters.

The group chart also gives $h\|B\|_\infty \le C\kappa$, using eq. 3.dcxcv. Hence eq. 3.dcc is valid at every torus plaquette. By eq. 3.dclxxxi with $p = 8$,

$$\|DB\|_8 \le C(\|F\|_8 + \|B\|_{16}^2) \le C(\kappa + \kappa^2). \tag{3.dccvi}$$

Apply the Sobolev estimate in eq. 3.dclxxx to each component of B, retaining its mean, which is bounded by $C\|B\|_{16}$. We obtain

$$\|B\|_\infty \le C\kappa. \tag{3.dccvii}$$

This is a finite-p estimate. It does not estimate a Riesz transform of F in L^∞. Return to eq. 3.dclxxxi, now with $p = 16$ and the bound $\|E(B)\|_{16} \le C\|B\|_\infty\|B\|_{16}$. Thus

$$\|DB\|_{16} \le C\kappa. \tag{3.dccviii}$$

On Q_6 the auxiliary connection before the last gauge agrees with U^{g_0}. Its current on Q_5 is therefore the gauge transform of the given actual current. Norms of real currents are unchanged by that unitary conjugation. Use Lem. 3.46, eq. 3.dccvii and eq. 3.dccviii; the full current stencil lies in the region of agreement. On a fixed cube between Q_4 and Q_5,

$$\|L_h B\|_{16} \le C(j + \kappa + \kappa^2). \tag{3.dccix}$$

The terms are respectively the actual current, BDB, the transport term BF, and the bounded sine correction h^3F^3. Since $h \le 1$ and $\kappa \le \varepsilon_*$, the last two are bounded in the indicated fixed volume. The extension's current near its cutoff does not enter this estimate.

Let ψ be a fixed scalar cutoff supported in Q_4 and equal to one on Q_3. Its sampled first and second differences are uniformly bounded, including when $h = 1$. The discrete product rule, eq. 3.dccviii, and eq. 3.dccix give

$$\|L_h(\psi B)\|_{16} \le C(j + \kappa).$$

All neighbours read by the product rule are inside Q_5. The constant mode is killed by D, so $D(\psi B) = DG_hL_h(\psi B)$. The order-minus-one bound in eq. 3.dclxxx proves

$$\|DB\|_{\infty,Q_2} \le C(\kappa + j). \tag{3.dccx}$$

The margin between Q_2 and Q_3 contains every difference stencil even at $h = 1$. This is the step that obtains a bounded first derivative. It uses the current to control a Laplacian in L^{16}, then an order-minus-one kernel. It is not an endpoint consequence of $F \in L^\infty$ alone.

Use the exact equation eq. 3.dccii again, now pointwise on Q_1. If $h \le 1/1024$, the same estimate holds on $Q_{3/2}$: every first-difference base point read by eq. 3.dcciii then lies in Q_2. Equations eq. 3.dccvii, eq. 3.dccx, and eq. 3.dcciv give

$$\|L_h B\|_\infty \le C(j + \kappa(\kappa + j) + h^3\kappa^3) \le C(j + \kappa^2). \tag{3.dccxi}$$

Decrease ε_* once to absorb the harmless factor $1 + C\kappa$ into the constant multiplying j.

When $h \le 1/1024$, apply the mesh-uniform interior Hölder estimate of § 3.15 to the ordinary componentwise Laplacian, using eq. 3.dccvii and eq. 3.dccxi on $Q_{3/2}$. It yields $[DB]_{\alpha,Q_1} \le C_\alpha(\kappa+j)$. If its fixed interior radius is less than $64h$, then h has a fixed positive lower bound. In that finite coarse-mesh range, $|DB(x) - DB(y)| \le 2\|DB\|_\infty$ and $|x - y| \ge h$ prove the same Hölder bound directly from eq. 3.dccx.

The combined gauge is $g = e^{i\lambda}g_0$ on the original cube. It is real and its transformed links agree with e^{ihB} on the inner region. Undo eq. 3.dclxxviii. This gives every power of r in eq. 3.dclxxvi and eq. 3.dclxxvii. □

The smallness constant is chosen in a definite order: first fix the matrix logarithm polydisc and the side-32 torus; then the finite constants in Lem. 3.42 for $p = 8, 16$; then the contraction ball in eq. 3.dcxcvii; finally choose ε_* below the finitely many branch and absorption ceilings in the proof. None of these choices depends on the mesh, the value of j, or the connection outside the original cube.

3.16.7. *Comparison with the background regularity hypotheses.*

Corollary 3.47 (Local-scale regularity)**.** Suppose a cube of radius $8m\xi$ lies inside the action region, including every plaquette used by the current, and

$$\|F\|_\infty \le \alpha\xi^{-2}, \qquad \|J_A(U)\|_\infty \le \alpha\xi^{-3}, \qquad m^2\alpha \le \varepsilon_*. \tag{3.dccxii}$$

Here $m \ge 1$, $m\xi/\eta \in \mathbb{N}$, and $0 < \eta \le \xi$. There is a local gauge on the inner cube for which

$$\begin{aligned} \xi\|A\|_\infty &\le Cm\alpha, \\ \xi^2\|DA\|_\infty &\le C(1+m)\alpha, \\ \xi^3\|L_\eta A\|_\infty &\le C(\alpha + m\alpha^2), \\ \xi^{2+\beta}[DA]_\beta &\le C_\beta m^{-\beta}(1+m)\alpha \quad (0<\beta<1). \end{aligned} \tag{3.dccxiii}$$

■□

Proof. Use Prop. 3.41 with $r = m\xi$, $\kappa = \alpha\xi^{-2}$ and $j = \alpha\xi^{-3}$, then substitute. The first-order bounds are linear in m. The hypothesis $m^2\alpha \le \varepsilon_*$ is a cost of this proof and is retained. □

The corollary supplies the potential, gradient and Laplacian premises needed for the local Hölder deduction. It applies to the physical Wilson current. A separately prescribed auxiliary current satisfying a numerical bound cannot replace $J_A(U)$ in eq. 3.dccxii. At a boundary where some plaquettes of the bond stencil are absent, the exact equation has different terms; the statement above does not remove that boundary problem.

The connection may already have a block-axial or a Landau representative required by its global construction. The local gauge just constructed is an additional frame in which to estimate that connection. Passing a non-covariant gauge functional to this frame requires its own exact transformation law. In particular, the absolute soft contour gauge is only invariant under block-constant gauges. The present g is not claimed to be block constant.

3.16.8. *The coupled inverse and its analytic inputs.* For the actual constrained action, the field operator on its fixed slice is the Lagrangian Hessian

$$K_{\mathrm{L}} = F''(A) - D^2\langle\omega, q(A)\rangle, \qquad \mathcal{S} = \begin{pmatrix} K_{\mathrm{L}} & -q'(A)^\sharp \\ q'(A) & 0 \end{pmatrix}. \tag{3.dccxiv}$$

Here q is the actual nonlinear straightened block constraint, the pairing contains its physical row weights, and ω is the corresponding row multiplier. A uniform bound for $\mathcal{S}^{-1}$ uses the following quantitative ingredients:

(i) a quantitative inverse on the true constrained tangent space, including the boundary and seam conditions;
(ii) a controlled right inverse of the actual nonflat field rows, with their node factors and outward-row correction;
(iii) a relative bound for the constraint-curvature term paired with the actual multiplier, and for every specified independent-current completion;
(iv) the exact comparison of the chosen contour gauge with the gauge block used in the precision;
(v) control of the nonlocal subtraction that replaces the full Feynman term $D_U D_U^\sharp$ by $D_U R D_U^\sharp$.

The local gauge result proves an input to these estimates. Its derivative bounds do not by themselves settle the five requirements. The following single-scale results prove a separate projected-Landau gap and the corresponding real constrained saddle inverse in the homogeneous periodic energy spaces; see Prop. 3.56 and Cor. 3.51. They retain the actual constraint curvature. The proper strong response and its coefficient families on a given admitted reference are constructed in Props. 3.163 and 3.169. The forced trial residual is treated in Prop. 3.120, and Prop. 3.194 produces the required homogeneous trial from coarse curvature. The proper curvature-data construction is proved in

Prop. 3.124, its tower admission in Cors. 3.130 and 3.131, and the response on the full first-order regular-pair domain in Prop. 3.174. The local gauge theorem itself remains an estimate for the given connection.

3.17. **The Landau precision on one physical block scale.** The full Feynman form controls every component of the divergence. The background precision retains only its projection by R. Subtracting the omitted component is therefore a substantive step. We first prove a gap for that projected flat form, uniform as the fine mesh tends to zero. We then compare the actual nonflat coefficients with the flat ones in a common small physical gauge. The result includes the nonlinear constraint Hessian and an inverse on the actual simultaneous constraint and Landau slice.

There is one physical block scale throughout this section. All coarse sites have the same assigned level, all rows are present, and the ambient lattice is periodic. Thus there are no outward rows, layer seams or proper-region boundary terms. The assumptions below do not replace those cases by a periodic problem.

3.17.1. *Operators and pairings.* Set the coarse spacing to one and the fine spacing to $h = L^{-k}$, where $L \geq 3$ is a fixed odd integer and $k \geq 0$. The periodic side length is an integer $M \geq 2$. Fine site, bond and plaquette norms use measure h^4; coarse site and bond norms use counting measure, which is their physical volume measure at this scale. The symbol $\sharp$ denotes the transpose for these pairings, and is the Hermitian adjoint when taking norms of real coefficients on their complexification.

At a flat connection choose a parallel frame on the universal cover. The resulting boundary twists are unitary. Write D for the fine gradient, d for its curl and $D^\sharp$ for its divergence adjoint. Let q be the arithmetic average on each centered fine block, and let Q be the physical straight bond derivative:

$$(q\lambda)(y) = h^4 \sum_{x \in B(y)} \lambda(x), \qquad (Qv)_\mu(y) = h^5 \sum_{x \in B(y)} \sum_{t=0}^{h^{-1}-1} v_\mu(x + the_\mu). \tag{3.dccxv}$$

These are the flat derivatives of the actual site and straightened field maps. Telescoping proves the exact identity

$$QD = \mathrm{d}_c q, \tag{3.dccxvi}$$

where d_c is the gradient with unit coarse step. In particular the site map in this identity is an average, not evaluation at the block center.

Put

$$L = D^\sharp D, \quad G = (L + q^\sharp q)^{-1}, \quad F = Gq^\sharp, \quad \mathcal{M} = F^\sharp F, \quad R = 1 - F\mathcal{M}^{-1}F^\sharp. \tag{3.dccxvii}$$

The positive coefficient multiplying $q^\sharp q$ can be changed without changing the range of F, hence without changing R. Indeed, for any $z \in \ker q$, $GLz = z$. Consequently

$$\operatorname{ran} R = L(\ker q), \qquad qGR = 0, \qquad LGR = R. \tag{3.dccxviii}$$

For the first assertion, $F^\sharp Lz = qz = 0$. Conversely, if $F^\sharp f = 0$, then $z = Gf$ satisfies $qz = 0$ and $Lz = f$. The other two identities follow from the same calculation.

The omitted projector is not $q^\sharp q$. The factor G in eq. 3.dccxvii is important. It makes R the projection onto $L(\ker q)$, rather than onto $\ker q$.

3.17.2. *A gap for the flat projected form.*

Proposition 3.48 (A flat Landau gap uniform in the fine mesh)**.** There is a constant $c_0 > 0$, independent of k, M and the real flat holonomies, such that

$$\|\mathrm{d}v\|_2^2 + \|RD^\sharp v\|_2^2 + \|Qv\|_2^2 \geq c_0(\|Dv\|_2^2 + \|v\|_2^2). \tag{3.dccxix}$$

Here Dv denotes all componentwise fine first differences. The assertion holds for the three real colour components and for their complexification. ■

Proof. We include the fine-mesh limiting argument. It avoids substituting a fixed-L estimate for a bound uniform in L^k.

Translation by one coarse unit commutes with all the flat operators. Bloch decomposition reduces the problem to one coarse block with quasiperiodic phase $\theta \in [-\pi, \pi]^4$. It is enough to prove one bound for every such phase. The allowed phases of any finite periodic torus form a subset. A flat adjoint holonomy merely shifts these phases after simultaneous diagonalization of its commuting unitary translations. It therefore suffices to treat one complex colour component.

On a Bloch block the coarse site space has dimension one. The site inequality from the block variance proof gives

$$\|u\|_2^2 \le C(\|Du\|_2^2 + |qu|^2), \tag{3.dccxx}$$

uniformly in h and θ: it uses only internal edges of the block, so no boundary phase enters its constant. Hence $\|G\| \le C$, $\|F\| \le C$ and $\|DF\| \le C$. Also $\|F\| \ge c > 0$ uniformly. To verify the last assertion, choose a fixed smooth function φ supported strictly inside the cell and of integral one. For all sufficiently fine meshes its sample has $|q\varphi| \ge 1/2$ and $\|(L + q^\sharp q)\varphi\|_2 \le C$. Then

$$\tfrac{1}{2} \le |q\varphi| = |\langle F, (L + q^\sharp q)\varphi\rangle| \le C\|F\|_2.$$

For the finitely many remaining values of $h = L^{-k}$, $L + q^\sharp q$ has operator norm at most $16h^{-2} + 1$ and $\|q^\sharp 1\| = 1$, which gives the same conclusion with a fixed smaller constant. The argument is uniform in θ.

Write $P = 1 - R$. Since P has range spanned by F, summation by parts gives

$$\|PD^\sharp v\|_2 = \frac{|\langle F, D^\sharp v\rangle|}{\|F\|_2} \le \frac{\|DF\|_2}{\|F\|_2}\|v\|_2 \le C\|v\|_2. \tag{3.dccxxi}$$

The exact flat Hodge identity thus yields the uniform Gårding inequality

$$\|Dv\|_2^2 \le \|\mathrm{d}v\|_2^2 + \|RD^\sharp v\|_2^2 + C\|v\|_2^2. \tag{3.dccxxii}$$

There is no fine-spacing loss in eq. 3.dccxxi; the derivative falls on the smoothed column F, not on an arbitrary bounded site function.

Suppose there were no uniform mass gap. Choose Bloch data with $\|v_n\|_2 = 1$ and with the left side of eq. 3.dccxix tending to zero. Pass to a subsequence with $\theta_n \to \theta$. If the corresponding integers h_n^{-1} remain bounded, all the matrices vary continuously in a fixed finite-dimensional space. A zero-energy limit would then contradict the kernel calculation below, applied on that finite lattice. We may therefore assume $h_n \to 0$.

Inequality eq. 3.dccxxii bounds every first difference. Here is a useful precise compactness realization. Interpolate a Bloch grid function by its trigonometric polynomial with frequencies $\theta_n + 2\pi\ell$ in the Brillouin cube. Parseval gives the same L^2 norm, and $4h_n^{-2}\sin^2(h_n k_\mu/2) \ge c|k_\mu|^2$ gives a uniform continuum H^1 bound. After multiplication by $e^{-i\theta_n x}$ these are periodic H^1 functions on a fixed cell. Rellich compactness gives strong L^2 convergence of a subsequence. The piecewise constant interpolation has the same limit. Indeed discrete orthogonality gives, for $|z| \le Ch_n$,

$$h_n^4 \sum_x |u^{\mathrm{trig}}(x + z) - u^{\mathrm{trig}}(x)|^2 \le |z|^2 \|\nabla u^{\mathrm{trig}}\|_2^2.$$

Integrating z over one fine cell shows that its L^2 difference from the piecewise constant interpolation is at most $Ch_n\|Du\|_2$. The limiting field v consequently has norm one and lies in the quasiperiodic H^1 space.

Ordinary differences converge weakly to the continuum differences, as is seen first on a fixed finite set of Fourier modes and then by their uniform L^2 bound. Thus $\mathrm{d}v = 0$. The site averages

converge to integration over the cell. The straight rows converge to

$$(Q_\theta v)_\mu = \int_B \int_0^1 v_\mu(x + te_\mu)\, dt\, dx. \tag{3.dccxxiii}$$

To check this convergence, rewrite the discrete row as $h_n \sum_t qT_{th_n} v_n$ on piecewise constant fields. Replacing th_n by t changes the result by at most $Ch_n \|Dv_n\|_2$, using the translation estimate just proved. Strong L^2 convergence handles the remaining integral. It follows that $Q_\theta v = 0$.

The smoothed columns F_n converge strongly as well. The equation $(L_n + q_n^\sharp q_n)F_n = q_n^\sharp 1$, its uniform energy bound, and the same compactness give a strong L^2 limit F_θ. Testing against sampled smooth Bloch functions identifies its equation

$$\langle \nabla\phi, \nabla F_\theta \rangle + \overline{q_\theta \phi}\, q_\theta F_\theta = \overline{q_\theta \phi}. \tag{3.dccxxiv}$$

The continuum block Poincaré inequality proves uniqueness. The uniform lower bound on $\|F_n\|$ passes to the limit. Therefore their rank-one orthogonal projections converge in operator norm under these interpolations. Since $D^\sharp v_n$ is weakly bounded and $R_n D^\sharp v_n \to 0$, we obtain $R_\theta \nabla^\sharp v = 0$. The continuum projector satisfies the same range identity as eq. 3.dccxviii; its proof uses the continuum resolvent and is unchanged.

We finish by identifying the zero kernel, both in the continuum and on each fixed mesh. Fourier decomposition of a curl-free Bloch field gives $v = D\lambda$ unless $\theta = 0$, in which case $v = D\lambda + c$ with c a constant vector. In the continuum D means ∇ and $\lambda \in H^2$; on a fixed mesh this is the same finite Fourier calculation. For each nonzero frequency the curl condition says that its vector coefficient is a scalar multiple of the gradient symbol. The only zero gradient symbol is the constant mode at $\theta = 0$.

At $\theta = 0$ the row of a constant vector is that vector, while $QD\lambda = \mathrm{d}_c q\lambda = 0$. Hence $Qv = 0$ forces $c = 0$. At $\theta \neq 0$, some component of d_c is nonzero, so $QD\lambda = 0$ forces $q\lambda = 0$. At $\theta = 0$ we obtain the same normalization by subtracting a scalar constant from λ. Now $D^\sharp v = L\lambda$ belongs to $L(\ker q)$, and eq. 3.dccxviii gives

$$0 = RD^\sharp v = L\lambda, \qquad D\lambda = 0.$$

Thus $v = 0$, contradicting its unit norm. This proves a uniform mass gap. Combine that gap with eq. 3.dccxxii to obtain the full estimate eq. 3.dccxix. □

The compactness proof determines a positive constant for the fixed four-dimensional cell geometry. It gives no claimed numerical optimal value. Its quantifiers include all k and every Bloch phase, hence every compatible periodic volume and flat holonomy.

3.17.3. *Actual nonflat coefficients in a common gauge.* For a real connection U, let D_U be its covariant site gradient. Let q_U be the actual hierarchical site average on each assigned pillar and let Q_U be the derivative of the actual straightened nonlinear field constraint $\mathcal{Q}_U$. These are the operators whose coefficients were obtained from the finite recursion in § 3.13. In particular

$$Q_U D_U = d_{\mathcal{M}^k U} q_U. \tag{3.dccxxv}$$

The derivative of the plain nonlinear average has a different site readout and is not substituted here. Define

$$G_U = (D_U^\sharp D_U + q_U^\sharp q_U)^{-1}, \quad F_U = G_U q_U^\sharp, \quad R_U = 1 - F_U (F_U^\sharp F_U)^{-1} F_U^\sharp. \tag{3.dccxxvi}$$

These are the actual site resolvent and Landau projector, with the physical pairings already declared.

Lemma 3.49 (Comparison of the actual rows and projector)**.** Let V be real and flat, and suppose in a common parallel frame

$$U_b = e^{iha_b} V_b, \qquad \|a\|_\infty \leq \delta. \tag{3.dccxxvii}$$

For a fixed sufficiently small $\delta_L > 0$, the actual regular mean branches are admitted. Uniformly in k and M,

$$\begin{aligned} \|D_U - D_V\|_{2\to 2} &\le C\delta, \\ \|q_U - q_V\|_{2\to 2} + \|Q_U - Q_V\|_{2\to 2} &\le C_L\delta, \\ \|R_U - R_V\|_{2\to 2} &\le C_L\delta, \\ \|(R_U D_U^\sharp - R_V D_V^\sharp)v\|_2 &\le C_L\delta(\|D_V v\|_2 + \|v\|_2). \end{aligned} \tag{3.dccxxviii}$$

There is also a right inverse E_U of Q_U with

$$\|E_U b\|_2 + \|\nabla_U E_U b\|_2 \le C_L\|b\|_2, \qquad \sup_c \sum_{b \text{ fine}} h^4 |E_U(b,c)| \le C_L. \tag{3.dccxxix}$$

■

Proof. The first comparison follows from $\|\mathrm{Ad}(e^{iha}) - 1\| \le Ch\delta$. The actual internal path transports over physical length at most one differ from those at V by $C_L\delta$ in this frame. For hierarchical transports, the perturbation at intermediate physical scale ξ_s is at most $C_L\delta\xi_s$, and $\sum_{s<k}\xi_s \le 1/(L-1)$. The site-row comparison follows by averaging their transported coefficients.

For the field rows use the actual finite analytic recursion and its row and column estimates in § 3.13. Its one-step relative error at scale s is $C_L\delta\xi_s$; their geometric sum is bounded by $C_L\delta$. This bounds both Schur sums for the declared physical measures by that amount; the fine column density retains its factor h^4. Schur's test gives the second line of eq. 3.dccxxviii. This is the row comparison for Q_U, including the differentiated straightening factors. It is not an infinity-norm estimate for a transported straight main term in place of Q_U.

The flat site gap and the first two comparisons make $D_U^\sharp D_U + q_U^\sharp q_U$ coercive on the physical site H^1 space, uniformly in volume. Its form differs from the flat form by at most $C_L\delta$ times the two H^1 norms. The resolvent identity in the H^1–H^{-1} pairings gives

$$\|G_U - G_V\|_{L^2\to H^1} \le C_L\delta, \qquad \|F_U - F_V\|_{2\to H^1} \le C_L\delta.$$

The flat Gram lower bound used in the Bloch proof, or equivalently the normalized site Gram estimate already proved for the actual pillars, implies $F_V^\sharp F_V \ge cI$. Hence its inverse and the inverse at U are uniformly bounded, and their difference is $O_L(\delta)$. Substitute these facts in eq. 3.dccxxvi. This proves the projector comparison. Its last stated line now follows by inserting and subtracting $R_U D_V^\sharp v$. It loses no factor h^{-1} because the remaining divergence is measured by the physical first-difference norm of v.

For the right inverse, use the compact signed two-moment row seeds of § 3.13. At one assigned scale their sum E^0 satisfies $Q_V E^0 = 1$ exactly. Every column is supported inside a physical block, has bounded amplitude and bounded first differences, and has physical L^1 norm at most a fixed constant. Their supports and gradient collars have bounded overlap. The field-row comparison therefore gives

$$S_U = Q_U E^0 = 1 + B_U, \qquad \|B_U\|_{\ell^1\to\ell^1} + \|B_U\|_{\ell^\infty\to\ell^\infty} \le C_L\delta.$$

Reduce δ_L so each norm is less than $1/2$. Then $E_U = E^0 S_U^{-1}$ is an exact right inverse. The Neumann series is bounded in both coarse Schur norms. The seed energy and column L^1 bounds prove eq. 3.dccxxix. The finitely many small block sizes, including a bare identity row, have the explicit finite seed inverses given in that construction. All constants are therefore independent of k. □

3.17.4. *The physical constraint curvature and the full form.* For a row multiplier ω, pair the Hessian of the actual per-length nonlinear constraint with the physical coarse measure:

$$B_\omega(v,w) = \sum_c \langle \omega(c), D^2\mathcal{Q}_{U,c}(0)[v,w]\rangle. \tag{3.dccxxx}$$

The two-leg coefficient moment of the actual recursion gives

$$|B_\omega(v,w)| \le C_L\|\omega\|_\infty\|v\|_2\|w\|_2. \tag{3.dccxxxi}$$

This is eq. 3.dciii at physical scale one. No Gaussian sample is required to lie in a holomorphic chart: the estimate is a bound on the resulting finite bilinear coefficient matrix.

Let a now denote a fixed positive row weight, with $0 < a_* \le a(c) \le a^* < \infty$. It is unrelated to the potential a_b in eq. 3.dccxxvii. Define the real form

$$\begin{aligned}\mathcal{K}_{U,\omega}(v,w) = D^2A_W(U)[v,w] - B_\omega(v,w)\\ + \langle R_U D_U^\sharp v, R_U D_U^\sharp w\rangle + \langle Q_U v, aQ_U w\rangle.\end{aligned} \tag{3.dccxxxii}$$

The action derivatives use physical left variations $e^{ihtv}U$. Thus the action Hessian has its physical L^2 normalization; no unscaled bond Hessian is inserted in this formula.

Proposition 3.50 (Uniform nonflat precision in a common small gauge)**.** Fix L, a_*, a^*. There are $\varepsilon_L > 0$ and $c_L, C_L > 0$ independent of k, M and the flat holonomies such that the following holds. Suppose eq. 3.dccxxvii holds relative to a real flat connection V,

$$\|F(U)\|_\infty \le \kappa, \qquad \|\omega\|_\infty \le \gamma, \qquad \delta + \kappa + \gamma \le \varepsilon_L. \tag{3.dccxxxiii}$$

Then, with $\|v\|_X^2 = \|\nabla_U v\|_2^2 + \|v\|_2^2$,

$$c_L\|v\|_X^2 \le \mathcal{K}_{U,\omega}(v,v) \le C_L\|v\|_X^2. \tag{3.dccxxxiv}$$

Consequently its full penalized inverse has $L^2 \to L^2$ norm at most c_L^{-1} and $X^* \to X$ norm at most c_L^{-1}.

If U is a critical point of the actual constrained action and its full physical stationarity equation is $J_A(U) = Q_U^\sharp\omega$, then

$$\|\omega\|_\infty \le C_L\|J_A(U)\|_\infty. \tag{3.dccxxxv}$$

Thus a sufficiently small bound on this actual current supplies the multiplier hypothesis. Criticality and that full stationarity identity are hypotheses, not conclusions of the coercivity estimate. ■

Proof. Apply Prop. 3.48, with the positive lower row weight, to the flat reference. It controls both the flat component gradient and the physical mass norm. The two gradient norms at U and V are equivalent for small δ by Lem. 3.49.

For the Wilson term use the exact plaquette Hessian and its curl comparison from § 3.13. In the common frame its linearized plaquette curl differs from the flat curl by at most $C\delta\|v\|_2$. Its remainder is bounded by $C\kappa\|v\|_2\|w\|_2$, since the two insertions contribute h^2, the plaquette defect contributes κh^2, and their product is the physical measure h^4. Thus

$$\left|D^2A_W(U)[v,w] - \langle \mathrm{d}_V v, \mathrm{d}_V w\rangle\right| \le C(\delta+\kappa)\|v\|_{X_V}\|w\|_{X_V}. \tag{3.dccxxxvi}$$

The gauge and row comparisons in eq. 3.dccxxviii, followed by expansion of the two squares, bound their form differences by $C_L\delta\|v\|_{X_V}\|w\|_{X_V}$. Equation eq. 3.dccxxxi bounds the actual constraint curvature by $C_L\gamma$ times these norms. Choose ε_L so the total relative error is below one half of the flat lower constant. This proves both inequalities in eq. 3.dccxxxiv. The inverse estimates are the ordinary finite-dimensional energy estimates in the stated physical norms.

Finally apply the exact right inverse in Lem. 3.49 to the full stationarity equation: $\omega = E_U^\sharp J_A(U)$. The last bound in eq. 3.dccxxix, with fine measure h^4 and unit coarse measure, proves eq. 3.dccxxxv. There is no factor h^{-4} in that readout. □

This proves positivity of a form with the actual projected gauge. It does not infer that positivity by monotonicity from the Feynman form. The flat projected gap was proved separately, before the nonflat perturbation was made.

3.17.5. *The actual constraint and Landau slice.* The positive extension in eq. 3.dccxxxii gives an inverse for the actual constrained Hessian as well. Let

$$X_L = \{v : R_U D_U^\sharp v = 0\}, \qquad X_0 = X_L \cap \ker Q_U, \qquad K_L = D^2 A_W(U) - B_\omega. \tag{3.dccxxxvii}$$

Give both subspaces the covariant X norm. On X_0 the two penalty terms vanish exactly, so $K_L(v,v) \geq c_L \|v\|_X^2$.

Corollary 3.51 (A uniform saddle inverse on the actual slice)**.** Under the assumptions of Prop. 3.50, the equations

$$K_L u - Q_U^\sharp \mu = f \quad \text{on } X_L, \qquad Q_U u = b \tag{3.dccxxxviii}$$

have a unique solution $u \in X_L$, μ in the actual coarse field-row space, and

$$\|u\|_X + \|\mu\|_2 \leq C_L(\|f\|_{X_L^*} + \|b\|_2). \tag{3.dccxxxix}$$

The constant is independent of fine spacing and compatible volume. ■□

Proof. First construct a bounded right inverse taking values in X_L. The exact site range identities in eq. 3.dccxviii also hold at U. Thus

$$S_U = 1 - D_U G_U R_U D_U^\sharp, \quad S_U X \subset X_L, \quad Q_U S_U = Q_U. \tag{3.dccxl}$$

The last identity uses eq. 3.dccxxv and $q_U G_U R_U = 0$.

We verify the energy bound, rather than treating $D_U^\sharp$ as a bounded operator on bare L^2. For $\lambda = G_U R_U D_U^\sharp v$, the range identity gives $q_U \lambda = 0$ and $D_U^\sharp D_U \lambda = R_U D_U^\sharp v$. The site gap gives $\|\lambda\|_2 + \|D_U\lambda\|_2 \leq C_L \|v\|_X$. The covariant Hodge identity applied to $D_U\lambda$ yields

$$\|\nabla_U D_U \lambda\|_2 \leq C_L(\|D_U^\sharp D_U \lambda\|_2 + \kappa\|\lambda\|_2 + \sqrt{\kappa}\|D_U\lambda\|_2).$$

Indeed the covariant curl of $D_U\lambda$ is a shift commutator bounded by $C\kappa\|\lambda\|_2$, and the Hodge remainder is $C\kappa\|D_U\lambda\|_2^2$. These are precisely the complete periodic plaquette stencils, so there is no boundary term. This proves $\|S_U v\|_X \leq C_L\|v\|_X$. Therefore $E_L = S_U E_U$ is a bounded right inverse of Q_U into X_L.

Write $u = E_L b + z$, $z \in X_0$. The coercive equation

$$K_L(z,w) = f(w) - K_L(E_L b, w) \quad (w \in X_0)$$

has a unique solution and bounds $\|z\|_X$ by $C_L(\|f\|_{X_L^*} + \|b\|_2)$. Define the row vector μ by $\langle \mu, b' \rangle = (K_L u - f)(E_L b')$ for every row vector b'; abbreviate this as $\mu = E_L^\sharp(K_L u - f)$. The functional $K_L u - f$ annihilates X_0, and every $w \in X_L$ decomposes as $w - E_L Q_U w + E_L Q_U w$. Hence it equals $Q_U^\sharp \mu$ on X_L, proving eq. 3.dccxxxviii. The boundedness of K_L and E_L proves the multiplier estimate. The same kernel coercivity and the surjectivity of $Q_U|_{X_L}$ give uniqueness. □

3.17.6. *Complex perturbations and localization.* On a physical first-order complex ball about the real reference, the already established coefficient estimates for the actual means, site resolvent and Gram projector give a form perturbation of size $C_L s$, where

$$\|E\|_\infty + \|\nabla_U E\|_\infty + \|\zeta\|_\infty \leq s, \qquad U_E = e^{ihE} U, \quad \omega_E = \omega + \zeta. \tag{3.dccxli}$$

The first derivative of E controls the complex plaquette change; its presence cannot be replaced by a fixed unscaled bond radius. For s smaller than a fixed fraction of the real lower constant, the full penalized inverse exists holomorphically by the normalized Neumann series. This statement uses analytic bilinear transposes for the continued coefficients; it asserts no Hermitian positivity at a complex field. A constrained inverse on a moving complex slice additionally requires a uniformly bounded holomorphic identification of that slice. Only the real-slice saddle estimate and the full penalized complex inverse are asserted here.

The common-gauge assumption in eq. 3.dccxxvii has global scope. The local gauge construction supplies it on bounded patches. The following localization argument combines those frames without

requiring one global small gauge; Prop. 3.56 proves the real curvature-small homogeneous periodic case.

To patch the estimates, one needs an operator comparison for the nonlocal R_U. For a scalar partition function χ, its gauge term contains

$$R_U D_U^\sharp(\chi v) - \chi R_U D_U^\sharp v = R_U[D_U^\sharp, \chi]v + [R_U, \chi]D_U^\sharp v. \tag{3.dccxlii}$$

An L^2 bound for $[R_U, \chi]$ alone leaves a fine derivative on v. The required composed bound and the sharper quadratic IMS estimate are proved in Lem. 3.54 using the smoothed projector's physical cell operators. The comparison between the global site inverse and the periodic extension is proved in Lem. 3.55. Both arguments use the exact actual site rows and Gram matrix, independently of the fine vector inverse.

Together these results close a real nonflat periodic one-scale energy inverse with the actual nonlinear constraint curvature. Different assigned scales, proper-region boundary forms, a moving complex slice and the independent-current completion remain distinct tasks. The equality with a separately specified absolute contour precision and the stronger pointwise response norms also require their own estimates.

3.18. **Localization of the one-scale Landau precision.** The common-gauge theorem leaves a nonlocal comparison: a local frame does not change the global projector into a projector computed on that patch. We prove the comparison using only the site inverse and its Gram matrix. The proof uses physical L^2 operators between unit cells. It requires neither pointwise mixed-derivative estimates nor a fine vector inverse as an input.

We retain the periodic, single-assigned-scale setting of § 3.17. The coarse spacing is one, the fine spacing is $h = L^{-k}$, and every site and field row is present. All constants below can depend on L and on the fixed upper and lower row weights, but not on h or the compatible periodic volume. The real curvature ceiling is fixed below the one for the actual site gap and normalized Gram gap already proved in § 3.13.

3.18.1. *Derivative bounds for the smoothed projector.* Let B_y be the fine cell belonging to coarse site y. A fine bond is assigned to the cell of its initial site. Distances between cells are physical coarse lattice distances, denoted by $d(y, z)$. Write

$$A_s = D_U^\sharp D_U + q_U^\sharp q_U, \quad G = A_s^{-1}, \quad F = G q_U^\sharp, \quad \mathcal{M} = F^\sharp F, \quad P = F\mathcal{M}^{-1}F^\sharp = 1 - R. \tag{3.dccxliii}$$

Here the q_U and all transports are the actual ones. In this homogeneous setting the site gap and Gram gap read

$$\langle u, A_s u\rangle \geq \beta\|u\|_2^2, \qquad m_0 I \leq \mathcal{M} \leq m_1 I, \tag{3.dccxliv}$$

with fixed positive constants. The first bound is the current-free site gap. We justify the second bound here without importing a site lift that requires differentiated background regularity.

Lemma 3.52 (A current-free energy lift and Gram lower bound)**.** At a fixed sufficiently small real curvature ceiling there is a right inverse E of the actual site average with

$$q_U E = 1, \qquad \|Eb\|_{A_s}^2 := \|D_U Eb\|_2^2 + \|q_U Eb\|_2^2 \leq C_E^2\|b\|_2^2. \tag{3.dccxlv}$$

It follows that the actual Gram matrix satisfies

$$C_E^{-4} I \leq F^\sharp F \leq \beta^{-2} I. \tag{3.dccxlvi}$$

No bound on the physical current or on a derivative of a local potential is used. ■

Proof. The site rows use disjoint pillars. Since each coefficient is a unitary adjoint transport with averaging weight h^4, their physical transpose obeys $q_U q_U^\sharp = 1$. Thus $\|q_U\| = \|q_U^\sharp\| = 1$.

First work on a periodic volume large enough to embed the fixed surrounding cubes used below. For each unit pillar choose a real comb frame on such a cube. Its potential has supremum at most $C_L\kappa$, using curvature only. In this frame every actual hierarchical internal transport differs

from one by $C_L\kappa$: the admitted intermediate open transports have size $C_L\kappa\xi_s$, and the physical scales obey $\sum_{s<k}\xi_s \le 1/(L-1)$. This is the current-free pillar transport estimate. It does not differentiate that potential.

Choose one fixed nonnegative smooth scalar bump ϕ supported strictly inside the centered unit cell, with $\phi(0) > 0$. Its sampled mean $m_h = h^4\sum_x \phi(x)$ has a uniform positive lower bound for $h = L^{-k}$. For fine meshes this follows from its positive integral; the finitely many larger meshes contain the center site and have a positive mean as well. Its physical L^2 norm and the physical L^2 norm of its first differences are uniformly bounded, by the smooth bounds and the fixed support.

On the pillar at y put $u^{(y)}(x) = \phi(x-y)\psi_y$ in this local frame and zero outside. The actual site readout there is $N_y\psi_y$, where $\|N_y - m_hI\| \le C_L\kappa m_h$. Decrease the curvature ceiling so that N_y is invertible with $\|N_y^{-1}\| \le 2/m_h$. Choose ψ_y to give the prescribed b_y, including the unitary coarse-frame rotation. No other site row sees this bump, so the sum of these fields is an exact right inverse E.

The first-difference product rule in its local frame gives

$$\|D_U u^{(y)}\|_2 \le C\big(\|D\phi\|_2 + \kappa\|\phi\|_2\big)|\psi_y|.$$

Only the potential's supremum enters. The first-difference collars of the bumps have bounded overlap, including the finitely coarse meshes. Sum over pillars and use $q_U E b = b$ to obtain eq. 3.dccxlv.

For a smaller torus perform this construction on a sufficiently large finite periodic cover. Average the resulting right inverse over the finite deck group, acting simultaneously on its fine output and coarse input. Every conjugate remains a right inverse and has the same energy bound, since the pulled-back coefficients commute with those unitary deck translations. Their average has the same bound and preserves the deck-invariant subspaces. It therefore descends to an exact right inverse on the original torus. Its squared input and output norms acquire the same number of sheets, so the constant in eq. 3.dccxlv is unchanged. This proves the energy lift even when a surrounding comb cube does not embed in the original small torus.

Finally $A_sFb = q_U^\sharp b$, and hence

$$\|b\|_2^2 = \langle A_sFb, Eb\rangle \le C_E\|Fb\|_{A_s}\|b\|_2, \qquad \|Fb\|_{A_s}^2 = \langle q_U^\sharp b, Fb\rangle \le \|b\|_2\|Fb\|_2.$$

For $b \neq 0$ these imply $\|Fb\|_2 \ge C_E^{-2}\|b\|_2$. The upper bound follows from $\|G\| \le \beta^{-1}$ and $\|q_U^\sharp\| = 1$. This proves eq. 3.dccxlvi. □

Lemma 3.53 (Exponential cell bounds with one derivative)**.** There are positive constants s_0 and C with the following property. The operators F, D_UF and $\mathcal{M}^{-1}$ have exponentially decreasing cell L^2 operator entries. In particular

$$\begin{aligned} T &= PD_U^\sharp = F\mathcal{M}^{-1}(D_UF)^\sharp, \\ K_s &= D_UPD_U^\sharp = (D_UF)\mathcal{M}^{-1}(D_UF)^\sharp \end{aligned} \tag{3.dccxlvii}$$

are bounded on their physical L^2 spaces, with

$$\|1_{B_y}T1_{B_z}\| + \|1_{B_y}K_s1_{B_z}\| \le Ce^{-s_0d(y,z)}. \tag{3.dccxlviii}$$

Every fixed polynomial cell moment of these bounds is finite uniformly in the fine mesh and volume. ■

Proof. Let ψ be bounded and physically one-Lipschitz on fine sites, and give coarse sites the sampled values $\psi(y)$. For $w = e^{s\psi}$, the fine-gradient commutator satisfies

$$\|wD_Uw^{-1} - D_U\|_{2\to2} \le C|s| \qquad (|s| \le 1).$$

The difference of weights across one fine bond is $O(|s|h)$; division by h therefore leaves a physical constant. Similarly $w_fq_U^\sharp w_c^{-1} - q_U^\sharp$ has norm $O_L(|s|)$, since the actual site-row carrier has physical

diameter bounded independently of k, and its physical Schur sums are bounded. Conjugating the site form changes it by at most $C_L|s|$ times the two site H^1 norms.

By eq. 3.dccxliv, for fixed sufficiently small $|s|$ the conjugated inverse is bounded from H^{-1} to H^1. Consequently

$$\|w_f F w_c^{-1}\|_{2\to 2} + \|w_f D_U F w_c^{-1}\|_{2\to 2} \le C_L. \tag{3.dccxlix}$$

For the derivative term, differentiate the weighted output and subtract the first commutator above. Thus only one site energy derivative is used. The same conjugation argument and resolvent identity give $\|w_c \mathcal{M} w_c^{-1} - \mathcal{M}\| \le C_L|s|$. Its inverse is uniformly bounded for smaller $|s|$ by the real Gram lower bound. This is the weighted Gram argument, now keeping the energy derivative in eq. 3.dccxlix.

Insert these weighted bounds in eq. 3.dccxlvii. The adjoint factors use the same bounds with $-\psi$. Choose ψ to be a truncated distance from the input cell; its values differ from cell distance by at most a fixed constant. This gives eq. 3.dccxlviii. Homogeneous four-dimensional cells have at most $C(n+1)^3$ neighbors at distance between n and $n+1$. Summing against the exponential proves every fixed polynomial moment.

All weights can be chosen on the finite periodic lattice. Truncated distance functions have the required physical Lipschitz bounds, independently of its size. □

The derivative in $PD_U^\sharp$ has fallen on the smoothed column F. Its boundedness here is not an order-zero L^∞ projection estimate. All the bounds in the lemma are physical L^2 cell bounds.

3.18.2. *Cutoffs and the nonlocal IMS bound.* Choose a smooth partition $\{\chi_i\}$ on the covering lattice, with supports of radius CR_0, where $R_0 \ge 2$ is an integer. It may be chosen so that

$$\begin{aligned} &\sum_i \chi_i(x)^2 = 1, \qquad \sum_i |D\chi_i(x)|^2 \le C R_0^{-2}, \\ &\sum_i |\chi_i(x) - \chi_i(z)|^2 \le C \min\{1, |x-z|^2/R_0^2\}. \end{aligned} \tag{3.dccl}$$

For instance translate one fixed smooth bump at every integer coarse site, scale its radius by R_0, and divide by the square root of the sum of its squared translates. This sum is comparable with R_0^4. The construction is invariant under integer translations and has the bounds displayed, including

$$\sum_i \sup_{x\in B_y} |\chi_i(x) - \chi_i(y)|^2 \le C R_0^{-2}. \tag{3.dccli}$$

The number of summands can grow with R_0; their squared derivative sum has the stated bound. Cutoffs on fine bonds are sampled at their initial sites.

Lemma 3.54 (A composed commutator and an IMS estimate)**.** For these cutoffs,

$$\sum_i \|[P, \chi_i] D_U^\sharp v\|_2^2 \le C R_0^{-2} \|v\|_2^2. \tag{3.dcclii}$$

The projected gauge form satisfies the sharper quadratic identity with error

$$\left| \sum_i \|R D_U^\sharp(\chi_i v)\|_2^2 - \|R D_U^\sharp v\|_2^2 \right| \le C R_0^{-2} \|v\|_2^2. \tag{3.dccliii}$$

The sums use the corresponding finite periodic cutoff family. ■

Proof. Put $B_i = D_U^\sharp \chi_i - \chi_i D_U^\sharp$, with the site and bond multiplications understood. The local product rule and eq. 3.dccl give $\sum_i \|B_i v\|_2^2 \le C R_0^{-2} \|v\|_2^2$. The exact algebra is

$$[P, \chi_i] D_U^\sharp = [T, \chi_i] - P B_i, \qquad T = P D_U^\sharp. \tag{3.dccliv}$$

To estimate $[T, \chi_i]$, first replace each cutoff on each cell by its center value. The input and output remainders satisfy the vector-valued bound $C R_0^{-1}$ by eq. 3.dccli and the L^2 bound on T.

For the center values, the cell entry of the commutator is $(\chi_i(z)-\chi_i(y))T_{yz}$. Let $t_{yz}=\|T_{yz}\|$ and $\rho_{yz}=1+d(y,z)$. Cauchy–Schwarz with weights $\rho_{yz}t_{yz}$ gives

$$\sum_i \left\| \sum_z (\chi_i(z)-\chi_i(y))T_{yz}v_z \right\|^2 \le \left(\sum_z \rho_{yz}t_{yz}\right)\sum_z \frac{t_{yz}}{\rho_{yz}}\sum_i |\chi_i(z)-\chi_i(y)|^2\|v_z\|^2.$$

Sum over y. The first row moment and first column moment are finite by Lem. 3.53; eq. 3.dccl supplies $C\rho_{yz}^2/R_0^2$. This proves eq. 3.dcclii.

For the quadratic estimate use

$$\|RD_U^\sharp v\|_2^2 = \|D_U^\sharp v\|_2^2 - \langle v, K_s v\rangle. \tag{3.dcclv}$$

The first term is the form of the local operator $D_U D_U^\sharp$. Its double-cutoff identity has error at most $CR_0^{-2}\|v\|_2^2$: its coefficients have range at most $2h$ and size at most Ch^{-2}, while the squared cutoff difference is $O(h^2/R_0^2)$.

We give the double-cutoff estimate for the smoothing term in eq. 3.dcclv, since an operator norm of a commutator alone would not give its quadratic improvement. For a cell entry K_{yz} of K_s, the contribution is

$$\sum_i \chi_i|_{B_y}\, K_{yz}\, \chi_i|_{B_z} - K_{yz}.$$

Write $\chi_i|_{B_y} = c_i(y)+\delta_i^y$, where $c_i(y)=\chi_i(y)$. Use $\sum_i(\chi_i|_{B_y})^2=1$ on each cell to expand it as minus one half of

$$\begin{aligned} &\sum_i (c_i(y)-c_i(z))^2 K_{yz} \\ &+2\sum_i (c_i(y)-c_i(z))(\delta_i^y K_{yz} - K_{yz}\delta_i^z) \\ &+\sum_i ((\delta_i^y)^2 K_{yz} - 2\delta_i^y K_{yz}\delta_i^z + K_{yz}(\delta_i^z)^2). \end{aligned} \tag{3.dcclvi}$$

The first line is bounded by $Cd(y,z)^2R_0^{-2}\|K_{yz}\|$. For the second line use Cauchy–Schwarz in i, eq. 3.dccl and eq. 3.dccli. Its bound is $C(1+d(y,z))R_0^{-2}\|K_{yz}\|$. For the last line use the squared within-cell bounds at both ends. Its bound is $CR_0^{-2}\|K_{yz}\|$. Thus the entry is at most $CR_0^{-2}(1+d(y,z))^2\|K_{yz}\|$ in operator norm. The exponential second moment and Schur's test prove eq. 3.dccliii. □

The same partition gives

$$\left| \sum_i \mathcal{K}_{U,\omega}(\chi_i v, \chi_i v) - \mathcal{K}_{U,\omega}(v,v) \right| \le CR_0^{-2}\|v\|_2^2. \tag{3.dcclvii}$$

Here $\mathcal{K}$ is exactly eq. 3.dccxxxii. For the Wilson Hessian the coefficient matrix has range $O(h)$ and physical size $O(h^{-2})$, so the same local double-cutoff calculation applies. The operator $Q_U^\sharp a Q_U$ has bounded physical Schur sums and bounded physical carrier diameter, by the actual row and column estimates. The actual constraint Hessian has the same properties, with an extra factor $\|\omega\|_\infty$, by its two-leg moment. Their double-cutoff errors are therefore $O(R_0^{-2})$ as well. This proves eq. 3.dcclvii without replacing either row by a straight main term.

3.18.3. *Shielding the projector from the exterior.*

Lemma 3.55 (Local comparison of the actual projector)**.** Consider two real admitted single-scale connections whose operators have the uniform site, Gram and weighted energy bounds above. They may live on different periodic volumes, with the common cubes embedded injectively in both.

Suppose, in common local frames, the connections and their actual site-row carriers agree on Q_{24R}, where R exceeds a fixed carrier-diameter constant. For fine sources supported in Q_{2R},

$$\|1_{Q_{2R}}(R_1 - R_2)1_{Q_{2R}}\|_{2\to 2} \le Ce^{-cR}. \tag{3.dcclviii}$$

The compression is identified by their common fine coordinates. ■

Proof. We separate the fine resolvent and coarse Gram inverse comparisons. All cutoffs below are supported in the common region. Their commutators use only coefficient stencils in that region.

First let f, g be fine sources supported in Q_{10R}. Choose $\theta = 1$ on Q_{12R} and supported in Q_{16R}. The geometric resolvent identity, with the cutoff embedding the two spaces, is

$$\theta G_1 - G_2\theta = G_2(A_{s,2}\theta - \theta A_{s,1})G_1. \tag{3.dcclix}$$

The middle operator is supported in an annulus separated from the source region. Its $H^1 \to H^{-1}$ norm is bounded uniformly: the gradient commutator contains a physical derivative of θ; the site-row commutator has bounded range and physical Schur sums. There is no subtraction of different connection coefficients outside this annulus. The weighted site energy bound makes both resolvent solutions exponentially small in the annulus. Pairing the identity with g proves a compressed resolvent difference Ce^{-cR}. The same energy argument gives its localized one-derivative version. The distances can be decreased by a fixed row-carrier radius without changing this conclusion.

Now put $\Lambda = \{y : B_y \subset Q_{6R}\}$ for coarse nodes. For a node vector supported in Λ, its source $q^\sharp b$ is supported in Q_{7R}, and the two actual $q^\sharp$ agree there. The just-proved comparison controls the difference of the two F outputs in Q_{10R}. Their tails outside Q_{10R} are exponentially small by the weighted F bound. Therefore

$$\|1_\Lambda(\mathcal{M}_1 - \mathcal{M}_2)1_\Lambda\| \le Ce^{-cR}. \tag{3.dcclx}$$

This is an operator bound for arbitrary vectors supported in Λ, not a sum of separate entry errors over its volume.

Each compressed Gram matrix $\mathcal{M}_{j,\Lambda} = 1_\Lambda \mathcal{M}_j 1_\Lambda$ retains the lower bound m_0. Its inverse is bounded by m_0^{-1}, so the resolvent identity bounds the difference of their inverses by Ce^{-cR}. For a node source b supported in Q_{3R}, compare that finite compression inverse with the global inverse. The exact identity is

$$\mathcal{M}_{j,\Lambda}^{-1} b - 1_\Lambda \mathcal{M}_j^{-1} b = \mathcal{M}_{j,\Lambda}^{-1} 1_\Lambda \mathcal{M}_j 1_{\Lambda^c} \mathcal{M}_j^{-1} b. \tag{3.dcclxi}$$

The global inverse has an exponentially small tail on Λ^c, by the weighted Gram estimate. The other two operators on the right are uniformly bounded. It follows that the compressions of $\mathcal{M}_1^{-1}$ and $\mathcal{M}_2^{-1}$ to Q_{3R} differ by Ce^{-cR}.

Finally write $P_j = F_j \mathcal{M}_j^{-1} F_j^\sharp$. If f is supported in Q_{2R}, $F_j^\sharp f$ has an exponentially small tail outside Q_{3R}. Their retained parts differ by Ce^{-cR} from the fine resolvent comparison. Insert these facts and the compressed Gram-inverse comparison into $\langle f, P_j g\rangle$. All uncompressed factors have uniform norms. We obtain eq. 3.dcclviii, since $R_j = 1 - P_j$. □

Only the existing site and Gram estimates entered this comparison. No fine vector Green function or lower bound for the full precision has been used in its proof.

3.18.4. *The nonflat periodic gap from local curvature bounds.*

Proposition 3.56 (A one-scale curvature-small Landau inverse)**.** Fix L and the row bounds a_*, a^*. There are positive constants κ_*, γ_*, c and C, independent of k, periodic volume and real holonomy, with the following property. Let U be real on a compatible periodic fine lattice, with every site and field row at the same assigned physical scale one. Suppose

$$\|F(U)\|_\infty \le \kappa_* \quad \text{and} \quad \|\omega\|_\infty \le \gamma_*. \tag{3.dcclxii}$$

Use the actual straightened nonlinear constraint and actual projected Landau gauge in eq. 3.dcclxxxii. Then

$$c(\|\nabla_U v\|_2^2 + \|v\|_2^2) \le \mathcal{K}_{U,\omega}(v,v) \le C(\|\nabla_U v\|_2^2 + \|v\|_2^2). \tag{3.dcclxiii}$$

The full penalized inverse and the real actual-slice saddle inverse have the uniform physical bounds stated in Prop. 3.50 and Cor. 3.51, with no common global small-gauge assumption. ■

Proof. Choose an integer R large enough for the carrier margins below. It will be fixed before the curvature ceiling is chosen. First assume that the periodic side is at least $512R$. At every coarse center use the local comb gauge and extension construction on a cube of radius $32R$. Apply the scaled version of Lem. 3.45 with its radius parameter $4R$. The resulting auxiliary periodic potential, on a torus of side $128R$, satisfies

$$\|A_i^{\mathrm{ext}}\|_\infty \le CR\kappa, \qquad \|F_i^{\mathrm{ext}}\|_\infty \le C\kappa, \tag{3.dcclxiv}$$

and agrees with the locally gauged original field on Q_{24R}, provided $R^2\kappa$ is sufficiently small. The estimates use curvature only. The current derivative bounds of the Coulomb gauge theorem are not needed for this coercivity step.

All operators in the present precision transform covariantly under this local frame change. For the nonlinear straightened constraint, both its internal means and its endpoint means are bi-equivariant; therefore its output logarithm is conjugated at the coarse initial site. Its first and second derivatives transform accordingly. Transform ω by that same unitary conjugation. The physical pairings and its supremum bound are unchanged. This covariance statement concerns the projected Landau form. It makes no assertion about the separately specified absolute soft contour gauge.

Take the cutoffs in eq. 3.dccl supported in cubes of radius R by a fixed rescaling of their defining bump. For $v_i = \chi_i v$, the full stencil of $D_U^\sharp v_i$ is in Q_{2R}. Extend ω and the row weights to the local torus, keeping their actual values on every row whose carrier meets that support, and keeping $\|\omega\|_\infty \le \gamma$ and $a_* \le a \le a^*$. The Wilson, row-mass and constraint-curvature forms of v_i then agree exactly with those in the local extension. The only difference is the projector.

For a real orthogonal projector, $\|Rf\|^2 = \langle f, Rf\rangle$. Consequently Lem. 3.55, with $f = D_U^\sharp v_i$, gives

$$|\mathcal{K}_U(v_i,v_i) - \mathcal{K}_i^{\mathrm{ext}}(v_i,v_i)| \le Ce^{-cR}\|v_i\|_X^2. \tag{3.dcclxv}$$

The coefficient values outside the common region have not been subtracted in a bare fine operator norm.

Choose γ_* smaller than the multiplier ceiling of Prop. 3.50. Once R is fixed, choose κ_* so that $R^2\kappa_*$ admits eq. 3.dcclxiv and $CR\kappa_* + C\kappa_* + \gamma_*$ is below the common-gauge ceiling in that proposition. Each auxiliary periodic precision then has the same lower constant $c_1 > 0$. It follows that

$$\mathcal{K}_U(v_i,v_i) \ge (c_1 - Ce^{-cR})\|v_i\|_X^2. \tag{3.dcclxvi}$$

The usual covariant gradient double-cutoff calculation also gives

$$\left|\sum_i \|v_i\|_X^2 - \|v\|_X^2\right| \le CR^{-2}\|v\|_2^2. \tag{3.dcclxvii}$$

The covariant shifts are unitary. Their fine denominator cancels the squared cutoff difference just as in the local term of Lem. 3.54. Sum eq. 3.dcclxvi, use eq. 3.dcclvii and eq. 3.dcclxvii, and first choose R so that $Ce^{-cR} \le c_1/4$ and the total CR^{-2} error is at most $c_1/4$. The constants used for these choices are uniform throughout a fixed initial small curvature and multiplier ceiling. The subsequent decrease of κ_*, γ_* does not change them. This proves the lower bound in eq. 3.dcclxiii. The upper bound follows from the local Wilson Hessian bound, $\|RD_U^\sharp v\| \le \|D_U^\sharp v\|$, and the actual row and two-leg Schur bounds.

For a smaller periodic volume, choose an integer t such that its t^4-sheeted periodic cover has side at least $512R$. Pull back the connection, the actual rows and the row multiplier to that finite

cover. The local averaging formulas and covariant differences commute with pullback, including every periodic path multiplicity. The same is true of their physical transposes. The site inverse preserves the deck-invariant subspace and equals the pullback of the smaller site's inverse there, by uniqueness of its gapped equation. The same argument applies successively to F, its Gram matrix and its Gram inverse. Consequently the actual R and the full form descend exactly. For a lifted vector, the full form and its squared X norm both multiply by t^4. Divide the already proved estimate on the cover by that factor. This proves the original finite-volume estimate without selecting a global nearly flat gauge or restricting its torus holonomies.

The actual one-scale row lift is constructed from the same compact two-moment seeds in local unit-cell gauges. The actual row/column comparison makes its coarse error $O_L(\kappa)$ in both Schur norms; its Neumann correction gives the bounds in eq. 3.dccxxix. There are no adjacent-level leakage terms in this homogeneous setting. The gauge-restoration energy argument in Cor. 3.51 uses only the site gap, curvature bound and full periodic covariant Hodge identity. It is therefore valid here as well. Repeating that corollary's kernel coercivity argument proves the actual-slice saddle assertion. □

At a physical critical background whose full stationarity equation is $J_A(U) = Q_U^\sharp \omega$, the column L^1 bound for this actual row lift again gives $\|\omega\|_\infty \le C_L \|J_A(U)\|_\infty$. Thus sufficiently small physical curvature and actual current supply the two numerical ceilings. This does not construct the critical background or infer the equation from criticality on an unidentified slice.

Restoring a common physical coarse spacing ξ replaces the mass norm by $\xi^{-2}\|v\|_2^2$, the curvature ceiling by $\xi^2\|F\|_\infty \le \kappa_*$, and the multiplier ceiling by $\xi^3\|\omega\|_\infty \le \gamma_*$. The full inverse then has $L^2 \to L^2$ norm at most $C\xi^2$. These powers follow from the declared physical pairings and do not depend on the number of fine bonds in a coarse cell.

3.18.5. *The scope of the recovered inverse.* The new conclusion is a real, unweighted Wilson/Lagrangian precision estimate at one assigned physical scale, with its actual straightened field rows and actual projected Landau gauge. The proof of the nonlocal localization used only the separately proved site and Gram package. In particular it does not assume the vector inverse in order to prove that inverse.

Several tasks remain in the general background problem. Different assigned scales change the cell metric and carrier bookkeeping. Proper-region inner-boundary forms have boundary terms in both Hodge and shielding identities. Independent analytic-current completions contain additional specified defects. An identification with the absolute contour precision has its own gauge-restoration terms. None of these is an equality supplied by the periodic one-scale result. The nonlinear trial residual and construction of the many-level critical orbit remain separate from the coercive linear inverse established here.

3.19. **Weighted site inverses on the actual determining region.** The projected gauge removes site directions with zero assigned average. Its inverse must therefore control these directions in the physical units of their own pillars. We prove that control on the actual active graph, including the outer boundary. We also prove a normalized Gram gap from curvature alone. A differentiated local potential and a physical-current bound are not premises of this Gram estimate.

The setting is the separated aligned sequence of § 3.11. The fine spacing is $\eta = L^{-k}$ and the assigned spacings are $\xi_j = L^{j-k}$. The pillars B_y partition Ω_0; write $\xi(x) = \xi_y$ on B_y. Outside Ω_0 put $\xi(x) = \eta$ for geometric purposes only. All operators below act on the original graph of sites and bonds inner to Ω_0. The action still contains precisely its inner plaquettes.

Fine scalar and vector norms use measure η^4. Top norms use the physical measure ξ_y^4, and field-row norms use ξ_c^4. The actual own-pillar site average is denoted by q. Thus q is the transported average in eq. 3.cdlxxx, not an unextended endpoint readout. Its physical transpose satisfies $qq^\sharp = 1$.

Let

$$L_U = D_U^\sharp D_U, \qquad A_s = L_U + q^\sharp a_s q, \qquad a_s(y) = c_y \xi_y^{-2}, \qquad 0 < c_* \le c_y \le c^* < \infty. \quad \text{(3.dcclxviii)}$$

The physical gradient is $D_U\lambda(b) = \eta^{-1}(\lambda(b_-) - \mathrm{Ad}\, U_b \lambda(b_+))$. Adjoints in real positivity statements are the Hilbert adjoints for these physical measures. All the estimates extend to the complexification of the real coefficient spaces.

Assume the actual means are on their regular branches and

$$\left\| \frac{1}{i\eta^2} \log U(\partial p) \right\| \le \kappa\, \mathsf{L}(p)^{-2} \quad (p \subset \Omega_0), \qquad \text{(3.dcclxix)}$$

with a fixed sufficiently small κ. The branch admission and the estimates on the actual pillar transports follow from the current-free comb comparison already proved in § 3.13. Constants below depend only on L, c_*, c^* and fixed geometric collar choices. We specify the additional lower bound on the separation ρ below. There is no dependence on depth, compatible volume or number of components. Enlarge the fixed C_{geom} once so it exceeds four times the chosen comb-cube radius in units of a pillar side.

3.19.1. *A slowly varying length comparable to the assigned scale.* The assigned scale jumps by a factor L at a layer boundary. Differentiating that step function would introduce a fine-mesh loss. We instead use a comparable Lipschitz length.

Lemma 3.57 (A smooth comparison length)**.** For $0 < \delta < 1$, define on the ambient torus

$$\ell(x) = \inf_z \{\xi(z) + \delta\, \mathrm{dist}_\infty(x, z)\}. \qquad \text{(3.dcclxx)}$$

If $\rho\delta \ge 1$, then

$$L^{-1}\xi(x) \le \ell(x) \le \xi(x), \qquad |\ell(x) - \ell(z)| \le \delta\, \mathrm{dist}_\infty(x, z). \qquad \text{(3.dcclxxi)}$$

On the bare layer and outside the active region, $\ell = \eta$. For $w = \ell^{-1}$, neighboring fine sites satisfy

$$|D\ell| \le \delta, \qquad |Dw| \le C\delta\ell^{-2}. \qquad \text{(3.dcclxxii)}$$

The site and bond choices of ℓ are equivalent with absolute constants. The actual bond scale $\mathsf{L}(b)$ is comparable to them with a constant depending only on L. ■

Proof. The upper bound follows by choosing $z = x$. Infima of functions with Lipschitz constant δ have that same Lipschitz bound. Suppose x belongs to layer j. If z has assigned level at least $j-1$, then $\xi(z) \ge \xi_j/L$. If its level is at most $j-2$, it lies outside Ω_{j-1}, whereas $x \in \Omega_j$. Separation gives $\mathrm{dist}_\infty(x, z) > \rho\xi_{j-1}$, so the distance term is at least ξ_j/L. The same argument includes exterior points, assigned the bare scale. For $j = 0$, the global lower bound $\xi(z) \ge \eta$ suffices. This proves the lower bound. At a bare point the lower bound $\ell \ge \eta$ and the upper bound $\ell \le \eta$ coincide.

The inverse-weight estimate follows by subtracting reciprocals. Since $\ell \ge \eta$ and its change over one fine step is at most $\delta\eta$, neighboring values have a fixed ratio. Fine neighbors can have assigned levels differing by at most one: otherwise they would cross the separated intervening layer in physical distance at most η. Thus their assigned scales, and the largest corner scale used on a bond or plaquette, differ by at most a fixed power of L. This proves the last assertion. □

The comparison metric obtained by integrating ds/ℓ is equivalent to the actual layer metric within a factor L. We will choose δ small using fixed block constants, and then choose $\rho \ge \max\{C_{\mathrm{geom}}L, \delta^{-1}\}$. These choices precede the depth and volume.

3.19.2. *Current-free block inequalities and a weighted site lift.* The real block argument gives two useful versions of Poincaré:

$$\begin{aligned}\|\xi^{-1}u\|_2^2 &\le C(\|D_U u\|_2^2 + \|\Xi^{-1}qu\|_{\mathrm{top}}^2),\\ \|u\|_2^2 &\le C(\|\ell D_U u\|_2^2 + \|qu\|_{\mathrm{top}}^2).\end{aligned} \tag{3.dcclxxiii}$$

Here Ξ is multiplication by ξ_y on tops. The first is the actual weighted site gap. For the second multiply each pillar's block inequality by ξ_y^2 before summing, and use Lem. 3.57. Only internal pillar bonds are used in either estimate, so crossing energies have a favorable sign. Both inequalities use curvature and unitarity, not a current. In particular A_s is strictly positive on its finite site space.

Lemma 3.58 (A weighted right inverse for the actual site average)**.** There is a real linear map E with $qE = 1$ such that, for each fixed $r \in \{-1, 0, 1\}$,

$$\|\ell^{-r}E\psi\|_2 \le C_r\|\Xi^{-r}\psi\|_{\mathrm{top}}, \qquad \|\ell^{-r}D_U E\psi\|_2 \le C_r\|\Xi^{-r-1}\psi\|_{\mathrm{top}}. \tag{3.dcclxxiv}$$

The constants are uniform over all layers and proper determining regions in this setting. ■

Proof. For a nonbare pillar use a comb frame on a fixed surrounding cube of radius $C_L\xi_y$. Separation places the required cube inside $\Omega_{j-1} \subset \Omega_0$ when the pillar has level $j \ge 1$. Its local potential has supremum at most $C_L\kappa\xi_y^{-1}$. The actual hierarchical internal transports in the pillar differ from one by $C_L\kappa$, by their geometric sum over intermediate physical scales. This comparison differentiates no potential.

Choose a fixed nonnegative smooth scalar bump supported strictly inside the unit centered cube, positive at its center, and sample its rescaling by ξ_y. Its mean m_y on the pillar has positive lower and finite upper bounds independent of its level. Fine meshes converge to the positive integral; the finitely coarse possibilities contain the center site. The actual readout of this bump times a constant colour is N_y times that colour, with $\|N_y - m_y I\| \le C_L\kappa m_y$. Normalize by N_y^{-1} and the unitary top-frame rotation. This gives the prescribed site value exactly. No other site row reads this bump.

The bump has amplitude $O(|\psi_y|)$ and first differences $O(\xi_y^{-1}|\psi_y|)$ in its local frame. The connection term has the same first-difference bound. Its physical support volume is $O(\xi_y^4)$. Multiplying by ℓ^{-r} and using eq. 3.dcclxxi gives exactly the two powers of Ξ in eq. 3.dcclxxiv. For a bare pillar take its one site value: q_0 is the identity. The first-difference collar of this seed has the same estimates, since $\xi_y = \eta$ and all transports are unitary. The collars of different pillars have bounded overlap; adjacent scales have ratio at most L. Summing proves the claims.

If a surrounding cube does not embed in the original torus, make the construction on a sufficiently large finite periodic cover and average E over the deck group, acting on its fine output and coarse input simultaneously. Pullback preserves the actual rows, Ξ and ℓ. Deck translations are unitary in each of the weighted norms above. The averaged map retains $qE = 1$ and all the bounds, preserves deck-invariant vectors and descends. The squared norms have the same sheet-count factor. This also handles nontrivial flat holonomies without a global gauge choice. □

3.19.3. *The normalized layered Gram matrix from weighted energy.* Put

$$G = A_s^{-1}, \qquad F = Gq^\sharp\Xi^{-2}, \qquad M = F^\sharp F. \tag{3.dcclxxv}$$

The normalization is the one in the actual site projector. It retains the physical top measures; an unweighted array transpose would give a different matrix.

Proposition 3.59 (A current-free normalized Gram gap)**.** For a sufficiently small fixed δ and the corresponding separation above, there are $m_*, m^* > 0$ such that

$$m_* I \le M \le m^* I. \tag{3.dcclxxvi}$$

They are independent of depth, region shape and compatible volume. Only eq. 3.dcclxix and the actual site geometry are required; no physical current or differentiated local-gauge bound is a premise. ■

Proof. Write $u = Fb$ and introduce

$$\mathcal{E}_+(u) = \|\ell D_U u\|_2^2 + \|qu\|_{\mathrm{top}}^2. \tag{3.dcclxxvii}$$

By eq. 3.dcclxxiii, $\|u\|_2^2 \le C\mathcal{E}_+(u)$. Test $A_s u = q^\sharp \Xi^{-2} b$ against $\ell^2 u$. The exact discrete product identity is

$$\mathrm{Re}\langle D_U u, D_U(\ell^2 u)\rangle = \|D_U(\ell u)\|_2^2 - \sum_b \eta^4 |D\ell(b)|^2 \,\mathrm{Re}\langle u(b_-), \mathrm{Ad}\, U_b u(b_+)\rangle. \tag{3.dcclxxviii}$$

Using $|D\ell| \le \delta$ and one more product rule bounds this from below by $\frac{1}{2}\|\ell D_U u\|_2^2 - C\delta^2\|u\|_2^2$.

Within one pillar the variation of ℓ is at most $C\delta\xi_y$, and $\ell(y) \ge \xi_y/L$. Therefore

$$\|\Xi^{-2}\{q(\ell^2 u) - \ell(y)^2 qu\}\|_{\mathrm{top}} \le C_L \delta \|u\|_2. \tag{3.dcclxxix}$$

This follows by Cauchy–Schwarz for each unitary transported average and the disjointness of pillars. The site-mass term in the tested equation is consequently bounded below by $c\|qu\|_{\mathrm{top}}^2 - C_L\delta^2\|u\|_2^2$, after a fixed Young inequality. Moreover,

$$\|\Xi^{-2} q(\ell^2 u)\|_{\mathrm{top}} \le \|u\|_2, \tag{3.dcclxxx}$$

because $\ell(x) \le \xi_y$ at every site in B_y and the absolute transported averaging operator is an L^2 contraction. We obtain

$$c\mathcal{E}_+(u) \le \|b\|_{\mathrm{top}}\|u\|_2 + C_L\delta^2\|u\|_2^2.$$

Choose δ so the last term is absorbed using eq. 3.dcclxxiii. Then

$$\mathcal{E}_+(Fb) \le C\|b\|_{\mathrm{top}}\|Fb\|_2, \qquad \|Fb\|_2 \le C\|b\|_{\mathrm{top}}. \tag{3.dcclxxxi}$$

This proves the Gram upper bound.

For the lower bound test the same equation against $\varphi = E\Xi^2 b$, with the exact lift from Lem. 3.58. Then $q\varphi = \Xi^2 b$ and $\|\ell^{-1} D_U \varphi\|_2 \le C\|b\|_{\mathrm{top}}$. Thus

$$\begin{aligned} \|b\|_{\mathrm{top}}^2 &= \langle A_s Fb, E\Xi^2 b\rangle \\ &\le (\|\ell D_U Fb\|_2 + C\|qFb\|_{\mathrm{top}}) C\|b\|_{\mathrm{top}}. \end{aligned} \tag{3.dcclxxxii}$$

Here the site-mass pairing is $\langle qFb, c_y b\rangle_{\mathrm{top}}$, with no extra length factor. For $b \ne 0$, the last display gives $\mathcal{E}_+(Fb) \ge c\|b\|_{\mathrm{top}}^2$. Combine this with eq. 3.dcclxxxi to obtain $\|Fb\|_2 \ge c\|b\|_{\mathrm{top}}$. Squaring proves the lower bound in eq. 3.dcclxxvi. □

This argument uses first differences of a compact site lift. It does not estimate $L_U E$ by differentiating a local connection. In particular the weighted Gram normalization has not introduced a hidden current premise.

3.19.4. *Two weighted derivatives on the zero-average directions.* Let $\lambda \in \ker q$. At every bare top it vanishes exactly, because the assigned zero-level average is the identity. Therefore

$$\lambda = 0 \quad \text{on } \Omega_0 \setminus \Omega_1. \tag{3.dcclxxxiii}$$

The separation of Ω_1 from Ω_0^c supplies more than the fixed number of fine layers needed by the following stencils. This fact concerns the actual assigned averages. It would fail for an arbitrary clipped average on an arbitrary subset of a block.

Proposition 3.60 (A weighted inverse on $\ker q$)**.** For every site vector λ with $q\lambda = 0$,

$$\|\xi^{-2}\lambda\|_2 + \|\mathsf{L}^{-1}D_U\lambda\|_2 + \|\nabla_U D_U\lambda\|_2 \le C\|L_U\lambda\|_2. \tag{3.dcclxxxiv}$$

The last norm is the full component-gradient norm after zero extension across the action boundary. Every nonzero stencil in that norm lies inside the action region. Thus it is an estimate for the true inner site operator, not a replacement by a compressed periodic Laplacian. ■

Proof. Put $f = L_U\lambda$ and $w = \ell^{-1}$. Apply the site gap in eq. 3.dcclxxiii to $w\lambda$. Since $q\lambda = 0$,

$$\|\Xi^{-1}q(w\lambda)\|_{\mathrm{top}} \le C_L\delta\|\ell^{-2}\lambda\|_2. \tag{3.dcclxxxv}$$

Indeed subtract $w(y)q\lambda$ in each pillar and use $|w(x) - w(y)| \le C_L\delta\xi_y^{-1}$. Since $\ell \ge \xi/L$, the left mass term in the site gap controls $\|\ell^{-2}\lambda\|_2$ up to L. Choose δ smaller if necessary and absorb eq. 3.dcclxxxv. We get

$$\|\ell^{-2}\lambda\|_2^2 \le C\|D_U(w\lambda)\|_2^2. \tag{3.dcclxxxvi}$$

Apply the weighted Green identity with weight w to the equation $L_U\lambda = f$. The weight difference in eq. 3.dcclxxii gives

$$\|D_U(w\lambda)\|_2^2 \le \|f\|_2\|\ell^{-2}\lambda\|_2 + C_L\delta^2\|\ell^{-2}\lambda\|_2^2.$$

Absorb its last term using eq. 3.dcclxxxvi. This proves the weighted mass bound. The product rule then proves $\|\ell^{-1}D_U\lambda\|_2 \le C\|f\|_2$. The comparison length converts them into the first two terms of eq. 3.dcclxxxiv.

For the last term extend λ by zero outside Ω_0 and extend the connection there by arbitrary unitary bonds. By eq. 3.dcclxxxiii, all nonzero two-step stencils are still inner to Ω_0, where the extended Laplacian agrees with $L_U\lambda$. The covariant curl of $D_U\lambda$ is a commutator of two shifts. Its coefficient is a conjugate of $(\mathrm{Ad}\,U_p - 1)/\eta^2$, so

$$\|\mathrm{d}_U D_U\lambda\|_2 \le C_L\kappa\|\xi^{-2}\lambda\|_2. \tag{3.dcclxxxvii}$$

The covariant Hodge calculation in eq. 3.dlxviii, applied to this zero-extended gradient, therefore gives

$$\|\nabla_U D_U\lambda\|_2^2 \le C(\|f\|_2^2 + \kappa^2\|\xi^{-2}\lambda\|_2^2 + \kappa\|\mathsf{L}^{-1}D_U\lambda\|_2^2). \tag{3.dcclxxxviii}$$

The corner scales in each plaquette are comparable within a fixed power of L. The unknown exterior curvature contributes nothing: the relevant field and its shifted values vanish in its collar. Substitute the first two estimates to finish the proof. □

No derivative of the curvature, of a gauge potential or of a current has entered this estimate. The extra boundary control comes from the actual zero-level site constraints.

3.19.5. *Gauge restoration and the actual outward rows.* Define the actual orthogonal projector and restoration map by

$$R = 1 - FM^{-1}F^\sharp, \qquad \Lambda = GRD_U^\sharp, \qquad S = 1 - D_U\Lambda. \tag{3.dcclxxxix}$$

The exact range identities are

$$\operatorname{ran} R = L_U(\ker q), \qquad qGR = 0, \qquad L_UGR = R. \tag{3.dccxc}$$

For $qz = 0$, $GL_Uz = z$ and $F^\sharp L_Uz = \Xi^{-2}qz = 0$. Conversely $F^\sharp f = 0$ implies $qGf = 0$ and $L_UGf = f$. This proves the range statement and the two identities.

For a field v define

$$\|v\|_X^2 = \|\nabla_U v^0\|_2^2 + \|\mathsf{L}^{-1}v\|_2^2, \tag{3.dccxci}$$

where v^0 is zero on all bonds outside the active graph. This comparison norm counts its boundary jumps; it does not add those jumps to the action. Its value is independent of an arbitrary unitary extension of the connection: when a difference has two nonzero field values its connecting bond is active, and otherwise unitarity fixes the norm of its single nonzero value.

Corollary 3.61 (Uniform restoration on every assigned scale)**.** The actual maps satisfy

$$\|\xi^{-2}\Lambda v\|_2 + \|D_U\Lambda v\|_X + \|Sv\|_X \le C\|v\|_X, \qquad RD_U^\sharp S = 0, \qquad QS = Q. \tag{3.dccxcii}$$

Here Q is the derivative of the actual straightened field map on the literal determining rows, including its outward rows. ■□

Proof. The site vector $\lambda = \Lambda v$ obeys $q\lambda = 0$ and $L_U\lambda = RD_U^\sharp v$. Since R is orthogonal and the inner divergence is bounded by the component-gradient comparison norm, Prop. 3.60 proves the estimates. The identity $RD_U^\sharp S = 0$ follows from $RL_UGR = R$.

The actual field intertwining is

$$QD_U = \bar{D}_{\mathrm{ext}}q. \tag{3.dccxciii}$$

At an outward endpoint of a level-j row, $\bar{D}_{\mathrm{ext}}$ averages the L^4 assigned level-$(j-1)$ top values in that whole outer block, with their actual mean transports. That is the exact extension in eq. 3.cdlxxxiii; it is not evaluation at one outer top. Hence $QD_U\Lambda = \bar{D}_{\mathrm{ext}}qGRD_U^\sharp = 0$. This proves $QS = Q$ on the true row spaces. □

Let $a_f(c) = d_c\xi_c^{-2}$, with fixed positive lower and upper bounds for d_c, and give data the norm

$$\|b\|_Y^2 = \sum_c \xi_c^4 a_f(c)|b(c)|^2. \tag{3.dccxciv}$$

The actual field-row construction in eq. 3.dcv supplies a right inverse E_Q with $\|E_Qb\|_X \le C\|b\|_Y$. Its compact seeds and adjacent-level Neumann correction are the actual ones; they include the identity seeds in the bare boundary layer. The zero-extension part of X adds only a bounded bare mass term to their stated inner energy estimate. Thus $E_L = SE_Q$ is an exact right inverse into $X_L = \ker(RD_U^\sharp)$ and

$$QE_L = 1, \quad \|E_Lb\|_X \le C\|b\|_Y, \quad \inf_{\mu\neq 0}\sup_{v\in X_L\setminus\{0\}} \frac{|\langle \mu, Qv\rangle_{\mathrm{rows}}|}{\|\mu\|_{Y^*}\|v\|_X} \ge C^{-1}. \tag{3.dccxcv}$$

The dual norm is $\|\mu\|_{Y^*} = \|a_f^{-1/2}\mu\|_{2,\mathrm{rows}}$. For the last inequality choose a unit data vector attaining the dual norm and apply E_L. This proves the uniform inf-sup part of the actual saddle problem. Coercivity on its simultaneous constraint kernel is a separate estimate.

3.19.6. *The explicit current correction has an energy bound.* For an independent current define $\mathcal{W}_U[J]$ and the symmetric correction $\mathcal{B}_U[J]$ by their actual explicit formulas eq. 3.cdlxiv and eq. 3.cdlxvii. Suppose

$$\gamma_J = \sup_b \mathsf{L}(b)^3|J(b)| < \infty. \tag{3.dccxcvi}$$

Then

$$|\langle u, \mathcal{B}_U[J]v\rangle| \le C\gamma_J\|u\|_X\|v\|_X. \tag{3.dccxcvii}$$

Indeed, the endpoint formula for $\mathcal{W}$ and comparable corner scales give

$$|\langle u, \mathcal{W}_U[J]\lambda\rangle| \le C_L\gamma_J\|\mathsf{L}^{-1}u\|_2\|\xi^{-2}\lambda\|_2.$$

Insert this inequality and Cor. 3.61 in the four undifferentiated-current pairings of eq. 3.cdlxix. This proves eq. 3.dccxcvii without a derivative of J. Likewise the actual quadratic constraint form satisfies

$$|\langle Su, \Delta^{(2)}[\omega]Sv\rangle| \le C\gamma_\omega\|u\|_X\|v\|_X, \qquad \gamma_\omega = \sup_c \xi_c^3|\omega(c)|,$$

by its already proved two-leg moment. The particular induced readout is bounded in Prop. 3.73 and eq. 3.mviii; the exact canonical current identification is Prop. 3.90.

The results in this section supply the current-free normalized site Gram gap, two weighted derivatives on the actual zero-average site directions, a uniformly bounded field restoration, and the inf-sup part of the all-layer proper-region saddle problem. They use the literal outward-row

extension and the actual bare boundary collar. The remaining mass estimate is a vector estimate, not another site gap.

3.20. **A flat regular decomposition on the determining regions.** The missing vector estimate is a div–curl estimate with the actual changing-scale averages. The outward rows are useful here: their outer endpoint average is taken over an entire block whose smaller internal rows are also prescribed. Those rows control the fluctuations hidden by the endpoint average. We first quantify this fact and then glue local scalar primitives. The construction retains holes, disconnected components and the true bare boundary layer.

Use the sites, active bonds and inner plaquettes of § 3.19. Let V be a real flat connection on this active cubical complex. Flatness means every inner plaquette is one. There is no assumption that V is a global pure gauge. In a convex fine box it can be put in a parallel frame, so all calculations there reduce to ordinary cochains. All norms and transposes have their previously declared physical measures. The field-row map Q is the actual straightened derivative; at a flat field it is the transported straight average, not the derivative of the plain average.

3.20.1. *What the outward endpoint average controls.* Make a graph with the assigned tops as vertices. Two vertices are joined when their pillars share a face containing a fine bond. Their levels are equal or adjacent. Let σ_{yz} be the larger of their two scales and let T_{yz} be the parallel transport between them inside the local one- or two-scale rectangle. The local rectangle is contractible in the active region, so flatness makes this transport independent of its chosen path there.

For a fixed real r define

$$\begin{aligned} J_r(\psi) &= \sum_{y\sim z} \sigma_{yz}^{2-2r} |\psi(y) - \operatorname{Ad} T_{yz}\psi(z)|^2, \\ D_r(\psi) &= \sum_{c\in\mathbf{B}} \xi_c^{4-2r} |\bar{D}_{\mathrm{ext}}\psi(c)|^2. \end{aligned} \tag{3.dccxcviii}$$

Count each unoriented graph edge once. The coarse difference in the second line has the exact averaged outer endpoint from eq. 3.cdlxxxiv.

Lemma 3.62 (Completion of the outward rows)**.** For every fixed r,

$$C_{L,r}^{-1} J_r \le D_r \le C_{L,r} J_r. \tag{3.dccxcix}$$

The constants have no depth or volume factor. The comparison is local: a graph edge uses only its associated determining row and the smaller internal rows in its outer endpoint block. ■

Proof. An equal-level graph edge is precisely an interior determining row, so the two differences coincide, with their length factor. For a face between a level-j pillar at y and smaller pillars, put $s = \xi_j$. The determining outward bond from y ends at the level-j grid point z across that face. Its entire block $B^j(z)$ is outside Ω_j and inside Ω_{j-1}. It consists of L^4 assigned level-$(j-1)$ pillars. Every level-$(j-1)$ bond internal to this block is an actual prescribed interior row. This is the no-inward-row geometry; it is not true for an arbitrary clipped block.

Choose a parallel frame on the two-block rectangle. Write a for the value at y, b_w for the smaller top values in $B^j(z)$, and $\bar{b} = L^{-4}\sum_w b_w$. The outward difference is $s^{-1}(a - \bar{b})$. For every smaller pillar on the face,

$$|a - b_w|^2 \le 2|a-\bar{b}|^2 + 2|b_w - \bar{b}|^2, \qquad \sum_w |b_w - \bar{b}|^2 \le CL^2 \sum_{\substack{w\sim w' \\ \text{internal}}} |b_w - b_{w'}|^2.$$

The last inequality is the finite L-grid path Poincaré inequality. Multiply by s^{2-2r} and sum over at most L^3 face neighbors. The smaller row weights differ by the fixed factor L^{2-2r}. This bounds the corresponding graph edges by the outward row and the smaller internal rows.

Conversely choose one smaller face neighbor w_0. Write $a-\bar{b} = (a-b_{w_0}) + L^{-4}\sum_w (b_{w_0}-b_w)$ and join w_0 to w by coordinate paths in the smaller block. Their length is at most $4L$. Cauchy–Schwarz gives the reverse bound. The same argument works after restoring the unitary parallel transports.

A top belongs to only one such outer large block at the next level. That block can be read by at most eight incident outward rows. Internal smaller edges and all the constructed local paths therefore have bounded overlap depending only on L. Summing the local inequalities proves eq. 3.dccxcix. □

In particular the kernel of $\bar{D}_{\mathrm{ext}}$ is the space of parallel sections of this pillar adjacency graph. Such a section lifts to a parallel fine site field by the actual own-pillar transport, and conversely every parallel fine field gives such a section. For a fine edge crossing a pillar face, this follows in the same two-block rectangle used in the proof. Thus the averaged outward endpoint does not introduce an extra coarse gauge kernel. This statement retains the actual holonomy conditions on each connected component.

3.20.2. *A local box estimate with its actual boundary adjoint.*

Lemma 3.63 (Local cochain Hodge decomposition)**.** Let D be a convex rectangular fine box of diameter comparable to $s \geq \eta$, with bounded aspect ratio. For any one-cochain v on all its inner bonds there is a site potential λ and a remainder $r = v - D_V\lambda$ such that

$$\|r\|_{2,D} \leq Cs\|\mathrm{d}_V v\|_{2,D}, \qquad \|\nabla_V r\|_{2,D'} \leq C\|\mathrm{d}_V v\|_{2,D}, \tag{3.dccc}$$

where D' is any fixed smaller concentric box. The potential is unique up to a parallel constant after its gradient is chosen. It may be normalized to have zero transported average on a specified centered pillar of scale comparable to s in D. The constants include only the fixed size ratios. They are independent of the number of fine sites in the box. ■

Proof. Choose a parallel frame on D. Let d_0 be the fine gradient on the path-product graph and d_1 its plaquette coboundary. Project v orthogonally onto $\operatorname{ran}\mathrm{d}_0$ and write $r = v - \mathrm{d}_0\lambda$. The projection equation is $\mathrm{d}_0^\sharp r = 0$, with the full graph adjoint, including its boundary vertices. Also $\mathrm{d}_1 r = \mathrm{d}_1 v$.

Here the required graph Hodge estimate is elementary and includes the boundary condition. In one direction with $N+1$ vertices, $\mathrm{d}_0^\sharp\mathrm{d}_0$ is the path Neumann matrix, whereas $\mathrm{d}_0\mathrm{d}_0^\sharp$ on its N edges is the tridiagonal matrix with diagonal $2\eta^{-2}$ and off-diagonal $-\eta^{-2}$. Their nonzero eigenvalues are

$$4\eta^{-2}\sin^2\frac{\pi m}{2(N+1)} \quad (1 \leq m \leq N).$$

The scalar matrix also has its constant zero mode. Tensor products give the one-cochain Hodge matrix $\mathrm{d}_0\mathrm{d}_0^\sharp + \mathrm{d}_1^\sharp\mathrm{d}_1$: in component μ its longitudinal factor is the edge matrix and its other factors are the scalar path matrices. Every component therefore has a nonzero longitudinal frequency and a gap at least cs^{-2}. Its quadratic form is the sum of the interior component first differences, with additional nonnegative longitudinal boundary edge terms. This follows either from those tensor factors or by expanding their squares.

It follows that $\|r\|_2 \leq Cs\|\mathrm{d}_1 r\|_2$, and that every first difference whose stencil is inside D' is bounded by $C\|\mathrm{d}_1 r\|_2$. This proves eq. 3.dccc. Changing the parallel constant in λ does not change r. The transported average of a parallel constant is that constant, so it gives the stated normalization. The local Poincaré inequality with a mean on the specified positive-volume pillar has a constant depending only on the size ratio. It includes a bare one-site pillar, whose physical volume is then s^4 up to that same fixed ratio. Restore the parallel frame to finish. □

The divergence in this lemma is the full adjoint on the finite box graph. An interior formula that discards its boundary vertex equations would allow additional harmonic gradients and would not prove the lemma.

3.20.3. *Adaptive patches and removal of the bare boundary collar.* Let C_{row} be a fixed physical carrier-radius constant for the flat rows and their completed outer blocks. Fix an integer $K \geq 64L^2(C_{\mathrm{row}}+1)$ and require $\rho \geq CKL$. A box of radius $K\xi_y$ about an assigned top can meet at most two adjacent assigned levels. Indeed it cannot reach level $j-2$ from a level-j top without crossing distance $\rho\xi_j/L$, nor level $j+2$ without crossing distance $\rho L\xi_j$. It cannot meet both levels $j-1$ and $j+1$, because those sets are separated by more than $\rho\xi_j$. Increase the fixed constant in the separation for the enlarged boxes used below.

Consequently these boxes have uniformly bounded overlap, and there is a partition $\sum_y \chi_y = 1$ on Ω_0 with

$$\operatorname{supp}\chi_y \subset Q_{2\xi_y}(y), \quad |D\chi_y| \leq C_L\xi_y^{-1}, \quad |D^2\chi_y| \leq C_L\xi_y^{-2}. \tag{3.dccci}$$

One construction uses a fixed smooth bump equal to one on a centered pillar, supported in its doubled cube, and divides by the sum of all such bumps. Pillars are disjoint; every overlapping scale has ratio at most L. Hence that sum is bounded above and below and its first two differences have the stated bounds. The same counting proves bounded overlap for the larger $K\xi_y$ boxes and the pair neighborhoods below. Whenever cutoff overlaps are used, include their two-fine-step collars as well; the same estimates apply since $\eta \leq \xi_y$.

We first remove a genuine boundary difficulty. Choose $H > 10K$ fixed and a scalar cutoff θ, equal to one within $H\eta$ of Ω_0^c, zero beyond $2H\eta$, with $|D\theta| \leq C\eta^{-1}$. Choose $\rho > 4H + C_L$. Every bond read by $v_b = \theta v$ is a bare determining row, and $Q_0 v = v$ on these bonds. The elementary finite-difference bound gives

$$\|v_b\|_X \leq C\|Qv\|_Y, \quad \|\mathrm{d}_V(v-v_b)\|_2 + \|Q(v-v_b)\|_Y \leq C(\|\mathrm{d}_V v\|_2 + \|Qv\|_Y). \tag{3.dcccii}$$

For the curl use the product rule; its cutoff term has size $C\eta^{-1}|v|$ in the bare collar. For Q use its bounded physical row and column sums, or note that the higher row carriers miss this collar. The norm X includes zero-extension boundary jumps, also bounded by $\eta^{-1}|v|$ there.

Put $w = v - v_b$. It vanishes in this boundary collar. Every nontrivial box used to decompose w is therefore inside Ω_0: nonbare boxes are inside by separation, and a bare box meeting the exterior lies entirely in the zero collar. For such a zero box choose potential and remainder zero. This convention never requires a flat extension across a hole.

3.20.4. *Gluing the normalized local potentials.*

Proposition 3.64 (A regular decomposition with the actual averages)**.** For every field v on the true active bonds there is a site potential λ_0 with $q\lambda_0 = 0$ and a field r_0 such that

$$v = D_V\lambda_0 + r_0, \qquad \|r_0\|_X \leq C(\|\mathrm{d}_V v\|_2 + \|Qv\|_Y). \tag{3.dcccciii}$$

The constant is independent of all assigned levels and the topology of the determining region. ■

Proof. First suppose every ambient period exceeds $8K\max_y \xi_y$, so all local boxes below embed. A finite-cover argument at the end removes this temporary restriction. It suffices by eq. 3.dcccii to decompose w. For every nonzero patch apply Lem. 3.63 on $D_y = Q_{K\xi_y}(y)$, choosing

$$w = D_V\lambda_y + r_y, \quad q_y\lambda_y = 0, \quad \|r_y\|_{2,D_y} \leq C_L\xi_y\|\mathrm{d}_V w\|_{2,D_y}, \quad \|\nabla_V r_y\|_{2,D_y/2} \leq C_L\|\mathrm{d}_V w\|_{2,D_y}. \tag{3.dccciv}$$

The local normalization uses the actual pillar average.

We spell out the comparison of constants on overlaps. If the cutoff supports, including the stated fine collars, meet at y, z, their scales have ratio at most L and their centers have distance at most $8\sigma_{yz}$. A coordinate route and every completed outer block lie in a rectangle of radius $C(C_{\mathrm{row}}+1)\sigma_{yz}$ about these centers, with the dimensional constant absorbed by the choice of K above. Since the smaller radius of $D_y/2, D_z/2$ is at least $K\sigma_{yz}/(2L)$, this rectangle lies in their intersection. Choose a convex pair neighborhood there containing the two central pillars and the following finite row route. Follow a coordinate fine path from one pillar to the other and record the pillars it crosses.

There are at most C_L of them because their scales are comparable. Replace each unequal-level face crossing by its determining outward row and the internal smaller rows in its full outer block, as in Lem. 3.62. All these carriers fit in the pair neighborhood when K is chosen large enough.

For $f_a = q_a\lambda_z$, the local intertwining gives $\bar{D}_{\mathrm{ext}} f = Q(w - r_z)$ on that route. Since $f_z = 0$, the finite-route version of the completion lemma implies

$$|q_y\lambda_z| \le C_L \left\{ \sum_{c\in\mathcal{C}_{yz}} \xi_c^2 |Qw(c)|^2 + \|\mathrm{d}_V w\|^2_{2,D_y\cup D_z} \right\}^{1/2} =: C_L e_{yz}. \tag{3.dcccv}$$

For the remainder term used here, a physical flat row has L^2 functional norm at most $C\xi_c^{-2}$: its coefficient sum is bounded and its largest fine coefficient is at most $C(\eta/\xi_c)^4$. Thus $\xi_c|Qr_z(c)| \le C_L\|\mathrm{d}_V w\|_{2,D_z}$, by eq. 3.dccciv. Only C_L rows enter.

On the pair neighborhood let $h_{yz} = \lambda_y - \lambda_z$. Its derivative is $r_z - r_y$ and its average at y is $-q_y\lambda_z$. Poincaré with that central pillar mean gives

$$\|h_{yz}\|_2 \le C_L\sigma_{yz}^2 e_{yz}, \qquad \|D_V h_{yz}\|_2 \le C_L\sigma_{yz} e_{yz}. \tag{3.dcccvi}$$

In particular the constants of the primitives have been estimated; a bound only on their gradient would not suffice. The pair neighborhoods and their routes have bounded overlap, so

$$\sum_{y,z} e_{yz}^2 \le C_L(\|\mathrm{d}_V w\|_2^2 + \|Qw\|_Y^2), \tag{3.dcccvii}$$

where only intersecting cutoff supports are summed.

Glue $\lambda = \sum_y \chi_y\lambda_y$ and put $r = w - D_V\lambda$. The exact product rule writes r as the sum of the cutoff remainders $\chi_y r_y$ and terms $D\chi_y$ times shifted primitive differences. The latter are differences because $\sum_y D\chi_y = 0$. A second product rule for its gradient, Cauchy–Schwarz and bounded overlap give

$$\begin{aligned} \|r\|_X^2 &\le C_L \sum_y (\|\nabla_V r_y\|^2_{2,D_y/2} + \xi_y^{-2}\|r_y\|^2_{2,D_y}) \\ &\quad + C_L \sum_{y,z} (\sigma_{yz}^{-4}\|h_{yz}\|_2^2 + \sigma_{yz}^{-2}\|D_V h_{yz}\|_2^2). \end{aligned} \tag{3.dcccviii}$$

All shifts in this formula remain in the specified interior boxes. Equations eq. 3.dccciv through eq. 3.dcccvii prove $\|r\|_X \le C_L(\|\mathrm{d}_V w\|_2 + \|Qw\|_Y)$.

The glued potential need not have zero assigned average. On B_y, subtract λ_y in its average and use its zero normalization. The averaging functional has norm ξ_y^{-2}, so eq. 3.dcccvi gives

$$\sum_y |q_y\lambda|^2 \le C_L(\|\mathrm{d}_V w\|_2^2 + \|Qw\|_Y^2). \tag{3.dcccix}$$

This is a counting norm of the site values, with no missing physical volume factor: the factor ξ_y^{-2} cancels the ξ_y^2 in the potential overlap estimate.

At a flat field the exact bump lift of the site averages can be chosen with the additional estimate

$$qE = 1, \qquad \|D_V E\psi\|_X^2 \le C_L \sum_y |\psi_y|^2. \tag{3.dcccx}$$

To verify it, use the same normalized compact scalar profiles in parallel frames. Their second differences are $O(\xi_y^{-2}|\psi_y|)$ and their physical volume is $O(\xi_y^4)$. The weighted first differences have that same squared counting bound. For a bare seed the two differences cost at most $C\eta^{-2}$, canceled by its physical volume η^4 after squaring. The two-step collars have bounded overlap. A large finite cover and deck averaging handle any nonembedded surrounding box, preserving these energy estimates.

Set $\lambda_0 = \lambda - Eq\lambda$. Its average is zero exactly. Add $D_V Eq\lambda$ to r and then add back v_b to the remainder. Equations eq. 3.dcccix, eq. 3.dcccx and eq. 3.dcccii prove eq. 3.dccciii on this large volume.

For an arbitrary smaller compatible torus, pull back the entire determining geometry, flat connection and input field to a finite periodic cover whose periods exceed $8K \max_y \xi_y$. Construct the potential λ_0 there by the preceding proof. Average it over the deck group. The lifted input is invariant, and the actual D_V, q, Q, all physical weights and the boundary collar commute with these translations. The averaged potential still has $q\lambda_0 = 0$. Its remainder is the average of the translated remainders, so its X norm obeys the same bound by the triangle inequality. It descends to the original torus. The squared remainder norm and both squared input norms multiply by the number of sheets under pullback; these factors cancel. This proves the full regular decomposition at every compatible volume, not only the auxiliary site lift. The argument does not require trivial holonomy on the original active complex. □

The argument uses local primitives on convex boxes and compares their constants through the actual determining rows. It therefore also controls closed fields with periods around holes: their constants can be glued only if the corresponding row information is retained. No assertion that the region has trivial first cohomology was made.

3.20.5. *The flat layered Landau gap.*

Proposition 3.65 (Flat coercivity on the true determining region)**.** For the separated aligned determining geometry, with separation larger than the fixed constants above, there are $c, C > 0$ such that

$$c\|v\|_X^2 \le \|\mathrm{d}_V v\|_{2,\Omega_0}^2 + \|RD_V^\sharp v\|_2^2 + \|Qv\|_Y^2 \le C\|v\|_X^2. \tag{3.dcccxi}$$

This is uniform in depth, compatible volume, region topology and real flat holonomies. The curl includes exactly the inner plaquettes, the divergence is the adjoint on the true active graph, and R, Q are the actual assigned-site projector and straightened determining rows. ■

Proof. Take the regular decomposition of Prop. 3.64. Since $q\lambda_0 = 0$, the actual range identity gives

$$L_V\lambda_0 = RL_V\lambda_0 = RD_V^\sharp v - RD_V^\sharp r_0.$$

The divergence is bounded by the X norm. Apply Prop. 3.60 to conclude

$$\|D_V\lambda_0\|_X \le C(\|RD_V^\sharp v\|_2 + \|r_0\|_X).$$

Add the remainder bound. This proves the lower estimate in eq. 3.dcccxi. The upper estimate follows from the local curl and divergence bounds and the physical row and column bounds for Q. The boundary part of X is a comparison norm only; no boundary plaquettes or crossing action bonds have been added in deriving its lower bound. □

At a flat field the physical Wilson Hessian is precisely the inner curl form. Hence the theorem proves the full flat projected-Landau precision gap on the actual multi-level region. Together with the actual-slice inf-sup estimate in eq. 3.dccxcv, it gives the flat constrained saddle inverse in the declared energy and dual-energy norms. The passage to a nonflat regular field still requires coefficient comparison and localization of this same geometry. It is not a consequence of a site gap alone.

3.21. **A nonflat Landau inverse on the determining regions.** We retain the same determining geometry in every comparison. A local comparison field equals the identity outside its gauge patch, but it has the original assigned averages, determining rows and inner action region. No periodic inner operator is substituted at a seam or an action boundary.

Use the hypotheses and physical spaces of §§ 3.19 and 3.20. All row weights are positive and comparable with the assigned inverse scale squared. The estimates concern the unweighted inner

Wilson action and actual straightened constraint. Signed profiles and separately specified current completions are not included by a change of notation.

3.21.1. *A common weighted gauge.* Let V be real and flat on the active graph and write

$$U_b = e^{i\eta A_b} V_b, \qquad \varepsilon = \sup_b \mathsf{L}(b)|A_b|. \tag{3.dcccxii}$$

The reference may have nontrivial holonomy around holes or torus cycles. Site and row spaces, Ξ, and ℓ are the same for both backgrounds.

Lemma 3.66 (Actual weighted coefficient comparison)**.** For sufficiently small fixed ε and curvature ceiling,

$$\begin{aligned}
\|\ell(D_U - D_V)u\|_2 &\le C\varepsilon\|u\|_2,\\
\|(q_U - q_V)u\|_{\mathrm{top}} + \|\Xi^{-2}(q_U - q_V)(\ell^2 u)\|_{\mathrm{top}} &\le C\varepsilon\|u\|_2,\\
\|a_f^{1/2}(Q_U - Q_V)v\|_{2,\mathrm{rows}} &\le C\varepsilon\|\mathsf{L}^{-1}v\|_2,\\
\|(F_U - F_V)b\|_2 + \|\ell(D_U F_U - D_V F_V)b\|_2 &\le C\varepsilon\|b\|_{\mathrm{top}},\\
\|R_U - R_V\|_{2\to 2} &\le C\varepsilon,\\
\|(R_U D_U^\sharp - R_V D_V^\sharp)v\|_2 &\le C\varepsilon\|v\|_{X_V}.
\end{aligned} \tag{3.dcccxiii}$$

The Q's are the actual straightened rows, including their node derivatives and extended outward endpoint averages. ■

Proof. The first bound follows from $\|\operatorname{Ad}(e^{i\eta A}) - 1\| \le C\eta|A|$. Each actual row carrier meets at most two adjacent assigned scales. In a local parallel frame for V, intermediate transports at physical scale ξ_s change by at most $C_L\varepsilon\xi_s/\xi_c$. Their geometric sum is bounded uniformly in the number of averaging levels. Differentiating the finite actual recursion gives both physical Schur sums for the field-row difference. This is the row comparison proved in § 3.13, including the straightening factors. The site-row comparison is the same transported-average calculation. Inserting ℓ^2/ξ_y^2, or the field mass weights on a two-scale carrier, preserves its $O_L(\varepsilon)$ bound. Thus the first three lines have no fine-density loss.

For completeness, the normalized resolvent comparison uses the weighted energy, rather than an unscaled fine matrix norm. Put $u = (F_U - F_V)b$. Then

$$A_{s,U}u = (q_U^\sharp - q_V^\sharp)\Xi^{-2}b - (A_{s,U} - A_{s,V})F_V b. \tag{3.dcccxiv}$$

Test against $\ell^2 u$. The real part on the left controls $\|\ell D_U u\|_2^2 + \|q_U u\|_{\mathrm{top}}^2$, and this energy controls $\|u\|_2^2$, by the proof of Prop. 3.59. The first right-hand term is bounded by $C\varepsilon\|b\|_{\mathrm{top}}\|u\|_2$. In the gradient-form difference, the factor $D_U - D_V$ has size $C\varepsilon\ell^{-1}$, while $D_U(\ell^2 u)$ is bounded by $C\ell(|\ell D_U u| + \delta|u|)$. The alternative placement of that difference is bounded by $C\varepsilon\|\ell D_V F_V b\|_2\|u\|_2$. The two site-mass differences are bounded by $C\varepsilon(\|F_V b\|_2 + \|q_V F_V b\|_{\mathrm{top}})\|u\|_2$. The weighted column energy bounds therefore give $\|u\|_2 + \|\ell D_U u\|_2 \le C\varepsilon\|b\|_{\mathrm{top}}$. This proves the fourth line after adding the gradient coefficient difference.

Consequently $M_U - M_V = O(\varepsilon)$ in normalized top norm. The Gram gaps give the same bound for the inverse difference. Insert these in $R = 1 - FM^{-1}F^\sharp$. For the last line insert and subtract $R_U D_V^\sharp v$. The remaining adjoint coefficient difference is bounded by $C\varepsilon\|\mathsf{L}^{-1}v\|_2$. No derivative of the potential is used. □

Pair the Hessian of the actual per-length nonlinear constraint by

$$B_\omega(v,w) = \sum_{c\in\mathbf{B}} \xi_c^4 \langle \omega(c), D^2\mathcal{Q}_{U,c}(0)[v,w]\rangle, \qquad \gamma_\omega = \sup_c \xi_c^3|\omega(c)|. \tag{3.dcccxv}$$

Its actual two-leg moment bounds it by $C\gamma_\omega\|\mathsf{L}^{-1}v\|_2\|\mathsf{L}^{-1}w\|_2$. Define the physical form

$$\begin{aligned} K_{U,\omega}(v,w) &= D^2A_{\Omega_0}(U)[v,w] - B_\omega(v,w) \\ &\quad + \langle R_U D_U^\sharp v, R_U D_U^\sharp w\rangle + \langle Q_U v, a_f Q_U w\rangle_{\mathrm{rows}}. \end{aligned} \tag{3.dcccxvi}$$

Action derivatives use physical left variations $e^{i\eta tv}U$.

Corollary 3.67 (Coercivity in the common gauge)**.** There are fixed $c_1, \epsilon_1 > 0$ such that

$$K_{U,\omega}(v,v) \geq c_1\|v\|_{X_U}^2 \quad \text{if} \quad \varepsilon + \kappa + \gamma_\omega \leq \epsilon_1. \tag{3.dcccxvii}$$

■□

Proof. The exact Wilson plaquette Hessian differs from its transported linearized curl form by $C\kappa$ times the mass norms. The curl differs from its flat reference by $C\varepsilon$ times the mass norm: each prefix transport changes by $O(\eta\varepsilon/\mathsf{L})$ before division by η. This is a local inner-plaquette calculation, including the action boundary. Thus

$$|D^2A_{\Omega_0}(U)[v,w] - \langle \mathrm{d}_V v, \mathrm{d}_V w\rangle| \leq C(\varepsilon+\kappa)\|v\|_{X_V}\|w\|_{X_V}.$$

Use Lem. 3.66 for the gauge and row squares and the two-leg estimate for the actual constraint form. Absorb the total error in Prop. 3.65. The two component-gradient norms are equivalent by their coefficient comparison. This proves eq. 3.dcccxvii. □

3.21.2. *Weighted cell kernels and IMS.* Let ψ be bounded and one-Lipschitz for $d_\ell = \inf\int ds/\ell$, with its sampled top values. Conjugate site and top vectors by $e^{s\psi}$. A derivative commutator has size $C_L|s|\ell^{-1}$, and a normalized site row changes by $C_L|s|$. Testing the conjugated equation against $\ell^2 u$ perturbs the coercive weighted-energy form by $C_L|s|$. The preceding weighted proof consequently gives, for a fixed $|s| \leq s_0$,

$$\|e^{s\psi_f}Fe^{-s\psi_t}\|_{2\to 2} + \|e^{s\psi_f}\ell D_U F e^{-s\psi_t}\|_{2\to 2} \leq C. \tag{3.dcccxviii}$$

The difference from the original column is $O(|s|)$. Therefore the conjugated normalized Gram matrix changes by $O(|s|)$, and its inverse stays bounded by the Gram gap. It follows that the normalized smoothing operator

$$\widetilde{K}_s = \ell D_U P D_U^\sharp \ell = (\ell D_U F)M^{-1}(\ell D_U F)^\sharp, \qquad P = 1 - R, \tag{3.dcccxix}$$

has exponential physical cell-L^2 entries. This is not a pointwise fine-kernel assertion.

The actual layer metric has cell growth bounded by $C_L(1+t)^5 e^{8t\log L/\rho}$. Since $d_\ell \geq d$, choose the separation so that

$$8\log L/\rho < s_0/2. \tag{3.dcccxx}$$

All fixed polynomial row and column moments of the smoothing entries are then bounded. This explicitly pays for the growth of the changing-scale metric.

Choose a large fixed R. A useful partition can be constructed without assumptions on boundary density. Select tops greedily in decreasing scale so that balls of radius $R\xi_y/(20L)$ are disjoint and the selection is maximal. If $\rho \geq C_L R$, intersecting enlarged balls have only adjacent scales. The larger balls of radius $R\xi_y$ cover the region, allowing a fixed enlargement for fine cells. Packing the smaller disjoint balls gives overlap at most C_L. Take bumps equal to one on the covering balls and supported in their doubles, and normalize their squared sum. They satisfy

$$\begin{aligned} &\sum_i \chi_i(x)^2 = 1, \qquad \sum_i |D\chi_i(x)|^2 \leq C_L R^{-2}\xi(x)^{-2}, \\ &\sum_i |\chi_i(x) - \chi_i(z)|^2 \leq C_L \min\{1, d_\ell(x,z)^2/R^2\}, \\ &\sum_i \sup_{x\in B_y} |\chi_i(x) - \chi_i(y)|^2 \leq C_L R^{-2}. \end{aligned} \tag{3.dcccxxi}$$

The pair difference follows by integrating the vector of cutoff gradients along a metric path. The supremum bound follows from bounded overlap and the bounded metric diameter of a pillar.

Lemma 3.68 (IMS for the actual layered form)**.** For the true inner form,

$$\left|\sum_i K_{U,\omega}(\chi_i v,\chi_i v)-K_{U,\omega}(v,v)\right|\le CR^{-2}\|\mathsf{L}^{-1}v\|_2^2. \tag{3.dcccxxii}$$

The same error bounds the difference between $\sum_i \|\chi_i v\|_X^2$ and $\|v\|_X^2$. ■

Proof. Split the gauge form as $\|D_U^\sharp v\|_2^2-\langle v,D_UPD_U^\sharp v\rangle$. For the smoothing term put $u=\ell^{-1}v$ and use eq. 3.dcccxix. The exact double-cutoff expansion eq. 3.dcclvi bounds its cell error by $CR^{-2}(1+d_\ell(y,z))^2$ times the entry norm, using the within-cell and pair estimates in eq. 3.dcccxxi. The exponential moments give $CR^{-2}\|u\|_2^2$ by Schur.

The remaining divergence matrix has fine range 2η and physical coefficients $C\eta^{-2}$. Its squared cutoff difference is $C\eta^2/(R^2\mathsf{L}^2)$, cancelling the fine denominator. The Wilson Hessian is a sum of the same kind of finite plaquette matrices, including at its true boundary. The row-mass and actual constraint-curvature matrices have bounded mass-normalized physical Schur sums and bounded carrier diameter in the layer metric. Their double-cutoff errors have the same mass bound. This proves eq. 3.dcccxxii. The local unitary-shift calculation gives the X identity. A zero-extension boundary jump has only one nonzero field value, so the identity $\sum_i \chi_i^2=1$ makes its contribution exact. No exterior cutoff extension or action bond is required. □

3.21.3. *Shielding on the same determining geometry.*

Lemma 3.69 (Localized comparison of the actual projector)**.** Let two admitted real backgrounds have the same determining geometry and agree in common local frames on $Q_{24R\xi_0}$. Suppose all assigned scales there are comparable to ξ_0 within a fixed factor depending on L. For its core $Q_{2R\xi_0}$,

$$\|1_{\mathrm{core}}(R_1-R_2)1_{\mathrm{core}}\|_{2\to2}\le Ce^{-cR}. \tag{3.dcccxxiii}$$

Only the current-free site and normalized Gram package is used. ■

Proof. Use a cutoff supported strictly inside the agreement cube in

$$\theta G_1-G_2\theta=G_2(A_{s,2}\theta-\theta A_{s,1})G_1. \tag{3.dcccxxiv}$$

The middle operator is a gradient or actual site-row commutator in a buffer annulus. Its energy-to-dual-energy norm has the physical scale. Weighted site estimates make the two solutions exponentially small there. The compressed fine inverse difference is consequently $C\xi_0^2e^{-cR}$, with the corresponding localized energy-derivative bound. No exterior connection coefficients have been subtracted.

Apply this comparison to $F=Gq^\sharp\Xi^{-2}$. All local scale factors are comparable to ξ_0, so its inverse-length factors cancel the site-inverse scale. The normalized cell-L^2 comparison is Ce^{-cR}. Compress the Gram matrices to an intermediate top box. Their difference is Ce^{-cR} in operator norm, by splitting the F outputs into the common interior and their tails. Both compressions retain the uniform positive gap.

Their inverse difference is therefore Ce^{-cR}. The exact compression-inverse identity eq. 3.dcclxi, applied to these normalized matrices with the physical top measures, compares each compressed inverse with its global inverse. The boundary term is exponentially small by the weighted Gram estimate. Insert this and the $F^\sharp$ source tails into $P=FM^{-1}F^\sharp$. This proves eq. 3.dcccxxiii. These are operator estimates on arbitrary supported vectors, not sums of separate errors over a patch volume. □

3.21.4. *Local fields and the bare action boundary.* For an interior patch of scale ξ_0, choose a comb frame on $Q_{32R\xi_0}$. At most two adjacent scales occur there when $\rho \geq C_L R$. The local potential is bounded by $C_L R\kappa\xi_0^{-1}$. Multiply it by a cutoff equal to one on $Q_{24R\xi_0}$, vanishing farther out while still inside Ω_0. On the original active graph put $U_i^{\mathrm{ext}} = e^{i\eta A_i^{\mathrm{ext}}}$, and put the identity connection elsewhere. The local logarithm and product rules give

$$\sup_b \mathsf{L}(b)|A_i^{\mathrm{ext}}(b)| \leq C_L R\kappa, \qquad \sup_p \mathsf{L}(p)^2|F_i^{\mathrm{ext}}(p)| \leq C_L(\kappa + R^2\kappa^2). \tag{3.dcccxxv}$$

The cutoff derivative contributes $(R\xi_0)^{-1}(R\kappa\xi_0^{-1})$, with the required curvature units. All its averages use the original geometry.

A nonbare center has the surrounding cube inside Ω_0 by separation. For a bare center either it is also interior, or its whole field support is in a bare outer collar. In that case $Q_0 v_i = v_i$, no higher row carrier reaches its support, and its nonlinear constraint Hessian vanishes. The true inner Wilson lower bound and positive gauge give

$$K_{U,\omega}(v_i, v_i) \geq (d_* - C\kappa)\eta^{-2}\|v_i\|_2^2 \geq c_b\|v_i\|_X^2. \tag{3.dcccxxvi}$$

Here $d_* = \inf_c \xi_c^2 a_f(c) > 0$ is the lower field-row weight. The last inequality uses only the fine finite-difference bound $\|\nabla v_i^0\|_2 \leq C\eta^{-1}\|v_i\|_2$. This is a boundary estimate for the actual inner action.

3.21.5. *The real all-layer precision and saddle inverse.*

Proposition 3.70 (Uniform nonflat precision on the determining region)**.** There are fixed positive constants ρ_*, κ_*, γ_*, c and C such that, for every depth, compatible volume and separated aligned determining geometry with $\rho \geq \rho_*$, a real field with

$$\sup_{p\subset\Omega_0} \mathsf{L}(p)^2|F_U(p)| \leq \kappa_*, \qquad \sup_{c\in\mathbf{B}} \xi_c^3|\omega(c)| \leq \gamma_* \tag{3.dcccxxvii}$$

satisfies

$$c\|v\|_X^2 \leq K_{U,\omega}(v, v) \leq C\|v\|_X^2. \tag{3.dcccxxviii}$$

The constants depend only on L and the positive weight bounds. The form uses the actual inner Wilson Hessian, projected gauge and nonlinear straightened-constraint curvature. ■

Proof. Initially pass to a finite periodic cover large enough for the patches. Fix the initial curvature ceiling for the site estimates. It gives fixed normalized column and Gram constants and a rate s_0. The flat gap gives a fixed lower constant. Choose R large enough that CR^{-2} and Ce^{-cR} are smaller than a quarter of the common-gauge lower constant. Choose ρ to satisfy the flat patch conditions, eq. 3.dcccxx, and every fixed $C_L R$ collar inequality. These prior constants are uniform above their initial separation thresholds. Finally reduce κ_*, γ_* so eq. 3.dcccxxv admits every local mean and meets eq. 3.dcccxvii. This order does not use the depth.

Use eq. 3.dcccxxi. For an interior $v_i = \chi_i v$, change to its local comb frame and retain ω, a_f on every carrier meeting its support. Extend those coefficients elsewhere with the same bounds. Fixed gauge changes conjugate the actual means, their first and second derivatives and the site projector by the matching unitary fine and top rotations. The forms are unchanged. All local terms of the comparison field therefore agree exactly on v_i. Only R differs.

For a real orthogonal projector, $\|Rf\|^2 = \langle f, Rf\rangle$. Apply Lem. 3.69 to $f = D_U^\sharp v_i$; its support is inside the core. Thus

$$|K_{U,\omega}(v_i, v_i) - K_{U_i^{\mathrm{ext}},\omega_i}(v_i, v_i)| \leq Ce^{-cR}\|v_i\|_X^2.$$

The extension has a global small weighted potential relative to the identity on the same active graph, so Cor. 3.67 applies. Boundary patches have the direct estimate eq. 3.dcccxxvi. Sum these local lower bounds and use Lem. 3.68; the chosen errors are absorbed. The upper bound follows

from the local Wilson coefficient estimate, orthogonality of R, physical row bounds and the two-leg constraint moment.

On a smaller original torus the actual local rows, physical transposes and Ξ commute with pullback to the finite cover. Uniqueness of the gapped site equation and then the normalized Gram equation makes their inverses preserve deck-invariant subspaces. Thus the actual projector descends exactly. The full form and the squared X norm have the same sheet-count factor. Divide the cover estimate by that factor. No global almost-flat gauge or topology restriction is used. □

Let $X_L = \ker(R_U D_U^\sharp)$ and $K_L = D^2 A_{\Omega_0}(U) - B_\omega$. On $X_L \cap \ker Q_U$, both positive penalties vanish exactly. Coercivity there and the true-row inf-sup eq. 3.dccxcv prove the actual saddle estimate

$$K_L u - Q_U^\sharp \mu = f \text{ on } X_L, \qquad Q_U u = b, \qquad \|u\|_X + \|\mu\|_{Y^*} \le C(\|f\|_{X_L^*} + \|b\|_Y). \tag{3.dcccxxix}$$

Explicitly write $u = E_L b + z$, solve the coercive kernel equation for z, and define $\langle \mu, b' \rangle = (K_L u - f)(E_L b')$. Every $w \in X_L$ splits as $w - E_L Q_U w + E_L Q_U w$, proving the full equation. Kernel coercivity and surjectivity give uniqueness. This bounds a linearized saddle with its actual small multiplier; it does not construct the critical background.

3.21.6. *Exponential energy kernels in the physical layer metric.* The energy inverse also has a quantitative spatial consequence. Let $G_f = K_{U,\omega}^{-1}$, and let 1_y restrict fields to an assigned cell. With d the original physical layer metric,

$$\begin{aligned} \|1_y G_f 1_z\|_{2\to 2} &\le C\xi_y \xi_z e^{-md(y,z)}, \\ \|1_y \nabla_U G_f 1_z\|_{2\to 2} &\le C\xi_z e^{-md(y,z)}, \\ \|1_y G_f \nabla_U^\sharp 1_z\|_{2\to 2} &\le C\xi_y e^{-md(y,z)}, \\ \|1_y \nabla_U G_f \nabla_U^\sharp 1_z\|_{2\to 2} &\le C e^{-md(y,z)}. \end{aligned} \tag{3.dcccxxx}$$

Here ∇_U is the comparison component gradient defining X; its adjoint in the last two lines is the corresponding typed force source. The norms include all physical cell measures. These are cell operator estimates, not pointwise bounds at the fine Green singularity.

To prove them, conjugate by $w = e^{s\psi}$ for bounded one-Lipschitz ψ in d_ℓ. A local derivative commutator has size $C|s|\ell^{-1}$, so the local Wilson and divergence forms change by $C|s|$ in the two X norms. The normalized nonlocal term in eq. 3.dcccxix has conjugated difference $C|s|$ in physical L^2 norm, by the weighted column and Gram resolvent arguments. The row and constraint terms have bounded normalized Schur sums and fixed metric carrier radius; their differences have that same $C|s|$ bound. Thus

$$\|w K_{U,\omega} w^{-1} - K_{U,\omega}\|_{X \to X^*} \le C|s|. \tag{3.dcccxxxi}$$

The real lower bound and a normalized Neumann series imply $\|w G_f w^{-1}\|_{X^* \to X} \le C$ for a fixed small s. Converting a force supported in B_z to the dual energy norm costs at most $C\xi_z$; converting output energy to $L^2(B_y)$ costs at most $C\xi_y$. Choose ψ to be a truncated distance from the source cell. Its within-cell oscillation is bounded. This proves the first two lines of eq. 3.dcccxxx. A gradient-adjoint source has dual norm bounded by its L^2 norm, with no scale factor. The same argument proves the last two lines, including the single-step support enlargement of that source.

After increasing the fixed separation once more so that $8 \log L/\rho < m/2$, the metric ball count makes their normalized row and column sums finite. In particular, if $\sup_z \xi_z \|f\|_{2,B_z} \le B$, then

$$\sup_y \left\{ \xi_y^{-1} \|G_f f\|_{2,B_y} + \|\nabla_U G_f f\|_{2,B_y} \right\} \le CB. \tag{3.dcccxxxii}$$

Indeed the scale factors in the first two kernel lines cancel these input and output amplitudes before the exponential cell sum is taken. This is an energy response estimate uniform in the number of fine bonds in each pillar.

3.21.7. *The scope of the current extension.* The explicit current bound eq. 3.dccxcvii also makes

$$\widetilde{K}[J,\omega] = D^2 A_{\Omega_0}(U) - \mathcal{B}_U[J] + D_U R_U D_U^\sharp + Q_U^\sharp a_f Q_U - S^\sharp \Delta^{(2)}[\omega] S \tag{3.dcccxxxiii}$$

coercive for sufficiently small $\sup_b \mathsf{L}(b)^3|J(b)|$ and $\sup_c \xi_c^3|\omega(c)|$. First apply eq. 3.dcccxxviii at zero multiplier and then absorb the two explicit relative form estimates. On X_L, S is the identity and every term in $\mathcal{B}_U[J]$ has a zero horizontal gauge factor. Its restriction to the constraint kernel is therefore the same Lagrangian Hessian when ω is the actual multiplier.

Prop. 3.90 identifies this completion with the specified canonical current correction. Prop. 3.73 supplies the induced readout, and Props. 3.92, 3.171 and 3.174 supply the strong outputs, carrier families and nonlinear response. The absolute soft contour precision retains its separate gauge-restoration comparison.

3.22. **Coarse Schur kernels and the induced current.** The current in the second explicit precision is obtained by transposing a minimizing field lift. An energy bound for that lift alone would charge the whole volume of a bounded current. We prove exponential energy columns and sum them in the actual layer metric. This gives the required weighted supremum bound for the induced row multiplier without a fine pointwise Green estimate.

Keep the real background and determining geometry of Prop. 3.70. Currents below may be complex, but the background is real. Every transpose in an analytic formula is the physical bilinear transpose. Hilbert norms use the complexification of the real physical spaces. Let

$$\begin{aligned} \mathcal{K}_0 &= D^2 A_{\Omega_0}(U) + D_U R D_U^\sharp + Q^\sharp a_f Q, \\ \mathcal{K}_J &= \mathcal{K}_0 - \mathcal{B}_U[J], \qquad G_J = \mathcal{K}_J^{-1}, \\ \gamma_J &= \sup_b \mathsf{L}(b)^3 |J(b)|. \end{aligned} \tag{3.dcccxxxiv}$$

This is the displayed symmetric current extension, with $\mathcal{B}_U[J]$ defined by its endpoint formula. It is not identified here with another named completion.

The field-row Hilbert space $\mathcal{H}_\Gamma$ has measure ξ_c^4. Write $\Xi_\Gamma b(c) = \xi_c b(c)$, so that $\mathcal{U}_\Gamma = \Xi_\Gamma^2$ is unitary from $\mathcal{H}_\Gamma$ to the counting ℓ^2 space. This is a field-row scale. At an outward row it is the row's assigned level, even when one endpoint is a top of the smaller level. It is not the site-scale operator Ξ.

3.22.1. *Weighted restoration for the current correction.*

Lemma 3.71 (Restoration with a spatial exponential weight)**.** For a fixed small $s_* > 0$, a bounded one-Lipschitz function ψ for d_ℓ, and $w = e^{s\psi}$, $|s| \le s_*$, the actual maps satisfy

$$\|\ell^{-2} w \Lambda w^{-1} v\|_2 + \|w D_U \Lambda w^{-1} v\|_X + \|w S w^{-1} v\|_X \le C \|v\|_X. \tag{3.dcccxxxv}$$

Consequently

$$\|w \mathcal{B}_U[J] w^{-1}\|_{X \to X^*} \le C \gamma_J. \tag{3.dcccxxxvi}$$

No derivative of J or second derivative of ψ is a premise. ■

Proof. For $\lambda = \Lambda w^{-1} v$, put $f = L_U \lambda = R D_U^\sharp w^{-1} v$. The weighted site projector bound and the first derivative commutator give $\|wf\|_2 \le C\|v\|_X$. Also $q\lambda = 0$, including the bare collar.

Repeat the zero-average proof with the combined scalar weight $h_w = \ell^{-1} e^{s\psi}$. Its relative first difference is bounded by $C(\delta + |s|)/\ell$. Subtract $h_w(y) q\lambda = 0$ in each pillar. The site-gap average error is now at most

$$C_L(\delta + |s|)\|\ell^{-2} w \lambda\|_2.$$

Testing $L_U \lambda = f$ against $h_w^2 \lambda$ and using the exact weighted Green identity proves

$$\|\ell^{-2} w \lambda\|_2 + \|\ell^{-1} w D_U \lambda\|_2 \le C \|wf\|_2,$$

after reducing s_*. The weighted Hodge calculation for $wD_U\lambda$ contains only first weight differences. Its divergence is wf plus $O(|s|\ell^{-1}wD_U\lambda)$; its curl is weighted curvature times λ, plus that same first-weight error. Thus its component-gradient energy is bounded by $C\|wf\|_2^2$. The zero-average bare collar keeps every nonzero stencil inside the action region, exactly as before. This proves eq. 3.dcccxxxv.

Use the four undifferentiated-current pairings in eq. 3.cdlxix. Apply the restoration bounds with both signs of s. The endpoint ratios of w in $\mathcal{W}_U[J]$ are bounded uniformly, since an endpoint step has bounded metric length. Each pairing is therefore bounded by $C\gamma_J\|u\|_X\|v\|_X$. This proves eq. 3.dcccxxxvi. □

The weighted inverse of $\mathcal{K}_0$ is bounded from X^* to X by the already proved field Combes argument. For sufficiently small γ_J, the normalized Neumann series using eq. 3.dcccxxxvi gives

$$\|wG_Jw^{-1}\|_{X^*\to X}\le C. \tag{3.dcccxxxvii}$$

This proof applies to complex currents. It invokes positivity only at the real zero-current reference; it asserts no Hermitian positivity at a complex current.

3.22.2. *The normalized coarse Schur matrix.* Define

$$S_J=QG_JQ^\sharp,\quad T_J=a_f^{1/2}S_Ja_f^{1/2},\quad T_J^{\mathrm{n}}=\Xi_\Gamma^2T_J\Xi_\Gamma^{-2}. \tag{3.dcccxxxviii}$$

The last matrix acts in counting orthonormal coordinates. Give a row c the metric location of one incident top at its row level. Its carrier is within bounded layer-metric distance of that top, and at most eight rows are assigned to a top. The row ball count has the same growth rate as the top count.

Proposition 3.72 (Uniform normalized Schur inverse)**.** For sufficiently small γ_J, S_J is invertible and

$$|(T_J^{\mathrm{n}})^{-1}(c,d)|\le Ce^{-md(c,d)}. \tag{3.dcccxxxix}$$

The constants are independent of depth, compatible volume and the current. After increasing the fixed separation so the metric growth rate is less than $m/2$, this inverse is bounded on counting ℓ^∞. Equivalently T_J^{-1} is bounded in the coarse amplitude norm $\sup_c\xi_c^2|\beta(c)|$. ■

Proof. At $J=0$, T_0 has a positive lower and upper bound. For the lower bound take the exact field-row right inverse E_Q, or $E_L=SE_Q$, and put $v=E_Qa_f^{-1/2}\beta$. Its $\mathcal{K}_0$ energy is at most $C\|\beta\|_{\mathcal{H}_\Gamma}^2$. Since $Qv=a_f^{-1/2}\beta$,

$$\begin{aligned}\|\beta\|_{\mathcal{H}_\Gamma}^2&=\langle Q^\sharp a_f^{1/2}\beta,E_Qa_f^{-1/2}\beta\rangle\\&\le C\langle\beta,T_0\beta\rangle^{1/2}\|\beta\|_{\mathcal{H}_\Gamma}.\end{aligned} \tag{3.dcccxl}$$

The upper bound follows from the energy inverse and the bounded trace $a_f^{1/2}Q:X\to\mathcal{H}_\Gamma$. No positivity of a different effective coarse action has been used.

Let w_Γ sample $e^{s\psi}$ at the row locations. The finite metric carriers of the actual rows imply

$$\|w_\Gamma a_f^{1/2}Qw^{-1}-a_f^{1/2}Q\|_{X\to\mathcal{H}_\Gamma}\le C|s|.$$

The weighted base inverse differs from its original value by $C|s|$. Its current perturbation differs from the weighted base inverse by $C\gamma_J$, by the same normalized Neumann series as in eq. 3.dcccxxxvii. Therefore

$$\|w_\Gamma T_Jw_\Gamma^{-1}-T_0\|\le C(|s|+\gamma_J). \tag{3.dcccxli}$$

Choose s and the current ceiling so the right side is less than half the real lower constant of T_0. The inverse of the conjugated T_J is uniformly bounded. Truncated distance weights give exponential entries between physical orthonormal row cells, which are precisely $(T_J^{\mathrm{n}})^{-1}(c,d)$. The cell count and a strict rate reserve give the stated ℓ^∞ bound. □

For reference, these normalizations imply the counting-array and physical-kernel bounds

$$|S_J^{-1}(c,d)|_{\text{array}} \le C\xi_c^{-3}\xi_d e^{-md(c,d)}, \qquad |S_J^{-1}(c,d)|_{\text{physical}} \le C\xi_c^{-3}\xi_d^{-3}e^{-md(c,d)}. \tag{3.dcccxlii}$$

Indeed $S_J^{-1} = a_f^{1/2}T_J^{-1}a_f^{1/2}$, and converting its array to a physical kernel divides by the input measure ξ_d^4. These are different conventions for the same operator.

3.22.3. *The actual minimizing lift and the induced multiplier.* Define the actual first-variant lift by

$$\widehat{H}_J = G_J Q^\sharp S_J^{-1}, \qquad Q\widehat{H}_J = 1. \tag{3.dcccxliii}$$

It has the useful normalized representation

$$\widehat{H}_J a_f^{-1/2} = G_J Q^\sharp a_f^{1/2} T_J^{-1}. \tag{3.dcccxliv}$$

The weighted inverse, trace and normalized Schur bounds make this a bounded weighted operator from $\mathcal{H}_\Gamma$ to X. In particular, with X_y the cell component energy and a fixed one-fine-step collar for its derivatives,

$$\|1_y\widehat{H}_J 1_c b\|_{X_y} \le Ce^{-md(y,c)}\|1_c b\|_Y. \tag{3.dcccxlv}$$

This follows directly by a distance weight centered at c; it does not require summing an unnormalized fine kernel.

Proposition 3.73 (The actual induced row current has its physical amplitude)**.** For the displayed first precision and its actual lift, put

$$\omega_J = \widehat{H}_J^\sharp J. \tag{3.dcccxlvi}$$

For sufficiently small γ_J,

$$\sup_c \xi_c^3|\omega_J(c)| \le C\gamma_J. \tag{3.dcccxlvii}$$

This holds uniformly over all assigned scales, determining regions and compatible volumes, including bare and outward rows. It is valid for complex independent currents on the stated small ball. ■

Proof. Fix a row c and a unit colour e. Take the datum $b^{(c)}(c) = \xi_c^{-1}e$, zero elsewhere. Its squared Y norm is $\xi_c^2 a_f(c)$, bounded above and below by fixed constants. Physical transposition gives the exact dimensional identity

$$\xi_c^3\langle e, \omega_J(c)\rangle = \langle \widehat{H}_J b^{(c)}, J\rangle_{\text{fine}}. \tag{3.dcccxlviii}$$

On a cell of side ξ_y, the current assumption implies

$$\|\xi J\|_{2,B_y} \le C_L\gamma_J. \tag{3.dcccxlix}$$

Indeed its physical volume is ξ_y^4, while the current amplitude is at most $C_L\gamma_J\xi_y^{-3}$. Pair cell by cell in eq. 3.dcccxlviii:

$$|\xi_c^3\langle e, \omega_J(c)\rangle| \le \sum_y \|\xi J\|_{2,B_y}\|\xi^{-1}\widehat{H}_J b^{(c)}\|_{2,B_y} \le C\gamma_J \sum_y e^{-md(y,c)}.$$

Increase the fixed separation, if necessary, so that its growth rate is below $m/2$. The last sum is uniform. Taking the three colour components proves eq. 3.dcccxlvii. For complex currents the same bilinear identity and absolute Cauchy–Schwarz bound apply; no current was replaced by the Wilson gradient. □

At a bare row this proof is still dimensionally exact. The cell force norm is $\eta\|J\|_2 = O(\gamma_J)$, the unit energy datum has size η^{-1}, and its physical row dual has size $\eta^3\omega$. No ultraviolet factor was discarded.

3.22.4. *Exponential coupled kernels.* The full penalized saddle equations have an explicit inverse:

$$\begin{aligned} \mathcal{K}_J u - Q^\sharp \mu = f, \qquad Qu = b, \\ u = (G_J - \widehat{H}_J Q G_J) f + \widehat{H}_J b, \\ \mu = S_J^{-1}(b - Q G_J f). \end{aligned} \tag{3.dcccl}$$

All normalized factors on the right have uniformly bounded weighted energy norms. The same holds for the resulting block inverse. It consequently has exponential cell kernels with the physical force and datum normalizations: a force cell costs $\xi_z \|f\|_{2,B_z}$, a datum row costs $\|b\|_{Y_c}$, a field output is measured in X_y, and a multiplier output is measured by $\xi_c^3 |\mu(c)|$, up to its fixed weight constant. In particular

$$\|u\|_X + \|\mu\|_{Y^*} \le C(\|f\|_{X^*} + \|b\|_Y). \tag{3.dcccli}$$

Removing the artificial row-mass term from the first equation changes its multiplier to $\mu - a_f b$, a local bounded map in Y^*. It preserves these bounds and spatial kernels.

For an independent current the minimizing lift is not asserted to lie in the Landau slice. Its exact off-current slice defect is the one in eq. 3.cdlxxiv. The coupled formula above keeps the displayed gauge penalty.

3.22.5. *The second explicit variant is admitted.* Now define, in the prescribed order,

$$\mathcal{K}_{1,J} = \mathcal{K}_J - S^\sharp \Delta^{(2)}[\omega_J] S, \qquad \omega_J = \widehat{H}_J^\sharp J. \tag{3.dcccllii}$$

The lift inside ω_J is the first-variant lift in eq. 3.dcccxliii. Replacing it by the lift formed from $\mathcal{K}_{1,J}$ would be a different definition.

The new term has energy norm at most $C\gamma_J$, by eq. 3.dcccxlvii and the actual two-leg constraint estimate. Its conjugated norm has that same bound: the row carrier has fixed metric radius and the two restoration factors have eq. 3.dcccxxxv. Thus a further fixed decrease of the current ceiling makes $\mathcal{K}_{1,J}$ invertible with the same weighted energy bounds. Its normalized coarse Schur matrix, lift and coupled inverse satisfy the same conclusions. This proves admission of the second displayed variant, rather than merely bounding a constraint form at a freely chosen small multiplier.

The maps are holomorphic in the independent current ball. The first inverse, normalized Schur inverse and lift are convergent analytic inverses there; the readout $\widehat{H}_J^\sharp J$ is consequently holomorphic, and so is the second precision and its inverse. The transpose is bilinear throughout this continuation.

3.22.6. *The physical slice.* If $J = j_U$ is the actual Wilson current, the inner-action Ward identity and the displayed current correction give

$$G_J D_U R = D_U G R. \tag{3.dcccliii}$$

The second subtraction annihilates $D_U G R$, since $S D_U G R = 0$, so this identity also holds for $G_{1,J} = \mathcal{K}_{1,J}^{-1}$. Transposition and $Q D_U G R = \bar{D}_{\mathrm{ext}} q G R = 0$ imply that its row lift is horizontal. With that second lift $\widehat{H}_{1,J}$, set $C_1 = G_{1,J} - \widehat{H}_{1,J} Q G_{1,J}$. Then $C_1 S^\sharp f$ is both horizontal and in $\ker Q$: the gauge cancellation is $(S D_U G R)^\sharp = 0$. These identities use the actual extended outward difference.

At an actual constrained critical point, full stationarity is $j_U = Q^\sharp \omega_{\mathrm{phys}}$. Since $Q\widehat{H}_J = 1$, it follows exactly that $\omega_J = \omega_{\mathrm{phys}}$. In particular the induced-current theorem bounds the actual multiplier at every layer if the actual current has its stated small amplitude. The full stationarity equation can be deduced from criticality on the actual horizontal constraint kernel: S preserves Q, and the Ward identity annihilates the gauge part $1 - S$. It is not inferred from criticality on an unidentified slice.

On horizontal vectors the Ward correction vanishes and $S = 1$. Therefore the second form restricts to the actual Lagrangian Hessian plus the row-mass term. The formula

$$u = C_1 S^\sharp f + \widehat{H}_{1,J} b$$

is the physical horizontal saddle solution, with its multiplier adjusted by $-a_f b$ when the row mass is removed. Its exponential energy kernels follow from the weighted factors already proved. No fine pointwise inverse has been used to obtain this physical slice statement.

The conclusions concern the two displayed current extensions. They do not identify another independently named $\Delta'_\pi[J]$, nor a signed-profile precision or the absolute contour precision. Fine pointwise and Hölder responses, moving complex background slices and the nonlinear trial-field construction remain further tasks.

3.23. **A complex inverse on the moving Landau slice.** The real proper-region estimate has a uniform complex neighbourhood. The relevant slice moves because its site projection and covariant divergence depend on the background. We first continue the site operators by a balanced physical form. We then solve a saddle problem with a fixed target space for both constraints. Its zero gauge datum is exactly the moving Landau condition.

The geometry, scales, bare collars and outward rows remain those of Prop. 3.70. Fix a real background U_0 with the curvature ceiling there reduced by a factor of two. All comparison norms and ℓ in this section refer to this fixed geometry and real frame. Write

$$U_A(b) = e^{i\eta A(b)} U_0(b), \qquad \|A\|_{\mathcal{T}} = \sup_b \ell(b)|A(b)| + \sup \ell^2 |\nabla_{U_0} A| < \varepsilon_*. \tag{3.dcccliv}$$

The comparison derivatives use the same zero-extension convention as X. At the bare boundary $\ell = \eta$, so the boundary step also has this scale. The field A may be complex. Transposes of analytic operators are bilinear physical transposes, not conjugate transposes.

3.23.1. *Uniform continuation of the actual site projection.* For this section use a subscript A for actual objects at U_A, and a subscript 0 for the real reference. In particular

$$\begin{aligned} A_{s,A} &= D_A^\sharp D_A + q_A^\sharp a_s q_A, \qquad G_{s,A} = A_{s,A}^{-1}, \\ F_A &= G_{s,A} q_A^\sharp \Xi^{-2}, \qquad M_A = F_A^\sharp F_A, \qquad R_A = 1 - F_A M_A^{-1} F_A^\sharp. \end{aligned} \tag{3.dccclv}$$

Here Ξ is the site-top scale, not the field-row scale.

Lemma 3.74 (Balanced complex site form)**.** For a fixed small ε_*, all operators in eq. 3.dccclv are defined and holomorphic throughout eq. 3.dcccliv. With physical Hilbert operator norms,

$$\|F_A - F_0\| + \|\ell D_0 (F_A - F_0)\| + \|M_A - M_0\| + \|R_A - R_0\| \le C\|A\|_{\mathcal{T}}. \tag{3.dccclvi}$$

Moreover $R_A^\sharp = R_A$, $R_A^2 = R_A$, and

$$q_A G_{s,A} R_A = 0, \qquad D_A^\sharp D_A G_{s,A} R_A = R_A. \tag{3.dccclvii}$$

No complex orthogonality is claimed. ■

Proof. The actual transport paths have length at most a fixed multiple of their row scale. On their carriers ℓ is comparable to that scale. Telescoping their transports therefore gives, for each of the finitely many powers $r = -2, -1, 0, 1, 2$,

$$\begin{aligned} \|\Xi^{-r}(q_A - q_0)\ell^r\| &\le C\|A\|_{\mathcal{T}}, \\ \|\ell(D_A - D_0)u\|_2 &\le C\|A\|_{\mathcal{T}}\|u\|_2, \\ \|a_f^{1/2}(Q_A - Q_0)v\|_{\mathcal{H}_\Gamma} &\le C\|A\|_{\mathcal{T}}\|\ell^{-1}v\|_2. \end{aligned} \tag{3.dccclviii}$$

The same estimates follow for the transposes. Physical carrier Schur sums, rather than the number of fine sites in a carrier, are used. The local logarithms remain in their common analytic charts after a fixed decrease of ε_*. The geometric sum of path lengths through successive homogeneous subscales is bounded by the final path scale. Thus these comparisons are independent of depth. They include completed outward rows; the identity $Q_A D_A = \bar{D}_{\mathrm{ext},A} q_A$ is kept with its actual extension.

Put

$$\|u\|_{Z_+}^2 = \|\ell D_0 u\|_2^2 + \|q_0 u\|_{\mathrm{top}}^2.$$

The real block inequality gives $\|u\|_2 \le C\|u\|_{Z_+}$. Consider the sesquilinear testing form $\mathfrak{a}_A(u,v) = \langle A_{s,A}u, \ell^2\overline{v}\rangle_{\mathrm{bil}}$. At the real reference its weighted Green identity, and the $O(\delta)$ average commutator, give

$$\mathrm{Re}\,\mathfrak{a}_0(u,u) \ge c\|u\|_{Z_+}^2, \qquad |\mathfrak{a}_A(u,v) - \mathfrak{a}_0(u,v)| \le C\|A\|_{\mathcal{T}}\|u\|_{Z_+}\|v\|_{Z_+}. \tag{3.dccclix}$$

Indeed a derivative error is $O(\varepsilon_*\ell^{-1}u)$ before multiplication by ℓ; the row errors in the test $q_A(\ell^2 v)$ are cancelled by the Ξ^{-2} site mass. The first differences of ℓ cost $O(\delta)$. Choose δ as in the real proof and then ε_* small. This is a coercive perturbation on Z_+.

The source for $F_A b$ obeys

$$|\langle q_A^\sharp \Xi^{-2} b, \ell^2\overline{v}\rangle_{\mathrm{bil}}| \le C\|b\|_{\mathrm{top}}\|v\|_{Z_+}.$$

Its difference from the real source is $O(\|A\|_{\mathcal{T}})$ in that dual norm. Solving the balanced form and subtracting the two equations proves the first two bounds in eq. 3.dccclvi. This also proves holomorphy by a form Neumann series on fixed spaces; the testing weight does not depend on A.

The real normalized Gram lower bound implies M_0^{-1} is uniformly bounded. The L^2 bound for $F_A - F_0$ gives $M_A - M_0 = O(\|A\|_{\mathcal{T}})$. Its inverse and the last bound in eq. 3.dccclvi follow by another Neumann series. The projection algebra in eq. 3.dccclv is bilinear algebra, so it continues exactly. Since $F_A^\sharp R_A = 0$ and $G_{s,A}^\sharp = G_{s,A}$, the first identity in eq. 3.dccclvii follows. Apply $A_{s,A}G_{s,A} = 1$ to obtain the second. □

3.23.2. *Two physical derivatives in the complex zero-average space.*

Lemma 3.75 (Complex gauge restoration)**.** If $q_A\lambda = 0$ and $D_A^\sharp D_A\lambda = f$, then

$$\|\ell^{-2}\lambda\|_2 + \|\ell^{-1}D_0\lambda\|_2 + \|\nabla_0 D_A\lambda\|_2 \le C\|f\|_2. \tag{3.dccclx}$$

Consequently

$$\Lambda_A = G_{s,A}R_A D_A^\sharp, \qquad S_A = 1 - D_A\Lambda_A \tag{3.dccclxi}$$

are bounded holomorphic maps in their physical norms, with $\|D_A\Lambda_A v\|_X + \|S_A v\|_X \le C\|v\|_X$. Exactly

$$R_A D_A^\sharp S_A = 0, \qquad Q_A S_A = Q_A. \qquad \blacksquare$$

Proof. Apply the real weighted site gap to $\ell^{-1}\lambda$. In its average term insert $q_A\lambda = 0$. The weight commutator costs $C\delta\|\ell^{-2}\lambda\|_2$, and eq. 3.dccclviii costs $C\varepsilon_*\|\ell^{-2}\lambda\|_2$. After absorption,

$$\|\ell^{-2}\lambda\|_2 \le C\|D_0(\ell^{-1}\lambda)\|_2.$$

Test the complex equation against $\ell^{-2}\overline{\lambda}$, and compare its bilinear form with the real reference form. The exact real weighted Green identity has an $O(\delta^2)\|\ell^{-2}\lambda\|_2^2$ error. Each complex derivative perturbation contributes at most

$$C\varepsilon_*(\|\ell^{-1}D_0\lambda\|_2 + \|\ell^{-2}\lambda\|_2)^2.$$

Absorb these errors using the preceding gap. The forcing pairing is bounded by $\|f\|_2\|\ell^{-2}\lambda\|_2$. This proves the first two terms of eq. 3.dccclx, without a derivative of A.

For the last term the C^1 part of the tube is used. Expanding the actual covariant stencils gives

$$\|(D_0^\sharp D_0 - D_A^\sharp D_A)\lambda\|_2 + \|\nabla_0(D_A - D_0)\lambda\|_2 \le C\varepsilon_*(\|\ell^{-1}D_0\lambda\|_2 + \|\ell^{-2}\lambda\|_2). \tag{3.dccclxii}$$

The differentiated transport difference contains $\nabla_0 A$, $AD_0\lambda$, and $A^2\lambda$. Its two explicit tube powers are respectively ℓ^{-2}, ℓ^{-1}, and ℓ^{-2}; there is no second derivative of A. The bare row is still the identity, so $\lambda = 0$ on the same bare collar. Extend it by zero and apply the real covariant Hodge estimate eq. 3.dlxviii to $D_0\lambda$. Every nonzero second-derivative stencil stays inside the inner-plaquette region. Equation eq. 3.dccclxii and the first two bounds give the final term in eq. 3.dccclx.

Take $\lambda = G_{s,A} R_A D_A^\sharp v$. Its source has norm at most $C\|v\|_X$, and eq. 3.dccclvii supplies both equations just used. The mass term of $D_A\lambda$ follows from the same comparison. This proves the norm assertions. All maps are holomorphic in the finite coordinate system, and their uniform estimates are in the displayed spaces. Finally eq. 3.dccclvii and the actual intertwining give the two exact identities. □

3.23.3. *The combined constraint has a fixed target.* Let $Z_0 = \operatorname{ran} R_0$, with its physical L^2 norm. Define the combined constraint on the fixed field space X by

$$\mathcal{C}_A u = (Q_A u,\ R_0 R_A D_A^\sharp u) \in Y \oplus Z_0. \tag{3.dccclxiii}$$

This use of R_0 fixes the target only. It does not replace the moving projection R_A.

Lemma 3.76. The reference constraint has a bounded exact right inverse

$$\mathcal{E}_0(b,\rho) = E_{L,0} b + D_0 G_{s,0} R_0 \rho. \tag{3.dccclxiv}$$

Also

$$\|\mathcal{C}_A - \mathcal{C}_0\|_{X \to Y \oplus Z_0} \le C\|A\|_{\mathcal{T}}. \tag{3.dccclxv}$$

Thus $\mathcal{C}_A$ is onto with a uniform right inverse. Its second component is zero if and only if $R_A D_A^\sharp u = 0$. ■

Proof. The first summand of eq. 3.dccclxiv has row value b and zero reference divergence. The second has zero row value by $Q_0 D_0 G_{s,0} R_0 = \bar{D}_{\mathrm{ext},0} q_0 G_{s,0} R_0 = 0$. Its reference projected divergence is ρ, by eq. 3.dccclvii. Its energy bound is the real zero-average restoration estimate. This proves the right-inverse assertion.

For the comparison use eq. 3.dccclviii, eq. 3.dccclvi, and $\|D_0^\sharp u\|_2 \le C\|u\|_X$. Then $\mathcal{C}_A\mathcal{E}_0 = 1 + O(\varepsilon_*)$; inverting it gives a right inverse for $\mathcal{C}_A$.

If $z \in \operatorname{ran} R_A$ and $R_0 z = 0$, then $z = (R_A - R_0)z$, so $z = 0$. Also $R_0 R_A : Z_0 \to Z_0$ is $1 + O(\varepsilon_*)$. It follows that $R_0 : \operatorname{ran} R_A \to Z_0$ is an isomorphism. Apply its injectivity to $z = R_A D_A^\sharp u$ to prove the final equivalence. □

3.23.4. *The full complex saddle and its actual slice.* Let ω be a complex field-row multiplier with $\gamma_\omega = \sup_c \xi_c^3 |\omega(c)|$, and define the displayed Lagrangian Hessian

$$H_{A,\omega} = D^2 A_{\Omega_0}(U_A) - \Delta_A^{(2)}[\omega]. \tag{3.dccclxvi}$$

No current extension is part of this definition.

Proposition 3.77 (Complex saddle inverse)**.** There are depth- and volume-independent constants $\varepsilon_* > 0$ and $\gamma_* > 0$. If eq. 3.dcccliv holds and $\gamma_\omega < \gamma_*$, the system

$$\begin{aligned} H_{A,\omega} u - Q_A^\sharp \mu - D_A R_A R_0 \nu &= f, \\ Q_A u = b, \qquad R_0 R_A D_A^\sharp u &= \rho \end{aligned} \tag{3.dccclxvii}$$

has a unique solution $(u, \mu, \nu) \in X \oplus Y^* \oplus Z_0$ and

$$\|u\|_X + \|\mu\|_{Y^*} + \|\nu\|_2 \le C(\|f\|_{X^*} + \|b\|_Y + \|\rho\|_2). \tag{3.dccclxviii}$$

Its inverse is holomorphic in (A, ω) on this fixed product of spaces. At $\rho = 0$ it gives exactly the actual moving-slice saddle

$$\begin{aligned} X_L(A) = \ker(R_A D_A^\sharp), \qquad Q_A u = b, \\ H_{A,\omega} u - Q_A^\sharp \mu = f \quad \text{on } X_L(A). \end{aligned} \tag{3.dccclxix}$$

This saddle has a unique (u, μ) with the bound

$$\|u\|_X + \|\mu\|_{Y^*} \le C(\|f\|_{X_L(A)^*} + \|b\|_Y). \tag{3.dccclxx}$$

■

Proof. At $A = 0, \omega = 0$, the real proper-region gap gives coercivity of $H_{0,0}$ on $\ker \mathcal{C}_0 = X_L(0) \cap \ker Q_0$. Its full form is bounded on X. The right inverse eq. 3.dccclxiv then proves invertibility of the full reference saddle. For completeness, lift (b, ρ), solve the coercive equation on $\ker \mathcal{C}_0$, and recover the multiplier by evaluating the remaining functional on the same right inverse. This gives the full estimate, with no positivity assertion off the constraint kernel.

The actual plaquette expansion implies

$$\|H_{A,\omega} - H_{0,0}\|_{X \to X^*} \le C(\|A\|_{\mathcal{T}} + \gamma_\omega). \tag{3.dccclxxi}$$

To check the scale, the curl-row difference is bounded by $C\|A\|_{\mathcal{T}}\|\ell^{-1}v\|_2$. The physical plaquette-curvature difference is bounded by $C\|A\|_{\mathcal{T}}\ell^{-2}$, from the first covariant differences of A and the quadratic A term. The exact Wilson Hessian remainder is consequently a bounded mass-form perturbation. This is a local inner-plaquette calculation and has no boundary integration-by-parts term. The actual constraint two-leg estimate bounds the last term of eq. 3.dccclxvi by $C\gamma_\omega$ in the two energy norms. Its carrier estimates continue on the same complex charts with a uniform constant.

The transpose of the second component of $\mathcal{C}_A$ is $D_A R_A R_0$, because R_A is bilinearly symmetric. The whole operator in eq. 3.dccclxvii therefore differs from the reference saddle by $C(\|A\|_{\mathcal{T}} + \gamma_\omega)$ between fixed energy and dual-energy product spaces. The reference inverse and a Neumann series prove the first assertion and eq. 3.dccclxviii. Every entry is holomorphic, so its inverse is holomorphic there as well. Complex orthogonality and a parametrization of the moving field slice were not used.

Set $\rho = 0$. The previous lemma identifies its kernel with the actual $X_L(A)$. Testing the first equation there eliminates ν, proving eq. 3.dccclxix. Conversely, any solution of that restricted equation extends to eq. 3.dccclxvii by a unique gauge multiplier. Indeed $R_0 R_A D_A^\sharp : X \to Z_0$ is onto, and the annihilator of its kernel is its transpose range. The transpose is injective by the same right-inverse bound. Full saddle uniqueness therefore proves restricted uniqueness.

For a functional given only on $X_L(A)$, take a norm-preserving Hahn–Banach extension to X. The full bound gives eq. 3.dccclxx. Another extension changes only the eliminated gauge multiplier, so the field and physical row multiplier are independent of that choice. Holomorphic dependence is asserted for fixed-space source extensions, and requires no holomorphic choice of Hahn–Banach extensions. □

On any smaller fixed tube, Cauchy's inequality gives uniform operator jets in the displayed physical energy norms, with factorial bounds in the tube norm and in γ_ω. The result permits complex backgrounds and the actual moving Landau constraint at all assigned scales on proper determining regions. Strong outputs and the regular-pair nonlinear response are proved in Props. 3.92, 3.171 and 3.174. The C^1 background tube itself does not bound the physical Wilson current; the current is independent in these constructions.

3.24. **Uniform discrete singular-integral estimates.** The strong proper-region estimates require finite-p bounds for second derivatives of a scalar inverse. Supremum-norm bounds for these order-zero operators would be false. We extend Lem. 3.42 to arbitrary masses and periods, and derive the local heat embeddings needed for the response bootstrap. The singular-integral proof itself was given in that lemma, including the physical measure at the last atomic scale; only its new uniform parameters are checked here.

Let $\Gamma_h = h\mathbb{Z}^4$, $0 < h \le 1$, with measure h^4 times counting measure. Put

$$D_i^+ u(x) = h^{-1}(u(x + he_i) - u(x)), \qquad D_i^- u(x) = h^{-1}(u(x) - u(x - he_i)), \qquad L_h = -\sum_i D_i^- D_i^+.$$

Either sign of a difference will be allowed below. The notation $\|D^2 u\|_p$ includes all ordered component second differences. Replacing a forward by a backward difference only shifts its location and

changes fixed constants. All convolutions use physical measure:

$$(k * u)(x) = h^4 \sum_y k(x-y)u(y).$$

3.24.1. *The scalar kernels.* Let p_t^h be the physical heat density of e^{-tL_h}. The differentiated heat estimates already proved in eq. 3.dclxxxv imply, for every derivative order $a \geq 0$ and every integer N,

$$|D^a p_t^h(x)| \leq C_{a,N}(t+h^2)^{-(4+a)/2}\left(1+\frac{|x|}{\sqrt{t+h^2}}\right)^{-N}. \tag{3.dccclxxii}$$

The quoted local-time estimate extends to $t \geq 1$ on the infinite lattice by the exact scaling

$$p_t^h(x) = t^{-2} p_1^{h/\sqrt{t}}(x/\sqrt{t}).$$

Each derivative contributes a further $t^{-1/2}$. The rescaled mesh is still at most one, so the same pointwise estimate holds at every time. They also give

$$\|D^a p_t^h\|_{r,h} \leq C_{a,r}(t+h^2)^{-a/2-2(1-1/r)}, \qquad 1 \leq r \leq \infty. \tag{3.dccclxxiii}$$

For finite r, sum eq. 3.dccclxxii with $Nr > 4$; the lattice sum of the spatial factor is at most $C(t+h^2)^2$. The case $r = \infty$ is the pointwise bound. Thus these estimates include $t < h^2$.

For $\mu \geq 0$, define the second-resolvent kernel

$$k_{ij}^{h,\mu}(x) = \int_0^\infty e^{-\mu^2 t} D_i^\sigma D_j^\tau p_t^h(x)\, dt, \qquad \sigma,\tau \in \{+,-\}. \tag{3.dccclxxiv}$$

This integral converges at every lattice point, including zero: the mesh is fixed at the lower limit, and the large-time derivative bound is Ct^{-3}. For compactly supported input, its convolution is the Fourier multiplier

$$\frac{\mathrm{d}_i^\sigma(\theta)\mathrm{d}_j^\tau(\theta)}{\lambda_h(\theta)+\mu^2}, \qquad \lambda_h(\theta) = 4h^{-2}\sum_i \sin^2(h\theta_i/2), \quad \theta \in [-\pi/h, \pi/h]^4.$$

At $\mu = 0, \theta = 0$ the value of the multiplier can be assigned arbitrarily on the infinite lattice. Since $|\mathrm{d}_i^\sigma \mathrm{d}_j^\tau| \leq \lambda_h$, its L^2 operator norm is at most one.

Lemma 3.78. Uniformly in h, μ, these kernels satisfy

$$|k_{ij}^{h,\mu}(x)| \leq C(|x|+h)^{-4}, \qquad |D_\ell^\pm k_{ij}^{h,\mu}(x)| \leq C(|x|+h)^{-5}. \tag{3.dccclxxv}$$

In particular, for a lattice displacement y,

$$h^4 \sum_{|x| \geq C_0(|y|+h)} |k_{ij}^{h,\mu}(x-y) - k_{ij}^{h,\mu}(x)| \leq C \tag{3.dccclxxvi}$$

with a fixed sufficiently large C_0. ■

Proof. The mass factor can be omitted for these upper bounds. For $|x| = r \geq h$, integrate eq. 3.dccclxxii with $a = 2$, using $v = t + h^2$. For $v \leq r^2$, its bound is $Cr^{-N}v^{N/2-3}$, which has integral Cr^{-4} if $N > 4$. For $v \geq r^2$, the integral of Cv^{-3} has the same bound. At $r \leq h$ the full integral is Ch^{-4}. Using $a = 3, N > 5$ proves the derivative bound.

Join 0 to y by coordinate steps. On the region in eq. 3.dccclxxvi, all intermediate arguments are comparable in distance to x. Telescoping the derivative bound gives

$$|k(x-y) - k(x)| \leq C|y|\,|x|^{-5}.$$

On the dyadic annulus with radius $2^q(|y|+h)$, the physical number of lattice points is at most $C[2^q(|y|+h)]^4$. Summing the resulting $C|y|/[2^q(|y|+h)]$ proves the assertion. For $y = 0$ the difference is zero. □

3.24.2. *Weak type and finite-p bounds.*

Proposition 3.79 (Uniform finite-p scalar estimate)**.** For every $1 < p < \infty$,

$$\|D_i^\sigma D_j^\tau (L_h + \mu^2)^{-1} f\|_{p,h} \le C_p \|f\|_{p,h}, \qquad \mu \ge 0. \tag{3.dccclxxvii}$$

At zero mass on a finite periodic lattice the inverse is the inverse on constants' orthogonal complement; the second derivative annihilates its constant mode. The same constant is valid on every compatible periodic torus, independently of its size. ■

Proof. Write T for the infinite-lattice multiplier above. The proof of eq. 3.dclxxxix uses only the L^2 bound and its displayed kernel cancellation condition. Both have just been established for T, with constants independent of h, μ. On the infinite lattice take dyadic cubes of sides $2^n h$, $n \ge 0$. For compactly supported input their averages eventually tend to zero, so the same maximal-cube decomposition exists at every positive threshold. A parent has sixteen children of equal physical volume, which improves its average bound to 16λ. The bad part of a single-site cube is zero. Its good-part estimate and the cancellation sum outside enlarged cubes are therefore exactly the two estimates in that proof. They give weak-$(1,1)$ with the same uniform parameters. The distribution-function interpolation and adjoint duality proved there apply unchanged, giving every $1 < p < \infty$. Density defines these operators on L^p. This step reuses the proof rather than assuming a continuous singular-integral theorem for the lattice.

For $\mu > 0$, the kernel is absolutely summable at every fixed mesh and mass: by eq. 3.dccclxxiii,

$$\|k^{h,\mu}\|_{1,h} \le C \int_0^\infty e^{-\mu^2 t} (t + h^2)^{-1}\, dt < \infty.$$

Periodize that kernel. To transfer its L^p bound, truncate a periodic input to an increasing union of fundamental cells. Absolute summability makes the truncated convolution converge to the periodic one on cells a large fixed distance from the truncation boundary. Apply the infinite-lattice estimate, divide its p-th power by the number of cells, and let the truncation grow. The proportion of boundary cells tends to zero; a further limit of the fixed distance removes the summable tail. This proves the same periodic bound. Finally, on any fixed finite torus let $\mu \downarrow 0$. Every nonconstant Fourier mode converges to the massless second-resolvent multiplier and the constant second-derivative mode is zero. The uniform bound passes to this finite matrix limit. □

No assertion at $p = \infty$ follows from this proposition. The proof uses cancellation of the bad parts, precisely where an absolute supremum-norm kernel estimate would fail.

3.24.3. *Interior second derivatives and the flat Hodge estimate.* Let $Q_r \subset Q_R$ be concentric physical fine boxes, and assume that their distance is large enough to contain the fixed two-step stencils. Choose a sampled smooth cutoff $\chi = 1$ on Q_r, supported strictly inside Q_R, with $|D\chi| \le C(R-r)^{-1}$ and $|D^2\chi| \le C(R-r)^{-2}$. Extend χu by zero to the full lattice. The product rule and Prop. 3.79 give the following precise local estimate.

Corollary 3.80. For $1 < p < \infty$,

$$\begin{aligned}\|D^2 u\|_{p,Q_r} \le C_p \{ &\|L_h u\|_{p,Q_R} + (R-r)^{-1} \|Du\|_{p,Q_R} \\ &+ (R-r)^{-2} \|u\|_{p,Q_R} \}.\end{aligned} \tag{3.dccclxxviii}$$

The constants are independent of the fine mesh. ■□

Proof. For a compactly supported v, $D_i^\sigma D_j^\tau v = D_i^\sigma D_j^\tau L_h^{-1} (L_h v)$ as Fourier multipliers. Thus $\|D^2 v\|_p \le C_p \|L_h v\|_p$. Apply this to $v = \chi u$. The exact product rule for $L_h(\chi u)$ has the term $\chi L_h u$, terms containing one difference of χ and one of u, and terms containing two differences of χ and a shifted value of u. Their indicated supports lie in Q_R. This proves the estimate. □

For a compactly supported one-cochain v, the flat operator identity

$$\mathrm{d}^\sharp \mathrm{d} + DD^\sharp = L_h I$$

and Prop. 3.79 imply

$$\|Dv\|_p \le C_p(\|\mathrm{d}v\|_p + \|D^\sharp v\|_p). \tag{3.dccclxxix}$$

To see the operator types, write $v = L_h^{-1}(\mathrm{d}^\sharp \mathrm{d}v + DD^\sharp v)$ and differentiate. Each resulting coefficient is a second-resolvent multiplier of Prop. 3.79; there are only finitely many component pairs. Applying this to χv proves

$$\|Dv\|_{p,Q_r} \le C_p\{\|\mathrm{d}v\|_{p,Q_R} + \|D^\sharp v\|_{p,Q_R} + (R-r)^{-1}\|v\|_{p,Q_R}\}. \tag{3.dccclxxx}$$

These are full-lattice cutoff identities. They impose no boundary condition on an arbitrary proper determining region.

3.24.4. *Finite-time heat Sobolev and Morrey estimates.* We state the strict-exponent estimates needed for bootstrap, so no critical endpoint is used implicitly. For compactly supported u put $g = L_h u$. The exact finite-time identity is

$$u = P_1 u + \int_0^1 P_t g \, dt, \qquad P_t = e^{-tL_h}.$$

Young's inequality and eq. 3.dccclxxiii give the following consequences.

Proposition 3.81. Let $1 < p \le q \le \infty$. If $2(1/p - 1/q) < 1$, then

$$\|u\|_q \le C_{p,q}(\|u\|_p + \|L_h u\|_p). \tag{3.dccclxxxi}$$

If $1/2 + 2(1/p - 1/q) < 1$, then

$$\|Du\|_q \le C_{p,q}(\|u\|_p + \|L_h u\|_p). \tag{3.dccclxxxii}$$

The constants are uniform in h. The corresponding interior estimates follow by a cutoff, with the commutator terms of eq. 3.dccclxxviii. ■

Proof. The fixed-time operators P_1, DP_1 map L^p to L^q with a uniform bound. The integrals of their variable-time bounds are respectively

$$\int_0^1 (t+h^2)^{-2(1/p-1/q)}\, dt, \qquad \int_0^1 (t+h^2)^{-1/2-2(1/p-1/q)}\, dt.$$

The stated strict inequalities make both integrals bounded independently of h. The cutoff product rule proves the local version. □

Proposition 3.82 (Uniform gradient Morrey estimate)**.** If $4 < p < \infty$, then

$$\|u\|_\infty + \|Du\|_\infty + [Du]_{1-4/p,\mathrm{loc}} \le C_p(\|u\|_p + \|L_h u\|_p). \tag{3.dccclxxxiii}$$

The seminorm uses distinct fine lattice points at physical distance at most one. This estimate also has the cutoff interior version. ■

Proof. The two supremum bounds follow from Prop. 3.81. For a fine lattice translation z, the derivative kernel obeys

$$\|\tau_z Dp_t^h - Dp_t^h\|_{p'} \le C_p \min\{(t+h^2)^{-1/2-2/p}, |z|(t+h^2)^{-1-2/p}\}.$$

The first bound follows from the triangle inequality and eq. 3.dccclxxiii; the second telescopes one further difference along a coordinate path. Distinct grid points have $|z| \ge h$. Split its time integral at $t = |z|^2$. The small-time contribution is at most $C_p|z|^{1-4/p}$, since $1/2 + 2/p < 1$. The large-time contribution has the same bound by direct integration. The fixed-time term $DP_1 u$ has increment at most $C_p|z|\|u\|_p$, which is bounded by $C_p|z|^{1-4/p}\|u\|_p$ for $|z| \le 1$. This proves the seminorm. Cutoffs again use only the displayed first and second difference commutators. □

For example, a bounded physical H^1 solution with a locally L^2 Laplacian first has gradient in L^3 and itself in every fixed finite L^q. A subsequent L^3 forcing bound gives a bounded solution and gradient in L^8. An L^8 forcing bound then gives bounded gradient. Once the equation supplies an L^p Laplacian bound at any chosen finite $p > 4/(1-\alpha)$, the same estimate gives a gradient Hölder exponent at least α. All these implications use strict inequalities in Prop. 3.81; none uses an endpoint singular-integral estimate.

3.24.5. *All finite-p derivatives in an admitted Coulomb gauge.* We finish with the coefficient estimate needed in the vector bootstrap. Suppose the real connection on a fine box has an admitted logarithmic Coulomb representative $U = e^{ihA}$ with

$$\|A\|_\infty \le C\kappa/R, \qquad D^\sharp A = 0, \qquad \|F_U\|_\infty \le \kappa/R^2,$$

on a box of size comparable to R, with the fixed larger collars of the current-free local gauge construction. The existence of that representative was proved in § 3.16; its initial construction does not use a current bound.

Corollary 3.83. For every finite $1 < p < \infty$, on a fixed smaller box,

$$\|DA\|_p \le C_p \kappa R^{4/p-2}. \tag{3.dccclxxxiv}$$

There is no derivative of curvature or physical current in this assertion. ■□

Proof. The exact four-link logarithm gives

$$F_U = \mathrm{d}A + E_h(A), \qquad \|E_h(A)\|_\infty \le C\|A\|_\infty^2.$$

Every nonlinear term has at least two factors hA before the factor h^{-2} is removed. This bound therefore requires no derivative of A. Insert this curl identity and $D^\sharp A = 0$ into eq. 3.dccclxxx, using an intermediate cutoff box. Its three terms are bounded by

$$C_p \left\{ \frac{\kappa}{R^2} R^{4/p} + \frac{\kappa^2}{R^2} R^{4/p} + R^{-1} \frac{\kappa}{R} R^{4/p} \right\}.$$

The fixed small curvature ceiling bounds κ^2 by a constant times κ. This proves the claim. In a complex C^1 tube about this real representative, the added potential and its first derivative have the same finite-p scale bounds by their stated supremum norms. This last observation changes coefficients only; it does not assert unitarity of the complex connection. □

3.25. **Strong responses on actual determining regions.** The proper-region energy inverse controls a local physical potential and its gradient. We upgrade it to pointwise and Hölder control for bounded forces. The proof uses finite L^p estimates in current-free Coulomb frames. Curvature is never differentiated. The actual Wilson stencil, the smoothed site projection and the nonlinear constraint Hessian enter the argument in different ways.

Keep the actual determining regions, outward rows and physical pairings of Prop. 3.70. Increase their fixed separation and interior collar constants once as specified below. All constants may depend on the fixed L, and Hölder constants may depend on their exponent. They are independent of depth, mesh, compatible volume and the shapes of the separated regions. The frozen geometry and positive row weights are used throughout.

3.25.1. *The forces and the local strong norm.* Let B_y be an assigned pillar of side ξ_y. At a nonbare pillar choose concentric interior boxes of radii fixed multiples of ξ_y. The largest multiple R_* is fixed after the actual carrier constant C_L, with enough room for four successive cutoffs and four complete carrier enlargements. Take ρ larger than their fixed geometric margin. These boxes meet at most two adjacent assigned levels; all their scales are comparable within a fixed L factor. Their fixed enlargements meet only a bounded number of assigned pillars. A nonbare such box lies inside the inner-plaquette region. The bare case has $\xi_y = \eta$ and will be estimated directly.

For finite $4 < p < \infty$ define

$$\mathcal{F}_p(f) = \sup_y \xi_y^{3-4/p} \|f\|_{p,B_y}, \qquad \mathcal{F}_\infty(f) = \sup_b \mathsf{L}(b)^3 |f(b)|. \tag{3.dccclxxxv}$$

Replacing B_y by any of the fixed boxes changes this norm by a constant. In particular $\mathcal{F}_p(f) \le C_p \mathcal{F}_\infty(f)$. The cell energy amplitude is

$$\mathcal{E}_{\text{cell}}(v) = \sup_y \{\xi_y^{-1} \|v\|_{2,B_y} + \|\nabla_U v\|_{2,B_y^+}\}. \tag{3.dccclxxxvi}$$

One fine-step collars are included in B_y^+.

The local gradient Hölder seminorm below is covariant: two values based at x, x' are compared after adjoint transport along a fixed coordinate-order path inside the same interior box. Denote it by $[\nabla_U v]^U_{\alpha,y}$. At bare cells it is the corresponding finite one-step collar seminorm on present stencils, with the zero-extension convention of X. No regularity across a rough exterior boundary or between disconnected components is asserted. Set

$$\|v\|_{\text{str},\alpha} = \sup_y \{\xi_y \|v\|_{\infty,B_y} + \xi_y^2 \|\nabla_U v\|_{\infty,B_y} + \xi_y^{2+\alpha} [\nabla_U v]^U_{\alpha,y}\}. \tag{3.dccclxxxvii}$$

This definition uses actual short transports and is unchanged by real site gauges. For complex connections in the tube considered below, the short transports and their inverses have uniformly bounded norms.

3.25.2. *Scalar finite-p regularity without a current bound.* On a nonbare box use the current-free local Coulomb construction. In its frame, with $r = \xi_y$,

$$U_\mu = e^{i\eta a_\mu}, \qquad \|a\|_\infty \le C\kappa r^{-1}, \qquad \|Da\|_q \le C_q \kappa r^{4/q-2} \quad (1 < q < \infty). \tag{3.dccclxxxviii}$$

These are the curvature-only conclusions of Cor. 3.83, not the additional current-dependent L^∞ derivative estimate. The largest box still has small scaled curvature after one fixed reduction of the curvature ceiling.

For site fields the exact covariant Laplacian is

$$(\Delta_U u)(x) = \eta^{-2} \sum_\mu \{2u(x) - \operatorname{Ad} U_\mu(x) u(x+\eta e_\mu) - \operatorname{Ad} U_\mu(x-\eta e_\mu)^{-1} u(x-\eta e_\mu)\}. \tag{3.dccclxxxix}$$

Rescale the box by r, putting $h = \eta/r$. In this frame it can be written exactly as

$$\Delta_U = L_h + \sum_\mu a_\mu^{(1)} D_\mu + b^{(0)}, \qquad \|a^{(1)}\|_\infty \le C\kappa, \qquad \|b^{(0)}\|_q \le C_q(\kappa + \kappa^2). \tag{3.dcccxc}$$

Forward and backward shifted terms may both occur in this notation. To verify it, put $E_+(x) = (\operatorname{Ad} e^{iha(x)} - 1)/h$ and define E_- with the inverse exponential. The difference from L_h is a first difference of u with coefficients $E_\pm = O(a)$, and a zeroth coefficient $-[E_+(x) + E_-(x - he_\mu)]/h$. Since $E_- = -E_+ + O(ha^2)$, that coefficient is bounded by first differences of a and Ca^2. This proves eq. 3.dcccxc with no second derivative of the connection.

Lemma 3.84 (Two local bootstraps)**.** In the rescaled fixed boxes the following estimates are uniform in h.

If $\Delta_U u = g$, $g \in L^p$, $4 < p < \infty$, and the outer H^1 norm of u is at most B, then, for $0 < \alpha < 1 - 4/p$,

$$\|u\|_\infty + \|Du\|_\infty + [Du]_\alpha \le C_{p,\alpha}(B + \|g\|_p). \tag{3.dcccxci}$$

The same conclusion holds if additional zeroth-order operators with complete carrier radius C_L have bounded local L^q-operator norms for every finite q, using the successive carrier enlargements.

If $\Delta_U u = D_U^\sharp F + g_0$, the same outer energy is at most B, $F \in L^p$, and g_0 is bounded by B in all the finite norms used, then

$$\|u\|_p + \|Du\|_p \le C_p(B + \|F\|_p). \tag{3.dcccxcii}$$

This is a finite-p divergence-source estimate. It includes no $p = \infty$ derivative assertion. ∎

Proof. All cutoffs have fixed physical derivative bounds. The energy heat embedding first gives $u \in L^3$. For example insert $u = P_1 u + \int_0^1 P_t D^\sharp Du\, dt$; the derivative heat bound is integrable from L^2 to L^3. Thus $b^{(0)}u \in L^2$, using $b^{(0)} \in L^6$. The other lower term is in L^2. The interior estimate Cor. 3.80 and the strict heat embeddings Prop. 3.81 give $D^2u \in L^2$, $Du \in L^3$, and $u \in L^{12}$ on a smaller box. Now $b^{(0)}u \in L^3$, using its L^4 bound. A second cutoff gives $D^2u \in L^3$, then $u \in L^\infty$ and $Du \in L^q$ for every $q < 12$. If $4 < p \le 8$, take $q = p$. If $p > 8$, first take $q = 8$ and use another cutoff to obtain $D^2u \in L^8$ and $Du \in L^\infty$. The final L^p equation and Prop. 3.82 prove eq. 3.dcccxci. The extra zeroth-order operator reads the already controlled outer L^q field at each step. The reserved complete carrier collars give exactly these larger input boxes. All Sobolev gains just used are strict.

For the divergence source, apply $D(L_h + 1)^{-1}$ to the cutoff equation. Its divergence term is $D(L_h + 1)^{-1}D^\sharp(\chi F)$, bounded on each finite L^q by Prop. 3.79. Its other terms have the order-minus-one heat bound. Starting with the outer energy, these terms are in L^2; the derivative output is in L^3, since $1/2 + 2(1/2 - 1/3) < 1$. The field output is also in L^{12}. The next lower terms are in L^3, giving derivatives in every $q < 12$, in particular $q = \min\{8, p\}$. The field becomes bounded. If $p > 8$, the order-minus-one terms from L^8 are bounded, while the divergence term is estimated in the requested finite L^p. This proves eq. 3.dcccxcii. Covariant $D_U^\sharp F$ differs from the flat divergence by the bounded product aF. Cutoff commutators add only F, Du, u on the outer boxes. No difference of F is estimated in an L^p norm. □

First-order Morrey, obtained from the same order-minus-one translation kernels, gives $\|a\|_{C^\beta} \le C_q(\|a\|_q + \|Da\|_q)$ for $q > 4$, $\beta = 1 - 4/q$. Thus the ordinary derivative estimate in eq. 3.dcccxci also controls the covariant derivative and its transported Hölder seminorm. The short transport differs from the identity by $C\kappa|x - x'|$ in these rescaled boxes.

3.25.3. *Strong columns of the actual site smoothing.* Let $F = G_s q^\sharp \Xi^{-2}$, $M = F^\sharp F$, $P = 1 - R = FM^{-1}F^\sharp$, with all the actual site definitions. A column input supported at top z is measured in its physical top Hilbert norm. Its unit vector has value $\xi_z^{-2}e$, not value e.

Proposition 3.85 (Strong normalized site columns)**.** For every $0 < \alpha < 1$, a fixed $m > 0$ can be chosen such that

$$\begin{aligned}\xi_y^2\|F1_z b\|_{\infty,B_y} + \xi_y^3\|D_U F1_z b\|_{\infty,B_y} \\ + \xi_y^{3+\alpha}[D_U F1_z b]^U_{\alpha,y} \le C_\alpha e^{-md(y,z)}\|1_z b\|_{\mathrm{top}}.\end{aligned} \tag{3.dcccxciii}$$

No current premise or differentiated curvature is used. ■

Proof. The weighted balanced site form and normalized Gram estimates give the column energy bounds

$$\|u\|_{2,\mathcal{D}_y} + \xi_y\|D_U u\|_{2,\mathcal{D}_y} \le Ce^{-m_0 d(y,z)}\|1_z b\|_{\mathrm{top}}, \qquad u = F1_z b,$$

on each fixed enlarged box. Its equation is

$$\Delta_U u = q^\sharp \Xi^{-2} 1_z b - q^\sharp a_s q u.$$

The actual adjoint $q^\sharp$ extends a top colour as its transported density on its own pillar. Its pointwise coefficient has norm one for real fields; there is no fine factor η^{-4}. Also $|qu(t)| \le C\xi_t^{-2}\|u\|_{2,B_t}$. Consequently the right side on $\mathcal{D}_y$ is bounded by $C\xi_y^{-4}e^{-m_0 d(y,z)}\|1_z b\|_{\mathrm{top}}$. The direct source vanishes unless the box meets the source pillar, where its normalized amplitude has this same bound. Scale u as $u' = \xi_y^2 u$. Its outer H^1 norm and bounded forcing have the same exponential bound. Apply Lem. 3.84 with a finite $p > 4/(1 - \alpha)$ and scale back.

At a bare cell the physical atomic inequality $|u(x)| \le \eta^{-2}\|u\|_{2,B_x}$ and its one-step neighbours prove the result directly, including the finite collar seminorm. This gives eq. 3.dcccxciii, after one fixed decrease of the exponential rate. □

Lemma 3.86 (The smoothed gauge term is a pointwise force)**.** If $\mathcal{E}_{\mathrm{cell}}(v) \le B$, then

$$\|D_U P D_U^\sharp v\|_{\infty,B_y} + \xi_y^\alpha [D_U P D_U^\sharp v]^U_{\alpha,y} \le C_\alpha \xi_y^{-3} B. \tag{3.dcccxciv}$$

The actual field row mass also satisfies

$$\|Q^\sharp a_f Q v\|_{\infty,B_y} \le C\xi_y^{-3} B, \qquad \|Q^\sharp \mu\|_{\infty,B_y} \le C\xi_y^{-3} \sup_c \xi_c^3 |\mu(c)|. \tag{3.dcccxcv}$$

■

Proof. Use physical orthonormal top columns F_z. Their derivative L^2 norm on B_y is at most $C\xi_y^{-1} e^{-md(y,z)}$. Thus

$$|(D_U F)^\sharp v(z)|_{\mathrm{ON}} \le \sum_y C\xi_y^{-1} e^{-md(y,z)} \|v\|_{2,B_y} \le CB.$$

The normalized Gram inverse has exponential orthonormal entries and bounded row sums. Its output coefficients are therefore $O(B)$. Sum the pointwise and Hölder derivative columns eq. 3.dcccxciii to obtain eq. 3.dcccxciv. The metric growth rate is chosen below half the retained exponential rate.

For an actual field row, $|Qv(c)| \le C\xi_c^{-2}\|v\|_{2,\mathrm{car}\, c} \le C\xi_c^{-1} B$. Its coefficient at a fixed fine bond is at most $C(\eta/\xi_c)^4$. The physical adjoint multiplies this by $\eta^{-4}\xi_c^4$; these factors cancel. Only a bounded number of rows with comparable scales meet that bond. Since $a_f(c) = O(\xi_c^{-2})$, both assertions in eq. 3.dcccxcv follow. This argument uses the full outward carrier, not an individual endpoint value. □

3.25.4. *The exact Wilson stencil in Feynman form.*

Lemma 3.87 (A bounded curvature kernel)**.** On every interior full stencil,

$$D^2 A_{\Omega_0}(U) + D_U D_U^\sharp = \Delta_U^{\mathrm{comp}} + \mathcal{Z}_U, \tag{3.dcccxcvi}$$

where Δ_U^{comp} is the site covariant Laplacian acting on each vector component. The operator $\mathcal{Z}_U$ reads at most a fixed two-fine-step collar and has row and column coefficient sums at most $C\kappa\xi_y^{-2}$ on a scale-ξ_y box. In particular it is a local bounded L^p zeroth-order operator, uniformly for $1 \le p \le \infty$. ■

Proof. Use the exact plaquette insertion eq. 3.dlxiii. After division by the physical fine measure, its remainder is a matrix on four plaquette edges with each coefficient bounded by $C|P_p - 1|\eta^{-2}$. Hence this part is a bounded finite-shift kernel of size $C|F_p|$.

Let $\mathfrak{d}_U = \mathrm{d}_U + E$, with the actual curl convention in eq. 3.dlxvi. Each coefficient of E is bounded by $C|P_p - 1|/\eta \le C\eta|F_p|$; each coefficient of d_U is $O(\eta^{-1})$. Expand

$$\mathfrak{d}_U^\sharp \mathfrak{d}_U - \mathrm{d}_U^\sharp \mathrm{d}_U = \mathrm{d}_U^\sharp E + E^\sharp \mathrm{d}_U + E^\sharp E.$$

Multiply these stencil coefficients directly. The first two products have coefficients $O(|F|)$ at a bounded number of shifted sites; the last is $O(\eta^2|F|^2)$. Taking a finite difference of F is unnecessary. This also proves the absolute column bounds, since each product has a bounded number of incoming as well as outgoing stencil entries.

Finally set $T_\mu = \mathrm{Ad}\, U_\mu\, \tau_\mu$. The off-diagonal entries of the exact covariant curl-divergence sum are

$$\eta^{-2}(T_\mu T_\nu^{-1} - T_\nu^{-1} T_\mu).$$

The two paths differ by an elementary plaquette. Their difference is a two-step shift times a conjugate of $\mathrm{Ad}\, P - 1$, with bounded transport coefficients. It has size $C|F|$. The diagonal entries equal $\sum_\nu \eta^{-2}(2 - T_\nu - T_\nu^{-1})$. Combining these three finite-stencil calculations proves eq. 3.dcccxcvi. All identities hold bilinearly on the complex continuation as well. □

The actual constraint Hessian $B_\omega = \Delta^{(2)}[\omega]$ has a different carrier, of radius $C_L\xi_y$. The two-leg moment eq. 3.dcii and its symmetric counterpart give

$$\|B_\omega v\|_{p,\mathcal{D}_y} \le C_L\gamma_\omega\xi_y^{-2}\|v\|_{p,\mathcal{D}_y^{+C_L}}, \qquad 1 \le p \le \infty. \tag{3.dcccxcvii}$$

Indeed its physical fine operator is $\eta^{-4}\sum_c \xi_c^4\omega(c)D^2q_c$. For a fixed fine leg the absolute other-leg sum is at most $C\gamma_\omega\xi_y^{-2}$; symmetry gives the column bound. Weighted Schur or Jensen gives eq. 3.dcccxcvii. Scales on a complete carrier are comparable. This proves an L^p-to-L^p estimate, not an energy-to-pointwise assertion.

3.25.5. *The penalized and full row-constrained strong inverses.* Write the actual displayed precision as

$$K_{U,\omega} = D^2A_{\Omega_0}(U) - B_\omega + D_URD_U^\sharp + Q^\sharp a_f Q. \tag{3.dcccxcviii}$$

The multiplier is small in $\gamma_\omega = \sup_c \xi_c^3|\omega(c)|$. This is the local constraint Hessian, with no independently named current completion.

Proposition 3.88 (The proper strong inverse)**.** For the real backgrounds of Prop. 3.70, and finite $4 < p < \infty$, $0 < \alpha < 1 - 4/p$,

$$\|K_{U,\omega}^{-1}f\|_{\mathrm{str},\alpha} \le C_{p,\alpha}\mathcal{F}_p(f). \tag{3.dcccxcix}$$

In particular bounded physical forces give this estimate for every $\alpha < 1$. For the full row saddle

$$K_{U,\omega}v - Q^\sharp\mu = f, \qquad Qv = b,$$

one has

$$\|v\|_{\mathrm{str},\alpha} + \sup_c \xi_c^3|\mu(c)| \le C_{p,\alpha}\{\mathcal{F}_p(f) + \sup_c \xi_c|b(c)|\}. \tag{3.cm}$$

Removing the row mass from the saddle adjusts its multiplier by $-a_f b$ and preserves this bound. ■

Proof. The exponential energy inverse gives $\mathcal{E}_{\mathrm{cell}}(v) \le C\mathcal{F}_p(f)$; on a pillar the dual forcing norm is $\xi_y\|f\|_2 \le C\mathcal{F}_p(f)$. Use $R = 1 - P$ and eq. 3.dcccxcvi to write its interior equation as

$$\Delta_U^{\mathrm{comp}}v = f + D_UPD_U^\sharp v - Q^\sharp a_f Qv + B_\omega v - \mathcal{Z}_U v. \tag{3.cmi}$$

The first two nonlocal smoothing and row terms have bounded pointwise amplitude $C\xi_y^{-3}\mathcal{E}_{\mathrm{cell}}(v)$. The remaining two operators have the local L^q bounds just proved. Set $v' = \xi_y v$, $f' = \xi_y^3 f$. Its outer H^1 bound is the cell energy amplitude and its scalar component equation is precisely the one in Lem. 3.84. The four nested boxes retain every constraint carrier at each step. That lemma proves the required local bound. Ordinary and covariant derivative seminorms are equivalent by eq. 3.dccclxxxviii and first-order Morrey. Bare cells follow directly from their physical atomic energy bounds; only a fixed number of neighbouring fine values is read there.

For the row saddle, the normalized Schur construction of Prop. 3.72 applies also to this precision: its proof uses its energy inverse, exact row lift and weighted coefficients, which eq. 3.dcccxcviii has. The array kernel of its unnormalized inverse is bounded by $C\xi_c^{-3}\xi_d e^{-md(c,d)}$. Thus its action on b has ξ^3 amplitude at most $C\sup_d \xi_d|b(d)|$. The other multiplier contribution is the transpose minimizing lift applied to f. Its energy column proof eq. 3.dcccxlviii requires only $\sup_y \xi_y\|f\|_{2,B_y} \le C\mathcal{F}_p(f)$; it gives the same pointwise multiplier bound. The row-lift columns and energy inverse give the corresponding cell energy estimate for v. Finally eq. 3.dcccxcv bounds $Q^\sharp\mu$ as an additional point force. Apply the same local bootstrap to obtain eq. 3.cm. □

3.25.6. *Finite-p restoration of a force.* The adjoint restoration is not bounded on fine L^∞ in general. The finite-p substitute is sufficient.

Proposition 3.89. For every finite $4 < p < \infty$,

$$\mathcal{F}_p(S^\sharp f) \le C_p \mathcal{F}_p(f). \tag{3.cmii}$$

In particular a bounded physical force has a restored force with the correct local amplitude in every finite L^p. ■

Proof. Put $u_s = G_s D_U^\sharp f$. The weighted site energy form gives the four cell bounds for G_s, $D_U G_s$, $G_s D_U^\sharp$ and $D_U G_s D_U^\sharp$, with scale factors $\xi_y\xi_z, \xi_z, \xi_y, 1$, respectively, times $Ce^{-md(y,z)}$. This is the same weighted form and distance argument as for the field energy inverse. The site gap supplies its mass term. Choose the fixed separation so the metric growth and one scale-ratio cost ξ_y/ξ_z together are less than half the retained decay rate. Since $\|f\|_{2,B_z} \le C\xi_z^{-1}\mathcal{F}_p(f)$, summing these columns gives

$$\|u_s\|_{2,\mathcal{D}_y} + \xi_y \|D_U u_s\|_{2,\mathcal{D}_y} \le C\mathcal{F}_p(f).$$

Its local equation is $\Delta_U u_s = D_U^\sharp f - q^\sharp a_s q u_s$. The average term is a bounded point forcing of amplitude $C\xi_y^{-4}\mathcal{F}_p(f)$. Rescale $u_s' = \xi_y^2 u_s$, $f' = \xi_y^3 f$, and apply the divergence-source estimate eq. 3.dcccxcii. It follows that

$$\xi_y^{3-4/p} \|D_U u_s\|_{p,\mathcal{D}_y} \le C_p \mathcal{F}_p(f).$$

Exactly $S^\sharp f = f - D_U R u_s$. Write $Ru_s = u_s - FM^{-1}F^\sharp u_s$. The just proved L^2 cell amplitude of u_s, the exponential F columns and the normalized Gram inverse bound its last derivative pointwise by $C\xi_y^{-3}\mathcal{F}_p(f)$. The displayed finite-$p$ derivative bound controls the other term. This proves eq. 3.cmii. The proof used the divergence-source Riesz estimate only at finite exponents. □

3.25.7. *Physical critical pairs and the actual Landau saddle.* Let $H_L = D^2 A_{\Omega_0}(U) - B_\omega$. At a physical critical pair full stationarity is $j_U = Q^\sharp \omega$. Let $\mathfrak{q}_U(a)$ be the actual nonlinear field-mean coordinate, with $D\mathfrak{q}_U(0) = Q$. Its actual gauge cocycle gives

$$D\mathfrak{q}_U(a)\Gamma_a\lambda = \mathrm{d}(a)\tau_a\lambda, \qquad \tau_0 = q, \quad \mathrm{d}(0) = \bar{D}_{\mathrm{ext}}.$$

The extended outward endpoint values are part of this identity. Fine gauge invariance and onto q give $\bar{D}_{\mathrm{ext}}^\sharp \omega = 0$. For $q\lambda = 0$, differentiate the displayed identity in an arbitrary field direction. Its $\mathrm{d}'(a)\tau_0\lambda$ term vanishes, and its $\mathrm{d}(0)\tau_0'\lambda$ term pairs to zero with ω. Differentiating the action Ward identity produces the same gauge-generator acceleration; full stationarity cancels it against the constraint term. Therefore

$$H_L D_U \lambda = 0 \qquad (q\lambda = 0). \tag{3.cmiii}$$

This proves the full bilinear null identity, not only its restriction to the constraint kernel.

Since $qG_sR = 0$, it follows that $K_{U,\omega} D_U G_s R = D_U R$. Its inverse and row lift have the physical horizontality identities used in § 3.22. For a force represented by f on the actual Landau slice, use the extension $S^\sharp f$; it agrees there with f. The full row saddle with this source is horizontal, by the Ward identity and $SD_U G_s R = 0$. It is consequently the actual physical Landau saddle after removal of the row mass. Prop. 3.89 and eq. 3.cm give

$$\|v\|_{\mathrm{str},\alpha} + \sup_c \xi_c^3 |\mu(c)| \le C_{p,\alpha}\{\mathcal{F}_p(f) + \sup_c \xi_c |b(c)|\}. \tag{3.cmiv}$$

There is no fine L^∞ assertion for $S^\sharp f$ in this argument.

3.25.8. *Complex tubes, small volumes and scope.* The same full penalized and full row-saddle strong estimates hold in a sufficiently small weighted C^1 tube eq. 3.dcccliv about a real reference. Here is the required weighted energy justification. Repeat the balanced site form with a spatial exponential weight on both the source and the output. Its normalized perturbation is $O(\varepsilon_*+|s|)$. The normalized F, Gram and R therefore have the same weighted bounds by Neumann series from the real reference. The full penalized field form changes by $C(\varepsilon_* + \gamma_\omega + |s|)$. Its real reference gap gives the complex weighted inverse and all cell energy estimates used above. This argument concerns the full penalized form; it does not assert a new spatial exponential theorem for an arbitrary moving-slice saddle.

Use the real reference's Coulomb frame on each interior box. The complex perturbation has bounded potential and first derivative in that frame by the C^1 tube. The logarithm of its product with the real reference links therefore has $\|a\|_\infty \le C(\kappa+\varepsilon_*)/\xi$ and $\|Da\|_q \le C_q(\kappa+\varepsilon_*)\xi^{4/q-2}$. All scalar and stencil estimates continue with these bounds. Their proofs use bilinear algebra and absolute estimates, not positivity at a complex background. The strong conclusions follow unchanged. For actual complex physical critical pairs the analytic Ward identity eq. 3.cmiii also proves eq. 3.cmiv. At an arbitrary noncritical complex pair only the full penalized and full row-constrained strong conclusions have been established here.

If a local box does not embed in a small torus, lift all actual rows, data and backgrounds to a finite periodic cover with periods exceeding $8R_*\max\xi$. Site and Gram inverses preserve the deck-invariant subspace by uniqueness; so do the full field and saddle inverses. The cover has a bounded number of sheets because the original periods contain at least two largest blocks. Apply the local estimates on the cover and descend. For a single column sum its finitely many deck images; each distance is at least the original torus distance, and one fixed decrease of the rate absorbs the bounded multiplicity.

We have proved bounded-force pointwise and local gradient Hölder responses on the actual multiscale determining regions. The estimates include the physical critical Landau saddle and small complex C^1 continuations in the ranges explicitly stated. They do not infer a pointwise singular Green kernel from a bounded-force bound. No nonlinear proper-region critical background or independently named current completion has been constructed by these linear estimates.

3.26. **An explicit current completion of the projected Hessian.** The invariant physical Hessian is determined by its Landau projection. Its continuation to an independent current requires an additional convention. We fix that convention by the second jet of an explicit finite gauge chain. This identifies the current correction with the symmetric form already estimated, fixes the order of its two propagator variants, and gives their bounded-force regularity on the actual determining geometry.

Throughout this section the action is the sum of inner Wilson plaquettes in Ω_0. Its physical current is j_U, its ambient Hessian is $\mathscr{D}_U$, and transposes use the physical bilinear pairings. All means are re-centered at U. The field row Q is the derivative per unit row length of the actual straightened mean; it is not the derivative of the plain mean. The site map q is its actual own-pillar gauge average, with the full extension at outward endpoints. Increase the fixed complete-carrier collars and separation once for the finitely many scalar stages below, independently of depth.

3.26.1. *The physical projection and its normalization.* Write

$$\begin{aligned} G_s &= (D^\sharp D + q^\sharp a_s q)^{-1}, \quad F = G_s q^\sharp \Xi^{-2}, \quad M = F^\sharp F, \\ R &= 1 - FM^{-1}F^\sharp, \qquad \Lambda = G_s R D^\sharp, \qquad S = 1 - D\Lambda. \end{aligned} \tag{3.cmv}$$

The site-top scale Ξ differs from the field-row length multiplication m. The intrinsic projection in this formula is precisely

$$R = 1 - G_s q^\sharp (q G_s^2 q^\sharp)^{-1} q G_s.$$

Indeed the two factors Ξ^{-2} in M cancel with its inverse in their indicated order. This is an algebraic normalization, not a different projection.

The exact identities are

$$\begin{gathered} qG_sR = 0, \qquad D^\sharp DG_sR = R, \qquad RD^\sharp S = 0, \qquad S^2 = S, \\ QD = \bar{D}_{\mathrm{ext}} q, \qquad QS = Q, \qquad SDG_sR = 0. \end{gathered} \tag{3.cmvi}$$

Thus $\lambda = \Lambda A$ is the unique zero-average parameter for which $RD^\sharp(A - D\lambda) = 0$. For uniqueness, $D^\sharp D$ maps $\ker q$ onto $\operatorname{ran} R$, with inverse G_sR. The site gap makes its kernel zero. These statements hold on the actual proper graph, including all outward rows and bare collars.

The physically projected Hessian is

$$\Delta^{\mathrm{phys}}_{\pi,U} = S^\sharp \mathscr{D}_U S. \tag{3.cmvii}$$

This determines the correction at $J = j_U$. It alone does not determine a linear map at every independent J: the physical current graph is not an open subset of the field-current product domain. We use the following finite-gauge convention to make that continuation explicit.

3.26.2. *A finite gauge chain fixes the independent-current map.* For a fine vector A put $\lambda_A = \Lambda A$. At a bond b, let $\ell_A = \lambda_A(b_-)$ and $r_A = \operatorname{Ad} U_b\, \lambda_A(b_+)$. Define the germ

$$\Pi_U(A)(b) = \frac{1}{i\eta} \log\left(e^{-i\ell_A} e^{i\eta A(b)} e^{ir_A}\right). \tag{3.cmviii}$$

It is the relative potential of the actual finite gauge transform by $e^{-i\lambda_A}$. Only its second jet at zero is used. Its first derivative is S. No nonlinear Landau-slice assertion about $\Pi_U(A)$ is needed or made.

Set $T_U[u,v] = D^2\Pi_U(0)[u,v]$, and define the independent-current completion by

$$\langle u, \Delta'_{\pi,U}[J]v\rangle := \langle J, T_U[u,v]\rangle. \tag{3.cmix}$$

This is our explicit convention for the named independent-current operator. It retains the physical projection eq. 3.cmvii. It does not assert uniqueness among all possible current-linear extensions with that physical value.

Proposition 3.90 (The explicit gauge-chain jet)**.** The completion eq. 3.cmix equals exactly $\mathcal{B}_U[J]$ of eq. 3.cdlxvii, for every independent real or complex current. More explicitly,

$$\begin{aligned} W_U[J]\lambda(b) &= \frac{i}{2}[\lambda(b_-) + \operatorname{Ad} U_b \lambda(b_+), J(b)], \\ \Delta'_{\pi,U}[J] &= W_U[J]\Lambda + \Lambda^\sharp W_U[J]^\sharp \\ &\quad - \tfrac{1}{2}\Lambda^\sharp(D^\sharp W_U[J] + W_U[J]^\sharp D)\Lambda. \end{aligned} \tag{3.cmx}$$

In particular it is symmetric and linear in J. The identity does not assume $D^\sharp J = 0$. ∎

Proof. The second-order logarithm of the three exponentials in eq. 3.cmviii has the quadratic part

$$\frac{i}{2}[A, \ell_A + r_A] - \frac{i}{2\eta}[\ell_A, r_A].$$

Polarizing gives

$$\begin{aligned} T_U[u,v](b) = \frac{i}{2}\{&[u(b), \ell_v + r_v] + [v(b), \ell_u + r_u]\} \\ &- \frac{i}{2\eta}\{[\ell_u, r_v] + [\ell_v, r_u]\}. \end{aligned} \tag{3.cmxi}$$

The apparent inverse spacing in the second term does not produce an inverse-spacing norm loss. Since $D\lambda = (\ell - r)/\eta$, cyclicity of the colour trace rewrites its pairing as

$$\begin{aligned}\langle J, T_U[u,v]\rangle &= \langle u, W_U[J]\Lambda v\rangle + \langle W_U[J]\Lambda u, v\rangle \\ &\quad - \tfrac{1}{2}\langle D\Lambda u, W_U[J]\Lambda v\rangle - \tfrac{1}{2}\langle W_U[J]\Lambda u, D\Lambda v\rangle.\end{aligned} \tag{3.cmxii}$$

For example $[\ell_u - r_u, \ell_v + r_v] + [\ell_v - r_v, \ell_u + r_u] = 2([\ell_u, r_v] + [\ell_v, r_u])$. This verifies both the factor one half and the physical power of η. Taking bilinear transposes in eq. 3.cmxii proves eq. 3.cmx. No divergence identity for the independent current was inserted. □

Gauge invariance gives $A_{\Omega_0}(e^{i\eta A}U) = A_{\Omega_0}(e^{i\eta\Pi_U(A)}U)$. The second chain rule at zero therefore proves

$$\mathscr{D}_U - \Delta'_{\pi,U}[j_U] = S^\sharp \mathscr{D}_U S. \tag{3.cmxiii}$$

This is an exact physical identity even when U is not constrained critical. It also follows from $\mathscr{D}_U D = W_U[j_U]$. The bilinear chain calculation extends holomorphically in the complex field. After taking its jet, the coefficient formula defines the form on all finite fine vectors; an arbitrary Gaussian vector need not lie in a nonlinear gauge chart.

The current dependence is now fully specified: $D_J\Delta'_{\pi,U}[J]\dot{J} = \Delta'_{\pi,U}[\dot{J}]$, and higher current derivatives vanish. A definition using a nonlinear gauge parameter with the same first jet could add a term pairing $D^\sharp J$ with its second gauge jet. Such a term is invisible on physical currents. Formula eq. 3.cmix fixes this freedom before using the physical current identity.

3.26.3. *The two named variants in their actual order.* Let $C_U^{(2)}$ be the quadratic part per unit length of the re-centered straightened field mean. Thus its group-angle logarithm has expansion $\mathsf{m}(QA + C_U^{(2)}(A) + \cdots)$. Define

$$\langle A, \Delta_U^{(2)}[\omega]A\rangle = 2\langle \omega, C_U^{(2)}(A)\rangle. \tag{3.cmxiv}$$

In particular this is the quadratic part of the straightened mean. With the row pairing held fixed, multiplication by m converts a multiplier for the group angle to a per-length multiplier. The conversion to the counting pairing is given below. All transposes below include the physical fine and row volumes.

The precise order is

$$\begin{aligned} G_0^{-1} &= \mathscr{D}_U + DRD^\sharp + Q^\sharp a_f Q, \\ G^{-1} &= G_0^{-1} - \Delta'_{\pi,U}[J], \\ H &= GQ^\sharp(QGQ^\sharp)^{-1}, \qquad \omega_* = H^\sharp J, \\ G_1^{-1} &= G^{-1} - S^\sharp \Delta_U^{(2)}[\omega_*]S, \\ H_1 &= G_1 Q^\sharp (QG_1Q^\sharp)^{-1}. \end{aligned} \tag{3.cmxv}$$

3.26.3.1. The same completion in the propagator convention. There is a sign change between the covariant forward difference used for the propagator construction and the infinitesimal gauge generator of eq. 3.cdlx. In per-length units the forward difference at a bond is $(r - \ell)/\eta$, while the gauge generator is $(\ell - r)/\eta$. The site inverse and R are unchanged by this change of sign when the site average and mass are the same. Hence the site parameter in the forward-difference formula for the projection is $v = -\lambda_A$. In that convention the current term is the pairing of J with

$$i[v_-, A] - i[A, v_+] - i[v_-, D_{\mathrm{forw}}v].$$

Here v_- and v_+ are its values at the initial and transported terminal endpoints and $D_{\mathrm{forw}}v = (v_+ - v_-)/\eta$. Substitution gives

$$i[A, \ell_A + r_A] - \frac{i}{\eta}[\ell_A, r_A],$$

which is $T_U[A,A]$ from eq. 3.cmxi. The fine pairing in both formulas is the trace pairing with the bare cell volume. By polarization the two current completions agree also for independent complex J on the common graph with the same site average. For the separate correction from the second derivative of the mean, let $C^{(2)}$ be the quadratic term in the per-length row coordinates of the same mean. The raw quadratic term is $\mathsf{m}C^{(2)}$. The raw lift is $\mathsf{H}\mathsf{m}^{-1}$, with counting-row transpose $\mathsf{m}^3\mathsf{H}^*$, where the star uses row volumes m^4. Therefore its contraction with the current is

$$(\mathsf{m}C^{(2)}(A), \mathsf{m}^3\mathsf{H}^*J)_{\mathrm{count}} = (C^{(2)}(A), \mathsf{H}^*J)_{\mathrm{vol}}.$$

The two corrections thus have the same readouts when the site maps, row maps, and local domains have been identified. The calculation is unchanged on the projected field with A replaced by SA.

The additional quadratic form is exactly

$$-2\langle J, HC_U^{(2)}(SA)\rangle = -\langle A, S^\sharp \Delta_U^{(2)}[H^\sharp J]SA\rangle.$$

Thus it is precisely the second explicit variant. The H inside the second subtraction is formed from the first G, and is not H_1. Nor does $C_U^{(2)}(SA)$ mean the second derivative of the nonlinear composite mean with $\Pi_U(A)$.

With the convention eq. 3.cmix, all operators in eq. 3.cmxv are the already constructed explicit operators. In particular Prop. 3.73 proves

$$\sup_c \xi_c^3 |\omega_*(c)| \le C\gamma_J, \qquad \gamma_J = \sup_b \mathsf{L}(b)^3 |J(b)|. \tag{3.cmxvi}$$

At a small full-current radius this already admits the second subtraction. The support refinement below requires only a small current on bonds touching Ω_1 for the propagators, while retaining the full γ_J in this all-row amplitude estimate.

3.26.4. *Gauge defects and the meaning of on shell.* Put $\delta J = J - j_U$. Equations eq. 3.cmvi and eq. 3.cmxiii give

$$\begin{aligned} G^{-1}DG_sR &= DR - \Delta'_{\pi,U}[\delta J]DG_sR, \\ G_{(1)}DR &= DG_sR + G_{(1)}\Delta'_{\pi,U}[\delta J]DG_sR. \end{aligned} \tag{3.cmxvii}$$

The parenthesized index permits either variant. The second subtraction vanishes on DG_sR because $SDG_sR = 0$. Transposition gives its corresponding row-lift slice defect. On shell that defect is zero; no independent current is declared divergence free.

Only bonds touching Ω_1 matter for the first current correction. Write J^{t} for its restriction to those bonds and $J^{\mathrm{b}} = J - J^{\mathrm{t}}$. Set

$$\gamma_{\mathrm{t}} = \sup_{b:\, b \text{ touches } \Omega_1} \mathsf{L}(b)^3 |J(b)|. \tag{3.cmxviii}$$

The supremum is zero if that set is empty. Indeed $\Lambda A \in \ker q$, so it vanishes at all bare sites. Both $W_U[\delta J]\Lambda A$ and $D\Lambda A$ therefore vanish on bonds with neither endpoint in Ω_1. Consequently equality $J = j_U$ on bonds touching Ω_1 already implies $\Delta'_{\pi,U}[J] = \Delta'_{\pi,U}[j_U]$. In particular $\Delta'_{\pi,U}[J] = \Delta'_{\pi,U}[J^{\mathrm{t}}]$.

The same support statement suffices for the second variant. Let P_0 restrict row data to the bare rows. The exact identity is $Q_0 H = P_0$. Thus a datum supported on a nonbare row lifts to a vector which must vanish on bare bonds. Under the touching-bond on-shell condition, $H^\sharp(J - j_U)$ has only bare row components. More precisely, because the physical fine and bare-row measures are both η^4, that same restriction identity gives

$$\omega_*[J] = \omega_*[J^{\mathrm{t}}] + \iota_0 J^{\mathrm{b}}, \tag{3.cmxix}$$

where ι_0 identifies a bare bond with its level-zero row. The actual bare mean is linear, so its quadratic part is zero. These components do not contribute to $\Delta_U^{(2)}[\omega_*]$. This retains the on-shell

condition on bonds touching Ω_1; it does not strengthen it to a condition on every bare bond. Applying eq. 3.cmxvi to J^{t} gives

$$\sup_{c\ \text{nonbare}} \xi_c^3 |\omega_*(c)| \le C\gamma_{\mathrm{t}}. \tag{3.cmxx}$$

Thus G, H, G_1, H_1 are independent of J^{b}. Their admission and strong bounds use only a small γ_{t}. Their all-row induced current still has the full γ_J bound eq. 3.cmxvi; arbitrary bare currents are not thereby bounded.

If U is an actual physical critical point with $j_U = Q^\sharp \omega_{\mathrm{phys}}$, then, for the full physical current, $H^\sharp j_U = \omega_{\mathrm{phys}}$. The physical Lagrangian Ward identity eq. 3.cmiii gives

$$S^\sharp(\mathscr{D}_U - \Delta_U^{(2)}[\omega_{\mathrm{phys}}])S = \mathscr{D}_U - \Delta_U^{(2)}[\omega_{\mathrm{phys}}].$$

Therefore the named second precision is exactly

$$G_1^{-1} = \mathscr{D}_U - \Delta_U^{(2)}[\omega_{\mathrm{phys}}] + DRD^\sharp + Q^\sharp a_f Q. \tag{3.cmxxi}$$

The same conclusion holds under the on-shell condition just described, since the remaining induced-current change is bare-only. This identifies the actual physical second variant with the local Lagrangian precision, not merely their restrictions to horizontal vectors.

3.26.5. *Regularity at independent currents.* The current form eq. 3.cmxii and weighted restoration give

$$|\langle u, \Delta'_{\pi,U}[J]v\rangle| \le C\gamma_{\mathrm{t}} \|u\|_X \|v\|_X.$$

Its conjugated version has that same bound. The first and second variants therefore have the energy and exponential cell estimates already proved, for a fixed ceiling on γ_{t}. Only the nonbare part of ω_* enters the second subtraction. For real pairs their forms are positive. On a small complex C^1 tube the inverses are bounded analytic perturbations of the real references. No physical-current identification is used for this admission.

We next justify the pointwise and gradient Hölder bounds for these independent-current variants. Keeping the apparent derivative of J out of this proof is essential. Use the normalized $\mathcal{F}_p$ and cell energy amplitudes of § 3.25. All exponents p in the following argument are finite.

Lemma 3.91 (Current terms from cell energy)**.** Suppose $\mathcal{E}_{\mathrm{cell}}(v) \le B$. Then

$$\mathcal{F}_8(\Delta'_{\pi,U}[J]v) \le C\gamma_{\mathrm{t}} B. \tag{3.cmxxii}$$

For a row current with small $\gamma_{\omega,\mathrm{nb}} = \sup_{c\ \text{nonbare}} \xi_c^3 |\omega(c)|$, the second form initially has the bound

$$\mathcal{F}_3(S^\sharp \Delta_U^{(2)}[\omega] S v) \le C\gamma_{\omega,\mathrm{nb}} B. \tag{3.cmxxiii}$$

Here $\mathcal{F}_3$ means the same local physical formula as eq. 3.dccclxxxv at exponent 3. If in addition the normalized local $W^{2,3}$ norm of v is bounded by B_1, both current terms belong to every finite $\mathcal{F}_p$, $p > 4$, with bound $C_p(\gamma_{\mathrm{t}} + \gamma_{\omega,\mathrm{nb}})(B + B_1)$. ∎

Proof. By the exact support identities, replace J by J^{t} and discard bare row components of ω. Set $\lambda = \Lambda v$. The local divergence has site L^2 amplitude of grade two bounded by B. Its P part has the pointwise grade-two bound: use $F^\sharp D^\sharp v = (DF)^\sharp v$, whose normalized top coefficients are $O(B)$, and sum the strong site columns. Thus $RD^\sharp v$ has the same local L^2 amplitude. Site energy columns, with one scale-ratio reserve, give

$$\xi_y^{-2} \|\lambda\|_{2,\mathcal{D}_y} + \xi_y^{-1} \|D\lambda\|_{2,\mathcal{D}_y} \le CB.$$

Since $\Delta_U \lambda = RD^\sharp v$, the first scalar L^2 cutoff step gives normalized $W^{2,2}$ bounds. The strict heat embeddings imply that λ lies in every finite local L^q and $D\lambda$ lies in local L^3, with the physical dimensions 1 and ξ^{-1}. The same L^3 bound holds for v.

Group the exact correction as

$$\Delta'_{\pi,U}[J]v = W_U[J]\lambda + DRG_s\{W_U[J]^\sharp(v - \tfrac{1}{2}D\lambda) - \tfrac{1}{2}D^\sharp W_U[J]\lambda\}. \tag{3.cmxxiv}$$

The site force $a = W_U[J]^\sharp(v - D\lambda/2)$ has local L^3 grade-four amplitude at most $C\gamma_{\mathrm{t}}B$. The site energy estimate for $G_s a$ and the scalar cutoff bootstrap through L^3 give a derivative in L^8 of grade three. Subtracting its P part costs a bounded pointwise derivative, by the site columns and the Gram inverse. Hence $DRG_s a \in \mathcal{F}_8$. The other force $h = W_U[J]\lambda$ is in every finite $\mathcal{F}_q$, because λ is in every finite local L^q. Exactly

$$DRG_s D^\sharp h = h - S^\sharp h.$$

The finite-q restoration bound eq. 3.cmii, at $q = 8$, controls this term and the direct h term. This proves eq. 3.cmxxii.

For the second form, $Sv = v - D\lambda$ has its normalized L^3 bound. The actual two-leg Schur estimate eq. 3.dcccxcvii gives $\Delta_U^{(2)}[\omega]Sv \in \mathcal{F}_3$. The force-restoration proof at this one exponent uses just its first divergence-source step: the local energy of $G_s D^\sharp h$ and $h \in L^3$ give $DG_s D^\sharp h$ in L^3 by finite-3 Riesz, while the remaining point source in L^2 has derivative in L^3. Its P part is bounded pointwise. This proves the finite-3 version of restoration and eq. 3.cmxxiii. No endpoint statement is needed.

Finally a local $W^{2,3}$ bound gives $v \in L^\infty$ and $Dv \in L^8$. Then $RD^\sharp v \in L^8$ at grade two, since its P part remains pointwise bounded. The scalar bootstrap for λ gives both λ and $D\lambda$ bounded at their physical grades. Thus h and a in eq. 3.cmxxiv are bounded. A bounded site force of grade four has $DRG_s a$ pointwise bounded at grade three, by the scalar point bootstrap and its smoothed P part. The divergence-source term is controlled in every finite $\mathcal{F}_p$ by force restoration. Also Sv is bounded, so the constraint term is bounded before the final finite-p restoration. This proves the last assertion. Every derivative has fallen on a resolvent or the unknown field, not on the independent current. □

Proposition 3.92 (Strong propagators under the explicit convention)**.** For $G_\bullet \in \{G, G_1\}$, finite $4 < p < \infty$ and $0 < \alpha < 1 - 4/p$,

$$\|G_\bullet f\|_{\mathrm{str},\alpha} \le C_{p,\alpha}\mathcal{F}_p(f). \tag{3.cmxxv}$$

Their full row saddles have the same strong bound with $\sup_c \xi_c|b(c)|$ added, and their multiplier has bounded ξ^3 amplitude. The ceiling on γ_{t} can be fixed independently of α. The result holds on real regular pairs and their small complex C^1 tubes. ■

Proof. The energy estimate supplies $\mathcal{E}_{\mathrm{cell}}(v) \le C\mathcal{F}_p(f)$ for either variant. Relative to G_0^{-1}, their equations have the first correction in $\mathcal{F}_8$ and, for the second variant, the second correction in $\mathcal{F}_3$, with $\gamma_{\omega_*,\mathrm{nb}} \le C\gamma_{\mathrm{t}}$ by eq. 3.cmxx. The local scalar vector proof of Prop. 3.88, stopped after its L^3 equation, therefore gives normalized $W^{2,3}$ control. All carrier enlargements are retained. The last part of Lem. 3.91 now puts both corrections in the requested finite $\mathcal{F}_p$. Applying the full G_0 strong estimate proves eq. 3.cmxxv. This is a bootstrap, not an α-dependent absorption of a singular-integral constant into the current radius.

For the full row saddle the already proved coarse Schur kernels bound its multiplier by the normalized force and row datum amplitudes. Its $Q^\sharp\mu$ is a bounded point force by eq. 3.dcccxcv. The same two stages prove its strong field estimate. Complex coefficients have the same absolute bounds in the real reference's local Coulomb frames, and the energy inverse there was obtained by Neumann series. The proof does not use Hermitian positivity at a complex pair. □

The argument can retain a small spatial exponential weight at every step. Its local oscillation on complete carriers is bounded, and the site, Gram and row sums retain a strict metric growth reserve. This gives the corresponding cell-to-cell strong response estimates. For a bounded source supported in B_z, the normalized output in eq. 3.dccclxxxvii on B_y is at most $C_\alpha e^{-md(y,z)}\xi_z^3\|f\|_{\infty,B_z}$. If desired

the factor $(\xi_z/\xi_y)^3$ is absorbed by one further fixed scale-ratio reserve to express the potential bound as $C\xi_y^2 e^{-m'd(y,z)}\|f\|_\infty$.

These bounded-force conclusions do not assert an L^∞ bound for $\Delta_U G$ or $\Delta_U G_1$. The divergence term containing $W_U[J]\lambda$ was controlled only in finite L^p, as required when no derivative of J is prescribed. Nor does a weighted cell estimate alone construct a background-local walk expansion. The canonical continuation eq. 3.cmix has been made explicit; identifying another independently prescribed off-current convention would still require its own formula comparison.

3.27. **Finite-carrier equations for the actual current inverse.** The projected gauge and current terms conceal global site inverses. We display an equivalent system whose coefficients are actual local rows and covariant differences. We prove its inverse on every source component, including divergence sources. This also identifies the cutoff estimates needed for a subsequent local parametrix. No compression of an indefinite system is presumed invertible.

Use the actual proper geometry, physical bilinear transposes and comparison length ℓ of §§ 3.19 and 3.26. The field is real and has the curvature ceiling of Prop. 3.70. Put

$$A_s = D^\sharp D + q^\sharp a_s q, \quad G_s = A_s^{-1}, \quad F = G_s q^\sharp \Xi^{-2}, \quad M = F^\sharp F, \quad P = FM^{-1}F^\sharp, \quad R = 1 - P. \tag{3.cmxxvi}$$

All site rows, field rows and constraint derivatives are the actual re-centered ones, including complete outward carriers. As permitted by the exact support identities in § 3.26, replace J by its restriction to bonds touching Ω_1, and discard bare components of the constraint multiplier. These changes leave both named field operators unchanged. They avoid putting irrelevant, possibly unbounded bare currents into auxiliary coefficients.

Write $W = W_U[J]$ for the actual local current map, and $C_\omega = \Delta_U^{(2)}[\omega]$ for the unprojected constraint Hessian. In the second named variant $\omega = H^\sharp J$ uses the first lift. The following arguments also allow an independent nonbare ω within the same small ceiling. Define

$$K_F = H_U^{\mathrm{W}} + DD^\sharp + Q^\sharp a_f Q, \qquad \Lambda = G_s R D^\sharp, \qquad S = 1 - D\Lambda. \tag{3.cmxxvii}$$

Here H_U^{W} is the actual inner Wilson Hessian. The two field operators are

$$K^{(0)} = K_F - DPD^\sharp - B_U[J], \qquad K^{(1)} = K^{(0)} - S^\sharp C_\omega S. \tag{3.cmxxviii}$$

They are respectively the first and second canonical named precisions. Their energy inverses follow from § 3.26; the retained current ceilings are sufficiently small fixed constants.

3.27.1. *Site source spaces with their physical weights.* Let X be the actual field energy space. Introduce the site norms

$$\begin{aligned} \|u\|_Y^2 &= \|u\|_2^2 + \|\ell Du\|_2^2, \\ \|\lambda\|_Z &= \|\ell^{-2}\lambda\|_2 + \|\ell^{-1}D\lambda\|_2 + \|\nabla_U D\lambda\|_2, \\ \|h\|_{Y^\flat} &= \sup_{u\neq 0} \frac{|\langle h, \ell^2 u\rangle|}{\|u\|_Y}, \qquad \|c\|_{T_{-2}} = \|\Xi^{-2}c\|_{\mathrm{top}}. \end{aligned} \tag{3.cmxxix}$$

The second differences and the bare collar have the convention of eq. 3.dcclxxxiv. In particular this is a norm on the actual finite site space, with the atomic convention at bare sites. $Y^\flat$ uses the displayed weighted pairing; it is not the unweighted dual norm of Y. If $h = h_0 + D^\sharp h_1$, the product rule gives

$$\|h\|_{Y^\flat} \le C\{\|\ell^2 h_0\|_2 + \|\ell h_1\|_2\}. \tag{3.cmxxx}$$

Thus a divergence source need not have a pointwise derivative. The constants below are uniform in the depth and compatible volume, and in all separated proper shapes under consideration.

Lemma 3.93 (A second-difference site lift)**.** There is a real linear right inverse E of q such that

$$qE = 1, \qquad \|Ec\|_Z \le C\|c\|_{T_{-2}}. \tag{3.cmxxxi}$$

Only the small curvature and the actual pillar geometry are required; no physical-current bound is used. ■

Proof. Use the compact interior bump construction of Lem. 3.58, with its exact transported normalization on each pillar. On the fixed surrounding cube of length $C_L\xi_y$, a current-free Coulomb frame has

$$\|a\|_\infty \le C_L\kappa\xi_y^{-1}, \qquad \|Da\|_2 \le C_L\kappa.$$

The second bound is the finite exponent $p = 2$ case of Cor. 3.83; using a smaller cube keeps its collar inside the nonbare determining region. For one profile of amplitude c_y, two covariant differences contain $D^2\phi$, $aD\phi$, $Da\,\phi$ and $a^2\phi$. Their L^2 sum is at most $C_L|c_y|$. The mass and first-gradient terms with weights ℓ^{-2}, ℓ^{-1} have that same bound. The normalization matrix is constant within that pillar, so differentiating the profile does not differentiate this matrix.

For the finitely coarse meshes use its center seed; its weighted second differences cost a fixed constant depending on L. At a bare top this is the one-site seed, and every difference cost is paid by its scale η. The complete derivative collars have bounded overlap. Summing the squared estimates gives $\sum_y |c_y|^2 = \|\Xi^{-2}c\|^2_{\mathrm{top}}$. If the surrounding frame cube does not embed, construct on a finite periodic cover and average the operator over simultaneous fine and top deck shifts. The right-inverse identity and these Hilbert norms descend, with the same sheet factor on both sides. □

Lemma 3.94 (Two actual site blocks)**.** The following systems have unique solutions:

$$A_s\lambda + v = f, \qquad A_s v - q^\sharp \Xi^{-2}\beta = h, \qquad q\lambda = c, \tag{3.cmxxxii}$$

$$A_s t + q^\sharp \Xi^{-2}\beta' = h', \qquad A_s\kappa - t = r, \qquad q\kappa = d. \tag{3.cmxxxiii}$$

Their respective estimates are

$$\begin{aligned} \|\lambda\|_Z + \|v\|_Y + \|\beta\|_{\mathrm{top}} &\le C\{\|f\|_2 + \|h\|_{Y^\flat} + \|c\|_{T_{-2}}\}, \\ \|\kappa\|_Z + \|t\|_Y + \|\beta'\|_{\mathrm{top}} &\le C\{\|r\|_2 + \|h'\|_{Y^\flat} + \|d\|_{T_{-2}}\}. \end{aligned} \tag{3.cmxxxiv}$$

■

Proof. The balanced test in eq. 3.dcclxxviii, with the same small Lipschitz constant for ℓ, proves

$$\|G_s h\|_Y \le C\|h\|_{Y^\flat}. \tag{3.cmxxxv}$$

Indeed its real left side controls $\|\ell Du\|_2^2 + \|qu\|^2_{\mathrm{top}}$, and the actual block Poincare estimate controls $\|u\|_2^2$. This argument applies to every source h, not only to a site-row column.

For the first block put $w = G_s h$. Its exact solution is

$$\begin{aligned} \beta &= M^{-1}\{F^\sharp(f - w) - \Xi^{-2}c\}, \\ v &= w + F\beta, \qquad \lambda = G_s(f - v). \end{aligned} \tag{3.cmxxxvi}$$

The identity $qG_sF = \Xi^2 M$ proves $q\lambda = c$. The normalized Gram gap and the balanced column estimate for F bound β, v. For the Z bound set $\lambda_0 = \lambda - Ec$. It has zero q average and

$$\Delta_U\lambda_0 = f - v - q^\sharp a_s c - \Delta_U Ec.$$

The right side has L^2 norm at most the first displayed source bound: $q^\sharp$ is the actual coisometric transported extension, $a_s \asymp \Xi^{-2}$, and Lem. 3.93 controls $\Delta_U Ec$. Apply eq. 3.dcclxxxiv and then add Ec.

For the second block put $w' = G_s h'$. The formulas are

$$\begin{aligned} \beta' &= M^{-1}\{F^\sharp(w'+r) - \Xi^{-2}d\}, \\ t &= w' - F\beta', \qquad \kappa = G_s(t+r). \end{aligned} \tag{3.cmxxxvii}$$

The same argument, with $\kappa - Ed$, proves its bound. These formulas prove existence and uniqueness as well. All transposes and weights are the physical ones; replacing the top transpose by a counting-array transpose would change the displayed normalizations. □

3.27.2. *The complete finite-carrier system.* Use $\epsilon = 0$ or 1 to select a variant, and set

$$\delta = \epsilon C_\omega(A - D\lambda), \qquad h_J(A,\lambda) = W^\sharp A - \tfrac{1}{2}\{D^\sharp W\lambda + W^\sharp D\lambda\} - D^\sharp\delta. \tag{3.cmxxxviii}$$

The unknowns are $A, \lambda, v, \beta, t, \kappa, \beta'$. With arbitrary sources they satisfy

$$\begin{aligned} K_F A - Dv - W\lambda - \delta - Dt &= f_0, \\ A_s\lambda + v - D^\sharp A &= g, \\ A_s v - q^\sharp\Xi^{-2}\beta &= h, \qquad q\lambda = c, \\ A_s t + q^\sharp\Xi^{-2}\beta' - h_J(A,\lambda) &= h', \\ A_s\kappa - t &= r, \qquad q\kappa = d. \end{aligned} \tag{3.cmxxxix}$$

In particular every coefficient is a covariant difference, an actual site or field row, a local current multiplication or an actual two-leg constraint coefficient. No inverse, projector or induced multiplier is hidden in this matrix. For the second variant its prescribed coefficient ω must still be constructed from the first lift.

Proposition 3.95 (Unfolding with all auxiliary sources)**.** For the actual real curvature-small pairs and current ceilings above, eq. 3.cmxxxix is invertible and

$$\begin{aligned} &\|A\|_X + \|\lambda\|_Z + \|\kappa\|_Z + \|v\|_Y + \|t\|_Y + \|\beta\|_{\mathrm{top}} + \|\beta'\|_{\mathrm{top}} \\ &\quad \le C\{\|f_0\|_{X^*} + \|g\|_2 + \|r\|_2 + \|h\|_{Y^\flat} + \|h'\|_{Y^\flat} + \|c\|_{T_{-2}} + \|d\|_{T_{-2}}\}. \end{aligned} \tag{3.cmxl}$$

At zero auxiliary sources its field block is precisely $(K^{(\epsilon)})^{-1}$. Appending $-Q^\sharp\mu$ to the field equation and $QA = b$ gives the full actual row saddle, with the additional bound $\|a_f^{-1/2}\mu\|_{\mathrm{rows}}$ on the left and $\|a_f^{1/2}b\|_{\mathrm{rows}}$ on the right. ■

Proof. At zero auxiliary sources, the primal block gives

$$v = PD^\sharp A, \qquad \lambda = \Lambda A.$$

The transpose block gives $t = RG_s h_J(A,\lambda)$. Since $\Lambda^\sharp = DRG_s$, the exact symmetric current formula in eq. 3.cmxxiv shows

$$W\lambda + DRG_s\{W^\sharp A - (D^\sharp W\lambda + W^\sharp D\lambda)/2\} = B_U[J]A.$$

The remaining two terms are $-\delta + DRG_s D^\sharp\delta = -S^\sharp\delta$. Thus the field equation is exactly eq. 3.cmxxviii, with the stated sign and order.

Here are the bounds needed for nonzero auxiliary sources. The actual undifferentiated current and two-leg coefficient estimates give

$$\begin{aligned} \|\ell W\lambda\|_2 &\le C\gamma_J\|\lambda\|_Z, \\ \|W^\sharp A\|_{Y^\flat} + \|W^\sharp D\lambda\|_{Y^\flat} + \|D^\sharp W\lambda\|_{Y^\flat} &\le C\gamma_J(\|A\|_X + \|\lambda\|_Z), \\ \|\ell C_\omega u\|_2 &\le C\gamma_\omega\|\ell^{-1}u\|_2. \end{aligned} \tag{3.cmxli}$$

The last line is a two-leg Schur estimate, not an L^2-to-supremum estimate. A fixed-leg row has physical density $(\eta/\xi)^4$; the transpose cancels it with $\eta^{-4}\xi^4$. Both absolute sums are bounded and all lengths on the carrier are comparable. This proves that line also for an arbitrary field u. The

middle line uses eq. 3.cmxxx for the term $D^\sharp W\lambda$, and differentiates no current. Also Dv and Dt have X^* norms at most $C\|v\|_2, C\|t\|_2$, by the full physical adjoint.

Solve the primal block for a given A. Subtract its zero-source solution to obtain remainders λ_e, v_e bounded by $\|g\|_2 + \|h\|_{Y^\flat} + \|c\|_{T_{-2}}$. Solve the transpose block next. Its part depending on A is exactly the one just eliminated. Its remaining source is bounded by the auxiliary source norms, using eq. 3.cmxli. The field equation therefore reduces to $K^{(\epsilon)}A = f_0 + f_e$, where $\|f_e\|_{X^*}$ has that same bound. The actual canonical energy inverse bounds A. The site-block estimates then bound every remaining variable. For the row saddle use its actual energy inverse instead; the right inverse of Q and its dual row norm give the displayed multiplier estimate. No stationarity or physical-current identity is used in this argument. □

Corollary 3.96 (Bilinear complex continuation)**.** The full augmented inverse and the appended row saddle continue holomorphically, with the same source and solution spaces, to a common smaller weighted C^1 tube about a real reference. The independent touching current and nonbare multiplier stay within their indicated complex ceilings. ■□

Proof. Use the fixed real physical norms. In the local coefficients, the weighted C^1 tube controls $A_s : Z \to L^2$ and $A_s : Y \to Y^\flat$; it controls $K_F : X \to X^*$ by its weak plaquette form. Actual transported row differences have both normalized sums bounded by $C\varepsilon$. The same holds for the actual two-leg coefficient by its analytic row bound. The current terms obey eq. 3.cmxli without a current derivative. Thus the augmented operator is a small holomorphic perturbation, between exactly the displayed spaces, of its real reference operator. A Neumann inverse proves the assertion. Complex transposes are bilinear throughout; no orthogonal positivity at a complex pair is used. □

3.27.3. *Slow cutoffs and the bare boundary.* Let χ be a real scalar cutoff on the actual sites. Use its endpoint average on fine bonds and its center or midpoint value on tops and rows. Suppose

$$|\chi| \le 1, \qquad |\ell D\chi| \le C/m, \qquad |\ell^2 \Delta_{\mathrm{graph}}\chi| \le C/m.$$

The last operator is the scalar Laplacian on the actual site graph. At an action boundary its first-difference boundary term satisfies this bound because $\ell \asymp \eta$ there. Inside a smooth patch a second-difference cutoff bound $C/(m^2\ell^2)$ is sufficient. Every complete row carrier must have χ oscillation at most C_L/m. Cutoffs on patches of radius $m\ell$ have this property when m Lip(ℓ) is sufficiently small.

Proposition 3.97 (The actual unfolded commutator)**.** Let $\mathbb{K}^{(\epsilon)}$ be eq. 3.cmxxxix, and multiply each component by the corresponding scalar cutoff. In the source and solution norms of eq. 3.cmxl,

$$\|[\mathbb{K}^{(\epsilon)}, \chi]\Psi\|_{\mathrm{source}} \le Cm^{-1}\|\Psi\|_{\mathrm{solution}}. \tag{3.cmxlii}$$

The same assertion holds with the actual row saddle appended. ■

Proof. The product rules and the complete row oscillation give

$$\begin{aligned}
\|[A_s, \chi]\lambda\|_2 &\le Cm^{-1}\|\lambda\|_Z,\\
\|[A_s, \chi]v\|_{Y^\flat} &\le Cm^{-1}\|v\|_Y,\\
\|[q, \chi]\lambda\|_{T_{-2}} &\le Cm^{-1}\|\lambda\|_Z,\\
\|[q^\sharp\Xi^{-2}, \chi]\beta\|_{Y^\flat} &\le Cm^{-1}\|\beta\|_{\mathrm{top}}.
\end{aligned}$$

For the first line the Laplacian product contains $D\chi\, D\lambda$ and $\Delta_{\mathrm{graph}}\chi\,\lambda$. For the second line use the weak gradient form and test against $\ell^2 u$; only $D\chi$ is required. For the two row lines use the actual positive absolute averaging weights and the displayed carrier oscillation. The site mass commutator satisfies the first two bounds by these same two row estimates.

The field form K_F has $\|[K_F, \chi]A\|_{X^*} \le Cm^{-1}\|A\|_X$. This follows in its actual weak plaquette and full-divergence forms, together with the two field-row sums. It uses no derivative of curvature. The

commutators of $D, D^\sharp$ cost $|D\chi|$, with exactly the powers of ℓ in the displayed spaces. In the current terms insert the same carrier oscillation in eq. 3.cmxli. For the constraint term, $\|\ell[C_\omega,\chi]u\|_2 \le C\gamma_\omega m^{-1}\|\ell^{-1}u\|_2$. The extra expression from $A - D\lambda$ is $-[D,\chi]\lambda$, whose weighted mass norm is at most $Cm^{-1}\|\lambda\|_Z$. Finally treat every $D^\sharp\delta$ as a divergence source using eq. 3.cmxxx. These bounds cover each row of the system.

At a rough action boundary use the actual full adjoints and retain every incident inner plaquette. The preceding weak product identities still hold. The atomic scale $\ell \asymp \eta$ pays every one-edge boundary difference. No continuation of a Coulomb frame across a missing plaquette is part of this estimate. □

3.27.4. *A positive factorization for the projected gauge.* The following alternative separates the actual nonlocal Landau gauge into local positive operators and a positive coarse inverse.

Proposition 3.98 (The actual gauge Schur matrix)**.** At a real curvature-small background put $G_F = K_F^{-1}$ and

$$\Theta = M - (DF)^\sharp G_F (DF). \tag{3.cmxliii}$$

There is a uniform $0 < \delta < 1$ such that

$$\delta M \le \Theta \le M. \tag{3.cmxliv}$$

For the zero-current Landau inverse one has exactly

$$G_0 = G_F + G_F DF\, \Theta^{-1} (DF)^\sharp G_F. \tag{3.cmxlv}$$

All quantities admit bounded holomorphic continuation to a common smaller weighted C^1 tube. ■

Proof. The actual zero-current precision is $K_0 = K_F - DFM^{-1}(DF)^\sharp$. It controls $c\|A\|_X^2$ by Prop. 3.70, while the local form K_F is at most $C\|A\|_X^2$. Thus $K_0 \ge \delta K_F$. For $T = G_F^{1/2} DF M^{-1/2}$ this says $1 - TT^\sharp \ge \delta$, hence $1 - T^\sharp T \ge \delta$. Conjugating by $M^{1/2}$ proves eq. 3.cmxliv. The normalized Gram gap makes it a genuine depth-uniform coarse gap. The resolvent multiplication identity for $K_F - DFM^{-1}(DF)^\sharp$ now gives eq. 3.cmxlv. Every transpose here uses the physical fine or top measure.

In a weighted C^1 tube the operators between the displayed fixed physical spaces are small holomorphic perturbations of their real reference values. This is the site/Gram continuation of § 3.23 and the actual current continuation. In particular the gaps supply bounded inverse maps there, and the algebraic identity continues by multiplication. Positivity is used only at the real reference. □

For a set of sites or bonds, the Dirichlet compression of A_s or K_F is obtained by zero extension into the actual form. Retain the full coefficient of every crossing row. Each positive compression inherits the corresponding global coercivity in its zero-extension energy norm. Its entries depend only on the complete carriers of rows meeting that set, together with the incident inner plaquettes. Thus its geometry-decision collar is finite. In contrast, deleting the exterior part of a row and asking for a new row right inverse can destroy its inf-sup constant; and an indefinite compression of eq. 3.cmxxxix is not asserted to satisfy eq. 3.cmxl.

The positive local walk construction in Prop. 3.102 supplies the local parametrix step with these crossing coefficients and a deterministic finite-decision cover. Coarse inverse carriers for M, Θ, the small current corrections and the first-lift readout must then be assembled in their actual physical normalizations. The present equations, estimates and factorization do not by themselves produce the enlarged independent cube-parameter family or identify a clean conditional history law.

3.28. **Local walks for the positive finite-carrier inverses.** An exponentially decreasing inverse need not have coefficients determined by local background data. We now construct such coefficients for the two positive finite-carrier operators that enter the projected precision. The local inverse in the construction is a genuine Dirichlet compression. No global inverse with a locally changed background is declared to be a local coefficient.

The geometry, physical measures and true outer boundary are those of § 3.19. All coarse rows retain their complete carriers. For a real regular connection set

$$\mathcal{A}_s = D_U^\sharp D_U + q^\sharp a_s q, \qquad \mathcal{A}_F = \mathscr{D}_U + D_U D_U^\sharp + Q^\sharp a_f Q. \tag{3.cmxlvi}$$

Here $\mathscr{D}_U$ is the actual inner Wilson Hessian. The second operator in eq. 3.cmxlvi is the full Feynman operator, before a site-projection subtraction or an independent-current correction. The result of this subsection does not assert locality of those nonlocal subtractions.

3.28.1. *The positive forms and the cell norms.* For site or vector fields write

$$\|u\|_{\mathcal{E}}^2 = \|D_U u\|_2^2 + \|\xi^{-1} u\|_2^2,$$

using the componentwise covariant gradient in the vector case. Its local version $\mathcal{E}_y$ uses B_y and one fine-step collar. For an L^2 source supported in B_z, put

$$\|f\|_{\mathcal{F}_z} = \xi_z \|f\|_{2,B_z}.$$

Thus $|\langle f, u\rangle| \le \|f\|_{\mathcal{F}_z} \|u\|_{\mathcal{E}_z}$. Cutoffs and collars have bounded overlap, so the squared global energy is equivalent to the sum of the squared cell energies. No pointwise Green-kernel norm is implicit in this notation.

There are uniform constants $c, C > 0$ with

$$c\|u\|_{\mathcal{E}}^2 \le \langle u, \mathcal{A}_\bullet u\rangle \le C\|u\|_{\mathcal{E}}^2, \qquad \bullet = s, F. \tag{3.cmxlvii}$$

For the site operator this is the actual block inequality eq. 3.dcclxxiii and the transported-average upper bound. For the vector operator,

$$\mathcal{A}_F = K_{U,0} + D_U P D_U^\sharp, \qquad P = 1 - R.$$

The real projection satisfies $0 \le P \le 1$. The lower bound follows from eq. 3.dcccxxviii, and the additional term is at most $\|D_U^\sharp u\|_2^2$, bounded by the component gradient and bare mass energy. The latter pays any one-edge term at a rough true boundary, where $\xi = \eta$. This proves the upper bound too. In particular this argument does not require the connection to be stationary.

Let E_D inject fields supported on any chosen fine set D, extending them by zero in the original field space. Define the compressed operator

$$\mathcal{A}_{\bullet,D} = E_D^\sharp \mathcal{A}_\bullet E_D. \tag{3.cmxlviii}$$

Its norm is the energy of that zero extension, including the artificial boundary gradients. Inequality eq. 3.cmxlvii holds on this space with exactly the same constants. For an averaging row crossing the boundary of D, its contribution is its full row evaluated on the zero extension. It is not replaced by a renormalized partial average. There is consequently no new partial-row inf-sup assertion.

The matrix eq. 3.cmxlviii depends only on the fine coefficients on D, their finite difference collar and the complete rows meeting that collar. It also retains which original inner plaquettes are present. These observations apply at the actual outer boundary as well as at an artificial interior boundary.

3.28.2. *A locally determined slow partition.* We give an explicit weighted cover so its choice does not depend on a globally ordered greedy selection. Fix a large integer m, and then choose the comparison-length slope $\delta > 0$ so $m\delta$ is sufficiently small. Increase the separation so $\rho\delta \geq C_L$. The comparison length ℓ is eq. 3.dcclxx. Put

$$r_x = m\ell(x)$$

at every ambient fine site x, including the geometric exterior. Initially suppose the periods are large enough that the balls below embed. The periodic case is handled after the construction.

Choose a fixed nonnegative smooth bump φ, equal to one on the ball of radius $1/4$ and supported in the ball of radius $1/2$. Define

$$\begin{aligned} \omega_x &= \eta^4 r_x^{-4}, \\ N(y) &= \sum_x \omega_x \varphi((y-x)/r_x)^2, \\ \chi_x(y) &= N(y)^{-1/2}\varphi((y-x)/r_x). \end{aligned} \tag{3.cmxlix}$$

The sums are ambient fine-site sums. The radius and weight at a centre are fixed while y varies.

Lemma 3.99 (Weighted slow partition)**.** Uniformly in mesh and depth,

$$\sum_x \omega_x \chi_x(y)^2 = 1, \qquad c \leq N(y) \leq C, \qquad |D^a \chi_x(y)| \leq C_a r_x^{-a} \quad (a = 0, 1, 2). \tag{3.cml}$$

Only centres with $|x - y| \leq C r_y$ contribute at y, and

$$\sum_{x:\, y \in B(x, 2r_x)} \omega_x \leq C. \tag{3.cmli}$$

The same holds with any fixed enlargement of those balls, after reducing $m\delta$. All assigned scales in an enlarged ball are comparable within a fixed L factor. ■

Proof. If $|x - y| \leq a r_x$ for a fixed enlargement factor a, then

$$|r_x - r_y| \leq m\delta |x - y| \leq am\delta r_x.$$

Thus r_x and r_y are comparable, and all such centres lie in a fixed multiple of the radius-r_y ball. Conversely, in a sufficiently small fixed multiple of that ball, the bump is bounded below. Since $\ell \geq \eta$, $r_y \geq m\eta$. Fine lattice counting in these balls therefore gives

$$\sum_{|x-y| \leq c r_y} \eta^4 r_x^{-4} \geq c, \qquad \sum_{|x-y| \leq C r_y} \eta^4 r_x^{-4} \leq C.$$

This proves the bounds for N and the weighted overlap. In particular the large number of fine centres has been canceled by their physical weights, rather than estimated by an unweighted count.

Differentiate N in y. Only the bump is differentiated; each derivative costs r_x^{-1}. The preceding comparisons give $|D^a N(y)| \leq C_a r_y^{-a}$ for $a \leq 2$. The product rule and its positive lower bound prove the cutoff estimates. The partition identity is its definition. Finally $\ell \leq \xi \leq L\ell$; the same Lipschitz comparison makes all assigned scales in these balls comparable. The separation excludes nonadjacent layers on the fixed enlargements used here. □

The partition is local in the geometry. Here is the point needed for exact carrier dependence. If x is assigned level j, a candidate in the infimum defining $\ell(x)$ at level at most $j - 2$ has distance at least $\rho\xi_{j-1}$. For $\rho\delta > 2L$, its distance term alone exceeds $2\xi_j$, so it cannot attain an infimum at most ξ_j. A candidate at level at least j cannot improve the zero-distance candidate x. Only the adjacent lower level need be examined, at distance at most ξ_j/δ. Consequently all radii and cutoff normalizations used near x are decided by the layer labels in

$$B(x, C_L \ell(x)/\delta). \tag{3.cmlii}$$

Increase $\rho\delta$ so this collar meets only the fixed adjacent levels. This may be a larger collar than the support of χ_x; it is included in the carrier. It cannot be omitted on the ground that ℓ is merely comparable to the assigned scale.

3.28.3. *Local inverse columns and the commutator.* Let D_x contain the fine variables in $B(x, 2r_x)$, intersected with the original field space. Its finite carrier collar is fixed large enough for the complete rows. Write

$$I_x = E_{D_x} \mathcal{A}_{\bullet, D_x}^{-1} E_{D_x}^{\sharp}.$$

The two outside injections specify zero extension; I_x is not an inverse on a newly defined determining region. In particular the original outward rows and bare boundary are still used in its compression.

Lemma 3.100 (Uniform local columns)**.** For some $s_0 > 0$,

$$\|1_y I_x 1_z\|_{\mathcal{F}_z \to \mathcal{E}_y} \le C e^{-s_0 d(y,z)}. \tag{3.cmliii}$$

The constants are independent of the patch radius, mesh and depth. The same conclusion holds for the small complex C^1 tubes of the actual regular real coefficients. ■

Proof. The unweighted energy estimate is eq. 3.cmxlvii restricted to the zero-extension subspace. For a spatial exponential weight whose logarithm changes by at most $s\eta/\xi$ on a fine step, the covariant product rule changes the principal form by at most $Cs\|u\|_{\mathcal{E}}^2$. The oscillation of that weight on one complete q or Q carrier is $O(s)$, so the same estimate holds for their positive row forms. The Wilson plaquette form has this bound by its exact finite insertion formula. Taking s small compared with c preserves coercivity. Multiplication by the weight preserves zero extension, so this argument is unchanged by the artificial Dirichlet boundary. Choose the logarithm to be the truncated distance from B_z in the actual comparison metric. The weighted energy bound gives eq. 3.cmliii.

For a complex tube, the local operator coefficients are analytic. The relative form estimates used in the complex inverse construction apply to the two local parts of eq. 3.cmxlvi. Restricting their inputs by zero extension cannot enlarge those relative bounds. A fixed sufficiently small tube is therefore a Neumann perturbation of every weighted real compression, with the same constants. No Hermitian positivity at the complex coefficient is asserted. □

Lemma 3.101 (Slow commutator)**.** The local columns also satisfy

$$\|1_y [\mathcal{A}_\bullet, \chi_x] I_x \chi_x 1_z\|_{\mathcal{F}_z \to \mathcal{F}_y} \le \frac{C}{m} e^{-s_1 d(y,z)} \tag{3.cmliv}$$

for a fixed $s_1 > 0$, whenever the indicated supports meet. Its output and input lie in fixed enlargements of the patch. Its coefficients depend only on the data there and in eq. 3.cmlii. ■

Proof. In a covariant second difference, the exact scalar product rule expresses the commutator as first differences of χ_x times first covariant differences of the unknown, and second differences of χ_x times the unknown. There is no derivative of a connection coefficient in this identity. The same rule for the curl-divergence principal form of the vector operator has these two types of terms. The remaining plaquette kernel has bounded finite-shift coefficients by Lem. 3.87; its commutator is smaller. After multiplying the output L^2 norm by ξ_y, these bounds are

$$\frac{C}{m} \{ \|D_U u\|_{2, \mathcal{D}_y} + \xi_y^{-1} \|u\|_{2, \mathcal{D}_y} \}.$$

Here $\mathcal{D}_y$ is a fixed cell enlargement, with comparable assigned scales.

For the averaging terms introduce the scalar cutoff value at the row's top. The row commutators are $q\chi_x - \chi_{x,\mathrm{top}} q$ and $Q\chi_x - \chi_{x,\mathrm{row}} Q$. Every difference of cutoff values on their complete carrier is at most C/m. Their normalized row and column bounds therefore give that same factor in the physical L^2 mass norm. The physical transpose is essential here; no fine row count remains. Use Lem. 3.100 and sum the finitely many neighbouring cells. This proves the displayed exponential

bound with a slightly smaller rate. The local matrices, cutoffs and complete row carriers give its stated exact dependence.

At a true boundary, the full interior curl-divergence identity need not hold. Every such stencil is in the bare layer. There $\xi = \eta$; each present matrix entry is $O(\eta^{-2})$, its cutoff difference is $O(1/m)$, and it reads only finitely many neighbours. Thus its force norm is bounded by C/m times the bare mass energy directly. This also covers a missing neighbour in the scalar graph Laplacian. No second-difference cancellation across an absent stencil is required. □

This proof works next to a rough outer boundary. The identity is the commutator of the original finite matrices and their diagonal cutoffs. Missing exterior plaquettes are not filled in, and no local Coulomb gauge across that boundary has been assumed.

3.28.4. *An exact local parametrix.* Define

$$\mathcal{M} = \sum_x \omega_x \chi_x I_x \chi_x. \tag{3.cmlv}$$

All sums are finite on a finite torus. The preceding bounds make them uniformly bounded between cell source and energy norms. Because χ_x has support strictly inside D_x, compression and extension give the exact identity

$$\mathcal{A}_\bullet \mathcal{M} = 1 + \mathcal{T}, \qquad \mathcal{T} = \sum_x \omega_x [\mathcal{A}_\bullet, \chi_x] I_x \chi_x. \tag{3.cmlvi}$$

To verify its main term, multiply by χ_x on the output of the compressed equation:

$$\chi_x \mathcal{A}_\bullet I_x \chi_x = \chi_x^2.$$

The weighted sum of these squares is one by eq. 3.cml. Thus there is no residual background-replacement term in eq. 3.cmlvi. The operator is exactly the same local operator inside each compression.

For a block operator $\mathcal{B}$ put

$$\|\mathcal{B}\|_{\mathrm{Sch,s}} = \max\left\{ \sup_y \sum_z e^{sd(y,z)} \|\mathcal{B}_{yz}\|, \sup_z \sum_y e^{sd(y,z)} \|\mathcal{B}_{yz}\| \right\},$$

using $\mathcal{F} \to \mathcal{F}$ for $\mathcal{T}$ and $\mathcal{F} \to \mathcal{E}$ for $\mathcal{M}$. The metric growth reserve is chosen below the column decay. Equations eq. 3.cmli and eq. 3.cmliv give

$$\|\mathcal{T}\|_{\mathrm{Sch,s}} \le C/m, \qquad \|\mathcal{M}\|_{\mathrm{Sch,s}} \le C. \tag{3.cmlvii}$$

Indeed for fixed output cell, all contributing centres have radius comparable to $m\xi_y$; their physical weights have bounded sum. Sum the remaining exponential input-cell kernel. The column calculation is the same with the two cells reversed. The local column estimate, rather than the total energy norm of a whole patch, prevents a spurious factor m^2.

3.28.5. *Carriers and absolute walk sums.* Choose a tagged mixed-scale carrier for each centre as follows. Include its inverse patch, the complete row collars and the geometric decision collar eq. 3.cmlii. Cover that set by the ambient $M_{\mathrm{c}}\xi_j$ cubes at its adjacent assigned levels. Choose the fixed carrier cube integer M_{c} large enough compared with C_L/δ, and then enlarge ρ once more compared with LM_{c}. Only a bounded number B_L of tagged cubes is needed. Connect intersecting or neighbouring cubes at the same or adjacent level. Their connected carrier has length at most B_L. This records geometry dependence as well as matrix endpoints. Include also the canonical tagged M_{c}-cube of each input and output assigned cell. Any short joining paths cost a fixed number of tags, absorbed into B_L. For multiplication use the shifted length $\sigma_c(X) = b_L + d_c(X)$, where $b_L \ge 1$ exceeds the fixed cost of joining two carriers at a common intermediate cell. Then

$$\sigma_c(X \vee Y) \le \sigma_c(X) + \sigma_c(Y).$$

Here $X \vee Y$ includes that short join if the original tagged vertex sets were disjoint despite meeting physically. This convention is equivalent, up to one fixed factor in the norm, to the usual connected tree length. Enlarge B_L once to bound σ_c for every local carrier.

Fix any desired finite carrier rate $\kappa_c > 0$. The parameters are chosen in this order: first κ_c and the operator gap constants, then m sufficiently large, then δ, M_{c}, and ρ as above. The bound on the number of carrier cubes is independent of their physical radius after this choice of M_{c}. Thus m can be chosen so

$$(C/m)e^{\kappa_c B_L} < 1/2. \tag{3.cmlviii}$$

The global curvature and complex tube ceilings are reduced only after these finitely many fixed parameters. None depends on depth or volume. The carrier cube size M_{c} is distinct from a prescribed history cube size. A statement giving only one history cube of separation must verify that its physical width meets the present ρ bound. That history-admission comparison is not part of the local inverse theorem.

Group local terms with the same carrier. For an operator-valued family $\mathbf{B} = (B_X)$, use the weighted Schur norm with the additional sum over X and factor $e^{\kappa_c \sigma_c(X)}$. The two estimates eq. 3.cmlvii then become

$$\|\mathbf{T}\|_{\mathrm{Sch,s},\kappa_{\mathrm{c}}} < 1/2, \qquad \|\mathbf{M}\|_{\mathrm{Sch,s},\kappa_{\mathrm{c}}} \le C_{\kappa_c}.$$

These are sums of norms of actual coefficients. No cancellation between distinct carriers was used to make the first estimate small.

Proposition 3.102 (Positive inverse walks on the determining geometry)**.** For $\bullet = s, F$, the actual inverse has the exact absolutely convergent walk representation

$$\mathcal{A}_\bullet^{-1} = \mathcal{M} \sum_{n \ge 0} (-\mathcal{T})^n. \tag{3.cmlix}$$

Its carrier coefficients are holomorphic functions of the local background data and the declared local determining geometry. They have a uniform weighted cell-to-cell energy norm with the carrier rate just fixed. This holds on the real regular domain and its specified small complex C^1 tubes. ■

Proof. The Schur family norm is submultiplicative. In a product, the intermediate cell joins the two contributing carriers. Use their recorded short join and the subadditive shifted length. Its carrier weight is at most the product of the two weights, and the output-input distance obeys the triangle inequality. Thus the absolute sum of walks of length n is at most $C_{\kappa_c} 2^{-n}$. This proves convergence in the stated family norm. Multiply eq. 3.cmlix by $\mathcal{A}_\bullet$ and use eq. 3.cmlvi. The product is the identity. The positive operator is invertible by eq. 3.cmxlvii, so the series is its inverse.

Each local factor is a finite matrix inverse of eq. 3.cmxlviii, multiplied by local cutoff and stencil coefficients. Its matrix entries, original row list and original plaquette list are determined inside its recorded carrier. Agreement of two determining geometries and backgrounds there therefore gives exactly the same coefficient. This is stronger than a shielding estimate. The local inverse is analytic on its uniformly admitted complex tube. Products are analytic on the union carrier, and the normally convergent walk sum is analytic there. □

3.28.6. *Small periods and the scope of the locality statement.* If the physical balls wrap around a small compatible torus, first lift the local coefficients and geometry to a finite periodic cover whose periods exceed the fixed patch and decision diameters. A bounded number of copies suffices, since the original periods contain the largest assigned blocks. The lifted positive form has the same uniform gap. Its inverse preserves the deck-invariant subspace by uniqueness. On that subspace it is exactly the original inverse.

There is no multiplicity loss in the Schur contraction. Before projecting a filed family, average it over the finite deck group, translating its carrier tags as well. This step is needed if the auxiliary carrier grid itself is not deck invariant. The total inverse is unchanged by uniqueness, and the

absolute family norm does not increase. The averaged family is now deck covariant. For a base source cell, its lift is the sum of its deck images. The projected kernel is the sum of the corresponding cover kernel columns. A base row sum is consequently bounded by the cover row sum; the column statement follows by deck covariance. Projection of a connected carrier cannot increase its length. The same absolute walk estimates descend with the same contraction constant. This argument concerns classical linear inverses; it does not identify a quantum partition function with the partition function on a larger cover.

The result constructs local walks for $\mathcal{A}_s^{-1}$ and the full Feynman inverse. The exact compositions with the site projection, coarse inverses and independent current are supplied by Prop. 3.112. Their strong output and cube-parameter versions are Prop. 3.171 and Lem. 3.172. The carriers retain their tagged mixed-scale determining geometry. Equality with a separately prescribed signed or absolute-contour precision requires that precision's gauge comparison.

3.29. **Local carrier expansions for the positive coarse inverses.** We invert the actual positive coarse matrices while retaining the full background carriers of their coefficients. The short part is cut by its recorded carrier, and its matrix powers are combined before taking absolute values. This permits the real Hilbert-space gap to control the absolute walk expansion. The local positive inverses used as inputs are those of Prop. 3.102; no compression of an indefinite projected operator is inverted here.

The initial carrier tile size is M_0. The final size M_c will be chosen larger after a finite short-power construction. Both are aligned powers of L. Neither is silently identified with a prescribed history size M_h. The result requires the separation specified below. Admission of a separately fixed history geometry remains a comparison to be checked.

3.29.1. *The actual matrices and their physical normalization.* Retain the physical site and field pairings. Put

$$\begin{aligned} G_s &= \mathcal{A}_s^{-1}, \qquad F = G_s q^\sharp \Xi^{-2}, \qquad M = F^\sharp F, \\ G_F &= \mathcal{A}_F^{-1}, \qquad \Theta = M - (D_U F)^\sharp G_F (D_U F). \end{aligned} \tag{3.cmlx}$$

Here $\mathcal{A}_s, \mathcal{A}_F$ are the positive local operators in eq. 3.cmxlvi. The coarse Hilbert norm is $\sum_y \xi_y^4 |b(y)|^2$. Thus

$$(Jb)(y) = \xi_y^2 b(y), \qquad \widehat{M} = JMJ^{-1}, \qquad \widehat{\Theta} = J\Theta J^{-1} \tag{3.cmlxi}$$

is the unitary passage to counting ℓ^2. All matrix families and ordinary transposes below refer to this normalization. Physical transposes have not been replaced by unweighted transposes in eq. 3.cmlx.

Both normalized real matrices have fixed positive gaps. The Gram bound is eq. 3.dcclxxvi. To check the second one define

$$K_L = \mathscr{D}_U + D_U R D_U^\sharp + Q^\sharp a_f Q.$$

The real lower bound in eq. 3.dcccxxviii and the upper energy bound for $\mathcal{A}_F$ give $K_L \geq \delta \mathcal{A}_F$ for one $\delta > 0$. Since

$$K_L = \mathcal{A}_F - (D_U F) M^{-1} (D_U F)^\sharp,$$

the operator $C = G_F^{1/2} (D_U F) M^{-1/2}$ obeys $I - CC^\sharp \geq \delta I$. Hence $\|C\|^2 \leq 1 - \delta$ and $I - C^\sharp C \geq \delta I$. Conjugating gives

$$\delta M \leq \Theta \leq M. \tag{3.cmlxii}$$

Consequently, for $A \in \{\widehat{M}, \widehat{\Theta}\}$,

$$aI \leq A \leq bI \tag{3.cmlxiii}$$

with $a, b > 0$ independent of depth, compatible volume and the sufficiently separated geometry. This is a real positivity argument; complex continuation will use a perturbation of these real matrices.

3.29.2. *Actual coefficient families before coarse inversion.* A tag is a complete ambient $M_0\xi_j$ cube with its level, row weights and determining-geometry data retained. A carrier X records a finite connected collection of tags and its fixed short joins. Its endpoints include the canonical assigned-level tags of the matrix input and output cells. Use the guarded length

$$\sigma_0(X) = b_L + d_{c,0}(X)$$

from the positive-walk construction, with b_L paying the fixed cost of joining two carriers at a common cell. In particular $\sigma_0(X \vee Y) \le \sigma_0(X) + \sigma_0(Y)$. The length is that of the recorded full carrier, not a shorter connection through unrecorded geometry. Each tag covers assigned cells of its declared level, or is one of the fixed short joining tags. This is the covering used for the positive-walk coefficients.

For a counting-normalized matrix family $\mathbf{A}$ write $A_{yz} = \sum_X A_{yz}[X]$ and set

$$\|\mathbf{A}\|_{s,\kappa,0} = \max\left\{ \sup_y \sum_{z,X} e^{sd(y,z)+\kappa\sigma_0(X)} \|A_{yz}[X]\|, \sup_z \sum_{y,X} e^{sd(y,z)+\kappa\sigma_0(X)} \|A_{yz}[X]\| \right\}. \quad \text{(3.cmlxiv)}$$

Joining full carriers in a matrix product and using the triangle inequality for d makes this an algebra norm. The identity has its canonical one-cell carrier and a bounded norm. Transpose preserves the norm. Replacing a family by half its sum with its transpose makes each coefficient symmetric under endpoint reversal without changing a symmetric total matrix or increasing the norm.

For every fixed finite κ_0, the actual matrices in eq. 3.cmlxi have families with

$$\|\mathbf{A}\|_{s_0,\kappa_0,0} \le K_{\kappa_0}, \qquad s_0 > 0. \quad \text{(3.cmlxv)}$$

Here M_0 and the primitive parameters are first chosen as in the positive-walk theorem. The constants are uniform above their separation threshold. We verify the physical scaling in this assertion, because an unweighted fine entry sum would not prove it.

A unit counting top datum is $b_z = \xi_z^{-2} e_z$ in physical coordinates. The source $q^\sharp \Xi^{-2} b_z$ has cell L^2 norm at most $C\xi_z^{-2}$, and thus source norm $C\xi_z^{-1}$ in the notation of Lem. 3.100. The actual site walk gives

$$\|1_y F b_z\|_2 \le C(\xi_y/\xi_z) e^{-sd(y,z)}, \qquad \|1_y D_U F b_z\|_2 \le C\xi_z^{-1} e^{-sd(y,z)}. \quad \text{(3.cmlxvi)}$$

These inequalities also hold as sums over the recorded walk carriers, at the prescribed rate. Applying a fine derivative uses the energy output of that walk and its already included one-step collar. Similarly the field walk gives

$$\|1_w G_F 1_v\|_{2\to 2} \le C\xi_w\xi_v e^{-sd(w,v)}. \quad \text{(3.cmlxvii)}$$

The local q injection and all complete row carriers are retained in these coefficients.

For $\widehat{M}_{yz}$, Cauchy–Schwarz inside each physical fine cell leaves the factor $\xi_w^2/(\xi_y\xi_z)$. For the subtracted term in $\widehat{\Theta}_{yz}$ it leaves $\xi_w\xi_v/(\xi_y\xi_z)$ on the chain y, w, v, z. The level barrier eq. 3.cdliii bounds each scale ratio by a fixed L factor times $\exp\{(\log L)d/\rho\}$. Reserve that small rate before summing the exponentially decreasing cell chains with eq. 3.cdlvi. This proves eq. 3.cmlxv, including its absolute row and column sums. No fine lattice count or power of η^{-1} is inserted. Carrier unions are the actual unions of the site, field and injection carriers; the shared fine cell and the guarded join preserve locality.

The positive-walk coefficients are uniformly holomorphic on their common small complex C^1 tubes. Their normally convergent family sums and the preceding bounded compositions give the same assertion for eq. 3.cmlxv. On a smaller tube, Cauchy's estimate gives a Lipschitz bound in that family norm. This supplies the complex perturbation used below; it supplies no independent enlarged physical cube domains.

3.29.3. *Short connected carriers and the actual local counting.* Fix finite integers m, R for the moment. On the final separation threshold, every old connected carrier of length at most R has metric diameter at most $C_L M_0(R+1)$. A chain of m of them is contained in a ball of radius

$$B = C_L M_0 m(R+1). \tag{3.cmlxviii}$$

Choose the final separation much larger than LB. A metric ball of radius $4B + C_L$ then meets at most two adjacent levels. Otherwise a path between its extreme levels would cross a complete intervening layer at cost at least ρ, contradicting the bound on the ball diameter. Its reached scales are comparable within L, and its physical extent is $C_L(B+1)\xi_y$. Counting their two grids proves

$$\#\{z : d(y,z) \le 2B + C_L\} \le C_L(1+B)^4. \tag{3.cmlxix}$$

This is a local consequence of separation. The global growth bound remains eq. 3.cdlvi.

The same observation proves the diameter assertion for the short carriers: neighboring complete old tiles have metric span $C_L M_0$, and fewer than $m(R+1)$ such steps cannot cross two separated interfaces. All tag eligibility, row lists and geometry decisions are read in the resulting ball, since those data were included in each old carrier. Cutoff by endpoint distance alone would not have this property.

Now choose a final aligned carrier tile size M_c, an L-power multiple of M_0, with

$$M_c \ge C_L(B+1), \qquad \rho \ge C_L L M_c. \tag{3.cmlxx}$$

The union of two radius-B balls with centers at distance at most B has a connected covering with at most B_L final tags, including its endpoint tags and short joins. The constant B_L depends on L and the fixed collar conventions, not on m, R. Write σ_c for the same guarded length on these final tags. Reblocking an old carrier into its aligned parents preserves its connections and satisfies $\sigma_c(\pi X) \le \sigma_0(X)$. Thus an old strong carrier weight pays the final one.

3.29.4. *Combine a short power before taking absolute values.* Fix the desired finite final carrier rate $\kappa > 0$ and use eq. 3.cmlxv at a stronger rate $\kappa_0 > \kappa + 1$. Choose $\beta \ge 2b$ and put

$$H = I - A/\beta, \qquad S = H_{\le R}, \qquad E = H_{>R}, \qquad H_{\le R} = \sum_{\sigma_0(X)\le R} H[X]. \tag{3.cmlxxi}$$

The families are first transpose symmetrized. This cutoff consequently preserves the real symmetry of S. Let $K_H \ge 2$ bound the old family norms of H, S, including the identity carrier. The tail obeys

$$\|\mathbf{E}\|_{s,\kappa,c} \le K_H e^{-(\kappa_0-\kappa)R} \quad (0 \le s \le s_0). \tag{3.cmlxxii}$$

Here and below the old coefficients are reblocked into final tags when the subscript c is used. This estimate cuts by the full old carrier; a long dependency with nearby matrix endpoints remains in the tail.

For R above a fixed threshold the unweighted tail operator norm is at most $a/(2\beta)$. Since $I/2 \le H \le (1 - a/\beta)I$ on the real domain,

$$0 \le S \le qI, \qquad q = 1 - a/(2\beta) < 1. \tag{3.cmlxxiii}$$

The real spectral estimate is used only at this point. The short matrix has propagation at most B/m. Therefore its mth power has range B, and eq. 3.cmlxix and the Hilbert row estimate give

$$\|S^m\|_{\mathrm{Sch}} \le C_L(1+B)^2 q^m. \tag{3.cmlxxiv}$$

Indeed the squared Hilbert–Schmidt norms of the blocks in one row sum to at most $r_0\|S^m\|_{2\to 2}^2$, where r_0 bounds the colour block dimension. Cauchy–Schwarz costs the square root of the number of reached cells. The same holds for columns. This estimate concerns the already combined matrix entries of S^m, not the absolute values of its individual paths.

Assign that combined entry $(S^m)_{yz}$ the symmetric full carrier covering the two radius-B balls of y, z. It includes every short path and its full coefficient dependencies. This is a local function:

agreement of backgrounds and recorded geometry on that carrier leaves all contributing short factors and their eligibility unchanged. Its final length is at most B_L. Denote the resulting family by $\mathbf{N}_m$. At any $0 < s \le s_0$ it has bound

$$\|\mathbf{N}_m\|_{s,\kappa,c} \le C_L e^{sB+\kappa B_L}(1+B)^2 q^m. \tag{3.cmlxxv}$$

In particular the cancellation used in eq. 3.cmlxxiv takes place inside a specified local coefficient before filing it.

Keep the remaining part as an actual word family:

$$\begin{aligned} E_j &= H^j - S^j = \sum_{i=0}^{j-1} S^i E H^{j-1-i}, \\ \|\mathbf{E}_j\|_{s,\kappa,c} &\le j K_H^j e^{-(\kappa_0-\kappa)R}. \end{aligned} \tag{3.cmlxxvi}$$

The family products file into their full joined carriers. Transpose symmetrization preserves the bound and the total operator E_j, since both H^j and S^j are symmetric. Define

$$\mathbf{T} = \mathbf{N}_m + \mathbf{E}_m. \tag{3.cmlxxvii}$$

Its total operator is exactly H^m.

Here is a noncircular choice of all parameters. First fix the real gaps and $\kappa_0 > \kappa + 1$, using the earlier fixed energy estimates. The comparison length slope and the carrier grids are auxiliary; changing them for the walks does not change the actual coarse matrices or their already proved gaps. Next construct the positive-walk inputs and their initial M_0, K_H, s_0. Choose a positive integer D so that $(\kappa_0-\kappa)D > 2\log K_H + 2$ and $R = Dm$ already meets the spectral-tail threshold for $m \ge 1$. Then choose m large. The quantity

$$\Gamma_m = 2C_L e^{\kappa B_L}(1 + C_L M_0 m(Dm+1))^2 q^m + m K_H^m e^{-(\kappa_0-\kappa)Dm} \longrightarrow 0. \tag{3.cmlxxviii}$$

Choose m with $\Gamma_m < 1/4$, put $R = Dm$, and choose M_c as in eq. 3.cmlxx. Set

$$s_* \le \min\{s_0/2, \log 2/(B+1)\}. \tag{3.cmlxxix}$$

Only then require

$$\rho \ge \max\{\rho_0, C_L L M_c, 64(\log L)/s_*\}. \tag{3.cmlxxx}$$

The last reserve allows later scale conversions and global cell sums at a smaller spatial rate. All gap and input-family bounds used to choose m are uniform above their initial separation threshold ρ_0. Thus no constant used earlier is being determined by the final geometry. The resulting family satisfies

$$\|\mathbf{T}\|_{s_*,\kappa,c} < 1/4 \quad \text{at real inputs.} \tag{3.cmlxxxi}$$

This retains the prescribed finite carrier rate κ through the enlarged carrier tile M_c. It does not retain a prescribed spatial rate: the positive s_* has been decreased explicitly.

3.29.5. *The exact coarse inverse and complex continuation.*

Proposition 3.103 (Carrier inverses of the actual positive coarse matrices)**.** For every prescribed finite carrier rate κ, choose the parameters and require the separation in eq. 3.cmlxxx. Then both M^{-1} and Θ^{-1} have exact local carrier families. In counting normalization these satisfy

$$\|\widehat{\mathbf{M}^{-1}}\|_{s_*,\kappa,c} + \|\widehat{\mathbf{\Theta}^{-1}}\|_{s_*,\kappa,c} \le C_\kappa. \tag{3.cmlxxxii}$$

The constants are uniform in depth, compatible volume and admitted geometry. The families are holomorphic on a common sufficiently small complex C^1 tube about the real regular domain and retain exact dependence on their recorded background and geometry. The physical inverses are obtained by $J^{-1}(\cdot)J$. ∎

Proof. For either normalized A, let $\star$ denote the joined-carrier product. The explicit inverse family is

$$\mathbf{B} = \beta^{-1} \left(\sum_{j=0}^{m-1} \mathbf{H}^{\star j} \right) \star \sum_{v \geq 0} \mathbf{T}^{\star v}. \tag{3.cmlxxxiii}$$

Its norm is at most $\beta^{-1} \sum_{j<m} K_H^j / (1 - \|\mathbf{T}\|)$. Every factor is an actual local coefficient and every product carries its full joined union. The sum is absolutely convergent in the family norm. At complete coefficients its operator is

$$\beta^{-1} \left(\sum_{j<m} H^j \right) (I - H^m)^{-1} = A^{-1}.$$

This follows from $(I - H) \sum_{j<m} H^j = I - H^m$; all factors are polynomials or convergent functions of H. The real positive matrix is invertible by eq. 3.cmlxiii. Final transpose symmetrization of the family preserves this inverse, its locality and its two-sided norm.

For complex continuation keep the same geometry, cutoff, powers and filing balls. Suppose

$$\|\mathbf{A}(z) - \mathbf{A}(z_0)\|_{s_0,\kappa_0,0} < \varepsilon, \tag{3.cmlxxxiv}$$

where z_0 is real. Tail projection is bounded in the input family norm. Each finite product is Lipschitz with bound $j(K_H + 1)^{j-1}$ times the perturbation. For the combined near entry, the same finite matrix identity bounds $S(z)^m - S(z_0)^m$ in unweighted Schur norm; its single local ball carrier costs at most $e^{s_* B + \kappa B_L}$. Thus $\|\mathbf{T}(z) - \mathbf{T}(z_0)\|_{s_*,\kappa,c} \leq C_{m,R,\kappa} \varepsilon$. Choose ε so this is below $1/4$. The actual input-family analyticity proved above then provides a common physical C^1 tube on which $\|\mathbf{T}(z)\| < 1/2$.

Series eq. 3.cmlxxxiii is normally convergent there. Its polynomial identity still makes it the inverse of the actual complex matrix. No Hermitian inequality is asserted at a complex input. Each filed coefficient is analytic in the data on its union carrier; the absolutely convergent sum over words with that recorded carrier preserves this local dependence. Banach Cauchy estimates give local derivative families on a smaller tube. Equivalently the actual first derivative has the local representation $-A^{-1}(DA)A^{-1}$, and higher derivatives are the finite ordered resolvent products. Their joined carriers retain every marked input. This proves the proposition. □

Corollary 3.104 (Positive row Schur inverses)**.** The same construction applies to an actual coarse matrix with a bounded number of row types per assigned cell, after its physical counting-unitary normalization, whenever its real gap and its actual input family satisfy eq. 3.cmlxiii and eq. 3.cmlxv. Its inverse then has the bound in eq. 3.cmlxxxii, with the corresponding finite constants and carrier tiles. In particular it applies to

$$\begin{aligned} T_0^{\mathrm{n}} &= \Xi_\Gamma^2 a_f^{1/2} Q G_0 Q^\sharp a_f^{1/2} \Xi_\Gamma^{-2}, \\ G_0 &= G_F + G_F (D_U F) \Theta^{-1} (D_U F)^\sharp G_F. \end{aligned} \tag{3.cmlxxxv}$$

There are at most eight rows per assigned top, as in eq. 3.dcccxxxviii. Small symmetric actual family-norm perturbations of this matrix have the same inverse construction through eq. 3.cmlxxxiv. ■□

Proof. Bundle the actual row species at each top; their colour block dimension is at most twenty-four. The counting proof and Hilbert row estimate use only this uniform bound. The row metric locations and their complete carriers differ from their incident tops by a fixed distance, so the same local and global geometry estimates apply.

For eq. 3.cmlxxxv, multiplication by $K_L = \mathcal{A}_F - (D_U F) M^{-1} (D_U F)^\sharp$ verifies the displayed Woodbury expression for $G_0 = K_L^{-1}$, using $\Theta = M - (D_U F)^\sharp G_F (D_U F)$. It is an exact composition of the local families already constructed. More explicitly, eq. 3.cmlxvi and eq. 3.cmlxvii give

$\|1_y G_F D_U F J^{-1} e_z\|_{\mathcal{E}_y} \le C e^{-s' d(y,z)}$. The only intermediate scale ratio is ξ_w/ξ_z, absorbed by the retained separation reserve. Thus G_0 has the same source-to-energy family grade as G_F. The normalized trace $\mathcal{Q}_n = \Xi_\Gamma^2 a_f^{1/2} Q$ is bounded from field energy to counting row space and has the actual finite complete carriers. Its counting-to-physical transpose is $\mathcal{Q}_n^\sharp = Q^\sharp a_f^{1/2} \Xi_\Gamma^{-2}$. The actual row density gives the stronger source bound

$$|(\mathcal{Q}_n^\sharp e_c)(b)| \le C \xi_c^{-3} \mathbf{1}_{\mathrm{carrier}(c)}(b), \qquad \xi_y \|1_y \mathcal{Q}_n^\sharp e_c\|_2 \le C. \tag{3.cmlxxxvi}$$

On its complete carrier ξ_y/ξ_c is bounded by a fixed L factor, and physical cell volume is ξ_y^4. Thus the injection belongs to the force-cell space already admitted by the positive walks. Composing these local injections with G_0 gives the actual input family of T_0^{n} in the proved norms. The real lower bound is the smooth-row-seed argument in eq. 3.dcccxl; the upper bound follows from the same trace and energy inverse bounds.

Choose the Gram and Θ inverse families at a stronger intermediate carrier rate first. Their finite compositions then meet the stronger input rate needed for this second application. This adds only finitely many parameter choices: the final carrier tile and separation are increased after both constructions. The proof does not identify that tile with M_h. Finally the perturbation assertion is precisely the common-tube argument above, once the stated actual family difference has been bounded. □

3.29.6. *Periods and the admission retained by the result.* For small periods choose one finite periodic cover large enough for all primitive and coarse patches and both aligned tile grids. Construct the primitive positive-walk families, their coarse compositions and the short-power inversion on that same cover, before any family projection. Thus the aligned M_0 to M_c parent relation is used before projected translated tag phases appear. The positive operators and actual coarse matrices descend on the deck-invariant subspace. If the auxiliary tile grid is not itself invariant under deck shifts, first average the final filed inverse family over the deck group, translating its carrier tags as well. Its total inverse is unchanged by uniqueness, and its family norm does not increase. The average is now covariant as a family.

Project its carriers with their level and periodic identification data retained. A base row sum is bounded by the cover row sum over all source lifts. By deck covariance the base column sum is bounded by the cover column sum over all output lifts. Connected projection cannot increase carrier length, and base distances cannot increase. Thus there is no sheet-count loss. Projected translated-cover tags are part of the recorded geometry; agreement with an independently prescribed history tiling is not assumed.

The result supplies actual local coefficients for the positive Gram and coarse Schur inverses on the admitted proper row graph. All complex bounds concern the common physical tube just proved. Independent enlarged cube input domains, separately prescribed indicator fillings, and hereditary tower comparisons require their own admission arguments. In particular a source history with only its stated M_h separation is not declared to satisfy eq. 3.cmlxxx by renaming M_h as the carrier size M_c.

3.30. **Local carriers for the named current propagators.** We assemble the actual site projection, gauge restoration and two current inverses in their prescribed order. The local positive walks are those of Prop. 3.102. All coefficients retain the actual complete rows, outward endpoints and inner plaquette set. The carrier cube size is M_{c}; it is independent of a prescribed history cube size M_{h}. Start with the actual real curvature-small reference and determining geometry of that positive-walk theorem. The final separation requirements of the finitely many coarse-inversion stages are retained. Complex coefficients range over their common weighted C^1 tube about this real reference. The independent current is specified below. The fixed positive site and field row weights are included in the declared local coefficient data.

Use the shifted connected carrier length σ_c, canonical endpoint tags and short joins of that construction. For a family $\mathbf{T} = (T_C)$ between normalized cell spaces, write

$$\|\mathbf{T}\|_{s,\kappa} = \max\Big\{ \sup_y \sum_{z,C} e^{sd(y,z)+\kappa\sigma_c(C)} \|1_y T_C 1_z\|, \\ \sup_z \sum_{y,C} e^{sd(y,z)+\kappa\sigma_c(C)} \|1_y T_C 1_z\| \Big\}. \tag{3.cmlxxxvii}$$

Only families whose carriers contain their matrix endpoints are used. Products retain the indicated short join at each intermediate cell. The carrier inequality and triangle inequality make this norm submultiplicative whenever the intermediate normalized spaces agree.

Site L^2 uses the physical fine measure. The field energy is X, its dual is X^*, and Z is the actual site second-difference space in eq. 3.cmxxix. Their local norms use the assigned cells and fixed derivative collars. Energy spaces and their duals are localized by a smooth square partition on fixed enlargements of the disjoint pillars. Each profile has first difference at most C_L/ξ_y, and their overlaps are bounded. Such a partition is obtained by normalizing squared smooth bumps on the pillar cubes; overlapping scales are comparable within L. Multiplication is bounded in the physical energy and dual norms by the product rule and the mass term. Inserting this square partition in a product proves the Schur multiplication estimate, with one fixed localization constant. Absorb that constant in the algebra norm used in eq. 3.cmlxxxvii. This avoids a sharp characteristic-function cutoff in a Sobolev space. Pure L^2 blocks may use disjoint cell ownership. All localization collars are retained in the carriers. Coarse spaces will always specify their physical-to-counting unitary. No unweighted array transpose is substituted for a physical transpose.

Fix a finite carrier rate $\kappa > 0$, and leave a strict reserve in the positive-walk spatial rate. Separation can be increased so that, for some sufficiently small ε_g,

$$\max\{\xi_y/\xi_z, \xi_z/\xi_y\} \le C_L e^{\varepsilon_g d(y,z)}. \tag{3.cmlxxxviii}$$

Indeed crossing one nonadjacent layer costs a fixed multiple of ρ in the layer distance, while its scale changes by a factor L; adjacent-cell ratios cost only C_L. The same choice leaves the cell-count reserve used below. All geometric constants, tile choices and rate reserves are fixed before the depth or volume.

3.30.1. *The output site factor has two differences.* The positive site parametrix uses the local output factor

$$L_x = \chi_x E_{D_x} A_{s,D_x}^{-1} E_{D_x}^{\sharp} \chi_x.$$

Its artificial Dirichlet boundary lies outside the support of χ_x, with room for every complete row and derivative collar. Let Z_y denote the local Z norm.

Lemma 3.105 (A current-free second-difference output)**.** For a source in the assigned cell B_z,

$$\|1_y L_x 1_z f\|_{Z_y} \le C \frac{\xi_z}{\xi_y} e^{-s_1 d(y,z)} \|f\|_{2,B_z} \tag{3.cmlxxxix}$$

whenever this local factor is present. The constants are uniform in the patch radius, mesh and depth. The coefficient retains the positive-walk carrier, enlarged only by its fixed derivative collar. No physical-current bound is a premise. ∎

Proof. Put $v = E_{D_x} A_{s,D_x}^{-1} E_{D_x}^{\sharp} \chi_x 1_z f$ and $u = \chi_x v$. The local positive column bound gives

$$\|v\|_{\mathcal{E}_y} \le C \xi_z e^{-s_0 d(y,z)} \|f\|_{2,B_z}.$$

Multiplication by χ_x preserves that bound on fixed cell enlargements. Consequently the mass and first-gradient parts of Z_y have the stated additional factor ξ_y^{-1}.

For the second differences the exact compressed equation, multiplied by the interior cutoff, gives in the original site space

$$\Delta_U u = \chi_x^2 1_z f + [A_s, \chi_x]v - q^\sharp a_s q u. \tag{3.cmxc}$$

There is no artificial boundary source on the support of this equation: its cutoff is separated from the compression boundary by more than the row range. The slow-commutator estimate bounds its middle term locally by $C\xi_y^{-1}\|v\|_{\mathcal{E}_{\mathcal{D}_y}}$. For the last term the actual transported row bound gives

$$\|q^\sharp a_s q u\|_{2,B_y} \le C\xi_y^{-2}\|u\|_{2,\mathcal{D}_y} \le C\xi_y^{-1}\|u\|_{\mathcal{E}_{\mathcal{D}_y}}.$$

The enlargement $\mathcal{D}_y$ has bounded metric radius and bounded cell count, with comparable scales. This uses the complete average; it is not a point evaluation of u.

At a nonbare cell take a cutoff on a fixed surrounding cube, equal to one on the derivative collar of B_y. That cube is inside the actual action region by separation. The unitary discrete Hodge calculation applied to its compactly supported scalar field gives

$$\|\nabla_U D_U u\|_{2,B_y} \le C\{\|\Delta_U u\|_{2,\mathcal{D}_y} + \xi_y^{-1}\|D_U u\|_{2,\mathcal{D}_y} + \xi_y^{-2}\|u\|_{2,\mathcal{D}_y}\}. \tag{3.cmxci}$$

In that calculation the curl of $D_U u$ is a finite curvature multiplication and shift of u, and the covariant Hodge commutator is curvature times $D_U u$. Thus it uses no derivative of curvature or of the current. It is the local compact-support version of the calculation in eq. 3.dcclxxxviii.

Insert eq. 3.cmxc and the column bound. The direct source term is present only on a bounded enlargement of B_z, where its scale ratio and distance are bounded. All other terms are at most $C(\xi_z/\xi_y)e^{-s_1 d(y,z)}\|f\|_2$, with a fixed smaller rate.

At a true outer boundary the assigned scale is η. Each first or second difference has finitely many neighbours and costs η^{-1} or η^{-2}. The atomic mass and gradient parts of the local energy therefore bound the same Z_y norm by $C\eta^{-1}\|v\|_{\mathcal{E}_{\mathcal{D}_y}}$. This proves the result without applying an interior Hodge identity across a missing plaquette. □

Proposition 3.106 (A local site inverse in the actual source norm)**.** The actual site inverse has an exact carrier family with

$$\|\mathbf{G}_s\|_{L^2\to Z;s,\kappa} \le C_\kappa, \qquad \|D\mathbf{G}_s\|_{L^2\to X;s,\kappa} \le C_\kappa \tag{3.cmxcii}$$

for a fixed $s > 0$. It is holomorphic on a common smaller complex C^1 tube about each actual real reference, in the fixed real physical norms. ■

Proof. In the positive-walk series replace only its leftmost output factor by eq. 3.cmlxxxix. An error factor between source cells, expressed in unweighted L^2 norms, carries the ratio $\xi_{\text{input}}/\xi_{\text{output}}$, since the positive source norm is $\xi\|f\|_2$. The output upgrade has exactly that same endpoint ratio. In every product these ratios telescope. The whole walk therefore pays ξ_z/ξ_y once, not once at every step. Use eq. 3.cmlxxxviii to absorb it by the fixed spatial-rate reserve. The positive error-family contraction and the shifted carrier weight are unchanged. Its absolute geometric sum proves the first bound. The local map $D : Z \to X$ proves the second.

For complex coefficients, use the real reference Hodge estimate. The weighted C^1 tube gives locally

$$\|(\Delta_U - \Delta_{U_0})u\|_2 \le C\varepsilon\{\xi_y^{-1}\|D_{U_0}u\|_2 + \xi_y^{-2}\|u\|_2\}.$$

This is the actual first-difference coefficient expansion used in the complex site comparison; it needs only the tube's first covariant difference. The local inverse column bound is already uniform on that tube. The preceding proof and the equivalence of its fixed norms therefore apply unchanged. Each factor is a local finite matrix inverse and a local differentiated cutoff, so it is holomorphic; normal convergence gives the claimed carrier-family holomorphy. □

Corollary 3.107 (A divergence source without its derivative)**.** The same family gives

$$\|\,\mathbf{G}_s D^\sharp\,\|_{X^*\to L^2;s,\kappa} \le C_\kappa. \tag{3.cmxciii}$$

This is the physical transpose of $D\mathbf{G}_s : L^2 \to X$. The source $D^\sharp f$ is interpreted by its variational pairing. The assertion requires no fine pointwise value of that divergence. ■□

3.30.2. *Variational sources and balanced site fields.* Let $\mathfrak{F}$ be the fine force space with local norm $\|f\|_{\mathfrak{F}_y} = \xi_y \|f\|_{2,B_y}$. It embeds in X^*, with a uniform constant. Use the site spaces $Y, Y^\flat$ of eq. 3.cmxxix.

Lemma 3.108 (Local positive factors accept variational sources)**.** The positive inverse families also have the bounds

$$\|\mathbf{G}_F\|_{X^*\to X;s,\kappa} + \|\mathbf{G}_s\|_{Y^\flat\to Y;s,\kappa} \le C_\kappa. \tag{3.cmxciv}$$

The site divergence $D^\sharp f$ can be used as a $Y^\flat$ source whenever $\|\ell f\|_2$ is finite. All these statements use physical variational pairings. ■

Proof. The local weighted energy proof in Lem. 3.100 applies to a functional supported on a cell, with its local energy-dual norm: testing the compressed equation and its exponential conjugate uses only that norm. Multiplying the functional by the smooth input cutoff is bounded by the same product rule. Thus the local inverse has energy-dual to energy columns with the same decay.

The commutator in Lem. 3.101 maps that energy output into the L^2 force space $\mathfrak{F}$. Consequently the rightmost error factor in a nonempty positive walk maps an energy-dual source to $\mathfrak{F}$; all subsequent error factors act in the already controlled $\mathfrak{F}$ space. The zero-error term is the local variational inverse itself. The same absolute geometric sum therefore proves the energy-dual inverse family. For the vector operator its energy space is X.

For the scalar operator the local unbalanced energy is $\mathcal{E}_y \simeq \xi_y^{-1} Y_y$. Its local dual, when used as a source in the unweighted equation, has norm $\xi_y^{-1} Y_y^\flat$, because $Y^\flat$ uses the pairing with $\ell^2 u$. Thus the scalar column picks up the ratio ξ_y/ξ_z. Absorb this once by eq. 3.cmlxxxviii, as in the preceding proof. This gives the second bound in eq. 3.cmxciv. Finally

$$|\langle D^\sharp f, \ell^2 u\rangle| = |\langle f, D(\ell^2 u)\rangle| \le C\|\ell f\|_2 \|u\|_Y.$$

No derivative of f, and hence no derivative of an independent current, is taken. □

3.30.3. *Physical top units, projection and restoration.* Write $J_s = \Xi^2$ for the unitary from the physical top space to counting ℓ^2, and define

$$\widehat{F} = F J_s^{-1} = G_s q^\sharp \Xi^{-4}, \qquad \widehat{M} = J_s M J_s^{-1} = \widehat{F}^\sharp \widehat{F}. \tag{3.cmxcv}$$

The transpose in the last equality is between counting top space and the physical fine site space. In particular $\widehat{F}^\sharp = J_s F^\sharp$; this formula keeps the physical top measure.

The actual injection $q^\sharp \Xi^{-4}$ maps a unit counting top datum to density $C\xi_z^{-4}$ on its pillar. Its local $Y^\flat$ norm is bounded, since that norm supplies ξ_z^2 and the physical L^2 norm supplies another ξ_z^2. Consequently eq. 3.cmxciv proves

$$\|\widehat{\mathbf{F}}\|_{\ell^2_{\mathrm{top}}\to Y;s,\kappa} + \|\widehat{\mathbf{F}}\|_{\ell^2_{\mathrm{top}}\to L^2;s,\kappa} + \|D\widehat{\mathbf{F}}\|_{\ell^2_{\mathrm{top}}\to \mathfrak{F};s,\kappa} \le C_\kappa. \tag{3.cmxcvi}$$

Equivalently its unabsorbed local L^2 column has factor ξ_y/ξ_z, as in eq. 3.cmlxvi. No fine-coordinate dimension appears in this normalization.

Choose the positive coarse inverse families of Prop. 3.103 at a stronger intermediate carrier rate $\kappa_1 > \kappa + 1$. Their input positive families are first obtained at another fixed rate $\kappa_2 > \kappa_1 + 1$. All ensuing finite reblockings use aligned parents and the recorded joins. Work on a common final periodic cover when needed, so the canonical tags for all products agree before descent. Until the row-Schur inversion, the following compositions are made at κ_1; their displayed weaker bounds at κ follow by decreasing the carrier weight.

Proposition 3.109 (Local projection and gauge restoration)**.** The actual maps

$$P = \widehat{F}\widehat{M}^{-1}\widehat{F}^{\sharp}, \quad R = 1 - P, \quad \Lambda = G_s R D^{\sharp}, \quad S = 1 - D\Lambda \tag{3.cmxcvii}$$

have exact local carrier families, with bounds

$$\begin{aligned} \|\mathbf{P}\|_{L^2\to Y;s,\kappa} + \|\mathbf{R}\|_{L^2\to L^2;s,\kappa} + \|\mathbf{R}\|_{Y\to Y;s,\kappa} &\le C_\kappa, \\ \|\mathbf{\Lambda}\|_{X\to Z;s,\kappa} + \|D\mathbf{\Lambda}\|_{X\to X;s,\kappa} + \|\mathbf{S}\|_{X\to X;s,\kappa} + \|\mathbf{S}^{\sharp}\|_{\mathfrak{F}\to\mathfrak{F};s,\kappa} &\le C_\kappa. \end{aligned} \tag{3.cmxcviii}$$

■

Proof. Compose eq. 3.cmxcvi with the actual normalized Gram inverse. Its transpose has $\widehat{F}^{\sharp} : L^2 \to \ell^2_{\mathrm{top}}$. This proves $P : L^2 \to Y$, hence the two bounds for R. The full physical divergence maps X locally into site L^2. Near a true boundary its one-edge term is paid by the bare mass. Composing it with R and eq. 3.cmxcii proves the bounds for $\Lambda, D\Lambda, S$.

For the last bound use the exact transpose

$$S^{\sharp} = 1 - DRG_sD^{\sharp}.$$

Here $D^{\sharp} : \mathfrak{F} \to Y^{\flat}$ is a variational divergence, $G_s : Y^{\flat} \to Y$, $R : Y \to Y$, and $D : Y \to \mathfrak{F}$. Every factor has the claimed local family. This is a weighted L^2 force assertion; it is not an L^∞ Riesz assertion. □

3.30.4. *The zero-current Landau inverse.* Put $\widehat{B} = D\widehat{F}$ and $\widehat{\Theta} = J_s\Theta J_s^{-1}$. The exact positive factorization is

$$G_0 = G_F + G_F\widehat{B}\widehat{\Theta}^{-1}\widehat{B}^{\sharp}G_F. \tag{3.cmxcix}$$

It is the physical identity eq. 3.cmxlv in counting top units. The injection $\widehat{B} : \ell^2_{\mathrm{top}} \to \mathfrak{F}$ is eq. 3.cmxcvi; its transpose acts from field energy into counting top space. Thus the positive coarse families and the positive field families prove

$$\|\mathbf{G}_0\|_{X^*\to X;s,\kappa_1} + \|\mathbf{G}_0\|_{\mathfrak{F}\to X;s,\kappa_1} \le C_{\kappa_1}. \tag{3.m}$$

There is no replacement of the actual projected precision by a periodic model in this argument. The inverse $\widehat{\Theta}^{-1}$ contains its complete geometry carrier.

3.30.5. *The first current precision and its lift.* Restrict J to bonds touching Ω_1, as allowed by the exact identities in § 3.26, and put

$$\gamma_{\mathrm{t}} = \sup_{b \text{ touches } \Omega_1} \mathsf{L}(b)^3|J(b)|.$$

We make this restriction before forming the coefficient families. It is a global exact identity for the field operator; an individual inverse coefficient need not vanish on bare bonds.

Lemma 3.110 (The actual current family is small)**.** The canonical symmetric correction has an exact carrier family with

$$\|\mathbf{B}_U[J]\|_{X\to\mathfrak{F};s,\kappa_1} + \|\mathbf{B}_U[J]\|_{X\to X^*;s,\kappa_1} \le C_{\kappa_1}\gamma_{\mathrm{t}}. \tag{3.mi}$$

No derivative of J is used. ■

Proof. For $\lambda = \Lambda v$, the exact grouped formula is

$$B_U[J]v = W[J]\lambda + DRG_sh_J, \qquad h_J = W[J]^{\sharp}v - \tfrac{1}{2}\{D^{\sharp}W[J]\lambda + W[J]^{\sharp}D\lambda\}. \tag{3.mii}$$

The actual endpoint current bounds in eq. 3.cmxli show that $W[J]\lambda$ is in $\mathfrak{F}$, with norm $C\gamma_{\mathrm{t}}\|\lambda\|_Z$, while h_J is in $Y^{\flat}$, with norm $C\gamma_{\mathrm{t}}(\|v\|_X + \|\lambda\|_Z)$. These bounds hold as carrier bounds: $W[J]$ reads one bond and its two endpoint values, and $D^{\sharp}W[J]\lambda$ is kept as a divergence. Compose with the families $G_s : Y^{\flat} \to Y$, $R : Y \to Y$ and $D : Y \to \mathfrak{F}$. Every union keeps the actual input current bond and both matrix endpoints. The two Schur sums give eq. 3.mi. □

The first named inverse is now defined without an implicit change of completion:

$$\begin{aligned}\mathbf{G} &= \mathbf{G}_0 \star \sum_{n\ge 0} (\mathbf{B}_U[J] \star \mathbf{G}_0)^{\star n},\\ \sum_C G_C = G &= (G_0^{-1} - B_U[J])^{-1}.\end{aligned} \tag{3.miii}$$

The right side denotes its joined-carrier family; at complete coefficients it is the displayed matrix inverse. Choose the touching-current ceiling so the $\mathfrak{F} \to \mathfrak{F}$ error norm is less than $1/4$. For a general X^* source only the rightmost current factor uses $B_U[J]G_0 : X^* \to \mathfrak{F}$; all subsequent loops use $\mathfrak{F} \to \mathfrak{F}$. This proves exact normally convergent families for

$$G : X^* \to X, \qquad G : \mathfrak{F} \to X,$$

and the same family norm for $G - G_0$ is $C_{\kappa_1}\gamma_{\mathrm{t}}$.

Let the physical field-row measure be ξ_c^4 and put

$$J_\Gamma = \Xi_\Gamma^2, \qquad \mathcal{Q}_n = J_\Gamma a_f^{1/2} Q, \qquad \mathcal{Q}_n^\sharp = Q^\sharp a_f^{1/2} J_\Gamma^{-1}. \tag{3.miv}$$

This trace maps X to counting row space; its transpose maps counting rows into $\mathfrak{F}$. The latter is the actual density computation in eq. 3.cmlxxxvi. Define

$$\mathcal{T}_0 = \mathcal{Q}_n G_0 \mathcal{Q}_n^\sharp, \qquad \mathcal{T}_J = \mathcal{Q}_n G \mathcal{Q}_n^\sharp. \tag{3.mv}$$

The real $\mathcal{T}_0$ has the actual SCR lower and upper gap. Apply Cor. 3.104 to its actual family, using the intermediate rate κ_1. It gives a family for $\mathcal{T}_0^{-1}$ at the final rate κ, with a final aligned tile size M_{c} and a positive final spatial rate. Reblock all preceding families into those parents and reduce their spatial rates to a common positive value. The actual difference satisfies

$$\|\mathcal{T}_{\mathbf{J}} - \mathcal{T}_{\mathbf{0}}\|_{s,\kappa} \le C_\kappa \gamma_{\mathrm{t}}.$$

A second fixed decrease of the current ceiling therefore gives the exact inverse family

$$\mathcal{T}_{\mathbf{J}}^{-\mathbf{1}} = \mathcal{T}_{\mathbf{0}}^{-\mathbf{1}} \star \sum_{n\ge 0} [-(\mathcal{T}_{\mathbf{J}} - \mathcal{T}_{\mathbf{0}}) \star \mathcal{T}_{\mathbf{0}}^{-\mathbf{1}}]^{\star n}. \tag{3.mvi}$$

At a complex background the same argument is centered at the real reference matrix and includes its small actual physical-tube family perturbation.

Write $\mathcal{S}_J = QGQ^\sharp$. The normalized first lift and the physical lift are

$$\mathcal{H}_J = G\mathcal{Q}_n^\sharp \mathcal{T}_J^{-1}, \qquad H = \mathcal{H}_J J_\Gamma a_f^{1/2} = GQ^\sharp \mathcal{S}_J^{-1}. \tag{3.mvii}$$

Thus $\mathcal{H}_J$ has a local carrier family from counting rows into X, and H has one from the physical datum norm $\|a_f^{1/2} b\|_{\mathrm{rows}}$. In particular $QH = 1$ as an exact identity of the sums. No termwise row or gauge identity has been assumed.

3.30.6. *The first-lift induced multiplier retains its root.* Let J_{full} be the original current and $J = J_{\mathrm{full}}^{\mathrm{t}}$ the restriction used above. Define

$$\omega^{\mathrm{t}} = H^\sharp J.$$

For a carrier coefficient H_C, retain the coefficient $\omega_{c,C}^{\mathrm{t}} = ((H_C)^\sharp J)(c)$. The carrier contains the row c, every paired fine current bond, and the background data read by H_C.

Proposition 3.111 (An absolutely rooted multiplier family)**.** The first-lift readout satisfies

$$\sup_c \sum_C e^{\kappa\sigma_c(C)} \xi_c^3 |\omega_{c,C}^{\mathrm{t}}| \le C_\kappa \gamma_{\mathrm{t}}. \tag{3.mviii}$$

The actual multiplier for the unrestricted bare current is

$$\omega_* = H^\sharp J_{\mathrm{full}} = \omega^{\mathrm{t}} + \iota_0 J_{\mathrm{full}}^{\mathrm{b}}. \tag{3.mix}$$

In particular its nonbare family obeys eq. 3.mviii. An all-row bound retains the full current radius $\gamma_{\mathrm{full}} = \sup_b \mathsf{L}(b)^3 |J_{\mathrm{full}}(b)|$. ■

Proof. Fix a unit colour e at row c and use $b^{(c)} = \xi_c^{-1} e_c$. Its physical datum energy is bounded above and below by constants. The exact physical transpose gives

$$\xi_c^3 \langle \omega^{\mathrm{t}}_{c,C}, e\rangle = \langle H_C b^{(c)}, J\rangle_{\mathrm{fine}}. \tag{3.mx}$$

Split the fine pairing by disjoint bond ownership cells. On such a cell $\|\xi J\|_2 \le C_L \gamma_{\mathrm{t}}$, while the mass part of X_y bounds $\|\xi^{-1} H_C b^{(c)}\|_2$. Therefore the left side of eq. 3.mviii is at most

$$C\gamma_{\mathrm{t}} \sum_{y,C} e^{\kappa \sigma_c(C)} \|1_y H_C b^{(c)}\|_{X_y} \le C_\kappa \gamma_{\mathrm{t}}.$$

This is the actual family column sum; no whole-volume energy norm is charged to J.

For the bare split use the exact identities $Q_0 H = P_0$ and equality of the physical fine and level-zero row measures, both η^4. They give $H^\sharp J^{\mathrm{b}}_{\mathrm{full}} = \iota_0 J^{\mathrm{b}}_{\mathrm{full}}$, as in eq. 3.cmxix. This is an identity of the complete sums. It does not assert that a nonbare column of each H_C vanishes at every bare bond. We used the touching restriction before forming these families for precisely this reason. The extra bare family is local and has amplitude bounded by the full γ_{full}, rather than by γ_{t}. □

3.30.7. *Only then form the second precision.* The actual bare re-centered mean is linear, so its second derivative is zero. Thus $\Delta^{(2)}_U[\omega_*] = \Delta^{(2)}_U[\omega^{\mathrm{t}}]$. Form the constraint family by joining each rooted coefficient $\omega^{\mathrm{t}}_{c,C}$ to the complete two-leg carrier of its actual row c. The physical two-leg bound then gives

$$\|\mathbf{C}_{\omega_*}\|_{X\to\mathfrak{F};s,\kappa} \le C_\kappa \gamma_{\mathrm{t}}, \qquad \mathbf{C}_{\omega_*} = \boldsymbol{\Delta}^{(\mathbf{2})}_{\mathbf{U}}[\omega_*]. \tag{3.mxi}$$

To check this normalization, the row coefficient has fixed-leg density $C_L \xi_c (\eta/\xi_c)^4$. Its physical transpose has factor $\eta^{-4}\xi_c^4$. Inserting $|\omega(c)| \le C\gamma_{\mathrm{t}} \xi_c^{-3}$ and the field mass norm gives $\|\ell C_{\omega_*} v\|_2 \le C\gamma_{\mathrm{t}} \|\ell^{-1} v\|_2$. Both leg sums are retained. There are only a bounded number of incident rows and neighbouring assigned cells. For a coefficient family, use the absolute sum eq. 3.mviii in place of the single multiplier bound. The row star changes the carrier length by a fixed constant and its input-output metric radius is bounded. This proves eq. 3.mxi as a family estimate.

By eq. 3.cmxcviii,

$$\mathbf{V}_* = \mathbf{S}^\sharp \star \mathbf{C}_{\omega_*} \star \mathbf{S}, \qquad \|\mathbf{V}_*\|_{X\to\mathfrak{F};s,\kappa} \le C_\kappa \gamma_{\mathrm{t}}. \tag{3.mxii}$$

The exact second inverse is consequently

$$\begin{aligned} \mathbf{G}_1 &= \mathbf{G} \star \sum_{n\ge 0} (\mathbf{V}_* \star \mathbf{G})^{\star n}, \\ \sum_C (G_1)_C = G_1 &= (G^{-1} - S^\sharp \Delta^{(2)}_U[\omega_*] S)^{-1}. \end{aligned} \tag{3.mxiii}$$

Decrease the same fixed touching-current ceiling so the force-space error has family norm less than $1/4$. As for the first inverse, a general X^* source is used only at the rightmost factor; subsequent loops take place in $\mathfrak{F}$. This proves carrier bounds for $G_1 : X^* \to X$ and $G_1 : \mathfrak{F} \to X$, and for $G_1 - G_0$ with the additional factor γ_{t}. The lift inside ω_* throughout this construction is H, not the following H_1.

Put

$$\mathcal{T}_1 = \mathcal{Q}_n G_1 \mathcal{Q}_n^\sharp, \qquad \mathcal{H}_1 = G_1 \mathcal{Q}_n^\sharp \mathcal{T}_1^{-1}, \qquad H_1 = \mathcal{H}_1 J_\Gamma a_f^{1/2}. \tag{3.mxiv}$$

The difference $\mathcal{T}_1 - \mathcal{T}_0$ has family norm $C_\kappa \gamma_{\mathrm{t}}$. Use the already constructed $\mathcal{T}_0^{-1}$ in the same Neumann formula as eq. 3.mvi. This constructs the exact $\mathcal{T}_1^{-1}$ and H_1 families without a new positivity assumption at an independent current. At complete coefficients,

$$H_1 = G_1 Q^\sharp (Q G_1 Q^\sharp)^{-1}, \qquad Q H_1 = 1.$$

3.30.8. *The assembled range and its limits.*

Proposition 3.112 (Named current carriers in physical norms)**.** Fix a finite carrier rate κ. Choose the two positive coarse-inversion stages, their carrier tiles, separation reserves and common complex C^1 tube as above. There is a fixed touching-current ceiling $\gamma_* > 0$ such that for $\gamma_{\mathrm{t}} < \gamma_*$ the actual canonical operators have exact, normally convergent local carrier families in the following spaces:

$$\begin{aligned} &G_s : L^2 \to Z, \quad R : L^2 \to L^2, \quad \Lambda : X \to Z, \quad S : X \to X, \\ &G_0, G, G_1 : X^* \to X, \qquad G_0, G, G_1 : \mathfrak{F} \to X, \\ &\mathcal{H}_J, \mathcal{H}_1 : \ell^2_{\mathrm{rows}} \to X, \qquad \mathcal{T}_J^{-1}, \mathcal{T}_1^{-1} : \ell^2_{\mathrm{rows}} \to \ell^2_{\mathrm{rows}}. \end{aligned} \tag{3.mxv}$$

Their family norms are bounded by fixed C_κ. The nonbare first-lift readout has the rooted bound eq. 3.mviii. The physical Schur inverses and lifts are obtained by the displayed row unitaries and weights, not by changing transposes.

Every coefficient is holomorphic in the actual local background and touching-current data on the stated tubes, with its complete recorded determining geometry. Agreement of these data on its carrier gives the same coefficient. The constants are independent of depth, compatible volume and admitted region shape. The current ceiling depends on the fixed carrier constants and has no Hölder-exponent dependence. ■

Proof. The successive constructions prove each assertion in the order $R, \Lambda, G_0, G, H, \omega_*, G_1, H_1$. All operations are finite local coefficients, positive coarse inverse families or normally convergent small-current series in the same joined-carrier algebra. Each product retains its endpoint tags and fixed short joins. The cutoff-source and scale-ratio reserves were paid before these compositions.

Only finitely many positive inversion and reblocking stages are used. Choose the intermediate carrier rates first, then their finite short-power parameters, the final aligned M_{c}, and the common separation threshold. Finally choose the common physical tube and the current ceiling smaller than the finitely many displayed Neumann bounds. None of these steps uses a Hölder exponent or a current derivative. The H^2 output argument uses only the L^2 unitary Hodge estimate.

For small periods carry out this entire finite construction on one sufficiently large compatible periodic cover with common tag grids. Average the final families over simultaneous deck shifts and project their recorded tags. The row and column absolute sums descend as in the positive and coarse walk proofs. Their shifted joins and physical unitaries are retained. Thus no sheet-count factor enters the bound or the current ceiling. □

For clarity, write $G_J = G$. The physical row Schur inverse is

$$(QG_iQ^\sharp)^{-1} = a_f^{1/2} J_\Gamma^{-1} \mathcal{T}_i^{-1} J_\Gamma a_f^{1/2}, \qquad i = J, 1.$$

In the physical angle and dual-current units $\beta = \Xi_\Gamma b$, $\nu = \Xi_\Gamma^3 \mu$, put $c_a = \Xi_\Gamma a_f^{1/2}$. The mass-removed coarse operator therefore has the exact normalized family

$$\Xi_\Gamma^3 \{(QG_iQ^\sharp)^{-1} - a_f\} \Xi_\Gamma^{-1} = c_a(\mathcal{T}_i^{-1} - 1)c_a. \tag{3.mxvi}$$

This identity retains the artificial mass subtraction. The separate lower bound on the actual outer plain-constraint fibre is proved in Prop. 5.45, with the full outer constraint curvature and the specified relative contour quadratic form. Its physical absolute-contour identification holds in an averaged-contour axial representative.

The result gives the canonical completion selected in § 3.26. The first-order output families on the same C^1 domain and their enlarged cube-parameter discs are constructed in Prop. 3.171 and Lem. 3.172. All prescribed histories must use the common separation and filing constants of these constructions. The positive clean conditional measure and its normalizer require their own measure and gauge identifications.

3.31. **Second derivatives in a discrete Hölder norm.** The bounded-force estimates give a first derivative in every fixed Hölder class below exponent one. For the nonlinear gauge slice, a site gauge requires two derivatives in the same Hölder class. This needs cancellation of the second-resolvent kernel, in addition to its finite-p bounds. We prove that estimate and then apply it to the actual site inverse.

Fix $0 < \alpha < 1$. On $h\mathbb{Z}^4$ write

$$\|f\|_{C_h^\alpha} = \|f\|_\infty + \sup_{0<|x-x'|\le 1} \frac{|f(x) - f(x')|}{|x - x'|^\alpha}.$$

Only distinct lattice points occur in the seminorm. First and second difference norms use all forward or backward component differences. On a fixed interior box the same notation restricts to present stencils. All statements concern the physical lattice measure of § 3.24.

3.31.1. *Cancellation of a truncated second kernel.* Let $k_h = D_i^\sigma D_j^\tau (L_h + 1)^{-1}$ denote its physical convolution kernel. In addition to Lem. 3.78, it obeys

$$|k_h(x)| + |Dk_h(x)| \le C_N |x|^{-N}, \qquad |x| \ge 1, \tag{3.mxvii}$$

for any fixed sufficiently large N. Indeed insert the heat representation and use eq. 3.dccclxxii with spatial power N, retaining the factor e^{-t}. For a derivative of order $a = 2, 3$, its integral is bounded by

$$C_N |x|^{-N} \int_0^\infty e^{-t} (t + h^2)^{(N-4-a)/2}\, dt.$$

Choose $N > 4 + a$. The integral is uniformly bounded for $h \le 1$. No exponential off-diagonal assertion is needed here.

Lemma 3.113 (Uniform truncated cancellation)**.** The kernel has total physical sum zero and

$$\left| h^4 \sum_{|x|\le r} k_h(x) \right| \le C, \qquad r \ge h. \tag{3.mxviii}$$

The constant is independent of the mesh and of the truncation radius. ■

Proof. For fixed h the massive kernel is absolutely summable by the heat L^1 estimate in Prop. 3.79. Each differentiated heat kernel has sum zero, so its time integral has sum zero as well. For the truncated sum split the time integral at r^2.

For $0 < t < r^2$, replace the sum over the ball by the negative sum over its complement. The differentiated heat tail bound eq. 3.dclxxxiv, in its infinite-lattice version used in § 3.24, gives

$$\left| h^4 \sum_{|x|\le r} D_i^\sigma D_j^\tau p_t^h(x) \right| \le C_N (t + h^2)^{-1} (1 + r/\sqrt{t + h^2})^{-N}.$$

Its integral is bounded uniformly: put $v = t + h^2$, use $h \le r$, and bound it by $C_N r^{-N} \int_0^{2r^2} v^{N/2-1}\, dv$. For $t \ge r^2$, use the physical volume Cr^4 of the lattice ball and the pointwise derivative bound Ct^{-3}. The integral is at most $Cr^4 \int_{r^2}^\infty t^{-3}\, dt = C$. The mass factor only improves both bounds. □

The same bound holds for a ball whose centre is displaced from the kernel centre by at most one quarter of its radius. Compare concentric inner and outer balls of comparable radii; their difference is an annulus on which the absolute kernel bound has total mass at most C. This observation will control the constant part of a nearby Hölder increment.

3.31.2. *The scalar Hölder multiplier.*

Proposition 3.114 (Uniform discrete Schauder multiplier)**.** For every $0 < \alpha < 1$,

$$\|D_i^\sigma D_j^\tau (L_h + 1)^{-1} f\|_{C_h^\alpha} \le C_\alpha \|f\|_{C_h^\alpha}. \tag{3.mxix}$$

The same constant holds on every compatible periodic torus. At fixed mesh the convolution is the ordinary absolutely summable massive kernel; no principal value is left undefined. ■

Proof. Write $Tf = k_h * f$. Cancellation gives

$$Tf(x) = h^4 \sum_w k_h(x - w)[f(w) - f(x)].$$

For $|w - x| \le 1$, use the Hölder increment and $|k_h(z)| \le C(|z| + h)^{-4}$. The sum over dyadic annuli is at most $C_\alpha [f]_\alpha$, including its last atomic annulus. For larger distances, eq. 3.mxvii gives $C\|f\|_\infty$. This proves the supremum bound.

For the seminorm let $\delta = |x - x'|$, $h \le \delta \le 1/16$. Since each translate of the kernel has sum zero,

$$Tf(x) - Tf(x') = h^4 \sum_w [k_h(x - w) - k_h(x' - w)][f(w) - f(x)].$$

Split the sum into $|w - x| \le 4\delta$ and its complement. In the near part, the first kernel is multiplied by a Hölder increment at its own centre, so its absolute sum is $C_\alpha [f]_\alpha \delta^\alpha$. For the second kernel write

$$f(w) - f(x) = [f(w) - f(x')] + [f(x') - f(x)].$$

The first term has the same bound. The constant second term is multiplied by the kernel sum over a ball of radius 4δ centred a distance δ from its kernel centre. Lem. 3.113 and the following annulus comparison bound that sum by C. Its contribution is therefore also at most $C[f]_\alpha \delta^\alpha$.

In the far part, telescope a coordinate path from x to x'. The derivative bound of Lem. 3.78 gives

$$|k_h(x - w) - k_h(x' - w)| \le C\delta |x - w|^{-5} \quad (4\delta < |x - w| \le 1).$$

Multiplication by $[f]_\alpha |x - w|^\alpha$ and dyadic summation gives

$$C_\alpha [f]_\alpha \delta \sum_{\delta \le r \le 1,\ \text{dyadic}} r^{\alpha - 1} \le C_\alpha [f]_\alpha \delta^\alpha.$$

At distance at least one, the differentiated far bound eq. 3.mxvii gives $C\delta\|f\|_\infty$, which is bounded by $C\delta^\alpha \|f\|_\infty$. For $1/16 < \delta \le 1$, the already proved supremum bound suffices. If the mesh exceeds $1/16$, this last case covers every distinct pair.

A periodic input extends to the infinite lattice with the same supremum norm and no larger local Hölder seminorm: the periodic distance is at most the distance of any chosen lifts. Convolution by the absolutely summable kernel is its periodic massive multiplier. Applying the estimate to that extension proves the periodic claim, independently of the periods. □

3.31.3. *Interior flat and covariant second derivatives.*

Corollary 3.115 (Interior second-derivative estimate)**.** Let $Q_r \subset Q_R$ be fixed nested physical boxes with room for the difference stencils. For their sampled smooth cutoffs,

$$\|D^2 u\|_{C^\alpha(Q_r)} \le C_{\alpha,r,R} \{\|L_h u\|_{C^\alpha(Q_R)} + \|u\|_{C^{1,\alpha}(Q_R)}\}. \tag{3.mxx}$$

The constant is uniform in the fine mesh. Here $\|u\|_{C^{1,\alpha}}$ means $\|u\|_\infty + \|Du\|_\infty + [Du]_\alpha$. ■□

Proof. Take a cutoff $\chi = 1$ on Q_r, with support a fixed distance inside Q_R. Extend $v = \chi u$ by zero. Then

$$D^2 v = D^2 (L_h + 1)^{-1} (L_h + 1) v.$$

The discrete product rule writes its source as $\chi L_h u$, χu, and finitely many shifted products of $D\chi, Du$ or $D^2\chi, u$. Their global C^α norms have the displayed bound. All neighbours lie in the indicated outer box, including at the support boundary. Apply Prop. 3.114. □

Suppose now that $U_0 = e^{iha_0}$ in a local real Coulomb frame and that

$$\|a_0\|_{C^{1,\alpha}} \le C_0. \tag{3.mxxi}$$

This is available at an actual small-curvature, small-current reference from eq. 3.dclxxvii; the current is used here, in contrast with the curvature-only estimates in § 3.25. The exact scalar decomposition eq. 3.dcccxc has first and zeroth coefficients bounded in C^α by a constant depending only on C_0. For the zeroth coefficient this uses one difference of a_0, not two: the divided difference of $(\operatorname{Ad} e^{iha_0} - 1)/h$ is an analytic bounded coefficient times a first difference of a_0.

Consequently a site solution $\Delta_{U_0} u = g$ satisfies

$$\|D^2_{U_0} u\|_{C^\alpha(Q_r)} \le C_{\alpha,C_0,r,R}\{\|g\|_{C^\alpha(Q_R)} + \|u\|_{C^{1,\alpha}(Q_R)}\}. \tag{3.mxxii}$$

Indeed its flat Laplacian is g minus the lower-order terms in that decomposition. Products preserve C^α, so Cor. 3.115 applies. Covariant second differences differ from ordinary second differences by terms containing $a_0 Du$, $Da_0 u$ and bounded quadratic coefficients in a_0. Their C^α norms have the same bound. All comparisons are in this fixed local frame; transported seminorms give the gauge-covariant statement.

3.31.4. *The proper site gauge inverse.* Fix the real proper reference and its actual q_0, G_s, F, M, R_0, with their physical site and top pairings. Assume the actual local curvature and current bounds needed for eq. 3.mxxi on the same complete interior collars. Let

$$\|\rho\|_{\alpha,2} = \sup_y \{\xi_y^2 \|\rho\|_{\infty,B_y} + \xi_y^{2+\alpha}[\rho]^{U_0}_{\alpha,y}\}. \tag{3.mxxiii}$$

At bare cells the finite collar convention of § 3.25 is used. The seminorm compares pairs across adjoining pillar faces on the same complete interior boxes used there. Taking it separately inside each pillar would admit discontinuous piecewise-constant sources and is not the norm in this statement. This same complete-box convention is used for the derivatives below. For a site gauge parameter define

$$\begin{aligned}\|\lambda\|_{Y^{2,\alpha}} = \sup_y \{&\|\lambda\|_{\infty,B_y} + \xi_y \|D_{U_0}\lambda\|_{\infty,B_y} \\ &+ \xi_y^2 \|D^2_{U_0}\lambda\|_{\infty,B_y} + \xi_y^{2+\alpha}[D^2_{U_0}\lambda]^{U_0}_{\alpha,y}\}.\end{aligned} \tag{3.mxxiv}$$

The short coordinate transports are those already fixed in the strong norm. A fixed increase of the separation absorbs the scale-ratio factors in the following sums.

Proposition 3.116 (Strong inverse of the proper site gauge block)**.** The actual projection and inverse satisfy

$$\|R_0\rho\|_{\alpha,2} \le C_\alpha \|\rho\|_{\alpha,2}, \tag{3.mxxv}$$

and

$$\|G_s R_0 \rho\|_{Y^{2,\alpha}} \le C_\alpha \|\rho\|_{\alpha,2}. \tag{3.mxxvi}$$

In particular, for $\rho \in \operatorname{ran} R_0$, $\lambda = G_s\rho$ is the unique site solution of

$$q_0\lambda = 0, \qquad R_0 \Delta_{U_0}\lambda = \rho. \tag{3.mxxvii}$$

The true bare-site zero condition is retained. ■

Proof. Use physical orthonormal top columns F_z. Prop. 3.85 gives their pointwise size $C\xi_y^{-2} e^{-md(y,z)}$ and their local Hölder size $C_\alpha \xi_y^{-2-\alpha} e^{-md(y,z)}$. A source with unit $\|\cdot\|_{\alpha,2}$ norm has L^2 norm at most C on each pillar. Thus its pairing with F_z is bounded by the uniformly summable exponential cell sum. The normalized Gram inverse preserves that coefficient bound. Summing the output columns

proves eq. 3.mxxv for $R_0 = 1 - FM^{-1}F^\sharp$. For the Hölder bound use the column increment before summing; no derivative of the input is taken.

Set $u = G_s R_0 \rho$. The exact site identity $q_0 G_s R_0 = 0$ gives

$$q_0 u = 0, \qquad \Delta_{U_0} u = R_0 \rho.$$

The second identity follows from $G_s^{-1} = \Delta_{U_0} + q_0^\sharp a_s q_0$. Its exponential energy columns give

$$\xi_y^{-2} \|u\|_{2,B_y} + \xi_y^{-1} \|D_{U_0} u\|_{2,B_y^+} \le C \|\rho\|_{\alpha,2}.$$

For example the G_s block carries $\xi_y \xi_z e^{-md(y,z)}$, while the source L^2 cell norm is bounded. The ratio ξ_z/ξ_y is absorbed by the retained metric rate. The derivative block is treated in the same way. The projection R_0 has already been included in the source bound.

Rescale a nonbare box by ξ_y. The bounded-force site bootstrap Lem. 3.84 gives its $C^{1,\alpha}$ norm, by choosing a finite $p > 4/(1-\alpha)$. Its source has the rescaled C^α bound from eq. 3.mxxv. The local reference potential has the fixed $C^{1,\alpha}$ bound eq. 3.mxxi. Apply eq. 3.mxxii on one further nested box. Rescaling back gives every power in eq. 3.mxxvi. At a bare cell the physical atomic energy inequality and a fixed two-step collar prove those same weighted bounds directly. In particular $q_0 u = 0$ forces the bare site values to be zero.

For uniqueness in eq. 3.mxxvii, suppose $q_0 \lambda = 0$ and $R_0 \Delta_{U_0} \lambda = 0$. Then $\Delta_{U_0} \lambda$ belongs to $\operatorname{ran} F$, say $\Delta_{U_0} \lambda = Fa$. Since $q_0 \lambda = 0$, $\lambda = G_s F a$. Applying q_0 gives $q_0 G_s F a = 0$. But

$$q_0 G_s F = \Xi^2 M$$

by $F = G_s q_0^\sharp \Xi^{-2}$ and $M = F^\sharp F$. The normalized Gram is invertible, so $a = 0$, then $\lambda = 0$. This proves uniqueness with the same actual boundary convention. □

The second-derivative site estimate uses the physical current of the reference to obtain its $C^{1,\alpha}$ local potential. It does not upgrade an arbitrary curvature-only connection to that class. Its purpose is the gauge restoration near an actual regular critical reference, while the curvature-only bounded-force field estimates remain as stated in § 3.25.

3.32. **Nonlinear continuation of an actual proper critical reference.** We construct a nearby constrained-critical orbit on the actual determining region, starting from a given real physical critical reference. The region, its outward rows and its bare boundary are kept. The reference is not inferred from small curvature of a datum. The input is an actual solution, and the conclusions are a uniform analytic continuation, full stationarity, literal constraint and local orbit uniqueness.

Use the proper geometry and norms of § 3.25. Write U_0 for the real reference, η for the fine spacing, ξ_c for a field-row length and ξ_y for a site-top length. Let q_0 be the actual own-pillar site average, Q_0 the actual straightened field derivative and R_0 the real site projection. The site Laplacian and inverse are $\Delta_0 = D_0^\sharp D_0$ and $G_s = (\Delta_0 + q_0^\sharp a_s q_0)^{-1}$. Assume the actual bounds

$$\sup_p \mathsf{L}(p)^2 |F(U_0)(p)| \le \kappa_0, \qquad \sup_b \mathsf{L}(b)^3 |j_0(b)| \le \gamma_0, \qquad j_0 = Q_0^\sharp \omega_0. \tag{3.mxxviii}$$

Here j_0 is the actual Wilson current. The last equality is full physical stationarity in the declared straightened coordinates. The induced-current theorem gives $\sup_c \xi_c^3 |\omega_0(c)| \le C\gamma_0$. We take κ_0, γ_0 below fixed sufficiently small ceilings, with the strict collar room required by the strong and Schauder estimates. The separation is fixed large enough for their metric growth and scale-ratio reserves.

For a fine increment use $\|A\|_{\mathrm{str},\alpha}$ of eq. 3.dccclxxxvii, with its derivatives and short transports measured in the fixed reference U_0. Set

$$\|f\|_{3,\infty} = \sup_b \mathsf{L}(b)^3 |f(b)|, \qquad \|b\|_{1,\infty} = \sup_c \xi_c |b(c)|, \qquad \|\nu\|_{3,\infty} = \sup_c \xi_c^3 |\nu(c)|. \tag{3.mxxix}$$

The same symbol on fine forces and row multipliers denotes their respective displayed index sets. The site source and gauge norms are eq. 3.mxxiii and eq. 3.mxxiv.

Put $U_A = e^{i\eta A}U_0$ and $F(A) = A_{\Omega_0}(U_A)$. Let $q(A)$ be the actual re-centered straightened per-length constraint, so $q(0) = 0$, $q'(0) = Q_0$. We use the sign convention $D_0\lambda = \eta^{-1}(\lambda_- - \operatorname{Ad} U_0\lambda_+)$; its gauge curve is $g_t = e^{it\lambda}$. All complex transposes below are physical bilinear transposes.

Proposition 3.117 (Uniform proper critical continuation)**.** There are $r, \rho, C > 0$, independent of depth, compatible volume and the shapes of the separated regions, such that the following holds for the actual reference eq. 3.mxxviii. For every literal mixed datum $V_B(c) = e^{i\xi_c B(c)}V_0(c)$ with $\|B\|_{1,\infty} < \rho$, there is a holomorphic pair $(A(B), \nu(B))$ satisfying

$$\begin{aligned} q(A(B)) &= b(B), \qquad R_0 D_0^\sharp A(B) = 0, \\ F'(A(B)) &= q'(A(B))^\sharp(\omega_0 + \nu(B)), \\ \|A(B)\|_{\mathrm{str},\alpha} + \|\nu(B)\|_{3,\infty} &\le C\|B\|_{1,\infty}. \end{aligned} \tag{3.mxxx}$$

The reduced datum $b(B)$, defined next, includes every outward factor. A specified admitted fine gauge gives a literal field with determining mean exactly V_B. It is holomorphically constrained critical, and real for real B. It is the unique nearby critical orbit modulo fine gauges equal to one at all assigned tops. Its actual curvature and current, and its physical multipliers, obey the uniform continuation bounds proved below. Localized analytic response holds in the fixed determining geometry. ■

3.32.1. *The reduced datum at an outward endpoint.* Write $a_j(A)$ for the raw level-j straightened coordinate and $v_j[A]$ for its actual node factor. For each assigned top y put $v[A](y) = v_{j_y}[A](y)$. At an outward end z of a level-j row, z is assigned level $j-1$, and

$$v_j[A](z) = v_{j-1}[A](z)\, u_{W_{j-1}}(a_{j-1}(A))(z), \qquad W_{j-1} = M^{j-1}U_0. \tag{3.mxxxi}$$

The last factor is the actual one-step factor of eq. 3.mcclii. Its entire input block consists of the prescribed level-$(j-1)$ interior rows. There is no missing inward datum in this formula.

For a row c, let $u_-(q)$ or $u_+(q)$ be this factor at a lower assigned outward endpoint, and one at a same-level endpoint. Only one endpoint is lower. Removing the assigned node factor from the actual plain-mean diagram gives exactly

$$\mathcal{T}(q)_c = u_-(q)\{e^{i\xi_c q_c}V_0(c)\}u_+(q)^{-1}, \qquad M_{\mathbf{B}}(U_A) = \mathcal{T}(q(A))^{v[A]}. \tag{3.mxxxii}$$

This is eq. 3.cccxliv with its endpoint order retained. Interior rows are unchanged. Each outward factor reads only finer interior rows, which are themselves unchanged by $\mathcal{T}$. Its inverse is therefore explicit:

$$e^{i\xi_c b_c(B)}V_0(c) = u_-(B)^{-1}\{e^{i\xi_c B_c}V_0(c)\}u_+(B). \tag{3.mxxxiii}$$

For an interior-to-outer orientation this says $e^{i\xi_c b_c} = e^{i\xi_c B_c}\operatorname{Ad}(V_0(c))u_+(B)$, as in the original outward diagram. Reversing the stored row uses the inverse group equation and requires no new convention.

The one-step block contains C_L coarse coordinates, has a uniform relative analytic branch at the real reference and reads adjacent scales with ratio L. Consequently

$$b(0) = 0, \qquad \|b(B)\|_{1,\infty} \le C_L\|B\|_{1,\infty}, \qquad \|D^m b(B)\| \le C_{L,m} \tag{3.mxxxiv}$$

on a fixed smaller raw-angle ball. These derivative bounds use the displayed amplitude norms and bounded complete carriers. The same holds for the inverse datum chart. There is no product over levels: the factors depend on interior rows, never on an already corrected outward row. In particular solving $q(A) = B$ on outward rows without this reduction would solve a different literal problem.

3.32.2. *A quadratic residual for the actual Wilson equation.* The actual proper reference has local Coulomb potentials a_0 with uniformly small $C^{1,\alpha}$ norm after rescaling each nonbare box by its assigned length. This uses both the physical curvature and the actual current in eq. 3.mxxviii; it is not a curvature-only $C^{1,\alpha}$ assertion.

Lemma 3.118 (The complete re-centered Wilson residual)**.** Let $J(A) = F'(A)$ denote the physical normal-coordinate gradient. Then

$$N_F(A) = J(A) - J(0) - F''(0)A, \qquad \|N_F(A)\|_{3,\infty} \le C\|A\|^2_{\mathrm{str},\alpha}. \tag{3.mxxxv}$$

It is uniformly holomorphic on a fixed strong ball and has Lipschitz constant Cr there from the strong norm to the weighted force norm. ■

Proof. On a nonbare box scale its length to one and set $h = \eta/\xi$. In its Coulomb frame the actual product has potential

$$P_h(A) = \frac{1}{ih}\log(e^{ihA}e^{iha_0}).$$

The actual gradient in this frame is $D_A P_h(A)^\sharp J_h(P_h(A))$, with J_h the flat-coordinate Wilson gradient. The exact current calculation of Lem. 3.184 gives $J_h(a) = \mathrm{d}^\sharp \mathrm{d}a + N_h(a)$, with uniformly analytic $N_h : C^{1,\alpha} \to L^\infty$. Use cutoffs on one larger box if necessary; the current on the inner full stencil is unchanged.

Here are the divided-difference bounds needed for the coordinate composition. Put $P_0 = D_A P_h(0)$ and $E_h(A) = P_h(A) - a_0 - P_0 A$. Matrix multiplication and its logarithm are analytic functions of ha_0, hA, and division by h removes their zero of first order. Taylor's formula gives

$$\begin{aligned} |E_h(A)| &\le Ch|A|^2, \qquad \|DE_h(A)\|_\infty \le Chr^2, \\ \|D^2E_h(A)\|_\infty &\le Cr^2, \qquad |D_A P_h(A) - P_0| \le Chr, \\ |D_A P_h(A) - P_0 - D_A^2 P_h(0)[A]| &\le Ch^2r^2. \end{aligned} \tag{3.mxxxvi}$$

Here D denotes ordinary spatial differences and the parameter derivatives are written D_A; r bounds the rescaled strong norm of A. To check the only delicate second-difference line, each leading term is hAD^2A or $h(DA)^2$. Differentiating a coefficient with respect to a_0 supplies another h, so the remaining terms contain $h^2A^2D^2a_0$, h^2ADADa_0 or higher such powers. The elementary bounds $\|D^2A\|_\infty \le Ch^{-1}\|DA\|_\infty$ and $\|D^2a_0\|_\infty \le Ch^{-1}\|Da_0\|_\infty$ cancel every apparent mesh loss. The first differences of E_h and its bounded second differences also bound its $C^{1,\alpha}$ norm by Cr^2.

Now Taylor-expand $D_A P_h(A)^\sharp J_h(P_h(A))$ at $A = 0$. The constant and linear terms are exactly $J(0)$ and $F''(0)A$. The flat second-order term acting on E_h is bounded by eq. 3.mxxxvi. The term $(D_A P_h(A) - P_0)^\sharp \mathrm{d}^\sharp \mathrm{d}(P_0 A)$ is also bounded: its first factor is $O(hr)$, while the second is at most $Ch^{-1}r$. Spatial differences of P_0 have an additional h for each derivative of its dependence on a_0; the same rough second-difference estimate covers them. The quadratic remainder of $D_A P_h$ times $J_h(a_0)$ has bound Ch^2r^2 times $Ch^{-1}\|Da_0\|_\infty + C$, and is bounded by Cr^2. All terms containing N_h are bounded by its uniform Banach derivative estimates and the $C^{1,\alpha}$ coordinate bounds. This proves the complete gradient residual, rather than merely an action remainder.

At a bare cell $\xi = \eta$, its rescaled mesh is one. There are only a fixed number of incident inner plaquettes. Their exact finite matrix gradient has uniformly bounded analytic derivatives at a unitary reference. Direct Taylor subtraction therefore gives the same quadratic estimate, including a rough action boundary; no exterior Coulomb frame is used.

Rescaling back multiplies the gradient by ξ^{-3}. The complete carriers have comparable assigned lengths, proving eq. 3.mxxxv. The same estimates hold for complex A on a fixed larger ball. Banach-valued Cauchy estimates and $N_F(0) = DN_F(0) = 0$ give its Cr Lipschitz bound. No second derivative of A or of a_0 was assumed. □

3.32.3. *Actual constraint coefficients in the physical norms.* Put $C_q(A) = q(A) - Q_0 A$. On each determining row the actual two-leg coefficient bound is

$$\sup_e \sum_{e'} \|\partial_e \partial_{e'} q_c(A)\| \le C_L \xi_c (\eta/\xi_c)^4. \tag{3.mxxxvii}$$

It is eq. 3.dcii on the admitted complex increment ball about this actual regular reference. All fine input scales on the complete carrier are comparable to ξ_c within a fixed L factor. The physical transpose uses $\eta^{-4}\xi_c^4$. Thus, for joint amplitude at most r,

$$\begin{aligned} \|C_q(A)\|_{1,\infty} &\le C r^2, \\ \|(q'(A) - Q_0)^\sharp \nu\|_{3,\infty} &\le C r \|\nu\|_{3,\infty}, \\ \|(q'(A) - Q_0 - q''(0)[A])^\sharp \omega_0\|_{3,\infty} &\le C \|\omega_0\|_{3,\infty} r^2. \end{aligned} \tag{3.mxxxviii}$$

For the first line sum the fixed-leg bound over the $O((\xi_c/\eta)^4)$ legs and insert $|A| \le Cr/\xi_c$. For the second line its column density is $Cr(\eta/\xi_c)^4$, which cancels the physical transpose factors. For the last line apply Cauchy's estimate to the actual Hessian in its fixed-leg sum norm on the larger weighted increment ball, then integrate twice. Its column remainder is $Cr^2(\eta/\xi_c)^4$. This is not an inference of an absolute coefficient sum from an arbitrary second directional bound. There are only a bounded number of rows reading a fixed fine leg. All three maps have the corresponding quadratic Lipschitz bounds.

3.32.4. *The joint proper fixed point.* Let $H_{L,0} = F''(0) - \langle \omega_0, q''(0) \rangle$. The actual physical reference and the strong row-saddle theorem give an inverse $\mathcal{L}_0^{-1}$ of

$$(H_{L,0} + D_0 R_0 D_0^\sharp) A - Q_0^\sharp \nu = f, \qquad Q_0 A = b, \qquad \|A\|_{\mathrm{str},\alpha} + \|\nu\|_{3,\infty} \le C(\|f\|_{3,\infty} + \|b\|_{1,\infty}). \tag{3.mxxxix}$$

Choose a finite $p > 4/(1-\alpha)$ in eq. 3.cm; its force norm is bounded by the displayed supremum norm. Removing the artificial row mass changes its multiplier by $-a_f b$, which has exactly the displayed amplitude. No moving-slice inverse is being presumed here.

The actual penalized equations are

$$F'(A) + D_0 R_0 D_0^\sharp A - q'(A)^\sharp(\omega_0 + \nu) = 0, \qquad q(A) = b(B). \tag{3.mxl}$$

After subtracting the base equation they are precisely

$$\begin{aligned} (A, \nu) = \mathcal{L}_0^{-1}\big(&- N_F(A) + (q'(A) - Q_0 - q''(0)[A])^\sharp \omega_0 \\ &+ (q'(A) - Q_0)^\sharp \nu, \quad b(B) - C_q(A)\big). \end{aligned} \tag{3.mxli}$$

The residual estimates give quadratic size Cr^2 and Lipschitz constant Cr in the joint strong/multiplier ball. Choose r below all primitive ceilings with $Cr < 1/4$, and then choose ρ so $C\|b(B)\|_{1,\infty} < r/4$. The radius-$r/2$ ball is invariant and contractive. Iteration constructs a unique holomorphic pair, real for real data, with

$$\|A(B)\|_{\mathrm{str},\alpha} + \|\nu(B)\|_{3,\infty} \le C\|B\|_{1,\infty}. \tag{3.mxlii}$$

It remains to remove the penalty and identify the literal fibre.

3.32.5. *Strong site extensions in the actual reference.*

Lemma 3.119 (Average and node extensions)**.** There are real linear maps E_q, E_n from assigned top values to site functions such that

$$q_0 E_q = 1, \qquad \mathrm{ev}_{\mathrm{tops}}\, E_n = 1, \qquad \|E_q c\|_{Y^{2,\alpha}} + \|E_n c\|_{Y^{2,\alpha}} \le C\|c\|_\infty. \tag{3.mxliii}$$

Top data zero on the bare tops give site functions zero on the bare layer. Their supports and fine derivative collars have bounded overlap and bounded layer-metric radius. ∎

Proof. In a nonbare pillar use a local Coulomb frame of U_0 on its fixed complete surrounding box. The rescaled reference potential has its $C^{1,\alpha}$ bound from the actual current. Choose a smooth scalar bump supported strictly inside the pillar and positive at its center. Its sampled scalar average m_y is bounded above and below: fine meshes tend to its positive integral, and the finitely many coarse possibilities contain the center. The actual transported average of this bump times a constant colour is a three-by-three matrix N_y with $\|N_y - m_y I\| \le C\kappa_0 m_y$. The actual hierarchical transports have this bound by their geometric sum of physical lengths in the small local potential. Normalize by N_y^{-1}, including the unitary top-frame rotation, to obtain E_q. Normalize the bump's center value instead to obtain E_n.

Each covariant first derivative is a profile derivative plus the local reference potential times the profile. Two derivatives contain only $D^2\phi$, $a_0 D\phi$, $Da_0\phi$ and $a_0^2\phi$, with the indicated rescaled bounds. Their α differences therefore use $Da_0 \in C^\alpha$, not $D^2 a_0$. This proves eq. 3.mxliii on the fine meshes. For the finitely many smaller ratios ξ_y/η, direct center seeds have bounded weighted differences of every fixed order. At a bare top use its one site. Adjacent assigned scales differ by at most L, so its fine collars cost only a fixed L factor. The supports are disjoint inside pillars; their derivative collars have bounded overlap. These estimates are covariant under the real local frame changes and hence give the global norms stated above. They also prove the assertion about bare zeros and carriers.

If a required surrounding box does not embed in a small periodic volume, pass first to a finite periodic cover where it does. Construct the two extension operators there, and average each operator over simultaneous fine and top deck shifts. The actual reference, its averages and node evaluation commute with these shifts, so both right-inverse identities are retained. The local strong norms are convex and invariant under the shifts; the averaged operator has the same bound. It descends on invariant data with the same supremum and Hölder bounds and with the same carriers modulo the torus. No sheet count enters these norms. This treats small volumes without a nonembedded frame or a limit in the volume. □

The site block needed for a finite gauge is now an actual bounded inverse:

$$\lambda \longmapsto (q_0\lambda, R_0\Delta_0\lambda), \qquad \lambda = E_q c + G_s\{\rho - R_0\Delta_0 E_q c\}. \tag{3.mxliv}$$

The second formula uses eq. 3.mxxvi; its braces are in $\operatorname{ran} R_0$ and have bounded $\|\cdot\|_{\alpha,2}$ norm. The exact identities $q_0 G_s R_0 = 0$, $\Delta_0 G_s R_0 = R_0$ prove both equations. Uniqueness is eq. 3.mxxvii. If c is zero at bare tops, this inverse retains bare site zeros.

3.32.6. *The actual assigned effective gauge and its outward extension.* For a fine gauge g define at assigned tops

$$h_A(g) = v[A^g]^{-1} g|_{\mathrm{tops}} v[A], \qquad h_A(e^{it\lambda}) = 1 + it\tau_A\lambda + O(t^2). \tag{3.mxlv}$$

Apply plain gauge covariance to eq. 3.mxxxii. It gives the exact nonlinear identity

$$\mathcal{T}(q(A^g)) = \mathcal{T}(q(A))^{h_A(g)}. \tag{3.mxlvi}$$

This identity fixes the coarse action being differentiated; it does not replace an outward endpoint by a chosen top value.

For completeness the one-step derivative can be computed directly. Put $T_{\mathrm{cur}} T_{\mathrm{ref}}^{-1} = e^{iz_x}$, $m = L^{-4}\sum_x z_x$ and $u = e^{im}$, and let $\mathcal{E}_z w = \int_0^1 \mathrm{Ad}(e^{itz}) w\, dt$. Then

$$\tau^{(1)}_{W,a}\lambda(y) = \mathrm{Ad}(u^{-1})\left[\lambda(y) + \mathcal{E}_m L^{-4}\sum_x \mathcal{E}_{z_x}^{-1}\{-\lambda(y) + \mathrm{Ad}(T_{\mathrm{cur}})\lambda(x)\}\right]. \tag{3.mxlvii}$$

The left derivative of the relative transport is $i(\lambda(y) - \mathrm{Ad}(T_{\mathrm{cur}})\lambda(x))$; differentiating $u(A^g)^{-1} g(y) u(A)$ gives the displayed sign. At $a = 0$ all z_x, m are zero and this is exactly the real transported site average. One-step analytic differentiation gives an error at most $C_L\|a\|_\infty$ in the site supremum norm. The actual raw radii on a top's hierarchy are bounded by $C_L r\xi_s/\xi_y$,

with summable ratios. Telescoping the products, whose zero-increment factors are unitary averages, therefore proves

$$\|\tau_A - q_0\|_{\infty\to\infty} \le Cr, \qquad \tau_0 = q_0. \tag{3.mxlviii}$$

At a bare top $\tau_A\lambda = \lambda$ exactly.

At an outward endpoint, the higher-level effective parameter is eq. 3.mxlvii applied to all assigned finer parameters on its complete outer block. At zero this is precisely $L^{-4}\sum_w \mathrm{Ad}(T_{z,w}(W_{j-1}))\lambda(w)$, the extension in eq. 3.cdlxxxiii. At nonzero A the evaluation terms and the logarithmic differentials in eq. 3.mxlvii must remain. In particular it is not an independently chosen site average at the new field.

Let Γ_A be the normal-coordinate fine gauge generator. Differentiating eq. 3.mxlvi gives

$$q'(A)\Gamma_A\lambda = \mathcal{D}_A\tau_A\lambda, \qquad \mathcal{D}_A = \big[D\mathcal{T}(q(A))\big]^{-1} d^{\mathrm{lit}}_{\mathcal{T}(q(A))}. \tag{3.mxlix}$$

The derivative is left trivialized in the declared per-length group coordinates, as is the literal gauge gradient. At zero, $\mathcal{D}_0 = \bar{D}_{\mathrm{ext}}$ by the preceding explicit outer-block derivative and eq. 3.cdlxxxvii. Most importantly, $\tau_A\lambda = 0$ makes the left side zero on every interior and outward row.

3.32.7. *Surjective gauge tests and removal of the penalty.* The exact fine generator is

$$\Gamma_A\lambda = \mathcal{E}_{\eta A}^{-1} D_{U_A}\lambda. \tag{3.ml}$$

Covariant product differences give

$$\|(\Gamma_A - D_0)\lambda\|_{\mathrm{str},\alpha} \le Cr\|\lambda\|_{Y^{2,\alpha}}. \tag{3.mli}$$

Terms have the forms $A\lambda$, $\eta A D_0\lambda$ and their first covariant differences. The second site derivative and its α seminorm in $Y^{2,\alpha}$ control exactly those differences. On a carrier $\eta \le \xi$ and the physical powers cancel. This argument uses the fixed reference transports, not ordinary differences of an uncontrolled frame change.

For $\rho \in \operatorname{ran} R_0$ with finite $\|\rho\|_{\alpha,2}$ put $\lambda_0 = G_s\rho$ and

$$\lambda_A(\rho) = \lambda_0 - E_q(\tau_A E_q)^{-1}\tau_A\lambda_0. \tag{3.mlii}$$

The inverse exists since $\tau_A E_q = 1 + O(r)$. Then $\tau_A\lambda_A = 0$ and $\|\lambda_A - \lambda_0\|_{Y^{2,\alpha}} \le Cr\|\rho\|_{\alpha,2}$. Bare zeros are exact: λ_0 is zero at bare sites; τ_A there is evaluation; and the bare-output block of $\tau_A E_q$ is the identity with no nonbare input. Thus the correction's bare coefficients and site values are zero.

The boundedness of R_0 on the site source space, its range identities, and eq. 3.mli give

$$\mathscr{R}_A\rho = R_0 D_0^\sharp \Gamma_A\lambda_A(\rho), \qquad \|\mathscr{R}_A - 1\|_{\alpha,2\to\alpha,2} \le Cr \quad \text{on } \operatorname{ran} R_0. \tag{3.mliii}$$

A smaller r makes this invertible. For every site source $\sigma \in \operatorname{ran} R_0$ there is consequently a gauge parameter with $\tau_A\lambda = 0$ and $R_0 D_0^\sharp\Gamma_A\lambda = \sigma$.

For the fixed point, $\rho = R_0 D_0^\sharp A$ belongs to this source space, with norm at most $C\|A\|_{\mathrm{str},\alpha}$. Test eq. 3.mxl against these actual gauge vectors. Gauge invariance of the inner-plaquette action kills $F'(A)$; eq. 3.mxlix kills the constraint term. Thus

$$\langle\rho,\sigma\rangle_{\mathrm{bil},\eta} = 0 \quad (\sigma \in \operatorname{ran} R_0). \tag{3.mliv}$$

This forces $\rho = 0$ also at complex data. Indeed R_0 is the real reference projection in Pauli coordinates, so the coefficientwise conjugate $\overline{\rho}$ remains in its range and source space. Taking $\sigma = \overline{\rho}$ gives its positive physical squared norm. We have not used the false implication that a complex vector with zero bilinear self-pairing must vanish. The full equations eq. 3.mxxx are now proved.

3.32.8. *An admitted literal representative and the local mean slice.* For assigned node values $w = e^{i\zeta}$ define the actual fine extension $\mathfrak{E}(w) = e^{iE_n\zeta}$. Its node values are exactly w and its logarithm has the stated $Y^{2,\alpha}$ bound. Set

$$\widehat{U}(B) = U_{A(B)}^{\mathfrak{E}(v[A(B)]^{-1})}. \tag{3.mlv}$$

The datum diagram and eq. 3.mxxxiii give $M_{\mathbf{B}}\widehat{U}(B) = V_B$ exactly.

The finite gauge map has the uniform estimate

$$\|A^{e^{i\lambda}} - A - D_0\lambda\|_{\mathrm{str},\alpha} \le C(\|A\|_{\mathrm{str},\alpha} + \|\lambda\|_{Y^{2,\alpha}})\|\lambda\|_{Y^{2,\alpha}}. \tag{3.mlvi}$$

To see the mesh cancellation, write $\mathrm{Ad}(U_0)\lambda_+ = \lambda_- - \eta D_0\lambda$ inside the exact three exponential factors defining A^g. At $\eta = 0$ their product is one. Dividing its logarithm by η is analytic in $\lambda, A, D_0\lambda$, uniformly down to zero mesh. Covariant product rules use only the site $Y^{2,\alpha}$ norm for its first field derivative and its Hölder increment. This proves the bound and its local Lipschitz version. Thus the specified literal gauge leaves the increment in a fixed admitted strong ball after one decrease of the continuation radius. No general complex gauge saturation of a domain is assumed.

For an arbitrary nearby literal field define the assigned mean-slice map

$$\lambda \longmapsto (i^{-1}\log h_A(e^{i\lambda}),\ R_0 D_0^{\sharp} A^{e^{i\lambda}}). \tag{3.mlvii}$$

Its derivative at zero reference and zero parameter is $(q_0\lambda, R_0\Delta_0\lambda)$, whose actual inverse is eq. 3.mxliv. The effective site comparison and finite gauge bound make its derivative a small perturbation on the same fixed spaces. The explicit contraction with that linear inverse solves

$$h_A(e^{i\lambda}) = v[A], \qquad R_0 D_0^{\sharp} A^{e^{i\lambda}} = 0, \qquad \|\lambda\|_{Y^{2,\alpha}} \le C\|A\|_{\mathrm{str},\alpha}. \tag{3.mlviii}$$

At bare tops its target is one and its first coordinate is exactly λ, so bare zeros are retained during this contraction. The resulting potential $\overline{A}$ has $q(\overline{A}) = b(B)$ if the original literal datum is V_B, by eq. 3.mxlvi. Its node factor equals the node values of the chosen gauge. The subsequent literal extension therefore differs from the original field by a gauge exactly one at every assigned top.

No tangent of the full q fibre is lost by imposing Landau gauge: eq. 3.mliii corrects its Landau component by a gauge vector in $\ker\tau_A$, and hence in $\ker q'(A)$. The action is constant along that vector. These two analytic maps consequently identify all tangent directions of the local $q = b(B)$ and literal fibres modulo that residual gauge group, and preserve the action. The inverse of the specified literal extension solves the mean-slice equations at a mean-slice point; uniqueness of their contraction proves this assertion, including the full group values. Differentiating the action identities proves physical constrained criticality of eq. 3.mlv at complex as well as real data.

3.32.9. *Competing multipliers and uniqueness of the critical orbit.* Smallness of a competing field does not by itself assert a small multiplier. Here the necessary readout is available from the actual row lift. Let H_0 be the zero-force physical row lift at U_0, so $Q_0H_0 = 1$. The normalized energy columns in eq. 3.dcccxlv and their proper strong upgrade in eq. 3.cm give a uniformly bounded right inverse from the row amplitude norm to the strong norm. The actual two-leg density estimate, paired with those columns, also gives, for some $s > 0$,

$$T_A = q'(A)H_0 = 1 + E_A^{\mathrm{row}}, \qquad \|E_A^{\mathrm{row}}\|_{\mathrm{Schur},s} \le Cr. \tag{3.mlix}$$

The matrix here is normalized by $b_c = \xi_c^{-1}\beta_c$; the Schur norm is the maximum of its absolute weighted row and column sums in $d(c,c')$, the layer distance between completed carriers. To verify the estimate, fix a fine leg in the variation of q' and use eq. 3.mxxxvii. The factor $(\eta/\xi_c)^4$ sums its fine carrier and the remaining increment gives r. The lift column has exponential cell energy decay. Carrier enlargement and its finitely many neighboring scales change this exponential by a fixed factor. The geometric growth and scale-ratio reserves leave a summable exponential after

taking both normalized sums. Thus this is an absolute matrix estimate, not merely an unweighted operator bound. A Neumann series in this Schur algebra proves that

$$\mathscr{H}_A = H_0 T_A^{-1}, \qquad q'(A)\mathscr{H}_A = 1 \tag{3.mlx}$$

has the same exponentially decaying energy columns.

Suppose now that a competing mean-slice point A is critical on $q = b(B)$. Surjectivity just proved supplies its unique multiplier ω. Write $\nu = \omega - \omega_0$. For the unit energy datum $b_{c,e} = \xi_c^{-1} e$, the physical coarse pairing gives the exact identity

$$\xi_c^3 \langle \nu(c), e\rangle = \left\langle F'(A) - F'(0) - (q'(A) - Q_0)^\sharp \omega_0, \mathscr{H}_A b_{c,e} \right\rangle. \tag{3.mlxi}$$

This identity does not estimate $F'(A)$ in the fine supremum norm. Instead integrate the actual Wilson Hessian along tA and use its weak plaquette form. On a rescaled complete cell of length ξ, its absolute pairing is bounded by

$$C\left\{ \|D_0 A\|_\infty \|D_0 w\|_1 + \xi^{-1}(\|A\|_\infty \|D_0 w\|_1 + \|D_0 A\|_\infty \|w\|_1) + \xi^{-2}\|A\|_\infty \|w\|_1 \right\}. \tag{3.mlxii}$$

Indeed differentiating a plaquette twice gives the two first covariant differences, and terms without such a difference carry its small plaquette curvature. The normal coordinate correction contains $\eta[A, w]$ in the first variation. Sum its difference by parts onto this product; $\eta \le \xi$ bounds the resulting terms by the same display. This calculation uses only first differences of A, w and the actual curvature along tA, never $D^2 A$ or an estimate of its current. It is also valid on a bare cell, where the rescaled plaquette is a fixed finite matrix product.

For $w = \mathscr{H}_A b_{c,e}$, the column energy bound and Cauchy–Schwarz on a physical cell give $\|w\|_1 \le C\xi^3 e^{-sd}$ and $\|D_0 w\|_1 \le C\xi^2 e^{-sd}$. Every term in eq. 3.mlxii is therefore $C\|A\|_{\mathrm{str},\alpha} e^{-sd}$. The other term in eq. 3.mlxi has force bound $C\gamma_0 \|A\|_{\mathrm{str},\alpha}\xi^{-3}$ and the same summable column estimate. Summing proves

$$\|\omega - \omega_0\|_{3,\infty} \le C\|A\|_{\mathrm{str},\alpha}. \tag{3.mlxiii}$$

This also holds for complex competing points, since all pairings used to obtain the identity are bilinear and the estimates use their absolute values.

Choose the fixed orbit neighborhood small enough that its mean-slice field and eq. 3.mlxiii lie in the contraction ball. Both solve the complete penalized system with zero penalty, so the fixed-point uniqueness identifies them. The literal-to-mean-slice construction then identifies the two literal representatives by a gauge exactly one at all assigned tops. This proves the claimed local orbit uniqueness, without assuming a multiplier bound for a competing critical point.

3.32.10. *Physical continuation bounds and localized response.* The normal and left gradients are related by $F'(A) = \mathcal{E}_{\eta A}^\sharp j(U_A)$. The exact stationarity equation and the fixed-leg density of q' give

$$\|j(U_{A(B)}) - j_0\|_{3,\infty} + \|F(U_{A(B)}) - F(U_0)\|_{2,\infty} \le C\|B\|_{1,\infty}. \tag{3.mlxiv}$$

Here the curvature norm is the assigned plaquette norm with weight L^2. For the current, subtract $Q_0^\sharp \omega_0$ from $q'(A)^\sharp(\omega_0 + \nu)$, use eq. 3.mxxxviii, and multiply by $\mathcal{E}_{\eta A}^{-\sharp} = 1 + O(\eta A)$. For curvature, differentiate the four exact link factors; their first differences give $D_0 A$ and their commutators give A^2, with weights L^{-2}. The bare calculation is the same finite product with $\mathsf{L} = \eta$. The specified literal gauge has supremum distance $C\|B\|$ from one. Its adjoint action changes the reference curvature and current by at most $C\|B\|\kappa_0$ and $C\|B\|\gamma_0$. Thus eq. 3.mlxiv holds for $\widehat{U}(B)$ as well. In particular its actual curvature and current ceilings increase by at most $C\|B\|_{1,\infty}$.

There are two different coarse multipliers. The one in eq. 3.mxxx belongs to reduced straightened coordinates. The derivative of the literal critical action $S(B) = F(A(B))$ is instead

$$D_B S = Db(B)^\sharp(\omega_0 + \nu(B)) =: \mu_B, \qquad \mu_{\mathrm{left}}(B) = \mathcal{E}_{\xi B}^{-\sharp}\mu_B. \tag{3.mlxv}$$

The inverse datum chart has bounded normalized row and column sums on its adjacent complete carriers. Physical transpose therefore maps amplitude three to amplitude three: normalized dual variables are ξb and $\xi^3\omega$. Consequently

$$\|\mu_B - Db(0)^\sharp\omega_0\|_{3,\infty} + \|\mu_{\mathrm{left}}(B) - Db(0)^\sharp\omega_0\|_{3,\infty} \le C\|B\|_{1,\infty}. \tag{3.mlxvi}$$

This formula retains the outward datum derivative at zero; it need not be the identity. The local fibre coordinates and stationarity proved above show that this is the actual literal physical multiplier, not only a formal action derivative.

For clarity, the localization claim can be stated without assuming an all-order absolute tree sum. Give each completed cell a weight $w = e^{s\psi}$, where ψ is one-Lipschitz in the proper layer metric. Use w times each local norm, and the same weight at the associated row. The base exponential saddle columns, followed by the local finite-p bootstrap, give eq. 3.mxxxix in these weighted spaces for a fixed small $s > 0$. Enlarging a carrier changes w by at most a fixed factor. The residual's derivative consequently has weighted operator norm Cr. No derivatives of the artificial weight are needed: this is a cellwise norm estimate with bounded carrier comparisons. The differentiated fixed point therefore gives

$$\|DA(B)[h]\|_{\mathrm{str},\alpha;w} + \|D\nu(B)[h]\|_{3,\infty;w} \le C\|h\|_{1,\infty;w}. \tag{3.mlxvii}$$

The same bound holds at every point of a smaller fixed complex datum ball. Banach Cauchy estimates there give one weighted input slot and any fixed number $m-1$ of unweighted slots, with constant C_m independent of depth and volume. The explicit datum reduction and literal extension have bounded carriers and the same estimates. Choosing ψ to be distance from any selected source carrier proves, for point data $c_1, \dots, c_m$, normalized response at an output cell y bounded by

$$C_m e^{-s\max_i d(y,c_i)}. \tag{3.mlxviii}$$

This establishes localized holomorphic response in the fixed proper geometry. It asserts neither an all-arity summed connected kernel bound nor existence of a proper reference from arbitrary curvature-only coarse data. All inverse constants, branch choices, and admitted complex increments used above are uniform around the given real physical reference. This completes the proof of Prop. 3.117.

3.33. **Critical correction of a regular trial on the actual proper region.** The reference in Prop. 3.117 can be a regular trial rather than a critical point. Its stationarity defect must then remain in the equation. This gives actual critical fields on the proper image of sufficiently regular fine trials. It does not supply a fine trial for every proposed curvature-only mixed datum. The fixed one-step face lift is an explicit example; its existence theorem was already proved in § 3.8.

Retain the determining geometry, actual inner action, complete carriers, true bare boundary, physical pairings and strong norms of § 3.32. The reference U_0 is real, with its actual curvature and Wilson current satisfying

$$\|F(U_0)\|_{2,\infty} \le \kappa_*, \qquad \|j_0\|_{3,\infty} \le \gamma_*. \tag{3.mlxix}$$

The fixed ceilings have strict room inside the bounds for the current-dependent local $C^{1,\alpha}$ regularity and the site Schauder estimates. An independent real row multiplier ω_0 may be chosen with $\|\omega_0\|_{3,\infty} \le \gamma_{\omega,*}$, the precision ceiling. Define the actual defect

$$e_0 = j_0 - Q_0^\sharp\omega_0, \qquad \delta = \|e_0\|_{3,\infty}. \tag{3.mlxx}$$

There is no stationarity assumption. In particular $\omega_0 = 0$ is allowed and gives $\delta = \|j_0\|_{3,\infty}$.

Proposition 3.120 (Correction of an actual regular trial)**.** There are positive δ_*, ρ, r, C, independent of depth, compatible volume and the separated geometry, such that a trial satisfying eq. 3.mlxix

with $\delta < \delta_*$ has the following property. Write $V_0 = M_{\mathbf{B}}(U_0)$ and let $V_B(c) = e^{i\xi_c B(c)} V_0(c)$, with $\|B\|_{1,\infty} < \rho$. There is a holomorphic pair $(A(B), \nu(B))$, real at real B, satisfying

$$\begin{aligned} q(A(B)) &= b(B), \qquad R_0 D_0^\sharp A(B) = 0, \\ F'(A(B)) &= q'(A(B))^\sharp(\omega_0 + \nu(B)), \\ \|A(B)\|_{\mathrm{str},\alpha} + \|\nu(B)\|_{3,\infty} &\le C(\delta + \|B\|_{1,\infty}). \end{aligned} \tag{3.mlxxi}$$

Here $F(A) = A_{\Omega_0}(e^{i\eta A} U_0)$ and $b(B)$ is the actual outward-corrected datum in eq. 3.mxxxiii. An admitted specified fine gauge gives a literal constrained-critical representative $\widehat{U}(B)$ with $M_{\mathbf{B}}\widehat{U}(B) = V_B$. It is the unique critical orbit in a fixed neighborhood of the trial, modulo gauges one at all assigned tops. The curvature and current differences from the trial have weighted bounds $C(\delta + \|B\|_{1,\infty})$. The literal multiplier and fixed-order localized analytic responses have the corresponding bounds of the critical-reference theorem. ∎

Proof. We give the changes to the model proof explicitly. None of the following uses an inverse at a desired, as yet unknown orbit. Put $H_{L,0} = F''(0) - \langle \omega_0, q''(0) \rangle$. The real gap in eq. 3.dcccxxviii and the full strong saddle in eq. 3.cm apply to $H_{L,0} + D_0 R_0 D_0^\sharp + Q_0^\sharp a_f Q_0$ without stationarity. Removing its row mass by the stated multiplier shift gives the bounded full inverse $\mathcal{L}_0^{-1}$ in eq. 3.mxxxix. In particular no physical Ward identity is used to turn a slice equation into a full equation at the trial.

Use the complete actual gradient residual $N_F(A) = F'(A) - j_0 - F''(0)A$. The local Coulomb regularity needed in Lem. 3.118 comes from eq. 3.mlxix, not from criticality. Its proof therefore gives exactly the same quadratic bound and Cr Lipschitz constant. The actual constraint densities in eq. 3.mxxxviii also hold about this real regular reference, including the $\omega_0 q'''$ remainder. Subtracting the trial equation now gives the following complete fixed point:

$$\begin{aligned} (A, \nu) = \mathcal{L}_0^{-1}\big(&- e_0 - N_F(A) + (q'(A) - Q_0 - q''(0)[A])^\sharp \omega_0 \\ &+ (q'(A) - Q_0)^\sharp \nu, \quad b(B) - C_q(A)\big). \end{aligned} \tag{3.mlxxii}$$

The new term is the actual force $-e_0$, not an omitted linear remainder. The nonlinear part has size Cr^2 and Lipschitz constant Cr in the same joint strong/multiplier ball. Choose r below the primitive ceilings with $Cr < 1/4$, then choose δ_*, ρ so that $C(\delta_* + \rho) < r/8$. The same explicit iteration is invariant and contractive and gives the last bound in eq. 3.mlxxi. Its affine source does not affect differentiated contraction estimates.

The actual node diagram, outward datum map, and effective gauge cocycle are reference identities. Thus eq. 3.mxlvi and eq. 3.mxlix remain exact. The site extensions and the bound in eq. 3.mxxvi use the actual trial current already assumed in eq. 3.mlxix. They give the same corrected gauge tests in $\ker \tau_A$ and the surjectivity in eq. 3.mliii. Testing the full penalized equation at the fixed point therefore gives eq. 3.mliv, with arbitrary test $\sigma \in \operatorname{ran} R_0$. Its conjugate test removes the penalty also at complex data. There is no extra e_0 in this gauge identity: it is the exact Wilson equation being tested, not only its Taylor part.

The admitted literal extension and the mean-slice construction in eq. 3.mlv and eq. 3.mlviii likewise use no trial stationarity. They identify full fibres modulo precisely the top-fixed gauge group, including bare zeros and outward factors.

For uniqueness use the FULL penalized row lift H_0 at the trial. It has $Q_0 H_0 = 1$, the strong bound and the normalized exponential energy columns. It need not be a horizontal physical lift. Those are all the properties used to construct the right inverse $\mathscr{H}_A = H_0(q'(A)H_0)^{-1}$. For a competing critical mean-slice point with multiplier $\omega_0 + \nu$, its correct readout is

$$\xi_c^3 \langle \nu(c), e \rangle = \langle e_0 + F'(A) - j_0 - (q'(A) - Q_0)^\sharp \omega_0, \mathscr{H}_A b_{c,e} \rangle, \qquad b_{c,e} = \xi_c^{-1} e. \tag{3.mlxxiii}$$

The new first pairing is bounded by $C\delta$: its fine force is at most $\delta \xi^{-3}$ on a cell and the column's physical L^1 norm is $C\xi^3 e^{-sd}$. The weak Wilson readout in eq. 3.mlxii bounds the other terms.

Thus

$$\|\omega-\omega_0\|_{3,\infty}\le C(\delta+\|A\|_{\mathrm{str},\alpha}). \tag{3.mlxxiv}$$

A smaller fixed orbit neighborhood and δ_* put this pair in the contraction ball. The mean-slice and fixed-point uniqueness prove the claimed orbit uniqueness.

Finally subtract $j_0=Q_0^\sharp\omega_0+e_0$ from the new stationarity equation. The transpose density, normal-to-left gradient differential and fine literal gauge give

$$\|j(\widehat U(B))-j_0\|_{3,\infty}+\|F(\widehat U(B))-F(U_0)\|_{2,\infty}\le C(\delta+\|B\|_{1,\infty}). \tag{3.mlxxv}$$

The literal parameter multiplier is exactly $Db(B)^\sharp(\omega_0+\nu(B))$, with the left-gradient conversion in eq. 3.mlxv. Its difference from $Db(0)^\sharp\omega_0$ has the same defect-plus-datum bound. The weighted differentiated contraction and fixed-order Cauchy argument in eq. 3.mlxvii are unchanged. This proves the proposition. □

3.33.1. *What is obtained on the real proper image.* For the fixed determining geometry let $\mathscr{T}_{\mathrm{reg}}$ be the real fine trials obeying eq. 3.mlxix, with $\|j(U)\|_{3,\infty}<\delta_*/4$. Define the actual image

$$\mathscr{I}_{\mathrm{reg}}=\{M_{\mathbf{B}}(U):U\in\mathscr{T}_{\mathrm{reg}}\}. \tag{3.mlxxvi}$$

This is a definition by the given fine fields, not a characterization using coarse plaquettes alone. For every $V\in\mathscr{I}_{\mathrm{reg}}$, choose an actual preimage U_0. Apply Prop. 3.120 with $\omega_0=0$ and $B=0$. It produces a critical fine field with literal mean V. Thus the constrained-critical fibre is nonempty at every point of this real image. The same preimage gives a holomorphic critical branch on a uniform complex raw-angle ball about V, without presupposing a critical reference there.

The image is nonempty and contains genuinely nonflat examples. For instance use the identity reference and a smooth one-colour potential supported in a complete homogeneous layer patch of scale ξ. A rescaled fixed profile with amplitude t/ξ has actual fine curvature at most $C|t|\xi^{-2}$ and Wilson current at most $C|t|\xi^{-3}$. These estimates follow directly from its two bounded physical differences and the exact plaquette current; no unknown inverse is used. A cutoff equal to one on the complete carriers of a coarse plaquette may contain $a_2(x)=t(x_1-y_1)H/\xi^2$, $a_1=0$, for one fixed colour H. There all factors commute. The actual means reduce to line averages plus contour gradients; the latter cancel around the coarse plaquette. Its mean plaquette is therefore e^{itH}, whereas the cutoff makes the trial the identity outside the patch. For sufficiently small nonzero t this is a nonflat point of the image in eq. 3.mlxxvi, with bounds independent of the layer depth. This example requires the indicated complete patch; flat trials require none.

There is also a precise local independence of the chosen trial. Suppose two real trials are related by

$$U_1^g=e^{i\eta C}U_0,\qquad \|C\|_{\mathrm{str},\alpha;U_0}<\varepsilon_r, \tag{3.mlxxvii}$$

where g is unitary and one at all assigned tops. Reduce the trial-current ceiling and working datum balls once. The branches built about these trials then agree as physical orbits on their common datum domain, and their total Wilson actions agree.

Indeed covariance of the reference transports identifies the strong norms before the real gauge g. The relative connection after it is the divided logarithm of $e^{i\eta A_1}e^{i\eta C}$. Its Banach product estimate gives strong norm at most $C(\|A_1\|_{\mathrm{str},\alpha}+\|C\|_{\mathrm{str},\alpha})$. This is the same mesh-uniform relative BCH calculation used in eq. 3.mlvi. Choose ε_r and the working radii so this and the specified literal extension fit the U_0 uniqueness neighborhood. Both fields are then critical with the same literal datum. The mean-slice map and the competing-multiplier bound in eq. 3.mlxxiv identify them modulo a top-fixed fine gauge. The slice and extension are holomorphic in the datum, so the identification is holomorphic on the overlap. Gauge invariance identifies the actions. These transitions satisfy the cocycle where the controlled overlaps meet: their quotient fixes an invertible connection and all assigned tops, and propagation along each connected fine component forces it to be the identity.

The construction is also covariant under simultaneous arbitrary real unitary fine gauge changes of a trial and its datum. They conjugate the actual full saddle and all relative coordinates isometrically. Its unique small fixed point consequently transforms covariantly; different admitted literal extensions differ by a top-fixed gauge. The action is invariant. No smallness of the gauge function itself is required when the reference is transformed at the same time.

These facts give existence on the real image and holomorphic descent on controlled trial overlaps. Lem. 3.122 and Prop. 3.124 identify arbitrary real regular critical preimages of the same mixed datum, including those not initially given in a common trial chart. Their action is consequently canonical on the stated regular class. The existence proof there constructs regular backgrounds for every sufficiently small curvature datum by datum descent and critical ascent; it does not require a separately smoothed fine trial for that datum.

3.33.2. *The fixed one-step face lift as a concrete corollary.* For comparison with the already proved first-step result, fix $\eta = 1/L$ and the actual centered blocks. Given any real coarse V, set $U_0(V) = I$ on internal fine links and $U_0(V)(x,\mu) = V_\mu(y)$ on every positive edge crossing from $B(y)$ to $B(y+e_\mu)$; use the inverse on reversed edges. All coordinate-order internal paths stay in their block. Each length-one coarse line used in the actual mean crosses exactly one face. Thus its internal contour transport is I, its line transport is $V_\mu(y)$, and its central reference line is the same matrix. In the notation of the actual mean,

$$T_{y,x}(U_0(V)) = I, \qquad Z_x(U_0(V)) = I, \qquad \mathcal{M}U_0(V) = V. \tag{3.mlxxviii}$$

The logarithm taken here is $\log Z_x = 0$, never a logarithm or a group root of $V_\mu(y)$. Large and central holonomies present no obstruction.

Every fine plaquette is I unless both of its directions cross their block faces. In that case its four ordered factors are exactly the coarse plaquette, up to the frame conjugation or orientation. With $\epsilon = \sup_P \|V(\partial P) - I\|$ this gives

$$\|F(U_0(V))\|_\infty \le CL^2\epsilon, \qquad \|j(U_0(V))\|_\infty \le CL^3\epsilon. \tag{3.mlxxix}$$

The current estimate uses the bounded number of incident fine plaquettes and their exact sine terms with prefactor η^{-3}. For the homogeneous one-step geometry the assigned scale is one. Choose $\epsilon < \epsilon_L$ so these two bounds meet the trial ceilings. The proposition, at $B = 0$ and $\omega_0 = 0$, then constructs the actual one-step critical orbit of mean V.

This recovers the existence in § 3.8 through the new full proper-region inverse. The bounded-area block-contour comparison there puts every competing fixed-L regular field of that same mean into this trial neighborhood after a node-fixed axial gauge; all fixed strong differences cost constants depending only on L. Local uniqueness therefore identifies this branch with that previously constructed background, including its compatible complex continuations. Here coarse chart closeness really does give the required fine comparison: if $V_1 = e^{iB}V_0$, the relative fine potential of their face lifts is $\eta^{-1}B$ on crossing edges and zero internally. Since $\eta = 1/L$ is fixed, its covariant strong norm is at most $C_L\|B\|_\infty$; ordinary finite differences cost at most fixed powers of L, with unitary reference transports. Thus shrinking the coarse working balls makes their overlaps controlled in the sense of eq. 3.mlxxvii. The face-lift corollary concerns one bare step at fixed L. The multilevel construction in Prop. 3.124 uses the regular solution of the preceding truncation as its next real trial, with the explicit L^3 current loss and the subsequent uniform reset.

Indeed an iterated face lift concentrates rather than spreads curvature. Take one positive coarse link $e^{i\theta H}$, all other links I, on a coarse torus of side at least two, with $H^2 = I$ and small nonzero θ. After j face lifts, $\eta = L^{-j}$ and a fine corner plaquette still equals $e^{i\theta H}$. A face edge adjacent to exactly that nontrivial plaquette has current $\pm\eta^{-3}\sin\theta H$; the other incident plaquettes are I. Its action variation is $\eta\sin\theta$, and the physical pairing is η^4. Thus at assigned scale one,

$$\|F(U_{\mathrm{face},j})\|_\infty \ge L^{2j}|\theta|, \qquad \|j(U_{\mathrm{face},j})\|_\infty \ge L^{3j}|\sin\theta|. \tag{3.mlxxx}$$

Fixed small coarse curvature consequently does not make these iterated trials uniformly regular. Prop. 3.124 instead descends the coarse datum and solves a critical equation at each ascent, applying Lem. 3.123 before the next level. This controls the current as well as curvature uniformly in the depth.

3.34. **Regular backgrounds on separated regions.** The preceding proper-region inverse and trial correction complete the ascent through the determining sets of eq. 3.cccvi. We give the matching argument, including the current bound at the action boundary and uniqueness among all regular critical configurations. These determining sets have no inward rows. Every outward row and its entire lower endpoint block remain part of the construction.

Fix $0 < \alpha < 1$ and the geometric floors needed for Props. 3.88 and 3.120. Use $L \geq 13$, and the aligned separated geometry of § 3.11.23; increase its floors below when choosing a fixed local regularity patch. Constants may depend on these choices and L, α, but not on depth, the number of pillars, the volume, or the shapes and number of components of the regions. They apply to every truncation with these floors.

For a real field define

$$\mathcal{R}(U) = \max\left\{ \sup_{p} \eta^{-2}\mathsf{L}(p)^2|\partial U(p) - 1|, \sup_{b \text{ full}} \mathsf{L}(b)^3|J(U)(b)| \right\}. \tag{3.mlxxxi}$$

Here the first supremum is over inner plaquettes, and a full bond has all six incident plaquettes in the action region. Write $\mathcal{R}_{\mathrm{mat}}$ for the same expression with $\eta^{-2}|D_U^{\mathsf{T}}\partial U|$ in place of $|J(U)|$. The scalar and traceless parts of a plaquette give

$$\mathcal{R}(U) \leq \mathcal{R}_{\mathrm{mat}}(U), \qquad \mathcal{R}_{\mathrm{mat}}(U) \leq r + \tfrac{3}{2}r^2 \quad \text{if } \mathcal{R}(U) \leq r. \tag{3.mlxxxii}$$

Indeed write $\partial U(p) = c_p 1 + i\eta^2 F_p$. The scalar divergence at a full bond is a sum of three opposite differences of c_p, with $1 - c_p = |\partial U(p) - 1|^2/2$. Use $\eta \leq \mathsf{L}(b) \leq \mathsf{L}(p)$ on each incident plaquette. This proves the second inequality; the traceless part proves the first. Thus either regularity convention gives the other after a fixed reduction of its ceiling.

3.34.1. *The strong norm of a regular comparison.* The following observation permits use of the strong nonlinear chart for competitors initially specified only by curvature and current. It is an elliptic consequence of those currents, not an embedding of a discrete C^1 ball into a uniform $C^{1,\alpha}$ ball.

For a comparison based at U_0, write

$$\mathcal{N}_1(A) = \max\left\{ \|\mathsf{L}A\|_\infty, \|\mathsf{L}^2\nabla_{U_0}A\|_\infty \right\}. \tag{3.mlxxxiii}$$

This is the weighted first-order supremum norm used in the slice absorption estimate. The energy space X retains its physical L^2 definition in the inverse estimates.

Lemma 3.121 (Improvement of a comparison in the slice)**.** Let U_0 and $U_1 = e^{i\eta A}U_0$ be real, with $\mathcal{R}(U_i) \leq \varepsilon$. Suppose that $R_0 D_0^\sharp A = 0$ and $a = \|\mathsf{L}A\|_\infty$ is below a fixed small ceiling. Let $b = q(A)$ be the actual straightened per-length row coordinate. Put $\beta = \sup_c \xi_c|b_c|$. Then

$$\mathcal{N}_1(A) + \|\mathsf{L}^3 D_0^\sharp D_0 A\|_\infty + \|A\|_{\mathrm{str},\alpha} \leq C(\varepsilon + \beta). \tag{3.mlxxxiv}$$

The ceiling on a and the constant are uniform in the geometry. No critical equation for U_0 or U_1 is required. ∎

Proof. The covariant D_0 in $D_0^\sharp D_0 A$ is the curl, whereas the D_0 in the Landau condition is the site derivative. We first use the actual first-order slice calculation eq. 3.ccclxxiv. Its boundary and normalization checks are useful here. The exact differentiated plaquette product gives, at a full bond,

$$\mathsf{L}(b)^3|D_0^\sharp D_0 A(b)| \leq C\varepsilon + C_L a\mathcal{N}_1(A).$$

One obtains this by subtracting the two currents in their common normal coordinate. The normal-to-left logarithmic differential is $1 + O(\eta A)$. The linear curvature correction has size $C\varepsilon a$, and the nonlinear remainder has size $C_L a\mathcal{N}_1(A)$. These estimates use first covariant differences of A. In the paired plaquette terms a difference of a product places one difference on one factor; terms with two undifferenced potentials carry the remaining mesh factor. The elementary inequality $\eta|\nabla A| \le C \sup|A|$ on the same stencil pays that factor. It gives no inverse power of the fine mesh.

At a nonfull inner bond the bond and every bond in its action plaquettes belong to the bare determining layer. There ηA is the bare row coordinate. The at most six plaquette stencils therefore give the bound 24β for $\mathsf{L}^3|D_0^\sharp D_0 A|$. On an outward row the additional factor reads only the prescribed interior rows of the preceding level, so its group angle is at most $C_L\beta$. The same conclusion holds if β is initially measured after the triangular datum change eq. 3.mxxxiii.

For clarity, use the actual derivative $Q_0 = q'(0)$ in the completed operator, rather than an auxiliary comb derivative. The actual two-leg bound gives

$$\sup_c \xi_c|q(A)_c - Q_0 A(c)| \le Ca^2, \qquad \|\mathsf{L}^3 Q_0^\sharp a_f Q_0 A\|_\infty \le C(\beta + a^2).$$

In the second estimate $a_f(c)$ has size ξ_c^{-2}. The physical transpose density cancels the fine counting measure on the complete row carrier. This is precisely the fixed-leg estimate used in eq. 3.dcccxcv; it includes the outer block of an outward row.

Let $K_0 = K_{U_0,0}$ be eq. 3.dcccxcviii with zero independent multiplier. The exact identity in the slice is

$$K_0 A = D_0^\sharp D_0 A + \Delta'_{U_0} A + Q_0^\sharp a_f Q_0 A, \qquad \|\mathsf{L}^3 \Delta'_{U_0} A\|_\infty \le C_L \varepsilon a. \tag{3.mlxxxv}$$

To verify the last bound directly, the difference between the quadratic Wilson plaquette and the covariant curl square has coefficients bounded by $C\eta^{-2}|\partial U_0 - 1|$ on its finite four-link stencil. It therefore contributes at most $C\varepsilon \mathsf{L}^{-2}$ times the local maximum of $|A|$. Adjacent scales cost at most L.

The first-order consequence of Prop. 3.88 applied to this equation gives

$$\mathcal{N}_1(A) \le C_1(\varepsilon + \beta) + C_2(\varepsilon + a)\mathcal{N}_1(A).$$

Every norm is finite on the finite lattice, so absorption is legitimate before any uniform estimate has been obtained. Choose ε, a so that the last coefficient is at most $1/2$. This proves the first-order bound and then the displayed bound on $D_0^\sharp D_0 A$. Now the entire right side of eq. 3.mlxxxv has weighted supremum at most $C(\varepsilon + \beta)$. Its $\mathcal{F}_p$ norm has the same bound for each fixed finite $p > 4/(1-\alpha)$. Apply eq. 3.dcccxcix again to obtain the strong bound in eq. 3.mlxxxiv. □

Lemma 3.122 (Comparison of arbitrary regular fibres)**.** Suppose

$$\mathcal{R}(U_i) \le \varepsilon, \qquad \sup_c |M_{\mathbf{B}} U_1(c)(M_{\mathbf{B}} U_0(c))^{-1} - 1| \le \delta.$$

For $\varepsilon + \delta$ below a fixed ceiling, there are a real A and a unitary gauge g such that

$$U_1^g = e^{i\eta A} U_0, \qquad R_0 D_0^\sharp A = 0, \qquad g(y) = v[A](y) \quad (y \text{ a top}), \qquad \|A\|_{\mathrm{str},\alpha} \le C(\varepsilon + \delta). \tag{3.mlxxxvi}$$

The reduced datum is the literal datum difference under eq. 3.mxxxiii. In particular, equal literal data give $q(A) = 0$ on every row. If also $A = 0$, g belongs to $\mathcal{G}_{\mathrm{res}}$. ■

Proof. First use the preliminary gauge constructed in eq. 3.cccxix–eq. 3.cccxx. Here are the uniformity and seam checks. With $W_n = \mathcal{M}^n U_1$ and $W_{0,n} = \mathcal{M}^n U_0$, descend from $g_k = 1$ by

$$g_n = 1 \text{ on the level-}n \text{ tops}, \qquad g_n(x) = W_{0,n}(y,x)^{-1} g_{n+1}(y) W_n(y,x) \quad (x \in B(y)).$$

The blocks of the next level and the current tops partition the current member, and the zero-length contour equals one. Thus these formulas extend one another and define one fine gauge $\widehat{g}$. They align all the averaged internal contours and fix every final top.

In a double block the reconstructed link differs from its coarse mean by C_L times the local plaquette defect, by eq. 3.cccxviii. The coefficient of a difference of the coarse means is one, since all other factors are unitary. At a seam the inward comparison link belongs to the corridor of the outward row one level above. The latter's outer block lies in the lower layer; its internal paths use at most $2(L-1)$ prescribed lower rows. Telescoping gives an endpoint defect at most $2(L-1)\delta + C_L\varepsilon$. The other endpoint is already aligned. Along a deeper pillar the remaining errors sum as $C_L\varepsilon \sum_{n<j} L^{2(n-j)}$. Consequently every carried link, at every intermediate level, has relative group error at most $C_L(\varepsilon+\delta)$. This argument also controls the omitted inward comparison links; no inward constraint is added.

Ascend through the truncations, keeping these same two fine configurations; the fields are not replaced by their averages. If $\mathsf{L}^{(j)}$ is a truncated assigned scale in physical units, then $\mathsf{L}^{(j)} \le \mathsf{L}^{(k)}$ on each bond and plaquette. Both fields consequently have truncated regularity at most ε at every stage, directly from their full-depth bounds. The real gauge corrections preserve these bounds. At an intermediate truncation the carried-link estimate just proved bounds its literal datum difference by $C_L(\varepsilon+\delta)$; the triangular outward change therefore gives $\beta_j = \sup \xi|q^{(j)}(A)| \le C_L(\varepsilon+\delta)$. At the last truncation its bound is instead $C_L\delta$. At depth zero the relative logarithm has $\mathcal{N}_1$ norm at most $C_L(\varepsilon+\delta)$, because the mesh in its units is one. At the next level replace A^- by $A_0 = LA^-$; its $\mathcal{N}_1$ norm increases by at most L^2, and $\eta A_0 = \eta^- A^-$ leaves all old node factors unchanged. Apply the real site correction following eq. 3.ccclxxii. Prescribe the correction factor one at old tops and $v_{j-1}[A_0]^{-1}v_j[A_0]$ at promoted tops, with the multiplication order of the node diagram. The latter defect is at most $C_L\|\mathsf{L}A_0\|_\infty$. The site equation has the actual weighted inverse and fixes the Landau condition; its nonlinear remainders have the quadratic bounds displayed there. It therefore gives $\|\mathsf{L}A\|_\infty \le C_L\mathcal{N}_1(A_0)$ and the required normalized node values.

The first-order absorption in Lem. 3.121 now resets $\mathcal{N}_1(A)$ to $C_*(\varepsilon+\delta)$ at each level. Choose C_* first, then $\varepsilon+\delta$ so that $C_L L^2 C_*(\varepsilon+\delta)$ lies below the fixed site and slice ceilings. This is one choice for all levels, rather than an iteration of the factor L^2. On the last truncation the normalized datum is the actual literal difference, because $\widehat{g}$ fixes both endpoints of each determining row. The explicit triangular outward formula gives $\sup \xi|q(A)| \le C_L\delta$, including a reversed outward orientation. The strong conclusion follows from Lem. 3.121. If $\delta = 0$, all interior coordinates vanish, hence every outward node factor and outward coordinate also vanish. Finally $v[0] = 1$, and every node of the present determining set is a top, so $A = 0$ makes g a residual gauge. □

3.34.2. *Resetting the current as well as curvature.*

Lemma 3.123 (Uniform regularity reset)**.** There are a fixed geometric floor, a ceiling $a_c > 0$, and C_R such that every real constrained-critical field with $\mathcal{R}(U) \le a_c$ satisfies

$$\mathcal{R}(U) \le C_R\varepsilon \tag{3.mlxxxvii}$$

when all its ordinary and mixed datum test holonomies are within ε of one. The same conclusion, with twice the constant, holds for $\mathcal{R}_{\mathrm{mat}}$ after decreasing the ceiling. The constants are uniform over all truncations. ■

Proof. We apply the local calculation underlying eq. 3.ccclxxv, with the actual flat inverse and nonlinear equation now available. We record the admission checks needed for that application. First fix the auxiliary flat construction and its collar radius ρ_c. Next choose a fixed patch parameter $\sigma \ge L^2\rho_c$ that is a power of L, and only then increase the ambient separation floor to $\rho \ge L^2\sigma$. Put $R = 20L\sigma$ in layer distance. The auxiliary nested cubes described in the local datum-profile calculation have positive-level union inside $Q_{\sigma-2}$, a complete two-level inner patch, and a flat bare exterior. The local real gauge on Q_σ uses curvature and the actual full current. It gives potential, first difference, and flat second-order reads at most $C\sigma r_R(x)$ in local units. Its domain condition is $C\sigma a_c < c$.

Extend that potential by zero on the auxiliary torus. All cutoff plaquettes belong to the bare layer, so their physical current, including the terms created by the cutoff, has weighted bound $C\sigma r_R(x)$. In the positive-level members there was no cutoff. A constrained variation on the auxiliary geometry patches to a constrained variation of the original field: a row of lower assigned endpoints is fixed by induction in the actual averaging recursion; a row with an endpoint at its reading level is an auxiliary row and has its entire double block inside Q_σ. The outward endpoint block is included in this assertion. Every changed action plaquette is also inside Q_σ. The extended field is therefore critical at its own auxiliary datum.

Apply the local datum transport gauge and the real site correction relative to the identity. They give a preliminary slice potential of small weighted supremum and a datum of size $b_f \le C\sigma^2 r_R(x)$. The zero extension's current and Lem. 3.121 put this slice potential in the strong ball, with size $r_f \le C\sigma^2 r_R(x)$. Thus the actual flat case of Prop. 3.117 applies, and its local uniqueness identifies the auxiliary critical field with that solution. The required conditions are $C\sigma^2 a_c < \min(r, \rho)$, with the fixed flat chart radii r, ρ. They can be imposed after σ is fixed.

Split the flat saddle's exponentially decaying datum sum at distance $\sigma/8$. In the inner part a row at y has size $C(1 + |y - x|)v_R(x)$ in the datum transport gauge; its outward factor is one because its entire lower internal contour has been prescribed. The linear distance is summable against the kernel. The outer part contributes $Ce^{-c\sigma} b_f$. The exact nonlinear force and row remainders contribute Cr_f^2. The multiplier readout uses the same split and contributes $Ce^{-c\sigma}(b_f + r_f)$ outside and Cr_f^2 nonlinearly. At the full bond the exact equation $F'(A) = q'(A)^\sharp \omega$ reads only the complete carriers of nearby rows. Its physical transpose density bounds the current by that multiplier estimate. The first-order field estimate bounds curvature. Thus both readouts obey

$$r_U(x) \le K v_R(x) + \{C\sigma^2 e^{-c\sigma} + C\sigma^4 a_c\} r_R(x).$$

The original and auxiliary local scales differ by at most L at the readouts; this fixed factor is included in C, K. Bare points away from the next member use prescribed plaquettes directly and the sum of their six current terms.

Choose σ so the exponential term is at most $1/4$, and then a_c so the quadratic coefficient is at most $1/4$ and all preceding local and strong chart conditions hold. The result is $r_U(x) \le K v_R(x) + r_R(x)/2$. Let $P(x) = \sup_\tau e^{-\delta d_{\mathbf{B}}(x,\tau)} |\partial V(\tau) - 1|$, where $e^{\delta R} = \sqrt{2}$. For $t > 0$ maximize $r_U(x)/(P(x)+t)$. At a maximizing point its value S satisfies $S \le \sqrt{2}K + S/\sqrt{2}$; this uses $P(y)+t \le \sqrt{2}(P(x)+t)$ in the patch. Letting t decrease to zero gives $r_U \le 5KP$. The bounded displacement from a corner to the base of a readout changes this constant by a fixed factor. Since $P \le \varepsilon$, this proves eq. 3.mlxxxvii. Finally, eq. 3.mlxxxii gives the current assertion. □

3.34.3. *The mixed datum and the critical ascent.*

Proposition 3.124 (The regular proper background)**.** For the geometry fixed above there are $a_0, a_1, C_B > 0$, independent of depth and volume, with $C_B a_1 < a_0$, such that every real mixed datum whose ordinary and mixed test holonomies are within $\varepsilon \le a_1$ of one has a constrained-critical configuration $U(V)$ with

$$M_{\mathbf{B}} U(V) = V, \qquad \mathcal{R}_{\mathrm{mat}}(U(V)) \le C_B \varepsilon. \tag{3.mlxxxviii}$$

Any two critical configurations for the same datum with $\mathcal{R} \le a_0$ lie in the same residual gauge orbit. The orbit is covariant under real gauges of the datum. ■

Proof. The constants are chosen in the following order. Fix the inverse and site bounds, the regularity patch and its ceiling a_c, and a reset constant $B \ge \max(12, 2C_R)$. Fix the one-level trial-correction constant C_A below. Finally choose a_0 below the comparison, multiplier and PNC uniqueness ceilings, and choose a_1 so that

$$C_\downarrow a_1 < \varepsilon_\downarrow, \qquad C_A L^3 B C_\downarrow a_1 < \min(a_c, a_0), \qquad 6L^3 B C_\downarrow a_1 < \min(\kappa_*, \gamma_*, \delta_*), \tag{3.mlxxxix}$$

as well as the fixed RTC datum radius. Here $\kappa_*, \gamma_*, \delta_*$ are its real-trial ceilings. Enlarging a fixed constant or shrinking a_1 makes the inequalities strict without changing their uniformity.

Descend the datum by eq. 3.cdxxvi–eq. 3.cdxxvii. The two-radius argument there gives $V^{(n)}$ at every truncation, with all tests bounded by $\varepsilon' = C_{\downarrow}\varepsilon$ and with the new one-step mismatch at most ε'. There is no accumulated factor: a deep new plaquette is one or a single old coarse plaquette; every seam comparison uses the smaller-radius tests within six current-level spacings of the seam. Separation keeps that neighborhood outside the part changed at the previous descent.

At depth zero every inner bond is prescribed. The field $V_0^{(0)}$ is therefore critical in its singleton fibre. Its curvature is at most ε' and its full current is at most $6\varepsilon'$, in bare units. This starts the bound $\mathcal{R}_{\mathrm{mat}} \le B\varepsilon'$.

Suppose $U^{(n-1)}$ has that bound. Regard the same physical field on the determining set of depth n. An assigned scale increases by at most L. The curvature radius increases by at most L^2 and the full-current radius by at most L^3. Hence its new radius is at most $\alpha_0 = L^3 B\varepsilon'$. The inner action has not changed. A nonfull inner bond has a missing incident plaquette. Its endpoints and every bond in its remaining inner plaquettes are within 3η of Ω_0^c. The separation $\mathrm{dist}_\infty(\Omega_0^c, \Omega_1) > R_1\mathfrak{M}\eta > L^2\eta$ puts this entire stencil outside Ω_1. Every local scale on it is therefore η, and all its inner edges are bare determining rows. The exact current is a sum of at most six traceless plaquette terms with prefactor η^{-3}, so

$$\|J_{\Omega_0}(U^{(n-1)})\|_{3,\infty} \le 6\alpha_0. \tag{3.mxc}$$

This checks the actual force required by RTC, including the action boundary. A bound only on full bonds would not suffice for that application.

On a double block of a new row, every level-$(n-1)$ bond read by its one-step average lies in the last determining member of the old truncation. Exact local recursion gives

$$\mathcal{M}^n U^{(n-1)}(c) = \mathcal{M}(V_{n-1}^{(n-1)})(c).$$

All older row values remain the target values. Define the literal per-length displacement by $\xi_c B_c = i^{-1}\log(V_n^{(n)}(c)(\mathcal{M}^n U^{(n-1)}(c))^{-1})$ on new rows and by zero on older rows. Then $\|B\|_{1,\infty} \le (\pi/2)\varepsilon'$. At a new outward endpoint the reduced datum factor reads only older interior displacements, which are zero. Thus eq. 3.mxxxiii gives the same raw new displacement, with the exact orientation order.

Apply Prop. 3.120 with this actual real trial and independent row multiplier zero. Its defect is the force in eq. 3.mxc; the forced equation retains this entire defect. The physical continuation bounds give a real critical field at the target literal datum, of provisional regularity at most $C_A\alpha_0$. The literal node restoration is unitary, so it preserves curvature and the full current; the outward factors cancel exactly. The inequalities eq. 3.mlxxxix put this output inside a_c. Apply Lem. 3.123 at its new datum to return to $\mathcal{R}_{\mathrm{mat}} \le B\varepsilon'$. This completes the induction with one radius for all n. At the final level apply the reset once more using the original datum radius ε, obtaining eq. 3.mlxxxviii with a fixed C_B.

For uniqueness take any two real regular critical fields of the same datum. The action boundary obeys the same bound as eq. 3.mxc, so either field is an admitted physical critical reference after choosing a_0 sufficiently small. Apply Lem. 3.122 to the pair with $\delta = 0$. It gives a strong mean-slice point of size at most Ca_0 and zero reduced datum. This is a coordinate on the actual constraint fibre by the gauge parametrization in eq. 3.mlviii; constrained criticality therefore gives its full multiplier equation. The competing-multiplier readout eq. 3.mlxi bounds its multiplier displacement by Ca_0. Choose a_0 so both components lie in the fixed PNC contraction ball. They and the zero solution have the same datum; uniqueness gives $A = 0$. The last assertion of Lem. 3.122 makes the gauge residual. Covariance follows by transforming the entire actual equation and then applying this uniqueness. □

Corollary 3.125 (A fixed physical continuation ball)**.** For every orbit in Prop. 3.124, choose a real representative U_0. There is $r_G > 0$, uniform in the datum, depth and admitted geometry, such

that for $V_B(c) = e^{i\xi_c B_c} V(c)$ and $\|B\|_{1,\infty} < r_G$ there is a holomorphic literal critical representative $\widehat{U}(B)$ with $M_{\mathbf{B}}\widehat{U}(B) = V_B$. In its prescribed Landau coordinate,

$$\|A(B)\|_{\mathrm{str},\alpha} + \|\omega(B) - \omega(0)\|_{3,\infty} \le C\|B\|_{1,\infty}, \qquad \|F(\widehat{U}(B))\|_{2,\infty} + \|J_{\Omega_0}(\widehat{U}(B))\|_{3,\infty} \le C(\varepsilon + \|B\|_{1,\infty}). \tag{3.mxci}$$

The physical bounds hold on complete fine plaquette and current stencils. Their fixed-order localized responses are those of eq. 3.mlxviii. The construction requires no hereditary complex-domain hypothesis. ■□

Proof. The regular real reference just constructed has bounded inner current by the bare-boundary calculation above and bounded induced multiplier by the actual row lift. All hypotheses of Prop. 3.117 therefore hold, uniformly. Its literal node extension and eq. 3.mlxiv prove the assertion after one fixed decrease of its datum radius. If desired choose r_G so the right side of eq. 3.mxci is below any fixed small regularity ceiling on the full complex ball. This decreases a fixed radius and introduces no dependence on depth or ε.

For real B in this ball the output is a real regular critical orbit. Wherever it also lies in the real datum domain of Prop. 3.124, that theorem's uniqueness identifies it with the real background there. Thus these are continuations of the canonical real orbit. Changing a real representative conjugates the equations isometrically. On a common real datum neighborhood the physical orbits agree by uniqueness; their holomorphic germs, and their actions, agree by the identity theorem. No choice of a distant complex gauge representative is needed for these local continuations. □

3.35. **Uniform continuation on every tower.** The tower tests require a gain that is quadratic in the complex potential radius. A first derivative estimate alone gives a linear loss. The argument below obtains the quadratic gain from three real configurations, after constructing a common complex neighborhood on which the normalized curvature and current are bounded. The real background theorem supplies the three real estimates; the constrained Euler equation supplies the crude complex current estimate. The independent current of a regular pair does not enter this construction. This is the tower argument underlying Bałaban's construction [B87], with the determining rows, complex gauges and parameter neighborhoods specified here.

3.35.1. *The two erosion rules.* We use the two cube rules fixed in the localization geometry. Write $M_1 = M_1'$ in this subsection. All continuation and restriction statements below use determining sets with all inward rows removed. The box-datum lemma also states the variant in which those rows are retained. All lengths in the next lemma are in bare units. Thus level q has spacing L^q. The constants satisfy

$$L \ge 13, \qquad R_1 \ge L, \qquad M_1 \ge R_1\mathfrak{M}, \qquad M \ge L^4 M_1.$$

The background floors are imposed before these inequalities. The localization size M is then increased, if necessary; no constant chosen below depends on the cutoff, the tower depth or a coupling.

Lemma 3.126 (Protected stencils)**.** Let Z be a union of level-$(p-1)$ localization cubes, and construct either of its two towers $Z = \Omega_0 \supset \cdots \supset \Omega_p$. For every $n \le p$, the truncation at n is an admitted separated sequence with no inward determining rows. Its two-layer inset $Z^{\ominus 2}$ has distance at least ML^{p-1} from Ω_n^c. Every bare site within distance $4L^n$ of that inset lies in a free level-n pillar. In particular every plaquette and every full six-plaquette current stencil based in the inset lies in these pillars. ■

Proof. Let w_q be an upper bound on the distance from a discarded point to Z^c. The discarded cube at the next step either meets the old complement or fails its distance test. Its side plus the

rejected gap therefore gives

$$w_0 = 0, \qquad w_{q+1} - w_q = \begin{cases} R_1 \mathfrak{M} L^q + \mathfrak{M} L^{q+1}, & \text{first rule,} \\ 2M_1 L^{q+1}, & \text{second rule.} \end{cases}$$

Nested grids ensure that this argument also applies to a cube not contained in the previous member. Summing the geometric series gives $w_q \le ML^{p-3}$ for either rule and all $q \le p$. Consequently

$$\mathrm{dist}_\infty(x, \Omega_q^c) \ge \mathrm{dist}_\infty(x, Z^c) - ML^{p-3}. \tag{3.mxcii}$$

The two-layer inset has distance at least $2ML^{p-1}$ from Z^c, which proves the asserted distance bound.

Both rules give unions of the required aligned cubes and gaps greater than $R_1 \mathfrak{M} L^{q-1}$ at level q. At a nonempty localization the level-n torus side has at least $M/L \ge L^3 R_1 \mathfrak{M}$ sites, so the torus floor for the background theorem is satisfied. The statements remain valid when a member is empty; for the full torus every member is the full torus.

If x is within $4L^n$ of the inset, its level-n block has distance at least $(M/L - 5)L^n > L^{n-1}$ from Ω_n^c. Its only determining node is its top: a lower row has an endpoint outside Ω_n and length at most L^{n-1}, whereas a level-n node is a block center. This proves that the pillar is free. A current stencil reaches at most two bare steps, which are contained in the stated halo. □

For a separated sequence of depth n, put $\eta = L^{-n}$. Let $\ell_-(b)$ be the smaller assigned physical pillar length at the endpoints of b, and take the minimum over the corners for a plaquette. A level-q determining row and its full double-block read have $\ell_- \ge L^{q-1-n}$. This follows because the read has radius at most $3L^{q-n}/2$, while the preceding layer gap is greater than $R_1 \mathfrak{M} L^{q-1-n}$. Define

$$\|w\|_- := \sup_b \ell_-(b)|w(b)|. \tag{3.mxciii}$$

The real and imaginary Hermitian parts of w have no larger norm.

3.35.2. *A common complex ball.* The next lemma uses only the real mixed background and its physical datum chart, proved in Prop. 3.124 and Cor. 3.125. In particular, neither result is applied with a hereditary core as an input. We write b for a raw row logarithm: the datum is $e^{ib}V_0$. Thus $\|b\|_\infty$ is the norm called $\|B\|_{1,\infty}$ when $b(c) = \xi_c B(c)$.

Throughout this subsection, $\mathcal{H}_{\mathbf{B},V_0}(b)$ means the Landau component of the physical chart with *literal* datum $e^{ib}V_0$. Its evaluation includes the triangular conversion eq. 3.mxxxiii: if $\mathfrak{n}_{\mathbf{B},V_0}(b)$ denotes that converted per-length datum, the solved equation is $q(H) = \mathfrak{n}_{\mathbf{B},V_0}(b)$. The completed representative is

$$\Phi_{\mathbf{B},V_0}(b) = (e^{i\eta H}U_0)^{u_b}, \qquad M_{\mathbf{B}}(e^{i\eta H}U_0) = (e^{ib}V_0)^{u_b^{-1}|_{\mathrm{nodes}}}.$$

Thus H alone need not have the literal datum. Every estimate in the literal variable b includes this conversion and the node factors. Their finite carriers and uniform derivatives are part of eq. 3.mxxxiv and eq. 3.mlxvii.

Lemma 3.127 (Uniform analytic readouts)**.** There are $a_*, r_*, C_* > 0$, depending only on the fixed background geometry, with the following property. Let U be real and satisfy

$$|\partial U(P) - 1| < e_0 \eta^2 \ell_-(P)^{-2}, \qquad e_0 < a_*. \tag{3.mxciv}$$

For every $\|w\|_- < 2r_*$ the averaged datum $M_{\mathbf{B}}(e^{i\eta w}U)$ and its continued critical configuration $\Phi_U(w)$ exist and depend holomorphically on w. For a plaquette or full current stencil contained in free pillars of one level q, put $\xi = L^{q-n}$. Then

$$\left|\eta^{-2}\xi^2(\partial\Phi_U(w) - 1)\right| \le 1, \qquad |\xi^3 J_\eta(\Phi_U(w))| \le 1. \tag{3.mxcv}$$

The same r_* works for every depth, compatible volume, real base U satisfying eq. 3.mxciv, and either tower rule. ■

Proof. The actual averaging estimates give constants $a_{\mathrm{av}} > 0$ and $C_Q \geq 1$, independent of the number of iterations, such that

$$b(w) = \frac{1}{i}\log\{M_{\mathbf{B}}(e^{i\eta w}U)M_{\mathbf{B}}(U)^{-1}\}, \qquad \|b(w)\|_\infty \leq C_Q L\|w\|_- \tag{3.mxcvi}$$

is holomorphic when $L^2 e_0 < a_{\mathrm{av}}$ and $L\|w\|_- < a_{\mathrm{av}}$. Indeed the order-$q$ read has $L^q\eta|w| \leq L\|w\|_-$ and curvature at most $L^2 e_0 L^{-2q}$ on its whole double block. Apply Prop. 3.6 on that block. At level zero the formula is $b(w) = \eta w$. The same averaging estimate shows that the real datum $V_0 = M_{\mathbf{B}}(U)$ has every ordinary and mixed test bounded by $C_V e_0$. On a mixed plaquette the coarse bond reconstructed from the lower rows equals the actual next average, by the semigroup identity. No new seam defect occurs.

Choose a_* so $C_V a_*$ lies below the real datum ceiling. The real background theorem supplies its normalized critical representative U_0 and the bound

$$\eta^{-2}\ell(P)^2|\partial U_0(P) - 1| + \ell(b)^3|J_\eta(U_0)(b)| \leq C_B e_0. \tag{3.mxcvii}$$

Here and below a current is evaluated only at a full stencil; boundary forcing is retained in the regional equations used to construct U_0. The physical chart gives $W(b) = e^{i\eta H(b)}U_0$, holomorphic for $\|b\|_\infty < r_B$, with weighted potential and first differences at most $C_H\|b\|_\infty$. It also bounds the change of multiplier by $C_H\|b\|_\infty$.

We record why its current is bounded on that entire ball. If T_H is the differential of the exponential coordinate in left-trivialized physical variables, the equation after removal of the gauge penalty is

$$T_H^\sharp J_\eta(W(b)) = q'(H)^\sharp(\omega_0 + \nu(b)). \tag{3.mxcviii}$$

The adjoints in this identity include the physical row and bond densities. The actual row-transpose estimate and the multiplier readout of eq. 3.mlxvi give

$$\|\ell^3 J_\eta(W(b))\|_\infty \leq C(e_0 + \|b\|_\infty).$$

At $H = 0$, the multiplier also has a uniform bound. If R_0 is the actual right inverse of the retained row map, apply $R_0^\sharp$ to $J_\eta(U_0) = q'(0)^\sharp\omega_0$. The weighted row-lift estimate in eq. 3.mlx gives $\|\omega_0\|_{3,\infty} \leq C\|\ell^3 J_\eta(U_0)\|_\infty \leq CC_B e_0$. Thus both the base multiplier and its increment have the claimed bounds. The bondwise factors $T_H^{\pm 1}$ are bounded because $\eta|H| \leq \ell|H|$ is small. Thus this estimate comes from the Euler equation and its multiplier bound. It does not require a second-difference estimate inferred from the first-difference norm. The plaquette product similarly gives $\eta^{-2}\ell^2|\partial W(b) - 1| \leq C(e_0 + \|b\|_\infty)$.

We must also control the prescribed normalized representative. Fix a free pillar with top y at level t and let $h_y = \sup_{B^t(y)} \xi_y|H|$. For a level-$q$ segment of the ascent to y, retain its real reference contour as a unitary factor. Its condition number is 1; it need not be the identity. The actual relative contour logarithm is bounded by $C_\Gamma C_Q L^{q-t+1}h_y$. To see the scale, apply the actual mean estimate to the order-q double block: its relative row logarithm is at most $C_Q L^{q-t}h_y$. A one-step comb has at most $4(L-1)$ bonds; the finite product logarithm on the fixed chart costs at most twice their sum. Indeed, factoring each bond into its relative factor and its real reference factor rewrites the relative contour as a product of unitary conjugates of the relative factors. The same bound for the mean contour follows from the contraction defining that contour. The node factor at the top is a product of exponentials with logarithms bounded by $C_N L^{q-t}h_y$. These are the one-step site corrections in Prop. 3.6: at level q, their normalized input is $L^{q-t}h_y$, and the inverse derivative of the site mean is bounded on its fixed chart. The constant C_N depends on L, including the finite comb length, and not on q or t. The real reference node factors are unitary as well. Telescope these factors through the ascent; their conjugations preserve all norms in the estimates just obtained. Each ascent meets a level at most once. Therefore the gauge u_b which takes $W(b)$ to the prescribed

representative satisfies

$$|u_b(x)^{\pm1}| \le \exp\left(C_G h_y \sum_{q=0}^{t-1} L^{q-t}\right) \le e^{C_G h_y/(L-1)}, \qquad \Phi_U(w) = W(b(w))^{u_{b(w)}}. \tag{3.mxcix}$$

The identity holds at real data by the defining axial normalization and at complex data by holomorphy. All factors use the actual averaging contours and node correction. Choose $r_G \le r_B/2$ so $\|b\|_\infty < r_G$ makes the condition number of this gauge at most 2. Conjugation then costs at most 2 in both readouts. The entire stencil lies in the indicated free pillars, so it has no additional boundary term.

Finally choose, in this order,

$$r_* \le \frac{1}{4L}\min\{a_{\mathrm{av}}, r_G/C_Q\}, \qquad C(a_* + 2C_Q L r_*) \le \frac{1}{2}. \tag{3.mc}$$

The constants in the second inequality are the fixed plaquette, Euler-current and gauge constants just obtained. Decrease a_* and r_* once to satisfy it. Equations eq. 3.mxcvi, eq. 3.mxcviii and eq. 3.mxcix now prove eq. 3.mxcv throughout the ball $\|w\|_- < 2r_*$. □

3.35.3. *The quadratic gain.*

Lemma 3.128 (Three real configurations)**.** Let F be a matrix-valued holomorphic map on $\|w\| < 2r$, bounded there by 1, and write $w = w_{\mathrm{r}} + iw_{\mathrm{i}}$ with $\|w_{\mathrm{r}}\|, \|w_{\mathrm{i}}\| \le \|w\| \le r$. Then

$$|F(w)| \le |F(w_{\mathrm{r}})| + |F(w_{\mathrm{i}})| + |F(0)| + \frac{3\|w\|^2}{2r^2}. \tag{3.mci}$$

■

Proof. For each of the three vectors, the one-variable Taylor series on $|z| < 2r/\|w\|$ bounds the remainder after degree one by $\|w\|^2/(2r^2)$. Apply the triangle inequality to $F(w) - F(w_{\mathrm{r}}) - iF(w_{\mathrm{i}}) + iF(0)$. Its constant and linear terms vanish. The estimate follows by duality with norm-one linear functionals on the matrix space. □

Proposition 3.129 (Uniform tower bound)**.** There are $a_T > 0$ and $K_T \ge 1$ such that, on every separated sequence of depth n, the assumptions

$$\begin{gathered} |\partial U(P) - 1|, \quad |\partial(e^{i\eta f}U)(P) - 1| < e_0\eta^2\ell_-(P)^{-2}, \\ \|f\|_- < e_1, \qquad e_0 + e_1 < a_T, \end{gathered} \tag{3.mcii}$$

with U real imply, at every free level-q full read,

$$|\partial\Phi_U(f) - 1| < K_T(e_0 + e_1^2)\eta^2\xi^{-2}, \qquad |J_\eta(\Phi_U(f))| < K_T(e_0 + e_1^2)\xi^{-3}, \quad \xi = L^{q-n}. \tag{3.mciii}$$

The bounds also hold for $f = 0$. The configuration is holomorphic in f on the fixed ball of Lem. 3.127, regardless of whether the second curvature bound in eq. 3.mcii holds at other points of that ball. ■

Proof. Transport the four increments of a plaquette to its initial corner using U. Their sizes are at most $\eta e_1/\ell_-(P)$. The remainder after the linear term in their exponential product is bounded by $C\eta^2 e_1^2\ell_-(P)^{-2}$. The closing transport changes the linear curl by at most $C\eta|f|\,|\partial U - 1|$. Hence the two curvature bounds imply

$$\ell_-(P)^2|\mathrm{d}_U f(P)| \le C(e_0 + e_1^2).$$

Real unitary transports commute with taking Hermitian real and imaginary parts. Applying the same product estimate to f_{r} and f_{i} proves

$$|\partial(e^{i\eta f_\sigma}U) - 1| \le C_P(e_0 + e_1^2)\eta^2\ell_-^{-2}, \qquad \sigma \in \{\mathrm{r}, \mathrm{i}\}.$$

Choose $a_T \le \min\{r_*, a_*/(2C_P), 1/2\}$ and reduce it, if necessary, to meet the real datum ceiling for these two configurations. Their averaged data are regular, and the continued real fields are

their unique regular critical orbits by Prop. 3.124 and Cor. 3.125. Thus each normalized readout at $0, f_{\mathrm{r}}, f_{\mathrm{i}}$ is bounded by $C_R C_P(e_0 + e_1^2)$. Here the prescribed gauges at real points are unitary, so they preserve the norm.

Apply Lem. 3.128 to the two readouts in eq. 3.mxcv. It gives eq. 3.mciii with, for example, $K_T = 1 + 4C_R C_P + 2r_*^{-2}$. This number has no depth or real-base dependence. In the notation of eq. 3.ccclxxxvii, one may take $M = 1$ and $r_0 = r_*$ after the harmless conversion from a ball to the displayed two-variable polydisc. In particular $M/r_0^2 = r_*^{-2}$ is a fixed constant. No running normalization is part of this radius. □

Write $\mathcal{Q}_j(X; e_0, e_1, c_0)$ for the simple triple class of eq. 3.ccviii, with independent current bound c_0. We display this third radius in $\mathcal{C}_j$ when it differs from the curvature radius.

Corollary 3.130 (Both towers at every truncation)**.** Let $(U, A', \mathbf{J})$ satisfy the simple conditions on a level-j domain X, with radii (e_0, e_1, c_0) and $e_0 + e_1 < a_T$. For every tower address (p, Z), both erosion rules and $1 \le n \le p$, the tower at the actual averaged datum exists. In level-n units, on $Z^{\ominus 2}$ it satisfies

$$\begin{aligned} |\partial U_n(Z; \mathbf{U}) - 1| &< K_T(e_0 + e_1^2) L^{-(j-p)} L^{-2p}, \\ |J^{(n)}(U_n(Z; \mathbf{U}))| &< K_T(e_0 + e_1^2) L^{-(j-p)} L^{2(n-p)}. \end{aligned} \tag{3.mciv}$$

The same is true for the real factor. Consequently

$$e_0 \le b_0, \quad e_1 \le b_1, \quad K_T(e_0 + e_1^2) \le b_0 \quad \Longrightarrow \quad \mathcal{Q}_j(X; e_0, e_1, c_0) \subset \mathcal{C}_j(X; b_0, b_1, c_0). \tag{3.mcv}$$

The third radius denotes the independent current bound; when omitted it equals the first radius. ■□

Proof. First take $j = p$ and put $\lambda = L^{n-p}$. On changing from level p to level n, the potential becomes $\lambda A'$, the physical curvature gains λ^2, and the real gauge clauses remain valid. The simple input therefore satisfies eq. 3.mcii with $\epsilon_0 = \lambda^2 e_0$ and $\epsilon_1 = \lambda e_1$. The minimum length is at most 1, so these uniform bounds imply its weighted bounds. By Lem. 3.126, every required output stencil lies in free top pillars, where $\xi = 1$. The output curvature has factor $\lambda^2 L^{-2n} = L^{-2p}$, and its level-$n$ current has factor $\lambda^2 = L^{2(n-p)}$.

For a general address of a level-j domain, restriction to level p has simple radii $\mu e_0, \mu e_1$ with $\mu = L^{-(j-p)}$. The curvature and first difference actually gain μ^2, which is stronger than the radius asserted here. Apply the preceding result and use $\mu e_0 + \mu^2 e_1^2 \le \mu(e_0 + e_1^2)$. The independent current restricts by μ^3 and has no role in a tower's physical current. This proves every clause, without summing over addresses. For a bounded localization domain in $\mathbb{R}^4$, choose a torus large enough to contain it and every determining read injectively. The regional action, row constraints and normalization paths are then exactly the same finite objects. Uniqueness makes the tower independent of the chosen torus, so all assertions hold on that domain with the same constants. □

3.35.4. *Membership throughout parameter discs.*

Corollary 3.131 (Uniform parameter domains)**.** Let U be fixed and let $A'(z), \mathbf{J}(z)$ be holomorphic on an open polydisc D, with their values satisfying the simple bounds at fixed radii (e_0, e_1, c_0) throughout D. If the inequalities of eq. 3.mcv hold, the whole image of D lies in the target core. All tower maps are holomorphic on D, with the common bounds displayed in eq. 3.mciv. A strict inequality $K_T(e_0 + e_1^2) < b_0$ supplies a common strict margin, also on every smaller closed polydisc. No radius loss depending on the number of tower addresses is required. ■□

Proof. The mean and background compositions in Lem. 3.127 are defined before testing core membership. The simple bounds put each composition in its fixed ball at every point of D. Cor. 3.130 therefore applies pointwise with the same constants, and composition proves holomorphy. Taking

the supremum over all addresses preserves this common bound; no uniform strict margin is inferred merely from individual strict inequalities. The real factor and its real gauges are fixed; the independent current uses its own assumed simple inequality. □

Proposition 3.132 (Outer substitutions)**.** Fix $c_\natural \geq 1$. There is $c_O > 0$, depending only on the fixed geometry and $c_\natural$, such that the following holds. Suppose $0 < b_0 \leq b_1$, $b_1^2 \leq c_\natural b_0$, and $b_0 + b_1$ is below a fixed ceiling. Start from a triple in the core with radii $(b_0/2, b_1/2, c_0)$ and put $\mathbf{U} = e^{i\eta_j A'} U$. If

$$\max\{|A|, |\nabla_{\mathbf{U}} A|\} < c_O b_0, \qquad |\mathbf{J}'| < b_0,$$

then $(e^{i\eta_j z A}\mathbf{U}, \mathbf{J}')$ belongs to the core at radii (b_0, b_1, b_0) for every $|z| \leq 1$, with the real factor U and the principal Baker–Campbell–Hausdorff potential. The same conclusion is uniform in any other parameters for which the stated bounds hold on their whole polydisc. ■

Proof. Choose $R = 64K_T(1 + c_\natural)$, and then choose $c_O \leq (16C_O R)^{-1}$, where C_O bounds the first-order product and plaquette estimates on the fixed logarithm chart. For $|z| < R$,

$$A'_z = (i\eta_j)^{-1} \log(e^{i\eta_j z A} e^{i\eta_j A'})$$

is holomorphic. The product rule, including the change from $\nabla_{\mathbf{U}}$ to ∇_U, gives $|A'_z|, |\nabla_U A'_z| < b_1$ and $|\partial(e^{i\eta_j A'_z} U) - 1| < b_0 \eta_j^2$. Indeed the additional terms are bounded by $C_O(Rc_O b_0 + (Rc_O b_0)^2 + Rc_O b_0 b_1)$; reducing the fixed ceiling makes their sum smaller than $b_0/4$. The real gauges remain those of U. Thus the simple conditions hold on this entire auxiliary disc, although membership in the target hereditary core has not yet been used there.

At an address (p, Z) divide the two tower readouts by the factors $L^{-(j-p)}L^{-2p}$ and $L^{-(j-p)}L^{2(n-p)}$, respectively. By Cor. 3.130 each resulting holomorphic function $F(z)$ is bounded on $|z| < R$ by $K_T(1 + c_\natural)b_0$. Schwarz applied to $F(z) - F(0)$ gives

$$|F(z) - F(0)| \leq \frac{2K_T(1 + c_\natural)b_0}{R} \leq b_0/32 \qquad (|z| \leq 1).$$

The old core gives $|F(0)| < b_0/2$. This proves every complex tower test with room. Its real counterpart is unchanged. The simple conditions and $|\mathbf{J}'| < b_0$ complete the inclusion, uniformly over the auxiliary parameters. □

3.35.5. *Inner window substitutions.* We now prove the core inclusion for the actual inner window map. This application uses the upper triple's tower clause. That clause has either been retained from the upper core or has been established by Cor. 3.131; it is not an assumption used to prove Prop. 3.129.

Let $\square$ be the union of the 2^4 level-k localization cubes meeting a vertex, and let $\square_0 = \square^{\sim 5}$ be a box with an injective torus lift. Put $N = k + 1$, $\nu = N - j \geq 1$ and $\sigma = L^{-\nu}$. Let $\mathbf{B}^+$ be the first-rule tower of $\square_0$ to depth N, and $\mathbf{B}_j$ its truncation at j. For a dimensionless fine logarithm a write $\mathsf{q}_q(a) = i^{-1} \log \mathcal{M}^q(e^{ia})$ on the fixed near-trivial chart. This is the plain nonlinear mean logarithm; it is not the linear row map or the straightened constraint. The upper triple has radii $\alpha_\iota = (1 + 2\beta_s)A_\iota$, $0 < \beta_s \leq 1/4$. We require its simple conditions on $\square_0$ and its tower clause at $(N, \square_0)$. Thus its upper tower has curvature less than $\alpha_0 L^{-2N}$ and level-N current less than α_0 on $\square^{\sim 3}$.

Lemma 3.133 (The box datum)**.** For $M(\alpha_0 + \alpha_1)$ below a fixed ceiling, there are holomorphic maps at fixed real U, denoted V and v, with

$$M_{\mathbf{B}^+}(\mathbf{U}) = V^v, \qquad \|\log V\|_\infty \leq C_\square M\alpha_0, \qquad |v(x)|\,|v(x)^{-1}| \leq e^{C_\square M\alpha_1}. \tag{3.mcvi}$$

They use only the bare field on $\square_0$, agree under restriction from a larger domain, and are real at real potentials. The node gauge is normalized at the prescribed base point. The construction and these bounds also hold when all inward rows of the determining set are retained. At fixed real base

and fixed radii, the domain defined by the simple conditions and the upper tower clauses is open. Both maps commute with translations of the aligned box and with its injective lifts between tori. Under a real gauge h, their values become $h(y_0)Vh(y_0)^{-1}$ and $h(x)v(x)h(y_0)^{-1}$, respectively. ■

Proof. Choose the prescribed comb from the base point to each site and write $g(x)$ for its transport in $\mathbf{U}$. The comb has length at most CM in level-N units. Along it the real transports have norm one, while a complex increment has norm and inverse norm at most $e^{L^{-N}|A'|}$. Consequently $\mathrm{cn}(g) \le e^{CM\alpha_1}$. In the comb gauge, a bare link is the product of the plaquettes in the strip between the two combs. There are at most CML^N such plaquettes. Their transports have the preceding condition-number bound, and their defects are at most $\alpha_0 L^{-2N}$. The product bound therefore gives $|\mathbf{U}^g(b) - 1| \le CM\alpha_0 L^{-N}$, after reducing the fixed ceiling. This argument uses the complex curvature clause as well as the real base and the potential bound.

Write $\mathbf{U}^g = e^{ia}$ on this chart. Every order-q determining read has $L^q \sup|a| \le CM\alpha_0$, since $q \le N$. The actual near-trivial averaging estimate therefore gives $V = M_{\mathbf{B}^+}(\mathbf{U}^g)$ and $\|\log V\|_\infty \le C_\square M\alpha_0$. Complex covariance of each mean is valid on this same chart: the condition number above times its input diameter remains below the mean radius. Set $v = g^{-1}$ at the nodes. This proves the exact identity in eq. 3.mcvi. Paths, products, logarithms and means are holomorphic on the stated domain. Their definitions prove the restriction, reality and base-point assertions. Every determining row, including an inward row if it is retained, has its full read inside the box. The same construction therefore proves the asserted variant. All simple inequalities are strict inequalities between continuous functions on finitely many cells. For a fixed real base, each admitted tower clause uses a fixed holomorphic datum chart; its defining raw-log inequalities and its curvature and current inequalities are also strict and continuous. There are only finitely many truncations and cells on this box. Their intersection is open, including when it is empty because a real-base clause fails. This is a finite-dimensional openness argument; it makes no assertion of a uniform margin over increasing cutoffs. The common bounds used for parameter discs were established separately in Cor. 3.131. Finally the prescribed comb transports obey $g^h(x) = h(y_0)g(x)h(x)^{-1}$. This gives the displayed gauge laws; their paths and all averaging reads are preserved by the indicated translations and injective lifts, which proves the remaining claims. □

Lemma 3.134 (Restriction of a constrained critical field)**.** Let $\mathbf{B}$ be the determining set of a separated sequence $\Omega_0 \supset \cdots \supset \Omega_m$, with all inward rows removed. Let $Z \subset \Omega_m$, and let $\mathbf{B}'$ be the determining set of a separated sequence $Z = \Omega'_0 \supset \cdots \supset \Omega'_{m'}$, $m' \le m$, also with all inward rows removed. A critical field for $\mathbf{B}$ restricts to a critical field for $\mathbf{B}'$ at its induced datum. The same assertion holds when $\mathbf{B}'$ is the determining set of a truncation of the original sequence. ■

Proof. Write W for the old field and extend a competing field W' on Z by W outside the inner bare bonds of Z, obtaining $\widetilde{W}'$. Suppose $M_{\mathbf{B}'}W' = M_{\mathbf{B}'}W$. We first show $M_{\mathbf{B}}\widetilde{W}' = M_{\mathbf{B}}W$. Put $\Omega'_{m'+1} = \varnothing$. For $0 \le q \le m'$, let $P(q)$ assert equality of the two order-q averages at every row whose full blocks lie in Ω_0 and whose endpoints both avoid Ω'_{q+1}.

For $q = 0$, a changed bare bond satisfying this condition belongs to the frozen set Γ'_0, so $P(0)$ holds. Suppose $P(q-1)$ holds. If a row in $P(q)$ has an endpoint in Ω'_q, this endpoint is in $\Omega'_q \setminus \Omega'_{q+1}$. Separation puts its other endpoint and its complete read inside Ω'_{q-1}, and the row belongs to Γ'_q, including when it points outward. Its average is thus fixed by the induced datum. Otherwise both endpoints are outside Ω'_q. Since Ω'_q is a union of aligned level-q cubes, both predecessor blocks avoid Ω'_q. The one-step mean reads only these blocks, where $P(q-1)$ applies. This proves $P(q)$. At $q = m'$ all order-m' averages agree. The averaging semigroup therefore gives equality at every higher-order old row as well. An old row of order $q < m'$ has an endpoint outside $\Omega_{q+1} \supset Z$. Its length is L^q, whereas Ω'_{q+1} has distance greater than L^q from Z^c. Neither endpoint can lie in Ω'_{q+1}, so $P(q)$ also covers every such old row.

Every bare plaquette meeting Z^c has each corner within one bare step of Z^c. Separation therefore puts any of its inner bonds in Γ_0', where $W' = W$. All action terms crossing the boundary are constant along a constrained variation of W'. Extending that variation consequently preserves the old datum, and its old action derivative equals its regional action derivative. The first is zero by criticality, which proves the assertion.

For a truncation at depth t, the retained rows below t are unchanged. Its top rows include every order-t row needed in the read of an old row of order greater than t; this follows from the separation collar and the nested block identity. Equality of the top averages and the semigroup preserve these old rows. The underlying region is unchanged, so there is no crossing action term. The same variation argument applies. □

Proposition 3.135 (Inner substitution on its full domain)**.** There are fixed $M_0, C_I, c_2, c_3 > 0$ and, for $M \geq M_0$, a ceiling $a_I(M) > 0$ with the following property. Let $A_0, A_1, a_0, a_1 > 0$, and suppose

$$A_0 + A_1 < a_I(M), \quad a_0 + a_1 < a_I(M), \quad C_I M a_0 < a_1, \quad A_0 \leq 2\sqrt{1+\nu}\, a_0. \tag{3.mcvii}$$

Put $C_V = 2C_\square$. First allow an independent complex raw datum b_V on $\mathbf{B}^+$ in the ball

$$\|b_V\|_\infty < C_V M A_0, \tag{3.mcviii}$$

and set $H_+ = \mathcal{H}_{\mathbf{B}^+,1}(b_V)$. For the physical application, take the upper input and the data of Lem. 3.133 and put $b_V = i^{-1}\log V$. Equation eq. 3.mcvi places this datum in eq. 3.mcviii, since $\alpha_0 \leq 3A_0/2$. For a complex field A in level-k units and a raw row correction b' on $\mathbf{B}_j$, suppose $\|A\|_\infty < c_2 A_0$ and $\|b'\|_\infty < c_3 a_0$. Define the actual raw fine potential and row datum by

$$a = (i)^{-1}\log(e^{iL^{-k}A}e^{iL^{-N}H_+}), \qquad b(c) = \mathsf{q}_q(a)(c) \quad (c \in \Gamma_q), \qquad H = \mathcal{H}_{\mathbf{B}_j,1}(\tau b + b'), \quad 0 \leq \tau \leq 1. \tag{3.mcix}$$

The application of $\mathcal{H}$ in this formula includes $\mathfrak{n}_{\mathbf{B}_j,1}(\tau b + b')$ before the reduced saddle equations are solved. All maps are defined and holomorphic on the full indicated open balls in (b_V, A, b'). Whenever b_V comes from an upper input as above, their output on $\square^{\sim 3}$ satisfies

$$|H|, |\nabla H| < a_1/2, \qquad |\partial e^{iL^{-j}H} - 1| < a_0 L^{-2j}/8, \qquad |J^{(j)}(e^{iL^{-j}H})| < a_0/16. \tag{3.mcx}$$

For every $X \subset \square^{\sim 2}$ of level $j-1$, the triple $(1, H, J^{(j)}(e^{iL^{-j}H}))|_X$ lies in the core with radii $(a_0/4, a_1/2, a_0/4)$. These estimates and inclusions hold throughout every auxiliary parameter polydisc on which the upper input and the two increment bounds hold. The same inclusion holds on a larger level-$(j-1)$ domain whose complete simple-condition stencils lie in $\square^{\sim 3}$ and which is contained in the top member of the truncated window tower. ■

Proof. We give the choices and the estimates in their dependency order. The physical datum chart and eq. 3.mlxvii give constants $C, r, s > 0$ for the flat reference on every tower. Its potential, first difference, normalized curvature and physical current have Lipschitz constant C on $\|d\|_\infty < r$. If a change of d is supported at distance at least D from an output cell, its contribution is at most Ce^{-sD} times its supremum. For the current this follows by differentiating eq. 3.mxcviii; the weighted row-transpose density sums the differentiated multiplier columns. It therefore retains the same exponential bound. The output cell and the full current stencil are included in the distance convention.

We first admit the full product of input balls, without requiring b_V to arise from an upper field. On a complete order-q read of the truncated tower, the upper assigned length is at least L^{q-1-N}, by the geometry used in eq. 3.mxcvi. The weighted flat response therefore gives

$$L^q \sup_{\mathsf{P}^q(c)} |L^{-N}H_+| \leq C\|b_V\|_\infty, \qquad c \in \Gamma_q, \quad q \leq j.$$

The product logarithm and the actual mean on this read consequently give, uniformly for $0 \leq \tau \leq 1$,

$$\|\tau b + b'\|_\infty \leq C(\|b_V\|_\infty + L^{j-k}\|A\|_\infty) + \|b'\|_\infty.$$

Since $j \le k$, decreasing $a_I(M)$ makes the right side smaller than $r/2$ on the entire product of the three stated balls, and keeps all preceding raw logarithms and means in their fixed charts. This proves the full-domain holomorphy. Only the sharper readout estimates below use the actual upper input.

For any one of these readouts $R_x(d)$, with $R_x(0) = 0$, Cauchy on the fixed datum ball gives

$$|R_x(d) - DR_x(0)d| \le C\|d\|_\infty^2/r. \tag{3.mcxi}$$

For a split $d = d_{\mathrm{in}} + d_{\mathrm{out}}$, where the outer part is at distance D, apply this bound only to d_{in} and use the preceding Lipschitz estimate for d_{out}. This gives

$$|R_x(\tau d) - \tau R_x(d)| \le C(e^{-sD}\|d_{\mathrm{out}}\|_\infty + \|d_{\mathrm{in}}\|_\infty^2/r), \qquad 0 \le \tau \le 1. \tag{3.mcxii}$$

These are estimates of the actual current map, so no bound on second differences of an arbitrary complex input is being assumed.

Let $q_0 = \log M_{\mathbf{B}_j}(e^{iL^{-N}H_+})/i$. The uniform row bounds give

$$\|q_0\|_\infty \le CMA_0, \qquad \|q_{\mathrm{in}}\|_\infty \le CMA_0\sigma, \qquad \sup_{\square^{\sim 4}} |H_+| \le CMA_0.$$

Here an inner row is one whose complete double block is in $\square^{\sim 4}$. Such a row is a level-j row, by Lem. 3.126; its raw input has the extra factor $L^{j-N} = \sigma$. Every other row is at distance $D \ge cML^{\nu-1}$ from $\square^{\sim 3}$ in the level-j metric. This follows by stopping a path at the edge of $\square^{\sim 4}$; before that edge all pillar lengths are one in level-j units.

Choose M_0 so, for every $M \ge M_0$ and $\nu \ge 1$,

$$CMe^{-scML^{\nu-1}} \le 10^{-3}L^{-2\nu}. \tag{3.mcxiii}$$

Such a choice exists: with $t = L^{\nu-1} \ge 1$, the quantity t^2e^{-scMt} decreases for $scM \ge 2$. Next choose $a_I(M)$ so $CMa_I < r/8$ and $CM^2a_I/r < 10^{-3}$, as well as all preceding logarithm and gauge ceilings. Combining eq. 3.mcxi and eq. 3.mcxii then gives, for curvature and current,

$$|R_x(\tau q_0) - \tau R_x(q_0)| \le A_0\sigma^2/100. \tag{3.mcxiv}$$

For the potential and first difference they give the simpler bound $CMA_0\sigma$.

We identify the endpoint $\tau = 1$. Restriction of the upper critical field to the truncated sequence is critical at its full induced datum by Lem. 3.134. Real regular uniqueness identifies the restricted orbit. The equality extends to the common complex datum ball by holomorphy. Restoring the upper node gauge and both axial normalizations gives

$$e^{iL^{-j}\mathcal{H}_{\mathbf{B}_j,1}(q_0)} = (U_N(\square_0;\mathbf{U}))^G.$$

The local products in eq. 3.mxcix, the corresponding product for the flat nested tower, and eq. 3.mcvi give $\mathrm{cn}(G) \le \exp(CM(A_0 + A_1)) \le 4/3$ after decreasing $a_I(M)$. We check the real-base change in this identification. Write $V_0 = M_{\mathbf{B}^+}(U)$ and $e^{ib^U}V_0 = M_{\mathbf{B}^+}(\mathbf{U})$; the mean estimate gives $\|b^U\|_\infty \le C\alpha_1$. At the point in question use the polar factorization $v = v_re^{i\mu}$, with v_r unitary and $\|\mu\|_\infty \le CM\alpha_1$. Conjugating the exact datum identity by v_r^{-1} gives

$$e^{i\widetilde{b}^U(c)}V_0'(c) = e^{i\mu(c_-)}e^{ib_V(c)}e^{-i\mu(c_+)}, \quad V_0' = V_0^{v_r^{-1}}, \quad \widetilde{b}^U = \mathrm{Ad}(v_r^{-1})b^U.$$

Thus $V_0' = e^{ib_0}$ with b_0 real and $\|b_0\|_\infty \le CM(A_0 + A_1)$. The four logarithms $b_0, \widetilde{b}^U, \mu, b_V$ lie in a fixed smaller datum ball. At real values, global regular uniqueness gives both change of base and covariance under the node gauge. The two sides are holomorphic functions of the small logarithms on that ball, so the identity theorem gives these identities for their complex values. The polar factorization is used only to locate the given point in the common charts; no holomorphic dependence of its unitary factor is claimed. This proves the required base change and the endpoint identification on the actual domain. The upper tower clause and $\alpha_0 \le 3A_0/2$ now imply

$$L^{2j}|\partial e^{iL^{-j}\mathcal{H}_j(q_0)} - 1| \le 2A_0\sigma^2, \qquad |J^{(j)}(e^{iL^{-j}\mathcal{H}_j(q_0)})| \le 2A_0\sigma^3.$$

On their fixed common logarithm chart the actual means obey

$$|\mathsf{q}_q(a)(c) - \mathsf{q}_q(a_{\mathrm{ref}})(c)| \le CL^q \sup_{\mathsf{P}^q(c)} |a - a_{\mathrm{ref}}|. \tag{3.mcxv}$$

Here $a_{\mathrm{ref}} = L^{-N} H_+$. Apply Cauchy to the segment $a_{\mathrm{ref}} + z(a - a_{\mathrm{ref}})$ on the radius $c_{\mathrm{av}}/(2L^q \sup|a - a_{\mathrm{ref}}|)$, where the mean is bounded by $C_Q c_{\mathrm{av}}$; Schwarz applied after subtracting its value at zero gives the displayed inequality. The product logarithm in eq. 3.mcix satisfies $|a - a_{\mathrm{ref}}| \le CL^{-k}|A|$. Consequently $\|b - q_0\|_\infty \le CL^{j-k}\|A\|_\infty$. Choose c_2, c_3 so

$$C\|\tau(b - q_0) + b'\|_\infty < a_0/128$$

in every readout norm above. This is a fixed choice, because $2\sqrt{1+\nu}L^{-(\nu-1)} \le 2\sqrt{2}$. It also puts the entire datum in $\|d\|_\infty < r/2$, and so proves the asserted holomorphy on the full product of increment balls. The resulting curvature and current bounds are, respectively,

$$(2 + 1/100)A_0\sigma^2 + a_0/128, \qquad 2A_0\sigma^3 + A_0\sigma^2/100 + a_0/128.$$

Since $2\sqrt{1+\nu}L^{-2\nu} < 0.0168$ and $2\sqrt{1+\nu}L^{-3\nu} < 0.0013$, these are smaller than $a_0/8$ and $a_0/16$. For the potential and first difference the bound is $CMA_0\sigma + a_0/128$. Choose C_I large enough that eq. 3.mcvii, together with $2\sqrt{1+\nu}L^{-\nu} < 0.218$, makes it smaller than $a_1/2$. This proves eq. 3.mcx.

It remains to check every nested tower, without paying the large constant K_T against this fixed fractional radius. Let (p, Z) be any address of X, use either rule, and truncate at $n \le p$. The whole Z is inside the top member of the level-j window tower. The constraint and boundary-action checks in Lem. 3.134 identify its normalized critical continuation with $(e^{iL^{-j}H}|_Z)^w$. All its mean logarithms have size at most $C_Q \sup_Z |H|$ and are inside the common datum ball by the choice of a_I. Its normalization gauge is the product of its free ascent contours. At level q their logarithms are bounded by $CL^{q-j}\sup_Z|H|$; summing $q < n$ and including the top node factor yields $\mathrm{cn}(w) \le e^{Ca_1} < 6/5$. Thus the identity with its bound is

$$U_n(Z; e^{iL^{-j}H}) = (e^{iL^{-j}H}|_Z)^w, \qquad \mathrm{cn}(w) < 6/5. \tag{3.mcxvi}$$

For real window data this is the residual-gauge identity from regular uniqueness. Holomorphy on the common product ball proves it for all the present complex parameters.

By Lem. 3.126, the full stencils of every inset lie in Z. Conjugation and the change from level j to level n therefore give

$$|\partial U_n - 1| < \frac{6}{5}\frac{a_0}{8}L^{-2j} < \frac{a_0}{4}L^{-(j-p)}L^{-2p},$$
$$|J^{(n)}(U_n)| < \frac{6}{5}\frac{a_0}{16}L^{-3(j-n)} < \frac{a_0}{4}L^{-(j-p)}L^{2(n-p)}.$$

The second comparison uses $2j - p - n \ge 0$; the first uses $p \le j$. The real factor is 1, whose towers are identically 1. Its gauge clause holds with zero potential. The independent current chosen in the statement is the physical current already bounded in eq. 3.mcx. Thus every core clause holds on the whole parameter domain, including every nested tower and both rules. The proof used $X \subset \square^{\sim 2}$ only to put its complete simple stencils in $\square^{\sim 3}$ and its nested regions in the top member. This proves the last assertion as well. □

3.36. **Tower response in the ambient metric.** We now prove the weighted tower comparison used at settled elements of a deeper sequence. The determining tower is fixed during this comparison. The ambient sequence specifies the input norm and the output weights; it does not change the constrained problem.

Let $1 \le p \le i$. Take separated sequences $(\Omega'_q)_{q\le p}$ and $(\Omega^\flat_q)_{q\le i}$, with $\Omega'_q \subset \Omega^\flat_q$ for $q \le p$, and put $Z = \Omega'_0$. The determining set $\mathbf{B}'$ has all inward rows removed. All rows, outward endpoint blocks and physical pairings retain their actual conventions. Work in level-p units, so $\eta = L^{-p}$ and $\xi_q = L^{q-p}$. Assume each torus period contains at least 6ρ level-p sites, where $\rho = R_1\mathfrak{M}$. All volume-uniform assertions below refer to this admitted class. Sup distances between torus points

mean their shortest periodic sup distance. At a bare site x, let $\lambda'(x)$ and $\lambda^\flat(x)$ be the largest indices of the two sequences containing x, with value zero outside their first members, and set

$$\xi'(x) = \xi_{\lambda'(x)}, \qquad l(x) = \xi_{\lambda^\flat(x)}, \qquad s(x) = l(x)/\xi'(x) \geq 1.$$

Write d' and $d^\flat$ for the corresponding weighted path metrics, with density the reciprocal of the assigned scale. For a real field U and a complex potential v on the inner bare bonds of Z, define

$$\epsilon_U(P) = \eta^{-2} l(x_P)^2 |\partial U(P) - 1|, \qquad n_b(v) = \max\{l(x_b)|v(b)|,\ l(x_b)^2 |\nabla_U v|_Z(b)\}. \tag{3.mcxvii}$$

Only differences whose entire three-bond read lies in Z enter the last term. Put $n(v) = \max_b n_b(v)$ and

$$E_U^\delta(x) = \max_P e^{-\delta d^\flat(x,x_P)} \epsilon_U(P), \qquad E_v^\delta(x) = \max_b e^{-\delta d^\flat(x,x_b)} n_b(v).$$

The maxima use positively oriented inner plaquettes and bonds. An output element is an inner plaquette or a full bond, the latter meaning that all six incident plaquettes are inner. Its base x is settled if $\lambda'(x) = \min\{\lambda^\flat(x), p\}$. For the normalized critical continuation $\Phi_U(v) = \Phi_{\mathbf{B}', M_{\mathbf{B}'} U}(b(v))$, with the literal raw datum $b(v)$ of eq. 3.mxcvi, define

$$\mathcal{G}_U(v; P) = \eta^{-2} l(x_P)^2 (\partial \Phi_U(v)(P) - 1), \qquad \mathcal{G}_U(v; b) = l(x_b)^2 \xi'(x_b) J_\eta(\Phi_U(v))(b). \tag{3.mcxviii}$$

In particular, the current weight is $l^2 \xi'$. These are the weights of eq. 3.cdxix.

Lemma 3.136 (The two metrics and local scales)**.** Let $\rho = R_1 \mathfrak{M}$ be the separation in layer units. For bare sites x, y and for a row c of level q, with a bond b in its complete read, one has

$$\begin{aligned} d'(x,y) &\geq d^\flat(x,y), \qquad d'(x,y) \geq |x-y|_\infty/\xi_p, \\ s(x)/s(y) &\leq L^{2+2d'(x,y)/\rho}, \qquad d'(y_c, x_b) \leq C_L, \qquad \xi_q \leq L\xi'(x_b). \end{aligned} \tag{3.mcxix}$$

If $|x-y|_\infty \leq 3l(x)$, then $L^{-1} l(x) \leq l(y) \leq L l(x)$ and $d^\flat(x,y) \leq C_L$. If x is settled and $|x-y|_\infty > l(x)$, then $s(x) \leq 1 + d'(x,y)$. ■

Proof. Containment of the sequences gives $\xi' \leq l$ at every point of each bare path. Also $\xi' \leq \xi_p$. Integrating these two pointwise inequalities and taking infima proves the first two statements. To pass between layers with indices $a < a'$, a connected path must cross each intermediate collar. The collar at level q has width greater than $\rho\xi_q$ and path density ξ_q^{-1}, so each intermediate layer costs at least ρ. Thus the change of either index is at most one plus the corresponding weighted distance divided by ρ. Apply this to both indices and use $d^\flat \leq d'$ to bound their difference, giving the ratio of s in the display.

The complete order-q read lies in the preceding member; its endpoint blocks have diameter $O(\xi_q)$ and every site there has scale at least ξ_{q-1}. A path through these blocks has weighted length C_L. This includes the lower block at an outward endpoint and proves both row-read bounds. A ball of radius $3l(x)$ cannot cross two ambient layers, since the intervening collar is wider than that ball once $\rho \geq 6L$. Its scales differ by at most L, and coordinate paths in that ball give the local distance estimate. Finally, at a settled site either $s(x) = 1$ or $\lambda'(x) = p$. In the latter case, a far point satisfies $d'(x,y) \geq |x-y|_\infty/\xi_p > l(x)/\xi_p = s(x)$. □

Lemma 3.137 (Physical readouts on the fixed datum ball)**.** There are fixed $r, C, m > 0$ such that, at every sufficiently regular real datum V_0, the following holds on $\|b\|_\infty < r$. At any output element with base x, define $F_x(b)$ as the curvature or current readout normalized by $\xi'(x)$ rather than $l(x)$ in eq. 3.mcxviii. Then F_x is holomorphic, $|F_x(b)| \leq C$, and

$$|F_x(b_1) - F_x(b_0)| \leq C \max_c e^{-m d'(x,y_c)} |b_1(c) - b_0(c)|. \tag{3.mcxx}$$

The constants cover full current stencils across layer seams. ■

Proof. Use Cor. 3.125 on a smaller fixed raw datum ball. The naturalizer and literal node extension are included, as specified before Lem. 3.127. Apply eq. 3.mlxvii with the cell weight $e^{-md'(x,\cdot)}$. It is an admitted Lipschitz weight. The potential, first difference and multiplier derivatives therefore have the right-hand side of eq. 3.mcxx in their physical local norms.

For the current differentiate the literal constrained Euler identity eq. 3.mxcviii. The terms differentiate the local bond factor, the actual constraint transpose and the multiplier. The transpose retains its physical row density and the full outward endpoint read. Its weighted row sum, and the corresponding sum of its differentiated coefficients, are bounded by the actual carrier estimates. Thus the current derivative has the same bound in grade three. This step uses the equation of the continued critical field; it does not bound second differences of an arbitrary C^1 potential. Plaquette products give the curvature derivative in grade two.

The representative's gauge must also be differentiated. On each free ascent its factors have logarithmic differences bounded by CL^{q-t} times the weighted local difference of the Landau fields, as in eq. 3.mxcix. Each factor's complete read lies within distance C_L of the pillar top, and the site being normalized has distance at most 2 from that top. The geometric sum over $q < t$ is bounded. The top node factor has the same local bound from the site-mean inverse. Real reference factors are unitary; telescoping relative factors preserves the bounds. Consequently the gauge derivative and its inverse have the same weighted bound. Differentiating conjugation of the physical readout adds its already bounded value times this gauge derivative.

Every corner in a full stencil is at most two bare steps from x. Its assigned scale differs from $\xi'(x)$ by at most L. Absorb these fixed grade-two and grade-three factors and the bounded metric displacement in C. This treats seams without omitting any plaquette. At a nonfull boundary bond no full-current claim is made; the regional Euler equation still retains its boundary terms. Integrating the derivative along the segment from b_0 to b_1 proves the display. □

Proposition 3.138 (Ambient tower bound and response)**.** There are fixed $a, K_a, K_T, \delta > 0$ such that the following holds, uniformly in both sequences, their depths and volume. Suppose $\max_P \epsilon_U(P) < a$ and $n(w) < a$. The actual tower configuration and its full current exist. For fixed U they are holomorphic on this open convex set of potentials. At each settled output element with base x,

$$|\mathcal{G}_U(w)| \le K_a(E_U^\delta(x) + E_w^\delta(x)) \le 1. \tag{3.mcxxi}$$

For $n(w_0), n(w_1) < a$,

$$|\mathcal{G}_U(w_1) - \mathcal{G}_U(w_0)| \le K_T E^\delta_{w_1 - w_0}(x). \tag{3.mcxxii}$$

These statements include both erosion rules and their truncations whenever they are contained in the indicated ambient sequence. ■

Proof. Fix a common smaller rate m for Lem. 3.137 and the real local profile proved in Lem. 3.123. These primitive estimates are uniform above their fixed geometric floors. Increase the separation floor so that $4 \log L/\rho \le m/16$, and put $\delta = m/8$. Choose all amplitude ceilings after these choices.

The input norm $\|w\|_-$ of eq. 3.mxciii is at most $n(w)$. The averaged real datum is regular by the same full-read argument as eq. 3.mxcvi. That equation and Prop. 3.124 and Cor. 3.125 therefore define the tower holomorphically on a fixed ball $\|w\|_- < r_w$. Its current is a polynomial in its bond matrices and inverses on a full stencil. The conditions $n(w) < a$ are finitely many continuous linear seminorm bounds, so their intersection is open and convex.

We first need a spatial real estimate. If W is real with $\sup \epsilon_W$ small, the actual mean on a tested mixed plaquette τ of level q has defect bounded by

$$|\partial M_{\mathbf{B}'} W(\tau) - 1| \le C \max_{P \text{ in its complete read}} s(x_P)^{-2} \epsilon_W(P). \tag{3.mcxxiii}$$

Indeed the averaging estimate costs ξ_q^2 times the physical fine curvature, and $\xi_q \le L\xi'(x_P)$ on that read. At a mixed seam the reconstructed lower row is the actual next average by the semigroup

identity, so the same calculation includes the outward endpoint block. At level zero this is the prescribed plaquette itself. The read has d'-diameter C_L. In detail, the local real profile obtained from the proof of Lem. 3.123 is

$$r_{U(V)}(x) \le C \max_{\tau} e^{-md'(x,\tau)} |\partial V(\tau) - 1|. \tag{3.mcxxiv}$$

Here r is the maximum of the grade-two curvature and grade-three full-current norms at x in the tower scales, and τ ranges over the actual ordinary and mixed datum tests. To recall the implication, that proof gives $r(x) \le K v_R(x) + \frac{1}{2} \sup_{d'(x,y)\le R} r(y)$. For $P(x) = \max_\tau e^{-md'(x,\tau)} |\partial V(\tau) - 1|$ choose m small enough that $e^{mR} \le \sqrt{2}$. Maximizing $r(x)/(P(x) + t)$, $t > 0$, gives $S \le \sqrt{2}K + S/\sqrt{2}$, hence $r \le 5KP$ as $t \downarrow 0$. This proves the displayed profile with the stated metric and tested datum, including its local full-current readout. Multiply that profile by $s(x)^2$ and use Lem. 3.136 in eq. 3.mcxxiii. The factor $s(x)^2/s(x_P)^2$ costs at most $L^4 e^{4(\log L) d'(x,x_P)/\rho}$, which the reserved decay absorbs. Hence the real critical field at $M_{\mathbf{B}'}W$ has ambient readout at x bounded by $C E^\delta_W(x)$.

Write $F(w) = \mathcal{G}_U(w; x)$ and $s = s(x)$. For a real g the plaquette product expansion gives

$$\epsilon_{e^{ing}U}(P) \le 2\epsilon_U(P) + C \max_{b \subset P} n_b(g),$$

when $n(g)$ is small. The quadratic remainder is bounded by $Cn(g)^2$ and is absorbed into the displayed linear term. Real global uniqueness matches the critical field just estimated to the real point of the fixed chart. Consequently

$$|F(g)| \le C(E^\delta_U(x) + n(g)), \qquad |F(0)| \le C E^\delta_U(x). \tag{3.mcxxv}$$

On the whole complex ball $\|w\|_- < r_w$, the crude bound from Lem. 3.137 is $|F(w)| \le Cs^2$.

Choose a scalar cutoff on positive bonds, equal to one for $|x_b - x|_\infty \le l(x)$, equal to zero beyond $2l(x)$, and linear in between. Denote it by $\chi(b)$ and extend the resulting form by its reversal convention. Call a bond near if $|x_b - x|_\infty \le 3l(x)$, and write $n_{\mathrm{near}}(v) = \max_{b \text{ near}} n_b(v)$. The discrete product rule and Lem. 3.136 give

$$n(\chi v) \le C_\chi n_{\mathrm{near}}(v), \qquad \|\chi v\|_- \le C_L s^{-1} n(\chi v), \qquad n_{\mathrm{near}}(v) \le e^{C_L \delta} E^\delta_v(x). \tag{3.mcxxvi}$$

For the first inequality the cutoff difference is at most $\eta/l(x)$; adjacent ambient scales differ by at most L. For the second, if $s > 1$ the output is in the tower's top layer and every tower scale is at most $\xi_p = l(x)/s$; if $s = 1$, use $\ell_- \le l$. These arguments also hold when a difference stencil meets the edge of the support.

Apply Lem. 3.128 to χw on the fixed $\|\cdot\|_-$ ball. Its remainder is at most $Cs^2 \|\chi w\|_-^2 / r_w^2 \le Cn(\chi w)^2 / r_w^2$. Together with eq. 3.mcxxv, this proves

$$|F(\chi w)| \le C(E^\delta_U(x) + n_{\mathrm{near}}(w)) \tag{3.mcxxvii}$$

on a smaller fixed amplitude ball. This is the two-power gain for near forcing; it uses no positive travel distance.

For far forcing let $u = w - \widetilde{w}$ vanish on $|x_b - x|_\infty \le l(x)$. Cauchy on the actual averaging chart gives, row by row,

$$|b(w)(c) - b(\widetilde{w})(c)| \le C\xi_q \sup_{b \in \mathsf{P}^q(c)} |u(b)| \le C \max_{b \in \mathsf{P}^q(c)} s(x_b)^{-1} l(x_b) |u(b)|.$$

Put $D = d'(x, x_b)$. A contributing bond is far, so $s \le 1 + D$. The scale ratio and row-read distance give

$$s^2 e^{-md'(x,y_c)} s(x_b)^{-1} \le C(1 + D) e^{-(m - 2\log L/\rho)D} \le C e^{-2\delta D}. \tag{3.mcxxviii}$$

Apply eq. 3.mcxx, remembering that $F = s^2 F_x$, and then use $d' \ge d^\flat$. We obtain

$$|F(w) - F(\widetilde{w})| \le C E^\delta_u(x). \tag{3.mcxxix}$$

The bound is uniform on a common smaller fixed ball. Combining it with eq. 3.mcxxvii, with $\widetilde{w} = \chi w$, proves eq. 3.mcxxi on $n(w) < \bar{a}$ after fixing K_a and reducing $\bar{a}$.

For the Lipschitz assertion put $v = w_1 - w_0$ and split the difference through $w_0 + \chi v$. The far bracket is bounded by eq. 3.mcxxix. Put $n_v = C_\chi n_{\mathrm{near}}(v)$ and $E = E^\delta_U(x) + E^\delta_{w_0}(x)$. If $n_v = 0$, the near bracket vanishes. Otherwise set $R = 1 + E/n_v$. For $|z| \le R$,

$$n(w_0 + z\chi v) \le a + n_v + E \le (3 + 2C_\chi)a.$$

Choose $a < \bar{a}/(4 + 2C_\chi)$. The a priori estimate applies on a neighborhood of this disc and bounds $F(w_0 + z\chi v)$ by $K_a(2E + n_v)$. Schwarz's lemma, after subtracting the value at zero and testing by norm-one linear functionals, bounds the near bracket by

$$\frac{2K_a(2E + n_v)}{1 + E/n_v} \le 4K_a n_v \le CE^\delta_v(x).$$

Together with the far bracket this proves eq. 3.mcxxii. Finally choose $2K_a a \le 1$. No constant counts cells, addresses or truncations. □

Corollary 3.139 (Covariance and reality)**.** For a real gauge u on Z, let $\widetilde{u}$ equal $u(y)$ throughout each pillar with top y. Then the domain and bounds above are unchanged by $(U, w) \mapsto (U^u, \mathrm{Ad}(u)w)$, and

$$\Phi_{U^u}(\mathrm{Ad}(u)w) = \Phi_U(w)^{\widetilde{u}}.$$

The output readout is conjugated by $\widetilde{u}(x)$. For w^* obtained by taking the adjoint at positive bonds,

$$\Phi_U(w^*) = \Phi_U(w)^{\mathfrak{c}}, \qquad \mathfrak{c}(X) = (X^{-1})^*, \qquad |\mathcal{G}_U(w^*)| = |\mathcal{G}_U(w)|.$$

For Hermitian w the tower is its real normalized critical representative. Also $\Phi_1(0) = 1$ and $\mathcal{G}_1(0) = 0$. ■□

Proof. Unitary covariance preserves the input plaquette and derivative norms. The actual means transform by u at their nodes. The prescribed axial normalization extends this node gauge constantly on each pillar, so real uniqueness gives the first identity at real potentials; holomorphy gives it on the complex domain. Plaquettes and the complete current transform by conjugation at the stated base. The anti-holomorphic involution $\mathfrak{c}$ fixes the real group. The map $w \mapsto \mathfrak{c}(\Phi_U(w^*))$ is holomorphic and agrees with $\Phi_U(w)$ on the real slice, which proves the second identity. The current is changed to its adjoint. For $X \in \mathrm{SL}(2, \mathbb{C})$, $X^{-1} - 1 = \mathrm{adj}(X - 1)$; the two singular values of the adjugate are interchanged. Thus $|X^{\mathfrak{c}} - 1| = |X - 1|$ in the operator norm, proving the curvature norm assertion as well. Finally the identity configuration has identity averages, zero action derivative and the prescribed normalization; real uniqueness identifies it with $\Phi_1(0)$. Its current and curvature vanish. □

Corollary 3.140 (Localizations and deeper top metrics)**.** Let the ambient sequence extend from depth i to depth $t \ge i$, with the same members through i. Write d_t for its metric and $y_t(x)$ for the top of the pillar containing x. The bounds of Prop. 3.138 remain valid with $d_t(y_t(x), y_t(x_b))$ in the input envelopes, after multiplying their constants by $e^{4\delta}$.

For a region X made of localization cubes at level zero, let $X_{[q]}$ be its second-rule erosion and put $\Omega'_q = \Omega^\flat_q \sqcap X_{[q]}$, $q \le p$. Assume $M_1 \ge \rho$ and $M \ge L^3 M_1$. Retain the preceding period floor at depth p; a period floor at the deeper level t implies it. Define $X^{\wr p}$ as the union of the level-$(p-1)$ localization cubes whose two-layer enlargements lie in X. The new sequence is separated. Every output element whose base is within $2\xi_p$ of $X^{\wr p}$ is settled, provided its full stencil belongs to Ω'_0. Thus the same weighted response holds for these actual localized towers. ■□

Proof. Extending the ambient sequence only increases assigned scales, so $d_t(x, z) \le d^\flat(x, z)$. Each site is within weighted distance 2 of its own top. The triangle inequality gives

$$d_t(y_t(x), y_t(z)) \le d^\flat(x, z) + 4,$$

which proves the first assertion for both envelopes.

All intersections are taken cube-wise on nested aligned grids. Their next-member gaps are bounded below by the minimum of the two original gaps, so the localized sequence is separated. For the second-rule erosion, every discarded point at stage q is within distance $2M_1\sum_{r=1}^{q}\xi_r < 3M_1\xi_p$ of X^c. This is the rejected-cube induction in Lem. 3.126, which only requires that the initial region be a union of the aligned smaller cubes. On the other hand, $\operatorname{dist}_\infty(X^{\wr p}, X^c) \geq 2M\xi_{p-1}$. A point within $2\xi_p$ of that inset therefore has distance at least $(2M/L-2)\xi_p > 3M_1\xi_p$ from X^c. It lies in every $X_{[q]}$ through p. At such a bare site, membership in Ω'_q is exactly membership in $\Omega^\flat_q$, so the tower index is the ambient index truncated at p. This is precisely the settled condition. If the coarse partition is absent the inset is empty and the assertion about its neighborhood is vacuous. The assumption on the full stencil retains all physical current terms at the localized boundary. □

Corollary 3.141 (Multiplicative shifts in the ambient norm)**.** Let (U, w) satisfy the hypotheses of Prop. 3.138, and put $\mathbf{U} = e^{i\eta w}U$. For a complex shift A, set

$$u_b = \max\{l(x_b)|A(b)|,\ l(x_b)^2|\nabla_{\mathbf{U}}A|_Z(b)\}, \qquad w \boxplus A = (i\eta)^{-1}\log(e^{i\eta A}e^{i\eta w}).$$

Let u_b^+ be the maximum over b and its forward translates that occur in a difference stencil in Z. After decreasing the fixed ceiling a, the conditions $u_b \leq (256L^2)^{-1}$ and $n_b(w) + 2u_b^+ < a$ imply

$$|\mathcal{G}_U(w \boxplus A) - \mathcal{G}_U(w)| \leq 2e^{\delta}K_T \max_b e^{-\delta d^\flat(x,x_b)}u_b \tag{3.mcxxx}$$

at every settled output. The shifted tower exists on this whole domain and has the literal product $e^{i\eta A}\mathbf{U}$ as the input to its actual average. ■□

Proof. Take $a \leq a_* = (64L^2)^{-1}$. The product logarithm lies in the fixed chart since $\eta|w| \leq a_*$ and $\eta|A| \leq a_*/4$. Write $\log(e^Xe^Z) = X + Z + \ell(X, Z)$. The absolutely convergent logarithm and exponential series on $|X|, |Z| \leq 1/16$ give the derivative bound $8(|X| + |Z|)$ for the remainder. As $\ell(0, Z) = 0$, Schwarz in X also gives $|\ell(X, Z) - \ell(X, Z')| \leq 16|X||Z - Z'|$. Apply these estimates to neighboring bonds after transporting them to the same base by U. The product rule gives

$$l(b)|(w \boxplus A - w)(b)| \leq (1 + 10a_*)u_b,$$

and the first-difference bound has coefficient at most

$$(1 + 10a_*)(1 + 2.1a_*L) + 16a_* < 2$$

times $\max(u_b, u_{b^+})$. Here the term $2.1a_*L$ comes from replacing $\nabla_U A$ by $\nabla_{\mathbf{U}}A$: $|1 - \mathrm{Ad}(e^{i\eta w})| \leq 2.1\eta|w|$, and neighboring scales differ by at most L. Thus $n_b(w \boxplus A - w) \leq 2u_b^+$ and the stated margin puts $w \boxplus A$ in the same domain as w. The product identity holds at positive bonds by exponentiating the logarithm and at reverse bonds by inversion. Apply eq. 3.mcxxii. Adjacent bare sites have ambient distance at most one, so replacing u_b^+ by u_b in the weighted maximum costs at most e^{δ}. This proves the bound and the actual-input assertion. □

3.37. **Changing the localization of a tower.** The response in Prop. 3.138 keeps the determining sequence fixed. We next compare the two constrained problems obtained when a localization is cut and its tower is rebuilt. All determining sets in this subsection have their inward rows removed. A common row means a common geometric row with its complete averaging read; a common free pillar includes its prescribed normalization paths.

Use the notation of § 3.36. Let $\widehat{\mathbf{B}}$ be a finer localization of $\mathbf{B}'$, of the same depth p, inside the same ambient sequence of depth i. Its first member is $\widehat{Z} \subset Z$. Finer means that its members are contained in the old members, so that the exact restriction identity eq. 3.cccxciii applies. Denote their weighted metrics by d' and $\widehat{d}$. Both dominate $d^\flat$. An old interior or outward row cannot become inward in the finer sequence: an old layer endpoint stays outside the next old member and hence outside the next finer member. Thus removing all finer inward rows does not remove an old retained row needed by the restriction identity. Write $\mathcal{C}$ for the new determining rows together with rows whose complete reads have changed. Enlarge $\mathcal{C}$ once to include the complete preceding

blocks of its outward rows and the old pillars containing its endpoints; call the resulting carrier $\mathcal{C}^+$. It is a set of reads, not a new set of constraints.

For $W = e^{i\eta w}U$, where U is real, set

$$A(z) = \max\{\epsilon_U(P), n_b(w) : x_P = z \text{ or } x_b = z\}, \qquad A_\delta(y) = \max_z e^{-\delta d^\flat(y,z)} A(z). \tag{3.mcxxxi}$$

Empty maxima are zero. Complete stencils can instead be based at any of their corners, at a fixed change in the constant. All amplitudes below lie below one fixed smaller ceiling than the ceilings of Cor. 3.125 and Prop. 3.138.

Lemma 3.142 (Alignment of the induced and literal data)**.** There are fixed $C, r, \delta > 0$ with the following property. The literal finer datum of W and the datum induced by the old critical continuation can be put in one physical datum chart on $\widehat{\mathbf{B}}$, holomorphically in w. In this chart their logarithms $b_0(w), b_1(w)$ satisfy $\|b_\nu(w)\|_\infty < r$, and

$$b_1(c) - b_0(c) = 0 \quad (c \notin \mathcal{C}), \qquad |b_1(c) - b_0(c)| \le C A_\delta(y_c) \quad (c \in \mathcal{C}). \tag{3.mcxxxii}$$

The segment joining these logarithms stays in the chart. At an unchanged free pillar, the normalized finer continuations at its two endpoints are respectively the new and the old tower in exactly the same node frame. ■

Proof. First fix the gauges and the real centre of the chart. On every old pillar of height j put the ascent gauge equal to one at its top. If $W_q = \mathcal{M}^q W$ and $T_q(z, x; W_q)$ is the actual level-q mean contour from the parent centre z, its descent is

$$g_j = 1, \qquad g_q(x) = g_{q+1}(z) T_q(z, x; W_q), \quad q < j. \tag{3.mcxxxiii}$$

This is the contour convention of eq. 3.cccxvii; reverse orientations use inverses. The contour from a centre to itself is one, so the descent extends the gauge already assigned at coarser sites. Covariance of the actual mean makes the internal mean contours one. At layer tops set $g = 1$. The pillar partition makes these assignments compatible. Thus every retained old datum is unchanged, including an outward row and its lower endpoint block. The old critical field is already in the canonical old normalization, so its own descent is the identity on its old pillars. We do not apply the raw input's gauge to that critical field. The two separately normalized fields still have the same old datum.

For completeness this construction is uniform on the complex ball. Retain each real contour as a unitary factor. The relative level-q factor on a pillar of height j has logarithm bounded by CaL^{q-j} when $n(w) < a$. The local mean estimate and its site correction give this bound exactly as in eq. 3.mxcix. Moving the real factors to the right and telescoping the relative factors gives

$$\|g(w)g(0)^{-1} - 1\| \le Ca \sum_{q<j} L^{q-j} \exp\left(Ca \sum_{q<j} L^{q-j} \right), \qquad \|g(w)^{\pm 1}\| \le e^{Ca/(L-1)}.$$

The real factors have norm one. All means and inverses are holomorphic on their fixed charts, so this also proves the holomorphy and the uniform condition number of the descent. There is no factor raised to the number of levels.

We spell out the comparison of new rows. At a real input the two fields just normalized have the same old datum. In an old pillar of height j, averaging a curvature bound at level $q < j$ gives $C\epsilon L^{2(q-j)}$. The finite contour reconstruction eq. 3.cccxvii–eq. 3.cccxx therefore compares both fields to the same face lift of their old datum. For the complex version use relative products, rather than unitary invariance. If E_q is the normalized bond discrepancy at the current level, its exact factorization has the form

$$E_q = (\widehat{W}_q \widehat{W}_{q+1}^{-1})(\widehat{W}_{q+1} L_{q+1}^{-1}).$$

Here L_{q+1} denotes the face lift in the common old frame. The two level-$(q+1)$ quantities in this identity mean their actual face lifts evaluated at the same level-q bond as $\widehat{W}_q$. Thus all three factors

belong to one matrix fibre. The one-step lift consistency proved with eq. 3.cccxcvi gives this exact product. For matrices near one, $|AB-1| \le |A-1| + (1+|A-1|)|B-1|$. Consequently

$$e_q \le C\epsilon_q + (1+C\epsilon_q)e_{q+1}, \quad \epsilon_q \le C\epsilon L^{2(q-j)}, \qquad \prod_{q<j}(1+C\epsilon_q) \le \exp\left(\frac{C\epsilon}{L^2-1}\right). \tag{3.mcxxxiv}$$

The endpoint discrepancy at an old retained row is zero. At an inward link, use the outward link one level higher and its complete preceding block, as in the inward calculation following eq. 3.cccxx. This adds a fixed number of lower internal links, not another constraint. Summing eq. 3.mcxxxiv proves a uniform raw discrepancy.

Here the local ϵ can be replaced by $CA_\delta(y_c)$. For the real critical field this follows from eq. 3.mcxxiv and the complete-read datum estimate eq. 3.mcxxiii. For its complex variation, differentiate the physical datum chart from the real input to w. The raw variation of a level-q input row is bounded by $C\xi_q|w|$ on its complete read, hence by $Cn(w)$. The weighted potential and first-difference response in eq. 3.mlxvii, and the differentiated literal pillar factors from the proof of Lem. 3.137, give the same weighted estimate for the continued field. A plaquette in an old pillar of height j then has defect bounded by $CA_\delta(y_c)L^{2(q-j)}$ at level q. Indeed its physical curvature is bounded in scale ξ_j; averaging to level q multiplies it by ξ_q^2. The complete reconstruction reads remain within bounded d'-distance of that pillar top. The ambient metric is smaller, and the convolution of the two weighted suprema uses a smaller fixed δ. This proves the local form of the estimate in eq. 3.mcxxxiv. The input field obeys the same curvature bound by the plaquette product formula and the C^1 norm in eq. 3.mcxvii. No second difference of the input is used.

Take as finer real reference the regular critical orbit at the finer datum of $U^{g(0)}$. Its existence follows from Prop. 3.124. At $w=0$ the preceding descent places the datum induced by the old real critical field within $C\sup A$ of this reference datum. The variation of either endpoint datum on the whole complex ball is Ca: this follows for the input from the mean derivative and the bound on g, and for the old field from its physical chart and the fact that a new-row scale does not exceed the old scale on its complete read, up to a single factor L. Choose $C(\sup\epsilon_U + a) < r/4$. Both endpoints then lie in the concentric half-ball of one finer physical chart. Their logarithms and the segment joining them lie in that chart. The logarithm has bounded derivative there, so the raw discrepancy just proved gives eq. 3.mcxxxii.

Every common old row is fixed exactly by the gauge at its endpoints, and the two fields have identical old data. Thus their finer data, and their raw logarithms relative to the same reference, agree on common rows. Passing to the reduced datum does not produce a chain of new supports: eq. 3.cccxiv says that its off-diagonal part reads interior columns into outward rows and that $TD^{-1}T = 0$. Hence differentiation or subtraction enlarges the changed support by at most one complete preceding block.

It remains to identify the endpoints as fields, rather than merely as data. For real w, restriction of the old critical field is critical at its induced finer datum by eq. 3.cccxciii. Its regularity improves in the finer scales. Global regular-fibre uniqueness from Lem. 3.122 identifies it with the finer physical chart at that datum. These identities extend to the whole complex ball by the identity theorem: both sides, their constraints, and all normalization factors are holomorphic there, and the real ball is a uniqueness set. Only the transformed raw data are admitted to this common physical chart. No invariance of a hereditary core under complex gauge transformations is asserted or required.

Finally the gauge eq. 3.mcxxxiii fixes every old top. Gauge covariance of a normalized finer representative uses the pillar-constant extension of its values at the new tops. On a common free pillar its top is an old top, so this extension is exactly one. Normalizing the old restriction on the finer sequence also acts identically on such a pillar: its mean contours there already have the prescribed old normalization. These statements hold as matrix identities also at complex data. In

particular the readout frames agree; a bound on the condition number alone is not used to identify their magnitudes. □

Proposition 3.143 (Comparison after a geometric cut)**.** Suppose the complete read at x lies in unchanged free pillars and is settled for both towers. Work in the units of the earlier ambient level i, so that $l \le 1$. Suppose every changed complete datum carrier is at physical distance at least $D \ge 2$ from x. Let $\mathcal{R}(z) > 0$ be an allowance profile satisfying

$$A(z) \le C_0 \mathcal{R}(z), \qquad \mathcal{R}(z) \le C_1 \mathcal{R}(x) e^{\kappa d^\flat(x,z)}, \quad \kappa < \delta/4. \tag{3.mcxxxv}$$

Then the difference of the two matrix readouts in eq. 3.mcxviii, in their prescribed common frame, is at most

$$C C_0 C_1 e^{-c\delta D} \mathcal{R}(x). \tag{3.mcxxxvi}$$

The assertion includes full current stencils across a common layer seam. All constants are independent of depths and volume. ■

Proof. Apply Lem. 3.137 on the finer chart of Lem. 3.142. Its proof includes the reduced datum, the literal node correction and the full physical Euler current. Thus it gives the native estimate

$$|\widehat{F}_x - F_x| \le C \max_{c \in \mathcal{C}^+} e^{-m\widehat{d}(x,y_c)} A_\delta(y_c). \tag{3.mcxxxvii}$$

All source measures are included in its weighted row sum. There is no cardinality factor for the collar. The support enlargement described above is why the distance hypothesis uses the complete carrier.

The ambient readout is the native one times $s(x)^2$ for both curvature and current. At a settled read either $s = 1$ or $\xi'(x) = \xi_p$. In the latter case $\widehat{d}(x, y_c) \ge |x - y_c|_\infty / \xi_p \ge D/\xi_p$ and $s = l/\xi_p \le 1/\xi_p$, so $s \le 1 + \widehat{d}(x, y_c)$. A fixed part of the exponential in eq. 3.mcxxxvii pays s^2. This is the extra scale argument needed for a changed geometry; the current retains the weight $l^2 \xi'$.

For any c, z, the triangle inequality and $\widehat{d} \ge d^\flat$ give

$$d^\flat(x, z) \le \widehat{d}(x, y_c) + d^\flat(y_c, z).$$

Insert eq. 3.mcxxxv in the definition of A_δ. Reserve one part of each exponential for this triangle inequality and another part for the factor s^2. The remaining decay is at most $\exp(-c\delta \widehat{d}(x, y_c)) \le \exp(-c\delta D)$, since $\xi_p \le 1$ in the present units. This proves eq. 3.mcxxxvi. The frame identities in Lem. 3.142 make it a matrix difference estimate. The reverse triangle inequality therefore gives the same estimate for the change of magnitude. The full-stencil convention and the physical Euler derivative in Lem. 3.137 cover a common seam without discarding any incident plaquette. □

The distance hypothesis in Prop. 3.143 holds for the geometric restriction of an earlier address. Here $p \le r \le i$, its nominal layer is at most i, and the read retains its r-window. The accumulated erosion reach through level q is at most $C_L M_1 \xi_q$, by the geometric sum in the proof of Lem. 3.126. Therefore a new row at level $q \le p$ is within $C_L M_1 \xi_p$ of the deleted set. Its complete backward read adds $C_L \xi_p$. An old parent pillar adds at most $C_L \xi_i = C_L$ in physical distance, even when its new endpoint has smaller assigned scale. Thus increasing the fixed cut halo from $3M_1$ to $C_{\mathrm{cut}} M_1$ puts every changed complete carrier at distance at least D when the read starts at distance $D + C_{\mathrm{cut}} M_1$ from the deleted set. There is no additional spacing assumption on discarded components. Indeed every old erosion test and complete read has a connected backward carrier contained in Z when it is retained. Follow such a carrier from the retained component toward a point removed by the component selection. If it stays in $Z \cap R$, it connects the two components, contrary to their definition. Its first exit must therefore meet $Z \setminus R$. A contact with Z^c was already an old boundary and cannot cause a new membership change. The same geometric bound on the backward carrier therefore locates the changed row near the actual cut, also in this case. The constant C_{cut} is chosen before the localization floor M is increased. Equivalently retain the original hypothesis $D + 3M_1$

and increase K_{cl} in $D \geq K_{\mathrm{cl}}M$ so that $(C_{\mathrm{cut}}-3)_+M_1 \leq D/2$. Then every changed complete carrier has distance at least $D/2$, which changes only the fixed decay rate in eq. 3.mcxxxvi. This argument uses physical distance and $l \leq 1$. It does not assert that an arbitrary Euclidean collar has bounded width in the ambient weighted metric.

3.38. **A selected nonlinear extension at an independent current.** We specify a nonlinear auxiliary action whose Hessian is the canonical second precision. Its small constrained solution agrees with the physical critical germ at physical inputs. Away from those inputs it is a selected independent-current extension, not a claim of Wilson stationarity.

The distinction between two output fields is essential. The auxiliary potential has first derivative H_1 in the reduced datum. Its finite gauge completion has first derivative SH_1. These agree at a physical pair, where the second lift is horizontal, and can differ off current. Literal outward data also retain their actual datum differential.

3.38.1. *The fixed chart and the strong spaces.* Fix $0 < \alpha < 1$, and a given real regular reference U_* on the actual determining geometry, with strict room in its curvature and physical-current ceilings. The reference need not be critical. Its local real Coulomb potentials have the scaled $C^{1,\alpha}$ bounds supplied by the current-regular construction. All choices of local frames and node extensions in this subsection are made in this fixed reference chart.

Let

$$U = e^{i\eta E}U_*, \qquad \|E\|_{\mathrm{str},\alpha;U_*} < \varepsilon.$$

This is a weighted $C^{1,\alpha}$ tube, stronger than the C^1 tube used for energy inverses. Let $\mathcal{A}_\alpha$ be the fine strong space in eq. 3.dccclxxxvii, measured in U_*, and let $\mathcal{Y}_{2,\alpha}$ be the site gauge space in eq. 3.mxxiv. Its source space $\mathcal{C}_{\alpha,2}$ is eq. 3.mxxiii, including comparisons across neighbouring pillar faces on complete boxes. Use the amplitude norms

$$\|f\|_{\infty,3} = \sup_b \mathsf{L}(b)^3|f(b)|, \quad \|h\|_{\infty,4} = \sup_x \xi_x^4|h(x)|, \quad \|b\|_{\mathcal{B}} = \sup_c \xi_c|b(c)|, \quad \|\nu\|_{\mathcal{W}} = \sup_c \xi_c^3|\nu(c)|. \tag{3.mcxxxviii}$$

The site scale and the field-row scale retain their different assigned levels at outward endpoints. The finite-p force norm is $\mathcal{F}_p$ of § 3.25.

The independent fine current has a full amplitude $\|J\|_{\infty,3} < \gamma_*$. This is convenient for the updated-current assertion below. The actual Wilson current j_U at a complex U is not assumed to have this bound. Its possibly large value will be subtracted only as an exact linear centering. All transposes and scalar pairings in an analytic formula are physical bilinear ones.

For this fixed U, retain its actual D, q, Q, G_s, R and put

$$\Lambda = G_sRD^\sharp, \qquad S = 1 - D\Lambda. \tag{3.mcxxxix}$$

Define G, H, ω, G_1, H_1 in the order of § 3.26, with $\omega = H^\sharp J$. In particular the lift in ω is the first lift. Their completed operators are uniformly admitted on a smaller tube by the energy and strong inverse results already proved.

3.38.2. *Completed restoration and the two adjoint bounds.*

Lemma 3.144 (Strong restoration on the stronger tube)**.** After shrinking the fixed $C^{1,\alpha}$ tube,

$$\|\Lambda A\|_{\mathcal{Y}_{2,\alpha}} \leq C_\alpha \|A\|_{\mathcal{A}_\alpha}. \tag{3.mcxl}$$

This map is holomorphic in the background on that tube. It is an estimate for the completed operator. It does not assert a $C^{2,\alpha}$ bound for every individual inverse-walk coefficient. ∎

Proof. The first physical divergence maps $\mathcal{A}_\alpha$ into $\mathcal{C}_{\alpha,2}$. The site projection R preserves that source space. Indeed its nonidentity part is $\widehat{F}\widehat{M}^{-1}\widehat{F}^\sharp$. The completed site-column point and first-difference bounds, the normalized coarse inverse and their exponential sums control its scaled C^α norm. All normalizations are those of eq. 3.cmxcv.

For $\lambda = G_s R D^\sharp A$, the exact identities $qG_sR = 0$ and $\Delta_U G_s R = R$ give

$$q\lambda = 0, \qquad \Delta_U \lambda = RD^\sharp A.$$

The zero-average energy estimate supplies its weighted mass and first differences. The local scalar bootstrap supplies the $C^{1,\alpha}$ bound needed on the right of the interior Schauder estimate. On a fixed complete nonbare box the coefficients of Δ_U have their C^α bounds because the potential is $C^{1,\alpha}$. At a true boundary the bare values of λ are exactly zero and the remaining atomic stencils have their scaled bounds. The proof of eq. 3.mxxvi therefore gives eq. 3.mcxl.

For completeness, at complex U compare with the real reference Laplacian. Its difference is a first-order term with C^α coefficients and a zeroth-order term with C^α coefficients, bounded by the $C^{1,\alpha}$ tube. Apply the real Schauder estimate to this equation, using the already controlled $C^{1,\alpha}$ solution norm. No $C^{2,\alpha}$ conclusion from merely C^1 coefficients is used. The energy inverse is holomorphic, and the uniform stronger bound on the smaller tube gives its holomorphy in these displayed Banach norms. □

Lemma 3.145 (Adjoints in their actual force classes)**.** For site and field forces respectively,

$$\begin{aligned} \|\Lambda^\sharp h\|_{\infty,3} &\le C\|h\|_{\infty,4}, \\ \mathcal{F}_p(\Lambda^\sharp D^\sharp f) &\le C_p\|f\|_{\infty,3}, \qquad 4 < p < \infty. \end{aligned} \tag{3.mcxli}$$

The completed operators in these assertions are holomorphic on a common smaller physical tube. There is no endpoint $p = \infty$ in the second line. ■

Proof. The balanced site family $G_s : Y^\flat \to Y$ gives bounded local Y norm for $u = G_s h$: a site force of amplitude four has uniformly bounded local $Y^\flat$ norm. Its equation is

$$\Delta_U u = h - q^\sharp a_s q u.$$

The average qu is bounded by its cell L^2 norm, and $q^\sharp$ extends the transported density. Thus the rescaled forcing is bounded. The scalar L^2-to-L^3-to-L^8 bootstrap gives $\xi_y^2|u| + \xi_y^3|Du| \le C\|h\|_{\infty,4}$. It uses no derivative of h. Subtract $Pu = \widehat{F}\widehat{M}^{-1}\widehat{F}^\sharp u$. Its normalized coarse coefficients are bounded by the column sum, then by the coarse ℓ^∞ inverse. The completed F columns and their first differences give that same point bound for Pu, DPu. Since $\Lambda^\sharp = DRG_s$, this proves the first line.

For the second line use the exact identity

$$\Lambda^\sharp D^\sharp f = f - S^\sharp f.$$

The finite-p force-restoration estimate of § 3.25 applies to this variational divergence source. It gives the stated bound without a pointwise differentiation of f. Its proof on the complex physical tube uses the same energy inverse and local finite-p bootstrap. The absolute estimates are uniform on a smaller tube; the analytic inverse formulas give the claimed holomorphy. □

3.38.3. *The finite gauge chain as a strong analytic map.* For $A \in \mathcal{A}_\alpha$, set $\lambda = \Lambda A$ and define

$$\Pi_U(A)_b = \frac{1}{i\eta} \log\{e^{-i\lambda(b_-)} e^{i\eta A_b} e^{i\,\mathrm{Ad}(U_b)\lambda(b_+)}\}, \qquad \Xi_U(A) = \Pi_U(A) - SA. \tag{3.mcxlii}$$

This is the finite chain already selected for the canonical current convention. Its associated fine gauge is $e^{-i\lambda}$. The exact equality

$$\mathrm{Ad}(U_b)\lambda(b_+) = \lambda(b_-) - \eta D\lambda(b)$$

makes division by η uniform: at $\eta = 0$ the three factors multiply to one. Thus Π_U is an analytic local function of $A, \lambda, D\lambda$, with

$$\Pi_U'(0) = S, \quad \|\Pi_U(A)\|_{\mathcal{A}_\alpha} \le C_\alpha \|A\|_{\mathcal{A}_\alpha}, \quad \|\Xi_U(A)\|_{\mathcal{A}_\alpha} \le C_\alpha \|A\|_{\mathcal{A}_\alpha}^2. \tag{3.mcxliii}$$

Covariant product rules use at most two site differences of λ. They prove these bounds also for the gradient Hölder seminorm. If $RD^\sharp A = 0$, then $\lambda = 0$ and $\Pi_U(A) = A$ exactly. No claim that $\Pi_U(A)$ lies in that slice for a general A is made.

The physical adjoint of the derivative requires more than an abstract strong-space norm. Pair a local function of $A, \lambda, D\lambda$ with a bounded force f. Differentiating its three arguments gives the exact form

$$f_0 + \Lambda^\sharp f_\lambda + \Lambda^\sharp D^\sharp f_d. \tag{3.mcxliv}$$

Here f_0, f_d are field forces of amplitude three and f_λ is a site force of amplitude four. Endpoint injections cost only the fixed number of incident bonds. The two bounds in eq. 3.mcxli therefore control this expression in finite $\mathcal{F}_p$, without differentiating f.

In particular $\langle J, \Xi_U''(0)[u,v]\rangle = \langle u, B_U[J]v\rangle$ by the exact finite-chain calculation. Subtracting that quadratic jet in eq. 3.mcxliv gives

$$\mathcal{F}_p\{\nabla\langle J, \Xi_U(A)\rangle - B_U[J]A\} \le C_{p,\alpha}\|J\|_{\infty,3}\|A\|^2_{\mathcal{A}_\alpha}. \tag{3.mcxlv}$$

Indeed the remaining three local coefficients are quadratic in the scaled variables $\xi A, \lambda, \xi D\lambda$. Their amplitude grades are exactly those just listed. This proves the residual in a force norm; an arbitrary Banach adjoint is not presumed to preserve pointwise bounds.

3.38.4. *The centered Wilson residual at a complex base.* Put $F_U(A) = A_{\Omega_0}(e^{i\eta A}U)$, $j_U = F_U'(0)$, and $H_U = F_U''(0)$. The centering in the next lemma is essential even when U belongs to the complex strong tube.

Lemma 3.146 (Uniform residual after linear subtraction)**.** On a fixed strong ball in the displayed tube,

$$\|F_U'(A) - j_U - H_U A\|_{\infty,3} \le C_\alpha \|A\|^2_{\mathcal{A}_\alpha}. \tag{3.mcxlvi}$$

The difference is uniformly holomorphic in U, A. Its Lipschitz constant on a radius-r strong ball is at most $C_\alpha r$. No amplitude bound for j_U at that complex base is used. ■

Proof. On a nonbare box rescale its physical length to one, and write $h = \eta/\xi$. In a fixed real reference frame the complex base has potential a_0 with bounded $C^{1,\alpha}$ norm. The relative logarithm is

$$P_h(A) = \frac{1}{ih}\log(e^{ihA}e^{iha_0}).$$

Let $P_0 = D_A P_h(0)$ and $E_h(A) = P_h(A) - a_0 - P_0 A$. The divided finite matrix expansion gives

$$\|E_h(A)\|_\infty + \|DE_h(A)\|_\infty \le Chr^2, \qquad \|D^2E_h(A)\|_\infty \le Cr^2,$$

and

$$\|D_A P_h(A) - P_0\| \le Chr, \qquad \|D_A P_h(A) - P_0 - D_A^2 P_h(0)[A]\| \le Ch^2r^2.$$

The second spatial difference contains hAD^2A and $h(DA)^2$; when a coefficient is differentiated in a_0, it supplies an additional h. The elementary bounds $\|D^2A\|_\infty \le Ch^{-1}\|DA\|_\infty$ and $\|D^2a_0\|_\infty \le Ch^{-1}\|Da_0\|_\infty$ therefore remove every apparent mesh loss.

In flat logarithmic coordinates the exact Wilson gradient is $\mathrm{d}^\sharp\mathrm{d}\,a + N_h(a)$, with the uniform local analytic remainder proved in the actual Wilson calculation. The relative gradient is $D_A P_h(A)^\sharp\{\mathrm{d}^\sharp\mathrm{d}P_h(A) + N_h(P_h(A))\}$. Subtract its constant and linear terms before estimating. The term with $\mathrm{d}^\sharp\mathrm{d}E_h$ is bounded by Cr^2. The first derivative difference $O(hr)$ times $\mathrm{d}^\sharp\mathrm{d}(P_0A)$ is bounded by $Chr\,h^{-1}r$. The remaining quadratic derivative coefficient $O(h^2r^2)$ multiplying the rough base $\mathrm{d}^\sharp\mathrm{d}a_0$ is bounded by $Ch^2r^2h^{-1}\|Da_0\|_\infty$. All terms containing N_h obey its uniform Banach derivative bounds. This is the same relative-BCH calculation as in Lem. 3.118; it uses the complex base's first differences, rather than a bound for its second-difference current.

At a bare cell the rescaled mesh is one and the finite incident plaquette calculation gives the same analytic Taylor estimate. Rescaling the gradient back multiplies it by ξ^{-3}. All estimates hold

on a larger complex ball. Their uniform holomorphy and the double zero give the stated Lipschitz bound by Cauchy. □

3.38.5. *An explicit auxiliary action with the second precision.* Write $\mathfrak{q}_U(A)$ for the actual re-centered straightened per-length field constraint of PNC; this notation distinguishes it from the site average q. Put

$$N_q(x) = \mathfrak{q}_U(x) - Qx, \qquad r_U(A) = \mathfrak{q}_U(\Pi_U(A)). \tag{3.mcxlvii}$$

Then $r_U(0) = 0$ and $r'_U(0) = Q$, because $QS = Q$. The latter equality uses the actual extended outward endpoint intertwining.

Define the following scalar function on the fine strong ball:

$$\begin{aligned}\mathcal{L}_{U,J}(A) = F_U(A) - F_U(0) - \langle j_U, A\rangle \\ - \langle J, \Xi_U(A)\rangle - \langle \omega, N_q(\Pi_U(A))\rangle \\ + \tfrac{1}{2}\langle RD^\sharp A, RD^\sharp A\rangle_{\mathrm{site}} + \tfrac{1}{2}\langle r_U(A), a_f r_U(A)\rangle_{\mathrm{rows}}.\end{aligned} \tag{3.mcxlviii}$$

Every pairing is bilinear, including the gauge penalty. Thus this is a holomorphic scalar function at complex inputs. The multiplier $\omega = H^\sharp J$ is fixed when differentiating in A.

Lemma 3.147 (The exact auxiliary Hessian)**.** The action satisfies

$$\mathcal{L}'_{U,J}(0) = 0, \qquad \mathcal{L}''_{U,J}(0) = G_1^{-1}. \tag{3.mcxlix}$$

■

Proof. The first derivative of the nonlinear current term is zero. Its second derivative is $-B_U[J]$, by the finite-chain identity. Since $N_q(0) = DN_q(0) = 0$, the constraint term contributes $-S^\sharp \Delta_U^{(2)}[\omega]S$. The projection identities $R^\sharp = R$, $R^2 = R$ make the penalty contribution $DRD^\sharp$, also on the complex bilinear continuation. Finally $r'_U(0) = Q$ makes the row-mass contribution $Q^\sharp a_f Q$. Together with H_U, these are exactly the terms in the canonical G_1^{-1}. □

This formula selects a nonlinear off-current convention. Equality of its second jet with the canonical precision does not identify it with a separately prescribed nonlinear core map.

Lemma 3.148 (Constraint and composed-adjoint residuals)**.** For $\|A\|_{\mathcal{A}_\alpha} \le r$, the actual constraint obeys

$$\begin{aligned}\|r_U(A) - QA\|_{\mathcal{B}} &\le Cr^2, \\ \mathcal{F}_p\{(r'_U(A) - Q)^\sharp \nu\} &\le C_{p,\alpha} r \|\nu\|_{\mathcal{W}}, \\ \mathcal{F}_p\{r'_U(A)^\sharp a_f r_U(A) - Q^\sharp a_f Q A\} &\le C_{p,\alpha} r^2, \\ \mathcal{F}_p\{\nabla\langle \omega, N_q(\Pi_U(A))\rangle - S^\sharp \Delta_U^{(2)}[\omega] S A\} &\le C_{p,\alpha}\|\omega\|_{\mathcal{W}} r^2.\end{aligned} \tag{3.mcl}$$

These are estimates for finite $4 < p < \infty$. ■

Proof. The actual fixed-leg two-leg estimate and its analytic remainder, in eq. 3.mxxxvii and eq. 3.mxxxviii, give

$$\|N_q(x)\|_{\mathcal{B}} \le C\|x\|^2_{\mathcal{A}_\alpha}, \qquad \|(D\mathfrak{q}_U(x) - Q)^\sharp \nu\|_{\infty,3} \le C\|x\|_{\mathcal{A}_\alpha}\|\nu\|_{\mathcal{W}}.$$

After subtracting its linear term, $DN_q(x)^\sharp \omega$ has a force remainder bounded by $C\|\omega\|_{\mathcal{W}}\|x\|^2_{\mathcal{A}_\alpha}$. These are actual column-density estimates on the common admitted row ball, also when the base is in the smaller complex tube. Physical row transposition cancels the fine density $(\eta/\xi_c)^4$; the recentering and complete outward carriers are retained.

The value estimate follows by $\Pi_U(A) = SA + O(A^2)$ and $QS = Q$. For the adjoints use

$$r'_U(A)^\sharp = \Pi'_U(A)^\sharp D\mathfrak{q}_U(\Pi_U(A))^\sharp.$$

The operator $S^\sharp$ is bounded only in the finite force class needed here. The difference $\Pi'_U(A)^\sharp - S^\sharp$ has bound $C_{p,\alpha} r$ from a bounded amplitude-three force into $\mathcal{F}_p$. This is eq. 3.mcxliv, before subtracting its quadratic jet, together with Lem. 3.145. In particular this argument never differentiates the fine force supplied to that adjoint.

The identity $S^\sharp Q^\sharp = Q^\sharp$ cancels the constant part in the multiplier residual. The remaining two factors give its second line. For the row mass, use $a_f \asymp \Xi_\Gamma^{-2}$, so it converts datum amplitude one into multiplier amplitude three, and insert the first two lines. For the last line put $x = \Pi_U(A)$ in the subtracted DN_q estimate and apply the same adjoint formula. Terms with $\Xi_U(A)$ and $\Pi'_U(A) - S$ are quadratic after their displayed leading term is removed. □

Combining the preceding lemmas, set

$$N_{\mathcal{L}}(A) = \mathcal{L}'_{U,J}(A) - G_1^{-1} A.$$

For fixed small current ceiling they give

$$\mathcal{F}_p(N_{\mathcal{L}}(A)) \le C_{p,\alpha} \|A\|^2_{\mathcal{A}_\alpha}. \tag{3.mcli}$$

All maps in this subtracted expression are uniformly holomorphic on a larger strong ball. Their double zeros consequently give Lipschitz constant $C_{p,\alpha} r$ there.

3.38.6. *The quantitative auxiliary germ.* Let $b_U(B)$ be the explicit actual outward datum in eq. 3.mxxxiii, relative to the present base U. Its finite complete carriers and analytic bounds hold on a smaller common tube. In particular $b_U(0) = 0$ and $\|b_U(B)\|_{\mathcal{B}} \le C_L \|B\|_{\mathcal{B}}$. Choose a finite $p > 4/(1-\alpha)$.

Proposition 3.149 (A selected independent-current critical germ)**.** There are fixed positive radii $\varepsilon, \gamma_*, r, \rho$ such that the following holds on this fixed determining geometry. For U in the smaller displayed $C^{1,\alpha}$ tube, $\|J\|_{\infty,3} < \gamma_*$, and $\|b\|_{\mathcal{B}} < \rho$, the system

$$\mathcal{L}'_{U,J}(A) - r'_U(A)^\sharp \nu = 0, \qquad r_U(A) = b \tag{3.mclii}$$

has a unique solution in a fixed joint strong ball. It is holomorphic in U, J, b, real at real data, and

$$\|A\|_{\mathcal{A}_\alpha} + \|\nu\|_{\mathcal{W}} \le C_{p,\alpha} \|b\|_{\mathcal{B}}. \tag{3.mcliii}$$

The radii and constants are independent of depth, compatible volume and admitted proper-region shape. At $b = 0$, $A = \nu = 0$ for every admitted pair, and the exact first derivatives are

$$D_b A(0) = H_1, \qquad D_b \nu(0) = (Q G_1 Q^\sharp)^{-1}, \qquad D_b(\Pi_U(A))(0) = S H_1. \tag{3.mcliv}$$

For literal data set $b = b_U(B)$; the first and third derivatives then acquire the right factor $D b_U(0)$. ■

Proof. The actual second-variant row saddle has the strong inverse

$$G_1^{-1} A - Q^\sharp \nu = f, \qquad QA = d, \qquad \|A\|_{\mathcal{A}_\alpha} + \|\nu\|_{\mathcal{W}} \le C_{p,\alpha}\{\mathcal{F}_p(f) + \|d\|_{\mathcal{B}}\}.$$

This is the completed canonical inverse, with its row mass retained. It is valid for an independent current and the common complex background tube. Write this inverse as $\mathfrak{I}_{U,J}$. Then eq. 3.mclii is exactly the preconditioned fixed point

$$(A, \nu) = \mathfrak{I}_{U,J}\big(-N_{\mathcal{L}}(A) + (r'_U(A) - Q)^\sharp \nu, \quad b - (r_U(A) - QA)\big). \tag{3.mclv}$$

The displayed residual estimates give size $C_{p,\alpha} r^2$ and Lipschitz constant $C_{p,\alpha} r$ in a radius-r joint ball. Choose r so this constant is less than $1/4$, then ρ so its affine input is less than $r/4$. Iteration is invariant and contractive. Its bound, realness and holomorphy prove the first assertions. All choices are made after fixing α, p, but use only the fixed uniform constants. The current ceiling can be kept inside the already admitted energy ceiling; shrinking the strong and datum radii pays the finite-p nonlinear constants.

At zero data the zero solution is exact. Every nonlinear residual on the right has zero first derivative there. Differentiation therefore gives the full canonical row-saddle response, $A' = G_1Q^\sharp(QG_1Q^\sharp)^{-1}b'$ and $\nu' = (QG_1Q^\sharp)^{-1}b'$. Applying $\Pi_U'(0) = S$ proves the third identity. Composition with the actual outward datum proves its last assertion. The argument uses the preconditioned residual map; no uniform boundedness of an un-subtracted second-order gradient on a $C^{1,\alpha}$ ball is presumed. □

Denote $C(B) = \Pi_U(A(b_U(B)))$. It satisfies $\mathfrak{q}_U(C(B)) = b_U(B)$ and lies on the same fine gauge orbit as the auxiliary field. Off current it is not asserted to satisfy the fixed linear Landau condition. In particular the auxiliary identity $D_bA(0) = H_1$ is not an identity for this completed field. A specified literal node extension adds its further fine gauge vector to the first jet.

3.38.7. *Exact matching at physical pairs.*

Proposition 3.150 (The auxiliary germ equals the physical germ)**.** Suppose the base is an admitted real physical pair,

$$J = j_U = Q^\sharp\omega_{\mathrm{phys}}. \tag{3.mclvi}$$

Then $\omega = H^\sharp J = \omega_{\mathrm{phys}}$. For reduced datum $b = b_U(B)$, let $(A_{\mathrm{phys}}, \nu_{\mathrm{phys}})$ be the physical PNC germ in the Landau slice fixed at the present base U, namely $R_UD_U^\sharp A = 0$. The reference U_* fixes comparison frames, norms and the node extension; this slice and Λ_U vary with the base U. On a common smaller datum ball,

$$A = A_{\mathrm{phys}}, \qquad C = A_{\mathrm{phys}}, \qquad \nu = \nu_{\mathrm{phys}} + a_fb. \tag{3.mclvii}$$

The same identity holds on the already proved holomorphic physical critical graph wherever the displayed strong and current admission holds. ■

Proof. The row identity $QH = 1$ identifies the first-lift readout with the physical multiplier. Fine gauge invariance gives $F_U(A) = F_U(\Pi_U(A))$ for the exact finite chain. Its first Ward identity gives $\langle j_U, D\Lambda A\rangle = 0$. Substitute $J = j_U = Q^\sharp\omega$ in eq. 3.mcxlviii. The three linear and current terms combine exactly as

$$-\langle j_U, A\rangle - \langle j_U, \Pi_U(A) - SA\rangle - \langle \omega, \mathfrak{q}_U(\Pi_U(A)) - Q\Pi_U(A)\rangle = -\langle \omega, r_U(A)\rangle.$$

Thus the scalar action is

$$\begin{aligned}\mathcal{L}_{U,j_U}(A) = {} & F_U(\Pi_U(A)) - F_U(0) - \langle \omega, r_U(A)\rangle \\ & + \tfrac{1}{2}\langle RD^\sharp A, RD^\sharp A\rangle + \tfrac{1}{2}\langle r_U(A), a_fr_U(A)\rangle.\end{aligned} \tag{3.mclviii}$$

This is an equality of scalar functions on the admitted ball, not merely a second-jet comparison.

The physical solution has $RD^\sharp A_{\mathrm{phys}} = 0$, hence $\Pi_U(A_{\mathrm{phys}}) = A_{\mathrm{phys}}$. Its complete stationarity equation is

$$F_U'(A_{\mathrm{phys}}) = D\mathfrak{q}_U(A_{\mathrm{phys}})^\sharp(\omega + \nu_{\mathrm{phys}}).$$

Differentiate eq. 3.mclviii at this field. The pullback of the first two nonconstant terms is $r_U'(A_{\mathrm{phys}})^\sharp\nu_{\mathrm{phys}}$; the gauge penalty is zero and the row mass adds $r_U'(A_{\mathrm{phys}})^\sharp a_fb$. Therefore this field, with $\nu = \nu_{\mathrm{phys}} + a_fb$, solves eq. 3.mclii. The fixed-point uniqueness proves eq. 3.mclvii.

For continuation along an already admitted holomorphic physical graph, consider the fixed-slice equation, constraint and full physical stationarity just proved as fine-coordinate functions of its graph parameters and B. They are holomorphic and vanish on the real parameter neighbourhood. The identity theorem therefore makes them vanish on the common complex neighbourhood. This proves the stated physical continuation without presuming a new reference or stationarity for an arbitrary off-current input. □

3.38.8. *A bounded updated current and a literal field.* Before the final literal gauge correction put

$$\begin{aligned} U_{\mathrm{new}} &= e^{i\eta C(B)}U, \qquad \omega_{\mathrm{new}} = \omega + \nu - a_f b_U(B), \\ J_{\mathrm{new}} &= \mathcal{E}^{-\sharp}_{\eta C(B)} \left\{ D\mathfrak{q}_U(C(B))^{\sharp}\omega_{\mathrm{new}} + (J - Q^{\sharp}\omega) \right\}. \end{aligned} \tag{3.mclix}$$

Here $\mathcal{E}_z v = \int_0^1 \mathrm{Ad}(e^{itz})v\,dt$. The last expression is an explicit independent-current selection at off-current inputs. Its coefficient ω_{new} is in the displayed original re-centered row coordinates; it is not declared to be a new first-lift readout at the output pair.

Proposition 3.151 (The updated independent current)**.** The update is holomorphic and satisfies

$$J_{\mathrm{new}}(0) = J, \qquad \|J_{\mathrm{new}} - J\|_{\infty,3} \le C_{p,\alpha}\|B\|_{\mathcal{B}}, \qquad \|J_{\mathrm{new}}\|_{\infty,3} \le \|J\|_{\infty,3} + C_{p,\alpha}\|B\|_{\mathcal{B}}. \tag{3.mclx}$$

Its field and curvature increments have the corresponding strong and weighted bounds. At a physical input it is exactly the left Wilson current of U_{new}. ■

Proof. The actual transpose column bounds send row amplitude three into fine force amplitude three. The difference $(D\mathfrak{q}_U(C) - Q)^{\sharp}\omega$ has norm at most $C\|\omega\|_{\mathcal{W}}\|C\|_{\mathcal{A}_\alpha}$. Also $\|\nu - a_f b_U(B)\|_{\mathcal{W}} \le C\|B\|_{\mathcal{B}}$ and $\mathcal{E}^{-\sharp}_{\eta C} - 1$ has point norm at most $C\eta|C|$. Subtract the value at $C = 0, \nu = 0, B = 0$ in eq. 3.mclix. These estimates and eq. 3.mcliii prove eq. 3.mclx. The curvature increment follows from the four exact link factors and the first difference of C. No actual Wilson derivative of U_{new} has been used in this off-current estimate.

At a physical input eq. 3.mclvii gives $\omega_{\mathrm{new}} = \omega + \nu_{\mathrm{phys}}$, and the residual $J - Q^{\sharp}\omega$ vanishes. The physical normal and left gradients satisfy

$$F'_U(C) = \mathcal{E}^{\sharp}_{\eta C} j(U_{\mathrm{new}}).$$

Insert physical stationarity from the preceding proof. Equation eq. 3.mclix is then exactly $J_{\mathrm{new}} = j(U_{\mathrm{new}})$. □

Finally choose the fixed real-reference node extension E_n of Lem. 3.119. Apply it to the small logarithm of the assigned factor $v_U[C(B)]^{-1}$, and exponentiate to obtain the admitted fine gauge $g(B)$. Its node values are exactly that factor; its bare values are one. The literal output pair is

$$\widehat{U}(B) = U_{\mathrm{new}}^{g(B)}, \qquad \widehat{J}(B)_b = \mathrm{Ad}(g(B)_{b_-})J_{\mathrm{new}}(B)_b. \tag{3.mclxi}$$

The actual outward diagram gives $M_{\mathbf{B}}\widehat{U}(B) = V_B$. The finite gauge estimate in eq. 3.mlvi preserves the strong field bound, and its bounded point adjoint preserves the current bounds. At physical inputs this is the physical critical orbit and its actual current in the specified literal gauge.

All choices here belong to the fixed reference chart. A different admissible node extension differs by a fine gauge one at assigned tops, but no global off-current chart-descent or externally specified core convention has been identified by this construction.

3.38.9. *Fixed-order spatial response.* The global strong estimates above can retain a spatial weight. Let $w_y = e^{s\psi(y)}$, where ψ is one-Lipschitz in the proper layer metric and $s > 0$ is fixed sufficiently small. Weighted strong and source norms multiply each local norm by w_y. Hölder seminorms are still taken on the same complete local boxes; the artificial weight is not differentiated.

Proposition 3.152 (Localized jets of the completed germ)**.** The first datum response satisfies

$$\|D_B A[h]\|_{\mathcal{A}_\alpha;w} + \|D_B\nu[h]\|_{\mathcal{W};w} \le C_{p,\alpha}\|h\|_{\mathcal{B};w}. \tag{3.mclxii}$$

The same conclusion holds for the completed field and the literal output pair in their displayed strong and current norms. For each fixed m, the normalized response at an output cell y to point data at $c_1, \dots, c_m$ is bounded by

$$C_m e^{-s\max_i d(y,c_i)}. \tag{3.mclxiii}$$

These constants are uniform in depth and volume. No all-arity absolute connected-kernel bound is asserted. ■

Proof. The actual completed energy inverses, normalized row kernels and readouts have exponential cell bounds. Their local scalar and finite-p upgrades retain the cell weight, since the ratios w_z/w_y are bounded on each fixed complete neighbouring box. This proves the weighted canonical strong row-saddle inverse. For $\Lambda : \mathcal{A}_\alpha \to \mathcal{Y}_{2,\alpha}$, use its weighted zero-average energy estimate and the completed source equation $\Delta_U \lambda = RD^\sharp A$. The projection R preserves the weighted C^α source norm by its completed exponential column sums. The local Schauder estimate then gives the weighted site bound. The same argument gives the weighted first adjoint bound in eq. 3.mcxli; the finite-p force-restoration proof gives the weighted second bound.

In the derivative of every local residual keep one perturbation slot weighted and all background slots in the unweighted small ball. All point products, complete row moments and the adjoint factory have the same bounds with that one weight. The derivative of the preconditioned fixed-point map therefore has weighted norm at most $C_{p,\alpha} r < 1/2$. Its derivative equation gives eq. 3.mclxii. The explicit outward datum, finite gauge chain, node extension and current update have the same one-slot weighted bounds.

These estimates hold on a fixed larger complex datum ball. Cauchy in its other slots gives the corresponding mixed derivative with one weighted slot and $m-1$ unweighted slots. Choose ψ to be distance from the carrier of any selected point datum. Symmetry of the m derivative slots allows the farthest such datum to be selected. This proves eq. 3.mclxiii. The use of the completed source equation in the Schauder step is essential; no individual walk term was assigned a $C^{2,\alpha}$ site bound. □

This construction supplies a strong analytic auxiliary germ, a literal mean-preserving field/current update, and exact physical matching on the admitted critical graph. It keeps the H_1, SH_1 and outward datum first jets distinct. It does not construct a reference from arbitrary curvature-only coarse data, identify an externally prescribed off-current core convention, or establish full proper history domains and deletion laws. Independent enlarged cube inputs and a complete nonlinear connected-carrier norm remain separate requirements.

3.39. **Local biharmonic formulas for strong gauge restoration.** The completed restoration $G_s R$ gains two Hölder derivatives on a Hölder source. A decomposition through second-order site inverses need not have that property term by term: an individual averaging-mass source can jump at a pillar face. We obtain a different local decomposition from a fourth-order positive operator. Its auxiliary Laplacian absorbs that jump before the final Schauder estimate.

The construction uses the actual site graph, own-pillar averages and physical measures. It neither clips an average at an artificial boundary nor replaces the determining region by a periodic model. The final carrier scale is chosen separately from a prescribed history scale.

3.39.1. *The operator and its physical normalization.* Retain the geometry, comparison length ℓ, assigned site scales ξ, and actual covariant site Laplacian $\Delta = D_U^\sharp D_U$ of § 3.19. Let q take values in raw top coordinates equipped, in this section, with counting measure. Its adjoint for that measure is

$$q^* = q^\sharp \Xi^{-4}. \tag{3.mclxiv}$$

The sharp on the right is the physical top transpose. Consequently the local fourth-order operator is

$$K_4 = \Delta^2 + q^* q = \Delta^2 + q^\sharp \Xi^{-4} q, \qquad G_4 = K_4^{-1}. \tag{3.mclxv}$$

At real unitary coefficients its form is

$$a_4(u,v) = \langle \Delta u, \Delta v \rangle_{\mathrm{fine}} + \langle qu, qv \rangle_{\mathrm{count}}.$$

All analytic continuations use the bilinear form. In particular q^* in an analytic formula is not a conjugate transpose of a complex coefficient.

Use the graph norm

$$\|u\|_{V_2}^2 = \|\Delta u\|_2^2 + \|\ell^{-1} D_U u\|_2^2 + \|\ell^{-2} u\|_2^2. \tag{3.mclxvi}$$

It controls the full covariant second differences on complete interior stencils. At the true boundary we use the original graph and the bare atomic norm; the latter pays any omitted-neighbour stencil. Local norms $V_{2,y}$ use a fixed complete enlargement of the assigned cell.

Lemma 3.153 (A site lift with two energy derivatives)**.** For real curvature-small admitted data there is a local right inverse E of the actual q such that

$$qE = 1, \qquad \|Eb\|_{V_2} \le C\|b\|_{\ell^2(\mathrm{count})}. \tag{3.mclxvii}$$

The support and collars of its seeds have bounded overlap. No physical-current bound is needed for this energy estimate. ■

Proof. Use the normalized compact bump seed from Lem. 3.58 in a local real Coulomb frame on a complete nonbare cube of side $C_L\xi_y$. Its normalization matrix is the actual transported bump average and is uniformly invertible. The supremum of the connection is $C_L\kappa/\xi_y$. The finite-p Coulomb estimate, used at $p = 2$, gives $\|Da\|_2 \le C_L\kappa$ on the rescaled cube. These facts follow from curvature alone by Cor. 3.83.

For a seed of value b_y, the ordinary second differences of the scalar bump have L^2 norm $C|b_y|$. Expanding the covariant second differences adds $aD\chi$, $(Da)\chi$ and $a^2\chi$, with the same bound. No pointwise bound for Da is used. The first and zeroth weighted norms in eq. 3.mclxvi have that bound by the existing site-lift estimates. Its Laplacian is a sum of these second differences. The physical support volume is $O(\xi_y^4)$, so all four powers of length cancel against the two derivatives. This is precisely a counting-top bound, rather than a physical-top bound.

For the finitely coarse meshes, including a bare pillar, use the original transported atomic seed. Here η/ξ_y has a fixed positive lower bound; each second difference is bounded by $C\eta^{-2}$ times its value and the physical volume is $O(\xi_y^4)$. Thus the same bound holds. For fine meshes the compact bump and its two-step collar lie inside its pillar. The bounded overlap and adjacent scale comparability sum these estimates.

If a frame cube does not embed, perform this entire seed construction on a sufficiently large periodic cover and average over simultaneous deck shifts. The exact equation $qE = 1$, the quadratic energy bound and its support description descend. This is an averaging of linear maps, not an assertion about partition functions on different volumes. □

Proposition 3.154 (The full fourth-order gap)**.** For the same real data,

$$c\|u\|_{V_2}^2 \le a_4(u,u) \le C\|u\|_{V_2}^2. \tag{3.mclxviii}$$

The constants are uniform in depth, compatible volume and admitted proper-region shape. The corresponding zero-extension Dirichlet compressions have the same gap in the energy of their zero extension. ■

Proof. Write $u = u_0 + Equ$, where $qu_0 = 0$. The exact zero-average estimate eq. 3.dcclxxxiv gives

$$\|u_0\|_{V_2} \le C\|\Delta u_0\|_2 \le C\{\|\Delta u\|_2 + \|qu\|_{\mathrm{count}}\}.$$

Use Lem. 3.153 for the other term. This proves the lower bound. Jensen's inequality for the transported average gives

$$\sum_y |qu(y)|^2 \le C \sum_y \xi_y^{-4} \|u\|_{2,B_y}^2 \le C\|\ell^{-2}u\|_2^2,$$

which proves the upper bound. For the component second derivatives apply the zero-average Hodge estimate eq. 3.dcclxxxviii to u_0, and the seed bound to Equ. The bare mass controls its two-step outer collar.

If E_D injects site fields supported in a fine set D, the compression is $E_D^\sharp K_4 E_D$. Its form is $\|\Delta E_D u\|_2^2 + \|qE_D u\|_{\mathrm{count}}^2$. It retains the full Laplacian and every full crossing row evaluated on the zero extension. Restricting the already proved inequality proves the assertion; there is no partial-row inf-sup assumption. □

For the strong estimates fix $0 < \alpha < 1$ and a class $\mathscr{U}_{\alpha,C_0}$ of these real connections whose local Coulomb potentials on the complete boxes have scaled $C^{1,\alpha}$ norm at most C_0. The current-regular references used for the nonlinear germ of § 3.38 belong to this class. Write $\mathcal{A}_\alpha$ for the fine strong space of eq. 3.dccclxxxvii. We work on a fixed smaller weighted $C^{1,\alpha}$ tube about a chosen member. All constants below may depend on α, C_0, L. Curvature alone is the premise of eq. 3.mclxviii; it is not being asserted to give these stronger coefficient bounds. On the complex tube, the Banach norms are measured in the fixed real-reference comparison frames. Covariant norms at the varying base are uniformly equivalent on that tube.

3.39.2. *A source class suited to a local fourth-order inverse.* The source must allow two derivatives of a Hölder function without requiring their pointwise bound. Let $\mathcal{S}_\alpha$ consist of finite site arrays with a representation

$$\rho = \Delta f + D_U^\sharp g + h, \tag{3.mclxix}$$

where

$$\|f\|_{\alpha,2} = \sup_y \{\xi_y^2 \|f\|_{\infty,B_y^+} + \xi_y^{2+\alpha}[f]_{\alpha,B_y^+}^U\}, \quad \|g\|_{\infty,3} = \sup_b \mathsf{L}(b)^3 |g(b)|, \quad \|h\|_{\infty,4} = \sup_x \xi_x^4 |h(x)|.$$

The Hölder comparisons use complete boxes across pillar faces. The norm $\|\rho\|_{\mathcal{S}_\alpha}$ is the infimum of the sum of these three norms over all representations. Local source spaces $\mathcal{S}_{\alpha,z}$ use representations supported on a fixed complete enlargement of B_z. The output space is $Y^{2,\alpha}$ of eq. 3.mxxiv. The source norms at nearby bases are uniformly equivalent to the fixed-reference norm. Indeed $(\Delta_U - \Delta_*)f$ can be put in $D_*^\sharp g_1 + h_1$ using bounded values of f and the connection difference and its first difference. Likewise $(D_U^\sharp - D_*^\sharp)g$ is a grade-four point source. Thus this comparison does not differentiate the Hölder function f or the bounded field g.

Smooth localization is bounded on this source class. Indeed $\chi\Delta f = \Delta(\chi f) - [\Delta, \chi]f$. The last commutator can be put in the form $D^\sharp g_\chi + h_\chi$, with coefficients $D\chi$ and $D^2\chi$ multiplying shifted values of f. It differentiates no f. The same product rule applies to $\chi D^\sharp g$. Consequently a fixed smooth cell partition localizes $\mathcal{S}_\alpha$ with bounded overlap and bounded norm. We use this localization for block products; sharp characteristic functions are not multipliers on the Hölder part of the source space. Use its bounded reconstruction and restriction maps in the cell-block norm. Multiplying that norm by their fixed bound makes the product norm submultiplicative; this constant is absorbed in the local commutator constant below.

Every local source is a bounded functional on V_2. For example, for a unit source at scale ξ_z,

$$\|f\|_2 \le C, \qquad \|g\|_2 \le C\xi_z^{-1}, \qquad \|h\|_2 \le C\xi_z^{-2}.$$

Pair these with $\Delta u, D_U u, u$, respectively. Thus its dual energy norm is uniformly bounded.

Lemma 3.155 (Strong local fourth-order columns)**.** Let $I_D = E_D(E_D^\sharp K_4 E_D)^{-1} E_D^\sharp$. On a region strictly inside its artificial boundary,

$$\|I_D \rho_z\|_{Y^{2,\alpha}(B_y^+)} \le C_\alpha e^{-s_0 d(y,z)} \|\rho_z\|_{\mathcal{S}_{\alpha,z}}. \tag{3.mclxx}$$

The interior collar contains all stencils and all complete averaging carriers used in the estimate. The same statement holds at a true boundary with the bare atomic convention. It holds uniformly on the stated smaller complex $C^{1,\alpha}$ tube. ■

Proof. First obtain a weighted energy estimate. A raw metric distance is only Lipschitz and is insufficient for this fourth-order step. Smooth the truncated distance from B_z at the local ℓ scale using the weighted smooth partition of Lem. 3.99, with a fixed radius multiple in place of m. If its squared normalized weights are $p_x(y)$, use $\psi(y) = \sum_x p_x(y) d(x, B_z)$. On an overlapping support, centre distances differ by at most C_L. Subtract one such constant before differentiating $\sum_x p_x = 1$. The cutoff derivative estimates, extended through order three by the same bump calculation, give

$$|\psi - d(\cdot, B_z)| \le C_L, \quad |D\psi| \le C_L \ell^{-1}, \quad |D^2\psi| \le C_L \ell^{-2}.$$

For $w = e^{s\psi}$, the exact covariant product rule therefore gives

$$\|w^{-1}\Delta(wu) - \Delta u\|_2 \le C_L|s|\{\|\ell^{-1} D_U u\|_2 + \|\ell^{-2} u\|_2\}.$$

The oscillation of $\log w$ on a full averaging carrier is $O_L(s)$. The conjugated bilinear form $a_4(w^{-1}u, wv)$ differs from $a_4(u, v)$ by at most $C_L|s|\|u\|_{V_2}\|v\|_{V_2}$. Choose s below the gap constant. Multiplication by w preserves the zero-extension space, so the weighted estimate holds for every compression. It gives

$$\|I_D \rho_z\|_{V_{2,y}} \le C e^{-s_1 d(y,z)} \|\rho_z\|_{\mathcal{S}_{\alpha,z}}. \tag{3.mclxxi}$$

There is no endpoint scale ratio: both the source dual norm and the site H^2 norm are dimensionless in four dimensions.

Now put $u = I_D \rho_z$ and $v = \Delta u - f$. On a complete interior box its exact equation is

$$\Delta v = D_U^\sharp g + h - q^* q u, \qquad \Delta u = f + v. \tag{3.mclxxii}$$

The local L^2 norm of v is controlled by eq. 3.mclxxi and the local norm of f. The actual average satisfies $|qu(y)| \le C\xi_y^{-2}\|u\|_{2,B_y}$. The physical density of q^* then bounds q^*qu by $C\xi_y^{-4}$ times the nearby energy. Complete carriers change only fixed local constants.

Rescale this box to length one. The finite-p divergence-source bootstrap used in Prop. 3.89 bounds v in $W^{1,p}$ for every finite p, using its local L^2 norm, g in L^∞, and the bounded scalar forcing. First testing its equation against a squared-cutoff multiple of v supplies the local H^1 bound from those same quantities. On the complex tube use the real-reference comparison in this energy test. Lower-order covariant coefficients have their admitted bounds. Choose $p > 4/(1-\alpha)$. Morrey gives a C^α bound for v; no derivative of g is taken. Thus $f + v$ is a C^α source for the second equation. The scalar finite-p bootstrap first gives the lower $C^{1,\alpha}$ norm of u, and the interior Schauder estimate eq. 3.mxxii gives its full $C^{2,\alpha}$ norm. Its coefficient premise is exactly the displayed $C^{1,\alpha}$ background class. Applying this on fixed nested boxes preserves the exponential factor in eq. 3.mclxxi.

At the true boundary $\xi = \eta$. Each required cell has a fixed number of sites; its rescaled values, differences and Hölder seminorm are bounded directly by its local H^2 energy. No boundary Schauder theorem for a rough region is being inferred. Artificial boundaries are outside all boxes on which the strong assertion is used.

Finally, a small complex C^1 perturbation changes $\Delta : V_2 \to L^2$ and $q : V_2 \to \ell^2(\text{count})$ by a uniformly small operator. The weighted forms and every compression remain invertible by Neumann series. The stronger $C^{1,\alpha}$ tube bounds the coefficients of the two scalar equations above in their required Hölder norms. The same estimates then prove eq. 3.mclxx there. This uses bilinear perturbation of real positivity, rather than complex Hermitian positivity. □

The gain in this lemma explains the fourth-order choice. The discontinuous density q^*qu is a source for v, so v is Hölder. It is not a discontinuous source for the final Schauder equation $\Delta u = f + v$.

3.39.3. *A strong local walk for G_4.* Use the actual weighted cover eq. 3.cmxlix, with $r_x = m\ell(x)$ and $m\delta$ sufficiently small. Its smooth bump proof gives, for every fixed order, $|D^a \chi_x| \le C_a r_x^{-a}$. Take a compression patch D_x of radius $4r_x$ and retain all full crossing stencils, rows and geometry-decision

collars. The support of χ_x and its fixed complete cell enlargements lie strictly inside that artificial boundary. Put $I_x = I_{D_x}$.

Lemma 3.156 (The fourth-order cutoff error)**.** The error columns obey

$$\|[K_4, \chi_x] I_x \chi_x\|_{\mathcal{S}_{\alpha,z} \to \mathcal{S}_{\alpha,y}} \le \frac{C_\alpha}{m} e^{-s_2 d(y,z)}. \tag{3.mclxxiii}$$

Their representations and coefficients read only the declared local compression data and collars. ■

Proof. For a local strong output u, use the exact identity

$$[\Delta^2, \chi]u = \Delta[\Delta, \chi]u + [\Delta, \chi]\Delta u.$$

The first term has the source form Δf_χ, where $f_\chi = [\Delta, \chi]u$. The covariant product rule and $u \in Y^{2,\alpha}$ show

$$\|f_\chi\|_{\alpha,2} \le C_\alpha m^{-1} \|u\|_{Y^{2,\alpha}}.$$

Only one cutoff derivative is needed for the leading term; higher cutoff derivatives are smaller. For the second term apply summation by parts to the commutator itself, rather than differentiating Δu. For any site array w, the exact two-shift product rule writes

$$[\Delta, \chi]w = D_U^\sharp g_\chi(w) + h_\chi(w),$$

where the coefficients are shifted values of $D\chi$ and $D^2\chi$ times transported values of w. Explicitly, for the orientation $D_U w(b) = \eta^{-1}(w_- - T_b w_+)$, take

$$g_\chi(w)(b) = (D\chi)(b)(w_- + T_b w_+), \qquad h_\chi(w)(x) = -(\Delta_{\mathrm{scalar}}\chi)(x) w(x).$$

The scalar Laplacian uses exactly the edges of the original graph. With $w = \Delta u$, its weighted L^∞ size has grade two. Therefore

$$\|g_\chi(\Delta u)\|_{\infty,3} + \|h_\chi(\Delta u)\|_{\infty,4} \le C_\alpha m^{-1} \|u\|_{Y^{2,\alpha}}.$$

This step contains no third difference of u. The summation-by-parts formula holds on the original graph, so missing outer edges do not create an unrecorded boundary term.

For $[q^* q, \chi]u$, insert the cutoff value at the row top. Each difference on the complete row carrier is at most C_L/m. The actual physical row and column density bounds give a point source of grade four with norm at most $C_L m^{-1} \|u\|_{Y^{2,\alpha}}$. Bare identity rows commute exactly. At a rough true boundary all remaining fourth-order entries have size $O(\eta^{-4})$; the atomic output and a cutoff difference give the same source bound directly.

Apply Lem. 3.155 on the cutoff support and its collars. Input localization is bounded on $\mathcal{S}_\alpha$, and all local weights are comparable. This proves eq. 3.mclxxiii with a smaller decay rate. Every formula uses the original compressed matrix and its declared finite carriers. □

Define

$$\mathcal{M}_4 = \sum_x \omega_x \chi_x I_x \chi_x, \qquad T_4 = \sum_x \omega_x [K_4, \chi_x] I_x \chi_x. \tag{3.mclxxiv}$$

Compression and the strict support collar give $\chi_x K_4 I_x \chi_x = \chi_x^2$ as exact finite operators. Thus

$$K_4 \mathcal{M}_4 = 1 + T_4. \tag{3.mclxxv}$$

In smooth cell blocks, use the maximum of the row and column sums with weights $e^{sd(y,z)}$. The weighted overlap eq. 3.cmli and the local column estimates give

$$\|\mathcal{M}_4\|_{\mathcal{S}_\alpha \to Y^{2,\alpha};\mathrm{Sch},s} \le C_\alpha, \qquad \|T_4\|_{\mathcal{S}_\alpha \to \mathcal{S}_\alpha;\mathrm{Sch},s} \le C_\alpha/m.$$

For fixed output or input cell the sum of contributing centre weights is bounded. The remaining cell sum is exponential. A whole-patch energy norm, which would introduce a power of m, has not replaced these column estimates.

File each local matrix on its full carrier as in the positive-walk construction. Include its patch, full crossing-row collar, local geometry-decision collar, canonical input/output tags and fixed short joins.

For sufficiently large carrier cube integer M_0, each local factor uses at most B_L tags, independently of m. With $\sigma_0(X) = b_L + d_{c,0}(X)$, include the short join in $X \vee Y$, so $\sigma_0(X \vee Y) \le \sigma_0(X) + \sigma_0(Y)$. For any prescribed finite rate κ_1, choose parameters in the order

$$\alpha, C_0, \kappa_1; \quad m; \quad \delta; \quad M_0; \quad \rho, \qquad (C_\alpha/m)e^{\kappa_1 B_L} < 1/2.$$

Here $m\delta$ is small, $M_0 \ge C_L/\delta$, and ρ pays the full collars and metric growth reserve. Only after these choices shrink the common complex tube. There is no dependence on depth or volume.

Proposition 3.157 (Strong fourth-order inverse carriers)**.** The exact series

$$G_4 = \mathcal{M}_4 \sum_{n\ge 0} (-T_4)^n \tag{3.mclxxvi}$$

is absolutely convergent in the strong carrier Schur norm from $\mathcal{S}_\alpha$ to $Y^{2,\alpha}$, at the chosen carrier rate. Every coefficient is holomorphic on its complete local background $C^{1,\alpha}$ tube and depends only on its declared local determining data. ■

Proof. The source-to-source family norm of T_4 is less than $1/2$. Joined carrier products are submultiplicative, so the series converges with the asserted output norm. Multiplication by K_4 and eq. 3.mclxxv identify its sum with the positive energy inverse. Every local inverse is a finite Dirichlet matrix with its full coefficients. Its uniform complex invertibility and the strong column estimate give holomorphy into the displayed local Banach spaces. Every output factor includes its smooth cutoff, so zero extension across its artificial boundary preserves $Y^{2,\alpha}$.

The coefficient difference under a small $C^{1,\alpha}$ background change is bounded in the same family norm. Choose the final tube inside a slightly larger common complex tube on which the weighted strong local columns and cutoff commutators have the bounds just proved. Their finite matrices are holomorphic, and the source-space equivalence identifies their input spaces with fixed Banach spaces. Banach-valued Cauchy on the larger tube therefore bounds the change of each cutoff local column and error column by $C\varepsilon$, with the same exponential weight and full local carrier. The weighted centre sum and the convergent walk preserve this bound in the absolute family norm. This argument uses strong regularity only on the output and commutator collars. It does not assign a global $Y^{2,\alpha}$ norm to an uncut Dirichlet solution across its artificial boundary. □

3.39.4. *A positive coarse inverse and the exact restoration.* Define on counting top space

$$M_4 = qG_4q^* = qG_4q^\sharp \Xi^{-4}. \tag{3.mclxxvii}$$

This is already in counting normalization. It is not obtained by treating a physical transpose as an ordinary transpose.

Lemma 3.158 (The fourth-order coarse gap)**.** At real coefficients,

$$cI \le M_4 \le I. \tag{3.mclxxviii}$$

It has an actual local carrier family in counting coordinates at any required fixed intermediate rate. On a smaller common complex tube its inverse has the local carrier family supplied by Prop. 3.103. ■

Proof. Set $u = G_4q^*b$. Then $a_4(u,u) = \langle b, M_4 b\rangle$. The exact lift E gives

$$\|b\|^2 = \langle q^*b, Eb\rangle = a_4(u, Eb) \le C a_4(u,u)^{1/2}\|b\|.$$

This proves the lower bound. For the upper bound, $K_4 \ge q^*q$; equivalently $\|qK_4^{-1/2}\| \le 1$, giving $M_4 \le I$.

A unit counting top source q^*e_z has actual point density at most $C\xi_z^{-4}$ on its complete pillar, so it is a unit grade-four h source. The strong fourth-order family sends it to a dimensionless site field; the actual output average q is bounded on that field. Its input/output columns therefore have

no missing physical length factors. Composition gives the actual family of M_4, with exponentially summable cell blocks and its declared full carriers. The number of top species is fixed.

Apply the positive coarse inversion algorithm of Prop. 3.103 to this matrix. Request the primitive family at a stronger intermediate rate first. Full-carrier truncation, combined short powers and reblocking then give the inverse at the requested final rate, with its finite parameter and separation choices. The family perturbation bound proved above supplies its common complex tube. No indefinite local compression is used. □

For a site source f, put

$$\mathcal{T}_4 f = G_4 \Delta f - G_4 q^* M_4^{-1} q G_4 \Delta f. \tag{3.mclxxix}$$

The notation $\mathcal{T}_4$ is distinct from the parametrix error T_4.

Proposition 3.159 (Exact constrained least squares)**.** For every site source on the finite graph,

$$\mathcal{T}_4 f = G_s R f. \tag{3.mclxxx}$$

Consequently

$$\Lambda = \{G_4 - G_4 q^* M_4^{-1} q G_4\} \Delta D_U^{\sharp}. \tag{3.mclxxxi}$$

These equalities hold also on the admitted bilinear complex continuation. ■

Proof. At real data the unique minimum of $\|\Delta\lambda - f\|_2^2$ subject to $q\lambda = 0$ exists by the zero-average gap. Adding $\|q\lambda\|^2$ to the normal operator does not change its constrained equation. Its saddle system is

$$K_4\lambda + q^*\mu = \Delta f, \qquad q\lambda = 0.$$

Its Schur elimination is exactly eq. 3.mclxxix. The residual $\Delta\lambda - f$ is orthogonal to $\Delta \ker q$. The actual identities $qG_sR = 0$ and $\Delta G_s R = R$ show $\operatorname{ran} R \subset \Delta \ker q$. The zero-average gap makes Δ injective on $\ker q$, and both spaces have dimension equal to the site dimension minus the top dimension. Hence $\Delta \ker q = \operatorname{ran} R$. Since $\Delta\lambda$ belongs to that range, the normal equation gives $\Delta\lambda = Rf$. Its zero-average solution is $G_s R f$.

All matrices are holomorphic on the common complex tube and their admitted inverses remain defined. Alternatively replace real orthogonality by the identities of the bilinear projection R. Either argument extends the exact formula. Finally set $f = D_U^{\sharp} A$. □

Proposition 3.160 (Local strong restoration coefficients)**.** Fix α, C_0 and a finite requested carrier rate κ. After the fixed parameter, separation and tube choices above, $G_s R$ has an exact local family from $\mathcal{C}_{\alpha,2}$ to $Y^{2,\alpha}$, and Λ has an exact local family from $\mathcal{A}_\alpha$ to $Y^{2,\alpha}$. In smooth complete cell blocks these families satisfy

$$\max\left\{ \sup_y \sum_{z,X} e^{sd(y,z)+\kappa\sigma_c(X)} \|\mathcal{T}_{4,yz}[X]\|_{\alpha,2\to Y^{2,\alpha}}, \sup_z \sum_{y,X} e^{sd(y,z)+\kappa\sigma_c(X)} \|\mathcal{T}_{4,yz}[X]\|_{\alpha,2\to Y^{2,\alpha}} \right\} \le C_{\alpha,\kappa}. \tag{3.mclxxxii}$$

The analogous bound holds for Λ with the strong field norm in its input. Each coefficient, including its extended output, has the stated $C^{2,\alpha}$ site norm. Its holomorphic dependence reads only its full local background and determining-data carrier. The constants are uniform in depth and volume. ■

Proof. The injection $f \mapsto \Delta f$ is bounded from $\mathcal{C}_{\alpha,2}$ to $\mathcal{S}_\alpha$ by definition. Compose the strong fourth-order family, the local q, q^* maps and the counting inverse M_4^{-1} in eq. 3.mclxxix. Keep full endpoint tags and short joins in each product. Their carrier norm is submultiplicative and the stronger intermediate rate pays the one coarse reblocking. The final carrier cube integer M_c and its separation are chosen after these finitely many operations. Smooth cell localization preserves the displayed source and output spaces. This proves the first family bound.

The first divergence maps the strong field space to the complete-box source space $\mathcal{C}_{\alpha,2}$. Its finite stencils give a bounded local input family. Applying eq. 3.mclxxxi proves the second assertion,

without taking a second difference of A. The exact identities identify the sums with the original restoration. A coefficient of this new family need not obey $q\lambda = 0$ by itself. Its strong bound comes from the fourth-order local equation, not cancellation of its averaging-mass source. □

All primitive patches, frame cubes and the aligned coarse inversion grids are first constructed on one sufficiently large periodic cover. Perform the entire matrix construction there before any carrier projection. The exact operators commute with deck shifts; average their families over those shifts and descend. The $C^{2,\alpha}$ local norms are convex, and lifted endpoint sums cover the original Schur sums, so no sheet factor is introduced. Retain every lifted carrier before projection, including winding paths and the full data collar. Bilinear matrix identities and inverses restrict to the deck-invariant sector by uniqueness. This proves the small-period case for both the primitive and coarse stages.

The new family supplies coefficientwise strong restoration in the fixed $C^{1,\alpha}$ chart. It does not upgrade the old second-order walk coefficients, or remove the need to construct connected coefficient families for the nonlinear germ of § 3.38. Independent background cube domains, complete history admission and an externally prescribed off-current core convention remain separate.

3.40. **Strong local families for the current row saddle.** The energy families alone do not give a strong Banach space in which to expand a nonlinear equation. We construct the required strong families in the same prescribed order as the current precisions. The fourth-order restoration in Prop. 3.160 is used for its new local $C^{2,\alpha}$ site coefficients. No such conclusion is assigned to the older second-order site pieces.

Fix $0 < \alpha < 1$, a finite $p > 4/(1-\alpha)$, and a finite requested carrier rate κ. Let U_* be a real current-regular reference on an admitted determining geometry, with strict curvature and current margins. Use a fixed smaller weighted $C^{1,\alpha}$ tube $U = e^{i\eta E}U_*$. The independent current J is bounded in its full amplitude norm. The actual j_U need not have that bound at a complex input. All spaces and their comparison frames are fixed at U_*.

Write $\mathcal{A}_\alpha$ for the strong field space of CNG, $\mathcal{Y}_{2,\alpha}$ for its site gauge space, and $\mathcal{F}_p$ for the force space of § 3.25. More explicitly the force cell norm is

$$\|f\|_{\mathcal{F}_{p,y}} = \xi_y^{3-4/p}\|f\|_{L^p(B_y^+)}. \tag{3.mclxxxiii}$$

The cell enlargements include all stencils and row collars. A family norm is the maximum of its weighted row and column Schur sums between these normalized cell Banach spaces, with the joined full carriers of eq. 3.cmlxxxvii. Use a smooth local partition to reconstruct strong fields from their cell slots. Its profiles, differences and Hölder differences have their scaled bounds on complete collars. The product rule bounds this reconstruction independently of the number of fine coordinates. The fixed reconstruction constant is included in the algebra norm.

3.40.1. *The positive field primitive on finite-p forces.* Let

$$K_F = H_U + D_U D_U^\sharp + Q_U^\sharp a_f Q_U$$

be the actual positive field operator of the local-walk construction, before the site projection and current terms. Its local compressions retain all crossing coefficients. For a compression domain D, let I_D be its inverse and let χ be an output cutoff separated from its artificial boundary.

Lemma 3.161. For a force supported in an assigned cell z, the local output factors satisfy

$$\|\chi I_D \chi\, 1_z f\|_{\mathcal{A}_{\alpha,y}} \le C_{p,\alpha} e^{-sd(y,z)}\|f\|_{\mathcal{F}_{p,z}}. \tag{3.mclxxxiv}$$

The constants are independent of compression size, fine mesh, depth and region shape. The same estimate holds on a common smaller complex tube. ■

Proof. For $p > 2$, the normalized force cell embeds in the L^2 force cell, by the physical volume $O(\xi_z^4)$. The positive local inverse therefore has its proved exponential energy column with this source norm. On a complete nonbare box, the exact Wilson curl–divergence stencil rewrites the equation as a component covariant Laplacian with a bounded curvature multiplication and fixed neighboring shifts. The row-mass term has locally bounded point forcing: the complete average of the solution is bounded by its energy cell L^2 norm, and the actual transpose extends its bounded density. Its bound uses the physical column density, not a point evaluation of that energy function.

In a real local Coulomb frame the potential and its first derivative are bounded by the stated current regularity. In the complex tube they have the same bounds with fixed larger constants. The component covariant Laplacian is the ordinary discrete Laplacian plus first-order and zeroth-order terms with these bounded coefficients. This follows by pairing the positive and negative covariant shifts before dividing by the fine spacing. There is no derivative of curvature in that identity.

Apply the local finite-p estimates on nested complete boxes. Energy gives H^1; the first scalar estimate gives $W^{2,2}$, then strict embeddings give gradient L^3. The next estimate gives $W^{2,3}$, hence gradient L^8 and field L^∞. If $p \le 8$, apply the final p estimate there. For $p > 8$, use the $p = 8$ estimate first, then the desired finite p. The bounded row forcing and curvature terms remain controlled at every stage. The original force belongs to each lower exponent used on these fixed boxes. Morrey then gives the gradient Hölder exponent $\alpha < 1 - 4/p$.

All outer energy terms retain the exponential column weight, and neighboring box weights differ by a fixed factor. The source is present only near z. This proves the displayed column estimate with a fixed smaller rate. The output cutoff avoids the artificial boundary. At a true bare boundary the scale is η; the finite atomic stencils and the local mass bound give the same scaled estimate directly. Multiplication by the smooth output cutoff preserves the strong norm. □

The slow commutator in the positive parametrix maps this strong output into $\mathcal{F}_p$, with norm $C_{p,\alpha}/m$. Its derivative terms are $D\chi\, Dv$ and $\Delta\chi\, v$; the row commutator has the complete row difference $\chi(x) - \chi(\mathrm{row})$. The slow-cutoff bounds and the actual row moments give the same m^{-1} after rescaling. The finite-shift curvature terms also have that bound: their commutators contain one cutoff difference and a bounded curvature coefficient. No derivative of the input force is used. Choose m large after fixing p, α, κ. The weighted-overlap sum in the local-walk proof then makes this an absolute contraction on $\mathcal{F}_p$. Exactly as in eq. 3.cmlix, the resulting identity is

$$\mathbf{G}_F = \mathbf{M} \sum_{n \ge 0} (-\mathbf{T})^{\star n}, \qquad \|\mathbf{G}_F\|_{\mathcal{F}_p \to \mathcal{A}_\alpha} \le C_{p,\alpha,\kappa}. \tag{3.mclxxxv}$$

The determining geometry and background data of every coefficient are retained. This is a new strong parametrix; it need not use the same local pieces as a previously fixed energy representation.

3.40.2. *The two site adjoints without a current derivative.* For a site field u, use the normalized first-order cell norm

$$\|u\|_{\mathcal{V}_{p,y}} = \xi_y^{2-4/p} \|u\|_{L^p(B_y^+)} + \xi_y^{3-4/p} \|D_U u\|_{L^p(B_y^+)}.$$

For a site source write $\rho = D_U^\sharp f + h$, where f is a field force and h a site force. Give this source the infimum norm of

$$\|f\|_{\mathcal{F}_{p,y}} + \xi_y^{4-4/p} \|h\|_{L^p(B_y^+)}$$

over such representations on the complete source collar. The divergence is variational. Its values need not be bounded fine functions.

Lemma 3.162. The actual completed site inverse has local families from this source class into $\mathcal{V}_p$. It also has a local family with

$$\sup_y \{\xi_y^2 |G_s h|_y + \xi_y^3 |D_U G_s h|_y\} \le C \|h\|_{\infty,4} \tag{3.mclxxxvi}$$

in the corresponding weighted row/column norms. Both assertions hold on the common complex tube. ■

Proof. For the local positive site compression, first use its variational energy column with the displayed source representation. In the interior it solves

$$\Delta_U u = D_U^\sharp f + h - q^\sharp a_s q u.$$

The last term has bounded point amplitude of grade four, from the local energy norm and the actual complete row. For the divergence term, the uniform finite-p double-gradient resolvent in Prop. 3.79 gives the local first-order estimate. To see its lower-order bootstrap explicitly, put each potential-times-gradient term into a divergence of potential times u, plus a bounded first potential derivative times u. Initially $u \in H^1$, hence $u \in L^3$. The first-order estimate at 3 gives $u \in W^{1,3}$, hence $u \in L^8$. If $p \le 8$, finish at p at this stage: the outer field belongs to L^p, as do the prescribed sources. If $p > 8$, its next application at 8 gives $u \in L^\infty$. The final finite-p estimate then uses f, h at their prescribed exponent and the now bounded lower-order coefficients. All cutoffs are nested inside the artificial compression boundary.

The scalar commutator maps $\mathcal{V}_p$ to a site L^p force of grade four with norm C_p/m. Thus the same positive parametrix, now in this source class, converges after fixing m large enough. Multiplication of a variational source by a cutoff is legitimate: $\chi D^\sharp f$ is $D^\sharp(\chi f)$ plus a bounded one-step cutoff difference times f. This stays in the declared source class, without Df. This proves its exact local family assertion.

For the second assertion take $f = 0$ and bounded h. The scalar second-order bootstrap, with the same bounded row forcing, gives W^{2,p_0} for any fixed $p_0 > 4$ on the output support. Morrey gives u and its gradient bounded with the displayed scaling. The slow commutator takes that output into a bounded site force, so its positive local series converges in this class as well. True bare cells use their atomic stencils throughout. □

Let $\widehat{F} = G_s q^\sharp \Xi^{-4}$ and let $\widehat{M}$ be the actual counting site Gram matrix of eq. 3.cmxcv. Its inverse has the CCI local family. The injection $q^\sharp \Xi^{-4}$ has point density of grade four. Thus eq. 3.mclxxxvi gives local families for $\widehat{F}, D\widehat{F}$ with site amplitude grades two and three. Pairing a grade-two site field with a grade-two column costs no cell volume: $\xi^{-2}\xi^{-2}\xi^4 = 1$. Hölder's inequality gives the same statement in $\mathcal{V}_p$. Consequently the exact projection

$$P = \widehat{F}\widehat{M}^{-1}\widehat{F}^\sharp, \qquad R = 1 - P$$

has a bounded family on $\mathcal{V}_p$ and on the displayed bounded site-amplitude/gradient space. All scale ratios are absorbed once in the fixed spatial reserve, as in NCW.

The exact adjoint formulas now give the coefficient families

$$\begin{aligned} &\Lambda^\sharp = DRG_s : \{\text{site amplitude four}\} \longrightarrow \{\text{field amplitude three}\}, \\ &S^\sharp = 1 - DRG_s D^\sharp : \mathcal{F}_p \longrightarrow \mathcal{F}_p. \end{aligned} \tag{3.mclxxxvii}$$

In particular a bounded field force may be used in the second line by its cell L^p norm. The exact identity $\Lambda^\sharp D^\sharp = 1 - S^\sharp$ also gives the explicit family bound

$$\|\mathbf{\Lambda}^\sharp \mathbf{D}^\sharp f\|_{\mathcal{F}_p} \le C_{p,\kappa} \|f\|_{\infty,3}. \tag{3.mclxxxviii}$$

There is no assertion that $S^\sharp$ maps arbitrary bounded forces to bounded forces. This is the finite-p distinction required by the nonlinear adjoint factory.

3.40.3. *Woodbury before either current correction.* Use the new strong family for G_F and the actual positive coarse family for $\widehat{\Theta}^{-1}$. The Woodbury identity is

$$G_0 = G_F + G_F(D\widehat{F})\widehat{\Theta}^{-1}(D\widehat{F})^\sharp G_F.$$

The map $D\widehat{F}$ has field-force amplitude three by the preceding site lemma. Its transpose maps a strong field to counting top amplitudes: the physical pairing on a cell costs $\xi^{-3}\xi^{-1}\xi^{4}=1$. The two exponential Schur sums make these maps bounded between their cell spaces. It follows that

$$\|\mathbf{G}_0\|_{\mathcal{F}_p\to\mathcal{A}_\alpha}\le C_{p,\alpha,\kappa}. \tag{3.mclxxxix}$$

For $\lambda=\Lambda v$, the new fourth-order restoration gives a family in $\mathcal{Y}_{2,\alpha}$. Hence $S=1-D\Lambda$ has a family on $\mathcal{A}_\alpha$. The exact grouped current formula eq. 3.mii can be written as

$$B_U[J]v=W[J]\lambda+\Lambda^\sharp\Big(W[J]^\sharp v-\tfrac12 W[J]^\sharp D\lambda\Big)-\tfrac12\Lambda^\sharp D^\sharp W[J]\lambda.$$

The first term is a bounded field force of grade three. The two site forces in parentheses have amplitude four. The last term uses a bounded grade-three field force inside its variational divergence. Equation eq. 3.mclxxxvii therefore proves

$$\|\mathbf{B}_U[J]\|_{\mathcal{A}_\alpha\to\mathcal{F}_p}\le C_{p,\alpha,\kappa}\gamma,\qquad \gamma=\|J\|_{\infty,3}. \tag{3.mcxc}$$

Every product keeps the actual current bond, both matrix endpoints, and all background and geometry data read by its factors. No derivative of J is present.

Choose γ below a fixed smaller positive ceiling, after p,α and the carrier constants have been chosen. Then

$$\mathbf{G}=\mathbf{G}_0\sum_{n\ge0}(\mathbf{B}_U[J]\mathbf{G}_0)^{\star n}$$

converges normally as a family from $\mathcal{F}_p$ to $\mathcal{A}_\alpha$. Its complete sum is the first canonical inverse. The difference $G-G_0$ has the additional factor $C\gamma$.

3.40.4. *The two row inversions and the multiplier root.* The normalized field-row trace and its transpose are those in eq. 3.miv. The trace maps a strong field into counting row amplitudes, by its complete average and outward row completion. The transpose of a unit counting row has fine-force density of grade three, as in eq. 3.cmlxxxvi. The positive family for $\mathcal{T}_0^{-1}$ is already available. Since $G-G_0$ has strong family norm $C\gamma$, its normalized row-Schur difference has that same small bound. The exact Neumann formula eq. 3.mvi therefore gives the first row inverse and the first lift as strong families.

For the induced $\omega_*=H^\sharp J$, retain the original fine-current bond in each paired coefficient. The exact physical duality calculation eq. 3.mx gives its absolute rooted bound

$$\sup_c\sum_C e^{\kappa\sigma_c(C)}\xi_c^3|\omega_{c,C}|\le C\gamma.$$

The bare readout is the separately retained local term described in NCW. Its bare second constraint jet is identically zero. Thus the actual $\Delta_U^{(2)}[\omega_*]$ has a family from $\mathcal{A}_\alpha$ into bounded forces of amplitude three. This uses both absolute leg sums of the actual row coefficient, its fixed-leg fine density and the physical transpose. No individual invariant Taylor piece is used.

Compose it with S and $S^\sharp$. The second correction then has family norm

$$\|\mathbf{S}^\sharp\mathbf{\Delta}_{\mathbf{U}}^{(\mathbf{2})}[\omega_*]\mathbf{S}\|_{\mathcal{A}_\alpha\to\mathcal{F}_p}\le C\gamma.$$

A further fixed decrease of the current ceiling admits the second Neumann series for G_1. Its row matrix differs from the same positive $\mathcal{T}_0$ by $C\gamma$. The second row inverse and H_1 follow by the same exact formula. The readout in this second correction has throughout used the first lift H.

Proposition 3.163. Fix the stated α,p,κ. On a smaller common $C^{1,\alpha}$ background tube and for a sufficiently small full independent current radius, the canonical second-variant row saddle has an exact local family

$$\mathfrak{I}_{U,J}:\mathcal{F}_p\oplus\mathcal{B}\longrightarrow\mathcal{A}_\alpha\oplus\mathcal{W}. \tag{3.mcxci}$$

Its row/column carrier norm is bounded by a fixed constant. The same construction gives strong families for $G_F,G_0,G,G_1,\ H,H_1,S,\Lambda$ and the two adjoint maps in eq. 3.mclxxxvii, in their stated

spaces. Each coefficient is holomorphic in its actual local background and independent current data, with its complete determining geometry. The constants and radii are independent of depth, compatible volume and admitted region shape. ■

Proof. The preceding constructions give the displayed maps in their prescribed order. The full row saddle is obtained from the exact formulas

$$A = G_1 f + H_1(d - QG_1 f), \qquad \nu = (QG_1 Q^\sharp)^{-1}(d - QG_1 f),$$

with the physical normalizations of eq. 3.mxvi. They give precisely the datum amplitude one and multiplier amplitude three in $\mathcal{B}, \mathcal{W}$. Products use the smooth cell reconstruction and the joined-carrier algebra. Local primitive matrices, the new restoration families and all coarse inverses are holomorphic on their common smaller tube. Normal convergence preserves this property. Only finitely many strong primitive, coarse inversion and current Neumann stages are used, so the final tube and ceiling are positive.

When a cover is needed, use one common sufficiently large cover for all stages, keep full lifted carriers, and deck-average before projection. The strong local cell norms and physical measures descend with the same row/column bounds, as in NCW. No quantum-volume identity is involved. □

The current ceiling in this strong family construction may depend on the fixed p, α and carrier rate. The larger energy ceiling from the completed inverse theorems is unchanged. The proposition supplies the stated stronger local representation on its smaller domain; it makes no endpoint Riesz assertion and no history-deletion identification.

3.41. **Connected families for the selected nonlinear germ.** The strong inverse families let us expand the selected nonlinear equation itself. Each coefficient below retains the background carrier of every inverse factor, the actual row carrier of every nonlinear vertex, and all datum inputs. Their absolute connected norm is summable over every Taylor degree. This is stronger than a separate bound for each fixed derivative.

We use the auxiliary action and current convention of § 3.38. The resulting sum is that selected germ, rather than a new equation with generic local coefficients. Its physical identification is inherited from the exact scalar identity proved there.

3.41.1. *Inputs, cell slots and carrier norms.* Fix $0 < \alpha < 1$, a finite $p > 4/(1-\alpha)$, and a finite requested carrier rate κ. Choose the primitive families at a stronger fixed rate κ_+. The final rate reserve will pay fixed local stars, supported-slot collars and the base-variable Taylor expansion. All choices precede depth and volume.

Let $\mathfrak{D}$ be a smaller common domain of pairs $P = (U, J)$ supplied by Props. 3.160 and 3.163. The background is in a weighted $C^{1,\alpha}$ tube about a fixed given real current-regular U_*. The full independent-current amplitude is smaller than the strong-family ceiling. That ceiling may depend on α, p, κ_+; the larger energy ceiling is not asserted to suffice for this representation. Comparison frames, norms and the node extension are fixed at U_*. At each present base U, its restoration, constraint and fixed-for-this-germ slice are the actual $\Lambda_U, \mathfrak{q}_U$, and $R_U D_U^\sharp A = 0$.

Use the spaces

$$\mathcal{X} = \mathcal{A}_\alpha \oplus \mathcal{W}, \qquad \mathcal{Z} = \mathcal{F}_p \oplus \mathcal{B}, \qquad x = (A, \nu). \tag{3.mcxcii}$$

Here the fine strong norm, row datum amplitude one and multiplier amplitude three are those of § 3.38. The local norm is the maximum of its field and row components at the assigned cell. There are only a fixed number of row species at a top.

Local field slots are supported strong fields from a fixed smooth cell partition with complete collars. Their reconstruction has bounded overlap and is bounded from the maximum slot norm into the global strong norm. Use the corresponding bounded smooth localization for force slots. Row data and currents, which are not differentiated, can use bounded ownership slots. Multiply

block norms by the fixed reconstruction bound so their carrier product norm is submultiplicative. None of these slots counts its individual fine coordinates.

A carrier X records full tags, short joins, local determining geometry and complete coefficient read sets. Its length is $\sigma_c(X) = b_L + d_c(X)$, with the guarded joining rule used in the linear families. Every multilinear coefficient has its output tag, all input tags and its primitive coefficient carriers in X. For an m-linear family $\mathbf{V}_m$, use

$$\|\mathbf{V}_m\|_\kappa = \max\Big\{ \sup_y \sum_{z_1,\dots,z_m,X} e^{\kappa\sigma_c(X)} \|V_{m;y,\mathbf{z}}[X]\|, \max_i \sup_{z_i} \sum_{y,\mathbf{z}_{\neq i},X} e^{\kappa\sigma_c(X)} \|V_{m;y,\mathbf{z}}[X]\| \Big\}. \tag{3.mcxciii}$$

The operator norm inside the sum uses the displayed local Banach slots and is the supremum on a fixed larger common local base domain $\mathfrak{D}_X^+$. Thus the supremum is taken separately for each coefficient before the carrier sum. The final base domain has a strict margin inside these larger domains. It is a norm of divided Taylor coefficients $D^m/m!$. Ordered products and joins of such families obey the usual substitution bound: sum each internal cell index using a row sum, or the distinguished column sum along its path to a selected input. The remaining factors use row sums. The resulting norm is at most the product of the factor norms. Fixed local stars cost C_κ^m at an m-linear vertex.

Lemma 3.164 (Uniform local-data majorants)**.** The required strong linear families can be chosen with bounded coefficientwise local-data supremum norms as just defined. Their constructive bounds are positive majorants uniform on the larger common base domain. The same convention applies to every tree and nonlinear coefficient below. ■

Proof. The local Dirichlet columns and cutoff errors have bounds $Ce^{-sd(y,z)}$ and $(C/m)e^{-sd(y,z)}$, respectively, uniformly on their entire larger local coefficient tube. Taking each coefficient's local-data supremum therefore preserves those bounds. The weighted centre count and the positive word sums for the primitive walks give the sum of these suprema, rather than only a bound at one common global base point.

There is one extra count in a positive coarse inversion. A combined near power $H_<^m$ has 2-operator norm at most $(q')^m$ uniformly on the larger complex tube. Its entrywise local-data supremum is consequently at most $C(q')^m$. The declared endpoint balls have radius B_m, so the number of possible endpoints in a row or column is at most $C(1+B_m)^4$. Hence its stronger supremum family bound is

$$C(1+B_m)^4(q')^m.$$

The smaller $(1+B_m)^2(q')^m$ Hilbert-row estimate at one base point would not by itself justify moving the supremum through the endpoint sum. The larger polynomial above suffices: B_m grows polynomially in m, and the contraction is exponential. Use it in every positive coarse short-power stage, choose the long cutoff $R = Dm$ as before, and increase m until both this term and the positive long-word bound are small in the carrier norm. Reblocking still uses a bounded number of final tags for the combined entry. This only enlarges fixed carrier and separation choices.

The remaining coarse words, inverse Neumann series, rooted multiplier pairings and strong-current perturbations are bounded by products of these positive coefficient majorants. Choose the strong current radius after their constants. This proves the claim through the finitely many primitive, Gram, gauge-Schur and row-Schur stages. No exchange of a pointwise base supremum with an infinite carrier sum is presumed. All later analytic local vertex bounds hold uniformly on a larger local strong ball, so their Cauchy majorants and ordered tree bounds have the same property. □

The strong linear inputs are the exact families

$$\mathfrak{I}_P : \mathcal{Z} \longrightarrow \mathcal{X}, \quad \Lambda : \mathcal{A}_\alpha \longrightarrow Y^{2,\alpha}, \quad S : \mathcal{A}_\alpha \longrightarrow \mathcal{A}_\alpha, \tag{3.mcxciv}$$

and the adjoint families

$$\Lambda^\sharp : \{h\}_{\infty,4} \longrightarrow \{f\}_{\infty,3}, \quad S^\sharp : \mathcal{F}_p \longrightarrow \mathcal{F}_p, \quad \Lambda^\sharp D^\sharp : \{f\}_{\infty,3} \longrightarrow \mathcal{F}_p. \tag{3.mcxcv}$$

They are precisely eq. 3.mcxci, eq. 3.mclxxxvii and eq. 3.mclxxxviii, with the new strong site restoration. The first-lift readout $\omega = H^\sharp J$ has its rooted local family with multiplier amplitude $C\|J\|_{\infty,3}$. The original fine-current input stays in each such carrier. It is not replaced by a second-lift readout.

3.41.2. *The actual nonlinear vertices.* For this fixed P put

$$\Pi(A) = \Pi_U(A), \quad \Xi(A) = \Pi(A) - SA, \quad N_q(a) = \mathfrak{q}_U(a) - Qa, \quad r(A) = \mathfrak{q}_U(\Pi(A)).$$

These are the finite gauge chain and actual re-centered straightened constraint already defined. The local formula for Π is analytic in $A, \Lambda A, D\Lambda A$. Its divided fine-mesh factor is removed before taking norms. Prop. 3.160 controls the site argument through two Hölder derivatives, coefficient by coefficient. Use $S = 1 - D\Lambda$ from this same restoration family when expanding the finite chain. Its local linear jet $A - D\Lambda A$ is then removed before filing the quadratic remainder Ξ.

Lemma 3.165 (Local analytic vertices without fine counts)**.** The local Wilson, constraint, finite-gauge and adjoint remainders used below have exact multilinear carrier families with coefficients bounded by $C_0 C_1^m$ at degree m. Their coefficients are holomorphic in their actual local (U, J) data. The constants are uniform in depth and fine mesh. ■

Proof. The centered Wilson remainder $F'_U(A) - j_U - F''_U(0)A$ is a local map from the strong ball into bounded grade-three forces. Its uniform holomorphic bound is Lem. 3.146. Its constant and linear jets are subtracted before expansion. This is essential when the actual j_U at a complex base is large. Each plaquette expression reads a fixed fine stencil, hence a fixed number of complete strong cell slots. Banach-valued Cauchy and polarization give C^m, independently of the dimensions of those slots.

The actual mean and its adjoint satisfy the holomorphic bounds in Lem. 3.148. An assigned row and its complete carrier read at most a fixed number of adjacent-scale cells. Proper separation excludes a second intervening level. The fixed-leg fine density in eq. 3.mxxxvii cancels the inverse fine volume in the physical transpose. Thus the bound for $DN_q(a)^\sharp \nu$ is a bounded point-force bound before any adjoint restoration is applied. Cauchy on its uniform larger strong ball gives all higher Taylor degrees in the same point-force class. It does not count fine coordinates or postulate a new higher-leg density estimate.

The finite gauge function uses local point products of the scaled variables $\xi A, \Lambda A, \xi D\Lambda A$. Its local power series has C^m coefficients. Substitute the strong linear families in eq. 3.mcxciv; each occurrence brings its entire data carrier. A point derivative paired with a bounded fine force has exactly the factory

$$f_0 + \Lambda^\sharp h_\lambda + \Lambda^\sharp D^\sharp f_d.$$

Its local fields f_0, f_d have grade three and h_λ has site grade four. Endpoint injection uses only the fixed graph degree. The families in eq. 3.mcxcv give the required finite-p output bounds. No derivative is taken of the paired force or of J.

These local analytic maps involve a fixed number of strong cell slots. Substitution of an m-linear coefficient attaches at most a fixed multiple of $m+1$ linear factors or fixed local stars. Their family norms and the guarded joins change C^m by another fixed exponential in m. The rooted family of $\omega(P)$, where it occurs, attaches its full base-current carrier and has bounded norm. This proves the assertion in eq. 3.mcxciii. □

We next make explicit which force classes enter these adjoints. The identity $QS = Q$ gives

$$r(A) - QA = Q\Xi(A) + N_q(\Pi(A)). \tag{3.mcxcvi}$$

Use this equality before filing coefficients. It avoids a linear cancellation between different restoration carriers. Similarly,

$$\begin{aligned}(r'(A) - Q)^\sharp \nu = \Xi'(A)^\sharp Q^\sharp \nu + S^\sharp DN_q(\Pi(A))^\sharp \nu \\ + \Xi'(A)^\sharp DN_q(\Pi(A))^\sharp \nu.\end{aligned} \tag{3.mcxcvii}$$

Every input to $S^\sharp$ or the nonlinear adjoint factory on the right is a bounded grade-three force, supplied by the actual row columns. A generic $\mathcal{F}_p$ force is not passed to a factory that requires a point force.

For the constraint-current term put $C_\omega = \Delta_U^{(2)}[\omega]$, the actual second field-mean jet. Its residual has the exact expression

$$\begin{aligned}&\nabla\langle \omega, N_q(\Pi(A))\rangle - S^\sharp C_\omega SA \\ &\quad = S^\sharp\{C_\omega \Xi(A) + DN_q(\Pi(A))^\sharp \omega - C_\omega \Pi(A)\} + \Xi'(A)^\sharp DN_q(\Pi(A))^\sharp \omega.\end{aligned} \tag{3.mcxcviii}$$

The braces are bounded point forces and are quadratic in A. In the row mass use

$$r'(A)^\sharp a_f r(A) - Q^\sharp a_f QA = (r'(A) - Q)^\sharp a_f r(A) + Q^\sharp a_f (r(A) - QA). \tag{3.mcxcix}$$

The weight $a_f \asymp \xi^{-2}$ converts datum amplitude one into multiplier amplitude three. The current term is the exact finite-chain adjoint remainder after its quadratic scalar jet $\langle A, B_U[J]A\rangle/2$ is removed. Its factory inputs are again bounded forces. These formulas show both the quadratic zeros and the valid source classes coefficient by coefficient. There is no L^∞ Riesz claim.

3.41.3. *An exact tree solution of the selected equation.* Let $N_{\mathcal{L}}(A) = \mathcal{L}_P'(A) - G_1^{-1}A$ be the subtracted auxiliary gradient of eq. 3.mcli. Define

$$L_P b = \mathfrak{I}_P(0, b), \qquad \mathfrak{N}_P(A, \nu) = \mathfrak{I}_P \begin{pmatrix} -N_{\mathcal{L}}(A) + (r'(A) - Q)^\sharp \nu \\ -(r(A) - QA) \end{pmatrix}. \tag{3.mcc}$$

The actual preconditioned equation is

$$x = L_P b + \mathfrak{N}_P(x). \tag{3.mcci}$$

It is exactly eq. 3.mclv. In particular its multiplier and curvature terms have not been replaced by model vertices.

Lemma 3.166 (The nonlinear vertex family)**.** On a fixed smaller joint strong ball,

$$\mathfrak{N}_P(x) = \sum_{m \geq 2} V_{m,P}[x^m], \qquad \|\mathbf{V}_{m,P}\|_{\kappa_+} \leq C_0 C_1^m. \tag{3.mccii}$$

The series and carrier sums converge absolutely on a uniform ball. The linear family L_P has bounded row and column carrier norm. ∎

Proof. Use the explicit remainder formulas above and Lem. 3.165. The source tuple is in $\mathcal{Z}$: its first component is in finite $\mathcal{F}_p$, and its second has datum amplitude one. Apply the exact strong row-saddle family $\mathfrak{I}_P$. All constant and linear jets were removed before filing. For example eq. 3.mcxcvi has two manifestly quadratic terms; the second line of eq. 3.mcxcvii has at least one field factor and one multiplier factor. The removed finite-chain scalar quadratic jet has exactly the canonical $B_U[J]$ coefficient. Consequently no degree-zero or degree-one representative is left for cancellation between carriers.

Linear primitive family products and the local C^m coefficient bounds give the displayed bound, after enlarging C_1. Full base coefficient carriers are retained in every product. The linear assertion is the same row-saddle theorem with zero force source. □

Proposition 3.167 (All-degree connected datum families)**.** For $P \in \mathfrak{D}$ and sufficiently small $\|b\|_{\mathcal{B}} < r_b$, the selected auxiliary solution has an exact expansion

$$x(P,b) = \sum_{n\ge 1} \sum_{\mathbf{z},X} x_{n;\mathbf{z}}[X](P_X)[b_{z_1},\dots,b_{z_n}], \qquad \sum_{n\ge 1} r^n \|\mathbf{x}_n(P)\|_{\kappa_+} \le Cr, \quad 0 < r \le r_b. \tag{3.mcciii}$$

Every X is connected and contains its output, all datum inputs and all background/current read sets. Each coefficient is holomorphic in those local base data. Constants and radii are uniform in depth, compatible volume and admitted shape. ■

Proof. Expand eq. 3.mcci in ordered rooted trees. A leaf carries the exact linear family $L_P b$; an internal vertex of arity $m \ge 2$ carries $V_{m,P}$. Join every edge carrier at its shared cell, including the fixed short join when physical overlap does not mean an identical tagged vertex. The result contains every primitive base read set, not merely the geometric hull of the datum leaves. An n-leaf tree has at most $n-1$ internal vertices. All fixed stars and linear factors can therefore be paid by a fixed exponential in its degree.

To prove convergence, replace every coefficient by its family norm. The positive scalar majorant is

$$t = C_{\mathrm{lin}} r + \sum_{m\ge 2} C_0 C_1^m t^m. \tag{3.mcciv}$$

Choose r_b so this map preserves $[0, 2C_{\mathrm{lin}} r_b]$ and its derivative there is less than $1/2$. Iteration from zero gives a convergent positive series with $t(r) \le 2C_{\mathrm{lin}} r$. Its coefficients are the sums of all ordered tree weights. The substitution estimate for eq. 3.mcxciii bounds the actual coefficient families by this series. There is no extra factorial: the vertices were divided Taylor coefficients, and the ordered tree sum is precisely the expansion of the powers in eq. 3.mcciv.

Normal convergence permits substitution in the actual equation. Its sum lies in the strong uniqueness ball of Prop. 3.149, so it equals that already constructed selected solution. This establishes the claimed identity, not only a formal series solution. The local holomorphy of every primitive and normal convergence give the final assertion. □

For functions rather than multilinear kernels, group all terms with the same carrier. If $x_y(P,b) = \sum_X x_{y,X}(P_X, b_X)$, then

$$\sup_y \sum_X e^{\kappa_+ \sigma_c(X)} \sup_{\|b_X\|<r} \|x_{y,X}(P_X,b_X)\|_{\mathcal{X}_y} \le Cr. \tag{3.mccv}$$

This is an absolute connected family bound for the whole analytic function. It does not follow merely by summing separate fixed-order decay estimates.

3.41.4. *The literal datum and the field/current outputs.* The explicit inverse outward chart $b_U(B)$ in eq. 3.mxxxiii is local and analytic with uniform C_L^m Taylor bounds. Its outward factor reads only finer interior rows, which are unchanged by that chart. There is no recursive product over already corrected outward data. A one-step complete carrier therefore supplies its connected family, including its actual linear derivative $Db_U(0)$. Substitute that family in eq. 3.mcciii, and reduce the literal radius r_B so its absolute image lies in the reduced datum ball. The same tree majorant then gives eq. 3.mccv with B and a changed fixed constant.

Put $C = \Pi_U(A)$ and $\delta\omega = \nu - a_f b_U(B)$. The point-force identity for the new current can be written without cancellation of its degree-zero terms:

$$\begin{aligned} J_{\mathrm{new}} - J &= (\mathcal{E}_{\eta C}^{-\sharp} - 1)J \\ &\quad + \mathcal{E}_{\eta C}^{-\sharp}\{DN_q(C)^\sharp \omega + D\mathfrak{q}_U(C)^\sharp \delta\omega\}. \end{aligned} \tag{3.mccvi}$$

Here $\omega = H^\sharp J$ still uses the first lift. Every term on the right has positive datum degree. Its row transpose inputs have the actual fine fixed-leg density, so its output is a point force of amplitude three. No adjoint-restoration map is needed after this point-force expression. The divided exponential differential is a bounded local analytic point function.

Proposition 3.168 (Connected families for the literal update)**.** The following outputs have exact holomorphic connected families satisfying a bound Cr of the form eq. 3.mccv on $\|B\|_{\mathcal{B}} < r \le r_B$:

$$A, \quad C, \quad \nu, \quad \delta\omega, \quad J_{\mathrm{new}} - J, \quad \widehat{A}, \quad \widehat{J} - J.$$

Field outputs have the strong norm, multiplier outputs amplitude three, and current outputs point-force amplitude three. The literal potential is defined by $e^{i\eta\widehat{A}}U = \widehat{U}$, with the fixed node extension E_n. Every coefficient retains all actual datum and base read sets. ■

Proof. For $C = SA + \Xi(A)$, substitute the connected solution family in the local finite-gauge series and its strong linear factors. Its degree-one term is $SH_1 b$; higher terms are controlled by the same analytic majorant. The multiplier increment is an exact linear combination of known families. Equation eq. 3.mccvi and the point-force row bounds prove the current assertion by local analytic substitution. Multiplication of two such normalized families has its usual joined-carrier bound.

For literalization let $\zeta = E_n(i^{-1}\log v_U[C]^{-1})$, and $g = e^{i\zeta}$. The actual assigned node factor is a local analytic function of the fine strong field on its complete physical star. At an assigned scale ξ_y, its relative one-step factors at intermediate scale ξ_t have variations bounded by $C_L r\xi_t/\xi_y$ on a uniform complex strong ball. The actual path-row bounds for the group mean give this estimate before any fine coordinates are counted. Since $\sum_{t\le j_y}\xi_t/\xi_y \le L/(L-1)$, their product is uniformly analytic there. The union of its nested read sets lies in a star of radius $C_L\xi_y$, by the same geometric sum. Cauchy gives its C^m strong-cell coefficients. These are the actual relative node products, including the outward one-step factor of eq. 3.mxxxi. The fixed-reference E_n has local $Y^{2,\alpha}$ seeds. Thus ζ has a connected strong site family with norm Cr. The exact relative potential is

$$\widehat{A}_b = (i\eta)^{-1}\log\{g_- e^{i\eta C_b} U_b g_+^{-1} U_b^{-1}\}.$$

It is an analytic local function of $C, \zeta, D\zeta$; its apparent fine-mesh division cancels before expansion, as in eq. 3.mcxliii. The same local substitution proves the strong field assertion. Finally use

$$\widehat{J} - J = \mathrm{Ad}(g_-)(J_{\mathrm{new}} - J) + (\mathrm{Ad}(g_-) - 1)J.$$

This is a local point product of families already controlled and a bounded base current. Its constant datum jet is zero before filing. The bare node values are one and no boundary gauge term is added. □

3.41.5. *Joint base and datum Taylor coefficients.* The preceding coefficients are holomorphic in their local base data. We now record an absolute all-degree statement in those base variables too. The variables are supported Banach slots, not independent raw fine-bond discs.

Fix an interior pair $P_0 = (e^{i\eta E_0}U_*, J_0)$ of the smaller common domain. Let e_y be supported strong field slots and j_y bounded current ownership slots. Write

$$E = E_0 + \sum_y \mathcal{R}_y e_y, \qquad J = J_0 + \sum_y \mathcal{O}_y j_y. \tag{3.mccvii}$$

The smooth reconstruction $\mathcal{R}_y$ has complete collars and bounded overlap. The ownership $\mathcal{O}_y$ is bounded in the point-current norm and need not be smooth, since no current is differentiated. Normalize the slot radii by fixed positive fractions of their admitted tube and current margins.

Every tag and its read collar meet only their own and adjacent assigned levels when $\rho \gg LM_c$. A tag then meets at most $C_L M_c^4$ assigned strong cells. In particular, after one fixed collar enlargement, the number n_X of base slots read by a full carrier satisfies

$$n_X \le C_{L,M_c}\sigma_c(X). \tag{3.mccviii}$$

Each slot may contain arbitrarily many fine coordinates. This count does not count them. The enlargement and its canonical input tags are charged in the initial rate choice. The tag graph has bounded degree D_L. At most C_{L,M_c} assigned output cells belong to one tag. Thus a radius-j tag ball has at most $C_{L,M_c} D_L^{j+1}$ possible output cells. Reserve also a rate $a > \log D_L + 1$ for columns anchored at a base input, in addition to the supported-collar and Taylor rate costs.

Proposition 3.169 (Joint all-degree connected expansion)**.** On fixed sufficiently small reconstructed base slot and literal datum balls, all outputs in Prop. 3.168 have an exact joint Taylor expansion with absolutely summable connected coefficients at the requested final rate κ. If n is the datum degree and ℓ the combined base-slot degree, then

$$\sum_{n\ge 1,\ell\ge 0} r^n \delta^\ell \|\mathbf{F}_{n,\ell}\|_\kappa \le Cr. \tag{3.mccix}$$

Every coefficient contains its output, all datum input cells, all selected base input cells and its full determining-geometry carrier. All coefficients in this display are divided Taylor coefficients in their indicated Banach slots. ■

Proof. Begin with the datum family at a reserved rate larger than the final rate, including the one-time supported-slot collar enlargement. For a fixed carrier its coefficient function is holomorphic on a product of n_X normalized Banach slot balls of a common positive radius. Bounded-overlap reconstruction maps that product into the admitted global strong/current tube. Locality means slots outside its complete read set do not enter.

Multivariate Cauchy and dimension-free polarization bound the coefficient of total base degree ℓ, for each distribution of degrees among slots, by C^ℓ times its larger-ball bound. The polarization factor at degree j is at most $j^j/j! \le e^j$, so it is included in C^ℓ. Summing those distributions on a radius-δ_1 ball costs

$$(1 - C\delta_1)^{-n_X} \le \exp(C'\delta_1 C_{L,M_c}\sigma_c(X)), \qquad C\delta_1 < 1/2.$$

Choose δ_1 so this exponent fits the reserved Taylor rate, while leaving the additional rate a unused. Take the actual base radius $\delta \le \delta_1/2$. The coefficient of degree ℓ then gains $(\delta/\delta_1)^\ell$, so summing the degreewise supremum norms is justified by a geometric series. The original datum row and datum-column norms retain their bounds.

For a column anchored at a base slot w, every contributing carrier contains its tag. The unused rate gives $e^{-a\,\mathrm{dist}_{\mathrm{tag}}(y,w)}$ for its possible output cell y. The tag-ball count above makes the sum over those outputs finite, uniformly in volume and depth. Use the degreewise datum row bound for the remaining carrier and input sums. The same geometric factor sums the base degrees. This proves the base-column bound as well, and hence eq. 3.mccix in the row/column norm, not only its row half. All selected slots meet the declared carrier collar, which already contains their input tags and short joins. Thus the resulting joint Taylor carrier is connected. There is no fine-coordinate Cauchy count.

Every output increment under consideration vanishes at $B = 0$ for every admitted base pair. Its datum expansion was filed with that zero removed before the base expansion. Consequently every resulting joint coefficient has $n \ge 1$, as asserted. Normal convergence identifies the joint series with the same selected germ and literal update. □

3.41.6. *Physical identification and the scope retained.* At a physical pair $J = j_U = Q^\sharp \omega_{\mathrm{phys}}$, the sums are the physical PNC solution and its actual left current after the specified literal gauge. This follows from uniqueness and Props. 3.150 and 3.151; the tree construction does not impose a new physical equation. The first reduced-datum derivatives remain

$$D_b A(0) = H_1, \qquad D_b C(0) = SH_1.$$

Literal data add the actual right factor $Db_U(0)$, and the literal node gauge adds its fine gauge vector to the field derivative. Off current these jets are not equated.

For small periods construct every primitive, local vertex and tree on one common cover before projecting its carrier. Retain each lifted input and every winding join. The absolute tree norm is bounded by the same cover sums. Projection solves the actual finite periodic fixed-point equation, and strong contraction uniqueness identifies its sum. Deck averaging preserves the supported strong bounds and the original physical units. No assertion about a quantum partition function on different volumes is used.

The result supplies all-degree connected families for the selected off-current germ on its fixed strong reference chart. It does not identify a different prescribed core convention, manufacture a reference from arbitrary curvature-only coarse data, or prove history/tower admission or deletion laws. Its local background domains are the stated strong Banach domains. The auxiliary cube-Gaussian discs of a separate formation construction are not being declared independent fine-background discs.

3.42. **The decoupled response on the regular-pair domain.** The equations in eq. 2.x use a field and its first covariant differences. Their background domain is consequently the regular-pair domain of §3.12.1.1. We construct the response on that domain. The Hölder estimates below are estimates on inverse outputs; no Hölder seminorm of the complex background perturbation is assumed. The construction follows the constraint straightening and replacement of propagators in Bałaban's background and cluster arguments [B85e, B88a]. We give the first-order estimates, the independent-current completion, and the complete carrier convention needed for the present averages.

Fix the blocking factor and the separation constants of the positive local inverse construction. All its complete gauge cubes, row reads, outward completions, and local compression collars are retained. A pair is written $\mathbf{U} = e^{i\eta E}U$, $\mathbf{J}$, where U is real, its local gauge potentials and first differences have the regular-pair bounds, and

$$\sup_y\{\xi_y|E|_y + \xi_y^2|\nabla_U E|_y\} < a_*, \qquad \sup_y \xi_y^3|\mathbf{J}|_y < \gamma_*.$$

The curvature bounds in that domain are imposed as well. The positive constants a_*, γ_* are fixed after the finitely many proof carrier rates and minimal tile floors. After the auxiliary inverse construction and its proof rates have been fixed, the local column estimates are uniform in every larger admissible final tile size and separation. Enlarging those final parameters does not decrease these numerical radii: the already constructed families remain available, and reblocking into larger aligned tiles does not increase their carrier norms. At each such choice the regular-pair hypothesis is imposed on that choice's complete gauge cubes and collars. A pair admitted for smaller cubes is not thereby admitted for larger cubes. Obtaining the required local potentials from curvature alone can require a smaller curvature ceiling after the cube size has been fixed. The radii above are independent of depth, fine mesh, compatible volume, and admitted region shape on the resulting regular-pair domain. The current is an independent variable.

Use the physical fine pairing and counting measure on raw rows. A raw row is a logarithmic angle, without division by its length. If Q denotes the per-length row of the propagator construction, set $\widehat{Q} = \mathsf{m}Q$. In this subsection the letters H, H_1, Σ denote the raw-coordinate maps

$$H = \mathsf{H}\mathsf{m}^{-1}, \qquad H_1 = \mathsf{H}_1\mathsf{m}^{-1}, \qquad \Sigma = (\widehat{Q}G\widehat{Q}^\sharp)^{-1}.$$

Thus $\widehat{Q}^\sharp \mathsf{a}\widehat{Q} = Q^\sharp a_f Q$ defines the raw row mass a. A sharp denotes the transpose in these pairings, continued complex linearly. We reserve S for the fine projection $1 - D\Lambda$. Every field derivative, row, site inverse and current completion in this subsection is evaluated at the complex configuration $\mathbf{U}$. Thus an unsubscripted D means $D_{\mathbf{U}}$, and d means its covariant curl. The real factor U is used only to describe the input chart.

For a positive weight satisfying $w(y)/w(z) \le e^{\theta d(y,z)}$, put

$$\begin{aligned} \|A\|_{1,w} &= \sup_y w(y) \max\{\xi_y |A|_y, \xi_y^2 |\nabla_{\mathbf{U}} A|_y\}, \\ |f|_{3,w} &= \sup_y w(y) \xi_y^3 |f|_y, \qquad |b|_w = \sup_c w(y_c)|b(c)|. \end{aligned} \tag{3.mccx}$$

All differences and suprema use complete assigned stencils. The fixed neighborhood radius r_0 below includes these stencils and the complete nonlinear row read. Local scale ratios on this neighborhood are bounded by constants depending only on the blocking factor. The positive inverse decay leaves a fixed admissible $\theta > 0$ after these ratios have been absorbed.

3.42.1. *Strong outputs without a stronger background domain.*

Lemma 3.170. Fix $p = 8$ and $\alpha = 1/4$. The positive field inverse G_F has complete local carrier families from cell L^p forces to $\mathcal{A}_\alpha$, and from covariant divergences of cell C^α tensors to $\mathcal{A}_\alpha$. The assertions hold on the complex C^1 tube just specified. The second assertion uses the infimum norm on representations $f = D^\sharp F + g$, with F of grade two in C^α and g of grade three in cell L^p. ■

Proof. The positive field primitive and its local compressions are those of Lem. 3.161. Its proof uses the background potential and its first difference only. Indeed, after pairing the oppositely oriented covariant shifts, its equation in a local gauge is the ordinary component lattice Laplacian plus first-order and zeroth-order terms with bounded coefficients. The Wilson correction is a bounded curvature term with fixed neighboring shifts. The complete row mass is bounded point forcing from the local energy norm. None of these statements differentiates the curvature. The finite-p bootstrap in eq. 3.dcccxci therefore proves the first assertion on this C^1 tube, coefficient by coefficient.

Here is the additional divergence estimate. In unit cell coordinates let P_t be the flat lattice heat operator. The kernel of $\nabla(L_h + 1)^{-1}\nabla^\sharp$ is $\int_0^\infty e^{-t}\nabla\nabla^\sharp P_t\, dt$. The sum of each differentiated heat kernel is zero. In its action on F, subtract $F(x)$ before taking absolute values. The heat derivative bounds already proved in the discrete estimates give

$$\sum_z |\nabla^2 P_t(x,z)|\,|F(z) - F(x)| \le C t^{-1+\alpha/2}[F]_\alpha \quad (t \ge h^2).$$

For $t < h^2$, the same estimate with t replaced by h^2 and the length of that interval gives a bounded contribution. For the difference between outputs at points a distance r apart, split the integral at r^2. The near part is at most $Cr^\alpha[F]_\alpha$; the far part uses one further heat derivative and is bounded by

$$Cr[F]_\alpha \int_{r^2}^1 t^{(\alpha-3)/2}\, dt + Cr\|F\|_\infty \le Cr^\alpha \|F\|_{C^\alpha}.$$

The mass controls the integral beyond one. These estimates are uniform down to $r = h$.

Apply the finite-p divergence bootstrap eq. 3.dcccxcii first. It bounds $A, \nabla A$ in L^8, and A in the supremum norm on a smaller box. Every lower-order term is now an L^8 force. Its flat inverse has gradient in C^β for every $\beta < 1/2$. The preceding heat calculation handles the prescribed divergence. The difference between a covariant and an ordinary divergence is a bounded potential times F. Cutoff commutators involve only $F, A, \nabla A$ on the next reserved collar. This proves the interior $C^{1,\alpha}$ estimate without a derivative of F or a Hölder bound on a derivative of the background.

For completeness the estimate also gives a family, not only a bound on its complete sum. Use the same local parametrix M and commutator $T = K_F M - 1$ as in the positive field construction. For an initial divergence source, the just proved estimate bounds Mf strongly. The slow cutoff commutator maps that output to a cell L^8 force with norm C/m. Hence $G_F f = Mf - G_F T f$, where the second G_F is its already convergent L^8-force family. This is an exact normally convergent carrier formula. All collars used in the estimate are recorded in its carrier. At a true bare cell the spacing equals the cell scale and the same estimates are finite atomic estimates. The cover and deck averaging prescription of Prop. 3.112 preserves these bounds. □

Proposition 3.171. On the preceding regular-pair domain, the raw maps H, H_1, Σ, and

$$\mathcal{G} = G_1 - G_1 D R D^\sharp G_1 - H_1 \widehat{Q} G_1$$

have exact complete local carrier families with exponentially summable norms from raw row amplitudes, or bounded grade-three forces, to the first-order norm in eq. 3.mccx. The same assertion holds for H_0, $\widetilde{G} = G_0 - H_0 \widehat{Q} G_0$, and $P_1 = D(1-R)D^\sharp$, the last as a map from first-order fields to grade-three forces. The physical transpose $H^\sharp$ maps bounded grade-three forces to raw row amplitudes with the corresponding column bound. Every coefficient is holomorphic in the background and independent current on its complete local domain. ■

Proof. We specify the changes needed in the strong-family argument; the positive coarse inversions and complete current coefficients remain those of Prop. 3.112. The site inverse, its normalized columns, and the two adjoint families in eq. 3.mclxxxvii use only bounded background potentials and first differences. Their proofs through eq. 3.mclxxxix therefore apply on the present domain. The positive field factor is supplied by Lem. 3.170. Woodbury's formula gives $G_0 : \mathcal{F}_8 \to \mathcal{A}_\alpha$.

For $v \in \mathcal{A}_\alpha$, set $\lambda = \Lambda v = G_s R D^\sharp v$. The source $D^\sharp v$ is bounded of grade two. The projection $R = 1 - \widehat{F}\widehat{M}^{-1}\widehat{F}^\sharp$ preserves this bounded source class, since its column factor is bounded and its two absolute cell sums converge. The bounded-source site bootstrap gives bounded $\lambda, D\lambda$, of grades zero and one. This is the only regularity of Λv needed here. In particular $Sv = v - D\lambda$ is bounded of grade one.

The exact grouped first current correction is

$$B_{\mathbf{U}}[J]v = W[J]\lambda + \Lambda^\sharp\{W[J]^\sharp v - \tfrac{1}{2} W[J]^\sharp D\lambda\} - \tfrac{1}{2}\Lambda^\sharp D^\sharp W[J]\lambda. \tag{3.mccxi}$$

Here $W[J]\lambda(b) = \frac{i}{2}[\lambda(b_-) + \mathrm{Ad}(\mathbf{U}(b))\lambda(b_+), J(b)]$. The first term is a bounded grade-three force. The source of $\Lambda^\sharp$ in braces is bounded of grade four; the last term uses the finite-p bound eq. 3.mclxxxviii. Thus $B_{\mathbf{U}}[J] : \mathcal{A}_\alpha \to \mathcal{F}_8$ has family norm $C\gamma_*$, with no difference of J. The Neumann series for G converges in these spaces after a fixed decrease of γ_*.

The first row inverse and H now follow from eq. 3.mvi, using the actual bounded row injection. Its multiplier $H^\sharp J$ keeps the fine-current root and has eq. 3.mviii. The second row jet has both absolute leg sums and the physical fine density of eq. 3.mxi; it consequently maps a bounded grade-one field to a bounded grade-three force. Since Sv is bounded, and $S^\sharp$ is bounded on $\mathcal{F}_8$,

$$\|\mathbf{S}^\sharp \boldsymbol{\Delta}^{(\mathbf{2})}[\mathbf{H}^\sharp \mathbf{J}]\mathbf{S}\|_{\mathcal{A}_\alpha \to \mathcal{F}_8} \le C\gamma_*.$$

The second Neumann series and second row inversion give G_1, H_1. This argument does not assert that S preserves $\mathcal{A}_\alpha$ on the larger background domain.

It remains to check $G_1 D R D^\sharp G_1$. For a bounded grade-three source f, the first G_1 has $C^{1,\alpha}$ output. Therefore $D^\sharp G_1 f$ is a grade-two C^α site source. The potential in $D^\sharp$ is Lipschitz, which suffices for this conclusion. The same column formula for R, whose site columns have bounded first derivatives, preserves this C^α class. The final $G_1 D$ is bounded on it by Lem. 3.170, followed by Woodbury and the two small-current series. These are coefficientwise estimates in the joined-carrier algebra. They prove the asserted family for $\mathcal{G}$. Likewise $P_1 = D\widehat{F}\widehat{M}^{-1}\widehat{F}^\sharp D^\sharp$ uses only bounded first derivatives of the site columns and of the input field.

Finally each cell pairing for $H^\sharp f$ costs $\xi^{-1}\xi^{-3}\xi^4 = 1$. The absolute column sum proves the raw row bound. The raw-coordinate changes are precisely the physical weights in eq. 3.mxvi. All series above converge normally on the same smaller C^1 tube. They preserve their input and output collars, the read of the independent current, and the specified coarse Gram inverses. This proves both holomorphy and locality. □

3.42.2. *Cube parameters and local nonlinear coefficients.* Choose the partition P after the inverse tiles. Each coefficient is assigned its complete connected hull before intersection with Ω_0. Use the following cell convention. For a tile q at scale ξ_q, its j-enlargement has width $j\mathfrak{M}\xi_q$, $0 \le j \le 5$,

and its uncut hull has width $5\mathfrak{M}\xi_q$. A cell hull is the union of these tile hulls. Choose the fixed lower floor for $\mathfrak{M}$ so that output and input supports fit the one-enlargements and each successive nonlinear read fits the next enlargement. The local inverse compressions and their bootstrap collars are recorded within the five-enlargements; fixed reblocking, before choosing the partition, achieves this for the finitely many proof primitives. This is a choice of the common geometric hierarchy, not a deletion of rows from a previously specified sequence. All factors in a product, including output cutoffs, endpoint columns, current roots, and short joins, are retained. Let $\mathsf{P}(C)$ be the cubes meeting this hull. Use face connectivity, adding the finitely many intervening cubes at a face or corner when necessary. This convention is fixed before introducing the parameters.

Lemma 3.172. For $\bar\kappa \ge 0$, the partition side can be chosen so that each family in Prop. 3.171 defines

$$O_Y(s) = \sum_{\mathsf{P}(C)\subset Y} \Big(\prod_{\Delta\in\mathsf{P}(C)} s(\Delta)\Big) O_C, \qquad |s(\Delta)| \le e^{\bar\kappa}, \tag{3.mccxii}$$

with normally convergent row and column sums. For one constant $b = Ce^{C\bar\kappa} \ge 2$, all its weighted operator norms needed in eq. 3.mccx are at most b, including $H_s^\sharp$ and $P_{1,s}$. A fixed positive exponential decay rate remains. No algebraic inverse identity is claimed for the decorated maps at $s \ne 1$. ■

Proof. Choose a target side ℓ_*, a compatible power of L, at least $C(1+\bar\kappa)$ times the largest fixed tile size and the inverse of a reserved carrier rate. Write p for the side of the present cubic torus in the same units. First suppose $p \ge 3\ell_*$, and take $\ell_{\mathsf{P}} = \ell_*$. Cover the connected hull by balls whose radius is a fixed fraction of ℓ_{P}. Starting at an endpoint, charge a new ball to the first tree segment leaving the previous ball. Disjoint charged segments have total length at most a fixed multiple of the hull tree length $\ell(C)$. Each ball meets at most a dimension-dependent number of partition cubes, and the fixed carrier thickness is smaller than that radius. Hence

$$\#\mathsf{P}(C) \le C_0 + C_1\ell(C)/\ell_{\mathsf{P}}.$$

The factor $e^{\bar\kappa\#\mathsf{P}(C)}$ consumes at most half the reserved rate and leaves $e^{C_0\bar\kappa}$.

If $p < 3\ell_*$, use instead the coarsest compatible partition, of side $p/2$. The declared tori have equal sides that are twice a power of L, so this partition has exactly 2^4 cubes. For every carrier,

$$\#\mathsf{P}(C) \le 16, \qquad \Big|\prod_{\Delta\in\mathsf{P}(C)} s(\Delta)\Big| \le e^{16\bar\kappa}.$$

No spatial decay is spent in this case. When primitive patches require a cover, first construct and deck-average the undecorated families on one common cover, as in Prop. 3.112. Retain the full lifted carrier and its collars when projecting it. Only then assign the product of parameters, once for each distinct base cube meeting the projected complete hull. Several lifted cubes or carrier visits can therefore use the same base parameter only once in this coefficient. This prescription is squarefree in the independent base parameters; it is not substitution of tied variables into an already decorated cover monomial. The parameter factor is unchanged by a deck translation, and its absolute bound multiplies the descended row and column majorants without a sheet-count factor.

The projected complete hull still contains every projected input, output and nonlinear-read collar. It is connected, and its cube set is face connected with the fixed joins. Thus a coefficient surviving an active mask lies in one wall component. The five-enlargement stocks are formed from these complete projected hulls; a neighborhood path lifts starting at its chosen carrier lift, so their successive-read inclusions and component separation persist. This changes only the auxiliary parameter partition and supplies no identification with a separately prescribed history partition.

In either case the remaining row and column sums are the convergent carrier sums of Prop. 3.171. Increasing the fixed constant in $b = Ce^{C\bar\kappa}$ to include 16 proves the asserted weighted bounds. Normal convergence on smaller polydiscs proves holomorphy. The coefficient prescription on complete

common lifts proves transport compatibility whenever the parameter partitions and recorded geometry are carried as well. Gauge averaging and averaging with complex conjugation preserve the invariant coefficient norms and the complete sums. □

For a union Z of active cubes, define $\mathbb{Y}^j(Z)$ to be the assigned sites in the j-enlargements of tiles belonging to cells X whose complete uncut hull has $\mathsf{P}(X) \subset Z$. This is the five-enlargement convention for $j = 0, \dots, 5$, rather than erosion of $\Omega_0 \cap \bigcup Z$. The chosen floor gives $N_{r_0}(\mathbb{Y}^j(Z)) \subset \mathbb{Y}^{j+1}(Z)$ for $j \le 4$, and puts all these reads in $\Omega_0 \cap \bigcup Z$. This preserves the actual determining sequence and the indicated $\mathbb{Y}^1, \mathbb{Y}^2, \mathbb{Y}^3$ stocks. Retain nonlinear rows and the plaquette pairing at $\mathbb{Y}^2(Y)$, force values at $\mathbb{Y}^1(Y)$, and first-difference reads at $\mathbb{Y}^3(Y)$. A retained nonbare force has all six incident plaquettes. At a true bare boundary its cell scale is the fine spacing. Zero extension outside these stocks defines the local maps.

Write $\mathcal{Q}(A) = \widehat{Q}A + C(A)$ for the actual iterated straightened logarithmic average and

$$F(A) = \sum_{p \subset \Omega_0 \cap \bigcup Y} a_p(e^{i\eta A}\mathbf{U}), \qquad V(A) = F'(A) - F'(0) - F''(0)A, \qquad \Delta' = F''(0) - \mathrm{d}^\sharp \mathrm{d}.$$

Only the retained force values enter V, Δ'. The associated bilinear form is $\mathcal{E}(A, B) = (\mathrm{d}A, \mathrm{d}B)_\eta$ on the retained plaquette stock, with the same complete read convention.

Lemma 3.173. There are fixed $c, C > 0$ such that, for $\|A_i\|_1 \le r < c$, the local maps just defined obey

$$\begin{aligned} |C(A)|_y &\le C \max_{N_{r_0}(y)} |A|_{\mathrm{sc}}^2, & |C(A_1) - C(A_2)|_w &\le Cr\|A_1 - A_2\|_{1,w}, \\ |C_A'^\sharp z|_{3,w} &\le Cr|z|_w, & |(C_{A_1}' - C_{A_2}')^\sharp z|_{3,w} &\le C\|A_1 - A_2\|_{1,w}|z|, \\ |V(A)|_{3,w} &\le C\|A\|_1\|A\|_{1,w}, & |V(A_1) - V(A_2)|_{3,w} &\le Cr\|A_1 - A_2\|_{1,w}. \end{aligned} \tag{3.mccxiii}$$

The same bounds hold with the weight on the multiplier in the second line. The linear maps $\widehat{Q}, \widehat{Q}^\sharp, \Delta'$ have the corresponding fixed local bounds. All maps are holomorphic in the actual local data, covariant, real on real data, and determined by their complete reads. In particular $C(0) = C_0' = V(0) = V'(0) = 0$. ■

Proof. For the row maps use the complete iterated-average expansion in Prop. 3.6, with its absolute row and column sums and physical transpose. Iteration is in raw angles; the factors of successive row lengths form the geometric sums in that proof. The first nonlinear coefficient is quadratic. Cauchy's formula on a fixed larger amplitude ball gives the two derivative bounds; one uses the absolute fixed-leg column sum for the transpose. Local weights on a complete read differ by at most $e^{\theta r_0}$, which is incorporated in C.

We record the cancellation needed for the action bound. Rescale one assigned cell to unit size and denote its fine mesh by h. For a plaquette transport the four field values to its base, writing them as $\alpha_1, \dots, \alpha_4$, with the last two reversed. Set

$$ih^2\Theta_p = \log(e^{-ih\alpha_3}e^{-ih\alpha_4}e^{ih\alpha_1}e^{ih\alpha_2}), \qquad \Theta_p = (\mathrm{d}A)(p) + N_p.$$

The exponential and logarithm series on a fixed small ball give $|N_p| \le Cs^2$, $|N_p'| \le Cs$, where $s = \max|A|$. Also $|\Theta_p| \le C(g + s^2)$, where $g = \max|\nabla_{\mathbf{U}}A|$, and $h|\Theta_p| \le Cs$. Write the background plaquette as $c_p 1 + ih^2 F_p$. The two by two determinant identity gives $|c_p - 1| \le Ca_*^2h^4$. Expansion of its trace against $e^{ih^2\Theta_p}$ shows that the plaquette derivative is

$$h^4\langle F_p + c_p\Theta_p + M_p, \delta\Theta_p\rangle, \qquad |M_p| \le Ch^4(|\Theta_p|^3 + a_*|\Theta_p|^2).$$

At a nonbare force bond pair the opposite incident plaquettes before dividing by h. Transported differences of their N_p's are bounded by $Chsg$, by the bound on N_p'. The terms $(c_p - 1)\mathrm{d}A$ are bounded after the two differences by Ca_*s, using their h^4 factor. The terms M_p are bounded after

one difference by $Cs^2(s+a_*)$, using $h|\Theta_p| \le Cs$. The remaining differentiated N_p factor costs $Cs(a_*+g+s^2)$. Therefore

$$|F'(A)-F'(0)-\mathrm{d}^\sharp \mathrm{d}A| \le Cs(a_*+\max\{s,g\}).$$

Put $n=\max\{s,g\}$ and $w(z)=F'(zA)-F'(0)-z\mathrm{d}^\sharp \mathrm{d}A$. The preceding estimate gives $|w(z)| \le C|z|s(a_*+|z|n)$ whenever $|z|s \le c_0$, for a fixed $c_0>0$. The case $s=0$ is immediate. For $s>0$, Cauchy's formula on $|z|=R$ bounds the linear coefficient by $Cs(a_*+Rn)$. Letting $R \downarrow 0$ gives $|\Delta' A| \le Ca_* s$, without a first-difference term. To bound the higher coefficients, take $T=c_0/s \ge 2$, decreasing the field radius if necessary. Cauchy's formula gives $|[z^q]w| \le CsT^{1-q}(a_*+Tn)$, so

$$|V(A)| \le 2Cs(a_*/T+n) \le Csn.$$

The last inequality uses the fixed bound for a_* and $s \le n$. This proves the bounds on Δ' and V. At a bare cell $h=1$; direct absolute estimates on the at most six plaquettes give the same bounds. The small amplitude ball is fixed before the mesh is chosen. Polarization, or Cauchy's formula along the segment between two fields on a ball twice as large, gives the Lipschitz estimate. Rescaling supplies the factors ξ_y, ξ_y^2, ξ_y^3. All identities used are local products, convergent logarithms, and physical transposes. They prove the final assertions as well. □

3.42.3. *The two fixed points and the current.* Write $C''[a]=(\mathrm{d}C_2)_a$, where C_2 is the quadratic part of C. For $\|a\|_1 < c/b$ define

$$D_s(a)=C(a-H_sD_s(a)), \qquad T_s(a)=a-H_sD_s(a), \qquad E_s(a)=C'_{T_s(a)}{}^\sharp(1+H_s^\sharp C'_{T_s(a)}{}^\sharp)^{-1}. \tag{3.mccxiv}$$

The first equation is solved by iteration from zero. Define, for every parameter value, the following particular nonlinear force:

$$\begin{aligned} W_s(a) = &-[E_s(a)-C''[a]^\sharp]H_s^\sharp \mathbf{J} - \widehat{Q}^\sharp(\Sigma_s-\mathsf{a})D_s(a) \\ &- E_s(a)(\Sigma_s-\mathsf{a})\widehat{Q}a + E_s(a)(\Sigma_s-\mathsf{a})D_s(a) \\ &+ V(T_s(a)) - E_s(a)H_s^\sharp V(T_s(a)). \end{aligned} \tag{3.mccxv}$$

This is the force specified in eq. 3.cdxlv. Its definition uses bounds on the decorated maps, and does not use $\widehat{Q}H_s=1$ or a slice identity at general s.

Proposition 3.174. There are fixed $c, C>0$ such that the following holds with $r=cb^{-8}$, where b is the constant of Lem. 3.172. For $|B'|<r$, the equations

$$a=H_{1,s}B'+A_1, \qquad A_1=-\mathcal{G}_s W_s(a), \qquad \mathbf{H}=T_s(a) \tag{3.mccxvi}$$

and eq. 3.mccxiv have a unique solution in $\|A_1\|_1 \le br$, $|D_s| \le Cb^2r^2$. They are exactly eq. 2.x in raw coordinates. The solution is holomorphic jointly in $s, B', \mathbf{U}, \mathbf{J}$ on the indicated open domains and satisfies

$$\|\mathbf{H}\|_{1,w} \le Cb|B'|_w, \qquad \|A_1\|_{1,w} \le Cb^5|B'||B'|_w, \qquad \|\delta\mathbf{H}\|_{1,w} \le Cb|\delta B'|_w. \tag{3.mccxvii}$$

Here the difference is between two data in the same ball. The field and its datum derivative also satisfy the corresponding $\mathcal{A}_{1/4}$ output estimates, with a fixed larger power of b.

For $\|A\|_1 \le 4br$, put

$$\begin{aligned} \langle H_s^{\circledast}(A)(c), e\rangle &= \mathcal{E}(A, H_s e_c) + (\Delta' A + V(A), H_s e_c)_\eta, \\ \Psi_{s,A} &= (1+H_s^\sharp C'_A{}^\sharp)^{-1}, \qquad \omega_s(A) = \Psi_{s,A}[H_s^\sharp \mathbf{J} + H_s^{\circledast}(A)], \\ j_s(A) &= f_A^{-1}(\widehat{Q}+C'_A)^\sharp \omega_s(A) - \widehat{Q}^\sharp H_s^\sharp \mathbf{J}, \end{aligned} \tag{3.mccxviii}$$

where $f_A(b)=\sum_{n\ge 0}(-i\eta\,\mathrm{ad}_{A(b)})^n/(n+1)!$. Then the reduced response is $\mathfrak{h}_Y=(\mathbf{H}, j_s(\mathbf{H}))$. It vanishes at $B'=0$, and

$$|j_s(\mathbf{H})|_{3,w} + |P_{1,s}\mathbf{H}|_{3,w} \le Cb^4|B'|_w, \qquad |\delta j_s(\mathbf{H})|_{3,w} \le Cb^4|\delta B'|_w. \tag{3.mccxix}$$

The shifted independent current is $\mathbf{J} + j_s(\mathbf{H})$. ■

Proof. Let $\|a\|_1 \le \varepsilon < c/b$. By Lem. 3.173, the first equation of eq. 3.mccxiv maps the row ball $|D| \le C\varepsilon^2$ into itself and has Lipschitz constant $Cb\varepsilon < 1/4$. Its solution satisfies

$$|D_s(a)|_w \le C\varepsilon\|a\|_{1,w}, \quad |\delta D_s|_w \le C\varepsilon\|\delta a\|_{1,w}, \quad \|T_s(a)\|_1 \le 2\varepsilon, \quad \|\delta T_s\|_{1,w} \le 2\|\delta a\|_{1,w}.$$

Differentiation of that fixed-point equation gives the last expression in eq. 3.mccxiv. In particular $E_s(a) = C''[a]^\sharp + O(b\varepsilon^2)$ as a map from raw multipliers to bounded forces. To check the quadratic error, write $C'_{T_s(a)} - C''[a] = (C'_{T_s(a)} - C''[T_s(a)]) + C''[T_s(a) - a]$. The first term is quadratic by Cauchy's formula and the second is bounded by $Cb\varepsilon^2$; the inverse in eq. 3.mccxiv differs from one by $Cb\varepsilon$. The same decomposition gives its difference bound $Cb\varepsilon\|\delta a\|_{1,w}$.

Insert these bounds into each of the six terms of eq. 3.mccxv. The first term uses $|H_s^\sharp \mathbf{J}| \le b\gamma_*$. The result, with a fixed larger C, is

$$|W_s(a)|_{3,w} \le Cb^2\varepsilon\|a\|_{1,w}, \qquad |\delta W_s|_{3,w} \le Cb^2\varepsilon\|\delta a\|_{1,w}. \tag{3.mccxx}$$

For $\|A_1\|_1 \le br$, $|B'| \le r$, take $\varepsilon = 2br$. The outer map in eq. 3.mccxvi has norm at most Cb^5r^2 and Lipschitz constant at most $Cb^4r < 1/4$. It maps the stated ball into itself. Banach's theorem proves existence and uniqueness. Subtracting the two fixed-point equations in a weighted norm and absorbing the same constant proves eq. 3.mccxvii. The iterates are holomorphic and converge uniformly on smaller polydiscs, proving joint holomorphy. The construction remains valid on each local chart of the regular-pair domain; agreement on overlapping charts follows from the identical equations and contraction uniqueness. For the stronger output assertion, let $\|\cdot\|_{\mathcal{A}_{1/4};w}$ be the norm in eq. 3.dccclxxxvii, with each complete local-box norm multiplied by $w(y)$. The weight is not differentiated. The strong families in Prop. 3.171, with the parameter summation of Lem. 3.172, give

$$\|H_s z\|_{\mathcal{A}_{1/4};w}, \quad \|H_{1,s} z\|_{\mathcal{A}_{1/4};w} \le b_{\mathrm{str}}|z|_w, \qquad \|\mathcal{G}_s f\|_{\mathcal{A}_{1/4};w} \le b_{\mathrm{str}}|f|_{3,w}, \qquad b_{\mathrm{str}} \le Cb^q$$

for a fixed exponent q. These are first-gradient Hölder output norms on the same background domain. For two data in the indicated ball, subtract the exact equations:

$$\delta\mathbf{H} = H_{1,s}\delta B' - \mathcal{G}_s\delta W_s - H_s\delta D_s.$$

The weighted fixed-point estimates already give $\|\delta a\|_{1,w} \le Cb|\delta B'|_w$. With $\varepsilon = 2br$, the force and straightening bounds yield

$$|\delta W_s|_{3,w} \le Cb^4 r|\delta B'|_w, \qquad |\delta D_s|_w \le Cb^2 r|\delta B'|_w.$$

Consequently $\|\delta\mathbf{H}\|_{\mathcal{A}_{1/4};w} \le Cb^q|\delta B'|_w$, since $r = cb^{-8}$. Taking one datum to be zero bounds the field itself. At every datum in the open ball, divide by the size of a complex datum increment and pass to the derivative. Holomorphy and continuity of the finite-lattice norm give the same bound for the datum derivative on that whole ball.

For the current, the physical gradient pairing gives

$$|H_s^{\circledast}(A)|_w \le Cb(\|A\|_{1,w} + \|A\|_1\|A\|_{1,w}).$$

In the curl term the two first differences have grades two and two, so the cell factor is $\xi^{-2}\xi^{-2}\xi^4 = 1$. The other terms use the field/force pairing and its column sum. The same estimate holds for differences. Since $\|H_s^\sharp C'^\sharp_A\| \le Cb\|A\|_1 < 1/4$, $\Psi_{s,A}$ is uniformly bounded. It follows that

$$|\omega_s(A) - H_s^\sharp\mathbf{J}|_w \le Cb(1 + b\gamma_*)\|A\|_{1,w}.$$

The exponential series gives $\|f_A^{-1}\| \le 2$ and $\|f_A^{-1} - 1\| \le C\eta|A|$, with the corresponding difference bounds. Expand the last line of eq. 3.mccxviii around $A = 0$; its constant term cancels. The local transpose estimate and the preceding multiplier bound give $|j_s(A)|_{3,w} \le Cb^3\|A\|_{1,w}$, and the same Lipschitz estimate. Combine this with eq. 3.mccxvii and the bound on $P_{1,s}$. □

Proposition 3.175. The response of Prop. 3.174 is real on real data and real parameters, gauge covariant, and compatible with common-lift transport. If Z is a union of wall components of $\{\Delta : s(\Delta) \neq 0\}$, its restriction is the response formed using precisely those ambient carrier coefficients whose complete cube sets lie in Z. The field is supported over $\mathbb{Y}^1(Z)$; its reduced current and projected field are supported over $\mathbb{Y}^2(Z)$. All their complete native reads lie over $\mathbb{Y}^3(Z)$. A component with no retained datum has zero response.

If every cube meeting any complete cell hull has parameter one, all three fields equal their all-one fields on that same sequence. At the all-one point their complete sums are independent of the auxiliary carrier tiling and parameter partition. They commute with an identification of two regional sequences that preserves all local field, row, gauge, and geometry data in common lifts. For a general proper mask the derivative of the first response at zero is $H_{1,s}$; its linearized constraint is $\widehat{Q}H_{1,s}B'$. ■

Proof. The cube set of a surviving coefficient is face connected and contains its complete input and output read collars. It therefore lies in one active wall component. Formula eq. 3.mccxii splits into a direct sum over these components. The local maps in Lem. 3.173 cannot cross their reserved collars. Starting both iterations at zero proves the same direct-sum assertion for every iterate and hence for its limit. The transpose, curl pairing, and local row derivative in eq. 3.mccxviii use at most the next two reads; this proves the asserted support of the current. The field, current and projected field vanish on a component with zero datum because the zero iterate remains zero there. An indicator of a component consequently creates no commutator on these complete native reads.

Every coefficient and local operation is real, covariant and transported with its complete determining data. The same iteration argument proves these properties. At the specified one-valued cubes every coefficient weight is one and the local structure is the whole-sequence structure. Absolute summation gives the actual operators, independently of their expansion. Their physical formulas and local structure commute with the stated identification; uniqueness identifies the responses. Finally $D_s(0) = D_s'(0) = W_s(0) = W_s'(0) = 0$, which gives the last assertion by differentiation. Thus restriction to an active component identifies its restricted ambient solve. Identification with a different sequence's solve requires the indicated equality of determining data and operators. □

3.42.4. *Physical identification and the coupled response.*

Proposition 3.176. Let Y contain every complete cell hull of the determining sequence, let $s = 1$ there, and let U_0 be a real critical reference in the regular class, with its physical current $J(U_0) = \widehat{Q}^\sharp\lambda_0$. Use the closed noninward determining stock, including the complete preceding block at every outward row. For the response of Prop. 3.174,

$$\mathcal{Q}(\mathbf{H}) = B', \qquad RD^\sharp\mathbf{H} = 0, \qquad F'(\mathbf{H}) = (\widehat{Q} + C'_{\mathbf{H}})^\sharp\omega_1(\mathbf{H}). \tag{3.mccxxi}$$

Consequently $J(U_0) + j_1(\mathbf{H}) = J(e^{i\eta\mathbf{H}}U_0)$ and $P_{1,1}\mathbf{H} = DD^\sharp\mathbf{H}$. Every sufficiently small critical field on this slice is this response at its own reduced datum. The node factors and the change between ordinary and reduced outward data are the actual ones of eq. 3.cccxi–eq. 3.cccxiii. They depend only on the complete corresponding row or top pillar, are holomorphic on a fixed larger small ball, and commute with restriction and common-lift transport on those reads. ■

Proof. The current identification in Prop. 3.90 identifies the second precision on the Landau slice with the Hessian of the constrained action. In particular $\mathcal{G}$ is its inverse on $\mathcal{K} = \ker\widehat{Q} \cap \ker RD^\sharp$, and H, H_1 lie in the Landau slice, with $\widehat{Q}H = \widehat{Q}H_1 = 1$. The first equation of eq. 3.mccxiv therefore gives the first two equations of eq. 3.mccxxi. Differentiate the straightening map. On $\mathcal{K}$ its derivative maps bijectively onto $\ker(\widehat{Q} + C'_{\mathbf{H}}) \cap \ker RD^\sharp$, since it differs from the identity by $O(b\|a\|_1)$. Here is the exact action calculation. Write $\Delta_\pi = F''(0) - B_{U_0}[J]$, $\Delta_1 = \Delta_\pi - S^\sharp\Delta^{(2)}[H^\sharp J]S$, and $V_0(T) = F(T) - F(0) - (J,T) - \frac{1}{2}(T, F''(0)T)$. For a on the Landau slice, put

$D = D_1(a)$, $T = a - HD$, and $P = \Sigma - \mathsf{a}$. The exact identities $\Delta_\pi H = \widehat{Q}^\sharp P$, $H^\sharp \Delta_\pi H = P$, and $\frac{1}{2}(a, \Delta^{(2)}[H^\sharp J]a) = (C_2(a), H^\sharp J)$ give

$$F(T) = F(0) + (J, a) + \tfrac{1}{2}(a, \Delta_1 a) + \mathcal{R}(a),$$
$$\mathcal{R}(a) = -(D - C_2(a), H^\sharp J) - (\widehat{Q}a, PD) + \tfrac{1}{2}(D, PD) + V_0(T). \tag{3.mccxxii}$$

The pairings in each term are its physical fine or raw row pairing. The first two identities follow from $G^{-1}H = \widehat{Q}^\sharp \Sigma$ and $RD^\sharp H = 0$. The current completion vanishes bilinearly on two Landau fields, so $F''(0)$ may be replaced by Δ_π in their pairing. Differentiating eq. 3.mccxxii, using $(\mathrm{d}D)^\sharp_a = E_1(a)$, yields precisely

$$(F'(T), (\mathrm{d}T)_a \delta) = (J + \Delta_1 a + W_1(a), \delta)$$

for every slice vector δ. The six derivatives of $\mathcal{R}$ are the six terms of eq. 3.mccxv, with their displayed signs. On $\mathcal{K}$, Δ_1 is the precision inverted by $\mathcal{G}$, while $\Delta_1 H_1 B'$ and J annihilate $\mathcal{K}$. Thus the outer fixed-point equation makes $F'(\mathbf{H})$ annihilate the constrained tangent space.

Finite-dimensional annihilator duality now gives $F'(\mathbf{H}) = (\widehat{Q} + C'_{\mathbf{H}})^\sharp \lambda + DR\phi$. Pairing with H gives $\lambda = \omega_1(\mathbf{H})$, because $H^\sharp DR = 0$ and $H^\sharp F'(\mathbf{H}) = H^\sharp J + H^{\circledast}(\mathbf{H})$. The remaining gauge multiplier vanishes. Here no uniform second-derivative gauge estimate is needed: on each fixed finite lattice, the infinitesimal gauge directions with trivial reduced site averages map onto $\operatorname{ran} R$ under $RD^\sharp$ at zero, by the actual site inverse. Surjectivity persists in some neighborhood of zero. The action is invariant on those directions and their reduced constraint derivative is zero. For real data in this neighborhood, pair with a gauge direction whose $RD^\sharp$ value is $R\phi$. The result is $\|R\phi\|_2^2 = 0$. The residual is holomorphic throughout the uniform datum ball and vanishes on its real neighborhood of zero; the identity theorem extends its vanishing to that ball. This proves the last equation in eq. 3.mccxxi.

Differentiating $e^{i\eta A}$ shows directly that $F'(A) = f_A J(e^{i\eta A} U_0)$. Inserting this in eq. 3.mccxviii gives the physical current identity. The projection identity follows from $RD^\sharp \mathbf{H} = 0$. Conversely a sufficiently small critical field on this slice has a unique multiplier, since $(\widehat{Q} + C'_A)H = 1 + C'_A H$ is invertible. Reversing the straightening calculation makes it a solution of the same two contractive equations; uniqueness identifies it.

The node statements follow from the exact iterated averaging identity: each elementary node factor is the small logarithmic mean of finitely many transports in its block. Its convergent exponential and logarithm formulas read only that block. Successive factor errors have the length factors of the averaging iteration; their geometric sum gives the fixed ball and bound in eq. 3.cccxiii. At an outward row the only additional factor is the node factor of its preceding lower block. Solving the triangular near-identity change of raw data gives eq. 3.cccxii, with the same complete read. These identities also prove the covariance and transport assertions. □

Proposition 3.177. The actual coupled system eq. 3.cdxliv, using the families of Lem. 3.172, has a unique small solution when $\gamma + |B'| \le cb^{-3}$, where $\gamma = |\mathbf{J}|_3$. In the centered multiplier variable $\mu = \lambda - H_s^\sharp \mathbf{J}$, it satisfies

$$\|T\|_{1,w} + |\mu|_w \le Cb|B'|_w.$$

It has the same holomorphy, weighted locality, covariance and transport properties on its own complete reads. The component restriction and support assertions apply to T and $\mu + \mathsf{a}B'$. The centered multiplier μ itself retains the local term $-\mathsf{a}B'$ outside the active components. At $B' = 0$, $T = \mu = 0$. At the whole-sequence all-one point specified in Prop. 3.176, over a physical critical reference and on the same closed determining stock, it is the same sufficiently small constrained critical field as Prop. 3.176, after the prescribed change of raw data. Equality of the two decoupled systems away from $s = 1$ is not asserted. ■

Proof. Substitute $\lambda = H_s^\sharp \mathbf{J} + \mu$ into eq. 3.cdxliv. Its constant term is $(H_{0,s}B', (\Sigma_s - \mathsf{a})B')$. On the ball $\|T\|_1 + |\mu| \le Cbr$, the remaining terms have Lipschitz constant at most $Cb^2(\gamma + r)$, by

Lem. 3.173. Decrease c so this is below $1/4$; the ball is invariant. The weighted subtraction proof and normally convergent iterates give all the quantitative and analytic assertions. Carrier supports give transport as in Prop. 3.175. For the component assertion retain the local multiplier term. Writing $N_T = V(T) - C_T'^{\sharp}\lambda$, the second equation is

$$\mu + \mathsf{a}B' = \Sigma_s[B' - C(T)] + H_s^{\sharp}N_T + \mathsf{a}C(T).$$

The first two terms have their indicated carrier supports, and the last reads only the next complete row neighborhood of the support of T. The reserved collars separate these reads between active wall components. For a component carrier $\mathbb{Y}_i = \mathbb{Y}^1(c_i)$, the field T_i is supported over $\mathbb{Y}_i$, and $\mu_i + \mathsf{a}B'$ is supported on its first complete row neighborhood. The datum is read on that neighborhood: the local term $-\mathsf{a}B'$ also enters $C_{T_i}'^{\sharp}\lambda_i$. These reads lie in $\mathbb{Y}^2(c_i)$. If T_i, μ_i are obtained using the maps of component i, with the same full datum, the fixed-point equations and uniqueness give

$$T = \sum_i T_i, \qquad \mu = -\mathsf{a}B' + \sum_i (\mu_i + \mathsf{a}B').$$

In verifying these equations, λ and its component counterpart agree on every row read by $C_T'^{\sharp}$ in that component; their common remaining term is $-\mathsf{a}B'$. Thus the assertion also holds when no carrier survives: then $T = 0$ and $\mu = -\mathsf{a}B'$. For the physical identification start with the already constructed response of Prop. 3.176 on the common smaller datum ball. Put $K_T = \widehat{Q} + C_T'$ and $R_T = V(T) - C_T'^{\sharp}\lambda$. Its Euler equation and slice condition give

$$G_0^{-1}T = \widehat{Q}^{\sharp}[\lambda - \lambda_0 + \mathsf{a}\widehat{Q}T] - R_T.$$

The exact identities $\widetilde{G}\widehat{Q}^{\sharp} = 0$ and $\widetilde{G}G_0^{-1} = 1 - H_0\widehat{Q}$ give the first coupled equation. For the second, pair the Euler equation with H. Since H, T are Landau fields, their current-completion pairing vanishes, and

$$H^{\sharp}F''(0)T = H^{\sharp}\Delta_{\pi}T = (\Sigma - \mathsf{a})\widehat{Q}T, \qquad H^{\sharp}J = \lambda_0.$$

It follows that $\lambda = H^{\sharp}J + (\Sigma - \mathsf{a})\widehat{Q}T + H^{\sharp}R_T$, the second coupled equation. Contraction uniqueness identifies the two fields on this common ball. The coupled field's Euler, constraint, and Landau residuals are holomorphic on its entire connected datum ball and vanish on that smaller ball. The identity theorem proves their vanishing throughout the coupled ball. The smaller independent-current radius is part of this coupled-system statement and is not imposed on Prop. 3.174. □

Corollary 3.178. At the trivial reference, the Landau representative of a sufficiently small regular box minimizer is $\mathbf{H}(1; b(B))$, with its actual ordinary row logarithm B, the reduced datum $b(B)$ of eq. 3.cccxi, and its actual node factors. Suppose a common local gauge gives $|B(c)| \le C\xi_c a_0$ on the full determining stock. Then the potential has size Ca_0, and its first differences have size Ca_0/ξ_y at cell y. On complete stencils its normalized first differences have a fixed Hölder bound. The datum derivative has exponentially screened first-order and Hölder output bounds. Node factors on a complete common top pillar have a bounded holomorphic chart and the corresponding Cauchy difference bound. ■□

Proof. The trivial reference is critical and has zero current. The characterization in Prop. 3.176 applies to the small Landau representative. The outward datum change reads one preceding block only, with neighboring row lengths in a fixed ratio; it preserves the profile $\xi_c a_0$. Integrate the weighted datum derivative in eq. 3.mccxvii from zero. The ratio ξ_c/ξ_y is absorbed in the fixed reserved exponential rate of the assigned-cell kernels. This gives the two displayed sizes. The strong output conclusion following eq. 3.mccxvii gives the Hölder estimate. For the node factors use their complete-read holomorphic formula and fixed larger ball in Prop. 3.176; Cauchy's formula gives the difference bound on a smaller ball. These statements apply to the actual ordinary datum, and include its outward change of variables. □

Corollary 3.179. The response hypotheses in eq. 2.x are supplied by Props. 3.174 to 3.176 on the original first-order regular-pair domain, with the fixed small radii above. For any function F whose complete core contains $(\rho_X P) \dotplus \nu\mathfrak{h}_Y$ on its stated ν-disc, eq. 2.xi is a holomorphic interpolation with the wall splitting and rooted alternating-bracket properties stated there. A change of units multiplies its potential by L^{-m} and its current by L^{-3m}, as in eq. 2.viii. Core inclusion is checked for the particular function and its full auxiliary disc; it is separate from solving the response equations. ■□

Proof. The first assertion collects the preceding results in the specified raw-coordinate convention. The shift is the small-branch product in eq. 2.ix; its logarithm is holomorphic on the indicated core. Composition gives the interpolation. The support and splitting proof in Prop. 3.175 gives the vanishing required for the alternating brackets, and the Cauchy calculation following eq. 2.xiii applies on the complete disc. The last assertion is the physical scaling of a potential and a current. □

3.43. **A strong field–multiplier inverse at every homogeneous depth.** The energy gap of the homogeneous Landau precision can be upgraded to the strong physical norm needed for a nonlinear background equation. We prove that upgrade for the actual flat rows, uniformly in the number of fine bonds in a block. The argument uses the already proved energy gap, but assumes no supremum-norm vector inverse. In particular an order-zero singular integral is not applied at the L^∞ endpoint.

Fix the odd blocking factor $L \geq 3$ and an integer depth $k \geq 0$. The coarse spacing is one, the fine spacing is $h = L^{-k}$, and the periodic side is an integer $M \geq 2$. A real flat reference may have arbitrary commuting unitary holonomies. Work in its parallel frame on the universal cover. Fields are then quasiperiodic; all local norms and matrix products below are understood in that frame. Fine pairings use h^4 times counting measure and coarse pairings use counting measure. The transpose $\sharp$ always uses those pairings. Complexification is linear, with the analytic bilinear transpose, rather than conjugating a complex parameter.

Fix $0 < \alpha < 1$. For a fine field or site function define

$$\begin{aligned} \|u\|_{X_\alpha} &= \|u\|_\infty + \|Du\|_\infty + [Du]_{\alpha,\mathrm{loc}}, \\ \|u\|_{Y_\alpha} &= \|u\|_\infty + \|Du\|_\infty + \|D^2u\|_\infty + [D^2u]_{\alpha,\mathrm{loc}}, \\ \|u\|_{Z_\alpha} &= \|u\|_\infty + [u]_{\alpha,\mathrm{loc}}. \end{aligned} \tag{3.mccxxiii}$$

All products of forward differences of the indicated order are included. The local seminorm compares pairs at physical distance at most one, using parallel transport of the flat reference. Differences at larger distances are controlled by the supremum term. This definition is independent of a choice of fundamental cube, up to fixed constants.

Let q be the actual flat site average, Q the actual flat straightened field row, and D, d the flat gradient and curl. Thus

$$(q\lambda)(y) = h^4 \sum_{x \in B(y)} \lambda(x), \qquad (Qv)_\mu(y) = h^5 \sum_{x \in B(y)} \sum_{t=0}^{h^{-1}-1} v_\mu(x + the_\mu), \quad QD = \mathrm{d}_c q. \tag{3.mccxxiv}$$

The formula is in the parallel frame; the corresponding transports are restored on the quotient torus. The plain nonlinear group average has a different derivative before straightening.

3.43.1. *Smooth exact right inverses of the actual rows.*

Lemma 3.180 (Uniform physical row seeds)**.** There are exact right inverses E_q, E_Q with

$$qE_q = 1, \qquad QE_Q = 1, \qquad \|E_q b\|_{Y_\alpha} + \|E_Q c\|_{Y_\alpha} \leq C_\alpha(\|b\|_\infty + \|c\|_\infty). \tag{3.mccxxv}$$

Their columns have fixed physical support and bounded overlap. They also have uniformly bounded physical H^1 energy from coarse ℓ^2. These constants are independent of h, M and flat holonomy. ■

Proof. Write $h = 1/n$, with n odd. Choose an even nonnegative smooth function φ supported in $(-1/4, 1/4)$ and positive on $[-1/8, 1/8]$. Set

$$m_0 = h\sum_j \varphi(jh), \qquad m_2 = h\sum_j (jh)^2\varphi(jh). \tag{3.mccxxvi}$$

For $n \geq 9$ both have uniform positive lower bounds. Their limits are the positive integrals; the finitely many earlier values are positive, since $\pm h$ lie in the positive core. They have uniform upper bounds by the fixed support. The scalar site bump $m_0^{-4}\prod_\nu \varphi(x_\nu - y_\nu)$ is an exact column of E_q. Different pillars have disjoint supports.

For a positive coarse bond $c = (y, y + e_\mu)$ put

$$(E_Q c)_\mu(x) = (a_h + b_h(x_\mu - y_\mu))\prod_\nu \varphi(x_\nu - y_\nu), \qquad a_h = m_0^{-4}, \quad b_h = \frac{1-h}{2m_0^3 m_2}, \tag{3.mccxxvii}$$

with all other components zero, and multiply by the desired colour. Only the two rows c and $(y - e_\mu, y)$ can see this seed. At longitudinal coordinate $s = x_\mu - y_\mu$, their exact weights in eq. 3.mccxxiv are respectively $s + (1+h)/2$ and $(1-h)/2 - s$, with the common physical fine weight h^4. This follows by counting the possible starting points of a straight segment. In one dimension, at site index $j \in [-(n-1)/2, (n-1)/2]$, the two counts are $j + (n+1)/2$ and $(n-1)/2 - j$.

The transverse mean of the seed is m_0^3. Its longitudinal zeroth and first moments are $a_h m_0$ and $b_h m_2$. The two row values are therefore one and zero, exactly. The coefficients a_h, b_h are uniformly bounded, so these signed smooth seeds and all their fixed-order physical differences are uniformly bounded.

For the finitely many $n < 9$, use direct seeds. The site row has the right inverse $q^\sharp$. For a field row use its middle positive face bond, at longitudinal index $(n-1)/2$ and transverse block centres. Its coefficient in that row is h^4, and its coefficient in the preceding row is zero. Put h^{-4} times the prescribed colour there. This includes $n = 1$, when $Q = 1$. All finite-difference norms are bounded by a fixed constant in this finite range of n. On a two-cell periodic direction the two positive coarse bonds remain distinct, so the same count applies.

The columns have bounded support and overlap. Squaring their pointwise first-difference bounds and summing with physical measure proves the $\ell^2 \to H^1$ energy bounds. Their continuous smooth profiles, or the fixed coarse-mesh difference bounds, prove eq. 3.mccxxv. Unit parallel transports preserve all these estimates and give the construction for arbitrary flat holonomy. □

3.43.2. *The heat regularization used in the strong norm.* For a field u write $\|u\|_{2,\mathrm{cell},\infty} = \sup_y \|u\|_{L^2(B_y)}$. Let $L = -\sum_\mu D_\mu^- D_\mu^+$ and $P_t = e^{-tL}$. The differentiated heat kernels in the discrete Hölder estimate give, for $0 < t \leq 1$,

$$\|D^j p_t\|_1 \leq C_j (t + h^2)^{-j/2}. \tag{3.mccxxviii}$$

At $t = 1$ their physical L^2 norms on successive unit cells have summable tails. The differentiated fixed-time Fourier bound gives a uniform pointwise bound, and the proved heat-tail estimate gives arbitrarily fast decreasing cell L^1 tails. Apply $\|k\|_{L^2(C)}^2 \leq \|k\|_\infty \|k\|_{L^1(C)}$ with a sufficiently high tail power. Equivalently use the pointwise estimate eq. 3.dclxxxv directly. Consequently $D^j P_1$ maps the cell-L^2 supremum norm to L^∞, uniformly in the mesh. Twisted torus kernels are periodizations with unitary phases; their absolute bounds are dominated by the same infinite-lattice sums.

Lemma 3.181 (Strong estimates from a bounded scalar Laplacian)**.** If $Lu = g$, then

$$\|u\|_{X_\alpha} \leq C_\alpha(\|u\|_{2,\mathrm{cell},\infty} + \|g\|_\infty). \tag{3.mccxxix}$$

If $g \in Z_\alpha$, then

$$\|u\|_{Y_\alpha} \leq C_\alpha(\|u\|_{2,\mathrm{cell},\infty} + \|g\|_{Z_\alpha}). \tag{3.mccxxx}$$

The same estimates hold with a fixed small exponential spatial weight, using local cell seminorms for the weighted norms. ■

Proof. The exact finite-time identity is

$$u = P_1 u + \int_0^1 P_t g \, dt. \tag{3.mccxxxi}$$

The first term and all its fixed derivatives are controlled by the cell-L^2 norm. The zeroth and first derivatives of the integral are controlled by eq. 3.mccxxviii, since the power $1/2$ is integrable. For a grid translation of length $a \le 1$, the first-derivative kernel difference is bounded in L^1 by

$$C \min\{(t+h^2)^{-1/2},\ a(t+h^2)^{-1}\}. \tag{3.mccxxxii}$$

Use the first estimate directly and the second by telescoping one more difference. Splitting the time integral at $t = a^2$ gives $Ca(1+\log(1/a)) \le C_\alpha a^\alpha$. This proves eq. 3.mccxxix.

For two derivatives use the cancellation of $D^2 p_t$ against a constant. Subtract $g(x)$ inside its convolution. The spatial α moment of the differentiated heat bound gives

$$\|D^2 P_t g\|_\infty \le C_\alpha (t+h^2)^{\alpha/2-1} \|g\|_{Z_\alpha}. \tag{3.mccxxxiii}$$

This is integrable because $\alpha > 0$. A translated difference is bounded by twice that expression. When $a \le \sqrt{t+h^2}$ one can instead use $Ca(t+h^2)^{(\alpha-3)/2}\|g\|_{Z_\alpha}$: telescope an extra kernel difference and use its cancellation, subtracting the forcing at a fixed endpoint. The translation is within the diffusion length, so its moment has the same scale. Thus the minimum of these two bounds is valid. Splitting again at a^2 gives $C_\alpha a^\alpha$. This proves the last seminorm in eq. 3.mccxxx. The other terms were already bounded.

We specify why a small exponential weight causes no mesh loss. For a constant vector s, conjugation by $e^{s\cdot x}$ replaces the symbol of L by its value at $\xi + is$. Its real part is bounded below by $c|\xi|^2 - C|s|^2$ on the Brillouin cube. The same differentiated Fourier calculation as for the heat tails, now on the shifted contour, gives

$$\|e^{s\cdot x} D^j p_t(x)\|_1 \le C_j (t+h^2)^{-j/2} \quad (0 < t \le 1,\ |s| \le s_0), \tag{3.mccxxxiv}$$

with the analogous fixed-time cell-L^2 and increment bounds. One can see the uniformity directly from $\mathrm{Re}\,\lambda_h(\xi+is) = 2h^{-2}\sum_\mu\{1-\cosh(hs_\mu)\cos(h\xi_\mu)\}$: the negative part is at most $C|s|^2$, and every derivative of the remaining multiplier has the same scaled Gaussian bound. The estimate for $e^{s|x|_1}$ follows by summing the sixteen sign choices. A one-Lipschitz spatial weight has ratio at two points at most $e^{s|x-y|_1}$, so the same convolution estimates apply. Local Hölder differences read weights at neighboring cells, whose ratios are bounded. This proves the weighted assertion. □

3.43.3. *Strong smoothing by the actual site projector.* Put

$$G_s = (L + q^\sharp q)^{-1}, \quad F = G_s q^\sharp, \quad M_s = F^\sharp F, \quad P = F M_s^{-1} F^\sharp, \quad R = 1 - P. \tag{3.mccxxxv}$$

The proved site/Gram energy package gives exponential physical cell-L^2 entries of G_s, F, DF, M_s^{-1}. These statements are applicable here before any vector inverse is constructed.

For coarse $b \in \ell^\infty$, weighted cell summation gives $\|Fb\|_{2,\mathrm{cell},\infty} + \|qFb\|_\infty \le C\|b\|_\infty$. The actual scalar equation is

$$L(Fb) = q^\sharp (b - qFb). \tag{3.mccxxxvi}$$

Here $q^\sharp$ extends a coarse colour as a constant in its cell in the parallel frame, so its supremum norm is unchanged. Thus Lem. 3.181 proves

$$\|Fb\|_{X_\alpha} \le C_\alpha \|b\|_\infty, \qquad \|DFb\|_{Z_\alpha} \le C_\alpha \|b\|_\infty. \tag{3.mccxxxvii}$$

The weighted forms follow from the same scalar equation and eq. 3.mccxxxiv.

The adjoints $F^\sharp, (DF)^\sharp$ map fine cell-L^2 supremum norms to coarse ℓ^∞: sum their exponentially decreasing cell operator entries. Composing these estimates gives

$$\begin{aligned} \|Pu\|_{X_\alpha} &\le C_\alpha \|u\|_{2,\mathrm{cell},\infty}, \\ \|DPD^\sharp u\|_\infty &\le C \|u\|_{2,\mathrm{cell},\infty}. \end{aligned} \tag{3.mccxxxviii}$$

In the second line use the exact factorization $DPD^\sharp = (DF)M_s^{-1}(DF)^\sharp$. No absolute pointwise kernel bound was inferred from a cell-L^2 bound. Strong estimates were obtained only after the scalar heat regularization. In particular R is bounded on Z_α.

If $\rho \in \operatorname{ran} R \cap Z_\alpha$, the exact range identities give $qG_s\rho = 0$ and $LG_s\rho = \rho$. Using the site energy bound and eq. 3.mccxxx,

$$\|G_s\rho\|_{Y_\alpha} \le C_\alpha \|\rho\|_{Z_\alpha}. \tag{3.mccxxxix}$$

This estimate will be used for gauge restoration, rather than an unbounded point-evaluation map on an energy space.

3.43.4. *The full vector inverse and its multiplier.* Define the actual flat penalized operator

$$H_0 = \mathrm{d}^\sharp \mathrm{d} + DRD^\sharp, \qquad K_0 = H_0 + Q^\sharp Q. \tag{3.mccxl}$$

The uniform flat Landau gap proves $\langle v, K_0 v\rangle \ge c(\|Dv\|_2^2 + \|v\|_2^2)$. Conjugating by a small physical exponential weight changes this form by at most $C|s|$ times its two H^1 norms. For the nonlocal part use the weighted factorization in eq. 3.mccxxxviii; for Q use its fixed carrier and physical row/column bounds. The normalized Neumann series in H^1, H^{-1} therefore gives exponential cell-L^2 bounds for K_0^{-1}, uniformly in h and the torus. In particular, for $u = K_0^{-1} f$,

$$\|u\|_{2,\mathrm{cell},\infty} \le C\|f\|_\infty. \tag{3.mccxli}$$

Flat Hodge theory is the exact operator identity $\mathrm{d}^\sharp \mathrm{d} + DD^\sharp = LI$. Hence

$$Lu = f + DPD^\sharp u - Q^\sharp Q u. \tag{3.mccxlii}$$

The second term is bounded by eq. 3.mccxxxviii. The row Q maps the cell-L^2 supremum norm to coarse ℓ^∞. Its transpose maps coarse ℓ^∞ to fine ℓ^∞: in eq. 3.mccxxiv each fine coefficient is at most h^4 and each fine bond occurs in a fixed number of coarse rows. The physical transpose divides by h^4, giving a bounded density, not h^{-4} growth. Therefore the right side of eq. 3.mccxlii is bounded by $C\|f\|_\infty$. Apply Lem. 3.181 to obtain the actual strong inverse

$$\|K_0^{-1} f\|_{X_\alpha} \le C_\alpha \|f\|_\infty. \tag{3.mccxliii}$$

The argument with spatial weights gives its exponentially weighted version as well.

Proposition 3.182 (Uniform strong joint inverse)**.** For every bounded fine f and coarse b, the system

$$H_0 A - Q^\sharp \omega = f, \qquad QA = b \tag{3.mccxliv}$$

has a unique solution, and

$$\|A\|_{X_\alpha} + \|\omega\|_\infty \le C_\alpha(\|f\|_\infty + \|b\|_\infty). \tag{3.mccxlv}$$

It satisfies the corresponding small exponentially weighted estimate. The constants are uniform in homogeneous depth, volume and flat holonomy. This is an inverse for the penalized equations; removing their gauge penalty in a nonlinear solution requires a separate Ward argument. ∎

Proof. Put $S = QK_0^{-1}Q^\sharp$. It has a uniform positive lower bound. Indeed the actual lift E_Q from Lem. 3.180 has bounded K_0 energy, and

$$\|b\|_2^2 = \langle K_0 K_0^{-1} Q^\sharp b, E_Q b\rangle \le C\langle b, Sb\rangle^{1/2}\|b\|_2. \tag{3.mccxlvi}$$

The upper bound follows from the full gap and the bounded row norm. The weighted vector inverse and the fixed row carrier show that the conjugate of S differs from S by $O(|s|)$ in operator norm. Its inverse is therefore exponentially decreasing on the coarse lattice, and S^{-1} is bounded on coarse ℓ^∞ and its small weighted variants.

Set

$$\sigma = S^{-1}(b - QK_0^{-1} f), \qquad A = K_0^{-1}(f + Q^\sharp \sigma), \qquad \omega = \sigma - b. \tag{3.mccxlvii}$$

Direct substitution gives eq. 3.mccxliv. The coarse bound on σ and eq. 3.mccxliii give eq. 3.mccxlv. Uniqueness follows from positivity of S and K_0. Every step retains the full multiplier; no supremum bound for a field-only inverse is used in its place. □

There is one more strong inverse needed for the nonlinear gauge argument. On site functions the map

$$\lambda \longmapsto (q\lambda, RL\lambda) : Y_\alpha \longrightarrow \ell^\infty(\text{coarse sites}) \oplus (\operatorname{ran} R \cap Z_\alpha) \tag{3.mccxlviii}$$

has the explicit inverse

$$\lambda = E_q c + G_s\{\rho - RLE_q c\}. \tag{3.mccxlix}$$

The braces belong to $\operatorname{ran} R$; $qG_sR = 0$ and $LG_sR = R$ prove both equations. The bounds for E_q, R and eq. 3.mccxxxix prove a uniform Y_α bound. If both outputs vanish, $\lambda \in \ker q$ implies $L\lambda \in \operatorname{ran} R$, so $L\lambda = 0$ and the site gap gives $\lambda = 0$. This establishes uniqueness in the stated strong space, including nontrivial flat holonomy.

3.44. **An actual homogeneous Wilson background on a fixed coarse ball.** The strong inverse just proved permits a construction of the nonlinear background itself. The data are arbitrary small coarse bond logarithms relative to a flat connection. Their smallness is measured in the physical coarse supremum norm; neither volume nor fine depth reduces the radius. This class includes spatially varying noncommuting coarse connections. It is smaller than the class of all small-curvature coarse fields on arbitrarily large tori. Proper regions and hereditary decoupling domains are separate questions.

Keep the setting of § 3.43. Let V_0 be a real flat coarse connection and choose a flat fine extension W_0 with $M^k W_0 = V_0$. To obtain one, choose commuting logarithms of its four holonomies, use their constant fine representatives, and extend the coarse node gauge to the fine nodes. Gauge covariance of the actual mean, and its value on identical transported paths, give the equality. Use the parallel frame of W_0 and put

$$U_A = e^{ihA}W_0, \qquad V_B = e^{iB}V_0, \qquad \mathscr{A}_h(A) = \sum_{p>0} \left(1 - \tfrac{1}{2}\operatorname{tr} U_A(p)\right). \tag{3.mccl}$$

The sum uses each positive unoriented plaquette once. The colour pairing is $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr}(XY)$. Thus the flat quadratic action is $\frac{1}{2}\|\mathrm{d}A\|_2^2$ with fine measure h^4. Complex parameters mean holomorphic continuation to $\mathrm{SL}(2,\mathbb{C})$, without inserting a real-part operation.

Proposition 3.183 (A depth-uniform homogeneous critical orbit)**.** Fix L and $0 < \alpha < 1$. There are $r, \rho, \delta, C > 0$, independent of depth, volume and flat holonomies, such that for $\|B\|_\infty < \rho$ there is a holomorphic pair $(A(B), \omega(B))$ with

$$\begin{aligned} q_h(A(B)) &= B, \qquad RD^\sharp A(B) = 0, \\ \mathscr{A}_h'(A(B)) &= q_h'(A(B))^\sharp \omega(B), \\ \|A(B)\|_{X_\alpha} + \|\omega(B)\|_\infty &\le C\|B\|_\infty. \end{aligned} \tag{3.mccli}$$

The actual straightened nonlinear average q_h is defined below; R, D are the fixed flat site projector and gradient. The pair is real for real B. A specified holomorphic fine gauge converts U_A into $\widehat{U}(B)$ satisfying the literal constraint $M^k\widehat{U}(B) = V_B$. For real data this is a critical point of the actual Wilson action on that constraint manifold.

Among node-fixed gauge orbits possessing a representative $U = e^{ihA}W_0$ with $\|A\|_{X_\alpha} < \delta$, its orbit is the only constrained-critical orbit. The representative with $q_h(A) = B$ and $RD^\sharp A = 0$ has a strict local action minimum on that simultaneous slice. The literal representative is a local minimum modulo node-fixed gauges. All maps have fixed absolute complex radii inside the actual logarithm charts and exponentially localized response bounds. ■

The fixed reference gauge distinguishes the orbit during construction. It is not identified with a background-dependent Landau gauge. Its penalty will be removed by the exact nonlinear Ward identity.

3.44.1. *The actual nonlinear row and its physical transpose.* On a one-step lattice with spacing set to one, let $T_{yx}(W)$ be the specified internal mean transport and put

$$\begin{aligned} u_W(a)(y) &= \exp\left[L^{-4}\sum_{x\in B_L(y)}\log\{T_{yx}(e^{ia}W)T_{yx}(W)^{-1}\}\right], \\ F_W(a)(c) &= \frac{1}{i}\log\left[u_W(a)(c_-)^{-1}M(e^{ia}W)(c)u_W(a)(c_+)M(W)(c)^{-1}\right]. \end{aligned} \tag{3.mcclii}$$

The transports use the actual Federbush mean. The auxiliary u_W is the explicit exponential of averaged relative logarithms. Set $a_0 = hA$, $v_0 = 1$, $W_s = M^s W_0$, and define

$$a_{s+1} = F_{W_s}(a_s), \qquad v_{s+1} = v_s|_{\mathrm{next}} u_{W_s}(a_s), \qquad q_h(A) = a_k, \qquad v_h(A) = v_k. \tag{3.mccliii}$$

At depth zero $q_1(A) = A$ and $v_1 = 1$. Solving the second line of eq. 3.mcclii for the plain mean, and applying its gauge covariance inductively, proves

$$M^k(U_A) = \left(e^{iq_h(A)}V_0\right)^{v_h(A)}, \qquad U^g(c) = g(c_-)U(c)g(c_+)^{-1}. \tag{3.mccliv}$$

In particular the order of the two factors $v_s u_s$ is fixed.

All W_s are one in the parallel frame. The finite analytic coefficient estimates proved for this actual recursion give

$$\|D_a F_1(a)\|_{\mathrm{row}} \le L(1 + C_L\|a\|_\infty), \qquad \|D_a F_1(a)\|_{\mathrm{col}} \le L^{-3}(1 + C_L\|a\|_\infty). \tag{3.mcclv}$$

These are counting-coordinate sums of colour matrix norms. At zero they are the exact straight-row values. The analytic chart is $\|a\|_\infty < c_L/L$. Put $\xi_s = hL^s$. A first-exit argument gives

$$\|a_s\|_\infty \le 2r\xi_s \quad (0 \le s \le k), \qquad \sum_{s<k}\xi_s \le (L-1)^{-1}. \tag{3.mcclvi}$$

Indeed $p_s = \|a_s\|_\infty/\xi_s \le 2r$ implies $p_{s+1} \le p_s(1 + C_L\xi_s p_s)$. Choose $2C_L r/(L-1) < \log 2$ and multiply. The first exit contradicts this product. Decrease r to keep every raw input strictly inside its chart. Since $\|u_{W_s}(a_s) - 1\| \le C_L\|a_s\|_\infty$, also

$$\|v_h(A) - 1\|_\infty \le C_L r, \qquad \sup_y \|v_h(A)(y)\|\|v_h(A)(y)^{-1}\| \le e^{C_L r}. \tag{3.mcclvii}$$

These are complex-ball bounds; unitary reference twists preserve them.

The actual row calculation and its fixed-input-leg Hessian bound give on this entire fine supremum ball

$$\begin{aligned} q_h(0) = 0, \qquad q_h'(0) = Q, \\ \|q_h'(A) - Q\|_{\mathrm{row}} \le C_L\|A\|_\infty, \qquad \|q_h'(A) - Q\|_{\mathrm{col}} \le C_L h^4\|A\|_\infty, \\ \sup_b \sum_{b'} \|\partial_b\partial_{b'} q_{h,c}(A)\| \le C_L h^4. \end{aligned} \tag{3.mcclviii}$$

Every fine input affects a fixed number of coarse row carriers. The last line is eq. 3.dcii at output scale one. Its fine density follows from one inner column sum, one inner row sum and the geometric sum over the differentiated scale. Integrate that bound from zero to A for the column error; sum over the $O(h^{-4})$ row inputs for the row error. A bound on second directional derivatives alone would not justify this step.

Writing $C_h(A) = q_h(A) - QA$, one obtains

$$\|C_h(A)\|_\infty \le C_L\|A\|_\infty^2, \qquad \|(q_h'(A) - Q)^\sharp\omega\|_\infty \le C_L\|A\|_\infty\|\omega\|_\infty. \tag{3.mcclix}$$

The fine component of the transpose is $h^{-4}\sum_c (q_h'(A)-Q)^{\mathsf{T}}_{c,b}\omega(c)$. The column density in eq. 3.mcclviii cancels this h^{-4}. The same moment gives, on the joint radius-r ball,

$$\begin{aligned}\|C_h(A)-C_h(\widetilde{A})\|_\infty &\le C_L r\|A-\widetilde{A}\|_\infty,\\ \|(q_h'(A)-Q)^\sharp\omega-(q_h'(\widetilde{A})-Q)^\sharp\widetilde{\omega}\|_\infty &\le C_L r(\|A-\widetilde{A}\|_\infty+\|\omega-\widetilde{\omega}\|_\infty).\end{aligned} \tag{3.mcclx}$$

These estimates use the full finite-lattice constraint.

3.44.2. *The complete Wilson residual.* Let $J_h(A)$ be the normal-coordinate physical gradient: $\mathscr{A}_h'(A)[a] = h^4\sum_b\langle J_h(A)(b),a(b)\rangle$. The field equation needs this entire gradient, including its coordinate differential, rather than an action remainder alone.

Lemma 3.184 (Uniform quadratic Wilson residual)**.** The map $N_h(A)=J_h(A)-\mathrm{d}^\sharp\mathrm{d}A$ is holomorphic from a fixed X_α ball to fine ℓ^∞. On a smaller ball,

$$\|N_h(A)\|_\infty \le C\|A\|_{X_\alpha}^2, \qquad \|N_h(A)-N_h(\widetilde{A})\|_\infty \le Cr\|A-\widetilde{A}\|_{X_\alpha} \tag{3.mcclxi}$$

when both arguments have norm at most r. The constants are uniform in mesh and volume. ■

Proof. For $U_A(p)=e^{ih^2F_p(A)}$, the exact four-factor logarithm gives

$$F(A)=\mathrm{d}A+E_h(A), \qquad \|E_h(A)\|_\infty\le C\|A\|_\infty^2, \qquad \|DE_h(A)\|_\infty\le C\|A\|_\infty\|DA\|_\infty. \tag{3.mcclxii}$$

The linear logarithm is $h^2\mathrm{d}A$. Each remaining monomial contains at least two factors hA, so division by h^2 leaves a uniformly analytic function of the four neighbouring A values. Its derivative is bounded by $C\max|A|$. Subtract the same function at neighbouring plaquettes to prove the last bound. No D^2A is used.

The exact physical left current is eq. 3.dcci. Its covariant difference differs from the ordinary difference by at most $C\|A\|_\infty\|F\|_\infty$. Its sine factor obeys

$$W_h(F)=h^{-2}\sin(h^2F)=F+S_h(F), \quad |S_h(F)|\le Ch^4|F|^3, \quad |D^-S_h(F)|\le Ch^3\|F\|_\infty^3. \tag{3.mcclxiii}$$

The last estimate uses two bounded values, not a derivative of F. It follows that

$$J_h^{\mathrm{left}}(A)=\mathrm{d}^\sharp\mathrm{d}A+R_h(A), \qquad \|R_h(A)\|_\infty\le C(\|A\|_\infty\|DA\|_\infty+\|A\|_\infty^3+h^3\|F\|_\infty^3)\le C\|A\|_{X_\alpha}^2. \tag{3.mcclxiv}$$

No zero-divergence assumption is required: the ordinary divergence of the linear plaquette curl is $\mathrm{d}^\sharp\mathrm{d}A$ itself.

The exact change from normal to left coordinates is

$$\mathcal{E}_{hA}a=\int_0^1 \mathrm{Ad}(e^{ithA})a\,dt, \quad J_h(A)=\mathcal{E}_{hA}^\sharp J_h^{\mathrm{left}}(A), \quad \|\mathcal{E}_{hA}-1\|\le Ch\|A\|_\infty. \tag{3.mcclxv}$$

Although $\mathrm{d}^\sharp\mathrm{d}$ is not uniformly bounded from X_α to ℓ^∞, the elementary fine difference bound gives $\|\mathrm{d}^\sharp\mathrm{d}A\|_\infty\le Ch^{-1}\|DA\|_\infty$. The explicit h in the coordinate differential cancels that loss. This proves the first bound in eq. 3.mcclxi for the complete gradient. The same holomorphic estimates hold on a larger fixed complex ball. Banach-valued Cauchy estimates bound its second derivative uniformly on a smaller ball. Integrating that derivative, using $N_h(0)=DN_h(0)=0$, proves the stated Cr Lipschitz estimate. No coordinate-count factor enters this operator-norm argument. □

3.44.3. *The actual joint field–multiplier contraction.* Let $\mathcal{L}_0^{-1}(f,b)$ be the inverse in Prop. 3.182, with norm at most C_0. The actual penalized system

$$J_h(A)+DRD^\sharp A-q_h'(A)^\sharp\omega=0, \qquad q_h(A)=B \tag{3.mcclxvi}$$

is exactly equivalent to

$$(A,\omega)=\mathcal{T}_B(A,\omega):=\mathcal{L}_0^{-1}\big(-N_h(A)+(q_h'(A)-Q)^\sharp\omega,\,B-C_h(A)\big). \tag{3.mcclxvii}$$

For $\|(A,\omega)\| = \|A\|_{X_\alpha} + \|\omega\|_\infty \le r$, the preceding estimates imply

$$\|\mathcal{T}_B(x) - \mathcal{T}_B(\widetilde{x})\| \le C_1 r \|x - \widetilde{x}\|, \qquad \|\mathcal{T}_B(0)\| \le C_0 \|B\|_\infty. \tag{3.mcclxviii}$$

Choose r below all primitive ceilings and $C_1 r \le 1/4$, and $\rho \le r/(4C_0)$. The radius-$r/2$ ball is invariant and contractive. Iteration from zero gives a unique pair there. The same Lipschitz inequality identifies it with any solution in the radius-r ball, and

$$\|A(B)\|_{X_\alpha} + \|\omega(B)\|_\infty \le 2C_0 \|B\|_\infty. \tag{3.mcclxix}$$

The holomorphic iterations converge locally uniformly on the fixed complex polydisc. Their limit is holomorphic; reality and uniqueness give reality for real B. An unknown orbit or inverse is not a hypothesis of this construction.

3.44.4. *The effective coarse gauge and removal of the penalty.* The nonlinear straightened average has its own effective coarse gauge. One cannot obtain its Ward identity by replacing the flat site map by a convenient average. For $g = e^{-it\lambda}$ define

$$h_A(g) = v_h(A^g)^{-1} g|_{\mathrm{coarse}} v_h(A), \qquad h_A(e^{-it\lambda}) = 1 - it\tau_A\lambda + O(t^2), \tag{3.mcclxx}$$

where $U_{A^g} = U_A^g$. The exact factorization proves

$$e^{iq_h(A^g)} V_0 = (e^{iq_h(A)} V_0)^{h_A(g)}, \qquad q_h'(A)\Gamma_A\lambda = \Gamma^c_{q_h(A)}\tau_A\lambda. \tag{3.mcclxxi}$$

The symbols Γ_A, Γ^c_B are normal-coordinate infinitesimal fine and coarse gauge generators, with the same sign convention.

We derive estimates for these actual maps. Write $\mathcal{E}_z v = \int_0^1 \mathrm{Ad}(e^{itz}) v\, dt$. At one step in the flat frame put $T_{yx}(e^{ia}) = e^{iz_x}$, $m = L^{-4}\sum_x z_x$ and $u = e^{im}$. Direct differentiation gives

$$\tau_a^{(1)}\lambda(y) = \mathrm{Ad}(u(y)^{-1})\Big[\lambda(y) + \mathcal{E}_m L^{-4} \sum_{x \in B_L(y)} \mathcal{E}_{z_x}^{-1}\{-\lambda(y) + \mathrm{Ad}(T_{yx}(e^{ia}))\lambda(x)\}\Big]. \tag{3.mcclxxii}$$

Indeed the left logarithmic derivative of $T_{yx}(U^g) = g(y)T_{yx}(U)g(x)^{-1}$ is $i[-\lambda(y) + \mathrm{Ad}(T_{yx}(U))\lambda(x)]$. Convert it by $\mathcal{E}_{z_x}^{-1}$, average, and differentiate $u(A^g)^{-1}g(y)u(A)$. This gives the formula, including its sign. At zero it is exactly the one-step site average $q^{(1)}\lambda$. The finite one-step logarithm domain therefore gives

$$\|\tau_a^{(1)} - q^{(1)}\|_{\infty\to\infty} \le C_L \|a\|_\infty, \qquad \|\tau_a^{(1)}\|_{\infty\to\infty} \le 1 + C_L\|a\|_\infty. \tag{3.mcclxxiii}$$

The effective gauge parameters compose through successive scales. Telescoping that product and using eq. 3.mcclvi yields

$$\|\tau_A - q\|_{\infty\to\infty} \le C_L \|A\|_\infty. \tag{3.mcclxxiv}$$

This retains the node evaluation terms in eq. 3.mcclxxii; no fine L^2 point-evaluation bound is inferred from it.

The fine generator has the exact expression

$$\begin{aligned} \Gamma_A\lambda &= \mathcal{E}_{hA}^{-1} D_{U_A}\lambda, \\ (D_{U_A}\lambda)_\mu(x) &= h^{-1}\{\mathrm{Ad}(U_A(x,\mu))\lambda(x + he_\mu) - \lambda(x)\}. \end{aligned} \tag{3.mcclxxv}$$

It implies

$$\|(\Gamma_A - D)\lambda\|_{X_\alpha} \le C\|A\|_{X_\alpha}\|\lambda\|_{Y_\alpha}. \tag{3.mcclxxvi}$$

Its zeroth-order difference consists of $A\lambda$ and $hAD\lambda$ terms. One fine difference gives terms of the forms $DA\lambda$ and $AD\lambda$, together with $hDAD\lambda$ and $hAD^2\lambda$. Their analytic coefficients are uniformly bounded. The discrete product rule for the local Hölder seminorm proves the stated last seminorm as well. The inverse of $\mathcal{E}_{hA}$ is uniformly analytic on the chosen chart.

For $\rho \in \operatorname{ran} R \cap Z_\alpha$ put $\lambda_0 = G_s\rho$. The strong site inverse gives $q\lambda_0 = 0$, $RL\lambda_0 = \rho$ and $\|\lambda_0\|_{Y_\alpha} \le C\|\rho\|_{Z_\alpha}$. Since $\tau_A E_q = 1 + O(r)$, define

$$\lambda_A(\rho) = \lambda_0 - E_q(\tau_A E_q)^{-1}\tau_A\lambda_0. \tag{3.mcclxxvii}$$

Then $\tau_A\lambda_A = 0$ and $\|\lambda_A - \lambda_0\|_{Y_\alpha} \le Cr\|\rho\|_{Z_\alpha}$. Boundedness of R on Z_α and eq. 3.mcclxxvi show that

$$\mathscr{R}_A\rho = RD^\sharp\Gamma_A\lambda_A(\rho), \qquad \|\mathscr{R}_A - 1\|_{\operatorname{ran} R\cap Z_\alpha} \le Cr. \tag{3.mcclxxviii}$$

After decreasing r this is invertible by a Neumann series. Consequently for every such ρ there is a site function with

$$\tau_A\lambda = 0, \qquad RD^\sharp\Gamma_A\lambda = \rho. \tag{3.mcclxxix}$$

For the real fixed point choose $\rho = RD^\sharp A$. Test eq. 3.mcclxvi against the actual gauge vector $\Gamma_A\lambda$ from eq. 3.mcclxxix. Wilson gauge invariance annihilates its current term. The constraint term vanishes by eq. 3.mcclxxi and $\tau_A\lambda = 0$. The remaining equality is

$$0 = \langle\rho,\rho\rangle_{2,h}. \tag{3.mcclxxx}$$

Thus $RD^\sharp A = 0$ and the full unpenalized stationarity equation holds. The identity extends to the complex data ball by holomorphy from its real open part. This proves the equations in eq. 3.mccli.

The same argument with arbitrary λ gives

$$(\Gamma^c_B)^\sharp\omega = 0. \tag{3.mcclxxxi}$$

Indeed τ_A is onto with right inverse $E_q(\tau_A E_q)^{-1}$. The coarse frame term arising on differentiation of eq. 3.mccliv therefore contributes no extra physical force.

3.44.5. *A literal trial lift and admission of its fine gauge.* There is an explicit bounded extension of coarse node values to fine site functions. Choose fixed scalar smooth bumps supported inside the pillars and equal to one at their centers, transporting colours in the flat frame. Their supports are disjoint. Sample them on the fine lattice. At the finitely many coarse meshes use center Kronecker bumps, whose physical derivatives have a fixed bound. This gives

$$\mathrm{ev}_{\mathrm{coarse}}\, E_n = 1, \qquad \|E_n\nu\|_{Y_\alpha} \le C\|\nu\|_\infty. \tag{3.mcclxxxii}$$

For $w = e^{i\nu}$ close to one define $\mathfrak{E}(w) = e^{iE_n\nu}$. It is unitary for real ν, holomorphic for complex ν, and has exactly the prescribed node values.

The finite fine gauge transformation satisfies

$$\begin{aligned} A^{e^{-i\lambda}} &= A + D\lambda + \mathcal{B}_h(A,\lambda),\\ \|\mathcal{B}_h(A,\lambda)\|_{X_\alpha} &\le C(\|A\|_{X_\alpha} + \|\lambda\|_{Y_\alpha})\|\lambda\|_{Y_\alpha}. \end{aligned} \tag{3.mcclxxxiii}$$

The dependence is uniformly analytic. To check the apparent h^{-1}, write $\lambda(x + he_\mu) = \lambda(x) + hD_\mu\lambda(x)$ before taking the product and logarithm. At $h = 0$ the numerator is exactly one. Its logarithm divided by h therefore extends analytically to $h = 0$. Differences of this finite formula use only $A, DA, \lambda, D\lambda, D^2\lambda$ and their stated local seminorms. Its first-order terms are $A + D\lambda$. Taylor's formula proves eq. 3.mcclxxxiii and the corresponding local Lipschitz bound, uniformly in the mesh.

For any small A put

$$\widehat{U}_A = U_A^{\mathfrak{E}(v_h(A)^{-1})}. \tag{3.mcclxxxiv}$$

Its literal mean is $e^{iq_h(A)}V_0$. Its normal coordinate has X_α norm at most $C\|A\|_{X_\alpha}$ by eq. 3.mcclvii, eq. 3.mcclxxxii and eq. 3.mcclxxxiii. Choose the fixed-point radius below the primitive admission ceiling divided by this fixed constant. Then the original field, its literal representative, the specified fine gauges and all intermediate raw logarithms stay inside the actual complex charts. This proves admission for this gauge without assuming arbitrary complex gauge saturation of the domain.

A literal section can also be constructed independently of the action. On the range of the actual smooth seed E_Q, solve

$$A_{\mathrm{tr}} = E_Q\{B - C_h(A_{\mathrm{tr}})\}. \tag{3.mcclxxxv}$$

The seed bound and eq. 3.mcclx make this a supremum-norm contraction. Its solution has $q_h(A_{\mathrm{tr}}) = B$. Since it is E_Q applied to a bounded coarse vector, its whole Y_α norm is at most $C\|B\|_\infty$. Equation eq. 3.mcclxxxiv gives an actual literal trial lift. Its physical curvature is at most $C\|B\|_\infty$. Before the last gauge its physical current has the same bound by the exact current formula and the Y_α estimate. Gauge covariance preserves this bound for real data; the bounded condition number of the specified complex gauge gives it in the complex case. This supplies a genuine deep trial field with physical derivative bounds, rather than assuming the required trial lift.

3.44.6. *The mean-gauge slice and physical criticality.* For λ small in Y_α define

$$\mathscr{S}_A(\lambda) = \big(i\log h_A(e^{-i\lambda}),\ RD^\sharp A^{e^{-i\lambda}}\big). \tag{3.mcclxxxvi}$$

At $(A,\lambda) = (0,0)$ its derivative is $(q\lambda, RL\lambda)$, with explicit bounded inverse eq. 3.mccxlix. On a small $X_\alpha \times Y_\alpha$ ball that derivative differs from this fixed map by at most $C(\|A\|_{X_\alpha} + \|\lambda\|_{Y_\alpha})$. For the first component differentiate the exact h_A at the current point: its derivative is τ_{A^g} composed with the bounded near-identity exponential gauge differential and a near-identity conjugation. Use eq. 3.mcclxxiv. For the second use $RD^\sharp : X_\alpha \to Z_\alpha$ and eq. 3.mcclxxxiii. These bounds give the claimed operator norm between the precise strong spaces.

The contraction with linear part eq. 3.mccxlix therefore solves

$$h_A(e^{-i\lambda}) = v_h(A), \qquad RD^\sharp A^{e^{-i\lambda}} = 0, \qquad \|\lambda\|_{Y_\alpha} \le C\|A\|_{X_\alpha}. \tag{3.mcclxxxvii}$$

Explicitly subtract $\mathscr{S}_A(0)$ and move the quadratic remainder to the right after applying that inverse. The Lipschitz constant is $C(\|A\|_{X_\alpha} + \|\lambda\|_{Y_\alpha})$. A ball of radius $C'\|A\|_{X_\alpha}$ is invariant and contractive for a fixed smaller input radius. This proves existence, analytic dependence and uniqueness in this gauge chart.

If $M^k U_A = V_B$, write $g = e^{-i\lambda}$ for this solution and $\overline{A} = A^g$. Exact factorization gives

$$q_h(\overline{A}) = B, \qquad RD^\sharp \overline{A} = 0, \qquad v_h(\overline{A}) = g|_{\mathrm{coarse}}. \tag{3.mcclxxxviii}$$

The last equality follows from $v_h(\overline{A})^{-1} g|_{\mathrm{coarse}} v_h(A) = v_h(A)$. No coarse stabilizer is divided out. Consequently $\widehat{U}_{\overline{A}}$ differs from U_A by a gauge exactly equal to one at every coarse node.

These maps establish the critical-point correspondence with all tangent directions included. The map $A \mapsto \widehat{U}_A$ sends the entire local $q_h = B$ manifold into the literal fibre and preserves the action. Conversely eq. 3.mcclxxxvii sends the local literal fibre into $q_h = B$ and preserves the action. At a mean-slice point the two maps compose to the identity up to a node-fixed gauge: the inverse of its specified extension gauge solves eq. 3.mcclxxxvii, and uniqueness in the small gauge chart identifies it. Differentiating these action identities proves both directions of the critical-point correspondence. In particular the constructed fixed point is a literal constrained-critical orbit.

3.44.7. *Competing multipliers and orbit uniqueness.* Let A be any small mean-slice critical point. Its actual derivative is onto, with the exact right inverse

$$T_A = q_h'(A)E_Q = 1 + O(r), \qquad E_A = E_Q T_A^{-1}, \qquad q_h'(A)E_A = 1. \tag{3.mcclxxxix}$$

The error has coarse row and column sums at most Cr, by eq. 3.mcclviii and the seed carriers. Thus its Neumann inverse is bounded in both Schur norms. Columns of E_A have uniformly bounded physical $W^{1,1}$ norms and H^1 energy: sum the column ℓ^1 bound of T_A^{-1} against the corresponding fixed seed-column bounds.

Since $q_h'(A)$ is onto, constrained criticality gives a unique ω with $J_h(A) = q_h'(A)^\sharp\omega$. To bound it without an assumed D^2A estimate, evaluate the first variation on $E_A e_c$. The exact current identity, summed by parts once, gives

$$|\mathscr{A}_h'(A)[a]| \le C\|F(A)\|_\infty(\|D(\mathcal{E}_{hA}a)\|_{1,h} + \|A\|_\infty\|\mathcal{E}_{hA}a\|_{1,h}). \tag{3.mccxc}$$

The measure is h^4. This is summation by parts of the covariant difference of $W_h(F)$, whose supremum is at most $C\|F\|_\infty$. Its transport differs from the flat one by $Ch\|A\|_\infty$. Moreover $\|\mathcal{E}_{hA}\| \le C$ and $\|D\mathcal{E}_{hA}\| \le Ch\|DA\|_\infty$. The seed column bounds and eq. 3.mcclxii therefore imply

$$\|\omega\|_\infty \le C\|A\|_{X_\alpha}. \tag{3.mccxci}$$

There is no fine point-mass readout or hidden h^{-1} here.

A literal critical connection in the radius-δ chart maps to a mean-slice critical point of norm at most $C\delta$. Choose δ so that its field and the multiplier in eq. 3.mccxci have joint norm less than r. Decrease ρ further so that the literal constructed representative has strong norm below $\delta/2$. They solve eq. 3.mcclxvii, since the fixed gauge penalty vanishes. Contraction uniqueness identifies them with $(A(B), \omega(B))$. Equation eq. 3.mcclxxxviii then identifies the original connection with $\widehat{U}(B)$ modulo a node-fixed gauge. This proves the stated orbit uniqueness.

3.44.8. *A real minimum and the effective coarse action.* Keep $\omega = \omega(B)$ fixed and set $\mathscr{L}(A) = \mathscr{A}_h(A) - \langle\omega, q_h(A) - B\rangle$. On a fixed small strong ball,

$$\mathscr{L}''(A)[a,a] + \|RD^\sharp a\|_2^2 + \|q_h'(A)a\|_2^2 \ge c(\|Da\|_2^2 + \|a\|_2^2). \tag{3.mccxcii}$$

Here is the comparison with the proved flat gap. The exact twice-varied plaquette estimate eq. 3.dccxxxvi uses physical left variations. Normal variations are $\mathcal{E}_{hA}a$; their difference from a has coefficient $O(h\|A\|_\infty)$ and first-difference coefficients $O(h\|DA\|_\infty)$ and $O(h\|A\|_\infty)D$. The change in the curl form is at most $Cr\|a\|_{H^1}^2$. The second coordinate differential contributes a local coefficient $O(h)J_h^{\mathrm{left}}(A)$, bounded by Cr: the exact current formula gives $h\|J_h^{\mathrm{left}}(A)\|_\infty \le C\|DA\|_\infty + Ch\|A\|_{X_\alpha}^2$. The two-leg moment bounds the constraint curvature by $C\|\omega\|_\infty\|a\|_2^2$. Finally physical Schur applied to eq. 3.mcclviii gives $\|q_h'(A) - Q\|_{2\to 2} \le Cr$. Thus the entire left side differs from $\langle a, K_0 a\rangle$ by at most $C(r + \|\omega\|_\infty)\|a\|_{H^1}^2$, proving eq. 3.mccxcii after a fixed decrease.

Let $A_* = A(B)$ and let $A_* + a$ be another real mean-slice point with the same $q_h = B$. Then $RD^\sharp a = 0$. The equality of the two constraints and the two-leg moment imply, for $A_t = A_* + ta$,

$$\|q_h'(A_t)a\|_2 \le C\|a\|_\infty\|a\|_2. \tag{3.mccxciii}$$

Indeed $D^2q_h(A)[a,\cdot]$ has physical L^2 operator norm at most $C\|a\|_\infty$: the fixed-leg moment supplies its column density $Ch^4\|a\|_\infty$, and summing over a row's inputs supplies row sum $C\|a\|_\infty$. Apply physical Schur, then integrate in t. Since $\mathscr{L}'(A_*) = 0$, Taylor's formula gives

$$\mathscr{A}_h(A_* + a) - \mathscr{A}_h(A_*) \ge c'(\|Da\|_2^2 + \|a\|_2^2) \tag{3.mccxciv}$$

on a smaller fixed ball. The mean-gauge slice preserves action and is a local node-fixed orbit section of the literal fibre. This proves the minimum assertion.

Define $S_c(V_B) = \mathscr{A}_h(A(B))$ and continue holomorphically. It is locally coarse-gauge invariant. A sufficiently small coarse gauge has the fine extension eq. 3.mcclxxxii. The transformed critical field remains in the strong chart and is critical for the transformed literal coarse data. Orbit uniqueness identifies it with the new critical orbit, and fine gauge invariance identifies their actions. Thus the coarse invariance required when subtracting $S_c \circ M^k$ is a conclusion of this construction.

3.44.9. *Localized analytic response and the exact domain supplied.* Every carrier in the depth-k recursion has bounded physical diameter: the radii grow by a fixed multiple of ξ_s, and $\sum_{s<k}\xi_s \le (L-1)^{-1}$. The nonlinear Wilson term is local on the fine stencil. Therefore all the residual and transpose Lipschitz bounds above remain valid with a small fixed exponential weight, with constants multiplied by at most $e^{C_L s}$. For the Wilson residual use its local stencil and the same product estimates; for q_h and its transpose use their actual finite carriers and eq. 3.mcclviii. Write $w_z(x) = e^{sd(x,z)}$, where d is physical periodic distance and $s > 0$ is small enough for the weighted inverse of Prop. 3.182. Here $\|w_z u\|_{X_\alpha}$ denotes the cellwise weighted norm of u and its indicated local derivatives and seminorms; it does not mean differentiating the product $w_z u$.

Decrease r so that the derivative of the nonlinear fixed-point map has norm below $1/2$ in these weighted spaces as well. Differentiating eq. 3.mcclxvii gives the explicit Neumann inverse of $1 - D\mathcal{T}_B$. Thus

$$\|w_z DA(B)b\|_{X_\alpha} + \|w_z D\omega(B)b\|_\infty \le C\|w_z b\|_\infty. \tag{3.mccxcv}$$

The constant is uniform on a smaller unweighted complex data ball, and in z, k, M. The same proof shows analytic dependence on the unweighted parameter B with values in the weighted operator space. Banach-valued Cauchy differentiation in the remaining $m-1$ directions consequently gives

$$\begin{aligned}\|w_z D^m A(B)[b_1,\dots,b_m]\|_{X_\alpha} + \|w_z D^m\omega(B)[b_1,\dots,b_m]\|_\infty \\ \le C_m\|w_z b_1\|_\infty \prod_{j=2}^m \|b_j\|_\infty.\end{aligned} \tag{3.mccxcvi}$$

Any input can occupy the first slot by symmetry. For unit inputs at coarse bonds $y_1,\dots,y_m$, choosing $z = y_j$ for each j gives decay $C_m e^{-s\max_j d(x,y_j)}$. Since $\max_j d(x,y_j) \ge m^{-1}\sum_j d(x,y_j)$, every fixed jet has all spatial moments, with constants uniform in depth and volume. The literal representative has the same property: v_h has uniformly bounded analytic derivatives and bounded carriers by its explicit formula; E_n is local; and the gauge change is uniformly analytic in the stated strong norms.

These estimates provide an actual fixed-radius homogeneous background, its multiplier, a literal trial lift, a local minimum, local coarse gauge invariance and localized analytic response. They do not provide independent fine decoupling domains, a hereditary tower, mixed-scale proper-region inversion, or extension to every curvature-only regular coarse field. Those are separate model estimates, not conclusions hidden in the domain of this proposition.

3.45. **The second jet and fixed-leg density of the actual hierarchy.** The straightened nonlinear row in eq. 3.mccliii is obtained by composing the actual one-step rows. This fact is essential for its fixed-fine-leg moment. We compute the one-step second jet before estimating the composition. A single mean over a block of side L^k would be a different nonlinear map and does not replace this computation.

For Hermitian logarithms write $[X,Y]_{\mathrm{H}} = i[X,Y]$; thus the quadratic term in $i^{-1}\log(e^{iX}e^{iY})$ is $\frac{1}{2}[X,Y]_{\mathrm{H}}$. At one output row let t_x, t'_x, ℓ_x denote the degree-one logarithms of its source internal transport, target internal transport, and straight length-L line. A bar denotes the normalized average over the L^4 starting sites. Subscripts two denote homogeneous degree-two terms. Define

$$\mathrm{Cov}_{\mathrm{H}}(a,b) = \overline{[a_x, b_x]_{\mathrm{H}}} - [\bar a, \bar b]_{\mathrm{H}}.$$

Lemma 3.185 (The actual one-step quadratic coefficient)**.** The quadratic coefficient of the literal plain mean is

$$\begin{aligned}(M_{\log})_2 = \bar t_2 + \bar\ell_2 - \bar t'_2 \\ + \frac{1}{2}\overline{[t,\ell]_{\mathrm{H}} - [t,t']_{\mathrm{H}} - [\ell,t']_{\mathrm{H}}}.\end{aligned} \tag{3.mccxcvii}$$

For the actual straightened map F_1 in eq. 3.mcclii, its linear coefficient is the straight row $Q_1 = \bar{\ell}$, and

$$(F_1)_2 = \bar{\ell}_2 + \frac{1}{2}\{\mathrm{Cov}_{\mathrm{H}}(t,\ell) - \mathrm{Cov}_{\mathrm{H}}(t,t') - \mathrm{Cov}_{\mathrm{H}}(\ell,t')\}. \tag{3.mccxcviii}$$

In particular all second jets of the internal transports cancel from the straightened row. ■

Proof. For a normalized group mean, the logarithm of its output through degree two is the average of the input logarithms. Indeed its defining equation is the average of $\log(e^{iz_x}e^{-im}) = 0$; the degree-two commutator containing $m_1 = \bar{z}_1$ averages to $[\bar{z}_1, \bar{z}_1] = 0$. This proves the assertion for the internal implicit mean. The outer operation is explicitly the exponential of the average of closed logarithms, so its logarithm is that arithmetic average exactly. Its second jet has the same property without replacing it by an implicit mean.

Let c be the logarithm of the central straight reference path. The actual closed inputs of the outer mean are $T_x e^{i\ell_x}(T'_x)^{-1}e^{-ic}$. Expand their logarithms and then multiply the output by e^{ic}, as in the actual definition of M. The terms containing c_1 cancel: the averaged commutator with $-c_1$ is cancelled by the final product with e^{ic_1}. The linear result is $\bar{t}+\bar{\ell}-\bar{t}'$, and its quadratic coefficient is eq. 3.mccxcvii. This also checks that the central reference path has not been silently discarded before taking a jet.

The straightening is $e^{-i\bar{t}_{\log}}M(e^{ia})e^{i\bar{t}'_{\log}}$, where the barred logarithms include their second coefficients. Its first coefficient is $\bar{\ell}$. The two endpoint exponentials cancel $\bar{t}_2 - \bar{t}'_2$ exactly. Their additional quadratic commutators are

$$\frac{1}{2}\{-[\bar{t},\bar{\ell}]_{\mathrm{H}} + [\bar{t},\bar{t}']_{\mathrm{H}} + [\bar{\ell},\bar{t}']_{\mathrm{H}}\}.$$

Combining them with the averaged commutators gives eq. 3.mccxcviii. □

Lemma 3.186 (One-step rows and one fixed input leg)**.** On a raw fine-angle ball $\|a\|_\infty < c/L$, with $c > 0$ universal, the actual one-step map has

$$\begin{aligned} \|DF_1(a)\|_{\mathrm{row}} &\le L + CL^2\|a\|_\infty, \\ \|DF_1(a)\|_{\mathrm{col}} &\le L^{-3} + CL\|a\|_\infty, \\ \sup_b \sum_{c,b'} \|D^2F_1(a)_{c;b,b'}\| &\le CL. \end{aligned} \tag{3.mccxcix}$$

The constants are independent of $L \ge 3$, compatible volume and the number of path-mean inputs. ■

Proof. Every coordinate-order path has length at most CL and contains a fixed fine bond at most once. Its first derivative therefore has absolute row sum CL; with one input bond fixed its norm is bounded by a constant. Its second derivative with one fixed leg and the other leg summed is at most CL. These statements remain valid on the indicated complex ball, since products of complex transports have norm at most $e^{CL\|a\|_\infty}$.

All averaging weights are normalized. The inverse in the implicit group-mean equation is within a fixed neighborhood of the identity on this ball. Implicit differentiation once and twice preserves the preceding bounds: terms have either one second path derivative or two first derivatives, with the fixed leg in one of them. Products of separate normalized averages obey the same bounds. Only a fixed number of coarse rows can use a given internal bond or a given crossing bond. This proves the last line of eq. 3.mccxcix. Without a fixed leg, the same calculation gives the absolute row Hessian bound CL^2.

At zero the exact derivative is Q_1, with row norm L and column norm L^{-3}. Integrate the two Hessian bounds on the segment from zero to a. This gives the first two lines. The column estimate can be written

$$L^{-3}(1 + CL^4\|a\|_\infty).$$

Thus uniform path-row bounds do not assert a uniform nonlinear column constant after factoring out L^{-3}. □

Proposition 3.187 (Hierarchical fixed-fine-leg density)**.** Let $h = L^{-k}$, and let q_h be the actual straightened hierarchy of eq. 3.mccliii, with physical fine potential as input. There are universal $c, C > 0$ such that, for $\|A\|_\infty < cL^{-3}$,

$$\sup_b \sum_{c,b'} \|D^2 q_h(A)_{c;b,b'}\| \le CL^3 h^4. \tag{3.mccc}$$

Consequently

$$\|q_h'(A) - Q_h\|_{\mathrm{col}} \le CL^3 h^4 \|A\|_\infty. \tag{3.mccci}$$

For each fixed L these bounds are uniform in depth. They require no projection of the multiplier onto a divergence-free subspace. ■

Proof. At zero the chain rule is the exact identity

$$D^2 q_h(0)[u,v] = h^2 \sum_{t=0}^{k-1} Q_1^{k-1-t} D^2 F_1(0)[Q_1^t u, Q_1^t v]. \tag{3.mcccii}$$

The powers denote the actual linear row at its successive grids. The factor h^2 comes from the two physical inputs $a_0 = hA$. Write $B_L = \sup_b \sum_{c,b'} \|D^2 F_1(0)_{c;b,b'}\|$. Propagate the fixed inner leg by the column bound L^{-3t}, the other inner leg by the row bound L^t, and the output by the remaining column bound $L^{-3(k-1-t)}$. The resulting bound is

$$\begin{aligned} h^2 B_L \sum_{t=0}^{k-1} L^{-3(k-1-t)} L^{-3t} L^t &= h^4 \frac{B_L L^3}{L-1}(1-h) \\ &\le CL^3 h^4. \end{aligned} \tag{3.mcccciii}$$

For $k = 0$ the Hessian is zero. The central-edge contribution in the one-step row is included in B_L; it is not estimated as a density of order L^{-4}.

At a nonzero field, the second-order chain rule has the same sum with the actual Jacobian products on its three branches and $D^2 F_1(a_t)$ at its vertex. The row majorant gives $\|a_t\|_\infty \le 2\xi_t \|A\|_\infty$, where $\xi_t = hL^t$. The row products, divided by their powers of L, are bounded by $\exp(C\|A\|_\infty \sum_t L\xi_t) \le e^{C\|A\|_\infty}$. The analogous column products are bounded by

$$\exp\left(CL^4 \|A\|_\infty \sum_{t<k} \xi_t \right) \le e^{CL^3 \|A\|_\infty}.$$

This is a fixed constant on the stated ball. The same estimate applies to every subproduct, and the one-step fixed-leg Hessian remains CL. Thus eq. 3.mcccciii, multiplied by fixed constants, proves eq. 3.mccc. Integration in A proves eq. 3.mccci. □

The factor L^3 in this absolute estimate can be necessary. For the symmetric coordinate-order contours, choose different axes μ, ν and a background, on the two relevant blocks, of the form

$$A_\mu(X) = rX_\nu \sigma_2, \qquad A_\lambda = 0 \quad (\lambda \ne \mu).$$

Extend it periodically with supremum at most Cr; the extension can be smooth with uniformly bounded physical derivatives. Vary only the positive fine ν-bond from the source block center, with physical variation $\delta A = \sigma_1$. Put $x \in \{-(L-1)/2, \dots, (L-1)/2\}^4$ and $u = x/L$. The first internal-transport variation at that bond has weight

$$w_x = \frac{\mathbf{1}_{\{x_\nu > 0\}}}{d(x)}, \qquad d(x) = \#\{\lambda : x_\lambda \ne 0\}.$$

Indeed the ν-segment must be first among the nonzero coordinate segments. The target transport does not contain this fine bond. The background has $\ell_x = r u_\nu \sigma_2$ and $t'_x = (r/2) u_\mu u_\nu \sigma_2$. Their averages vanish. The covariance formula eq. 3.mccxcviii gives

$$D^2 q_{1/L}(0)[\delta A, A]_\mu(0) = -\frac{r}{L} a_L \sigma_3, \qquad a_L = L^{-4} \sum_x w_x \frac{x_\nu}{L} = \frac{(L+1)(L^4-1)}{32L^5}. \tag{3.mccciv}$$

The $u_\mu u_\nu$ term cancels by reflection in x_μ. In the remaining sum there are $\binom{3}{j}(L-1)^j$ choices with exactly j nonzero other coordinates. Use

$$\sum_{j=0}^{3} \frac{\binom{3}{j}(L-1)^j}{j+1} = \frac{L^4-1}{4(L-1)}$$

and sum $x_\nu = 1, \dots, (L-1)/2$ to obtain eq. 3.mccciv. Since $a_L \to 1/32$, a bound written as $C_L h^4 \|A\|_\infty$ cannot in general have C_L bounded as L grows. This example does not contradict its uniformity in k for fixed L.

3.45.1. *Uniform row amplitudes and a radius that may depend on L.* For completeness the row majorant has a different dependence on L. Normalized path and mean expansions give higher absolute row coefficients $(CL)^m$ at one step. Their proof uses products of at most CL edge factors and normalized averages, and has no factor equal to the number of sites in the block. The combined linear row is exactly L. Thus the positive majorant is

$$b_{t+1} \le L b_t + \frac{C(L b_t)^2}{1 - C L b_t}.$$

For a partial output spacing $\xi = h L^s$, $\sum_{t<s} L\xi_t \le L\xi/(L-1) \le 3\xi/2$. The first-exit product, the node-frame product and the final plain logarithm therefore give the following bound for the divided degree-m Taylor row coefficients $T^{(m)} = D^m/m!$:

$$\sum_{b_1,\dots,b_m} \|T^{(m)}_{h,s;b;b_1\dots b_m}\| \le C^m \xi^m, \tag{3.mcccv}$$

with C independent of $L \ge 3$. The physical row support has radius $C\xi$, by the same sum. This statement applies to the literal mean as well as to the straightened row and its node correction.

The free inverse and its smooth seeds in § 3.43 use the direct rows eq. 3.mccxxiv at $h = 1/N$. Their constants are uniform over odd N; they do not depend on a factorization $N = L^k$. The Wilson residual and the smooth node extension also have such mesh-uniform constants. The nonlinear transpose constant in eq. 3.mccci may instead grow as L^3. Decrease the nonlinear ball after fixing L, so that its contribution to the fixed-point Lipschitz bound is at most one half. The solution estimate is still

$$\|A(B)\|_{X_\alpha} + \|\omega(B)\|_\infty \le 2\|\mathcal{L}_0^{-1}\|\,\|B\|_\infty.$$

Its amplitude constant is uniform in L; its allowed radius need not be. The actual node product, its local extension and finite-gauge formula preserve this uniform amplitude estimate for the literal field. Equation eq. 3.mcccv, stopped one level before the output, consequently gives

$$\|\log p_j^{\mathrm{lit}}(e^{iB})\|_\infty \le \frac{C}{L}\|B\|_\infty. \tag{3.mcccvi}$$

Here $p_j^{\mathrm{lit}} = M^{j-1} U_j$ is the literal partial mean, before the one-step axial gauge, and B lies in the smaller ball just specified. The axial representative need not have this raw angle bound: a flat constant Abelian input can become a field with identity internal links and full output angle on the face-crossing links. For a complete gauge-invariant physical function, the two representatives give the same value on their admitted overlap. Apply that complete identity before composing any noninvariant local representatives with p_j^{lit}. A fixed analytic-radius shrink in these literal coordinates can be paid by choosing a sufficiently large blocking integer once. A full iteration still

requires the representative identity and the local analytic-family and source-domain bounds at its buffered radii.

3.46. **Descent between real flat reference charts.** The homogeneous background was constructed on a fixed complex bond ball about each real flat connection. We now show that these physical background orbits agree wherever the balls overlap. A comparison of coarse flat connections must retain their holonomies. A globally periodic pure gauge is generally unavailable; the construction below uses a smooth gauge on the universal cover with its exact two-sided deck law. It selects no logarithm or root of a global holonomy.

The physical coarse torus has integer side $M \geq 2$, and the fine spacing is $h = L^{-k}$, $k \geq 0$. Coarse links have unit length. All connections in the first geometric construction are real SU(2) connections. Complexification will be used only after a uniformly close real flat fine extension has been produced. Constants depend on L, $0 < \alpha < 1$ and a fixed scalar bump, but not on depth, volume or holonomy. Bond-ball distances use a unitary-invariant matrix operator norm; it is equivalent to the Pauli norm in the strong background estimates.

3.46.1. *Parallel frames with their based holonomies.* For a flat connection on a periodic lattice, choose its parallel frame on the universal cover, based at the coarse node zero. It is convenient to write the links as

$$W(x, x') = p(x)p(x')^{-1}, \qquad p(0) = 1. \tag{3.mcccvii}$$

Thus $p(x)$ is the inverse of the path transport from zero to x. Let H_ν denote the actual positively oriented holonomy from zero to Me_ν. Periodicity and flatness give

$$p(x + Me_\nu) = p(x)H_\nu^{-1}. \tag{3.mcccviii}$$

Indeed the two parallel potentials for the translated connection differ by a constant right factor, determined at zero. The four H_ν commute, since the fundamental cycles commute and the connection is flat. They need not be close to the identity.

Let V_0, V_1 be real flat coarse connections, and let W_0 be any real flat fine extension of V_0. On coarse nodes the potential p_0 of W_0 also represents V_0: every path in a flat averaging row has the same endpoint transport, so $M^k W_0 = V_0$ is that transport. Choose the coarse parallel potential p_1 for V_1. In the parallel frame of W_0, put

$$g_z = p_0(z)^{-1}p_1(z), \qquad \widetilde{V}_1(z, z + e_\mu) = g_z g_{z+e_\mu}^{-1}. \tag{3.mcccix}$$

Here a tilde denotes the frame change p_0^{-1}, which makes W_0 the identity on the cover. With the actual based holonomies $H_{i,\nu}$, equations eq. 3.mcccviii give exactly

$$g_{z+Me_\nu} = H_{0,\nu} g_z H_{1,\nu}^{-1}. \tag{3.mcccx}$$

The order of these factors matters. In particular g_z need not be periodic or globally close to one.

Suppose

$$\sup_c \|V_1(c) - V_0(c)\| \leq \varepsilon. \tag{3.mcccxi}$$

The two parallel-frame factors are unitary, so $\|\widetilde{V}_1(c) - 1\| \leq \varepsilon$. Consequently $\|g_z - g_{z+e_\mu}\| \leq \varepsilon$. Any fixed-radius unit star of nodes has

$$\|g_z - g_{z'}\| \leq C_* \varepsilon \tag{3.mcccxii}$$

by a coordinate path with a fixed number of edges. This estimate uses no path going around the torus. In particular it does not bound $H_{1,\nu} - H_{0,\nu}$ by a volume-independent multiple of ε.

3.46.2. *A smooth quaternion interpolation respecting both deck factors.* Choose a nonnegative smooth product bump ρ, supported in $(-1,1)^4$ and positive on $[-3/4,3/4]^4$. On $\mathbb{R}^4$ set

$$\eta_z(X)=\frac{\rho(X-z)}{\sum_{w\in\mathbb{Z}^4}\rho(X-w)},\qquad M_g(X)=\sum_z \eta_z(X)g_z. \tag{3.mcccxiii}$$

Only a fixed number of terms are nonzero. The denominator has uniform positive lower and finite upper bounds. Thus $\sum_z \eta_z = 1$, $|\partial^r \eta_z| \le C_r$ and $\eta_z(y)=\delta_{zy}$ at the integer nodes. The last assertion uses the open support: every other bump and all its derivatives vanish at that boundary point. Also $\eta_{z+Mn}(X+Mn)=\eta_z(X)$.

Lemma 3.188 (Normalized quaternion interpolation)**.** For a fixed small ε, the formula

$$g(X)=\frac{M_g(X)}{\sqrt{\det M_g(X)}} \tag{3.mcccxiv}$$

defines a smooth SU(2) function, using the positive real scalar square root. It has $g(z)=g_z$ and the exact deck law

$$g(X+Me_\nu)=H_{0,\nu}g(X)H_{1,\nu}^{-1}. \tag{3.mcccxv}$$

For $1\le r\le 3$,

$$\|\partial^r g\|_\infty \le C_r\varepsilon. \tag{3.mcccxvi}$$

These bounds also hold for the right-translated derivatives. ■

Proof. Every g_z has real quaternion form $a_zI+ib_z\cdot\sigma$, with $a_z^2+|b_z|^2=1$. Real scalar averaging preserves that form. For a matrix $M=aI+ib\cdot\sigma$ with real coefficients, $M^*M=(a^2+|b|^2)I$ and $\det M=a^2+|b|^2$. At a given X fix one active node z_0. All other active nodes belong to a fixed star, so $\|M_g(X)g_{z_0}^{-1}-1\|\le C\varepsilon$ by eq. 3.mcccxii. Hence $\det M_g\ge(1-C\varepsilon)^2>0$. Formula eq. 3.mcccxiv is therefore globally smooth and lies in SU(2).

For $r\ge 1$, subtract g_{z_0} in the differentiated partition sum, since $\sum_z \partial^r\eta_z=0$. It follows that $\|\partial^r M_g\|\le C_r\varepsilon$ for $r\le 3$. Differentiating the determinant and its positive scalar square root gives the same bounds for their derivatives. Their uniform positive lower bound proves eq. 3.mcccxvi. No derivative of a globally selected root node is taken: on each star z_0 is fixed, and the resulting uniform estimates cover the whole space.

Normalization by this determinant is equivariant under constant left and right SU(2) multiplication. Both determinant factors are one, and the positive square root is unchanged. Reindexing eq. 3.mcccxiii by a deck shift and using eq. 3.mcccx proves eq. 3.mcccxv. Cardinality gives $g(z)=g_z$. Finally right translation by the unitary $g(X)^{-1}$ preserves the derivative norms. □

Only a positive real number is square-rooted in this argument. There is no choice of a continuous matrix logarithm of a holonomy, including at a central holonomy, and no choice of an h-dependent group root.

Proposition 3.189 (A uniformly close flat fine extension)**.** There is a real flat fine extension W_1' of V_1 such that

$$W_1'=e^{ihC_h}W_0,\qquad \|C_h\|_{X_\alpha(W_0)}\le C_\alpha\varepsilon. \tag{3.mcccxvii}$$

It is periodic in the original frame, has the actual holonomies of V_1, and has $M^kW_1'=V_1$ exactly. ■

Proof. In the W_0 parallel frame define $\widetilde{W}_1'(x,x+he_\mu)=g(x)g(x+he_\mu)^{-1}$. Its plaquettes are one by telescoping. The deck law is conjugation by $H_{0,\nu}$, so undoing the p_0^{-1} frame gives an ordinary periodic fine connection.

Every flat path from a coarse node y to $y+e_\mu$ has transport $g_yg_{y+e_\mu}^{-1}$ in this frame. Its actual contour means therefore give that same matrix. Identical transported inputs have their exact common mean even when that common value is far from one: the defining relative products are

one, and unitary covariance translates their regular branch. Thus the full average is exactly V_1. Its coarse cycle products are the fine cycle products, so the based fine holonomies are $H_{1,\nu}$.

Set $F_{h,\mu}(X) = g(X)g(X + he_\mu)^{-1}$. The integral difference formula and eq. 3.mcccxvi imply $\|\partial^r(F_{h,\mu} - 1)\| \le Ch\varepsilon$ for $r = 0, 1, 2$. For $r = 2$ this uses the third derivative of g; products of lower derivatives are also bounded by $C\varepsilon$. The logarithm near one consequently satisfies

$$\left\| \partial^r \frac{1}{ih} \log F_{h,\mu} \right\|_\infty \le C\varepsilon, \qquad r = 0, 1, 2.$$

Sampling this continuous expression on fine sites bounds its physical first differences and their local α seminorm, for every $\alpha < 1$. The frame change is unitary and converts ordinary differences into the flat covariant differences of W_0. This proves eq. 3.mcccxvii. The logarithm is traceless anti-Hermitian, so C_h is real. □

3.46.3. *Independence of the chosen real flat fine extension.*

Lemma 3.190 (Node-fixed equivalence of flat extensions)**.** If W_1 is any other real flat fine extension of the same literal coarse connection V_1, then there is a periodic unitary fine gauge f, equal to one at every coarse node, with $W_1^f = W_1'$. Applying f simultaneously to a field and its reference preserves its reference-covariant strong norm. ■

Proof. Let $T_1(0, x)$ and $T'(0, x)$ be their path transports on the cover and set $f(x) = T'(0, x)^{-1}T_1(0, x)$. Flatness shows directly on every edge that $W_1^f = W_1'$. On a coarse path both transports are the same product of the links of V_1. Hence f is one at all coarse nodes. Equality of the based holonomies makes f periodic: their left deck factors cancel in its displayed formula. The argument uses equality of every coarse link. Four equal holonomies alone would not force the gauge to be one at all coarse nodes.

For $U = e^{iha}W_1$, one has $U^f = e^{ih\,\mathrm{Ad}(f)a}W_1^f$. The unitary frame change intertwines every reference-covariant difference and parallel comparison. Thus its supremum and local Hölder norms are preserved. No ordinary fine derivative of this possibly large gauge is estimated or required. Unitary covariance also preserves the admitted complex mean branches. □

3.46.4. *Complex uniqueness inside a flat-reference strong chart.* We record the complex uniqueness statement needed on overlaps. It is stronger than equality obtained by analytic continuation from a possibly empty real overlap.

Lemma 3.191 (Complex critical-orbit uniqueness)**.** There are fixed $\delta_c, \rho_c > 0$, uniform in depth, volume and real flat reference, with the following property. Let $V = e^{iB}V_0$, $\|B\|_\infty < \rho_c$. A complex literal constrained-critical connection with a representative $U = e^{ihA}W_0$, $\|A\|_{X_\alpha(W_0)} < \delta_c$, is in the node-fixed complex gauge orbit of the constructed background. The comparison gauge is holomorphic in any holomorphic family of such representatives. ■

Proof. The canonical local branch is holomorphically critical: its real stationarity identities, written using the analytic right inverse, extend to its complex parameter ball. This is continuation inside one chart about a real center, not a claim about a real overlap.

The actual mean-slice contraction eq. 3.mcclxxxvii is an analytic contraction between the displayed complex Banach spaces. It sends U to $\overline{A}$ with $q_h(\overline{A}) = B$ and $RD^\sharp\overline{A} = 0$. Its group equation fixes the complete effective coarse gauge, so the subsequent literal correction differs from U by an exactly node-fixed gauge, including at reducible coarse connections.

The two analytic fibre maps in the proof of eq. 3.mcclxxxviii identify all tangent directions and preserve the action. Thus holomorphic constrained criticality implies $J_h(\overline{A}) = q_h'(\overline{A})^\sharp\omega$. The actual derivative is onto with right inverse eq. 3.mcclxxxix. The multiplier readout eq. 3.mccxc uses absolute values and the physical $W^{1,1}$ seed-column bound, so it remains valid for complex fields and gives $\|\omega\|_\infty \le C\|\overline{A}\|_{X_\alpha}$. This estimate uses no positivity.

Choose δ_c so the resulting pair lies in the joint contraction ball of eq. 3.mcclxvii. Its reference Landau penalty is zero by the imposed slice. It therefore solves that actual fixed-point equation and equals its unique complex solution. The explicit mean-slice gauge and literal node extension give the claimed comparison. They depend holomorphically on the representative. Real positivity was used to prove a minimum, but is not used in this uniqueness argument. □

3.46.5. *Changing the reference in a strong complex neighbourhood.* In the smooth relative frame of Prop. 3.189, an increment a measured relative to W_1' has the same strong norm as $\mathrm{Ad}(g^{-1})a$ in the trivial frame. Product differences and eq. 3.mcccxvi give

$$\|a\|_{X_\alpha(W_0)} \le C_\alpha \|a\|_{X_\alpha(W_1')}. \tag{3.mcccxviii}$$

Only the first two bounded derivatives of g enter. Its values may range throughout the unitary group. Locally its variation is $O(\varepsilon)$, and unitary conjugation does not change the zeroth-order size.

The pointwise matrix map

$$Z_h(a,C) = \frac{1}{ih}\log(e^{iha}e^{ihC}) \tag{3.mcccxix}$$

extends analytically to $h=0$ with value $a+C$. Its nonlinear part is h times a uniformly analytic quadratic expression. The discrete product rule and the Banach algebra estimate for X_α give, on a fixed small ball,

$$\|Z_h(a,C_h)\|_{X_\alpha(W_0)} \le C_\alpha(\|a\|_{X_\alpha(W_1')} + \varepsilon). \tag{3.mcccxx}$$

There is no h^{-1} loss: the logarithm in eq. 3.mcccxix vanishes to first order in h.

Let $\mathcal{B}_\rho(V_f)$ be the coarse complex ball $V = e^{iB}V_f$, $\|B\|_\infty < \rho$, about a real flat V_f. If two such balls intersect, their real centers satisfy

$$\|V_1 - V_0\|_\infty \le 2e^\rho \rho \le 3\rho \tag{3.mcccxxi}$$

for a fixed small ρ. Choose ρ below the homogeneous existence radius, the real interpolation threshold, and the complex uniqueness radius divided by the constants in eq. 3.mcccxx. All these choices are uniform.

Take the V_1-based background and use the real unitary node-fixed gauge of Lem. 3.190 to put its flat fine reference at W_1'. Its increment has $X_\alpha(W_1')$ norm at most $C\rho$. Equations eq. 3.mcccxx and eq. 3.mcccxxi put that actual connection inside the W_0 complex uniqueness chart, with strong norm at most $C'\rho < \delta_c$. Its coarse mean is still V, and it remains holomorphically constrained critical. Lem. 3.191 identifies it with the V_0-based background orbit.

3.46.6. *Holomorphic orbit and action descent.*

Proposition 3.192 (Uniform descent over real flat centers)**.** For a fixed $\rho > 0$ independent of M, k, the physical homogeneous background orbits and their total actions descend to the open set

$$\mathcal{D}_\rho^{\mathbb{C}} = \bigcup_{V_f \text{ real flat}} \mathcal{B}_\rho(V_f). \tag{3.mcccxxii}$$

The total action is a single holomorphic function there. The local background representatives have holomorphic node-fixed transition gauges, with the cocycle identity on triple overlaps. On real data these gauges are unitary. The descended orbit is covariant under all real coarse gauge transformations, and its total action is invariant under them. ■

Proof. The preceding argument gives equality of the physical orbits on each complex overlap directly, without a real-overlap identity theorem. The transition gauge is explicit: apply the fixed real comparison gauge f, then the actual holomorphic mean-slice gauge for the resulting W_0 representative, then the canonical literal extension at its mean-slice solution. Their product has value one at every coarse node. The exact group equations in the mean-slice construction prove this, not merely a first-order calculation. After factoring out the fixed real comparison gauge, the remaining site gauge has a logarithm of $Y_\alpha(W_0)$ norm at most $C\rho$. The comparison formula is analytic on a fixed

larger neighborhood of the inner overlap: choose an intermediate coarse radius so its background and BCH image still lie inside δ_c, and then choose the atlas radius much smaller. Cauchy's formula on that intermediate ball gives uniform parameter jets of the transition in these relative norms. The possible thinness of the final overlap therefore introduces no volume- or depth-dependent constant.

The node-fixed gauge action is free on a connected fine graph. Indeed a stabilizing gauge is transported from its value at one coarse node along every fine edge; that value is one. Consequently the transition gauge between two given literal representatives is unique. If $U_i = U_j^{t_{ij}}$, this implies $t_{02} = t_{01}t_{12}$ on triple overlaps. This argument does not require irreducibility of the coarse holonomies. Local representatives and these holomorphic transitions give the asserted holomorphic orbit descent. Fine Wilson trace invariance gives equality of their holomorphic total actions, so ordinary function gluing gives a single holomorphic action. Only its local strong representatives are asserted; a globally chosen gauge with the same strong bound is not needed.

For a real coarse gauge u, the center V_f^u is again real flat, and $V^u(V_f^u)^{-1} = u(VV_f^{-1})u^{-1}$ at each initial node. Thus the ball is carried isometrically to the ball about V_f^u. Extend u by any unitary fine site values with the prescribed coarse values, and transform a field and its flat reference together. Their reference-covariant strong norms are preserved exactly. Their literal means transform by u and their actions are unchanged. Complex critical-orbit uniqueness at the transformed reference identifies the transformed branch with the descended branch. Two choices of the fine extension differ by a node-fixed gauge, so the orbit statement is independent of that choice. This proves the last assertions, including arbitrary real coarse gauges. □

The domain eq. 3.mcccxxii is a union of small complex balls about real flat connections. It is not asserted to contain all curvature-only regular data or to be saturated under arbitrary complex gauges. On an overlap the equality is the nonlinear total identity $S_0(B_0(V)) = S_1(B_1(V))$. Taylor polynomials about the two centers need not agree term by term. This descent therefore does not identify local fixed-reference Taylor pieces, their filing, their contour cutoffs, or older fine-core activity transports. Those require their own coordinate and domain comparisons.

3.47. **A homogeneous critical atlas from coarse curvature alone.** We construct the regular trial which was an input to Prop. 3.120. The coarse connection need not be close to a flat connection in a global bond gauge. Its plaquettes are small, but its holonomies and topological sector are unrestricted. A smooth quaternionic line bundle interpolates the coarse connection. Sampling its connection gives fine trials with bounded physical curvature and current at every depth.

The setting is homogeneous. Coarse spacing is one, the fixed blocking integer is the odd number L, and $h = L^{-k}$ is the fine spacing. The coarse torus has compatible integer periods at least two. Write V_{ij} for transport from the head j to the tail i, and $V_{ji} = V_{ij}^{-1}$. Assume

$$\|V(\partial p) - 1\| \le \kappa \quad \text{for every coarse plaquette } p. \tag{3.mcccxxiii}$$

For small angles this norm and the admitted plaquette logarithm are equivalent. All smallness choices below are independent of h, the periods and the number of coarse cells. Constants may depend on L and on the fixed regularity exponents and collars.

3.47.1. *Local transports without logarithms of coarse links.* We work first on the universal covering lattice $\mathbb{Z}^4$. Lift V periodically. Distinct lifted vertices remain distinct indices, even if they have the same image on a small torus. This convention is important on a two-cell period: its two oppositely placed crossing edges can have different holonomy.

A bounded closed edge word in a box of fixed diameter can be reduced by interchanging adjacent coordinate steps and canceling inverse pairs. An interchange inserts a conjugate of one plaquette. A word of bounded length requires a bounded number of such interchanges. Since all transports are unitary,

$$\|V(\gamma) - 1\| \le C_R\kappa \tag{3.mcccxxiv}$$

for every contractible edge word of length and diameter at most R. This is a finite word estimate on the covering lattice. It is not a bound for a noncontractible torus loop.

One may fix short paths with reverse paths paired. We specify a symmetric choice to retain lattice covariance. For two vertices in a fixed bounded box, take the at most 4! coordinate-order paths, traversing each nonzero coordinate displacement in one block. Let their backward transports be W_γ and set

$$M_{ij} = \frac{1}{n_{ij}} \sum_\gamma W_\gamma, \qquad U_{ij} = M_{ij} (\det M_{ij})^{-1/2}, \qquad U_{ii} = 1. \tag{3.mcccxxv}$$

An average of real quaternion matrices is quaternionic and $M_{ij}^\sharp M_{ij} = (\det M_{ij}) I_2$. Here $\sharp$ on a 2 by 2 matrix means its quaternionic adjugate; on real quaternion matrices it is the Hermitian transpose. The paths differ by bounded contractible loops. Thus M_{ij} is within $C_R\kappa$ of any one W_γ, and its determinant is positive and close to one. The square root in eq. 3.mcccxxv is the positive scalar root. It is not a root or logarithm of a coarse group element. Only overlapping partition indices are used in the projection below. Auxiliary paths to an anchor are used only on the fixed read boxes. Fix their maximum radius after the required mean and regularity collars, and choose κ afterward. No estimate for unbounded-length paths is asserted.

This construction is equivariant under left and right unitary multiplication, and under reversal. Indeed the reversed path average is $M_{ij}^\sharp$, its determinant is unchanged, and $U_{ji} = U_{ij}^\sharp = U_{ij}^{-1}$. For adjacent vertices there is one path, so U_{ij} is exactly the given coarse link. Coordinate isometries permute the path multiset. In particular the choice is translation and hypercubic covariant. The short loop bound also gives, for three vertices in a fixed box,

$$\|U_{ij} - U_{iy} U_{yj}\| \le C_R \kappa. \tag{3.mcccxxvi}$$

To see this for the normalized averages, first use eq. 3.mcccxxiv for one chosen path in each factor, then use their $C_R\kappa$ difference from the averages.

3.47.2. *A smooth rank-one projection.* Choose an even nonnegative smooth function φ, supported in $(-1, 1)$, positive on $[-1/2, 1/2]$, with $\varphi(0) = 1$. For $i \in \mathbb{Z}^4$ put

$$f_i(x) = \prod_{\mu=1}^4 \varphi(x_\mu - i_\mu), \qquad s_i(x) = \frac{f_i(x)}{(\sum_j f_j(x)^2)^{1/2}}, \qquad \psi_i = s_i^2. \tag{3.mcccxxvii}$$

The denominator is uniformly bounded below. The s_i are smooth, including at their support boundaries; all their fixed derivatives are uniformly bounded. At most sixteen indices are active at any point, $\sum_i s_i^2 = 1$, and $s_i(j) = \delta_{ij}$. Along a coarse edge only its two endpoints can be active.

Use these indices as quaternionic coordinates and form

$$P_{ij}(x) = s_i(x) s_j(x) U_{ij}. \tag{3.mcccxxviii}$$

This is a bounded self-adjoint quaternionic operator on $\ell^2(\mathbb{Z}^4; \mathbb{H})$, but every calculation below is finite: its nonzero block at a point has at most sixteen quaternionic rows. Derivatives on a fixed small neighborhood use a fixed finite union of those indices. No infinite determinant or infinite-dimensional Gaussian is introduced.

Fix an anchor y and a physical box of fixed radius R about it. Define, throughout that box,

$$u_y(x)_i = s_i(x) U_{iy}. \tag{3.mcccxxix}$$

The anchor need not be active. Its transports have bounded length in a fixed larger box. The exact identities are $u_y^\sharp u_y = I_2$ and $u_y^\sharp \mathrm{d} u_y = 0$; the latter follows from $\sum_i s_i \, \mathrm{d} s_i = 0$, since each U_{iy} is independent of x. By eq. 3.mcccxxvi,

$$\|P - u_y u_y^\sharp\|_{C^m} \le C_{m,R} \kappa \tag{3.mcccxxx}$$

for every fixed m. The matrix norm has no volume factor, because the active block is bounded in size.

Choose κ below the finitely many margins used below. At each point the spectrum of P lies in disjoint small neighborhoods of zero and one. Define

$$\Pi(x) = \frac{1}{2\pi i} \int_{|z-1|=1/4} (z - P(x))^{-1}\, dz. \tag{3.mcccxxxi}$$

The contour resolvent is computed on the finite active block; the integral on its zero complement is zero. The resolvent identity, or its Neumann expansion about $u_y u_y^\sharp$, proves

$$\Pi^2 = \Pi = \Pi^\sharp, \quad \operatorname{rank}_{\mathbb{H}} \Pi = 1, \quad \|\Pi - u_y u_y^\sharp\|_{C^m} \le C_{m,R}\kappa. \tag{3.mcccxxxii}$$

For example every differentiated resolvent is a finite sum of products of bounded resolvents and derivatives of P; a difference from the comparison projector contains at least one $O(\kappa)$ difference. This proves the derivative estimate without a spectral-eigenvector choice. The quaternionic rank is one, equivalently its complex rank is two.

The ranges $E_x = \operatorname{ran} \Pi(x)$ form a smooth quaternionic line bundle on the cover. Put on it the connection $\nabla = \Pi \mathrm{d}$. A normalized local frame on the whole fixed box is

$$v_y = \Pi u_y \, (u_y^\sharp \Pi u_y)^{-1/2}. \tag{3.mcccxxxiii}$$

The middle 2 by 2 matrix is a positive scalar multiple of I_2: it is a self-adjoint quaternion of size one. It differs from I_2 by $O(\kappa)$. Thus $v_y^\sharp v_y = I_2$, $v_y v_y^\sharp = \Pi$, and $v_y - u_y = O(\kappa)$ in every fixed local C^m norm. Its connection matrix and its curvature satisfy

$$A_y = v_y^\sharp \mathrm{d} v_y, \qquad \|A_y\|_{C^m} + \|\mathrm{d}A_y + A_y \wedge A_y\|_{C^{m-1}} \le C_{m,R}\kappa. \tag{3.mcccxxxiv}$$

Indeed $u_y^\sharp \mathrm{d} u_y = 0$ cancels the order-one terms from the partition. The remaining terms have a factor $v_y - u_y$ or its derivative. The matrices A_y are traceless anti-Hermitian. Local frames have exact unitary transition functions $v_z^\sharp v_y$. We have not chosen a global smooth frame, which need not exist.

3.47.3. *Exact edge interpolation and periodic descent.* On a coarse edge (y, z) the two-index block of P is exactly a projection. Its unit frame is

$$u = (s_y I_2,\ s_z U_{zy})^{\mathsf{T}}.$$

Its connection $u^\sharp \mathrm{d} u$ is zero along that edge. At the endpoints this frame is e_y and $e_z U_{zy}$, respectively. Backward connection transport therefore sends $e_z \zeta$ to $e_y U_{yz} \zeta$. Consequently

$$\text{backward transport in } (E, \nabla) \text{ along } (y, z) \text{ in canonical endpoint frames} = V_{yz}. \tag{3.mcccxxxv}$$

This checks the orientation against the lattice convention. It also explains why a potentially large V_{yz} requires neither a logarithm nor a fractional power.

For each coarse-period translation t, let S_t shift the ambient vertex indices. The construction is exactly equivariant:

$$P(x + t) = S_t P(x) S_t^{-1}, \qquad \Pi(x + t) = S_t \Pi(x) S_t^{-1}. \tag{3.mcccxxxvi}$$

Thus the bundle and connection descend under $(x, w) \sim (x + t, S_t w)$. At every coarse vertex their canonical frame e_y descends as well. This gives an ordinary rank-one quaternionic bundle over the coarse torus. The use of a covering-coordinate Hilbert space is only a uniform way to retain distinct local indices; all its bundle charts and connection computations are finite-dimensional.

Equivalently, any fixed finite collection of read stars can first be placed in a sufficiently large compatible finite cover. The finite active-block formulas are the same there. One descends the equivariant bundle and connection. One does not average projections or compare partition functions on different volumes. In particular no short torus loop is silently declared contractible: the loop filling in eq. 3.mcccxxiv always occurs between distinct lifted indices.

3.47.4. *Fine trials and the actual Wilson-current cancellation.* At the fine vertices choose arbitrary unit quaternionic frames of E, with frame e_y at the coarse vertices. A choice is possible on this finite set even when the bundle is nontrivial. Define W_h on a fine edge by backward parallel transport of ∇ between those frames. This is a real SU(2) lattice connection. Changing the fine frames changes it by a fine gauge equal to one at coarse vertices.

On every fixed physical read box use the smooth frame v_y. In this frame $A_y = ia_y$, where a_y is Hermitian and has all fixed C^m norms bounded by $C_m\kappa$. The parallel transport equation and its integral iteration give fine links $W_h(x,\mu) = e^{iha_{h,\mu}(x)}$ with

$$\|a_h\|_{C^m} \le C_m\kappa. \tag{3.mcccxxxvii}$$

Here differences or derivatives of the sampled function have physical units. To justify division by h, the transport matrix is $1 + \int_0^h A_y(x + te_\mu)\,dt + O(h^2\kappa^2)$. Its admitted logarithm vanishes at $h = 0$, so its logarithm divided by h is smooth with these bounds. Differentiating the integral equation proves every fixed spatial derivative estimate. This is a local frame statement, not a global small potential for the lattice connection.

The same equation applied to a square gives, uniformly in the local box,

$$\|W_h(\partial p) - 1\| \le C\kappa h^2, \qquad \|\partial_x^m(W_h(\partial p) - 1)\| \le C_m\kappa h^2. \tag{3.mcccxxxviii}$$

The missing linear term follows either by multiplying the four integral expansions or by cancellation of the two oppositely oriented segments in each direction. Their second-order coefficient is the connection curvature. The integral remainder and all its fixed derivatives are bounded by $C_m\kappa h^3$. These estimates follow from eq. 3.mcccxxxiv; no field equation is assumed.

It is essential to pair the plaquettes in the current. Put $\mathcal{S}(P) = (P - P^{-1})/(2i)$. For the positive bond $b = (x, x + he_\mu)$, the two plaquettes in the $\mu\nu$ plane contribute, in the tail frame,

$$h^{-3}\left\{\mathcal{S}(P_{\mu\nu}(x)) - \mathrm{Ad}(W_h(x - he_\nu, \nu)^{-1})\mathcal{S}(P_{\mu\nu}(x - he_\nu))\right\}. \tag{3.mcccxxxix}$$

This follows by varying the first link in the upper plaquette and the inverse third link in the lower one, then using cyclicity of the trace. It is the actual physical left Wilson current normalization. The translated difference of the two $\mathcal{S}$ terms is $O(\kappa h^3)$ by eq. 3.mcccxxxviii. The transporting link in this smooth frame is $1 + O(\kappa h)$; its additional commutator is $O(\kappa^2 h^3)$. Summing the three planes proves

$$\|F(W_h)\|_\infty + \|j(W_h)\|_\infty \le C\kappa, \tag{3.mcccxl}$$

with physical fine spacing h. The statement is gauge covariant, so arbitrary vertex frames cause no loss. Bounding each summand before this pairing would instead give the false bound $C\kappa/h$ for this purpose.

3.47.5. *The complete hierarchical mean has a uniform error.* The actual k-fold mean of a coarse edge reads a fixed physical star of radius R_L, independent of k: the successive stencil radii sum as a geometric series. Choose the preceding local boxes larger than that star, including every internal contour and a fixed collar. Use its tail vertex y as anchor. The local smooth frame at its head is $e_z U_{zy}$, so the central coarse edge is 1 in this frame, by eq. 3.mcccxxxv.

Extend a_h from the read star by a fixed smooth cutoff on a larger box. Its supremum and fixed physical derivatives remain $C_L\kappa$. On a small period one can do this on a large compatible cover; the mean's read set is kept before descent. The actual mean is local and gauge covariant, so this extension does not alter the edge under consideration.

Apply the absolute plain-mean row majorant eq. 4.cccxlv, with physical output length $\xi_s = hL^s$. It gives

$$\|\log M^s(e^{iha_h})\|_\infty \le C_L\kappa\xi_s, \qquad 0 \le s \le k \tag{3.mcccxli}$$

on every intermediate read edge needed for this final edge. The majorant also controls the actual internal node factors and closed logarithms. Choose κ below its fixed raw input ceilings; thus every

actual mean operation is on its intended analytic branch. At $s = k$ the local output is $1 + O(\kappa)$, while its central original edge was exactly 1. Transforming the endpoints back gives

$$\sup_c \left\| \frac{1}{i} \log\{M^k(W_h)(c)V(c)^{-1}\} \right\| \le C_L \kappa. \tag{3.mcccxlii}$$

No factor C_L^k has been iterated. This estimate uses the full actual plain mean, not a straight-line linear replacement. Its gauge covariance also shows that the answer is independent of the arbitrary fine vertex frames with canonical coarse endpoints.

Proposition 3.193 (Actual smooth homogeneous trials)**.** For sufficiently small coarse curvature in eq. 3.mcccxxiii, every $h = L^{-k}$ has a real fine trial $W_h(V)$, defined up to a gauge one at coarse nodes, with eq. 3.mcccxl and eq. 3.mcccxlii. Its local smooth-frame potentials have all fixed C^m bounds $C_m\kappa$. Its connection has exactly the original coarse-edge transports before hierarchical averaging. All actual hierarchical mean charts are admitted. The constants are uniform in depth and compatible volume. ∎

3.47.6. *A holomorphic comparison which has no fine-frame jump.* Fix a real coarse centre V_*. Write nearby coarse data as $V_B = e^{iB}V_*$, with the raw coarse bond supremum norm of B small. Products along every fixed short path are holomorphic in B. In complex formulas $\sharp$ is the analytic quaternionic adjugate, followed by block transpose; it does not conjugate a parameter. The normalized averages in eq. 3.mcccxxv use the scalar branch near determinant one. Reversal and $P^\sharp = P$ remain algebraic identities. The fixed contour in eq. 3.mcccxxxi therefore defines a holomorphic idempotent Π_B, with

$$\|\Pi_B - \Pi_*\|_{C^m} \le C_m \|B\|_\infty. \tag{3.mcccxliii}$$

All these norms are local finite-block norms and their supremum over physical boxes. The constants do not count the coarse volume. In particular $\Pi_B(y) = \Pi_*(y) = e_y e_y^\sharp$ at coarse vertices.

For two close idempotents put

$$T_B = \{\Pi_B\Pi_* + (1-\Pi_B)(1-\Pi_*)\}\{1 - (\Pi_B - \Pi_*)^2\}^{-1/2}. \tag{3.mcccxliv}$$

The inverse square root is its norm-convergent binomial series. Direct multiplication of the two idempotents shows

$$T_B^\sharp T_B = 1, \qquad T_B\Pi_* = \Pi_B T_B, \qquad T_B(y) = 1, \qquad \|T_B - 1\|_{C^m} \le C_m\|B\|_\infty. \tag{3.mcccxlv}$$

For the first equality, if $S = \Pi_B\Pi_* + (1-\Pi_B)(1-\Pi_*)$, then $S^\sharp S = 1 - (\Pi_B - \Pi_*)^2$; the latter commutes with both idempotents and with S. Real data make T_B unitary. These formulas remain valid on the smaller complex ball by the same algebra and series.

Choose the fine frames of the deformed trial to be $T_B(x)e_x^*$, where e_x^* are the fixed frames of $W_h(V_*)$. Pulling its connection back to E_* gives

$$\nabla_B^{\mathrm{al}} = \nabla_* + C_B, \qquad C_B = \Pi_* T_B^\sharp(\mathrm{d}T_B)\Pi_*\big|_{E_*}, \qquad \|C_B\|_{C^m} \le C_m\|B\|_\infty. \tag{3.mcccxlvi}$$

This follows by applying $T_B^\sharp \Pi_B \mathrm{d}$ to $T_B s$ for a section s of E_*. It compares connections, not discontinuously chosen vertex frames.

The relative edge transport of these two smooth connections is $1 + O(h\|B\|)$ in the tail reference frame. The integral transport equation and its differentiated equations give an admitted logarithm

$$W_h(V_B)(b)W_h(V_*)(b)^{-1} = e^{ihE_h(B)(b)}, \qquad \|E_h(B)\|_{\mathrm{str},\alpha;W_h(V_*)} \le C_\alpha\|B\|_\infty. \tag{3.mcccxlvii}$$

The local reference connection has its uniform C^m bounds, so ordinary derivatives of the relative transport give the stated covariant differences and Hölder seminorm. Division by h is removable in this relative equation, just as in eq. 3.mcccxxxvii. Both the map and its fixed-order derivatives are holomorphic uniformly in h. A new arbitrary anchor choice at each B would not justify this

estimate: it could create a jump of order B/h in relative coordinates. The prescribed aligned frames avoid that jump.

The construction is local in the coarse data on each fixed physical box: P reads finitely many short paths, and the functional calculus and T_B act pointwise on its finite active block. Thus its parameter estimates do not hide a volume-dependent global gauge choice.

3.47.7. *Critical correction at every curvature-small datum.* Fix the real trial $W_h(V_*)$, with approximate multiplier zero. By eq. 3.mcccxl it satisfies the actual regularity and defect hypotheses of Prop. 3.120, with defect at most $C\kappa$. The homogeneous determining region has no artificial inner boundary; the actual site and field rows are admitted by eq. 3.mcccxli. All fixed inverse and regularity collars can be chosen before decreasing κ.

Let $V_*^{\mathrm{tr}} = M^k W_h(V_*)$. For target $V_B = e^{iB}V_*$, its literal datum relative to the trial is

$$B^{\mathrm{tr}}(B) = \frac{1}{i}\log(V_B(V_*^{\mathrm{tr}})^{-1}), \qquad \|B^{\mathrm{tr}}(B)\|_\infty \le C(\kappa + \|B\|_\infty). \tag{3.mcccxlviii}$$

This logarithm is close to the identity even when the individual coarse links are not. Apply the actual forced fixed point of RTC to this datum. It retains the source $-j(W_h(V_*))$, removes its gauge penalty by the exact gauge tests, and uses the actual mean slice and literal correction. It gives a holomorphic critical orbit with mean exactly V_B. In its trial chart its field and multiplier are bounded by $C(\kappa + \|B\|_\infty)$. In particular a critical orbit at V_* has actual physical curvature and current at most $C\kappa$. No critical reference or globally small coarse potential was assumed in obtaining it.

3.47.8. *Identification on overlaps and the resulting atlas.* Consider two real centres $V_*, \widetilde{V}_*$ whose sufficiently small complex datum balls overlap. Their raw relative coarse distance is at most $C\rho$, where ρ is the common radius. The real unitary map in eq. 3.mcccxliv aligns their trial bundles. It is the identity at coarse nodes. Fine frames chosen differently from this alignment are related by a real fine gauge one at those nodes. After that fixed gauge, their real trials differ by at most $C\rho$ in the covariant strong norm of eq. 3.mcccxlvii.

Both complex critical germs on the overlap differ from their respective trials by at most $C(\kappa+\rho)$. Reference strong norms are equivalent under the above small real connection change. Choose ρ and then κ so that their combined bound lies inside the larger RTC uniqueness chart of the first trial. They have the same literal coarse datum. The actual complex mean-slice and competing-multiplier uniqueness of RTC therefore identify their orbits modulo fine gauges one at coarse nodes. No positivity of a complex square is needed in this step.

These orbit identifications are holomorphic. Indeed the fixed-reference mean-slice maps and the specified literal correction used in RTC are holomorphic. They send the two germs to the same unique slice solution; composing their admitted gauges gives the transition gauge, with identity coarse-node values. On a triple overlap this describes the same orbit, irrespective of which sequence of Kato maps was used. Kato maps themselves need not obey a multiplicative cocycle. Alternatively both local trial frames can be placed in the common bundle $E(V)$ by $T(\Pi(V), \Pi(V_*))$; ordinary frame changes there obey the exact cocycle. The node-fixed fine gauge action is free: an element fixing the fine connection is propagated from its identity value at a coarse node along every fine edge. Thus the transition gauges for specified local representatives also obey that cocycle. The total Wilson action is unchanged by all these gauges.

Changing the bounded short-path policy does not change this orbit. Its used transports differ by $O(\kappa)$ by eq. 3.mcccxxiv; hence the two projections differ by $O(\kappa)$ in every fixed local derivative norm. Their Kato identification is the identity at coarse nodes and their trial connections differ by $O(\kappa)$ in the strong norm. The same RTC uniqueness argument identifies the two critical corrections. The partition and all finite read collars remain fixed in this comparison.

The interpolating construction is equivariant under real coarse gauge transformations. For $g = (g_i)$, the ambient constant diagonal map $G = \mathrm{diag}(g_i)$ gives $P(V^g) = GP(V)G^{-1}$, and similarly for Π and its connection. At coarse nodes $Ge_i = e_i g_i$, exactly the required coarse gauge. Coordinate isometries likewise permute the chosen path averages and partition. Arbitrary fine frame choices only introduce the gauges already factored out. RTC uniqueness thus gives real coarse gauge and lattice covariance of the total critical orbit and action.

Proposition 3.194 (A physical homogeneous curvature atlas)**.** For sufficiently small κ, every real homogeneous coarse field satisfying eq. 3.mcccxxiii has an actual constrained-critical fine orbit at every $h = L^{-k}$. Its literal mean is the given coarse field, and its physical fine curvature and current are bounded by $C\kappa$, uniformly in depth and volume. Large holonomies and nontrivial bundle topology are allowed.

There is a fixed positive datum radius about every such real field on which the orbit has a holomorphic germ. These germs identify on overlaps as above. The total action therefore defines a holomorphic function on their union, with real coarse gauge and lattice covariance. On real curvature-small data,

$$|S_h(V)| \le C\kappa^2 \mathcal{V}. \tag{3.mcccxlix}$$

Local fixed-order analytic responses have the uniform localized bounds of RTC in each chosen real trial chart. ■

Proof. The trial proposition supplies all the real reference hypotheses of RTC and the small literal datum error. Its actual forced fixed point gives the critical orbit and its physical bounds. The overlap argument identifies the germs, so at any real curvature-small datum the estimate obtained by using that datum itself as centre is the estimate for the glued orbit. Its fine plaquette action is at most $Ch^4\kappa^2$; summing $O(h^{-4}\mathcal{V})$ plaquettes proves eq. 3.mcccxlix. Localized responses follow from the RTC fixed point on the same reference, composed with the pointwise analytic datum eq. 3.mcccxlviii. The deck-equivariant geometric construction gives the same statements at small periods without any quantum-cover comparison. □

The uniqueness asserted here is the controlled trial-atlas uniqueness. It does not classify arbitrary disconnected fine critical sectors. The result removes the need to assume a regular trial for homogeneous curvature-small data; it does not construct a mixed-scale proper trial, an independent-current history core, or a positive conditional measure.

3.48. **Constant diagonal weights on the curvature atlas.** The curvature atlas supplies smooth fine trials without a global small-potential assumption. We keep those trials and change the six plaquette coefficients. The change of principal part is bounded from $W^{2,p}$ to L^p. We first retain that stronger field norm in the joint inverse, and then solve the weighted equation about the given trial.

Coarse spacing is one, $h = L^{-k}$, and the homogeneous periods are compatible with the fixed odd L. Let

$$\mathcal{A}_h^a(U) = \sum_{\mu<\nu} a_\mu a_\nu \sum_x \left(1 - \tfrac{1}{2}\operatorname{tr} U_{\mu\nu}(x)\right), \qquad a = (a_1, \dots, a_4). \tag{3.mcccl}$$

The parameter a is constant in space. Its complex domain is a polydisc about 1; the real weights in that disc are positive. The group mean, its straightening, the site restoration and the literal node correction do not change.

3.48.1. *A joint inverse with two L^p derivatives.* Fix $0 < \alpha < 1$ and a finite $p > 4/(1-\alpha)$. Use complete enlarged unit cells C_y^+, with all stencils and averaging collars. In the fixed comparison

frames of a real regular reference put

$$\begin{aligned}\|f\|_{\mathcal{F}_p} &= \sup_y \|f\|_{L^p(C_y^+)},\\ \|A\|_{\mathcal{W}_p} &= \sup_y \sum_{r=0}^{2} \|D^r A\|_{L^p(C_y^+)}.\end{aligned} \tag{3.mcccli}$$

The fine measure is h^4, and all mixed covariant second differences are included. The fixed enlargement can accommodate the finitely many nested cutoffs used below. Discrete Morrey gives

$$\|A\|_{\mathcal{A}_\alpha} \le C_{p,\alpha}\|A\|_{\mathcal{W}_p}. \tag{3.mccclii}$$

These are cell Banach norms, not sums over fine coordinates. On a sufficiently small $C^{1,\alpha}$ tube the second-difference norms at the present base and at the fixed real reference are equivalent: their difference contains only the connection difference, its first difference, and at most one difference of the field.

Lemma 3.195 (The strong row saddle in $W^{2,p}$)**.** On the homogeneous restriction of the domain of Prop. 3.163, the field output of the full row saddle satisfies

$$\|A\|_{\mathcal{W}_p} + \|\mu\|_\infty \le C_{p,\alpha}(\|f\|_{\mathcal{F}_p} + \|b\|_\infty). \tag{3.mcccliii}$$

It retains a sufficiently small exponential weight on the cell norms. The constants are uniform in depth and compatible volume. The present base may have a rough unweighted Wilson current; it must be in the stated strong tube, and the independent current has the smaller strong-family bound. ■

Proof. We give the second-derivative step in the strong column proof, rather than inferring it from $C^{1,\alpha}$ regularity. For the positive field primitive, its local energy columns and coarse row bounds first control the local H^1 norm and the multiplier. A localized $\mathcal{F}_p$ force is in the physical L^2 force class because $p > 2$ and these cells have bounded physical volume.

After adding the full divergence term, the exact Wilson curl–divergence stencil is a component covariant Laplacian plus bounded curvature multiplications and fixed neighboring shifts, as in Lem. 3.161. The row-mass forcing is bounded pointwise by the energy cell norm of the complete average. The physical transpose has its actual fine density, so no energy field is evaluated at a point. In the local real comparison frame the scalar equation has bounded first-order and zeroth-order coefficients. On the complex tube the same coefficient bounds hold. Pairing opposite shifts before dividing by h gives these bounds without differentiating curvature. Possible normal-coordinate current coefficients carry h; the plaquette bound controls them. No pointwise estimate for the present unweighted current is used.

Apply the local scalar estimates on nested complete boxes. Energy gives H^1. The 2 estimate gives $W^{2,2}$, hence gradient L^3 by a strict four-dimensional embedding. The 3 estimate then gives $W^{2,3}$, hence gradient L^8 and bounded field. If $p \le 8$, apply the final p estimate. If $p > 8$, first apply the 8 estimate, obtaining bounded gradient, and then the desired finite p estimate. Retain the D^2A term in that last estimate, before the Morrey step in the strong-field proof. Each prescribed force belongs to the lower exponents used in this finite bootstrap. The cutoff commutators contain only the controlled outer field and first differences.

The slow positive parametrix has the same C_p/m force error when its output slot is supplied with this additional $W^{2,p}$ norm. Its local output factors have just been bounded in that norm. Thus its positive series still converges after increasing the fixed m. In the Woodbury correction the field output again has a positive field inverse on the left. The current corrections map strong fields to $\mathcal{F}_p$, by eq. 3.mcxc and the subsequent second-constraint estimate. Their physical adjoints use the finite-p force classes of eq. 3.mclxxxvii. Consequently the current Neumann series retains the $W^{2,p}$ output: apply this improved leftmost field factor, and use the already proved strong bounds for

its remaining factors. There is no assertion that the restoration projection by itself preserves three derivatives.

The row lift has the same leftmost field factor. The normalized coarse row inverse bounds its multiplier in ℓ^∞, while $Q^\sharp$ sends a bounded row multiplier to the bounded fine force supplied by its physical column density. The full row-saddle formula in eq. 3.mcxci therefore gives eq. 3.mcccliii.

For a spatial weight use the energy columns first and apply the finite sequence of scalar estimates on each output cell. Weights on neighboring enlarged cells have bounded ratio. No artificial weight is differentiated. The other primitive and coarse factors have their two exponential Schur sums. This proves the weighted assertion. □

We shall use this lemma at a smooth trial U_0, with independent current zero. Write

$$H_0 = F_0''(0), \qquad F_0(A) = \mathcal{A}_h^1(e^{ihA}U_0).$$

Remove the artificial row mass from the saddle by its exact multiplier shift. If the mass-added saddle returns (A,μ) with $QA = b$, put $\nu = \mu - a_f b$. Then the inverse of

$$\mathcal{L}_0\begin{pmatrix} A \\ \nu \end{pmatrix} = \begin{pmatrix} (H_0 + DRD^\sharp)A - Q^\sharp\nu \\ QA \end{pmatrix} \tag{3.mcccliv}$$

has the same bound eq. 3.mcccliii. Here a_f is bounded above and below, since coarse spacing is one. This is the full saddle; no horizontality of its trial lift is assumed.

3.48.2. *The principal perturbation and the weighted trial force.* Let V_* be real with $\|V_*(\partial p)-1\| \le \kappa$. Choose the smooth trial

$$U_0 = W_h(V_*) \tag{3.mcccclv}$$

of Prop. 3.193. Its local potentials have all fixed C^m bounds $C_m\kappa$, and its ordinary Wilson current and curvature are bounded by $C\kappa$. All finite read collars are chosen before κ is decreased. Holonomies and bundle topology are not restricted.

Put

$$F_0^a(A) = \mathcal{A}_h^a(e^{ihA}U_0), \qquad j_0^a = (F_0^a)'(0), \qquad H_{0,a} = (F_0^a)''(0).$$

Lemma 3.196 (Bounds at the unchanged trial)**.** On a fixed complex weight polydisc,

$$\begin{aligned} \|j_0^a\|_\infty &\le C\kappa, \\ \|(H_{0,a} - H_0)A\|_{\mathcal{F}_p} &\le C|a-1|\|A\|_{\mathcal{W}_p}, \\ \|(F_0^a)'(A) - j_0^a - H_{0,a}A\|_{\mathcal{F}_p} &\le C\|A\|_{\mathcal{W}_p}^2. \end{aligned} \tag{3.mcccclvi}$$

The last remainder has Lipschitz constant Cr on a radius-r ball. The maps are holomorphic on fixed larger product domains, uniformly in h and volume. ■

Proof. For each plane multiply the exact paired-current formula eq. 3.mcccxxxix by $a_\mu a_\nu$. The two plaquettes retain the same constant coefficient. Their transported difference is $O(\kappa h^3)$, including the commutator from the connecting link. Division by h^3 and summation over the three planes incident to a bond give the first bound. A bound on separate unpaired plaquettes would lose a factor h^{-1}.

Each plane Hessian at this smooth trial is a second-difference operator with bounded principal coefficients and bounded lower-order coefficients in the fixed local frames. This also follows directly by differentiating its paired plaquette expression: covariant second differences act on the variation, and differences of trial coefficients have the bounds eq. 3.mcccxxxvii. Thus its L^p output is bounded by the three terms in eq. 3.mcccli. The difference of the weights is $a_\mu a_\nu - 1$. There are six planes, which proves the second bound. It is not a bounded perturbation on $C^{1,\alpha}$ alone.

For the final bound apply the centered Wilson residual calculation Lem. 3.146 separately to those six planes. Constant coefficients $a_\mu a_\nu$ change only its constants. Subtract both the constant

and the full linear jet before estimating. The remainder is bounded pointwise by $C\|A\|^2_{\mathcal{A}_\alpha}$. Use eq. 3.mccclii and the cell volume to obtain its $\mathcal{F}_p$ bound. Cauchy on the larger strong ball gives the Lipschitz estimate. □

Define $\mathcal{L}_{0,a}$ by eq. 3.mcccliv with $H_{0,a}$ in place of H_0. The perturbation affects only its first component. By the second bound in eq. 3.mccclvi,

$$\|\mathcal{L}_0^{-1}(\mathcal{L}_{0,a}-\mathcal{L}_0)\|_{\mathcal{W}_p\oplus\ell^\infty\to\mathcal{W}_p\oplus\ell^\infty}\le C|a-1|. \tag{3.mccclvii}$$

Choose the weight disc so that this quantity is at most $1/4$. Its Neumann series gives the actual $\mathcal{L}_{0,a}^{-1}$, with uniform bound and a sufficiently small exponential cell weight. This choice is independent of depth and volume.

3.48.3. *The forced weighted equation.* Let $V_*^{\mathrm{tr}} = M^k U_0$. For target $V_B = e^{iB}V_*$, use the actual literal offset and its reduced datum

$$\beta_*(B) = i^{-1}\log\{V_B(V_*^{\mathrm{tr}})^{-1}\},\qquad \mathsf{b}_*(B) = b_{U_0}(\beta_*(B)),\qquad \|\mathsf{b}_*(B)\|_\infty \le C_L(\kappa+\|B\|_\infty). \tag{3.mccclviii}$$

Here β_* is eq. 3.mcccxlviii, and b_{U_0} is the actual datum map eq. 3.mxxxiii. The relative product inside the logarithm is near one. Neither individual coarse transport needs a small logarithm.

Write $\mathfrak{q}_0(A)$ for the actual re-centered straightened constraint at U_0, and put

$$C_q(A) = \mathfrak{q}_0(A) - QA,\qquad N_a(A) = (F_0^a)'(A) - j_0^a - H_{0,a}A.$$

The weighted counterpart of the forced RTC equation, with approximate multiplier zero, is

$$\begin{pmatrix} A \\ \nu \end{pmatrix} = \mathcal{L}_{0,a}^{-1}\begin{pmatrix} -j_0^a - N_a(A) + (\mathfrak{q}_0'(A)-Q)^\sharp\nu \\ \mathsf{b}_*(B) - C_q(A)\end{pmatrix}. \tag{3.mccclix}$$

Its affine source is the weighted trial current. It has not been removed by a centering convention.

Lemma 3.197 (A weighted correction of the smooth trial)**.** After fixed reductions of the weight disc, the curvature ceiling and a datum radius ρ, eq. 3.mccclix has a unique solution in a fixed $\mathcal{W}_p\oplus\ell^\infty$ ball. It is holomorphic in a, B, real at real data, and

$$\|A(a,B)\|_{\mathcal{W}_p} + \|\nu(a,B)\|_\infty \le C(\kappa+\|B\|_\infty). \tag{3.mccclx}$$

It solves the full weighted critical equation after the gauge penalty is removed. The specified literal correction gives mean exactly V_B. ■

Proof. The actual constraint estimates give

$$\|C_q(A)\|_\infty \le C\|A\|^2_{\mathcal{A}_\alpha},\qquad \|(\mathfrak{q}_0'(A)-Q)^\sharp\nu\|_\infty \le C\|A\|_{\mathcal{A}_\alpha}\|\nu\|_\infty.$$

These are the physical column-density bounds in Lem. 3.148, before any restoration adjoint. The fixed-L hierarchical density, not a single direct large-block contour, supplies their uniformity in depth. Use eq. 3.mccclii for the field arguments and the cell volume for the force. Together with eq. 3.mccclvi, the nonlinear tuple in eq. 3.mccclix has size Cr^2 and Lipschitz constant Cr on a joint radius-r ball.

Choose r inside the strong mean, site and gauge-chart margins so that the inverse bound times this Lipschitz constant is below $1/4$. Include the SCF background tube among these margins. The trial qualifies by its unweighted bounds from CTA. Choose ρ, κ afterward so the inverse times the two affine sources has norm below $r/8$. Require also $C(\kappa+\rho)$ below the smaller independent-current ceiling of that strong-family tube. Iteration from zero is invariant and contractive. Normal convergence on the larger product domain proves holomorphy and eq. 3.mccclx. The construction used the inverse at the given trial; no inverse at the desired critical field was assumed.

The equation before penalty removal is exactly

$$(F_0^a)'(A) + DRD^\sharp A = \mathfrak{q}_0'(A)^\sharp\nu,\qquad \mathfrak{q}_0(A) = \mathsf{b}_*(B).$$

The actual gauge cocycle, site extensions and gauge-test surjectivity in RTC depend on the trial geometry and its strong regularity. They do not depend on the plaquette coefficients. The trial already has the needed unweighted regularity. Test this equation on the actual fine gauge directions in the kernel of its effective coarse gauge. The first term vanishes by gauge invariance of $\mathcal{A}_h^a$, and the constraint term vanishes by the exact intertwining. The site surjectivity used in eq. 3.mliii therefore annihilates the penalty. For complex parameters use the same range and bilinear nondegeneracy argument as RTC, not positivity of a complex square. Thus $RD^\sharp A = 0$, and the full equation is

$$(F_0^a)'(A) = \mathfrak{q}_0'(A)^\sharp \nu.$$

The actual mean slice and literal node extension are unchanged. They convert the reduced datum in eq. 3.mccclviii to the prescribed literal mean V_B. The extension has its fixed strong site bound and gives a strong literal field of size $C(\kappa + \|B\|)$. No $W^{2,p}$ claim for an arbitrary choice of literal extension is needed. □

At the literal output, denote by j_a the weighted left current and by ω_a its physical row multiplier in the declared coarse coordinates. The row-coordinate conversion is the actual derivative of the datum map and the normal-to-left conversion of eq. 3.mlxv; the trial multiplier ν is not silently identified with this covector. The actual transpose density and bounded divided exponential differential give

$$j_a(U_h^a(V_B)) = Q^\sharp \omega_a, \qquad \|\omega_a\|_\infty + \|j_a(U_h^a(V_B))\|_\infty + \|F(U_h^a(V_B))\|_\infty \le C(\kappa + \|B\|_\infty). \tag{3.mccclxi}$$

Here Q is re-centered at the output and the physical fine units are retained. The first equality follows from actual weighted stationarity and the admitted coordinate conversions. The current estimate uses that equality, not an estimate of a second difference of the corrected potential. Its unweighted Wilson current is not asserted pointwise bounded.

Lemma 3.198 (Uniqueness on a strong trial neighborhood)**.** For the same fixed smaller domains, the weighted critical orbit is unique among orbits with a sufficiently small $C^{1,\alpha}$ representative relative to the trial and with the same literal datum. No $W^{2,p}$ assumption on a competing representative or point bound on its unweighted current is required. ■

Proof. Put a competing orbit into the actual trial mean slice by the unchanged site construction. Its relative potential has strong norm at most $C\varepsilon$. Let $\mathcal{H}_{0,a}$ be the full row lift of $\mathcal{L}_{0,a}$. It satisfies $Q\mathcal{H}_{0,a} = 1$. The inverse

$$\mathscr{H}_A = \mathcal{H}_{0,a}(\mathfrak{q}_0'(A)\mathcal{H}_{0,a})^{-1}$$

exists on a small strong ball, by the actual row estimates. Its normalized columns have the exponential energy bounds of the full trial lift.

Pair stationarity with $\mathscr{H}_A b_{c,e}$. The trial term j_0^a gives $C\kappa$. For the remaining term use the weak Wilson readout in eq. 3.mlxii, separately in each weighted plane. Its integration by parts pairs the first field difference with the energy column. Thus it is bounded by $C\|A\|_{\mathcal{A}_\alpha}$; no derivative of the actual current is evaluated pointwise. Consequently

$$\|\nu\|_\infty \le C(\kappa + \|A\|_{\mathcal{A}_\alpha}).$$

This is the weighted version of eq. 3.mlxxiii.

Every finite-array competing field has a finite $W^{2,p}$ norm, though it has not yet been uniformly bounded. Apply the exact inverse $\mathcal{L}_{0,a}^{-1}$ to its equation. The nonlinear forces are bounded in $\mathcal{F}_p$ by its strong norm and the multiplier bound just obtained. It follows that

$$\|A\|_{\mathcal{W}_p} + \|\nu\|_\infty \le C\{\kappa + \|B\|_\infty + \varepsilon^2 + \kappa\varepsilon\}.$$

Choose the strong uniqueness neighborhood and then the working ρ, κ so this lies inside the larger contraction ball. The competing field is therefore the same slice solution. The slice and literal extension identify precisely the top-fixed gauge orbit. All these norm estimates and inverses hold on the smaller complex domains, so the same uniqueness applies there. □

3.48.4. *The weighted physical atlas.*

Proposition 3.199 (A diagonal deformation of the curvature atlas)**.** There are a curvature ceiling $\kappa_* > 0$, a weight polydisc $\mathcal{A}_+$ about 1, and a datum radius $\rho > 0$, independent of depth and compatible volume, with the following properties. Every real coarse V with $\|V(\partial p) - 1\| \le \kappa \le \kappa_*$ has an actual weighted constrained-critical fine orbit $U_h^a(V)$. For real positive a in a fixed smaller cube, its literal mean is V, and

$$\|F(U_h^a(V))\|_\infty + \|j_a(U_h^a(V))\|_\infty + \|\omega_a(V)\|_\infty \le C\kappa. \tag{3.mccclxii}$$

It has a jointly holomorphic germ in a and $\|B\|_\infty < \rho$ about each real coarse centre $V_B = e^{iB}V$. The germs identify as physical orbits on controlled overlaps and define a holomorphic total action

$$S_h^a(V) = \mathcal{A}_h^a(U_h^a(V)).$$

For real data,

$$0 \le S_h^a(V) \le C\kappa^2\mathcal{V}. \tag{3.mccclxiii}$$

Large holonomies and nontrivial bundle topology are allowed. At fixed a there is real coarse gauge, translation and reflection covariance; axis permutations also permute a. ■

Proof. The smooth trial of CTA is independent of a. Lem. 3.196 supplies its weighted force, and Lem. 3.197 corrects the actual datum error. At the real centre $B = 0$, both errors are $O(\kappa)$. Equation eq. 3.mccclxi gives eq. 3.mccclxii.

For overlaps use the same Kato alignment of the trial bundles as CTA. The aligned trial connections differ by $C\rho$ in every fixed local derivative norm, by eq. 3.mcccxlvi and eq. 3.mcccxlvii. The two weighted corrections have strong size $C(\kappa + \rho)$. Choose these constants inside the common strong uniqueness neighborhood of Lem. 3.198. The two solutions have the same literal datum and weighted equation, so their orbits coincide. The actual mean slices and literal gauges are holomorphic, giving holomorphic transitions. Their top-fixed cocycle follows from the free top-fixed gauge action, exactly as in CTA. No multiplicative cocycle for the Kato map itself is being assumed.

Real fine-frame changes and coarse gauge changes conjugate the full equations. The CTA short-path and partition policy is equivariant under coordinate isometries. The weighted action has the stated joint permutation covariance, and the other constructions are unchanged. Uniqueness therefore proves all the covariance claims.

Finally a fine plaquette has $1 - \frac{1}{2}\,\mathrm{tr}\,U_p \le Ch^4\kappa^2$. The weights are bounded and positive, and there are $O(h^{-4}\mathcal{V})$ plaquettes. This proves eq. 3.mccclxiii. Use the datum itself as real centre for this estimate. No global bond gauge close to a flat connection enters the proof. □

The corrected orbit remains in a fixed smaller $C^{1,\alpha}$ tube of its smooth trial. Thus the existing off-current families may be evaluated there with a sufficiently small independent current, in particular with j_a. They are the operators of their stated convention. This observation does not identify an unweighted canonical precision with the weighted physical Hessian. The latter is treated next.

3.48.5. *The physical weighted Hessian and its row mass.* Fix a real point of the atlas, put $U = U_h^a(V)$, and re-center the actual rows and derivatives there. Let $\mathscr{D}_a$ be the full normal Hessian of $\mathcal{A}_h^a$, and write $\Delta^{(2)}[\omega_a]$ for the actual second constraint jet paired with its physical multiplier. Define

$$\mathscr{H}_{L,a} = \mathscr{D}_a - \Delta^{(2)}[\omega_a]. \tag{3.mccclxiv}$$

All these derivatives are in the declared straightened normal coordinates. Gauge invariance identifies the scalar action with the literal coarse action, as in NEG.

We record the gauge-current term as well. Let $B_U[j_a]$ be the finite-gauge curvature operator of the canonical convention. The first inverse and its first row lift are

$$G_a = \{\mathscr{D}_a - B_U[j_a] + DRD^\sharp + Q^\sharp a_f Q\}^{-1}, \qquad H_a = G_a Q^\sharp (QG_aQ^\sharp)^{-1}. \tag{3.mccclxv}$$

The second inverse is formed afterward:

$$\mathscr{G}_a^{-1} = \mathscr{D}_a - B_U[j_a] + DRD^\sharp + Q^\sharp a_f Q - S^\sharp \Delta^{(2)}[H_a^\sharp j_a] S. \tag{3.mccclxvi}$$

The lift in the multiplier readout is the first lift. The row covariance in the effective Hessian below will use the second inverse.

Lemma 3.200 (The physical identity and energy gap)**.** For the real atlas points and the fixed real weight cube, the two inverses in eq. 3.mccclxv–eq. 3.mccclxvi exist with uniform energy bounds. Moreover

$$\mathscr{G}_a^{-1} = \mathscr{H}_{L,a} + DRD^\sharp + Q^\sharp a_f Q, \qquad c\|v\|_{\mathcal{E}}^2 \le \langle v, \mathscr{G}_a^{-1} v\rangle \le C\|v\|_{\mathcal{E}}^2. \tag{3.mccclxvii}$$

Here the physical homogeneous unit-scale norm is

$$\|v\|_{\mathcal{E}}^2 = \|D_U v\|_{2,h}^2 + \|v\|_{2,h}^2,$$

equivalent to the energy norm in the nonflat row-saddle theorem. The fine Lagrangian form also satisfies

$$\langle v, \mathscr{H}_{L,a} v\rangle \ge c\|\mathrm{d}_U v\|_{2,h}^2 - C\kappa\|v\|_{\mathcal{E}}^2, \qquad |\langle v, \mathscr{H}_{L,a} w\rangle| \le C\|v\|_{\mathcal{E}}\|w\|_{\mathcal{E}}. \tag{3.mccclxviii}$$

■

Proof. The Wilson form calculation is applied plane by plane. The real coefficients $a_\mu a_\nu$ lie between fixed positive constants. Its leading curl form is therefore comparable to $\|\mathrm{d}_U v\|_2^2$. The actual curvature terms have bound $C\|F(U)\|_\infty\|v\|_{\mathcal{E}}^2$. They are retained in $\mathscr{D}_a$. The finite-gauge current operator has bound $C\|j_a\|_\infty\|v\|_{\mathcal{E}}^2$. The constraint-curvature form has bound $C\|\omega_a\|_\infty\|v\|_{\mathcal{E}}^2$, by the actual two-leg estimate. All three small quantities are $O(\kappa)$ by eq. 3.mccclxii.

The geometric curl, restoration and row estimate underlying eq. 3.dcccxxviii is unchanged. Combining it with these plane estimates proves the gap for the first variant. The actual coarse row lift and its physical energy seeds give an invertible row Schur matrix. Since $QH_a = 1$ and $j_a = Q^\sharp \omega_a$, its readout is exactly

$$H_a^\sharp j_a = \omega_a.$$

The small second correction therefore preserves the energy gap.

For its exact form use weighted gauge invariance before simplifying any term. If Π is the finite gauge chain, then $\mathcal{A}_h^a(e^{ih\Pi(A)}U) = \mathcal{A}_h^a(e^{ihA}U)$. Twice differentiating gives the same finite-chain identity as in ACC, with the weighted physical current:

$$\mathscr{D}_a - B_U[j_a] = S^\sharp \mathscr{D}_a S.$$

The physical stationarity and the exact mean gauge cocycle give the Lagrangian Ward identity used in eq. 3.cmxxi. Its algebra is unchanged by constant plaquette weights: its first gradient is $Q^\sharp \omega_a$, and the coarse Ward covector annihilates the induced coarse gauge. Consequently

$$S^\sharp\{\mathscr{D}_a - \Delta^{(2)}[\omega_a]\}S = \mathscr{H}_{L,a}.$$

Substitution in eq. 3.mccclxvi proves the equality in eq. 3.mccclxvii. Thus the gauge-current term was used in the identity; it was not discarded on the ground that the background is nearly flat.

The plane form estimate and the actual multiplier bound now give eq. 3.mccclxviii. The same geometric curl–restoration–row estimate, with κ decreased, proves the two energy inequalities in eq. 3.mccclxvii. They concern the actual full constrained precision. No positive fine Dirichlet compression has been identified with it. □

Let

$$\mathcal{H}_a = \mathscr{G}_a Q^\sharp (Q\mathscr{G}_a Q^\sharp)^{-1}. \tag{3.mccclxix}$$

The actual row seeds, their energy bound and eq. 3.mccclxvii imply

$$Q\mathcal{H}_a = 1, \qquad RD^\sharp \mathcal{H}_a = 0, \qquad \|\mathcal{H}_a z\|_{\mathcal{E}} \le C\|z\|_{2,c}.$$

For completeness, the row-seed identity $QE_Q = 1$ gives an inf-sup lower bound for $Q^\sharp$ from coarse L^2 to $\mathcal{E}^*$. The bounded row map gives its upper bound. The energy inverse then makes $Q\mathscr{G}_a Q^\sharp$ comparable to the coarse identity, with no fine rank factor. The middle identity follows from the physical gauge test, as in NEG.

Twice differentiate the actual constrained graph. Its first derivative in the fixed horizontal slice is $\mathcal{H}_a$, and its second derivative satisfies the differentiated constraint. The first fine action derivative is $Q^\sharp\omega_a$. The chain rule therefore gives

$$\Gamma_{h,a} := D^2 S_h^a(V) = \mathcal{H}_a^\sharp \mathscr{H}_{L,a} \mathcal{H}_a = (Q\mathscr{G}_a Q^\sharp)^{-1} - a_f. \tag{3.mccclxx}$$

The second equality follows by adding the artificial row mass to the differentiated stationarity equation and using $Q\mathcal{H}_a = 1$ and horizontality. Subtracting a_f is essential. Positivity of the mass-added Schur matrix alone would not give the next estimate.

3.48.6. *The effective curl and the outer Gaussian.* The covariant flux comparison Lem. 5.38 is independent of the action coefficients:

$$\|\mathrm{d}_V Q v\|_{2,c} \le C_L\{\|\mathrm{d}_U v\|_{2,h} + \kappa\|v\|_{2,h}\}. \tag{3.mccclxxi}$$

The local gauge proof there needs curvature, the actual normalized row moments and complete carriers, not an unweighted current bound. Alternatively the constructed trial frames and the strong correction provide the same small local potential for this atlas. The comparison retains the flat positive flux-row norm before its $O(\kappa)$ error. It does not count fine plaquettes.

Insert $v = \mathcal{H}_a z$ in eq. 3.mccclxviii and use eq. 3.mccclxxi. The lift energy bound absorbs the error. Together with the upper form bound this gives

$$\Gamma_{h,a}[z,z] \ge c\|\mathrm{d}_V z\|_{2,c}^2 - C\kappa\|z\|_{2,c}^2, \qquad |\Gamma_{h,a}[z,w]| \le C\|z\|_{2,c}\|w\|_{2,c}. \tag{3.mccclxxii}$$

This is a bound on the mass-removed physical Hessian.

Use the actual one-step outer graph

$$V(u) = e^{i\Phi_V(u)}V, \qquad \mathsf{C}_V = D\Phi_V(0), \qquad \Phi_V(u)|_{\mathrm{free}} = u.$$

Let $K_G(V)$ be its relative contour-gauge Hessian. Define the complete composed action and outer precision by

$$K_{A,h,a}(V) = \mathsf{C}_V^\sharp \Gamma_{h,a}\mathsf{C}_V + \langle \omega_a, D^2\Phi_V(0)\rangle, \qquad K_{h,a,\tau}(V) = \tau K_{A,h,a}(V) + K_G(V). \tag{3.mccclxxiii}$$

The second term is the outer nonlinear constraint contribution. It is separate from the inner $\Delta^{(2)}[\omega_a]$ in eq. 3.mccclxiv. Centering the outer action by its linear free-coordinate jet changes neither term.

Proposition 3.201 (A gap throughout the weighted real atlas)**.** On a fixed smaller real weight cube and for sufficiently small κ, the actual outer precision obeys

$$c_L I \le K_{h,a,\tau}(V) \le C_L I \tag{3.mccclxxiv}$$

for real τ in a fixed interval about one. Constants are independent of the inner depth and compatible volume. ∎

Proof. The finite outer chart has $\|\mathsf{C}_V\|_{2\to 2} \le C_L$, $\|\mathsf{C}_V u\|_2 \ge \|u\|_2$, and its second derivative paired with a bounded covector has norm at most $C_L\|\omega_a\|_\infty$, by eq. 5.cclv. Thus eq. 3.mccclxxii gives

$$K_{A,h,a}(V)[u,u] \ge c\|\mathrm{d}_V\mathsf{C}_V u\|_2^2 - C_L\kappa\|u\|_2^2, \qquad \|K_{A,h,a}(V)\| \le C_L.$$

The geometric outer curl–gauge inequality eq. 5.cclvii has no action coefficient in it. Combine it with this estimate. The positive real coefficients and τ in a compact positive interval retain a fixed multiple of that geometric gap. Decrease κ to absorb its negative mass error. This proves the lower bound. The finite chart, gauge and effective Hessian bounds give the upper bound.

For a nonaxial real V, the exact node-fixed block gauge is a real orthogonal change of free coordinates, and it carries the relative contour gauge to the axial one. Thus the estimate holds in the literal frame as well. No global trivialization of the coarse bundle is used. □

3.48.7. *Complex sectors and the response retained.*

Proposition 3.202 (A uniform sector near every real centre)**.** Fix a real atlas centre (a_0, V_0, τ_0) in smaller weight and τ intervals. There are positive radii, uniform in that centre, depth and compatible volume, such that the physical analytic continuation satisfies

$$K_{h,a,\tau}(e^{iB}V_0) = K_0^{1/2}(I+E)K_0^{1/2}, \qquad K_0 = K_{h,a_0,\tau_0}(V_0), \qquad \|E\| \le \tfrac{1}{2}, \tag{3.mccclxxv}$$

when $|a-a_0|, \|B\|_\infty, |\tau-\tau_0|$ lie in those radii. The matrices are complex symmetric; the inverse and the determinant branch are holomorphic. ■

Proof. Use the common trial chart at V_0. The weighted fixed point and multiplier are holomorphic on larger uniform product domains. Their Cauchy estimates give uniform first variations in the strong field and row norms. The actual row column moments and restoration comparisons then bound their fine energy-operator variations. In particular, after identifying the fixed real comparison spaces, the change of $\mathscr{H}_{L,a} + DRD^\sharp + Q^\sharp a_f Q$ as an operator $\mathcal{E} \to \mathcal{E}^*$ is at most

$$C\{|a-a_0| + \|B\|_\infty\}.$$

The Wilson estimate here is a derivative form estimate with bounded coefficient variations; it does not estimate a rough unweighted current. The multiplier term uses its actual point-small analytic ω_a. The unchanged row and gauge families have their uniform complete-column bounds on the same strong tube.

The real gap in eq. 3.mccclxvii therefore gives a complex energy inverse by a Neumann series. Its row Schur inverse has the uniform inf-sup margin proved above. The exact resolvent identity and eq. 3.mccclxx give

$$\|\Gamma_{h,a}(e^{iB}V_0) - \Gamma_{h,a_0}(V_0)\|_{2,c\to 2,c} \le C\{|a-a_0| + \|B\|_\infty\}.$$

Subtracting the fixed artificial mass does not alter this variation bound. The finite outer chart, its second derivative, the physical covector and the relative gauge have uniform analytic bounds as well. Consequently the difference of the two matrices in eq. 3.mccclxxv is bounded by $C_L(|a-a_0| + \|B\| + |\tau-\tau_0|)$. Choose the radii below half the real lower gap. This proves eq. 3.mccclxxv.

It follows, for example, that the Hermitian real part of this matrix is at least $K_0/2$. This is a sector estimate, not positivity of an analytic bilinear square. The finite-dimensional log branch of $I+E$ is its convergent power series, fixing the determinant and normalized Gaussian branches. The radii came from operator norms, so they do not shrink with volume. □

The construction also retains the following fixed-order response. For a small exponential cell weight $w = e^{s\psi}$, with ψ one-Lipschitz, its trial-chart first variation satisfies

$$\|D_B A[b]\|_{\mathcal{W}_p;w} + \|D_B \nu[b]\|_{\infty;w} \le C\|b\|_{\infty;w}. \tag{3.mccclxxvi}$$

Differentiate eq. 3.mccclix. Its weighted linear inverse is bounded, and its weighted nonlinear derivative is a contraction after a fixed further reduction of the working ball. The actual mean vertices have fixed physical carriers, so the weight ratios within them are bounded. The same fixed-order Cauchy argument as RTC bounds mixed derivatives with one weighted datum slot and the other slots unweighted, by constants C_n. The literal node extension and physical weighted current inherit their strong and point-force versions. The geometric sum of the intermediate physical path lengths keeps their read stars uniform in depth.

The parameter a is global. Cauchy on its larger fixed polydisc controls its derivatives and their mixed datum responses; no spatial location is assigned to a weight derivative. In particular at a real curvature centre the first weight variation of the trial-chart field and the physical multiplier is

$O(\kappa)$. These are fixed-order localized responses. An all-degree connected carrier expansion for this weighted nonflat atlas is not inferred from them.

The results concern the homogeneous curvature atlas and its actual physical Gaussian. They construct neither a mixed-scale trial from arbitrary proper data nor independent background cube discs, a history tower, or a quantum identity between covers.

3.49. **Constant diagonal weights in the selected connected germ.** We deform the selected off-current equation, including its actual row inversions, by constant diagonal plaquette weights. The principal change requires two finite-p derivatives of the auxiliary field. It does not require two derivatives of the base potential or of its input perturbations. The finite gauge chain and the literal output retain their $C^{1,\alpha}$ regularity. These distinctions permit the physical weighted curvature atlas to enter the construction.

3.49.1. *Fixed strong base data and the stronger output space.* Fix $0 < \alpha < 1$, a finite $p > 4/(1-\alpha)$, and a finite requested carrier rate κ. Work on an admitted SCF and NGC determining geometry with its complete carriers, separation reserves and true bare boundary. The reference U_* is a given real current-regular field with strict margins. The base has the form

$$U = e^{i\eta E} U_*,$$

in a smaller common $C^{1,\alpha}$ tube, and J is a bounded independent current. The full amplitude of J, rather than only its touching restriction, is used here. No point bound for the actual unweighted or weighted Wilson current at the present complex base is assumed. All comparison frames, smooth reconstructions and the literal node extension are fixed at U_*.

The constant parameter is $a = (a_1, \dots, a_4)$, near 1. Replace each $\mu\nu$ Wilson plaquette coefficient by $a_\mu a_\nu$. Write

$$F^a_U(A) = \mathcal{A}^a_{\Omega_0}(e^{i\eta A} U), \qquad j^a_U = (F^a_U)'(0), \qquad H^a_U = (F^a_U)''(0). \tag{3.mccclxxvii}$$

The normalization is that of eq. 3.mcccl and the actual normal coordinates of CNG. The averaging map, site restoration, row mass a_f, and finite gauge chain are unchanged. In particular a_f is not a component of a. Every adjoint below is the physical bilinear transpose; analytic parameters are not conjugated.

Use the force space $\mathcal{F}_p$, field strong space $\mathcal{A}_\alpha$, row datum space $\mathcal{B}$, and multiplier space $\mathcal{W}$ of SCF. On a complete enlarged scale-ξ_y cell set

$$\|A\|_{\mathcal{W}_{p,y}} = \sum_{r=0}^{2} \xi_y^{1+r-4/p} \|D_*^r A\|_{L^p(B_y^+)}. \tag{3.mccclxxviii}$$

All mixed second differences are included. The measure is the physical fine measure. Fixed further cell enlargements accommodate the finite nested estimates. On a bare cell the rescaled mesh is one and this is a finite atomic norm. Uniform discrete Morrey and fixed smooth reconstruction give

$$\mathcal{W}_p \longrightarrow \mathcal{A}_\alpha, \qquad \|A\|_{\mathcal{A}_\alpha} \le C \|A\|_{\mathcal{W}_p}. \tag{3.mccclxxix}$$

Changing D_* to the covariant difference at a base in the stated tube gives an equivalent norm: the difference of the second differences contains only the connection difference, its first difference, and at most one difference of A. It uses no second difference of E.

Put $\mathcal{X}_p = \mathcal{W}_p \oplus \mathcal{W}$ and $\mathcal{Z} = \mathcal{F}_p \oplus \mathcal{B}$. Family norms are the row and each-input column sums of eq. 3.mcxciii, using these cell Banach spaces. They include the supremum of each coefficient on its larger common local base domain before the carrier sum. Full tags, short joins and read collars are charged with $\sigma_c(X) = b_L + d_c(X)$. Choose the primitive families at a stronger rate κ_+, reserving the costs of supported base slots and their output counts. The constant a is a global analytic parameter; it is not a spatial input tag.

3.49.2. *The plane Hessian on a complex strong base.*

Lemma 3.203 (A local principal perturbation)**.** Let $V_a = H^a_U - H^1_U$. It has a local holomorphic coefficient family satisfying

$$\|\mathbf{V}_a\|_{\mathcal{W}_p\to\mathcal{F}_p;\kappa_+} \le C \max_\mu |a_\mu - 1|. \tag{3.mccclxxx}$$

This is a coefficientwise local-data supremum bound on the complex strong tube. The base variable is E in $C^{1,\alpha}$, not in $W^{2,p}$. No current derivative or point bound for j^a_U is required. ■

Proof. Rescale a nonbare cell to length one, with mesh $h = \eta/\xi_y$. Apply the exact four-edge insertion formula eq. 3.dlxiii to one orientation plane. In the same normalization as eq. 3.dcccxcvi, its assembled Hessian is

$$H_{U,\mu\nu} = \mathrm{d}^\sharp_{U,\mu\nu}\mathrm{d}_{U,\mu\nu} + Z_{U,\mu\nu}. \tag{3.mccclxxxi}$$

Here $\mathrm{d}_{U,\mu\nu}$ is the corresponding component of the covariant curl; Z is a finite-shift kernel on a complete two-step collar, with both coefficient sums at most $C\sup_p |P_p - I|/h^2$.

Indeed the exact plaquette insertion differs from the covariant curl by a matrix E_p of size $C|P_p - I|/h$. Expand

$$(\mathrm{d} + E_p)^\sharp(\mathrm{d} + E_p) - \mathrm{d}^\sharp\mathrm{d} = \mathrm{d}^\sharp E_p + E_p^\sharp\mathrm{d} + E_p^\sharp E_p.$$

Multiply their finite stencil entries, whose curl entries have size C/h. The first two products have size $C|P_p - I|/h^2$; the last is smaller on the stated plaquette domain. The remaining insertion term has that same bound. This is multiplication of shifted matrices, not differentiation of the curvature. The incoming and outgoing stencil counts are bounded. The argument works for each plane before any planes are added.

In the complex strong tube the rescaled plaquette satisfies $|P_p - I| \le Ch^2$. Each term of $\mathrm{d}^\sharp_{U,\mu\nu}\mathrm{d}_{U,\mu\nu}A$ is a transported mixed second covariant difference, up to bounded lower-order shifts. It is bounded in cell L^p by the displayed $W^{2,p}$ norm. Thus $H_{U,\mu\nu} : \mathcal{W}_p \to \mathcal{F}_p$ has a uniform local bound. Restoring the scale gives $\xi_y^{3-4/p}$ on its force, exactly the second derivative term in eq. 3.mccclxxviii.

This reasoning also explains the harmless normal-coordinate current coefficients. The single-link current is a fixed sum of plaquette insertions divided by h^3, so $h|j_U| \le C\sup_p |P_p - I|/h^2$, even if $|j_U|$ is not bounded. Terms with that coefficient are bounded finite-shift terms; no derivative of it is taken. On true bare stencils the rescaled mesh is one and the exact finite matrices give the same conclusion.

Relative to the fixed real frame, the first covariant difference changes by a bounded local multiplication and shift depending analytically on E. Its first difference depends only on E, DE. Consequently the second-difference operator in this proof is holomorphic from the strong base ball into $\mathcal{W}_p \to \mathcal{F}_p$, with a uniform bound on its larger local domain. Fixed smooth slot reconstruction uses derivatives only through order two on the field slot and through order one on the base slot.

Finally

$$V_a = \sum_{\mu<\nu}(a_\mu a_\nu - 1)H_{U,\mu\nu}.$$

There are six planes. Assign the bounded local stars, endpoint tags and actual background collar before taking the coefficient norm. Their local-data suprema obey the bounds just proved, so their finite row and column sums give eq. 3.mccclxxx. □

3.49.3. *Two derivatives in the unweighted primitive families.*

Lemma 3.204 (A coefficientwise stronger output)**.** The unweighted families needed for G_0, the positive normalized row inverse $\mathcal{T}_0^{-1}$, and the strong kinematic maps may be chosen so that

$$\|\mathbf{G}_0\|_{\mathcal{F}_p\to\mathcal{W}_p;\kappa_+} \le C. \tag{3.mccclxxxii}$$

The same local-data supremum convention holds. Kinematic maps have the spaces stated in SCF; in particular $\Lambda : \mathcal{A}_\alpha \to \mathcal{Y}_{2,\alpha}$ and $S : \mathcal{A}_\alpha \to \mathcal{A}_\alpha$. No $W^{2,p}$ assertion about S is made. ■

Proof. Retain the second differences at the output of the local positive field inverse in Lem. 3.161. Its energy column and complete row average bound first give local H^1. The component Laplacian equation then gives $W^{2,2}$, gradient L^3, next $W^{2,3}$, and gradient L^8 with bounded field. If $p \le 8$, finish at p; for $p > 8$, use 8 first and then p. All cutoffs lie in a fixed larger complete output collar. Coefficient bounds in the comparison frame use only its $C^{1,\alpha}$ potential. The complex tube has the same bounds. Before Morrey, this proves

$$\|\chi I_D \chi\, 1_z f\|_{\mathcal{W}_{p,y}} \le C e^{-sd(y,z)} \|f\|_{\mathcal{F}_{p,z}}.$$

This does not assert global $W^{2,p}$ regularity for an uncut Dirichlet solution across its artificial boundary.

The slow commutator still maps this output into $\mathcal{F}_p$ with norm C_p/m: its terms contain $D\chi\, Dv$, $D^2\chi\, v$, complete row differences, and finite-shift curvature terms with one cutoff difference. None uses a third output difference. The weighted centre sums in the positive local walk therefore give a normally convergent family $\mathbf{G}_F : \mathcal{F}_p \to \mathcal{W}_p$. The column and error bounds are uniform on the larger local coefficient domains, so taking their suprema before summing leaves the same positive majorants.

The actual Woodbury identity is

$$G_0 = G_F + G_F(D\widehat{F})\widehat{\Theta}^{-1}(D\widehat{F})^\sharp G_F.$$

Keep the improved output at the leftmost G_F. The site column $D\widehat{F}$ is a bounded force of amplitude three. Its transpose on a strong field has a bounded counting-row pairing, since $\xi^{-3}\xi^{-1}\xi^4 = 1$. All remaining factors use their existing strong or energy bounds. This proves eq. 3.mccclxxxii without differentiating a projection kernel. Unit counting field-row injection has force density of amplitude three by eq. 3.cmlxxxvi, so the improved output also applies at each required row source.

For completeness the positive coarse inversions retain the stronger local-data convention of Lem. 3.164. A combined short power of length m has each entry at most $C(q')^m$ uniformly on the larger coefficient tube. Its declared endpoint ball has polynomially many possible endpoints. Summing their separate suprema costs $C(1 + B_m)^4(q')^m$, not the smaller Hilbert-row bound available at one common base point. Choose the finite power and long cutoff using this polynomial; exponential decay still makes the norm small. Reblocking and the prescribed rate reserves proceed before any weight or current radius is chosen. The exact positive coarse matrices and their gaps have not changed. This proves the required coefficientwise bound through $\widehat{\Theta}^{-1}$ and $\mathcal{T}_0^{-1}$. □

3.49.4. *Weighted inverses in the prescribed order.* Write $B_U[J]$ for the unchanged finite-gauge current operator, and let $S = 1 - D\Lambda$. Define complete operators in the following order:

$$(G_0^a)^{-1} = H_U^a + DRD^\sharp + Q^\sharp a_f Q, \tag{3.mccclxxxiii}$$

$$(G^a)^{-1} = (G_0^a)^{-1} - B_U[J], \qquad H^a = G^a Q^\sharp (QG^a Q^\sharp)^{-1}, \tag{3.mccclxxxiv}$$

$$\omega^a = (H^a)^\sharp J, \qquad C_a = S^\sharp \Delta_U^{(2)}[\omega^a] S, \tag{3.mccclxxxv}$$

$$(G_1^a)^{-1} = (G^a)^{-1} - C_a, \qquad H_1^a = G_1^a Q^\sharp (QG_1^a Q^\sharp)^{-1}. \tag{3.mccclxxxvi}$$

The first lift inside ω^a is fixed before the second inverse is formed.

Proposition 3.205 (Weighted full row-saddle families)**.** On a fixed smaller base tube, weight polydisc and full independent-current ball, the operators above have exact holomorphic local families. The field inverses map $\mathcal{F}_p$ to $\mathcal{W}_p$, and the row lifts map $\mathcal{B}$ to $\mathcal{W}_p$. The second full saddle has a bounded family

$$\mathfrak{I}_{U,J}^a : \mathcal{Z} \longrightarrow \mathcal{X}_p, \qquad (G_1^a)^{-1} A - Q^\sharp \nu = f, \quad QA = d. \tag{3.mccclxxxvii}$$

Its coefficientwise local-data supremum norm at the reserved rate is bounded uniformly in depth, compatible volume and admitted shape. The rooted multiplier family has amplitude at most $C\|J\|_{\infty,3}$. ∎

Proof. First use the genuinely principal family from Lem. 3.203:

$$\mathbf{G}_0^a = \mathbf{G}_0 \star \sum_{n\ge 0} (-\mathbf{V}_a \star \mathbf{G}_0)^{\star n}. \tag{3.mccclxxxviii}$$

Its loop is on $\mathcal{F}_p$; the field output of each G_0 is $\mathcal{W}_p$. Choose the larger weight polydisc so its loop norm is below $1/4$. The series is normally convergent and its complete sum is the exact inverse in eq. 3.mccclxxxiii. Moreover $\mathbf{G}_0^a - \mathbf{G}_0$ has norm $C|a-1|$. This step does not assume invertibility of a new local weighted compression.

SCF gives $\mathbf{B}_U[J] : \mathcal{A}_\alpha \to \mathcal{F}_p$ of norm $C\gamma$, $\gamma = \|J\|_{\infty,3}$, with no derivative of J. Use eq. 3.mccclxxix at its input. Then

$$\mathbf{G}^a = \mathbf{G}_0^a \star \sum_{n\ge 0} (\mathbf{B}_U[J] \star \mathbf{G}_0^a)^{\star n} \tag{3.mccclxxxix}$$

converges after decreasing γ. It retains the stronger output at its leftmost factor.

The physical row normalization is kept explicit:

$$\mathcal{Q}_n = \Xi_\Gamma^2 a_f^{1/2} Q, \qquad \mathcal{Q}_n^\sharp = Q^\sharp a_f^{1/2} \Xi_\Gamma^{-2}. \tag{3.mcccxc}$$

The first map has counting-row output, and the second is its counting-to-physical transpose. Put $\mathcal{T}_0 = \mathcal{Q}_n G_0 \mathcal{Q}_n^\sharp$. Its inverse is already constructed. For $\mathcal{T}^a = \mathcal{Q}_n G^a \mathcal{Q}_n^\sharp$ the complete local families satisfy

$$\|\mathcal{T}^{\mathbf{a}} - \mathcal{T}_{\mathbf{0}}\| \le C(|a-1| + \gamma).$$

Decrease the weight disc and then the current radius so the following exact family converges:

$$(\mathcal{T}^{\mathbf{a}})^{-\mathbf{1}} = \mathcal{T}_{\mathbf{0}}^{-\mathbf{1}} \star \sum_{n\ge 0} [-(\mathcal{T}^{\mathbf{a}} - \mathcal{T}_{\mathbf{0}}) \star \mathcal{T}_{\mathbf{0}}^{-\mathbf{1}}]^{\star n}. \tag{3.mcccxci}$$

Thus H^a is obtained by actual row inversion, not by inferring a new gap from a completed operator bound. The physical lift is $G^a \mathcal{Q}_n^\sharp (\mathcal{T}^a)^{-1} \Xi_\Gamma^2 a_f^{1/2}$.

Pair its current coefficients in that order. A datum $b_c = \xi_c^{-1} e$ at one row obeys

$$\langle H^a b_c, J\rangle = \xi_c^3 \omega_c^a \cdot e.$$

The same absolute row and column sums as in eq. 3.mx prove

$$\sup_c \sum_X e^{\kappa_+ \sigma_c(X)} \xi_c^3 |\omega_{c,X}^a| \le C\gamma.$$

Keep the original fine-current bond in each carrier. The complete bare contribution may be retained here since the full current is small; its second constraint jet is zero. This is the first-lift readout throughout.

The actual two-leg row estimate gives $\Delta_U^{(2)}[\omega^a]$ as a bounded point-force family on strong fields. Composing with the SCF $S, S^\sharp$ families gives $\mathbf{C}_a : \mathcal{A}_\alpha \to \mathcal{F}_p$ of norm $C\gamma$. Consequently

$$\mathbf{G}_1^a = \mathbf{G}^a \star \sum_{n\ge 0} (\mathbf{C}_a \star \mathbf{G}^a)^{\star n} \tag{3.mcccxcii}$$

has the required $W^{2,p}$ output. Its normalized row matrix differs from $\mathcal{T}_0$ by $C(|a-1|+\gamma)$. Apply eq. 3.mcccxci with that difference to construct its exact inverse and then H_1^a.

In physical row spaces write $\mathcal{S}_1^a = Q G_1^a Q^\sharp$. The full saddle is exactly

$$\begin{aligned} A &= G_1^a f + H_1^a (d - Q G_1^a f), \\ \nu &= (\mathcal{S}_1^a)^{-1} (d - Q G_1^a f). \end{aligned} \tag{3.mcccxciii}$$

The row injection is a force of amplitude three; its normalized density is the one used in the preceding step. These identities prove the spaces and bounds. Each Neumann series is dominated by products of positive coefficientwise suprema on the larger local domains. There is no exchange of a global supremum with a conditionally convergent carrier sum. □

The order of choices is important. Fix the rates, p, α, reference margins and all carrier geometry first. Establish the stronger primitive constants. Choose the weight polydisc from the principal and row loop bounds, then the independent-current radius from both current loops and both row inversions. Choose the nonlinear datum and supported base radii last. None depends on fine mesh, depth or volume. These choices do not identify the carrier tile with a prescribed history cube or assert that a one-history-cube separation supplies these larger collars.

3.49.5. *The centered weighted auxiliary equation.* Keep the actual finite gauge map Π_U of eq. 3.mcxlii, and put

$$\Xi(A) = \Pi_U(A) - SA, \quad N_q(c) = \mathfrak{q}_U(c) - Qc, \quad r(A) = \mathfrak{q}_U(\Pi_U(A)). \tag{3.mcccxciv}$$

The inputs to Π are $A, \Lambda A, D\Lambda A$. The identity $\mathrm{Ad}(U_b)(\Lambda A)_+ = (\Lambda A)_- - \eta D\Lambda A$ cancels its apparent division by η before expansion. BHR and SCF therefore give $\Pi : \mathcal{W}_p \to \mathcal{A}_\alpha$, with $\Pi'(0) = S$ and $\Xi(A) = O(\|A\|^2_{\mathcal{W}_p})$ in the strong norm. Two differences of Π would require a third site difference of ΛA. No such estimate is part of this construction.

With ω^a fixed in the A derivative, define

$$\begin{aligned}\mathcal{L}^a_{U,J}(A) = {} & F^a_U(A) - F^a_U(0) - \langle j^a_U, A\rangle - \langle J, \Xi(A)\rangle \\ & - \langle \omega^a, N_q(\Pi_U(A))\rangle + \tfrac{1}{2}\langle RD^\sharp A, RD^\sharp A\rangle \\ & + \tfrac{1}{2}\langle r(A), a_f r(A)\rangle.\end{aligned} \tag{3.mcccxcv}$$

It is a finite-dimensional holomorphic scalar at each finite lattice. Its uniformly bounded nonlinear force families are obtained after centering, as proved next. The actual j^a_U is subtracted even if it is rough.

Lemma 3.206 (The Hessian and its subtracted force family)**.** The exact first two auxiliary jets are

$$(\mathcal{L}^a_{U,J})'(0) = 0, \qquad (\mathcal{L}^a_{U,J})''(0) = (G^a_1)^{-1}. \tag{3.mcccxcvi}$$

Let $N^a_{\mathcal{L}}(A) = (\mathcal{L}^a_{U,J})'(A) - (G^a_1)^{-1}A$. The source tuple

$$\mathcal{R}^a(A,\nu) = \begin{pmatrix} -N^a_{\mathcal{L}}(A) + (r'(A) - Q)^\sharp \nu \\ -(r(A) - QA)\end{pmatrix} \tag{3.mcccxcvii}$$

has an exact analytic local family into $\mathcal{Z}$ with divided coefficient bounds $C_0 C_1^m$ for every degree $m \geq 2$. Constant and linear jets are absent in each filed representative. The coefficientwise larger local-data norms are uniform on the weight polydisc. ■

Proof. The current quadratic jet is precisely $B_U[J]$, because Π is the selected finite chain. The constraint-current jet is $S^\sharp \Delta^{(2)}_U[\omega^a]S$, since $N_q(0) = DN_q(0) = 0$. The penalty contributes $DRD^\sharp$, and $r'(0) = QS = Q$ gives $Q^\sharp a_f Q$ from the row mass. The remaining Hessian is H^a_U. These are exactly the terms in eq. 3.mccclxxxvi, with their stated signs.

For the Wilson force perform the subtraction separately in each orientation plane, pairing the two incident plaquettes on an output bond before estimating. The relative logarithmic calculation of Lem. 3.146 applies plane by plane. It gives the local point-force bound

$$\|(F^a_U)'(A) - j^a_U - H^a_U A\|_{\infty,3} \leq C\|A\|^2_{\mathcal{A}_\alpha}. \tag{3.mcccxcviii}$$

Here the constant and linear jets were removed before filing the paired local force, not canceled between separately estimated carriers. To recall the rough-base step, in a unit cell write $P_h(A) =$

$(ih)^{-1}\log(e^{ihA}e^{iha_0})$, with a_0 bounded in $C^{1,\alpha}$. Its remainder after the linear A jet has first spatial difference $O(hr^2)$ and second difference $O(r^2)$. The apparent terms hAD^2A and $h^2A^2D^2a_0$ use $\|D^2A\|_\infty \le Ch^{-1}\|DA\|_\infty$ and the corresponding estimate for a_0. The linearization difference multiplying D^2A has its factor h. Thus no second base derivative or base-current amplitude survives. Multiplication by the six bounded constants $a_\mu a_\nu$ preserves the bound and holomorphy. A bounded point force enters $\mathcal{F}_p$ with its physical cell normalization.

For completeness, the remaining force identities must keep their valid source types. The constraint value is

$$r(A) - QA = Q\Xi(A) + N_q(\Pi(A)). \tag{3.mcccxcix}$$

For the multiplier force one has

$$\begin{aligned}(r'(A) - Q)^\sharp\nu = \Xi'(A)^\sharp Q^\sharp\nu + S^\sharp DN_q(\Pi(A))^\sharp\nu \\ + \Xi'(A)^\sharp DN_q(\Pi(A))^\sharp\nu.\end{aligned} \tag{3.mcd}$$

The row transposes in this expression are bounded point forces by their actual fixed-leg density. A derivative of Ξ, paired with such a force, has the exact adjoint-factory form

$$f_0 + \Lambda^\sharp h + \Lambda^\sharp D^\sharp f_d.$$

Its inputs have grades three, four and three, respectively. SCF bounds the first two pointwise and the last in finite $\mathcal{F}_p$. It does not put a generic finite-p force into a factory that requires a point force.

Let $C_\omega = \Delta_U^{(2)}[\omega^a]$. The subtracted constraint-current gradient is exactly

$$\begin{aligned}&\nabla\langle\omega^a, N_q(\Pi(A))\rangle - S^\sharp C_\omega SA \\ &\quad = S^\sharp\{C_\omega\Xi(A) + DN_q(\Pi(A))^\sharp\omega^a - C_\omega\Pi(A)\} + \Xi'(A)^\sharp DN_q(\Pi(A))^\sharp\omega^a.\end{aligned} \tag{3.mcdi}$$

The braces are point forces quadratic in A. The outside $S^\sharp$ uses its finite-p bound; the last term uses the same point-force factory. The row-mass remainder is

$$r'(A)^\sharp a_f r(A) - Q^\sharp a_f QA = (r'(A) - Q)^\sharp a_f r(A) + Q^\sharp a_f(r(A) - QA). \tag{3.mcdii}$$

Since $a_f \asymp \xi^{-2}$, its datum input has the required multiplier amplitude three. Finally remove the exact quadratic scalar jet $\langle A, B_U[J]A\rangle/2$ from the nonlinear current term before its factory expansion. Its residual is quadratic in A, with size $C\gamma\|A\|^2$. The gauge penalty is exactly linear in the gradient and has no remaining nonlinear vertex.

All these expressions have at least two joint factors of (A,ν) after the displayed subtractions. Their complete read sets and local strong bounds are the NGC local vertices. The only new local operation, the weighted Wilson sum, has been verified above. Uniform Banach Cauchy bounds on a larger strong ball, followed by the embedding from $\mathcal{W}_p$, give divided coefficients $C_0C_1^m$ into $\mathcal{Z}$. Substitution and the SCF factory bounds preserve their coefficientwise local-data sums. This proves the claim with the arity zeros already removed in each representative. □

3.49.6. *The actual selected germ and its connected trees.* The weighted auxiliary equation is

$$(\mathcal{L}^a_{U,J})'(A) - r'(A)^\sharp\nu = 0, \qquad r(A) = b. \tag{3.mcdiii}$$

Using eq. 3.mcccxciii, it is exactly

$$x = L^a_P b + \mathfrak{N}^a_P(x), \qquad L^a_P b = \mathfrak{I}^a_{U,J}(0,b), \qquad \mathfrak{N}^a_P = \mathfrak{I}^a_{U,J}\mathcal{R}^a, \quad x = (A,\nu). \tag{3.mcdiv}$$

The nonlinear family is a series $\sum_{m\ge2} V^a_{m,P}[x^m]$ with $\|\mathbf{V}^a_{m,P}\|_{\kappa_+} \le C_0C_1^m$. Its linear output space is $\mathcal{X}_p$, while all finite-gauge vertices use the included strong field.

Proposition 3.207 (All-degree weighted selected families)**.** On smaller uniform base, weight and independent-current domains, there is a unique small solution of eq. 3.mcdiii with auxiliary field in $\mathcal{W}_p$. It is holomorphic and has exact connected datum families satisfying

$$x(P,a,b)=\sum_{n\ge1}\mathbf{x}_n^a(P)[b^n],\qquad \sum_{n\ge1}r^n\|\mathbf{x}_n^a\|_{\kappa_+}\le Cr\quad(0<r\le r_b). \tag{3.mcdv}$$

Every coefficient retains all output, input and base read sets, including those of the inverse factors. The supremum on the larger local base and weight domains is included inside this family norm. ■

Proof. The exact fixed point is a contraction on a sufficiently small $\mathcal{X}_p$ ball. For the connected representation expand it in ordered rooted trees. A leaf is the exact family $L_P^a b$, and an internal vertex of arity $m\ge2$ is $V_{m,P}^a$. An n-leaf tree has at most $n-1$ internal vertices. Join its carriers at each shared internal cell, retaining the canonical endpoint tags and fixed short joins. This includes every background read set, not just the tree joining datum positions.

The positive scalar majorant is

$$t=C_{\mathrm{lin}}r+\sum_{m\ge2}C_0C_1^m t^m. \tag{3.mcdvi}$$

Choose r_b so its map preserves $[0,2C_{\mathrm{lin}}r_b]$ and its derivative there is below $1/2$. Iteration from zero gives the convergent sum of all ordered tree weights, bounded by $2C_{\mathrm{lin}}r$. Row sums bound every internal index except along a distinguished input path, where the corresponding column sums apply. The guarded carrier union is submultiplicative. It follows that the actual divided coefficient norms are bounded by this positive series. No additional factorial enters.

The coefficientwise local-data bounds of the linear families and vertices were established before these products. Thus the same positive majorant bounds the sum of their separate suprema. Normal convergence permits substitution into the actual equation. Its sum is in the contraction ball and hence is the selected solution, not merely a formal expansion. □

3.49.7. *Literal inputs and the bounded current update.* For literal coarse data use the actual map $b=b_U(B)$ of eq. 3.mxxxiii, including $Db_U(0)$. Its explicit outward inverse reads the prescribed finer interior rows without a depth product. The NGC local analytic family for this map is unchanged by a. Substitute it into eq. 3.mcdv, decreasing the literal radius so its absolute image lies in the reduced ball. Put

$$C=\Pi_U(A),\qquad \delta\omega=\nu-a_f b_U(B).$$

The selected point-current update is

$$J^a_{\mathrm{new}}=\mathcal{E}^{-\sharp}_{\eta C}\{D\mathfrak{q}_U(C)^\sharp(\omega^a+\delta\omega)+J-Q^\sharp\omega^a\}. \tag{3.mcdvii}$$

Here $\mathcal{E}$ is the same divided exponential differential as in CNG. Its zero datum jet is J. For filing the increment use the exact expression

$$\begin{aligned}J^a_{\mathrm{new}}-J&=(\mathcal{E}^{-\sharp}_{\eta C}-1)J\\&\quad+\mathcal{E}^{-\sharp}_{\eta C}\{DN_q(C)^\sharp\omega^a+D\mathfrak{q}_U(C)^\sharp\delta\omega\}.\end{aligned} \tag{3.mcdviii}$$

Every displayed summand has positive datum degree and point-force amplitude three. In particular there is no termwise cancellation of large bare currents in these estimates; the present independent current is bounded in its full amplitude norm.

Let g be the actual node correction with the fixed extension E_n, and define the literal output by

$$\widehat{U}=(e^{i\eta C}U)^g=e^{i\eta\widehat{A}}U,\qquad \widehat{J}=\mathrm{Ad}(g_-)J^a_{\mathrm{new}}. \tag{3.mcdix}$$

The node argument has a connected $\mathcal{Y}_{2,\alpha}$ family. At assigned scale ξ_y, the actual intermediate node factors have size $C_Lr\xi_t/\xi_y$; their absolute sum is bounded by C_Lr, since $\sum_{t\le j_y}\xi_t/\xi_y\le L/(L-1)$. Their read stars obey the same geometric bound. Thus literalization has no C^k accumulation. Its

divided logarithm is a local analytic function of $C, \zeta, D\zeta$, where $g = e^{i\zeta}$. One field difference uses at most two site differences.

Consequently $C, \widehat{A}$ have strong connected families; $\nu, \delta\omega$ have multiplier families, and $J^a_{\mathrm{new}} - J, \widehat{J} - J$ have point-force families. All have a function-family bound Cr with the same carrier convention as eq. 3.mcdv. The auxiliary field A retains $\mathcal{W}_p$. No $W^{2,p}$ bound for the general C or literal $\widehat{A}$ is inferred from this construction.

3.49.8. *Joint supported-base and global-weight coefficients.* Fix an interior base pair $P_0 = (e^{i\eta E_0} U_*, J_0)$ and an interior weight a_0. Reconstruct base changes exactly as in NGC:

$$E = E_0 + \sum_y \mathcal{R}_y e_y, \qquad J = J_0 + \sum_y \mathcal{O}_y j_y,$$

using supported strong $C^{1,\alpha}$ field slots and bounded current ownership slots. No $W^{2,p}$ condition is imposed on e_y. Write $a = a_0 + \epsilon$, with four global complex components.

Proposition 3.208 (Joint connected Taylor bounds)**.** On fixed smaller slot, literal datum and weight balls, every output increment just listed has an exact joint divided Taylor expansion with

$$\sum_{n\geq 1,\ell,t\geq 0} r^n \delta^\ell \zeta^t \|\mathbf{F}_{n,\ell,t}\|_\kappa \leq Cr. \tag{3.mcdx}$$

Here n is datum degree, ℓ is supported base-slot degree and t is total degree in the four global weight variables. The norm has output and all spatial input row/column bounds; there is no spatial column assigned to a global weight derivative. ■

Proof. For a full carrier X, bounded-overlap reconstruction of its base slots lies in the larger local strong/current domain. The number of slots it reads is $n_X \leq C_{L,M_c}\sigma_c(X)$, by the complete collar counting of eq. 3.mccviii. Their Banach-valued Cauchy and polarization bounds cost at most C^ℓ for each distribution of degree. Summing these distributions at a smaller slot radius costs $(1-C\delta_1)^{-n_X}$, bounded by $\exp(C'\delta_1 C_{L,M_c}\sigma_c(X))$. Choose δ_1 inside the reserved carrier-rate budget and take the actual $\delta \leq \delta_1/2$. The resulting factor $(\delta/\delta_1)^\ell$ justifies summing degreewise supremum norms.

For a column anchored at a supported base input, reserve an additional rate larger than $\log D_L + 1$, where D_L is the bounded degree of the tag graph. A possible output at tag distance j then has its factor $e^{-\upsilon j}$, where $\upsilon > \log D_L + 1$, which sums against the bounded-degree output count. This is the same base-column estimate as in NGC; it does not count fine coordinates within a Banach slot.

The coefficient functions are jointly holomorphic in the four global weight variables on a larger fixed polydisc. Weight Cauchy costs

$$\sum_{\beta\in\mathbb{N}^4} (\zeta/\rho_a)^{|\beta|} = (1 - \zeta/\rho_a)^{-4}$$

for a sufficiently small ζ. This is a fixed factor, independent of the carrier and its size. It requires no extra spatial tag. Combining these bounds with the datum tree majorant proves eq. 3.mcdx. Every output increment vanishes at $B = 0$ for every base and weight, with that zero removed before filing; hence $n \geq 1$. Normal convergence identifies the joint series with the same selected germ and literal update. □

3.49.9. *Physical restriction through the curvature atlas.* Physical matching uses a supplied weighted physical critical pair and its controlled local physical branch. It is not a producer for arbitrary proper coarse data. Suppose first that

$$J = j^a_U = Q^\sharp \omega_{\mathrm{phys}} \tag{3.mcdxi}$$

is within the independent-current radius. Then $QH^a = 1$ gives $\omega^a = \omega_{\mathrm{phys}}$ from the first lift. Weighted fine gauge invariance implies the exact scalar identity

$$\begin{aligned}\mathcal{L}^a_{U,j^a_U}(A) = F^a_U(\Pi_U(A)) - F^a_U(0) - \langle \omega_{\mathrm{phys}}, r(A)\rangle \\ + \tfrac{1}{2}\langle RD^\sharp A, RD^\sharp A\rangle + \tfrac{1}{2}\langle r(A), a_f r(A)\rangle.\end{aligned} \tag{3.mcdxii}$$

Indeed $F^a_U(\Pi_U(A)) = F^a_U(A)$, $QS = Q$, and the two centered linear/current terms combine to $-\langle \omega_{\mathrm{phys}}, Q\Pi_U(A)\rangle$. The nonlinear constraint term supplies its remainder. This is an exact weighted identity, not a comparison of Hessians alone.

Proposition 3.209 (Admission and matching of WCA physical pairs)**.** Restrict the WCA weight polydisc, curvature ceiling and datum radius by the fixed constants of this subsection. Every real physical WCA pair, and its smaller complex physical continuation, lies in the present strong-base and independent-current domain through its smooth real CTA trial. At that pair the selected weighted germ and its literal update coincide with the continued physical WCA orbit and actual weighted left current. The all-degree connected and joint bounds above apply to these local perturbation germs. ■

Proof. At a real coarse centre the CTA trial U_* has uniform local all-C^m potentials and unweighted curvature/current $O(\kappa)$. It is a valid real current-regular reference for SCF, BHR and NGC, even with large holonomies and nontrivial bundle topology. WCA puts its literal weighted orbit in a strong tube of this trial of size $C\kappa$, and proves $j_a(U) = Q^\sharp_U \omega_a$ with both point amplitudes $C\kappa$. On a smaller coarse complex ball the bounds are $C(\kappa+\rho)$. Choose these below the present strong and current margins and use the intersection of the two weight polydiscs. The unweighted current of the weighted orbit need not be point-small. The hypotheses concern its strong distance from the smooth real trial, which WCA supplies.

Consider the supplied physical WCA branch about this pair. The unchanged actual mean-slice chart puts it in $RD^\sharp A_{\mathrm{phys}} = 0$, with $\mathfrak{q}_U(A_{\mathrm{phys}}) = b_U(B)$, by its actual assigned node correction. This is the kinematic chart of PNC and NGC on the strong tube. Its site estimates use the bounded local $C^{1,\alpha}$ coefficients and completed site families available there. The finite gauge is the actual chart gauge; the equation $q\lambda = 0$ alone is not being treated as an exact top-fixed finite gauge. The given branch has $\|A_{\mathrm{phys}}\|_{\mathcal{A}_\alpha} \le C\|B\|_{\mathcal{B}}$, by WCA's localized strong response and this gauge chart.

Its multiplier difference is small in amplitude as well. Pair its weighted equation with $H^a_1(\mathfrak{q}'_U(A_{\mathrm{phys}})H^a_1)^{-1}$. This is an actual nearby row right inverse. Compare with the base equation, integrating by parts plane by plane in the Wilson difference. Only first field differences and the exponentially summable energy columns of this lift occur. The variation of the right inverse has size $C\|A_{\mathrm{phys}}\|_{\mathcal{A}_\alpha}$, and the base weighted current is bounded. Thus the multiplier difference has bound $C\|A_{\mathrm{phys}}\|_{\mathcal{A}_\alpha}$. This is the weak readout used in RTC and WCA, applied to the difference from the critical base; there is no nonstationary defect here.

There is no assumed $W^{2,p}$ estimate for this competing branch. Each finite array has a finite such norm. Its physical linearized full precision is the one in eq. 3.mccclxxxvi: the weighted finite-chain and Lagrangian Ward identities are exactly those of Lem. 3.200. Apply $\mathfrak{I}^a_{U,j^a_U}$ to its equation after subtracting this linearization. The weighted centered Wilson residual and the actual second constraint remainder give force bound $C\|A_{\mathrm{phys}}\|^2_{\mathcal{A}_\alpha}$; the multiplier product has the same bound by the weak readout. The datum remainder is quadratic. More explicitly, put $N_F(A) = (F^a_U)'(A) - j^a_U - H^a_U A$ and $C_\omega = \Delta^{(2)}_U[\omega_{\mathrm{phys}}]$. In the physical slice the source and row unknown are

$$\begin{aligned} f &= -N_F(A) + DN_q(A)^\sharp \omega_{\mathrm{phys}} - C_\omega A + DN_q(A)^\sharp \delta\omega_{\mathrm{phys}} - Q^\sharp a_f N_q(A), \\ d &= b_U(B) - N_q(A), \qquad \nu = \delta\omega_{\mathrm{phys}} + a_f b_U(B). \end{aligned} \tag{3.mcdxiii}$$

Every force summand is quadratic with the just established weak multiplier bound. The last force is the retained artificial row-mass correction; it has not been omitted. Consequently

$$\|A_{\mathrm{phys}}\|_{\mathcal{W}_p} + \|\delta\omega_{\mathrm{phys}}\|_{\mathcal{W}} \le C\{\|b_U(B)\|_{\mathcal{B}} + \|A_{\mathrm{phys}}\|^2_{\mathcal{A}_\alpha}\}.$$

For sufficiently small B this lies inside the auxiliary contraction ball. On its exact slice $\Pi_U(A_{\mathrm{phys}}) = A_{\mathrm{phys}}$. Differentiate eq. 3.mcdxii: the physical pair solves the auxiliary system with $\nu = \delta\omega_{\mathrm{phys}} + a_f b_U(B)$. The penalty derivative vanishes there. Contraction uniqueness identifies the auxiliary and physical slice solutions. The actual node gauge then identifies their literal representatives.

At the physical pair, eq. 3.mcdvii reduces to the actual normal-to-left differential conversion of the weighted stationarity equation. It is therefore the weighted left current of the updated field. Its gauge transform in eq. 3.mcdix is the actual literal weighted current. All arguments use algebraic identities and uniform norm estimates on the smaller complex domains, so holomorphic continuation retains the matching. No unrelated weighted PNC producer outside the supplied WCA branch was imported. □

Off current the first reduced-datum jets are

$$D_b A(0) = H_1^a, \qquad D_b C(0) = S H_1^a.$$

They are not equated. Literal data add $Db_U(0)$ and the actual fine node-gauge derivative. On a physical WCA overlap, both sums represent the same weighted physical orbit by the exact matching, so their total action and actual current agree under the declared gauge transition. Individual filed coefficients are not asserted gauge invariant or reference independent.

For small periods perform the primitive construction, coarse inverse, nonlinear vertices and tree products on one sufficiently large common cover before any carrier projection. Retain lifted data and winding joins, then deck-average the classical coefficient families and descend their sums. The norms have the same positive majorants, and uniqueness identifies the descended sum with the finite periodic selected equation. This is descent of classical operators and critical germs, not an equality of quantum partition functions on different covers.

This result supplies the weighted all-degree selected germ and its physical WCA restriction. It constructs neither a trial for arbitrary proper coarse data nor independent background cube discs. It proves no weighted nonflat formation/tower/deletion statement, and it does not identify a separately prescribed off-current core convention merely from agreement of its quadratic precision.

4. Effective interactions and large fields

We now organize one integration step and its iteration. The local estimates separate the relevant terms from the remainder, while the history sum controls regions in which the small-field expansion is unavailable. The source-dependent construction must preserve both normalization and the stated analytic domains. The estimates below specify where each of these requirements enters the induction.

4.1. **Separated correlations from local source bounds.**

4.1.1. *From local source estimates to separated correlation limits.* Local analytic energy estimates and a bound on the residual finite-volume integral imply bounds on separated correlations. This section proves that implication. Its hypotheses remain to be established for four-dimensional gauge theory: exact transport of a measure alone gives neither a partition-function bound nor decay of the dependence on older scales.

4.1.1.1. Finite-dimensional starting point. Let $\mathcal{P}$ be a finite set of plaquettes, let $G = \mathrm{SU}(2)$, and assign one variable $U_b \in G$ to each positively oriented bond. Opposite orientations carry $U_{-b} = U_b^{-1}$. The product of normalized Haar measures is denoted by dU. Write

$$a_p(U) = 1 - \tfrac{1}{2} \operatorname{Re} \operatorname{tr} U_p,$$
$$A(U) = \sum_{p \in \mathcal{P}} a_p(U).$$

Since $0 \le a_p \le 2$, the probability measure

$$d\mu_\beta(U) = Z_\beta^{-1} e^{-\beta A(U)} dU,$$
$$Z_\beta = \int e^{-\beta A(U)} dU$$

is well defined for every real β. Given bounded real functions f_i on plaquette barycentres, set $X_i = \sum_p f_i(x_p) a_p(U)$ and

$$\Gamma(z) = \int e^{\sum_i z_i X_i} d\mu_\beta.$$

The bound $|X_i| \le 2 \sum_p |f_i(x_p)|$ implies uniform domination on every compact subset of $\mathbb{C}^n$. Differentiation under the integral therefore proves that Γ is entire. Since $\Gamma(0) = 1$, it has a holomorphic logarithm in a neighbourhood of the origin, and the joint cumulant is defined by

$$\langle X_1; \dots; X_n \rangle = \partial_{z_1} \cdots \partial_{z_n} \log \Gamma(0).$$

The neighbourhood obtained this way can depend on the number of plaquettes.

If T is a fixed linear operation on finite measures which preserves their total mass, then $T(e^{\sum z_i X_i} \mu_\beta)$ has total mass $\Gamma(z)$. For a finite sequence of such operations the same conclusion follows by induction. If T is specified only on gauge-invariant densities, all insertions in this assertion must be gauge invariant. Neither linearity nor mass preservation supplies a uniform analytic bound on the resulting density.

4.1.1.2. A zero-free disc from a normalized bound.

Lemma 4.1. Let $c_i > 0$ and put $\|z\|_c = \sum_i c_i |z_i|$. Suppose h is holomorphic on $\|z\|_c < 1$, $h(0) = 1$, and $|h| \le Q$, where $Q \ge 1$. Then

$$|h(z) - 1| \le (Q+1) \|z\|_c.$$

In particular h is nonzero on $\|z\|_c < (Q+1)^{-1}$. On the smaller ball $\|z\|_c < [2(Q+1)]^{-1}$ its principal logarithm satisfies $|\operatorname{Log} h| \le \log 2$. ■

Proof. For $z \ne 0$ write $z = tv$, where $t = \|z\|_c$ and $\|v\|_c = 1$. The function $(h(\zeta v) - 1)/(Q+1)$ maps the unit disc into its closed unit disc and vanishes at zero. The Schwarz lemma gives the first estimate. Strictness on the stated open balls follows by using slightly smaller compact discs and taking limits. For $|u| \le 1/2$, the logarithmic series gives

$$|\operatorname{Log}(1+u)| \le \sum_{k \ge 1} \frac{|u|^k}{k} = -\log(1 - |u|) \le \log 2.$$

The same series defines a single holomorphic branch on that ball. □

Corollary 4.2. Suppose $\Gamma = e^{\mathcal{E}} h$ near zero, where $\mathcal{E}(0) = 0$ and the lemma applies to h. Then

$$|\partial_{z_1} \cdots \partial_{z_n} (\log \Gamma - \mathcal{E})(0)| \le \log 2 \, [2(Q+1)n]^n \prod_i c_i. \qquad \blacksquare\square$$

Proof. Use Cauchy's integral formula on a polydisc with radii strictly smaller than $[2(Q+1)nc_i]^{-1}$. The sum of the weighted radii is less than $[2(Q+1)]^{-1}$. Each variable is differentiated once, so no factorial occurs in this application of the formula. Let the radii increase to their stated values. □

The dependence on Q is material. For example, $\cosh(Vz)$ is the moment-generating function of a random variable taking the values V and $-V$ with equal probabilities. It is at most e^V on the unit disc, but has a zero at $i\pi/(2V)$. Thus an exponential-in-volume estimate alone cannot give a volume-independent zero-free disc.

4.1.1.3. A precise local-energy hypothesis. Fix $L > 1$, a finite physical volume V, and a scale index $s \ge 0$, with physical scale $a_s = L^{-s}$. A local domain D has an integer size $m(D) \ge 0$ and a dependence region D^+. Assume there are constants $C_0, C_1, \kappa_{\text{ent}}$ such that

$$\operatorname{diam} D^+ \le C_0 a_s (m(D) + 1),$$

and, for every fixed compact test support B, the number of domains of size m whose dependence region meets B is at most

$$C_1 V a_s^{-4} e^{\kappa_{\text{ent}} m}.$$

The factor V is a finite-volume allowance; a more local counting estimate can replace it when available.

For cutoff K let $e_{K,s,D}(w)$, $0 \le s < K$, be holomorphic on the source ball

$$\sum_i |z_i| \|f_i\|_* / r < 1,$$

$$w = \sum_i z_i f_i,$$

where $r > 0$ is independent of K. Suppose

$$e_{K,s,D}(0) = 0,$$

$$|e_{K,s,D}(w)| \le A e^{-\kappa m(D)},$$

$$\kappa > \kappa_{\text{ent}}.$$

Finally suppose $e_{K,s,D}$ depends on w only through its values in D^+. These statements are the local-energy hypothesis. They include analyticity, an entropy gap, and geometric control of the dependence region.

Proposition 4.3. Let $n \ge 2$ and suppose the supports of $f_1, \dots, f_n$ have pairwise distance at least $d > 0$. Under the local-energy hypothesis,

$$\left| \partial_{z_1} \cdots \partial_{z_n} \sum_{s<K} \sum_D e_{K,s,D}(w) \Big|_{z=0} \right| \le CAV \max\{1, d^{-4}\} (n/r)^n \prod_i \|f_i\|_*.$$

The constant is independent of K; it can depend on L, C_0, C_1 and the entropy gap. ■

Proof. If D^+ misses one support, the energy is independent of the corresponding variable, and the mixed derivative vanishes. Every contributing domain therefore meets both the first and second supports. Consequently

$$m(D) \geq d/(C_0 a_s) - 1.$$

Cauchy's formula with radii approaching $r/(n\|f_i\|_*)$ bounds each derivative by

$$A(n/r)^n \prod_i \|f_i\|_* \, e^{-\kappa m(D)}.$$

A zero test function makes the assertion immediate, so these radii may be assumed finite. The entropy estimate and a geometric series in m bound the sum at scale s by a constant times

$$AV a_s^{-4} \exp(-cd/a_s),$$

with a larger harmless constant when d/a_s is bounded. It remains to estimate $\sum_{s\geq 0} L^{4s} e^{-cdL^s}$.

For $0 < d \leq 1$, let s_* be the largest integer with $dL^{s_*} \leq 1$. The sum over $s \leq s_*$ is at most $CL^{4s_*} \leq Cd^{-4}$. Writing $s = s_* + j$ in the remaining sum bounds it by

$$Cd^{-4} \sum_{j\geq 1} L^{4j} e^{-c' L^j},$$

which converges by the ratio test. For $d > 1$, replace d by 1 in the exponential. This proves the stated uniform bound. At each fixed cutoff the domain sums converge normally on smaller source balls, permitting termwise differentiation there. Only the separated mixed-derivative estimate is asserted uniformly as the number of scales grows; the undifferentiated energy sums need not converge in that limit. □

If in addition $\Gamma_K = e^{\mathcal{E}_K} h_K$, where $\mathcal{E}_K = \sum_{s,D} e_{K,s,D}$ and $|h_K| \leq Q(V)$ with $h_K(0) = 1$, the corollary above gives a cutoff-independent bound on every separated cumulant. This conclusion remains a finite-volume conclusion when Q depends on V.

4.1.1.4. What a comparison of cutoffs must prove. Uniform boundedness gives compactness only after equicontinuity and a suitable countable test class have been supplied. Convergence along a prescribed whole sequence requires more. One sufficient condition is a summable comparison: for every fixed separated tuple,

$$|S_{K+1}(f_1, \ldots, f_n) - S_K(f_1, \ldots, f_n)| \leq C(f_1, \ldots, f_n)\epsilon_K,$$

$$\sum_K \epsilon_K < \infty.$$

The triangle inequality then makes the sequence Cauchy. Its tail is bounded by $C \sum_{k\geq K} \epsilon_k$.

A local proof of this comparison can split the scales at $K/2$. At $s \leq K/2$, it is enough to compare paired local energies with a bound

$$|e_{K+1,s,D} - e_{K,s,D}| \leq C(1+s)\epsilon_K e^{-\kappa m(D)}.$$

At $s > K/2$, the individual energy bounds already give a tail controlled by $\sum_{s>K/2} L^{4s} e^{-cdL^s}$. This tail is summable in K. The factor $1 + s$ in the paired bound is harmless because $\sum_s (1 + s) L^{4s} e^{-cdL^s} < \infty$ for each $d > 0$. The residual integrals must also be compared on a common zero-free source ball. If both residuals are within $1/2$ of 1, then the logarithmic difference is bounded by

twice the difference of the residuals, using the line segment between their values and the derivative $1/z$ of the logarithm. Cauchy's formula converts this bound into a cumulant comparison.

These are consequences of an actual paired estimate. A bound for two covariant constant-source lists does not automatically prove the estimate for position-dependent arrays. Similarly, ordinary holomorphy in an unweighted norm does not create geometric factors for the age of an inherited activity. Those factors must be present in a separately established weighted estimate or in the domain and bound of a weighted holomorphic extension.

4.1.1.5. A conditional tuning lemma. Let F_K be a real differentiable renormalized-coupling coordinate on an interval I_K, and suppose

$$2/3 \le F_K'(\beta) \le 4/3$$
$$(\beta \in I_K).$$

If F_K takes values on both sides of x_*, the intermediate value theorem gives a unique $\beta_K \in I_K$ with $F_K(\beta_K) = x_*$. Moreover,

$$\tfrac{2}{3}|\beta - \beta'| \le |F_K(\beta) - F_K(\beta')| \le \tfrac{4}{3}|\beta - \beta'|.$$

Both inequalities follow from the mean value theorem on the interval between the two points. The inverse coordinate is therefore Lipschitz with constant $3/2$ on its image. If $F_K(\beta) = \beta - \sum_{j\le K} b_j(\beta)$, the derivative hypothesis follows from $\sum_j \sup_{I_K} |b_j'| \le 1/3$. Existence of the interval, the crossing of the target, and the derivative estimate remain hypotheses to be established by the construction.

The conclusions of this section concern separated correlation functionals. They do not, without additional estimates, provide extensions across coincident points, reflection-positive distributions on the full test-function algebra, rotation covariance, or a characterization of the resulting scalar observable sector as a Yang–Mills theory. Those are additional mathematical steps with their own hypotheses.

4.2. **A pointed Hessian and its action counterterm.** For one local interaction, we identify the marginal quadratic term of its pointed total composed with the free response defined below. This term determines the correction to the Wilson coupling; the renormalisation step must subtract it with that interaction's own coefficient and action normalization before estimating the nonlinear remainder. The calculation treats each base point separately. The bulk sum is a convolution, although an individual pointed total need not be. The nonlinear window remainder and its field-domain estimates are not consequences of this quadratic calculation.

The pointed object and the analytic hypotheses. Work in coarse lattice units on $\mathbb{Z}^4$. An independently oriented bond (x, μ) starts at x and ends at $x+e_\mu$. A real field $B_\mu(x)$ is a traceless Hermitian matrix. Choose a color basis t_a, $1 \le a \le 3$, orthonormal for $\langle A, C\rangle = \frac{1}{2}\operatorname{Tr}(AC)$. Thus $B = \sum_a B^a t_a$. Complexification is bilinear, not sesquilinear. The normalization gives the quadratic Wilson action

$$A(e^{itB}) = \frac{t^2}{2}\sum_p |\mathrm{d}B(p)|^2 + O(t^3) \tag{4.i}$$

on every finite set of plaquettes. Here d is the oriented lattice curl and each coordinate-plane plaquette is counted once.

An individual member G has the pointed pair total, in its exponential chart,

$$\mathcal{G}_z(e^{i\eta A}, J) = \sum_{X \ni z} G(X; (e^{i\eta A}, J), z), \qquad z \in \mathbb{Z}^4. \tag{4.ii}$$

The bare spacing η can differ from the coarse unit spacing. There are no stored spectators in a regular member. Assume the following analytic and geometric properties of this actual family.

The sum in eq. 4.ii and its first two derivatives converge normally in the neighborhoods where they are used. Every summand is local in its pair variables and invariant under all local gauge transformations for which the arguments remain in its declared domain. In particular it is invariant under constant adjoint rotations. Let $\mathcal{P}(B)$ be the actual background pair formed from a sufficiently small finitely supported coarse field, with $\mathcal{P}(0) = (1, 0)$. It is twice differentiable, is gauge covariant, and has first response, expressed in this potential chart,

$$hB = D\mathcal{P}(0)B = (HB, D_1^* D_1 HB). \tag{4.iii}$$

The precise linear map H is part of the background construction; it is not replaced by a convolution chosen for this proof. Put $F_z(B) = \mathcal{G}_z(\mathcal{P}(B))$. Assume the pointed totals are covariant under unit translations and the full hypercubic group, including reflections. These are properties of one individual member at all its base points. They do not require two different members chosen at different sites to be related.

Finally assume an absolute exponential Hessian moment bound. With $\delta^a_{x,\mu}$ the field of color a at one coarse bond, let

$$\mathcal{K}_z^{\mu\nu;ab}(x, y) = \sum_{X \ni z} D^2 G(X, (1, 0), z)[h\delta^a_{x,\mu}, h\delta^b_{y,\nu}]. \tag{4.iv}$$

All derivatives of a pair function are taken in the (A, J) chart of eq. 4.ii, including its factor η. For some $m > 0$ assume

$$\sum_{\mu,\nu,a,b} \sum_{x,y\in\mathbb{Z}^4} e^{m(|x-z|+|y-z|)} |\mathcal{K}_z^{\mu\nu;ab}(x, y)| \le C_m. \tag{4.v}$$

The corresponding derivative representation on finitely supported fields is assumed, as is usual when the normally convergent localized Hessians are composed with the free response. This bound and the actual background/core hypotheses are model inputs here. In particular, this section does not infer a uniformly bounded coefficient extractor from a potential norm on a shrinking field ball.

We first justify that eq. 4.iv really is the Hessian of F_z. Constant adjoint invariance makes the first derivative of each summand of eq. 4.ii zero at $(1, 0)$: each coefficient of a potential or current coordinate would be an invariant linear functional on the adjoint color space, which has no fixed vector. This applies to its finitely many local coordinates and hence to every direction supplied by h. The normal convergence permits summation. In the second-order chain rule the term containing $D\mathcal{G}_z(1, 0)D^2\mathcal{P}(0)$ is therefore zero. Consequently eq. 4.iv, with the displayed actual h, is exactly $D^2F_z(0)$.

The exponential moment hypothesis has a concrete sufficient localized form. Suppose the analytic core contains a usable pair norm ball of radius a_{core}, and on it $|G(X, \cdot, z)| \le ge^{-\kappa d(X)}$. Suppose also

$$\|h\delta^a_{x,\mu}\|_{1;X} \le Ce^{-\delta \operatorname{dist}(x,X)}, \qquad \#\{X \ni z : q \le d(X) < q + 1\} \le Ce^{\kappa_D q}, \qquad \kappa > \kappa_D,$$

and $\operatorname{diam}(X) \le C_M(d(X) + 1)$, with fixed localization cube size. These are the core, response and entropy conditions used in quadratic extraction. Two-variable Cauchy gives

$$|D^2G(X, 0, z)[h\delta_x, h\delta_y]| \le Cga_{\mathrm{core}}^{-2} e^{-\kappa d(X)} e^{-\delta(\operatorname{dist}(x,X)+\operatorname{dist}(y,X))}.$$

Choose $m < \delta/2$ so small that $2mC_M < (\kappa - \kappa_D)/2$. Since $z \in X$, inserting the weight in eq. 4.v costs at most $e^{2mC_M(d(X)+1)}$ after retaining the two response distances. The remaining sums over x and y cost at most $C(d(X) + 1)^8$: enclose X in a box of diameter $C_M(d(X) + 1)$ and sum the exponentially decreasing distances outside it. The rooted domain count now gives

$$C_m \le Cga_{\mathrm{core}}^{-2} \sum_{q\ge 0} (q + 1)^8 e^{-(\kappa-\kappa_D-2mC_M)q} \le C' ga_{\mathrm{core}}^{-2}. \tag{4.vi}$$

Thus eq. 4.v need not be introduced as a new physical estimate when these localized hypotheses have already been supplied. The factor a_{core}^{-2} is retained. The coordinate normalization and near-zero core inclusions needed to use this Cauchy estimate are treated in § 4.23.

Color, symmetry, and the local gauge Ward identity. Constant adjoint invariance implies

$$\mathcal{K}_z^{\mu\nu;ab}(x,y) = \delta_{ab} K_z^{\mu\nu}(x,y), \qquad K_z^{\mu\nu}(x,y) = \frac{1}{3}\sum_{a=1}^{3} \mathcal{K}_z^{\mu\nu;aa}(x,y). \tag{4.vii}$$

For a direct proof, fix the spatial indices and view the color coefficients as a 3×3 complex matrix. Adjoint rotations through π about the color axes force its off-diagonal entries to zero, and rotations interchanging the axes make its diagonal entries equal. The adjoint action of SU(2) contains these rotations. This also proves the assertion for complex members. Equality of mixed derivatives gives

$$K_z^{\mu\nu}(x,y) = K_z^{\nu\mu}(y,x). \tag{4.viii}$$

For a real member all these coefficients are real. Thus its bulk kernel is self-adjoint in addition to being bilinearly symmetric. No positivity of a Hessian is assumed.

For a finitely supported Lie-algebra-valued site function χ, write $(\mathrm{d}\chi)_\mu(x) = \chi(x+e_\mu) - \chi(x)$. Gauge covariance of $\mathcal{P}$ and gauge invariance of the member give $F_z(B^g) = F_z(B)$, where B^g is the logarithmic coarse bond field after a small gauge transformation. Choose the gauge convention $U(x,\mu)^g = g(x)^{-1}U(x,\mu)g(x+e_\mu)$, so the derivative at the identity of $g = e^{it\chi}$ is $\mathrm{d}\chi$. Differentiate this identity once in t and once in a finitely supported field direction C. The mixed derivative of the gauge coordinate itself is multiplied by $DF_z(0) = 0$. Therefore

$$D^2F_z(0)[\mathrm{d}\chi, C] = 0. \tag{4.ix}$$

The other sign convention for gauge transformations gives the same zero identity. The argument initially uses real directions and extends complex bilinearly. It uses neither a noncompact gauge transformation nor a polynomial field in a nonlinear field domain.

Polynomial extensions justified by absolute moments. For fields of polynomial growth define the extended Hessian by

$$\mathcal{H}_z(B,C) = \sum_{\mu,\nu,a}\sum_{x,y} K_z^{\mu\nu}(x,y) B_\mu^a(x) C_\nu^a(y). \tag{4.x}$$

Every such sum converges absolutely by eq. 4.v. Compact cutoffs converge to it by dominated convergence, independently of the cutoffs. This defines an extension of the quadratic jet, not a value of the nonlinear member at an unbounded polynomial field.

The Ward identity extends to polynomial gauge parameters and polynomial second arguments. To prove it, let χ be a polynomial of degree q on sites and let $\chi_R(x) = \chi(x)\vartheta(x/R)$ with a fixed smooth compact cutoff equal to one near zero. Its forward differences converge pointwise to $\mathrm{d}\chi$, and satisfy a bound

$$|\mathrm{d}\chi_R(x)| \le C_\chi(1+|x|)^{\max(q-1,0)} \quad (R \ge 1).$$

Indeed away from the cutoff annulus the usual polynomial difference has this bound; on the annulus the extra difference of the cutoff costs R^{-1} and $|x|$ is comparable to R. Apply eq. 4.ix with a single-bond second argument and pass to the limit using eq. 4.v. Summing against any polynomial second argument is then legitimate by absolute convergence. The radius of the nonlinear gauge chart may depend on R: only its derivative at zero was used for each fixed R.

In particular the linear polynomial $\chi(x) = x_\mu t_a$ gives the separate row and column zeroth moments

$$\sum_x K_z^{\mu\nu}(x,y) = 0, \qquad \sum_y K_z^{\mu\nu}(x,y) = 0. \tag{4.xi}$$

The second identity follows also from eq. 4.viii. Both hold for each fixed remaining edge and pair of directions, not only after a full trace.

For the moment at $z = 0$ set, initially in lower bond-start coordinates,

$$T_{\mu\nu,\kappa\lambda} = \sum_{x,y} K_0^{\mu\nu}(x,y) x_\kappa y_\lambda. \tag{4.xii}$$

For a symmetric matrix S, the quadratic gauge polynomial $\chi(x) = \frac{1}{2}\sum_{\mu,\kappa} S_{\mu\kappa}x_\mu x_\kappa t_a$ has forward difference

$$(\mathrm{d}\chi)_\mu(x) = \sum_\kappa S_{\mu\kappa}x_\kappa + \tfrac{1}{2}S_{\mu\mu}.$$

Use the polynomial Ward identity with second argument $C_\nu(y) = y_\lambda t_a$ at one direction ν. The constant term vanishes by eq. 4.xi. Since S is arbitrary symmetric, one obtains

$$T_{\mu\nu,\kappa\lambda} = -T_{\kappa\nu,\mu\lambda}, \qquad T_{\mu\nu,\kappa\lambda} = -T_{\mu\lambda,\kappa\nu}. \tag{4.xiii}$$

The second identity follows by interchanging the two Hessian arguments and repeating the first. These are exactly the two antisymmetries which remove symmetric gradients.

The bulk sum and its second moment. Unit translation covariance of this individual member gives $K_z^{\mu\nu}(x,y) = K_0^{\mu\nu}(x-z, y-z)$. The bulk quadratic kernel is

$$\Pi^{\mu\nu}(x,y) = \sum_z K_z^{\mu\nu}(x,y) = \pi_{\mu\nu}(y-x), \qquad \pi_{\mu\nu}(u) = \sum_x K_0^{\mu\nu}(x, x+u). \tag{4.xiv}$$

All sums converge absolutely. In fact, for $0 < m' \le m$,

$$\sum_u e^{m'|u|}|\pi_{\mu\nu}(u)| \le \sum_{x,y} e^{m'(|x|+|y|)}|K_0^{\mu\nu}(x,y)| < \infty. \tag{4.xv}$$

For fixed x, y, translation of z in eq. 4.xiv is a sub-sum of the same absolute double sum. This proves both the bulk formula and every rearrangement below. Only this bulk object is a convolution. The original $K_0(x,y)$ continues to have two independent spatial arguments.

The zeroth and first bulk moments vanish by eq. 4.xi. Its second moment is

$$\begin{aligned} P_{\mu\nu,\kappa\lambda} &:= \sum_u \pi_{\mu\nu}(u)u_\kappa u_\lambda \\ &= \sum_{x,y} K_0^{\mu\nu}(x,y)(y_\kappa - x_\kappa)(y_\lambda - x_\lambda) \\ &= -T_{\mu\nu,\kappa\lambda} - T_{\mu\nu,\lambda\kappa}. \end{aligned} \tag{4.xvi}$$

The pure $y_\kappa y_\lambda$ term is zero after summing x by its row identity, and the pure $x_\kappa x_\lambda$ term is zero after summing y. This is the reason for both minus signs; no pointwise convolution identity for K_0 has been used.

The Fourier series $\widehat{\pi}_{\mu\nu}(k) = \sum_u \pi_{\mu\nu}(u)e^{ik\cdot u}$ is holomorphic for sufficiently small $|\operatorname{Im} k|$ by eq. 4.xv. In particular $\partial_\kappa\partial_\lambda\widehat{\pi}_{\mu\nu}(0) = -P_{\mu\nu,\kappa\lambda}$. This optional Fourier description concerns the bulk sum alone.

Bond midpoints and the hypercubic tensor. Under reflections, lower bond starts undergo a direction-dependent integer displacement. To apply tensor symmetry without dropping that displacement, move first to bond-midpoint coordinates $\widetilde{x} = x + e_\mu/2$ and $\widetilde{y} = y + e_\nu/2$. The value of T does not change: expanding $\widetilde{x}_\kappa\widetilde{y}_\lambda$ adds only terms of the forms $\sum_{x,y} K_0 y_\lambda$, $\sum_{x,y} K_0 x_\kappa$ and $\sum_{x,y} K_0$, all zero by eq. 4.xi. The displacement u changes by the constant $(e_\nu - e_\mu)/2$. Because the bulk zeroth and first moments are zero, its second moment P is also unchanged. Thus the lower-start extractor and the midpoint extractor agree exactly.

The midpoint bond sets are $\mathbb{Z}^4 + e_\mu/2$. A signed permutation w with $we_\mu = s_\mu e_{\sigma(\mu)}$ maps this set to the midpoint set of direction $\sigma(\mu)$; a negative orientation contributes the sign s_μ to a one-form. Pointed covariance at the fixed base point zero and the ordinary transformation of the midpoint coordinates therefore give

$$T_{\sigma(\mu)\sigma(\nu),\sigma(\kappa)\sigma(\lambda)} = s_\mu s_\nu s_\kappa s_\lambda T_{\mu\nu,\kappa\lambda}. \tag{4.xvii}$$

There is no affine correction left in this formula; the preceding moment calculation proved that it vanished.

We now classify this tensor directly. By eq. 4.xiii it is zero if $\mu = \kappa$ or $\nu = \lambda$. Reflecting one coordinate in eq. 4.xvii shows that each coordinate label must occur an even number of times among

the four indices of a nonzero entry. The two antisymmetric pairs must consequently be the same unordered pair. The only possibilities are $T_{\mu\mu,\kappa\kappa}$ and $T_{\mu\kappa,\kappa\mu}$ for $\mu \neq \kappa$. Permutations make all the first values equal to a number b; eq. 4.xiii makes the second values $-b$. Hence

$$T_{\mu\nu,\kappa\lambda} = b(\delta_{\mu\nu}\delta_{\kappa\lambda} - \delta_{\mu\lambda}\delta_{\kappa\nu}). \tag{4.xviii}$$

Full reflections are used here: four distinct indices, including a possible alternating tensor, are excluded by a single reflection. This is not merely an argument using proper rotations.

Combining eq. 4.xvi with eq. 4.xviii gives

$$P_{\mu\nu,\kappa\lambda} = b(\delta_{\mu\lambda}\delta_{\kappa\nu} + \delta_{\mu\kappa}\delta_{\lambda\nu} - 2\delta_{\mu\nu}\delta_{\kappa\lambda}), \qquad b = P_{\mu\nu,\mu\nu} \quad (\mu \neq \nu). \tag{4.xix}$$

Thus the extractor is explicitly $b[G] = \sum_u \pi_{\mu\nu}(u)u_\mu u_\nu$ for any distinct μ, ν. It has no extra color factor, because eq. 4.vii used the average of the three diagonal color Hessians. The quadratic term of the bulk symbol is $b(|k|^2\delta_{\mu\nu} - k_\mu k_\nu)$; higher lattice derivative terms have not been set to zero.

For fixed actual h, the extractor is complex linear in the member and ignores additive constants. It is continuous in the exponential Hessian moment norm: the absolute second moment is bounded by that norm times $\sup_{r\geq 0} r^2 e^{-mr}$, with a harmless choice of a smaller exponential weight. Normal convergence in that norm also permits termwise extraction and preserves holomorphy in a member parameter. These are Hessian-norm statements, and make no assertion that the Hessian norm is uniformly bounded by a potential norm on coupling-dependent shrinking cores.

The affine quadratic jet and the tent action. Let $B_\mu(x) = a_\mu + \sum_\kappa C_{\mu\kappa}(x-z)_\kappa$, using midpoint coordinates or lower starts. The difference is a constant one-form, which the Ward identity annihilates. eq. 4.xi removes every occurrence of a, and eq. 4.xviii proves

$$\frac{1}{2}\mathcal{H}_z(B,B) = \frac{b}{2}\sum_{\mu<\nu} |C_{\nu\mu} - C_{\mu\nu}|^2. \tag{4.xx}$$

For complex fields the squared norm means the bilinear color pairing; for real fields it is the ordinary squared norm. The identity follows by expanding $\sum_{\mu,\kappa}|C_{\mu\kappa}|^2 - \sum_{\mu,\nu}\langle C_{\mu\nu}, C_{\nu\mu}\rangle$ into unordered pairs. In particular the axial affine choice $\ell_\mu(x) = \sum_{\kappa<\mu}(x-z)_\kappa F_{\kappa\mu}$ has curl F and quadratic jet $\frac{b}{2}\sum_{\mu<\nu}|F_{\mu\nu}|^2$.

To identify this with the member's own action counterterm, one more normalization of the actual free response is needed. Assume that on an affine coarse field ℓ of constant curl it reproduces the same affine bare potential modulo a lattice gradient, and its current component is zero. This is an affine-response statement about the actual averaging/background problem, not a consequence of the tensor classification. For this normalization argument assume also that, at each bare bond in a bounded coarse region, the matrix elements of H in the coarse input bond decay exponentially in its distance from that region. Their constants must be uniform in its finitely many bare bonds at fixed η. This is a precise sufficient free-response input. It makes $H(\ell\vartheta(\cdot/R))$ converge at every such bond to an absolutely convergent polynomially weighted matrix sum, by dominated convergence. Thus its affine reproduction statement is an equality of well-defined extended first responses, with no nonlinear affine field inserted outside its analytic domain.

This input also gives the moment bound needed to apply the tensor theorem to the pointed action itself. Its Hessian is a finite sum over the bare plaquettes in a unit tent of products of first differences of H. The exponential bounds at their finitely many bonds give a product bound $C_\eta e^{-m_H(|x-z|+|y-z|)}$. Summing in the two input bonds with any smaller exponential weight proves eq. 4.v for this action. This absolute convergence statement does not assert that C_η has a uniform cutoff bound; any such bound is a further quantitative input.

There is a useful stronger scope. Write $\mathrm{d}_\eta H(p) = \eta^{-1}\sum_{b\in\partial p} H(b)$ with oriented signs. If the actual response satisfies the physical first-difference bound

$$|\mathrm{d}_\eta H \delta^a_{x,\mu}(p)| \leq Ce^{-m_H|x-z|} \qquad (h_z(p) > 0),$$

uniformly in η, then the moment constant for the pointed action is uniform in η as well. Indeed, its kernel contains

$$\sum_p \eta^4 h_z(p)(\mathrm{d}_\eta H\delta_x)(p)\cdot(\mathrm{d}_\eta H\delta_y)(p).$$

The exact quadrature proved below makes the weighted plaquette sum one in each orientation. Consequently this kernel is bounded by $Ce^{-m_H(|x-z|+|y-z|)}$, and any smaller exponential moment is summable uniformly. This explains precisely why physical first differences, rather than identity-only bounds on bare values, remove the possible C_η loss.

Let $\eta = 1/N_0$ for an integer $N_0 \geq 1$, and use the midpoint tents

$$h_z(p) = \prod_{r=1}^{4}(1-|p_r - z_r|)_+, \qquad A_z(U) = \sum_{\mu<\nu}\sum_{p_{\mu\nu}} h_z(p)A_p(U). \tag{4.xxi}$$

For each fixed plaquette orientation,

$$\eta^4 \sum_{p_{\mu\nu}} h_z(p) = 1. \tag{4.xxii}$$

Here its barycenters have half-mesh shifts in directions μ,ν and zero shifts in the other directions. The one-dimensional unshifted identity is $N_0^{-1}[1+2\sum_{r=1}^{N_0-1}(1-r/N_0)]=1$. For the half-mesh shift it is $2N_0^{-1}\sum_{r=0}^{N_0-1}(1-(r+1/2)/N_0)=1$. Multiplication of these four identities proves eq. 4.xxii, including the coarse case $N_0=1$. Also $\sum_z h_z(p)=1$, since the two neighboring one-dimensional tents sum to one in each coordinate.

For a bare affine potential of curl F, the first variation of a plaquette holonomy in the chart $U_b = e^{it\eta A_b}$ is $it\eta^2 F_{\mu\nu}$. eq. 4.i, gradient invariance and eq. 4.xxii therefore give

$$\frac{1}{2}D^2(A_z\circ\mathcal{P})(0)[\ell,\ell] = \frac{1}{2}\sum_{\mu<\nu}|F_{\mu\nu}|^2. \tag{4.xxiii}$$

The affine-response hypothesis was used exactly in this line. The same pointed tensor theorem applies to this pointed action, so eq. 4.xx identifies its extractor as $b[A_z]=1$. Consequently

$$D^2(F_z - b[G](A_z\circ\mathcal{P}))(0)[\ell,\ell] = 0 \tag{4.xxiv}$$

for every affine constant-curvature field. Polarization gives the corresponding mixed affine identity. This proves the exact extractor-to-counterterm match with the normalization $A(e^{iB}) = \frac{1}{2}|\mathrm{d}B|^2 + O(B^3)$.

Selecting different members at different sites. The proof can be applied separately to each member $G^{[z]}$ chosen at a site. Within that member the family at all hypothetical base points is translation covariant, so its own bulk extractor $b[G^{[z]}]$ is defined by eq. 4.xiv and eq. 4.xix. Its pointed total at the actual site z is canceled by that number times A_z, by eq. 4.xxiv. No translation covariance of the array $z\mapsto G^{[z]}$ has been used.

This distinction is necessary for an inhomogeneous scalar source: one may freeze it to a different constant-source member at each site, but must use the coefficient of that same member at that site. Equality of two unpointed totals alone does not identify their separate site weights. Likewise the classification above does not prove a nonlinear third-order or Hölder remainder. That step still requires the actual localized response, its uniform field-domain inclusion and its stated analytic derivative bounds on the window.

4.3. **The nonlinear remainder in a pointed window.** The algebraic cancellation of a pointed Hessian concerns the second derivative at the identity. Its use at a nonlinear background requires a local Taylor estimate. We prove that estimate from the background hypotheses, retaining the domain assertions that make the compositions meaningful. In particular, the datum is large compared with its value close to the base point. A Taylor bound using only its global supremum would lose the required power of the scale ratio.

The background inputs and their coordinate radii. Work first in units in which the coarse spacing of a single-scale window is one. The background package A1–A3 is used in the following explicit form. On a box with collars there is a holomorphic potential/current representative $\Phi_\Lambda(B)$, with $\Phi_\Lambda(0) = 0$, for the actual constrained critical orbit. Its datum domain contains

$$\mathcal{B}_{r_B} = \{B : \text{determining bonds} \longrightarrow \mathfrak{su}(2)^{\mathbb{C}} : \sup_c |B(c)| < r_B\}. \tag{4.xxv}$$

Here $B(c)$ is a dimensionless logarithm of a group-valued coarse bond. This is a product of balls at bonds. It has no coupled derivative restriction. The orbit representative, its current, and the response estimates hold throughout this complex domain. Core and tower memberships under restriction and substitution are also part of A1; they are not inferred from the formal existence of a logarithm. All differentiations and coordinate sums below are first carried out on finite boxes. Their uniform bounds permit the stated coherent infinite-volume limits; no representation of a derivative on an infinite ℓ^∞ space by coordinate derivatives is assumed.

For clarity, the output norm over a top of local spacing ξ_y is

$$Z_y = (\xi_y H, \xi_y^2 \nabla H, \xi_y^3 J), \qquad |Z_y| = \max\{\xi_y|H|,\ \xi_y^2|\nabla H|,\ \xi_y^3|J|\}. \tag{4.xxvi}$$

The first-response bound supplied by A3 is

$$|\partial_{B(c)} Z_y(B)[e]| \le C e^{-md(y,y_c)}|e|, \qquad B \in \mathcal{B}_{r_B}, \tag{4.xxvii}$$

together with the stated spatial Hölder bound for ∇H. The distance is the local-scale path distance. On the central homogeneous part of the window, eq. 4.xxvii uses level-j distances and the output weights are one. Constants may depend on the fixed window size, the decay rate and the Hölder exponent, but not on the bare mesh, the torus or the number of preceding steps.

In the central part of the old box the top spacing is one in the old units. The finer tops in the tower collar are handled by the weighted row sum for the actual response. We compare affine data on this same determining set, with its prescribed seam rows.

Choose a fixed basis of the three-dimensional complex colour space. If $\sup|B| \le r_B/2$, varying one basis coordinate by $|s| < c_{\mathrm{col}} r_B/2$ stays in eq. 4.xxv, where $c_{\mathrm{col}} > 0$ depends only on this choice of norm and basis. Thus there is an admissible coordinate Cauchy radius independent of the number of bonds. The factors ξ_y in eq. 4.xxvi are output weights and commute with this Cauchy estimate. If instead a physical potential datum is written $a(c) = B(c)/\xi_c$, its coordinate radius is $c_{\mathrm{col}} r_B/(2\xi_c)$, and $\partial_{a(c)} = \xi_c \partial_{B(c)}$. Contracting a second derivative against $a(c)a(c')$ cancels both input factors $\xi_c, \xi_{c'}$. No inverse local spacing is lost by using dimensionless bond coordinates.

Lemma 4.4 (Product decay for the second response)**.** On $\mathcal{B}_{r_B/2}$, the first-response hypothesis implies

$$|\partial_{B(c)}\partial_{B(c')} Z_y(B)[e, e']| \le \frac{C}{r_B} e^{-\frac{m}{2}(d(y,y_c)+d(y,y_{c'}))}|e|\,|e'|. \tag{4.xxviii}$$

The same conclusion holds for the local first-order pair norm on a domain X, with $d(y, y_c)$ replaced by $\mathrm{dist}(c, X)$. ■

Proof. Apply the coordinate Cauchy estimate to $\partial_{B(c)} Z_y(B)[e]$, using a slightly smaller fixed fraction of the radius just specified. This gives $C r_B^{-1} e^{-md(y,y_c)}|e|\,|e'|$. The finite-dimensional holomorphic map has symmetric second derivatives, so the same argument interchanging c, c' gives the corresponding bound with c'. The smaller of two positive bounds is at most their geometric mean, which proves eq. 4.xxviii. Taking the supremum over the output bonds, first differences and current components gives the stated local norm version. Colour indices cost only a fixed factor. □

This argument uses response decay at every complex datum in a ball. Decay of the linear response at $B = 0$ alone is insufficient.

Weighted polynomial data and the local expansion. Fix a base point z, a scale ratio $0 < \sigma \le 1$, and a window whose distance from z to its boundary is comparable with σ^{-1}. Write a for a radius of a regular member, a_{out} for the output field radius and

$$\rho = a_{\mathrm{out}}/a \ge 1, \qquad a \le C_0 r_B, \qquad a_{\mathrm{out}} \le c_0, \qquad 0 < \alpha \le 1. \tag{4.xxix}$$

The constant c_0 is chosen small enough for the matrix charts. A fixed constant C_{ax} in the axial datum estimate below is understood. The local Taylor argument requires the full axial datum in $\mathcal{B}_{r_B/2}$. The central rows have bound $C_{\mathrm{ax}} a_{\mathrm{out}}\sigma$; the remote rows require their own absolute bound in the same chart. The gain σ on the central rows does not bound the remote ones. Once both bounds are below $r_B/2$, the entire segment from zero lies in this polydisc.

For the flat-reference windows, this absolute domain condition has a choice uniform in the depth. Use a fixed continuation radius $r_{\mathrm{abs}} > 0$, smaller if necessary than the common row and node-gauge radii, as r_B. Fix the window geometry first. The full induced datum of the upper background has bound $C_M A_0$, whereas its central rows have the additional factor σ, as in the proof of Prop. 3.135. A fluctuation of amplitude A_{fl} changes a level-q raw row by at most $CL^{q-k}A_{\mathrm{fl}}$, with $q \le j \le k$. First omit the independent raw correction and call the physical datum D. If $A_0, A_{\mathrm{fl}} \le a_{\mathrm{abs}}$, its full norm is at most $C_M a_{\mathrm{abs}}$. Construct the old axial change from this uncorrected datum. Its comb in the settled layer has $O(\sigma^{-1})$ links of logarithmic size $O(C_M a_{\mathrm{abs}}\sigma)$, so the node gauge has logarithm $O(C_M a_{\mathrm{abs}})$. Extend that logarithm with a cutoff supported inside the settled layer. On every row the transformed datum is the product of its original value and its two endpoint gauge factors.

For an independent raw correction E, hold this gauge g fixed with respect to E. It may still depend holomorphically on the physical input. In additive datum coordinates its action is the rowwise map

$$\mathcal{G}_g(D)(c) = \frac{1}{i}\log\Big[g(c_-)e^{iD(c)}g(c_+)^{-1}\Big].$$

The endpoint factors and their inverses have bounded norms, and these products lie in a fixed smaller logarithm chart. The bounded derivatives of the exponential and logarithm therefore give

$$\|\mathcal{G}_g(D+E) - \mathcal{G}_g(D)\|_\infty \le C\|E\|_\infty.$$

There is no comb-length factor in this rowwise estimate. With the prescribed correction norm below a_{abs}, the full axial datum consequently satisfies

$$\|\widehat{B}\|_\infty \le C_{\mathrm{win}}(M)a_{\mathrm{abs}} \le r_{\mathrm{abs}}/4. \tag{4.xxx}$$

This is absolute admission of the correction disc. No central factor σ is assigned to an arbitrary correction, and the axial gauge is not reconstructed from that correction. Choose a_{abs} after M and r_{abs}, also below the ceiling $a_I(M)$ of Prop. 3.135. All contour logarithms and the complex gauge covariance then stay in their fixed charts. The constants do not depend on the depth: only the comb length contains σ^{-1}, and its link size supplies the compensating σ. This is an absolute-domain check for the actual admitted window maps. It does not enlarge a regular member's prescribed core: its membership and the local Taylor ball of radius a remain separate requirements. It also does not imply the proportional remote-datum bound Ca_{out} used below. That bound is needed to retain a_{out} in the exponentially small response from the remote rows. If only an absolute remote bound is available, its response must instead be retained as a separate tail. For another family with radii r_B that vary with age, the full-datum inclusion remains a separate hypothesis; the central age estimate alone does not establish it. A harmless enlargement of the radius ratio to at least one is allowed. For a domain X containing z, let $D = 2 + \operatorname{diam} X$, in the present units.

For the combined background and fluctuation we need, on the central rows before choosing the old axial gauge, a logarithm B with

$$|B| \le Ca_{\mathrm{out}}\sigma, \quad |\nabla B| \le Ca_{\mathrm{out}}\sigma^2, \quad |\nabla B(x) - \nabla B(x')| \le Ca_{\mathrm{out}}\sigma^{2+\alpha}|x-x'|^\alpha. \tag{4.xxxi}$$

Here is the passage from the common-frame field estimate to these row bounds. Write a for the bare bond logarithm of the product field, with $\eta_k^{-1}\mathsf{a}$ bounded by Ca_{out} in $C^{1,\alpha}$ in level-k units. Translate the central double-block carriers, including the one-row translates used for the differences, to a common carrier. The actual order-j row map on a smaller absolute sup ball has first and second derivative norms bounded by CL^j and CL^{2j}, by directional Cauchy estimates. An old step has length σ in upper units. The sup and first-difference estimates therefore give the first two bounds in eq. 4.xxxi. For two rows r old spacings apart, applying the first derivative to the difference of the increments costs $Ca_{\mathrm{out}}\sigma^{2+\alpha}|r|^{\alpha}$. The change of the point at which that derivative is applied costs $Ca_{\mathrm{out}}^2\sigma^4|r|$ by the second derivative bound. The latter fits the former on $|r| \le C/\sigma$.

The decoupled field estimate is covariant for the input connection. To use it here we must verify the spatial regularity of that connection in the same frame as the box field on the complete central carriers. A bound on the gauge condition number controls the shifted link sizes. The first differences also use the change of that frame along the links. The localization uses the relative frame and its restoring identity.

All statements in this section require the actual pair and its restrictions to remain in the member's prescribed core. The small chart ball used for Taylor's formula must be contained in that core. The radius a is a fixed fraction of its curvature radius. Throughout the admitted core $|\partial\mathbf{U}(p)-1| \le Ca\eta_j^2$, so its own tent action obeys $|A(h_z,\mathbf{U})| \le Ca^2$, by the determinant identity and exact tent plaquette count. This is a bound on the whole core, not just its contained chart ball.

In the axial gauge about z, the datum has a local form

$$\widehat{B}(c) = \ell_F(c) + r(c), \qquad \ell_{F,\mu}(x) = \sum_{\kappa<\mu} (x-z)_\kappa F_{\kappa\mu}, \tag{4.xxxii}$$

where $|F| \le Ca_{\mathrm{out}}\sigma^2$, and

$$|\widehat{B}(c)| \le Ca_{\mathrm{out}}\sigma^2(1+|c-z|), \qquad |r(c)| \le Ca_{\mathrm{out}}\sigma^{2+\alpha}(1+|c-z|)^2. \tag{4.xxxiii}$$

These estimates are for the central rows. On the remote rows the full axial datum is bounded by Ca_{out} on the admitted box chart. The distance of those rows from an inner domain is at least c/σ in the local-scale metric.

Here are details of eq. 4.xxxii–eq. 4.xxxiii. Fixing the transports along a coordinate comb to the identity expresses a remaining bond as a product of at most $C(1+|c-z|)$ transported plaquette holonomies. With constant plaquette strengths, the linear part of this product is exactly ℓ_F. eq. 4.xxxi shows that a plaquette logarithm differs from its value at z by at most

$$C\big[a_{\mathrm{out}}\sigma^{2+\alpha}(1+|c-z|)^{\alpha} + a_{\mathrm{out}}^2\sigma^3(1+|c-z|)\big].$$

The second term includes differentiation of the non-Abelian commutator. Conjugating a plaquette logarithm by a comb transport adds at most $Ca_{\mathrm{out}}^2\sigma^3(1+|c-z|)$. Since a_{out} is bounded and $\sigma^{1-\alpha} \le 1$, both errors are bounded by $Ca_{\mathrm{out}}\sigma^{2+\alpha}(1+|c-z|)$. Summing along the strip proves the second bound in eq. 4.xxxiii.

For completeness, the finite-product estimate used here follows by differentiating $P(t) = \prod_i e^{tX_i}$ twice: if $\sum_i |X_i| \le c$, then $\sup_{0\le t\le 1}|P''(t)| \le C(\sum_i |X_i|)^2$. The analytic logarithm in this fixed matrix ball has bounded first two derivatives, so

$$\left|\log\prod_i e^{X_i} - \sum_i X_i\right| \le C\left(\sum_i |X_i|\right)^2.$$

Its error here is at most $Ca_{\mathrm{out}}^2\sigma^4(1+|c-z|)^2$, which is absorbed in eq. 4.xxxiii. The unexpanded product bound gives the first bound of eq. 4.xxxiii. All transports stay in a fixed matrix ball, because the original comb has length $O(\sigma^{-1})$ and its original bond logarithms are $O(a_{\mathrm{out}}\sigma)$.

We shall use the elementary weighted sum

$$\sum_{c\in\mathbb{Z}^4} e^{-\mu\,\mathrm{dist}(c,X)}(1+|c-z|)^k \le C_{\mu,k}D^{k+4}, \qquad k\ge 0. \tag{4.xxxiv}$$

Indeed $|c-z|\le D+\mathrm{dist}(c,X)+C$. The shell at distance between q and $q+1$ is contained in a box with at most $C(D+q+1)^4$ bonds. Bounding each shell by this larger number and summing $e^{-\mu q}(D+q+1)^{k+4}$ proves eq. 4.xxxiv. Only a polynomial in D is spent. Every power needed below is fixed.

Let $\widehat{B}^\circ$ agree with $\widehat{B}$ on the central rows and vanish on the remote ones. The weighted first-response bound along the segment between the two data gives

$$\|\Phi_\Lambda(\widehat{B})-\Phi_\Lambda(\widehat{B}^\circ)\|_X \le C a_{\mathrm{out}} e^{-c/\sigma}.$$

The support is remote from X in the tower metric. We use the homogeneous sums on the central data.

Integrate the first response on the segment $t\widehat{B}^\circ$, and then use eq. 4.xxvii, eq. 4.xxxiii and eq. 4.xxxiv. With a fixed integer s, increased below when necessary, this gives

$$\|\Phi_\Lambda(\widehat{B}^\circ)\|_X \le CD^s a_{\mathrm{out}}\sigma^2, \qquad \|D\Phi_\Lambda(0)r\|_X \le CD^s a_{\mathrm{out}}\sigma^{2+\alpha}. \tag{4.xxxv}$$

The remote first-response bound gives the same first bound for $\Phi_\Lambda(\widehat{B})$. Taylor's formula with integral remainder for Φ_Λ, followed by eq. 4.xxviii and the product of the two weighted sums, gives

$$\|\Phi_\Lambda(\widehat{B}^\circ)-D\Phi_\Lambda(0)\widehat{B}^\circ\|_X \le \frac{CD^s}{r_B}a_{\mathrm{out}}^2\sigma^4. \tag{4.xxxvi}$$

The use of eq. 4.xxxiv in eq. 4.xxxvi is essential. Using only $\sup|\widehat{B}|=O(a_{\mathrm{out}}\sigma)$ would produce the wrong power σ^2 in its right-hand side.

The finite-box linear response. Take the depth-j truncation of the upper window tower. Fix a constant curvature tensor F in one real colour direction. Use the affine potential in eq. 4.xxxii on the bare lattice, and on every inner bare bond take $U_t=\exp(i\eta_j t a_F)$. Prescribe the actual averages of this field on every determining row, including the finer rows of the tower. Every inner bond of a plaquette crossing the exterior is a fixed level-zero row, since the first inward layer is separated from that exterior. Variations therefore act only at bonds where the full current is defined. There the opposite abelian plaquette contributions to the Wilson current cancel. Thus U_t is critical for its own datum. On the fixed finite box it is regular for small real t, and regular uniqueness identifies the box orbit at that datum.

On a level-q row c, the first datum is $L^{q-j}a_F(c_-)$ in level-j units. The centered block average reproduces the affine transverse coordinates and the longitudinal line average leaves $a_{F,\mu}$ unchanged. The actual composition gives the same result at every scale, as in Lem. 5.17. The literal first average gives the same affine datum. In an endpoint block, a contour integral from the centre has terms linear in a coordinate displacement or bilinear in two distinct displacements. Each has zero centered mean. Thus the endpoint contour terms cancel in the linearized literal average and only the straight corridor term remains. Differentiating the orbit identity gives the linear pair L_F of this box, equal to the affine field modulo a fine gradient, with zero linear current. Linearity gives the same conclusion for complex colours. The pointed Hessian kills that gradient, since the activity is gauge invariant and its first derivative at the identity is zero.

On the central rows of level j, the affine datum is ℓ_F of eq. 4.xxxii in the same linear frame. Extend this central datum by zero on the other rows in the expression below. If X is at distance at least cR from those other rows, with $R\asymp\sigma^{-1}$, the difference from the full affine datum is bounded by $C|F|(1+R)^s$ and supported at that distance. The first-response estimate on the same tower, using the uniform weighted sum over tops, gives

$$\|D\Phi_\Lambda(0)\ell_F-L_F\|_X \le C|F|(R+1)^s e^{-cR}. \tag{4.xxxvii}$$

The central sum with its linear growth and this outer sum also give $\|L_F\|_X \le CD^s|F|$. The comparison uses the literal affine datum on the prescribed finite tower.

For $R \asymp \sigma^{-1}$, $(R+1)^s e^{-cR} \le C_\alpha \sigma^\alpha$. Combining eq. 4.xxxii, eq. 4.xxxv–eq. 4.xxxvii and the first-response bound on the remote datum, we obtain

$$\begin{aligned} \Phi_\Lambda(\widehat{B}) = L_F + E_X, \qquad \|L_F\|_X \le CD^s a_{\mathrm{out}}\sigma^2, \\ \|E_X\|_X \le CD^s \left[a_{\mathrm{out}}\sigma^{2+\alpha} + r_B^{-1}a_{\mathrm{out}}^2\sigma^4\right]. \end{aligned} \tag{4.xxxviii}$$

Domains too large for the interior condition in eq. 4.xxxvii are put into the exponentially long tail. In applications their tree length is at least $c/\sigma - C$.

Taylor's formula for the actual member. Let G_X denote a regular pointed member on X, in its exponential potential/current chart. Assume its supremum on that core is $g_X = \mathtt{g}e^{-\kappa d_j(X)}$, and that its chart contains the ball of radius a. A fixed change of constants allows the use of a ball of radius $2a$ and estimates on its half ball; this is the specified room in the field domains. Constant gauge transformations act by the adjoint representation on both chart components. There is no invariant linear functional on $\mathfrak{su}(2)$, so $DG_X(0) = 0$. In particular the second derivative of the background does not enter the Hessian of this composition:

$$D^2(G_X \circ \Phi_\Lambda)(0)[B, B'] = D^2G_X(0)[D\Phi_\Lambda(0)B, D\Phi_\Lambda(0)B']. \tag{4.xxxix}$$

This assertion concerns regular members without stored colour spectators.

Proposition 4.5 (Pointed nonlinear window remainder)**.** Under the background and domain hypotheses above, for the near domains there is a fixed integer s_1 such that

$$\begin{aligned} G_X(\Phi_\Lambda(\widehat{B})) - G_X(0) = \tfrac{1}{2}D^2G_X(0)[L_F, L_F] + \mathcal{R}_X, \\ |\mathcal{R}_X| \le C\mathtt{g}e^{-\kappa d_j(X)}D^{s_1}\left[\rho^2\sigma^{4+\alpha} + \rho^3\sigma^6\right]. \end{aligned} \tag{4.xl}$$

The same estimate holds after absorbing the member's own tent-action counterterm, with its own Hessian subtracted. ■

Proof. Put $t = D^s\rho\sigma^2$, increasing s to cover every polynomial factor in eq. 4.xxxv–eq. 4.xxxviii. First suppose t is less than a sufficiently small fixed constant. eq. 4.xxxv then places the composed pair in the half chart ball of G_X. Cauchy's formula and Taylor's theorem give

$$\left|G_X(P) - G_X(0) - \tfrac{1}{2}D^2G_X(0)[P,P]\right| \le Cg_X(\|P\|_X/a)^3, \qquad \|D^2G_X(0)\| \le Cg_X/a^2. \tag{4.xli}$$

In eq. 4.xxxviii, divide by a, use $a/r_B \le C_0$, and expand the difference of the two quadratic forms. The product of L_F/a and the first error term is bounded by $CD^{2s}\rho^2\sigma^{4+\alpha}$. The product with the second error term is bounded by $CD^{2s}\rho^3\sigma^6$. The square of the first error is smaller than the first of these bounds. The square of the second is bounded by Ct^4, which is at most a constant times t^3 in this case. The mixed product is bounded by the sum of the squares. Finally the Taylor remainder in eq. 4.xli is at most Cg_Xt^3. Increasing s_1 proves eq. 4.xl.

If t exceeds that fixed small constant, use the actual core bound $|G_X(P) - G_X(0)| \le 2g_X$, rather than Taylor's formula at P. The Hessian estimate still bounds its polynomial value at L_F by Cg_Xt^2. For $t \ge c > 0$, $1 + t^2 \le C_ct^3$, and this proves eq. 4.xl after increasing s_1 again. Thus large radius ratios do not impose an upper bound on the age.

The coefficient of a member obeys $|b(G)| \le C\mathtt{g}/a^2$ by its Hessian moments. The tent action on its fixed local support has norm at most Ca^2. Consequently its counterterm is another holomorphic family of size $C\mathtt{g}$, with bounded support. Apply the same proof to this family and subtract. The affine field and its curvature in the two quadratic terms are the same. □

The exact own-counterterm identity in eq. 4.xxiv of the pointed Hessian cancellation theorem now applies to this last, normalised family: its complete sum of pointed curvature tensors is zero. A sum cut to the window is the complete sum minus the omitted domains. The omitted domains

have $d_j(X) \geq c/\sigma - C$. If the number through a cube of length q is at most $Ce^{\kappa_D q}$ with $\kappa > \kappa_D$, polynomial factors in $D \leq CM(d_j(X)+2)$ are summable. Thus eq. 4.xl, summed over near domains, gives

$$C\mathsf{g}[\rho^2\sigma^{4+\alpha} + \rho^3\sigma^6], \tag{4.xlii}$$

and completing the tensor sum costs at most $C\mathsf{g}\rho^2\sigma^4 e^{-c/\sigma}$. If a far domain is not contained in any non-wrapping chart, its value has only the norm bound and contributes an additional $C\mathsf{g}e^{-c/\sigma}$. Such a term has no justified factor proportional to the actual curvature: a flat field around a hole or a torus cycle need not be a pure gauge. It must be retained when a field-sensitive tail estimate is claimed.

For the radius estimate used in a window, $\rho \geq 1$ permits both exponentially small tails to be absorbed in eq. 4.xlii. Moreover

$$\rho^2\sigma^{4+\alpha} + \rho^3\sigma^6 \leq 2\rho^3\sigma^{4+\alpha}, \qquad \rho \geq 1, \quad \sigma \leq 1, \quad \alpha \leq 1. \tag{4.xliii}$$

There are $O(\sigma^{-4})$ level-j base points in one window of the later scale. Its remaining factor is therefore $C\mathsf{g}\rho^3\sigma^\alpha$. With $\rho^2 \leq C(1+n-j)$ and $\sigma = L^{-(n-j)}$, this is the summable ordinary weight $C\mathsf{g}(1+n-j)^{3/2}L^{-\alpha(n-j)}$. The central part of the input-domain condition holds at every sufficiently old age. Indeed

$$\frac{C_{\mathrm{ax}}a_{\mathrm{out}}\sigma}{r_B} \leq C_{\mathrm{ax}}C_0\rho\sigma \leq C(1+n-j)^{1/2}L^{-(n-j)}.$$

Choose a fixed age $a_\star$ beyond which this is at most $1/2$. The remote datum must independently lie in the same half ball. For the flat-reference windows this follows, at every age, from eq. 4.xxx; it is not a consequence of the displayed age gain. At each of the finitely many younger ages, the given core membership bounds the member and its own counterterm by $C\mathsf{g}(1+\rho^2\sigma^4)$. Here ρ is uniformly bounded and $\sigma \geq L^{-a_\star}$, so this is bounded by a fixed constant times $\mathsf{g}\rho^3\sigma^{4+\alpha}$. This last argument uses the ordinary age bound; it is not valid for an arbitrary unrelated pair (ρ,σ). Each site may carry a different Euclidean member: cancellation and the estimate are performed at that site with that member's own coefficient.

Proposition 4.6 (Vacuum-only window bound)**.** For an analytic gauge-invariant member without stored colour spectators, no lattice covariance is needed to obtain

$$|G_X(\Phi_\Lambda(\widehat{B})) - G_X(0)| \leq C\mathsf{g}e^{-\kappa d_j(X)}D^{s_2}\rho^2\sigma^4. \tag{4.xliv}$$

After the domains in a window have been summed, the bound is $C\mathsf{g}\rho^2\sigma^4$, with the same qualifications on remote, non-wrapping charts and young ages as above. ■

Proof. Gauge invariance under constant colour rotations again gives $DG_X(0) = 0$, independently of any spatial symmetry. By eq. 4.xxxv, with $t = D^s\rho\sigma^2$, the local pair lies in the half chart ball whenever $t < c$. Cauchy's estimate for the second-order Taylor remainder then gives Cg_Xt^2. For $t \geq c$, use the core bound $2g_X \leq C_c g_X t^2$. This proves eq. 4.xliv, and the domain entropy absorbs its fixed polynomial in D. A remote domain not admitting a chart contributes only $C\mathsf{g}e^{-c/\sigma}$; it is bounded by $C\mathsf{g}\rho^2\sigma^4$ for $\rho \geq 1$. If the central input-disc condition is unavailable at finitely many young ages, while the remote condition is supplied as above, the core bound is absorbed since $\rho \geq 1$ and $\sigma \geq L^{-a_\star}$. Finally the $O(\sigma^{-4})$ sites in the window cancel σ^4, leaving the ordinary weight ρ^2 for a defect or a remainder. □

4.4. **Localized complex Gaussian comparison.** The finite regular expansion uses a normalized Gaussian functional on a local set of fluctuation coordinates. Its auxiliary Gaussian can have as many coordinates as the surrounding volume. We prove its comparison bound without paying for that auxiliary dimension. The argument permits sharp, measurable cutoffs; it never shifts their arguments into the complex domain.

The local matrices and the normalized functional. All matrix norms below are operator norms for the Euclidean complex space. Transposition is bilinear transposition, not conjugate transposition. Let $C = C^{\mathsf{T}}$ be an invertible complex m-by-m matrix and let Γ be a complex m-by-n matrix, where $m \geq 1$ and $n \geq 0$. Assume real reference matrices $C_0 = C_0^{\mathsf{T}} > 0$ and Γ_0, and one constant $K \geq 1$, such that

$$\|C_0\| + \|C_0^{-1}\| + \|\Gamma_0\| \leq K, \qquad \|C - C_0\| + \|\Gamma - \Gamma_0\| \leq \varepsilon. \tag{4.xlv}$$

The number ε will be smaller than a constant depending only on K. No dimension enters that choice. Define

$$D = C + C\Gamma\Gamma^{\mathsf{T}}C, \qquad D_0 = C_0 + C_0\Gamma_0\Gamma_0^{\mathsf{T}}C_0. \tag{4.xlvi}$$

The real matrix D_0 is positive definite. In the small neighborhood in eq. 4.xlv, normalized Gaussian densities are defined by

$$d\mu_D(b) = (2\pi)^{-m/2}(\det D)^{-1/2}e^{-b^{\mathsf{T}}D^{-1}b/2}\,db, \qquad b \in \mathbb{R}^m. \tag{4.xlvii}$$

The square root is the holomorphic continuation of the positive root at the reference. We justify both convergence and normalization below.

The auxiliary representation of the localized functional is

$$\mathcal{J}_{C,\Gamma}[F] = \int d\mu_{I_n}(\xi)\, e^{-\xi^{\mathsf{T}}\Gamma^{\mathsf{T}}C\Gamma\xi/2} \int d\mu_C(b)\, e^{b^{\mathsf{T}}\Gamma\xi}F(b). \tag{4.xlviii}$$

For $n = 0$ the outer integral is absent. An integrand F need only be Borel measurable on $\mathbb{R}^m$. Suppose

$$|F(b)| \leq F_0 e^{\rho|b|^2}, \qquad F_0 \geq 0, \quad \rho \geq 0. \tag{4.xlix}$$

Proposition 4.7 (Local Gaussian comparison)**.** There are positive constants $\varepsilon_0(K), \rho_0(K)$ and A_K, independent of m, n, such that eq. 4.xlv, $\varepsilon \leq \varepsilon_0$ and $\rho \leq \rho_0$ imply absolute convergence of eq. 4.xlviii,

$$\mathcal{J}_{C,\Gamma}[F] = \int F\, d\mu_D, \qquad \mathcal{J}_{C,\Gamma}[1] = 1, \qquad |\mathcal{J}_{C,\Gamma}[F]| \leq F_0 e^{A_K(\varepsilon+\rho)m}. \tag{4.l}$$

The last estimate also bounds $\int |F|\,|d\mu_D|$. In particular its exponential cost uses the local dimension m, even when the auxiliary dimension n grows with the full volume. ∎

Accretivity, integration and the determinant branch.

Proof. The joint precision in eq. 4.xlviii is the symmetric block matrix

$$Q = \begin{pmatrix} C^{-1} & -\Gamma \\ -\Gamma^{\mathsf{T}} & I_n + \Gamma^{\mathsf{T}}C\Gamma \end{pmatrix}. \tag{4.li}$$

At the real reference it factors as

$$Q_0 = L_0^{\mathsf{T}} \begin{pmatrix} C_0^{-1} & 0 \\ 0 & I_n \end{pmatrix} L_0, \qquad L_0 = \begin{pmatrix} I_m & -C_0\Gamma_0 \\ 0 & I_n \end{pmatrix}. \tag{4.lii}$$

Both $\|L_0\|$ and $\|L_0^{-1}\|$ are at most $1 + K^2$. Thus $Q_0 \geq c_K I$ for a positive c_K. The resolvent identity for $C^{-1} - C_0^{-1}$, followed by expanding the products in eq. 4.li, gives $\|Q - Q_0\| \leq A_K\varepsilon$ for small ε; the constants depend only on K. For real (b, ξ), consequently,

$$\mathrm{Re}(b,\xi)^{\mathsf{T}}Q(b,\xi) \geq (c_K/2)(|b|^2 + |\xi|^2).$$

Choose $\rho_0 < c_K/8$. This proves absolute convergence of the joint integral with eq. 4.xlix, hence Fubini's theorem applies. It does not yet give the desired dimension bound; taking its absolute value before integrating ξ would pay unnecessarily for n.

Set $A = I_n + \Gamma^{\mathsf{T}}C\Gamma$. It is a small perturbation of the real positive matrix $I_n + \Gamma_0^{\mathsf{T}}C_0\Gamma_0$. Integrate ξ first. The normalized Gaussian exponential formula gives the factor

$$(\det A)^{-1/2} \exp\{b^{\mathsf{T}}\Gamma A^{-1}\Gamma^{\mathsf{T}}b/2\}.$$

This formula for a complex linear force does not require a contour change. One proof normalizes the complex Gaussian first and differentiates its exponential transform in the force. Integration by parts, justified by accretivity, gives $\nabla_\ell G = A^{-1}\ell G$, with $G(0) = 1$, and hence $G(\ell) = e^{\ell^{\mathsf{T}} A^{-1}\ell/2}$.

For completeness, the normalization used in that proof follows without a complex change of variables. On an accretive symmetric precision neighborhood the real-space integral is holomorphic by a fixed Gaussian majorant. It equals $(\det A)^{-1/2}$ at every real positive symmetric matrix there by real orthogonal diagonalization and the one-dimensional real Gaussian integral. Successive one-variable identity theorems in the independent symmetric matrix entries extend the equality to the complex neighborhood, with the root continued from its positive value. The real symmetric matrices have a full set of real coordinates for this argument; no identity is being extended from a lower-dimensional background graph.

The elementary inverse and determinant identities are

$$\begin{aligned} D^{-1} &= C^{-1} - \Gamma(I_n + \Gamma^{\mathsf{T}} C\Gamma)^{-1}\Gamma^{\mathsf{T}}, \\ \det D &= \det C \, \det(I_n + \Gamma^{\mathsf{T}} C\Gamma). \end{aligned} \tag{4.liii}$$

The inverse identity follows by multiplication with eq. 4.xlvi. For the determinant identity, eliminate either off-diagonal block in $\left(\begin{smallmatrix} I_m & -C\Gamma \\ \Gamma^{\mathsf{T}} & I_n \end{smallmatrix}\right)$; this proves $\det(I_m + C\Gamma\Gamma^{\mathsf{T}}) = \det(I_n + \Gamma^{\mathsf{T}} C\Gamma)$. The roots in eq. 4.liii agree at the real reference, and agree throughout the connected small perturbation neighborhood by continuity and nonvanishing. The remaining b integral is precisely eq. 4.xlvii. This proves the first identity in eq. 4.l, including for Borel F, since no complex argument of F was introduced. The same normalization argument gives $\int d\mu_D = 1$.

The local-dimension estimate. The reference bounds imply $D_0 \geq C_0$, $\|D_0\| \leq A_K$ and $\|D_0^{-1}\| \leq K$. Expanding eq. 4.xlvi yields

$$T = D_0^{-1/2}(D - D_0)D_0^{-1/2}, \qquad \|T\| \leq A_K\varepsilon =: \eta. \tag{4.liv}$$

Shrink ε_0 so that $\eta \leq 1/4$. Then $D^{-1} = D_0^{-1/2}(I+T)^{-1}D_0^{-1/2}$ and $\|(I+T)^{-1} - I\| \leq \eta/(1-\eta)$. For the real symmetric matrix

$$R = D_0^{1/2}(\operatorname{Re} D^{-1} - 2\rho I_m)D_0^{1/2}$$

we obtain $\|R - I_m\| \leq A_K(\varepsilon + \rho) =: u$. Shrink the constants further so that $u \leq 1/2$. Taking the absolute value of the *marginal* density, and then integrating the real positive Gaussian, gives

$$\int e^{\rho|b|^2}|d\mu_D(b)| = |\det(I+T)|^{-1/2}(\det R)^{-1/2} \leq (1-\eta)^{-m/2}(1-u)^{-m/2}. \tag{4.lv}$$

Every singular value of $I+T$ is at least $1-\eta$, and every eigenvalue of R is at least $1-u$. Finally $-\log(1-v) \leq 2v$ for $0 \leq v \leq 1/2$ proves eq. 4.l. □

Sharp failures, holomorphic parameters, and scope. Partition the real coordinates into disjoint bond blocks. If a set P_0 of distinct blocks fails the tests $|b_\beta| < \mathfrak{b}$, then $\sum_{\beta\in P_0}|b_\beta|^2 \geq |P_0|\mathfrak{b}^2$. Therefore, for $\gamma \geq 0$ with $\rho + \gamma/2 \leq \rho_0$, eq. 4.lv gives

$$\int \prod_{\beta\in P_0} \mathbf{1}_{\{|b_\beta|\geq\mathfrak{b}\}} e^{\rho|b|^2}|d\mu_D| \leq \exp\{A_K(\varepsilon + \rho + \gamma/2)m - \gamma\mathfrak{b}^2|P_0|/2\}. \tag{4.lvi}$$

This is a product-of-failures estimate, with one Gaussian tail per distinct tested bond. Multiplication by additional indicators only improves the bound.

If the matrices and the potential part of an integrand depend holomorphically on external parameters, and the sharp cutoff sets remain fixed, the same estimates give a parameter-independent Gaussian majorant on each smaller parameter neighborhood. Fubini and Morera then prove holomorphy of the integrals. No holomorphy of a characteristic function in its real field argument is asserted or required.

The rescaled structure $(\tau^{-1}C, \tau^{1/2}\Gamma)$ has effective covariance $\tau^{-1}D$. For real τ in a fixed compact subinterval of $(0,\infty)$, rescale the real reference in the same way; all constants remain uniform. A small complex displacement of τ is included in the matrix perturbation.

In the regular step, let $N_c(Z)$ be the number of coarse localization cubes. The local free-bond color-coordinate count obeys $m \le c(LM)^4 N_c(Z)$. Consequently eq. 4.l supplies the factor $\exp\{A_K(LM)^4(\varepsilon+\rho)N_c(Z)\}$, and eq. 4.lvi supplies the failed-cutoff tail. If a Gaussian comparison instead uses the symbol $N(Z)$ for the local coordinate count itself, its bound is $\exp\{A_K(\varepsilon+\rho)N(Z)\}$, with no second factor $(LM)^4$. These are two conventions for the same local cost. The matrix gap and kernel closeness are the actual A2–A3 inputs; this section proves the resulting finite-dimensional integration estimate.

Completeness is separate. Equality of this local functional with a marginal of one whole-torus Gaussian requires the actual completeness identity for its effective covariance. In the source construction that identity is supplied on complete physical pairs and their physical-data continuation. Arbitrary independent-current cores need only their own local holomorphic functionals and the bounds proved here. This theorem does not manufacture a global covariance on those cores, verify nonlinear background/core estimates, or establish a Gaussian law for actual averaging disintegrations.

4.5. **Local interactions and their source response.** The local part of a renormalization step has several distinct operations. A function of a fluctuation is first localized, its Gaussian integral is expanded, the connected logarithm is pointed at bonds, and an old gauge-invariant function is rewritten on a new constrained background. These operations have different domains. Their order is part of the definition of the step.

We work at a fixed finite cutoff and volume until uniform estimates have been established. A local pair $P = (U, J)$ consists of the admitted potential data and current data on the specified local domain. Those coordinates may be independent. A complete physical pair is denoted p_W; its current is the one induced by the actual constrained background belonging to the datum W. At such pairs the complete Gaussian structure reproduces the physical fluctuation integral. An identity on the set of pairs p_W is not an identity on an open set of independent pairs P.

We use three fluctuation coordinates. The free Gaussian coordinates are B; the constrained physical fluctuation is $\phi(B)$; and the background response supplies the actual field $U_P(B)$. Their typical sizes have the form

$$g = \theta^{-1/2}, \qquad \alpha = g\,A(\theta), \qquad \delta = s\alpha, \qquad 0 < s \le s_\star. \tag{4.lvii}$$

The analytic size $A(\theta)$, the cutoff in the coordinates B, and the physical radius δ are separate quantities. In particular a bound on a cutoff box in B does not furnish a holomorphic continuation to a ball of radius α/g. The larger ball used below is obtained from the actual unscaled constraint chart and the admitted datum polydiscs of the background response.

At a fixed real reference θ, two scalar-source presentations are useful. In the variable-profile presentation the real action weight x, the calibration, Gaussian, determinant branches and cutoffs are fixed; the profile is carried as an older potential member. Its new source energy is the difference of the localized fluctuation functions. In a constant-shift presentation one may instead change the action weight and compensate the free action, numerical terms and scalar counterterm. Differentiating that presentation changes more than the older potential. We use the first presentation for the insertion and source-gradient estimates in this section. Numerical or constant-shift identities require their own reference-change rule.

For clarity, the primitive data needed for the local argument are the following.

(i) The actual local Gaussian matrices have a uniform real reference gap and a small complex perturbation on every admitted local core and every enlarged decoupling disc. Their local

scalar-coordinate dimension is at most $c(LM)^4$ times the coarse-cube count. The auxiliary dimension may be the volume of the surrounding torus.

(ii) The background and window maps are jointly holomorphic on the specified datum and current domains, including all actual local substitutions, fine brackets and fluctuation amplifications used below. Their first-response, compatible-boundary and spatial regularity bounds are uniform in cutoff and volume.

(iii) Fine restrictions are hereditary; at all interpolation parameters equal to one they reproduce the original local function. A component separated from the declared root and containing a differenced cube gives zero. The actual coarse collar is a fixed-width union of neighboring cubes, rather than a rectangular bounding box of a thin domain.

(iv) The actual pointing is formed from its free-potential, cutoff and interaction legs. The interaction leg uses its actual bond densities. We prove a sufficient explicit convention below, and distinguish it from an identification with a separately prescribed pointing.

These are matrix, map and support assertions. They do not assume the activity bound, its source derivative, the many-scale recurrence, or the old-origin Hessian concluded below.

The roles of the six constructions are successive. The unmarked expansion first provides a spare potential slot. Cauchy's formula in that slot gives derivatives in the norm of the inserted potential. The pointing argument then differentiates the actual bond density. Window-first normalization supplies the old potential direction with its small radius factor before any fine bracket is applied. The source-gradient recurrence and the larger fluctuation-domain argument finally propagate the first jet and bound the old Hessian.

There is one common normalization issue throughout. If H is an old action member, the new fluctuation direction is

$$v(P,B) = H(U_P(B)) - H(U_P(0)). \tag{4.lviii}$$

The omitted value $H(U_P(0))$ is carried by the old background action. It is not subtracted a second time from the logarithmic fluctuation integral. Equally, when a new scalar coefficient is extracted from a pointed member, it is that member's own tent action that is subtracted. A coefficient obtained from a different pointing or a different background readout cannot be substituted by name.

All norm bounds in this section are on fixed admitted local cores. When a complex parameter is varied, a cutoff remains fixed at its real reference value and is a function of real integration coordinates. The reference measure, geometric indices and filing rule are also fixed. A moving cutoff or changing history fibre must first be expressed over a common reference operation before a norm-holomorphy argument applies. The last section records precisely how the local estimates interact with a finite history and its retained inner Wilson weight.

4.5.1. *The actual unmarked expansion.* We derive the activity and local logarithm bounds from the finite Mayer, cutoff and coarse-decoupling formulas. The Gaussian estimate is used at a potential budget larger than the physical one. No logarithm or nonvanishing assertion is needed at that larger budget. Only after the activities have been estimated do we apply the proved hard-core polymer theorem.

All bounds below are local on the specified complex potential/current cores. At complete physical pairs the completeness identity also identifies their gas with the regular fluctuation integral. No global Gaussian on an arbitrary independent-current pair is assumed.

The actual formulas and their analytic inputs. Fix a finite four-dimensional periodic cube lattice, with fine localization cubes of side M and coarse cubes of side LM. Both grids tile the torus. Lengths d_f and d_c are tree lengths in fine and coarse cube units. There are a fixed finite number of potential types: the free, intrinsic, already formed older, and optional inserted potential groups may each be combined within their type before the Mayer expansion. Thus this type count is independent of the number of older levels or source profiles. A tagged fine domain Y includes its

type in its name; geometry ignores this tag. For the actual local potentials assume

$$|V(Y;P,B)| \le a_Y e^{-\lambda_V d_f(Y)} \left(1 + M^{-4} \sum_{b \subset Y} |B(b)|^2\right), \qquad a_Y \le a,$$
$$a = \sum_{\text{types}} \|\mathbf{V}^{\text{type}}\|_{\lambda_V}. \tag{4.lix}$$

The potentials are holomorphic on their declared complex fluctuation boxes and external cores. If a potential is defined only on a box, the corresponding sharp cutoff is kept wherever that potential occurs. The free potential, when present, is defined on all real fluctuation coordinates.

Let $\mathbf{D}$ be a selected set of tagged domains. Write Y_0 for the union of the free-bond sets requiring a field box among those selections, and let P_0 be a set of distinct failed-cutoff bonds outside Y_0. The exact integrand is

$$F_{\mathbf{D},P_0} = (-1)^{|P_0|} \chi_{Y_0} \prod_{b \in P_0} (1 - \chi_b) \prod_{Y \in \mathbf{D}} (e^{V(Y)} - 1), \qquad \chi_b = \mathbf{1}_{\{|B(b)| < \mathfrak{b}\}}. \tag{4.lx}$$

The cutoff product on Y_0 makes every selected factor well defined. Outside that product its integrand is set to zero. This convention introduces no complex argument into a characteristic function.

For any set S of fine cubes, let $\mathcal{N}_f(S)$ consist of S and its fixed first layer of neighbors in the given cubical adjacency. Define $S^\sharp$ to be the union of the coarse cubes containing at least one fine cube of $\mathcal{N}_f(S)$. This is a local cube union, not a rectangular bounding box. A bond uses its incident fine cubes in this definition. The collar preserves unions. A fixed number of additional neighboring cube layers is permitted only if it is part of the assigned collar in every structural identity; its fixed width is then included in the constants.

For the index $\iota = (\mathbf{D}, P_0)$, let $Z_0(\iota)$ be its actual fine support, including the globally defined selected potentials and the failed bonds. Its coarse collar is $Z_0^\sharp$. For $Z \supset Z_0^\sharp$, the actual terms and activities are

$$T(Z,\iota) = \prod_{\Delta \in Z \setminus Z_0^\sharp} \int_0^1 ds_\Delta \, \partial_{s_\Delta} \, \mathcal{J}(Z_0; s_Z)[F_{\mathbf{D},P_0}],$$
$$H(Z) = \sum_\iota T(Z,\iota). \tag{4.lxi}$$

Here $s_Z = 1$ on $Z_0^\sharp$, has the displayed parameters on the remaining cubes of Z, and is zero elsewhere. Polymers Z are nonempty wall-connected coarse domains. The empty selection contributes only the gas's leading one. It is omitted from H: on a nonempty Z, its term has a derivative of the normalized constant functional $\mathcal{J}[1] = 1$, and is zero. Disconnected regions factor into their wall-components rather than being counted again as single polymers.

The local functional $\mathcal{J}$ has the actual matrix representation and reference bounds of the local Gaussian comparison theorem. Uniformly on independent coarse parameter discs

$$|s_\Delta| < R_s, \qquad R_s > 2, \tag{4.lxii}$$

its matrix perturbation is at most ε, its gap constant is K, and its local real scalar-coordinate dimension obeys

$$m(Z_0; s_Z) \le c_m (LM)^4 N, \qquad N = \#Z. \tag{4.lxiii}$$

The auxiliary Gaussian dimension is unrestricted. For an integrand bounded by $F_* e^{\rho |B|^2}$, including distinct failed tests P_0, the proved comparison gives

$$|\mathcal{J}[F]| \le F_* \exp\Big\{ A_K(\varepsilon + \rho + \gamma/2)m - \gamma \mathfrak{b}^2 |P_0|/2 \Big\} \tag{4.lxiv}$$

when $\gamma > 0$, $\varepsilon \le \varepsilon_0(K)$, and $\rho + \gamma/2 \le \rho_0(K)$. This is the marginal local-coordinate estimate; no factor using the exterior auxiliary dimension is allowed. The potentials, matrices and integrals are jointly holomorphic in the scalar potential couplings and s, with the fixed cutoff sets just described.

The remaining structural hypotheses are the actual heredity, normalized-functional and wall-factorization identities. They identify eq. 4.lxi with the finite expansion on complete physical pairs and define coherent local terms on every local complex core. These hypotheses concern the assigned operators and supports. They are not consequences of a numerical Gaussian bound.

The two geometric counts. Let $Y^\sharp$ be the locally defined coarse collar above. All neighborhood widths are fixed independently of L and of the diameter or shape of Y. There is a fixed geometric constant C_h such that

$$\#Y^\sharp \le C_h \left(1 + \frac{d_f(Y)}{L}\right). \tag{4.lxv}$$

To prove this, take a fine covering tree for Y, with length within one of its infimum, and traverse every edge twice. After adding a bounded initial segment, this gives a closed walk of length at most $C(d_f(Y)+1)$ passing within a fixed fine-cube distance of every cube of Y. Split the walk into segments of length at most L. Each segment and its fixed fine collar meet a bounded number of coarse cubes, independent of $L \ge 2$. The number of segments is at most $C(1 + d_f(Y)/L)$. This proves eq. 4.lxv, also on the periodic graph. A failed bond has a coarse collar with at most a fixed number of cubes. Therefore, writing

$$r = \#\mathbf{D}, \qquad v = \#P_0, \qquad h = \#(Z \setminus Z_0^\sharp),$$

and increasing C_h once,

$$N \le C_h \left[r + \frac{1}{L} \sum_{Y \in \mathbf{D}} d_f(Y) + v + h \right]. \tag{4.lxvi}$$

There is no term proportional to the surrounding volume.

We also use the rooted fine-domain entropy bound. For every $\xi > \kappa_f$,

$$S_\xi := \sup_{\text{fine cube } q} \sum_{\text{tagged } Y \ni q} e^{-\xi d_f(Y)} < \infty. \tag{4.lxvii}$$

One direct counting proof rounds a covering tree to the bounded-degree graph of fine cubes, traverses its edges twice starting at q, and records which neighboring cubes are selected in Y. The walk has length at most $C(d_f(Y)+1)$. Its choices and the selection flags are bounded by a fixed constant to that power. This proves exponential rooted entropy, uniformly on the torus. The fixed number of type tags changes that constant only. In particular, if $\mathcal{Y}(Z) = \{Y : Y^\sharp \subset Z\}$,

$$\sum_{Y \in \mathcal{Y}(Z)} e^{-\xi d_f(Y)} \le c_f L^4 S_\xi N. \tag{4.lxviii}$$

This follows by charging a domain to any one of its fine cubes and summing over all fine cubes in Z; overcounting only enlarges the bound. The number of free bonds whose collars fit in Z is at most $n_b N$, where

$$n_b \le c_b (LM)^4. \tag{4.lxix}$$

The bond count in eq. 4.lxix and the scalar-coordinate dimension in eq. 4.lxiii are distinct counts. Neither is counted twice.

A Gaussian-only budget and repeated Schwarz. Choose a positive number W_0, independent of cutoff and volume, small enough for a Gaussian tilt, and choose $\lambda_0 > \kappa_f$. This is not yet the physical potential budget. For every selected tagged domain introduce its own coupling ζ_Y. For $a > 0$ use the radii

$$\mathcal{R}_Y = \frac{W_0}{a} e^{(\lambda_V - \lambda_0) d_f(Y)}. \tag{4.lxx}$$

We will require $a \le W_0/2$, so every radius exceeds one. On the product of these discs, the local density bound in eq. 4.lxvii gives

$$\sum_{Y\in\mathbf{D}} |\zeta_Y V(Y;P,B)| \le W_0 \sum_{Y\in\mathbf{D}} e^{-\lambda_0 d_f(Y)} \left(1 + M^{-4} \sum_{b\subset Y} |B(b)|^2\right) \le C_p W_0 L^4 N + C_p W_0 M^{-4} |B|^2. \tag{4.lxxi}$$

The constant C_p includes the finite tags and their rooted sums, but no number of selected domains or global volume. The quadratic term uses the rooted sum at each bond. The constant term uses the fine-cube count in eq. 4.lxviii.

Choose W_0, γ so that

$$C_p W_0 M^{-4} + \gamma/2 \le \rho_0(K). \tag{4.lxxii}$$

For complex ζ_Y the Mayer product is bounded by $2^r \exp(\sum_Y |\zeta_Y V(Y)|)$. Apply eq. 4.lxiv with the tilt in eq. 4.lxxi. Uniformly on both parameter polydiscs its integral is at most

$$2^r e^{C_0 N} p^v, \qquad p = e^{-\gamma \mathfrak{b}^2/2}, \quad C_0 = C_p W_0 L^4 + A_K c_m (LM)^4 (\varepsilon + C_p W_0 M^{-4} + \gamma/2). \tag{4.lxxiii}$$

The failed tests are distinct bond blocks, so the tail is p^v. The Gaussian theorem was applied only to a fixed real integration space with measurable cutoffs. There is no zero-freeness assertion on this larger coupling polydisc.

The integral vanishes whenever any selected $\zeta_Y = 0$. Repeated Schwarz on the polydisc consequently divides the bound by $\prod_Y \mathcal{R}_Y$ at $\zeta_Y = 1$. For clarity, division by $\prod_Y \zeta_Y$ has removable singularities; apply the maximum principle on smaller polydiscs and let their radii increase to eq. 4.lxx. This proves the product factor without a factorial. Mixed coarse Cauchy on circles centered at $s_\Delta \in [0,1]$, of radii tending to $R_s - 1$, then gives the literal term bound

$$|T(Z,\mathbf{D},P_0)| \le e^{C_0 N} \left(\frac{2a}{W_0}\right)^r e^{-(\lambda_V - \lambda_0)\sum_{Y\in\mathbf{D}} d_f(Y)} p^v (R_s - 1)^{-h}. \tag{4.lxxiv}$$

The coupling derivatives and the coarse derivatives commute by joint holomorphy and the same Gaussian domination. If $a = 0$, all nonempty Mayer selections are zero. Apply the same estimate to the cutoff-only terms, with $r = 0$; this is also the limiting interpretation of eq. 4.lxxiv.

The activity bound, including its linear amplitude. Fix a desired activity rate $\Lambda > 0$. For wall-connected coarse polymers there is a fixed c_ℓ such that

$$\ell(Z) := d_c(Z) + 5 \le c_\ell N. \tag{4.lxxv}$$

A cube-adjacency spanning tree proves $d_c(Z) \le CN$; the added five is absorbed because $N \ge 1$. Choose

$$K_* = 2 + c_\ell \Lambda, \qquad \Theta = C_h (C_0 + K_*). \tag{4.lxxvi}$$

Choose one $\sigma_0 > \kappa_f$, and impose the explicit rate conditions

$$\lambda_V - \lambda_0 \ge \Theta/L + \sigma_0, \qquad \log(R_s - 1) \ge \Theta. \tag{4.lxxvii}$$

Using eq. 4.lxvi to pay for $e^{(C_0+K_*)N}$, eq. 4.lxxiv becomes

$$|T(Z,\mathbf{D},P_0)| \le e^{-K_* N} \prod_{Y\in\mathbf{D}} \left(\frac{2a}{W_0} e^{\Theta} e^{-\sigma_0 d_f(Y)}\right) \prod_{b\in P_0} (p e^{\Theta}). \tag{4.lxxviii}$$

The unused factor $((R_s - 1)^{-1} e^{\Theta})^h$ is at most one. Every selection has thereby paid its own part of the hull cost. This is the place where the coarse hull estimate retains the favorable $d_f(Y)/L$.

At fixed Z, dropping the restriction $P_0 \cap Y_0 = \emptyset$ only enlarges the nonnegative sum. All selected domains and failed bonds still have collars inside Z, so eq. 4.lxviii–eq. 4.lxix give

$$\begin{aligned}\sum_{\substack{\mathbf{D},P_0\\(\mathbf{D},P_0)\neq(\emptyset,\emptyset)}} \prod_{Y\in\mathbf{D}} u_Y \prod_{b\in P_0} q &\le \prod_{Y\in\mathcal{Y}(Z)} (1+u_Y) \prod_{b:b^\sharp\subset Z} (1+q) - 1 \\ &\le e^{\tau N} - 1,\end{aligned} \tag{4.lxxix}$$

where

$$\begin{aligned} u_Y &= (2a/W_0)e^{\Theta} e^{-\sigma_0 d_f(Y)}, \qquad q = pe^{\Theta}, \\ p_\star &= n_b p, \qquad \tau = e^{\Theta}\left[\frac{2c_f L^4 S_{\sigma_0}}{W_0} a + p_\star\right]. \end{aligned} \tag{4.lxxx}$$

The products are finite in finite volume. In particular no factor from bonds outside Z has been inserted. The subtraction of one is essential: it retains a selected potential or failed cutoff, rather than creating a unit amplitude for every Z.

Take the physical smallness conditions

$$a \le W_0/2, \qquad \tau \le 1. \tag{4.lxxxi}$$

For $N \ge 1$, $e^{\tau N} - 1 \le \tau N e^{\tau N}$ and $N \le e^N$. eq. 4.lxxviii–eq. 4.lxxxi prove

$$\begin{aligned} |H(Z)| &\le e^{-K_* N}(e^{\tau N} - 1) \le \tau e^{-(K_*-2)N} \\ &\le C_\tau (a + p_\star) e^{-\Lambda \ell(Z)}, \\ C_\tau &= e^{\Theta} \max\left\{\frac{2c_f L^4 S_{\sigma_0}}{W_0}, 1\right\}. \end{aligned} \tag{4.lxxxii}$$

This is an estimate for the actual activities eq. 4.lxi. Their smallness has not been assumed. The larger Gaussian budget W_0 was used only to bound the analytic integrals; the physical budget in eq. 4.lxxxi is smaller and is what makes the activity amplitude at most one.

The actual union filing and the local logarithm. Choose a desired logarithmic output rate $\mu > 0$, and set $g_* = \Lambda - \mu$. Let $N \le C_v(d_c(Z) + 1)$, and let the number of coarse polymers containing a fixed cube with $m \le d_c(Z) < m + 1$ be at most $C_D e^{\kappa_D m}$. These geometric bounds follow from bounded-degree cube covering walks as above. There is a fixed c_{adj} such that polymers incompatible with Z contain at least one cube in a set of at most $c_{\mathrm{adj}} N$ cubes: use Z and its wall neighbors.

For these counting estimates use the positive norm majorants

$$h_Z = \sup_{P_Z} |H(Z; P_Z)|, \qquad \widehat{w}_Z = h_Z e^{\mu \ell(Z)},$$

where the supremum is over the declared local core of Z, not over restrictions of one global field. At a fixed core pair P_X on X, actual activities are used only for polymers $Z \subset X$, with their values obtained by the assigned hereditary restrictions. Their weighted absolute values are bounded by $\widehat{w}_Z$. The bound with amplitude $\tau \le 1$ in eq. 4.lxxxii gives

$$\sup_q \sum_{Z \ni q} \widehat{w}_Z e^{\#Z} \le \tau\, C_D e^{2C_v} e^{-5g_*} \sum_{m\ge 0} e^{-(g_* - \kappa_D - C_v)m} =: \tau B_*. \tag{4.lxxxiii}$$

Choose the rates so that

$$g_* > \kappa_D + C_v, \qquad c_{\mathrm{adj}} B_* \le 1. \tag{4.lxxxiv}$$

The explicit factor e^{-5g_*} allows the last inequality by increasing the rate gap. Only the upper bound $\tau \le 1$ is needed here. It follows that

$$\sum_{Z' : Z' \not\sim Z} \widehat{w}_{Z'} e^{\#Z'} \le c_{\mathrm{adj}} N \tau B_* \le N. \tag{4.lxxxv}$$

This verifies, for the actual activities, the criterion of the already proved hard-core tree theorem, with the root weight $\#Z$. In particular it verifies that criterion on every hereditary subsystem $Z \subset X$, without assigning actual fields outside X.

If two polymers are incompatible, their covering trees can be joined at a common cube or wall with length at most five in four dimensions. For a connected incompatibility list this gives

$$\ell\left(\bigcup_{i=1}^{n} Z_i\right) \le \sum_{i=1}^{n} \ell(Z_i). \tag{4.lxxxvi}$$

Indeed a spanning tree of the incompatibility graph has $n-1$ joins, and $d_c(\cup_i Z_i) \le \sum_i d_c(Z_i) + 5(n-1)$. Thus the union filing has the same exponential weight.

Define the actual local logarithmic term by

$$E(X) = \sum_{n\ge 1} \frac{1}{n!} \sum_{\cup_i Z_i = X} \phi(Z_1, \dots, Z_n) \prod_i H(Z_i), \tag{4.lxxxvii}$$

where ϕ is the connected hard-core graph coefficient. For a fixed cube $q \in X$, mark a vertex whose polymer contains q. This only enlarges the absolute series. Symmetry of the ordered list changes its factor $1/n!$ into $1/(n-1)!$. When dropping the union restriction, retain $Z_i \subset X$ for the actual terms at P_X. Use eq. 4.lxxxvi, and then the rooted tree theorem for that subsystem, dominated by the numerical majorants $\widehat{w}$. Only the subsequent nonnegative majorant sum may be enlarged to all geometric polymers containing q. The result is

$$\begin{aligned} e^{\mu \ell(X)} |E(X)| &\le \sum_{\substack{Z \subset X \\ Z \ni q}} \widehat{w}_Z e^{\#Z} \le \tau B_* \\ &\le C(a + p_\star). \end{aligned} \tag{4.lxxxviii}$$

There is no factor $\#X$ or ambient volume in this local estimate. It proves the unmarked potential-ball majorant used in the finite insertion theorem.

Uniform convergence on smaller external and coupling domains follows from these same positive majorants. The actual local functions are holomorphic, so their union-filed sum is holomorphic on each declared core. At complete physical pairs, the complete-Gaussian and wall-factorization identities identify the gas with the finite regular integral; the hard-core theorem then gives $\exp(\sum_X E(X)) = \Xi \neq 0$. On general local cores the activity and logarithm bounds do not require that global Gaussian interpretation.

Parameter choices and the available insertion slot. All constants above concern fixed geometry and the actual matrix/core package. One must verify eq. 4.lxii–eq. 4.lxiv and eq. 4.lxxii, eq. 4.lxxvii, eq. 4.lxxxi, eq. 4.lxxxiv together. For example, it suffices that the matrix package control $A_K c_m (LM)^4 \varepsilon$ by a fixed constant on the chosen decoupling discs. Choose W_0 so its two contributions to C_0 are bounded, and choose positive γ of order at most $(LM)^{-4}$ to bound its contribution. Then C_0 is bounded independently of exterior volume. One can choose the target rate gap, the coarse parameter radius and λ_V by eq. 4.lxxvi–eq. 4.lxxvii, and finally make a small and $\mathfrak{b}$ large to obtain eq. 4.lxxxi. This lists explicit sufficient inequalities; it does not assert them for a matrix family whose constants have not been controlled on the required enlarged discs.

For an explicit insertion budget, put

$$A_a = \frac{2 c_f L^4 S_{\sigma_0}}{W_0}, \qquad a_* = \min\left\{ \frac{W_0}{2}, \frac{1}{4 e^{\Theta} A_a} \right\}, \qquad p_\star \le \frac{1}{4 e^{\Theta}}.$$

Then every total norm $a \le a_*$ has $\tau \le 1/2$ and satisfies eq. 4.lxxxi, uniformly above this cutoff threshold. If the base total potential norm is at most $a_*/2$, reserve one additional potential slot of norm $w_\star \le a_*/2$. The entire slot stays within eq. 4.lxxxi and hence within eq. 4.lxxxii–eq. 4.lxxxviii. This proves the local uninserted majorant uniformly on that slot without assuming a zero-free

activity domain. The constants may depend on the fixed localization parameters and rate choices, but not on cutoff, torus size, old-level count or the auxiliary Gaussian dimension.

4.5.2. *A small potential direction and its first jet.* We keep the calibration, the real action weight, the Gaussian structure, the cutoffs and the background maps fixed. The result below concerns the finite regular fluctuation integral. It does not enlarge the source domain of a large-field history operation. This distinction permits a small potential direction to be used at its actual norm. Localized terms and their insertion estimates are defined on the specified complex local cores, with potential and current independent there. Their sum is identified with a complete Gaussian integral only at physical complete background pairs $P = p_W$. An identity on that graph is not extended to arbitrary independent currents.

The fixed-reference integral. Let $\theta \geq \theta_{\min}$, $g = \theta^{-1/2}$, and $\tau = x/\theta$, with $|\tau - 1| < \sigma_\star$. The fixed small number $\sigma_\star$, the allowed external pairs and their Gaussian gap are restricted by the ordinary Gaussian comparison and potential allowance below. In particular the decoupled precisions stay in their coercive regime and the free potential leaves the declared allowance; no assertion is made for arbitrary positive x/θ. The sourced run considered here keeps $x > 0$ fixed. At a physical complete background pair $P = p_W$, in the fluctuation coordinates B on the non-pivot bonds, the reference measure is the normalized Gaussian with covariance $\tau^{-1}C(P)$. The same identity holds on a holomorphic continuation in physical background data wherever that continuation and the ordinary Gaussian bounds have been supplied. It is not an identity on every complex core pair. Put

$$\Xi(P,z) = \int d\mu_{\tau^{-1}C(P)}(B) \prod_b \chi_b(B) \exp\left\{ \sum_Y V(Y;P,B) + z \sum_Y v(Y;P,B) \right\}. \tag{4.lxxxix}$$

The parameter z enters only the potentials. In particular the covariance, the cutoff thresholds and the Gaussian determinant do not depend on z. Here the cutoffs are sharp indicators $\chi_b(B) = \mathbf{1}\{|B(b)| < \mathfrak{b}\}$ on free bonds. For a bond set A, write $\chi_A = \prod_{b\in A} \chi_b$ and $\chi_A^c = \prod_{b\in A}(1-\chi_b)$. Thus the latter denotes a product of complements. After the old action at the background has been carried separately, the new fluctuation increment is

$$\mathcal{E}(P,z) = \mathcal{N}[\theta;x](P) + \log \Xi(P,z). \tag{4.xc}$$

The field-independent Gaussian number includes $-\frac{3}{2}|\mathfrak{X}| \log \tau$, as well as the Haar-coordinate, scaling and gauge-fixing counts. The localised normalizer $\mathbf{N}^\circ$ and the free-potential family $\mathbf{G}_\tau$ also use the fixed (θ, x). All these quantities have zero z-derivative. Moving z into x, and differentiating the resulting Gaussian determinant, would describe a different presentation.

For an inserted exponent H, the potential actually inserted is its fluctuation difference. Thus, when the fibre field is $U_P(B)$ and $U_P(0)$ is its background, at a physical complete pair,

$$\sum_Y v(Y;P,B) = H(U_P(B)) - H(U_P(0)). \tag{4.xci}$$

Consequently the background subtraction in the step functional is already present in eq. 4.lxxxix; it must not be subtracted a second time. Equivalently, the raw fibre integral with the additional exponent $zH(U_P(B))$ has the extra factor $e^{zH(U_P(0))}$ outside eq. 4.lxxxix. That factor belongs to the separately carried old action, rather than to the newly generated cluster activities.

The ordinary Gaussian expansion used here. We state the analytic inputs at the level at which they are applied. On an arbitrary complex local core pair P, the actual localized Gaussian functionals $\mathcal{J}(Z_0; s; P)$, potentials and union-filed terms below are defined by their local rules. We do not assume that the complete values for different Z_0 all arise as marginals of one global Gaussian at

that P. A potential has norm

$$|V(Y;P,B)| \le \|\mathbf{V}\|_\lambda e^{-\lambda d(Y)} \left(1 + M^{-4} \sum_{b\subset Y} |B(b)|^2\right) \tag{4.xcii}$$

on its specified complex field box. The potential may be defined only there: the cutoff product on its domain is kept in every term in which it occurs. The ordinary Gaussian comparison estimate bounds a localized decoupled functional by $F_0 e^{CuN(Z)}$ when its integrand is bounded by $F_0 e^{\rho|B|^2}$, with $u = \widehat{\varepsilon} + \rho$. Here $N(Z)$ counts the affected real scalar Gaussian coordinates, of order $(LM)^4 N_c(Z)$, where $N_c(Z)$ is the number of coarse cubes. In the cube-count convention the same bound is $F_0 e^{C(LM)^4 u N_c(Z)}$; the dimension factor is counted once. The Gaussian reference gap, radii and tilt are chosen so these factors are bounded per coarse cube. The decoupling parameters have a fixed enlarged complex disc. These are the ordinary background/Gaussian inputs; no source derivative is contained in them.

Choose a potential allowance $w_\star > 0$, independent of the cutoff and volume, small enough for the following ordinary expansion. The base potentials use at most a fixed number of slots, and leave one slot available for an additional potential of norm at most $w_\star$. The same Gaussian comparison, rooted domain count and decoupling majorant are valid throughout this allowance. The convergent hard-core expansion gives local logarithmic terms $E(X)$ and bounds

$$|E(X)| \le C_\star e^{-\lambda_{\mathrm{out}} d(X)}. \tag{4.xciii}$$

The constant is uniform in all allowed external pairs, decoupling parameters and potentials. This is the ordinary uninserted polymer majorant, not an assumed differentiability bound for a renormalization map.

Here is the explicit finite expansion to which that majorant applies. Let $\mathbf{D}$ be a set of selected interaction domains. Write Y_0 for the union of their free-bond sets on which a field box is required; a globally defined free potential does not contribute to this union. Let $P_0 \subset \mathfrak{X} \setminus Y_0$ be a set of failed fluctuation cutoffs. The indicator identities $\chi_b^2 = \chi_b$ and $\chi_b(1-\chi_b) = 0$ give exactly $\prod_b \chi_b = \chi_{Y_0} \sum_{P_0 \subset \mathfrak{X}\setminus Y_0} (-1)^{|P_0|} \chi^c_{P_0}$. Before decoupling, its integrand is

$$F_{\mathbf{D},P_0}(z) = (-1)^{|P_0|} \chi_{Y_0} \chi^c_{P_0} \prod_{Y\in\mathbf{D}} (e^{V(Y)+zv(Y)} - 1). \tag{4.xciv}$$

The empty choice $(\mathbf{D}, P_0) = (\emptyset, \emptyset)$ supplies the leading one in the gas and is omitted from its activities. All polymer domains Z are nonempty and wall-connected, so no $H(\emptyset)$ is defined. Disconnected regions are products of their wall-component activities, rather than additional single polymers. Equivalently, for a nonempty Z the empty choice has at least one decoupling derivative of the normalized constant functional $\mathcal{J}[1] = 1$ and contributes zero. The index Y carries one of the fixed potential-type tags used in the unmarked expansion. Its cutoff requirement is fixed by its support and type: a boxed type uses the common indicators on all free bonds in that support, as in the definition of Y_0, while the globally defined free type requires none. Older levels and source origins do not add cutoff types. Before forming the product, sum all origins having the same support and type into that type's potential. This grouping is performed before taking absolute majorants, and the norm in eq. 4.xcii is the norm of the grouped potential. Indeed, for finitely many origins on that same support,

$$e^{\sum_o V_o} - 1 = \sum_{\emptyset \ne I \subset \{o\}} \prod_{o\in I} (e^{V_o} - 1).$$

Linearity of the local Gaussian functional and of the decoupling operations therefore preserves the complete integral and its union filing. The estimate is applied to the grouped left-hand side; it does not require an absolute majorant for the separate terms on the right. An individual origin can still be selected as the marked direction v, without adding a Mayer type. The geometric support

and its tree length ignore the fixed type tag. For each support index $\iota = (\mathbf{D}, P_0)$ let $Z_0(\iota)$ be its support. The decoupled activities are formed by

$$\begin{aligned} T(Z,\iota;z) &= \prod_{\Delta\subset Z\setminus Z_0(\iota)^\sharp} \int_0^1 ds_\Delta\, \partial_{s_\Delta}\, \mathcal{J}(Z_0(\iota);s_Z)[F_{\mathbf{D},P_0}(z)], \\ H(Z,z) &= \sum_\iota T(Z,\iota;z). \end{aligned} \tag{4.xcv}$$

In this formula s_Z is one on $Z_0(\iota)^\sharp$, is the displayed free parameter on $Z\setminus Z_0(\iota)^\sharp$, and is zero outside Z. The normalized Gaussian functional $\mathcal{J}$ is independent of z. At a physical complete pair $P = p_W$, its complete-structure identity is

$$\mathcal{J}(Z_0;1)[F] = \int d\mu_{\tau^{-1}C(P)}(B)\, F(B).$$

This physical identity, together with interpolation of every remaining coarse-cube parameter and the ordinary decoupling property, is what identifies the gas with eq. 4.lxxxix. Two polymers are compatible when they have neither a cube nor a wall in common. Multiplicativity over their wall-components yields $\Xi = \sum_{\Gamma\ \text{compatible}} \prod_{Z\in\Gamma} H(Z)$, with empty product one, at a physical complete pair. On general local cores, the same local formulas define activities and their union-filed terms without asserting one global Gaussian representation. Those logarithmic terms are

$$E(X,z) = \sum_{n\ge1} \frac{1}{n!} \sum_{\cup_{i=1}^n Z_i = X} \rho^T(Z_1,\dots,Z_n) \prod_{i=1}^n H(Z_i,z). \tag{4.xcvi}$$

Here $\rho^T(Z_1,\dots,Z_n)$ is the sum over connected graphs on $\{1,\dots,n\}$ of the products of their edge weights; an edge has weight -1 for an incompatible pair and zero otherwise. For $n = 1$, $\rho^T = 1$. Every sum is first formed in finite volume; the logarithmic series and its differentiated series are justified by the same absolute majorants.

Proposition 4.8 (The inserted-potential estimate)**.** Let $\varepsilon = \|\mathbf{v}\|_\lambda$. Under the ordinary local Gaussian/polymer hypotheses above, on every specified complex local core pair,

$$|\partial_z E(X,z)|_{z=0}| \le C\varepsilon e^{-\lambda_{\mathrm{out}} d(X)}. \tag{4.xcvii}$$

At a complete physical pair $P = p_W$, its sum over X is the normalized expectation

$$\partial_z \log \Xi(P,z)|_{z=0} = \frac{\int d\mu \prod_b \chi_b\, e^{\sum_Y V(Y)} \sum_Y v(Y)}{\int d\mu \prod_b \chi_b\, e^{\sum_Y V(Y)}}. \tag{4.xcviii}$$

There is no volume factor in the local estimate in eq. 4.xcvii. ∎

Proof. If $\varepsilon = 0$, every inserted potential vanishes on its domain and the assertion is immediate. Otherwise take the actual auxiliary potential $z\mathbf{v}$ on $|z| < w_\star/\varepsilon$. This is a legitimate potential in the actual local Mayer and decoupling expressions, regardless of whether it would be an admissible physical source for a history construction. The ordinary Gaussian comparison and the polymer criterion hold on this disc. Every integrand and every activity is holomorphic there; Gaussian domination and the parameter-independent majorant give holomorphy of the union-filed series eq. 4.xcvi. This local argument does not require one global Gaussian at a general independent-current core pair. Apply Cauchy's estimate on $|z| = w_\star/(2\varepsilon)$ to eq. 4.xciii. It gives eq. 4.xcvii with constant $2C_\star/w_\star$, which depends only on fixed ordinary allowances.

The same argument gives a norm derivative, which is useful when the actual source family of potentials is nonlinear. On the potential space with its fixed declared domains, joint holomorphy on every two-dimensional affine plane makes the directional derivative complex linear: the degree-one term of $E_X(\mathbf{V} + z_1\mathbf{h}_1 + z_2\mathbf{h}_2)$ is the sum of its two degree-one terms. Cauchy's estimate makes this

linear map bounded. For $\|\mathbf{h}\|_\lambda < w_\star/2$, apply one-variable Cauchy to $t \mapsto E_X(\mathbf{V}+t\mathbf{h})$ on the disc $|t| < w_\star/\|\mathbf{h}\|_\lambda$. Summing its Taylor coefficients of degree at least two gives

$$|E_X(\mathbf{V}+\mathbf{h}) - E_X(\mathbf{V}) - DE_X(\mathbf{V})[\mathbf{h}]| \le CC_\star \frac{\|\mathbf{h}\|_\lambda^2}{w_\star^2} e^{-\lambda_{\mathrm{out}} d(X)}.$$

All constants are uniform in the external core pair, so this is also the remainder bound in the unpointed supremum norm $\|F\|_{\mathrm{out}} = \sup_{X,P} e^{\lambda_{\mathrm{out}} d(X)} |F_X(P)|$. It does not assert a pointed or stronger rooted norm bound. Uniform local Lipschitz bounds on smaller potential balls give the unique continuous extension to the potential-norm completion. The displayed remainder passes to that extension and proves Fréchet differentiability there as well. If an actual ordinary source family satisfies $\mathbf{V}(w) = \mathbf{V}(0) + w\dot{\mathbf{V}} + o(|w|)$ in that norm, the remainder bound proves its chain rule $\partial_w E_X(\mathbf{V}(w))|_0 = DE_X(\mathbf{V}(0))[\dot{\mathbf{V}}]$. Pointwise holomorphy of $\mathbf{V}(w)$ on its ordinary source disc, with a uniform potential-norm bound there, gives its norm Taylor expansion by the same uniform Cauchy estimates. Thus the auxiliary linear direction computes the actual first source jet; the nonlinear physical source family need not exist on the large auxiliary-potential disc.

This is also the marked-activity derivative of the explicit expansion. The derivative of eq. 4.xciv replaces precisely one selected factor by

$$v(Y) e^{V(Y)} \prod_{\substack{Y' \in \mathbf{D} \\ Y' \neq Y}} (e^{V(Y')} - 1). \tag{4.xcix}$$

Domains with $V(Y) = 0$ are included: their unmarked term is zero, but their derivative need not be zero. In a term with a marked potential the cutoff on its domain is retained. Derivatives commute with eq. 4.xcv by the same Gaussian comparison and with eq. 4.xcvi by absolute convergence. Thus this proof differentiates the actual finite interaction formula, not a hypothetical normed map without a specified integral. The pinned cluster estimate is what keeps the local Cauchy bound in eq. 4.xciii independent of volume.

At a complete physical pair, the sum rule for the ordinary cluster expansion gives $\exp(\sum_X E(X,z)) = \Xi(P,z) \neq 0$. Differentiating it and the finite integral proves eq. 4.xcviii. The denominator is this one regular Gaussian integral; no division by the full sourced partition function or by a history mass is used. □

The actual scalar action direction. Let $H_u(U) = A(u,U)$, where u is a bounded function on bare plaquettes. It is filed by assigning a plaquette to the union of the fixed cubes containing its corners. In the background field domain, $|\partial U(p) - 1| \le C\alpha\eta^2$. The identity $a_p(U) = \frac{1}{2}\det(1 - \partial U(p))$ gives a raw local action member of norm at most $CM^4\alpha^2 \sup|u|$. It has no coupling counterterm at this stage.

More precisely, at a complete physical pair the potential in eq. 4.xci is a first difference of this action. Suppose the actual background response and its decoupled substitutions stay in their stated field domains and shift a plaquette by at most $C\delta\eta^2$, where $0 < \delta \ll \alpha$. Along this shift, the plaquette remains at distance at most $C\alpha\eta^2$ from the identity. The derivative of the determinant of a 2×2 matrix is bounded by a constant times its size, so

$$|a_p(U_P(B)) - a_p(U_P(0))| \le C\alpha\delta\eta^4. \tag{4.c}$$

This proof works at complex pairs. It uses the actual curvature and response bounds, not positivity. On a local core the same determinant bound applies to the two actual substituted plaquette matrices whenever the declared response bounds give these curvature and difference estimates; it does not require a global fibre map there.

We give the quantitative fine-response localization separately from the coarse Gaussian decoupling in eq. 4.xcv. For a plaquette p, let $X_0(p)$ be the union of its fixed corner cubes. Its number of cubes is at most a dimensional constant C_{base}, and its tree length is zero. Write $\mathcal{U}(P')$ for the bare configuration used by the actual background evaluation of a pair P', and set $F_p(P') = u(p) a_p(\mathcal{U}(P'))$.

Thus this local function is explicitly a function of a pair. The ordinary response-bracket hypotheses used here are the following. On each $Y \supset X_0(p)$ the same response rule gives

$$\Phi_{p,Y}(s,\nu) = F_p(P|_{X_0(p)} \dot{+} \nu\, \mathfrak{h}_Y(s;B)), \qquad |s_\Delta| < R = e^{\bar\kappa}, \quad |\nu| < \rho,$$

holomorphic jointly on these independent discs, uniformly in the allowed complex external fields and fluctuation box, with

$$\rho \geq c_R \alpha/\delta > 1, \qquad \sup|\Phi_{p,Y}| \leq C\alpha^2\eta^4|u(p)|.$$

Here $c_R > 0$ can contain the fixed loss $e^{-C\bar\kappa}$. In particular all substituted pairs must stay in their actual cores on the whole joint polydisc, not just for real interpolation parameters. The operation $\dot{+}$ shifts the potential and current by the actual response. These are quantitative background and membership hypotheses; the following Cauchy argument does not prove them.

For disjoint fine-cube sets K, N, define the mixed bracket by

$$\Psi[K|N] = \int_{[0,1]^K} \prod_{\Delta\in K} \partial_{s_\Delta} \Psi(s_K + 1_N)\, ds_K,$$

with all other parameters zero. Its scalar-action potential is the bracket of $\Psi(s) = \Phi_{p,Y}(s,1) - \Phi_{p,Y}(s,0)$ with $K = Y\setminus X_0(p)$, $N = X_0(p)$. At $\nu = 0$ the field is the unshifted restricted background. Schwarz's lemma in ν first gives $\sup_s|\Psi(s)| \leq 2C\rho^{-1}\alpha^2\eta^4|u(p)|$. For every $s_\Delta \in [0,1]$, a coordinate Cauchy circle has radius arbitrarily close to $R-1$. Iterated Cauchy therefore bounds the bracket by

$$2C\rho^{-1}\alpha^2\eta^4|u(p)|(R-1)^{-|K|} \leq C_R\alpha\delta\eta^4|u(p)|e^{-(\bar\kappa-1)|K|},$$

where $\bar\kappa \geq 1$. There is no factorial in this estimate, because each independent cube parameter is differentiated once. The ordinary rooting property says that a component containing a differenced cube and missing $X_0(p)$ makes the bracket zero. For the present choice $N = X_0(p)$, every component missing the base has such a differenced cube. No assertion about an arbitrary remote component consisting only of undifferenced parameters is being made. The local response rule agrees with the response built on the active wall-component. Between local ambient domains on which the substitutions are defined it has hereditary consistency, $\Phi_{p,Y}(s,\nu) = \Phi_{p,\Lambda}(s\,\mathbf{1}_Y,\nu)$, with the base parameters set to one. At a complete physical pair on the full lattice its endpoints are

$$\Phi_{p,\Lambda}(1,1) = u(p)a_p(U_P(B)), \qquad \Phi_{p,\Lambda}(1,0) = u(p)a_p(U_P(0)).$$

Apply the finite fundamental theorem of calculus in every fine-cube parameter outside $X_0(p)$, expanding evaluation at one as evaluation at zero plus the integral of the derivative. The resulting sum of brackets equals $\Phi_{p,\Lambda}(1,1) - \Phi_{p,\Lambda}(1,0)$. At a complete physical pair this is exactly $u(p)[a_p(U_P(B)) - a_p(U_P(0))]$. Wall-component consistency identifies each bracket with the corresponding $\Phi_{p,Y}$; rooting removes the disconnected terms. Thus the construction gives a uniformly bounded local extension on every declared core and localizes the actual difference in eq. 4.xci at complete physical pairs.

For a surviving bracket, the ordinary domain geometry gives $d(Y) \leq C_{\mathrm{geo}}(|K|+1)$. At fixed Y there are at most $CM^4\eta^{-4}(|K| + C_{\mathrm{base}})$ possible plaquette anchors. Choose the fixed surplus $\bar\kappa$ so that $\bar\kappa - 1 > C_{\mathrm{geo}}\lambda + 2$, increasing it further if a subsequent rooted-domain entropy sum is included. Sum the displayed bracket bound over its plaquette anchors and use that a linear polynomial times $e^{-2|K|}$ is bounded. This proves

$$\|\mathbf{v}_u\|_\lambda \leq CM^4\alpha\delta\, \sup|u| = CM^4\alpha^2 s\, \sup|u|, \qquad s = \delta/\alpha. \tag{4.ci}$$

The factors η^4 and η^{-4} cancel exactly. The required membership and response estimates on all decoupling substitutions are background inputs. The bracket argument neither creates those inclusions nor replaces them by bounds at real fields.

Combining eq. 4.xcvii with eq. 4.ci proves a one-step unpointed source response of size $C\alpha^2 s\sup|u|$ for the actual local regular family on every declared complex core. Its physical sum

is the source response of the fixed-reference regular integral. The Gaussian normalizer contributes zero derivative. Its interaction vacuum value contributes $\partial_z E(X,(1,0),0)$, which is the source vacuum energy and obeys the same bound. Subtracting it leaves a factor at most two in the estimate.

Own counterterms and the remaining iteration. For a constant source, once the actual bond-density pointing is applied, the derivative member is Euclidean. Its own coefficient is $\partial_z b[E_z]|_0 = b[\partial_z E_z|_0]$, because the free response and the coefficient extractor are fixed and complex linear. The counterterm to remove from the derivative action at site y is therefore this number times $A(h_y, U)$. A coefficient belonging to another member, or a derivative of a Gaussian presentation with a different weight, would not supply that identity.

There are two further tasks before the one-step estimate is a complete multilevel source-response bound. First, renewing the actual pointing uses three bond-indexed interpolation legs. Differentiating a normalized leg expectation produces both the derivative of its leg insertion and a connected expectation of that insertion with $\sum_Y v(Y)$. A root-plus-insertion cluster bound and the path's admissible field room must control those terms before the same estimate is asserted for each pointed term.

Second, the old regular derivative members and the nonconstant source defects have to be inserted again with their own coefficients. A constant-source derivative of size $\alpha_j^2 s_j$ can be normalized by the pointed Hessian theorem. Write $\alpha_H > 0$ for the fixed spatial Hölder exponent, as distinct from an analytic field radius. On a non-wrapping local chart, retaining the *true* radius ratio $\rho_j = \delta_{\mathrm{out}}/\alpha_j$ in the near-domain Taylor proof, before any enlargement to $\max\{1, \rho_j\}$, would give after the fine-site count the contribution

$$C\alpha_j^2 s_j \left[\left(\frac{\delta_{\mathrm{out}}}{\alpha_j} \right)^2 L^{-\alpha_H (n-j)} + \left(\frac{\delta_{\mathrm{out}}}{\alpha_j} \right)^3 L^{-2(n-j)} \right]. \tag{4.cii}$$

Its quadratic part sums to $C\delta_{\mathrm{out}}^2 s_{\mathrm{out}}$ when the ordinary flow and age weights apply. For $\alpha_{\mathrm{out}} \asymp x_{\mathrm{out}}^{-1/2} (\log x_{\mathrm{out}})^q$ and $\delta_{\mathrm{out}} \asymp x_{\mathrm{out}}^{-1/2} (\log x_{\mathrm{out}})^{p_0}$, this is

$$C x_{\mathrm{out}}^{-1} (\log x_{\mathrm{out}})^{3p_0 - q}. \tag{4.ciii}$$

This arithmetic is conditional on that homogeneous local refinement and does not establish the derivative recurrence. In particular the defects must retain the factor $\alpha_j^2 s_j$ as well as the scaled Lipschitz gain. A fixed-disc source bound alone does not provide it. Marked and remote terms need their own uniform fixed-disc bounds. Field-independent tails around holes or torus cycles cannot be assigned an extra ρ_j^2 when $\rho_j < 1$; they require a non-wrapping chart or a separately small class estimate.

Identification with the fixed-reference source family. In the variable-profile presentation the actual regular family has the decomposition

$$\mathcal{E}^{\,w} = N^{\circ} + G_\tau + F^{\,w}, \qquad \mathcal{E}^{\,w} - \mathcal{E}^{\,0} = F^{\,w} - F^{\,0}. \tag{4.civ}$$

Here x, θ, the Gaussian matrices, cutoffs and numerical normalizers are fixed; the variable profile is an older member. Thus N° and G_τ have no source derivative. If its actual potential family is differentiable in the accepted potential norm, the uniform norm-Taylor proof of eq. 4.xcvii applies to each unpointed local logarithm:

$$\begin{aligned} \dot{\mathcal{E}} &= \dot{F}, \\ \partial_w E_X(V(w))\big|_0 &= DE_X(V(0))[\,V'(0)\,]. \end{aligned} \tag{4.cv}$$

The derivative of the pointed density is computed in the next section; it includes both terms of eq. 4.cxvi. These are identities for the fixed presentation, rather than derivative rules for one that also changes its reference data.

For two auxiliary directions of norms A and ε, one spare potential family suffices:

$$\|\lambda a + zv\|_{\lambda_V} \le |\lambda|A + |z|\varepsilon < w_\star. \tag{4.cvi}$$

The two directions are combined in this one slot; they do not increase the allowed number of potential types. The resulting auxiliary Cauchy polydisc is a domain of the local potential expansion. It need not be a domain of physical Wilson sources or of history operations.

4.5.3. *Renewing the actual bond pointing.* An estimate for an unpointed localized logarithm is not by itself an estimate for a particular pointing of that logarithm. We retain the actual three interpolation legs and differentiate their bond densities. For a source filed as an older weight member, the Gaussian structure, free potential, cutoffs and numerical normalizers are fixed. Thus the first, free-potential leg and the second, cutoff leg are independent of this source. Only the interaction leg contributes below. This choice of presentation is essential to the statement.
The finite local logarithm and its actual locality. Fix a finite regular fluctuation integral, its Gaussian reference, its cutoffs and all geometric and external data. Fix nested full complex fluctuation boxes

$$\mathcal{B}_0 = \{B : \sup_b |B(b)| < \mathfrak{b}_0\}, \qquad \mathcal{B}_1 = \{B : \sup_b |B(b)| < (1+\nu)\mathfrak{b}_0\}, \qquad \mathfrak{b}_0 \ge 1, \quad \nu > 0.$$

The ordinary spare-potential estimate below accepts potentials holomorphic and bounded in its growth norm on $\mathcal{B}_0$; all its actual Gaussian cutoff supports lie there. The larger box $\mathcal{B}_1$ is reserved for the one coordinate Cauchy estimate on an inserted potential. Both boxes and the external-pair and decoupling domains are fixed once for this step. Ordinary inputs whose cutoffs fill half their analytic box leave this room by choosing an intermediate $\mathcal{B}_0$. This uses that existing room once, and is not a new shrinking at each source derivative. A local potential family has the Gaussian growth norm

$$|V(Y;P,B)| \le \|\mathbf{V}\|_\lambda e^{-\lambda d(Y)} Q_Y(B), \qquad Q_Y(B) = 1 + M^{-4}\sum_{b\subset Y} |B(b)|^2. \tag{4.cvii}$$

There are finitely many bond variables but arbitrarily many of them as the volume grows. The norm and all constants used below are those of the ordinary regular expansion, uniform in that number. A fixed finite number of potential types can be combined by addition of their norms.

Let $E_X(\mathbf{V})$ denote the actual union-filed logarithmic term of the local Gaussian functionals, Mayer expansion and connected-cluster expansion. These local objects are defined on their independent-current pair cores. Away from complete physical data, their wall-component functionals need not be margins of one global Gaussian measure. Normalized global Gaussian expectations will be used only at a complete physical pair $P = p_W$, or by holomorphic continuation in the physical datum W where the same complete construction is defined. We use the following consequence of its uninserted estimate, as established for the fixed-reference integral in § 4.5. At every base system used below there is an available potential allowance $w_\star > 0$ such that

$$|E_X(\mathbf{V}+\mathbf{U})| \le C_\star e^{-\lambda_{\mathrm{out}} d(X)}, \qquad \|\mathbf{U}\|_\lambda < w_\star. \tag{4.cviii}$$

The actual finite local Gaussian functionals and the absolutely convergent logarithmic series are holomorphic in every finite collection of potential parameters on this allowance. A single additional family is enough: several auxiliary directions are added together in $\mathbf{U}$, with the sum of their norm loads less than $w_\star$.

The locality needed here is part of this actual expansion, not a choice made after taking its logarithm. A contribution filed at X uses only selected interaction domains whose enlarged union is in X. In particular, if a family $\mathbf{a}_b$ vanishes except on domains containing $b^\sharp$, then

$$DE_X(\mathbf{V})[\mathbf{a}_b] = 0, \qquad D^2E_X(\mathbf{V})[\mathbf{a}_b, \mathbf{v}] = 0 \quad \text{if } b^\sharp \not\subset X. \tag{4.cix}$$

The assertions include domains with a zero unmarked potential. In a Mayer derivative their zero factor can be replaced by the nonzero inserted factor, and the marked domain must then be included in the union. Omitting such terms would invalidate both locality and the insertion identity.

Lemma 4.9 (Two derivatives of the actual localized term)**.** For potential families $\mathbf{a}, \mathbf{v}, \mathbf{c}$, the allowance in eq. 4.cviii gives

$$\begin{aligned} |D^2 E_X(\mathbf{V})[\mathbf{a}, \mathbf{v}]| &\le \frac{16 C_\star}{w_\star^2} \|\mathbf{a}\|_\lambda \|\mathbf{v}\|_\lambda e^{-\lambda_{\mathrm{out}} d(X)}, \\ |D E_X(\mathbf{V})[\mathbf{c}]| &\le \frac{2 C_\star}{w_\star} \|\mathbf{c}\|_\lambda e^{-\lambda_{\mathrm{out}} d(X)}. \end{aligned} \tag{4.cx}$$

These derivatives preserve eq. 4.cix. ■

Proof. If either direction in the first estimate has zero norm, the claim is immediate. Otherwise put the single auxiliary family $\zeta \mathbf{a} + z\mathbf{v}$ in the available slot. On the bidisc with radii $w_\star/(4\|\mathbf{a}\|_\lambda)$ and $w_\star/(4\|\mathbf{v}\|_\lambda)$ its total norm is at most $w_\star/2$. Apply the two-variable Cauchy formula to the actual holomorphic term in eq. 4.cviii. For the second estimate use radius $w_\star/(2\|\mathbf{c}\|_\lambda)$. The support assertion follows from the union filing before differentiation. No radius for a full history construction has been enlarged in this argument. □

Thus a separate second-mark estimate is not needed if the actual ordinary bound is already uniform on the full potential allowance in eq. 4.cviii. If only a one-direction derivative bound were known, without this allowance and locality, the lemma would not follow from it.
The interpolation paths and the differentiated density. Let $\mathbf{V}_t$, $0 \le t \le 1$, be the actual interaction leg, with local bond directions $\mathbf{a}_{t,b}$ satisfying

$$\partial_t \mathbf{V}_t = \sum_b \mathbf{a}_{t,b}. \tag{4.cxi}$$

Every summand is supported on domains containing $b^\sharp$. At a finite volume the sum is finite. We assume that the ordinary interpolation construction supplies the uniform bound

$$\sup_b \int_0^1 \|\mathbf{a}_{t,b}\|_\lambda \, dt \le A. \tag{4.cxii}$$

This is a property of the actual unmarked legs and their admissible field domains. It is not inferred merely from an endpoint norm. Uniform ordinary bounds and field room are required throughout the path, with the same allowance in eq. 4.cviii. The third-order vanishing of the intrinsic $V^{G,A}$ makes its t^{-2} radial path regular at zero. It does not by itself identify the actual bond split. We do not replace that split by an Euler decomposition or any other pointing to infer eq. 4.cxii.

For the inserted source let

$$\mathbf{V}_t(s) = \mathbf{V}_t + s\mathbf{v}_t, \qquad \mathbf{a}_{t,b}(s) = \mathbf{a}_{t,b} + s\mathbf{c}_{t,b}, \qquad \partial_t \mathbf{v}_t = \sum_b \mathbf{c}_{t,b}, \tag{4.cxiii}$$

with the same bond support and with $\mathbf{v}_0 = 0$. It is enough that these are the first source derivatives of holomorphic paths; the affine notation in s makes the formula easier to read. Suppose

$$\sup_{0 \le t \le 1} \|\mathbf{v}_t\|_\lambda \le C_v \varepsilon, \qquad \sup_b \int_0^1 \|\mathbf{c}_{t,b}\|_\lambda \, dt \le C_c \varepsilon. \tag{4.cxiv}$$

Choose $|s| C_v \varepsilon < w_\star/4$. The remaining allowance in eq. 4.cviii bounds the first derivative in direction $\mathbf{a}_{t,b} + s\mathbf{c}_{t,b}$ by a constant times $\|\mathbf{a}_{t,b}\|_\lambda + |s|\|\mathbf{c}_{t,b}\|_\lambda$. This is integrable by eq. 4.cxii and eq. 4.cxiv. On a smaller source disc the mixed Cauchy estimate similarly dominates its source derivative. Consequently the actual path integral is holomorphic there and may be differentiated under the integral.

The actual interaction-leg term carrying bond b is

$$E_s^\bullet(X,b)=\int_0^1 DE_X(\mathbf{V}_t(s))[\mathbf{a}_{t,b}(s)]\,dt. \tag{4.cxv}$$

Its source derivative contains two terms:

$$\dot E^\bullet(X,b)=\int_0^1 \Big\{D^2E_X(\mathbf{V}_t)[\mathbf{a}_{t,b},\mathbf{v}_t]+DE_X(\mathbf{V}_t)[\mathbf{c}_{t,b}]\Big\}\,dt. \tag{4.cxvi}$$

The first is the connected root-plus-source term. The second is the direct change of the leg insertion. Neither can be dropped. eq. 4.cx, eq. 4.cxii and eq. 4.cxiv prove

$$|\dot E^\bullet(X,b)|\le C_\star\left(\frac{16AC_v}{w_\star^2}+\frac{2C_c}{w_\star}\right)\varepsilon e^{-\lambda_{\mathrm{out}}d(X)}. \tag{4.cxvii}$$

It is zero unless $b^\sharp\subset X$.

Here is the exact normalized-expectation interpretation on a complete physical pair $P=p_W$. Write $\langle\cdot\rangle_t$ for the normalized, possibly complex, fixed-reference integral with interaction $\sum_Y V_t(Y)$. The denominator is nonzero by the same ordinary expansion. With $a_{t,b}=\sum_Y a_{t,b}(Y)$, $v_t=\sum_Y v_t(Y)$ and $c_{t,b}=\sum_Y c_{t,b}(Y)$,

$$\partial_s\langle a_{t,b}(s)\rangle_{t,s}\big|_{s=0}=\langle c_{t,b}\rangle_t+\langle a_{t,b}v_t\rangle_t-\langle a_{t,b}\rangle_t\langle v_t\rangle_t. \tag{4.cxviii}$$

This follows by differentiating its numerator and denominator. It requires no positivity. On these complete data, summing eq. 4.cxvi over X gives eq. 4.cxviii by the physical sum rule for the actual connected expansion. The disconnected product cancels; an unnormalized insertion would give the wrong formula and could carry a volume factor. On an arbitrary independent-current core, eq. 4.cxvi remains the local derivative identity and bound. It is not identified there with a covariance under an unavailable global Gaussian measure.

The older-weight path and its field room. An older weight enters through its fluctuation difference $v(Y;P,B)$, so $v(Y;P,0)=0$. Its interaction path is $v_t(Y;P,B)=v(Y;P,tB)$, and

$$c_{t,b}(Y;P,B)=\sum_{a=1}^{3}B_b^a\partial_{B_b^a}v(Y;P,tB). \tag{4.cxix}$$

The intrinsic path $t^{-2}V^{G,A}(tB)$ belongs to the terms with the action or gauge-fixing inverse-coupling coefficient. It must not be applied to this older weight. In particular, no artificial vanishing of its linear and quadratic fluctuation terms is assumed.

We spell out sufficient field room for eq. 4.cxiv. Suppose v is holomorphic and obeys eq. 4.cvii with norm ε on $\mathcal{B}_1$, for the actual external pairs and decoupling parameters. We prove the bound on the entire $\mathcal{B}_0$ required by eq. 4.cviii, not merely on real integration points. For real $t\in[0,1]$, vary one coordinate at tB by a complex number of radius

$$r_b=c_\nu(1+|B_b^a|),$$

where $c_\nu>0$ is a sufficiently small fixed fraction of the available room and the fixed color norm. This disc lies in $\mathcal{B}_1$, uniformly for $B\in\mathcal{B}_0$ and in t,b and the number of variables. Moreover its growth factor is bounded by $C_\nu Q_Y(B)$: the one changed coordinate adds at most $C_\nu(1+|B_b^a|^2)$ to the sum defining Q_Y, with M fixed. Cauchy's formula therefore gives

$$|B_b^a\partial_{B_b^a}v(Y;P,tB)|\le C_\nu\varepsilon e^{-\lambda d(Y)}Q_Y(B). \tag{4.cxx}$$

Also $Q_Y(tB)\le Q_Y(B)$, so $\|\mathbf{v}_t\|_\lambda\le\varepsilon$ on the actual cutoff box. Summing the three color coordinates in eq. 4.cxix proves eq. 4.cxiv. The original B is used in this majorant, so no $1/t$ or $\mathfrak{b}_0^2$ loss occurs as t tends to zero. Thus the differentiated root is a potential on exactly $\mathcal{B}_0$, while the undifferentiated source used its one fixed larger box.

A complex disc for a common radial parameter alone would not justify eq. 4.cxx. The individual bond legs require the independent-coordinate field room just used. Nor does an endpoint inserted-potential norm bound the variation of an arbitrary interpolation path. The prescribed path and its ordinary bounds in eq. 4.cxii remain necessary.

Midpoint tents, entropy, and the exact sum rule. Use the actual geometric tents at bond midpoints. The collars in the union filing are assumed large enough that $h_z(b) \neq 0$ implies $z \in b^\sharp \subset X$ for a nonzero bond-root term. The fixed block ratio L gives

$$\sum_{z\in X} h_z(b) = 1, \qquad \sum_b h_z(b) \leq 4L^4. \tag{4.cxxi}$$

These geometric identities are source independent. The renewed pointed source family is therefore

$$\dot{F}(X,z) = \sum_b h_z(b)\dot{E}^\bullet(X,b), \qquad \sup_{X,z} e^{\lambda_{\text{out}} d(X)} |\dot{F}(X,z)| \leq C\varepsilon. \tag{4.cxxii}$$

The constant depends on L, the fixed ordinary allowances, A and the field room, and is independent of volume and field-variable count. No additional source-position factor is spent in this supremum norm.

For completeness, the geometric root cost also remains harmless when an application sums domains through z. If that count at length m is at most $C_D e^{\kappa_D m}$ and $\lambda_{\text{out}} > \kappa_D$, then a smaller output rate costs only

$$4L^4 C\varepsilon \sum_{m\geq 0} (m+1)^q e^{-(\lambda_{\text{out}}-\kappa_D)m} < \infty$$

for every fixed polynomial power q. The two-variable Cauchy estimate has already summed all cluster multiplicities and choices of source domains through the ordinary rooted majorant. No second unsummed count of them is justified or needed. Additional geometric position sums cost a polynomial only when the actual domain geometry supplies that bound. This does not permit unrecorded extra domain types or an exponent loss larger than the available gap.

Finally sum eq. 4.cxv over b at a fixed localization domain X. The chain rule and eq. 4.cxi give $E_X(\mathbf{V}_1(s)) - E_X(\mathbf{V}_0(s))$. Because $\mathbf{v}_0 = 0$, its source derivative is $DE_X(\mathbf{V}_1)[\mathbf{v}_1]$. On every declared local core, eq. 4.cxxi therefore proves the algebraic identity

$$\sum_{z\in X} \dot{F}(X,z) = DE_X(\mathbf{V}_1)[\mathbf{v}_1].$$

At a complete physical pair p_W, the physical Gaussian sum rule then gives

$$\sum_{X,z} \dot{F}(X,z) = \sum_X DE_X(\mathbf{V}_1)[\mathbf{v}_1] = \partial_s \log \Xi(\mathbf{V}_1 + s\mathbf{v}_1)\big|_0. \tag{4.cxxiii}$$

The same formula is valid on the permitted analytic continuation of the physical datum W. This is not continuation from the physical graph to an arbitrary independent current. All sums are finite-volume sums followed by absolutely convergent cluster sums. Coherent infinite-volume extensions of the local families use the same uniform local bounds and companion limits, when supplied.

If the actual reference, union filing and interpolation paths are covariant under lattice isometries and a constant source has that same covariance, these identities also hold after transporting X, b, P. The midpoint tent then gives $\dot{F}(\mathsf{r}X, \mathsf{r}z; \mathsf{r}P) = \dot{F}(X,z;P)$. The same reasoning preserves gauge invariance and reality. These properties are inherited from the actual construction, not inferred from a scalar norm estimate. A nonconstant source does not acquire Euclidean covariance merely from eq. 4.cxxii.

If the source entered a Gaussian structure or the cutoff leg, additional derivatives would be required. With a fixed reference, a cutoff root is in general a boundary functional $N_{t,b}$, not the bounded function $\partial_t\chi/\chi$. Its differentiated normalized density is $N_{t,b}(ve^V)/\Xi - (N_{t,b}(e^V)/\Xi)\langle v\rangle$, plus any direct variation of $N_{t,b}$. If the reference measure also varies, its score contributes to the

denominator derivative as well. The retained Gaussian tail would have to be proved under insertion. The fixed-reference older-weight presentation avoids those additional terms here.

The conclusion is one finite-step pointed insertion estimate. It does not differentiate a full history class, prove closure of the response under iteration, supply the spatial defect gain of a nonconstant profile, or establish the clean-history Gaussian disintegration needed for an interacting two-point law.

The three legs and the normalized intrinsic path. We make the finite structural issue in eq. 4.cxii explicit. The free potential is a compensation of the fixed gauge quadratic form. If $Q^G = \sum_Y Q^G_Y$ and the forms are symmetric in the free real coordinates, put

$$q_b(Y;B) = \tfrac{1}{2}B(b)\cdot(Q^G_Y B)(b), \qquad \sum_b q_b(Y;B) = Q^G_Y(B). \tag{4.cxxiv}$$

The dot product here is bilinear, so the formula has its holomorphic meaning on a complex field box. When the actual row bounds give $\|\mathbf{q}_b\|_{\lambda_V} \le A_G$, the path $-u(1-\tau)Q^G$, $0 \le u \le 1$, has root $-(1-\tau)q_b$. The inserted-potential estimate gives its pointed bound as soon as this globally defined free family and its path fit in the ordinary potential budget. In particular one needs the actual smallness of $|1-\tau|A_G$; this is not a statement for arbitrary positive τ. The gauge form has not been multiplied by the variable action weight.

The cutoff leg is next, before the intrinsic and older potentials have been turned on. Its only interaction is this globally defined quadratic free potential. Each sharp test is brought from infinity to its final position. Its boundary estimate is proved below.

The third leg turns on the Jacobian and older functions by $V_Y(tB)$. The gauge and action remainders are normalized differently:

$$V^{\mathrm{int}}_{t,Y}(B) = t^{-2}V_Y(tB), \qquad V_Y(0) = DV_Y(0) = D^2V_Y(0) = 0. \tag{4.cxxv}$$

The three vanishings concern the intrinsic remainder after its quadratic part has been put in the free Gaussian. Its actual bond derivative includes the derivative of the factor t^{-2}. The diagonal path in eq. 4.cxxv alone does not specify how that derivative is distributed among bonds.

Here is a fully specified sufficient convention. It is a construction of an admissible local pointing; its identification with any separately prescribed pointing remains an exact finite structural question. For every intrinsic domain Y, let $\mathfrak{B}_Y$ be its actual free bonds and choose fixed, field-independent numbers

$$\vartheta_{Y,b} \ge 0, \quad \vartheta_{Y,b} = 0 \ (b \notin \mathfrak{B}_Y), \quad \sum_{b\in\mathfrak{B}_Y} \vartheta_{Y,b} = 1. \tag{4.cxxvi}$$

The choices are covariant under the geometric symmetries of the assigned domains. Equal shares are possible on any nonempty such domain. If there are no free variables, the intrinsic member with the vanishings in eq. 4.cxxv is zero and is omitted. Define

$$\begin{aligned} T_Y(\mathbf{t}) &= \sum_b \vartheta_{Y,b}t_b, \\ \widetilde{V}_Y(\mathbf{t};B) &= T_Y(\mathbf{t})^{-2}V_Y((t_bB(b))_b), \\ a_{t,b}(Y;B) &= t^{-2}B(b)\cdot D_bV_Y(tB) - 2\vartheta_{Y,b}t^{-3}V_Y(tB). \end{aligned} \tag{4.cxxvii}$$

At $\mathbf{t} = t\mathbf{1}$, the second line is exactly eq. 4.cxxv. Differentiating there gives the third line, including its normalization term. Summing over bonds gives $\sum_b a_{t,b}(Y;B) = \partial_t[t^{-2}V_Y(tB)]$. This is an independent-bond interpolation formula; no replacement by an Euler pointing has been made.

We prove its uniform norm and domain assertions. Use the nested boxes in eq. 4.cvii, and suppose that V_Y is holomorphic on the larger one with potential norm at most ε. Fix B in the smaller complex box. A fixed radial disc $|z| < 1 + c_\nu$ maps zB into the larger box. On that disc $Q_Y(zB) \le C_\nu Q_Y(B)$. The function $z \mapsto V_Y(zB)$ has a zero of order at least three. The maximum

principle applied to $V_Y(zB)/z^3$ therefore gives

$$|V_Y(tB)| \le C_\nu \varepsilon e^{-\lambda_V d(Y)} t^3 Q_Y(B), \qquad 0 \le t \le 1. \tag{4.cxxviii}$$

To bound a coordinate derivative, leave a fixed further margin in the radial disc and vary one color coordinate by a complex w with $|w| < c_\nu(1 + |B_b^a|)$. The resulting point is still in the larger box, and $Q_Y(zB + we_{b,a}) \le C_\nu Q_Y(B)$. Coordinate Cauchy gives a bound $C_\nu \varepsilon e^{-\lambda_V d(Y)} Q_Y(B)$ on $B_b^a D_{b,a} V_Y(zB)$, uniformly on a fixed disc of radius greater than one. This last function of z vanishes to order at least two. Applying the same maximum-principle argument and summing the three colors proves

$$|B(b) \cdot D_b V_Y(tB)| \le C_\nu \varepsilon e^{-\lambda_V d(Y)} t^2 Q_Y(B). \tag{4.cxxix}$$

Consequently every summand in the last line of eq. 4.cxxvii is bounded in the original potential norm, uniformly down to $t = 0$, and

$$\sup_b \int_0^1 \|\mathbf{a}_{t,b}\|_{\lambda_V}\, dt \le C_\nu \varepsilon. \tag{4.cxxx}$$

The apparent negative powers of t have been canceled by the actual orders of vanishing.

The independent parameters have a common admitted neighborhood at every $t > 0$. Choose $c_\nu > 0$ smaller if necessary and take $\max_b |t_b - t| < c_\nu t$. Nonnegative shares imply $|T_Y(\mathbf{t}) - t| < c_\nu t$, so $|T_Y(\mathbf{t})| \ge t/2$. The scaled coordinates remain in the larger fluctuation box. Applying the radial argument to $(t_b/t)B(b)$ gives $\|\widetilde{\mathbf{V}}(\mathbf{t})\|_{\lambda_V} \le C_\nu t\varepsilon$. Thus the potential path also stays inside its accepted slot if its initial allowance is chosen small enough. There is no singular parameter domain being asserted at $t = 0$; the integrable diagonal limit is the one just proved.

Each density is zero unless b belongs to its actual domain. Its root therefore survives the actual union filing. Fixed covariant shares introduce neither field dependence nor a new read outside that domain. For Jacobian and older families the unnormalized path $V_Y(tB)$ uses only the coordinate Cauchy bound in eq. 4.cxx; no third-order assumption is needed for its bond derivative.

For a separately prescribed independent-bond convention, write its normalization density as $R_b(Y; B)$. The same proof applies if its exact derivative is $t^{-2}B(b) \cdot D_b V_Y(tB) - 2t^{-3} R_b(Y; tB)$, with $\sum_b R_b = V_Y$, the same support and norm bound, and third-order vanishing of each R_b. Those are concrete finite identities and estimates to verify. They do not follow from the single equality $\sum_b a_{t,b} = \partial_t V_t$: arbitrarily large terms with zero bond sum would leave that equality unchanged. In particular, the construction in eq. 4.cxxvii does not silently identify inherited coefficients, own scalar readouts or operation identities with a different convention.

The moving sharp cutoff has an integrable rooted tail. We next prove the boundary estimate for the actual cutoff leg. The cutoffs act on three real coordinates of each tested free bond, with the Euclidean Lie-algebra norm. A cutoff on a further linear image would require its corresponding uniformly nondegenerate marginal bound; it is not identified with a free coordinate by name. Put

$$\chi_t(B(b)) = \mathbf{1}_{\{t|B(b)|<\mathfrak{b}\}}, \qquad 0 \le t \le 1, \quad \mathfrak{b} \ge 1. \tag{4.cxxxi}$$

At zero the cutoff is one. As a measure derivative for $t > 0$, $-\partial_t \chi_t = |B(b)|\delta(\mathfrak{b} - t|B(b)|)$. For every fixed $B(b)$ away from the final boundary its integral over $0 < t < 1$ is exactly $\mathbf{1}_{\{|B(b)|>\mathfrak{b}\}}$. This is the one-dimensional change of variables $u = t|B(b)|$. It proves the total-variation identity without dividing by a characteristic function.

For use with complex Gaussian matrices we require a uniform majorant before integrating in t. Let D be the local covariance after the auxiliary coordinates have been integrated out as in eq. 4.l. Its real reference and perturbation obey the same uniform gap bounds as there. If $|F(B)| \le F_0 e^{\rho|B|^2}$, define

$$S = \mathrm{Re}(D^{-1}) - 2\rho I, \qquad Z_{\mathrm{abs},\rho} = |\det D|^{-1/2} (\det S)^{-1/2}. \tag{4.cxxxii}$$

For the permitted ρ and perturbation size, $S>0$, both S and S^{-1} have uniform operator bounds, and

$$|d\mu_D(B)|\, e^{\rho|B|^2} = Z_{\mathrm{abs},\rho}\, d\mu_{S^{-1}}(B), \qquad Z_{\mathrm{abs},\rho} \le e^{A_K(\widehat{\varepsilon}+\rho)m}. \tag{4.cxxxiii}$$

This follows by writing the density on the real contour. The bound is the normalized local comparison bound, with the local scalar-coordinate dimension m. No factor depending on the auxiliary dimension has appeared.

Let P be a set of q other distinct failed free-bond tests. For a sufficiently small fixed $\gamma>0$,

$$\prod_{a\in P} \mathbf{1}_{\{|B(a)|\ge \mathfrak{b}\}} \le e^{-\gamma q\mathfrak{b}^2/2} \exp\{\tfrac{\gamma}{2}\sum_{a\in P}|B(a)|^2\}. \tag{4.cxxxiv}$$

Writing Π_P for the orthogonal projection onto their $3q$ real coordinates, the new real precision is $S-\gamma\Pi_P$. Choose $\gamma\|S^{-1}\|\le 1/2$. It is positive, its covariance has uniform upper and lower bounds, and its normalization ratio is at most $e^{C_K\gamma q}$. To verify the last claim, diagonalize $\gamma S^{-1/2}\Pi_P S^{-1/2}$. There are at most $3q$ nonzero eigenvalues, each at most $1/2$, and $-\log(1-u)\le 2u$ on that interval. Thus the logarithm of the ratio is at most $3K'\gamma q$. This cost counts failed coordinates, rather than all coordinates outside the local Gaussian.

The marginal of the root block $B(b)$ in this tilted normalized real Gaussian is three dimensional, even if the full local dimension is large. Its radial density is bounded by $C_K u^2 e^{-c_K u^2}$. Indeed its covariance is a principal block of the full covariance, with the same uniform eigenvalue bounds. The usual three-dimensional Gaussian density is therefore at most $C_K e^{-c_K|x|^2}$; integration over the sphere of radius u gives the assertion. Consequently, uniformly in all admitted complex matrix parameters,

$$\begin{aligned}\Big|\mathcal{J}\Big[|B(b)|\delta(\mathfrak{b}-t|B(b)|)\,F(B)\prod_{a\in P}\mathbf{1}_{\{|B(a)|\ge\mathfrak{b}\}}\Big]\Big| \\ \le C_K F_0 e^{A_K(\widehat{\varepsilon}+\rho)m + C_K\gamma q} e^{-\gamma q\mathfrak{b}^2/2}\mathfrak{b}^3 t^{-4} e^{-c_K\mathfrak{b}^2/t^2}.\end{aligned} \tag{4.cxxxv}$$

This display uses the auxiliary-integrated local functional. For a boundary distribution it follows either by polar coordinates in $B(b)$, or by first using a thin positive shell in that coordinate and taking its limit. The Gaussian density and the free-potential factors are smooth; the remaining cutoffs are bounded Borel functions. Fubini and the displayed integrable majorant justify that limit.

The singular-looking factor is integrable. The substitution $u=\mathfrak{b}/t$ gives

$$\int_0^1 \mathfrak{b}^3 t^{-4} e^{-c_K\mathfrak{b}^2/t^2}\,dt = \int_{\mathfrak{b}}^\infty u^2 e^{-c_K u^2}\,du \le C_K e^{-\gamma\mathfrak{b}^2/2}, \tag{4.cxxxvi}$$

after reducing γ if necessary. For example, bound $u^2e^{-c_Ku^2/2}$ by its finite maximum and integrate the remaining Gaussian tail; with $\mathfrak{b}\ge 1$ the extra polynomial is absorbed by a still smaller fixed exponent. The bound in eq. 4.cxxxv is uniform before the t integration. It therefore also bounds an integral whose integrand is a supremum over the admitted matrix and external parameters. This constructs the boundary envelope with the supremum inside the integral.

We spell out its application to the literal finite expansion. During this leg only the globally defined quadratic free family is present, so its selected interaction domains require no small-field box Y_0. Expand $\prod_b \chi_{t_b}$ into failed tests. Differentiation in a tested bond b selects terms whose failed set contains b, replaces its factor $1-\chi_{t_b}$ by the boundary measure, and leaves every other failed factor at threshold at least $\mathfrak{b}$. Use eq. 4.cxxxv for that root. The independent coupling Schwarz estimates for the selected free-potential Mayer factors, the enlarged coarse-decoupling Cauchy estimates and the coarse collar count are exactly those used in eq. 4.lxxi–eq. 4.lxxxii. The root bond is part of the actual support $Z_0(\mathbf{D},P_0)$. Thus no remote root is created.

For precision, the Schwarz argument is a holomorphic one and never squares the boundary measure. At fixed $t > 0$, insert independent couplings λ_Y in the selected free-potential Mayer factors. The resulting boundary functional is holomorphic on the discs in eq. 4.lxx, with the unsquared boundary factor kept inside that linear functional. Indeed the larger Gaussian-only budget bounds its integrand by $2^{\#\mathbf{D}} e^{CW_0 L^4 N} e^{\rho|B|^2}$; eq. 4.cxxxv gives a common boundary majorant on those discs. Differentiation or a contour integral in the couplings is therefore justified by this same majorant. The functional vanishes when any selected $\lambda_Y = 0$. Apply the scalar Schwarz lemma successively in these variables, and then Cauchy's formula in the coarse parameters. Writing $r = \#\mathbf{D}$, $v = |P_0| \geq 1$, $h = |Z \setminus Z_0^\sharp|$, $p = e^{-\gamma \mathfrak{b}^2/2}$, and $g_{\mathrm{bd}}(t) = \mathfrak{b}^3 t^{-4} e^{-c_K \mathfrak{b}^2/t^2}$, this gives

$$\begin{aligned} |\partial_{t_b} T_{Z,\mathbf{D},P_0}| \leq C_K g_{\mathrm{bd}}(t) e^{C_0 N + C_K \gamma(v-1)} \left(\frac{2a}{W_0}\right)^r \\ \times e^{-(\lambda_V - \lambda_0)\sum_{Y \in \mathbf{D}} d_f(Y)} p^{v-1} (R_s - 1)^{-h}. \end{aligned} \tag{4.cxxxvii}$$

The zero tests and the boundary factor have no coupling derivative and are not subjected to an L^2 Cauchy–Schwarz inequality. The coarse parameters change the Gaussian matrices, not the real boundary coordinate. The formula in eq. 4.cxxxvi integrates g_{bd} to at most $C_K p$, restoring p^v in the unmarked estimate in eq. 4.lxxiv. The factor $e^{C_K \gamma(v-1)}$ costs at most e^{CN}, since $\gamma(LM)^4$ is bounded. Enlarge C_0 by that fixed constant before choosing K_*, Θ and the strict rates in eq. 4.lxxvi–eq. 4.lxxvii. Thus the same literal hull and product sums apply with an integrable root envelope throughout.

Allocate the same strict surplus of spatial and decoupling rates as in the unmarked proof. The normalization cost in eq. 4.cxxxiii is bounded per coarse cube when $(LM)^4(\widehat{\varepsilon} + \rho + \gamma)$ is bounded. The rank-tilt cost $e^{C_K \gamma q}$ is included in this allocation, since $q \leq C(LM)^4 \#Z$. The failed-test position sum has at most $C(LM)^4$ choices per coarse cube. Every selected cutoff term still has its root failure. In place of an unmarked linear amplitude this gives

$$\begin{aligned} p_{\mathrm{cut}} &= C_K (LM)^4 e^{\Theta} e^{-\gamma \mathfrak{b}^2/2}, \\ \int_0^1 \sup_P |\partial_{t_b} H_Z(\mathbf{t}; P)|_{\mathbf{t}=t\mathbf{1}}\, dt &\leq C p_{\mathrm{cut}} e^{-\lambda_2 (d_c(Z)+5)}. \end{aligned} \tag{4.cxxxviii}$$

Here Θ is the rate expenditure of eq. 4.lxxviii–eq. 4.lxxx; all physical smallness conditions include this factor. The supremum includes the local decoupling parameters needed in forming this activity. The estimate follows by the same nonnegative sum over selected domains and failed sites as before, with one site pinned to b. Pinning removes a position sum; any polynomial number of possible roots inside a domain is absorbed by the already retained strict rate gap. An empty choice cannot contribute a root.

Finally differentiate the absolutely convergent cluster series once. In each connected cluster mark the polymer that is differentiated. The rooted tree bound sums all its remaining polymers to at most $e^{a(Z)}$, where a is the size weight of the hard-core criterion. Sum its marked activity with eq. 4.cxxxviii, spending the strict difference between λ_2 and the retained output rate. Positivity of these majorants permits Tonelli in t and in the cluster sum. We obtain

$$\int_0^1 \sup_P |\partial_{t_b} E_X(\mathbf{t}; P)|_{\mathbf{t}=t\mathbf{1}}\, dt \leq C p_{\mathrm{cut}} e^{-\lambda_{\mathrm{out}} d(X)}, \quad \partial_{t_b} E_X = 0 \quad (b^\sharp \not\subset X). \tag{4.cxxxix}$$

At a fixed pair on X, only the hereditary subsystem of polymers contained in X is used. When the union condition is dropped in the positive bound, only the uniform local norm majorants are extended, not actual activity functions outside their declared cores. Its integral is the cutoff-leg density $E_\chi(X, b)$. Summing in b recovers the difference between the all-one and all-zero cutoff endpoints. Midpoint tents preserve its root as in eq. 4.cxxi–eq. 4.cxxiii.

All these estimates are local on their admitted independent-pair cores. The complete sum is identified with the physical cutoff integral only at p_W and its admitted continuation in W. No

global Gaussian on arbitrary independent currents, and no estimate for a history's outer stored-variable integral, has been used.

4.5.4. *Window-first ordinary formation.* The estimate in this section is an ordinary formation estimate. It concerns the actual functions obtained by rewriting old terms on box backgrounds, before their fine-cube bracket localization. The outer potential/current pair may have independent components. This causes no substitution of an arbitrary pair into a theorem about critical backgrounds: the inner window function first constructs its own constrained box background and its own current. The hypotheses below retain the domain and response assertions needed to perform that construction on the whole complex parameter set. We fix the window-first rule below as part of the localization: at one old base point, its near pointed sum is one window member before any fine bracket is applied.

The maps and geometric inputs. Fix a step from level k to level $k+1$. For an old level $j \le k$, put

$$a = k - j, \qquad \sigma_j = L^{-a}, \qquad \rho_j = \max\{1, \alpha_k/\alpha_j\}, \qquad s_k = \delta_k/\alpha_k. \tag{4.cxl}$$

The radii and the ordinary flow satisfy $\rho_j \le C(1+a)^{1/2}$. In particular

$$C_{\mathrm{age}} := \sup_{a\ge 0} C(1+a)^{1/2}L^{-a} < \infty, \qquad \rho_j\sigma_j \le C_{\mathrm{age}}. \tag{4.cxli}$$

Here δ_k is the tested fluctuation amplitude, whereas α_k is the analytic field radius. The ratio in eq. 4.cxl is an ordinary full-radius ratio. No small-field gain is being deduced by replacing a smaller ratio with one.

We spell out the ordinary background and geometry inputs. They must concern the actual assigned maps, not arbitrary functions with similar notation.

First, let P range over the local complex core used to form the new potentials. Let $\mathbf{H}_k(P; s, B')$ be the actual response to the tested free fluctuation coordinates, with the indicated fine-cube parameters. The plain nonlinear iterated block average is denoted $\mathcal{M}^j$. On a window $\square$, with a fixed partition function $\zeta_\square$, the datum used for an inner term is the restriction of

$$\mathcal{V}_{j,\square}(P; s, t, \nu) = M_{\mathbf{B}_j}\Big(e^{\,i\eta_k(t+\nu\zeta_\square)\mathbf{H}_k^{\mathrm{win}}(P;s,B')}\,U_+(P)\Big), \qquad 0 \le t \le 1. \tag{4.cxlii}$$

The notation $U_+(P)$ denotes the exponential representative of the upper box. The response $\mathbf{H}_k^{\mathrm{win}}$ is transported to that same frame. The original node frame is recovered at the two endpoints of each row. The factor η_k converts a potential in level-k units to a bare bond logarithm. The map $M_{\mathbf{B}_j}$ reads the ordinary iterated average at each determining row, at the level of that row. The formula in eq. 4.cxlii uses the nonlinear average. Replacing it by its differential would not give the window identity. The box-background map applied to this datum returns a pair

$$P^{\mathrm{box}}_{j,\square}(P; s, t, \nu) = \big(U^{\mathrm{box}}_{j,\square}(\mathcal{V}_{j,\square}), J^{\mathrm{box}}_{j,\square}(\mathcal{V}_{j,\square})\big) \tag{4.cxliii}$$

in which J^{box} is the current of that constrained critical configuration. The ordinary maps include their prescribed node and gauge representatives.

Second, all these maps are holomorphic on the full independent fine-parameter discs $|s_\Delta| < R = e^{\bar\kappa}$. The unamplified substitutions have a fixed positive amount of datum, chart and core room. If $r_{B,j}$ is the dimensionless datum radius, the background package supplies $\alpha_j/r_{B,j} \le C$. Its response and nonlinear-average estimates bound the change caused by the ν term, in every relevant normalized datum, chart and core norm, by

$$C_R|\nu|\,\delta_k\sigma_j/\alpha_j \le C_R|\nu|s_k\rho_j\sigma_j. \tag{4.cxliv}$$

The constant C_R includes the fixed loss from the enlarged fine-parameter discs. These are estimates for the actual nonlinear average and response on their assigned analytic domains. For example, in a datum norm the estimate follows by the uniform first-response bound, composed with the derivative of the actual average, and integration along the perturbation. For the inner terms it must also hold

for the other core inequalities and the tower substitutions. The outer and removed terms use their own directional estimates below. Existence and holomorphy on the corresponding datum balls are part of the background package.

4.5.4.1. Restriction and the two substitutions. Use the two fixed tower rules on the admitted localization domains, with inward rows removed. Here L is odd, $L \geq 13$, the geometric sizes are powers of L, $R_1 \geq L$, $M_1 \geq R_1\mathfrak{M}$, and $M \geq L^4 M_1$. The localized tower estimates use the separated-region floors and the fixed continuation and averaging radii of the background construction. In particular they use $\rho = L^{-2}R_1\mathfrak{M} \geq \mathfrak{r}^{\mathrm{s}}$, the last-level torus width of at least 6ρ sites, and the floors $\mathfrak{M} \geq \mathfrak{m}_{\mathcal{M}}$ and $R_1\mathfrak{M} \geq \max\{\mathfrak{r}_{\mathcal{M}}, \mathfrak{r}^{\mathrm{prop}}\}$. The corresponding decay rates satisfy

$$\delta_{0,\mathrm{red}} R_1\mathfrak{M} \geq 8(4\log L + \log 2), \qquad \mathring{\delta}_{0,\mathrm{b}} R_1\mathfrak{M} \geq 512 \log L, \qquad \delta_*^{\natural} R_1\mathfrak{M} \geq 18(4\log L + \log 2).$$

For nested localization domains $X \subset Y$ of levels $j-1$ and $j'-1$, set $\lambda = L^{-(j'-j)}$. Restriction from Y to X maps the core at radii (b_0, b_1, c_0) into the old core at radii $(\lambda b_0, \lambda b_1, \lambda^3 c_0)$. In the simple clauses of eq. 2.v, the curvature and the first difference of the potential gain λ^2, the potential gains λ, and the independent current gains λ^3. The tower of any admitted $Z \subset X$ is computed from the same bare product in both presentations. Its curvature and current, read at their own levels as in eq. 2.vii, inherit the ratio λ in the hereditary ceiling. If $j' = k+1$, $m = k+1-j \geq 1$ and $A_i \leq 2\sqrt{1+m}\, a_i$ for $i = 0, 1$, this restriction sends the radii $((1+2\beta_{\mathrm{s}})A_0, (1+2\beta_{\mathrm{s}})A_1, A_0)$ into $(a_0/3, a_1/3, a_0/3)$ for $0 < \beta_{\mathrm{s}} \leq 1/4$. Indeed $(1+2\beta_{\mathrm{s}})2\sqrt{1+m}\, L^{-m} \leq 1/3$.

For the outer substitution in level-j units, take the old triple in $\mathcal{U}_j^{\circ}(X; b_0/2, b_1/2, c_0)$, with $c_0 > 0$, $0 < b_0 \leq b_1$, $b_1^2 \leq c_{\natural} b_0$, $c_{\natural} \geq 1$ and $b_0 + b_1 \leq a^{\sharp}$. Choose $a^{\sharp} < a_T$ and below the fixed ceiling of Prop. 3.132. Take $K_{\mathrm{red}}^{\sharp} \geq K_T$ from Cor. 3.130, enlarged to cover the fixed substitution constants. If the potential shift satisfies

$$\max\{|\mathbf{A}|, |\nabla_{\mathbf{U}}\mathbf{A}|\} < \frac{b_0}{128 K_{\mathrm{red}}^{\sharp}(1+c_{\natural})}$$

on the internal bonds and differences, and $|\mathbf{J}'| < b_0$, the triple of $(e^{i\eta_j \mathbf{A}}\mathbf{U}, \mathbf{J}')$ obtained by eq. 2.ix lies in $\mathcal{U}_j^{\circ}(X; b_0, b_1, b_0)$. In fact, multiply the shift by z with $|z| \leq R = 4K_{\mathrm{red}}^{\sharp}(1+c_{\natural})$. Write $\beta_0 = b_0/32$. The shift and its first difference with transport by $\mathbf{U}$ have size at most β_0 on this disc. Transport by the real base U changes the difference by at most $\beta_0/4$. The local logarithm estimate gives bounds below b_1 for the potential and its first differences. In the plaquette product estimate, the change relative to the old plaquette is bounded by $\eta_j^2(2\beta_0 + 14\beta_0^2 + 4\eta_j\beta_0 b_0)$. Multiplying back by the old plaquette gives a change at most $3\beta_0\eta_j^2$, within the curvature margin. For a tower of level $p \leq j$ put $\lambda_p = L^{-(j-p)}$. Each normalized curvature or current readout of that tower is holomorphic in z and bounded on the disc by $K_{\mathrm{red}}^{\sharp}(1+c_{\natural})\lambda_p b_0$. Its value at zero has size less than $\lambda_p b_0/2$. Schwarz's lemma applied to the difference from its value at zero gives a change at $z = 1$ of at most $\lambda_p b_0/2$. This checks the hereditary clauses for both rules and every truncation. The real base is unchanged and the independent current is tested separately.

The inner window estimate starts with the flat-reference continuation $e^{i\mathsf{a}_0}$ of the upper window datum, in the node frame of the window. For a fluctuation $\mathbf{A}$ in level-k units put

$$\mathsf{a} = \frac{1}{i}\log(e^{iL^{-k}\mathbf{A}}e^{i\mathsf{a}_0}), \qquad B(c) = \frac{1}{i}\log \mathcal{M}^q(e^{i\mathsf{a}})(c), \quad c \in \Gamma_q.$$

Here Γ_q is the level-q row set of the old truncated window tower. The actual nonlinear average is used. If B_0 is the same datum at $\mathbf{A} = 0$, the logarithm bound and holomorphy of the average on its absolute row disc give $|B(c) - B_0(c)| \leq (9/2)C_Q L^{q-k} \sup|\mathbf{A}|$. Indeed the bond logarithm changes by at most $(9/8)L^{-k}|\mathbf{A}|$, and the Schwarz estimate on the line joining the two bond logarithms costs $4C_Q L^q$. The bound holds for complex fluctuations in the admitted ball. This also controls the whole datum with its actual amplitude. Write $d = A_0 + \sup|\mathbf{A}|$, where A_0 is the upper curvature radius on the complete box. The upper comb estimate gives $\|B_V\|_{\infty} \leq C_M A_0$. The restriction

identity and the weighted response of the flat upper background then give $\|B_0\|_\infty \le C_M A_0$ on every row of the old tower. Since $q \le j \le k$, the preceding increment estimate yields

$$\|B\|_\infty \le C_M d, \qquad \|B\|_{\mathrm{central}} \le C_M d L^{j-k}. \tag{4.cxlv}$$

The second bound uses that every central row has level j; its upper background contribution has the stronger factor $L^{j-(k+1)}$. These constants depend on the fixed geometry, not on either depth. The same first bound holds after the cutoff axial change: its comb has $O_M(L^{k-j})$ links of size $O_M(dL^{j-k})$, hence its logarithm has size $O_M(d)$. Its cutoff extension has that same supremum, and the endpoint product on every row stays within $C_M d$ of the identity. Lower rows outside its support retain their original values. For the central axial bound require, in addition, the complete upper field's first-difference bound of size $C_M d$ on the settled layer, as in the window theorem. Its level-j plaquette defects are consequently at most $C_M d L^{2(j-k)}$. An axial link is a product of at most $C_M L^{k-j}$ such plaquette factors, with bounded transports. It is therefore within $C_M d L^{j-k}$ of the identity on the central carrier, where the cutoff equals one and the complete comb rectangles remain in the settled layer. This proves the central bound after the axial change; it uses that first-difference estimate as well as the datum bound. On the outer cutoff transition only the global $C_M d$ bound is asserted here. Thus the remote response has the proportional factor d; the fixed continuation ceiling supplies only its domain. An independent correction B_{err} contributes $C\|B_{\mathrm{err}}\|_\infty$ separately. In particular, the auxiliary correction ball used for Cauchy estimates does not change the amplitude of the actual evaluation, where $B_{\mathrm{err}} = 0$.

The window continuation at the datum $\tau B + B_{\mathrm{err}}$, $0 \le \tau \le 1$, gives the old curvature, potential and current margins $(a_0/4, a_1/2, a_0/4)$ in its normalized core. Use the actual upper input and literal-datum convention of Prop. 3.135. Choose $M_\flat \ge M_0$, $a^\flat < a_I(M)$ and $2B_3^2\mathsf{K} \ge C_I$ from that proposition, and take its increment constants c_2, c_3 or smaller ones. The required bounds are $M \ge M_\flat$, $A_0 + A_1 \le a^\flat$, $a_0 + a_1 \le a^\flat$, $2B_3^2\mathsf{K}Ma_0 < a_1$, and $A_0 \le 2\sqrt{1+m}\, a_0$. Its fluctuation has size below $c_2 A_0$ and the complex correction has raw norm below $c_3 a_0$. These are local input bounds on the entire window. In eq. 4.cxlii, the fluctuation includes $(t + \nu\zeta_\square)\mathbf{H}_k$ in the same node frame. To use the window estimate, identify its upper continuation and determining rows with the ones in that formula. Check the bounds on the transported fluctuation and any correction on the full retained ν-disc and all independent fine-parameter discs. In particular, if the transported response has $\|\mathbf{H}_k\|_\infty \le C_R\delta_k$ on the fine-parameter discs, then the window input on $|\nu| < c_\nu/s_k$ is at most $C_R(s_k + c_\nu\|\zeta_\square\|_\infty)\alpha_k$. When α_k/A_0 is bounded, small s_k and c_ν make this less than $c_2 A_0$. The same row derivative bound on the admitted window gives a datum increment of size $C|\nu|L^{q-k}\delta_k$ relative to $\nu = 0$. This is the datum estimate in eq. 4.cxliv after dividing by the old radius. Its use for the actual maps requires the comparison just described. The other core readouts must also obey their bounds throughout the disc. The outer estimate starts with a shift already bounded in its required norm. The removed substitution likewise requires its own joint background and current bounds.

4.5.4.2. The window at the radii of the run. Let x_i be the inverse couplings and use the running radii $\alpha_\iota(x) = C_\iota x^{-1/2}(\log x)^q$, $\iota = 0, 1$. For $m = k + 1 - j$ set $A_\iota = (1 + \beta_{\mathrm{s}})\alpha_\iota(x_k)$ and $a_\iota = \min\{\alpha_\iota(x_{j-1}), \alpha_\iota(x_j)\}$. The drift bounds are $-c_5 \le x_i - x_k \le \bar{B}(k-i)$. If $x_i \ge x_k$, the logarithmic factor in $\alpha_\iota(x_k)/\alpha_\iota(x_i)$ is at most one, and the remaining factor is at most $\sqrt{1+k-i}$ by $\bar{B}\gamma^2 \le 1$ and $x_k \ge \gamma^{-2}$. If $x_i < x_k$, the logarithmic ratio is at most $1 + c_5\gamma^2$, using $\log x_i \ge 1$. The choice $(1 + c_5\gamma^2)^q \le 1 + \beta_{\mathrm{s}}$ therefore gives $A_\iota \le 2\sqrt{1+m}\, a_\iota$. Both minima are attained at the same index, so $a_1/a_0 = C_1/C_0$. The running ceiling gives $a_1 \le 1$ and both radius sums at most $a^\flat \le a^\sharp$. Thus $a_1^2 \le (C_1/C_0)a_0$, and the choice of C_1/C_0 gives $2B_3^2\mathsf{K}Ma_0 < a_1$.

For the inner substitution take the window of 2^4 level-k cubes with a common vertex and put $\square_0 = \square^{\sim 5}$. Use the rule-$\mathfrak{r}_1$ tower of $\square_0$ to depth $k+1$ and its truncation to depth j, with the same convention on inward rows at both depths. The upper determining data $M_{\mathbf{B}^+}(\mathbf{U})$ on the first tower

have a comb representative V and a node gauge v, with

$$M_{\mathbf{B}^+}(\mathbf{U}) = V^v, \qquad |V-1| < K_\square M \alpha_0^+, \qquad |v|\,|v^{-1}| < e^{K_\square M \alpha_1^+}, \qquad \alpha_\iota^+ = (1+2\beta_{\mathrm{s}})A_\iota.$$

The comb construction uses the upper bare plaquette and potential bounds and the $\mathfrak{r}_1$ tower clause for this box. It also uses the fixed smallness ceiling on $(LM)^4(\alpha_0^+ + \alpha_1^+)$. The representative is holomorphic at fixed real base. Its node gauge extends by the tower's prescribed site rule. The tower at the actual upper datum is the extended gauge transform of the flat-reference continuation at $B_V = i^{-1}\log V$.

4.5.4.3. The ordinary response and its current. At the ordinary step on the constant sequence, the raw source of the response is the corrected fluctuation logarithm at level k. In the free variables of eq. 4.cxcii it is $B' = z - h_W d$. The constant-sequence datum conversion is the identity. On the tested fluctuation box the contraction gives $\|B'\|_\infty \le C\delta_k$. On each admitted local restriction use the same source on the corresponding rows, extended by zero outside its support. For the assigned response settings take $C\delta_k < \varepsilon_{\mathrm{D}} e^{-64\kappa}$.

Write $\mathbf{H}_k$ for the potential of the decoupled response and j_k for its increment of current, both in level-k units. The bounds for the solution on the independent parameter discs give

$$\|\mathbf{H}_k\|_\infty + \|\nabla_{\mathbf{U}}\mathbf{H}_k\|_\infty + \|\mathsf{j}_k\|_\infty \le C_R \|B'\|_\infty.$$

For the current use the difference estimate between B' and zero. At a local layer these are bounds in that layer's units; the assertion here uses the constant sequence. The potential and current maps are holomorphic at fixed real base and have the same restriction law on the interior of active wall components, with their own support collars. Outside the components retain the explicit current of the reduced system. The current estimate is for the current from the decoupled equations. A second difference of a decoupled potential cannot be estimated from the first-difference bound.

The weighted estimate for the current follows in the same fixed pair and tower, with the switches held fixed. For two sources B_1', B_2', let $\mathbf{H}_i$ be their potentials and use the multiplier of eq. 3.mccxviii. Put

$$Z_s = \widehat{Q}^\sharp H_s^\sharp \mathbf{J}, \qquad K_i = (\widehat{Q} + C'_{\mathbf{H}_i})^\sharp \omega_s(\mathbf{H}_i), \qquad R_i = \mathsf{f}(-i\xi\,\mathrm{ad}_{\mathbf{H}_i})^{-1}.$$

The shifted current in Prop. 3.174 is $\mathbf{J}_i^\flat = \mathbf{J} - Z_s + R_i K_i$. Since Z_s is independent of the source, its difference is

$$\mathbf{J}_2^\flat - \mathbf{J}_1^\flat = R_2(K_2 - K_1) + (R_2 - R_1)K_1.$$

The local transpose bound and the multiplier estimate in eq. 3.mccxviii give a bounded unweighted current norm for K_i and the weighted Lipschitz bound for $K_2 - K_1$. Indeed, subtract the two multiplier formulas, use the resolvent identity for $\Psi_{s,\mathbf{H}_i}$, and then the first-difference bounds for $H_s^{\circledast}$ and $C'_{\mathbf{H}}$, as in the proof of eq. 3.mccxix. The inverse factors R_i are bounded on the common bond disc, and their difference costs $C\xi|\mathbf{H}_2 - \mathbf{H}_1|$. Thus the second term, in the current units at a site y, is at most $C\xi_y|\mathbf{H}_2 - \mathbf{H}_1|$, since $\xi \le \xi_y$. This is the weighted potential difference bound with the same spatial weight. Thus a source difference supported away from the full read has the same exponential gain in the current norm as in the potential norm, on the admitted complex switch disc. This applies to the cutoff change within one old box after including the stencils of its converted rows. The weighted Hölder difference of the potential on that same tower follows by using the local first-difference Hölder norm in the propagator estimate of the response theorem. It retains the same source difference and spatial weight. The identification with the physical current is at a real critical point at full switches. Its stencil is the native action's stencil before restriction to the read.

Consider the outer evaluation obtained by shifting the restricted pair by the ordinary response with multiplier λ. In level-j units its potential shift and current are

$$\mathbf{A}_{j,\lambda} = \lambda\sigma_j \mathbf{H}_k|_X, \qquad \mathbf{J}_{j,\lambda} = L^{-3(k+1-j)}\mathbf{J}_+|_X + \lambda\sigma_j^3 \mathsf{j}_k|_X,$$

where $\mathbf{J}_+$ is the input current in level-$(k+1)$ units. The shift in the bare exponential is unchanged because $\eta_j\sigma_j = \eta_k$. The covariant difference of the shift gains σ_j^2 since the transport uses the same bare bond field. If the upper triple is in the upper core used in the two-level restriction, the first current on the right is within $a_0/3$. The potential, its covariant first difference, and the current increment are all bounded by $C_R|\lambda|\sigma_j\delta_k$. On $|\lambda| < c\,a_0/(C_R\sigma_j\delta_k)$, with c chosen below the explicit outer shift ceiling and the remaining current margin, the outer substitution therefore belongs to the old core. The running radius comparison makes this disc contain $|\lambda| < c_\nu/s_k$ for a fixed small c_ν. This checks the outer shift that enters the ordinary bracket of the full response. The extra distance gain in a window derivative, and the two differences of responses in a removed term, require estimates for those distinct shifts on their full carriers.

For real physical data with all fine parameters one, the same response satisfies the exact constraint and slice equations. Its potential produces the critical configuration at the prescribed datum up to the node factors, and its completed current is the actual current of that configuration. Thus the two endpoints of the ordinary interpolation are the physical backgrounds, after the stated node normalization. At intermediate multipliers the pair can be off the critical graph. The current slot in the displayed formula is retained there.

The upper frame on the full window box. Let $\mathbf{B}^+$ be the upper tower of $\square_0$ and $\mathbf{B}_j$ its depth-j truncation, with the same convention on inward bonds. The bracket domain Y contains $\square_0$. The upper input is in the ordinary core at $((1+2\beta_{\mathrm{s}})A_0, (1+2\beta_{\mathrm{s}})A_1, A_0)$; its hereditary clause supplies the tower of $\square_0$ and its truncations used in the comb construction. The window uses the upper tower configuration at its comb datum V^v. Put $B_V = i^{-1}\log V$. The comb estimate gives $\|B_V\|_\infty \le CMA_0$ and $\mathsf{k}(v) \le e^{CMA_1}$ at every node, where $\mathsf{k}(g) = \|g\|\|g^{-1}\|$. Its prescribed extension $\tilde{v}$ has the same bound at every site.

Apply the flat-reference Landau pair to the continuation $\Phi_{\mathbf{B}^+,\mathbf{1}}(B_V)$ on a fixed datum disc inside the continuation and response radii. The identity holds on every inner bare bond of $\square_0$:

$$\Phi_{\mathbf{B}^+,\mathbf{1}}(B_V) = \big(e^{i\eta_{k+1}\mathbf{H}_{k+1}(B_V)}\big)^{u_{B_V}}.$$

The derivative bound in this theorem has one inverse local length ξ_y^{-1} for the potential on the pillar at y. Integrate it on the line from 0 to B_V. Each top is an endpoint of a bounded number of determining rows, and the exponential sum over tops is uniform. Consequently

$$\sup_y \xi_y \sup_{b \text{ over } y} |\mathbf{H}_{k+1}(B_V)(b)| \le C\|B_V\|_\infty.$$

In the ascent-gauge lemma the form $\mathbf{H}_{k+1}$ therefore lies in its weighted small ball, by weak coupling. The gauge factor of the Landau pair equals $h[\mathbf{H}_{k+1}]$ of that lemma. At real data the two factors have the same prescribed values at all tops and both satisfy the ascent identities, which determine their remaining values. The equality continues holomorphically on the datum disc. In particular $\mathsf{k}(u_{B_V}) < 3/2$ on the whole box. Thus the upper tower configuration equals $E^{g_\square}$, where

$$E = e^{i\eta_{k+1}\mathbf{H}_{k+1}(B_V)}, \qquad g_\square = \tilde{v}\,u_{B_V}, \qquad \sup_{\square_0} \mathsf{k}(g_\square) \le C.$$

Only the local-length bound for the background potential is used here. A uniform upper-level bound for that potential on the finest layers is unnecessary.

Use the assigned ordinary decoupled response on its own admitted active region, then restrict it to $\square_0$ using the full response reads. The fine parameters vanish off the active cells. The pivot-completed datum B' is supported at level k, with $\|B'\|_\infty \le C\delta_k$. The constant-sequence bound above uses local lengths one. On an admitted layered response chart the support estimate on the fine-parameter discs gives

$$\xi_y \sup_{b \text{ over } y} |\mathbf{H}_k(P; s, B')(b)| \le C\delta_k e^{-\theta d_{\mathbf{B}}(y,\Sigma_k)}, \qquad \theta = \mathring{\delta}_0/8.$$

Here Σ_k consists of the tops at which the datum is supported. Their local lengths in level-k units are at least L^{-1}, by the geometry of the level-k rows. If there is no such top the response is zero. The scale comparison in the same separated sequence gives $\xi_{y'}/\xi_y \le Le^{(\mathring{\delta}_0/16)d_{\mathbf{B}}(y,y')}$. Take y' in Σ_k at minimum distance. The loss from dividing by ξ_y is therefore paid by the support exponential. This proves $\sup|\mathbf{H}_k(P;s,B')| \le C\delta_k$ in level-k units also on the fine layers of the box. The proof uses the support estimate for the whole datum, without summing the responses over its nonzero sites.

For an assigned shift in the upper tower frame, transport it by $\mathrm{Ad}(g_\square^{-1})$ at the initial site of each bond. Its bound remains $C\delta_k$ in the exponential frame on the whole box. A shift supplied directly in the exponential frame uses the same bound. The window condition $\|\mathbf{A}\|_\infty < c_2A_0$ then has fixed room, since $\delta_k/A_0 \to 0$. This comparison concerns the two representatives of the upper tower. The fluctuation originating at the input pair uses the relative frame below.

The rows in this frame. For $c \in \Gamma_q$ in the truncated tower, every bare bond of its full two-block carrier lies in Ω_{q-1} of the upper tower for $q \ge 1$. The local lengths there are at least $L^{q-1-(k+1)}$ in upper-level units. This is the carrier assertion of the tower lemma, applied to the truncated sequence, whose Ω_{q-1} is the same as in the upper tower. It yields

$$L^q \sup_{b \text{ in carrier}(c)} |\eta_{k+1}\mathbf{H}_{k+1}(b)| \le CLMA_0.$$

For $q = 0$ the corresponding estimate follows directly from the local-length bound at a bare bond. If $\mathbf{A} = (t+\nu\zeta_\square)\mathbf{H}_k^{\mathrm{win}}$, then $L^q \sup|\eta_k\mathbf{A}| \le CL^{q-k}(1+|\nu|\|\zeta_\square\|_\infty)\delta_k$. In particular all the row discs stay within the absolute average radius on $|\nu| < c_\nu/s_k$ for small fixed c_ν. On this disc the transported shift also satisfies $\|\mathbf{A}\|_\infty < c_2A_0$ on the whole box.

These checks identify the row of the actual shifted upper tower with the row of $e^{i\eta_k\mathbf{A}}E$, transformed at its two endpoints by $g_\square$. To check the regular branch, all intermediate row carriers are inside the same two centered blocks. The row estimate keeps their values close to the identity in the exponential frame. Hence every relative contour matrix and every argument of its local mean lies in the prescribed logarithm chart. Conjugating the relative matrices multiplies their distances from the identity by at most the fixed condition number of $g_\square$. The same charts remain valid in the actual frame. In fact for a relative contour diameter d, smallness ensures $\mathsf{k}(g_\square)d < \omega_{\mathrm{c}}$ and $4\mathsf{k}(g_\square)d < \omega_{\mathrm{c}}'$. The transported mean therefore lies in the ball that characterizes the unique normalized mean. The relative logarithm is transported by the same conjugation. Applying this at each intermediate scale gives the claimed endpoint rule on the full carrier, including the outer determining rows. Thus no outer-row datum correction is needed for the unamplified path of this upper box construction.

On the exponential chart write $B(t,\nu)$ for the family of logarithms of these transported row values. The old continuation uses $\tau = 1$ in its datum $\tau B + B_{\mathrm{err}}$. Use $B(t,0)$ and add $B(t,\nu) - B(t,0)$ in the correction slot, which is zero at $\nu = 0$. The bare bond logarithm changes by at most $C|\nu|\eta_k\delta_k$, so the order-q row estimate gives a datum change at most $C|\nu|\delta_kL^{q-k}$. Taking the supremum over all $q \le j$ bounds the correction by $C|\nu|\delta_kL^{j-k}$. Decrease c_ν so that this is less than half the old correction radius c_3a_0 on the common disc, using $\alpha_kL^{j-k}/a_0 \le C$ uniformly in $j \le k$. On an addressed old domain in $\square^{\sim 2}$ the window theorem's normalized core readouts and the two hereditary tower readouts are holomorphic and bounded on that correction disc. Their Cauchy estimate gives eq. 4.cxliv. The independent input current is used by the decoupled response; the output current of the inner box is its constrained current.

At a physical real critical endpoint use the nested restriction on the constant-layer box, or on its truncated sequence. The box rows determine every ambient row affected by extension by the unchanged exterior. Any bond inside the box in a crossing plaquette is a fixed row of level zero. Hence such an extension preserves the ambient constraints and changes only the action of the inner plaquettes. The restricted field is critical, and regular uniqueness identifies its orbit with the box

orbit over its own datum. This applies at the upper and old critical endpoints with their respective towers. At other points the box reading is the prescribed continuation from the row family. The two removed terms use their assigned endpoints, response differences and core estimates.

The lower rows at the edge of the window box. Write $D_q(c)$ for the bare read set of the order-q average at a row $c \in \Gamma_q$. It is contained in the two centered blocks of side L^q with centers at the endpoints of c. Indeed the one-step contour averages, pivot and two block transports use only the double block. At the next finer level each site of those blocks is replaced by its centered block of children. Iterating gives the stated read set. In particular the mask is determined by the actual row, including those on the lower layers of the truncated tower. The second-kind holonomies are read from these rows by the tower rule.

Fix the background input, the node frame, the fine parameters and t as ν varies. A row whose $D_q(c)$ misses the support of the bond direction $\zeta_{\square}\mathbf{H}_k^{\mathrm{win}}$ is constant in ν.

The frame used for the fluctuation. Let E be the chosen representative of the upper box background and let $\mathbf{U}$ be the configuration of the input pair. The relative smoothing construction relates these two configurations. The single-lattice window construction in [B87, (3.24)–(3.29)] instead evaluates the response at the upper box field extended by $\mathbf{U}$ outside and carries the independent current. In that extension use the upper tower in the representative of its prescribed datum. The separation of its first two members leaves a bare rim inside the box. Each bare bond there is a determining row at level zero and hence already has the value of $\mathbf{U}$. Therefore a plaquette crossing the box face uses exactly the original links, including its links on the inner side. The same check applies to a stencil of fixed bare reach inside this rim. If one uses the real tower of the input real factor for the extension's real factor, the two also agree on the rim and the relative bond logarithms there are the original ones.

The same splice preserves every order-q row whose full double block crosses the box face. More generally, write W_i for the members of the inserted tower, with W_0 the box and $W_i = \varnothing$ beyond its depth. Consider rows with their full double blocks in the ambient first member. The average of the splice at a row c is the original one if c is retained in the inserted determining set, or if neither endpoint of c lies in W_{q+1}. For a retained row this follows from the prescribed datum and its full carrier in the box. For the second case, argue by induction over q. At level zero a bond that is not inner to the box is unchanged, and an inner bond of this kind is fixed by the bare datum. At positive level, a row with an endpoint in W_q is an inner retained row, since its endpoints avoid W_{q+1} and the layer gap fits its double block inside W_{q-1}. If both endpoints are outside W_q, their order-q blocks are outside W_q by alignment. Each input row in the one-step average then has no endpoint in W_q, so its value at the preceding level is unchanged. This proves equality on the prescribed nonlinear branch. The same induction works on the admitted complex row charts, because each unchanged read has the same full input. If a double block crosses the box face, an endpoint in W_{q+1} would put that block inside the box by the layer gap. This proves the asserted crossing-row equality.

For an auxiliary hereditary tower use the criterion on every row of its actual truncated determining set, including the rows at its top. If it holds, the full real and complex input data for that tower agree with the original data, so its chosen continuation and its current read are unchanged. This covers, for example, a containing tower of the same depth and the same monotone rule. For an arbitrary straddling tower a lower determining row can be wholly inside an active member of the inserted tower and use a changed average. Such hereditary reads still require admission and estimates. For the relative-frame rule write

$$w_0(x) = E(\Gamma_{y_0,x})^{-1} a_{\square} \mathbf{U}(\Gamma_{y_0,x}), \qquad w = (\mathcal{K}_{E,\mathbf{U}} + 1)^{-1}[(\theta_{\square} + \Delta\theta_{\square})w_0].$$

Here the anchor $a_{\square}$ fixes the frame at the base point, and $\mathcal{K}_{E,\mathbf{U}}$ is the mixed covariant operator on the frame. The inner fluctuation uses $\mathrm{Ad}(w)$ on the support where the frame is invertible. The regularity estimate for this frame concerns the input connection expressed in the box coordinates.

It is this estimate, together with the covariant response bounds, that is needed for the ordinary first-difference Hölder estimate. The comparison of the two representatives of the upper tower above uses the bound on their condition number.

At a real critical reference in the nested geometry, restriction to the box is critical at its own full tower datum. By regular uniqueness, $E = \mathbf{U}^g$ there for a real gauge g, with anchor $a_\square = g(y_0)$. Transport along any comb path then gives $w_0(x) = g(x)$. For the mixed connection $E(x,y)g(y)\mathbf{U}(x,y)^{-1} = g(x)$. It follows that $(\mathcal{K}_{E,\mathbf{U}} + 1)(\theta_\square g) = (\Delta\theta_\square + \theta_\square)g$ with the prescribed cutoff boundary values. Uniqueness of the smoothing equation gives $w = \theta_\square g$. Hence its adjoint action is the one of the physical gauge wherever $\theta_\square \neq 0$. This comparison applies to the whole real path of shifts from that critical reference on those carriers, since the frame is fixed before varying the fluctuation direction.

If a determining row c of level q has its complete backward carrier in that chart, its changed datum is

$$\mathcal{M}^q(\mathbf{U}^g)(c) = g(c_-)\mathcal{M}^q(\mathbf{U})(c)g(c_+)^{-1}.$$

One must use both node factors at every such row, including the lower rows. The old constrained configuration at the transformed full datum is read in its corresponding orbit. Holomorphic continuation of this identity uses normalized frame representatives and the regular logarithm charts on the complete carrier. In particular the real quaternionic normalization of w may be replaced in a complex chart by a local analytic determinant normalization. The adjoint action itself only requires invertibility. The endpoint identity does not supply invertibility where the cutoff is zero.

For complex inputs, keep the operator on the same box and with the same boundary condition as at the real reference. Put $\mathcal{K}_0 = \mathcal{K}_{E_0,\mathbf{U}_0}$, $G_0 = (\mathcal{K}_0+1)^{-1}$, $\Delta\mathcal{K} = \mathcal{K}_{E,\mathbf{U}} - \mathcal{K}_0$, and $F = (\theta_\square + \Delta\theta_\square)w_0$. Subtract the equation for $\theta_\square g$ at the real reference. The identity is

$$w - \theta_\square g = (I + G_0\Delta\mathcal{K})^{-1}G_0\big[F - F_0 - \Delta\mathcal{K}(\theta_\square g)\big].$$

To obtain a uniform complex frame from this identity, one can bound $G_0\Delta\mathcal{K}$ by $1/2$ in the sup operator norm on the box. On a carrier with $\theta_\square \geq \theta_* > 0$, a sup bound $\theta_*/(4\sup\|g^{-1}\|)$ over the whole box for $G_0[F - F_0 - \Delta\mathcal{K}(\theta_\square g)]$ gives $\|(\theta_\square g)^{-1}(w - \theta_\square g)\| \leq 1/2$ there. The inverses in the formula and on the carrier then follow by convergent Neumann series, holomorphically and with a uniform condition number on that carrier. We must prove the operator and residual estimates on the admitted complex input sections, with a matched reference in the perturbation range and bounds uniform in the lattice spacing. The first-difference Hölder estimate also uses the relative-frame regularity in the box connection. Near a vanishing cutoff the assigned outer reads are part of the localization rule.

The cutoff on the baseline field. In the single-lattice formula both the baseline and the extra window direction are cut off. To compare the baseline with the one in eq. 4.cxlii, let $\chi_\square$ be one on $\square^{\sim 3}$ and supported inside the box. With the same fluctuation field in the same box frame, join the two baseline choices by

$$\mathbf{A}_v = \big(t[\chi_\square + v(1-\chi_\square)] + \nu\zeta_\square\big)\mathbf{H}_k^{\mathrm{win}}, \qquad 0 \leq v \leq 1.$$

With fixed room in the sup bound for the window field, this path stays in the inner substitution disc. On a level-q row the derivative of the plain logarithmic datum in v has size at most $C_R\delta_k L^{q-k}$. It vanishes whenever the complete double block lies in the plateau of $\chi_\square$, since there the bare product is independent of v. For an inner read contained in $\square^{\sim 2}$, all remaining source rows are separated from the read by at least cML^{k-j} in the old weighted distance, after reserving the fixed stencil reach. Indeed the distance through the physical margin is at least cML^k in bare units and the old tower uses spacings at most L^j. Integrating the weighted source estimate along the path gives the old potential and its first differences an error bounded by

$$C_R\delta_k L^{j-k}\exp(-cML^{k-j})$$

in old units on the inner read. For the constrained current in this same exponential representative, first pass the plain row difference through the local naturalizer. Its stencil enlarges the source support by a fixed reach and preserves the distance bound with a smaller constant. At the flat reference the current of the reconstructed field is $\mathsf{f}(-i\xi\,\mathrm{ad}_{\mathbf{H}})^{-1}\mathbf{F}^{\flat}$ at full old switches. The reduced first variation $\mathbf{F}^{\flat}$ has the same weighted source estimate as $\mathbf{H}$. Taking the difference of this product, the inverse acts bond by bond and its difference is bounded by the local difference of $\mathbf{H}$. This gives the same bound for the normalized constrained current on the inner read. This argument treats the baseline-cutoff change inside the fixed old box. The window identity retains the removed difference and uses the assigned domain and box pair on each of its other segments.

The change of localization and its sum rule. For an old member F, denote its original domain response along the source interpolation by $P^{\mathrm{dom}}(t,u)$. For a near window the source's two removed segments go from P^{dom} to its prescribed intermediate pair and from that pair to the frame-dependent inner pair P^{in}. Calling the intermediate pair P^{mid}, at fixed path parameter the restoring identity is

$$\partial_u F(P^{\mathrm{dom}}) = \partial_u F(P^{\mathrm{in}}) + \partial_u\big[F(P^{\mathrm{dom}}) - F(P^{\mathrm{mid}})\big] + \partial_u\big[F(P^{\mathrm{mid}}) - F(P^{\mathrm{in}})\big].$$

Use the fundamental theorem of calculus on each removed segment before differentiating at zero. The assigned current is carried as a component of the pair throughout the interpolation. Each removed direction uses its own full carrier and complex disc.

For the common pointed sum, apply this identity to each member and to its own tent, then add before the fine bracket. For the resolver on the prescribed box, include the complete box reads of the frame and the old inner carriers in the base. This uses a fixed number of current cubes. The locality estimate uses this base for the entire relative-frame construction. For the source's local fields, restriction to the wall component of the base gives the same field on that component. Once the whole frame read set is in the base, the hereditary restriction in the source rule therefore gives the same inner function on the smaller ambient domain. The finite resummation of the brackets recovers the physical sum, and the two differences restore its original domain response. This argument permits a common-base grouping of the complete localization and preserves its summed physical action difference.

For eq. 4.cxlii we must use the smoothed frame on the full carriers where it is admitted and the assigned outer-layer readings elsewhere. The cutoff can vanish on outer carriers. The frame, old endpoint data and both removed directions must therefore come from one localization rule on the bracket's actual complex domains. The same choice is required at both depths in the comparison.

It follows from eq. 4.cxli–eq. 4.cxliv and the fixed room that there is a constant $c_\nu > 0$, independent of j,k, such that the admitted inner substitutions have the common disc

$$|\nu| < \mathcal{R}_\nu = c_\nu/s_k. \tag{4.cxlvi}$$

Indeed choose c_ν so that $C_R c_\nu C_{\mathrm{age}}$ is smaller than half of every fixed room. eq. 4.cxliv then keeps each substitution inside its assigned domain. This argument verifies an inclusion using the actual input-domain estimates; it does not extend functions to unions of complex gauge orbits. We take $s_k < c_\nu/2$, so that $\mathcal{R}_\nu > 2$.

Third, the critical box-background maps in eq. 4.cxliii obey the weighted first-response, Hölder and compatible boundary-comparison estimates used in the nonlinear pointed window theorem. The complete fields on eq. 4.cxlvi, including the unperturbed field, have output size at most $C\alpha_k$. The central axial datum is bounded by $C\alpha_k\sigma_j$. Its half-datum-ball condition holds when $C\rho_j\sigma_j$ is sufficiently small, which fails at most at a fixed finite set of young ages by eq. 4.cxli. The remote datum has only an absolute $C_M\alpha_k$ bound. Use the fixed continuation radius and the once-for-all ceiling eq. 4.xxx to put the whole axial datum in its half ball. This ceiling is imposed on the upper input, the transported fluctuation and any raw correction throughout the discs of eq. 4.cxlvi, with the room already required in eq. 4.cxliv. The window stock satisfies it by reducing $a^{\flat} < a_I(M)$

after fixing M; no bound on the number of old levels is used. At the exceptional young ages the actual core memberships and the whole-core bound for a member's own tent action remain available. Finite boxes are treated first; their linearized boundary comparison is uniform in the bare mesh. Periodic or distant tails are retained with their stated exponential lengths.

Finally, assume the actual window identities and elementary localization geometry. The partition $\sum_{\square} \zeta_{\square} = 1$ is inserted along the direction of differentiation. For complete physical pairs the old action difference is the sum over windows of $\int_0^1 \partial_\nu|_0$ of its window evaluations eq. 4.cxlii–eq. 4.cxliii, plus the prescribed outer and removed terms. The same local formulas define their extensions at every local complex core. Equality with a global physical fibre integral is asserted only at complete physical pairs.

An inner window contains at most $C_M \sigma_j^{-4}$ old base points. At a fixed base point the number of old domains of tree length between m and $m+1$ is at most $C_D e^{\kappa_D m}$. Diameters are bounded by $C_M(1+m)$, and the input decay rate has a fixed surplus over κ_D. Thus every fixed polynomial diameter factor is summable. An outer field has the additional decay from its distance to the window, while a removed field has the factor e^{-cM/σ_j}. On the assigned single-lattice charts the bounds in the member's local pair norm on the complete $X^{\sim}$ are

$$C_R \delta_k \sigma_j \begin{cases} e^{-\delta_F \mathrm{d}(X,\square)}, & \text{outer cutoff direction,} \\ (1+|w|)e^{-\delta_F M/\sigma_j}, & \text{each removed segment direction.} \end{cases} \tag{4.cxlvii}$$

Here w is the removed term's direction parameter with $|w|\delta_k$ small. For the directional estimates we use admission in the old core on the respective parameter discs, with a fixed fraction of its radius available along each segment. Long input domains retain their original tree decay: one fraction supplies e^{-c_κ/σ_j}, and the remaining fraction supplies the declared output spatial rate. Here $c_\kappa > 0$ comes from tree length in M-cubes and the allocated input rate. When these long-domain tails are written together with removed-response tails as e^{-cM/σ_j}, the fixed c is allowed to depend on M and the input rate; it is not asserted independent of them. These are geometric and response hypotheses, not bounds on the value of an old member.

Renormalize the critical inner function first. Let an old regular member have norm g_j, be gauge invariant and Euclidean as a pointed total, and have its own coefficient b_j. At every base point remove its vacuum value and $b_j A(h_z, \cdot)$ before bracket interpolation. The pointed Hessian theorem fixes this coefficient by the unit normalization of the same tent action. Site-frozen members may differ at different base points: the argument uses each member's own hypothetical Euclidean family and its own coefficient. No translation invariance of that array is required.

For $j < k$, here is the quantitative window estimate on eq. 4.cxliii. The axial field near a base point has linear constant-curvature part of size $C\alpha_k \sigma_j^2$. The Hölder and nonlinear response errors have sizes $C\alpha_k \sigma_j^{2+\alpha_H}$ and $C r_{B,j}^{-1} \alpha_k^2 \sigma_j^4$, respectively. Cauchy's second and third derivatives of a member cost $C g_j/\alpha_j^2$ and $C g_j/\alpha_j^3$. The linear derivative vanishes by constant-color invariance. The quadratic value on the affine part is canceled on the whole pointed total at that base point by the member's own tent action. No coefficient is assigned to an individual domain X. The individual Taylor remainders have a fixed polynomial in $D = 2 + \mathrm{diam}_j X$ times their original spatial decay; the domain count above sums these polynomials. Multiplying the remaining errors therefore gives, at one base point after this domain sum,

$$C g_j \left[\rho_j^2 \sigma_j^{4+\alpha_H} + \rho_j^3 \sigma_j^6 \right]. \tag{4.cxlviii}$$

This is the local Taylor argument on the constrained critical box pair, with its uniform weighted response sums. When its local Cauchy-size parameter is large, the whole-core bound and the own-action bound give the same estimate after increasing that summable polynomial, since $1+T^2 \le CT^3$ for $T \ge 1$. Any fixed finite set of young ages failing the central half-datum-ball test is handled by the same whole-core bounds; their scale ratios have a fixed positive lower bound. The remote

absolute-domain condition is retained separately. The resulting constant is independent of the number of old levels.

Now sum the $C_M\sigma_j^{-4}$ base points. Because $\rho_j \geq 1$, $0 < \alpha_H \leq 1$, and $\sigma_j \leq 1$, the renormalized inner window function has bound

$$Cg_j\rho_j^3\sigma_j^{\alpha_H}. \tag{4.cxlix}$$

Truncating the pointed total at the enlarged window contributes only the long-domain tail already set aside in the geometry. It does not replace the pointed total by an unnormalized individual term.

A defect or an aged remainder has no coupling counterterm. After its value at the identity is removed, constant-color invariance still eliminates the linear derivative, provided there are no fixed color spectators. The vacuum-only window argument gives at one base point, after the same polynomial domain sum, $Cd_j\rho_j^2\sigma_j^4$, where d_j is that origin's norm. The large local-size case uses $1 \leq CT^2$ for $T \geq 1$. After the base-point sum its inner bound is

$$Cd_j\rho_j^2. \tag{4.cl}$$

Stored-field boundary origins are not included in this statement. Their spectators and their hull geometry need the separate boundary formation estimate.

At $j = k$ no age improvement is needed. The vacuum-normalized old member together with its own action has whole-core norm at most Cg_k, by the coefficient bound $|b_k| \leq Cg_k/\alpha_k^2$ and the core bound on its tent action. The actual fluctuation change has size $C\delta_k$ within a core with fixed analytic room $c\alpha_k$. Cauchy's inequality therefore gives Cs_kg_k, with its original spatial decay. There is no fine-volume age factor at this level. This is the term with weight ω_1; no cancellation between different current-level domains is used for it.

The one small factor and the outer terms. Fix $t \in [0,1]$ and the fine parameters. Subtract the window function's value at $\nu = 0$. It vanishes there and is holomorphic on eq. 4.cxlvi. The bounds in eq. 4.cxlix–eq. 4.cl hold on that whole disc. Schwarz's lemma, or Cauchy's formula for its derivative at zero, therefore supplies precisely one factor $1/\mathcal{R}_\nu \leq Cs_k$. The actual window formula uses that derivative and its integral over t; the interval has length one. Thus an old regular inner term contributes $Cs_kg_j\rho_j^3\sigma_j^{\alpha_H}$, and a vacuum-only origin contributes $Cs_kd_j\rho_j^2$. No second amplification factor is extracted later.

The outer direction and the two removed segments. Use the potential and the reduced current together in the norm for the old member. The unamplified domain field has size $C\sigma_j\delta_k$, and the directions obey eq. 4.cxlvii. These core estimates can omit the Hölder quotient. Use the current specified by the interpolation along each segment.

For the outer direction, at an admitted unperturbed point with fixed old core margin, the normalized shift stays in that margin on an affine multiplier disc of radius $ca_0(\sigma_j\delta_k)^{-1}e^{\delta_F\mathfrak{d}(X,\square)}$. Cauchy's estimate therefore gives the outer smallness and the distance factor together. If the window is near X but X leaves $\square^{\sim 2}$, use the spare old-domain length factor instead. For a removed segment use $|w| \leq 1$. On the admitted core the difference between its endpoint values of a member of size g_j is bounded by

$$Cg_je^{-\kappa d_j(X)}\frac{\sigma_j\delta_k}{a_0}e^{-\delta_F M/\sigma_j}.$$

This follows by integrating the old-field derivative along that segment, with the fixed fraction of the old radius available for Cauchy's inequality. A Cauchy estimate in w on the circle $|w| = 1/2$ gives the same bound for the directional derivative at zero. The two endpoint differences telescope at real physical arguments, with the exact domain and box endpoints of the window identity. The fine-cube brackets use these bounds on the entire parameter discs. This argument needs the bounds on the assigned segment directions and endpoints. The unamplified determining rows use the preceding comparison. After the fine-site count the bound is

$$Cs_kg_j\rho_j\sigma_j^{-3}e^{-cM/\sigma_j}. \tag{4.cli}$$

The same reasoning applies to a vacuum-only origin with g_j replaced by d_j. For $0 < \sigma \le 1$ and $\rho \ge 1$,

$$\rho\sigma^{-3}e^{-cM/\sigma} \le C\rho^3\sigma^{\alpha_H}, \qquad \rho\sigma^{-3}e^{-cM/\sigma} \le C\rho^2. \tag{4.clii}$$

Indeed the suprema of $\sigma^{-3-\alpha_H}e^{-cM/\sigma}$ and $\sigma^{-3}e^{-cM/\sigma}$ are finite. Any fixed additional polynomial from spatial summation is absorbed in the same way, by decreasing the exponential rate once. Long-domain tails are treated with their retained tree decay and the same elementary inequality. This proves the ordinary outer and removed bounds without applying the pointed cancellation to an arbitrary pair (U, J).

The sum of the removals on a capped window. For the native near unit whose own padded box fits the box-read class, one can estimate the two removed terms together. This class has $\nu \ge 3$ and its axial-datum majorant is strictly smaller than the old box radius $\alpha_{3,j}$. Use the same capped determining set at both endpoints. At the inner endpoint use the averages of the cutoff layered field. At the other endpoint splice the inner prescribed row datum with the rows read from the domain response. In the splice the exact nonlinear relative average $Q_\star$ uses both endpoint factors. The cutoff mismatch vanishes on each full double block in the plateau. For the splice itself, the capped and old determining sets agree on the complete row reads wherever the splice weight is nonzero. The old response has its prescribed row datum there, so the two relative logarithms being spliced are equal in the same node frame. Outside the splice the datum is read from the response itself. Thus on the complete capped set the second endpoint is the box restriction of the domain response over its own datum. This identification uses all rows of the splice, with the old current read in the same chart.

For a constituent T of the old member the two endpoint differences telescope to $G = T(P^1) - T(P^0)$, where P^0 is the inner endpoint and P^1 the spliced endpoint. Take the comb logarithms B^0, B^1 in the common old box. The cap estimate gives $\|B^0\|_\infty < 1.42\alpha_{3,j}/L$ and

$$\|B^1 - B^0\|_\infty \le \tfrac{1}{2}\alpha_{3,j}e^{-\delta D_v/8}.$$

Here D_v is the old distance of the box read to the cutoff change, and the native near-window geometry gives $D_v \ge ML^{n_0-j}/5$, with n_0 the window level. This is a multiple of M/σ_j when the window levels agree. On the admitted old section use $B^0 + \lambda(B^1 - B^0)$ for $|\lambda| < e^{\delta D_v/8}$. If T has majorant n_v there, its difference from the value at $\lambda = 0$ is bounded by $2n_v$ and vanishes there. Schwarz's estimate at one gives $|G| \le 2n_v e^{-\delta D_v/8}$. On the native removed-parameter disc of radius $r_\square^{\mathrm{rem}}$ this holds for the whole holomorphic difference. Cauchy's inequality gives

$$|\partial_u G(0)| \le 2n_v (r_\square^{\mathrm{rem}})^{-1} e^{-\delta D_v/8}.$$

The same bound applies after integration of the real interpolation parameter. This is a single-run estimate of the combined jet. To use it for the actual brackets, match their two endpoint jets with these capped reads on one admitted old-core section and use their actual removed-parameter radius.

Fine brackets and the resulting formation estimate. We now make the base of an inner bracket explicit. For $j < k$ and one old base point z, let $\square = \square_k(z)$ be its assigned current window and $R_z = X_k'(z)$ its fixed enlarged box. There are only a fixed number of direction-partition windows near this base window. Group their directions by linearity before applying the box-background replacement, and choose R_z to contain their fixed enlargements. In eq. 4.cxlii their sum is the inner cutoff $\zeta_\square$; the remaining directions are outer terms. This is part of the fixed window-first rule. The derivative at zero is linear in these directions, so this grouping preserves the partition identity without changing any incoming pointed member. Let $\mathsf{A}_{j,z}(X, \cdot)$ be the fixed action-family filing

with total $A(h_z^{(j)},\cdot)$, and put $\widetilde{G}_{j,z} = G_{j,z} - b_{j,z}\mathsf{A}_{j,z}$. The one inner window member is the function

$$\mathcal{W}_{j,z}(P') = \sum_{\substack{X\ni z\\ X\subset\square^{\sim 2}}} [\widetilde{G}_{j,z}(X, P^{\mathrm{box}}_{j,z}(P'), z) - \widetilde{G}_{j,z}(X,(1,0),z)]. \tag{4.cliii}$$

Here P' is a local potential/current pair of the current resolved level k, on R_z. The map $P^{\mathrm{box}}_{j,z}$ first forms its actual nonlinear averaged datum and then takes the old-level constrained critical box pair of that datum. The same box-background map is used for every X in this sum. The own tent action is included in the near sum; its fixed support lies inside the assigned enlarged window. The declared domain of this entire member is R_z, and its fine-cube base is $B_z = \mathrm{Hull}_{\pi_k} R_z$. Its number of current fine cubes is bounded by a fixed constant C_{win}, independent of j,k,z. This is a presentation of an inner member before localization; we do not apply different old-X bracket operators to its summands.

Let $\mathcal{W}_{j,z,Y}(s,t,\nu)$ mean this one function evaluated by the assigned local response and the datum in eq. 4.cxlii on $Y \supset B_z$. The inner potential contributed by this member is

$$V^{\mathrm{in}}_{j,z}(Y) = \int_0^1 [\,\partial_\nu \mathcal{W}_{j,z,Y}(\,\cdot\,,t,\nu)|_{\nu=0}]\,[\,Y\setminus B_z \mid B_z\,]\,dt. \tag{4.cliv}$$

The brackets in this formula are linear in the whole function in eq. 4.cliii. Thus its cancellation and bound in eq. 4.cxlviii are established before this single bracket is estimated. Different z's need not have a common base: there is no cancellation between them, and their bounds can be added. Hereditary restriction identifies the bracket on Y with the bracket of the same ambient interpolation function whose parameters outside Y are zero. Finite fundamental theorem of calculus therefore gives, at a complete physical pair,

$$\sum_{Y\supset B_z} V^{\mathrm{in}}_{j,z}(Y;p_W) = \int_0^1 \partial_\nu \mathcal{W}_{j,z,\Lambda}(1,t,\nu;p_W)\big|_{\nu=0}\,dt.$$

Every local term uses only the pair on Y and the input members on its declared domain $B_z \subset Y$. This proves chart-coherent locality and the physical sum identity for this fixed window-first rule. Summing the window identities and the retained outer and removed origins recovers the old action difference. No equality of individual output domains with a different filing is required. The current-level origins $j = k$, and the far and removed origins, retain their own declared bases and the bounds already proved. For vacuum-only near origins the same window-first grouping may be used without a marginal subtraction.

The assigned maps are hereditary on active wall-components, and complete at all parameters equal to one. For an interpolation function Ψ, its bracket is

$$\Psi[K|N] = \int_{[0,1]^K} \prod_{\Delta\in K} \partial_{s_\Delta}\Psi(s_K + 1_N)\,ds_K. \tag{4.clv}$$

For eq. 4.cliv, $K = Y\setminus B_z$, $N = B_z$. For an origin retaining its own base X_0, use $K = Y\setminus X_0$, $N = X_0$. A component carrying a differenced cube and missing the base makes this bracket zero. Each coordinate has the disc $R = e^{\bar\kappa}$, so mixed Cauchy costs $(R-1)^{-|K|} \le e^{-(\bar\kappa-1)|K|}$, with no factorial. The common ν-disc and its one derivative commute with these brackets by joint holomorphy. The preceding amplitude bounds therefore retain this additional spatial factor.

Here is a quantitative rate allocation. For a retained old domain X and its current fine hull X_0, put $m = \#(Y\setminus X_0)$. The actual cube geometry and rooting give a fixed constant C_h such that

$$d_k(Y) \le C_h[d_j(X) + m + 1]. \tag{4.clvi}$$

To see this, view a tree covering X on the old lattice in current units and cover it by current fine cubes. Its hull has a covering tree of length at most $C(d_j(X)+1)$. Each component of the

additional cubes meets that hull, or its bracket vanishes. A rooted spanning forest adds at most a fixed number of edges per additional cube. This proves eq. 4.clvi. For an inner window member the same argument and $\#B_z \le C_{\mathrm{win}}$ give $d_k(Y) \le C_h(m+1)$, increasing C_h by a fixed factor if necessary.

Split the old input rate into positive parts $\kappa_{\mathrm{sum}} + \kappa_{\mathrm{far}} + \kappa_{\mathrm{ret}} \le \kappa$. Choose $\kappa_{\mathrm{sum}} > \kappa_D$ with enough strict gap to sum the fixed polynomial diameter factors, and use κ_{far} for the long-domain tail in eq. 4.cxlvii–eq. 4.clii. The retained old-domain factor is $e^{-\kappa_{\mathrm{ret}} d_j(X)}$. Put $\beta = \bar{\kappa} - 1$. Choose an output rate λ and a number $\epsilon > 0$ with

$$C_h(\lambda + \epsilon) < \min\{\kappa_{\mathrm{ret}}, \beta\}. \tag{4.clvii}$$

All rates are chosen once, sufficiently large for the subsequent ordinary polymer criterion. Combining the retained old factor with $e^{-\beta m}$ and eq. 4.clvi gives $Ce^{-(\lambda+\epsilon)d_k(Y)}$. For inner members the bounded base needs only the β part of this inequality. The number of possible current-window anchors at fixed Y is polynomial in $\#Y$; the old bases within one such window have already been counted by $C_M \sigma_j^{-4}$. The bound $\#Y \le C(d_k(Y)+1)$ and the remaining ϵ absorb this polynomial. This is a pointwise output-domain estimate; any later sum over output domains spends its separately declared entropy reserve.

The function $\rho_j^3 \sigma_j^{\alpha_H}$ is bounded by

$$\omega_{k+1-j} := C(1+k-j)^{3/2} L^{-\alpha_H(k-j)}, \qquad \sum_{n \ge 1} \omega_n < \infty. \tag{4.clviii}$$

All inner age functions use the same enlarged fine-parameter domains and the same ν-disc. The outer and removed functions use the enlarged fine-parameter domains with their respective directional bounds. At fixed j, z, eq. 4.cliii–eq. 4.cliv provide the common-root function before any bracket. The resulting bounds for distinct base points and levels are then added absolutely. No bound for a sum of differently based bracket operators has been inferred from a sup norm of their unbracketed sum. The uniform bounds justify passage through the bracket integrals and to limits of the old-level sum.

We have proved the following formation estimate. Let g_j be regular-member norms, d_j defect norms and r_j aged-remainder norms. Under the explicit background, window and geometry inputs above, the actual old-origin potential satisfies

$$\|\mathbf{V}_k^{\mathrm{old}}\|_\lambda \le C s_k \left[\sum_{j \le k} \omega_{k+1-j} g_j + \sum_{j \le k} \rho_j^2 d_j + \sum_{j \in \mathcal{J}_k} \rho_j^2 r_j \right]. \tag{4.clix}$$

These are local bounds on every specified complex core. At complete physical pairs, finite fundamental theorem of calculus, the window partition, and hereditary resummation of eq. 4.clv identify their sum with the actual old action difference. There is no assertion of a single global off-graph Gaussian or a global torus background for an arbitrary local pair.

Adding a scalar Wilson-weight direction contributes at most $CM^4 s_k \alpha_k^2 \|u\|_\infty$, by the separate determinant/response-bracket argument. The formation rules are linear in their origins, so eq. 4.clix applies to their first source derivatives. For a frozen regular member that direction is $f(p_z)\mathcal{L}_j[1]$ with its own $b_j'(0)$; the remaining direction is the source defect. The finite potential-ball Taylor remainder justifies the chain rule even when the actual ordinary source families are nonlinear. This gives the formation inequality used in the linear source bootstrap, with no additional unexplained source-response norm assumption.

What remains as input is the actual joint complex window/core room, nonlinear averaged-datum estimates, Hölder response and compatible boundary comparison, and the outer/removed field locality for the assigned maps. None follows from a scalar stability bound. The result does not establish those background estimates, replace them by a formal graph identity, or address a large-field history operation.

Direct windows for the ordinary regular terms. In the ordinary constant-sequence part of the step one can localize the regular family by composing its members with the fixed box read before taking brackets. The complete box is then part of the term's carrier. The carrier and the pointing determine where the finished term is filed, as described below.

Write $(\mathbf{A}_s, \mathbf{K}_s)$ for the actual decoupled potential and current increment, in level-k units. On the ordinary constant-sequence core the response estimate gives

$$\|\mathbf{A}_s\| + \|\nabla_{\mathbf{U}}\mathbf{A}_s\| + \|\mathbf{K}_s\| \le C_R\delta_k \qquad (|s_\Delta| < R).$$

Fix the completed row source before varying the switches. The base of the response uses the independent current. For a fixed upper real factor U define the shifted pair by

$$P_{s,\lambda} = (e^{i\lambda\eta_k\mathbf{A}_s}\mathbf{U},\, \mathbf{J} + \lambda L^3\mathbf{K}_s)$$

in level-$(k+1)$ units. The potential and its first difference cost $L|\lambda|\|\mathbf{A}_s\|$ and $L^2|\lambda|\|\nabla_{\mathbf{U}}\mathbf{A}_s\|$ in these units. The current increment costs the stated L^3. Use upper target radii a fixed enlargement of the incoming radii, below the fixed window ceiling. On $|\lambda| < c/s_k$ the three changes are within the resulting reserve, for small fixed c. For the simple clauses this follows from the logarithm product and plaquette estimates. For the hereditary clauses use the proof of the outer substitution with the actual fractional room: put the same shift on a larger auxiliary amplitude disc, on which the simple bounds at most double, and apply the tower estimates there. Their holomorphic curvature and current readouts have the depth factors of eq. 2.vii throughout that disc. Schwarz then bounds their change by the reserved fraction of the same factors. The real factor is fixed, and the independent current is checked by its own affine formula. This admits the shifted upper pair on the full prescribed box, uniformly in the fine switches.

For inherited levels $1 \le j \le k$, far terms and a fixed number of the youngest ages restrict the original pair to the actual old domain and use the direct old substitution. Put $a = k - j$ and $\sigma_j = L^{-a}$. Its potential, first difference and current increment are $\lambda\sigma_j\mathbf{A}_s$, $\lambda\sigma_j^2\nabla_{\mathbf{U}}\mathbf{A}_s$ and $\lambda\sigma_j^3\mathbf{K}_s$ in old units. The base current is $L^{-3(a+1)}\mathbf{J}$. The two-level restriction leaves the old half-core room and the outer substitution admits a disc of radius $ca_{0,j}/(\sigma_j\delta_k)$. Far domains have the length loss $d_j(X) \ge c/\sigma_j - C$, which pays for the old base-point count. For the finitely many young ages this count is bounded directly.

For the remaining near terms use the same fixed window box, with its tower truncated at depth k. The comb map at level k applies to this determining set in the window box. Use the full averaged datum at $P_{s,\lambda}$, including the outer rows, and reconstruct its depth-k critical configuration in the flat-reference Landau chart. Its induced depth-j datum is formed on the truncation of that same tower. The nested continuation identity identifies the depth-j reconstruction with the depth-k field in its corresponding orbit. Read the field and its actual current in the admitted exponential representative. Thus the input current is used in making the shift and the window reads the current of its own reconstruction.

Take the window radii $a^\flat_{\iota,j} = \min\{a_{\iota,j}, c_\flat a_{\iota,k+1}\}$ with fixed $c_\flat > 0$. If b_ι are the admitted radii of the shifted upper pair, the running bounds give

$$\max_\iota \frac{b_\iota}{a^\flat_{\iota,j}} \le C(\sqrt{1+a} + c_\flat^{-1}).$$

Consequently the requirements $CM(b_0 + b_1)\sigma_j \le c' a^\flat_{0,j}$ and $CMb_0 \le c'$ of the window map hold from a fixed age onward, at weak coupling. In level-j units the simple bounds of that representative on the settled interior are at most $CMb_0\sigma_j$. Use the same bound for the curvature and current, which have additional factors of σ_j under rescaling. The tower estimate for simple data then admits both hereditary rules at the old window radii, since the age can absorb its fixed constant as well. On each of the two physical constant-sequence endpoints the depth-k box sequence is finer than the ambient one, and the depth-j box sequence is finer again. The no-inward restriction argument

applies at each restriction. Criticality at the induced datum and regular uniqueness then give the original activity value by real gauge invariance. On the inner read the six incident plaquettes of each current stencil are inside the box, so the constrained current of the restriction equals the ambient one in the same frame. Use the full-switch current identity and its node factors for the shifted physical endpoint. This gives the window identity at both endpoints of the bracket.

For the regular gain use the spatial estimate of the reconstructed upper field. The flat-reference Landau map in this box has the weighted first-datum-derivative bound of the walk estimate, including the Hölder seminorm of its first difference. Integrate this bound on the line from zero to the complete comb datum. The weighted sum over row sites is uniform. On the settled upper interior the local lengths equal one in upper units, and the reference connection is trivial. For the Hölder seminorm take cutoffs equal to one on the stencils of a fixed smaller neighborhood of each upper cell. Close points use a common cutoff, and the first-difference sup norm covers the remaining separations. The result is the ordinary C^{1,α_H} bound there, of size CMb_0. The central old double blocks are contained in this interior with their translated stencils. Applying the nonlinear row derivative estimates to this field gives eq. 4.xxxi with $\sigma = \sigma_j$. Take the root-z axial gauge on the settled old node lattice and extend it by the exponential of its logarithm times a cutoff supported inside that layer. The cutoff is one on the inner carriers and changes by $O(\sigma_j)$ per old link. The comb uses $O(\sigma_j^{-1})$ links of logarithmic size $O(CMb_0\sigma_j)$, so its logarithm is $O(CMb_0)$. The axial link identity also bounds its variation across one link by $O(CMb_0\sigma_j)$. The transition links therefore stay inside the regular logarithm chart, and the lower rows outside its support retain their complete original data. Choose the fixed ceiling on $b_0 + b_1$ to satisfy eq. 4.xxx. This admits the entire axial datum, including the remote rows, in the fixed continuation half ball uniformly in j, k. Define the window function by the old Landau continuation at this full axial datum. The central top rows have size $O(CMb_0\sigma_j)$; all remote rows have size $O(CMb_0)$, and are at old weighted distance at least c/σ_j from the inner read. The weighted Landau estimates for the potential and its first and second-order reads therefore give simple bounds $CMb_0(\sigma_j + e^{-c/\sigma_j})$ on the complete old stencils. The same bound holds for the physical current, by the full-switch first-variation formula. The tower estimate for these simple bounds admits both hereditary rules on the old window core, after increasing the fixed age threshold. This is the representative used in the Taylor calculation. The axial change is a real gauge at the physical endpoints, where invariance gives the same values. Use the resulting affine curvature for every member at the same base point and for its own plaquette tent. The old complete pointed subtraction and the retained domain tails then give the bound

$$Cg_j\,(\rho_j^\flat)^3\sigma_j^{4+\alpha_H}, \qquad \rho_j^\flat = \max\{1, \max_\iota b_\iota/a_{\iota,j}^\flat\},$$

before the bracket and the base-point count. A vacuum-only family uses its quadratic bound $C(\rho_j^\flat)^2\sigma_j^4$ times its own norm.

At a base point sum the near window functions before forming the bracket. Pin every switch cell meeting the full box, so a cut leaves all rows of this read in the active component of that base. The switch locality of the ordinary response then gives locality of the composed function on the same cut. At $\lambda = 0$ the function is independent of the switches. The alternating difference therefore vanishes at zero and Schwarz on the common amplitude disc gives the factor Cs_k. The fine-cell Cauchy factors pay for the added connected cells, and the fixed enlargement of the base costs a fixed constant. Summing the alternating differences gives the full shifted value minus the unshifted value of the same window function. The window identity identifies both values on the physical pair. Finally L^{4a} old base points cancel σ_j^4, leaving the summable Hölder gain for the regular family and the vacuum-only loads of eq. 4.clix. This supplies the ordinary regular bound on a source data line when the same direct window and pointing are used for all its systems.

The boundary assignment is made after formation. A cluster carrier contains the complete carriers of its vertices. If a window vertex uses a box crossing Λ^c, every cluster that contains it has a carrier outside Λ, even when its pointing lies in Λ^0. Its original endpoint contribution is then

in the boundary cluster family. The domain bound for that family must use the complete window read and its boundary pair. For the ordinary step, put $t = k+1$. The gap from Λ_t to Ω_t^c is at least $2\mathsf{a}_t = 2LM\check{R}_t$ in level-k units. A window box meeting Λ_t has diameter $12M$ in these units. The gap puts the whole box and any fixed enlargement for the response reads inside Ω_t, at weak coupling. This uses the ordinary sequence before preparations. A prepared stage can have $\Lambda_t = \Omega_t$.

On the ordinary crossing box the post-step pointwise and gauge bounds are those of the top layer t, in the upper units. They supply the simple core bounds on its full reads. Apply the tower estimate for simple data at fixed larger radii to obtain the hereditary core there. Complete the free source at the pivots on the actual level-k double blocks and set it to zero on the remaining bonds. Separation from Ω_k^c puts these double blocks in Ω_k. Use the response of this layered label on π_k, including when its active component extends past the window. The shift from this response uses the enlarged fluctuation bound $C_{\mathcal{A}}\delta_k$, including its collar coordinates. In the preceding upper substitution and the old window calculation the common amplitude disc can therefore be reduced to $c/(C_{\mathcal{A}}s_k)$. The box norms and old-age estimates remain the same, while the bracket now gains $CC_{\mathcal{A}}s_k$. The bound $C_{\mathcal{A}} \le C\varrho^{\mathrm{st}}(\theta_k)$ on the enlarged box is sufficient, since $\varrho^{\mathrm{st}}(\theta_k)s_k$ tends to zero.
4.5.4.4. The ordinary conditional Gaussian. Put $\mathsf{A} = \mathsf{C}^{\mathsf{T}}\Delta^{(k)}\mathsf{C}$ on the free fluctuation coordinates. Write I for the integration bonds, which have an endpoint in Λ_t, and E for the remaining free bonds with an endpoint in Ω_t. Let a denote the retained scaled exterior field. Completing the square at fixed a gives covariance $\tau_k^{-1}\mathsf{A}_{\mathrm{II}}^{-1}$ and mean $-\mathsf{A}_{\mathrm{II}}^{-1}\mathsf{A}_{\mathrm{IE}}\mathsf{a}$. The common ratio multiplies both blocks of the action and cancels from the mean. For a local integration set W, let P_W be the projection onto its integration coordinates. With the same guarded switches $\hat{s}$ in every factor, put

$$C = (P_W\mathsf{A}_{\mathrm{II}}(\hat{s})P_W)^{-1}, \quad \Gamma = P_W\mathsf{A}_{\mathrm{II}}(\hat{s})(1-P_W)R,$$
$$m = \mathsf{M}\mathsf{a}, \qquad\qquad \mathsf{M} = C(P_W - \Gamma R^{\mathsf{T}})\mathsf{A}_{\mathrm{IE}}(\hat{s}).$$

Here R is the inverse root in the switched walk expansion. The signed convention gives mean $-m$. At the full switches the block inverse identity gives $C(P_W - \Gamma R^{\mathsf{T}}) = P_W\mathsf{A}_{\mathrm{II}}^{-1}$, so this is the restriction of the same conditional mean. The local covariance and coupling to the auxiliary Gaussian are rescaled as $(\tau_k^{-1}C, \tau_k^{1/2}\Gamma)$.

The weighted row bounds for M and its difference from the real reference M_{r} have a common rate. Denote the small prefactor in the difference bound by ε_*. At a crossing bond the threshold is $\mathfrak{b}_b = \theta_{\mathrm{c}}\mathfrak{b}$; at other integration bonds it is $\mathfrak{b}$. The exterior datum is the sum of a real field obeying the three tiers and a complex correction. The collar bound is $\mathfrak{b}/C_{\mathrm{col}}$, the rim bound is $\theta_{\mathrm{c}}\mathfrak{b}$, and the bound beyond the rim is $C\mathfrak{b}$. The correction, of size at most $\varepsilon_{\mathsf{a}}\mathfrak{b}$, starts at distance $W_t - O(1)$ from the integration bonds, where $W_t = LM\check{R}_t$. Splitting the row sum at the collar width ϖ_T and at W_t gives

$$\frac{|m(b)|}{\mathfrak{b}_b} \le C\left(\frac{1}{C_{\mathrm{col}}\theta_{\mathrm{c}}} + e^{-c\varpi_T} + \frac{1+\varepsilon_{\mathsf{a}}}{\theta_{\mathrm{c}}}e^{-cW_t}\right).$$

The same estimate for the difference from M_{r} on the real exterior field has the factor ε_*. Since that reference mean is real,

$$\frac{|\operatorname{Im} m(b)|}{\mathfrak{b}_b} \le C\varepsilon_* + C\theta_{\mathrm{c}}^{-1}\varepsilon_{\mathsf{a}}e^{-cW_t} =: \varepsilon_\mu.$$

Choose the collar threshold and width as in the real step, then use the width of the rim to absorb the far-field enlargement. This gives $|\operatorname{Re} m(b)| \le \theta_\mu\mathfrak{b}_b$ with $\theta_\mu < 1/2$. The guard and small radii of the switched structure give $\varepsilon_\mu \le c(LM)^{-2}$.

To separate the real cutoff factors, fix the selected Mayer vertices and let Y_0 contain the free bonds on which they require a field box. In the signed convention above $B = \xi - m$ with ξ centered. On a bond of Y_0 split the displaced indicator into

$$\mathbf{1}_{\{|\xi_b - m_b| < \mathfrak{b}_b\}} = h_b(|\xi_b|) + \left[\mathbf{1}_{\{|\xi_b - m_b| < \mathfrak{b}_b\}} - h_b(|\xi_b|)\right],$$

where $h_b = 1$ up to $\mathfrak{b}_b/2 - 1$, decreases linearly to zero at $\mathfrak{b}_b/2$, and is zero beyond. The taper is fixed as the mean varies. At real data the bracket lies between zero and one, since the support of the taper is inside the displaced ball. Its support satisfies $|\xi_b| \geq \mathfrak{b}_b/2 - 1$. Outside Y_0 use one minus the failed indicator, whose failure implies $|\xi_b| \geq (1 - \theta_\mu)\mathfrak{b}_b$. Thus each selected remainder of a cutoff and each selected failure has a Gaussian tail cost. Summing these choices restores the physical product of cutoffs before taking the polymer logarithm.

For complex parameters continue the integral over the displaced ball by

$$f \longmapsto \int_{|A_b|<\mathfrak{b}_b} f(A_b + m_b)\, dA_b.$$

Here f is the common integrand in that centered coordinate. The taper integral keeps its real support fixed. The failure integral is the full integral minus the ball integral, and only coordinates outside the selected boxes use that full integral. To estimate a continued difference, freeze the integrand and subtract the ball integral with m_b replaced by $\operatorname{Re} m_b$. The difference integrates over the side swept by the real sphere as the imaginary part of m_b is turned on. Every point of that side has real radius at least $(1 - \theta_\mu)\mathfrak{b}_b$. The remaining difference is the real tapered shell or real complement above. This permits the Gaussian tail estimate on the difference itself. Joint holomorphy follows from the fixed-domain formulas by domination on the admitted boxes.

For the sharp radial tests, integrate the Gaussian auxiliaries and let Q be the marginal precision. It is complex symmetric, with a uniform real floor, bounded norm and small imaginary part. Write $m = u + iv$ and choose a smooth radial function ψ with values in $[0, 1]$, equal to one up to $3/4$ and zero from $7/8$ onwards. In each integration coordinate deform the real contour by

$$z_b = x_b - iv_b\psi(|x_b|/\mathfrak{b}_b).$$

Apply the deformation to the physical ball and complement integrals. The fixed taper is integrated on its real support. The deformation fixes the test sphere, is the identity on its complement, and stays inside the complex ball of each boxed holomorphic factor. Stokes' theorem therefore applies on each product piece of the radial tests. The Gaussian majorant permits passage to infinity on unrestricted coordinates, where the free quadratic potential is entire. The exterior test factor is constant in these integration coordinates.

Put $y = x + u$ and $w_b = v_b[1 - \psi(|x_b|/\mathfrak{b}_b)]$. Where $w_b \neq 0$ one has $|y_b| \geq |x_b| - |u_b| \geq \mathfrak{b}_b/4$, so $|w_b| \leq 4\varepsilon_\mu|y_b|$. It follows that

$$\operatorname{Re}\left((y + iw)^{\mathsf{T}} Q(y + iw)\right) \geq y^{\mathsf{T}}(\operatorname{Re} Q)y - C(\varepsilon_\mu^2 + \varepsilon_\mu\|\operatorname{Im} Q\|)|y|^2.$$

The complex Jacobian on one bond is $1 - iv_b \cdot \nabla[\psi(|x_b|/\mathfrak{b}_b)]$, with modulus at most $e^{C\varepsilon_\mu^2}$. For the normalized Gaussian comparison use the real precision obtained by subtracting this quadratic loss from $\operatorname{Re} Q$. The trace-log estimate between the resulting normalization and that of Q costs at most the dimension times $C(\|\operatorname{Im} Q\| + \varepsilon_\mu^2)$. With the Jacobian this is $O((LM)^{-4})$ per variable, since the guarded comparison gives $\|\operatorname{Im} Q\| = O((LM)^{-4})$. Each selected tail is estimated with a small centered quadratic tilt. The tapered shell starts at $\mathfrak{b}_b/2 - 1$, and the failed cutoff and the displaced side start at $(1 - \theta_\mu)\mathfrak{b}_b$ in the real part of the centered coordinate. The shifted-side bound pays a factor $C(1 + |v_b|)\exp(C|v_b|^2)$ on that bond in addition to the real Gaussian tail. The bound on ε_μ and the weak-coupling floor on the thresholds absorb this factor into the tail, leaving $e^{-c\mathfrak{b}_b^2}$ per selected bond. The free quadratic vertex is estimated using its own local support. On a fine cube Y its bound after the translation is $C|\tau_k - 1|(M^4\mathfrak{b}^2 + \sum_{b\in Y}|y_b|^2)$. Its Mayer estimate uses weight $C_M|\tau_k - 1|\mathfrak{b}^2$ in a boundary formation. Choosing the constant in that weight large enough makes the centered quadratic tilt $O((LM)^{-4})$ per variable after summing the local free cubes. The boundary ratio disc keeps this weight small. In formations contained in Λ_t the mean is zero, and the centered estimate gives the sharper interior weight. The fixed real contour gives holomorphy by local domination.

4.5.4.5. Differences of displaced tests. An exterior selector of the ordinary integral stays in the real retained kernel. In the unchanged collar its fluctuation coordinate is the original one. On the cutoff zone a sharp bond test at threshold at most δ_k is unchanged by the real substitution. The test against an approximate background is instead read using the localized backgrounds on its retained cubes. These identities apply before assigning switches to the analytic terms. In particular a sharp selector is not evaluated on a complex field. For the ordinary step on the standing part with its frames, sum out the passengers before forming the conditional integral. The new tests force the carried gauge indicators and the old plaquette tests on the integration region to equal one. The block-field tests read the fixed upper datum. The adjacent approximate bond factors that read the integration variables also equal one there: the ball test bounds the deviation from the global background by δ_k, and the interior comparison to the test background leaves room below the threshold $2\delta_k$. The remaining factors of those tests are carried by the real exterior kernel. This removal is an identity on the support of the whole term. It uses both the internal ball tests and the retained fluctuation factors in the unchanged part of the strip. At pivot bonds use the local constraint reconstruction on the full double block. This gives back the value in the retained test. With the zone equations the old whole-term description puts the characteristic functions and constraint distributions in the real kernel as the analytic pair varies. The conditional integral thus has precisely the product over $\mathfrak{X}_{\mathrm{I}}$ of its ordinary and crossing balls for the conditional integral.

For an integrated selector whose domain does move, write D_b for its nominal real ball and let ϕ_{b,ζ_b} be a transport of the ball, with $\phi_{b,0}$ the identity. At a real displacement it gives the tested domain $D_b(\zeta_b) = \phi_{b,\zeta_b}(D_b)$. Suppose the transport and its spatial derivative continue on the displacement disc, with their images in the common holomorphic read. For the calculation hold the Gaussian and the selected potentials fixed as a common integrand. On the coordinates in S take the difference between the transported and nominal domains. Expanding the product gives the alternating sum over $T \subset S$, in which exactly the balls of T are transported. At the physical real arguments the result is the integral of the common integrand times

$$\prod_{b\in S} (\mathbf{1}_{D_b(\zeta_b)} - \mathbf{1}_{D_b})$$

and the unchanged selectors. The product vanishes unless each bond of S is in its own difference of test domains. A change of the integrand between endpoints belongs to a separate analytic difference.

To continue this identity, use the pullback by ϕ_{b,ζ_b} of the holomorphic integrand and volume form to the fixed ball. Equivalently, for $b \in S$, integrate over the chain swept by ∂D_b under $\phi_{b,r\zeta_b}$, $0 \le r \le 1$, with boundary orientation giving the difference of the two balls. This follows by Stokes' theorem in that coordinate, and iteration gives one chain per moved test. A difference of complementary selectors has the opposite sign and the same chain. In the bounded logarithm chart the real strip shift and its first derivative have sizes $O(|H|)$ and $O(|H|/\delta_k)$, respectively, before scaling the field. In the plain zone the group multiplier itself continues holomorphically. In a cutoff zone any occurrence of the smooth cutoff in a transport formula must stay at the real parameter of the chain.

For transports whose whole disc gives a displacement $o(\mathfrak{b}_b)$ and bounded first derivatives in the scaled coordinates, the chain of a nominal sphere has real coordinate within $o(\mathfrak{b}_b)$ of that sphere, and its imaginary coordinate is $o(\mathfrak{b}_b)$. Thus the centered real coordinate on the chain has norm at least $\mathfrak{b}_b/3$, by the tier bound. In the other coordinates use the ordinary Gaussian contour above. The additional imaginary part on the chains costs an arbitrarily small multiple of $\sum_{b\in S} \mathfrak{b}_b^2$ in the quadratic estimate, by the uniform operator bound and the real floor of Q. To see this, apply the quadratic inequality to the cross term between the additional imaginary part and the ordinary imaginary part, using the bound $4\varepsilon_\mu|y|$ for the latter. The cross term with the real vector has coefficient $\|\operatorname{Im} Q\|$. Together they cost $O((LM)^{-4})|y|^2$ and a fixed multiple of the squared norm

of the additional imaginary part. The chain volume is at most a fixed polynomial in $\mathfrak{b}_b$ per bond. These costs leave $e^{-c\mathfrak{b}_b^2}$ for each moved test. The carrier includes the test bond and the reads of its transport. Summing the mixed differences over S at their physical endpoints recovers the full transported product by inclusion and exclusion.

For the product balls of the ordinary step one can bound the large switch discs by recentering before forming these differences. Use the prescribed radius $\rho = (\log \theta_k)^{6r+1}$ for the new switches. Choose the constant in the enlarged fluctuation box $C_{\mathrm{form}}\rho\mathfrak{b}_b$ after the absolute row bound for the mean. This fixed enlargement fits in the reader estimates because $C_{\mathrm{form}}\rho\delta_k/\alpha_{\iota,k} \to 0$. It changes the bound on the cubic potential only by a fixed factor, since $(C_{\mathrm{form}}\rho)^3\theta_k^{-1/2}\mathfrak{b}^3 \to 0$. The later stage therefore has the prescribed switch radius available for its Cauchy estimate. Inherited spectator switches keep their original discs. In a local Gaussian activity let m_μ be the signed mean with each new near field entering as $\mu_{b'}\mathsf{a}(b')$. The boxed factors in the centered coordinate y read $B = y - m_\mu$ and keep the same analytic arguments throughout the following calculation. Write their product, including the free factors, as $H_\mu(y)$. At a physical real point the success domain in y_b is $D_b + m_b$, and the failure domain is its complement. Replace each success by the centered success plus its translated-domain difference. Replace each failure by the centered failure minus that same difference. The integrand H_μ stays fixed in each of these operations.

For a row b of the mean split off the exterior bonds within weighted distance $D = C_0(1 + \log \rho)$ of b, using the distance of the kernel bound. Use only columns of the local mean. The weighted row estimate makes the contribution from the other bonds, after amplification by ρ, smaller than $c(LM)^{-2}\mathfrak{b}_b$ if C_0 is sufficiently large. For the close near bonds N_b keep their real values as spectators. In the translated-domain difference for b, use the shifts

$$\zeta_{b,T} = m_b\big(\mathsf{a}_{N_b}\mathbf{1}_T,\, (\mu\mathsf{a})_{N_b^c}\big), \qquad T \subset N_b.$$

Here only switched coordinates in N_b^c carry multipliers, and far analytic fields use their given arguments. Take the sum of the resulting swept chains with coefficients

$$\prod_{b' \in T} \mu_{b'} \prod_{b' \in N_b \setminus T} (1 - \mu_{b'}).$$

The tier estimate for the real close fields still holds when some are set to zero. The far tail estimate and the comparison to the real kernel give $|\operatorname{Re}\zeta_{b,T}| \le \theta'\mathfrak{b}_b$ with $\theta' < 1$, and $|\operatorname{Im}\zeta_{b,T}| \le C(LM)^{-2}\mathfrak{b}_b$. Indeed the unamplified far data outside the real tiers begin at distance $W_t - O(1)$ and give the same exponentially small error as in the mean estimate. The translated boundary thus remains at distance at least $(1 - \theta')\mathfrak{b}_b$ from zero in its real coordinate. The chain estimate for the centered Gaussian applies on the whole switch disc.

In the ordinary step the distance in the kernel bound controls the distance in level-k units between these top-layer bonds. Thus the number of close bonds is at most $C(D+1)^4$. The sum of the absolute interpolation coefficients costs at most $(1 + 2\rho)^{C(D+1)^4}$ for one translated test. Its logarithm is $O((1 + \log \rho)^5) = o(\mathfrak{b}^2)$. This spends only part of the tail charge for that test, even when many exterior switches enter the same row. An original failed test thus retains a tail factor, whether read as a centered failure or as a swept boundary. Differences of the mandatory success tests of a boxed vertex are on the union of the boxed reads. Count each bond in that union once, even when several vertices read it. Summing the choices over a cube adds at most $(1+e^{-c\mathfrak{b}^2})^{C(LM)^4}$ to the local bound.

The recentered integrand is admitted on these chains. At a mandatory boxed coordinate $|y_b| \le C\mathfrak{b}_b$, while $|m_\mu(b)| \le C\rho\mathfrak{b}_b$ by the absolute row bound. The formation-box constant accommodates the sum in the enlarged fluctuation box. The free quadratic factors on unrestricted coordinates have the centered tilt and a local cost bounded by $C_M|1 - \tau_k|\rho^2\mathfrak{b}^2$. This tends to zero in the boundary tube, where $|1 - \tau_k| \le C(\log \theta_k)^{-2q_{\mathrm{t}}}$ and $q_{\mathrm{t}} \ge q > \mathfrak{p}_0 + \deg \rho$; here $\deg \rho$ is the power of $\log \theta_k$ defining the switch radius. The same bound applies to the boxed quadratic correction.

The other vertices use their enlarged-box estimates. The estimates are uniform on the Gaussian decoupling discs, so the usual Cauchy bounds for those switches still apply.

At $\mu \equiv 1$ each chain sum selects the original shift. The affine substitution $y = B + m$ puts each original cutoff ball on the translated chain $D_b + m_b$. In a complementary test use full space minus the ball. Where the domain is full space the remaining factors are entire in that coordinate, and the Gaussian floor permits the affine contour to return to the real space. This recovers the original local activity before resummation, also at the admitted complex arguments. At $\mu_{b'} = 0$ the analytic arguments and all chain sums forget $\mathsf{a}(b')$. At $\mathsf{a}(b') = 0$ the two corresponding endpoints agree and their weights add to one, so the result forgets $\mu_{b'}$. The continuations are integrals over fixed real parametrizations and are jointly holomorphic with the real spectators fixed. They factor over wall components because the cut mean has zero columns off its component and all the selectors are coordinate factors. On the cutoff zone the smooth function in the real strip map and its Jacobian still use the retained real variable. For a mixed-history operation before passengers have been removed, or with additional integrated selectors, we must first reduce those selectors to the same product or transport their actual domains.

The window is inside Ω_t, while a cluster containing it can also have other carriers outside Ω_t. Use the new cluster estimate at formation, since the long-carrier estimate for an inherited boundary term requires support on both sides of Ω_t. When a window is unavailable, the corresponding origin still needs its assigned boundary or direct read with the same physical endpoint identity.

For the source comparison fix the label, Gaussian centre and retained exterior data. For the regular origins let $D_E(X)$ be the age-weighted sum of local differences of the inherited pointed families on the full reads of X. Vary each member with its companion and its own tent coefficient. The defects, aged remainders and local Wilson weights have the corresponding loads in eq. 4.clix, with the actual layer radii for Wilson terms off the regular region. Write $D(X)$ for the sum of these loads. The crossing vertex bound just proved is linear in the old origin, so a changed vertex has bound

$$C\varrho^{\mathrm{st}}(\theta_k) s_k D(X)\, e^{-r_V d_k(Y)}$$

on its actual carrier Y. In a selected rooted cluster interpolate its input arrays on the common data line. Differentiation selects one changed vertex; the derivative of its Mayer factor is $e^V \dot V$, with the same uniform sup bound on V as in the unmarked expansion. The absolute rooted summation at that vertex, with the unused decay for the other branches, gives

$$|\Delta B_{\mathrm{cross}}(X)| \le C\varrho^{\mathrm{st}}(\theta_k) s_k D(X)\, e^{-\kappa_{\mathbf{T}} d^\sharp_{k+1}(X)}.$$

The same proof bounds a fixed partial sum of complete clusters, by the same positive majorants. In particular one can select the clusters with a crossing origin by its record in the formation list. It applies to the source line over the changed slots, with the full pointed member retained at each slot. The small factor is uniform over the enlarged box. The endpoint identity uses all filed pieces at their actual physical switches.

The changed crossing clusters are used a second time in the ordinary strip transfer. Write $E^\sharp$ for their complete formed crossing sum and let $\mathbf{S}$ be the sum of the nonempty strip-displacement brackets. It is linear in the input potential, with each bracket formed on the actual enlarged strip carrier. The strip input is $E^\sharp + \mathsf{K}^{\mathrm{kp}} + \mathsf{B}^\times + \mathsf{B}^\tau$ minus the local Haar logarithms. The strip output is $\mathbf{S}$ of this input plus its strip Jacobian differences. At fixed geometry, fields, θ_k and scalar center x_k, a change of the regular input arrays or of the spatial Wilson weight enters through the formed vertices. Indeed the nonconstant part $x_k(\cdot) - x_k - \zeta_k(\cdot)$ is a Wilson potential, while the scalar center enters the Gaussian ratio. Thus the direct crossing output changes by $\Delta E^\sharp$, and the strip output changes by $\mathbf{S}[\Delta E^\sharp]$. The separate form and normalizer outputs and the strip Jacobians agree on this comparison. A change of the scalar center requires their own ratio and form estimates. Changes of old boundary or standing factors in the same formula use their own loads and majorants.

In each nonempty strip bracket, subtract the evaluations before summing the bracket. A displacement direction has size controlled by the localized response from the strip to its localization boundary. Cauchy's estimate on the transfer disc bounds that difference by the input norm times the displacement divided by its available radius. The exponential in the strip-to-boundary distance pays the enlargement factor of the strip disc and the ratio of the field radius to the fluctuation radius. The total strip term uses its own absolute majorant on the enlarged data line. This yields the strip source slope in its own decay norm. The direct crossing slope and the slope of the displaced output are assigned on their respective carriers.

For a carrier $X \subset \Lambda$, the ordinary formula is used with its actual vertices. It is filed in the regular family at $z \in \Lambda^0$, and in the edge sum $E^\partial(X)$ at $z \notin \Lambda^0$. At most $CM^4(1 + d_{k+1}(X))$ such sites can point into X, by the connected cube count. The spare unit in the pointed rate pays this factor. Thus the edge sum has the ordinary bound with one unit of decay spent, and its source difference has the corresponding data-line slope: the common $\mathbf{N}^\circ$ and $\mathbf{G}_\tau$ cancel before the sum. The Gaussian normalizer outside Λ is a separate term. This assignment retains the terms of the exponent with each complete carrier counted once.

Reconstructing the field of an old near term. On a box of the size of a level-$(\ell - 1)$ cube about its base point, a local level-ℓ triple has curvature, potential and current radii b_0, b_1, b_2. Its box reading uses the full box in the triple's domain and requires that the radii have small sum. Take the averaged datum of the product field in the axial gauge of that box and solve the old constrained critical problem for that datum [B87, (3.26)–(3.27)]. Evaluate the old term at the reconstructed field and at its current. This is a holomorphic function of the input field on the box. It ignores the independent input current. Its reconstructed potential has positive Hölder exponent. The absolute smallness conditions are $CMb_0 \le c$ and $CMb_0 L^{-a} \le c a_{0,j}$, where $a = \ell - j$ is the age and $a_{0,j}$ is the old curvature radius. Independence of the input splitting uses the stronger condition $CM(b_0 + b_1)L^{-a} \le c a_{0,j}$. At a matched real critical endpoint the box reading agrees modulo gauge by the restriction and uniqueness argument above.

On upper triples with $b_0 + b_1 \le C\alpha_k$ the old half-core condition holds at every sufficiently large age, since $\alpha_k / a_{0,j} \le C\sqrt{1+a}$. For the finite set of younger ages, use the old core interpolation when the whole old carrier is in Ω_j. For old term radii $a_{\iota,j} \ge \alpha_{\iota,j}$, the step uses at most $\phi_j^+ \alpha_{\iota,j}$, with $1 - \phi_j^+ \ge c_0/2$. The allowed disc in the dimensionless amplitude of the old response has radius

$$\left[C(1-\phi_j^+)^{-1} \frac{\delta_k}{\alpha_*} e^{C\varkappa} e^{-\delta_* \mathsf{d}_k(X)} \right]^{-1}, \qquad \alpha_* = \min_\iota \alpha_{\iota,k}.$$

Here $\mathsf{d}_k(X)$ is the response distance of the old carrier. At these ages the number of old base points per window is bounded by a fixed constant. The ordinary core estimate then supplies the corresponding young-age bound without a Hölder improvement. The dimensionless amplitude in this estimate is the old term's own interpolation parameter.

This use of the reconstruction requires an admitted triple on the complete box in eq. 4.cxlii. For the ordinary bracket the full-box frame gives the row comparison for the unamplified path and its cutoff perturbation. To use the positive Hölder bound we also need the prescribed reconstruction and the compatible boundary comparison on the entire addressed old domain.

Own readouts and the order of localization. The scalar readout is taken before the old member is evaluated on the new box background. We record the resulting algebra because it fixes both the normalization and the first jet.

At old level j and base point z, write $A_{j,z}$ for the member's assigned tent action. Let b_j be its pointed scalar readout with the fixed free-response kernel and fixed chart. Its definition is a linear combination of second derivatives at the identity with absolutely convergent moment weights. Hence it is complex linear and continuous in that Hessian-moment norm. Its action normalization

is $b_j(A_{j,z}) = 1$, as in the pointed quadratic classification. Set

$$\mathcal{R}_{j,z}G = G - G(0) - b_j(G)A_{j,z}. \tag{4.clx}$$

The zero denotes the identity point of the potential chart. The own coefficient and own action are held together in this definition. The quadratic classification proves the cancellation for this pair; it is not a cancellation between a coefficient and a tent action taken from unrelated families.

Fix j, z and its current window $R_{k,z}$. Let $\mathcal{N}_{j,z}$ be the near domains retained in eq. 4.cliii, and let $b_j(G)$ be the coefficient of the complete incoming pointed member, extracted before this truncation. Write Φ_{G_X}, $\Phi_{G_X(0)}$ and $\Phi_{A_{j,z}}$ for evaluations through its one common constrained box map. The bracket $\mathcal{L}_{Y;R_{k,z}}$ has the fixed declared root $R_{k,z}$; it differentiates each selected fine-cube parameter once and takes the prescribed fluctuation difference. In finite volume linearity gives exactly

$$\begin{aligned}
&\mathcal{L}_{Y;R_{k,z}}\left[\sum_{X\in\mathcal{N}_{j,z}}(\Phi_{G_X} - \Phi_{G_X(0)}) - b_j(G)\Phi_{A_{j,z}}\right] \\
&= \sum_{X\in\mathcal{N}_{j,z}} \mathcal{L}_{Y;R_{k,z}}[\Phi_{G_X} - \Phi_{G_X(0)}] \\
&\quad - b_j(G)\mathcal{L}_{Y;R_{k,z}}[\Phi_{A_{j,z}}].
\end{aligned} \tag{4.clxi}$$

For an infinite absolutely convergent old-domain sum, the same identity follows by uniform convergence on a slightly larger interpolation polydisc and Cauchy's formula. The own counterterm occurs once with the complete member's coefficient. The domains omitted by the near truncation remain in the separately estimated tail; no scalar coefficient is assigned to an individual domain.

This is the precise use of window-first normalization. It allows the pointed cancellation to be bounded before applying one bracket at the fixed j, z. Different z's may have different roots; their bounds are summed afterwards with the site and age counts of eq. 4.clv. No bound on a sum of functions is applied to a sum of different bracket functionals. Nor does eq. 4.clxi assert that the newly formed bracket member has scalar readout $b_j(G)$. That readout belongs to the old member before localization.

If $G(w)$ is a differentiable source family in this fixed presentation, continuity and linearity give

$$\partial_w \mathcal{R}_{j,z}G(w)|_0 = \dot{G} - \dot{G}(0) - b_j(\dot{G})A_{j,z}. \tag{4.clxii}$$

Thus a derivative of the formed member loses no normalization merely because its scalar coefficient has already been extracted. The derivative has its own coefficient, computed with the same free response. This is the rule used in the linear bootstrap.

If the presentation instead moves its readout or own action, the complete elementary product rule is

$$\partial_w[b_j^w(G^w)\,A_{j,z}^w]\Big|_0 = \{\,\dot{b}_j(G^0) + b_j^0(\dot{G})\,\}A_{j,z}^0 + b_j^0(G^0)\dot{A}_{j,z}. \tag{4.clxiii}$$

In this display the superscript zero denotes source parameter zero; G^0 is a member, rather than its value at the identity chart point. The distinction matters in a reference-change argument. eq. 4.clxii uses the fixed-presentation case; it cannot remove either extra term of eq. 4.clxiii.

4.5.5. *The many-scale first-source recurrence.* This section isolates a linear consequence of the ordinary local formation bounds. It uses the inserted-potential and renewed-pointing estimates at one step, rather than a larger domain of physical sources. In particular, it distinguishes the derivative of a newly generated activity from the old background action that is carried to the next scale. The formation and domain hypotheses below concern the actual local maps. Their statement is not an assertion that a nonlinear history construction satisfies them.

Meshes, norms, and the fixed ordinary run. Work first on a finite periodic lattice. Let $h_j = L^j h_0$ be the physical mesh of level j, with $L > 1$. A source profile is a bounded Lipschitz function on the physical torus; complex profiles are allowed by complexification. Put

$$\|f\|_{\sharp,j} = \max\{\|f\|_\infty, C_M h_j \operatorname{Lip} f\}. \tag{4.clxiv}$$

At level j, a pointed localization domain X contains its base point z. Its enlarged source region $X^\sim$ and the fixed plaquette point p_z satisfy

$$\sup_{p\in X^\sim} |p - p_z|_{\mathrm{tor}} \le C_M h_j (1 + d_j(X)). \tag{4.clxv}$$

Distances and diameters in this section use the torus metric. The family norm with rate γ is the supremum of $e^{\gamma d_j(X)}|F(X,P,z)|$ over the actual specified complex field domain and all X, z. Companion families, when present, are included in the same norm. A bound with rate $\kappa + 1$ therefore leaves one unit of rate for a diameter.

The ordinary run fixes all Gaussian structures, background maps, thresholds, label spaces and reference operations. Sources change local exponent members only. Let

$$\alpha_j = \theta_j^{-1/2}(\log\theta_j)^q, \qquad \delta_j = \theta_j^{-1/2}(\log\theta_j)^{p_0}, \qquad s_j = \delta_j/\alpha_j, \qquad q > p_0 \ge 0. \tag{4.clxvi}$$

Fixed multiplicative constants in these definitions can be included. Assume the ordinary flow bounds, with constants independent of the number of levels,

$$c(k-j) - C_0 \le \theta_j - \theta_k \le C(k-j) + C_0 \qquad (j \le k). \tag{4.clxvii}$$

All θ_j are sufficiently large. These bounds imply, with a uniform constant C_F, that for $1 \le j \le k$,

$$\begin{aligned} \alpha_{j-1}^2 s_{j-1} &\le C_F \alpha_k^2 s_k, \\ \varrho_{k,j}^2 \alpha_{j-1}^2 s_{j-1} &\le C_F \alpha_k^2 s_k, \qquad \varrho_{k,j} = \alpha_k/\alpha_j, \\ \varrho_{k,j}^2 &\le C_F(1 + k - j). \end{aligned} \tag{4.clxviii}$$

Indeed the functions $t^{-1}(\log t)^{2q}$ and $(\log t)^{p_0-q}$ are decreasing for sufficiently large t. The bounded displacement C_0 in eq. 4.clxvii changes their ratios by at most a constant. The second line also uses the bounded consecutive ratio α_{j-1}/α_j. The upper flow bound and $\theta_k \ge 1$ prove the last line. These observations also give $s_j \le C_F s_k$ when $j \le k$.

Locality supplies the source-gradient gain. Let E_j^w be an ordinary sourced regular family, holomorphic in a neighborhood of $w = 0$. Define its first derivative by

$$\mathcal{L}_j[f] = \frac{d}{dt} E_j^{tf}\Big|_{t=0}, \qquad A_j = \sup_{f\neq 0} \frac{\|\mathcal{L}_j[f]\|_{\kappa+1}}{\|f\|_{\sharp,j}}. \tag{4.clxix}$$

The supremum is taken over the permitted profiles and the tori in the run. At a fixed finite lattice all differentiations are in finite-dimensional source coordinates. Locality means that the value indexed by X depends only on the values of f on $X^\sim$. To apply the following argument it suffices that the source class admit bounded Lipschitz truncations of a profile and constants, as does the class defined by eq. 4.clxiv. No density claim about an infinite-dimensional algebraic dual is used.

The derivative defect at a base point is exactly

$$\mathcal{D}_j[f](X,P,z) = \mathcal{L}_j[f](X,P,z) - f(p_z)\mathcal{L}_j[1](X,P,z). \tag{4.clxx}$$

We claim

$$\|\mathcal{D}_j[f]\|_\kappa \le C_M A_j h_j \operatorname{Lip} f. \tag{4.clxxi}$$

To prove it, fix X, z, and subtract $f(p_z)$ from f. On $X^\sim$ the resulting profile has modulus at most $C_M h_j(1 + d_j(X)) \operatorname{Lip} f$. For real f, clip its values outside an interval twice this large, using a Lipschitz scalar map that is the identity on that interval and has Lipschitz constant at most a fixed constant. For a complex profile clip the real and imaginary parts separately. The resulting global profile g agrees with $f - f(p_z)$ on $X^\sim$ and satisfies $\|g\|_{\sharp,j} \le C_M h_j(1 + d_j(X)) \operatorname{Lip} f$. Locality

and linearity give $\mathcal{D}_j[f](X,P,z) = \mathcal{L}_j[g](X,P,z)$. Apply eq. 4.clxix and $(1+d)e^{-d} \le 1$ for $d \ge 0$ to obtain eq. 4.clxxi. Smooth profiles may instead use a smooth clipping map with the same fixed bounds. This proof applies uniformly at every external complex field point in the family norm. For a source measured at level $m \ge j$, it gives in particular

$$\|\mathcal{D}_j[f]\|_\kappa \le C A_j L^{-(m-j)} \|f\|_{\sharp,m}. \tag{4.clxxii}$$

Thus the derivative defect retains the response amplitude A_j and the scale-gradient gain together. The precise one-step inputs. For clarity we list the ordinary inputs needed for the bootstrap. The first two are estimates for local formation and the finite regular gas, at fixed background maps; they are not estimates for complex history masses.

The formation of potentials is linear in each origin. When a source is differentiated at zero, the regular-origin direction at a site is $f(p_z)\mathcal{L}_j[1]$, accompanied by its own coefficient, and the remaining direction is eq. 4.clxx. We require the following ordinary formation estimate on the actual localized complex pairs:

$$\begin{aligned}\|\dot{\mathbf{V}}_k[f]\|_\lambda \le C s_k \Big[&\alpha_k^2 \|\dot{u}_k[f]\|_\infty + \sum_{j\le k} \omega_{k+1-j} A_j \|f\|_\infty \\ &+ \sum_{j\le k} \varrho_{k,j}^2 \|\mathcal{D}_j[f]\|_\kappa + \sum_{j\in\mathcal{J}_k} \varrho_{k,j}^2 \|\dot{R}_j[f]\|_\kappa \Big].\end{aligned} \tag{4.clxxiii}$$

Here $\omega_n \le C(1+n)^C L^{-\alpha_H n}$ with $\alpha_H > 0$, and $\mathcal{J}_k$ contains the aged remainder origins that actually enter the regular part. Younger remainders remain in the remainder class. The factor s_k is the fluctuation-amplitude/field-radius ratio. In a formulation with additional positive powers of θ_k^{-1} in this factor, they are bounded by a constant times s_k at weak coupling. The current scalar-origin term in eq. 4.clxxiii follows more sharply from its fluctuation-difference estimate $C\alpha_k\delta_k\|\dot{u}_k[f]\|_\infty$. All substitutions in eq. 4.clxxiii must lie in their actual specified cores; a norm inequality does not create those inclusions. In particular, a proof of the pointed Hessian cancellation and nonlinear window estimate on constrained critical backgrounds alone does not prove eq. 4.clxxiii for independent potential/current pairs used in localization. Passing from those background estimates to all the actual localized pairs remains part of this ordinary formation input.

The inserted-potential estimate for the actual finite Mayer/decoupling series, followed by its actual renewed bond pointing, gives

$$\|\mathcal{L}_{k+1}[f]\|_{\kappa+1} \le C_P \|\dot{\mathbf{V}}_k[f]\|_\lambda. \tag{4.clxxiv}$$

Its proof uses the ordinary spare-potential slot. The auxiliary potential parameter is not a physical source in a history box. The normalized Gaussian, all reference measures, and $\mathbf{N}^\circ, \mathbf{G}_\tau$ have zero source derivative. These facts are essential to eq. 4.clxxiii–eq. 4.clxxiv.

The ordinary constant-source construction supplies numbers $b_j'(0)$ with $\sum_j |b_j'(0)| \le B_0$, uniformly in the cutoff. Its fixed tents and stage factors satisfy

$$\dot{u}_k[f](p) = f(p) + \sum_{j\le k} b_j'(0)\varphi_j(p) \sum_y h_y^{(j)}(p) f(p_y), \qquad \|\dot{u}_k[f]\|_\infty \le (1+B_0)\|f\|_\infty. \tag{4.clxxv}$$

The last bound uses $0 \le \varphi_j \le 1$, $h_y^{(j)} \ge 0$, and $\sum_y h_y^{(j)} = 1$. The coefficient of $\mathcal{L}_j[1]$ is exactly $b_j'(0)$: the free response and coefficient extractor are fixed and complex linear. Thus no coefficient of another member enters eq. 4.clxxiii, even when $f(p_z)$ varies with the site.

Finally suppose the ordinary remainder families are bounded by $C_R\theta_j^{-p}e^{-\kappa d_j(X)}$, with $p > 3$, on the fixed physical source disc $|t| < r/\|f\|_{\sharp,j}$ around zero. Cauchy's inequality at half that radius proves

$$\|\dot{R}_j[f]\|_\kappa \le (2C_R/r)\theta_j^{-p}\|f\|_{\sharp,j}. \tag{4.clxxvi}$$

This is the only source-disc estimate needed for the small remainder class. It is not used to produce an α_j^2 factor for regular activities.

The regular family in a history. Let Λ be the small-field region of the intermediate label and Λ^0 its set of admitted base points. A pointed term at (X, z) is filed in the regular family when $X \subset \Lambda$, $z \in \Lambda^0$ and all its vertices are regular [B88b, §3]. The fluctuation expansion places a fine vertex with domain Y in this term only when the hull of $Y^\sim$ in the level-$(k+1)$ grid lies in X. Thus its complete determining set is interior. The boundary, standing-operation and exterior-quadratic vertices vanish there, and the Gaussian has no exterior mean. The local Gaussian and threshold structure is that of the whole-lattice rule. At a fixed local field the normaliser, free potential and Gaussian ratio use the unsourced parameters (θ_k, x_k). The physical source enters the local potentials.

The quadratic gauge correction also has a regular part and a rim part. In the free fluctuation coordinates let K_A and K_G be the Wilson and gauge-fixing forms at the reference weight. With $\tau_k = x_k/\theta_k$, the actual precision is $\tau_k K_A + K_G$. The reference Gaussian of precision $\tau_k(K_A+K_G)$ therefore requires the potential $(\tau_k - 1)B^{\mathsf{T}} K_G B/2$, as in eq. 4.ccxxiv. The gauge coordinates eq. 4.ccxvii use only bonds inside their blocks. Thus this form splits into fine-cube terms $(\tau_k - 1)q_Y(B)$, where q_Y is the sum of the local gauge squares assigned to Y. The field B includes the stored exterior coordinates. Completing the square for the reference precision puts its mixed block in the conditional mean. Substitute the shifted integration coordinate into the entire $q_Y(B)$, including its mixed terms. A physical source in this presentation enters the Wilson potential at fixed τ_k.

When the coarse hull of Y with its neighbors lies in Λ, use the relative quadratic bound of the ordinary Gaussian leg. In the remaining cubes use the bounded rim vertex. On the real cut the local row bound and the number of bonds in a fine cube give

$$|(\tau_k - 1)q_Y(B)| \le C|1 - \tau_k| M^4 \mathfrak{b}_k^2.$$

The centered relative quadratic estimate allows a fixed disc about $\tau = 1$. At the rim use the disc of radius $2(\mathfrak{t}_k + \bar{c})/\theta_k$, where $\mathfrak{t}_k$ is the real weight tolerance and $\bar{c}$ the bounded weight displacement allowed by the tube. The choices of the tube pay for $\mathfrak{b}_k^2$, also on the enlarged field box used in the rim estimates. The two assignments sum to the original quadratic correction, and the choice between them reads the collared hull. Every quadratic vertex in a regular output satisfies the interior hull test, so it uses exactly the ordinary relative quadratic term. The same test applies in the ordinary recent-remainder contribution. Rim vertices enter the boundary expansion with their own bounds. The global cancellation of the marginal coefficient is the uncut Gaussian identity eq. 4.ccxxvii.

An old remainder of level j is aged here when $M\check{R}_j \le L^{k+1-j}$. It can enter a regular term and then contributes to the marked part of that term. Its origin carries the squared-radius weight $\varrho_{k,j}^2$. The weight ω_{k+1-j} applies to regular origins with their own pointed subtraction. Terms containing a recent remainder go to the new remainder on the one-layer inset of Λ and to the boundary family elsewhere. Terms with all vertices regular that are based at $z \notin \Lambda^0$ use the edge rule.

On the complete reads of a regular term, the sourced array uses the regular member for the constant $w(p_y)$ at each old base point y, its own Wilson coefficient, and the defect at y. The Wilson shift is the actual plaquette array with the stage factors and tents. The summable slopes of the scalar increments and the local defect and remainder bounds put this array in the ordinary source box about the same real centre. All its interior potentials, including those for a spatially varying shift, use the same local rule on every label. Therefore the term filed at (X, z) is the same function of the same local array as in the whole-lattice rule. Its field argument can vary with the history. For constant sources the defects vanish and the same pointing gives the Euclidean family with its source-dependent Wilson coefficient. A constant shift c is included as a Wilson-weight vertex of the rule at (θ_k, x_k).

To compare two arrays at this centre, join them inside the ordinary box. The difference of the local vertex potentials is linear in the input differences, with the four weights in eq. 4.clxxiii, before differentiation. In the convergent fluctuation expansion, differentiate along the line by inserting that difference at one vertex. The Gaussian terms are fixed, and the inserted-potential bound and the renewed bond pointing give the output rate $\kappa_{\mathbf{T}}$, where $\kappa_{\mathbf{T}} = (1+4\beta_{\mathrm{s}})\kappa \geq \kappa+1$. Equivalently the finite difference is bounded by $C\vartheta_k^{\sharp} e^{-\kappa_{\mathbf{T}} d_{k+1}(X)}$ times the weighted distance of the two arrays, where

$$\vartheta_k^{\sharp} = s_k + CM^4\theta_k^{-1/2}\mathfrak{b}_k^3 + \theta_k^{-1/4} \leq Cs_k$$

at weak coupling. Apply this comparison to $w = tf$ and to zero. The difference quotients of the four inputs give the actual shift derivative, the frozen regular derivative with its own coefficient, the defect derivative, and the aged remainder derivative. This gives the input to eq. 4.clxxviii on the common regular family once the vertex estimates on the assigned complex domains hold.
The absolute bound needed on a source data line. In the ordinary regular region $X \subset \Lambda$, retain the Gaussian compensation, cutoff and interaction legs in that order. Use the same pointing for the step and its constant-source companion. The comparison of two arrays at fixed Gaussian proves a difference bound for their interaction legs. To recover the absolute bound first sum the interaction vertices on an admitted line. For the regular, defect and aged-remainder norms in eq. 4.clix, the enlarged box gives, respectively, a fixed bound, $CL^{-(k+1-j)}$, and $C\theta_j^{-p}$. The three sums are therefore bounded by constants times

$$\sum_{a\geq 0}\omega_{a+1}, \qquad \sum_{a\geq 0}(1+a)L^{-(a+1)}, \qquad \sum_{a\geq 0}(1+a)(\theta_k+ca)^{-p}.$$

The first uses the positive Hölder gain and the third is $O(\theta_k^{2-p})$. In the Wilson direction the weight norm is at most $3r$; its vertex costs at most $CM^4 s_k \alpha_k^2\, 3r$. Add the intrinsic and coarea vertex bounds on the enlarged field box. They cost at most $CM^4\theta_k^{-1/2}\mathfrak{b}_k^3$, using $\mathfrak{b}_k \geq 1$.

The coordinate Cauchy bounds on the old and coarea legs and the normalized intrinsic pointing in eq. 4.cxxx give these bounds for the integrated bond roots in eq. 4.cxii. Insert each root as in eq. 4.cxv. The rooted cutoff term in eq. 4.cxxxix contributes at most $Ce^{-c\mathfrak{b}_k^2}$, on its own leg. Midpoint tents cost a bounded number of bond roots at a pointing, so retain an output rate $\kappa' \geq \kappa+1$. The free quadratic root eq. 4.cxxiv separately gives $C_{\mathbf{M}}|1-\tau_k|e^{-\kappa' d_{k+1}(X)}$. Consequently the three legs, after subtracting their common normalizer $\mathbf{N}^{\circ}$, give

$$|E(X,\cdot,z) - \mathbf{N}^{\circ}(X,\cdot,z)| \leq \left(C\vartheta_k^{\sharp} + C_{\mathbf{M}}\mathfrak{t}_k/\theta_k\right) e^{-\kappa' d_{k+1}(X)}.$$

Take the tube and weak-coupling ceiling so that the parenthesis is at most $1/2$. For the ordinary regular origins use the direct window construction on their complete bases throughout the enlarged data line. The other formation rules need their own local domains and bond-root estimates. Use the identical local pointing for the regular member in the sourced and unsourced arrays.
The induction and its constants.

Proposition 4.10 (Linear regular-response bootstrap)**.** Under eq. 4.clxiv–eq. 4.clxxvi, suppose the initial regular list is empty. There are $M_0 < \infty$ and $s_\star > 0$, independent of the number of levels, cutoff and volume, such that $s_j \leq s_\star$ for all j implies

$$A_j \leq M_0\alpha_{j-1}^2 s_{j-1} \qquad (j \geq 1). \tag{4.clxxvii}$$

■

Proof. Divide eq. 4.clxxiii by $\|f\|_{\sharp,k+1}$. Use eq. 4.clxxii with $m = k+1$, eq. 4.clxxiv–eq. 4.clxxvi, and the monotonicity $\|f\|_{\sharp,j} \le \|f\|_{\sharp,k+1}$. This gives

$$A_{k+1} \le C_1 s_k \left[\begin{array}{l} (1+B_0)\alpha_k^2 + \sum_{j\le k} \omega_{k+1-j} A_j \\ + C \sum_{j\le k} \varrho_{k,j}^2 A_j L^{-(k+1-j)} + C_r \sum_{j\le k} \varrho_{k,j}^2 \theta_j^{-p} \end{array} \right]. \tag{4.clxxviii}$$

There is no sum over the number of source labels in this inequality. The norms of linear directions have already combined them into f. The flow bound and eq. 4.clxviii imply

$$\sum_{j\le k} \varrho_{k,j}^2 \theta_j^{-p} \le C \sum_{a\ge 0} (1+a)(\theta_k + ca)^{-p} \le C_p \theta_k^{2-p} \le C_p \alpha_k^2. \tag{4.clxxix}$$

To check the middle bound, compare the sum with its first term and the integrals of $(\theta+ct)^{-p}$ and $t(\theta+ct)^{-p}$ on $[0,\infty)$. They are fixed constants times θ^{1-p} and θ^{2-p}, respectively; the latter integral is finite exactly when $p > 2$. The final bound uses $p > 3$, $q \ge 0$, and large θ_k.

Assume eq. 4.clxxvii through level k. Both sums involving A_j in eq. 4.clxxviii are at most

$$C M_0 \alpha_k^2 s_k \left(\sum_{n\ge 1} \omega_n + \sum_{n\ge 1} L^{-n} \right)$$

by eq. 4.clxviii. Consequently, for constants C_2, C_3 independent of k,

$$A_{k+1} \le \alpha_k^2 s_k (C_2 + C_3 M_0 s_k).$$

Choose $M_0 \ge 2C_2$ and $s_\star \le (2C_3)^{-1}$. The right-hand side is at most $M_0 \alpha_k^2 s_k$, as required. At the first step there are no old regular members, so the same argument starts the induction. The direct bottom-level scalar potential, if it occupies a separate initial slot, must satisfy the same fixed-reference fluctuation estimate used in eq. 4.clxxiii. □

A consequence on the near, non-wrapping window class. Let the final fluctuation size be $\delta_n = \alpha_n s_n$, and consider only terms for which the actual local window chart, the pointed affine comparison and its complex domain inclusions hold without a field-independent wrapping or holonomy tail. In addition, the axial and Hölder bounds of the nonlinear window proof must hold for the *complete* near background-plus-fluctuation pair with output size $a_{\mathrm{out}} \le C\delta_n$. A bound on its fluctuation increment alone is insufficient: an unrestricted background of size α_n would require that larger size in the radius ratio. The intended near class is the one on which the actual small-field tests impose a fixed multiple of δ_n on the complete field. Here the complete box includes the upper comb, every lower-row double block and the cutoff frame. The retained real profile in this class is bounded by $C\delta_n$ on that box and on the enlarged neighborhoods used by its gauge clauses. Then eq. 3.cdiii, with its local gauge estimate and tower restrictions, allows the real upper curvature radius $A_0 = C_M \delta_n$. The real reference has zero potential slot; for its perturbations use $A_1 = C_M \delta_n$ and the already required complex-domain inclusions. Those inclusions, rather than the real profile alone, admit the varied potential and independent current. The unamplified transported fluctuation is at most $C_R \delta_{n-1} \le C\delta_n$, together with its first difference on the settled box; the last inequality follows from the adjacent flow bound. Apply eq. 4.cxlv and its central axial estimate with $k = n-1$. It gives $C_M \delta_n$ on the whole raw and axial datum, including the remote rows, and $C_M \delta_n L^{j-n}$ on the central rows, absorbing one factor L in the constant. At depth $j = n$ use the upper datum itself. The far response therefore has size $C_M \delta_n e^{-cL^{n-j}}$, with the same homogeneous amplitude as the central estimate. This verifies the remote-amplitude requirement for this complete near class. It does not apply to a box whose tests control only an inner read or to an independent physical correction of larger size. For ordinary formation on the enlarged disc $|\nu| < c_\nu / s_k$, the fluctuation bound in

the inner substitution is instead $C\alpha_k$. The same calculation therefore supplies the required whole-datum and remote bound $C_M\alpha_k$ on that disc. No δ_k bound on the amplified ordinary disc is used. Keep the true ratio $\rho_j = \delta_n/\alpha_j$; it need not exceed one. The actual window-domain input includes $\alpha_j/r_{B,j} \le C$, where $r_{B,j}$ is the independent-coordinate datum radius, and the absolute full-datum inclusion of eq. 4.xxx. The central half-datum-disc condition holds uniformly at all ages: with $\sigma = L^{-(n-j)}$, the flow gives

$$\frac{C_{\mathrm{ax}}\delta_n\sigma}{r_{B,j}} \le C\rho_j\sigma \le Cs_n(1+n-j)^{1/2}L^{-(n-j)}.$$

Choose $s_\star$ smaller by a fixed factor so that the right-hand side stays below the required half-disc constant. The remote rows are admitted by the separate fixed absolute ceiling, with no age gain asserted for them. On the complete near class just specified, the preceding whole-window calculation supplies their proportional amplitude $C\delta_n$. Their response is therefore $C\delta_n e^{-c/\sigma}$ in the pointed-window estimate. This amplitude comes from the complete input and eq. 4.cxlv; the fixed ceiling supplies its domain. For other classes without that proportional estimate, the remote response remains a separate tail outside the homogeneous near contribution. This uses the near-domain proof for $\rho_j > 0$ itself and avoids the young-age core-only fallback, which would lose a ρ_j^2 gain when $\rho_j < 1$. After the four-dimensional fine-site count, the renormalized constant-member contribution at age $a = n - j$ is bounded by

$$CA_j\|f\|_\infty \left[\left(\frac{\delta_n}{\alpha_j}\right)^2 L^{-\alpha_H a} + \left(\frac{\delta_n}{\alpha_j}\right)^3 L^{-2a}\right]. \tag{4.clxxx}$$

The value at the identity and this member's own $b_j'(0)A(h_z,\cdot)$ are removed here. By eq. 4.clxxvii–eq. 4.clxviii, $A_j/\alpha_j^2 \le Cs_n$ and $\delta_n/\alpha_j \le Cs_n(1+a)^{1/2}$. The sum of eq. 4.clxxx is therefore at most $C\delta_n^2 s_n\|f\|_\infty$.

For the defect, the vacuum-only window estimate instead gives

$$C\|\mathcal{D}_j[f]\|_\kappa \left(\frac{\delta_n}{\alpha_j}\right)^2 \le C\delta_n^2 s_n h_j \operatorname{Lip} f. \tag{4.clxxxi}$$

Since $h_j = h_n L^{-(n-j)}$, its sum is at most $C\delta_n^2 s_n\|f\|_{\sharp,n}$. Thus these near regular and source-gradient terms together have size $C\delta_n^2 s_n\|f\|_{\sharp,n}$ per final window unit. With the radii in eq. 4.clxvi, the prefactor is $\theta_n^{-1}(\log\theta_n)^{3p_0-q}$. The gain is obtained because the newly generated derivative member already has amplitude $\alpha_j^2 s_{j-1}$.

No field-independent far tail is included in eq. 4.clxxx–eq. 4.clxxxi. Such a tail cannot acquire a factor ρ_j^2 merely from a source derivative. Recent large-field terms, marked remainder windows and torus or hole windings still require their own ordinary source-disc bounds, domain comparisons or separately small-class estimates. Nor do the local implications in this section verify the nonlinear formation hypotheses for every history. They show exactly how the α^2 normalization and the source-gradient gain propagate once those ordinary local hypotheses and the finite pointed estimate hold. The formation interface and the marked alternatives. For the fixed window-first rule, the regular terms of eq. 4.clxxiii are the linear application of eq. 4.clviii–eq. 4.clix. The regular direction is the source derivative of its near member with its own coefficient, as in eq. 4.clxii; the defect is eq. 4.clxx. The common-background and common-base rule in eq. 4.cliii–eq. 4.cliv is kept while taking that derivative. The bare scalar direction is eq. 4.ci, whose $\alpha_k\delta_k = \alpha_k^2 s_k$ factor was obtained before the finite gas expansion.

For the regular near class the small physical radius of the complete field is essential. At age $a = k - j$, use the true ratio δ_k/α_j until the homogeneous window estimate has been applied. The complete background-plus-fluctuation datum, rather than just the increment, obeys that small-radius bound. The all-age half-disc admission in eq. 4.clxxx then permits the quadratic and cubic

gains for ratios less than one. Replacing that ratio by one before estimating the window would lose the desired source normalization. The full-radius amplification and its subsequent Schwarz factor are the ones proved in the window formation argument.

The vacuum-only defect terms use the quadratic window estimate without a scalar-action cancellation. Marked remainder and remote classes require a different bound on their actual source domain. One sufficient finite condition is joint holomorphy and the ordinary estimate

$$\sup_{\substack{|w|\|f\|_{\sharp,j}<r_0\\ P\ \text{admitted}}} |R_j^{wf}(X,P,z)| \le B_j e^{-\kappa d_j(X)} \quad\Longrightarrow\quad \|\dot R_j[f]\|_\kappa \le \frac{2B_j}{r_0}\|f\|_{\sharp,j}. \tag{4.clxxxii}$$

Here r_0 is an actually admitted source radius independent of X, P, z; it is not inferred from a real-source estimate. The norm here is the pointwise exponentially weighted member norm $\sup_{X,P,z} e^{\kappa d_j(X)}|\dot R_j[f](X,P,z)|$, not a rooted domain sum at that same exponent. Cauchy's formula on a circle of radius $r_0/(2\|f\|_{\sharp,j})$ proves the implication. The reference frame, source rule, cutoff and read domain remain fixed on that circle. If the remainder includes a history operation, the corresponding bound is needed in its complete kernel norm, with its inner Wilson weight; a bound at one read datum is insufficient.

There is no extra α_j^2 on the right of eq. 4.clxxxii. Any power or logarithmic loss in using that bound must be carried into the marked age sum. For example, if a later covariance calculation has a geometric volume cost L^{8a}, the elementary sufficient rarity bound is

$$q_a \le C\varepsilon_{\rm bad}\, q^a, \quad L^8 q < 1 \quad\Longrightarrow\quad \sum_{a\ge 0} L^{8a} q_a \le \frac{C\varepsilon_{\rm bad}}{1-L^8 q}. \tag{4.clxxxiii}$$

The geometric sum is immediate; the point is its actual weight and quantifiers. An unweighted small bad-class probability does not supply this age moment. A differentiated one-zone mass test does not supply it either. It is a separate undifferentiated estimate on the allowed classes.

Remote and wrapping terms whose bound is independent of the local field cannot be assigned an invented factor $(\delta_k/\alpha_j)^2$. They must either be excluded by an actual non-wrapping near chart or retained with their own fixed-disc and age estimate. Thus the source recurrence proved above has a concrete formation route under its map hypotheses, while its marked and remote alternatives remain separate finite estimates to verify.

4.5.6. *The old-origin Hessian on the larger run window.* This section proves a conditional ordinary estimate. Its inputs are the complex domains and first-response estimates of the actual background maps, the pointed quadratic cancellation, and the window formation rule. The desired Hessian estimate is not one of the inputs. In particular, an estimate on the Gaussian cutoff box alone is not used as an analytic continuation theorem. We construct the larger fluctuation domain from the unscaled constraint chart and the datum domains of the background maps.

The conclusion concerns the regular old-origin part of a step exponent. The conditional exterior-force estimate, boundary origins with stored fields, and the identification of a clean-class conditional measure are separate issues.

Physical and Gaussian coordinates. Fix a step from level k to level $k+1$. Write

$$g_k = \theta_k^{-1/2}, \qquad \alpha_i \asymp \theta_i^{-1/2}(\log\theta_i)^q, \qquad \mathfrak{b}_k \asymp (\log\theta_k)^p, \qquad \delta_k = g_k\mathfrak{b}_k. \tag{4.clxxxiv}$$

Here $q > p$, and α_i is the smaller of the usable physical radii in the datum, potential, curvature and current conditions. Their fixed ratios are included in the constants. A finite alteration of an index in α_i, as needed for adjacent input layers, has the same meaning. We assume the ordinary flow gives

$$\rho_i := \max\{1, C\alpha_k/\alpha_i\} \le C_\rho(1+k-i)^{1/2}, \qquad \sigma_i = L^{-(k-i)}, \qquad i \le k. \tag{4.clxxxv}$$

Constants depend on the fixed block and collar choices, but not on the number of old levels, the cutoff or the volume. In particular, $C_{\mathrm{a}} := \sup_{i\le k} \rho_i\sigma_i < \infty$.

The free Gaussian variable y_b has three real color coordinates at each nonpivot bond of the level-k lattice. The physical bond logarithm is of size $g_k y_b$. Thus a physical analytic radius α_k corresponds to a Gaussian radius of order α_k/g_k, rather than to α_k. For complex variables we use the Euclidean norm of the three complex components at a bond and its supremum over bonds.

Let I be the finite box of free bonds under consideration. Its outside free coordinates and other exterior data are fixed. Every real free coordinate appearing in the regular fields is bounded by $C_{\mathrm{w}}\mathfrak{b}_k$. The number of coordinates satisfies

$$m_I = 3\#I \le C(rL^{-a_*} + D)^4 \le C(r + D)^4. \tag{4.clxxxvi}$$

This count will not enter the localized estimate below. It is relevant to the weaker estimate available without spatial locality.

The primitive analytic inputs. We specify the domain inputs used in the proof. These are statements about assigned maps and their first responses, not bounds on the Hessian of an old activity.

For the actual one-step average, at its actual background $V^{\mathrm{bg}}(W)$, put

$$q_W(a) = \frac{1}{i}\log\Big[\mathcal{M}(e^{ia}V^{\mathrm{bg}}(W))W^{-1}\Big] = LQ_W a + \mathcal{R}_W(a). \tag{4.clxxxvii}$$

The logarithm is taken on its assigned local branch. The fixed-pivot construction supplies linear maps C_W and h_W with

$$LQ_W\mathsf{C}_W = 0, \qquad LQ_W h_W = 1, \qquad \|\mathsf{C}_W\|_{\infty\to\infty} + \|h_W\|_{\infty\to\infty} \le C_L. \tag{4.clxxxviii}$$

The map C_W is the pivot elimination map, not a Gaussian covariance. The maps in eq. 4.clxxxvii–eq. 4.clxxxviii are the ones used in the run. Uniformly on an unscaled physical ball $\|a\|_\infty < r_0$, the remainder is holomorphic and obeys

$$\|\mathcal{R}_W(a)\|_\infty \le C_0\|a\|_\infty^2, \qquad \|D\mathcal{R}_W(a)\|_{\infty\to\infty} \le C_0\|a\|_\infty. \tag{4.clxxxix}$$

Here r_0, C_0, C_L are independent of the box size. The same bounds, with fixed enlarged constants, apply to the assigned local charts on the independent fine-parameter discs $|s_\Delta| < e^{\bar\kappa}$. At physical parameters the local charts agree with the run chart by their restriction and completeness identities.

The ordinary background package supplies holomorphic maps on their declared datum balls and their declared core slices. The datum radii satisfy $r_{B,i} \ge c\alpha_i$. At zero fluctuation all substituted data and pairs have a fixed positive fraction ϵ_0 of room in these domains. This room includes the tower conditions, node representatives, current bounds, and the inner, outer and removed substitutions. These are local complex assertions; they do not assert the existence of a global Gaussian measure for independent currents.

We use the following precise first-response form of that package. Measure each defining datum or core displacement at old level i in units of its allowed radius. Along an admitted physical perturbation of supremum size u, its derivative in any of these normalized quantities is bounded by

$$C_R\sigma_i\alpha_i^{-1}u. \tag{4.cxc}$$

The bound holds throughout the declared complex domains, for all the fine parameters above. It is the integrated first-response and nonlinear-average estimate: the actual average, not its differential, is used to construct the datum. The constant C_R includes the fixed enlarged parameter discs and the chart derivative. For outer and removed derivatives there are additionally their assigned exponential distance factors. As usual, a compatible boundary comparison and the weighted Hölder response estimates are assumed for the actual box backgrounds.

Finally, the field maps are hereditary on active wall components. After the parameters outside a domain Y are set to zero, a local potential uses only the free coordinates in Y. A fixed collar can

instead be included in Y throughout; it changes the constants below. The number of domains of tree length in $[m, m+1)$ containing a fixed cube is at most $C_D e^{\kappa_D m}$, and

$$\operatorname{diam} Y \le C_M(1 + d_k(Y)). \tag{4.cxci}$$

The spatial rates have the strict reserves needed to sum these domains and the fixed polynomial diameter factors.

The actual chart on the larger ball. For clarity we give the contraction argument behind the radius claim. Put $z = g_k \mathsf{C}_W y$. The nonlinear correction is the solution of

$$d = \mathcal{R}_W(z - h_W d), \qquad a_W(y) = z - h_W d, \qquad V_W(y) = e^{i a_W(y)} V^{\mathrm{bg}}(W). \tag{4.cxcii}$$

Choose $r_1 > 0$ so small that $2r_1 < r_0$ and $4C_0 C_L r_1 < 1/2$. For $\|z\|_\infty \le r_1$, the right side maps the ball $\|d\|_\infty \le 4C_0\|z\|_\infty^2$ to itself. Indeed $\|z - h_W d\|_\infty \le 2\|z\|_\infty$, and eq. 4.clxxxix then gives that bound. Its Lipschitz constant in d is at most $2C_0 C_L r_1 < 1/2$. Iteration from zero therefore converges uniformly to the unique solution, including locally uniformly in all complex parameters. Uniform limits of the holomorphic iterates are holomorphic. Differentiating the fixed-point equation, or its uniformly convergent derivative iteration on a smaller ball, gives

$$\|a_W(y)\|_\infty \le C g_k \|y\|_\infty, \qquad \|D_y a_W(y)\|_{\infty\to\infty} \le C g_k. \tag{4.cxciii}$$

Moreover, $q_W(a_W(y)) = 0$, because $LQ_W z = 0$ and $LQ_W h_W d = d$. This is the actual constraint, with its nonlinear average. Uniqueness shows that this chart agrees on the operational cutoff box with the chart already used in the run.

There is consequently a fixed $c_* > 0$ for which all the old substitutions are defined and holomorphic on

$$\|y\|_\infty < R_k, \qquad R_k = c_* \alpha_k / g_k. \tag{4.cxciv}$$

Here and below fixed exterior coordinates may be retained with norm at most $C_{\mathrm{w}}\mathfrak{b}_k$. To prove the assertion, first make α_k small enough that eq. 4.cxciii stays in its physical chart. For any fixed complex y in eq. 4.cxciv, follow the segment ty, $0 \le t \le 1$. On the portion where a composition of assigned background maps is admitted, eq. 4.cxc and eq. 4.cxciii bound each normalized displacement by

$$C_R g_k \|y\|_\infty \sigma_i/\alpha_i \le C_R c_* \rho_i \sigma_i \le C_R c_* C_{\mathrm{a}}. \tag{4.cxcv}$$

Choose c_* so that the last expression is less than $\epsilon_0/4$. The segment cannot reach a domain boundary: every defining inequality still has at least half of its initial room. Atomic background maps are already given on those domains, so this is an inclusion argument for a composition of assigned holomorphic maps, not an assertion that a branch continues through an unknown singularity. The same argument gives the required joint holomorphy with the fine parameters. It also applies to local independent-current formulas without postulating an off-graph global Gaussian integral.

The window multiplier used to differentiate an inner direction can range on a fixed disc $|\nu| < c_1/c_*$, where c_1 is chosen from the same room estimates and $c_* < c_1/4$. In fact its normalized increment is bounded by $C_R|\nu| c_* \rho_i \sigma_i$. The unamplified part costs only $C_R c_* C_{\mathrm{a}}$; choose c_1 so that their sum still spends less than half the room. Thus this disc contains $\nu = 1$ and permits the window derivative at zero.

Since $q > p$, increasing the coupling floor gives $C_{\mathrm{w}}\mathfrak{b}_k < R_k/4$. Two coordinate Cauchy circles with radii cR_k, centered anywhere on the real product window, therefore fit in eq. 4.cxciv. There is no factor depending on m_I in this inclusion. No indicator function or cutoff is analytically continued. Only the activity functions are evaluated on the larger ball; the real measure, Gaussian scaling and all numerical choices of the run remain fixed.

An amplitude bound before the Gaussian-growth norm. Let the regular member at old level i have norm e_i, with $e_i \le E_0$. The norms include its infinite companion if one is required. Each pointed total has local gauge invariance, constant color invariance and the full site symmetry needed for the pointed Hessian theorem. At every old base point subtract its vacuum and its own extracted coefficient times its own tent action. The tent has unit quadrature; the action convention is $A(e^{iB}) = \frac{1}{2}|\mathrm{d}B|^2 + O(B^3)$. This is the regular part of the decomposition: its counterterms have been transferred to the running Wilson coefficient. A profile deficit is retained as a lag or boundary origin, rather than silently subtracted from the measure. The clean interior has no such deficit in the canonical decomposition with redrawn profiles. Thus the affine quadratic term cancels with coefficient one in that normalization. We do not replace the pointed object by a bulk convolution or assign an action coefficient to an individual domain.

We recall explicitly the value estimate supplied by this cancellation. Write $D_X = 2 + \mathrm{diam}_i X$. On an inner box background the affine part and its error have sizes bounded, with fixed powers of D_X, by

$$\alpha_k \sigma_i^2, \qquad \alpha_k \sigma_i^{2+\alpha_H} + r_{B,i}^{-1} \alpha_k^2 \sigma_i^4, \qquad 0 < \alpha_H \le 1. \tag{4.cxcvi}$$

These estimates concern the complete substituted field, not just its fluctuation increment. They hold on the whole disc just constructed. Second and third derivatives of a member cost $Ce_i\alpha_i^{-2}$ and $Ce_i\alpha_i^{-3}$, with its original tree decay. Its linear derivative at the trivial pair is zero by constant color invariance.

Set $T = CD_X^s \rho_i \sigma_i^2$. When T is small, Taylor's formula, the cancellation of the affine quadratic value on the pointed total, and $r_{B,i} \ge c\alpha_i$ bound the remaining terms by $Ce_i D_X^{s'}[\rho_i^2 \sigma_i^{4+\alpha_H} + \rho_i^3 \sigma_i^6]$. For example, the product of the affine part and its error, divided by α_i^2, gives exactly these two powers. Squares of the error and cubic terms are bounded by the same expression, since their normalized sizes are bounded by $C(T\sigma_i^{\alpha_H} + T^2)$ and CT^3. If T is not small, whole-core bounds for the member and its own action, and the comparison affine quadratic value, give $Ce_i(1+T^2) \le Ce_i C_T T^3$. The fixed powers of D_X are summable against the tree decay. A fixed finite set of young ages failing the central half-datum-ball test is also covered by the whole-core bound: their σ_i have a fixed positive lower bound. The separate remote-datum admission is retained. This is precisely the reason to use the ordinary full-radius ratio $\rho_i \ge 1$ here.

At one base point, after its domain sum, the bound is therefore

$$Ce_i \left[\rho_i^2 \sigma_i^{4+\alpha_H} + \rho_i^3 \sigma_i^6\right]. \tag{4.cxcvii}$$

A current window contains at most $C\sigma_i^{-4}$ old base points. Its value bound is at most $Ce_i\rho_i^3\sigma_i^{\alpha_H}$. Terms extending beyond that window retain an exponentially small tree-length tail. Outer and removed terms retain their actual response factors. Even the cruder amplitude $Ce_i\sigma_i^{-4}e^{-c/\sigma_i}$, when only a long-domain value bound is used, is bounded by $Ce_i\rho_i^3\sigma_i^{\alpha_H}$. This does not give such a tail a square of the actual small curvature; it is a full analytic-radius estimate.

It is important where the inner estimate is localized. At each old base point form one window function from the near pointed sum, its own action counterterm, and one common critical box map. This entire function has one fixed current-scale base, of uniformly bounded size. Only then apply the fine brackets of the window formation rule. Thus the cancellation in eq. 4.cxcvii occurs before a single common bracket. No sup norm of a sum is used to bound a sum of differently based bracket operators.

On the independent fine discs the bracket with m differenced cubes costs at most $(e^{\bar\kappa}-1)^{-m}$, with no factorial. The derivative at $\nu = 0$ costs the reciprocal of its fixed disc, hence only a fixed constant. For outer origins retaining their original base, split the old input rate into the domain-sum, long-tail and retained parts. If Y is the resulting current domain, the hull and rooting geometry gives $d_k(Y) \le C_h(d_i(X) + m + 1)$; for a bounded window base it gives $d_k(Y) \le C_h(m+1)$. Choose the retained rates and $\bar\kappa - 1$ larger than $C_h(\lambda + \varepsilon)$, after spending the old rooted entropy and diameter

powers. The remaining ε absorbs the polynomial number of window anchors at a fixed Y. This proves the following pointwise analytic bound for the actual chosen window filing:

$$|V^{\mathrm{old}}_{k,i}(Y;y)| \le Ce_i\omega_{k+1-i}e^{-\lambda d_k(Y)}, \qquad \|y\|_\infty < R_k, \quad \omega_{a+1} = C(1+a)^{3/2}L^{-\alpha_H a}. \tag{4.cxcviii}$$

The current origin $i = k$ uses its direct whole-core Cauchy bound and $\omega_1 = O(1)$; it needs no age improvement. The proof was carried out on the larger ball using a physical fluctuation amplitude at most $c_*\alpha_k$, rather than the tested amplitude δ_k. It does not use the final potential bound with a Gaussian-growth factor $1 + M^{-4}\sum_{b\subset Y}|y_b|^2$. Using that weaker envelope on a radius R_k would introduce R_k^2 and erase the desired Cauchy gain.

Let

$$\mathcal{E}_k = \sum_{i\le k}\omega_{k+1-i}e_i \le CE_0. \tag{4.cxcix}$$

Absolute convergence is uniform on every smaller ball in eq. 4.cxciv, by eq. 4.cxcviii. It justifies summing old levels before taking two coordinate derivatives. All estimates can first be made with finitely many old levels and on a finite torus. Their constants do not change in either limit.

Cauchy, rooted counting and the weighted precision gap. Write $P^{\mathrm{old}}_{k,I}(y) = \sum_Y\sum_{i\le k}V^{\mathrm{old}}_{k,i}(Y;y)$, with exterior variables fixed. Additive terms depending only on the exterior can be omitted. By two coordinate Cauchy formulas, including the factor 2! for a repeated coordinate, eq. 4.cxcviii gives

$$\left|\partial_{y_b^a}\partial_{y_{b'}^{a'}}\sum_{i\le k}V^{\mathrm{old}}_{k,i}(Y;y)\right| \le C\mathcal{E}_kR_k^{-2}e^{-\lambda d_k(Y)} \tag{4.cc}$$

throughout the real product window. The derivative is zero unless both bonds belong to the declared domain, including its fixed collar. For such a domain, $d(b,b') \le C_M(1+d_k(Y))$. Choose $\mu > 0$ so small that $\lambda - \mu C_M > \kappa_D$, retaining a strict gap. Root the domains at a cube containing b; the fixed number of possible root cubes costs only a constant. Then

$$\sum_{Y\ni b,b'}e^{-\lambda d_k(Y)} \le Ce^{-\mu d(b,b')}\sum_{m\ge 0}e^{-(\lambda-\mu C_M-\kappa_D)m} \le C'e^{-\mu d(b,b')}. \tag{4.cci}$$

Consequently the full old-activity Hessian, in the actual Gaussian coordinates, satisfies

$$\|D_bD_{b'}P^{\mathrm{old}}_{k,I}(y)\| \le C\mathcal{E}_k\left(\frac{g_k}{\alpha_k}\right)^2e^{-\mu d(b,b')} \le CE_0(\log\theta_k)^{-2q}e^{-\mu d(b,b')}. \tag{4.ccii}$$

Color matrix norms cost a fixed factor only. The count of bonds at distance in $[r, r+1)$ is at most $C(1+r)^3$. Thus for every $0 < \mu' < \mu$, both weighted row and column sums of the Hessian are at most $C\mathcal{E}_k(g_k/\alpha_k)^2$. In particular, Schur's inequality gives

$$\|T_\phi D^2P^{\mathrm{old}}_{k,I}(y)T_\phi^{-1}\|_{2\to 2} \le C\mathcal{E}_k(g_k/\alpha_k)^2, \qquad |\phi(b)-\phi(b')| \le \mu' d(b,b'), \tag{4.cciii}$$

where T_ϕ multiplies the three coordinates at b by $e^{\phi(b)}$. This estimate is uniform in I.

For completeness, suppose the actual free precision A, after the prescribed constraint and gauge fixing, has a gap $A \ge a_-\mathbf{1}$ and uniform exponentially summable entries. This is a separate free-background input, concerning the constrained pullback Hessian in the coordinates C_W, not the ambient curl Hessian. Its Dirichlet compression A_{OO} has the same gap. For small Lipschitz weights its conjugation difference has norm at most $C\mu'$. Hence the conjugated inverse of A_{OO} has norm at most $2/a_-$ when $C\mu' < a_-/2$, by the Neumann series. Applying this to $\phi(b) = \mu' d(b,b')$ bounds each inverse entry by $(2/a_-)e^{-\mu' d(b,b')}$. This proof is uniform in O. It follows by two exponentially summable convolutions that

$$K_I = ((A^{-1})_{II})^{-1} = A_{II} - A_{IO}A_{OO}^{-1}A_{OI}$$

has exponentially summable entries at a possibly smaller rate. Furthermore, $\langle v, K_Iv\rangle = \inf_w\langle(v,w), A(v,w)\rangle \ge a_-\|v\|^2$. Thus the inverse marginal covariance, with its actual Schur correction, retains the free gap. Its conjugation difference also has norm at most $C\mu'$: use

$|e^{\phi(b)-\phi(b')}-1| \le \mu' d(b,b') e^{\mu' d(b,b')}$ and the row and column sums. Choose μ' so that this norm is at most $a_-/4$. Then

$$\mathrm{Re}\langle v, T_\phi (K_I - D^2 P^{\mathrm{old}}_{k,I}) T_\phi^{-1} v\rangle \ge \left(\frac{3a_-}{4} - C\mathcal{E}_k (g_k/\alpha_k)^2\right) \|v\|^2. \tag{4.cciv}$$

For a sufficiently large coupling floor the old activities cost at most $a_-/4$, leaving a weighted gap of at least $a_-/2$. Other intrinsic or remainder Hessians must be charged separately. The reference precision is held fixed while differentiating in y; no derivative of a covariance with respect to y is introduced.

Remainders and the weaker finite-box alternative. A holomorphic aged remainder with no fixed color spectators has zero linear term at the trivial pair, but has no action counterterm. Its vacuum-only window bound has weight ρ_i^2 instead of ω_{k+1-i}. The same proof gives eq. 4.ccii with $\mathcal{E}_k$ augmented by $\sum_i \rho_i^2 r_i$, provided its actual maps have the same domains and locality. If $r_i \le C\theta_i^{-s}$, $s > 2$, and $\theta_i \ge \theta_k + c(k-i) - C_0$, use the actual running radii to bound $\rho_i^2 \le C\theta_i/\theta_k$. Indeed the squared radius ratio is bounded by this expression times $(\log\theta_k/\log\theta_i)^{2q}$, which stays bounded on the admitted flow. The maximum with one has the same bound. Thus

$$\sum_i \rho_i^2 r_i \le C\theta_k^{-1} \sum_{a\ge 0} (\theta_k + ca - C_0)^{1-s} \le C_s \theta_k^{1-s}. \tag{4.ccv}$$

For the last inequality use the integral comparison with $\theta_k - C_0 \ge \theta_k/2$. This optional conclusion does not cover a boundary term retaining stored color variables or a term for which only an integrated norm is known.

If only a bound on the whole finite-box analytic function is available, a useful but weaker statement remains. Suppose $|P(y)| \le CE_0(r+D)^A$ on a Gaussian polydisc of radius $c\alpha_k/[g_k(r+D)^B]$. The same coordinate Cauchy estimate and $m_I \le C(r+D)^4$ give

$$\|D^2 P\|_{2\to 2} \le CE_0 (r+D)^{A+2B+4} (g_k/\alpha_k)^2. \tag{4.ccvi}$$

Indeed each matrix entry has the bound without the last power $(r+D)^4$, and the row and column contain at most m_I entries. For a direct volume bound $A=4, B=0$, this is $CE_0(r+D)^8(\log\theta_k)^{-2q}$. When $r + D \le C(\log\theta_k)^{2p}$, it tends to zero if $q > 8p$. It preserves an unweighted gap. It does not imply eq. 4.cciii at a fixed positive spatial weight: conjugation could cost $e^{\mu' \operatorname{diam} I}$. The rooted localization in eq. 4.cxcviii–eq. 4.cci proves the stronger weighted conclusion; counting window coordinates alone would not suffice.

The old-activity Hessian estimate and its scope. Under the stated ordinary analytic map hypotheses and the reviewed pointed and window inputs, the old regular-activity contribution to the localized Hessian is bounded by $CE_0(\log\theta_k)^{-2q}$, uniformly over old ages and box sizes. The analytic radius is in the physical-to-Gaussian coordinate conversion in eq. 4.cxciv. A bare operational cutoff domain, or only a Gaussian-growth potential bound on that domain, would not prove it. The two required primitive verifications are the uniform unscaled pivot estimates in eq. 4.clxxxviii–eq. 4.clxxxix and the all-domain first-response and membership estimates in eq. 4.cxc, including towers and outer substitutions. They belong to the actual ordinary background analysis and have not been proved for the Yang–Mills maps merely by this deduction.

The statement has no automatic consequence for the change in force when exterior data are changed. For that one needs, separately, uniform estimates such as $\|D_b(P_e - P_{e_0})\| \le C\mathfrak{b}_k e^{-cd(b,O)}$ on the actual compatible exterior and stored data. A Hessian bound in the variables of I does not bound derivatives with respect to the exterior parameters. Nor does it identify a clean-class restriction with a Gaussian fibre kernel. Field-independent remote or holonomy bounds without the analytic domain assertion used above remain separate tails; they receive neither an invented curvature factor nor an unjustified Cauchy estimate.

An analytic criterion for the old exterior force. The larger fluctuation ball controls derivatives in its free coordinates. A useful further reduction states precisely what function-level locality would also control a change of exterior datum. This reduction concerns regular old members, not a conditional class probability or a boundary measure.

Let $O_{e,Y}(y)$ be the actual old-origin member on a fixed run window, where e is a compatible exterior datum. Assume both e and a comparison datum e_0 give admitted functions on the same full complex ball $\|y\|_\infty < R$, with the same local filing. Let O be the exterior region where they differ. Suppose the actual background/boundary comparison yields the amplitude difference

$$|O_{e,Y}(y) - O_{e_0,Y}(y)| \le \varepsilon_\partial e^{-\lambda d(Y)} e^{-c\,\mathrm{dist}(Y,O)} \quad (\|y\|_\infty < R). \tag{4.ccvii}$$

This is a uniform function bound on the entire analytic ball. A comparison on the real cutoff box alone is not this premise.

Assume the actual supports have diameter at most $C_h(1 + d(Y))$, their free-coordinate counts are at most $C_h(1 + d(Y))^q$, and the number containing a fixed coordinate with $m \le d(Y) < m + 1$ is at most $C_D e^{\kappa_D m}$. When b belongs to Y,

$$\mathrm{dist}(b, O) \le C_h(1 + d(Y)) + \mathrm{dist}(Y, O). \tag{4.ccviii}$$

Choose $0 < c' < c$, $c'' > 0$ and a strict $\eta > 0$ so that $\lambda - C_h(c' + c'') > \kappa_D + \eta$. Such choices exist when $\lambda > \kappa_D$. On $\|y\|_\infty < R/2$, coordinate Cauchy in the common larger ball gives

$$\left| D_b \sum_Y [O_{e,Y} - O_{e_0,Y}](y) \right| \le \frac{C\varepsilon_\partial}{R} e^{-c'\mathrm{dist}(b,O)}. \tag{4.ccix}$$

Indeed only supports containing b contribute. For each such support Cauchy costs $2/R$. Using eq. 4.ccviii, its exponential weight is at most $e^{c'C_h} e^{-(\lambda - c'C_h)d(Y)} e^{-c'\mathrm{dist}(b,O)}$. The rooted count is then a convergent geometric sum. The estimates are uniform in finite volume, so the absolutely convergent sum and its derivative have the same limit on every smaller ball.

Two coordinate derivatives, with the same reasoning, also give the weighted row bound

$$\sum_{b'} e^{c''\mathrm{dist}(b,b')} \left| D_b D_{b'} \sum_Y [O_{e,Y} - O_{e_0,Y}](y) \right| \le \frac{C\varepsilon_\partial}{R^2} e^{-c'\mathrm{dist}(b,O)}. \tag{4.ccx}$$

For coincident coordinates the second Cauchy estimate has the factor 2!; for different coordinates use two independent discs. Both are bounded by a fixed constant times R^{-2}. For a given Y, summing b' costs at most its polynomial coordinate count times $e^{c''C_h(1+d(Y))}$. The strict rate inequality above leaves $e^{-\eta d(Y)}$, which absorbs that polynomial. This proves the asserted row bound, with constants independent of the number of window variables.

For the run-window radius $R = c\alpha_k/g_k$, these are explicit Cauchy costs in that same radius. If only R/r^4 has been admitted, the costs become r^4/R and r^8/R^2; they are not discarded. The remaining model question for eq. 4.ccix is the actual common-domain amplitude comparison in eq. 4.ccvii, with its compatible exterior, stored fields, gauges and relative boundary data. The interior Hessian estimate alone does not furnish that comparison. Even this regular old-origin force bound would leave the other terms of a clean conditional exponent, its boundary contributions and its normalized class law to be identified.

4.5.7. *Finite histories and the local formation norm.* The local estimates above are conditional on all admitted stored and boundary data. A finite history operation is an additional integral, not an extra name for the local Gaussian functional. We retain its read variables and its inner Wilson weight explicitly.

Let ω range over the discrete finite history records. Write ξ for the stored variables integrated by this history operation and P for the remaining admitted read data. Suppose the operation has been expressed over one parameter-independent positive reference measure $d\mu_\omega(\xi)$. Its complex integrand is $K_\omega(\xi; P)$. If a source derivative also inserts a local function $A_\omega(\xi; P)$, the relevant

estimate is on the product $K_\omega A_\omega$. One cannot first replace K_ω by its partition function and then take a supremum of A_ω on different fibres.

For $c \geq 0$, fixed history volume v_ω, and fixed read cores $\mathcal{C}_\omega$, define

$$\|K\|_c = \sum_\omega \int^* d\mu_\omega(\xi)\, e^{cv_\omega} \sup_{P \in \mathcal{C}_\omega} |K_\omega(\xi; P)|. \tag{4.ccxi}$$

The upper integral permits a supremum whose measurability has not been established; the same core and reference operation are used on both sides of every estimate. If $M_\omega(\xi)$ is a common positive envelope and

$$\sup_{P \in \mathcal{C}_\omega} |K_\omega(\xi; P) A_\omega(\xi; P)| \leq M_\omega(\xi)\, \varepsilon e^{-\lambda d(X)}, \qquad \sum_\omega \int d\mu_\omega\, e^{cv_\omega} M_\omega \leq C_H, \tag{4.ccxii}$$

then the norm of the localized insertion is at most $C_H \varepsilon e^{-\lambda d(X)}$. This follows directly from monotonicity of the upper integral and the sum over records. It requires no exchange of a supremum and an integral.

The inner Wilson factor belongs inside K_ω. On a real fibre it may supply positive action suppression, while its complex continuation must be controlled by the kept precision and the actual action reserve. It is not replaced by Haar measure alone. The weighted history argument supplies a route to the envelope in eq. 4.ccxii; the unmarked Gaussian polymer bound, by itself, supplies only a local bound after the history has been fixed.

For a finite list of records and a finite-dimensional source parameter, one may derive derivatives under this operation as follows. Suppose each kernel is pointwise holomorphic on a source disc, the operations and read cores are fixed, and a jointly measurable common envelope bounds its field supremum on every smaller source circle. Cauchy's formula, Tonelli's theorem for that envelope, and dominated convergence then give the kernel derivative and its norm bound. For an infinite record sum one needs the corresponding summable envelope or Banach-norm holomorphy in eq. 4.ccxi. Pointwise partition holomorphy alone does not establish that assertion.

Thus the finite local formation proof and the history-mass proof have distinct conclusions. The former gives the derivative in a small potential direction, with the actual bond rooting and own counterterm. The latter must pay the norm over stored fields and the sum of history records. Their composition is justified by a common envelope such as eq. 4.ccxii, not by dropping the retained Wilson factor or replacing an integral of a supremum by a pointwise polymer estimate.

4.6. **The actual Gaussian mismatch and its zero coefficient.** The action weight and the reference gauge weight need not coincide. Their ratio contributes a Gaussian term of order $|\tau - 1|$, which can be much larger than g. This term has an exact total at a physical constrained-critical background. The argument uses the actual contour gauge and the complete Hessian on the nonlinear averaging fibre. It explains why this Gaussian term is retained separately in an absolute-radius formation estimate.

4.6.1. *The averaging fibre and its residual gauges.* Consider one step on a periodic four-dimensional lattice, with odd block size $L \geq 3$. Let n be the number of coarse sites. There are $L^4 n$ fine sites. The symmetric contour average $\mathcal{M}$ is the map in eq. 2.iv. Its means and logarithms are taken on their admitted small branches. Fix a coarse field W.

Write $\mathcal{S}(V)$ for the physical action being expanded on the fine-step field V. At the first step it is the Wilson action. At a homogeneous later step it is the actual effective background action $A(U_k(V))$. In the latter case its gauge covariance, differentiability, and the asserted critical background are hypotheses concerning that actual background; the algebra below does not construct that background.

Let V_0 be a constrained-critical point on $\mathcal{M}(V) = W$. Use the actual nonpivot chart

$$V(u) = e^{i\Phi(u)} V_0, \qquad \mathcal{M}(V(u)) = W, \qquad \Phi(0) = 0, \quad D\Phi(0) = C, \tag{4.ccxiii}$$

where $u \in E \cong \mathbb{R}^d$ is the list of free nonpivot colour coordinates. One three-colour pivot is eliminated for each coarse bond. Hence

$$d = 12(L^4 - 1)n. \tag{4.ccxiv}$$

The chart is assumed to cover a neighborhood of the critical point, with the actual branch fixed. No assertion about a large history chart is made here.

A residual site gauge is a function v equal to one at every coarse node. Gauge covariance of the contour formula gives $\mathcal{M}(V^v) = \mathcal{M}(V)$, so these gauges preserve the actual fibre, including its nonlinear constraints. Their infinitesimal parameters form

$$\mathfrak{h} = \{\lambda : \lambda(y) = 0 \text{ at every coarse node } y\}, \qquad r := \dim \mathfrak{h} = 3(L^4 - 1)n. \tag{4.ccxv}$$

For a site $x \in B(y)$, let $T_{y,x}(V)$ be the actual symmetric contour transport from the block node y to x. It obeys

$$T_{y,x}(V^v) = v(y)T_{y,x}(V)v(x)^{-1}. \tag{4.ccxvi}$$

Indeed every path has this endpoint transformation, and uniqueness of the admitted symmetric mean gives the same law for that mean. Choose the block-axial representative with $T_{y,x}(V_0) = 1$. Such a representative is obtained by the single residual gauge $v(x) = T_{y,x}(V_0)$, with $v(y) = 1$. Define the actual local gauge coordinate

$$\ell_x(V) = \frac{1}{i} \log T_{y,x}(V), \qquad x \neq y, \qquad \ell(V_0) = 0. \tag{4.ccxvii}$$

These contours lie within their blocks and use no pivot crossing a block interface.

Let $T : \mathfrak{h} \to E$ be the tangent to the residual gauge orbit in the free coordinates of eq. 4.ccxiii, and let $G = D_u \ell(V(u))|_0$. At $v_t(x) = e^{it\lambda(x)}$, eq. 4.ccxvi gives $T_{y,x}(V_0^{v_t}) = e^{-it\lambda(x)}$. Therefore the exact linear identity is

$$GT = -I_{\mathfrak{h}}. \tag{4.ccxviii}$$

It is independent of the depth of the homogeneous background and does not require an estimate for an inverse Hessian.

4.6.2. *The complete physical Hessian.* Put $S(u) = \mathcal{S}(V(u))$. Constrained criticality means $DS(0) = 0$. The physical action Hessian is

$$K_A = D_u^2 S(0). \tag{4.ccxix}$$

It contains the second derivative of the constraint chart. To make this explicit, write $q(a) = \frac{1}{i} \log[\mathcal{M}(e^{ia}V_0)W^{-1}]$, $P = Dq(0)$, and take the actual pivot right inverse h, with $Ph = 1$. Differentiating $q(\Phi(u)) = 0$ gives

$$D^2\Phi(0)[u, v] = -hD^2q(0)[Cu, Cv]. \tag{4.ccxx}$$

The ambient action gradient satisfies $D\mathcal{S}(0) = P^{\mathsf{T}}\mu$, since it annihilates $\ker P$. Thus

$$K_A[u, v] = D^2\mathcal{S}(0)[Cu, Cv] - \langle \mu, D^2q(0)[Cu, Cv] \rangle. \tag{4.ccxxi}$$

The second term is part of the action Hessian, rather than a small potential to be discarded before the Gaussian split.

Lemma 4.11 (Gauge null directions of the constrained Hessian)**.** The complete Hessian satisfies $K_A T = 0$. Let the quadratic reference gauge fixing be $K_G = G^{\mathsf{T}} a G$, where $a > 0$ on $\mathfrak{h}$. Then

$$E = \ker G \oplus \operatorname{ran} T, \qquad K_A \operatorname{ran} T = 0, \qquad K_G \ker G = 0. \tag{4.ccxxii}$$

The two summands need not be orthogonal in the original free Euclidean coordinates; they are orthogonal for both quadratic forms in the displayed sense. ∎

Proof. The chart transports the residual gauge action to a local action $u \mapsto h_\lambda(u)$ on the exact fibre. Gauge invariance gives $S(h_{t\lambda}(u)) = S(u)$. Differentiate first in t, then in a free direction v, at $t = 0, u = 0$. The term involving the derivative of the gauge tangent is multiplied by $DS(0) = 0$. The remaining term is $K_A[v, T\lambda] = 0$. This uses the Hessian of S, including eq. 4.ccxxi.

For any $u \in E$, write $u = (u + TGu) - TGu$. The first term lies in $\ker G$ by eq. 4.ccxviii; the second is a gauge tangent. If $T\lambda \in \ker G$, then $0 = GT\lambda = -\lambda$, so the sum is direct. The assertions about K_G follow from its Gram form. □

When V_0 is a real constrained minimum, $K_A \geq 0$. The reference Gaussian requires

$$K := K_A + K_G > 0. \tag{4.ccxxiii}$$

This nondegeneracy is retained explicitly. The gauge argument does not prove positivity on the physical sector $\ker G$. The actual reference gauge-fixing function has zero first derivative at V_0, because all its contour variables equal one. Its Hessian is exactly $G^{\mathsf{T}} a G$; no chart curvature term is missing from that gauge Hessian.

4.6.3. *The determinant and the compensating expectation.* Let $\tau = x/\theta$ with $\mathrm{Re}\,\tau > 0$, and retain the reference gauge weight θ. The complete quadratic exponent is

$$Q_\tau = \tau K_A + K_G, \qquad -\tfrac{1}{2} u^{\mathsf{T}} Q_\tau u = -\tfrac{\tau}{2} u^{\mathsf{T}} K u + \tfrac{\tau-1}{2} u^{\mathsf{T}} K_G u. \tag{4.ccxxiv}$$

All determinant roots and $\log \tau$ below are continued from $\tau = 1$ in the right half-plane.

Proposition 4.12 (The actual mismatch normalizer)**.** Under the preceding physical hypotheses,

$$\det Q_\tau = \tau^{d-r} \det K, \tag{4.ccxxv}$$

$$\int e^{-u^{\mathsf{T}} Q_\tau u/2}\, du = \tau^{-(d-r)/2} \int e^{-u^{\mathsf{T}} K u/2}\, du, \tag{4.ccxxvi}$$

$$\int \exp\left(\tfrac{\tau-1}{2} u^{\mathsf{T}} K_G u\right) d\mu_{(\tau K)^{-1}}(u) = \tau^{r/2}. \tag{4.ccxxvii}$$

Here $d\mu_{(\tau K)^{-1}}$ is the normalized full Gaussian on E. It has no fluctuation cutoff. The two normalizing contributions are separately $-\frac{d}{2}\log\tau$ and $\frac{r}{2}\log\tau$; they do not cancel. ■

Proof. Choose bases in the two sectors of eq. 4.ccxxii, and combine them into one invertible change of coordinates S, independent of τ. In these coordinates

$$S^{\mathsf{T}} K_A S = \begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix}, \qquad S^{\mathsf{T}} K_G S = \begin{pmatrix} 0 & 0 \\ 0 & B \end{pmatrix}, \qquad A > 0,\ B > 0.$$

The dimensions are $d - r$ and r. The factors $(\det S)^2$ cancel in the determinant ratio, giving eq. 4.ccxxv. Real Gaussian integration gives eq. 4.ccxxvi for real $\tau > 0$. For complex τ in the right half-plane, $\mathrm{Re}\, Q_\tau$ is positive:

$$\mathrm{Re}\, Q_\tau = (\mathrm{Re}\,\tau) K_A + K_G \geq \min(\mathrm{Re}\,\tau, 1) K.$$

This gives absolute convergence and holomorphic integration locally uniformly in that half-plane, so the same identity continues there. Equivalently integrate the two displayed blocks directly. Normalizing the Gaussian with precision τK contributes $\tau^{d/2}$; multiplication by eq. 4.ccxxvi gives eq. 4.ccxxvii. □

Corollary 4.13 (The mismatch has zero physical coefficient)**.** Suppose the localized mismatch family $\mathbf{G}_\tau$ is the actual free-potential leg of the Gaussian expansion, so its sum on a complete physical pair is the logarithm of eq. 4.ccxxvii. Then

$$\sum_X \sum_{z \in X} \mathbf{G}_\tau(X, \mathbf{p}_W, z) = \tfrac{r}{2} \log \tau. \tag{4.ccxxviii}$$

This total is independent of W. When the local Hessians converge normally and use the actual physical response and pointing, their bulk polarization and coefficient satisfy

$$\Pi^{\mathbf{G}_\tau} = 0, \qquad b[\widehat{\mathbf{G}}_\tau] = 0. \tag{4.ccxxix}$$

■□

Proof. The right side of eq. 4.ccxxvii is positive at real $\tau > 0$. Hence the logarithm selected from that real value is $r \log \tau/2$, which gives the sum rule. Differentiate it twice along the physical background map. The normally convergent sum of local second derivatives equals the second derivative of the constant total. At the trivial pair a first derivative of a scalar gauge-invariant family vanishes by constant colour rotations. Thus the extra first-derivative term from differentiating the background map vanishes, and the second derivative is precisely the bulk polarization computed with its physical linear response. It is zero. The second-moment readout defining the coefficient therefore gives zero as well. Compatible finite-volume localizations pass this equality to the companion family by normal convergence; no estimate of an individual local summand by zero is asserted. □

The sum rule extends to admitted complex physical data by holomorphic continuation of the actual background map from its real domain. It does not make the total constant on the whole independent-current pair core. That core contains directions transverse to the physical background manifold.

4.6.4. *Two essential qualifications.* First, the nonlinear constraint term in eq. 4.ccxxi cannot be removed from this argument. Even on a circle $q(p) = |p|^2 - 1 = 0$, the invariant action $\mathcal{S}(p) = |p|^2$ is constant on the rotation orbit. Its constrained action Hessian is zero, while its ambient Hessian is $2I$. The constraint curvature subtracts exactly that tangent contribution. The same chain rule accounts for the gauge null directions on the averaging fibre.

Second, replacing the physical current by an independent current changes that constrained Hessian. For example take

$$K_A = \operatorname{diag}(1, 0), \qquad K_G = \operatorname{diag}(0, 1), \qquad \widetilde{K}_A = \operatorname{diag}(1, e).$$

The physical determinant has the power in eq. 4.ccxxv, while the changed form has $\det(\tau\widetilde{K}_A + K_G) = \tau(1 + \tau e)$. Its compensating expectation is

$$\tau^{1/2} \left(\frac{1+e}{1+\tau e} \right)^{1/2},$$

which depends on the defect e. This shows why the actual independent-current completion and its defect must remain in the local Gaussian family, even though its physical total has zero coupling coefficient.

Finally, inserting sharp fluctuation tests into eq. 4.ccxxvii changes the identity. The test contribution belongs to the cutoff leg of the non-Gaussian formation remainder. The exact Gaussian mismatch is defined before those tests are turned on.

4.7. **Absolute-radius formation for the first homogeneous step.** We prove an absolute-radius formation estimate for the first homogeneous Wilson step. The precision, contour gauge, nonlinear pivot chart, coarea density and sharp fluctuation cutoff are the actual ones. A local covariance construction below supplies the enlarged cube-parameter discs required by the local expansion. Thus no decoupling-domain assertion is assumed for this step. The construction does not extend a later nonlocal effective Hessian by treating it as a finite-range Wilson matrix.

4.7.1. *The range of fields and the exact completion.* Fix the odd block size L. Work on periodic lattices which admit the L-blocks and the localization grids chosen below; their side lengths are at least ten localization cubes. All constants can depend on L, the requested output rate and the fixed localization choices, and are uniform in the volume.

Group delta integrals here mean the regular coarea disintegration of the actual analytic averaging map. Fix a sufficiently small absolute ϵ_{reg} and restrict this disintegration to fields whose fine plaquettes in every double block are below that ceiling. The logarithms are the regular branches of the pivot construction. No property of an arbitrary Borel completion outside this regular set is used.

The actual first-step chart of § 3.7 solves

$$q_U(a) = \frac{1}{i}\log[\mathcal{M}(e^{ia}U)\mathcal{M}(U)^{-1}] = 0, \qquad a = \Phi_U(u), \qquad \Phi_U(u)|_{\mathrm{nonpivot}} = u. \tag{4.ccxxx}$$

Its unscaled free radius is $r_u = c_u L^{-9}$. Its linear part is C_U, and its nonlinear pivot values use only their defining double blocks. On a fixed smaller radius, $\|\Phi_U(u)\|_\infty \le C_L\|u\|_\infty$, and all local derivatives through the orders used below have uniform bounds depending only on L. These are the explicit joint chart estimates proved in that section, not estimates for an unspecified many-level constraint.

Here is a gauge-covariant current completion on the entire local tube used in this proof. Put $J_A(U) = DA_U(0)$, and define

$$S_{U,J}(u) = A(e^{i\Phi_U(u)}U) - A(U) + \langle J - J_A(U), \Phi_U(u)\rangle - \langle J, C_U u\rangle, \tag{4.ccxxxi}$$

$$G_U(u) = \sum_{x\neq y(x)} a_x \left[1 - \tfrac{1}{2}\,\mathrm{Tr}\{T_{y(x),x}(e^{i\Phi_U(u)}U)T_{y(x),x}(U)^{-1}\}\right], \qquad a_x = 1. \tag{4.ccxxxii}$$

These are the actual unit contour-gauge weights in canonical Gaussian units, held fixed as τ varies. The relative contour product is holomorphic and conjugates at the block node under a simultaneous gauge change. Its value at zero is one. It uses no crossing pivot bond. Consequently both expressions have zero linear part, and

$$K_A(U,J) = D^2 S_{U,J}(0), \qquad K_G(U) = D^2 G_U(0), \qquad K(U,J) = K_A(U,J) + K_G(U). \tag{4.ccxxxiii}$$

On the real block-axial locus this is exactly the first-step matrix in eq. 3.cxxx, with its nonlinear constraint curvature term. The relative gauge has the same Hessian there as the absolute contour gauge. Away from that locus eq. 4.ccxxxii specifies the completion; no equality with a differently defined completion is presumed.

Choose fixed positive ϵ_0, j_0 strictly inside the real smallness bounds of eq. 3.cxxx. At a real reference pair suppose

$$\sup_p |1 - \partial U(p)| < \epsilon_0, \qquad \|J\|_\infty < j_0. \tag{4.ccxxxiv}$$

A node-fixed unitary contour gauge makes U block axial. The relative completion is covariant under this change, so the actual first-step estimates give, in the free coordinates,

$$\gamma I \le K \le BI, \quad \mathrm{range}\, K \le R_K, \quad \|K\|_{\mathrm{Sch}} \le B_{\mathrm{Sch}}, \qquad \gamma > 0. \tag{4.ccxxxv}$$

Increase $R_K = C_L L$ so it also bounds the radius of the background and current variables read by each coefficient. The Schur norm is the maximum of the absolute row and column sums of the three-by-three colour blocks. These constants depend only on L and the chosen strict real ceilings.

For $\mathbf{U} = e^{iE}U$, $\mathbf{J} = J + Z$, the same finite local chart gives the explicit perturbation estimate

$$\|K(\mathbf{U},\mathbf{J}) - K(U,J)\|_{\mathrm{Sch}} \le C_L(\|E\|_\infty + \|Z\|_\infty). \tag{4.ccxxxvi}$$

For completeness the relative-gauge version has this estimate because it is a fixed finite product of the admitted means, inverses and logarithms. In local unitary frames its arguments lie in the joint chart of radius cL^{-5}; a coefficient reads at most C_L variables and has bounded derivatives there.

Summation over their bounded incidence gives the Schur estimate. Thus the radius is absolute, rather than divided by the number of lattice sites.

For $\tau = x/\theta$, let $Q_\tau = \tau K_A + K_G$. Taking $|\tau - 1|$ sufficiently small adds at most $C_L|\tau - 1|$ to eq. 4.ccxxxvi. We shrink all these radii explicitly in the construction below. They remain independent of $g = \theta^{-1/2}$.

At an actual real constrained-critical background, with $J = J_A(U)$ and $T_{y,x}(U) = 1$, one has

$$S_{U,J}(u) = A(e^{i\Phi_U(u)}U) - A(U), \qquad G_U(u) = \mathrm{GF}(e^{i\Phi_U(u)}U). \tag{4.ccxxxvii}$$

The final linear term in eq. 4.ccxxxi vanishes because $C_U^{\mathsf{T}} J_A(U) = 0$. At nonphysical independent currents it is an auxiliary completion, which was written explicitly to avoid an unaccounted term of order g^{-1} in the linear expansion.

4.7.2. *A local covariance construction from the actual gap.* We give the construction with Q_τ denoted by K_c, and its chosen real reference in eq. 4.ccxxxv by K_r. Choose a real $a \geq 2B$, and write

$$H_r = I - K_r/a, \quad H_c = I - K_c/a, \qquad r = 1 - \gamma/a < 1. \tag{4.ccxxxviii}$$

The spectrum of H_r lies in $[0, r]$. Finite propagation and Cauchy–Schwarz in each matrix row give

$$\|H_r^j\|_{\mathrm{Sch}} \leq C_L(1 + j)^2 r^j. \tag{4.ccxxxix}$$

Indeed a row of H_r^j has at most $C_L(1 + j)^4$ nonzero blocks, and its squared row norm is bounded by a fixed colour factor times r^{2j}. The column estimate is the same. This estimate retains the cancellations inside each matrix power.

We will also use the local version of the real gap. There is a fixed radius $R_{\mathrm{gap}} = C_L$, independent of volume and of the set D, with the following property. If the real pair is given and obeys the strict ceilings on the R_{gap}-neighborhood of D, its actual Dirichlet coefficient matrix $K_{r,D}$ on free bonds of D satisfies $\gamma I_D \leq K_{r,D} \leq B I_D$. To check this, repeat the partition-of-unity proof of eq. 3.cxxx for a field supported on D. Only the fixed-size frame patches meeting its support are used. Taking R_{gap} larger than their diameters and the coefficient margin makes all these patches available in the given data. Axialize the real reference by its internal means first; the relative completion makes this a unitary change of the free coordinates. The subsequent block-constant frames, IMS error and flat comparison are then precisely those in that proof. No extension of a real field through the holes of D or across its exterior is required.

The estimate eq. 4.ccxxxix concerns one common real pair. The rooted activity norm below takes each local coefficient's data supremum before summing its endpoints and carriers. We retain a positive majorant for that stronger norm. For an entry of the already combined power, the same local compression and its spectral gap give $\|(H_r^j)(b, b')\| \leq r^j$, uniformly over the real data on its complete carrier. There are at most $C_L(1 + j)^4$ possible endpoints in a row or column. Put

$$p_j^+ = C_L(1 + j)^4 r^j, \qquad P_*^+ = \sum_{j \geq 0} p_j^+ < \infty. \tag{4.ccxl}$$

These are sums of coefficientwise local-data suprema. The smaller polynomial in eq. 4.ccxxxix cannot be used after moving separate suprema through the endpoint sum.

Fix a desired cube-parameter radius $S > 2$. Choose an integer m, to be made large, and put

$$T = H_c^m. \tag{4.ccxli}$$

One entry of T depends only on the matrix coefficients within distance $(m + 1)R_K$ of its two endpoints. Put $R_c = (m + 1)R_K + R_{\mathrm{gap}}$. Assign it the symmetric carrier consisting of both radius-R_c balls and all localization cubes they meet. Choose the fine cube size $M = M(m)$ to be the smallest power of L at least $16R_c$. Then $16R_c \leq M < 16LR_c$, so it still grows only linearly in m, is odd, and is compatible with the centered grids and power-of-L tori. For a root set W, its guard $W^\sharp$ is the union of all coarse cubes containing a fine cube at fine-cube wall distance at most

eight from a cube incident to W. This definition preserves unions and is used in every restriction and factorization below. Every short prefix H_c^j, $j < m$, starting at a root bond, together with its real-gap data collar, is contained in that root guard. The number c_s of coarse cubes in a short-step carrier is then bounded independently of m, volume and field.

Set $\eta_m^+ = p_m^+$. We also need a positive bound for the finite complex perturbation. For each actual finite-range coefficient, take the supremum of its norm over its common local real-reference and complex-pair domain. The bounded number of coefficients meeting a bond, and the finite local chart estimate eq. 4.ccxxxvi, give a fixed positive Schur envelope B_+ for both K_r and K_c on an initial smaller tube, and a difference envelope of Schur norm

$$\delta_K^+ \le C_L(\|E\|_\infty + \|Z\|_\infty + |\tau - 1|).$$

These envelopes take the supremum separately for each raw entry. They use only its full finite carrier. Define $C_H = 1 + B_+/a$. The telescoping polynomial identity, with these positive entry envelopes multiplied and summed, gives for every fixed $j \ge 1$

$$\|\mathbf{H_c^j} - \mathbf{H_r^j}\|_{\mathrm{Sch,sup}} \le \Delta_j^+ := (\delta_K^+/a) j C_H^{j-1}. \tag{4.ccxlii}$$

Set $\Delta_0^+ = 0$. Here the left side means the positive word majorant after taking each local coefficient supremum. Thus the estimate is stronger than a Schur bound at one common pair. Choose the fixed absolute tube and τ disc small enough that $\Delta_m^+ \le r^m$ and

$$\sum_{j=1}^{m-1} \Delta_j^+ \le r^m. \tag{4.ccxliii}$$

Both are quantitative finite inequalities, imposed after m, a, B_+ have been fixed. Their radii are positive and independent of g, volume and the number of sites.

For the combined power, the real entry bound and the positive difference envelope give a coefficientwise Schur majorant at most $p_j^+ + \Delta_j^+$. In particular the short step has bound $2\eta_m^+$, and the sum of all complex prefixes has bound $P_*^+ + 1$. The sum of their difference envelopes is at most $r^m \le \eta_m^+$. These bounds may be used before every carrier sum even when different coefficients are evaluated at different admitted local data.

Attach one parameter s_Δ to each coarse cube. Multiply an entire combined entry $T(b, b')$ by $\prod_{\Delta \in \mathrm{car}(b,b')} s_\Delta$, and call the resulting matrix T_s. We combine the polynomial entry before inserting this parameter, rather than separating its mutually cancelling fine paths. For $|s_\Delta| < S$,

$$\|T_s\|_{\mathrm{Sch}} \le 2S^{c_s}\eta_m^+ =: q_m^+. \tag{4.ccxliv}$$

All its coefficients are holomorphic local functions of the actual pair. Its carrier contains every background variable used by the entry.

Let $c_j = 4^{-j}\binom{2j}{j}$, the coefficients of $(1-z)^{-1/2}$. For a finite root set of free bonds W, hold the parameters at one on its fixed guard $W^\sharp$, and set

$$R_W(s) = a^{-1/2}\chi_W \sum_{j=0}^{m-1} H_c^j \sum_{k \ge 0} c_{km+j} T_s^k. \tag{4.ccxlv}$$

The prefix is local inside the root guard. The tail is a chain of the short-step carriers. Since $0 < c_j \le 1$, the positive prefix bound $P_*^+ + 1$ and eq. 4.ccxliv show that this series converges in Schur norm as soon as $q_m^+ < 1$.

At all parameters equal to one, grouping the absolutely convergent binomial series by j mod m gives

$$R_W(1) = \chi_W K_c^{-1/2}, \qquad D_W(s) := R_W(s) R_W(s)^{\mathsf{T}}, \qquad D_W(1) = \chi_W K_c^{-1} \chi_W^{\mathsf{T}}. \tag{4.ccxlvi}$$

Transposition is bilinear; no conjugation is inserted in this analytic covariance. The binomial identity is legitimate in the convergent coefficient algebra. In particular its square is K_c^{-1}, and its transpose is itself before the cube parameters are inserted.

Define the real short-prefix reference

$$R_{0,r} = a^{-1/2} \sum_{j<m} c_j H_r^j, \qquad D_{0,W} = \chi_W R_{0,r}^2 \chi_W^{\mathsf{T}}.$$

Since $H_r \geq 0$,

$$a^{-1} I_W \leq D_{0,W} \leq \gamma^{-1} I_W. \tag{4.ccxlvii}$$

Its definition uses only the fixed guard of W. Equations eq. 4.ccxliii and eq. 4.ccxlv give, uniformly on all the enlarged parameter discs,

$$\|R_W(s) - \chi_W R_{0,r}\|_{2\to 2} \leq a^{-1/2} \left[\eta_m^+ + (P_*^+ + 1) \frac{q_m^+}{1 - q_m^+} \right] =: \zeta_m^+. \tag{4.ccxlviii}$$

Consequently $\|D_W(s) - D_{0,W}\| \leq C_{a,\gamma} \zeta_m^+ + (\zeta_m^+)^2$, independently of the local and auxiliary dimensions. As $m \to \infty$, this tends to zero exponentially, while $M(m)$ grows only linearly. We may therefore choose m so large that

$$q_m^+ < 1/4, \qquad C_{a,\gamma} \zeta_m^+ + (\zeta_m^+)^2 \leq c_* (LM(m))^{-4}. \tag{4.ccxlix}$$

Here c_* is any prescribed small Gaussian comparison constant. The order is: fix the desired rates and S; choose m and $M(m)$; then choose the absolute tube and τ disc by eq. 4.ccxlii–eq. 4.ccxliii.

Proposition 4.14 (Admitted local Gaussians for the actual first step)**.** On the resulting fixed absolute tube, the functions

$$\mathcal{J}_W(s)[F] = \int_{\mathbb{R}^{3\#W}} F(b)\, d\mu_{D_W(s)}(b) \tag{4.ccl}$$

are normalized holomorphic local Gaussian functionals for $|s_\Delta| < S$. They obey the local Gaussian comparison of Prop. 4.7, with perturbation at most $c_*(LM)^{-4}$ and with $\Gamma = 0$. At complete physical data and $s = 1$, they are the actual marginals of the first-step Gaussian with precision Q_τ. When parameters vanish outside a union Z containing $W^\sharp$, they read no pair data outside Z, and they factor over the wall components of Z. ■

Proof. The covariance is symmetric and within the bound eq. 4.ccxlix of the real positive matrix $D_{0,W}$. In particular its real part is uniformly positive after decreasing c_*. The Gaussian integral and its determinant branch are therefore defined on the whole indicated polydisc, are normalized, and are holomorphic with measurable sharp functions allowed in F. Apply Prop. 4.7 with $C = D_W$, $C_0 = D_{0,W}$, $\Gamma = \Gamma_0 = 0$. Only $3\#W$ coordinates occur in this comparison. The possibly much larger column space of R_W is not charged as a Gaussian dimension. Completeness is eq. 4.ccxlvi.

If parameters outside Z are zero, each surviving short step has its full carrier in Z. Every prefix from a root in W is already in its guard. Thus every coefficient and intermediate column used by that row lies in the same wall component as the root. A short-step carrier is a connected cube union containing both endpoints; it cannot join two different components without reading a zero parameter. Prefix carriers are connected and lie in the guard. Rows rooted in different components therefore have disjoint column supports. Their bilinear covariance blocks are zero, proving Gaussian factorization. The same description proves dependence only on pair data in Z and the required hereditary restriction identity for the fixed root support and its held guard: computing from the restricted pair in Z agrees with any larger ambient computation whose parameters are zero outside Z. At complete parameters the usual nested marginal identities follow from eq. 4.ccxlvi. □

This is a localization of the actual covariance, fixed once for all tori in the admitted class. It need not equal another walk expansion term by term. Its full Gaussian marginals, nonlinear integration variables, and complete physical integral are exactly the original ones.

The estimates and local recipes hold also for pairs given only on the output domain. To see the domain point explicitly, retain only entries whose full carrier is contained in that domain. If a retained entry has first endpoint b, the real data on its radius-R_c ball are present. Compress the actual matrix to the radius-mR_K ball about b. The preceding local gap applies to this compression. The row at b of its m-th H power is the same polynomial row as the uncompressed one: none of its length-m paths can leave that ball. This proves the pointwise estimate eq. 4.ccxxxix for that entire common-data row before any inadmissible entries are deleted. For coefficientwise local-data suprema, apply the same compression to each entry separately and count its possible endpoints; this gives eq. 4.ccxl. The positive finite-range perturbation envelopes give eq. 4.ccxlii on these local domains as well. Deleting entries can only decrease either positive majorant, and symmetry gives the corresponding column bounds. For the root prefix and $D_{0,W}$, use one common Dirichlet compression to the mR_K-neighborhood of W; its needed real-data collar is inside $W^\sharp$. This preserves eq. 4.ccxlvii. Thus the local estimates are on the actual local pair domains with these collars, rather than only on restrictions of globally extendible real fields.

The same construction localizes the Gaussian logarithms. For the actual matrix K_c, the identity

$$\log K_c = (\log a)I - \sum_{\ell\ge 1} H_c^\ell/\ell \tag{4.ccli}$$

converges in Schur norm after grouping $\ell = km + j$, $0 \le j < m$, and omitting $\ell = 0$. Its diagonal trace at a bond is expanded as the finite prefix H_c^j followed by k actual short steps. The carrier is the union of their full carriers. It is connected and has tree length at most $C_0 + c_s k$. Multiplication by $e^{\kappa d_c(X)}$ therefore costs at most $e^{\kappa C_0} e^{\kappa c_s k}$. Choose S and m above with this additional rate reserve. Each word is bounded by the product of the positive prefix and short-step envelopes before taking the supremum on its own union carrier. Summing those positive words proves that the coefficientwise absolute rooted sum is bounded by

$$C \sum_{j<m} (p_j^+ + \Delta_j^+) \sum_{k\ge 0} (2\eta_m^+ e^{\kappa c_s})^k \le C(P_*^+ + 1),$$

with the finitely many $k = 0, j > 0$ terms included. This proves locality and an absolute, volume-independent rooted norm for the actual Gaussian log normalizer. The value $\log J_U(0)$ adds its finite local pivot terms and fixed numerical powers; its nonconstant part has the same type of bound by the joint pivot chart.

Use the same localization on a τ disc four times larger than the disc retained in the conclusion. Applying it to $-\frac{1}{2}\log\det Q_\tau + \frac{1}{2}\log\det K + \frac{d}{2}\log\tau$ gives the mismatch family. Its terms vanish at $\tau = 1$, and the Cauchy bound in this actual larger disc proves their rooted bound $C|\tau - 1|$. Its physical total remains exactly eq. 4.ccxxviii. The construction does not classify that term as a g-small vertex.

The Gaussian normalizer is pointed at each diagonal free bond in eq. 4.ccli; each zero-field pivot factor is pointed at that pivot. Colour traces are invariant under the local adjoint frames. Their midpoint tents give the pointed normalizer, with the same rate reserve.

For the mismatch use its actual gauge-quadratic interpolation, which also fixes its bond density:

$$\begin{aligned} Q_{\tau,u} &= \tau K - u(\tau - 1)K_G, \qquad 0 \le u \le 1, \\ g_{\tau,b}(P) &= \frac{\tau - 1}{2} \int_0^1 \mathrm{tr}[K_G Q_{\tau,u}^{-1}](b,b)\, du. \end{aligned} \tag{4.cclii}$$

Indeed $Q_{\tau,0} = \tau K$ and $Q_{\tau,1} = Q_\tau$. Differentiating its Gaussian determinant gives $\sum_b g_{\tau,b} = \mathcal{G}_\tau$, with the same normalization as above. The integrand is the expectation of the actual insertion $(\tau - 1)B(b) \cdot (K_G B)(b)/2$. All $Q_{\tau,u}$ stay in the same admitted local tube after a fixed decrease of the retained τ disc. Their inverse series uses $a^{-1}\sum_{\ell\ge 0} H_{\tau,u}^\ell$, grouped by the same short powers. Marking its one finite-range K_G factor preserves the carrier bound. Hence eq. 4.cclii has local

pointed terms of norm $C|\tau-1|$. This identifies the pointing by the defining interpolation, including in the noncommuting curved case.

4.7.3. *The actual coarea density and the intrinsic vertices.* Write $g=\theta^{-1/2}$, $\mathfrak{b}=A_1(\log\theta)^p\geq 1$, and keep the cutoff $\chi=\prod_b \mathbf{1}_{\{|B(b)|<\mathfrak{b}\}}$ fixed at this real reference. Choose g so small that

$$16C_L g\mathfrak{b}<r_u. \tag{4.ccliii}$$

Every argument $\Phi_U(tgB)$, $|t|\leq 4$, and every local old-member substitution below then lies in the actual unscaled graph chart. Cube parameters do not change these local graphs: their complete double-block support is kept in each selected potential's guard. They change the local Gaussian functional outside that guard by the construction just proved.

Let p_c be the three pivot coordinates of row c. Eliminating the exact constraint in these coordinates gives the relative density

$$J_U(u)=\prod_b j(\Phi_U(u)(b))\prod_c \det[D_{p_c}q_U(\Phi_U(u))]^{-1}\,,\qquad j(a)=\left(\frac{\sin\sqrt{a\cdot a}}{\sqrt{a\cdot a}}\right)^2. \tag{4.ccliv}$$

The factor g^d and the fixed Haar and gauge normalizers are independent of B. We use the raw logarithmic constraint q_U throughout. A per-length constraint would require its compensating factor L^{-3} for every coarse bond; it must not be inserted a second time into the present raw convention. At real arguments the pivot determinant has its positive identity sign. Its reciprocal in eq. 4.ccliv is the holomorphic continuation, with no complex absolute determinant. Each pivot factor reads only its double block.

On the uniform unscaled radius, the logarithm of each factor of $J_U(u)/J_U(0)$ is holomorphic, vanishes at zero and has derivative bounded by C_L. For the pivot factor this follows by differentiating $-\operatorname{Tr}\log[D_pq(u)(D_pq(0))^{-1}]$, using the actual pivot lower bound and joint chart radius. For Haar it follows from eq. 4.dlvii. Hence its local logarithmic vertices satisfy a linear bound $C_Lg\sum_{b\in S}|B(b)|$ on their fixed supports.

Subtract the full quadratic parts, and define

$$\begin{aligned}P_g^A(B)&=-g^{-2}\{S_{U,J}(gB)-\tfrac12 g^2B^{\mathsf T}K_AB\},\\ P_g^G(B)&=-g^{-2}\{G_U(gB)-\tfrac12 g^2B^{\mathsf T}K_GB\},\\ P_g^J(B)&=\log[J_U(gB)/J_U(0)].\end{aligned} \tag{4.cclv}$$

Every Wilson plaquette, relative contour gauge term and current-completion term is a holomorphic function of a fixed finite list of free coordinates, on an unscaled radius. Its first three derivatives are uniformly bounded there by the actual chart estimates. Taylor's integral formula consequently gives, for each such local term,

$$|g^{-2}R_3(gB)|\leq C_Lg\left(\sum_{b\in S}|B(b)|\right)^3\leq C_Lg\mathfrak{b}\sum_{b\in S}|B(b)|^2. \tag{4.cclvi}$$

The second inequality uses the fixed bound on $\#S$ and the actual cutoff $|B(b)|\leq\mathfrak{b}$. The coefficients are uniform on the absolute external pair tube, not just at $U=1$. The current-completion correction is included before this Taylor expansion. The same estimates, with a fixed increase in C_L, hold on nested complex fluctuation boxes $|B_b^a|<\nu(1+\mathfrak{b})$, for a fixed $1<\nu<9/8$, and on their fixed radial margins. The factor sixteen in eq. 4.ccliii leaves this room for the independent-bond pointing construction.

Group these terms into their actual fine localization cube carriers, enlarged by the fixed double-block collar. For $Q_Y(B)=1+M^{-4}\sum_{b\subset Y}|B(b)|^2$, the finite support and incidence counts give

$$|V_g^{\mathrm{int}}(Y;P,B)|\leq C_{L,M}g\mathfrak{b}\,e^{-\lambda_V d_f(Y)}Q_Y(B),\qquad V_g^{\mathrm{int}}=\tau P_g^A+P_g^G+P_g^J. \tag{4.cclvii}$$

Indeed each elementary support meets a bounded number of fine cubes, so its exponential carrier cost is fixed; a fine cube contains $O(M^4)$ elementary terms, and a free coordinate enters only C_L of them. All these finite factors are included in $C_{L,M}$. The same bound holds for $|\tau-1|$ on its fixed admitted disc.

4.7.4. *Admitted old functions and ordinary source profiles.* Let a local input $S_{\mathrm{old}}=\sum_X f_X$ be holomorphic on fixed pair balls of radii $a_{H,U},a_{H,J}>0$ around the indicated field and current, with bounds B_X and

$$\sup_b \sum_{X\ni b} e^{\lambda_H d_f(X)} B_X \le H_0. \tag{4.cclviii}$$

Use the same physical input function before and after the chart substitution. To include its current argument, define the following explicit pair map:

$$\begin{aligned} U_P(u) &= e^{i\Phi_U(u)}U, \\ \Psi_P(u) &= \big(U_P(u), J + J_A(U_P(u)) - J_A(U)\big), \qquad P=(U,J). \end{aligned} \tag{4.cclix}$$

The fine Wilson gradient reads only its adjacent plaquettes. Its local derivative is uniformly bounded on these fixed unscaled tubes. Hence $\|J_A(U_P(u))-J_A(U)\|_\infty \le C_L\|u\|_\infty$. This is the actual gradient variation, rather than a current held fixed during a physical field variation. The map is jointly gauge covariant, equals P at zero, and at a physical pair maps exactly to $(U_P(u), J_A(U_P(u)))$. Put $a_H=\min(a_{H,U},a_{H,J})$, and require $16C_L g\mathfrak{b}<a_H$ as well as eq. 4.ccliii. The exact old potential is

$$V_X^{\mathrm{old}}(B) = f_X(\Psi_P(gB)) - f_X(P). \tag{4.cclx}$$

Its second value stays in the carried background action. The local substitution reads the fixed double-block enlargement of X, including the adjacent plaquettes needed for its current arguments. The analytic path $t\mapsto f_X(\Psi_P(tgB))$ is admitted up to $|t|<c\min(r_u,a_H)/(g\mathfrak{b})$. The one-variable Cauchy difference on that actual path gives

$$|V_X^{\mathrm{old}}(B)| \le C_L \frac{g\mathfrak{b}}{\min(r_u,a_H)} B_X.$$

The enlargement of a domain has at most a fixed multiple of its tree length plus one. Choose λ_H larger than that multiple of λ_V, with a strict incidence/entropy reserve. Refiling by that enlargement and summing eq. 4.cclviii then gives the actual potential budget

$$\|V^{\mathrm{old}}\|_{\lambda_V} \le C_{L,M,a_H} H_0 g\mathfrak{b}. \tag{4.cclxi}$$

All input origins with the same carrier are summed before this estimate; their number is not counted as a number of independent Gaussian variables or potential types.

This includes the ordinary profile presentation of a Wilson source $\sum_p w(p)a_p$, with w on fixed complex coefficient discs. Each term is a finite local holomorphic word, and eq. 4.cclviii holds with $H_0\le C\|w\|_\infty$. The Gaussian, chart, reference gauge weight and sharp cutoff stay fixed as this source is varied. A constant-shift presentation which instead changes τ uses its separate Gaussian mismatch and numerical factor.

4.7.5. *The three output families and the exact first-step law.* Let $\mathcal{N}^\circ$ be the localization just constructed of the vacuum-subtracted coarea and Gaussian normalizer at $\tau=1$:

$$\mathcal{N}^\circ(P) = \log\frac{J_U(0)}{J_1(0)} - \frac{1}{2}\log\frac{\det K(U,J)}{\det K(1,0)}. \tag{4.cclxii}$$

Its vacuum factors stay in the exact numerical normalization. Let $\mathcal{G}_\tau$ be the free Gaussian mismatch defined before cutoffs, as in Prop. 4.12. Define the last family from the actual normalized integral

$$\mathcal{F}_g(P) = \log\int \chi(B)\exp\{V_g^{\mathrm{int}}(P,B)+V^{\mathrm{old}}(P,B)\}\,d\mu_{Q_\tau^{-1}}(B). \tag{4.cclxiii}$$

This is a definition of the output, not an assumed bound.

Proposition 4.15 (Actual first-step absolute-radius formation)**.** Choose the rates and localization sizes as above. There is a coupling floor θ_*, independent of volume, such that eq. 4.cclxiii is nonzero and has an actual union-filed logarithm $F_g(X,P)$ on the full fixed absolute pair tube. Its terms satisfy

$$|F_g(X,P)| \le C\left[(1+H_0)g\mathfrak{b} + e^{-c\mathfrak{b}^2}\right] e^{-\kappa d_c(X)}. \tag{4.cclxiv}$$

The same estimate, after a fixed decrease of the spatial rate, holds for the constructed bond-rooted pointing and its midpoint tent sum. The constants include the fixed absolute radii, but contain no inverse power of g. The output splits into the actual Gaussian family, the Gaussian mismatch, and this remainder. In particular $F_g = O(g(\log\theta)^p)$. ■

Proof. The preceding construction supplies every Gaussian matrix, enlarged coarse-parameter disc, normalization, hereditary restriction and wall factorization used in the unmarked local expansion of § 4.5. The local coordinate set for an index is taken to contain all free bonds of its full fine support Z_0, not merely the bonds on which its integrand changes. Thus background supports of the selected potentials, including a background factor constant along the fibre, lie in the held guard as well. Adding these integrated coordinates does not change a Gaussian expectation of that integrand. The local coordinate count is at most $C(LM)^4\#Z$ for a polymer Z. There is no auxiliary-volume factor. The potentials are the literal local functions in eq. 4.cclv and eq. 4.cclx, with their actual cutoff supports. Equations eq. 4.cclvii and eq. 4.cclxi give $a_g \le C(1+H_0)g\mathfrak{b}$.

Apply the proved finite Mayer, cutoff and coarse interpolation estimates to these data. Their activity bound has amplitude $C(a_g + p_*)$, where $p_* \le C(LM)^4 e^{-c\mathfrak{b}^2}$. The rooted connected-cluster bound and union filing then give eq. 4.cclxiv. All its numerical smallness conditions hold after increasing θ_*, because $g\mathfrak{b} \to 0$ and $e^{-c\mathfrak{b}^2} \to 0$. This uses that $A_1(\log\theta)^p$ tends to infinity; choose $p > 1/2$, so its cutoff tail is also $O(g\mathfrak{b})$.

The Jacobian and old functions are rooted along $B \mapsto tB$. For the cubic action and gauge remainders use precisely the normalized path $t^{-2}P_g^{A,G}(tB)$, with the covariant shares and independent bond derivatives of eq. 4.cxxvii. In particular the bond insertion includes the derivative of t^{-2}; it is not the unnormalized Euler insertion. The admitted chart has a fixed radial margin by eq. 4.ccliii. The order-three vanishing and the local unscaled derivative estimates used in eq. 4.cclvi, followed by eq. 4.cxxx, give the rooted budgets with the same $g\mathfrak{b}$ factor, uniformly down to $t = 0$. The cutoff leg is the actual moving-threshold boundary term of eq. 4.cxxxix. It supplies its Gaussian tail and retains its bond root. Thus the previously proved rooted expansion applies to these literal legs; no pointing is inferred merely from an unpointed norm. Summing their midpoint tents costs a polynomial domain factor, absorbed by the reserved rate. □

Here is the exact physical normalization. Let U be the actual first-step constrained-critical background, in its block-axial representative. In this formula $S_{\mathrm{old}}(U)$ denotes the physical restriction $S_{\mathrm{old}}(U, J_A(U))$. Apart from the external coarse indicator, the bare small-field integral equals

$$e^{-xA(U)+S_{\mathrm{old}}(U)}\, C_{\mathrm{num}}(\theta) g^d J_U(0)(2\pi)^{d/2}(\det K)^{-1/2}\tau^{-d/2}\exp\{\mathcal{G}_\tau + \mathcal{F}_g\}. \tag{4.cclxv}$$

More explicitly, let n_f, n_c be the numbers of positive fine and coarse bonds, and let $n_s = N_f - N_c$ be the number of node-fixed gauge sites. In the raw convention of eq. 4.ccxxx,

$$C_{\mathrm{num}}(\theta) = \sigma_0^{n_f - n_c} z(\theta, \varepsilon_\theta)^{-n_s}, \qquad d = 3(n_f - n_c), \tag{4.cclxvi}$$

where σ_0 is Haar density at zero and z is the normalized group integral of the single-site contour gauge with its retained test. Indeed the fine Haar densities contribute $\sigma_0^{n_f}$, and each coarse group delta, written in raw logarithms, contributes σ_0^{-1}. At the trivial field the raw pivot derivative is $L^{-3}I_3$, so $J_1(0) = L^{9n_c}$. Equivalently one may use the per-length pivot derivative $L^{-4}I_3$, its factor L^{12n_c}, and the delta conversion L^{-3n_c}; their product is the same. Equation eq. 4.ccliv proves the coordinate identity; the centered action and gauge agree with the physical ones by eq. 4.ccxxxvii.

The quadratic normalization is Prop. 4.12. Every factor depending on U is included, in particular $J_U(0)$.

The first-step formation budget can therefore be closed in the prescribed order. The construction gives constants C_N, C_G, C_F, independent of the allowed old-input size N_*, such that

$$\|\mathbf{N}^\circ\|_+ \le C_N, \qquad \|\mathbf{G}_\tau\|_+ \le C_G|\tau-1|, \qquad \|\mathbf{F}_g\|_+ \le C_F\{(1+N_*)g\mathfrak{b} + e^{-c\mathfrak{b}^2}\}.$$

Choose $E_0 \ge 2C_N$, then the old-input ceiling N_*; decrease $|\tau-1|$ so that its term is at most $E_0/4$; finally increase the coupling floor so that the last term is at most $E_0/4$. This gives $\|\mathbf{N}^\circ + \mathbf{G}_\tau + \mathbf{F}_g\|_+ \le E_0$ for the proved first-step range, without putting E_0 or N_* into the preceding Gaussian or nonlinear chart choices.

Corollary 4.16 (Ordinary source jets in the proved first-step range)**.** Insert the literal profile $\sum_p w(p)a_p$ on the fixed source ball $\|w\|_\infty < \rho$. For $\|w\|_\infty \le \rho/2$, $n \ge 1$, and arbitrary profiles $h_1, \dots, h_n$,

$$|D_w^n F_g(X; w)[h_1, \dots, h_n]| \le C_\rho g\mathfrak{b}\, n!(2e/\rho)^n e^{-\kappa d_c(X)} \prod_{i=1}^n \|h_i\|_\infty. \tag{4.cclxvii}$$

The same estimate holds for the constructed pointing with its reserved spatial rate. Only the source plaquette vertices whose complete carriers are filed inside X occur in that derivative. ■□

Proof. The source potential is the actual admitted old difference already considered; its norm on the full source ball is at most $C\rho g\mathfrak{b}$. All Gaussian, chart and cutoff arguments stay fixed. The Mayer integrals and convergent cluster sum are therefore holomorphic in these source coefficients, with eq. 4.cclxiv uniformly on that ball. For nonzero h_i, use the independent complex directions $w + \sum_i z_i h_i$ on $|z_i| < \rho/(2n\|h_i\|_\infty)$. They lie in the admitted source ball. The multivariate Cauchy formula bounds the mixed derivative by $C_\rho g\mathfrak{b}(2n/\rho)^n \prod_i \|h_i\|_\infty e^{-\kappa d_c(X)}$. Use $n^n \le e^n n!$. A zero direction gives zero. The same argument applies to the analytic rooted terms. Their source support is literal: an activity filed inside X uses only selected potential carriers inside X, and its cluster union stays there. No source variable outside that set is read. □

4.7.6. *Physical tests and the largest proved range.* The preceding fixed absolute domain is the analytic domain of the filed functions. It is not the support of every original external small-field indicator. We first check that this regular coarea fibre contains exactly the graph used above on the free fluctuation box. Every middle pivot belongs to an elementary plaquette whose other three bonds are free; choose a fixed perpendicular direction. Odd $L \ge 3$ keeps every other middle pivot out of that plaquette. Let V be in the regular coarea set, with average $W = \mathcal{M}(U)$, and with its free bond logarithms relative to U at most $\delta = g\mathfrak{b}$. Solving that plaquette product for its pivot bond, and subtracting the corresponding expression for U, gives

$$\max_{\text{pivots}} \|V(b)U(b)^{-1} - I\| \le 2\epsilon_{\mathrm{reg}} + C\delta. \tag{4.cclxviii}$$

Here the other three factors are unitary and their differences are at most $C\delta$; the two plaquette products differ by at most $2\epsilon_{\mathrm{reg}}$. These estimates are unchanged by their left and right transports. The pivot fixed-point map has derivative at most $C_L\|a\|_\infty$ in a fixed full-field ball $r_p = c_p L^{-5}$, as follows from its quadratic constraint remainder and pivot inverse. Choose ϵ_{reg} so small that eq. 4.cclxviii puts all full logarithms in that uniqueness ball, and then take δ below its fixed small floor. The equation $q_U(a) = 0$, with its fixed free values, has exactly the chart solution there. Thus $V = e^{i\Phi_U(gB)}U$ and its pivot values obey the sharper $C_L g\mathfrak{b}$ bound. Conversely the graph stays in the regular coarea set when the background curvature is below $\epsilon_{\mathrm{reg}}/2$ and $g\mathfrak{b}$ is small enough. This proves the full change of variables on the stated regular fibre, rather than just an injective parametrization of a subset.

For the homogeneous tame operator itself, take real coarse data with $\|1 - \partial W\|_\infty < \epsilon_W$, where the fixed absolute $\epsilon_W > 0$ is small enough for § 3.8 and eq. 4.ccxxxiv. No shrinking curvature

condition is needed for that operator. Its additional tests are the contour-gauge tests at threshold $\varepsilon_\theta = A_0 g(\log\theta)^p$. The background contour means are exactly one, and those means read only internal free bonds. The product/mean derivative estimate therefore gives their deviations at most $C_L g\mathfrak{b}$, independently of the background curvature. Choose $A_0 > 2C_L A_1$. These indicators are then identically one. The nonpivot fluctuation field in eq. 4.ccxxx is exactly gB, so its prescribed test is exactly χ. This proves eq. 4.cclxv on the fixed absolute physical background range.

If this factor is used in a history with extra plaquette indicators, a sufficient common-test range is smaller. Namely require $\|1-\partial W\|_\infty < c_L\varepsilon_\theta$. The actual background fine curvature is then at most $C_L c_L \varepsilon_\theta$, and the graph changes fine plaquettes by at most $C_L g\mathfrak{b}$. Taking c_L and A_1/A_0 sufficiently small puts their sum below half the indicated thresholds. Those additional tests are identically one on the same box. Their identity has not been asserted on the larger absolute physical range.

Any such coarse external indicator remains outside this formula. It cannot be analytically continued as an indicator on a fixed absolute ball: a fixed nonzero real curvature eventually exceeds $A_0 g(\log\theta)^p$. The absolute-radius output is instead the analytic extension of the displayed fluctuation formula, with that external indicator factored out. The physical background continuation in the first-step theorem supplies its composition on a fixed sufficiently small complex coarse-datum ball. Its current is $J_A(U(W))$, so it lies in the physical completion just used. Source profiles vary on their independent fixed discs while the tests keep their real reference values.

The proof covers this actual first bare homogeneous step, arbitrary compatible volume and real flat holonomy, and its specified absolute complex tube. For a later step the effective action $A(U_k(V))$ is nonlocal. Its Hessian does not have the finite propagation used in eq. 4.ccxxxix and the short-step carrier construction. Extending this result to all homogeneous levels requires a proved localized effective-precision expansion and the corresponding unscaled nonlinear response domains. Those are the earliest additional operator and domain inputs; they are not replaced by assuming the desired activity norm.

4.8. **Connected trees for the actual homogeneous effective action.** The homogeneous background of Prop. 3.183 has an expansion whose spatial rate is independent of its Taylor order. We derive it from the actual Wilson and averaging equations. This supplies local analytic terms for $S_k(V) = A(U_k(V))$; localization by a single large ball would not give the connected carriers below.

The coarse spacing is one and $h = L^{-k}$. Use the parallel frame of the fixed flat reference W_0, and coarse coordinates $V_B = e^{iB}V_0$. Fine pairings use h^4 times counting measure. The constants in this section are independent of h, compatible volume and flat holonomy. The localization size, chosen after a requested rate, is allowed to enter them. No independent fine-current core or hereditary nonlinear tower is asserted.

Gauge invariance holds for the complete physical sum S_k. Its individual fixed-reference Taylor terms need not be gauge invariant. An application requiring individually invariant core members needs a gauge-covariant localization in addition to the analytic filing proved here.

4.8.1. *The actual cell equation and its analytic vertices.* Write $x = (A,\omega)$, with the strong norm $\|A\|_{X_\alpha} + \|\omega\|_\infty$ of eq. 3.mccxxiii. The actual equation is

$$x = \mathcal{L}_0^{-1}\{\iota B + \mathcal{R}_h(x)\}, \qquad \mathcal{R}_h(A,\omega) = \begin{pmatrix} -N_h(A) + (q_h'(A) - Q)^\sharp\omega \\ -[q_h(A) - QA] \end{pmatrix}. \tag{4.cclxix}$$

Here $\iota B = (0,B)$, not a fine point-mass insertion. Each component in the first row is an actual fine forcing with the physical density appropriate to $\sharp$. The second row is a coarse bond field. In particular $\mathcal{R}_h(0) = D\mathcal{R}_h(0) = 0$.

Choose a smooth partition $\sum_y \chi_y = 1$ on physical unit cells, with a fixed support enlargement and uniformly bounded derivatives through order three. Sample it on the fine lattice. The discrete

product rule, including the local Hölder difference, gives

$$\|\chi_y A\|_{X_\alpha} \le C\|A\|_{X_\alpha(\mathrm{star}\, y)}. \tag{4.cclxx}$$

Indeed $D(\chi A) = (D\chi)A + \chi(\cdot+h)DA$; one further difference uses only bounded $\chi, D\chi, D^2\chi$, A, DA and the local Hölder seminorm of DA. All distances are physical, so there is no negative power of h in this multiplication bound.

Let $E_y x$ consist of $\chi_y A$ and the coarse multiplier components assigned to cell y. Then $x = \sum_y E_y x$, with bounded local overlap. Let I_z inject the fine forcing on cell z and its assigned coarse forcing. The fine input norm here is L^∞, so characteristic cell restriction of that forcing needs no derivative. The actual block propagator

$$\mathcal{G}_{yz} = E_y \mathcal{L}_0^{-1} I_z \tag{4.cclxxi}$$

satisfies

$$\|\mathcal{G}_{yz}\| \le C_G e^{-s_0 d(y,z)}, \qquad G_s := \sup_y \sum_z e^{s d(y,z)} \|\mathcal{G}_{yz}\| < \infty \quad (0 < s < s_0). \tag{4.cclxxii}$$

This follows directly from the weighted joint inverse Prop. 3.182, with the weight centered on the input cell. The constants include the fixed support of E_y. The inverse retains both the field and the multiplier.

There is a fixed physical radius $R_0 = C_L$ such that the output $\mathcal{R}_{h,z}$ reads only the components $E_y x$ with $d(y,z) \le R_0$. This assertion concerns the actual nonlinear functions in eq. 4.cclxix. For the Wilson term it follows from its fine plaquette stencil and its exact normal-coordinate differential. The factor h in that differential cancels the possible h^{-1} from a second fine difference, as in eq. 3.mcclxv. For q_h, the successive carrier radii are bounded by $C_L \sum_{j<k} hL^j \le C_L$. In its physical transpose only coarse rows whose carriers meet the output fine bond occur. The fixed-leg column density Ch^4 in eq. 3.mcclviii cancels the h^{-4} in $\sharp$. Thus the same locality and uniform norm statement holds for $(q_h' - Q)^\sharp \omega$.

On a fixed local strong ball the resulting Taylor expansion is

$$\mathcal{R}_{h,z}(x) = \sum_{m\ge 2} \sum_{y_1,\dots,y_m \in \mathcal{N}(z)} T_{h;z;y_1\dots y_m}[x_{y_1}, \dots, x_{y_m}], \tag{4.cclxxiii}$$

where $\mathcal{N}(z) = \{y : d(y,z) \le R_0\}$. The coefficient includes $1/m!$. Uniform Banach Cauchy differentiation on the larger admitted local ball, followed by polarization, gives

$$\sum_{y_1,\dots,y_m \in \mathcal{N}(z)} \|T_{h;z;y_1\dots y_m}\| \le C_R r_0^{2-m}. \tag{4.cclxxiv}$$

To see the constants explicitly, the number of possible cells in each slot is at most $D_0 = \#\mathcal{N}(z)$, independent of h. The norm of $m!^{-1} D^m \mathcal{R}$ is at most $C(e/r)^m$ on a smaller ball. Absorb D_0^m, the localization multiplication bound and the fixed two-component choices into r_0. No fine-coordinate count is made. This proves eq. 4.cclxxiv from the actual local residual, rather than postulating an analytic vertex family for the desired output.

4.8.2. *The rooted expansion and its connected carriers.* Substitute eq. 4.cclxxiii into eq. 4.cclxix repeatedly. A leaf is the actual coarse input $B(c)$ propagated by $\mathcal{G}$. An internal vertex has arity $m \ge 2$, coefficient T_h, and an outgoing propagator. Order its children in their Taylor slots. The resulting ordered rooted trees are exactly the formal substitution expansion; the Taylor factorial is already in T_h and must not be inserted again.

Assign a carrier to each term as follows. An edge $\mathcal{G}_{yz}$ is carried by a fixed coordinate-order shortest unit-cell path from y to z. A local vertex is carried by its radius-R_0 cell star, including the short paths to all its input components. The root support and leaf coarse bonds are included.

The resulting carrier is a connected cube union containing every input value read by the term. Its physical tree length is bounded by

$$\ell(\mathcal{T}) \le C_0 \sum_{\text{propagator edges } e} (1 + d_e) + C_1 \#\mathcal{T}. \tag{4.cclxxv}$$

All local stars have fixed size, so C_0, C_1 are independent of arities, mesh and depth.

Choose a positive t so small that the propagator distance cost in eq. 4.cclxxv uses at most $s_0/2$ of its decay. The remaining constant per edge and per input slot is absorbed by a fixed decrease of r_0 and increase of G_s, C_R. With these modified constants, the absolute sum of trees rooted at one cell, charged by $e^{t\ell(\mathcal{T})}$, is bounded by the small solution of

$$U = G_*\|B\|_\infty + G_* C_* \frac{U^2}{1 - U/r_*}. \tag{4.cclxxvi}$$

For example choose $u_* < r_*/4$ and $8G_*C_*u_* < 1$, and then $G_*\rho_* < u_*/4$. The right side maps $[0, u_*/2]$ into itself and is contractive there. Its solution is at most $2G_*\|B\|_\infty$ for $\|B\|_\infty < \rho_*$.

This scalar estimate bounds the actual tree sum: at a fixed root sum each outgoing propagator over its endpoint using eq. 4.cclxxii; then sum its local child-cell choices using eq. 4.cclxxiv; finally repeat from its children. The resulting recursion is exactly dominated by eq. 4.cclxxvi. All sums are absolute. A tree with n leaves has at most $n - 1$ internal vertices, since every internal arity is at least two. Hence its bounded-star and zero-length-edge costs introduce only a fixed exponential per leaf, already included in that majorant.

The summed tree series solves the actual fixed point equation and lies in the uniqueness ball of Prop. 3.183. It therefore equals its constructed background. The spatial rate t has not been divided by the number of leaves.

For a torus, expand each flat propagator in its lifted, twisted infinite-lattice entries. These entries have the same exponential cell bound. Their absolutely convergent periodization solves the periodic joint system; uniqueness identifies it with eq. 4.cclxxi. This can also be obtained by taking the local limit of its uniformly exponentially bounded finite-volume inverse and then periodizing. The nonlocal linear site projector is included in that inverse, not in a purported local nonlinear vertex. Each lifted propagator is given its path on the cover before projection to the torus. Summing its endpoint and period image is the single infinite-lattice sum in eq. 4.cclxxii. Thus the carrier and entropy bounds are uniform in volume and flat twists, including wrapping trees.

4.8.3. *The actual Wilson action on these backgrounds.* Let $a_{h,z}(A)$ be the sum of the actual fine plaquette actions assigned to physical cell z, each plaquette assigned once. The exact plaquette logarithm has the form $U_A(p) = e^{ih^2 F_p(A)}$, with $\|F\|_\infty \le C(\|DA\|_\infty + \|A\|_\infty^2)$. Hence on a fixed local strong ball

$$|a_{h,z}(A)| \le C\|A\|^2_{X_\alpha(\text{star } z)}. \tag{4.cclxxvii}$$

Indeed each plaquette contributes at most $Ch^4\|F\|^2$, and the number assigned to one physical cell is at most Ch^{-4}. The same bound holds on the complex ball by the analytic cosine estimate, without a real part. The local action has no constant or linear Taylor coefficient.

Expand this actual cell action in its local strong variables, and substitute the convergent background trees in every slot. The outer action vertex has arity at least two. Its Banach Cauchy coefficients have the same form of bound as eq. 4.cclxxiv. Consequently, on a smaller fixed ball, the sum of resulting action trees rooted at z satisfies

$$\sum_{\mathcal{T} \text{ rooted at } z} e^{t\ell(\mathcal{T})} \sup_{\|B\|_\infty < \rho} |s_{h,\mathcal{T}}(B)| \le C\rho^2, \qquad 0 < \rho \le \rho_*/2. \tag{4.cclxxviii}$$

The literal fine gauge of the homogeneous background preserves Wilson action exactly. Therefore this series is the actual effective action $S_k(V_B)$, rather than the action of an approximate or separately chosen lift.

Fix any desired $\kappa > 0$. Choose an odd localization size M, a power of L, large enough that a path of length d meets at most $C(1 + d/M)$ localization cubes and $C\kappa/M < t/4$. For each action tree file its full carrier into its actual union X of these cubes. Constants per tree edge cost $e^{C\kappa}$ and are absorbed by a further fixed decrease of ρ_*, since the number of edges is bounded by a fixed multiple of its leaves. Write the resulting local function as $s_{k,M}(X; B_X)$.

Proposition 4.17 (Uniform local analytic effective action)**.** There is a fixed positive ρ_κ, independent of homogeneous depth and compatible volume, such that on $\|B\|_\infty < \rho_\kappa$

$$S_k(e^{iB}V_0) = \sum_X s_{k,M}(X; B_X), \qquad \sup_Q \sum_{X \ni Q} e^{\kappa d_M(X)} \sup_{\|B\|_\infty < \rho} |s_{k,M}(X; B_X)| \le C_M \rho^2 \tag{4.cclxxix}$$

for $0 < \rho \le \rho_\kappa$. Each term is holomorphic only in its indicated coarse bond coordinates and is zero to order two at the flat datum. The representation is formed from the actual fixed-point coefficients and their connected propagator carriers. ■

Proof. Only incidence at a cube different from the outer action root needs an additional check. If the carrier reaches that cube, its charged length is at least the physical distance from the action root to that cube, up to a fixed cell margin. Reserve half the exponential weight in eq. 4.cclxxviii to pay this distance. Summing action roots then costs at most $C \sum_z e^{-ctd(z,Q)} \le C_M$, including the $O(M^4)$ possible roots inside Q. The remaining weight pays $e^{\kappa d_M(X)}$ and the fixed tree costs just described. This proves the bound. Absolute normal convergence proves holomorphy and the sum identity. Every leaf is an actual input coarse bond in the filed carrier, so the indicated local coordinate dependence is literal. Each action tree has at least two leaves. Its zero of order two survives filing. □

4.8.4. *Marked leaves and the actual effective Hessian.* Taylor coefficients can be marked before filing. Fix a coarse bond b and mark one leaf there. Transfer the root to that leaf by reserving an exponential weight for the path to the action root. Sum the remaining root positions as in the preceding proof. Marking a second leaf at b' charges its distance to b by the same connected tree. The number of ordered choices of marked leaves is at most $n(n-1)$ at order n; its polynomial factor is summable on a fixed smaller analytic ball. Since each coarse cell has a fixed finite number of bond and colour coordinates, these are absolute kernel sums, not only multilinear operator-norm estimates.

In particular the actual Hessian admits a local symmetric family $H_{k,X}(B)$, obtained by differentiating the filed action terms and, if necessary, symmetrizing each carrier term. For a fixed smaller rate and radius,

$$\begin{aligned} S_k''(B) &= \sum_X H_{k,X}(B), \\ \sup_b \sum_{X \ni b} e^{\kappa d_M(X)} \sum_{b'} \|H_{k,X}(B)(b, b')\| &\le C_M, \\ \sup_b \sum_{X \ni b} e^{\kappa d_M(X)} \sum_{b'} \|[H_{k,X}(B) - H_{k,X}(0)](b, b')\| &\le C_M \|B\|_\infty. \end{aligned} \tag{4.cclxxx}$$

All rows and field variables read by a term lie in its full carrier. The column estimates are identical by symmetry. More finely one can keep the physical tree length in these estimates, with a fixed positive rate, by reserving it before passing to the M-grid.

This conclusion uses the all-order rooted expansion of the actual residual. Fixed-order max-distance response estimates alone would not justify the absolute marked-leaf sums or the unaltered spatial rate here.

4.9. **Absolute-radius formation at homogeneous depth.** We now use the actual effective-action expansion of Prop. 4.17. The external variables in this section are coarse bond logarithms in its fixed ball about a specified flat reference. This proves a homogeneous physical range and its local analytic extension in those coordinates. Independent fine-current cores and proper-region background towers are separate domains. Choose the frozen flat coarse reference V_0 in its outer block-axial representative, so its actual internal contour means are one. Every flat holonomy class has such a representative: the node-fixed gauge $T_{y,x}(V_0)$ sets those means to one and preserves the next coarse field. This is a fixed reference choice before the Taylor expansion. Consequently the later axializing gauge of a nearby field is small in that same fixed coordinate ball. A parallel-frame calculation alone would not replace this condition on the actual absolute contour gauge. The norms below are analytic spatial and pointing bounds. The fixed-reference filing does not establish individual gauge invariance of its terms, so it does not by itself give the termwise PH marginal cancellation or transport of older gauge-invariant core families. The torus periods are divisible by the chosen localization sizes and exceed ten such cubes. The estimates are uniform on this compatible volume class.

4.9.1. *The uniform flat fluctuation gap.* Let Q_h be the actual flat straight row in eq. 3.mccxxiv, with $h = 1/n$. Its coarse curl is an actual positive surface average of the fine curl:

$$\begin{aligned}(\mathrm{d}_c Q_h a)_{\mu\nu}(y) &= h^6 \sum_{x\in B(y)} \sum_{r,s=0}^{n-1} (\mathrm{d}_h a)_{\mu\nu}(x + rhe_\mu + she_\nu) \\ &=: (T_h d_h a)_{\mu\nu}(y), \qquad \|T_h F\|_2 \le \|F\|_{2,h}.\end{aligned} \tag{4.cclxxxi}$$

The identity follows by telescoping the difference of the two straight rows. The sum of its positive weights in one coarse row is $h^6 n^6 = 1$. A fixed fine plaquette has exactly n^2 possible choices of (r, s); for each choice its starting point belongs to exactly one block. Thus its column sum is $h^6 n^2 = h^4$. Jensen's inequality proves the contraction, with the physical fine measure. Parallel unitary transports in a flat reference preserve this argument, including arbitrary flat twists.

Let $\mathcal{H}_k = S_k''(0)$. The derivative of the actual background at zero minimizes the flat curl energy subject to $Q_h a = v$. Indeed its physical stationarity after removal of the gauge penalty is $\mathrm{d}_h^\sharp \mathrm{d}_h a = Q_h^\sharp \omega$, so the energy cross term against any other lift with the same row value is zero. The literal fine gauge does not change that energy. Equation eq. 4.cclxxxi and the smooth exact row seed Lem. 3.180 consequently give

$$\|\mathrm{d}_c v\|_2^2 \le \langle v, \mathcal{H}_k v\rangle = \inf_{Q_h a = v} \|\mathrm{d}_h a\|_{2,h}^2 \le C\|v\|_2^2. \tag{4.cclxxxii}$$

For the upper bound insert its exact lift $E_Q v$, whose physical H^1 energy is bounded uniformly in h. No inverse of a coarse point-mass readout is used.

Let C_0 be the actual one-step pivot inclusion at the coarse flat field, using the plain outer average and its middle pivots. Let $K_{G,0}$ be its actual contour-gauge quadratic form. Therefore

$$K_{k,0} := C_0^{\mathsf{T}} \mathcal{H}_k C_0 + K_{G,0} \ge C_0^{\mathsf{T}} \mathrm{d}_c^\sharp \mathrm{d}_c C_0 + K_{G,0} \ge \gamma_0 I, \qquad \|K_{k,0}\| \le B_0. \tag{4.cclxxxiii}$$

The last lower bound is the proved actual first flat precision gap; the outer constraint here has not been replaced by the straight row. The bounds on $C_0, K_{G,0}$ depend only on L. Thus γ_0, B_0 are independent of the inner homogeneous depth.

4.9.2. *The complete nonlinear effective precision.* For a coarse background $V = e^{iB_0} V_0$ inside a fixed smaller ball, let $V(u) = e^{i\Phi_V(u)} V$ be the actual one-step outer pivot graph. All its unscaled radii and support bounds are those already used in § 4.7. Define the centered composed action

$$\begin{aligned}\mathscr{S}_{k,V}(u) &= S_k(V(u)) - S_k(V) - D_u[S_k(V(u))]_{u=0} u, \\ K_{A,k}(B_0) &= D^2 \mathscr{S}_{k,V}(0), \qquad K_k(B_0) = K_{A,k}(B_0) + K_G(V).\end{aligned} \tag{4.cclxxxiv}$$

Use the explicit relative contour gauge of eq. 4.ccxxxii for its auxiliary extension. At an actual axial constrained-critical V, the subtracted linear functional is zero, so this is precisely the physical fluctuation action.

The composition in eq. 4.cclxxxiv includes both inner response terms in D^2S_k and the outer constraint-curvature term $DS_k\,D^2\Phi_V$. To control it, substitute the finite local holomorphic map

$$B(u) = i^{-1}\log[V(u)V_0^{-1}]$$

in the actual action trees. Its coefficients have a fixed physical star and uniform Cauchy bounds depending only on L. The marked-leaf proof of eq. 4.cclxxx therefore gives a symmetric local operator family $K_{k,X}(B_0)$, with every matrix endpoint and every background input in its carrier. For a fixed $t_0 > 0$, its absolute physical-length Schur sums obey

$$\begin{aligned} K_k(B_0) &= \sum_X K_{k,X}(B_0), \\ \sup_b \sum_{X\ni b} e^{t_0\ell(X)} \sum_{b'} \|K_{k,X}(B_0)(b,b')\| &\le C_K, \\ \sup_b \sum_{X\ni b} e^{t_0\ell(X)} \sum_{b'} \|[K_{k,X}(B_0) - K_{k,X}(0)](b,b')\| &\le C_K\|B_0\|_\infty. \end{aligned} \tag{4.cclxxxv}$$

Here X is retained as a connected physical unit-cell carrier, before the larger filing grid is introduced. Retain its lifted cover tree as a tag, and use its charged length as $\ell(X)$; tags with the same cubical union can be combined after the norm sum. This also retains winding lengths before projection. The transpose column sums satisfy the same bounds. Local contour gauge terms are included as additional fixed-star terms.

These inequalities follow from the actual tree coefficients, not from a hypothesis on the desired precision. A marked second derivative contributes at most a polynomial in its leaf number; summability follows on the smaller ball. The first-derivative factor in the outer curvature term vanishes at the flat field and is $O(\|B_0\|)$. Thus after decreasing the fixed real ball,

$$\gamma I \le K_k(B_0) \le BI \tag{4.cclxxxvi}$$

uniformly in depth. Complex differences have the corresponding Schur Lipschitz bound on a fixed smaller ball. The same carrier estimates hold for $Q_{k,\tau} = \tau K_{A,k} + K_G$, for τ in a fixed disc about one.

4.9.3. *Admitting cube parameters for the nonlocal precision.* We construct the local Gaussians rather than declaring the nonlocal matrix finite-range. Set $H = I - Q_{k,\tau}/a$, with $a \ge 2B$. The identity has a one-cell carrier. Take the actual carrier expansion induced by eq. 4.cclxxxv, and write

$$H = H_{\le R} + H_{>R}, \qquad H_{\le R} = \sum_{\ell(X)\le R} H_X. \tag{4.cclxxxvii}$$

Every entry of $H_{\le R}$ reads only fields within radius $R + C_L$ of its endpoints and has that finite matrix range. For real data symmetrize each carrier term if necessary. Its tail has Schur norm at most Ce^{-t_0R}. For large R, eq. 4.cclxxxvi therefore implies

$$0 \le H_{\le R} \le r'I, \qquad r' = 1 - \gamma/(2a) < 1. \tag{4.cclxxxviii}$$

The lower bound uses $I - Q/a \ge I/2$ and a tail smaller than $1/4$. All constants are uniform.

Fix the requested coarse parameter radius $S > 2$. For a connected physical carrier, round it to localization cubes of side M. The geometric covering-tree bound gives

$$S^{\#X_M} \le S^{C_0}\exp\{C_0(\log S)\ell(X)/M\}.$$

Choose M large enough that the rate in this exponent is below $t_0/4$. In the operator-family norm charging this cube weight, eq. 4.cclxxxv then gives constants $C_H, c > 0$ such that

$$\|H\|_{\mathrm{car}},\ \|H_{\le R}\|_{\mathrm{car}} \le C_H, \qquad \|H_{>R}\|_{\mathrm{car}} \le C_H e^{-cR}. \tag{4.cclxxxix}$$

The norm is the maximum absolute row/column sum of all its carrier terms, with the displayed weight. Operator composition files into the union of two carriers sharing their intermediate bond. Thus these unions are connected, their weights are submultiplicative, and this norm is submultiplicative as well.

The Gaussian parameters themselves are placed on the coarse LM-cube grid of the UM expansion. The displayed M-cube carrier weight is stronger than the needed LM-cube weight. On the fixed held root guard use $\widehat{s}_\Delta = 1$, and use $\widehat{s}_\Delta = s_\Delta$ elsewhere in every carrier weight below.

Choose an integer m, and then $R = Dm$, where the fixed D is large enough that $cD > 2\log(1 + C_H) + 2$. Use the exact identities

$$\begin{aligned} H^m &= (H_{\le R})^m + E_m, \\ E_j &= H^j - (H_{\le R})^j = \sum_{i=0}^{j-1} (H_{\le R})^i H_{>R} H^{j-1-i}. \end{aligned} \tag{4.ccxc}$$

The second line defines an actual filed carrier series. It gives, simultaneously for $j \le m$,

$$\|E_j\|_{\mathrm{car}} \le j C_H^j e^{-cR}. \tag{4.ccxci}$$

Combine all paths inside the entries of $(H_{\le R})^m$ before assigning their parameters. At one common real background the Hilbert-row estimate of eq. 4.ccxxxix gives a pointwise Schur bound with the factor $(1+m(R+C_L))^2$. For a local coefficient family, take the supremum separately for each entry over its full real read set. Denote the maximum of this row sum and its transpose column sum by $\|\cdot\|_{\mathrm{Sch},+}$. The appropriate bound is

$$\|(H_{\le R})^m\|_{\mathrm{Sch},+} \le C(1 + m(R + C_L))^4 (r')^m. \tag{4.ccxcii}$$

Indeed each matrix entry is at most $(r')^m$ by eq. 4.cclxxxviii, and a row has at most $C(1 + m(R + C_L))^4$ possible endpoints. Every permitted local real read set extends by zero relative to V_0 to admitted global real data. Thus the entry estimate remains valid when its supremum is taken independently of the other entries. A supremum outside the whole row sum would not establish this bound.

For the uncombined families H_X, their tails and the words defining E_j, use the sums of individual carrier suprema on a common larger complex ball. The positive Taylor-tree majorants proving eq. 4.cclxxviii and eq. 4.cclxxx give precisely these stronger bounds. Their products are bounded by the positive products of those majorants. Thus C_H in eq. 4.cclxxxix and eq. 4.ccxci can be chosen for these norms as well.

Take M to be the smallest power of L at least $C_L m(R + C_L)$. The combined near entries then have connected ball carriers meeting a bounded number of M-cubes. Their cube weights cost only S^{C_L}. Weight the already filed E_m terms individually by their full carriers, and weight each combined near entry by its full ball carrier. Denote their sum by T_s, symmetrizing its carrier terms if needed. At $s = 1$, $T_1 = H^m$ exactly. On all independent $|s_\Delta| < S$,

$$\|T_s\|_{\mathrm{Sch}} \le C S^{C_L} (1 + m(R + C_L))^4 (r')^m + m C_H^m e^{-cR} =: q_m \longrightarrow 0. \tag{4.ccxciii}$$

Here $M = O_L(m^2)$. Its appearance in the volume budget is polynomial, whereas both terms on the right decrease exponentially.

For each prefix $j < m$, similarly put $A_{j,s} = (H_{\le R})^j + (E_j)_s$. Only the near prefix is left unweighted; its row from a root is contained in the fixed held root guard. The tail prefix is weighted by its actual full carriers. Choose the guard to contain every radius-$m(R + C_L)$ root ball,

using the same fixed eight-cell collar convention as before, with constants increased once. With $c_\ell = 4^{-\ell}\binom{2\ell}{\ell}$, define

$$R_W(s) = a^{-1/2}\chi_W \sum_{j<m} A_{j,s} \sum_{v\ge 0} c_{vm+j} T_s^v, \qquad D_W(s) = R_W(s)R_W(s)^{\mathsf{T}}. \tag{4.ccxciv}$$

At complete parameters this is exactly $\chi_W Q_{k,\tau}^{-1/2}$, by the grouped binomial identity. Thus its covariance is the actual Gaussian marginal. Every prefix-tail and short-step carrier contains its endpoints and every external coordinate read by that coefficient. Zero outside a wall domain therefore gives literal local dependence and wall factorization, with fixed-root heredity.

For the real comparison use $R_{0,r} = a^{-1/2}\sum_{j<m} c_j (H_{\le R,r})^j$. Its compression has covariance at least $a^{-1}I$, by eq. 4.cclxxxviii. The sum of the Schur norms of its prefixes is at most a polynomial in R. Equations eq. 4.ccxci and eq. 4.ccxciii therefore make $D_W(s)$ uniformly $o(M^{-4})$-close to that real covariance by increasing m. Finally shrink the fixed complex coarse-field and τ balls, after m, R, M have been fixed. To justify this step coefficientwise, let δ_H^+ be the positive carrier bound for the difference between the complex near matrix and the matrix at its real data. Nested-ball Cauchy estimates on the actual Taylor-tree majorants give $\delta_H^+ \le C\delta$, where δ is the tube size. A positive word expansion of the telescoping identity gives, for $1 \le j \le m$,

$$\Delta_j^+ \le j C_H^{j-1} \delta_H^+.$$

This is the same finite-power estimate as eq. 4.ccxlii; enlarge $C_H \ge 1$ once if necessary. Choose the tube so that $\sum_{j=1}^m \Delta_j^+ \le (r')^m$. This pays the whole prefix sum and the m-th power perturbation. Increasing the fixed constant in eq. 4.ccxciii absorbs the latter term, and the covariance comparison remains $o(M^{-4})$. No bound polynomial in m for this coefficientwise perturbation is required. These radii are positive and independent of depth and coupling. Local data in this coarse product ball extend by zero relative to V_0, so no extra curvature-extension premise is needed. This proves the local Gaussian comparison, enlarged independent cube discs and exact structural identities needed by the homogeneous unmarked expansion for the actual nonlocal effective precision.

4.9.4. *The actual intrinsic and ordinary-source vertices.* Fix $g = \theta^{-1/2}$ and $\mathfrak{b} = A_1(\log\theta)^p$, with $p > 1/2$. Keep the real product cutoff $\chi(B) = \prod_b \mathbf{1}_{|B(b)|<\mathfrak{b}}$ fixed throughout. Take θ so large that $16C_L g\mathfrak{b}$ is less than both the outer pivot radius and the margin in the actual coarse-data ball just proved. Then $V(gB)$, its normalized radial paths and the needed complex fluctuation boxes all remain in that ball. In particular $U_k(V(gB))$ is the actual admitted homogeneous background on the entire box; its existence has not been replaced by an activity-norm assumption.

Compose each of the actual action terms in eq. 4.cclxxix with the outer graph. Subtract its constant, linear and quadratic Taylor terms in u, before summing. The global linear sum is the centering term in eq. 4.cclxxxiv, and the global quadratic sum is the complete $K_{A,k}$. Each composed member is holomorphic on a fixed ℓ^∞ free-coordinate ball and reads only the fixed chart collar of its filed carrier. If its admitted bound is B_X, directional Cauchy Taylor remainder gives

$$|g^{-2}R_{3,X}(gB)| \le CB_X g \max_{b\in X^+} |B(b)|^3 \le CB_X g\mathfrak{b} \sum_{b\in X^+} |B(b)|^2. \tag{4.ccxcv}$$

The first inequality includes the fixed analytic-radius cost; the maximum is over its actual chart support. The second inequality is valid on the kept cutoff and on its fixed complex margin after a fixed change of constants. Thus the actual Gaussian-growth potential norm has amplitude $C_M g\mathfrak{b}$, uniformly in depth. No derivative count proportional to the fine lattice size is present.

The outer relative contour gauge, Haar and pivot coarea factors are the literal one-step factors of eq. 4.ccxxxii and eq. 4.ccliv, on the present coarse lattice. Their supports and unscaled derivative bounds depend only on L, so their intrinsic remainders have the same amplitude. The zero-field coarea value $J_V(0)$ stays in the Gaussian normalizer.

There is an actual all-depth ordinary source class in this range. Take fine plaquette profiles $w(p)$, with $\|w\|_\infty < \rho_w$ on independent fixed coefficient discs, and put

$$H_{k,w}(V) = \sum_{p>0} w(p) a_p(U_k(V)). \tag{4.ccxcvi}$$

Its unit-cell action bound is $C\rho_w \|A\|^2_{X_\alpha}$, by the same h^4 plaquette count as eq. 4.cclxxvii. Apply the actual action-tree construction with these weights at its outer action vertices. It produces a genuinely local analytic coarse-source family with norm $C_M\rho_w$, uniformly in depth and in the number of fine plaquettes. The literal fine gauge preserves each plaquette trace separately, so this family represents precisely eq. 4.ccxcvi.

Its interaction is the exact difference $H_{k,w}(V(gB)) - H_{k,w}(V)$. Cauchy differentiation on the already admitted coarse-data margin gives its potential amplitude $C_M\rho_w g\mathfrak{b}$. The background, Gaussian, gauge weight and sharp cutoff have not been moved when inserting this ordinary source. Additional inputs already supplied as local holomorphic coarse-data families with incidence norm H_0 obey the same difference bound $C_M H_0 g\mathfrak{b}$. This last optional statement does not assert that arbitrary older fine-core families have already been transported into that class.

4.9.5. *The formed homogeneous families and their physical range.*

Proposition 4.18 (Actual homogeneous formation near a flat reference)**.** There are fixed positive coarse-data and τ radii and a coupling floor, independent of homogeneous depth and compatible volume, for which the actual regular homogeneous step with action xS_k, contour-gauge weight θ and the ordinary sources eq. 4.ccxcvi has an analytic filed decomposition $\mathbf{N}^\circ_k + \mathbf{G}_{k,\tau} + \mathbf{F}_k$. For every fixed requested output rate its constructed pointing satisfies

$$\|\mathbf{N}^\circ_k\|_+ \le C_N, \qquad \|\mathbf{G}_{k,\tau}\|_+ \le C_G|\tau - 1|, \qquad \|\mathbf{F}_k\|_+ \le C_F\{(1 + \rho_w + H_0)g\mathfrak{b} + e^{-c\mathfrak{b}^2}\}. \tag{4.ccxcvii}$$

Here $H_0 = 0$ unless the optional already filed coarse-input class is used. The constants are independent of depth. The normalizer and mismatch are retained as Gaussian terms, not counted in $\mathbf{F}_k$. ■

Proof. The local covariance construction eq. 4.ccxciv supplies all enlarged cube discs, Gaussian comparison, normalization, fixed-support heredity and wall factorization needed in the actual UM expansion. In each functional take all free bonds of its full fine support, so its held guard also contains any background inputs constant along that fibre. Its scalar dimension is at most $C(LM)^4 \# Z$. The comparison error can be chosen below $c_*(LM)^{-4}$, without an auxiliary-volume factor.

The actual interaction estimates just proved give the UM potential budget $C_M(1 + \rho_w + H_0)g\mathfrak{b}$. The unchanged sharp cutoff gives the Gaussian tail $Ce^{-c\mathfrak{b}^2}$. The already proved finite Mayer, connected cluster and union-filing theorems therefore give the stated $\mathbf{F}_k$ bound. The normalized cubic, ordinary radial and moving- threshold pointings are exactly those used in the first-step proof; their analytic margins and budgets are now uniform in depth.

For the Gaussian normalizer, use the actual carrier expansion of $\log Q$, grouping $\log Q = (\log a)I - \sum_{j<m,v\ge 0,vm+j>0} H^j(H^m)^v/(vm+j)$. The near combined powers and the filed tails have precisely the summable cube weights proved above. Root the trace at its actual diagonal bond and add the local zero-field pivot terms. This proves the $\mathbf{N}^\circ_k$ bound after vacuum subtraction. The same construction on a larger τ disc gives the mismatch bound; its pointing is its actual resolvent insertion as in eq. 4.cclii. □

The actual source functions are holomorphic on independent fixed fine profile discs. The same argument as Cor. 4.16 therefore gives

$$|D^n_w F_k(X; w)[h_1, \dots, h_n]| \le Cg\mathfrak{b}\, n!(2e/\rho_w)^n e^{-\kappa d_c(X)} \prod_j \|h_j\|_\infty$$

on the half-radius source ball, with the reserved rate for pointed terms. There is no factor counting fine plaquettes or source profiles in this bound.

4.9.6. *The actual nested physical step.* To identify the complete physical step, take $W = e^{iB_+}V_{0,+}$, where $V_{0,+} = M(V_0)$ and $\|B_+\|_\infty$ is sufficiently small, and form its actual depth-$k+1$ background $U_+(W)$. Choose the flat fine extensions compatibly at the two depths, so the intermediate flat reference is the partial average of that same flat fine field. The partial averaging recursion and its raw bound eq. 3.mcclvi put $V = M^k U_+(W)$ in the coarse-data ball above. Its node-fixed one-step axial gauge is also there after a fixed decrease of the outer radius. The required gauge extension is the bounded local node extension from the homogeneous background construction. More explicitly, truncating that recursion at its intermediate spacing $1/L$ gives $\|a_k\|_\infty + \|v_k - I\|_\infty \le C_L\|B_+\|_\infty/L$. On rescaling this spacing to one, the fine potential is A/L and its first difference is DA/L^2. The one-step axial node values have size $C_L\|B_+\|_\infty$; their actual smooth fine extension has the same bound in the required strong gauge norm. Thus the intermediate field and its axial representative remain within $C_L\|B_+\|_\infty$ of the fixed strong orbit chart, uniformly in k.

This intermediate field is an actual constrained-critical point for S_k under $M(V) = W$. Indeed every fine variation with $D(M^k)\delta U = 0$ also has $D(M^{k+1})\delta U = 0$. Thus $U_+(W)$ is critical on its M^k-fibre. Both it and the constructed $U_k(V)$ are in the fixed strong regularity class, after rescaling the intermediate coarse spacing by the fixed factor L. Orbit uniqueness in Prop. 3.183 identifies their actions. To check outer stationarity, lift any admitted curve $V(t)$ with $M(V(t)) = W$ by its actual background $U_k(V(t))$. Then $M^{k+1}U_k(V(t)) = W$ exactly. At $t = 0$ its background is the same critical orbit as $U_+(W)$, so $\partial_t A(U_k(V(t)))|_{t=0} = 0$. This proves that $DS_k(V)$ annihilates the outer constraint tangent. The linear centering in eq. 4.cclxxxiv therefore vanishes physically.

On $|B| < \mathfrak{b}$, the actual $V(gB)$ stays in the same fixed coarse-data ball, so all inner backgrounds exist on one admitted chart. The outer regular-fibre anchor, coarea density and contour-gauge test removal are the literal one-step proofs already given; their constants do not involve k. The exact physical normalization is therefore eq. 4.cclxv with $A(U)$ replaced by $S_k(V)$, the present complete precision, and its actual zero-field pivot factor. At these critical axial pairs gauge invariance of the actual effective action supplies the orbit-null action sector. Thus the physical mismatch total is again $(r/2)\log\tau$, with the explicit numerical $\tau^{-d/2}$ retained separately.

This physical range is a fixed bond ball about a frozen flat reference. Extra history plaquette indicators still require their own threshold margin. The proof does not extend the local functions across arbitrary flat-reference changes, to all curvature-only fields, to independent fine currents, or to proper-region and history towers. Those conclusions are not prerequisites for the homogeneous near-flat step proved here.

4.10. **Gauge invariance of the complete homogeneous pointing.** The fixed-reference Taylor terms in Prop. 4.17 need not be individually gauge invariant. A complete physical pointed density can nevertheless be constructed invariantly: specify its insertions at the original fine plaquettes before making the Taylor-tree filing. This distinction is needed for its quadratic Ward identity. The result below concerns the homogeneous near-flat range already proved, with the literal fine plaquette sources and no untransported older fine-core inputs. Here V is the input coarse background of that one-step integral. The actual output datum W enters through the critical intermediate background $V = p_k(W)$, considered explicitly below.

4.10.1. *Invariant functions before filing.* For each original fine plaquette put

$$f_{k,p}(V) = a_p(U_k(V)), \qquad f_{k,p;V}(u) = f_{k,p}(e^{i\Phi_V(u)}V). \tag{4.ccxcviii}$$

These are the actual functions, before decomposition into connected carriers. For every sufficiently small coarse site gauge g whose two data lie in the admitted ball, the literal node extension and

orbit uniqueness of Prop. 3.183 give

$$f_{k,p}(V^g) = f_{k,p}(V). \tag{4.ccxcix}$$

Indeed a fine gauge with those node values takes the critical field to the same regular orbit as $U_k(V^g)$. A fine gauge leaves the trace of each plaquette, at its fixed position p, unchanged. This is an assertion about the nonlocal function $f_{k,p}$, not its local Taylor pieces.

Write $(O_g u)(b) = \mathrm{Ad}(g(b_-))u(b)$ on outer free bonds. The actual pivot equation, its relative coarse logarithm and its free-identity convention give

$$\Phi_{V^g}(O_g u) = O_g \Phi_V(u), \qquad f_{k,p;V^g}(O_g u) = f_{k,p;V}(u). \tag{4.ccc}$$

The first identity follows by covariance and uniqueness of the same small pivot graph. At real unitary g, O_g is a real block-orthogonal map on the actual integration space. It preserves each retained cutoff.

Let $R_{3,p;V}(u)$ be the third-order remainder of $f_{k,p;V}$, subtracting its own constant, linear and quadratic jets. Each homogeneous jet transforms under the same linear map O_g. Consequently $R_{3,p;V^g}(O_g u) = R_{3,p;V}(u)$. This gives invariant cubic remainders before their carrier expansion.

The other functions have the same property directly. Every relative contour-gauge trace conjugates at its node. The Haar density is a function of the invariant colour pairing of Φ. A pivot matrix transforms on its two sides by adjoint matrices of determinant one; its determinant ratio and chosen logarithm are invariant. The ordinary source $\sum_p w(p) f_{k,p}$ is invariant with its fine profile held fixed.

4.10.2. *The actual normalization shares and insertions.* For each fine plaquette p, choose nonnegative fixed numbers $\vartheta_{p,b}$ on outer free bonds with

$$\sum_b \vartheta_{p,b} = 1, \qquad \vartheta_{p,b} = 0 \quad \text{if } d(\mathrm{mid}\, p, \mathrm{mid}\, b) > C_L. \tag{4.ccci}$$

One choice is equal shares on the nearest free-bond midpoints. Their distance is uniformly bounded because each outer L-block contains free bonds. This choice is covariant under the geometric symmetries preserving the actual pivot pattern. It is independent of the field and its Taylor-carrier presentation.

Put $P_{p,V}(B) = -\tau g^{-2} R_{3,p;V}(gB)$. The normalized intrinsic interpolation and its actual bond insertion are

$$\begin{aligned} P_{p,V;t}(B) &= t^{-2} P_{p,V}(tB), \\ a_{p,t,b}(V;B) &= t^{-2} B(b) \cdot D_b P_{p,V}(tB) - 2\vartheta_{p,b} t^{-3} P_{p,V}(tB). \end{aligned} \tag{4.cccii}$$

The shares are specified at the original plaquette, before any carrier filing. For independent parameters the defining function is

$$\left(\sum_b \vartheta_{p,b} t_b\right)^{-2} P_{p,V}((t_b B(b))_b).$$

Its derivative at $t_b = t$ is exactly the displayed insertion. Thus its sum over b is the derivative of the normalized diagonal path, including its normalization term. Nonnegative shares keep the denominator at least $t/2$ on $\max_b |t_b - t| < ct$, $c < 1/2$. All the field arguments remain in the proved common chart.

Equation eq. 4.ccc and invariance of the colour pairing show, for every bond separately,

$$a_{p,t,b}(V^g; O_g B) = a_{p,t,b}(V;B). \tag{4.ccciii}$$

The same construction is applied to the original relative gauge remainders with shares on their fixed local free supports. The Haar, pivot and ordinary-source legs use their ordinary radial insertions $B(b) \cdot D_b V(tB)$, which satisfy the same identity.

There is a uniform rooted bound for these actual insertions. Retain each p as the outer action-tree tag. Its local analytic bound is Ch^4, before summing the fine plaquettes. The proof of eq. 4.ccxcv then gives its intrinsic amplitude $Ch^4 g\mathfrak{b}$. A derivative rooted at b marks a leaf in that actual tree. The exponential root-transfer estimate sums these terms over the plaquette's physical cell positions. The normalization root in eq. 4.cccii is within C_L of p by eq. 4.ccci. It has the same carrier budget. Finally there are $O(h^{-4})$ fine plaquettes per physical cell, cancelling their h^4 bound. The order-three radial vanishings and eq. 4.cxxx therefore give the same $Cg\mathfrak{b}$ integrable rooted norm as the homogeneous formation theorem, with a fixed rate reserve. This step uses the absolute marked-tree sums, rather than differentiating only an unpointed output estimate.

4.10.3. *Invariant normalized expectations.* The complete precision of the actual composed action and relative contour gauge transforms on the shared domain as $Q_\tau(V^g) = O_g Q_\tau(V) O_g^{\mathsf{T}}$. For real unitary gauges, Gaussian change of variables by O_g, the fixed product cutoff and eq. 4.ccciii therefore give invariance of each normalized bond insertion expectation. Every interpolation partition function is zero-free in the ordinary cluster domain proved above. Its canonical logarithm and expectation are thus unambiguous there. The identities are holomorphic in the external data and extend on their common complex domain.

The cutoff leg is invariant as well. Its one-bond moving threshold and its boundary surface measure depend only on $|B(b)|$, which the real orthogonal map $O_g(b)$ preserves. Its rooted Gaussian tail estimate is the already proved one. The free mismatch leg has its actual density $(\tau - 1)\int \mathrm{tr}(K_G Q_{\tau,u}^{-1})(b,b)du/2$; the diagonal colour trace is invariant.

Let $\mathcal{E}_b^F(V)$ be the integral of the cutoff and interaction insertions just specified. The exact path identities give $\sum_b \mathcal{E}_b^F = \mathcal{F}_k$. Use the fixed midpoint tents for the output base point:

$$F_{k,z}^{\mathrm{phys}}(V) = \sum_b h_z(\mathrm{mid}\, b)\mathcal{E}_b^F(V). \tag{4.ccciv}$$

The tents h_z are on the outer output-node lattice, of spacing L in the present input units. After rescaling, the same z is the output unit-lattice basepoint in the composition below. This complete physical density is gauge invariant. Its local analytic filing obtained from the actual insertions has the uniform bound $Cg\mathfrak{b}$ of eq. 4.ccxcvii. The individual filed functions still need not be gauge invariant.

Proposition 4.19 (The physical quadratic Ward identity)**.** At a trivial flat reference, the complete density eq. 4.ccciv has zero first derivative in coarse bond logarithms. Its quadratic kernel annihilates a coarse gradient in either argument:

$$D^2 F_{k,z}^{\mathrm{phys}}(0)[\mathrm{d}_c\lambda, a] = 0. \tag{4.cccv}$$

Its second spatial moments are bounded by $Cg\mathfrak{b}$, uniformly in homogeneous depth. ■

Proof. A constant colour conjugation fixes the trivial field. An invariant linear functional in one adjoint colour is zero, so $DF_{k,z}^{\mathrm{phys}}(0) = 0$. Differentiate the proved local gauge identity in a small gauge parameter and then in the field. The linear gauge variation at zero is $\mathrm{d}_c\lambda$. The nonlinear variation multiplies the vanished first derivative. This proves eq. 4.cccv; symmetry gives the second argument. One may first use finite-support gauges. The exponential marked-tree bounds then extend the identity to the stated finite-moment class.

The filed insertion bounds are holomorphic on a fixed external coarse-coordinate ball. Two Cauchy derivatives cost its fixed inverse radius squared. A carrier has at most $C(LM)^4(1 + d_c(X))$ outer coarse bond coordinates. Summing all derivative pairs and multiplying by their squared spatial distances therefore costs only a fixed polynomial in $d_c(X)$, with an M-dependent constant and no fine depth factor. The reserved spatial rate makes that polynomial summable. Thus the second moments have the asserted $g\mathfrak{b}$ bound. In particular an affine curvature probe may be inserted in this quadratic kernel by its absolutely convergent spatial moments. □

4.10.4. *Composition with the actual critical background.* Let $p_k(W)$ be the intermediate coarse field $M^k U_{k+1}(W)$, in its actual one-step axial gauge, as constructed in the physical-range argument above. Define the complete output density, now in output coarse units, by

$$\mathscr{F}_{k,z}(W) = F_{k,z}^{\mathrm{phys}}(p_k(W)). \tag{4.cccvi}$$

This is the density for the actual output datum W. Gauge invariance makes it independent of the admitted intermediate representative.

Corollary 4.20 (Physical homogeneous marginal bound)**.** At the trivial flat reference and zero ordinary profile, the complete output densities eq. 4.cccvi retain the quadratic Ward identity and the uniform rooted second-moment bound $Cg\mathfrak{b}$. Their infinite-lattice companion has the physical Yang–Mills quadratic coefficient

$$|b_k^{\mathrm{phys}}| \le Cg\mathfrak{b}, \tag{4.cccvii}$$

with the Wilson normalization of eq. 4.i. This coefficient belongs to the complete physical pointed density. ■□

Proof. The map p_k is analytic on the fixed output ball. Its actual first response $H_k = Dp_k(I)$, written in potential coordinates, has uniform exponential row and column moments. Indeed it is the localized response eq. 3.mccxcv followed by a partial actual average and its local axial gauge. Each partial row has bounded physical carrier and bounded directional norm by eq. 3.mcclvi. There are only L^4 intermediate cells per output cell. Thus summing its responses costs a fixed L factor, not a fine-depth factor.

Write $\mathcal{K}_{k,z}$ for the quadratic kernel already proved for $F_{k,z}^{\mathrm{phys}}$. The first jet vanishes, so the second-order chain rule is exactly

$$D^2\mathscr{F}_{k,z}(I)[a,b] = \mathcal{K}_{k,z}[H_k a, H_k b]. \tag{4.cccviii}$$

No $D^2 p_k$ term remains. The kernel moments on the right are bounded by composing the two exponentially localized responses. For example the squared distance of an output bond from z is bounded by a constant times the squared distance to its intermediate bond plus that intermediate bond's squared distance from z. Sum the former with the exponential response moment and the latter with the rooted moment of $\mathcal{K}_{k,z}$. This is the pointed root-transfer bound and retains $Cg\mathfrak{b}$.

If W is changed by a sufficiently small output site gauge, actual orbit covariance of U_{k+1} shows that the two intermediate fields are gauge equivalent, with the prescribed output node values. The proved invariance of $F_{k,z}^{\mathrm{phys}}$ therefore makes $\mathscr{F}_{k,z}$ invariant. Its first jet is zero and the same differentiated gauge identity gives the output Ward identity.

We check the physical affine normalization using the actual successive map. By eq. 5.clxxvi, its first response to an affine output field of constant curl is the intermediate affine potential modulo a lattice gradient. That identity follows from the actual nesting in § 4.9.6, the full Wilson affine response, and then its literal partial mean. The literal and straightened means are kept distinct; their frame correction is an intermediate gradient. The vanished first jet removes the nonlinear response, and the Ward identity removes both gradient legs. Thus eq. 5.clxxvii applies to this complete invariant density. Its physical Wilson normalization is unchanged by the intermediate change of units.

The original plaquette functions, the nearest-midpoint shares and the midpoint tents are covariant under translations and the full hypercubic group, including reflections. Orbit uniqueness preserves this covariance in the complete composed density. The quadratic tensor classification eq. 4.xviii–eq. 4.xix therefore applies to this invariant kernel with its proved moments. It is applied to the complete kernel, using the Ward and symmetry identities just established; no termwise invariant-core hypothesis is inferred from its Taylor filing. The extractor is a fixed linear second-moment functional, so the bound in eq. 4.cccvii follows from the composed moment bound. Absolute

lifted-carrier and cluster convergence gives the infinite-lattice companion as the limit of compatible tori, with these derivatives and moments. □

The quadratic identity belongs to the complete invariant density. It is not applied separately to the fixed-flat Taylor terms. A termwise invariant core family, an older-family transport theorem or a full fine-current marginal identification still requires its corresponding additional argument.

4.11. **An even-coupling improvement of the intrinsic formation bound.** The first absolute formation estimate uses the size of the cubic fluctuation vertex. The normalized integral has an additional cancellation: changing the sign of the fluctuation coupling can be compensated by changing the sign of every Gaussian variable. We use that exact symmetry before estimating the resulting local activities. It improves their intrinsic smallness by one power of the coupling.

Keep the homogeneous law and its actual Gaussian structure from Prop. 4.18. In this subsection the cutoff radius $\mathfrak{b}$ is fixed independently of a complex parameter g. The parameters τ, the external background and any admitted ordinary coarse input are also held fixed. Only at the end do we set $g = \theta^{-1/2}$ and $\mathfrak{b} = A_1(\log\theta)^p$. The explicit numerical Jacobian and Gaussian normalizer have already been factored as in eq. 4.cclxv; their logarithms are not included in the intrinsic function considered here.

4.11.1. *The analytic deformation at zero coupling.* Write the exact normalized fluctuation integral in the form

$$\mathcal{Z}(g;V,H,\mathfrak{b}) = \int \chi_{\mathfrak{b}}(B) e^{P(g;V,H,B)}\, d\mu_{Q_\tau(V)^{-1}}(B), \qquad \mathcal{F}(g) = \log \mathcal{Z}(g). \tag{4.cccix}$$

The Haar and pivot ratio in P is the actual one. The action and relative gauge sectors are their exact centred functions minus their quadratic Gaussian parts. The ordinary input is $H(V(gB)) - H(V)$. The free coordinate graph $V(u)$ and all precision localizations are independent of g.

Every divided remainder in this formula is analytic at $g = 0$. For example a centred action remainder of order three has the convergent expansion

$$g^{-2} R_3(gB) = \sum_{n\geq 3} \frac{g^{n-2}}{n!} D^n R_3(0)[B,\dots,B].$$

The ordinary density and source terms are analytic functions of gB, with their constant values subtracted. Thus $P(0;V,H,B) = 0$. The cutoff is a fixed real even function of each $B(b)$.

The absolute formation proof remains valid on the complex disc

$$|g| < r_g = c_*/\mathfrak{b}. \tag{4.cccx}$$

Here $c_* > 0$ is fixed below all intrinsic-vertex and analytic-chart ceilings. Its proof estimates the vertices by $C|g|\mathfrak{b}$; positivity of g was not used. All Gaussian covariances are held fixed on their already admitted complex background domain. The cluster series is therefore a holomorphic, zero-free logarithm on this disc, with the same fixed spatial rate reserve. This gives holomorphy in the norm of the local activity family, rather than only pointwise finite-volume holomorphy.

At zero coupling,

$$\mathcal{F}(0) = \log \mu_{Q_\tau(V)^{-1}}(\chi_{\mathfrak{b}}).$$

Its actual wall expansion is exponentially small in $\mathfrak{b}^2$. With a fixed slightly smaller spatial rate, its unpointed and canonical pointed norms are bounded by

$$Ce^{-c\mathfrak{b}^2}. \tag{4.cccxi}$$

Polynomial wall factors have been absorbed by decreasing c. This is the wall estimate of the formation proof, and does not bound a whole-volume cutoff probability away from zero.

4.11.2. *Exact sign symmetry of the filed functions.* The exact integrand satisfies

$$P(-g;V,H,-B)=P(g;V,H,B). \tag{4.cccxii}$$

Indeed every occurrence of the actual group field has argument gB, which is unchanged. A prefactor g^{-2} is also unchanged. The subtracted quadratic Gaussian form is even in B. The ordinary input depends on the same group field, and the Haar and pivot functions have that same argument. The even Gaussian density and cutoff therefore give

$$\mathcal{F}(-g)=\mathcal{F}(g). \tag{4.cccxiii}$$

The equality uses the zero-free branch through $g=0$, so there is no undetermined logarithm constant.

We need this identity for the local family, where an arbitrary refiling could obscure cancellation. Use the actual Taylor and carrier filing of the homogeneous formation theorem. That filing commutes with the involution $(g,B)\mapsto(-g,-B)$: each Taylor monomial keeps its degree in g and its degree in the fluctuation variables, and its carrier is independent of their signs. An intrinsic degree-n monomial has the factor $g^{n-2}B^n$, whose sign is $(-1)^{2n-2}=1$. An ordinary degree-n term has g^nB^n, also with sign one.

Mayer exponentiation, multiplication, the fixed Gaussian weakening, connected extraction and union filing all commute with this involution. Any partial Gaussian integral has the same symmetry under negation of all its Gaussian coordinates. Consequently every resulting filed activity can be chosen even in g. This statement concerns sign parity; it does not assert gauge invariance of individual fixed-reference terms.

The canonical bond pointings have the same parity. For an intrinsic plaquette function $P_g(B)$, its normalized insertion is

$$t^{-2}B(b)\cdot D_bP_g(tB)-2\vartheta_{p,b}t^{-3}P_g(tB).$$

Equation eq. 4.cccxii changes both the vector and its derivative sign in the first term; the second term is unchanged. Ordinary radial insertions satisfy the same identity. A moving cutoff threshold and its surface measure are even in $B(b)$. Thus the actual pointed expectation, its integral in the interpolation parameter, and its carrier filing are even as well.

4.11.3. *The improved absolute bound.*

Proposition 4.21 (Quadratic intrinsic formation smallness)**.** Let $\mathbf{F}(g)$ denote the actual unpointed or canonical pointed homogeneous intrinsic activity family, with the spatial rate just fixed. For $|g|\le c_*/(2\mathfrak{b})$,

$$\|\mathbf{F}(g)\|\le C\{\,|g|^2\mathfrak{b}^2+e^{-c\mathfrak{b}^2}\,\}. \tag{4.cccxiv}$$

The constants are independent of the inner fine depth and volume. The same statement holds on the smaller fixed external and ordinary source discs of the formation theorem. ■

Proof. The activity-valued function $\mathbf{F}(g)-\mathbf{F}(0)$ is holomorphic and even on eq. 4.cccx. On a fixed slightly smaller concentric disc its norm is bounded by a constant, since the original estimate is $C|g|\mathfrak{b}$ plus the wall tail. Use the Cauchy circle of radius $R=3r_g/4$. For the stated $|g|\le r_g/2$, $|g|/R\le 2/3$.

Its Taylor coefficients of odd order vanish, and its constant coefficient is zero. Cauchy's estimate in the activity norm gives

$$\|\mathbf{F}(g)-\mathbf{F}(0)\|\le C\sum_{n\ge1}(|g|/R)^{2n}\le C|g|^2\mathfrak{b}^2.$$

Add eq. 4.cccxi. The same proof applies to each pointed norm because the preceding parity argument held before those norms were estimated. □

For the physical choice $g = \theta^{-1/2}$, $\mathfrak{b} = A_1(\log\theta)^p$, $p > 1/2$, this reads

$$\|\mathbf{F}\| \le C\theta^{-1}(\log\theta)^{2p} + C\exp[-c(\log\theta)^{2p}]. \tag{4.cccxv}$$

The tail is smaller than any fixed inverse power of θ after increasing its threshold, because $2p > 1$. This is a stronger bound than estimating the cubic vertex alone. It is still a bound on the specified homogeneous model.

Two external field derivatives cost a fixed inverse radius squared. The reserved spatial rate absorbs their coordinate incidence and second spatial moments, exactly as in Prop. 4.19. For zero ordinary profile, or an input already proved to satisfy the same complete physical gauge identities and the Euclidean symmetries required for a scalar coefficient in Cor. 4.20, that complete physical quadratic coefficient consequently satisfies

$$|b_k^{\mathrm{phys}}| \le C\{\theta^{-1}(\log\theta)^{2p} + e^{-c(\log\theta)^{2p}}\}. \tag{4.cccxvi}$$

This uses the complete invariant pointed density and its actual critical substitution. It is not a coefficient readout from separate noninvariant Taylor pieces.

4.11.4. *The first nonzero fluctuation coefficient.* The same argument identifies the algebraic coefficient without a numerical two-loop claim. Write the actual exponent as

$$P(g;B) = gP_1(B) + g^2P_2(B) + O(g^3).$$

Then P_1 is odd and P_2 is even. Let $\langle\cdot\rangle_\chi$ be the normalized Gaussian expectation with its retained even cutoff. Expansion of the normalized integral gives

$$\mathcal{F}(g) = \mathcal{F}(0) + g^2\left\{\langle P_2\rangle_\chi + \tfrac{1}{2}\langle P_1; P_1\rangle_\chi\right\} + O(g^4). \tag{4.cccxvii}$$

Here the semicolon is the connected two-variable expectation. Its subtracted product is zero because $\langle P_1\rangle_\chi = 0$. The $O(g^4)$ is in the smaller activity norm, bounded by $C|g|^4\mathfrak{b}^4$.

For precision, let $\mathscr{S}_V(u)$ be the complete centred composed action on the actual pivot graph, and let $\mathscr{G}_V(u)$ be the actual relative contour gauge. Their quadratic parts are the two parts of $Q_\tau(V)$. Let $\mathscr{H}_V(u) = H(V(u))$ and $\mathscr{J}_V(u) = \log J_V(u)$. All derivatives below are at zero. The two polynomials in eq. 4.cccxvii are

$$\begin{aligned} P_1(B) &= -\frac{\tau}{6}D^3\mathscr{S}_V[B^3] - \frac{1}{6}D^3\mathscr{G}_V[B^3] + D\mathscr{J}_V[B] + D\mathscr{H}_V[B], \\ P_2(B) &= -\frac{\tau}{24}D^4\mathscr{S}_V[B^4] - \frac{1}{24}D^4\mathscr{G}_V[B^4] + \frac{1}{2}D^2\mathscr{J}_V[B^2] + \frac{1}{2}D^2\mathscr{H}_V[B^2]. \end{aligned} \tag{4.cccxviii}$$

Repeated arguments are denoted by B^n. In particular the action vertices contain derivatives of the nonlinear pivot constraint; the relative gauge vertices are those of the literal trace gauge. Neither can be replaced by an ambient unconstrained action or a squared logarithmic gauge.

Without the cutoff, eq. 4.cccxvii is evaluated by Gaussian pairings. Each contribution of $\langle P_1; P_1\rangle$ has a line joining the two vertices: their separate odd expectations vanish. Thus its connected carrier is rooted without a second volume factor. The cutoff correction has the already proved wall smallness. Equation eq. 4.cccxvii therefore specifies the next fluctuation coefficient as an actual connected expression, but does not assign it a universal numerical beta-function value.

4.12. **Gaussian comparison for the classical-pullback homogeneous law.** We compare the actual regular homogeneous one-step measure with its quadratic Gaussian reference at the trivial flat background. The observables are fine Wilson plaquettes evaluated on the classical background $U_k(V)$. Their quadratic parts are resummed into the Gaussian precision. The source derivatives are then rooted at their original fine plaquettes. This gives an error proportional to source support mass, rather than the whole lattice volume.

This measure is a classical-pullback one-step model. It does not include an arbitrary older-core Gibbs weight, history conditioning or independent fine quantum fluctuations. The Gaussian

observables below are quadratic functions of a Gaussian field; their joint law need not itself be a multivariate Gaussian.

4.12.1. *The actual normalized fibre measure.* Fix the compatible periodic geometry of Prop. 4.18, inner depth k, and $h = L^{-k}$. Let τ be real in the proved fixed interval about one, and put $g = \theta^{-1/2}$, $\mathfrak{b} = A_1(\log\theta)^p$, $p > 1/2$. Work on the regular coarea branch of the actual outer fibre $M(V) = 1$. The background is exactly one. Its actual graph is

$$V(gB) = e^{i\Phi_1(gB)}, \qquad \Phi_1(gB)|_{\text{free}} = gB, \qquad \chi(B) = \prod_b \mathbf{1}_{|B(b)|<\mathfrak{b}}. \tag{4.cccxix}$$

The entire regular fibre on this free box is this graph, by the proved pivot anchor and uniqueness. The contour-gauge indicators equal one there. Their reference weights, the gauge normalizers and the real box remain fixed as sources vary.

Let $J_1(u)$ be the literal Haar and raw pivot coarea density in eq. 4.ccliv. The normalized law $\nu_{k,g}$ is the pullback to B of

$$\frac{1}{Z_{k,g}} \mathbf{1}_{\text{retained tests}}(V) e^{-\tau S_k(V)/g^2 - \mathrm{GF}(V)/g^2} \delta_G(M(V))\, dV. \tag{4.cccxx}$$

The delta denotes the stated regular Haar coarea disintegration. All field-independent gauge, Haar and g^d factors cancel in this normalized measure; no determinant depending on a source is discarded.

Write $K_0 = Q_{k,\tau} = \tau K_{A,k}(0) + K_G(1)$, $C_0 = K_0^{-1}$, and $\mu_0 = \mu_{C_0}$. For this subsection only, write C for the free-to-full pivot tangent $D\Phi_1(0)$, to distinguish it from the covariance C_0. Define the exact remainder

$$P_g(B) = -g^{-2}\left\{\tau S_k(V(gB)) + \mathrm{GF}(V(gB)) - \tfrac{1}{2} g^2 B^{\mathsf{T}} K_0 B\right\} + \log \frac{J_1(gB)}{J_1(0)}. \tag{4.cccxxi}$$

There is no linear term: the flat action, contour gauge and their first derivatives vanish. Consequently the literal normalized identity is

$$d\nu_{k,g}(B) = \frac{\chi(B) e^{P_g(B)}}{\mathcal{Z}_g(0)}\, d\mu_0(B), \qquad \mathcal{Z}_g(0) = \mu_0[\chi e^{P_g}] > 0. \tag{4.cccxxii}$$

The preceding formation proof supplies the actual local vertices of P_g, with norm $Cg\mathfrak{b}$, and the complete admitted Gaussian structure. All constants are uniform in k, h and compatible volume.

4.12.2. *Actual quadratic source operators.* For a fine plaquette p, put $f_{k,p}(V) = a_p(U_k(V))$. It has value and first derivative zero at one. Let $\mathcal{H}_k$ be the actual linear fine background response in physical potential units. Its literal gauge correction is a gradient and does not change its linear curvature. Define

$$\mathcal{D}_{k,p} B = (\mathrm{d}_h \mathcal{H}_k \mathsf{C} B)(p).$$

The fine Wilson normalization then gives

$$\frac{1}{2} D^2[f_{k,p}(V(u))]_{u=0}[B,B] = B^{\mathsf{T}} A_p B, \qquad A_p = \frac{h^4}{2} \mathcal{D}_{k,p}^{\mathsf{T}} \mathcal{D}_{k,p}. \tag{4.cccxxiii}$$

The nonlinear outer chart contributes no extra quadratic term, because $Df_{k,p}(1) = 0$. The second derivative of the inner background is likewise multiplied by the vanished first fine plaquette variation. The remaining jet is exactly the displayed linear-curvature Gram matrix.

The actual weighted strong response eq. 3.mccxcv gives exponentially summable coarse bond kernels for $\mathcal{D}_{k,p}$. In particular, for a fixed $s > 0$,

$$\sum_{b,b'} e^{s(d(b,p)+d(b',p))} \|A_p(b,b')\| \le C h^4. \tag{4.cccxxiv}$$

Keep the original p and its two propagator paths as its carrier tag. This is an actual source-rooted operator family: each carrier contains p and its matrix endpoints. The same bound holds

with a fixed connected-carrier exponential weight, by the marked-tree construction. There is no fine-coordinate dimension factor in eq. 4.cccxxiv.

For profiles w_i, $1 \le i \le r \le 4$, with $\|w_i\|_\infty \le 1$, put

$$\begin{aligned} H_i(V) &= \sum_p w_i(p) f_{k,p}(V), \quad X_i(B) = g^{-2} H_i(V(gB)), \\ A_i &= \sum_p w_i(p) A_p, \qquad\qquad Q_i(B) = B^{\mathsf T} A_i B. \end{aligned} \tag{4.cccxxv}$$

Equivalently $Q_i = \frac{1}{2} H_i''(1)[\mathsf{C}B, \mathsf{C}B]$. Summing eq. 4.cccxxiv over fine plaquettes near a fixed coarse bond cancels the physical h^4 with their $O(h^{-4})$ density. It proves a uniform weighted Schur bound $\|A_i\|_{\mathrm{Sch,s}} \le C$. Summing its absolute entries instead gives $\|A_i\|_{\mathrm{tr}} \le C h^4 \sum_p |w_i(p)|$.

The exact source remainder is

$$R_{p,g}(B) = g^{-2} f_{k,p}(V(gB)) - B^{\mathsf T} A_p B, \qquad X_i = Q_i + R_{i,g}, \quad R_{i,g} = \sum_p w_i(p) R_{p,g}. \tag{4.cccxxvi}$$

Retaining its original fine tag in the action-tree proof gives a rooted potential norm at most $C h^4 g\mathfrak{b}$ for each $R_{p,g}$, with a fixed reserve in the carrier rate. Its zero of order three follows from the own value, first jet and quadratic subtraction. Its chart is the same fixed admitted chart as eq. 4.cccxix. Thus the large factor g^{-2} in the observable has been treated by its actual Taylor decomposition; it is not inserted into a bounded- source theorem as an old coefficient.

4.12.3. *Exact quadratic resummation.* For $t = (t_1, \dots, t_r)$, define

$$K_t = K_0 - 2 \sum_i t_i A_i, \qquad C_t = K_t^{-1}, \qquad \mu_t = \mu_{C_t}, \qquad R_g(t) = \sum_i t_i R_{i,g}. \tag{4.cccxxvii}$$

The uniform Schur bounds and the actual gap of K_0 give a fixed complex polydisc $|t_i| < 4\rho$ on which the real part of K_t is positive and its actual carrier norms are uniform. Decrease ρ further, if necessary, in the near/tail and combined-power construction of eq. 4.ccxciv. It supplies precisely the local Gaussian structures for every K_t, including enlarged independent cube discs, normalization and walls. The radius $\rho > 0$ is independent of h, k, g, volume and profiles satisfying the displayed supremum bound. At most four profiles are used, so no shrinking radius in their number is hidden here.

Let

$$\mathcal{Z}_g(t) = \mu_t[\chi e^{P_g + R_g(t)}], \qquad E_g(t) = \log \mathcal{Z}_g(t), \qquad L_G(t) = -\frac{1}{2} \log\det\left(I - 2C_0 \sum_i t_i A_i \right). \tag{4.cccxxviii}$$

Choose the determinant branch zero at $t = 0$, and the cluster-log branch for E_g. The actual nonlinear source vertices remain of norm $C g\mathfrak{b}$ on this disc. Formation therefore makes $\mathcal{Z}_g(t)$ zero-free. Gaussian completion, with all its normalizers retained, proves

$$\log \mathbb{E}_{\nu_{k,g}} e^{\sum_i t_i X_i} = L_G(t) + E_g(t) - E_g(0), \qquad \log \mathbb{E}_{\mu_0} e^{\sum_i t_i Q_i} = L_G(t). \tag{4.cccxxix}$$

The determinant here is essential. It is the source-dependent ratio of the normalized Gaussians before the residual interaction is expanded.

4.12.4. *Source-rooted differentiation of the actual expansion.* We prove a source-mass bound, not an unpointed error times the volume. Introduce independent fine coefficients ζ_p, with $K_\zeta = K_0 - 2\sum_p \zeta_p A_p$ and source remainder $\sum_p \zeta_p R_{p,g}$. The preceding argument is uniform on a fixed product ball $\|\zeta\|_\infty < \zeta_*$: the uniform Schur bound uses the h^4 density sum already proved. At each filed carrier combine the fine-plaquette contributions into one source-potential type. Keep p as the mark when differentiating. The type count is then uniform in the fine mesh. Choose the retained ρ

so that $16\rho < \zeta_*$; the four-profile polydisc is then inside this product ball. For each original fine plaquette, the complete covariance has the exact derivative

$$\partial_{\zeta_p} C_\zeta = 2C_\zeta A_p C_\zeta. \tag{4.cccxxx}$$

Its source tag and its h^4 mass are retained in every term. At complete parameters the exact normalized differentiation identity is

$$\partial_{\zeta_p} E_g = \left\langle R_{p,g} + B^{\mathsf{T}} A_p B \right\rangle_{g,\zeta} - \mathrm{Tr}(C_\zeta A_p),$$

where the brackets have density $\mathcal{Z}_g(\zeta)^{-1}\chi e^{P_g + \sum_q \zeta_q R_{q,g}}\, d\mu_\zeta$. Thus the Gaussian quadratic normalizer is subtracted exactly.

This tag can also be retained through the local covariance construction. Truncate by the full connected carrier, including the original p, both matrix endpoints and their propagator paths. A term with nearby matrix endpoints and a distant source root belongs to the long-carrier tail. Endpoint distance alone does not define this truncation. Symmetrize each real carrier contribution under transpose, retaining its p tag. The source-rooted exponential entry bound above and the physical h^4 density sum give a uniformly small Schur tail. Thus the short-carrier real precision differs from the full gapped precision by that small tail; the near-power spectral argument still applies. More explicitly, reserve two copies of a small length weight in the tagged source family. Since its carrier joins b to p, the tail then obeys

$$\sum_{b',\,\gamma:\ell(X_\gamma)>R} \|A_{p,\gamma}(b,b')\| \le Ch^4 e^{-cR} e^{-cd(b,p)}.$$

Summing p uses $h^4 \sum_p e^{-cd(b,p)} \le C$, so the full source tail has Schur norm Ce^{-cR}, uniformly on the source product ball. Transposition gives the column estimate. In this full-carrier algebra, differentiating a near power inserts exactly one $2A_p/a$; differentiating its tail telescoping formula does the same. For example

$$\partial_{\zeta_p} H^m_{\le R} = \sum_{j<m} H^j_{\le R}(2A_{p,\le R}/a) H^{m-1-j}_{\le R}.$$

Its full combined ball carrier contains the original p. The global absolute entry sum of that marked matrix is Ch^4, with constants allowed to depend on the fixed m, R, M. For products, this follows from $\|UE_pV\|_{\mathrm{entry},1} \le \|U\|_{\mathrm{Sch}} \|E_p\|_{\mathrm{entry},1} \|V\|_{\mathrm{Sch}}$. The marked geometric series costs $\sum_{v\ge1} vq^{v-1} < \infty$. The prefix derivatives have the same estimate. Consequently the filed local covariance $D_{W,Z}$, with root coordinate set W, held root guard, and all cube parameters zero outside its wall domain Z, has a source derivative $\dot D_{W,Z,p}$ with

$$\|\dot D_{W,Z,p}\|_{\mathrm{entry},1} \le Ch^4, \qquad \|D^{-1}_{W,Z}\| \le C, \tag{4.cccxxxi}$$

uniformly on the full coarse parameter discs. If the local wall domain does not contain the source root p, this derivative is zero: every retained carrier contains that root. The same statements hold for the explicit source-potential derivative $R_{p,g}$.

At complete parameters the actual covariance restriction instead has derivative

$$\partial_{\zeta_p}(\chi_W C_\zeta \chi_W^{\mathsf{T}}) = 2\chi_W C_\zeta A_p C_\zeta \chi_W^{\mathsf{T}},$$

which can be nonzero for a source outside W. The vanishing above concerns the filed $D_{W,Z}$ for $p \notin Z$; it does not assert vanishing of this complete restriction for $p \notin W$.

Here the entry norm is the sum of colour-block norms. It bounds trace norm up to the fixed colour factor. Thus the derivative of the normalized marginal Gaussian density is its density times

$$\frac{1}{2}\{B^{\mathsf{T}} D^{-1}_{W,Z} \dot D_{W,Z,p} D^{-1}_{W,Z} B - \mathrm{Tr}(D^{-1}_{W,Z} \dot D_{W,Z,p})\}, \tag{4.cccxxxii}$$

whose absolute value is at most $Ch^4(1+|B|^2)$. There is no factor equal to the local or exterior Gaussian dimension in this bound. Its expectation against the normalized constant functional is exactly zero. In particular an empty selection cannot contribute a source-dependent vacuum activity.

Use a fixed small extra Gaussian tilt $\delta \asymp (LM)^{-4}$ to bound $1+|B|^2 \le C_\delta e^{\delta|B|^2}$. The Gaussian comparison charge is then only a fixed constant per polymer cube, absorbed by the reserved spatial rate and cube-Cauchy radius. It retains the same small amplitude as the selected integrand: every nonempty selection has either an actual remainder potential of size $Cg\mathfrak{b}$ or a failed-cutoff tail of size $Ce^{-c\mathfrak{b}^2}$. Polynomial boundary factors can be absorbed by a fixed decrease of c. Differentiating a potential instead inserts the actual $R_{p,g}$, whose norm is already $Ch^4 g\mathfrak{b}$. Apply the proved inserted-potential estimate to that literal norm.

It follows, by the same finite Mayer and coarse-interpolation formulas, that the actual activity derivative with its source root satisfies

$$|\partial_{\zeta_p} H_g(Z;\zeta)| \le Ch^4\eta(g)e^{-\lambda d_c(Z)}\mathbf{1}_{\{p\in Z\}}, \qquad \eta(g) = g\mathfrak{b} + e^{-c\mathfrak{b}^2}. \tag{4.cccxxxiii}$$

Membership includes the fixed assigned carrier collar. A fixed boundary multiplicity is absorbed in C. The extra tilt used above is chosen in the ordinary Gaussian margin before the final spatial rates; it does not depend on g, h, k or source support.

Differentiate the convergent connected-cluster logarithm. Mark the differentiated activity and retain its original p. The rooted cluster theorem bounds all polymers attached to it. Summing its own polymers containing the fixed coarse cube of p uses their reserved exponential rate. Hence

$$|\partial_{\zeta_p} E_g(\zeta)| \le Ch^4\eta(g). \tag{4.cccxxxiv}$$

This is the source-rooted argument; the estimate is not obtained by first bounding $|E_g|$ by a volume term and then differentiating.

For $\zeta_p = \sum_i t_i w_i(p)$, eq. 4.cccxxxiv therefore proves

$$|\partial_{t_i} E_g(t)| \le C\eta(g)h^4 \sum_p |w_i(p)|. \tag{4.cccxxxv}$$

Integrating on the straight path from zero to t also gives

$$|E_g(t) - E_g(0)| \le C\eta(g)\sum_i |t_i| h^4 \sum_p |w_i(p)|.$$

4.12.5. *Cumulants and their Gaussian leading terms.*

Proposition 4.22 (Uniform comparison for classical-pullback observables)**.** For $1 \le n \le 4$ and the profiles of eq. 4.cccxxv,

$$\left|\kappa_{\nu_{k,g}}(X_{i_1},\dots,X_{i_n}) - \kappa_{\mu_0}(Q_{i_1},\dots,Q_{i_n})\right| \le C_n\eta(g)\, m_h(w), \qquad m_h(w) = h^4\sum_{p,i}|w_i(p)|. \tag{4.cccxxxvi}$$

The constants and the analytic source disc are independent of homogeneous depth, mesh, volume and source support. For the unscaled actual observables $H_i(V)$, the corresponding error is $C_n g^{2n}\eta(g)m_h(w)$. ∎

Proof. Use the exact normalized identity eq. 4.cccxxix. Bound one source derivative of its error by eq. 4.cccxxxv. The remaining at most three derivatives follow from Cauchy on the fixed larger source disc. Repeated indices give only their fixed factorials. This proves eq. 4.cccxxxvi. Finally $H_i = g^2 X_i$ exactly, so multilinearity of cumulants gives the factor g^{2n}. □

4.12.6. *The source-rooted even-coupling refinement.* Keep the physical retained threshold $\mathfrak{b}$ fixed and replace the coupling in the remainder by an auxiliary complex γ. In particular $R_{p,\gamma}(B) = \gamma^{-2} f_{k,p}(V(\gamma B)) - B^{\mathsf{T}} A_p B$, with its removable value at zero. Write $E_{\gamma,\mathfrak{b}}(\zeta)$ for the resulting logarithm. The Gaussian precision K_ζ, the profiles and the real cutoff are held fixed. The divided remainders have removable values at $\gamma = 0$. The actual chart and local potential bounds hold on $|\gamma| < c_*/\mathfrak{b}$, with c_* fixed below their uniform ceilings, exactly as in eq. 4.cccx. This is an analytic deformation of the retained-cutoff integral; it makes no change to the original real tests at the physical value $\gamma = g$.

The source-rooted proof above therefore holds throughout that disc, with

$$|\partial_{\zeta_p} E_{\gamma,\mathfrak{b}}| \le C h^4 \{|\gamma|\mathfrak{b} + e^{-c\mathfrak{b}^2}\}.$$

Its h^4 tag is unchanged. At zero coupling only failed-cutoff activities remain, so $|\partial_{\zeta_p} E_{0,\mathfrak{b}}| \le C h^4 e^{-c\mathfrak{b}^2}$.

Moreover the actual source remainders satisfy $R_{p,-\gamma}(-B) = R_{p,\gamma}(B)$. The Gaussian, cutoff and the matrix construction are independent of γ. The exact sign argument of eq. 4.cccxii thus applies with every original source tag retained. The chosen cluster logarithm, and its ζ_p derivative, are even in γ. Subtract the zero-coupling derivative. This holomorphic function has a double zero at the origin. Cauchy on a fixed smaller concentric circle, followed by the order-two Schwarz bound, gives

$$|\partial_{\zeta_p} E_{g,\mathfrak{b}}| \le C h^4 \eta_2(g), \qquad \eta_2(g) = (g\mathfrak{b})^2 + e^{-c\mathfrak{b}^2}, \qquad g\mathfrak{b} < c_*/4. \tag{4.cccxxxvii}$$

This step uses the uniform source-rooted bound, so it retains h^4 rather than introducing a whole-volume factor. Profile summation and the same fixed source-disc Cauchy argument show that the log-MGF comparison and eq. 4.cccxxxvi hold with η_2 in place of η. The unscaled observables retain their exact factor g^{2n}. The preceding η estimate is its weaker corollary when $g\mathfrak{b} < 1$.

For clarity the leading Gaussian law is entirely explicit:

$$\begin{aligned} \mathbb{E}_{\mu_0} Q_i &= \operatorname{Tr}(C_0 A_i), \\ \operatorname{Cov}_{\mu_0}(Q_i, Q_j) &= 2 \operatorname{Tr}(C_0 A_i C_0 A_j), \\ L_G(t) &= \sum_{n \ge 1} \frac{2^{n-1}}{n} \operatorname{Tr}\left[\left(C_0 \sum_i t_i A_i\right)^n\right]. \end{aligned} \tag{4.cccxxxviii}$$

Equivalently, the actual linear fine-curvature covariance is $\Gamma_k(p,q) = \mathcal{D}_{k,p} C_0 \mathcal{D}_{k,q}^{\mathsf{T}}$. The first two observable formulas are

$$\mathbb{E} Q_i = \frac{h^4}{2} \sum_p w_i(p) \operatorname{Tr} \Gamma_k(p,p), \quad \operatorname{Cov}(Q_i, Q_j) = \frac{h^8}{2} \sum_{p,q} w_i(p) w_j(q) \operatorname{Tr}[\Gamma_k(p,q)\Gamma_k(q,p)].$$

Thus a mixed Gaussian quadratic cumulant is $\frac{2^{n-1}}{n} \sum_{\sigma \in S_n} \operatorname{Tr}(C_0 A_{i_{\sigma(1)}} C_0 A_{i_{\sigma(2)}} \cdots C_0 A_{i_{\sigma(n)}})$. For coincident observables this is $2^{n-1}(n-1)! \operatorname{Tr}[(C_0 A_i)^n]$. These are quadratic Gaussian cumulants, including their generally nonzero higher orders.

Since $p > 1/2$, $\eta_2(g) \to 0$ and the cutoff term is eventually $O((g\mathfrak{b})^2)$. The log-MGF and cumulant errors tend to zero whenever $\eta_2(g) m_h(w) \to 0$, including bounded source masses and masses growing by a fixed power of $\log(1/g)$. Their reference is the actual depth- k quadratic Gaussian law. All bounds retain the actual constraint, Haar/coarea density and fixed tests of this concrete classical-pullback model.

4.13. **Symmetry of the cutoff fibre integral.** The cutoff is imposed in free bond coordinates. Its lattice symmetry therefore requires more than symmetry of the group average. For the middle pivots used here, an isometry permutes the free bonds and acts orthogonally on their colour coordinates. We prove this and the covariance of the resulting integral.

Let $L = 2m + 1$, and use the centered blocks and the pivot policy of § 3.3. For a positive coarse bond $c = (y, y + Le_\mu)$, its pivot is

$$b_0(c) = (y + me_\mu, y + (m + 1)e_\mu). \tag{4.cccxxxix}$$

Write Π for these bonds and E for their complement among positive bonds, with all three colour coordinates on each bond. A real fluctuation cutoff in this subsection means

$$\chi_{\mathfrak{b}}(B) = \prod_{b \in E} \mathbf{1}_{\{\langle B(b), B(b)\rangle < \mathfrak{b}^2\}}. \tag{4.cccxl}$$

Thus each factor is a colour ball. Complex polydiscs used for estimates are separate analytic domains.

Let $\mathsf{r}(x) = Rx + a$, where R is a signed permutation matrix and a is a coarse lattice translation. On oriented bonds define $(\mathsf{r}U)(\mathsf{r}b) = U(b)$. We consider isometries between compatible tori, or of the infinite lattice. On a cubic torus this includes the full hypercubic group.

Lemma 4.23 (The action on free coordinates)**.** The isometry preserves the pivot set, with positive orientations restored. For a background U, define the linear map r_U on bond coordinates as follows. Let b^{r} be the positive representative of the image unoriented bond, and let $\varepsilon_{\mathsf{r}}(b)$ be its image orientation sign. Put

$$(\mathsf{r}_U a)(b^{\mathsf{r}}) = \begin{cases} a(b), & \varepsilon_{\mathsf{r}}(b) = 1, \\ -\operatorname{Ad}(U(b)^{-1})a(b), & \varepsilon_{\mathsf{r}}(b) = -1. \end{cases} \tag{4.cccxli}$$

Then

$$\mathsf{r}(e^{ia}U) = e^{i\mathsf{r}_U a}\mathsf{r}U. \tag{4.cccxlii}$$

For real U, r_U permutes bonds and acts by an orthogonal matrix on each colour space. Its restrictions $\mathsf{r}_U^E, \mathsf{r}_U^\Pi$ are well defined, and $\chi_{\mathfrak{b}}(\mathsf{r}_U^E B) = \chi_{\mathfrak{b}}(B)$. ■

Proof. If $Re_\mu = e_\nu$, the image of eq. 4.cccxxxix is the middle pivot of the image coarse bond. If $Re_\mu = -e_\nu$, the new positive coarse bond starts at $\mathsf{r}y - Le_\nu$. Its middle pivot has endpoints $\mathsf{r}y - (m + 1)e_\nu, \mathsf{r}y - me_\nu$, the reversed endpoints of the image pivot. Thus Π, and hence E, are preserved.

On a reversed bond,

$$(e^{ia(b)}U(b))^{-1} = U(b)^{-1}e^{-ia(b)} = e^{-i\operatorname{Ad}(U(b)^{-1})a(b)}U(b)^{-1}.$$

This proves eq. 4.cccxlii. For real U, its adjoint action preserves the colour pairing; the minus sign also preserves it. The product cutoff is therefore unchanged. The same assertion would not follow for a box in fixed individual colour components. □

Lemma 4.24 (Covariance of the mean and graph)**.** The actual one-step mean satisfies $\mathcal{M}(\mathsf{r}U) = \mathsf{r}\mathcal{M}(U)$. For its admitted graph,

$$\Phi_{\mathsf{r}U}(\mathsf{r}_U^E u) = \mathsf{r}_U \Phi_U(u). \tag{4.cccxliii}$$

The identity is holomorphic wherever both graph arguments lie in the joint chart. ■

Proof. The isometry sends the coordinate-order path multiset from y to x to the one from $\mathsf{r}y$ to $\mathsf{r}x$, preserving its weights. The internal implicit mean therefore has the same covariance, by uniqueness.

The outer operation is the exponential of the average of closed logarithms, followed by multiplication by the central path. It also respects reversal. To check this directly, write its central transport as C and its closed inputs as Z_x, so $\mathcal{M}(U)(c) = e^{\overline{\log Z}}C$. For the reversed coarse bond the closed inputs are $C^{-1}Z_x^{-1}C$. Their average logarithm is $-\operatorname{Ad}(C^{-1})\overline{\log Z}$; multiplication by C^{-1} gives $(e^{\overline{\log Z}}C)^{-1}$. This uses the actual outer operation.

Put $W = \mathcal{M}(U)$. The relative raw constraint q_U transforms as

$$q_{\mathsf{r}U}(\mathsf{r}_U a) = \mathsf{r}_W q_U(a).$$

For a reversed coarse bond this is $-\operatorname{Ad}(W(c)^{-1})q_U(a)(c)$; it follows by conjugating the inverse relative coarse product before taking its admitted logarithm. The right side of eq. 4.cccxliii thus solves the transformed constraint with the transformed free values. Its small-graph uniqueness proves the identity for real data. Joint analyticity extends it on the common complex chart. □

Proposition 4.25 (Covariance of the regular transfer)**.** Use the actual regular coarea component, unit contour-gauge weights, the middle pivots eq. 4.cccxxxix, and the cutoff eq. 4.cccxl. Assume the action and input pointings are covariant under r. Then the transfer and its complete pointings are covariant under r. This applies to the canonical homogeneous step with action S_j, and to an ordinary input with the same covariance. In particular the scalar quadratic readout is available for that model on the infinite companion. ■

Proof. For the physical background action, reflecting a critical orbit gives a critical orbit with reflected datum. The strong norm, regularity bounds and averaging constraint are preserved. Orbit uniqueness in Prop. 3.183 identifies the two orbits. Each original plaquette trace, and hence S_j, has the required covariance. Any difference between chosen literal representatives is a node-fixed gauge and is removed at the level of the complete function.

For a fixed real background, use $u' = \mathsf{r}_U^E u$ as free coordinates. This is an orthogonal change with absolute determinant one. Equation eq. 4.cccxliii maps the graph and the product colour balls exactly. Regular plaquette bounds and the coarea fibre are preserved.

The coordinate coarea factor is preserved as well. Its pivot derivative transforms by r_W on its rows and $(\mathsf{r}_U^{\Pi})^{-1}$ on its columns. For each reversed coarse row the corresponding pivot is also reversed: the two factors of determinant minus one cancel. Their adjoint factors have determinant one. Row and pivot permutations cancel in the same way. The reciprocal pivot determinant on its identity branch is therefore unchanged. The Haar density $j(a)$ depends only on the invariant colour pairing. Consequently the full coarea density in eq. 4.ccliv is carried to itself.

The internal transports and their reference products are permuted by the isometry. Their relative contour trace, tame test and unit-weight sum are unchanged. The centered action and its full quadratic Hessian transform by the same free orthogonal map. Thus the Gaussian precision transforms as $K' = \mathsf{r}_U^E K(\mathsf{r}_U^E)^{-1}$, and its normalized real Gaussian measure is carried to the new one. This proves covariance of the integral, including its actual cutoff, Haar factor, pivot factor and numerical Gaussian normalization.

The block axial choice also commutes with the isometry. If $s_U(x) = T_{y,x}(U)$ and $s_U(y) = 1$, then $s_{\mathsf{r}U}(\mathsf{r}x) = s_U(x)$. Hence $(\mathsf{r}U)^{s_{\mathsf{r}U}} = \mathsf{r}(U^{s_U})$. This uses the symmetric contour mean, not a choice of one ordered axial path.

For the pointings, choose their insertions before carrier filing. Original plaquettes and old anchors are permuted, and midpoint tents are covariant. On a reversed free bond, the vector and its derivative are contracted by the same orthogonal map, so the radial and normalized cubic insertions are preserved. The cutoff surface and its colour surface measure are also preserved. For a Gaussian trace, each diagonal colour block is conjugated and its free bond is permuted. The pivot normalizer factors have the covariance already proved. Normalized expectations and their interpolation integrals therefore give covariant complete pointed functions.

The real identities extend to the admitted complex external-field domain by holomorphy. This extension does not change real integration contours by a complex orthogonal map. The auxiliary localization grid need not be preserved by every translation; it files an already defined complete pointing and does not change its sum. Its individual pieces need not have the full covariance. □

4.14. **Physical transport of homogeneous invariant densities.** The subtraction appropriate to a classical pullback is a multiple of its actual effective action. We prove a local analytic irrelevance estimate for the remaining complete physical density. The proof uses the literal intermediate mean, including its node frames. Its localized analytic representatives are not required to be separately gauge invariant.

We work on the fixed coarse bond ball about the trivial flat field, with compatible periodic volumes, and on its infinite lattice companion. All assertions about the companion concern normally convergent pointed analytic functions. They do not require an infinite volume probability measure. Fix $0 < \beta < \alpha < 1$. Constants below may depend on L, α, β, a fixed analytic input radius, and the localization size. They are independent of the two depths and the compatible volume.

4.14.1. *Literal intermediate coordinates.* Use final coarse spacing one. For $0 \le s \le k$ put

$$h = L^{-k}, \qquad \xi = hL^s = L^{s-k}, \qquad W = e^{iw}, \qquad U_k(W) = e^{ih\widehat{A}_k(w)}.$$

Here U_k is the literal critical representative of Prop. 3.183. Define

$$V_{s,k}(W) = M^s U_k(W), \qquad B_{\xi,b}(w) = \frac{1}{i} \log V_{s,k}(W)(b). \tag{4.cccxliv}$$

The bond b on the left is indexed in old coarse units; its physical midpoint in final units is ξm_b. The logarithm will be shown to stay in its actual chart. This definition does not replace M^s by the straightened linear average.

We first isolate a bound for the actual mean acting on an arbitrary small fine field. Write

$$\mathscr{P}_{h,s}(A) = \frac{1}{i} \log M^s(e^{ihA}).$$

For a homogeneous polynomial row, its absolute row norm is the sum over its fine input bonds of the norms of its colour multilinear coefficients. Ordered coefficients include their Taylor factorial once.

Lemma 4.26 (Absolute rows of the full plain mean)**.** There are $c, C > 0$ such that the degree m row coefficient $T^{(m)}_{h,s;b}$ of $\mathscr{P}_{h,s}$ satisfies

$$\sum_{b_1,\dots,b_m} \|T^{(m)}_{h,s;b;b_1\dots b_m}\| \le C^m \xi^m, \qquad m \ge 1. \tag{4.cccxlv}$$

Its fine inputs lie within physical distance $C_L\xi$ of the output bond. On $\|A\|_\infty < c$,

$$\|D\mathscr{P}_{h,s}(A)\|_{\mathrm{row}} \le C\xi, \qquad \|D^2\mathscr{P}_{h,s}(A)\|_{\mathrm{row}} \le C\xi^2. \tag{4.cccxlvi}$$

The same conclusions, with zero constant term, hold for the logarithm of the node frame in the partial factorization. For the row assertion alone C, c can be chosen uniformly over $L \ge 3$, as in eq. 3.mcccv; the nonlinear fixed-leg estimate has the different dependence proved in Prop. 3.187. ■

Proof. Use the exact recursion and factorization eq. 3.mccliii–eq. 3.mccliv, stopped at s. The one-step stencil is finite for fixed L. Its actual analytic chart therefore gives absolute Taylor row bounds K_L^m for the terms of degree $m \ge 2$. The derivative of the straightened row at zero must first be combined: its row norm is exactly L, by eq. 3.mcclv. Taking the absolute values of its separate contour and node-frame contributions before this cancellation would give an incorrect depth factor.

The positive formal series majorizing the raw angle rows can consequently be chosen to satisfy

$$b_0(z) = hz, \qquad b_{t+1}(z) = Lb_t(z) + \frac{K_L^2 b_t(z)^2}{1 - K_L b_t(z)}. \tag{4.cccxlvii}$$

Increasing K_L also majorizes the zero-constant one-step node increment by $K_L b_t/(1 - K_L b_t)$. Let $\xi_t = hL^t$, and evaluate these positive series at $z = c_0/\xi$. A first-exit argument gives

$$b_t(z) \le 2\xi_t z \quad (0 \le t \le s), \qquad \sum_{t<s} \xi_t \le \frac{\xi}{L-1}.$$

Indeed, below this bound the quotient $b_{t+1}/(Lb_t)$ is at most $1 + C_L b_t$. The product of these quotients is at most $\exp(C_L z \sum_{t<s} \xi_t)$, less than two when c_0 is fixed sufficiently small. All inputs then stay strictly in the one-step analytic chart.

The product defining v_s, and the product in reverse order defining v_s^{-1}, have positive row majorants bounded by

$$\prod_{t<s}(1-K_L b_t(z))^{-1}.$$

They differ from one by at most $C_L c_0$. Apply the convergent exponential and logarithm series to the exact product

$$M^s(e^{ihA})(b) = v_s(b_-)\, e^{ia_s(b)}\, v_s(b_+)^{-1}.$$

Its zero-constant absolute logarithm majorant at $z = c_0/\xi$ is at most $C_L c_0$. Positivity of the majorant gives the coefficient bound $C_L c_0^{1-m}\xi^m$, which implies eq. 4.cccxlv. This is an absolute coefficient sum over fine inputs; it is stronger than a Cauchy bound for a function of an unspecified number of fine coordinates.

The one-step support enlargement at stage t has physical size at most $C_L\xi_t$. Its total size is at most $C_L\sum_{t<s}\xi_t \le C_L\xi$. Differentiating the absolutely convergent majorant on a fixed smaller supremum ball proves eq. 4.cccxlvi. The proof for $\log v_s$ uses the same product majorant. Restriction of a node row to a coarser node set does not increase any of these row sums. □

The literal representative also has a uniform connected expansion. This step is needed because the action trees can use a mean-slice representative by gauge invariance, whereas the intermediate field in eq. 4.cccxliv uses the literal one.

Lemma 4.27 (Connected expansion of the literal field)**.** On a fixed coarse ball, $\widehat{A}_k(w)$ has uniform strong-cell norm $C\|w\|_\infty$ and an analytic connected-tree family rooted at each output fine cell, with a fixed positive physical length rate. After choosing M large enough for a requested cube rate and decreasing the field radius, its rooted M-cube norm is at most Cr on $\|w\|_\infty < r$. The same family bound holds for the entire field vector on one localization cube, with its maximum strong-cell norm. For a fixed requested rate its amplitude constant can be chosen independent of M and L, with an allowed radius that may depend on L. ■

Proof. The mean-slice field $A_k(w)$ has the strong-cell trees of eq. 4.cclxix and eq. 4.cclxxii. The prescribed literal field is eq. 3.mcclxxxiv. Its node function $v_h(A)$ reads only a fixed physical star: the support sum in the preceding proof is bounded at full spacing one. That proof also gives its uniform analytic supremum bounds on a fine supremum ball.

The actual extension E_n in eq. 3.mcclxxxii is local and bounded from coarse supremum norm to Y_α. Hence $\lambda(A) = E_n(i^{-1}\log v_h(A))$ is a local analytic map into Y_α with norm $C\|A\|_\infty$. A fixed physical star contains only a fixed number of strong cell slots. Banach Cauchy in those slots gives coefficients C^m; it does not introduce the number of fine sites.

Finally use the uniformly analytic finite-gauge map eq. 3.mcclxxxiii. Its apparent h^{-1} is removed before expanding by writing $\lambda(x+he_\mu) = \lambda(x) + hD_\mu\lambda(x)$. The logarithm of the remaining product vanishes at $h=0$, so division by h is analytic and bounded in the stated strong cell norms. Substitution of the field trees into these local analytic maps attaches fixed physical stars and vertices of arity at least one. Their number is bounded by a fixed multiple of the leaves after combining the linear maps. Their coefficients and the fixed per-leaf carrier costs are absorbed by a smaller input radius. The physical exponential length reserve in the field trees pays the stars. This proves both the connected family and the strong bound for the actual literal representative.

For the vector assertion perform the same actual fixed-point expansion with macroblocks from the outset. Let $\mathcal{E}_Q$ be a smooth macroblock partition, whose physical derivatives are bounded uniformly for $M \ge 1$, supported in the $M/10$ enlargement of its block. The product slots are supported strong fields on these enlargements; their bounded overlap reconstructs the global field. Let $\mathcal{I}_{Q'}$ restrict the fine and coarse forcings to a macroblock. The target norm is the maximum strong unit-cell norm on the support of the output slot. Work first with the infinite-cover inverse.

The unit-cell inverse estimate gives

$$\|\mathcal{E}_Q\mathcal{L}_{0,\infty}^{-1}\mathcal{I}_{Q'}\| \le Ce^{-s_0\,\mathrm{dist}(Q,Q')/2}, \qquad \sup_Q \sum_{Q'} e^{\kappa d_M(Q,Q')}\|\mathcal{E}_Q\mathcal{L}_{0,\infty}^{-1}\mathcal{I}_{Q'}\| \le C_\kappa. \tag{4.cccxlviii}$$

For the first estimate take the supremum over target cells after summing the exponential unit-cell kernel over source cells in Q'. There is no factor M^4. For the second, touching macroblocks give only a fixed number of terms; other terms have physical separation at least $cM(d_\infty(Q,Q')-C)_+$. Choose M larger than $C\kappa/s_0$. On a torus periodize these cover blocks, retaining the image $Q'+nP$ as a separate label before any carrier assignment. Give each such block a fixed path on the cover joining its two macroblocks. The sum over source blocks and period images is exactly the cover sum just bounded. Short endpoint distance in the quotient does not remove a long image path. Projection of this normally convergent tree construction solves the actual periodic fixed-point equation; contraction uniqueness identifies its sum with the finite-volume background.

The actual residual in eq. 4.cclxix, restricted to Q, reads only a fixed number of neighboring macroblocks when M exceeds its physical carrier radius. Its target is a forcing supremum norm. Its analytic bound is $C_L\|x\|^2$, independent of M: pointwise locality and the physical transpose bound are used before taking the supremum. Banach Cauchy counts these neighboring macroblocks, not their fine coordinates. The actual ordered-tree majorant therefore gives a macroblock field family of size $2C_\kappa r$ after reducing the radius below its nonlinear ceiling. The free constant in eq. 4.cccxlviii is uniform in L, since its direct rows depend on $N=L^k$, as explained after eq. 3.mcccv. Growth in C_L is paid only by the allowed radius. The actual node extension and finite-gauge map are local analytic maps on the same macroblock spaces. Their uniform path-row bounds give the stated literal-field vector family. Its amplitude constant is independent of M, L. □

Proposition 4.28 (Actual smooth intermediate fields)**.** There is a fixed $r_*>0$ such that eq. 4.cccxliv is admitted on $\|w\|_\infty<r_*$, at every pair of depths. For $\|w\|_\infty<r\le r_*$ there is a continuous connection $a_\xi(w)$, holomorphic in w, such that

$$B_{\xi,b}(w) = \xi\, a_\xi(w)(\xi m_b), \qquad \|a_\xi(w)\|_{C^{1,\alpha}} \le Cr. \tag{4.cccxlix}$$

For any fixed requested cube rate, with M and the radius chosen accordingly, each raw coordinate also has an actual local analytic family

$$B_{\xi,b}(w) = \sum_{X\ni q(\xi m_b)} B_{\xi,b,X}(w_X), \qquad \sum_X e^{\kappa_0 d_M(X)}\|B_{\xi,b,X}\|_r \le C\xi r. \tag{4.cccl}$$

The norm uses the final M-cube lattice; $q(x)$ is a cube containing x. There is also a Banach-valued version of eq. 4.cccl: all old bond coordinates in one old M-cube can be grouped into a single vector with its supremum norm. Its rooted family has the same bound $C\xi r$. ■

Proof. Apply Lem. 4.26 to the literal field of Lem. 4.27. The raw angle and all intermediate raw inputs are bounded by $C\xi r$ and $Cr\xi_t$. The node products remain in their actual charts. A fixed reduction of r_* proves admission.

For the spatial assertion write $A=\widehat{A}_k(w)$. Translation by ξe_ν preserves every grid in the partial hierarchy. The actual mean commutes with this translation. Put $\delta_\nu A=\tau_{\xi e_\nu}A-A$. The first two derivatives of the full plain mean give

$$D^\xi_\nu(B_\xi/\xi) = \xi^{-2}\int_0^1 D\mathscr{P}_{h,s}(A+t\delta_\nu A)[\delta_\nu A]\,dt. \tag{4.cccli}$$

The fine physical strong norm gives $\|\delta_\nu A\|_\infty \le C\xi r$, so the right side is bounded by Cr. At an old-grid displacement y of physical size $d\le 1$,

$$\|\tau_y\delta_\nu A-\delta_\nu A\|_\infty \le C\xi r d^\alpha.$$

The points on the two interpolation segments satisfy $\|\tau_y(A+t\delta_\nu A)-(A+t\delta_\nu A)\|_\infty \le Crd$: use their convex-combination expression. Equation eq. 4.cccxlvi therefore bounds the difference of eq. 4.cccli by

$$Crd^\alpha + C\xi r^2 d \le Crd^\alpha.$$

The zeroth supremum of B_ξ/ξ is Cr. The exact staggered interpolation Prop. 5.22, at mesh ξ, now gives eq. 4.cccxlix. It is a fixed linear interpolation and hence preserves holomorphic parameter dependence.

To prove the family assertion, keep the absolute degree m raw row before attaching external-field carriers. Its fine inputs lie within $C_L\xi$ of the root. Join those m points to the root by a physical tree of length at most $C_L m\xi$, and substitute their actual literal-field trees. The coefficient sum is bounded by $C^m\xi^m$; the field trees contribute $(Cr)^m$. Charging the join costs only $e^{C\kappa_0 m}$ after passage to M-cubes, and a physical length cost bounded by $e^{Cm\xi}$. These are per-leaf costs, absorbed by a smaller fixed radius. Sum $\sum_{m\ge1}(C\xi r)^m$. This proves eq. 4.cccl. There is no separate rounding cost at each of the arbitrarily many intermediate averaging stages.

For the grouped assertion, the image of an old M-cube has side $\xi M \le M$. Its raw output vector reads the fine field in a fixed enlargement of that image. Apply Banach Cauchy to the actual plain-mean map on the fixed number of neighboring macroblock strong spaces. The map is analytic for an input supremum c/ξ, and its whole output vector has supremum Cc there, by eq. 3.mcccv. Its multilinear coefficients are consequently bounded $C^m\xi^m$, with only a fixed macroblock-slot factor. Compose the whole-cube literal-field families of the preceding lemma. This proves the vector bound without counting the old bond coordinates in that cube. □

Lemma 4.29 (All-depth physical nesting)**.** On a fixed smaller ball, in common physical units,

$$U_s(V_{s,k}(W)) \sim U_k(W), \qquad S_s(V_{s,k}(W)) = S_k(W). \tag{4.ccclii}$$

The first equivalence is a fine gauge equal to one at the old coarse nodes. The old-unit fine mesh is L^{-s}. ■

Proof. Every variation tangent to an M^s-fibre is tangent to the M^k-fibre, since $M^k = M^{k-s}\circ M^s$. Thus $U_k(W)$ is critical on the former fibre. In old physical units its literal potential is $\xi\widehat{A}_k$, its first derivative is $\xi^2 D\widehat{A}_k$, and its local Hölder derivative has the corresponding additional ξ^α. Its strong norm is at most $C\xi r$. The old datum is admitted by Prop. 4.28. Orbit uniqueness in Prop. 3.183, including its competing-multiplier bound eq. 3.mccxci, identifies the two orbits for real data. Analytic continuation gives the equality of their plaquette traces and actions for complex data. Wilson action in four dimensions is unchanged by this change of physical units. This proves eq. 4.ccclii directly; no product of one-step uniqueness constants is used. □

4.14.2. *Complete pointed inputs and their quadratic coefficient.* An old pointed family is a normally convergent representation

$$f_z(B) = \sum_{X\ni q(z)} f_{z,X}(B_X), \qquad \sup_z \sum_X e^{\kappa d_M(X)}\|f_{z,X}\|_\rho \le \epsilon. \tag{4.cccliii}$$

Here X is a connected set of old M-cubes, $d_M(X) = |X|-1$, and $\|\cdot\|_\rho$ is the supremum on the old bond product ball $\|B\|_\infty < \rho$. Assign each positive bond to its starting-site cube. The complete f_z is normalized by $f_z(0) = 0$, invariant under all sufficiently small admissible site gauges, and covariant under translations of the anchor. The individual $f_{z,X}$ need not have these invariances.

On finite tori we use the canonical physical family and its infinite-lattice companion, with the coupling parameters held fixed during the comparison. Write $\Pi_P f_z^\infty(B) = f_z^\infty(B^{\mathrm{per}})$ for evaluation at the periodic lift of B. This is a function of the finite bond variables; it is not a quantum expectation on a larger torus. The compatibility property needed below is

$$f_z^P - \Pi_P f_z^\infty = \mathcal{W}_z^P, \qquad \|\mathcal{W}_z^P\|_{\rho,\kappa/2} \le C\epsilon e^{-c\kappa P/M}. \tag{4.ccccliv}$$

Here P is the shortest old physical period. The notation on the second line means an actual rooted analytic family with the stated norm. Its terms retain a finite-period-tail label. For the canonical families this property is proved below from their lifted operator and cluster carriers. Gauge invariance on one finite torus alone would not imply it.

Lemma 4.30 (Ward moments of the complete density)**.** The infinite companion has zero first jet. Its Hessian $K_z = D^2 f_z^\infty(0)$ has, for some $m > 0$, an absolute two-leg moment

$$\sum_{b,c} e^{m(|m_b - z| + |m_c - z|)} \|K_z(b,c)\| \le C\epsilon, \qquad K_z[\mathrm{d}\lambda, a] = K_z[a, \mathrm{d}\lambda] = 0. \tag{4.ccclv}$$

The Ward identity extends to polynomial λ, a. Define

$$a_\mu^F(x) = \frac{1}{2}\sum_\nu F_{\nu\mu}(x_\nu - z_\nu), \qquad q_f(F) = \frac{1}{2} K_z[a^F, a^F]. \tag{4.ccclvi}$$

If the complete pointing is covariant under the full hypercubic group, including reflections, then

$$q_f(F) = \frac{b_f}{2}\sum_{\mu<\nu}\langle F_{\mu\nu}, F_{\mu\nu}\rangle, \qquad |b_f| \le C\epsilon. \tag{4.ccclvii}$$

The colour pairing is bilinear on the complexified algebra. ■

Proof. The first derivative is a colour vector at each bond. Invariance under constant colour rotations forces each vector to be zero. Differentiate the local gauge identity once in the gauge parameter and once in a bond direction. The term containing a derivative of the gauge-coordinate transformation is multiplied by this vanished first jet. The remaining identity is the Ward identity for the Hessian.

A carrier contains at most $CM^4(1 + d_M(X))$ old bond and colour coordinates. Two Cauchy derivatives cost this coordinate-pair count and a fixed inverse power of ρ. Every coordinate in a rooted carrier is within $CM(1 + d_M(X))$ of its anchor. Choose m small relative to κ/M; the exponential and polynomial costs are paid by a reserved fraction of the rate in eq. 4.cccliii. This proves the moment bound. Cut off polynomial gauge parameters and arguments and pass to their limit using this absolute moment. No unbounded polynomial gauge transformation is applied to the nonlinear function.

Constant colour rotations make the quadratic colour pairing scalar. Independent axis reflections kill terms joining different curvature planes, since the two planes have different reflection characters. Axis permutations make the remaining six coefficients equal. The definition of a^F has no diagonal term $F_{\mu\mu}$; using a bond midpoint instead of its initial site therefore does not change it. The moment bound proves the estimate for its scalar coefficient. □

4.14.3. *An actual classical marginal representative.* Let $\theta_z(X) = \prod_{\lambda=1}^4 (1 - |X_\lambda - z_\lambda|)_+$, periodized on a torus. On the old lattice with fine spacing $h_s = L^{-s}$, put

$$e_{s,z}(B) = \sum_{p>0} \theta_z(m_p) a_p(U_s(e^{iB})). \tag{4.ccclviii}$$

The point m_p is the original fine plaquette midpoint in old physical units.

Lemma 4.31 (Exact representative and normalization)**.** The complete $e_{s,z}$ is gauge invariant and has an analytic rooted family with a bound uniform in s. It has zero value and first jet at the flat datum, and

$$\sum_z e_{s,z}(B) = S_s(e^{iB}), \qquad q_e(F) = \frac{1}{2}\sum_{\mu<\nu}\langle F_{\mu\nu}, F_{\mu\nu}\rangle. \tag{4.ccclix}$$

Moreover

$$\sum_z e_{s,z}(B_\xi(w)) = S_k(e^{iw}) \tag{4.ccclx}$$

on the actual common ball. Its finite periodic point function is exactly the evaluation of its classical infinite companion at periodic data. ■

Proof. The first identity follows from $\sum_z \theta_z = 1$. Each original fine plaquette trace is gauge invariant as a function of its actual critical orbit, by eq. 4.ccxcix. The action trees keep the factor h_s^4 at that original plaquette. Summing the plaquettes in the fixed support of θ_z therefore gives a bounded rooted family, by eq. 4.cclxxviii; there is no fine-count loss.

For every plaquette orientation the exact identity

$$h_s^4 \sum_{p \text{ in that orientation}} \theta_z(m_p) = 1 \tag{4.ccclxi}$$

holds. To see it in one coordinate, write the fine index as $Nn + r$, $0 \le r < N = h_s^{-1}$. For each fractional shift, including the half shift of a plaquette midpoint, the sum over integer translates of the one-dimensional tent is one. The N residue classes therefore sum to N. Take the product of the four identities.

The actual affine first response is the sampled affine fine potential modulo a fine gradient, by eq. 5.clxxiii. Every plaquette has zero first variation at the flat field, so no second background response contributes to its quadratic jet. The fine quadratic action is $h_s^4\langle F_{\mu\nu}, F_{\mu\nu}\rangle/2$. Equation eq. 4.ccclxi gives the second identity in eq. 4.ccclix.

Classical critical fields lift to periodic covers: the lifted field and multiplier solve the same local critical equations and constraints and stay in the strong uniqueness class. Orbit uniqueness identifies them with the constructed background. Each fixed plaquette trace is consequently unchanged by that lift. The compact tent, with its periodic sum, then gives the last assertion. This is a statement about a classical function, not about a Gaussian determinant or a quantum cutoff integral. Finally use eq. 4.ccclii to obtain eq. 4.ccclx. □

For a scalar input set

$$r_{s,z} = f_{s,z} - b_s e_{s,z}, \qquad b_s = b[f_s^\infty]. \tag{4.ccclxii}$$

It has complete first jet zero, curvature readout zero, and input norm at most $C\epsilon_s$. The subtracted coefficient multiplies S_k exactly after nesting. It is not replaced by a coarse Wilson action that only approximates S_k.

4.14.4. *A large analytic carrier bound from the actual map.*

Lemma 4.32 (Coarse block composition)**.** Suppose $f_z(0) = Df_z(0) = 0$ and eq. 4.ccliii holds. For a fixed range $\xi \le \xi_{\max} \le 1$, choose a target rate $\kappa_0 > 0$ and a field radius $r > 0$ satisfying

$$C\kappa_0\xi_{\max} + C_{\kappa_0}\xi_{\max} r/\rho < \kappa/4, \qquad C_{\kappa_0}\xi_{\max} r/\rho < 1/4. \tag{4.ccclxiii}$$

The actual composition $f_z(B_\xi(w))$ then has a rooted final-scale analytic family with

$$\|f_z \circ B_\xi\|_{r,\kappa_0} \le C\epsilon r^2 \xi^2. \tag{4.ccclxiv}$$

All constants are independent of the depths. Using whole-cube Banach groups, the conditions in eq. 4.ccclxiii contain no coordinate-count factor M^4. ■

Proof. Replace each local representative by

$$f_{z,X}(B_X) - f_{z,X}(0) - Df_{z,X}(0)[B_X].$$

Its complete sum is unchanged. The removed constant and linear sums vanish; their absolute convergence follows from Cauchy and a rate reserve.

Treat the variables in each old M-cube as one complex Banach vector with its supremum norm. There are $n = |X| = 1 + d_M(X)$ such groups. Multivariate Cauchy and polarization give a coefficient bound $\|f_{z,X}\|_\rho (e/\rho)^m$ at total degree m, for each distribution of degrees among the groups. To verify the polarization factor, insert independent scalars in the j slots of a homogeneous polynomial.

Cauchy on circles of radius $1/j$ bounds its multilinear norm by $j^j/j! \le e^j$ times its polynomial norm. Multiplying over the groups gives the stated factor, independently of their dimensions.

The grouped actual intermediate family in Prop. 4.28 has norm $d = C_{\kappa_0}\xi r$ for each group. The Taylor terms of degree at least two are therefore bounded by

$$C\|f_{z,X}\|_\rho\, n^2(\xi r/\rho)^2 \exp(C_{\kappa_0}\xi r n/\rho). \tag{4.ccclxv}$$

Indeed sum the degree distributions with the product generating function $(1-ed/\rho)^{-n}$, subtract its degrees zero and one, and use Taylor's formula for that scalar function.

The image of the connected old cube carrier can be joined in the final cube lattice with at most $C(1+\xi d_M(X))$ edges. Scale a spanning curve of length at most $CMd_M(X)$ and cover its $O(\xi M)$ neighborhood by final M-cubes; a curve of length ℓ needs at most $C(1+\ell/M)$ such cubes. This joins all grouped output roots at once, including all coordinate positions in each old cube image. Union this join with their actual intermediate families. Extra fixed costs per Taylor slot are absorbed in C_{κ_0}. The geometric rate charge is $\exp(C\kappa_0(1+\xi d_M(X)))$.

The two exponential costs in this expression and eq. 4.ccclxv fit in the input rate by eq. 4.ccclxiii. The remaining polynomial n^2 uses another fixed rate reserve. Summing the old family gives eq. 4.ccclxiv. This filing is an exact analytic substitution; its pieces read only the final bond variables in their indicated connected carriers. □

4.14.5. *Canonical physical inputs and finite-period tails.*

Lemma 4.33 (The actual canonical input class)**.** The complete physical normalizer, mismatch and interacting pointings obtained from Prop. 4.18 belong to the class eq. 4.cccliii on a fixed output coarse bond ball, after subtracting their vacuum values. Their rooted norm bounds can be taken respectively as

$$C, \qquad C|\tau-1|, \qquad C\{(g\mathfrak{b})^2 + e^{-c\mathfrak{b}^2}\}. \tag{4.ccclxvi}$$

For the interacting term this assertion concerns the homogeneous classical-pullback model with zero ordinary profile and the physical pointing of Cor. 4.20. These canonical families satisfy eq. 4.ccccliv. ■

Proof. The intrinsic functions are local analytic families by Prop. 4.18. Their physical arguments are the actual intermediate data $p_j(W)$, not independent fine currents. The p_j in the physical pointing is in its one-step axial gauge. On a fixed sufficiently small output ball both it and $p_j^{\mathrm{lit}}(W) = M^{j-1}U_j(W)$ are admitted, and the prescribed node-fixed gauge connects them. Apply gauge invariance to the complete pointed function first:

$$f_z^{\mathrm{phys}}(p_j(W)) = f_z^{\mathrm{phys}}(p_j^{\mathrm{lit}}(W)).$$

The identity extends holomorphically on this admitted overlap by the joint chart identity. The local representatives below are composed with p_j^{lit}; they are not individually replaced by gauge invariance. To see a full analytic carrier bound after this substitution, group each intrinsic localization cube as one Banach vector. In output units its side is M. The actual one-level-before-output map has whole-cube family norm $C\rho/L$, by Prop. 4.28. Apply the block Cauchy argument in Lem. 4.32, retaining the constant and linear degrees when necessary. The old intrinsic cube has side LM in its input units; after output rescaling the geometric join costs $C(1+d_M(X))$, with only fixed L constants. Request a larger intrinsic rate first, then choose a fixed smaller output radius so the block Cauchy charge fits that rate. The bound is linear in the intrinsic rooted norm. There is no count of its $(LM)^4$ individual coordinates.

The complete interacting pointing is gauge invariant by eq. 4.ccc and Cor. 4.20. For the normalizer use its exact definition eq. 4.cclxii. Point $\log J_U(0)$ at its pivot factors and $-\frac{1}{2}\operatorname{Tr}\log K$ at its diagonal free bonds, with the midpoint tents and the stated vacuum subtraction. Under a real coarse gauge the actual free coordinates transform by the block-orthogonal matrix O_g of eq. 4.ccc. Thus $\log K$ conjugates by O_g, and the colour trace of each diagonal block is invariant. The pivot

determinant transforms on both sides by adjoint matrices of determinant one. This proves invariance of the complete pointed normalizer before any carrier filing. For the mismatch use its exact pointing eq. 4.cclii. The diagonal colour trace of $K_G Q_{\tau,u}^{-1}$ has the same covariance. These arguments also give the vanished first jets by global colour rotations.

The matrix-log localization is eq. 4.ccli, with the actual nonlocal near/tail construction eq. 4.ccxc and eq. 4.ccxciv in place of a finite-range precision. Group powers before attaching their parameters and retain every precision carrier in the diagonal trace. The absolute rooted logarithm series gives the normalizer bound. The mismatch interpolation gives its factor $|\tau - 1|$. The interacting estimate is the physical even-coupling bound eq. 4.cccxv. Taking a vacuum difference changes these bounds by at most a fixed factor. The original midpoint pointing, the symmetric middle pivot pattern and the actual orbit map preserve translations and the full hypercubic group. The scalar readout is therefore applicable.

We give the finite-period argument explicitly. Retain the lifted propagator paths in the background trees, all precision carriers in the grouped matrix powers, and the full assigned carriers of the normalized local Gaussian functionals and Mayer clusters. In particular, the precision cutoff is by the full lifted connected carrier Γ, containing both matrix ends and every external background input, *before* periodic projection. Its charged length is $\ell(\Gamma)$. The stronger input estimate is already eq. 4.cclxxxv: the absolute Schur sum carries $e^{t_0 \ell(\Gamma)}$, by the HET connected-tree estimate and marked leaves. Symmetrize a real term with its transposed lifted carrier, and cut $\ell(\Gamma) \le R$. A long winding carrier never enters the short precision because its projected endpoints happen to be close.

The covering estimate used in HAF also holds for cubes on the cover:

$$S^{\#\Gamma_M^{\mathrm{cover}}} \le S^C \exp\{C(\log S)\ell(\Gamma)/M\}.$$

Thus its stronger lifted weight is paid by the displayed exponential input norm. The exact telescoping tails retain that weight. Combine each short power on the cover before projection; its finite-range Hilbert-row proof is unchanged, since the real cover gap follows by local limits of the uniform quadratic-form gap. The resulting short-step ball includes all its external background support. For the finite recipe lift the auxiliary cube parameters and the fixed held root guards periodically: parameters of period images are identified, while their full lifted carriers remain separately charged. The cover operator is then deck covariant; its row and column Schur sums descend without a sheet factor. A nonwrapping object sees the same single guard and wall recipe as its companion. The grouped root and the matrix-log series therefore have their absolute lifted carrier bounds as well. Subsequent absolute Gaussian and cluster estimates use submultiplicative carrier weights and an upper bound on the number of integration variables. The lifted weight is still submultiplicative, and identifying period images only reduces that number of variables. Retaining these labels throughout the absolute estimates gives the same rooted canonical-family bound with a reserved lifted rate.

For this comparison the support of a common object includes its entire fixed held root guard and the data and trace collars, also before projection. These are fixed-layer enlargements of the connected cube support. Their size is at most a fixed multiple of its size. Request that larger rate in the formation estimate first and reserve the needed fraction for this enlargement. A rooted object contained in a common embedded old neighborhood of radius $P/8$ is calculated by exactly the same local formulas on the torus and on its lift. For a Gaussian trace, a nonzero endpoint period image is a long object and is not put in this common part. For a local Gaussian marginal the comparison uses the same fixed root, guard and wall domain; it is not a comparison between different choices of held root guards. The normalized finite-dimensional integral, cutoff, source pointing and cluster coefficient of a common object are consequently identical.

Objects not paired in this way have connected carriers reaching distance cP from their root, or carrying a period winding. Their charged lifted cube size is at least cP/M. The stronger weight just described pays this size even when projection identifies some of its cubes. Reserve half of the

exponential input rate. The absolute remaining rooted sum is bounded by $C\epsilon e^{-c\kappa P/M}$. Projecting a lifted carrier to the torus only decreases the required cube-tree length. This constructs the difference family in eq. 4.cccliv, including its finite-period-tail label. Cauchy derivatives and any fixed rooted spatial moments use a further rate reserve and obey the same exponential comparison with a smaller c.

This proves a comparison of pointed analytic functions and their local expansions. The finite determinant and cutoff integral are not identified with those on a larger cover, which would introduce extra quantum fluctuation modes. □

4.14.6. *A slow-field estimate from a small local gauge.*

Lemma 4.34 (Complete slow-field expansion)**.** Let f_z^∞ obey the input norm and complete gauge identity above. For a continuous connection with $\|a\|_{C^{1,\alpha}} \le r$ put

$$B_\xi(b) = \xi a(\xi m_b).$$

On a fixed smaller radius, uniformly in $0 < \xi \le 1$,

$$\left| f_z^\infty(B_\xi) - \xi^4 q_f(F_a(\xi z)) \right| \le C\epsilon r^2 \xi^{4+\alpha}, \quad F_{\nu\mu} = \partial_\nu a_\mu - \partial_\mu a_\nu + i[a_\nu, a_\mu]. \tag{4.ccclxvii}$$

The same estimate holds uniformly on holomorphic external-field balls of such connections. ■

Proof. For sufficiently small ξ, take $R = R_0 + D\log(1/\xi)$, with D large compared with the inverse rooted spatial rates. In old coordinates $y = x - z$, let a smooth cutoff equal one on $|y| \le R$ and zero on $|y| \ge 2R$. Set

$$A_\mu = a_\mu(\xi z), \qquad g(z+y) = \exp\left(i\xi \chi_R(y) \sum_\mu A_\mu y_\mu \right).$$

Its exponent is bounded by $C\xi r R$, and its old first difference by $C\xi r$. For ξ below a fixed threshold, $\xi R \le 1$. The original and transformed bond logarithms both have supremum at most $C\xi r$, strictly inside the original analytic ball. For real connections the gauge is unitary. The equality $f_z^\infty(B_\xi^g) = f_z^\infty(B_\xi)$ extends jointly and holomorphically while both arguments remain in that ball. In particular it applies to a holomorphic connection $a(w)$. No arbitrary complex gauge saturation is used.

Within the region where the cutoff is one, second-order BCH, with the stated gauge convention, gives

$$B^g_{\xi,\mu}(z+y) = \xi^2 \Big\{ \sum_\nu \partial_\nu a_\mu(\xi z)(y_\nu + \tfrac{1}{2}\delta_{\nu\mu}) + \frac{i}{2} \sum_\nu y_\nu [A_\nu, A_\mu] \Big\} + E_\mu(y),$$
$$|E_\mu(y)| \le Cr\xi^{2+\alpha}(1+|y|)^3. \tag{4.ccclxviii}$$

To check the commutator, the three leading exponents are $i\xi Y, i\xi A_\mu, -i\xi(Y + A_\mu)$, where $Y = \sum A_\nu y_\nu$. Their BCH bracket is $\frac{1}{2}\xi^2 [Y, A_\mu]_{\mathrm{H}}$. The continuous first-order Taylor remainder costs $r\xi^{2+\alpha}(1+|y|)^{1+\alpha}$; the cubic BCH terms are bounded by $Cr^3\xi^3(1+|y|)^3$, which fits the displayed error since $\alpha < 1$.

The leading polynomial is $\xi^2(a^F + \mathrm{d}\chi)$, where

$$\chi(y) = \frac{1}{4} \sum_{\mu,\nu} (\partial_\nu a_\mu + \partial_\mu a_\nu)(\xi z) y_\mu y_\nu. \tag{4.ccclxix}$$

Its exact forward difference contributes $\frac{1}{2}\partial_\mu a_\mu$, so this identity includes the staggered midpoint term. The Ward identity removes $\mathrm{d}\chi$ in both Hessian arguments.

Expand each local representative through degree two at zero, and sum. The complete constant and first terms vanish. For a carrier contained in the inner patch the transformed angle is bounded by $C\xi^2 r(1 + M d_M(X))$. Block Cauchy bounds its cubic remainder by $\|f_{z,X}\|_\rho \xi^6 r^3$ times a fixed polynomial in $1 + d_M(X)$. The exponential Taylor charge can be bounded using the global supremum

$C\xi r$; it is paid by a fixed rate reserve. Summing such carriers gives $C\epsilon\xi^6 r^3$. A carrier exiting the patch has $d_M(X) \geq cR/M$. Its cubic remainder, estimated using the global angle bound, is $C\epsilon\xi^3 r^3 e^{-cR/M}$ after reserving rate and coordinate-count polynomials. Choose D so that this is also bounded by $C\epsilon\xi^6 r^3$.

For the quadratic term use eq. 4.ccclv. Inside the patch, replacing either argument by the error in eq. 4.ccclxviii costs $C\epsilon r^2\xi^{4+\alpha}$; the fixed spatial polynomials are summable against the two-leg exponential moment. Outside the patch, compare the globally bounded transformed field with the leading polynomial using the exponential tail of that moment. Increasing D pays this error as well. The remaining quadratic term is exactly $\xi^4 q_f(F_a(\xi z))$. Since $\xi^6 r^3 \leq r^2\xi^{4+\alpha}$, the estimate follows for small ξ.

For ξ above the fixed threshold the ordinary degree-two Cauchy bound and $|q_f(F)| \leq C\epsilon r^2$ prove the same statement with a larger fixed constant. All estimates hold in the complex norm, so they are uniform on the indicated holomorphic external-field ball. □

Apply this lemma to the actual interpolant in eq. 4.cccxlix. For the remainder eq. 4.ccclxii the curvature readout is zero. It has the complete bound

$$\|r^{\infty}_{s,z}(B_\xi(\,\cdot\,))\|_r \leq C\epsilon_s r^2\xi^{4+\alpha}. \tag{4.ccclxx}$$

On a finite torus its old shortest period satisfies $P_{\text{old}} \geq 2/\xi$. By eq. 4.cccliv and Lem. 4.32, the difference from this periodic companion has a large carrier norm bounded by

$$C\epsilon_s r^2\xi^2 e^{-cP_{\text{old}}}. \tag{4.ccclxxi}$$

For small ξ this is at most the right side of eq. 4.ccclxx. For $\xi \geq \xi_0 > 0$, the large bound $C\epsilon_s r^2\xi^2$ suffices, with a constant depending on ξ_0. The finite composed remainder therefore has the same small supremum estimate. Equivalently the logarithmic gauge patch can be used in the finite old torus whenever $8R < P_{\text{old}}$, which holds below a fixed ξ_0. No nonperiodic polynomial gauge is applied to that finite torus.

4.14.7. *Block Möbius interpolation without a volume loss.* The small complete supremum in eq. 4.ccclxx is stronger than the large norm eq. 4.ccclxiv, but is not itself a local-family estimate. We now construct one by an exact block decomposition.

Lemma 4.35 (Rooted block interpolation)**.** Let $F(0) = 0$ be holomorphic on a bond ball $\|w\|_\infty < r$. Suppose it has a connected family rooted at a cube Q_0, with norm at most E at rate $\kappa_0 > 0$, and $\|F\|_r \leq \delta$. Fix $0 < \vartheta < 1$. There are $t > 0, a > 0$, independent of the number of blocks, for which F has an actual rooted local family on $\|w\|_\infty < tr$, satisfying

$$\|F\|_{tr,a} \leq C\delta^{\vartheta} E^{1-\vartheta}. \tag{4.ccclxxii}$$

The construction is linear in F. It preserves holomorphic auxiliary parameters. ■

Proof. For a finite set S of final M-cube groups let w_T keep the groups in $T \subset S$ and set the others to zero. Define

$$F_S(w_S) = \sum_{T\subset S} (-1)^{|S\setminus T|} F(w_T). \tag{4.ccclxxiii}$$

On a finite torus inclusion-exclusion gives $F = \sum_S F_S$ exactly; $F_\varnothing = 0$. Each F_S reads only the indicated groups.

Put $u = t/(1-t)$. For a fixed w_S of norm at most tr, write each group as tv_Q. The holomorphic function $F(\sum_{Q\in S} z_Q v_Q)$ has supremum δ on $|z_Q| < 1$, since the groups are disjoint. The alternating sum in eq. 4.ccclxxiii keeps only the Taylor terms of positive degree in every group. Cauchy in these scalars therefore gives

$$b_S := \|F_S\|_{tr} \leq \delta u^{|S|}. \tag{4.ccclxxiv}$$

There is one scalar per block, not one per bond or colour coordinate.

For a local representative G_X of the given large family, its S-projection vanishes unless $S \subset X$, and obeys the same estimate with $\|G_X\|_r$ in place of δ. Let d_S be the length of a minimal Steiner cube tree joining S and Q_0. Then $d_S \le d_M(X)$ whenever $S \subset X$. Choose $0 < \kappa' < \kappa_0$ and t small enough that $\log(1+u) + \kappa' < \kappa_0$. Summing the projections of the large family gives

$$\sum_S e^{\kappa' d_S} b_S \le \sum_X \|G_X\|_r e^{\kappa' d_M(X)} (1+u)^{|X|} \le CE. \tag{4.ccclxxv}$$

Only triangle inequalities are used here; cancellations between representatives are used solely in the complete bound eq. 4.ccclxxiv.

Choose $a < (1-\vartheta)\kappa'$, and put $\lambda = ((1-\vartheta)\kappa' - a)/\vartheta > 0$. Hölder's inequality, applied to $b_S^{1-\vartheta} b_S^{\vartheta}$, gives

$$\sum_S e^{a d_S} b_S \le \delta^{\vartheta} (CE)^{1-\vartheta} \left\{ \sum_S u^{|S|} e^{-\lambda d_S} \right\}^{\vartheta}. \tag{4.ccclxxvi}$$

The last sum is finite for a fixed sufficiently small t, at every positive λ. Here is a bound that does not assume a large enough rate for an unweighted animal count.

For $n = |S|$, the minimum metric spanning tree on $S \cup \{Q_0\}$ has length at most $2d_S$: traverse a Steiner tree and shortcut the resulting walk. Consequently

$$e^{-\lambda d_S} \le \sum_{\mathcal{T} \text{ on } \{0,\dots,n\}} \prod_{\{i,j\} \in \mathcal{T}} e^{-\lambda |Q_i - Q_j|_1 / 2}.$$

Allow repeated positions and divide by $n!$ to overcount the sets. For a fixed labelled tree, sum its leaves successively. Each leaf costs at most $B_\lambda = \sum_{x \in \mathbb{Z}^4} e^{-\lambda |x|_1/2} < \infty$. The number of labelled trees is $(n+1)^{n-1}$. Thus

$$\sum_S u^{|S|} e^{-\lambda d_S} \le 1 + \sum_{n \ge 1} \frac{u^n}{n!} (n+1)^{n-1} B_\lambda^n < \infty \quad \text{if } euB_\lambda < 1. \tag{4.ccclxxvii}$$

On a periodic cube lattice the sum of the metric exponential is no larger than this infinite-lattice sum, by choosing nearest representatives of the period classes. The same constant is therefore valid.

File F_S on a fixed minimal Steiner tree joining S to Q_0. A fixed geometric tie rule suffices; it depends on no field or coupling parameter. Equations eq. 4.ccclxxvi and eq. 4.ccclxxvii prove eq. 4.ccclxxii. For the infinite lattice, the projections can be interchanged with the given large family by the absolute bound eq. 4.ccclxxv. Each local representative has the finite inclusion-exclusion identity, so their sum still reconstructs F. This proves both normal convergence and the exact identity there. Every operation is linear and the estimates are uniform on smaller auxiliary polydiscs, proving the last assertion. □

For the actual remainder take

$$\vartheta = \frac{2+\beta}{2+\alpha}. \tag{4.ccclxxviii}$$

Combining eq. 4.ccclxx and eq. 4.ccclxiv gives, per old anchor,

$$\|r^{\infty}_{s,z} \circ B_\xi\|_{tr,a} \le C(\epsilon_s r^2 \xi^{4+\alpha})^{\vartheta} (\epsilon_s r^2 \xi^2)^{1-\vartheta} = C \epsilon_s r^2 \xi^{4+\beta}. \tag{4.ccclxxix}$$

The equality of exponents uses $2 + (2+\alpha)\vartheta = 4 + \beta$. The reconstructed complete function remains gauge invariant. This conclusion does not assert invariance of its Möbius pieces.

4.14.8. *A compatible finite-volume filing.* We retain the finite-period information when making the improved filing. Merely replacing every finite-volume coefficient by a short minimal carrier could hide its winding dependence.

First perform the construction in Lem. 4.35 on the infinite companion. Evaluate its normally convergent local family at periodic final data and project its full carrier to the torus. Its analytic supremum does not grow: identifying some of its input variables still gives values inside the same

product ball. Its required cube length does not grow either. Keep its lifted carrier label before projection.

Classical lift compatibility identifies the actual finite B_ξ with the periodic restriction of its infinite counterpart. The difference of the two transported quantum densities is therefore the composed old finite-period difference. By eq. 4.ccclxxi it has large norm $C\epsilon_s r^2\xi^2 e^{-c_w P_{\mathrm{old}}}$. Put $\gamma_{\mathrm{per}} = Mc_w$; the prescribed period exponent is measured in M-cube units, as $e^{-\gamma_{\mathrm{per}}P_{\mathrm{old}}/M}$. Both complete remainders have the small supremum in eq. 4.ccclxx, so their difference has it as well. Apply block interpolation to this difference. It gives

$$C\epsilon_s r^2\xi^{4+\beta}e^{-(1-\vartheta)c_w P_{\mathrm{old}}} \tag{4.ccclxxx}$$

for its minimal-carrier family.

Retain its finite-period-tail label and enlarge each of these carriers by a fixed shortest period cycle through the root. This is a legitimate enlargement of a carrier: it alters no function and adds no field variables. If $P_{\mathrm{fin}} = \xi P_{\mathrm{old}}$ is the shortest final period, the extra length is at most $C(1 + P_{\mathrm{fin}}/M)$. Choose $a > 0$ also to satisfy

$$Ca\xi_{\max} < (1-\vartheta)\gamma_{\mathrm{per}}/2. \tag{4.ccclxxxi}$$

The added weight is paid by eq. 4.ccclxxx. The remaining rooted norm is bounded by

$$C\epsilon_s r^2\xi^{4+\beta}e^{-c'_w P_{\mathrm{old}}} = C\epsilon_s r^2\xi^{4+\beta}e^{-(c'_w/\xi)P_{\mathrm{fin}}}.$$

Thus all coefficients with genuine finite-period dependence remain explicitly in long carriers. The short part is the projected companion construction. The output again satisfies a compatibility estimate of the form eq. 4.cccliv, now with the improved size. This argument compares functions through their absolutely convergent local expansions; it does not add or remove quantum modes.

4.14.9. *Summing old anchors and the exact counterterm.*

Proposition 4.36 (Actual homogeneous physical irrelevance)**.** Let $f_{s,z}$ be complete gauge-invariant, translation-covariant physical pointed densities on the old coarse ball, with full hypercubic covariance, input bound ϵ_s, and the compatible companion property above. This includes the actual canonical inputs of Lem. 4.33. Set $b_s = b[f_s^\infty]$ and $r_{s,z} = f_{s,z} - b_s e_{s,z}$. For every fixed L there are fixed positive output radius r_{out}, rate a, and constant C such that

$$\sum_z f_{s,z}(B_\xi(w)) = b_s S_k(e^{iw}) + \sum_y \mathcal{R}_{s\to k,y}(w), \qquad \sup_y \|\mathcal{R}_{s\to k,y}\|_{r_{\mathrm{out}},a} \le C\epsilon_s r^2 L^{-\beta(k-s)}, \tag{4.ccclxxxii}$$

where $r_{\mathrm{out}} = tr$, with the fixed large-ball radius r chosen in eq. 4.ccclxiii. Each complete output density is gauge invariant and has zero first jet and zero companion curvature coefficient. It has an actual compatible local analytic representation; its individual representatives need not be gauge invariant. ∎

Proof. With the final unit tents define the complete output density by the exact formula

$$\mathcal{R}_{s\to k,y}(w) = \sum_{z\ \mathrm{old}} \theta_y(\xi z)\, r_{s,z}(B_\xi(w)). \tag{4.ccclxxxiii}$$

Only old anchors within fixed final distance of y occur. Transfer their roots to y by bounded cube joins. By eq. 4.ccclxxix and the compatible finite-period construction, the rooted norm is at most

$$C\epsilon_s r^2\xi^{4+\beta}\sum_z \theta_y(\xi z).$$

Because ξ^{-1} is an integer, the same residue-class tent computation as eq. 4.ccclxi gives

$$\xi^4\sum_z \theta_y(\xi z) = 1. \tag{4.ccclxxxiv}$$

This proves the bound in eq. 4.ccclxxxii. It also proves the uniform finite-period-tail bound after the old-anchor sum, with exponential rate in P_{fin}.

The tents sum to one in y. Sum eq. 4.ccclxxxiii over y, substitute the definition of $r_{s,z}$, and apply eq. 4.ccclx. This proves the exact identity in eq. 4.ccclxxxii.

Under a small final coarse gauge, actual orbit covariance of U_k changes $V_{s,k}$ by an old-lattice site gauge. The complete old functions are invariant, and the tents are geometric scalars. Thus each complete output density is invariant. Its first jet is zero, either by the chain rule or global colour rotations.

For the companion curvature coefficient test a final affine potential of constant curvature. The actual affine response eq. 5.clxxiii, followed by the literal partial mean, is the old affine potential with its length factor, modulo an old gradient. The node-frame correction is included as in eq. 5.clxxvi. The old constant term is another polynomial gradient. The remaining old curvature is $\xi^2 F$. The second chain rule contains no second background response because $Dr_{s,z}(0) = 0$. The two-leg Ward identity removes all gradient terms and gives

$$\frac{1}{2} D^2(r_{s,z} \circ B_\xi)(0)[a^F, a^F] = \xi^4 q_{r_s}(F) = 0.$$

All polynomial substitutions are justified by the exponential moments of the actual response and old Hessian. Multiplication by the fixed root tents and their finite anchor sum preserves zero. Translation and hypercubic covariance of the complete density are preserved since ξ^{-1} is an integer. The compatible filing was constructed above and is an exact representation of this same complete function. □

Corollary 4.37 (Uniform sum of proper old scales)**.** If $0 \le s < k$ and $\epsilon_s \le E_0$ on common parameter domains, their transported remainders satisfy

$$\sup_y \left\| \sum_{s<k} \mathcal{R}_{s\to k,y} \right\|_{r_{\mathrm{out}},a} \le C r^2 \sum_{s<k} \epsilon_s L^{-\beta(k-s)} \le \frac{C r^2 E_0}{L^\beta - 1}. \tag{4.ccclxxxv}$$

Their complete sum is invariant, has zero companion curvature coefficient, and has the same finite-period compatibility. The total classical counterterm is exactly $(\sum_{s<k} b_s) S_k$. ■□

Proof. Sum the positive rooted bounds. The age is at least one, so the geometric series is $\sum_{n\ge 1} L^{-\beta n} = (L^\beta - 1)^{-1}$. Normal convergence preserves the identities and holomorphy. The finite-period comparison has at least the final-period exponential at every old scale, so its sum has the same property. □

4.14.10. *Rate and analytic-radius bookkeeping.* For arbitrary fixed odd L, the proof gives a fixed positive output rate and one fixed smaller field ball. The loss is independent of the number of old scales. One can prescribe an output rate $a > 0$ in advance for proper old ages. Choose κ_0, κ' with $(1-\vartheta)\kappa' > a$, then choose t using eq. 4.ccclxxvii. Since $\xi_{\max} = 1/L$, the conditions eq. 4.ccclxiii and eq. 4.ccclxxxi are met for all proper old ages by taking L sufficiently large relative to these fixed rates and a fixed ratio r/ρ. The old-carrier geometric charge contains $\xi d_M(X)$, and the finite-period padding charge contains $P_{\mathrm{fin}} = \xi P_{\mathrm{old}}$. Both improve under this coarsening. Write $\gamma'_{\mathrm{per}} = M c'_w$. The remaining finite-period tail is $e^{-\gamma'_{\mathrm{per}} P_{\mathrm{fin}}/(M\xi)}$. To retain a prescribed compatibility exponent $\gamma_{\mathrm{per,target}}$ in M-cube units, impose $L\gamma'_{\mathrm{per}} \ge \gamma_{\mathrm{per,target}}$, in addition to the padding reserve. These conditions contain no M: the common M^{-1} length conversion cancels. They compensate the fixed $1-\vartheta$ interpolation loss for every proper age. Thus a chosen carrier rate and a chosen finite-period compatibility exponent can both be retained under sequential proper transport when the blocking integer is large enough. Choose M afterwards to realize the requested macroblock field rate.

The grouped family bound in Prop. 4.28 has an amplitude constant independent of M, L, while its allowed radius can depend on L. Thus eq. 3.mcccvi gives the necessary buffer in literal coordinates:

an outer data ball of radius R maps inside a working input ball of radius $tR/4$ if $C/L < t/4$. Fluctuation substitution additionally requires $g\mathfrak{b} < tR/4$, with the fixed chart constants included. These inequalities can be imposed in the order: rates and t, blocking integer, localization size, common small physical radius, then coupling floor. The nonlinear radius need not be uniform as L grows.

The theorem proves the physical homogeneous transport estimate from actual intermediate maps and explicit canonical inputs. A complete RG trajectory additionally requires formation and substitution on the buffered domains at every step. No transport theorem for arbitrary independent fine-current cores, proper history domains or changing nonlinear towers is asserted here.

4.15. **Formation at the literal centre.** The literal intermediate map decreases bond angles by a factor L^{-1}. We use that decrease to evaluate the old interaction on its original analytic ball while allowing a larger ball of output data. The relative contour gauge gives an exact change to the axial step on their real overlap. Holomorphy on the larger ball will follow from the literal integral itself.

4.15.1. *The input and parameter domains.* Fix the middle pivots and real product colour balls of Prop. 4.25. Choose a carrier rate $\kappa > 0$. An input h consists of pointed functions

$$h_z(B) = \sum_{X \ni q(z)} h_{z,X}(B_X), \qquad \|h\|_{r,\kappa} = \inf_{\{h_{z,X}\}} \sup_z \sum_X e^{\kappa d_M(X)} \|h_{z,X}\|_r. \tag{4.ccclxxxvi}$$

Here $B = i^{-1} \log V$ is in old coarse units, and X is a connected set of old M-cubes. The complete h_z is gauge invariant, normalized at $B = 0$, and covariant under old-lattice translations and the hypercubic group. Its companion coefficient eq. 4.ccclvii is zero. Take the Banach quotient of local analytic families by representations whose complete function is zero at every anchor. The infimum in eq. 4.ccclxxxvi is its quotient norm. Write $\mathfrak{H}_{r,\kappa}$ for the closed subspace with the stated identities, and put

$$H(V) = \sum_z h_z(i^{-1} \log V). \tag{4.ccclxxxvii}$$

The sum is finite on a torus. We also write $h_z(V) = h_z(i^{-1} \log V)$. The conditions on the complete functions are closed and linear: evaluation, the Ward moment extractor and covariance identities are continuous in this norm. Complex variations of h below are in this space. Reality is required only when a real transfer is asserted.

For families over compatible periods, include an infinite companion and take the supremum of these spatial norms. Fix $\gamma_{\mathrm{per}} > 0$ in cube units and also require

$$\|h\|_{r,\kappa,\gamma_{\mathrm{per}}} = \max\left\{ \sup_P \|h^P\|_{r,\kappa}, \|h^\infty\|_{r,\kappa}, \sup_P e^{\gamma_{\mathrm{per}} P/M} \|h^P - \Pi_P h^\infty\|_{r,\kappa/2} \right\} < \infty. \tag{4.ccclxxxviii}$$

The notation Π_P is periodic evaluation, as in eq. 4.cccliv. Write $\mathfrak{H}_{r,\kappa,\gamma_{\mathrm{per}}}$ for this space. The same norm notation will be used for output pointings before their scalar coefficient is subtracted.

Let $t \in (0, 1/8)$ be the fixed shrink in Lem. 4.35. We shall use the radii

$$r = tR. \tag{4.ccclxxxix}$$

Choose an intrinsic output rate κ_{int} larger than a fixed geometric multiple of $\kappa + \gamma_{\mathrm{per}}$. The whole-cube intermediate bound in Prop. 4.28 has an amplitude C_p, independent of L, M on an allowed smaller ball. Choose an odd L so large that

$$\frac{64 C_p}{Lt} < \min(1, \kappa_{\mathrm{int}}), \qquad \frac{C \kappa_{\mathrm{int}}}{L} < \frac{\kappa}{8}. \tag{4.cccxc}$$

The fixed constants include the rate and padding reserves in the transport theorem. Choose M afterwards for those rates and for the covariance construction. Then choose $R > 0$ inside the common background and graph radii. These choices are independent of depth and volume. The nonlinear radius may depend on L.

Fix a real reference coupling θ_0, and set

$$g_0=\theta_0^{-1/2},\qquad \mathfrak{b}_0=A_1(\log\theta_0)^p,\quad p>1/2,\qquad x=\theta_0\tau. \tag{4.cccxci}$$

Choose $\delta>0$ so that $|\tau-1|<8\delta$ lies in the admitted Gaussian disc, and retain $|\tau-1|<\delta$ for the Lipschitz conclusions. The reference gauge weight θ_0, all tame thresholds, and the real cutoff $\chi_{\mathfrak{b}_0}$ stay fixed as x varies. Numerical powers of g_0 and Gaussian determinants are separated below.

4.15.2. *The literal graph and old arguments.* At stage j, define

$$V_*(W)=\mathsf{p}_j(W):=M^jU_{j+1}(W),\qquad W=e^{iw},\quad \|w\|_\infty<R. \tag{4.cccxcii}$$

The output spacing is one; the old input spacing is $1/L$ in these units. Thus p_j is the map p_{j+1}^{lit} in eq. 3.mcccvi. It is not put in its one-step axial gauge. The actual graph about $V=V_*(W)$ is

$$Z_V(u)=e^{i\Phi_V(u)}V,\qquad M(Z_V(u))=W,\qquad \Phi_V(u)|_E=u,\qquad \Lambda_V(u)=i^{-1}\log Z_V(u). \tag{4.cccxciii}$$

All logarithms here are on the admitted branches.

Lemma 4.38 (Admission on the two radii)**.** For R as above,

$$\|\log V_*(W)\|_\infty\le C_pR/L<r/64.$$

There is $C_{\mathrm{dom}}=C_{\mathrm{dom}}(L)$ such that the actual graph and logarithm satisfy

$$\|\Lambda_V(u)-\Lambda_V(0)\|_\infty+\|\Phi_V(u)\|_\infty\le C_{\mathrm{dom}}\|u\|_\infty.$$

Consequently every argument of h in eq. 4.cccxciii stays in its old r-ball for all nested fluctuation boxes used by the formation theorem, provided

$$64C_{\mathrm{dom}}|\gamma|\mathfrak{b}_0<r \tag{4.cccxciv}$$

and the fixed graph ceiling holds. These domains are holomorphic in w,γ. At the real physical value $\gamma=g_0$, the retained regular fibre is precisely this graph on the free box. ■

Proof. The first assertion is the literal mean bound and eq. 4.cccxc. The graph estimate is the joint local pivot estimate in eq. 4.ccxxx, followed by the logarithm of $e^{i\Phi_V(u)}V$. Its constants do not involve j. Increase C_{dom} to include the fixed nested box factors and radial margins of that theorem. The two inequalities leave a fixed margin inside the old r-ball. In particular the same old function is evaluated before and after substitution.

At real W, the background curvature is below the fixed regular ceiling when R is small. The actual three-free-edge pivot anchor eq. 4.cclxviii puts every other point of that regular coarea fibre, with its free values in the box, inside the pivot uniqueness ball. It therefore has the graph value. Conversely the graph remains in the regular coarea set by the same curvature and free-angle bounds. The argument uses its fixed absolute regular ceiling, not an extra history plaquette threshold. The background, graph and logarithm are jointly holomorphic, and every argument of the old functions remains in their declared ball. This proves the stated complex domain directly. □

The contour tests can be removed on the physical box. Set $\varepsilon_0=A_0g_0(\log\theta_0)^p$ with the same retained test convention as the homogeneous step. The relative argument is $T_{y,x}(Z_V(g_0B))T_{y,x}(V)^{-1}$. Internal contours use no middle pivot. The normalized path/mean derivative bound therefore puts this argument within $C_Lg_0\mathfrak{b}_0$ of one, uniformly in the background. Choose $A_0>2C_LA_1$. The tests are then identically one on $\chi_{\mathfrak{b}_0}$, exactly as in the proof of eq. 4.cclxv. The analytic deformation in γ below is defined after this removal. The remaining real integration box does not vary.

4.15.3. *The change to axial gauge.* For a real V, define

$$s_V(x) = T_{y,x}(V) \quad (x \in B(y)), \qquad s_V(y) = 1, \qquad V^{\mathrm{ax}} = V^{s_V}.$$

This is a unitary gauge equal to one at the output nodes. Put

$$G_V(u) = \sum_{x \neq y(x)} \left[1 - \frac{1}{2}\,\mathrm{tr}\{T_{y(x),x}(Z_V(u))T_{y(x),x}(V)^{-1}\}\right]. \tag{4.cccxcv}$$

Lemma 4.39 (Exact axial comparison on the real overlap)**.** Whenever both old arguments are admitted, the literal-centre integral with gauge G_V equals the axial physical step. The free-coordinate change is $(O_V u)(b) = \mathrm{Ad}(s_V(b_-))u(b)$. It preserves the real product colour balls. ■

Proof. The endpoint gauge law gives the exact identities

$$T_{y,x}(V^{s_V}) = 1, \qquad T_{y,x}(Z^{s_V}) = T_{y,x}(Z)T_{y,x}(V)^{-1}. \tag{4.cccxcvi}$$

Thus the absolute contour weight and its tame test at Z^{s_V} are the relative ones at Z, including their numerical normalizer. The actual graph uniqueness and joint gauge identity give

$$\Phi_{V^{\mathrm{ax}}}(O_V u) = O_V \Phi_V(u), \qquad Z_{V^{\mathrm{ax}}}(O_V u) = Z_V(u)^{s_V}. \tag{4.cccxcvii}$$

For real V, O_V is block orthogonal and has determinant one. It preserves the free box, Haar/coarea density and regular plaquette bounds. It is independent of the integration variable u.

Gauge invariance gives $S_j(Z^{s_V}) = S_j(Z)$ and $h_z(Z^{s_V}) = h_z(Z)$ on their admitted overlap. Apply these identities to the complete functions h_z, before local filing. Equations eq. 4.cccxcvi and eq. 4.cccxcvii now identify the two coarea integrals by a real orthogonal change of free coordinates. No old function is evaluated outside its own ball in this comparison. □

For fixed L this overlap contains a neighborhood of $W = 1$. On the full R-ball the literal integral below is defined without evaluating h on the axial representative. Its holomorphy there will not require complex gauge saturation of the old domain.

4.15.4. *The integral and its Gaussian factors.* Let $d\lambda_W$ denote regular coarea disintegration of normalized Haar measure on $M(Z) = W$. Restrict it to the retained free box about $V_*(W)$, and use the contour tests just removed. Define

$$\mathcal{I}_j(W; x, h) = z(\theta_0, \varepsilon_0)^{-n_s} \int_{M(Z)=W}^{\mathrm{reg}} \chi_{\mathfrak{b}_0}(u/g_0)\, e^{-xS_j(Z)+H(Z)-\theta_0 G_V(u)}\, d\lambda_W(Z), \quad V = V_*(W). \tag{4.cccxcviii}$$

The notation for χ means the graph free coordinates of eq. 4.cccxciii. The omitted contour tests equal one on this real support. Here n_s is the number of old sites other than output nodes. This definition uses only the retained regular component. For complex data we use its literal graph pullback displayed below, with the same fixed real integration variables B.

Criticality and nesting give

$$S_j(V_*(W)) = S_{j+1}(W), \qquad D_u S_j(Z_V(u))|_{u=0} = 0. \tag{4.cccxcix}$$

Define the matrices

$$K_A(V) = D_u^2 S_j(Z_V(u))|_0, \quad K_G(V) = D^2 G_V(0), \quad K_\tau(V) = \tau K_A(V) + K_G(V), \quad K(V) = K_1(V). \tag{4.cd}$$

They include the complete inner background and outer constraint-curvature derivatives. The gap and full lifted carrier bounds are those of eq. 4.cclxxxiv and eq. 4.cclxxxv. We do not replace a precision carrier by the distance between its matrix endpoints.

Let $J_V(u)$ be the exact coarea density eq. 4.ccliv. Put

$$R_V^A(u) = S_j(Z_V(u)) - S_j(V) - \tfrac{1}{2}u^{\mathsf{T}} K_A(V) u,$$
$$R_V^G(u) = G_V(u) - \tfrac{1}{2}u^{\mathsf{T}} K_G(V) u,$$

$$P_{\gamma,\tau,h}(V,B) = -\tau\gamma^{-2}R^A_V(\gamma B) - \gamma^{-2}R^G_V(\gamma B) + \log\frac{J_V(\gamma B)}{J_V(0)} + H(Z_V(\gamma B)) - H(V). \tag{4.cdi}$$

The divided remainders have removable values at $\gamma = 0$. The graph, precision and cutoff are independent of γ. Define

$$Z_j(\gamma; V, \tau, h) = \int \chi_{\mathfrak{b}_0}(B) e^{P_{\gamma,\tau,h}(V,B)}\, d\mu_{K_\tau(V)^{-1}}(B). \tag{4.cdii}$$

For complex data the normalized Gaussian uses the determinant branch from positive real data. The local expansion below fixes the zero-free logarithm.

Lemma 4.40 (Exact normalization at the literal centre)**.** Let n_f, n_c count positive old and output bonds, $d = 3(n_f - n_c)$, and $r_{\mathrm{g}} = 3n_s$. At $V = V_*(W)$,

$$\det K_\tau(V) = \tau^{d-r_{\mathrm{g}}} \det K(V). \tag{4.cdiii}$$

For the physical value $\gamma = g_0$, the coarea integral factors exactly as

$$\begin{aligned} \mathcal{I}_j(W; x, h) &= C_{\mathrm{num}}(\theta_0) g_0^d (2\pi)^{d/2} J_V(0) (\det K_\tau(V))^{-1/2} \\ &\quad \times e^{-xS_{j+1}(W)+H(V)} Z_j(g_0; V, \tau, h), \\ C_{\mathrm{num}}(\theta_0) &= \sigma_0^{n_f-n_c} z(\theta_0, \varepsilon_0)^{-n_s}. \end{aligned} \tag{4.cdiv}$$

Here σ_0 is Haar density at zero. All determinant branches start at positive real data and $\tau = 1$. ■

Proof. The graph change of variables has exactly the density eq. 4.ccliv. Rescaling $u = g_0 B$ contributes g_0^d. The fine Haar and coarse delta densities give $\sigma_0^{n_f-n_c}$, as in eq. 4.cclxvi. The action is centered by eq. 4.cccxcix. Subtract its full quadratic part and that of the relative gauge. Gaussian integration with precision K_τ gives eq. 4.cdiv, including the extracted value $H(V)$.

The action Hessian annihilates the tangent to node-fixed gauge orbits, by constrained criticality and gauge invariance. For the relative gauge at a nonaxial V, the orbit derivative of its contour logarithm at an internal site is $-\mathrm{Ad}(T_{y,x}(V))\lambda(x)$. This is an invertible real orthogonal change of the gauge parameter. It gives the same form decomposition as eq. 4.ccxxii, with gauge dimension r_{g}. The complete precision is positive on real data by HAF. Applying Prop. 4.12 after this parameter change proves eq. 4.cdiii. Equivalently conjugate to the axial graph using eq. 4.cccxcvii. Both arguments retain the full constraint curvature in K_A. The identities continue to the admitted complex domain. □

Define

$$N_j(W) = \log\frac{J_{V_*(W)}(0)}{J_1(0)} - \frac{1}{2}\log\frac{\det K(V_*(W))}{\det K(1)}. \tag{4.cdv}$$

It is independent of x, h. Equation eq. 4.cdiii puts the entire remaining power of τ in a scalar factor. At the identity the raw pivot derivative is $L^{-3}I_3$, so $J_1(0) = L^{9n_c}$. Consequently the factor outside the field-dependent terms is

$$C^0_j(\theta_0, x) = C_{\mathrm{num}}(\theta_0) g_0^d (2\pi)^{d/2} J_1(0) (\det K(1))^{-1/2} \tau^{-(d-r_{\mathrm{g}})/2}. \tag{4.cdvi}$$

In particular no logarithm of this factor is included in the intrinsic remainder.

4.15.5. *Admission and filing of the old source.* For an intrinsic construction before composing with $V_*(W)$, use the centered extension

$$s_V(u) = S_j(Z_V(u)) - S_j(V) - DS_j(V)[D\Phi_V(0)u]. \tag{4.cdvii}$$

It is the coarse version of the completion in eq. 4.ccxxxi. It has zero first jet. Its Gaussian Hessian is the full composed Hessian, including the outer constraint curvature. At $V = V_*(W)$, its last term is zero and it equals the action difference in eq. 4.cdi. This extension supplies local functions on an old background ball; it does not alter the physical integral.

Lemma 4.41 (The ordinary source budget)**.** Suppose $\|h\|_{r,\kappa} \le H_*$. On the intrinsic background ball $\|i^{-1}\log V\|_\infty < r/8$, its source in eq. 4.cdi has an actual potential family with budget

$$\|\mathbf{V}^h_\gamma\|_{\kappa_{\rm int}} \le C_{L,M,r} H_* |\gamma| \mathfrak{b}_0. \tag{4.cdviii}$$

The input functions and their arguments are the ones in eq. 4.ccclxxxvi; no extension of their domain is used. ■

Proof. For each representative use exactly

$$V^h_{z,X}(V,B) = h_{z,X}(\Lambda_V(\gamma B)) - h_{z,X}(\Lambda_V(0)). \tag{4.cdix}$$

On the nested cutoff boxes the path $\zeta \mapsto h_{z,X}(\Lambda_V(\zeta\gamma B))$ is admitted for $|\zeta| < cr/(C_{\rm dom}|\gamma|\mathfrak{b}_0)$. Its supremum is bounded by $\|h_{z,X}\|_r$. One-variable Cauchy, with the constant value subtracted, gives

$$|V^h_{z,X}| \le C \frac{C_{\rm dom}|\gamma|\mathfrak{b}_0}{r} \|h_{z,X}\|_r.$$

This bound treats all the coordinates of X at once.

The local substitution reads only the actual double-block enlargement of X. After changing to output units, its carrier has at most $C(1 + d_M(X)/L)$ output M-cube edges. Cover the scaled old spanning curve and its fixed output-size collar, as in Lem. 4.32. Thus the requested rate cost fits the old one by eq. 4.cccxc.

We also convert the pointed norm to an incidence norm before applying HAF. If an input carrier contains a bond in cube Q, its size is at least the cube distance from $q(z)$ to Q. Reserve part of the old rate and sum the anchors:

$$\sup_Q \sum_{z, X \ni Q} e^{\kappa d_M(X)/2} \|h_{z,X}\|_r \le C_{\kappa,M} H_*.$$

Indeed each old cube contains $O(M^4)$ unit anchors and the remaining sum is a convergent metric exponential. This factor affects the source budget, not its analytic domain. Combine all origins with the same carrier before estimating its potential. The source has no additional Gaussian coordinates and introduces no number of potential types proportional to the volume. These estimates prove eq. 4.cdviii. □

Choose a fixed ceiling $4H_* + 1$ for the norm of h. The actual interaction budgets of HAF and the preceding lemma are small on a disc

$$|\gamma| < c_{H_*}/\mathfrak{b}_0. \tag{4.cdx}$$

The positive constant c_{H_*} is below the graph and old-source ceilings and the UM potential ceiling. It can depend on H_*, L, M, r. The Gaussian structure and its domains were chosen before H_*; only this coupling ceiling depends on that norm. Increasing the reference coupling floor makes g_0 lie in one half of this disc.

4.15.6. *Formation on the larger output ball.*

Lemma 4.42 (A local family on the two radii)**.** The normalized integral eq. 4.cdii, with the centered extension for intrinsic filing, is zero free. Its logarithm is a holomorphic family on eq. 4.cdx, the larger τ-disc, and the $4H_* + 1$ input ball. After composing with $V_*(W)$, it has a local family on $\|w\|_\infty < R$, with any fixed output rate chosen above. The constants are independent of j and volume. ■

Proof. Use the actual covariance construction eq. 4.ccxciv. It truncates full lifted precision carriers, including external background inputs, before projection. Held root guards contain the entire support of each graph, potential and required data collar. The matrix dimension of a local functional remains at most $C(LM)^4 \# Z$. The old source changes its potential but not this Gaussian structure.

The action, relative gauge and coarea budgets are those proved in HAF. Equation eq. 4.cdviii supplies the ordinary input budget on its actual domain. The UM, Mayer and connected cluster proof therefore gives the intrinsic analytic family on eq. 4.cdx. This is the existing proof applied with the specified source, rather than an assumption on its desired output. It fixes a logarithm by its convergent local series. No whole-volume lower bound for a cutoff probability is used.

The intrinsic carrier is on output localization cubes, of side LM in old input units. Group its external background coordinates as one Banach vector per such cube. Its image in output units has side M. Use the Banach-valued proof for all old bond midpoints in this final cube: the target is their supremum norm, the pure mean row bound holds for all of them, and only the same fixed neighboring macroblocks are read. Their number does not enter that proof. The literal whole-cube family has norm C_pR/L, by Prop. 4.28. Its ratio to the intrinsic background radius $r/8$ is at most $8C_p/(Lt)$. The block Cauchy argument of Lem. 4.32, retaining all degrees, therefore gives a charge $\exp(CC_p|X|/(Lt))$. The intrinsic rate reserve pays it by eq. 4.cccxc. This counts cube groups, not their $(LM)^4$ individual background coordinates. The bounded root joins and rate reserve give a bounded composition operator on the local family norm.

The map $V_*(W)$ is independent of γ, τ, h, so this composition preserves their holomorphy and any norm-valued derivative estimate. On physical data the centered extension equals the action difference by eq. 4.cccxcix. Thus the resulting family is the logarithm of the literal integral already defined, after its stated Gaussian factors have been removed. □

4.15.7. *Pointing the old source.* Write $\langle A\rangle_{\gamma,V,\tau,h}$ for the normalized expectation obtained from eq. 4.cdii by inserting A. For one old anchor set

$$\Delta h_{z,V,\gamma}(B) = h_z(Z_V(\gamma B)) - h_z(V). \tag{4.cdxi}$$

This definition precedes every local representation of h_z. Let $\mathscr{F}^0_{j,y}(W;\gamma,\tau)$ be the HPT pointing with $h = 0$, composed with the literal centre using complete gauge invariance on the real overlap. Extend it to the R-ball by the literal formula. Define

$$\begin{aligned}\mathscr{F}^{\mathrm{raw}}_{j,y}(W;\gamma,\tau,h) &= \mathscr{F}^0_{j,y}(W;\gamma,\tau)\\ &\quad + \sum_{z\ \mathrm{old}} \theta_y(z/L)\int_0^1 \langle \Delta h_{z,V,\gamma}\rangle_{\gamma,V,\tau,\ell h}\, d\ell, \qquad V = V_*(W).\end{aligned} \tag{4.cdxii}$$

The tents use output units. The sum over y is $\log Z_j(\gamma;V,\tau,h)$: differentiate that logarithm along ℓh and use the partition of unity of the tents.

Lemma 4.43 (A marked source family)**.** The pointings eq. 4.cdxii have local analytic representations on the R-ball. Their complete functions are gauge invariant and hypercubic covariant. They are even in γ. The part depending on h vanishes at $\gamma = 0$. ■

Proof. Introduce independent ordinary source parameters ζ_z, multiplying $\Delta h_{z,V,\gamma}$. The incidence estimate in Lem. 4.41 is uniform on a fixed product disc in these parameters: their number adds no Gaussian coordinates or potential types. Differentiate the convergent connected cluster expansion in ζ_z. Each resulting object contains one old-source insertion rooted at z, with its entire original carrier and the graph collar retained. The marked vertex has budget $C|\gamma|\mathfrak{b}_0$ times the old rooted norm. Sum its unmarked branches by the same convergent cluster bound. This constructs the local family of the expectation in eq. 4.cdxii without a whole-volume factor. The old origins are combined on their assigned carriers before forming the local Gaussian functionals. Root transfer by $\theta_y(z/L)$ costs only fixed L, M constants.

Under a small real site gauge the actual graph changes by the same linear adjoint map in its free variables as in eq. 4.ccc. Both $Z_V(\gamma B)$ and V remain in the old ball. Gauge invariance of the complete h_z therefore preserves eq. 4.cdxi. The Gaussian, cutoff and coarea change by the corresponding real orthogonal coordinates. The normalized expectation is unchanged. The same

is true for the $h = 0$ pointing, so each complete function in eq. 4.cdxii is invariant. At the output, orbit covariance of the actual background induces exactly such an old site gauge between the two centres. The argument uses complete functions before local representatives are composed. Joint holomorphy extends the identity on the admitted complex domain.

Lattice covariance follows from Prop. 4.25. The middle-pivot set and real colour balls in that proposition are the ones fixed here. The old insertions are covariant by the input assumption, and their tents transform with their anchors. Thus this assertion concerns the actual integral, not a subsequent average of pointings.

Finally every occurrence of the group field has argument γB. Hence the exact transformation $(\gamma, B) \mapsto (-\gamma, -B)$ preserves the exponent and all insertions, including their Taylor subtractions. The centered Gaussian and fixed cutoff are even. The HAF filing commutes with this transformation; each local Gaussian integration and connected extraction therefore gives an even function of γ, as in FEC. At zero coupling $\Delta h_{z,V,0} = 0$ exactly. The normalized empty Gaussian selection has zero derivative in an old source. These facts prove the last assertion at the level of the marked family. □

Lemma 4.44 (Even-coupling bound on the output family)**.** On a fixed smaller τ-disc, $\|h\|_{r,\kappa} \le 4H_* + 1$, and $|\gamma| \le c_{H_*}/(2\mathfrak{b}_0)$, the pointed logarithm satisfies

$$\|\mathscr{F}_j^{\mathrm{raw}}(\gamma, \tau, h)\|_{R,\kappa} \le C_{H_*}\{|\gamma|^2\mathfrak{b}_0^2 + e^{-c\mathfrak{b}_0^2}\}. \tag{4.cdxiii}$$

The constants are uniform in depth and volume. ■

Proof. The original HAF estimate and Lem. 4.41 give the norm bound $C_{H_*}|\gamma|\mathfrak{b}_0 + Ce^{-c\mathfrak{b}_0^2}$ on eq. 4.cdx, including the marked family of Lem. 4.43. The composition on the R-ball is a bounded operator on this norm by Lem. 4.42. At zero coupling the logarithm is the cutoff wall contribution $\log \mu_{K_\tau(V)^{-1}}(\chi_{\mathfrak{b}_0})$. Its pointed norm is bounded by $Ce^{-c\mathfrak{b}_0^2}$, and it is independent of h.

Subtract this zero-coupling value. The remaining activity-valued function is analytic and even, with a double zero at the origin. The norm-valued Cauchy argument of Prop. 4.21 on a fixed inner circle of eq. 4.cdx gives $C_{H_*}|\gamma|^2\mathfrak{b}_0^2$. All these bounds are for a rooted local family; no extensive error is differentiated. Adding the cutoff wall term proves eq. 4.cdxiii. □

4.15.8. *Companions and the quotient norm.* In eq. 4.ccclxxxviii, the difference representations retain their lifted carriers and finite-period-tail labels, as in the construction following eq. 4.ccclxxx. The infimum is over compatible representations with those labels. The formula compares complete functions; it does not discard the labels when an infimum is approached.

Lemma 4.45 (Preservation of the period norm)**.** Fix $\gamma_{\mathrm{per}} > 0$ in cube units. The rate choices and then L can be made so that the bound eq. 4.cdxiii also holds in the full norm $\|\cdot\|_{R,\kappa,\gamma_{\mathrm{per}}}$ for input norm at most $4H_* + 1$. The same exponent is retained at every step. ■

Proof. The old shortest period is LP when the output period is P. Decompose the old input as

$$h^{LP} = \Pi_{LP}h^{\infty} + w^{LP}.$$

Retain the entire old carriers and the finite-period-tail labels of w^{LP} in the potential eq. 4.cdix. The source incidence estimate is linear in its norm and gives its extra factor $e^{-\gamma_{\mathrm{per}}LP/M}$. Marking this source in the connected expansion retains that factor. The normalized empty selection has zero old-source derivative, and parity makes the resulting difference even in γ with value zero at the origin. Cauchy on eq. 4.cdx therefore supplies its $|\gamma|^2\mathfrak{b}_0^2$ factor as well.

For the remaining difference between the finite integral and its companion, compare the full lifted operator, Gaussian and cluster objects before projection, as in Lem. 4.33. The precision has the $e^{t_0\ell(\Gamma)}$ bound eq. 4.cclxxxv. The near cutoff is by its whole lifted carrier. Finite auxiliary parameters and held guards are lifted periodically with image parameters identified. Include all guards

and data collars in the common-object support. Nonwrapping objects therefore have identical coefficients. Every unmatched primitive Gaussian object or connected cluster pays a fixed cube-unit period exponential. Request an intrinsic rate large enough to make that exponent larger than the prescribed γ_{per}.

These estimates hold uniformly on the auxiliary coupling disc. At zero coupling the only wrapping terms are cutoff wall terms, with their $e^{-c\mathfrak{b}_0^2}$ factor. The nonconstant difference in γ is even and has a double zero. The same norm-valued Cauchy argument therefore supplies $|\gamma|^2\mathfrak{b}_0^2$ with the period factor retained.

In refiling the old source tail, its period LP leaves an exponential reserve for fixed carrier and guard padding at output period P. Choose L so that the remaining exponent is at least γ_{per}. All exponents are in M-cube units; M cancels from this choice. The same estimates apply to the vacuum value at $W = 1$. This proves the full-norm assertion. □

Define the vacuum-subtracted pointings at the physical coupling by

$$F_{j,y}(W; x, h) = \mathscr{F}^{\mathrm{raw}}_{j,y}(W; g_0, \tau, h) - \mathscr{F}^{\mathrm{raw}}_{j,y}(1; g_0, \tau, h), \qquad \mathbf{F}_j(x, h) = (F_{j,y})_y. \tag{4.cdxiv}$$

Their sum is the field-dependent remainder. The subtracted total is part of a scalar vacuum factor.

The definition eq. 4.cdxii depends only on the complete h_z. Two local representations of the same pointings give the same $\mathbf{F}_j$ in the quotient. All estimates above can be applied to representatives whose norms approach the input quotient norm. On any finite-dimensional affine subspace of inputs, choose local representatives for the base and directions. The cluster construction is holomorphic in their coefficients. Uniform local bounds, Cauchy estimates and polarization give a norm-convergent Taylor series on a smaller input ball. Thus $\mathbf{F}_j$ is a holomorphic map of the quotient Banach spaces, including their period component.

4.15.9. *The formation estimate and its derivatives.*

Proposition 4.46 (Formation with a fixed radius buffer)**.** Choose the rates, $t, L, M, R, r = tR$ as above. Fix $H_* < \infty$. There is a coupling floor such that for every reference θ_0 above that floor, the literal regular step is defined on $\|w\|_\infty < R$, $|x/\theta_0 - 1| < \delta$, and $\|h\|_{r,\kappa,\gamma_{\mathrm{per}}} \le H_*$. Its nonconstant Gaussian term is N_j from eq. 4.cdv, independent of x, h. The map $\mathbf{F}_j$ is holomorphic on fixed larger relative coupling and profile domains and satisfies

$$\begin{aligned} \|\mathbf{F}_j(x,h)\|_{R,\kappa,\gamma_{\mathrm{per}}} &\le C_{H_*}\epsilon(\theta_0), \\ \|D_h\mathbf{F}_j(x,h)[a]\|_{R,\kappa,\gamma_{\mathrm{per}}} &\le C_{H_*}\epsilon(\theta_0)\|a\|_{r,\kappa,\gamma_{\mathrm{per}}}, \\ \|\partial_x\mathbf{F}_j(x,h)\|_{R,\kappa,\gamma_{\mathrm{per}}} &\le C_{H_*}\epsilon(\theta_0)/\theta_0, \qquad \epsilon(\theta_0) = \theta_0^{-1}(\log\theta_0)^{2p}. \end{aligned} \tag{4.cdxv}$$

Its complete pointings are gauge invariant, translation covariant and hypercubic covariant, and are normalized at the flat field. For real data the map is real. The constants are independent of depth and compatible volume. ■

Proof. At $\gamma = g_0$, eq. 4.cdxiii and Lem. 4.45 give the first bound, including the vacuum subtraction. The cutoff tail is absorbed in $\epsilon(\theta_0)$ after increasing the floor, because $2p > 1$. No cutoff varies in this estimate.

The same bound holds on $\|h\| < 4H_* + 1$ and the larger fixed τ-disc. The quotient construction permits Cauchy in any input direction on a fixed norm margin. It gives the h derivative with its direction norm retained. Cauchy on the larger τ-disc gives $C_{H_*}\epsilon(\theta_0)$ for $\partial_\tau\mathbf{F}_j$. Since $\tau = x/\theta_0$, $\partial_x = \theta_0^{-1}\partial_\tau$. This proves all three estimates. Neither an indicator nor the input h is rescaled by x.

Gauge invariance and lattice covariance are Lem. 4.43; the cutoff symmetry is Prop. 4.25. Their definitions on complete pointings make the identities independent of the chosen local representatives. For real data the integral is positive and its coarea density has the identity sign. Its logarithm and source interpolation are real on the selected branch.

Finally eq. 4.cdiv and eq. 4.cdiii give N_j with no remaining field-dependent τ power. The scalar factors are outside the remainder. Every domain and norm provider is uniform in j and volume, as required. □

For two inputs on the retained margins, integrating the derivative bounds gives

$$\|\mathbf{F}_j(x,h) - \mathbf{F}_j(\widetilde{x},\widetilde{h})\| \le C_{H_*}\epsilon(\theta_0)\left\{|x-\widetilde{x}|/\theta_0 + \|h-\widetilde{h}\|_{r,\kappa,\gamma_{\mathrm{per}}}\right\}. \tag{4.cdxvi}$$

The norm on the left is the output norm in eq. 4.cdxv. It controls the local interaction rather than an extensive logarithm.

4.15.10. *Transport on the same field ball.* For $i<k$, let $\mathsf{p}_{i,k}(W)$ be the literal partial mean of the actual orbit at depth k, expressed on the old unit lattice. Put $\xi = L^{-(k-i)}$, and define the linear transport of complete pointings by

$$(T_{i,k}h)_y(W) = \sum_z \theta_y(\xi z)\, h_z(\mathsf{p}_{i,k}(W)). \tag{4.cdxvii}$$

The sum means the periodized sum on a torus. Its lifted carriers are retained in the norm.

Corollary 4.47 (A common field ball for transport)**.** Choose $0<\beta<\alpha<1$. After increasing the fixed L, the rate and radius choices above give

$$\|T_{i,k}h\|_{r,\kappa,\gamma_{\mathrm{per}}} \le C_T L^{-\beta(k-i)}\|h\|_{r,\kappa,\gamma_{\mathrm{per}}}, \qquad i<k. \tag{4.cdxviii}$$

The constants are uniform in the depths and compatible periods. The transports obey $T_{j,k}T_{i,j} = T_{i,k}$. ■□

Proof. Apply Prop. 4.36 with its large test ball equal to R, its input radius equal to $r = tR$, and its output radius equal to $tR = r$. The literal intermediate map has size $C_{\mathsf{p}}\xi R$ on the large test ball, by Prop. 3.187 and the actual intermediate-field construction in the transport proof. For every proper age, $\xi \le L^{-1}$. The inequalities eq. 4.cccxc therefore place the entire intermediate vector family inside the old r-ball. They also pay the big-composition Cauchy and carrier-rate costs. Thus no shrink is iterated with $k-i$. The Ward–Möbius estimate and old-anchor count give eq. 4.cdxviii. The infimum over local representations passes through this linear bound.

The period estimate uses $P_{\mathrm{old}} = P_{\mathrm{new}}/\xi$. The dimensionless retained exponent from eq. 4.ccclxxx is multiplied by ξ^{-1} in output cube units. The large-L choice retains γ_{per}, including the carrier padding reserve. It is uniform for every proper age.

For the semigroup identity, actual nesting identifies successive literal partial means up to a gauge on the old lattice. Apply complete gauge invariance before choosing local representatives. For integer ξ^{-1}, the nodal hat identity is

$$\sum_z \theta_y(\xi z)\theta_z(s) = \theta_y(\xi s). \tag{4.cdxix}$$

Indeed both sides are multilinear on each old unit cube, and they agree at its vertices; the knots of the left-hand coarse hat are aligned with old vertices. Periodization preserves the identity. Substitution in eq. 4.cdxvii now proves the semigroup law. It is an identity of complete functions, not of filed pieces. □

4.15.11. *The exact physical step.* Let $e_{j+1,y}$ be the action pointing from the actual fine-plaquette hats. It has sum S_{j+1}, companion coefficient one and a uniform norm on the fixed field ball. For a compatible family a, write $b[a]$ for the scalar quadratic coefficient of its infinite-volume companion, defined by its Ward kernel. This is a bounded linear functional in the norms used here: Cauchy controls the quadratic jet, and the reserved spatial rate controls its second moment.

Choose the complete Gaussian pointings $\mathbf{N}_j$ with sum N_j as in the homogeneous Gaussian construction. Define

$$b_j = b[\mathbf{N}_j], \qquad\qquad n_j = \mathbf{N}_j - b_j e_{j+1},$$

$$r_j(x,h) = b[\mathbf{F}_j(x,h)], \qquad f_j(x,h) = \mathbf{F}_j(x,h) - r_j(x,h)e_{j+1}. \tag{4.cdxx}$$

All pointings in these differences are normalized at the flat field. The infinite-volume scalar coefficient of n_j, f_j is zero. The norm of n_j is bounded uniformly in j, and b_j, n_j are independent of x, h. For r_j, f_j, the three bounds eq. 4.cdxv hold with a changed fixed constant, both on the larger output ball and after restriction to the working ball.

Corollary 4.48 (Closure of one regular physical step)**.** On the domains of Prop. 4.46, there is a field-independent, holomorphic nonzero factor $C_j(x,h)$ such that

$$\mathcal{I}_j(W;x,h) = C_j(x,h)\exp\left\{-x_+ S_{j+1}(W) + \sum_y (h_+)_y(W)\right\}, \qquad \begin{cases} x_+ = x - b_j - r_j(x,h), \\ h_+ = T_{j,j+1}h + n_j + f_j(x,h). \end{cases} \tag{4.cdxxi}$$

The new profile belongs to the same quotient space on radius r, with the same spatial and period exponents. Its norm satisfies

$$\|h_+\|_{r,\kappa,\gamma_{\mathrm{per}}} \le C_T L^{-\beta}\|h\|_{r,\kappa,\gamma_{\mathrm{per}}} + C_N + C_{H_*}\epsilon(\theta_0). \tag{4.cdxxii}$$

■□

Proof. By the partition of unity for the hats,

$$H(\mathsf{p}_j(W)) = \sum_y (T_{j,j+1}h)_y(W).$$

The exact factorization eq. 4.cdiv, eq. 4.cdv and eq. 4.cdxiv therefore give

$$\mathcal{I}_j(W;x,h) = C_j(x,h)\exp\{-xS_{j+1}(W) + H(\mathsf{p}_j(W)) + N_j(W) + \sum_y F_{j,y}(W;x,h)\}.$$

Here C_j is the scalar eq. 4.cdvi times $\exp\{\sum_y \mathscr{F}^{\mathrm{raw}}_{j,y}(1;g_0,\tau,h)\}$. The zero-free normalized integral makes this factor nonzero on the retained domains. For real data it is positive.

Insert eq. 4.cdxx and $\sum_y e_{j+1,y} = S_{j+1}$. This proves eq. 4.cdxxi with its stated signs. Gauge invariance and hypercubic covariance hold for every complete term. The exact action counterterm and the Ward readout give zero companion scalar coefficient for h_+. The period estimates and the common-radius transport give membership and eq. 4.cdxxii. □

The integral in this result is the regular step with action xS_j, the literal Haar and coarea densities, the relative contour gauge and the fixed free-bond colour-ball cutoff. Its old input is a complete coarse-field interaction h. The result supplies a fixed-radius producer for this model. It makes no identification with an arbitrary transported interaction on independent fine-current cores or a conditional history measure.

4.15.12. *A derivative of constant diagonal weights.* Let $a = (a_1, \dots, a_4)$ range over a fixed complex polydisc about 1, and set

$$S^a_j(V) = \mathscr{A}^a_{h_j}(U^a_j(V)), \qquad \mathsf{p}^a_j(W) = M^j U^a_{j+1}(W). \tag{4.cdxxiii}$$

The action, actual orbit and literal representative are those of Prop. 5.47. The mean, middle pivots, relative contour gauge and frozen free cutoff are unchanged. For fixed a, the required lattice symmetries are translations and independent coordinate reflections. Axis permutations act jointly on a and the fields.

Write $\mathfrak{H}^{\mathrm{ref}}_{r,\kappa,\gamma_{\mathrm{per}}}$ for the quotient space with gauge invariance, these reflection and translation identities, flat-field normalization, and zero coefficient in each of the six curvature planes. Its norm is eq. 4.ccclxxxviii. At $a = 1$, the hypercubic subspace with zero scalar coefficient is contained in this space. The six readouts are the $b_{\mu\nu}$ of § 5.15. Independent reflections exclude cross-plane quadratic curvature terms, so these readouts exhaust the quadratic curvature part. They are continuous by the same Cauchy and Ward moment bounds used above.

Put

$$K_\tau^a(V) = \tau D_u^2 S_j^a(Z_V(u))|_0 + D^2 G_V(0).$$

Define $\mathbf{F}_j^a$ by eq. 4.cdxiv with the weighted action and centre. Define N_j^a by eq. 4.cdv with K_1^a, p_j^a, and use its complete Gaussian pointings $\mathbf{N}_j^a$ as above.

Proposition 4.49 (The first weight jet of formation)**.** On a smaller fixed weight polydisc, the literal formation construction is jointly holomorphic in a, x, h, with input $\|h\|_{\mathfrak{H}^{\mathrm{ref}}} \le H_*$. Its formation $\mathbf{F}_j^a(x,h)$ satisfies the three bounds eq. 4.cdxv uniformly in a, and also

$$\|D_a \mathbf{F}_j^a(x,h)[e]\|_{R,\kappa,\gamma_{\mathrm{per}}} \le C_{H_*}\epsilon(\theta_0)|e|. \tag{4.cdxxiv}$$

The nonconstant Gaussian $\mathbf{N}_j^a$ is independent of x, h. Its norm and first weight derivative are uniformly bounded. Covariance under axis permutations is a joint identity in a, h; full hypercubic covariance at $a = 1$ is recovered for hypercubic h. ∎

Proof. We first verify the providers needed uniformly on a weight disc. The principal change in the actual joint equation is the constant local operator $K_a - K_1$ of eq. 5.ccxci. The exponentially weighted $\mathcal{W}_p$ joint inverse in Lem. 5.46 therefore gives an a-dependent inverse by a convergent Neumann series on a fixed weight disc. Each factor has a fixed physical carrier; retain the connected chain of those carriers in the series. Its exponentially weighted cell sum is uniform on a larger weight disc.

Smooth unit-cell localization is bounded on $\mathcal{W}_p$ by the product rule through two differences. The remaining actual nonlinear vertices have arity at least two and uniform Banach Cauchy bounds, by eq. 5.ccxciii and the unchanged constraint recursion. The ordered-tree proof of Prop. 4.17 thus applies with this inverse. Its scalar majorant and its connected-tree length charge are uniform in a. This is an all-arity carrier bound, not just a separate bound for each fixed jet. Multiplying each original fine-plaquette density by $a_\mu a_\nu$ preserves its h_j^4 bound and the cancellation of the fine plaquette count. Consequently the full lifted precision family, including every background leg and winding label, has the uniform bound required in eq. 4.cclxxxv.

The actual free form has the lower and upper bounds eq. 5.ccxcvii. The unchanged contour gauge gives the complementary gauge gap. Equivalently use the relative-form perturbation argument in Lem. 5.49 for the physical Gaussian. The full composed constraint curvature is retained. On a smaller complex product domain the normalized inverse is a Neumann continuation of the real gap. The HAF near/tail construction therefore works uniformly, always cutting by the whole lifted carrier before projection.

The free inverse and its local L^p upgrade have constants for the odd mesh parameter N, independent of its factorization $N = L^j$. Their linear amplitude is therefore uniform in L. The nonlinear radius may shrink with L. Use this amplitude in eq. 4.cccxc when choosing L, and decrease the common R afterward. The HFM plain-row bound then gives $\|\mathsf{p}_j^a(W)\| \le CR/L$ uniformly on the larger weight disc. This verifies the literal old-source admission, rather than assuming it for the deformed orbit.

All algebraic identities for the relative gauge, the joint graph, Haar and coarea densities are independent of the action. Weighted criticality and nesting give eq. 4.cccxcix with S_j^a, p_j^a. The gauge-null Hessian and relative gauge rank give the same scalar power $\tau^{d-r_{\mathrm{g}}}$. Thus the Gaussian defined with K_1^a, after vacuum subtraction, remains independent of x, h.

The old-source budget uses analytic norms and gauge invariance, not axis permutations or scalar isotropy. It holds on $\mathfrak{H}^{\mathrm{ref}}$. The intrinsic exponent remains invariant under $(\gamma, B) \mapsto (-\gamma, -B)$. All activity, source-rooting and period estimates above are uniform on the larger weight disc, giving eq. 4.cdxv. Cauchy in the fixed weight margin gives eq. 4.cdxxiv. It also bounds the first weight jet of $\mathbf{N}_j^a$.

Finally the middle-pivot and colour-ball identity of Prop. 4.25 commutes with reflections and with simultaneous permutation of a. The actual weighted orbit has that covariance by uniqueness. Source interpolation before filing gives the stated covariance of the complete pointings. No invariance of an individual filed term is used. □

For a fixed old input h in $\mathfrak{H}^{\mathrm{ref}}$, define $T_j^a h$ by eq. 4.cdxvii with $\xi = L^{-1}$ and p_j^a. The big-composition estimate already gives a uniform norm on the larger weight disc: the whole old-coordinate family is $CR/L < r/64$; the coarsened old carrier rate pays the output rate; and the one-step old-anchor count is the fixed L^4. The constant may depend on this fixed L. Keeping the old period labels uses $P_{\mathrm{old}} = LP$ exactly as above. Hence Cauchy in a gives

$$\left\| D_a T_j^a h[e]\Big|_{a=1} \right\|_{r,\kappa,\gamma_{\mathrm{per}}} \le C_L|e|\,\|h\|_{r,\kappa,\gamma_{\mathrm{per}}}. \tag{4.cdxxv}$$

Every plane readout of $T_j^a h$ is zero on this weight domain: the actual affine response eq. 5.ccxcvi is independent of a modulo an actual gradient. The complete first jet vanishes, and Ward annihilates the gradient in the quadratic jet. Differentiating this exact identity proves that the profile in eq. 4.cdxxv also has six zero readouts. It is a bounded profile; no power of g_0 is claimed for it.

At the isotropic point, take $\sum_\mu e_\mu = 0$. Let $e^{\mathrm{str}}_{j+1,e}$ be the complete fine-action pointing

$$(e^{\mathrm{str}}_{j+1,e})_y(W) = \sum_{p\|\mu\nu} (e_\mu + e_\nu)\theta_y(x_p)\, a_p(U_{j+1}(W)). \tag{4.cdxxvi}$$

Its six coefficients are $e_\mu + e_\nu$, and its sum is $D_a S^a_{j+1}(W)[e]\Big|_{a=1}$. Indeed the derivative of the orbit makes no contribution to the total first variation at fixed mean, by actual constrained criticality. The coefficient claim follows from the exact fine-hat quadrature and the affine response.

Use the transfer index j and put $b^a_{j,\mu\nu} = b_{\mu\nu}[\mathbf{N}^a_j]$. This is the increment from depth j to $j+1$, indexed $j+1$ in the cumulative sum in the Gaussian comparison. Joint permutation covariance gives numbers u_j, v_j such that $\partial_{a_\lambda} b^a_{j,\mu\nu}|_1 = u_j$ for $\lambda \in \{\mu,\nu\}$ and equals v_j otherwise. Put $m_j = u_j - v_j$. Then

$$D_a \mathbf{N}^a_j[e]\Big|_{a=1} = m_j e^{\mathrm{str}}_{j+1,e} + q_{j,e}, \qquad \|q_{j,e}\|_{r,\kappa,\gamma_{\mathrm{per}}} \le C|e|, \qquad b_{\mu\nu}[q_{j,e}] = 0. \tag{4.cdxxvii}$$

The normalization of this Gaussian is the canonical contour and coarea normalization of Prop. 5.50. That proposition identifies

$$m_j = \frac{11}{3\pi^2}\log L + O_\sigma(L^{-\sigma j}) \qquad (0<\sigma<1).$$

The uniform norm in eq. 4.cdxxvii follows from weight Cauchy and the uniform action-pointing norm. The six-plane readouts of its difference vanish by the definition of m_j. Formation and its first weight jet can likewise be projected onto the six readouts and their zero-readout complement, retaining the $C_{H_*}\epsilon(\theta_0)$ bounds.

These statements differentiate a constant diagonal deformation on its actual common domain. They do not assert closure of a finite anisotropic trajectory or admit a spatially varying weight.

4.16. **The first weight derivative with a frozen centre.** The weighted formation step moves its critical centre. A stress insertion in a fixed regular kernel does not. We compare their first derivatives on the actual fibre. A node-fixed gauge aligns the relative contour factors exactly. The remaining difference is a surface source on the fixed free colour balls. Its Gaussian tail is small in the rooted local norm, including normalization and the finite-period companions.

This is a first-derivative result. A fixed-centre weighted kernel need not have the uniform small formation disc of the moving-centre construction: its off-centre linear force can have size $O((a-1)/g_0)$. No such disc is asserted here.

4.16.1. *The two actual integrals.* Use the homogeneous regular model and parameter choices of Prop. 4.46. In particular

$$g_0 = \theta_0^{-1/2}, \qquad b_0 = \mathfrak{b}_0 = A_1(\log\theta_0)^p, \quad p > \tfrac{1}{2}, \qquad x = \theta_0\tau.$$

The reference θ_0, colour-ball radius, middle pivots and all tame thresholds are fixed. The old input satisfies $\|h\| \le H_*$ in the compatible quotient norm of eq. 4.ccclxxxviii. For the analytic estimates retain the larger bounded h ball and τ disc used in the formation proof. All old interactions below are complete gauge invariant functions on their original field ball.

Fix real output data W first, and put

$$V_a = \mathsf{p}_j^a(W), \qquad V_0 = V_1, \qquad Z_0(B) = Z_{V_0}(g_0 B).$$

The index zero on V_0 means the frozen isotropic centre. It is not the depth index. Let $\mathcal{I}^{\mathrm{mov}}(a)$ be the actual weighted LCF integral with centre V_a, action S_j^a, relative contour factor G_{V_a}, and the product colour balls in its own free coordinates. Define the frozen-centre integral on the same Haar-disintegrated fibre by

$$\mathcal{I}^{\mathrm{fr}}(a) = z(\theta_0,\varepsilon_0)^{-n_s}\int_{M(Z)=W}^{\mathrm{reg}} \chi_{b_0}(B_0(Z))e^{-xS_j^a(Z)+H(Z)-\theta_0 G_{V_0}(Z)}\,d\lambda_W(Z), \qquad Z = Z_0(B_0(Z)). \tag{4.cdxxviii}$$

Here the old lattice spacing is one and the raw free angle is g_0B_0. The measure and graph in eq. 4.cdxxviii are frozen. The regular/tame tests have their strict admission margin and equal one on the retained support. For a real weight near one both charts lie in this same regular component. The size of that real weight interval may depend on g_0b_0. Only its derivative at one will be used.

At $a = 1$, write their common value as $\mathcal{I}_0$ and the corresponding normalized expectation as $\langle\cdot\rangle_0$. The envelope identity for the actual critical action gives

$$D_a \log\mathcal{I}^{\mathrm{fr}}(1)[e] = -x\langle E_{j,e}\rangle_0, \qquad E_{j,e}(Z) = D_aS_j^a(Z)[e]\Big|_1 = \sum_z E_{j,e,z}(Z). \tag{4.cdxxix}$$

The pointings on the right are the original fine-plaquette stress pointings, evaluated on the isotropic actual orbit. The derivative of that orbit pairs to zero by constrained criticality at fixed literal mean.

4.16.2. *Exact contour alignment and the surface velocity.* At an old site x in the output block rooted at y, write $T_a(x) = T_{y,x}(V_a)$, $T_0(x) = T_{y,x}(V_0)$. Define

$$s_a(x) = T_0(x)^{-1}T_a(x), \qquad \widetilde{V}_a = V_a^{s_a}, \qquad C_a(b) = i^{-1}\log(\widetilde{V}_a(b)V_0(b)^{-1}). \tag{4.cdxxx}$$

Then $s_a(y) = 1$, $C_1 = 0$, and $T_{y,x}(\widetilde{V}_a) = T_0(x)$. Put $c_{e,b} = D_aC_a(b)|_1[e]$. The logarithm in this definition is only needed near its zero at $a = 1$.

Lemma 4.50 (The exact frozen-coordinate cutoff)**.** For real admitted arguments, the change $Z \mapsto Z^{s_a}$ identifies every factor in $\mathcal{I}^{\mathrm{mov}}(a)$ with its frozen-centre counterpart except its free colour-ball indicator. In frozen coordinates that indicator is

$$\prod_{b\in\mathcal{F}} \mathbf{1}_{\{|F_{a,b}(B)|\le b_0\}}, \qquad F_{a,b}(B) = \frac{1}{g_0 i}\log(e^{ig_0B_b}e^{-iC_a(b)}). \tag{4.cdxxxi}$$

Here $\mathcal{F}$ is the actual set of free middle-pivot coordinates. At $|B_b| = b_0$, with $n_b = B_b/b_0$,

$$n_b \cdot D_aF_{a,b}(B)|_1[e] = -g_0^{-1}n_b \cdot c_{e,b}. \tag{4.cdxxxii}$$

■

Proof. The endpoint law for a node-fixed old gauge is $T_{y,x}(Z^s) = T_{y,x}(Z)s(x)^{-1}$. Consequently

$$T_{y,x}(Z^{s_a})T_0(x)^{-1} = T_{y,x}(Z)T_a(x)^{-1}.$$

This proves $G_{V_0}(Z^{s_a}) = G_{V_a}(Z)$, including the same numerical contour normalization and the removed tame tests. Both the action S_j^a and the complete old interaction are gauge invariant on

the real admitted overlap. Haar measure and its coarea disintegration on $M(Z) = W$ are unchanged, because the gauge is one at the output nodes. Thus no graph-Jacobian or coarea term is left over. Equivalently, the two graph densities transform as the pullbacks of this same intrinsic measure.

The joint graph identity sends the old free angle u_b to $\mathrm{Ad}(s_a(b_-))u_b$ relative to $\widetilde{V}_a$. This is a real orthogonal colour change with determinant one, so it preserves each product ball. A field in the frozen graph has free link $e^{ig_0B_b}V_0(b)$. Its relative free angle about $\widetilde{V}_a$ is exactly the logarithm in eq. 4.cdxxxi. Pivot coordinates follow by the uniqueness of the actual graph.

For the last assertion differentiate $i^{-1}\log(e^{iX}e^{-itY})$ at $t = 0$. Its derivative is $-Y$ plus terms containing commutators with X. Pairing with X kills every such term by invariance of the colour inner product. The convergent logarithm series therefore gives $\langle X, D_t i^{-1}\log(e^{iX}e^{-itY})|_0\rangle = -\langle X, Y\rangle$. Take $X = g_0B_b$, $Y = c_{e,b}$, and divide the paired identity by $g_0^2 b_0$. This proves the exact normal velocity, with no polynomial approximation to the group chart. □

The centre jet has a compatible local analytic family with

$$\|\mathbf{c}_e\|_{R,\kappa_+,\gamma_{\mathrm{per},+}} \le C|e|, \qquad c_{e,b}(1) = 0. \tag{4.cdxxxiii}$$

Indeed the weighted literal intermediate fields have their all-degree connected bounds on a larger weight disc. Cauchy gives the centre derivative. The finite L-step contours in eq. 4.cdxxx attach fixed local stars and point products. Their constants can depend on the fixed L, M, R, but not on the inner depth. The identity at $W = 1$ follows because every weighted trivial background is the identity. This construction uses only the background matrices. It never evaluates h on a complex gauge transform outside its old ball.

Define a radial surface functional at one free bond by integrating its three colour coordinates over $S^2_{b_0}$, with ordinary area measure, and all other coordinates over their retained balls. In the Gaussian pullback of LCF put

$$\mathbf{m}_b(W; x, h) = \frac{\displaystyle\int_{|B_b|=b_0} n_b \prod_{d\neq b} \chi_{b_0}(B_d) e^{P_{g_0,\tau,h}(V_0,B)} d\mu_{K_\tau(V_0)^{-1}}(B)}{Z_j(g_0; V_0, \tau, h)}. \tag{4.cdxxxiv}$$

The numerator notation means the surface functional just defined, not a product of an ambiguous boundary value of χ_{b_0} with a delta distribution. The full coarea and Haar densities are included in P. The scalar boundary source is

$$\beta_{e,b} = g_0^{-1} c_{e,b} \cdot \mathbf{m}_b. \tag{4.cdxxxv}$$

Proposition 4.51 (Exact total first-jet comparison)**.** For real admitted data,

$$D_a \log \mathcal{I}^{\mathrm{mov}}(1)[e] - D_a \log \mathcal{I}^{\mathrm{fr}}(1)[e] = \sum_{b\in\mathcal{F}} \beta_{e,b}. \tag{4.cdxxxvi}$$

The right side has the local analytic continuation and bounds proved below. This continuation does not differentiate a sharp indicator at a complex weight. ■

Proof. Use Lem. 4.50 to write both integrals in the frozen fibre measure. Their action derivative is identical, and their old interaction, relative gauge and measure are the same. The product of colour balls has first shape derivative

$$\sum_{b\in\mathcal{F}} g_0^{-1}(n_b \cdot c_{e,b})\, \delta(|B_b| - b_0) \prod_{d\neq b} \chi_{b_0}(B_d).$$

Intersections of two sphere boundaries have codimension at least two and contribute no first shape derivative. The formula follows by the one-variable thin-shell change of variables in each three-dimensional colour group. The finite graph and strict chart margin justify differentiation under these compact integrals. Divide by their common value $\mathcal{I}_0$. The exact LCF Gaussian factorization cancels its scalar prefactors, giving eq. 4.cdxxxiv and eq. 4.cdxxxvi. □

4.16.3. *A normalized rooted sphere estimate.*

Lemma 4.52 (One surface mark in the actual local expansion)**.** There are fixed $c, C, K > 0$ such that the complete sources eq. 4.cdxxxv have compatible holomorphic local families, rooted at their marked free bonds, with

$$\|\boldsymbol{\beta}_e\|_{R,\kappa,\gamma_{\mathrm{per}}} \le C_{H_*}|e|g_0^{-1}(1+b_0)^K e^{-cb_0^2}. \tag{4.cdxxxvii}$$

The norm includes the original centre-jet carriers, old-source carriers, coarea terms, normalization and the period companions. It is independent of total volume and inner depth. It holds on the smaller complex τ, h, W domains retained from LCF. ■

Proof. We verify the additional local functional estimate and then its normalization. Use the actual filed Gaussian matrices of eq. 4.ccxciv. Use its finer comparison error ε_D and Gaussian tilt δ_1, both at most $c_*(LM)^{-4}$. Its scalar dimension is $d \le C(LM)^4 \# Z$. Relative to the positive real comparison covariance D_0, the determinant and inverse estimates give

$$|d\mu_D(B)| \le e^{C\varepsilon_D d} e^{C\varepsilon_D |B|^2}\, d\mu_{D_0}(B).$$

Reweighting this positive Gaussian by $e^{(\delta_1 + C\varepsilon_D)|B|^2}$ changes its normalization by at most $e^{C(\delta_1+\varepsilon_D)d} \le e^{C'\# Z}$ and leaves its covariance between fixed positive constants. The constants are uniform on the enlarged covariance-parameter discs and local complex background domains. Replacing the three-dimensional integral at b by its sphere now uses the positive Gaussian's three-coordinate marginal and gives

$$\int_{|B_b|=b_0} |n_b| e^{\delta_1|B|^2} |d\mu_D(B)| \le e^{C\# Z}(1+b_0)^2 e^{-c_2 b_0^2}. \tag{4.cdxxxviii}$$

The factor is the area of a three-colour sphere. Remaining Gaussian moments and any fixed local parameter derivatives give only further polynomial factors, or a fixed exponential per wall cube. They do not introduce Gaussian coordinates from the inner fine hierarchy.

The integration set of this marked local functional is fixed. Gaussian densities, graph vertices and source functions are holomorphic in their actual parameters and have the same tilted domination. Thus the marked functional is holomorphic, and the usual covariance-parameter Cauchy estimates apply. Its full guard includes the marked coordinate, every potential support, the centre-jet coefficient carrier and all background read collars. A wall factorization is made only after these complete guards are included. If separated components remain, the marked functional is the marked root component times the ordinary functionals of the others.

The actual action/coarea interaction and bounded old source have the small budgets of Lems. 4.41 and 4.42 at the physical g_0. Their existing connected expansion uses the tilted Gaussian bound, parameter holomorphy and this wall factorization. The estimate eq. 4.cdxxxviii verifies these same requirements with one marked functional. It supplies an additional root factor $\eta_\partial = Cg_0^{-1}(1 + b_0)^K e^{-c_2 b_0^2}$. The centre coefficient is expanded first using eq. 4.cdxxxiii; its entire carrier is attached to that root. The reserved intrinsic rate pays its fixed guard enlargement. The sum over old free bonds in an output cube is at most $C(LM)^4$, a fixed local factor.

For completeness normalization is obtained by a single formal source, not a global lower bound for an integral. Give the boundary functional a parameter ζ with $\zeta^2 = 0$. The finite partition expression is $Z + \zeta Z_\partial$. The coefficient of ζ in its connected logarithm is exactly Z_∂/Z. In the actual polymer expansion every such connected term has one marked object and otherwise ordinary objects. At the marked coordinate the surface functional replaces its physical cutoff: an ordinary cutoff factor at that same coordinate is excluded. Larger nested auxiliary cutoffs equal one on the marked sphere by the retained admission margin. Thus no product of a radial delta with an unspecified boundary value of its own indicator is formed. This exclusion does not enlarge any absolute activity bound. The ordinary small activity majorant is ε, within the retained cluster ceiling. Its rooted tree sum

is bounded by

$$C\eta_\partial \sum_{n\ge 0}(C\varepsilon)^n \le C'\eta_\partial.$$

All original source origins are combined on their full carriers before this sum, exactly as in LCF. This proves the local bound for the normalized expectation, without estimating an extensive $1/|Z|$. Since the unmarked logarithm is the selected LCF branch, the formal degree-one identity equals the actual ratio in eq. 4.cdxxxiv. There is no product of two delta distributions at one colour coordinate.

The intrinsic output is composed with the actual literal centre on the large R-ball by the same whole-cube Banach composition of Lem. 4.42. It is a bounded operator on the rooted family norm; its rate cost and the old anchor incidence count are already paid by the blocking buffer. The surface root creates no extra background coordinates. The old input quotient is respected because the complete integral and marked expectation depend only on the complete h_z; take the infimum over their compatible representations.

Finally retain lifted carriers and finite-period labels before projection. In a companion difference, telescope the primitive matrices, centre factor, vertices and old-source families. One factor carries the existing period tail or a retained winding carrier. The marked Gaussian derivative still obeys eq. 4.cdxxxviii: $D\log d\mu_D$ is a trace term and a quadratic form in B, whose moments add only a fixed polynomial in b_0 and wall size. A slight decrease of c_2 pays the former; the spatial rate reserve pays the latter. Thus the period interpolation and padding of Lem. 4.45 retain γ_{per}. Construct all Gaussian and centre families on one common large cover, with the full periodic held guards, before descent. No comparison of quantum partition functions on different volumes is made. These estimates prove the full compatible norm in eq. 4.cdxxxvii. □

Before multiplication by $c_{e,b}$ and g_0^{-1}, the same marked construction gives a compatible local vector family for $\mathbf{m}_b$, with the corresponding sphere-tail bound and fixed W jets. Its exponential moments justify the affine product-rule probes used below; scalar gauge invariance is required only after pairing with the centre jet.

The construction also proves bounds for any fixed number of τ and bounded old-input derivatives, by Cauchy on their retained larger domains. Differentiation in x adds θ_0^{-1} per derivative. No derivative of the frozen reference cutoff radius is taken. Because $2p>1$, for every fixed N the right side of eq. 4.cdxxxvii is at most $C_{H_*,N}\theta_0^{-N}|e|$ after increasing the reference floor.

4.16.4. *Complete pointings and physical normalization.* Choose output hats θ_y in output units and define the complete pointing

$$(\mathcal{B}_{j,e})_y(W)=\sum_{b\in\mathcal{F}}\theta_y(m_b/L)\,\beta_{e,b}(W). \tag{4.cdxxxix}$$

Here m_b is the old bond midpoint. The finite local anchor count changes only constants. The hats sum to one, so this pointing has total $\sum_b\beta_{e,b}$.

Lemma 4.53 (Symmetry and the flat value)**.** The complete functions in eq. 4.cdxxxix are gauge invariant and jointly covariant under lattice symmetries acting on the weight direction e, old input h and fields. They vanish at $W=1$. These assertions precede local filing. ■

Proof. Let an output gauge induce old gauges k_a on V_a and k_0 on V_0. Their output-node values are the same. The contour endpoint law gives $s'_a=k_0s_ak_a^{-1}$. Hence $\widetilde{V}'_a=\widetilde{V}_a^{k_0}$: alignment to the frozen contour frame removes the weight-dependent gauge. Thus $C_a(b)$ and its first weight jet transform by the same frozen $\mathrm{Ad}(k_0(b_-))$ as the free colour variable. The sphere and its normal transform by this real orthogonal map. The complete old interaction, action and coarea factors are invariant on their admitted overlap. Each complete scalar surface source is therefore invariant. Holomorphy extends this identity on the stated complex domain.

Translations and reflections use the fixed middle-pivot identity of Prop. 4.25. For a reversed stored bond the centre jet and free colour variable have the same sign and adjoint twist. Joint axis permutation acts on e, h and the weighted orbit simultaneously. Their sphere pairing and the hats therefore have the stated covariance. At $W = 1$, $c_{e,b} = 0$, so the complete source vanishes exactly. □

Proposition 4.54 (A frozen-kernel stress jet with a rooted tail)**.** The moving weighted logarithmic first jet minus $\mathcal{B}_{j,e}$ is a compatible pointed representation of the actual frozen-kernel first jet eq. 4.cdxxix, after the same flat-field normalization. Moreover

$$\|\mathcal{B}_{j,e}\|_{R,\kappa,\gamma_{\mathrm{per}}} \le C_{H_*}|e|\theta_0^{1/2}(1+b_0)^K e^{-cb_0^2}. \tag{4.cdxl}$$

The same type of bound holds for any fixed x, h jet on smaller retained domains. Constants are uniform in depth and compatible volume. ■

Proof. The total identity is Prop. 4.51; the local family and normalization are Lem. 4.52. The symmetry lemma supplies the complete function identities required by the profile space. The flat value of the error is zero, so vacuum subtraction does not alter it. Select the frozen output pointings by subtracting this rooted family from the known moving weight-jet pointings. Their total is exactly the fixed-kernel first jet. This selection does not assert equality to every independently chosen per-anchor filing of a stress insertion. □

4.16.5. *The affine readout and the regular stress equation.* For the present homogeneous literal convention the boundary source has a further cancellation. It is not needed for its superpolynomial norm bound.

Corollary 4.55 (Zero boundary curvature readouts)**.** All six quadratic curvature readouts of $\mathcal{B}_{j,e}$ vanish. ■□

Proof. At $W = 1$, the frozen normalized law is invariant under global colour conjugation, also for complex h in the complete gauge invariant input class. Its vector surface moment in eq. 4.cdxxxiv consequently vanishes. We already know $c_{e,b}(1) = 0$.

The exact affine response of eq. 5.ccxcvi is independent of the constant weight modulo a gradient. Both literal means are the same prescribed affine datum, so that gradient can be chosen zero at the final output nodes: its node values are constant and the flat constant can be subtracted. Taking the actual partial mean gives an old-site gauge equal to one at the outer nodes. Contour alignment in eq. 4.cdxxx is unique on such an orbit. It therefore removes this gradient and gives

$$D_W c_{e,b}(1)[B^F] = 0$$

for every affine curvature probe B^F. These polynomial probes are interpreted in the exponentially summable infinite companion kernels, as in the affine response proof.

Differentiate the exact product $\beta_{e,b} = g_0^{-1} c_{e,b} \cdot \mathbf{m}_b$ twice in two affine directions. Its constant factors vanish at the flat field, and each remaining mixed product contains one of the displayed zero derivatives. Thus its entire quadratic affine form is zero. The absolute local family bound permits the hat and companion sums. Complete gauge invariance and the Ward readout then give all six stated zero coefficients. □

In a response $-tE_{j,e} + v$ of the fixed isotropic kernel, the weight direction is $a = 1 + (t/x)e$. Hence the actual output jet is the moving-weight one minus $(t/x)\mathcal{B}_{j,e}$. Using the selected pointings above, the stress recurrence of eq. 5.ccclxv has the same scalar leading coefficient; its profile forcing is changed from $A_j[e]$ to $A_j[e] - \mathcal{B}_{j,e}$. The change is in the six-zero-readout space and is superpolynomially smaller than the permitted formation error. Even without the affine cancellation, its readouts are bounded by eq. 4.cdxl and give only a superpolynomial scalar correction after division by x. The coupling-weighted Volterra estimates therefore retain their bounds.

This comparison proves only the first constant-weight jet of the stated homogeneous regular kernel with bounded old input. It neither assigns holomorphy to a sharp indicator nor establishes small formation on a fixed-centre weight disc. The frame, regular fibre, cutoff and normalization in the frozen response remain the isotropic ones.

4.17. **Large-field histories and induction.**

4.17.1. *Large-field regions, their weights, and induction.* Convergence of the regular expansion in § 4.5 requires that the plaquette variables of the backgrounds lie within $\varepsilon_k \eta_k^2$ of $\mathbf{1}$, with $\varepsilon_k = g_k p_0(g_k)$, and that the fluctuation fields stay below $\delta_k = (A_1/A_0)\varepsilon_k$. The two thresholds carry the same power of the logarithm and differ by a small fixed ratio. Nothing guarantees these conditions. For this reason each step inserts decompositions of unity $1 = \sum \prod \chi \prod \chi^c$ built from local tests. At a place where a test fails, no expansion is made and the fields there are stored. The weights present at that place then supply a factor $e^{-c\, x_k \varepsilon_k^2} = e^{-c\, p_0(g_k)^2}$. For a failed plaquette test this weight is the Wilson weight. When an approximate fluctuation test fails, the weights present are the Wilson weight together with the kept gauge-fixing exponents. When a fluctuation test, a collar test or a rim test fails, the weight is only the Gaussian form of the stored fluctuation field. (The failure of one of the sharper tests of the large-field operation $\mathbf{R}$ of §4.17.1.3 yields only $e^{-p_0(g_k)}$: 2a, 2c.) In this way the density becomes a sum over histories h, collected according to their labels $\mathcal{S} = (\Omega_j, \Lambda_j, S_j)_{j \le k}$, which consist of the regions of all levels. Beside the regions, a label carries the marks left by the operations $\mathbf{R}$, one for each island and for each drop-out that stands §4.17.1.1; the labels are thus labels with marks. The exponential form given below holds for sums of histories and not for an individual history (2a, 2d).

Proposition 4.56 (inductive description; ultraviolet stability)**.** Assume the primitive hypotheses $\mathcal{H}_{\mathrm{step}}$ of the joint step, on the actual histories and domains used below. Suppose that the constants are chosen in the prescribed order, beginning with the exponents and a large L and ending with a small γ §4.17.1.45. Suppose also that the torus is admissible up to level k, and up to K for the stability bounds; that is, $\ell L^{K-j} \ge 2L^{11} M \check{R}_j$ for $j \le k$, respectively for $j \le K$. This is a lower bound on ℓ, which is chosen after M and g. When $\ell \in \mathfrak{L}(g)$ and the datum is a first-passage datum, the condition follows from the upper flow bound up to K § 5.2.2. Assume further that $g_j \le \gamma$ and $\lambda_\sharp(j-i) - c_5 \le x_i - x_j \le B_\sharp(j-i)$ for $i \le j \le k$ (up to K for the stability bounds). When g is small, § 5.2.1 provides these conditions for any first-passage datum; they are conditions on how the couplings flow: β has no upper bound, and $K = K(\beta, g)$ is determined together with it. Then $\rho_k = (\mathbf{RT})^k \rho_0$ is defined, and

$$\rho_k(V_k) = \sum_{\mathcal{S}} \mathbf{T}'_k(\mathcal{S}) \big[\chi_{k,\mathcal{S}} \exp A_{k,\mathcal{S}}(U_k(\mathbf{V}))\big],$$

$$A_{k,\mathcal{S}} = -A(x_k(\cdot), U) + \mathbf{E}_k + \mathbf{R}_k + \mathbf{B}_k - \mathcal{E}_{k,\mathcal{S}},$$

$$\|E^{(j)}\|_{j,\kappa} \le E_0,$$

$$\|R^{(j)}\|_{j,\kappa} \le g_j^{\kappa_0},$$

$$|B^{(j)}(X,\cdot)| \le \mathsf{B}\, e^{-\kappa d_j^\natural(\mathcal{S};X)},$$

$$\mathsf{M}(i,X) \le e^{-(2-c')p_0(g_i)},$$

where $-\mathcal{E}_{k,\mathcal{S}}$ is the constant term of the action of the label and B is a constant fixed before γ (2d, 2e); $d_j^\natural$ is the hull length of § 4.17, and $\mathsf{M}(i,X)$ is the mass of a component X that, at some level $i \le k$, has passed all those tests of the operation $\mathbf{R}$ that come before the last one. Such a

component is either healed at that level or an island, which remains in the label. The mass is defined as follows. Let e^{Φ_ω} be the integrand of the operation $\mathbf{T}'_i(X)$ belonging to the component. This integrand contains the Wilson factor of the core (2b, 2c), and its exponent has the form $\Phi_\omega(\xi;\mathbf{p}) = \Phi^\circ_\omega(\xi) + \sum_f \Phi_{\omega,f}(\xi;\mathbf{p})$, which is Borel in ξ, analytic in $\mathbf{p}$ and real at real pairs; each factor $\Phi_{\omega,f}$ depends on ξ and on $\mathbf{p}$ only on a support $R_f \subset X^\sim$ § 4.17. Let ω run over the local histories of X, each with a frame in the hypercubic group W_4. Let ξ denote the stored variables and μ_ω their measure. Let $\mathbf{p} = (\mathbf{U},\mathbf{J})$ denote the complex pairs of the space $U^c_i(X^\sim,\alpha_{0,i},\alpha_{1,i})$, which contains the pairs at which a term evaluates the factor. Let $\mathsf{v}(\omega)$ be the space-time volume of the patch, which counts $\log x_j$ for each stored site of level j §4.17.1.40. Then

$$\mathsf{M}(i,X) := \frac{1}{|W_4|}\sum_\omega e^{c\,\mathsf{v}(\omega)} \int \sup_{\mathbf{p}} \left|e^{\Phi_\omega(\xi;\mathbf{p})}\right| \mu_\omega(d\xi).$$

The operation of a drop-out contains no Wilson factor of its own. For a drop-out the same bound holds in a modified form, with two changes. First, the Wilson weight over X of the carrying term is included with the operation; the carrying term is a term of some level $k \geq i$ in which X stands. Second, the supremum is taken over the evaluations at ξ. These are the real configurations that are compatible with ξ, regular layer by layer, and at which every test listed by ω, with the exception of the test whose failure made X a drop-out, takes its recorded value (the full list of these conditions is given in §4.17.1.40). This form follows from the induction and is not one of its clauses. The mass bound holds for every $c \geq 0$ in the weight $e^{c\mathsf{v}}$ and for every small $c' > 0$ in its exponent, provided the ceiling on the couplings is chosen after both c and c'; the reserve constant c_{r} of §4.17.1.45 is one such c. Apart from the core factors of the islands in Z §4.17.1.3, the operation $\mathbf{T}_k(Z)$ contains no Wilson weight. The factor coming from a failed plaquette test sits in $e^{-A(x_k(\cdot),U_k)}$, which is the function the operation acts on. In the sum over the labels, every component C of Z_k keeps a factor $e^{-c\,p_0(g_k)-\kappa_1|C|_k}$. The constant $\kappa_1 > 0$ outweighs the number C^n of connected sets of n cells that contain a given cell §4.17.1.40. Each term is a non-negative measure. Neither $E^{(j)}$ nor $R^{(j)}$ carries a label index. On non-wrapping domains with matching step data they are the values, at the lifts of the domains in those units, of a single family on $\mathbb{Z}^4$. (The constant $\mathcal{E}_{k,\mathcal{S}}$, by contrast, depends on the label and on the torus.) Finally, let $\chi_k = \chi_k(T_\ell)$ denote the product, taken over every cube of the torus, of the tests $\chi_k(\square)$. Then, uniformly, there is a constant E_- such that

$$\chi_k\, e^{-x_k A(U_k(V_k))-E_-|\mathbb{T}^{(k)}|} \leq \bar\rho_k,$$
$$\int \bar\rho_k\, dV_k \leq e^{C(1+\log x_k)|\mathbb{T}^{(k)}|},$$
$$\int \rho_k\, dV_k = Z_{K,\ell,\beta}. \qquad \blacksquare$$

The theorem is the case $\theta = x(\beta)$ of an induction carried out along an arbitrary frame θ (the frame setting of § 5.2). That induction runs on all admissible tori simultaneously. The tube is a clause of the class $\mathfrak{C}_k(\theta)$ of §4.17.1.45, and the step by itself does not yield this clause again (Prop. 4.65). For complex field arguments, each term depends on a triple $(U, A', \mathbf{J})$, in which U is G-valued and $A', \mathbf{J}$ are complex. For a term of level j, the pair of a triple is $\mathbf{p} = (\mathbf{U},\mathbf{J})$ with $\mathbf{U} = e^{i\eta_j A'} U$, and the triple is called a splitting of its pair. The domains of analyticity are defined by conditions on triples. One of the statements propagated by the induction is that a term takes the same value at any two splittings of one pair. The splitting is therefore a coordinate and not an argument, and a term is a function of the pair. (In the regular setting of § 2.2 the values coincide on a complex gauge orbit, with a margin left in the radii. Here what is asserted concerns two splittings of a single pair, and it holds on the domain itself.)

The core of the theorem is due to Bałaban: Comm. Math. Phys. 119 (1988) 243, Theorem 1 and Corollary 3, supplemented by 122 (1989) 175 and 122 (1989) 355, Theorem 1. For the idea of $\mathbf{R}$, he cites Gallavotti and co-workers as the closest earlier work. He states, for one torus, that neither the regular terms nor the part of the remainders produced by the fluctuation integral depend on the region Λ_{k+1} of the step (119, pp. 278–279). He also names a second choice of the thresholds and sets it aside; in that choice only the logarithm is evaluated on a bound of the flow known a priori (p. 246). The upper bound he gives is pointwise, $\varrho_k(V_k) \le e^{E_+ |T_\eta|}$, stated for his density and his volume ((2.50)), and Prop. 4.56 gives the integrated form in its place. Besides what is quoted here from his papers, this step contains the following:

- a proof of that independence and a range for it, covering the remainders in their entirety, simultaneously on all admissible tori and cutoffs, by means of one family on $\mathbb{Z}^4$ (for the counterterm, the paper 119, p. 281, uses "$\mathbf{E}^{(j)}(U_j, z)$ defined on the whole lattice $L^{-j}Z^d$");
- the positivity of each term, as a clause of the statement (in 122 (1989) 175, p. 176, Bałaban announces that the densities $\rho(Z'', V)$ are positive);
- measure and formula as two separate layers;
- the class taken as a set of presentations along a frame;
- the flow inequalities, hypotheses of Prop. 4.56, obtained as conclusions of the induction, carried out jointly with the flow § 5.2.1;
- covariance of the R-terms under translations only, term by term, and covariance of their totals by way of measures;
- the mass, with the complex supremum inside the integral and with a reserve weight;
- the hull length;
- the three possible fates of a component, two of which are recorded as marks of the label (in 122 (1989) 175, pp. 181–183, Bałaban removes from the region on which $\mathbf{R}$ acts any component in which a test of some stage fails, and in 122 (1989) 355, (1.102), he retains the operation $\mathbf{T}_k(Y)$ where the last test fails), and the two sequences of regions attached to a label;
- an order of the constants that contains no circle;
- exactness for each partition of the histories.

The route also makes use of the following:

- the collar and the rim of §4.17.1.1, which ensure that in the interior the mean of the conditional Gaussian measure stays under the thresholds set by the tests;
- the factor depending on two levels in the sum over the scales of the R-terms §4.17.1.45;
- the terms $V(Y)$, which are not of small amplitude, with the consequent removal of the empty family before the exponentiation § 4.17;
- in §4.17.1.40, a single count over the genealogy of a component, which replaces the double induction of 122 (1989) 355, pp. 384–387.

In that induction, the case in which two regions join rests on a sentence on p. 387. That sentence states that after n_1 further hull operations, where n_1 is a fairly small number, for instance $n_1 < 10$, the domain obtained from the region whose wait is the longer of the two contains the domain obtained from the other region. The count never compares two such regions. A joining of regions costs it no separate payment, because the count charges the joining to a created cell §4.17.1.40. The later steps rely on the theorem and on its sourced form §5.20.1.1, which contains the insertion formula.

4.17.1.1. The tests, the nested regions, and the content of a history. **The tests of step** $t = k + 1$ are applied to a term of ρ_k whose live region is $Y = \mathcal{Z}_k$. First, outside a margin of Y, the new field V_t undergoes the plaquette test $\chi_t(\square)$, and the cubes where this test fails make up P_t. The factor $\mathfrak{G}_k$ is inserted away from $P_t \cup Y$. By the Faddeev–Popov argument, its integral over the gauge group of the blocks equals 1. Next comes the approximate fluctuation test $\chi'_k(\square) =$

$\mathbf{1}\{|V_k(\mathcal{M}^k U_{t,\square}(V_t))^{-1} - \mathbf{1}| < 2\delta_k\}$. This test is evaluated at the bonds of $\square^{\sim 2}$ other than the pivot bonds § 4.5, and the third test below is evaluated at the same bonds. Here one takes from the orbit $U_{t,\square}(V_t)$ an element whose k-th average lies in the block axial gauge, and every such element gives the same tested number. The cubes where this test fails make up Q_t. These sets determine Ω_t, the region on which $U_t(\mathbf{V})$ and $\mathcal{A}_k$ exist. The backgrounds and the propagators on sequences of regions are taken from § 3.11, § 3.12.1, together with their regularity and their decay in the local scale. A label's sequence starts at $\Omega_0 = \Omega_1^{\sim}$, the first region enlarged by one layer of level-1 cells, and not at T_ℓ; this is required by Prop. 5.70 and Lem. 5.72 in §§ 5.20 and 5.22. After that the test $\chi^{(k)}(\square)$ is made, and its failures make up R_t. Hulls $(', \hat{\cdot})$ and layers are taken in level-t cells. At $t = 1$, P_0' is the level-one hull of failures of the bare density test. Its indicator on a level-zero density cube $\square$ requires $\sup_{p \subset \square^{\sim}} |\partial U(p) - \mathbf{1}| < \varepsilon_0$. Use the indicator and its complement in the positive partition by P_0. The first plaquette tests avoid $P_0'^{\sim 1}$ and the first gauge draw avoids $P_1'^{\sim 1} \cup P_0'^{\sim 2}$. The boxing map bx replaces each component contained in a cube of $O(1)$ cells by its bounding box, and repeats this until nothing changes. With these conventions,

$$\Omega_t^c = Q_t'^{\sim 2} \cup P_t'^{\sim 4} \cup H_t, \qquad H_t = \begin{cases} \widehat{Y}^{\sim 8}, & t \geq 2, \\ P_0'^{\sim 5}, & t = 1, \end{cases}$$

$$Z^{[0]} = \mathsf{bx}\big((\Omega_t^c)^{\sim 2} \cup R_t'^{\sim}\big),$$

$$Z^{[n+1]} = \mathsf{bx}\big(Z^{[n]} \cup (\text{cells of the failures of round } n)^{\sim}\big),$$

where the rounds are those of the recursion described below. The set $Z_t = \Lambda_t^c$ is the final $Z^{[n]}$, and $S_t = (\Omega_t \setminus \Omega_t^{\sim -1}) \cup R_t$. The numbers of layers are those of Bałaban. If the boxing map is omitted, $Z^{[0]}$ coincides with his Λ_t^c in the form that the display (3.20) gives it, that is, before he replaces the small components by rectangular parallelepipeds (Comm. Math. Phys. 119 (1988) 243, (3.16), (3.20), p. 269). The order of these numbers matters: an older set receives more layers, and each move is made one layer inside the region of the move before it. The creation set $\mathcal{W}_t^{\wedge}$ is the union of the hulls of every failure of the step: P_t', Q_t', R_t', the cells of the collar bonds that fail, and the rim cells that fail. At $t = 1$ it also includes P_0'. If nothing is created, the step reduces to the free evolution

$$\Phi_t(Y) = \mathsf{bx}(\widehat{Y}^{\sim 10});$$

in general

$$Z_t \subset \widehat{Y}^{\sim 10} \cup (\mathcal{W}_t^{\wedge})^{\sim O(1)}$$

up to boxing

That is, one coarsens, adds a margin and boxes; this is the $S(Z)$ of §4.17.1.40. The size $\check{R}_k$ is the smallest power of L that is not less than $(\log \min_{i \leq k} x_i)^r$, with $r \geq 2$. Because it is a power of L, all partitions are nested. Because it is formed from the running minimum, it is non-increasing (Bałaban forms it with g_k itself), and therefore $t \mapsto t - \mathsf{N}_t$ is strictly increasing. Moreover $\check{R}_k \leq L\check{R}_{k+1}$, and two consecutive drops never occur, because x changes by $O(1)$ per step and γ is small. A test cube $\square$ has side of order $M_2\check{R}_t$. Here M_2 is a power of L with $L^2 M_2 \leq M$ and $c\,\delta_0 M_2 \geq 1$. The last condition is a floor on M_2, and δ_0 is fixed together with L, which comes earlier. Hence a test cube lies well inside a cell of its own level.

What factorises, and why. The tests χ_t and χ_k' are built from localised backgrounds. They depend on V_t, V_k only within $O(1)$ test cubes. The third test, and also the collar and rim tests, depend on $\mathcal{A}_k = \frac{1}{i}\log(V_k(V_k^{\mathrm{bg}})^{-1})$, which is formed with the true averaged background on Ω_t.

These tests are therefore local in the new variable $\mathcal{A}_k$, but not in (V_k, V_t). Once $\mathcal{A}_k$ has become the variable, the surviving χ'_k (on S_t) compare $e^{i\mathcal{A}_k}V_k^{\mathrm{bg}}$ with $\mathcal{M}^k U_{t,\square}$. They therefore depend on V_t over the whole of Λ_t. Four facts together ensure that everything attached to Z_t factorises over its components:

- The backgrounds are localised.
- The substitution on the strip §4.17.1.3 is made. After it, every indicator kept in $\Omega_t \cap \Lambda_t^c$ depends on V_t in Λ_t^c alone, apart from the factors that equal 1 on the supports.
- The conditional quadratic form is cut §4.17.1.3.
- The pivot values are dealt with as follows. An old test $\chi_k(\square)$ of a cube $\square \subset \Omega_t^c$ adjacent to Ω_t also depends on V_k at pivot bonds § 4.5 inside Ω_t. There V_k is not a stored variable, and through $e^{i\mathcal{A}_k}V_k^{\mathrm{bg}}$ a test of one component depends on the fields stored in another. At such a bond, V_k is given the value that solves, for the pivot bond, the averaging constraint $\mathcal{M}(V_k)(c) = V_t(c)$ of the single coarse bond c. Bałaban (122 (1989) 175, (1.3)) retains only those old tests whose cubes lie outside $\Omega_t^{\sim}$; the tests that he leaves out equal 1 on the supports by Lem. 4.57 (a), whose margin is this layer.

Inside **R**, the tests are built from backgrounds localised in the component, never from the background of the torus. The induction carries the following property: each indicator stored in $\mathbf{T}_k(Z)$ depends only on those field variables that are localised in Z.

Totality. By definition, a test is the indicator function of the set where all objects occurring in it are defined and where its inequalities hold. These objects are the localised minimal orbit, which must exist and be a single orbit, the contour variables, and the logarithms. Consequently $\chi^c = 1$ wherever some object is undefined. A factor of $\mathfrak{G}_k$ equals 1 where $V(y,x)$ is undefined, by a separate convention for that factor. Hence $\int \mathfrak{G}_k(V^u)\,du = 1$ for every V, and each insertion $1 = \sum\prod\chi\prod\chi^c$ is an identity that needs no assumption. The tests are not empty conditions. The averaging constraint transports smallness: $|\partial V_k - \mathbf{1}| < 2\varepsilon_k$ implies $|\partial V_t - \mathbf{1}| < CL^2\varepsilon_k$. Therefore $U_{t,\square}(V_t)$ exists on the carried supports, and the new test further restricts a field that is regular already (Comm. Math. Phys. 119 (1988) 243, p. 246 at the first step, p. 264 at a general step). The cost is that χ^c contains configurations that are not large in any quantitative sense. These configurations must also produce the small factor §4.17.1.40.

For the convention just stated, normalization can be checked one site at a time. Let $D_{y,x}$ be the set where the prescribed mean $V(y,x)$ exists. Uniqueness of the mean and its biunitary covariance show that $D_{y,x}$ is gauge invariant and that, when the block centers are fixed, $V^u(y,x) = V(y,x)u(x)^{-1}$ on $D_{y,x}$. On this set the integral of the normalized site factor over $u(x)$ is 1 by Haar invariance. Off $D_{y,x}$ the site factor is identically 1, so its integral is again 1. Each noncentral site belongs to one block, and the Haar variables $u(x)$ are independent. Integrating the product therefore gives $\int \mathfrak{G}_k(V^u)\,du = 1$ for every V. This uses the sitewise extension: a defined site retains its prescribed weight even if a different site's mean is undefined.

Collar and rim: this device is an addition to the step of Comm. Math. Phys. 119 (1988) 243, Sect. 3. The stored part of the fluctuation field moves the mean of the conditional Gaussian measure of §4.17.1.3. The thresholds δ_k are stated in the unscaled normalisation, and in that normalisation $\mathcal{A}_k|_{\Lambda_t}$ has the mean

$$\mathbf{m} = -\,C^{(k)}(\Lambda_t)\,\Delta^{(k)}\,\mathcal{A}_k|_{\Omega_t\setminus\Lambda_t}$$

This map is linear, so in terms of the scaled fields A, A' of §4.17.1.3 it is the same map, with each threshold divided by g_k; only $|\mathbf{m}|$ enters. This shift must remain well below the least threshold among the inner tests; that threshold is $\theta_c\delta_k$ on bonds that cross $\partial\Lambda_t$, and δ_k further in. A recursion therefore keeps enlarging the region until, along its boundary, $\mathcal{A}_k$ satisfies two sharper tests:

- $|\mathcal{A}_k| < \delta_k/(4\Theta_1)$ on the collar, the set of stored bonds whose distance from Λ_t, measured in level-k units, is at most ϖ_T. With a qualifying word added, the same name serves for three further objects. In §4.17.1.3, the collar of a stage means the shell over which that stage integrates. In §4.17.1.40, the waiting collar is the set that keeps $X \cap \Omega_{m_\flat}$ apart from the floor. In § 4.17, §4.17.1.45, the collar of a component X means the enlarged component $X^{\sim}$, that is, X together with one layer of M-cubes of its level.
- $|\mathcal{A}_k| < \theta_c \delta_k$ on the rim, the set of cells of level t of Z_t that touch Λ_t.

At greater distance, all that is known is $|\mathcal{A}_k| < C\delta_k$. Every round examines bonds and cubes that no earlier round examined. Hence the recursion terminates, and the admissible outcomes it produces constitute a decomposition of unity. The kernel of $\mathbf{m}$ is bounded, with exponential decay. The contributions of the collar, the rim and the remaining part give

$$|\mathbf{m}| \le C(\Theta_1^{-1} + \theta_c e^{-c\varpi_T} + e^{-cM\check{R}_t})\delta_k < \tfrac{1}{2}\theta_c\delta_k$$
$$\text{if } \theta_c\Theta_1 \ge C,\ \varpi_T \ge C,\ \gamma \text{ small.}$$

The rim fraction $\theta_c \in (0,1)$, the collar width ϖ_T and the collar factor Θ_1 serve this purpose.

Lemma 4.57 (the new tests imply the old; the form reproduces itself)**.** (a) On $\Omega_t^{\sim}$, that is, on Ω_t enlarged by a margin of one layer of cells, the indicators in $\mathfrak{G}_k$ and the tests $\chi_k(\square)$ that the term carries all equal 1. This holds on the support of the new tests together with the tests and averaging constraints that the term still carries alongside them. (b) The test $\chi^{(k)}$ is made only on cubes $\square \subset \Omega_t^{\sim -1}$. For such a cube, if $\square \not\subset R_t$, then $\chi^{(k)}(\square)$ implies $\chi'_k(\square)$. Consequently, within Ω_t, the test χ'_k remains only on S_t. (c) After the step, the tests that stand outside the integral operations are exactly $\chi_{t,\mathcal{S}} := \chi_t(\Omega_t) = \prod_{\square\subset\Omega_t} \chi_t(\square)$. They are formed with the action sequence defined below, and with χ_t^c on an island less a margin. ■

For (a): up to a gauge transformation,

$$U_{k,\square} = \exp\left[i\eta_k \mathbf{H}_{k,\square}(\tfrac{1}{i}\log[V_k(\mathcal{M}^k U_{t,\square'})^{-1}])\right] U_{t,\square'}$$

. Here $\mathbf{H}_{k,\square}$ is the localised response $\mathbf{H}_k$ of § 3.11, so the argument has the form displayed only off a boundary layer of the cube; inside that layer the argument is built from the lift of V_k and is bounded by $C\varepsilon_k$. It follows that

$$|\partial U_{k,\square} - \mathbf{1}| \le (1+c)\varepsilon_t\eta_t^2 + C(\delta_k + e^{-c\,\delta_0 M_2 \check{R}_k}\varepsilon_k)\eta_k^2 \le (CL^{-2} + CA_1/A_0 + Ce^{-\check{R}_k})\varepsilon_k\eta_k^2 < \varepsilon_k\eta_k^2.$$

Three mechanisms appear repeatedly:

- Curvature scales by the factor $\eta_t^2/\eta_k^2 = L^{-2}$.
- The ratio $\delta_k/\varepsilon_k = A_1/A_0$ is small.
- The artificial boundary conditions of localised minimisers act only through $e^{-c\delta_0\,\mathrm{dist}}$ (1b, 1c). On cubes whose side is proportional to $\check{R}_k \approx (\log x_k)^r$, this quantity is smaller than every power of g_k, because $r > 1$. In the present estimate it needs to beat only constants (Bałaban, Comm. Math. Phys. 119 (1988) 243, (3.7)); on the strip §4.17.1.3 it must beat a negative power of g_k (ibid., p. 269, below (3.22), where on S_{k+1} the bound $O(1)\varepsilon_k \exp(-R_k)$ is obtained, "hence by any positive power of g_k").

For (b):

$$|V_k^{\mathrm{bg}}(\mathcal{M}^k U_{t,\square})^{-1} - \mathbf{1}| \le C\tfrac{A_0}{A_1}e^{-c\check{R}_k}\delta_k < \delta_k$$

((3.17)–(3.19) of the same paper). The factor 2 between the two thresholds gives exactly the room needed for this. For (c): a test cube of step t serves also as a test cube of the density of level t,

carrying the same background and the same threshold. The old tests inside Ω_t^c pass into the integral operation. The same ladder of implications holds inside **R** §4.17.1.3. On the support of the tests of a stage, the following tests equal 1: those of the preceding stage, and those of the single stored step whose fields the stage integrates (the plaquette tests of its two levels, the gauge indicators, the approximate fluctuation tests). Further, consider any point at which a term is evaluated: the triple $(U, 0, J(U))$, where U is the background of the label, at fields where no characteristic function of the term vanishes. At such a point, $\chi_q(\square)(V_q) = 1$ holds on every cube $\square$ of level q that lies in the layer $\Omega_{q+1}^{\sim} \setminus \Omega_{q+1}$ and not in a drop-out. Also at such a point, the current of the bare field, formed from the plaquettes inside Ω_0 only, stays within the bound that the domains of analyticity impose on their current (the size $|\mathbf{J}|$ is smaller than the first radius of the domain, measured in the units of the local scale). This bound holds also at the bonds of Ω_0 that have a plaquette leaving Ω_0.

Proposition 4.58 (geometry)**.** The regions are nested, $T_\ell \supset \Omega_1 \supset \Lambda_1 \supset \cdots \supset \Lambda_k$, with gaps of at least $cM\check{R}_j$ in level-j units. Every large-field region and every creation set is a union of cells of its own level. Both statements hold for the regions that the steps produce (119, p. 256). Each operation **R** has stage sequences and prepared sequences (namely the regions that the label has after m of its stages, and after all N_k of its stages; see §4.17.1.3), on which the backgrounds of §4.17.1.3, § 4.17 are formed, and drop-outs keep nests. In these sequences and nests the boxes are only unions of M-cubes, and the only lower bound on the gaps is one M-cube of the later level:

$$\mathrm{dist}_\infty(Z_j, \Omega_{j+1}) \geq LM,$$
$$\Omega_{j+1} \subset \Lambda_j^{\sim -1},$$
$$Z_j^{\sim} \cap \Omega_{j+2} = \emptyset$$
(level-j units; layers of M-cubes of level j),

the first inequality is an equality at the active indices $k_0(t) + 2 \leq j < t$ of the upper part of a nest prepared at level t, where $\Lambda_j = \Omega_j$. Here $k_0(t)$ is the change-of-scale index in eq. 4.cdxliv. At $j = k_0(t) + 1$ the gap is instead $w_j - w_{j+1}$ in bare units, which still gives the displayed lower bound but can be larger; the reason is that a step places eight layers around the old region, and the preparation removes most of them again. Consequently the estimates of § 3.11, § 3.12.1 must hold with a separation of only one M-cube. A smallness $e^{-c\check{R}_j}$ from a separation is available only at regions produced by steps. In each decomposition exactly one outcome equals 1, and hence $\rho_k \, dV_k = \sum_h \tau_h$ holds exactly. A history records every outcome: P_t, Q_t, R_t, the collars and rims, and the outcomes inside **R**. It determines the label, the creations and the ages. There are two kinds of creation:

- Creations by step t are the cells $\mathfrak{c}$ of $\mathcal{W}_t^{\wedge}$, which are $M\check{R}_t$-cubes of level t. They are counted one cell at a time, and each carries one factor $e^{-c\, p_0(g_t)^2}$.
- Creations by an operation are its marked components (next paragraph); the tests of an operation are decompositions of unity component by component. Each marked component counts as one creation and carries one factor $e^{-p_0(g_t)}$, which must pay for $\mathsf{N}_t + 1$ levels of one small box.

All later statements of locality rest on this. ∎

Three fates, marked components, two sequences of regions (Bałaban declines "a systematic description of all possible cases": 122 (1989) 175, p. 183). We set $t_\flat := t - \mathsf{N}_t$. In each component X of its class §4.17.1.3, the operation **R** of level t prepares a nest $Z_l''(X)$, $t_\flat < l \leq t$. The nest consists of increasing boxes in $\Omega_l^c \cap X$. Each box is what remains of the large-field region of scale l after the stage that integrates the stored fields of level $l - 1$. A component has one of three fates:

- *Healed* (every test passed): $\Omega_l, \Lambda_l, S_l \mapsto \Omega_l \cup X, \Lambda_l \cup X, S_l \setminus X$ for $l \le t$, and X leaves the label.
- *Island*, also called second class (only the last test fails, namely χ_t of the new field on X less a margin): the same replacement is made for $l < t$ and for Ω_t, S_t, but Λ_t is left unchanged. Thus $X \subset Z_t$ remains, as a component without inner regions. On it, $\mathbf{T}'_t(X)$ is kept together with its variables, its measure and its factors § 4.17.
- *Drop-out* (some test fails after $n \le \mathsf{N}_t$ stages): X remains in Z_t and retains, as its integral operation §4.17.1.3, the operations carried out up to that point. Moreover

$$\Omega_l \cap X = X \setminus Z''_l(X)$$
$$(t_\flat < l \le t_\flat + n),$$
$$\Lambda_l \cap X = X \setminus Z''_l(X),$$
$$S_l \cap X = \emptyset$$
$$(\text{the same } l,\ l < t),$$
$$\text{all other entries unchanged, so } \Lambda_t \cap X = \emptyset.$$

  (For $n = \mathsf{N}_t$, then, the entry $S_t \cap X$ keeps its previous value; on the support of the tests of the last stage, the tests retained on it equal 1, by the ladder of implications inside $\mathbf{R}$ that is stated in the argument for Lem. 4.57.)

Islands and drop-outs are the *marked components*. A mark persists until a later healed component or island swallows it. The word also denotes two other objects: the mark $\Theta^{(k)}$ below, which is a weight on histories, and a marked term of a cluster expansion §4.17.1.45. A drop-out's mark keeps the regions $\Omega_l^\circ = \Omega_l \cap X$, and likewise $\Lambda_l^\circ, S_l^\circ$, which X possessed before the operation. For $0 \le \boldsymbol{\nu}(X) \le n(X)$ we set

$$\Omega_l^{\boldsymbol{\nu}} \cap X = X \setminus Z''_l(X)\ (t_\flat < l \le t_\flat + \boldsymbol{\nu}(X)),$$
$$\Omega_l^{\boldsymbol{\nu}} \cap X = \Omega_l^\circ\ (t_\flat + \boldsymbol{\nu}(X) < l \le t_\flat + n(X)).$$

The choice $\boldsymbol{\nu} = n$ gives the *determining sequence*, which describes the field in its present state. It governs the determining sets, $U_k(\mathbf{V})$ and the propagators; the analyticity spaces involve all the $\Omega^{\boldsymbol{\nu}}$. The choice $\boldsymbol{\nu} = 0$ gives the *action sequence*, the regions for which the terms were constructed. It governs the index sets of $\mathbf{E}_k, \mathbf{R}_k, \mathbf{B}_k$, the profiles φ_j and hence $x_k(\cdot)$, the carried tests, and the constants $\mathcal{E}_{k,\mathcal{S}}$. No single sequence can play both roles. A stored region in a determining position requires variables that have already been integrated. A stage region in the sums re-sorts terms that were made for other regions. Three facts make the device workable. *Marks wait*: a mark of level s lying inside a class component of level t satisfies $s < t_\flat$. Hence standing marks that share a part are nested, their windows $(s_\flat, s]$ are disjoint, and at any one place and level at most one drop-out mark is active. Next, each $\Omega^{\boldsymbol{\nu}}$ is the geometric label obtained by stopping the operations after $\boldsymbol{\nu}$ stages. Its steps, classes and nests coincide with those of the history at hand, and so Prop. 4.58 applies to it: *the later geometry does not depend on how far an interrupted operation had progressed.* Finally, a term created at a moment μ (following the step, or the operation, of a level $j \le t$) retains permanently the space in which it was made:

$$\Omega_l^{\boldsymbol{\nu}_\mu}(\mathcal{S}_t) = \Omega_l(\mathcal{S}^\mu) \cup Z_{[j,t]}$$
$$(l \le j),$$

where $\mathcal{S}_t$ denotes the level-t label as it stands once the operation $\mathbf{R}$ has been performed, $\boldsymbol{\nu}_\mu = n$ on each mark older than μ and is 0 on each later mark, $\mathcal{S}^\mu$ is the label at that moment, and $Z_{[j,t]}$ is the union of all components that were healed or became islands at levels $j, \dots, t$.

4.17.1.2. Changing stage indices. Keep the recorded step data and the three fates fixed, and change the stage indices on the drop-outs within their allowed ranges. To form the new top regions of level s, the recursion uses the recorded step data and the top stored region from level $s-1$. Changing the stage index leaves that region fixed. At each level the class and the nests for the next operation are also fixed. Indeed, a difference at an earlier index l in a candidate component can only be carried by a standing drop-out of level s with $l < s < t$. If l is in the eligibility window of level t, then s is in that window as well. The drop-out lies in the candidate component by nesting, so that component is ineligible in both labels. All eligible components therefore have the same traces through their windows. Their nests are determined by these traces and the grids.

The dependence on the indices can be checked one cell at a time. Fix a level l in a later label and take q in a common cell refinement of the regions being compared. Look for a standing drop-out X of level s with $q \subset X$ and $s_\flat < l \le s$. There is at most one such drop-out because standing marks that meet have disjoint windows. If there is none, membership of q in the level-l regions is unchanged. Otherwise let m be the replacement stage index of X. For the Ω entry, the membership is given by

$$q \subset \Omega_l^{\mathrm{new}} \quad \Longleftrightarrow \quad \begin{cases} q \not\subset Z_l''(X), & l \le s_\flat + m, \\ q \subset \Omega_l(\mathcal{S}'_s), & l > s_\flat + m. \end{cases} \tag{4.cdxli}$$

For $l < s$ the same rule holds for Λ_l with $\Lambda_l(\mathcal{S}'_s)$ in the second line. At $l = s$ the drop-out keeps q outside Λ_s.

These assertions follow together by induction over the operations. At a step, the earlier entries and standing marks are copied. At an operation, a healed component or an island overwrites the older entries on its region, and any mark it swallows ceases to stand. Outside the class no entries change. On a drop-out the new entries through stage m are the complements of the corresponding prepared boxes, and the earlier rule supplies all entries outside its window. The eligibility check above shows at each induction step that the recorded components and their fates still define the same choices of regions. This proves eq. 4.cdxli, including the case of zero stages. It also explains why the action and determining sequences can use different indices on the same records. In using this identity for an integral, one must still use the measures, variables and test factors carried by that integral.

The resulting labels obey the stage version of the nesting rules. A fully prepared box need not be a component of the stored region, and at the current operation one uses the prepared margins in place of the ten-layer margin of an ordinary step.

Locality in the history. If the large-field regions of two histories never touch, the two histories combine into a single history $h^1 \cup h^2$. There are two reasons for this. (i) *A margin beats a reach*: two regions made of level-t cells that do not touch are separated by at least one cell. One cell exceeds the largest range of a bond test, of a pivot bond and of a collar (in the case of the collar, this is the floor $M \ge \varpi_T$ of §4.17.1.45). Taking hulls and adding layers commute with forming unions, and boxing depends on a single component only. A step therefore acts on each component separately. (ii) The class of $\mathbf{R}$ depends on the past: it depends on every object whose creation falls in the window $(t_\flat, t]$, or which carries a mark at a level in $[t_\flat, t)$. Moreover, the recursion is not monotone: a later region may cover a healed place once more. *Window lemma*: no creation within the window of t has been healed. Indeed, healing at a level $i < t$ something created at step s requires $s \le i - \mathsf{N}_i$, and for something marked at level s it requires $s < i - \mathsf{N}_i$. On the other hand $i - \mathsf{N}_i < t - \mathsf{N}_t \le s$, since $\mathsf{N}_t = \check{R}_t$ is non-increasing. Within the window, therefore, the creations and the marks of each history still lie in that history's own region. It follows that each history is the superposition of its clusters, and that clusters may be interchanged between histories.

Each cluster has a root. The root is of one of two kinds: a component that remains live at the last level, or a healed component, considered at the level where its healing took place. The cluster is made up of the components of earlier levels that flow into the root, together with everything the history records in them. Each component, at whatever level, lies in the cluster of precisely one root, namely the last of its successors; thus nothing recorded about the healed components is lost.

A hard-core gas; the standing part. Erasing clusters of healed components from a history yields a history with the same label. The *standing part* $h^\flat$ of h is what remains after the clusters of all its healed components are removed. For a string s of outcomes of the next step one has $(h,s)^\flat = (h^\flat, s)^\flat$. Let $\bar h$ be a history of level k in which no component is healed at any level. In the rest of this paragraph each history is taken without the clusters healed below level k. It stands for the class of histories that yield it once these clusters are erased. Consider the histories that reduce to $\bar h$ once one erases the clusters belonging to those components that were healed at level k. They are in bijection with the pairs $(\mathfrak{X}, (\varsigma_X)_{X\in\mathfrak{X}})$. Here $\mathfrak{X}$ is a finite collection of possible healed components which touch neither one another nor $Z_k(\bar h)$. Further, $\varsigma_X \in \mathcal{H}^1(X)$ ranges over the histories of X alone in which X is healed and nothing is healed before it, and the choices for different components are independent. All these histories share one label. An individual τ_h is a measure produced by the moves, and it has no exponent of the form in Prop. 4.56. The part of $R^{(k)}$ produced by $\mathbf{R}$ arises only through the sum over $(\mathfrak{X},\varsigma)$, a sum which $\mathbf{R}$ itself performs (under the exact-operation hypothesis and the locality hypothesis). This sum is a polymer sum. The reasons are the gas property and two facts of locality in the history: $\mathbf{T}'_k(X)$ is the same in all the summands containing X, and a healed patch leaves no trace outside itself § 4.17.

Traces and charts. Let W be a region, and let W_i^+, the *window* of level i, denote the enlargement of this region by a single layer of level-i cells. The trace of a label is

$$\mathrm{tr}_W \mathcal{S} := \big(\Omega_i^c \cap W_i^+,\ Z_i \cap W_i^+,\ S_i \cap W_i^+\big)_{i\le k},$$

with two additions. The first addition is the marks of the label. They enter collectively rather than individually. For every level, every kind of mark and, for drop-outs, every number of stages carried out, the trace records the union of all components carrying such a mark, together with the unions of their stored regions and of their nests. Hence the trace does not show which pieces in W_i^+ belong to the same component. The second addition consists of three separate sets of stored bonds of level $i-1$ contained in W_i^+: those in the collar, those at a rim cube, and those in the cut-off zone of the strip. The cut-off zone consists of those bonds of the strip that lie near the set where the fluctuation field is large and carry no kept small-field condition of that field; there the substitution of §4.17.1.3 uses the cut-off ϑ_{co}. Where a bond carries a kept condition, the variables are left unchanged, while on the remaining bonds of the strip the substitution proceeds with no cut-off. The terms depend on these three sets, and the regions in W_i^+ do not fix them: to decide if a cube is a rim cube one has to look at its neighbours, and Ω_i^c enters up to two layers away. The trace $\mathrm{tr}_W h$ of a history is formed in the same way from everything that h records, including the labels of its earlier moments. It is finer: knowing the trace of a label does not fix the traces of the histories that carry it. A *chart* $\iota : W \to W'$ is a translation whose vector belongs to the lattice of level-t cells. It acts either within one torus or, through a common lift of a non-wrapping region, from one torus and cutoff to another. It leaves invariant the lattices $\mathbb{T}^{(i)}$, $i \le t$, and also the partitions of all levels up to t into cubes and cells. Neither a shift by a single site nor a mere isometry of the windows does so. Suppose that $\iota\,\mathrm{tr}_W h = \mathrm{tr}_{W'} h'$, that the increments and the earlier terms in W correspond, and that the bare weight corresponds on the window W_t^+ (this window contains $W^{\sim O(1)}$, since the cells of level t are $M\check R_t$-cubes). Then the two histories produce corresponding terms with domains in W, including their natural domains of definition. Here a term is regarded as a partial map on the fields on W, and one of two corresponding terms is defined exactly where the other is. The integration of V_k over the box is the move (3) of §4.17.1.3 and forms part of the

main step of **R**. Consider the part of the regular terms on a domain of level m that this integration takes. It depends only on the trace through the levels $i \geq m$, whatever the values of k, of $i-m$, of the size of the domain and of the volume. The standing operations, that is the $\mathbf{T}'_i(X)$ of the islands carried by a term, are transported likewise, measures included and one factor at a time §4.17.1.45. For the analyticity spaces more is needed. The spaces of two corresponding terms are mapped onto one another if two further conditions hold. First, the bonds of the bare lattice carrying their fields must form corresponding sets. This is needed because whether a bond leaving the domain ends in $\Omega_1^{\sim}$ is decided outside the window; for a domain in Ω_1 there is no condition. Second, at each level, two unit cubes of the bare lattice lying in the domain must belong to a common drop-out mark of that level if and only if their images do. This is needed because a domain that is not convex may cut a mark into two pieces; where no mark meets the domain there is no condition. On the infinite lattice no sum over labels is formed. The only terms formed there are those of the never-large label, the label at which no region stands; domain by domain, these terms contain the sums over healed components that **R** performs (under the exact-operation hypothesis and the locality hypothesis). A term whose window W_t^{+} lies in Λ_t coincides with the corresponding term of that label (the terms belong to the standing part $h^{\flat}$, and everything that this part records lies in Z_t: 2e). For a non-wrapping domain and corresponding step data under the chart, the lifted term is the same function, and in this way $E^{(j)}$ and b_j are computed § 5.2.2. It is this that links the geometry to § 5.2.2 and §5.20.1.2.

The net form of the exponent. Each function $\mathfrak{t}$ of the exponent has a fixed place, given by its level, its domain and the way it is made. The label enters only through the sorting into $\mathbf{E}_k, \mathbf{R}_k, \mathbf{B}_k$, the profile and one number:

$$A_{k,\mathcal{S}}(U) = -\beta A(U) + \sum_{\mathfrak{t}} \mathfrak{t}(U) + \mathrm{const}(\mathcal{S}),$$

because

$$x_k(\cdot) + \sum_{j \leq k} b_j \varphi_j(\cdot) = \beta$$

and $c \mapsto A(c,U)$ is linear. Consequently a change of label changes no local term.

The law of the histories. The set $\mathfrak{H}_k$ of histories of level k is finite. A run has the datum $\mathfrak{d} = (\beta, m, g)$ and takes place on T_{ℓ_m}, $\ell_m = 2L^m$. The mass of a history $h \in \mathfrak{H}_k$ is $\mathfrak{t}(h) := \tau_h(\mathcal{V}_k) \geq 0$. Since the steps are exact §4.17.1.45, $\sum_{h \in \mathfrak{H}_k} \mathfrak{t}(h) = Z_{K,\ell,\beta}$ at each level. The law of the histories of the last level, and in the same way the law of the histories of any level, is

$$\tilde{\nu}_{K,\ell,\beta}(\mathsf{E}) := Z_{K,\ell,\beta}^{-1} \sum_{h \in \mathsf{E}} \mathfrak{t}(h),$$

where events of histories are written in sans-serif letters, $\mathsf{E}_k \subset \mathfrak{H}_k$. The creation set of step j is $\mathcal{W}_j^{\wedge}(h)$, as before. Without the superscript, $\mathcal{W}_K(g)$ denotes the first-passage window. The live region $\mathcal{Z}_k(h)$ is $Z_k(h)$ with the components healed at step k removed; it is the input region of step $k+1$. The event that the cell $\mathfrak{c}$ is live at level k, that is $\mathfrak{c} \subset \mathcal{Z}_k$, is written $\mathfrak{lv}_k(\mathfrak{c})$. For a cell z of level n, the survival event $\mathcal{E}_z^{(n)}(k)$ is the event that $z \subset \mathcal{Z}_i$ for $n \leq i \leq k$. For the flow and history statements below, we continue the run past K. For these statements we fix an anchor coupling $\gamma_\star \in (0, \gamma_J]$, which is determined by the constants up to c_a and by γ_J, and put $X_\star = \gamma_\star^{-2}$. The anchor level is $K_\star = K(\beta, \gamma_\star)$; it goes to infinity as β does. In these statements the reference coupling obeys the ceiling $g \leq \gamma_\star$, and the bounds on the flow hold up to $K_\star$. Also, at a given β, the run passes the window of g no later than the window of $\gamma_\star$, $K(\beta, g) \leq K_\star$; this follows from $g \leq \gamma_\star$ and the monotonicity of the first passage.

Lineages and marks. Let X be a component of $Z_k(h)$, at any level, that is either live at level k or healed at step k; a component live at $k = K$ is a terminal component. Its lineage regions are $\mathsf{Z}_k(X) := X$ and, going down the levels, $\mathsf{Z}_j(X) :=$ the union of all components that are live at level j and contained in $\mathsf{Z}_{j+1}(X)$. Its lineage consists of these components (j, C) together with the creations (j, W) inside the $\mathsf{Z}_j(X)$. The weight of a level-j cell is $\mathsf{r}_j := M^4 \check{R}_j^5$. We write $|C|_j$ for the number of level-j cells in C. For comparison, $\#_j(Y)$ counts the sites of $\mathbb{T}^{(j)}$ in Y, and a cell contains $(M\check{R}_j)^4$ of them. Finally, for a component W of the set that step j creates in the lineage, $d'_j(W)$ is its tree length measured in these cells; the marks set by the operations **R** do not enter Dep. Then

$$\mathrm{Vol}_k(X; h) := \sum_{(j,C) \text{ in the lineage}} 2\,\mathsf{r}_j\,|C|_j,$$

$$\mathrm{Dep}_k(X; h) := \sum_{(j,W) \text{ created in the lineage}} c\,p_0(g_j)^2 (d'_j(W) + 1).$$

The mark at level k has two dials, $\mathsf{w} \in [0, 1)$ and $c_a \geq 0$, the second also called the age constant. The mark and the end mark $\Theta := \Theta^{(K)}$ are

$$\Theta^{(k)}(h) := \sum_{X \text{ live at level } k} \left[\mathsf{w}\,\mathrm{Dep}_k(X; h) + c_a\,\mathrm{Vol}_k(X; h)\right] \geq 0.$$

The mark keeps back the fraction w of the small factor of each creation in such a lineage. It also charges $2c_a$ for every cell, multiplied by the weight of the cell, and for every level. The constant c in Dep is a single number. It is chosen after the fractions of the fluctuation threshold, namely those at which a step evaluates its tests: $1/(4\Theta_1)$ on the collar and θ_c on the rim. It is subject to $c \leq c' \min\{\Theta_1^{-2}, \theta_c^2\}$, where c' is chosen after (A_1, A_0) and contains the ratio $(A_1/A_0)^2$ of δ_k^2/g_k^2 to $p_0(g_k)^2$. No ceiling on g can take the place of this choice, since such a test has a deposit and a gain that are both of degree $2\mathfrak{p}_0$ in $\log x_j$.

Three ages are attached to a component C that is live at level k. (1) The age in the label is determined by the oldest region of the label within C: $\mathrm{age}^{\mathrm{lab}}(C) = k - \min\{t : C \cap Z_t \neq \emptyset\}$, where Z_t denotes the large-field region of the label $\mathcal{S}(h)$ at scale t. (2) The age of the lineage is $\mathrm{age}_k(C; h) = k - j_0$, with j_0 the earliest level that has a non-empty lineage region $\mathsf{Z}_j(C)$. The events $\mathfrak{lv}_k^{=a}(\mathfrak{c})$ and $\mathfrak{lv}_k^{\geq a}(\mathfrak{c})$ say that at level k the cell lies in a live component whose lineage has age a, respectively at least a. (3) The seed age is k minus the seed level. The seed level is the least t such that C contains, as a point set, some region created at step t. The corresponding events are written $\mathfrak{sd}_k^{=a}(\mathfrak{c})$. The first age is at most the second, and the second is at most the third. The second and the third can differ. Suppose that a region was healed at some earlier step and that a later live region again covers its place. Such a region counts for the seed level but is not part of the lineage. If the seed level is defined using the regions that no step before k has healed, the two ages coincide. Payment is made for the age of the lineage alone §4.17.1.45. Age has to be paid for with creations. In a lineage with n creations, those made by operations included, a creation made at step t satisfies $k - t \leq Cn(\mathsf{N}_t + \log n)$. Here a creation is a created cell or a mark. By the waiting bound and the count of §4.17.1.40, this bound holds for every lineage; hence an old lineage contains many creations. These objects enter §4.17.1.45 and §§ 5.20 and 5.22. Through the age series of §§ 5.20 and 5.22 they underlie both the two-point law §6.1.1.1–§6.1.1.11 and the four-point law Thm. A(c)(5), namely clauses (c)(3), (c)(4), (c)(5).

History data, effective densities, and large-field operations. **[1.6] The sequences of a label.** Let $\boldsymbol{\nu}' \leq \boldsymbol{\nu}$. Then $\Omega_l^{\boldsymbol{\nu}}$ is a region for π_l. The chain $\Omega_0 \supset \Omega_1^{\boldsymbol{\nu}} \supset \cdots \supset \Omega_s^{\boldsymbol{\nu}}$ holds, and any two consecutive boundaries in it lie at least one M-cube of the coarser level apart, which is the separation that

minimisers and propagators require. Further, $\Omega_l^{\nu'} \subset \Omega_l^{\nu}$ and $\Omega_l^n = \Omega_l$, so that *the entries make up the determining sequence*. The action regions are unions of coarse cubes,

$$T_\ell \supset \Omega_1^{\mathsf{a}} \supset \Lambda_1^{\mathsf{a}+} \supset \Lambda_1^{\mathsf{a}} \supset \Omega_2^{\mathsf{a}} \supset \cdots \supset \Lambda_s^{\mathsf{a}},$$
$$\Omega_{l+1}^{\mathsf{a}} \subset (\Lambda_l^{\mathsf{a}})^{\sim -1},$$
$$Z_s^{\mathsf{a}} = Z_s,$$

and the correspondence (entries, stored regions) $\leftrightarrow$ (action regions, nest) gives two presentations of one and the same datum.

4.17.1.3. Stored variables and integration in a large-field patch. **Measure and formula: two layers.** At the level of measures, a step acts as a finite sequence of moves on states $\sigma = (h, \xi)$, where h records the outcomes obtained up to that point and ξ the variables stored up to that point. Each move is given by positive kernels $\Pi^{(o)}$, finitely many and one for each outcome o, such that $\sum_o \Pi^{(o)}(\sigma, \cdot)$ has total mass one. A move is one of four kinds. It may be a test, that is, an indicator whose outcome is added to h. It may be a map. It may be a draw, that is, the image of a probability law, as for gauge fixings, hypercubic frames $r \in W_4$ and reference laws. Or it may delete a stored variable, which amounts to an integration. Everything expressed by a formula, such as Gaussian weights, Jacobians, expansions, resummations or the replacement of a test by 1, only recasts an integrand in another form. A gauge fixing is a draw as a move. For it to be the factor that the formula writes, the measure must be gauge invariant. Let ν_0 be any complex bare measure, let $\mu_0 \geq 0$, and let $\mathfrak{w}$ be any bounded Borel function of the bare field. Denote by K_h the kernel of the history h, obtained by composing the kernels $\Pi^{(o)}$ of its moves along its outcomes. Then

$$\tau_h[\nu_0] = \int \nu_0(dU)\, \mathsf{K}_h(U, \cdot),$$
$$\sum_h \mathsf{K}_h(U, \mathcal{V}_k) = 1 \ \forall U,$$
$$\sum_h |\tau_h|(\mathcal{V}_k) \leq |\nu_0|(\mathcal{V}_0),$$
$$\big|\tau_h[\mathfrak{w}\mu_0]\big| \leq \sup|\mathfrak{w}|\, \tau_h[\mu_0].$$

By totality (2a; a reference law whose normalisation does not lie in $(0, \infty)$ is not drawn, and in that case the component drops out; this case does not arise on the supports, see §4.17.1.45, "Reality and positivity"), every kernel is defined at every field, and no smallness condition is needed. They depend on the lattices and the constants. Of the frame θ (2e: the real sequence from which the choices are formed; in the present case $\theta = x$, and it is only through this frame that the kernels then vary with β; it is unrelated to the hypercubic frames r), they use only the gauge-fixing law, the coupling of the reference law, the cube sizes and the thresholds. Apart from their dependence through the frame, the kernels do not depend on any measure, on β, on a source, on any term of an effective action, or on whether some series converges. Exactness and positivity therefore come at no cost, because a move does nothing but split mass. The sourced densities come from $\nu_0 = \rho_0^w\, dU$ with the same kernels §5.20.1.1, and runs whose choices are frozen are runs of that same procedure § 5.2.3. The estimates of §4.17.1.40–§4.17.1.45 add only one further statement: the formulas are formed on the supports and there coincide with the measures. *Passengers*: divide the stored variables into those, $\mathfrak{v}$, that a move uses and the remaining ones, $\mathfrak{q}$. In step $t = k + 1$ the operation $\mathbf{T}$ uses the fields of level k and, from the lower levels, only the data of its backgrounds; the operation $\mathbf{R}$ uses variables only in the components of its class and next to them. Any variable not used by a move of the step is carried along unchanged,

$$\Pi^{(o)}((h,(\mathfrak{v},\mathfrak{q})),\cdot)=\Pi^{\flat(o)}((h,\mathfrak{v}),\cdot)\otimes\delta_{\mathfrak{q}}$$

, in which $\Pi^{\flat(o)}((h,\mathfrak{v}),\cdot)$ is a measure on the states $((h,o),\mathfrak{v}')$. This is the locality of a step, in measure-theoretic form.

T on a label (Bałaban, Comm. Math. Phys. 119 (1988) 243, Sect. 3) comprises the following, carried out on the regions fixed by the tests of §4.17.1.1. First the multi-scale background $U_{j+1}(\mathbf{V})$ is formed, on the determining set $\mathbf{B}[\Omega_\bullet]$ § 3.11; the third test of §4.17.1.1 is formed with this background. Next, a change of variables is made on the strip. Then a conditional Gaussian integral is taken over Λ_{j+1}. Finally, the logarithm of this integral is split into E, R and B. The operation sits between § 4.5, in which no regions occur, and the operation **R**.

The strip. The tests kept on $S_{j+1}\subset\Omega_{j+1}\setminus\Lambda_{j+1}$ depend on V_{j+1} non-locally, by way of V_j^{bg}. For $j\geq 1$ use the joined background on $Z_{j+1}\sqcap\Lambda_j$ with $Z_{j+1}=\Lambda_{j+1}^c$, and denote its average by V_j^{loc}. At the first step use the one-part localization. Then

$$V_j^{\mathrm{bg}}(V_j^{\mathrm{loc}})^{-1}=e^{i\mathbf{H}^{(j)}},$$

$$|\mathbf{H}^{(j)}|\leq C\varepsilon_j e^{-\check{R}_j};$$

$$V_j(b)=W' e^{-i\vartheta_{\mathrm{co}}(W')\mathbf{H}^{(j)}(b)}e^{i\mathbf{H}^{(j)}(b)}\,V_j^{\mathrm{loc}}(b),$$

The last line is a bondwise change of variables with $W'=e^{i\mathcal{A}'(b)}$. In it, ϑ_{co} is a smooth cut-off in $|\log W'|$. It equals zero for angles at most $4\delta_j/3$ and one for angles at least $5\delta_j/3$. This cut-off is applied on the cut-off zone of the strip (§4.17.1.1); at the remaining bonds where the substitution is made, $\vartheta_{\mathrm{co}}\equiv 1$. Consequently, on the cut-off zone, a test whose threshold does not exceed δ_j is the same function of the new variable as of the old one, while a test at $2\delta_j$ is now made against the localised background. Because $|\mathbf{H}^{(j)}|\ll\delta_j$, a contraction argument shows that the map on a bond is a diffeomorphism of G. Its Jacobian equals 1 wherever ϑ_{co} is locally constant, and it contributes to the action as a boundary term. Once the substitution has been made, the factors of the kept tests at those bonds next to Λ_{j+1} where no substitution takes place are omitted. On the supports these factors equal 1, and where $\vartheta_{\mathrm{co}}<1$, the $2\delta_j$ check relative to either strip background passes by the angle bound below. Moreover, at a pivot bond of Ω_{j+1} the value of V_j is taken from the constraint $\mathcal{M}(V_j)=V_{j+1}$. The strip tests after substitution have the support described below. Factorization of the integral operations also uses the locality of the integration domain and the full kernel.

Write $W=V_j(b)V_j^{\mathrm{bg}}(b)^{-1}$ and $H=\mathbf{H}^{(j)}(b)$. In the strip change of variables the backgrounds are held fixed while the bond variables are changed. The displayed substitution gives

$$W=W'\exp(-\mathrm{i}\vartheta_{\mathrm{co}}(W')H).$$

For $W\in\mathrm{SU}(2)$ let $\varphi(W)\in[0,\pi]$ be the angle of its eigenvalues $e^{\pm\mathrm{i}\varphi(W)}$. In particular, $\varphi(-\mathbf{1})=\pi$. Off $-\mathbf{1}$ the angle is the norm of $(1/\mathrm{i})\log W$ with the intrinsic logarithm. On the cut-off zone the cut-off vanishes for $\varphi(W')\leq 4\delta_j/3$ and equals one for $\varphi(W')\geq 5\delta_j/3$. Here $\delta_j\leq 1$, and we use the strip bound with $|H|\leq\delta_j/8$. The triangle inequality for the angle gives $\varphi(W)\geq\varphi(W')-|H|$. If $\varphi(W')<4\delta_j/3$, the substitution leaves $W=W'$. Otherwise

$$\varphi(W)\geq(4/3-1/8)\delta_j=29\delta_j/24.$$

Consequently, for $0<a\leq 29/24$, $\varphi(W)<a\delta_j$ and $\varphi(W')<a\delta_j$ are equivalent. If the test passes, $W=W'\neq-\mathbf{1}$. The strict inequality gives the same conclusion at the upper allowed threshold. This proves the assertion for the products and differences of sharp bond tests whose bonds are all either in the cut-off zone or unchanged. At unchanged bonds the test is literally the old one. On

the cutoff zone where $\vartheta_{\mathrm{co}} < 1$, the upper triangle inequality gives $\varphi(W) < (5/3 + 1/8)\delta_j < 2\delta_j$. Also $V_j(V_j^{\mathrm{loc}})^{-1} = W' e^{i(1-\vartheta_{\mathrm{co}})H}$, so its angle has the same upper bound. For an approximate test based on the test cube's own background, use the comparison with the localized strip background below.

4.17.1.4. The strip Jacobian. For fixed H write $\vartheta = \vartheta_{\mathrm{co}}$ and $\Psi(W') = W' \exp(-i\vartheta(W')H)$. The map is smooth on the whole group because the cutoff is constant near both ends of the angle chart. For given W its inverse solves $W' = W \exp(i\vartheta(W')H)$. With the bi-invariant metric this is a contraction when $\|d\vartheta\|_\infty |H| < 1$. The cutoff derivative costs C/δ_j, so the real shift bound is used with $C|H|/\delta_j \le 1/2$.

Measurability of the partial inverse holds on the full unique-solution set. The graph $\Psi_H(W') = W$ is closed in the variables (H, W, W'), and the last variable ranges over the compact group. Its projection is closed. The parameters with two different solutions are the union, over $n \ge 1$, of the projections with two solutions at distance at least $1/n$. Each of these projections is closed by compactness of the two group variables. The unique-solution set is therefore Borel. Intersecting the graph with W' in a closed subset of the group shows that the recovered value is Borel there. A fixed failure output on its complement is measurable.

When a change of measure requires a diffeomorphism, also require injectivity, surjectivity and nonvanishing Jacobian everywhere. Injectivity is the complement of the same countable family of collision projections, now over H. The image is compact, so surjectivity is equivalent to containing a countable dense set of target values. Nonvanishing of the Jacobian everywhere is open by compactness. This gives a Borel diffeomorphism domain. The inverse is jointly continuous on it, since any converging sequence of parameters and target values has inverse subsequential limits solving the same unique equation.

In left-trivialized tangent coordinates the differential is

$$X \longmapsto \mathrm{Ad}(\exp(i\vartheta H))X - iH\, d\vartheta_{W'}(W'X).$$

Factor out the adjoint, which has determinant one and fixes iH. The remaining rank-one determinant gives the Haar Jacobian for dW in the variable W' is

$$\mathcal{J}_b(W') = 1 - d\vartheta_{W'}(W'iH) > 0.$$

Its logarithm vanishes off the cutoff shell. At fixed real W' it is holomorphic in H on the smaller complex ball where $C|H|/\delta_j < 1$. Here the real differential in the last display is extended complex linearly.

For the local expansion take a decomposition $H = \sum_Y H_Y$ in which the carriers all contain the bond read and meet a boundary cube. Write $\ell(Y)$ for the size of a shortest connecting tree in the cell grid, including the fixed read. If $\sum_Y e^{\mu\ell(Y)}|H_Y|$ is available and its product with C/δ_j is less than $1/2$, expand $\log(1-u) = -\sum_{n\ge1} u^n/n$ with $u = d\vartheta_{W'}(W'iH)$. Put a product on the union of its carriers. Because they share the read, the size of a tree for their union is no more than the sum of their tree sizes. The absolute sum at weight $e^{\mu\ell}$ is then at most $2(C/\delta_j)\sum_Y e^{\mu\ell(Y)}|H_Y|$. The result still has a boundary cube on every carrier. It uses the holomorphic shift expansion on the admitted pair domain, while the cutoff variable itself stays real.

Where $\vartheta_{\mathrm{co}} = 1$, the substitution gives $V_j(b) = W' V_j^{\mathrm{loc}}(b)$. In an approximate fluctuation test at such a bond, insert this expression for $V_j(b)$. The localized datum of a test cube $\square$ uses only the level-$(j+1)$ bonds with both ends in $\square^{\sim 4}$. Agreement on those bonds fixes both whether the local background is defined and its value in the test. To see that these bonds lie in the retained region, put $t = j+1$ and use level-j units. The test cube has side $L^2 M_2 \check{R}_t$, its layers have width $LM_2\check{R}_t$, and a level-t cell has side $LM\check{R}_t$. A bond being substituted has an end z in the strip $(\Omega_t \setminus \Omega_t^{\sim -1}) \cup R_t$. If the bond occurs in the test on $\square$, its other end, or z itself, lies in $\square^{\sim 2}$. Each point of $\square^{\sim 4}$ is thus at distance at most $(L^2 + 6L)M_2\check{R}_t + 1 < LM\check{R}_t$ from z, using $L \ge 4$ and $L^2 M_2 \le M$. If z lies in the outer layer of Ω_t, its one-cell enlargement lies in $(\Omega_t^c)^{\sim 2}$. If it lies in R_t,

that enlargement lies in the one-cell enlargement of its hull. Both regions belong to the completed stored region Z_t, so $\square^{\sim 4} \subset Z_t$. The strip construction here requires V_j^{loc} to depend on V_t only through the bonds meeting $\Omega_t \cap Z_t$. These restrictions determine the transformed test including its zero value when the local background is undefined. The statement concerns this change of variables. Independence under a later integration requires the corresponding property of that integration's domain and kernel.

4.17.1.5. The local strip background. For a step $t = j + 1 \geq 2$ use the joined region $W = Z_t \sqcap \Lambda_j$, with Z_t a part at level t and Λ_j a part at level j. Here we use the step geometry in which their four-layer collars of M_1-cubes lie in the corresponding ambient layers, with the separation of the collars required for the join. The one-cell enlargement of the proposed strip is in Z_t, and the top torus has at least 6ρ sites in each direction. Use either admitted first member of the joined sequence. If the level-j curvature in W is at most ε' and $V_t = \mathcal{M}(V_j)$ on the level-t bonds of W, the joined test bound is $\max\{2L^2, \mathsf{k}_L\}\varepsilon'$ for sufficiently small ε'. At level j the tested holonomy is the one of V_j. At level t both types of bond in a test give the same values of $\mathcal{M}(V_j)$, using the exact constraint for the given bonds and the same average for the mixed ones. Its plaquette deviation is at most $L^2\varepsilon' + C_L(\varepsilon')^2$. The face lift transfers this test bound to the joined datum. Thus the joined datum has a real regular critical orbit under the background smallness and torus conditions.

Normalize this orbit by setting its averaged contour transports from the strip block centers equal to one at the level-j sites that are free to be gauged. A site that is a node of the joined determining set already has its gauge fixed. The centers themselves are nodes. The residual gauge relating two normalized representatives therefore equals one at the free sites whose contour transports were normalized as well. The level-j averages on the strip bonds, whose blocks lie in the joined first member and whose endpoints are such sites or nodes, are consequently independent of the representative. This gives V_j^{loc} from the retained real datum. For a union of components of W the same argument restricts to that union. The determining bonds and contour blocks used here belong to individual components.

4.17.1.6. The shift in the block axial frames. For a strip bond b, include in its read set the double block of the level-j average and the contour blocks used to fix the gauges of its two endpoints. Let D_b be the least distance to W^c on the bare averaging carriers of this set, in the ambient weighted distance, also including the order-q blocks of the fixed sites at their own levels $q \in \{j, t\}$. Consider a read on which the full datum is in the real comparison class of eq. 3.cdii, with test radius ε_1. Use a read set inside the common interior of the full and joined sequences. The two sequences use the same block centres and fix the same endpoints there. For each of these fixed sites, use the node comparison of the real collar estimate. In particular these sites must be outside the wide collars.

Write the restriction of the full field as E^w, where $E = e^{i\eta f} U_W^\circ$ in the notation of the collar comparison. The level-j fields $\mathcal{M}^j E$ and $\mathcal{M}^j U_W^\circ$ differ on the read set by at most

$$e_b = C\varepsilon_1 e^{-\delta_0 D_b/2}.$$

Indeed the bare field has plaquette bound $C\varepsilon_1 L^{-2j}$ on these blocks, since both layers are of level at least j. The relative perturbation, in the units of level j, is bounded by $C\varepsilon_1 e^{-\delta_0 D_b/2}$ by eq. 3.cdii. Apply the local average estimate of Prop. 3.6 to the full double blocks, using the bound for both the straightened angle and its endpoint factors. The same estimate at the fixed sites gives $|w(z) - 1| \leq Ce_b$ there, by $w(z) = v_q[f](z)^{-1}$. The order-q spacing is at most L times that of the level-j blocks. The bound for f is measured at the corresponding tops, at weighted distance $O(1)$ from these blocks. The contour means of the two fields in the comparison frame differ by at most $C_L e_b$, since their transports are products of bounded length and the derivative of the local mean is bounded on its small-diameter chart.

For a free site x with centre y write $T_E(y, x)$ for the contour mean of $\mathcal{M}^j E$. The normalization of the full field uses $w(y)T_E(y, x)w(x)^{-1}$ at x, by gauge covariance. Its composition with w is therefore $w(y)T_E(y, x)$. The joined field uses $T_{U_W^\circ}(y, x)$. The two factors differ by at most $C_L e_b$

since y is fixed and $w(y)$ is already close to one. At a fixed endpoint the only comparison factor is $w(x)$. Taking the product at the two endpoints of b proves

$$|V_j^{\mathrm{bg}}(b)V_j^{\mathrm{loc}}(b)^{-1}-1|\le C_L e_b.$$

The intrinsic logarithm has the same bound up to a constant. There is no need to estimate the unconstrained values of w.

On the strip read sets with a full layer of level-t cells between their bond and Z_t^c, the read carriers use only $O(L^t)$ of this margin. For the part of W^c coming from Λ_j^c, use the nested margin of Ω_t inside Λ_j, with the same allowance for the read carriers. The largest layer in the ambient sequence is t, so a physical margin $cM\check{R}_tL^t$ gives $D_b\ge cM\check{R}_t$ in the weighted distance. This yields the small real shift for such reads, with the prefactor given by the real datum radius. Taking the fixed M large enough gives the cutoff bound at small coupling. For each use we must check the full carriers and the common fixed-site clause. The real endpoint estimate does not supply the holomorphic expansion of the shift on an independent current domain used to transfer the boundary terms.

The datum in the response equation must be distinguished from the physical change of the averages. In the real collar comparison write $\widehat{U}=(U|_W)^g$ for the aligned restriction and $\widehat{V}=M_{\mathbf{B}^W}(\widehat{U})$. The alignment gives

$$\widehat{V}=e^{iB_1}V^W,\qquad B_1(c)=\frac{1}{i}\log\Big[\mathcal{M}^q(U^{v_a})(c)V_q^W(c)^{-1}\Big]$$

on the positively oriented collar bonds, and $B_1=0$ on the remaining determining bonds. Here c has level q and belongs to the collar of the part a. The physical increment that changes $\widehat{V}$ into V^W is therefore $b=-B_1$. For the response built at $\widehat{U}$, its row input is $B'=b_\natural$, with the conversion formed using $\widehat{V}$. At an interior row this is b; at an outward row it includes the endpoint factor of that conversion. In particular, the support of B' may include the outward rows whose endpoint blocks meet the support of b.

This conversion is holomorphic wherever its stated pair and logarithm bounds hold, and satisfies

$$|b_\natural|_y\le C_{\natural,\mathrm{s}}\max_{y'\in N_L(y)}|b|_{y'}.$$

It identifies the row input once an admitted physical defect b is given. To apply the complex response theorem uniformly, the aligned pair must lie in its joined local class and the converted input must satisfy

$$\|b_\natural\|_\infty<\varepsilon_{\mathrm{D,s}}e^{-32\varkappa}.$$

The real collar alignment supplies a real defect of size $C\varepsilon_1$.

4.17.1.7. The input on a fixed real collar chart. Fix the real configuration U used in the collar comparison. Let g_0 be its real alignment, equal to the supplied v_a on each wide collar Q_a and to one elsewhere. Keep g_0 fixed when the complex perturbation varies. On a part of level $j_a=j$, use level-j units and write

$$\mathbf{U}=e^{i\eta_jA'}U,\qquad \|A'\|_\infty\le\alpha_1,\qquad \eta_j=L^{-j}.$$

The full averaging carriers considered below lie in the collar region where the real plaquette bound is $|\partial U-1|\le C\varepsilon_1\eta_j^2$. Take $C\varepsilon_1$ and α_1 below the fixed averaging ceilings. Since g_0 is real, the same bounds hold after its action, with A' replaced by $\mathrm{Ad}(g_0)A'$.

The complex averaging formula, applied with real reference U^{g_0}, gives, for $0\le q\le j$,

$$\mathcal{M}^q(\mathbf{U}^{g_0})(c)=v_q(c_-)e^{ia_q(c)}\mathcal{M}^q(U^{g_0})(c)v_q(c_+)^{-1},$$

where the functions are holomorphic in A' and

$$|a_q|\le 2L^{q-j}\alpha_1,\qquad |v_q^{\pm1}-1|\le\frac{8}{Lr_*}L^{q-j}\alpha_1.$$

For $q = 0$ the site factors equal one. The central average is unitary. Subtracting it in the preceding formula and using the three displayed bounds therefore gives

$$\left|\mathcal{M}^q(\mathbf{U}^{g_0})(c) - \mathcal{M}^q(U^{g_0})(c)\right| \le C_L L^{q-j}\alpha_1.$$

This estimate includes the endpoint factors. Its constant is independent of j and q.

Form the joined datum $V^W[\mathbf{U}]$ by the same rule as at the real center: use the unchanged common rows and the face lifts of the top averages on the collar rows. At a collar row, the lift is either one or the value of a single top averaged bond. The estimate at $q = j$ consequently gives

$$|V^W[\mathbf{U}](c) - V^W[U](c)| \le C_L\alpha_1.$$

Set $\widehat{V}[\mathbf{U}] = M_{\mathbf{B}^W}((\mathbf{U}|_W)^{g_0})$. On the collar rows define

$$b[\mathbf{U}](c) = \frac{1}{i}\log\Big[V^W[\mathbf{U}](c)\widehat{V}[\mathbf{U}](c)^{-1}\Big].$$

At the center, $\widehat{V}[U] = e^{iB_1}V^W[U]$, so this logarithm is $-B_1$. For sufficiently small $C\varepsilon_1 + \alpha_1$, its argument stays in a fixed series-logarithm ball around one, and

$$\|b[\mathbf{U}]\|_\infty \le C_L(\varepsilon_1 + \alpha_1), \qquad b[U] = -B_1.$$

Indeed both matrix factors differ by $O(\alpha_1)$ from their unitary center values, and the center product is e^{-iB_1}. Inversion and the series logarithm are uniformly bounded and holomorphic on this smaller ball.

On a common row the alignment is one at both endpoints, by the collar geometry. Gauge covariance under the fixed real g_0 thus gives $\widehat{V}[\mathbf{U}] = V^W[\mathbf{U}]$ there. Set $b = 0$ on those rows. This proves exact support on the collar rows and the identity

$$e^{ib[\mathbf{U}]}\widehat{V}[\mathbf{U}] = V^W[\mathbf{U}].$$

For a closed set of response rows, the converted datum $B' = b_\natural$, formed at $\widehat{V}[\mathbf{U}]$, is holomorphic and has norm at most $C_{\natural,\mathrm{s}}C_L(\varepsilon_1 + \alpha_1)$. Its carriers include the endpoint blocks read by this conversion. The response radius is met when this bound is smaller than $\varepsilon_{\mathrm{D},\mathrm{s}}e^{-32\varkappa}$.

The current remains an independent variable. First form the restricted pair by keeping $\mathbf{J}$ at the bonds whose incident plaquettes all remain inside the joined first member, and by using the current of the restricted action only on its free boundary. Then apply the fixed real gauge g_0 to both members of this pair. The defect above does not depend on $\mathbf{J}$. Thus its holomorphy imposes no equation $\mathbf{J} = J(\mathbf{U})$ in the interior. The response still requires this restricted and gauged pair to belong to its joined local class.

4.17.1.8. A target defined by the complex configuration. The tower transports give a choice of target that depends only on the complex configuration. The response can remain based at the original restricted pair.

On a wide collar Q_a of level j, write $\mathbf{W}_q = \mathcal{M}^q\mathbf{U}$. For a bare site x let x_q be its ancestor at level q, so that $x \in B^q(x_q)$, and put $y = x_j$. Define

$$g[\mathbf{U}](x) = \mathsf{T}_{\mathbf{U}}(y, x) := \mathbf{W}_{j-1}(y, x_{j-1})\mathbf{W}_{j-2}(x_{j-1}, x_{j-2})\cdots\mathbf{W}_0(x_1, x).$$

Set $g[\mathbf{U}] = 1$ off the collars and at their top sites; for $j = 0$ the product is empty. This is the supplied tower transport, using the averaged contour variables of the same averaging scheme. Its factors depend only on the configuration in $B^j(y)$. On their averaging domain they are holomorphic functions of $\mathbf{U}$ itself. At a real configuration U this is exactly the descending collar alignment g_0.

The tower estimate is uniform in its depth. In the local representation $\mathbf{U} = e^{i\eta_j A'}U$ used above, with $\|A'\|_\infty \le \alpha_1$, the supplied product formula writes

$$E(x) := g[\mathbf{U}](x)g_0(x)^{-1} = \prod_{q=0}^{j-1} e^{Z_q(x)}, \qquad \sum_q |Z_q(x)| \le z := 2\kappa_6 C''_{\mathrm{mav}}L^{-1}\alpha_1 \le \tfrac{1}{8}.$$

The order of the factors is the one in the tower product. Thus $|E^{\pm1}-1| \le e^z - 1 \le C_L\alpha_1$. The real matrices g_0 are unitary, so these bounds also give uniform bounds on $g[\mathbf{U}]$ and its inverse. Every full averaging carrier used here is included in the collar carrier, as in the preceding chart estimate.

Let

$$V_* = M_{\mathbf{B}^W}(\mathbf{U}|_W), \qquad T = V^W[\mathbf{U}], \qquad R[\mathbf{U}] = T^{g[\mathbf{U}]^{-1}}.$$

The last expression is just the action on the endpoints of the determining rows. Define the physical input at the original restricted pair by

$$b_{\mathrm{can}}(c) = \frac{1}{i}\log\bigl(R[\mathbf{U}](c)V_*(c)^{-1}\bigr).$$

This expression is holomorphic and has the same uniform size as the previous chart input. To check this, set $\widehat{V}_0 = V_*^{g_0}$. At a row c, its logarithm's argument equals

$$g_0(c_-)^{-1}\bigl[E(c_-)^{-1}T(c)E(c_+)\widehat{V}_0(c)^{-1}\bigr]g_0(c_-).$$

The preceding chart calculation bounds $T\widehat{V}_0^{-1} - 1$ by $C_L(\varepsilon_1+\alpha_1)$. The bounds on $E^{\pm1}$, together with the bounded matrix factors in that chart, give the same bound for the bracket minus one. On the fixed logarithm ball this proves

$$\|b_{\mathrm{can}}\|_\infty \le C_L(\varepsilon_1+\alpha_1), \qquad e^{ib_{\mathrm{can}}}V_* = R[\mathbf{U}].$$

At both endpoints of a common row the tower gauge is one, by the same collar geometry used for the real alignment. There $T = V_*$ and $b_{\mathrm{can}} = 0$. Its possible nonzero rows are therefore collar rows. Form $B'_{\mathrm{can}} = (b_{\mathrm{can}})_\natural$ at the original restricted pair. On a closed determining row set, the conversion is holomorphic and gives

$$\|B'_{\mathrm{can}}\|_\infty \le C_{\natural,\mathrm{s}}C_L(\varepsilon_1+\alpha_1).$$

At a collar row c of level $q < j$, let $y_\pm$ be the level-j ancestors of $c_\pm$. Its physical input is determined by the bare configuration in $B^j(y_-)\cup B^j(y_+)$: these are the two tower blocks, the lower averaging blocks lie inside them, and the face lift reads either one or their common top bond. Both blocks lie in Q_a. The converted input also reads the neighboring physical rows and the endpoint blocks specified by the conversion. For a localized term these complete carriers must lie in its admitted enlarged domain. The response is defined when the displayed norm fits its radius and the original restricted pair is admitted to the joined class. The response is evaluated directly at that pair.

This target also gives the required gauge transformation law. For a real gauge u, define $\bar{u}(x) = u(y)$ in a top block $B^j(y)$ of a collar and $\bar{u} = u$ elsewhere. Covariance of the averaged contour variables cancels the intermediate factors in the tower product and gives

$$g[\mathbf{U}^u] = \bar{u}\,g[\mathbf{U}]u^{-1}.$$

The face lift is one inside a top block and takes the value of one top bond across its face. Hence its target transforms by $T[\mathbf{U}^u] = T[\mathbf{U}]^{\bar{u}}$. The same identity holds on common rows, whose endpoints in a collar are top sites. Consequently

$$R[\mathbf{U}^u] = R[\mathbf{U}]^u, \qquad b_{\mathrm{can}}[\mathbf{U}^u] = \mathrm{Ad}(u)b_{\mathrm{can}}[\mathbf{U}].$$

Covariance of the conversion and of the response applies at the unchanged restricted pair, with every interior current entry still independent. The resulting response is thus a holomorphic function of the pair and is covariant for all admitted complex cube parameters. Its definition is identical in overlapping real factorizations, since every factor in the input is a function of $\mathbf{U}$ itself.

At a real center, $g[U] = g_0$ and $\widehat{V}_0 = e^{iB_1}V^W[U]$. It follows that

$$b_{\mathrm{can}}[U] = \mathrm{Ad}(g_0^{-1})(-B_1).$$

The converted inputs obey the same identity, with the conversion on each side taken at its own base pair. Real-gauge covariance of the response therefore identifies this response with the earlier aligned response, conjugated by g_0^{-1}. On a full read star outside the collars, $g_0 = 1$, so their

covariantly formed dressed reads agree in the same fixed frame. For other read frames the explicit frame factors are retained.

4.17.1.9. The rows and their complete carriers. Use the rows prescribed by the decoupled response theorem: the positively oriented interior and outward bonds of the joined determining set $(\mathbf{B}^{W})^{\flat}$. These rows form a closed set. Indeed, if a row of level q has outer endpoint z, every positive level-$(q-1)$ bond joining sites of $B(z)$ is an interior row of the joined set. These are exactly the lower rows required by the conversion $b \mapsto b_{\natural}$.

Let $\mathfrak{k}$ be an active wall-component of the parameter partition $\mathsf{P} = \pi[\mathsf{w}]$, and put $Y = \bigcup \mathfrak{k}$. Work in level-t units. Every collar top level satisfies $j \le t$, so its block has side $\xi_j = L^{j-t} \le 1 \le \mathsf{w}$. Both side lengths are powers of L on the same centered grids. The nesting of standard partitions has the following consequence: if a bare lattice site x lies in Y, then the entire level-j block containing x lies in Y. To see this, the site belongs to one parameter cube, since lattice sites do not lie on its boundary walls. That cube is a union of complete level-j blocks. Its unique such block containing x is therefore contained in it.

Write $\mathbb{Y}^{\circ}_{\mathfrak{k}}$ for the sites carried by the component's walk terms. The response uses its source only at rows at $N_L(\mathbb{Y}^{\circ}_{\mathfrak{k}})$. The conversion at such a row uses physical inputs only at rows at $N_L^2(\mathbb{Y}^{\circ}_{\mathfrak{k}})$. The endpoints of these rows belong to $N_L^3(\mathbb{Y}^{\circ}_{\mathfrak{k}})$: for an interior row the neighboring endpoint has an adjacent block; for an outward row this is the outer-end assertion $B(z) \subset N_L(y_c)$. The margin lemma places the blocks of all sites in $N_L^4(\mathbb{Y}^{\circ}_{\mathfrak{k}})$ in the interior of Y.

Consequently every endpoint just listed lies in Y, and the partition argument places its complete collar top block in Y. Thus every tower transport and every top average used by a required physical input is read inside Y. The lower averaging carriers and the endpoint-block carriers of the conversion are inside Y by the same argument. The complete canonical input carriers thus lie in the active component even when a top block is larger than the joined row's own scale.

The original collar geometry is retained in this argument. At a noncommon row, its two endpoints lie in the wide collar of one part, in the original layer j, and the complete ancestor blocks lie in that collar. The original and joined region traces therefore specify the common-row classification, this original level and the tower blocks used in the source. When these traces agree on Y, the canonical inputs read by the component agree there as well. The margin comparison for the response then applies. This uses the original traces in addition to the joined traces; the latter alone need not identify the collar top level.

4.17.1.10. The source radius. Let K_B denote the constant in the canonical input bound, including the conversion and the fixed changes of local units, so that

$$\|B'_{\mathrm{can}}\|_{\infty} \le K_B(\alpha_0 + \alpha_1).$$

Suppose that the joined pair has the fixed configuration radii required by the response theorem and current radius at most $K_J\alpha_0$. In the strip domain $\alpha_0 + \alpha_1 \le ce^{-C\varkappa}$, choose $C \ge 32$ and

$$0 < c < \min\left\{c_0, \frac{\varepsilon_{\mathrm{D,s}}}{2\max\{K_B, K_J, 1\}}\right\},$$

where c_0 is smaller than the averaging and physical logarithm ceilings used in the canonical construction. Then both the converted source and the current radius are smaller than $\frac{1}{2}\varepsilon_{\mathrm{D,s}}e^{-32\varkappa}$. The constants are chosen after the geometric scales and before the final small-coupling threshold, as in the strip domain. Both radii enter the boundary size. Taking the infimum over the admissible radii gives

$$\|B'_{\mathrm{can}}\|_{\infty} \le K_B\mathfrak{n}_W(P).$$

The input has no cube-parameter dependence. Its radius and carrier bounds consequently hold throughout the parameter polydisc of the response theorem.

4.17.1.11. The retained current and the free boundary. The joined sequence is finer than the original sequence. At a site where its layer is q and the original layer is j, one has $q \le j$ and $\xi_q \le \xi_j$. Consequently the original pointwise bounds on curvature, the complex potential and its covariant difference, and the independent current imply the corresponding joined pointwise bounds at every retained stencil. This assertion concerns the pointwise clauses; local gauges are a separate clause of the pair class.

At a free-boundary bond replace the current by the first variation of the restricted plaquette action, as prescribed above. For any set of retained plaquettes the supplied first-variation formula is

$$J^\flat(\mathbf{U}) = D_{\mathbf{U}}^{\mathsf{T}} F_{\mathbf{U}}, \qquad F_{\mathbf{U}}(p) = \frac{\partial \mathbf{U}(p) - \partial \mathbf{U}(p)^{-1}}{2i\xi^2}.$$

Each bond receives at most six plaquette contributions. The scaled curvature bound implies $|F_{\mathbf{U}}(p)| \le C\alpha_0 \xi^{-2}$, and the one-bond transports in $D_{\mathbf{U}}^{\mathsf{T}}$ are uniformly bounded by the complex-potential ceiling. It follows that $|J^\flat(\mathbf{U})(b)| \le C\alpha_0 \xi^{-3}$. A free-boundary bond has a missing incident plaquette within two bare spacings. Separation of the first two joined regions therefore places its initial site in joined layer zero. Since $\xi_0 = \xi$, the bound reads

$$\xi_0^3 |J^\flat(\mathbf{U})(b)| \le C\alpha_0.$$

Together with the retained-current bound this gives the strip current radius used above.

4.17.1.12. The local gauges on the joined cubes. For a triple in the structured domain on the full carrier, the pointwise and good-gauge clauses in §2.2.4.1 give the original enlarged local class. Restricting to the active parameter union keeps the same splitting and all the inner stencils. The gauges of the joined class are then obtained from the original gauges. We use the enlarged class $\mathfrak{P}_t^\natural$, whose gauge cubes of index q are all the cubes of side at most $12\mathfrak{M}\xi_q$ that are unions of $\mathfrak{M}$-cubes at that level and lie in $\mathcal{L}_q \cup \mathcal{L}_{q+1}$. Such a cube need not meet $\mathcal{L}_q$.

First recall the supplied truncation argument. Removing the last member from a separated sequence carries $\mathfrak{P}_{k+1}^\natural$ into $\mathfrak{P}_k^\natural$ with the same radii and the same decomposition $\mathbf{U} = e^{i\xi A'} U$. All scaled inequalities can be read in the same original units throughout. For the gauge clause, only a cube q of index $k-1$ which meets the former last region Ω_{k+1} needs attention. Its side satisfies

$$s(\mathsf{q}) \le 12\mathfrak{M}\xi_{k-1} < \mathfrak{M}\xi_k.$$

Cover it by a cube q' of side $2\mathfrak{M}\xi_k$, made of 2^4 cubes at scale $\mathfrak{M}\xi_k$. If $x \in \mathsf{q} \cap \Omega_{k+1}$, every point of q' is within $2\mathfrak{M}\xi_k$ of x. Separation therefore puts q' in $\Omega_k = \mathcal{L}_k \cup \mathcal{L}_{k+1}$. The original gauge clause at q' gives the gauge on q, with

$$\xi_{k-1}|A| \le L^{-1} a_0, \qquad \xi_{k-1}^2 |\nabla^\xi A| \le L^{-2} a_0,$$

where a_0 is its original gauge radius. Thus no loss accumulates under successive truncations.

The same argument applies to the local class on an active union Y of parameter cubes. Here it is necessary to choose q' inside Y. This is possible on the actual nested grids. Put $m = \mathfrak{M}\xi_k$; then m divides w and $\mathsf{w} \ge 2m$. The parameter cubes whose interiors meet q form a rectangular block of cubes, all belonging to Y. In each coordinate the interval of q, whose length is less than m, meets at most two m-grid intervals. If it meets only one, include an adjacent interval within that rectangular block. The block has at least two such intervals in every coordinate. The product of the chosen two-interval unions is the required cube $\mathsf{q}' \subset Y$. It still has side $2m$, so the separation argument is unchanged. The original local gauge clause is therefore applicable to q'.

Now let $\widetilde{\mathsf{q}}$ be any gauge cube of index q for the joined sequence, contained in Y. Fineness gives

$$\widetilde{\mathsf{q}} \subset \mathcal{L}_q^W \cup \mathcal{L}_{q+1}^W \subset \Omega_q^W \subset \Omega_q.$$

Truncate the original sequence at q. In that sequence Ω_q is the last layer, so $\widetilde{\mathsf{q}}$ is one of its gauge cubes of index q. The preceding local truncation argument supplies a real gauge on a cube containing $\widetilde{\mathsf{q}}$, with $\xi_q|A|, \xi_q^2|\nabla^\xi A| \le a_0$. Restrict this gauge to $\widetilde{\mathsf{q}}$ itself. This is an allowed choice of

the cube in the joined gauge clause; it lies in $\Omega_0^W \cap Y$, and every derivative stencil retained there is unchanged. Thus the joined local-gauge clause holds with the original radius.

Together with the pointwise and free-boundary-current estimates above, this proves joined-pair admission from the original enlarged local class on $\Omega_0 \cap Y$. The argument uses the actual standard parameter cubes and the original local-class decomposition. It does not require a holomorphic choice of the real local gauges.

4.17.1.13. Localization with the bond read fixed. Suppose the strip datum $B'[P]$ has been supplied on the joined rows, holomorphically on the required pair domain, with $\|B'[P]\|_\infty \le \beta < \varepsilon_{\mathrm{D,s}} e^{-32\varkappa}$. Its values are fixed while the cube parameters are varied. Include the full local masks of this datum in its carriers, and let $\mathfrak{B}$ be the cubes carrying its possible nonzero rows. Thus these are boundary cubes, with the prescribed fixed enlargement when the row conversion uses an endpoint block. Use the admitted joined pair throughout.

Let N_b be a wall-connected set of parameter cubes, where adjacency means sharing a face. Include enough local margin that the full averaging and frame read at b, with every read stencil, lies in the interior of its union. Hold the parameters on N_b at one and vary those on $S = \mathsf{P} \setminus N_b$. Write

$$F_b(s; P) = \mathsf{Rd}_b(P; \mathbf{H}(s; B'[P])), \qquad R = e^{\varkappa}.$$

Here the read vanishes at zero response and is holomorphic on the response ball. If its bound in the local response norm is C_{rd}, the response estimate gives

$$\sup_{|s_\Delta|<R} |F_b(s; P)| \le A, \qquad A = C_{\mathrm{rd}} C_{\mathrm{D,s}} e^{16\varkappa} \beta.$$

All suprema below use the same admitted pair domain.

For $K \subset S$ define the mixed difference

$$H_b[K; P] = \sum_{A_0 \subset K} (-1)^{|K|-|A_0|} F_b(\mathbf{1}_{A_0}; P).$$

The fixed parameters on N_b are understood in every summand. Finite Möbius inversion gives $F_b(\mathbf{1}_S; P) = \sum_{K \subset S} H_b[K; P]$. The response restricted to a wall component uses only the pair and datum in that component. Since N_b is wall-connected, the read uses its one component. If $K \cup N_b$ has another component, pair the terms in the mixed difference by switching one cube of that component; they cancel. If the component of N_b meets no cube of $\mathfrak{B}$, its datum is zero and its response is zero. Therefore

$$H_b[K; P] = 0 \quad \text{unless } K \cup N_b \text{ is wall-connected and meets } \mathfrak{B}.$$

This reasoning uses the full component locality of the response, not just decay of its kernel.

For $1 < r < R$, the exact Cauchy formula for the difference is

$$H_b[K; P] = \frac{1}{(2\pi i)^{|K|}} \int_{|z_\Delta|=r,\, \Delta \in K} F_b(z_K, 0_{S\setminus K}; P) \prod_{\Delta \in K} \frac{dz_\Delta}{z_\Delta(z_\Delta - 1)}.$$

For $K = \emptyset$ this means $H_b[\emptyset; P] = F_b(0; P)$. Taking $r \uparrow R$ gives

$$\|H_b[K]\| \le A(R-1)^{-|K|}.$$

Each parameter is differenced once, so there is no factorial. The norm here is a supremum before summation.

Let W_K contain $K \cup N_b$ and all its local read and datum masks. Use a fixed bounded enlargement, and write $\ell(W)$ for the tree length on the output cube grid. Keep the geometric comparison explicit:

$$\ell(W_K) \le c_h(|K| + n_b), \qquad n_b = |N_b|.$$

The constant c_h includes the change between the parameter grid and the output grid. The rooted cube count gives constants C_b, c_{e}, independent of the torus and the number of levels, such that at

most $C_b e^{c_{\mathrm{e}} n}$ wall-connected sets $K \cup N_b$ have $|K| = n$. Here C_b depends on the bounded read star. Put

$$n_* := \min\{|K| : K \cup N_b \text{ is wall-connected and meets } \mathfrak{B}\}, \qquad q_a := \frac{e^{c_{\mathrm{e}} + a c_h}}{R-1}.$$

If no such set exists, all terms vanish. Otherwise, for $q_a < 1$,

$$\sum_K e^{a\ell(W_K)} \|H_b[K]\| \le A C_b e^{a c_h n_b} \frac{q_a^{n_*}}{1-q_a}.$$

Group the terms having the same W_K. Their sum depends only on the fields in W_K and retains the boundary cube and the whole read star. On each local pair domain where the component equations and the supplied datum have been admitted with the same bounds, this sum is holomorphic and obeys the same absolute estimate. Locality alone would not extend a bound to local data outside those domains. Reality and covariance of the datum, response and read pass through the mixed differences.

For $\varkappa \ge 1$, one may use $\log(R-1) \ge \varkappa - 1$. For example, when $\varkappa > 1 + c_{\mathrm{e}}$, the choice $a = (\varkappa - 1 - c_{\mathrm{e}})/(2c_h)$ gives $q_a \le e^{-(\varkappa - 1 - c_{\mathrm{e}})/2}$. The strip margin supplies the lower bound on n_*; it must be measured after the read and datum masks have been included. The factor $e^{16\varkappa}$ in A and the fixed read-star cost remain in the displayed estimate and can then be absorbed by that margin. If the parameter partition is the level-t M-cube partition, the response floor is $\mathring{\delta}_0 M \ge 4352\varkappa$, in addition to its other size conditions. Choosing $\varkappa$ proportional to $\mathring{\delta}_0 M$ within this floor gives a tree rate of that order. This final choice does not supply the missing holomorphic datum or its norm bound.

4.17.1.14. The background of an approximate test on the cutoff zone. Let b be a bond of the cutoff zone that occurs in the approximate test of $\square'$. Use bare length units and set $w = M_2 \check{R}_t L^t$, so that the test cubes have side Lw and each of their enlargement layers has width w. By the definition of the cutoff zone, an endpoint of b is within $2w + L^j$ of a test cube in R. Since b is tested at $\square'$, every point of $C = \square'^{\sim 4}$ is within $(L+8)w + 2L^j$ of R. The region R lies inside the one-cell inset of Ω_t. The one-cell enlargement of its hull lies in Z_t. The side of a level-t cell is $M\check{R}_t L^t$. Choose the fixed M with enough room that the preceding distance leaves at least four M_1-layers in both of these margins. For $t \ge 2$ also use the one-cell margin of Ω_t inside Λ_j. This is a margin in level-j cells, with side $M\check{R}_j L^j$. The cube sizes at these consecutive levels are comparable, so the same fixed choice of M leaves room in this margin. For $t = 1$ only the part Z_1 is used. Thus C lies in the top layer of the joined strip sequence.

In that top layer C is a one-part collared region at level t, with collar width M_1 and the full first member. Its joined sequence is the tower of C, since all the members of the strip sequence contain C. The top rows on C carry the prescribed V_t. The new collar rows are formed by the same lifts of those top rows as in the definition of the test background. This identifies the joined datum with the datum of $U_{t,\square'}$. In particular its regular orbit is defined when the strip datum belongs to the real comparison class.

The test reads b inside $\square'^{\sim 3}$, where its endpoint contour blocks are wholly in the top layer of the tower. In both sequences their centres are nodes and the other level-j sites of these blocks are free. The block axial frames therefore fix the same sites in the comparison. The read carriers have distance at least $w - CL^t$ from C^c, which exceeds $3M_1 L^t$. The top spacing in bare units is L^t. Apply the real collar estimate eq. 3.cdii and the endpoint calculation for the strip shift to the strip background and its localization to C. It follows that

$$\varphi(V_j^{\mathrm{loc}}(b) V_{j,\square'}^{\mathrm{bg}}(b)^{-1}) \le C\varepsilon_1 e^{-cM_2 \check{R}_t},$$

with ε_1 the real test radius of the joined strip datum. On the strip support it is bounded by $C_L \varepsilon_j$. The fixed scale floors and then the coupling ceiling give a bound $\delta_j/8$ for this angle. Use also $|H| \le \delta_j/8$ for the strip shift. Where $\vartheta_{\mathrm{co}} < 1$, the angles of both $V_j(b) V_{j,\square'}^{\mathrm{bg}}(b)^{-1}$ and $W' V_j^{\mathrm{loc}}(b) V_{j,\square'}^{\mathrm{bg}}(b)^{-1}$

are at most $(5/3 + 1/8 + 1/8)\delta_j < 2\delta_j$. Their group distances from one are no larger than their angles. Where $\vartheta_{\mathrm{co}} = 1$ the two products coincide. Hence on the cutoff zone the approximate bond factor can be written through W' and the two localized backgrounds on this support.

What is frozen and what is integrated. On Ω^c_{j+1}, step $j+1$ stores V_j together with its constraint. On Ω_{j+1} the action is expanded around $U_{j+1}(\mathbf{V})$. The only variable integrated there is $A = \mathcal{A}^{\mathrm{s}}_j|_{\Lambda_{j+1}}$, conditionally on the stored field $A' = \mathcal{A}^{\mathrm{s}}_j|_{\Omega_{j+1}\setminus\Lambda_{j+1}}$. The measure is Gaussian with covariance $C^{(j)}(\Lambda_{j+1})$, which is the inverse of the restriction of the form $\Delta^{(j)}$ to the bonds of Λ_{j+1}, and its linear term is $-\langle A, \Delta^{(j)} A'\rangle$. The coupling flows only where this integration is carried out, $\varphi_{j+1} \in C^\infty_0(\Lambda_{j+1})$, and it stops on the whole of Z_{j+1}, including $\Omega_{j+1} \setminus \Lambda_{j+1}$. Once the square is completed, the conditional weight appears as a factor inside $\mathbf{T}_k(Z) = \prod_{j<k} \mathbf{T}^{(j)}$, with the regions cut by the component Z ((2.21), (3.23) there; this operation is the product shown for a component on which $\mathbf{R}$ has never acted, and in general the operation is defined by a recursion over the marks):

$$\mathbf{T}^{(j)} = \int dV_j\big|_{\Omega^c_{j+1}} \delta(\mathcal{M}(V_j)V_{j+1}^{-1})\, \mathfrak{G}_j\, \chi\chi^c\, z^{(j)} \int dA'\, \chi\chi^c\, e^{-\frac12\langle A', [\Delta^{(j)} - \Delta^{(j)} C^{(j)}(\Lambda_{j+1})\Delta^{(j)}]A'\rangle}.$$

This operation contains no Wilson weight. That weight belongs to $\exp A_{k,\mathcal{S}}$, the function on which the operation acts; $z^{(j)}$ is explained below. The old R-terms, $\mathbf{R}'_j = \sum_{i\le j_1} \mathbf{R}^{(i)}$, appear alongside $\mathbf{E}_j$ in the regular action $\widetilde{\mathbf{E}}_j$ of the step §4.17.1.45. Here $\mathbf{R}^{(i)} = \sum_X [R^{(i)}(X,\cdot) - R^{(i)}(X,\mathbf{1})]$ is the summand of level i of $\mathbf{R}_j$, and j_1 is the index k_1 of § 4.17 at level j. The young ones, $\mathbf{R}''_j = \sum_{j_1<i\le j} \mathbf{R}^{(i)}$, are handled separately. Put $\delta F := F(U_j) - F(U_{j+1})$. Attach a parameter t to the terms present only next to Z_j, and a parameter s to $\mathbf{R}''_j$ ((3.23)–(3.28) and pp. 270–271 there). Let $z^{(j)} = \prod_c z^{(j)}(c)$ be the factors that come from eliminating the delta functions. In $\mathbf{T}^{(j)}$ their product runs over those coarse bonds c of Ω_{j+1} that lie outside Λ_{j+1}; under the logarithm it runs over those of Λ_{j+1}. Then

$$\log\Big[z^{(j)}\int dA\, \chi^{(j)} e^{-\frac12\langle A,\Delta^{(j)}A\rangle - \langle A,\Delta^{(j)}A'\rangle + \cdots}\Big] - \tfrac12\langle A', \Delta^{(j)} C^{(j)}(\Lambda_{j+1})\Delta^{(j)} A'\rangle$$

$$= \int_0^1 dt\, \langle \delta\mathbf{B}_j - \delta A(x_j(\cdot) - x_j, \cdot) + \cdots\rangle_{s=1,t} + \int_0^1 ds\, \langle \delta\mathbf{R}''_j\rangle_{s,t=0} + \Big(\log[\cdots] - \tfrac12\langle A', \Delta^{(j)} C^{(j)}(\Lambda_{j+1})\Delta^{(j)} A'\rangle\Big)_{s=t=0}.$$

In this identity $\langle\,\cdot\,\rangle_{s,t}$ denotes the normalised mean with respect to the measure that the parameters s, t interpolate. The left-hand side is the unnormalised integral. Its second term is present since the square, once completed, was moved into $\mathbf{T}^{(j)}$. Inside the integral over t, the dots stand for two kinds of terms. Those of the first kind are the terms of the strip, namely the Jacobians together with the differences of all quantities that depend on a shifted variable. Those of the second kind are the differences, taken between the two backgrounds, of the exponents of the integral operations carried by the term. The coupling difference carries the sign it has in $A_{j,\mathcal{S}} = -x_j A - A(x_j(\cdot) - x_j, \cdot) + \ldots$. A cluster expansion as in § 4.5 is applied to each part. The Gaussian normaliser is contained in the last part; it is that part of $E^{(j+1)}$ which carries b^0_{j+1}. On a label it is a determinant: that of a multi-scale operator, compressed to Λ_{j+1} and taken in the background of that label; so by definition it is non-local. It is converted into local terms,

$$\sum_c \log z^{(j)}(c) - \tfrac12 \log\det\big(\mathsf{C}^{\mathsf{T}}\Delta^{(j)}\mathsf{C}\big|_{\Lambda_{j+1}}\big) = \text{number} + \sum_X \mathbf{N}(X, U_{j+1}),$$

$$|\mathbf{N}(X)| \le C e^{-cM d_{j+1}(X)},$$

which Bałaban does by means of a resolvent integral and walks § 4.5, whereas here it is done by sums with alternating signs, taken over the sub-regions $Y \subset X$, whose terms are the diagonal entries of the logarithm of the operator built on Y. Möbius inversion yields the sum rule, and the

decay is due to a sign-reversing involution, which cancels everything except the chains of walks that visit all of X. Only the last two parts give terms with domain $X \subset \Lambda_{j+1}$. These terms are built from walks and fields inside X, so the regions do not affect them. They are of two kinds. The first is $R^{(j+1)}$; it arises from the part with s, to which what $\mathbf{R}$ leaves is added § 4.17. The second is $E^{(j+1)}$, which comes from the last part, where $\mathbf{E}_j$ stands with the old R-terms. These terms are the same for every label. For the regular terms this holds for every such X. For the remainders it holds for X one cell inside Λ_{j+1}, the only place where a label shows them, and this is the margin of §4.17.1.45. Thus the young R-terms go into $R^{(j+1)}$, and the old ones go into $E^{(j+1)}$ and b_{j+1} (2d, 2e). Everything else makes up $B^{(j+1)}$; among it is the part of the quadratic form that is rejected (next paragraph). It amounts to $O(1)$ for each M-cube of level $j+1$ next to Z_{j+1}; this cost is covered by §4.17.1.40, and for that reason no renormalisation of it is required.

What a one-step operation keeps. The conditional form is a Schur complement on the free coordinates. In the following display the factors of $\Delta^{(j)}$ act as its free-coordinate precision $\mathsf{C}^{\mathsf{T}}\Delta^{(j)}\mathsf{C}$. A chain is either a precision walk or a product of a precision walk, a walk of the inverse of the integrated compression, and a precision walk. These chains have connected carriers. A chain is kept if the cells of level $j+1$ that it meets intersect $\Omega_{j+1} \cap Z_{j+1}$ in a non-empty connected set and remain within one layer of that set; a kept chain is then counted with the component $\mathfrak{k}$ of $\Omega_{j+1} \cap Z_{j+1}$ that contains this set:

$$\langle A', [\Delta^{(j)} - \Delta^{(j)} C^{(j)}(\Lambda_{j+1}) \Delta^{(j)}] A' \rangle = \sum_{\mathfrak{k}} \langle A', \Delta^{\mathrm{loc}}_{\mathfrak{k}} A' \rangle - 2 \sum_{X} \mathsf{B}^{\times}(X),$$

$$|\mathsf{B}^{\times}(X)| \le C p_0(g_j)^2 e^{-cM\check{R}_{j+1}} e^{-\kappa d_{j+1}(X)},$$

$$\langle A', \Delta^{\mathrm{loc}}_{\mathfrak{k}} A' \rangle \ge c \sum_{b \text{ in } \mathfrak{k}} |A'(b)|^2 :$$

(the bound on $\mathsf{B}^{\times}$ is for stored fields in range, $|A'(b)| \le C p_0(g_j)$, which they are on the support of the kept tests) a rejected chain has length at least a fixed multiple of the cell side in fluctuation units, up to a fixed additive constant, and the size $p_0(g_j)^2$ of the stored field squared is outweighed by $e^{-cM\check{R}_{j+1}}$. The rule of Bałaban, by which the form "couples only the fields A_j in the same component" (119, p. 258), fails to be local: whether two bonds belong to a common component is settled at a distance from them, and two histories having the same trace can differ on it. The kept form is not a Schur complement but a truncated series. Its lower bound is therefore not inherited and has to be proved. The proof localises to small cubes (Ismagilov–Morgan–Simon) and compares with a genuine Schur complement. *Carriage*: in an integral operation that a term carries, the tests, the gauge factors and the delta functions depend on no background, whereas the forms $\Delta^{\mathrm{loc}}_{\mathfrak{k}}$ and the factors $z^{(j)}$ do. These forms and factors are given as functions of a pair $(\mathbf{U}, \mathbf{J})$ near $\mathfrak{k}$, and they are constructed once and for all. They are evaluated at the current background, which in a term of level k is $U_k(\mathbf{V})$. The background changes with every later step and with every stage of $\mathbf{R}$. Each change produces differences

$$-\tfrac{1}{2} \langle A', [\Delta^{\mathrm{loc}}_{\mathfrak{k}}(U^{\sharp}) - \Delta^{\mathrm{loc}}_{\mathfrak{k}}(U^{\flat})] A' \rangle$$

, where $U^{\flat}$ is the background after the move and $U^{\sharp}$ the one before it, both written in the new variables. Localisation turns these differences into boundary terms. For a piece of a kept form whose carrier is R, the oscillation with respect to the pair is bounded by $C p_0(g_j)^2 e^{-\kappa d^{\flat}_{j+1}(R)}$, measured in the hull length of the carrier; this bound is large, being the square of a field at its threshold. The difference of such a piece at the two backgrounds of a stage of level i is localised as well, and each

of its localised pieces, on a domain Y, has modulus at most $C(\log x_i)^{-(q-3\mathfrak{p}_0-Cr)}e^{-c\,d_k^{\natural}(Y)}$. This is a negative power, because q is chosen after $\mathfrak{p}_0$ and r, with $q > 3\mathfrak{p}_0 + Cr$.

4.17.1.15. The comparison at an admitted real reference. Fix the step and use level-j units. Let A be the complete free-variable precision of the step after eliminating the pivot bonds, including the projected gauge and current terms. The free-variable space is the space on the nonpivot bonds meeting Ω_{j+1}. The integration set I consists of the free bonds meeting Λ_{j+1}. Equivalently it is the set of free bonds meeting the blocks of the level-$(j+1)$ sites in Λ_{j+1}. The stored set E is its complement among the free bonds of the step. In particular a free bond with one end in the integration region is assigned to I.

Take a real reference in the admitted class of the fluctuation theorem, on the entire determining sequence, with these conditioning sets. The complete precision has a lower bound $\gamma\,\mathrm{Id}$ there, with $\gamma > 0$ fixed. The Schur complement

$$S = \mathsf{A}_{EE} - \mathsf{A}_{EI}(\mathsf{A}_{II})^{-1}\mathsf{A}_{IE}$$

then satisfies $S \geq \gamma\,\mathrm{Id}$. Indeed, its quadratic form is the minimum of the full quadratic form over the integrated coordinates with the stored coordinates fixed. For a stored component D the compression of S to its free bonds therefore has the same bound. When the integration set is empty the second term of S is zero.

We can also control the difference between this compression and the chain sum assigned to D. For the precision and for the covariance $(\mathsf{A}_{II})^{-1}$, the absolute sum of the length-m walk kernels is bounded entrywise by

$$C\,2^{-m-1}e^{-\delta|b_- - b'_-|_1}.$$

Their absolute row and column sums are at most $C'2^{-m-1}$, since the endpoints range over the unit lattice in these units. For the three factors of a Schur chain, sum first over the internal bond indices. The resulting row and column bounds are at most the products of the three bounds. At total length n the number of choices of the three lengths is $(n+1)(n+2)/2$. The tails above length r thus have absolute row and column sums at most $C''e^{-cr}$. The same bound for a single precision walk follows directly from its length estimate.

Each walk cell has a carrier of bounded diameter in these units, with a bound fixed by the walk construction. The carrier of a nonzero chain of total length n has diameter at most $C_{\mathrm{cell}}(n+3)$, since successive carriers meet at their common bond. Let W be the side of a level-$(j+1)$ large-field cell, measured in level-j units. Use the closed cells for the retention test, with a common corner counting as contact. Disjoint closed cells of this grid are at sup distance at least W. Thus if the carrier has sup diameter less than W, all the cells it meets touch one another. A chain contributing to the compression on D meets D, since its carrier contains both endpoints of each of its exterior bonds. The cells it meets in the stored strip then form a connected nonempty set in D, and the other cells it meets are at most one touching layer away. Such a chain is retained for D. Thus every omitted term in this compression belongs to the length tail $n \geq W/C_{\mathrm{cell}} - 3$. The tail estimate and the matrix Schur test give

$$\|\Delta_D^{\mathrm{loc}} - S|_D\|_{2\to 2} \leq Ce^{-cW/C_{\mathrm{cell}}}. \tag{4.cdxlii}$$

When this error is at most $\gamma/2$, the kept form at this reference is bounded below by $(\gamma/2)\,\mathrm{Id}$. This argument applies in the full real reference class and on the literal integration and stored coordinate sets just described.

To use this bound when the pair is given only near the stored component, we must compare on small cubes with admissible real references on which the same short chain terms are defined. Such a comparison must match the region traces, the integrated free bonds, and the real pair on the carriers, in a common frame for the retained coordinates. The walk locality identities then match the short terms. The estimate eq. 4.cdxlii controls length tails in an admitted comparison. If K is

the kept matrix in the retained coordinates and $\sum_\nu \chi_\nu^2 = 1$, the kernel of $K - \sum_\nu \chi_\nu K \chi_\nu$ is

$$\tfrac{1}{2} K(b,b') \sum_\nu (\chi_\nu(b) - \chi_\nu(b'))^2.$$

For cutoffs of bounded overlap varying on length h, the factor multiplying $K(b,b')$ is at most $Ch^{-2} d(b,b')^2$. The exponential kernel bounds give uniform row and column sums with this second distance moment. The matrix Schur test bounds the error by Ch^{-2}. Thus patch lower bounds with a fixed margin give the complete lower bound once the comparisons on the actual localized histories have been constructed in the same coordinates.

4.17.1.16. Comparison in complex coordinates. The precision and inverse walk terms are holomorphic in the background and current restricted to their carriers, on the piecewise local classes of the fluctuation expansion. Two realizations give the same coefficient for a decorated walk with the same cell families and parent map if the region and fluctuation-structure traces agree on its common padded hull $[\![\omega]\!]$, at every level. Match the actual free-coordinate and conditioning masks on all their reads in that hull. The first-member trace then gives the same field carrier $[\omega] = [\![\omega]\!] \cap \Omega_0$. The pair on this carrier must be the same, in the same coordinates, and belong to both local classes. This applies factor by factor to a Schur chain.

Suppose the actual later read and its comparison are expressed in one chart about a coercive real reference. Normalize the chart to radius 1 in the background and current norms. If every carrier restriction belongs to its local class throughout the chart and the absolute walk sums obey the uniform row and column bounds with summable length tails there, the kept matrix $Q(z)$ is holomorphic and uniformly bounded in operator norm. Cauchy's estimate along the radial line to z then gives $\|Q(z) - Q(0)\|_{2\to 2} \le C\theta$ for $\|z\| \le \theta \le \frac{1}{2}$. If $Q(0) \ge c\,\mathrm{Id}$, it follows that $\mathrm{Re}(a^{\mathsf{T}} Q(z) a) \ge (c - C\theta)\|a\|_2^2$ on the real retained coordinates. For the localized histories we must construct the real comparisons in these coordinates and verify the common chart with a uniform fraction of the available radius for the prepared and substituted data, including the independent current. The existence of each coefficient on its open carrier class alone does not give that fraction.

4.17.1.17. A real comparison by cutting off the potential. Keep the separated determining sequence of the step which formed the strip, including its original free-coordinate set and its integration mask. Work in level-j units. On an enlarged comparison cube about a retained bond suppose there is a real gauge in which the background is $e^{i\eta_j A}$, with $|A|$ and $|\nabla^{\eta_j} A|$ at most a, and with the transported independent real current bounded by a in absolute norm in level-j units. Take the cube and its whole stencil collar strictly inside Ω_j, so that only layers $j, j+1$ occur there in the forming sequence. The retained form and its comparison must use the same pivots and the background-relative contour quadratic term of the fluctuation operator. In that prescription the real gauge acts on free coordinates by the adjoint at the initial endpoint, an orthogonal change in the counting pairing.

Choose a real cutoff χ equal to one on the middle cube and zero in a neighborhood of the outer boundary, with first difference bounded by C/h for a cube of side $O(h)$. Extend χA and χJ by zero in this gauge on the whole original torus. The first difference of χA is at most $Ca(1 + h^{-1})$. In a plaquette product the linear sum of its bond logarithms is η_j^2 times the discrete curl of χA, so its norm is bounded by $C\eta_j^2 a(1 + h^{-1})$. The product remainder is bounded by $C\eta_j^2 a^2$. Thus the real reference $(e^{i\eta_j \chi A}, \chi J)$ has plaquette defects at most $C\eta_j^2(a + a^2)$, including at the cutoff. It has a single small logarithm on the torus. The scaled potential and its first difference cost at most a further factor L^2 on the adjacent upper layer. The independent current costs at most L^3 there. Outside the collar the reference is flat with zero current. Choosing a small after the comparison width makes this pair satisfy the real curvature, gauge and current conditions of the fluctuation theorem on the original geometry. The current can be cut independently because that theorem takes a real current as a separate argument.

Extend the local site gauge arbitrarily on the torus and transform the reference back. The real pair conditions are invariant under this gauge action, so the reference is still admitted and agrees with the actual read on the middle cube in its original frame. The global reference uses the same region traces and conditioning masks there. Every chain whose endpoints are tested in the inner cube and whose length is less than its buffer to the middle boundary divided by C_{cell}, up to the fixed additive constant, has its carrier in the common region. Its terms agree by carrier locality. On the local read the absolute walk estimates must hold on all carriers of the kept form, including the long ones. With these estimates the remaining length tails in both forms have operator norm at most $C\exp(-ch/C_{\mathrm{cell}})$. Choose h to make these errors and the partition error smaller than the fixed reference gap, with the cutoff support fitting inside the original collar. Thus this cutoff supplies a positive real comparison on the patch once its gauge, mask and current conditions hold. The partition estimate above combines such comparisons with constants uniform in the depth. For a later read, the two-layer bounds must be verified in the forming sequence and in its retained frames, including the enlarged cubes about the external bonds. These comparisons use the reference-weighted form. A change in the ratio of physical and reference weights uses the gauge compensation in eq. 4.d.

4.17.1.18. Reading a retained strip in its original domain. Let D be a component of $\Omega_t \cap Z_t$ as formed by step $t = j + 1$. A later record retains this strip only if no intervening healing, island operation or re-made level of a drop-out has replaced its field. Write $D^{\sim}$ and $D^{\oplus}$ for its one- and two-cell enlargements on the level-t partition. For a retained strip the trace on $D^{\oplus}$ is unchanged in the following sense. The level-t region is the one formed by step t and the regions of levels $1, \dots, t-1$ contain $D^{\oplus}$. This applies to the determining and action readings of level t, with every stage index. The local structure has no radius lift there and uses only the indices $t-1, t$. The retained factors use the pivots, integration mask, and gauge of step t. A factor of the quadratic form reads the field $\mathcal{A}^{\mathrm{s}}_{t-1}$ in these same coordinates. The associated elimination factor depends only on the background.

Take $P = (U, A', \mathbf{J})$ in the structured space of the level-t reading on an address containing the full carriers of the read. Use the pointwise and gauge clauses of §2.2.4.1, including layer zero at the first strip. At its formation this is the reading on either side of the step-t operation that retains the strip. For a later label use the level-t reading for the retained factor on its original carrier. Use the larger of the running radii at $t-1$ and t for each quantity and set $\bar{a}_i$ to their uniform upper bounds over the chosen coupling range, for $i = 0, 1, 2$. The third radius is for the independent current. The pointwise bounds on curvature, potential and current in these two layers cost at most $L^3(1 + \beta_{\mathrm{s}})\bar{a}_i$ in the fixed strip domain. Indeed the pointwise clause of the layered triple assigns a bond, plaquette or difference by its base point. The strip domain may assign it by another of its sites. Consecutive boundaries are separated by more than the diameter of such a stencil, so the two local spacings differ by at most L. The largest power in these norms is the third, for the current.

In level-$(t-1)$ units the level-t cells have side $LM\check{R}_t$. The gauges required in the strip domain are those on the cubes of the walk expansion, of width of order M_1 in their local units, and on the enlarged tiny cubes, of width of order M/L^2 in level-$(t-1)$ units. The latter are required at the endpoints of the external bonds as well as of the integration bonds. A required cube contained in the upper region fits in a gauge cube of width of order M in level-t units inside the upper trace. One can place the containing cube within the block intervals met by its coordinate projections. For a required cube reaching the lower side of the boundary, the containing cube has width of order M in level-$(t-1)$ units. Its collar lies in $D^{\oplus}$ and hence in the level-$(t-1)$ region. The stipulated hierarchy of the walk and cell widths gives room for both choices. These are cubes of the gauge clause of the layered triple, whose bound for $U = e^{i\eta A}$ in that gauge is $\xi_n|A|, \xi_n^2|\nabla A| \le CM\bar{a}_0$. Converting the gauge estimate to the other layer costs at most L^2. At $t = 1$ the level-zero lattice has unit spacing in these units. Axial transport in its local cube gives $|A| \le C\,\mathrm{side}\,\bar{a}_0$ from the plaquette bound, and the unit-spacing first difference is bounded by twice the potential bound.

Choose the fixed radii a_i^{loc} inside the positive strip class $\mathcal{O}^+$ of the original kept operator. Its gauge radius includes the fixed small radius of the walk cubes. The remaining contributions to the radii tend to zero with the coupling ceiling. It suffices to impose

$$L^3(1+\beta_{\mathrm{s}})\bar{a}_i < a_i^{\mathrm{loc}} \quad (0 \le i \le 2), \qquad CL^2M\bar{a}_0 < a_0^{\mathrm{loc}}.$$

For an external elimination bond also require the small averaging chart on its full read set. The two-layer potential and gauge bounds give a chart size $C(L,M)(\bar{a}_0+\bar{a}_1)$ in the upper units, so the separate condition is $4C(L,M)(\bar{a}_0+\bar{a}_1) < r_{\mathrm{P}}(L)$, the pivot chart radius. The floors on M, including that required by the decay rate of the kept chains, precede the choice of these coupling ceilings.

These checks admit the read to the fixed domain of the original kept operator and of its elimination factors. On the positive strip class the kept-form estimate bounds the real part in its original free coordinates, before multiplication by a coupling ratio. In a real comparison the potential may now be cut off in the gauges just obtained. Choose the comparison width h with its interaction collar inside the two-cell trace. If γ is the original real reference gap, take the error in eq. 4.cdxlii at most $\gamma/2$ and the sum $Ce^{-ch/C_{\mathrm{cell}}} + Ch^{-2}$ of the comparison and partition errors at most $\gamma/4$. The free bonds and masks have not changed, so this leaves a lower bound $\gamma/4$ for the old conditioned form in its original coordinates. On its admitted complex domain, the walk and elimination coefficients are holomorphic. For the actual substitutions we must verify the stated layered bounds on these carriers. For a normalized Gaussian with a changed coupling use the compensation in eq. 4.d. 4.17.1.19. The amplified substitution on carried strip factors. Let n be the level of the main step and $j+1<n$ the level at which a retained strip was formed. Consider a quadratic or elimination factor of that strip which is carried through step n. The carriage rule puts its carrier X outside the one-layer enlargement of the region being healed. The carrier is in the exterior region of the healing, and the structure in which the factor is read counts no stage in a component meeting its sites. These are the served targets of the main step. The strip coordinates and the pivots on X are those at its formation.

Let $P=(U,A',\mathbf{J})$ belong to the complex space of the output structure on a saturated localization Y containing the enlarged carrier and the response collars. For the admitted inner variables ξ use the formed inner data $\varsigma[P;\xi]$ of the main step. The belt map gives the reduced increment $\beta = \mathfrak{b}^Y(s;P,\varsigma[P;\xi])$ of the row averages on the belts, with $\|\beta\|_\infty \le \varepsilon_\beta/2$ throughout the complex cube polydisc $|s(\Delta)| < e^{\bar{\kappa}}$. For each wall component w of the localization use the response with cube parameters $s\mathbf{1}_w$ and form

$$(\mathbf{A},\mathbf{K}) = \sum_w \lambda_w(\mathbf{H},\mathrm{j})(s\mathbf{1}_w;\beta)[\mathfrak{p}^{\mathrm{e}}(P)].$$

Here $\mathfrak{p}^{\mathrm{e}}(P)$ is the restriction to the local exterior response class, with the current of that class. Level this shift on the exterior pillars with the pins at the sites of the stored data. Write $\mathfrak{h}^\delta$ for the result restricted to the bonds of X. The argument of the retained factor is $(P|_X) \dotplus \mathfrak{h}^\delta$, with the potential composed by the group product and the current increment added. Its stored fluctuation argument is held fixed.

The response estimate gives decay from the belts for the potential, its first difference and the current in their local units. The response of a wall is supported inside that wall, so independent multipliers cost $\max_w|\lambda_w|$. Pinning increases the bound by a fixed factor C_δ. The wall carriers lie strictly inside Y, so the shift vanishes on the target boundary outside the interior of Y. Put $\alpha_{*,n} = \min_{\iota=0,1}\alpha_{\iota,n}$ and let Σ denote the belts. The bound in the profile used by the target clauses is

$$|\mathfrak{h}^\delta|_b \le c_{\mathrm{sh}}\alpha_{*,n}\, C_\delta a(X) \max_w |\lambda_w| e^{-\delta_1 d_{\bar{\mathbf{B}}}(x_b,\Sigma)}.$$

Here $|\cdot|_b$ takes the maximum of the potential, its first difference and the current, multiplied by the first, second and third powers of the local spacing, respectively. The constant c_{sh} is the fixed

profile allowance in the substitution bound. The gap of a lower clause with index m' absorbs the profile in the units of that clause. In particular, if m is the exterior index at the base point, the gap comparison for $m' \le m$ is

$$L^{-(m-m')}\alpha_{*,n}e^{-\delta_1 d_{\mathbf{B}}(x_b,\Sigma)} \le Ce^{-c\delta_1 M\check{R}_n}\mathsf{gp}(m'),$$

where $\mathsf{gp}(m')$ is the gap supplied by the descent of the layer bounds and the fixed gap constant is included in C. With this profile the substitution into the served target preserves its tower, potential and one-scale clauses whenever $C_\delta a(X)\max_w|\lambda_w| < 1$. The free rim and wide shell used in that comparison hold for both classes of the main step. The stored sites of a served target are outside the boxes, so its pinned transports are among those admitted by the comparison.

To bound $a(X)$, split the response decay between the pointwise profile and the distance from the belts to X. A path crosses a distance at least $cM\check{R}_n$ in level-n units before it can reach the exterior. It then crosses the intervening layers, each of scaled width at least ρ. If n_X is the largest level for which X meets the interior of the healed exterior region $\bar{\Omega}_{n_X}$, this gives

$$2a(X) \le \frac{Ce^{C\bar{\kappa}}\varepsilon_\beta}{\alpha_{*,n}}\exp\bigl[-c\delta_0 M\check{R}_n - c\delta_0\rho(n-1-n_X)_+\bigr].$$

Thus for $a_X = 2C_\delta a(X)$ the whole disc $|\lambda| < 2/a_X$ is available when every λ_w equals λ. Since $\check{R}_n$ grows at least quadratically in the logarithm of the coupling parameter, the belt factor absorbs $\alpha_{*,n}^{-1} \lesssim g_n^{-1}$ uniformly in n. In particular $a_X < 1$ at small coupling.

On these discs the substituted triple lies in the layered space of the factor on its original carrier. The strip-domain comparison above then applies to its holomorphic walk coefficients and elimination factors in the original masks. At $s = 0$ or $\lambda = 0$ the argument is $P|_X$. At the physical point $s = \mathbf{1}$, $\lambda = 1$ the reduced belt datum is the true one. The pinned response gives the pair of a single representative of the prepared critical orbit, restricted to each served carrier. Its transports agree with those of the healed background at the stored sites. This proves the amplified read for the carried strip factors. Factors read inside the healing require the localized inner substitutions.

The idea of R (Comm. Math. Phys. 122 (1989) 175, (0.3)–(0.4)). We split $Z = Z' \cup Z''$. Following that paper (p. 176), Z'' is the union of the components "for which the small factors connected with large field control some number of next steps", and Z' is the union of all the other components (the rule by which the present construction sorts the components into these two sets is given under "Where it acts", below). The density is a sum $\rho(V) = \sum_Z \rho(Z,V)$ over the large-field regions Z ((0.2) of that paper), and $\rho(Z'',V)$ denotes the density belonging to the region Z''. Then

$$(\mathbf{R}\rho)(V) = \sum_Z \rho(Z'',V)\,\frac{\int dV|_{Z'}\,\rho(Z,V)}{\int dV|_{Z'}\,\rho(Z'',V)},$$

$$\int \mathbf{R}\rho = \int \rho.$$

In Z' the law of the field is replaced by the small-field law, and the mass is preserved. This idealised form is not linear in ρ. The form actually used is linear, because its reference law contains the Wilson weight and nothing else.

Where it acts. Let $k_\flat = k - \mathsf{N}_k \ge 1$ (*depth*); this restriction excludes nothing at $k = \mathsf{N}_k$, since by the second condition below a member of the class is then a component of Z_k in which no step has created anything, and no such component exists. A component X of Z_k belongs to the class when three conditions hold. First, for $k_\flat \le i \le k$ the set $X \cap Z_i$ is a single component of Z_i and is small, that is, contained in a cube of $O(1)$ level-i cells (*single boxes*, and at $i = k$ *size*). Second, for $k_\flat < i \le k$, step i creates nothing in it, and it does not absorb a second component, $X \cap Z_i = \Phi_i(X \cap Z_{i-1})$ (*quiet*). Third, it has not been acted on by any operation whose level lies in $[k_\flat, k)$, the lower end point included (*quarantine*). Bałaban imposes two conditions, that the

component be small and quiet (ibid., p. 177); to these two, the exclusion of a joining is added here as a separate condition, named explicitly. Two small quiet boxes that join within the window form a small component without creation, yet its trace one scale lower has two components: a joining does not count as a creation. Let H_i denote the hull, in level-i cells, of $X \cap Z_{i-1}$, and let $B^{[r]}$ be the box B with each face pushed outward by r such cells. Then

$$\Omega_i^{c} \cap X = H_i^{[8]},$$

$$Z_i \cap X = H_i^{[10]}$$

$$(k_\flat < i \le k):$$

the boxes are nested, the two of each level being concentric, and the number of cells along a side of X is comparable to the fixed point of the free evolution §4.17.1.40. As long as $k \le \mathsf{N}_k$ the class is empty, so $\mathbf{R}$ does nothing at the first levels. For $k > \mathsf{N}_k$ the class may be non-empty: the component obtained by free evolution from a single isolated cell created at step $k - \mathsf{N}_k$, with no intervening creation, joining or operation before level k, belongs to it. Up to level k the torus is admissible, $\frac{1}{2}\ell L^{K-i} \ge CM\check{R}_i$ for $i \le k$, as the opening of Prop. 4.56 requires. For a first-passage datum, $x_K \le g^{-2} + B_\sharp$, this condition follows from $\ell \in \mathfrak{L}(g)$ by way of $\check{R}_i \le L^{K-i}\check{R}_K$, a consequence of the upper flow bound; it is a lower bound on ℓ. Hence no cell, no box and no neighbourhood of a box wraps around the torus.

4.17.1.20. Restriction to a component. Let C be a union of components of the final region Z_i of a step, and let Y be its input region. The hull, layer, and boxing operations in this step restrict to C. Indeed, each cube of the hull of Y meets an input component. Their union is connected and belongs to Z_i, so the cube and the input component belong to the same final component. The enlargement of a cube by any of the layers used in the recursion is also connected and lies in Z_i. It therefore belongs to C precisely when that cube belongs to C. Finally, when a small component is replaced by its box, the box lies in Z_i and is connected to the component it replaces. The replacement can thus be made before or after restricting to C. Applying this argument at each boxing pass and test round shows that the restricted output is the one obtained from $Y \cap C$ and the recorded failures in C. In particular, if C contains no creation of that step, the output on C is given by the eight-layer hull and by boxing the ten-layer hull of $Y \cap C$, respectively.

4.17.1.21. Tracing an eligible component. Write $\mathcal{S}'_s$ and $\mathcal{S}_s$ for the labels just before and after the operation of level s. We use the nesting of Prop. 4.58 on these labels. At a fixed earlier index i, the regions Z_i decrease as these labels advance from $\mathcal{S}'_i$. A step leaves earlier entries alone, and the three rules for an operation only enlarge their integrated regions. Fix $h = k_\flat \le i \le k$ and a cell of $G_i = Z_i(\mathcal{S}'_k) \cap X$. For $i \le s < k$ the cell lies in a component of $Z_s(\mathcal{S}'_s)$. Healing that component would remove the cell from Z_i. Making it an island would have the same effect if $i < s$. In the remaining cases an operation on that component would give it a mark. If the mark ceased to stand, a later healing or island would have integrated the cell at level i. If it still stands, the marked component lies in the component X that contains the cell. This contradicts the exclusion of marks at levels $h, \dots, k-1$. Thus no intermediate class component contains the cell. Its memberships in the regions at i, and at $i-1$ when $i \ge 2$, are the same in $\mathcal{S}'_k$ as in $\mathcal{S}'_i$.

The entire component of $Z_i(\mathcal{S}'_i)$ through that cell survives in $Z_i(\mathcal{S}'_k)$. To see this, at each intermediate operation the whole component still lies in the same component of the top region as the chosen cell, by nesting and connectedness. The operation leaves that top component alone. Induction proves the assertion, and the surviving component is contained in X. Hence G_i is a union of components of the original output of step i. For $i > h$, the step's input restricted to G_i is G_{i-1}. The absence of creations for $h < i \le k$ now lets us use the preceding restriction argument at each of those steps. All these steps have $i \ge 2$, since $h \ge 1$. The argument allows the first step's creation set to be defined separately.

For clarity, measure lengths in bare units and set $a_i = M\check{R}_i L^i$, the side of a level-i cell. For a box B, write $B^{[s]}$ for its enlargement by a distance s in the sup norm in its lift. A negative s means moving each face inward by $|s|$. The oldest component G_h is a nonempty box by the small-component boxing rule and the survival just proved. Inductively its successor is a box. Its hull H_i on the next grid satisfies $G_{i-1} \subset H_i \subset G_{i-1}^{[a_i]}$, by rounding the endpoints in each coordinate to the level-i grid. A box is fixed by the boxing operation. Thus the two restricted outputs have the form

$$O_i := \Omega_i^c \cap X = H_i^{[8a_i]}, \qquad G_i = H_i^{[10a_i]}, \qquad G_k = X. \tag{4.cdxliii}$$

Here and below Ω_i, Z_i without a label refer to $\mathcal{S}'_k$. Each H_i is nonempty by eligibility, which requires a predecessor through the waiting window. In particular, X has at least 21 cells along each side. With the small-box cutoff of 100 cells per side, it has at most 100.

4.17.1.22. The boxes used in preparation. Let $u_i = ML^i$ be the side of an M-cube at level i in bare units. In the scale choice used for the preparation, let $k_0 + 1$ be the last index with $a_{k_0+1} = u_k$, with $h < k_0 \le k - 2$. Thus $a_i \le u_k$ up to $k_0 + 1$, and $a_i \ge Lu_k$ thereafter. The construction uses the following distances from the original small-field region. For $h < i \le k_0$ take distance $5a_i$ from Ω_i. For $k_0 < i \le k$ take distance $5u_k + w_i$ from Ω_{k_0+1}, where

$$w_{k_0+1} = 2u_k, \qquad w_k = 0, \qquad w_i = \sum_{q=i+1}^{k} u_q \quad (k_0 + 2 \le i < k).$$

In each case $Z_i''(X)$ is the union of cells of X all of whose points have at least the indicated distance. The change of scale allows the later boxes to stay on the scale u_k.

For $0 < s \le 8a_i$, the points of X at distance at least s from Ω_i form $O_i^{[-s]}$. The interior of O_i is disjoint from Ω_i. Every face of O_i lies in the closure of the part of Ω_i between O_i and G_i, since G_i extends two cells beyond O_i in every direction. Thus the distance inside O_i is the least distance to one of its faces, which gives the erosion formula. It follows from eq. 4.cdxliii that

$$Z_i''(X) = \begin{cases} H_i^{[3a_i]}, & h < i \le k_0, \\ H_{k_0+1}^{[3u_k - w_i]}, & k_0 < i \le k. \end{cases} \tag{4.cdxliv}$$

The first box is a union of level-i cells. For the second, the faces of H_{k_0+1} lie on the grid with side u_k. Since L is odd, those faces also lie on the grid with side u_i, and $3u_k - w_i$ is a multiple of u_i. The second box is therefore a union of M-cubes of level i. These grid facts justify passing from the set of points at the specified distance to the union of cells in the definition.

The earlier boxes satisfy $H_i \subset Z_i''(X) \subset O_i \subset G_i$. The next hull contains G_i, so the nest increases through the change of scale. At the later levels the centre is fixed, and w_i decreases from $2u_k$ to zero. The boxes remain inside O_{k_0+1}, which itself is contained in all later O_i. Thus $G_h \subset Z_{h+1}''(X) \subset \cdots \subset Z_k''(X)$. The separation from G_h to the exterior of $Z_{h+1}''(X)$ in X is at least $3a_{h+1}$. At early stages, $Z_i''(X)$ is separated from the exterior of $Z_{i+1}''(X)$ in X by at least $7a_i$, since $G_i = H_i^{[10a_i]}$ is already inside the next box. For $k_0 < i < k$, the separation is $w_i - w_{i+1}$. In particular the late separations have the scale of the M-cubes, which explains the smaller margins used for the prepared backgrounds.

With the cutoff of 100 cells per side, $G_{k_0+1} = H_{k_0+1}^{[10u_k]}$ has sides at most $100u_k$. Hence the sides of H_{k_0+1} are at most $80u_k$, and those of $Z_k''(X)$ are at most $86u_k$. The central integration below adds one more layer of M-cubes. Its box therefore has sides at most $88M$ in level-k units.

The operation. (0) *The frame sum*, described first although it belongs inside (3). In each component the identity $1 = |W_4|^{-1} \sum_{r \in W_4} 1$ is inserted after the integration over the box and before the tree gauge of (3). Consequently no frame sum is attached to a component that leaves at (2) or during the stages. In the summand of r, every object of (3), (4) and § 4.17 that belongs to X is formed in the frame r: the reference law with its denominator, the two trees, the test on

B, and the last-shell test. If a component leaves after the tree gauge, its frame sum stays with it, and an island keeps, in each summand, the cut-off and the delta function belonging to the frame of that summand. Formed instead in a single frame r_0, these make the bracket of X integrate to $|W_4|^{-1}\sum_r \mathfrak{a}_r\, \mathsf{D}_X^{(r_0)}/\mathsf{D}_X^{(r)}$ instead of $|W_4|^{-1}\sum_r \mathfrak{a}_r$. Here $\mathfrak{a}_r$ denotes that part of the summand of r which is independent of the box, and $\mathsf{D}_X^{(r)}$ denotes the denominator of $\mathfrak{s}_X$ formed in the frame r. The reason for keeping the frame sum is that neither the test on B nor the last-shell test is gauge invariant. Each of them enters only by way of decompositions of unity performed within a single gauge-fixed measure. Let m be the measure before the tree gauge, and let $\mathsf{a}_r, \mathsf{b}_r$ be its parts that pass, respectively fail, these tests in the frame r. Then $\mathsf{a}_r + \mathsf{b}_r = \mathsf{m}$ for every r, and hence $|W_4|^{-1}\sum_r(\mathsf{a}_r + \mathsf{b}_r) = \mathsf{m}$. With the drop-outs kept instead in a single fixed frame r_0, the total is $\mathsf{m} + (|W_4|^{-1}\sum_r \mathsf{a}_r - \mathsf{a}_{r_0})$. The average is taken so as to obtain covariance under the lattice isometries; trees break this covariance. (1) *Stages*: for $j = k_\flat, \dots, k-1$, one value after another, the stored variables of the outer shell of $Z_{j+1} \cap X$ are integrated out as in a small-field step. Before this, tests at the sharper threshold $\delta'_j := g_j A_1 p_1(g_j)$ are made, formed with backgrounds localised in X. What remains is a box Λ^{pr} of side at most CM: it is the set $Z''_k(X)$ of the nest of §4.17.1.1 enlarged by one more layer of M-cubes, and this layer is the shell of level k. In it the levels $k_\flat, \dots, k$ are stacked in thin shells, the shell of level j carrying the field V_j. Together these fields form the datum V'' attached to the prepared determining set $\mathbf{B}''$; its background is U''_k. At the innermost position lies the core $C_{k_\flat} = X \cap Z_{k_\flat}$, whose side, measured in the unit of level k, is at most $CM\check{R}_{k_\flat} L^{-\mathsf{N}_k}$. (2) The test asks whether the exterior field V_k admits an extension into Λ^{pr} whose curvature is less than $2\varepsilon_k$. If an exterior field satisfies $|\partial V_k - \mathbf{1}| < \varepsilon$, then it does admit an extension, one with $|\partial V_k - \mathbf{1}| < CM^2\varepsilon$. To see this, set the field equal to $\mathbf{1}$ inside after choosing an axial gauge on the surface $\partial^+\Lambda^{\mathrm{pr}}$. Hence "no regular extension" means that some exterior plaquette has $|\partial V_k - \mathbf{1}| \geq cM^{-2}\varepsilon_k$, and this pays as in §4.17.1.40 (p. 193 there). (3) V_k is integrated over Λ^{pr}. Taken together, the parts from the test (2) up to the division into the two classes § 4.17 form the *main step* of the operation; the stages that precede it and the exponentiation that follows it are counted separately. This integration is the first move to alter the density. Integrals against any bounded F that is independent of V_k on the box are left unchanged by it. Level by level, from the top down, it resolves the bare chain $\prod_j \int dV_j\, \delta(\mathcal{M}(V_j)V_{j+1}^{-1})$, $\int dV_i(c)\, \delta(\mathcal{M}(V_{i-1})(c)V_i(c)^{-1}) = 1$. What remains is a free Haar integral,

$$\chi_k\, \chi_{k,\Lambda} \int dV_{k_\flat}\big|_{C_{k_\flat}} \chi\, \mathbf{T}_{k_\flat}(C_{k_\flat}) \int dV''\big|_{\mathbf{B}''\cap(\Lambda^{\mathrm{pr}}\setminus C_{k_\flat})}\, \chi''_k\, e^{A''_k(U''_k)}$$

Here $\chi_{k,\Lambda}$ is the test of (2); χ stands for the plaquette tests of level $k_\flat$ carried by the term on $\Omega_{k_\flat} \cap C_{k_\flat}$; and χ''_k, A''_k are the tests and the action that (1) leaves. The integral is regarded as a density that is constant in V_k on the box. The field that is expanded is V''. It is expanded around the averages of the minimiser U_0 below, each average taken at its own level:

$$V'' = e^{ig_k B} V_0 \ \text{ on the fluctuation bonds } \mathbb{B}_0 = \mathbf{B}'' \cap (\text{shells}),$$

$$V_0 = (\mathcal{M}^j U_0 \text{ on the shell of level } j)_{k_\flat \leq j \leq k}.$$

Thus B is a multi-scale field on bonds of every level from $k_\flat$ to k, with a single scaling g_k and a single test $|B| < \delta'_k/g_k$. This is the field B in the prepared-component operation. The gauge of this field is fixed along a tree T_0 in $\mathbb{B}_0$, whose bonds come from all these lattices, one lattice for each shell. For two nested boxes $P_1 \supset P_2$, the tree is made up of axial contours which leave from a single initial site and go around P_2. The family of trees has two properties (pp. 195–196 there). First, the family is stable. Suppose one enlarges P_2 to a box $P'_2 \subset P_1$ and deletes the contour of each site of $P'_2 \setminus P_2$; then what is left is the tree of $P_1 \setminus P'_2$. Bałaban describes this property

as important, and boxes are needed because of it. Second, the tree of a shell uses up the gauge freedom at every site of the shell except one. For this reason, for each inner box on which a gauge is fixed, one external bond crossing its boundary is added, and one more is added for the outermost box. Through these extra bonds the trees of the shells become a single tree, and it fixes the gauge entirely, with Jacobian one. The core operation $\mathbf{T}_{k_\flat}(C_{k_\flat})$, which contains the failed tests, is at no point unfolded. One further test, the *last-shell test*, compares with the vacuum and constrains $V_{k_\flat}$ between the core and $\mathbb{B}_0$. It requires

$$|\partial U_{k_\flat,\square}((\mathbf{1},V_{k_\flat}))-\mathbf{1}|<\tfrac{1}{2}\varepsilon_{k_\flat}\eta_{k_\flat}^2$$

on the test cubes of that region, where $(\mathbf{1},V_{k_\flat})$ equals $\mathbf{1}$ on the side of $\mathbb{B}_0$. (4) In the run with its own choices, $\theta_k=x_k$. For each term τ, its mass $m_\tau=\int\tau\,dV_k|_{\Lambda^{\mathrm{pr}}}$ multiplies the reference law. Of the two factors of this product, the level-k variable $V_k':=V_kV_\Lambda^{-1}$, where V_Λ is the element of the critical orbit of Prop. 4.59 that equals $\mathbf{1}$ on the tree of δ_{ax}, occurs only in the reference law: in its numerator as the argument, and in its denominator as the variable of integration. Let $U_{k,X}$ denote the background localised in X, and let $\chi(\Lambda^{\mathrm{pr}})$ be the cut-off $\mathbf{1}\{|\frac{1}{i}\log V'(b)|<M_0\varepsilon_k$ on the bonds of the box$\}$, in which M_0 is a constant chosen large after M. Let δ_{ax} be the tree gauge of $\mathbb{T}^{(k)}\cap\Lambda^{\mathrm{pr}}$, taken on a tree that contains all the sites of this set together with the external bond of T_0 into the box; that bond is part of the tree because, without it, a constant gauge transformation of the box is left unfixed. Let ϕ_X be a cut-off function that equals 1 well inside X. Then

$$\mathfrak{s}_X(V_k)=\frac{\delta_{\mathrm{ax}}(V_k')\,\chi(\Lambda^{\mathrm{pr}})(V_k')\,e^{-x_kA(\phi_X,U_{k,X}(V_k'V_\Lambda))}}{\int dV'|_{\Lambda^{\mathrm{pr}}}\,\delta_{\mathrm{ax}}(V')\,\chi(\Lambda^{\mathrm{pr}})(V')\,e^{-x_kA(\phi_X,U_{k,X}(V'V_\Lambda))}}.$$

If one of the tests of (1)–(3) fails in a component, that component drops out §4.17.1.1: it remains live, and the failure counts as a creation. This can happen to the same component repeatedly. Each occurrence counts as a creation and is followed by a wait of at most N_k+1 steps.

The purpose of the stages, and the reason for a sharper threshold. Continuing the field by least action costs a constant factor: for exterior data with $|\partial V_k-\mathbf{1}|<\varepsilon$ one has $|\partial V_\Lambda-\mathbf{1}|\le C(M)\varepsilon$, and $|\partial U_0-\mathbf{1}|\le C(M)\varepsilon_k\eta_k^2$ near the box. For every value of the coupling, the stages reduce the component from side $M\check{R}_k$ to a box of side $O(M)$. Hence $C(M)$ is a constant fixed before γ. On the shell of level j it is overcome by the factor $L^{-(k-j)}$ by which a level-k field scales when viewed from level j. On the outermost shells it is overcome by $\check{R}_k^{-c}$, because $1+\log_L\check{R}_i$ levels are compressed into the box, namely the levels from the last level i with $L^{k-i}=\check{R}_i$ up to level k. Consequently U_0 belongs to the domain of the carried tests. It also belongs to the analyticity domains of the terms, insofar as the domain of a term lies inside the component or at a distance from the box; the remaining terms are treated in § 4.17. In the words of 122 (1989) 175, p. 195, "it is the reason why we have done the preliminary integrations". As in Lem. 4.57, the tests made at a stage have to imply the earlier ones. Here this must happen $\mathsf{N}_k=\check{R}_k$ times in succession, and the losses together amount to a fixed fraction. If two backgrounds satisfy the same constraints on a region, their difference is the response to the difference of their data, and this response decays. Let $U_{k,X}^{(n)}$ denote the background after n stages, localised in X. Put $j=k_\flat+n$, and let β_{s} be a fixed small fraction, written β by Bałaban. Then, on the support of the tests of stage $n+1$ and on the shell of level $i\le j$,

$$|\partial U_{k,X}^{(n)}-\mathbf{1}|\le|\partial U_{k,X}^{(n+1)}-\mathbf{1}|(1+c\beta_{\mathrm{s}}2^{-(j-i)})+c\beta_{\mathrm{s}}2^{-(j-i)}\varepsilon_i\eta_i^2,$$

where the error term is

$$C\delta'_j e^{-\delta_0 \operatorname{dist}} \eta_i^2 \le C\,\tfrac{\delta'_j}{\varepsilon_j}(1+(j-i)^{1/2}) e^{-c(M/M_1)(j-i)}\varepsilon_i\eta_i^2$$

. For M large, the decay $e^{-c(M/M_1)(j-i)}$ across the shells dominates $1+(j-i)^{1/2}$, and $2^{-(j-i)}$ remains. The ratio $\delta'_j/\varepsilon_j = (A_1/A_0)(\log x_j)^{-(\mathfrak{p}_0-\mathfrak{p}_1)}$ dominates the fixed constants, and at this point smallness of A_1/A_0 suffices as well. After n stages the test requires $|\partial U^{(n)}_{k,X} - \mathbf{1}| < (1-\beta_{\mathrm{s}}(1-2^{-(j-i+1)}))\varepsilon_i\eta_i^2$ on that shell. On the outermost shells, where the collars of the stages overlap, this requirement is relaxed by a factor $L_\flat^2$ for each overlap, where $L_\flat$ is an integer with $2 \le L_\flat < \frac{1}{2}L$ that is chosen together with L; such a choice requires $L \ge 5$. Let $k_\sharp + 1$ be the last of the levels at which an $M\check{R}$-cube, measured in units of level k, has side precisely M; in formulas, $M\check{R}_{k_\sharp+1}L^{k_\sharp+1-k} = M$. The collars of the stages with $j \le k_\sharp$ are disjoint, and the collars of the later stages overlap. On the shell i the factor is $L_\flat^{2(i-k_\sharp-1)_+}$. It depends on the shell only, not on the stage, and it becomes larger with each successive shell until it reaches $\check{R}_k^c$. In the region not covered by the shells, the bound is expressed in the scale j of the stage and equals $(1-\frac{1}{2}\beta_{\mathrm{s}})\,c_l\,L_\flat^{2e}\varepsilon_j\eta_j^2$, where $c_l = 3$ adjacent to Ω_k at the last stage and $c_l = 1$ elsewhere. Far from the box, the curvature of the least-action continuation U_0 can be as large as $(2+c)\varepsilon_k\eta_k^2$, and the test of the last stage must be implied there as well. Here $e = 0$ for $j \le k_\sharp$. For $j > k_\sharp$ one takes $e = (j-l-1)_+$ on $\Omega_l \setminus \Omega_{l+1}$, $k_\sharp < l \le j$, and $e = j - k_\sharp - 1$ between the shells and $\Omega_{k_\sharp+1}$. This is the form of 122 (1989) 175, (1.24), where the level is denoted k_0 and the base L_0. On a layer $l < j$ this bound remains below $\varepsilon_l\eta_l^2$, the scale of that layer. In passing from the scale $j+1$ to the scale j one gains L^{-2} and pays $L_\flat^2$. On plaquettes read by both stages, the later threshold leaves a margin for the geometrically decaying response. The crossing plaquettes of the prepared interface require the comparison below. The sharper threshold enters in two places. In the first, the new fluctuation test has to imply the old fluctuation tests of step $j+1$, at δ_j and $2\delta_j$. This holds because $CL\delta'_j = CL(\log x_j)^{-(\mathfrak{p}_0-\mathfrak{p}_1)}\delta_j \le \frac{1}{2}\delta_j$ for γ small, and it puts no new condition on the constants chosen earlier. In the second, at the main step, δ'_k has to remain a small fraction of $\varepsilon_{k_\flat}$, which is the curvature threshold of the core level. In this estimate the thresholds of two levels may differ by $\varepsilon_j/\varepsilon_i \le C(1+(j-i)^{1/2})$, the bound valid for every pair of levels §4.17.1.45. Restrict to the common support of the carried tests χ_k, the tests of (2) and (3), and the tests and delta functions that stand under the core operation; the latter make the data stored in the core regular. On this set, and provided $C(M)\check{R}_k^{-c} \le c$ on the outermost shells, the final regularity threshold follows on its inset, with the seam implications discussed below. The equality does not reach ∂X: adjacent to X^c, the background localised in X is constrained bond by bond on the bare lattice. There the test therefore demands $|\partial V_k - \mathbf{1}| < C\varepsilon_k\eta_k^2$ of the block field itself, while the carried tests provide only $2\varepsilon_k$. All of this holds under the condition

$$C\,\mathsf{N}_k^{1/2}\,\frac{p_1(g_{k_\flat})}{p_0(g_{k_\flat})} + C(M)\,L^{-\mathsf{N}_k}\mathsf{N}_k^{1/2} \le c\,:$$

the first term is δ'_k measured relative to $\varepsilon_{k_\flat}$, including the allowance $\mathsf{N}_k^{1/2}$. Since $\mathsf{N}_k = \check{R}_k \approx (\log x_k)^r$, it is a condition on the exponents, $\mathfrak{p}_1 < \mathfrak{p}_0 - Cr$, together with a ceiling on g_k. The second term is a level-k field viewed from level $k_\flat$, so N_k is taken large.

4.17.1.23. The prefinal reads at the old wall. Let $q = j+1 < m$ and $a_q = M\check{R}_qL^q$ in bare units. Write S_X for the level-j average of the prospective background localized to X, in the frame of the approximate collar test. In an uncompressed stage the new prepared face is five level-q cells from the old Ω_q wall. At $q = k_0+1$ it is seven cells away. For $q > k_0+1$ the prepared box is outside Ω_{k_0+1}, hence outside Ω_j; the ordinary separation from Ω_j^c to Ω_q is at least eight level-q cells. At every prefinal stage this last separation also bounds the clearance to the older determining layers on the exterior side. The clearance on the interior side from the wall to Ω_{q+1} is at least eight level-$(q+1)$

cells. Since $a_{q+1} \geq a_q$, the four-cell wall neighborhood lies in $\Omega^c_{q+1} \cap X$, which is inside the two-cell top inset of X. The two groups of ordinary approximate tests have bases in $(\Omega_q^{\sim 2} \setminus \Omega_q) \cap X$ and $(\Omega_q \setminus \Lambda_q^{\sim}) \cap X$. Their bases lie within two level-q cells of the wall, since in this quiet component Z_q is the two-cell enlargement of $\Omega^c_q \cap X$. The four test layers of the localization, with the endpoint blocks of every bond it reads, add less than one level-q cell. This follows from $L^2 M_2 \leq M$ and the fixed block reach. Thus the complete reads lie within three cells of the wall, with clearance from the new prepared face, the older core, the next ordinary layer and X^c.

Use this region with an additional fixed comparison band on each side, of total reach less than four cells from the wall. The ordinary global center at step q and the preceding stage center have the same rows there: the level-q rows on the inside and the level-j rows on the outside. Earlier promotions change only rows beyond the original Ω_j wall, with the associated one-step completions there. The comparison band misses those rows and their blocks by the ordinary separation. The two problems agree on the corresponding localized datum with these two scales. The joined comparison and averaging on the full endpoint blocks bound the difference of their axial readings by $C_L \varepsilon_\star e^{-cM \check{R}_q}$. Use the ordinary strip shift on the bonds on which that substitution is made. Its physical real bound on those bonds is $C_L \varepsilon_j e^{-c\check{R}_q}$.

The prospective stage has level q throughout these test cubes and their enlargements. The localized test-cube datum is the lift of the same V_q there. Its background has the same axial read at the common nodes. Comparing it with the prospective background on the full $\square'^{\sim 4}$ box gives

$$|S_X(b)(V^{\mathrm{bg}}_{j,\square'}(b))^{-1} - \mathbf{1}| \leq C_L \varepsilon_\star e^{-cM_2 \check{R}_q}.$$

The distance from the endpoint blocks to the comparison boundary is a fixed fraction of a test layer, in level-q units. This is the local comparison for the actual test centre, with V_q fixed on its whole halo.

At a bond with neither endpoint in Ω_q the relative collar test gives $2\delta'_j$ against S_X. For a bond read by the completed ordinary strip test, compare instead the ordinary center to the preceding-stage center and then to S_X by the real fixed-target path of §4.17.1.27. Across the old wall the coarse row joining the two centres is an outward row of the old member and a retained row after the promotion. Its value is $V_q(c)$ throughout the path. The two centres are old nodes of levels j and q. The physical increment of this outward row is zero. Its reduced increment still reads the changing internal links of the exterior block. Use that increment in the response and restore both node values as in the two-block calculation below. The restoration of the node gauges in the one-step average leaves the identity at the old level-j node and v_j^{-1} at the level-q node, with $|\mathsf{v}_j - \mathbf{1}| \leq C_L \delta'_j$. The additional exterior contour of the ordinary strip frame uses only internal bonds of the exterior block. This block is entirely in the new exterior collar, and those nonpivots are tested there. The regular mean therefore bounds that contour change by $C_L \delta'_j$. This gives the comparison in the inherited frame with both endpoint factors included. At an old-wall crossing the physical link itself is controlled by the completed ordinary test at δ_j, including the original real strip substitution. The new collar test need not read that crossing. Any adjacent factor on an old ordinary Gaussian bond was removed by the ordinary whole-term identity, with its original cutoff and the retained strip factors. That bond is absent from the free variables of the prepared collar. The sum of the center errors is at most $C_L \delta'_j + C_L \varepsilon_\star e^{-cM_2 \check{R}_q} = o(\delta_j)$, so both groups of ordinary approximate factors equal one on their complete reads. The argument uses the direct exterior-only collar mask.

At the first stage put $h = m_\flat$. The surviving level-h density tests at this first stage have bases in $Z_h \cap \Omega_h \cap X$. The stage-zero guard recovered in §4.17.1.25 implies the other bottom density tests outside $\Omega_q^{\sim}$, by localization on their own complete four-layer boxes. The bases outside Z_h lie in Λ_h, two level-h cells from Ω^c_h, and the exclusion of $\Omega_q^{\sim}$ keeps the full boxes in $\Omega_h \setminus \Omega_q$. The comparison with the pure level-h test tower there costs at most $C\varepsilon_\star e^{-c\check{R}_h}$ in level-h units, within

the $\beta_{\mathrm{s}}/2$ margin of the stage-zero bound. The retained bottom tests read the frozen core: its gap to the first prepared face is at least $3a_q$ and their halos reach less than a_q.

4.17.1.24. The comparison of the centers at the last stage. Put $j = m-1$, let $Y = Z''_m(X)$, and write

$$K = (\Omega_m^c \setminus Y) \cap X.$$

At the last stage the approximate collar test compares the actual V_j to the j-fold average S of $U_{m,X}^{(\mathsf{N}_m)}$ on its nonpivot mask. In the direct graded rule the mask consists of bonds with an endpoint in K and neither endpoint in Ω_m. The incident-bond convention additionally tests crossings into Ω_m. In the direct rule those crossings use the completed ordinary strip test. Denote the corresponding preceding-stage average by S^-, the ordinary strip center by $V_j^{\mathrm{bg,loc}}$, and the ordinary test-cube center by $V_{j,\square'}^{\mathrm{bg}}$. We use the actual averaging constraints and keep the preceding-stage guard during these comparisons.

The ordinary approximate tests in the fifth and sixth groups have bases in $(\Omega_m^{\sim 2} \setminus \Omega_m) \cap X$ and $(\Omega_m \setminus \Lambda_m^{\sim}) \cap X$, respectively. Their tested bonds are those of $\square'^{\sim 2}$ other than the pivots. Their centers are formed on $\square'^{\sim 4}$. The interior bases are at least one level-m cell from the exterior of X. The exterior bases lie in O_m and within two level-m cells of Ω_m. Each added test layer has width $M_2\check{R}_m$ in level-m units. The complete reads therefore remain at distance at least $cM\check{R}_m$ from ∂X, including the blocks of their end points. The prepared core Y lies in $\Omega_{k_0+1}^c \cap X$, which lies in the old live region $Z_{m-1} \cap X$. By eq. 4.cdxliii this old region lies at least eight cells inside Ω_m^c. Thus a test-cube box can meet the old wall and stays clear of the prepared seam and the tower collar at ∂X. The prospective joined sequence has top level m throughout the test-cube box.

On that box the actual top rows of $U_{m,X}^{(\mathsf{N}_m)}$ have values V_m. Its joined datum for the test-cube localisation is therefore exactly the one induced by the lift of V_m. The localised background is the ordinary test-cube background. For a real datum below the ceiling of the joined comparison, eq. 3.cdi and eq. 3.cdii compare these backgrounds in their common block axial reading. In the units of level m the read is at distance $cM_2\check{R}_m$ from the boundary of its test-cube box. Consequently

$$|S(b)(V_{j,\square'}^{\mathrm{bg}}(b))^{-1} - \mathbf{1}| \le C\varepsilon_\star e^{-cM_2\check{R}_m}$$

on the stated footprint, with $\varepsilon_\star$ a bound on the small regular datum. This application reads the ordinary coarse density test and the retained rows. It does not require the discarded sharp bound at the artificial bare rim of X.

For this real datum bound, the preceding-stage support already controls the tested holonomies before the last enlargement. Levels below j keep their rows. The determining set of level j loses rows. Its second-kind rows still use the unchanged layers $j-1, j$, so the remaining tested plaquettes have their old holonomies. For a tested plaquette of level m with a corner in a carried density cube, that test bounds its actual V_m holonomy. The density cubes on $\Omega_m^{\sim 4}$ cover the plaquettes near the old seam. In the other case the averaging blocks of a newly tested plaquette are in the preceding member of level j, by the last prepared separation, and outside the old Ω_m. Here the fine plaquettes are formed from the old determining rows, and the exact one-step average gives the coarse bound. At the exterior tower a tested holonomy is either the identity or the lifted top holonomy. These are bounds at the respective levels of the datum. They do not divide the top plaquette deviation by the square of the fine spacing.

To compare S^- with $V_j^{\mathrm{bg,loc}}$, use the region between an inner and an outer box. For the outer box take X inset by a quarter of a level-m cell, rounded on the grid of M_1-cubes at that level. The old exterior inside X is a box separated from X^c by two cells by eq. 4.cdxliii. For the excluded inner box take $(\Omega_m^{\sim 2})^c \cap X$, with this enlargement in level-m cells. The outer boundary band is in the old Ω_m, and the inner boundary band is in $\Lambda_j \setminus \Omega_m$. The earlier stages alter rows only beyond the inner band, towards Z_j. The new gauge draws of the last stage start outside $\Omega_m^{\sim 3}$. On the annulus both backgrounds therefore have the same regions and the same determining rows, including the

top data used on its two boundary bands. Join at level m on the outer band and at level j on the inner band. Both localisations give the same datum. For a read of either group with both ends in Ω_m, use the localized strip center only on the actual strip read and substitution mask. For a read unaffected by that substitution use the ordinary global center. It has the same determining rows on the annulus. The block axial frame on the full read blocks is common, and the distance to the annulus boundary is of order $M\check{R}_m$. With $L_j(b)$ denoting the applicable ordinary center, the comparison gives

$$|L_j(b)(S^-(b))^{-1} - \mathbf{1}| \le C\varepsilon_\star e^{-c\check{R}_m}.$$

For the change from S^- to S, use the finer determining set at the averages of the prospective background, then change the freed level-j rows to their actual values. The two constraint sets are nested at the induced datum. A critical point for the larger constraint set is critical for the smaller set there. The common rows, including the exterior tower rows, do not change. On a nonpivot freed row the approximate collar test bounds the increment by $2\delta'_j$. For a freed pivot, both blocks of its coarse row are outside the old Ω_m. Each nonpivot in the double block is controlled by the collar test or belongs to the common fine rows. This includes the rows crossing the prepared seam, whose outside block is in K. The two fine fields have the same V_m average on that coarse row. The contour argument in §4.17.1.27 places the actual pivot in the small branch, and inversion of the averaging equation bounds its increment by $C(L)\delta'_j$. A pivot crossing the old Ω_m seam is absent from the finer determining set in the convention excluding inward bonds. The resulting plain datum increment is supported in the collar. The real response supplies the reduced datum and the node factors, including any factors at a seam. On the common axial reads inside the old Ω_m the response bounds $S^-(b)S(b)^{-1} - \mathbf{1}$ by $C(L)\delta'_j$. This argument is made after the approximate collar test and before the sharper collar split.

On a read of either group in the old Ω_m, add these three bounds to the carried or completed fluctuation bound δ_j, with the ordinary strip shift on its actual mask. The exponentially small terms and $C(L)\delta'_j$ fit in the remaining δ_j of the approximate threshold $2\delta_j$. A nonpivot read with an end point outside the old Ω_m has that end point in K. For a bond with neither endpoint in Ω_m, use the collar test and the test-cube comparison to obtain $2\delta'_j + C\varepsilon_\star e^{-cM_2\check{R}_m} < 2\delta_j$. Both tests exclude the geometric pivots on this overlap.

For a crossing of the old wall in the direct graded rule, the full exterior endpoint block lies in K, and the interior endpoint is in the first stored ordinary strip layer. The completed strip test there gives the bound at δ_j in the original plain coordinate, without changing the physical link or its soft gauge factor. Compare only the centres L_j and S on the full two blocks by the mean-contour axial reading used below for the global collar chart. The exterior contour of the ordinary frame is determined by the internal bonds of the exterior block. The direct collar test does read these internal bonds. Its bound controls that contour without assuming an additional collar bound on the crossing itself. On the common retained centre pins the comparison gives

$$|L_j(b)S(b)^{-1} - \mathbf{1}| \le C_L\delta'_j + C_L\varepsilon_\star e^{-c\check{R}_m}$$

in the inherited frame. Together with the test-cube error and the original strip bound this is less than $2\delta_j$. The two-block comparison below uses the actual retained centre nodes and the original exterior contour. When using the incident-bond convention and the crossing itself lies in its mask, the direct collar bound applies in that convention's prescribed frame.

For the inset split the finer path supplies the local collar response at a fixed admitted prospective datum. The clearances above place the complete footprints of the two groups in its comparison region, including the interior endpoints of tests based outside Ω_m. The inherited strip quadratic form uses the original strip variables.

4.17.1.25. The preceding guard on the prepared seam. The stage-zero plaquettes lie in $\Omega_h \setminus \Omega_{h+1}$. At any stage with $h < j < m$, every prepared shell lies in Ω_m^c. The deep exterior clause is in $\Omega_{i_0+1}^c$,

where $i_0 + 1 \le j$, and the other exterior clauses are in $\Omega_l \setminus \Omega_{l+1}$ with $l \le j$. Consequently every earlier guard reads only $X \cap \Omega_m^c$, which is two level-m cells inside X by eq. 4.cdxliii. The artificial bare rim occurs only in the final guard.

Consider the implication from stage m to $j = m - 1$ on the ordinary real collar chart. On a shell of index $i < j$ already present at j, the geometric factor and curvature scale are unchanged. The later coefficient is smaller by $\beta_{\mathrm{s}} 2^{-(j-i+2)}$. The weighted response from the changed collar rows is at most

$$C_L \frac{\delta'_j}{\varepsilon_j}(1 + \sqrt{j-i}) e^{-c(M/M_1)(j-i)} \varepsilon_i L^{-2i}.$$

Here the ratio of thresholds contributes $1 + \sqrt{j-i}$ and the shell separation gives the exponential. Choose M/M_1 so their product is bounded by a constant times $2^{-(j-i)}$, then use δ'_j/ε_j to fit the reserve. Any overlap factor on this fixed shell is the same at both stages and is at least one. On a plaquette wholly in the new shell $Z''_m \setminus Z''_j$ the factor is unchanged and the coefficient decreases from $1 - \beta_{\mathrm{s}}/2$ to $1 - 3\beta_{\mathrm{s}}/4$. Outside both nests the ratio of the two thresholds is at most

$$\frac{\varepsilon_m}{\varepsilon_j} \frac{\max\{L_\flat^2, 3\}}{L^2}.$$

The factor 3 occurs at $l = j$. The adjacent reference couplings differ by a bounded increment, so their threshold ratio tends to one in the small-coupling range. These exterior comparisons have a fixed fractional margin. On admitted real data the collar response fits these reserves, by its weighted bound on the old shells and the small ratio δ'_j/ε_j in the exterior. All these reads remain in the inset.

A plaquette can cross a face of Z''_m while lying wholly in $\Omega^c_{k_0+1} \setminus Z''_j$. Choose a normal and a tangent edge away from the corners of the face, using the half-spacing faces of the centered blocks. The separation of the prepared boxes puts the whole plaquette outside Z''_j, and the clearance from Ω_{k_0+1} puts it inside the old deep exterior region. The preceding guard reads it. The final shell and exterior clauses both require containment, so neither reads it. It is also outside the later original layers used by the other exterior clauses.

At such a plaquette write $T_j(p)$ for the preceding threshold and $E_j(p)$ for the bound on the curvature change from the collar response, in bare plaquette units. We must show that the prospective curvature is below $T_j(p) - E_j(p)$ on the support on which the preceding guard is removed. The small-paying-energy estimate below gives this comparison on the part of the support to which it applies. In the two joined sequences the plaquette has the old and new scales j and m under the largest-corner convention. The weighted profile at p is bounded, in either sequence, by

$$C\left(\varsigma_* + \check{R}_m^2 \varepsilon_m e^{-c\delta_{\mathrm{c}} M \check{R}_m}\right) = o(\varepsilon_j).$$

Indeed the old rows are read from the compatible field, and the new rows use its retained averages. The distance to the floor uses the same prepared-window separation. The two bare curvature bounds gain respectively L^{-2j} and L^{-2m}. Their sum is $o(T_j(p))$ and bounds the change in any common real frame. This proves the preceding seam test on the low-energy branch.

In an inset split let χ_{in} be the literal final clauses restricted to plaquettes whose corners are all in X with one full level-m cell layer removed. The preceding reads omitted by these clauses are the deep exterior plaquettes straddling $\partial Z''_m$. Put $F_j = L_\flat^{2(j-k_0-1)}$ and let $\kappa_\times$ test each of these plaquettes at $(1 - 3\beta_{\mathrm{s}}/4) F_j \varepsilon_j L^{-2j}$ in the prospective background. The datum for that test is held fixed in the collar integration. If χ_{in} is the final inset test, the complementary factor of their product is

$$1 - \chi_{\mathrm{in}} \kappa_\times = (1 - \chi_{\mathrm{in}}) + \chi_{\mathrm{in}}(1 - \kappa_\times).$$

Both terms belong to one regularity stop before the collar integration. Under the payment hypotheses a failure in the second term is impossible at small paying energy by the profile estimate above.

The threshold leaves a reserve $\beta_{\mathrm{s}}F_j\varepsilon_j L^{-2j}/4$ for the old-chart response. On the passing branch the fixed-target response estimate below fits that reserve throughout the admitted real collar box, so the preceding seam test is one there. This implication uses the reserve and the collar response. The small-paying-energy argument concerns failure of $\kappa_\times$.

There is also a direct bound for these crossings on the admitted real inset chart at the final stage. Translate a crossing plaquette p a bounded number of bare spacings into an exterior cell at one of its outside corners, keeping its orientation. The translated plaquette lies in the deep exterior and in the inset. The separation from Z_j'', the old wall, and the outer boundary lets both plaquettes lie in the smaller part of a fixed-size level-j regularity cube. Its enlargement is in the prospective member of level j. The local gauge on this cube applies across the level-m face. In that common gauge the difference of the linear plaquette curls is at most $C\varepsilon_\star L^{-(2+\alpha)j}$ by the first-difference Hölder bound. The quadratic and higher terms in the products of the four links vary by at most $C\varepsilon_\star^2 L^{-3j}$, by the bound on their first differences. The inset test bounds the defect of the translated plaquette by

$$(1-\tfrac{1}{2}\beta_{\mathrm{s}})L_\flat^2 F_j\varepsilon_m L^{-2m}.$$

Divide the three bounds by $F_j\varepsilon_j L^{-2j}$. The last displayed bound has ratio at most $(1-\beta_{\mathrm{s}}/2)(L_\flat/L)^2(1+o(1))$. The two errors tend to zero, since $j=m-1$ and $m\ge \check R_m$, with $\check R_m$ growing at least as the square of the logarithm of the inverse coupling. This gives the reserve of $\kappa_\times$ once the real prospective datum has been admitted. The fixed-target response then transfers it to the preceding guard.

An earlier transition has its own crossing reads. In particular, with $h=m_\flat$, the first prepared face lies strictly in $\Omega_h\setminus\Omega_{h+1}$. A bare plaquette astride that face is tested at stage zero. Neither containment clause of the next guard tests it. The final-stage Hölder estimate uses $j=m-1$ to make its translation error small compared with the tested threshold. At the first face that argument does not supply the reserve.

For a transition from j to $j+1$, use the actual list of preceding reads. At a newly omitted crossing the preceding coefficient is $(1-\beta_{\mathrm{s}}/2)F_j\varepsilon_j L^{-2j}$, with $F_j=1$ in the uncompressed range and $F_j=L_\flat^{2(j-k_0-1)_+}$ thereafter. In a split with an added crossing test, impose $(1-3\beta_{\mathrm{s}}/4)F_j\varepsilon_j L^{-2j}$ on the fixed prospective datum. This leaves $\beta_{\mathrm{s}}F_j\varepsilon_j L^{-2j}/4$ for the old response. The first face has the separation from the bottom core and the old wall needed for the local paying-energy comparison. On the complete crossing carriers the small-energy argument bounds the prospective defect by $C(\sigma+\check R_m^2\varepsilon_m e^{-cM\check R_m})L^{-2j}$, with $\sigma=c_*\delta_j'\check R_m^{-4}$. This fits the crossing test. Its failure on the otherwise passing prospective branch belongs to the same regularity stop on the old prefix. On its passing branch eq. 4.cdxlv fits the reserve on the whole admitted real collar chart and gives the preceding inequality there.

At the first transition put $h=m_\flat$, $q=h+1$ and $Y=Z_q''$. The original stage-zero guard tests plaquettes contained in $\Omega_h\setminus\Omega_q$. All of these lie in the inset of X. For one contained in Y, the first prepared shell tests it at $(1-3\beta_{\mathrm{s}}/4)\varepsilon_h L^{-2h}$. For one contained in Y^c, use the new deep exterior bound $(1-\beta_{\mathrm{s}}/2)\varepsilon_q L^{-2q}$. Here $h<k_0$, so $i_0=h$ at the new stage and the exterior power of $L_\flat$ is zero. A plaquette with corners on both sides uses the added crossing test at the first shell threshold. In all three cases the new bound leaves at least $\beta_{\mathrm{s}}\varepsilon_h L^{-2h}/4$ below the old bound. For the exterior case use $\varepsilon_q/\varepsilon_h=1+o(1)$ and L^{-2}.

The passing prospective guard fixes an admitted real orbit at the outgoing datum. Hold that datum fixed and use §4.17.1.27 with target V_q on the full approximate collar ball. The old curvature can change by at most $C_L\delta_h' L^{-2h}$ on the first face and on the adjacent layer-h stencils. On the covered layer the same weighted comparison applies to its full stencils, using the old units there. This error fits the reserve. Thus the old stage-zero guard equals one on the real chart for the passing graded branch. The earlier value of that guard is still used in assigning the incoming stage-zero outcome. The removal in the collar integral follows from the comparison on its whole real chart.

For an operation using these additional tests through the whole window, assign a newly added crossing of $\partial Z''_{s+1}$ to shell s. Retain it in the subsequent guards with the same coefficient as that shell. Its first coefficient is $1-3\beta_{\mathrm{s}}/4$. From the transition j to $j+1$ with $j>s$, its coefficient drops by $\beta_{\mathrm{s}}2^{-(j-s+2)}$. The stencil at this old face has scale at least L^s, and the separated faces give the same weighted distance to the newly changed rows as for the adjoining shell, up to a bounded stencil enlargement. For each intervening layer the path crosses a width of at least a fixed multiple of M in its own units. The decaying form of eq. 4.cdxlv therefore gives a defect change bounded by

$$C_L\delta'_jL^{-2s}e^{-c(M/M_1)(j-s-1)}.$$

The adjacent frame comparison costs at most $C(1+\sqrt{j-s})$ in comparing ε_j with ε_s. Choose the shell separation and then the coupling floor as in the original ladder. The change fits the decrease of the coefficient at every transition. Thus the already assigned crossings are carried with the earlier shell tests until the main integration.

For comparison of the last-stage change alone with the literal final test, let r be the product of its omitted indicators on the same prospective datum. The literal continuation is $\chi_{\mathrm{in}}r$, and the inset continuation is $\chi_{\mathrm{in}}\kappa_\times$. Their two differences are the sectors $\chi_{\mathrm{in}}r(1-\kappa_\times)$ and $\chi_{\mathrm{in}}(1-r)\kappa_\times$. One may refine each literal branch by multiplying it by $\kappa_\times$ and by $1-\kappa_\times$, leaving that branch's kernel and stage count on both pieces. Summing the two pieces recovers the original labelled measure. To change the continuation rule itself one must compare the two kernels on the difference sectors. The later implication for $\kappa_\times$ is on the main-step support. An earlier stop belongs to a different branch.

The inset test can also define the split of a separate operation from its first formation. Put $G=\chi_{\mathrm{in}}\kappa_\times$ on the real pre-collar fibre, counting an undefined tested background as a failure. Use the old prefix kernel on $G=0$ and the admitted continuation moves on $G=1$. Both are positive kernels with unit mass when written as the real moves, before their weights are expressed by the collar integral. Thus the two branches exhaust the incoming measure. A bare insertion multiplies the incoming bare measure before applying the same kernels. For example a curvature source replaces $\rho_0(U)dU$ by $e^{A(w,U)}\rho_0(U)dU$ and uses the tests and reference laws of the unsourced run. The stage count on $G=0$ is the preceding count, and failure of the product makes a single mark. The mark's own mass omits its failing test. A later containing fibre can use that failure for one payment when the complete witness is available there. The move identities give an exact weighted integral with labels assigned by G. To identify its positive branch with the localized stage integral, we must account for every retained test on its actual chart. An indicator on the fixed outgoing datum can stay outside the collar logarithm, as the final top-density factor does. An indicator on an integration variable can stay in the expansion if it is fixed in the real integration coordinates throughout the parameter domain, has a fixed carrier, and accepts the smaller coordinate ball used in its Gaussian tail estimate. For example, on a plain strip bond with reference S_{o}, an approximate factor with cube centre D tests $|e^{ig_ja_b}S_{\mathrm{o}}D^{-1}-\mathbf{1}|<2\delta_j$. If the centre error is at most $c\delta_j$ with $c<1$, the factor accepts the ordinary ball $|g_ja_b|<\delta_j$. This is a one-bond factor when the centres are fixed on that fibre. A test I_S with active slots S that accepts the product ball of radius h has

$$1-I_S\le\sum_{b\in S}\mathbf{1}_{\{|z_b|\ge h\}}.$$

The absolute Gaussian tilt on its carrier therefore gives a tail factor $|S|e^{-\gamma_2h^2}$ times the normalization and mean costs of the tilt. For several fixed selectors I_α, use their complete carriers in every binary cut, and let S_α be their sets of active coordinates. Suppose that at most ν of the sets S_α contain any one bond, counting each selector only once. For a family A of failed selectors choose one bond from each S_α. Then

$$\prod_{\alpha\in A}(1-I_\alpha)\le\sum_{\substack{b_\alpha\in S_\alpha\\ \alpha\in A}}\prod_{b\in\{b_\alpha:\alpha\in A\}}\mathbf{1}_{\{|z_b|\ge h\}}.$$

The set of bonds on the right has size at least $|A|/\nu$. Apply the stage Gaussian tilt once per distinct bond. On a stage chart with the real-contour Gaussian bound this gives the tail

$$\left(\prod_{\alpha\in A}|S_\alpha|\right)e^{-ch^2|A|/\nu}$$

in addition to the Gaussian normalization cost and the bounds on the analytic vertices. In the tilt the positive quadratic form is diagonal on the selected coordinates, so its size per coordinate is independent of the multiplicity. If no active coordinate enters a selector, carry its real value outside the Gaussian estimate.

For a bond selector attached to an original test cube, the sets of tested bonds lie in the two test layers of that cube. A bond can belong to only a fixed number of such enlarged cubes. Restrictions to the integration mask can decrease this number. The product of a repeated selector with itself can be counted once. For a single stage, the cardinality of each such set grows at most as a fixed power of $\check{R}_m$. At the stage fluctuation threshold h is a fixed fraction of δ_j'/g_j, so the factor $\log|S_\alpha|$ is absorbed by ch^2/ν at weak coupling. This count uses only the original test-cube reads. If the inverse chart introduces new active coordinates, use the resulting supports and prove the overlap estimate on them.

Fixed Borel selectors in these coordinates require no continuation as functions of z. Keep them on the real contour, and use the locally uniform integrable Gaussian majorant for holomorphy in the analytic parameters. Their compatibility with the factorization uses their full carriers. For a preceding curvature guard on the active collar, we must either show it holds on the complete chart or put it in this fixed-domain form and prove that it accepts the reference ball. A bound only on the support of the literal prospective guard gives neither property for the inset rule. If its domain moves with the analytic parameters, the real tail estimate leaves the transported integral and its bound to be proved.

For the remaining inset-positive evaluations use the real collar path at fixed prospective datum. The seam estimate concerns the prepared interface inside X. The face-lift calculation at ∂X supplies its own restriction.

4.17.1.26. The lower density and contour tests on the inset. In the ordinary step $t=j+1$ use the query rule in which a collar bond has both ends in the explored region of its round. Since that region only grows, a bond crossing the final small-field boundary $\partial\Lambda_t$ has not been queried in the collar. Its inner threshold in the ordinary mean estimate is the rim threshold $\theta_c\delta_j$. In a rule that queries incident crossings as well, the accepted ball must fit all the resulting tests, including $\delta_j^\flat$ on a queried pass. The stored coordinates in either case are the complement of the actual integrated nonpivots.

The ordinary labelled fluctuation split retains the signs of the queried outcomes. For explored bonds E^{T}, failure set I, explored rim cells E^{H}, and failed rim set $\mathsf{R}^{\mathrm{bord}}$, its factors include

$$\prod_{b\in I}\chi_b^1\prod_{b\in E^{\mathrm{T}}\setminus I}\chi_b^0\prod_{c\in\mathsf{R}^{\mathrm{bord}}}(1-\mathrm{lo}(c))\prod_{c\in E^{\mathrm{H}}\setminus\mathsf{R}^{\mathrm{bord}}}\mathrm{lo}(c).$$

Keep also the failed-cube factors and the unexplored bond factors of the ordinary split. An eligible component of the collar operation misses every ordinary creation set at the unfolded levels. A failed queried bond puts its endpoint cells in that set. A failed rim test puts its rim cell there, and a failed cube puts its coarse hull there. Their incident reads and the cube test halos lie in the same enlarged ordinary component as the creation. By nesting, such a component reaching the stage component would put its creation inside it. Thus only the passing queried factors occur on the eligible collar carriers. The failed factors in the frozen core keep their original signs.

Apply the test implications on the successful real support in their original order. Use the prospective regularity and approximate collar tests, the exact retained averages, and the ordinary completed fluctuation factors. These last factors control the inherited strip variables read by the belt tests,

including the old-wall crossings. The sharp exterior mask concerns a different stock of nonpivots. Remove the preceding regularity, ordinary density, contour, and two approximate belt groups on the full reads where the corresponding implications have been proved. Keep the final top density, the soft gauge density, the carried fluctuation factors, and the separate collar and rim factors. Keep any other positive guard on its inverse read until its own implication is established. An active guard belongs inside the exact real collar integral, while a fixed-datum factor can be taken outside. The sharp collar split follows these comparisons.

To apply the ordinary fluctuation factors, let L_b and G_b be the localized and global ordinary strip centres and put $e^{iH_b} = G_b L_b^{-1}$. Write $W' = e^{iA'_j(b)}$ for the real strip coordinate after the ordinary cutoff substitution. The physical link on a substituted bond is

$$V_j(b) = W' e^{i(1-\vartheta(W'))H_b} L_b, \qquad V_j(b) G_b^{-1} = W' e^{-i\vartheta(W')H_b}.$$

In the unchanged zone use the original coordinate. On the plain strip $\vartheta = 1$. In the cutoff zone $\vartheta = 0$ through angle $4\delta_j/3$ and $\vartheta = 1$ from $5\delta_j/3$ onward. If $|H_b| \le \delta_j/8$, the hard angle tests at thresholds at most $29\delta_j/24$ are unchanged. They agree below the start of the cutoff, and above it the change of angle is at most $\delta_j/8$, so both fail. For an approximate test with cube centre D, put $X_b = L_b D^{-1}$. When $\vartheta < 1$ and $|H_b| + |X_b - \mathbf{1}| < \delta_j/3$, both $W'X_b$ and $W' e^{i(1-\vartheta)H_b} X_b$ have chord distance less than $2\delta_j$ from the identity. When $\vartheta = 1$ the two arguments are equal. This proves the approximate test identity on the cutoff mask with its centre reserve.

Let $h < j < m$ and $q = j + 1$. The lower density cubes have bases in $(\Omega_j^{\sim 4} \setminus \Omega_q^{\sim}) \cap X$. The full four-layer halo of a base extends by less than one level-j cell. The eight-cell gap to Ω_{j-1}^c and the exclusion of $\Omega_q^{\sim}$ put this halo in $\Omega_{j-1} \setminus \Omega_q$. It lies in the top inset of X because it lies in $\Omega_q^c \cap X$. It misses the old prepared box Z_j''. For $j \le k_0$ this follows from its five-cell distance to Ω_j. At $j = k_0 + 1$ the distance is seven cells. At later levels Z_j'' is outside Ω_{j-1}. Hence the whole density box is in the old prepared layer j. Use that layer's actual determining rows at the old target.

For $q \le k_0 + 1$, the halo can meet the new prepared box Z_q''. A plaquette wholly inside that box is in the new shell and is tested at $(1 - 3\beta_{\mathrm{s}}/4)\varepsilon_j L^{-2j}$. This includes any such plaquette crossing the original Ω_j wall. A plaquette wholly outside the box uses the deep exterior bound $(1 - \beta_{\mathrm{s}}/2)\varepsilon_q L^{-2q}$, since $i_0 = j$. A plaquette astride the new face is wholly in $\Omega_j \setminus \Omega_q$. Indeed that face is five level-q cells from the old Ω_q wall, or seven when $q = k_0 + 1$, and the gap to Ω_j^c is eight cells. The preceding guard therefore reads this plaquette and the new crossing test supplies the shell bound on it. In all three cases the new bound leaves a fixed fractional margin below $\varepsilon_j L^{-2j}$.

For $q > k_0 + 1$, the halo misses both prepared boxes. When $j = k_0 + 1$ both faces are at least five level-j cells from Ω_j, and for $j > k_0 + 1$ both boxes are outside Ω_{j-1}. The halo is now in the prospective layer q. On plaquettes wholly in one of the two original layers at the Ω_j wall the exterior bounds give

$$|\partial U^+(p) - \mathbf{1}| \le (1 - \beta_{\mathrm{s}}/2) \max\{3, L_\flat^2\} \varepsilon_q L^{-2q}.$$

For an astride plaquette translate it a bounded number of bare spacings into the original layer j in the same prospective comparison cube. In a common real gauge eq. 3.cdxxxiii and eq. 3.ccclxvii bound the change in the curl by $C\varepsilon_\star L^{-(2+\alpha)q}$. The variation of the terms of order at least two in the plaquette logarithm is $O(L^{-3q})$. In the compressed range, $q \ge m - O(\log \check{R}_m)$, while $m \ge \check{R}_m$. With the frame comparison in the waiting window, both errors are $o(\varepsilon_j L^{-2j})$. Since $L_\flat < L/2$, the resulting bound has a fixed fractional margin here as well.

The fixed-target real response transfers the resulting bound to the preceding background at a cost $C_L \delta'_j L^{-2j}$. On the full density localization box the preceding background has the actual level-j determining datum. Its localization with the lifted V_j on the pure level-j tower is precisely the background used by the density test. The inner plaquettes of the test have the three-small-cube margin to the boundary of that localization. The joined comparison bounds the additional error in

level-j units by $C\varepsilon_\star e^{-c\check{R}_j}$. Both errors fit the fractional reserve, so the original lower density test passes on the inset-positive real collar chart.

The contour indicators of the old exterior belt use the order-q blocks based between the first and third layers outside Ω_q. Those blocks lie entirely in the collar. The new gauge insertion starts beyond the third layer, on the remaining exterior collar blocks outside Z_q''. All their internal links are tested exterior nonpivots. There are no finer determining nodes in these blocks after promotion, so the prospective axial frame has trivial mean contours. On the local approximate collar ball the links differ from their prospective axial reference by $O(\delta_j')$, also after changing the stage centre. The mean contour map on the full block has a bounded derivative, so these indicators pass at their original ε_j threshold. The soft density on the same contours stays in the integral.

At a prefinal stage the level-q density test has bases in $(\Omega_q^{\sim 4} \setminus \Omega_{q+1}^{\sim}) \cap X$. Its small-cube halo and the fixed row stencils fit in $\Omega_q^{\sim 5}$ and miss Ω_{q+1}, since $a_{q+1} \geq a_q$. They are in the promoted level-q member. Thus the factor reads the fixed V_q and may stand outside that collar integration. At the following executed stage of the same component its bases are precisely the lower density bases, on the actual halo just treated. On an intervening stop it keeps its real value on the recorded outgoing graph. The final top density remains with the fixed top field.

Thus on the passing graded collar branch the old lower density and contour factors can be removed by identities of real integrands. The ordinary completed fluctuation factors keep their original strip coordinates, and the new sharp factor uses the free exterior mask. The bottom-core and fixed-outgoing selectors stay on their real arguments. This reduction uses the original chart and the crossing clauses of the graded rule. The next estimates must admit the remaining analytic weights on the complex collar section with these masks.

At the final stage set $j = m - 1$. For an old approximate cube test whose halo crosses Ω_m, distinguish the actual collar mask and the inherited strip mask as in the two belt comparisons. For the remaining old approximate tests meeting the small-field integration region, apply the implication inside the original ordinary-step integral, using its Gaussian cutoffs on the variables integrated there. The later stage has no independent copy of those variables. If an earlier integral already filed a factor involving those tests, use its actual resulting factor until it has been identified on the retained coordinates.

4.17.1.27. The real finer datum and the collar path. At a transition from level j to $q = j + 1$, put $Y = Z_q''$ and let K be its exterior collar in $\Omega_q^c \cap X$. Fix an admitted prospective real datum and its regular critical configuration U^+ in the frame of the approximate collar test. Choose the common stored rows in that frame, which stays fixed in the comparison. The prospective datum is held fixed while the finer variables vary in the collar. Use the fixed small regularity ceiling of the background theorem, with room for constants depending on L. Let V^0 be the datum obtained by reading U^+ on the preceding determining set. All common rows have their prescribed values. The two joined sequences have the same bare domain and agree except for the enlargement at level q. An old level-j inward bond has an end in the old Ω_q, hence also in the enlarged member. It is absent from the new no-inward determining set. The lower inward masks agree. This is the nested restriction condition for the no-inward determining sets. The old curvature and current scales are no larger, so U^+ is regular also in the old units. A tested plaquette of the old datum at level $q \geq 1$ has a corner in the old member at level q. Its full order-q carrier lies in the member at level $q - 1$ by the gap between the members. Its holonomy is the plaquette of $\mathcal{M}^q U^+$, including the one-step completions of the second-kind bonds. The average bound gives an old datum radius of at most a fixed multiple of the regularity ceiling of U^+. At level zero use the bare curvature bound. Also, the constraints of the old set at V^0 imply those of the new set, since the new rows are obtained by composition on their old fine blocks. It follows that U^+ is critical for the old set at V^0.

Keep the inherited coarse rows at their prescribed values. For each newly imposed coarse row c, both endpoint blocks are outside the old Ω_q, by alignment on the level-q grid. They are inside the

old member of level j. At the first stage the new face is at least three level-q cells from the bottom core. At subsequent stages the gap between the prepared faces exceeds the width of the full double block. Every nonpivot in the double block either meets K and is read by the approximate collar test, or belongs to the common fine rows on the other side of ∂Y. The actual fine field and the induced fine field $\mathcal{M}^j U^+$ therefore differ by at most $2\delta'_j$ on the nonpivots in the frame of that test. Their common coarse value is $W = V_q(c)$.

To check the actual pivot's branch, use the straight corridor product q as its coordinate, with every other fine bond fixed. In each of the two logarithmic cases of the real averaging formula the coarse value is

$$F(q) = \exp_q\left(\sum_a \alpha_a \log_q H_a\right), \qquad \sum_a \alpha_a = 1 - L^{-3}.$$

Here $\exp_q, \log_q$ are the round exponential and principal logarithm on SU(2), with the group logarithm transported by right translation from the identity. The anchors H_a are the transports avoiding the middle bond. On the regular branch their end pieces are the averaged block contours. On the other logarithmic branch use the literal contours. The L straight transports through the middle bond give zero in the logarithmic sum, which accounts for L^{-3}. If neither logarithmic case applies, $F(q) = q$.

Write e for the regularity radius of U^+. The level-j plaquette defects of its induced field on the double block are $O_L(e)$, by the average bound in the member of level j. Sweeping the fixed contour stencils shows that their transports lie within $C(L)e$ of the central transport, and hence within $C(L)e$ of W by the regular averaging formula. Changing the nonpivots by $2\delta'_j$ moves each literal transport by at most $C(L)\delta'_j$. The averaged end contours have the same bound, since each stays in its regular mean chart and reads no pivot. Thus $d(H_a, W) \le C(L)(e + \delta'_j)$ in either formula. If $F(q) = W$ in a logarithmic case, the tangent sum has norm less than π. Its norm therefore equals $d(q, W)$, and the triangle inequality gives

$$d(q, W) \le (1 - L^{-3})\big(d(q, W) + C(L)(e + \delta'_j)\big).$$

The remaining case gives $q = W$ directly. This puts the actual pivot within $C(L)(e + \delta'_j)$ of the induced pivot, since the other bonds of the corridor are already close. Taking the fixed regularity ceiling sufficiently small puts the full double block in the regular case of the averaging formula. In a fixed double-block gauge of the induced field, the derivative in the middle bond is then close to its flat value $L^{-3}I$. The regular one-row inverse gives the sharper bound $C(L)\delta'_j$ for the pivot increment using the prescribed coarse value and the tested nonpivots.

Interpolate the nonpivot logarithms and solve the pivot equation along the interpolation, keeping W fixed. Each double block contains only its own pivot. The regular inverse is uniform on the small one-row chart, so the row logarithms and their derivatives along the path are $O_L(\delta'_j)$. The endpoints are the induced and the actual old datum. The common rows stay fixed. This path lies in the real continuation polydisc about V^0. The old critical orbit therefore exists along the path, with initial orbit that of U^+. This construction starts at the fixed prospective orbit and applies to the full real product ball on which the released nonpivots satisfy the approximate collar bound. For a point of the ball use the regular implicit inverse at the induced field. Its uniform radius and derivative bound give the required pivot values on the whole ball. This constructs the old datum without using the preceding guard on that ball.

Write the old orbit on this path as $(e^{i\mathcal{H}(B)}U^+)^{u_B}$ in bare units, with B the plain determining logarithms relative to V^0. The real Landau response to a row at its top y_c has value and covariant first-difference bounds with factors ξ_y^{-1} and ξ_y^{-2}, respectively, times $Ce^{-ad_{\mathbf{B}}(y,y_c)}$. Here the metric and the tops belong to the preceding joined sequence. The derivative is taken in the plain row logarithm, so its input is the velocity just bounded by $C_L\delta'_j$. The top sum at this rate is bounded uniformly on the separated sequences, and at most eight rows use any one top. Thus on the old

level-j plaquette carriers at the new prepared face,

$$\xi_j\|\mathcal{H}\| + \xi_j^2\|\nabla_{U^+}\mathcal{H}\| \le C_L\delta'_j, \qquad \xi_j = L^j.$$

This estimate sums in the actual metric, with the same bound at the first stage and at the last stage. If the affected stencil is a distance D from the changed row tops, keep half the rate in the sum. The bound at a stencil whose local scale is at least ξ_s is then $C_L\delta'_j e^{-aD/2}$ in the norm with weights ξ_s, ξ_s^2.

The scaled plaquette estimate on this complete stencil gives the following bound for the difference of defects between the old field U^- and the prospective field. The real node gauge preserves the defect norm.

$$\Big|\,|\partial U^-(p) - \mathbf{1}| - |\partial U^+(p) - \mathbf{1}|\,\Big| \le C_L\delta'_j L^{-2j}. \tag{4.cdxlv}$$

For small coupling this is less than $\beta_{\rm s}F_j\varepsilon_j L^{-2j}/4$, the reserve left by the prospective crossing test.
4.17.1.28. The global real collar coordinates. Fix the prospective real datum with its global and local critical orbits admitted. Set $q = j+1$. Write $S_{\rm g} = \mathcal{M}^j U_m^{(n+1)}$ for the averaged global background of the integral, and $S_X = \mathcal{M}^j U_{m,X}^{(n+1)}$ for the center of its approximate collar test. The two joined problems have the same determining rows and data in the interior of X; the difference comes from the outer joining collar. Use the prescribed block axial readings with the same retained node data. On the full blocks these readings include the node factors of the real response.

In bare units let $a_m = M\check{R}_m L^m$. The collar K lies in $\Omega_q^c \cap X \subset O_m$ of eq. 4.cdxliii, which is two level-m cells inside X. Include the order-j average carriers of the fine rows, the full double blocks of the new coarse rows, and those of any inherited coarse row whose averaging stencil reads a changed fine bond. These finite stencils add only CL^m to the distance from K. The old groups of tests at the original Ω_q seam use their full four-layer test-cube enlargements. Before the final stage the preceding wall comparison gives their clearance. At the last stage use the corresponding level-m calculation. For the collar coordinates themselves the full endpoint stencils have the stated clearance at every stage by the prepared-face separations. Their prospective scales are q and, across ∂Y, j. The separation from the preceding prepared face, or from Z_h at the first stage, keeps these carriers out of finer layers. Weighted lengths to the joining collar are at least $cM\check{R}_m$, since no layer has spacing greater than L^m. The real joined comparison, followed by averaging on these blocks, gives

$$\sup_b \Big|S_{\rm g}(b)S_X(b)^{-1} - \mathbf{1}\Big| \le C\varepsilon_\star e^{-cM\check{R}_m} = o(\delta'_j)$$

on the collar read set. The last equality uses the frame drift and $\check{R}_m \ge (\log(\theta_m - c_5))^r$, with $r \ge 2$. The same bound is smaller than δ'_j times any fixed negative power of $\log\theta_j$.

For the same actual nonpivot value the two coordinates are related by the exact product

$$V_j S_X^{-1} = (V_j S_{\rm g}^{-1})(S_{\rm g}S_X^{-1}).$$

Consequently the global sharp cutoff at δ'_j implies the local approximate cutoff at $2\delta'_j$. On the positive local approximate branch the global nonpivot increment is bounded by $3\delta'_j$, which fits the same regular chart before the sharp split. The actual middle bond in each double block is the same in both charts, since the other physical bonds are fixed and the unique regular pivot solves the same constraint with target $V_q(c)$.
4.17.1.29. The sharp guard in the two charts. On the nonpivot bonds put $H = (1/i)\log(S_{\rm g}S_X^{-1})$ and write $V_j = e^{ig_jB_{\rm g}}S_{\rm g} = e^{ig_jB_X}S_X$. Then

$$e^{ig_jB_{\rm g}} = e^{ig_jB_X}e^{-iH}.$$

In each chart let h be the density of Haar measure in exponential coordinates, with the same normalization. Right translation gives on the free bonds

$$dB_{\mathrm{g}} = \prod_b \frac{h(g_j B_X(b))}{h(g_j B_{\mathrm{g}}(b))}\, dB_X.$$

The scale factors $g_j^{\dim G}$ cancel bond by bond. The pivot densities are evaluated on the same solved physical fields. They multiply both sides of the Haar identity, since the retained row targets and the graph of the regular inverse are the same in the two charts.

On the free collar variables the global sharp guard in the first chart is the fixed product ball $\max_b |g_j B_{\mathrm{g}}(b)| < \delta_j'$. In the second chart it is the indicator of this same inequality after substitution. The local sharp guard tests $\max_b |g_j B_X(b)| < \delta_j'$. Either guard implies the local approximate test on these bonds by the centre comparison. To compare the two sharp guards, use the uniform derivative of the logarithm on the regular small chart. For $h_X = C \sup_b |H(b)| = o(\delta_j')$,

$$\big|\max_b |g_j B_{\mathrm{g}}(b)| - \max_b |g_j B_X(b)|\big| \le h_X.$$

In particular, where the guards disagree both maxima lie within h_X of δ_j'. This band already has a bond of size at least $\delta_j'/2$ in either chart, the size used in the sharp-test energy estimate. On the common real fibre the four products of the two guards and their complements sum to one. In changing charts each product travels with its density. A comparison of the labelled operations must also account for the two disagreement terms and the variables their branches keep.

On the failed global branch the cutoff substitution can be checked bond by bond. Write W' for its new coordinate, and use the cutoff ϑ at scale δ_j'. The physical local and global readings are respectively

$$W' e^{i(1-\vartheta)H} \quad \text{and} \quad W' e^{-i\vartheta H}.$$

Here H is the fixed shift of the two prospective centres on the real fibre. The joined estimate gives $|H| \le \delta_j'/8$. Below the start of the cutoff, at $4\delta_j'/3$, the global reading equals W'. Above that start its angle is at least $4\delta_j'/3 - |H| > \delta_j'$. Thus its sharp failure is exactly the complement of the sharp ball in W'. For the approximate local reading the two chord tests in the old and new coordinates agree where $\vartheta = 1$. Where $\vartheta < 1$ both pass by $5\delta_j'/3 + |H| < 2\delta_j'$. Apply this identity on the actual transformed mask, with the derivative of the cutoff map in its Haar factor and with unchanged coarse targets for the pivots.

The domain of a test and the set of independent collar coordinates must be distinguished. The direct graded rule tests a sharp ball on its free exterior nonpivots. The inherited strip coordinates, including the old-wall crossings, keep their ordinary tests in their original chart. Its local sharp failure keeps that prefix before the shift. Under the incident-bond convention, the sharp mask can also include a bond crossing into the old Ω_m. Such a bond belongs to the inherited ordinary chart. Write S_{o} for its reference. The global sharp factor there tests the logarithm of $(V_j S_{\mathrm{o}}^{-1})(S_{\mathrm{o}} S_{\mathrm{g}}^{-1})$, with V_j read in the inherited chart. There is no new integration coordinate for that bond. The whole sharp indicator is the free-ball indicator times the product of the sharp readings on the other bonds of its mask. The ordinary strip bounds at scale δ_j do not by themselves imply these readings at scale δ_j'.

For the final collar, on a crossing of the old Ω_m wall this inherited bond lies in the ordinary plain strip. The bases of both passed and failed ordinary cube tests are inset by a full level-m cell. In level-j units this cell has side $LM\check{R}_m$. The read reaches at most $2LM_2\check{R}_m + 1$ beyond the base, including the incident endpoint convention. This is less than one cell by the choice of M. Hence these masks do not read the crossing bond. Its inside endpoint is in the stored boundary strip. The ordinary bond map on this plain part is just the expression of the physical link relative to the local strip centre, with Haar measure. The explored collar factors belong to the ordinary interior test

mask, and the ordinary rim test excludes the stored strip bonds. The newly completed fluctuation test and the inherited approximate test can still read the crossing on their stated masks.

There is a direct way to keep an additional sharp read on an inherited free bond. Let ρ_b be the smallest retained ordinary upper threshold at that bond, including its factors in the cube and rim tests. Suppose that on its actual tested fibre

$$\left|\tfrac{1}{i}\log(S_{\mathrm{g}}S_{\mathrm{o}}^{-1})\right| + \delta'_j < \rho_b.$$

On the global sharp ball the triangle inequality for the group angle makes all these ordinary upper factors at b equal to one. On this plain strip use the physical group variable with its Haar density. With the ordinary centre fixed in the inherited coordinates being changed, holding the fresh variables fixed makes the re-centering a right translation bond by bond. Thus the new sharp read can be written as a fixed ball in that inherited coordinate, with the exact change in its conditional density. The completed tests on the other bonds remain factors of the original integral. For application we must check the displayed centre bound on the complete crossing stencils, the fixed-centre property on this joint fibre, and the analytic bounds of the resulting density on the strip discs. In particular the difference of the ordinary and prospective centres, rather than the difference of the two prospective centres, enters this bound.

On the ordinary reads with both ends inside Ω_m the preceding comparison gives this difference at order $O_L(\delta'_j)$ in a common axial frame. That size fits any of the retained ordinary thresholds, which are fixed fractions of δ_j. The estimate for the approximate test on a crossing used a direct test of V_j against the prospective centre. It leaves a frame comparison to check for the present re-centering at a crossing. The node factors at both ends of that bond must be those of the inherited coordinate.

Let $x_- = b_-$, $x_+ = b_+$, and let y_-, y_+ be the centres of their two complete level-m blocks. Write $T_W(y, x)$ for the one-step mean contour of a level-j field W in such a block. Its axial reading of the crossing is

$$\mathcal{R}_b(W) = T_W(y_-, x_-)W(b)T_W(y_+, x_+)^{-1}.$$

For any gauge u the covariance of the mean contour gives

$$\mathcal{R}_b(W^u) = u(y_-)\mathcal{R}_b(W)u(y_+)^{-1}.$$

Thus the comparison can be made through the factors at the two block centres. If the two complete read blocks lie in the common joined annulus with its matching rows, the joined comparison controls the real potentials there. At the old wall of the quiet component the two blocks and their order-j carriers are within CL^m of the wall. The outer band is separated from that wall by the quiet-box margin, and the inner boundary lies past the enlargement by two cells of the old member. These margins place the two blocks in the common interior of the annular comparison. The old exterior block lies in the old member of level j, and the clearance from the prepared box puts both blocks in the new member of level m. Orient the crossing from the exterior block to the interior block and write $c = \langle y_-, y_+\rangle$ for the coarse bond between their centres. It is an outward row of the old member at level m, so it is retained even when the fine inward rows are omitted. It becomes an interior row after promotion of the exterior block and has the same value $V_m(c)$. The old exterior centre is a node of layer j by its internal level-j rows. The old interior centre is a node of layer m. Both complete blocks lie inside the common part of the two joined sequences away from their joining bands. Their rows and the levels of these nodes are therefore unchanged in the annular comparison. The average factorization on the full two-block carrier controls the level-j links and the factors at the two centres. At the interior centre of the old sequence the restoring gauge has value $w(y) = v_{j+1}(y)^{-1}$. In the averaged comparison its product with the level-j factor is $w(y)v_j(y) = \mathsf{v}_j(y)^{-1}$, since $v_{j+1}(y) = v_j(y)\mathsf{v}_j(y)$. The exterior centre lies in the old layer j, even though it lies on the level-m lattice. There the product is the identity, as it is at the exterior fine node before auxiliary axialization. The one-step means have uniformly bounded derivatives on the same regular blocks. Applying the last identity to both comparisons with the joined reference shows

that the axial readings of the ordinary and preceding backgrounds differ by at most $C_L\varepsilon_\star e^{-cM\check{R}_m}$ on that carrier. For the comparison with S_{g}, start the real response on the old determining set at the datum induced by the prospective field, in its prescribed frame. The physical row increment $\beta(c)$ is zero throughout the interpolation. The reduced increment $\beta_{\natural}(c)$ can be nonzero because its outward-row conversion reads the changing internal links of the exterior block. Use the response to $\beta_{\natural}$ and restore at each old node y by $v(y)^{-1}$. The datum of the uncorrected response is the desired physical datum turned by v, so this restoration fixes the prescribed value of c and returns both node frames to the inherited ones. At the two centres the average factorization now leaves the identity at the old layer-j node and v_j^{-1} at the old layer-m node, as in the annular comparison. The latter is $O_L(\delta_j')$ from the identity by the local response on the full carrier. This gives the same bound for the difference of the two axial readings. Use a fixed residual gauge for the response to start at the chosen prospective representative. Both restored node values are then prescribed for the whole path.

The original strip frame has a further contour at the exterior endpoint. The ordinary joined strip has nodes at all level-j sites of that block. Its level-j internal bonds keep the actual values of V_j, which determine the old contour from the block centre to the endpoint. The axial condition is imposed at the nonnodes of the ordinary interior blocks. An internal path has no pivot, since each pivot lies in a corridor between distinct blocks. The interior endpoint is axial already. On the prospective block the contour is the identity. The difference of the exterior contour from the identity is therefore bounded by $C_L \max_e |V_j(e)S_{\mathrm{g}}(e)^{-1} - \mathbf{1}|$, where e runs only over the internal bonds of that exterior block. For the native collar mask the changed exterior block next to the old wall lies wholly in the active collar. The members consist of whole blocks, and the clearance from the old wall to the prepared face is greater than the width of this block, also at compressed levels. Every internal contour bond of the exterior block is therefore in the passed approximate test. That test and the comparison of the two prospective centres bound the maximum by $C_L\delta_j'$. This uses the internal paths even when the sharp test on the crossing fails. Undoing the exterior contour gives the desired bound in the inherited frame on the old-wall crossings just described. If using a sharp mask with additional inherited reads, we must check their complete carriers and admit the resulting conditional density in the complex stage space. In this auxiliary axial comparison the stored fields keep their original values. In a physical re-centering of a crossing nonpivot, solve its old coarse constraint for the pivot at the same fixed target. The native exterior block bound and the inherited ordinary bounds put the two-block read in the original regular pivot chart, where the change of the solved pivot is bounded by a constant times the change of the nonpivots. Keep its elimination factor and the Jacobian of the original strip substitution. For a crossing in the ordinary plain strip the old rim factor excludes that strip bond. A completed ordinary fluctuation test can read it, at threshold δ_j; its factor at the crossing is then one on the re-centred sharp ball by the centre bound above.

The induced datum argument also applies to the global critical background itself. Read the global prospective configuration on the global old determining set. Its old units are no larger than its prospective units. As in the localized argument, the complete tested carriers give an admitted old datum and the nested constraints give criticality at that datum. Interpolate the nonpivots of the global chart and solve the corresponding pivots. The modified rows are at level j; the point response sum in level-m units is bounded uniformly in the number of components and in the torus. Thus the real global first background in the collar integral is the regular old orbit on this path. The localized old guard can be checked at each of these physical row values by the localized path estimate. Its value was not used to continue the old orbit along the chart.

At the final stage the inherited strip is kept in the ordinary coordinates. Some of its rows lie close to ∂X, where the estimate comparing S_{g} with S_X need not hold. An inherited double block that reads a changed collar bond was already included in the interior comparison. The ordinary and preceding global backgrounds on the rest of the strip can be compared on a common annulus.

Use an outer box obtained by enlarging X by half a level-m cell and rounding on the top tower grid. Remove the inner box $(\Omega_m^{\sim 2})^c \cap X$. The outer band is in the unchanged top layer, including the neighboring cells outside X. The inner band is in the old layer j. The earlier stages change rows over the old components inside $G_{m-1} \subset H_m$. Even their full row carriers stay inside the removed box by the six-cell gap. The two boundary bands have the widths required for the joined comparison. At the physical collar value, the ordinary and preceding global problems read the same rows and data throughout this annulus. Its interior contains the ordinary strip rows touching the old Ω_m and their full double blocks, including those reaching across ∂X. This gives the comparison of the ordinary orbit in its own gauge on those stencils. The inherited Schur complement and the retained rim and collar indicators are evaluated in the original variables of the strip.

4.17.1.30. The real endpoint triples. At a stage of the depth-m problem set $q = j + 1$ and put $e_l = \check{R}_l^2 \varepsilon_l$. This allows the complete averaging footprints at the original exterior walls of a compressed stage. The tests with sharper thresholds keep those thresholds.

The input at a point of a history includes the fields of the operations still carried by it. A datum plaquette uses the actual determining links and, for a completed bond, the average of the lower field on its full double block. These values agree with the fields in the tests by the retained averaging constraints. In the convention excluding inward rows these plaquettes exhaust the datum tests. For each test place in a stopped component use its least-level departure. If the component is untouched, or the full footprint misses the inset of its stage test region, the ordinary density test supplies the bound. The same ordinary test is used at untouched higher levels and on the unchanged action top. An inner footprint in a tested shell uses the last positive stage test at that departure. At a restored core the bound comes from the retained reference cutoff and the surrounding ordinary tests.

For a departure at the sharp collar split after n integrations, put $j' = i_\flat + n$, where i is the level of that operation. The retained positive stage test is $\chi_i^{(n+1)}$. A level-j' test in its inset with a corner in the collar uses also the retained approximate bound at $2\delta'_{j'}$. In the translated coordinates this bound is on the reconstructed field of the localized collar chart. The failed sharp indicator is carried as a separate factor. The ordinary test supplies a footprint outside the inset. At the promoted level the retained coarse row is fixed by its averaging constraint. In the direct exterior collar rule, a retained determining link of level j' has no endpoint in the next old member. A completed bond uses both full blocks outside the corresponding next member. Thus the determining reads use the collar bound on its exterior slots and the retained values on the other slots. The original strip tests remain the suppliers for the old-wall fluctuation reads.

These choices use the tests carried at the relevant moment. Before the first stage use the untouched datum. While a positive stage test is carried it supplies the ordinary assignment. After removal and before integration the collar assignment uses the new positive stage test and the approximate collar test on the old collar variables. The new ordinary assignment uses the new positive stage test on the promoted datum. A standing departure below the current waiting window keeps its earlier determining reads and averaging completions. Gauge moves conjugate the holonomies. A refresh with the inner datum retained leaves its test values in that inner slot. Where restoration replaces the field, the tests of the restored datum provide the new bound.

The witnesses on the complete footprints give $2\psi_l \varepsilon_l$ for the datum defect. Here ψ_l is $O(1)$ for an ordinary witness or a restored core and is bounded by $C(\log \theta_l)^r$ on the remade shells. The latter bound uses $L_\flat^2 < L$ and the compressed interval of length at most $\log_L \check{R}_{k_0+1}$, with the frame comparison of the running scales. Thus each test gives a bound Ce_l directly. Fix an entering point on these retained tests and on the constraints with coarse target V_q. This is the input for varying the collar. To remove the preceding guard on an enlarged support, use the separate comparison in §4.17.1.25.

Let U_{in} denote the regular global old configuration at that input. The datum defects are below the fixed real existence ceiling at weak coupling. Apply the real profile estimate in its layer form.

In bare units it controls $L^{2\lambda(p)}|\partial U(p)-\mathbf{1}|$ and $L^{3\lambda(b)}|J^0(U)(b)|$ at each corner or endpoint by the profile there, where λ of a stencil is its largest corner layer. For the datum bounds just obtained the profile has the form

$$C\sup_l e_l \exp\bigl[-\delta\, d_{\mathbf{B}}(x,\mathcal{L}_l)\bigr], \qquad \mathcal{L}_l=\Omega_l\setminus\Omega_{l+1}.$$

The place of a tested level-l plaquette is in $\mathcal{L}_l$, and its corners lie in layers $l-1,l$. From layer s the path to this read crosses the intervening layers, giving distance at least $c(|l-s|-2)_+$. The frame bounds make e_l/e_s grow at most polynomially in $|l-s|$. Hence the profile in layer s is at most $C'e_s$. The zeroth layer uses the bare datum tests. Apply the same profile on both stage sequences.

In the prospective datum all holonomies below level j are unchanged. At level j, the new first-kind row set is a subset of the old one, and the second-kind completions still use the same level-$(j-1)$ layer. At level zero there is no completion from a finer layer. Above level q the original row values and masks are retained. For a completion at level $q+1$, its full two-block read stays in the old Ω_q, by the old gap from Ω_{q+1} to Ω_q^c. Thus the enlargement at q does not change that completion. At higher levels neither layer of the completion changes.

Every tested holonomy at level q equals the plaquette of $\mathcal{M}^qU_{\mathrm{in}}$ on the relevant complete stencil. A determining row that met the old Ω_q keeps its value. Each newly imposed row has both blocks outside the old Ω_q and within the old level-j member. Its retained constraint reads the old determining V_j on these two blocks. For a second-kind bond the two blocks are outside the prospective member at q and in the preceding layer, so they also use the old level-j values. In either case the identity with the average follows from the actual retained constraint or its one-step completion. For each tested plaquette the order-q carrier stays inside the unchanged level-j member by the separation of the prospective sequence. The level-j profile and the adjacent-scale conversion give bare curvature at most Ce_jL^{-2j} on that carrier. Averaging there gives $C(L)e_j\le C'(L)e_q$ for the datum defect at the promoted level. The artificial face lift of the localized problem does not enter this global calculation. We have thus obtained the levelwise datum estimate for the fixed prospective global orbit and, by the weighted maximum, its normalized curvature and current bounds with Ce_s in layer s.

For the varied old datum, begin at the averages of this prospective configuration on the old determining set. For an old level-l tested plaquette, $l\ge 1$, its order-l carrier is inside the old member of level $l-1$. The prospective scale on this carrier is at least $l-1$. In changing its units to $l-1$ the powers $L^{-2(s-l+1)}$ absorb the frame variation in e_s/e_{l-1}, so averaging gives a defect bounded by $C(L)e_{l-1}\le C'(L)e_l$. At level zero the bare profile gives the estimate directly. Now vary the actual old collar coordinates in the regular fibre chart. Only level-j rows move, by $O_L(\delta_j')$ including the reconstructed pivots. Only tests of levels j and q can read these rows, either directly or through a one-step completion. On their full double blocks the regular average and the product of the four group factors change by $O_L(\delta_j')$. This is smaller than the datum allowance Ce_l at these adjacent levels. The old orbit on the varied fibre has the same layerwise bound from the weighted maximum. These are bounds for the two endpoints from the entering datum profile of the history. In using them through the stages, take the profile from the actual retained tests at each entrance, as in the levelwise datum bound. Repeatedly multiplying the adjacent comparison constant would not give a uniform profile.

Consider now any stage clause of scale s for either real endpoint. In a prepared layer and in an untouched higher layer its reading set is in the actual member at scale s. In the compressed clause of an original layer i, the assigned scale is $s=j$ for the old stage and $s=q$ for the prospective one. On the remade component the actual member at that scale contains the reading set, since the prepared box is outside Ω_{k_0+1}. For a reading on an unchanged part, use at least its original scale i. The loss of scale $s-i$ is at most $\log_L \check{R}_{k_0+1}$, by the compression identity. Thus one can use the actual member at scale $a\le s$ with $a=s$ except for this possible compression loss. Use the same

calculation at an earlier address whose determining regions are contained in the endpoint sequence, with the compression interval of that address.

For the pointwise bounds at scale s, changing from the scale a costs at most $L^{3(s-a)}$. This accounts for the physical current, whose scale power is three, as well as the curvature, whose power is two. At the plain triple $A' = 0$, so the two potential clauses hold identically. For each of the tower clauses, the actual level-p tower on a localization domain is formed by intersecting the stage sequence, truncated at p, with its localization tower. Hence it is subordinate, even if p exceeds the layer of the read. The restriction identity eq. 3.cccxciii identifies its real background at the own averages with a gauge copy of the restriction. It therefore has the same curvature and the same current at every full current stencil. In level-p units the current gains $L^{3\mathsf{p}}$ relative to bare units. This gives the scale required in the tower clause, since its allowed factor $L^{s-\min(\mathsf{p},s)}$ is at least one. The curvature factor in that clause simplifies to L^{-2s}.

For a local gauge cube of side $C_{\mathrm{g}}ML^s$ wholly in the actual member of level a, use the gauge estimate at $b = \max\{0, a-1\}$. The halo of thickness $8\rho L^b$ lies in Ω_b by separation, or in the ambient bare torus at $a = 0$. The powers $L^{-2(l-b)}$ and $L^{-3(l-b)}$ for its actual layers l absorb the frame drift. Thus the curvature and current in those units are bounded by Ce_a on their complete halo stencils. The side in level-b units costs $O(ML^{s-b})$, and conversion of the potential and its first differences costs at most $L^{2(s-b)}$. The normalized gauge bound is $CML^{3(s-b)}e_a$. Use the torus side condition $2C_{\mathrm{g}}ML^s + 16\rho L^b < \mathsf{s}_T$. The fixed curvature ceiling and the product of the gauge radius in level-b units with Ce_a fit the smallness requirements at weak coupling. Restrict the resulting gauge to the prescribed read set.

The frame bounds across the waiting window give $\check{R}_{k_0+1} \le C(\log\theta_s)^r$ on the compressed interval, and $e_a \le Ce_s$ there. Consequently the pointwise and tower bounds, normalized at the nominal scale, are at most $Cg_s(\log\theta_s)^{\mathfrak{p}_0+5r}$, and the gauge bounds are at most the same quantity times M. The stage choice $q_{\mathrm{rad}} \ge \mathfrak{p}_0 + 7r + 1$ for $\alpha_{0,s} = C_0 g_s(\log\theta_s)^{q_{\mathrm{rad}}}$, and the positive coefficient floor $1 - 4\beta_{\mathrm{s}}$ for the choice $0 < \beta_{\mathrm{s}} < 1/4$ in the allowed window, put the plain triples inside half the radii for small coupling. At earlier hereditary addresses use their wide subordinate sequences, the full gauge cubes, and their fixed positive coefficient floor and nonnegative lifts. The same bounds fit the smaller adjacent running radius, with its bare-level value at level zero, and the half radius of the fixed local regularity clause. This verifies the real clauses on the specified carriers.

4.17.1.31. The complex substitutions in the collar integral. Keep the real factor U fixed and use independent complex potential and current coordinates. The source for the old critical problem is the nonlinear datum increment on the full double blocks of the collar. Solve the regular one-row inverse at the applicable background. For a free field B this gives the completed logarithmic datum $B' = g_j \mathsf{C} B - \mathsf{h}\widetilde{\mathsf{D}}(g_j \mathsf{C} B)$. Take its corresponding reduced datum for the reduced response. On the complex chart the inverse gives $\|B'\|_\infty \le C_L g_j \|B\|_\infty$, so the source in the box $\|B\|_\infty < Cp_0(g_j)$ is within the small response bound. Using the nonlinear row is necessary for the equality at the physical endpoint.

For the current increment the reduced first variation has a local linear term in addition to its supported parts. Write $D_{\mathrm{lin}}B' = \hat{Q}^{\mathsf{T}}\mathsf{a}B'$ for this row term. The supported bracket of the raw response is $\mathbf{F}^\flat + D_{\mathrm{lin}}B'$. Gate each row of the linear term by the product of the switches in a connected set containing its complete source and output stencils. Include the plain-data and parent carriers in this set. The neighbor margins and the one-step row charts give bounded size on the aligned standard grain. Denote the resulting linear term by $D_{\mathrm{lin,s}}B'$. It agrees with $D_{\mathrm{lin}}B'$ at full switches, and a nonzero row term lies in one active wall component. In the current formula use

$$\mathbf{J}_s = \mathsf{f}(-i\xi\ \mathrm{ad}_{\mathbf{H}})^{-1}\big[\mathbf{J} + (\mathbf{F}^\flat + D_{\mathrm{lin}}B') - D_{\mathrm{lin,s}}B'\big].$$

The raw formula gives exactly the current of the perturbed real configuration at full switches. At a cut the supported bracket and the gated linear term use only their respective components. The inverse f^{-1} acts bond by bond, so the increment from $\mathbf{J}$ is supported on those components. The

product of the bounded number of switches costs a fixed factor in the current bound. Use this continuation to form the component increments j_c alongside the localized potential increments. In the weighted current estimate write $D_{\mathrm{lin,s}} = D_{\mathrm{lin}}G_s$, where G_s multiplies each source row by its carrier product. The increment from $\mathbf{J}$ is the inverse f applied to $\mathbf{F}^\flat + D_{\mathrm{lin}}(I - G_s)B'$, plus $(\mathsf{f}^{-1} - I)\mathbf{J}$. The local row bound for $\hat{Q}^{\mathsf{T}}$ and the fixed switch cost bound the added row term in cubic current units. The small exponential chart bounds the last term by the weighted response times the current radius. The gauge turn has the same estimate for the additional factor $\mathrm{Ad}(u) - I$.

For a wall component $\mathfrak{w}$, evaluate the response with cube switches $s\mathbf{1}_{\mathfrak{w}}$. Call its potential $\mathbf{H}_{\mathfrak{w}}$ and its current increment $\mathsf{j}_{\mathfrak{w}}$, where the increment is the current in the displayed formula minus the input $\mathbf{J}$. Use the same component parameter in both output sums,

$$\mathbf{A} = \sum_{\mathfrak{w}} \lambda_{\mathfrak{w}} \mathbf{H}_{\mathfrak{w}}, \qquad \mathbf{K} = \sum_{\mathfrak{w}} \lambda_{\mathfrak{w}} \mathsf{j}_{\mathfrak{w}}.$$

The independent input current is kept in its original slot. The component parameter multiplies the response after solving the equations. This is the interpolation used in the common radius budget.

Use one levelled triple on the union of the carriers of the inherited reads, restricting it to each reader's address. If $W = e^{i\eta_t \mathbf{A}}\mathbf{U}$ is the unlevelled configuration, take $u(y) = v[P; \mathbf{A}](y)^{-1}$ at the pillar top y, using the same switched potential $\mathbf{A}$ that defines W. At a stored site a of this pillar take

$$u(a) = \tau[\mathbf{U}](a)^{-1} u(y) \tau[W](a).$$

Indeed $\tau[W^u](a) = u(y)\tau[W](a)u(a)^{-1} = \tau[\mathbf{U}](a)$. Write $\mathbf{A}^\circ$ for the relative logarithm in $W^u = e^{i\eta_t \mathbf{A}^\circ}\mathbf{U}$. Use the same analytic gauge in the current substitution

$$\mathbf{J} + \mathbf{K}^\circ = \mathrm{Ad}(u)\,(\mathbf{J} + \mathbf{K})\,.$$

At the physical endpoint the corrected configuration and current agree with those of the actual old critical orbit in its adapted frame. The transport identity keeps the spectators of each earlier term in their prescribed frames. Each inherited term uses its own attachment. The new crossing test reads the fixed prospective datum and adds no pin.

On the integrated strip fields the substitution in an inherited potential uses the scalar multiplier $\lambda_{\mathfrak{w}} s(\square_b)$. It multiplies the far argument and serves as the switch for an unchanged near argument. The real cutoffs act in their integration charts. Keep the inherited Schur form and take the difference of its two evaluations as a potential on these discs. Its two evaluations use the same stored variables. The shift of the local free coordinate has a separate source radius of the form $c(\log \theta_j)^{-e_y}\delta'_j$. In the local sharp rule, keep the free collar coordinate B_X in its original real product ball $|g_j B_X| < \delta'_j$. For a free shift y use the global increment

$$g_j B_{\mathrm{g}}(B_X, \mathsf{y}) = \frac{1}{i}\log\!\left(e^{ig_j B_X} e^{-i\mathsf{y}}\right)$$

on each released nonpivot, and then complete the pivot at the fixed coarse target. For the physical value of y, take $e^{-i\mathsf{y}} = V_X V_{\mathrm{g}}^{-1}$ with V_X, V_{g} the local and global centres in the same frame. This is the orientation for left link logarithms. Its bound is $C\varepsilon_t(\log\theta_t)^C e^{-cM\check{R}_t}$ in a component of level t. In its waiting window θ_j and θ_t are comparable with one logarithm. The exponential therefore puts the actual shift inside the source disc at every stage, including the bottom. On a complex section its source is the localized shift map formed on that section.

On the real free-source disc the trial global angle is at most $\delta'_j + |\mathsf{y}|$ in this same ball. Combining with the actual centre comparison gives the approximate bound $2\delta'_j$ on its full free mask. The old completion and rim tests act on the original strip coordinates. At fixed admitted prospective datum the guard implications, including the crossing reserve, apply on this entire real ball. This permits the use of the same integration domain for all the real trial shifts. At the physical shift it gives the original local sharp integral. The source is then continued in the integrand while retaining those real integration coordinates.

On the passing local-sharp branch the smooth stage Haar factor has a separate calculation. Write $z = g_j B_X$ and $z_{\mathsf{y}} = g_j B_{\mathrm{g}}(B_X, \mathsf{y})$. For the Haar density h normalized at zero, the change from dz_{y} to dz has logarithm

$$L(z, \mathsf{y}) = \log h(z) - \log h(z_{\mathsf{y}}).$$

Indeed right translation preserves Haar. The logarithms are taken on the fixed small chart where h is nonvanishing. Its evenness gives $L(0, \mathsf{y}) = -\log h(-\mathsf{y}) = O(|\mathsf{y}|^2)$. The difference $L(z, \mathsf{y}) - L(0, \mathsf{y})$ vanishes at either zero argument. Integrate the mixed derivative along the two segments in the product-logarithm chart to bound it by $C|z||\mathsf{y}|$. On the fixed real ball $|z| < \delta_j'$ and the independent shift disc this is at most $C(\delta_j')^2 (\log \theta_j)^{-e_y}$. On an enlarged complex section use the actual bound $g_j B_*$ for $|z|$ on that section.

When two moves use the same free link, let z_0, z_1, z_2 be its successive logarithms in their actual order. The two smooth log factors add to $\log h(z_0) - \log h(z_2)$. Subtract its centre at $z_0 = 0$, with the intermediate value $z_1(0)$ induced by the first map. Keep the centre once on the shift carrier. After substituting the localized shifts its series has two or more shift factors for a single zero-centred translation, while the difference has at least one. For the composition use the ordered product-logarithm chart and the same rooted union of the shift carriers. A cutoff Jacobian between the smooth changes is carried as its own factor.

For the global sharp branch keep B_{g} as the variable of the fixed sharp ball on the free nonpivots. Its real local approximate test on those bonds is then identically one. In comparing with the analytic integral having a free shift source, the candidate for this fixed ball is the source value zero with its integration variable identified with B_{g}. The physical integral with a local sharp guard instead uses B_X as the variable of the fixed ball and the actual shift in the analytic factors. The cutoff stays fixed in both representations on the free coordinates. In the direct graded rule the sharp mask is confined to the free exterior nonpivots. The inherited crossing coordinates remain in the ordinary strip integral with their completed tests. A test on the broader incident mask gives a separate real kernel. If the full sharp mask also reads an inherited bond, we must show that its exact pulled-back indicator is among the fixed real factors of the analytic integral, or prove it equals one on the support being integrated. In the expansion with Gaussian tails each real cutoff uses its prescribed carrier and accepts the corresponding smaller ball. Freezing the real datum alone leaves those carrier and tail checks to be made. The failed global branch uses its own real strip map with the transported tests. Where that map has a cutoff, its derivative enters the density in addition to the Haar factors.

4.17.1.32. Substituting the physical shift in the stage logarithm. Keep the shift y independent while forming the analytic logarithm of the collar integral on the fixed real domain. On the admitted graded section the physical value at an exterior nonpivot b is

$$H[P](b) = \sum_W H(b; W; P|_W).$$

Each summand reads a decoupled response to the localized difference of data. Its full carrier W joins b to the boundary of the localization. At an ordinary strip the resulting lower bound on the length is $d_j(W) + 1 \geq \check{R}_j$ in the strip units. In a stage it is $d_t(W) + 1 \geq \check{R}_t$ in the component units. In the respective units the pieces have bound $C(\alpha_{0,j} + \alpha_{1,j}) e^{-cM(d(W)+1)}$. This construction requires the actual endpoint centres and their decoupled reads on the common graded section. The scalar annular bound on their difference alone would give no such carrier assignment.

Let $F_X(\mathsf{y})$ be a carrier term of the free-shift logarithm. For a shift piece indexed by $a = (b, W)$ write $h_a = H(b; W; P|_W) e_b$. The contribution using a finite nonempty set A of shift pieces is

$$\sum_{J \subset A} (-1)^{|A|-|J|} F_X \left(\sum_{a \in J} h_a \right).$$

Assign it to $X\cup\bigcup_{(b,W)\in A}W$, with the fixed enlargements of the stage reads. Each nonzero difference uses only the shift slots read by F_X. If the full deformation $\mathsf{y}=\sum_{a\in A}t_a h_a$ is admitted for $|t_a|<R_a$, $R_a>1$, Cauchy's estimate bounds this mixed difference by the bound for F_X on the deformation disc times $\prod_{a\in A}(R_a-1)^{-1}$. Indeed the alternating sum selects exactly the Taylor terms of positive degree in every t_a.

The exponential bounds on the response pieces and their lower length allow part of the decay to be spent on the larger union carrier. In the stage sum use the rooted closing load

$$\sup_{\Delta}\sum_{a=(b,W)\colon c(b)=\Delta}\frac{e^{Cr_{\natural}\varrho_{\mathrm{a}}\#W}}{R_a-1},$$

where $c(b)$ is the anchor cell of the shift slot and the exponent is the graded carrier price. Denote this load by $\varepsilon_{\mathrm{cl}}$. The dilation radii must fit in the free-shift disc and satisfy $CN^*\varepsilon_{\mathrm{cl}}\le 1$, where N^* bounds the number of anchor cells of one component. Use the native stage collars and the island hulls as exempt boxes, with the stage separation, bounded hull multiplicity and adjacent grades differing by at most one. The separated-component packing estimate applies to the tree of X with its prescribed trace enlargement. Include the full anchor reads in checking this enlargement. If the tree Γ is counted at level m and the operation is at level t, the weighted count of anchors, with weight $\epsilon^{(D-1)_+}$ at wall distance D, is bounded by

$$N^*\left(1+\frac{4|\Gamma|L^{m-t}}{\mathsf{N}}+\frac{|\Gamma|+1}{L\mathsf{N}}\right),\qquad 0<\epsilon\le c.$$

The weight is one for a directly read anchor at wall distance at most one. A fixed larger wall distance costs only a fixed inverse power of ϵ. In component units this gives $CN^*(1+d(X)/\mathsf{N})$ for the directly read anchors. For graded trees keep the indicated scale conversion when charging the length. For fixed X the sum over subsets of shift pieces is bounded by the product of one plus their weighted Cauchy costs. Its logarithm is at most the sum of those costs. The anchor count thus gives a bounded prefactor and a loss $O(1/\mathsf{N})$ in the spatial rate. In summing over terms refiled at the same enlarged carrier, use the anchored tree attachment count and the summed-to-pinned norm estimate of the stage expansion. The spatial rate losses add over the N stages, whereas the bounded prefactors multiply individual stage amplitudes before those amplitudes are summed.

Keep $F_X(0)$ on X. The nonempty anchored differences sum to $F_X(H[P])-F_X(0)$ by telescoping, first for finitely many pieces and then by absolute convergence in the closing norm. The resulting terms, with these enlarged carriers, are the ones passed to the next stage and to the source oscillation norm. The integration domain and the real factors in the stage law stay those used in forming the free-shift integral.

If the inherited crossing coordinates are re-centred, write $\mathbf{z}$ for the old combined strip and collar coordinates and $\widetilde{\mathbf{z}}$ for the coordinates with fixed sharp balls. The ordinary centre can depend on the new collar variables even when it is fixed in the inherited variables being translated. Its first derivative in those collar variables contributes to the linear part T of $\mathbf{z}$ in the new coordinates. For old precision D, the corresponding quadratic becomes $\mathsf{T}^{\mathsf{T}}D\mathsf{T}$, with any constant shift accounted for in the mean and the scalar. If $D\ge\gamma I$ and T has a bounded real inverse, the new real floor is at least $\gamma\|\mathsf{T}^{-1}\|^{-2}$. Its covariance is $\mathsf{T}^{-1}D^{-1}\mathsf{T}^{-\mathsf{T}}$. These identities show where the crossing correction enters the Gaussian estimates. In particular the linear mixing need not carry a small coupling factor. We must bound its local matrix entries and the nonlinear remainder on the declared sections, including the cut and switched sections, before using the stage expansion for this re-centred density.

Use the ordinary endpoint frames for the substitution in the strip coordinate. In physical logarithmic units write $h(\beta)=\frac{1}{i}\log(S_{\mathrm{g}}S_{\mathrm{o}}(\beta)^{-1})$, where $\beta=g_jB$. The response in reduced row coordinates has a coupling-independent analytic radius. To use that radius here, suppose the completed row map, the ordinary strip reference and the two endpoint readouts give this same fixed-radius property for h on the section, say $\|\beta\|_\infty<R$, with $\|h(\beta)-h(0)\|_\infty\le CR$ there. Cauchy's estimate

along the line through β then gives

$$\|h(\beta) - h(0) - \mathsf{L}\beta\|_\infty \le 2CR^{-1}\|\beta\|_\infty^2, \qquad \|\beta\|_\infty \le R/2,$$

where $\mathsf{L} = Dh(0)$. The logarithm of $e^{ig_j a_{\mathrm{new}}} e^{ih(g_j B)}$ gives the old strip coordinate. Bound all the combined coordinates by T, including the ordinary slots left untouched. Take $g_j T \le R/2$ and the product logarithm in the small chart. If $\|h(0)\|_\infty \le g_j d$, then after subtracting $a_{\mathrm{new}} + \mathsf{L}B + h(0)/g_j$ the remaining scaled coordinate r satisfies

$$\|r\|_\infty \le C g_j (R^{-1}T^2 + T(T+d)).$$

The first term is the Cauchy remainder and the second comes from the product logarithm. The remainder is supported on the re-centred slots. If D has bounded absolute row sums, the change from this remainder in the quadratic is bounded by $CN_{\mathrm{cr}}[(T+d)\|r\|_\infty + \|r\|_\infty^2]$, where N_{cr} counts those slots. The log of the Haar quotient costs at most $CN_{\mathrm{cr}} g_j^2 (T+d)^2$ in the small exponential chart, since the log density has zero first derivative at the identity. For fixed R and logarithmic T, d these remainders are small per crossing. Localization still requires the complete carriers and decay of the coefficient maps.

For example, a uniform derivative bound $|\partial_c h_b| \le Ce^{-\mu d(b,c)}$ on the fixed source disc gives a second derivative bound on its half-disc by Cauchy in one coordinate. Interchanging the coordinates gives both exponentials, so that their geometric mean bounds the same derivative:

$$|\partial_{c'}\partial_c h_b| \le \frac{2C}{R} e^{-\mu[d(b,c)+d(b,c')]/2}.$$

The second-order remainder of h_b is bounded by C/R times the square of the corresponding weighted sum of $g_j|B_c|$. Divide by g_j in the scaled coordinate and include the product logarithm as above. The source radius is needed here for the centre map and its derivatives. The old entries are still evaluated within their original strip discs.

On a fixed real ordinary chart the input to this centre map can be formed in physical units. Solve the one-row pivot equations on their fixed regular charts. On the joined sequence a row reads a retained row, or the lift of a named parent row. The latter is the identity or the oriented parent link, so it is holomorphic in the parent link on the same fixed geometry. Take logarithms relative to the ordinary datum at zero and form the reduced row input. These local operations preserve a fixed small disc where the regular row and lift charts have uniform room. Use the holomorphic critical family of the ordinary joined sequence at its admitted real datum. Read the averaged endpoint contours in the ordinary strip frame. This gives the centre germ at that fixed real reference, using the real identification of the critical family at its physical data. The annular comparison supplies its value at zero. To use this construction on cuts, each parent row and the complete endpoint stencils must remain in the prescribed carrier, and the ordinary reference itself must be provided by the compatible cut rule. Exponential real comparison of two references does not give the exact local dependence required of that rule.

For the ordinary localized branch at depth $q = j+1$ the strip join is formed on $W_q = Z_q \cap \Lambda_j$, with the corresponding cell intersections. Use $W_1 = Z_1$ at the first ordinary step. For $q > 1$ this region lies in the original Ω_j. The earlier preparations change members at levels at most j only within the old live sets contained in Z_j. Those changes miss W_q. The preceding sequence has yet to change the members at levels q and above. Thus the joined row masks agree, including which collar rows are lifts. Their values on the real fibre come from the actual inherited and completed physical datum. With that datum and the same first-member action they give the same localized critical orbit.

Form the input at the supplied global prospective pair restricted to the first member of the ordinary join. The ordinary join is finer than the prospective sequence: each joined member is contained in the original member of its level, and the stage enlarges the latter. An inward bond omitted from the joined set has an endpoint in the next joined member. That endpoint lies in

the next prospective member, so the bond is also absent from the prospective no-inward set. The removed-row condition in the restriction argument is therefore satisfied. Extend variations at fixed joined averages by the unchanged exterior configuration. The original averages remain fixed, and every interior bond of a plaquette straddling the true first member is fixed by a joined bare row. The action difference of the extension equals the difference of the restricted action. Thus on the real critical slice the restriction is critical at its own averaged datum. The complete boundary blocks use their original collar levels on the prospective pair.

Use the tower transport of the complex configuration on these alignment collars. Let D_* be the averaged datum of the unaligned restriction on the ordinary joined rows, and let D_o be the target ordinary joined datum in its aligned frame, formed from the physical collar variables and their retained rows. With $g = g[\mathbf{U}]$, the source at the unaligned response pair is

$$b_o(c) = \frac{1}{i}\log\bigl(D_o^{g^{-1}}(c)D_*(c)^{-1}\bigr).$$

The restricted pair used in the response keeps the independent current on complete stencils and the restricted-action variation on its true incomplete stencils. On the fixed regular physical row charts the inverse pivot map changes the prescribed rows by at most $C_L\|\beta\|_\infty$. The aligned zero-increment comparison and the bound on the complex tower transport then give $\|b_o\|_\infty \le C_L(\varepsilon_1 + \alpha_1 + \|\beta\|_\infty)$ on those charts. Here ε_1 bounds the real datum and α_1 the scaled complex potential on the alignment collars. Where the ordinary row targets are formed on this same chart, a fixed small physical source radius and small coupling put this input in the reduced response disc. The real response with the endpoint factors identifies the ordinary localized centre at full parameter values.

In the physical row conversion use the nonlinear one-row correction at $V = \mathcal{M}^j\mathbf{U}$. It is holomorphic at complex V in the admitted double-block chart and keeps the coarse value $\mathcal{M}(V)$ exactly. Only the rows released by the stage are replaced by that completed field. The inherited inward crossing and its ordinary pivot do not add coordinates to the no-inward ordinary datum. Its other actual rows and its lifted parents retain their prescribed reads. If r_{piv} is the physical radius of the regular inverse, take a smaller fixed radius inside r_{piv} so that the bound for b_o, multiplied by the constant of reduction, is below half the native response radius. The terms $\varepsilon_1 + \alpha_1$ fit the other half of this allowance at weak coupling. The fixed disc here is for the centre map. The inherited actions are evaluated on their own running discs.

On the prospective pair take the running bounds on the full ordinary read and on the original parent blocks of the artificial rows. Use ε_* for the real plaquette radius in layer units, α_* for the complex potential and first difference, and γ_* for the independent current. The full tower averaging estimate puts each parent and endpoint average within $C_L\alpha_*$ of its unitary value in the same frame, uniformly in the tower depth. After real gauge on a full one-step block, the logarithms of the real reference are $O_L(\varepsilon_*)$. The complex change and a fixed reduction of the physical source disc therefore keep the pivot and contour maps inside the one-step analytic charts. In particular the complete pivot derivative is near its flat invertible value on these discs. This supplies the uniform chart room for the fixed-size collar source. It also bounds the derivative of the endpoint mean on the output of the response, after reducing the source disc by the fixed response constant. For the complex gauge identities use the one-step covariance on each complete double block. In its small reference gauge let t bound the link logarithms and let $\varkappa_{\mathrm{end}}$ bound the condition numbers of the gauge at the coarse endpoints. The condition is

$$\varkappa_{\mathrm{end}}\bigl(e^{4(L-1)t} - 1\bigr) < \omega_*.$$

The tower transport bound keeps $\varkappa_{\mathrm{end}}$ bounded. The running radii and a fixed reduction of the source disc put t below the resulting threshold. This checks covariance for the actual complex contours and pivots. For the collared first member form the alignment and parent averages on their full original collars before restricting the gauge to the chosen first member.

Put $b_* = \varepsilon_{\mathrm{D,s}} e^{-32\varkappa}$, the reduced source and current radius of the switched response. Choose the physical collar radius so that its contribution to $\|(b_o)_\natural\|_\infty$ is at most $b_*/4$. In addition, take it small enough that the response estimate times the derivative bound for the endpoint read fits in half the fixed contour and product-logarithm chart. The source from the base is bounded by $C_L(\varepsilon_* + \alpha_*)$, including the conversion. The restricted pair has current radius at most $C_L(\gamma_* + \varepsilon_* + \alpha_*)$: use the independent current on its full stencils and the variation of the restricted plaquettes on the true boundary. For the latter estimate the complex curvature bound and the bare-layer units give the three powers of the bare spacing in the current norm. At sufficiently small running radii the base part and the current thus fit below $b_*/4$ and $b_*/2$, respectively. The joined gauge condition follows by the local truncation and restriction of the enlarged gauge cubes. These are bounds on one fixed-real-factor section, with independent current. They allow the whole fixed collar source disc before introducing the old running argument multipliers.

For the ordinary joined class the pointwise bounds on the prospective pair pass to the smaller layer spacings. Use the complete prospective gauge cubes as in the ordinary strip restriction. If a cube of the join lies in a coarser prospective layer, cover it by the allowed larger cube in the two adjacent prospective layers and restrict its real gauge. This uses the enlarged family of §2.2.4.1 on the prospective carrier. For the ordinary response at depth $q = j + 1$ use the join of the strip and the original sequence on $Z_q \cap \Lambda_{q-1}$, with parts of levels $q, q-1$. At $q = 1$ the join has the single part Z_1. Use the first member of the actual ordinary operation. The full first member and the version trimmed to the collared first member give different boundary actions. On a fixed joined section use the independent current on complete stencils. At an incomplete stencil of the first member use the first variation of the restricted action. There are at most six retained plaquettes at a bond, and such a boundary bond is in the bare layer of the separated join. Thus its bare current is bounded by the curvature radius there. This is the same restricted current prescription as in the ordinary strip comparison. The joined response uses the torus and native walk floors at this depth. The ordinary kept strip quadratic uses the separate conditioned one-step walks. The full-depth collar form uses the native full-depth operators.

When a comparison changes the first member itself, the change of its boundary current is local. For nested first members $D^- \subset D^+$, let $J_{p,\mathbf{U}}$ denote the contribution of a single retained plaquette to the first variation in bare units. At a bond of D^- the physical currents differ by the sum of $J_{p,\mathbf{U}}$ over plaquettes incident to that bond and retained only in D^+. This is an identity for the first variations of the two actions. It extends holomorphically on the full plaquette stencils. When transporting the independent increment as well, subtract this sum from the current on D^+; the increment relative to the physical current is preserved. In the bare current norm a single subtracted plaquette contributes at most $C|\partial\mathbf{U}(p) - \mathbf{1}|$. Indeed the first-variation formula is the traceless part of the holonomy transported to the bond, with the bare factor ξ^{-3} canceled by the current norm. If the admitted bound for this plaquette is a_ℓ in the prospective layer-ℓ curvature units, this contribution is at most $C a_\ell L^{-2\ell}$. Sum over the actual omitted plaquettes, at most six at any bond. The layer-zero bound suffices for a changed stencil in that layer; the factor L^{-2} is available when its bound comes from layer one. The running curvature ceiling therefore gives the same small fixed current radius for either first-member choice. On a fixed first member, binary switches preserve its true boundary current and its original plaquette list.

Use the fixed joined sequence in the switches of the reduced response. The plain datum is formed once by the row formula above. The cut removes operator terms on this geometry and keeps the original boundary current. In an active wall component the response reads the reduced datum at the first neighborhood of its response sites. Plain-data conversion reads through the second neighborhood and the outer ends of those rows through the third. The four neighborhood margin contains the complete stencils. For the walk terms use the closed padded hull of the trail. It includes the output blocks and their four-neighborhood with a one-bare-spacing stencil margin. The switch carrier includes every closed cell meeting that hull. Thus a true-boundary current read

on a surviving term uses only the incident retained plaquettes of the original join inside the active cells. Only these used current entries enter the trace comparison.

For this comparison use an aligned standard switch grid at depth q, with side a multiple of the order-q block and with the native response floors. The comparison uses the same grid and native tags in both settings. If the guard is filed on a coarser nested stage grid, take the parent image of those tags. The lift is the identity when the fine endpoints have the same parent. Otherwise it is the directed link between their two parents. The fine endpoints lie in the full blocks of those parents. A released inner parent that is a pivot lies in the corridor between two order-q blocks. These are the blocks containing the fine endpoints of its lift. The four-neighborhood margin of a surviving row includes both endpoint cells, hence those complete blocks by grid nesting. For a held parent keep its actual prescribed link and its carrier. The outer parents are level-q links and keep their coarse targets. A parent block can extend outside the joined first member. Use the original prospective pair on that part of the active cell. Thus the trace comparison uses the pair and the row and parent masks on the full active carrier in addition to the joined traces on the first member.

On a generating carrier use the fixed-point equations on the surviving walk terms. For a component $\mathfrak{k}$ let $Y_{\mathfrak{k}}^{\circ}$ be the sites over its walk terms. The potential response is supported over $Y_{\mathfrak{k}}^{\circ}$ and the shifted multiplier $\lambda + \mathsf{a}B'$ at rows of its first neighborhood. In the component equations one can set B' to zero at the unused rows: the margin statement gives the same response and the same supported first variation. Every local nonlinearity and every weighted difference in the contraction proof then uses the stencils in the four-neighborhood margin. The surviving walk bounds are bounds on these carriers. Thus the contraction runs on the supplied local pair and its complete stencils. This comparison requires the enlarged gauges on the carrier of each surviving term.

Keep the cells of both endpoint blocks of the crossing at switch value one. They are the same cell or adjacent cells, so the read belongs to one active component. For the perturbed averaged field T, let v be the accumulated node factor. At a retained node put $u(y) = g(y)v(y)^{-1}$. The order of the factors restores the aligned datum from the pulled-back datum turned by v. For an auxiliary axial read at a nonnode x in the block of y, set $u(x) = u(y)T(y,x)$. This makes the contour of T^u the identity. If the read instead has a stored transport $T_0(y,x)$ in the aligned inherited frame, use

$$u(x) = T_0(y,x)^{-1}u(y)T(y,x).$$

Indeed $T^u(y,x) = u(y)T(y,x)u(x)^{-1} = T_0(y,x)$. When the data have been aligned by g, use the aligned transport $T_0(y,x) = g(y)T_{0,\mathrm{old}}(y,x)g(x)^{-1}$. At a retained exterior fine node use its own node correction, since it is a top of the old sequence. The ordinary exterior contour can enter as an auxiliary link read and is undone on return to the inherited chart. These formulas are local on the two complete endpoint blocks. To compare two cuts, match the restricted pair on the surviving walk hulls and their four-neighborhood margin. Match also the ambient parent and pivot reads there. The latter equality gives the same plain source on the rows of the conversion and hence the same reduced source on the rows read by the fixed-point equations. The true-boundary plaquettes are those of the original first member on both cuts. Thus the two component equations have the same coefficients, source and restricted current. Uniqueness in the small response ball identifies the resulting read. At full switches the real response and the same restoration identify the ordinary centre. The pair, plain source and endpoint maps above therefore give the cut rule on the fixed standard partition. On the common graded cells we use the native walk construction below, with the same complete row traces.

Distinguish the fields integrated by the stage from the datum that sets the prospective base. The stage keeps the prospective average fixed while the collar nonpivots and their completed rows vary. These motions enter the response as source coordinates at a fixed pair. On its admitted section a variation of one physical collar coordinate β_c changes a bounded set S_c of plain rows. This includes a lift if its named parent uses that released coordinate. Let S_c^{+} include the finite row neighborhood used by reduction. The regular inverse and the one-step seam conversion give a reduced increment

of size $C_L|\Delta\beta_c|$ supported on S_c^+. For a crossing b, take its two complete endpoint blocks as the read set K_b. Apply the weighted response difference bound with weight $e^{-\mu d(\cdot,K_b)}$. It bounds the increment on K_b by $C_L e^{C\varkappa} e^{-\mu d(K_b,S_c^+)}|\Delta\beta_c|$. The mean contour and relative logarithm are holomorphic on fixed smaller charts there. Dividing by the increment gives

$$|\partial_{\beta_c} h_b| \le C e^{-\mu d(y_b,y_c)}.$$

Both endpoint blocks lie in the adjacent-scale geometry. The endpoint change of units and the number of such coordinates at a pillar are bounded in terms of L. The symmetric distance sum gives weighted row and column bounds at a reduced rate. The switch product for the two endpoint cells costs at most a fixed factor. In the scaled logarithm the factor g_j from $\beta_c = g_j B_c$ cancels the output factor $1/g_j$. The constants in this derivative bound are uniform as the prospective pair ranges over the joined section used for the ordinary joined response. The retained determining values used in that section are the prospective averages of the same complex pair. If another entry uses those values in a different role, verify its admission on its original domain.

At the real prospective datum with the free increment zero, the actual rows of D_o and D_*^g agree. This is an analytic chart anchor. The sharp and other retained tests remain in the real integral on their original arguments. On a released row, this uses the zero nonpivot increment and uniqueness of the pivot at the prospective coarse value. The plain logarithmic difference is then carried by the original artificial boundary rows. In particular if the alignment writes $D_*^g = e^{iB_1} D_o$, the source at the unaligned pair is $-\operatorname{Ad}(g(c_-)^{-1})B_1(c)$. The finite neighbourhood used by reduction keeps it in the boundary collar. The crossing reads lie inside the original quiet window away from this collar. The weighted response bound therefore gives exponential smallness of the zero-increment change there, uniformly in the switches with the endpoint guard fixed. This uses the prospective frame for the zero response, where the exterior contour of the induced datum is the prospective axial contour. For the complex version let W_* be the level-j average of the supplied prospective configuration before one-step axialization. The prospective datum on the real fibre is the stock of retained determining rows, including the top coarse targets. The prospective background has precisely these averages. Define the target on every common row of level q by the corresponding average $\mathcal{M}^q\mathbf{U}$. At a released row use W_* in the regular pivot inverse, with coarse target $\mathcal{M}(W_*)$. At an artificial row use the prescribed lift of the same parent value. Put $q = j+1$. At the inner face of the ordinary join the level-j parent carriers lie near $\partial\Lambda_j$. For $q \le k_0$ the prepared box begins five coarse cells inside the old exterior; for $q = k_0+1$ it begins seven. The quiet exterior extends eight cells beyond the hull of G_j. Hence the full inner alignment collar and its parent blocks lie inside Z_q''. The margin before inserting this band is at least one coarse cell. Their lifts use the held level-j rows. For $q > k_0+1$, the prepared box is inside the old exterior of level k_0+1. The carried box G_j includes the two-cell enlargement of that exterior. The inner band at $\partial\Lambda_j$ is now in the released collar. Its lifts use the completed level-j field there. At depth one there is no inner band. The parents on the outer face are level-q rows and use the fixed coarse averages. If a free increment is expressed in the prospective axial frame, pull it back by the adjoint of the inverse gauge factor at the start of its bond. The gauge fixes the coarse centres, so the one-step covariance gives the same coarse target in both frames on the admitted double block. The one-row inverse is zero at zero free increment. This gives the induced rows there in the same frame on the entire section.

These are all the rows of the joined target. On the real fibre the common values agree with the actual retained datum, the released values are the completed collar datum, and the lifts give the ordinary joined datum by its defining rule. The complete averaging blocks of the common rows stay in the join. A common row meeting a join collar of level r has that collar's level. If its endpoint in the original layer is in the full collar band, that band is in the original layer r. An endpoint beyond the band is at distance at least $\mu_{\mathrm{col}} L^r$ from the collar endpoint, where $\mu_{\mathrm{col}} \ge L^2$ is the joining width. The row would then have level at least $r+2$. The separation of the original members would put its collar endpoint in the member at level $r+1$, which is disjoint from the collar. The

alignment fixes both endpoints of a row at level r. This uses the same geometric test for complex and real fields.

A held determining row in the analytic target is read as the prospective average. That value equals its prescribed group datum on the real fibre. A stored value can also occur as a spectator of an old factor. Keep that separate occurrence in its original slot, including when it is Borel. Thus the common rows agree at zero collar increment, even as the prospective pair varies. Only the artificial boundary rows can supply a zero-increment source. The finite reduction stencil enlarges that support once. The current on the original restricted action stays in the base of the response and is not a separate input increment.

For a crossing oriented from the old exterior into the old interior, let τ_-^*, τ_+^* be the two prospective mean contours of W_*, from the block centres to the fine endpoints. Use $S_{\mathrm{g}} = \tau_-^* W_*(b)(\tau_+^*)^{-1}$ as the prospective analytic link. If the response is zero, the ordinary read with inverse node factors gives $W_*(b)(\tau_+^*)^{-1}$, since the exterior fine site is an old node. Express the ordinary read in the common frame by multiplying on the left by the same τ_-^*, with the inverse prescription for the reversed bond. At zero response it is now exactly S_{g}. On the real prospective axial chart $\tau_-^* = 1$, so the frame choice leaves the physical ordinary read unchanged. This continuation uses only the two complete blocks. It is applied to the link read and does not change the independent current in the response. With targets formed from the same complex pair and the same stored data, the tower transport, pivot completion, response and mean contours are functions of that pair. Two splittings of the pair give the same read by uniqueness of the regular response on the admitted domain.

With this common-frame prescription and the row identification, the zero-increment change on complex sections comes only from the original join boundary. The uniform decay bound for the switched response gives $\|h(0)\|_\infty \le C(\varepsilon_* + \alpha_*)e^{-cM\check{R}_q}$ at the guarded crossings, where the complex bound uses both the real curvature and complex potential radii. The old-wall read is at distance of order $a_q = M\check{R}_q L^q$ from either ordinary alignment band, by the eight-cell inner clearance and the two-cell outer clearance. At the fixed depth q this gives the indicated weighted distance after including the complete blocks. Here $\check{R}_q$ grows like $(\log\theta_j)^r$, with $r > 1$, and $g_j^{-1} = \theta_j^{1/2}$. After division by g_j the error is still smaller than any fixed inverse power of $\log\theta_j$. Thus the offset fits the smaller crossing tier in the Gaussian mean.

For the substitution in the inherited density, pull the prospective link back to the ordinary frame. For the orientation above this gives $S_{\mathrm{g}}^{\mathrm{o}} = (\tau_-^*)^{-1} S_{\mathrm{g}}$. The ordinary link is $S_{\mathrm{o}}^{\mathrm{o}}$, before the multiplication by τ_-^*. Reverse either link by inversion for the opposite orientation. In the displayed orientation the relative logarithm in the ordinary frame is conjugate to the common-frame logarithm by $(\tau_-^*)^{-1}$. This factor depends on the background pair and is fixed as the independent collar coordinates vary. A restored interior transport common to both centres cancels from their relative group element. The new coordinate a_{new} is in the ordinary frame. If a coordinate was first introduced in the common frame, conjugate it by $(\tau_-^*)^{-1}$ when pulling back the physical link. This pullback is the identity at the real axial anchor. The old coordinate is given by the small logarithm of

$$e^{ig_j a_{\mathrm{old}}} = e^{ig_j a_{\mathrm{new}}} S_{\mathrm{g}}^{\mathrm{o}} (S_{\mathrm{o}}^{\mathrm{o}})^{-1}.$$

Here the reconstructed nonpivot values are used in the inverse pivot map at the original coarse target. At fixed collar and coarse data the Lebesgue Jacobian is $\mathrm{j}(g_j a_{\mathrm{new}})/\mathrm{j}(g_j a_{\mathrm{old}})$, where j is the Haar density in exponential coordinates. Keep it with the reconstructed row in the inherited integral.

The old-wall crossing is in the far part of the plain strip. At its formation, with $q = j + 1$, the collar of Λ_q is separated from the old wall by the ordinary coarse layers. The ordinary rim masks exclude the strip. The failed-cube cutoff halos have their bases inset by one coarse cell and stay inside Ω_q even with the two smaller halo layers and a bond reach. Thus none of the three birth near masks contains the crossing. The condition uses those original masks even if a later stage

tests the crossing sharply. Its endpoints belong to adjacent level-q blocks, one on either side of the old wall. A level-m row can read this fine bond only if its double block contains both endpoints. The only such row joins the two block centres. Its inverse pivot uses the full double block and the prescribed coarse target. On the quiet geometry both blocks are clear of the old integration collar and of the cut-off test enlargement. Their stored bonds are in the ordinary boundary strip, away from the rim tier. The fine exterior links are ordinary determining values. Thus the local inverse pivot and its Jacobian on this double block read far stored variables and prescribed group values. A direct boundary read of this coordinate and its occurrence in the conditional polynomial of the strip use the old far analytic argument. Evaluate both at a_{old}. The near strip fields and the real island arguments keep their original coordinates. The ordinary completion includes this nonpivot bond with its stored threshold $\mathbf{1}_{\{|g_j a_{\mathrm{old}}|<\delta_j\}}$. The direct sharp mask tests the free exterior collar nonpivots and leaves this crossing in the ordinary strip integral. If an enlarged mask also tests the crossing, its new factor is $\mathbf{1}_{\{|g_j a_{\mathrm{new}}|<\delta'_j\}}$. On its passing branch the triangle inequality for the real angle gives $|g_j a_{\mathrm{old}}| \le |g_j a_{\mathrm{new}}| + |h|$, with h in the ordinary frame. Hence the old factor is one if $\|h\|_\infty + \delta'_j < \delta_j$. Use this inequality on the whole fixed real collar box. The uniform row derivative of h, integrated along the segment from 0 to $g_j B$, gives $\|h\|_\infty \le C_L \delta'_j + \|h(0)\|_\infty$ on that box. The ratio δ'_j/δ_j and the boundary estimate give the inequality at weak coupling, including on the edge of the box.

For the simultaneous change, hold the collar coordinates and the coarse targets fixed. The ordinary centre depends on the exterior determining rows and the coarse targets. The prospective centre has the same fixed targets, and its finer determining set excludes the old inward crossings. Thus neither centre uses a crossing nonpivot being changed. Translate all those nonpivots by their respective right factors. For each coarse row solve one pivot with all the nonpivots of its double block at their new inverse values. Two rows can share internal links, which are free or prescribed. Each row uses only its own pivot. This gives the simultaneous inverse and the product density.

The new ordinary completion tests the stored coordinate. In the component-local run, on a plain strip bond this is the original logarithm relative to its local strip centre. Let d_{lg} bound the angle between this centre and the old global centre on the actual test domain. The global relative angle is at most $\delta_j + d_{\mathrm{lg}}$ on a local pass and at least $\delta_j - d_{\mathrm{lg}}$ on a local failure. Use this reserve when removing the ordinary approximate bond test: the additional angle from the global centre to the approximate centre, together with d_{lg}, must be less than δ_j. A failure-energy estimate using the global angle can use the lower bound $\delta_j - d_{\mathrm{lg}}$. A factor with the literal global sharp threshold keeps its own indicator where it is not implied. In the enlarged mask, on the new sharp ball its original relative angle is at most $\delta'_j + \|h\|_\infty + d_{\mathrm{lg}}$. If this is less than δ_j, both old completion indicators are one on that passing support. The bound uses the ordered product of the two centre shifts in the same real frame. Carry the actual local completion indicator on both branches of the component-local operation. For the direct mask keep the crossing in the native strip slot a_{old} with its ordinary ball. The sharp balls then apply only to the exterior collar variables. If a crossing outside the sharp mask is translated for a chart calculation, use the inverse translation inside the old indicator. The new radius gives no bound on that untested coordinate. Only in the enlarged mask, where the crossing also passes the new sharp ball, is the old completion redundant by the displayed inequality. On the passed ordinary cube mask, the approximate test uses two different supplies of the δ_j bound on its nonpivot reads. A bond with no endpoint in Λ_{j+1} has the bound from its recorded split indicator. A bond with an endpoint there has it from the cutoff of the ordinary Gaussian integral. On this mask the strip change is the ordinary one. The centre reserve above puts both groups below the approximate threshold $2\delta_j$. The second group is outside the large-field operation. Thus removal of the approximate factor is an identity on the support of the whole term, with those exterior cutoffs present. Earlier regularity tests are removed where the corresponding stage comparison gives their value one. This product uses one choice of sharp centre. A comparison of both sharp masks also

retains their two disagreement sectors. A version of the operation retaining any of those tests still carries its original inverse argument. On a local sharp failure retain the untransformed prefix with its local indicator. On a global sharp failure use the transported real tests and density.

With $q = j + 1$, the inherited free mask comes from $\Omega_q^{(j)*}$. The stage test uses $(\Omega_q^c \setminus Z_q'')^{(j)*} \cap Z$. The star uses incident bonds and deletes the pivots for the corresponding region. The overlap therefore consists of crossings between Ω_q and $\Omega_q^c \setminus Z_q''$, subject to the stage's restriction to Z and both pivot exclusions. Keep only the crossings present in the selected stored strip stock. These are the additional inherited reads in the broader sharp mask.

For a quiet inherited crossing in the broader incident mask, the approximate pass also removes the ordinary upper cutoff on the sharp-failure branch. The approximate chord test on that mask gives angle less than $2\arcsin(\delta_j')$ in the local stage frame. The quiet-crossing estimate bounds the separation of the local and ordinary centres by $C_L\delta_j'$ using the internal approximate passes. It applies on both outcomes of the sharp test on this crossing. Thus its ordinary angle is less than $2\arcsin(\delta_j') + C_L\delta_j' < \delta_j$ at weak coupling. The plain ordinary upper indicator is one throughout the approximate-pass support for this crossing. In an ordinary cube test this removes the crossing's factor from the product. Any complement of the whole cube test still uses the other factors.

For the global failure change of variables, the raw identity transports the approximate pass to the chord ball in each new relative coordinate W_b' and keeps the sharp threshold at angle δ_j'. Order the finite tested mask and partition the complement of the sharp pass by its first failing bond. In each part the earlier bonds lie in the sharp balls, the failing bond has angle in $[\delta_j', 2\arcsin(\delta_j'))$, and the later bonds lie in the approximate balls. These are fixed real factors on this mask. The free V_j collar stock uses the determining bonds in the exterior, which have no endpoint in Ω_q. The crossing has one integration slot in the inherited stock even when it enters two tests. Its cutoff Jacobian and the solved pivot of its row remain in the same density. This partition concerns the approximate-pass part of the global failure kernel. The other real factors retain their native arguments.

The density correction here is taken at the actual enlarged radii. Let A and B_* bound the scaled strip and collar arguments, including their multipliers when they occur in the substitution. Write $H_* = C(B_* + d)$, so $|a_{\mathrm{old}} - a_{\mathrm{new}}| \le H_*$. The normalized Haar density has a nonvanishing analytic extension on the fixed small chart and its logarithm has zero linear term. The log of the free-slot quotient is therefore at most $Cg_j^2(A + H_*)H_*$ in absolute value. For the pivot factor compare the full coarea densities at the old and new nonpivot arguments with the same coarse target, in the original row chart. In Haar frames each is the inverse determinant of the derivative in the solved pivot. On a reduced fixed physical disc this derivative is invertible and its log determinant has bounded first derivatives. Insert all translated nonpivots of the double block before evaluating the inverse. The change of the solved link, and the log ratio of the row densities, are bounded by $C_L g_j H_*$. The fixed number of fine links of a double block is included in C_L. Count the row once. The unchanged density and the ratio give the original coarea factor at its inverse argument. If the density is written with the centre determinant and the linearizing derivative, their product with the pivot Haar factor represents this same coarea density. To check the normalization, differentiate $\widetilde{\mathsf{Q}}_V(\psi_V(B)) = LQ_VB$ at fixed nonpivots in the unconstrained pivot direction, then set $Q_VB = 0$. For one private row c at its pivot p, this gives

$$(D_p\widetilde{\mathsf{Q}}_V)(\psi_V(B))D_p\psi_V(B) = L\mathsf{m}_c(V).$$

On the fibre the output logarithm is at the coarse target, so its derivative at the identity is the identity. The first factor on the left is the actual row derivative in Haar frames composed with the pivot exponential derivative. The latter has determinant $h(B_p')$, where $B' = \psi_V(B)$. Taking determinants identifies the coarea factor as

$$|\det(L\mathsf{m}_c(V))|^{-1}|\det D_p\psi_V(B)|\,h(B_p').$$

The analytic continuation uses the determinant sign fixed at the real anchor. In the convention using m_c for elimination, the factor L^{-3} is the row's fixed output normalization. It cancels in a ratio with the same row. The full coarea factor is evaluated once at the fully composed inverse argument.

The double block is the fixed root of this ratio. Its crossing increments read the same two complete blocks of the row. On a face with this root active, the restrictions to its active component give the same nonpivot list, target and pivot solve as the full face calculation. For a parent read outside the joined first member, use the completed collar value for a released link and the original prospective average for a held one. Include that parent carrier and the neighboring rows of the naturalization in the read before restricting the switches. These parent values are common to the source and the row chart. Form the response from the completed prospective collar first. Its restriction to the face uses the same source values at the rows of the response. Then use its output in the pivot solve at the inverse strip arguments, with the target held fixed. When the crossing gate is off the two evaluations of the determinant coincide. The difference of their logs is one holomorphic rooted read. Apply the rooted Cauchy estimate with the double block fixed and attach the response carriers at this root. The one-slot Haar quotients use their own endpoint roots.

The recentered free crossing is plain. Its group map is a right translation at fixed collar variables, so it has unit Haar Jacobian before passing to exponential coordinates. The cutoff derivative occurs on the original ordinary transition shell. At a fixed real shell argument it is the rank-one determinant in the ordinary shift. Its log is of size $C(\varepsilon_j/\delta_j)e^{-c\check{R}_{j+1}}$ with the original carrier decay. Its old switched version holds the shell argument fixed and multiplies the difference from the zero argument by the old near-field switch. This costs its switch radius. A varying smooth cutoff argument in another composed map requires its own real treatment.

On a fixed real part of the global sharp-failure domain the new cutoff has scale δ'_j. Hold its transformed argument W'_b real and write the inverse factor, up to the fixed right translation, as $W'_b \exp[-i\vartheta(W'_b)H_b]$. Its Haar Jacobian is

$$J_b = 1 - d\vartheta_{W'_b}(W'_b iH_b).$$

In left-trivialized coordinates factor out the adjoint of the exponential. It has determinant one and fixes iH_b, leaving a rank-one perturbation. The derivative of the cutoff is supported in the transition annulus and has norm at most C/δ'_j. Thus the principal log J_b is holomorphic in the shift on the independent shift disc once $C|H_b|/\delta'_j < 1/2$. Its absolute bound there is $2C|H_b|/\delta'_j$. This continuation leaves the real cutoff argument and the fixed selectors in the Haar integral. In the inherited frame the old relative link is $W'_b e^{-i\vartheta(W'_b)H_b} S_{g,b} S_{o,b}^{-1}$, where $S_{g,b}$ is the prospective global centre and $S_{o,b}$ the old strip centre. If a frame map is needed, transport W'_b, H_b together and keep ϑ evaluated at the original real W'_b. For fixed W'_b, its logarithm is holomorphic in H_b and the admitted relative centre logarithm h_b of $S_{g,b}S_{o,b}^{-1}$. The bound in angular units is $C(2\arcsin\delta'_j + |H_b| + |h_b|)$. Use the independent shift margin for H_b and the original strip margin for the other arguments. The old pivot is then evaluated at these inverse arguments on its original chart.

If the admitted shift is expanded in carriers that all contain the same complete bond read, take its weighted absolute tree sum $\|H_b\|_\mu$. Each product in the series for $\log J_b$ has a carrier given by the union of its factors, whose tree length is at most the sum of their lengths. The weighted logarithm sum is therefore at most $2C\|H_b\|_\mu/\delta'_j$ when $C\|H_b\|_\mu/\delta'_j \le 1/2$. At the physical quiet crossing the carrier of the shift also reaches the distant boundary, so its spare decay pays this inverse radius. At fixed prospective centres the Jacobians multiply on the free slots. If an ordinary crossing centre reads the new collar coordinates, form the collar change first. Its reconstruction does not read those old crossing coordinates, and the full derivative is triangular.

The fixed physical radius for the relative logarithm gives $\|a_{\mathrm{old}}\|_\infty \le C_L(T+d)$ on the smaller section above. Here $T = O(p_0(g_j))$ bounds the unamplified strip and collar fields. Form the inverse

on this box. Its use in a switched far argument must fit the old radius $\varrho_{\mathrm{st}}\mathfrak{b}_b$, with $\mathfrak{b}_b$ of order $p_0(g_j)$ in these units. The direct field and the displacement will use separate fractions of this radius. The response of the triple uses its own margin. The reductions concern only slots integrated in the stage. Each carried far slot has its full output radius. The Gaussian operators use their own cut parameters.

Let $\Phi(s,\lambda)$ be a switched weight with the inverse arguments. Its perturbation at fixed switches is $\Phi(s,\lambda)-\Phi(s,0)$, where zero sets all amplitude parameters to zero. This is the difference to which the rooted estimate with a zero at vanishing amplitude applies. For one factor per wall-component take a mixed difference using a separate amplitude for each component. Cauchy's formula then gains one inverse radius for each such amplitude, once the common cut satisfies component deletion. A component carrying only the fixed join boundary can affect a difference in a nonlinear weight even when it carries no collar amplitude. In the rooted estimate include that boundary among the source cells and use its own small weight for such a component. Compare the zero-amplitude value with the prescribed reference value in a separate term. When comparing two far arguments at the same pair and fixed remaining arguments, Cauchy's estimate along their line bounds the difference by the old envelope times the displacement divided by the remaining radius. In particular a residual shift $h(0)/g_j$ in this comparison is paid by the boundary offset, with any multiplier on its slot included in the displacement.

On the aligned flat stage partition, let K_b be the cells containing the two complete endpoint blocks. It consists of one cell or two face neighbors. Use the same K_b wherever b is read. Write s^{K_b} for the parameters with the switches of K_b set to one and put $\eta_b(s)=\prod_{\Delta\in K_b} s_\Delta$. An interpolation of the far coordinate is

$$a_b(s)=\frac{1}{ig_j}\log\Big(e^{ig_j a_{\mathrm{new}}(b)}e^{i\eta_b(s)h_b(s^{K_b},g_jB)}\Big).$$

Here h_b is the admitted response read in the ordinary frame, using the original row and parent traces. At the full evaluation this is the inverse coordinate. If a cell of K_b is inactive, the correction vanishes. When both endpoint cells are active, K_b lies in one wall-component, and the guarded response reads that component. Thus no correction uses a different component's parameters or collar increments. The coarse pivot for this crossing uses the same blocks. At fixed collar increment the interpolated group map is a right translation, so the same density quotient applies. The bound on the product of the at most two switches costs a fixed factor. The preceding argument for the far radius absorbs it in the common switch margin.

Every inherited weight uses these common far coordinates at the crossings it reads. To estimate the inherited weights, split the field argument into the original coordinate and its displacement. For an entry F with actively integrated far and near slots f,n, put $\varsigma_b=\lambda_{\mathfrak{w}}s(\square_b)$ on its active component and zero otherwise. On an unchanged far slot let $a_{\mathrm{new}}(b)$ be its original coordinate and set $\Delta_b=0$. On a moved far slot put $\Delta_b=a_b(s)-a_{\mathrm{new}}(b)$. Use the joint interpolation

$$F\big(P_s;(\varsigma_b a_{\mathrm{new}}(b)+\tau_{\mathfrak{w}}\Delta_b)_{\mathsf{f}},(\varsigma_b)_{\mathsf{n}},(a_b)_{\mathsf{n}},\mathfrak{q}\big).$$

Here a_b on a near slot is its real value in the original chart. A moved crossing has a far slot. When all parameters are one, the far argument is exactly its value in the original chart. On a binary cut the displacement uses its endpoint gate, so is zero when either endpoint cell is inactive. Its read includes $a_{\mathrm{new}}(b)$, even at $\lambda_{\mathfrak{w}}=0$, because this coordinate enters the product logarithm. Choose the owner cell $\square_b$ from K_b, so this read is also guarded by the endpoint switches. The direct argument uses the owner's switch. The argument $\mathfrak{q}$ contains all the spectators and their inherited switches, which keep their values and radii. At zero switch a near entry is independent of its Borel field. At zero field it is independent of that switch. Thus the empty cut gives the original zero-slot value, also for the near entries. Use the unmultiplied physical coordinates for the integrated arguments of the Gaussian reference and its coordinate change. For the analytic interaction formed from a difference of strip quadratics use the corresponding far variables and the switched near variables $\mu_b a_b$ on all

its spectator legs. In a rooted read the original domain contains an endpoint of every crossing it reads. The extra endpoint cells therefore attach as face neighbors to those root cells. This is the attachment estimate on the flat grain, with at most eight choices of extra cell at each root cell. The old history trace is kept at the reader's address. The stage profile below controls admission of the pair, and the switched chain and double-block estimates control the native operators with several crossing slots. A newly composed map that varies an old near-field argument must supply its own real coordinate and near switch. The old switch property applies to spectators in their original slots.

The displacement part of an inherited term can be present at $a_{\mathrm{new}} = 0$, since the transformed argument then contains the collar field. The value at $\tau = 0$ depends on a_{new} and is estimated with the ordinary strip amplitudes. Let $\varrho_{\mathrm{st}}\mathfrak{b}_b$ be the old far radius. Even on the smaller enlarged disc $|a_{\mathrm{new}}(b)| < c\varrho_{\mathrm{st}}\mathfrak{b}_b$, the bound $g_j\varrho_{\mathrm{st}}\mathfrak{b}_b = o(1)$ keeps the product logarithm on its fixed physical disc. Differentiating it in its second factor gives

$$|\Delta_b(s)| \le C_L|\eta_b(s)h_b(s^{K_b}, g_jB)|/g_j \le C'_L\mathfrak{b}_b.$$

Here the last inequality uses the collar field on its stage box and the bound on the boundary offset. For the finite difference estimate the physical free strip box has $|a_{\mathrm{new}}(b)| \le C_0\mathfrak{b}_b$. Reserve at most a fixed fraction of the old radius for each of $C_0e^{\varkappa}|\lambda_{\mathfrak{w}}|$ and $C'_L|\tau_{\mathfrak{w}}|$, in units of $\mathfrak{b}_b$. In particular the two amplitude discs are simultaneously available, after intersecting the first with the pair-admission disc. For an estimate with an enlarged a_{new}-disc, the unshifted term instead uses its bound $e^{\varkappa}|\lambda_{\mathfrak{w}}|\,|a_{\mathrm{new}}(b)|/\mathfrak{b}_b$ in this same radius budget. The large strip-field disc and the large direct amplitude cannot both be spent on the same slot. Spectator slots keep their own discs.

Suppose an inherited term is bounded on the old joint disc by b_X. For a set of k components take the mixed difference in their displacement multipliers between zero and one. The Taylor coefficients that survive have positive order in every one of these multipliers. Summing the Cauchy bounds in those variables bounds the mixed difference by

$$b_X(c'\varrho_{\mathrm{st}} - 1)^{-k}.$$

This uses one supremum of the whole term, including when it reads several bonds of a component. In a coefficient whose displacement adds an endpoint cell, that cell must be active because η_b contains its switch. It attaches to the old support at the original endpoint. A component sourced only by the fixed boundary offset uses the offset estimate.

In the switched estimate keep the unchanged strip arguments and their old real selectors fixed while expanding in the new switches. Use the pair amplitude and the displacement amplitude separately. Let G be the difference between the evaluations with displacements $(\tau_{\mathfrak{w}}\Delta_b)_{\mathfrak{f}}$ and with zero displacements. Both evaluations use the same switched pair, the same direct strip argument and the same near dials. The fixed joined pair is part of the base. Any zero-increment displacement from the join enters G with the same multiplier $\tau_{\mathfrak{w}}$ as the rest of Δ_b.

Keep the original rooted read N fixed. For new switch support K, use the finite difference

$$\sum_{J\subset K}(-1)^{|K|-|J|}G(1_{N\cup J}).$$

The amplitude of a wall component is inherited by the pieces obtained on cutting it, with the original anchors retained. The finite difference is the integral of a mixed first derivative in the switches of K, over their unit intervals. Estimate this derivative on the spatial switch discs while keeping the two families of amplitudes on their joint discs. If both amplitudes of a new wall component vanish, the binary cut can delete its new cells. The pair increment is zero there and so are the direct amplified arguments and the guarded displacements of that component. Pairing the subsets with and without a new cell of this component makes the finite difference zero. The fixed read is kept in every summand. The difference also vanishes when all the displacement amplitudes are zero. Consequently its Taylor series in the amplitudes has at least one positive degree per new

wall component and a positive displacement degree somewhere. The unchanged strip dependence is summed by the original strip block expansion.

Let $a = (R_{\mathrm{pair}} - 1)^{-1}$ and $u = (R_\tau - 1)^{-1}$, where the radii refer to the two amplitude discs. Let z be the ratio from Cauchy's estimate in the spatial switches, with the exponential margin of their original discs. For k wall components the sum of the absolute amplitude coefficients of the finite difference at physical amplitudes is bounded by

$$b_X z^{|K|}\, ku(1+a)[a + u(1+a)]^{k-1}.$$

For a component marked with a displacement the degree sum is $u(1+a)$. At every other component sum the degrees with at least one of the two amplitudes present. Mark one of the k displacement components for this bound. It has an original crossing anchor in X, since the old reader contains a crossing endpoint and its displacement vanishes unless the endpoint gate is active. An endpoint added by the gate is one face step from that anchor. The other components attach as arms. Their connected sets, weighted by the spatial switch factor, have a sum bounded by $Ce^{-\nu D(a_0, X)}$ at a source anchor a_0. Here $D(a_0, X)$ is the length of a shortest attaching cell chain. The attachment length is included in the decay weight of the new domain. The old packing bound on the sum over source anchors then gives a factor at most

$$\exp\left(CN^*(a+u) \left[1 + \frac{d^\sharp(X) + 1}{\mathsf{N}_i} \right] \right)$$

for the unmarked components. The amplified pair radius here uses the stage gas choice $q - \mathfrak{p}_0 - 5r > 9r$. Together with the old strip radius this makes $N^*(a+u)$ small. Charge the displayed factor and the distance estimate below by reserving a further fixed multiple of $1/\mathsf{N}_i$ in the stage rate. Over the full waiting window this is a fixed loss.

For a pin in the new domain, let C' be the component of the marked original anchor. The old anchored tree and its attached arms place the pin at distance at most $CM(\mathsf{N}_i + d^\sharp(X) + \sum_W |W|)$ from C', in stage units. The same bound holds for the distance from C' to a component C met by the new domain. Reserve c/N_i of the rate and part of the spatial switch margin to extract the corresponding exponential in the distance divided by $M\mathsf{N}_i$. At fixed C', sum the marked original anchors using $J^\Sigma_{\mathsf{A}}(C')$. Earlier stage terms give the additional $CN^*\mathsf{N}_i$ from their old pinned bound. Lower-level strip reads are counted at their original endpoints, where their long-domain estimate still applies. Summing the distance weight over C' by the packing of the treated components gives, for both the new pinned norm and the new sum over domains meeting C^+, the bound

$$Cu \sup_{C'} \big(J^\Sigma_{\mathsf{A}}(C') + CN^*\mathsf{N}_i \big).$$

This uses the old summed-anchor norm for the marked component even when the new pin lies in the original domain. Replacing that sum by a count of possible anchors in a domain would cost a further waiting-depth factor.

To enter the difference in the primary potentials, subtract its value at $B = 0$ in the actual collar integration variables, at fixed pair and unchanged strip arguments. Every surviving block difference then uses at least one new collar block, and the old collar ratio $\hat{q}$ applies. The factor $\hat{q}N^*$ in the component bound is small, so this insertion fits the old $\mathsf{N}_i u N^* \mathsf{N}_i$ bound after summing the stages. At zero free source the subtracted offset has the exponentially small bound from the joined comparison. If the independent free shift y moves the physical zero, its extra displacement on the free-source disc is at most $C|\mathsf{y}|/g_j$, by the same fixed physical chart estimate. This part uses the free-source closure bound, with the small ratio $|\mathsf{y}|/(g_j \mathfrak{b}_b)$.

Count hot blocks among the current integration variables. At stage $j+1$ these are the level-j strip and collar nonpivots. Assign each coordinate block to the level-$(j+1)$ M-cube of one of its endpoints in the stage collar. If both endpoints are available, use a fixed convention. In the direct graded rule take the new sharp product on the free exterior mask and keep the inherited crossings

in their original strip coordinates. For the broader re-centred rule use the product on the passing branch above, where the smaller crossing ball makes the transported old test redundant. The two disagreement sectors of a comparison of sharp masks keep their real indicators and are treated separately. An earlier stored spectator or a solved pivot adds no block. Each owner cube has at most $C(LM)^4$ slots. The quiet box at level $j+1$ contains at most $C\check{R}_{j+1}^4 \le CN^*$ owner cubes in its collar.

At fixed shift source the smooth Haar difference at a free collar bond reads just its own variable block. The centre term reads none. Thus the alternating block expansion of the difference has the usual single active-block factor $\hat{q}_v$ and at most e^{2a_1} from the hot-block charge. The shifts formed from fixed stage data add their rooted carriers to the localization and no current integration block. This potential reads the same shift source as the translated field. Add its envelope to the primary potentials while keeping the one closing index for that source and carrier. The bounded number of free slots of a cell then adds a bounded amount to the hot-block activity, using the smooth Haar bound above. In the stage tail estimate the weight of order $(\mathsf{N}^*)^{-1}$ at the sharp threshold gives the factor $\exp[-c(\log\theta_t)^{2\mathfrak{p}_1-4r}]$, which pays the existing activity of order $(\log\theta_t)^{4r}$ under the exponent choices of the step. A failed sharp branch uses the separate cutoff Jacobian in its real history kernel. On the quiet branch the product over at most $C\mathsf{N}^*$ transformed slots has log bounded by the sum of their shift bounds divided by δ_j'. The quiet collar decay pays this sum as well as the factor N^*. This costs only a bounded density multiplier. A space-time weight on that history uses the survival reserve.

Use this ownership for the gas and keep the graded tree for the term's carrier. In the flat last stage the two cell systems coincide. If a graded carrier cell meets new hot slots from a treated component, their number is at most the total CN^* for that component. The separation of the treated boxes puts the slots of at most one treated component at such a cell, including the fixed endpoint enlargement. Hence the new hot-block factor on the added cells is bounded by $\exp(Ca_1N^*\sum_W|W|)$. With $a_1=\hat{q}$ it fits the spatial switch margin, also in the earlier stages. The bound on the ratio of adjacent grades gives bounded degree for the graded cell chains. Its volume conversion $L^{4(\mu-j-1)}$ is used for an old coarse pin, where the path to the anchor pays for it. Count only the assigned integration blocks in the gas. The Gaussian cost below uses the same count, with the error from each guard weighted by the number of slots it protects.

There is a compatible grading for the switched walks of the stage response. On the part below the next layer use the level-$(j+1)$ M-cubes. Above it, on the prospective layer $\mu > j+1$, use the level-μ M-cubes. The stage regions are unions of these nested cells and consecutive faces are separated by a full cell of the larger level. Small relative enlargements of the cells meet at most adjacent grades. They have bounded overlap, by separation of nonadjacent layers and by the cube count within each grade.

For a walk use the union of the closed hulls of its constituent cells. Let G be a graded cell of side $M\xi_\mu$ that meets one such hull. The hull's diameter is at most $C_L\mathfrak{M}\xi_{n_*}$, with n_* its largest constituent level. If $n_* > \mu+1$, its level-n_* constituent would lie in the region at that level at distance at least $M\xi_{n_*}$ from G. For sufficiently large M this is impossible. Thus the tree that visits the hull meets a small relative enlargement of G. Unless that tree lies wholly in the enlargement with twice the width, it spends weighted length at least c_LM there: before exiting it uses spacings at most $\xi_{\mu+1}$. Sum over these enlargements by bounded overlap. The cells whose larger enlargements contain the whole tree are bounded in number by the same overlap. The initial hull, when it is counted separately from the visited hulls, meets only a bounded number of cells by its diameter and the same level comparison. Hence, for the walk length $|\Gamma|_{\mathbf{B}}$, the number of distinct graded switches is at most

$$C_L + C_L|\Gamma|_{\mathbf{B}}/M.$$

The switch loss therefore fits in the native walk decay after taking M large compared with the switch radius. Apply the weighted walk summation to the individual scalar monomials and their partial

sums. The cost $e^{C_L \varkappa}$ is fixed and the rate loss is $C_L \varkappa / M$. In particular, if the native absolute sum is bounded with weight $e^{\delta |\Gamma|_{\mathbf{B}}}$, the sum after switching at rate δ' is bounded by the same constant times $e^{C_L \varkappa}$ for $\delta' + C_L \varkappa / M \le \delta$. This follows termwise from the distinct-tag count. Replacing old tags by their parent set can only decrease the number of factors on this common polydisc. The prefactor $e^{C_L \varkappa}$ changes the fixed source and current ceilings in the contraction. Reduce these ceilings before choosing the small-coupling range.

Use an auxiliary standard partition at the top of the native tower to check the torus floors for the unswitched walk setting. On the actual graded grid attach to each native term the product over all closed graded cells meeting its padded hull. For a test of a native kernel, apply the graded count to each tree joining the two test sites and visiting the trail, including the initial hull as above. If the native test has rate ϑ, this gives

$$|s^{\omega}| e^{-\vartheta d_{\mathbf{B}}(\omega; y, y')} \le e^{C_L \varkappa} e^{-(\vartheta - C_L \varkappa / M) d_{\mathbf{B}}(y, y')}.$$

Here the distance in the exponent on the left is the infimum over the test trees. The native sum over terms starting in the test window gives the same estimate for every partial sum with the native window and mass constants. Use this for $\mathsf{H}_0, \tilde{\mathsf{G}}, \mathsf{H}, \hat{\mathsf{S}}$ and, when used for the potential read, the walk datum of $D(1 - \mathsf{R}) D^{\mathsf{T}}$. Take M large so that the rate loss fits inside the native spare decay in each test. The layer-ratio estimate and the weighted count of bonds over each site then give the same operator types for the four maps of the reduced system, including the transpose of H, with a larger constant. The contraction and its weighted source estimate use these operator types and the unchanged local structure. This gives the switched response on the smaller fixed source and current discs, with holomorphy by uniform convergence of the fixed-point iteration.

For a face of the switch polydisc, keep the switches in the allowed cells complex and put the others at zero. Define wall adjacency by a common face of positive three-dimensional measure, including a subface at a change of grade. At any point of the padded hull the closed cells containing that point form a wall-connected set. Indeed the small coordinate orthants at the point belong to these cells, and consecutive orthants across a coordinate plane give either the same cell or a common wall. The union of the cells containing the point contains a neighborhood of the point. Since the hull is connected, its full tag set is wall connected: different wall components of the tag set would partition the hull into disjoint closed sets, since all tags containing any given point belong to one component.

A term survives on the face only when all its tags lie in a single allowed wall component. The native margin puts the four-neighborhood of each kernel site inside the padded hull. Thus, if $\mathbb{Y}_{\mathfrak{k}}$ is the set of sites whose four-neighborhood cubes lie in the interior of the union of $\mathfrak{k}$, the surviving terms and their transposes are carried by $\mathbb{Y}_{\mathfrak{k}}$. The interiors for distinct components are disjoint. In particular the three-neighborhood of $\mathbb{Y}_{\mathfrak{k}}$ misses every other carrier set. Apply the splitting of the reduced system over separated carriers. The potential and the supported current bracket split. In the multiplier the diagonal term $-\mathsf{a} B'$ is kept separately, as in the splitting identity for the bracket. Use the ambient row types and true-boundary stencils throughout. In the native exterior collar the two parent blocks of a completed level-j row are in the old layer j, at distance $O(L \xi_j)$ from the row. Include both parent blocks and the pivot in the source read. A lifted row with an additional parent outside the raw margin uses the connected carrier of that parent stencil as well. Gate the corresponding source contribution by the tags of its complete stencil. The extra switches cost a fixed factor in the source bound, and at full switches the source is the original completed source. At full switches the native terms sum to the original four operators and the full response is identified by uniqueness.

To compare two placements of a cut on the same torus, use a common union R of the graded cells in native settings with the same scales and walk parameters. Take the layer traces $\Omega_n \cap R$ on the common finest native grid. Choose the graded cell widths on the nested grids of the native setting. Each graded cell, each layer member and each padded native hull is then a union on the

finest grid. Equality of the traces on R therefore gives equality on each padded hull contained there. For a surviving atom the identity on its hull is then a native shift. Locality in the sequence identifies its index and coefficient at matching pair data on that hull. Repeat the comparison from the other setting, so the sums have exactly the same indices. The common graded grid gives the same tags to the matched hulls. For a marked conditional walk include its conditioning set in the trace and use the same compression mask on the matched endpoints.

For the reduced equations the same comparison identifies the tops, levels, row types and outward endpoints throughout the padded read. A site or a row is assigned by the layer memberships of its sites, with both ends covered by the margin. Thus the same actual averaging blocks, pivot reads and retained plaquettes enter the local structure. Compare the pair and the independent current on these stencils, including the true first-member plaquette lists. The reduced source agrees when the plain source agrees on the full stencils of its naturalization and completion. These are the data needed to apply uniqueness to the component equations on R. The comparison is an identity with the switches in R still complex and those outside R set to zero.

In a preparatory integration the changed member at level $q = j + 1$ removes the released fine rows from the level-j determining set and adds the coarse target rows. A retained coarse row on the old face can change from outward to interior. Both versions of such a row may carry the same grade-q tag. The outward equation reads the seam factor of its full outer block, even at zero plain increment on that row. The interior equation uses its plain increment. At the changed seam form each response in its own native setting. In the old problem the new coarse value is the fixed target of the pivot equation, and the completed fine rows supply the old plain source. In the prospective problem use its actual new row set. In the ordinary centre construction the restriction to the fixed join uses the induced ordinary datum and its own action. It follows from the critical restriction on that join, together with the endpoint restoration above. The ordinary depth-q operators and the full-depth collar operators retain their own native settings.

Along a fixed sequence of stages the regions grow and the base level increases. An old graded cell lies in a current cell because the current grade there is at least its old grade and all grids are nested. For an inherited walk take the parent image of its actual tag set, using its original boundary ownership. On a binary cut the old tags are all active exactly when all their parents are active. The full-switch coefficient is unchanged. For a walk formed with a closed-hull tag rule the same image is the set of current cells meeting that hull, since each is the union of its old cells. Apply this identity with the original hull. Additional endpoint or response stencils are filed as new attachments. The new monomial uses each graded cell once. The per-slot switches of the carried fields keep their own arguments. For the ordinary join, which ends at level $j + 1$, this retagging starts from its admissible standard partition. The full-depth collar Hessian and its roots use their native walk terms on the same active cells.

For the shear $\mathsf{T} = I + \mathsf{S}$, $\mathsf{S}(a, B) = (\mathsf{L}B, 0)$, one has $\mathsf{S}^2 = 0$ and $\mathsf{T}^{-1} = I - \mathsf{S}$. On a full component preserved by this shear the Gaussian determinant is unchanged, and the covariance is the old covariance multiplied by $I-\mathsf{S}$ and its transpose. Weighted row and column bounds for L and the old covariance give the corresponding decay for this product by the triangle inequality for the distance. A compression to a set preserved by the shear is a congruence of the compressed precision. Use $\mathsf{T}z + d$ as the affine reference of the inverse map, with z the new coordinates. Here L is the derivative in B at zero and d is the value of the inverse at zero. Write $r(z)$ for the remaining part. It includes the term linear in the new strip coordinate from its product with the offset, bounded by $Cg_j|a_{\mathrm{new}}(b)||d_b|$. If the old quadratic is $Q(v) = \frac{1}{2}v^{\mathsf{T}}Dv - \ell^{\mathsf{T}}v + c$, then its new precision and linear coefficient are

$$D' = \mathsf{T}^{\mathsf{T}}D\mathsf{T}, \qquad \ell' = \mathsf{T}^{\mathsf{T}}(\ell - Dd).$$

The new constant is $c + \frac{1}{2}d^{\mathsf{T}}Dd - \ell^{\mathsf{T}}d$. The rest of the change in Q is

$$r(z)^{\mathsf{T}}[D(\mathsf{T}z + d) - \ell] + \tfrac{1}{2}r(z)^{\mathsf{T}}Dr(z).$$

Include the sum of the one-slot logarithmic Haar quotients and one inverse-pivot Jacobian per changed coarse row.

Suppose the derivative L has a walk expansion on the same grain as the old precision, with both endpoints in each term's carrier. In $D' = D + \mathsf{S}^{\mathsf{T}}D + D\mathsf{S} + \mathsf{S}^{\mathsf{T}}D\mathsf{S}$, join successive carriers at their common index. Their union is connected, its length is bounded by the sum of the lengths, and the weighted masses multiply. The same joining applies to the quadratic remainder. For $r^{\mathsf{T}}Dr$ the two remainder carriers attach at the two ends of the same old precision term. The weighted row sum therefore bounds the mass per crossing. Use the actual products of switches in both expressions. A shared cell can appear more than once in such a product.

For the derivative carrier expansion, fix the complete endpoint carrier of the crossing and expand $\mathsf{L}_b(s;c)$ in the other switches. To read a coefficient with support Y, set to zero every switch outside Y and the endpoint carrier. The face identity then makes the resulting function independent of any switch separated from the endpoint carrier. It is also independent of the source coordinate if its complete stencil is outside that component. This gives the required vanishing of the corresponding coefficients. On a uniform switch disc of radius R_s, a bound $Ce^{-\mu d(b,c)}$ on the derivative gives, by Cauchy, the absolute sum at $|s| \le r_s < R_s$ over monomials with support exactly Y bounded by

$$Ce^{-\mu d(b,c)} \left(\frac{r_s}{R_s - r_s}\right)^{|Y|}.$$

Indeed each cube of Y has a positive degree, and summing those degrees gives the displayed geometric factor. Choose the raw response disc larger than the disc for the expansion, and use its corresponding source radius. The degree of the graded wall graph is bounded in terms of L, since neighboring grades differ by at most one. A connected set attached to the endpoint carrier is visited by a walk around a spanning tree, of length at most twice its size plus a fixed endpoint term. The resulting exponential count is paid by the switch margin. Choose the margin before M. The same argument applies to r_b, since subtracting the affine reference preserves the cut dependence and the bound of order $g_j(T+d)^2$ on the larger section. For d_b the terms that remain connect the two-block read to the join boundary. Use this same grain for the wider guards.

For the restricted inverse on W, take an intermediate set W_1 of variables containing W with a buffer. Use a guard beyond W_1 on which the switches are one, and the old real reference pair. Set the exterior switches to zero for the reference operators. Write P_W, P_1 for the projections onto W, W_1. The old local-positivity bound supplies a floor on the compression $D_{1,0}$ to W_1. Its congruence by $I + P_1\mathsf{S}_0P_1$ has a uniform floor, since its off-diagonal part is square-zero. To compare its W-compression with $A_{W,0} = P_W\mathsf{T}_0^{\mathsf{T}}D_0\mathsf{T}_0P_W$ on W, use

$$\mathsf{T}_0P_W = P_1\mathsf{T}_0P_W + (I - P_1)\mathsf{S}_0P_W.$$

The difference of the two compressions contains the last term at least once. Its norm is exponentially small in the buffer by the weighted bounds. This gives the real floor on W for a sufficient buffer, also when W cuts across the shear.

At the same reference pair, a term in the complex cut perturbation must reach from W outside the guard. The joined carriers, including those of S, bound this perturbation by $Ce^{-\nu d_{\mathrm{cell}}(W,G^c)}$ in the weighted row and column norm, with G the guard. The full perturbation E_W adds the change from the admitted real pair to the actual pair on this guard. At fixed switches the change of $\mathsf{T}^{\mathsf{T}}D\mathsf{T}$ is bounded by $C(\|\Delta D\|_\nu + \|\Delta\mathsf{S}\|_\nu)$, where $\|\cdot\|_\nu$ denotes the maximum of the weighted row and column sums. Expand the difference in the three factors and use the uniform norms on each section. The real floor and the exponential kernel bound give the weighted inverse of $A_{W,0}$ at a reduced rate by the local-resolvent estimate. Use

$$(A_{W,0} + E_W)^{-1} = \sum_{q\ge 0}(-A_{W,0}^{-1}E_W)^q A_{W,0}^{-1}$$

when the weighted norm of the product is less than one.

Use the old real Gaussian reference, namely the pair $(U, 0)$ with the same real factor U. On the joined response carrier use the same true-boundary prescription there, so at an incomplete stencil its current is the restricted variation at U. Compare the shear on the section with coordinates $q = (A', \mathbf{J})$, with norm $\|q\|_*$ the supremum of the layer-scaled potential, its first covariant difference and the independent current. The gauges on the complete joined cubes depend only on U. Fix their ceiling and a fixed upper bound for the real curvature with room in the raw response class. On a small fixed ball $\|q\|_* < r_*$, the plaquette perturbation estimate bounds the change of scaled curvature by $C_L(r_* + r_*^2)$, using adjacent-layer units on the full stencils. The independent current and the restricted variation on the true boundary fit the raw current radius by the same choice of ceilings. Reduce r_* to leave room also for the source conversion and the endpoint logarithm on the fixed physical collar disc.

On this section, for $\|q\|_* < r_*/2$, apply Cauchy's estimate to the line $(U, zA', z\mathbf{J})$ at fixed switches. The uniform weighted bound for the shear coefficients on $|z| < r_*/\|q\|_*$ gives

$$|\mathsf{L}_{bc}(U, A', \mathbf{J}) - \mathsf{L}_{bc}(U, 0, 0)| \le C\,\frac{\|q\|_*}{r_*} e^{-\mu d(y_b, y_c)}.$$

Summing at a smaller weight gives the required $\Delta\mathsf{S}$ bound. The old Gaussian comparison supplies ΔD on its own domain. The running radii fit inside this fixed response ball at weak coupling. The targets along this line use the averages, completed rows and parent lifts above. At $z = 0$ these give real targets in the same fixed frame. The old spectator occurrences keep their values.

Transport the old admitted decoupled covariance factor $R_0(\hat{s})$ by $R_1(\hat{s}) = (I - \mathsf{S}(\hat{s}))R_0(\hat{s})$, using the same guarded switches as in the precision. For the compression to W write A_W for the transformed precision on W and B_W for its block from the complement into W. Take $C_W = A_W^{-1}$ and $\Gamma_W = B_W R_1$, extending B_W by zero on W. The product has the same carrier joining and guard comparison. At full evaluation $R_1 R_1^{\mathsf{T}} = (D')^{-1}$, and block inversion gives its marginal on W as

$$C_W + C_W \Gamma_W \Gamma_W^{\mathsf{T}} C_W.$$

The full normalizer uses $\det D' = \det D$.

For the cut identities choose the guard as a union of fixed guards G_b of individual active slots. The connected cell set G_b contains every cell meeting the closed owner cell enlarged by a fixed fraction of its side divided by L. Add the prescribed endpoint and parent cells. For two cells with touching closures the grades differ by at most one. Subdivision on the finer of those grids shows that only $O_L(1)$ cells meet this enlargement. With M larger than the fixed resolvent patch width, it contains the local buffers of every cutoff that is nonzero at b. All masks come from the ambient stage. For a set W of integrated slots put $G(W) = \bigcup_{b \in W} G_b$. If Z contains $G(W)$, every G_b lies in the wall-component of Z containing its owner. Thus, for each such component Z_α, with W_α the slots of W owned there,

$$G(W_\alpha) = G(W) \cap Z_\alpha.$$

Use the same ambient lattice partition and the original slot mask in each component. Every surviving precision and root atom retains its full parent and patch reads. Its indices therefore agree with the indices in the component restriction. In particular A_W splits into the blocks A_{W_α} at a binary cut, and its inverse is taken block by block. The same split for $B_W R_1$ uses independent noise slots in the distinct components.

With $F = \Gamma_W$, use the normalized joint Gaussian in the active variable x and standard auxiliary noise y with quadratic exponent

$$-\tfrac{1}{2}(x - m - C_W F y)^{\mathsf{T}} A_W (x - m - C_W F y) - \tfrac{1}{2} y^{\mathsf{T}} y.$$

With n_x, n_y the real dimensions of the two vectors, the density factor is

$$(2\pi)^{-(n_x+n_y)/2} \exp(\tfrac{1}{2} \operatorname{Tr} \log A_W).$$

Both factors multiply over the components. Unused noise integrates to one. Here $\log A_W$ is the matrix logarithm of the actual switched compression, taken on the branch fixed by its positive real reference. The guarded perturbation is smaller in norm than the real floor, so A_W stays strictly accretive on the cut polydisc. Its logarithm there is holomorphic and agrees with the blockwise logarithm on a binary cut. This normalization uses the inverse A_W^{-1} and its own determinant. The independently switched full root supplies $F = B_W R_1$. Away from full switches the definition only uses that factor and its carrier bounds. Integrate out y to obtain the normalized cut measure $\mathcal{J}(W; s)$. At full switches it is the actual Gaussian marginal, by the identity for R_1.

For a cell set $\mathcal{K}$ outside $G(W)$, take the alternating sum

$$\nu(W, \mathcal{K}) = \sum_{\sigma \subset \mathcal{K}} (-1)^{|\mathcal{K} \setminus \sigma|} \mathcal{J}(W; \mathbf{1}_{G(W) \cup \sigma}).$$

Summing over $\mathcal{K}$ gives the full marginal by inclusion and exclusion. For fixed $\mathcal{K}$, the preceding guard equality factors the sum over wall-components of $G(W) \cup \mathcal{K}$. These are the two cut identities used by the gas. The determinant in $\mathcal{J}$ is a conditional normalization. The direct normalizer of the full stage is filed separately. The root identity with the inverse precision is used at full switches.

For the Gaussian cost on a graded cell Δ use its physical side ℓ_Δ and put $q = j + 1$. The ordinary kept form and the centre response of the strip use towers through level q. The new collar form uses the full prospective stage sequence. Its walks start at active slots in Z_q, with a fixed endpoint enlargement. Inside a quiet component the old Ω_{q+1} is separated from Z_q by at least ca_q, where $a_q = M \check{R}_q L^q$. This member is unchanged by the present stage. For an earlier stage the active slots are also far inside X, clear of the top joining collar. At the last stage the top level itself is q. Thus, until a walk has traveled a distance ca_q from an active slot in an earlier stage, it cannot use a layer with spacing greater than L^q. If it reaches another treated box, use the same estimate up to the first exit from the inset of its own box.

Put the guard at one on the cells touching the cells of the active set and include the fixed carrier reach of the chosen switched root. Use the local collar root below on the full stage tower and form the kept-strip root on its ordinary level-q grid. Take the parent image of the strip's fixed guard in the common cells. Adjacent grades differ by at most one, so around each active cell this guard covers a physical neighborhood of width at least $c\ell_\Delta / L$. Reserve the fixed one-step carrier reach inside this neighborhood. For q below the last stage the next original member stays unchanged on X. The active slots are in the carried G_q, which has the quiet gap of at least $8a_{q+1}$ to that member. Since $a_{q+1} \geq a_q$, this gives the upper clearance through the compressed stages. At the last stage there is no higher member. On the lower side the prepared separation gives room ML^q for the local buffers. For a patch crossing the exterior of the treated box use the separation of the prospective member at level q from the complement of the member at level $q - 1$.

Use the padded hulls in the escape estimate. A native atom whose constituent cells are all of levels at most q stays within $C_L \mathfrak{M} L^q$ of the tree point that meets it. If it has a constituent of higher level, follow its connected cell family to the first such constituent. The bounded family of earlier cells puts the tree point within $C_L \mathfrak{M} L^q$ of the higher member. The initial cell is anchored to the active index and has the same carrier bound. After allowing these fixed reaches, an off-guard term must already have paid weighted length

$$d_\Delta = c_L L^{-q} \min\{\ell_\Delta, a_q\}.$$

Use the native weighted lengths for the absolute walk sums. The displayed bound extracts an exponential for a term which leaves the guard, with room left for the row and column estimates. The number n_Δ of actual active slots is at most $C_L L^{-4q} \min\{\ell_\Delta, a_q\}^4$: use the volume of the cell and, separately, the size of the level-q quiet box. In the native stage grid the active slots and their local guards are at base grade q. If a later stage retags the formed term, retain the estimate at

the time of its formation. For each cell with active slots, $u = L^{-q}\min\{\ell_\Delta, a_q\} \geq M$. Consequently $n_\Delta e^{-cd_\Delta} \leq C_L u^4 e^{-c'u}$ is uniformly small once M is large.

For the shear, compare the response at s with the response at s_G, obtained by setting the switches outside the guard to zero. Keep the same ambient pair and stencils. Fix an active anchor b in Δ and use the weight $w_b(y) = e^{-\eta d_{\mathbf{B}}(y_b,y)}$, with $0 < \eta \leq \theta_0$ in the contraction estimate and η small compared with the physical source-decay rate. In each difference of primitive operators, a surviving summand visits a tag outside the guard. Join the test tree for a summand read at y to a path from y_b to y. The first native cell meets the test window at y, so its padding is covered by the same initial allowance. The first-escape bound applies to the joined tree with its padded carrier. The weight pays for the path to y; a fixed part of the native rate pays for the rest of the escape. The remaining rate gives the ordinary site sums. Thus for each primitive map O and bounded input f, in the raw scaled norms,

$$\|(\mathsf{O}(s) - \mathsf{O}(s_G))f\|_{w_b} \leq Ce^{-cd_\Delta}\|f\|_\infty.$$

The source prepared before switching is the same in both equations. If a source contribution is gated, its change has the same weighted bound, since its complete stencil meets the exterior. The change in the initial multiplier $\lambda^0 = \mathsf{KJ}$ satisfies the operator difference bound as well.

Subtract the two reduced fixed-point equations in the coordinates $(T, \lambda - \lambda^0)$. The first difference, between two iterates of the same map, costs at most half their distance in the weighted norm. At the second solution the change of map costs Ce^{-cd_Δ}, by the preceding bound and the local weighted estimate for $\mathsf{C}_T'^{\mathsf{T}}\Delta\lambda^0$. This proves the bound for the solution difference in that norm. The local endpoint read at d can lose the factor $e^{\eta d_{\mathbf{B}}(y_b,y_d)}$. Use Cauchy on the smaller fixed physical source disc to bound the difference of its derivatives. For the shear the rescaling of the input by g_j cancels the division of the output by g_j. Independently, the weighted source estimate bounds either derivative by $Ce^{-\mu d_{\mathbf{B}}(y_d,y_c)}$ for an input at c. The geometric mean of the two bounds gives, before summing,

$$\left|\mathsf{L}_d(s;c) - \mathsf{L}_d(s_G;c)\right| \leq Ce^{-c_0 d_\Delta} w_b(y_d)^{-1/2} e^{-\frac{\mu}{2} d_{\mathbf{B}}(y_d,y_c)}.$$

For the transposed kernel use $w_b(y_d)^{-1/2} \leq w_b(y_c)^{-1/2} e^{\frac{\eta}{2} d_{\mathbf{B}}(y_d,y_c)}$. The estimate holds on the larger switch disc, so the same escape factor can be retained when taking Cauchy coefficients that use an exterior switch. For a product difference, telescope the factors. At the changed factor the preceding bound leaves the inverse weight at its entry site. The triangle inequality bounds this inverse weight by the product of $e^{\eta d_{\mathbf{B}}/2}$ along the preceding legs from the anchor. The unused decay on those legs pays this product, with room for the row and column sums. The inverse root uses the local-resolvent carriers below.

This also gives a comparison away from a changed seam. Use a common R with matching traces and complete source and endpoint stencils as above. In each setting restrict the switches to R; the restricted read at b is the same in the two settings. For each unrestricted read subtract its own restricted read by the weighted estimate. The error is the sum of the two escape bounds, each in its own weighted distance. The derivative comparison uses the same physical input coordinate and the same Cauchy disc on both sides. A carrier that reaches a changed layer, boundary action or completion stencil can instead contribute an unmatched term. In the conditional strip form use the native kept and rejected chains at each evaluation, including the crossing and coupling terms in the interaction. At the collar use the actual pivot and seam curvatures in the native Hessian below.

To see the normalization cost, set $A = C_W^{-1}$ and $F = \Gamma_W$. In the representation with auxiliary standard Gaussian noise the joint precision is

$$K = \begin{pmatrix} A & -F \\ -F^{\mathsf{T}} & I + F^{\mathsf{T}} C_W F \end{pmatrix}, \qquad \det K = \det A.$$

At the real reference it has a uniform floor by the floor of A, the bounded norms, and the triangular change of variables. Let e_b be the sum of the absolute row perturbations of A and F at active slot b. With bounded absolute row and column sums, $C_W - C_{W,0} = -C_W(A - A_0)C_{W,0}$ and the three-factor expansion of $F^{\mathsf{T}}C_W F - F_0^{\mathsf{T}}C_{W,0}F_0$ bound the entrywise sum of $K - K_0$ by $C\sum_b e_b$. This also bounds its trace norm. The uniform operator perturbation is small on the admitted section. Let P be the projection onto the x coordinates. After multiplying the absolute density by $e^{\gamma_2|x|^2}$, the real precision is $\operatorname{Re} K - 2\gamma_2 P$. In the centered case the logarithm of its total mass is

$$\tfrac{1}{2}\operatorname{Re}\operatorname{Tr}\log A - \tfrac{1}{2}\log\det(\operatorname{Re} K - 2\gamma_2 P).$$

At the reference the two determinants agree. For a positive reference H_0 of uniform floor and a small perturbation E, the series for $\operatorname{Tr}\log(I + H_0^{-1}E)$ is bounded in absolute value by $C\|E\|_1$, where $\|\cdot\|_1$ is the trace norm. Apply this to A and to the real precision. The projection has trace n_x, so the tilt cost is at most $C\gamma_2 n_x$. For complex m first translate the real x by $\operatorname{Re} m$. In the real exponent the imaginary mean then contributes a constant bounded by $C\|\operatorname{Im} m\|^2$ and a real linear term with norm bounded by $C\|\operatorname{Im} m\|$. The tilt contributes the constant $\gamma_2\|\operatorname{Re} m\|^2$ and a linear term of norm $2\gamma_2\|\operatorname{Re} m\|$. Completing the square using the uniform floor gives the additional cost $C\|\operatorname{Im} m\|^2 + C\gamma_2\|\operatorname{Re} m\|^2$. The offset on the physical section is exponentially small. A free source that moves the mean can be restricted to a disc on which $N^*\|m\|_\infty^2$ is small, leaving room for the physical shift.

The guard and pair comparisons give $e_b \le C(e^{-cd_\Delta} + \epsilon_{\mathrm{pair}})$ for b in Δ, with the switch loss included in the constant. The running pair radii give $N^*\epsilon_{\mathrm{pair}} = o(1)$ on the joint Gaussian section. Use the reference coupling of the current scale in this Gaussian. The difference from the position-dependent Wilson weight is in the interaction. In particular the separate disc for ratios in the incoming terms does not enter e_b. If the Gaussian itself is rescaled by a ratio τ, the same argument charges $Cn_\Delta|\tau - 1|$ in each occupied cell, so that option requires the corresponding occupancy bound on its radius. Take γ_2 a fixed small multiple of $1/N^*$. After integrating out y, the tilted total variation of the cut marginal is bounded by the tilted absolute mass of the joint density. Thus in any cut component the logarithm of this bound is at most

$$C\sum_\Delta n_\Delta\big(e^{-cd_\Delta} + \epsilon_{\mathrm{pair}} + \gamma_2 + \|\operatorname{Im} m\|_{\infty,\Delta}^2 + \gamma_2\|\operatorname{Re} m\|_{\infty,\Delta}^2\big),$$

where the sum is over its occupied cells. Each summand is bounded, and a hot coordinate at scale $(\log\theta_j)^{\mathfrak{p}_1}$ gains $\exp[-c(\log\theta_j)^{2\mathfrak{p}_1-4r}]$. This pays the stage bound $\exp(C(\log\theta_j)^{4r})$ because $2\mathfrak{p}_1 > 8r$. The determinant of the full active reference still uses the old stage normalizer. These bounds use the native response and local-resolvent families in the fixed ambient stage. For a change of ambient tower the corresponding completed rows and native carriers must agree on the component where the comparison is made.

In the complete stage integral the reference is the sum of the homogeneous retained strip quadratic and the homogeneous collar quadratic. The collar linear term in the constant-weight Wilson part vanishes by stationarity on the fixed averaging fibre. The remaining interactions, including cross-component strip reads and changes of the background or coupling, are in the potentials. Unintegrated near fields occur there with their original real values and switches. Thus $\ell = 0$ for this reference on the complete integration vector. The new mean is $-\mathsf{T}^{-1}d = -d$, since d has only strip components. On a guarded read take the continuation $-P_W d(\hat{s})$. It has the same carrier as the offset and fits the smaller sharp threshold on the crossing slots. At full switches it is the mean of the full Gaussian restricted to W. The covariance formula accounts for the complementary coordinates. The real sharp tests still act in the specified fixed integration charts. All Taylor terms of the potentials, including any linear terms, remain in the interaction.

On the admitted joint switched section write the exact active inverse as $\mathsf{T}z + d + r(z)$. With T bounding all arguments on their actual enlarged discs, the nonlinear Gaussian correction is

$$-r(z)^{\mathsf{T}} D(\mathsf{T}z + d) - \tfrac{1}{2} r(z)^{\mathsf{T}} D r(z).$$

The product-logarithm and fixed-radius source estimates give $\|r\|_\infty \le C g_j T^2$. With the same endpoint gates for the exact and affine coordinates, r_b vanishes if either endpoint cell is inactive, and a surviving cut coefficient reads only the wall component of those cells. Cauchy's bound on the switched disc sums these coefficients with an exponential weight on their attached cells. The weighted row sums of D and T then bound the displayed terms, filed at a changed strip slot, by $C(g_j T^3 + g_j^2 T^4)$ in the weighted sum. For the term with two remainders use the row bound at one of its slots and join the other slot's carrier along the same matrix chain. This requires only one sum over the changed anchors. The Haar and pivot terms have the rooted bounds just proved. For the norm pinned at any cell of a joined carrier, keep a portion of the native spatial switch and walk decay. The distance from that cell to the changed anchor is at most the number of cells in the connected carrier, with a fixed enlargement for the endpoint blocks. Use the reserved decay to sum over anchor cells, whose face graph has bounded degree. Multiplication by the slot count of an owner cube, or by the logarithmic slot count over the component, is harmless since $g_j T^3$ is smaller than any fixed inverse logarithmic power.

For a mixed strip term with a fixed exterior slot, substitute only its active argument and keep the original switch on any near exterior argument. Its affine change involves $\mathsf{S}z + d$, so estimate it using the old chain bound. In the physical stage the selected strip masks are whole kept components. The kept form therefore has no mixed block against the fixed exterior components. Rejected chains and the ratio term are still boundary interactions. Use their formation rule chain by chain. Under the original local keeping predicate, rejection costs a cell of travel. This gives $e^{-c\delta_\star W_t}$ on the enlarged strip box before the remaining carrier decay. The ratio term uses kept chains, possibly short ones, with a factor $|1 - \tau_k|$. Both estimates absorb the two enlarged field radii. A Gaussian cut inside a kept component uses the full marginal formula for the omitted variables.

The stage bounds allocate separate margins for the pointwise reads, the tower reads and the core. The independent current increment uses the pointwise margin. The tower current, formed by averaging and solving the auxiliary problem for the substituted configuration, has its own estimate. On the graded interior windows, Prop. 3.138 and Cor. 3.140 apply to settled elements and control the changes by the weighted potential and its first difference. For an address read to depth p, its nonempty level-$(p-1)$ localization cube forces at least $M/L \ge L^3 \rho \ge 6\rho$ sites per torus period at that depth. This supplies the period floor in the ambient estimate. The full current stencil and the complete row reads remain those of that fixed address.

Here is the budget on a fixed inherited address. Write $\mu(x)$ for the layer of the preceding response sequence and $\ell(x) \le \mu(x)$ for the layer of the earlier address, truncated at its reading depth when required. Put $q = j + 1$, and write $D_n(x \mid l) = |l - q|$ inside a treated component and $(t - q) + (t - l)$ outside the treated components of this stage of level t. The decrease in the coefficient of the radius at a fixed index l is

$$\frac{\mathsf{g}_t(l)}{2\mathsf{N}_t} + \beta_{\mathrm{R}} 2^{-|l-q|} \mathbf{1}_{\mathcal{Z}_n}(x) \;\ge\; \gamma_\star 2^{-D_n(x|l)}, \qquad \gamma_\star = \frac{\theta_{\mathrm{s}} \beta_{\mathrm{s}}}{2\mathsf{N}_t}.$$

For a fixed tower address its left-hand read and its lift are unchanged. Multiply this coefficient gap by $L^{i-r} L_\flat^{2e_{\sigma''}(x)} \alpha_{0,\ell(x)}^{\sigma'' | \sigma_n}$ for the slack in normalized tower units. Where the layer changes from j to q, the ratio of the two scale weights gives a fixed fractional margin, including the lift gained there. The stage transition uses the same unfinished top radius.

At an inherited core the restriction also keeps the left-hand read and its radius number. The coefficient in the restricted bound is the later coefficient at the core's index. The gap at that index supplies the core margin throughout the original domain, even outside the treated component. In

the room estimate let $\mathsf{v}(b)$ be the pointwise gap profile in the input length $\mathsf{L}(b)$. For a core of scale ξ_i the shift uses the profile $\xi_i \mathsf{v}(b)/\mathsf{L}(b)$. The layered profile comparison bounds it by a fixed multiple of the core margin, with the core's radius profile included. This is the core condition in the same room estimate that handles the pointwise and tower clauses. Thus the stage amplitude bound also gives admission to the inherited graded cores.

The weighted path from the changed zone to x must traverse the intervening layers. On leaving a treated component it first reaches the original top layer t, since the quiet exterior of level t is inset. Thus, with $d^-(x, \Sigma')$ the preceding weighted distance, $D_n(x \mid \mu(x))$ is at most a fixed constant plus $d^-(x, \Sigma')/\varrho_0$. Here ϱ_0 is the layer traversal floor. The pinned response, in its own normalized units, has profile bounded by

$$s_1(x) = C g_j p_0(g_j) |\lambda_{\mathfrak{w}}| |B|_{\mathfrak{w}} e^{-\delta_1 d^-(x,\Sigma')}.$$

The constant includes the turn and switch costs, and $\delta_1 \le \delta_{\mathbf{H}}/2$. At the earlier address the shift and its first difference are bounded by $L^{-(\mu-\ell)} s_1$ in that address's units. The pointwise plaquette and current comparisons have at least this scale gain.

The settled tower estimate convolves this last bound with the tower decay in the earlier ambient distance d^{a}. One has $d^{\mathrm{a}} \ge d^-$. The difference of the layer losses $\mu - \ell$ at two points is at most $2 + 2d^{\mathrm{a}}/\varrho_0$, by counting crossings in each sequence. It follows that this convolution, in the form of the weighted supremum in the tower estimate, is at most a constant times $L^{-(\mu(x)-\ell(x))} s_1(x)$ when $\delta_1 + 2 \log L/\varrho_0 \le \delta_{\mathrm{T}}$. The required joint tower-class admission uses the same small shift on the full localization as in the estimate below.

Divide by the gap at $\ell(x)$. The bound

$$D_n(x \mid \ell(x)) \le D_n(x \mid \mu(x)) + \mu(x) - \ell(x)$$

leaves the factor $(2/L)^{\mu(x)-\ell(x)}$. It pays the part of the adjacent-coupling comparison coming from lowering the index to $\ell(x)$. In the running radius $C_\iota \theta_l^{-1/2} (\log \theta_l)^{q_{\mathrm{rad}}}$ the function of θ_l decreases at weak coupling. For an earlier index the logarithmic factor in the radius is at least as large, and the bounded drift of θ_l costs at most the square root of one plus the level difference. For the remaining comparison the bounded running increments cost at most $C(1 + \sqrt{|j - \mu(x)|})$. The factor $e^{-\delta_1 d^-/2}$ pays this factor and the factor $2^{D_n(x|\mu(x))}$, since $\delta_1 \varrho_0 \ge 2$. This leaves the distance factor at rate $\delta_1/2$.

For an inherited domain X, write $\mathsf{d}_n(X)$ for its distance in cubes from the changed layer and $|B|_{\mathfrak{w}}$ for the supremum on component $\mathfrak{w}$ divided by the box radius. The stage profile budgets the increment by

$$\mathsf{a}_n^\delta |\lambda_{\mathfrak{w}}| |B|_{\mathfrak{w}} < e^{\delta_\star \mathsf{d}_n(X)}, \qquad \mathsf{a}_n^\delta = O\big((\log \theta_j)^{-(q_{\mathrm{rad}} - \mathfrak{p}_0 - r)}\big).$$

Here q_{rad} is the exponent of the running radii. The power comes from $\gamma_\star^{-1} = O(\mathsf{N}_t)$, $\mathsf{N}_t = O((\log \theta_j)^r)$, and the ratio $g_j p_0(g_j)/\alpha_{0,j}$. The physical values $\lambda_{\mathfrak{w}} = 1$ satisfy this inequality for small coupling. For the common substitution take the intersection of these discs over the actual readers. The uniform disc obtained at $\mathsf{d}_n = 0$ is sufficient for joint admission. On the readerwise larger discs the same common formula is restricted to that reader. The fixed analytic disc of the ordinary joined centre is used for its own unamplified source. The stage discs are also limited by $|\lambda_{\mathfrak{w}}| e^{\varkappa} < c_{L,\varkappa} \varrho_{\mathrm{st}}$, where $|s| < e^{\varkappa}$ and ϱ_{st} is the strip radius in units of its field threshold. The fixed factor $c_{L,\varkappa}$ leaves the margin for the inverse strip arguments. The physical value remains interior for small coupling.

The real layer allowance $e_l = \check{R}_l^2 \varepsilon_l$ is used in admitting the base pair. The largest compressed conversion for current or local gauges costs $O(\check{R}_l^3)$. Thus it requires the $\mathfrak{p}_0 + 5r$ logarithmic power of the real endpoint estimate. The stage floor $q_{\mathrm{rad}} \ge \mathfrak{p}_0 + 7r + 1$ leaves room also for a sum over the N_i stages. The response amplitude in the displayed profile inequality is determined by the free collar box and the running radius. It uses the actual collar increment, rather than the bound for the base pair.

4.17.1.33. Domains of the earlier entries. Use the graded formation rule from the bare density onward. A tower address consists of an earlier structure σ'', a localization Z of level i, and a depth $1 \le \mathsf{p} \le i$. At level zero the direct bare reads use the pointwise clause. The tower of an earlier address is the meet of the localization tower with the sequence of σ'', through depth p. For $\mathsf{p} \le r \le i$ remove two layers of level-$(r-1)$ M-cubes from Z to form $Z^{\wr r}$. Read a plaquette or a current stencil when all its corners lie in this window. Its base point then lies in the layer of the tower of index $\min\{\mathsf{p}, \ell\}$, where $\ell = \min\{m_{\sigma''}(x), i\}$ is the nominal layer. Indeed $Z^{\wr r} \subset Z^{\wr \mathsf{p}}$. The two-layer localization comparison for $M \ge L^3 M_1$ gives agreement of the tower and earlier region memberships through p on that window, where M_1 is the cube size used in the localization. In bare units the curvature and current bounds use the respective factors $L^{-2\ell}$ and $L^{-2\ell-\min\{\mathsf{p},\ell\}}$. Both are multiplied by the layer radius with its lift and ambient coefficient, and by L^{i-r}. The tower itself and the ambient coefficient thus play different roles under restriction.

4.17.1.34. The allowance and the stage gap. The profile hypothesis in Prop. 3.143 follows from the actual radius clauses at an inherited stage address. We give the comparison, including its dependence on the stage. Fix that address, its ambient structure σ'', its level i and its window index r. Put $m(z) = \min\{m_{\sigma''}(z), i\}$. This is the nominal ambient layer, denoted by ℓ above; the native settled tower index is $\min\{\mathsf{p}, m(z)\}$. The clause radius and its stage gap are indexed by the nominal layer. Abbreviate by α_m the radius $\alpha_{0,m}^{\sigma''|\sigma_n}$ selected by the bare-level, unfinished-top and adjacent-index conventions. Let $e(z) = e_{\sigma''}(z)$. Choose a fixed $C_\alpha \ge 1$ large enough to include the ratio of the potential radius to the curvature radius, the adjacent-index comparisons and the change of base point within a complete stencil. Here we use the actual running radius system of the stage formation, whose components have the same logarithmic power, or a fixed absolute system. The ratio of its two components is fixed in either case; taking the minimum of neighboring running radii preserves that ratio. Thus C_α is independent of the stage, coupling and depth. Define

$$B(z) = (1 + \beta_{\mathrm{s}}) C_\alpha L_\flat^{2e(z)} \alpha_{m(z)}. \tag{4.cdxlvi}$$

The pointwise clauses on this inherited address give $A(z) \le B(z)$, with A as in eq. 3.mcxxxi. The independent current does not enter A. At a real stage endpoint the stronger tested allowance $\check{R}_l^2 \varepsilon_l$, including its compressed-scale conversion, also fits this bound by the endpoint calculation.

Fix $0 < \kappa < \delta/4$, where δ is a common rate in Lem. 3.142 and Prop. 3.143. Decrease the common stage rate δ_0 if necessary so that $0 < \delta_0 \le \delta$. Set

$$R_\kappa(x) = \max_z e^{-\kappa d^\flat(x,z)} B(z). \tag{4.cdxlvii}$$

This is a deterministic function of the structure and its radii. The triangle inequality gives $R_\kappa(z) \le e^{\kappa d^\flat(x,z)} R_\kappa(x)$, and $A \le B \le R_\kappa$. Thus eq. 3.mcxxxv holds with $C_0 = C_1 = 1$ and $\mathcal{R} = R_\kappa$, without an additional profile assumption.

For the stage profiles one also has

$$R_\kappa(x) \le C_B B(x), \tag{4.cdxlviii}$$

where C_B is fixed before the coupling ceiling. Here is the layer comparison that proves this assertion. Let $d = d^\flat(x,z)$ and let ϱ_0 be the common traversal floor of the inherited determining sequence and its original regularity ancestor. The latter sequence is coarser; truncating the inherited sequence at i only increases its metric. Counting the intervening separated layers in each sequence gives

$$|m(z) - m(x)| \le 2 + 2d/\varrho_0, \qquad |e(z) - e(x)| \le 4 + 4d/\varrho_0.$$

Indeed a path between layers differing by h crosses at least $h-2$ complete intervening collars. Apply this to the two layer functions whose difference is e. The endpoint conventions for the radius move its selected index by at most one at either end.

For the running function $\theta^{-1/2}(\log\theta)^{q_{\mathrm{rad}}}$, choose the coupling ceiling so that $\log\theta > 2q_{\mathrm{rad}}$. This function decreases with θ. The frame bounds eq. 2.xx then give, for radius indices a, b, $\alpha_a/\alpha_b \le$

$C(1+\sqrt{|a-b|})$. To check this in the direction in which the radius grows, use $\theta_b \le \theta_a + \bar{B}|a-b| + c_5$; the square-root factor bounds the power of θ, while the logarithmic factor decreases. The opposite direction costs only the fixed correction c_5 at the weak-coupling ceiling. The minimum of neighboring radii obeys the same estimate. Consequently

$$\frac{B(z)}{B(x)} \le CL_\flat^8(1+\sqrt{4+2d/\varrho_0})e^{8(\log L_\flat)d/\varrho_0} \le C_B e^{(8\log L_\flat+1)d/\varrho_0}.$$

Choose the fixed geometric floor so that $(8\log L_\flat+1)/\varrho_0 \le \kappa/2$. Multiplication by $e^{-\kappa d}$ and maximization over z prove eq. 4.cdxlviii. This choice enlarges only the fixed geometric constants.

Write $\mathsf{gp}_n(x,m)$ for the deterministic margin between the two stage bounds at the read. The coefficient decrease computed above gives, at a fixed inherited index,

$$\mathsf{gp}_n(x,m) \ge \gamma_\star 2^{-D_n(x|m)} L^{i-r} L_\flat^{2e(x)} \alpha_m, \qquad \gamma_\star = \frac{\theta_{\mathrm{s}}\beta_{\mathrm{s}}}{2\mathsf{N}_t}. \tag{4.cdxlix}$$

Define the number used in the cut estimate by

$$a_{\star,n,x} = \min\left\{1, \frac{\gamma_\star 2^{-D_n(x|m)}}{C_B(1+\beta_{\mathrm{s}})C_\alpha}\right\}. \tag{4.cdl}$$

It is positive and depends only on the address, the stage and the read. Combining eq. 4.cdxlvi, eq. 4.cdxlviii and eq. 4.cdxlix gives

$$R_\kappa(x) \le a_{\star,n,x}^{-1}\mathsf{gp}_n(x,m).$$

The radius and lift cancel, and $L^{i-r} \ge 1$ improves the bound. No lower bound for $a_{\star,n,x}$ independent of the stage and the read is asserted. By Prop. 3.143, a cut satisfying its complete-carrier separation changes either settled read by at most

$$K_{\mathrm{D}} a_{\star,n,x}^{-1} e^{-c\delta_0 D}\mathsf{gp}_n(x,m). \tag{4.cdli}$$

In particular, to spend at most a fraction $\vartheta \in (0,1)$ of that margin it suffices to impose

$$D \ge \max\left\{K_{\mathrm{cl}}M, \frac{1}{c\delta_0}\log\frac{2K_{\mathrm{D}}}{\vartheta a_{\star,n,x}}\right\}. \tag{4.cdlii}$$

The same criterion applies simultaneously to finitely many reads by taking the largest required distance. In particular the dyadic coefficient loss is paid by the displayed distance when such simultaneous margin preservation is required.

4.17.1.35. A geometric cut of a settled tower clause. In the units of the earlier address, cut Z by a union R of level-t cubes and keep the component Z_R of $Z \cap R$ containing the base x of the read. Components here join also at corners. Suppose that x is at distance at least $\mathsf{D}+3M_1$ from $Z \setminus R$, with $\mathsf{D} \ge K_{\mathrm{cl}}M$ as in the localization bound. The cut tower is finer by monotonicity of the layer construction under enlargement of the complement. The complete-carrier argument following Prop. 3.143 locates every changed datum read within $C_{\mathrm{cut}}M_1$ of the actual cut $Z \setminus R$, including the old parent pillars and outward endpoint blocks. It also treats discarded components by first exit through the actual cut; it requires no extra spacing assumption between such components. Increase K_{cl}, once, so that $(C_{\mathrm{cut}}-3)_+M_1 \le \mathsf{D}/2$. Every changed complete carrier is then at distance at least $\mathsf{D}/2$ from the read. The retained r-window, complete stencil, free pillars, nominal layer m and native tower index $\min\{\mathsf{p}, m\}$ are unchanged there.

For the same admitted pair, Prop. 3.143 and eq. 4.cdli bound the change in the magnitude of either settled read by

$$K_{\mathrm{D}} a_{\star,n,x}^{-1} e^{-c\delta_0 \mathsf{D}}\mathsf{gp}_n(x,m).$$

The positive factor $a_{\star,n,x}$ is the explicit number in eq. 4.cdl; it depends on the stage and the read. The bound of the clause depends on the location, indices and structures, so it has the same right-hand side before and after this cut. An available margin larger than the displayed error survives

the restriction. In particular eq. 4.cdlii is sufficient for preserving a prescribed fraction of the stage gap. This proves the row and domain comparison with its stated margin and distance condition. A composed reader uses the stencils and switches of its actual presentation.

4.17.1.36. Shifting the field in a fixed tower. On these windows the tower comparison estimates the recomputed current itself. Write $\mathbf{U} = e^{i\eta A'}U$ and $W^u = e^{i\eta(A'\boxplus\mathbf{A}^\circ)}U$ using one real factor U and the local logarithm of the shifted pair. When both potentials lie in the admitted tower class, Prop. 3.138 and Cor. 3.140 give the normalized current increment

$$L^{2\ell+\min\{\mathsf{p},\ell\}}|J^0_{\mathsf{p},Z}(W^u)(b) - J^0_{\mathsf{p},Z}(\mathbf{U})(b)| \le C\max_{c\in\mathsf{Z}^+} e^{-\delta_{\mathrm{T}}d^\flat(y_b,y_c)}\max\{l(c)|v(c)|, l(c)^2|\nabla_U v|_{\mathsf{Z}}(c)\},$$
$$v = A'\boxplus\mathbf{A}^\circ - A'.$$

Here J^0 is in bare units, and Z^+ consists of the positively oriented inner bare bonds of the localized first member. Each y_c is the earlier ambient pillar top of the base point of c. Use that sequence's distance $d^\flat$ and local length l. The comparison allows the base point of the full stencil within $2\xi_{\mathsf{p}}$ of Z^{lp} when the whole stencil is in the localization's bond set. The agreement of the region memberships on that halo covers stencils that straddle an ambient layer boundary. The same estimate holds for the plaquette increment with weight $L^{2\ell}$.

Use the real factor U of the input on the whole localized first member. The pinning estimate bounds the relative logarithm of $W^u\mathbf{U}^{-1}$ and its covariant first differences with this same real factor. Use the resulting $\mathbf{A}^\circ$ in the product $A'\boxplus\mathbf{A}^\circ$. The small product chart gives a single holomorphic choice on each fixed-real-factor section. In particular the real factor of the recomputed tower stays U.

The ambient lengths of the earlier address are at most the lengths at the input, including at the base points of the full stencils. The pointwise input radii therefore bound $\mathsf{n}_c(A')$ and the normalized curvature of U on the entire first member. This uses the full localization for the radius test, while its inset windows specify the output stencils. Let $\mathsf{u}(c)$ be the maximum of the earlier weighted value and first-difference bounds for the relative shift, and let $\mathsf{u}^+(c)$ include the neighboring bonds of the product estimate. Apply the common radius budget on these complete carriers. The turn cost is fixed in terms of L, M, and the bound on u there is $CL^{-(\mu-\ell)}s_1$. The same bound, with the neighboring maximum, is small uniformly as the coupling ceiling decreases. Thus the base norms can be put below $a_{\mathrm{T}}/2$ and the shift satisfies

$$\mathsf{u}(c) \le (256L^2)^{-1}, \qquad \mathsf{n}_c(A') + 2\mathsf{u}^+(c) < a_{\mathrm{T}}$$

throughout the section. The geometric floor of the tower theorem is the separation floor for its fixed localization, chosen before the coupling ceiling. The shift estimate now gives the recomputed tower on this same mask. Fit the resulting normalized increment in the available window margin, with L^{i-r} at window r, by the preceding budget.

Give each boundary term, potential away from the strip, stage correction and interaction term the structure of its formation. Carriage retains its level, localization domain, function and space. When a stored coordinate is integrated, its occurrence in the prospective evaluation is set to zero and its switch is discarded. Before the first stage the sum of earlier stage corrections is zero. At each subsequent stage keep the original domain and stored attachment of each correction already formed. The old evaluation of the sum uses the levelled triple restricted to each of those domains. The prospective evaluation uses its own triple and sets the strip field being integrated to zero. The difference of the two evaluations enters the new collar integral. Lifting a lower boundary term to a larger support for the expansion leaves its original argument domain in place.

On the common stage profile bound, the levelled substitution sends the stage-$(n+1)$ space into every earlier graded space. The prospective evaluation uses restriction from the output structure. At each slot the formation structure precedes its own level's output structure, which in turn precedes the current output. For a carried slot take the standard reduction of the outgoing history on its

original domain. This reduction has the completed trace of the healed label on the domain's windows, treats no component on its enlarged read, and keeps the recorded drop-outs there. If the enlarged read misses the treated components, one may use the history itself. The traces on the smaller windows agree for all these choices. Restrict the reduction to the old level before taking the term and its space. The first members of its tower clauses are contained in the original domain. Intersecting the same read of the old sequence with the same localization tower gives the same clause on each such first member. Identify the original first member, the determining rows including the one-step completions, and the pairs of tower and window indices before comparing radii. The order of layers and the distance comparison in the profile estimate are comparisons within that common labelled sequence. For a stored region transported to the new label they apply after the address has been matched: on its full read the sequences agree, and the tower, window, lifts and address radii persist. Pointwise descent between two output labels by itself gives only the pointwise part of this comparison. A term whose old hull has been absorbed uses the new filed term, with its own formation domain.

For example, a kept-chain summand of the ordinary ratio correction that reads a far strip bond has the form

$$-\tfrac{1}{2}(\tau_l - 1)\sum_{b,b'} a_b^{\mathsf{T}} D_\omega(P)_{bb'} a_{b'}.$$

Here l is its original fluctuation level and D_ω is the kept chain in the ordinary conditional form. On its full carrier in the original two-cell strip trace, the regular-pair comparison applies to the pinned pair with the original free rows. It uses pointwise potential and current estimates. Its far arguments are the joint strip coordinates bounded above, and its near arguments use the old multipliers. The row estimate on the enlarged field disc is $C|\tau_l - 1|(\varrho_{\mathrm{st}}\mathfrak{b})^2$ times the old carrier decay, within the tube bound. A rejected long chain uses its own pair domain on its complete carrier.

For an inherited cluster or strip-transfer entry that uses an earlier activity, keep the two blocks and parent transports of the strip at its original step, as well as every tower address $(\sigma'', i, Z; \mathfrak{p}, r)$ of its formed argument. For instance, in a boundary cluster $E^\sharp(X)$ a vertex made from an incoming boundary term uses its ordinary potential on Y, with $Y^\sharp \subset X$. Recover the ordinary trace on the one-layer windows of X at each level. It specifies the old cuts, tiers and which input potential has this support. The ordinary fluctuation substitution in that potential uses the determining rows and gauge cubes of $Y^\sharp$. Its fluctuation variables are those of the ordinary formation, distinct from the free variables of the later collar.

At the original step k, a vertex of species V^B is made from localized differences $f(P_{\mathrm{o}}; \mathfrak{q}) - f(P; \mathfrak{q})$ of an incoming boundary term on its original domain, say D. In both evaluations the stored arguments $\mathfrak{q}$ are the same. The ordinary substitution in P_{o} uses the corrected ordinary fluctuation at $\mathcal{M}^k\mathbf{U}$. With the real factor U fixed, the substituted link and the independent current are

$$e^{i\eta A''(b)} U(b) = u(b_-)\, e^{i\eta\lambda_{\mathrm{o}} H(b)} \mathbf{U}(b)\, u(b_+)^{-1},$$
$$\mathbf{J}'' = \mathrm{Ad}(u)(\mathbf{J} + \lambda_{\mathrm{o}}\mathsf{j}).$$

Here u is the ordinary pinned levelling map and j the reduced current increment of this response. The identity of the second line with the physical current holds at the full real arrival. The complex tower clauses use the recomputed tower current of the shifted potential on their own windows.

For admission at this inner substitution, the ordinary step supplies the coefficient gap $c_0\, 2^{-(k+1-n)}$ at the index n of each earlier clause, including when the reference radius is the smaller of two adjacent radii. Use that gap times the clause's radius as the margin in the weighted potential estimate. For a tower clause the comparison of profiles includes the length ratio between the tower and the input sequence, and the exponential in their distance from the stencil. The separation of successive layers pays the geometric gap ratio and the square-root ratio of the running radii. This gives the pointwise, tower and core room for the ordinary response. Write r_D for this room after

the reduction for the pinned map. For an ordinary field box bounded by $\mathsf{C}_{\mathcal{A}}\delta_k$, use the separate amplitude radius given by

$$R_{\mathrm{o},D}^{-1} = \frac{C}{r_D}\,\mathsf{C}_{\mathcal{A}}\frac{\delta_k}{\alpha_*}e^{C\varkappa-\delta_*\mathsf{d}_k(D)}.$$

The bound applies to the whole original graded section, with $R_{\mathrm{o},D} > 2$ on the enlarged ordinary formation box at small coupling. Here one uses its enlarged $\mathsf{C}_{\mathcal{A}}$ in the radius formula. The far stored fields in the vertex retain their original radii.

In the closed cluster the integration over the original ordinary B_I still uses its fixed real cutoff balls and its original conditional Gaussian, with the mean determined by B_E. At a moved crossing it is only the external far slot of B_E that takes the reconstructed coordinate. When the outer substitution lies in the original graded formation section and these external slots lie in its field box, the original absolute majorant of the cluster applies on the whole integration section. The outer stage margin is used for these membership tests. The subsequent ordinary substitution uses the gap and amplitude of its original step, uniformly over the section just reached.

For the first earlier-stage correction that reads this strip, carry the closed first-stage function, including the carriers of its original source closing. The second stage uses the common stage triple on that full formation domain, including its inherited cores. The old evaluation uses the reconstructed far field, and the prospective evaluation sets that integrated field to zero. The original source closing is already part of the carried function. In this correction the far slot is a stored-field argument. Its parent targets are read through the pair.

For a tower-current read use the common real factor and the earlier weighted potential bounds on all of Z for both the base and the pinned input. The six incident plaquettes and the halo of the stencil must be in the graded read of the same address. The new filing hull carries the resulting term while these evaluations use their original tower. In a newly composed operator we must also check the joint read of the retained datum if it enters the strip inverse in a separate role. The one-step forms and elimination factors use their separate pair domains.

4.17.1.37. Carrying the formation reads through a cut. A formed summand has a fixed base read. Keep the part of its original structure, row masks and stored attachments used there. An earlier normalizer or closed cluster term enters once with its formed function. The closed carrier includes the attachments of the earlier source closing. The current substitution acts on this closed function. It changes the pair and the currently integrated variables, on the rooted arms of the current bracket. In particular the zero-increment value at the fixed base has no new spatial switch. A bracket using the old summand keeps its complete base, and its response differences are formed on that base with the new source arms.

An old quadratic or pivot factor instead uses its one-step trace. For the strip Schur form keep the original chain classification into kept and rejected terms. The rejected chains and the coupling correction retain their original filing in the exponent. At each full evaluation their sum with the kept form is the original conditional quadratic. Retag a kept chain by the parent image of its actual old tags, as in the grading argument. Its binary-cut criterion is inherited through that image. A solved row uses the original target and its complete double block, including the prescribed group links. Keep its determinant with that row.

At a stored spectator the base includes its original site, top, and full pillar. In an actual stage structure a stored endpoint at level l is a top or has its parent at level $l+1$. The complete level-$(l+1)$ block is separated from level $l+2$. Its membership at level $l+1$, and hence its pillar and transport, persist while this field remains stored. In a genuinely finer geometric localization the same site may become a top. The transport in that new localization cannot be used for the earlier occurrence just from the depth bound.

Use the fixed stage geometry in the binary cuts of the response. For each original top y keep the complete dependence set of the pin, including the averaged contour, the level-one neighbor margin, and the stencils of its turning cells. Use the padded hull of each response walk, which

contains the blocks of the output tops and of their four successive neighbors. The node factor and contour use internal links of the original block. The turn uses its child blocks. Include the actual two-bare-step stencil at the exceptional level-one pins. There the fixed ambient eligibility condition puts the nearby pillars at levels at most one. For an actual stored field of level zero the full parent block is separated from layer two by the storage margin. The complete pin and turn stencils are over the two-neighborhood of the original top. If a response at an output top z meets this read, the symmetry of the neighborhood puts the whole read in the four-neighborhood of z. This is the neighborhood in the padded hull.

Tag every closed graded cell meeting the padded hull. At each point of a surviving hull the cells that contain that point contain a neighborhood in their union. Their star is connected through positive-area faces by nested grid refinement. The connected hull is therefore inside one active wall component, also at faces between grades. For inherited walks use the parent image of their actual old tags. Pulling back binary switches along the parent map gives exactly the old binary test, even when the deduplicated complex monomial uses fewer factors. Keep the old closed functions as filed. For a new operator without an admitted native walk presentation we must still verify this padded-hull condition on the actual pin and output stencils.

On a binary cut a nonzero increment on a pin carrier then has that whole carrier in one active wall component. Otherwise the increment on the carrier is zero. On the original pillar form

$$p_y(a) = \tau[\mathbf{U}](a)^{-1} v[P;\mathbf{A}](y)^{-1} \tau[e^{i\eta\mathbf{A}}\mathbf{U}](a).$$

If the response is zero there, $p_y(a) = 1$. Otherwise the formula uses only the unchanged pair on the fixed carrier and the increment of the one wall component. On the child turning cell use $u = e^{-i\mu}$ with

$$\mu(x) = \mathrm{bp}_a(x) T_a[U](x) \tfrac{1}{i} \log\bigl(p_y(a)^{-1}\bigr),$$

using the original bump and real-factor transport. The turn is supported in the inner third of that cell. The pinned response therefore splits into the responses of the wall components, with disjoint nontrivial gauge factors. The cut restriction agrees on each active pin because its carrier, data and response traces agree.

Choose the pinned tops in the ambient geometry once. Work on sections where all needed tops are levelled on the whole polydisc. The local pin estimate is $|p_y^{\pm 1} - 1| \le c_{\mathrm{m}}(L)\varepsilon_y(\mathbf{A})$. The turned pair costs at most $\mathsf{C}_{\mathrm{m}}\|(\mathbf{A},\mathbf{K})\|_y^+$, where the norm includes the neighboring level-one pillars. Check this bound against the room of every earlier reader on the complete turn polydisc. On that polydisc gauge invariance at real turns and holomorphy give independence from the auxiliary bump or real-factor representation when the values at the stored sites agree. On a fixed real-factor section the transport equality extends from its open real slice to the admitted complex branch because both contour means there are holomorphic. Equality of pairs gives the same point pins. To identify the full turned triples compare their real frames on the turning cells as well, or use the invariance of the term on the admitted turn polydisc. Throughout this argument the stored values are fixed. In particular a Borel near value stays in its original slot and an inherited scalar switch keeps its old radius. Only an actively integrated far field is multiplied by $\lambda_{\mathfrak{w}} s$; for an actively integrated near field this multiplier is its scalar switch and the value itself is fixed. The recentered crossings use the far arguments already checked on their two blocks. A new reader which uses a different near argument must supply the corresponding real coordinate and switch rule.

For an island factor keep its standing history, integration measure and whole exempt hull. At fixed internal real data take the old and new pair evaluations of its analytic exponent. The internal real weight is the same in those evaluations. The island hull is one fixed read, with new response arms at its attachment. The cut factorization is applied at fixed island data, and the original island integral is taken once on its carrier. This uses the old bounds for the analytic exponent and the mass on that carrier.

Once the Gaussian and potential estimates on the common discs are available, the localized logarithm defines the new correction in its own formation space. On a fixed support, the accumulated correction is read from the last contributing stage by restriction to the spaces of its summands. The construction of the incoming terms begins with the bare density as described below.

The ordinary boundary terms, the lower boundary terms other than pointed sums and the earlier stage terms use only their declared strip slots. Such a slot is read from a domain containing one of its endpoints. A term formed at level l reads stored fields of levels strictly below l. Contact with an endpoint at the integration level can therefore create an attachment through the pair response while leaving the stored arguments as they were.

For the direct exterior-only mask consider the complete blocks of a new collar pivot. Both endpoint blocks lie outside the old Ω_q, whereas an old strip fluctuation slot at the integrated level has an endpoint in Ω_q. An inherited pivot that reads the changed collar uses its original ordinary strip component, whose affected blocks are in the old-wall comparison. The kept strip reference is integrated by whole components, and its mixed blocks across distinct components vanish. Keep separate strip factors and long-chain corrections on their original carriers. A determining value has its own row coordinate, even when it forms a reference on a block with a stored fluctuation.

The successful completed strip tests are products of bond tests. Split each product between its active and retained coordinates. On an active bond use the smallest of its passing thresholds. On the retained bonds keep the real indicators in the carried kernel. A failed rim product belongs to the component of its failure, so it is carried in the frozen part or the separate stopped branch. These splits give the real cutoffs of the passing conditional integral.

In the analytic difference of strip quadratics, write each chain on its carrier as

$$-\tfrac{1}{2}\sum_{b,b'}\eta_b^{\mathsf{T}}(D_Y^- - D_Y^+)_{bb'}\eta_{b'}, \qquad \eta_b = \begin{cases} a_b, & b \text{ far}, \\ \mu_b a_b, & b \text{ near}. \end{cases}$$

The multiplier belongs to the original slot in each occurrence, including both occurrences on the diagonal. This preserves the carrier and its cut rule. Inherited analytic functions keep the switches of their own slots. At $\mu_b = 0$ these functions and the new interaction forget a_b. At $a_b = 0$ they forget μ_b. The Gaussian reference and its determinant use the integrated coordinates themselves. The retained real kernel carries its own selectors. The surviving far fields keep their full original discs, since their occurrences in analytic functions and in the new polynomial are separate from the determining rows that form the response. The switch identities pass through the connected expansion once its old analytic readers are admitted on their formation domains.

4.17.1.38. The first stage and the physical density. At level zero the array contains only the bare inverse coupling, and the density is $e^{-\beta A(U)}dU$. There are no carried analytic entries at this level. At a step of level t, form the boundary terms of $\mathbf{T}$ as the sum of the localized normalizer, the cluster terms assigned to the boundary, the covariance and weight corrections, and the localized strip shift. Inside the small-field region their triple argument lies in the enlarged level-t core with radii $(1+\beta_{\mathrm{s}})\alpha_{\iota,t-1}$. On the other supports it lies in the graded space of the completed label just after $\mathbf{T}$. This structure precedes the structure at stage zero and has the same radius numbers there. The boundary construction gives holomorphy in the triple, the far fields and the switches on those spaces. On the cutoff zone it uses the real stored field; its Jacobian difference there is the difference from the zero-field value multiplied by the switch.

For a newly external plain far bond let $R_{\mathrm{out}}(b)$ be its declared carried radius. Choose the ordinary formation box on these bonds to contain the discs of radii $R_{\mathrm{in}}(b) = 2R_{\mathrm{out}}(b)$. This is a choice of the fixed prefactor C_1^{fl} of the current far-field box, made before the weak-coupling ceiling. Keep the real collar and rim bounds and all inherited stored-field discs at their prescribed sizes.

Indeed, in the guarded conditional Gaussian, the external-to-mean operator is a sum of kernels with a uniform exponential row bound. Newly external far bonds are at distance at least $cW_t - O(1)$

from the integrated bonds, where $W_t = LM\check{R}_t$. Their contribution to the absolute row sum, divided by the integrated threshold, is bounded by

$$C\theta_{\mathrm{c}}^{-1} \sup_{b \text{ far}} \frac{R_{\mathrm{in}}(b)}{\mathfrak{b}_k} e^{-c' W_t}.$$

It tends to zero even on the larger discs. Thus the far part of the mean leaves the margin supplied by the real collar and rim. The enlarged-box potential bounds apply with the chosen fixed prefactor. In particular the ordinary fluctuation bounds include the cubic error on the enlarged box and the ratio error bounded by a constant times $|1-\tau_k| \sup_b R_{\mathrm{in}}(b)^2$. The same tube bound and the same choice of logarithmic powers make these errors small for the cluster sum. Apply the uniform activity majorant on this formation box before making the strip substitution.

In the plain zone write $H(b;W)$ for a piece of the strip shift with carrier W. It includes both endpoint blocks and reaches the join boundary. On the pair sections of its formation,

$$\sum_{W} \sup |H(b;W)| \le C(\alpha_{0,k} + \alpha_{1,k}) e^{-c\check{R}_t}.$$

The same bound with a reduced $c > 0$ holds with a fixed prescribed carrier weight in the sum, by choosing M large at that rate. For a simultaneous transfer put

$$a_{\mathrm{pre}}(b;\mathbf{z}) = \frac{1}{ig_k} \log \left(e^{ig_k a_{\mathrm{post}}(b)} e^{-i\sum_W z_W H(b;W)} \right).$$

The physical map is recovered at $z_W = 1$. On fixed amplitude discs $|z_W| < R$, $R > 1$, the product logarithm gives a uniform displacement bound $CRg_k^{-1} \sum_W \sup |H(b;W)|$. The exponential in $\check{R}_t$ makes this smaller than $R_{\mathrm{in}}(b) - R_{\mathrm{out}}(b)$. This holds on all plain far slots together. Thus the transferred cluster can be evaluated on the full output discs and the transfer differences can be estimated on the z_W discs. On the cutoff zone use the separate real-field switch, with the smooth cutoff evaluated at the real field of that zone.

At full switches the strip correction is the change of the localized exponent under the strip substitution, together with the logarithm of its Jacobian. To see this, let $F_X[S]$ denote the combined term on X with the strip substitution turned on at the bonds of S. Include in F_X the retained quadratic term, the cluster and covariance terms that read the shifted coordinates, and the negative of the localized Haar log term in the strip formula. If W is the set of eligible switches, the sum of the differences with nonempty index set is

$$\sum_{\emptyset \ne S \subset W} \sum_{T \subset S} (-1)^{|S|-|T|} F_X[T] = F_X[W] - F_X[\emptyset].$$

File a difference on the union of the old domain and the shift carriers. This changes the index of the term in the sum and leaves its value there unchanged. In real coordinates $a \mapsto a'$ the pullback of the incoming Haar density is the Haar density at a' times the strip Jacobian. Keep the Haar density at a' in the reference factors and sum the log strip Jacobian separately on its carriers. At the zero stored field the cutoff is constant near zero, so the strip map is the identity there and its log Jacobian is zero.

At a physical argument U the ordinary step $t = k+1$ gives the exponent for the completed label $\mathcal{S}^+$.

$$\begin{aligned} \mathsf{A}^+(U) = A_{k,\mathcal{S}}(U) + \mathsf{n}_k + \sum_{z \in \Lambda^0} \sum_{z \in X \subset \Lambda} E^{(k+1)}(X,U,z) \\ + \sum_{X \subset \Lambda^{\sim -1}} R''^{(k+1)}(X,U) + \sum_X \left[E^{\partial}(X,U) + B_{\mathbf{T}}^{(k+1)}(X;U,\underline{\mathfrak{f}}) \right]. \end{aligned}$$

Indeed on $X \subset \Lambda$ the bond pointing weights sum to one. Points outside Λ^0 give E^{∂}, and remainder terms outside $\Lambda^{\sim -1}$ go to the boundary cluster family. Outside Λ the normalizer and cluster terms enter the boundary family directly. The form corrections and the strip differences just described are

added once. Use the reference convention with covariance $\tau_k^{-1} A_{\mathrm{II}}^{-1}$ on the integrated bonds. The change in the gauge quadratic relative to this convention belongs to the interaction. The scalar in this identity is

$$\mathsf{n}_k = n'(\theta_k)\#(\mathfrak{X}_{\mathrm{I}} \sqcup \mathfrak{X}_{\mathrm{E}}) + n''(\theta_k)\#(\Omega \cap \mathbb{T}^{(k+1)}) + \big(n''' - \tfrac{3}{2}\log \tau_k\big)\#\mathfrak{X}_{\mathrm{I}}.$$

The three counts refer to fluctuation bonds, gauge blocks and integrated bonds, respectively. They give the Haar value and field rescaling, the gauge normalization, and the Gaussian constant including the ratio correction. These counts depend on the label. The localized normalizer accounts for $-\frac{1}{2}\log\det A_{\mathrm{II}}$ and the log inverse determinants of the interior eliminated rows. The exterior row factors belong to the retained operation. If the local normalizer is centered at the trivial field, put the removed constant in the scalar part before recentering the vacuum energy. Rewriting the coupling and vacuum energy repartitions A^+. The tests of the completed label and the reference measures of its operations multiply this exponent in the real measure.

For the ordinary output, evaluate a stored factor in an adapted representative of the native determining set. The geometry of an ordinary stored bond of level ι puts its endpoints at tops or at sites in a pillar with top of level $\iota + 1$. Its parent block is separated from the next higher layer. Use the endpoint at which the stored factor is represented. If it is a top, the residual gauge group fixes it. Otherwise let a be the site, $y(a)$ its native pillar top, and $\tau[U](a)$ its contour transport. A gauge that is $\tau[U](a)$ at each such stored site and the identity elsewhere sets those transports to one and fixes the nodes. This uses the original pillars at all the stored reads. If U and U^w are two such representatives,

$$1 = \tau[U^w](a) = w(y(a))\,\tau[U](a)\,w(a)^{-1} \quad \Longrightarrow \quad w(a) = 1.$$

The background and its physical current transform together. Covariance of the factor therefore gives the same value on both representatives with its stored arguments held fixed. For a kept operation that reads a root, the same conclusion uses the determining-node condition on that root. When a cut or a test tower changes a pillar, use the native evaluation for the stored factor.

For an earlier island (I, i) the root is a level-i site in $I^{\sim} \subset \Lambda_i^+ \subset \Omega_i$, which is disjoint from Ω_{i+1}. For $i = k$ this uses the separation of the newly appended Ω_{k+1} from the standing island. Thus the root lies in layer i of the ordinary output. If preparation acts on the component containing I, quarantine gives $i < h_t$ and the prepared sequence agrees with the original up through h_t. This leaves both layer membership tests at i and $i + 1$ unchanged. Outside the acted components the sequence was left fixed. In the healed comparison the surviving spectator roots are outside the added region Z, where their layers are unchanged. A level-i site in layer i is a determining node. For a current main operation, the entrance bond has its outer endpoint in the component and outside the prepared core, by the clearance of the main box. That endpoint is a node at the current top level. These checks fix the roots in the corresponding residual gauge groups. A crossing fluctuation whose initial point is outside its box transforms at that point. The exterior starts in an earlier operation lie on the same single-layer shell inside $I^{\sim}$. In the current operation they lie on the outer level-t shell outside Z_t''. They are fixed nodes by the same layer argument. The residual action fixes the group-valued drop-out frame variables directly.

Lower boundary terms formed in the same manner keep their original graded spaces. The stage-origin terms have the graded spaces of their respective stages. The terms from exponentiating an earlier large-field operation use its output space. They are the interaction with no chosen component together with the additional localized remainder. The boundary parts of the pointed regular sums are formed on their single-scale cores. In the next ordinary step take the boundary and stored-operation potentials by subtracting the two evaluations at a common pinned substitution, with the stored arguments fixed. This subtraction is analytic on the graded step domain and preserves the far-field and switch radii of the entering terms. The regular, remainder and Wilson potentials use the plain substitution on their respective domains, since their arguments carry no stored fields.

These are the argument spaces of the formation rules. When using a reduced history as a donor, take its member with the space in which that member was formed.

For real evaluation, fix the outgoing and retained data on an admitted labelled fibre of a stage. Write a for the active old strip fields and B for the free collar variables. Let $F_-(a,B)$ denote the old evaluation of the entries being transferred, and let $F_+(a)$ denote their prospective evaluation at the same strip values. In both evaluations the switch of each integrated strip field is one and the spectator fields and switches are preserved. Each earlier correction has already closed its own response source against its stage shift. The fixed exterior exponent is $F_+(0)$, with zero in each integrated strip slot in the original coordinates and its switch removed. The zero of a translated integration coordinate can give a different stored value. For the remaining measure $d\Gamma$, containing the tests, chart Jacobians and Gaussian factors, the exact identity is

$$e^{F_+(0)} \int \exp\{[F_-(a,B) - F_+(a)] + [F_+(a) - F_+(0)]\}\, d\Gamma = \int e^{F_-(a,B)}\, d\Gamma.$$

The first bracket uses the same-strip response potentials. The second is the active strip dependence already present at the prospective background. It stays inside the integral. Use the estimates for the prospective strip entries at fixed background and include them once. In the joint strip substitution above, the empty cut gives $F_+(0)$ and the full cut gives $F_-(a,B)$. Its difference therefore includes both brackets. The kept strip quadratic has its separate Gaussian decomposition below. At this stage put $q = j+1$. The active variables of $d\Gamma$ are the level-j stored fluctuation on the selected old strip and the free nonpivots of the newly opened collar. The old slots meet Ω_q, have no endpoint in Λ_q, and exclude the original pivots. Take the whole stored mask of each selected component of $\Omega_q \cap Z_q$ contained in the treated X. Other stored variables are fixed, including deep core variables whose factors can still read the moving background. On the active strip keep the ordinary completion, collar and rim indicators on their original masks. Where several apply to the same slot their conjunction tests at the least of their thresholds. The free sharp collar ball acts on the new nonpivots. An additional sharp test in the inherited strip uses its own transported chart as above. The prospective regularity guard depends on the fixed datum and remains outside the collar logarithm. This includes the prospective crossing tests wherever they are used. The tests inside the integral are in their original real slots, with their thresholds set by the frozen frame. Completing a Gaussian square changes the reference exponent and its normalization. The domain of a hard cutoff stays in its original coordinate.

On a common parameter section, write the integral as $\int 1_E(a,B)\, F(P,\mathsf{y};a,B)\, d\mu(a,B)$, where E contains the fixed real cutoffs and F contains the chart and exponent factors. At each point of E, holomorphy of those factors gives holomorphy of F in the section parameters. A common integrable absolute majorant on smaller parameter discs then lets the Cauchy formula pass through the integral. This gives the analytic integral on that section, evaluated on the real data selected by the external guard. Its nonvanishing and localized logarithm use the connected expansion and its separate estimates below.

To identify $d\Gamma$ in the two representations, use the same reference scale θ_j for the Gaussian, the gauge draw, the thresholds and the fluctuation units. Put the difference from the running action weight in the interaction.

For the collar form put $q = j+1$ and use the full preceding determining sequence with its closed no-inward row set. The exterior collar is $K_q = (\Omega_q^c \setminus Z_q'') \cap X$. The direct graded rule releases level-j rows with an endpoint in K_q and neither endpoint in the old Ω_q. The added level-q coarse rows have the same endpoint rule on their own grid. Alignment by complete level-q blocks puts both blocks of any such row outside the old Ω_q. They lie inside the preceding member of level j. In fact for $q \le k_0$ the collar extends at most five coarse cells into the old exterior and for $q = k_0+1$ it extends seven. The eight-cell quiet gap keeps the blocks away from the old G_j, also at the first stage. For $q > k_0+1$ the separation between successive prepared faces is at least one M-cell of level

q. This gives the same conclusion for the remade preceding member. The parent and double-block reaches are $O(L^q)$ and fit these margins. Thus the nonlinear source is tight in the preceding metric.

In a double block meeting the prepared face, the internal fine links on the prepared side are held, and the links in the active block and those crossing that face are released. The regular inverse solves the new coarse row for its unique middle pivot. A row with no endpoint in K_q cannot read any released link in its full double block. A crossing into the old Ω_q belongs instead to the ordinary strip. Any change of its old pivot through the exterior block is handled in the inherited double-block chart with the old coarse target and the old coarea factor. At the overlapping stages use the actual preceding determining datum. In particular the level-j rows completed by the last stage already have its averaged values. Among the exterior coarse deltas, use exactly the newly added rows at this stage. The exterior stock has neither endpoint in the old Ω_q, since the ordinary step already eliminated rows meeting that member. This leaves the old-wall coarse row with one elimination factor and no additional exterior delta.

The base of the collar form is the prospective real background at the induced datum. This background is critical on the preceding fibre since that fibre at the induced datum is contained in the prospective fibre. Write z for the physical level-j nonpivot perturbation, $b(z)$ for the plain row increment in that sequence, and $\beta(z) = b(z)_\natural$ for its reduced logarithm. The retained plain rows have increment zero. On the free rows and pivots use the exact one-step graph

$$\mathsf{C}_V z - \mathsf{h}_V \widetilde{\mathsf{D}}_V(\mathsf{C}_V z).$$

The pivot contribution to the second derivative of this graph is $-2\mathsf{h}_V C_V^{(2)}(\mathsf{C}_V\cdot, \mathsf{C}_V\cdot)$. The moving level-$j$ rows of the old set are interior there. The naturalizer can also change an old outward row c of level q whose outer block reads the moving fine rows, even though $b(c) = 0$. For its outer end x write $\ell_x = (1/i)\log\theta_x(b)$ for the seam factor on the complete internal links of that block. Then $\beta(c)$ is $\mathrm{Ad}(V(c))\ell_x$ if $x = c_+$, and $-\ell_x$ if $x = c_-$. No internal link in that block is a coarse pivot, since each pivot is between two blocks. Reuse the same factor at an outer end shared by several rows, with the orientation of each row. Thus the pivot curvature belongs to the released level-j pivot rows and the seam curvature to the old level-q rows that read the moving block. A newly added coarse row is used in the fixed-target pivot equation and has no coordinate in β. The naturalizer reads plain increments once. It does not propagate a reduced increment into a further level.

Write $F(A)$ for the Wilson action at $e^{i\xi A}U_0$ in the reference normalization, and $\mathcal{Q}(A)$ for the straightened row logarithm. Let $A(z) = \mathbf{H}(\mathbf{1}; \beta(z))$ be the real response of the preceding problem. Its node gauge restores exactly the plain datum and preserves F. Put $K = \mathcal{Q}'(0)$. On the real critical pair $F'(0) = K^{\mathsf{T}}\lambda_0$. For $P_B = A'(0)[B]$, differentiate $\mathcal{Q}(A(z)) = \beta(z)$ twice to obtain

$$(F\circ A)''(0)[B,C] = (P_B, F''(0)P_C)_\xi - \big(\lambda_0, \mathcal{Q}''(0)[P_B, P_C]\big)_{\mathrm{rows}} + \big(\lambda_0, \beta''(0)[B,C]\big)_{\mathrm{rows}}.$$

Here the reduced row pairing is the counting pairing, and the two terms with λ_0 are distinct. The inner constraint curvature has the minus sign from eliminating A''. The outer derivative uses the signed pivot and seam terms of β''. These terms remain even though the first variation on the fixed-target collar fibre vanishes.

Put $\nu(z)(c) = \xi_c^{-1}\beta(z)(c)$ and write $(\cdot,\cdot)_{\mathbf{B}}$ for the pairing with row weights ξ_c^4. On the real response slice the first two terms are the constrained form with $\mathsf{G}_1^{-1} - Q^{\mathsf{T}}\mathsf{a}Q$. Indeed the gauge projection term vanishes on the slice and the current correction of G_1^{-1} is the quadratic constraint term there. The linearized response in these units is $P_B = \mathsf{H}_1\nu'(0)[B]$: it solves the linear constrained variation at the prescribed row $\nu'(0)[B]$, with the same slice condition. Using $Q\mathsf{H}_1 = I$ and $\mathsf{G}_1^{-1}\mathsf{H}_1 = Q^{\mathsf{T}}\mathsf{S}_1$, where $\mathsf{S}_1 = (Q\mathsf{G}_1Q^{\mathsf{T}})^{-1}$, gives the Wilson form

$$\big(\nu'(0)[B], (\mathsf{S}_1 - \mathsf{a})\nu'(0)[C]\big)_{\mathbf{B}} + \big(\lambda_0, \beta''(0)[B,C]\big)_{\mathrm{rows}}.$$

Multiply by θ_j and insert g_jB, g_jC for the two physical perturbations. The scalar factor is $\theta_j g_j^2 = 1$.

On the complex section with independent current use this operator formula for the continuation. For the outer term take $\lambda_0(c) = \xi_c^3(\mathsf{H}^{\mathsf{T}}\mathbf{J})(c)$, using the same linear right inverse H as in the inner current correction. On the real critical slice $J = \hat{Q}^{\mathsf{T}}\lambda_0$ and $\hat{Q} = \mathrm{diag}(\xi_c)Q$. In the row pairing with weights ξ_c^4, this gives $\mathsf{H}^{\mathsf{T}}J = \xi_c^{-3}\lambda_0$, and $\mathsf{H}_1^{\mathsf{T}}J$ gives the same value there. The uncorrected and corrected right inverses can give different current contractions off the critical slice. The two attachments of $\nu'(0)$ to the native walk family of S_1 use bounded local stencils at levels $j, j+1$. The subtraction a is a local row term. In the last summand attach the native multiplier walk $\mathsf{H}^{\mathsf{T}}\mathbf{J}$ at the row of β''. In each case the walk and the complete local stencils meet at a common row. The bounds are absolute in the weighted walk norm and apply to the actual full-depth form.

The switched fixed-source derivative used for a centre response is obtained from the reduced equations. At zero source the multiplier is $\lambda_s^0 = \mathsf{H}(s)^{\mathsf{T}}\mathbf{J}$, in the counting coordinates of that system. Writing $\mathsf{m}(c) = \xi_c$, the counting right inverse is the weighted right inverse times m^{-1}. The Schur operator in the multiplier equation is $\mathsf{m}^3\mathsf{S}\mathsf{m}^{-1}$ in place of the weighted S. The diagonal conversions keep each native walk carrier. With $T_\lambda v = (\mathsf{C}''(0)[v,\cdot])^{\mathsf{T}}\lambda$, the derivative of the potential there is

$$\partial_{B'}\mathbf{H}(s;0) = \big(I - \widetilde{\mathsf{G}}(s)T_{\lambda_s^0}\big)^{-1}\mathsf{H}_0(s).$$

Expand the inverse by the small-current bound of the contraction. Each T_λ uses the complete local row stencils and the walk for the multiplier attached at that row. At a nonzero admitted source, differentiate the whole contraction and use its Neumann inverse on the potential and multiplier increments. The carrier is formed by joining the surviving walks and local row stencils at shared entries. At full switches on a real critical datum the linearized response on the constraint slice is $B' \mapsto \mathsf{H}_1\mathsf{m}^{-1}B'$, the response used above in weighted units. This identification does not replace the derivative on the switched section.

On the prospective axial blocks the soft density adds the Gram of the linearized mean contours. The carried gauge factors cover the old band. The gauge draws beyond that band cover the remaining active blocks. Their reference weight θ_j cancels the g_j^2 of the two scaled derivatives. Write its derivative in a changed gauge as the derivative of the relative contour $T(z)T(0)^{-1}$. A fixed real gauge conjugates this relative contour at the block centre and hence preserves the Gram. In the no-inward sequence a noncentral site of a complete active prospective block is free: the level-j rows from the preceding layer into that block are the omitted inward rows. If a density factor also reads a contour at a retained node in another mask, keep its exact factor in the interaction unless its variation on the collar graph is established. Constancy on a fixed critical orbit alone does not give its constancy along the varying datum.

The flat floor uses the actual plain tangent v_j of $\mathcal{M}^j U^-$, after the row and node conversion. Its value at an active nonpivot is $g_j B$, and its linear coarse row increment is zero on the fixed targets. At the flat pair the multiplier term and the inner constraint correction vanish. The Wilson Hessian controls the level-j curl of v_j. Indeed the linear one-step flux map on full carriers has row and column sums L^2, L^{-2}. Its ℓ^2 bound is one in dimension four, so one may iterate on the full backward union without a depth loss.

On a complete double block, zero circulations make a fine one-form a site difference. Zero mean contours make its site function constant on each block. A zero coarse row makes the constants equal. The resulting finite-dimensional bound controls the fine norm by the squares of the circulations, mean contours and coarse row. For a collar slot internal to one block use the single-block part. For a collar crossing use both endpoint blocks. At the prepared inner face the other block lies in the preceding layer j and its internal rows are held, so their tangent is zero. The coarse row across that face remains fixed. The crossing into the old Ω_{j+1} belongs to the inherited strip. Thus all the active collar slots have their full curl carriers and the required contour terms. Sum the block bound with bounded overlap and use $\theta_j g_j^2 = 1$. This gives the flat floor uniformly in the full depth, and coordinate compressions inherit it by extending the test vector by zero.

For the real comparison use the full form on the fixed collar geometry. Partition the active slots by real multipliers χ_ν with $\sum_\nu \chi_\nu^2 = 1$, supported in cubes of side $O(RL^{j+1})$ with bounded overlap, for a large fixed R. For a kernel $D(b,c)$ the partition error has matrix entries proportional to

$$D(b,c)\sum_\nu(\chi_\nu(b)-\chi_\nu(c))^2.$$

The row and column bounds give an error $C(R^{-2}+e^{-cR})$. To see this, use the first differences of the cutoffs for distances at most RL^{j+1}, and use the first escape length in the native metric at larger distances. Near an active slot the quiet collar and the preceding prepared separation leave a cell of clearance for the full buffered patch in layers $j, j+1$. Fix the native walk grain $\mathfrak{M}$, take R large at that grain, then take M large enough to put the buffer and its full stencils inside the clearance. Use one real gauge on the complete comparison cube, which meets the preceding layer j. For $j \geq 1$ the quantitative gauge clause gives, for $U^u = e^{i\mathsf{v}}$,

$$L^j|\mathsf{v}| + L^{2j}|\delta\mathsf{v}| \leq CML_\flat^{2e}\alpha_{0,j}.$$

The radius tends to zero after the fixed geometric choices, also with the allowed overlap factor. At $j=0$ the patch has fixed size in bare lattice units. A tree gauge on this box bounds the links and their first differences by a fixed constant times the bare plaquette bound, by filling its fundamental cycles inside the box.

Conjugate the whole compressed patch by the same real gauge, acting orthogonally on the active bond variables. Compare short terms with their values at the trivial pair by interpolating its link logarithms and its independent current from zero. Evaluate each short term on its full carrier, using the fixed local admission radii and Cauchy's estimate. Sum with the native weighted walk majorant. The multiplier term vanishes at the trivial pair and has the same small bound. A term leaving the buffer pays the first-escape cost even if its matrix indices coincide. Until escape it takes at least $cR/\mathfrak{M}$ native cell steps, which gives the exponential tail from the walk majorant. The floor follows by perturbing the flat form on each patch and applying the partition bound to the active nonpivot vectors. In applying the flat estimate to a localized vector, lift that vector through the collar graph at the fixed targets. Cutting off the previously lifted tangent need not preserve those constraints. This comparison also gives a real floor for the analytic operator at the admitted small real pairs with independent current, including the zero-current reference.

A carrier-restricted comparison uses the complete double blocks of its active slots and the same gauge and coarse target rows. If the guard includes the full local stencils and the surrounding buffer, the terms it deletes have the exponential first-escape bound. Their operator norm is below a fixed fraction of the real floor by the Schur estimate. This supplies local positivity for the guarded precision on these carriers.

Use the local resolvent construction for the inverse root. Take the partition above by translates on the level-j grid, with supports of side $O(RL^{j+1})$, normalized by the sum of the squares of the translated bump. Its normalizing sum is fixed by the grid and independent of the active mask. Around each nonempty support take an enlarged active patch and a further buffer, both of width of order RL^{j+1}. Write D_ν for the compression to the active slots of that patch of the sum of the unswitched precision terms with complete carriers in its buffer. Use the same rule after any prescribed compression of the strip. The real floor and the small complex pair perturbation give the local resolvent $G_\nu(x) = (D_\nu + x)^{-1}$ for $x \geq 0$. The weighted conjugation estimate bounds its rows and columns by $C/(x+\gamma)$ at a reduced rate, with $\gamma > 0$ fixed. For two active indices in a patch the distance is at least a fixed multiple of their physical separation in level-$(j+1)$ units. A path staying in the two-layer neighborhood uses at most that spacing, while a path leaving it first pays a length larger than the diameter of the patch in those units. Summing the entrywise resolvent bound over the level-j slots gives uniform row and column bounds, including on the compressed patches.

Set

$$G_{\mathrm{p}}(x) = \sum_{\nu} \chi_\nu G_\nu(x) \chi_\nu, \qquad (D + x) G_{\mathrm{p}}(x) = I + E(x).$$

With local matrices extended by zero the error has the two families

$$[D, \chi_\nu] G_\nu(x) \chi_\nu, \qquad \chi_\nu (D - D_\nu) G_\nu(x) \chi_\nu.$$

In the first use the absolute walk sum of D, placing the cutoff difference on each term. The square-sum of the cutoffs and the ℓ^2_ν bound on their translated differences give C/R after the weighted row sum. More explicitly, $\sum_\nu |\chi_\nu(d)||\chi_\nu(b) - \chi_\nu(c)|$ is at most $C \min\{1, |b - c|/(RL^{j+1})\}$ by Cauchy–Schwarz on the translates. Use the common entrywise resolvent bound before this sum. In the second, a discarded term with both endpoints in the local matrix must leave the buffer, which costs Ce^{-cR}. These estimates use the same complete carriers for D and its local inverse. The local inverse itself has the buffered patch as its carrier. All patches with active slots fit inside the base-layer neighborhood of the collar. Choose M larger than their fixed size in one-step units. Then a patch meets only a bounded number of the aligned base cells, so its switches cost $e^{C\varkappa}$. A small exponential weight on the physical patch diameter also costs a fixed factor, by choosing its rate of order $1/R$. With the remaining native walk decay for D, the absolute majorant of $E(x)$ is bounded by

$$\frac{Ce^{C\varkappa}}{x + \gamma} (R^{-1} + e^{-cR}).$$

Use R sufficiently large for this bound to be less than $1/2$ uniformly in x, then use the indicated floor on M.

Expand $G_{\mathrm{p}}(x)(I + E(x))^{-1}$ in these terms. The products join their carriers at common indices. Tag each resulting term by the distinct graded cells meeting the joined carrier. At binary cuts the surviving terms use only their active components. At full switches the sum is the resolvent of the physical precision. The integral with weight $x^{-1/2}/\pi$ gives the inverse-root family, since $(x + \gamma)^{-1}$ is integrable against this weight. The pair comparison uses

$$G_\nu(p; x) - G_\nu(p_{\mathrm{r}}; x) = -G_\nu(p; x)(D_\nu(p) - D_\nu(p_{\mathrm{r}}))G_\nu(p_{\mathrm{r}}; x).$$

It bounds the difference by $C\epsilon_{\mathrm{pair}}/(x + \gamma)^2$ on the same buffered mask. Telescope each resolvent product in one factor at a time. The remaining geometric sum is bounded by $\sum_{n \geq 1} n\, 2^{-(n-1)}$. Integration gives the pair bound $C\epsilon_{\mathrm{pair}}$ for the switched root on the same carriers. The guard error uses the separate first-escape bound on its deleted carriers.

For the log use the resolvent representation with kernel $(1 + x)^{-1}(D - I)(D + x)^{-1}$; this keeps the cancellation at large x before integrating. Each inverse or log uses the same locally formed precision and actual retained masks. The trace for the direct stage term is taken on the full compressed precision before the shear, since its determinant is preserved by that shear. Attach each diagonal of a log term at its actual active slot, with the whole carrier of the term. Subtract its value at the trivial pair using the same mask and buffer, and use the pair comparison of the resolvent products for the subtracted term. This trace supplies the direct stage normalizer. The normalized cut measure uses the separate determinant of A_W. This defines a switched root by its carrier terms. The full-switch identity is the one used in the Gaussian marginal formula.

On the stage carriers that admit a common-scale box cover, the subtracted trace can be summed over stages. Write $\ell = \log \theta_i$ at the top of the waiting window and choose $H = A \log \ell$. Split the trace expansion by carrier length in its native units. The terms of length greater than H spend part of the rooted decay to gain e^{-cH}. In the native local-gauge walk use the sum of its itinerary length and its number of factors for this split. The itinerary exponential pays for the first quantity and the spare walk-factor decay pays for the second. Use the actual stage anchors in the split.

For a short term, suppose the complete carrier is in a filled nonwrapping box of side at most CH in the stage units, with an admissible cover by overlapping gauge boxes of that scale. The

local potential, first difference and current have size bounded by $\mathfrak{a} = C\theta_i^{-1/2}\ell^C$ there, using the running radii in the waiting window. At small coupling $H\mathfrak{a}$ is within the gluing radius. Gluing the real gauges across the cover costs the side ratio in the potential and its first difference. Rotate the independent complex perturbation and current in the same gauges. This gives one gauge on the whole carrier, with the complete pair bounded by $CH^C\mathfrak{a}$ in its local coefficient norms. The trace is unchanged by this change of gauge.

Compare with the trivial pair on the same buffer and masks. The local resolvent comparison, or the telescoping estimate on the factors of the log walk, bounds the subtracted diagonal by its summable majorant times $CH^C\mathfrak{a}$. In the walk formula the closing loop stays in the filled box. Its transport is also within $CH^C\mathfrak{a}$ of the identity, by multiplying its links in that gauge. A fixed power of the walk length is paid for by the rate reserve. The same bound holds when a short term fits in just one of the original gauge boxes.

The uncompressed stages supply such a region for their interior roots on the real support. At stage level $q \le k_0$, the old component of level $q-1$ lies inside the hull whose eight-cell enlargement is Ω_q^c in the class box. The prepared Z_q'' removes five of those layers. The roots of the collar outside Z_q'' therefore have at least three level-q cells to the older core, before the fixed carrier padding. The lower re-made regions lie in that core. In the other direction use the quiet gap to the unchanged upper member. If the padded root neighborhood stays in the class box, a box of side ca_q around it, with fixed additional padding, uses only the two neighboring units. The ordinary current and curvature bounds on that box give the small real gauge by the large-box gauge estimate, with loss bounded by a fixed power of $\check{R}_q$. The two units cost a fixed factor of L. A carrier leaving this neighborhood pays $e^{-c\check{R}_q}$ in the first-escape bound. Thus these uncompressed roots can be summed on the physical real sections. On a complex section the same gluing applies when its real factor has the native small gauge clauses on the padded cover. The independent current is then estimated in its own coefficient norm.

In the local-gauge walk expansion, join consecutive transition sites by a path γ_a inside the padded localization $X_a^{\sim}$ of the factor. Close the path inside X_0. The localizations used here are connected unions of a bounded number of boxes, so each segment uses a bounded number of coordinate-monotone paths. The comparison of the two gauges at an overlap uses a path in the retained overlap. This must be checked for any overlap clipped by the stage mask. The complete closing loop γ lies in $X_0 \cup \bigcup_{a\ge 1} X_a^{\sim}$.

Let u_a be the gauge of the a-th factor. After conjugating the transition product by the root gauge, its factors have the form $u_a(x_a)^{-1}u_a(x_{a+1})$. Write this as $W_aU(\gamma_a)$, where W_a is a conjugate of $U^{u_a}(\gamma_a)^{-1}$. If the clause bound holds along the path, and $s(b)$ is the local scale of the link, then

$$\|W_a - I\| \le C\mathfrak{a}\sum_{b\in\gamma_a}\frac{\eta}{s(b)}.$$

Include the closing factor in this estimate. Telescoping the products of unitaries bounds the difference from $U(\gamma)$ by the sum of these errors. The adjoint closing factor differs from the identity by at most twice this sum plus $2\,\mathrm{dist}(U(\gamma), Z(G))$. The box diameter bounds the physical path length. For the weighted sum use the smallest scale met by the path. On a complete connected cover of $O(H)$ graded cells with adjacent grades differing by at most one, the ratio of the largest and smallest scales is at most L^{CH}. The errors of the local gauges then cost only a fixed power of ℓ times $\mathfrak{a}$.

For the walk comparison one can also use a chain of clause boxes in which only consecutive boxes intersect. Use this when the complete closing path runs through such a chain and the successive grades differ by at most one. Adjust the constant gauges in order along the chain so that the two gauges match at one site of each overlap. On an overlap the change of the relative gauge along a path is bounded by the sum of the link deviations in the two local gauges. The coordinate paths in a clause box have physical length at most a fixed multiple of its side. If $\xi_{\min}$ and $\xi_{\max}$ are

the smallest and largest units of the chain, moving the transition to the point of the closing path costs at most $C\mathfrak{a}\,\xi_{\max}/\xi_{\min}$. The paths within the boxes obey the same bound for their transports. Multiplying these factors bounds the closing loop by a fixed power of H times $\mathfrak{a}\,\xi_{\max}/\xi_{\min}$. For a short term through $O(H)$ boxes, this size ratio is at most L^{CH}. The remaining log-walk factors use their original local gauge estimates. This proves the short-term bound also for these chains, with an extra factor at most ℓ^{C_A}.

There is a second closing-loop estimate for the compressed region. Suppose its complete closing loop is carried between the class box and the strictly inset old box of level k_0. On that full shell the prepared grades are at least k_0. With the real curvature bound on the shell, each bare plaquette defect is at most $C\mathfrak{a}\eta^2/\xi_{k_0}^2$. Choose each coordinate path in a box by moving in one coordinate at a time. A walk with p nonroot factors then has at most $C(p+1)$ straight pieces. The local path errors also obey the weighted estimate above with $s(b) \geq \xi_{k_0}$. A padded box of diameter $C_L\mathfrak{M}\xi_i$ therefore costs at most $C\mathfrak{a}\mathfrak{M}\xi_i/\xi_{k_0}$ per factor. The rectangular shell loop estimate bounds the holonomy defect by the number of straight pieces times the square of the outer side in bare units times this plaquette bound. The two bare mesh factors cancel. The remaining scale cost is bounded by

$$C(M\check{R}_i)^2(\xi_i/\xi_{k_0})^2 \leq C(M\check{R}_i)^2(L\check{R}_{k_0+1})^2.$$

Here $\mathsf{s}_{k_0+1} = M$ gives the last ratio. The result is a fixed power of ℓ times $\mathfrak{a}$, with the walk-length factors paid as above. Thus a short loop on this full shell is also covered by the subtracted trace estimate, even when it has no thin-chain cover. This use of the shell requires the plaquette bound between the two boxes, including its filling plaquettes. It applies on the real support and to restrictions of stage pairs on the whole class domain with this bound on their real factor. For a supremum on a locally defined pair space, one must show that the admitted pair has such an extension or bound its closing transport on the filed domain itself. The local gauge clauses can hold on an annular carrier for a flat connection with noncentral holonomy.

In the native weighted distance there is a lower escape bound for these compressed collars. At $q = k_0 + 1$, the prepared box removes seven of the eight outer cells from the old hull enlargement. To reach the old core from the active collar costs at least one of these cells. At $q \geq k_0 + 2$, a connection from outside Z_q'' to the old core crosses the band between Z_{k_0+2}'' and Z_{k_0+1}''. Its width is $2M - \mathsf{w}_{k_0+2}$ in top units, at least cM. That band has grade $k_0 + 1$. By the definition of weighted length, crossing it costs at least $cM\xi_i/\xi_{k_0+1} = cM\check{R}_{k_0+1}$. Fixed root padding can be reserved inside this width. This proves confinement away from the old core for short connections charged their native weighted length. For a native walk whose padded factors are unions of a bounded number of proper-scale localization boxes, the same escape also controls the closing paths. These are the native localizations of [B85c, Section 3C and Theorem 3.10]. Indeed, with the native grain $\mathfrak{M}$ fixed, the physical diameter of any one padded factor is at most $C_L\mathfrak{M}\xi_i$. Choose the stage M large enough that this is less than a fixed fraction of the width of the protecting band. If a padded factor meets the old core, every itinerary through that factor must penetrate to the inner part of the band or cross it. The cost is again at least $cM\check{R}_{k_0+1}$, including the first compressed stage where the protecting width is one cell. In the native walk distance the itinerary visits the localizations in order, using the weighted contour metric of [B84b, (2.45)–(2.46)]. Each possible itinerary pays this cost, even when the two matrix indices coincide. All closing paths stay in the padded factors, so no additional escape estimate on their lengths is needed for this confinement.

For the directly compressed collar precision this check can be made in each factor of the composite resolvent. Its active slots are on the outer side of the prepared box, up to the fixed one-step stencil. Every local inverse reads only its buffer, which meets such a slot and has diameter $CR\xi_q$. With M chosen after R, the buffer cannot reach the old core. Between local inverses each precision factor is compressed to active slots. If such a factor reaches the core, its own native walk or tree must pay the inner escape from an active endpoint. Use the native tree distance for the Schur and multiplier

kernels. Its tree meets the constituent localizations. The endpoint condition supplies the root visit when that localization is omitted from the tree convention. The outer slots are within fixed reach of the first and last localizations. Thus if an escape occurs on a multiplier branch, it is still charged from an active endpoint. Keep the attached row stencils in the fixed padding of this test. The bounds on the remaining factors sum by the parametrix majorant, with the count of choices of an escaping factor paid by the geometric series. This argument uses the collar mask before the shear. A factor with additional active slots inside the old core would require a different estimate.

For a buffered inverse use the real gauge on its full buffer. The pair comparison for the local resolvent replaces its kernel there by the kernel at the trivial pair with the same mask. The vacuum matrix block between two bond slots is scalar on colour, by constant gauge invariance and irreducibility of the adjoint representation. When the block is written back in the original gauge, its scalar factor multiplies $\mathrm{Ad}_{u(b_-)^{-1}u(c_-)}$. Join b_- to c_- by a coordinate path in the buffer. At shared slots the segments join without additional paths. In a diagonal trace they add the buffered inverses' segments to the native precision segments in the closed path. The matrix errors are summed by the telescoping resolvent comparison. For the remaining closed path use the shell estimate when the complete loop and its filling are admitted there.

On the outer side, a wall-connected chain from a collar bond to a cell touching the class boundary has at least $2\mathsf{N}-1$ cells. When the padded carrier of the diagonal is filed at such a bond, reaching the boundary costs $Ce^{-c\mathsf{N}}$ at the reduced rate, using part of the filed cell decay. This controls the outer escape for these collar roots. For the other active slots one must check their own outer margin.

For these terms in the pinned sum the number of stages and the normalized anchor count are bounded by a fixed power ℓ^{C_*}. Thus these short terms and the long tail together cost at most

$$C\ell^{C_*}\big(H^C\ell^{C_A}\mathfrak{a}+e^{-cH}\big)$$

at the reduced rate. Choose A after the anchor count. This bounds their contribution to the summed direct norm uniformly. The scalar stage numbers remain separate from this subtraction. The compressed stages may have anchors over the same cells. The outer M-cell layer of $\Omega^c_{k_0+1}\cap X$ lies outside each Z''_q in the compressed range and is inside Ω^c_q by nesting. Thus even a fixed pin can meet all these stages, and the absolute sum needs a gain beyond the count of its anchor layers. For other short carriers crossing the graded interfaces we must still control the closing transport on the complete admitted reads. A chain that closes around a hole can require filling plaquettes beyond those reads.

For the strip part use the kept chains of the external Schur form with their original keep rule. The source supplies their positive floor after the chain selection. The original Schur complement already eliminated the ordinary inner integration variables. File each kept chain with its original strip component, including its one-layer carrier margin. This margin includes the full read of the middle resolvent in a Schur chain. In the class box X, let H_q be the level-q cell hull of $Z_{q-1}\cap X$. The complement $\Omega^c_q\cap X$ is its eight-layer enlargement. The strip component lies in $\Omega_q\cap Z_q\cap X$, so its one-layer enlargement is still separated from H_q by at least seven level-q cell widths. At the first compressed stage this excludes the old hull. At later stages the first prepared box $Z''_{k_0+1}\cap X$ is already contained in $Z_{q-1}\cap X$. A new buffered inverse at an active strip slot has fixed reach in level-q units and stays in this clearance. Thus the kept strip precision and its buffered inverse avoid the old core on their complete physical reads.

At a nonterminal ordinary strip root the same one-scale argument uses the two quiet gaps. Let $a_q=M\check{R}_q\xi_q$. The root lies in $\Omega_q\cap Z_q\cap X$, up to the fixed endpoint reach. Its distance to Z''_q is at least $5a_q$ before compression and $7a_q$ at the first compressed stage, before subtracting that reach. At a later compressed stage the old inset is contained in $Z_{q-1}\cap X$, giving the $8a_q$ gap of the original class geometry. In the other direction $Z_q\cap X$ is at least $8a_{q+1}$ from the unchanged Ω_{q+1}. These roots are also away from the class wall, by the same separation at the top level. The native

towers of the direct strip operator stop at q. Thus the native itinerary and bounded padding put a short trace in a box of side $CH\xi_q$, inside the single grade q of the prospective physical sequence. The real profile and the box gauge give the same small-field estimate. Use the prospective strip quadratic for this direct term. Its difference from the old quadratic is carried in the interaction.

For the terminal strip, this also gives a one-scale box for the short trace on the physical pair. At $q = i$ the active roots are in the original $\Omega_i \cap X$. The prepared inset $Z_i'' \cap X$ is contained in $Z_{i-1} \cap X$, which is separated from these roots by eight top cell widths. The exterior top coarse cells of $\mathcal{P}_i$ touching X belong to the clean region and hence to Ω_i. In the global prepared sequence the first exterior layer stays unchanged. Thus, on either side of the wall of X, a neighborhood of width $cM\check{R}_i\xi_i$ about each strip root is entirely in the global grade i. For a short word of length at most H, the native itinerary, its bounded padding per factor, and any local inverse buffers put the entire trace in a box of side $CH\xi_i$. The real profile controls all plaquettes of that box. The box gauge estimate then bounds the short trace by its majorant times a fixed power of H times $\mathfrak{a}$, as above. Use this on the global physical pair, retaining the original operator masks and native weights. If the strip mask is compressed further at an admitted guarded binary switch, the comparison still uses the fixed physical pair and the surviving carriers. The box filling estimates their closing transport. Its plaquettes are used in the bound and the coefficient continues to read its own carrier.

The chain functions and their read sets are fixed at formation. Evaluate them on the supplied old or prospective pair in the common stored coordinates. An endpoint transport needed for this evaluation fits in one additional layer. At the stage that consumes a crossing slot, use its prescribed endpoint conversion, since the preparation can change its determining row. A response used to construct a prospective pair contributes its own carriers when the coefficient is composed with that response. The original keep rule applies to the coefficient's reads on the supplied pair.

On the native stage grid these margins remain below the next higher member, by the quiet separation of the carried Z_q. Rejected chains keep their old carriers and their separate boundary terms. Compress this kept precision to the strip slots integrated in the stage, carrying mixed terms with fixed exterior slots in the interaction. For a further guarded subset W take the compression

$$D_W(s) = P_W\Big(\sum_{\varpi \text{ kept at formation}} s^{\operatorname{par}(\operatorname{tags}(\varpi))} D_\varpi \Big) P_W.$$

Here par takes each original cell to its parent on the current grid. This compression retains the keep decision made for the original Schur form. Run the local resolvent construction on this compression at the ordinary level-$(j+1)$ grain, then map the old tags to the common graded cells. The root is formed after compression. A row restriction of the ambient root factors the marginal covariance. A square compression of the ambient root need not factor either the marginal or the conditional covariance. Use the root of the compressed precision here and take the direct sum of the two roots of the reference, then apply the shear transport.

Keep the inherited conditional strip form at its prescribed base and put its complete difference at the two backgrounds in the interaction once. In common stored coordinates let D_- and D_+ be the old and prospective kept precisions, and write $a = (a_{\mathrm{I}}, a_{\mathrm{E}})$ for the active and retained slots. The exact split is

$$\begin{aligned} -\tfrac{1}{2}\langle a, D_- a\rangle = & -\tfrac{1}{2}\langle a_{\mathrm{I}}, D_{+,\mathrm{II}} a_{\mathrm{I}}\rangle - \tfrac{1}{2}\langle a_{\mathrm{E}}, D_{+,\mathrm{EE}} a_{\mathrm{E}}\rangle \\ & - \langle a_{\mathrm{I}}, D_{+,\mathrm{IE}} a_{\mathrm{E}}\rangle - \tfrac{1}{2}\langle a, (D_- - D_+) a\rangle. \end{aligned}$$

Use the first term for the integrated reference Gaussian and keep the second in the prospective retained factor. The last two terms go in the interaction. The rejected chains and coupling-ratio terms are boundary entries. Evaluate their old and prospective values with the other boundary entries. In the physical stage the active strip is a union of whole kept components. The kept prospective mixed block against unintegrated components then vanishes. A subsequent Gaussian cut uses the marginal of the integrated reference. Chains coupling distinct original components

are included among the rejected chains. Together with the kept form and the ratio term they give the original conditional weight at each background. Keep the Haar density, the derivative of the linearizing correction, the row elimination factor, and the scalar vacuum contribution in the assembled weight. In particular the row elimination factor is in the collar integral and the vacuum term is in its prefactor. The fixed-ball identification requires these identities and the same retained strip tests on the same active bond sets.

This identity allows the graded expansion to form its own localized logarithm. For equality of the physical kernels one starts with the same incoming density on each labelled fibre and verifies the same measure transformation there. The incoming kept strip in the real kernel has its unnormalized quadratic and its inherited row factors. When the complete reference is written as a probability Gaussian, its full logarithmic prefactor in Lebesgue coordinates is

$$n_{\mathrm{free}}\nu_{\mathfrak{g}} - \tfrac{1}{2}\operatorname{Tr}\log D + \log z_{\mathrm{col}}.$$

Here n_{free} counts the free Lie-algebra slots of both integrals and $\nu_{\mathfrak{g}} = (\dim \mathfrak{g}/2)\log(2\pi)$ for ordinary Lebesgue measure on a slot. The factor z_{col} consists of the new collar pivot factors in their declared pairing convention. Keep the old strip row factors in their original places. The unnormalized quadratic becomes the normalized Gaussian with this one restoration of its mass. If a previously normalized quadratic is rewritten with a new precision, the determinant factor is the ratio of the two masses. The same rule applies when a coupling difference is transferred between the precision and the potential.

For the Haar delta the derivative at the reference pivot is $L\mathsf{m}_c$. The full coarea away from that point uses the nonlinear chart factors. In the convention that files $|\det \mathsf{m}_c|^{-1}$ in z_{col}, the factor $L^{-\dim \mathfrak{g}}$ per row is in the scalar of the same operation. The flat pivot is $\mathsf{m}_c(\mathbf{1}) = L^{-4}I$, so their product is $L^{3\dim \mathfrak{g}}$ per new row. This is the same factor as the inverse determinant of $L\mathsf{m}_c$ in Haar frames. Count the scalar once at the row's owner. The normalized cut determinant in $\mathcal{J}$ is separate from this full prefactor.

The shear preserves the full determinant, and the affine mean translation has no extra scalar after completing the homogeneous reference square. In the direct logarithm subtract the value at the trivial pair from the field-dependent determinant and new pivot factors. File their trivial values with the fixed Gaussian number of that stage and the collar constant $E_0^{(j)}$.

The new pivot logarithm has a local small-field bound. At the reference row centre its Haar differential $L\mathsf{m}_c$ is the pivot coefficient in the relative logarithmic frame. Conjugations at the two endpoint frames have determinant one. In the small gauge on the complete double block write this coefficient as $L^{-3}I + E_c$. The pivot estimate gives $\|E_c\| \le CL^2\varepsilon$ with ε the local chart size. The subtraction at the trivial field is therefore $-\operatorname{Tr}\log(I + L^3E_c)$, whose modulus is at most $C_L\varepsilon$ on the small branch. This estimate uses only the complete double block. The running physical row bound makes $\varepsilon \le C\theta_i^{-1/2}\ell^C$ on the waiting window, so the polynomial count of stages and anchors gives a uniform summed bound for these local pivot terms. The same bound holds on admitted complex sections with this row bound. The inherited strip coarea keeps its original carrier.

The remaining direct terms and the cluster logarithm are functions on their own formed carriers. A term meeting several components occurs once, through the maximum of their executed stage counts. The constants count the slots and rows of each executed component once.

In the convention with the Wilson-weight adjustment in the stage term, the expression also contains $A(x^{(n+1)} - x^{(n)}, \cdot)$. For a component stopped after n integrations all earlier terms are reevaluated at the common background U, with integrated stored slots set to zero. The compensations then combine with the current Wilson exponent as

$$-A(x^{(n)}, U) + \sum_{\nu=0}^{n-1} A(x^{(\nu+1)} - x^{(\nu)}, U) = -A(x^{(0)}, U).$$

This is the Wilson exponent that the stopped component keeps when its stage logarithms are filed separately. The scalar parts are carried with the component factors. At each prefix apply the real operations to the joint measure in the variables retained by that prefix. Use the same labelled tests and chart in the physical integral and in its graded representation. In the local sharp convention, the sharp collar test is made at the local centre before the shift. Its failure keeps the actual prefix with its failure indicator and the still untransformed variables of that stage. If the global sharp convention is used, the failure branch instead uses the transported test and its transported density. In either case the failed stage contributes no collar integral or stage logarithm. These two choices need separate labelled identifications wherever the tests split the fibre differently.

For a fixed output standing part $\mathfrak{o}'$, let $\mathrm{Str}(\mathfrak{o}')$ consist of the real strings whose output has that standing part. For such a string write K_ς for the product of its real kernels and $\pi_{\mathfrak{o}'}$ for the marginal onto the retained variables. The measure of the standing part is

$$\sum_{\varsigma\in\mathrm{Str}(\mathfrak{o}')}(\pi_{\mathfrak{o}'})_*(\hat\tau_{\mathfrak{o}(\varsigma)}K_\varsigma),$$

where $\hat\tau_{\mathfrak{o}(\varsigma)}$ is the incoming measure already summed over its standing part. Once the stage identity holds for each applied real kernel, it holds under continuation by a later kernel and under this marginal and sum.

The continuation uses the completed determining rows, the field on them and the stored data of the components that still stand. On a healed X, the lower rows have been replaced by top rows, including the top bonds meeting X. The restored field is read on these rows just as it is elsewhere. The integration over the healed history is the identity on all remaining coordinates. At a fixed value of these coordinates the same geometric rules give the next tests and operations. The class window cannot read a birth or mark erased by an earlier healing. In particular, the next normalized reference draw is computed from the current exterior field, the prescribed coupling and frame, the tree slice and the local cutoffs and Wilson weight. Its denominator is the integral of that same local weight on the slice. No term in the incoming healed bracket enters this denominator. Repeating this argument gives the same positive continuation on the common retained space. The healed terms remain part of the incoming density to which this continuation is applied. This permits summation over the healed histories before continuation and exponentiation of the resulting bracket. Keep the measure, carrier and internal coordinates of an island that still stands. An absorbed island is removed from the list of standing islands at filing, after its real operation has been included. Equality here is an identity on the retained variables of the construction, with any physical insertions carried by the corresponding kernels. The coupling coefficient still comes from the whole-lattice small-field construction of $\mathbf{T}$, with its own Taylor expansion. The analytic bounds still use the declared arguments of every graded term, including the evaluations at F_+.

The domains are part of the data carried by the induction. At a new level a boundary entry of $\mathbf{T}$ uses the space of the completed ordinary label. A stage entry uses the spaces of its contributing executed stages, restricted to its formed domain. A main-step entry uses the output space. The regular remainder and pointed entries use their single-scale cores. For the exceptional top boundary read of a pointed entry the restriction lies in the core with radii $\bar a_\iota$; the boundary function is given on that core. This read uses its own core radius. Kept forms and elimination logs use the regular pair on their birth carriers. A standing island keeps its birth core, its measure and its inner factors. When it stands inside a new drop-out, restrict its outer arguments to the admitted smaller space.

For an older entry the output structure is restricted to its original level, carrier and stored argument list. Each earlier tower address is still a test of the output structure, with the right-hand radius for that read. The inclusion into the entry's space uses the radius comparison on that address, including on the interior of a fixed-real-factor section. In a re-made member the retained field is evaluated by its actual row map, with its original bond kind where carried. This re-reads

the same function. The boundary and stage entries are formed on the graded domains from their first occurrence.

To verify the real induction on these domains, fix the outer list and the retained real variables of the outcome. The ordinary sum rule gives the incoming exponent of the first stage. At each executed stage the full-switch evaluation, with the same cutoffs and row coordinates, gives the physical collar integral. At the main step use the real restoration and class split on that retained measure. Composition and the marginal on the output variables give the sum over the real strings with the specified standing part. In a healed region the equality is for this sum. For a standing island the same inner measure and factors are compared at equal outer reads. These identities are evaluations on the real history kernels. The analytic estimates for the new entries use their formation spaces. The prospective regularity guard is the bare plaquette test of the real background, with the crossing clauses of the chosen rule. Its split precedes the evaluation of the analytic collar logarithm.

If instead one imports the ordinary boundary total formed on the whole-layer tower domain, its first use in a collar is already at the first active stage. A stage correction imported with that domain first enters the earlier-stage sum at the second active stage. Each such read would require an extension of that entry or bounds on its original tower stencils. An evaluation using that broader condition on a stencil outside the comparison window requires a bound for

$$L^{2\ell+\min\{\mathsf{p},\ell\}}\left|J^0_{\mathsf{p},Z}(W^u)(b)-J^0_{\mathsf{p},Z}(\mathbf{U})(b)\right|$$

by the available tower margin at nominal layer ℓ and depth p, uniformly on the complex sections and strip discs in use. Here $J^0_{\mathsf{p},Z}$ is the tower current in bare units. The exponent follows from the conversion $J^{[\mathsf{p}]}=L^{3\mathsf{p}}J^0$ and the layer factor $L^{\ell-\min\{\mathsf{p},\ell\}}(L^{\ell-\mathsf{p}})^{-3}$ in the allowed current. The broader condition is a requirement on the domains of those particular terms. To see the importance of the domain for the current bound, at a fringe where the full stencil is fixed by bare determining rows, a one-link perturbation of size $c\alpha_{0,\ell}L^{-2\ell}$ at the identity has a bare current of the same order. For $\mathsf{p}\ge\ell$ its normalized current is of order $c\alpha_{0,\ell}L^{\ell}$, even if the independent current coordinate is zero. Using the tower's own local layer a for the current estimate supplies the weight L^{3a}; converting it to the required one costs $L^{2\ell+\min\{\mathsf{p},\ell\}-3a}$. There is a direct comparison for some such reads. Suppose the weighted value and first difference of the levelled potential change are at most $Ae^{-\gamma d^-(b,\Sigma)}$, where Σ is the source collar and d^- is the ambient weighted distance. Use the tower itself as the ambient sequence for the tower comparison at the common real factor. If its distance dominates d^- on the carriers, the comparison gives

$$CAL^{2\ell+\min\{\mathsf{p},\ell\}-3a}e^{-\min\{\gamma,\delta_{\mathrm{T}}\}d^-(x,\Sigma)}.$$

Indeed, taking the finer scales only decreases the weighted input norm. In the maximum of the tower estimate the two distance factors combine by the triangle inequality. This bound fits the margin when the scale loss is covered by the radius budget or by clearance from the source. At a bare fringe $a=0$ the loss can grow with the depth of the ambient sequence, even at a fixed distance from the source in ambient units. The response estimate for second differences also allows a scale loss in its own ambient layer. Its zero-layer estimate applies at zero in that sequence. The auxiliary tower can have layer zero at a site in a coarse layer of the response sequence. On such a bare fringe the recomputed current is the Wilson current of the shifted links on its full stencil. The physical endpoint identity for this current uses the equations at the full parameter values. Those equations cannot be substituted term by term into the walk series with independent switch parameters. At such a read the argument requires an additional estimate on the decoupled response, or an analytic construction on the settled domains.

4.17.1.39. The crossing test at the main integration. Use the exterior, extendibility, fluctuation and last-shell tests of the main step, together with its regular core and seam data. On this support

the joined datum V'' is $\varepsilon_\star$-regular, so the regular critical orbit $U''_{m,X}$ exists by the real main-step admission assertion [B89a, (1.89)–(1.98)], which uses these inputs before the crossing test is evaluated.

Put $h = m_\flat$. The frozen core is a nonempty component of Z_h, while the input large region at level zero is empty. Thus $h \geq 1$. Work for the moment in the original step-h frame. In level-$(h-1)$ units a coarse cell of this step has side $LM\check{R}_h$. The side of the test cube enlarged by four test margins, with both endpoint blocks allowed for in its averaging reads, is at most

$$(L^2 + 8L)M_2\check{R}_h + 4L < LM\check{R}_h.$$

Here $L^2M_2 \leq M$ and $L \geq 13$. If the enlargement straddles $\partial\Omega_h$, its entire read lies within one coarse layer on either side. The pre-operation region Z_h contains $(\Omega_h^c)^{\sim 2}$, so the connected read lies in the component that meets the exterior at the seam. For the seam of the frozen component this component survives the operation of level h. A healed component would have disappeared, and the class rule excludes a drop-out or a second-class mark at h itself. Operations on disjoint components preserve its entries. The preceding input region is at least eight coarse layers from Ω_h when $h > 1$. For $h = 1$ the initial rule gives the same test buffers with the empty input region and the five-layer margin at P'_0.

Read the seam field of level $h-1$ from the retained one-step factor of $\mathbf{T}_h$ on this core. In the exterior of Ω_h this factor integrates the level-$(h-1)$ links with their averaging delta. In the interior it stores the nonpivot fluctuations, with the pivots fixed by the averaging equation. On the strip read the links through its substitution. The resulting field v satisfies $\mathcal{M}(\mathsf{v}) = V_h$ on the coarse reads. The lower determining rows are unchanged by preparation, and the level-h rows on the frozen core read the original V_h. Consequently the seam fields extend these rows in the original frame of the retained operation.

The two-layer buffer around the approximate failures and the three-layer gauge buffer give the passing approximate test and the passing coarse plaquette test on the test bases in $\Omega_h^{\sim 2}$. The ordinary step estimate on $\Omega_h^{\sim}$ therefore gives $|\partial\mathsf{v} - 1| < \varepsilon_{h-1} + \varepsilon_h$ and the contour bound $CL\delta_{h-1} < \varepsilon_{h-1}$ on the seam reads. These are statements about the retained fields in their original frame. Under a simultaneous change of gauge, lift the level-h transformation constantly on its blocks. The average transforms at the block nodes, and the contour and the relative link against the axial center are conjugated. The latter uses uniqueness of the center on the tested bonds in that axial gauge.

This gives an identity of the retained factors on an ordinary seam carrier whose moved sites belong to the joint integration. The complete lower double blocks lie on the Ω_{h-1} side of its outer boundary, since the old large region Z_{h-1} is separated from $\partial\Omega_h$ by eight coarse layers. The complete level-h blocks for the lift fit in that margin. There are no determining rows below level $h-1$ on these blocks. The level-$(h-1)$ rows are read through the retained Haar factors, the solved pivots or the strip inverse, as above.

Fix the exterior of the main step. Its reference transformation is then fixed. On a union of these complete blocks on which every moved lower coordinate is carried by the joint law, lift its restriction constantly on each block. Change variables in the retained Haar integral and conjugate the free logarithmic coordinates at their initial sites. The Haar and Lebesgue reference measures are preserved. At a row c, the argument of the old delta becomes

$$\mathcal{M}(\mathsf{v}^u)(c)(V_h^u(c))^{-1} = u(c_-)\,[\mathcal{M}(\mathsf{v})(c)V_h(c)^{-1}]\,u(c_-)^{-1}.$$

The delta is unchanged. Evaluate the old indicators and weights on the inverse reads of this same change, including the original strip and reconstructed pivots. Thus their joint integral gives the transported factor at the new V_h with the guards still in place. This applies where the full changed carrier is within the joint law and has no moved genuinely fixed lower endpoint or older mark with independent stored data.

There is also a comparison of the density factors that only reads the level-h field. In the first prepared box the inner cut leaves one coarse $(h+1)$-layer about the hull of the old core. For an old density cube in $\Omega_h \cap Z_h$, the fourfold enlargement fits inside that layer because $4M_2\check{R}_h L^h + L^h < M\check{R}_{h+1}L^{h+1}$, including its bond reads. The last-shell artificial datum keeps the level-h links on this whole test box, in its compensating site gauge. Lifting that gauge to the localized tower identifies the two backgrounds up to gauge and gives the same tested plaquette norms. Thus the passing half-threshold implies the old density factor on these cubes. This implication uses only the literal level-h test reads. In the full last-shell split it also applies to a tested box crossing $\partial\Omega_h$: inside the old core the level-h field is still a target of the retained graph there. When the split is restricted to the inset whose whole test box lies in Ω_h, keep the old density factors at the other cubes. The small-energy estimate for the inset continues to use that full-box condition.

One can keep the old frame on the ordinary seam without changing the reads of the two outer tests. The seam and its full level-h test halo lie strictly inside the inner cut. All genuine rows below level h of the frozen core have their endpoints in Z_h, by the no-inward convention. Choose a site gauge equal to one on those endpoints and the seam halo, and equal to the mixed compensator off an interior neighborhood of them. Use this only to change coordinates of the retained datum. Every endpoint of a fluctuation row lies outside the change, so the mixed reference and the relative fluctuation reads on $\mathbb{B}_0$ keep their values. On the artificial last-shell field the additional gauge is supported strictly inside the kept side, with a bond margin to the cut. The bonds set to one therefore remain one, and the whole artificial datum changes by a site gauge. The last-shell indicator is unchanged. With this gauge the old lower rows and the seam averages are read in their original frame.

For the guarded integral, let τ be the mixed compensator and u the gauge just chosen, both fixed by the exterior and the frame. Write $Y = V_h^u$ for the new target. The old factor reads $V_h = Y^{u^{-1}}$, and a factor written in the mixed frame reads $V_h^\tau = Y^{\tau u^{-1}}$. The change from the mixed to the reset target uses only Haar links in the interior of the inner cut. In the old core these are the dV_h targets of its retained operation. Outside the old core and inside this cut the level-h links belong to the prepared level-h member, since they are in Ω_h and clear of Ω''_{h+1}. They lie in the second Haar integral before the split. At the common face use the original ownership of the links, so the two stocks remain disjoint.

Keep the lower integration variables ξ in their original charts. A native factor $F(V_h, \xi)$, including its stored gauge factor, its contour, its reconstructed pivot and its indicator, is then read as $F(Y^{u^{-1}}, \xi)$. The averaging delta in that factor is

$$\delta\big(\mathcal{M}(\mathsf{v})(Y^{u^{-1}})^{-1}\big) = \delta\big((\mathcal{M}(\mathsf{v}))^u Y^{-1}\big)$$

row by row, by conjugation at the initial endpoint. This transports the output graph and leaves the lower rows and gauge factors in their original variables. Haar measure on the changed targets is preserved at each fixed exterior. Factors that read the mixed target keep the argument $Y^{\tau u^{-1}}$, with the same rule for all their other reads. Thus this change is made with the guards still in the integral.

In the prepared datum use the transformed value at both ends of each retained row. The old lower rows are fixed by u at both endpoints. The level-h rows in the inner cut are the targets just described, and all fluctuation rows and the outer join retain their mixed-frame values. Consequently the prepared datum is the original prepared datum transformed by u on its full row set. Gauge covariance of the averages and of the regular critical orbit identifies the prepared guard with its original read, including its domain of definition. The change of variables does not use a passing value of that guard.

At an earlier face $\partial Z''_{s+1}$, with $m_\flat \le s \le m-2$, the adjoining shell scales are $s, s+1$. The first shell is $Z''_{m_\flat+1} \cap \Omega_{m_\flat}$. Consider a crossing whose corners are all in the inset $\Omega_{m_\flat} \cap X^\natural$ of the

completed main-step estimate. The separation of successive prepared faces is more than the bare plaquette diameter, so every corner has one of these two profiles. For $s \le m-2$ the completed coefficient on shell s satisfies

$$1-\beta_{\mathrm{s}}(1-2^{-(m-s+1)}) \le 1-7\beta_{\mathrm{s}}/8.$$

On shell $s+1$, the unmultiplied threshold is at most $(L_\flat/L)^2(\varepsilon_{s+1}/\varepsilon_s)$ times $F_s\varepsilon_s L^{-2s}$. The ratio of consecutive radii is $1+o(1)$, and $L_\flat < L/2$, so this too lies below $(1-3\beta_{\mathrm{s}}/4)F_s\varepsilon_s L^{-2s}$ for small coupling. The completed main-step estimate on the inset bounds the curvature by the maximum of these corner thresholds, including for astride plaquettes. It therefore implies the added crossing test there. This uses the completed corner estimate. At the prefinal passing stage one still uses the separate prospective test and the real collar response.

We use the frame of the main fluctuation test and put $h = m_\flat$, $Y = Z''_m(X)$, and

$$I_h = \Lambda^{\mathrm{pr}} \cap ((\Omega''_{h+1})^{\sim 2})^c.$$

The fluctuation bonds $\mathbb{B}_0$ extend from the outer box to this inner cutoff. They include the bonds crossing $\partial\Lambda^{\mathrm{pr}}$ and exclude those crossing the boundary of I_h towards its interior.

We first identify the reads of the tested holonomies of V''. A determining row of level $i < m$ in the prepared sequence has both ends in Y, since neither end can lie in the prepared member of level m. A row of level $i \ge h+2$ has an end outside $Z''_i(X)$. Its length is L^i in bare units, less than the separation of $Z''_i(X)$ from $Z''_{i-1}(X)$. For the early separations this follows from $7a_{i-1} > L^i$, and for the compressed ones from the gap of at least one M-cube of level i. Such a row cannot reach $Z''_{h+1}(X)$. A row of level $h+1$ has one end outside $Z''_{h+1}(X)$. Its other end can enter that box by at most L^{h+1}, so it stays on the fluctuation side of the two-cell enlargement. Thus a genuine row missed at the inner cutoff has level at most h. A top row beyond Λ^{pr} instead has its exterior value $V_m = V_\Lambda$.

At a tested plaquette of level q, the reads are the determining rows on that plaquette and, for each bond completed by one-step averaging, the two full blocks of level-$(q-1)$ sites at its ends. The latter blocks lie wholly in the preceding layer in the convention excluding inward bonds. Consequently a test that reads an inner row missed by $\mathbb{B}_0$ has $q \le h+1$. All its corners lie in $Z''_{h+1}(X)$ or within CL^{h+1} of it. The shell between $Z''_{h+1}(X)$ and $Z''_{h+2}(X)$ has width at least $7a_{h+1}$ and spacing L^{h+1}. Every path from this read to a crossing plaquette at ∂Y must cross that shell. After the footprint allowance its weighted length is at least $cM\check{R}_m$. These holonomies can therefore be estimated by $\varepsilon_\star$ with the weight $e^{-c\delta_{\mathrm{c}}M\check{R}_m}$ in the local regularity estimate.

For the remaining interior holonomies compare with the averages of the basic background U_0 in its mixed axial–Landau gauge. The restriction of this datum to $\mathbb{B}_0$ is the actual center V_0 of the main fluctuation test. On $\mathbb{B}_0$ the relative bond logarithms have norm less than δ'_m, and on the exterior top rows the relative bond equals the identity. All lower-level plaquette carriers lie within CL^m of Y. The same holds for a top plaquette using a completed bond, because that bond reads the layer $m-1$. The outer M-layer of Λ^{pr} and the clearance of this box from the joining collar put these carriers in the top layer of the tower for U_0. By Prop. 4.59 the top datum V_Λ has defect at most $C(M)\varepsilon_m$, using the unchanged exterior and its outer tests off the box. The lift to the tower preserves this bound on the tested holonomies. Regularity on that tower gives $C(M)\varepsilon_m L^{-2m}$ in bare plaquette units on the full backward carriers in its top layer. The real change to the mixed gauge preserves this curvature bound. Averaging on a level-q carrier gives the bound $C(M)\varepsilon_m L^{-2(m-q)}$ for the baseline holonomy. A top holonomy with only top rows uses the plaquette estimate of V_Λ directly. The one-step relative average on a completed bond changes it by at most $C(L)\delta'_m$. Multiplication of the four unitary factors gives a defect of at most $C(M)\varepsilon_m + C(L)\delta'_m$ for each of these actual tested holonomies. The remaining tests at the artificial join are the identity or lifted top plaquette holonomies. Their defects obey the top datum bound.

Use the test-by-test profile eq. 3.ccclxxviii in eq. 3.ccclxxix at a corner of a crossing plaquette p. Here $\check{R}_m$ is formed from the running minimum of the frame, which is at least $\theta_m - c_5$, and the exponent of its logarithm is at least 2. Thus $\varepsilon_\star e^{-c\delta_{\rm c} M \check{R}_m} = o(\varepsilon_m)$. The normalized curvature at p is at most $C(M)\varepsilon_m$. Since p has a corner in the prepared member of level m, its bare curvature satisfies

$$|\partial U''_{m,X}(p) - \mathbf{1}| \le C(M)\varepsilon_m L^{-2m}.$$

Finally, $L^{m-k_0-1} = \check{R}_{k_0+1}$ gives $F_j = L_\flat^{-2}\check{R}_{k_0+1}^{2\log_L L_\flat} \to \infty$. With $\varepsilon_m/\varepsilon_j \to 1$, the last bound is strictly below the threshold of $\kappa_\times$ for small coupling. This removes the crossing factor in the main integration on its stated real support, before the reference-law restoration of the free top field.

How a patch recovers its coupling. Each stage gives back to the collar it integrates the increments that this collar did not receive. After m stages the Wilson term is formed with

$$x^{(m)}(\cdot) := \beta - \sum_{i\le k} b_i\, \varphi[\Lambda_i^{(m)}](\cdot),$$

$$\Lambda_i^{(m)} = \Lambda_i \cup (X \setminus Z_i'') \ (k_\flat < i \le k_\flat + m),$$

$$\Lambda_i^{(m)} = \Lambda_i \text{ otherwise}$$

where $\varphi[\Lambda]$ is the profile of the region Λ. The difference $A(x^{(m)} - x^{(0)}, \cdot)$ is carried separately, as a Wilson piece § 4.17. In this way, at each bare plaquette the weights of all the Wilson terms in the exponent still sum to $-\beta$ §4.17.1.1. Hence $x_k'' = x^{(\mathsf{N}_k)}$ coincides with x_k on the part of X outside the box, and equals $x_k + O(1+k-j)$ on the shell of level j. This is what makes the three Wilson terms in the prepared-component operation cancel; two of them carry the constant x_k.

What a patch retains. A component that has passed every test carries the operation below. It is (1.71) of 122 (1989) 355, in which the summation extends over r and over all sequences of regions contained in X, with the prefactor $1/(2^d d!)$ in front. The notation is that of § 4.17, and χ' denotes the test $|B| < \delta'_k/g_k$.

$$\mathbf{T}'_k(X)F = \frac{1}{|W_4|}\sum_{\text{histories in } X,\, r}\int dB\, \delta_{rT_0}(B)\,\chi' \int dV_{k_\flat}\,\chi\,\mathbf{T}_{k_\flat}(C_{k_\flat})\, e^{-\frac12\|\phi^{1/2}D\mathsf{H}B\|^2 - A(x_k''\phi_{\rm c}, U''(e^{ig_kB}, V'')) + V(X^{\sim}, U_k) + \text{numbers}}\, F.$$

The final test is $\chi_k(X^{\sim -6})$, evaluated after the draw of the new V_k. Write H for the level-k hull of the preceding region in X. In the quiet boxed nest, $\Omega_k^c \cap X = H^{\sim 8}$ and $X = H^{\sim 10}$, with the layers taken in the level-k partition. Thus $X^{\sim -6} = H^{\sim 4}$ supplies the test bases left by the carried $\chi_k(\Omega_k^{\sim 4})$. On the passing branch the two tests give the ordinary level-k density bound on X. The localized-background comparison on the margin gives the extendibility implication of Comm. Math. Phys. 122 (1989), p. 378: the new V_k gives an extension whose localized bare plaquettes on $\Omega_k^c \cap X$ are strictly below the $2\varepsilon_k\eta_k^2$ threshold of $\chi_{k,\Lambda}$.

In the recorded frame both V_k and V_Λ are the identity on the box tree and its entrance bond. For each non-tree bond, close it by the tree path. Its value in each configuration is determined by the corresponding cycle product. Express the cycle products by plaquettes with their transports. The box has fixed size in level-k units, so the number of plaquettes used for each bond is bounded in terms of M. The density and minimizer bounds give $|\log(V_k V_\Lambda^{-1})(b)| \le C(M)\varepsilon_k < M_0\varepsilon_k$, by the choice of M_0. Hence $\chi_{k,\Lambda}$ and $\chi(\Lambda^{\rm pr})$ both equal one on the passing support. Merge the two positive density tests into the test of the enlarged region.

For the complete invariant density, the tree constraint can now be removed. Hold the exterior and the entrance root fixed. Recursion along the tree writes box Haar measure as the product of Haar measure on the box gauge transformations and Haar measure on the non-tree links of the slice. The recursive substitutions on the tree and the left and right translations on the other

bonds preserve Haar measure. The gauge transformations lift constantly on each level-k pillar. They preserve the exterior, the stored variables and the inner quadratic and Wilson factors, and transform the outer background covariantly. The localized interaction factors and the remaining tests are invariant. The integral of this invariant extension against box Haar therefore equals the slice integral. If a field in the original gauge is carried, transport it by the same Haar gauge draw. The healed X is removed from the label and its internal history is summed in the bracket § 4.17. For an island the final test fails. The two positive cutoffs and the tree constraint then remain, and X retains

$$\chi_k^{c}\,\chi_{k,\Lambda}\,\chi(\Lambda^{\mathrm{pr}})\,\delta_{\mathrm{ax}}(V_k V_\Lambda^{-1})\,\mathbf{T}'_k(X)$$

as a single block under its frame sum, for as long as it stands ((1.72), (1.104)). The cut-off $\chi(\Lambda^{\mathrm{pr}})$ is absent from (1.102). A drop-out after n stages retains the following: the one-step operations of the levels that were not reached; the bare integrals $\int dV_l\,\delta(\mathcal{M}(V_l)V_{l+1}^{-1})$ of the levels already treated; the core; and the functions present together with the failing factor $1-\chi$. The terms produced up to that point are added to the boundary terms.

Proposition 4.59 (minimiser in a box; Comm. Math. Phys. 122 (1989) 175, Proposition 1)**.** There exist a radius $a_\Lambda > 0$ and a constant $C(M)$, both chosen after M and before γ, with the following property. Suppose the exterior data satisfy $|\partial V_k - \mathbf{1}| < \varepsilon$ with $C(M)\varepsilon \le a_\Lambda$. In the application $\varepsilon \le C\varepsilon_k$, so this is a ceiling on γ. Take the class of continuations with $|\partial V_k - \mathbf{1}| < a_\Lambda$ at those level-k plaquettes whose corners all lie within distance 1 of Λ^{pr}. On this class the background $U_{k,X}(V_k)$ exists. Within the class, the map $V_k|_{\Lambda^{\mathrm{pr}}} \mapsto A(U_{k,X}(V_k))$, which is a function of V_k on the level-k bonds having an end point in Λ^{pr}, has exactly one critical gauge orbit V_Λ. This orbit is the set of its minima in the class, it is analytic in the data, and it satisfies $|\partial V_\Lambda - \mathbf{1}| \le C(M)\varepsilon$. Bałaban states it for the function on fields of mild regularity and for ε sufficiently small, with the bound $CM^5\varepsilon$. The orbit of this class is the one used in (3) and (4). Moreover, $U_0 := U_{k,X}(V_\Lambda)$ satisfies $J(U_0) = 0$ in Λ^{pr}. ■

The proof is a contraction argument in a ball whose radius depends on M, which is why a class of continuations is fixed. In axial gauge the Hessian is bounded below by $c(M) > 0$; Bałaban obtains the lower bound cM^{-5}. Indeed, the Hessian is at least $c\|\partial B\|^2$ less a small multiple of $\|B\|^2$. In a tree gauge one has the Poincaré inequality

$$\sum_b |B(b)|^2 \le C(M) \sum_p |\partial B(p)|^2$$

(122 (1989) 355, (1.7)–(1.9), with CM^5). The corresponding inequality for a single bond of the nest, $|B(b)|^2 \le C(M\check{R}_k)^C \sum_p |\partial B(p)|^2$, yields the factor of the test in (3). It is (1.78) there, with every bond and every plaquette taken in its own lattice. The power of $\check{R}_k$ has two origins: the side of a shell in its own lattice, which is at most $CML\check{R}_k$ sites, and the number $\mathsf{N}_k + 1$ of shells, where $\mathsf{N}_k = \check{R}_k$. The first form serves the level-k variable V'_k and the uniqueness of V_Λ. The form for one bond serves the multi-scale field B.

A gauge for the configuration generated in a healing. The configuration generated in a healing must belong to the analyticity spaces. To this end it is put in the form $e^{i\eta_k A^{\mathrm{g}}}U$, where A^{g} and its covariant differences $\nabla_U A^{\mathrm{g}}$ (∇_μ^U as in the Notation) are smaller than $(1 + c\beta_{\mathrm{s}})\alpha_{1,k}$ § 4.17. On boxes of side $O(1)$ around the sites y there are local gauges $\hat{u}_y$, with potentials $A^{(y)}$. They are glued together by a partition of unity (χ_y) taken in the quaternions,

$$u^{\mathrm{g}}(x) = \mathcal{N}\Big(\sum_y \chi_y(x)\hat{u}_y(x)\Big),$$

$$\mathcal{N}(m) = (\det m)^{-1/2} m,$$

and then $A^{\mathrm{g}} = \sum_y \chi_y A^{(y)}$ is a mean whose weights sum to 1. This holds apart from quadratic terms and from terms comparable to the differences $\hat{u}_y \hat{u}_{y'}^{-1} - \mathbf{1}$ between local gauges at neighbouring sites. In $\nabla_U A^{\mathrm{g}}$, the term $\sum_y (\partial\chi_y) A^{(y)}$ has weights summing to 0. It is therefore comparable in size to the differences between the potentials of neighbouring sites, which have a leading part in common. The square root is the branch on $|\det m - 1| < 1$ that takes the value 1 at $\det m = 1$. The mean in the display lies in this domain. It is the product of one local gauge $\hat{u}_y$ with $\mathbf{1}$ plus an average of the small differences $\hat{u}_{y'} \hat{u}_y^{-1} - \mathbf{1}$ between local gauges at neighbouring sites. The real linear span of SU(2) is $\mathbb{H}$. There $\det m = |m|^2$, and $\mathcal{N}(m) = m/|m| \in G$ for $0 < |m| < \sqrt{2}$. Wherever the root is defined, $\mathcal{N}$ is holomorphic and bi-covariant: $\mathcal{N}(u_1 m u_2) = u_1 \mathcal{N}(m) u_2$ whenever $u_1, u_2 \in G^c$. Hence the same construction applies to complex fields.

[2.2] Finer sequences: nested constraint sets, nested critical configurations.

Suppose that $\mathbf{S}'$ is finer than $\mathbf{S}$ and that U^* is a real configuration on Ω_0, and put $V := M_{\mathbf{B}}(U^*)$ and $V' := M_{\mathbf{B}'}(U^*|_{\Omega_0'})$. Given a configuration U' on Ω_0', we write $\widetilde{U'}$ for the configuration obtained by extending it by U^*.

(a) Whenever $M_{\mathbf{B}'}(U') = V'$, one has $M_{\mathbf{B}}(\widetilde{U'}) = V$; moreover, the difference $A_{\Omega_0}(\widetilde{U'}) - A_{\Omega_0'}(U')$ is independent of U'.

(b) Suppose that $M_{\mathbf{B}'}(U') = V''$, where $V'' = V'$ holds on the bare member Γ_0'. Then $(\mathcal{M}^q \widetilde{U'})(c) = V''(c)$ holds for each bond $c \in \Gamma_q \cap \Gamma_q'$, and the difference of the two actions takes the same value.

(c) When U^* is a regular critical configuration for $(\mathbf{B}, V)$, its restriction $U^*|_{\Omega_0'}$ is a regular critical configuration for $(\mathbf{B}', V')$.

[2.4] Comparison of two backgrounds, at real and at complex parameters.

Suppose that $\mathbf{S}, \mathbf{S}'$ agree on a region W which is collared in $\mathbf{S}$. Let p be a bare plaquette lying well inside W, in the unit cell y, and set $D := d_{\mathbf{B}}(y, W^c)$. Let f_1, f_2 be holomorphic functions on $\{\|w\| < r\}$ whose modulus is at most $\frac{1}{2}\mathsf{m}^2 \eta^4 \xi(p)^{-4}$. Suppose further that for each real w there exist real data V_w on $\mathbf{B}$ and V_w' on $\mathbf{B}'$ that are both ε_1-regular, with $\varepsilon_1 \le \mathsf{m}$, and equal on W, and for which $f_1(w) = a_p(U_{\mathbf{B}}(V_w))$ and $f_2(w) = a_p(U_{\mathbf{B}'}(V_w'))$. Then

$$|f_1 - f_2| \le C \varepsilon_1^2 \, e^{-\delta_0 D} \, \eta^4 \xi(p)^{-4} \text{ at real } w,$$

$$|f_1 - f_2| \le C \, \mathsf{m}^2 \, e^{-\frac{2}{3}\delta_0 D} \, \eta^4 \xi(p)^{-4} \text{ for } \|w\| \le \tfrac{1}{5} r.$$

[2.5] Averages of a complex configuration near 1.

Let $\mathbf{A}$ be a bare 1-form with values in $\mathfrak{g}^c$, and suppose that $s := \sup |\mathbf{A}|$ is sufficiently small. Then, for $0 \le n \le k$ and for each bond c of $\mathbb{T}^{(n)}$, the complex average is defined, and

$$\mathcal{M}^n(e^{i\eta \mathbf{A}})(c) = e^{i \mathcal{Q}_n(\mathbf{1}; \mathbf{A})(c)},$$

$$|\mathcal{Q}_n(\mathbf{1}; \mathbf{A})(c)| \le C \, L^n \eta \, s = C \, \xi_n \, s,$$

where $\mathcal{Q}_n(\mathbf{1}; \mathbf{A})(c)$ is holomorphic in $\mathbf{A}$ and real when $\mathbf{A}$ is real. It depends only on the values of $\mathbf{A}$ in the two blocks $B^n(c_\pm)$, so the supremum may be taken over these blocks alone. (Here the map $\mathcal{Q}_n(\mathbf{1}; \cdot)$ is formed with the single spacing $\eta = \eta_k$ for every $n \le k$.)

[2.10] The basic and the glued background from the exterior field alone.

Let V^{e} be a configuration on $\mathsf{C}(Z) \setminus E$ with values in G, with ε small and with $|\partial V^{\mathrm{e}}(p') - \mathbf{1}| < \varepsilon$ at every plaquette made up of bonds of this configuration. Let G be an arbitrary gauge tree.

First part (the mixed gauge). The objects V_Λ, U_0, U_0°, B^Λ, $\mathbb{A}_0$, w, $\tilde{w}$, t_0, $\tilde{U}_0 = (U_0^\circ)^{\tilde{w}^{-1}}$ and $V_0 = M_{\mathbf{B}''}(\tilde{U}_0)|_{\mathbb{B}_0}$ are well defined and real. They are determined by V^{e} and G, and they satisfy

$$|\partial V_\Lambda - \mathbf{1}| < C\varepsilon;$$

$$|\mathbb{A}_0|, |\nabla^\eta \mathbb{A}_0|, |D_{\mathbf{1}}^* D_{\mathbf{1}} \mathbb{A}_0| \le C\varepsilon \text{ on } \Lambda^-;$$

$$|\mathsf{t}_0 - \mathbf{1}| \le C\varepsilon;$$

$$|V_0(c) - \mathbf{1}| \le C L^{i-k}\varepsilon \ (c \in \mathbb{B}_0 \text{ of level } i < k).$$

Second part (the quadratic form). Assume in addition that $(\mathbf{1}, V^{\mathrm{in}})$ is ε-regular on $\mathbf{B}^{\mathrm{in}}$. Then the glued datum V^0 is $C\varepsilon$-regular, and U^0, Φ_{V^0}, $\mathcal{H}_{\mathbf{B}^{\mathrm{f}},V^0}$ and H_1 are defined. The operator H_1 is complex-linear and takes real values on real increments. In the unit cell y it obeys $\xi|\mathsf{H}_1\mathsf{f}|$, $\xi^2|\nabla_{U^0}\mathsf{H}_1\mathsf{f}| \le \mathsf{h}(y;\mathsf{f})$. Finally, for real f one has $\mathsf{q}_0(\mathsf{f}) = \sum_p \eta^4 \zeta(p) |(D_{U^0}\mathsf{H}_1\mathsf{f})(p)|^2 \ge 0$.

The minimiser on Λ^{pr} and the inner background. **[1.6] Regular data.** (a) *Stability.* Suppose that $|\partial V(p') - \mathbf{1}| < \varepsilon$, that on the double blocks $B(b_-) \cup B(b_+)$, where b runs over the outer bonds of p', the configuration V is moderately regular, and that b is real and $|\mathsf{b}|_\infty = s$ is small. Under these conditions $|\partial(e^{i\mathsf{b}}V)(p') - \mathbf{1}| < \varepsilon + K_{\mathrm{s}} s$. (b) *Averages of a regular configuration.* Let $\Omega'_\bullet$ be finer than $\Omega_\bullet$, and let $U \in \mathfrak{U}(\mathbf{B};\varepsilon)$. Then $M_{\mathbf{B}'}(U)$ is $C\varepsilon$-regular. The local form is as follows: whenever $|\partial U(p) - \mathbf{1}| \le \alpha\eta^2 \xi_j^{-2}$ holds in the blocks $B^j(x)$ around the corners of a plaquette p' at level j, one has $|\partial(M_{\mathbf{B}'}U)(p') - \mathbf{1}| \le 2\alpha$. (c) *Lifts.* Each holonomy of $V^\downarrow = M_{\mathbf{B}_k(\Omega)}(Q_k^{s*}V)$ is either $\mathbf{1}$ or a plaquette variable of V. Moreover $(e^{i\mathsf{b}}V)^\downarrow = e^{i\mathsf{b}^\downarrow}V^\downarrow$. Here $\mathsf{b}^\downarrow(b) = \mathsf{b}(c)$ when b crosses from the block of c_- into the block of c_+, and the lifted increment takes the value 0 on bonds lying inside a block.

[1.14] The minimiser over Λ^{pr}. Take a geometry of the main step, and let $0 < \varepsilon \le \varepsilon'$ with $C\varepsilon' \le a_\Lambda$. Fix a configuration V^{e} on $\mathsf{C}(Z) \setminus E$ satisfying $|\partial V^{\mathrm{e}}(p') - \mathbf{1}| < \varepsilon$ at the plaquettes made up of these bonds. Let V' be an extension of it to $\mathsf{C}(Z)$ that satisfies $|\partial V'(p') - \mathbf{1}| < \varepsilon'$ on $S(P)$. We denote by $\mathcal{V}(V^{\mathrm{e}})$ the set of extensions V for which $|\partial V(p') - \mathbf{1}| < a_\Lambda$ on $S(P)$.

(o) There is always an extension for which $\varepsilon' = 4N_{\mathrm{sh}}\varepsilon$, with the inequality taken in its non-strict form.

(a) On $\mathcal{V}(V^{\mathrm{e}})$ the configuration $U_{k,Z}(V)$ is defined. The map $V \mapsto A_Z(U_{k,Z}(V))$ is real-analytic and is invariant under gauge transformations at the sites of P.

(b) The critical points of this map form a single orbit. The minimum value is attained on that orbit and nowhere else. The orbit contains exactly one element $V_\Lambda = V_\Lambda(V^{\mathrm{e}})$ that is equal to $\mathbf{1}$ on Tr.

(c) On $S(P)$ one has $|\partial V_\Lambda(p') - \mathbf{1}| < (1 + 4K_\Lambda)\varepsilon'$. Every bare plaquette p of Z, lying in the cell y of $\mathbf{B}^Z$, satisfies

$$|\partial U_{k,Z}(V_\Lambda)(p) - \mathbf{1}| \le |\partial U_{k,Z}(V')(p) - \mathbf{1}| + K_5\, \varepsilon'\, e^{-\delta_0 d_{\mathbf{B}^Z}(y,\Lambda^{\mathrm{pr}})}\, \vartheta_y.$$

(d) *Continuation.* If ε' is reduced further, a unique map $\mathsf{b} \mapsto B_\Lambda(\mathsf{b}) \in \mathcal{X}^{\mathbb{C}}$ exists with the following properties. It is defined on the complex increments b on $\mathsf{C}(Z) \setminus E$ with $|\mathsf{b}|_\infty < a_\Lambda$. It satisfies $|B_\Lambda|_\infty \le r_\Lambda$. At every such increment, its value $B_\Lambda(\mathsf{b})$ is a critical point, in the directions of $\mathcal{X}^{\mathbb{C}}$, of the function

$$\delta \mapsto \sum_p \left[a_p(\mathsf{W}_{V_\Lambda^\downarrow}(\mathsf{b}^\downarrow + \delta)) - a_p(\mathsf{W}_{V_\Lambda^\downarrow}(\mathsf{b}^\downarrow)) \right]$$

. This map is holomorphic and satisfies $B_\Lambda(0) = 0$, and

$$|B_\Lambda(\mathsf{b})|_\infty \le 2K_\Lambda |\mathsf{b}|_\infty,$$

$$V_\Lambda(e^{i\mathsf{b}}V^{\mathrm{e}}) = e^{iB_\Lambda(\mathsf{b})}V_\Lambda(V^{\mathrm{e}}) \text{ on } E \text{ for real small } \mathsf{b}.$$

4.17.1.40. Rare patches: gains per creation and costs per site.

Lemma 4.60 (one factor per creation)**.** The following holds on the support of a term, at step $t = k+1$. (i) When the plaquette test of step t fails at $\square$, each real U that has the stored averages satisfies $\sum_{p \text{ near } \square} a_p(U) \geq c\,\varepsilon_t^2$. (ii) When the approximate fluctuation test fails, this sum, increased by the part of GF_k located at the blocks near $\square$, is bounded below by $c\,\delta_k^2$. (In what follows, $\mathrm{GF}_j(X;V_j)$ denotes the part of $\mathrm{GF}_j(V_j)$ located at the blocks in X.) (iii) One has $\langle B, \Delta^{(k)}B\rangle \geq c\|B\|^2$, and the bound extends to the kept conditional forms §4.17.1.3. Consequently a failed fluctuation test yields

$$\tfrac{1}{2}\langle \mathcal{A}_k^{\mathrm{s}}, \Delta^{(k)}\mathcal{A}_k^{\mathrm{s}}\rangle \geq c\,\delta_k^2/g_k^2$$

. The same is true of a failed collar test or rim test, because the threshold of either differs from δ_k by a constant factor, namely $1/(4\Theta_1)$ or θ_c §4.17.1.1. ■

For (i) and (ii) the contraposition also establishes existence of the objects read by the tests. For (iii), once the actual background is admitted, use the group angle, including the value π at the cut of the logarithm. Let d be the tested threshold, including the fixed collar or rim fraction where appropriate. On the cutoff zone an angle below d lies where the cutoff is zero, so the substitution leaves it unchanged. At the other substituted bonds take the strip comparison bound $C\varepsilon_k e^{-\check{R}_k}$ below $d/2$, using the fixed lower bound on d/δ_k. The triangle inequality then shows that an old angle at least d requires a new angle at least $d/2$. This includes an undefined old logarithm. The energy payment uses the coercive form in the retained free coordinates of that test. Since the coupling at that place is x_{t-1}, every created cell produces the factor $e^{-c\,p_0(g_t)^2}$. Inside $\mathbf{R}$ the tests work with the threshold δ_j', and the gauge trees they use have length at most $C\check{R}_j^C$. When a bond variable is large in a tree gauge, all that follows is that some plaquette exceeds $c\check{R}_j^{-C}\delta_j'$. For this reason these tests give no more than $e^{-c\check{R}_j^{-C}p_1(g_j)^2} \leq e^{-p_0(g_j)}$, and this inequality requires $2\mathfrak{p}_1 - Cr > \mathfrak{p}_0$.

The energy of the actual averages. The small-action alternative admits a depth-uniform estimate even when the bare curvature does not satisfy eq. 3.xii. The additional ingredient is an ℓ^1 estimate for the *local quadratic remainder*, followed by its propagation through the positive flux matrices. We give the argument with its exact carrier. All plaquette sums count each independent orientation once. For a real SU(2) field write

$$a_p(U) = 1 - \tfrac{1}{2}\operatorname{Tr} U_p = \tfrac{1}{2}\|1 - U_p\|_{\mathrm{op}}^2.$$

All group logarithms used in the proof are the branches near the identity. The actual total measurable average retains its prescribed fallback outside those branches.

Lemma 4.61 (Local quadratic flux remainder)**.** Fix the odd block size L. There are $d_L > 0$ and $C_L < \infty$ with the following property. For a coarse plaquette C, let $D(C)$ be the rectangular site box formed by its four centered fine endpoint blocks. If the fine plaquette logarithms $F_p = -i\log U_p$, for plaquettes internal to $D(C)$, satisfy $\max|F_p| \leq d_L$, all averages used by C are on their regular branches, and

$$F_C^+ = \sum_p T_{C,p}O_{C,p}F_p + R_C, \qquad |R_C| \leq C_L \sum_{p\subset D(C)} |F_p|^2. \tag{4.cdliii}$$

Here T is the positive scalar flux matrix of eq. 3.v, with row sum L^2 and column sum L^{-2}, and the color factors are orthogonal. After periodic identifications, a collapsed coefficient may be a sum of these transported terms; its norm is still dominated by the corresponding scalar coefficient with multiplicity. ■

Proof. The union of the four endpoint blocks is already a rectangular site box. In the two directions of C it contains two adjacent blocks, and in each transverse direction it contains one. Every internal coordinate-order contour and every joining straight segment in the four actual averages stays in this box. The fine plaquettes in the sheet formula eq. 3.v also have all four corners there.

Choose the coordinate comb from the initial corner of C. Its paths stay in this rectangle. The cycle made by a remaining fine edge and its two comb paths is filled by commuting coordinate steps inside the same rectangle. Every inserted elementary plaquette is therefore internal to $D(C)$; no extra layer is needed. The unitary product inequality and the small logarithm give, in this real gauge, $U_b = e^{ia_b}$ and

$$\|a\|_{\ell^2(D(C))} \le C_L \|F\|_{\ell^2(D(C))}. \tag{4.cdliv}$$

Indeed each edge cycle uses a finite number of internal plaquettes, bounded for fixed L, and the resulting finite incidence matrix has bounded row and column sums. This proves the estimate directly, as in eq. 3.viii.

Decrease d_L so that these finite lists of a_b's belong to one common small analytic ball. The local implicit mean and the coarse plaquette logarithm are then defined there. The fine logarithm has linear part $\mathrm{d}a$, and the coarse logarithm has linear part $\mathrm{d}_c Q a$. Their second derivatives on the smaller ball are bounded finite matrices. Their remainders are consequently bounded by $C_L\|a\|_2^2$. The identity $\mathrm{d}_c Q a = T\mathrm{d}a$ follows from eq. 3.iv–eq. 3.v: the endpoint gradient cancels and the remaining sum is the positive sheet flux. Substituting the fine Taylor remainder, using eq. 4.cdliv, and returning to the original real frame proves eq. 4.cdliii. All frame changes are unitary, even when open transports or global holonomies are large.

On a periodic lattice the calculation is made on the finite lifted rectangle and then projected. Repeated images are retained in the finite incidence count. The row and column sums of T are exactly the periodic counts in eq. 3.vi. This is an identity of actual classical averaging functions, not an equality of quantum measures on different covers. □

For any family of such rectangles, each fine plaquette occurs in at most C_L of them, with the same bound for their lifted multiplicities. Thus, if all fine plaquettes read by a selected set of coarse plaquettes belong to a set P, the restricted transported flux matrix A and remainder satisfy

$$\|A\|_{\ell^2\to\ell^2} \le 1, \qquad \|A\|_{\ell^1\to\ell^1} \le \rho := L^{-2} < 1, \qquad \|R\|_{\ell^1} \le C_L \|F|_P\|_{\ell^2}^2. \tag{4.cdlv}$$

For the first inequality use the row sum L^2, the column sum L^{-2}, and Cauchy–Schwarz as in eq. 3.vii. Deleting rows or columns only decreases these positive majorants. The last inequality sums the actual local quadratic bound.

We next specify the backward carriers. Let S be a finite set of plaquettes at level t. For $C \in S$ and $0 \le j \le t$, let $D_j(C)$ be the union of the four centered blocks of side L^{t-j} on the level-j grid, centered at the corners of C. Put

$$P_j(S) = \bigcup_{C\in S} \{p \text{ at level } j : \text{ every corner of } p \text{ lies in } D_j(C)\}. \tag{4.cdlvi}$$

The union is taken without multiplicity. We write $\mathcal{N}_0(S) = P_0(S)$: a bare plaquette is included exactly when all its corners lie in the four blocks $B^t(y)$ belonging to one member C of S. Also $S \subset P_t(S)$.

These masks are closed under all the required one-step readouts. If $C' \in P_{j+1}(S)$, choose a final C whose $D_{j+1}(C)$ contains its four corners. The centered odd-L block tiling decomposes $D_j(C)$ into the children of the sites in $D_{j+1}(C)$. Consequently the four child blocks at the corners of C' lie in $D_j(C)$, and every fine plaquette in their rectangle belongs to $P_j(S)$. The same assertion for a bond uses its two child blocks. It proves locality and regular-branch admission for the actual averages on the cone, including their internal means, without any assumption on the exterior field. Periodic masks can contain additional plaquettes after projection; the inclusion and the restricted

matrix bounds remain valid. They use sets of projected plaquettes and the periodic scalar matrix entries, rather than charging an energy once per image.

Proposition 4.62 (Small total energy on an exact averaging cone)**.** There is $e_*(L) > 0$, independent of depth, compatible volume, real holonomy and the exterior field, such that every real U and every finite S satisfy

$$\sum_{p\in\mathcal{N}_0(S)} a_p(U) \geq \min\left\{ \tfrac{1}{4}\sum_{C\in S} \|1-\partial\mathcal{M}^t U(C)\|_{\mathrm{op}}^2, \tfrac{1}{2}e_* \right\}. \tag{4.cdlvii}$$

When the left side is less than $e_*/2$, all the actual means required by this cone are on their regular branches. ■

Proof. The empty set is immediate. Write $E = \sum_{p\in P_0(S)} a_p(U)$. Choose the local threshold $d_L \leq 1$ in the lemma smaller if necessary. For $0 \leq s \leq d_L$,

$$s^2 \leq \tfrac{9}{4}(1-\cos s).$$

For example $1-\cos s \geq s^2/2 - s^4/24 \geq (11/24)s^2$ proves this for $s \leq 1$. Set

$$b_L = \min\left\{ \tfrac{3}{4}d_L, \frac{3(1-\rho)}{32C_L} \right\}.$$

Choose $E_L > 0$ so small that

$$E_L < 1-\cos d_L, \qquad \tfrac{9}{4}E_L < b_L^2, \qquad e_* = 2E_L.$$

All constants are chosen before the depth or the volume.

Suppose first $E < E_L$. Each fine plaquette in $P_0(S)$ has its small logarithm, and, with F_0 restricted to that mask,

$$a_0^2 := \|F_0\|_2^2 \leq \tfrac{9}{4}E < b_L^2. \tag{4.cdlviii}$$

If $a_0 = 0$, the local remainder and flux identities inductively give zero logarithmic curvature on every cone mask. The flat local means are admitted, so the result follows.

Otherwise construct successive actual averages on the masks. While $a_j = \|F_j|_{P_j(S)}\|_2 \leq (4/3)a_0$, the local threshold is respected, so the next cone means and their plaquette logarithms exist by the lemma. Their exact equations are $F_{j+1} = A_jF_j + R_j$, with eq. 4.cdlv. The color matrices may depend on the actual fields; only their stated norms are used. Unrolling these identities gives

$$F_n = A_{n-1}\cdots A_0F_0 + \sum_{j=0}^{n-1} A_{n-1}\cdots A_{j+1}R_j. \tag{4.cdlix}$$

An empty product is the identity. The first term has ℓ^2 norm at most a_0. For a remainder propagated through $n-1-j$ matrices, use first its ℓ^1 norm:

$$\|A_{n-1}\cdots A_{j+1}R_j\|_2 \leq \|A_{n-1}\cdots A_{j+1}R_j\|_1 \leq \rho^{n-1-j}C_La_j^2.$$

Thus the induction hypothesis gives

$$a_n \leq a_0 + \frac{C_L}{1-\rho}\left(\tfrac{4}{3}a_0\right)^2 \leq \tfrac{7}{6}a_0 < \tfrac{4}{3}a_0. \tag{4.cdlx}$$

The middle inequality uses the specified choice of b_L. This proves the strict bootstrap, and at the same time admits every required local mean. No unregularized exterior plaquette has entered the induction.

For a real small logarithm, $\|1-e^{iF}\|_{\mathrm{op}} \leq |F|$. Since $S \subset P_t(S)$,

$$\sum_{C\in S} \|1-\partial\mathcal{M}^t U(C)\|_{\mathrm{op}}^2 \leq a_t^2 \leq \tfrac{16}{9}a_0^2 \leq 4E.$$

This proves the first alternative. If instead $E \ge E_L = e_*/2$, the minimum in eq. 4.cdlvii is at most E without any regularity assumption. This proves the assertion for all real configurations under the total measurable averaging convention. □

4.17.1.41. The plaquette and approximate fluctuation witnesses. Fix a test cube $\square$ of step $t = k+1$, and put $D = \square^{\sim 4}$. Use the separated M_1-tower of D, with the geometry and smallness floors of the real tower estimates. The test partitions are aligned, $\check{R}_t \le \check{R}_k \le L\check{R}_t$ are powers of L, and there are at least 6ρ top sites in each torus direction. The third enlargement is inside the top member D_t. In bare units one layer has width $w = M_2\check{R}_t L^t$. On the third enlargement the distance from the complement of D is at least w, and the part of D omitted by D_t is within $2M_1L^t$ of that complement. A level-t site of the third enlargement is therefore a node. The noncentral sites of its level-k block are free to be gauged: any node from a lower row is within $2M_1L^t + L^{t-1} < w$ of the complement of D. The blocks themselves lie in the third enlargement by alignment of the grids.

For the plaquette test, let S_t be the level-t plaquettes with all corners in D. Use the Wilson action on $\mathcal{N}_0(S_t)$ from Prop. 4.62. A real evaluation used to pay for this test must reproduce V_t on the bonds of S_t. If the action there is less than $c\varepsilon_t^2 < e_*/2$, then

$$|\partial V_t - \mathbf{1}| < 2\sqrt{c}\,\varepsilon_t \qquad \text{on } S_t.$$

Indeed, since the action is less than $e_*/2$, the minimum in eq. 4.cdlvii must be attained by its squared-defect term. The lifted datum of the tower, obtained by lifting V_t and then taking the determining averages, is ε_1-regular for $\varepsilon_1 = 3\sqrt{c}\,\varepsilon_t$. For the real orbit estimate use the test's regularity radius ε_{v}, with $\varepsilon_1 \le a_1$ and $B_3\varepsilon_1 \le \varepsilon_{\mathrm{v}} \le a_0$. The real orbit exists and its curvature in the top member is less than $B_3\varepsilon_1 L^{-2t}$ in bare units. The plaquette test on $\square^{\sim}$ passes when $3B_3\sqrt{c} < 1$. This also rules out an undefined orbit under the same small-action bound.

For the approximate fluctuation test, also include the level-k plaquettes in the full double blocks used in the one-step rigidity estimate for each tested bond, including the plaquettes crossing the common face. Call their union S_k. Let E be the Wilson action on the union of the two backward carriers. Require that the compatible real field reproduce V_k on the bonds of S_k and that $\mathcal{M}(V_k) = V_t$ at the corresponding top rows. Let $G_\square$ be the unweighted part of the soft axial gauge action containing the contour variables used by these comparisons. At birth the approximate test is placed one coarse-cell layer inside the region of the gauge factors. Since $M \ge L^2M_2$, three test-cube layers fit inside this margin, so the block centers of the third enlargement carry those factors. If $E + G_\square < c\delta_k^2 < e_*/2$, the same energy argument bounds the level-k and level-t plaquette defects by $2\sqrt{c}\,\delta_k$ on their respective sets. The tower estimate now applies with $\varepsilon_1 = 3\sqrt{c}\,\delta_k$, again with $\varepsilon_1 \le a_1$ and $B_3\varepsilon_1 \le \varepsilon_{\mathrm{v}} \le a_0$. The two endpoint blocks needed for a bond in $\square^{\sim 2}$ lie in the top region with their averaging carriers. Averaging the bare background to level k bounds the plaquette defects of $\mathcal{M}^k U_{t,\square}$ on these blocks by $2B_3\varepsilon_1 L^{-2}$. For $B_3\varepsilon_1 \le L^2 c_2^{\mathrm{elem}}$ the contour means of this background exist. The small plaquette bound on V_k ensures the existence of its corresponding contour means as well.

In the chosen block axial representative the background contour means equal $\mathbf{1}$. For the means of V_k, the identity $1 - \frac{1}{2}\,\mathrm{Re}\,\mathrm{tr}\,H = \frac{1}{2}|H - \mathbf{1}|^2$ bounds each contour defect by $(2G_\square)^{1/2}$. On the top bond of a double block the background has average V_t by the determining constraint and the identity that averaging the lift recovers V_t. The block centers are fixed in the axial normalization. The other field has average V_t there by the recorded one-step constraint. By Lem. 3.8 their relative bond defect is at most $C\sqrt{c}\,\delta_k$ on the tested nonpivot bonds. Choosing c smaller makes this less than $2\delta_k$, so the approximate test passes. The contour existence just checked also makes this deduction valid with the sitewise extension of the gauge factor at undefined means.

These two contrapositives apply to every real evaluation with the specified averaging and gauge readouts, with constants independent of depth. To use them in a history integral, the recorded compatible rows must cover these carriers, the gauge action must be the one retained on the required blocks, and the paying sets must have the ownership and floor exclusion in eq. 4.div. The true

fluctuation, collar and rim tests additionally use the coercivity and domination in the actual stored coordinates.

4.17.1.42. Ordinary creation witnesses in a later drop-out. The paying region of an ordinary creation is separated from older large fields when the creation occurs. In a sufficiently later operation it belongs to the protected floor; the new payment of that operation is taken outside the floor. These are different exclusions.

Consider a failed P_t- or Q_t-test on a cube $\square$, put $j = t-1$, and let $\mathfrak{c}$ be the level-t cell containing $\square$. The birth geometry gives

$$\mathfrak{c}^{\sim} \subset \Omega_t^c \cap D_{\mathrm{birth}},$$

where D_{birth} is its component of Z_t and the enlargement is by one layer of level-t cells. Containment in Ω_t^c uses the two layers around Q_t' and the four around P_t'; when $t > 1$, the birth margin also places this neighbourhood inside Λ_j. It also separates it from the older live region and its marks.

The exact averaging cones used above fit in this neighbourhood. Indeed, $M \geq L^2 M_2$, the partitions are aligned, and $L \geq 3$. Thus $\square^{\sim 4} \subset \mathfrak{c}^{\sim}$. Both sets are unions of complete level-t blocks. For a plaquette whose corners lie in $\square^{\sim 4}$, all four blocks in eq. 4.cdlvi therefore lie in $\mathfrak{c}^{\sim}$. The double blocks for the approximate test lie in the third enlargement, so the same conclusion holds for their level-j ancestor cones. Every paying bare plaquette has all its corners in $\mathfrak{c}^{\sim}$; no enlargement of the paying field's domain is hidden in this use of the averaging estimate.

Suppose this creation is carried into a later drop-out X without being absorbed into a healed component or an island. Every ordinary step leaves the older entries unchanged. If an operation of level m contains this creation, its quiet-window condition gives $t \leq m_\flat$. A drop-out from that operation re-makes only entries of levels $l > m_\flat$. It therefore leaves both Ω_j and Ω_t, and hence the determining rows of level j on $\mathfrak{c}^{\sim}$, unchanged. Its stages integrate stored fields only at levels at least $m_\flat$, so the birth field V_j, its gauge factors, and its one-step averaging constraints are retained. This argument applies at every later drop-out on the same branch.

Consequently, for a compatible real evaluation on X, the required identities are

$$\mathcal{M}^j U = V_j \quad \text{on the protected birth rows,} \qquad \mathcal{M}(V_j) = V_t \quad \text{on the retained step-}t\text{ rows.}$$

Together with locality and the semigroup identity, these give $\mathcal{M}^t U = V_t$ on every top row used in the birth test. At $t = 1$, the first identity uses the stored bare field wherever the action field is completed by that field. The recorded gauge readouts in the Q_t case are likewise the original ones. The exception for an earlier main-step drop-out, whose later constraint reads the density argument rather than its integrated datum, does not replace these protected birth rows. A mark older than the birth is outside its paying neighbourhood; a mark containing the birth has its re-made levels strictly above t.

The floor containment has the corresponding precise meaning. For a class component u containing the old creation, the geometric statement $\mathfrak{c}^{\sim} \subset \mathsf{Fl}(u)$ contains the exact cones just checked. It separates this old Wilson payment from the fresh payment of u, which uses no plaquette with a corner in $\mathsf{Fl}(u)$. The same floor rule places each older marked component enlarged by two layers of unit cubes at its own level inside the later floor, so that a fresh payment which uses that neighborhood in the older window cannot spend the later window's complement. At a fixed birth level, one recorded witness per creation cell has its carrier within two layers of that cell. A fixed plaquette or retained gauge-factor site can therefore enter only a bounded number of the same-level witnesses. Divide the available Wilson and gauge coefficients by that multiplicity. For distinct untouched pieces the source of the shares is the disjoint union of their paying neighborhoods. The action sequence still assigns the old payment its preceding-level weight at birth; at later reads the stated lower bound is a fixed multiple of θ_t. The local history must retain the failed birth indicator, the stated one-step constraints, and, for Q_t, the specified gauge factors. The compatible evaluation must reproduce these protected rows on the whole neighbourhood. Under these explicit readout conditions, the two contrapositives above apply uniformly to the later evaluating field. Once an

island absorbs the creation, its payment belongs instead to the island's mass; this argument does not charge it again. Nor does it supply the separate witness needed for the later operation's own failed test.

For an ordinary plaquette or approximate fluctuation witness, the protected birth rows give the same readout in the three places where its payment is used. At a point of a later term, they are defining constraints of the current background. For a drop-out's mass they are imposed by eq. 4.cdlxx. The retained one-step constraints then give the required next average in both cases. The failed birth test is retained in either variant of the mass: the excluded test is the later operation's own root test. At the first step the convention is explicit: V_0 is the stored bare field and Γ_0 consists of bare bonds with both endpoints in Ω_1^{c}. Hence the birth neighbourhood already lies in this prescribed bare readout.

The third place is the real inner Wilson factor of a completed main integration. Let its level be i. The quiet window puts every ordinary birth belonging to its outer history at a level $t \le i_\flat$. Its paying neighbourhood lies in the old core. On that core the localised inner determining set agrees with the prepared set. Its datum is the retained V^{in} on the inner rows, whereas the new increment is supported on rows with no endpoint in the core. Thus the protected level-$t-1$ rows still carry V_{t-1}, and the retained step-t constraints still produce V_t. The first-step bare completion is understood as above.

Here the representative matters for literal averaging identities. The real Landau form used to write the inner Wilson action is gauge-equivalent to a member of the constrained orbit; an arbitrary member of that gauge orbit need not reproduce the fixed stored datum. Use the adapted constrained representative in the preceding identities. This leaves the Wilson sum unchanged, since each plaquette holonomy changes by conjugation, and the gauge-fixing variables in the certificate remain their stored values. The prescribed adaptation acts on the read, not on those stored variables. No equality of every historical coarse field is needed: the protected birth rows and their one-step constraints suffice.

For an island, this paying inner Wilson factor is a function of ξ alone. Taking the supremum over a complex exterior pair does not replace it by a Wilson action at that pair. These observations settle the averaging and recorded-value part of the ordinary witness read. Admission of the read to an analytic core and the one-sided bounds on its kept forms and elimination factors are separate requirements in eq. 4.dxxv.

For a failed ordinary fluctuation cube, a collar creation or a failed rim cube, use the kept Gaussian form of the birth step $t = j + 1$. The split places the failed fluctuation cubes in R and adds the cells of the collar creations and the failed rim cubes to the exploration set. The enlargements in the split put the test bonds in Z_t. They are kept nonpivots of the birth step, and the quiet-window condition $j < t \le i_\flat$ keeps them below the integrations of the containing operation.

Write $a' = g_j \mathcal{A}_j^{\mathrm{s}}$ for the retained coordinate in angular units. An explored collar bond is in the union of the passed cube masks, where the strip substitution leaves the coordinate fixed. The rim mask excludes the ordinary strip $\mathfrak{S}$. The failed fluctuation cubes have their masks in the enlargement of R, so on these masks the substitution either leaves the coordinate fixed or uses the cutoff $\theta_j(a')$. The same alternatives hold on the rim mask, since the plain zone is contained in the bonds meeting $\mathfrak{S}$. On the ball below the corresponding test threshold, $\theta_j(a') = 0$, and the exact product logarithm gives $\mathcal{A}_j = a'$. The logarithm also exists throughout that ball. Thus a failed cube or rim test, or a collar creation with $\chi_b^1 = 1$, has a bond with $|a'| \ge c\delta_j$, where $c > 0$ is the smallest fixed threshold fraction. The coercive reserve of the kept form pays $c'\delta_j^2/g_j^2$ at that original bond.

No integration is undone. For the superrenormalisable model [B82b, Sect. 3.A, Prop. 3.1], the conditional integrations are reversed at the end. This recovers a single positive quadratic form over many scales, and the factors of all the characteristic functions are then extracted from it simultaneously. Here the operations $\mathbf{T}_k(Z)$ keep the integrals, and every factor is obtained at the

place where it sits. It comes from the Wilson weight, through the display above; or from the Wilson weight together with the kept gauge-fixing exponents (Lem. 4.60 (ii)); or from the conditional form, whose lower bound is c. (Taken as a whole, the conditional form inherits the bound because it is a Schur complement; the kept part of this form is treated in §4.17.1.3.) With this, the cost of the stored Gaussian integrals is $e^{C\#_j(Z_j\cap\Omega_j)}$ (122 (1989) 355, p. 383).

One action and many claimants. Gains are added up in the ledger that follows, and this addition is legitimate: nothing is spent twice and the integrals that remain converge. The Wilson, gauge and stored Gaussian terms are assigned once to the witnesses used in the ledger. A *piece* is a pair (t, D) in which D is a component of the region produced by step t on which no $\mathbf{R}$ has acted up to level k. We write $\mathsf{w}_{t,D}$ for the weight of its one-step operation. This weight consists of the gauge-fixing factors with their normalisers, the Jacobians that arise when the averaging constraint is solved for the pivot bonds, and the conditional Gaussian. We write $\xi_{t,D}$ for the variables that belong to the piece. Let the shares of the action be as defined in (1), and extend U_k beyond Ω_0 by means of the stored bare field. Then at each point of the support of the term of a history one has $A(x_k(\cdot), U_k) \geq A_{\mathrm{cr}} := \sum_{(t,D)} x_{t-1} \sum_{p\in\text{share of }(t,D)} a_p(U_k) \geq 0$, so that the only part of the action spent is the Wilson sums over the shares. Moreover

$$e^{-A_{\mathrm{cr}}} \prod_{(t,D)} \mathsf{w}_{t,D} \leq \exp\Big[- c \sum_{\text{created cells } (t,\mathfrak{c}) \text{ in the pieces}} p_0(g_t)^2 \Big] \prod_{(t,D)} \bar{\mathsf{w}}_{t,D}(\xi_{t,D}),$$

$$\int \bar{\mathsf{w}}_{t,D} \leq e^{C\#_t(D)},$$

where each $\bar{\mathsf{w}}_{t,D}$ is a Borel function of the variables of its own piece only (the background no longer enters it), and $C = C(L)$ carries no $\log x$ per site. (1) *The action is divided among the pieces, not held in common.* The share of a piece consists of the bare plaquettes that lie within one cell of those of its cells at which a plaquette test or an approximate fluctuation test has failed. Distinct pieces have disjoint shares, because the tests of a step are carried out several cells away from the hull of the older region. On a share one has $x_k(\cdot) = x_{t-1}$, since the older profiles have completed their transition and the newer ones have not yet begun it. Every other plaquette contributes ≥ 0. (2) *The tests concern averages of the configuration that appears in the action.* The tests were built from localised backgrounds of step t, whereas the Wilson term contains U_k. The two are connected as follows. The identity $\mathcal{M}^{t-1}U_k = V_{t-1}$ on Γ_{t-1} is the constraint that defines the background, and on the same set $V_t = \mathcal{M}(V_{t-1})$ is imposed by the corresponding delta function. To apply the witnesses of Lem. 4.60 to U_k, the retained rows at the read must contain the averaging and contour readouts on their full carriers. This coverage is needed in addition to the identities on the retained rows. (3) *Each exponent is split into two halves*: one half is used for payment and the other is kept for the integration. The only normaliser that depends on the coupling is integrated against half of its own exponent,

$$\int du\, e^{-\frac{1}{2}x(1-\frac{1}{2}\operatorname{Re}\operatorname{tr} u)} \leq C \int du\, \mathbf{1}\{|u-\mathbf{1}| < \epsilon\}\, e^{-x(1-\frac{1}{2}\operatorname{Re}\operatorname{tr} u)}$$

$$(\epsilon \leq 1,\ x\epsilon^2 \geq 1),$$

and both sides are of order $x^{-3/2}$. For a gauge factor at level j, keeping half its exponent leaves the full denominator $z(\theta_j, \varepsilon_j)$ in place. Dropping the numerator cutoff bounds its integral by $z_\infty(\theta_j/2)/z(\theta_j, \varepsilon_j)$, where z_∞ is the untruncated integral. The displayed estimate bounds this ratio uniformly. On the cut-off zone of the strip the shift §4.17.1.3 vanishes for small $\mathcal{A}'$; consequently a large tested field is a large variable of integration. (The approximate test requires GF for the following reason. Take a gauge transformation at a site of $\mathbb{T}^{(t-1)} \setminus \mathbb{T}^{(t)}$. It moves V_{t-1} away from $\mathcal{M}^{t-1}U_{t,\square}(V_t)$, the average of the localised background §4.17.1.1, which is in the block axial gauge

§ 3.1.1. It does so at no cost in action, since it alters neither the action nor V_t.) (4) *Within a patch on which the* $\mathbf{R}$ *of level* m *acts*, a test that fails may not be paid for by the large field from which the patch arose. Whatever is older than the wait was paid for by plaquettes that lie in the *floor*. The floor is made up of those cells of level $m_\flat$ of the old core, $\Omega^{c}_{m_\flat} \cap X$, which do not touch $\Omega_{m_\flat}$; this old core is a part of the core $C_{m_\flat} = X \cap Z_{m_\flat}$ of §4.17.1.3, because $\Lambda_i \subset \Omega_i$. The gauge-fixing factors and stored Gaussians that paid for it are all of levels below $m_\flat$. Let X be a drop-out, and consider stored variables with two properties: each test carried by X takes its recorded value, and the plaquette variables of the datum in the old core, taken layer by layer (V_l on Γ_l), are within $C\check{R}_l^2\varepsilon_l$ of $\mathbf{1}$. (These two properties hold on the supports. They also hold at the points of evaluation that occur below in the mass of a later drop-out that contains X. Within the mass of X itself, no use is made of the test whose failure turned it into a drop-out.) For every real U compatible with stored variables of this kind, one has

$$c\,x_m \sum_{p\subset X^+,\,\text{no corner in the floor}} a_p(U) + \sum_{j=m_\flat}^{m-1} x_j \mathrm{GF}_j(X;V_j) + c\sum_{j=m_\flat}^{m-1} \|\mathcal{A}^{\mathrm{s}}_j\|^2_X \;\geq\; (\log x_m)^{2\mathfrak{p}_1 - Cr} \;\geq\; p_0(g_m),$$

where X^+ denotes X enlarged by two layers of unit cubes of level m and U is compatible on X^+. For the stops with the full witness calculation below, smallness of the left-hand side forces the tested quantity below its threshold. At a final regularity stop this calculation applies to the inset and its prepared-seam test. In every case the bridge runs along the same chain. It starts from the action of a compatible configuration and passes to the plaquettes of the stored fields (by the energy of the averages). From there it passes to the regularity of the background used by the test, and then to the bond variables. That last link comes either from the rigidity of a single averaging step or from loops in the shells. In a shell of side $s \leq CM\check{R}_m$, a loop composed of n straight pieces has a deviation from $\mathbf{1}$ that is bounded by $C\mathsf{n}s^2$ times the plaquette bound, and this is where the loss $\check{R}_m^{-C}$ comes from. The waiting collar screens off the unknown large field in the floor; this collar is what separates the floor from the region $X \cap \Omega_{m_\flat}$ in which the operation acts, and its width is at least one cell in the local units. Where the operation acts, the background of any stage has curvature at most $C(\varsigma + \check{R}_m^2\varepsilon_m e^{-\delta_0 M\check{R}_m})\eta_j^2$ on the layer j, which is the local scale. Here ς is the size of the plaquettes of the stored fields that follows from the smallness of the left-hand side. A drop-out uses no plaquette that has a corner in its floor. By contrast, each plaquette used beneath it lies inside the floor, and each gauge-fixing factor and stored Gaussian used beneath it is of a level below $m_\flat$. The windows of nested operations are disjoint.

4.17.1.43. Filling a loop on one shell lattice. Consider two lattice boxes of side at most s, in units of their common lattice spacing, with the inner box strictly inside the outer one in each direction. Use the lattice sites of the outer box outside the inner box, with their bonds and plaquettes. Suppose each side of the inner box has at least one layer of sites beyond it in the outer box. A closed path in this shell with n straight pieces contracts across at most Cns^2 plaquettes of the shell, counted with multiplicity.

Indeed, the shell is the union of the eight rectangular slabs beyond the faces of the inner box. Each straight piece lies in one slab. Consecutive pieces have a common endpoint in the intersection of their two slabs. Choose once a lattice point in each nonempty slab intersection. Join every change of slab to the chosen point for that pair by a coordinate path in the intersection, inserting the path and its reverse. Within each slab, replace the portion between successive chosen points by fixed coordinate paths through one fixed point of the slab. Each replacement is a loop of length at most Cs in a rectangular box of side at most s. Sweeping its edges to the fixed point one coordinate at a time fills it with at most Cs^2 plaquettes.

It remains to contract the loop of slab labels. The nerve has the eight slabs as vertices and has a simplex for each selection with at most one of the two sides in any coordinate. It is the join of four

pairs of points, hence is simply connected. In this fixed finite nerve, choose paths from one vertex to each vertex and fix a contraction of the loop associated with each edge and the two chosen paths. Any loop of n edges contracts by at most Cn triangle moves. To realize one triangle move in the shell, take a point in the common intersection of its three slabs. Connect it to the three pairwise reference points inside their intersections. The triangle then splits into three loops in the individual slabs. Each has length $O(s)$ and costs $O(s^2)$ plaquettes by the same rectangular sweep. This gives the stated area.

For a real unitary field each plaquette move changes the based loop transport by a conjugate of a plaquette transport or its inverse. Thus the loop defect is bounded by Cns^2 times the maximum plaquette defect on the filling.

For the main-step bond test write $\mathsf{W}_l = \mathcal{M}^l U$ and $\mathsf{W}_l^0 = \mathcal{M}^l \tilde{U}_0$ in the prescribed shell gauge. They must give the tested datum and its reference in the same frame at b. Suppose the fields agree on the shell's gauge tree and exterior bonds. Set $\varsigma_* = c_* \delta'_m \check{R}_m^{-4}$ with $c_* > 0$ fixed. Suppose the stored datum bound is at most ς_*. The screening term is at most ς_* for small coupling, since $\check{R}_m^6 (\varepsilon_m/\delta'_m) e^{-\delta_0 M \check{R}_m}$ tends to zero. For W_l use the small paying energy on the full backward carriers to bound its shell plaquette defects by $C\varsigma_*$. For W_l^0 the basic reference needs its own curvature estimate on these carriers. Where its bare curvature is bounded by $C\varsigma_* L^{-2l}$ there, the averaging estimate gives the same $C\varsigma_*$ bound. On its prescribed top rows the corresponding estimate uses the top datum of the reference. The stage screening estimate above does not establish this reference bound on its own. For a test and return on the same level, take a loop ℓ_b that traverses b in its chosen orientation and otherwise uses these common bonds on the level-l lattice. Writing the common transports before and after b as P and Q, the relative loop holonomy is

$$(P\mathsf{W}_l(b)Q)(P\mathsf{W}_l^0(b)Q)^{-1} = P\mathsf{W}_l(b)\mathsf{W}_l^0(b)^{-1}P^{-1}.$$

Thus the norm of the tested defect is bounded by the sum of the two loop defects. If the chosen loop has a bounded number of straight pieces in a shell of side $s \le CM\check{R}_m$ in level-l units, the stated filling bound gives $CM^2\check{R}_m^2\varsigma_*$ for the defect and, after changing the constant, for its small logarithm. This is below δ'_m when $\check{R}_m$ is sufficiently large. In applying this estimate we must find the filling in the charged shell, with its area counted with multiplicity and its backward carriers outside the old floor. The required rows and the common transports must come from the actual drop-out datum.

4.17.1.44. The datum tested before a stage is integrated. Consider stage $n+1$ of the operation of level m. Put $j = m_\flat + n$. At that point the old determining set is the stage-n set. The only member of the sequence that changes is the one of index $j+1$, which grows from Ω_{j+1} to the prepared member. The level-j determining rows that remain are a subset of the old rows. The margin from the next region leaves the higher row read sets unchanged. The level-$(j+1)$ rows with an end in the old region are already old determining rows. The new rows have both ends outside Ω_{j+1} and at least one end in the part being added. The two order-$(j+1)$ blocks of each such row lie outside Ω_{j+1} by grid alignment. The prepared separation puts both blocks inside the old member at level j, so all their level-j bonds are old determining rows. The averaging delta on the new row is still present when the stage is tested.

Let U be the real field of a compatible evaluation reproducing the old rows on these full carriers. Its level-$(j+1)$ averages on the new rows give the prospective datum by the retained delta and the composition of the averages. The other prospective rows have their old values. In particular the test can be compared with U before the collar variables have been eliminated.

For the no-inward determining set, each tested holonomy in the prospective sequence is a plaquette of $\mathcal{M}^q U$ at its test level q. A bond that is not a determining row has both ends in the layer just below q. Its two blocks lie in that layer, where the determining rows supply the complete one-step average. Thus this identification also holds at a seam. Upon joining to the tower of the component,

an interior holonomy is unchanged. A holonomy on the joining collar is trivial or equals a top plaquette holonomy of the original datum. The top data of this collar are outside the old floor.

If the bare carrier of a tested plaquette avoids the floor, the paying energy bounds its defect by $C\varsigma_*$. Otherwise its level is less than $m_\flat$ and its blocks lie in the old core. The floor does not touch $\Omega_{m_\flat}$, and the adjacent blocks at the outward rows of that level reach only the touching layer. The old plaquette defect is bounded by the regularity of the recorded old datum. This bound is at most $C\check{R}_m^2\varepsilon_m$, using the comparison of the thresholds at earlier levels. The joined datum is therefore regular at a radius tending to zero. On the part of the stage test at weighted distance at least $cM\check{R}_m$ from the old carriers, the local regularity estimate gives

$$|\partial U_{m,X}^{(n+1)}(p) - \mathbf{1}| \le CL^{-2\lambda(p)} \left(\varsigma_* + \check{R}_m^2\varepsilon_m e^{-c\delta_{\mathrm{c}}M\check{R}_m}\right).$$

Here $\lambda(p)$ is the local level of the joined stage sequence at p. This first establishes existence of the tested orbit and then bounds its curvature.

In the interior away from the joining collar, the old prepared shells have their stated levels. Before the compressed part of the nest the exterior region has at least the level prescribed by the test. In the compressed part the loss in level is at most $1 + \log_L(C\check{R}_m)$. Thus changing the curvature units to those of any of the stage thresholds costs at most $C\check{R}_m^2$. The factor $\check{R}_m^{-4}$ in ς_* and the exponential in the old-carrier term suffice for these interior thresholds. This is the inset used in the implication for carried tests. In the outer joining collar the tower can descend to the bare scale. There a bound in its local units gives the top-scale threshold only with an additional bound on the lifted top datum. When the stage tests that collar, we must estimate its failure separately.

At the final stage the literal regularity test includes bare plaquettes of $X \cap \Omega_m$ with the bound $3(1 - \frac{1}{2}\beta_{\mathrm{s}})\varepsilon_m L^{-2m}$. Near the outer boundary of the localized background, an inner bare bond of its level-zero layer is a determining bond and carries the face lift of V_m. Choose a plaquette tangent to a flat outer face, in that bare layer, crossing two faces of the level-m blocks. All four of its bonds are determining bonds. Its plaquette variable is exactly the level-m plaquette variable of V_m at the four block centers. The first-member convention that keeps only a collar of the level-one member still leaves a layer of bare determining bonds. It does not give the factor L^{-2m} on this lifted plaquette.

This mismatch occurs even for a top datum obtained by averaging a low-action bare field. Work in level-m units and put $\eta_m = L^{-m}$. In a fixed box containing the four order-m blocks, take an abelian potential $A_1(x) = ax_2T$, $A_\nu = 0$ for $\nu \neq 1$, where T has eigenvalues $1, -1$. Continue the potential smoothly to zero outside a slightly larger fixed box at the rim, keeping it away from the old core. Sample it on links by $U(b) = \exp(i\eta_m A_\nu(x_b))$. All bare plaquette angles are bounded by $C|a|\eta_m^2$, and their number in the support is at most $C\eta_m^{-4}$. Hence the bare Wilson action on the support is at most Ca^2.

In the affine region a centered averaging step keeps the same affine potential with the next spacing. The formula for the contour mean reduces to arithmetic averaging of its logarithms on the common abelian arc. On a bond in direction 1 the contours at opposite ends are translates with equal logs, and the sum of straight path logs averages x_2 to the coordinate of the block center. On a bond in direction 2, translating the contour changes its logarithm by a fixed multiple of $x_1 - y_1$, where y is its block center. That term averages to zero. Translations in the other two directions change none of the logs. The construction is local on the four order-m blocks, so the same calculation iterates to their top plaquette. Its defect is $2|\sin(a/2)|$.

Taking $a = 8\varepsilon_m L^{-2m}$ for large m therefore fails the literal rim bound, although the available Wilson exponent of this local perturbation is only $C\theta_m\varepsilon_m^2 L^{-4m}$. Thus the Wilson energy alone cannot pay for this failure. We must obtain the additional rim estimate from the other factors of the retained operation.

The same scale discrepancy affects the proposed fresh payment on the retained real support. Fix the top coupling in a plateau of the rounded cube size that contains the whole waiting window,

and put $h = m - \mathsf{N}_m$. Both N_m and the sides of the fresh carriers in level-m units are then fixed as m varies. Create one cell at h by a pure gauge, constant on each order-$(h-1)$ block. Give one block whose centre is outside the level-h lattice a gauge value u_x with

$$2\delta_{h-1} < |u_x - 1| < \varepsilon_{h-1},$$

and give the other blocks the identity. Before step h the gauge is constant within each fixing block and the tested fields are flat. At step h a nonpivot bond from x inside its level-h block fails the approximate fluctuation test. Its gauge-fixing factor still has positive support, and its cost belongs to level $h-1$. Place the failing footprint in one birth cell. With all subsequent tests passing, its single successors stay within the class size bound. Indeed if a_j is the side of a cell at level j, their sides satisfy $s_j \le s_{j-1} + 22a_j$, with $a_j = La_{j-1}$ on the plateau. The creation enters the class at m, at the lower end of the waiting window.

Multiply this field by the commuting perturbation at the rim above, with its support disjoint from the birth footprint. Use its actual successive averages as the retained fields. The lower creation is unchanged, and from level h upward the pure gauge restricts to the identity. The real averaging bound gives plaquette defects at level j of size $C|a|L^{-2(m-j)}$. The joined regular backgrounds exist at radius $C|a|$. On the full double blocks used for the retained relative bond tests, the contour variables and coarse rows in the prescribed frames differ by $O(|a|)$. The one-step comparison and the actual strip and collar translations therefore keep the retained fluctuation coordinates of size $O(g_j^{-1}|a|)$ on the fresh carriers. At the birth the failed bond is outside the strip. The earlier stage tests read away from the outer joining collar and at joined layers at least h, where the bare curvature is bounded by $C|a|L^{-2h}$. Since $m-h$ is fixed, these tests pass for large m.

The fresh gauge and Gaussian payments are $O(|a|^2)$, since there are boundedly many fresh variables at fixed top coupling. Together with the Wilson bound above this tends to zero for $a = 8\varepsilon_m L^{-2m}$, while the last stage test fails. The gauge payment at the birth is already owned below the fresh window. Thus the pointwise certificate must use the joined layer of the rim plaquette, which is zero here, even though the plaquette lies in the original region Ω_m. The inset estimate applies on the inset, and the ordinary rim fluctuation tests have their separate kept-form payments. For the negative branch of the full stage test we must supply a different payment or an accounting that carries this branch without a fresh gain.

For an internal seam at level $l > m_\flat$, the common return traverses the two adjacent shell trees and their single entrance. Indeed, contracting each shell tree gives the chain of nested shells. When the tested bond is itself the entrance, its defect vanishes. Otherwise the resulting cycle uses bonds only at levels l and $l-1$.

We use the following reduction at a compatible real evaluation. The tested bond and the original common return must read the retained averages of the same bare field U, in the specified frame, and the reference values must read the averages of $\widetilde{U}_0$ in that frame. The exterior top datum used to form $\widetilde{U}_0$ must also equal the top averages of U on the exterior bonds of the component. These are the exterior averages used in the real component evaluation. A restored current-stage datum after the main integration does not identify an earlier kept determining bond with the average of the evaluation field. At such a use one needs the retained-row identity for that bond. The extra fine bonds introduced below are computed averages of the evaluation fields on their full carriers and need no stored constraints of their own.

At a descendant drop-out (m, Y), the field used by the main-step bond test is the datum retained in the frame of that drop-out. The two fields carried on a top bond meeting the central box have different roles. The free argument V_m enters the averaging delta functions of step $m+1$. On a determining bond the earlier variable $V''(b)$ is still the datum of the determining set. The same prepared datum is retained on the finer shell rows. Before the central integration, a re-made level instead keeps its bare averaging delta functions. If the drop-out leaves at extendibility, there has been no central integration and these are the constraints still present in the innermost box.

For a descendant whose bond test was formed, the original shell rows are therefore read from the retained variables of the frame operation. The later real Wilson background is constrained at those rows. Indeed, the drop-out keeps its determining regions at departure for as long as it stands. If an intervening island has closed the branch, the outer mass reads that island's factor instead. Otherwise the enlarged component of the descendant lies in the old core of a containing operation, whose stages leave those determining rows in place. In the broader waiting convention a full re-entry at the lower end of a window has the same distinction: the coarse bonds whose averaging reads the free box argument are deep inside the first retained core, and are not data of the first stage. The top exterior of the child's central box remains on the prepared top grid, with its original exterior datum. Thus the bare field of the compatible evaluation reproduces V'' on the tested bond and on the return tree, and it reproduces the child's exterior top datum on its exterior determining bonds. The tree condition compares these values with V_0 in the child's recorded frame.

This check concerns the test of $V''V_0^{-1}$. A test or delta function that reads the free top argument would need its own comparison. The determining datum gives no equality between that argument and the kept prepared value on the same bond. When a descendant's failure is charged in a later mass, use the evaluation on the containing component and the child's charged carriers in Y^+. An evaluation in the mass of the child itself is only defined on Y, and its own failure is omitted in that version of the mass bound.

For the selected waiting rule the child's enlargement lies in the containing floor. Its paying region consists of the plaquettes of Y^+ with no corner in the child's own floor. In a completed ancestor the inner Wilson cutoff is one there. At a stopped ancestor use its carrying Wilson weight on the same region. Use the level-m weight in Y and the bound of the first ordinary successor on the added collar. The latter weight is bounded below by $c\theta_{m+1} \geq c'\theta_m$. The annular carriers in the shell comparison are contained in this paying region. The averaging energy bound, followed by the bond and last-shell comparisons, forces both shell tests to pass if that Wilson share is less than $c\theta_m(\delta'_m \check{R}_m^{-4})^2$. This argument uses the child's prepared rows as reproduced by the ancestor's compatible field in the child's original frame.

These Wilson shares are disjoint. The share of a descendant lies in the floor excluded from the share of a containing mark. For incomparable marks use the separate components at the larger level. The smaller enlargement is inside its ordinary successor and hence inside its later component, and the larger enlargement stays clear of that component by the cell separation. At equal levels the enlarged components are already disjoint.

At main-step entry itself, the prepared real background has the averages V'' on $\mathbf{B}''$, so it gives the test and tree reads in the prepared frame.

For the inner Wilson read, let the child Y stand as a drop-out inside a component Z whose main step is at level i. The waiting rule gives $m < i_\flat$, and Y^+ lies in the protected old core of Z. The parent's inner determining set agrees there with its prepared set. Its inner datum retains the child's determining datum; the parent's new increment has no endpoint in that core. Hence, whenever the parent's real inner background has been formed, its adapted constrained representative reproduces every retained child shell row whose full carrier lies in Y^+. In particular, it reproduces the same V'', the same tree values, and the same exterior V^{e} in the preceding main-step comparison. If the child stopped during preparation, this assertion concerns its actual stopped determining list. The unperformed stages do not supply additional rows. As with the ordinary witnesses, replacement of a real Landau form by this constrained representative leaves its paying Wilson sum unchanged.

Write $P_1 = Z''_{l+1}$ for $l < m$ and $P_1 = \Lambda^{\mathrm{pr}}$ for $l = m$. Write $P_2 = Z''_{l-1}$ for $l - 1 > m_\flat$ and $P_2 = I_2$ otherwise. These boxes are aligned on the level-$(l-1)$ grid. A level-l bond of the outer shell with both ends outside Z''_l has both order-l endpoint blocks in $P_1 \setminus Z''_l$. At either crossing, the inner endpoint lies in the first row of level-l sites of Z''_l. Its block is contained in the one-block skin of Z''_l. The prepared margin between P_2 and the boundary of Z''_l exceeds one level-l block. This follows from the early and late nesting margins, and from the lowest-shell width when $P_2 = I_2$.

Thus every endpoint double block needed for a level-l bond of the cycle is in $P_1 \setminus P_2$. By the same alignment, the bare backward carriers of the level-$(l-1)$ plaquettes used in that annulus stay in the annulus. In particular they avoid the floor in I_2 and lie in the charged part of X^+. The small-paying-energy estimate therefore bounds the curvature of the evaluation field throughout these carriers after averaging to level $l-1$. This use requires the energy on the full charged region and an evaluation field on those carriers. Control of the stored bond values alone would give neither.

For the reference field, take the box Z in the main-step geometry to be the component X in the drop-out estimate. The reference uses the exterior restriction of the top averages on this component. The endpoint blocks of each exterior top plaquette avoiding the free bonds are outside Λ^{pr}. By alignment with the top grid they lie in X if all its corners lie there. For a reference read that reaches across ∂X, use the full blocks in X^+ and their original exterior values. These carriers lie in the charged region of the containing evaluation. The paying energy bounds these exterior top plaquettes by $C(L,M)\varsigma_*$. The extension and reference-minimizer estimates put the same bound on the top plaquettes touching a free bond. A free bond has an end in Λ^{pr}, so each of its incident top plaquettes lies within one top unit of the box, in the minimizer's test set. All other plaquettes are unchanged exterior plaquettes. For $l < m$ the annulus lies in Λ^-, where the basic-reference potential and first derivative bounds give the corresponding bare curvature bound. For $l = m$ use the regular basic-reference tower on Z. The full carrier of the prepared box and its first exterior layer lies inside its top inset. The potential bound on Λ^- alone would only treat the part of that carrier in Λ^-. In both cases the local averaging estimate now gives the required curvature bound of $C(L,M)\varsigma_*$ at level $l-1$ throughout the annulus.

For either field, replace each oriented level-l bond c by the straight level-$(l-1)$ path with the same endpoints. The one-step averaging formula gives the level-l value of c as the straight-path product multiplied by a group element at distance at most $C(L,M)\varsigma_*$ from the identity. This estimate uses precisely the level-$(l-1)$ plaquettes of its endpoint double block. Gauge covariance gives the comparison at the same endpoints in the original frame, including the entrance and tested crossing. For a reversed orientation use the inverse product. Telescoping the unitary products bounds the error on a cycle with $O(s)$ bonds by $C(L,M)s\varsigma_*$, where s is the outer side length in level-$(l-1)$ units. Here $s \le C(L,M)\check{R}_m$.

The resulting cycle lies in the single-grid annulus $P_1 \setminus P_2$. In the prescribed shell tree a contour from the root has at most five straight pieces. The return within one shell is obtained by concatenating two such contours, reversing one of them and removing backtracking. With the entrance and the tested crossing this gives a bounded number of straight pieces on the fine grid as well. The filling therefore has area at most $C(L,M)s^2$. The two original cycle products are each at distance at most $C(L,M)(s+s^2)\varsigma_*$ from the identity. The comparison along the original common tree bounds $V'(b)$ and its small logarithm by $C(L,M)c_*\check{R}_m^{-2}\delta'_m$. This is the comparison in the retained frame. Equality of the auxiliary straight-path products of the two fields is unnecessary.

For a bond crossing the outer boundary of Λ^{pr}, the outermost tree return, the outer entrance, and the return in the first exterior layer are already on the top grid. The single-grid argument applies with this exterior layer included in P_1, provided the full top carriers lie in the charged region and the regular reference inset as above. The prepared test set excludes bonds with an endpoint in I_2, so there is no further crossing at the inner cut.

The last-shell test uses the field obtained by replacing part of the lowest prepared shell by the identity. Put $h = m_\flat$, and let $\square$ be one of its test cubes. Its full reading region $\square^{\sim 4}$ lies in $Z''_{h+1} \cap \Omega_h$. All level-h bonds of this region are therefore determining rows of the prepared set. The artificial datum keeps the original value on bonds with an end in I_2 and sets the remaining bonds to $\mathbf{1}$. Each replaced bond belongs to $\mathbb{B}_0$ and has level h.

Under the small-paying-energy assumption, each original top plaquette of the reading region has defect at most $C\varsigma_*$. Indeed, its full order-h carrier lies in the region of the compatible evaluation and avoids the floor in Ω_h^c, since the reading region and the prepared regions are unions of order-h

blocks. The preceding test χ' has passed. In the retained frame, $V_h'' = e^{ig_m B} V_0$ on every replaced bond, so its distance from $\mathbf{1}$ is at most $\delta'_m + CL^{h-m}\varepsilon_m$ by the basic-reference bound. Replacing the factors of any plaquette one at a time gives the uniform bound

$$C\varsigma_* + 4\delta'_m + CL^{h-m}\varepsilon_m$$

on the plaquettes of the artificial datum. The two ratios δ'_m/ε_h and $L^{h-m}\varepsilon_m/\varepsilon_h$ tend to zero with the coupling under the threshold and waiting choices. Linear regularity of the tower on $\square^{\sim 4}$ then bounds the bare curvature in the top inset $\square^{\sim}$ by less than $\frac{1}{2}\varepsilon_h L^{-2h}$. Thus the last-shell test also passes under small paying energy. This argument uses both the original averages and the bond bound in the retained frame at the artificial cut.

The cutoff of the reference law is distinct from the test on B. After the central integration the term is constant in the free top argument on the box. In each frame the tree constraint and the two shell tests are imposed on the retained variables before a reference field is drawn. The cutoff in the subsequent draw restricts $V'_m = V_m V_\Lambda^{-1}$ to its reference ball. Its complement carries zero mass under that law. There is no failure event for that cutoff on the free Haar argument.

On the support reaching the draw, the exterior plaquette bounds and extendibility give a regular minimizer V_Λ. The reference ball is in the regularity domain of $U_{m,X}(V'_m V_\Lambda)$ for small coupling. In fact the plaquette bound at V_Λ is $C\varepsilon_m$, and the four factors of a plaquette increase this bound by at most $4M_0\varepsilon_m$ on the ball. The centre $V'_m = \mathbf{1}$ belongs to the reference tree slice. The numerator is positive on a relative neighbourhood of that centre in the slice, which has positive product Haar measure on the free bonds. It is bounded by 1 there and on the rest of its support after the tree delta functions have been evaluated, since the real Wilson action with its nonnegative cutoff is nonnegative. Thus $0 < \mathsf{D}_X \le 1$ in the same frame as the numerator. The provision for an invalid denominator defines the move off this support and needs no mark payment on it.

At fixed frame use the same product Haar on the reference tree slice for every exterior. The numerator is jointly Borel in the exterior and the free slice variables. Its integral, and the integral with any bounded Borel insertion, are Borel in the exterior. For the denominator, the continuation of V_Λ in the admitted exterior chart and the regular background continuation give pointwise continuity of the Wilson weight at fixed V'_m. The reference cutoff in these coordinates is fixed and the weight is bounded by one. Dominated convergence gives relative continuity of D_X on this chart. Dividing by its positive value defines the normalized Borel draw.

The later plaquette split does read the restored field. Its failing branch is an island. The restored V_m is the top datum of the healed determining set there. In a containing mass its failed plaquette test is evaluated on these top averages of the compatible field. The test cube lies inside the eroded island, with its reading region inside the island. By the linear regularity estimate, failure implies a top plaquette defect at least $c\varepsilon_m$ in that reading region. The energy of the averages transfers this defect to the bare carrier in the containing Wilson action. With its local weight bounded below by $c\theta_m$, the resulting exponent is at least $c\theta_m\varepsilon_m^2$. This is the island event in the outer genealogy. The island's own operation mass is formed before the later plaquette split and does not count that event.

The cost. Carrying Z_n through step n incurs three costs. The first is a factor $e^{C\log x_{n-1}}$ for each site of $\mathbb{T}^{(n)}$ lying in $T_\ell \setminus \Omega_n$. The logarithm in question is $-\frac{3}{2}\log x_{n-1}$ times the number of free bonds less the number of gauge-fixed sites. It is produced where the fluctuation field is introduced and rescaled, $\mathcal{A}_{n-1} = g_{n-1}\mathcal{A}^{\mathrm{s}}_{n-1}$, which takes place on the whole of Ω_n, whereas the field is integrated only on the smaller set $\Lambda_n \subset \Omega_n$; it is also produced where the gauge is fixed, namely on the region in which the step's plaquette tests were passed, and this region contains Ω_n. The normalisation removes it everywhere. The second is a factor e^C for each site of $\mathbb{T}^{(n)}$ lying in the rest of Z_n enlarged by a layer. It comes from $\mathbf{E}_n, \mathbf{R}_n, \mathbf{B}_n$ and from the stored Gaussians, which are written in the scaled variable. The third is the entropy of the subregions. Altogether, for each component Z of Z_n, the cost is $\exp[CM^4\check{R}_n^5(d'_n(Z)+1)]$, where d'_n is the tree length measured in

cells. Of the five powers of $\check{R}_n$, four account for volume and one for $\log x_{n-1} \le C\check{R}_n$. In the count below, the number $|Z_n|_n$ of level-n cells of the region takes the place of Bałaban's linear size d'_n, because a region that consists of a single cell has $d'_n = 0$ and still incurs a cost.

Proposition 4.63 (the count, the ledger, the credit)**.** Let a history and one of its components be given. Then the count and the ledger displayed below are valid. If moreover the component is neither healed nor marked, the credit inequality displayed below is valid for it as well. In all of these the constants depend neither on the history nor on the number of steps. The ledger and the credit are statements of Bałaban (Comm. Math. Phys. 122 (1989) 355, (1.79)–(1.81), (1.89)), here with volumes counted in cells. ∎

The proof does not follow his induction. It rests instead on counting once, globally, along the whole genealogy of a component.

Shrinking and waiting. We denote by $S(Z)$ the free evolution of §4.17.1.1, that is, the hull of Z in the next partition together with $O(1)$ further layers, boxed. Every step that is not a drop of the cube size $\check{R}$ divides linear sizes, counted in cells, by the factor L. When a step is a drop, the partitions before and after it coincide, so that nothing shrinks; and of two consecutive steps, at most one is a drop. Consequently

$$d'_n(S^{n-j}Z) \le CL^{-\nu} d'_j(Z) + C,$$

$$\nu \ge \lfloor (n-j)/2 \rfloor \quad \text{the number of the steps } j+1, \dots, n \text{ that are not drops}$$

(this display replaces the scaling of the tree length in Bałaban, Comm. Math. Phys. 122 (1989) 355, pp. 384–385, which in the present notation, with $Z^{(n-j)}$ denoting the hull of Z in level-n cells, reads: $d'_n(Z^{(n-j)}) \le L^{-\frac{1}{2}(n-j)} d'_j(Z) \le 2^{-(n-j)} d'_j(Z)$ over $n-j>1$ steps, and $d'_n(S^{n-j}Z) \le C\, d'_n(Z^{(n-j)})$, or $\le C$ when the right-hand side vanishes). The added layers do not pile up, because every change of scale contracts what the earlier levels added. It follows that within $\mathsf{k} \le \mathsf{N}_j + C\log(d'_j(Z)+2)$ steps a component Z of Z_j does one of three things, provided the history continues up to level $j+\mathsf{k}$: it meets a creation, it joins another component, or it enters the class of $\mathbf{R}$.

Why a quiet component is bound to enter the class. Shrinking makes the component small, but smallness alone does not place it in the class. Consider a quiet line, that is, the sequence of components v_i, of the levels i, into which a single component is carried as long as nothing is created in it and nothing joins it. Such a line develops deterministically, $v_i = \Phi_i(v_{i-1})$, where Φ_i is the rule of the step for a lone component, and its side in cells obeys $\alpha_{i+1} \le \alpha_i + C$, $\alpha_{i+2} \le \alpha_i/L + 2C$. This is an affine contraction over two levels; its fixed point is less than half of the threshold of smallness, so that room is left for a drop. Apart from the depth $k_\flat \ge 1$, which is a matter of the arithmetic of the cube sizes, a line of this kind acquires on its own precisely the conditions of the class §4.17.1.3. Moreover, a condition that fails on account of an event in the window also fails at every intermediate level, because $n \mapsto n - \mathsf{N}_n$ is strictly increasing. Hence the carrier of the event cannot have been healed in between. The waiting bound is founded on this property of the class (in Bałaban, a single sentence, p. 385).

The count. An unhealed component of level $n-1$ is a predecessor of the component (n, C) of Z_n that contains its hull. For a component u_*, the genealogy $\mathrm{Gen}(u_*)$ is the tree formed by all $u = (n(u), C(u))$ that equal u_* or are iterated predecessors of it. It is the same object as the two of §4.17.1.1. The cluster of a root is its genealogy together with everything that the history records within it. The lineage of a live component is made up of the members of its genealogy, regarded as regions Z_j, together with the creations in them. None of these three contains a component that was healed earlier and whose place a later region covers again; such a component is the root of a cluster of its own. Write $\mathsf{n}_t^{\mathrm{cr}}(u_*)$ for the number of created cells of level t in $\mathrm{Gen}(u_*)$. Write $\mathsf{n}_t^{\mathrm{mk}}(u_*)$ for the number of its marked components of level t other than u_* itself. These are the drop-outs

and islands, that is, the creations by an operation of §4.17.1.1, and they are counted separately here because, for each of them, the count uses only the factor e^{-p_0}, which is all that a drop-out contributes. Set $\mathsf{T}_t := C\check{R}_t$; this number bounds the length of a wait and does not increase with t. Then for non-increasing costs $\hat{\mathsf{r}}_1 \geq \hat{\mathsf{r}}_2 \geq \cdots \geq 0$,

$$\sum_{u \in \mathrm{Gen}(u_*)} \hat{\mathsf{r}}_{n(u)}\, |C(u)|_{n(u)} \;\leq\; C \sum_t \mathsf{n}_t^{\mathrm{cr}}(u_*)\, \mathsf{T}_t \hat{\mathsf{r}}_t + C \sum_t \mathsf{n}_t^{\mathrm{mk}}(u_*)\, \mathsf{T}_t \hat{\mathsf{r}}_t.$$

In particular, for each history and without any bound on the number of steps,

$$\sum_{n \leq k} \#\{\text{cells of } Z_n\} \leq C \sum_{\text{creations of both kinds, at steps } t} (\mathsf{N}_t + 1).$$

The proof consists of three moves, and nothing is carried through an induction over the levels. (1) Contraction. The cells are counted by way of the occupied cells of a component, which are the hulls of the created cells beneath it. These form a skeleton that commutes with taking hulls, and the component never lies more than $O(1)$ cells away from it. Over a fixed number s_0 of levels, at least half of the occupied cells of a component that has many of them disappear, whereas a light component (one with at most C of them) retains at most C. Let o_n be the number of occupied cells of level n in $\mathrm{Gen}(u_*)$, and let l_n be the number of its light components of level n. Then

$$\mathsf{o}_n \leq \tfrac{1}{2}\mathsf{o}_{n-s_0} + C\, l_{n-s_0} + \sum_{n-s_0 < t \leq n} \mathsf{n}_t^{\mathrm{cr}}$$

(quantities of levels ≤ 0 are 0),

whence

$$\sum_n \hat{\mathsf{r}}_n \mathsf{o}_n \leq C \sum_n \hat{\mathsf{r}}_n l_n + C \sum_t \hat{\mathsf{r}}_t \mathsf{n}_t^{\mathrm{cr}},$$

where the second inequality is obtained from the first by multiplying by $\hat{\mathsf{r}}_n$ and summing, since the costs are non-increasing. (2) Stretches. A member of $\mathrm{Gen}(u_*)$ is called an event if it contains a created cell of its own level, or comes after a marked component, or has two or more predecessors (in which case it is a joining). The genealogy is a disjoint union of stretches. Each stretch starts at an event and continues through components that have a single predecessor and contain nothing new. A stretch that starts at level t has at most T_t light members. Indeed, after $O(1)$ levels a light component is contained in a cube of $O(1)$ cells, and once a further N_t quiet levels have passed it has entered the class, in which it is healed or else marked. (3) For each of the three kinds, the events are paid for through an injective assignment, and no created cell is charged more than twice. An event that contains a created cell is paid by that cell, the successor of a mark is paid by that mark, and a joining is paid by a created cell at a leaf of the tree. For joinings, select at each member that has predecessors one of them as its first predecessor. To a joining v, assign the leaf reached by starting from a predecessor of v other than the first and always moving to the first predecessor. No two joinings receive the same leaf, since on the way up from that leaf the first member entered otherwise than from its first predecessor is v (the leaves of a tree outnumber its branchings). Moreover, the leaf lies at a lower level, at which $\mathsf{T}_t\hat{\mathsf{r}}_t$ is no smaller (because $\mathsf{T}_t = C\check{R}_t$ is non-increasing in t), and each leaf contains a created cell. Thus a joining requires no payment of its own.

The ledger. In the numerical ledger assign gain $\mathsf{g}_t = c\, p_0(g_t)^2$ to each created cell and gain $\mathsf{p}_t = p_0(g_t)$ to each mark. To apply the ledger on a history we must verify the witnesses of Lem. 4.60 on their paying carriers. Against these stands, for every cell and every level, the cost $\mathsf{r}_n^{\mathrm{c}} = C \log x_n\, (M\check{R}_n)^4$ found above, which is bounded by the non-increasing sequence $\hat{\mathsf{r}}_n = C\mathsf{r}_n = CM^4\check{R}_n^5$; here r_n is the weight of a cell, §4.17.1.1. With these choices,

$$\mathsf{F}(u_*) := \sum_t \mathsf{g}_t \mathsf{n}_t^{\mathrm{cr}}(u_*) + \sum_t \mathsf{p}_t \mathsf{n}_t^{\mathrm{mk}}(u_*) - \sum_{u \in \mathrm{Gen}(u_*)} \mathsf{r}^{\mathrm{c}}_{n(u)} |C(u)|_{n(u)}$$
$$\geq\ 2 \sum_t \mathsf{p}_t \mathsf{n}_t^{\mathrm{cr}}(u_*)\ \geq\ (2-c)\, p_0(g_{n(u_*)}),$$

provided that $\mathsf{g}_n \geq 2\mathsf{p}_n + C\mathsf{T}_n \hat{\mathsf{r}}_n$ and $\mathsf{p}_n \geq C\mathsf{T}_n \hat{\mathsf{r}}_n$. An event costs $\mathsf{T}_n \hat{\mathsf{r}}_n \leq CM^4 \check{R}_n^6 \approx (\log x_n)^{6r}$. This cost is made up of $\log x_n \leq C\check{R}_n$ for each site, $(M\check{R}_n)^4$ sites in a cell and a wait of length $\check{R}_n$. Since $\mathfrak{p}_0 \geq 6r+1$, the quantity $p_0(g_n) = A_0(\log x_n)^{\mathfrak{p}_0}$ exceeds this cost, and the extra power of the logarithm absorbs the constants. The final inequality holds for the following reason. The genealogy has a leaf, and so it has a created cell at some level $t \leq n(u_*)$. Moreover $p_0(g_t) \geq (1-c)p_0(g_{n(u_*)})$, because, up to a bounded error, the flow is monotone: $x_t \geq x_{n(u_*)} - C$. When volumes are counted in cells, $e^{-\mathsf{F}(u_*)}$ coincides, up to two modifications, with the quantity whose supremum is taken on the right-hand side of (1.79). First, the $2p_0(g_j)$ that Bałaban reserves for each created region is taken from $c\,p_0(g_j)^2$ (the first of the two conditions). Second, Bałaban's cost of a cell, which in the notation of this text reads $CM^4\check{R}_j^5$, is replaced by the finer form $C\log x_j\,(M\check{R}_j)^4$. Components that consist of a single cell are included: such a component costs $\mathsf{r}^{\mathrm{c}}_n$ at every level, and this cost is paid through the light members.

The credit, as a corollary. Let $z = (j, Z)$ be a component that is neither healed nor marked. We write its factor as $e^{-\mathsf{F}(z)} = e^{-\kappa_j(Z) - 2p_0(g_{j(Z)})}$, where $j(Z)$ is the level of the first creation occurring in its genealogy. We extend the history of z by the fictitious continuation $S(Z), \dots, S^{\mathsf{k}}(Z)$. Nothing is created in this continuation, and it runs at the levels $n \leq K$, where the flow bounds are available; beyond K it is cut off. We mark the last member of this continuation and apply the ledger to the extended history. The extended history has the same created cells as z, and apart from that last member it has the same marked components. The ledger therefore gives

$$\mathsf{F}(z) - \sum_n \mathsf{r}^{\mathrm{c}}_n |S^{n-j}Z|_n \geq 2 \sum_t \mathsf{p}_t \mathsf{n}_t^{\mathrm{cr}}(z) \geq 2\mathsf{p}_{j(Z)},$$

which is to say

$$\kappa_j(Z)\ \geq \sum_{n=j+1}^{\min\{j+\mathsf{k},K\}} \mathsf{r}^{\mathrm{c}}_n \,|S^{n-j}Z|_n,$$

that is, (1.80), in which Bałaban has K in place of k and the summand

$$O(1)M^d R_n^{d+1} d'_n(S^{n-j}(Z)),$$

where R_n is his size, built from g_n itself. Thus the credit $\kappa_j(Z)$ pays for the entire wait. By the shrinking, the sum is bounded by $CM^4L^5\check{R}_j^6(d'_j(Z)+1)$ ((1.81)).

Who pays. At a creation, the deposit $c\,p_0(g_j)^2(d'_j+1)$ has degree $2\mathfrak{p}_0$ in $\log x_j$. It exceeds the total cost $CM^4L^5\check{R}_j^6(d'_j+1)$, whose degree is $6r$, so that per cell p_0^2 covers $\check{R}_j^6$. Now suppose a test fails inside $\mathbf{R}$. All that is known of the old credit is $\kappa \geq 0$, so the renewed wait of a small region, $CM^4\check{R}_j^6$, must be paid by the new factor $e^{-p_0(g_j)}$ on its own. For this reason that factor has to be at least as good as e^{-p_0}. A joining neither pays anything nor releases anything.

Joining: Bałaban's induction and the count. Bałaban proves (1.80) by induction on j and, within a step, by induction on the number of regions that join (pp. 384–387). When X and Y join to form Z, he sets free the reserve $2p_0(g_{j(X)}) + 2p_0(g_{j(Y)}) - 2p_0(g_{j(Z)}) \geq c\,p_0(g_j)$ to pay for the renewed wait. This step relies on the claim of p. 387 that, once S has been applied n_1 further times, the domain built from Y includes the domain built from X; there n_1 is called a rather small

number, with $n_1 < 10$ given as an instance. The count uses no containment of this kind. For the hull operation of this part, whether such a containment holds depends on where the two regions lie relative to the grids. In a single coordinate the hull map is $\mathsf{h}(y) = \lfloor (y + \frac{1}{2}(L-1))/L \rfloor$, and S, taken with ten layers, maps an upper end y to $\mathsf{h}(y)+10$. Set $y_{m_*} = 10$ and $y_m = L(y_{m+1} - 10) + \frac{1}{2}(L-1)$. Then

$$\mathsf{h}(y_m) = y_{m+1} - 10,$$
$$\mathsf{h}(y_m + l) = y_{m+1} - 9$$
$$(0 \le m < m_*,\ 1 \le l \le L).$$

Suppose that Y ends at y_0 and that X, a block of side 5, begins at $y_0 + 1$. The two regions touch. The upper end of $S^m(Y)$ is y_m and the upper end of $S^m(X)$ is $y_m + 1$ (for $m = 0$ it is $y_0 + 5$), so $S^m(X) \not\subset S^m(Y)$ for every $m \le m_*$, where m_* is arbitrary. At each of these steps the two partitions have ratio L, so that none of the steps is a drop. The count never compares $S^m(X \cup Y)$ with $S^m(Y)$.

Proposition 4.64 (the mass bound; where the small factor stands)**.** Consider a component X that has passed every test of the operation $\mathbf{R}$ of some level $i \le k$ before the last decomposition $1 = \chi_i + \chi_i^c$ (a healed component or an island). The operation of §4.17.1.3 on it is written

$$\mathbf{T}_i'(X) F\,(\mathbf{p}) = \frac{1}{|W_4|} \sum_{\omega} \int e^{\Phi_\omega(\xi;\mathbf{p})} F\, \mu_\omega(d\xi),$$
$$\mathbf{p} = (\mathbf{U}, \mathbf{J}) \text{ a pair on } X^{\sim},$$

where ω ranges over the local histories of X, that is, the results of every decomposition of unity inside X together with a frame in W_4 of the gauge tree, ξ ranges over the stored variables, and e^{Φ_ω} contains the core Wilson factor $\exp[-A(x_i'' \phi_{\mathrm{c}}, U''(\xi))]$. (a) For a healed component or an island, in which case $\mathsf{k} = 0$,

$$\mathsf{M}(i, X) = \frac{1}{|W_4|} \sum_{\omega} e^{c\,\mathsf{v}(\omega)} \int \sup_{\mathbf{p} \in U_i^c(X^{\sim}, \alpha_{0,i}, \alpha_{1,i})} \left| e^{\Phi_\omega(\xi;\mathbf{p})} \right| \mu_\omega(d\xi) \le e^{-(2-c')p_0(g_i)},$$
$$\mathsf{v}(\omega) = \sum_{u \text{ listed by } \omega} \log x_{n(u)}\, \#_{n(u)}(C(u)) :$$

here the coupling is taken at the level of the action, and the current coupling then follows from $p_0(g_i) \ge (1-c')p_0(g_k)$; the supremum stands inside the integral; and a weight growing exponentially with the space-time volume of the patch is paid for in advance. It is this bound that the induction carries forward. Bałaban has (1.89) instead, at each fixed pair ((1.73)). (b) Take a single history and a single component Z of Z_k on which no $\mathbf{R}$ has acted. For every value of the exterior variables (in total variation),

$$\left\| \mathbf{T}_k(Z) \big[\chi_{k,\mathcal{S}}\, e^{-A(x_k(\cdot), U_k)} \big] \right\| \le \prod_{\text{created cells } (t,\mathfrak{c}) \subset Z} e^{-c\, p_0(g_t)^2} \cdot \exp\Big[C \sum_{t \le k} \#_t(Z_t \cap Z) \Big],$$

with each factor taken at the coupling of its own step. Bounds of this kind do not multiply over the components, since the background couples the components. The pointwise majorant given above does multiply (Tonelli). Consequently, after integration and for an entire label, there exists a history h carrying the label $\mathcal{S}$ for which the bound below holds, where $(t, \mathfrak{c})$ ranges over the created cells of h lying outside the marked X. (The created cells are not determined by the label: a row of three failing cells and that row with its middle cell removed yield the same regions.)

$$\int \mathbf{T}'_k(\mathcal{S})\,[\chi_{k,\mathcal{S}}\, e^{-A(x_k(\cdot),U_k)}]\,dV_k \le e^{C\sum_j \#_j(Z_j)} \prod_{(t,\mathfrak{c})\text{ outside the marked }X} e^{-c\,p_0(g_t)^2} \prod_{(i,X)\text{ marked, outermost}} \mathsf{M}(i,X).$$

(The bound holds a fortiori when each site carries $C(1+\log x_j)$ and each created cell carries $e^{-c\,p_0(g_t)}$. This logarithmically weighted form is required by the rarity hypothesis and by the source construction.) (c) Consider the sum over the labels, with the costs paid by Prop. 4.63 and the nested sequences inside each component summed. In this sum every component C of Z_k keeps the factor $\exp[-c\,p_0(g_k) - \kappa_1 \#_k(C)]$, in which the volume is counted in sites (and so, a fortiori, the factor with $\kappa_1|C|_k$, in which it is counted in cells; this is the form in the introduction to this step). Since $d_k(C) \le \#_k(C)$, this per-site factor includes the term $-\kappa_1 d_k(C)$, which is what p. 387 asserts for an arbitrary region. The decay is one further cost $O(1)$ per site; like the other costs, it is paid for by the events (Prop. 4.63). If failed plaquette tests have produced a component, then $\mathbf{T}_k(Z)1 \le \exp[C\sum_{t\le k} \#_t(Z_t \cap Z)]$ (119, (2.21), and p. 264, where it is said of the Wilson term that it "yields the small factors for large field characteristic functions"). ■

The bound has to reproduce itself. (i) *Supremum under the integral.* In a term of level $k \ge i$, the factor remaining on X is evaluated at $(U_k, J(U_k))|_{X^\sim}$, and U_k depends on the integration variables of every component. Such an integral is not controlled by a bound at each fixed pair, and bounds that are merely uniform in the other variables do not multiply. For example, take $\mathbf{1}\{|\xi_1 - \xi_2| < \epsilon\}$ on $[0,T]^2$: each integral in one variable is $\le 2\epsilon$, whereas the double integral is about $2\epsilon T$. What does multiply is a majorant given by a Borel function that depends only on the variables of X. The integral of the supremum itself is understood as an upper integral. Such a majorant can be obtained because only two factors depend on the pair: $\Phi_\omega = \Phi^\circ_\omega(\xi) + [-\frac{1}{2}q_X + V(X^\sim)](\xi;\mathbf{p})$, where $q_X = \|\phi^{1/2} D\mathsf{H}B\|^2 \ge 0$ is formed at a real background, and $|V(X^\sim)| \le C\,\#\{\text{stored sites}\}$, with every site counted at its own level. The real background here is the critical configuration of the prepared sequence, taken with $B = 0$ and at the datum of the main step, and it is built with the G-valued factor of the triple. Through that factor, q_X depends on the splitting, whereas the exponent Φ_ω, taken in its entirety, is determined by the pair (see clause (vi) of the class of §4.17.1.45). (ii) *A fixed set.* If the supremum is taken over the arguments that occur at level k, the bound is not stable, since the next step replaces U_k by U_{k+1}. The supremum is therefore taken over a larger fixed set, built only from the data of the levels $\le i$. This set contains every later argument, since on the supports the following bounds hold, with nothing lost in the age $k-j$ of the layer $\Omega_j \setminus \Omega_{j+1}$,

$$|\partial U_k - \mathbf{1}| < C\check{R}_j^2 \varepsilon_j \eta_j^2,$$

$$|J(U_k)| < C\check{R}_j^2 \varepsilon_j$$

$$\text{on } \Omega_j \setminus \Omega_{j+1} \text{ (the current in level-}j\text{ units)},$$

$$C\check{R}_j^2 \varepsilon_j < \alpha_{0,j} \text{ for all } j \ (q \text{ is chosen after } \mathfrak{p}_0 \text{ and } r, \text{ the ceiling on the couplings after } L)$$

(the last inequality holds since the quotient is a negative power of $\log x_j$, since $q > \mathfrak{p}_0 + 2r$, and since the couplings stay below a ceiling; moreover, a layer at index distance a has an influence that decays like $e^{-(a-1)_+}$, while its threshold grows like $(1+a)^{1/2}$). Hence the mass seen at level k does not exceed the mass of level i, and a single step **RT** gives back the bound with unchanged constants. (iii) *The weight* $e^{c\,\mathsf{v}(\omega)}$ has as its only cost a larger constant in the ledger. Since it carries $\log x_j$ per site, it absorbs every perturbation of size $O(1)$ per site of the patch, such as the factor e^{Ψ} of the sources, §5.20.1.2. In the weighted rarity hypothesis, the weight covers whatever the label still exhibits inside a drop-out, since the small factors produced there have already been consumed by the mass; this is why the masses are required with the weight raised. The argument requires a

volume that counts each pair (level, cell) only a bounded number of times. A volume that counts the past again at every drop-out may exceed the total of all the gains; an example is a row of cells all created at the first step, with a drop-out at each completed wait. (iv) *Drop-outs.* The factors kept by a drop-out contain no Wilson action, and no inner background has been formed; the kept factors are evaluated at the background of whichever later term contains X. On the other hand, the configuration enters the Wilson term of the carrying term, and the failed plaquette tests listed by ω, which are the tests of the creations in X, bound plaquettes formed from averages of this configuration. That weight, over the bare plaquettes all of whose corners lie in X, is therefore counted with the operation. The pair is replaced by an *evaluation* at ξ. By this we mean a real U on X with four properties. First, it is compatible with ξ, that is, $\mathcal{M}^j U = V_j(\xi)$ on the bonds of Γ_j in X. Second, it is regular layer by layer, in the same sense as on the supports: on the layer j one has $|\partial U - \mathbf{1}| < C\check{R}_j^2 \varepsilon_j \eta_j^2$, while the plaquette variables of the stored fields differ from $\mathbf{1}$ by less than $C\check{R}_j^2\varepsilon_j$. Third, at this configuration each test listed by ω takes its recorded value, with the exception of the test whose failure made X a drop-out; here the fields not formed by ξ, including V_i on X, are read as averages of U. Fourth, $(U, J(U))|_{Y^\sim} \in U_m^c(Y^\sim, \alpha_{0,m}, \alpha_{1,m})$ for each island (m, Y) in X. Together with this configuration, the supremum also runs over further free parameters subject to one-sided bounds. These are the weight υ of the Wilson term, with $\upsilon \geq c\, x_t$ on Z_t, and the kept forms and Jacobians, which are evaluated at a background §4.17.1.3, with form $\geq c$ and Jacobian $\leq C$:

$$\frac{1}{|W_4|} \sum_{\omega} e^{c\,\mathsf{v}(\omega)} \int^{*} \sup_{\text{evaluations at } \xi} e^{-A(\upsilon \mathbf{1}_X, U)}\, (\text{kept factors})\, \mu_\omega(d\xi) \leq e^{-(2-c')p_0(g_i)}$$
$$(\upsilon \geq c\, x_t \text{ on } Z_t).$$

(Over the empty set the supremum is 0.) Take a term of some level $k \geq i$ in which X occurs, and a point of its support. Put $\upsilon = x_k(\cdot)$ and take the forms and Jacobians at the background of that term. To apply the mass bound we must verify membership of this later read in the compatible fibre, using its original rows, carried tests, and form and density bounds. On prepared rows the read uses the retained datum, with the restored free outgoing field in its separate slot. For a drop-out, the left-hand side is its mass $\mathsf{M}(i, X)$. Where X has no sum over frames, nothing depends on the frame listed by ω, by the identity $1 = |W_4|^{-1} \sum_r 1$. This bound does not use the test whose failure made X a drop-out. The gain $p_0(g_i)$ of that test is reserved for the renewed wait, and it is drawn on once X occurs as a mark within a subsequent component (Prop. 4.63). The supremum is taken over a set fibred over ξ. Over a product set it selects a flat U, at which $e^{-A} = 1$, and the factor is lost. In a toy case, take $\Phi = -T(1 - \cos p)$ together with the test $|\xi| \geq \epsilon$: its mass is $e^{-cT\epsilon^2}$ on the graph $p = \xi$, but approximately 1 over the product set. For this reason each inequality of Lem. 4.60 is required to hold at every configuration of this set, and not merely at backgrounds. These inequalities do hold for all real configurations having the prescribed averages. Nothing is assumed inductively at a drop-out; the inductive bound is the one for the islands. (v) *Islands inside X.* The count and the ledger are carried out on the outer genealogy of X, that is, on the genealogy cut off at the islands (m, Y) in X. Each such island is a minimal node, and all of its cells, of which there are $O(1)$, are counted as created cells of level m. These cells are paid for in the same way as created cells. The factor χ_m^c of the new field on Y less a margin is present in the term. Consequently there is a test cube of level m in which the test fails at $V_m(\xi)$; this is the test χ_m of a step, and not one of the sharper tests of the stages. Lem. 4.60 (i) then extracts the gain $c\, p_0(g_m)^2$ from the core Wilson factor of X, on plaquettes that are used by no other gain; when X is a drop-out, the gain is extracted instead from the Wilson weight that (iv) counts together with its operation, and this holds at every evaluation. This test does not enter the mass of Y, since that mass is formed before the last decomposition. One evaluates the factor of Y at a real pair

of $U^{c}_{m}(Y^{\sim},\alpha_{0,m},\alpha_{1,m})$, where it is bounded by its majorant. Since v is additive, the sum over the local histories of Y separates and is at most $\mathsf{M}(m,Y)\le 1$. Every leaf of the cut genealogy contains a created cell, and the ledger leaves $2\sum_t \mathsf{p}_t\mathsf{n}^{\mathrm{cr}}_t$, just as it does on the full genealogy.

4.17.1.45. The induction closed: the order of the constants, the sum over histories, exactness. **The class.** Let a frame $\theta_0,\dots,\theta_k$ be fixed (see the frame setting of § 5.2). Each choice of level j that depends on the coupling is made with θ_j. These choices are the thresholds, the radii, the scaling $\mathcal{A}_j=\theta_j^{-1/2}\mathcal{A}^{\mathrm{s}}_j$, the gauge-fixing weight, the Gaussian forms of the stages of $\mathbf{R}$ and the reference laws. The cube size and the waiting depth $\check{R}_j=\mathsf{N}_j$ are made instead with the running minimum $\min_{i\le j}\theta_i$ (a frame can go up by as much as c_5, and taking the minimum ensures that $\check{R}_j$ does not increase). Consequently the tests, the regions, the histories, the domains of analyticity and the measures of the integral operations depend on θ only, and on no entry of the density. In the bounds of this part we put $g_j:=\theta_j^{-1/2}$, so that $g_j\le\gamma$ is the floor $\theta_j\ge\gamma^{-2}$ of a frame (at $\theta=x$ this is the $g_j=x_j^{-1/2}$ of Prop. 4.56). In this way the circle "regions depend on couplings, couplings on activities, activities on regions" is broken. The elements of $\mathfrak{C}_k(\theta)$ are presentations rather than densities: the families are not determined by the density, and locality in the history is a property of how the terms are placed. Each element is an array, specified simultaneously on every admissible torus and cutoff,

$$\mathfrak{P}=\Big((x_j);\,(E^{(j)},\widehat{E}^{(j)},b_j);\,(R^{(j)},\widehat{R}^{(j)});\,(B^{(j)}_{\mathcal{S}}(X));\,(\Phi_\omega)\Big)_{j\le k},$$

$$b_j=x_{j-1}-x_j=b[\widehat{E}^{(j)}]$$

$$(1\le j\le k;\ \text{at } j=0 \text{ the one entry } x_0),$$

whose density $\rho[\mathfrak{P}]$ has the form given in Prop. 4.56. In it, $\widehat{E}^{(j)},\widehat{R}^{(j)}$ denote single families on $\mathbb{Z}^4$ that coincide with the families on the torus on lifts of non-wrapping domains; $b[\cdot]$ denotes the coefficient of § 4.2; and $\Phi_\omega=\Phi^\circ_\omega(\xi)+\sum_f\Phi_{\omega,f}(\xi;\mathbf{p}|_{R_f})$ are the exponents of the integral operations attached to the islands. Membership in the class requires the following. (i) *the tube*: $|\mathsf{z}_j|\le\mathsf{t}_j\approx c\,\theta_j(\log\theta_j)^{-C}$, where $\mathsf{z}_j=\theta_j-x_j$. This relates the frame, which is given, to the weights of the array, and the choice of the frame ensures it. The tube is wide in absolute terms, but narrow enough that the discrepancy between the weight and the choices, regarded as a potential, is small: $|\mathsf{z}_j|\alpha_{0,j}\delta_j\le c(\log\theta_j)^{-c}$. (ii) The bounds of Prop. 4.56 on $E^{(j)},R^{(j)},B^{(j)}$ hold, together with analyticity, reality, and invariance under a single fixed gauge action on the stored data. A boundary term is analytic as well in the switches μ_b that belong to those of its stored bonds lying close to a test, and in those of its stored fields that lie far from every test. Any two splittings of one pair give each term the same value. The boundary terms, terms away from the small-field region and collar corrections are supplied on the graded spaces assigned to their slots, including the formation addresses used for their reads. A carried entry retains its formation space, carrier and stored arguments. The standing-island factors use their birth cores. If the tests of a term take the value 1, then each boundary term and each exponent has in its domain the triple $(U,0,J(U))$ formed from the background U. (iii) *identification*: $E^{(j)}$ coincides with the family produced from the earlier entries, in each of its two chains (see below), by the small-field step at the pair (θ_{j-1},x_{j-1}), carried out on the whole lattice. (iv) *lattice covariance* holds in the forms stated below. Above all, the pointed totals are covariant under every isometry of $\mathbb{T}^{(j)}$, unit translations included (119, (2.29)). (v) *coherence*: whenever a chart §4.17.1.1 identifies, scale by scale, what two histories record within one cell of X, it identifies the terms at X. This single clause contains locality in the history, covariance under translations and independence of the torus. For a standing operation, the chart identifies μ_ω together with the Borel set carrying it, as well as its set of factors and Φ°_ω. It also identifies the factors one by one: $\Phi_{\iota\omega,\iota f}(\iota\xi;\iota\mathbf{p})=\Phi_{\omega,f}(\xi;\mathbf{p})$, for arbitrary components and histories,

provided that the chart ι transports both the record of the history at R_f and ξ on R_f. (vi) For an island, μ_ω is a Borel measure, positive and defined on a Borel set; Φ_ω depends in a Borel way on ξ, is analytic in $\mathbf{p}$, and takes real values whenever the pair is real. Moreover, gauge invariance holds pointwise in ξ and for each factor separately. The bounds of *mass* (Prop. 4.64, with a reserve constant $c_{\mathrm r} \ge 0$ in the weight) and of *oscillation* § 4.17 hold, the coupling in them being taken at the level at which the island was made. The class imposes no bound on the exponent itself, which is large and negative, and no clause concerns a drop-out. (vii) Each term, taken together with its stored variables, is a non-negative measure $\hat\tau_h$. The form and the bounds alone do not give a class that a step preserves. The gain $L^{-c(n-j)}$ of the regular terms does not follow from $\|E^{(j)}\|_{j,\kappa} \le E_0$, and neither do $|\mathbf{E}_k| \le C\sum_n |\Gamma_n| + C\sum_j \#_j(Z_j^+)$ (below) and the input of the next step, which depend on that gain. Clause (iv) cannot be omitted, for the following reason. Take the tents $h_z^{(j)}$ of § 4.5; then $\mathsf{Q}_z := c\,\alpha_{0,j}^{-2} A(h_z^{(j)}, \cdot)$ defines an activity whose norm does not exceed 1, and

$$\sum_z \mathsf{Q}_z = c\,\alpha_{0,j}^{-2} A(U),$$

$$\alpha_{0,j}^{-2} = c\,\theta_j (\log \theta_j)^{-2q} :$$

this is the inverse coupling θ_j itself, up to the factor $c(\log\theta_j)^{-2q}$. For a regular family, such a part is removed by the counterterm $-b_j A$ together with the covariance of the pointed totals. That covariance makes the second-order part at each site equal to the multiple b_j of the action at that site. Take the format of the complex class below, in which each site has its own member. In that format one can find arrays which obey all the bounds of clause (ii) and which, at each level, add no less than the same multiple of the action, with no part of it decaying as the age grows. Such arrays are ruled out by covariance of the pointed totals, already when it is required only for unit translations.

Proposition 4.65 (the step preserves the class)**.** Suppose the following:

(i) $\mathfrak{P} \in \mathfrak{C}_k(\theta)$;
(ii) θ_{k+1} prolongs the frame;
(iii) all $g_j \le \gamma$, and they lie below a ceiling that depends on the reserve constant $c_{\mathrm r}$ and does not increase with it;
(iv) the next weight $x_{k+1} := x_k - b[\widehat{E}^{(k+1)}]$ lies in the tube (here $\widehat{E}^{(k+1)}$ is formed first, by clause (iii), at the pair (θ_k, x_k), and its formation involves no choice of level $k+1$).

Then each move of step $k+1$ is defined. The identity $\mathbf{RT}\rho[\mathfrak{P}] = \rho[\mathfrak{P}']$ holds separately for each standing part §4.17.1.1, every part keeping its stored variables, and $\mathfrak{P}' \in \mathfrak{C}_{k+1}(\theta)$, the constants being unchanged. Moreover, $\mathfrak{P}'$ prolongs $\mathfrak{P}$. An array belongs to the class of level $k+1$ exactly when its cut at k belongs to the class of level k and, in addition, it satisfies the conditions of level $k+1$. Finally, the array whose single entry is $x_0 = \beta$, with $|\beta - \theta_0| \le \mathfrak{t}_0 \le c\,\theta_0 (\log\theta_0)^{-C}$, belongs to $\mathfrak{C}_0$. *Not returned*: the hypothesis that the next weight belongs to the tube. At every step this hypothesis is provided by how the frame is chosen. For $\theta = x$ the deviation is 0, and the requirement is that x_{k+1} prolong the frame, which is the content of the flow inequalities § 5.2.1 and of the nesting of the cube sizes, a consequence of the upper flow inequality § 5.2.2. For the reference line it is supplied by the drift bound of § 5.2.3, below. (In Bałaban, Comm. Math. Phys. 119 (1988) 243, the second remark after Theorem 1 concerns the space of "all densities satisfying the conditions of the inductive assumption", and these conditions include (2.29) and (2.32), together with (2.33), which follows from (2.32).) ∎

Why the induction closes. (a) Room: fix a fraction $\beta_{\mathrm s}$. The terms produced by **T** §4.17.1.3 are defined on the radii $(1+\beta_{\mathrm s})\alpha_{i,k}$ and decay at the rate $(1+4\beta_{\mathrm s})\kappa$. Moreover, $e^{-\kappa d_j(X)} \le e^{-c\kappa d_j(X)} e^{-(1+3\beta_{\mathrm s})\kappa d_k(\widehat X)}$ for $j<k$, where $\widehat X$ denotes the union of all level-k cubes meeting X, taken

without any added layer. The operation **R** uses up this room and nothing beyond it. (b) Drift: for $m < n$ one has $g_m \le (1+c)g_n \le C(n-m)^{1/2}g_m$. This does no harm over the N_k levels that **R** spans, because every threshold changes only by powers of $\log x_k$. (c) Order. Every constant is of one of four kinds:

- free upward: its range has no upper bound (the odd integers for L, the powers of L for M);
- free downward: its range is some $(0, c]$;
- confined to a window;
- determined by earlier constants: a function of those alone.

The text states finitely many conditions on the constants. Each condition is assigned to the last constant on which it depends, after every determined constant in it has been expressed through the earlier ones. For every admissible choice of the earlier constants, the condition is then, at that constant, of one of three kinds:

- a floor on a constant free upward, which holds from some value onward;
- a ceiling on a constant free downward, which holds on some $(0, c]$;
- an endpoint of a non-empty window.

A condition that follows from such conditions is not counted separately, with one exception: the requirement that a window be non-empty. That requirement is a floor or a ceiling on some earlier constant, and it counts at that earlier constant. Two further stipulations apply:

- No equality between thresholds is imposed, since every assertion made with a threshold remains true when the threshold is lowered.
- Some families of conditions are indexed by a quantified number, for instance the floor $c_a \ge C(b)\log L$, $C(b) \to \infty$, with an integer $b \ge 1$. No single c_a satisfies this for all b. Such a family belongs to the statement in which the number is quantified, and it is imposed only at finitely many values of that number, which are fixed before the constant.

Thus, apart from the constant it bounds, each of these conditions depends only on earlier constants, in the order

$$(r, \mathfrak{p}_0, \mathfrak{p}_1, q, \kappa_0, \beta_{\mathrm{s}}, \theta_{\mathrm{s}}, \theta_c) \prec L \prec (M_1, R_1) \prec \kappa \prec (\bar\kappa, \kappa_1) \prec M_1' \prec M_2 \prec (\Theta_1, \varpi_T) \prec M$$

$$\prec \sigma_\star \prec a_\Lambda \prec (A_1, A_0, C_0, C_1, \alpha_1, \alpha_0, E_0, M_0, \alpha', \mathsf{B}) \prec (\underline\lambda, c_5, \bar B) \prec \gamma \prec (\varrho, c_a) \prec (g, \ell, \beta, K)$$

Some of these constants require comment:

- $\underline\lambda = \frac14 c_0$ and $c_5 = 2c_2$, where c_2 is the constant in eq. 5.xviii, are fixed together with L; in the display they appear as members of the triple that defines the frames.
- $\bar\kappa$ is the exponent of the decoupling parameters of § 4.5. The factor $e^{-\bar\kappa}$ of a link of the cluster expansion comes from the decay of the covariance across an M-cube.
- $\delta_0 M \gg \bar\kappa \gg L\kappa$: rates are chosen before the cube sizes M_1', M_2, M. These are floors on $\bar\kappa$ and then on M, both chosen after L and κ.
- M_1 depends on L alone, and R_1 depends on L and M_1.
- $\Theta_1 \ge C/\theta_c$ and $\varpi_T \ge C$, and both are free upward.
- $M \ge \max\{\varpi_T, L^2 M_2\}$ and $A_0 \gg A_1$.
- The later ceilings on g may depend on c_a and on the dial $\mathsf{w} \in [0, 1)$. The dial is a fraction that depends on no constant and is fixed together with the exponents.
- The reserve constant $c_{\mathrm{r}} \ge 0$ of (vi) is free upward and comes before the ceiling that depends on it.

Induction along this order now gives two facts, because finitely many floors hold together from some value onward, finitely many ceilings hold together on some $(0, c]$, and the finitely many endpoints of a window leave it non-empty owing to the conditions that count at earlier constants. Every admissible choice made up to a given position can be extended, and no later condition places

a restriction on a constant chosen earlier. Adding one more floor or ceiling of the same kind leaves both facts true. The choice is therefore not circular, and a condition arising in the construction fits the order if, at the last of the constants on which it depends, it is of the kind named. A bound may contain a number that a statement returns, or a number given as a function of a whole group of constants. Any such number must be fixed before the constant that it bounds; the cycle cut below is a case of this. A representative condition on M or on γ asserts that e^{-cM}, $e^{-c\check{R}_k}$ or a power of $\log x_k$ dominates constants fixed earlier. *The supremum form*: a condition imposed on all couplings of all runs reduces to a single ceiling on γ, namely $\sup_{x\geq\gamma^{-2}} h(x)\leq c$ with $h\to 0$, because the scale appears in it solely by way of x_k (122 (1989) 175, p. 186: the conditions "do not depend on any scale"). *The cycle that is cut*: $\lambda_\sharp$ and $B_\sharp$ depend on γ through $\sup|\mathsf{q}|$ (here q_i is the part of the increment b_i contributed by the remainders; see below), whereas the ceiling γ depends on the flow constants. Frames are therefore defined with constants $(\underline{\lambda}, c_5, \bar{B})$ fixed before γ: $\underline{\lambda}(j-i)-c_5\leq\theta_i-\theta_j\leq\bar{B}(j-i)$ for $i\leq j$. Hence a frame with first member θ_0 has length at most $(\theta_0-\gamma^{-2}+c_5)/\underline{\lambda}$, in the same way that K is bounded in terms of β in § 5.2.1. In addition, $\underline{\lambda}\leq\lambda_\sharp$ and $B_\sharp\leq\bar{B}$. *No caps*: there is no upper bound on $L, M, \kappa, q, \mathfrak{p}_0, c_a$, on the cutoff, the volume, an age or a number of steps, and no lower bound on a coupling. *The role of each exponent*:

(i) $\mathfrak{p}_0-\mathfrak{p}_1>Cr$, for comparing the ratio δ'_j/δ_j with the square root of the waiting depth, and for the accumulation of the stages (§4.17.1.3, § 4.17);
(ii) $2\mathfrak{p}_1-Cr>\mathfrak{p}_0$, so that a failed stage test still gives e^{-p_0};
(iii) $\mathfrak{p}_0>3Cr$, so that p_0 dominates $\kappa\check{R}^4$ and the cost $\check{R}^6$ of a wait, and so that the window left for $\mathfrak{p}_1$ by the two preceding conditions is non-empty;
(iv) $q-\mathfrak{p}_0>Cr$, so that $\mathsf{N}_k\mathsf{s}_k\to 0$;
(v) $q>C\mathfrak{p}_0$, so that the product of an amplitude and an oscillation of degree $2\mathfrak{p}_0$ remains bounded;
(vi) $\kappa_0>C$, for the sum over the scales.

(d) *No circle.* New terms are built from old ones, and the old ones are bounded with the same constants that the new ones must satisfy. The circle is avoided because an old term enters again in one of three ways:

- unchanged;
- through a difference, which carries a small factor. In $\mathbf{T}$ this factor is s_k, the factor that the Cauchy estimate gives for a first difference in the fluctuation field § 4.17.
- as a whole, in the main step of $\mathbf{R}$. In that case there are two possibilities. Either the domain of the term must stretch across $c\check{R}_kL^{k-j}$ cubes of the term's own level $j<k$, in which case its decay makes the term small. Alternatively, the domain lies inside an enlarged component (a collar $X^\sim$ of § 4.17). There only a volume bound $C\sum_{j\leq k}\#_j(Z_j\cap X)$ is available, each scale counted in its own unit, and its constant is chosen later. The list is given below under *Boundary terms.*

Uniformity of the input. A term of level j is defined on the radii at which it was created, and step $n+1$ evaluates it up to the output radius of that step. Let $w_{n,j}$ be the square of the ratio of these two radii. Let $\mathsf{f}_\nu\approx\nu^CL^{-c\nu}$ be forgetting weights; in § 4.5 they are written ω_ν. Then

$$w_{n,j}\approx\frac{\theta_j}{\theta_n}\Big(\frac{\log\theta_n}{\log\theta_j}\Big)^{2q}\leq C(n+2-j),$$

$$\sum_{j\leq n}\mathsf{f}_{n+1-j}\|E^{(j)}\|_{j,\kappa}+\sum_{j\leq n}w_{n,j}\|R^{(j)}\|_{j,\kappa}\leq N_\star$$

uniformly in n and in the frame. This is everything that the small-field step requires of its input. The two factors in the approximation above are never unfavourable at the same time: the upper flow bound gives $\theta_j/\theta_n\leq 1+(n-j)$, and the lower one gives $\theta_n\leq\theta_j+c_5$. The regular terms are

of size $O(1)$ at each level and gain $L^{-c(n-j)}$ through their two counterterms. The remainders gain nothing; they are summed by means of the coupling. *Regular terms*: let $\tau_k = x_k/\theta_k$, and let E_{N} be the bound on the localised Gaussian normaliser. Then

$$\|E^{(k+1)}\|_{k+1,\kappa} \le E_{\mathsf{N}} + C\big(|1-\tau_k| + \delta_k + \mathsf{s}_k N_\star + e^{-c\,p_0(g_k)^2}\big) \le E_0,$$

so E_0 is determined by the one-loop normaliser alone, and C depends neither on E_0 nor on $N_\star$.

The new regular remainder. At output level t, separate the fluctuation clusters which contain a vertex from a remainder of origin j with $M\check{R}_j > L^{t-j}$. Charge each of these clusters to one such vertex. In the absolute cluster bound with one distinguished insertion, this vertex has size $C\mathsf{s}_{t-1} w_{t-1,j} g_j^{\kappa_0}$, using the induction bound of the input. The root sum over the other vertices has a uniform constant at the rate of the fluctuation move. The sum over j must include both the number of origins and their radius losses.

Indeed, for $r = t-j$ the flow gives $x_t - C \le x_j \le x_t + C(r+1)$. Since $\check{R}_j$ is polynomial in $\log x_j$, the inequality $L^r < M\check{R}_j$ implies

$$L^r \le C(1+\log x_t)^C (1+\log(1+r))^C.$$

The last factor on the right is bounded by $CL^{r/2}$. Hence the contributing ages are $O(1+\log\log x_t)$. On this range $g_j^{\kappa_0} \le C g_t^{\kappa_0}$ and $w_{t-1,j} \le C(r+1)$. The fluctuation contribution in the regular remainder range therefore has norm at most

$$C\mathsf{s}_{t-1}(1+\log\log x_t)^2 g_t^{\kappa_0} \le \tfrac{1}{2} g_t^{\kappa_0}.$$

The negative power of $\log x_t$ in s_{t-1} pays for the sum. Use the final rate κ on the domains one cell layer inside the output regular region, with the ordinary contribution supplied on the regular range of the appropriate donor when the filing changes. There the exponentiation of healed histories supplies the other contribution, of norm at most $e^{-p_0(g_t)}$. The domains outside this range are filed as boundary terms. Since $e^{-p_0(g_t)} \le g_t^{\kappa_0}/2$, the new regular remainder has the same bound $g_t^{\kappa_0}$ as the input remainders at their respective levels.

Boundary terms: B is the maximum of the bounds coming from their origins § 4.17, and none of these bounds depends on B. This holds because data bounded by B enter the step in four ways:

- In $\mathbf{T}$, they enter as differences in the fluctuation field, with $C\mathsf{B}\mathsf{s}_k \le 1$.
- In the stages, they enter as differences in the collar field, taken one block of bonds at a time. Each block that is switched off contributes a negative power of $\log\theta_k$, with an exponent larger than the one belonging to (number of stages)$\times$(M-cubes of level k of a component).
- In the main step, outside the enlarged components, they enter either through the difference under the substitution, or as undifferenced functions of a level $j<k$. In the second case the domain must extend over $c\check{R}_k L^{k-j}$ cubes, and $\mathsf{B}e^{-c\check{R}_k} \le 1$.
- Inside an enlarged component there is no smallness, and they enter only through the volume bound on $V(X^\sim)$. The constant of this bound is chosen after B, and the bound is paid by the ledger of §4.17.1.40.

(e) *No growth with the level.* A carried term is literally unchanged: it has the same function, the same domain of analyticity and the same tree length. The bounds of a standing island refer to the level at which it was made and do not involve k. Indeed, the supremum in the mass bound is taken over a fixed set, built from the data of the levels up to that level, and this set contains the arguments of all later levels. Hence the mass at level k does not exceed the mass at the level at which the island was made (§4.17.1.40, "The bound has to reproduce itself", (ii)). The sizes that are returned are maxima rather than sums.

4.18. **Uniform weighted masses of histories.** A large-field factor is an integral operation with a memory. Its arguments include the fields that a later operation is allowed to read; its integration variables include the fields that an earlier operation has retained. These two lists need not describe the same region. We give the lists separate types, specify the mass of the operation, and prove the counting and domination implications needed to propagate that mass. In particular, a supremum over complex arguments is placed inside the integral. An estimate of a partition function at each fixed argument does not establish this stronger assertion.

We prove the geometric counts, scalar reserve estimates and their consequences. Their application to the interacting construction requires the background domains, kept quadratic forms, changes of variables and localized energy bounds stated below. A flat quadratic comparison supplies a special case of the matrix estimates; it does not establish their current-dependent, layered versions.

4.18.1. *The records carried by an operation.* Fix a finite four-dimensional periodic lattice. Work in bare coordinates: the bonds of level j have length $a_0 L^j$, where L is an odd integer. The torus and all partitions are chosen compatibly. A region is a union of partition cells, with a fixed half-open convention when a unique cell address is required. Closed cells are used for collars. Passing from one convention to the other changes only a fixed number of neighboring cells in each estimate below.

Let $\theta_j > 1$ be the reference inverse couplings, and write

$$\ell_j = \log \theta_j, \qquad g_j = \theta_j^{-1/2}, \qquad R_j = \check{R}_j, \qquad b_j^{\mathrm{cell}} = a_0 L^j M R_j. \tag{4.cdlxi}$$

Here M and every auxiliary partition size are fixed powers of L. The symbol g_j in this section uses the reference weight; the physical coupling is $x_j^{-1/2}$ when $x_j \neq \theta_j$. The partitions have cell side b_j^{cell}. The size sequence has $R_{j+1}/R_j \in \{1, L^{-1}\}$, and two size drops are not consecutive. Thus a passage between levels either coarsens these cells by L, or leaves their physical partition unchanged. Bounds are required uniformly in the height and volume, with the fixed geometric parameters held fixed.

A label of height s contains regions

$$\mathcal{S} = (\Omega_j, \Lambda_j)_{1 \le j \le s}, \qquad \Lambda_j \subset \Omega_j, \quad \Omega_{j+1} \subset \Lambda_j, \quad Z_j = \Lambda_j^c. \tag{4.cdlxii}$$

There is an initial collar $\Omega_0 = \Omega_1^{\sim}$; there is no independently generated large-field region of level zero. The state after a step and the state after its large-field operation are different labels. A history records their ordered succession, rather than a single final sequence with an inferred past. This distinction is necessary when an operation heals a region and a later creation covers the same physical location.

The operation also records a class of components and, in each component, the stages that have actually been completed. A component in the dynamic class has small size, the required depth, a quiet preparation window, quarantine from earlier marks in that window including its lower endpoint, and the single-box traces required by the local background construction. The dynamic class contains every eligible component. A static representation may deliberately use a smaller class. Waiting estimates below concern the dynamic rule; they would be false for an arbitrarily withheld class.

For an eligible component X of level t, let $t_\flat$ be the bottom of its preparation window and let $Z_j''(X)$ be its prepared cores. After $n(X)$ stages, with some components $\mathcal{F}$ completely removed, the determining entries have the form

$$\begin{aligned} N_j^{n,\mathcal{F}} &= \bigcup \mathcal{F} \ \cup \bigcup_{\substack{X \notin \mathcal{F} \\ t_\flat < j \le t_\flat + n(X)}} (X \setminus Z_j''(X)), \\ \Omega_j^{n,\mathcal{F}} &= \Omega_j \cup N_j^{n,\mathcal{F}}, \qquad \Lambda_j^{n,\mathcal{F}} = \Lambda_j \cup N_j^{n,\mathcal{F}}. \end{aligned} \tag{4.cdlxiii}$$

The full prepared sequence uses every permitted stage. In the construction its consecutive determining boundaries have a collar of at least one coarse M-cube. This separation, and the joint admission of all interpolated data to the background cores, are geometric and analytic conditions on these actual entries. Merely naming the entries in eq. 4.cdlxiii does not prove either condition.

There is also an action sequence. At a level redone by a drop-out it lists what is still stored, rather than what has just been prepared:

$$T \supset \Omega_1^{\mathsf{a}} \supset \Lambda_1^{\mathsf{a}+} \supset \Lambda_1^{\mathsf{a}} \supset \Omega_2^{\mathsf{a}} \supset \cdots \supset \Lambda_s^{\mathsf{a}}, \qquad Z_s^{\mathsf{a}} = Z_s. \tag{4.cdlxiv}$$

The extra symbol $\Lambda_j^{\mathsf{a}+}$ denotes the specified collared support of the action weights. Determining and action sequences can coincide at recent levels and differ at older ones. Equality at the last level does not permit replacing one by the other everywhere.

The rules for reading these records are as follows. Class membership, the traced old cores and the old integration regions use the stored label entries Z_j. Backgrounds, their data, determining sets, measures and gauge nests use the determining entries. Density tests and the last-shell test use the action regions. Coupling and vacuum weights use their specified action supports. The integral operation uses the retained operations and variables actually listed by the mark. These rules fix five distinct uses of the same geometry. They are part of the meaning of a read, not five interchangeable descriptions of a region.

4.18.1.1. A full re-entry in the broader waiting convention. This example allows a mark at the lower endpoint of the preparation window. The dynamic class used below includes that endpoint in its quarantine. The reason for this separation is visible without an estimate. Denote the old central box by P°, its old remaining core by N°, the old stored domain by the complement of O°, and put $W^\circ = O^\circ \setminus N^\circ$. In a full re-entry the field on W° is the current datum of the old level. It is not a field of the preceding level: those preceding-level variables have already been integrated. If the earlier failure preceded central integration, the field on N° is supplied by its still-present averaging delta function. If the earlier failure occurred during or after central integration, a bond touching P° can instead carry a free Haar argument and a retained datum used by the determining sets.

Substituting the stored region at a determining place would in the first case ask for an integrated-out variable on W°. Substituting the determining region at a test place would in general assert the presence of tests that the term no longer carries. Both errors can occur before any smallness estimate is attempted. Conversely, an argument that preserves the five lists above can check a read by a finite inspection of its variable and test lists. Its remaining analytic question is admission of those data to the required cores.

4.18.1.2. Gauge factors at a retained seam. Each retained gauge factor includes a tag specifying its gauge prescription and its actual background. At the retained inward seam one uses the relative prescription; away from that seam one uses the prescription specified by the corresponding operation. The group element in such a factor must be evaluated by that tag. An absolute gauge factor at a field and a relative factor at a field and background are different functions. A bound for one cannot identify the other. Removing every gauge factor on a bond with an endpoint in the next determining region is a possible different convention, but using it requires a new exact operation identity. No such removal is made by an estimate below.

4.18.2. *Local histories and their two argument spaces.* Write $\mathcal{F} = W_4$ for the finite group of gauge-tree frames and $h = |\mathcal{F}|$. A component that reaches the final decomposition, and hence can be healed or made an island, carries an operation

$$\mathbf{T}_i'(X)F(\mathbf{p}) = \frac{1}{h}\sum_{\omega}\int_{S_\omega} e^{\Phi_\omega(\xi;\mathbf{p})}F\,d\mu_\omega(\xi). \tag{4.cdlxv}$$

The local history ω is a discrete record. It lists the outer frame, the outcomes of the decompositions in its outer genealogy, and recursively a local history for each inner island. The variable ξ lists the variables left free after every surviving averaging delta function has been used as the definition

$V_{j+1}(c) = \mathcal{M}(V_j)(c)$. Its reference measure is a product of normalized Haar measures and Lebesgue measures. Gaussian, chart, elimination and gauge normalizers remain in the integrand; they are not silently changed into probability measures of mass one.

The support S_ω records the tests actually carried by that summand. It is a Borel set when the finite background maps used to define the tests are Borel. At this point Borel dependence of actual multi-level backgrounds is required; continuity of a formal expression in a named background is not a substitute for that property.

The external argument is a pair $\mathbf{p} = (\mathbf{U}, \mathbf{J})$ on $X^\sim$, where $X^\sim$ is the fixed coarse collar, not a rectangular bounding box. Its admitted complex core is denoted $\mathcal{U}_i(X^\sim)$. It is the core with the actual current normalization and substitution rules of the construction. An arbitrary complex current array is not declared admissible by this notation.

To describe the splitting used in the main step, present a pair by

$$\mathfrak{z} = (U, a', \mathbf{J}), \qquad \mathbf{U} = e^{ia'}U, \qquad U \in G^{\mathrm{bonds}}. \tag{4.cdlxvi}$$

Here a' is the dimensionless bond logarithm. If a rescaled potential A' is used instead, its lattice spacing factor must be included in a'. A typical layer bound for a bare bond is $|a'| < \hat{\alpha}_{1,j}L^{-j}$ and that for a bare plaquette is $|\partial\mathbf{U} - 1| < \hat{\alpha}_{0,j}L^{-2j}$. The additional current and local-gauge conditions are retained in the admitted core. These displayed layer inequalities alone do not replace them.

The complete exponent has the special decomposition

$$\Phi_\omega(\xi; \mathbf{p}) = \Phi^\circ_\omega(\xi) - \tfrac{1}{2}q_X(\xi; \mathfrak{z}) + V_X(\xi; \mathfrak{z}), \qquad q_X \geq 0. \tag{4.cdlxvii}$$

The inequality $q_X \geq 0$ here is on the real stored integration fibre. The glued background defining q_X is built at the G-valued factor U, and is real. This is the reason for its nonnegativity, including when the represented pair is complex. The two last summands need not be functions of the pair separately. Their sum is required to take the same value at any two admitted triples representing the same pair, and to be holomorphic in that pair. Cauchy estimates apply to that sum, not to an arbitrarily selected real quadratic summand.

4.18.2.1. Continuing the main-step quadratic form. Keep the real quadratic term in eq. 4.cdlxvii. Its continuation as a function of the pair can be constructed at the admitted glued datum. Let V be the real datum of the joined set of X, using the prescribed inner rows and the exterior averages, and let $V_\gamma = e^{i\gamma}V$. Write the continuation of its critical representative in the form $\mathbf{L}_V(\gamma)^{u_\gamma}$, where $\mathbf{L}_V(\gamma)$ is in the Landau chart about the real critical representative of V. Use the on-shell pair $(\mathbf{L}, J(\mathbf{L}))$, $\mathbf{L} = \mathbf{L}_V(\gamma)$. Let N_{V_γ} be the differential at zero of the map from raw row data to reduced data at V_γ. It is the identity on interior rows. At an outward row c it adds $\mathrm{Ad}(V_\gamma(c))\dot{v}_{c_+}(f)$ when c_+ is the outer endpoint and subtracts $\dot{v}_{c_-}(f)$ when c_- is the outer endpoint. Here $i\dot{v}_z(f)$ is the first derivative at zero of the one-step node factor at z with datum direction f on its block. For a raw datum direction f, put $f^\sharp = \mathrm{Ad}(u_\gamma^{-1})N_{V_\gamma}f$, with the node gauge acting at the initial endpoint of each row.

With the cutoff ζ^{co}_X of the main step and the walk expansion $H_1 = \sum_\omega H_{1,\omega}$ at this pair, define

$$t_{\omega\omega'}(\gamma; V)[f] = \sum_p \eta^4 \zeta^{\mathrm{co}}_X(p)\big\langle (D_{\mathbf{L}}H_{1,\omega}f^\sharp)(p), (D_{\mathbf{L}}H_{1,\omega'}f^\sharp)(p)\big\rangle.$$

Here $D_{\mathbf{L}}$ is the covariant exterior difference and the pairing is bilinear. The ordered pair of walks is counted once. This uses the fixed regular-pair radii of the walk expansion. If a_{D} is their ceiling and K_{S} the profile-to-response constant, choose the real regularity ceiling ε_{w} and the datum radius c_{w} so that

$$C(L, \varrho_0)M_1\varepsilon_{\mathrm{w}} \leq a_{\mathrm{D}}/4, \qquad CK_{\mathrm{S}}c_{\mathrm{w}} \leq a_{\mathrm{D}}/4, \qquad Cc_{\mathrm{w}} \leq \varepsilon_{\mathrm{w}}/2.$$

The layer gaps of the joined set must give the weighted summation for these walks. In particular the exponential decay after the spacing comparison must absorb L^{d/ϱ_0} at distance d. On $|\gamma|_\infty < c_{\mathrm{w}}$, the profile and response bounds admit the pair on the enlarged walk carriers, and the node gauge and reduced datum give $|f^\sharp|_\infty \leq C|f|_\infty$.

At a plaquette based in a cell of spacing ξ_y, the walk estimate gives parameter-independent weights for the differentiated lifts. Summing their termwise suprema with these weights gives $C\xi_y^{-2}|f|_\infty$. Its square costs $C\xi_y^{-4}|f|_\infty^2$, while the cell has at most $6(\xi_y/\eta)^4$ plaquettes based in it. Thus, for the n cells carrying the cutoff, the sum of the suprema of the ordered-pair terms is at most $K_{\mathrm{w}}\mathsf{n}|f|_\infty^2$. The sum $q^{\mathrm{w}}_{0,X}$ is holomorphic on the datum disc. At a real datum the covariance of each walk term removes the gauge u_γ and identifies the sum with the real main-step form there. This uses the differential of the physical response, namely $H_1 N_{V_\gamma}$ in the reduced row coordinates. Recenter at $e^{ib}V$ for real $|b|_\infty < c_{\mathrm{w}}/4$ and $|\gamma|_\infty < c_{\mathrm{w}}/2$. The two formulas agree at real data, so their holomorphic continuations agree on the common datum disc after the logarithmic change of center. This uses the same glued datum in both splittings, including the node and inner-row conventions.

For an actual triple, the real glued datum must be $\varepsilon_{\mathrm{w}}/4$-regular, and its increment from the exterior pair must satisfy $|\gamma|_\infty \le C\alpha_{1,t} < c_{\mathrm{w}}/4$. Schwarz's lemma on the complex line through γ gives

$$\left|\tfrac{1}{2}\theta_t(q^{\mathrm{w}}_{0,X} - q_{0,X})(g_tB)\right| \le \frac{CK_{\mathrm{w}}}{c_{\mathrm{w}}}\,\theta_t\mathsf{n}(\delta_t')^2\alpha_{1,t}.$$

Here $|g_tB|_\infty < \delta_t'$ and $q_X = \theta_t q_{0,X}(g_tB)$ in the real presentation. The coupling ceiling imposes $C\max\{2\alpha_{0,t},\varepsilon_1\} \le \varepsilon_{\mathrm{w}}/4$, where ε_1 bounds the input datum tests, and makes the last bound at most 1. The absolute sum in the exponent is at most $\frac{1}{2}K_{\mathrm{w}}\theta_t\mathsf{n}(\delta_t')^2$. The same ceiling requires this to be at most $(\log\theta_t)^{2\mathfrak{p}_0}/10$ using $\mathsf{n} \le C\check{R}_t^5$. If the analytic form is separated from the exponent, the corresponding remainder is $V_X + \frac{1}{2}\theta_t(q^{\mathrm{w}}_{0,X} - q_{0,X})(g_tB)$. For the modulus bound keep the original nonnegative real term. This comparison concerns the form of the main step at its glued datum. The older conditioned forms carried by the history have separate free coordinates, row restrictions and elimination factors.

An exact consequence worth isolating is

$$\sup_{\mathbf{p}\in\mathcal{U}_i(X^\sim)} |e^{\Phi_\omega(\xi;\mathbf{p})}| \le e^{\Phi^\circ_\omega(\xi)+C_V N_0(\omega)} \quad \text{if} \quad \sup_{\mathfrak{z}} |V_X(\xi;\mathfrak{z})| \le C_V N_0(\omega). \tag{4.cdlxviii}$$

All representatives in the supremum must obey the same bound. The function N_0 counts the stored sites at their own levels, with the fixed finite multiplicities belonging to the operation. Proving the complete localized bound on V_X is an analytic formation task. Once it is proved, eq. 4.cdlxviii is a pointwise envelope before integration; it is stronger information than a pointwise polymer logarithm.

4.18.2.2. The retained inner factors in the complete exponent. In the fixed real reference system, take a component X of the main step at level t and write its kept exponent in eq. 4.cdlxvii as $\sum_R T^1_R$. Its carriers are in the enlargements of older islands or of strips of levels at most $h_t = t - \mathsf{N}_t$, where N_t is the number of preparatory stages. The stored bonds are in the inner core. The enlargement costs at most one cell of level h_t, and the next two shells have widths at least $M\check{R}_t\xi_{h_t}$ and $LM\check{R}_t\xi_{h_t}$. The scale comparison $\check{R}_{h_t} \le L\check{R}_t$ therefore puts every carrier inside Z''_{h_t+2}. Write $l(x)$ for the layer index of the prepared sequence and put $n_R = \max_{x\in R} l(x)$ and $u_R = (h_t - 1 - n_R)_+$. In particular $n_R \le h_t + 1$.

Let $U_\star$ be the adapted real background of the datum $\mathbf{V}_\star$ on the component determining set. This datum equals V_X^{in} on the inner rows and the identity on the other rows. Its regularity follows from the kept tests: the half-threshold test at the top of the prepared nest and the tests inside its last preparatory operation give the levelwise bounds on the holonomies of $(\mathbf{1}, V_X^{\mathrm{in}})$. The real data V_X^{in} remain in that range while the box variable B varies on $|g_tB| < \delta_t'$. Write $q(\xi)$ for the pair of $U_\star$ and use the pinned representative at the stored sites. For every kept factor φ_R, use its original target space, with a bound $b_R^\sharp$ on the admitted spectator range. In the reference evaluation the spectator values are held fixed at their stored values. The factors are invariant, have the tie on real

triples with the same pair and have their data at the prescribed adapted sites. Their structures precede the prepared structure.

To form the new argument, apply the exterior row map $E_X^\natural$ to the decoupled response to the formed belt datum $\mathfrak{b}^Y(s;P,\varsigma[P;\xi])$. Here P is in the space of the output structure on the localization Y and $|s(\Delta)|<e^{\bar\kappa}$. On $X^\sim$ the output space has the single level t, so the endpoint blocks of the tower rows lie in its one-scale core. For a tower row c, write $T_c(W)$ for the average of W transported along its two combs to the tops $y_\pm$. Put $U^+=e^{i\eta\mathbf{A}}\mathbf{U}$ for the exterior shift, $v_\pm=v[P;\mathbf{A}](y_\pm)$ for its site factors and $u_-=\mathbf{U}(\mathrm{cb}_{c_-})$. Then

$$e^{i\gamma(c)}=u_-^{-1}v_-^{-1}T_c(U^+)v_+T_c(\mathbf{U})^{-1}u_-.$$

The set read by this product consists of the two combs, the blocks of their tops and the double block of c. It is at distance at least $cM\check R_t$ from the belt. If ϕ is the supremum of $|\mathbf{A}|$ on this set, the product with $z\mathbf{A}$ is holomorphic and within $1/2$ of the identity for $|z|<r_{\mathrm E}/\phi$, by the site factor and relative average bounds. The product is the identity at zero. Schwarz's lemma bounds its logarithm at $z=1$ by $\phi/r_{\mathrm E}$. The response bound for ϕ therefore gives $|\gamma|\le Ce^{C\bar\kappa}\varepsilon_\beta e^{-c\delta_0 M\ell_t^r}$, where $\ell_t=\log\theta_t$. The logarithm and the composed maps are holomorphic at fixed real factor throughout the given cube polydisc.

Form $\mathbf{V}$ on the component rows using $e^{i\gamma}\mathcal{M}^m(\mathbf{U})$ on the tower rows of level m, $\mathcal{M}^t(\mathbf{U})$ on the remaining exterior rows, the formed box averages with their left factor e^{ig_tB} on the box rows, and V_X^{in} on the inner rows. Put

$$B^\natural=\frac{1}{i}\log(\mathbf{V}^{\hat w}\mathbf{V}_\star^{-1}),$$

where $\hat w$ is the comb of $\mathbf{U}$ at the exterior nodes and the identity at the nodes of the core. It acts on the datum at the nodes. The low box rows have both ends in the core and there

$$e^{iB^\natural(c)}=e^{ig_tB(c)}e^{i\nu_P(c)},\qquad |\nu_P(c)|\le C_1'L^{m-t}a_C,\qquad a_C=\max_\iota\alpha_{\iota,t}.$$

The factor $e^{i\nu_P}$ is the continuation of the box average in the exterior datum. At an entrance row the closed contour through the gauge tree and the exterior path has $O(M^2)$ plaquettes. Its curvature bound and the exterior comb bound give the bound $\epsilon_{\mathrm{out}}=C_1'M^2\check R_ta_C$ on the remaining rows, including the small complex correction to the comb. On the inner rows $B^\natural=0$. Thus the whole formed datum is in the continuation ball $\|B^\natural\|_\infty<r_{\mathrm B}$.

Use the Landau response $(\mathcal{H}_\star,\mathsf{j}_\star)$ at $\mathbf{V}_\star$, transported to the adapted anchor. Its current increment is $J(e^{i\eta\mathcal{H}_\star(B^\natural)}U_\star)-J(U_\star)$. On the bonds of R define the actual argument by

$$\mathfrak{S}_R(\lambda;\mathfrak{z},\xi)=\mathfrak{p}[U_\star]|_R\mathbin{\dot+}(\lambda\mathcal{H}_\star(B^\natural),\lambda\mathsf{j}_\star(B^\natural))^\delta|_R.$$

The superscript denotes levelling with all the stored sites pinned. Both increments are multiplied before levelling. The normalized potential, first difference and current of the unlevelled response are bounded by a constant times the datum profile. The current bound follows from the continued current difference and the deviation of the response gauge from the identity, since the scaled current is bounded in the chart.

For $h_t\le m\le t$ let $D_m(x)$ be the prepared weighted distance to the sites outside the innermost region I_2 with layer index at least $m-1$. Convolving the row bounds with the decaying response gives the slowly varying profile

$$s_\star(x)=C\left[\delta_t'e^{-\theta_\flat D_{h_t}(x)}+a_C\sum_{m=h_t}^{t-1}L^{m-t}e^{-\theta_\flat D_m(x)}+\epsilon_{\mathrm{out}}e^{-\theta_\flat D_t(x)}\right].$$

Separation gives $D_m(x)\ge\varrho_0(m-2-l(x))_+$. Sum the middle series using $Le^{-\theta_\flat\varrho_0}\le1/2$. Since $\mathsf{N}_t\gtrsim\ell_t^r$, both $a_CL^{3-\mathsf{N}_t}$ and $\epsilon_{\mathrm{out}}e^{-\theta_\flat\varrho_0(\mathsf{N}_t-3)}$ are smaller than a fixed multiple of δ_t' at small coupling. Thus for $l(x)\le h_t+1$ the profile is at most $C\delta_t'e^{-\theta_\flat\varrho_0(h_t-1-l(x))_+}$.

The anchor has half the radius of each target clause. Its scaled curvature and current follow from the profile of the regular datum. Local gauges follow on the required cubes from the curvature bound, using one finer layer for cubes touching the boundary. On a subordinate tower the restricted anchor is critical over its own averages. The tower therefore has the same curvature and current up to gauge. This also applies to the towers of the single-scale cores after truncation. These checks use separation of the component sequence, which can have holes near its exterior boundary.

The available shift is $\kappa_{\mathrm{R}} \min_\iota \alpha_{\iota,l(x)}$ in prepared units for a slowly varying profile. For a target index $m \le l(x)$ the shift gains at least $L^{-(l(x)-m)}$ in its own units. The radii change by at most a factor $C(1+l(x)-m)^{1/2}$, so the rescaled profile spends only a fixed small part of the pointwise radii. The gauges of the unchanged real factor still apply. For each subordinate tower the local Lipschitz bound on its curvature and current takes the supremum of the rescaled profile with an exponential distance penalty. The tower decay rate exceeds $\theta_\flat + 2\log L/\varrho_0$, which absorbs the variation of both layer indices and of the profile. The same bounds apply to the single-scale cores and their towers. This gives the stated allowance for the pinned shift, including the fixed cost of levelling. The allowance also puts its local potential below the levelling threshold on the neighboring tops.

Finally, the running radii satisfy $\min_\iota \alpha_{\iota,l} \ge c(1+t-l)^{-1/2} \min_\iota \alpha_{\iota,t}$, and $1+t-l \le (2+\mathsf{N}_t)(1+(h_t-1-l)_+)$. The ratio $\delta'_t/\min_\iota \alpha_{\iota,t}$ is a constant times $\ell_t^{\mathfrak{p}_1-q}$, where q is the radius exponent. Absorbing the remaining square root into the layer exponential shows that the normalized pinned shift on R costs at most $|\lambda| a_R/2$ of the allowance, with

$$a_R = C_* \ell_t^{-(q-\mathfrak{p}_1-r/2)} e^{-\delta_\flat \varrho_0 u_R} \le C_* \ell_t^{-(q-\mathfrak{p}_1-r/2)} (4L^8)^{-u_R}.$$

In particular the argument stays in the target space for $|\lambda| < 2/a_R$ uniformly on the cube polydisc and the admitted spectator range. It is holomorphic with the real factors held fixed. It reads $q(\xi)|_R$ at zero. At the real full parameters the belt and the box averages give the actual prepared datum. The pinned response has its current and the prescribed transports on the stored sites. The adapted representative on R can be taken from the prepared orbit because its pillars inside the core are the same as the component pillars. Its plain triple satisfies the target bounds by the real profile comparison, and the tie identifies its value with the pinned value. Thus $T_R^1 = \varphi_R(\mathfrak{S}_R(1;\mathfrak{z},\xi))$.

Set $\delta_R = T_R^1 - \varphi_R(q(\xi)|_R)$. Schwarz's lemma, applied to the difference along this substitution disc, gives $\sup|\delta_R| \le a_R b_R^\sharp$.

For the absolute bound of a kept member, split off the part of each island factor that depends only on its stored variables. The remaining factor vanishes at the trivial triple, for each value of those variables, so its supremum is bounded by its oscillation. A strip factor consists of the quadratic walk sum and the logarithms of its external pivot Jacobians. On the enlarged spectator discs the quadratic sum costs the square of the radius multiplier $(\log\theta_{j-1})^{6r+1}$ at the birth level j. For each pivot, the full logarithm has modulus bounded by $9\log L + c_{\mathrm{P}}$ on the pivot chart. In the convention $\log z^{(j-1)} = -\mathfrak{p}_{c'} + 9\log L$ the second term is the scalar from the pivot normalization. Its carrier has one or two adjacent cubes and there are at most $8M^4$ such pivots per carrier. These bounds give, for each level-j carrier,

$$b_R^\sharp \le B_\Phi^\sharp \widehat{\ell}_j^{\,\mathfrak{p}^\sharp} e^{-\kappa d_j^{\mathrm{in}}(R)}, \qquad \mathfrak{p}^\sharp = 2\mathfrak{p}_0 + 12r + 2, \qquad \widehat{\ell}_j \le \ell_t(1+\log(2+t-j)).$$

Here $\widehat{\ell}_j = \max\{\log\theta_j, \log\theta_{j-1}\}$ and $B_\Phi^\sharp$ absorbs the strip and island constants. If the member is divided into identical copies, the bounds are divided by the number of copies, so their sum is still bounded by $b_R^\sharp$.

Write $z_j(X)$ for the number of side-M cubes in level-j units in the action large-field region of X, and put $Q = 4L^8$. The levels $j \ge h_t - 1$ contain at most $C\ell_t^{5r}$ such cubes in total. A kept carrier has a cube in the enlarged action large-field region of its level. Counting through that cube with the entropy bound, including the bounded multiplicity of exempt island hulls, controls the contribution

from these levels by $CB_\Phi^\sharp \ell_t^{5r+\mathfrak{p}^\sharp}$, since the bounded frame increments give $\theta_{j-1} \le \theta_t + C(\mathsf{N}_t + 2)$ throughout this window.

For $p = h_t - j \ge 2$, the possible rooting cubes are in the level-$(j+1)$ action large-field region. It is contained in the inner region with at most $C\ell_t^{4r}$ cubes of level h_t, so the rooting count at level j costs at most $C\ell_t^{4r} L^{4p}$. The root lies outside a member of the prepared sequence with index $i'+1 \le h_t$. Here i' is j, $j+1$, or the level $i < h_t$ of a drop-out having $i - N_0(i) \le j < i$, where $N_0(i)$ is the number in the old drop-out index $k_0(i) = i - N_0(i)$. If $n_R \ge i' + 2$, the carrier must cross a prepared gap and $d_j^{\mathrm{in}}(R) \ge \frac{1}{2} L^{n_R - j}$. The exempt hulls stay on the lower side of the gap and can reduce the length by at most two level-j cube units. Summing over $m = n_R - j$ in this case costs, after entropy,

$$\sum_{m \ge 2} e^{-cL^m} Q^{-(p-1-m)_+} \le C Q^{-(p-1)}.$$

Otherwise, in the two unmarked cases $n_R \le j + 2$ and $u_R \ge (p-3)_+$. In the drop-out case $u_R \ge (p-2-N_0(i))_+$, so the extra cost is at most $Q^{N_0(i)} \le L^9 \rho_i^9$, where $\rho_i = \check{R}_{j'}$ at $j' = i - N_0(i) + 1$. The frame bound $\theta_{j'} \le \theta_t + C(t - j')$ and $N_0(i) \le 2 + r \log_L \log \theta_{j'}$ give $t - j' \le 2(t-j)$ at small coupling, since $t - j \ge \mathsf{N}_t + 2$ absorbs the logarithmic delay. Consequently

$$\rho_i \le C[\ell_t (1 + \log(1 + \mathsf{N}_t))(1 + \log(1+p))]^r.$$

In all three cases the geometric factor from L^{4p} and Q^{-p} sums against every fixed power of $1 + \log(1+p)$. Since $\mathsf{N}_t \le C\ell_t^r$, summation gives

$$\sum_R b_R^\sharp (4L^8)^{-u_R} \le K_A' B_\Phi^\sharp \ell_t^{\mathfrak{p}^\sharp + 13r} (1 + \log \ell_t)^{9r + \mathfrak{p}^\sharp}.$$

It follows that

$$\sum_R \sup |\delta_R| \le C_* K_A' B_\Phi^\sharp \ell_t^{5r+E+12r+2} (1 + \log \ell_t)^{9r+\mathfrak{p}^\sharp}, \qquad E = 2\mathfrak{p}_0 + \mathfrak{p}_1 + \tfrac{17}{2} r - q.$$

With $E + 12r + 2 < 0$, the coupling ceiling and the geometric bound $N_0(\omega) \ge 2^{-5} M^4 \ell_t^{5r}$ make this at most $N_0(\omega)$. All suprema here are over the common admitted range. Use $b_R^\sharp$ on this range, including the enlarged discs of the one-step quadratic factors.

Let V_X^{col} be the collar sum before the change of inner reads. It consists of the localization of the action difference, the small Wilson sum $\mathfrak{v}(X^\sim)$ and the Wilson weight correction W_X. Use the count $N_0(\omega) = M^4 \sum_{j \le t} z_j(X)$ from the action regions.

In the action difference, cancel each carried trunk with its partner on the new side before estimating the localized sum. The localized differences of a carried entry of size b_e and amplitude a_e have total grouped bound $a_e b_e$ on a collar. Indeed each difference over switched cubes is bounded by $a_e b_e$ times exponential decay in the number of cubes needed to connect its support to its base. Summing the connected extensions uses the cube entropy and the chosen decoupling radius. The same estimate for the nontrunk pieces of an unpaired entry can be absorbed into its bound Cb_e, since $a_e \le 1$. An unpaired entry is counted with its whole trunk, and a plain entry with its whole value. The kept trunks have the separate estimate just proved.

At level t there are at most six ordinary entries on a domain and at most $3^4 z_t(X)$ collar cubes through which to root them. For $j < t$ the index condition on an ordinary entry gives a cube in a one-layer enlargement of the action large-field region of level j. A root of a domain in this collar belongs to the enlargement of X's own trace, by separation of the class components. There are at most $3^4 z_j(X)$ such cubes. The level-j island hulls in the collar have at most $3^4 \mathsf{m} z_j(X)$ cubes in total, where m bounds their multiplicity. Counting domains with their length decay therefore costs at most $C(1 + \bar{c}) z_j(X)$ for the ordinary entries. This count includes the unpaired plain numbers of the gained domains, namely the pointed values and the remainder values at the trivial pair. The

large quadratic strip terms of level t are carried outside the collars. The older strip factors with base in the collar occur among the kept entries.

For the regular members, let m be their level and n the last action layer that their carrier meets. A member with $n < t-1$ is rooted in the action complement of level $n+1$. That complement in X offers at most $L^{4(n+1-m)} z_{n+1}(X)$ possible level-m roots. The carrier has length at least $2L^{n-m-3} - 2$, so after paying the domain entropy and one power of the length for the member norm, the remaining decay sums over $m \le n$ against the root count. The last layer uses the collar cubes as roots instead. This gives another $C(1+\bar{c})\sum_j z_j(X)$. Thus the action difference has the required absolute majorant, with every grouped piece, unmatched trunk and unpaired plain number counted.

In the small Wilson sum, the unpaired functions are the local correction (including the analytic reference normalizer) and the interior Wilson comparison with the actual exterior row increment. Their combined supremum is at most $C(1+\bar{c})\ell_t^{5r}$. The two unsuppressed Wilson differences are supported in at most $C\check{R}_t^4$ and $C\check{R}_t^5$ unit cells, respectively. The difference of weights times the squared field bound is $O(1+\bar{c})$ on these cells. This follows from the tube width $C\theta_t \ell_t^{-2q_t}$, $q_t \ge q$, and the field size $C\theta_t^{-1/2}\ell_t^q$ on the prepared window. For the localized Wilson forms, the regularity of the real inner data is bounded by $C\ell_t^r \varepsilon_t$ on the retained tests, as required by the box construction. The other comparisons have small factors $e^{-c\delta_0 M\check{R}_t}$ or $L^{-\mathsf{N}_t}$, or the Taylor loss $\theta_t^{-1/2}$, which absorb the polynomial logarithmic factors. For the comparison in the deep inner region the cells are summed with their weighted-distance decay. The layer gap absorbs the possible increase of the weight there. The analytic normalizer is uniformly bounded on its box. These estimates are on the complex increment ball used in the formed Wilson comparison.

The remaining Wilson entries are matched exterior-cell terms. On this collar their total localized difference is bounded by $C(1+\bar{c})\check{R}_t^4 \ell_t^{2q+2r}$ times the pinned exterior response amplitude, and that amplitude is bounded by $C\theta_t^{1/2} e^{-c\delta_0 M\check{R}_t}$. This term is bounded by $C(1+\bar{c})\ell_t^{5r}$ at small coupling. There is at least one level-j label cell in the trace for each $h_t \le j \le t$. Each such cell contains $\check{R}_j^4 \ge 2^{-4}\ell_t^{4r}$ side-M cubes, and there are $\mathsf{N}_t + 1 \ge \ell_t^r/2$ levels. Hence $\sum_j z_j(X) \ge 2^{-5}\ell_t^{5r}$. Thus the small Wilson sum is bounded by $C(1+\bar{c})\sum_j z_j(X)$.

Finally, W_X is the sum over plaquettes in X of the Wilson density times the difference between the new weight and θ_t. The density is bounded by $C\eta_t^4 \alpha_{0,t}^2$, and each level-t cube has at most $CM^4\eta_t^{-4}$ such plaquettes. The correction therefore costs $CM^4 z_t(X)\alpha_{0,t}^2(|x_{t-1} - \theta_t| + \bar{c})$. The adjacent-level flow and the tube bound make the last product of the radius squared and the weight difference $O(1+\bar{c})$. This estimate holds after splitting the correction into its local plaquette carriers as well.

Adding the three majorants gives $\sum_\lambda \sup|v_\lambda| \le C_{\mathrm{col}}(1+\bar{c})N_0(\omega)$ for the collected collar terms on their common range. The separate Gaussian counting constant keeps its assignment in the reference normalization. The exact assignment is

$$V_X = V_X^{\mathrm{col}} + \sum_R \delta_R, \qquad \Phi_\omega^\circ = \Phi_\omega^{\mathrm{rest}} + \sum_R \varphi_R(q(\xi)|_R).$$

Here $\Phi_\omega^{\mathrm{rest}}$ contains the retained inner Wilson factor and all other terms depending on ξ alone. This preserves the exponent and proves $|V_X| \le [C_{\mathrm{col}}(1+\bar{c})+1]N_0$. The pair-dependent part of the exponent remains $-q_X/2 + V_X$, with its given holomorphy and independence from the representation of the pair. The bound gives eq. 4.cdlxviii. The integral of $e^{\Phi_\omega^\circ}$ uses the real coordinate estimates below.

4.18.2.3. The pair reader on a formed island. On the output core of the formed island kit, write $f = g_t B$ and $\zeta_X = \zeta_{0,X} - \zeta_{1,X}$. Use the generated Wilson configuration $W^{\mathrm{f}}(b,f)$ of the prepared component. Its datum is the lift of the outer average of the pair, the perturbed collar datum $e^{if}V_0[d;V^{\mathrm{e}}]$, and the retained inner datum V_X^{in}. At fixed real stored variables these give the same

complex datum for all splittings of the pair. On the real slice the glued representatives are related by a gauge equal to one at the stored pins. The Wilson trace is invariant there, and the tied outer factors also have equal values there. On the two prepared leaves use a complex parameter z taking the real interpolations to the desired splitting at $z = i$. For each real collar parameter the two holomorphic reads agree for real z, so they agree at i by the identity theorem on the neighborhood of $[0, i]$. Continuing in the collar parameters gives equality on the common formed core. The prepared leaves use the same pinned data and margins.

Define the weight-free quadratic form

$$q^{\natural}_{0,X}(f; P) = \frac{d^2}{ds^2}\bigg|_{s=0} A(\zeta_X, W^{\mathrm{f}}(b, sf)).$$

It reads the pair P at fixed stored variables. No entry of the source system enters it. The holomorphic remainder

$$T_6(b, f) = -\theta_t \left[A(\zeta_X, W^{\mathrm{f}}(b, f)) - A(\zeta_X, W^{\mathrm{f}}(b, 0)) - \tfrac{1}{2} q_{0,X}(f)\right]$$

is bounded by one on the admitted radius $5\delta'_t$ under the prepared Wilson estimate and coupling ceiling. For $\|f\|_\infty \le \delta'_t$, its second coefficient along sf, bounded by Cauchy on $|s| = 1$, gives

$$\left|\tfrac{1}{2}\theta_t (q^{\natural}_{0,X} - q_{0,X})(f)\right| \le 1.$$

Before transferring the kept trunks to $q(\xi)$, denote the collar interaction by V^{col}_X. The new pair reader is

$$V^{\natural}_X = V^{\mathrm{col}}_X + \sum_R \delta_R + \tfrac{1}{2}\theta_t (q^{\natural}_{0,X} - q_{0,X})(f).$$

Thus $-\theta_t q^{\natural}_{0,X}(f)/2 + V^{\natural}_X$ plus the trunks read at $q(\xi)$ gives exactly the old exponent. The formed-core bounds use the carrier $X^{\sim}$, which already contains the prepared box and the gauge tree. They apply on the output island core to the individual pieces with the stored variables fixed. In the sourced-minus-zero difference the common quadratic correction cancels, leaving the Wilson source on the inner field and $V^{\natural,w}_X - V^{\natural,0}_X$, with the inherited exponents read at their admitted real pairs. This is eq. 4.dxvi on that core. The absolute cost and the source oscillation use its carrier bounds. 4.18.2.4. The factor bound at the birth of an island. On the admitted formed core, split each collar member into its value at the fixed anchor and the change from that value, using the prescribed carrier of the member. In the old core the anchor reads the stored variables alone. File its value in Φ°_{ω}. The oscillation of the change costs at most twice the amplitude on the anchored substitution disc. The factor $(4L^8)^{-u_R}$ at depth u_R pays for the L^{4u_R} root count and its fixed multiplicities. The additional negative power of $\ell_t = \log \theta_t$ absorbs the inherited sizes. Outside the old core, the waiting window contains at most ℓ_t^{Cr} local cells, counted at their own levels. The regular terms use the pointed sums with their two subtractions, and the Wilson terms use their net weights. The collar oscillations therefore use the single-term collar bound with constant b_V, independently of the inherited oscillation constant.

For the kept contributions use the bound on $\sum_R |\delta_R|$ on the common formed range. Their total oscillation is at most twice that bound. Its logarithmic factor, by the choice of q, makes the cost at most $\mathsf{B}_\Phi \ell_t^{2\mathfrak{p}_0}/4$ when the coupling ceiling is chosen after the strip and island constants. The new quadratic and its displayed correction cost at most $CA_1^2 \check{R}_t^5 \ell_t^{2\mathfrak{p}_1} + C$. Use the prescribed tied expression for these pieces. The hierarchy between $\mathfrak{p}_0$, $\mathfrak{p}_1$ and the waiting exponent makes their oscillation at most another quarter. Use the whole carrier $X^{\sim}$ for the quadratic and the retained changes formed through the island's full reader. That carrier is exempt in the output label and has hull length zero. The smaller carriers of the local collar members keep their own decay.

Choose $\mathsf{B}_\Phi \ge 2\mathsf{b}_V$ and keep the preceding quarters as reserves. The new island then has the same real oscillation bound per carrier as the incoming islands. For an island that stands, its exponent,

stored variables and carriers remain those of its birth operation. Restricting its arguments preserves the old oscillation bound. These two cases return a maximum over the formed and carried entries. The bound applies on the formed cores and uses the admission and mass bounds of the actual histories.

4.18.2.5. Why the old Wilson factor is retained. The main action expansion is made only on the small-field part $\zeta_0(1-\zeta_1)$, about the real glued background. On the old core the factor

$$\exp\{-A(x''\zeta_1\mathbf{1}_X, U''(\xi))\} \tag{4.cdlxix}$$

is left intact. The inner background $U''(\xi)$ is real and is formed from the prepared determining sequence, with its old innermost datum and its new fluctuation datum. It is distinct from the glued background used in q_X. The factor in eq. 4.cdlxix is independent of the external pair. Its plaquettes include the old failed tests that have survived in the record. Replacing it by a quadratic Taylor expression throughout the old core would remove the positive reserve used below.

The other factors in e^{Φ° are chart densities, constants, kept gauge and elimination factors, stored Gaussian weights and the inner operations evaluated at real pairs. Their normalization and their domains will be used individually. The phrase "the complete exponent" always includes all of them.

The restored law is combined with the old Wilson action before estimating its normalization. Fix one frame and all exterior data of a component X at level i. Write s for its retained old variables and v for its restored box field, with the tree condition imposed. Set

$$S_{\mathrm{old}}(s) = A(x'', U_i''(s)),$$
$$S_{\mathrm{ref}}(v) = \theta_i A(\zeta_0, U_{i,X}(v)), \qquad S_0 = \theta_i A(\zeta_0, U_0).$$

The denominator has the form $D_X^{-1} = e^{S_0 + I_X - E_X}$, where I_X is the negative logarithm of the centered reference integral after extracting its counting factor e^{E_X}. To combine its center with the Wilson actions, write $w = x''$, $\theta = \theta_i$, $W'' = U_i''$, $W^0 = U_0$, $W^Z = U_{i,X}(v)$, and let $\bar W$ be the completed background. In a succession of completed components these are the backgrounds before and after the current replacement. Use $w_{\mathrm{fr}} = \theta$ on the component and $w_{\mathrm{fr}} = w$ off it. The replacements telescope over the components.

The three parts of the old weight use $1-\zeta_0$, $\zeta = \zeta_0 - \zeta_1$, and ζ_1, with $\zeta_0 = 1$ on the support of ζ_1. Write $\Delta_a(P,Q) = A(a,P) - A(a,Q)$ for weighted action differences. The exterior part of the remainder is

$$\Delta_{w(1-\zeta_0)}(\bar W, W'') + A((w_{\mathrm{fr}} - w)(1-\zeta_0), \bar W) + \theta\,\Delta_{\zeta_0}(\bar W, W^Z).$$

On the middle part put $W_{\mathrm{g}}^0 = W^{\mathrm{f}}(b,0)$ and $W_{\mathrm{g}}^f = W^{\mathrm{f}}(b,f)$, the two glued backgrounds with the same exterior increment b and with fluctuation increments 0 and $f = g_i B$. Its remainder, including the center of the reference law, is

$$\theta A(\zeta_1, W^0) + \theta\,\Delta_\zeta(W^0, W_{\mathrm{g}}^0) - (\theta\,\Delta_\zeta(W_{\mathrm{g}}^f, W_{\mathrm{g}}^0) - q_X/2)$$
$$+ \theta\,\Delta_\zeta(W_{\mathrm{g}}^f, W'') - A((w-\theta)\zeta, W'').$$

The bracket contains the Taylor remainder after the quadratic form $q_X/2$ is extracted. The inner part keeps $S_{\mathrm{core}} = A(w\zeta_1, W^{\mathrm{in}}(f))$. For the difference on this part insert the inner background $W^{\mathrm{in}}(\gamma^\flat)$. Its datum on the rows adjacent to the core is $e^{if}e^{iv_0}$ relative to the inner center, where v_0 is the continued basic datum with $|v_0| \le CL^{-\mathsf{N}_i}\mathsf{s}$. Here s bounds the fields and increments. On the remaining rows, including the joining rows, keep f. The two errors are

$$\Delta_{w\zeta_1}(W^{\mathrm{in}}(f), W^{\mathrm{in}}(\gamma^\flat)), \qquad \Delta_{w\zeta_1}(W^{\mathrm{in}}(\gamma^\flat), W'').$$

The graded row response in the first and the screened background comparison in the second bound them by $C\|w\|_\sharp \mathsf{N}_i \mathsf{s}^2 \mathsf{n}$ times $L^{-\mathsf{N}_i}$ and $e^{-c\delta_0 M \check R_i}$, respectively. Here n bounds the number of remade pillars and $\|w\|_\sharp$ divides the weight on layer j by $1 + i - j$. This places the response to the complex outer variables in the remainders and leaves the retained core at its real configuration.

Let w' be the actual outgoing weight, so that $S_{\mathrm{new}} = A(w', \bar{W})$. Add $\mathsf{W}_X = A(w' - w_{\mathrm{fr}}, \bar{W})$ to the remainder when completing the action. This difference of weights is supported in the replaced component. It has the positive sign since $-A(w_{\mathrm{fr}}, \bar{W}) = -A(w', \bar{W}) + \mathsf{W}_X$. The profiles give $w' = w$ off the components. At the unsourced center, the weight in a completed component is the current bulk weight before the next increment. The preceding bounds on the localized terms include this correction. Summing the three differences and W_X gives P_X and the identity

$$-S_{\mathrm{ref}} - S_{\mathrm{old}} + S_0 = -S_{\mathrm{new}} - \tfrac{1}{2} q_X - S_{\mathrm{core}} + P_X.$$

The exact product of the old weight and the restored law is

$$e^{-S_{\mathrm{old}}} \frac{\chi_X e^{-S_{\mathrm{ref}}}}{D_X} = e^{-S_{\mathrm{new}}}\, \chi_X e^{-q_X/2 - S_{\mathrm{core}}} e^{P_X + I_X - E_X}.$$

This is an identity in (s, v) before either variable is integrated. It remains valid with the actual subsequent averaging graph attached, including at a prescribed terminal raw field. No integration over that terminal field is used to cancel S_0.

The factor $e^{-S_{\mathrm{new}}}$ belongs to the new label's action. The operation retains the core Wilson factor and q_X together with the produced terms, scalar constants and its other carried factors. Thus the original old action is used once in this rewrite. One must not also keep it as a separate reserve, or append the old factor D_X^{-1} to the rewritten operation. The terms $P_X + I_X$ are included once in the complete remainder; E_X is included once in the scalar normalization at its original levels.

For example, on an admitted real fibre suppose the complete remainder bound gives $P_X + I_X \le C N_0$, the scalar count gives $|E_X| \le C V^\circ$, and $q_X, S_{\mathrm{core}} \ge 0$. Then the factor left after $e^{-S_{\mathrm{new}}}$ has been assigned to the new action is bounded by $e^{C V^\circ}$ times the other retained reference factors. The new action itself is nonnegative on a real carrying configuration, so it can also be retained or bounded above by one in an estimate of the complete positive measure. If some of the core energy is used for a failed-test payment, its unspent portion is the reserve in this estimate; the same energy is not available twice.

This handles the nonflat normalization within the complete Wilson product whenever those actual identities and bounds apply. It supplies no upper bound of this size for the isolated probability density $\chi_X e^{-S_{\mathrm{ref}}}/D_X$. The reference integration at fixed terminal output still requires the retained Haar or chart coordinates, including the treatment of tree-fixed pivots. The cancellation changes none of those coordinates or constraints.

4.18.2.6. Drop-outs use a fibre. A drop-out has no inner background formed for its own main operation. Its kept factors have no new core Wilson term. Its mass therefore includes the Wilson factor from the carrying term, on the bare plaquettes whose corners lie in X. The argument is an evaluation $\mathfrak{e} = (e, U, v, Q_1, Q_2, \dots)$ in a set $\mathcal{K}^\sharp(\omega, \xi)$, with the following meaning:

(i) U is real on X and compatible with the stored variables. The exterior datum e supplies the portions of the carried test reads outside X, in the same physical frame and on the same averaging graph;
(ii) every carried test has its recorded value on its complete read, using the reconstructed fields of ξ, the prescribed averages of U and the exterior datum where the test reaches beyond X;
(iii) the real pair restricted to every inner island lies in that island's admitted real core;
(iv) the complete kept quadratic forms and elimination factors obey their stated one-sided lower and upper bounds;
(v) $v(p) \ge c\theta_j$ on the action layer that owns p.

Compatibility means the actual identities

$$\mathcal{M}^j U(b) = V_j(\xi)(b) \quad \text{on every determining bond whose two blocks lie in } X. \tag{4.cdlxx}$$

At level zero these are identities with the stored bare bonds. Layer regularity is imposed both on U and on the relevant stored averages. It cannot be inferred merely from the realness of U.

There are two variants. The mass of the drop-out itself excludes the test whose failure made it a drop-out. A later component may restrict the evaluation set by including that test. It can charge the root when the full paying carrier supplies a witness. The mass of the earlier drop-out leaves this root uncharged.

Write the evaluated factor as

$$\mathsf{E}_{\omega}(\xi;\mathfrak{e}) = e^{-A(v\mathbf{1}_X,U)}\mathsf{K}_{\omega}(\xi;\mathfrak{e}). \tag{4.cdlxxi}$$

For a fixed exterior the actual kernel keeps the exterior tests on their inverse reads. For the upper integral one may take the supremum over admitted exterior values after expressing their kernels in the common free coordinates. The shell and restored tree charts use relative Haar in a fixed recorded frame. The exterior changes the reconstruction and its weight, and these enter the pointwise envelope. The Wilson share in eq. 4.cdlxxi is still the share on the plaquettes owned by X. The set of evaluations varies with ξ. For example, take a probability measure supported on $\epsilon \le |\xi| \le 2\epsilon < 1$, and a factor $e^{-T(1-\cos p)}$. On the compatible graph $p = \xi$ its integral is at most $e^{-cT\epsilon^2}$. If $p = 0$ is admitted independently of ξ, the supremum of the factor is one. For $p = i\eta$ it is $e^{T(\cosh\eta - 1)}$. Neither replacing the fibre by a product nor complexifying the real Wilson argument preserves the same small factor.

4.18.3. *Upper integrals and the strong real induction norm.* For a nonnegative function f on a Borel measure space define

$$\int^{*} f\,d\mu = \inf\left\{\int F\,d\mu : F \text{ Borel},\ 0 \le f \le F\right\}. \tag{4.cdlxxii}$$

All inequalities are pointwise, including on sets on which a parameter-dependent supremum has not been proved Borel. The value of the supremum over an empty evaluation fibre is zero. Monotonicity, positive homogeneity and subadditivity follow directly: majorize each summand by a Borel function within an arbitrary positive error and add the majorants. For a countable sum use errors of size $2^{-n}\delta$ and monotone convergence of the sum of the Borel majorants. These facts do not authorize interchanging an upper integral with an arbitrary parameter supremum.

For an ordinary component put

$$m_{\omega}^{\mathbb{C}}(\xi) = 1_{S_\omega}(\xi) \sup_{\mathbf{p}\in\mathcal{U}_i(X^{\sim})} |e^{\Phi_\omega(\xi;\mathbf{p})}|, \qquad m_{\omega}^{\mathbb{R}}(\xi) = 1_{S_\omega}(\xi) \sup_{\mathbf{p}\in\mathcal{U}_i(X^{\sim})\ \text{real}} e^{\Phi_\omega(\xi;\mathbf{p})}. \tag{4.cdlxxiii}$$

For a drop-out use the supremum of E_ω over its compatible fibre, including the specified variant of the tests. The weighted mass, for a nonnegative function v, is

$$\mathsf{M}_c(i,X;v) = \frac{1}{h}\sum_{\omega} e^{cv(\omega)} \int^{*} m_\omega\,d\mu_\omega. \tag{4.cdlxxiv}$$

An artificial average $h^{-1}\sum_r 1 = 1$ is inserted when the operation does not otherwise depend on a root frame. This gives the same normalization in both classes. The integrals over inner components have their own factors h^{-1}; one does not cancel them a second time in the outer count.

When the core is a fixed finite-dimensional separable space, the ordinary supremum in eq. 4.cdlxxiii is Borel under the stated separate measurability and continuity assumptions. Indeed, choose a countable dense subset of the core. Continuity in the pair identifies its supremum with the supremum over that subset. A countable supremum of Borel functions is Borel. The fibre of eq. 4.cdlxxi need not have such a fixed dense set. We will dominate it by an explicit Borel envelope instead of assuming a measurable optimizing choice.

The input needed in the induction is the real version of eq. 4.cdlxxiv, with the supremum already inside the integral. Uniform fixed-argument bounds are weaker. On the finite probability space $\{1,\dots,N\}$ with uniform measure, let $f(\xi,p) = N1_{\{\xi=p\}}$. For every fixed p, $\int f(\xi,p)d\mu = 1$, while $\int \sup_p f(\xi,p)d\mu = N$. The example is measurable and positive. Thus the issue is not cured by positivity, finite volume or measurability of a partition function.

There is a parallel analytic example. On $\xi \in \{1, \dots, N\}$, let the parameter space be the disjoint union of N unit discs and set $f(\xi, z) = N$ on the disc indexed by ξ, zero on the others. The functions are holomorphic on that parameter space. Holomorphy alone does not interchange supremum and integral. In the application one must use the explicit structure of the complete exponent, such as eq. 4.cdlxvii–eq. 4.cdlxviii, and a common integrable envelope.

A real argument of an inner operation can depend on the outer variables. If it also depends on a variable integrated within that inner operation, its factor is nevertheless bounded pointwise by $m^{\mathbb{R}}$, provided it remains in the admitted real core. The strong real mass norm is precisely what permits this domination. A mere fixed-read estimate would instead require a read-before-sampling condition, absent from that weaker norm. These are two different ways of making a kernel estimate applicable; they must not be conflated.

Finally, upper integrals are used only after a Borel majorant has been constructed on the full product of stored variables. We will apply ordinary Tonelli to that majorant. Bounds for fibrewise upper integrals alone are not a substitute for such a construction.

4.18.4. *Genealogies and a volume that does not recount the past.* A node is a component $u = (n, C)$ of the region made at level n. Unless it is healed, it has one successor, the component at level $n+1$ that contains its propagated hull. Predecessors of a node are disjoint components at the preceding level. The genealogy $\mathcal{G}(x)$ of $x = (i, X)$ consists of x and all its iterated predecessors. A component healed earlier and covered again later is the root of a different genealogy. It is not put back into this tree because its physical location has been reused.

Cut $\mathcal{G}(x)$ at every clearing island below x. The cut tree $\mathcal{G}^\circ(x)$ includes these cut nodes as leaves; their own trees are the inner genealogies. A keeping mark records a drop-out without clearing its earlier history. Its older nodes are not copied every time that mark is subsequently carried.

Write $|C|_n$ for the number of level-n cells in C and define

$$\begin{aligned} w_n &= \ell_n (M R_n)^4, \qquad \mathrm{vol}(u) = w_{n(u)} |C(u)|_{n(u)}, \\ V(x) &= \sum_{u \in \mathcal{G}(x)} \mathrm{vol}(u), \qquad V^\circ(x) = \sum_{u \in \mathcal{G}^\circ(x)} \mathrm{vol}(u), \\ v^\natural(x) &= V^\circ(x) + \sum_{y \text{ inner}} v^\natural(y). \end{aligned} \tag{4.cdlxxv}$$

This is a sum of sites times levels, with the reference logarithm at the level of each site. It is not the volume of the final component multiplied by its age. Nor is it a sum of all successive presentations of its accumulated past.

Lemma 4.66 (Bounded multiplicity)**.** For every finite genealogy,

$$\begin{aligned} v^\natural(x) &= \sum_{u \preceq x} m_x(u)\, \mathrm{vol}(u), \\ m_x(u) &= \begin{cases} 2, & u \neq x \text{ is a clearing node}, \\ 1, & \text{otherwise}, \end{cases} \qquad V(x) \le v^\natural(x) \le 2V(x). \end{aligned} \tag{4.cdlxxvi}$$

■

Proof. Expand the recursion in eq. 4.cdlxxv. A node not on a cut boundary belongs to exactly one outer tree in that expansion. A clearing node below x is the last node of its parent's outer tree and the first node of its own outer tree. No third tree meets it. The root belongs only to its own outer tree. This accounts for every term in the finite expansion and proves the formula. □

For site counts without the logarithm put

$$N^\circ(x) = \sum_{u \text{ listed with multiplicity}} (M R_{n(u)})^4 |C(u)|_{n(u)}. \tag{4.cdlxxvii}$$

Thus $\ell_{\min} N^\circ \le v^\natural$, where $\ell_{\min} = \min_{j\le i} \ell_j$. A fixed-width read collar of one coarse M-layer has at most twice the sites of its owning MR_j-cells once R_j exceeds a fixed dimensional threshold. To see this, enlarge each owning cell separately and sum: its volume is multiplied by at most $(1+C/R_j)^4$, while overlaps only reduce the union. Lattice endpoint conventions change C, not the conclusion. Finite bond directions and color components are included in the per-site bound on a term. Consequently the site budget for such reads can be chosen so that

$$n^\sharp(x) \le 2N^\circ(x), \qquad n^\sharp(x) \le \frac{2}{\ell_{\min}} v^\natural(x). \tag{4.cdlxxviii}$$

The ownership of each read collar is important. Adding its whole past again at each keeping mark would not satisfy this argument.

The part of the earlier label that remains inside a drop-out has volume

$$V_{\mathrm{in}}(i, X; \mathcal{S}) = \sum_{j<i} \ell_j \, \#_j(Z_j(\mathcal{S}) \cap X). \tag{4.cdlxxix}$$

The volume in the left member is the volume of the retained label. On the compatible inner histories the volume rule gives $V_{\mathrm{in}} \le v^\natural(\omega_X)$. The restored action and stored regions also enter the load. Compare their larger volume to V_{in} using the waiting window. At an uncompressed level its retained box is either the old quiet box or the prepared box $H_j^{[3a_j]}$ in eq. 4.cdxliv. It contains at least 7^4 cells of its level. The compressed end of the window has only $O(\log_L \check{R}_i)$ levels. Thus a fixed fraction of the window gives this lower count, while the restored count is at most a fixed multiple of 100^4 at each level. The weights $M^4 \check{R}_j^4 \log \theta_j$ of those cells are comparable up to $O(L^4)$ by bounded drift in the window and rounding to powers of L. Outside the window the entries coincide. This bounds the restored load by $C_L V_{\mathrm{in}}$, while the compatible mass pays with the retained volume in eq. 4.cdlxxix.

4.18.4.1. Why an arbitrary larger volume is not available. Bounded multiplicity is a substantive restriction on a useful weight. For an algebraic example, let a first set of N creations have gain NP^2 and volume Nw. Let r later keeping marks each have gain P. A volume that recounts that first set at every mark is at least $(r+1)Nw$. Choose N so that $cNw > 2P$. Then

$$c(r+1)Nw - NP^2 - rP \longrightarrow +\infty \quad (r \longrightarrow \infty). \tag{4.cdlxxx}$$

The initial gain is finite and cannot pay for this repeatedly copied volume. Additional new volume at the marks only worsens the problem. This example concerns the inference from a gain budget to a volume weight; it does not assert that a particular field realizes every abstract record. It already proves that the scalar ledger alone cannot dominate an unbounded number of copies of the past. The correct comparison is eq. 4.cdlxxvi, or another volume with a proved fixed multiplicity bound.

4.18.5. *Contraction, light stretches and the weighted count.* We now prove the geometric counting statement used in the reserve calculation. We state its local hypotheses precisely so that its application can be checked independently of the analysis of fields. The cell graph at every level has bounded degree. The propagation of an unmarked component takes its next-scale hull, adds a fixed number of layers, and boxes only components whose diameter in cells is bounded by a fixed threshold. A new component is a union of such propagated components and its new creation cells, with these local collars. Equivalently, one can construct an occupied skeleton $O(u)$ from the creation cells below u, so that

$$\begin{gathered} O(u) \subset C(u) \subset O(u)^{[e]}, \qquad O(u) \text{ is } h_0\text{-connected}, \\ |C(u)|_{n(u)} \le C_e \nu(u), \quad \nu(u) = |O(u)|_{n(u)}. \end{gathered} \tag{4.cdlxxxi}$$

Here e, h_0, C_e are fixed. The relation between two levels is the actual cell parent map, with the bounded collars just described. There are no consecutive size drops.

For clarity, the skeleton uses unique cell addresses. The actual closed-cube hull may meet several cells at a boundary. Include that bounded ambiguity in e. The sum of the radii of earlier fixed collars, expressed in the current cell units, is bounded: over every two levels the older radii contract by L^{-1}. Its bound is a geometric series. This proves the fixed-collar part of eq. 4.cdlxxxi under the stated local propagation rule. Boxing a bounded-size component adds only another fixed collar. Taking unions does not increase these constants. Connectivity holds if a merger is made by the specified bounded-distance touching rule. A rectangular hull of a large thin domain would not obey these properties and is not the propagation used here.

Lemma 4.67 (Coarse images of connected sets)**.** Let A be h_0-connected, with N cells, in the fine cell grid. At a coarsening by a factor $q \geq 1$, its occupied coarse hull satisfies

$$|\mathsf{H}_q A| \leq C_0(1 + h_0 N/q). \tag{4.cdlxxxii}$$

The same bound holds with a fixed additional coarse collar by increasing C_0. ■

Proof. A spanning tree of the connected cell graph has $N - 1$ edges. Traverse every tree edge in both directions. Join the cell centres by line segments along this walk. Its total length in the sup norm is at most $2h_0(N - 1)$. Split it into consecutive pieces of length at most q. Each piece lies in a sup-norm ball of radius q about its first point. Such a ball, together with the width of one fine cell, meets a fixed number of coarse q-cells in four dimensions. There are at most $1 + 2h_0N/q$ pieces. Their union contains every visited cell centre and hence the required cell hull, up to its one fixed collar. Counting this collar changes the same fixed dimensional number. □

Choose a fixed number s_0 of levels so large that the net coarsening, at least $L^{\lfloor s_0/2 \rfloor}$, makes the coefficient of N in eq. 4.cdlxxxii at most $1/4$. Choose a fixed $\nu_\star \geq 4C_0$. Propagating a component with $\nu(u) > \nu_\star$ through s_0 levels without new material then leaves at most $\nu(u)/2$ occupied cells. For a component with at most $\nu_\star$ cells, it leaves at most a fixed constant. Call such components light. The estimate concerns occupied cells; a large component need not become light in s_0 levels.

Let o_n be the total number of occupied cells in a fixed genealogy at level n, and let l_n be its number of light nodes there. Unions at mergers reduce the count. A creation in the intervening s_0 levels contributes at most a fixed number of cells to their coarse image. Thus

$$o_n \leq \tfrac{1}{2} o_{n-s_0} + C_1 l_{n-s_0} + C_1 \sum_{n-s_0<t\leq n} n_t^{\mathrm{cr}}, \tag{4.cdlxxxiii}$$

with quantities at nonpositive levels set to zero. At a cut leaf of an outer genealogy its bounded number of boundary seed cells is included among n_t^{cr}. The reserve for those seeds is the last test outside the inner mass, not a second use of the inner component's old gain.

Let r_n be any nonnegative non-increasing sequence. Multiply eq. 4.cdlxxxiii by r_n and sum over the finite levels. The shifted first sum is at most half of $\sum_n r_n o_n$, because $r_{m+s_0} \leq r_m$. The light-node sum is at most $C_1 \sum_n r_n l_n$. A creation of level t occurs in at most s_0 summands, with weights no larger than r_t. Rearrangement therefore gives

$$\sum_n r_n o_n \leq 2C_1 \sum_n r_n l_n + 2C_1 s_0 \sum_t r_t n_t^{\mathrm{cr}}. \tag{4.cdlxxxiv}$$

There is no initial-height term. This is the first reason the estimate is uniform in the number of levels.

4.18.5.1. Waiting only for the light part. A quiet node contains no new creation, has one predecessor and does not immediately follow a mark. Along a quiet stretch the side lengths in cells satisfy, for a fixed A,

$$\alpha_{n+1} \leq \alpha_n + A, \qquad \alpha_{n+2} \leq L^{-1}\alpha_n + 2A. \tag{4.cdlxxxv}$$

Iteration on either parity gives $\alpha_{n+2m} \leq L^{-m}\alpha_n + 2A/(1 - L^{-1})$. The smallness threshold in the dynamic class is chosen strictly above this fixed point and the intervening one-step enlargement. A light node has uniformly bounded side length by eq. 4.cdlxxxi: an h_0-connected set with at most

$\nu_\star$ cells has a spanning path of bounded diameter. After a fixed further number b_0 of quiet levels, the component is small and is a single box evolving by the prescribed free rule.

The preparation depths satisfy $\bar N_n \le R_n + 1$, with a non-increasing majorant, and the lower preparation endpoints are strictly increasing. These are the admitted size-profile conditions of the dynamic construction. After at most another $\bar N_t + 3$ quiet levels from a light node of level t, the entire preparation window lies after the event that began its stretch. Quietness, quarantine including the lower endpoint, depth and single-box traces then all hold. The full dynamic class acts on the component. Hence a stretch begun at level t contains at most

$$T_t = b_0 + \bar N_t + 3 \le C_T R_t \tag{4.cdlxxxvi}$$

light nodes. The numbers T_t can be taken non-increasing. Before these light nodes a stretch can have many heavy nodes; they have already been controlled by eq. 4.cdlxxxiv. Replacing this argument by a uniform bound on the whole stretch would incorrectly discard the time needed to shrink a large component.

There is a uniform terminal alternative on a small torus. If a light component or its fixed required collar has no embedded cell chart, some periodic cell side is at most a fixed constant $C_\star = C_\star(h_0, \nu_\star, e)$, enlarged for the other fixed collars. Otherwise the bounded-diameter connected lift of that component and its collar would embed. After q further levels the same periodic cell side is divided by at least $L^{\lfloor q/2 \rfloor}$. It must remain at least one at every admitted partition. Thus at most $2\lceil \log_L C_\star \rceil + 1$ levels remain. Increase the fixed constant in eq. 4.cdlxxxvi to include them. The proof then stops at the terminal level; it does not claim existence of a nonwrapping box background for a wrapped component.

There are three ways a stretch begins: a node contains a creation of its own level, it follows a keeping mark, or it is a merger with at least two predecessors. Charge the first kind to one of its new creations and the second to its preceding mark. To handle a merger, choose at every non-leaf node a first predecessor. At a merger choose one non-first predecessor and descend through first predecessors to a leaf. Assign that leaf to the merger.

This assignment is injective. If two merger nodes are incomparable, their leaf sets are disjoint. If one lies below the other, a path from the assigned leaf to its merger first enters that merger from a non-first predecessor. A descent by first predecessors from the higher node cannot follow that same branch through it. Thus it cannot reach the same assigned leaf. A leaf contains a creation; one of its creation cells pays for the assigned merger. A creation can pay for its own start and for at most one merger. This fixed multiplicity is the second reason the count has no dependence on height.

Because $T_n r_n$ is non-increasing, paying at the lower level of an assigned creation or mark only increases the available weight. Summing the at most T_t light nodes on every stretch and then using eq. 4.cdlxxxi and eq. 4.cdlxxxiv proves

$$\boxed{\sum_{u \in \mathcal{G}(x)} r_{n(u)} |C(u)|_{n(u)} \le C_2 \sum_t T_t r_t n_t^{\mathrm{cr}}(x) + C_2 \sum_t T_t r_t n_t^{\mathrm{mk}}(x).} \tag{4.cdlxxxvii}$$

The root's own mark is excluded from n_t^{mk}. It has not yet supplied a renewed wait. The constants depend on the fixed local geometry and L, not on the number of levels, the component size or the torus volume.

4.18.6. *The scalar reserve and its coupling threshold.* Let G_t be the gain per creation, P_t the gain per mark and c_t the cost per cell at level t. Choose a non-increasing majorant $r_t \ge c_t$. Define

$$\mathfrak{F}(x) = \sum_t G_t n_t^{\mathrm{cr}}(x) + \sum_t P_t n_t^{\mathrm{mk}}(x) - \sum_{u \in \mathcal{G}(x)} c_{n(u)} |C(u)|_{n(u)}. \tag{4.cdlxxxviii}$$

If

$$G_t \ge 2P_t + C_2 T_t r_t, \qquad P_t \ge C_2 T_t r_t, \tag{4.cdlxxxix}$$

then eq. 4.cdlxxxvii gives

$$\mathfrak{F}(x) \geq 2 \sum_t P_t n_t^{\mathrm{cr}}(x). \tag{4.cdxc}$$

Every nonempty genealogy has a leaf and therefore a creation. If $P_t \geq (1+\beta_0)^{-1} P_i$ for $t \leq i = n(x)$, the last bound is at least $2(1+\beta_0)^{-1} P_i$. The reserve is taken from the creation gain; a mark in this conservative calculation can spend all its credit on the later wait.

Here are sufficient quantitative choices. Assume R_t is comparable to ℓ_t^r, with fixed $r \geq 1$, and put

$$P_t = p_0(g_t) = A_0 \ell_t^p, \qquad G_t = c_G P_t^2, \qquad r_t = C_3 c_1 M^4 R_t^5, \qquad p \geq 6r+1. \tag{4.cdxci}$$

Here $c_G > 0$ is a fixed gain constant, distinct from the bare spacing a_0 in eq. 4.cdlxi. The actual cost $c_1 \ell_t (M R_t)^4$ is bounded by this r_t, since $\ell_t \leq C R_t$. Also $C_2 T_t r_t \leq C_4 c_1 M^4 \ell_t^{6r}$. Thus eq. 4.cdlxxxix follows from a lower bound

$$\ell_t \geq K(1+c_1), \qquad c_G A_0 \ell_t \geq 4, \tag{4.cdxcii}$$

after increasing a fixed K. To verify this explicitly, divide the cost bound by P_t. The quotient is at most $(C_4 M^4/A_0) c_1 \ell_t^{-(p-6r)}$, at most a prescribed small constant once eq. 4.cdxcii holds. Divide the remaining condition $c_G P_t^2 \geq 2P_t + C_2 T_t r_t$ by P_t and use the second condition, increasing K if necessary. Only one extra logarithmic power is needed. In particular the lower threshold may be affine in the coefficient of the volume weight.

For reference sequences with $\theta_t \geq \theta_i - c_5$ for $t \leq i$, one has $\log \theta_t \geq \log \theta_i + \log(1 - c_5/\theta_i)$. For a sufficiently high coupling floor the ratio of its p-th power to ℓ_i^p is at least $(1+\beta_0)^{-1}$, uniformly in $t \leq i$. This proves the comparison used after eq. 4.cdxc, without assuming exact monotonicity of the reference couplings.

There is a useful consequence for an unfinished component $z = (j, Z)$. Here unfinished means that its root is neither healed nor marked. Continue it freely for q levels, stopping before a new creation, merger or completed wait. The genealogy and its gains are unchanged; only the later cell costs are added. Apply eq. 4.cdxc to the extended genealogy. If t_0 is its first creation level, the sum of its creation credits is at least P_{t_0}. Therefore

$$\kappa(z) := \mathfrak{F}(z) - 2P_{t_0} \geq \sum_{n=j+1}^{j+q} c_n |S^{n-j} Z|_n \geq 0. \tag{4.cdxciii}$$

The extension is cut at the terminal level if necessary. It is important to stop before the completed wait: withholding an eligible component would change the dynamic construction. The formula shows exactly what portion of the old reserve is available to pay for the future quiet evolution.

For comparison, if a separately admitted continuation begins with a just-marked root, that mark becomes a non-root mark of the extended genealogy. Its gain is then added once: the corresponding available expression is $\mathfrak{F}(z) + P_j - 2P_{t_0}$, not the unmarked expression in eq. 4.cdxciii. This observation does not assert the admission of such a continuation or spend the same mark a second time.

4.18.7. *The entropy of the outer records.* The outer indices include choices of regions and tests, not just a choice of frame. We give a counting argument that includes their positions. Make a graph whose vertices are cell addresses together with their levels. Join neighboring cells at one level and join each cell to its parent at the next level. The degree is bounded by

$$D \leq L^4 + 3^4 + 2. \tag{4.cdxciv}$$

At a size drop the parent has only one child in this address graph; otherwise it has at most L^4. On a torus some neighboring addresses can coincide, which only decreases the bound. The outer component is rooted at a fixed cell of X. Its active cell addresses form a connected set by the hull and merger rules. If a local convention requires a bounded additional collar, include it in the active record and its already specified volume.

In any graph of degree at most D, the number of connected vertex sets of size N containing a fixed root is at most

$$D^{2(N-1)}. \tag{4.cdxcv}$$

Indeed, put a fixed order on the neighbors. For a connected set choose its canonical rooted spanning tree, and walk around that tree in depth first order, traversing each edge twice. The walk has length $2(N-1)$ and its visited set determines the original set. Each step has at most D choices. This count includes possible birth levels and translations of an old creation within a coarse component; they are encoded in the path down the address graph. There is no extra factor equal to the torus volume or to the total height.

For a fixed active set A, the remaining discrete record has at most

$$\exp\{bV^\circ(A)\} \tag{4.cdxcvi}$$

possibilities, for a fixed b. Here is the finite coding condition behind this estimate. A cell of level n contains $O((MR_n)^4)$ sites and bonds. Each local decomposition has a fixed finite list of outcomes, and a fixed finite number of such tests is attached to each site or bond. Specify these outcomes by letters from a fixed finite alphabet. The stage place and the local failure type have at most a fixed power of R_n+1 possibilities per node; their logarithm is at most $C\ell_n$ since $R_n \asymp \ell_n^r$. Tree frames are treated by their normalized averages below. Lists of owned seam factors, node entrances and retained-variable types use the same finite local addresses, not arbitrary new continuous parameters. These facts prove eq. 4.cdxcvi by multiplying the local counts. A proposed history representation with another discrete choice must include its code and its cost here before this estimate is applied.

There are $1+f(\omega^\circ)$ independently recorded outer frames, including the artificially inserted root frame when needed. Their normalized sum is at most one for each non-frame record. Since $V^\circ(A) \geq |A|$ for $\ell_n \geq 1$ and $MR_n \geq 1$, choose $c_{\mathrm{e}} \geq b + 2\log D + \log 2$. eq. 4.cdxcv–eq. 4.cdxcvi give

$$\begin{aligned}\sum_{\omega^\circ} h^{-1-f(\omega^\circ)} e^{-c_{\mathrm{e}} V^\circ(\omega^\circ)} \\ \leq \sum_{N\geq 1} D^{2(N-1)} e^{-(c_{\mathrm{e}}-b)N} \leq \sum_{N\geq 1} 2^{-N} = 1.\end{aligned} \tag{4.cdxcvii}$$

The outer component is nonempty, which is why the sum starts at one. An empty component instead represents the identity operation and has no rare-mass assertion. This exact upper bound of one will prevent an unwanted constant from accumulating at every induction step.

The coding condition is a condition on the literal finite construction. It must count every tested bond and every stage place. The connected-set argument then proves the uniform entropy bound; it is not an assumption that arbitrarily chosen discrete histories already have small total mass.

4.18.8. *Normalizer bounds with their complex branches.* We next isolate scalar and finite-matrix estimates that enter the integrable envelope. This keeps them distinct from the analytic problem of identifying the actual kept matrix.

4.18.8.1. The truncated gauge normalizer. Use the matrix operator norm and the sharp radial cutoff $|u-1|<\epsilon$, $0<\epsilon\leq 1$. If the class angle is $r\in[0,\pi]$, then $|u-1| = 2\sin(r/2)$. Thus this cutoff is exactly the monotone interval $0\leq t<t_\epsilon$ in $t = 1-\cos r$, with $t_\epsilon = \epsilon^2/2$. The same proof applies to a cutoff explicitly known to be such an interval with $t_\epsilon \geq c\epsilon^2$. Put

$$z(y,\epsilon) = \int_G 1_{\{|u-1|<\epsilon\}} e^{-y(1-\frac{1}{2}\operatorname{tr} u)}\, du. \tag{4.cdxcviii}$$

Haar measure is normalized to one. This is an entire function of y. For real $x\geq 1$ and $x\epsilon^2 \geq 1$ it satisfies $cx^{-3/2} \leq z(x,\epsilon) \leq Cx^{-3/2} \leq C$; also $z(x,\epsilon)\leq 1$. The lower bound follows by integrating a Lie-algebra ball of radius $c_0 x^{-1/2}$: it lies in the cutoff, the exponential is bounded below there, and the Haar chart density is bounded below. For the upper bound use $1-\cos r \geq cr^2$ on $0\leq r\leq \pi$

and the radial Haar density. Both estimates have constants uniform in the cutoff satisfying the displayed condition.

The required complex branch can also be proved uniformly, without assuming that an integral of positive real values remains nonzero after complexification. Write $t = 1 - \frac{1}{2}\operatorname{tr} u = 1 - \cos r$. The radial Haar density is a constant times $t^{1/2}(2-t)^{1/2}\,dt$, $0 < t < 2$; the cutoff restricts this to $0 < t < t_\epsilon$. Thus at $y = a + ib$, $a > 0$, $z(y,\epsilon)/z(a,\epsilon)$ is the expectation of e^{-ibt} under a probability density proportional to $t^{1/2}e^{-at}w(t)$, where $w(t) = (2-t)^{1/2}1_{\{t<t_\epsilon\}}$ is nonnegative and non-increasing.

For completeness, decreasing reweighting lowers increasing moments. If ρ is a probability, f increases and w decreases, then

$$\operatorname{Cov}_\rho(f,w) = \tfrac{1}{2}\iint (f(s)-f(t))(w(s)-w(t))\,d\rho(s)d\rho(t) \le 0.$$

Divide by $\rho(w) > 0$ to obtain $\rho(fw)/\rho(w) \le \rho(f)$. Apply this with the gamma density proportional to $t^{1/2}e^{-at}$ and $f(t) = t^2$. Two integrations by parts show its second moment is $15/(4a^2)$. Since $\cos v \ge 1 - v^2/2$ for every real v, it follows that

$$\operatorname{Re}\frac{z(a+ib,\epsilon)}{z(a,\epsilon)} \ge 1 - \frac{15b^2}{8a^2}.$$

On $|y-\theta| \le \theta/2$, elementary geometry gives $|b|/a \le 1/\sqrt{3}$. Indeed this inequality is equivalent to $3b^2 \le a^2$, whose worst ratio on that disc occurs at its tangent from the origin. It also follows by maximizing $[(\theta/2)^2 - (a-\theta)^2]/a^2$ for $\theta/2 \le a \le 3\theta/2$. Hence the last real part is at least $3/8$. In particular $z(y,\epsilon) \ne 0$ throughout that disc. When $\theta\epsilon^2 \ge 1$, the real bounds, with their threshold changed by a fixed factor if necessary, give $|z(y,\epsilon)|^{-1} \le C\theta^{3/2}$ there.

Here the lower floor can be seen directly. The fixed norm gives $t_\epsilon \ge c_{\mathrm{cut}}\epsilon^2$. Choose a fixed $0 < c_0 < \min(1, c_{\mathrm{cut}}/2)$. The interval $0 < t < c_0/\theta$ lies inside the cutoff because $\theta\epsilon^2 \ge 1$. On it $(2-t)^{1/2} \ge 1$ and $e^{-at} \ge e^{-3c_0/2}$ for $a \le 3\theta/2$. Integration of $t^{1/2}$ on this interval proves $z(a,\epsilon) \ge c\theta^{-3/2}$. Combining it with the $3/8$ real-part bound proves the asserted inverse estimate. The cutoff is frozen at its real reference value throughout: neither ϵ nor its characteristic function is varied with y. Decreasing reweighting alone would not supply this lower floor for an arbitrarily vanishing cutoff.

There is a unique logarithm on the open disc agreeing with the real logarithm at θ. The closed-disc bounds follow by continuity. Thus a kept inverse gauge normalizer costs at most $C + \frac{3}{2}\ell_j$ in the logarithmic budget even on this complex disc. This proves the normalizer part of the parameter extension; it does not assert a parameter extension for the other factors.

4.18.8.2. The pivot factor. On an admitted one-step contour chart the three-color pivot matrix is the per-length averaging derivative m_c. Its identity value is $L^{-4}I$; the raw logarithmic pivot derivative instead has identity value $L^{-3}I$. In the per-length convention it obeys

$$|L^{12}\det \mathsf{m}_c - 1| \le \tfrac{1}{2}, \qquad \ell_c = 12\log L - \log(L^{12}\det \mathsf{m}_c), \qquad e^{\ell_c} = (\det \mathsf{m}_c)^{-1}. \tag{4.cdxcix}$$

The logarithm is its power series about one, so $|\ell_c - 12\log L| \le C$. On the real admitted chart the determinant has the positive identity sign, and this equals the inverse absolute determinant for the per-length averaging coordinates. The actual real operation also retains the conversion factors from raw averaging and Haar coordinates and its other chart Jacobians. There is no holomorphic absolute-value determinant. The local averaging-chart theorem supplies eq. 4.cdxcix once its small-field and embedding hypotheses hold. Its use at every retained layer additionally requires that the actual iterated averages exist and fall in those local charts; a one-step statement alone does not verify that multi-level premise.

Each scalar coordinate, bond direction and local gauge variable is charged to its owning site. The inverse gauge normalizers in eq. 4.cdxcviii, the pivot factors in eq. 4.cdxcix, and the rescaling

powers of $g_j = \theta_j^{-1/2}$ therefore have combined modulus at most $\exp(CV^\circ)$, provided their number at level j is at most a fixed multiple of the stored sites there. Chart Jacobians with uniform local upper and lower bounds have the same property. The identity value of the pivot determinant fixes its power L^{12}; counting the same three colors again in this factor would be an error.

4.18.8.3. The normalization numbers at their own levels. The numerical parts in the collar sum are distinct from the real normalization number retained by an operation. In the notation of the step, the latter is $e_X = n_X - n_G(X)\log(\theta_t^{-3/2}\sigma_0)$, where $n_G(X)$ is the number of integration bonds of the reference law and σ_0 is the Haar density at the chart origin. Here n_X subtracts the normalization of the label with X adjoined to its regions and all marks in X removed, including its own, from the current label normalization. It includes the stage and drop-out numbers. Its ordinary step part is the sum, for $1 \le j \le t$, of $N^{(j-1)}(\Omega_j^{\mathsf{a}}, \Lambda_j^{\mathsf{a}+}) - N^{(j-1)}(\Omega_j^{\mathsf{a}} \cup X, \Lambda_j^{\mathsf{a}+} \cup X)$, with the action region and collared integration region of eq. 4.cdlxiv. In this formula,

$$N^{(j)}(\Omega,\Lambda) = (\log\sigma_0 - \tfrac{3}{2}\log\theta_j)|\Omega^{(j)*}| \\ - (L^4-1)\log z(\theta_j)\,|\Omega|_{j+1} + (\nu_{\mathfrak{g}} - \tfrac{3}{2}\log(x_j/\theta_j))|\Lambda^{(j)*}|.$$

The three counts are respectively the fluctuation bonds, the level-$(j+1)$ sites and the nonpivot integration bonds of level j. Here z is the single-site gauge normalizer and $\nu_{\mathfrak{g}}$ is the logarithm of the standard unnormalized Gaussian integral on $\mathfrak{g}$. The real normalization stays in Φ°_ω or in the identical normalization of the reference measure. The retained powers of θ_j are charged at level j in the volume eq. 4.cdlxxv.

For complex normalization weights y_j with $|y_j - x_j| \le \bar{c}$, while the bare and net Wilson weights stay fixed, let $\mathfrak{g}_j(X)$ count the integration bonds of level $j-1$ added when X is joined to $\Lambda_j^{\mathsf{a}+}$. The exact scalar difference is

$$e_X[y] - e_X[x] = \tfrac{3}{2}\sum_{j=1}^{t} \mathfrak{g}_j(X)\left(\log\frac{y_{j-1}}{\theta_{j-1}} - \log\frac{x_{j-1}}{\theta_{j-1}}\right).$$

Indeed the stage and drop-out numbers here use only reference weights and the first two terms of $N^{(j)}$. In the ordinary step difference, only the last logarithm changes. The logarithms use their common branch near the real positive weights, with $|x_j - \theta_j| + \bar{c} < \theta_j/2$ at every level. If the added bonds are covered by one layer around the $z_j(X)$ owned M-cubes, then $\mathfrak{g}_j(X) \le 4\cdot 3^4 M^4 L^4 z_j(X)$. Each enlarged cube accounts for at most M^4L^4 sites of level $j-1$, and there are four positive bonds per site. Using the bound 2 for the logarithm derivative on the common half-disc and $N_0 \ge M^4\sum_j z_j(X)$ gives

$$|e_X[y] - e_X[x]| \le 972 L^4 \bar{c}\,(\min_{0\le j<t}\theta_j)^{-1} N_0(\omega).$$

In the sourced comparison this difference is added to the new excess in eq. 4.dxvi. The real $e_X[x]$ can carry logarithms of the reference weights, which are counted at their own levels in eq. 4.cdlxxv.

4.18.8.4. The complete quadratic form. The real physical weight x and the reference weight θ have different roles. Write $\tau = x/\theta$. If K_A is the physical action part and K_G is the reference gauge part of the actual free-coordinate Hessian, the complete quadratic exponent uses

$$Q_\tau = \tau K_A + K_G, \qquad K = K_A + K_G, \qquad e^{-B^{\mathsf{T}}Q_\tau B/2} = e^{-\tau B^{\mathsf{T}}KB/2} e^{(\tau-1)B^{\mathsf{T}}K_G B/2}. \tag{4.d}$$

Both matrices are in the actual free coordinates of the constraint chart. The last factor compensates for the use of τK in the normalized reference Gaussian. For unnormalized input the prefactor is the Gaussian mass of τK. With a normalized incoming precision D_0 it is the ratio of that mass to the Gaussian mass of D_0. Using Q_τ directly instead gives its own Gaussian mass and omits the compensating potential.

Here is a sufficient full half-disc estimate. Suppose K_A, K_G are real symmetric, $K_G \ge 0$, $K_A \ge -\varepsilon I$ and $K \ge \gamma I$ with $\varepsilon \le \gamma$. For $|\tau - 1| \le 1/2$, put $s = \operatorname{Re}\tau$. If $s \le 1$, $sK_A + K_G = sK + (1-s)K_G \ge$

$\gamma I/2$. If $s \geq 1$, $sK_A + K_G = K + (s-1)K_A \geq (\gamma - \varepsilon/2)I$. Therefore

$$\operatorname{Re} Q_\tau \geq \gamma I/2. \tag{4.di}$$

If the matrices have complex perturbations with $\frac{3}{2}\|\delta K_A\| + \|\delta K_G\| \leq \gamma/4$, the same argument gives $\operatorname{Re} Q_\tau \geq \gamma I/4$. This is a matrix estimate to be checked on the actual joint core. If instead a free evaluation directly specifies a complete complex form Q with $\operatorname{Re} Q \geq \gamma I$, that one-sided bound is sufficient by itself.

One must not multiply two independently accretive complex objects and retain the same conclusion. In one dimension, $K = 1 + 100i$ and $\tau = 1 + i/2$ have respectively positive real part and $|\tau - 1| = 1/2$, whereas $\operatorname{Re}(\tau K) = -49$. Nor should the compensating potential in eq. 4.d be bounded by its modulus while independently replacing $\operatorname{Re}\tau$ by its worst possible value: their real coefficients are correlated. Keeping the complete form avoids both losses.

4.18.9. *The integrable reference envelope.* We spell out the additional coordinate statement needed to integrate the stored Gaussian reserve. For each outer index there are finitely many real Lebesgue blocks $b_1, \dots, b_s$, together with normalized Haar variables. A permitted reference change of coordinates has triangular form $A_j = F_j(b_j; b_{j+1}, \dots, b_s, U)$ in a chosen order. For fixed later variables it is injective on its integration chart, and its inverse Jacobian is bounded by $J_0^{d_j}$, where d_j is the number of scalar coordinates in that block. It can be a translation, in which case this Jacobian is one. The actual stored-coordinate identity or its explicitly bounded chart Jacobian is required; the mere use of the symbol A_j does not provide it.

Suppose the unspent part of the complete forms gives at least $\beta \sum_j |A_j|^2$ in the exponent, for a fixed $\beta > 0$. Dropping restrictions only enlarges a nonnegative integral. Change the variables one block at a time, starting with the block whose later variables are held fixed. Each integral is at most the full Euclidean Gaussian integral. Induction over the blocks gives

$$\int \exp\{-\beta \sum_j |A_j|^2\}\, d\mu^\circ \leq J_0^{\sum_j d_j} (\pi/\beta)^{\frac{1}{2}\sum_j d_j} \leq e^{C_5 V^\circ}. \tag{4.dii}$$

The last inequality uses the actual coordinate count $\sum_j d_j \leq C N_0^\circ \leq C V^\circ$. Haar variables contribute at most one after positive restrictions are dropped. If the coordinate image is only a subset, the same upper bound holds. Surjectivity onto a product is not needed.

For a more general actual reference map it is enough to construct directly a nonnegative Borel function g_{ω° with this integral bound and with the pointwise domination used below. Such a construction is a real integration estimate. It cannot be replaced by a statement that a complex Gaussian at each fixed external argument has a small partition function.

For a completed main operation whose core contains no earlier marks, the reference coordinates can be read directly from its integral. The one-step operations in that core integrate the stored strip fields $\mathcal{A}_j^{\mathrm{s}}$ against Lebesgue measure. Changing the background changes their quadratic forms and elimination factors, but does not change these integration variables. The main operation also integrates a field B on the shells, with the single scaling g_t. The variable B stays fixed when the external pair changes, even though the reconstruction of the group field uses the background. Its tree gauge sets the tree bonds to zero and has Jacobian one; on the remaining bonds the carried test is

$$\chi'(B) = \mathbf{1}_{\{|B| < r_t\}}, \qquad r_t = \delta_t'/g_t = A_1 p_1(g_t).$$

Here the averaging constraints can be eliminated from the top down: the outgoing coarse field is integrated in the main operation, and integration of its delta function sets it equal to the average of the preceding field. The remaining group variables carry normalized Haar measure. Chart densities and normalizers remain among the factors estimated separately above.

Suppose the complete kept forms leave the reserve $\beta\sum_j|\mathcal{A}_j^{\mathrm{s}}|^2$ after the event gains have been extracted. If there are n_A free strip bonds and n_B free shell bonds, a reference function is

$$g_\omega(B,\mathcal{A}^{\mathrm{s}})=\chi'(B)\exp\Bigl(-\beta\sum_j|\mathcal{A}_j^{\mathrm{s}}|^2\Bigr),\qquad\int g_\omega\,d\mu^\circ\le[C(1+r_t)]^{3n_B}(\pi/\beta)^{3n_A/2}.\tag{4.diii}$$

Indeed, enlarge the domain to the product of the cutoff balls for B, the full Euclidean spaces for the strip fields, and the full compact groups. The three factors are respectively a finite volume, ordinary Gaussian integrals, and a Haar mass of one. This estimate is uniform in the background and uses no change of coordinates for the strip fields.

If N_P pivot-elimination factors J_c remain in the integrand relative to this product measure, use the Borel majorant $J_*^{N_P}g_\omega$, provided the actual pivot charts satisfy $|J_c|\le J_*$. For the convention $J_c=|\det\mathsf{m}_c|^{-1}$, the stated regular-chart estimate is $J_*=2L^{12}$. On an admitted full row the least singular value of $\mathsf{m}_c(V)$ is at least $L^{-4}(1-\kappa_9L^3\varpi)$, where ϖ bounds the plaquette defect of its chart. The determinant bound follows for $\kappa_9L^3\varpi\le1-2^{-1/3}$. In a fixed common real frame change u, the coefficient transforms as

$$\mathsf{m}_c(V^u)=\mathrm{Ad}(u(c_-))\,\mathsf{m}_c(V)\,\mathrm{Ad}(u(b_0(c)_-))^{-1}.$$

Both rotations preserve its absolute determinant. The physical Haar derivative is $L\mathsf{m}_c$, with inverse density $L^{-3}|\det\mathsf{m}_c|^{-1}$. The carried elimination factor above uses its stated normalized convention. This calculation keeps u fixed in the pivot derivative and requires the same admitted row and pivot. The local count $N_P\le CV^\circ$ absorbs this additional factor. Factors already estimated separately are not inserted a second time.

To express the bound in terms of V°, let $n_{B,j}$ count the free shell bonds on the lattice of level j, so that $n_B=\sum_{j\le t}n_{B,j}$. The local coordinate count is $n_A+\sum_{j\le t}\ell_jn_{B,j}\le CV^\circ$; strip bonds of level j belong to their one-step operation at level $j+1$. The fixed lattice ratio is included in this constant. Since $r_t\le C\ell_t^{\mathfrak{p}_1}$ and $\ell_t\le2\ell_j$ for $j\le t$, the logarithm of the right-hand side of eq. 4.diii is at most CV°. Thus this class has an explicit Borel reference function. Keep χ' when discarding the main quadratic form. The event payments in this bound use the retained Wilson and gauge weights and the strip forms with their stated reserve. For a drop-out with fixed outgoing field we must still identify the reference coordinates with that constraint in place.

A drop-out that leaves after the central box integration has a different reference measure from one that leaves during a stage or at the extendibility test. The central integration has already integrated the incoming top field. The fresh Haar field left in the box is a new variable; holding that field fixed does not restore a constraint on the old incoming field. The frame operation retained by a late drop-out includes this completed integration and its tree gauge. Its old averaging chain is therefore eliminated from the top down, just as in the completed main operation.

Suppose the core contains no earlier marks, and retain the shell variables in their original group coordinates. The non-tree shell bonds then carry normalized Haar measure, also when the test χ' fails. The tree gauge has Jacobian one. Thus this case does not need the cutoff volume in eq. 4.diii. Let n_j count the retained gauge normalizers at level j, let n_A count the free stored strip bonds, and let N_P count any remaining pivot factors. Under the actual reserve and chart bounds stated above, a Borel majorant for the reference part is

$$J_*^{N_P}\prod_j\mathfrak{z}(\theta_j,\varepsilon_j)^{-n_j}\exp\Bigl(-\beta\sum_j|\mathcal{A}_j^{\mathrm{s}}|^2\Bigr).$$

Indeed the gauge exponentials and the remaining characteristic functions are at most one. Integration in the old stored coordinates gives the bound

$$J_*^{N_P}\prod_j\mathfrak{z}(\theta_j,\varepsilon_j)^{-n_j}(\pi/\beta)^{3n_A/2}.$$

The scalar estimate $\mathfrak{z}(\theta_j,\varepsilon_j)^{-1} \le C\theta_j^{3/2}$, together with the own-level coordinate count, bounds this by e^{CV°. Scalar factors already included in the normalization estimate are counted only once. This reference bound applies to the late exits at the shell-field or last-shell tests. Merely having completed all preparatory stages is insufficient: an exit at extendibility still precedes the central integration and retains its fixed-output constraints. The complete interaction estimate, the energy payments and admission of the actual evaluation remain separate requirements.

One fixed-outgoing-field case is covered by the regular fibre charts. Consider a drop-out before any stage has re-made a level, with no earlier marks. Suppose every averaging constraint in its retained operations is already consumed by the regular pivot elimination, so that no factor $dV_j\,\delta(\mathcal{M}(V_j)V_{j+1}^{-1})$ on Ω_{j+1}^{c} remains. The outgoing field V_t is held fixed. For each retained step the exact fibre chart solves $\mathcal{M}(V_j)=V_{j+1}$ for the pivot variables; its free strip coordinates are the stored $\mathcal{A}_j^{\mathrm{s}}$. Thus these constraints need not be removed by integrating V_t. The reference integral is already an iterated Lebesgue integral in the free strip coordinates, with the coarea and chart factors in the integrand.

Suppose the actual pivot charts have the bound $|J_c| \le J_*$, the remaining densities have the separately established scalar bound, and the complete forms leave $c_{\mathrm{res}}\sum_j |\mathcal{A}_j^{\mathrm{s}}|^2$ after the required payments, with $c_{\mathrm{res}}>0$ independent of the outgoing field. If there are N_P retained pivot factors and n_A free strip bonds, the reference part is bounded by the Borel function

$$J_*^{N_P}\exp\Bigl(-c_{\mathrm{res}}\sum_j |\mathcal{A}_j^{\mathrm{s}}|^2\Bigr), \qquad \int J_*^{N_P} e^{-c_{\mathrm{res}}\sum_j |\mathcal{A}_j^{\mathrm{s}}|^2}\,d\mathcal{A}^{\mathrm{s}} = J_*^{N_P}(\pi/c_{\mathrm{res}})^{3n_A/2}.$$

Restricting this function to the actual chart only decreases its integral. The local coordinate count bounds the result by e^{CV°, uniformly in the fixed outgoing field for which these same chart and reserve bounds hold. The estimate concerns the reference part; the complete interaction bound and admission of the actual evaluation remain separate requirements.

If an unresolved averaging factor survives, positivity alone does not give this conclusion. Its normalization integrates the outgoing field, whereas the drop-out keeps that field fixed. Nor does disintegration into conditional probability measures remove the problem: their base measure is the actual pushforward of the fine measure, whose density relative to outgoing Haar measure has not thereby been bounded.

A remaining Haar averaging factor can also be estimated when its carried support lies in one of the ordinary regular fibre charts. Keep its outgoing field W fixed and write the exact chart in unscaled free coordinates a as

$$V_W(a) = e^{i\psi_W^{\mathrm{lin}}(\mathsf{C}_W a)}p_W, \qquad \mathcal{M}(V_W(a)) = W.$$

The fixed-fibre coarea formula on this chart gives

$$\int_D F(V)\,\delta(\mathcal{M}(V)W^{-1})\,dV = \int_{D_W} F(V_W(a))\rho_W(a)\,da.$$

The complete chart density ρ_W includes the pivot elimination factors, the determinant of the nonlinear linearising map, the Haar densities, and the fixed powers of L from the group delta functions. Suppose the chart is admitted on $|a(b)|<r_L$, where r_L is fixed, and its stated local estimates give $\rho_W \le C_L^N$, with N the number of fine bonds in these double blocks. There are at most N free bonds, so

$$\int_D \delta(\mathcal{M}(V)W^{-1})\,dV \le C_L'^N.$$

The bounded coordinate box supplies the volume bound for this Haar factor. The variables here are unscaled. If scaled variables are used instead, their scaling Jacobian must remain in ρ_W.

This covers the averaging factors on the belt $\Omega_t^{\sim}\setminus\Omega_t$ wherever the two retained step-t tests pass and the step-t averaging constraint is imposed on the full double blocks. The plaquette and approximate fluctuation tests put the free bonds in the chart of their prescribed background. In

that chart the constraint solves the pivot with contraction defect $O(L^{-3})$. The bound uses these tests at the birth of the factor, in its original gauges. For the remaining Haar factors the contour formula gives a direct bound whenever the averaging pivot is one of the integrated bonds. A regular extension of the coarse datum only supplies a single candidate fine field.

The total contour formula gives a useful description of the remaining one-row problem. Fix all fine bonds except the middle corridor bond of a coarse bond c, and put $q = U[c]$. Haar measure in the middle bond becomes Haar measure in q. For each of the L sites on the central coordinate axis of $B(c_-)$, the contour family has a single member, and cancellation of the backtracking portions gives

$$U(\Gamma)U([x, x'])U(\Gamma')^{-1} = U[c].$$

These terms contribute zero to the logarithmic sum. Every other joining path avoids the middle bond, as do both endpoint contour families. Thus, on the literal-contour branch, the row has the form

$$F(q) = \exp\left(\sum_{r=1}^{N_c} \alpha_r \log(H_r q^{-1})\right) q, \qquad \alpha_r > 0, \quad \sum_r \alpha_r = 1 - L^{-3}, \qquad N_c \le (4!)^2(L^4 - L),$$

where the H_r depend only on the other fine bonds. Its intrinsic logarithms are defined except at the finitely many values $q = -H_r$. Moreover the norm of the exponent is at most $\pi(1 - L^{-3})$. On the regular branch the same form holds with the averaged endpoint contours in place of the literal contours and at most $L^4 - L$ terms. Their values are again independent of the middle bond.

4.18.9.1. Haar measure for a single averaging pivot. Identify $G = \mathrm{SU}(2)$ with the unit round three-sphere. Write $\exp_q$ and $\log_q$ for its Riemannian exponential and principal logarithm. The formula at a fixed set of nonpivot bonds is, on either logarithmic branch,

$$F(q) = \exp_q(w(q)), \qquad w(q) = \sum_i \alpha_i \log_q H_i, \qquad s = \sum_i \alpha_i = 1 - \beta, \qquad \beta = L^{-3}.$$

Indeed right translation to the identity sends $\log_q H_i$ to the intrinsic group logarithm of $H_i q^{-1}$. Each H_i and α_i is fixed while the pivot varies. Consider this formula on the complement of all the antipodes of its anchors, before restricting it to the set on which its branch is used.

For a tangent vector v with $r = |v| < \pi$, set $H(v) = P_v + r \cot r\,(1 - P_v)$, with $H(0) = 1$. In the polar metric $dr^2 + \sin^2 r\, d\omega^2$, this is the Hessian at q of the half squared distance to $\exp_q v$. The logarithm is the negative gradient of this squared-distance function in the base point. Differentiate $\log_q F(q) = w(q)$ to obtain

$$B(w)\, dF_q = H(w) - \sum_i \alpha_i H(v_i), \qquad v_i = \log_q H_i, \qquad B(w) = D_2 \log_q(F(q)).$$

Use parallel transport back to q to compare the tangent spaces. In these coordinates $B(w)$ is positive with radial eigenvalue one and transverse eigenvalue $|w|/\sin|w|$, as follows by differentiating the exponential in polar coordinates.

The matrix function $1 - H(v)$ is convex in quadratic-form order. The cotangent expansion [DLMF, formula 4.22.3] gives

$$1 - H(v) = 2 \sum_{n \ge 1} \frac{|v|^2 I - v v^{\mathsf{T}}}{\pi^2 n^2 - |v|^2}.$$

For a unit vector e each summand tested against e is a positive multiple of $|P_{e^\perp} v|^2/(\pi^2 n^2 - |v|^2)$. The quadratic-over-linear function is convex and decreasing in its positive denominator. Its composition with the linear numerator vector and the concave denominator is therefore convex. Jensen with weights α_i/s gives

$$H(w) - \sum_i \alpha_i H(v_i) \;\; \ge \;\; H(w) - sH(w/s).$$

Writing $r = |w|$, the radial eigenvalue of the lower bound is β. The transverse eigenvalue divided by the transverse eigenvalue of $B(w)$ is $\sin(\beta r/s)/\sin(r/s) \geq \beta$, using concavity of sine on $[0, \pi]$. At $r = 0$ this holds by continuity. Thus the determinant of dF_q is positive and at least $\beta^3 = L^{-9}$.

The multiplicity is at most one for this formula with its fixed anchors. Combine coinciding anchors and let p be one of their antipodes, with weight a_p. As q approaches p along the unit tangent direction ω, the map tends to

$$\exp_p(b_p + \pi a_p \omega), \qquad b_p = \sum_{i:H_i \neq -p} \alpha_i \log_p H_i.$$

The limit is uniform in ω. The ball of vectors of centre b_p and radius πa_p lies inside $|v| < \pi$, since $|b_p| + \pi a_p \leq \pi s$. Replace each puncture by its sphere of directions and attach a ball mapping by $u \mapsto \exp_p(b_p + \pi a_p u)$ for $|u| \leq 1$. Give the attached ball the orientation of the original geodesic coordinates about p. Its interior map preserves orientation. The resulting continuous map of a three-sphere has degree one. To see this, multiply every weight by $t \in [0, 1]$ and use the same construction. At $t = 0$ the map collapses each attached ball to its puncture and agrees with the identity outside these collapses, hence has degree one. For a target off the finitely many cap boundary spheres, the preimages are finite and each has local degree $+1$. Degree additivity [Hat02, Proposition 2.30] therefore bounds the number of original preimages by one. Two original preimages of a target on a cap sphere would give two for an open set of nearby targets, by the local diffeomorphism property. This proves the same bound there.

The total row is a restriction of at most two such injective maps on measurable branch sets and is the identity in the corridor product on the remaining cut set. That set is finite for fixed nonpivots, by the literal-contour formula. Thus the pushforward of normalized pivot Haar measure has density at most $2L^9$ relative to normalized outgoing Haar measure. In particular, for every nonnegative normalized approximate identity δ_ϵ and every outgoing value W,

$$\int_G \delta_\epsilon(F(q)W^{-1})\, dq \leq 2L^9.$$

This upper bound is uniform even at an exceptional output. An exact Borel choice of fibre kernel is available. For each smooth branch take the inverse pivot on the image of its actual Borel branch set, with weight $|\det dF|^{-1}$. The map from the pair of nonpivot data and pivot to the pair of nonpivot data and outgoing value is a Borel injection on each such set. The branch sets and the graph of this map are definable, by the same finite formulas as for the real averaging map. Projection gives the image and the graph of the inverse, so both are Borel by cell decomposition. The Jacobian weight is Borel by the displayed derivative formula. Add the two kernels, using the zero measure on an empty branch. The resulting kernel has mass at most $2L^9$ for every target and gives the exact joint measure by change of variables on each branch. The finite fallback set has zero conditional incoming Haar measure and so contributes zero to this version of the kernel. For the version retaining the pointwise fallback branch, write $D_c(a)$ for its finite pivot set at nonpivots a. Since the row is the identity in the corridor product there, add $\mathbf{1}_{D_c(a)}(W)\delta_W(dq)$ to the kernel. This keeps the exact joint Haar measure and increases the pointwise bound to $2L^9 + 1$. The choice of version has to be identified if a prescribed target comes from a singular lower graph. Equality of the integrated joint measures does not identify their pointwise values on that graph. On a later gauge slice one must also check the actual order and gauge action of the stored operation.

The row kernel is covariant under the real site gauges of the row. The corridor product, the anchors and the output are all multiplied by the same two endpoint factors. The logarithmic branches and their cut sets are preserved. Left and right multiplication preserve Haar volume, so the inverse Jacobian weights are unchanged. In particular the kernel can be used before taking a gauge quotient when that is the order of the operation and the quotient acts on the corresponding joint measure.

4.18.9.2. Gauge draws with an outgoing value held fixed. Before discarding any variables, realize one partial history on the bare field, the frames and the fresh fields of the reference draws. Keep the temporary Haar variables of the reference boxes as separate coordinates until they are overwritten. The history weight is the product of its tests, gauge factors and reference factors, each evaluated on the field at the time of insertion.

Let u be a real gauge admissible for the specified draw. It is constant on blocks already gauged and equals one near the retained drop-outs and the retained second-class slices. At time s, use $u^{(s)} = 1$ on the inner boxes that will be remade after s, and $u^{(s)} = u$ elsewhere. On the bare field use $u^{(0)}$, and on a fresh field use $u^{(s)}$ for the time of its draw. The frames are unchanged. The temporary Haar coordinates can be transformed separately, since they carry no earlier test and are subsequently overwritten. The lifted change of variables preserves the bare Wilson measure, the fresh Haar factors and the retained slice measures. At a retained slice the lift is the identity.

Induct over the moves using covariance of the averages. Outside the boxes already remade, the current field transforms with the mask at the time of the read. On replaced rows later moves take the fresh field at its own level, and earlier factors retain their original arguments. Earlier blocks are either contained in a later inner box or disjoint from it, so the mask is still constant on each previously gauged block. The plaquette tests and the relative bond tests keep their norms. The inner tests use the transported tree or a slice on which the gauge is the identity. Reference normalizers transform with their defining integrals. Thus the lift preserves the history weight and intertwines the gauge action on its stored variables.

In a preparatory stage first impose the next-background regularity test. Then insert the new level-j gauge factors at the coarse centers in $((\Omega^{\sim 3}_{j+1})^c \setminus Z''_{j+1}) \cap Z$ whose blocks do not yet carry the factor. The other collar factors are retained from their earlier steps. The lift uses these actual block groups and the normalizer of each factor at its insertion.

For the ordinary draw the gauges fix all coarse centers. Its raw outgoing average W is therefore unchanged. Insert any bounded Borel $F(W)$ in the invariant history measure. Changing variables in the normalized gauge draw gives the same weighted integral as inserting the gauge factor as a product. This proves the identity of outgoing measures, also in the presence of any other prescribed coordinates fixed by the draw. On the pass support the contour variables used in the factor are defined. Outside that support the total kernel uses its separate normalized convention at undefined contours. A pointwise bound on a singular prescribed fibre still uses the corresponding branch or graph estimate.

On a row with a free Haar pivot the ordinary draw has a stronger fixed-output identity. Hold the nonpivot coordinates fixed before the draw, and write $q = V[c]$ for the straight corridor product. Both coarse endpoints are fixed by the gauge. Thus q, the coarse-endpoint contour anchors in its row map, and the criterion for the averaging branch are unchanged under the draw. The contour density of the draw uses only internal block links and is independent of q. At a prescribed output W, the pivot inverse in q therefore has the same branches and densities before and after the draw, using a common version at exceptional targets. Equivalently each normalized group approximation to the averaging delta has identical arguments on the two sides. The gauge integration in these coordinates is the normalized draw on the nonpivot data. For a later ordinary draw the old large region is excluded by the five-layer buffer. The blocks read by these contours and the neighboring endpoint blocks are in the old small-field region with its positive density tests. Their inherited plaquette bound implies that the contour means in the normalized density are defined. Other prescribed coordinates moved by the draw remain in the joint graph. An older reference-tree gauge that moves coarse endpoints requires its own row calculation.

In a nonwrapping one-step domain a pivot crosses the face between the two endpoint blocks of its row. No other coarse double block contains both ends of that fine bond. Thus, if every constrained row has a free integrated pivot, fixing all nonpivots gives a product of single-pivot maps. The preceding bound is then $(2L^9)^N$ for N constrained rows, uniformly in all the outgoing values. In a

stack of such integrals one may integrate the pivots from the finest level upward, keeping all still unintegrated coarse variables fixed. This supplies the Haar part of a positive reference bound on those stacked domains. The actual kernel requires a check that each pivot is present in its stated integration measure, that no pivot is also used as a separately stored coordinate, and that the stated charge counts the participating rows.

In the completed main-box operation the incoming averaging constraints have already been integrated when the frame tree is imposed. An older island can instead carry a restored field with a tree factor into a later averaging step. An interior middle bond can belong to that tree. For example, in the frame of a box comb whose last coordinate change is in direction one, the interior bonds in that direction are tree bonds. For a later row with a pivot on such a bond we must identify another free coordinate or work with an exact gauge quotient of the joint averaging law. A tree gauge moving the two coarse endpoints also acts on the prescribed output of that row.

The next ordinary block gauge is fixed at the coarse centers, so by covariance its insertion preserves that raw average. In addition it is applied outside the enlargement of the older input region. To see the role of the endpoint condition, take identity exterior and the identity box minimum. The restored cutoff bounds every bond in an interior double block by $C\varepsilon_i$ from identity. The average of its row is then within $C_L\varepsilon_i$ of identity. A Haar draw of an independent old gauge at one coarse endpoint instead gives that row a Haar marginal. Thus integrating those old gauges would change the raw law on this fibre. The Wilson weights can change the density inside its support; an estimate for the raw output still needs the variables of its actual chart.

4.18.9.3. Averaging the restored box in its tree slice. Fix the frame of one restored box $P = \Lambda_i^{\mathrm{prop}}$ and hold its exterior fixed. The box is a union of M-cubes of level i. Since M is a positive power of L on the nested centered grids, P is a union of whole level-$(i+1)$ blocks. Use a lift of the box, so that each coordinate line of blocks in it is a finite interval. Take direction one last in the chosen frame. In the restored variables $V_i' = V_i V_\Lambda^{-1}$ the tree fixes all interior direction-one bonds. It fixes just one boundary-crossing bond, the entrance. The other restored bonds with an endpoint in P carry product Haar measure. A row of the first ordinary average can read one of these variables only if at least one of its endpoint blocks lies in P.

For the derivative at the identity use dimensionless fine bond logarithms A. Let e_μ denote a fine unit vector and let

$$\varphi(y) = L^{-4} \sum_{x \in B(y)} \bar{A}(y, x),$$

where $\bar{A}(y, x)$ is the average of the path integrals along the symmetric contours in $B(y)$. For a row $c = (y, y')$ the derivative in the coarse bond logarithm is

$$(\mathbf{Q}^{\mathrm{lin}} A)(c) = \varphi(y) - \varphi(y') + L^{-4} \sum_{x \in B(y)} A([x, x + y' - y]).$$

In particular, when both blocks are in P and c has direction one, the straight-path term vanishes on the tree slice. Choose in each block $B(y)$ the fine bond $b_y = (y + e_1, y + e_1 + e_2)$. It lies strictly off the first coordinate face supporting the tree's direction-two bonds. Its coefficient ω_y in $\varphi(y)$ is positive. Indeed, the contour to $y + e_1 + e_2$ in order $1, 2$ uses it with positive sign, giving at least $(2L^4)^{-1}$, and every other contour using that bond has the same sign. No other block contour uses b_y.

On each direction-one line through P, leave the selected bond in the first block among the fixed variables. For each interior direction-one row, select $b_{y'}$ in its right block. Order the rows along the line. Each such row reads the current and at most the preceding selected bond. The diagonal derivative is $-\omega_{y'}$ on each color. For each remaining row, select a free fine bond crossing the face between its two blocks. In an interior transverse row the middle crossing bond is off the first coordinate face, hence free. In a boundary row the face has L^3 crossing fine bonds, at most one of which is the entrance. Every chosen face-crossing bond belongs to just that row among the double

blocks and is absent from the endpoint contours. Its coefficient in the straight-path average is L^{-3}. Order these rows after the interior direction-one rows. The square derivative in the selected variables is thus block lower triangular, with these nonzero diagonal blocks.

The same order gives a real chart on the restoration cutoff. Fix all unselected coordinates. On the full double block of a row take a real gauge of the fixed reference V_Λ with links close to identity, using the comb estimate in the admitted small-plaquette domain. The full blocks reach at most L fine spacings outside P. The prepared box is at distance at least $4M$ from Ω_{k_0+1} in level-i units and is inside $X \cap \Omega_i^c$, two coarse cells from X^c. Thus every full double block lies inside the region of the retained exterior tests and the extendibility comparison. The box minimum bounds the plaquettes meeting P by $C(M)\varepsilon_i$; on the remaining plaquettes it agrees with the prescribed exterior, whose retained density tests give the small-plaquette bound. On the cutoff the restored logarithms have size at most $M_0\varepsilon_i$. Conjugating the logarithms at their initial sites preserves the balls and fixes their zeros on the old tree. The output gauge acts by left and right translations at the coarse endpoints. Both changes preserve the corresponding Haar measures.

In the row's gauge the complete field on its double block is uniformly near identity. The analytic averaging formula therefore applies throughout the selected coordinate ball. The derivative of the row logarithm in its selected variable differs from its scalar diagonal derivative at the identity by at most $C(L, M)\varepsilon_i$. Take the coupling ceiling small enough that this error is less than half the absolute diagonal coefficient. On a convex relative-logarithm ball, integration of the derivative along the segment between two inputs bounds the difference of their outputs below by half the coefficient times the input distance. Thus the row map in this coordinate is injective with inverse Jacobian bounded by a fixed C_L for Haar measures. Fixed site conjugations and output translations return the bound in the original frame.

The row dependencies used for the triangular order are exact double-block dependencies of the nonlinear map. An interior direction-one row sees its two selected block links only. The remaining rows can also read previously selected block links, and each reads only its own selected face-crossing link. Use the one-row change of variables in this order to reconstruct the selected coordinates at a raw output. Equivalently, apply change of variables in the resulting triangular chart. The Jacobian product and the remaining normalized Haar coordinates give the bound $C_L^N\, dW$ on the joint output of the N rows meeting the restored box. This is a bound for the Haar slice restricted to the cutoff, before applying its density. The inverse on the chart and zero mass off its image give a Borel version, with the bound at every output. The joint graph over the unselected coordinates and the exterior is Borel, and the inverse is taken on that graph. All these changes hold with the actual fixed exterior values in the fibre.

After combining the restoration law with its old Wilson weight, use this chart with the core factor, the real form and the scalar prices assigned as above. The bound covers the whole first-step row stock of one restored box at fixed exterior. Its use in a subsequent constraint requires the joint-coordinate check below.

4.18.9.4. Disjoint row charts and their outgoing fields. At a fixed level i, let $P_a \subset X_a$ be the boxes restored in the distinct second-class components of the label, whose tree conditions remain in the output. Every double block used by a row meeting P_a lies inside X_a. Indeed $P_a \subset X_a \cap \Omega_i^c$ with the two-cell buffer of eq. 4.cdxliii, and a double block reaches at most L fine spacings past P_a. The X_a are distinct components on the same grid. Thus their first outgoing row sets are disjoint. Conditioning on the complement of the restored bonds fixes all the exterior coordinates in each box chart. In the ordinary step $i+1$ the complement of the new determining member contains the hull of the carried region Z_i enlarged by eight layers. The block-gauge draw is outside its five-layer enlargement. All the first outgoing double blocks of P_a lie inside $X_a \subset Z_i$, so they lie on the actual raw averaging rows and outside the next gauge draw. This applies even when the successor merges with another component. The waiting rule leaves these rows in place, including when $i+1$ is the

ordinary bottom of the later operation. The remaining rows have both blocks outside all restored boxes and use no restored coordinate.

This also identifies the input coordinates at the outer face of each second-class component. The top inner member of the prepared nest lies in P_a. Central integration consumes its last bare averaging graph and the old top bonds meeting P_a. After the draw the determining list contains every level-i bond meeting X_a. The tree constraint is only on bonds meeting P_a. Outside those bonds the output carries the original level-i Haar coordinates with the old weights; there is no remaining averaging constraint there from inside the island. In particular the physical exterior that supplies the centre of the restored law is read in the top coordinates of $X_a \setminus P_a$. The earlier core variables stay in the island integral.

Since X_a is a union of whole level-$(i+1)$ blocks, a row that reads a fine bond meeting X_a has an endpoint block there. If its two blocks miss P_a, its middle fine bond meets X_a and avoids P_a. It is one of these ordinary Haar inputs, even when the other block is outside X_a. The next-step enlargement contains the complete double block in the bare region. The one-cell separation of distinct same-level components prevents any such row from having a pivot in two components. The rows touching P_a and the remaining rows touching X_a together number at most $8\#_{i+1}(X_a)$. Charge them to the actual successor of the island, which contains their full blocks. Use the strong real bound for the old island integral at its admitted argument. The tree condition on the fresh restored field is outside that integral and belongs to the first outgoing chart.

On any additional bare row having an actual free Haar pivot, use its middle face bond. That bond crosses between its own two blocks, so a row of a box chart cannot read it. Conversely an ordinary row with both blocks outside all boxes reads none of their selected coordinates. First solve the ordinary rows outside the boxes at fixed unselected coordinates. These row maps do not use the fresh box fields. Reconstruct the ordinary pivots and fix that resulting physical exterior. Then use the simultaneous chart for each disjoint box stock. The reference field of a box may depend on the reconstructed exterior, so it is fixed at this second step. Fubini's theorem gives the product of the inverse-density bounds in this order. Any retained weights are evaluated at the resulting inverse graphs. In each recorded frame write the fresh field as $V'V_\Lambda(e)$ at its reconstructed exterior e. The reference tree fixes V' on the same bonds for every such e in that frame, and right multiplication preserves the relative Haar measure on the non-tree coordinates. Bound the outgoing density uniformly in e on this relative chart. If an outside input to one of the ordinary rows comes from a lower graph, keep it in that joint measure while reconstructing e. The frame and denominator are those of the original summand. The actual tests restrict the domain of the chart. The number of rows touching a box is at most eight times its number of coarse blocks. The successor contains the propagated component and hence these row blocks. Charge the rows to its sites at level $i+1$. The disjointness at one level and the waiting rule for successive marks count each assigned row once. Thus the product of the first-box chart constants has an exponential cost in the corresponding site-level count of eq. 4.cdlxxv.

There is a different first chart for a component that stops at the shell-field or last-shell test, after central integration and before reference restoration. Central integration has consumed the old box field, including the bonds crossing its boundary. The resulting term has unrestricted fresh Haar coordinates on the bonds with an end in the box. Its frame tree acts on the old shell variables. After bounding the retained weights, the normalized Haar integral over that tree is at most one at fixed exterior root. Any older marked core retains its own joint integral and reference bound.

At this fixed exterior the reference used to define the old shell fluctuation depends only on the old exterior of the prepared box. The tree coordinates on that fluctuation have the rooted-tree Jacobian one. They are disjoint from the fresh Haar pivots of the outgoing rows, so the combined coordinate change has this unit tree factor times the raw row Jacobian. This permits weights to be evaluated on the reconstructed old shell without imposing any tree condition on the fresh pivots.

The box is a union of full level-$(i+1)$ blocks. A first outgoing row of level $i+1$ that reads the box has at least one of its two blocks there, so its middle fine bond has an end in the box. It is one of the free Haar coordinates, even for a row crossing the box face. At fixed exterior the corresponding one-row bounds therefore apply jointly to all these rows. A row whose two blocks lie outside the box reads none of those coordinates. The same count of at most eight rows per coarse block pays for the first outgoing density, using the unrestricted Haar row bound instead of the regular restored chart. The reference-law slice enters only on the branch that reaches restoration.

At a shell or last-shell stop the same order recovers the exterior of the prepared integral. A density cube based in the component uses four test layers of width $M_2\check{R}_i$ in level-i units, and at most one further bond. This reaches less than one level-i cell, of side $M\check{R}_i$, since $L^2M_2 \le M$. The next full double blocks add at most $2L$ in these units. The hierarchy $M \ge L^2M_2$ and $M_2 = L^2M_1$ gives $4M_2\check{R}_i + 1 + 2L < M\check{R}_i$, so the resulting read still lies within one old cell of the component. An old cell is no wider than a new one by $\check{R}_i \le L\check{R}_{i+1}$. The five- and eight-layer enlargements in the next ordinary step therefore put these rows in the raw region outside the gauge draw. Solve the ordinary rows with both blocks outside the box first. Their inverses return the selected exterior pivots. Keep the other exterior coordinates among the unselected variables or in their lower joint graph. A value shared by two tests is reconstructed once. At this recovered exterior solve the remaining rows, each with its free box pivot. The prepared datum and the frame variables are read in the old integral at this exterior. Use the same values for the original positive tests and for the prepared-datum support condition.

Keep the exact inverse charts when using the resulting bound. Write t for the selected variables and s for the unselected ones. At fixed s and raw output w, let $t = \psi(s, w)$ be the reconstructed variables on the image, with zero density off the image. If a one-row change has several branches, keep the corresponding sum of inverse densities. A nonnegative Borel test or weight depending on t is evaluated at these reconstructions. The bound by a product Haar measure on w can be used with an envelope uniform over the reconstructions. This applies in particular to the uniform upper bounds for the complete Wilson product, after the old, new and core factors have been assigned their stated roles.

For the next raw averaging constraint the input is the current-level field. On its full double blocks this field is assembled from w and the other current-level coordinates that actually remain fixed or integrated. A later bare average can use w as a dominated Haar coordinate when these other arguments are fixed by s or supplied by their own joint chart. Older variables in test and energy factors are still evaluated by the inverse charts. The ordinary and prepared averaging deltas both take the physical field at their own level as input. The tree condition at an older island remains in that island's chart. An old fine variable read by a carried weight therefore stays in the inverse kernel for its outgoing row. If an operation prescribes such a fine value as a target in its own right, we must use its joint graph with the outgoing row. The available pivot choices are then those left free under both prescriptions.

The waiting rule protects the first outgoing scale of an old island. A later operation containing it remakes only scales above $i + 1$, so its retained variables below the first outgoing field remain in the old core. The averaging constraint at $i + 1$ reads the restored argument V_i and produces the raw field V_{i+1}; the older determining datum V_i'' is in a separate slot. This assigns the first raw rows once, without giving a second Haar variable to a stored datum. For the continued stack use the row assignment in §4.18.9.5, with the joint outgoing law of the earlier operation on the actual read set. Along a consecutive bare chain whose current-level rows satisfy the joint chart condition, iterate the conditional estimate from the first outgoing field to the last one. The Jacobian product then bounds the chain, with its actual stored weights read at the inverse charts.

The intermediate fields must be read from the retained operations. In an unmarked core, the field of level j has its stored values on Γ_j; inside Ω_j^c its values are the averages of the preceding field, imposed by the preceding averaging kernel. They are not additional independent Haar variables.

The same reading applies when the earlier drop-out left before completing its preparatory stages. In the broader waiting convention, at a full re-entry the remade shell instead carries the old current-level datum; the remaining core carries either the preceding average or the fresh argument from the central integration, according to the recorded place of departure. A fresh argument and a retained determining datum on the same bond remain distinct variables.

Successive graph kernels can be compressed exactly. If x denotes the variables of an earlier operation, $u = A(x)$ its output, and p the other retained arguments, then, for every nonnegative Borel function H,

$$\int \delta_{A(x)}(du) \int \delta_{B(u,p)}(dw) H(x,u,w) = H\big(x, A(x), B(A(x),p)\big).$$

Here B uses the full field assembled from its stored and generated entries. Consequently the compressed terminal constraint is the constraint on this assembled field. It becomes $\mathcal{M}^2(V_{j-1}) = V_{j+1}$ only when all entries used by the second average are supplied by the first. This identity follows by substitution in Dirac measures and preserves the original integration variables and their measure.

For a fixed-output mass estimate, this substitution does not supply Haar measure in a generated pivot. The one-row estimate applies directly only to a pivot that is actually integrated against Haar measure, with the other row arguments held fixed. To apply it to a pivot supplied by an earlier operation one first needs a bound for that operation's joint output density on the variables used. Likewise, a pivot crossing a retained seam must be assigned to its actual integration or reconstruction.

The prepared-stage Haar factor integrates the determining bonds of level j in the nest Z''_{j+1}. Those bonds lie in or meet the shell $Z''_{j+1} \setminus Z''_j$. After the collar change the remaining delta rows are the inner bonds of level $j+1$ of Z''_{j+1}. A row crossing the prepared face belongs to the collar chart below. For a row deep in Z''_j its lower inputs have to be read through the preceding operation. At the outside seam, the middle fine bond of a crossing row may itself cross into Ω''_{j+1}. The no-inward determining set then omits that bond. In an ordinary one-step operation such a row, where it is in the admitted chart, belongs to the pivot reconstruction for the fluctuation step. At a re-made level the distinction is between the old wall and the new prepared face. The following decomposition identifies which reconstruction each uses.

Use the preparatory integral of [B89a, (1.60), (1.71)–(1.72)]. At a preparatory stage put $Y = Z''_{j+1}$ and $C = (\Omega^c_{j+1} \setminus Y) \cap Z$. Divide the rows of the step in this component into three sets. Let R_{ord} be the rows meeting the old Ω_{j+1}, whose pivots are in the ordinary fluctuation and strip reconstruction. Among the rows with both endpoints outside that region, let R_{col} be the ones meeting C, and R_{in} the ones with both endpoints in Y. A row crossing ∂Y belongs to R_{col} by the separation from the old Ω_{j+1}. The regions are unions of complete level-$(j+1)$ blocks. The full double block of a row in R_{in} is contained in Y, and that of a row in R_{col} lies outside Ω_{j+1}. Such a row meets Ω^c_{j+1}, which is contained in the complement of the top determining member. The two-layer buffer of that complement in Z keeps the full double block in the same component. The old-wall and preceding-face margins put it in the level-j member of the preceding stage. All its links are then in the non-inward level-j datum before the promotion. The double block misses the earlier inner face. At the first stage the collar uses at most five of the eight layers separating the old members. At a compressed stage the spacing of the prepared faces is at least one level-$(j+1)$ M-cube, which exceeds the double-block reach. Thus a middle bond selected from this collar stock is a level-j integration coordinate. It is either an old exterior Haar datum or a target left with its carried density after the preceding collar integration. The earlier inner graph has its targets wholly in Z''_j and does not solve this pivot. The soft gauge insertion is a density on this stock. Rows carried outside the component keep their own reconstruction.

Use the collar correction on R_{col},

$$V_j = \exp i[g_j \mathsf{C} B_j - \mathsf{h}\widetilde{\mathsf{D}}(g_j \mathsf{C} B_j)] V^{(j)}.$$

The map C keeps the nonpivot increments, and h acts only on its assigned pivots. The pivot of a crossing row lies on the common face of its endpoint blocks, outside every double block for R_{in}. A row of R_{col} uses no pivot of R_{ord}, since that pivot lies within or crosses into the old Ω_{j+1}. An ordinary row can read the collar field in its exterior block. Its middle bond enters Ω_{j+1} and is inward also in the promoted determining list. Its value was solved by the original strip reconstruction at the prescribed coarse row. In a sharp failure, first recover the collar variables in the branch where the approximate test passed. At fixed prospective datum the two stage centres are independent of the changed collar bonds, since those bonds are omitted from the new determining list. The extra joined datum of the localized centre lies over the old top member and does not reinstate those bonds. The old strip centre may read the collar field. Read it at the reconstructed exterior and use the original inverse at the same prescribed coarse target. The two sets of pivot derivatives are triangular in this order. Keep the old elimination and strip Jacobian factors, and evaluate each surviving old indicator on its complete original carrier. The collar chart supplies only the density for its own rows. On inner fine bonds of Y the field still consists of the retained shell values and those of the earlier operation.

Here is the exact factorization on any carried support on which this collar chart is valid. Denote those remaining variables by s, their measure by $d\nu(s)$, and all the actual collar coordinates by a. Write $\Psi_W(s,a)$ for the reconstructed field at the prescribed coarse datum W. Its full coarea density, with the old densities and tests, is $J_W(s,a)$. If F is the remaining integrand, the collar integration gives

$$I_W(s) = \int F\big(\Psi_W(s,a)\big) J_W(s,a)\, da.$$

The surviving operation is therefore, with unsolved constraints still understood as their original graph kernels,

$$\int I_W(s)\, \delta(\mathcal{M}(V_j(s))W^{-1})\big|_{R_{\mathrm{in}}}\, d\nu(s).$$

Indeed the local coarea formula consumes each collar row once, and the remaining row constraints are independent of a by the double-block containment. All dependence on the prescribed crossing-row values stays in I_W, including the elimination factors and nonlinear Jacobians. No extra Haar variable has been adjoined. The delta factor for each solved row has already been integrated in the chart, together with its coarea factor.

4.18.9.5. The seam after successive stages. At the first stage the prepared-box separation from $Z_{m_\flat} \cap X$ puts the full row blocks near the new face outside the bottom operation. At each subsequent stage the preceding completed collar leaves its level-j graph constraints only on rows with both endpoints in Z''_j. The separation from Z''_j to the new face exceeds the reach of the full row blocks at that face. Thus the input field there is read from the level-j coordinates of the retained law or from the outgoing side of the preceding chart. The earlier inner graph is integrated in its own variables. A read through an inverse chart adds no second constraint. The three-way split at this stage leaves the next inner graph on the rows wholly in Z''_{j+1}. This proves the row assignment for every prefix of completed stages.

A failure of the regularity or approximate collar test, or of the completed ordinary fluctuation test, precedes the collar pivot change. A failure of the sharp collar test occurs after it. In this branch retain the collar nonpivots and the exact translated density, including the factors for the eliminated pivots. The determining list can still be the one before the stage. The stage has no integrated normaliser because its collar integral has yet to be performed.

The row assignment uses the collar chart on its actual real support. We must verify its admission on each support where it is used. A bound for the surviving inner integral also needs the joint law of the old core arguments from their retained operation.

4.18.9.6. The joint law of the retained core rows. Fix the history and its outgoing field W. Unfold each older drop-out into the ordinary and stopped-stage operations it retains, including the remaining averaging graphs. Each unfolding lowers the level of the mark. The evaluation in the upper integral varies the weights and compatible real backgrounds at the stored values. The graph equations in those variables and their target W are fixed throughout the supremum.

Write $h = t - \mathsf{N}_t$ for the bottom of a class window of level t, and $G = Z_h \cap X$ for its core. Every mark in G has level strictly below h. For $h > 0$, the last move in the core is thus the ordinary operation $\mathbf{T}^{(h-1)}$ on the output of the lower operation. It keeps the density test at level h. The unresolved raw targets at that step have their two endpoints in $\Omega_h^c \cap G$. At its birth the component contains two cell layers around $\Omega_h^c \cap G$, so the complete double blocks of these targets are in the ordinary core. In the unmarked part of each block a level-$(h-1)$ bond with an endpoint in Ω_{h-1} is an original Haar datum. There is no inward higher endpoint on this block. A bond wholly in Ω_{h-1}^c is a raw output of the preceding operation. This supplies the ordinary bottom row bound from the joint measure of that lower operation and the original Haar slots. At $h = 0$ there is only the bare Haar input with its Wilson weight and tests.

In the lower operation use the field belonging to each older mark at its recorded departure. For a partial stop, use the original top determining data and the surviving lower graph for the other rows. A fully prepared stop at extendibility has its new top datum on the remade shell and the surviving bare graph in the inner box. Central integration comes later. A stop after that integration and before restoration has the fresh Haar slots and free first pivots described above. In a closed island the fresh field is on its reference tree slice, so the field at that level can be singular with respect to product Haar. The simultaneous first outgoing chart supplies the bound for those rows. The fixed exterior includes the actual retained arguments of each such chart. On a fresh slot used only by the next graph, a same-bond datum retained in the old determining set is a separate variable.

The unrestricted fresh field of a late open stop is carried upward by averaging. To see which later reads can prescribe it, let the mark be born at level i. While it stands, it remains a component of its original Z_i with all entries through level i unchanged. In the input Z_{r-1} of a later ordinary step $r > i$ it lies in the old large region. The ordinary step places Ω_r at least eight coarse cells from that input region. The new plaquette tests are inserted outside the four-layer enlargement of its hull. The approximate tests and their gauge reads are outside the five-layer buffer. Even the complete four-margin test cube with its endpoint blocks has reach less than one coarse cell. These tests, and the density tests with bases in $\Omega_r^{\sim 4}$, miss the fresh box. The immediately carried test of level i uses the separate clearance of the central box from the original level-i density reads. At subsequent ordinary steps the preceding density tests use the same ordinary buffer argument at their own level.

If a later eligible component contains this standing mark, its preparation begins at a level strictly above i. Inside the mark the lower determining entries are therefore the frozen ones. A retained row at a bond address shared by the fresh field still reads its old slot. This argument uses the eight-cell gap only on the original ordinary walls. The prepared walls use their own, possibly compressed, gaps. Outgoing averages can use the fresh slot, and a boundary weight reading it stays in the joint integral. Thus these generated fields remain in the joint measure up to the next average. The bound for their reconstructed row law still has to hold at the actual exterior datum.

These cases identify the measure used at each row of the bottom operation. For an induction through older marks, use unnormalized joint upper measures with these Haar coordinates and reference slices before the first outgoing average. Then use the outgoing row bound for the whole read set of the next operation. The prescribed-output estimate requires this domination with the actual fixed exterior and with all weights read on the reconstructed fibres. If a generated slot is

later used as a spectator, keep it in this joint bound unless a bound for a fixed value of that slot has been proved separately. Dividing a joint upper bound by the spectator's marginal gives no uniform conditional bound without a lower estimate for that marginal. The scalar mass of an older mark, obtained by integrating its outgoing field, does not give that domination.

Write A for the fluctuation coordinates with kept Gaussian forms and a for any free nonpivots of a translated, unintegrated sharp stop, in unscaled logarithmic units. Let v be the group coordinates and ζ the old variables of the inner islands.

For an open subtree of pre-main stops, the stopped factor consists of the Wilson and gauge weights, scalar stage and orientation terms, the elimination factors, the kept quadratics, and the old island factors. Completed stage corrections are in the carrying boundary exponent. After allocating the event payments, the residual Wilson and gauge numerators are bounded by one. The gauge denominators cost at most $C\theta_l^{3/2}$ per normalized factor at level l. On the admitted real chart each ordinary pivot determinant inverse is at most eL^{12}, and the kept forms give a common Gaussian reserve $\beta > 0$. At an admitted real pair use a Borel upper majorant $m_a(\zeta_a)$ for the old island, taking its integral as close to the earlier upper-integral norm as needed.

For a sharp stop the passed approximate test in the translated chart gives $|e^{ia_b} - \mathbf{1}| < 2\delta'_{j(b)}$ on its free collar bonds, including when the sharp test fails. In the principal chart this gives $|a_b| < C\delta'_{j(b)}$. At fixed centre and coarse datum the pre-translation coordinate is also $O(\delta'_{j(b)})$ on the admitted chart. The density on this box keeps the Haar factors and the exact derivative of the linearizing correction. In the correction the range is on the pivots. Ordered after the nonpivots, the pivot blocks are independent because each double block uses only its own pivot. The correction has zero first derivative at the origin and uniformly bounded second derivative in the regular one-row chart. Its Jacobian therefore costs at most a constant per pivot on this small box. Include the Jacobian of the cutoff translation on its own bonds as well. The translated free coordinates need only the volume of their compact box.

Thus on admitted inverse charts of this subtree the complete reference weight has the pointwise bound

$$H_\omega(A, a, \zeta, v) \le C_\omega e^{-\beta \sum_b |A_b|^2} \prod_y m_y(\zeta_y) \prod_{b \text{ at sharp stops}} \mathbf{1}_{\{|a_b| < C\delta'_{j(b)}\}}.$$

Here $\log C_\omega$ is bounded by the sum of the stage and orientation scalar costs and the gauge, pivot, Haar and cutoff density costs just listed, counted at their own levels. Old chart factors are read at their reconstructed arguments. The tree and cutoff of a fresh restored field stay in its coordinate measure. For earlier stops in the main step use the retained measures described below.

At an old island, a later raw row uses the fresh field in the restored box. The old island variables occur in the weights. The pair-uniform envelope allows us to integrate those old variables first and keep the fresh field and its exterior as inputs to the next average.

Start at the lowest remaining scale. Carry a joint upper measure for the full current field read by the next double blocks, adjoining its original stored Haar coordinates. For a fresh restored box use its first-average chart on the whole row stock at fixed exterior. On each other bare row use its middle bond when this is a Haar coordinate of the assembled field. A double block reads no other selected middle bond at the same scale. Rows already solved in the ordinary or preparatory collar charts keep their coarea factors. Use the remade masks in §4.18.9.5. A complete inner double block at a prepared level $j+1$ lies in Z''_{j+1}. For $j > h$ in the completed prefix, each of its level-j bonds with an endpoint outside Z''_j is an original datum coordinate at level j, including a bond straddling that earlier face. Both of its endpoints are outside the new determining member at level $j+1$, so the inward mask does not remove it. A bond with both endpoints in Z''_j is an output of the surviving bare averaging graph at the preceding level. These two sets give all the inputs to the next double block. Once the preceding graph has a joint bound on its whole output, adjoin the original Haar datum coordinates in the same conditional integral. This supplies the next row

bound with no seam coordinate left unaccounted for. The first graph still requires the joint output of the bottom operation, and a stopped level uses the determining mask at departure.

A variable still read by a weight is evaluated at its inverse graph. Every additional argument of the next constraint must be in the same joint measure or fixed among the unselected coordinates of that chart.

More explicitly, fix the genuine exterior e, let v be the selected current Haar inputs and let s be the unselected coordinates with reference measure ν_e. Use the lower joint density $k_e(v,s)$ against $dv\,\nu_e(ds)$. If $\kappa_{e,s}(W,dv)$ is the inverse kernel of the new row chart and f_e is its remaining weight, the composed density is

$$\int \nu_e(ds) \int \kappa_{e,s}(W,dv)\, k_e(v,s) f_e(v,s,W).$$

If the inverse kernel has mass at most B and the product of the two densities on its branches is at most $E_e(s)$, this is bounded by $B \int E_e\, d\nu_e$. This calculation uses the branch versions defined by the original graph equations. If a generated spectator z is itself a prescribed raw target, include its generating row in the same complete lower row stock as the surviving outputs v. If the enlarged list admits a joint inverse chart with independent pivot coordinates at the complete fixed exterior, the inverse-density sum gives a kernel $k_e(v,z,s)\,dv\,\nu_e(ds)$ at the specified target z. Use the chart's actual branch version and its density bound. For example, when v,z are distinct targets of one bare averaging operation with their original middle links still free under Haar, their complete double blocks do not read one another's middle links. Hold the common nonpivots fixed and take the product of the corresponding inverse measures. There is an additional regular case when the spectator z occupies the middle link of the outgoing bare row. Suppose the lower joint law at prescribed z leaves another link b' of the same corridor free under Haar. Restrict to the part of this row with small plaquette defects on the entire double block. Choose the comb starting at the initial top and traversing the row direction first. Every comb crossing the face crosses it at the fixed middle link z, so the frame is independent of b'. The comb estimate puts the links in a small logarithmic box in this same frame for all the admissible b'. Write $f(a)$ for the logarithm of the output in these frames, where a is the logarithmic coordinate of b'. The derivative in a at the flat field is $L^{-3}1$. Indeed, the L^4 straight paths of the average traverse that link for precisely L choices of the starting point, and the averaged contours are internal to each endpoint block. At fixed L, analyticity gives $\|D_a f - L^{-3}1\| \le L^{-3}/2$ throughout a sufficiently small box. Choose a convex coordinate ball in that box containing the admissible free-link values. Integration on a segment gives

$$|f(a) - f(a')| \ge \tfrac{1}{2} L^{-3} |a - a'|.$$

The row thus has one inverse branch on the image of the ball and an inverse Haar density bounded by C_L. Use this density on the actual test subset, evaluated on the inverse, with value zero off its image. The other rows do not read b', since it joins only these two endpoint blocks. Index the constraints by positive orientations and read the reverse as the inverse on the graph. This replaces just the conflicted pivot, with a geometric choice of alternative and the same reference Haar measure before taking the exterior supremum. The lower joint density $k_e(b',z,s)$ at z remains part of the integral, with the old pivot recovered for every old weight that reads it. The alternate solve uses no division by a marginal at z. A tree-fixed alternative or the intermediate logarithmic branch outside the regular box requires its own chart. On the straight-path fallback the output is independent of b'. If a different link b'' of the original straight path still has a free Haar coordinate, hold the other inputs fixed and solve that link instead. The row is $Pb''Q = W$, so its inverse is $P^{-1}WQ^{-1}$ with Haar Jacobian one. Read the original fallback predicate at that inverse. Other rows can read b'', but the fallback row does not read their face pivots. Solve the fallback row first and then the other pivots at the reconstructed inputs. If the straight path is already fixed, its output stays on the prescribed graph. Fix the free-coordinate reference measure separately for each geometry, frame and branch choice.

A relation between prescribed outputs belongs to the graph measure. Adding its generating equation to the list gives no second free output or independent Haar factor. Fix the pivot assignment by the geometry and the frame. The reference measure of the remaining free variables is then the same for the different exterior values. Put the inverse-density factors and the weights read at reconstructed values inside the pointwise supremum on those free variables. Rows whose targets are only used to reconstruct the exterior keep their target variables in the integral. If all inputs of a row are already prescribed, its target remains on the deterministic graph. This argument applies to the complete protected first-average stock of a restored box.

For an ordinary successor of a collar-kind departure of level i, keep the old top density halo through the first outgoing row changes. The old top link of the stopped prefix is either a value of its original determining list or the output of its surviving $\mathcal{M}(V_{i-1})$ constraint. Include the full lower double block of the latter row, as well as the first outgoing blocks. This read lies within the two-layer enlargement of the old component on the next grid, inside the ordinary successor. When the component later enters a containing preparation this read is in its floor. The new prepared rows leave these old graphs in place.

For a halo link used only in evaluating a retained test, use the value $z = \zeta(v, s)$ of that old graph in the weight of the joint row integral. In the notation above the corresponding operation is

$$\int \nu_e(ds) \int \kappa_{e,s}(W, dv)\, k_e(v, s) f_e(v, s, W, \zeta(v, s)).$$

Equivalently the conditional integrand has the graph measure $\delta_{\zeta(v,s)}(dz)$ in that read slot. The test retains this value until it has entered the weight. Its presence uses the original inverse rows and carries no new Haar density or integration volume. In particular, the translated collar variable of the stopped history remains the variable in its recorded reference measure, as does the birth fluctuation in the kept Gaussian. If a new constraint solves one of these variables, insert its inverse value and chart density before taking the supremum and omit its integration volume.

For a prescribed exterior slot outside that stock, classify its generating row at the stage where it was formed, using §4.18.9.5. A promoted collar row uses the original collar chart on its complete double block. An ordinary wall row reads its old inverse at its coarse datum. A row of the surviving inner graph uses the lower joint law. If the backward closure of the read leaves the stored row sets already supplied, we must supply that joint exterior read or keep its relation on the actual graph.

The geometric closure along consecutive bare rows has an exact description. Start with a level-q row and stop the closure at level $u < q$. Each first inverse reads fine links with both endpoints in the union of the two endpoint blocks. For every such link, the next inverse reads only the two blocks of its endpoints. The union of the child blocks of all sites in a parent block is the same parent block in the finer lattice. Induction thus keeps the level-u reads in the two order-$(q - u)$ blocks of the original row. This applies also to the nonpivots required by the inverse, which lie in its full double block. If one of those slots comes from a restored box or a collar change, stop the bare closure there and include the stock of that actual chart. Test and weight carriers are included separately at their own levels.

For an ordinary row outside Ω_q, apply the no-inward rule to its level-$(q - 1)$ inputs. A link meeting Ω_{q-1} is a datum there, since both endpoints are outside Ω_q. A link with both endpoints outside Ω_{q-1} uses its actual lower raw graph if that graph is still carried. The two-cell enlargement of Ω_{q-1}^c in Z_{q-1} keeps its full double block in the same predecessor component. The next growth contains the predecessor in its successor. If the upper row reads several predecessors, keep their inputs together in the same conditional integral. When the recursion reaches the first outgoing step of a closed island, the read ends at its physical field. Use the restored first-average stock and the ordinary Haar inputs at its face. On a region absorbed by healing, the new determining list has dropped the old lower members. Read the current datum there and include the healed contribution

through its weight. For a standing drop-out use its departure list and continue only through the rows it retains.

On a carrier clear of older replaced boxes, compatibility with a real bare field propagates through the same graph. On every complete double block the determining inputs agree by admission and the other inputs agree by the lower rows. Applying the average proves agreement at the new target. For a carrier meeting a box replaced at a late stop, the branch uses its fresh top argument. The nested block inclusion above locates this exception using the whole backward carrier. The older integrated datum stays in the inner weight of the box.

If a later weight reads an older branch or any of its remaining variables, keep that argument in the joint integral, or include the weight in the lower envelope before taking the marginal. An earlier generated coordinate can be supplied as a selected Haar input only after its joint density has been obtained by integrating the lower rows. The test and weight functions still use the inverse values of any lower arguments they read. In the prepared seam, first solve the inner raw rows, then the collar rows, then the ordinary rows that can read solved collar values. This order keeps the actual one-sided dependencies. An inner pivot can enter the collar centre through a retained determining row. Reconstruct it before using the collar chart. In the no-inward convention the ordinary pivot at the old member is absent from the fine determining list of the collar centre, since it has an endpoint in the enlarged member. Thus this pivot is solved only at the last step of the order.

At fixed output the compatible set is read at the inverse of each branch. The pointwise envelope bounds the upper integral in these coordinates before the unselected variables are integrated. Use the Borel inverse-density sum with zero mass off the chart images. For the outputs admitted to the independent charts, iteration on full read sets gives the joint outgoing density bound

$$C_\omega(\pi/\beta)^{3n_A/2} \prod_{b \text{ with remaining } da_b} (C\delta'_{j(b)})^3 \prod_y \int m_y \, d\mu_y \, (2L^9)^{N_{\mathrm{raw}}} (C_L+1)^{N_{\mathrm{alt}}} C_L^{N_{\mathrm{box}}}.$$

Here n_A counts independent Gaussian coordinates still carrying dA after the row solves. A fixed or reconstructed slot has its actual chart weight and no volume factor. For each pivot choice, count the remaining volumes on that branch. Sum the resulting bounds if the branches have different free coordinates. The counts N_{raw}, N_{alt} and N_{box} are for bare rows using the middle link, bare rows using the alternate charts and first-average rows of restored boxes, respectively. The alternate charge allows a regular branch of bound C_L and a straight-path fallback of bound one. Only rows still to be solved enter these counts. Use $2L^9+1$ in place of $2L^9$ for a middle-link row with the pointwise fallback version. Deterministic read slots stay on their graph. The already consumed ordinary and collar coarea factors are in C_ω. Fixed exterior slots keep their prescribed values. The row assignment charges each price at its own level.

For an earlier open operation, start with the full blocks read by the selected outgoing rows and take the closure under lower reads. A row in that closure is either genuine exterior data, a retained Haar datum, part of a restored box's first-average stock, or an output of lower rows. Include the whole stock when using a restored box chart. Keep any unused outputs until all weights that read them have been included. An earlier promoted shell coordinate was the target integrated against Haar in the corresponding averaging integral. Its density already includes the elimination of the old pivot. Use that density in the joint integral, with the old pivot recovered for any weight that still reads it.

At an extendibility stop all preparatory stages have been completed, and the bare averaging rows inside the prepared members survive. The last such row has its target in the old field V_i on Z_i''. This is the field to be used in the next ordinary average. The extendibility indicator reads only the part of V_i outside the central box, which contains Z_i''. The infimum in that test gives a predicate on the exterior. Its trial extensions do not appear as variables or constraints in the retained kernel.

When forming the density of this history, take its old top field as the argument of the inner graph. At the next raw average use the same top field in the outgoing Haar integral. Where the old

field is a target of the inner graph, include its earlier joint density in that integral. The other top bonds have their original Haar factors. Thus the inner graph is evaluated at its prescribed target before that target is used as an input of the next row. Lower coordinates read by the old tests stay with their inverse branches. In particular the negative extendibility indicator is evaluated on the reconstructed top exterior in this joint change of variables. The fine inputs retain their integration and constraints from the earlier graph.

At a shell or last-shell stop the central integration has eliminated the prepared chain. Its old terminal variables and the variables of the old core are disjoint. Indeed, write h for the bottom of this component's waiting window. The shell gauge ends at the two-cell enlargement of the prepared Ω_{h+1}. The prepared inner box at $h+1$ has three cell layers around the hull of the bottom core. After the two layers have been removed, the remaining cell layer still contains the complete bottom-core reads. The frame tree therefore acts only on the retained shell. At fixed reconstructed exterior, the change from the prepared group variables to their relative values preserves Haar measure. Solving the frame tree leaves normalized Haar on the free shell bonds. Use this reference measure and the separate fresh box Haar measure. The shell and last-shell failures are separated before the operation that assigns the reference-law density. Their fresh output therefore carries bare normalized Haar. For a component that remains in the class, combine the reference-law density with the old Wilson weight by the pointwise identity following eq. 4.cdlxix. Assign the new action to the carrying term and keep the core Wilson weight for the ledger. The scalar and remainder costs enter at their own levels. This cancellation holds before imposing any new averaging row, so it also holds on the prescribed-output branch. The subsequent Haar bound still uses the retained tree and averaging coordinates. Keep all earlier variables still read by the weights in the lower joint integral.

Consequently a late node adds a normalized shell factor and the free first-outgoing pivots to the reference measure, with the old lower charts and branch tests retained. If the middle bond of an ordinary bottom row is itself a lower graph output, the joint outgoing bound at that lower level supplies its Haar domination together with the other inputs. At a closed island use the simultaneous first-outgoing chart on its restored slice. The pointwise envelope can be formed before those row integrations.

The upper-integral fibre allows one-sided evaluation coordinates for the quadratic forms and pivot factors. Write q_j for the complete kept quadratic at step j, including its coupling ratio. In the stopped weight this is $\tau_j\langle A_j, K_j A_j\rangle$, where K_j is the sum of the kept chain contributions on the original free-coordinate space. Use $\operatorname{Re} q_j \ge \gamma_{\mathrm{p}}\|A_j\|^2$ and the upper bound eL^{12} for a carried elimination factor. For an actual field read in the original positive strip domain $\mathcal{O}^+$, the local kept form has a real-part lower bound $c\gamma_0$. With the original masks and the same free coordinates, the absolute walk sums give $\|K_j\| \le C_{\mathrm{w}}$. For the ordinary strip the ratio is formed at its original level j. On eq. 4.dxvii, the radius bound $t_j \le C\theta_j(\log\theta_j)^{-2q_{\mathrm{t}}}$ gives

$$|\tau_j - 1| \le C(\log\theta_j)^{-2q_{\mathrm{t}}} + \bar{c}/\theta_j.$$

The radius estimate follows from adjacent frame comparability. At the last level use monotonicity of the tube radii and the preceding bound, or the single-level tube condition. Choose the inverse coupling floor so that twice the right-hand side times C_{w} is at most $c\gamma_0/2$. This leaves at least $c\gamma_0/2$ for the complete kept form throughout the ordinary strip ratio disc.

The ordinary normalization may instead keep the form at $\tau_j = 1$ and place its ratio correction in the boundary potential. The gauge compensation in eq. 4.d also has to be accounted for in the complete exponent. On the stored boxes the compensation costs at most $C|\tau_j - 1|M^4 p_0(g_j)^2$ per cube. This stays bounded on the tube because $q_{\mathrm{t}} > \mathfrak{p}_0$. On the enlarged stored-field discs the squared field bound adds $(\log\theta_j)^{12r+2}$. The same cost tends to zero there with the hierarchy $q_{\mathrm{t}} > \mathfrak{p}_0 + 6r + 1$. If a compensation is included in the form, check the lower bound for the resulting complete form. Preparatory and main restoration Gaussians use θ_j. In those operations the change

of coupling belongs to the Wilson potential. An abstract one-sided evaluation is admitted by the lower bound on the complete q_j itself.

For the event payments, let S_{W} be the real carrying Wilson action on the bare plaquettes with all corners in X, and let G_j be the sum of the kept nonnegative gauge costs at level j. In the prepared component the Wilson weight is the running weight times the core cutoff. The cutoff is one on the one-layer hull of the old core, which contains the charged plaquette footprints. The running weight there is a convex combination of the stage weights. At an event of level t it can run for at most $\check{R}_t$ more levels before freezing. The bounded comparison over this range gives a coefficient at least $c\theta_t$ on the event's plaquettes. Indeed, an earlier vertex of the running profile has reference weight at least $\theta_t - c_5$, and a later vertex can lose at most $\bar{B}\check{R}_t$ before freezing. At a final preparatory stage the extra vertex is $x_s + b_{s-1}$. Its frame value is $\theta_s + \theta_{s-2} - \theta_{s-1}$, so the adjacent frame comparison gives the same lower bound after allowing the three relative tube errors. The cutoff leaves all these bounds intact on the charged footprint. Set

$$E_{\mathrm{pay}} = a_\sharp \sum_t p_0(g_t)^2 n_t^{\mathrm{cr}} + \sum_t p_0(g_t) n_t^{\mathrm{mk}},$$

where the counts concern the creations and marks paid in this component. A drop-out's own mass omits its failed root test. Charge it in a later waiting window when that component supplies its root witness on the compatible inverse branch. With all the assigned certificates on that branch, the event ledger gives

$$S_{\mathrm{W}} + \sum_j \theta_j G_j + \tfrac{1}{4}\gamma_{\mathrm{p}} \sum_j \|A_j\|^2 \geq E_{\mathrm{pay}}.$$

Use the same real branch for the Wilson action and the witnesses, and the same stored coordinates for the gauge costs and the kept quadratics. The part of the exponent involving these three weights satisfies

$$e^{-S_{\mathrm{W}} - \sum_j \theta_j G_j - \frac{1}{2}\sum_j \operatorname{Re} q_j} \leq e^{-E_{\mathrm{pay}}} e^{-\frac{1}{4}\gamma_{\mathrm{p}} \sum_j \|A_j\|^2}.$$

The bound is pointwise before taking the supremum over admitted evaluations. It leaves $\beta = \gamma_{\mathrm{p}}/4$ in the reference Gaussian majorant above. The normalized shell measure needs no further scalar factor. The old island factor uses its pair-uniform Borel majorant at the admitted inner core. Any reference-law denominator belongs with the corresponding Wilson product and restoration measure.

For the application to an actual later term, use its current background at each factor. On the full carrier of a retained strip, the original-domain comparison gives the coercive bound using the original masks and the two adjacent layer radii. A carried determinant uses the full two blocks of the original elimination row and the small-curvature bound there. Keep this determinant separate from the inverse-density factor of a raw row change. The restriction to each inner island's complete core gives the argument for its majorant.

For the layer estimate at a point of a birth carrier, separate the datum plaquettes of the current background according to their complete test footprints. The retained ones with the prepared-datum support bound have defect at most Ce_q at level q. Use the original positive density bounds on the unchanged exterior tests. Where these tests cover a weighted neighborhood of radius D about the carrier, the profile eq. 3.ccclxxviii–eq. 3.ccclxxix gives

$$Ce_s + C\varepsilon_\star e^{-\delta D}$$

in actual layer s, with $\varepsilon_\star$ the fixed admission ceiling on the rest of the datum. The first term follows by the exponential decay across separated layers and the frame bound on e_q/e_s. When a waiting collar separates the carrier from those other tests, its weighted width of order $M\check{R}$ absorbs the second term. At carriers reaching the exterior halo we must include the positive tests of that halo in the read.

Assign the tests of the current carrying datum before taking any marginal. At an unchanged ordinary place use its original density cube and the full lower blocks for each completed bond. A

passed test supplies a local configuration with the stated averages and a curvature bound. The averaging estimate then bounds the holonomy of the datum plaquette. For a plaquette in a stopped component use its recorded departure place and the full union of its corner blocks. When this union lies in the tested inset, use the last applicable stage guard. On a collar departure include the approximate test on the collar links. A footprint extending outside that inset uses the original density cube on its complete exterior read. The ordinary defect bound is of order ε_q. For a remade shell it has the factor $C_L L_\flat^{2(q-k_0-1)_+}$, and a collar departure has the factor $c_\mathrm{f} L_\flat^{2(q-k_0)_+}$, with k_0 the cutoff between the uncompressed and compressed stages. These powers are bounded by $C(\log \theta_q)^r$ by the frame and the stage widths.

Carry the resulting holonomy bound through the actual later moves. A gauge draw, a frame change or a tree gauge conjugates the holonomy. A late Haar refresh leaves the old determining datum in the inner slot. The fresh field is used by the outgoing rows. A reference-law draw changes the datum itself. At an island its reference has the exterior bounds of the passed extendibility test and of the surrounding density cubes. On the restored box, comparing the four links with the reference and using the retained cutoff gives the bound $C(M)\varepsilon_i$. If a determining plaquette is deleted, the newly made layer uses its own regularity guard. In each case the certificate concerns the data that enter the current background.

In a quiet containing component that heals, the old top test enlarged by four cells covers the outer six cells of the component, since the complement of the top member is inset by two cells. The final positive test on the six-cell inset supplies the interior. Their merger gives the top test of the new domain. The reverse tree gauge preserves the resulting plaquette bounds. The full original localized read of each test is used even where it crosses the two sets of test places.

On a joint inverse branch these tests act on the assembled datum of the current background. Their supports give the input of the weighted profile in its current determining sequence. Restrict that profile to each complete birth carrier before applying the strip, pivot or island bound. In using the upper integral for a later term we must verify the full inverse reads of these tests and the energy inequalities for its recorded event witnesses.

The coordinate count is local. A localized Gaussian marginal with m scalar coordinates has an $\exp(Cm)$ cost from its proved comparison estimate, regardless of the number of exterior auxiliary coordinates used to construct it. Here $m \le C(LM)^4 N_c$, where N_c counts the relevant coarse cubes. If the count already denotes m, the factor $(LM)^4$ is not inserted again. In particular the reference cost does not acquire an outside-volume multiplier.

The analytic and geometric conditions needed for an outer component can now be stated without assuming its desired mass. On every carried support and on every argument in the supremum, require:

(i) the exact exponent identity in eq. 4.cdlxvii, or the evaluated identity in eq. 4.cdlxxi, with the correct retained Wilson weight;
(ii) the complete bound $|V_X| \le C_V N_0$ on the admitted triples, including the actual source and counterterm normalization;
(iii) real admissible reads for the inner factors, whose differences from substituted complex reads have been included in V_X;
(iv) the scalar normalizer estimates just proved and the actual kept-form coercivity and reference-coordinate domination;
(v) a disjoint ownership of the action, gauge and Gaussian pieces used to pay for the event witnesses.

The last requirement has the quantitative meaning

$$A(\upsilon, U) + \sum_j \theta_j G_j(\xi) + \beta_1 \sum_j |A_j|^2 \ge \sum_e \Gamma_e, \qquad \Gamma_e = c_\mathrm{w} \theta_{t(e)} a_e^2, \tag{4.div}$$

leaving the positive reserve β of eq. 4.dii unspent. For ordinary components U is the real inner background; for a drop-out the inequality must hold for every compatible real evaluation in its fibre. A bound only at one preferred background would not justify the supremum in its mass.

These are finite-domain conditions, uniform in the admitted parameters and in the outer index. Each has specific mathematical content. In particular eq. 4.div requires actual witnesses at the stated levels, while eq. 4.dii requires an integral in the actual stored coordinates. We next derive the mass from them; no mass estimate is included in this list of conditions.

4.18.9.7. Which coupling pays for an old event. There is a useful profile check in eq. 4.div. Suppose a plaquette first enters the large-field region at level t. The action rule stops its running weight within $C_R R_t$ further levels. On its owned support the retained weight is a convex combination of inverse couplings with indices no later than $t + C_R R_t$. Earlier couplings are at least $\theta_t - c_5$, and a later coupling in this range is at least $\theta_t - C_6 R_t - c_5$ by the bounded one-step drift. For a sufficiently high floor, $R_t = O(\ell_t^r) = o(\theta_t)$, and hence

$$\upsilon(p) \geq c\theta_t. \tag{4.dv}$$

This argument uses the action supports. Continuing the weight all the way to an arbitrarily later level could make it much smaller, and the same conclusion would not follow from bounded drift alone. The freeze rule is thus a local support statement, not an informal appeal to approximate monotonicity.

4.18.9.8. Elementary witness estimates. For $W \in \mathrm{SU}(2)$ in the operator norm, $1 - \frac{1}{2}\operatorname{tr} W = \frac{1}{2}|W - 1|^2$. A failed kept gauge test $|W - 1| \geq a$ therefore supplies the gain $\theta a^2/2$. A failed physical fluctuation test of size a has $|A^{\mathrm{s}}| \geq \theta^{1/2} a$, and its reserved Gaussian form supplies $c\gamma\theta a^2$.

For a plaquette certificate, suppose the actual tested quantity has the following bound on the failure event, with nonnegative coefficients T_{ep}. Weighted Cauchy–Schwarz then proves

$$\begin{aligned} a \leq \sum_{p \in P_e} T_{ep} |\partial U(p) - 1|, \qquad &\Longrightarrow \\ \sum_{p \in P_e} \upsilon(p)\bigl(1 - \tfrac{1}{2}\operatorname{tr} \partial U(p)\bigr) \geq \frac{a^2}{2\sum_{p \in P_e} T_{ep}^2/\upsilon(p)}&. \end{aligned} \tag{4.dvi}$$

For an ordinary loop identity such a certificate follows by writing the loop as a product of conjugated plaquettes and telescoping the product; all other unitary factors have norm one. For a nonlinear block average the certificate and its uniform weighted row bound must instead be proved for that actual average. A flat linear row estimate alone does not supply it. Similarly, a failed existence or extendibility test needs a quantitative implication saying that all smaller local witnesses would put its data inside the admitted background chart. Its contrapositive then provides a witness. Failure of an unquantified existence assertion is not itself an energy lower bound.

For an outer mass, assign the Wilson part of a mark witness on u to the plaquettes of $C(u)^+$ with no corner in its floor. Use this assignment when the outer Wilson action contains that share. A stopped root can use this halo when it is read in a containing operation. For a mark at level m, $C(u)^+$ adds two layers of *unit cubes of level* m, of total thickness $2L^m$ in bare units. Distinct components at this level are separated by a coarse cell of side $a_m = M\check{R}_m L^m > 4L^m$, so their enlargements are disjoint. For incomparable marks at different levels, follow the earlier enlargement in the ordinary hull to the level of the later mark. It is contained in the earlier mark's successor there. That component and the later mark are separated by a coarse cell of the later level, which exceeds the thickness of the later enlargement.

If a standing mark v of level m lies below a later eligible mark u of floor level h, then $m < h$. The ordinary step into level $m+1$ contains $C(v)^+$ in the one-cell enlargement of the hull of Z_m. If $h = m + 1$, the eight-cell buffer of the ordinary step therefore puts the full h-cell carrier of $C(v)^+$ in the floor. If $h > m + 1$, the neighborhood has already entered Z_{h-1}, whose hull again receives

eight cells before reaching Ω_h. The floor entries on the containing component agree with those of this ordinary step. In both cases the earlier share lies inside the floor. The same argument applies to Wilson paying halos at birth levels $t < h$, which are contained in the ordinary Z_t at birth. For a plaquette or approximate-fluctuation birth at h, the cell lies in P'_h or Q'_h, with the level-zero plaquette births also included in the first step. Its one-cell paying halo leaves at least one coarse cell of the original two-cell buffer to Ω_h. Thus it too is in the floor. A fresh ordinary test outside a live component and its margin has its Wilson share outside the carried halo.

The stored gauge terms are assigned by their original level and contour. Assign the Gaussian payment from the scalar reserve $\frac{1}{4}\gamma_{\mathrm{p}} \sum_b |A_b|^2$ of the complete form, using the actual failed coordinate. Nested mark windows are disjoint by the quarantine at the lower end of the waiting window. A creation at the first level h of a later window uses gauge and fluctuation coordinates of the step from $h-1$ to h, below the coordinates $j \geq h$ used by the mark. Incomparable nodes have disjoint spatial shares in any common layer. Within a one-step share, divide by the bounded multiplicity of its cell witnesses before assigning the gains.

Cut the outer genealogy at its inner islands. Their earlier witnesses have already been used in the inner masses. For a still-standing second-class island Y of level m, the negative density test is on its restored level-m field. Its determining datum includes every top bond meeting Y. The original entries through m on Y survive, and any later eligible window containing this island starts above m. On the full interior carrier of that test, the retained constraints therefore identify the level-m averages of a compatible real evaluation with the restored field.

Choose a density cube C in the six-cell inset of Y where the test fails. The regularity criterion uses C enlarged by four test layers, of side $M_2\check{R}_m$ in level-m units. Completing the averaging blocks of a plaquette on that carrier costs at most one more such unit. Since $4M_2\check{R}_m + 1 < M\check{R}_m$, all these blocks stay in Y. If every level-m plaquette defect on the carrier were less than $c\varepsilon_m$, the local regularity criterion would give the required orbit with regularity radius $B_3 c\varepsilon_m$. Choose $B_3 c < 1$ and the weak-coupling floor so that $c\varepsilon_m \leq a_1$ and $B_3 c\varepsilon_m \leq \varepsilon_{\mathrm{v}} \leq a_0$, with the torus condition of the regularity criterion. The curvature on the tested inset is then strictly within its threshold. Thus a failure, including undefinedness, gives a plaquette defect of size at least $c\varepsilon_m$. The four-block certificate eq. 4.cdlvii gives bare Wilson energy at least $c'\varepsilon_m^2$ in Y. The action gives this region weight x_m at birth and includes a still-standing second-class Y in the level-m action region. The core taper is one on the whole old-core carrier just located. Use the lower stopped-weight comparison eq. 4.dv there and $x_m \asymp \theta_m$. This gives $c_0\theta_m\varepsilon_m^2 = c_0 P_m^2$ in the carrying Wilson action.

The inner core Wilson action is on its prepared inverse field. These are distinct action factors in the exponent identity. If the outer ledger counts the island as b creation cells and also as a mark, both credits come from the fresh-field witness. Here b is uniformly bounded. From an energy $c_0 P_m^2$ reserve $a_\sharp b P_m^2$ for the creation cells, with $a_\sharp b \leq c_0/2$, and use $P_m \leq c_0 P_m^2/2$ for the mark at weak coupling.

For the untouched one-step pieces, the tests and the averaging constraints give the witness estimates for every compatible real evaluation. Each created cell has a witness within two cells, and a plaquette, gauge factor or fluctuation coordinate is used by at most 5^4 of these witnesses. Divide the available energy by this multiplicity. The plaquette witnesses use the Wilson action on $N_{t,D}$; the approximate fluctuation witnesses use that action and the gauge-fixing energy; the remaining fluctuation, collar and rim witnesses use the scalar Gaussian reserve at their actual failed slots. The threshold comparisons therefore give

$$\frac{1}{2}A_{\mathrm{cr}} + \frac{1}{2}\sum_{(t,D)} \theta_{t-1} G_{t,D} + \frac{\gamma_{\mathrm{p}}}{4} \sum_{(t,D)} \sum_{b \in \mathfrak{X}^{\mathrm{s}}_{t,D}} |\mathcal{A}^{\mathrm{s}}_{t-1}(b)|^2 \;\geq\; c \sum_{(t,\Delta)\in \mathrm{Cr}} P_t^2 .$$

Here $\mathfrak{X}^{\mathrm{s}}_{t,D}$ is the stored strip mask of the piece and A_{cr} is the Wilson sum over the disjoint shares. The sums run over the selected untouched pieces, Cr is their set of created cells, and the actual kept-form coercivity and coupling comparisons are those used in the one-step witness estimates.

The unused halves of the Wilson, gauge and Gaussian energies remain available for the reference integral.

The smallest stated mark witnesses can have size $cR_t^{-4}\min_{j\text{ in window}}\delta'_j$. With $\delta'_j \asymp g_j\ell_j^{p_1}$ and the bounded comparison of reference couplings over a window of $O(R_t)$ levels, this gives

$$\Gamma_e \geq c\ell_t^{2p_1-8r} \quad \text{for a mark witness.} \tag{4.dvii}$$

The comparison uses bounded drift over that window, not over the entire history. Assume $2p_1-8r > p$ and $p > 6r$. Then this gain exceeds both a fixed multiple of the later cell costs and $2P_t+1$, after the coupling floor is raised. Creation witnesses of size cg_tP_t have gains bounded below by cP_t^2 and obey the same conclusion. For example the two strict exponent inequalities are compatible with the other hierarchy $p_1 \leq p-4r-1$ once p is sufficiently large. This arithmetic does not establish the witness certificates or their ownership; it proves what those certificates would pay for.

The root witness of a drop-out needs a separate comment. Its certificate can use X^+, whereas that drop-out's own evaluation has a field only on X. Its root gain cannot then be claimed in its own mass. A later containing component can use the gain if its record includes the failed last test and its compatible field supplies the witness on the full paying carrier in X^+. Restricting the earlier mass to this additional test can only decrease it. Comparing the old creation reserve gives the sharper mass bound below without an additional root gain.

4.18.10. *The pointwise envelope and the mass induction.* The reference-coordinate maps used here are fixed by the stored record. They may depend on other stored variables, including inner variables, but not on a parameter still inside the supremum. An argument-dependent translation that makes each separate Gaussian integral small would not bound the integral of the supremum.

Under the conditions stated above, split the positive outer action, gauge and Gaussian energy into two parts. Use the first to pay $\sum_e \Gamma_e$ in eq. 4.div. Use the second to construct the nonnegative Borel reference envelope of eq. 4.dii, or use eq. 4.diii for the completed operation with unmarked core. Bound the scalar normalizers and the complete local energy by $e^{C_7V^\circ}$. For every inner operation replace its actual real admissible read pointwise by its strong real majorant in eq. 4.cdlxxiii. The result is

$$\boxed{m_\omega(\xi) \leq h^{-f(\omega^\circ)}e^{C_7V^\circ-\sum_e\Gamma_e}g_\omega(\xi^\circ;\xi_{\mathrm{in}})\prod_y \frac{m^{\mathbb{R}}_{\omega'_y}(\xi'_y)}{h}, \quad \sup_{\xi_{\mathrm{in}}}\int g_\omega\, d\mu^\circ \leq e^{C_5V^\circ}.} \tag{4.dviii}$$

The same formula is required over the compatible evaluation fibre in a drop-out. The envelope is constructed before that supremum: every evaluation satisfies the same energy certificates, and every complete free form satisfies the same coercivity bound. Its stored coordinates, reference measure and normalizers have been identified as just specified. These facts, not a pointwise gas expansion at each evaluation, prove eq. 4.dviii.

The displayed uniform conditional integral is enough even if the envelope depends on inner variables. When the actual reference construction makes it a function of outer variables alone, this condition follows immediately. In the more general case it follows from the uniform triangular calculation in eq. 4.dii. Its constant must be independent of all inner histories and all their variables. An estimate only after averaging some other exterior law would not be the estimate in eq. 4.dviii.

We prove explicitly the induction step. Fix the outer index ω°. Its inner islands, their levels and their domains are fixed. The remaining histories are the independent local records ω'_y. The retained operations have the corresponding product reference measures, as in eq. 4.cdlxv, and $v^\natural(\omega) = V^\circ(\omega^\circ) + \sum_y v^\natural(\omega'_y)$ exactly. Suppose the earlier strong real masses with coefficient c are at most one. Multiply eq. 4.dviii by the weight and use its Borel envelope. Tonelli allows the outer variables to be integrated first. The bound on that integral is uniform in the inner variables. The remaining nonnegative integrals and sums separate into the earlier masses. Each is at most one.

We obtain

$$\frac{1}{h}\sum_{\omega \text{ over } \omega^\circ} e^{cv^\natural(\omega)} \int^* m_\omega \, d\mu_\omega \le h^{-1-f(\omega^\circ)} e^{(C_5+C_7+c)V^\circ - \sum_e \Gamma_e}. \tag{4.dix}$$

No measurable choice of a maximizing pair or evaluation has entered this calculation. No interchange of supremum and integral was made. The earlier masses include their own frame factors and their own histories, exactly once.

The scalar reserve is applied with $c_1 = C_5 + C_7 + c + c_{\mathrm{e}}$. The counting estimate applies to the outer tree cut at its inner islands. Its cut leaves have their bounded number of boundary seeds. Their last tests lie outside the inner masses and supply the assigned creation gains. Under the witness hierarchy in eq. 4.dvii, or directly under the strengthened scalar conditions, the coefficient of each event can pay its allocated part of $c_1 V^\circ$ and leave the specified reserve. In particular it is sufficient to have

$$\sum_e \Gamma_e - c_1 V^\circ \ge \sum_{e \text{ creation}} (2P_{t(e)} + 1). \tag{4.dx}$$

For the stronger event-by-event form one leaves $2P_{t(e)} + 1$ also at every mark whose witness is available on this outer tree. This stronger form follows from the actual witness powers, not from treating a conservative credit of size P_t as if it were a larger gain. The weaker form in eq. 4.dx already suffices for the mass.

Here is the exact comparison that removes a possible constant loss. For $t \le i$ the bounded-error monotonicity of the reference flow gives, by the mean value theorem,

$$P_t \ge P_i - C_8 \frac{\ell_i^{p-1}}{\theta_i}.$$

The error tends to zero. Raise the common floor so that it is at most $1/4$. The outer tree has a creation leaf, and its reserved $2P_t + 1$ is then at least $2P_i$. Thus, for any fixed $0 < a_\star \le 2$, eq. 4.dix–eq. 4.dx and eq. 4.cdxcvii yield

$$\boxed{\mathsf{M}_c(i, X; v^\natural) \le e^{-a_\star p_0(g_i)}.} \tag{4.dxi}$$

This argument proves the same constants at the new level; it does not produce a factor e at every step. It proves the complex-pair mass from the previous strong real masses. Restricting the pair supremum to the real core recovers that induction hypothesis at the new level. At the first level there are no earlier islands, so the same proof starts with an empty product equal to one.

For a drop-out the same proof holds with every complete form in its evaluated fibre and with its retained Wilson weight. The version including its own failed last test is a restriction and has no larger mass. The reserve argument with the extra unit per old creation also gives the exact e^{-2P_i} bound when that value is desired. Its proof does not use an unavailable root witness on the smaller domain X.

4.18.10.1. The induction has a stronger input than stability. For emphasis, the induction statement is

$$\frac{1}{h}\sum_{\omega} e^{cv^\natural(\omega)} \int^* \sup_{\text{admitted real reads}} e^{\Phi_\omega} \, d\mu_\omega \le 1, \tag{4.dxii}$$

not a bound on an integral at each fixed read. It is also not a bound on a final Gibbs partition function. The real statement is propagated by the pointwise structural envelope, while its complex counterpart is an output of the same step. The proof uses the strong norm exactly where the inner histories are separated in eq. 4.dix. Stating this quantifier prevents an otherwise hidden strengthening in the induction.

4.18.11. *Perturbations paid for by the stored-site reserve.* The same volume weight pays for an explicitly bounded source excess. Suppose, on the full domain used in the supremum,

$$|\Psi_\omega| \le \bar f\, n^\sharp(\omega). \tag{4.dxiii}$$

If $c\ell_j \ge 2\bar f$ for all relevant levels, eq. 4.cdlxxviii gives $|\Psi_\omega| \le c v^\sharp(\omega)$. Therefore, for a bounded insertion F,

$$\frac{1}{h}\sum_\omega \int^* \sup\bigl|F e^{\Phi_\omega+\Psi_\omega}\bigr|\, d\mu_\omega \le e^{-a_\star p_0(g_i)} \sup|F|. \tag{4.dxiv}$$

This is pointwise domination by the already weighted mass. It requires the same normalizers, reference measures, test supports and argument domains on both sides. A source that moves a sharp test or changes the reference measure must first be rewritten in that fixed representation; it is not covered by eq. 4.dxiii by name.

The weight may also be changed. If $0 \le v' \le v^\sharp$, monotonicity gives the mass bound with v'. If an owned window volume satisfies $v^{\mathrm{win}} \le 2v^\sharp$, use the original mass with coefficient $2c$ to bound its weight with coefficient c. This comparison must have a fixed multiplicity. The formula in eq. 4.cdlxxx explains why there is no analogous statement for arbitrary repeated-past volumes.

One can verify eq. 4.dxiii from local carrier information. The following criterion records the constant terms as well as the oscillations. At level j, let $\mathcal{A}_j(\omega)$ be its owned stored sites, with the fixed read collars assigned as in eq. 4.cdlxxviii. Let $\psi_f(z;\cdot)$ have carrier R_f of that level, and let a reference argument p_f^0 in its admitted carrier core be fixed. Suppose

$$|\psi_f(z;p_f^0)| + \sup_{p\ \text{admitted}} |\psi_f(z;p)-\psi_f(z;p_f^0)| \le b_f(z), \qquad \sup_{x\in\mathcal{A}_j}\sum_{f:x\in R_f}\frac{b_f(z)}{|R_f|} \le \bar f. \tag{4.dxv}$$

Every carrier here has a nonempty owned site set R_f; using a coarser root convention instead changes the displayed density bound by its proved local counting constant. Sum first over the sites of R_f, then over the carriers. Absolute convergence or finite sums gives $\sum_f \sup|\psi_f| \le \bar f \sum_j |\mathcal{A}_j|$. The total source excess consequently satisfies eq. 4.dxiii, with its fixed dimensional factors included in $\bar f$.

For a usual rooted decay criterion, assume each carrier has a chosen owned root and tree length $d(R)$, and its total rooted weight is at most $Be^{-\kappa d(R)}$. Assume, at any fixed owned site x, that at most $C_0C_1^m$ carrier/root choices of length m contribute with $x \in R_f$, and that $\kappa > \log C_1$. This is an incidence count at x, not merely a count of carriers whose original root was x. Then the per-site sum is bounded by $BC_0/(1-C_1e^{-\kappa})$. A polynomial root multiplicity is absorbed by a strictly smaller retained exponent: for $\varepsilon > 0$, $m^q e^{-\varepsilon m}$ has a finite maximum, obtained by differentiating $t^q e^{-\varepsilon t}$. This proves the carrier criterion with explicit rate expenditure. It does not verify the amplitudes of the actual formed members.

The constant term in eq. 4.dxv cannot be omitted. A function constant in its field arguments has zero oscillation and can have arbitrarily large parameter-dependent value. In the actual source class its value is fixed by the vacuum, scalar-action and normalizer readouts. Those readouts must be bounded along with the oscillation. Gauge invariance does not bound an arbitrary constant.

For the actual recursion let V_X^{col} be the interaction before re-evaluating the kept inner factors. Write the exponent of an inner source perturbation as its stored-variable part plus its carrier functions ψ_R. The two systems use the same response map for a kept trunk. Its contribution to the difference is

$$T_R^1[\psi_R] = \psi_R(q(\xi)|_R) + \delta_R[\psi_R].$$

The first term rejoins the stored-variable part of that inner exponent at its admitted real pair. Keep the second term in the new excess. The resulting identity is

$$\begin{aligned}\Psi_\omega &= \Psi_\omega^{\mathrm{new}} + \sum_y \Psi_{\omega'_y}(\xi'_y; \mathbf{q}_y(\xi)),\\ \Psi_\omega^{\mathrm{new}} &= A(\zeta''(z)\zeta_1 \mathbf{1}_X, U''(\xi)) + V_X^{\mathrm{col},z} - V_X^{\mathrm{col},0} + \sum_R \delta_R[\psi_R].\end{aligned} \tag{4.dxvi}$$

The reference measures, the bare stored fluctuation slots and the reference quadratic q_X are common to the two systems. The source profile ζ'' multiplies the core cutoff ζ_1. Where that cutoff is nonzero the real inner datum gives curvature $C\check{R}_l^2 \varepsilon_l \eta_l^2$ on layer l. At weak coupling the square of this bound is $O(\eta_l^4)$. The $O(\eta_l^{-4})$ bare plaquettes per site of that layer give a cost $C_W \bar{c} N_0$ when $|\zeta''| \le \bar{c}$. The bare layer uses the uniform Wilson bound.

Apply the formed collar bound to each of $V_X^{\mathrm{col},z}$ and $V_X^{\mathrm{col},0}$. Their absolute bounds include the unpaired plain values at the trivial pair and give $(c_V - 1)(1 + \bar{c})N_0$ for each. This application uses the full common field and stored-slot ranges of the two systems. For $\delta_R[\psi_R]$, subtract the value at $q(\xi)|_R$ and use the same response disc as for the unsourced kept factor. Its bound is the product of the response amplitude and the admitted carrier bound for ψ_R. A term constant in the pair cancels in this difference. The weighted carrier sum for the kept factors therefore applies to these differences with the bound for the source carriers. In particular the resulting sum is at most $2(1+\bar{c})N_0$ once the weak-coupling ceiling is chosen with that carrier bound fixed. The power margin is the one in the kept-trunk estimate, before division by $N_0 \ge 2^{-5} M^4 \ell_t^{5r}$. Hence

$$|\Psi_\omega^{\mathrm{new}}| \le [C_W \bar{c} + 2c_V(1 + \bar{c})]N_0.$$

Add the older complete exponents at their own sites. The additivity of the fresh and older counts takes the maximum of their coefficients. For the source induction we must check the admitted carrier bounds of the actual generated members on the enlarged source lines, including the stored switches. The argument uses these bounds and the full interaction bound on the same domains.

4.18.12. *Complex parameters with the inner Wilson weight fixed.* The preceding proof has a parameter version, but its domain must be specified before taking suprema. Fix the real reference sequence θ and the real physical weight profile x. Freeze all characteristic cutoffs, stored-coordinate charts, reference measures and the inner Wilson weight at those real data. Let the complex parameters y_j satisfy

$$|x_j - \theta_j| \le t_j, \qquad |y_j - x_j| \le \bar{c}, \qquad \tau_j = y_j/\theta_j. \tag{4.dxvii}$$

Here t_j is the radius of the frozen tube at level j. The ordinary strip uses the ratio disc of radius $2(t_j + \bar{c})/\theta_j$ about 1. This radius tends to zero uniformly as the inverse-coupling floor increases. In particular it lies within the half-disc used for the scalar logarithms. The actual ordinary presentation determines where y_j occurs. In particular its complete free-coordinate quadratic form is the one prescribed by eq. 4.d, including a compensation if a normalized $\tau_j K_j$ Gaussian is used. The weight in eq. 4.cdlxix is still the specified real weight x''. Putting a complex weight on a large real Wilson variable and estimating it as if it were the same positive reserve is a different operation.

For a complete matrix on a fixed real free-coordinate space there is a determinant calculation on a half-disc. Under the real hypotheses preceding eq. 4.di, define $B = K^{-1/2} K_A K^{-1/2}$. Since $K_G \ge 0$, one has $B \le I$. Since $K_A \ge -\gamma I$ and $K \ge \gamma I$, one has $B \ge -\gamma K^{-1} \ge -I$. Thus $\|B\| \le 1$ and

$$Q_\tau = K^{1/2}[I + (\tau - 1)B]K^{1/2}, \qquad \mathrm{Re}\, Q_\tau \ge K/2, \qquad \|Q_\tau^{-1}\| \le 2/\gamma. \tag{4.dxviii}$$

The matrix logarithm of $I + (\tau - 1)B$ is its uniformly convergent power series on $|\tau - 1| \le 1/2$. Consequently the determinant and its square root have coherent analytic branches agreeing with the positive branches at one. In d free scalar coordinates, the absolute determinant ratio is between

2^{-d} and $(3/2)^d$. The normalized Gaussian at each fixed real K has total variation at most $3^{d/2}$: its determinant costs $(3/2)^{d/2}$ and replacing the real precision K by $K/2$ costs $2^{d/2}$ on integration.

For a supremum over varying backgrounds, this last fixed-matrix integral is not the argument. If $\gamma I \le K \le \Gamma I$ uniformly on those backgrounds, eq. 4.dxviii and the determinant upper bound instead give the common pointwise envelope $C^d\Gamma^{d/2}e^{-\gamma|b|^2/4}$ in their same stored coordinates. Its integral is at most $(C\Gamma/\gamma)^{d/2}$. Alternatively one uses the proved local comparison with a fixed positive reference matrix. This additional uniformity is precisely what places a parameter supremum inside eq. 4.dii. The classical flat gap is one realized matrix case. A one-step nonflat estimate can verify the same criterion by bounding the negative part of K_A in the actual free coordinates; neither case alone verifies it on every retained layer.

At complex parameters the pair-independent factor of the exponent need no longer be real. The proof uses its modulus, with the real positive energy and the fixed reference envelope just constructed. Suppose the actual local formations and read maps are jointly holomorphic on eq. 4.dxvii, with their cores admitted throughout, and their complete energy bound is $C_V(1+\bar{c})N_0$ there. The scalar gauge branch is supplied by eq. 4.cdxcviii; the pivot branches by eq. 4.cdxcix; the complete Gaussian forms by eq. 4.dxviii where its real matrix hypotheses apply, and by the original-strip estimate for the kept chains on their admitted reads. All their costs are then bounded in modulus by the same type of e^{CV° factor.

Apply the mass proof to the norm with the parameter supremum included from the start. Its real part refers to real *field reads*, while the parameters can be complex:

$$m_\omega^{\mathbb{R},\mathcal{D}}(\xi) = \sup_{y\in\mathcal{D}} \sup_{\mathbf{p}\ \text{real admitted}} |e^{\Phi_\omega(\xi;\mathbf{p},y)}|. \tag{4.dxix}$$

The parameter domains at earlier levels are the corresponding prefix domains of the same reference frame and initial weight. This prefix condition is essential for using the earlier norm in eq. 4.dix. It is not a conclusion of separate holomorphic presentations with different frozen frames.

Uniform primitive estimates give eq. 4.dviii with these parameter suprema, so the same induction proves eq. 4.dxi uniformly in y. The floor is raised for the new local-energy and Gaussian cost constants. If the former is $C_V(1+\bar{c})$, its stated dependence is preserved in the floor. This is a proof from uniform finite domination. It is not analytic continuation of an inequality known only on the real parameter slice.

The positivity used to pay for old failed plaquettes is exactly why the inner Wilson weight was frozen. In a variable-profile source presentation, physical Wilson perturbations are instead carried as specified older potential members. The equality of that presentation with an original numerator and denominator is an exact formation identity with its own counterterm rule. A complex parameter in eq. 4.dxvii cannot be declared a physical source in every factor merely because both families agree at zero.

4.18.12.1. Holomorphy of the actual integrals. Let $\mathcal{D}_0$ be a compact parameter polydisc strictly inside the admitted domain. The pointwise integrands are holomorphic there, all reference measures are parameter independent, and the mass proof supplies one integrable Borel majorant on the full product of stored variables. Integrate the one-variable contour identity of each integrand around a triangle in any coordinate; Fubini is justified by that majorant. The integrated function is continuous by dominated convergence and has zero contour integrals. Morera's theorem gives holomorphy in that coordinate. The same bound and the iterated Cauchy formula give joint holomorphy and its power series. Countable sums of histories are handled by the summable common majorant. It is the reference measure and not just the notation $d\mu$ that must be independent of the parameter in this argument.

For a polydisc with radii $r_1, \dots, r_m$ and a bounded holomorphic numerator insertion F, Cauchy's formula gives

$$|\partial_1^{a_1} \cdots \partial_m^{a_m} I(0)| \le (\prod_{\nu=1}^{m} a_\nu! r_\nu^{-a_\nu}) e^{-a_\star P_i} \sup |F|. \tag{4.dxx}$$

This differentiates the complete numerator with its actual normalizers. A normalized expectation additionally differentiates its denominator. It is not obtained by differentiating only the newly inserted local member.

For example, if an actual one-zone perturbation has the uniform bound

$$|\Psi_\omega(z)| \le C|z| p_n n^\sharp(\omega), \qquad |z| < r_n, \qquad r_n \le \frac{c\ell_{\min}}{2Cp_n}, \tag{4.dxxi}$$

and all its fields remain in the same admitted domains, eq. 4.dxiv and eq. 4.dxx prove the corresponding complex mass and derivative estimate. This specifies a concrete formation estimate to test: the coefficient C must be uniform in the histories, stored fields, levels, zones and allowed reads. A bound for one chosen background or a source function before it is formed is not eq. 4.dxxi.

For several perturbations, a per-site criterion can contain $\sum_\nu |z_\nu| p_\nu$ or a smaller local sum, depending on their supports. The admitted product radii must satisfy that actual criterion. A one-source disc alone does not authorize an equally large product disc for arbitrarily many overlapping sources. Disjoint local supports and a proved local carrier bound can supply such a product disc; its justification is the local bound, not one-variable Cauchy.

4.18.13. *Consequences for stored volume, ages and marked integrals.* Fix $c > 0$ at which the mass bound holds. Any subcollection of histories with $v^\natural \ge T$ has unweighted mass at most

$$e^{-a_\star P_i - cT}. \tag{4.dxxii}$$

This follows pointwise from $e^{cv^\natural} \ge e^{cT}$ and monotonicity of the upper integral. More generally the power series of the exponential gives

$$\frac{1}{h} \sum_\omega \int^* (v^\natural(\omega))^q m_\omega \, d\mu_\omega \le q! c^{-q} e^{-a_\star P_i}, \qquad q = 0, 1, \dots. \tag{4.dxxiii}$$

For $q = 0$ use the unweighted mass directly. For $q > 0$ use $(cv)^q/q! \le e^{cv}$. Combining this with eq. 4.cdlxxviii gives the site-count moments with c^{-1} replaced by $2/(c\ell_{\min})$.

A path of length A in the recorded genealogy contains a node at every one of those levels. Since each such node has at least one cell, $v^\natural \ge \ell_{\min} A$. Therefore its recorded age has exponential tails with exponent $c\ell_{\min}$. For any fixed $b > 0$ the age-weighted sum with factor L^{bA} is finite once $c\ell_{\min} > b \log L$, by summing the geometric series from eq. 4.dxxii. This age is the length inside the recorded genealogy. The number of later levels for which an already formed mark is carried is not part of its unchanged local record; that later duration is controlled by the live-component reserve and the global label comparison, not by silently increasing its local volume at every read.

These bounds control stored counts, not every unbounded function of a stored field. Small mass alone would allow an integrand concentrated at field magnitude T with a second moment of order T^2 times that mass. Field moments need the Gaussian reserve or their own pullback estimate. If the pointwise proof retains $e^{-\beta|A|^2}$, then

$$|A_b|^q e^{-\beta|A|^2} \le \left(\frac{q}{e\beta}\right)^{q/2} e^{-\beta|A|^2/2} \le \beta^{-q/2} \sqrt{q!}\, e^{-\beta|A|^2/2}. \tag{4.dxxiv}$$

The maximum of $t^q e^{-\beta t^2/2}$ occurs at $t^2 = q/\beta$; the factorial estimate follows from $q^q \le e^q q!$. Run the same reference calculation with $\beta/2$. Its extra factor $2^{d/2}$ is another local volume cost, independent of q. With the floor chosen for that unspent reserve, the marked mass has the prefactor in eq. 4.dxxiv. For a physical observable the coefficients and the map from its field to A_b must still

be tracked, including their powers of the scale ratio. A small bad-event probability alone does not give this marked estimate.

These consequences also explain what the volume weight can contribute to a large-field extraction remainder. It supplies arbitrary fixed exponential weights in the recorded age after the corresponding floor is chosen. It supplies polynomial stored-coordinate moments when a spare Gaussian fraction is kept. It does not identify a resummed positive conditional measure, its boundary terms, or its exterior force. Those are different estimates in an interacting covariance calculation.

4.18.14. *Formation, reading and loading at later levels.* We make the quantifiers of transfer explicit. A factor is formed at level i, with a definite local history, stored-coordinate space, support and normalizer. For each later level $k \geq i$ at which its mark stands in a term, let $\mathrm{Read}_{k,i,X}$ denote the actual argument supplied to that same factor. The coverage assertion needed is

$$\begin{cases} \xi_X \in S_{\omega_X}, \quad \mathrm{Read}_{k,i,X} \in \mathcal{U}_i(X^{\sim}) \text{ and is real}, & o = 2, \\ \mathrm{Read}_{k,i,X} \in \mathcal{K}^{\sharp}(\omega_X, \xi_X), & o = \mathsf{d}, \end{cases} \tag{4.dxxv}$$

for every admissible later history and every configuration on its carried support. It is a pointwise statement, before taking an exterior expectation. In the drop-out case it includes every average in eq. 4.cdlxx, every recorded test, the prescribed variant of the last test, the inner-island cores and all one-sided matrix and Jacobian bounds. In the island case its reality is what allows the strong real induction in the new formation.

The finite geometric part can be checked as an invariant of the construction. A later operation outside the component does not change its stored variables. An operation containing it acts above the levels protected by its waiting floor, and its determining entries use the actual re-entry field described after eq. 4.cdlxiv. The action entries keep the tests with their recorded ownership. Restricting a history before that later operation consequently retains the same local history and the same operation. This proves the record statement under the specified local separation and window rules. Admission of the resulting background to the analytic core, and the actual matrix bounds in eq. 4.dxxv, are the remaining analytic part of coverage; the invariant does not prove them by itself.

4.18.14.1. A stop at the prospective regularity test. With n stages completed, let $j = i_\flat + n$. A stop before the collar gauge and the removal keeps the ordinary prefix of index n. Where the preceding guard is already present, it remains a passing test of this prefix. For a datum plaquette of level q, take the union of the endpoint blocks at that level. Read the ordinary density test at its original cube if $q > j$, if the top action region there is untouched, or if this footprint leaves the region covered by the preceding stage test. On the covered footprints use the passing guard $\chi_i^{(n)}$. An untouched prefix reads its original density tests throughout. The old core keeps its own tests and the averaging constraints remain in the original rows.

The ordinary-prefix real admission applies with these tests and with the old determining list. It gives the layer profile of the old background, and the half-radius bounds on the corresponding earlier spaces, including those of carried drop-outs. This uses the tests of the prefix without the newly failed test. The stage has added neither a gauge draw nor a pivot elimination. Its one-step factors remain in their birth coordinates and there is no normalizer from the interrupted stage. For a later evaluation of this stopped fibre, the full footprints of these tests and the same original pivot and walk carriers must occur in the compatible read.

4.18.14.2. The datum at an early prepared seam. A complete test footprint can meet both sides of a prepared face, even when the bare guard uses plaquettes contained in its individual shells. Use the coarse datum at the preceding averaging fibre to control such a footprint. Consider promotion to level $q = j + 1$ in the uncompressed range $q \leq k_0$. In the quiet geometry $G_j \subset H_q$, and the new face is the boundary of $H_q^{[3a_q]}$, with $a_q = M\check{R}_q L^q$. Every new top test based on this promotion has a row outside the new inner box. Its complete blocks extend only $O(L^q)$ inward, so the footprint

remains outside G_j. A footprint meeting the original exterior wall of Ω_q is read on the full halo of the original level-q density test, before that test is removed. The other new footprints lie in the old exterior $\Omega_q^c \setminus Z_j''$. At the first promotion their full carrier is in the region $\Omega_{i_\flat} \setminus \Omega_{i_\flat+1}$ of the stage-zero test. At later promotions the preceding exterior guard reads the whole carrier. The local preceding background reproduces the old determining level-j rows there. Thus the guard bounds the curvature of that background by $C\varepsilon_j L^{-2j}$ on the full carrier. The new coarse rows are its averages by the original constraints. The plaquette estimate gives $C_L\varepsilon_j \le C'_L e_q$, including a plaquette that straddles the new prepared face. The surviving tests of level j use a subset of the previous rows and the same lower completions. The lower tests are unchanged. If level $q+1$ is present, each tested plaquette there has a first-kind edge meeting the original Ω_{q+1}. Its whole carrier is within $O(L^{q+1})$ of that member. The quiet gap of $8a_{q+1}$ keeps this carrier inside the original Ω_q. Its completion therefore uses the same rows as before. Higher tests read neither the changed member nor its immediate completions. Each promotion uses its own positive tests and a single averaging bound. Carry the new coarse holonomy bound as a support condition before marginalizing the old fibre.

4.18.14.3. The datum at a compressed prepared seam. For $q = j+1 > k_0$, take a level-q test whose complete footprint meets $\partial Z_q''$. The footprint extends at most $2L^q$ from this face. The separation from Z_j'' is at least ML^q. The face is at least $5ML^i$ from the old Ω_{k_0+1}, and the original exterior has its clearance from ∂X. Thus the entire footprint lies in

$$X \cap (\Omega_{k_0+1}^c \setminus Z_j''),$$

in the layer of index j before promotion. At the first compressed transition $j = k_0$ this is the single exterior region of the preceding guard. At the other transitions it is its deep exterior region. In both cases its bound is at most $F_j\varepsilon_j L^{-2j}$, with $F_j = L_\flat^{2(j-k_0-1)_+}$.

The local preceding background has the actual old level-j datum on this footprint. Every first-kind link of the new test is promoted there, and every second-kind link uses the same retained level-j blocks. Their common coarse values are exactly the one-step averages. Hence the test holonomy is the level-q plaquette of the averages of this local preceding background. The guard on the full bare carrier and the averaging estimate give a defect at most $C_L F_j \varepsilon_j$. Since $L_\flat < L$ and $L^{i-k_0-1} = \check{R}_{k_0+1}$, this is at most $C_L e_q$. The preceding guard is available before deletion, so the bound can be carried in the coarse support after the integration. This also applies to the last prepared face before the final collar integration.

At a new level-q footprint away from the prepared face, use the same calculation when it is contained in a single preceding exterior region. The factors in those clauses are at most F_j. A remaining new footprint meets an original face $\partial\Omega_l$, with $k_0+1 \le l \le q$. Consecutive original walls of levels $l, l+1$ are separated by at least $8a_{l+1}$, so the footprint meets only one wall level. Use the density test at that face before it is removed.

Put $u = ML^i$ in bare units. Start with a tube of width $3u/2$ about the original face of level l. Its level-l plaquettes are covered by the original density test with its full halo. Before deletion this gives $|\partial V_l - \mathbf{1}| \le C\varepsilon_l$ there. For each subsequent promotion to level $p > l$ reduce the width by $4L^p$ to keep complete backward averaging carriers. The total width lost is less than $8L^i$, so more than u remains. These tubes stay clear of every prepared inner face and every other original face. In a later promotion at level p the tube is outside the original Ω_p and inside the preceding prepared member. Thus its level-p field is the exact average of the retained level-$(p-1)$ field there.

On the joint graph before eliminating these averaging constraints, the level-q field on the remaining tube is the iterated average of the same V_l. Any frame changes act at the block endpoints and preserve its plaquette defect. Apply the full-carrier estimate of Prop. 3.4 once at depth $q-l$. This gives a bound

$$CL^{2(q-l)}\varepsilon_l \le C'\check{R}_q^2\varepsilon_q.$$

The inequality uses $L^{i-k_0-1} = \check{R}_{k_0+1}$ and the frame comparison within the compressed interval. At weak coupling the coarsest scaled input is inside the averaging radius. This bound is a condition on the coarse holonomies in the tube. Insert it on the joint graph and carry it through the exact pushforwards. When $q = l$ the bound follows directly from the original positive density test. In this way the quantitative datum bound uses the original density test at its actual level of formation.

At a final regularity stop, $n = \mathsf{N}_i - 1$. If the broader waiting convention admits the component at the lower endpoint of a later window, it is a partial re-entry. Its level-i density test uses its original action region at that level, which agrees there with the determining region. The top field is the old datum on Γ_i. For an unprescribed level-i bond c with both ends in Ω_i^c, the surviving one-step delta gives

$$V_i(c) = \mathcal{M}(V_{i-1})(c).$$

On a protected footprint containing the two full blocks of c, the later read has the same level-$(i-1)$ rows. Composition of the averages therefore gives this same value of $V_i(c)$ in the later read. The ordinary density tests and the local stage guard are evaluated from the resulting old datum on their own footprints. Deeper remade levels still use their separate determining lists. For a strip operator this argument applies when the later structured address also contains its complete birth carriers. The footprint needed for an energy witness can be smaller than that operator address.

On a fixed incoming prefix and outgoing field, the old and inset versions of the early stop have the same fibre after omitting the root's negative indicator in its own mass. The gauge factors, averaging graphs, retained measure and weight are those of the common prefix. A later read with the new failure imposed restricts this common fibre. Any absolute reference bound on the root-omitted fibre thus also bounds the restricted read. The uniform joint-core bound on the common fibre and the fresh payment for the new failure are separate estimates.

4.18.14.4. A stop between removal and collar integration. Consider a component X of the operation of level i with n completed collar integrations, and put $j = i_\flat + n$. Stage $n+1$ first imposes the localized regularity test $\chi_i^{(n+1)}$, the collar gauge and the approximate closeness test χ'_j at $2\delta'_j$. It completes the fluctuation tests of step $j+1$. The removal uses the retained averaging deltas and the density tests at the top and the old core, together with the lower and higher one-step tests left in the term. The sharper collar radius $CL\delta'_j < \delta_j/2$ supplies the fluctuation implications on the covered exterior bonds. The completed ordinary tests supply the bounds on inherited strip bonds. For the regularity implication the error on a shell of level $l \le j$ is bounded by $C\delta'_j e^{-c(M/M_1)(j-l)}$ in its curvature units, which fits the gap $c\beta_{\mathrm{s}}2^{-(j-l)}\varepsilon_l$ in the ladder. For the missing crossing reads use the comparison of §4.17.1.25. Remove an old indicator only when it equals one on the complete chart used for its removal. Any retained active indicator is part of the real collar integral. The sharp collar split at δ'_j follows the removal.

At the collar gauge insertion, the group is pinned at the next coarse sites, so the coarse row remains fixed. Use Prop. 3.23 on the incoming law before selecting the subsequent gauge-dependent outcomes. The group is supported beyond the reach of the old retained factors. For the prospective regularity test use covariance of the critical orbit. This gives the invariance needed to use the soft gauge factor against the incoming law at this insertion. Its support is in complete blocks inside the component's exterior buffer. A link moved across the prepared inner face belongs to the collar, while the exterior of the component is fixed. The later tests are weights on the resulting variables.

In the row chart this can also be seen directly. At a site x with contour $V_j(y,x)$ from its coarse centre, the draw of a new contour value q_x uses $u(x) = q_x^{-1}V_j(y,x)$. Its input contour lies in a single block and reads no corridor pivot. At fixed nonpivots and gauge draw, each selected pivot changes by fixed group translations preserving Haar, and the averaging row has its original target. Carry the resulting contour values in the branch kernel for the tests that read them. A prescribed contour value would be a graph relation with the gauge draw.

When the component leaves at that split, the retained determining list still has index n. Its tests include the stage-$(n+1)$ regularity test and the approximate collar test at the old level j, together with the failing sharp test. In the local sharp convention this test is formed at the localized next-stage background, so its failure is a condition on the variables of this component. In the global sharp convention use the exact transported test and density of the failed branch. In the variant of the fibre that omits the root's failing test the positive tests are still required. The same list supplies the passing branch up to integration. For comparison one may form the prospective ordinary stage-$(n+1)$ datum with the same variables and the retained averaging constraints. The two determining sequences differ only at $j+1$, by the collar. The prospective read introduces no additional stored variables.

For a datum plaquette use the smallest-level drop-out containing its place. Take its full footprint, the union of the endpoint blocks at its test level. An ordinary density test is read at its original action cube if the footprint leaves the region of the stage test, if its level exceeds $j+1$, or if it belongs to the untouched top action region. The old core also keeps its own tests. Within the covered part use $\chi_i^{(n+1)}$, supplemented by χ'_j for a level-j plaquette touching the collar. The density cube and the footprint are taken in the prescribed partitions of the test. The current mask of an old exponent does not set the place of this guard.

On footprints covered by the literal clauses, the prospective guard gives the curvature bounds used in the block averages. At a prepared or original wall use the coarse holonomy bound from the preceding promotion calculation. The prospective datum has the layer bounds Ce_q on all these reads. The unchanged older rows use the bounds retained in their own prefix. For the old datum on the collar, where the complete double block is in this comparison, the retained deltas give the pivots from the nonpivot values and the prospective coarse row values. The pivot comparison at equal coarse values costs a fixed factor depending on L. Thus the approximate test changes the old level-j holonomy bound by at most $C_L\delta'_j$, by telescoping the plaquette product. The changed lower rows leave their prescribed coarse values fixed. In particular the corresponding level-$(j+1)$ holonomies agree with their prospective values, also when an old edge reads the complete one-step average. For the no-inward determining sets the global and localized next-stage averages differ on the collar by at most

$$C\mathsf{F}e_i e^{-\delta_0 M\check{R}_i} \le \delta'_j/2.$$

Apply this comparison on the complete collar blocks inside the quiet component. The collar lies in O_{j+1}. If $j+1<i$ this is inside G_{i-1}, whose clearance from ∂X is $10a_i$. At $j+1=i$ the clearance of O_i is $2a_i$. The endpoint blocks and the original small test-cube halos together reach less than a_i past this carrier by the cell-size hierarchy. This gives the weighted margin of order $M\check{R}_i$ in the comparison. On a passing sharp branch the two centre comparisons fit $2\delta'_j$. At a sharp stop the retained local approximate test gives $2\delta'_j$ in local coordinates, hence at most $3\delta'_j$ in global coordinates. Use the transported density for the global sharp stop.

For the energy estimates at the approximate and sharp collar stops, consider a bond crossing the old Ω_{j+1} wall. Let y_- and y_+ be the centres of its exterior and interior blocks. The interior block is in the first ordinary strip cell. Both the passed and the failed ordinary cube masks start at least one full cell inside the wall, and their enlarged reads reach at most $2LM_2\check{R}_{j+1}+1$ in level-j units. The complete interior block, including its internal bonds, therefore lies in the plain zone. Its internal bonds are nonpivots and are kept by the original ordinary split. On them the exact plain-zone map gives

$$V_j(e) = \exp\bigl(ig_j\mathcal{A}_j^{\mathrm{s}}(e)\bigr)S_{\mathrm{o,loc}}(e),$$

where $S_{\mathrm{o,loc}}$ is the original localized strip centre. The full block lies in the joined interior, and its fine sites other than y_+ are free for the ordinary axial gauge. Hence the mean contours of $S_{\mathrm{o,loc}}$ from y_+ are the identity. If the kept Gaussian reserve gives $\max_e|g_j\mathcal{A}_j^{\mathrm{s}}(e)|\le\sigma$ on these links, telescoping the contour products and using the derivative bound of the mean gives $\max_{x\in B(y_+)}|V_j(y_+,x)-\mathbf{1}|\le C_L\sigma$.

At the exterior block the retained soft gauge weight gives the same bound if its gauge exponent, divided by its inverse coupling, is at most σ^2.

On the complete double block, the curvature bounds below give defects of size $C\sigma$ for V_j and the prospective local centre. Axialize V_j with the two mean contours, fixing the two block centres. Its coarse row is still $V_{j+1}(\langle y_-, y_+\rangle)$, the target of the original ordinary pivot constraint. The prospective centre is axial on both blocks and has that same target. Double-block rigidity compares each axialized crossing to the common coarse value with error $C_L\sigma$. Undoing the auxiliary gauge costs another $C_L\sigma$, by the two contour estimates. Thus the principal angle on this crossing is bounded by $C'_L\sigma$ in the local chart. On a full history the global angle is at most this bound plus $C\sup_b|H(b)|$, where H is the stage centre shift. Consequently either failure indicator on the full history is bounded above by the local indicator that some tested angle is at least $\delta'_j/2$. Use this upper bound in a local mass, where the evaluation need only give the localized centre. The labelled integral still uses the actual sharp test. The crossing estimate uses the stored ordinary coordinates on the interior block even when the guard itself reads only the crossing.

Write $q = j+1$, and take the one or two full order-q blocks of a tested collar bond together with the backward carriers for their averages. The bond has an endpoint in $(\Omega_q^c \setminus Z''_q)\cap X$. The blocks and their carriers reach only $O(L^q)$ past this set. The gap from Z''_q to Z''_j is at least ML^q, also in the compressed nest. At the first stage use the gap from $G_{i_\flat}$ to the first prepared face and $\Omega^c_{i_\flat}\cap X \subset G_{i_\flat}$. Thus all these carriers have prospective layer at least j. The clearance of O_q from ∂X is at least $2a_i$. After the $O(L^q)$ enlargement this leaves the carriers in the unchanged interior of the join. In particular the old Ω_i wall at the last stage is separated from the outer tower collar.

Every tested datum plaquette whose backward carrier touches the floor is below level $i_\flat$, and its blocks are in the protected core. To reach it from a sharp carrier one must cross the band from $G_{i_\flat}$ to the first prepared face. That band has thickness of order $a_{i_\flat+1}$ in layers at most $i_\flat$, which gives weighted distance at least $cM\check{R}_i$ even at a later compressed stage. The artificial tower collar at ∂X is also this far in the weighted metric. The datum tests inside this weighted neighborhood use the exact old or promoted averages on their complete carriers. Their bare Wilson shares are in the charged enlargement and outside the floor, so small energy gives defect $C\sigma$ there. Use the real admission bound $C\check{R}_i^2\varepsilon_i$ on the other tests. The weighted regularity estimate now gives bare defect

$$C\big(\sigma + \check{R}_i^2\varepsilon_i e^{-cM\check{R}_i}\big)L^{-2j}$$

on the sharp carriers. The averaging estimate gives the same bound without L^{-2j} for the prospective level-j field. The exponential term and the stage centre shift are $o(\sigma)$ for $\sigma = c_*\delta'_j\check{R}_i^{-4}$, uniformly in the waiting window.

For the actual V_j at an old wall crossing, the interior free links belong to the old ordinary fibre. Use the original localized strip centre on the component of $Z_q\cap\Lambda_j$ in X, with the one-part join on Z_1 at the first step. The strip admission uses the original density bounds and exact averaging rows on this domain. Reproduce those rows and the density tests from the retained birth data in the compatible evaluation. This admits the joined centre at the fixed ordinary radius without requiring an extension of the evaluation to the torus.

A tube of width ca_q about ∂O_q lies deep in this join, stays outside G_j, and misses the next original wall. Its layers are j and q. The level-j datum reads on its exterior are old determining values, and the level-q reads are the old coarse targets and the exact averages of the exterior rows. Their full backward carriers lie in the charged enlargement and avoid the floor. The datum away from the tube has the ordinary admission bound, and leaving the tube costs weighted distance $cM\check{R}_q$. The weighted estimate therefore gives level-j plaquette defects of size $C\sigma$ for the localized strip centre on the full double block.

On the interior and crossing nonpivots the exact plain-zone formula and the small retained Gaussian coordinates bound the change from this centre by $C\sigma$. The exterior internal links are

determining and are identical for the two fields. The centre and the actual field have the same coarse target there by the joined row identity. Use the straight-corridor estimate for the actual middle bond with the centre curvature and the nonpivot changes both of order σ. This puts the actual solution in the regular branch, where the one-row inverse gives the bound $C_L\sigma$ for its pivot. If the stopped branch has translated a retained slot, use its original value from the recorded inverse map for the Gaussian reserve. Telescoping each plaquette gives the required bound for the actual field, including plaquettes using the pivot.

If the sharp bond does not cross the old wall, its endpoint blocks are in the old exterior and their fine links are preceding determining rows. The compatible field supplies the full V_j plaquettes directly, so the Wilson share gives their bound. At the inner prepared face the inner block remains in the determining layer. The two fields have identical internal links and hence identical mean contours there. When the two auxiliary axial gauges are undone, this common contour factor cancels in the norm of the relative link. Only the contour errors on the collar side enter. On every collar block the soft weight is either inherited from the old exterior belt or inserted on the remaining blocks of this stage. This covers all its contours. For a bond internal to a single block use the single-block clause of rigidity. The coarse target for a crossing of two blocks is the common retained target.

At the approximate split, use the test in the original coordinates, against $V_X^{(j)}$. At this moment the collar gauge has been inserted and the old ordinary strip still carries the coordinates in the preceding calculation. The complete coarse average is its original constraint. The completion of the ordinary fluctuation tests takes place later. The bound on the two block contours here follows from the small paying energy in the old strip and the collar gauge. Together with the curvature and rigidity estimates it gives

$$|V_j(b)(V_X^{(j)}(b))^{-1} - \mathbf{1}| \le C_L\sigma$$

on the original approximate mask. With $\sigma = c_*\delta_j'\check{R}_i^{-4}$ this lies below $2\delta_j'$. On a sharp stop use the original values recovered from the recorded translation, and apply the angle bound in the chart of its sharp test.

The paying Gaussian and gauge variables in these two estimates are of level $j \ge i_\flat$. Their weights and the Wilson coefficients on the charged enlargement are comparable to the corresponding ones at level i. If the sum of the fresh Wilson, gauge and kept Gaussian shares is at most $c\theta_i\sigma^2$, the individual bounds used above follow after reducing c. Hence a failure at either collar test forces the opposite inequality. In a containing mass these shares use the child's full carriers in its enlargement outside its own floor and its original level-j slots. For a containing operation whose waiting window starts at h, an older standing mark of level $m < h$ is in the retained Z_{h-1} and in the innermost box G_h. Its witness extends only two unit layers of level m beyond the mark. The hull and eight-layer enlargement at step h still contain the witness after removing the cells in contact with Ω_h. Thus the child's bare plaquettes lie in the containing operation's floor. The containing operation spends only plaquettes with no corner in that floor. Its gauge and Gaussian shares start at h, above the corresponding slots used by the child. This separates payments through nested waiting windows.

The newly inserted ordinary completion uses the stored fluctuation coordinate with threshold δ_j. The ordinary change of variables has already been made, and the symbol $\mathcal{A}_j$ in the completion denotes its resulting coordinate. The masks extend from the first strip cell by the two small test-cube layers and fit in the retained two-cell strip. A failed completion therefore has a stored bond with $|g_j\mathcal{A}_j^{\mathrm{s}}| \ge \delta_j$. The kept Gaussian reserve pays for this event on the local fibre. Tests inserted before the ordinary change of variables keep their original pullbacks, as in the ordinary birth payment.

These bounds give regularity on the two stage sequences. On directly guarded old collar reads the layer factor at level j is $O(1)L_\flat^{2(j-k_0(i))_+}$. On the full datum use the bound Ce_l at level l. The real profile estimate then bounds the scaled curvature and current on the retained carriers. The

adjacent-layer conversions and the compression loss are those of the real endpoint comparison, so the bounds fit the half radii of the earlier structures. Local gauges follow from the curvature bounds on their full cubes. The restricted real background is critical over its own averages on subordinate towers, so the tower and core conditions use the same bounds. In particular an inner island leaf is read in its original real core.

The stopped operation keeps the old one-step integrals of levels $l = j + 1, \dots, i$, the bare nest averaging integrals of levels $l = i_\flat + 1, \dots, j$, and the old core. The factors present at departure multiply this product, including the current test and any collar gauge already inserted. At a sharp-test stop in the global convention the first kept one-step integral uses the transported collar coordinates, with the corresponding pivot and Jacobian factors. A sharp failure in the component-local convention keeps the untransformed prefix with its local test. The earlier stops use the original pivot representation. Only completed integrations contribute stage Gaussian normalizers and terms to the stage boundary sum. At each domain D that sum is filed once, with stages up to the maximum completed index of the components meeting D and with already integrated stored fields set to zero. The stopped action keeps the Wilson weight from before the stages. The Wilson differences from completed stages cancel when one returns to that weight and are excluded from this boundary sum.

For its norm separate each completed stage term into its direct normalizer and its localized cluster term. At a fixed carrier, the cluster bounds cost at most $2\mathsf{N}_i\epsilon_{\mathrm{comp}}$ at the final rate. The conversion to pinned norms and the closing of carriers spend $O(1/\mathsf{N}_i)$ in the rate at each stage. The count uses the separation of the stage components. The total loss is therefore bounded independently of N_i. The direct part must use the sum of its pinned normalizer norms, not a single-stage supremum multiplied by N_i. In particular the coefficient for the retained stage entry is bounded by $\bar{D}_1^{\Sigma} + 2\mathsf{N}_i\epsilon_{\mathrm{comp}}$, where $\bar{D}_1^{\Sigma}$ is this summed direct norm. This gives the stage amplitude in the standing-list bound once the direct sum is bounded on the actual graded cells. The identification of each Gaussian determinant with its own stage integral supplies the direct terms for that sum.

For a sourced excess taken at the same evaluated pair, the direct terms cancel before the pinned estimate is used. Keep the stage prefix, masks, stored data and reference Gaussian the same in the two systems. The direct term is then the same function of the pair, including its subtraction at the trivial pair. All system entries occur in the potentials and the resulting closed clusters. Thus the difference of the retained stage sums uses only the cluster majorants on that common chart, including for graded cells. The absolute real bound on the standing term still uses the direct sum.

A later connected coefficient of a stored Gaussian piece also admits a comparison at varying pairs. Suppose the piece has been formed with its local holomorphic majorant ν_e on the larger fixed pair domain. Along the interpolation $t \mapsto p(t)$, let δ be the size of the pair displacement in the local coefficient norms, divided by their admitted radii. Restrict the piece to the complex line of the displacement. On a disc of radius c/δ its supremum is bounded by ν_e, so its change up to $t = 1$ is bounded by $C\nu_e\delta$. For a stored quadratic form $Q_e(y, y)$ read at $y = \mathrm{Ad}_g\, x$, a change from y_1 to y_2 gives the two terms $Q_e(y_2 - y_1, y_2) + Q_e(y_1, y_2 - y_1)$. The bilinear majorant and the bounded adjoint actions give the corresponding linear bound in the size of the relative rotation. When a connected coefficient reads the piece through a mixed derivative in a nonempty set of its switches, that derivative kills the value at the initial pair. Cauchy's estimate then applies to the increment. This argument applies to an arrested Gaussian piece, or a survivor piece meeting its arrested collar, when its formation gives the stated local majorant. The undifferentiated term in the standing sum still requires the absolute bound.

For each completed stage the number of integration bonds is bounded by $C(L)$ times the number of sites of its collar and strip at that stage's scale. Distinct drop-outs sharing a re-made level are disjoint, so this count charges the completed stage numbers once at that level. Changing the departure place within the preparatory stage at fixed n keeps the regions, the integration list and the action carrier. The particular tests and any inserted gauge are recorded in the local history.

There are at most $4\mathsf{N}_i + 6$ possible places for an operation at level i, including the second class, while the marked label uses at most $\mathsf{N}_i + 2$ indices. This is the stage-place part of the count in eq. 4.cdxcvi.

4.18.14.5. A stop at extendibility. At extendibility all preparatory integrations have been completed. Use the prepared datum with the holonomy bounds carried from those integrations and the original exterior and core tests. The real profile estimate therefore applies on the prepared read. The negative extendibility test depends on its exterior argument. At this stop the central integration is still pending. The old top values and the bare averaging chain in the core keep their original places. A later read replays the prepared support from those values and constraints. The frame operation and the new box Haar field enter only at the subsequent stops.

For its failure in a containing mass, use the old exterior top values on the complete carriers in the child's enlarged component. These carriers are outside the central box and its floor. If their Wilson share is E, the averaging energy bound controls the exterior top plaquette defects by $C\sqrt{E/\theta_i}$. The axial extension on the boundary of the central box gives a top field with the same exterior and defects at most $C(M)\sqrt{E/\theta_i}$. In the top inset the localized critical configuration for this extension has bare curvature bounded by this number times CL^{-2i}. The required region $\Omega_i^c \cap X$ lies in that inset by the two-cell clearance. Thus the recorded negative extendibility test forces $E \geq c(M)\theta_i\varepsilon_i^2$. This witness uses the original exterior argument of the test.

4.18.14.6. The tests retained at a late stop. A shell or last-shell failure occurs after the final preparatory guard and the central integration. It retains the positive tests preceding the failure. Use the prepared datum for the regularity comparison and the fresh outgoing field for the later averaging rows that read it. The removal that uses the two passing shell tests takes place only on the continuing branch. In the old prepared estimate the final stage test is used with the original top density tests and the old core tests.

For the inset operation, a level-q tested plaquette of the prepared sequence in X, with $q < i$, has its place in $Z_i'' \subset O_i$ by nesting. Its footprint is the union of the four order-q blocks. From its place this extends by at most $3L^q/2$, so the clearance in eq. 4.cdxliii puts the full footprint inside the spatial inset. On the no-inward list every corner is outside the prepared Ω_i: the first-kind ends are outside Ω_{q+1} and the second-kind ends are outside Ω_q. Thus all four corners are in Z_i''. The final box is a union of aligned level-i blocks, and each order-q block lies in its containing order-i block. The footprint is therefore in Z_i'' in this geometry. The top tests in the original top action region keep their halos, including any reads outside X.

A lower footprint can still meet a seam of an earlier shell. The literal shell bound is on bare plaquettes wholly contained in that shell. The comparison test at the last prepared seam supplies a different set of plaquettes. For real half-radius admission on an inverse branch, use the real endpoint comparison. Before marginalizing the central integral, keep its prepared variables. In the framed integral these are $V'' = V'V_0$ on the changed coordinates, with V_0 formed from the retained exterior. This substitution recovers the same datum in each frame summand. On its admitted incoming fibre the old lower holonomy bounds and the exact top averaging relation give the prepared datum bounds Ce_q of the endpoint comparison. Carry these bounds as a support condition on the prepared datum. For each actual exterior e, the indicator of the bounds depends only on e and the prepared variables. It equals one before the old averaging deltas are integrated, on the admitted entering fibre. The pushforward therefore assigns full mass to the set where this indicator is one. Insert the indicator in the retained kernel to use this density version at each exterior. The real layer estimate and the subordinate tower criterion then give the half-radius membership on its earlier read sets.

On a later read, use the prepared first-kind links directly, including the old top determining values on bonds meeting the box. A second-kind edge is completed from its retained lower double block. This replays each prepared holonomy on the complete read set. The promotion relation on the original fine rows was used to prove its bound before integration. Its consequence on the

prepared datum is the support condition just described. The fresh box field continues to enter the outgoing averaging rows. The original exterior density tests near Ω_i retain their own halo. In the quiet window the central box lies in $\Omega^c_{k_0+1}$, with $k_0+1 \le i-1$, inside the predecessor's hull. The eight-cell growth leaves at least four cells between it and $\Omega_i^{\sim 4}$. A complete density-test read uses less than one more cell. These tests therefore read the retained top exterior even after the central draw. The shell test itself reads the retained framed fluctuation, and the last-shell test reads the prescribed configuration with the retained bottom shell field on the core side of the cut and the identity on the shell side. It keeps the original values on bonds crossing the inner cut, according to the convention in the last-shell comparison.

The first outgoing chart recovers this exterior before it uses the free box pivots. Keep the old tests and support indicator in that joint measure. If a later row change makes an exterior argument a function of its outputs and remaining coordinates, compose each test with that function on the actual inverse branch. An earlier exterior row is read from a fixed coordinate, a retained variable, or an output of its original graph. Take the closure of the test reads under these graphs before taking a marginal. This preserves the support indicator even at a generated exterior. We must check the joint row bound at the resulting read set. A later estimate that uses a sharper threshold at an earlier shell seam must supply that bound on the full footprint.

For the analytic read of the kept one-step factors use X enlarged by one level-i cell layer. The walk carrier of a kept strip from step $l < i$ lies inside X. A strip still carried from step i can have a carrier extending to that full one-cell enlargement. Once all preparatory stages are complete the top strip no longer remains and the one-M-cube collar suffices. In either case the pivot factor uses the two endpoint blocks in its original birth partition. Their regularity follows from the corresponding pointwise bounds and the adjacent-level radius comparison. The walk and pivot bounds are then taken on their original masks. The larger analytic read does not extend the paying Wilson plaquettes or the integration coordinates of the component. The fibre on X receives only the resulting values of the forms and Jacobians, with their uniform one-sided bounds.

4.18.14.7. The same halo at later levels. Include the old top density reads and their first outgoing double blocks in the read set B_{read} of a mark of level i. For a still-carried top strip include its one-cell analytic read as well. The old cells are nested in the new grid and are no wider than the new cells. Thus B_{read} lies within two layers of the level-$(i+1)$ hull of X. This is inside the eight-layer growth and away from the new gauge draw. It lies in the ordinary successor of the mark. At subsequent ordinary steps, a full double block of a row reading the preceding component lies in its hull enlarged by one cell. The same growth includes these rows in the raw region of the successor. Their outputs use the carried lower graph and the current ordinary inputs.

For a later containing preparation let h be its bottom level. The window rule gives $i < h$. If $h = i+1$, the initial two-layer bound puts B_{read} strictly inside the old core of the new operation, even after removing the cells touching Ω_h. If $h > i+1$, apply the eight-layer inclusion at h to the successor at $h-1$, which still contains B_{read}. In both cases the original reads are in the floor. The preparation uses the old outputs in its bottom core and modifies only higher layers. The carried chart and its inverse factors therefore stay at their original levels. A healing that absorbs the component instead supplies the new determining datum on the healed part. The old graphs there remain in the integrated weight of the healing.

4.18.14.8. The next ordinary strip. The larger coarse bounds at the original walls are attached to footprints within the carried components. At the next ordinary step the complement of Ω_t contains the hull of Z_{t-1} and its eight-layer enlargement. The old wall tubes are inside that hull. The ordinary test and strip carriers use the exterior halo of this growth rule. On that halo the ordinary positive tests retain their original thresholds and use their complete averaging footprints. For the one- or two-cell neighborhood of Ω_t, the complete small test and contour carriers cost less than one more cell. The no-inward convention also puts the places of all tests of level below $t-1$ outside

Ω_{t-1}, hence inside Z_{t-1}. The near part of the datum profile uses levels $t-1$ and t, with comparable thresholds.

In the real profile near the ordinary strip, the part of the datum carrying the enlarged wall bound is at weighted distance at least $cM\check{R}_t$. Indeed the path across the guarding cells uses layers of spacing at most L^t, and the fixed read halos leave a cell margin. By eq. 3.ccclxxix its contribution to the normalized curvature and current is at most $C\varepsilon_\star e^{-c\delta M\check{R}_t}$. This is smaller than ε_{t-1} at weak coupling. The ordinary tests in the remaining part of the profile give the usual $C\varepsilon_{t-1}$ bound near the strip. For the localized strip at $t \geq 2$, the join is formed on $Z_t \cap \Lambda_{t-1}$. This domain is disjoint from Z_{t-1}, which contains the carried wall tubes. Use the two-field construction of the joined datum there. Its top values satisfy the retained $\mathcal{M}(V_{t-1}) = V_t$ constraints on the complete blocks, and its lower values use the original positive density cubes meeting the join. Their bounds give the local datum radius of order ε_{t-1} on the full joined test carriers. At $t = 1$ the strip has the one-part join and there are no earlier marks.

One can formulate the transfer without identifying two unnecessarily large sets of physically realizable arguments. Let $\mathcal{W}_k(i, X, o)$ be the admitted reads at level k, obtained by imposing the creation rule and all retained prefix constraints up to that level. Then

$$\mathcal{W}_k(i, X, o) \subset \mathcal{W}_{k'}(i, X, o) \subset \mathcal{W}_i(i, X, o), \qquad i \leq k' \leq k. \tag{4.dxxvi}$$

Actual read images are contained in these sets by eq. 4.dxxv. If one defines $\mathcal{W}_k$ instead as those images themselves, the same nesting additionally needs the exact prefix projection to commute with the local read. Label truncation alone is not a proof of that stronger statement. The admitted-set version already gives the required mass transfer.

Restriction of a supremum and reduction of either its volume or its coefficient can only reduce its upper integral. Hence for $0 \leq c' \leq c$ and $0 \leq v' \leq v^\natural$,

$$\mathsf{M}^\sharp_{c'}(\mathcal{W}_k; v') \leq \mathsf{M}_c(i, X; v^\natural) \leq e^{-a_\star P_i}. \tag{4.dxxvii}$$

There is no new factor for the number of later levels. If an inductive class permits a slack constant, for example $e\, e^{-a_\star P_i}$, that same slack is preserved for its old marks; it is not multiplied once per read. Newly formed marks have the sharper bound proved in eq. 4.dxi. A run started with those formed bounds retains them at all later levels as long as its weights, domains and presentations remain in their admitted tubes.

If a mark is absorbed into a new outer component, its local record occurs inside the new local history. Its weighted mass is then used in eq. 4.dix, not loaded again as a second independent outer mark. This is the operational version of the multiplicity identity in eq. 4.cdlxxvi.

4.18.14.9. Field readings and compensators. Gauge covariance acts on stored variables as well as external triples. Denote that action by $\rho_\omega(u)$. It conjugates the variables rooted in a tree by the transformation at its root and acts on an outside-starting stored bond at its actual starting site. Let R_ω be this finite list of sites. A permitted compensator $c[U]$ is Borel, obeys the actual chart conditions, and is the identity on R_ω. Then $\rho_\omega(c[U])$ is the identity on every stored variable. The compensated tied expression

$$\Phi^\diamond_\omega(\xi; \mathfrak{z}) = \Phi^\circ_\omega(\xi) - \tfrac{1}{2} q_X(\xi; c[U] \star \mathfrak{z}) + V^\diamond_X(\xi; \mathfrak{z}) \tag{4.dxxviii}$$

therefore needs no change of integration variable depending on ξ. Its formation identity must say that the whole tied sum takes the same value for equivalent triples. Separate invariance of the two summands is not required.

On every admitted fixed U chart the dependence on the complex potential and current is holomorphic, and the same envelope bounds the compensated read. Borel dependence on U follows from its Borel compensator and background maps. These claims require the actual compensator with these fixed sites and core admission. Gauge covariance for a fixed global transformation would

not alone construct it. This is why the read rule retains the sites on which the stored-variable action depends.

On the chart of a healed label one can construct this compensator on the ordinary towers and the rooted trees. Let $a(x)$ be the top of the tower of x, or the root of its box if that tower starts in a treated box. For each x, take $c[\mathbf{U}](x)$ to be the product of the averaged transports from $a(x)$ to x. The path first follows the box tree to the top, when present, and then the combs in the successive layers of the tower. The product is empty at a top outside the boxes and at every root. At the real points of the chart the factors are measurable group variables, and on the admitted complex chart they are holomorphic. Endpoint covariance cancels the transformations at the intervening sites, giving

$$c[\mathbf{U}^u](x)u(x) = u(a(x))c[\mathbf{U}](x).$$

The compensator gauges each comb and tree transport to one. For a tree variable the action on the stored coordinate is at the root. For an outside-starting variable whose start is a tower top the action is at that top. Both kinds of coordinates are therefore fixed in this reading. A term that reads another stored site requires that site to be checked separately in R_ω.

Apply $c[U]$ to the real arguments, and $c[\mathbf{U}]$ to the pair $(\mathbf{U}, \mathbf{J})$, with the current conjugated. At fixed U the real change is fixed. The compensated pair depends only on the original pair, so a tied slice expression is still tied on its admitted reads at fixed stored variables. Under a real gauge change the two normalized presentations differ by $u \circ a$. The invariance of the slice expression under these root and top rotations gives the joint invariance of the read expression. For a local term the carrier must contain the full paths just used. In particular a carrier that reads a box tree must contain that tree as well as the affected towers.

The analytic bound applies on the preimage of the original chart. This restriction matters even for flat complex links. Take a constant imaginary potential in one direction on a local order-j tower, with bare spacing L^{-j} in level-j units and with $U = 1$. The real compensator is then one. The complex comb puts the transport across two adjacent tops on the bare bond joining their towers. A logarithmic potential of size α before this change has size $L^j\alpha$ on that bond afterward. Hence the small potential bound in the original chart does not give the same bound on every compensated pair. At a frozen seam we must also check the endpoint constraints on the unchanged lower rows. The rooted product on the healed label supplies the gauge correction only on the admitted chart just described.

4.18.15. *Causal restrictions and the product of healed histories.* We give a finite algebraic statement that explains when a sum of histories is a gas. Spatial separation by itself is not its hypothesis. The determining reads, action reads and reference normalizers must all be included.

An elementary move has a discrete type, a finite list of input slots, a finite list of output slots and an integral kernel for the variables it removes. A test is a move with a finite outcome list and no integral. Every slot has a level, region and type, such as a group field, free Gaussian coordinate, scalar constant or background chart. Slots are versioned. The dependency relation directs an edge from a move producing a slot version to each later move reading it. It also orders every write/write, remove/read and priority conflict consistently with a legal execution. We require every pair left incomparable to be independent in the sense below; producer-to-reader edges alone do not imply this property. The relation includes the discrete reads deciding class membership and the background box in which a critical configuration is formed.

Let $R(\gamma)$ be the closed read support of a group of moves γ: include their written variables, every input they read, every relevant action plaquette and every wall and gauge factor they use. If a move evaluates a background on a whole box, that box is included. A carrier cannot be reduced to a short random-walk segment merely because a displayed kernel term has a short segment. In particular q_X in eq. 4.cdlxvii has carrier $X^\sim$; its separate walk terms use a minimizer on all of X and are not local functions of their short walk supports.

The exact locality hypotheses for the finite statement are as follows. Local functions and reference kernels obey hereditary restriction: forming a move on a larger admitted box and setting outside decoupling parameters to zero gives the move formed on the smaller box. Its normalizers and counterterm readout obey the same identity. Two independent groups have disjoint closed read supports and no common decoupling wall. Their incoming exterior slots agree with the reference carried history, and all parameters stay in the common admitted domains.

Lemma 4.68 (Order and restriction)**.** Under these hypotheses incomparable independent moves commute, and a hereditary subcollection has the same outputs on its slots whether executed alone or inside the larger history. ■

Proof. For two integral moves, the inputs held fixed by one contain no variable written or removed by the other. Their integrands multiply, their reference measures are product measures and their output substitutions concern disjoint slots. Tonelli proves commutation for positive real kernels. For complex kernels the common Borel envelope in eq. 4.dviii supplies the absolute integrability for Fubini. A test has the same value before and after an independent move, since every slot it reads has been included in its closed support. This also proves commutation for a test and an integral move.

Two linear orders of a finite dependency partial order differ by adjacent swaps of incomparable entries. To see this, take the first entry of the second order and move it forward in the first. Every entry passed is incomparable: an ancestor must precede it in both orders, and a descendant cannot precede it in the first. Remove this entry and continue by induction. The commutation just proved makes the operation independent of the chosen linear order.

For restriction, take the first retained move. Every input it needs is either fixed exterior data or produced by a retained move, by heredity. It therefore has the same value as in the larger history. Determinism of its test and formation formula gives the same outcome and output. Induct on the remaining moves. Hereditary consistency identifies their formation on the restricted boxes with their restrictions from the larger boxes. This proves the statement for functions and integral operations. □

A minimizer on a larger domain generally changes with exterior data. Locality of the bare action does not make that minimizer a function of an arbitrarily smaller carrier. The actual localized background functions, interpolation endpoints and hereditary identities must be used in this lemma. Naming a carrier is not such an identity.

Fix the outer history $\bar{H}$ and the real data that the completed operation leaves unintegrated. In a history over $\bar{H}$ the healed components of the current level t form a compatible family $\mathfrak{X}$. Restricting to a subfamily deletes the entire ancestral cluster of each omitted component. Conversely, disjoint admissible component histories that do not touch the standing region can be united with $\bar{H}$. This follows from the history restriction and union rules in §4.17.1.1. The output label and marks are common to the resulting histories. The sums over the histories and frames of $C \in \mathfrak{X}$ are part of the same local integral operation $\mathbf{T}'_t(C)$ in every family that contains C.

These are statements about the records. In the integral, each record uses its actual test support. In particular, on the first-class support $\chi_t(C^{\sim -6}) = 1$ the two reference-law cutoffs in C are one. For the cutoff of the reference fluctuation this uses the bound by a fixed multiple of $M^6\varepsilon_t$ and the choice of its radius $M_0\varepsilon_t$ above that bound. The exterior small-field test incorporates $\chi_t(C^{\sim -6})$, and the reverse gauge-fixing identity removes the reference tree factor relative to the variables of C. The second-class support retains the complementary test, both cutoffs and the tree factor. The operations for the standing components thus act on the following sum, with their variables kept fixed while summing over the healed components.

Write $\mathcal{Y}(\mathfrak{X})$ for the domains, other than single collars, that contain at least one collar of a healed or second-class component and contain every collar that they meet in a cube. Here the collar is the whole enlargement $C^{\sim}$. Write $V_{\mathfrak{X}}(Y)$ for the interaction on such a domain. The single-collar terms

are included in the component operations. The completed action is common to the families over $\bar{H}$. To compare the interactions, take the local lists before assigning terms to slots. Localisation is linear within each of the pointed, unpointed and Wilson classes. The Wilson pieces with profile-dependent weights first use one common cover, on which the level-t filing profile φ_t cancels by the Wilson identity. The terms made by $\mathbf{T}$ and the stages are compared before filing.

For a complete localized carrier missing the enlarged collars of the deleted first-class components, use the comparison obtained by erasing their histories. It preserves the local comparison before filing. The input members on the domain must correspond, the strip shifts must be determined by the zone and label, and the standard rules for the levels must apply. Under these conditions the local operators, fluctuation potentials and stage terms agree as partial maps. In particular the stage region is read on the enlarged domain and its preparatory entries and nests are read in their declared level windows. The Wilson pieces using the final filing profile were combined above. The comparison keeps the common functions even when the final mask changes their slots.

The same argument applies to a deleted second-class component when the formal history space permits reassigning its class for this comparison. The ordinary and stage functions are formed before the final class test. Give the component the first-class designation in the comparison of those unfiled functions, apply the erasure identity, then file under the actual output mask. This uses the order of formation and the formal admissibility of the reassigned history. The real histories in the integral retain their actual tests.

More generally, suppose $\mathfrak{X} \subset \mathfrak{X}'$ and Y is admitted for both families and meets none of the added collars. Every entry with base in Y then has the same carrier, domain and value: the earlier entries agree by coherence, the response evaluations use the same components and pinned sites in Y, and the kept factors have carriers inside the collars of the components that remain. Thus $V_{\mathfrak{X}'}(Y) = V_{\mathfrak{X}}(Y)$ on its common domain. The requirement that Y be admitted for both families is used here. Adding a component can change the admissible resummation domains.

For $Y \in \mathcal{Y}(\emptyset)$ let $Y^{\mathfrak{X}}$ be Y together with each healed collar sharing a cube with Y. The collars are disjoint, so this is a domain in $\mathcal{Y}(\mathfrak{X})$. Set

$$W_{\mathfrak{X}}(Y') = V_{\mathfrak{X}}(Y') - \sum_{Y \in \mathcal{Y}(\emptyset)\colon Y^{\mathfrak{X}} = Y'} V_{\emptyset}(Y).$$

If Y' contains no healed collar, the only preimage is Y' and the difference vanishes by the preceding comparison. Each other difference depends on the family only through the healed components in Y' and their local histories. The subtracted terms are independent of the variables integrated by the healed operations. Grouping by $Y^{\mathfrak{X}}$ and taking their common exponential outside these operations gives

$$\begin{aligned} &\sum_{\mathfrak{X}} \prod_{C \in \mathfrak{X}} \mathbf{T}'_t(C) \exp\left(\sum_{Y \in \mathcal{Y}(\mathfrak{X})} V_{\mathfrak{X}}(Y) \right) \\ &\quad = \exp\left(\sum_{Y \in \mathcal{Y}(\emptyset)} V_{\emptyset}(Y) \right) \left[1 + \sum_{\mathfrak{X} \neq \emptyset} \prod_{C \in \mathfrak{X}} \mathbf{T}'_t(C) \exp\left(\sum_{Y' \in \mathcal{Y}(\mathfrak{X})} W_{\mathfrak{X}}(Y') \right) \right]. \end{aligned} \tag{4.dxxix}$$

The component operations act on the exponentials following them, including the dependence on their histories and integration variables. This identity requires no division by a history density.

Expand the exponential inside the square bracket as a product of $1 + (e^{W_{\mathfrak{X}}(Y')} - 1)$. Make a graph with one vertex for each healed collar and each selected interaction domain. Two vertices are joined when their supports have coinciding or adjacent cubes outside the hulls of the fixed outer label. Each healed collar lies outside those hulls. The restriction and union rule for the histories and the agreement of the differences on retained domains give the product of the activities for the different connected components of this graph. The standing variables remain parameters during this factorisation. Each activity contains a healed component, since every selected nonzero

difference has a healed collar in its domain. The local terms $V_\emptyset$ remain in the boundary exponent, since their domains contain a second-class collar.

The identity is for the complete operation on its physical data. Local activity formulas can also be estimated on their admitted complex cores, including independent currents. Their estimates do not extend the global physical identity to independently varied currents. To take the logarithm of the square bracket and file its terms by their unions, use the polymer criterion after the integrable kernel majorants and the activity bounds have been established.

4.18.15.1. A kernel norm for composition. For a local kernel $K(\xi;p)$ define

$$\|K\|_c = \int^* e^{cv(\xi)} \sup_p |K(\xi;p)|\, d\mu$$

including its discrete sums and frame factors. It gives

$$\|\mathbf{T}_K F\|_\infty \le \|K\|_0 \|F\|_\infty$$

for bounded insertions on the full read space. If a second kernel is evaluated at an admitted read $q = q(\xi,p)$ and has stored variables η, then the additive weight and a common Borel envelope give

$$\int^* e^{c(v_1+v_2)} \sup_p |K_1(\xi;p)K_2(\eta;q(\xi,p))|\, d\mu(\xi)d\nu(\eta) \le \|K_1\|_c \|K_2\|_c. \tag{4.dxxx}$$

Bound the factors by their strong majorants and use Tonelli. The read must lie in the core of the second norm at every point of the first support. A real read needs only the strong real version of that norm. This is the kernel mechanism of eq. 4.dix, and shows why a fixed-read integral estimate is not the right norm for general composition.

4.18.16. *The loaded label sum and its analytic price.* At final level k let N_k be the number of lattice sites. In the site load put $D_m = \Omega_m \setminus \Omega_{m+1}$ for $m \ge 1$. At level zero use $D_0 = T_\ell \setminus \Omega_1$, counted on the bare lattice. This counts all bare sites of that layer, including the sites outside the collared first member. The determining sets of eq. 4.cdlxx still use their own first member. Put $Z_j^+ = ((\Lambda_j^{\mathsf{a}})^c)^\sim$. The required load is

$$\mathscr{L}_c(\mathcal{S}) = c\sum_{m=0}^{k} |D_m| + \sum_{j=1}^{k} (C_{\log}\ell_{j-1} + C_{\mathrm{loc}} + c)\#_j(Z_j^+), \qquad C_{\log}, C_{\mathrm{loc}} \ge 0. \tag{4.dxxxi}$$

Every count is at its indicated level. At the last level the outer remaining layer has at most N_k sites. Older layers may have been redone; their counts are not a single present-day volume.

The scalar site load has a direct estimate. Since Ω_{m+1} is aligned at level $m+1$, for $m < k$ one has

$$|D_m| \le \#_m(\Omega_{m+1}^c) = L^4 \#_{m+1}(\Omega_{m+1}^c) \le L^4 \#_{m+1}(Z_{m+1}^{\mathsf{a}}).$$

This uses inclusion of the action members in the determining members and applies also to the full bare count at $m = 0$. The last layer costs at most N_k. For the source share charge each dilated cell of Z_j^+ to a cell of Z_j^{a}; the one-layer enlargement costs a fixed multiplicity. The bounded drift gives $\log\theta_{j-1} \le C\log\theta_j$. Thus the total scalar load is bounded by $cN_k + B(c)\sum_{j\le k}\log\theta_j\, \#_j(Z_j^{\mathsf{a}})$ for a fixed coefficient $B(c)$. To use it in the product over histories, retain the ownership of each local action load. The geometric load comparison is checked by ownership of level-site addresses. Charge an unaltered layer site to its action cell, with one-step fine-to-coarse multiplicity at most L^4. Charge a site in a redone determining layer to the local history of the mark that remade it, retaining its original level. Prepared means are charged to their owned level cells, not to an integrated-out variable at a different level. The re-entry rule after eq. 4.cdlxiv specifies the actual datum read there. With nested marks, use the latest operation owning the remade slot; its inner record already accounts for the earlier history.

The precise finite geometric assertion is that each such read belongs to its owning cell collar, with a bounded covering multiplicity C_{d}. It involves the actual determining entries, action entries

and retained nests together. Once this assertion holds, summing the assigned addresses and using eq. 4.cdlxxvi–eq. 4.cdlxxviii bounds the load by current-level sites, live action-cell volume and marked volumes. No factor counts the number of later reads of a mark. In particular a coefficient of the following form suffices to pay the redone load:

$$b(c) = L^4(c + C_{\mathrm{sf}}) + C + c, \qquad c^\sharp(c) = c + CC_{\mathrm{d}}(b(c) + c_{\mathrm{e}}). \tag{4.dxxxii}$$

The fixed C includes the proved collar, scalar-number and normalizer prices, and C_{sf} bounds final unmarked local energy per site. Both expressions are affine in c. The terms $C_{\log}\ell_{j-1}$ and C_{loc} in eq. 4.dxxxi enter the cell price before the reserve is applied. Another price in an actual presentation must be inserted with its actual sign and magnitude, not omitted by this notation.

Separate the surviving real coarse Wilson factor. The unmarked localized energy costs $C_{\mathrm{sf}}N_k$ plus the assigned defect-site prices. Rescaling and the remaining Gaussian and chart factors have the local prices proved above. The vacuum subtraction uses the following count.

4.18.16.1. Subtracting the label vacuum number. The label density carries $e^{-\mathcal{E}_k(\mathcal{S})}$. Let $E_{(k)} = -\mathcal{E}_k(\mathcal{S}_\emptyset)$ be the vacuum number of the all-small-field prefix. Its subtraction leaves the scalar $-[E_{(k)} + \mathcal{E}_k(\mathcal{S})]$. Use the same one-step constants for the label and the all-small-field term, with the actual ranges in each.

For a region Y made up of level-j blocks, put

$$n_{j-1}(Y) = |Y^{(j-1)*}| - (L^4 - 1)\#_j(Y).$$

The first count consists of the incident free bonds after pivot removal. The second counts the sites of the normalized block gauge factors. In bare gauge dimension three, the Gaussian scale contributes $-\frac{3}{2}\log\theta_{j-1}$ per free bond and the gauge denominator returns the same amount per gauge site. The remaining normalizer terms, including the bounded ratio x_{j-1}/θ_{j-1}, are the regular part of the count. On the full torus $n_{j-1}(T_\ell) = 3(L^4 - 1)N_j$.

To compare a region with the torus, let $b_j(Y)$ count the bonds of level j with one end in Y. Counting bond endpoints outside Y shows that the number of bonds with both ends outside is $4\#_j(Y^c) - b_j(Y)/2$. The corresponding fine count uses $L^4\#_j(Y^c)$ sites and $L^3 b_j(Y)$ crossing bonds. Subtract the coarse pivots and then the gauge sites. This gives

$$n_{j-1}(T_\ell) - n_{j-1}(Y) = 3(L^4 - 1)\#_j(Y^c) - \tfrac{1}{2}(L^3 - 1)b_j(Y).$$

Since $b_j(Y) \le 8\#_j(Y^c)$, this difference is nonnegative and is at most $3(L^4 - 1)\#_j(Y^c)$. For $Y = \Omega_j$ the logarithmic term in $\mathcal{E}_k(\mathcal{S}) - \mathcal{E}_k(\mathcal{S}_\emptyset)$ is $-\frac{3}{2}\log\theta_{j-1}$ times that difference. Thus its sign in the density after vacuum subtraction is positive.

For the bounded activity terms, charge a missing base point in Z_j^+ by the rooted domain sum. Every other missing pointing lies in a domain meeting Z_j. Root there and sum the at most $CM^4(1 + d_j(X))$ pointings with the spare exponential decay. A missing remainder meets Z_j^+ because its range excludes the inner boundary layer. These sums cost $C\#_j(Z_j^+)$. The analogous source difference uses the same ranges, and the Gaussian and gauge counting factors cancel in that difference.

For the stage scalars, sum by the remade level j. At a fixed remade level the drop-out windows have disjoint spatial pieces. Their strip and collar bonds are counted in Z_j, at most $CL^4\#_j(Z_j)$ fine bonds in total. The normalized Gaussian spectrum at the trivial pair gives a bounded log contribution per integration coordinate. Each pivot contributes its fixed flat Jacobian, and the change of scale costs $C\log\theta_{j-1}$ per bond. This gives the stage price $(C\log\theta_{j-1} + C)\#_j(Z_j)$. Together these counts give the local price in eq. 4.dxxxi, including its logarithmic share.

For a later normalization $E_{(n)}$ of the same run, the full-lattice count gives

$$E_{(n)} - E_{(k)} = -\tfrac{9}{2}(L^4 - 1)\sum_{j=k}^{n-1}(\log\theta_j)N_{j+1} + \mathfrak{r}_k, \qquad |\mathfrak{r}_k| \le CN_k.$$

Here the bounded one-step costs sum geometrically because $N_{j+1} = L^{-4(j+1-k)}N_k$. The logarithmic sum is nonnegative before its minus sign, and is bounded by $C(1+\log\theta_k)N_k$ using bounded adjacent frame drift. In particular the prefix floor expressed with $E_{(n)}$ loses at most e^{-CN_k} compared with its expression using $E_{(k)}$.

Under these local prices, the modulus of each literal label term is bounded, before its last Haar integration, by its positive history envelopes, defect-site prices and $e^{C_{\mathrm{sf}}N_k}$. The remaining real Wilson factor is at most one. Normalized Haar integration is at most one after positive restrictions are dropped. For a drop-out restrict the sum to inner histories compatible with its marked label. The volume bound on these histories is $\mathsf{v} \geq V_{\mathrm{in}}$, where V_{in} is the inner volume of that label. If $c' = c^\sharp(c)$, then on these histories

$$e^{c\mathsf{v}} \leq e^{c'\mathsf{v}}e^{-(c'-c)V_{\mathrm{in}}}.$$

The restricted mass with weight c' is bounded by the full mass with that weight. The extra factor pays the redone load of the label. The full mass at weight c includes other histories with smaller inner volume, so the restriction precedes this estimate. For a second-class mark use its full estimate at c' by monotonicity. Unfinished live lineages use eq. 4.cdxciii, not an additional fictional marked mass.

Here is the final field-free summation. Root each connected carried component at a cell of the fixed M partition at level k. There are N_k/M^4 root addresses. Earlier healed components have already been resummed into local energies and are not independent roots in this carried forest. Counting them again would incorrectly introduce a bare-volume root factor.

At each root the connected histories have the bounded-degree address encoding of eq. 4.cdxcv and the local code in eq. 4.cdxcvi. Increase the cell price by a fixed amount and use the reserve at the correspondingly raised floor. The residual positive weights a_γ of carried components then have

$$\sum_{\gamma \text{ at } r} a_\gamma \leq 1. \tag{4.dxxxiii}$$

This is the connected-set sum in eq. 4.cdxcvii, with the specified load in the price and the marked masses already integrated. Histories leading to the same label are retained; the code pays for them. No one-to-one identification of labels with histories has been assumed.

More explicitly, let a_γ be the integrated positive envelope of the particular component record, after its marked operations have been dominated and its owned defect prices assigned. The per-record bound used here is $a_\gamma \leq h^{-f_\gamma}e^{-\sigma V_\gamma}$, with σ larger than its ancillary code constant plus $2\log D + \log 2$. It follows from the event reserve with this additional cell price only when the complete kept forms and stored coordinates of that literal term have the uniform domination already stated. They give a product envelope in the disjoint owned coordinates; bounded reads are first replaced by their strong majorants. Counting that bound by eq. 4.cdxcv–eq. 4.cdxcvii proves eq. 4.dxxxiii. Thus eq. 4.dxxxiii and eq. 4.dxxxv require the product-coordinate and analytic price estimates for each component record. Positivity of the final gas alone does not give these estimates.

Allowed collections are a subset of all finite collections of distinct component records. Dropping compatibility in a nonnegative sum proves

$$\sum_{\Gamma \text{ allowed}} \prod_{\gamma\in\Gamma} a_\gamma \leq \prod_\gamma (1+a_\gamma) \leq \exp\Big(\sum_\gamma a_\gamma\Big) \leq e^{N_k/M^4}. \tag{4.dxxxiv}$$

For countably many records first prove the finite inequality, then take the increasing limit. The loaded integral and label sum consequently satisfy

$$\sum_{\mathcal{S}} e^{\mathrm{Re}\,\mathcal{E}_k(\mathcal{S})}\tau^{\mathrm{abs}}_{\mathcal{S},c}(\mathcal{V}_k)e^{\mathscr{L}_c(\mathcal{S})} \leq e^{(c+C_{\mathrm{sf}}+M^{-4})N_k}. \tag{4.dxxxv}$$

Here $\tau^{\mathrm{abs}}_{\mathcal{S},c}$ is the literal measure with moduli and $e^{cv^{\natural}}$ at its outer marks. The term cN_k is the current-level part of eq. 4.dxxxi; older loads have been paid by the action-cell reserve or the excess marked weight.

Thus eq. 4.dxxxv is a conditional deduction from the exact normalized label representation, local analytic prices, read ownership and mass bounds. It is not a consequence of a final coarse gas alone. History summation forms the local energies and carried measures to which the prices apply. Their formation and common vacuum normalization must first be matched. Complex sources need uniform modulus prices on their admitted domains. For a real system, positivity of the remaining Wilson term is used exactly in the final Haar comparison.

4.18.17. *A weighted analytic extension from a prepaid real norm.* There is a second way to extend a parameter family. It does not require the pointwise excess estimate in eq. 4.dxxi. Its hypotheses are instead an ordinary complex bound in the actual kernel norm and a more strongly weighted bound along a real interval. We prove the statement, including its norm and domain requirements.

Fix the stored-variable spaces and allowed argument fibres $D_\omega(\xi)$. They may be ordinary cores or compatible evaluation fibres. They do not change with the parameter being continued. For a complex kernel family $F = (F_\omega)$ put

$$\|F\|_{\mathfrak{E}_c} = \frac{1}{h}\sum_{\omega}\int^{*} e^{cv(\omega,\xi)} \sup_{p\in D_\omega(\xi)} |F_\omega(\xi;p)|\, d\mu_\omega, \qquad c \ge 0. \tag{4.dxxxvi}$$

The weight is a fixed finite nonnegative Borel function. The history volume is its principal example; the proof does not require it to be integer valued. Reference measures, fibres and weight are fixed throughout the argument. Kernels with zero norm are identified. No measurable maximizing argument is required by the definition.

Lemma 4.69 (Completeness and truncation)**.** The finite-norm quotient in eq. 4.dxxxvi is a complex Banach space. Multiplication by a Borel scalar of modulus one is an isometry. Multiplication by $1_{\{v\le N\}}$, denoted P_N, has norm at most one, and $P_N : \mathfrak{E}_0 \to \mathfrak{E}_c$ has norm at most e^{cN}. ■

Proof. Homogeneity and the triangle inequality follow from their pointwise supremum versions and eq. 4.cdlxxii. Zero norm has a concrete meaning. If its supremum is f, choose Borel majorants $g_n \ge f$ with weighted integrals at most 2^{-n}. Fatou gives $\liminf g_n = 0$ almost everywhere, so $f = 0$ outside a Borel null set. Conversely, zero outside such a set has zero upper integral. The null set is common to every argument in the fibre. Thus the quotient is a normed space with one common equivalence relation for all arguments.

For completeness take a Cauchy sequence and a subsequence F_{n_j} whose consecutive norm differences are at most 2^{-2j}. On the countable disjoint union of the stored spaces choose Borel majorants b_j for the fibre suprema of these differences, with total weighted integrals at most 2^{-j}. Countably many errors in the separate history integrals can be allocated a further summable amount when choosing these majorants. Tonelli gives $\sum_j b_j < \infty$ outside one Borel null set. Remove also the null sets on which one of the countably many finite-norm kernels is unbounded on its fibre. On the complement the subsequence is uniformly Cauchy on every fibre, so it has a uniform pointwise limit $F_\omega(\xi;\cdot)$. Define it to be zero on the removed null set. Its difference from F_{n_j} is majorized by $\sum_{l\ge j} b_l$, whose weighted integral tends to zero. Thus the subsequence, and then the original Cauchy sequence, converges in norm. Uniform limits preserve continuous field dependence or holomorphy on fixed field cores when those properties are imposed. These give closed subspaces if desired. The multiplier and projection claims follow pointwise from their absolute values. □

We will also use continuity from below of this upper integral when the integrals have a finite bound. If $0 \le f_n \uparrow f$ and $\int^* f_n \le M$, choose Borel majorants $g_n \ge f_n$ with integrals at most $M + 1/n$. The functions $h_n = \inf_{k\ge n} g_k$ are Borel, increase, and satisfy $h_n \ge f_n$. Each has integral at most M, since it is bounded by every g_k with $k \ge n$. Measurable monotone convergence shows

that $h = \lim h_n$ majorizes f and has integral at most M. Monotonicity gives the reverse inequality needed for equality of the increasing upper integral limit. This applies on the disjoint union as well.

4.18.17.1. A criterion for ordinary Banach holomorphy. Suppose the actual representatives $K_\omega(z;\xi;p)$ are holomorphic in $z \in D_R$ at each fixed (ξ,p), and their $\mathfrak{E}_0$ norms are at most M for every z. This is a kernel bound, not just a bound on an integrated scalar partition function. Assume on each interior source circle $|z| = R_0 < R$ that

$$a_\omega(\xi,t) = \sup_{p\in D_\omega(\xi)} |K_\omega(R_0e^{it};\xi;p)| \tag{4.dxxxvii}$$

is jointly Borel. The reference measure and fibres are independent of z.

For a fixed separable field core the usual Carathéodory hypotheses give this measurability. For each fixed field argument, Borel dependence on ξ and continuous dependence on the source circle give joint measurability by piecewise-constant circle approximations. A countable dense set in the field core then makes its supremum a countable supremum of these jointly Borel functions. A fibre depending on ξ need not have a fixed dense set; in that case eq. 4.dxxxvii, or a direct common Borel envelope, must be checked on the actual evaluation rule.

Tonelli and the uniform ordinary norm give

$$\frac{1}{h}\sum_\omega \int \frac{1}{2\pi}\int_0^{2\pi} a_\omega(\xi,t)\,dt\,d\mu_\omega \le M.$$

Outside a Borel null set the circle average is finite. Cauchy's formula at each field argument therefore gives, simultaneously for every argument and $|z| \le r < R_0$,

$$\sup_{|z|\le r}\sup_p |K_\omega(z;\xi;p)| \le \frac{R_0}{R_0-r}\frac{1}{2\pi}\int_0^{2\pi} a_\omega(\xi,t)\,dt. \tag{4.dxxxviii}$$

The right side is a common Borel majorant with total integral at most $MR_0/(R_0-r)$. It was obtained first from the Cauchy kernel, and then from Tonelli; no uncontrolled supremum was interchanged with the stored-variable integral.

The coefficient formula $A_m = (2\pi)^{-1}\int K(R_0e^{it})R_0^{-m}e^{-imt}\,dt$ has fibre supremum at most R_0^{-m} times the same circle average. Thus $\|A_m\|_{\mathfrak{E}_0} \le MR_0^{-m}$. The Taylor remainder on $|z| \le r$ has norm at most $M(r/R_0)^{n+1}/(1-r/R_0)$ after degree n, by the geometric series in Cauchy's formula. It proves actual norm holomorphy and identifies its coefficients with the differentiated kernels outside one common Borel null set. The centre can be any point with an interior source circle, so the proof works throughout the domain.

This criterion applies when the ordinary complex mass is available before the weighted one. The ordinary complex kernel bound, pointwise kernel holomorphy and the stated measurability then supply the Banach premise of the following theorem. Scalar integral holomorphy alone supplies none of these kernel assertions.

4.18.17.2. An explicit harmonic barrier. Let $0 < a < R$, remove the segment $[-a,a]$ from the disc $D_R = \{|z| < R\}$, and for z outside the segment let $w(z)$ be the root of $w + w^{-1} = 2z/a$ with $|w| > 1$. There is exactly one such root: the two roots have product one, and a root on the unit circle would put z on the segment. The roots are simple off its endpoints. The unique choice is therefore holomorphic there by its local simple-root formula, and these choices agree on overlaps. Put

$$g(z) = \log|w(z)|, \qquad H = \operatorname{arcosh}(R/a), \qquad h_*(z) = 1 - g(z)/H. \tag{4.dxxxix}$$

The function g is positive and harmonic, locally the real part of a logarithm of its nonzero holomorphic root. It tends to zero on the segment. The identity for w and the triangle inequalities give

$$a\sinh g(z) \le |z| \le a\cosh g(z). \tag{4.dxl}$$

Thus $g \geq \operatorname{arcosh}(R/a)$ on the outer circle and $g \leq \operatorname{arsinh}(r/a)$ on $|z| \leq r$. In particular $h_* \leq 0$ on the outer boundary, $h_* = 1$ on the slit, and

$$h_*(z) \geq h_0 := 1 - \frac{\operatorname{arsinh}(r/a)}{\operatorname{arcosh}(R/a)} \quad (|z| \leq r). \tag{4.dxli}$$

Take $a = R/2$ and $r = R/4$. Then $h_0 > 1/2$: indeed $\cosh(2\operatorname{arsinh}(1/2)) = 3/2 < 2$. No assertion about a general harmonic-measure theorem is needed for these explicit bounds.

4.18.17.3. The weighted extension theorem. Suppose $F : D_R \to \mathfrak{E}_0$ is holomorphic in the Banach norm and $\|F(z)\|_{\mathfrak{E}_0} \leq M$ there. The case $M = 0$ is immediate; take $M > 0$ in the proof. Suppose also that, on the real segment $[-a, a]$, its same kernel classes belong to $\mathfrak{E}_C$ and obey $\|F(x)\|_{\mathfrak{E}_C} \leq M$. If $Ch_0 > c$, then

$$F : D_r \longrightarrow \mathfrak{E}_c \text{ is holomorphic}, \qquad \sup_{D_r} \|F(z)\|_{\mathfrak{E}_c} \leq M. \tag{4.dxlii}$$

The same M is retained. In particular, the choices $a = R/2$, $r = R/4$, $C = 2(c + 1)$ give the conclusion with an additional positive weight margin. If the two hypotheses have different constants, their maximum is a valid M.

Proof. First truncate to $v \leq N$. On any simply connected patch off the slit choose a holomorphic function b with $\operatorname{Re} b = h_*$. Then $e^{Cb(z)v} P_N F(z)$ is a holomorphic $\mathfrak{E}_0$ vector. To check the multiplier analyticity, its power series and all derivatives converge in operator norm on the truncation, since $v \leq N$. Its norm is

$$u_N(z) = \left\| e^{Ch_*(z)v} P_N F(z) \right\|_{\mathfrak{E}_0}. \tag{4.dxliii}$$

Different local harmonic conjugates differ by imaginary constants. Their multipliers differ by diagonal phases of modulus one. The norm in eq. 4.dxliii is consequently single valued even though a global harmonic conjugate around the slit need not exist.

The logarithm of the norm of a holomorphic Banach vector is subharmonic. A direct proof at a nonzero value chooses a norm-one complex linear functional attaining the norm there by Hahn–Banach, applies the scalar logarithmic submean inequality, and bounds its values on the circle by the vector norm. At a zero, use the value minus infinity and the same upper semicontinuity argument. Thus $\log u_N$ is globally subharmonic off the slit, because its local definitions agree.

On the slit the fixed truncation is continuous in the $\mathfrak{E}_0$ norm and h_* tends to one. Its boundary norm is at most the assumed real $\mathfrak{E}_C$ norm, hence at most M. On the outer circle $h_* \leq 0$; the ordinary bound and $v \geq 0$ give boundary upper limit at most M. For this fixed N the function is bounded above throughout the punctured disc, and the subharmonic maximum principle gives $u_N(z) \leq M$. This proof uses the norm-continuous extension of F across the real interval in $\mathfrak{E}_0$. Scalar partition holomorphy would not supply it.

At any point of D_r off the slit, $Ch_*(z) \geq Ch_0$. Monotonicity of the weight gives $\|P_N F(z)\|_{\mathfrak{E}_{Ch_0}} \leq M$. The continuity-from-below argument proves that $F(z)$ belongs to that weighted space, with the same bound. On $D_r \cap [-a, a]$ it follows directly from the assumed $\mathfrak{E}_C$ bound. The identification with the original $\mathfrak{E}_0$ class is exact, since these spaces have the same common-null-set identification and P_N exhausts the fixed finite weight.

Choose $\delta > 0$ with $c + \delta \leq Ch_0$. Uniformly on D_r one now has

$$\|(1 - P_N)F(z)\|_{\mathfrak{E}_c} \leq e^{-\delta N} \|F(z)\|_{\mathfrak{E}_{c+\delta}} \leq M e^{-\delta N}. \tag{4.dxliv}$$

Each $P_N F$ is holomorphic into $\mathfrak{E}_c$, because $P_N : \mathfrak{E}_0 \to \mathfrak{E}_c$ is bounded. Their uniform convergence on the smaller disc proves holomorphy of the limit there. One can see this directly by taking their Banach-valued Cauchy integrals on any interior circle: uniform norm convergence permits the limit in that formula and its differentiated formula. This proves eq. 4.dxlii. □

It follows in particular that

$$\|F^{(m)}(0)\|_{\mathfrak{E}_c} \leq m! M (4/R)^m, \qquad m = 0, 1, \ldots, \tag{4.dxlv}$$

for the quarter-disc choice. There is no factor growing with the number of histories, no reciprocal survival probability in the radius, and no extra constant in front of M. The cost is the explicitly larger prepaid real weight and a fixed radius loss. The coupling floor for the real mass remains affine in the requested final coefficient c.

When the formed kernels have their pointwise holomorphic representatives, these norm derivatives are their actual kernel derivatives. Here is a way to avoid separate exceptional sets at uncountably many parameters. The norm Taylor coefficients on an interior circle satisfy a summable geometric bound on any smaller circle. Choose Borel majorants for their fibre suprema with a summable total integral. Outside one Borel null set, the resulting power series converges uniformly on every smaller parameter disc and every fibre. At a countable dense set of parameters it agrees with the original kernel classes outside a further countable union of Borel null sets. Pointwise holomorphic uniqueness then makes it agree for all parameters there. Differentiating this series gives the actual differentiated kernels on that same full set. No point evaluation is asserted to be a bounded functional on $\mathfrak{E}_0$.

4.18.17.4. What would make this an actual source extension. For a fixed cutoff, use exactly its admitted field or evaluation fibres and the fixed history volume. The required ordinary family is the complete kernel, including its normalizers, not just the scalar integral of that kernel. Its $\mathfrak{E}_0$ holomorphy can be established from jointly measurable, pointwise holomorphic kernels with a common integrable fibre-supremum envelope on each compact parameter disc. The dominated Cauchy argument used before eq. 4.dxx, now in this complete normed space, proves that assertion. A proved norm power series is another sufficient verification. Pointwise holomorphy without this norm control is not the premise of eq. 4.dxlii.

The other premise is the strongly weighted real family over the same field core, uniformly for every real parameter in the stated interval. Real field values only on the physical-current graph would not be that premise. The theorem neither enlarges the current core nor proves admission of a noncovariant source array to the ordinary complex source class. A moving domain must first be replaced by a single common admitted domain with the two stated bounds. Here "real" concerns the source parameter. If the desired norm takes its supremum over a complex field core, the real-source premise already takes its supremum over that same complex field core.

If these checks hold with $M = e^{-a_\star P_i}$, the theorem proves the same weighted complex mass and its Cauchy bounds on the smaller source disc. It is thus an alternative to proving the pointwise one-zone estimate in eq. 4.dxxi. Each route has a finite, testable analytic input. The local-excess route uses actual carrier and read bounds. The interpolation route uses an ordinary complex kernel bound and arbitrarily prepaid strong real weights on the same admitted core. Neither route follows from compactness of source parameters or scalar partition holomorphy.

4.18.17.5. A dimension-free real-form corollary. Let $\mathcal{B}$ be a complex Banach source space with an isometric conjugation κ. Its real form is $\mathcal{B}_{\mathbb{R}} = \{u : \kappa u = u\}$. Thus every h has $h = h_1 + ih_2$ with real h_1, h_2 and $\|h_j\| \le \|h\|$. Finite coordinate arrays with their usual real-form supremum or sum norms have this property, with constants independent of the number of coordinates.

Suppose F is Fréchet-holomorphic from $\{\|h\| < R\}$ into $\mathfrak{E}_0$, bounded by M, and for every real u with $\|u\| < R$ it obeys the $\mathfrak{E}_C$ bound M on the same field fibres. For any $c \ge 0$ the prepaid coefficient $C = 2(c+2)$ suffices to give

$$\sup_{\|h\|<R/8} \|F(h)\|_{\mathfrak{E}_{c+1}} \le M, \qquad F : \{\|h\| < R/8\} \longrightarrow \mathfrak{E}_c \text{ is Fréchet-holomorphic.} \tag{4.dxlvi}$$

The number of source coordinates does not enter the constant.

Proof. Fix h in that smaller ball. If $h_2 = 0$, the asserted weighted bound is part of the real premise. Otherwise put $v = h_2/\|h_2\|$ and consider the single translated complex line $f(z) = F(h_1 + zv)$. For $|z| < R/2$ its source norm is less than $5R/8$, so its ordinary bound is M. For real $|z| \le R/4$ it is a real source of norm less than $3R/8$, and has the prepaid weighted bound. Apply eq. 4.dxlii with

ordinary radius $R/2$, real half-segment $R/4$ and target radius $R/8$. Since $h_0 > 1/2$, the chosen C is strictly more than sufficient for weight $c+1$. Evaluation at $z = i\|h_2\|$ gives the first assertion.

This estimate is uniform on the whole smaller source ball, not just on one line. Therefore $\|(1-P_N)F(h)\|_{\mathfrak{E}_c} \le Me^{-N}$ uniformly there. Each $P_N F$ is Fréchet-holomorphic into $\mathfrak{E}_c$ by its bounded linear projection from $\mathfrak{E}_0$. Their uniform norm limit is Fréchet-holomorphic. One direct verification uses the Cauchy formula on every complex line in an interior source ball and polarizes its homogeneous Taylor coefficients. Uniform convergence gives convergence of those coefficients; polarization gives convergence in each multilinear norm, and the Cauchy bounds make their Taylor series converge uniformly in a smaller ball. Its sum is the asserted limit. □

The derivative constants in this assertion can be made explicit as well. Write $\rho = R/8$. The m-th homogeneous Taylor polynomial at zero is bounded by $M\rho^{-m}$ on the unit source ball, by scalar Cauchy on each line. For its symmetric m-linear form, the coefficient of $z_1 \cdots z_m$ in its value at $\sum z_j h_j$ is $m!$ times that form. Cauchy's formula on m unit circles bounds this coefficient by m^m times the polynomial bound, since $\|\sum z_j h_j\| \le m$ when $\|h_j\| \le 1$. Consequently

$$\|D^m F(0)\|_{\mathcal{B}^m \to \mathfrak{E}_c} \le m^m M(8/R)^m \le m! M(8e/R)^m. \tag{4.dxlvii}$$

These are multilinear operator norms. No coordinatewise entropy or factor equal to the dimension of the source space has been introduced.

For a normalized excess with $F(0) = 0$, scalar Schwarz on every source line and Hahn–Banach scalarization give $\|F(h)\|_{\mathfrak{E}_c} \le (8M/R)\|h\|$ in the smaller ball. For a general kernel family the difference $F(h) - F(0)$ has the corresponding bound with $2M$. This is a weighted estimate on the complete kernel, not a bound on a chosen logarithm of that kernel. A logarithm requires its own zero-free normalization.

The corollary is useful precisely because its premises say which source arrays are allowed. If the real source bound is known only on a gauge-covariant subspace, the translated real point h_1 and direction v must both lie in that subspace. The theorem then extends its complexification, not the whole ambient array space. It does not create a noncovariant extension from data on the physical-current graph. If the ordinary complex family is available on a radius depending on an order parameter p_n, that same radius, with the fixed factor eight, appears in eq. 4.dxlvii; no larger radius has been supplied by interpolation.

For a genuine array extension one must verify all three statements: the ordinary kernel map is Fréchet-holomorphic and bounded in $\mathfrak{E}_0$ on its actual complex source ball; the real weighted bound holds on the full real ball used in the argument; and all those families use the same field/read fibres and fixed reference data. These are formation and domain assertions. Once they hold, the weighted extension, its source-dimension independence and its derivative bounds follow from eq. 4.dxxxvi–eq. 4.dxlvii.

4.18.18. *The finite estimates to be matched.* The weighted-history assertion requires the following estimates for finite operations. Their quantifiers are uniform in the height, compatible torus, component, local history, stored variables on its support, and the full joint argument domain. Compactness at a single cutoff is not that uniform statement.

First, every stored-coordinate formula must be the literal operation in eq. 4.cdlxv or eq. 4.cdlxxi, with its fixed measures, normalizers, tests and frame factors. Later reads must satisfy eq. 4.dxxv. The five region lists and retained seam prescriptions determine this check. Full re-entry must use the current datum on its remade shell, not an earlier field already integrated there.

Second, every prepared, localized and substituted datum must belong to its joint background/current core. Its complete kept forms must have the lower bound and reference domination in eq. 4.dii or eq. 4.diii on their respective coordinate domains, with local dimension counted once. Independent-current completion needs its projected gauge and other current terms. It does not follow from setting a coarse averaging constraint equal to the identity. The flat precision and one-step perturbation results verify restricted cases, not the whole all-layer assertion.

Third, the sum assigned to V_X needs eq. 4.cdlxviii on those cores for all supported stored fields. Its counterterm, vacuum and boundary conventions belong to this bound. Oscillation of isolated terms does not bound their constants. A pointwise KP logarithm does not supply the integrable supremum without its common envelope. The local formation and Gaussian theorems address parts of this check under their primitive matrix and background hypotheses.

Fourth, failed tests need their certificates and owned energy sets in eq. 4.div. The core Wilson term keeps its stopped real weight, and every compatible drop-out evaluation must preserve the certificates used under its supremum. A last test requiring an extra collar supplies its renewed gain only in a later read with that collar present.

For parameters these checks hold on one fixed-frame domain. The finite gauge-normalizer extension and the complete Gaussian half-disc criterion have been proved here. For a second-class component formed by the island kit, the pair reader gives the exact source excess on the admitted output core. The bound following eq. 4.dxvi uses the formed carrier estimates there. At other weighted source lines or drop-out fibres we must check eq. 4.dxiii on the actual read, for example by the carrier criterion eq. 4.dxv. eq. 4.dxxi is a precise one-zone test for a finer weighted source extension. Ordinary source compactness and holomorphy alone do not prove that test.

Alternatively the full kernel family can meet the explicit same-core ordinary complex and prepaid real norm hypotheses of eq. 4.dxlii or eq. 4.dxlvi. Those theorems then prove its weighted extension without the pointwise excess test. They do not assert those formation or domain premises for a source-array class outside the stated hypotheses.

The loaded global comparison also needs the literal label representation, common vacuum price, local unmarked energy price and finite read-ownership map. These are more specific identifications than the claim that a density is a sum of labels. Once matched, eq. 4.cdlxxv–eq. 4.dxxxv give the counting, reserve, integration and summation with constants uniform in height and volume. They retain the field, current and history types needed to apply those scalar consequences.

4.19. **Complex stage integrals with retained tests.** Preparatory integrations of the large-field operation integrate a stored fluctuation on a strip and a new fluctuation on the adjacent collar; see [B89a, B89b]. The earlier stage terms enter later integrals. The estimates below control this accumulation on one component. They also give a branch of the logarithm when the actual tested integral is close, pointwise in its density, to a positive reference integral. All coordinate maps and all retained factors are specified before this comparison is applied.

4.19.1. *The tested local integral.* Let $b \in \mathbb{R}^m$ be the actual free real coordinates of a finite stage chart. A parameter σ includes its geometry, real background, real stored spectators and retained tests. Fix

$$S_\sigma \subset [-T,T]^m, \qquad T \geq 1, \qquad Q_{0,\sigma} = Q_{0,\sigma}^{\mathsf{T}} > 0.$$

The set S_σ is Borel. It can be disconnected and can contain arbitrary carried sharp tests. On each complex parameter domain used below it is fixed. Put

$$I_{0,\sigma} = c_{0,\sigma} \int_{S_\sigma} e^{-b^{\mathsf{T}} Q_{0,\sigma} b/2 + V_{0,\sigma}(b)}\, db, \qquad 0 < I_{0,\sigma} < \infty, \tag{4.dxlviii}$$

where $c_{0,\sigma} > 0$ and $V_{0,\sigma}$ is real. No uniform lower bound on $I_{0,\sigma}$ is assumed.

For z in a complex polydisc $\mathcal{D}$, suppose the actual integral has the exact representation

$$I_\sigma(z) = c_{0,\sigma} \int_{S_\sigma} e^{-b^{\mathsf{T}} Q_{0,\sigma} b/2 + V_{0,\sigma}(b)} e^{H_\sigma(z,b)}\, db. \tag{4.dxlix}$$

Here H is Borel in the real variables and holomorphic in z, with a uniform bound on the polydisc. In an application it contains

$$H = \ell - \tfrac{1}{2} b^{\mathsf{T}} (Q - Q_0) b + (V - V_0) + \log(J/J_0). \tag{4.dl}$$

The term ℓ includes every change of a Gaussian normalizer. The factor J/J_0 includes the nonlinear constraint determinant, Haar densities and other chart Jacobians. These terms are retained even when the quadratic exponent alone is small. An analytic reciprocal determinant is used at complex arguments; an absolute determinant is used only at real arguments.

Proposition 4.70 (A tested integral in a disc about its reference)**.** If $\sup_{\sigma,z,b\in S_\sigma}|H_\sigma(z,b)|\le\delta\le 1/8$, then

$$\left|\frac{I_\sigma(z)}{I_{0,\sigma}}-1\right|\le e^{\delta}-1<1,\qquad \left|\log\frac{I_\sigma(z)}{I_{0,\sigma}}\right|\le 4\delta. \tag{4.dli}$$

The logarithm is the analytic branch defined by the power series about 1. For a polydisc with coordinate radii R_i, centred at z_*, and a multi-index $\alpha\neq 0$,

$$|\partial_z^\alpha \log I_\sigma(z_*)|\le 4\delta\,\alpha!\prod_i R_i^{-\alpha_i}. \tag{4.dlii}$$

Let $\nu_{0,\sigma}$ be the probability measure obtained by dividing the integrand in eq. 4.dxlviii by $I_{0,\sigma}$, and let $\nu_{z,\sigma}$ be its complex reweighting by e^H. Then

$$\int|d\nu_{z,\sigma}-d\nu_{0,\sigma}|\le 8\delta. \tag{4.dliii}$$

These bounds are uniform when the positive reference masses tend to zero. ■

Proof. By eq. 4.dxlix, the ratio is $w=\int e^H\,d\nu_0$. Thus $|w-1|\le r:=e^\delta-1$. For $\delta\le 1/8$, one has $r<1/4$, and $|\log w|\le r/(1-r)\le 4\delta$. Dominated integration gives holomorphy of the ratio. Its values remain in the simply connected disc $|w-1|<1$; hence this logarithm is coherent throughout $\mathcal{D}$. Apply the several-variable Cauchy formula on smaller concentric polydiscs, then increase their radii to R_i. The reference mass is independent of z, so its logarithm has no derivative in eq. 4.dlii.

The complex density relative to ν_0 is e^H/w. Therefore

$$\int\left|\frac{e^H}{w}-1\right|d\nu_0\le\frac{\int|e^H-1|d\nu_0+|w-1|}{|w|}\le\frac{2r}{1-r}\le 8\delta.$$

No inverse bound on the unnormalized mass has entered this proof. □

4.19.2. *The complete Gaussian and coarea factors.* The comparison can be verified using local matrices before taking any absolute values in an auxiliary Gaussian. In the notation of Prop. 4.7, the exact marginal has covariance $D=C+C\Gamma\Gamma^{\mathsf T}C$. Its auxiliary dimension need not be local. The following bound uses only the number m of free coordinates remaining in the marginal.

Lemma 4.71 (The Gaussian density ratio on a tested box)**.** Suppose

$$\|C_0\|+\|C_0^{-1}\|+\|\Gamma_0\|\le K,\qquad \|C-C_0\|+\|\Gamma-\Gamma_0\|\le\epsilon,$$

where C_0 is real symmetric positive definite and Γ_0 is real, and take $\epsilon\le\epsilon_*(K)$. Let $D_0=C_0+C_0\Gamma_0\Gamma_0^{\mathsf T}C_0$. The normalized marginal density ratio has an analytic logarithm

$$L_D(b)=-\tfrac12\log\frac{\det D}{\det D_0}-\tfrac12 b^{\mathsf T}(D^{-1}-D_0^{-1})b,\qquad \sup_{|b|_\infty\le T}|L_D(b)|\le C_K\epsilon m(1+T^2). \tag{4.dliv}$$

The bound is independent of the auxiliary dimension. ■

Proof. Expanding the products that define D gives $\|D-D_0\|\le C_K\epsilon$, since all rectangular factors are bounded in operator norm. The real matrix D_0 has $D_0\ge C_0\ge K^{-1}I$. For $E=D_0^{-1/2}(D-D_0)D_0^{-1/2}$, decrease ϵ_* so that $\|E\|\le 1/2$. The convergent matrix logarithm gives

$$\left|\log\frac{\det D}{\det D_0}\right|=|\operatorname{Tr}\log(I+E)|\le m\sum_{q\ge1}\frac{\|E\|^q}{q}\le C_K m\epsilon.$$

The resolvent identity gives $\|D^{-1} - D_0^{-1}\| \le C_K\epsilon$. Finally $|b|^2 \le mT^2$. The marginal identity in Prop. 4.7 must be applied before this calculation; a total-variation estimate on the unreduced joint Gaussian would charge its auxiliary dimension. □

Here is the corresponding finite coarea calculation. Suppose the actual constraints in a chart are solved for three-component pivot coordinates, and their real derivative matrices are $M_{0,a}(b)$, $1 \le a \le k$, with positive determinant. Suppose their complex continuations satisfy

$$E_a = M_a M_{0,a}^{-1} - I, \qquad \sup_{z,b} \|E_a\| \le \epsilon_a < 1/2. \tag{4.dlv}$$

Then the logarithm of the ratio of the inverse pivot determinants is

$$L_{\mathrm{piv}}(z,b) = -\sum_{a=1}^{k} \operatorname{Tr}\log(I + E_a), \qquad |L_{\mathrm{piv}}| \le 6\sum_{a=1}^{k} \epsilon_a. \tag{4.dlvi}$$

Indeed the power series used in the preceding proof applies to each three-by-three matrix. The identity powers of L in the per-length pivot convention cancel in the ratio. Their real normalization remains in I_0. Equation eq. 4.dlv requires the actual nonlinear pivot chart on the tested domain.

For SU(2), write $a \in \mathbb{C}^3$ in the orthonormal Pauli basis, using the bilinear square $a \cdot a$. Apart from its fixed normalizing constant, the analytic Haar density in exponential coordinates is

$$j(a) = \left(\frac{\sin\sqrt{a\cdot a}}{\sqrt{a\cdot a}}\right)^2.$$

The quotient at zero is defined by its power series. On $\|a\|_2, \|a_0\|_2 \le 1/4$, it has a nonzero analytic logarithm and

$$|\log j(a) - \log j(a_0)| \le C(\|a\|_2 + \|a_0\|_2)\|a - a_0\|_2. \tag{4.dlvii}$$

To see this, put $f(s) = \sum_{q\ge 0}(-1)^q s^q/(2q+1)!$. For $|s| \le 1/16$, its power series bounds $|f-1|$ away from one and bounds $|f'|$ by an absolute constant. Thus $2\log f(s)$ has bounded derivative. Integrate that derivative along the segment between $a_0 \cdot a_0$ and $a \cdot a$. Summing eq. 4.dlvii over the chart bonds accounts for all Haar density ratios. If the coupling is fixed, the rescaling $a = gb$ has the same Jacobian in the two integrals. A change of g contributes an additional explicit scalar factor.

For comparison, the bare Wilson Taylor remainder also has a direct local bound. For unitary background factors V_i and complex Lie algebra elements B_i, put

$$F(t) = g^{-2}\left(1 - \tfrac{1}{2}\operatorname{Tr}\prod_{i=1}^{4} e^{itgB_i}V_i\right), \qquad S_B = \sum_i \|B_i\|_{\mathrm{op}}.$$

Three differentiations and the multinomial theorem give $|F'''(t)| \le CgS_B^3 e^{gS_B}$ for $0 \le t \le 1$. The integral Taylor formula therefore yields

$$|F(1) - F(0) - F'(0) - \tfrac{1}{2}F''(0)| \le CgS_B^3 e^{gS_B}. \tag{4.dlviii}$$

On a bounded stage box its sum costs at most $Cg\,mT^3 e^{CgT}$ when each coordinate meets a bounded number of plaquettes. Composition with a nonlinear constrained response requires the first three derivatives of that actual response.

Proposition 4.72 (Response of the constrained density)**.** Let the actual constraint chart solve $F(z,b,y(z,b)) = 0$, where F is holomorphic and $M = \partial_y F$ is invertible throughout the admitted domain. Suppose the complete stage density on its fixed tested set is

$$a(z,b)\exp[-S(z,b,y(z,b))]\,(\det M(z,b))^{-1}.$$

All factors are holomorphic and have the uniform bounds needed to differentiate under the finite integral; a includes the Haar densities and numerical normalizers. A local analytic determinant

branch agrees with the positive coarea density at the real reference. With dots denoting differentiation at fixed b,

$$\dot y = -M^{-1}F_z,$$
$$\ddot y = -M^{-1}(F_{zz} + 2F_{zy}\dot y + F_{yy}[\dot y,\dot y]), \tag{4.dlix}$$
$$L = \partial_z \log a - S_z - S_y\dot y - \mathrm{Tr}(M^{-1}\dot M). \tag{4.dlx}$$

On a nonzero domain supplied by Prop. 4.70, the normalized complex stage expectation satisfies

$$\partial_z \log I = \langle L\rangle_z,$$
$$\partial_z^2 \log I = \langle \dot L\rangle_z + \langle L^2\rangle_z - \langle L\rangle_z^2, \tag{4.dlxi}$$
$$\partial_z\langle A\rangle_z = \langle \dot A\rangle_z + \langle AL\rangle_z - \langle A\rangle_z\langle L\rangle_z. \tag{4.dlxii}$$

The second determinant contribution in $\dot L$ is

$$\mathrm{Tr}(M^{-1}\dot M M^{-1}\dot M - M^{-1}\ddot M). \tag{4.dlxiii}$$

For successive triangular eliminations the traces in these formulas are summed over their pivot blocks. ■

Proof. Differentiate the exact constraint at fixed free coordinates. Its first derivative is $F_z + M\dot y = 0$; differentiating again gives eq. 4.dlix. Differentiate the complete logarithmic density, including the reciprocal determinant, to obtain eq. 4.dlx. Jacobi's formula and $\partial_z M^{-1} = -M^{-1}\dot M M^{-1}$ give eq. 4.dlxiii. The quotient rule after differentiation under the integral gives eq. 4.dlxi and eq. 4.dlxii. The tests do not produce a boundary term because they have not been varied. □

For example, suppose $\|M^{-1}\| \le K_M$, $\|F_z\| \le a$, $\|F_{zz}\| \le b$, $\|F_{zy}\| \le c$ and $\|F_{yy}\| \le d$, in the specified Euclidean and multilinear operator norms. Then

$$\|\dot y\| \le K_M a, \qquad \|\ddot y\| \le K_M(b + 2cK_M a + dK_M^2 a^2).$$

For a pivot block of dimension d_p, $|\mathrm{Tr}(M^{-1}\dot M)| \le d_p K_M \|\dot M\|$. These estimates identify which constraint derivatives a quantitative source bound needs. In particular the term $S_y\dot y$ is retained unless the actual stationarity equations cancel it. A derivative of a Gaussian potential at fixed chart alone would omit these contributions.

4.19.3. *Resetting the stored fields.* A near stored field is a real variable under a sharp test. Its reset is represented by a complex switch. Let a_v be such a field, and let μ_v range over $|\mu_v| < \Lambda_v$, where $\Lambda_v > 1$. The term at $\mu_v = 0$ is required to be independent of a_v; at $a_v = 0$ it is independent of μ_v. At a far stored field the same notation represents the radial parameter in the holomorphic substitution $a_v \mapsto \mu_v a_v$. The carried tests are kept at their real arguments in both cases.

Lemma 4.73 (Block differences)**.** Let f be holomorphic and bounded by B on $\prod_{v\in T}\{|\mu_v| < \Lambda_v\}$, together with its other holomorphic arguments. For $A \subset T$, set

$$f_A = \sum_{E\subset A} (-1)^{|A|-|E|} f(\mathbf{1}_E, \mathbf{0}_{T\setminus E}), \qquad q_v = (\Lambda_v - 1)^{-1}.$$

Then

$$f(\mathbf{1}_T) = \sum_{A\subset T} f_A, \qquad |f_A| \le B\prod_{v\in A} q_v, \qquad \sum_{A\neq\emptyset} |f_A| \le B\left(\exp\sum_{v\in T} q_v - 1\right). \tag{4.dlxiv}$$

If f is independent of a_v, μ_v, then $f_A = 0$ whenever $v \in A$. The differences preserve the full declared radii of the remaining stored fields and switches. ■

Proof. The finite inclusion-exclusion identity gives the first equality. For one variable and $1 < r < \Lambda_v$, Cauchy's formula gives

$$f(1) - f(0) = \frac{1}{2\pi i} \int_{|w|=r} \frac{f(w)}{w(w-1)}\, dw.$$

Apply this identity in each variable of A, with all other variables in T set to zero. The bound is $B \prod_{v\in A}(r_v - 1)^{-1}$; increase r_v to Λ_v. Summing this bound gives $B[\prod_v(1+q_v)-1]$, which is bounded by the last expression in eq. 4.dlxiv. Variables absent from the function have zero difference. The proof fixes the other arguments at arbitrary points of their full domains, so it spends none of their radii. □

For a family (f_e), let T_e be its set of read blocks and B_e its analytic bound. Its incidence norm on the integrated blocks is

$$J = \sup_{v\in T} \sum_{e: v\in T_e} B_e. \tag{4.dlxv}$$

If $\sum_{v\in T_e} q_v \le 1$ for every e, then

$$\sum_e \sum_{A\neq\emptyset} |(f_e)_A| \le 2 \sum_{v\in T} q_v \sum_{e: v\in T_e} B_e \le 2J \sum_{v\in T} q_v. \tag{4.dlxvi}$$

This counts blocks actually read by a term. It introduces no Gaussian coordinates on the rest of that term's domain.

A background displacement has the same estimate in an auxiliary parameter. If $f_e(P_e(\lambda))$ is holomorphic and bounded by B_e on $|\lambda| < a_e^{-1}$, with $a_e \le 1/2$, then

$$|f_e(P_e(1)) - f_e(P_e(0))| \le 2a_e B_e. \tag{4.dlxvii}$$

Consequently, if $P_e(1)$ is the old background read and $P_e(0)$ is the new one, the exact split is

$$\begin{aligned} f_e(P_e(1);a) - f_e(P_e(0);0) = {} & f_e(P_e(1);a) - f_e(P_e(0);a) \\ & + f_e(P_e(0);a) - f_e(P_e(0);0). \end{aligned} \tag{4.dlxviii}$$

The first line is bounded with eq. 4.dlxvii; the second uses eq. 4.dlxvi. A displacement that fixes the stored data at their carrying sites is needed for this identity to be the split of the actual operation.

4.19.4. *Uniform accumulation in a finite sequence of stages.* We give the quantitative conclusion in a form that can be applied to a fixed chain of actual tested charts. This is a local assertion for one component. Let $\theta > e$, $\ell = \log\theta$, and $r \ge 1$. Fix constants independent of the stage number, and assume

$$N \le C_N \ell^r, \qquad K_s \le C_s \ell^{4r}, \qquad m_n \le C_m \ell^{4r}, \qquad T_n \le C_T \ell^p. \tag{4.dlxix}$$

At most K_s stored blocks are reset at a stage. Put $\Lambda = \ell^{6r+1}$, and suppose every such block has this switch radius. The bounds hold on the full domains of the switches that remain after integration.

For each remaining stored datum, assume that setting its switch to zero makes both the complete reference kernel and the inserted integrand independent of that datum. Setting the datum to zero makes them independent of its switch. This includes the retained tests and any normalization of the reference kernel. Thus integration preserves the two independence properties defining a switch.

Let J_0 bound the incidence norm of the earlier terms. Suppose that each earlier stage contributes at most one function at each block, with norm at most one; its incidence norm before stage n is then at most $J_0 + n$. The admitted background displacement satisfies the separate budget

$$\sum_e a_{n,e} B_{n,e} \le C_a \ell^{4r-e}(J_0 + n), \qquad e \ge 6r + 1. \tag{4.dlxx}$$

This budget concerns the individual old functions and their response amplitudes. It does not assume a logarithm of the stage integral.

Assume that the sum of the Gaussian density ratio, the coarea and Haar ratios, and the intrinsic interaction remainder is bounded by

$$G_n \le C_g \theta^{-1/2} \ell^A \tag{4.dlxxi}$$

on the tested box. Here A is fixed. For example, eq. 4.dliv gives this form when $\epsilon_n \le C\theta^{-1/2}\ell^a$; then its contribution has $A = a + 4r + 2p$. Equations eq. 4.dlvi, eq. 4.dlvii and eq. 4.dlviii give the same form under the corresponding chart bounds. All these estimates must hold on the full remaining switch domains.

Proposition 4.74 (The logarithmic balance of the stages)**.** Suppose each stage has eq. 4.dxlix, with the old terms split by eq. 4.dlxviii, and the preceding bounds hold. There is a floor θ_*, depending only on the displayed constants, r, p, A and J_0, such that all the normalized stage integrals are nonzero for $\theta \ge \theta_*$. Their logarithms $C_n = \log(I_n/I_{0,n})$ satisfy

$$\sum_{n=0}^{N-1} \|C_n\|_\infty \le C\left[\theta^{-1/2}\ell^{A+r} + \ell^{-1} + J_0\ell^{-r-1}\right]. \tag{4.dlxxii}$$

In particular each $\|C_n\|_\infty \le 1$, so the stated incidence bound closes by induction whenever the next tested chart admits the preceding functions on their declared switch domains. The norm in eq. 4.dlxxii is a supremum over those domains, at fixed real spectators. ■

Proof. Increase the floor so that $q = (\Lambda - 1)^{-1} \le 2\ell^{-(6r+1)}$ and $K_s q \le 1$. The exact exponent change at stage n is bounded, by eq. 4.dlxvi and eq. 4.dlxvii, by

$$\delta_n = C_g\theta^{-1/2}\ell^A + 4C_s\ell^{-2r-1}(J_0 + n) + 2C_a\ell^{4r-e}(J_0 + n). \tag{4.dlxxiii}$$

For $n < N$, this tends to zero uniformly as $\theta \to \infty$. Choose the floor so that every $\delta_n \le 1/8$ and $4\delta_n \le 1$. Prop. 4.70 proves existence and the bound $\|C_n\| \le 4\delta_n$. This supplies the bound one required at the next stage, rather than using the final sum as an inductive assumption.

Sum eq. 4.dlxxiii. The stored-field term is at most

$$C\ell^{-2r-1}(NJ_0 + N^2) \le C\big[J_0\ell^{-r-1} + \ell^{-1}\big].$$

The response term is at most

$$C\ell^{4r-e}(NJ_0 + N^2) \le C\big[J_0\ell^{5r-e} + \ell^{6r-e}\big],$$

which has the same bound since $e \ge 6r + 1$. The intrinsic term sums to $C\theta^{-1/2}\ell^{A+r}$. These are eq. 4.dlxxii. The reference factors remain in the exact equality $\prod_n I_n = (\prod_n I_{0,n}) \exp(\sum_n C_n)$. They can be absorbed into vacuum constants only when they are independent of the retained pair. □

The power $6r + 1$ pays for two stage counts and one block count:

$$N\,\Lambda^{-1}\,K_s\,N \le C\ell^{r-(6r+1)+4r+r} = C\ell^{-1}.$$

This is the role of the enlarged stored-field domain. Its admission is a condition on the actual nonlinear averaging and background maps. Since $g = \theta^{-1/2}$, a fixed power of ℓ times g still tends to zero. This arithmetic alone does not prove that an asserted enlarged chart exists.

4.19.5. *Preservation of switches under integration.* For the induction just used, holomorphy is preserved by the integral and the logarithm under a common integrable bound. Independence at a zero switch requires a further condition on the actual kernel. For a remaining stored datum a_v, suppose that, when $\mu_v = 0$, the integration domain and the complete normalized reference kernel are independent of a_v, and that the non-reference integrand has the same independence. Then the normalized ratio I/I_0 and its logarithm are independent of a_v there. The unnormalized integral has this independence when its reference mass I_0 does as well. If $a_v = 0$ removes μ_v from the integration domain, the normalized reference kernel and the inserted integrand, the ratio and its logarithm are

independent of that switch. No smaller switch disc is needed: dominated holomorphic integration and the fixed logarithm branch apply at arbitrary points of the original disc.

This condition cannot be inferred from freezing the tests only in the complex variable. For example, on $0 < b < 1$, let $S_a = (0, a)$, $0 < a < 1$, and use normalized Lebesgue measure on S_a. The function $F(b) = b$ does not read a, but its expectation is $a/2$. Thus a spectator-dependent test can create dependence on that spectator even when the inserted function has none. In the stage application, any such dependence must either stay outside the integral as a retained factor or be included in the declared switch structure of the kernel.

4.19.6. *A common envelope and the scope of the local result.* The preceding non-vanishing proof is relative to a positive reference integral. A mass estimate for an upper integral of a supremum is a different consequence. Fix the geometry, the number m of free coordinates and its site count N_0. Here is a sufficient condition in these same coordinates. Suppose, uniformly in all real read parameters,

$$Q_{0,\sigma} \geq \gamma I, \qquad c_{0,\sigma} \leq C_0^m, \qquad V_{0,\sigma}(b) \leq B_0 N_0 + \tfrac{1}{4}\gamma |b|^2, \qquad m \leq c_0 N_0. \tag{4.dlxxiv}$$

The complete unnormalized density then has the Borel envelope

$$\sup_{\sigma, z} \left| \mathbf{1}_{S_\sigma} c_{0,\sigma} e^{-b^{\mathsf{T}} Q_{0,\sigma} b/2 + V_{0,\sigma}(b) + H_\sigma(z,b)} \right| \leq C_0^m e^{B_0 N_0 + \delta} e^{-\gamma |b|^2/4}. \tag{4.dlxxv}$$

Its integral is at most $\exp(CN_0 + \delta)$, by the Euclidean Gaussian integral. The supremum need not itself be measurable: the displayed Borel function bounds its upper integral. The common coordinates, the coercive bound and the determinant bound in eq. 4.dlxxiv must hold before the supremum is taken. Separate parameter-dependent changes of variables do not establish this envelope.

The local conclusions require the exact stage chart, the Gaussian marginal identity, the enlarged switch and displacement domains, and the stated kernel condition at zero switches. For simultaneous components, their interaction and the spatial filing of logarithmic terms require the local Gaussian and connected expansion estimates of § 4.5. The finite-component bound here supplies neither their decoupling matrix estimates nor a common chart for an arbitrary history.

Proposition 4.75 (Sharp tests can force a zero without switches)**.** Let w_0 be integrable on a real measure space, let $W = \int w_0 \neq 0$, and let $x \geq 0$ have an atomless push-forward of $|w_0|\, d\mu$. Then there are $c > 0$ and $\lambda \in [-\pi, \pi]$ such that

$$\int w_0 e^{i\lambda \mathbf{1}_{\{x > c\}}}\, d\mu = 0. \tag{4.dlxxvi}$$

■

Proof. Set $A(c) = \int_{x \leq c} w_0$ and $B(c) = W - A(c)$. Atomlessness makes both continuous in c, with $A(0) = 0$, and $A(c) \to W$ as $c \to \infty$. The continuous function $|A(c)| - |B(c)|$ is negative at zero and positive for sufficiently large c. At a zero it has $|A(c)| = |B(c)| > 0$, since $W \neq 0$. Choose $e^{i\lambda} = -A(c)/B(c)$. This gives eq. 4.dlxxvi. □

For a free three-color strip coordinate, $x = |\mathcal{A}^{\mathrm{s}}(b)|$ is atomless under an integrable Lebesgue density. Its norm is invariant under the adjoint action. The phase in eq. 4.dlxxvi can therefore be split among four local terms, each of modulus at most $\pi/4$, without changing analyticity in the background pair. The switch hypothesis rules out precisely this uncontrolled dependence on a tested stored field.

4.20. **Source formation and volume bounds.** A source bound requires three estimates: formation of local interactions, convergence of the polymer logarithm, and a bound on the resulting integral. A convergent polymer expansion supplies the second estimate. To apply it to a renormalisation construction one must first verify its activity bounds for the interactions that the construction

actually forms. The elementary implications below specify the ordinary source estimates needed for that verification.

Site-dependent members and the summation over old levels. Fix an odd integer $L \geq 3$. Let $(\theta_j)_{0\leq j\leq n}$ satisfy

$$\theta_j \geq Y, \qquad \lambda(n-j) - c_5 \leq \theta_j - \theta_n \leq B(n-j), \qquad Y \geq 2(c_5+1), \tag{4.dlxxvii}$$

where $\lambda > 0$ and $c_5 \geq 0$. A pointed family at level j is a collection $G(X,P,z)$, where X is a localisation domain, $z \in X$ is a base point and P belongs to a specified complex field domain. Its norm is

$$\|G\|_{j,\kappa} = \sup_{X,z,P} e^{\kappa d_j(X)} |G(X,P,z)|. \tag{4.dlxxviii}$$

Any companion families used to define its quadratic coefficient are included in this supremum. The field domains and restrictions are part of the data; holomorphy does not assert that a function extends beyond these domains.

At every site z, permit a different family $G_{j,z}$. Each such family comes with its own quadratic coefficient $b(G_{j,z})$. The hypothesis on an individual member is covariance of its *pointed* total, so that subtracting its coefficient times the tent action cancels the marginal quadratic term at that base point. No covariance relating the chosen members at different sites is required. This distinction permits a scalar source to be frozen at the value belonging to each site.

Here is a precise quantitative formulation of the input used in this argument. Put $\epsilon_{jn} = L^{-(n-j)}$, and let ρ_{jn} be the ratio between the field radius required at level n and the radius on which the member of level j is given, enlarged to be at least one. Suppose $\rho_{jn}^2 \leq C_\rho(1+n-j)$. Assume a linear localisation rule with the following estimates, including summability over its rooted domains. For one base point of level j, a normalised regular member contributes at most

$$C\,\|G_{j,z}\|_{j,\kappa}\,\rho_{jn}^3 \epsilon_{jn}^{4+\alpha} \qquad (0<\alpha<1) \tag{4.dlxxix}$$

to a window of level n, after its vacuum and quadratic subtractions. A member with its vacuum value alone subtracted contributes at most

$$C\,\|F_{j,z}\|_{j,\kappa}\,\rho_{jn}^2 \epsilon_{jn}^4. \tag{4.dlxxx}$$

The constants include the permitted loss of spatial decay and the count of domains through the base point. The rule must evaluate every member within its stated field domain for every allowed decoupling and window parameter. This domain inclusion is a separate hypothesis of eq. 4.dlxxix–eq. 4.dlxxx.

The factor ϵ_{jn}^4 has a direct dimensional meaning. A window of level n contains at most $C\epsilon_{jn}^{-4}$ base points of level j. Thus vacuum subtraction balances the number of base points in four dimensions, whereas the quadratic subtraction must supply the additional ϵ_{jn}^{α} to make old regular members summable.

Proposition 4.76 (Uniform ordinary input)**.** Suppose the regular members obey

$$\sup_z \|G_{j,z}\|_{j,\kappa} \leq G_\star.$$

Suppose defects and remainders obey, respectively,

$$\|D_j\|_{j,\kappa} \leq L^{-(n+1-j)}, \qquad \|H_j\|_{j,\kappa} \leq C_H \theta_j^{-p}, \qquad p>2. \tag{4.dlxxxi}$$

Under eq. 4.dlxxix–eq. 4.dlxxx, their combined ordinary input at level n is bounded by

$$CG_\star + C\sum_{a=0}^{n}(1+a)L^{-(a+1)} + CC_H\sum_{a=0}^{n}(1+a)(\theta_n+\lambda a - c_5)^{-p} \leq CG_\star + C + CC_H\theta_n^{2-p}. \tag{4.dlxxxii}$$

The constant is independent of n, the number of sites and the choice of the member at each site. ∎

Proof. The sum in eq. 4.dlxxix over the base points in a window. The inverse volume factor cancels ϵ_{jn}^4, leaving

$$CG_\star \sum_{a\ge 0} (1+a)^{3/2} L^{-\alpha a} < \infty.$$

This uses only the supremum of the member norms, so choosing a different covariant member at each site has no effect on the bound. The same cancellation in eq. 4.dlxxx gives the second sum in eq. 4.dlxxxii for defects. That sum is bounded by a convergent geometric series and its first derivative.

For remainders use eq. 4.dlxxvii and put $y = \theta_n - c_5 \ge \theta_n/2$. For $a \ge 0$, $1 + a \le C(y + \lambda a)$, since $y \ge 1$. Comparison with the integral of the decreasing function $(y + \lambda t)^{1-p}$ gives

$$\sum_{a\ge 0} (1+a)(y+\lambda a)^{-p} \le C\left(y^{1-p} + \int_0^\infty (y+\lambda t)^{1-p}\, dt \right) \le C y^{2-p}. \tag{4.dlxxxiii}$$

This proves the stated, deliberately nonoptimal, bound. □

For SU(2), a bounded scalar action weight supplies another admissible ordinary input. If $W \in SL(2,\mathbb{C})$, then

$$1 - \tfrac{1}{2} \operatorname{tr} W = \tfrac{1}{2} \det(1 - W). \tag{4.dlxxxiv}$$

On a field domain where each plaquette satisfies $|W_p - 1| \le \alpha_n \eta_n^2$, its $\operatorname{tr}(F^2)$ is bounded by $C\alpha_n^2 \eta_n^4$. A local cube contains at most $CM^4 \eta_n^{-4}$ plaquettes. Filing each plaquette in a domain containing its corners therefore gives a weight member of norm at most $CM^4 \alpha_n^2 \sup |u|$. This proof uses the determinant identity and the count of plaquettes; it does not use covariance of u. With $\alpha_n = O(\theta_n^{-1/2} (\log \theta_n)^q)$, any fixed bound on $\sup |u|$ is eventually admissible.

These summations do not prove eq. 4.dlxxix. That estimate contains the renormalisation assertion: covariance of the pointed Hessian, the correct counterterm, regularity of the background within the window and uniform membership after substitution must be proved for the actual formation rule. They cannot be replaced by a bound on the full total or by holomorphy in the source alone.

Exponential reserves and generated local perturbations. Let a local history ω have a positive measure μ_ω on S_ω, a reference exponent $\Phi_\omega(\xi; P)$, and a site count N_ω. Suppose its reserve volume V_ω is additive over the older histories swallowed by this history and satisfies

$$V_\omega \ge A N_\omega, \qquad \sum_\omega \int_{S_\omega} e^{c_\star V_\omega} \sup_P |e^{\Phi_\omega(\xi;P)}|\, d\mu_\omega(\xi) \le \varepsilon, \qquad c_\star \ge 1. \tag{4.dlxxxv}$$

The supremum in eq. 4.dlxxxv is assumed measurable, or is replaced by a measurable majorant throughout. If a holomorphic perturbation satisfies $|\Psi_\omega| \le f N_\omega$ and $A \ge f$, then for bounded F

$$\left| \sum_\omega \int F e^{\Phi_\omega + \Psi_\omega}\, d\mu_\omega \right| \le \varepsilon \|F\|_\infty. \tag{4.dlxxxvi}$$

Indeed, $|e^{\Psi_\omega}| \le e^{V_\omega}$, so take absolute values inside every integral and apply eq. 4.dlxxxv. If each integrand is holomorphic in the parameter and all reference measures and integration sets are parameter independent, a common integrable majorant on compact parameter sets also proves holomorphy by Fubini and Morera. This is an ordinary complex mass estimate. It requires no derivative with respect to the number of source zones.

Suppose a formation rule produces a new excess exponent in the explicit form

$$\Psi_\omega^{\mathrm{new}} = A(\widehat{c}\, \upsilon 1_X, U''(\xi)) + V^{\mathrm{col}}(X^\sim; \mathfrak{L}) - V^{\mathrm{col}}(X^\sim; \mathfrak{L}^0) + \sum_R \delta_R[\psi_R], \tag{4.dlxxxvii}$$

where $U''(\xi)$ is real, $0 \le \upsilon \le 1$ and $\sup|\widehat{c}| \le \bar{c}$. To apply eq. 4.dlxxxvi it is sufficient to prove the actual formation bounds

$$|A(\widehat{c}\,\upsilon 1_X, U''(\xi))| \le C_W \bar{c}\, N_\omega^0, \qquad |V^{\mathrm{col}}(X^\sim;\mathfrak{L})| + |V^{\mathrm{col}}(X^\sim;\mathfrak{L}^0)| + \sum_R \sup|\delta_R[\psi_R]| \le 2c_{\mathrm{int}}(1+\bar{c})N_\omega^0. \tag{4.dlxxxviii}$$

If $\Psi_\omega = \Psi_\omega^{\mathrm{new}} + \sum_{\omega'} \Psi_{\omega'}$ and $N_\omega = N_\omega^0 + \sum_{\omega'} N_{\omega'}$, these inequalities give

$$|\Psi_\omega| \le \max\{C_W\bar{c} + 2c_{\mathrm{int}}(1+\bar{c}), f\}\, N_\omega. \tag{4.dlxxxix}$$

The maximum, rather than a sum of the old and new constants, is what permits unbounded nesting. Preserving the carrier oscillation norm needed in the next formation step is an additional estimate; it does not follow from eq. 4.dlxxxix.

There is a genuine reason to generate local perturbations through this formation rule. A bound on an exponent and on the oscillation of each carrier does not bound arbitrary partial sums of carrier factors. On a one-point integration space, write $\Psi^\circ(\lambda) = -N\lambda$ and $\psi_1(\lambda; P) = N\lambda$, with a second, unused carrier if desired. Then $\Psi^\circ + \psi_1 = 0$, both carrier oscillations in P are zero, and the full mass is unchanged for every λ. Interpolating only ψ_1, while leaving Ψ° at its value at zero, instead gives the mass $e^{N\lambda}$. No bound uniform in N survives on a fixed complex disc. This example refutes that partial-interpolation argument, not a construction which defines its localized exponents recursively by eq. 4.dlxxxvii and proves eq. 4.dlxxxviii on the whole complex data box.

Local data lines in the source induction. Use the fixed real step and its source radius r. Construct the constant-source data first. If the radial increment bounds through level k sum to at most $1/3$, the running constant shift satisfies $|\zeta_k(\zeta)| \le 4|\zeta|/3$. The reserve $2r$ puts the shifts with $|\zeta| < r$ in the real weight tube. The doubled disc $|\zeta| < 2r$ instead uses the larger analytic range of the increment maps. The frame bound $\theta_{i+1} \le \theta_i + C$ carries the upper bound on the tube width at the next scale back to $C'\theta_i(\log\theta_i)^{-C_1}$ at scale i. Choose the tube constant smaller than the constant in the analytic range. The fixed extra reserve $2r/3$ is then available at weak coupling. Thus the previously formed shifts and remainders are admitted when computing $b_{k+1}(\zeta)$ on the doubled disc. The summable increment bound keeps the same $1/3$ bound for the enlarged list. The regular constant-source step supplies the covariant member with this coefficient, before the position-dependent source is introduced.

At level $k+1$ let W be a region containing the complete input reads of the output under consideration. For a source w and a constant comparison value ζ_0 in $D(0,r)$, let m be the source deviation on W, including the scaled Lipschitz bound at level $k+1$. Form the data line at ζ_0 by adding t times the difference from w. Move the site values and the entries with their complete one-layer enlargements inside W. For a boundary sum make this choice at each of its summands. Keep the reference entries outside W. The linear consistency rules are checked at these same indices.

The inherited slope bounds put the line in the enlarged data box for $|t|\mathsf{m} < r$. In detail, the site source has modulus at most $2r$. If $\mathsf{d}_{2r}[b_i]$ is the bound for $(b_i(\zeta) - b_i(0))/\zeta$ on $D(0,2r)$, the increment has Lipschitz constant at most $2\mathsf{d}_{2r}[b_i]$ on $D(0,r)$, by Cauchy's estimate for $b_i - b_i(0)$ on the doubled disc. Thus on the line its deviation is at most $3r\mathsf{d}_{2r}[b_i]$. The sum of the $\mathsf{d}_{2r}[b_i]$ is at most $1/3$, so the site source and the accumulated shifts fit the $3r$ bound. For an earlier defect at level i, the gradient scale is smaller by $L^{-(k+1-i)}$. The product $|t|\mathsf{m}/r$ is at most one. The other rooted entries use their bound at the constant source and $|t|$ times their inherited slope, on the same carrier.

Define the excess exponents on this line by applying the formation operations to the line. At the bare level there are no excess exponents. A region for the new grid is a region for the preceding grid, the new source class is contained in the preceding one, and the deviation at the earlier scale is at most m. Thus the earlier line is available on the same disc. The still-standing exponents are carried from it. At a new operation apply eq. 4.dlxxxvii and eq. 4.dlxxxviii to the data on the line.

The fresh count and inherited counts add, so the maximum in eq. 4.dlxxxix keeps the same total bound at every depth. This defines the line also when W cuts a large-field component. At $t = 0$ it gives the constant-source output. When the disc contains $t = 1$, the line there gives the sourced output on complete reads in W, by locality of the actual operations and of the lines at the inherited operations.

The carrier oscillation has its own bound on the data box. At a new island the common quadratic factors contribute zero to the source difference. The remaining factors are differences of the formed interaction pieces, with the whole island carrier $C^{\sim}$. Its length after exempting the island hull is zero. Normalize the sum of oscillations by $\ell_t^{2\mathfrak{p}_0}$, with the length weight of the carrier. The local pieces on $C^{\sim}$ cost $C(1+\bar{c})\check{R}_t^4$ before normalization, and the source-dependent Wilson pieces have a uniform bound. These contributions give a fixed $b_V^{\times}(\bar{c})$. The difference of an inherited factor is evaluated by the kept response at fixed stored variables. Subtract its value at the common real pair before taking the response difference. The weighted sum of these oscillations contributes at most $C_V^{\times}\vartheta(\theta_t)s$, where s is the admitted normalized bound of the old carriers and

$$\vartheta(\theta_t) = (A_1/C_0)(\log\theta_t)^{-(q-\mathfrak{p}_0)}.$$

This estimate sums over the individual pieces and therefore also bounds their grouped oscillations. Standing factors retain their carriers and restrict their old functions.

Choose $c_\times \geq 2b_V^{\times}(\bar{c})$. On the enlarged data line admit $s \leq 4c_\times$. With $8C_V^{\times}\vartheta(\theta_t) \leq 1$, the new normalized bound is at most $c_\times$. Thus the new carriers have the same room on the next data line. This argument concerns the oscillation at fixed stored variables. For the full source excess use the absolute estimate of eq. 4.dxvi, including the parts constant in the pair.

For a holomorphic output bounded by B on the line disc, Schwarz applied after subtracting its value at zero bounds the difference of the endpoint values by $2B\mathsf{m}/r$. If $\mathsf{m} \geq r$, the separate bounds at the two endpoints give the same inequality. If $\mathsf{m} = 0$, the two reads coincide. For carrier oscillations first subtract the common reference value in each factor. Suppose their sum of suprema on one carrier is bounded by B throughout the line. For any finite collection choose evaluation points and phases to make the endpoint differences nonnegative and arbitrarily close to their individual suprema. The corresponding phased sum is a single holomorphic function bounded by B on the line. Apply the same Schwarz estimate to it and take the suprema and then the increasing sums. This preserves the carrier oscillation slope separately from the slope of the total exponent.

For the regular pointed coefficient the defect needs the bound with unit slope and a spare unit of spatial decay. The Gaussian normalizer is common to the line and the constant source. After removing it, the cluster contribution and the correction for the real weight relative to the frozen one have combined bound

$$\left(C_V\vartheta_k^{\sharp} + C_{\mathbf{M}}\frac{\mathsf{t}_k}{\theta_k}\right)e^{-(\kappa+1)d_{k+1}(X)}.$$

Here $\vartheta_k^{\sharp}$ includes the small fluctuation-radius ratio, and t_k is the real weight tube radius. The ordinary absolute estimate gives this bound on the direct regular windows. For an interior pointing the hulls of the input vertices and the complete Gaussian guards are in the small-field region, so the bracket uses its ordinary legs without a stored exterior mean. The normalizer and the free ratio term use the same real reference in both members of the comparison. The constant source chosen first at each base supplies its own pointed subtraction. The local line uses the direct estimates on its complete bases. Any other origin entering the regular rule must satisfy its own carrier and admission bounds on this line. Taking the parenthesis at most $1/2$ gives the unit slope by the preceding Schwarz estimate. Compare with the constant $\zeta_0 = w(p_z)$ at the pointing z. The spare unit of decay pays for the diameter of the source read, leaving the factor $L^{-(k+1-i)}$ in a defect of level i when it is read at level $k+1$. The uniform box bound supplies the output slope. The input slopes only supply admission to that box.

Form the torus regular output of $\mathbf{T}$ before choosing its new extension. At each pointing z choose the Euclidean member at the constant value $w(p_z)$, with marginal coefficient $b_{k+1}(w(p_z))$. The pointed slope bounds the difference between the actual regular output and this member, admitting the defect of the extension. For estimates along the complex data line, interpolate the extension members and their coefficients affinely between the constant and sourced endpoints. On $|t|\mathsf{m} < r$ the member norm is at most $E_0 + |t|\mathsf{m}/r \le E_0 + 1$, and the summed coefficient deviation and site source obey the same $3r$ scalar bound. For the extension defect use the two regular outputs with the common normalizer removed and the affine change of the member. The first two terms are bounded by $\frac{1}{2}e^{-\kappa d}$ each and the affine change by $e^{-\kappa d}$. The defect therefore fits the enlarged box on the entire line. Applying $\mathbf{R}$ there returns the slope bounds for the new boundary and remainder terms by the same Schwarz estimate. Carried terms keep their previous slopes on their native reads.

For the actual sourced endpoint set

$$c_{k+1}(p; w) = \varphi_{k+1}(p) \sum_{y} h_y^{(k+1)}(p) b_{k+1}(w(p_y)).$$

Use the real-center update $x_{k+1}^{\natural} = x_k^{\natural} - b_{k+1}(0)\varphi_{k+1}$ for the position-dependent weight, and update the running shift by $\zeta_{k+1} = \zeta_k + c_{k+1} - b_{k+1}(0)\varphi_{k+1}$. Then at every plaquette

$$x_k^{\natural} - \zeta_k = x_{k+1}^{\natural} - \zeta_{k+1} + c_{k+1}.$$

Thus the pointed subtraction $-A(c_{k+1}, U)$ pays exactly for the change of the Wilson coefficient. The profiles, tests and integration kernels come from the frozen step throughout. On the admitted formation domains the exactness of $\mathbf{RT}$ with this extension gives the sourced output in the resummed standing classes with their retained variables, on the same source class for the next step.

What a pointwise polymer logarithm proves. Consider a finite hard-core polymer gas Ξ_y, depending on an exterior field y. Suppose its activities $h_y(C)$ satisfy a uniform Kotecký–Preiss criterion, including absolute convergence of the logarithmic series and the bound

$$|\log \Xi_y| \le \sum_{C} |h_y(C)| e^{a(C)} \le c\, N. \tag{4.dxc}$$

If the activities are real for real y, then $\Xi_y > 0$. One proof is that the absolutely convergent logarithmic series is real. Equivalently, replace every activity by $t h_y(C)$, $0 \le t \le 1$. The same criterion applies, so $\Xi_y(t)$ is real, continuous and nonzero. Its value at zero is one, proving positivity at one. Individual activities need not be positive. Moreover $e^{-cN} \le \Xi_y \le e^{cN}$.

If a final representation really has the form $Z = \int q(y)\Xi_y \, d\nu(y)$, with $q \ge 0$, then

$$e^{-cN} \int q \, d\nu \le Z \le e^{cN} \int q \, d\nu. \tag{4.dxci}$$

Thus even a pointwise polymer logarithm requires control of the remaining integral. It does not turn a restricted Haar integral into a probability integral. For example, take the empty gas and let q be the indicator of a Haar set of measure e^{-N^2}. The polymer criterion is automatic, but a lower bound e^{-CN} for Z is false.

In a representation that sums over standing labels, the useful upper estimate instead has the form

$$\sum_{\mathcal{S}} |e^{\mathcal{E}_{\mathcal{S}}(w)} \tau_{\mathcal{S}}^{w}|(\mathcal{V}_n) e^{W(\mathcal{S})} \le e^{cN_n}. \tag{4.dxcii}$$

Here the absolute value is total variation, W pays for missing vacuum contributions and the measures inside the retained integral operations are also majorised before they are integrated. A pointwise bound for one gas does not supply this sum: arbitrarily many positive labels can have the same individually bounded gas. The counting of labels and the exponential reserve in their operations are essential.

For the lower bound at zero source, positivity permits retaining the all-small-field label. Suppose its density, after vacuum normalisation, is at least $\chi(y)\exp[-xA(U(y)) - CN_n]$. Assume the ball in which every one of the $4N_n$ link variables has distance less than s from the identity is contained in $\{\chi = 1\}$, and that $A(U(y)) \le Cs^2N_n$ there. Normalized Haar measure on SU(2) assigns this link ball mass at least $(cs^3)^{4N_n}$. It follows directly that

$$e^{-E}Z(0) \ge \exp[-(C + 4\log(1/(cs^3)) + Cxs^2)N_n].$$

Taking $s = c_0x^{-1/2}$ gives $e^{-E}Z(0) \ge \exp[-(C + 6\log x)N_n]$. The terminal background estimates needed here are proved in the calculation on a product of link balls. No zero-free statement for the sourced full partition function is needed.

4.21. **Weighted domains and scale forgetting.** A multiscale argument needs several analyticity statements with different meanings. Holomorphy of a finite-dimensional partition function is the weakest of them. Holomorphy of a renormalization step on a class of effective interactions is stronger. Geometric forgetting of old interactions is stronger still. The latter can follow from an analytic extension with appropriate domains and bounds, but the weights in that extension must themselves be justified.

There is an additional distinction for large-field regions. A small probability of retaining an old region can appear as a small factor in the bound on an analytic function. It does not in general permit the analytic input radius to grow by the reciprocal of that probability. This section defines a suitable input space and states the local inequalities needed for continuum reconstruction. Their validity for the Yang–Mills step remains to be proved.

4.21.1. *The ordinary input space.* Fix a cutoff, a reference sequence of inverse couplings $\theta_0, \dots, \theta_k$, a complete set of labels and tests, and hereditary complex domains $\mathcal{U}_j(X; r_{0,j}, r_{1,j})$. A point of such a domain is a pair $(\mathbf{U}, \mathbf{J})$; the current $\mathbf{J}$ is an independent complex variable. A regular background gives the special pair $(U, J(U))$. An identity verified at these special pairs is not automatically an identity at all pairs in the complex domain.

One convenient Banach space of pointed local families has the norm

$$\|E\|_{j,\kappa} = \sup_{X,z} e^{\kappa d_j(X)} \sup_{\mathbf{p}\in\mathcal{U}_j(X;r_{0,j},r_{1,j})} |E(X,\mathbf{p},z)|.$$

When families have infinite-lattice companions, include their norms in the maximum. Gauge invariance, the prescribed translation covariance, and compatibility with restriction are closed linear conditions. Holomorphy is preserved under convergence in this supremum norm on the declared domains. These facts specify a closed space of analytic families once the domains themselves have been fixed. The existence and uniform size of those domains remain analytic inputs.

A regular Euclidean member comprises a family G, its companion, and its scalar quadratic coefficient $b(G)$. It has covariant pointed totals on the prescribed regular backgrounds and carries its own counterterm. A spatially varying input can assign a different such member $G_z^{(j)}$ to every base point z. This does not require the array $z \mapsto G_z^{(j)}$ to be translation invariant. Its norm is $\sup_z \|G_z^{(j)}\|_{j,\kappa}$. Each assigned member must retain its own coefficient and the required pointed covariance; covariance of the sum of all base points is a weaker condition.

The complete input also contains a bare plaquette weight u, defect families $D^{(j)}$, remainders $H^{(j)}$, boundary terms B, and excess exponents Ψ in stored large-field operations. A useful product box has inequalities of the following form:

$$\|u\|_\infty + \sum_{j\le k} \sup_z |b(G_z^{(j)}) - b_j^{\mathrm{ctr}}| \le C_u, \qquad \sup_z \|G_z^{(j)}\|_{j,\kappa} \le E_0,$$
$$\|D^{(j)}\|_{j,\kappa} \le L^{-(m-j)}, \qquad \|H^{(j)}\|_{j,\kappa} \le C_H \theta_j^{-\kappa_0/2}, \qquad m \in \{k, k+1\}.$$

Here b_j^{ctr} denotes the coefficient of the chosen real centre. Boundary terms are measured against specified positive majorants $\overline{B}_{j,h}(X)$:

$$\|B\|_{\mathrm{bdry}} = \sup_{j,h,X} |B_{j,h}(X)|/\overline{B}_{j,h}(X).$$

For a stored history ω let $N(\omega)$ count its stored sites, each at its own scale. Excess exponents require both a bound on $|\Psi_\omega|/N(\omega)$ and a carrier-by-carrier oscillation bound. If $\Psi_\omega = \Psi_\omega^0 + \sum_f \psi_f$, the latter can be expressed as

$$\sup_{\omega,R} \overline{M}_\omega(R)^{-1} \sum_{f: R_f = R} \operatorname{osc} \psi_f < \infty.$$

The carrier majorant and the history size are part of the definition, not consequences of holomorphy.

A source trajectory fits this format by freezing the source separately at each base point. Schematically,

$$G_z^{(j)} = E^{(j),\zeta}\big|_{\zeta = w(p_z)},$$
$$D^{(j)}(X,\cdot,z) = E^{(j),w}(X,\cdot,z) - E^{(j),w(p_z)}(X,\cdot,z).$$

For a Lipschitz source, a local freezing estimate can give $\|D^{(j)}\| \le CL^{-(K-j)}\|w\|_{\mathrm{Lip}}$. This smallness is a property of the source trajectory. An arbitrary defect in an ordinary box has only the size stipulated by that box.

This organization resolves a possible misconception. Global noncovariance does not by itself invalidate a construction with sources: Euclidean members can be frozen site by site. The remaining task is to prove the asserted bounds for the full array, including its noncovariant defects, its variable Wilson weight, and its stored-history operations. Scalar marginal cancellation for a globally covariant list alone does not complete that task.

4.21.2. *Three kinds of weights.* Let e_j measure a difference of regular families and let r_j measure a difference of remainders. The regular irrelevance weights have the form

$$\omega_\nu \le C\nu^4 L^{-\sigma(\nu-1)},$$
$$\sigma > 0.$$

They represent decay after the vacuum and the applicable marginal quadratic terms have been removed. In contrast, an unrenormalized remainder can enter with a squared ratio of analyticity radii,

$$w_{n,j} = \left(\frac{r_{0,n}}{r_{0,j}}\right)^2 \asymp \frac{\theta_j}{\theta_n}\left(\frac{\log\theta_n}{\log\theta_j}\right)^{2q}.$$

These weights account for the small size of the remainder. Geometric forgetting and any further symmetry cancellation require separate estimates.

A typed tangent list may also have a marginal amplitude z_j and an increment b_j. Define

$$z_n^\sharp = \sum_{j\le n} L^{-(n-j)}(z_j + z_{j-1}),$$

$$z_{-1} = 0,$$

and a load

$$p_n(\delta) = \sum_{j\le n} \omega_{n+1-j} e_j + \sum_{j\in\mathcal{J}_n} w_{n,j} r_j + \gamma_n \left(z_n^\sharp + \sum_{j\le n} L^{-c(n-j)} b_j \right),$$

$$\gamma_n \asymp \theta_n^{-1} (\log \theta_n)^C.$$

The set $\mathcal{J}_n$ consists of remainder slots still present at level n in the chosen filing convention. If a remainder is re-filed into a regular family, it must be removed from this set and counted in that family instead. A recent-level window can have a separate term without geometric age decay. This bookkeeping cannot be suppressed by changing notation.

The survival age of a large-field zone has a different role. For $0 < q_a < 1$ the desired integrated bound involves

$$\Sigma_k(\delta) = \sum_{n\le k} (k-n+1) q_a^{(k-n-1)_+} p_n(\delta).$$

The factor $k-n+1$ counts the steps at which the zone can be used. The survival factor $q_a^{(k-n-1)_+}$ belongs to an output estimate on the relevant histories. It need not, and generally cannot, be put into a single enlarged analytic input radius.

4.21.3. *What weighted holomorphy actually proves.*

Proposition 4.77 (A segment estimate)**.** Let X, Y be complex Banach spaces, let $C \subset X$ be convex, and let p be a continuous seminorm. Suppose F is holomorphic on

$$\mathcal{T} = \{a + h : a \in C,\, p(h) < \epsilon\},$$

$$\epsilon > 0,$$

and, for every $a \in C$ and $p(h) < \epsilon$, one has $\|F(a+h) - F(a)\|_Y \le B$. Then, for $a, b \in C$,

$$\|F(b) - F(a)\|_Y \le (B/\epsilon) p(b-a). \qquad \blacksquare$$

Proof. Put $h = b - a$. At every $a_t = a + th$, $0 \le t \le 1$, the map $z \mapsto F(a_t + zh) - F(a_t)$ is holomorphic on $|z| < \epsilon/p(h)$ and has norm at most B. The Banach-valued Cauchy formula gives $\|DF(a_t)h\| \le Bp(h)/\epsilon$. Integrating over t proves the inequality. If $p(h) = 0$, the line is entire and bounded in oscillation. Every bounded linear functional on Y makes it a bounded scalar entire function, so Liouville's theorem and separation by the dual imply that it is constant. □

The seminorm in this proposition is an assumption. Applying it with $p(h) = \sum_j \omega_{k+1-j} \|h_j\|$ produces the displayed weights because they already occur in the domain. It does not derive them from an ordinary product box.

A differential bound is a weaker sufficient input. If $\|DF(a)h\| \le Cp(h)$ holds on the segment joining two actual trajectories, integrating this inequality proves the same comparison without constructing a full weighted tube around every off-trajectory point. Thus a derivative estimate along these segments suffices.

Proposition 4.78 (A marked slot)**.** Suppose $F(a,m)$ is jointly holomorphic when $p(a-a_0)<r_a$ and $\|m\|<r_m$, is bounded there by B, and satisfies $F(a,0)=0$. For $p(a-a_0)\le r_a/2$ and $\|m\|\le r_m/2$,

$$\|D_aF(a,m)h\|\le\frac{CB}{r_ar_m}\,p(h)\|m\|.$$ ■

Proof. Cauchy's formula first in the complex line $a+zh$, and then in the line tm, bounds the mixed derivative by $CBp(h)\|m\|/(r_ar_m)$ on the reduced domains. Integrate that derivative from $m=0$ to m. The derivative in a vanishes at $m=0$ because $F(a,0)$ is identically zero. The argument uses two complex lines and introduces no factor equal to the dimension of either input space. □

This explains how a small marked-slot size can multiply the sensitivity to other data. It does not establish the required radii, the marked-slot vanishing identity, or a uniform bound on the enlarged domain.

4.21.4. *Exact examples separating the assertions.*

Proposition 4.79 (Ordinary analyticity does not imply forgetting)**.** Fix $0<q<1$ and any positive numbers s_k. There are entire scalar functions F_k bounded by s_k on the unit polydisc of $\mathbb{C}^{k+1}$ for which no constant C independent of k satisfies

$$|F_k(x)-F_k(y)|\le Cs_k\sum_{j=0}^{k}q^{k-j}|x_j-y_j|$$

on the polydisc of radius $1/4$. ■

Proof. Take $F_k(x)=s_kx_0$, $x=0$, and $y=(1/4,0,\dots,0)$. The inequality would give $1\le Cq^k$. This is impossible as $k\to\infty$. The same example is holomorphic on every weighted tube, but its supremum on a tube of fixed weighted radius grows like s_kq^{-k}. Thus the missing issue can be a uniform bound, even when the extension itself is entire. □

This example is a logical separation, not a model satisfying the gauge-theory power-counting identities. It shows exactly why analyticity and a small one-step factor alone do not establish those identities.

Proposition 4.80 (Rarity does not enlarge the complex radius)**.** Let $0<q<1$, let a be a nonnegative integer, and let an event have probability q^a. For $\delta>0$ its excess generating function can be

$$R_a(\lambda)=q^a(e^{\lambda\delta}-1).$$

On $|\lambda|\delta<c$ it has a bound C_cq^a, and $R_a'(0)=q^a\delta$. There is no age-independent bound C_cq^a on the larger disc $|\lambda|q^a\delta<c$. ■

Proof. The first bound follows from $|e^z-1|\le e^{|z|}+1$. At the positive point $\lambda=cq^{-a}/(2\delta)$ the ratio $R_a(\lambda)/q^a$ equals $e^{cq^{-a}/2}-1$, which tends to infinity. The event model is a genuine two-point probability space, and its generating function is entire. □

The correct large-field strategy is therefore a bounded per-zone radius together with a rare-history factor in the bound. A demand for a bounded extension on a single tube measured only by the aged load would be stronger than what the argument needs and can fail even in this elementary example.

There is a separate reason for the Euclidean-member and marginal-slot conditions. Let

$$H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

$$M_\varepsilon = I + \varepsilon H.$$

For real $|\varepsilon| < 1$, take independent centered Gaussian vectors $X_1, \dots, X_m$ with precision matrix M_ε. The normalized block average $m^{-1/2} \sum_i X_i$ has exactly the same covariance M_ε^{-1}, hence exactly the same precision matrix. The traceless quadratic perturbation $\varepsilon x_1 x_2$ survives this blocking unchanged, although its scalar trace coefficient is zero. The Gaussian integrals are holomorphic and uniformly coercive on each fixed complex disc $|\varepsilon| < r < 1$.

This example explains why a scalar counterterm cannot remove an arbitrary anisotropic quadratic perturbation. It does not contradict the use of a scalar counterterm in the Euclidean class. It also explains why a typed extension must supply all of its prescribed tensor counter-insertions before claiming that the remaining part is irrelevant.

Finally, identities on backgrounds do not determine an arbitrary extension to independent currents. The polynomial $P(z, j) = j - z^2$ vanishes on every pair (z, z^2) and is nonzero elsewhere. An analogous cancellation known only on a graph of background pairs needs an additional argument before it can be used throughout a domain where the current is independent.

4.21.5. *Two local inequalities to establish.* The first test concerns the small-field potentials before exponentiation. Let $\dot{V}_{k,j}$ be the variation of the actual localized fluctuation potential caused by one regular family at level j. Include its own scalar counterterm, or the complete tensor counter-insertions for a typed family. A suitable test is

$$\|\dot{V}_{k,j}[\delta E]\|_{\mathrm{pot}} \le C \varkappa_k \omega_{k+1-j} \|\delta E\|_{j,\kappa}. \tag{4.dxciii}$$

The supremum in the potential norm must range over the complex pairs and decoupling parameters actually used inside the fluctuation integral, not just real backgrounds. One possible norm is the least M such that

$$|\dot{V}(Y; \mathbf{p}, B)| \le M e^{-\lambda d_k(Y)} \left(1 + M_0^{-4} \sum_{b \subset Y} |B(b)|^2 \right).$$

Here M_0 is the fixed block-size parameter. The complex field domains, the allowed fluctuation cut, and the enlarged decoupling discs must be specified. Defects and remainders have their own test with squared-radius weights in place of ω. A variable Wilson weight is a marginal slot; it does not receive an irrelevance weight merely by being written as a local family.

For ordinary site-frozen Euclidean members, a regular-output dependence estimate already has this intended structure after the cluster expansion. The additional typed claim requires the same cancellation at the complex pairs in the integrand, with the full tensor removed. A background extraction formula alone is insufficient for that extension. Proving eq. 4.dxciii for these typed inputs is a concrete local task.

The second test concerns a single zone in a large-field operation. Fix a zone n and a step k' at which it is used, where $n \le k' \le k$. Let $\lambda_{n,k'}$ multiply only that use of a direction of load $p_n(\delta)$. Isolate the normally convergent subseries of histories affected by this dial, and denote the resulting output by $R_{k;n,k'}(\lambda)$. The required axis estimate has the form

$$\sup_{|\lambda| p_n(\delta) < c} \|R_{k;n,k'}(\lambda)\|_{k,\kappa} \le C \epsilon_k q_a^{(k-n-1)_+}. \tag{4.dxciv}$$

The constants c, C are independent of the levels, the history, and the positions of the sources. The source is allowed to be complex on the declared source ball. The factor ϵ_k is the small mass assigned to the output under consideration. For an unintegrated boundary term it must be replaced by that

term's actual majorant; its residual dependence is subsequently paid under the relevant mass, rather than being assigned a false termwise survival bound.

Cauchy's formula applied to eq. 4.dxciv gives

$$\|\partial_\lambda R_{k;n,k'}(0)\|_{k,\kappa} \le (C/c)\epsilon_k q_a^{(k-n-1)_+} p_n(\delta).$$

Summing over the $k-n+1$ uses of that zone, and then over n, gives $(C/c)\epsilon_k \Sigma_k(\delta)$. This proves the differentiated age estimate from a precise bounded complex-mass statement. The proof needs no age-dependent increase of the analytic radius.

A complete verification must establish eq. 4.dxciv under the moduli and suprema used by the operation, including its reserve weights. Undifferentiated rarity at the unperturbed source alone does not establish it: the dial changes the integrand on the rare histories. Conversely, a full off-trajectory analytic tube is not necessary if these axis estimates and the needed first and mixed derivatives have been proved directly on the actual trajectories.

4.21.6. *What remains conditional and what can proceed separately.* The ordinary product-box theorem, a weighted dependence estimate for its regular output, and the finer marked and zone-wise statements are different inputs. Coefficient readout uses the physical radii of the chain, as proved in § 4.23: fixed radii give a uniform norm, and running radii carry the inverse-square factor already recorded by the radius weights. Near-zero core inclusion is supplied by the stated tower bounds. The remaining finer assertions concern actual marked and typed evaluations in complex windows, with their proper radius normalization. When a linearized class includes these properties among its hypotheses, contraction rows derived from that class remain conditional on the class hypothesis. Such a deduction does not turn an ordinary complex-box bound into the weighted or zone-wise estimates.

For a prescribed whole-sequence continuum limit, the smallest useful replacement is an actual comparison of the two tuned trajectories: a step inequality, domination of the observable evaluations by the comparison distance, and a terminal-integral inequality with summable defects. These three estimates can be proved directly. Establishing a large weighted analytic class is one route to them, not a logical necessity.

Finite-cutoff source identities do not require this finer statement. Fixed-spacing DLR compactness and closedness of reflection positivity do not require it either. Subsequential separated-correlation compactness follows from its own cutoff- and volume-uniform source bounds. Those bounds remain substantial analytic requirements; their volume-free proof can use undifferentiated age-resolved rarity without proving the differentiated zone estimates above. Similarly, tuning at one fixed cutoff can follow from its own holomorphic slope and target-crossing estimates without establishing a comparison of consecutive cutoffs.

The proposed summable matched-cutoff comparison and the resulting whole-sequence tuned convergence use the stronger comparison route. The typed tangent contractions, scale-uniform memory estimates, and differentiated anisotropy method for rotations also require their corresponding finer inputs. Estimates for several insertions at nonzero source require joint bounds beyond a single first derivative at zero. An independent vanishing-flux theorem would provide a different route to rotations and should be assessed on its own hypotheses.

Thus the unresolved point is a specific family of uniform estimates on the declared inputs, not a general objection to sources or a proof that the construction fails. The potential test and the one-zone complex-mass test locate two places where a proof can be supplied, sharpened, or tested by an explicit counterexample.

4.22. **A reserve estimate for one large-field zone.** The one-zone estimate must bound a complexly perturbed integral; rarity at a real centre alone does not suffice. This section proves a reserve lemma, verifies the direct Wilson-weight channel on regular zones, and tests whether carrier oscillation and a prepaid end mark supply the remaining perturbation bound. The result is partial. The missing statement can be expressed as a norm bound for an explicit local evaluation map. The reserve argument alone does not prove that statement for every formed copy of a direction.

4.22.1. *The integral before taking a logarithm.* Keep the reference sequence, tests, history labels, changes of variables and reference kernels fixed. A local operation is then represented on a disjoint union of stored-variable spaces by

$$I(\lambda, p) = \sum_{\omega} \int_{S_\omega} e^{\Phi_\omega(\lambda, \xi; p)}\, \mu_\omega(d\xi),$$

where each μ_ω is positive and independent of λ. A harmless finite frame average can be included in these measures. The parameter p ranges over the external complex-pair domain. A single zone-use parameter has an index (n, k'): it multiplies the chosen direction where the stored zone n is used at step k'. Only histories containing that use are affected.

Before re-exponentiation, the substitution of an input list by a line of lists is affine in the exponent when its evaluation maps have been frozen. Its perturbation has the concrete form

$$L_{\omega,n,k'}[\delta](\xi, p) = \sum_{e \in \mathcal{E}(\omega,n,k')} \epsilon_e\, \delta F_e\big(\kappa_{\omega,e}(\xi, p)\big) + \delta c_\omega(\xi). \tag{4.dxcv}$$

The index e records the entry, its counter-insertions, its carrier, and its particular occurrence. The signs or fixed coefficients are ϵ_e. The formed argument $\kappa_{\omega,e}$ must lie in that entry's original analytic domain. The last term records scalar parts, including the values subtracted at the trivial field and their contributions to the vacuum bookkeeping. Constants cannot be omitted from this map merely because the observable ultimately uses a connected derivative.

For this affine use one has $\Phi_\omega(\lambda) = \Phi_\omega(0) + \lambda L_{\omega,n,k'}[\delta]$. After local histories or polymer clusters have been summed, the resulting effective families are generally nonlinear in λ. Their control requires a convergent marked expansion; their logarithms need not remain affine. If the formed arguments themselves are changed with the input, their derivatives are additional terms in eq. 4.dxcv; they require their own domain and norm bounds.

4.22.2. *What an exponential reserve proves.*

Proposition 4.81 (A one-zone reserve lemma)**.** Let the fixed affected event E_a consist of histories of age at least a in a measurable disjoint union of history spaces with positive reference measure. Assume that the integrands are measurable in the stored variables and holomorphic in λ for almost every stored configuration. Suppose $W_\omega(\xi) \geq 0$ is measurable, and assume a measurable majorant M_ω satisfies

$$\sup_p e^{\operatorname{Re}\Phi_\omega(0,\xi;p)} \le M_\omega(\xi),$$

$$\sum_\omega \int_{E_a} M_\omega e^{cW_\omega}\, d\mu_\omega \le B\varepsilon q^a,$$

where $c, C > 0$ and $0 < q < 1$. Put $R = c/(2Cp_n)$, interpreted as infinity when $p_n = 0$. Suppose the perturbation is defined and holomorphic on $|\lambda| < R$, and throughout that disc

$$\sup_p |\Phi_\omega(\lambda,\xi;p) - \Phi_\omega(0,\xi;p)| \le C|\lambda| p_n W_\omega(\xi). \tag{4.dxcvi}$$

Then on $|\lambda| p_n < c/(2C)$ the integral over E_a is bounded in modulus by $B\varepsilon q^a$, and is holomorphic. Its difference from its value at zero is bounded by $2B\varepsilon q^a$. Its first derivative at zero is at most

$$(2CB/c)\varepsilon q^a p_n.$$

The same conclusions hold with an additional insertion which is measurable in the stored variables, holomorphic in λ on the same disc, and uniformly bounded there, multiplied by that uniform bound. A fixed bounded insertion is a special case. ■

Proof. The modulus of the perturbed integrand is at most $M_\omega e^{cW_\omega/2}$. The assumed positive majorant is integrable over the disjoint union of histories. Dominated integration, or Morera's theorem and Fubini, proves holomorphy on the open disc. The mass estimate follows from the same domination. The difference is bounded by the sum of the two masses. Cauchy's formula on circles of radius approaching $c/(2Cp_n)$ gives the derivative bound. If $p_n = 0$, the hypothesis makes the perturbation independent of λ, and the derivative is zero. □

Finite upper integrals can replace the measurable majorants in this proposition: choose measurable majorants with integrals arbitrarily close to their upper integrals, apply the argument, and take the infimum. The supremum over external pairs must remain inside the history integral.

A reserve can also produce the age factor. If $W \ge a$ on E_a and

$$\sum_\omega \int M_\omega e^{(c_{\mathrm{pert}}+c_{\mathrm{age}})W}\, d\mu_\omega \le B\varepsilon,$$

then a perturbation paid by $c_{\mathrm{pert}}W$ leaves the estimate $B\varepsilon e^{-c_{\mathrm{age}}a}$. Thus a per-zone radius independent of the age is compatible with geometric forgetting. The perturbation budget and age budget must be allocated before the coupling ceiling is selected.

This argument is effective only after eq. 4.dxcvi has been proved. An oscillation bound for the unperturbed factors does not establish it.

4.22.2.1. The ordinary boundary terms in the zone allowance. Fix a history and time. Count each live first-group boundary term once at its stored level e, with one term per formation domain at that level. Its assigned anchor is a level-e cube in the native anchor region. For a term whose domain misses the live strips, the dial majorant is

$$C_{99} e^{-p_0(g_e)} e^{-(1+\beta/2)\kappa d_e(X)}.$$

At a fixed anchor Q the ordinary rooted domain sum is bounded by K_0 at rate $\kappa - 1$. Since each domain X contains Q, the summed majorant there is at most $C_{99}K_0 e^{-p_0(g_e)}$.

Write $\Gamma_l^\natural$ for the native level-l strata, and use $N_l(S) = \ell_l^{-4}\,\mathrm{vol}(S \cap \Gamma_l^\natural)$ for the volume in units of one native lattice cell. The anchor geometry puts Q entirely in strata with $l \le e$. Since its side is $M\ell_e$, its contribution to the anchor count is

$$1 = M^{-4} \sum_{l \le e} L^{-4(e-l)} N_l(Q).$$

The level-e anchor cubes are disjoint. Thus for anchors contained in S, the summed majorant is at most

$$C_{99}K_0M^{-4}e^{-p_0(g_e)}\sum_{l\le e}L^{-4(e-l)}N_l(S).$$

This uses the native stratum of each cell even when it is finer than the anchor. In a convention counting native M-cubes, absorb their volume factor into the fixed coefficient.

The lower flow window gives $x_e=g_e^{-2}\ge X-2c_2+\lambda(K-e)$ for a run with $x_K\ge X$, including $e=K$. For $p_0(g_e)=A_0(\log x_e)^{p_0}$ with $p_0>1$ we obtain the history-independent bound

$$\sum_{e\le k}e^{-p_0(g_e)}\le\sum_{t\ge0}\exp(-A_0[\log(X-2c_2+\lambda t)]^{p_0}).$$

For large X the series is dominated by the summable inverse squares of $X-2c_2+\lambda t$, so it tends to zero with the coupling ceiling. This gives the all-level allowance for ordinary first-group terms. On domains meeting live strips the size estimate and the counting norm are different. To use a small sum over the levels we must sum the relative estimates over the actual levels of storage. The age estimate below also has a version using a uniform level bound.

4.22.2.2. The relative count in the strip allowance. For the relative comparison family the formation estimate supplies a size $c_1^\flat(g_e)\le Cg_e^{\kappa_0}$, $\kappa_0\ge7$, after removing the relative length weight. Its level sum is bounded by

$$C\sum_{t\ge0}(X-2c_2+\lambda t)^{-\kappa_0/2},$$

which tends to zero with the ceiling. This is a sum of weighted suprema over production domains, before rooting. At an exterior anchor, the rooted relative count converts each supremum to a sum of values. For an activity bound the native site expansion must also supply the rate margin converting those values to activities. The site anchor must stay outside the excused islands with the prescribed separation.

There is a different count for the volume bound in a region Y. When the filed domain is contained in Y and contains an excused island, assign it to the first whole island I it contains. The relative rooted sum at I is at most $1+3^dC_{\mathrm{fib}}N_e(I)$, with $N_e(I)$ the number of its level-e cubes. Thus at each level its cost is proportional to the number of native determining cells of I. Summing over islands of that level uses their disjointness, and summing the size coefficients then uses the level sum. This pays the volume bound on the admitted whole-island reads. A single interior anchor for an activity would lack this whole-island volume.

For the admitted one-run strip terms filed on their tree domains, use the unweighted rooted sum for terms formed by $\mathbf{R}$, after subtracting their declared reference values. Their formation decay $(1+\beta/2)\kappa$ pays the count of domains over each tree at rate λ_θ and leaves at least c_d+1 for the tree count. The resulting bound at a tree root is $Cc_1^\flat(g_e)$. This gives a true-read bound at the formation level e. A direct read at a level-e cube uses a tree root within one layer. A pocket read uses a root within two cells of its strip. There are a bounded number of possible roots for pockets of side at most one. The disjoint larger pockets contribute at most 909^d choices, each with at most $(s_e+7)^d$ roots for side s_e. For a pocket born at $m\le e$, $s_e\le101\check R_mL^{-(e-m)}\le101\check R_e$, by the lower bound on the radius after each step. Thus the reading-cube sum for this class of $\mathbf{R}$-formed terms is at most $C(1+\check R_e)^dc_1^\flat(g_e)$. Since $\check R_e$ is polylogarithmic in inverse coupling and $\kappa_0\ge7$, the bound is at most Cx_e^{-2} at large x_e. The lower flow window therefore also sums it over levels.

For the uniform bound at a fixed level there is a direct count for the dialled family. Use the individual first-group strip domains X before combining their values. At birth such an X contains a union of whole excused strips and has a nonempty part outside them. Its filed skeleton $A(X)$ is chosen from a shortest relative tree. It contains X outside the strips, and each included strip touches it. The term reads $A(X)$ and the attached strips, including the older strips read through a touching cube. The separation of the new strips puts one of these off-strip cubes in the regular

part. The filing rule keeps a cube of the skeleton in $(Z_e^{\boxplus})^{(1)}$ of the current label for a term still stored at level e.

On the admitted age bidisc use the first-group strip majorant $C_{99}\alpha^{1/3}e^{-(1+\beta/2)\kappa d_{\mathrm{rel}}(X)}$. The strip amplitude α is fixed. The skeleton has $d_e(A) \le d_{\mathrm{rel}}(X)$, and there are at most $e^{\lambda_\theta(d_e(A)+1)}$ constituents over one A. Summing the individual majorants over that fibre leaves

$$C_{99}e^{\lambda_\theta}\alpha^{1/3}e^{-(1+\beta/4)\kappa d_e(A)}$$

when $\beta\kappa/4 \ge \lambda_\theta$. This bound also applies to any selected subfamily, so the witness marks may be kept on the individual terms. At a filed cube, the rooted ordinary tree sum gives a pile bounded by $C\alpha^{1/3}$. On the admitted relative-format reads of $\mathbf{T}$, the preparatory move, and the large-field move, the same filed skeleton and attached strips are used for marked and dialled terms. The native cube-to-cell attribution charges this pile to the volume and activity bounds. This gives the uniform part of the age estimate without a summable strip amplitude.

To use the count on every actual mixed history we must admit its fluctuation and preparatory readers on the enlarged age bidisc with the increased history weight. For an inherited crossing in the prepared sharp mask, the quiet-crossing calculation gives the real domains on the approximate pass. On a broader reader we must still account for each pulled-back indicator, and show that the densities on the fixed domains have the required holomorphic majorants. The skeleton sum applies once those readers and majorants have been identified.

4.22.2.3. Carrying the age bound through earlier terms. Compare two lists on the same lattice and scaffold. Give each use of a list entry in a production formula its own parameter. A parameter of class $z = (n, k')$ is bounded using the load of zone n. The diagonal holomorphy proof extends to these separate parameters when it charges each direct use by its absolute size, and reads a previously produced object only through its value and size bound. Indeed the size allowance of every direct input is the same on its own disc. In the order of formation, the earlier inputs retain their admitted suprema, so the same composition estimate gives the next holomorphic object. In this product of discs one may turn off a set of uses without moving the other parameters.

An occurrence means a stored object or a copy of its formula inside an object that read it. The outer domain contains the domain of the copy. For a boundary term of level e its witnessing components are those of Z_e whose enlarged sets meet that domain. A frozen term uses its arrested component. A use with $n < k'$ starts with the geometric witness containing its anchor in $\Omega_n \backslash \Omega_{n+1} \subset Z_{n+1}$. A top-zone use starts in the own set of the object that created it. A fluctuation or an arrest deletes no witness. At healing, a carrier of a use with a deleted geometric witness meets the operated set at its anchor. If an object loses its last witnessing component, the carrier meets the operated set through the domain of that object. The absorption rule places these uses in the newly produced terms. All other uses keep their live witnesses, including those read through copies. This gives a unique current assignment along each occurrence, with witness level $n + 1$ for a surviving geometric witness and level e for a surviving owner of level e. An occurrence carrying a copy need not have the same level as its witness.

In a healed component X_i, the histories that contain a direct old use of zone n must have $Z_{n+1} \cap X_i \neq \varnothing$. For a non-top use this follows from the zone read in the enlarged component and the filling rule for its older cells. An inner top use at level $k' = n$ is supported on $Z_{k'+1} \cap X_i$. A deleted geometric witness must itself be present in Z_{n+1} of the component history. In all these cases the first index of that history is at most $n + 1$. For an absorbed leaf with witness at level e, the component history must contain the witness in Z_e. This also applies if the leaf occurs inside a carrier, because that carrier's formation history includes the leaf's. At operation j these facts give ages at least $j - n - 1$ and $j - e$, respectively.

For a first-group boundary term or a second-group remainder φ formed by $\mathbf{R}_j$, fix a class $z = (n, k')$ of the source uses, and compare the value of φ before and after turning off its own uses. These are the direct old uses of the healed histories, the direct top uses at formation, and the uses

whose witnesses are lost at the operation. Let $\mathfrak{B}_\varphi$ be its holomorphic majorant and $0 < q \le 1/2$ the age rate. The history decomposition partitions the uses into one group of direct and low-level uses and groups of nested leaves of levels $n+2 \le e < j$. A nested leaf in such a group loses its last witness inside a healed component of this operation. The history-absorption rule is needed also for occurrences inside another stored term: a carrier containing such a leaf contains its witnessing anchor and is absorbed with that component.

Turn off the level groups one at a time. For a group of level e, replace a leaf value by its value with the selected uses off plus ν times its change. If the carried bound at level e is $C_*\mathfrak{B}_\psi q^{(e-n-1)_+}$ for each such leaf ψ, take

$$|\nu| \le R_e = [C_*\varepsilon_e^{\mathrm{all}} q^{(e-n-1)_+}]^{-1},$$

where $\varepsilon_e^{\mathrm{all}}$ is the per-cell pile for all such leaves. A zero pile requires no interpolation. This choice makes the changed part at most $\mathfrak{B}_\psi/\varepsilon_e^{\mathrm{all}}$. The summed changed parts therefore cost one additional unit of the per-cell allowance at a direct read with the stored-family count. An intermediate term, once formed, enters subsequent steps by its own majorant.

At the same time, multiply the sub-sums of healed histories of age at least $j-e$ by μ, leaving the younger histories unchanged. The bound on the age tail permits $|\mu| \le q^{-(j-e)}$ at the price of doubling the component input. At $\mu = 0$ the level-e leaves in this group cannot occur: the last-witness condition requires age at least $j-e$ in the component that absorbs the leaf. The term on this face is therefore independent of ν. At $\mu = 1$ the difference between $\nu = 1$ and $\nu = 0$ is the required change for this group. On an admitted bidisc with bound $\mathfrak{B}_\varphi$, subtract both axis values and add the value at the origin. The resulting function vanishes on both axes and has supremum at most $4\mathfrak{B}_\varphi$. Schwarz's estimate in the two variables therefore bounds the group change by

$$4\mathfrak{B}_\varphi\, q^{j-e}/R_e = 4C_*\varepsilon_e^{\mathrm{all}}\mathfrak{B}_\varphi q^{j-n-1}.$$

Here the native formation estimate admits both parameters by two separate charges. First use the enlarged constant in the exponential volume bound for stored values and in the site activity, allowing one more unit of the per-cell stored-value bound at each direct read. The history tail estimate is taken at this enlarged constant, so it is uniform in ν. Then split the sequence sum in each healed component. At $|\mu| \le q^{-(j-e)}$ the two pieces together cost at most twice the unsplit component majorant. This factor is paid in the product over components. At a component formed at step j, the native geometric factor is at most $e^{2\kappa(100\check{R}_j)^4}$. For the native growth number $\beta_0^{\mathrm{pr}} \le 1/7$, the component history bound is at most $e^{1-7p_0(g_j)/4}$ before the split. Thus the doubled bound leaves $e^{-3p_0(g_j)/2}$ by the native coupling ceiling, which implies

$$-p_0(g_j)/4 + 2 + 2\kappa(100\check{R}_j)^4 \le 0.$$

The next formation sum costs a factor $e^{p_0(g_j)/2}$ and retains the original $e^{-p_0(g_j)}$ coefficient per component. This proves the budget on one age disc and one leaf-level disc in the same receiving cluster, with the other interpolations at their endpoints. It uses the admitted per-use formation and the history tail bound at the larger volume and activity allowances.

After the level groups have been turned off, the remaining uses require a healed history of age at least $a = (j-n-1)_+$, except for direct top-level uses when $a = 0$. The single age split at a gives $2\mathfrak{B}_\varphi q^a$ by the same zero-face argument, or by the trivial difference bound at $a = 0$. Summing the levels gives the factor $2 + 4C_*\varepsilon_{\mathrm{tot}}^{\mathrm{all}}$ in front of $\mathfrak{B}_\varphi q^a$. Here $\varepsilon_{\mathrm{tot}}^{\mathrm{all}}$ bounds the sum of the level piles uniformly. Thus $C_* = 4/(1 - 4\varepsilon_{\mathrm{tot}}^{\mathrm{all}})$ closes the induction when the total pile is at most $1/16$.

4.22.2.4. The carry with a uniform bound at each level. There is a second way to close the age estimate. Suppose the actual leaf families have level piles $\varepsilon_e^{\mathrm{all}} \le B$ on the enlarged formation and history domains. Put $A = 1 + 4B$. Choose $0 < q_0 \le q$ with $2Bq_0 \le 1/3$, and set $q' = q_0/A$. Use the history weight increased by $\log(q/q')$. A history of age at least s has volume at least s, so the weighted mass bound gives the share $(q')^s$. The weight is fixed before the coupling ceiling. We must have the formation bounds and the same level-pile bound B at this weight.

With $a = (j-n-1)_+$ use the inductive bound $D_a \mathfrak{B}_\varphi(q')^a$, $D_a = 2A^a$. At a nested leaf of age $b = e-n-1 \geq 1$, the inverse leaf radius is $\varepsilon_e^{\mathrm{all}} D_b(q')^b$, at most $2Bq_0^b \leq 1/3$. The same two-parameter estimate bounds its level group by $4\varepsilon_e^{\mathrm{all}} D_b \mathfrak{B}_\varphi(q')^a$. The direct contribution is at most $2\mathfrak{B}_\varphi(q')^a$. The geometric sum gives

$$2 + 4B \sum_{b=1}^{a-1} D_b \leq D_a$$

with the sum empty for $a \leq 1$. This closes the induction, since $D_a(q')^a = 2q_0^a \leq 2q^a$.

For a term of the second group, its domain is disjoint from the surviving large-field regions and the enlargements that would make the domain first-group. A class use with a surviving geometric witness would put its anchor in one of those regions. A copied occurrence with a surviving witness would put its anchor in the surviving region or its prescribed enlargement, which is excluded for a second-group domain. Thus every class use of a second-group term belongs to its own set, and the carried bound controls the full change of that class. At a native zero-index cluster containing a live strip Y_h and no healed constituent, the only own uses are the direct top uses at formation. There is then no age loss in bounding their change by twice the strip majorant. Uses copied from a surviving witness are still assigned to that witness level. This special case gives a leaf difference bound and still requires the strip count for summation.

4.22.2.5. Reading the transported uses at the later component. Let X be a component of Z_k on which the operation acts, and put $G = \mathbf{T}'_k(X)F$ for a fixed bounded insertion F. In each history summand of G, the filling property takes an older anchor in $X^\sim \cap Z_{n+1}$ into X. A surviving geometric witness at that anchor therefore requires the summand's Z_{n+1} to meet X. An earlier owned use of witness level $e < k$ instead has an anchor of its owner in $Z_{e+1} \cap X^\sim$. The same filling property puts that anchor in X, and its live witnessing component lies in the same surviving component as the anchor. The history containing the occurrence must contain its witness already at e. Hence a use of witness level e requires age at least $k-e$ in the summand. Direct inner top uses and constants attached to a zone satisfy the direct bound $b = (k-n-1)_+$ by the index of the region at which they occur.

To turn off class $z = (n, k')$, first telescope over the own sets of live leaves of levels $n+2 \leq e < k$. These sets are disjoint as sets of uses, and the surviving witness assignment is made in each summand's own history. For a fixed level use the disc of radius R_e above. Its change is supported on histories of age at least $k-e$. The absolute tail majorant for this subfamily, with the extra stored-value unit included, is $\mathrm{e}\,\mathfrak{E}_k q^{k-e} \sup|F|$, where $\mathfrak{E}_k = \exp[-2p_0(g_k)/(1+\beta_0^{\mathrm{pr}})]$. It applies even though different summands use different points of the product of discs, since the tail bound sums the individual majorants. Schwarz's estimate in the single leaf parameter gives $2\mathrm{e}\,\mathfrak{E}_k q^{k-e} \sup|F|/R_e$. The powers $q^{k-e}q^{e-n-1}$ combine to q^{k-n-1}.

The remaining direct uses, geometric uses, and own sets of carriers or leaves at levels at most $n+1$ require age at least b. Their difference costs at most twice the age-b tail bound. Adding the telescope therefore bounds the change of class z by

$$2\mathrm{e}\,\mathfrak{E}_k q^b \sup|F|\,(1 + C_* \varepsilon_{\mathrm{tot}}^{\mathrm{all}}).$$

For the uniform level bound, repeat the telescope with q' and the age-dependent leaf radii. Writing $a = (k-n-1)_+$, it gives

$$2\mathrm{e}\,\mathfrak{E}_k (q')^a \sup|F| \left(1 + B \sum_{d=1}^{a-1} D_d\right) \leq 2\mathrm{e}\,\mathfrak{E}_k q^a \sup|F|,$$

because $1 + B\sum_{d=1}^{a-1} D_d \leq A^a$. On the diagonal of the product of discs this is the difference from setting the class parameter to zero. Apply the one-variable disc estimate in each class in succession. At zone n there are $k-n+1$ classes (n, k'), each with the inverse disc radius given by the zone

load divided by the fixed load allowance. Thus the change between the endpoints is controlled by the load sum with weights $(k-n+1)q^{(k-n-1)_+}$.

4.22.3. *A common reserve for many zone uses.* One-zone bounds do not by themselves give a common polydisc for an arbitrarily large number of zone uses. The reserve also needs a capacity bound. The following precise version distinguishes the two issues.

Proposition 4.82 (A capacity criterion for joint parameters)**.** Let $\mathcal{Z}$ be a finite set of zone-use indices and let i label finitely many directions at each index. Their nonnegative loads are $p_{z,i}$. Use the measurable stored spaces and fixed reference measures of the preceding proposition, and assume that all phase coefficients are measurable in the stored variables. Fix nonnegative measurable history charges $W_{\omega,z}$, a finite family of nonnegative measurable action pieces $S_{\omega,b}$, and nonnegative coefficients $a_{\omega,z,b}$, with

$$\sum_{z\in\mathcal{Z}} W_{\omega,z} \le W_\omega,$$

$$\sum_{z\in\mathcal{Z}} a_{\omega,z,b} \le 1$$

for every b.

Suppose the affine perturbation is

$$\Phi_\omega(\vec{\lambda}) - \Phi_\omega(0) = \sum_{z,i} \lambda_{z,i} L_{\omega,z,i},$$

$$|L_{\omega,z,i}| \le p_{z,i}\left(CW_{\omega,z} + \sum_b a_{\omega,z,b} S_{\omega,b}\right).$$

Assume a majorant at zero obeys, on a fixed event E,

$$\sum_\omega \int_E M_\omega \exp\left(cW_\omega + \sigma \sum_b S_{\omega,b}\right) d\mu_\omega \le B_E,$$

$$\sup_p e^{\operatorname{Re}\Phi_\omega(0,\xi;p)} \le M_\omega,$$

where $c,\sigma>0$. Choose $0<r<\min\{c/(2C),\sigma/2\}$. Then the integral over E is jointly holomorphic and bounded by B_E on the product of weighted balls

$$\mathcal{D}_r = \left\{\vec{\lambda} : \sum_i |\lambda_{z,i}| p_{z,i} < r \text{ for every } z\right\}.$$

All statements use fixed reference measures, fixed events and fixed evaluation maps; if additional parameter dependence is present, the integrands must instead be given holomorphically on this domain with the same phase bound. ∎

Proof. For $\vec{\lambda}\in\mathcal{D}_r$, the triangle inequality and the two capacity inequalities give

$$|\Phi_\omega(\vec{\lambda}) - \Phi_\omega(0)| \le Cr\sum_z W_{\omega,z} + r\sum_b \left(\sum_z a_{\omega,z,b}\right) S_{\omega,b} \le (c/2)W_\omega + (\sigma/2)\sum_b S_{\omega,b}.$$

Thus a single integrable majorant works on the entire parameter domain, independently of $|\mathcal{Z}|$ and the number of directions. Dominated integration proves separate holomorphy and local boundedness, and the finite-dimensional Hartogs theorem gives joint holomorphy. Equivalently, one can apply Morera's theorem successively on each compact polydisc. □

If a designated parameter affects only an event E_a whose majorant has $B_{E_a} \le B\varepsilon q^a$, setting that parameter to zero changes the whole integral by at most $2B\varepsilon q^a$, uniformly in the other parameters on $\mathcal{D}_r$. At their common zero, Cauchy's formula gives the first derivative bound $B\varepsilon q^a p_{z,i}/r$. To see this without subtracting a large unaffected part, differentiate only the integral on E_a; outside E_a the integrand is independent of the designated parameter. For m distinct directions at the same index, Cauchy's formula on circles with radii tending to $r/(mp_{z,i})$ gives

$$\left|\partial_{\lambda_{z,1}} \cdots \partial_{\lambda_{z,m}} I_E(0)\right| \le B_E (m/r)^m \prod_{i=1}^{m} p_{z,i}.$$

Here zero loads mean that the corresponding affine perturbations vanish, and those derivatives are zero. Higher mixed bounds require an appropriate majorant for the history event on which all marked uses occur; they do not acquire a product of rarity factors from holomorphy alone.

The capacity assumptions have content. Without them, let a single history have $W = 1$ and let each of N indices contribute $L_z = p_z = 1$. Every separate perturbation is bounded by $|\lambda_z|W$, but choosing all $\lambda_z = r/2$ gives a total exponent change $Nr/2$. A fixed reserve in W cannot pay this uniformly in N. A proof for the actual formation must therefore assign all simultaneous uses to a common site and action budget, in addition to bounding each separate direction.

4.22.4. *A channel which can be bounded directly.* For $U \in \mathrm{SL}(2,\mathbb{C})$, the determinant identity

$$\det(U - I) = \det U - \operatorname{tr} U + 1 = 2 - \operatorname{tr} U$$

gives $a(U) = \frac{1}{2}\det(U - I)$. With the operator matrix norm,

$$|a(U)| \le \tfrac{1}{2}\|U - I\|^2. \tag{4.dxcvii}$$

For real $U \in \mathrm{SU}(2)$ this is an equality, since the two singular values of $U - I$ coincide.

Consider a regular zone expressed in its own units, with bare mesh η_n, containing N_n unit cells. Suppose the configurations used in the actual substitutions satisfy $\|U_p - I\| \le \alpha_n \eta_n^2$ on that zone. There are at most $C\eta_n^{-4}$ plaquettes per unit cell. Hence

$$|A(\delta c, U)| \le C\alpha_n^2 \|\delta c\|_\infty N_n. \tag{4.dxcviii}$$

Consequently, if the zone load includes $p_n \ge \alpha_n^2 \|\delta c\|_\infty$, this direct Wilson-weight contribution to eq. 4.dxcv is bounded by Cp_nN_n. If the reserve volume dominates the number of such stored cells, it satisfies eq. 4.dxcvi.

The usual scale $\alpha_n^2 \asymp \theta_n^{-1}(\log \theta_n)^{2q}$ has precisely the form of the coefficient multiplying a marginal amplitude in the load. This calculation uses only the stated curvature regularity at the actual formed arguments. It does not prove that every substitution has that regularity; that is a separate inclusion requirement.

4.22.5. *An unrestricted bare zone needs action reserve.* There is a genuine limitation to a reserve measured only by stored sites. On a bare region where the real field is unrestricted, a plaquette can have $U_p = -I$, and then $a_p = 2$. A direction which changes a local Wilson weight by one has an exponent variation of order one at such a configuration. If its proposed load is

$$p_0 = \beta^{-1}(\log \beta)^C$$

and the site reserve is of order $W = N \log \beta$ on a fixed finite region, then

$$\frac{|\delta\Phi|}{p_0 W} \geq c \, \frac{\beta}{(\log \beta)^{C+1}}$$

for an appropriate fixed configuration and fixed N. This diverges as $\beta \to \infty$. Such configurations are obtained, for example, by setting one bond to $-I$ and the other bonds of a small patch to I. The number of altered plaquettes remains bounded. Thus eq. 4.dxcvi with this site reserve cannot hold pointwise on the entire unrestricted bare-field domain.

This does not refute the complex mass estimate. The Wilson factor strongly suppresses those configurations. For a bare insertion $\mathfrak{b}$ satisfying a pointwise action bound

$$|\mathfrak{b}| \leq C_b \|k\|_\infty A_{\text{available}}(U),$$

one instead has

$$|\lambda \mathfrak{b}| \leq \frac{C_b |\lambda| \|k\|_\infty}{\beta} S_{\text{available}}(U),$$

$$S_{\text{available}} = \beta A_{\text{available}}.$$

At a radius $|\lambda| p_0 < c$, with $p_0 \geq C_b \beta^{-1} (\log \beta)^C \|k\|_\infty$, this spends at most $c(\log \beta)^{-C}$ of the available Wilson action. It is a fractional action estimate, not a site-volume estimate.

A correct reserve lemma can include both budgets. If the majorant has the stronger reserve $e^{cV + \sigma S_{\text{available}}}$ and

$$|\Phi(\lambda) - \Phi(0)| \leq C|\lambda| p_n V + \eta(\lambda) S_{\text{available}},$$

$$C|\lambda| p_n \leq c/2,$$

$$|\eta(\lambda)| \leq \sigma/2,$$

the mass proof above applies on the intersection of these parameter domains. A derivative estimate proportional to p_n requires this intersection to contain a disc of radius c'/p_n with a fixed $c' > 0$. One sufficient condition is $|\eta(\lambda)| \leq C_{\text{act}} |\lambda| p_n$; otherwise the Cauchy radius is the smaller action-controlled radius. To use this statement one must identify action genuinely left available after the failed-test payments, gauge fixing, and other insertions. It cannot be spent twice. For several active zone-use parameters, their fractions must have a bounded sum on each shared action piece. The existence of a site-volume reserve alone proves neither of these action-allocation facts.

4.22.6. *What the carrier bounds do not control.* Write a standing exponent as

$$\Phi_\omega(\xi; p) = \Phi^0_\omega(\xi) + \sum_f \phi_{\omega,f}(\xi; p|_{R_f}).$$

The carrier condition bounds the oscillation of each factor in its external pair argument. Choose a permitted anchor $p^0_\omega(\xi)$ and split

$$\phi_{\omega,f}(\xi; p) = \phi_{\omega,f}(\xi; p^0_\omega) + \{\phi_{\omega,f}(\xi; p) - \phi_{\omega,f}(\xi; p^0_\omega)\}.$$

Oscillation controls the braced expression. It controls neither the scalar anchor value nor the dependence of either expression on an input-list parameter. Cauchy's estimate in the pair variables is not Cauchy's estimate in that new input parameter.

A simple exact example separates these statements. Let $\pi_a \downarrow 0$ be a sequence of putative loads, let one stored history have volume $W_a \geq a$, and let its formed exponent be

$$\Phi_a(x;p) = \log \varepsilon - (c+s+1)W_a + xW_a,$$
$$s > 0.$$

It is entire, independent of p, and its carrier oscillation is zero. At $x = 0$ it has the prepaid mass bound $e^{\Phi_a(0)+cW_a} \leq \varepsilon e^{-sa}$. The same bound, with room, holds throughout a fixed ordinary disc $|x| \leq 1/2$. It fails on a disc $|x|\pi_a < c'$: at $x = c'/(2\pi_a)$ the positive exponent eventually overwhelms every fixed reserve coefficient.

This is a counterexample to deriving the enlarged-radius estimate from carrier oscillation and an ordinary reserve alone. It is not a counterexample to the Wilson operation. Even if the incoming direction has size π_a, the example can be produced by an abstract local formation map which multiplies that direction by π_a^{-1}. All input sizes are then small, while the output exponent has the displayed variation. Locality and holomorphy do not exclude this map; a uniform norm bound on formation must exclude it. Exact preservation of a represented integral, by itself, also supplies no such bound on a particular decomposition into exponents and vacuum factors.

4.22.7. *A sufficient structural bound on the actual formation map.* The needed additional statement is precise. For the actual index set and formed arguments in eq. 4.dxcv, prove

$$|L_{\omega,n,k'}[\delta](\xi,p)| \leq C p_n(\delta) V_\omega(\xi) \quad \text{for every admissible } (\xi,p), \tag{4.dxcix}$$

with a separate fractional-action term when the bare zone is unrestricted. The norm must include scalar anchors, not only oscillations. The constant must be independent of the age, the level, and the history. If a parameter changes the arguments as well, prove the corresponding derivative bound throughout its proposed disc and integrate it to obtain eq. 4.dxcvi.

One verifiable route to eq. 4.dxcix is a carrier incidence estimate. Let the charged stored cells be a finite set $\mathcal{C}_\omega(\xi)$ with $|\mathcal{C}_\omega(\xi)| \leq V_\omega(\xi)$, and decompose the scalar part as $\delta c_\omega = \sum_{c\in\mathcal{C}_\omega} \delta c_{\omega,c}$. Assign each formed occurrence to these cells and give nonnegative weights $w_{e,c}$ with $\sum_c w_{e,c} = 1$. Suppose the actual evaluations stay in their original domains and

$$\sum_e |\epsilon_e| w_{e,c} \, \|\delta F_e\|_{\text{evaluation}} + |\delta c_{\omega,c}| \leq C p_n(\delta)$$

at every stored cell c. Summing over the cells proves eq. 4.dxcix. The evaluation norm here is the norm after the specified unit change and counter-insertions. Using a norm at an earlier level without those transformations can lose exactly the scale weights under investigation.

A condition on the sum of sizes of the original direction terms in a unit cell is helpful, but it is not yet this incidence statement. The proof must show that substitutions and repeated occurrences preserve that bound. The formation rules distinguish copies still anchored in a live component, copies whose last anchor is removed now, and copies inside older already formed terms. The first kind is not a small new remainder. The last kind must be summed with its own norm and history mass. Counting its occurrence does not permit its bound to be reused without payment. In particular a factor obtained only after summation of histories is not a pointwise bound on the unsummed exponent.

For the deletion of a class born at level n, use the weighted mass on the class disc with survival ratio q_*. Suppose a term formed at level e is read in a component that remains live until level $j > e$. Its own deletion bound has age factor q_*^{e-n-1} when $e \geq n+2$. The later mass is restricted to histories that carry this anchor from e to j, giving the factor q_*^{j-e}. At healing the anchor is removed and the two factors combine to q_*^{j-n-1}. Keep the size of the older term in this product. A term carried by a component left standing at its production costs a fixed amount per level. The

genuine small remainders also carry the small factors of their producing levels, which sum along the flow.

More precisely, let A_j be the constant after extracting $q_*^{(j-n-1)_+}$ at level j. The per-level sum over the older reads, including the size ϵ_e of their terms, gives

$$A_j \le C + C \sum_{e=n+2}^{j-1} \epsilon_e A_e.$$

Here $\epsilon_e \le \epsilon + \sigma_e$, with the fixed bound ϵ for the standing-component terms and with $\sum_e \sigma_e \le C'$ for the genuine remainders. Iterating the positive inequality bounds A_j by a constant times $\prod_{e=n+2}^{j-1}(1 + C\epsilon_e)$, which is at most a constant times $(1 + C\epsilon)^{(j-n-2)_+}$. Taking $q_* = q_a/(1 + C\epsilon)$ in the weighted mass, by adding $\log(1+C\epsilon)$ to the reserved exponent per age level, recovers $q_a^{(j-n-1)_+}$ with a fixed constant. This calculation uses the mass of the actual histories on the disc. A small new remainder has its domain away from the surviving region. Since a copied term's domain is contained in the domain that carries it, such a remainder contains no copy with an anchor still live there.

For the fresh ordinary crossing terms there is a direct incidence bound at fixed history and stored fields. Assign each complete marked cluster to one visited step cube in the strip $\Omega_t \setminus \Lambda_t$, when its crossing carrier has such a free cube. Retain this assignment while summing the complete clusters. The rooted domain count, with the spare decay in the marked crossing estimate, bounds the sum at a fixed free cube by a constant times the crossing factor and the supremum of its local input loads. For a cluster assigned to an exempt hull H, the corresponding rooted sum costs at most $C\#H$. If the whole hull is charged, distribute this bound over its $\#H$ cells. An exempt hull whose cells are not all charged needs the actual refiling estimate. This argument counts one fresh crossing occurrence once. An old member carried by restriction retains its carrier bound, and repeated reads in later substitutions still require their own incidence count.

If "size in a unit cell" is defined to mean the sum of the evaluated sizes of all formed occurrences, including their scalar parts and the indicated unit changes, then the incidence estimate is already part of that definition. Under that stronger interpretation the reserve implication is proved. The substantive remaining task is to show that directions generated by the actual renormalization step meet that definition with the proposed load. A condition on the unformed entries cannot be silently replaced by this stronger condition.

4.22.8. *Passing the bound through a logarithmic sum.* Suppose that a finite polymer system has activities $z_X(\lambda)$ with a common Kotecký–Preiss majorant r_X, and that the corresponding pinned bound is e^{a_X}. Choose the logarithm defined by the connected expansion, equal to zero at the zero activity vector. Then a useful marked estimate is

$$|\log Z(z(\lambda)) - \log Z(z(0))| \le \sum_X e^{a_X} |z_X(\lambda) - z_X(0)|,$$

To prove it, join the two activity vectors by their straight segment. Every point on that segment has modulus at most r_X coordinatewise. The absolutely convergent connected expansion gives $|\partial_{z_X} \log Z| \le e^{a_X}$ by the pinned bound. Differentiate along the segment and integrate. Thus a summable one-zone age bound on the changed activities passes to the logarithm. No logarithm of an individual possibly zero history integral is required.

For a countably infinite system, apply the same estimate first to finite restrictions of the same activity and incompatibility system. If $\sum_X e^{a_X} |z_X(\lambda) - z_X(0)| < \infty$, the integrated pinned series defines a relative cluster logarithm and satisfies this bound, independently of the finite exhaustion by dominated convergence of the rooted expansions. Writing two separate infinite-volume logarithms requires convergence of both unmarked global series; a common local majorant alone does not

ensure that. This deduction has two inputs: the activity majorant must remain valid on the complex parameter disc, and the changed activities must satisfy the marked bound after the actual formation and copy bookkeeping. A convergence theorem for polymers does not prove those inputs. The exponential reserve can pay a controlled perturbation of an integrand; it cannot manufacture control of the perturbation map.

A recursive treatment of old copied terms can be sufficient instead of a single global phase estimate. Fix $0 < q_0 < q < 1$ and $C > 0$. For example, suppose an already justified local operation estimate gives nonnegative bounds, for $j \geq n+1$,

$$A_{j,n} \leq C q_0^{j-n} + C \sum_{n<e<j} \epsilon_e q_0^{j-e} A_{e,n},$$
$$0 \leq \epsilon_e \leq \epsilon.$$

Divide by q_0^{j-n}. A discrete Gronwall induction gives

$$A_{j,n} \leq C q_0^{j-n} \prod_{n<e<j} (1 + C\epsilon_e) \leq C\{q_0(1+C\epsilon)\}^{j-n}.$$

The induction follows by expanding the product, or using $\prod_{e<j}(1+C\epsilon_e) = 1+\sum_{e<j} C\epsilon_e \prod_{r<e}(1+C\epsilon_r)$. Thus choosing $q_0 \leq q/(1+C\epsilon)$ preserves the desired rate q. No small sum of the ϵ_e is needed. This pays growth of output bounds after local integration. It does not pay an exponentially amplified perturbation inside an unsummed exponent. To apply this alternative one must establish the local parameterized operation bound at each new loss of an anchor and the stated copying recurrence. The sufficient formation test above is not claimed to be the only possible proof route.

The present test therefore proves the reserve implication and the direct regular-zone Wilson channel. It shows why an unrestricted bare channel requires available action, and isolates the additional uniform formation estimate for the remaining entries. The specified carrier oscillation, ordinary presentation bounds, and end-mark definitions do not alone establish that estimate. A proof of it for the actual local substitutions, with their scalar parts and copy incidence, would complete the proposed one-zone argument.

4.23. **Coordinates for quadratic extraction.** The quadratic coefficient of an effective interaction is defined in the physical block field. The field used to normalize a fluctuation Gaussian is a different coordinate. Keeping these coordinates distinct explains both the fixed radii of the unmarked chain and the squared radius ratios in the marked chain. The required inclusion near the trivial field follows from the background and tower estimates already used to define the analytic field domains. It is not an additional loss caused by the number of blocking steps.

All constants below may depend on the fixed blocking and localization parameters. They are independent of the level, torus size and number of earlier levels. The non-Abelian background and tower estimates are explicit hypotheses in the statements that use them.

4.23.1. *Physical fields, Gaussian fields and the coefficient.* At level j, write the configuration in the trivial-field chart as

$$\mathbf{U} = e^{i\eta_j A'},$$
$$v = (A', J),$$
$$\|v\|_{1,j} = \max\{\|A'\|_\infty, \|\nabla_j A'\|_\infty, \|J\|_\infty\}. \tag{4.dc}$$

Here $0 < \eta_j \le 1$ is the bare spacing in level-j units and ∇_j is a difference quotient in those units. The current J is an independent analytic variable. In particular, A' in this chart is a physical potential; no coupling multiplies it by definition.

The block datum of the background is $V = e^{iB}$. Let $p_j(B)$ be its configuration-current pair in a fixed gauge near zero. Derivatives of its configuration entry are taken in the chart in eq. 4.dc. Suppose

$$p_j(0) = (\mathbf{1}, 0), \qquad Dp_j(0)B = \mathfrak{h}_j B = (H_j B, D^* D H_j B). \tag{4.dci}$$

The datum B is again physical. Its continuation radius r_{B} is a radius in e^{iB}, chosen independently of the coupling. For a gauge-invariant local interaction E, the first derivative at the trivial pair vanishes: invariance under constant gauge transformations makes it an invariant linear functional on the semisimple Lie algebra, and that functional is zero. Consequently

$$D^2(E \circ p_j)(0)[B_1, B_2] = \operatorname{Hess} E(0)[\mathfrak{h}_j B_1, \mathfrak{h}_j B_2]. \tag{4.dcii}$$

The possible second derivative of p_j contributes nothing because it is multiplied by $DE(0)$.

The polarisation kernel is the kernel of eq. 4.dcii. Its second spatial moments define the curvature tensor $T[E]$, and, in the invariant scalar channel, the action coefficient $b(E)$. These definitions use derivatives with respect to B. For the Wilson functional of the block datum itself, use the normalization

$$b(A_{\mathrm{block}}) = 1, \tag{4.dciii}$$

whose quadratic part is $\frac{1}{2}\|\mathrm{d}B\|^2$. For a Wilson family evaluated after the background map, the analogous normalization requires the linear-response identities in that map. Likewise $T[\mathsf{A}[c]] = c$ for a constant allowed curvature tensor c is the prescribed counter-insertion normalization. Neither identity for pulled-back families is proved merely by the coordinate change below.

In a fluctuation integral one instead sets $\mathcal{A}_j = g_j \mathcal{A}_j^{\mathrm{s}}$. If $\widetilde{E}_j(B^{\mathrm{s}}) = E(p_j(g_j B^{\mathrm{s}}))$, the chain rule gives

$$\begin{aligned} D^2 \widetilde{E}_j(0) &= g_j^2 D^2(E \circ p_j)(0), \\ T_{\mathrm{s}}[E] &= g_j^2 T[E], \\ b_{\mathrm{s}}(E) &= g_j^2 b(E). \end{aligned} \tag{4.dciv}$$

Thus a fixed probe in the Gaussian variable may fit inside a shrinking physical domain. It extracts the rescaled coefficient in eq. 4.dciv. Recovering the coefficient of the physical action requires division by g_j^2.

For example, impose the response-decay and localization hypotheses of the amplitude proposition below, and suppose that the usable physical radius is a_j and the family norm is at most M_j there. Its radius in B^{s} is comparable to a_j/g_j. The Cauchy radius factor in the amplitude bound then transforms as

$$|g_j^2 b(E)| \le C M_j (g_j/a_j)^2, \qquad |b(E)| \le C M_j a_j^{-2}. \tag{4.dcv}$$

The coupling cancels from the coordinate conversion. The same calculation holds if the physical radius is written g_j^{γ}: the canonical radius is $g_j^{\gamma-1}$, and the physical coefficient bound remains $C M_j g_j^{-2\gamma}$. The change of variables creates no extra loss and removes no loss already present in the physical radius.

4.23.2. *A near-zero ball lies in the hereditary core.* We state precisely the background information needed for this local inclusion. A core at radii (a_0, a_1) is described by the following simple conditions and their tower counterparts. A triple is (U, A', J), with U real and $\mathbf{U} = e^{i\eta_j A'}U$. The real and complex curvatures are smaller than $a_0\eta_j^2$; the real factor has the prescribed local regular gauges; $|A'|, |\nabla_U A'| < a_1$; and $|J| < a_0$. The tower counterparts require the corresponding curvature and current bounds for every allowed restricted tower inside the domain.

By Cor. 3.130, if the simple conditions hold at sufficiently small radii (ϵ_0, ϵ_1), the tower configurations exist holomorphically. Their curvature and current, in the local units of their layer, satisfy the defining tower bounds with radius

$$K_{\mathrm{T}}(\epsilon_0 + \epsilon_1^2). \tag{4.dcvi}$$

The estimate includes the stated change-of-unit factors and holds for both tower rules, every restricted domain and every depth. Its common analytic ball and physical-current bound are proved in Lem. 3.127 and Prop. 3.129.

Proposition 4.83 (Near-zero core inclusion)**.** For sufficiently small a_0, a_1 there is a constant $c > 0$, independent of j, such that

$$\|v\|_{1,j} < r_j,$$
$$r_j = c\min\{a_0, a_1, 1\}$$
$$\Longrightarrow$$
$$(\mathbf{1}, A', J) \text{ belongs to the core at } (a_0, a_1). \tag{4.dcvii}$$

■

Proof. The plaquette product in the exponential chart obeys

$$|\partial e^{i\eta_j A'} - \mathbf{1}| \le C\eta_j^2(\|\nabla_j A'\|_\infty + \|A'\|_\infty^2) \le C\eta_j^2(r_j + r_j^2). \tag{4.dcviii}$$

This follows by expanding a plaquette product: the linear term is the discrete curl and each nonlinear term contains at least two bond potentials. On a fixed small chart the remaining exponential series is bounded by a uniform multiple of the quadratic term. In particular, there is no inverse power of η_j in eq. 4.dcviii. This is the reason for including the first difference quotient in eq. 4.dc.

Choose

$$\epsilon_0 = \frac{a_0}{4K_{\mathrm{T}}},$$
$$\epsilon_1 = \min\{a_1/2, \sqrt{a_0/(4K_{\mathrm{T}})}\}.$$

Taking c smaller if necessary ensures $C(r_j + r_j^2) < \epsilon_0$ and $r_j < \min\{\epsilon_0, \epsilon_1\}$. The real configuration is $\mathbf{1}$, so its curvature and local gauge potential vanish. All simple conditions therefore hold at these reduced radii. Moreover, $K_{\mathrm{T}}(\epsilon_0 + \epsilon_1^2) \le a_0/2$. The tower estimate gives every hereditary clause at the desired radii, and $|J| < r_j < a_0$. This proves eq. 4.dcvii. □

A bound only on $\|A'\|_\infty$ would be different: arbitrary oscillation on the bare lattice can give $\|\nabla_j A'\|_\infty$ of order $\eta_j^{-1}\|A'\|_\infty$. The ball relevant to eq. 4.dcvii is the C^1 ball, not that weaker ball.

For completeness, suppose the continued background in eq. 4.dci is holomorphic for $\|B\|_\infty < r_{\mathrm{B}}$ and satisfies

$$\|\mathcal{H}_j(B)\|_{C^1} + \|J(e^{i\eta_j \mathcal{H}_j(B)})\|_\infty \le C_{\mathrm{B}}\|B\|_\infty. \tag{4.dcix}$$

Then its triple belongs to the core whenever

$$\|B\|_\infty < \min\{r_{\mathrm{B}}/2, r_j/(2C_{\mathrm{B}})\}. \tag{4.dcx}$$

The same conclusion for a first-order response uses the derivative of eq. 4.dcix. eq. 4.dcvii–eq. 4.dcx give a fixed datum radius when (a_0, a_1) are fixed, and a radius comparable to $\min\{a_0, a_1\}$ when they run. The hypotheses are exactly the background continuation, its uniform first-order bounds and the hereditary tower estimate. No analyticity radius in the normalized Gaussian variable is substituted for this physical datum radius.

4.23.3. *Cauchy's estimate, response decay and entropy.* Work first with the infinite-lattice companion. Its torus versions use the uniform response bounds and the lift convergence supplied by the background hypotheses. Consider pointed local families satisfying

$$|E(X, p, z)| \le Me^{-\kappa d(X)}$$

on the core, and assume that the near-zero ball there has usable radius a. Assume also the uniform response decay

$$\|\mathfrak{h}_j \delta_c\|_{1,j;X} \le Ce^{-\delta \operatorname{dist}(c,X)}. \tag{4.dcxi}$$

The distances in this section are measured in the unit of level j. For a fixed root cube, suppose the number of domains with tree length between m and $m+1$ is at most $Ce^{\kappa_{\mathrm{D}} m}$, while the number of pointed sites in such a domain is at most $C(m+1)^4$. These are the usual localization entropy and volume bounds.

Proposition 4.84 (The amplitude bound with its radius)**.** If $\kappa > \kappa_{\mathrm{D}}$ with a fixed positive margin, the polarisation sums converge absolutely and

$$\begin{aligned} |\Pi_{\mu\nu}(u)| &\le Ca^{-2}Me^{-\delta_1|u|}, \\ |b(E)| + \|T[E]\| &\le Ca^{-2}M, \end{aligned} \tag{4.dcxii}$$

whenever the scalar coefficient or the stated summed tensor is defined by the prescribed covariance conditions. The constants are uniform in the level and torus volume. A tensor at a fixed base point satisfies the same bound before any isotropy assertion is made. ∎

Proof. The two-variable Cauchy formula on a bidisc inside the ball gives

$$|\operatorname{Hess} E(X, z)[v, w]| \le Ca^{-2}Me^{-\kappa d(X)}\|v\|_{1,j;X}\|w\|_{1,j;X}.$$

Combining this with eq. 4.dcxi bounds the contribution of one term by

$$Ca^{-2}Me^{-\kappa d(X)}e^{-\delta(\operatorname{dist}(c,X)+\operatorname{dist}(c',X))}. \tag{4.dcxiii}$$

If the localization cubes have fixed side M_0, then

$$|c_- - c'_-| \le \operatorname{dist}(c, X) + \operatorname{dist}(c', X) + CM_0(d(X) + 1).$$

A sufficiently small fixed fraction of the exponents in eq. 4.dcxiii supplies $e^{-\delta_1|c_- - c'_-|}$ for a fixed $\delta_1 > 0$. Choose that fraction so that the remaining localization rate stays strictly above κ_{D}. The remainder is summable. Indeed, root a domain at a cube closest to c. Summing its pointed sites costs at most a polynomial in $m+1$, and summing its shapes costs $e^{\kappa_{\mathrm{D}} m}$. These costs are absorbed by the remaining $e^{-\kappa_{\mathrm{rem}} m}$, where $\kappa_{\mathrm{rem}} > \kappa_{\mathrm{D}}$. Summing the root location is bounded by

the remaining exponential in its distance from c. This proves the first inequality in eq. 4.dcxii and absolute convergence.

The second moment defining $b(E)$ is bounded by $Ca^{-2}M\sum_u |u|^2 e^{-\delta_1|u|}$, a finite constant times $a^{-2}M$. For a tensor at a fixed base point, the two moment factors are bounded by polynomials in the domain diameter and the two response distances. Those polynomials are absorbed by a further fixed fraction of the same exponentials. This proves the tensor bound. Identifying an invariant tensor with the scalar action coefficient uses the separate Ward and lattice-symmetry identities; it is not needed for the amplitude estimate. □

In particular, the extraction norm is uniform on fixed physical radii. On the running radii

$$a_j \asymp \theta_j^{-1/2}(\log\theta_j)^q,$$
$$q \geq 0,$$

the correct general bound has the factor $a_j^{-2} \asymp \theta_j(\log\theta_j)^{-2q}$. One can express uniformity instead as $a_j^2|b(E)| \leq C\|E\|$, provided that this normalization is retained in all later estimates.

4.23.4. *What the squared radius ratios pay for.* Let a remainder born at level j have size r_j on physical radius a_j and have its vacuum value removed. There is no linear term. Assume a gauge-invariant window estimate in which the quadratic curvature factors each scale by $L^{-2(n-j)}$, while a unit volume at level n contains $L^{4(n-j)}$ sites of level j. After these geometric factors cancel, evaluating the quadratic part on a background of radius a_n costs

$$w_{n,j}r_j,$$
$$w_{n,j} = (a_n/a_j)^2. \tag{4.dcxiv}$$

This is the relative quadratic evaluation weight. Extracting the physical coefficient from the resulting family at level n costs a_n^{-2}, so

$$a_n^{-2}w_{n,j}r_j = a_j^{-2}r_j. \tag{4.dcxv}$$

Charging another factor a_j^{-2} on top of the left side would count the same shrinking input radius twice. Omitting the displayed readout factor a_n^{-2} would instead confuse an evaluated potential with its physical quadratic coefficient. The full nonlinear remainder estimate also requires its actual backgrounds to lie in the original analytic domains and the separate control of large domains. eq. 4.dcxiv records the radius factor in that estimate; it provides no continuation outside those domains.

For example, let $r_j \leq C\theta_j^{-\kappa_0/2}$, $\kappa_0 > 4$, and suppose $\theta_j \geq \theta_n + \lambda(n-j) - C_0$ with $\lambda > 0$ and a sufficiently large common lower bound on θ_n. Integral comparison gives

$$\begin{aligned}\sum_{j\leq n} w_{n,j}r_j &\leq Ca_n^2\sum_{j\leq n}\theta_j^{1-\kappa_0/2}(\log\theta_j)^{-2q} \leq C\theta_n^{1-\kappa_0/2},\\ a_n^{-2}\sum_{j\leq n} w_{n,j}r_j &\leq C\theta_n^{2-\kappa_0/2}(\log\theta_n)^{-2q}.\end{aligned} \tag{4.dcxvi}$$

The first loss of one power of θ comes from summing the older levels; the second comes from quadratic extraction. This is the usual g^{κ_0-4} bound, with its logarithmic improvement. The two losses are already present in eq. 4.dcxvi.

The unmarked chain uses fixed physical radii, so eq. 4.dcxii preserves its small interaction bound. The marked chain uses running radii, and its stronger smallness pays the readout loss in eq. 4.dcxvi. These facts establish the radius bookkeeping under the stated background, response and entropy hypotheses. They do not establish every later power-counting assertion. Cancellation in a complex window, removal of every required tensor marginal, weighted sensitivity to earlier members, and differentiated bounds for a large-field zone remain their own estimates. Neither the physical-to-Gaussian coordinate change nor the local core inclusion supplies those conclusions.

5. Couplings, sources and continuum limits

The coupling recursion selects the physical scale and the tuned sequence. Uniform source estimates then bound connected correlations at separated supports. We track their dependence on the number of insertions before taking subsequences, and extend the limiting functionals to Schwartz tests flat on the coincidence diagonals. This gives the quantitative test-space estimates needed for §7.

5.1. **First passage for a marginal flow.** We derive the numerical consequences of a coupling recursion. Its realization by a renormalization step, consistency between tori and cutoffs, and estimates with nonconstant sources remain analytic hypotheses. An individual increment of the recursion is allowed to be negative.

5.1.1. *Hypotheses and first passage.* Fix numbers $c_0 > 0$, $C_0 \geq 0$, $h > 0$, and $0 \leq \epsilon < c_0/4$. For each initial value $\beta \geq h + B$ consider a real run with $x_0 = \beta$. Assume that a next step exists whenever $x_j > h$, and that every such step has the form

$$x_j = x_{j-1} - b_j,$$
$$b_j = b_j^0 + r_j.$$

For every interval of defined steps assume

$$\left| \sum_{l=i+1}^{j} (b_l^0 - c_0) \right| \leq C_0,$$
$$|r_j| \leq \epsilon$$
$$\text{if } x_{j-1} \geq h.$$

All constants are independent of β and of the length of the run. The existence of the next step is an assumption: an inequality about hypothetical increments cannot establish that an underlying analytic construction is defined. Set

$$\lambda = c_0 - \epsilon > 0,$$
$$\Lambda = c_0 + \epsilon,$$
$$B \geq c_0 + C_0 + \epsilon.$$

The one-step case of the interval estimate gives $b_j \leq B$. This is an upper bound, not an assertion $b_j \geq 0$.

Define

$$K(\beta) = \min\{k \geq 0 : x_k \leq h + B\},$$

$$\mathcal{W}_k = \{\beta \geq h + B : K(\beta) = k\}.$$

The minimum is initially a possible stopping index; its finiteness will be proved. At the endpoint $\beta = h + B$ it equals zero. When $\beta > h + B$ it is positive, so the last increment is an actual step of the run.

Proposition 5.1 (Block drift and a protected terminal window)**.** The first-passage time is finite. For defined segments up to that time,

$$\lambda(j - i) - C_0 \leq x_i - x_j \leq \Lambda(j - i) + C_0$$
$$(0 \leq i \leq j \leq K).$$

If $K > 0$, then $h < x_K \leq h + B$. Moreover

$$\max\left\{0, \frac{\beta - h - B - C_0}{\Lambda}\right\} \leq K \leq \frac{\beta - h + C_0}{\lambda}. \qquad \blacksquare$$

Proof. Summing the recursion over a segment gives

$$x_i - x_j = c_0(j - i) + \sum_{l=i+1}^{j} (b_l^0 - c_0) + \sum_{l=i+1}^{j} r_l.$$

The two assumed bounds give the block drift estimate. If the run never reached $h + B$, all its values would remain above h, so it could be extended for every finite number of steps. The lower drift bound would then imply $x_j \leq \beta - \lambda j + C_0$, contradicting $x_j > h + B$ for sufficiently large j. Thus first passage occurs.

For $K > 0$, minimality gives $x_{K-1} > h + B$. Since $b_K \leq B$, $x_K = x_{K-1} - b_K > h$. The upper endpoint follows from the definition of K. The extra width B is exactly what prevents the final step from overshooting the lower reference value h.

Finally $\beta - x_K \leq \Lambda K + C_0$ and $x_K \leq h + B$ give the lower bound on K. Conversely $\lambda K - C_0 \leq \beta - x_K < \beta - h$ gives the stated upper bound for $K > 0$. The endpoint case $K = 0$ also satisfies both inequalities. □

In particular $\beta \to \infty$ forces $K(\beta) \to \infty$. Conversely a sequence with $K \to \infty$ has $\beta \to \infty$, by the lower drift estimate and the terminal lower bound. Since these stopping times are integers, the sets $\mathcal{W}_k$ are nonempty for infinitely many different k. Nothing here says that every one is nonempty, that one is an interval, or that the terminal values in it coincide. Such assertions would require additional information about the dependence of the analytic recursion on β.

For example, $b_j^0 = c_0 + (-1)^j q$, with $q > c_0$ and $r_j = 0$, satisfies the interval hypothesis with $C_0 = q$. Every other increment is negative. Nevertheless the block drift is positive and the preceding proof applies. Replacing block drift by a silently assumed monotonicity would discard this perfectly admissible numerical example.

5.1.2. *Vanishing mean error and the leading logarithm.* Assume in addition that

$$|r_j| \leq \omega(x_{j-1}),$$
$$\omega_*(M) := \sup_{x \geq M} \omega(x) \longrightarrow 0$$
$$(M \to \infty),$$

where ω is nonnegative on $[h, \infty)$. The earlier global bound by ϵ remains in force. No monotonicity of ω is needed.

Lemma 5.2 (Time spent below a fixed inverse coupling)**.** For $M \geq h$, the number N_M of steps $1 \leq j \leq K$ with $x_{j-1} \leq M$ is bounded by

$$N_M \leq 1 + \frac{M - h + C_0}{\lambda}.$$

This bound does not depend on the initial value. ■

Proof. There is nothing to prove if no such step occurs. Otherwise let i be the smallest index in $\{0, \dots, K-1\}$ with $x_i \leq M$. Every counted step has its preceding index between i and $K-1$, so $N_M \leq K - i$. The block drift from i to K gives $\lambda(K-i) - C_0 \leq x_i - x_K < M - h$. The displayed, slightly weaker bound therefore follows. Excursions back above M cause no problem: all subsequent indices are included in this bound on the entire remaining time. □

Proposition 5.3 (Leading first-passage asymptotics)**.** With h and the other constants fixed,

$$\frac{1}{K(\beta)} \sum_{j=1}^{K(\beta)} |r_j| \longrightarrow 0,$$

$$\frac{\beta}{K(\beta)} \longrightarrow c_0$$

$$(\beta \to \infty).$$

If $a_K = L^{-K}$ for a fixed $L > 1$, then $\beta / \log(a_K^{-1}) \to c_0 / \log L$. ■

Proof. Split the steps according to whether $x_{j-1} \leq M$. The lemma gives

$$\frac{1}{K} \sum_{j=1}^{K} |r_j| \leq \frac{\epsilon}{K} \left(1 + \frac{M - h + C_0}{\lambda}\right) + \omega_*(M).$$

First let $\beta \to \infty$ at fixed M, using $K \to \infty$, and then let $M \to \infty$. This proves the first limit. Summing the whole recursion gives

$$\beta = x_K + c_0 K + e_K + \sum_{j=1}^{K} r_j,$$

$$|e_K| \leq C_0.$$

The terminal value belongs to a fixed bounded interval, so division by K proves the second limit. Finally $\log(a_K^{-1}) = K \log L$ exactly. □

If the analytic one-loop calculation establishes $c_0 = 11N^2 \log L / (12\pi^2)$ in the convention $\beta = g_0^{-2}$, this arithmetic yields $\beta / \log(a_K^{-1}) \to 11N^2/(12\pi^2)$. The one-loop coefficient is an input to this implication, not a consequence of the stopping argument. The result is uniform over any family of runs satisfying the same constants and error envelope.

5.1.3. *A quantitative optional error bound.* Suppose $h \geq e$, $p \geq 0$, and $\omega(x) \leq C x^{-1/2} (\log x)^p$. Then

$$|\beta - c_0 K| \leq h + B + C_0 + C' \sqrt{\beta + C_0 + 2}\, [\log(\beta + C_0 + 2)]^p.$$

To prove it, partition $[h, \infty)$ into unit intervals. If two preceding indices $i < j < K$ have values in the same interval, then $x_i - x_j < 1$ and the lower drift gives $j - i < (1 + C_0)/\lambda$. A unit interval

is therefore visited at most $J = 2 + \lceil (1 + C_0)/\lambda \rceil$ times. Also $x_i \le \beta + C_0$ by the lower drift from zero. Summing the supremum of the error envelope in each visited interval consequently gives

$$\sum_{j=1}^{K} |r_j| \le CJ[\log(\beta + C_0 + 2)]^p \sum_{n=0}^{\lceil \beta + C_0 - h \rceil} (h+n)^{-1/2}.$$

The last sum is at most a constant times $\sqrt{\beta + C_0 + 2}$, by comparison with the integral of $x^{-1/2}$. Substitute this bound in the summed recursion. For fixed h this is an $O(\sqrt{\beta}(\log \beta)^p)$ error. It supplies no second logarithmic coefficient, convergence rate for observables, or summable comparison between neighboring cutoffs.

5.1.4. *Depth envelopes and sums over scales.* The same block estimate gives useful bounds measured from the terminal scale. For $0 \le s \le K$, apply it between $K - s$ and K to obtain

$$h + \lambda s - C_0 < x_{K-s} \le h + B + \Lambda s + C_0.$$

This estimate records the possible bounded backward excursions without requiring the inverse couplings to be monotone. It also distinguishes the fixed reference value h from the terminal value x_K, which is allowed to range over its window.

For an explicit scale-summation corollary suppose $h \ge \max\{1, 2C_0\}$. Then the lower bound becomes $x_{K-s} \ge h/2 + \lambda s$. Write $g_j = x_j^{-1/2}$. For every $q > 2$, comparison with a decreasing integral gives

$$\sum_{j=0}^{K} g_j^q \le \sum_{s=0}^{\infty} (h/2 + \lambda s)^{-q/2} \le (h/2)^{-q/2} + \frac{(h/2)^{1-q/2}}{\lambda(q/2 - 1)} \le C(q, \lambda) h^{1-q/2}.$$

The bound is uniform in the number of steps. The first term of the sum is bounded separately, and the integral accounts for all later terms; this explains both the strict condition $q > 2$ and the displayed dependence on the reference inverse coupling.

At the marginal exponent $q = 2$ the integral instead grows logarithmically. Using the two depth envelopes gives, for example,

$$\frac{1}{\Lambda} \log\left(1 + \frac{\Lambda(K+1)}{h + B + C_0}\right) \le \sum_{j=0}^{K} g_j^2 \le \frac{2}{h} + \frac{1}{\lambda} \log\left(1 + \frac{2\lambda K}{h}\right).$$

For the lower bound integrate the decreasing function $(h + B + C_0 + \Lambda s)^{-1}$ from zero to $K + 1$. For the upper bound separate its counterpart's first term and integrate from zero to K. These elementary estimates are often the numerical part of a renormalization argument: sufficiently high powers of the running coupling are summable, whereas the marginal power is not. They cannot replace the analytic cancellation or renormalization that produces those powers in a particular model.

The extra lower bound on h belongs only to this corollary. For a smaller fixed h, separate the finitely many terminal depths $s < 2C_0/\lambda$, bound them using $x_j > h$, and apply the same integral comparison to the remaining depths. The resulting constants may depend on h and C_0.

5.1.5. *An exact change of lattice units.* There is also an identity that requires no coupling-flow assumption. Let $\Lambda_{K,\ell} = (L^{-K}\mathbb{Z}/\ell\mathbb{Z})^4$ be a compatible finite lattice and let b be an integer with $K - b \ge 0$. Dilation $D_b x = L^b x$ maps it bijectively to $\Lambda_{K-b, L^b \ell}$. Transfer a bond variable by $U'_{D_b e} = U_e$. The map bijects positively oriented bonds and plaquettes, preserves their incidence,

and satisfies $U'_{D_b p} = U_p$. Product normalized Haar measure is therefore unchanged, as is the Wilson action $A(U) = \sum_p [1 - \frac{1}{2} \operatorname{Re} \operatorname{tr} U_p]$.

For a test function on T_ℓ, put $f^{[b]}(x') = f(L^{-b}x')$ on $T_{L^b\ell}$. Plaquette barycenters obey $x(D_b p) = D_b x(p)$, so

$$\mathcal{O}_{K-b,L^b\ell}(f^{[b]})(U') = \sum_p f(L^{-b} D_b x(p)) a_p(U) = \mathcal{O}_{K,\ell}(f)(U).$$

No spacing prefactor appears in this observable. Consequently the two Wilson partition functions at the same β are equal, and all joint moments and cumulants satisfy the exact identity

$$S^T_{n;K,\ell,\beta}(f_1, \dots, f_n) = S^T_{n;K-b,L^b\ell,\beta}(f_1^{[b]}, \dots, f_n^{[b]}).$$

This is one finite-dimensional integral written in two units, including the same number of sites in each direction. It is not a comparison between different bare measures.

For $\|f\|_\sharp = \max(\|f\|_\infty, c_M \operatorname{Lip} f)$ the exact norm transformations are

$$\|f^{[b]}\|_\infty = \|f\|_\infty,$$
$$\operatorname{Lip} f^{[b]} = L^{-b} \operatorname{Lip} f.$$

Thus $\|f^{[b]}\|_\sharp \le \|f\|_\sharp$ for $b \ge 0$, and it is at most $L^{|b|}\|f\|_\sharp$ for $b < 0$. All support distances and support radii are multiplied by L^b. The same bond bijection carries a fixed-spacing DLR specification to its dilated specification, since it carries each finite integration set and all plaquettes meeting that set to their counterparts. It therefore also transports infinite-lattice Gibbs states and their local moment identities.

To identify this unit change with two initial portions of a renormalization run requires more: the blocking rules, gauges, auxiliary choices and extraction of increments must commute with the bond bijection. Likewise, torus-independent increments require a separately defined common local construction. Neither property follows from the scalar recursion or the exact dilation identity proved here.

5.2. **The running coupling and first-passage arithmetic.**

5.2.1. *The run, its coefficient, and the analytic requirements.* The numerical running coupling and the auxiliary frame have different roles. This distinction is needed even for a real bare coupling. A frame fixes the tests, cubes, Gaussian references, coordinate choices and large-field operations. The weight of the Wilson action is another entry of the construction. A derivative with the frame fixed differentiates that weight; it does not differentiate a characteristic function whose threshold is moving. We first state the actual analytic requirements on these objects. The arithmetic consequences below are conditional theorems under those requirements.

Fix odd $L > 1$, the block parameter M, and the remaining geometric constants. In the ordinary-source application use its actual $M = L^{m_0}$, fixed collar layers and compatible cofinal tori; admissibility below refers to those assigned lattices. The group is SU(2); in the coefficient formulas we retain $N = 2$. The pointed families, hereditary complex potential/current cores, midpoint tents, companions and Euclidean list norms are those of § 2.2. In particular a pointed family of level j has terms $E(X, P, z)$, where X is a connected localization domain and z is a unit lattice site in X. Its norm is a supremum in z, not a sum in z. The inequality

$$|E(X, P, z)| \le \mathcal{A} e^{-\kappa d_j(X)} \tag{5.i}$$

is required on its declared core. Summing domains requires the separate rooted-domain entropy estimate. A companion agrees on every common nonwrapping lift, with the required collar, for all

admissible tori with the same frame prefix and initial weight. Local analyticity does not establish that compatibility. We write $E^{(j)}$ for a torus family and $\widehat{E}^{(j)}$ for its common infinite-lattice companion. The coefficient of the run is extracted from the latter. A regular list member $G^{(j)}$ has a companion $\widehat{G}^{(j)}$; neither is the additional one-step Gaussian family G_τ. The family created at step $k+1$ has pointed level $k+1$, although its creation choices are those of level k.

Write $P_\circ = (\mathbf{1}, 0)$, and let $\mathfrak{h}_\infty$ be the infinite-volume first response of the background potential/current pair at $P_\circ$. Its existence, gauge covariance, affine reproduction modulo gauge and exponential decay belong to the background hypotheses. For the composed physical Hessian, average over an orthonormal color basis t_a, $1 \le a \le 3$, and set

$$\begin{aligned} \Pi(x,y) &= \frac{1}{3}\sum_{a=1}^{3}\sum_{z}\sum_{X \ni z} D^2E(X, P_\circ, z)[\mathfrak{h}_\infty \delta_x^a, \mathfrak{h}_\infty \delta_y^a], \\ b(E) &= \sum_{u \in \mathbb{Z}^4} \Pi_{\mu\nu}(u) u_\mu u_\nu, \qquad \mu \neq \nu. \end{aligned} \tag{5.ii}$$

The second line is used only after translation covariance and independence of the choice $\mu \neq \nu$ have been justified. Hessians are in the pair exponential chart, not in an unspecified coordinate system. Their directions have norm $\max\{\|A\|_\infty, \|\nabla A\|_\infty, \|J\|_\infty\}$. For $a' = \min\{a_1, c, a_0/C\}$, a ball of radius ca' in this norm must lie in the core. A ball controlled by $\|A\|_\infty$ alone does not supply the uniform mesh normalization.

The coefficient and pointed-cancellation theorem in § 4.2 applies under its full response, summability and symmetry hypotheses. Its consequences used here are absolute convergence, vanishing zeroth and first moments, and

$$\begin{aligned} \sum_u \Pi_{\mu\nu}(u) u_\rho u_\sigma &= b(E)(\delta_{\mu\rho}\delta_{\nu\sigma} + \delta_{\mu\sigma}\delta_{\nu\rho} - 2\delta_{\mu\nu}\delta_{\rho\sigma}), \\ |\Pi(x,y)| &\le C(a')^{-2}\mathcal{A} e^{-c|y_- - x_-|}, \qquad |b(E)| \le C(a')^{-2}\mathcal{A}. \end{aligned} \tag{5.iii}$$

The extraction is linear, holomorphic in admitted parameters and real on real families. It depends on the total, so two coherent localizations of the same physical functional have the same coefficient. A constant total has coefficient zero. At absolute radii the factor $(a')^{-2}$ is a constant depending on M; at running radii it can cost $Cx_j(\log x_j)^{-2q}$. Neither the kernel nor its coefficient depends on which admitted radii were used to estimate it.

Pointed tensors and their period averages. There is a useful distinction between the Hessian of the total and that of a pointed total. Write $K_{\hat z}(x,y)$ for the contribution of the base point $\hat z$ to $\Pi(x,y)$, including the color average and the sum of domains in eq. 5.ii. Assume here the actual pointed Ward identities and absolute moments through order two, uniformly over one period, and covariance under simultaneous translations by $M\mathbb{Z}^4$. These are pointed response hypotheses; the bulk estimate in eq. 5.iii alone does not imply them. Set $\mathcal{C}_M = \{0, \dots, M-1\}^4$ and

$$\begin{aligned} T_{\mu\nu,\rho\sigma}(\hat z) &= \sum_{\substack{x \text{ of direction } \mu \\ y \text{ of direction } \nu}} (x_- - \hat z)_\rho (y_- - \hat z)_\sigma K_{\hat z}(x,y), \\ \overline{T} &= M^{-4}\sum_{\hat z \in \mathcal{C}_M} T(\hat z). \end{aligned} \tag{5.iv}$$

If only the total is invariant under all unit lattice isometries, the conclusion is

$$\overline{T}_{\mu\nu,\rho\sigma} = b(E)(\delta_{\mu\nu}\delta_{\rho\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho}) + c_{\mathrm{top}}\epsilon_{\mu\rho\nu\sigma}. \tag{5.v}$$

In particular an invariant total does not determine $T(\hat z)$ at an individual base point.

Here is the algebra behind this assertion. At the trivial field the first derivative of each scalar pointed total vanishes by constant color invariance. The twice differentiated gauge identity therefore makes its Hessian vanish when either argument is a gradient. One may insert linear and quadratic

gauge polynomials by cutoff and the assumed absolute moments. Linear polynomials give zero row and column sums in each bond direction. A quadratic polynomial $(x-\hat{z})_\mu(x-\hat{z})_\rho$ gives antisymmetry of T under $\mu \leftrightarrow \rho$; the column identity gives antisymmetry under $\nu \leftrightarrow \sigma$. For a repeated index the extra constant from the discrete derivative vanishes by the linear identity. These same zero sums show that replacing lower bond endpoints by midpoints does not change T.

To identify its symmetric part, sum the bulk second moment over one period cell and reindex the M-translates of $\hat{z}$. Absolute moments justify every interchange. For each fixed relative bond position there is exactly one such translate with its lower endpoint in $\mathcal{C}_M$. Translation invariance of the bulk kernel consequently gives

$$\sum_u \Pi_{\mu\nu}(u)u_\rho u_\sigma = -\overline{T}_{\mu\nu,\rho\sigma} - \overline{T}_{\mu\nu,\sigma\rho}. \tag{5.vi}$$

Indeed the $x_\rho x_\sigma$ and $y_\rho y_\sigma$ terms in $(x-y)_\rho(x-y)_\sigma$ vanish by the row and column zero sums; the two cross terms give the right-hand side. Subtract the Maxwell tensor in eq. 5.v, using eq. 5.iii. The residual is antisymmetric under the three transpositions $\mu \leftrightarrow \rho$, $\nu \leftrightarrow \sigma$, and $\rho \leftrightarrow \sigma$. These generate all permutations of four positions, so it is a fully alternating tensor. In four dimensions it is a multiple of $\epsilon_{\mu\rho\nu\sigma}$, proving eq. 5.v.

If the pointed totals themselves are scalar covariant under every unit translation and reflection, $T(\hat{z})$ is independent of $\hat{z}$, and one reflection kills its alternating coefficient. This is the pointed-symmetry case used by the local cancellation theorem. With only period covariance, a local subtraction instead requires its actual period/window argument; it cannot be inferred from the invariant total. The layered estimate below retains these pointed hypotheses.

The pointed Wilson family assigns the action to the same tents as the regular member. Its norm is Ca_0^2, so subtracting its own extracted coefficient preserves a constant multiple of the family norm. This is the subtraction used below. The stronger statement that the remaining translation-invariant quadratic kernel is divisible by three lattice derivatives, with exponentially decaying quotient kernels on a fixed polyannulus, is a separate analytic target when that representation is used. The moment identity alone is not presented as a proof of such division.

Frame, weight, tube and box. The functions used for the auxiliary choices are

$$\begin{aligned} \varepsilon(y) &= A_0 y^{-1/2}(\log y)^{\mathfrak{p}_0}, & \delta(y) &= A_1 y^{-1/2}(\log y)^{\mathfrak{p}_0}, \\ \delta'(y) &= A_1 y^{-1/2}(\log y)^{\mathfrak{p}_1}, & \alpha_\iota(y) &= C_\iota y^{-1/2}(\log y)^{q}, \\ R(y) &= \min\{L^b : L^b \geq (\log y)^r\}, & q &> \mathfrak{p}_0 + 2r, \quad \mathfrak{p}_1 < \mathfrak{p}_0. \end{aligned} \tag{5.vii}$$

The fluctuation scaling is $y^{-1/2}$. The frame also fixes the gauge-fixing weights, the scalings in the large-field stages and their main step, the stage Wilson constants, the Gaussian forms, and both the coefficient and cutoff in the reference laws. Every occurrence of a coupling is assigned either to such a choice or to the action weight, never to both. Cube sizes and waiting depths use the running minimum of the frame.

For a real positive-weight run and a terminal reference $g > 0$, put $h = g^{-2}$. The three real scales are

$$\mathfrak{g}_j = \theta_j^{-1/2}, \qquad g_j = x_j^{-1/2}, \qquad g = h^{-1/2}. \tag{5.viii}$$

The first scales a fluctuation in the fixed-frame step; the second is the actual running coupling on a real positive-weight run; the third is the terminal reference coupling. In a self-calibrated run $\theta_j = x_j$, the first two coincide. On an admitted narrow real tube they are uniformly comparable. Bounds for the creation domains, cutoffs and remainders use the frame scale; numerical sums along the actual flow use g_j.

A finite reference sequence $\theta_0, \dots, \theta_n$ is real, above its floor, and obeys

$$\lambda_\flat(j-i) - c_5 \leq \theta_i - \theta_j \leq \bar{B}(j-i), \qquad \check{R}_j = R(\min_{i\leq j}\theta_i), \quad \check{R}_j \leq L\check{R}_{j+1}. \tag{5.ix}$$

The lower slope $\lambda_\flat$ is the same slope used in the background and history estimates; in the weak triple it is $c_0/4$. The frame is a real reference sequence, not an independently varied list of complex action weights. In the run along this sequence,

$$x_0^\theta = \beta, \qquad x_{j+1}^\theta = x_j^\theta - b_{j+1}^\theta, \qquad z_j = \theta_j - x_j^\theta, \qquad \tau_j = x_j^\theta/\theta_j. \tag{5.x}$$

The frame and weights are separate lists. Put $\mathsf{t}^\sharp(y) = \varpi y(\log y)^{-2q_\mathrm{t}}$. A tube has positive, nonincreasing radii t_j, with $\mathsf{t}_j \le \mathsf{t}^\sharp(\theta_{j+1})$ for $j < n$. For a one-member sequence require $\mathsf{t}_0 \le 2\mathsf{t}^\sharp(\theta_0)$. The constants ϖ, q_t are fixed before the coupling ceiling, with $2\varpi \le \min\{\varpi_0, \sigma_\star\}$ and $q_\mathrm{t} \ge q_0$. The pair is alive through k if its coefficients through k are defined and $|z_j| < \mathsf{t}_j$. It has room r' when $|z_j| + r' \le \mathsf{t}_j$ at every required level.

The box of level k is the full inductive array: weights, regular families with companions and coefficients, remainders, boundary terms and standing large-field operations. It includes the bounds $|E^{(j)}| \le E_0 e^{-\kappa d_j}$, $|R^{(j)}| \le \theta_j^{-\kappa_0/2} e^{-\kappa d_j}$, the covariance and positivity clauses, identification with the two chains below, and the drift bounds. Its reserve constant c_m, also denoted c_r, enters the large-field mass bound; increasing it shrinks the box and does not change the step. An admissible torus must satisfy

$$\tfrac{1}{2}\ell L^{K-i} \ge C(L) M \check{R}_i \qquad (i \le k). \tag{5.xi}$$

All box statements are simultaneous statements on these tori and on their common companions. The positivity of a label measure $\tau_\mathcal{S} = \sum_{\mathcal{S}(h)=\mathcal{S}} \tau_h$ must not be confused with the numerical ratio τ_j.

The history part of each output box has a stronger requirement than small real masses. For the next operation write $i = k+1$, and let X be a component in either of the two classes with a pair-core representation at that level. Use its actual local histories ω, positive reference integration laws μ_ω, Borel supports S_ω, and exponents Φ_ω. All level normalizations remain in these assigned laws. For a nonnegative integrand f, the upper integral is

$$\int^* f\, d\mu = \inf\left\{ \int F\, d\mu : F \text{ is Borel and } F \ge f \right\}. \tag{5.xii}$$

It is used because the core supremum need not have been proved Borel. The required complex mass clause is

$$\frac{1}{|W_4|} \sum_\omega e^{c_\mathrm{m} v^\natural(\omega)} \int_{S_\omega}^* \sup_{\mathbf{p} \in U_i^c(X^\sim)} |e^{\Phi_\omega(\xi;\mathbf{p})}|\, \mu_\omega(d\xi) \le e^{-a_\star p_0(\mathfrak{g}_i)}. \tag{5.xiii}$$

Here W_4 is the finite hypercubic symmetry group, $v^\natural$ is the assigned history volume, $U_i^c(X^\sim)$ is the declared complex pair core, and p_0 is the fixed positive large-field logarithmic penalty. The history measure and integration coordinates are fixed before taking the parameter supremum. This is an integral of a supremum over the core; a pointwise real bound or a small pointwise polymer logarithm does not prove it.

The actual split is $\Phi_\omega = \Phi_\omega^\circ - \frac{1}{2}q_X + V_X$. At real weights Φ_ω° is real, Borel and independent of the pair. The reference quadratic form q_X may depend on the real factor of the triple presenting the pair. Only the combination $-\frac{1}{2}q_X + V_X$ is a function of that pair. A proof must retain this tie. An extra perturbation is absorbed by the reserve only after its actual support estimate $|\Psi_\omega| \le \bar{f}\, n^\sharp(\omega)$ and $c_\mathrm{m}\mathsf{l}_j \ge 2\bar{f}$ have been verified at every relevant level; $n^\sharp$ is the assigned site count and l_j its volume-comparison weight.

Drop-outs have a separate requirement, with their assigned supports Ξ_ω and compatible evaluation fibers $\mathfrak{K}^\sharp(\omega,\xi)$:

$$\frac{1}{|W_4|} \sum_\omega e^{c_\mathrm{m} v^\natural(\omega)} \int_{\Xi_\omega}^* \sup_{\mathfrak{e} \in \mathfrak{K}^\sharp(\omega,\xi)} |\mathsf{E}_\omega(\xi;\mathfrak{e})|\, \mu_\omega(d\xi) \le e^{-a_\star p_0(\mathfrak{g}_i)}. \tag{5.xiv}$$

The majorant E_ω is the one assigned to that evaluation, not a new product-core extension. When its own $1-\chi$ test is counted, the required gain is instead $e^{-2p_0(\mathfrak{g}_i)}$, with its root counted as an

additional event. A later label must read an evaluation in this same fiber at every point of its support. For a later occurrence of a second-class component, its stored variables must lie in S_ω and its later real pair in $U_i^c(X^\sim)$. Those read-out statements are additional geometric requirements. Complex quadratic forms in a drop-out evaluation are admitted only with the specified positive lower bound on their real part.

Complex variation of Gaussian ratios and stage constants additionally requires the actual inner Wilson factor to retain its assigned weight. Discarding that factor, changing its weight without a new comparison, or bounding the two tied summands separately does not establish eq. 5.xiii. The distinction is also relevant to the gauge convention at seams. If inward seam bonds are retained, an absolute contour gauge is invariant only under the corresponding seam-tied site gauges. A relative contour gauge uses its declared boundary reference. An omission convention that removes every bond with an endpoint in the next region can eliminate these seams, but its exact operation and compensators must then be checked. These are history-formation requirements, separate from the scalar flow arithmetic.

The sourced construction also fixes what a read-out means. At a formed level $j = K - u$, set

$$\begin{aligned} \mathsf{D}^{w}_{[u]}(Y) &= \sum_{z\in Y} E^{(j),w}(Y, P_\circ, z) + R^{(j),w}(Y, P_\circ), \\ e_{[u]}(Y; w) &= \mathsf{D}^{w}_{[u]}(Y) - \mathsf{D}^{0}_{[u]}(Y). \end{aligned} \tag{5.xv}$$

The zero-source run in this difference starts from the same real center and frame. The terminal read-out is the actual number $\mathsf{I}_k(w)$ at $k = K - s_\circ$, where $s_\circ$ is the stop depth, and the exact normalized terminal factor is $\widetilde{Z}^\circ(w) = \mathsf{I}_k(w)/\mathsf{I}_k(0)$. The common vacuum convention counts the removed Gaussian numbers once. Neither these read-out identities nor the terminal ratio supplies the complex local-energy or history majorants.

The Gaussian and one-loop requirements; statement 3.3. Let $\Phi_l = \log Z^{(l-1)} - \log Z^{(l-1)}(\mathbf{1})$. The localized Gaussian family is required to satisfy

$$\Phi_l(V) = \sum_X \sum_{z\in X} N^{(l)}(X, P_V, z), \qquad |N^{(l)}| \le E_{\mathsf{N}} e^{-\delta_{\mathsf{N}} M d_l(X)}, \qquad b(N^{(l)}) = b_l^0. \tag{5.xvi}$$

The constants depend only on L. No coupling occurs in these terms. Nonwrapping terms agree on all common lifts, and their pointed totals are covariant under every unit lattice isometry.

For precision, the localization meant here is the following actual one. Set $k = l - 1$, let $A_Y = C^{(k)}(Y)^{-1}$ be the supplied fluctuation form constructed inside Y and compressed to free bonds at depth cM, and define

$$\begin{aligned} f_b(Y) &= \operatorname{tr}(\log A_Y)(b, b), \\ E_{\mathsf{N}}(X; b) &= \sum_{\substack{Y\subset X \\ b \text{ at depth } cM \text{ in } Y}} (-1)^{|X|-|Y|} f_b(Y). \end{aligned} \tag{5.xvii}$$

The free-bond contribution to $N^{(l)}$ is $-[E_{\mathsf{N}}(X; P, b) - E_{\mathsf{N}}(X; P_\circ, b)]/2$, smeared by the midpoint tents. The pivot contribution is the logarithm of its actual elimination factor minus its value at $P_\circ$, smeared by the same tents. The required spectral gap and complex continuation of A_Y, the locality of its walk expansion and the c/M radius ceiling are part of this Gaussian-localization input. Finite Gaussian integration estimates alone do not prove them.

The distinction between an on-shell pair and a critical pair matters. On shell, the explicit current form expresses gauge covariance. Identifying it with the Hessian on a constrained fibre also requires the multiplier equation and actual criticality. The four tempered localization properties used in the one-loop analysis are its sum formula, common lifts, exponential domain decay on the shrinking supremum polydisc of radius $c\eta_l^2$, and invariance under constant gauge transformations. Uniform Hessian Cauchy estimates instead use the pair-direction norm preceding eq. 5.iii. For the canonical proper fluctuation precision, § 5.13 supplies the physical identification, uniform complex spectral

gap and complete local inverse and logarithm families used in eq. 5.xvii. The operator is explicitly eq. 5.cclxxviii: its inner correction uses the first lift, while its outer constraint curvature uses the second lift. The two readouts agree at a physical critical pair. Its relative contour quadratic form agrees with the absolute contour Hessian in the averaged-contour axial representative. An independent signed or counterterm-modified matrix requires its own comparison; zero-fold averaging alone would not justify one.

Input B3 requires, with the fixed momentum convention $(2\pi)^{-4}dq$,

$$\begin{aligned} c_0 &= \frac{11N^2}{12\pi^2}\log L, \qquad \left|\sum_{l=1}^{k} b_l^0 - kc_0\right| \le c_2, \\ |b_{l+1}^0 - b_l^0| &\le C_\alpha L^{-\alpha l} \quad (0<\alpha<1). \end{aligned} \tag{5.xviii}$$

Thus $b_l^0 \to c_0$. The standard convention is $\beta = 2N/g_{\mathrm{std}}^2$, giving the coefficient $11N/(24\pi^2)$ for g_{std}^{-2} per factor L. The numerical value in eq. 5.xviii requires the actual lattice trace calculation. It is not deduced by quoting a continuum perturbative coefficient.

The one-loop calculation has three steps: compose the actual Gaussian integrals, compare their averaging constraints with a positive mass, and evaluate the massive lattice coefficient. The proofs below concern the constructed homogeneous physical background. The complex-current localization requirements in eq. 5.xvii remain separate.

A tempered functional is real analytic near the trivial field on arbitrarily large tori, gauge and lattice-isometry invariant, normalized at that field, and has a plane-wave Hessian per unit volume converging uniformly in momentum to a bounded holomorphic symbol on a fixed strip. For such functionals the background substitution is required to preserve the coefficient. If $\widehat{\pi}^F$ is the limiting bond Hessian symbol with lower-endpoint Fourier convention, its midpoint form obeys

$$\begin{aligned} e^{i(p_\mu - p_\nu)/2}\widehat{\pi}^F_{\mu\nu}(p) &= b[F](|p|^2\delta_{\mu\nu} - p_\mu p_\nu) + O(|p|^4), \\ \widehat{\pi}^{F\circ p_k}(p) &= L^2\widehat{\pi}^F(p/L) + O(|p|^3). \end{aligned} \tag{5.xix}$$

The second formula uses the actual background response and uniform analytic control of its aliased modes. Its factor $L^2 = L^{-1}L^{-1}L^4$ records two form scalings and the volume ratio. The first formula follows from the moment identity and midpoint lattice symmetries.

For the actual Gaussian densities, coefficient preservation is proved in §5.6. The first jet of each invariant density vanishes, so its pullback Hessian uses only the first background response. That response reproduces an affine field modulo a gradient; the two-leg Ward identity removes the gradient, and the physical density normalization supplies the scale cancellation above. The successive maps use the actual nested criticality identity; see eq. 5.clxxvii.

The constrained Gaussian logarithms satisfy $\Psi_k = \Phi_k + \Psi_{k-1}\circ p_k$ by Cor. 5.7. This identity uses the complete constrained Hessian, including the curvature of the nonlinear constraint, before integrating in stages. Coefficient preservation then gives $\sum_{l\le k} b_l^0 = b[\Psi_k]$. The finite Schur representation uses

$$\mathbb{S}(X,Q) = \begin{pmatrix} X & Q^{\mathsf{T}} \\ Q & 0 \end{pmatrix}. \tag{5.xx}$$

Here Q is the covariant form average, R is the covariant site average, D is the covariant gradient, $\Delta = D^{\mathsf{T}}D$ is the site Laplacian, Δ^{W} is the Wilson-action Hessian on forms, and $\mathcal{K} = \Delta^{\mathrm{W}} + DD^{\mathsf{T}}$, $\Xi = \mathcal{K} - DR^{\mathsf{T}}RD^{\mathsf{T}} + Q^{\mathsf{T}}Q$. On the admitted real critical backgrounds with nondegenerate physical Gaussians, Prop. 5.6 gives the representation

$$\begin{aligned} \Psi_k = &-\tfrac{1}{2}\log|\det\mathbb{S}(\Xi - Q^{\mathsf{T}}Q + \mathrm{j}[J,\mu], Q)| \\ &+ \log|\det\mathbb{S}(\Delta, R)| - (\text{the same expression at } \mathbf{1}). \end{aligned} \tag{5.xxi}$$

Here $U = U_k(V)$ is the actual critical background, $J = \nabla_{\mathrm{fine}}A(U) = J_{\eta_k}(U)$, and $\mu = \nabla_{\mathrm{block}}(A\circ U_k)(V)$. The convention is the bilinear color pairing $\langle B, C\rangle = \frac{1}{2}\mathrm{tr}(BC)$; fine-bond inner products

have weight η_k^4, and block-bond inner products have unit weight. Explicitly,

$$\begin{aligned}\frac{d}{dt}A(e^{i\eta_k tB}U)\Big|_0 &= \eta_k^4\sum_b\langle J(b),B(b)\rangle,\\ \frac{d}{dt}(A\circ U_k)(e^{itC}V)\Big|_0 &= \sum_c\langle\mu(c),C(c)\rangle.\end{aligned}\tag{5.xxii}$$

All gradients and adjoints in eq. 5.xxi use those fixed conventions. The form $\mathfrak{j}[J,\mu]$ is fixed by second variation of the actual constrained background problem. To identify it, let $q(a)$ be the recentered logarithmic form of the straightened constraint at U, with $q(0)=0$ and $Dq(0)=Q$. The constrained Hessian on $\ker Q$ is

$$A''(U)-\langle\mu,D^2q(0)\rangle;$$

this is eq. 5.c. It is this fibre Hessian that enters the saddle determinant.

The third-order momentum statement concerns the determinant correction. Let C_k be the difference of the normalized vector logarithms with and without the multiplier term, evaluated at the actual $\mu=\nabla_{\mathrm{block}}(A\circ U_k)$. Then Prop. 5.8 proves

$$\widehat{\pi}^{C_k}(p)=O_k(|p|^3),\qquad b[C_k]=0.$$

In that proof the mixed field–multiplier Hessian has order $|p|$ by the gauge Ward identity, while the derivative of the actual multiplier has order $|p|^2$. Their composition has no quadratic coefficient. This does not give a pointwise third-order bound for the multiplier matrix. For complex continuation the real absolute determinants in eq. 5.xxi are replaced by coherent logarithmic branches on a domain where the constrained forms remain nondegenerate. The physical identity does not define an independent-current extension.

The mass calculation uses finite determinants before any thermodynamic limit. Work here on the unit bare lattice. For an expanding box $\Lambda_R=[-R,R]^4\cap\mathbb{Z}^4$, let ι_R^0 and ι_R^1 be zero extension from its sites and its positive bonds with both endpoints in the box. Put

$$\begin{aligned}U_f(x,1)=\exp(-ifx_2t_3),\quad U_f(x,\nu)=I\quad(\nu\neq1),\quad \tfrac12\operatorname{tr}(t_3^2)=1,\\ \Delta_{f,R}=\iota_R^{0*}\Delta_{U_f}\iota_R^0,\qquad \mathcal{K}_{f,R}=\iota_R^{1*}\mathcal{K}_{U_f}\iota_R^1.\end{aligned}\tag{5.xxiii}$$

These are Dirichlet compressions of the scalar and vector operators appearing in eq. 5.xxi, before its averaged-constraint corrections. The vector Wilson Hessian is defined by second variation against finitely supported bond fields, so no infinite total action is differentiated. The oriented 12-plaquette is $\exp(ift_3)$. Adjoint charged colors therefore carry fluxes $2f$ and $-2f$, while the third color is neutral.

For each fixed $z>0$ and finite box, take the real logarithms near $f=0$, where the massive matrices are positive, and set

$$\begin{aligned}\mathcal{D}_R(f,z)&=\log\det(\Delta_{f,R}+zI)-\tfrac12\log\det(\mathcal{K}_{f,R}+zI),\\ \widetilde{\beta}(z)&=\lim_{R\to\infty}\frac{1}{|\Lambda_R|}\frac{\partial^2}{\partial f^2}\mathcal{D}_R(f,z)\Big|_{f=0}.\end{aligned}\tag{5.xxiv}$$

Prop. 5.9 proves existence of this limit and independence of the lattice Følner exhaustion with the stated zero-extension compression. The identification with the bulk action coefficient uses the actual affine response and physical mass rescaling in eq. 5.clxxix. The order of the limits is the derivative at fixed positive mass, then volume, then $z\downarrow0$; no infinite determinant is differentiated.

The regulator comparison requires the finite-volume normalized Hessian kernels of this determinant pair to converge to the translation-invariant quadratic form denoted $\mathcal{L}(z)$. The bounded-remainder comparison concerns $\mathcal{L}^\sharp = \mathcal{L}(1) - \mathcal{L}'(1) + \mathcal{L}''(1)/2$, uniformly on the prescribed external-momentum strip. The resulting formulas for $N = 2$ are

$$\begin{aligned} \widetilde{\beta}(z) &= \frac{11}{6\pi^2}\log(1/z) + O(1), \\ |z\widetilde{\beta}'(z)| + |z^2\widetilde{\beta}''(z)| &\le C, \\ \sum_{l=1}^{k} b_l^0 &= \widetilde{\beta}(\eta_k^2) - \eta_k^2\widetilde{\beta}'(\eta_k^2) + \tfrac{1}{2}\eta_k^4\widetilde{\beta}''(\eta_k^2) + O(1). \end{aligned} \tag{5.xxv}$$

All field expansions are at fixed positive mass before taking that mass to zero. The vector form need not be nonnegative without its mass; the small-field hypothesis must make the massive form coercive.

The first two lines follow from Prop. 5.9. The third is eq. 5.cxcviii, combined with Gaussian composition and coefficient preservation. Its uniform $O(1)$ term comes from Prop. 5.21, which includes the averaged constraints and the gauge determinants. After the zero-coefficient multiplier correction has been removed, the difference of the two Hessian symbols is bounded on a fixed complex momentum neighborhood. If its bound is C on $\{|p_\nu| < \delta\}$, Cauchy's estimate bounds each mixed second derivative at zero by $C\delta^{-2}$. Both C and δ are independent of the depth. Thus the error compares the full cumulative normalizer with the massive determinant; it is not a sum of one bounded error per scale.

The stronger geometric convergence uses the Taylor triple $\mathfrak{J}_k = (\Delta_k, S_3^{(k)}, S_4^{(k)})$ of $\mathcal{S}_k = A \circ U_k$. With $\Lambda(B,\upsilon) = S(B) + \langle \upsilon, m(B)\rangle$, $\mathbb{G} = (D^2\Lambda(0))^{-1}$, $V_i = D^i\Lambda(0)$ and $\dot{z} = \mathbb{G}(0,C) = (HC, -\Delta_{k+1}C)$, the identity is

$$\begin{aligned} \Phi''_{k+1}(C,C') = {}& \tfrac{2}{3}\langle C, C'\rangle - \tfrac{2}{3}\langle HC, HC'\rangle \\ & - \tfrac{1}{2}\,\mathrm{Tr}(\mathbb{G}V_4[\dot{z},\dot{z}']) + \tfrac{1}{2}\,\mathrm{Tr}(\mathbb{G}V_3[\dot{z}]\mathbb{G}V_3[\dot{z}']). \end{aligned} \tag{5.xxvi}$$

Here $\eta_k = L^{-k}$ in the mass comparison, $S(B) = \mathcal{S}_k(e^{iB}) + \mathrm{GF}_k(e^{iB})$, $m(B) = i^{-1}\log\mathcal{M}(e^{iB})$, $\Delta_k = \mathcal{S}_k''(\mathbf{1})$, and $S_j^{(k)}$ is its degree-j Taylor form in the same chart. The vector HC is the linear constrained background response. Prop. 5.34 derives eq. 5.xxvi with the actual constraint and gauge vertices. Prop. 5.32 proves $\|\mathfrak{J}_{k+1} - \mathfrak{J}_k\| \le C_\alpha L^{-\alpha k}$ in the stated weighted kernel norms, and Prop. 5.36 supplies the uniform continuity of the Gaussian polarization in those norms. Their combination is Cor. 5.37, which proves the second line of eq. 5.xviii on the homogeneous physical germ. The bounded cumulative law alone would not give convergence of individual increments. These estimates preserve the geometric comparison used at equal depth in the stronger matched-cutoff route.

The two-coupling Gaussian factor. At level k, write the fibre form as $\Delta = \Delta_A + \Delta_{\mathrm{GF}}$. The frame fixes the field scaling and gauge-fixing weight; the action weight supplies $\tau\Delta_A + \Delta_{\mathrm{GF}}$. The actual finite-dimensional hypothesis is that Δ_A vanishes on the block gauge directions and Δ_{GF} acts only on those directions, with a positive nondegenerate form on each complementary integration space. Under this splitting the two ordinary Gaussian integrals factor, giving

$$\begin{aligned} \int dB\,\delta(QB)e^{-\langle B,(\tau\Delta_A+\Delta_{\mathrm{GF}})B\rangle/2} &= \tau^{-n_\perp/2}Z^{(k)}(V), \\ \int d\mu_{\tau^{-1}C^{(k)}}(B)e^{-(1-\tau)\langle B,\widehat{G}B\rangle/2} &= \tau^{n_{\mathcal{O}}/2}, \\ n_{\mathcal{O}} = 3\#(\mathbb{T}^{(k)} \setminus \mathbb{T}^{(k+1)}), &\quad n_\perp = 3|\mathfrak{X}_k| - n_{\mathcal{O}}, \quad l = k+1. \end{aligned} \tag{5.xxvii}$$

Here Q is the one-step average on the fibre, not the k-fold average, $\mathfrak{X}_k$ is the free-bond set, and $\widehat{G}$ is the gauge-fixing form in those coordinates. For positive real τ, the formula follows by rescaling only the non-gauge coordinates. Analytic continuation to the right half-plane uses the branch of $\log\tau$ real on the positive axis. Both dimensions are independent of the background. In particular $n_\perp$ is a dimension, while $d_l(X)$ is a domain metric. The two quantities are never denoted by the same unqualified symbol.

Consequently the total of the extra Gaussian family G_τ is $n_{\mathcal{O}}\log\tau/2$. Its physical second derivatives vanish and $b(G_\tau)=0$. Its norm may nevertheless be $K_{\mathbf{M}}|1-\tau|e^{-(1+c)\kappa d}$, with no small-coupling gain. This cancellation is needed to avoid a spurious accumulation of slopes of order θ_k^{-1}. The formula in eq. 5.xxvii presupposes the actual gauge splitting; it is not a consequence of a bound for a general complex Gaussian.

The complete regular family has the form $E^{(k+1)}=N^{(k+1)}+G_{\tau_k}+F$. Its physical step functional, up to field-independent normalizations, is

$$\begin{aligned}\mathcal{E}^{(k+1)}[\theta_k;x;S](V) = {}& \log\Bigl(\mathbf{T}_k^{\mathrm{sf}}[\theta_k]\,e^{-xA(U_k(\cdot))+S(U_k(\cdot))}\Bigr)(V)\\ &+xA(U_{k+1}(V))-S(U_{k+1}(V)).\end{aligned}\tag{5.xxviii}$$

The operator uses the whole torus as its small-field region, and S is the action of the incoming list. Equality between this physical functional and localized totals is asserted on their admitted backgrounds. It is not an identity of a global Gaussian at arbitrary independent-current pairs. The local formulas on such pairs have their own complex cores.

Lists, the two chains, and the increment as a function. A list contains regular families $G^{(j)}$ with their own action counterterms and unpointed remainders $H^{(j)}$. At the output radii $\bar a_\iota$ its required load is

$$\begin{aligned}\mathcal{N}(\mathfrak{f}) &= \sum_{j=1}^{k}\omega_{k+1-j}\|G^{(j)}\| + \sum_{j\ \text{aged}}[\bar a/a_j]^2\|H^{(j)}\|,\\ \omega_n &= (n+1)^{3/2}L^{-\alpha_{\mathrm{H}}(n-1)},\\ |F(X)| &\le K_{\mathbf{M}}\{\delta(\theta_k)+\mathsf{s}\mathcal{N}(\mathfrak{f})+e^{-c\theta_k\delta(\theta_k)^2}\}e^{-(1+c)\kappa d(X)}.\end{aligned}\tag{5.xxix}$$

The radius ratio is the maximum of the two coordinate-radius ratios; $\mathsf{s}=\delta(\theta_k)/\bar a_0$. The first, second and third terms pay respectively for the interaction beyond quadratic order, old-family differences, and failed Gaussian tests. The parameter choices used to infer a power bound for the last term must actually satisfy that power bound.

The fluctuation variable B ranges over nonpivot bonds. Its actual fibre map $B\mapsto V'(B)$ satisfies $\mathcal{M}(V'(B))=V$; the full fluctuation exponent $B'(B)$, including its pivot completion, is analytic and vanishes at $B=0$. It is not identified with the free array B. Old-family amplitude estimates have to hold on this actual map and its declared joint complex domains.

Likewise, in a localized Gaussian deformation with protected collar $Z_0^\sharp$, the independent coarse parameters s_Δ belong only to cubes outside that collar. The collar is a fixed-width cover and neighboring layers, not a rectangular box filled around a thin domain. Removing the protection can destroy invertibility of the compressed form. The unmarked potential-ball theorem applies after its actual matrix comparison, enlarged parameter discs, hereditary filing and smallness conditions are verified. It does not construct those domains. Local bounds on independent-current cores do not become global Gaussian identities away from complete physical pairs.

The regular part has a summable age weight because its own affine quadratic action is removed before localization. A remainder has no such counterterm, so its squared-radius load is not given an age decay. §§ 4.3 and 4.5 prove the corresponding finite implications under their actual background, bond-split, potential, core and boundary hypotheses. Their general polynomial age factors do not automatically give the exact exponent 3/2 in eq. 5.xxix. One must either verify that sharper estimate or allocate a smaller positive exponential rate and prove the resulting comparison.

A Euclidean list has one companion, the correct coefficients of its regular members, pointed totals covariant under the admitted unit lattice isometries, and invariant remainder totals on the declared covariance class. Its radii of covariance obey

$$e_{j'} \le L^{2(j'-j)} e_j \qquad (j \le j'). \tag{5.xxx}$$

Closure means that the actual output again has these properties. Restriction to smaller cores does not prove covariance. The backgrounds of fluctuating fields must first lie in a common covariance class on an open real fluctuation ball; continuation then requires a connected analytic domain on which both expressions exist. The actual domain and naturality hypotheses remain part of closure. For an isometry r, the Gaussian covariance assertion uses the actual orthogonal change of free coordinates O_{r} with $V'_{\mathsf{r}W}(\mathsf{O}_{\mathsf{r}}B) = \mathsf{r}V'_W(B)$. This is not a walk-expansion operator. Translation and cover locality of the supply rules do not by themselves give the full isometry covariance of the new family: that also uses the Euclidean covariance of the entire incoming list and its aged remainders.

For the remainder total, use the all-small-field term on its invariant open real window. The positive step kernels carry the term to its image under the lattice isometry. The localized density format is continuous on this window, so equality of the measures gives equality of the densities at the admitted backgrounds. Divide by the regular factors and inherited remainder factors whose totals are already invariant on the two background windows of the step. The remaining exponential is that of the new remainder total. It is real there, so equality of the exponentials gives equality of the totals. This is the induction at the actual backgrounds for the unsourced run and for invariant real source data. On an admitted connected complex family with the same density identity, the difference of the two totals is continuous with values in $2\pi i\mathbb{Z}$. It vanishes at the real anchor and hence throughout the family. For a holomorphic constant source, equality on the real source interval also gives this continuation.

At the age gate the next lattice's isometries preserve the old partition, so the group available for the total includes all the isometries required at the insertion step. The termwise translation covariance uses the translations preserving the actual partition. The companion and matching index sets are the other inputs for inserting the family in the regular list.

The unmarked chain uses absolute radii and starts with the empty list. The marked chain uses running radii $\check{\alpha}_{\iota,j} = \min\{\alpha_\iota(\theta_{j-1}), \alpha_\iota(\theta_j)\}$ and inserts a remainder only when $M\check{R}_j \le L^{m+1-j}$ at step $m+1$. This age is distinct from the waiting depth of a large-field operation. Before that age the remainder lies outside the regular exponent and contributes to the remainder and boundary outputs.

Restriction from absolute to running radii must restrict the same actual functions. Under that compatibility the marked difference cancels N and G_τ. The required recurrence is

$$\begin{aligned} u_{m+1} &\le K_{\mathbf{M}\mathsf{s}}(\theta_m) \left(\sum_{i=1}^{m} \omega_{m+1-i} u_i + C_{\mathfrak{h}} \theta_m^{-p} \right), \\ p &= (\kappa_0 - 2)/2 > 1, \\ \sum_{j \le m} \frac{\alpha_0(\theta_m)^2}{\check{\alpha}_{0,j}^2} \theta_j^{-\kappa_0/2} &\le C_{\mathfrak{h}} \theta_m^{-p}, \\ C_{\mathfrak{h}} &\le C \left(1 + \frac{1}{\lambda_{\mathfrak{b}}(p-1)} \right). \end{aligned} \tag{5.xxxi}$$

The lower flow bound pays one inverse-coupling power in this sum. With a constant frame its load can grow with the number of levels. Smallness of $K_{\mathbf{M}\mathsf{s}}$, relative to the summable kernel and $C_{\mathfrak{h}}$, closes a weighted supremum induction. This is an actual marked-chain requirement, not a conclusion of an unmarked finite source derivative.

The resulting increment requirement, with independent deviations before imposing their recursion, is

$$\begin{aligned} b_{k+1} &= b^0_{k+1} + f_{k+1}(z_0,\dots,z_k) + q_{k+1},\\ \sup|f_{k+1}| &\le C\theta_k^{-1/2}(\log\theta_k)^{\mathfrak{p}_0} =: \epsilon_k,\\ \sup|q_{k+1}| &\le C_v\theta_k^{-p_R} =: \epsilon'_k, \qquad\qquad p_R = (\kappa_0-4)/2. \end{aligned} \tag{5.xxxii}$$

For f, the supremum is on the whole product $|z_i| < \sigma_\star\theta_i$, with $0 < \sigma_\star \le 1/2$, and requires neither a lower flow bound nor remainders. For q, the frame obeys the weak flow bounds, $|z_i| < \varpi_0\theta_i(\log\theta_i)^{-2q_0}$ with $\varpi_0 \le \sigma_\star/2$, and the admitted aged remainder data range over their stated holomorphic parameter domain. Their bounds are $4\theta_j^{-\kappa_0/2}e^{-\kappa d_j}$, with eq. 5.xxx. The functions are real on real data. Constants are independent of the frame length, torus, cutoff and admitted parameter; the weak triple affects their threshold. These are bounds at every formed step $j \ge 1$ of the box:

$$\begin{aligned} |b^{\mathfrak{A}}_j - b^0_j| &\le \epsilon(\theta_{j-1}), \qquad & |b^{\mathfrak{R}}_j| &\le \epsilon'(\theta_{j-1}),\\ \omega_*(y) &= \sup_{v\ge y}\{\epsilon(v)+\epsilon'(v)\}. \end{aligned} \tag{5.xxxiii}$$

Here ϵ, ϵ' are the two functions in eq. 5.xxxii. The function ω_* is their least nonincreasing majorant on the admitted half-line. Calling a run alive does not replace the separate drift bounds at its earlier levels by a bound only at its last level.

There is also a weaker sufficient marked-multiplier estimate. If the marked exponent has norm $\mathfrak{n}_k \le C\mathfrak{g}_k^{\kappa_0-5}$ and the actual step is holomorphic in its multiplier on a disc of radius comparable to $1/\mathfrak{n}_k$, Schwarz yields

$$|E^{(k+1),\mathfrak{R}}| \le C\mathsf{s}(x_k)\mathfrak{n}_k e^{-(1+c)\kappa d}, \qquad |b^{\mathfrak{R}}_{k+1}| \le C\mathfrak{g}_k^{\kappa_0-7}. \tag{5.xxxiv}$$

This preserves an alternative sufficient error for the real flow; it does not prove the stronger exponent in eq. 5.xxxii. If this weaker error is used in a tuning argument, its actual power must replace p_R in the marked bare-disc estimate, and the summability condition $p_R > 1$ must be checked anew.
Well-founded induction and statement 3.1. The substantive construction input says that, for an admitted box and frame prefix through k, $E^{(k+1)}$ and its coefficient are formed before any choices using θ_{k+1}. They use the creation radii $(1+\beta_{\mathrm{s}})\alpha_\iota(\theta_k)$. The terminal covariance radius is $e_{k+1} = c\varepsilon(\theta_k)$, so it also needs no next frame entry. Once the computed next weight is admitted, the actual **RT** operation closes the next box, consistently on every admissible torus and its companion. The standing-history kernel identity is $(m(\kappa\otimes 1))|_{\mathsf{C}} = (m|_{\mathsf{A}})\kappa$ for a measure on $\mathsf{A}\times\mathsf{B}$ and a kernel from A to C. Here κ denotes a kernel, not a decay rate. This identity preserves the history sum; it does not supply its quantitative mass or locality bounds.

Choose $c_5 = 2c_2$, $C_\beta \ge c_0 + 2c_2 + 1$, and first fix

$$\lambda_\flat = c_0/4, \quad \bar B = C_\beta + 1, \qquad \bar\varsigma = C'\gamma^{\kappa_0-8}, \quad \lambda_\sharp = c_0/2 - \bar\varsigma, \quad B_\sharp = C_\beta + \bar\varsigma. \tag{5.xxxv}$$

The coupling ceiling is chosen so $\bar\varsigma \le \min\{c_0/4, 1\}$. The error exponent and other parameters must permit this ceiling; $\kappa_0 > 8$ is a sufficient choice for the displayed power of γ. An error estimate making that ceiling impossible cannot be used in this parameter selection. The marked-chain constants are computed at the weak triple before this choice. Then choose $Y_H = \gamma_H^{-2}$ above every required floor, with $\epsilon(y) + \epsilon'(y) \le \min\{c_0/4, 1\}$ and $\epsilon'(y) \le \bar\varsigma$ for $y \ge Y_H$. In particular $\gamma_H \le (\gamma^{-2} + c_5 + 1)^{-1/2}$. Increasing a reserve constant beyond $[0,1]$ is allowed only after its stated floor is increased.

Put $s_k = kc_0 - \sum_{l\le k} b^0_l$; then $|s_k| \le c_2$. For any formed prefix in the weak class,

$$\begin{aligned} x_i - x_j - (j-i)c_0 &= s_i - s_j + \sum_{i<l\le j}(b_l - b^0_l),\\ \tfrac34 c_0(j-i) - c_5 \le x_i - x_j &\le B_\sharp(j-i). \end{aligned} \tag{5.xxxvi}$$

The last upper bound uses the individual coefficient bound $|b^{\mathfrak{A}}| \le C_\beta$, $|b^{\mathfrak{R}}| \le \bar{\varsigma}$. The lower bound is stronger than the strong triple, which in turn implies the weak triple. Above a further floor, a bounded one-step drop changes the smallest power $R(y)$ by at most one factor L. Thus the computed next weight extends the frame while it remains above the floor. This is the numerical part of breaking the induction circle. The actual formation and box-closing input has not been proved by that telescoping calculation.

For $h = g^{-2} \ge Y_H$, define

$$K(\beta, g) = \min\{k : x_k(\beta) \le h + B_\sharp\}, \qquad \mathcal{W}_k(g) = \{\beta \ge h + B_\sharp : K(\beta, g) = k\}. \tag{5.xxxvii}$$

The conditional flow statement 3.1 is that the run exists through this index, has the same coefficients on all admissible tori, and

$$\begin{aligned} \lambda_\sharp(j-i) - c_5 \le x_i - x_j \le B_\sharp(j-i) \qquad & (i \le j \le K), \\ h < x_K \le h + B_\sharp, & \\ (\beta - h)/B_\sharp - 1 \le K \le 1 + (\beta - h - B_\sharp + c_5)/\lambda_\sharp. & \end{aligned} \tag{5.xxxviii}$$

At $\beta = h + B_\sharp$, $K = 0$; the terminal inequality still holds. For $K > 0$, the previous value exceeds $h + B_\sharp$ and the last decrement is at most $B_\sharp$, proving its strict lower endpoint. The lower drift forces finite exit. The two bounds for K follow by applying the upper drift to K and the lower drift to $K - 1$. Arbitrarily large β give arbitrarily large K, hence infinitely many nonempty $\mathcal{W}_k$. No stepwise positivity of b_k was used.

At depth s, write $x_{[s]} = x_{K-s}$. The exact envelopes are

$$h + \max\{0, \lambda_\sharp s - c_5\} < x_{[s]} \le h + B_\sharp(s+1). \tag{5.xxxix}$$

Put $\underline{x}_s = h + \lambda_\sharp s - c_5$, $\bar{x}_s = h + B_\sharp(s+1)$, and $g_-(g) = (g^{-2} + B_\sharp)^{-1/2}$. A diagonal subsequence of any first-passage sequence with $K \to \infty$ therefore converges at every fixed depth, to values in the closed versions of these intervals. Its couplings converge as well, with terminal value in $[(h + B_\sharp)^{-1/2}, g]$. A whole-sequence limit along tuned runs requires the later matched-cutoff input; compactness alone gives no such conclusion.

The arithmetic section, § 5.1, proves the leading logarithm from vanishing mean errors, including bounded backward excursions. In the present notation its uniform form is

$$\begin{aligned} |x_{[s]} - x_{[0]} - s c_0| \le \Delta_g(s), \qquad & \Delta_g(s)/s \longrightarrow 0, \\ \left| \frac{\beta}{K \log L} - \frac{11N^2}{12\pi^2} \right| \le \frac{\Delta_g(K) + h + B_\sharp}{K \log L} & \longrightarrow 0. \end{aligned} \tag{5.xl}$$

Here $K \ge 1$, $a_K = L^{-K}$. Denote the displayed relative error bound by δ_K, distinct from the fluctuation threshold δ_j. The function Δ_g can be chosen nondecreasing. Under the stated power/log error bound, with the marked exponent sufficient for the same estimate, one may take $\Delta_g(s) \le C + C(h + c_0 s/2)^{1/2}(\log(h + c_0 s/2))^{\mathfrak{p}_0}$. The smaller ceiling $\gamma_{\mathrm{d}} \le \gamma_H$ imposes the monotonicity and envelope floors for this quantitative bound; its constant is independent of g and s. For fixed g the relative error is $O_g(K^{-1/2}(\log K)^{\mathfrak{p}_0})$. For example $p_R > 1$, used below, suffices for the marked sum. The qualitative limit needs only an error tending to zero.

For the uniform depth statement, apply the arithmetic argument to the segment from $K - s$ to K. Its shifted one-loop partial sums have the same bound $2c_2$. Use the envelope ω_* of eq. 5.xxxiii on the self-calibrated run. One may explicitly choose

$$\Delta_g(s) = 2c_2 + \sum_{u=0}^{s-1} \omega_*(h + \max\{0, \lambda_\sharp u - c_5\})\,. \tag{5.xli}$$

This is independent of the initial coupling and cutoff, nondecreasing, and sublinear by Cesaro averaging. Integral comparison for the square-root/log envelope gives the quantitative bound above.

The sharper two-sided logarithmic comparison retained in statement 3.1 is

$$\begin{aligned} h - 2c_2 + \tfrac{1}{2}c_0(K-k) < x_k \le h + B_\sharp + 2c_2 + \tfrac{3}{2}c_0(K-k),\\ |x_k - x_K - c_0(K-k)| \le \Delta_g(K-k). \end{aligned} \tag{5.xlii}$$

These are the logarithmic inequalities since $\log(L^k a_K)^{-1} = (K-k)\log L$. The upper slope in the first line uses the uniform error bound $|b_l - b_l^0| \le c_0/2$ and the partial-sum bound, not an assumption on the sign of each one-loop increment.

Scale comparisons and layered evaluation. These estimates retain the distinction between a numerical flow sum and an estimate on an actual field. Write $t_p(x) = x^{-1/2}(\log x)^p$, $p \ge 0$. If $-c \le x_m - x_n \le B(n-m)$ for $m < n$, then, above a floor depending on a freely chosen $\beta_0 > 0$,

$$\begin{aligned} t_p(x_n) &\le (1+\beta_0)(n-m)^{1/2} t_p(x_m),\\ t_p(x_m) &\le (1+\beta_0)^2 (n-m)^{\beta_0} t_p(x_n),\\ \check{R}_j/\check{R}_{j+1} &\in \{1, L\},\\ \sum_{j=0}^{k} g_j^p &\le C_p \left(1 + \frac{1}{\lambda_\sharp (p-2)}\right) g_k^{p-2} \qquad (p > 2). \end{aligned} \tag{5.xliii}$$

The last line also assumes the lower drift bound. The first two follow by comparing the square roots and logarithms above their monotonicity floor; a bounded backward excursion is absorbed by the chosen $1+\beta_0$. The third uses the running minimum and the bounded one-step drop. For the last, use $x_j \ge x_k - c_5 + \lambda_\sharp(k-j)$, absorb c_5 by the floor, and compare the decreasing power with its integral. The exponent $p = 2$ is excluded: that integral is logarithmic.

For the field estimate fix an admitted label through k, with layers $\Gamma_n = \Omega_n \setminus \Omega_{n+1}$, $\Omega_0 = T_\ell$ and $\Omega_{k+1} = \emptyset$. The determining sequence starts at $\Omega_1^\sim$. Let Λ_j^+ be Λ_j together with its level-j islands. Base points are the unit sites lying one layer of $M\check{R}_j$-cells inside this set. Let Y be any set of these points and $\mathsf{Y}_n = \mathsf{Y} \cap \Gamma_n$. Here Γ_n is a geometric region difference; the bond record of a determining sequence is a separate object. Write $\#_n(D) = \#(\mathbb{T}^{(n)} \cap D)$. This is a site count, equal to volume in level-n units when D is a union of those unit cubes. It is not a count of localization cells. The cardinality $\#\mathsf{Y}_n$, by contrast, counts the original level-j base points. Remainder norms at a general fixed frame use $\mathfrak{g}_j^{\kappa_0}$; in the self-calibrated array $\mathfrak{g}_j = g_j$, as in the displayed layered bounds.

Assume the actual window/core and boundary hypotheses for these backgrounds, Euclidean regular members of norm at most E_0, remainders of norm at most $g_j^{\kappa_0}$, and the stated boundary-domain decay. In particular the actual boundary terms, indexed by their recorded origins o, must satisfy

$$|B_{\mathcal{S}}(j, X, o)| \le C_B e^{-\kappa d_j^\natural(\mathcal{S}; X)}. \tag{5.xliv}$$

The metric is the one in eq. 2.xxiv of § 2.2: the minimizing tree represents each chosen standing-island hull by one member, rather than filling it with all of its cubes. It is not ordinary d_j-decay

across an island. The retained layered requirements are

$$\begin{aligned}
&\left|\sum_{z\in\mathsf{Y}}\left\{\sum_{\substack{X\ni z\\ X\subset\Lambda_j^+}}[E^{(j)}(X,U_k,z)-E^{(j)}(X,\mathbf{1},z)]-b_jA(h_z,U_k)\right\}\right|\\
&\qquad\le E_1\sum_{n=j}^{k}(1+n-j)L^{-\beta_T(n-j)}L^{-4(n-j)}\#\mathsf{Y}_n,\\
&\sum_{\substack{X\in\mathcal{R}_j\\ X\cap\Omega\neq\emptyset}}|R^{(j)}(X,U_k)-R^{(j)}(X,\mathbf{1})|\\
&\qquad\le R_1g_j^{\kappa_0}\sum_{n=j}^{k}(1+n-j)\#_n(\Gamma_n\cap\Omega^{\sim}).
\end{aligned}\tag{5.xlv}$$

Here $0<\beta_T\le\alpha_{\mathrm{H}}<1$ is fixed, $\mathcal{R}_j$ consists of domains outside Z_j^+, and Z_j^+ is the stored large-field region enlarged by one cell layer. The set Ω is a union of level-k localization cubes and $\Omega^{\sim}$ has one level-j localization layer added. The constant R_1 in this display is not the geometrical gap parameter. Constants may depend on β_T, L, N and the construction parameters, but not on j,k,K,ℓ, the label or the positions.

For clarity, the local estimate behind the first line, when its actual background hypotheses have been verified, is

$$\begin{aligned}
&\left|\sum_{X\in\mathcal{X}}[G(X,U,z)-G(X,\mathbf{1},z)]-b(G)A(h_z,U)\right|\\
&\qquad\le C\mathsf{g}\left\{(\varepsilon_z/a_0)^2\sigma^{4+\beta_T}+e^{-\kappa\min\{\mathsf{D},c/\sigma\}/2}\right\},\qquad \sigma=L^{-(n-j)}.
\end{aligned}\tag{5.xlvi}$$

The member has norm g on radii $a_0\le a_1$, with $a_0\le 1$; the datum is ε_z-regular near z, $\varepsilon_z\le c$, and $\varepsilon_z\sigma\le ca_0$. Every term is evaluated in its actual core. The collection $\mathcal{X}$ contains all domains within D layers of z. With complete near coverage to distance c/σ, the right side is bounded by $C\mathsf{g}(1+(\varepsilon_z/a_0)^2)\sigma^{4+\beta_T}$. The local cancellation and nonlinear remainder sections supply the analytic implication under their full hypotheses, including the boundary-response compatibility and the own tent counterterm. Checking those hypotheses on this layered background remains part of the actual construction.

The tail in eq. 5.xlvi has no factor $(\varepsilon_z/a_0)^2$. For example a wrapping holonomy term, multiplied by its allowed exponentially small domain weight, can have zero nonwrapping companion and coefficient zero while its normalized value is nonzero on a flat torus field. A curvature amplitude cannot control that topological contribution.

To pass to eq. 5.xlv one also needs the actual geometry: $\varepsilon_z\le C\check{R}_n^2\varepsilon_n\le c\alpha_{0,n}$, the comparison $\alpha_{0,n}\le C(2+n-j)^{1/2}\check{\alpha}_{0,j}$, and the available collar depth. The condition $q>\mathfrak{p}_0+2r$ pays the first comparison. The factor $L^{-4(n-j)}$ is used entirely by converting level-j sites to level-n sites. For remainders the corresponding near-domain Cauchy bound is

$$|R^{(j)}(X,U_k)-R^{(j)}(X,\mathbf{1})|\le C(d_j(X)+3)^2L^{-4(n-j)}\left(\frac{\check{\alpha}_{0,n-1}}{\alpha_{0,j}}\right)^2g_j^{\kappa_0}e^{-\kappa d_j(X)}.\tag{5.xlvii}$$

Far domains use their norm, not this amplitude estimate. The second line of eq. 5.xlv also holds with $1+n-j$ replaced by $\varpi_{jn}=\max\{1,(x_j/x_n)^2\}$. Both dominate the required squared-radius ratio up to fixed constants. The two resulting summations give $|\mathbf{R}_k(U_k)|\le C\sum_n g_n^{\kappa_0-6}\#_n(\Gamma_n)$.

The regular counterterm in eq. 5.xlv is the sum of its own tents. An arbitrary smooth weight equal to one on Ω differs from that sum on its collar; it produces a new action term without an

age gain. For the actual profile φ_j, $\varphi_j h_z = h_z$ at every base point. The remaining rim terms obey $0 \le A(\varphi_j h_y, U_k) \le C\alpha_{0,j}^2$. The needed geometrical count is $\sum_j \#_j(Z_j^+) \le C\sum_n \#_n(\Gamma_n)$. It includes the stored collars of healed components, not just the re-done small-field regions. It is an actual window/history count and is not proved by a local Cauchy estimate.

Here $\mathbf{R}_k$ denotes the sum of the normalized remainder differences on the label. The net regular action $\mathbf{E}_k$ includes its profile counterterms and rim terms. With those counts and boundary sums, $|\mathbf{E}_k(U_k)| \le C\sum_n \#_n(\Gamma_n)$. For admitted complex perturbations of total coefficient size $\bar{c}$, regular norms at most $E_0 + 1$, remainder norms at most $4g_j^{\kappa_0}$, and the same boundary hull bounds, the sum of the elementary moduli is at most $C(1+\bar{c})\sum_n \#_n(\Gamma_n)$. This is a pointwise estimate on the admitted arrays; it is not an assertion that arbitrary such arrays are produced by the RG map. Its constants and evaluation floor need no new dependence on $\bar{c}$. The variable action coefficient is

$$x_k(y) = \beta - \sum_{j\le k} b_j \varphi_j(y), \qquad |x_k(y) - x_k| \le B_\sharp \#\{j \le k : \varphi_j(y) < 1\}. \tag{5.xlviii}$$

It equals x_k one coarse cell inside all the small-field regions. Let $A_{k,\mathcal{S}}$ be the ACTUAL label exponent and $\mathcal{E}_k$ its once-counted field-independent number. The sign convention is $\mathcal{E}_k = -E_{(k)}$, where $E_{(k)}$ is the accumulated log vacuum normalizer of eq. A.lxvi. Thus the constant part of the exponent is $E_{(k)}$. The exact density format, which is a construction input, identifies the sum of the elementary regular, remainder, rim and boundary terms with the expression on the left of

$$|A_{k,\mathcal{S}}(U_k) + \mathcal{E}_k + A(x_k(\cdot), U_k)| \le C(1+\bar{c})\sum_n \#_n(\Gamma_n). \tag{5.xlix}$$

Their modulus sum proves this consequence on the admitted backgrounds. Defining an exponent by a bound would not prove the required exact density format.

5.2.2. *A common companion and the torus floor.*
Statement 3.4 and its compatibility hypotheses. For each fixed frame and weight prefix there must be a single family $\widehat{E}^{(j)}(\widehat{X}, P, z)$ on the infinite level-j lattice, and likewise a companion for $R^{(j)}$. For every admissible (ℓ, K'), $K' \ge j$, and every label, it agrees with the torus term when the projection of $\widehat{X}$ is injective with its margin and lies one full level-j cell layer inside Λ_j. A sufficient margin places $\widehat{X}$ in a rectangle with sides less than half the corresponding torus side. The admissibility used here is

$$\ell L^{K'-i} \ge 2L^{11} M \check{R}_i \qquad (i \le j). \tag{5.l}$$

The family has the same analytic bounds and cube-translation covariance on every such lift. Its coefficient b_j is a function only of the frame and weights through level $j-1$.

Uniqueness follows because every bounded domain, collar included, embeds in arbitrarily large admissible tori. Existence requires more. First, the walk, propagator, localized background and gauge choices must be made once by a rule natural under covering maps, cube translations and the later dilations. Independently chosen objects on separate tori cannot be compared. Second, each local formation formula must use only its supplied neighborhood, earlier terms restricted there and the common numbers x_i, b_i. The induction is simultaneous over tori of unbounded side; it does not define b_j using only domains that happen to fit a particular torus.

Third, formation must be local in the history trace on the prescribed one-layer windows X_i^+. A chart identifying these traces, the bare weight, prefix coefficients and incoming local terms identifies the newly formed term on the common local core. In the one-layer interior the standing history has empty trace: healed components have already been absorbed into local terms. This actual standing-history property is what makes that interior term independent of the label. It is not implied by an inequality for the modulus of an activity.

The order $E^{(j)}, b_j, x_j, R^{(j)}$ avoids a circular definition. In particular the new large-field weight x_j does not enter the preceding extraction of b_j. This compatibility is an input to the full interacting

construction. The finite formulas proved elsewhere can verify pieces of it; their holomorphy alone does not prove it.

A locality-preserving choice of the supplied kernels. There is a conditional construction of a natural supply rule that also explains a restriction on gauge representatives. Suppose the real frame prefix and numerical parameters have first been fixed, and the actual finite-world construction has supplied local coefficients $\mathsf{K}_\omega[\mathfrak{s}; P]$, the declared joint complex charts and their uniform bounds, and its exact operator or functional sum rule. A setting $\mathfrak{s}$ records the geometric and regional data; its continuous field variables remain in P. Assume that the finite-cover identity for the total and the actual filing of lifted walks hold. Those are properties to check for the supplied linear algebra, not conclusions about an arbitrary collection of small functions.

Let $\mathcal{T}_{\mathfrak{w}}$ be the finite translation group of a world under which the class of settings is closed. Average by

$$\mathsf{K}^{\flat}_{\omega}[\mathfrak{s}; P](z, z_1) = \frac{1}{|\mathcal{T}_{\mathfrak{w}}|} \sum_{a \in \mathcal{T}_{\mathfrak{w}}} \mathsf{K}_{\tau_a \omega}[\tau_a \mathfrak{s}; \tau_a P](z + a, z_1 + a). \tag{5.li}$$

The norm and decay bounds are convex, so averaging preserves them. If two settings have the same trace on the supplied reading neighborhood, each pair of summands reads the same translated trace; the averaged terms therefore remain local. Covariance follows by reindexing the finite group. A single representative of each translation orbit would not do this: its translating displacement can depend on data outside the local hull.

For gauge orbits, the representative is instead fixed by an actual comb tree of a specified box and its entrance bond, with the root outside the box. Put the gauge equal to the identity outside and propagate its values from the root along that tree. Every tree bond then becomes the identity. This determines the gauge uniquely. Under a site gauge u, each root-to-site transport changes to $u(o)\, U[o, x]\, u(x)^{-1}$. Consequently $\mathrm{rep}(\mathcal{O}^u) = \mathrm{rep}(\mathcal{O})^{u^\natural}$, where $u^\natural = u$ outside the box and $u^\natural = u(o)$ inside. The entrance bond and the exterior root are part of its reading data. Selecting a representative by a global orbit ordering would not prove this local law.

The margins needed for this construction are also part of the input. Write ξ_l for lattice spacing and w_l for the assigned cell side at level l. A region Ω_i used to form a cover or cutoff has level at least i. Its cutoff makes its whole transition within distance less than one of its own cell sides from its complement. A walk cell of index n has size $O(M_1 \xi_n)$; write $\mathsf{q}^{\boxplus}$ for its enlargement by $5M_1\xi_n$ in the maximum norm. If n^* is the largest index with $\mathsf{q}^{\boxplus} \not\subset \Omega_0$, require $CM_1\xi_{n^*} < \mathsf{w}_{l(\Omega_0)}$; the condition is absent when no such cell occurs. Each layer enlargement adds exactly one layer of the assigned cubes, and inclusion, touching and component rules read the region within its one-cell margin. Finally each representative box, with its required neighborhood, fits in a closed cube of side less than one quarter of the side of each compared world. These conditions turn graded reading sets into the asserted local ones. Averaging alone does not verify them.

Push the averaged coefficients from a cofinal sequence of covering worlds to a fixed world and take a diagonal normal-family limit on their common finite-dimensional complex charts. Discrete settings, walks and matrix entries are countable; a countable compact exhaustion of those charts suffices. The relevant bounds are uniform before the limit. The cover filing may give a multiplicity at most $|\omega| + 1$. Its first moment is summable when the rooted walk entropy and $(CM_1^{-1/2})^{|\omega|}$ have a strict margin. Dominated convergence then preserves the exact sum rule. This uses the actual cover identity and the same declared cores throughout; it is not an extension of a physical identity to independent currents. Resolvents and square roots can subsequently be formed from the limiting basic coefficients on their supplied spectral domains. There is no need to diagonalize an uncountable list of resolvent parameters. This argument constructs coherent choices conditional on the uniform matrix, chart, reading and filing inputs just stated.

Under these inputs, the coefficient of the physical totals, taken in the large-torus limit, equals the companion coefficient in eq. 5.ii. Use the all-small-field totals for their unit-lattice symmetries;

a truncated label total has no such invariance in general. On nonwrapping domains the Hessians agree by compatibility and the torus response converges to $\mathfrak{h}_\infty$. Fix first a probing length R_{pr}, measured in level-j units and unrelated to the cube-size function $R(y)$. Take the torus side to infinity at this fixed length; once it exceeds $2R_{\mathrm{pr}}$, the relevant bounded domains are lifts. Only afterwards send R_{pr} to infinity. Domains escaping the probe carry the supplied tree-length tail, with rate after entropy $(\kappa - \kappa_{\mathcal{D}})(R_{\mathrm{pr}}/(2M) - 2)$. The rooted entropy is absorbed by the chosen κ; response decay controls remote differentiations. Dominated convergence therefore passes the differentiated invariant totals to the companion kernel. This proves the sum-rule identification, including that a constant total has coefficient zero. It requires the uniform response convergence and domain bounds just stated. The response limit need exist along a coherent admitted family of representatives, not along every possible gauge choice on separate tori. An oscillating fine gauge fixing the unit-lattice sites can destroy such a derivative limit. If two admitted limits differ by $(\mathrm{d}\lambda_B, 0)$, the differentiated gauge Ward identity annihilates that direction in the Hessian at $P_\circ$; the coefficient is therefore unchanged. This independence does not prove existence of the all-level response limit.

Wrapping domains have $d_j(X) \geq c\ell L^{K-j}/M$. Their exponential decay is used directly and does not define an additional renormalized coefficient. For a self-calibrated run, the cube ratio shows that eq. 5.l at the terminal level implies it at all preceding levels. Since $x_K \leq h + B_\sharp$, it suffices that

$$\ell \geq \ell_\star(g) := 2L^{11}MR(g^{-2} + B_\sharp). \tag{5.lii}$$

Write $\mathfrak{L}(g)$ for the compatible torus sides above this floor. The same floor works on the reference line, whose terminal value is h. The side is selected after g; the cutoff K remains unrestricted. This statement is a uniform construction requirement, not an assertion of infinite-volume uniqueness.

5.2.3. *Frozen choices, slopes, and tuned bare couplings.* The reference line for a run of length K is

$$\theta_k = h + c_0(K - k), \qquad h = g^{-2}. \tag{5.liii}$$

Equal depth in runs of lengths K and $K+1$ gives equal frame numbers, lattices, partitions, radii and cube sizes. It does not give equal tests, since their backgrounds live over different bare lattices, and it identifies no labels or histories between those runs. The line beginning at depth s is the reference line of $g_{[s]} = (h + c_0 s)^{-1/2}$.

For fixed choices, every inserted characteristic function is the same real Borel function of its integration fields. The actual operations act linearly on the initial density. If their required dominated analytic estimates hold, applying them to $e^{-(\beta-\zeta)A}$ gives holomorphic measures and local families in ζ. No derivative of a cutoff is taken. Continuity of the real run follows from the actual fixed-frame input-list Lipschitz estimate while the arrays retain their tube room. It is continuity, not a contraction. When each bare coupling chooses its own frame, the tests and domains can change discontinuously and this argument does not apply. For a fixed-frame remainder comparison, both real centers β, β' must be alive through the compared prefix. The interpolated aged remainder list is

$$H_\upsilon^{(j)} = R_{\beta'}^{(j)} + \upsilon(R_\beta^{(j)} - R_{\beta'}^{(j)}). \tag{5.liv}$$

At their common frame domains, the individual norm bounds give $\|H_\upsilon^{(j)}\| \leq 4\theta_j^{-\kappa_0/2}$ for $|\upsilon| < 3/2$. Its companion, covariance radius and invariance properties must be those required by the admitted list slot. The holomorphic list estimate then applies to this actual interpolation; an unrelated real center or an unspecified change of the earlier histories cannot be substituted for it.

The exact decompositions of unity and normalized gauge factors imply that the bare measure, partition function and observable integrals are independent of admitted auxiliary choices. Intermediate densities after a large-field operation need not coincide. Comparing their extracted couplings

requires E4. The retained self-calibrated versus reference-run target is

$$\max_{k\le K}|x_k(\beta)-x_k^{\natural}(\beta)|\le Cg(\log g^{-2})^{3\mathfrak{p}_0+1}\le 1. \tag{5.lv}$$

It is required for first-passage data with $\varrho\ge B_\sharp+1$ and $0<g\le\gamma_{U'}\le\gamma_U$. Threshold and large-field changes contribute at most $C\theta_k^{-(\kappa_0-4)/2}$ per step, and gauge-fixing changes at most $C\theta_k^{-3/2}(\log\theta_k)^{3\mathfrak{p}_0+1}$. Proving those actual comparisons remains necessary. Equality of bare integrals does not prove eq. 5.lv.

The small-field coupling polydisc has relative radius $\sigma_\star\theta_j$. The large-field tube is narrower. On a level-j unit cube, an action-weight change has size

$$|x-\theta_j|\alpha_{0,j}^2\le C\mathfrak{t}_j\theta_j^{-1}(\log\theta_j)^{2q}. \tag{5.lvi}$$

A logarithmic tube factor at least $(\log\theta_j)^{-2q}$ keeps this bounded. A fluctuation difference gains a further $\delta_j/\alpha_{0,j}$. This explains the different roles of the full small-field polydisc and the large-field tube. It does not justify extending the entire interacting history construction to the larger polydisc. Complex bare couplings in the latter construction are obtained near admitted real runs through the fixed-source analytic domain.

Statement 3.5: the weaker marked-disc slope route. For this paragraph require $p_R>1$. The unmarked term has the independent-variable bound in eq. 5.xxxii. Along the actual run the marked term is allowed a different analytic input: on a disc in the bare variable about each admitted real centre, of radius at least $c_\beta\varrho$, it is holomorphic and bounded by ϵ_k'. Bounds are uniform on centres whose required prefixes retain the prescribed tube room. On a smaller concentric disc, Cauchy therefore gives $C\epsilon_k'/\varrho$. This hypothesis includes the actual remainder dependence on that bare variable. It does not assert that this dependence is a function of independent deviations alone.

On the admitted constant-source disc this bound has a native frozen-step calculation. Form $E^{(k+1)}$ on the whole lattice with the real fluctuation Gaussian, its bond cut and the source vertex. The input list consists of the renormalized regular families and the vacuum-subtracted remainders that pass the age test. Form $P^{\flat(k+1)}$ with the same Gaussian, cut and running shifts, keeping just the recursively unmarked regular families. Put $Q^{(k+1)}=E^{(k+1)}-P^{\flat(k+1)}$.

At the frozen shift ζ_k, write $s=1-\zeta_k g_k^2$. The quadratic part of the source vertex is $-\frac12(s-1)\langle B,\Delta B\rangle$; together with the real Gaussian it gives the Gaussian of the unmarked recursion at ratio s. Its other vertices are the remaining terms of the same rewritten interaction. On a common cut and common radii the renormalized inputs agree by induction, up to field-free constants, with the unmarked list evaluated at the running shifts. Those constants do not enter the marginal coefficient. Hence the unmarked output supplies the one-loop and f_{k+1} terms in eq. 5.xxxii, and the marked coefficient is $\beta[Q^{(k+1)}]$.

In the frozen integral, split each inherited regular group by linearity of its own marginal subtraction. The groups from $Q^{(j)}$, together with the admitted remainders, are exactly the marked inputs. The first-stage contour estimate bounds their total weight by the small fluctuation factor times the sum of the renormalized Q loads and the aged remainder load. Removing these inputs leaves the unmarked integral. The second-stage cluster comparison bounds $\|Q^{(k+1)}\|$ by a fixed multiple of that marked weight. In the supremum form of eq. 5.xxxi, choose the coupling small enough that this multiple times the fluctuation factor and the total regular-group kernel is at most $1/2$, and the remainder load has coefficient at most $1/2$ as well. The remainder weight here is comparable to x_j/x_{k+1}; its sum uses the smallness at birth. Since both the earlier supremum and that load are bounded by the current envelope, the new Q lies in the same envelope, of size $C\theta_k^{-(\kappa_0-2)/2}$. Extracting the coefficient on the admitted running radius costs $C\theta_k$, giving the required $C\theta_k^{-p_R}$ on the bare-variable disc. This calculation requires holomorphic inherited remainders and the identification of the actual frozen production with these lists, including its cut and radii.

For the inherited remainders, the native step uses two bounds on the same constant-source disc. The contribution from summing healed components has weighted norm at most $Ce^{-p_0(g_{k+1})}$ on domains away from the one-layer enlargements of the live regions. Separate the empty-family term before taking the polymer logarithm. A polymer disjoint from all second-class strips contains a healed operation. The terms meeting those enlargements go into the boundary list and use the relative hull bound there. The complete boundary slot also retains its empty-family term with the ordinary amplitude.

The second contribution comes from the ordinary fluctuation integral. Keep its normalized reference Gaussian and its cut fixed as in [B88b, (3.26)–(3.29)]. Use two potential parameters. The first multiplies the boundary terms, including the changes of the stored operations, the weight variation near the large-field region, the bounded rim part of the quadratic gauge correction, and the change-of-variables terms of the rim. The second multiplies the recent remainders evaluated as differences between the pre-integration and post-integration backgrounds. For an origin j, first subtract the vacuum value of $R^{(j)}$. The vertex in the interpolation is the difference $R^{(j)}(U_k) - R^{(j)}(U_{k+1})$, evaluated on the two actual backgrounds in the variables of this integral. The boundary expectation varies the first parameter with the second at one. The recent-remainder expectation varies the second with the first at zero. The logarithm with both parameters zero uses the aged, vacuum-subtracted remainders with the regular vertices. The Gaussian normalizer belongs to the regular part of the step and the reference Gaussian used for these expectations has unit mass before the cut.

The two changes of the logarithm are the integrals of the corresponding normalized expectations. In the local expansion, the second change selects precisely the clusters with at least one marked recent vertex on the same cut. File its terms on the one-layer inset as $R''^{(k+1)}$. Its terms in the rim, and the first expectation's terms, go in the boundary family. The conditional Gaussian in the history can have a nonzero mean. With precision blocks A_{II} and A_{IE}, and exterior field a_{E}, the mean is $-\mathsf{A}_{\mathrm{II}}^{-1}\mathsf{A}_{\mathrm{IE}}a_{\mathrm{E}}$. The restriction to the centered term takes place after decoupling. In eq. 4.xcv every parameter off the carrier Z is zero. A walk meeting a cube off Z then has weight zero. The remaining Gaussian terms use the restricted form, its resolvent and the exterior columns carried by Z. A nonzero column must carry the whole closed bond segment of its exterior argument. For $Z \subset \Lambda_{k+1}$ there are no such exterior or crossing bonds. The localized mean therefore vanishes and every fluctuating bond of Z has the interior cutoff T_k. With the same reference scale and Gaussian factor the localized operators are those of the constant sequence of regions [B88b, p. 278]. In the logarithm each such Z, and the full collar of each selected potential, lie in the filing domain X. The marked coefficient is thus the centered coefficient at the same local pair and with the same restricted list of input potentials. Match the decoupling guard to compare individual summands. Regrouping the decoupling brackets gives the guard-independent coefficient. A term reading the rim uses the conditional formula and is filed at the boundary. The one-layer inset decides where the centered recent coefficient is filed.

The comparison uses a common localization and source presentation. For the presentation which instead puts the constant shift into the Gaussian, the comparison of marginal coefficients uses the identity of the totals up to the field-independent Gaussian normalization.

The recent input band is $J_{\mathrm{rec}}(k) = \{j \le k : M\check{R}_j > L^{k+1-j}\}$, restricted to existing remainder families. In particular it is defined at the input step, and includes the level k family when it fails the ageing test. A localized vertex from j is estimated first in its own box. In comb gauge the scaled bond logarithm there is bounded by $CD_Y t^2$ times the output radius, where D_Y is the diameter of the input domain plus the box margin in its own units. The first localization is at input level k, so $t = L^{j-k}$. Scale these bond logarithms by one complex parameter and use the admitted background map on the resulting disc. The first derivative at the trivial pair is an invariant linear functional for the adjoint action of the constant gauge group and vanishes by semisimplicity. The second-order Schwarz estimate bounds the vacuum-subtracted value by the input size times $CD_Y^2 t^4$ times the

squared radius ratio. The difference of the two background values has the same bound up to a fixed factor. If the scaling disc has radius at most one, use the input bound at the admitted endpoints. In the rooted sum the output tree length is at most t times the input tree length. The spare decay sums the diameter factor. A domain extending beyond the containing box has input tree length at least of order t^{-1}. For $j < k$, extract t^4 from its spare exponential tree decay and keep a positive tree rate for summation. At $j = k$ the input and root weights already leave a positive tree rate and $t^4 = 1$. In both cases the $O(t^{-4})$ fine anchors in a root cell are paid by the t^4 gain.

In the first localization of these marked vertices, fix a real fluctuation B on the cut $\sup|B| \le T_k$. The response estimates of [B88a, (1.17)–(1.22)] on the cut-parameter contours bound the undilated response on the read set by $K_R g_k \sup|B|$ and the part localized at a block by $K_R \eta g_k \sup|B|$. Here $\eta \le 1$ is the decay factor from the block to the read set and K_R is uniform on the output tube. The estimates apply also to complex B. If a is the amplitude of the localized part, the sufficient containment inequality is

$$(1 + \eta|a|)K_R g_k \sup|B| \le \alpha_{*,k},$$

where $\alpha_{*,k}$ is the response radius. The top layer uses the base and dilated parts separately within its analyticity margin. At older layers the shift gains the decaying transfer factors and uses the corresponding layer room. With $\theta_k = K_R g_k T_k/\alpha_{*,k} \le 1/3$, the disc in the rescaled amplitude ηa has radius $\theta_k^{-1} - 1 \ge 2$. If the vertex differs from its vacuum value there by at most N, its change from the undilated vertex is bounded by $2N$. The Schwarz estimate on the subdisc $|\eta a| \le 2$ bounds the change by $4N/(\theta_k^{-1} - 1) \le 6\theta_k N$. Each differentiated cut parameter, integrated over $[0, 1]$, contributes at most $e^{\kappa_1}(e^{\kappa_1} - 1)^{-2}$ by Cauchy's formula on $|\sigma| = e^{\kappa_1}$. The rooted sum therefore has the factor $\sigma_k \le C(\log x_k)^{\mathfrak{p}_0 - q}$ on the fixed real cut.

The second localization at zero exterior fluctuation keeps this cut in the Mayer factors. In eq. 4.xciv, Y_0 contains the free bonds read by each selected potential that requires a field box, including each marked one. The failed-cut bonds lie outside Y_0. For the conditioned Gaussian formula of [B88a, (2.5)–(2.6), (2.14)], let C_σ be the conditional covariance on the integration region and $\Gamma_\sigma X$ the exterior linear term, written in real auxiliary Gaussian coordinates X. The inner integral uses the real variable B, with Gaussian density of covariance C_σ and the multiplier $\exp(-\langle B, \Gamma_\sigma X\rangle)$. The cut-parameter contours complexify C_σ and Γ_σ. The potentials in the integrand are still evaluated at B with the factor χ_{Y_0}, and each Mayer contour multiplies its potential by a scalar. Take the potential bound on the support of χ_{Y_0} before enlarging the Gaussian domain for the modulus estimate. The determinant and Gaussian bounds then justify continuation on the same real integration space. Thus the small factor from the first localization is available on the contours used to form the ordinary Mayer activities.

Scale the marked activities together by a complex parameter. If their norm is ε_m, the unused activity margin gives a disc of radius proportional to ε_m^{-1} on which the localized output has the usual uniform decay bound. Subtract its value at zero and apply the Schwarz estimate at one. This makes the output proportional to the marked activity norm in the same weighted family. The bound for this contribution is therefore

$$C\sigma_k \sum_{j \in J_{\mathrm{rec}}(k)} w_{k,j} g_j^{\kappa_0},$$

where $w_{k,j}$ is the squared ratio with the enlarged radius of this step in the numerator. The amplitude bound uses the remainder's own analyticity domain for each read. The Mayer bound uses the same cut on every marked read and the activity margin uniformly on the admitted source disc. With the running-radius estimate $w_{k,j} \le C(k+2-j)$, the sum of these ratios is at most $C(1+\#J_{\mathrm{rec}}(k))^2$. To bound the band length, take $u = \lceil C(1+\log\log x_k)\rceil$ with the constant large enough. The flow gives $x_{k-u} \le x_k + C'(1+u)$ when $k \ge u$. Then $M\check{R}_{k-u} \le L^{u+1}$ by the logarithmic bound on $\check{R}$. Thus $k-u$ has already reached the ageing test. The grid rule $\check{R}_i \le L\check{R}_{i+1}$ implies $L^i\check{R}_i \le L^{k-u}\check{R}_{k-u}$ for $i \le k-u$, so those levels have reached it as well. If $k < u$, the whole list has length at most

u. The reference flow then gives $g_j^{\kappa_0} \le C g_{k+1}^{\kappa_0}$ throughout this band. For the native reference radii the sharper comparison is

$$w_{k,j} = \tfrac{9}{4}\,\frac{x_j}{x_k}\left(\frac{\log x_k}{\log x_j}\right)^{2q}.$$

Here the numerator uses the radius enlarged by $1+2\beta_s$ at input level k. The adjacent-scale room puts the actual output space with radii $(1+\beta_{\mathrm{pr}})\alpha_{k+1}$ inside this space, using $(1+\beta_{\mathrm{pr}})\alpha_{k+1} \le (1+\beta_s)\alpha_{k+1} \le \hat{\alpha}_k$ componentwise. Since $x_j = x_k + O(1+k-j)$ in the band, $w_{k,j} \le 3$ at small coupling, so this reference estimate costs only a single power of the band length. For $q > \mathfrak{p}_0$, even the running-radius bound with the squared count tends uniformly to zero after multiplication by the fluctuation factor.

Each contribution is at most $g_{k+1}^{\kappa_0}/2$ at weak coupling. Both bounds use the native enlarged remainder charts. In the recent term the running shifts are the functions of the single bare constant source on its disc. The marking parameter is independent of that source. The healed term is holomorphic by the majorized resummation, and the recent term by the fixed-domain fluctuation expansion. Restrict their output to the running remainder radii when taking the inherited norm. The sum has norm at most $g_{k+1}^{\kappa_0}$ on the constant-source disc, so the next frozen integral can use it as inherited data. For this application the actual formation and norm domains must agree with the native ones.

The inherited family also needs the common-lift and symmetry properties. Compare the partial maps on the same local field arguments, using the same reference sequence, local operator terms, weights, source, increments and inherited lists. For coherent input, the earlier lists are determined on their own read sets by the standing history. In this history the healed components have already been integrated out. The remaining large-field regions are nested into the current complement, so the standing trace on an interior window is empty. The trace uses one layer of the label cells at each relevant level. On a carrier with this window inside Λ_{k+1}, the local data are those of the all-small-field history. This identifies the inherited coefficients as functions on the common domain. An evaluation in the actual history uses its own local field argument. A common non-wrapping lift transports the walk terms, cutoffs and inputs by a translation preserving their grids. The cluster logarithm and the filing therefore give the same coefficient on the lift. At a real constant source, evaluate on the backgrounds of the common hereditary window of very regular block fields. For isometries preserving the level-$(k+1)$ label cubes, the density obtained by summing the histories with the all-small-field label and the regular integral with its aged inputs are invariant. In the quotient of these densities the carried background factors are invariant too, since the new label cubes are unions of the older ones. Remove those factors and normalize at the trivial field. The quotient is now the exponential of the vacuum-subtracted new total. Its exponents at a field and at its isometric image are real and therefore equal. Holomorphy extends that identity to the connected constant-source disc on the same backgrounds.

When a remainder of origin j first enters the regular integral at step $k \to k+1$, the ageing rule gives $M\check{R}_j \le L^{k+1-j}$. Both lengths are powers of L on the nested grids. Thus the origin cube side divides the spacing of a level-$(k+1)$ block in origin units. Coarse unit translations and the coordinate isometries preserve the origin cube partition. The invariance of the earlier total therefore suffices for its use in the regular integral of this step.

For the strip estimate in the healed contribution, measure lengths in localization cubes of side M. Work on the class of boxed components of Z separated by at least $\check{R}_k$ cubes.

First consider the plain values in the input of the strip estimate. For a value unit v with carrier K_v, adjoin the full components of Z sharing a cube with K_v to obtain its filing domain Y_1. This leaves $Y_1 \setminus Z = K_v \setminus Z$. A tree for K_v meets every cube of $Y_1 \setminus Z$, so $d_Z(Y_1) \le d_k(K_v)$. For an

exterior Y_1, fix a cube $\Delta \subset Y_1 \setminus Z$. Every carrier in the fibre contains Δ, giving the bound

$$e^{ad_Z(Y_1)} \left\| \sum_{v:\, Y_1(v)=Y_1} \mathrm{val}_v \right\| \le \sum_{v:\, K_v \supset \Delta} e^{ad_k(K_v)} \|\mathrm{val}_v\|, \qquad a = 31\kappa/20.$$

Thus one needs a weighted sum at a cube outside Z. This first filing spends no decay and introduces no factor for the size of a component of Z.

The current chain values filed at $m < k$ are confined to a single component of Λ_k^c, by connectedness and $\Omega_{m+1}^c \subset \Omega_k^c \subset \Lambda_k^c$. A carrier of this kind meeting Z stays in Z. For those filed at k, the stage norm at $8\kappa/5$ bounds the column at rate a by $C_0^\sharp$ per stage. The current chain therefore costs at most $N(g_k)C_0^\sharp$.

For the plain top-level values left by an earlier arrest, put $j = k - s$, $s \ge 1$. Their native carrier X meets Ω_k^c. If its coarse closure $\bar{X}$ meets both Z and its complement, it crosses the collar between Ω_k^c and Λ_k. Consequently

$$d_j(X) \ge (2\check{R}_k - 1)L^s, \qquad d_k(\bar{X}) \le L^{-s} d_j(X) + c_{\mathrm{tr}}.$$

Coarsening spends at most $31\kappa/60$ of the native rate $8\kappa/5$, leaving $13\kappa/12$. Use the resulting factor $e^{-\kappa \check{R}_k L^s}$ and sum the native weighted sizes over the L^{4s} fine cubes in Δ. The stored values are a subsum of their birth-stage norm, which already sums the formed terms over the components of that stage. In this count each formed term is taken once. The bound over all stages and ages is thus

$$C_0^\sharp e^{ac_{\mathrm{tr}}} \sum_{s \ge 1} N(g_{k-s}) L^{4s} e^{-\kappa \check{R}_k L^s},$$

where the sum uses only existing earlier levels. The reference flow gives $N(g_{k-s}) \le C(\log x_k + c_s)^{r_{\mathrm{pr}}}$ with $c_s = O(\log(2+s))$. The width bound $\check{R}_k \ge (\log x_k - 1)^{r_{\mathrm{pr}}}$ bounds this count by $C(2+c_s)^{r_{\mathrm{pr}}} \check{R}_k$ for $\log x_k \ge 2$. Since $\check{R}_k \ge 1$, one has $\check{R}_k e^{-\kappa \check{R}_k L^s} \le 2(e\kappa L^s)^{-1} e^{-\kappa L^s/2}$. This bounds the contribution uniformly in k and the coupling. At the final stage, a lower-filed term whose carrier reaches bonds adjacent to an arrested piece must also satisfy the collar test if it enters the exterior column. When its native carrier passes this test at level $\ell < j$, use the birth-stage norm at $4\kappa/5$ and coarsen from ℓ. The spare rate is at least $17\kappa/60$, so its stage contribution is bounded by $CL^{4t} e^{-17\kappa \check{R}_k L^t/60}$, with $t = k - \ell$. At this filing level there are at most t earlier birth steps and at most $C(2 + c_t)^{r_{\mathrm{pr}}} \check{R}_k$ stages at each step. The collar factor sums these terms by the same absorption.

The change-of-variables terms of an earlier arrest use their own filing level $\ell < k$. Set $t = k - \ell$. Their connected exterior domains X meet their stage band inside Ω_k^c, so $d_\ell(X) \ge (2\check{R}_k - 1)L^t$. After coarsening, a native pointwise rate of $4\kappa/5$ leaves $17\kappa/60$. In using this pointwise estimate we must count the birth blocks even when several terms have the same localization domain.

For terms carrying their stage-cover anchor in X on the filing grid, there is at most one term per cover block, and the bands of simultaneous arrests are disjoint. At a fixed birth step and stage the number of such blocks for X is therefore at most a fixed multiple of $1 + d_\ell(X)$, by the cube count for a tree meeting X. The same bound applies for anchors in a fixed-width enlargement on this grid. The share $\kappa/20$ of the spare rate sums the rooted domains with this additional linear factor. The remaining $14\kappa/60$ pays the collar, and the L^{4t} fine roots in the coarse cube give a stage bound of the form

$$CL^{4t} \exp(-\tfrac{14}{60} \kappa \check{R}_k L^t).$$

There are at most t possible earlier birth steps at this filing level. Their stage counts are bounded by $C(2 + c_t)^{r_{\mathrm{pr}}} \check{R}_k$. Thus the sum over $t \ge 1$ is uniform by the same exponential absorption as for the top-level values. For a cover on a different grid we must show that its localization still has this anchor count in filing units, or use its own rooted norm before summing the scales. For the bond-indexed terms of an interrupted change of variables, the bound on each bond term gives

its size on the admitted chart. We must also identify the localized carriers and show the required tree-length decay on them.

The two kinds of stored boundary terms use different carriers. The boundary term formed in the **T**-operation has a pointwise bound at rate at least 2κ on its native domain. At $j = k$, the spare rate $2\kappa - a = 9\kappa/20$ sums the domains through the fixed root. At an earlier level $j = k - s$, the same coarsening bound leaves $89\kappa/60$. Spend $\kappa/20$ on the rooted native domain count and use the collar to bound the rest. With $\kappa \geq 10 \log L$ and $L \geq 3$, the remaining $86\kappa/60$ pays the L^{4s} fine roots and leaves $e^{-\kappa \check{R}_k L^s}$ for the sum over ages.

The boundary term formed in the **R**-operation is stored on its relative label, whose coarse image serves as the filing carrier. At the filing site the term is from an earlier level $j = k - s$, since the new one is formed after filing. Its label meets an earlier strip or zone inside Ω_k^c. For an exterior value the label therefore crosses the same collar. On the label the stored rate is at least κ, and coarsening spends at most $31\kappa/60$. Of the remaining $29\kappa/60$, use $\kappa/20$ for the rooted label count and split the rest into two equal parts. The first pays the fine roots since $\frac{13}{60}\kappa L^s \geq 4s \log L$. The second gives the summable factor $e^{-13\kappa \check{R}_k L^s/60}$. The skeleton fibres are summed when the stored terms are formed. This estimate uses their resulting bounded prefactor and the decay on the labels.

For a chain term with a nonempty decoupling bracket, the original filing level supplies the rate before the bracket enlargement. At the current level the rate $8\kappa/5$ leaves $\kappa/20$ after allocating a to the original carrier. At a lower level $j < k$, coarsening leaves $4\kappa/5 - 31\kappa/60 = 17\kappa/60$ in level-j units. The fixed prefactor $e^{ac_{\mathrm{tr}}}$ goes into the constant paid by the inverse localization parameter of the bracket estimate. The added decoupling carriers are charged in that estimate.

For the terminal, renormalized, tail and boundary rows of the size table, keep the point anchor when taking the weighted domain sum. In a zone n, set $u = L^{j-n}$ for an anchor of level $j \leq n$. The $O(u^{-4})$ fine points in a zone cell use up the u^4 in a short piece or a local Wilson term. For a short boundary piece, the anchor must lie in a slab of thickness $O(1 + n - j)$ fine units around a zone face. The slab count is $O((2 + n - j)u^{-3})$, leaving $(2 + n - j)u$ after the size is inserted. For an old tail the lower bound on its diameter supplies u^5 from the spare decay, so the same fine-point count leaves u. The frozen core has the extra power $u^{\hat{\alpha}}$ with $\hat{\alpha} > 0$, and the defect row has $L^{-(k-j)/2}$. The renormalized remainder row uses its admitted weighted sum over levels after the u^4 and fine-point count cancel.

In the rooted column the weight $7\kappa/4$ spends at most $7\kappa/12$ in level-j units when $j < k$. At $j = k$ the ordinary rows present there have the improved rate 2κ. After the rooted domain count and diameter factors, keep $e^{-\delta\kappa d_j(X)/4}$ for the point anchors. For anchors in the current zone, use the per-cell scale sums and then sum over cells. There are at most $(2i+3)^4$ cells in a current cube shell of radius i, and a domain reaching the root from this shell has length at least $(i-1)_+$ in its native units. For a unit born at $j = k - s < k$ with anchor in an old zone, the domain must cross the collar as well. One can split the remaining exponential into two halves and use $d_j(X) \geq (2\check{R}_k - 1)L^s$ in one and the distance from the anchor to the root in the other. In current units there are at most $C(ML^s)^4(2i + 3)^4$ fine anchors in a shell of radius i, and the distance bound in fine units is at least $L^s(i-1)_+$. The shell sum is bounded by a constant times L^{4s}, which is summable in the ages against the retained collar factor. These estimates use the assigned sizes and their native anchor domains, including the domains of the subtracted copies.

The short singles and the Wilson terms across the boundary use an enlarged filing label $\mathrm{lab}_1 = K_v \cup \mathfrak{h}_v$, where $\mathfrak{h}_v$ covers the additional reads. Write c for the fixed bound on $d_k(\mathrm{lab}_1) - d_k(K_v)$. For these terms split the filed value into the uncut value, which is charged at the size in the table, and its difference from the cut value. The carrier estimate bounds the absolute difference by $\mathfrak{N}^\flat(v)$ on the admitted cut chart, including when all decoupling families are empty. To use this estimate we must verify that the auxiliary consumer lies in its filing domain and is admitted on the cut chart.

For the exposed units in this exterior column, their point lies in a level-k collar cell or in a level-k cell next to the treated boundary on the Ω_k side. The cell weight there is 3. For such a cell Δ', the sum of $\mathfrak{N}^\flat(v)e^{(a+9)d_k(K_v)}$ over its point anchors is bounded by $27\mathsf{K}^\flat$ by the per-cell estimate at $a+9 \le 8\kappa/5$. If the enlarged label contains the exterior root Δ, its connectedness gives $d_k(K_v) \ge (i-1-c)_+$, where i is the cube distance between Δ' and Δ. Therefore even at the activity weight $a+8$ the cost of summing the difference over the anchors is at most

$$27\mathsf{K}^\flat e^{(a+8)c}\sum_{i\ge 0}(2i+3)^4 e^{-(i-1-c)_+}.$$

The constant is uniform in the coupling and in the level. This sum uses the location of the exterior point anchors. At an anchor beside a foreign arrest window the per-cell estimate uses its window weight. The amplitude Cauchy estimate for a pointed piece supplies the extra factor there, even for its continuation with all subsequent cut families empty.

For the terms with cut families, keep a connected label A and a separate read set $\mathcal{R}$ at each link. If Δ is a cut cell, let $\rho(\Delta)$ be the smallest size of a connected cut family joining it to a source cell. Here source cells are the designated cells adjoining the exempt boxes. The first link for plain interior terms cuts only cells disjoint from $\Lambda^\sim$, and links two to four leave cells meeting Λ uncut. Each contributing connected component of a cut family contains a source cell and meets the reads, with a source cell at a box meeting the reads also counted. The latter condition allows the cell's current parent to be at distance up to $c_\rho\rho(\Delta)$ from A. For a read already in the label this distance is at most one. At an exempt box it is bounded by the box diameter. At the rim of a component already met by the label it is at most $100\check{R}_k+1$, and the nearest source box is at distance at least $9\check{R}_k$ from the rim. At the first plain interior link the read set is $X_v \cup \hat{X}_v$ in the treated component. Cells eligible there are current cells. Their parents are within one cube of K_v, or within the hub diameter plus one for a source cell seeing the reads across a hub. These bounds apply on the separated geometry. At each link we must check them for the actual reads, including reads along a live strip.

Apply the fundamental theorem of calculus separately in each nonexempt cut variable. The term with all cut families empty is the value at the zero cuts. In every other term use the mixed derivative in precisely the selected variables, integrated over their real intervals. Cauchy on the common product polydisc of radius e^{κ_1} gives a factor $e^{-(\kappa_1-1)}$ per selected variable, since the circle at a point of $[0,1]$ can have radius up to $e^{\kappa_1}-1$. For a boundary-type term use its supremum on this chart. For a pointed term one can subtract the comparison value, which is constant in the cut variables, and use the smaller difference bound. For an interior term with reads deep in the core, the depth estimate gives the additional factor $Ce^{-\delta_P\hat{D}_v/4}$, where $\hat{D}_v$ is the depth of the read hull. The mixed Cauchy argument requires one product domain for the whole cut family and the source. In the first two background comparisons of the terms with reads off Z, we must check this domain by the lower-layer membership induction on the actual auxiliary carrier, including any portion inside the treated component. The sealed cut discs for the later comparisons apply at their own background layers.

At a link with family y, the new label is the old label joined to the current parents of the selected cells. Add the connecting cubes to the label and across the exempt boxes used by the family. A connected part with n cells reaches the old label by a corridor of length at most $c_\rho n$, and the bridges across its boxes cost a bounded number of cubes per cell. Thus the new tree length is at most $d_k(A)+c_l|\mathsf{y}|$ with c_l fixed. To sum at weight r, spend $c_l r$ per selected cell, leaving $\lambda_r = \kappa_1 - 1 - c_l r$ for counting. A connected family through a cell at depth t in the grid has at least $t+1$ members if it reaches a source. If it sees the reads from a current parent at distance x, it also has at least x/c_ρ members. Thus the exponential reserve pays both the L^{4t} fine cells in a parent and the polynomial shell count in x, as well as the exponential count of connected families through each cell. With the lower bound on λ_r, the sum for one connected part is at most a small constant times the number

of cubes of A. The sum over unordered sets of parts is bounded by the exponential of that sum. The constant can be chosen so the resulting factor is at most $e^{d_k(A)+1}$.

For a sum rooted at a prescribed cube, distinguish whether that cube is in A or in the addition. In the first case, after summing the other parts, one more unit of the old-label weight supplies the distance from the root to the parent of the cell seeing the reads. The reach correction costs at most $c_\rho \rho(\Delta)$, which is paid by the connected-family reserve. In the second case, mark a connected part of size n responsible for the root. A parent of one of its cells lies within $c_\rho n$ of the root. There are at most $O((1+n)^4)L^{4n}$ choices of parent and fine cell, and exponentially many connected parts through that cell. One of its n cells sees the reads. The factor $e^{\rho(\Delta)}$ to recover the reader sum costs at most e^n, so the same reserve sums over the marked parts. Finally the readers seen from Δ have a label cube among $O((1+\rho(\Delta))^4)$ candidates. The factor $e^{-\rho(\Delta)}$ in their sum absorbs this count and reduces it to the ordinary rooted label norm. Write Q_r for the supremum over a root cube of the sum of the assigned sizes times $e^{r d_k(A)}$ over the labels A through that cube. Thus a link, including the empty family, satisfies

$$Q_r(\text{after the link}) \le (1 + C_\Sigma C_e)\, Q_{r+2}(\text{before the link}),$$

where C_Σ is the count for the two root positions and C_e bounds the number of nearby label cubes with its $e^{-\rho}$ weight.

After the last link, fill in the components of Z that meet the final label. This adds no cube off Z, and a tree for the label bounds $d_Z(Y_4)$. A nonempty cut term filed off Z has both a source cell at distance at least $9\check{R}_k - 1$ from Z^c and an exterior cube in its connected label. Its length is therefore at least $\check{R}_k - 2$. Spend one unit of the final weight to extract $e^{-(\check{R}_k - 2)}$. Summing in reverse link order gives the rooted bound at weight a from the input rooted norm at $a+9$ for the four links of the plain interior terms, and at $a+7$ for the three cut links of the plain exterior terms with no amplitude. Choose the cut radius with $\kappa_1 - 1 \ge c_l(a+9) + \lambda_*$, where λ_* the counting threshold. The plain interior input uses the shallow column and the depth-weighted column in the core. For the exterior input, the root of the intermediate sum may lie inside Z. Choose an exterior cube on the same initial label and spend one more unit of weight to sum its distance from that root. This uses the size column at $a+8$, including the exposed difference estimate on its enlarged label. The two bounds are a polynomial in $\log x_k$ times the collar gain.

For a compact point-anchored amplitude piece p, the input to the later links has label $K_p \cup \mathfrak{h}_v$ and reads the domain $Y_0(p)$. Keep its amplitude disc of radius $\rho_A = \mathsf{w}_L/(\eta_p \mathsf{f}_v)^{1/2}$ throughout these links, where η_p is the decay to the amplitude block and f_v the decay from the point's window to the source region. On the common product chart, amplitude Cauchy contributes $2/\rho_A$ to the dilated size, and the cut variables have their own factors $e^{-(\kappa_1 - 1)}$. The factor $\mathsf{f}_v^{1/2}$ pays the cubic window weight in the pointed size column. The factor $\eta_p^{1/2}$, with a reserve from the first cut family, sums the possible amplitude blocks in shells around the cells already reached. This sum is made over the pieces behind one coarse root, retaining their actual zone and cut-cell levels. It gives a bound proportional to w_L^{-1} for the input rooted norm. Three further links and the transfer to an exterior input root use at most seven more units of weight. The small factor is already in the amplitude column, including its empty continuation.

The chain and long terms grouped into bundles use the sum over blocks and fine cells in the term's own grain before the three later links. For a bundle the comparison disc acts on that sum. The small factor belongs to the resulting rooted bundle norm. For a wrapped term of level j, let A_v be its connected label and $K_v = [A_v]_k$. Its true read consists of A_v and the attached older strips, each touching A_v. An attached strip healed at this step lies in the core of a treated component, and the adjacent label cube lies in that component's inner box. A strip that stays live is separated from the treated components. Thus the true read meets Z exactly when K_v does. When filing a value, adjoin the treated components met by K_v. This already includes all the healed strips read

by the term. For analytic evaluation also include the live strips touching the label in the enlarged background domain. In the integration over a treated component, the fields on the older live strips are parameters. Two polymer factors that share only these parameters factorize in that integral. Hence the older live strips are background reads and the connected label is what enters the length and compatibility tests.

In the cut expansion, keep the actual read at each link. It consists of the original reads, the previously selected cells and their touched boxes, with the rim of the preceding domain as an overestimate. A cell on a live strip is eligible for a cut under the same exemption rule as the other cells. Its parent is within $101\check{R}_k + 1$ of the old label, while a family from that strip to a source has size at least $10\check{R}_k$. The analogous rim estimate allows one more cube. This pays the corridor from the cell to the label by $c_l|\mathsf{y}|$. The rest of the live strip need only be adjoined to the background read domain. We must check that the global chain stays in the joined chart for the stored wrapped term, using inherited tubes on live strips and containment on healed strips. Keep the twist in the common admitted source slot and take the stored norm on the complex orbit domain, since contour gauges can be complex. On that product chart the same mixed Cauchy bound applies to each tuple of the composed layer, current and gauge readers, including one whose only cut is at the first link. At the third layer the support test applies to the primitive current entering its inverse.

For a term with the whole read in a core, if all cut cells stay in Z, each connected family seeing the reads stays in the same component. Its parents and connecting cubes stay in that box. Induction over the four links then keeps the final label inside the component. Thus an exterior tuple of such a term triggers the core depth estimate, and its assigned size in the exterior sum is the stored size times $C_{3F}e^{-\delta_P d^\wedge(X_v,\mathsf{C}_C^c)/2}$, with X_v the true read and C_C the core. Use the unmodified stored size for the other wrapped terms. The depth factor is assigned only in the majorant for exterior tuples.

There is a separate count for the initial wrapped reads. Let $\mathsf{b}_k(s)$ bound the weighted label column at native level $k-s$, with the spare native rate for labels finer than the probing cell. If the read through a cell of level n comes from an attached strip of level m, count labels of level $j \geq m$ by a cube touching that strip. At level j there are at most $(s_j+3)^4$ such cubes, where s_j is the strip side in those units. Since $s_j \leq 101\check{R}_m$, this count has no factor for the fine volume of the strip. A native probe meets at most 2^4 live strips, all with $m \geq n$. For a read through the label itself, use the native meeting convention for unlike levels, which requires a common interior point. The levels $j \geq n$ have a bounded number of anchor cubes per level. At $j < n$ the label must reach from the probe to an older strip. With $t = n - j$ the spare rate bounds the fine-anchor count by a constant times $L^{4t}e^{-c(L^t/2-2)_+}$. These are the two parts of the read count for a fixed probe. On the native cut geometry, $k - n + 1 \leq \rho(\Delta)$. For a cell contained in a live strip, a minimal path to a source must leave the birth-scale collar of width $5\check{R}_m/2$. Before its first source it takes no hub jump, and in that collar each cell spans at most one birth unit. In particular $\check{R}_m \leq 2\rho(\Delta) + 2$. Multiplying the read count by $e^{-\rho(\Delta)}$ absorbs its powers of the age and strip side and bounds the reception sum by $C_{\mathsf{a}}(1)\mathsf{b}_k(0)$. For a source cell seeing the read through an exempt box, its contribution is instead bounded by $h_\sharp^4 Q^\star_{a+9}$, where $h_\sharp$ bounds the box side and $Q^\star$ is the rooted norm for terms meeting Z with the assigned sizes above.

The first-link family includes an empty choice for a term meeting Z. A term disjoint from Z enters through one of four successive background differences, each with a nonempty first family. Use the reception bound for the nonempty first families, with at most four copies. The next three links use the two-unit weight estimate. For an exterior final label the collar length is at least $\check{R}_k - 2$. For a label reaching a live strip this follows from its reach to Z, supplied by a source cell if the original term was disjoint. Otherwise the label reaches from an inner box of a treated component to an exterior cube. Spending one unit of weight yields the exterior rooted bound

$$e^{-(\check{R}_k-2)}(1+C_\Sigma C_e)^3\big[(1+4h_\sharp^4 C_\Sigma)Q^\star_{a+9} + 4C_\Sigma C_{\mathsf{a}}(1)\mathsf{b}_k(0)\big].$$

For $Q^\star$, labels joining a live strip to the treated region already have a collar length in their own units, which pays the number of fine roots. Shallow labels attached only to healed strips are counted through the core on the native grids, using the collar for levels finer than its rim level. Deep labels use the depth factor when summing the core cells. For the deep count, assign the depth to a cell whose closure contains a point realizing the least distance to the core rim and whose interior meets the read. Use half of its depth weight to pay the probe count, and sum the remaining half by the core shell estimate. Its prefactor is the area of the core rim box, with side at most $104L\check{R}_{h+1}+2$ in level-h units. Here h is the level of the cells on the inner side of the core rim. These counts bound $Q^\star_{a+9}$ by a polynomial in the stage count and the radius at the core rim. The weighted column, depth estimate and joined chart must hold for the actual stored terms in this format before using the bound.

In the total pre-filed family, separate the ordinary values, the brackets, and the source-free remainder of the Wilson expansion. The all-empty term of an ordinary exterior unit is a value. A plain interior term whose domain stays inside its component goes into the interior volume sum. The first-link amplitude term and its later nonempty continuations count in the same amplitude budget, with each tuple charged once. The nonempty wrapped terms have their own budget. The Wilson family here consists of the action remainders after removing the explicit source-weight term at fixed background. The background arguments remain those of the chain. File the treated-component cell terms at their native anchors, the terms depending on the whole component at that component, and the terms in Ω_k off Z at their exterior cubes. Each resulting cut tuple belongs to the Wilson family alone. Use the absolute exterior bound and the summed strip bound $1+c_I M^5$ for this family from the Wilson localization estimate in Bałaban, Comm. Math. Phys. 122 (1989), pp. 374–375, (1.66)–(1.68). The bound for the denominator term $I(U_0)$ in (1.11) is charged once per component. To use these bounds, the localized Wilson family must agree with the one in that estimate on the admitted backgrounds.

For the Wilson cell terms of a treated component, the local bound follows from

$$a(W) = -\tfrac{1}{2}\,\mathrm{tr}_{\mathrm{norm}}((W-1)^2W^{-1}),$$

where $\mathrm{tr}_{\mathrm{norm}}$ is the normalized trace. If the plaquette increment at a level-n cell is at most $\widehat{\theta}(\eta/\ell_n)^2$, the plaquette count cancels the fourth power of η/ℓ_n in the density. Here η is the fine spacing and ℓ_n the spacing of the cell's grid. The bound on a cell with action coefficients at most x_n is $6M^4x_n\widehat{\theta}^2$. In the core subtract the value at zero cut parameters with the real rim datum fixed. Plaquettes entirely in zone zero cancel because the two backgrounds have the same datum there. For the other plaquettes the linearized holonomy difference is a covariant first difference of the displacement around the plaquette, with a term proportional to $1-\mathrm{Ad}_W$. The latter has an extra curvature factor. The commutators in the product of the four bond exponentials are quadratic in the displacement. Thus if the scaled displacement bound is B at the cell, the plaquette difference is at most $CB(\eta/\ell_n)^2$. The quadratic density identity then bounds the cell oscillation by $CM^4x_n(\widehat{\theta}B+B^2)$. The admitted exponential depth estimate for B makes this sum proportional to the core boundary count. Off the core use twice the cell bound and count the preparatory shells. These two counts give a polynomial in $\log x_k$ for the Wilson cell descendants. Four links give those descendants leaving Z the same collar gain as other plain interior terms. This estimate for the cell descendants supplements the absolute bound for the full Wilson family.

Thus the exterior input norm is bounded by the sum of the value bound, the bracket bound, and the Wilson bound. Once their admission has been verified, the plain interior and exterior cut estimates take one half of the bracket budget, the amplitude terms take one half, and the wrapped terms take one, for a total bound of two.

Use the prescribed disjoint strips $C^\sharp$ of width $m=\lfloor(\check{R}_k-1)/2\rfloor$ about the separated components C of Z. A pre-filed domain Y_4 is wall-connected and meets Z in a nonempty union of full components. File its term on Y obtained by adjoining every strip that shares a cube with Y_4. This

partitions the terms into fibres and preserves the sum. The true reads are inherited from the terms in each fibre, including the enlargement of any wrapped read.

For an exterior output Y, filing adds no outside-strip cubes, so all preimages share a root cube there and a relative tree for Y_4 also meets every cube of $Y \setminus Z^\sharp$. Thus $d_\sharp(Y) \le d_Z(Y_4)$. A wall path in Y_4 from Z to the outside of $Z^\sharp$ visits an exterior cube touching Z and an exterior cube at distance m from Z. A relative tree meets both cubes. The distance to Z changes by at least $m-1$ between the two points, so the tree has length at least $m-1$. With $b = 3\kappa/2$ and $\lambda = \kappa/40$, the input rate $a = 31\kappa/20 = b + 2\lambda$ therefore gives

$$e^{-ad_Z(Y_4)} \le e^{-bd_\sharp(Y)} e^{-\lambda(m-1)} e^{-\lambda d_Z(Y_4)}.$$

To count preimages of length at most n, traverse a relative tree twice and cut the walk into arcs of length at most one. Each arc meets at most 3^4 cubes, so the cubes met by the tree form a touching-connected family of $O(n+1)$ cubes through the root. A depth-first walk on this family gives exponentially many choices in $n+1$. Select the outside cubes among them, then select the full components of Z incident to the selected cubes. Every component in the preimage is incident to an outside cube, by wall connectedness and separation of the components. Each outside cube is incident to at most 3^4 components. This count sums the last factor in the fibre when λ exceeds its entropy rate. Thus the exterior strip amplitude is at most $C\|\mathfrak{v}\|_a e^{-\lambda(m-1)}$, where $\|\mathfrak{v}\|_a$ is the weighted exterior input norm. The interior fibre of $C^\sharp$ is the sum of the inputs contained in that strip and uses a volume bound. With no prescribed exterior root the free-family sum at a single link costs $e^{d_k(A)+1}$. Applied backward four times starting at weight zero, it gives $e^{4(d_k(K_v)+1)}$ times the assigned plain size. The bound for a wrapped unit, with its value included, has the additional factor four for the possible first brackets. The four units of tree-length weight are available at the unit's native rate, with a fixed coarsening offset absorbed in the constant.

For the running chain choose one anchor in the filing-level stage region of each term. At a given stage and anchor the weighted sum is the corresponding stage-norm sum. At a cell of zone n, the bound on the number of such stages grows as $1 + \log(1 + \log x_n)$, using the running stage count and $x_n \ge x_k - 2c_2$. Charge the other ordinary terms at the old determining cells of their zones. A window cell coarsened by e levels has at least L^{4e} distinct old cells to charge. This absorbs its cubic window weight, bounded by $27\mathsf{L}_0^{6(e+1)}$, since $\mathsf{L}_0^2 \le L/3$. For a frozen chain make this charge at its own pre-arrest cells. A level-m stage anchor meeting its band contains a cell of the band grid, and distinct anchors at that level contain distinct band cells. If $j+1$ is the band level, the number of band cells is at most a constant times $\check{R}_{j+1}^4 \sum_X (d'_{j+1}(X) + 1)$, where X runs over the band-containing components. The number of possible filing levels is bounded by a constant times $\check{R}_{j+1}$. Thus the frozen volume allowance is a sum of $\check{R}_{j+1}^5 (d'_{j+1}(X) + 1)$ over the stages and components of that arrest. The count uses the actual native anchor and band inclusion for the stored terms.

An exterior bracket filed wholly inside $C^\sharp$ reaches a cube of $C^\sharp \setminus C$. Charge it there using the exterior rooted sum, which is at most one per cube for the plain exterior brackets. Add the ordinary value allowance on those strip cells and one Wilson allowance for the entire component. This bounds the sum over all inputs contained in the strip. The finite filing preserves holomorphy on the common admitted source tube. The exterior gain absorbs the polynomial logarithmic factors in the pre-filing bound, including those from the Wilson vertices. Since the width grows at least as a fixed positive multiple of $(\log x_k)^{r_{\mathrm{pr}}}$ with $r_{\mathrm{pr}} > 1$, its gain beats every fixed inverse power of x_k, also after taking the cube root of the amplitude.

To pass from the filed potential to the healed remainder, fix the standing part of the history. Let W be the union of the strips of components that stay as second-class regions in this operation. Their variables are parameters in the healed sum. Keep their tests, group delta factors and component integrals outside that sum. A term on a single strip is local to its own component and is included in that component's operation. For the other domains, write the exponential as $1 + (e^{V^\sharp(Y)} - 1)$

at each domain Y. In a summand the integrations run over the chosen healed components. Each interaction factor reads the integration variables of a chosen component only if its domain includes that component's strip. A connected class consists of the touching parts off W, also joined whenever one interaction domain reads more than one of those parts. Each chosen healed component belongs to exactly one class. The class factors therefore multiply at fixed exterior data. Conversely, classes with disjoint and nontouching parts off W can be assembled by taking their union, by the locality of the history format. This gives a hard-core gas whose empty family has value one. For a nonempty healed set the term with no exterior interactions still contains the component masses. The class activity sums the compatible internal choices of healed components and interaction domains with that class's domain. Use the component operation of § 4.17 with its original measure factors, tests and internal history sum. Its localized action on the indicated reads is needed for the factorization.

For the component mass in this argument, keep the stored integration coordinates real on the supports of their original tests. In the inner Wilson factor the field assembled from these coordinates is unitary, while the source changes its weight. Let c_j be the unshifted nonnegative coefficient including its cutoff, and c_j^w the shifted coefficient. On the admitted source tube $\operatorname{Re} c_j^w \geq (1-\theta_W)c_j$ with $\theta_W \leq 1/4$. Plaquette by plaquette, $a_p(U) \geq 0$ for the real stored field, so

$$\left|e^{-A(c_j^w,U)}\right| = e^{-A(\operatorname{Re} c_j^w,U)} \leq e^{-(1-\theta_W)A(c_j,U)}.$$

This comparison also applies inside the nested component operations. In particular each Wilson test gain retains the fraction $1-\theta_W$ of its real exponent. The exterior pair can be complex. The quadratic form depending on it uses its source-independent complex-form estimate. The other quadratics, tree constraints, Jacobian factors and tests keep their real integration data. The sourced strip interaction is bounded by the exponential of its supremum on the same chart. Its ordinary value budget is charged to the original cells, and the frozen contribution to the stage cells of its arrest. The strip collar adds a fixed volume factor. The additional weights of the inner real fluctuation integrals use their per-cell bounds. For an unmatched counterterm on the regular part of its cutoff collar, the action on a native cell has modulus at most $C\varepsilon(\gamma)^2$. The source bound on its coefficient is $b_+ + 2r_{msrc}/3$ on the disc of radius $2r_{msrc}$. Thus its modulus costs at most e per native cell once $(b_+ + 2r_{msrc}/3)C\varepsilon(\gamma)^2 \leq 1$. Include that cost in the existing per-cell allowance. This yields a majorant before integrating the stored coordinates.

For a new region detected by a refused rim bond at level j, keep the corresponding gain from the constrained quadratic form. If the full form has lower bound γ_0, so does its Schur complement on the retained coordinates: minimizing $\langle (y,x), \mathcal{A}(y,x)\rangle$ over the discarded coordinate y gives at least $\gamma_0\|x\|^2$. With $\mathfrak{b}_j = g_j^{-1}\delta_j$, the rim test forces $|x_b| \geq \mathfrak{b}_j/(4\Theta_1)$ at each refused bond b. Spend one quarter of the quadratic form on the test gains and retain the other quarter in the density. The resulting factor at each such bond is $e^{-\gamma_0\mathfrak{b}_j^2/(64\Theta_1^2)}$. Its first square root pays the sum over refused-bond sets. At fixed ordinary test region each set is counted once and the sum is bounded by a product of $1 + e^{-\gamma_0\mathfrak{b}_j^2/(128\Theta_1^2)}$ over rim bonds. The other square root gives an exponent of at least $\gamma_0\mathfrak{b}_j^2/(256\Theta_1^2)$ per covered cube, since each bond covers at most two. The ordinary fluctuation tests use disjoint witnessing coordinates in this split. The rim gain therefore joins the Wilson and ordinary fluctuation gains in the cascade bound, and its cost is charged at its own level.

The cascade charge estimate uses the cover by the coarsened predecessor regions and the current test witnesses, including the cubes of the refused bonds. Write $\mathsf{P}_n = p_0(g_n)$ and Q_n for the common gain per created cell after setting aside the refused-bond count. For a created region H of level n, the cell count satisfies $d'_n(H) + 1 \leq \sqrt{4}\,|H|_n$, where $|H|_n$ is its cell count. Split $\mathsf{Q}_n|H|_n$ into halves. One half gives the deposit $\mathsf{Q}_n(d'_n(H)+1)/(2\sqrt{4})$ and the other leaves $2\mathsf{P}_n$ when $\mathsf{Q}_n \geq 4\mathsf{P}_n$. In unrolling the charge recursion, a quiet successor spends its current charge from the balance. A re-created successor adds its new test gain P_n before spending its charge. A join collects the predecessor balances and the birth deposits and releases the older creation reserves other than the

one at the earliest birth of the joined region. Thus each member keeps the balance and one oldest reserve of $2\mathsf{P}_{j_0}$. With the positive balance on the admitted family and the internal history count paid in the per-level charges, the exponent retains that reserve at the final component. Compare $p_0(g_{j_0}) \geq p_0(g_k)/(1+\beta_0)$ along the reference flow. In the complex modulus estimate there is one remaining factor e, giving the bound

$$|\mathbf{T}'_k(X;w)F| \leq e\, e^{-2p_0(g_k)/(1+\beta_0)} \sup |F|.$$

The supremum is on the internal support at fixed external data. For a product of compatible healed component operations use this bound successively, provided the admitted chart and the support bounds are valid with the other integration variables fixed. The same integrable majorant permits holomorphic integration of the analytic factors. To apply this estimate we must identify the actual real Wilson arguments, witnesses, charge family and common complex chart for the component integral. These checks concern the stored histories that enter the gas.

Let A denote one activity domain and $d_{\mathrm{out}}(A) = d_W(A)$. The tree for A off W is formed from the relative trees of its chosen interaction domains and a path through the cube centers of each chosen healed strip. A path in a rectangular strip of N_h cubes has length at most N_h in cube units. Join these trees and paths along a spanning tree of the class graph, at cost at most $2\sqrt{4}$ per join. The number of cubes off W is bounded by the sum of the N_h's and a constant times $\sum_Y (1 + d_\sharp(Y))$. Thus in the input rate $(1+2\beta_s)\kappa$ one can allocate $(1+\beta_s)\kappa$ to the output length, half of $\beta_s\kappa$ to the cube count and the other half to counting the interaction domains. If $\alpha^\sharp$ is the exterior strip amplitude, spend one third of its logarithm per chosen interaction to pay the fixed join and cube-count costs. For a healed component the modulus bound pays for the strip, whose cost is at most $2\kappa(100\check{R}_k)^4$ per box. The positive coefficient $2/(1+\beta_0) - 3/2$ at $\beta_0 \leq 1/7$ gives the room for this cost and the sourced modulus prefactor. With the degree ceiling the remaining factor is $e^{-3p_0(g_k)/2}$ per chosen healed component.

At fixed healed components the remaining interaction weights have the form $u_Y = (\alpha^\sharp)^{2/3} e^{-\beta_s \kappa d_\sharp(Y)/2}$. Every interaction domain contains an exterior cube off the strips used by that summand. The rooted domain count bounds $\sum_Y u_Y$ by a constant times $(\alpha^\sharp)^{2/3}$ times the number of cubes off W in A. If the interaction family is nonempty, $\prod_Y (1+u_Y) - 1 \leq (\sum_Y u_Y) e^{\sum_Y u_Y}$. Keep a factor $(\alpha^\sharp)^{1/3}$ and bound the remaining prefactor by an exponential in the cube count. The same count pays the exponential from $e^{V^\sharp(Y)} - 1$. The resulting cube penalty is small compared with the allocated cube-count decay. A class containing a strip in W necessarily has a nonempty interaction family, so it has the cube-root factor.

A class disjoint from W contains at least one healed component, since each interaction meets a component of that summand. This includes the class consisting of one healed strip and an empty interaction family. Extract $e^{-p_0(g_k)}$ from the product of the component factors and use the remaining $e^{-p_0(g_k)/2}$ per component for counting. Each candidate box is specified by its lowest cell and four bounded integer side lengths. At fixed A, the number of unordered sets of q such boxes is bounded by the q-th power of the candidate count divided by $q!$. The resulting exponential in the number of cubes is paid by the remaining volume decay. Thus the activities have rate $(1+\beta_s)\kappa$ in d_{out}, with amplitude $e^{-p_0(g_k)}$ off W and $(\alpha^\sharp)^{1/3}$ when they meet W.

For the logarithm only connected incompatibility graphs contribute. On a connected cluster the relative trees join at the touching cubes off W, with at most $2\sqrt{4}$ of extra length per edge of a spanning tree. Reweight each activity by $e^{(1+\beta_s/2)\kappa(d_{\mathrm{out}}(A)+2\sqrt{4})}$. The remaining $\beta_s\kappa/2$ in the length pays the exponential count of domains through a cube and the exponential of their cube count in the polymer criterion [KP86]. Indeed, each incompatible domain contains an exterior cube touching one of the cubes of the given domain off W, and for each such root the weighted series is bounded by a constant times the maximum activity amplitude. For weak coupling that constant times the maximum is small, giving the criterion with size equal to the number of cubes off W. To bound

the logarithmic coefficient at a fixed union, mark a cube of the union off W. Each cluster has at least one polymer covering it. Summing the mark over the polymer indices changes the factor $1/n!$ into $1/(n-1)!$ with the marked polymer first. The rooted cluster majorant then sums the other polymers and leaves the weighted rooted sum of this first polymer. In particular, for a union disjoint from W its amplitude is $e^{-p_0(g_k)}$. The logarithm is absolutely convergent on the admitted complex domain and the output decay rate is $(1+\beta_s/2)\kappa$.

Classify the resulting domains using the one-layer enlargements of the standing large-field region and the newly retained second-class components. Away from them the output length is d_k and the prefactor is $Ce^{-p_0(g_k)}$. The spare $\beta_s\kappa/2$ pays the rooted sum at rate κ, giving the healed contribution to the inherited remainder bound. The other terms are new boundary terms. Their bound uses the relative length off W; the prefactor can be taken as a fixed multiple of the larger of the two activity amplitudes. To store them on relative labels, sum the domains having a given skeleton. The count contributes $e^{\lambda_\theta(d_k(A)+1)}$ for the skeleton A, and $\lambda_\theta \le \beta_s\kappa/4$ leaves the stored rate $(1+\beta_s/4)\kappa$. This last step uses the skeleton length comparison and the relative-format bound for the actual complex charts.

The dimension-free polydisc lemma, Lem. A.14, gives on the half polydisc

$$\sum_i r_i|\partial_i F| \le \tfrac{4}{3}\sup|F|, \qquad |F(z)-F(z')| \le \tfrac{4}{3}\sup|F|\max_i |z_i - z_i'|/r_i. \tag{5.lvii}$$

One proves it by sending a single unit disc through the point with coordinate disc automorphisms, choosing their phases to align the derivative terms, and applying the one-variable Schwarz inequality. Integrating the derivative bound on the segment gives the second formula. Neither bound contains the number of earlier levels. Neither implies age forgetting: the bounded function $F(z) = Bz_0/(\sigma_\star\theta_0)$ already has a nondecaying weighted derivative in its first coordinate.

Here is the needed forced derivative estimate, including the marked term. Let $b_i' = db_i/d\beta$, $M_i = \max_{0\le j\le i}|x_j'|$, with $x_0' = 1$. The frame is fixed, so $z_j' = -x_j'$. The tube constants and coupling floor put these evaluation points in the half polydisc required in eq. 5.lvii. Since $\min_{j<i}\theta_j \ge \theta_{i-1} - c_5 \ge \theta_{i-1}/2$ above the floor, eq. 5.lvii and the separate bare-disc Cauchy estimate imply

$$\begin{aligned} &|b_i'| \le a_i M_{i-1} + d_i, \\ &a_i \le C\theta_{i-1}^{-3/2}(\log\theta_{i-1})^{\mathfrak{p}_0}, \quad d_i \le C\varrho^{-1}\theta_{i-1}^{-p_R}. \end{aligned} \tag{5.lviii}$$

The θ^{-1} factor in a_i is the unmarked independent-polydisc gain. The factor $1/\varrho$ in d_i comes from the actual bare-variable disc and must not be replaced by $1/\theta_i$.

Put $V_i = 1 + \sum_{l\le i}|b_l'|$. The differentiated recursion gives $M_i \le V_i$, and hence

$$V_i \le (1+a_i)V_{i-1} + d_i, \qquad \sum_{i=1}^K |b_i'| \le e^{A_K}(1+D_K) - 1, \quad A_K = \sum_{i=1}^K a_i, \quad D_K = \sum_{i=1}^K d_i. \tag{5.lix}$$

Indeed iterating the scalar inequality gives $V_i \le \prod_{l\le i}(1+a_l) + \sum_{l\le i} d_l \prod_{m=l+1}^i (1+a_m)$; each product is at most e^{A_i}. No triangular estimate for the physical field response is hidden in this one-dimensional calculation.

On the reference line, integral comparison yields, uniformly in K,

$$A_K \le Ch^{-1/2}(\log h)^{\mathfrak{p}_0}, \qquad D_K \le C\varrho^{-1}h^{1-p_R}. \tag{5.lx}$$

The condition $p_R > 1$ is exactly what makes the second bound uniform over all depths for a fixed bare-disc radius. For the original full Lipschitz numbers one may now take $\mathsf{a}_{[K-i]} = (4/3)a_i + d_i$, once the uniform slope bound below is imposed. Thus

$$\sum_{s<K} \mathsf{a}_{[s]} \le C\{h^{-1/2}(\log h)^{\mathfrak{p}_0} + \varrho^{-1}h^{1-p_R}\}. \tag{5.lxi}$$

The sharper original $O(g(\log g^{-2})^{\mathfrak{p}_0})$ rate follows when $p_R \geq 3/2$, in particular in the intended large-κ_0 regime. For merely $p_R > 1$, retain the general forced rate displayed here. Choose h large enough that $e^{A_K}(1+D_K)-1 \leq 1/3$. Then $|x'_j - 1| \leq 1/3$ at every admitted prefix. On real intervals, $2/3 \leq x'_j \leq 4/3$. The statement is uniform only where the analytic inputs and room conditions hold. It does not identify different frames.

The stronger independent marked-coupling polydisc route in the comparison appendix remains available separately. Under its actual hypotheses the marked row has order $\theta^{-p_R-1}\log^C\theta$; it is not the premise used in eq. 5.lviii–eq. 5.lx. Both routes require their actual domains. In particular a fixed physical-source disc alone does not supply the unmarked independent-variable gain.

Canonical tube and conditional tuning theorem 3.2. For a general finite reference sequence of length $n \geq 1$, the largest permitted tube is

$$\mathsf{t}^*_j = \min_{1\leq t\leq \min\{j+1,n\}} \mathsf{t}^\sharp(\theta_t). \tag{5.lxii}$$

This is just the combination of all preceding upper bounds and the requirement that tube radii do not increase with the level. For $n = 0$ the largest radius is instead $2\mathsf{t}^\sharp(\theta_0)$. The exponent q_{t} and the coefficient ϖ are the ordinary tube constants, not an insertion rate. The floor is chosen with $h \geq e^{2q_{\mathrm{t}}}$ and $c_5/h \leq 1$ when the terminal comparison is used. A constant-tube criterion follows by setting $\mathsf{K}(a) = \max\{1, (4a/e)^a\}$, with $\mathsf{K}(0) = 1$. Maximizing $t^a e^{-t/4}$ gives $(\log y)^a \leq \mathsf{K}(a)y^{1/4}$ for $y \geq 1$. In particular $\mathsf{K}(2q_{\mathrm{t}}) = \max\{1, (8q_{\mathrm{t}}/e)^{2q_{\mathrm{t}}}\}$. If the frame floor is $x_\star \geq e$ and $\mathsf{K}(2q_{\mathrm{t}})t \leq \varpi x_\star^{3/4}$, then

$$t \leq \frac{\varpi y^{3/4}}{\mathsf{K}(2q_{\mathrm{t}})} \leq \varpi y(\log y)^{-2q_{\mathrm{t}}} \qquad (y \geq x_\star). \tag{5.lxiii}$$

Thus the constant sequence of radius t is a tube; its criterion uses the floor rather than the frame length. On the line in eq. 5.liii, set

$$\begin{gathered} \iota_k = \epsilon_k + \epsilon'_k, \qquad T_K = \varrho + c_2, \qquad T_k = T_K + \sum_{j=k}^{K-1} \iota_j, \\ F_k(\beta) = \theta_k - x_k^\natural(\beta) + s_k, \qquad \widehat{\mathsf{t}}_k = T_k + c_2 + 2R', \qquad R' = 4\varrho. \end{gathered} \tag{5.lxiv}$$

This is the canonical tube: in depth notation $T_{[s]} = T_{[0]} + \sum_{u<s}\iota_{[u]}$. The correction by s_k removes the bounded one-loop oscillation, giving $|F_{k+1} - F_k| \leq \iota_k$. The accumulated drift satisfies $\sum_{j=k}^{K-1}\iota_j \leq C\theta_k^{1/2}(\log\theta_k)^{\mathfrak{p}_0}$ in the stated power regime. For sufficiently large h, uniformly over K and k, this and the fixed additive terms are smaller than $\varpi\theta_{k+1}(\log\theta_{k+1})^{-2q_{\mathrm{t}}}$. The same holds at the terminal convention. Thus $\widehat{\mathsf{t}}$ is an admitted tube. The choice is made after ϱ, not by shrinking the prescribed offset range as K grows.

We spell out the domain premise needed by the interval argument. At a prefix satisfying $|F_j| \leq T_j$, the actual array is defined with room $2R' = 8\varrho$. The next coefficient is formed using that prefix before the new large-field operation. The uniform bounds and the marked bare-disc hypothesis above apply on a neighborhood of every such prefix. Whenever the computed next value satisfies $|F_{k+1}| \leq T_{k+1}$, the next array closes with the same room. These are the actual prefix formation/closure hypotheses, not consequences of the scalar intermediate value theorem.

Define $I_0 = \{\beta : |F_0(\beta)| \leq T_0\}$. Since $F_0 = \theta_0 - \beta$, this is a compact interval mapped decreasingly onto $[-T_0, T_0]$. Choose the floor so all its initial weights are admitted. Inductively suppose I_{k-1} has this property for F_{k-1}, and every earlier prefix lies in its required room. The preceding domain premise defines b_k, hence F_k, on a neighborhood of I_{k-1}, even before restricting its new output to the target window. eq. 5.lviii–eq. 5.lx make $F'_k = -x'_k$ strictly negative there.

At the left endpoint, $F_{k-1} = T_{k-1} = T_k + \iota_{k-1}$, so $F_k \geq T_k$. At the right endpoint the corresponding inequality is $F_k \leq -T_k$. Continuity supplies unique points with values T_k and $-T_k$, and

strict monotonicity orders them. Their intervening compact interval

$$I_k = \{\beta \in I_{k-1} : |F_k(\beta)| \le T_k\} \tag{5.lxv}$$

is mapped decreasingly onto the whole target interval. On it $|\theta_j - x_j| \le T_j + c_2$ for $j \le k$, leaving room $2R'$ in eq. 5.lxiv. The actual closure premise therefore permits the induction to continue. This proves existence and connectedness of the interval from the closure premise.

At K, $x_K = h + s_K - F_K$. Since $|s_K| \le c_2$, its image contains every x with $|x-h| \le \varrho$. Each such value has exactly one preimage in I_K. Let $\gamma_U \le \gamma_J \le \gamma_H$ impose the preceding conditions for the fixed ϱ, with γ_J also imposing the admitted source analytic domain. For every K and $0 < g \le \gamma_U$, the conditional tuning theorem is

$$x_K^\natural(\beta_K^\natural) = g^{-2}, \qquad \tfrac{1}{2}c_0K - c_5 \le \beta_K^\natural - g^{-2} \le \tfrac{3}{2}c_0K + c_5. \tag{5.lxvi}$$

The common companion makes this the same number on every admissible torus. Uniqueness has been proved in the admitted canonical interval, not among arbitrary disconnected presentations at all bare couplings. Also $|\beta_K^\natural - h - c_0K| \le 2c_2 + \sum_k \iota_k$. The single-run spacing estimate is

$$\tfrac{1}{2}c_0(j-i) - c_5 \le x_i^\natural - x_j^\natural \le \tfrac{3}{2}c_0(j-i) + c_5 \qquad (0 \le i \le j \le K). \tag{5.lxvii}$$

It follows by summing $|b_l - b_l^0| \le c_0/2$. It gives eq. 5.lxvi, but gives no bound between the tuned numbers for two different cutoffs.

The bare offset t and terminal coordinate x are distinct: $x_K^\natural(\beta_K^\natural - t) = h - t + O(c|t|)$ when the slope error is at most c. The reference frame is independent of β; the self-calibrated frame $\theta_j = x_j(\beta)$ and its frozen shifted run are other applications of the uniform prefix induction. For a constant source ζ, the frozen shifted weights are $x_j(\beta) - \zeta_j$, and their slope normalization is $x_j'(\beta)$. Identifying this derivative with a particular F3 witness still requires that witness's admitted run and comparison hypotheses.

A finite nontriangular comparison lemma. The stronger two-run route uses an estimate different from eq. 5.lix. Fix integers $n \ge 1$, $0 \le r_0 < n$, numbers $0 < \rho < 1$, $\Pi, \mathsf{M} \ge 0$, nonnegative y_w for $r_0 < w \le n$, and nonnegative D_u, $k_{u,w}$, h_u, h' for the finite indicated ranges. Put

$$S_w = \frac{\rho^{n-w+1}}{1-\rho} + (w - r_0)\Pi. \tag{5.lxviii}$$

Suppose, for $r_0 \le s < n$ and $r_0 \le u < n$,

$$\begin{aligned} y_{s+1} &\le \sum_{u=r_0}^{s} D_u, \\ D_u &\le \mathsf{M}(\rho^{n-u} + \Pi) + \sum_{w=u+1}^{n} k_{u,w} y_w, \\ \sum_{w=u+1}^{n} k_{u,w} S_w &\le h_u S_{u+1} + h'\Pi, \qquad \sum_{u=r_0}^{n-1} h_u + h' \le \tfrac{1}{2}. \end{aligned} \tag{5.lxix}$$

Then $y_{s+1} \le 2\mathsf{M}S_{s+1}$. Indeed $S_w > 0$ and increases with w. Let $Z = \max_{r_0 < w \le n} y_w/S_w$, a finite number. The forcing sum is at most $\mathsf{M}S_{s+1}$. The kernel sum is at most

$$Z\left(\sum_{u=r_0}^{s} h_u S_{u+1} + (s - r_0 + 1)h'\Pi\right) \le \tfrac{1}{2}ZS_{s+1}. \tag{5.lxx}$$

Taking the maximum gives $Z \le \mathsf{M} + Z/2$, as claimed. The sums couple a difference to later depths, so this is weighted-max absorption, not a triangular recursion. Its actual profile-kernel bound in eq. 5.lxix remains a substantive two-run input. Bounds on increments alone do not establish that input.

Statement 3.6: real presentations of the same measure. A real presentation over a frame is an actual inductive array whose weights remain in the tube. At each level it carries one Euclidean member, with one scalar coefficient common to all sites and no defect, and one shift c_j. These shifts obey the admitted $\bar{c}$ bound, with the frame floor chosen in terms of $\bar{c}$. Members, companions and coefficients are independent of the admissible torus and real at real fields. The all-small-field label is the label whose regions fill the torus. In its exponent the action coefficient is $x_j - \mathsf{c}_j$. The weight actually used in the Gaussian step is x_j, and $\mathsf{c}_j A$ is carried in the local action. A spatially varying source array with defects is not a presentation of this special kind.

Write $b_j = x_{j-1} - x_j$, let $\widehat{b}_j$ be the coefficient actually extracted at step j, and set $u = \mathsf{c}_0$. Coefficient bookkeeping gives

$$\mathsf{c}_j = u + \sum_{l=1}^{j} (\widehat{b}_l - b_l), \qquad x_j - \mathsf{c}_j = x_{j-1} - \mathsf{c}_{j-1} - \widehat{b}_j. \tag{5.lxxi}$$

In particular $\widehat{b}_j$ need not equal the planned decrement b_j when the shifts vary with the level.

The comparison hypothesis is equality of the all-small-field measures at every level, on every admissible torus, for two real presentations over the same frame. Require hereditary open windows $\mathfrak{O}_m \ni \mathbf{1}$ of block fields: their backgrounds lie in every needed core, all the relevant tests equal one, and $\mathcal{M}^i U_m(V) \in \mathfrak{O}_i$ when $V \in \mathfrak{O}_m$ and $i < m$. Also require the common connected analytic fluctuation domain on which continuation from such a window to the next actual integral is used. These are properties of the actual maps and tests.

Under these hypotheses the conclusions are

$$\begin{aligned} x_j - \mathsf{c}_j &= x'_j - \mathsf{c}'_j, \\ \widehat{b}_j &= \widehat{b}'_j, \\ \sum_X (E^{(j)} - E'^{(j)})(X, U) &= \tfrac{3}{2} |\mathfrak{X}_{j-1}| \log(x_{j-1}/x'_{j-1}). \end{aligned} \tag{5.lxxii}$$

The first line holds through level n, and the coefficient equality through $n+1$ only when both presentations admit that next step. The total identity is evaluated at backgrounds $U = U_m(V)$, $j \le m \le n$, of the common windows. On those backgrounds the remainder totals agree, and the vacuum constants agree. When the total suppresses z, $E^{(j)}(X, U) = \sum_{z \in X} E^{(j)}(X, P_U, z)$. No assertion identifies individual localized terms of the two presentations.

Here is the induction. At the bare level equality identifies the effective weight $\beta - u$, since the action is not a constant function. On an open small-field window the positive continuous densities agree everywhere if they agree almost everywhere, because product Haar measure has full support on the open window. Their logarithmic exponents are real, so their equality involves no integer ambiguity. Analytic continuation on the admitted fluctuation domain therefore identifies the next unnormalized integral. The normalized Gaussian in each presentation uses that presentation's own weight. Its normalization has the term $-3|\mathfrak{X}_{j-1}| \log(x_{j-1}/\theta_{j-1})/2$. It follows that the next regular totals differ by the field-independent constant displayed in eq. 5.lxxii. The common-companion sum rule makes their extracted coefficients equal. eq. 5.lxxi then identifies the next effective weights. The same physical formation identities on the hereditary windows identify remainder totals and the once-counted vacuum constants. This completes the induction. It uses all-torus compatibility to extract a coefficient from a constant total; one finite torus alone is insufficient.

Apply the result to a real β whose prefix along θ has room 2ϱ, and a real $|\zeta| < \varrho$ for which both constructed presentations are admitted. One presentation starts at $\beta - \zeta$ with no shift; the other starts at β with constant source ζ and running shift ζ_k. The exact identity is

$$x_k^{\theta}(\beta - \zeta) = x_k^{\theta}(\beta) - \zeta_k. \tag{5.lxxiii}$$

The slope bound keeps the running shift at most $4|\zeta|/3$ in its stated range. For $\theta = x(\beta)$, this describes the frozen shifted run, not the self-calibrated run of $\beta - \zeta$. For the reference line it identifies the two presentations of $x_k^\natural(\beta - \zeta)$.

Complex gluing also uses one frame. First restrict the real centres to the canonical interval I_K constructed above, or to one of its admitted prefix intervals. Each source disc is supplied by the actual analytic source construction. Where a real point in that disc also belongs to the canonical interval, eq. 5.lxxiii identifies its continuation with the real run. If two discs about such centres overlap, their real overlap contains an open interval in I_K: the segment between the centres lies in I_K, and the open real disc intervals overlap along that segment. For coincident centres at an endpoint, a one-sided open interval inside the nondegenerate I_K suffices. The identity theorem therefore identifies the coefficient functions on the full complex overlap.

The union carries a single holomorphic bare-coupling function. Remainder totals on common windows glue likewise; regular totals glue after accounting for their explicit Gaussian constant. This proves uniqueness of the supplied continuations, not their original bounds or radii.

For arbitrary admitted real centres, every overlap used for gluing must contain a common open interval of admitted real runs, or the prefix/source-disc construction must prove real admission there. This condition requires a separate proof when the admitted real set has disconnected components. Geometric intersection of two discs alone does not settle it. With the stated admission property the same identity theorem proof applies. Nothing here identifies runs made with different frames, individual localized terms about different centres, or functions on independent-current cores from equality only on a physical graph.

5.2.4. *Units, common prefixes and change of reference.*
Statement 3.7: the finite dilation identity. For every integer b with $K - b \geq 0$ and $L^b\ell$ in the compatible torus family, the map $D_b x = L^b x$ identifies the bare lattices at (K, ℓ) and $(K - b, L^b\ell)$. For negative b use the inverse of the corresponding positive relabeling. The exact result of § 5.1 is

$$\begin{aligned} Z_{K,\ell,\beta}(w) &= Z_{K-b,L^b\ell,\beta}(w^{[b]}), \\ S^T_{n;K,\ell,\beta}(\underline{f}) &= S^T_{n;K-b,L^b\ell,\beta}(\underline{f}^{[b]}), \quad f^{[b]}(x) = f(L^{-b}x). \end{aligned} \tag{5.lxxiv}$$

There is no spacing prefactor in $\mathcal{O}(f) = \sum_p f(x(p))a_p$. The map transports bonds, plaquettes and Haar variables bijectively; both sides are the same finite integral. The partition identity is entire in its finite source variables and bare coupling. Logarithmic derivatives use its nonzero domain. The same transport identifies periodic Gibbs states on the two infinite lattices. Radii and distances multiply by L^b, $\operatorname{Lip} f^{[b]} = L^{-b} \operatorname{Lip} f$, and $\|f^{[b]}\|_\sharp \leq \max\{1, L^{-b}\}\|f\|_\sharp$. The free bilinear form is invariant. This proves neither convergence nor agreement of two RG constructions.
Statement 3.8: what the step reads. Construction naturality is the additional assertion that step k reads only its actual tower through k, partitions, frame and weight prefixes, earlier outputs and fixed supply rules. Those rules are made for the lattice, not for its unit of length. For two presentations of the same bare lattice, their images under D_b therefore identify the histories and their measures, regions, healing data and local nonwrapping terms through level $K - b$, provided the prefixes agree. This is required for both fixed-frame and self-calibrated runs. Separate arbitrary choices on the two lattices would not establish it.

For a self-calibrated run this permits continuation from first passage at g to first passage at g', provided the prolonged run remains below the anchor ceiling and the rescaled torus is admissible. Its history law through the earlier level is unchanged. For reference runs, $\theta_{[r+s]}(g) = \theta_{[r]}(g_{[s]})$; the prefix at g, down to depth s, is the reference run at $g_{[s]}$ of length $K - s$, on $T_{L^s\ell}$. Its deviations, increments and tube are the same. The implication $\ell \in \mathfrak{L}(g) \Rightarrow L^s\ell \in \mathfrak{L}(g_{[s]})$ uses the stated torus floor and parameter choices. Here the needed scalar comparison is $R(y + c_0) \leq LR(y)$ for $y \geq h$ above the chosen floor; it is a comparison of the function R, whereas a general run uses its running-minimum values $\check{R}_j$.

Keep the beyond-first-passage scope separate. If $\beta \in \mathcal{W}_{K'}(g)$ and $K = K' + q$, $q \geq 0$, then eq. 5.lxxiv identifies the cutoff-K measure with the first-passage measure on $T_{L^q\ell}$. Estimates for it require $L^q\ell \in \mathfrak{L}(g)$. A torus volume factor becomes $\exp[nc_V(L^q\ell)^4]$, whereas a per-ball estimate has its radius and separation changed to L^qR, L^qd and no side dependence in its constant:

$$|S^T_{n;K,\ell,\beta}(\underline{f})| \leq A(L^qd; L^qR)B(L^qR)^n(n!)^{\varkappa_\star} \prod_i \|f_i\|_\sharp. \tag{5.lxxv}$$

The stated lower torus floor for that bound also applies. For $q < 0$, use the identity in reverse, require $L^q\ell \in \mathfrak{L}(g)$, and allow a norm cost at most $L^{|q|}$ per test. Applying this to tuned offsets requires the separate comparison $|K(\beta^\natural_K - t, g) - K| \leq C(\varrho)$. It does not follow from tuning at a frozen frame.

Statement 3.9: all four reference-change targets. Let

$$S^{[t]}_\ell[g] = \lim_K S^T_{\cdot;K,\ell,\beta^\natural_K(g)-t}, \qquad \mathsf{J} = [-\varrho/4, \varrho/4]. \tag{5.lxxvi}$$

Existence along the whole sequence, its common complex offset strip and moving-offset continuity require E3 and the terminal-coordinate comparison in appendix A.8. The strip width and the cutoff floor may depend on the fixed torus. The coordinates are

$$\Xi_K(t) = x^\natural_K(\beta^\natural_K - t), \qquad \Xi_{\infty,s}(t) = \lim_K x^\natural_{K-s}(\beta^\natural_K - t). \tag{5.lxxvii}$$

The depth limit requires the matched comparison. Terminal matching uses $n_K = \Xi^{-1}_{K+1} \circ \Xi_K$ on its admitted common strip. Its near-identity and summable-error properties remain E3 hypotheses, not consequences of the existence of individual tuning inverses. First choose ϱ large compared with $c_2, B_\sharp + 1$ and a bound for the tuned increments; then choose g small. The bound for those increments is a two-run comparison requirement fixed before ϱ, not a consequence of one-run spacing.

(i) At fixed depth s, the same bare coupling has two offset labels:

$$\begin{aligned} \widehat{t}_{K,s} &= \beta^\natural_{K-s}(g_{[s]}) - \beta^\natural_K(g), \\ \beta^\natural_K(g) - t &= \beta^\natural_{K-s}(g_{[s]}) - (t + \widehat{t}_{K,s}). \end{aligned} \tag{5.lxxviii}$$

Require the actual two-run estimate $|\widehat{t}_{K,s}| \leq C(c_2 + \sum_{K-s\leq j<K} \iota_j)$, its summable cutoff increments, and the comparison domain on which both presentations exist. The initial comparison requires $2c_2 + \sum_{K-s\leq j<K} \iota_j \leq \varrho$; a smaller fixed fraction of that bound keeps the offsets in the inner window. For g small in terms of the fixed s, these offsets lie in $\frac{1}{2}\mathsf{J}$ and converge to $\widehat{t}_s$. The whole-sequence moving-offset limit then gives

$$\begin{aligned} S^{[t]}_\ell[g](\underline{f}) &= S^{[t+\widehat{t}_s]}_{L^s\ell}[g_{[s]}](\underline{f}^{[s]}), \\ \widehat{t}_{s+s'}(g) &= \widehat{t}_s(g) + \widehat{t}_{s'}(g_{[s]}). \end{aligned} \tag{5.lxxix}$$

Here $t \in \frac{1}{2}\mathsf{J}$; each depth condition must hold at its displayed reference coupling. The condition on g is an upper bound, never a lower bound. The terminal coordinate on the right is the limiting depth coordinate $\Xi_{\infty,s}(t)$. Its existence belongs to the matched comparison.

(ii) For arbitrary sufficiently small g, g' and $t \in \mathsf{J}$, retain the stronger target that some $t' \in \mathsf{J}$ and integer b satisfy

$$S^{[t]}_\ell[g](\underline{f}) = S^{[t']}_{L^b\ell}[g'](\underline{f}^{[b]}). \tag{5.lxxx}$$

Its torus conditions are both $\ell \geq \ell_\star(g)$ and $\ell \geq L^{(|g'^{-2}-g^{-2}|+C)/\lambda_\sharp}\ell_\star(g')$. The sign of b is that of $g'^{-2} - g^{-2}$, up to a bounded ambiguity. The number of intervening passage levels lies between $|g'^{-2} - g^{-2}|/B_\sharp - 1$ and $(|g'^{-2} - g^{-2}| + C)/\lambda_\sharp$; $|b|$ differs from it by $O(1)$. This assertion needs recognition of the same bare run in both calibrated families, not merely eq. 5.lxxiv.

(iii) Choose once an anchor $0 < \gamma_\star \le \gamma_J$, sufficiently small in terms of the fixed large-field window dials w, c_a. For any proposed $\mathbb{R}^4$ limit at $g \le \gamma_\star$, retain the prolonged-run target: after a subsequence, $K(\beta_j, \gamma_\star) - K_j = s_\star$ is constant, of order $g^{-2} - \gamma_\star^{-2}$, and

$$S_n(\underline{f}) = S'_n(\underline{f}^{[-s_\star]}). \tag{5.lxxxi}$$

The anchor limit uses the same bare-coupling sequence. eq. 5.lxxxi is the exact image under the point map $x \mapsto L^{s_\star} x$, expressed by the test $f^{[-s_\star]}(x) = f(L^{s_\star}x)$. No extra spacing factor is inserted in the observable. This preserves the target for arbitrary defining sequences carrying the required construction and recognition hypotheses. It does not give an automatic mirror/cone theorem for unrestricted growing-torus limits. Dilation-invariant properties need be checked at only one reference coupling once this target is established.

(iv) The bare tuned increments $\alpha_K = \beta^\natural_{K+1} - \beta^\natural_K$ are a new notation, unrelated to the field radii. The proposed conclusion is

$$\alpha_K \longrightarrow c_0, \qquad S^{[t]}_\ell(\underline{f}) = S^{[t-bc_0]}_{L^b \ell}(\underline{f}^{[b]}) \quad (b \ge 1,\ t, t - bc_0 \in \mathsf{J}). \tag{5.lxxxii}$$

For completeness the shape convention is $\mathfrak{T}_{\mathbb{R}^4}(d;\vartheta)$: both profiles are nonnegative nonzero Lipschitz functions supported in radius-d balls, with support distance in $[d, 6d]$, and each obeys $\int f \ge \vartheta d^4 \|f\|_\infty$ and $\operatorname{Lip} f \le \|f\|_\infty/(\vartheta d)$. Clause (iv) retains the condition $\mathfrak{T}_{\mathbb{R}^4}(1;\vartheta) \ne \emptyset$ and $g \le \gamma_W(\vartheta)$. The proof also needs the actual uniform smooth-profile envelopes on the pairs used in its iteration; class nonemptiness by itself is not an envelope estimate. The comparison appendix proves this implication with the actual two-run depth/profile estimates, summable comparison errors and uniform two-point envelopes. Those envelopes are uniform in the offsets and tori needed for repeated blocking; the fixed shape class must be nonempty. Two distinct cluster increments would give an offset period by analyticity, and repeated blocking would contradict the uniform short-distance envelopes. The unique increment limit is then c_0 by the averaged one-loop law. The same argument yields $\beta^\natural_{K+b} - \beta^\natural_K \to bc_0$ for every fixed integer b. The exact input hypotheses remain in force; finite dilation and weak numerical inequalities do not establish them.

Vacuum constants and the interfaces used later. The vacuum recursion is eq. A.lxvi of appendix A.8. Each assigned step number, Gaussian local number and regular/remainder value at $P_\circ$ is counted once. Using the normalized N requires adding its removed local numbers; using N° does not. At fixed calibration these numbers have zero source derivative. They are not additional freely chosen normalizations.

The further vacuum estimates required are

$$\begin{aligned} \log(\mathsf{c}_k(\theta_k) Z^{(k)}(\mathbf{1})) &= \mathsf{N}^{(k)} + \sum_X \sum_{z \in X} \mathsf{n}^{(k+1)}(X, z), \\ \mathsf{n}^{\mathrm{tot}}_k &= 36 N_{k+1} \log L - \tfrac{1}{2} \log\det(\mathsf{C}^{\mathsf{T}} \Delta^{(k)} \mathsf{C})|_{\mathbf{1}}, \quad |\mathsf{n}^{\mathrm{tot}}_k| \le C N_{k+1}, \\ \mathsf{n}^{\mathrm{tot}}_0 &\le -3(L-1)^4 N_1 \log 2 < 0. \end{aligned} \tag{5.lxxxiii}$$

Here N_j counts sites, c_k is the actual change-of-variable normalization and $\mathsf{N}^{(k)}$ is the free-bond/coarse-site count, with $-3 \log \theta_k/2$ per free bond. The map C is the actual linear embedding of nonpivot coordinates into full constrained fluctuations; this determinant is on the free color coordinates. It is not the covariance $C^{(k)}$. The local numbers are Gaussian values before normalization, real and coupling independent, with common nonwrapping lifts and the Gaussian domain bound. Their coefficients vanish. The count alone is not the entire step number. The determinant estimates and first-step sign in eq. 5.lxxxiii remain actual Gaussian targets, beyond the once-only accounting identity.

The explicit leading vacuum requirement is

$$\left| E_{K,\ell,\beta} + \tfrac{3}{2}(N^2-1)(L^4-1)\sum_{j=0}^{K-1}(\log x_j)N_{j+1} \right| \le CN_0, \tag{5.lxxxiv}$$

uniform in the admitted runs, tori and cutoffs. It is not proved merely by naming a field-independent constant.

Choose the constants in the following order. Fix the geometric exponents and L, with the Gaussian constants and one-loop data; then choose κ above the domain entropy, the required lower block scales and M. Choose $\sigma_\star$, absolute radii below c/M, $K_{\mathbf{M}}$, and $E_0 \ge \max\{2E_{\mathsf{N}}, 4K_{\mathbf{M}}\sigma_\star\}$. Next choose $N_\star$, the ceiling on s, the unmarked constants, C_β and the weak triple. Compute the marked-load and tube constants at that triple before choosing γ, the strong triple, γ_H and the smaller drift ceiling. The source radius and comparison windows have their stated place before the final smallness of g. Only then are ℓ, β, K varied. Each claimed uniformity depends on respecting this order.

For the germ comparisons, let $\mathfrak{P}_n(d;\vartheta)$ consist of real Lipschitz tuples with disjoint supports in one ball of radius $5d$, satisfying $\operatorname{Lip} f_i \le \|f_i\|_\infty/(\vartheta d)$. For $n = 2, 3$, a modulus (C, r_0) for $\sigma \approx_n \sigma'$ means

$$|(\sigma - \sigma')(\underline{f})| \le Cr^{4-n}\prod_i \|f_i\|_\infty \quad (0 < r \le r_0, \quad \underline{f} \in \mathfrak{P}_n(r;\vartheta) \cap C^2). \tag{5.lxxxv}$$

A comparison for every C^1 tuple is stronger and implies this one on the C^2 subclass. The converse requires its own density argument with the stated shape bounds. Transitivity adds the constants and takes the smaller radius. If $(\mathsf{T}_b\sigma)(\underline{f}) = \sigma(\underline{f}^{[b]})$, the modulus becomes $(L^{(4-n)b}C, L^{-b}r_0)$, directly by substitution. This accounts for the factor in an anchor comparison and keeps the smoothness quantifiers of the short-distance input explicit.

Later applications use the following hypotheses. Beyond-first-passage data use eq. 5.lxxv; prolonged runs use statement 3.8; fixed-depth reference prefixes and terminal coordinates use eq. 5.lxxix. The germ/rigidity route in the comparison appendix also carries its own shape and second-difference hypotheses. In the anchor unit its correlation constants acquire $L^{-(4-n)s_\star}$. Blocking and distinct-member comparisons use eq. 5.lxxxii, not just a relabeling of K. The sub-unit cells of the angular route keep their separate moment, mirror and ultraviolet interfaces. The ordinary interacting source construction, clean-class two-point extraction, cited reconstruction theorem and full gauge-theory identification retain their stated hypotheses.

5.3. **Gauge changes in the physical Gaussian integral.** The Gaussian normalizer on an averaging fibre contains a constraint Jacobian and a gauge Jacobian. Both must be retained when a contour gauge is replaced by a background gauge. We prove the finite-dimensional identity with its compensating factor, then apply it to the actual constraint and the complete constrained Hessian. The hypotheses needed for this application are a differentiable physical background graph, its local gauge covariance, and a nondegenerate physical Gaussian. The identity does not construct that graph or an independent-current extension of its Hessian.

5.3.1. *Constrained Lebesgue integrals and saddle determinants.* All spaces in this subsection are finite-dimensional real inner-product spaces. Lebesgue measures and determinants are defined by their orthonormal coordinates. Thus the formulas also apply to physical lattice measures, after those measures have been put into the inner products. Let $Q : E \to F$ be onto, put $n = \dim E$, $m = \dim F$, and let $T = T^*$ be positive on $\ker Q$. Define

$$I(T, Q) = \int_E \delta_F(Qa)e^{-\langle a, Ta\rangle/2}\, da. \tag{5.lxxxvi}$$

The delta function is normalized by the Lebesgue measure of F. This convention is important when the constraint is changed.

Lemma 5.4. With

$$\mathbb{S}(T,Q)=\begin{pmatrix}T & Q^*\\ Q & 0\end{pmatrix},$$

one has

$$I(T,Q)=(2\pi)^{(n-m)/2}|\det\mathbb{S}(T,Q)|^{-1/2}. \tag{5.lxxxvii}$$

For every symmetric $H : F \to F$, the saddle determinant and the integral are unchanged by $T \mapsto T + Q^*HQ$. ■

Proof. Split $E = E_0 \oplus E_1$ orthogonally, where $E_0 = \ker Q$, and write $Q = (0, Q_1)$. The square matrix $Q_1 : E_1 \to F$ is invertible and

$$\delta_F(Qa)\,da=|\det Q_1|^{-1}\,da_0\,\delta_0(da_1).$$

Consequently the integral equals $(2\pi)^{(n-m)/2}(\det T_{00})^{-1/2}(\det QQ^*)^{-1/2}$. Eliminating the T_{00} block in the saddle matrix gives

$$\det\mathbb{S}(T,Q)=(-1)^m\det T_{00}\det(QQ^*).$$

The remaining two-by-two block has determinant $(-1)^m|\det Q_1|^2$, independently of its upper-left entry. This proves both assertions. In particular no positivity of T on all of E was used. □

5.3.2. *The relative gauge determinant.* Let H be a fine site space and K a coarse site space. Write $D : H \to E$, $\mathrm{d} : K \to F$ for the two infinitesimal gauge actions. Assume initially that D and d are injective. Let $R : H \to K$ and $Q : E \to F$ be onto and satisfy

$$QD=\mathrm{d}R. \tag{5.lxxxviii}$$

Let $B = B^*$ satisfy $BD = 0$. Assume that its restriction to $\ker Q$ is nonnegative and that its kernel there is exactly $D(\ker R)$. Put $h = \dim H$, $k = \dim K$, and $H_0 = \ker R$. A residual gauge coordinate is a linear map $G : E \to Z$, where $\dim Z = h - k$, such that

$$GD_0 : H_0 \longrightarrow Z \quad \text{is invertible,} \qquad D_0 = D|_{H_0}.$$

It follows that $B + G^*G$ is positive on $\ker Q$, so its constrained integral is finite. Define

$$\mathcal{J}(B,Q,R,G)=\sqrt{\det(RR^*)}\,|\det(GD_0)|I(B+G^*G,Q). \tag{5.lxxxix}$$

The determinant of GD_0 uses the induced inner product on H_0; it is not a coordinate determinant in an unnormalized list of site values.

Proposition 5.5 (The gauge and constraint change identity)**.** The quantity $\mathcal{J}$ is independent of G. If $T : E \to K$ and

$$Q'=Q+\mathrm{d}T, \qquad R'=R+TD \tag{5.xc}$$

are onto, with the same nondegeneracy on the physical quotient, then $\mathcal{J}(B,Q',R',G') = \mathcal{J}(B,Q,R,G)$ for every admitted G'. The factor $\sqrt{\det(RR^*)}$ in eq. 5.lxxxix is necessary with the delta normalization eq. 5.lxxxvi. ■

Proof. Choose any full gauge coordinate $C : E \to H$ for which CD is invertible, and consider

$$\mathcal{Z}(C,Q)=|\det(CD)|\int_{E\times K}\delta_F(Qa-\mathrm{d}\xi)e^{-\langle a,Ba\rangle/2-\|Ca\|^2/2}\,da\,d\xi. \tag{5.xci}$$

This is a finite Gaussian integral. Indeed, if $Qa = \mathrm{d}\xi$, choose $v \in H$ with $Rv = \xi$. Then $a - Dv \in \ker Q$, and the value of B is unchanged by this subtraction. Its only null directions on that constrained space are gauge directions. The full coordinate C controls them, while injectivity of d controls the remaining ξ variable.

To prove independence of C, choose a fixed complement $E = E_{\mathrm{tr}} \oplus \operatorname{ran} D$ and write $a = b + Dv$. The Jacobian of this fixed decomposition is a constant independent of C and Q. In the other variable replace ξ by $\zeta + Rv$. Then

$$Qa - \mathrm{d}\xi = Qb - \mathrm{d}\zeta, \qquad \langle a, Ba\rangle = \langle b, Bb\rangle.$$

The integral in v is

$$\int_H e^{-\|Cb+CDv\|^2/2}\, dv = (2\pi)^{h/2} |\det(CD)|^{-1}.$$

It cancels the prefactor in eq. 5.xci and is independent of b. Thus $\mathcal{Z}$ is independent of C.

It is also unchanged by $Q \mapsto Q + \mathrm{d}T$. In eq. 5.xci, make the triangular change $\xi \mapsto \xi + Ta$, holding a fixed. Its Jacobian is one, and it removes $\mathrm{d}Ta$ from the argument of the delta function. The action and gauge weight are unchanged.

It remains to express $\mathcal{Z}$ by a residual gauge coordinate, retaining all measure factors. Choose $c : F \to K$ such that cd is invertible, and take

$$Ca = (Ga, cQa) \in Z \oplus K.$$

Its target has dimension h. In the orthogonal splitting of H, use the right inverse $R^\dagger = R^*(RR^*)^{-1}$ and write

$$v = v_0 + R^\dagger \eta, \qquad v_0 \in H_0.$$

The Jacobian of $(v_0, \eta) \mapsto v$ is $(\det RR^*)^{-1/2}$. The matrix of CD in these coordinates is block triangular:

$$\begin{pmatrix} GD_0 & GDR^\dagger \\ 0 & cd \end{pmatrix}.$$

Therefore

$$|\det(CD)| = \sqrt{\det(RR^*)}\, |\det(GD_0)|\, |\det(cd)|. \tag{5.xcii}$$

For a fixed ξ, the constrained integral with $Qa = \mathrm{d}\xi$, action B, and gauge weight $\|Ga\|^2/2$ equals the one with $Qa = 0$. To see this choose

$$v_\xi = R^\dagger \xi - (GD_0)^{-1} GDR^\dagger \xi,$$

where the second term is regarded as a vector in H_0. Then $Rv_\xi = \xi$, $GDv_\xi = 0$, and translation $a \mapsto a + Dv_\xi$ preserves the action and residual gauge weight. On the constraint, $cQa = cd\xi$. Hence the remaining integral in ξ in eq. 5.xci is $(2\pi)^{k/2} |\det(cd)|^{-1}$. Combining it with eq. 5.xcii yields

$$\mathcal{Z}(C, Q) = (2\pi)^{k/2} \mathcal{J}(B, Q, R, G).$$

The two independence assertions for $\mathcal{Z}$ now give the proposition. □

The constraint change in eq. 5.xc alters only the coarse gauge component of Qa. Its effect on the residual fine gauge group is $R \mapsto R + TD$. Omitting the resulting coarea factor would make the invariance statement false, even though the quotient of physical variations is the same.

5.3.3. *The background-gauge formula.* Let $\Delta = D^*D$. Since D is injective on H_0, its compression Δ_{00} to H_0 is positive. Write

$$P_0 = 1 - R^*(RR^*)^{-1}R$$

for the orthogonal projection onto H_0, and choose the residual coordinate $G = P_0 D^* : E \to H_0$. Then

$$GD_0 = \Delta_{00}, \qquad G^*G = DP_0D^*.$$

The same saddle calculation as in Lem. 5.4 gives

$$|\det \mathbb{S}(\Delta, R)| = \det(RR^*) \det \Delta_{00}.$$

Consequently

$$\mathcal{J} = \frac{|\det \mathbb{S}(\Delta, R)|}{\sqrt{\det(RR^*)}}\, I(B + DP_0D^*, Q). \tag{5.xciii}$$

In particular, if $RR^* = 1$, its logarithm is

$$\log \mathcal{J} = -\tfrac{1}{2}\log|\det \mathbb{S}(B + D(1 - R^*R)D^*, Q)| + \log|\det \mathbb{S}(\Delta, R)| + \tfrac{n-m}{2}\log(2\pi). \tag{5.xciv}$$

The site operator in the second determinant is the constrained site Laplacian. The gauge form in the vector determinant is $D(1 - R^*R)D^*$. It need not be the projected Landau gauge form used to construct the physical background; the gauge-change identity justifies this change in the integral.

5.3.4. *The actual constrained Hessian.* We now specialize the algebra. Let $U(V)$ be an actual differentiable critical background on a neighborhood of a real coarse field V_0, satisfying the prescribed homogeneous constraint $\mathcal{M}^j U(V) = V$. Assume its orbit is locally unique, is gauge covariant, and has a nondegenerate Gaussian on its physical fibre. Put $U_0 = U(V_0)$ and

$$S_c(V) = A(U(V)).$$

The action S_c is gauge invariant. All derivatives in what follows use the actual logarithmic fine and coarse coordinates at U_0, V_0, with the physical measures declared for the construction.

Let $q_0(a)$ be the logarithm of the plain constraint:

$$\mathcal{M}^j(e^{i\eta a}U_0) = e^{iq_0(a)}V_0, \qquad q_0(0) = 0, \quad Dq_0(0) = Q_0.$$

At a constrained-critical point,

$$DA(U_0) = Q_0^*\mu, \qquad DS_c(V_0) = \mu. \tag{5.xcv}$$

The second equality follows from the first by differentiating the constraint along $U(V)$. The derivative of that graph is a right inverse of Q_0, so no inverse of $Q_0^*Q_0$ is implicit here.

Let $q(a)$ be the actual straightened constraint with its actual node factor $v(a)$, as in eq. 3.cccxliii. Thus the plain coarse field and $e^{iq(a)}V_0$ are related by that node gauge. Gauge invariance gives the exact nonlinear identity

$$S_c(\mathcal{M}^j(e^{i\eta a}U_0)) = S_c(e^{iq(a)}V_0). \tag{5.xcvi}$$

For the convention in which a site tangent acts by D, differentiation of the node factorization gives

$$Q = Q_0 + \mathrm{d}T, \qquad R = R_0 + TD, \qquad QD = \mathrm{d}R. \tag{5.xcvii}$$

Here $Q = Dq(0)$, R_0 is evaluation of a fine gauge parameter at the coarse nodes, and T is minus the derivative of the node factor in that convention. Equivalently the first identity defines its sign. The second identity also follows by differentiating the same factorization along a fine gauge transformation. The resulting R is the actual transported site average, not an independently chosen map. Its formula is the own-block average in the field intertwining identity.

Consider the gauge-invariant fine function

$$\widetilde{A}(W) = A(W) - S_c(\mathcal{M}^j W). \tag{5.xcviii}$$

Its first derivative vanishes at U_0, by eq. 5.xcv. Let B be its Hessian there. Gauge invariance at a critical point gives $BD = 0$. The chain rule in eq. 5.xcvi gives the complete formula

$$B = A''(U_0) - \langle \mu, D^2q(0)\rangle - Q^*H_cQ, \qquad H_c = D^2S_c(V_0). \tag{5.xcix}$$

The middle term denotes the symmetric form

$$(a, b) \longmapsto \langle \mu, D^2q(0)[a, b]\rangle.$$

Thus, on the linearized fibre $\ker Q$, the actual quadratic action is

$$B_{\mathrm{fib}} = A''(U_0) - \langle \mu, D^2q(0)\rangle. \tag{5.c}$$

It contains the curvature of the nonlinear constraint. The last term in eq. 5.xcix does not change its saddle determinant, by Lem. 5.4. This derivation defines the physical current and multiplier correction from the actual constraint; it is not a form selected to obtain a determinant identity.

The same construction with q_0 gives the plain-fibre Hessian. The full B is the same for both constraints, because it is the Hessian of the single function eq. 5.xcviii. This is the reason the constraint-change part of Prop. 5.5 applies.

5.3.5. *The two actual site Jacobians.* In fine physical measure η^4, node evaluation has

$$R_0 R_0^* = \eta^{-4} I. \tag{5.ci}$$

This is a field-independent factor. By contrast the transported site average is a coisometry:

$$R R^* = I. \tag{5.cii}$$

To verify the latter at one step, write its value as the normalized sum of the transported fine site values in a block. Each real adjoint transport is orthogonal, distinct blocks have disjoint site sets, and the fine measure divided by the coarse measure is precisely the reciprocal of the number of sites in a block. Multiplying the average by its physical adjoint therefore gives the identity. For a homogeneous composition, write the site average as the composition of its one-step averages. Each is a coisometry between its own physical site spaces, so their composition is a coisometry as well. This calculation does not require flatness. Here the straightened constraint is re-centered at the actual U_0, so its relative contour logarithms vanish at $a = 0$. Equation eq. 5.cii concerns that derivative. It does not identify the effective gauge derivative of a fixed-flat-reference nonlinear chart at a nonzero field with a coisometry.

The actual contour gauge on the plain fibre supplies a residual coordinate G_{ax}. Its derivative along a gauge parameter which vanishes at the coarse nodes is triangular in the levels, with field-independent determinant. Here is the local verification. A contour from a block node y to x transforms by its two endpoint gauges. At the axial representative its value is one, so its logarithmic derivative along $v_t = e^{it\lambda}$ is $\lambda(y) - \lambda(x)$, with the fixed sign convention. Group the noncoarse sites by their first parent in the nested block hierarchy. The map from site values to these parent differences is triangular with diagonal entries $\pm I_3$, since the final coarse-node values are zero. Its determinant has absolute value one in these unweighted site coordinates. Transported frames insert adjoint rotations of determinant one. Physical measures and fixed gauge weights insert only constants depending on the mesh and the numbers of sites. Therefore

$$\sqrt{\det(R_0 R_0^*)}\,|\det(G_{\mathrm{ax}} D|_{\ker R_0})| = c_{\eta,j}, \tag{5.ciii}$$

where $c_{\eta,j} > 0$ is independent of the background. For one step the endpoint derivative is exactly $GT = -I$ from eq. 4.ccxviii; the preceding triangular calculation is its composition over the hierarchy.

Let $\Psi_j(V)$ be the logarithm of the actual physical Gaussian normalizer on the plain fibre, minus its value at the trivial background. Its Haar factors at the centre are field independent in left exponential coordinates. The nonpivot elimination factor is precisely the constraint coarea factor in eq. 5.lxxxvi. The reference contour-gauge normalizations and the constant eq. 5.ciii also disappear from this normalized difference.

Proposition 5.6 (The physical saddle representation)**.** On every admitted actual background domain described above,

$$\begin{aligned}\Psi_j(V) = {} & -\tfrac{1}{2}\log\left|\det \mathbb{S}\left(A''(U_0) - \langle \mu, D^2 q(0)\rangle + D(1 - R^* R)D^*,\, Q\right)\right| \\ & + \log|\det \mathbb{S}(D^* D, R)| - (\text{the same at } V = 1).\end{aligned} \tag{5.civ}$$

The multiplier and the constraint curvature use their physical coordinate normalizations. The formula is a physical-pair identity. It does not specify how either determinant is extended off that manifold to an independent current. ■

Proof. At a coarse field with no infinitesimal stabilizer, d is injective. The fine gradient is injective as well: $D\lambda = 0$ implies $\mathrm{d}R_0\lambda = 0$, hence $R_0\lambda = 0$; a parallel fine field which is zero at a coarse node is zero on its component. The same observation makes D injective on $\ker R$ without the generic coarse-field restriction.

The physical Gaussian is nondegenerate on the plain fibre modulo its residual gauges. Because the two constraints differ by a coarse gauge component, their physical quotients are isomorphic: given a vector on one constraint space, correct its coarse gauge value by $D\lambda$, using surjectivity of the other site map. The value of B is unchanged. Thus the required quotient nondegeneracy holds for both Q_0, R_0 and Q, R. The same argument shows Q is onto: if $Q_0 a = f$, then $Qa = f + \mathrm{d}Ta$; choose λ with $R\lambda = -Ta$ and replace a by $a + D\lambda$.

Apply Prop. 5.5 to the common gauge-invariant Hessian B. On the plain side use G_{ax} and eq. 5.ciii. On the straightened side use the background gauge and eq. 5.cii. Formula eq. 5.xciv gives the two logarithms. Replace B by B_{fib} using eq. 5.xcix and Lem. 5.4. All remaining constants cancel in the difference at $V = 1$, giving eq. 5.civ.

The auxiliary integral eq. 5.xci is singular when d has a kernel, so it is not used at such a field. After its coarse-gauge factors have been canceled, both sides of eq. 5.civ are finite constrained Gaussian or saddle-determinant expressions. Their nondegenerate fibre forms vary continuously on the admitted background domain. Irreducible coarse connections are dense near the trivial field: two small loop holonomies with distinct colour axes have no common nonzero adjoint fixed vector. Perturbations of links can create such loops on the connected periodic four-dimensional lattice. Thus the finite final identity extends to the trivial field and to fields with stabilizers by continuity. No continuity of the uncanceled auxiliary integral is asserted. □

On an admitted complex background domain with coherent logarithm branches, use bilinear physical transposes and the logarithm branches continued from the real positive Gaussians. The determinant identity then continues holomorphically wherever the constrained forms remain nondegenerate. Absolute values belong only to the displayed real formula. In particular one cannot insert them in a holomorphic localization.

5.3.6. *Gaussian composition with nonlinear constraints.* We also record why Gaussian integration in stages uses the complete constrained Hessians. This identifies the algebraic composition needed before a one-loop coefficient is extracted. Let $M : E \to F$ and $N : F \to G$ be onto. Let $T = T^*$ be positive on $\ker(NM)$, and let $H : F \to E$ be its constrained minimizer:

$$MH = 1, \qquad \langle u, THb\rangle = 0 \quad (u \in \ker M).$$

Put $T_2 = H^*TH$. The change $a = u + Hb$, $u \in \ker M$, has Jacobian $(\det MM^*)^{-1/2}$; this follows first for the orthogonal right inverse of M, and then for H, whose difference from it is a shear into $\ker M$. The constraint becomes $Nb = 0$, and the exponent splits. Hence exactly

$$I(T, NM) = I(T, M)I(T_2, N). \tag{5.cv}$$

Only positivity on the indicated constraint spaces is needed. There is no additional determinant of N or M: both are already included by the two delta functions with their declared measures.

For the nonlinear application let m_1, m_2 be the actual successive constraint maps in their local coordinates. Let $x = x(y)$ be the first physical critical graph and y a critical point of $S_1(y) = F(x(y))$ under $m_2(y) = z$. Write

$$M = Dm_1(x), \qquad N = Dm_2(y), \qquad DF(x) = M^*\mu_1, \qquad DS_1(y) = N^*\mu_2.$$

Differentiation of the first graph gives $\mu_1 = N^*\mu_2$. The complete first-stage Hessian is

$$B_1 = F''(x) - \langle \mu_1, m_1''(x) \rangle.$$

With $H = Dx(y)$, differentiation of the criticality equation gives

$$MH = 1, \qquad \langle u, B_1 H b \rangle = 0 \quad (u \in \ker M), \qquad S_1''(y) = H^* B_1 H. \tag{5.cvi}$$

For the last equality, differentiate $m_1(x(y)) = y$ twice and pair with μ_1. Its acceleration term cancels the corresponding acceleration term in $D^2(F \circ x)$. This calculation uses the multiplier term in B_1.

The total constraint has second derivative

$$D^2(m_2 \circ m_1)[a, a] = m_2''[Ma, Ma] + N m_1''[a, a].$$

Therefore its complete Hessian is

$$B_{\mathrm{tot}} = B_1 - M^* \langle \mu_2, m_2'' \rangle M. \tag{5.cvii}$$

In the coordinates $a = u + Hb$, this is the direct sum of $B_1|_{\ker M}$ and $S_1'' - \langle \mu_2, m_2'' \rangle$.

Gauge fixing does not spoil this decomposition. Choose the actual first-stage axial representative throughout the first background graph. Its first gauge coordinate vanishes along that graph, so its derivative G_1 satisfies $G_1 H = 0$. The second-stage gauge coordinate acts on $b = Ma$. The total quadratic gauge weight is then $\|G_1 u\|^2 + \|G_2 b\|^2$, precisely the two gauge weights used by the hierarchical Gaussian integral. Apply eq. 5.cv to these nondegenerate gauge-fixed forms, explicitly taking

$$T = B_{\mathrm{tot}} + G_1^* G_1 + M^* G_2^* G_2 M.$$

Corollary 5.7. On an admitted homogeneous physical background graph, the normalized Gaussian logarithms obey

$$\Psi_j = \Phi_j + \Psi_{j-1} \circ p_j, \tag{5.cviii}$$

where p_j is the actual preceding critical background map and Φ_j is the actual one-step Gaussian logarithm. All constraint-curvature terms are included before the Gaussian integration. Field-independent measure constants cancel in the normalization. ■□

Proof. Use the preceding Hessian decomposition at each stage and integrate first in u, then in b. The first factor in eq. 5.cv is the earlier Gaussian normalizer evaluated at $p_j(V)$; the second is the new step normalizer. Normalize both at the trivial coarse field and take their real logarithms. This gives eq. 5.cviii. □

5.3.7. *The physical multiplier correction has zero coefficient.* The statement that a current correction is of third order in external momentum concerns its determinant functional. The matrix $-\langle \mu, q'' \rangle$ itself need not have that order. We prove the required functional statement on the actual background domain.

Fix a homogeneous depth j, and work near a flat reference with trivial coarse field for the coefficient extraction. For the following argument assume the actual critical background graph and its first two derivatives are exponentially local on this neighborhood. The weighted construction in Prop. 3.183, with eq. 3.mccxcv and eq. 3.mccxcvi, supplies this property on the constructed flat-reference neighborhood; no estimate uniform in j for the determinants in this argument will be required. Let

$$T(V) = A''(U(V)) + D(1 - R^*R)D^*.$$

Introduce an independent coarse bond covector ν, with the same coordinate normalization as μ, and put

$$\begin{aligned} L_j(V, \nu) = & -\tfrac{1}{2} \log \det \mathbb{S}(T(V) - \langle \nu, q_V'' \rangle, Q_V) \\ & + \tfrac{1}{2} \log \det \mathbb{S}(T(V), Q_V). \end{aligned} \tag{5.cix}$$

Here and below the logarithms are coherent local branches; their difference is zero at $\nu = 0$. The actual multiplier correction to the physical saddle logarithm is

$$C_j(V) = L_j(V, DS_c(V)).$$

Proposition 5.8. The bulk Hessian of C_j has symbol

$$\widehat{\pi}^{C_j}(p) = O_j(|p|^3) \qquad (p \to 0). \tag{5.cx}$$

Consequently $b[C_j] = 0$. The statement is exact at every admitted depth. It does not assert a pointwise $O(|p|^3)$ bound for the multiplier matrix. ∎

Proof. We first justify the local trace jets used here. At a fixed fine mesh, the flat background-gauge vector form restricted to $\ker Q$ is nondegenerate. A zero of its sum of nonnegative squares would satisfy

$$\mathrm{d}v = 0, \qquad Qv = 0, \qquad P_0 D^* v = 0, \quad P_0 = 1 - q^* q.$$

In each Bloch fibre a closed field is a gradient, apart from the parallel constant modes at zero frequency. The actual average detects those constant vector modes, so they vanish. For the gradient part $v = D\lambda$, the identity $QD = \mathrm{d}_c q$ gives $\mathrm{d}_c q\lambda = 0$. Subtract a parallel constant from λ if needed; it then has $q\lambda = 0$. Pairing $P_0 D^* D\lambda = 0$ with λ gives $D\lambda = 0$. Thus $v = 0$.

The finite Bloch matrices and their constraint maps are continuous on the compact Brillouin torus. Surjectivity of Q, supplied by the actual row seeds, and the preceding kernel argument imply invertibility of the full saddle matrix in every fibre. At this fixed mesh, its inverse is uniformly bounded over that torus. It extends to a small complex momentum strip by the resolvent identity. Fourier contour shifting therefore gives exponential decay of its coarse-cell kernels, with constants which may depend on j.

All other coefficients in eq. 5.cix have fixed physical carriers before the background is substituted. The exponentially local background derivatives and the resolvent identity show that its two trace derivatives have absolutely summable exponential moments. For example, differentiating a logarithm twice gives $\mathrm{Tr}(S^{-1}S'' - S^{-1}S'S^{-1}S')$; each differentiated coefficient has its declared local carrier or the exponentially decreasing background response. Products of these kernels retain an exponential moment after decreasing its rate. There are finitely many fine coordinates per cell at fixed j, so their internal trace is finite. The same estimates make finite-volume winding terms vanish exponentially. This proves existence of the bulk Hessian symbols and their analyticity in a momentum neighborhood of zero. It also justifies the Ward identities below by first using finitely supported gauge parameters and then taking these absolutely convergent Fourier sums.

The function L_j is invariant under simultaneous coarse gauge transformation of V and the covector ν. Indeed, the background orbit is gauge covariant, all fine and coarse tangent spaces transform by their adjoint rotations, and the actual constraint second derivative transforms with its covector pairing. Both saddle determinants are conjugated by these rotations, whose determinants are one. In addition,

$$L_j(V, 0) = 0, \qquad DL_j(1, 0) = 0. \tag{5.cxi}$$

The first equality is its definition. For the derivative in ν, invariance under constant colour rotations makes each coefficient an invariant linear functional on the adjoint colour space, hence zero. The derivative in V vanishes by the first equality.

Let K_j be the mixed V, ν Hessian kernel at $(1, 0)$. Differentiate the joint gauge identity in a gauge parameter and in ν. The term from transforming ν is multiplied by the zero first derivative in eq. 5.cxi. Thus

$$D^2_{V,\nu} L_j(1, 0)[\mathrm{d}_c\chi, \nu] = 0.$$

Translation covariance gives, in Fourier coordinates,

$$\widehat{\mathrm{d}}_c(-p)^{\mathsf{T}} \widehat{K_j}(p) = 0. \tag{5.cxii}$$

Since $\widehat{\mathrm{d}}_{c,\mu}(p) = -ip_\mu + O(|p|^2)$, its linear Taylor term forces $\widehat{K_j}(0) = 0$: every column is annihilated by every row vector p^{T}. Analyticity then gives

$$\widehat{K_j}(p) = O_j(|p|). \tag{5.cxiii}$$

The actual multiplier is $\mu(V) = DS_c(V)$, and $\mu(1) = 0$. Let P_j be the symbol of its derivative, that is the Hessian of S_c at the trivial field. Gauge invariance gives $P_j(0) = 0$ by the same zero-momentum Ward argument. In midpoint coordinates, inversion sends both bond variations to their negatives and p to $-p$. The Hessian symbol in those coordinates is therefore even. Its linear Taylor term vanishes, and the endpoint-to-midpoint phase does not change this conclusion after $P_j(0) = 0$. Hence

$$P_j(p) = O_j(|p|^2). \tag{5.cxiv}$$

These are the actual symmetries of the symmetric contour construction and its critical orbit.

Apply the second-order chain rule to C_j. The pure V, V Hessian of $L_j(V, 0)$ is zero. The term involving $D^2\mu$ is multiplied by $D_\nu L_j(1, 0) = 0$. The remaining terms are the two mixed kernels composed with P_j, and the ν, ν kernel composed with P_j on both sides. Their orders are respectively at least $|p|^3$ and $|p|^4$, by eq. 5.cxiii and eq. 5.cxiv. This proves eq. 5.cx. The quadratic Maxwell coefficient is therefore zero. □

The coefficient identity $\sum_{l\le j} b_l^0 = b[\Psi_j]$ additionally uses invariance of the coefficient under the actual map p_j. The saddle formula and the recursion alone do not supply that response estimate. Likewise, identification of the physical Hessian here does not give localization or common complex-current domains for a chosen off-shell completion. These analytic requirements remain distinct from the finite gauge-change and composition identities.

5.4. **The massive Wilson determinant in a constant field.** We compute the logarithmic coefficient directly on the unit lattice. This is a component of the small-field Gaussian renormalization method of [B87]. This calculation concerns the Wilson Hessian and the covariant site Laplacian, before an averaging constraint is introduced. It includes the neutral colour, the charged colours, and the order of the volume, field, and mass limits. The comparison with a constrained determinant is a separate step.

Write t_a for the Pauli matrices, so that $\frac{1}{2}\,\mathrm{tr}(t_a t_b) = \delta_{ab}$. A real bond variation is written $e^{ita_b}U_b$, with $a_b = \sum_a a_b^a t_a$. The Wilson action and its Hessian have the normalization

$$A(U) = \sum_p (1 - \tfrac{1}{2}\,\mathrm{tr}\,U_p), \qquad A''(1)[a, a] = \sum_p |(\mathrm{d}a)_p|^2.$$

Adjoints below use counting measure on the unit lattice and this Euclidean colour inner product. Put

$$(D_U\lambda)_\mu(x) = \lambda(x) - \mathrm{Ad}_{U_\mu(x)}\,\lambda(x + e_\mu), \qquad \Delta_U = D_U^* D_U, \qquad K_U = A''(U) + D_U D_U^*.$$

The opposite sign for both occurrences of D_U gives the same operators. We use a star for the real Hilbert adjoint and for its complexification in this calculation.

For $f \in \mathbb{R}$, fix the Abelian field

$$U_{f,1}(x) = e^{-ifx_2 t_3}, \qquad U_{f,\mu}(x) = 1 \quad (\mu \neq 1). \tag{5.cxv}$$

Its positively oriented $(1, 2)$-plaquettes are e^{ift_3}; all its other plaquettes are the identity. The connection itself grows in a logarithmic coordinate as $|x_2|$, whereas every plaquette is constant. We therefore define the determinant density by finite compressions, rather than by placing arbitrary flux on a periodic torus.

Let $\Lambda_R = [-R, R]^4 \cap \mathbb{Z}^4$. Site operators are compressed to sites in Λ_R; vector operators are compressed to positive bonds with both endpoints there. Thus, for example, $K_{f,R} = P_R K_{U_f} P_R^*$. Compression is taken after the full lattice operator has been formed. This is the Hessian for

variations extended by zero outside the indicated coordinates. It does not discard plaquettes meeting the boundary. For $z > 0$, define

$$\begin{aligned}\mathcal{L}_R(f,z) &= \log\det(\Delta_{f,R}+z) - \tfrac{1}{2}\log\det(K_{f,R}+z),\\ \mathcal{D}_R(f,z) &= |\Lambda_R|^{-1}\big(\mathcal{L}_R(f,z)-\mathcal{L}_R(0,z)\big).\end{aligned} \tag{5.cxvi}$$

The determinants here are over all three real colours. For each fixed positive z, the matrices in this formula are positive when $|f|$ is sufficiently small in terms of z.

Proposition 5.9. The functions $\mathcal{D}_R(\cdot,z)$ and their first three derivatives at zero have limits as $R\to\infty$, locally uniformly for z in compact subsets of $(0,\infty)$. The limiting density $\mathcal{D}(f,z)$ is a real, even, smooth function of real f near zero. Let

$$\widetilde{\beta}(z) = \partial_f^2 \mathcal{D}(0,z).$$

Then

$$\widetilde{\beta}(z) = \frac{11}{6\pi^2}\log\frac{1}{z} + C_* + E(z), \qquad |E(z)| + |zE'(z)| + |z^2E''(z)| \le C\sqrt{z}, \quad 0<z\le 1, \tag{5.cxvii}$$

for finite constants C_* and C. In particular,

$$\sup_{0<z\le 1}\left\{\left|\widetilde{\beta}(z) - \frac{11}{6\pi^2}\log\frac{1}{z}\right| + |z\widetilde{\beta}'(z)| + |z^2\widetilde{\beta}''(z)|\right\} < \infty. \tag{5.cxviii}$$

The same limits hold for any lattice Følner exhaustion, using the same zero-extension compression convention. ∎

The proof will give an exact Brillouin-zone formula at every positive mass. The coefficient is therefore obtained from the actual lattice Hessian, without replacing it by a continuum vector operator.

5.4.1. *Charged coordinates and magnetic translations.* The neutral coordinate is a^3. For the other two real coordinates put $w = a^1 - ia^2$. Direct multiplication of Pauli matrices gives

$$\mathrm{Ad}_{e^{i\theta t_3}}\, w = e^{2i\theta} w.$$

Consequently the charged covariant shifts in the background eq. 5.cxv are

$$(u_1 w)(x) = e^{-i\varphi x_2} w(x+e_1), \qquad (u_\mu w)(x) = w(x+e_\mu) \quad (\mu\neq 1), \qquad \varphi = 2f.$$

They are unitary, and

$$u_1 u_2 = e^{i\varphi} u_2 u_1, \qquad u_\mu u_\nu = u_\nu u_\mu \quad \text{if } \{\mu,\nu\}\neq\{1,2\}. \tag{5.cxix}$$

For $m\in\mathbb{Z}^4$ define

$$W_m = e^{-i\varphi m_1 m_2/2} u_1^{m_1} u_2^{m_2} u_3^{m_3} u_4^{m_4}.$$

The commutation relation proves, including negative powers,

$$W_m W_n = e^{i\varphi(m_1 n_2 - m_2 n_1)/2} W_{m+n}, \qquad W_m^* = W_{-m}. \tag{5.cxx}$$

If $a(q) = \sum_m a_m e^{im\cdot q}$, write $\mathrm{Op}_\varphi(a) = \sum_m a_m W_m$. This definition also applies to matrix coefficients, with the matrix acting on the vector index. Absolute Fourier convergence is sufficient.

Let $\mathbb{B} = [-\pi,\pi]^4$, with opposite faces identified, and write

$$\int_{\mathbb{B}} h(q)\, d\bar{q} = (2\pi)^{-4}\int_{[-\pi,\pi]^4} h(q)\, dq.$$

The normalized trace of a finite Weyl polynomial is

$$\tau\big(\mathrm{Op}_\varphi(a)\big) = \mathrm{tr}\, a_0 = \int_{\mathbb{B}} \mathrm{tr}\, a(q)\, d\bar{q}.$$

Indeed, W_m has a zero diagonal unless $m = 0$, and $W_0 = 1$. The same formula holds for absolutely summable coefficients by approximation. For two such symbols the zero Fourier coefficient of their product involves only $n = -m$. The phase in eq. 5.cxx is then one. Hence

$$\tau(\operatorname{Op}_\varphi(a)\operatorname{Op}_\varphi(b)) = \int_{\mathbb{B}} \operatorname{tr}(ab)\, d\bar{q}. \tag{5.cxxi}$$

In particular this trace is cyclic, also for matrix-valued symbols. It is a trace per lattice site, with the ordinary matrix trace retained on the fibre.

For later remainder estimates, use

$$\|a\|_s = \sum_{m\in\mathbb{Z}^4} (1+|m|)^s \|a_m\|, \qquad s = 0, 1, \dots.$$

For real φ, the phase in eq. 5.cxx has modulus one, so $\|a \star_\varphi b\|_0 \le \|a\|_0\|b\|_0$. The same argument and $(1+|m+n|)^s \le (1+|m|)^s(1+|n|)^s$ give the corresponding weighted product bound. The operator norm is at most $\|a\|_0$. These estimates are uniform in the real magnetic field. They do not assert holomorphy in a complex constant flux.

Taylor expansion of the phase in eq. 5.cxx gives

$$a \star_\varphi b = ab - \frac{i\varphi}{2}\{a,b\} - \frac{\varphi^2}{8}\{\{a,b\}\} + \mathcal{R}_3(a,b;\varphi), \tag{5.cxxii}$$

where subscripts denote ordinary momentum derivatives and

$$\{a,b\} = a_1 b_2 - a_2 b_1, \qquad \{\{a,b\}\} = a_{11}b_{22} - 2a_{12}b_{12} + a_{22}b_{11}.$$

For example,

$$\|\mathcal{R}_3(a,b;\varphi)\|_0 \le C|\varphi|^3 \|a\|_3\|b\|_3.$$

To check the signs, the product of $e^{im\cdot q}$ and $e^{in\cdot q}$ acquires the positive phase in eq. 5.cxx, while each pair of momentum derivatives contributes $i^2 = -1$. The remainder bound follows from the third-order Taylor remainder of a real phase and $|m_1 n_2 - m_2 n_1| \le |m||n|$. Weighted versions follow by the same summation.

5.4.2. *The exact charged Wilson Hessian.* We first derive the finite Weyl polynomial that is needed in the calculation. This also fixes the spin normalization. Let $p = (x;\mu,\nu)$ be a positive plaquette and let its background holonomy be $P = e^{if_{\mu\nu}t_3}$, where $f_{12} = f$ and the other $f_{\mu\nu}$ vanish. Put $\lambda = e^{2if_{\mu\nu}}$. Moving the four varied factors to the base point gives

$$P(t) = e^{itX_1}e^{itX_2}e^{itX_3}e^{itX_4}P,$$

where

$$\begin{aligned} X_1 &= a_\mu(x),\\ X_2 &= \operatorname{Ad}_{U_\mu(x)} a_\nu(x+e_\mu),\\ X_3 &= -\operatorname{Ad}_{PU_\nu(x)} a_\mu(x+e_\nu),\\ X_4 &= -\operatorname{Ad}_P a_\nu(x). \end{aligned}$$

The inverses on the third and fourth edges are responsible for both minus signs and both occurrences of P. In charged coordinates the four rows acting on $(w_1,\dots,w_4)$ are

$$C_1 w = w_\mu, \qquad C_2 w = u_\mu w_\nu, \qquad C_3 w = -\lambda u_\nu w_\mu, \qquad C_4 w = -\lambda w_\nu. \tag{5.cxxiii}$$

These are rows of operators; their adjoints reverse the order of operator multiplication.

For real three-vectors x, y, the Pauli product is $(x\cdot t)(y\cdot t) = (x\cdot y)1 + i(x\times y)\cdot t$. Differentiating $1 - \frac{1}{2}\operatorname{tr} P(t)$ twice gives

$$\cos f_{\mu\nu}\left|\sum_r X_r\right|^2 - 2\sin f_{\mu\nu}\sum_{r<s}(X_r\times X_s)_3.$$

For charged vectors, $(x \times y)_3 = -\operatorname{Im}(\overline{w}_x w_y)$. Thus the charged plaquette Hessian is the Hermitian form

$$H_{\mu\nu} = \cos f_{\mu\nu}\, S^*_{\mu\nu} S_{\mu\nu} + \frac{\sin f_{\mu\nu}}{i} \sum_{r<s} (C_r^* C_s - C_s^* C_r), \qquad S_{\mu\nu} = \sum_{r=1}^{4} C_r. \tag{5.cxxiv}$$

For example, the second summand evaluated on w is $2 \sin f_{\mu\nu} \sum_{r<s} \operatorname{Im}\langle C_r w, C_s w\rangle$, as required. There is no continuum approximation in eq. 5.cxxiv.

Put $D_\mu = 1 - u_\mu$. The exact charged vector operator is

$$K_\varphi = DD^* + \sum_{\mu<\nu} H_{\mu\nu} = \mathrm{Op}_\varphi(\sigma_\varphi), \qquad \Delta_\varphi = \sum_\mu D_\mu^* D_\mu = \mathrm{Op}_\varphi(\psi), \tag{5.cxxv}$$

where

$$\psi(q) = \sum_{\mu=1}^{4} (2 - 2\cos q_\mu).$$

Each entry of σ_φ is a finite sum of terms $c e^{i\varphi p/2} e^{im\cdot q}$, with integer p. At zero flux the ordinary curl–divergence identity gives

$$\sigma_0(q) = \psi(q) I_4. \tag{5.cxxvi}$$

This identity can also be checked entry by entry: the diagonal curl terms supply all $D_\nu^* D_\nu$ with $\nu \neq \mu$, the divergence term supplies the missing direction, and their off-diagonal entries cancel when the shifts commute.

We record the short computation that determines the first flux coefficient. Only the $(1,2)$ plane can change the flat cancellation. To evaluate the Weyl symbol at $q = 0$, one must first put each product into the order eq. 5.cxx; setting the shifts to one beforehand would lose the magnetic phase. For example the full off-diagonal entry is

$$\begin{aligned}\sigma_\varphi^{12}(q) = 1 - e^{-iq_2} - e^{iq_1} - e^{i(q_1-q_2)} \\ + e^{-i\varphi/2}\big(e^{i(q_1-q_2)} + e^{iq_1} + e^{-iq_2}\big) - e^{i\varphi/2}.\end{aligned}$$

The phases in the second line arise from putting products of the two noncommuting shifts into Weyl order. Its combined coefficient at $q = 0$ is

half-phase p	-1	0	1
$\sum_m c^{12}_{m,p}$	3	-2	$-1.$

Here the contribution of DD^* is combined with the two terms in eq. 5.cxxiv. These three numbers can be obtained by substituting the four rows eq. 5.cxxiii, using $\lambda = e^{i\varphi}$, and applying $W_m W_n = e^{i\varphi(m_1 n_2 - m_2 n_1)/2} W_{m+n}$ to each pair. The diagonal sums vanish, and the opposite off-diagonal entry is the complex conjugate. Hence

$$\sigma_\varphi(0) = \big(3e^{-i\varphi/2} - e^{i\varphi/2} - 2\big) E_{12} + \big(3e^{i\varphi/2} - e^{-i\varphi/2} - 2\big) E_{21}. \tag{5.cxxvii}$$

The entries with indices outside $\{1,2\}$ are zero in this formula. Write

$$\sigma_\varphi = \psi I_4 + \varphi \Sigma_1 + \varphi^2 \Sigma_2 + O(\varphi^3). \tag{5.cxxviii}$$

The remainder and all its momentum derivatives are bounded by a constant times $|\varphi|^3$. In particular,

$$\Sigma_1(0) = -2i(E_{12} - E_{21}), \qquad \operatorname{tr} \Sigma_1(0)^2 = 8, \qquad \Sigma_2(0) = -\tfrac{1}{4}(E_{12} + E_{21}). \tag{5.cxxix}$$

The finite Fourier expressions also give

$$\|\Sigma_1(q) - \Sigma_1(0)\| \le C|q|, \qquad \|\Sigma_2(q)\| \le C, \qquad q \in \mathbb{B}. \tag{5.cxxx}$$

Representatives in $[-\pi, \pi]^4$ are used in the first bound.

There is a useful independent identity for this normalization. The field eq. 5.cxv is critical under compactly supported variations: on each bond the two adjacent $(1,2)$-plaquettes contribute opposite copies of $\sin f\, t_3$, and the remaining plaquettes contribute zero. Gauge invariance, differentiated at this critical field, gives $A''(U_f)D_{U_f} = 0$. Consequently

$$K_\varphi D = D\Delta_\varphi. \tag{5.cxxxi}$$

It also follows by multiplication of the finite Weyl polynomials above. The spin value in eq. 5.cxxix was computed from the Hessian itself; it was not inferred from the Ward identity alone.

5.4.3. *The neutral sector and positivity at fixed mass.* In the third colour all adjoint phases disappear. The four insertions commute, so the only changed plaquette Hessian is the factor $\cos f$ in the $(1,2)$ curl. With ordinary commuting shifts let

$$c_{12}(q)a = (1 - e^{iq_2})a_1 + (e^{iq_1} - 1)a_2.$$

The neutral site and vector symbols are exactly

$$\Delta^0 = \psi, \qquad K_\varphi^0 = \psi I_4 + (\cos(\varphi/2) - 1)c_{12}^* c_{12} = \psi I_4 + \varphi^2 N_2 + O(\varphi^4), \quad N_2 = -\tfrac{1}{8} c_{12}^* c_{12}. \tag{5.cxxxii}$$

In particular

$$\mathrm{tr}\, N_2(q) = -\tfrac{1}{8}(|1 - e^{iq_1}|^2 + |1 - e^{iq_2}|^2). \tag{5.cxxxiii}$$

This term is bounded and vanishes quadratically at zero momentum. It is retained in the exact coefficient formula below.

We next justify that the logarithms used in the calculation exist. By eq. 5.cxxvi and the finite Fourier description, the coefficient norm of $\sigma_\varphi - \psi I_4$ is at most $C|\varphi|$. This is a comparison with the magnetic scalar Laplacian, not with the nonmagnetic operator in a globally small potential. Since $\Delta_\varphi \geq 0$,

$$K_\varphi \geq -C|\varphi|, \qquad K_\varphi + z \geq z/2 \quad \text{if } |\varphi| \leq cz. \tag{5.cxxxiv}$$

The same lower bound holds for every compression. In the neutral sector, for $\cos f \geq 0$, the sum of the weighted curl squares and the divergence square is nonnegative. Thus $K_\varphi^0 + z \geq z$. These observations prove positivity of all finite determinants in eq. 5.cxvi in a common real interval about zero, for each fixed $z > 0$.

5.4.4. *Volume limits and differentiation in the field.* Here is a trace argument that deals with the linearly growing potential. It will also justify differentiation of the determinant density without a complex constant-field continuation.

Lemma 5.10. Let $T_\varphi = \mathrm{Op}_\varphi(t_\varphi)$ be a Hermitian finite Weyl polynomial on d components. Suppose $t_0 = \psi I_d$, and its finite Fourier coefficients are smooth functions of real φ. For fixed $z > 0$, the normalized logarithms of the zero-extension compressions of $T_\varphi + z$ converge near $\varphi = 0$, together with any fixed finite number of real flux derivatives. Their limits are the cyclic trace of the logarithm defined by its convergent power series. The convergence is locally uniform in positive z. ∎

Proof. First restrict z to a compact positive interval. Choose $a > 8 + z$ throughout that interval and put

$$B_\varphi = I - a^{-1}(T_\varphi + z).$$

At zero flux the sum of the norms of its combined Fourier coefficients is

$$1 - (8 + z)/a + 8/a = 1 - z/a < 1.$$

Coefficients with the same lattice shift are combined before this norm is taken. Their smoothness and their finite number give, on a sufficiently small real interval, $\|b_\varphi\|_0 \leq r < 1$, uniformly in the restricted masses. Consequently

$$\log(T_\varphi + z) = (\log a)I - \sum_{n \geq 1} \frac{B_\varphi^n}{n} \tag{5.cxxxv}$$

converges in operator norm and in the absolutely summable coefficient algebra. The same series holds for each compressed matrix, with the same norm bound.

Expand the trace of a power as a sum over walks. A term has steps $m_1, \dots, m_n$ in a fixed finite set. It contributes to the trace only if $\sum_j m_j = 0$. For such a closed word, the product formula eq. 5.cxx gives a scalar phase depending on the word and φ, but not on its starting site. Its phase exponent has absolute value at most Cn^2. A finite compression keeps precisely the words whose intermediate coordinates remain in the compression. A fixed word is therefore lost only when its starting site lies within distance Cn of the boundary. The proportion of such sites tends to zero in a Følner exhaustion. The normalized trace of each fixed power tends to its zero-coefficient trace. For vector compressions with both endpoints in the box, the omitted boundary bonds also have zero density. Their omission changes only the same boundary set for each fixed word. The constant term in the normalized logarithm tends to $d \log a$, because the number of retained vector coordinates divided by the number of sites tends to d.

For completeness, the flux derivative estimates must use the combined coefficients of b_φ; expanding separately terms which cancel at zero flux would lose the bound $r < 1$. For a derivative of order j, differentiating coefficient factors gives at most j exceptional factors, with uniformly bounded coefficient norms. Differentiating the closed-word phase gives a factor at most $C_j n^{2j}$. Summing the other factors in absolute value gives, after increasing the constant,

$$\left| \frac{1}{|\Lambda_R|} \partial_\varphi^j \operatorname{Tr}(P_R B_\varphi P_R^*)^n \right| \le C_j (1+n)^{3j} r^{\max\{n-j,0\}}.$$

The same bound holds for the limiting zero-coefficient trace. The sum of these bounds divided by n converges. It justifies termwise differentiation, passage to the volume limit, and their interchange. The argument is uniform on compact positive mass intervals. It proves the assertion for each finite number of derivatives. □

The proof applies to the charged vector polynomial by eq. 5.cxxvi, and to the charged scalar polynomial directly. The neutral polynomial is an ordinary finite Fourier polynomial with the same flat symbol, so it satisfies the same conclusion. A constant colour rotation by it_1 takes U_f to U_{-f}, preserves the compression coordinates, and conjugates both real operators orthogonally. Hence every finite density $\mathcal{D}_R(f, z)$, and its limit, is even.

The cyclic trace in eq. 5.cxxxv will be denoted by $\tau \log(T_\varphi + z)$. This notation means the convergent series with its scalar $d \log a$ term. It is independent of a, since it is the limit of the finite logarithms. It also satisfies

$$\partial_z \tau \log(T_\varphi + z) = \tau (T_\varphi + z)^{-1}. \tag{5.cxxxvi}$$

One may differentiate the locally uniformly convergent series to prove this identity; cyclicity combines the n positions of a differentiated factor. The same argument proves the Jacobi identity for any bounded differentiable perturbation which preserves a positive gap.

5.4.5. *The scalar orbital coefficient.* Let

$$u(q, z) = (\psi(q) + z)^{-1}, \qquad a(q) = \psi_{11}(q) = 2 \cos q_1, \qquad b(q) = \psi_{22}(q) = 2 \cos q_2.$$

For each fixed $z > 0$, u and all its momentum derivatives have absolutely summable Fourier coefficients. For instance, $\psi + z$ stays away from zero on a sufficiently small complex momentum strip, so the Fourier coefficients decrease exponentially by shifting each integration contour. The strip may depend on z. Uniformity of that strip as $z \downarrow 0$ is neither asserted nor needed here.

Lemma 5.11. If $s_2(z)$ is the coefficient of φ^2 in $\tau \log(\Delta_\varphi + z)$, then exactly

$$s_2(z) = \frac{1}{24} \int_{\mathbb{B}} a(q) b(q) u(q, z)^2 \, d\bar{q}. \tag{5.cxxxvii}$$

■

Proof. The Poisson bracket $\{\psi, u\}$ is zero, because u is a function of ψ. The product expansion eq. 5.cxxii therefore gives a second-order parametrix for the scalar resolvent with symbols

$$r_0 = u, \qquad r_1 = 0, \qquad r_2 = \frac{1}{8} u\{\{\psi, u\}\}.$$

More explicitly,

$$(\psi + z) \star_\varphi (u + \varphi^2 r_2) = 1 + O_z(|\varphi|^3) \tag{5.cxxxviii}$$

in the absolutely summable coefficient norm. All required weighted norms are finite at fixed z, so the remainder estimate follows from eq. 5.cxxii.

This parametrix determines the actual inverse coefficient. Indeed, $\Delta_\varphi + z$ has a convergent inverse series in the same coefficient algebra, obtained from eq. 5.cxxxv by the geometric series. Its coefficient norm is locally bounded at fixed z. Multiplying eq. 5.cxxxviii by that inverse shows that the symbol of the actual resolvent differs from $u + \varphi^2 r_2$ by $O_z(|\varphi|^3)$. Thus no formal inverse is being substituted for the lattice resolvent.

Since $\psi_{12} = 0$, differentiation of u gives

$$r_2 = \frac{1}{4}\left\{u^4(a\psi_2^2 + b\psi_1^2) - u^3 ab\right\}.$$

Integration by parts on the momentum torus yields

$$0 = \int_{\mathbb{B}} \partial_2(a\psi_2 u^3)\, d\bar{q} = \int_{\mathbb{B}} (abu^3 - 3a\psi_2^2 u^4)\, d\bar{q},$$

and the same formula with indices exchanged. Consequently

$$\tau \operatorname{Op}_\varphi(r_2) = \int_{\mathbb{B}} r_2\, d\bar{q} = -\frac{1}{12}\int_{\mathbb{B}} abu^3\, d\bar{q}. \tag{5.cxxxix}$$

By eq. 5.cxxxvi, the right side is $s_2'(z)$. The derivative in z of the right side of eq. 5.cxxxvii is the same expression.

Both candidate coefficients tend to zero as $z \to \infty$. For the exact logarithm this follows from the series $\log(\Delta_\varphi + z) = \log z + \log(1 + z^{-1}\Delta_\varphi)$, whose flux derivatives converge uniformly for large z. For the integral it follows from boundedness of a, b and $u \le z^{-1}$. Equality of their derivatives and of their limits at infinity proves eq. 5.cxxxvii. □

The term s_2 is the orbital contribution of one charged scalar determinant. In particular it is already nonzero before the vector spin term is included. Its sign and factor come from the second magnetic bracket, not from a pointwise Taylor expansion of the symbol ψ.

5.4.6. *The vector logarithm and colour multiplicities.* We give the trace calculation with its non-commuting factors. It prevents a scalar logarithm expansion from inadvertently discarding the magnetic contribution. Put

$$B_\varphi = (\Delta_\varphi + z)I_4, \qquad \delta_\varphi = K_\varphi - \Delta_\varphi I_4 = \operatorname{Op}_\varphi(\varphi\Sigma_1 + \varphi^2\Sigma_2 + O(\varphi^3)).$$

At fixed positive z, the segment $B_\varphi + t\delta_\varphi$, $0 \le t \le 1$, has a positive gap for small real φ. The trace Jacobi identity gives

$$\tau \log(B_\varphi + \delta_\varphi) - \tau \log B_\varphi = \int_0^1 \tau\big((B_\varphi + t\delta_\varphi)^{-1}\delta_\varphi\big)\, dt.$$

The exact twice-iterated resolvent identity and cyclicity then give

$$\tau \log(B_\varphi + \delta_\varphi) - \tau \log B_\varphi = \tau(B_\varphi^{-1}\delta_\varphi) - \frac{1}{2}\tau(B_\varphi^{-1}\delta_\varphi B_\varphi^{-1}\delta_\varphi) + O_z(|\varphi|^3). \tag{5.cxl}$$

For example, the remaining resolvent term contains three copies of δ_φ, bounded inverse factors, and a trace on four components per site, so its absolute value is at most $C_z\|\delta_\varphi\|_0^3$.

The scalar inverse has first flux coefficient zero by Lem. 5.11. The exact two-factor identity eq. 5.cxxi therefore shows that the quadratic coefficient of the first term on the right of eq. 5.cxl

is $\int u \operatorname{tr} \Sigma_2 \, d\bar{q}$. The quadratic coefficient of the second term is obtained at zero flux, and equals $-\frac{1}{2} \int u^2 \operatorname{tr} \Sigma_1^2 \, d\bar{q}$. No first magnetic bracket contributes an additional cross term: the scalar inverse has no linear symbol correction, and the trace of a two-factor star product is the ordinary integral exactly. If $v_2(z)$ denotes the quadratic flux coefficient of the charged vector logarithm, we have proved

$$v_2(z) = 4s_2(z) + \int_{\mathbb{B}} u \operatorname{tr} \Sigma_2 \, d\bar{q} - \frac{1}{2} \int_{\mathbb{B}} u^2 \operatorname{tr} \Sigma_1^2 \, d\bar{q}. \tag{5.cxli}$$

For the neutral vector logarithm the corresponding coefficient is simply

$$v_2^0(z) = \int_{\mathbb{B}} u \operatorname{tr} N_2 \, d\bar{q}. \tag{5.cxlii}$$

There are two real charged colours. Equivalently, complexifying the real two-colour space gives sectors with fluxes φ and $-\varphi$. The real determinant of a positive complex-linear charged matrix is the square of its complex determinant. Thus the scalar charged contribution to the real logarithm is twice the one-sector scalar logarithm, and likewise for the charged vector. The neutral scalar logarithm is independent of f. The coefficient of φ^2 in the combination $\log\det\Delta - \frac{1}{2}\log\det K$ is therefore

$$2s_2 - v_2 - \tfrac{1}{2}v_2^0 = -2s_2 - \int u \operatorname{tr} \Sigma_2 \, d\bar{q} + \tfrac{1}{2} \int u^2 \operatorname{tr} \Sigma_1^2 \, d\bar{q} - \tfrac{1}{2} \int u \operatorname{tr} N_2 \, d\bar{q}.$$

Finally $\varphi = 2f$, and a quadratic φ coefficient is multiplied by eight when it is differentiated twice in f. Combining this fact with eq. 5.cxxxvii proves the exact formula

$$\boxed{\widetilde{\beta}(z) = 8 \int_{\mathbb{B}} \left\{ \frac{1}{2} u^2 \operatorname{tr} \Sigma_1^2 - \frac{1}{12} abu^2 - u \operatorname{tr} \Sigma_2 - \frac{1}{2} u \operatorname{tr} N_2 \right\} d\bar{q}.} \tag{5.cxliii}$$

All four functions of momentum in this formula are independent of z. The last term is the neutral-colour correction.

5.4.7. *Infrared integrals and mass derivatives.* We now estimate eq. 5.cxliii uniformly as the mass decreases. On the chosen representatives of the momentum torus,

$$c|q|^2 \le \psi(q) \le |q|^2, \qquad \psi(q) = |q|^2 + O(|q|^4) \quad (q \to 0). \tag{5.cxliv}$$

The lower bound follows, for example, by applying $|\sin(t/2)| \ge |t|/\pi$ on $[-\pi, \pi]$ to each term $2 - 2\cos t$.

Lemma 5.12. Put $I_2(z) = \int_{\mathbb{B}} (\psi + z)^{-2} \, d\bar{q}$. There is a finite c_2 such that

$$I_2(z) = \frac{1}{16\pi^2} \log \frac{1}{z} + c_2 + e_2(z), \qquad |e_2(z)| + |z e_2'(z)| + |z^2 e_2''(z)| \le C\sqrt{z}. \tag{5.cxlv}$$

If h is smooth on $\mathbb{B}$ and $h(0) = 0$, then $R_h(z) = \int h(\psi + z)^{-2} \, d\bar{q}$ has a finite limit at zero and

$$|R_h(z) - R_h(0)| + |z R_h'(z)| + |z^2 R_h''(z)| \le C_h \sqrt{z}. \tag{5.cxlvi}$$

For bounded smooth h, the same conclusion holds for $\int h(\psi + z)^{-1} \, d\bar{q}$. ∎

Proof. Choose a fixed ball $|q| < \rho < \pi$. Outside this ball the integrands and their first two mass derivatives are uniformly bounded. Inside the ball, subtract $(|q|^2 + z)^{-2}$ from $(\psi + z)^{-2}$. The difference is bounded by

$$C \frac{|q|^4}{(|q|^2 + z)^3}.$$

Its first and second mass derivatives are bounded respectively by $C|q|^4(|q|^2 + z)^{-4}$ and $C|q|^4(|q|^2 + z)^{-5}$. Radial integration in four dimensions gives an integrable limit for the difference, a bound $C(1 + |\log z|)$ for the integral of its first derivative, and C/z for that of its second derivative. For instance the first derivative is bounded by a constant times $\int_0^\rho r^7 (r^2 + z)^{-4} \, dr$; splitting at

$r = \sqrt{z}$ gives the asserted logarithm. The second is handled by the same split. Integrating the first derivative from zero to z bounds the difference from its limiting constant by $Cz(1 + |\log z|)$.

The remaining continuum radial integral is exact:

$$\int_{|q|<\rho} (|q|^2 + z)^{-2}\, d\bar{q} = \frac{1}{16\pi^2} \left\{ \log \frac{\rho^2 + z}{z} + \frac{z}{\rho^2 + z} - 1 \right\}.$$

Its nonsingular part has a finite limit and bounded first two derivatives on a sufficiently small positive interval. This proves eq. 5.cxlv, with the slightly weaker but convenient error $C\sqrt{z}$, since $z(1 + |\log z|) \le C\sqrt{z}$ for $0 < z \le 1$.

For the second assertion, smoothness and $h(0) = 0$ give $|h(q)| \le C_h|q|$ near zero. This makes the zero-mass integral absolutely convergent. The first derivative has absolute integral at most

$$C_h \int_0^\rho \frac{r^4}{(r^2 + z)^3}\, dr \le C_h z^{-1/2},$$

and the second at most $C_h \int_0^\rho r^4(r^2 + z)^{-4}\, dr \le C_h z^{-3/2}$. The estimates follow by scaling $r = \sqrt{z}\, s$, extending the resulting integrals to infinity, and using their integrable large-s tails. Their weighted bounds are therefore $C_h\sqrt{z}$. Integration of the first derivative from zero to z proves the remaining difference bound.

For the last assertion the zero-mass integrand is bounded by $C|q|^{-2}$, which is integrable in four dimensions. The first derivative has integral at most $C(1 + |\log z|)$, and the second at most C/z. The same argument again gives the stated, weaker $C\sqrt{z}$ error and weighted derivative bounds. □

Proof of Prop. 5.9. The volume and field-differentiation assertions follow from Lem. 5.10. Positivity and evenness were established above. Formula eq. 5.cxliii identifies the second field derivative exactly.

By eq. 5.cxxix, $\frac{1}{2} \operatorname{tr} \Sigma_1(0)^2 = 4$, whereas $a(0)b(0)/12 = 1/3$. Thus the coefficient of the only logarithmically divergent integral in eq. 5.cxliii is

$$8(4 - \tfrac{1}{3}) I_2(z) = \frac{88}{3} I_2(z).$$

The remaining numerator in front of u^2 vanishes at $q = 0$, by eq. 5.cxxx and continuity of a, b. Its contribution is covered by eq. 5.cxlvi. The terms with one factor of u have bounded smooth numerators and are covered by the last assertion of Lem. 5.12. This includes the neutral term; discarding it would change the finite constant, though not the logarithm.

All these remainders have finite zero-mass limits and the weighted derivative bounds in that lemma. Since

$$\frac{88}{3} \frac{1}{16\pi^2} = \frac{11}{6\pi^2},$$

equation eq. 5.cxlv proves eq. 5.cxvii. Differentiating its logarithmic term and using the weighted error estimates proves eq. 5.cxviii. □

5.4.8. *The second-order mass subtraction.* The mass comparison uses a Taylor polynomial evaluated at zero mass, not merely the determinant at the comparison mass. The preceding proof controls the two derivative terms as well. Define

$$\widetilde{\beta}^\sharp(z) = \widetilde{\beta}(z) - z\widetilde{\beta}'(z) + \tfrac{1}{2} z^2 \widetilde{\beta}''(z).$$

Corollary 5.13. For $c = 11/(6\pi^2)$,

$$\widetilde{\beta}^\sharp(z) = c \log(1/z) + C_* + \tfrac{3}{2} c + O(\sqrt{z}). \tag{5.cxlvii}$$

In particular, for fixed $L > 1$,

$$\widetilde{\beta}^\sharp(L^{-2k}) = \frac{11}{3\pi^2} k \log L + O(1), \qquad k \ge 0. \tag{5.cxlviii}$$

The successive differences of these bare massive coefficients converge to $11\log L/(3\pi^2)$ with error $O(L^{-k})$. ■□

Proof. For $c\log(1/z)$ the first derivative is $-c/z$ and the second is c/z^2. Its two subtraction terms thus add $c + c/2$. The constant contributes only itself, and eq. 5.cxvii bounds the error and both subtracted derivatives by $C\sqrt{z}$. This proves eq. 5.cxlvii. Substitution of $z = L^{-2k}$ proves eq. 5.cxlviii; subtraction at consecutive values of k gives the last assertion. □

The successive differences in this corollary are defined by the bare massive determinant. To identify them, up to a uniformly bounded total error, with a sum of constrained Gaussian increments requires the gauge-change identity, the Schur-complement composition, and the uniform comparison between the averaging regulator and the scalar mass regulator. The present calculation supplies the lattice logarithm and the mass-derivative estimates in that comparison. It does not replace those three identifications. Nor does it imply geometric convergence of the actual renormalization increments from a bounded cumulative comparison alone.

5.5. **The diagonal anisotropy of the massive Wilson coefficient.** The scalar logarithm does not determine the response to a traceless change of the four coordinate weights. We compute that response from the weighted lattice Hessian. The computation retains the lattice symbol until its infrared estimate. In particular it differentiates an actual family of Gaussian operators, rather than differentiating a previously extracted scalar asymptotic without a parameter bound.

Let $a = (a_1, \dots, a_4)$, initially real and close to $(1, 1, 1, 1)$, with all $a_\mu > 0$. Write $\mathsf{A} = \operatorname{diag}(a_\mu)$ on the vector index, and put

$$A_a(U) = \sum_{\mu<\nu} \sum_x a_\mu a_\nu \left(1 - \tfrac{1}{2}\operatorname{tr} U_{\mu\nu}(x)\right), \qquad \Delta_{a,U} = D_U^\sharp \mathsf{A} D_U. \tag{5.cxlix}$$

The colour inner product, covariant shifts, and physical normalization are those in § 5.4. The symbol A_a denotes an action; A is its constant diagonal weight. Set

$$\begin{aligned} K_{a,U} &= A_a''(U) + \mathsf{A} D_U D_U^\sharp \mathsf{A}, \\ \widetilde{K}_{a,U} &= \mathsf{A}^{-1/2} K_{a,U} \mathsf{A}^{-1/2}, \qquad \widetilde{D}_{a,U} = \mathsf{A}^{1/2} D_U. \end{aligned} \tag{5.cl}$$

Square roots are positive for real a; near $a = 1$ their analytic branches are fixed by that choice.

For a chosen coordinate plane $\{\mu, \nu\}$, use the constant Abelian field of eq. 5.cxv, with the axes renamed if necessary. Compress the full operators after they have been formed, exactly as in Prop. 5.9. Define the vacuum-subtracted determinant density from

$$\mathcal{L}_{a,R}(f, z) = \log\det(\Delta_{a,f,R} + z) - \tfrac{1}{2}\log\det(\widetilde{K}_{a,f,R} + z), \qquad z > 0, \tag{5.cli}$$

and let $\widetilde{\beta}_a^{\mu\nu}(z)$ be its second f-derivative per lattice site in the exhaustion limit. All three real colours are included.

Proposition 5.14. There is $\delta > 0$ such that $\widetilde{\beta}_a^{\mu\nu}(z)$ extends holomorphically to $\max_\lambda |a_\lambda - 1| < \delta$. On every fixed smaller polydisc,

$$\widetilde{\beta}_a^{\mu\nu}(z) = c\, d(a) a_\mu a_\nu \log(1/z) + C_{\mu\nu}(a) + E_{\mu\nu}(a, z), \qquad c = \frac{11}{6\pi^2}, \quad d(a) = \prod_{\lambda=1}^{4} a_\lambda^{-1/2}, \tag{5.clii}$$

where $C_{\mu\nu}$ is holomorphic. For every fixed multi-index I,

$$\sup_a \sum_{j=0}^{2} \left| z^j \partial_z^j \partial_a^I E_{\mu\nu}(a, z) \right| \le C_I \sqrt{z}, \qquad 0 < z \le 1. \tag{5.cliii}$$

For real a, this is the coefficient of the actual real weighted Wilson determinant. The field variable f is differentiated as a real variable at fixed $z > 0$. No holomorphic constant-flux family is asserted. ■

5.5.1. *The gauge and the field-coordinate change.* The additional term in eq. 5.cl is the square of the weighted divergence $D_U^\sharp \mathsf{A} v$. Its Faddeev–Popov operator is $D_U^\sharp \mathsf{A} D_U = \Delta_{a,U}$. For real a, change vector coordinates by $v = \mathsf{A}^{-1/2} w$. The Hessian and its gauge square become

$$\widetilde{K}_{a,U} = \mathsf{A}^{-1/2} A_a''(U) \mathsf{A}^{-1/2} + \widetilde{D}_{a,U} \widetilde{D}_{a,U}^\sharp. \tag{5.cliv}$$

The constant Jacobian of this field-coordinate change is independent of U. It disappears from the vacuum-subtracted density. Equivalently the vector mass in the original coordinates is $z\mathsf{A}$:

$$K_{a,U} + z\mathsf{A} = \mathsf{A}^{1/2} (\widetilde{K}_{a,U} + z) \mathsf{A}^{1/2}.$$

This equality also holds for the declared zero-extension compressions. It specifies the regulator used here; comparison with an averaging regulator must retain this choice.

Every constant Abelian field used here is critical for A_a under compactly supported variations. On a bond, the two adjacent plaquettes in each plane contribute opposite copies of the same weighted $\sin f_{\mu\nu}\, t_3$. Gauge invariance therefore gives

$$A_a''(U_f) D_{U_f} = 0, \qquad \widetilde{K}_{a,U_f} \widetilde{D}_{a,U_f} = \widetilde{D}_{a,U_f} \Delta_{a,U_f}. \tag{5.clv}$$

The second identity is also obtained directly from eq. 5.cliv. The spin coefficient below is computed from the Hessian; the Ward identity is an independent check on the chosen gauge.

5.5.2. *An exact weighted symbol.* It suffices to use the plane $(1,2)$. Keep the magnetic shifts and the Weyl product of eq. 5.cxx, with $\varphi = 2f$. Let $H_{\rho\sigma}$ be the exact charged plaquette Hessian in eq. 5.cxxiv. The actual charged operators are

$$\begin{aligned} \widetilde{K}_{a,\varphi} &= \mathsf{A}^{-1/2} \left(\sum_{\rho<\sigma} a_\rho a_\sigma H_{\rho\sigma} \right) \mathsf{A}^{-1/2} + \mathsf{A}^{1/2} D D^\sharp \mathsf{A}^{1/2}, \\ \Delta_{a,\varphi} &= \mathrm{Op}_\varphi(\psi_a), \qquad \psi_a(q) = \sum_{\rho=1}^{4} a_\rho (2 - 2\cos q_\rho). \end{aligned} \tag{5.clvi}$$

Both sides are finite Weyl polynomials. At zero flux, their curl and divergence terms cancel off the diagonal and supply the same missing diagonal direction. Thus

$$\widetilde{\sigma}_{a,0}(q) = \psi_a(q) I_4. \tag{5.clvii}$$

For example the μ-component of the weighted curl square contains $\sum_{\rho\neq\mu} a_\rho D_\rho^\sharp D_\rho$; the divergence square supplies $a_\mu D_\mu D_\mu^\sharp$. This verifies the diagonal identity as well as the off-diagonal cancellation.

Write the actual flux expansion as

$$\widetilde{\sigma}_{a,\varphi} = \psi_a I_4 + \varphi \Sigma_{a,1} + \varphi^2 \Sigma_{a,2} + O(\varphi^3). \tag{5.clviii}$$

Its coefficients are holomorphic in a, with uniformly bounded momentum derivatives on a small complex polydisc. Only the $(1,2)$ plaquette can change the flat cancellation. Relative to the unweighted calculation, its diagonal corrections in the 11 and 22 entries have multipliers a_2 and a_1; its off-diagonal corrections have multiplier $\sqrt{a_1 a_2}$. All other corrections vanish. This follows entry by entry from eq. 5.clvi: congruence divides the 11 entry of $a_1 a_2 H_{12}$ by a_1, and its 12 entry by $\sqrt{a_1 a_2}$. The weighted divergence has exactly the same factors in its corresponding entries. Consequently

$$\Sigma_{a,1}(0) = -2i\sqrt{a_1 a_2}(E_{12} - E_{21}), \qquad \mathrm{tr}\, \Sigma_{a,1}(0)^2 = 8 a_1 a_2, \tag{5.clix}$$

and

$$\|\Sigma_{a,1}(q) - \Sigma_{a,1}(0)\| \le C|q|, \qquad \|\Sigma_{a,2}(q)\| \le C. \tag{5.clx}$$

The same bounds hold after any fixed number of a-derivatives. The neutral colour has expansion

$$\widetilde{K}^0_{a,\varphi} = \psi_a I_4 + \varphi^2 N_{a,2} + O(\varphi^4),$$

where $N_{a,2}$ is a bounded finite Fourier polynomial, holomorphic in a. Its 11 and 22 corrections have the same respective weights a_2, a_1. This neutral term is retained in the finite part of the coefficient.

5.5.3. *The weighted orbital and spin terms.* For $z > 0$, put

$$u_a(q,z) = (\psi_a(q) + z)^{-1}, \qquad A_1(q) = 2a_1 \cos q_1, \quad A_2(q) = 2a_2 \cos q_2.$$

The scalar Weyl calculation in Lem. 5.11 applies with ψ replaced by ψ_a. Its mixed second derivative is still zero. The integration by parts has, for example, the exact form

$$0 = \int_{\mathbb{B}} \partial_2(A_1\, \partial_2\psi_a\, u_a^3)\, d\bar{q} = \int_{\mathbb{B}} (A_1 A_2 u_a^3 - 3A_1(\partial_2\psi_a)^2 u_a^4)\, d\bar{q}.$$

There is no change of magnetic phase, and no approximation of the lattice dispersion in this identity. The scalar quadratic flux coefficient is therefore

$$s_{a,2}(z) = \frac{1}{24} \int_{\mathbb{B}} A_1 A_2 u_a^2\, d\bar{q}. \tag{5.clxi}$$

The trace logarithm is handled by the same twice-iterated resolvent identity as eq. 5.cxl. At fixed real positive a, z, the segment between the scalar diagonal operator and the full vector operator is positive for sufficiently small real flux. Its quadratic term is

$$4s_{a,2} + \int u_a \operatorname{tr} \Sigma_{a,2}\, d\bar{q} - \tfrac{1}{2} \int u_a^2 \operatorname{tr} \Sigma_{a,1}^2\, d\bar{q}.$$

The scalar inverse has no first flux coefficient. The exact trace of a two-factor Weyl product is its ordinary momentum integral, so no first magnetic bracket has been omitted here. The two real charged colours and $\varphi = 2f$ give the same multiplicity eight as in eq. 5.cxliii. We obtain the exact formula

$$\begin{aligned} \widetilde{\beta}_a^{12}(z) = 8 \int_{\mathbb{B}} \{ &\tfrac{1}{2} u_a^2 \operatorname{tr} \Sigma_{a,1}^2 - \tfrac{1}{12} A_1 A_2 u_a^2 \\ &- u_a \operatorname{tr} \Sigma_{a,2} - \tfrac{1}{2} u_a \operatorname{tr} N_{a,2} \}\, d\bar{q}. \end{aligned} \tag{5.clxii}$$

Its existence as a real exhaustion limit follows from the closed-walk proof in Lem. 5.10. All step ranges remain fixed. The coefficients and their parameter derivatives are uniformly bounded on compact sets of positive a, z. Every fixed number of flux and parameter derivatives therefore has the same dominated closed-walk limit. For example a derivative of a finite power inserts at most a polynomial in its length, while the fixed-mass resolvent series has geometric decay. Boundary walks occupy a vanishing fraction of any declared Følner exhaustion. This justifies taking the real field derivative before the exhaustion limit.

For complex a close to one, $\operatorname{Re} \psi_a(q) \geq c|q|^2$ on the standard momentum representatives. The right side of eq. 5.clxii is consequently holomorphic in a at each $z > 0$. It defines the claimed extension from its real values. This extension concerns the weight parameters; it does not complexify the constant flux used in the closed-walk argument.

5.5.4. *A uniform anisotropic infrared integral.* We prove the parameter estimate needed to differentiate the logarithm. For a in a small complex polydisc put

$$I_{2,a}(z) = \int_{\mathbb{B}} (\psi_a + z)^{-2}\, d\bar{q}, \qquad H_a(t) = \int_{\mathbb{B}} e^{-t\psi_a(q)}\, d\bar{q}.$$

For $t \geq 1$, uniformly on that polydisc,

$$H_a(t) = \frac{d(a)}{16\pi^2 t^2} + O(t^{-3}). \tag{5.clxiii}$$

To verify the remainder, take a fixed small momentum ball. On it,

$$\psi_a(q) = Q_a(q) + O(|q|^4), \quad Q_a(q) = \sum_{\lambda} a_\lambda q_\lambda^2, \quad \operatorname{Re} Q_a(q) \geq c|q|^2.$$

The difference of the two exponentials is bounded by $Ct|q|^4 e^{-ct|q|^2}$. Its four-dimensional integral is $O(t^{-3})$. Outside the ball the lattice integral is exponentially small, and extending the Gaussian integral to $\mathbb{R}^4$ costs the same type of error. The latter integral is

$$(2\pi)^{-4}\int_{\mathbb{R}^4} e^{-tQ_a(q)}\,dq = \frac{d(a)}{16\pi^2 t^2}.$$

For complex a, this identity follows from the one-dimensional Gaussian integrals on their branches near the positive real axis. Every estimate is uniform on a slightly larger polydisc than the one on which derivatives will be taken.

Since $\mathrm{Re}(\psi_a + z) > 0$, Fubini gives

$$I_{2,a}(z) = \int_0^\infty te^{-tz}H_a(t)\,dt. \tag{5.clxiv}$$

Let $c_a = d(a)/(16\pi^2)$. Subtract $c_a t^{-2}$ only for $t \geq 1$. The remaining integrand $tH_a(t)$ is bounded near zero and is $O(t^{-2})$ at infinity. Thus

$$I_{2,a}(z) = c_a\log(1/z) + C_I(a) + R_I(a,z), \qquad \sum_{j=0}^{2}|z^j\partial_z^j R_I(a,z)| \leq C\sqrt{z}. \tag{5.clxv}$$

Here is an explicit control of the limit and its derivatives. For the subtracted remainder $r_a(t)$, $|r_a(t)| \leq C\min(t, t^{-2})$. Replacing e^{-zt} by one costs

$$\int_0^\infty |e^{-zt}-1|\,|r_a(t)|\,dt \leq Cz\{1+\log(1/z)\}.$$

After one or two z-derivatives, the corresponding estimates are

$$z\int_0^\infty te^{-zt}|r_a(t)|\,dt \leq Cz\{1+\log(1/z)\}, \qquad z^2\int_0^\infty t^2e^{-zt}|r_a(t)|\,dt \leq Cz.$$

The subtracted leading term uses $\int_1^\infty e^{-zt}dt/t$. Its difference from $\log(1/z)$ has a finite limit, an error $O(z)$, and weighted first and second derivative errors $O(z)$. These estimates prove eq. 5.clxv. The constant $C_I(a)$ is a normally convergent integral plus a holomorphic multiple of a fixed number, and is holomorphic. Cauchy's estimate on the larger polydisc proves all fixed a-derivative versions of eq. 5.clxv.

Two elementary remainder estimates complete the calculation. If $h_a(q)$ is holomorphic in a and $|h_a(q)| \leq C|q|$, then

$$\int_{\mathbb{B}} h_a(q)u_a(q,z)^2\,d\bar{q} = C_h(a) + R_h(a,z), \qquad \sum_{j=0}^{2}|z^j\partial_z^j R_h(a,z)| \leq C\sqrt{z}. \tag{5.clxvi}$$

The limit at zero is integrable. Split its difference from the positive-mass integral at $|q| = \sqrt{z}$. The inner radial bound is $C\int_0^{\sqrt{z}}dr = C\sqrt{z}$; the outer bound is $Cz\int_{\sqrt{z}}^1 r^{-2}dr \leq C\sqrt{z}$. For the weighted first derivative, use $Cz\int_0^1 r^4(r^2+z)^{-3}dr$; for the second use $Cz^2\int_0^1 r^4(r^2+z)^{-4}dr$. Both are $O(\sqrt{z})$. The region outside the unit ball has a uniform positive dispersion and obeys the same bound. Normal convergence makes C_h holomorphic. Cauchy again gives its parameter derivative versions.

If the numerator is merely bounded and there is one factor u_a, the zero-mass limit is also finite. Its error is bounded by $Cz\{1+\log(1/z)\}$, its weighted first derivative by the same quantity, and its weighted second derivative by Cz. These follow from the radial integrals with numerator r^3. They too imply the bound $C\sqrt{z}$. This includes the neutral term and the second vector-symbol jet.

Proof of Prop. 5.14. At $q = 0$, the combined numerator of u_a^2 in eq. 5.clxii is

$$8\left(4a_1a_2 - \frac{1}{3}a_1a_2\right) = \frac{88}{3}a_1a_2.$$

After this constant is subtracted, the numerator vanishes at least linearly in $|q|$, by eq. 5.clx. Apply eq. 5.clxv to its constant part, eq. 5.clxvi to its vanishing part, and the single-resolvent estimate to the remaining two terms. The only logarithm is therefore

$$\frac{88}{3}a_1a_2\frac{d(a)}{16\pi^2}\log(1/z) = c\,d(a)a_1a_2\log(1/z).$$

All finite constants are holomorphic in a, and all remainders have the stated mass and parameter derivative bounds. Renaming the two axes proves every plane case. □

5.5.5. *The actual tracefree diagonal response.* Let $e = (e_1, \dots, e_4)$, and put $a_\lambda(t) = 1 + te_\lambda$. The weight of the (μ, ν) plaquette is

$$a_\mu(t)a_\nu(t) = 1 + t(e_\mu + e_\nu) + O(t^2).$$

Thus its derivative at zero is exactly the prescribed diagonal anisotropy; the harmless quadratic choice of the weight family does not change that derivative. Define

$$\dot\beta_e^{\mu\nu}(z) = \frac{d}{dt}\Big|_{t=0}\widetilde\beta_{a(t)}^{\mu\nu}(z).$$

The derivative of the logarithmic weight is

$$\frac{d}{dt}\Big|_0 d(a(t))a_\mu(t)a_\nu(t) = e_\mu + e_\nu - \frac{1}{2}\sum_{\lambda=1}^{4} e_\lambda. \tag{5.clxvii}$$

Consequently, if $\sum_\lambda e_\lambda = 0$,

$$\dot\beta_e^{\mu\nu}(z) = c(e_\mu + e_\nu)\log(1/z) + \dot C_e^{\mu\nu} + O(|e|\sqrt{z}), \tag{5.clxviii}$$

with the same two weighted mass derivative estimates for the error. For an isotropic direction $e_\lambda = s$, the logarithmic derivative vanishes. This verifies directly that the tracefree diagonal channel and the scalar weight channel have different one-loop responses.

Apply the second-order mass subtraction to the weighted family:

$$\widetilde\beta_a^{\mu\nu,\sharp}(z) = \left(1 - z\partial_z + \tfrac{1}{2}z^2\partial_z^2\right)\widetilde\beta_a^{\mu\nu}(z).$$

Parameter differentiation commutes with this finite differential operator. By eq. 5.cliii,

$$\widetilde\beta_a^{\mu\nu,\sharp}(z) = c\,d(a)a_\mu a_\nu\log(1/z) + C_{\mu\nu}(a) + \tfrac{3}{2}c\,d(a)a_\mu a_\nu + O_a(\sqrt{z}), \tag{5.clxix}$$

uniformly with any fixed number of parameter derivatives.

Corollary 5.15. Let $h_k = L^{-k}$, with the fixed blocking integer L. In the tracefree diagonal direction,

$$\frac{d}{dt}\Big|_0 \widetilde\beta_{1+te}^{\mu\nu,\sharp}(h_k^2) = \frac{11}{3\pi^2}(e_\mu + e_\nu)k\log L + C_e^{\mu\nu} + O(|e|L^{-k}). \tag{5.clxx}$$

Its consecutive differences converge geometrically to $11(e_\mu + e_\nu)\log L/(3\pi^2)$. The analogous cumulative coefficient in the isotropic weight direction is bounded. ■□

Proof. Substitute $z = h_k^2$ in eq. 5.clxix, differentiate, and use eq. 5.clxvii. The two errors at consecutive depths are each bounded by a constant times L^{-k}. The coefficient of the logarithm is zero in the isotropic direction, so only its finite limit and the bounded error remain there. □

This corollary concerns the bare massive determinants just defined. For the actual blocked anisotropic increment one must also compare the weighted constrained determinant, its classical background, and its regulator uniformly in the weight parameter. The uniform parameter estimate here is the input that permits that comparison to be differentiated. A bounded scalar comparison without such a parameter domain would not suffice.

5.6. **Affine response and preservation of the coupling coefficient.** The coefficient of the action can be read from an affine field of constant curvature. The nonlinear background map is defined on a bounded field ball, so this readout uses the polynomial extension of its linear response. We construct that extension from the actual exponentially localized inverse and compute its value. No nonlinear background is evaluated at an unbounded affine field.

Use the homogeneous construction of Prop. 3.183 at the trivial flat reference. The coarse spacing is one and $h = L^{-j}$, with $j \geq 0$. The fixed flat fine gradient is D, its curl is d, and the actual straightened derivative is Q. The derivative of the constructed strong-gauge representative will be denoted by H_h. The derivative of the literal physical representative will be denoted by $\widehat{H}_h$. Gauge covariance relates them, but they are not identified as coordinate maps.

5.6.1. *Polynomial extension of the actual inverse.* Group fine coordinates into unit coarse cells. At a fixed depth, each cell has finitely many fine coordinates. The exponential kernel estimates for the joint inverse and its coarse multiplier imply that its matrix entries have every polynomial spatial moment. The following observation makes the uniqueness argument precise.

Lemma 5.16. Suppose a cell operator $\mathcal{L}$ and a two-sided inverse $\mathcal{G}$ have absolutely summable exponentially weighted row kernels, uniformly in the row. Then both act on fields of any fixed polynomial growth, their products agree with their usual matrix products there, and $\mathcal{L}\mathcal{G} = \mathcal{G}\mathcal{L} = 1$ on that space. In particular a polynomial-growth solution of $\mathcal{L}u = 0$ is zero. ■

Proof. For a field with $|u(z)| \leq C(1 + |z|)^m$, use

$$(1 + |z|)^m \leq (1 + |x|)^m (1 + |z - x|)^m$$

in the row defining the output at x. The exponential moment dominates the last polynomial, so the row converges absolutely. In a product of two kernels the corresponding double sum converges absolutely by the same inequality applied twice. Fubini therefore identifies it with the matrix product, whose identity was already known on finitely supported fields. □

Apply this lemma to the actual flat joint operator

$$\mathcal{L}_h(A, \omega) = \big((\mathrm{d}^\sharp \mathrm{d} + DRD^\sharp)A - Q^\sharp \omega,\, QA\big).$$

The site smoothing in R has exponential cell kernels, as do its first derivatives; the remaining coefficients have fixed physical carriers. The operator and its inverse therefore satisfy the lemma at each fixed h. Their physical weighted estimates also show that the extensions of their fixed-degree jets have constants independent of depth. The finite-dimensional coordinate count inside a cell is not used as a uniform bound.

5.6.2. *The actual response to constant curvature.* Fix a real antisymmetric tensor $F = (F_{\mu\nu})_{\mu,\nu=1}^4$, and one constant colour t, with $\frac{1}{2}\,\mathrm{tr}\, t^2 = 1$. On the physical fine and coarse lattices set

$$a_\mu(x) = \tfrac{1}{2} \sum_\nu F_{\nu\mu} x_\nu t, \qquad b_\mu(y) = \tfrac{1}{2} \sum_\nu F_{\nu\mu} y_\nu t. \tag{5.clxxi}$$

Their fine curl is the constant tensor $F_{\mu\nu} t$. Their fine divergence is zero, since $F_{\mu\mu} = 0$. In particular

$$\mathrm{d}^\sharp \mathrm{d} a = 0, \qquad D^\sharp a = 0.$$

Lemma 5.17. For every homogeneous depth, the actual straightened linear average satisfies $Qa = b$. ■

Proof. Write $n = h^{-1}$, which is odd. In the parallel frame, the actual row is eq. 3.mccxxiv:

$$(Qa)_\mu(y) = h^5 \sum_{x \in B_n(y)} \sum_{\ell=0}^{n-1} a_\mu(x + \ell h e_\mu).$$

The centred block has coordinate mean y. The line average shifts only the μ-coordinate, by $(1-h)/2$. Its coefficient in a_μ is $F_{\mu\mu}/2 = 0$. The normalization of the block and line weights is $h^5 n^5 = 1$. These facts give $Qa = b$ exactly.

The formula for the row also follows directly by composing the actual one-step flat derivatives. Balanced base-L digits run once through the centred block of n sites in each direction, while ordinary digits $0, \dots, L-1$ run once through the n successive longitudinal steps. Thus no replacement of the composed average by a continuum mean has occurred. □

Proposition 5.18. The polynomial extension of the actual strong-gauge linear response satisfies

$$H_h b = a, \qquad D\omega(0)b = 0. \tag{5.clxxii}$$

The literal physical response has the form

$$\widehat{H}_h b = a + D\lambda_b, \tag{5.clxxiii}$$

for a polynomial-growth fine gauge parameter λ_b. Its degree is bounded independently of depth. ■

Proof. Differentiate the actual joint fixed-point equations at zero. Their nonlinear residuals have zero first derivative, so $(H_h b, D\omega(0)b)$ solves

$$(\mathrm{d}^\sharp \mathrm{d} + DRD^\sharp)H_h b - Q^\sharp D\omega(0)b = 0, \qquad QH_h b = b.$$

The pair $(a, 0)$ solves these same equations by Lem. 5.17 and the zero divergence and curl-divergence identities above. It also satisfies the fixed Landau condition. Both pairs have polynomial growth. Their difference vanishes by Lem. 5.16, which proves eq. 5.clxxii.

The actual literal representative is obtained from the strong-gauge one by the specified fine gauge in eq. 3.mcclxxxiv. Differentiate that identity at zero. The derivative of a fine gauge is a fine gradient, giving eq. 5.clxxiii. Its gauge parameter is the composition of the derivative of the actual node factor, the local node extension, and the response H_h. The node factor has a fixed physical carrier and bounded analytic derivatives, while H_h has exponential cell kernels. These operators map an affine input to a polynomial-growth output of bounded degree. This argument extends the linear jets only. □

For a single $(1,2)$ curvature component, the symmetric potential in eq. 5.clxxi differs from the potential $a_1 = -F_{12}x_2 t,\ a_2 = 0$ by the gradient of a quadratic polynomial, with sign determined by the convention for D. Thus either potential gives the same gauge-invariant quadratic readout.

5.6.3. *The coefficient readout after substitution.* Use the pointed density convention of eq. 4.ii and eq. 4.v. Thus write $\mathcal{F}_h = \sum_z \mathcal{F}_{h,z}$, where each localized scalar density is gauge invariant and is normalized at the trivial field. Its Hessian kernel at an anchor z satisfies, at each fixed mesh,

$$\sum_{b,c} e^{m(\mathrm{dist}(b,z)+\mathrm{dist}(c,z))} \left\| D^2\mathcal{F}_{h,z}(1)[\delta_b, \delta_c] \right\| < \infty \tag{5.clxxiv}$$

for some $m > 0$. The anchors, physical volume normalization and compatible pointing are part of this convention. A sum of already localized members with the same absolute moment has the same meaning here.

Constant colour rotations force the first derivative of each density to vanish. Its Hessian annihilates fine gradients by the differentiated local gauge Ward identity. The two-leg moment eq. 5.clxxiv extends that identity to polynomial gauge parameters and polynomial second arguments. In particular the affine readout is taken in a density anchored at z, then summed with the prescribed density of anchors per physical volume. It is not the formal extensive pairing $\langle a, Ka\rangle$ of a translation-invariant bulk kernel with an unbounded affine field.

For each localized density, the second derivative of $\mathcal{F}_{h,z}(\widehat{U}(B))$ has no term involving the second derivative of the background: that term is multiplied by $D\mathcal{F}_{h,z}(1) = 0$. Its affine quadratic readout is consequently the rooted Hessian applied to $\widehat{H}_h b$ twice. Equation eq. 5.clxxiii and the polynomial

Ward identity replace both arguments by a. Sum the resulting density identities with the actual anchor weights; their partition of unity preserves the physical density. The action itself has density $\frac{1}{2}\sum_{\mu<\nu}|F_{\mu\nu}|^2$ in this normalization. We have proved

$$b[\mathcal{F}_h \circ \widehat{U}] = b_h[\mathcal{F}_h]. \tag{5.clxxv}$$

The subscript on the right specifies physical fine spacing h. In four dimensions this coefficient has no extra power of the blocking scale.

For a successive map this also requires the actual nested criticality identity, rather than an identification of that map with a direct Wilson lift. We give the argument explicitly. Work first in common physical units. Write $\widehat{U}^{(j)}$ for the full Wilson critical lift to the original fine lattice, $\widehat{U}^{(j-1)}$ for the preceding lift, and p_j for the one-step critical map of the actual effective action S_{j-1}. The common-chart semigroup identity proved in § 4.9.6 says

$$\widehat{U}^{(j)}(B) \sim \widehat{U}^{(j-1)}(p_j(B)),$$

where the equivalence is a fine gauge equal to one at the final coarse nodes. Apply the literal $(j-1)$-step mean. The right side is $p_j(B)$. The gauge on the left induces a gauge on the intermediate coarse lattice. Consequently, at the trivial field,

$$Dp_j(0)b = Q_{\mathrm{lit}}^{(j-1)}\widehat{H}^{(j)}b + \mathrm{d}\chi = a_{\mathrm{intermediate}} + d\widetilde{\chi}. \tag{5.clxxvi}$$

Here b is an affine final coarse potential with constant antisymmetric curvature. The symbol $a_{\mathrm{intermediate}}$ denotes its link variation in the same convention as p_j: it is ξa_{phys} at intermediate bond centres for raw link-log coordinates, where $\xi = 1/L$ in final coarse units, and it is a_{phys} for variations normalized per unit length. For the second equality, the direct fine response is the affine fine potential modulo a gradient by eq. 5.clxxiii. The straightened block-line mean reproduces that affine potential at its intermediate bond centres, with this length factor; see also eq. 5.24. Its literal mean differs by the coarse frame gauge of the exact plain-mean recursion. The linearized gauge covariance of the literal mean takes a fine gradient to an intermediate gradient. These observations give exactly the last term in eq. 5.clxxvi, without claiming that the literal derivative equals the straightened average.

The same polynomial-growth extension used in the direct affine response justifies these linear identities on the infinite lattice. In particular, for any compatible complete pointed gauge-invariant density $\mathcal{F}$, its vanished first jet removes D^2p_j, and its two-leg Ward identity removes both gradients in eq. 5.clxxvi. Thus

$$b[\mathcal{F} \circ p_j] = b[\mathcal{F}]. \tag{5.clxxvii}$$

This applies to the physical cumulative Gaussian density in its successive-depth identity. It concerns complete invariant densities, rather than arbitrary fixed-reference Taylor pieces.

Finally change physical units between the successive lattices. Two factors L^{-1} from the one-form variation and a factor L^4 from the volume convert its quadratic Hessian to $L^2\hat{\pi}(p/L)$; the two powers of momentum remove that L^2. Equation eq. 5.clxxv is the coordinate-free affine version of this cancellation. It establishes preservation of the coefficient; it does not require equality of full polarization symbols at nonzero momentum.

5.6.4. *Physical mass and the bare lattice formula.* For a physical mass parameter $s > 0$, let

$$\mathcal{L}_h(s, U) = \log\det(\Delta_{h,U} + s) - \tfrac{1}{2}\log\det(K_{h,U} + s),$$

with its trivial-field value subtracted and with the bulk trace convention of § 5.4. These are unconstrained site and Wilson-vector operators. The fine physical variations and inner products give

$$\Delta_{h,U} = h^{-2}\Delta_{\mathrm{bare},U}, \qquad K_{h,U} = h^{-2}K_{\mathrm{bare},U}, \qquad z = h^2 s. \tag{5.clxxviii}$$

The scalar factors h^{-2} in their determinants are field independent and disappear from the normalized logarithm.

In a constant physical curvature tF, the angle of a fine plaquette in bare units is h^2tF. There are h^{-4} bare sites per unit physical volume. Hence the coefficient per physical volume is obtained by multiplying $\frac{1}{2}\widetilde{\beta}(h^2s)(h^2tF)^2$ by h^{-4}. The two powers of plaquette area cancel this density exactly. Thus

$$b[\mathcal{L}_h(s,\widehat{U}(\cdot))] = \widetilde{\beta}(h^2s). \tag{5.clxxix}$$

For the constant field, the symmetric and linear potentials are related by an exact Abelian gauge. Each finite compression of the massive bare operators is conjugated by its unitary adjoint matrices. Their determinant densities therefore agree. Equivalently their quadratic jets agree by the polynomial Ward identity.

At a fixed positive mass, the massive resolvent and its mass derivatives have exponential kernels, locally uniformly in that mass. Differentiation therefore commutes with the second-moment coefficient readout. Applying it once and twice to eq. 5.clxxix, define

$$\mathcal{L}_h^\sharp = \mathcal{L}_h(1) - \partial_s\mathcal{L}_h(1) + \tfrac{1}{2}\partial_s^2\mathcal{L}_h(1).$$

Then

$$b[\mathcal{L}_h^\sharp \circ \widehat{U}] = \widetilde{\beta}(h^2) - h^2\widetilde{\beta}'(h^2) + \tfrac{1}{2}h^4\widetilde{\beta}''(h^2). \tag{5.clxxx}$$

This identity supplies the scale and normalization needed to use Cor. 5.13 in the physical Gaussian comparison.

5.7. **Uniform comparison of the averaging and mass regulators.** The massive coefficient and the finite gauge-change identity have different regulators. We now compare them at the actual homogeneous background, uniformly in the fine depth. The comparison includes the constraint Schur determinants. Its ultraviolet remainder contains three free resolvents, which are trace summable in four dimensions. The few averaging directions are controlled by rank, rather than by pretending that the averaging regulator is a scalar mass.

We use the background of Prop. 3.183 on a smaller fixed coarse ball. Fine scalar and vector spaces have physical measure h^4, where $h = L^{-j}$; coarse spaces have counting measure. At each actual background $U = \widehat{U}(B)$, re-center the fine coordinate as $e^{iha}U$. Write $D = D_U$, $\Delta = D^\sharp D$, and let $q = q_U$, $Q = Q_U$ be the actual site and straightened field derivatives at this re-centered zero. Thus $qq^\sharp = 1$ and $QD = \mathrm{d}_c q$. They are not the effective τ_A of the fixed-flat chart at a nonzero A. Let $A''_W(U)$ denote the actual physical Wilson Hessian and put

$$K = A''_W(U) + DD^\sharp, \qquad V_s = q^\sharp q, \qquad V_v = Q^\sharp Q - Dq^\sharp qD^\sharp. \tag{5.clxxxi}$$

The two averaging precisions are $\Delta + V_s$ and $K + V_v$. The latter is the background-gauge precision with its field-row penalty, rather than the Landau precision used to construct the background. Its gap therefore requires a separate proof.

5.7.1. *A uniform gap for the flat background gauge.* All operators in this subsection are at a flat reference. Put $P_0 = 1 - q^\sharp q$, and retain R_L for the previously proved Landau projector. Define

$$\Xi_0 = \mathrm{d}^\sharp\mathrm{d} + DP_0D^\sharp + Q^\sharp Q. \tag{5.clxxxii}$$

Lemma 5.19 (The projected scalar Poisson equation)**.** For $f \in \ker q$, the equations

$$q\lambda = 0, \qquad P_0\Delta\lambda = f \tag{5.clxxxiii}$$

have a unique solution and

$$\|\lambda\|_2 + \|D\lambda\|_2 + \|D^2\lambda\|_2 \le C\|f\|_2. \tag{5.clxxxiv}$$

The constants are uniform in mesh, volume and real flat holonomy. ∎

Proof. The actual site Poincaré inequality on $\ker q$ makes $\langle D\lambda, D\lambda\rangle$ coercive there. Its finite variational equation gives existence, uniqueness and $\|\lambda\|_{H^1} \le C\|f\|_2$. The missing site component is a multiplier: $\Delta\lambda = f + q^\sharp\zeta$. Read it with the smooth exact site lift E_q of Lem. 3.180. For every coarse b,

$$\langle \zeta, b\rangle = \langle D\lambda, DE_q b\rangle - \langle f, E_q b\rangle.$$

The physical seed energy yields $\|\zeta\|_2 \le C\|f\|_2$. Since $q^\sharp$ is an isometry, $\|\Delta\lambda\|_2 \le C\|f\|_2$. In the flat parallel frame the Fourier identity $\sum_{\mu,\nu}\|D_\mu D_\nu\lambda\|_2^2 = \|\Delta\lambda\|_2^2$ proves the last estimate. Twisted periodic phases do not change this identity. No point readout of a fine energy function is used. □

Proposition 5.20 (Uniform background-gauge gap)**.** There are $c, C > 0$ such that

$$c\|v\|_{H^1}^2 \le \langle v, \Xi_0 v\rangle \le C\|v\|_{H^1}^2. \tag{5.clxxxv}$$

On a fixed smaller ball of real actual backgrounds the same comparison holds for $K + V_v$ and for $\Delta + V_s$, relative to their free physical H^1 norms. All constants are independent of j, M. On its smaller complex continuation, the normalized precisions have uniform relative-form Neumann inverses; no Hermitian inequality is asserted at a complex background. ■

Proof. Use the proved flat Landau restoration to write $v = s + D\lambda$, where $R_L D^\sharp s = 0$ and $q\lambda = 0$. Its intertwining gives $Qs = Qv$ and its flat curl gives $\mathrm{d}s = \mathrm{d}v$. The Landau gap therefore implies

$$\|s\|_{H^1} \le C(\|\mathrm{d}v\|_2 + \|Qv\|_2).$$

The remaining scalar function satisfies $P_0\Delta\lambda = P_0 D^\sharp v - P_0 D^\sharp s$. Apply Lem. 5.19 to obtain

$$\|D\lambda\|_{H^1} \le C(\|P_0 D^\sharp v\|_2 + \|s\|_{H^1}).$$

Together these inequalities prove the lower bound. The upper bound follows from the actual flat row bounds and $\|P_0\| \le 1$.

For the actual background, its literal normal coordinate has fixed small X_α norm. The actual coefficient comparisons give $D_U - D_0 = O(r)$ as a zero-order L^2 operator and $q_U - q_0, Q_U - Q_0 = O(r)$ in physical L^2 row norms. For the site map this follows directly by summing its disjoint transported pillar averages; path lengths are bounded in physical units and its transport errors are $O(r)$. For the field map use the actual row/column density estimates. The exact Wilson Hessian comparison bounds its form error by $Cr\|v\|_{H^1}^2$. Expanding $D_U(1 - q_U^\sharp q_U)D_U^\sharp$ as a form gives the same error. Thus $K + V_v$ is a small form perturbation of Ξ_0. The scalar statement is the actual site gap, or the identical comparison with $\Delta_0 + q_0^\sharp q_0$. A fixed smaller r preserves both gaps uniformly. □

The gaps hold throughout the interpolation

$$M = K_* + 1, \qquad S_t = M + t(V - 1), \qquad 0 \le t \le 1, \quad (K_*, V) = (\Delta, V_s) \text{ or } (K, V_v). \tag{5.clxxxvi}$$

Indeed its form is the convex combination of $K_* + 1$ and $K_* + V$. At the flat reference both are uniformly comparable to $L + 1$; the just proved coefficient perturbation preserves this fact. This observation also supplies coherent logarithm branches near the real positive precisions.

5.7.2. *Fixed free normalization and the actual low-rank factors.* Set $G_0 = (L+1)^{-1}$ on the flat fine scalar space, with identity on colours and, for vectors, on components. All normalizations below use this fixed free operator, independent of the background. Define

$$B = G_0^{1/2} M G_0^{1/2}, \quad B_t = G_0^{1/2} S_t G_0^{1/2}, \quad A = G_0^{1/2} V G_0^{1/2}, \quad X = A - G_0. \tag{5.clxxxvii}$$

The gaps prove uniform bounds for B, B_t and their inverses; they continue to a fixed smaller complex physical background ball by their relative form Neumann series.

The actual rank factorization is explicit. In the scalar case $A = (qG_0^{1/2})^\sharp (qG_0^{1/2})$. In the vector case it is

$$A = (QG_0^{1/2})^\sharp (QG_0^{1/2}) - (qD^\sharp G_0^{1/2})^\sharp (qD^\sharp G_0^{1/2}). \tag{5.clxxxviii}$$

At a flat coarse Bloch fibre its rank is at most 3 in the scalar case and $12+3=15$ in the vector case. Its norm is uniformly bounded. Indeed Q, q have the actual physical row bounds, and $D_U^\sharp G_0^{1/2}$ is bounded since its flat part is a normalized derivative and its remaining part is zero order $O(r)$. The negative sign in the vector regulator causes no problem for this norm/rank assertion or for the interpolation gap.

We record why these bounds retain two actual background derivatives. The site transport has physical path length bounded by a geometric sum of scales. Its complex adjoint matrices and all fixed parameter derivatives have bounded norm. Summation over disjoint pillars gives bounded physical q row norms on a fixed complex fine potential ball. The actual straightened recursion gives bounded physical Q row norms on the same joint reference/increment ball. To check that this joint ball is uniform after re-centering, truncate the construction at scale s. Its formulas give both $\|a_s\|_\infty \le Cr\xi_s$ and $\|v_s - 1\|_\infty \le Cr\sum_{r'<s}\xi_{r'} \le Cr\xi_s$. Thus the plain reference $W_s = M^s U$ has raw logarithm of size $Cr\xi_s$ in the flat frame. The joint one-step reference/increment estimates therefore have summable errors at every remaining scale. Their actual row/column products give the physical Q bound on a fixed ball, even for a complex reference. This does not assume that the fixed-reference effective gauge derivative is a site average. Banach-valued Cauchy differentiation in the reference potential therefore bounds the first two derivatives of these row operators; it does not infer absolute entry sums from a general bilinear infinity-norm estimate. The fine derivative D_U has first parameter variation of zero order $O(\|\delta A\|_\infty)$ and second variation $O(h\|\delta A_1\|_\infty\|\delta A_2\|_\infty)$, directly from the exponential link.

The physical Wilson Hessian can be checked without an unspecified coefficient theorem. In one plaquette put

$$\mathcal{C}_U a(p) = h^{-1}\sum_{b\in\partial p}\epsilon_{p,b}\,\mathrm{Ad}(T_{p,b}(U))a_b,$$

where $T_{p,b}$ is the ordered prefix which moves that link insertion to the plaquette base. In a reversed edge its inverse-link insertion is included in that prefix. Thus $\delta U(p)U(p)^{-1} = ih^2\mathcal{C}_U a(p)$ and $\mathcal{C}_0 = d$. Differentiating the four link factors a second time splits their Hessian into a quadratic form in $\mathcal{C}_U a, \mathcal{C}_U b$, with bounded analytic plaquette coefficient, and a remainder with physical bound $Ch^4|F_p|\sum_{b\in\partial p}|a_b|\sum_{b\in\partial p}|b_b|$. To see the power, the two link insertions supply h^2; the part not in their squared total insertion pairs with the antisymmetric plaquette defect, of size h^2F_p. At the identity this part is zero. The symmetric total-insertion coefficient is the bounded matrix obtained from $U(p)+U(p)^{-1}$.

Every prefix contains at most four links. Consequently $\mathcal{C}_U - d$ is zero order of size $C\|A\|_\infty$, its first parameter derivative is zero order of size $C\|\delta A\|_\infty$, and its second derivative has bound $Ch\|\delta A_1\|_\infty\|\delta A_2\|_\infty$. The curvature formula $F = \mathrm{d}A + E_h(A)$ bounds its first two parameter jets by the corresponding X_α background norms. Differentiate the preceding finite plaquette decomposition. Since $\mathrm{d}G_0^{1/2}$ is bounded, it proves uniform bounds on the first two derivatives of $G_0^{1/2}A''_W(U)G_0^{1/2}$. The same statement for $DD^\sharp$ follows from the already stated zero-order parameter derivatives of D. If these calculations are instead performed in the fixed A chart, the extra second coordinate differential contains hJ; it is also bounded because $h\|\mathrm{d}^\sharp\mathrm{d}\delta A\|_\infty \le C\|D\delta A\|_\infty$ and the complete Wilson residual is uniformly analytic. We have therefore obtained the first two normalized M jets from the actual Wilson formula, without assuming the desired inverse. The localized response estimates eq. 3.mccxcv and eq. 3.mccxcvi supply exactly those directions after the physical background is substituted.

5.7.3. *The exact resolvent subtraction.* The following identity is finite dimensional and retains the full regulator V, which need not be positive. Put $G = M^{-1}$ and $W = V - 1$. The resolvent identity applied twice gives

$$S_t^{-1} = G - tGWG + t^2GWGWS_t^{-1}.$$

Integrate $\partial_t \log\det S_t = \mathrm{Tr}(S_t^{-1}W)$, then expand $W = V - 1$. One obtains exactly

$$\begin{aligned}&\log\det(K_* + V) - \log\det(K_* + 1) + \mathrm{Tr}\, G + \tfrac{1}{2}\,\mathrm{Tr}\, G^2\\ &\quad = \mathrm{Tr}(GV) + \mathrm{Tr}(G^2V) - \tfrac{1}{2}\,\mathrm{Tr}(GVGV)\\ &\qquad + \int_0^1 t^2\,\mathrm{Tr}(GWGWS_t^{-1}W)\,dt.\end{aligned} \tag{5.clxxxix}$$

With the fixed normalization eq. 5.clxxxvii, its last trace is

$$\mathrm{Tr}(B^{-1}XB^{-1}XB_t^{-1}X). \tag{5.cxc}$$

The three earlier terms become respectively $\mathrm{Tr}(B^{-1}A)$, $\mathrm{Tr}(B^{-1}G_0B^{-1}A)$ and $\mathrm{Tr}(B^{-1}AB^{-1}A)$. Every such term contains an actual finite coarse-rank factor. Expand the three $X = A - G_0$ factors in eq. 5.cxc. All terms except the pure G_0 term again contain A, and are bounded by its rank times the product of the already bounded operator norms. The remaining term is bounded by Schatten Hölder:

$$|\,\mathrm{Tr}(B^{-1}G_0B^{-1}G_0B_t^{-1}G_0)| \le C\|G_0\|_{\mathcal{S}_3}^3. \tag{5.cxci}$$

One must apply this reasoning per physical coarse cell, not with the rank of a full finite-volume matrix. The next argument does so and proves the two external momentum derivatives needed for the coefficient.

5.7.4. *Two external jets without an unbounded complex plane wave.* Work at the trivial background for the bulk Hessian. Coarse Bloch momentum is $\kappa \in [-\pi,\pi]^4$. At fine spacing $h = 1/n$ the free aliases in one cell have momenta $q = \kappa + 2\pi m$, with m in n consecutive centered integer values per coordinate. Their Laplacian symbol is

$$\lambda_h(q) = 4h^{-2}\sum_\mu \sin^2(hq_\mu/2). \tag{5.cxcii}$$

At real κ, $\lambda_h(q) \ge c|q|^2$ on the fine Brillouin cube. A fixed small complex shift z obeys

$$\mathrm{Re}(1 + \lambda_h(q+z)) \ge c(1 + |q|^2), \qquad |z| < s_1, \tag{5.cxciii}$$

after a uniform decrease of s_1. To verify it, the imaginary shift changes the real symbol by at most $C|\,\mathrm{Im}\, z|^2$; the real shift remains in a fixed slightly enlarged Brillouin cube, where the same sine lower bound holds. Thus the principal square root of $G_0(\kappa + z)$ is analytic on a fixed strip. In particular

$$\int_{[-\pi,\pi]^4} \|G_0(\kappa+z)\|_{\mathcal{S}_3}^3 \frac{d\kappa}{(2\pi)^4} \le C\int_{\mathbb{R}^4} (1 + c|q|^2)^{-3}\,dq < \infty. \tag{5.cxciv}$$

The sum over the aliases simply partitions the fine Brillouin cube. The fixed colour/component multiplicities are included in C.

Let two external coarse variations have momenta p and $-p$. At first take p real, and use the formal coefficient ring $\mathbb{C}[\varepsilon_1,\varepsilon_2]/(\varepsilon_1^2,\varepsilon_2^2)$. Substituting $B = \varepsilon_1 be^{ipx} + \varepsilon_2 ce^{-ipx}$ records the zeroth, two first and mixed second jets of each actual operator. This is a jet computation at zero, not an assertion that a complex plane-wave field lies in a globally bounded background domain.

Represent the jets on four graded fine-cell spaces indexed by $\varnothing, \{1\}, \{2\}, \{1,2\}$. Their momenta are $\kappa, \kappa+p, \kappa-p, \kappa$. A jet block from grade T to grade S is zero unless $T \subset S$, and otherwise is the derivative with the external indices in $S \setminus T$. Operator multiplication and inversion of these finite triangular matrices are exactly the product rule and inverse rule for the mixed two-jet. The free matrix $\mathbf{G}_0$ is diagonal in these four grades, with the four shifted free symbols just specified. Use it on both sides to normalize every jet matrix.

All normalized jet blocks of $\mathbf{B}, \mathbf{B}_t, \mathbf{A}$ are bounded uniformly in h, κ, t, on a fixed complex neighborhood of $p = 0$. Here are the details of this assertion. The localized first and second background jets yield absolutely convergent Fourier sums for their cell profiles when $|\,\mathrm{Im}\, p| < s/4$. Their strong

norms are uniformly bounded by eq. 3.mccxcvi: the exponential kernel bound dominates both external factors, after decreasing its rate. Applying the finite-carrier transport and plaquette formulas proved above gives bounded normalized q, Q, D, K jets. Every factor comparing two fine momenta is also bounded by eq. 5.cxciii. Thus all off-diagonal normalized blocks are bounded without a fine-volume factor.

For the diagonal inverses, the actual flat precision gaps give a uniform relative form inverse. Conjugation by a small physical exponential changes each local derivative form by $O(s)$ in the H^1, H^{-1} pairing, and changes each averaging row form by $O(s)$ because its physical carrier is bounded. Consequently the flat interpolation inverse remains bounded for the shifted complex Bloch momenta in a fixed strip. This proves uniform bounds for the diagonal blocks of $\mathbf{B}_t^{-1}$; the corresponding statement for $\mathbf{B}^{-1}$ is immediate from the free massive operator. Inverting a four-grade triangular matrix then uses a finite sum with at most two nontrivial jet steps. The inverses of the full jet matrices are therefore bounded uniformly as well.

The rank statement also survives with no derivative-count problem. Apply the jet functor to eq. 5.clxxxviii. It still factors through four copies of the coarse site/field spaces. The rank of $\mathbf{A}$ is at most 12 in the scalar case and 60 in the vector case. In this statement the jet of $Q^\sharp$ is the analytic derivative of that operator; it is not the ordinary transpose of the four-grade jet matrix, which would reverse grades. The actual factorization, and hence this finite-rank bound, remains valid for complex external momentum.

Apply eq. 5.clxxxix in the jet algebra. The mixed trace is the trace of its $(\{1,2\},\varnothing)$ block, not the ordinary trace of the entire triangular matrix. For real p, cyclic permutation of trace factors is valid after integrating in κ: it only reindexes κ by one of the real external momenta, and the coarse Brillouin torus and its fibre identifications are periodic. Thus the displayed normalized trace identities give the desired mixed bulk Hessian.

Each term containing $\mathbf{A}$ has uniformly bounded trace norm by its rank and the operator bounds. For the pure free term apply Schatten Hölder to its three diagonal free factors, and Hölder once more in κ if their momenta are different. Equation eq. 5.cxciv bounds the resulting integral uniformly. Projection onto one graded block cannot increase trace norm, so it also bounds the required mixed trace. Integration over $0 \le t \le 1$ preserves that bound.

Every expression just obtained is holomorphic in a fixed complex neighborhood of $p = 0$, by the same integrable bounds and analytic inverse formulas. The equalities first proved for real p extend there by the identity theorem. This orders the steps correctly: real cyclic reindexing is performed before analytic continuation; there is no unproved change of a complex Bloch integration contour. We have proved a uniform holomorphic bound for the mixed bulk polarization of the entire remainder in eq. 5.clxxxix. Cauchy's formula bounds each second external momentum derivative by a uniform constant.

For completeness the same construction justifies the bulk limit. At fixed h the gapped fine and coarse Bloch inverses have exponential cell kernels. Finite torus kernels are their periodic sums, and every differentiated trace above is a sum of products with absolutely convergent spatial moments. Terms winding around the torus vanish when its side tends to infinity. The bulk polarization is the Bloch integral used above. Its bounds are uniform in h, because rank and eq. 5.cxciv were uniform before taking this limit. A full finite-volume trace was never bounded by a fixed rank.

5.7.5. *The constraint Schur determinants.* Let $T_v = K - Dq^\sharp qD^\sharp$ and $T_s = \Delta$. For either $C = Q$ or $C = q$, elementary block elimination gives

$$|\det \mathbb{S}(T_*, C)| = \det(T_* + C^\sharp C)\, \det\{C(T_* + C^\sharp C)^{-1} C^\sharp\}. \tag{5.cxcv}$$

It can also be checked in orthogonal $\ker C$ coordinates. The field/site constraint is onto, and the first precision has the uniform gap proved above. Its coarse Schur matrix has uniform positive lower and upper bounds. For the lower bound use the actual exact smooth row lift E_C and its physical

energy:

$$\|b\|_2^2 = \langle (T_* + C^\sharp C)(T_* + C^\sharp C)^{-1} C^\sharp b, E_C b\rangle \le C\langle b, C(T_* + C^\sharp C)^{-1} C^\sharp b\rangle^{1/2} \|b\|_2.$$

The upper bound follows from the gap and the bounded row form. The actual nonflat lift is obtained from the flat smooth seeds by the row comparison and its coarse Neumann inverse; its energy bound is uniform on the present physical strong ball.

The Schur matrix and its inverse have a fixed holomorphic momentum strip. This follows either by weighted relative conjugation of the precision and its fixed-carrier rows, or directly by its normalized jet formula. Its mixed logarithmic trace derivative is $\mathrm{Tr}(S^{-1}S_{12} - S^{-1}S_1 S^{-1} S_2)$, with the appropriate shifted momenta and integrated cyclic convention. The fine inverses in these coefficients have already been bounded. There are only 12 coarse vector colours and 3 coarse site colours per cell. Hence the mixed Schur-log polarization is bounded holomorphically, uniformly in h. Cauchy's formula again bounds its second momentum coefficient. Coherent branches are fixed by the positive real Schur matrices; only the displayed real saddle identity uses absolute values.

5.7.6. *The uniform physical coefficient comparison.* Let

$$\mathcal{L}_h(s,U) = \log\det(\Delta_U + s) - \tfrac{1}{2}\log\det(K_U + s), \qquad \mathcal{L}_h^\sharp = \mathcal{L}_h(1) - \partial_s \mathcal{L}_h(1) + \tfrac{1}{2}\partial_s^2 \mathcal{L}_h(1), \tag{5.cxcvi}$$

with trivial-field values subtracted and bulk trace per physical coarse volume. Let Ψ_j be the actual physical homogeneous Gaussian logarithm of eq. 5.civ.

Proposition 5.21 (Averaging versus mass with uniform two-jet error)**.** On the constructed homogeneous background domain,

$$\left| b[\Psi_j] - b[\mathcal{L}_h^\sharp \circ \widehat{U}] \right| \le C, \qquad h = L^{-j}, \quad j \ge 0, \tag{5.cxcvii}$$

where C is independent of j. More precisely, the difference after removal of the physical multiplier term has a bulk Hessian bounded holomorphically on a fixed external momentum neighborhood. ■

Proof. Apply eq. 5.clxxxix to the scalar and vector precisions, with coefficients 1 and $-1/2$. Their left sides combine as the averaging logarithm minus $\mathcal{L}_h(1) - \partial_s \mathcal{L}_h(1) + \frac{1}{2}\partial_s^2 \mathcal{L}_h(1)$. Indeed the two added trace terms on that left side combine to $\partial_s \mathcal{L}_h(1) - \frac{1}{2}\partial_s^2 \mathcal{L}_h(1)$, since the first log-determinant derivative is the resolvent trace and the second is minus the square trace. Moving them to the right gives precisely eq. 5.cxcvi. The uniform jet argument bounds the second-momentum coefficient of both remainder terms. Equation eq. 5.cxcv adds only the two coarse Schur logarithms, whose coefficients were uniformly bounded. Finally Prop. 5.8 proves that the actual physical multiplier correction has coefficient zero. This is a statement about its determinant functional; no false pointwise third-order claim for the matrix $-\mu q''$ is used. The finite-dimensional coefficient readout b is a fixed linear combination of these second momentum derivatives. The Cauchy bounds therefore prove eq. 5.cxcvii. □

The actual affine-response identity and physical mass rescaling in eq. 5.clxxx identify the massive term:

$$b[\Psi_j] = \widetilde{\beta}(h^2) - h^2 \widetilde{\beta}'(h^2) + \tfrac{1}{2} h^4 \widetilde{\beta}''(h^2) + O(1). \tag{5.cxcviii}$$

This uses the actual coefficient-preserving background pullback, not an arbitrary smooth coarse interpolation. Combining the massive lattice coefficient eq. 5.clxviii yields the cumulative physical law

$$b[\Psi_j] = \frac{11}{3\pi^2} j \log L + O(1). \tag{5.cxcix}$$

By the exact physical Gaussian stage composition and coefficient preservation eq. 5.clxxv, its one-step increments $b_\ell^0 = b[\Phi_\ell]$ therefore satisfy

$$\sum_{\ell=1}^{j} b_\ell^0 = \frac{11}{3\pi^2} j \log L + O(1). \tag{5.cc}$$

The bounded error here is a cumulative comparison. It does not imply geometric convergence, or even convergence, of the actual individual increments: a difference of two bounded comparison errors need not tend to zero. The geometric error in the bare massive coefficient is a separate, stronger property of that particular regulator.

5.8. **Interpolation and consistency of the actual lattice mean.** To compare the classical effective actions at successive fine depths, we need trial fields on both meshes with controlled curvature and controlled coarse mean. The trial field must use the actual group average. This subsection supplies the interpolation and consistency estimates for that comparison.

Work at the trivial flat reference on the periodic physical torus $\mathbb{T}^4_M$, $M \geq 2$. This is the reference used to extract the Taylor coefficients. Put $h = L^{-j}$, with odd $L \geq 3$, and fix $0 < \beta < 1$. Constants can depend on L, β and the fixed small coordinate ball, but not on j or M. A fine bond field $a_\mu(x)$ is regarded as data at its physical midpoint $x + \frac{h}{2}e_\mu$. Write $\|a\|_{X_\beta}$ for the physical first-difference and local Hölder norm in eq. 3.mccxxiii. At the trivial reference all differences are ordinary differences. Every continuous connection below is periodic.

5.8.1. *An exact interpolation with one Hölder derivative.* We first give the scalar interpolation; apply it to each colour component on each of the four shifted midpoint grids. On a grid of spacing h, let a_x be the values and put

$$g_x = (D_1^+ a(x), \dots, D_4^+ a(x)).$$

Suppose $\|a\|_\infty + \|g\|_\infty + [g]_{\beta,\mathrm{loc}} \leq K$. For grid points x, y with $h \leq |x-y| \leq 1$, a coordinate path and telescoping give

$$|g_x - g_y| \leq CK|x-y|^\beta, \qquad |a_y - a_x - g_x \cdot (y-x)| \leq CK|x-y|^{1+\beta}. \tag{5.cci}$$

For the second inequality, each directed path increment is its forward grid difference. Compare that difference with its value at x; all intervening points are within $C|x-y|$ of x. There are at most $C|x-y|/h$ increments, each with an error $CKh|x-y|^\beta$. Negative increments use the forward difference at their other endpoint and obey the same estimate. For fixed multiples of the unit radius the supremum terms give the analogous bound after increasing C.

Choose a smooth nonnegative product bump ρ, supported in $(-1,1)^4$, positive on $[-3/4, 3/4]^4$, with $\rho(0) = 1$. On the periodic lift of the grid define

$$\eta_x(X) = \frac{\rho((X-x)/h)}{\sum_z \rho((X-z)/h)}, \qquad P_x(X) = a_x + g_x \cdot (X-x), \qquad \mathcal{I}_h a(X) = \sum_x \eta_x(X) P_x(X).$$

Only a fixed number of terms occur at each X. The denominator has positive lower and finite upper bounds, because a nearest grid point is at coordinate distance at most $h/2$. Consequently

$$\sum_x \eta_x = 1, \qquad |\partial^\ell \eta_x| \leq C_\ell h^{-|\ell|}.$$

At a grid point y, the only nonzero bump is the one at y. All derivatives of bumps centered at its neighboring grid points vanish there, since their arguments are on the boundary of the open support. Hence

$$\mathcal{I}_h a(y) = a_y, \qquad \nabla \mathcal{I}_h a(y) = g_y. \tag{5.ccii}$$

The periodic lift makes these definitions independent of a fundamental cube. A componentwise shifted grid is treated by the same formula.

Proposition 5.22 (Uniform interpolation of the fine connection)**.** There is a linear periodic interpolation $\mathcal{I}_h$ from midpoint bond data to smooth continuous connections such that

$$(\mathcal{I}_h a)_\mu(x + \tfrac{h}{2}e_\mu) = a_\mu(x), \qquad \|\mathcal{I}_h a\|_{C^{1,\beta}} \leq C_\beta \|a\|_{X_\beta}. \tag{5.cciii}$$

For a continuous $C^{1,\beta}$ connection A, its midpoint samples

$$(\mathcal{S}_h A)_\mu(x) = A_\mu(x + \tfrac{h}{2}e_\mu)$$

satisfy

$$\|\mathcal{S}_h A\|_{X_\beta} \le C_\beta \|A\|_{C^{1,\beta}}. \tag{5.cciv}$$

Both constants are independent of the physical volume and the number of fine sites in a unit cell. ■

Proof. For X within Ch of a grid point x, compare each polynomial in the interpolation formula with P_x. Equation eq. 5.cci gives

$$|P_y(X) - P_x(X)| \le CKh^{1+\beta}, \qquad |g_y - g_x| \le CKh^\beta$$

whenever $\eta_y(X) \neq 0$. Differentiating the partition formula once and twice therefore gives

$$|\nabla \mathcal{I}_h a(X) - g_x| \le CKh^\beta, \qquad |\nabla^2 \mathcal{I}_h a(X)| \le CKh^{\beta-1}. \tag{5.ccv}$$

The second bound is not a uniform second-derivative estimate; only its consequence for the first-derivative Hölder norm will be used.

If $|X - Y| \le h$, integrate the second bound to obtain

$$|\nabla \mathcal{I}_h a(X) - \nabla \mathcal{I}_h a(Y)| \le CKh^{\beta-1}|X - Y| \le CK|X - Y|^\beta.$$

If $h \le |X - Y| \le 1$, compare the two gradients with g_x, g_y at their nearest grid points and use eq. 5.cci. The same Hölder bound follows. Zeroth and first suprema follow directly from the local formula, proving eq. 5.cciii. For vector fields use the appropriate shifted grid for each component. This preserves reality of the colour coordinates.

For sampling, a forward difference is an average of the corresponding continuous derivative on a segment of length h. Its supremum and its translated Hölder difference are bounded by the continuous $C^{1,\beta}$ norm. This proves eq. 5.cciv. □

5.8.2. *Wilson action consistency at this regularity.* For a continuous Hermitian connection A, put

$$U_h(A)_\mu(x) = \exp(ihA_\mu(x + \tfrac{h}{2}e_\mu)), \qquad F_{\mu\nu}(A) = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu].$$

The norm of a Hermitian colour matrix is the one with $|X|^2 = \frac{1}{2}\operatorname{tr} X^2$. Let

$$\mathscr{E}(A) = \frac{1}{2}\int_{\mathbb{T}^4_M} \sum_{\mu<\nu} |F_{\mu\nu}(A)(x)|^2 \, dx.$$

Proposition 5.23 (Consistency of the Wilson energy)**.** On a fixed sufficiently small $C^{1,\beta}$ ball,

$$|A_h(U_h(A)) - \mathscr{E}(A)| \le C_\beta h^\beta M^4. \tag{5.ccvi}$$

Here A_h is the actual Wilson sum over fine positive plaquettes. In particular, for the same interpolated connection,

$$|A_{h/L}(U_{h/L}(A)) - A_h(U_h(A))| \le C_\beta h^\beta M^4. \tag{5.ccvii}$$

The constants are uniform in the two meshes. ■

Proof. Let c_p be the plaquette center. Write the four edge values of A as their affine Taylor polynomials at c_p plus errors bounded by $C_\beta h^{1+\beta}$. The signed sum of the four link exponents is then

$$ih^2(\partial_\mu A_\nu - \partial_\nu A_\mu)(c_p) + O(h^{2+\beta}).$$

The quadratic Baker–Campbell–Hausdorff term for constant edge values is $-h^2[A_\mu, A_\nu]$. Replacing those constant values by the edge Taylor values changes that term by $O(h^3)$. The cubic and higher terms are also $O(h^3)$, uniformly on the fixed connection ball. All four-factor logarithms are

on the small branch. Consequently, with the actual plaquette logarithm defined by $U_h(A)(p) = \exp(ih^2 F_{h,p})$,

$$|F_{h,p} - F(A)(c_p)| \le C_\beta h^\beta. \tag{5.ccviii}$$

This argument uses one Hölder derivative, not a bounded second derivative of A.

The exact Pauli trace gives

$$1 - \tfrac{1}{2}\operatorname{tr} e^{ih^2 F_{h,p}} = \tfrac{1}{2}h^4 |F_{h,p}|^2 + O(h^8 |F_{h,p}|^4).$$

The plaquette fields are uniformly bounded. Equation eq. 5.ccviii changes this value by at most $C_\beta h^{4+\beta}$ when $F_{h,p}$ is replaced by $F(A)(c_p)$. There are $6M^4 h^{-4}$ plaquettes. Finally $|F(A)|^2$ has a uniform C^β modulus, so its midpoint Riemann sum has error at most $C_\beta h^\beta M^4$. These estimates prove eq. 5.ccvi. Apply it on both meshes and subtract to obtain eq. 5.ccvii. □

5.8.3. *The first two actual mean vertices.* Consider one step from fine spacing H/L to spacing H, with $0 < H \le 1$. At the identity write its output raw bond logarithm as

$$m_L(a) = m_{L,1}a + m_{L,2}[a,a] + R_{L,3}(a), \qquad |R_{L,3}(a)| \le C_L \|a\|_\infty^3.$$

The formula is on the finite double-block carrier of an output row. Its coefficients and the remainder bounds are the actual analytic coefficients of the symmetric contour mean. The finite dimension depends on L; there is no fine-depth dependence in this one-step statement.

Lemma 5.24 (Midpoint affine reproduction and quadratic cancellation)**.** The actual linear mean applied to midpoint samples of an affine connection is its coarse midpoint value multiplied by H. The quadratic map $m_{L,2}$ vanishes on every constant connection. ■

Proof. At the identity, the derivative of a group mean is the arithmetic mean of its Lie-algebra inputs. Let $T_{y,x}$ be the internal transport from the coarse node y to a fine starting point x in its centered block, and let $t_{y,x}$ be its first logarithmic variation. The positively oriented row has linear variation

$$\frac{1}{L^4}\sum_{x\in B(y)} \left\{ t_{y,x} + \sum_{\ell=0}^{L-1} a_\mu(x + \ell H e_\mu/L) - t_{y+He_\mu, x+He_\mu} \right\}. \tag{5.ccix}$$

The entries a_μ here are fine link angles, sampled at their fine bond midpoints.

For an affine continuous connection, midpoint sums along each coordinate path integrate that affine connection exactly. Translating such a path by He_μ changes its integral by

$$H\sum_\nu (\partial_\mu A_\nu)(x_\nu - y_\nu),$$

independently of the order of its coordinate edges. The mean of $x - y$ over the centered odd block is zero. Thus the two internal transport contributions in eq. 5.ccix cancel after averaging. The remaining straight line samples have mean position $y + \frac{H}{2}e_\mu$: the starting block is centered at y, and the average of its L fine bond midpoints is the midpoint of the coarse edge. They therefore give exactly $HA_\mu(y + \frac{H}{2}e_\mu)$. This proves the linear assertion, including the plain transport factors.

For the second assertion set a_μ to a constant matrix in each direction. The input and output connections are then translation invariant. Central inversion of the lattice takes these constant fine links to their inverses. The actual symmetric contour mean commutes with this inversion, including reversal of the coarse bond. Its admitted output logarithm therefore satisfies

$$m_L(-a) = -m_L(a).$$

Its quadratic coefficient on a constant connection is zero. Polarization gives zero for two different constant connections as well. This is an exact symmetry of the actual mean, not a continuum replacement of its quadratic vertex. □

5.8.4. *One-step consistency for the nonlinear mean.*

Proposition 5.25. For a continuous connection A in a fixed small $C^{1,\beta}$ ball, one actual averaging step obeys

$$\left\| \frac{1}{i} \log \mathcal{M}(U_{H/L}(A)) - H\mathcal{S}_H A \right\|_\infty \le C_\beta H^{2+\beta}. \tag{5.ccx}$$

The logarithm is the small raw logarithm at the identity. The estimate is local on each actual double block and uniform in the volume. ■

Proof. Fix a coarse bond. Expand the continuous connection to first order at its midpoint. Over its double block, the remainder in A is $O(H^{1+\beta})$; the remainder in a fine link angle is consequently $O(H^{2+\beta})$, with L fixed. By Lem. 5.24, the linear vertex applied to the affine part gives precisely the coarse midpoint angle. It has a bounded finite row sum, so its Taylor remainder has the asserted size.

Split the fine angles as $a = a_{\mathrm{const}} + a_{\mathrm{var}}$, where $\|a_{\mathrm{const}}\| \le C_L H \|A\|_\infty$ and $\|a_{\mathrm{var}}\| \le C_L H^2 \|\nabla A\|_\infty$. The constant–constant quadratic term is zero. All other quadratic terms have size $O(H^3)$. The cubic analytic remainder has size $O(H^3)$ as well. Since $H^3 \le H^{2+\beta}$ for $0 < H \le 1$, these estimates prove eq. 5.ccx. The fixed small connection ball keeps every path mean and raw logarithm on its admitted branch. □

5.8.5. *Uniform propagation through the remaining levels.* The plain derivative of one step contains internal transport terms. Iterating a crude bound on that derivative could lose a factor at each level. We instead use the exact node factorization to control the full plain mean in fine physical coordinates.

Lemma 5.26. There are $r_0, C > 0$, independent of $h = L^{-j}$ and volume, such that

$$\mathscr{P}_h(a) = \frac{1}{i} \log \mathcal{M}^j(e^{iha})$$

is holomorphic on $\|a\|_\infty < r_0$ and

$$\|\mathscr{P}_h(a) - \mathscr{P}_h(\widetilde{a})\|_\infty \le C \|a - \widetilde{a}\|_\infty \tag{5.ccxi}$$

on its half ball. This statement uses no derivative bound for the fine potentials. ■

Proof. Use the actual recursion and node factors in eq. 3.mccliii. Its first-exit estimate uses only $\|a\|_\infty$, and gives

$$\|a_s\|_\infty \le C\xi_s \|a\|_\infty, \qquad \|v_s - 1\|_\infty \le C\xi_s \|a\|_\infty, \quad \xi_s = hL^s.$$

The geometric sum of the intermediate ξ_s's bounds the second expression. At the final level, eq. 3.mccliv writes the plain mean as a product of $e^{iq_h(a)}$ and the two node factors at its endpoints. Both have norm and distance from the identity bounded uniformly on this fine supremum ball. Reduce r_0 so their output has its small logarithm. This proves a uniform bound for the holomorphic map $\mathscr{P}_h$ on a slightly larger fixed ball.

For a fine direction b, apply the one-variable Banach-valued Cauchy estimate to $\mathscr{P}_h(a + zb)$ on that larger ball. It gives a dimension-independent bound $\|D\mathscr{P}_h(a)b\|_\infty \le C\|b\|_\infty$ on the half ball. Integrate it along the segment from a to $\widetilde{a}$ to obtain eq. 5.ccxi. □

Proposition 5.27 (Consistency of the full actual coarse mean)**.** For a continuous $C^{1,\beta}$ connection in a fixed sufficiently small ball,

$$\left\| \frac{1}{i} \log \mathcal{M}^{j+1}(U_{h/L}(A)) - \frac{1}{i} \log \mathcal{M}^j(U_h(A)) \right\|_\infty \le C_\beta h^{1+\beta}, \qquad h = L^{-j}. \tag{5.ccxii}$$

The two coarse outputs have spacing one, and both logarithms are raw logarithms near the trivial coarse field. ■

Proof. After the first step from h/L to h, write its actual output as $e^{ih\widetilde{a}_h}$. Equation eq. 5.ccx gives

$$\|\widetilde{a}_h - \mathcal{S}_h A\|_\infty \le C_\beta h^{1+\beta}.$$

Both potentials lie in the half ball of Lem. 5.26 after the initial continuous ball is reduced once. The remaining j steps are exactly the same actual mean on the two fine-h fields. Applying eq. 5.ccxi proves eq. 5.ccxii. For $j = 0$ the remaining mean is the identity and the argument is unchanged. □

Together, Props. 5.22, 5.23 and 5.27 give actual real trial connections for comparing successive effective actions. They require only the first Hölder derivative supplied by the homogeneous background construction. The exact repair of their small coarse-mean error, the variational comparison, and the passage from real action differences to localized Taylor kernels are the next steps.

5.9. **Geometric convergence of the actual classical Taylor vertices.** The comparison in this section concerns the actual minimized Wilson action, rather than a postulated continuum minimizer. Interpolation is used only to transfer a real trial field between two adjacent meshes. Analyticity then transfers a real energy comparison to the second, third and fourth Taylor vertices. This last passage needs a real-to-complex argument: Cauchy's inequality alone cannot turn a small bound on a real cube into a small complex derivative.

Work near the trivial flat reference, at physical coarse spacing one. Put $h_j = L^{-j}$, let $\widehat{U}_j(B)$ be the actual literal background of Prop. 3.183, and write

$$S_j(B) = A_{h_j}(\widehat{U}_j(B)), \qquad \mathcal{V}_{j,m} = D^m S_j(0), \quad m = 2, 3, 4. \tag{5.ccxiii}$$

The coarse torus has side $M \ge 2$ and physical volume $\mathcal{V} = M^4$. Its real coarse field coordinates are traceless Hermitian bond logarithms. A complex ball always means their holomorphic continuation. Fix an exponent $0 < \beta < 1$ and use the preceding background construction with X_β. Its coarse radius may be decreased by a fixed amount below.

We use the proved mesh-consistency facts in Props. 5.22, 5.23, 5.25 and 5.27: a periodic fine potential in X_β admits a periodic $C^{1,\beta}$ interpolation with the same midpoint values and a uniform norm bound; its sampled Wilson energy differs from $\frac{1}{2}\int \sum_{\mu<\nu} |F_{\mu\nu}|^2$ by $Ch^\beta \mathcal{V}$; and the actual one-step mean applied to the refined midpoint samples differs from the coarser midpoint link by $Ch^{2+\beta}$ in raw relative logarithm. All these constants are uniform in M, j on the fixed small norm ball. These are consistency estimates for the actual mean and Wilson action, not assumptions about a continuum critical field.

5.9.1. *Repairing a transferred trial field without losing derivatives.*

Lemma 5.28 (An exact repair of a small coarse mean error)**.** Let $U = e^{ihA}$ be a real fine trial with small X_β norm, and suppose its literal mean is $e^{i(B+e)}$, where $\|e\|_\infty$ is small. There is a literal trial U^{rep} with mean e^{iB} and

$$|A_h(U^{\mathrm{rep}}) - A_h(U)| \le C\|e\|_\infty \mathcal{V}. \tag{5.ccxiv}$$

The trial and its repair stay in a fixed small strong chart. All constants are uniform in mesh and volume. ■

Proof. Apply the actual mean-gauge slice map eq. 3.mcclxxxvii to the literal trial. Its result $\overline{A}$ has $q_h(\overline{A}) = B + e$ and $RD^\sharp \overline{A} = 0$, with $\|\overline{A}\|_{X_\beta} \le C\|A\|_{X_\beta}$. It has exactly the same action as U.

Use the fixed smooth exact field seed E_Q and solve

$$z = -e - \{C_h(\overline{A} + E_Q z) - C_h(\overline{A})\}, \qquad C_h = q_h - Q. \tag{5.ccxv}$$

The actual nonlinear row Lipschitz estimate makes its right side a contraction on a ball of radius $2\|e\|_\infty$ when the strong chart is fixed sufficiently small. It gives $q_h(\overline{A} + E_Q z) = B$ and $\|E_Q z\|_{Y_\beta} \le C\|e\|_\infty$. The specified node-gauge extension in eq. 3.mcclxxxiv converts this into a literal field U^{rep} with the desired mean, preserving action.

To estimate the change, integrate the first variation along $\overline{A}+tE_Qz$. Its curvature is uniformly bounded on the small strong chart. Equation eq. 3.mccxc uses only that curvature bound and the physical $W^{1,1}$ norm of E_Qz, not a second derivative of $\overline{A}$. The smooth seeds have bounded overlap, amplitude and first differences, so $\|E_Qz\|_{1,h}+\|DE_Qz\|_{1,h}\le C\mathcal{V}\|z\|_\infty$. The normal-to-left differential has the same estimate, by its explicit bounded coefficients and their first differences. This proves eq. 5.ccxiv. The actual gauge and seed bounds also keep the repaired field inside the strong chart. □

The correction E_Qz need not be horizontal. When comparing its energy with the minimum, apply the literal-to-mean slice once more. It preserves the action and puts the trial in $q_h=B$ and $RD^\sharp A=0$, where the proved chord minimum applies.

Proposition 5.29 (Real adjacent-mesh action comparison)**.** On a fixed sufficiently small real coarse ball,

$$\sup_{\|B\|_\infty<\rho}\frac{|S_{j+1}(B)-S_j(B)|}{\mathcal{V}}\le C_\beta h_j^\beta. \tag{5.ccxvi}$$

The constants are independent of M,j. ■

Proof. Interpolate the literal real background potential at spacing $h=h_j$ by the proved periodic $C^{1,\beta}$ interpolation, and sample that same potential at spacing h/L. The refined trial has a uniformly small X_β norm. The two Wilson energies differ by at most $Ch^\beta\mathcal{V}$, since each differs from the energy of this common continuous interpolation by that bound.

The one-step mean of the refined trial differs from the original spacing-h field by $Ch^{2+\beta}$ in raw link logarithm. In physical spacing-h potential coordinates this is $Ch^{1+\beta}$. Apply Lem. 5.26 to the remaining j actual mean steps. Their final logarithms therefore differ by at most $Ch^{1+\beta}$. Thus the refined trial has literal final mean $e^{i(B+e)}$ with $\|e\|_\infty\le Ch^{1+\beta}$. Repair it by Lem. 5.28 and compare its action with the actual spacing-h/L minimum. This gives $S_{j+1}(B)\le S_j(B)+Ch^\beta\mathcal{V}$.

For the reverse inequality interpolate the spacing-h/L background, and sample its interpolation on the spacing-h grid. One-step consistency compares its actual fine first mean with that coarse sample. The remaining j means again produce an error $Ch^{1+\beta}$. The same energy and repair estimates give $S_j(B)\le S_{j+1}(B)+Ch^\beta\mathcal{V}$.

All transferred potentials have norm at most $C_\beta\rho$. The plain-mean Lipschitz bound at zero also gives $\|e\|_\infty\le C_\beta\rho$, in addition to its mesh-dependent bound. This ensures uniform admission even for the initial coarse meshes, where $h^{1+\beta}$ need not be small by itself. Choose the real ball once so this bound, and its slice and repair images, lie inside the fixed minimum chart. The local minimum is then applicable in both directions; no global minimizer or continuum critical field is invoked. □

The uniform complex bound needed below is simpler. For the actual holomorphic background, $\|F\|_\infty\le C\rho$ and a plaquette action is at most $Ch^4\rho^2$ in absolute value. There are $O(h^{-4}\mathcal{V})$ plaquettes. Consequently on a fixed complex coarse ball,

$$|S_j(B)|/\mathcal{V}\le C,\qquad j\ge 0. \tag{5.ccxvii}$$

5.9.2. *Small real values imply small fixed-order Taylor jets.*

Lemma 5.30 (A quantitative real-to-jet estimate)**.** Fix $m\ge 1$ and a complex polydisc. Suppose f is holomorphic there with $|f|\le C$, and $|f|\le\varepsilon$ on a fixed smaller real cube centered at zero. Then

$$|\partial_1\cdots\partial_m f(0)|\le C_m\varepsilon(1+\log(1/\varepsilon))^{2m},\qquad 0<\varepsilon<\tfrac12. \tag{5.ccxviii}$$

The constant depends only on the two radii, m and C. ■

Proof. Rescale the real cube to $[-1,1]^m$ and choose $R>1$ so that the product of closed Bernstein ellipses of parameter R lies inside the holomorphy domain. Expand in product Chebyshev polynomials: $f(t)=\sum_{n\in\mathbb{N}^m}c_n\prod_i T_{n_i}(t_i)$. Real orthogonality gives $|c_n|\le 2^m\varepsilon$. Analyticity on the

ellipses gives $|c_n| \le C_m R^{-|n|_1}$: write $t_i = (w_i + w_i^{-1})/2$ and apply the Laurent coefficient estimate on $|w_i| = R$ in each variable.

The derivative at zero has weight $\prod_i |T'_{n_i}(0)| \le \prod_i n_i \le (1 + |n|_1)^m$. For $|n|_1 \le N$, the real bound therefore gives at most $C_m \varepsilon (1+N)^{2m}$ after counting the multi-indices. The tail is at most $C_m R^{-N}(1+N)^{2m-1}$, by summing the geometric series with this fixed polynomial weight. Choose N to be the smallest integer at least $\log(1/\varepsilon)/\log R$. Restore the fixed rescaling of the cube to prove the lemma. The differentiated series is absolutely convergent. No Cauchy estimate has been applied to a real-only supremum bound. □

Take real coarse profiles $b_1, \dots, b_m$ with supremum norm at most one and apply the lemma to

$$f(z) = \mathcal{V}^{-1}(S_{j+1} - S_j)\left(\sum_{i=1}^{m} z_i b_i\right).$$

The complex polydisc and real cube can be chosen independently of profiles, depth and volume, by taking $m \max_i |z_i|$ smaller than the fixed action ball. Equations eq. 5.ccxvi and eq. 5.ccxvii give, for $2 \le m \le 4$,

$$\frac{|D^m(S_{j+1} - S_j)(0)[b_1, \dots, b_m]|}{\mathcal{V}} \le C_m h_j^{\beta} (1+j)^{2m}. \tag{5.ccxix}$$

The finitely many initial j for which the smallness parameter exceeds $1/2$ are covered by the uniform complex Cauchy bound. Complex Fourier profiles are permitted as well: expand each e^{ipx} into its real cosine and sine profiles and use complex multilinearity, at a fixed cost at most 2^m.

5.9.3. *Uniform analytic strips for the actual action vertices.* We spell out the locality assertion needed to pass from a small symbol to a small kernel. Resolve a coarse bond into its cell location and one of its twelve component-colour indices. In the infinite-volume translation-covariant notation write the m-vertex as $V_{j,m}(x_1, \dots, x_m)$, retaining these finite indices. Fix the first location at zero and define its star-weighted norm by

$$\|V\|_{m,\sigma} = \max_{\text{component-colour indices}} \sum_{x_2,\dots,x_m} e^{\sigma \sum_{i=2}^{m} |x_i|_1} |V(0, x_2, \dots, x_m)|. \tag{5.ccxx}$$

Here infinite-volume notation refers to kernels of compactly supported variations, obtained from their local periodic limits. No finite total action on a nondecaying infinite field is assigned. One may sum the finite index maximum instead, changing only a fixed constant. On a torus use periodic distance. This norm also controls any weight given by the length of a shortest tree connecting the inputs, since that length is no larger than the displayed star.

Lemma 5.31 (Uniform exponential moments at fixed Taylor order)**.** For $m = 2, 3, 4$ there are $\sigma_0 > 0$ and C_m, independent of j, M, such that

$$\|V_{j,m}\|_{m,\sigma_0} \le C_m. \tag{5.ccxxi}$$

The infinite-volume kernels exist, and finite-torus kernels are their periodic sums in the $m-1$ relative input variables. Their symbols are consequently bounded holomorphic functions on a fixed polystrip in $4(m-1)$ complex momentum variables. ■

Proof. Split the Wilson action into unit-cell densities, assigning a plaquette by its base point. Each density is uniformly analytic in a small local X_β ball and has bounded analytic norm: the fine plaquette factor is $O(h^4 \|A\|^2_{X_\beta})$, and only $O(h^{-4})$ plaquettes lie in one physical cell. Its derivatives of every fixed order are therefore bounded multilinear local maps, uniformly in h. Their carriers have bounded physical diameter.

For a background jet with unit inputs at $y_1, \dots, y_a$, eq. 3.mccxcvi bounds its cell-X_β norm at z by $C_a e^{-s \max_i |z - y_i|_1}$. This holds also for the literal background, as proved there. In the chain rule for an order-m cell-action jet, each partition of the m inputs contributes a product of such bounds.

For each block of size a, its maximum is at least a^{-1} times its sum. Since $a \le m$, every term is bounded by $C_m e^{-(s/m)\sum_i |z-y_i|_1}$, with a fixed change of the constant for the bounded action carrier. Summing action cells gives

$$|V_{j,m}(y_1,\dots,y_m)| \le C_m \sum_z e^{-(s/m)\sum_i |z-y_i|_1}. \tag{5.ccxxii}$$

With $y_1 = 0$, the triangle inequality implies $\sum_{i=2}^m |y_i|_1 \le (m-1)\sum_i |z-y_i|_1$. Split the exponent into two halves. One half pays a star weight of any size below $s/(2m(m-1))$; the other is summable in z and all relative inputs after a further fixed decrease of that weight. For example choose $\sigma_0 < s/(4m(m-1))$ and sum first in each y_i and then in z. This proves the moment bound, uniformly on periodic tori as well.

We justify the infinite-volume and periodization assertions at the level of these actual jets. At zero field the joint inverse has its translation-covariant Bloch representation and exponential cell kernels. Its finite periodic kernel is the periodic sum of the infinite one. Differentiate the actual fixed-point equation at order a. Its order-a unknown is the same linear joint inverse applied to a finite sum of products of lower-order jets and the finite-carrier derivatives of the actual residual. Induction for $a \le m$ shows that each such jet is an absolutely convergent rooted-tree kernel expression. Its leaves are the distinct external differentiation slots. Once a lift of an internal vertex is chosen, the images of the input slots in each child subtree are independent. Expanding a periodic propagator chooses the lift of its child vertex; continuing down the tree then gives the independent periodic images of all external slots. There is no closed internal cycle imposing an extra image constraint. Thus periodic summation commutes with these multilinear contractions; this is not the generally false identity that periodization commutes with an ordinary pointwise product in one shared argument. The exponential bounds justify all changes of summation order.

Apply the same finite-order chain rule to the cell action and sum its cells. Its action root joins those child trees, still without a closed cycle. Fixing the first external input lift removes the common translation; the remaining $m-1$ external images are independent. This proves existence of the infinite-volume action vertex and exact periodization in each relative input, not just a subsequential compactness claim.

Finally Fourier series are absolutely convergent when the imaginary parts of the $m-1$ independent external momenta are sufficiently small. The moment bound gives a uniform bound on a smaller closed polystrip. Cell positions rather than bond midpoints can be used in this Fourier convention; changing that convention introduces only fixed bounded analytic phases. □

Translation covariance now turns eq. 5.ccxix into a pointwise bound for the difference of vertex symbols. Choose complex unit component- colour profiles with allowed momenta $p_1,\dots,p_m$ whose sum is zero. The volume-normalized m-linear functional is exactly its vertex symbol at those momenta. It follows that

$$\sup_{p_2,\dots,p_m \text{ real}} |\widehat{V}_{j+1,m}(p_2,\dots,p_m) - \widehat{V}_{j,m}(p_2,\dots,p_m)| \le C_m h_j^{\beta} (1+j)^{2m}. \tag{5.ccxxiii}$$

Initially the momenta are on a finite-torus grid. Pass to arbitrarily large tori using the just proved periodization, then use continuity and density of the grids. This establishes the bound for every real momentum, with no volume-dependent Fourier inversion loss.

5.9.4. *Interpolation into a smaller complex momentum strip.* We need an elementary several-variable refinement of the three-lines principle. Let F be periodic and holomorphic in $|\operatorname{Im} p_i| < a_0$, $1 \le i \le d$, and bounded by C there. Suppose $|F| \le \varepsilon$ on the real torus. Choose $0 < a < a_0' < a_0$. Apply the three-lines bound in one variable, keeping the remaining variables real, between its real axis and the line of imaginary part a_0' or $-a_0'$. This gives $C^{1-\theta}\varepsilon^{\theta}$ for $|\operatorname{Im} p_1| \le a$, where $\theta = 1 - a/a_0'$. Periodicity permits the argument on a bounded cylinder, or equivalently by the annulus maximum principle. Keeping this first variable in its reduced strip, repeat in the second

variable. Its real-axis bound is now the improved preceding bound, whereas its outer-line bound is still C. After all d variables,

$$\sup_{|\operatorname{Im} p_i|\le a} |F(p)| \le C^{1-\Theta}\varepsilon^{\Theta}, \qquad \Theta = (1-a/a_0')^d. \tag{5.ccxxiv}$$

This argument uses the uniform bound on the full larger polystrip at each step. Smallness on the real torus alone is not assumed to imply the same smallness throughout that larger strip.

Apply eq. 5.ccxxiv to the vertex-symbol difference. The larger-strip bound is supplied by Lem. 5.31; the real bound is eq. 5.ccxxiii. Since $d = 4(m-1)$,

$$\sup_{|\operatorname{Im} p_i|\le a} |\Delta\widehat{V}_{j,m}(p)| \le C_m\{h_j^{\beta}(1+j)^{2m}\}^{(1-a/a_0')^{4(m-1)}}. \tag{5.ccxxv}$$

Again the finitely many initial indices are absorbed into C_m. Fourier contour shifting on this reduced polystrip gives

$$|\Delta V_{j,m}(0,x_2,\dots,x_m)| \le C_m\{h_j^{\beta}(1+j)^{2m}\}^{\Theta_m} e^{-a\sum_{i=2}^m |x_i|_1}, \quad \Theta_m = (1-a/a_0')^{4(m-1)}. \tag{5.ccxxvi}$$

Shift each scalar momentum coordinate with the sign prescribed by the corresponding relative lattice coordinate. Periodicity makes the opposite contour faces cancel. The fixed analytic midpoint phases, if used, change only C_m.

Proposition 5.32 (Convergence of the actual classical Taylor triple)**.** For every $0<\alpha<1$ there are $\sigma_\alpha>0$ and C_α such that, for $m=2,3,4$,

$$\|V_{j+1,m} - V_{j,m}\|_{m,\sigma_\alpha} \le C_\alpha L^{-\alpha j}. \tag{5.ccxxvii}$$

The limits $V_{\infty,m}$ exist in these weighted kernel norms and

$$\|V_{\infty,m} - V_{j,m}\|_{m,\sigma_\alpha} \le C_\alpha L^{-\alpha j}. \tag{5.ccxxviii}$$

The bounds hold uniformly on compatible finite tori by periodic summation. They concern the actual Wilson background Taylor vertices; no continuum minimizing field has been assumed. ■

Proof. Choose $\alpha<\beta<1$, and then choose the reduced strip $a>0$ so small that $\beta(1-a/a_0')^{12}>\alpha$. This gives $\beta\Theta_m>\alpha$ for every $m=2,3,4$. Take $\sigma_\alpha = a/2$. Summing eq. 5.ccxxvi with this weight costs only the convergent geometric lattice sum $\sum_{x_2,\dots,x_m} e^{-(a/2)\sum|x_i|_1}$. Its remaining polynomial in j is absorbed by the strict exponent margin $\beta\Theta_m-\alpha>0$. This proves eq. 5.ccxxvii. The geometric series in j converges in the Banach kernel norm, proving eq. 5.ccxxviii.

The periodization proved above preserves the weighted bound: periodic relative distances are no larger than their lifted distances, and the absolute value of a periodic sum is bounded by the sum of absolute values. Thus no finite-volume factor enters the result. □

The classical Taylor triple now has a genuine geometric mesh comparison. Passing this estimate through a Gaussian polarization requires a uniform Lipschitz estimate for that polarization in the same kernel norms. That is a separate analytic step. The cumulative one-loop comparison alone did not prove it, and no convergence of individual Gaussian increments is asserted here.

5.10. **The actual one-step Gaussian polarization.** The Gaussian coefficient of a step depends only on the second, third and fourth Taylor forms of its input effective action. Here the constraint is the actual plain group average and the gauge is the actual symmetric contour gauge. We derive this dependence, including the Haar and coarea factors, and prove a volume-independent Lipschitz bound. The resulting implication from convergence of the actual Taylor triples to convergence of the actual increments is quantitative.

Fix an odd step size L. The input lattice has unit spacing; it is not the original fine lattice of spacing L^{-k}. There are finitely many input coordinates, depending only on L, in one outer

coarse cell. Use orthonormal Pauli colour coordinates and product Lebesgue measures for group logarithms. Write

$$S(B) = S_k(e^{iB}), \qquad m(B) = i^{-1}\log \mathcal{M}(e^{iB}), \qquad m(0) = 0, \quad Q = Dm(0).$$

The background for all Taylor forms in this section is the trivial field. Let $\ell_x(B)$ be the actual contour coordinate from the node of a block to its nonnode site x, and put

$$\begin{aligned} F_g(B) &= \sum_x \left(1 - \tfrac{1}{2}\operatorname{tr} T_x(e^{iB})\right) \\ &= \sum_x \left(1 - \cos|\ell_x(B)|\right). \end{aligned} \tag{5.ccxxix}$$

This is the canonical $\tau = 1$ Gaussian gauge convention. The proof also permits fixed gauge weights in a fixed positive compact interval. Coupling powers and field-independent reference-gauge constants have been factored out; the two-coupling mismatch is a separate term. On complex coordinates the trace formula is its analytic continuation; the cosine expression is its power series in the bilinear colour square of ℓ_x.

5.10.1. *The actual free effective action supplies the gap.* We first record why the free input of every homogeneous depth has the lower bound used below. In the physical fine spacing $h = 1/n$, the actual straight derivative is

$$(Q_h a)_\mu(y) = h^5 \sum_{x\in B_n(y)} \sum_{r=0}^{n-1} a_\mu(x + rhe_\mu).$$

Telescoping the two coarse differences gives

$$(\mathrm{d}_c Q_h a)_{\mu\nu}(y) = h^6 \sum_{x\in B_n(y)} \sum_{r,s=0}^{n-1} (\mathrm{d}_h a)_{\mu\nu}(x + rhe_\mu + she_\nu) =: (T_h d_h a)_{\mu\nu}(y). \tag{5.ccxxx}$$

These weights are positive. A row sum is $h^6 n^6 = 1$. A fixed fine plaquette has n^2 choices of (r, s), each with exactly one starting block, so its column sum is $h^6 n^2 = h^4$. Jensen's inequality therefore proves $\|T_h F\|_2 \le \|F\|_{2,h}$, with constant one.

The plain derivative differs from this straight derivative by the actual coarse gradient of the linear node factor. Their curls agree. The physical derivative of the actual background minimizes fine curl energy at its fixed mean: its stationarity gives zero energy cross term against every zero-mean variation. A literal fine node gauge preserves this energy. The exact smooth row lift of Lem. 3.180, followed by the specified node gauge when passing to the plain mean, has uniform physical gradient energy. Consequently

$$\mathrm{d}^\sharp \mathrm{d} \ \le T_2 := S_k''(0) \ \le CI. \tag{5.ccxxxi}$$

The lower bound is the actual flux contraction applied to the minimizing lift. The upper bound inserts the exact smooth trial. It is uniform in the inner homogeneous depth, including depth zero. No continuum coercivity estimate is substituted for these actual averaging identities.

Lemma 5.33 (The fixed one-step contour gap)**.** Let $K_g = D^2 F_g(0)$. On compatible tori, uniformly in volume,

$$\mathrm{d}^\sharp \mathrm{d} + K_g + Q^\sharp Q \ge c_L I. \tag{5.ccxxxii}$$

The actual plain row Q is onto with a bounded right inverse depending only on L. Thus, whenever $\mathrm{d}^\sharp \mathrm{d} \le T_2$ and $\|T_2\| \le M$,

$$\mathbb{L}(T_2) = \begin{pmatrix} T_2 + K_g & Q^\sharp \\ Q & 0 \end{pmatrix} \tag{5.ccxxxiii}$$

has a uniformly bounded inverse. Only positivity on $\ker Q$ is required. ∎

Proof. The actual pivot chart supplies the onto row and its bounded local pivot right inverse at the trivial field. All its coefficients and inverse pivot coefficients belong to one fixed L-block geometry. For the gap use the Bloch decomposition by these blocks. The finite matrices of d, K_g, Q are continuous on the compact Bloch torus.

Suppose the three squared terms vanish in one fibre. A closed fine field is a gradient, except for constant vector modes at zero Bloch frequency. This follows also from the fine Fourier modes $p = (\theta + 2\pi r)/L$: at every nonzero mode the closedness equation says its vector is parallel to the fine gradient symbol. At zero frequency write the field as $D\lambda + c$. The actual plain average of its gradient is the coarse gradient of the node values, and its average of c is Lc. In that zero-frequency block the former gradient is zero, so $Qu = 0$ gives $c = 0$.

For the remaining gradient, contour covariance gives

$$D\ell(0)D\lambda(x) = \lambda(y) - \lambda(x).$$

The contour term therefore forces λ to be constant inside each block. Its plain row is the coarse gradient of that block value. Zero coarse gradient makes the field itself zero, at both zero and nonzero Bloch frequencies. Every Bloch matrix has zero kernel, so compactness proves eq. 5.ccxxxii. In particular $T_2 + K_g$ is uniformly coercive on $\ker Q$. The right inverse and the kernel equation prove the bounded full saddle inverse, by lifting the datum and recovering the multiplier from the remaining functional. □

5.10.2. *The global group-coordinate normalizer.* Define the actual Lagrangian in full input group coordinates and coarse multiplier coordinates by

$$\Lambda_C(B, v) = S(B) + F_g(B) + \langle v, m(B) - C\rangle. \tag{5.ccxxxiv}$$

Its stationary branch is $z(C) = (B(C), v(C))$, with $z(0) = 0$. Its Hessian at zero is $\mathbb{L}(T_2)$. The implicit function theorem therefore gives the actual local branch; the already constructed homogeneous background is this branch after its specified residual contour gauge. One can see the gauge removal directly: along a fine gauge equal to one at coarse nodes both S and m are constant. The derivative of ℓ_x in those gauge directions is an invertible endpoint map near zero. Stationarity of eq. 5.ccxxxiv thus implies $\ell(B(C)) = 0$: the derivative of the actual trace weight in its ℓ_x variable is $(\sin|\ell_x|/|\ell_x|)\ell_x$, whose scalar factor is nonzero on the chosen small branch. Hence $F_g = DF_g = 0$ at the centre, and this branch is physically constrained critical for S. At every such centre its gauge Hessian is $D\ell^\sharp D\ell$, also the Hessian of the quadratic logarithmic gauge. Thus their Gaussian log normalizers, after subtraction at zero, agree along this physical branch. All vertices below use the literal trace function eq. 5.ccxxix.

Let $j_f(B)$ and $j_c(C)$ be the products of the actual SU(2) exponential Haar densities. Factor out the classical weight $e^{-S(B(C))}$. The Gaussian log normalizer in these global coordinates is

$$\begin{aligned}\Phi(C) = {} & \log j_f(B(C)) - \log j_c(C) \\ & - \tfrac{1}{2}\log\left|\det D_z^2\Lambda_C(z(C))\right| - (\text{the same at } C = 0).\end{aligned} \tag{5.ccxxxv}$$

The omitted Gaussian powers and contour normalization are independent of C. At real C the saddle determinant has a fixed nonzero sign. Its local analytic logarithm means the logarithm of its ratio to the value at zero, continued from one.

Here is why this formula is the actual constrained Gaussian. In global logarithms the exact conditional measure has density

$$j_f(B)\, j_c(C)^{-1}\delta(m(B) - C)\, dB.$$

Its tangent constraint is $Dm(B(C))v = 0$. The quadratic form is the complete Lagrangian Hessian, including $v(C)D^2m$, plus the actual contour Hessian. The constrained Gaussian identity Lem. 5.4 replaces its quadratic integral and its coarea determinant by the full saddle determinant

in eq. 5.ccxxxv. In particular a coordinate-dependent coarea factor has not been discarded. Using left exponential fluctuation coordinates instead would transfer the centre Haar factor into this determinant; retaining just one of these changes would alter the answer.

5.10.3. *All vertices and the colour cancellation.* Put

$$\mathbb{G} = \mathbb{L}(T_2)^{-1}, \quad ZC = \mathbb{G}(0, C), \quad HC = (ZC)_B, \quad V_r = D_z^r \Lambda_0(0) \quad (r = 3, 4). \tag{5.ccxxxvi}$$

Differentiating stationarity gives

$$z'(0)C = ZC = (HC, -\Delta_{\mathrm{next}} C).$$

The sign of the multiplier follows from the plus sign in eq. 5.ccxxxiv. More explicitly, set $S_{\mathrm{next}}(C) = S(B(C))$. Its Hessian is $\Delta_{\mathrm{next}} = H^\sharp T_2 H$. Differentiating the zero contour coordinate gives $K_g H = 0$; pairing the linear stationary equation with HC' then gives $\upsilon'(0)C = -\Delta_{\mathrm{next}} C$.

For clarity, if $z_i = (a_i, \alpha_i)$, the full vertices are

$$\begin{aligned} V_3[z_1, z_2, z_3] &= (D^3 S + D^3 F_g)[a_1, a_2, a_3] + \sum_{i=1}^{3} \langle \alpha_i, D^2 m[a_j, a_l] \rangle, \\ V_4[z_1, z_2, z_3, z_4] &= (D^4 S + D^4 F_g)[a_1, a_2, a_3, a_4] + \sum_{i=1}^{4} \langle \alpha_i, D^3 m[a_1, \dots, \widehat{a}_i, \dots, a_4] \rangle. \end{aligned} \tag{5.ccxxxvii}$$

In the first sum $\{j, l\}$ is the complement of $\{i\}$. The constraint and gauge terms are fixed one-step terms; they are present before taking a Gaussian. There is no multiplier at zero, so no additional $D^4 m$ term occurs.

The scalar

$$z \longmapsto \log |\det D_z^2 \Lambda_0(z)|$$

is invariant under every constant colour rotation. Indeed S, F_g, m have their actual covariance, and the simultaneous adjoint rotation of all field and multiplier coordinates is orthogonal with determinant one. Its differential at zero is an invariant linear functional on a direct sum of adjoint colour copies. Each copy has no invariant linear functional. Therefore, in every field and multiplier direction v,

$$\mathrm{Tr}(\mathbb{G} V_3[v]) = 0. \tag{5.ccxxxviii}$$

This is a finite-volume identity, before any trace per volume or infinite-volume limit.

Proposition 5.34 (Actual Gaussian Hessian formula)**.** The Hessian of eq. 5.ccxxxv is

$$\begin{aligned} \Phi''(0)[C, C'] = {} & \tfrac{2}{3} \langle C, C' \rangle - \tfrac{2}{3} \langle HC, HC' \rangle \\ & - \tfrac{1}{2} \mathrm{Tr}(\mathbb{G} V_4[ZC, ZC']) \\ & + \tfrac{1}{2} \mathrm{Tr}(\mathbb{G} V_3[ZC] \mathbb{G} V_3[ZC']). \end{aligned} \tag{5.ccxxxix}$$

It depends on S only through $(D^2 S(0), D^3 S(0), D^4 S(0))$. ∎

Proof. The second derivative of a matrix logarithm is the trace of its inverse times its second derivative, minus the trace of the two first derivatives separated by inverse factors. Along $z(C)$, the first matrix derivative is $V_3[ZC]$; the second is $V_4[ZC, ZC'] + V_3[z''(0)[C, C']]$. The last term has zero trace by eq. 5.ccxxxviii, in its full augmented direction. Thus no unknown second background response survives.

In the present Pauli normalization the exact one-bond Haar density is

$$j(B) = \left(\frac{\sin |B|}{|B|} \right)^2, \qquad \log j(B) = -|B|^2/3 + O(|B|^4).$$

Its first derivative is zero and its second derivative is $-2/3$ times the colour inner product. This gives the two Haar terms in eq. 5.ccxxxix, including their signs. The complete saddle contains

all the constraint and contour contributions in eq. 5.ccxxxvii. Combining the terms proves the formula. □

5.10.4. *A norm for the actual Taylor triple.* Assign input bonds, contour coordinates and coarse multipliers to their outer L-cells. The number of coordinates per cell is $N_L < \infty$. For a scalar symmetric r-tensor W, define

$$\|W\|_{s,r} = \max_{1\le j\le r} \sup_{i_j} \sum_{(i_l)_{l\neq j}} e^{s\,\mathfrak{t}(i_1,\dots,i_r)} |W(i_1,\dots,i_r)|. \tag{5.ccxl}$$

Here $\mathfrak{t}$ is the least cell-tree length joining the indicated cells, and sums include bond and colour indices. For $r = 2$ this is the usual maximum absolute weighted row and column Schur norm. On a torus the least torus tree length is used; retained lifted kernels can instead use their cover tree lengths.

Use the derivative triple $\mathfrak{J} = (T_2, T_3, T_4)$, with $T_r = D^r S(0)$, and norm

$$\|\mathfrak{J}\|_s = \sum_{r=2}^{4} \|T_r\|_{s,r}. \tag{5.ccxli}$$

If S_3, S_4 denote the homogeneous Taylor forms, their polarizations have the additional fixed factors 3! and 4! in T_3, T_4. This fixes the convention without changing the asserted Lipschitz property.

Assume $\|\mathfrak{J}\|_s \le M$. The actual action-tree expansion supplies such norms on its smaller fixed ball. The fixed one-step constraint and gauge vertices have finite carriers and bounded norms of every fixed order. Thus the augmented V_3, V_4 satisfy the same kind of bounds, depending only on M, L, s.

Lemma 5.35 (Inverse kernels uniformly over the input triples)**.** On the class $\|\mathfrak{J}\|_s \le M$, $T_2 \ge \mathrm{d}^\sharp\mathrm{d}$, there is a fixed $t > 0$ such that $\mathbb{G}$ and Z have bounded weighted Schur norms at rate t. For two such triples,

$$\|\mathbb{G} - \mathbb{G}'\|_{t,2} + \|Z - Z'\|_{t,2} \le C\|T_2 - T_2'\|_{s,2}. \tag{5.ccxlii}$$

Constants are independent of the torus volume and the inner homogeneous depth. ■

Proof. The unweighted inverse is uniform by Lem. 5.33. For a bounded one-Lipschitz cell function ψ, the weighted Schur norm of the coefficients gives

$$\|e^{a\psi}\mathbb{L}(T_2)e^{-a\psi} - \mathbb{L}(T_2)\| \le C(M, L, s)|a|$$

for $|a| \le s/2$. A normalized Neumann series gives uniformly bounded inverses for a fixed nonzero a. Truncated distance weights yield exponential cell entries of $\mathbb{G}$. At a fixed smaller rate their row and column sums are bounded: cell growth is polynomial of degree four, with N_L internal coordinates. This also proves the bounds on Z, a fixed column inclusion of $\mathbb{G}$.

Only the field-field T_2 block differs between the two full saddle matrices. The exact resolvent identity is

$$\mathbb{G} - \mathbb{G}' = -\mathbb{G}\begin{pmatrix} T_2 - T_2' & 0 \\ 0 & 0 \end{pmatrix}\mathbb{G}'.$$

Weighted Schur norms are submultiplicative. This proves eq. 5.ccxlii and its column statement for Z. □

5.10.5. *Rooted trace contractions have no volume cost.* Let $\Pi(\mathfrak{J})$ be the coarse kernel of eq. 5.ccxxxix. All statements below concern absolute coordinate sums; an operator norm alone would not suffice.

For the quartic trace, fix one external coarse coordinate c. Sum first its response column Z_{ic}, then the quartic vertex rooted at i, then the second external response row. The internal loop kernel is bounded by its Schur norm. The distance from c to the other external coordinate is no greater

than the first response distance, the vertex tree length and the last response distance combined. Hence, at a common smaller rate,

$$\|\mathrm{Tr}(\mathbb{G}V_4[Z\,\cdot, Z\,\cdot])\|_{t,2} \le \|\mathbb{G}\|_{t,2}\|V_4\|_{t,4}\|Z\|_{t,2}^2. \tag{5.ccxliii}$$

For the pair of cubic vertices, root the first vertex at that first response index. Sum one intervening Green kernel, then the second cubic vertex rooted at its endpoint, then the second external response. The other loop kernel is bounded pointwise by its Schur norm. This gives

$$\|\mathrm{Tr}(\mathbb{G}V_3[Z\,\cdot]\mathbb{G}V_3[Z\,\cdot])\|_{t,2} \le \|\mathbb{G}\|_{t,2}^2\|V_3\|_{t,3}^2\|Z\|_{t,2}^2. \tag{5.ccxliv}$$

The scalar tensor norms already sum all internal coordinate indices. No free vacuum component remains once the first external coordinate is fixed. In particular neither estimate has a volume factor. The Haar contraction is a direct weighted matrix product of $H^\sharp$ and H.

Proposition 5.36 (Lipschitz Gaussian polarization)**.** For actual input actions whose Taylor triples satisfy the preceding boundedness and lower bound, there are $t > 0$ and $C < \infty$, independent of depth and volume, such that

$$\|\Pi(\mathfrak{J}) - \Pi(\mathfrak{J}')\|_{t,2} \le C\|\mathfrak{J} - \mathfrak{J}'\|_s. \tag{5.ccxlv}$$

Each polarization itself has bounded weighted norm. The full constraint and gauge vertices in the formula are fixed actual one-step vertices. ■

Proof. The augmented differences of V_3, V_4 are precisely the field vertices $T_3 - T_3'$, $T_4 - T_4'$. All other vertex terms are fixed. Subtract the two formulas eq. 5.ccxxxix by replacing one factor at a time. For a propagator or response difference use eq. 5.ccxlii. For a vertex difference use its rooted tensor norm. Each remaining contraction is bounded by eq. 5.ccxliii or eq. 5.ccxliv. The Haar difference uses the same weighted product bound. These are finitely many terms with fixed degrees, which proves the assertion. □

These estimates also pass to compatible bulk kernels: every rooted contraction has an absolute exponential majorant. Period images are summed with that same majorant, and winding contributions tend to zero along a growing compatible volume. Thus eq. 5.ccxlv holds for the common infinite-lattice companions whenever the input triples have those actual companions.

5.10.6. *The implication for actual increments.* For the invariant physical total, the colour-averaged second-moment readout of the coefficient is continuous in this norm. Explicitly its standard off-diagonal form is

$$b(\Pi) = \sum_{u\in\mathbb{Z}^4} u_\mu u_\nu\, \overline{\Pi}_{\mu\nu}(u), \qquad \mu \ne \nu,$$

after the actual Ward and lattice symmetries have identified this form. The bar here denotes the orthonormal colour average, not complex conjugation. Since $|u|^2 \le C_t e^{t|u|}$, absolute moments give

$$|b(\Pi(\mathfrak{J})) - b(\Pi(\mathfrak{J}'))| \le C_t\|\mathfrak{J} - \mathfrak{J}'\|_s. \tag{5.ccxlvi}$$

The coefficient convention is the physical one, with the outer coarse lattice rescaled to unit spacing.

Corollary 5.37. The actual homogeneous Taylor triples have common bulk companions and, by Prop. 5.32, for each $0 < \alpha < 1$ obey the following bound at some fixed positive rate s_α:

$$\|\mathfrak{J}_{k+1} - \mathfrak{J}_k\|_{s_\alpha} \le C_\alpha L^{-\alpha k}, \qquad 0 < \alpha < 1. \tag{5.ccxlvii}$$

Then their actual Gaussian increments $b_{k+1}^0 = b[\Phi_{k+1}]$ satisfy, with a fixed index shift absorbed into the constant,

$$|b_{k+1}^0 - b_k^0| \le C_\alpha L^{-\alpha k}, \qquad |b_k^0 - c_0| \le C_\alpha L^{-\alpha k}, \qquad c_0 = \frac{11}{3\pi^2}\log L. \tag{5.ccxlviii}$$

■□

Proof. The difference bound follows from eq. 5.ccxlvi. Its tail is a geometric series, so the increments converge with the displayed rate to some constant. The exact physical stage composition and the proved massive lattice comparison give eq. 5.cc. Dividing that cumulative identity by the number of steps and taking the Cesaro limit identifies the constant as $11 \log L/(3\pi^2)$. This last argument uses the lattice coefficient, not a quoted continuum beta coefficient. □

The actual formula and its uniform Taylor-triple Lipschitz estimate complete the geometric increment argument on the homogeneous physical germ. The rooted star norm of Prop. 5.32 dominates eq. 5.ccxl, after the fixed L-cell grouping and the fixed Taylor-factor conventions, and therefore supplies eq. 5.ccxlvii. A formal response expansion alone does not establish that mesh comparison. A bounded cumulative determinant error alone would likewise not give geometric successive-increment convergence.

5.11. **The nonflat effective curl bound and the outer Gaussian gap.** The positive massive row Schur matrix does not, by itself, give a gap after its artificial row mass is subtracted. We prove the needed physical comparison. It gives the actual outer contour-gauge gap near a regular nonflat critical reference, uniformly in the inner fine depth.

This subsection uses a homogeneous inner torus, with fine spacing $h = L^{-k}$ and intermediate coarse spacing one. The real fine reference U is an actual constrained-critical field of literal mean $V = M^k U$, with

$$\|F(U)\|_\infty \le \kappa, \qquad \|j_U\|_\infty \le \gamma. \tag{5.ccxlix}$$

The current is the physical Wilson current in these units. The actual hierarchical means are on their regular branches. All ceilings below are fixed sufficiently small. Nontrivial holonomies are allowed. We construct statements on the critical germ provided by Prop. 3.117. We do not infer existence of this reference from coarse curvature alone.

Let Q be its re-centered, straightened, per-length field derivative. Let ω be its physical multiplier, so $j_U = Q^\sharp \omega$. The actual readout gives $\|\omega\|_\infty \le C\gamma$. Write S for the local physical critical action as a function of the intermediate coarse field. The straightened datum and literal datum differ by their admitted node gauge. The action is invariant under that gauge, so their scalar action germs coincide in the declared coarse normal coordinates.

5.11.1. *The flat flux row with its physical norm.* We first give the exact flat identity used in the comparison. Put $N = h^{-1}$. For fine translations τ_μ, define

$$A_h = h^4 \sum_{x \in B_h(0)} \tau_x, \qquad S_{\mu,h} = h \sum_{t=0}^{N-1} \tau_\mu^t.$$

The symbol A_h here is a block average, rather than an action. The set $B_h(0)$ is the actual centered N^4-site block; in one coordinate its indices are $-(N-1)/2, \dots, (N-1)/2$ for the odd $N = L^k$ used here. The flat field row is $Q^0_\mu = A_h S_{\mu,h}$, sampled at coarse bond origins. The coarse difference obeys

$$\tau_\mu^N - 1 = S_{\mu,h} D^+_{\mu,h}.$$

All these flat translations commute. Consequently

$$(\mathrm{d}_c Q^0 v)_{\mu\nu} = A_h S_{\mu,h} S_{\nu,h} (\mathrm{d}_h v)_{\mu\nu} =: Q^2_{\mu\nu} (\mathrm{d}_h v)_{\mu\nu}. \tag{5.ccl}$$

The actual fixed-L hierarchy has this same linear row: multiplying the block averages combines the base-L digits of the averaging sites and line positions, giving the displayed N-site formula.

Every weight of $Q^2_{\mu\nu}$ is nonnegative, its row sum is one, and its counting column sum is h^4. The latter identity follows by summing over all translated coarse blocks; each fine site has total block-average weight h^4, and each $S_{\mu,h}$ preserves that total weight. Jensen's inequality therefore gives

$$\|Q^2 f\|_{2,c} \le \|f\|_{2,h}. \tag{5.ccli}$$

Thus the flat flux row has its physical norm without a factor counting fine plaquettes. The identity is local. It may be used in a flat parallel frame on a cover without choosing a logarithm of a global holonomy.

5.11.2. *A curvature-small covariant flux comparison.*

Lemma 5.38 (Actual coarse curl controlled by fine curl)**.** For the actual rows at the real reference eq. 5.ccxlix,

$$\|\mathrm{d}_V Q v\|_{2,c} \le C\|\mathrm{d}_U v\|_{2,h} + C\kappa\|v\|_{2,h}. \tag{5.cclii}$$

The constants are uniform in k and volume. ■

Proof. For each coarse plaquette choose a fixed surrounding physical cube containing all four complete row carriers. The current-free local gauge construction gives a fine potential of size $C\kappa$ on this cube, after decreasing the curvature ceiling by its fixed radius factor. Its literal coarse mean has bond angles at most $C_L\kappa$: the actual path lengths and normalized mean majorants are bounded in these physical units. Both fine and coarse comparisons are made in this same real frame.

The actual coefficient comparison eq. 3.dcccxiii then gives $Q = Q^0 + \mathcal{R}_Q$ on these rows, with $\|\mathcal{R}_Q v\|_{2,c} \le C_L\kappa\|v\|_{2,h}$ on their fixed enlarged carriers. There is no derivative of this row error at the fine scale: the following difference acts on the coarse unit lattice. Its L^2 norm is bounded by a fixed constant. Also $\mathrm{d}_V - \mathrm{d}_c = O_L(\kappa)$ as a coarse matrix, while $(\mathrm{d}_U - \mathrm{d}_h)v = O_L(\kappa)v$ on the fine cube in this frame. Insert these three comparisons in eq. 5.ccl and use eq. 5.ccli. They give the asserted bound for that plaquette and its fixed neighbours.

The gauge transformations are unitary, so these local norm bounds are covariant. A fine cell occurs in only a fixed number of the surrounding coarse cubes. Summing their squared estimates proves eq. 5.cclii. No single global flat frame has been selected. □

5.11.3. *The effective Hessian, with its constraint curvature.* Let

$$H_L = \mathscr{D}_U - \Delta_U^{(2)}[\omega], \qquad G_1^{-1} = H_L + D_U R D_U^\sharp + Q^\sharp a_f Q.$$

This is the actual physical second precision by eq. 3.cmxxi. Its row lift is

$$\mathcal{H} = G_1 Q^\sharp (Q G_1 Q^\sharp)^{-1}.$$

It is horizontal, $Q\mathcal{H} = 1$, and $\|\mathcal{H}a\|_{\mathcal{E}} \le C\|a\|_{2,c}$. The reference coarse spacing is one, so all row weights a_f are bounded above and below.

Proposition 5.39 (Physical nonflat effective curl bound)**.** The actual critical action germ satisfies

$$\begin{aligned} D^2S(V)[a,a] &\ge c\|\mathrm{d}_V a\|_{2,c}^2 - C(\kappa+\gamma)\|a\|_{2,c}^2, \\ |D^2S(V)[a,b]| &\le C\|a\|_{2,c}\|b\|_{2,c}. \end{aligned} \tag{5.ccliii}$$

Its exact Hessian is

$$D^2S(V) = \mathcal{H}^\sharp H_L \mathcal{H} = (QG_1Q^\sharp)^{-1} - a_f. \tag{5.ccliv}$$

These are local statements at the given physical critical reference. ■

Proof. Differentiate the actual critical graph in its fixed reference Landau slice. Its first derivative is $\mathcal{H}$, by the physical row-saddle identity. Twice differentiating its constraint gives

$$QA''[a,b] + q''(0)[\mathcal{H}a, \mathcal{H}b] = 0.$$

The first derivative of the fine action is $Q^\sharp\omega$. The second chain rule for the action therefore gives $D^2S = \mathcal{H}^\sharp H_L \mathcal{H}$. Equivalently the differentiated stationarity equation is

$$H_L\mathcal{H} = Q^\sharp D^2 S.$$

The horizontality of $\mathcal{H}$ annihilates its gauge penalty. After adding the row mass, multiplication by G_1, then by Q, proves the second identity in eq. 5.ccliv. Thus the artificial mass is subtracted in the stated order.

The actual Wilson form estimate and the constraint-curvature estimate give

$$\langle v, H_L v\rangle \geq (1 - C\kappa)\|\mathrm{d}_U v\|_2^2 - C(\kappa+\gamma)\|v\|_{\mathcal{E}}^2.$$

Insert $v = \mathcal{H}a$. The lift bound controls the last term by $C(\kappa+\gamma)\|a\|_2^2$. Since $Qv = a$, Lem. 5.38 implies

$$\|\mathrm{d}_U v\|_2^2 \geq c\|\mathrm{d}_V a\|_2^2 - C\kappa^2\|v\|_2^2.$$

Absorb its last term by the same lift bound. This proves the lower inequality. The fine Lagrangian form is bounded in energy and the lift is bounded, which proves the upper inequality and its polarized form. All its bounds use the actual fine Hessian and multiplier; an ambient unconstrained Hessian has not been substituted. □

5.11.4. *The actual outer plain-fibre precision.* Now use the actual one-step outer pivot chart based at V:

$$V(u) = e^{i\Phi_V(u)}V, \qquad \mathsf{C}_V = D\Phi_V(0), \qquad \Phi_V(u)|_{\mathrm{free}} = u.$$

Its dimensionless outer block size is the fixed L. The first-step chart estimates give

$$\|\mathsf{C}_V\|_{2\to 2} \leq C_L, \qquad \|\mathsf{C}_V u\|_2 \geq \|u\|_2, \qquad \|\langle \mu, D^2\Phi_V(0)\rangle\|_{2\to 2} \leq C_L\|\mu\|_\infty. \tag{5.cclv}$$

These are finite one-step bounds, independent of the inner depth. Let $K_G(V)$ be the actual relative trace-contour Hessian of eq. 4.ccxxxii. On a block-axial real reference it is the absolute contour Hessian.

At this physical reference the coarse gradient is ω. The full centred composed-action Hessian on free outer coordinates is therefore

$$K_A(V) = \mathsf{C}_V^\sharp D^2 S(V)\mathsf{C}_V + \langle \omega, D^2\Phi_V(0)\rangle. \tag{5.cclvi}$$

The second term cannot be omitted. It is the nonlinear plain-constraint contribution to the outer fibre Hessian. Subtracting the linear centering of the action has no effect on this equation.

Lemma 5.40 (The geometric outer curl-gauge gap)**.** On the real block-axial, small-curvature outer reference,

$$\|\mathrm{d}_V\mathsf{C}_V u\|_2^2 + \langle u, K_G(V)u\rangle \geq c_L\|u\|_2^2. \tag{5.cclvii}$$

■

Proof. Apply the proved actual first-step gap eq. 3.cxxx with zero independent current. Its action part is $\mathsf{C}_V^\sharp \mathscr{D}_V^{\mathrm{Wilson}}\mathsf{C}_V$. The exact plaquette upper estimate bounds it by

$$C_L\|\mathrm{d}_V\mathsf{C}_V u\|_2^2 + C_L\|F(V)\|_\infty\|\mathsf{C}_V u\|_2^2.$$

The outer curvature is $O_L(\kappa)$ by the local mean comparison in Lem. 5.38. Use eq. 5.cclv and decrease κ to absorb this last mass term into half the first-step gap. The resulting inequality is eq. 5.cclvii. This uses the first-step geometry and gauge, rather than treating the nonlocal effective action as a finite-range Wilson action. □

Proposition 5.41 (Nonflat late-depth outer Gaussian gap)**.** For fixed sufficiently small κ, γ, the actual outer precision satisfies

$$c_L I \leq \tau K_A(V) + K_G(V) \leq C_L I \tag{5.cclviii}$$

for real τ in a fixed interval about one. The constants are independent of the inner depth and volume. ■

Proof. Insert $\mathsf{C}_V u$ in eq. 5.ccliii. Equations eq. 5.cclv and eq. 5.cclvi give

$$\langle u, K_A(V)u\rangle \geq c\|\mathrm{d}_V\mathsf{C}_V u\|_2^2 - C_L(\kappa+\gamma)\|u\|_2^2.$$

Add the gauge term and use Lem. 5.40. Choose the two ceilings small enough to absorb the error. The upper bound follows from the effective Hessian and the finite outer chart and gauge

bounds. Finally changing τ near one costs at most $C_L|\tau-1|\|u\|_2^2$, so a fixed interval preserves both inequalities. □

5.11.5. *The explicit independent-current completion.* We specify the operator to which the complex assertion applies. For a field-current pair $P=(\mathbf{U},\mathbf{J})$ in a small tube about the preceding real physical pair, form the canonical G,H,G_1 in eq. 3.cmxv, in that order, and put

$$\begin{aligned}\omega_P &= H(P)^\sharp\mathbf{J}, \qquad \Gamma_P=(Q_{\mathbf{U}}G_1(P)Q_{\mathbf{U}}^\sharp)^{-1}-a_f,\\ \mathcal{K}_A(P) &= \mathsf{C}_{\mathbf{V}}^\sharp\Gamma_P\mathsf{C}_{\mathbf{V}}+\langle\omega_P,D^2\Phi_{\mathbf{V}}(0)\rangle,\\ \mathcal{K}_\tau(P) &= \tau\mathcal{K}_A(P)+K_G(\mathbf{V}), \qquad \mathbf{V}=M^k\mathbf{U}.\end{aligned} \tag{5.cclix}$$

The lift inside ω_P is the first lift. The row covariance inside Γ_P uses the second inverse. At the physical pair, eq. 5.ccliv and eq. 5.cclvi identify this operator exactly with the physical precision just estimated.

The actual current inverse, normalized row inverse and readout bounds show

$$\|\mathcal{K}_\tau(P)-\mathcal{K}_1(U,j_U)\|_{2\to2}\le C_L\{\,\varepsilon+\gamma+\gamma_{\mathbf{J}}+|\tau-1|\,\}. \tag{5.cclx}$$

Here ε is the weighted C^1 tube radius about the real regular fine reference and $\gamma_{\mathbf{J}}$ its independent-current amplitude. To check this estimate, perturb the actual inverse and its normalized row trace in energy norms, then invert the row matrix using its real coarse gap. The row weights are bounded in the present unit coarse spacing. The first-lift readout has its already proved supremum bound. Finally the actual plain-mean, pivot and relative gauge coefficients have the uniform one-step derivative bounds eq. 5.cclv. Their background changes are $O_L(\varepsilon)$ by the admitted hierarchical row and node-frame majorants. These facts give each term in eq. 5.cclx; there is no fine operator-norm comparison at scale h^{-2}.

After reducing the fixed radii, the right side is less than half the real gap. Thus $\mathcal{K}_\tau(P)$ is holomorphic and invertible on a common complex tube, with uniform inverse norm. For real symmetric pairs in that tube it has the same positive lower bound. No Hermitian positivity of the complex matrix is asserted.

This proves a matched explicit outer Gaussian precision on physical regular critical germs and the stated off-current extension. Away from the block-axial locus the relative gauge in eq. 4.ccxxxii specifies the extension. An independently prescribed absolute-gauge completion, signed profile, enlarged cube domain or history-conditioned matrix still requires its own exact comparison.

5.12. **Local Gaussian domains for the canonical nonflat coarse form.** We now admit independent cube parameters for the actual canonical coarse Gaussian near a real physical critical pair. The real spectral gap and the local coefficient expansion are separate inputs. Here they are supplied by Props. 3.112 and 5.41. The construction below also records the background domain on which its local coefficients are holomorphic.

Use a homogeneous inner hierarchy of arbitrary depth, with inner coarse spacing one. The outer step has the fixed blocking integer L. Fix a real physical critical pair $P_*=(U_*,j(U_*))$ satisfying the small curvature and current requirements of Prop. 5.41. Use its actual coarse datum $V_*=M^kU_*$, its one-step plain pivot chart, and the stated relative contour gauge. Choose a block axial representative when identifying the physical absolute gauge factor. All real transposes are extended bilinearly to the complex domains.

For $P=(U,J)$ near P_*, let Γ_P,ω_P and $K_\tau(P)$ be the explicit canonical extension in eq. 5.cclix. In particular ω_P uses the first lift and Γ_P uses the second inverse. We retain the $\langle\omega_P,\Phi_V''(0)\rangle$ term in the outer action Hessian. The following construction uses this specified matrix. An independent-current extension of its local Gaussians does not identify a separately prescribed history or signed-profile precision.

5.12.1. *The two uniform premises.* Assign the free outer fine-bond coordinates to their starting unit cells. A cell has at most a fixed number d_L of real coordinates; this number does not involve the inner fine mesh. The plain pivot map and its first two derivatives read a fixed L-dependent neighborhood. After an aligned reblocking, the named families therefore give an exact expansion

$$K_\tau(P) = \sum_C K_C(P,\tau), \qquad \sup_y \sum_{z,C} e^{\kappa_0\sigma_0(C)} \|1_y K_C 1_z\| + \sup_z \sum_{y,C} e^{\kappa_0\sigma_0(C)} \|1_y K_C 1_z\| \le B_0. \quad (5.\text{cclxi})$$

Here C is a connected carrier of fixed cubes of side M_0, with its full determining data, endpoints and short joins. The shifted length σ_0 is the carrier length of the positive-walk construction. The constant B_0 and M_0 are fixed before depth and volume. The exponent κ_0 can be chosen at any required fixed finite value by making the choices in Prop. 3.112 first. The permitted current ceiling and complex tube are retained from those choices.

At real physical data and τ near one, the geometric gap gives

$$c_L I \le K_\tau(P_*) \le C_L I. \quad (5.\text{cclxii})$$

After a fixed decrease of the background and current tube, the same real bounds, with changed constants, hold for every real admitted P. This follows from the actual family perturbation bound; it does not require the nearby independent current to be a Wilson gradient. Complex perturbations have the same operator-norm bound. The variable τ ranges in a fixed disc about one.

For completeness the locality in eq. 5.cclxi includes the row inverse in Γ_P. It is supplied by eq. 3.mxvi, with its actual mass subtraction. The outer chart factors and $\Phi_V''(0)$ enlarge the carrier by only a fixed star. The rooted family for ω_P retains its fine-current input bond by Prop. 3.111. Thus the current term has a genuine local carrier, as does the row-Schur term. A completed kernel decay bound alone would not prove eq. 5.cclxi.

5.12.2. *An admissible local background tube.* The background tube is a weighted C^1 tube in the actual fine relative field A, with an independent weighted current ball for $J - J_*$. For the homogeneous geometry the assigned physical cell scale is one. Given a finite carrier, retain a fixed complete data collar. A local C^1 field on that collar can be extended to a global relative field by a smooth cutoff which equals one on the read set. The cutoff is formed at the unit-cell scale, rather than the inner fine mesh. It satisfies

$$|\chi| + |D\chi| \le C_L.$$

Consequently

$$\|\chi A\|_{C^1} \le C_L \|A\|_{C^1(\text{collar})}. \quad (5.\text{cclxiii})$$

To construct such a cutoff, take the distance in the unit-cell graph from the read set, truncate it in a collar of fixed width, and interpolate its values by the fixed smooth partition on unit cells. Bounded overlap gives the displayed derivative bound. Make it zero before the outer edge of the supplied collar. Extension by zero then adds no fine-mesh jump. Multiplication of the independent current by a bounded cutoff preserves its current norm and takes no derivative of that current.

Choose the local radius smaller than the global tube radius divided by the constant in eq. 5.cclxiii. Every such local datum has an admitted extension relative to the fixed real reference. Exact determining-data locality in Prop. 3.112 makes the coefficient independent of which extension is used. This proves a local holomorphic domain for each coefficient. These are C^1 field domains; they are not independent scalar fine-bond discs with a mesh-free radius. The cube parameters introduced below are independent of these field variables. The geometry is the fixed homogeneous geometry in this statement. No extension theorem for an arbitrary partly specified proper history is used.

5.12.3. *Full-carrier truncation and a combined short power.* Fix a requested cube-parameter radius $S > 2$. Choose $b > 2(C_L + 1)$, and set

$$H = I - K_\tau/b.$$

Symmetrize each carrier coefficient by its transpose. Its endpoints and determining data are unchanged, its row/column family norm does not increase, and its sum is still the symmetric matrix H. For a positive integer R, put

$$H_< = \sum_{\sigma_0(C)\le R} H_C, \qquad H_> = H - H_<. \tag{5.cclxiv}$$

The tail has Schur norm at most $Ce^{-\kappa_0 R}$. For large R, real data satisfy

$$0 \le H_< \le qI, \qquad q < 1, \tag{5.cclxv}$$

by eq. 5.cclxii. The lower bound uses $H \ge I/2$ and a tail smaller than $1/4$. Shrink the common complex tube once more if necessary. Then $\|H_<\|_{2\to 2} \le q' < 1$ throughout it, for a fixed q'.

Every carrier in $H_<$ has diameter at most $CM_0(R+1)$. For $j \ge 1$, combine the matrix entries of $H_<^j$ before taking absolute values. Its matrix range is at most

$$B_j = CM_0 j(R+1).$$

The number of unit-cell coordinates in a ball of radius B_j is at most $C_L(1+B_j)^4$. A Hilbert row has norm at most $(q')^j$; Cauchy–Schwarz on that row, and then on columns, therefore gives

$$\|H_<^j\|_{\mathrm{Sch}} \le C_L(1+B_j)^2(q')^j. \tag{5.cclxvi}$$

The data read by this combined entry are retained in the two radius-B_j endpoint balls. Each short word has its whole carrier in those balls: its successive carriers are connected and have total diameter at most the sum of their declared ranges. The balls can be joined by a fixed short path. This is the reason for truncating by the full carrier in eq. 5.cclxiv. Truncation by endpoint distance would not justify this data claim.

The exact long-word remainder is

$$E_j = H^j - H_<^j = \sum_{i=0}^{j-1} H_<^i H_> H^{j-1-i}. \tag{5.cclxvii}$$

Keep its actual joined-carrier family. At a fixed reserved carrier rate its norm is bounded by

$$\|\mathbf{E}_j\| \le CjB_H^j e^{-cR}. \tag{5.cclxviii}$$

The constants are independent of the inner depth. Choose $R = Dm$, with $cD > 2\log(1+B_H) + 2$. Next choose an aligned cube side M_{G} larger than a fixed multiple of B_m. It may be taken to be a power of L. The combined short entries now meet a bounded number of these cubes, with a bound independent of m.

For the long terms, reblocking does not lose their rate. An old connected carrier can be visited by a curve of length at most $CM_0\sigma_0(C)$. Covering that curve and its CM_0 collar by M_{G}-cubes gives

$$\#C_{\mathrm{G}} \le C_L\{1 + (M_0/M_{\mathrm{G}})\sigma_0(C)\}. \tag{5.cclxix}$$

Choose M_{G} also so that the coefficient of $\sigma_0(C)$ in $(\log S)\#C_{\mathrm{G}}$ fits in the reserved old carrier rate. Thus weighting a full carrier by independent parameters of modulus at most S preserves eq. 5.cclxviii. No fine-background coordinate count enters eq. 5.cclxvi: the coordinates counted there are the outer free Gaussian coordinates at spacing one. The fine background is read as determining data.

5.12.4. *The actual local covariance.* Let $\mathbf{T}$ be the family obtained by combining and filing $H^m_<$ as above and adding the long-word family E_m. Its sum is H^m. Multiply each term by the product of its declared cube parameters, and denote the resulting matrix by T_s. On the held root guard replace each parameter by one. For $j < m$, similarly put

$$A_{j,s} = H^j_< + (E_j)_s.$$

The near prefix is held without parameters. Its rows from a root wall read only its fixed radius-B_m guard. The long prefix keeps its complete carrier parameters. The preceding bounds give

$$\sup_{|s_\Delta|<S} \|T_s\|_{\mathrm{Sch}} \le C_L S^{C_L}(1+B_m)^2(q')^m + CmB_H^m e^{-cDm} =: q_m \longrightarrow 0. \tag{5.cclxx}$$

Its convergence is exponential up to a polynomial in m.

For a finite root coordinate set W, let χ_W denote coordinate restriction, not a fluctuation cutoff. With $c_n = 4^{-n}\binom{2n}{n}$, define

$$\begin{aligned} R_W(s) &= b^{-1/2}\chi_W \sum_{j<m} A_{j,s} \sum_{v\ge 0} c_{mv+j} T_s^v, \\ D_W(s) &= R_W(s) R_W(s)^{\mathsf{T}}. \end{aligned} \tag{5.cclxxi}$$

The series converges normally in the weighted row/column norm after m is fixed large enough. Prefix norms grow only polynomially in R, by eq. 5.cclxvi and eq. 5.cclxviii. The binomial coefficients satisfy $0 < c_n \le 1$. At complete parameters the ordinary matrix identity is

$$R_W(1) = \chi_W K_\tau(P)^{-1/2}, \qquad D_W(1) = \chi_W K_\tau(P)^{-1}\chi_W^{\mathsf{T}}. \tag{5.cclxxii}$$

Indeed $H^{mv+j} = H^j(H^m)^v$, so grouping the convergent binomial series by its residue class modulo m gives the first identity. Its square gives the second. This identity is asserted for the complete matrices. Their filed summands need not commute.

For real comparison data put

$$R_{0,*} = b^{-1/2}\sum_{j<m} c_j H^j_{<,*}, \qquad D_{0,W} = \chi_W R_{0,*}^2 \chi_W^{\mathsf{T}}.$$

By eq. 5.cclxv,

$$b^{-1} I \le D_{0,W} \le C_L I \tag{5.cclxxiii}$$

for every W. The lower bound follows by zero-extending a vector in W: $R_{0,*} \ge b^{-1/2} I$. There is no Dirichlet inverse of an indefinite saddle here.

The long prefixes and the $v \ge 1$ terms in eq. 5.cclxxi are bounded by a polynomial in m, R times $q_m + mB_H^m e^{-cDm}$. Their norms can therefore be made smaller than any prescribed fixed inverse power of M_{G}, since $M_{\mathrm{G}} = O_L(M_0 m^2)$. Finally reduce the common complex field, current and τ tubes so that the finite held prefixes have the same perturbation budget. Their coefficient Lipschitz bounds follow from Cauchy on the local tube in eq. 5.cclxiii. All choices are uniform in depth and volume. For example they can ensure

$$\sup_{|s_\Delta|<S} \|D_{0,W}^{-1/2}(D_W(s) - D_{0,W})D_{0,W}^{-1/2}\| < \frac{1}{4} \tag{5.cclxxiv}$$

simultaneously for every root coordinate set W. The finer $o(M_{\mathrm{G}}^{-4})$ budget is available for per-cube determinant estimates.

Proposition 5.42. For every fixed $S > 2$, the canonical nonflat outer precision specified above has the local Gaussian extension eq. 5.cclxxi. It is holomorphic on independent cube discs $|s_\Delta| < S$ and on the common local C^1 background, independent-current and τ tubes. Its complete covariance is the actual marginal eq. 5.cclxxii. Every local covariance has the uniform complex sector eq. 5.cclxxiv. Its coefficients read their full declared data carriers. Setting parameters outside a wall to zero gives

literal wall dependence and fixed-root heredity, with the held guard retained as part of that wall. These assertions are uniform in the homogeneous inner depth and compatible volume. ■

Proof. The preceding construction proves the identities, bounds and normal convergence. A term which survives zero parameters outside a wall has its whole parameter carrier there, in addition to the fixed held guard. Its local coefficient reads only those data. Two wall systems which agree on these complete read sets therefore give the same coefficient. This proves heredity. Take components after adding their complete held guards and read sets, merging components whose guarded sets meet. Separated components then use disjoint coordinate sets and read sets; no surviving joined carrier connects them. Their Gaussian covariance is block diagonal, and their normalized Gaussian integrals factor.

On small tori perform the finite construction on one common sufficiently large cover. Retain all lifted data carriers before truncation and all short power endpoint balls before projection. Average families over deck shifts before descent. Periodic auxiliary parameters are identified on their images, and held guards are their full deck-periodic inverse images. Their full lifted carrier weights remain charged. The row/column Schur bounds descend without a sheet factor, as in Cor. 3.104. Descend the unrestricted root matrix before restricting its coordinates to W. Indeed let I be isometric pullback from the base to the periodic sector of the cover. Periodicity of parameters, guards and averaged families gives $R_{\mathrm{cov}} I = I R_{\mathrm{base}}$, and the periodic sector is reducing for these matrices and their transposes. Thus

$$I^{\mathsf{T}} R_{\mathrm{cov}} R_{\mathrm{cov}}^{\mathsf{T}} I = R_{\mathrm{base}} R_{\mathrm{base}}^{\mathsf{T}}.$$

Only then restrict the base coordinates to W. At complete parameters the classical operators and their positive matrix functions restrict to this sector, so this covariance is the actual finite matrix marginal. A covariance restricted to one chosen lift of W would not establish that assertion. This is an operator identity, not a comparison of quantum partition functions on different volumes. □

5.12.5. *Normalized complex Gaussian integrals.* The sector just proved also specifies the determinant branch and the normalization used in local Gaussian estimates. For $d = \dim W$, define

$$d\mu_{D_W}(u) = (2\pi)^{-d/2} \det(D_W)^{-1/2} e^{-u^{\mathsf{T}} D_W^{-1} u/2}\, du, \qquad u \in \mathbb{R}^d. \tag{5.cclxxv}$$

Continue the square root from the positive real comparison matrix $D_{0,W}$. Writing $D_W = D_{0,W}^{1/2}(I + E) D_{0,W}^{1/2}$, with $\|E\| < 1/4$, gives

$$|\det(I+E)|^{-1/2} \le (4/3)^{d/2}, \qquad \mathrm{Re}\, u^{\mathsf{T}} D_W^{-1} u \ge c_L |u|^2 \quad (u \in \mathbb{R}^d).$$

The inverse estimate follows from the norm-convergent series for $(I+E)^{-1}$ and eq. 5.cclxxiii. Consequently

$$\int_{\mathbb{R}^d} |d\mu_{D_W}| \le C_L^d, \qquad \int_{\mathbb{R}^d} d\mu_{D_W} = 1. \tag{5.cclxxvi}$$

The second equality is first the real Gaussian identity. The same dominating Gaussian on every smaller sector permits holomorphic continuation to the declared domain. Polynomial moments and fixed parameter derivatives follow by differentiating that dominated integral, or by Cauchy on a smaller sector. The empty selected expectation is exactly one.

The factor C_L^d is local: there are at most $C_L M_{\mathrm{G}}^4$ free coordinates per localization cube of a wall. Thus it costs a fixed exponential per wall cube, with no factor involving the total torus or the inner fine coordinate count. This is the normalization needed when a local Gaussian is used in a connected polymer integral. It does not assert zero-freeness of every sharply truncated whole-volume Gaussian integral. The normalized cutoff expansion and its interacting vertices must still be formed with their actual frozen cutoffs.

The result proves the Gaussian parameter-domain statement for the matrix and reference domain stated here. § 5.13 proves the proper-hierarchy version for eq. 5.cclxxviii, retaining its second-lift

outer current readout. The two off-current formulas agree on physical critical pairs; they are not identified elsewhere. Nonlinear core membership is established by the actual tower and substitution maps in Cors. 3.130 and 3.131.

5.13. **The fluctuation precision on a proper inner hierarchy.** The mass removed from a row Schur matrix is essential for the fluctuation covariance. We prove its lower bound on the actual outer constraint, including the second derivative of that constraint. The inner hierarchy may have proper regions and mixed scales. The current remains an independent variable until the final physical identification.

Work in level-k units. The inner sequence ends at Ω_k; its top row length is one. Retain an additional union Ω_{k+1} of complete L-blocks, with its prescribed separation from Ω_k^c. Put $\Theta' = \Omega_{k+1} \cap \mathbb{T}^{(k+1)}$ for the block centres and $\Theta = B(\Theta')$ for their level-k sites. Let X be the level-k bonds touching Θ, and $\check{X}$ their nonpivot bonds. All pivot double blocks and contour paths are retained. The separation floor below puts this entire stock inside the interior top rows of the inner sequence. Write ι for extension from X by zero to its independent row space. Thus $\iota^\sharp \iota = 1$ in counting coordinates; there is no mixed-scale measure factor on these length-one rows. The field and row pairings elsewhere retain their physical weights. The marked geometry may retain inward bonds, but Q and its row space use only the independent interior and outward rows.

Use a real regular reference and the common weighted C^1 complex tube of Props. 3.112 and 3.171. The independent current satisfies its weighted supremum bound. The enlarged regularity clause includes every complete cube needed in the two adjacent layers, including cubes lying wholly in the upper member. Choose all proof tiles first, then a common larger separation floor. The ceilings are independent of the fine mesh, depth, volume, conditioning set and all larger admitted geometries.

5.13.1. *The precise form and its two multiplier readouts.* Write $V = \mathcal{M}^k \mathbf{U}$ on the complete outer stock and let q_V be its raw one-step constraint logarithm. The actual pivot graph has

$$q_V(\Phi_V(u)) = 0, \qquad \Phi_V(u)|_{\check{X}} = u, \qquad C_V = D\Phi_V(0).$$

Let h_V be the pivot right inverse of $q_V'(0)$ and write $C_V^{(2)} = \frac{1}{2} q_V''(0)$. This superscript distinguishes the quadratic row map from the pivot lift C_V. The graph derivative satisfies

$$\Phi_V''(0)[u, v] = -2 h_V C_V^{(2)}(C_V u, C_V v). \tag{5.cclxxvii}$$

Indeed the free coordinates have zero second derivative, and twice differentiating the constraint determines all pivot coordinates by its invertible pivot block. Every map in this identity reads its complete one-step double block and has a bounded row and column norm depending only on L.

Let $b_V w(y, x)$ be the derivative of the relative contour logarithm

$$\frac{1}{i} \log\{(e^{iw} V)(y, x) V(y, x)^{-1}\}.$$

Set $G_V^{(2)} = b_V^\sharp b_V$, with one unit weight for every noncentral site in a retained block. This is the relative contour quadratic form. It is the Hessian of the absolute trace-contour functional when the reference averaged contours equal one.

The ordered inner operators are exactly eq. 3.cmxv. In particular put

$$\begin{aligned}\Sigma_1 &= (QG_1Q^\sharp)^{-1},\\ \omega_* &= H^\sharp\mathbf{J},\\ \omega_{1*} &= H_1^\sharp\mathbf{J},\\ E_V^{(2)}[\omega](w,z) &= 2\langle\omega, \iota h_V C_V^{(2)}(w,z)\rangle,\\ \Delta_{\mathrm{out}} &= \iota^\sharp\Sigma_1\iota - a_{\mathrm{top}} - E_V^{(2)}[\omega_{1*}],\\ A_Y &= e_Y^\sharp C_V^\sharp(\Delta_{\mathrm{out}} + G_V^{(2)})C_V e_Y.\end{aligned} \tag{5.cclxxviii}$$

Here e_Y extends the free coordinates on a conditioning set $Y = \check{X}(B(\Theta''))$, $\Theta'' \subset \Theta'$, by zero. Its retained blocks and collars are those of the conditioning geometry. The constant top row mass is $a_{\mathrm{top}} = a_0 I$; for the unit mass convention it is I. The term inside G_1 uses ω_*; the outer constraint term uses ω_{1*}. These are distinct away from physical currents.

5.13.2. *A lower bound before the outer gauge is added.*

Lemma 5.43 (The top-supported Schur form)**.** At every admitted real pair, for a top row field w supported on X,

$$\langle w, (\iota^\sharp\Sigma_1\iota - a_{\mathrm{top}})w\rangle \geq c\|\mathrm{d}_V w\|^2_{2,\mathcal{P}_X} - C(\alpha_0+\gamma_0)\|w\|_2^2. \tag{5.cclxxix}$$

Here w is extended by zero off X, and $\mathcal{P}_X$ is the complete finite stock of level-k plaquettes used by the outer block curl estimates. The form on the left has operator norm at most a fixed constant. ■

Proof. Put $v = H_1\iota w$. The actual proper row lift gives $Qv = \iota w$ and $\|v\|_{\mathcal{E}} \leq C\|w\|_2$. The equality $G_1^{-1}v = Q^\sharp\Sigma_1\iota w$ gives

$$\langle w, (\iota^\sharp\Sigma_1\iota - a_{\mathrm{top}})w\rangle = \langle v, (\mathscr{D}_U - \Delta'_{\pi,U}[J] - S^\sharp\Delta_U^{(2)}[H^\sharp J]S + DRD^\sharp)v\rangle. \tag{5.cclxxx}$$

This is the variational identity for the complete Schur matrix, followed by its exact row-mass subtraction. It uses neither criticality nor horizontality of v.

At the real reference R is an orthogonal projection, so the last term is nonnegative. The Wilson plaquette form and the current form estimates of § 3.13 and eq. 3.cmx bound the right side of eq. 5.cclxxx below by

$$c\|\mathrm{d}_U v\|_2^2 - C(\alpha_0+\gamma_0)\|v\|_{\mathcal{E}}^2.$$

The multiplier $H^\sharp J$ has the required row-weighted supremum bound. The second constraint jet and S have their bounded energy forms, so its term has the stated $C\gamma_0$ bound. No derivative of the independent current occurs.

Every plaquette in $\mathcal{P}_X$ and every carrier of its four top rows lies wholly in Ω_k. There the assigned scale is one. The local proof of Lem. 5.38 therefore gives

$$\|\mathrm{d}_V Qv\|_{2,\mathcal{P}_X} \leq C\|\mathrm{d}_U v\|_2 + C\alpha_0\|v\|_{\mathcal{E}}.$$

To see why the proper exterior changes nothing, choose a fixed physical cube around each of these complete plaquettes. In its real small gauge the actual top row differs from the flat flux row by $O(\alpha_0)$ in both physical row and column norms. The flat identity eq. 5.ccl and its Jensen bound have no depth factor. The coarse derivative has unit spacing. The cubes have bounded overlap; all norms outside them on the right may simply be kept. Thus the estimate does not replace the proper inner inverse by a homogeneous inverse. Since $Qv = \iota w$, it proves eq. 5.cclxxix. The upper bound follows from the proper row inverse and the constant mass on the supported top rows. □

Lemma 5.44 (The outer geometric inequality)**.** For every retained block family and its full pivot stock,

$$\|u\|_2^2 \leq C_L\{\|\mathrm{d}_V C_V e_Y u\|^2_{2,\mathcal{P}_X} + \|b_V C_V e_Y u\|_2^2\}. \tag{5.cclxxxi}$$

On the free subspace whose reconstructed field vanishes on each retained box tree, the same inequality holds with the second term omitted. ■

Proof. First take a flat frame. In a block, compare a path to a site with each coordinate-order path to that site. Successive plaquette swaps express their difference as a sum of at most C_L plaquette curls. Hence an internal bond is its mean contour-potential difference plus a sum of these curls. Squaring and summing bounds all internal bonds by the two terms on the right. In the tree version use the unique tree path instead; its potential is zero.

Across a block face, a plaquette with one transverse step compares neighboring crossing bonds. Its remaining two bonds are internal and already bounded. The transverse face grid is connected and has at most L^3 sites. Thus all crossing values differ from a common face value by controlled curls and internal bonds. The plain mean of that face is its common crossing value plus controlled internal path terms. It vanishes because $q_V'(0)C_V e_Y u = 0$, including each boundary pivot row. This controls the common value. Outside retained blocks the extended field has zero internal bonds, so the same argument applies to a boundary face. A fixed bond or plaquette enters at most C_L such block comparisons. The free restriction of $C_V e_Y u$ is $e_Y u$, proving the claim at a flat reference.

For a real small-curvature reference choose the regular real gauge on each block and its neighboring double blocks. Their sizes depend only on L. The plain row, pivot lift, relative contour derivative and coarse curl differ from their flat counterparts by $C_L\alpha_0$ in the finite row and column norms. The gauge changes are unitary. Absorb this perturbation in half the flat inequality and sum the same bounded-overlap comparisons. Vanishing on a box tree is preserved by these linear unitary coordinate changes. □

Proposition 5.45 (Proper fluctuation covariance)**.** For fixed sufficiently small ceilings, the exact matrix eq. 5.cclxxviii satisfies

$$c_L I \le A_Y \le C_L I \tag{5.cclxxxii}$$

at every admitted real pair. The same bounds hold on the strict tree-zero subspace when the gauge term is omitted. The inverses extend holomorphically to a common weighted C^1 complex background tube and independent-current ball, with uniform inverse norms. All conclusions are uniform in the proper inner hierarchy and the conditioning set. ■

Proof. The outer second-jet bound and the proper readout estimate give $\|E_V^{(2)}[H_1^\sharp J]\| \le C_L\gamma_0$. Insert $w = C_V e_Y u$ in eq. 5.cclxxix, add the relative gauge, and use Lem. 5.44. The pivot lift is bounded and its free restriction is the identity. Decreasing $\alpha_0 + \gamma_0$ absorbs the negative mass term. The same proof on the tree-zero subspace omits the gauge and uses the second part of that lemma. All upper bounds follow from the bounded factors in eq. 5.cclxxviii.

The proper energy and normalized row inverse bounds, together with their local coefficient families, imply

$$\|A_Y(\mathbf{U},\mathbf{J}) - A_Y(U,J_0)\|_{2\to 2} \le C_L(\|A'\|_{C^1_{\text{weighted}}} + \|\mathbf{J} - J_0\|_{\text{current}}).$$

The same holds on the tree-zero free coordinates. For the row Schur factor use its normalized inverse difference identity; for the two readouts retain their different lifts. The outer pivot and contour coefficients have fixed finite ranges. Restriction and zero extension have norm one, so the bound is independent of Y. It is a relative energy comparison, not a fine-lattice operator comparison at scale η^{-2}. A perturbation below half the real gap proves the assertion. □

5.13.3. *Local inverse families and logarithms.* The locality assertion requires more than the completed operator gap. We record how the existing carrier construction applies to this particular matrix. The mass-removed family eq. 3.mxvi, the rooted family of $H_1^\sharp J$, and the finite outer factors give an exact connected-carrier expansion of A_Y. For this second-lift readout repeat the proof of Prop. 3.111 with the named family of H_1 in place of H: its physical transpose pairs a unit top-row seed with the fine current, and the family column bound sums the cellwise energy pairings. It gives

$C\gamma_0$ with no whole-volume current norm. The bare restriction identity follows from $Q_0H_1 = P_0$, so that its separate bare term is retained exactly. The precision family's row and column sums, charged by a fixed exponential carrier rate, are bounded. Each term retains the complete inner regions, outer blocks, pivot rows and the trace of Y on its read set. These are precisely the data needed to multiply two terms.

The inverse construction of Cor. 3.104 can now be applied on the outer free coordinates, whose spacing is one and whose multiplicity per cell is bounded by C_L. Here are its domain and counting checks in this setting. Choose $b > 2C_L$ and write $H_Y = I - A_Y/b$. On a smaller common tube it has Hilbert norm at most $q < 1$. Truncate its family by full carrier length at R, giving $H_< + H_>$. The omitted family has norm Ce^{-cR}. Combine $H_<^m$ before taking absolute values. Its matrix range is $O(mR)$ in physical units, and hence

$$\|H_<^m\|_{\mathrm{Sch}} \le C_L(1+mR)^2 q_1^m, \qquad q_1 < 1. \tag{5.cclxxxiii}$$

The complete data of every contributing short word lie in the corresponding endpoint balls. Enlarge the separation floor so that these balls, including their fixed carriers, remain inside Ω_k. This is possible after m, R have been fixed, independently of depth. In these balls the inner assigned scale is one; in the extended outer sequence the only additional scale is L. Consequently the parameter count in a reblocked endpoint ball is bounded independently of the number of finer layers. The arbitrary boundary of Y changes its recorded zero-extension mask, not that count.

The same collar gives the required local background domains. Relative to the fixed global real reference, extend a local complex potential from a short-word read ball by multiplying it by a smooth cutoff equal to one on all reads and zero before the edge of its supplied collar. Its first difference is bounded in unit scale. The entire cutoff collar lies in Ω_k, so all its weights have scale one; multiplication preserves the weighted C^1 norm within a fixed factor. Extend by zero outside the collar. A bounded cutoff preserves the independent-current norm without taking a current derivative. Decrease the local complex radii by that factor. This supplies an admitted global extension for estimating each combined short coefficient. Its value is independent of the extension by exact determining-data locality. Long coefficients retain the local domains of their original complete NCW carriers. This does not require an extension of an arbitrary partially specified proper history.

Keep the long-word identity

$$H_Y^m - H_<^m = \sum_{i=0}^{m-1} H_<^i H_> H_Y^{m-1-i}$$

as a family with its full joined mixed-scale carriers. Its norm is at most $CmB_H^m e^{-cR}$. Take $R = Dm$ with D fixed large, then m large. Reblock after these choices. The short-power term meets only a bounded number of new cubes; the long carriers pay for their cube parameters from a reserved part of their old carrier rate. Thus, for every fixed requested parameter radius $S > 2$, the filed mth power has family norm less than $1/2$ on $|s_\Delta| < S$. The finitely many prefixes are bounded. The exact identity

$$A_Y^{-1} = b^{-1}\Big(\sum_{j=0}^{m-1} H_Y^j\Big)(I - H_Y^m)^{-1}$$

then gives a normally convergent local inverse family. This is the inverse of the compressed precision, not a marginal of an inverse taken before conditioning.

For the logarithm use the same grouped words in

$$\log A_Y = (\log b)I - \sum_{j=1}^{m} H_Y^j \sum_{v\ge 0} \frac{(H_Y^m)^v}{mv + j}. \tag{5.cclxxxiv}$$

At complete parameters this is the convergent real spectral logarithm and its holomorphic continuation. At other parameters the formula specifies its filed local extension. Its diagonal trace at a bond is a sum of closed joined carriers containing that bond. The same family norm bounds it, so the bound per outer cell is uniform. Differences between two regions or conditioning sets cancel term by term until a carrier meets their differing read sets. Parameter walls and fixed-root heredity follow by setting to zero every term whose complete carrier leaves the wall. No coefficient is evaluated on an unrecorded extension of the background.

For normalized Gaussian integrals use the binomial grouped series eq. 5.cclxxi, with A_Y in place of the homogeneous matrix. The two premises of that construction have now been proved: eq. 5.cclxxxii and the complete carrier family just described. Its combined short words read the same top-scale endpoint balls; its long words retain the same mixed-scale carriers. Its held real prefix is at least $b^{-1/2}I$, and its tail can be made arbitrarily small before decreasing the common complex tube. Consequently it gives the uniform local complex sector eq. 5.cclxxiv and the normalized Gaussian identity eq. 5.cclxxvi for every coordinate selection inside Y. At complete parameters its covariance is the selected marginal of A_Y^{-1}. The conditioning by Y has already occurred. The determinant branch is continued from the same positive real precision. These assertions do not identify two independently chosen off-parameter extensions of a logarithm. For small periods perform the construction on one common compatible cover and retain all lifted read balls before projection. Average the filed families over deck shifts, then descend the complete operator before any coordinate selection. The row and column sums descend without a sheet factor, as in Cor. 3.104; the reducing-sector calculation in the proof of Prop. 5.42 identifies the descended covariance with the actual base matrix inverse.

5.13.4. *Identification at a physical critical pair.* Suppose now that U is a regular constrained-critical field of the inner hierarchy and $J = j_U = Q^\sharp\omega$. Then $H^\sharp J = H_1^\sharp J = \omega$, since both lifts are right inverses of Q. The physical Ward identity makes H_1 the horizontal derivative of the actual critical graph, and the differentiated constraint gives its Lagrangian Hessian. Thus the second derivative of the critical action with respect to the supported top datum is $\iota^\sharp\Sigma_1\iota - a_{\mathrm{top}}$, as in eq. 5.ccliv, now on the proper hierarchy. Composing with the actual outer pivot graph adds $\langle\omega, \iota\Phi_V''(0)\rangle$. By eq. 5.cclxxvii this is exactly the negative $E_V^{(2)}[H_1^\sharp J]$ term in eq. 5.cclxxviii. Adding the specified relative contour Hessian and restricting free coordinates gives A_Y exactly. In an averaged-contour axial representative it is also the absolute-contour physical Gaussian precision. The equality continues on the corresponding physical holomorphic chart. No identification with an independently prescribed signed or counterterm-modified precision is used.

5.14. **An actual diagonal deformation of the homogeneous backgrounds.** A change of a plaquette weight changes the principal part of the background equation. Its derivative contains second differences of the potential. We therefore construct the deformed background in a finite-p second-order space before differentiating the hierarchy. The parameter in this section is a constant diagonal weight, not a spatially varying source.

Fix the odd blocking integer L, coarse spacing one, fine spacing $h = L^{-j}$ and a compatible periodic volume. Real flat references with arbitrary holonomy are permitted. Use their parallel frames and physical pairings as in § 3.43. For $a = (a_1, \dots, a_4)$ near 1 set

$$\mathscr{A}_h^a(U) = \sum_{\mu<\nu} a_\mu a_\nu \sum_x \left(1 - \tfrac{1}{2}\operatorname{tr} U_{\mu\nu}(x)\right). \tag{5.cclxxxv}$$

No factor depending on h is inserted into this four-dimensional Wilson action. Its normal-coordinate physical gradient is denoted $J_h^a(A)$ when $U = e^{ihA}W_0$. The actual group mean, its straightening q_h, and its node factor v_h are unchanged.

5.14.1. *A second-order inverse with physical cell norms.* Let C_y be unit cells, with a fixed enlargement C_y^+ for difference stencils. For $p > 4$ put

$$\begin{aligned} \|f\|_{\mathcal{F}_p} &= \sup_y \|f\|_{L^p(C_y^+)}, \\ \|A\|_{\mathcal{W}_p} &= \sup_y \sum_{r=0}^{2} \|D^r A\|_{L^p(C_y^+)}. \end{aligned} \tag{5.cclxxxvi}$$

All mixed forward differences and the physical measure h^4 are included. Any of the fixed larger cells gives an equivalent norm. Uniform discrete extension and Morrey estimates imply

$$\|A\|_{X_\alpha} \le C_{p,\alpha} \|A\|_{\mathcal{W}_p}, \qquad 0 < \alpha < 1 - 4/p. \tag{5.cclxxxvii}$$

One direct verification is to use the piecewise multilinear extension of DA on each shifted grid. Its $W^{1,p}$ norm is bounded by the grid $W^{1,p}$ norm, and its nodal values are exact. The continuum Morrey estimate on an enlarged unit cube bounds the nodal increments. Applying the same argument to A, and summing the finitely many staggered components, proves eq. 5.cclxxxvii. It has no mesh or volume factor. Local frames on a fixed periodic cover deal with cells that cross the fundamental boundary; unitary deck factors preserve the estimates.

Lemma 5.46 (The actual joint inverse on local L^p forces)**.** For the actual flat rows and $H_0 = \mathrm{d}^\sharp \mathrm{d} + DRD^\sharp$, the system

$$H_0 A - Q^\sharp \omega = f, \qquad QA = b \tag{5.cclxxxviii}$$

has a unique solution satisfying

$$\|A\|_{\mathcal{W}_p} + \|\omega\|_\infty \le C_p(\|f\|_{\mathcal{F}_p} + \|b\|_\infty). \tag{5.cclxxxix}$$

The same bound holds with a sufficiently small exponential weight on the cell norms. Constants are independent of depth, volume and real flat holonomy. ∎

Proof. Retain $K_0 = H_0 + Q^\sharp Q$ and $S = QK_0^{-1}Q^\sharp$. The energy cell kernels of K_0^{-1} and the exponential coarse inverse of S were proved in Prop. 3.182. A force in $\mathcal{F}_p$, $p > 2$, has bounded local L^2 norm. Resolvent energy bounds between unit cells and their exponential sums therefore give

$$\|K_0^{-1} f\|_{2,\mathrm{cell},\infty} + \|QK_0^{-1} f\|_\infty \le C_p \|f\|_{\mathcal{F}_p}.$$

For the second term, each physical row is supported in a fixed unit star and has uniformly bounded L^2 density. There is no point evaluation of an energy function. The exact formulas

$$\sigma = S^{-1}(b - QK_0^{-1} f), \quad A = K_0^{-1}(f + Q^\sharp \sigma), \quad \omega = \sigma - b$$

bound the multiplier and the cell L^2 norm of A.

The component scalar equation is exactly

$$LA = f + DPD^\sharp A - Q^\sharp QA + Q^\sharp \sigma, \qquad P = 1 - R. \tag{5.ccxc}$$

The actual factorization $DPD^\sharp = (DF) M_s^{-1} (DF)^\sharp$ in eq. 3.mccxxxviii maps cell L^2 to L^∞. This uses the scalar smoothing of its columns already proved there. The actual row density gives $\|Q^\sharp c\|_\infty \le C\|c\|_\infty$; the division by h^4 in the physical transpose cancels the row coefficient $O(h^4)$. Thus all terms in eq. 5.ccxc other than f are bounded pointwise by the asserted right side of eq. 5.cclxxxix. Here the local finite-p estimate Cor. 3.80 has outer $W^{1,p}$ terms, so an explicit nested bootstrap is needed. The energy kernels first bound the outer H^1 norm. The scalar equation and its local $p = 2$ estimate give $W^{2,2}$; the strict four-dimensional embeddings give $DA \in L^3$ and A in every fixed finite L^q. Its next $p = 3$ estimate gives $W^{2,3}$, hence $DA \in L^8$ and $A \in L^\infty$. If $p \le 8$ the final p estimate now applies. If $p > 8$, first apply the $p = 8$ estimate to obtain $DA \in L^\infty$, and then apply the desired finite p estimate. Choose a fixed enlargement large enough for all these successive cutoffs. All other sources in the scalar equation were already bounded; $f \in L^p$ belongs

to each lower exponent used on these fixed cells. This proves the field bound without an endpoint singular-integral estimate.

For the weighted assertion use the existing exponential energy kernels first and the scalar estimate separately on each cell. Weights on neighboring enlarged cells differ by a fixed factor. No derivative is taken of a truncated distance weight, and no pointwise estimate is inferred directly from a cell L^2 estimate. Uniqueness is the same K_0, S energy argument as before. □

Let $K_a = \mathrm{d}^\sharp W_a \mathrm{d}$, where $(W_a F)_{\mu\nu} = a_\mu a_\nu F_{\mu\nu}$. Constant weights give the elementary bound

$$\|(K_a - K_1)A\|_{\mathcal{F}_p} \le C|a-1|\|A\|_{\mathcal{W}_p}. \tag{5.ccxci}$$

Indeed this is a sum of finitely many second differences with coefficients $a_\mu a_\nu - 1$. This is the place where a mere X_α solution space would not suffice.

5.14.2. *The weighted equation is solved, rather than postulated.* The complete normal Wilson gradient has the decomposition

$$J_h^a(A) = K_a A + N_h^a(A). \tag{5.ccxcii}$$

On a fixed complex X_α ball and a fixed weight polydisc,

$$\|N_h^a(A)\|_\infty \le C\|A\|_{X_\alpha}^2, \quad \|N_h^a(A) - N_h^a(\widetilde{A})\|_\infty \le Cr\|A - \widetilde{A}\|_{X_\alpha}. \tag{5.ccxciii}$$

These are consequences of the actual plaquette proof of Lem. 3.184, applied separately to each of the six orientations before summing. To make the parameter dependence explicit, the four-factor logarithm is $F = \mathrm{d}A + E_h(A)$ with $DE_h = O(A\,DA)$; the sine remainder has rough one-difference bound $O(h^3\|F\|_\infty^3)$. Each is multiplied by the constant $a_\mu a_\nu$. The normal-to-left differential is $1 + O(hA)$, and its product with $K_a A$ is bounded by $Ch\|A\|_\infty h^{-1}\|DA\|_\infty$. Thus no unbounded second difference occurs in the nonlinear remainder. These finite expressions are jointly holomorphic in A, a on a larger fixed ball; Cauchy estimates there give the Lipschitz bound in eq. 5.ccxciii.

Set $C_h = q_h - Q$, and let $\mathcal{L}_0^{-1}$ be the inverse of Lem. 5.46. The actual penalized equation is equivalent to the fixed point

$$\begin{pmatrix} A \\ \omega \end{pmatrix} = \mathcal{L}_0^{-1} \begin{pmatrix} -(K_a - K_1)A - N_h^a(A) + (q_h'(A) - Q)^\sharp \omega \\ B - C_h(A) \end{pmatrix}. \tag{5.ccxciv}$$

The constraint in this equation is the entire actual recursion. In particular its physical transpose uses the fixed-input-leg $C_L h^4$ moment of eq. 3.mccc; no directional Hessian estimate is substituted for a column-density estimate. Its nonlinear terms have the bounds eq. 3.mcclx.

Proposition 5.47 (Holomorphic anisotropic critical backgrounds)**.** Fix $p > 4$ and $0 < \alpha < 1 - 4/p$. There are $\varepsilon, r, \rho > 0$, depending on these choices and L but not on j or the volume, such that for $|a-1| < \varepsilon$, $\|B\|_\infty < \rho$, the actual mean fibre has a holomorphic critical orbit $U_j^a(B)$. A unique mean-slice representative and its full multiplier satisfy

$$\|A_j^a(B)\|_{\mathcal{W}_p} + \|\omega_j^a(B)\|_\infty \le C_p\|B\|_\infty. \tag{5.ccxcv}$$

The prescribed literal representative has actual mean $e^{iB}V_0$. For real a, B the field is real and is the local constrained minimum in a fixed strong chart, modulo gauges equal to one at the coarse nodes. The orbit and total action descend across the overlapping real-flat charts of Prop. 3.192. ■

Proof. In the joint norm $\|A\|_{\mathcal{W}_p} + \|\omega\|_\infty$, the Lipschitz constant of eq. 5.ccxciv on radius r is at most $C_p|a-1| + C_{p,L}r$, by eq. 5.cclxxxvii and eq. 5.ccxci and eq. 5.ccxciii. Choose ε first so the first term is at most $1/8$, then r so the second term is at most $1/8$, and finally $C_p\rho < r/4$. The fixed point is invariant and contractive. The iteration from zero converges locally uniformly on the fixed complex product ball. It constructs the jointly holomorphic pair and proves eq. 5.ccxcv; it does not assume the desired background or its inverse.

The fixed point first solves the equation with its gauge penalty retained. The actual identities eq. 3.mcclxx and eq. 3.mcclxxi still hold because the group mean has not changed. For $\tau_A\lambda = 0$,

gauge invariance of the weighted action gives $\langle J_h^a(A), \Gamma_A \lambda \rangle = 0$, while the constraint force also vanishes in that test. The site construction used in the penalty-removal proof of Prop. 3.183 makes $\lambda \mapsto RD^\sharp \Gamma_A \lambda$ onto $\operatorname{ran} R \cap Z_\alpha$. Its correction uses the actual τ_A, the unchanged site inverse $G_s R$, and the smooth E_q. Our field lies in its required X_α ball by eq. 5.cclxxxvii. Therefore the remaining penalty annihilates every vector in that range and is zero. For complex fields this is a range and bilinear nondegeneracy argument, not an inference from a possibly zero complex square. All tests are finite-array gauge directions. They need not satisfy a uniform $\mathcal{W}_p$ norm for this exact identity to apply.

The literal node correction uses sampled smooth profiles with uniform bounds of every fixed order, as in eq. 3.mcclxxxii. Its node values are $O(B)$ and its logarithm has bounded local $W^{3,p}$ norm. The discrete product and exponential rules hence preserve the field $W^{2,p}$ bound. It gives the claimed actual literal mean by the exact plain/straightened factorization, not by equating their derivatives.

For real weights in this smaller ball, $W_a \geq cI$. On the linear slice and $\ker Q$, the weighted quadratic form plus the unchanged gauge penalty is a relative $O(|a-1|)$ perturbation of the proved positive flat form. The actual nonlinear Hessian and multiplier terms add a relative $O(r)$ form error. The constrained chord argument in the isotropic proof therefore retains a positive lower bound. This proves the local minimum and local orbit uniqueness.

For clarity, uniqueness is also valid against an admitted X_α competing critical field. Put it into the actual mean slice by the existing site construction. Its finite-array equation, eq. 5.ccxc and the constant principal perturbation give $\|A\|_{\mathcal{W}_p} \leq C_p(\|B\|_\infty + r^2) + C_p \varepsilon \|A\|_{\mathcal{W}_p}$. The competing multiplier is bounded by the same smooth-column weak readout used in eq. 3.mccxci; the weighted Wilson form changes its constant only by a fixed factor. Absorb the last term, shrink r, and use the contraction's larger uniqueness ball. Thus no extra $W^{2,p}$ assumption on competing orbits is hidden in this argument. The geometric flat-reference transitions preserve the equation and the original strong bounds. The same upgrade and uniqueness identify their total orbits on overlaps, proving the asserted descent. □

5.14.3. *Localization, affine probes and actual free coercivity.* The preceding inverse also gives a localized analytic response, including a derivative of the weight. Here the weight is one global parameter: no decay in a fictitious location of a is claimed. The flat inverse with K_a in place of K_1 is its Neumann series in the bounded operator $(K_a - K_1) : \mathcal{W}_p \to \mathcal{F}_p$. The weighted cell bound in Lem. 5.46 makes that series convergent at a fixed positive spatial rate. Smooth multiplication by the unit-cell partition is bounded on $\mathcal{W}_p$ by the discrete product rule through order two. The actual Wilson and constraint residuals are supported in fixed physical stars and have the same uniform Banach Cauchy bounds as in eq. 4.cclxix. Expanding this equation with the a-dependent inverse gives the same ordered rooted trees as there. Their propagator sums are uniform on a slightly larger weight polydisc. The scalar majorant for arity at least two converges after decreasing ρ once. Consequently every fixed-order background jet, and its first a derivative on a smaller polydisc, has exponential cell moments uniform in j. This proves the required response localization from the actual equation, rather than assigning a small global inverse to an arbitrary short carrier.

At the trivial flat reference let $A_\mu^F(x) = \frac{1}{2} \sum_\nu F_{\nu\mu} x_\nu$, with constant antisymmetric F. Then

$$K_a A^F = 0, \qquad D^\sharp A^F = 0, \qquad Q_h A^F = B^F. \tag{5.ccxcvi}$$

The first identity differentiates a constant weighted curl; the second is antisymmetry, and the third is the exact centroid count of the actual straight row. Exponential inverse kernels act on polynomial-growth data, so uniqueness of the extended joint inverse identifies its affine response with A^F and multiplier zero. The literal response adds only the actual node-frame gradient. This remains true after one a derivative: the transverse affine response is independent of a. Complete invariant densities with zero first jet therefore have coefficient-preserving direct and successive pullbacks,

by the argument of eq. 5.clxxvi and eq. 5.clxxvii. Individual fixed-reference Taylor pieces are not asserted invariant.

For the Taylor vertices and Fourier coefficient assertions below, take the trivial flat reference, as in the isotropic classical comparison. The existence and descent theorem above continues to allow arbitrary real flat holonomy. Put $S_j^a(B) = \mathscr{A}_{h_j}^a(U_j^a(B))$ and $T_{j,m}^a = D_B^m S_j^a(0)$. Positivity of the actual flux averages in eq. 5.ccxxx gives, for real a,

$$c\,\mathrm{d}^\sharp\mathrm{d} \le T_{j,2}^a \le CI, \tag{5.ccxcvii}$$

uniformly in depth. Indeed $W_a \ge cI$ bounds the fine curl energy below, the literal and straight means have the same curl, and the smooth exact trial lift bounds it above. The weighted actual background minimizes that fine energy at its mean. This establishes the free gap needed for the Gaussian comparison below without assuming an effective precision.

5.14.4. *Mesh comparison with a genuine parameter derivative.*

Proposition 5.48 (Anisotropic Taylor convergence and its weight jet)**.** For each $0 < \sigma < 1$ there are a sufficiently small real weight cube $\mathcal{A}_\sigma$ about 1 and $s_\sigma > 0$ such that, for $m = 2, 3, 4$,

$$\sup_{a\in\mathcal{A}_\sigma}\left\{\|T_{j+1,m}^a - T_{j,m}^a\|_{m,s_\sigma} + \sum_{\lambda=1}^4 \|\partial_{a_\lambda}(T_{j+1,m}^a - T_{j,m}^a)\|_{m,s_\sigma}\right\} \le C_\sigma L^{-\sigma j}. \tag{5.ccxcviii}$$

The norm is the rooted exponential kernel norm eq. 5.ccxx. The actual bulk companions and their first weight jets thus converge in that norm. Their finite torus versions are the corresponding periodic multilinear kernels. ∎

Proof. Choose $\beta < 1$ close enough to one and p so large that $\beta < 1 - 4/p$. The preceding construction and literal correction give a uniform X_β bound. For real weights the interpolation and actual mean estimates Props. 5.22 and 5.27 are unchanged. The energy estimate is simply the sum of its six orientation estimates multiplied by $a_\mu a_\nu$. The exact row-seed repair of Lem. 5.28 is unchanged as a field construction; its first-variation bound is multiplied by bounded weights. The weighted local minimum proved above is applicable in both directions of the two-mesh trial comparison. Hence, uniformly for real a in a fixed cube and real B in a fixed smaller ball,

$$|S_{j+1}^a(B) - S_j^a(B)|/\mathcal{V} \le C_\beta h_j^\beta. \tag{5.ccxcix}$$

The complex actions on the larger product ball are uniformly bounded per volume: each plaquette contributes Ch_j^4, and there are $O(h_j^{-4}\mathcal{V})$ plaquettes.

Fix m real source profiles of supremum at most one, a real weight centre a_* in a smaller cube, and one coordinate λ. Apply Lem. 5.30 to the $m+1$ variables of

$$f(t,z) = \mathcal{V}^{-1}(S_{j+1}^{a_*+te_\lambda} - S_j^{a_*+te_\lambda})\left(\sum_{i=1}^m z_i b_i\right).$$

All its real values on the fixed cube are small by eq. 5.ccxcix, and the enclosing complex polydisc is uniformly bounded. The resulting mixed derivative is bounded by $Ch_j^\beta(1+j)^{2(m+1)}$. The version without t gives the same statement for the undifferentiated vertex. This is a joint real-to-jet argument; Cauchy's inequality has not been applied to a real-only bound on a.

Cosine and sine profiles give the same bound for the vertex symbols at real external momenta. The localized background trees above, the physical plaquette density h_j^4, and the $O(h_j^{-4})$ plaquettes per unit cell give a uniform larger holomorphic momentum strip for these symbols. Its first weight jet has the same bound by Cauchy on the larger weight polydisc. For a fixed m each external distance is dominated by a fixed multiple of the root distances of the background jets; hence the exponential moments used here require no all-arity claim.

Apply the iterated strip interpolation argument of Prop. 5.32 to each symbol and its weight derivative. On a sufficiently smaller strip it gives the power $\theta = (1 - s/s_0)^{4(m-1)}$ of the real-momentum bound. Fourier contour shifts on a slightly larger strip than the output one bound its absolute exponentially weighted kernel sum. Choose β and then $s > 0$ so that $\beta\theta > \sigma$; the fixed powers of $1 + j$ are absorbed in that strict margin. This proves eq. 5.ccxcviii for all three orders at their common smaller rate. The parameter cube and the original holomorphic radii may depend on these choices, but not on depth.

Finally the differentiated fixed-point trees commute with periodic summation in their distinct external slots. After choosing a lift of an internal vertex, each child input image is independently summed. There is no extra closed-loop image constraint in these classical rooted trees. Thus the periodic multilinear kernels are the exact periodizations used above; no assertion that periodization commutes with ordinary pointwise products is needed. □

The conclusion supplies an actual differentiated classical hierarchy. It does not construct a proper history from arbitrary curvature-only coarse data, admit independent enlarged cube parameters, or produce a spatially varying anisotropy. Those are different domain questions.

5.15. **The actual anisotropic Gaussian increment.** We compare the constrained Gaussian of the weighted classical hierarchy with the weighted massive lattice determinant. Both the comparison and the convergence of successive increments retain a weight derivative. These are separate steps: a uniformly bounded cumulative regulator error alone does not imply convergence of its successive differences.

Keep the homogeneous domain of Prop. 5.47, using the trivial flat reference for the Taylor coefficients and their bulk symbols. Put $\mathsf{A} = \mathrm{diag}(a_\mu)$, acting on the vector index. For a complete invariant pointed functional, let $b_{\mu\nu}$ be its quadratic coefficient in the single plane $\{\mu, \nu\}$, in the physical convention of Prop. 5.14. At fixed diagonal weights there are six such coefficients. Axis permutations also permute the weights; scalar isotropy is not assumed at $a \neq 1$.

5.15.1. *The weighted physical gauge and its uniform gap.* At the actual background $U = U_j^a(B)$, re-center group coordinates at zero and let $q = q_U, Q = Q_U, D = D_U$ be the actual re-centered rows and gradient. Thus $qq^\sharp = 1$ and $QD = \mathrm{d}_c q$. These site rows are not the effective fixed-reference map τ_A away from zero. Define

$$\begin{aligned} \Delta_a &= D^\sharp \mathsf{A} D, & K_a &= (\mathscr{A}_h^a)''(U) + \mathsf{A}DD^\sharp\mathsf{A}, \\ \widetilde{K}_a &= \mathsf{A}^{-1/2} K_a \mathsf{A}^{-1/2}, & V_s &= q^\sharp q, \\ V_v &= \mathsf{A}^{-1/2} Q^\sharp Q \mathsf{A}^{-1/2} - \mathsf{A}^{1/2} D q^\sharp q D^\sharp \mathsf{A}^{1/2}. \end{aligned} \tag{5.ccc}$$

The scalar averaging precision is $\Delta_a + V_s$. In original vector coordinates it is $(\mathscr{A}_h^a)'' + \mathsf{A}D(1 - q^\sharp q)D^\sharp\mathsf{A} + Q^\sharp Q$; its congruent version is $\widetilde{K}_a + V_v$. The gauge is $(1 - q^\sharp q)D^\sharp\mathsf{A}$, with scalar Faddeev–Popov operator Δ_a on the corresponding constrained site variables. These choices are tied to one another.

The real positive branches of $\mathsf{A}^{1/2}$ extend holomorphically to a fixed complex polydisc about 1. Their coordinate Jacobian is field independent and disappears on vacuum subtraction. In the original vector coordinates the massive regulator is $s\mathsf{A}$, not sI. After congruence it is sI, as in eq. 5.cliv.

Lemma 5.49 (Uniform weighted averaging and interpolation inverses)**.** For real a and real actual backgrounds on a fixed sufficiently small domain, both averaging precisions are uniformly comparable to the free physical H^1 form. The same holds for

$$M = K_* + 1, \qquad S_t = M + t(V - 1), \quad 0 \le t \le 1, \quad (K_*, V) = (\Delta_a, V_s) \text{ or } (\widetilde{K}_a, V_v). \tag{5.ccci}$$

On a smaller complex weight/background domain these normalized operators have uniform relative-form inverses. Their weighted Bloch inverses are holomorphic in a fixed external momentum strip. No positivity assertion is made at complex parameters. ■

Proof. At $a = 1$ and a flat reference these are the background-gauge and site gaps of Prop. 5.20. Their upper forms have the same H^1 bound. Since the weights are constant,

$$|\langle v, (K_a - K_1)v\rangle| \le C|a-1|\|v\|_{H^1}^2$$

at a flat reference; the corresponding statement for Δ_a is immediate. The Q and $qD^\sharp$ terms have bounded relative forms, so their weighted changes satisfy the same estimate. The actual background has uniformly small X_α norm by eq. 5.ccxcv and eq. 5.cclxxxvii. Exact plaquette and transport comparisons give an additional relative $C\rho$ error, with the actual row densities retained. Taking $|a-1| + \rho$ sufficiently small preserves both gaps. The massive endpoint has the same gap; every real interpolation is their convex combination.

For complex parameters normalize by the fixed free $(L_h + 1)^{-1/2}$. The error from the real flat operator is small in operator norm, so a Neumann series proves the stated inverse bound. A small exponential physical cell conjugation changes each derivative form and each finite-carrier row form by $O(s)$. Its normalized inverse is again a Neumann series, uniformly in depth. This is the fixed-strip assertion. It uses relative-form estimates and the actual re-centered rows, not a complex orthogonal decomposition. □

5.15.2. *The regulator remainder is bounded with its weight jet.* Let $G = M^{-1}$ and $W = V - 1$. The exact integrated resolvent Taylor identity of eq. 5.clxxxix reads

$$\begin{aligned}\log\det(K_* + V) - \log\det(K_* + 1) + \operatorname{Tr} G + \tfrac{1}{2}\operatorname{Tr} G^2 = {}& \operatorname{Tr} GV + \operatorname{Tr} G^2 V - \tfrac{1}{2}\operatorname{Tr} GVGV \\ &+ \int_0^1 t^2 \operatorname{Tr}(GWGWS_t^{-1}W)\,dt.\end{aligned} \tag{5.cccii}$$

It holds first in finite dimensions, with coherent log branches; the polarized bulk traces below then give its volume limit. Use the *fixed isotropic* free comparison $G_0 = (L_h + 1)^{-1}$, with the identity in colour and vector indices, and put

$$B = G_0^{1/2} M G_0^{1/2}, \quad B_t = G_0^{1/2} S_t G_0^{1/2}, \quad C = G_0^{1/2} V G_0^{1/2}, \quad X = C - G_0. \tag{5.ccciii}$$

The integral trace is then $\operatorname{Tr}(B^{-1}XB^{-1}XB_t^{-1}X)$. In particular anisotropy has not been hidden in a parameter-dependent free reference whose differentiated trace might have a large rank.

For the vector case set $E = Q\mathsf{A}^{-1/2}G_0^{1/2}$ and $F = qD^\sharp \mathsf{A}^{1/2} G_0^{1/2}$. Then exactly

$$C = E^\sharp E - F^\sharp F. \tag{5.ccciv}$$

The scalar factorization is $C = (qG_0^{1/2})^\sharp (qG_0^{1/2})$. At a real Bloch momentum there are 12 vector rows and 3 site rows per physical coarse cell. The normalized vector C has rank at most 15, and the scalar one at most 3. The physical transposes and the constant congruence are part of these formulas.

All factors in eq. 5.ccciii and eq. 5.ccciv, and their first two background jets, are uniformly bounded on a fixed complex weight polydisc and momentum strip. Here are the estimates behind that assertion. The exact Wilson Hessian is a sum of finitely many derivative forms with bounded plaquette coefficients. Their two background derivatives are controlled by the X_α bounds and localized jets established above. Multiplication by $a_\mu a_\nu$ does not change their fine difference orders. The actual q, Q transports have uniformly bounded physical path length; their complete column densities and two differentiated moments are those of eq. 3.mccc. Also D' is zeroth order. Normalizing every derivative form on each side by $G_0^{1/2}$ therefore bounds all these jets without a fine-volume factor. Holomorphy on a larger weight polydisc gives the same statements after one a derivative. The diagonal inverse bounds come from Lem. 5.49.

To bound the polarized trace, use the four-grade algebra recording the two external momenta $p, -p$, as in the proof of eq. 5.cxciv. Its diagonal fine momenta are $\kappa, \kappa+p, \kappa-p, \kappa$. The normalized matrix inverses are finite triangular inverses with at most two strict jet steps. The jet of eq. 5.ccciv still factors through four copies of the coarse rows, hence has rank at most 60 in the vector case and 12 in the scalar case. Analytic transposes are taken before the jet construction; they do not reverse the four grades. Each term with a C factor thus has bounded trace norm by rank times the bounded operator norms.

The only remaining terms in the last line of eq. 5.cccii have three G_0 factors. For a fixed small complex shift z,

$$\int_{[-\pi,\pi]^4} \|G_0(\kappa+z)\|_{\mathcal{S}_3}^3 \frac{d\kappa}{(2\pi)^4} \le C \int_{\mathbb{R}^4} (1+c|q|^2)^{-3}\, dq < \infty. \tag{5.cccv}$$

This is the exact alias partition of the fine Brillouin cube. Schatten Hölder, followed by Hölder in κ for differently shifted factors, bounds their mixed traces. The bounded normalized anisotropic factors B^{-1}, B_t^{-1} do not affect this estimate. For real external p, cyclic trace identities are applied after integrating κ, where their shifts are legitimate torus reindexings. Holomorphic continuation then proves the identities and bounds for complex p. No complex contour reindexing is assumed. The bounds are uniform on a larger weight polydisc, so Cauchy's inequality legitimately bounds one weight derivative of the whole remainder.

The actual saddle factors must also be retained. For an onto constraint C_0 and T_* equal to the unconstrained site or background-gauge precision, block elimination gives

$$|\det \mathbb{S}(T_*, C_0)| = \det(T_* + C_0^\sharp C_0)\, \det\{C_0(T_* + C_0^\sharp C_0)^{-1} C_0^\sharp\} \tag{5.cccvi}$$

for real parameters. A constant congruence gives the vector version of eq. 5.ccc. The actual smooth row lifts and their H^1 energy bound give uniform upper and lower bounds on these coarse Schur matrices. They have 12 or 3 rows per cell, and the preceding weighted inverse estimates bound their mixed log derivatives on the same strip. Their branches are fixed by positivity on the real domain and then continued. Thus their plane coefficients and first weight jets are bounded as well. These are coarse finite ranks per fibre; no assertion of fixed rank for the full finite-volume trace is used.

5.15.3. *Current corrections and coefficient-preserving pullbacks.* The finite physical Gaussian gauge-change identity eq. 5.civ is algebraic for any invariant action. Apply it to $\mathscr{A}_h^a$ with the weighted divergence just used. Its actual multiplier correction is the Hessian term $-\langle \mu, q'' \rangle$ on the straightened fibre. It is not assumed to be third order in momentum as a matrix. As in that finite identity, coarse stabilizer factors are canceled before taking the flat limit or a holomorphic continuation. The remaining fine and coarse Schur expressions are the gapped ones estimated above. Thus no logarithm of an uncanceled singular coarse Faddeev–Popov factor is continued through the flat field.

For completeness its coefficient vanishes for every weight. Write the joint determinant ratio as $L_a(V,\nu)$, with $L_a(V,0) = 0$. Global colour covariance gives the full first ν derivative zero at $(1,0)$. The mixed V,ν Hessian $K_a(p)$ obeys its gauge Ward identity $d(p)^\sharp K_a(p) = 0$; analytic expansion at $p = 0$ gives $K_a(0) = 0$. The actual multiplier is the derivative of the classical coarse action. Its linear response $P_a(p)$ is $O(p^2)$: gauge covariance gives its zero constant term, and independent coordinate reflections in midpoint coordinates remove its odd linear term. Constant diagonal weights preserve these reflections. The quadratic jet of $L_a(V, \nu_a(V))$ is consequently $O(p^3)$, since its terms are a product of $K_a = O(p)$ and $P_a = O(p^2)$, or two P_a factors. This proves $b_{\mu\nu}[L_a \circ \nu_a] = 0$ for every plane. The exact identity on the weight polydisc also has zero first weight derivative. It is the functional argument of Prop. 5.8, with its hypotheses now verified for the weighted action.

The direct affine-response identity eq. 5.ccxcvi proves coefficient preservation for the weighted critical pullback. The successive version uses actual nesting, not identification with a direct bare

Wilson lift. A full-depth weighted critical field is critical on every smaller actual mean fibre. Its partial literal mean lies in the common small chart by the same mean-domain estimates. Local orbit uniqueness identifies it with the successive critical construction. Differentiating this identity on an affine probe gives the intermediate affine field, with its raw link-length factor, plus an actual gradient. The complete functional's vanished first jet removes the second derivative of the critical map; its two-leg Ward identity removes that gradient. This is precisely the literal/straightened distinction in eq. 5.clxxvi and is unchanged by a.

Define the vacuum-subtracted weighted massive physical logarithm

$$\mathcal{L}_h^a(s,U)=\log\det(\Delta_{a,U}+s)-\tfrac12\log\det(\widetilde{K}_{a,U}+s),\qquad(\mathcal{L}_h^a)^\sharp=\mathcal{L}_h^a(1)-\partial_s\mathcal{L}_h^a(1)+\tfrac12\partial_s^2\mathcal{L}_h^a(1). \tag{5.cccvii}$$

Let Ψ_j^a be the actual cumulative physical Gaussian logarithm of this weighted hierarchy in the canonical contour gauge. Gauge-change invariance makes it equal to the weighted background-gauge expression after the physical coarea factors are included. The preceding trace and Schur estimates give

$$b_{\mu\nu}[\Psi_j^a]=\widetilde{\beta}_a^{\mu\nu,\sharp}(h_j^2)+E_j^{\mu\nu}(a),\qquad\sup_{j,a}(|E_j^{\mu\nu}(a)|+|\partial_aE_j^{\mu\nu}(a)|)\le C. \tag{5.cccviii}$$

The error is holomorphic on a fixed smaller weight polydisc. The mass substitution is $z=h_j^2s$ even in the weighted case: both scalar and congruent vector physical operators are h_j^{-2} times the bare ones. The plaquette area h_j^2 squared cancels the site density h_j^{-4} in its plane coefficient. The left side of eq. 5.cccii combines to the averaging logarithm minus $\mathcal{L}_h^a(1)-\partial_s\mathcal{L}_h^a(1)+\frac12\partial_s^2\mathcal{L}_h^a(1)$; the signs here follow from $(\log\det)'=\operatorname{Tr}G$ and $(\log\det)''=-\operatorname{Tr}G^2$.

5.15.4. *The differentiated Gaussian polarization is continuous.* The actual one-step contour gauge remains $\sum_x(1-\frac12\operatorname{tr}T_x)$. Its cubic and quartic vertices, the actual plain mean vertices, Haar terms and coarea convention in eq. 5.ccxxxix are unchanged. Only the input Taylor triple $\mathfrak{J}_j^a=(T_{j,2}^a,T_{j,3}^a,T_{j,4}^a)$ has changed. The real gap eq. 5.ccxcvii and the fixed contour gap retain uniform full saddle inverses; replacing the lower constant one by $c>0$ only changes their constants. The rooted trace proof of Prop. 5.36 therefore applies on this weighted class and its smaller complex neighborhood.

Here is the extra derivative estimate needed, beyond continuity of the undifferentiated formula. Let Π_j^a be the actual Gaussian polarization. For a weight direction put a dot over its derivative. If $\mathbb{G}$ is its full field/multiplier saddle inverse and Z its actual response column, then

$$\dot{\mathbb{G}}=-\mathbb{G}\begin{pmatrix}\dot{T}_2&0\\0&0\end{pmatrix}\mathbb{G},\qquad\dot{Z}=\dot{\mathbb{G}}\,I_{\text{datum}}. \tag{5.cccix}$$

The augmented $\dot{V}_3,\dot{V}_4$ are exactly the differentiated action vertices. The constraint and literal trace-gauge vertices are fixed. Differentiate the two rooted contractions in eq. 5.ccxliii and eq. 5.ccxliv, using eq. 5.cccix, and differentiate the Haar matrix product. Each resulting expression has one dotted factor, bounded in the same smaller weighted Schur norm. Subtracting the formulas for two depths, one factor at a time, gives

$$\begin{aligned}\|\Pi_{j+1}^a-\Pi_j^a\|_{t,2}+\|\dot{\Pi}_{j+1}^a-\dot{\Pi}_j^a\|_{t,2}\le C(&\|\mathfrak{J}_{j+1}^a-\mathfrak{J}_j^a\|_s\\&+\|\dot{\mathfrak{J}}_{j+1}^a-\dot{\mathfrak{J}}_j^a\|_s).\end{aligned} \tag{5.cccx}$$

Uniform bounds for the undifferenced dotted triples follow from their holomorphic weight ball. Rooting the first external leg before summing the loop contractions removes any volume factor, exactly as in the original polarization proof. The complete field and multiplier first derivative of the log determinant still vanishes by colour, so no omitted $V_3[z'']$ term is introduced by the weight derivative.

Proposition 5.50 (The actual diagonal one-loop law)**.** Let $b^a_{j,\mu\nu}$ be the plane coefficient of the actual successive homogeneous Gaussian increment, in the same normalization as eq. 5.cc. For every $0 < \sigma < 1$, on a fixed sufficiently small real weight cube,

$$\begin{aligned} b^a_{j,\mu\nu} &= \frac{11}{3\pi^2} d(a) a_\mu a_\nu \log L + O_\sigma(L^{-\sigma j}), \\ \partial_{a_\lambda} b^a_{j,\mu\nu} &= \frac{11}{3\pi^2} \partial_{a_\lambda}\{d(a) a_\mu a_\nu\} \log L + O_\sigma(L^{-\sigma j}). \end{aligned} \tag{5.cccxi}$$

In particular, for $a(t) = 1 + te$ with $\sum_\lambda e_\lambda = 0$, the actual anisotropic increment obeys

$$\left.\frac{d}{dt}\right|_0 b^{1+te}_{j,\mu\nu} = \frac{11}{3\pi^2}(e_\mu + e_\nu) \log L + O_\sigma(|e| L^{-\sigma j}). \tag{5.cccxii}$$

The leading derivative in the isotropic weight direction is zero. ∎

Proof. Apply eq. 5.cccx to Prop. 5.48. The plane readout is a fixed second-momentum derivative, or equivalently a second absolute moment, and is continuous in the resulting exponential kernel norm. Hence $b^a_{j,\mu\nu}$ and each first weight jet have successive differences $O_\sigma(L^{-\sigma j})$. Their geometric tails converge at that rate to limits, uniformly on the smaller real weight cube.

Exact physical Gaussian composition and the weighted successive affine coefficient preservation proved above give

$$\sum_{\ell=1}^{j} b^a_{\ell,\mu\nu} = b_{\mu\nu}[\Psi^a_j].$$

It can be differentiated at finite j on the common weight domain. The weighted massive law eq. 5.clxix and eq. 5.cccviii show that these two cumulative quantities are respectively $2c\, d(a) a_\mu a_\nu j \log L + O(1)$ and its weight derivative with an $O(1)$ error. Dividing by j and taking Cesaro limits identifies the already proved successive limits and their jets. Since $2c = 11/(3\pi^2)$, this proves eq. 5.cccxi. Finally $\partial_t[d(1+te)(1+te_\mu)(1+te_\nu)]_{t=0} = e_\mu + e_\nu - \frac{1}{2}\sum e_\lambda$, which gives eq. 5.cccxii and the isotropic statement. □

The result concerns the actual homogeneous Gaussian increments and their constant diagonal anisotropy. It uses neither an unproved interacting cutoff symmetry nor a termwise invariant local activity decomposition. It supplies no independent fine-core or history-tower admission and no arbitrary proper-domain extension.

5.16. **A tuned trajectory for the homogeneous regular steps.** The step estimates have to be solved with conditions at opposite ends of the scale interval. The ultraviolet profile is zero, whereas the coupling is prescribed at the infrared end. We solve this problem without requiring the sum of the profile formation bounds to be small. That sum need not converge. The summable quantity is the coupling derivative of the formation map.

Throughout this subsection the integral operations are the homogeneous regular steps on the literal buffered domains. The frozen cutoffs, relative contour gauge, physical pointings and Gaussian normalization are the ones in Prop. 4.46. The large-field complementary operations are not included in the trajectory constructed here.

5.16.1. *The spaces and exact transport.* Fix the blocking integer and localization cubes after the rate choices in Prop. 4.36. Use the same field radius r and carrier rate at every scale, with the larger formation ball and its literal buffer retained. Let $\mathscr{B}_j$ be the space of complete pointed analytic families $h = (h_z)$ at scale j which have zero value, zero first jet and zero companion curvature coefficient. They are gauge invariant and covariant under translations and the hypercubic group. The norm is the infimum of the rooted analytic carrier norms over their compatible local representations. Include the fixed finite-period compatibility norm. All families have their infinite companions and their finite torus versions on the declared compatible volumes.

This infimum is used to identify representations of the same complete pointed functions. The space is completed in that norm. An absolutely summable sequence of representations gives a representation of its sum; hence this completion consists of normally convergent pointed analytic functions, with the same identities. In particular no invariance of each individual local representative is required. Write $H = \sum_z h_z$ for its total finite-volume function.

Let $T_{i,k}$ be the physical transport of eq. 4.ccclxxxiii, with no new Gaussian integration. For $i = k$, set $T_{i,i} = 1$. The common-radius construction gives fixed $C_T \geq 1$ and $0 < \lambda < 1$ such that

$$\|T_{i,k}\|_{\mathscr{B}_i \to \mathscr{B}_k} \leq C_T \lambda^{k-i}, \qquad i \leq k. \tag{5.cccxiii}$$

One may take $\lambda = L^{-\beta}$ with the fixed exponent in the transport theorem. The same bound holds for the compatibility component of the norm, with its fixed cube-unit period exponent.

The exact semigroup identity is

$$T_{j,k} T_{i,j} = T_{i,k}. \tag{5.cccxiv}$$

There are two points in this assertion. First, all-depth classical nesting identifies the two literal intermediate fields up to an old site gauge. The complete h_z is invariant, so its two compositions agree. Second, the pointing weights themselves have an exact refinement identity. For $\xi = L^{j-k}$ and a point t in the intermediate coordinate system,

$$\sum_z \theta_y(\xi z)\theta_z(t) = \theta_y(\xi t). \tag{5.cccxv}$$

In one dimension the right side is piecewise linear with all knots at integers, since ξ^{-1} is an integer. It equals its linear interpolant from those nodal values. Taking the product in four coordinates proves eq. 5.cccxv. The periodic sums give the same identity on compatible tori. The absolute local bounds justify interchanging these sums. Thus eq. 5.cccxiv is an identity of complete pointed functions, even if two chosen filings differ.

Fix a terminal scale N and choose $\lambda < q < 1$. Define an equivalent norm at scale j by

$$\|h\|_{j,*} = \sup_{0 \leq n \leq N-j} q^{-n} \|T_{j,j+n} h\|_{\mathscr{B}_{j+n}}. \tag{5.cccxvi}$$

Then

$$\|h\|_{\mathscr{B}_j} \leq \|h\|_{j,*} \leq C_T \|h\|_{\mathscr{B}_j}, \qquad \|T_{j,j+1} h\|_{j+1,*} \leq q \|h\|_{j,*}. \tag{5.cccxvii}$$

The last inequality follows directly by shifting n to $n+1$ in eq. 5.cccxvi and using eq. 5.cccxiv. Its constants do not depend on the horizon N. This removes the harmless prefactor C_T from a one-step induction.

5.16.2. *The exact step and its two formation derivatives.* Let $\mathcal{N}_j$ be the vacuum-subtracted physical Gaussian pointing for the step from j to $j+1$. Put

$$b_j = b[\mathcal{N}_j^{\infty}], \qquad n_j = \mathcal{N}_j - b_j e_{j+1},$$

where e_{j+1} is the actual classical pointing in eq. 4.ccclviii. The Gaussian comparison and the canonical input bounds give

$$\|n_j\|_{j+1,*} \leq B_n, \qquad |b_j - b_\infty| \leq B_b \lambda_b^j, \qquad b_\infty = \frac{11}{3\pi^2} \log L > 0, \quad 0 < \lambda_b < 1. \tag{5.cccxviii}$$

In particular the individual b_j are bounded. Positivity of every initial b_j is unnecessary.

The scalar power in the physical Gaussian mismatch is kept in the vacuum normalization. Consequently the nonconstant $\mathcal{N}_j$ and b_j do not acquire a coupling derivative from that power. The remaining dependence is in the formation map $\mathcal{F}_j(x, h)$. For a fixed frame value θ_j, its cutoff and fluctuation scaling are frozen. Write

$$\epsilon(t) = t^{-1} (\log t)^m, \qquad m = 2p > 1,$$

after increasing the coupling floor. The Gaussian cutoff tail is absorbed in this bound; $p > 1/2$ makes $\exp[-c(\log t)^{2p}]$ smaller than every fixed negative power of t.

For every fixed profile bound H_*, the buffered formation theorem gives, on slightly larger profile and coupling domains,

$$\begin{aligned} \|\mathcal{F}_j(x,h)\| &\le C_{H_*}\epsilon(\theta_j),\\ \|D_h\mathcal{F}_j(x,h)\| &\le C_{H_*}\epsilon(\theta_j),\\ \|\partial_x\mathcal{F}_j(x,h)\| &\le C_{H_*}\epsilon(\theta_j)/\theta_j. \end{aligned} \tag{5.cccxix}$$

Here $|x/\theta_j - 1|$ is smaller than a fixed constant and $\|h\|_{j,*} < H_* + 1$. All norms on the right concern the complete compatible family. The constants include the fixed equivalence in eq. 5.cccxvii. They are uniform in j, N and volume.

Let $\mathcal{P}_{j+1}$ subtract the companion curvature coefficient times e_{j+1}, and put

$$r_j(x,h) = b[\mathcal{F}_j(x,h)^\infty], \qquad f_j(x,h) = \mathcal{P}_{j+1}\mathcal{F}_j(x,h).$$

The coefficient readout and this projection are bounded on the fixed-radius spaces, by the complete two-leg Ward moment. Thus eq. 5.cccxix also holds for r_j, f_j, with changed constants.

Up to its positive scalar vacuum factor, the actual regular step is

$$e^{-xS_j+H} \;\longmapsto\; e^{-xS_{j+1}+H\circ p_j^{\mathrm{lit}}+\mathcal{N}_j+\mathcal{F}_j(x,h)}. \tag{5.cccxx}$$

Extracting the two curvature coefficients gives the exact recursion

$$\begin{aligned} x_{j+1} &= x_j - b_j - r_j(x_j,h_j),\\ h_{j+1} &= T_{j,j+1}h_j + n_j + f_j(x_j,h_j). \end{aligned} \tag{5.cccxxi}$$

The coupling is independent of the torus because its readout uses the infinite companion. The finite-period corrections remain in h_{j+1}. No finite quantum determinant has been replaced by the determinant on a periodic cover.

5.16.3. *Forward stability for a prescribed coupling sequence.* Choose

$$H_* \ge 1 + \frac{2B_n}{1-q}. \tag{5.cccxxii}$$

Fix a sufficiently large real number X, and define the nominal frame

$$a_j = X + \sum_{\ell=j}^{N-1} b_\ell, \qquad a_N = X. \tag{5.cccxxiii}$$

The summable errors in eq. 5.cccxviii imply

$$|a_j - \{X + b_\infty(N-j)\}| \le B_a, \qquad c(X+N-j) \le a_j \le C(X+N-j) \tag{5.cccxxiv}$$

once $X \ge 2B_a + 2$. Freeze $\theta_j = a_j$ throughout the construction. For a fixed small $\delta > 0$ consider coupling sequences in

$$\mathcal{D}_N = \{x : \ |x_j - a_j| \le \delta a_j, \quad 0 \le j \le N\}.$$

Use its slightly enlarged open polydisc for holomorphic estimates. The finite forward recursion in eq. 5.cccxxi, with $h_0 = 0$, defines $h_j(x)$. For X large enough,

$$\|h_j(x)\|_{j,*} \le H_* \quad (0 \le j \le N). \tag{5.cccxxv}$$

Indeed the next norm is at most $qH_* + B_n + C_{H_*}\epsilon(a_j)$. By eq. 5.cccxxii, the first two terms leave at least $(1-q)H_*/2$ of room, and the last term fits there when the coupling floor is fixed. This is an induction in the complete compatible family, so the gauge, symmetry and periodic identities are retained.

The dependence on the coupling sequence has a stronger bound than the profile size itself. Put

$$k_j = \epsilon(a_j)/a_j, \qquad D = \|x - \widetilde{x}\|_\infty.$$

For the two forward solutions, eq. 5.cccxix gives

$$d_{j+1} \le \{q + C_{H_*}\epsilon(a_j)\}d_j + C_{H_*}k_jD, \qquad d_j = \|h_j(x) - h_j(\widetilde{x})\|_{j,*}.$$

Choose X so the bracket is at most some $q_1 < 1$. For earlier $i < j$, the envelope in eq. 5.cccxxiv and monotonic decrease of $(\log t)^m/t^2$ for large t give $k_i \le Ck_j$. The bounded error B_a in that envelope changes only the fixed constant. Hence

$$d_j \le CD\sum_{i<j} q_1^{j-1-i}k_i \le Ck_jD. \tag{5.cccxxvi}$$

No sum of $\epsilon(a_j)$ over all scales appears in this stability estimate.

5.16.4. *The backward coupling contraction.* The terminal condition is $x_N = X$. Summing the first recursion in eq. 5.cccxxi backward suggests the map

$$(\Phi x)_j = a_j + \sum_{\ell=j}^{N-1} r_\ell(x_\ell, h_\ell(x)), \qquad (\Phi x)_N = X. \tag{5.cccxxvii}$$

We prove it is an invariant contraction on $\mathcal{D}_N$.

The following two elementary estimates are uniform in N, j:

$$\begin{aligned} \sum_{\ell=j}^{N-1} \epsilon(a_\ell) &\le C\{1 + (\log a_j)^{m+1}\}, \\ \sum_{\ell=0}^{N-1} k_\ell &\le C\frac{(\log X)^m}{X}. \end{aligned} \tag{5.cccxxviii}$$

To prove them, use eq. 5.cccxxiv to compare with the integer sequence $X + cn$, allowing fixed constant changes in its argument. The first integral is bounded by $\int_X^{Ca_j}(\log t)^m\,dt/t$. For the second, integration by parts bounds $\int_{cX}^{\infty}(\log t)^m\,dt/t^2$ by $C(\log X)^m/X$. For a noninteger $m > 1$, the same integration by parts uses $m(\log t)^{m-1}/t^2$; choosing X large absorbs it in half of the original integral. The finite first terms are covered by the same bounds.

The first line and eq. 5.cccxix give

$$|(\Phi x)_j - a_j| \le C_{H_*}\{1 + (\log a_j)^{m+1}\}.$$

Since $(1 + \log^{m+1} t)/t$ tends to zero, this is at most $\delta a_j/2$ for all $a_j \ge cX$, once X is fixed large enough. Thus the backward map is strictly inside its relative box.

For two sequences, the coupling and profile slopes give

$$\begin{aligned} |r_\ell(x_\ell, h_\ell(x)) - r_\ell(\widetilde{x}_\ell, h_\ell(\widetilde{x}))| &\le Ck_\ell D + C\epsilon(a_\ell)d_\ell \\ &\le Ck_\ell D. \end{aligned}$$

The second line of eq. 5.cccxxviii therefore proves

$$\|\Phi x - \Phi\widetilde{x}\|_\infty \le K(X)\|x - \widetilde{x}\|_\infty, \qquad K(X) \le C\frac{(\log X)^m}{X} < \frac{1}{2}. \tag{5.cccxxix}$$

For each finite N, the relative box is a complete closed subset of the finite supremum-norm space. Its diameter may grow with N; neither its contraction constant nor the floor does.

Proposition 5.51. For every sufficiently large infrared inverse coupling X and every finite horizon N, the actual homogeneous regular recursion eq. 5.cccxxi has a unique trajectory in the displayed weak-coupling box with

$$h_0 = 0, \qquad x_N = X.$$

It is real for real data, has $\|h_j\|_{j,*} \le H_*$, and satisfies

$$\begin{aligned} c(X+N-j) \le x_j \le C(X+N-j), \\ x_0 = X + b_\infty N + O\{1+(\log(X+N))^{m+1}\}. \end{aligned} \tag{5.cccxxx}$$

All constants and the coupling floor are independent of N and compatible volume. The finite-volume profiles are the actual profiles of the stated integral operations, with their compatible companions. ■

Proof. The invariant contraction has one fixed point. Its forward profiles satisfy eq. 5.cccxxv. Conversely every trajectory with the two stated end conditions has those forward profiles and satisfies eq. 5.cccxxvii. This proves uniqueness in the specified box. The real maps preserve real sequences, so iteration from the real nominal frame gives real trajectories. The relative box and eq. 5.cccxxiv give the first line of eq. 5.cccxxx. The sum of $b_j - b_\infty$ is bounded by eq. 5.cccxviii; the first line of eq. 5.cccxxviii bounds the sum of the interaction corrections. This gives its second line. Finally the exact identity eq. 5.cccxx at each step identifies the recursively formed density and its positive real normalizer. This identification uses the actual formation map, not only the abstract inequalities of a hypothetical flow. □

5.16.5. *Analytic tuning in a frozen presentation.* Keep the frame and all its cutoffs fixed at a real reference endpoint X_0. Replace the terminal condition by $x_N = X_0 + \zeta$, where $|\zeta| < cX_0$ for a fixed small $c > 0$. The backward map becomes

$$(\Phi^\zeta x)_j = a_j(X_0) + \zeta + \sum_{\ell=j}^{N-1} r_\ell(x_\ell, h_\ell(x)).$$

For small enough c, the same relative box and slightly larger coupling discs are invariant. Its contraction constant is unchanged. Iteration converges normally in ζ, and the forward profile maps are holomorphic. Thus the tuned trajectory is holomorphic on that disc.

Let a prime denote its endpoint derivative. The proof of eq. 5.cccxxvi gives

$$\|h_j'\|_{j,*} \le Ck_j \sup_\ell |x_\ell'|.$$

Differentiate the backward equation and use the same summable slopes. Then

$$\sup_j |x_j'| \le 1 + K(X_0) \sup_j |x_j'|, \qquad \sup_j |x_j' - 1| \le \frac{K(X_0)}{1 - K(X_0)}.$$

After increasing the floor,

$$\frac{3}{4} \le x_j' \le \frac{5}{4} \quad \text{on a smaller real interval,} \qquad \|h_j'\|_{j,*} \le Ck_j. \tag{5.cccxxxi}$$

The bare tuning is consequently strictly increasing in that real interval. These derivatives concern one frozen presentation. They do not differentiate its sharp cutoff or recompute it from the moving endpoint.

The complex bound for $x_j' - 1$ on a fixed smaller disc also yields, for every integer $s \ge 2$,

$$|x_j^{(s)}| \le C_s \frac{(\log X_0)^m}{X_0^s}, \tag{5.cccxxxii}$$

uniformly in the horizon. Apply Cauchy's estimate to $x_j' - 1$, with disc radius a fixed multiple of X_0. Its supremum is $O((\log X_0)^m / X_0)$, and each additional derivative costs one factor X_0^{-1}. This proves eq. 5.cccxxxii.

The construction controls the regular operations on the stated common domains. It is a coupling and profile trajectory, not a bound for the mass or source moments of all complementary histories. Those quantities require the positive history estimates used in the full construction.

5.17. **Universality for bounded ultraviolet profiles at a fixed horizon.** We keep one finite horizon, one family of regular maps and one frozen cutoff presentation. The ultraviolet irrelevant profile may now be any member of a bounded ball. Retuning the bare coupling to the same infrared value makes the final profiles exponentially close. This is a comparison within that same finite problem; it does not compare different horizons or construct an infinite-horizon flow.

5.17.1. *The fixed presentation and a bounded initial ball.* Use the complete compatible quotient spaces and physical regular maps of Prop. 5.51. Fix

$$\lambda < q < q_1 < q_2 < 1. \tag{5.cccxxxiii}$$

The adapted norm eq. 5.cccxvi makes the exact transport have norm at most q. Its equivalence constant C_T is independent of the horizon. The ultraviolet space $\mathscr{B}_0$ consists of the same gauge-invariant, hypercubic-covariant, normalized complete pointings, with zero first jet and zero companion curvature coefficient. All finite-period versions and their companions are included. An input with a nonzero marginal coefficient is not part of the irrelevant ball considered here.

Fix $R_{\mathrm{uv}} < \infty$. We shall allow $\|h_0\|_{\mathscr{B}_0} < R_{\mathrm{uv}} + 2$ and prove the estimates on the smaller ball $\|h_0\| \le R_{\mathrm{uv}}$. Take the common profile ceiling H_* large enough that

$$H_* \ge 2C_T(R_{\mathrm{uv}} + 2) + 1 + \frac{2B_n}{1-q}. \tag{5.cccxxxiv}$$

The enlarged formation domains are chosen with this ceiling. All constants below may depend on this fixed ball and on the fixed rates, but not on depth, horizon or compatible volume.

Choose a real reference endpoint X_0 above the coupling floor, and fix the entire presentation at

$$a_j = X_0 + \sum_{\ell=j}^{N-1} b_\ell, \qquad \theta_j = a_j, \quad a_N = X_0. \tag{5.cccxxxv}$$

These are scalar inverse-coupling frames, not diagonal weight parameters. In particular every cutoff, fluctuation scaling, gauge normalization and map stays fixed when h_0 or the endpoint is varied. Set

$$e_j = \frac{(\log a_j)^m}{a_j}, \qquad k_j = e_j/a_j, \qquad e_* = \frac{(\log X_0)^m}{X_0}. \tag{5.cccxxxvi}$$

Increase the floor when necessary. The HRT envelopes give $a_j \asymp X_0 + N - j$ and

$$\sum_{j<N} k_j \le C\frac{(\log X_0)^m}{X_0}, \qquad \sum_{j<N} e_j k_j \le Ce_*^2, \qquad \sup_j e_j k_j \le Ce_*^2/X_0. \tag{5.cccxxxvii}$$

For the second estimate compare with $\int_{cX_0}^{\infty} (\log t)^{2m} t^{-3}\, dt$. One integration by parts and a larger floor bound this integral by $C(\log X_0)^{2m}/X_0^2$. The last estimate is the corresponding supremum bound. We will also use

$$k_i \le Ck_j \ (i<j), \qquad e_\ell \le Ce_j\{1 + (\ell - j)/X_0\} \ (j \le \ell \le N). \tag{5.cccxxxviii}$$

Indeed the bounded Gaussian increments give $a_j \le a_\ell + C(\ell - j)$, while $a_\ell \ge cX_0$. For large arguments the logarithmic ratio is bounded when $a_\ell \le a_j + O(1)$; the bounded error in the nominal envelope covers the exceptional initial signs. Thus $e_\ell/e_j \le Ca_j/a_\ell$, which proves the second inequality. The same envelope proves the first one by monotonic decrease of $(\log t)^m/t^2$.

On the enlarged fixed domains the exact regular recursion is

$$x_{j+1} = x_j - b_j - r_j(x_j, h_j), \qquad h_{j+1} = T_j h_j + n_j + f_j(x_j, h_j). \tag{5.cccxxxix}$$

The input ceiling above and Prop. 4.46 give, uniformly there,

$$|r_j| + \|f_j\| \le Ce_j, \quad |\partial_x r_j| + \|\partial_x f_j\| \le Ck_j, \quad \|D_h r_j\| + \|D_h f_j\| \le Ce_j. \tag{5.cccxl}$$

Norms are the appropriate adapted norms. Fixed equivalence constants have been included in C. The scalar coefficients use the infinite companion, so the coupling sequence is independent of the torus. Finite-period differences remain in the profiles.

Proposition 5.52 (Analytic tuning with a nonzero initial profile)**.** For every fixed R_{uv} there is a sufficiently large floor X_0, independent of N, such that eq. 5.cccxxxix has a unique trajectory in the HRT relative coupling box with

$$h_0 = H, \qquad x_N = X_0 + \zeta, \quad \|H\| < R_{\mathrm{uv}} + 2, \quad |\zeta| < cX_0. \tag{5.cccxli}$$

It is jointly holomorphic in H, ζ, real for real data, and has $\|h_j\|_{j,*} \le H_*$. At a real endpoint in this presentation its coupling envelope and bare law are those of HRT, with constants uniform on the input ball. ■

Proof. For a prescribed coupling sequence in the fixed relative box, solve the profile equation forward starting from H. Its first norm is at most $C_T(R_{\mathrm{uv}} + 2)$. The induction

$$\|h_{j+1}\|_{j+1,*} \le qH_* + B_n + Ce_j \le H_*$$

is valid by eq. 5.cccxxxiv and a larger floor. The map is holomorphic on the enlarged coupling and input domains. For two sequences with the same initial H, its difference obeys

$$d_{j+1} \le q_1 d_j + Ck_j \|x - \widetilde{x}\|_\infty, \qquad d_0 = 0,$$

after making $Ce_j \le q_1 - q$. Therefore $d_j \le Ck_j \|x - \widetilde{x}\|_\infty$, exactly as in eq. 5.cccxxvi.

The backward map is

$$(\Phi^{H,\zeta} x)_j = a_j + \zeta + \sum_{\ell=j}^{N-1} r_\ell(x_\ell, h_\ell(x, H)), \qquad (\Phi^{H,\zeta} x)_N = X_0 + \zeta.$$

The same sum of k_j as in HRT makes its supremum-norm contraction constant less than $1/2$, uniformly in H. Its correction from a_j is at most $|\zeta| + C\{1 + (\log a_j)^{m+1}\}$. Choose c small and then the floor large so this is strictly inside the relative box at every j. The fixed-point iteration is invariant and locally normally convergent in H, ζ. It constructs the claimed holomorphic tuning map. No implicit inverse without a uniform bound is being assumed. Every other trajectory with these end data has the same forward profile and hence is the same fixed point. Reality follows by iteration. The bounded initial profile changes only the common formation constants in the proof of eq. 5.cccxxx. □

5.17.2. *The exact mixed-end linear inverse.* Linearize along any trajectory in the smaller complex domains. Write $u_j = \delta x_j$ and $v_j = \delta h_j$, and allow additive step sources $\sigma_j \in \mathbb{C}$, $\tau_j \in \mathscr{B}_{j+1}$. The exact linear system is

$$\begin{aligned} u_{j+1} &= A_j u_j - R_j v_j + \sigma_j, \\ v_{j+1} &= D_j v_j + F_j u_j + \tau_j, \qquad u_N = \xi, \quad v_0 = \eta, \end{aligned} \tag{5.cccxlii}$$

where

$$A_j = 1 - \partial_x r_j, \quad R_j = D_h r_j, \quad D_j = T_j + D_h f_j, \quad F_j = \partial_x f_j.$$

Thus $|A_j - 1|, \|F_j\| \le Ck_j$, $\|R_j\| \le Ce_j$, and $\|D_j\| \le q_1$. All products below use the actual operators; the symbols D_j here are scale-profile maps, not lattice derivatives.

For $j \le \ell$ put

$$P_{j\ell} = \prod_{t=j}^{\ell} A_t^{-1}, \qquad U_{ji} = D_{j-1} \cdots D_{i+1} \quad (i < j), \qquad U_{i+1,i} = 1. \tag{5.cccxliii}$$

The empty backward product is one. By eq. 5.cccxxxvii, the A_j do not vanish and $|P_{j\ell}| \le C$. Also $\|U_{ji}\| \le q_1^{j-1-i}$. Solving one equation backward and the other forward gives exactly

$$\begin{aligned} u_j &= u_j^0 + (\mathcal{B}v)_j, \qquad u_j^0 = P_{j,N-1}\xi - \sum_{\ell=j}^{N-1} P_{j\ell}\sigma_\ell, \\ v_j &= v_j^0 + (\mathcal{K}u)_j, \qquad v_j^0 = D_{j-1}\cdots D_0\eta + \sum_{i<j} U_{ji}\tau_i, \\ (\mathcal{B}v)_j &= \sum_{\ell\ge j} P_{j\ell}R_\ell v_\ell, \quad (\mathcal{K}u)_j = \sum_{i<j} U_{ji}F_i u_i. \end{aligned} \tag{5.cccxliv}$$

In the sums with $\ell \ge j$ the upper limit is $N-1$; at $j = N$ those sums vanish. The first homogeneous forward product at $j = 0$ is one.

For later use define positive scalar kernels

$$B^+_{j\ell} = Ce_\ell \mathbf{1}_{j\le\ell<N}, \qquad K^+_{ji} = Ck_i q_1^{j-1-i}\mathbf{1}_{i<j}. \tag{5.cccxlv}$$

They majorize the two exact operator kernels. In unweighted coupling supremum norm their product satisfies

$$\|B^+K^+\|_{\infty\to\infty} \le C\sup_j \sum_{\ell\ge j} e_\ell \sum_{i<\ell} q_1^{\ell-1-i}k_i \le C\sum_\ell e_\ell k_\ell \le Ce_*^2 =: \vartheta < \tfrac12. \tag{5.cccxlvi}$$

The middle step uses $k_i \le Ck_\ell$. This is the mixed-end loop. It contains both formation derivatives. A sum of e_j alone, which could grow with the horizon, has not been used as a smallness parameter.

Consequently the exact positive Green envelopes are

$$\begin{aligned} |u| &\le \sum_{n\ge0}(B^+K^+)^n(|u^0| + B^+|v^0|), \\ |v| &\le |v^0| + K^+\sum_{n\ge0}(B^+K^+)^n(|u^0| + B^+|v^0|). \end{aligned} \tag{5.cccxlvii}$$

The notation $|v_j|$ means its adapted norm. Every term is nonnegative and each intermediate scale is explicitly summed. The series converges absolutely in the coupling supremum norm; the profile output then follows from the forward kernel. This also proves uniqueness of eq. 5.cccxlii without a sign or self-adjointness assumption on its complex coefficients.

These formulas give useful estimates for arbitrary finite step sources. In fact

$$\begin{aligned} \|u\|_\infty &\le C\Big\{|\xi| + \sum_{i<N}|\sigma_i| + e_0\|\eta\| + \sum_{i<N} e_{i+1}\|\tau_i\|\Big\}, \\ \|v_j\| &\le q_1^j\|\eta\| + Ck_j\|u\|_\infty + \sum_{i<j} q_1^{j-1-i}\|\tau_i\|. \end{aligned} \tag{5.cccxlviii}$$

For the first line, a profile impulse at step i contributes at most $C\sum_{\ell\ge i+1} e_\ell q_1^{\ell-1-i} \le Ce_{i+1}$ to $B^+|v^0|$. The initial profile contributes Ce_0, by eq. 5.cccxxxviii. The scalar direct sources have exactly their backward sums in eq. 5.cccxliv. Apply $(1-\vartheta)^{-1}$ to finish. The second line is the forward equation and $k_i \le Ck_j$.

For example a scalar impulse at i has direct envelope $C\mathbf{1}_{j\le i}$; a profile impulse at i has direct envelope $q_1^{j-1-i}\mathbf{1}_{j\ge i+1}$. Their complete responses are those vectors propagated by eq. 5.cccxlvii. This explicitly retains the backward leakage caused by retuning the terminal coupling. No symmetric exponential bound for an arbitrary scale impulse is claimed.

5.17.3. *Exponential dependence on the initial irrelevant profile.* For the initial-profile response set $\xi = 0$, $\sigma = \tau = 0$ in eq. 5.cccxlii. Introduce

$$\|u\|_E = \max_j \frac{|u_j|}{e_j q_2^j}, \qquad \|v\|_Q = \max_j q_2^{-j}\|v_j\|. \tag{5.cccxlix}$$

The terminal $u_N = 0$ causes no difficulty in this norm. The backward kernel has the bound

$$\|\mathcal{B}v\|_E \le C\|v\|_Q \sup_j \sum_{n\ge 0} q_2^n \frac{e_{j+n}}{e_j} \le C_B\|v\|_Q, \tag{5.cccl}$$

where sums stop at the finite horizon. The last inequality follows from $e_{j+n}/e_j \le C(1+n/X_0)$, not from a false monotonicity in j. For the forward kernel,

$$\|\mathcal{K}u\|_Q \le C\|u\|_E \sup_j \sum_{i<j} q_2^{-1}(q_1/q_2)^{j-1-i} k_i e_i \le C_K e_*^2 \|u\|_E. \tag{5.cccli}$$

Here eq. 5.cccxxxvii even gives an additional factor X_0^{-1}, which is not needed. The loop norm $C_B C_K e_*^2$ is less than $1/2$ after the same fixed floor increase. Since $\|v^0\|_Q \le \|\eta\|$, solving the two inequalities proves

$$|u_j| \le C e_j q_2^j \|\eta\|, \qquad \|v_j\|_{j,*} \le C q_2^j \|\eta\|. \tag{5.ccclii}$$

The proof applies uniformly to complex trajectories on the smaller product domains. It is an inverse estimate with opposite end conditions, rather than an estimate obtained by evolving the scalar coupling forward.

Proposition 5.53 (Finite-horizon exponential universality)**.** Let $H, \widetilde{H}$ lie in the smaller bounded initial-profile ball, and use the same endpoint, horizon and frozen presentation. Then

$$\begin{aligned} |x_j(H) - x_j(\widetilde{H})| &\le C e_j q_2^j \|H - \widetilde{H}\|_{\mathscr{B}_0}, \\ \|h_j(H) - h_j(\widetilde{H})\|_{j,*} &\le C q_2^j \|H - \widetilde{H}\|_{\mathscr{B}_0}. \end{aligned} \tag{5.ccliii}$$

In particular the analytic bare tuning map has $\|D_H x_0\| \le C e_0$, and $\|D_H h_N\| \le C q_2^N$. These constants are independent of the horizon and compatible volume. ■

Proof. The jointly holomorphic trajectory can be differentiated in an arbitrary Banach direction η of the initial profile. Its derivative solves exactly eq. 5.cccxlii with the initial-response data just used. The initial adapted norm costs at most $C_T\|\eta\|_{\mathscr{B}_0}$. Apply eq. 5.ccclii and integrate this derivative along the straight segment between H and $\widetilde{H}$. That segment stays in the retained Banach ball. This proves the nonlinear difference bounds, not only stability of a formal linear recurrence. □

5.17.4. *All initial-profile jets in one frozen endpoint disc.* The derivative estimates above hold on a fixed larger H ball and a fixed larger disc $|\zeta| < cX_0$. Banach-valued Cauchy and polarization on strict inner margins therefore give, for integers $r \ge 1$, $s \ge 0$,

$$\begin{aligned} \|D_\zeta^s D_H^r x_j\| &\le C_{r,s} e_j q_2^j X_0^{-s}, \\ \|D_\zeta^s D_H^r h_j\|_{j,*} &\le C_{r,s} q_2^j X_0^{-s}. \end{aligned} \tag{5.ccliv}$$

The constants include the fixed input-ball margin and the usual finite-order factorials. One way to obtain them is to apply Cauchy to the already bounded first H derivative, taking $r-1$ further profile derivatives and s endpoint derivatives. Thus every profile jet retains the exponential factor. There is no count of its microscopic input coordinates.

For completeness, pure endpoint derivatives have the HRT bounds uniformly on this initial ball. The direct scalar product differs from one by at most $C\sum k_j$; the profile feedback contributes at

most Ce_*^2 by eq. 5.cccxlvi. For endpoint variation $\xi = 1$, $\eta = \sigma = \tau = 0$, eq. 5.cccxlviii consequently gives

$$\sup_j |\partial_\zeta x_j - 1| \le C\frac{(\log X_0)^m}{X_0}, \qquad \|\partial_\zeta h_j\|_{j,*} \le Ck_j. \tag{5.ccclv}$$

On a smaller real interval the first scalar derivative lies between 3/4 and 5/4. For $s \ge 2$, Cauchy applied to $\partial_\zeta x_j - 1$ yields

$$|\partial_\zeta^s x_j| \le C_s(\log X_0)^m X_0^{-s}, \qquad \|\partial_\zeta^s h_j\|_{j,*} \le C_s k_j X_0^{1-s}. \tag{5.ccclvi}$$

These are derivatives of the single presentation eq. 5.cccxxxv. They do not differentiate a sharp cutoff or replace its reference frame by one recomputed from the moving endpoint.

5.17.5. *The actual finite-volume interaction and law.* For real profiles and the same real endpoint $X = x_N$, let $\mathcal{V}$ be a fixed final coarse volume. The complete finite-volume interaction is $\mathcal{H}_N^H(V) = \sum_z h_{N,z}^H(V)$. There are $O(\mathcal{V})$ unit anchors. The local analytic norm controls the supremum of each complete pointing on the working field ball. Thus

$$\sup_V |\mathcal{H}_N^H(V) - \mathcal{H}_N^{\widetilde{H}}(V)| \le C\mathcal{V}q_2^N\|H - \widetilde{H}\|_{\mathscr{B}_0}. \tag{5.ccclvii}$$

The exact regular-step identity of LCF and the tuning construction identify these with the actual finite-volume effective interactions of the stated regular integral operations. Scalar vacuum factors may differ and do not affect a normalized final law.

To state such a comparison precisely, fix a common real final integration domain $\mathcal{D}$ inside the working ball and a fixed positive finite measure ν there, with nonzero total mass. Neither depends on the ultraviolet profile. Product Haar measure restricted to a fixed smaller real bond ball is one choice. Define

$$d\mathbb{P}_H(V) = Z_H^{-1}e^{-XS_N(V)+\mathcal{H}_N^H(V)}\mathbf{1}_{\mathcal{D}}(V)\,d\nu(V). \tag{5.ccclviii}$$

The real continuous densities have finite positive normalizers. If $\delta = C\mathcal{V}q_2^N\|H - \widetilde{H}\|$, then their likelihood ratio is between $e^{-2\delta}$ and $e^{2\delta}$: the unnormalized ratio is between $e^{-\delta}$ and e^{δ}, as is the ratio of the normalizers. Therefore, with the convention $\|\mathbb{P} - \mathbb{Q}\|_{\mathrm{TV}} = \frac{1}{2}\int |d\mathbb{P} - d\mathbb{Q}|$,

$$\|\mathbb{P}_H - \mathbb{P}_{\widetilde{H}}\|_{\mathrm{TV}} \le \min\left\{1, \quad \exp(C\mathcal{V}q_2^N\|H - \widetilde{H}\|_{\mathscr{B}_0}) - 1\right\}. \tag{5.ccclix}$$

The fixed multiplicative constant absorbs the factor two. This is a finite-volume bound on one common final domain. It is not uniform over growing volumes, and it proves neither clustering nor consistency between horizons. No complementary-history measure has been introduced in this comparison.

5.18. **Decay of the diagonal stress response in the regular trajectory.** The diagonal one-loop coefficient is useful only after the irrelevant part of the response has been included. We derive that coupled response from the weighted regular step and prove its decay. The memory term is estimated with the running coupling weight. An unweighted smallness bound on that term would not give the result.

Use the actual regular trajectory $(x_j, h_j)_{0\le j\le N}$ of Prop. 5.51, with terminal coupling $x_N = X$. Its frame, real cutoff and all integral conventions remain fixed during the response. Write b_j for its Gaussian scalar increment and m_j for the tracefree diagonal Gaussian increment in eq. 4.cdxxvii. The index in both symbols is the transfer from j to $j+1$. The actual comparison proves

$$|b_j - b_\infty| + |m_j - b_\infty| \le C\lambda_0^j, \qquad b_\infty = \frac{11}{3\pi^2}\log L > 0, \quad 0 < \lambda_0 < 1. \tag{5.ccclx}$$

5.18.1. *The stress representative and its exact transport.* Let $\mathcal{E} = \{e \in \mathbb{R}^4 : \sum_\mu e_\mu = 0\}$, with its Euclidean norm. For $e \in \mathcal{E}$ define the complete pointing

$$E_{j,e,z}(V) = \sum_{p \| \mu\nu} (e_\mu + e_\nu)\theta_z(m_p)a_p(U_j(V)). \tag{5.ccclxi}$$

The sum uses the original fine plaquettes in scale-j units. The actual critical orbit here is isotropic. These pointings are linear in e, gauge invariant, and covariant under simultaneous axis permutations and permutation of the entries of e. They have uniform local norm and plane readouts

$$b_{\mu\nu}[E_{j,e}] = e_\mu + e_\nu.$$

These assertions follow from the fine density and exact hat quadrature, with the actual affine response.

Their total is the derivative of the weighted classical action:

$$\sum_z E_{j,e,z}(V) = D_a S_j^a(V)[e]\Big|_{a=1}. \tag{5.ccclxii}$$

Indeed differentiation of the critical orbit at fixed literal mean is tangent to its constraint fibre. Pairing that derivative with the physical action gradient gives zero. Only differentiation of the plaquette weights remains. This is a total-action identity. It does not identify the derivative of every separately chosen weighted action pointing.

The particular choice eq. 5.ccclxi has exact isotropic transport:

$$T_{j,j+1}E_{j,e} = E_{j+1,e}. \tag{5.ccclxiii}$$

All-depth nesting identifies the original fine plaquette traces in the two expressions. The scale change and eq. 5.cccxv identify their pointing weights. Both equalities are applied to the complete functions before any local filing. This avoids an unnecessary irrelevant term of size equal to the stress coefficient itself.

5.18.2. *The zero-readout response space.* Use the reflection-invariant profile space of the weighted formation theorem, with all six plane readouts zero. Let $\mathscr{V}_j$ consist of bounded linear maps $v : \mathcal{E} \to \mathfrak{H}_j^{\mathrm{ref}}$ which are covariant under the axis permutation action. Its norm is the operator norm in the same compatible quotient norm as HRT. The identity

$$\mathcal{E} \ni e \longmapsto (e_\mu + e_\nu)_{\mu<\nu}$$

is the stress embedding.

The transport bound used in HRT also holds on the reflection space with six zero readouts. To see this, repeat the actual slow-field estimate in Lem. 4.34. Reflections kill every cross-plane curvature pairing. The six zero readouts therefore make its entire quadratic curvature form zero. The remaining proof uses only that cancellation, complete gauge invariance, analytic carrier bounds and the companion property. Its block interpolation and anchor sum are unchanged. Thus the same-radius estimate and exact complete-function semigroup hold, with fixed changed constants. Use the horizon-adapted norm on this space as in eq. 5.cccxvi. We may therefore fix

$$\|T_j\|_{j,*\to j+1,*} \le q < 1, \qquad T_j = T_{j,j+1}. \tag{5.ccclxiv}$$

The constants do not depend on N.

For clarity, the stress readout of a permutation-covariant linear response is one scalar. The stabilizer of the plane $\{\mu,\nu\}$ makes the coefficient of e_μ, e_ν equal, and makes the two complementary coefficients equal. The condition $\sum e_\mu = 0$ consequently reduces its readout to $c(e_\mu + e_\nu)$. Axis permutations make c the same for all planes. Reflections exclude cross-plane terms. A scalar isotropic readout of this response is zero. This elementary argument also shows that the scalar coupling has no first-order variation in this channel.

5.18.3. *The response equations from the actual step.* At scale j, perturb the exponent by

$$-t_j \sum_z E_{j,e,z} + \sum_z v_j(e)_z.$$

Here t_j is scalar and $v_j \in \mathscr{V}_j$. The leading-action perturbation is obtained by differentiating the constant weight

$$a = 1 + (t_j/x_j)e.$$

Equation eq. 5.ccclxii makes this exactly the stated first-order perturbation of the actual exponent. The output leading action is represented by $E_{j+1,e}$ using the same identity. All subsequent zero-readout terms are assigned to v_{j+1}. Thus no equality of individual weighted action pointings is being imposed.

Let $\mathcal{P}^{\mathrm{ref}}$ subtract the six plane readouts times the corresponding original fine-action plane pointings. It is a bounded linear map on the fixed-radius spaces. Differentiate the exact weighted regular step at $a = 1$ and the actual pair (x_j, h_j). Its moving centre is compared with the fixed isotropic kernel by Prop. 4.54. The actual fixed-kernel jet is the moving jet minus the rooted sphere family $\mathcal{B}_{j,e}$. By Cor. 4.55 this correction has six zero readouts. The pointings below are the selected fixed-total pointings of that comparison; no independent per-anchor filing is identified from equality of totals. Equations eq. 4.cdxxiv and eq. 4.cdxxvii give the exact linear system

$$\begin{aligned} t_{j+1} &= \left(1 - \frac{m_j}{x_j} + d_j\right) t_j + L_j v_j, \\ v_{j+1} &= T_j v_j + \mathcal{E}_j v_j + \frac{t_j}{x_j} A_j. \end{aligned} \tag{5.ccclxv}$$

The operators on the right are derivatives of the specified formation map, not independent assumptions about a scalar recursion. Their definitions are as follows. If the stress readout of $D_a \mathbf{F}_j^a[e]|_1$ is $\mu_j(e_\mu + e_\nu)$, then $d_j = -\mu_j/x_j$. The map L_j is minus the stress readout of $D_h \mathbf{F}_j[v]$. The map $\mathcal{E}_j$ is $\mathcal{P}^{\mathrm{ref}} D_h \mathbf{F}_j$. Finally

$$\begin{aligned} A_j[e] = \; & D_a T_j^a h_j[e]\Big|_1 + q_{j,e} \\ & + \mathcal{P}^{\mathrm{ref}} \, D_a \mathbf{F}_j^a[e]\Big|_1 - \mathcal{B}_{j,e}. \end{aligned} \tag{5.ccclxvi}$$

The first term has six zero readouts by eq. 4.cdxxv. The middle term is the actual Gaussian remainder in eq. 4.cdxxvii. This identifies the full forcing and its normalization. The final term is the actual sharp-cutoff boundary correction. Its compatible norm is superpolynomially small in the frozen coupling reference, so it preserves the stated forcing bound. The signs in the first line come from extracting the positive Gaussian logarithm from a negative action exponent.

Write

$$\epsilon_j = \frac{(\log x_j)^m}{x_j}, \qquad k_j = \epsilon_j / x_j.$$

The fixed-frame envelopes in HRT make these quantities comparable to their versions with a_j. The actual derivative bounds give

$$|d_j| \le C k_j, \qquad \|L_j\| + \|\mathcal{E}_j\| \le C\epsilon_j, \qquad \|A_j\| \le C_{H_*}. \tag{5.ccclxvii}$$

In particular the old profile's weight derivative is only bounded; it has no extra power of the fluctuation coupling. Its factor t_j/x_j in eq. 5.ccclxv is retained. No spatially varying weight is used in these identities.

5.18.4. *Eliminating the irrelevant response.* Start with the pure bare stress perturbation $v_0 = 0$. Choose the infrared floor so $q + C \sup_j \epsilon_j \le q_1 < 1$. The second equation and its actual operator bounds give the causal estimate

$$\|v_j\|_{j,*} \le C \sum_{i<j} q_1^{j-1-i} \frac{|t_i|}{x_i}. \tag{5.ccclxviii}$$

This follows by successively substituting that equation and bounding each product of its profile operators by q_1 per step. The constant is independent of horizon. Substitution into the first equation exhibits the memory bound

$$C\epsilon_j \sum_{i<j} q_1^{j-1-i} \frac{|t_i|}{x_i}.$$

The denominator here belongs to the old index i. It cannot be replaced by a claim of smallness for the unweighted memory sum.

5.18.5. *The coupling weight and the diagonal product.* Set

$$w_j = x_j/x_0, \qquad Y_j = t_j/w_j.$$

The actual scalar recursion is $x_{j+1} = x_j - b_j - r_j$, with $|r_j| \le C\epsilon_j$. Consequently

$$\frac{1 - m_j/x_j + d_j}{w_{j+1}/w_j} = 1 + \frac{b_j - m_j + r_j + x_j d_j}{x_{j+1}}. \tag{5.ccclxix}$$

This exact identity uses no monotonicity of the trajectory. For a large enough floor, the bounded Gaussian increments and small formation correction make x_{j+1} comparable to x_j. By HRT and eq. 5.ccclx,

$$\sum_{j<N} \frac{|b_j - m_j| + \epsilon_j}{x_{j+1}} \le \frac{C}{X} + C\frac{(\log X)^m}{X}. \tag{5.ccclxx}$$

The geometric numerator in the first part has bounded sum, and all $x_j \ge cX$. For the second part use the second scale sum in eq. 5.cccxxviii. Thus, for $i < k$, the actual diagonal product obeys

$$\prod_{i\le j<k} \left|1 - \frac{m_j}{x_j} + d_j\right| \le C\,\frac{x_k}{x_i}. \tag{5.ccclxxi}$$

Take absolute values in eq. 5.ccclxix, use $1 + u \le e^u$ for nonnegative u, and multiply. The factors w telescope. Early increments of either sign are harmless; they were only required to be bounded.

5.18.6. *The weighted memory estimate.* The identity

$$\frac{w_i}{x_i} = \frac{1}{x_0}$$

is the decisive cancellation. Divide the first response equation by w_{j+1}, insert eq. 5.ccclxviii, and use that identity. It gives

$$|Y_{j+1}| \le (1 + \eta_j)|Y_j| + C\frac{\epsilon_j}{x_{j+1}} \sum_{i<j} q_1^{j-1-i}|Y_i|, \qquad \eta_j = C\frac{|b_j - m_j| + \epsilon_j}{x_{j+1}}. \tag{5.ccclxxii}$$

The sum of the coefficients $\eta_j + \epsilon_j/x_{j+1}$ is bounded by eq. 5.ccclxx and becomes small with the infrared floor.

For completeness put $M_j = \max_{i\le j} |Y_i|$. The geometric age sum in eq. 5.ccclxxii is at most $M_j/(1 - q_1)$. Hence

$$M_{j+1} \le M_j \left\{1 + \eta_j + \frac{C\epsilon_j}{(1 - q_1)x_{j+1}}\right\}.$$

Multiplication and eq. 5.ccclxx give $M_j \le C|t_0|$ uniformly in j, N. Returning to eq. 5.ccclxviii then gives

$$\|v_j\|_{j,*} \le C\frac{|t_0|}{x_0}\sum_{i<j} q_1^{j-1-i} \le C\frac{|t_0|}{x_0}.$$

Both bounds use the coupling weight throughout the memory sum.

Proposition 5.54. The first tracefree diagonal response of the actual regular trajectory, with bare coefficient t_0 and no bare irrelevant response, satisfies

$$|t_j| \le C\frac{x_j}{x_0}|t_0|, \qquad \|v_j\|_{j,*} \le C\frac{|t_0|}{x_0}. \tag{5.ccclxxiii}$$

The constants and infrared floor are independent of horizon and compatible volume. For $|t_0| \le C_0\sqrt{x_0}$ and fixed sufficiently large $X = x_N$, its final nonconstant insertion tends to zero with bound

$$|t_N| + \|v_N\|_{N,*} \le C_X x_0^{-1/2}. \tag{5.ccclxxiv}$$

Since $x_0 = X + b_\infty N + O(\log^{m+1}(X+N))$, this is $O_X(N^{-1/2})$. ■

Proof. The preceding causal response construction and weighted estimates prove eq. 5.ccclxxiii. At $j = N$, $x_N = X$. Insert the stated bound on t_0 and the actual bare coupling law from HRT. The classical stress pointing has uniformly bounded norm, so these coefficient and profile bounds also control the resulting nonconstant insertion on the final working field ball. □

This proves the diagonal stress decay for the stated regular integral operations. It uses the actual weighted Gaussian law and the actual formation derivatives. It is a first-response statement; it does not claim a finite anisotropic trajectory or a variable-weight domain. The complementary-history insertions in a full rotation argument require their own source and mass estimates.

5.19. **A scalar insertion and nonvanishing in the regular sector.** Endpoint tuning gives an exact insertion identity when the entire regular presentation is held fixed. We use that identity to prove a positive variance bound which is uniform in the ultraviolet depth on a fixed terminal torus. The measure is the positive product of the specified regular kernels, capped in one common coarse field domain. It is not identified with the full Wilson measure.

The proof has two ingredients. The flat flux contraction makes every effective classical action nonconstant on the same two coarse patches. The endpoint derivative of the terminal profile is small, so the transferred scalar insertion retains this separation. Positivity then gives both patches a depth-independent probability.

5.19.1. *One frozen presentation and its exact density.* Fix a terminal torus in the compatible volume class of § 4.9 and Prop. 4.46. Its integer period $\ell \ge 2$ is divisible by the chosen localization sizes and exceeds ten cubes at each of those sizes. These sizes and the volume floor are fixed before the endpoint coupling is varied. Fix a real endpoint X above the coupling floor. For each ultraviolet depth N, take the HRT presentation at this reference endpoint. Keep every frame θ_j, fluctuation scale, relative contour factor, free colour-ball cutoff, branch and domain fixed. Different depths have different finite lattices, but the estimates are uniform in N. No derivative below compares different presentations.

Vary only the terminal coupling to $X + t$, with $|t| < cX$ in its analytic tuning disc. Write

$$\alpha_N(t) = x_0^{(N)}(X+t), \qquad H_N(t,V) = \sum_z h_{N,z}^{(N)}(X+t,V), \qquad S_N(V) = A(U_N(V)). \tag{5.ccclxxv}$$

The ultraviolet profile is zero. The classical action S_N is independent of t. All intermediate coupling and profile dependences are formed densities; they are not extra parameters of the frozen kernels.

The exact positive composition of these kernels has, relative to normalized terminal product Haar measure, the density

$$\mathcal{K}_N(t,V) = e^{C_N(t)} \exp\{-(X+t)S_N(V) + H_N(t,V)\}. \tag{5.ccclxxvi}$$

The scalar C_N includes every numerical, Gaussian-vacuum and profile-vacuum factor in the exact regular steps. It is real at real t, with its analytic logarithm fixed there. It need not be bounded uniformly in N.

Let $\mathbb{Q}_N$ be the positive chain measure formed from those frozen kernels and their Haar/coarea measures, before the ultraviolet factor $e^{-\alpha_N(t)S_0}$ is inserted. It is independent of t. Equivalently, use the successive deterministic mean constraints and their positive regular weights. This description keeps precisely the retained regular component and its contour factors. On every finite lattice the measure is finite and all moments of the initial Wilson action S_0 are finite.

Choose below one common real raw-coarse domain Ω in the terminal field ball, independent of N, t. Define the capped regular probability

$$d\mathbb{P}^{\mathrm{reg}}_{N,t} = \frac{\mathbf{1}_{\{V\in\Omega\}} e^{-\alpha_N(t)S_0}}{\mathcal{Z}^{\mathrm{reg}}_N(t)}\, d\mathbb{Q}_N. \tag{5.ccclxxvii}$$

Its terminal marginal is

$$d\nu_{N,t}(V) = \frac{\mathbf{1}_\Omega(V)}{Z_N(t)} e^{-(X+t)S_N(V)+H_N(t,V)}\, dV, \qquad \mathcal{Z}^{\mathrm{reg}}_N(t) = e^{C_N(t)} Z_N(t). \tag{5.ccclxxviii}$$

For real t near zero these are positive. Conditional expectations at a specified V mean the canonical coarea fibre ratios. This version is defined throughout the admitted coarse chart, including $V = 1$.

The HRT bounds, on fixed smaller tuning discs, give

$$\frac{3}{4} \le \alpha_N'(0) \le \frac{5}{4}, \qquad \|h_N(0)\| \le H_*, \qquad \|\partial_t h_N(0)\| \le C\frac{\epsilon(X)}{X}, \quad \epsilon(X) = X^{-1}(\log X)^m, \quad m > 1. \tag{5.ccclxxix}$$

For every fixed $s \ge 2$, Cauchy on the same frozen discs also gives

$$|\alpha_N^{(s)}(0)| \le C_s \frac{(\log X)^m}{X^s}, \qquad \|\partial_t^s h_N(0)\| \le C_s \frac{\epsilon(X)}{X^s}. \tag{5.ccclxxx}$$

The profile assertion follows by applying Cauchy to its first derivative on a fixed smaller disc. These are exactly endpoint derivatives in eqs. 5.cccxxxi and 5.cccxxxii; none differentiates a sharp cutoff.

5.19.2. *The scalar conditional identity and centering.* Put

$$T_N = \alpha_N'(0) S_0, \qquad O_N(V) = S_N(V) - \partial_t H_N(0,V). \tag{5.ccclxxxi}$$

Proposition 5.55 (Exact scalar action insertion)**.** For the capped regular measure,

$$\mathbb{E}^{\mathrm{reg}}_{N,0}[T_N \mid V] = O_N(V) - C_N'(0). \tag{5.ccclxxxii}$$

Consequently its centered form is

$$\mathbb{E}^{\mathrm{reg}}_{N,0}[T_N - \mathbb{E}^{\mathrm{reg}}_{N,0} T_N \mid V] = O_N(V) - \nu_{N,0}(O_N). \tag{5.ccclxxxiii}$$

This is a constant scalar action insertion. It makes no assertion about a spatially varying local source. ∎

Proof. At fixed V, differentiate the actual fibre composition defining $\mathcal{K}_N$. Its only endpoint-dependent ultraviolet factor is $e^{-\alpha_N(t)S_0}$, since $h_0 = 0$ and $\mathbb{Q}_N$ is frozen. Finite-dimensional dominated differentiation gives

$$\partial_t \log \mathcal{K}_N(0,V) = -\alpha_N'(0)\mathbb{E}[S_0 \mid V].$$

Differentiating eq. 5.ccclxxvi gives $C_N'(0) - S_N + \partial_t H_N(0)$. These identities prove eq. 5.ccclxxxii. The cap does not change a fibre conditional at an admitted $V \in \Omega$. Integrate that equality against

the terminal marginal and subtract it. This cancels the complete numerical vacuum derivative $C_N'(0)$, proving eq. 5.ccclxxxiii. The argument did not remove a vacuum term from the uncentered identity. □

Corollary 5.56 (Conditional variance and scalar information)**.** The exact decomposition is

$$\mathrm{Var}_{\mathbb{P}^{\mathrm{reg}}_{N,0}}(T_N) = \nu_{N,0}\{\mathrm{Var}(T_N \mid V)\} + \mathrm{Var}_{\nu_{N,0}}(O_N). \tag{5.ccclxxxiv}$$

The normalized endpoint scores are respectively $-T_N + \mathbb{E}T_N$ and $-O_N + \nu_{N,0}O_N$. Thus the two variances are the corresponding scalar Fisher informations, and coarse conditioning cannot increase them. ■□

Proof. Differentiate the normalized densities in eq. 5.ccclxxvii and eq. 5.ccclxxviii. The centered conditional identity identifies their scores. Apply the elementary conditional-variance formula. All variables have finite moments on each finite regular graph. □

5.19.3. *The same two action patches at every depth.* Let $n_b = 4\ell^4$ be the number of stored positive terminal bonds. In the normalization $\frac{1}{2}\operatorname{tr} T^2 = 1$, fix a real Abelian colour direction T, and the coarse field

$$B_2(x) = T\cos(2\pi x_1/\ell), \qquad B_\mu(x) = 0 \quad (\mu \neq 2). \tag{5.ccclxxxv}$$

Its supremum is one and its curl is nonzero. The period-two case in the next formula is a kinematic check; the regular-measure conclusions use only the compatible torus fixed above. Write

$$q_\ell = \|\mathrm{d}_c B\|_2^2 = \begin{cases} 4\ell^4, & \ell = 2, \\ 2\ell^4 \sin^2(\pi/\ell), & \ell \geq 3. \end{cases} \tag{5.ccclxxxvi}$$

The difference acts on stored oriented positive bonds, so the two parallel links on a period-two circle remain distinct lattice variables.

Lemma 5.57 (A uniform positive quadratic direction)**.** There are $R_0 > 0$ and $M_\ell < \infty$, independent of N, such that

$$S_N(1) = DS_N(1) = 0, \qquad D^2S_N(1)[B,B] \geq q_\ell, \qquad \sup_{\|A\|_\infty < R_0} |S_N(e^{iA})| \leq M_\ell. \tag{5.ccclxxxvii}$$

For real admitted fields $S_N \geq 0$. ■

Proof. At the identity the actual fine response to coarse datum B satisfies the literal linear mean constraint. The difference between the plain and straightened rows is a coarse gradient, so their curls agree. The exact flat identity and contraction eq. 5.ccl, eq. 5.ccli give

$$\|\mathrm{d}_c B\|_{2,c} \leq \|\mathrm{d}_h A^{\mathrm{lin}}\|_{2,h}.$$

The fine Wilson Hessian in this direction is exactly $\|\mathrm{d}_h A^{\mathrm{lin}}\|_{2,h}^2$ in the displayed colour normalization. Its first jet is zero, so the second derivative of the background map adds no acceleration term. This proves the middle assertion in eq. 5.ccclxxxvii. No positivity of a mass-added coarse Schur matrix is being used for this conclusion.

The actual local action trees of Prop. 4.17 give a normally convergent analytic action on a fixed complex coarse ball. Each original fine density has its physical h^4 weight; the fine plaquette count therefore cancels in a unit coarse cell. Summing its rooted pointings on the terminal torus bounds the total by $C_A\ell^4$ on a smaller fixed radius R_0. Take this as M_ℓ, enlarging it to include $N = 0$ if necessary. The field is real at real data, and its actual Wilson plaquette terms are nonnegative. Their sum proves the last assertion. □

Choose once

$$0 < r_\Omega \leq \min\{r/4, R_0/4, \pi/4\}, \qquad \Omega = \{V = e^{iA} : \max_b |A_b| < r_\Omega\}. \tag{5.ccclxxxviii}$$

The radius is fixed before N, X vary. Let

$$\begin{aligned} 0 < a &\le \min\{r_\Omega/4, R_0/2, q_\ell R_0^3/(8M_\ell)\},\\ d_\ell &= q_\ell a^2/4, \qquad L_\ell = 2M_\ell/R_0,\\ 0 < \delta_\ell &\le \min\{r_\Omega/4, a/8, d_\ell/(8L_\ell)\}. \end{aligned} \tag{5.ccclxxxix}$$

These constants are positive and do not depend on ultraviolet depth or endpoint coupling. Define product patches in raw coordinates by

$$P_0 = \{e^{iA} : \max_b |A_b| < \delta_\ell\}, \qquad P_1 = \{e^{iA} : \max_b |A_b - aB_b| < \delta_\ell\}. \tag{5.cccxc}$$

Both lie in Ω, and they are disjoint.

Lemma 5.58 (Uniform separation on fixed patches)**.** For every N,

$$\inf_{P_1} S_N - \sup_{P_0} S_N \ge \tfrac{3}{4} d_\ell. \tag{5.cccxci}$$

■

Proof. The scalar function $z \mapsto S_N(e^{izB})$ is bounded by M_ℓ on $|z| < R_0$. Its divided coefficients have bounds $M_\ell R_0^{-m}$. For $a \le R_0/2$, the terms of degree at least three have absolute sum at most $2M_\ell a^3/R_0^3 \le q_\ell a^2/4$. Its quadratic term is at least $q_\ell a^2/2$. Hence $S_N(e^{iaB}) \ge d_\ell$, uniformly in depth.

On the real r_Ω-ball, Cauchy in any unit coarse Banach direction gives $\|DS_N\| \le L_\ell$: a complex disc of radius $R_0/2$ stays inside the action ball. The line between a patch point and its centre is in that real ball. The variation of S_N on either patch is therefore at most $L_\ell \delta_\ell \le d_\ell/8$. Subtracting the two errors proves eq. 5.cccxci. □

For normalized Haar measure on SU(2), the log-coordinate density is

$$j(A) = (2\pi^2)^{-1}(\sin|A|/|A|)^2.$$

On $|A| \le \pi/2$ it is at least $2/\pi^4$. Thus the same two patches have the explicit product-Haar lower bound

$$\mathrm{Haar}(P_i) \ge m_\ell := \left(\frac{8}{3\pi^3}\delta_\ell^3\right)^{n_b} > 0. \tag{5.cccxcii}$$

Their dimension is $3n_b = 12\ell^4$, not an ultraviolet dimension. The common cap has Haar mass

$$m_\Omega = \left\{\frac{r_\Omega - \frac{1}{2}\sin(2r_\Omega)}{\pi}\right\}^{n_b} > 0.$$

5.19.4. *Uniform positive probabilities and variance.* The terminal pointing bounds imply

$$|H_N(0,V)| \le \ell^4 H_* =: H_\ell, \qquad |\partial_t H_N(0,V)| \le C\ell^4 \frac{\epsilon(X)}{X} \quad (V \in \Omega). \tag{5.cccxciii}$$

Choose X_ℓ above the HRT floor so that, for all $X \ge X_\ell$, the second quantity is at most $d_\ell/8$. This is possible because $\epsilon(X)/X \to 0$. Then the transferred observable has separation

$$\inf_{P_1} O_N - \sup_{P_0} O_N \ge d_\ell/2. \tag{5.cccxciv}$$

All choices of patches precede this coupling threshold.

Since $0 \le S_N \le M_\ell$ on the cap, its terminal normalizer obeys

$$m_\Omega e^{-XM_\ell - H_\ell} \le Z_N(0) \le m_\Omega e^{H_\ell}. \tag{5.cccxcv}$$

In particular each patch probability is at least

$$\nu_{N,0}(P_i) \ge p_{X,\ell} := m_\ell e^{-XM_\ell - 2H_\ell} > 0. \tag{5.cccxcvi}$$

These estimates do not use or bound the possibly extensive vacuum factor e^{C_N}.

Proposition 5.59 (Finite-volume regular nonvanishing)**.** For the fixed compatible terminal torus above and $X \geq X_\ell$, uniformly over all ultraviolet depths,

$$\mathrm{Var}_{\nu_{N,0}}(O_N) \geq c_{X,\ell} := p_{X,\ell}^2 d_\ell^2/4 > 0, \qquad \mathrm{Var}_{\mathbb{P}_{N,0}^{\mathrm{reg}}}(T_N) \geq c_{X,\ell}. \tag{5.cccxcvii}$$

Consequently $\mathrm{Var}_{\mathbb{P}_{N,0}^{\mathrm{reg}}}(S_0) \geq (16/25)c_{X,\ell}$. ■

Proof. For independent terminal samples V, V', $\mathrm{Var}(O_N) = \frac{1}{2}\mathbb{E}[(O_N(V) - O_N(V'))^2]$. Restrict to $P_0 \times P_1$ and $P_1 \times P_0$. The two probabilities and eq. 5.cccxciv give the first lower bound. The second follows from eq. 5.ccclxxxiv. Finally $T_N = \alpha_N'(0)S_0$ and $\alpha_N'(0) \leq 5/4$. This proves the last bound. The explicit constant may be small as X or ℓ changes; only uniformity in ultraviolet depth is asserted. □

5.19.5. *Generating functions and endpoint contact terms.* The fixed presentation also gives exact higher scalar identities. For sufficiently small t, the conditional coarea kernels satisfy

$$\begin{aligned}\log \mathbb{E}_{N,0}^{\mathrm{reg}}[e^{-(\alpha_N(t)-\alpha_N(0))S_0} \mid V] = C_N(t) - C_N(0) - tS_N(V) \\ + H_N(t,V) - H_N(0,V).\end{aligned} \tag{5.cccxcviii}$$

Every logarithm is its germ at $t = 0$. The global identity is

$$\mathbb{E}_{N,0}^{\mathrm{reg}} e^{-(\alpha_N(t)-\alpha_N(0))S_0} = e^{C_N(t)-C_N(0)} \frac{Z_N(t)}{Z_N(0)}. \tag{5.cccxcix}$$

Both follow by changing only the ultraviolet exponential in the same finite integral. In particular no moving-frame or cutoff derivative is hidden in these formulas.

For a genuine bare scalar source s, let $t = \psi_N(s)$ solve $\alpha_N(t) = \alpha_N(0) + s$. On a common smaller complex disc this inverse exists: the HRT complex derivative is uniformly close to one. Substitution in eq. 5.cccxcviii is the conditional cumulant-generating function of $-S_0$ for that source.

Let $\kappa_r(S_0 \mid V)$ denote its ordinary conditional cumulants. With $B_{q,r}$ the exponential partial Bell polynomials, differentiation gives

$$\sum_{r=1}^{q} (-1)^r \kappa_r(S_0 \mid V) B_{q,r}(\alpha_N'(0), \ldots, \alpha_N^{(q-r+1)}(0)) = C_N^{(q)}(0) - \mathbf{1}_{\{q=1\}} S_N(V) + \partial_t^q H_N(0,V). \tag{5.cd}$$

The convention is determined by $\partial_t^q [m(t)^r/r!]_{t=0} = B_{q,r}(m'(0), \ldots)$. For example the second and third identities are

$$\begin{aligned}(\alpha_N')^2 \,\mathrm{Var}(S_0 \mid V) - \alpha_N'' \mathbb{E}[S_0 \mid V] &= C_N'' + \partial_t^2 H_N, \\ -(\alpha_N')^3 \kappa_3(S_0 \mid V) + 3\alpha_N'\alpha_N'' \,\mathrm{Var}(S_0 \mid V) - \alpha_N''' \mathbb{E}[S_0 \mid V] &= C_N''' + \partial_t^3 H_N.\end{aligned} \tag{5.cdi}$$

All quantities here are evaluated at $t = 0$. The contact terms involving higher bare tuning derivatives cannot be dropped.

There is a useful vacuum-free version. The profiles and action vanish at the flat field for every endpoint. Subtract eq. 5.cccxcviii at $V = 1$ to obtain

$$\log \frac{\mathbb{E}[e^{-sS_0} \mid V]}{\mathbb{E}[e^{-sS_0} \mid 1]} = -\psi_N(s) S_N(V) + H_N(\psi_N(s), V) - H_N(0,V). \tag{5.cdii}$$

The reference fibre is the canonical coarea version, not an arbitrary value assigned to a conditional expectation on a null event. This exact ratio removes C_N at every order. For instance

$$\mathrm{Var}(T_N \mid V) - \mathrm{Var}(T_N \mid 1) = \partial_t^2 H_N(0,V) + \frac{\alpha_N''(0)}{\alpha_N'(0)} O_N(V). \tag{5.cdiii}$$

The endpoint bounds therefore control this field-dependent difference by $C_\ell(\log X)^m/X^2$, uniformly in depth. They do not bound its vacuum value. More generally Cauchy for the inverse source and eq. 5.ccclxxx give, for each fixed $r \geq 2$,

$$\sup_{V\in\Omega} |\kappa_r(S_0 \mid V) - \kappa_r(S_0 \mid 1)| \leq C_{\ell,r}\frac{(\log X)^m}{X^r}. \tag{5.cdiv}$$

Indeed $\psi_N'(0) = 1/\alpha_N'(0)$, while $|\psi_N^{(r)}(0)| \leq C_r(\log X)^m/X^r$ for $r \geq 2$. The latter follows by inverting the near-identity analytic tuning on a disc of radius cX and applying Cauchy to its derivative minus its value at zero. Composing the profile first-derivative bound gives the smaller profile contribution. Thus the estimate concerns coarse-field variation of the conditional cumulant, rather than a uniform ultraviolet cumulant bound.

Finally, direct differentiation of the total normalizer gives the consistency identities

$$\begin{aligned}
\partial_t \log \mathcal{Z}_N^{\mathrm{reg}}(0) &= -\mathbb{E}T_N,\\
\partial_t^2 \log \mathcal{Z}_N^{\mathrm{reg}}(0) &= \mathrm{Var}(T_N) - \alpha_N''(0)\mathbb{E}S_0,\\
\partial_t^2 \log Z_N(0) &= \mathrm{Var}_{\nu_{N,0}}(O_N) + \nu_{N,0}(\partial_t^2 H_N).
\end{aligned}$$

They retain the same endpoint contacts as eq. 5.cd and agree with the conditional-variance decomposition.

5.19.6. *A nondegenerate fixed-coarse subsequence.* The common terminal domain also permits an ultraviolet subsequence without changing the coarse cutoff or volume. Only finite-dimensional coarse functions and probabilities are extracted.

Reduce the common endpoint disc once, and write its radius as $T = c_1X$. The complex HRT slope bound gives

$$\sup_{|t|<T} |\alpha_N'(t) - 1| \leq K(X) < \tfrac{1}{4}, \qquad \sup_{|t|<T} \|\partial_t h_N(t)\| \leq C\epsilon(X)/X. \tag{5.cdv}$$

After decreasing c_1 or increasing the coupling floor, integration along the endpoint segment gives $\|h_N(t)\| \leq H_* + 1$. Put $\delta\alpha_N(t) = \alpha_N(t) - \alpha_N(0)$. Then $|\delta\alpha_N(t)| \leq (5/4)|t|$. The bare value $\alpha_N(0)$ itself is not bounded in this argument.

Use the finite colour coordinates $A = (A_{b,c}) \in \mathbb{C}^{3n_b}$, and the polydisc

$$\mathcal{D} = \{(t, A) : |t| < T,\ |A_{b,c}| < 2r_\Omega \text{ for all } b, c\}. \tag{5.cdvi}$$

Its field coordinates obey $\max_b |A_b| < 2\sqrt{3}r_\Omega < \min(r, R_0)$. The closure of the real cap Ω is strictly inside this field domain. Thus $S_N(e^{iA})$, $H_N(t, e^{iA})$, $\partial_t H_N(t, e^{iA})$, and $\delta\alpha_N(t)$ have uniform holomorphic bounds on $\mathcal{D}$, after using a slightly smaller endpoint disc for the displayed derivative. Their bounds depend on the fixed ℓ, X, not on N.

Lemma 5.60 (Finite-dimensional compact extraction)**.** There is a common subsequence $N_k \to \infty$ and holomorphic limits $S_\infty, H_\infty, \delta\alpha_\infty$, locally uniform on $\mathcal{D}$, with

$$S_{N_k} \longrightarrow S_\infty, \quad H_{N_k} \longrightarrow H_\infty, \quad \partial_t H_{N_k} \longrightarrow \partial_t H_\infty, \quad \delta\alpha_{N_k} \longrightarrow \delta\alpha_\infty. \tag{5.cdvii}$$

They are real on real arguments, $\delta\alpha_\infty(0) = 0$, and $3/4 \leq \delta\alpha_\infty'(0) \leq 5/4$. ∎

Proof. This is a normal-family argument on the $1 + 3n_b$ complex coordinates just displayed. On every smaller closed polydisc, the uniform supremum bounds and coordinate Cauchy formulas bound all first coordinate derivatives. The scalar functions are therefore uniformly bounded and equicontinuous on that compact finite-dimensional set. Apply Arzelà–Ascoli to the finite tuple of functions, then use a countable nested exhaustion of $\mathcal{D}$ and a diagonal subsequence. The uniform limits are holomorphic by the finite-dimensional Cauchy integral formula. That formula also identifies the limit of $\partial_t H_N$ with $\partial_t H_\infty$.

Reality passes under uniform convergence. The tuning vanishes at zero, and its first derivative converges by the one-variable Cauchy formula. The real slope interval eq. 5.ccclxxix passes to the limit. No normal-family theorem on an infinite-dimensional field space is being invoked. □

Proposition 5.61 (A nondegenerate coarse ultraviolet subsequence)**.** On this subsequence, the terminal regular probabilities converge in total variation to

$$d\nu_\infty(V) = \frac{\mathbf{1}_\Omega(V)}{Z_\infty} e^{-XS_\infty(V)+H_\infty(0,V)}\, dV. \tag{5.cdviii}$$

The observable $O_\infty = S_\infty - \partial_t H_\infty(0)$ is real analytic near the cap and satisfies

$$\inf_{P_1} O_\infty - \sup_{P_0} O_\infty \geq d_\ell/2, \qquad \mathrm{Var}_{\nu_\infty}(O_\infty) \geq c_{X,\ell} > 0. \tag{5.cdix}$$

In fact $O_{N_k} \to O_\infty$ uniformly on $\overline{\Omega}$, and the variances converge to the displayed positive variance. ■

Proof. The field closure of the cap lies in a compact subpolydisc. Hence both the density exponents and O_{N_k} converge uniformly there. The positive normalizer bounds eq. 5.cccxcv pass to Z_∞, so its value is nonzero. The unnormalized real densities converge uniformly; division by the convergent positive normalizers proves total-variation convergence. The cap is the same for every depth, with fixed product Haar measure. Its boundary has Haar measure zero.

The patch inequalities pass under uniform convergence, and both limiting patch masses are at least $p_{X,\ell}$. The same two-sample variance proof yields eq. 5.cdix. Alternatively bounded uniform convergence of O_{N_k}, together with total variation, gives convergence of its first two moments and hence of its variance. Analyticity and reality are those of the extracted functions. □

There is also a common complex endpoint disc on which normalized coarse source functionals converge. Set

$$D_\ell = C\ell^4\epsilon(X)/X, \qquad t_\ell = \min\{T/2, \log(3/2)/(M_\ell + D_\ell)\}.$$

For $|t| < t_\ell$, the density relative to its positive value at zero differs from one by at most $1/2$, since

$$|-tS_N(V) + H_N(t,V) - H_N(0,V)| \leq |t|(M_\ell + D_\ell).$$

Integrating against $\nu_{N,0}$ gives

$$|Z_N(t)/Z_N(0) - 1| < 1/2. \tag{5.cdx}$$

Thus these capped normalizers are zero free on this smaller, possibly ℓ-dependent disc. Uniform convergence on compact subsets gives locally uniform convergence of normalized coarse expectations of bounded fixed probes, and of their endpoint derivatives. The same holds for bounded holomorphic probes depending on t. This argument does not claim zero freeness on the full tuning disc $|t| < T$.

Corollary 5.62 (Limit of the vacuum-free conditional source ratios)**.** On a common smaller bare-source disc, $\psi_{N_k} \to \psi_\infty$ locally uniformly, where $\delta\alpha_\infty(\psi_\infty(s)) = s$. The logarithms in eq. 5.cdii converge locally uniformly in (s,V) to

$$-\psi_\infty(s)S_\infty(V) + H_\infty(\psi_\infty(s),V) - H_\infty(0,V). \tag{5.cdxi}$$

Their derivatives give the corresponding limiting coarse-field variations of conditional cumulants. No convergence of C_N, its derivatives, or the uncentered conditional source functions is asserted. ■□

Proof. Write $\delta\alpha_N(t) = t + r_N(t)$. On the smaller common disc, $r_N(0) = 0$ and $\sup|r_N'| \leq K(X) < 1/4$. For $|s| < T/2$, the map $t \mapsto s - r_N(t)$ is an invariant contraction on $|t| \leq 2T/3$. It gives the common inverse ψ_N. The same applies to the limit. Uniform convergence of r_{N_k} on that closed smaller disc and the contraction bound imply locally uniform convergence of the inverses. Substitute in the exact vacuum-free identity. Its analytic functions and their derivatives converge by Cauchy on compact subdomains, proving the result. □

This is a subsequential limit at one fixed coarse cutoff and finite volume. The terminal coarse observable is nondegenerate in that limit. The vacuum factor is neither needed nor extracted. No consistency between different terminal volumes, horizons or coarse cutoffs has been proved by this compactness argument.

5.19.7. *What a positive history mixture would add.* Suppose a separately established positive decomposition on the same ultraviolet observable space contains this capped regular probability with weight p_N:

$$\mu_N = p_N \mathbb{P}^{\mathrm{reg}}_{N,0} + (1 - p_N)\rho_N.$$

For any common square-integrable observable, the variance formula for its component flag implies $\mathrm{Var}_{\mu_N}(T_N) \geq p_N \mathrm{Var}_{\mathbb{P}^{\mathrm{reg}}_{N,0}}(T_N)$. Hence a genuine history floor $p_N \geq p_* > 0$ would give $\mathrm{Var}_{\mu_N}(T_N) \geq p_* c_{X,\ell}$. The corresponding S_0 bound has the factor $16/25$.

No such history floor is proved here. The regular kernel product, its gauge weights and its terminal cap are part of the measure in this subsection. There is no asserted full Wilson-law identification, volume-uniform lower bound, continuum Schwinger field or local-source transfer identity. The results are the exact scalar normalization, uniform finite-volume regular nonvanishing, and the nondegenerate fixed-coarse subsequence just constructed. They do not construct a limit of the full ultraviolet measures.

5.20. **Sources, local energies, and separated correlation bounds.**

Source domains and the exact running-shift identity. This section fixes the source variables used in the conditional estimates. The identities are finite-dimensional; the multiscale bounds for the effective energies stated later require separate estimates.

Fix K, a torus T_ℓ, an odd $L > 1$, and positive constants M, c_M with $c_M \geq M$. The depth s corresponds to physical spacing L^{-s}; level j of a run of length K corresponds to depth $K - j$. A source triple is

$$w = (c, (f_1, \ldots, f_n), (z_1, \ldots, z_n)),$$
$$w(x) = c + \sum_i z_i f_i(x),$$

where f_i are real Lipschitz functions, c, z_i are complex, and the family may be empty. Put

$$\|f\|_{\sharp,s} = \max\{\|f\|_\infty, c_M L^{-s} \operatorname{Lip} f\},$$
$$\|w\|_{\flat,s} = |c| + \sum_i |z_i| \|f_i\|_{\sharp,s}.$$

The class $\mathcal{W}^{(s)}_r$ means the triples with $\|w\|_{\flat,s} < r$. It is a union of finite-dimensional source balls, one for each fixed real family. Holomorphy on this class means holomorphy in (c, z) on each such ball. The number $\|w\|_{\flat,s}$ belongs to the chosen triple and can change when its function is represented differently. The physical partition function and every source-local energy are required to agree for two triples that have the same values on all bare plaquettes on which that quantity depends. This representation-independence is a separate part of the source format.

Always $\|w\|_\infty \leq \|w\|_{\flat,s}$ and $c_M L^{-s} \operatorname{Lip} w \leq \|w\|_{\flat,s}$. Thus, if $m = K - s$ and $j \leq m$,

$$c_M a_{K-j} \operatorname{Lip} w = L^{-(m-j)} c_M a_{K-m} \operatorname{Lip} w \leq L^{-(m-j)} \|w\|_{\flat,s},$$
$$a_t = L^{-t}.$$

This is the exact scale gain used when a source is frozen on an earlier block. It follows from the norm definition, not from an analyticity theorem.

Here is an explicit algebraic version of the running profiles. Let P_K be the finite set of bare plaquettes, with barycentres $x(p)$. For each level j choose a finite set Y_j of base points, anchors $p_y \in P_K$, and nonnegative functions $h_y^{(j)}$ on P_K such that

$$\sum_{y \in Y_j} h_y^{(j)}(p) = 1,$$

$$h_y^{(j)}(p) \neq 0 \implies |x(p) - x(p_y)| \leq C_h M a_{K-j}.$$

Choose also profiles $0 \leq \varphi_j(p) \leq 1$. The grid, anchors, partition and profiles are fixed independently of the source throughout the analytic family. They are supplied by the geometric construction; the identities below hold for every finite choice satisfying these conditions.

Let b_j be holomorphic on $D(0,\rho)$ and define

$$d_j = \sup_{0<|z|<\rho} \frac{|b_j(z) - b_j(0)|}{|z|},$$

$$\sum_{j \leq k} d_j \leq \frac{1}{3}.$$

For triples whose function has $\|w\|_\infty < \rho$, set

$$x_k(p) = \beta - \sum_{j \leq k} b_j(0)\varphi_j(p),$$

$$c_j(p;w) = \varphi_j(p) \sum_{y \in Y_j} h_y^{(j)}(p) b_j(w(x(p_y))),$$

$$\zeta_k(p;w) = w(x(p)) + \sum_{j \leq k} \varphi_j(p) \sum_{y \in Y_j} h_y^{(j)}(p)[b_j(w(x(p_y))) - b_j(0)].$$

Then the exact bookkeeping identity is

$$x_k(p) - \zeta_k(p;w) = \beta - w(x(p)) - \sum_{j \leq k} c_j(p;w).$$

To prove it, expand both sides and use $\sum_y h_y^{(j)} = 1$ in each constant term $b_j(0)\varphi_j$. Consequently the Wilson weights in a sourced exponent can be added or subtracted without changing their total bare coefficient. This assertion concerns that total coefficient; it does not assert equality of two effective measures unless the exact integral transport has also been proved.

The same definitions give

$$|\zeta_k(p;w) - w(x(p))| \leq \tfrac{1}{3}\|w\|_\infty,$$

$$|\zeta_k(p;w)| \leq \tfrac{4}{3}\|w\|_\infty.$$

Each value is holomorphic on every source ball with supremum norm below ρ, because only finitely many evaluations and compositions occur. It depends on the source only inside distance $C_h M a_{K-k}$ from p, including the value at p itself; here $a_{K-j} \leq a_{K-k}$ for $j \leq k$.

For a constant source $w \equiv z$, in a region where every $\varphi_j(p) = 1$, this reduces to

$$\zeta_k(z) = z + \sum_{j\le k}[b_j(z) - b_j(0)],$$

$$\zeta_0(z) = z,$$

$$\zeta_{k+1}(z) = \zeta_k(z) + b_{k+1}(z) - b_{k+1}(0).$$

Differentiating at zero gives $\zeta_k'(0) = 1 + \sum_{j\le k} b_j'(0)$ and $|\zeta_k'(0) - 1| \le 1/3$. The derivative bound follows from the definition of d_j by taking the limit at zero. For spatially varying sources, freezing the function w itself costs at most $\operatorname{Lip} w$ times the anchor distance. To freeze $b_j \circ w$ one must work on a smaller source disc. Indeed $h_j(z) = [b_j(z) - b_j(0)]/z$ extends holomorphically to zero and is bounded by d_j. For $|z| \le \rho' < \rho$, Cauchy's estimate gives

$$|b_j'(z)| = |h_j(z) + zh_j'(z)| \le d_j \frac{\rho}{\rho - \rho'}.$$

Thus on the half-disc the additional factor is at most $2d_j$. Apply the mean value integral on the segment between the two source values to bound their composed difference. The displayed depth identity then supplies the factor $L^{-(m-j)}$, with the fixed anchor constant $C_h M/c_M$. The radial bound d_j alone is not a Lipschitz bound on the whole open disc.

All smallness, radius and representation-independence conditions used here must be retained when invoking the sourced RG format. They do not follow from the entire dependence of the bare lattice partition function alone.

5.20.1. *Transporting the observables: sources, analyticity, and bounds independent of the cutoff and of the volume.* The measure $\mu_{K,\ell,\beta}$ is described at the unit scale in §§ 3.1, 4.17 and 5.2, and the present step gives the corresponding description of its correlations. Because $\mathcal{O}(f) = A(f, U)$, a source $w = c + \sum_i z_i f_i$ acts as a complex shift of the bare inverse coupling that depends on position:

$$\rho_0^w = e^{-A(\beta - w, U)},$$

$$Z(w) = \int \rho_0^w \, dU,$$

$$S^T_{n;K,\ell,\beta}(f_1, \dots, f_n) = \partial_{z_1} \cdots \partial_{z_n} \log Z(w)\big|_{w=0}.$$

The renormalisation steps are applied to ρ_0^w. Whatever the source leaves behind is gathered into local numbers, the source energies $e_j(X; w)$ of § 5.20. We set $\mathcal{E}(w) := \sum_{1\le j\le K} \sum_{X\in\mathcal{D}_j} e_j(X; w)$, $\widetilde{Z}(w) := e^{-E-\mathcal{E}(w)} Z(w)$ and $C_g := C(1 + \log g^{-2})$.

Proposition 5.63 (The construction with sources)**.** Let $0 < g \le \gamma_J$. For each first-passage datum (K, β), each $\ell \in \mathfrak{L}(g)$ and each source satisfying $\|w\|_\flat < 2\varrho$, the pair $(E, (e_j(X; \cdot)))$ forms a system of localised source energies. That is, each $e_j(X; w)$ is holomorphic in (c, z) (analyticity). If the f_i are real, it takes real values whenever (c, z) is real. It depends on w only through its values on $X^{\sim 1}$, and it is zero at $w = 0$ (locality). Moreover, uniformly,

$$|e_j(X; w)| \le Ce^{-\kappa_e d_j(X)},$$

$$|\widetilde{Z}(w)| \le e^{C_g \ell^4},$$

$$\widetilde{Z}(0) \ge e^{-C_g \ell^4}$$

(these are, in order: the exponential bound, whose rate $\kappa_e := c\kappa$, $c < 1$, exceeds the entropy rate of the localisation domains through a cube; upper stability; the floor). The same conclusions

hold along a reference run, with every choice frozen at θ_k, as long as $x_k^\natural$ remains in a tube around θ_k with a margin of $C\varrho$. For each K this requirement is a condition on β only. When $g \le \gamma_U$, it is met on the interval I_K of § 5.2.3, an interval that contains the tuned $\beta_K^\natural$. The conclusions also hold for the run stopped at any level k, where the sums are stopped at $j \le k$, the vacuum constant $E_{(k)}$ of the first k steps replaces E, and $C(1+\log x_k)$ per site of $\mathbb{T}^{(k)}$ replaces $C_g\ell^4$ §5.20.1.2. ■

The localized energies in the source transfer. At the trivial pair P_0, let Δ_w denote the difference from the unsourced term. The energy of a domain at level j is

$$e_j(X;w) = \sum_{z\in X\cap\mathbb{T}^{(j)}} \Delta_w E^{(j)}(X,P_0,z) + \Delta_w R^{(j)}(X,P_0), \qquad e_0 = 0.$$

The normalizer and the ratio contribution computed with the fixed real weight cancel in this difference. The source dependence of the coupling shift is already in the sourced regular term. On the fixed source ball of radius r, apply the source-line estimate with reference source zero. On the trivial pair the regular difference is bounded by $Cr^{-1}\|w\|_{\flat,K-j}e^{-(\kappa+1)d_j(X)}$ per pointing. The remainder difference has bound $Cr^{-1}\|w\|_{\flat,K-j}\theta_j^{-\kappa_0/2}e^{-\kappa d_j(X)}$. The at most $CM^4(1+d_j(X))$ pointings use the extra unit of decay, giving

$$|e_j(X;w)| \le CM^4 r^{-1}\|w\|_{\flat,K-j}e^{-\kappa d_j(X)}.$$

The retained source-energy rate can be chosen between the rooted entropy rate and κ. At a witness stop the separated-source bound uses only $j \le k_0$. Coarser origins can read both supports and remain in the source derivatives if the run is continued. If $\mathcal{S}_\emptyset$ denotes the all-small-field label, the vacuum bookkeeping gives

$$E + \sum_{j,X} e_j(X;w) = -\mathcal{E}_K(\mathcal{S}_\emptyset;w).$$

Thus the same E and the sum of these local energies are the constants removed in $\widetilde{Z}$. For a label $\mathcal{S}$, write $\mathsf{T}_j(\mathcal{S};w)$ for its share of these differences. The constant after normalization is

$$-E - \sum_{j,X} e_j(X;w) - \mathcal{E}_K(\mathcal{S};w) = -[E+\mathcal{E}_K(\mathcal{S};0)] - \sum_j [\mathsf{T}_j(\mathcal{S}_\emptyset;w) - \mathsf{T}_j(\mathcal{S};w)].$$

A missing pointing or remainder is rooted at a cube of Z_j^+ by its own domain or its base point. Sum the trivial-pair difference with its rooted decay, including the at most $CM^4(d_j+1)$ pointings per domain. The source share therefore costs $C\#_j(Z_j^+)$. The Gaussian, Haar, and gauge constants at the fixed real weight give the unsourced logarithmic price on the same missing-site count. These are the scalar prices in eq. 4.dxxxi. The weighted product majorant of the literal label terms is required when summing the remaining factors.

Use $s = K - j$ and put

$$V_{a,\ell} = E, \qquad U_{a,\ell}(W) = 0, \qquad e_{a,\ell,s,X}(W) = e_{K-s}(X;w).$$

Here w is the value of the physical source W at the prescribed plaquette points. At $s = K$ the last entry is zero. The source appears through these values and the values of the same function at the running source points, so the entries are functions of W. The radius in the all-state transfer is ϱ with the corresponding source norm, or any smaller fixed radius under the norm comparison of § 5.20. The two stability estimates of Prop. 5.63 then give eq. 5.cdxxxv with this assignment.

The actual domains at that scale are connected unions of cubes of side Ma_s. Their one-layer enlargement contains every source read, with one fixed additional reach for the barycentre of a bare plaquette incident to that enlargement. Use the compatible tori $\ell = 2L^m$ at sufficiently large allowed sides. The cube count in each direction is $2L^{m-m_0+s}$, so the indexing exists at all actual scales. The length $d_j(X)$ of the rooted estimate uses a cube-edge tree within X. The length $m(X)$ used in § 5.23 allows the tree to leave X, and hence $m(X) \le d_j(X)$. The bound for e_j therefore holds with $m(X)$. Choose the retained rate above the rooted closed-walk count for these cube

unions as in eq. 5.cdxlvi. Each union has its original single domain index. This gives the indexing and source range needed for the geometric estimates of D1. The uniform upper bound and floor use the sourced history estimates on the original source ball.

Proposition 5.64 (Correlation bounds: Thm. A (c)(1)–(2))**.** Take g, K, β, ℓ as above and $n \geq 2$. Let the f_i be real Lipschitz functions whose supports lie at mutual distances $\geq \mathsf{d} > 0$. For the second and the third bound, let each support also be contained in a ball of radius R. Then there exists a side $\ell_G \in \mathfrak{L}(g)$, depending only on g, for which

$$|S_n^T| \leq C_n(\mathsf{d}) e^{n c_V \ell^4} \prod_i \|f_i\|_\sharp;$$

$$|S_n^T| \leq C_n^\infty(\mathsf{d}; R) \prod_i \|f_i\|_\sharp \ (\ell \geq Cn(R+1));$$

$$|S_n^T| \leq A(\mathsf{d}; R) B(R)^n (n!)^{\varkappa_\star} \prod_i \|f_i\|_\sharp \ (\ell \geq \ell_G),$$

uniformly, where every constant may also depend on g, $C_n^\infty(\mathsf{d}; R)$ does not increase with d, and one may take $A(\mathsf{d}; R) = \exp(\max\{1, \mathsf{d}^{-5}\})$, after choosing $\varkappa_\star = k + 1$ with the fixed graded exponent k of Prop. 5.76. The sharper form proved there is $[2B(R)]^n (n!)^k \max\{1, \mathsf{d}^{-5n}\} \prod_i \|f_i\|_\sharp$. A polynomial separation prefactor independent of n would require an additional estimate; it is not asserted here. The second and third bounds also hold for $S_n^T[\nu]$ in each periodic Gibbs state ν at β on $\mathbb{L}_K$, with no condition on the volume. ■

Prop. 5.70 is the third main result. Under the law of the histories, the probability that a cell is live, or that it is created, is $\leq Ce^{-cp_0(g_k)}$, uniformly in the volume. The short-distance laws rest on this result. §6.1.1.2 treats the two-point law, clauses (c)(3), (c)(4). It uses these bounds on a single cell and, from their age-resolved form (Prop. 5.71), the age series at a rate $b \geq 8$. The four-point argument for Thm. A(c)(5) requires the series at a rate $b > 16$.

The same tools also give two further results that are used later. The first is the one-point transfer (eq. 5.cdxliv), $|\nu(\mathcal{O}(f)) - \nu'(\mathcal{O}(f))| \leq C_1(R)\|f\|_\sharp$, which compares periodic Gibbs states taken at the same first-passage datum (K, β). With it, one can centre a pure phase at the means of that periodic Gibbs state whose decomposition contains the phase, the coordinates of § 7.7 become compact, and the one-point numbers $\mathfrak{c}_\xi$ of Thm. B(3)(ii) are bounded. It also enters the growth constants of (b)(v), so that a single set of constants covers these components as well. The second consists of the insertion bound and the plate bound (Props. 6.23 and 6.24), which are the a priori estimates on which Prop. 6.26 rests.

Thm. A(a)(ii) uses only the bounds of Prop. 5.64. Thm. A(a)(iii) (for the re-centring), § 6.16, § 6.16 and § 7.7–§ 7.9 also use the one-point transfer in eq. 5.cdxliv. The tuned couplings need not be first-passage data. Prop. A.15 and appendix A.8 apply Prop. 5.63 along reference runs, and this gives the tuned limits the torus bound only. The format underlying Prop. 5.63 is the starting point of §6.1.1.1–§6.1.1.2. The difficulty lies in two places. The first is Prop. 5.63: the source term is marginal, since a step leaves it no smaller, just as it leaves the action no smaller; it takes complex values, so positivity is no longer available; and it must be carried through the large-field operations. The second is Prop. 5.70, in which the ratio of the stability bounds loses $e^{C_g \ell^4}$. Everything else (Cauchy's estimates, reflection positivity, the chessboard estimate) is classical, except for the bookkeeping in n.

5.20.1.1. The curvature source with fixed reference data. **Objects.** The norm $\|f\|_\sharp = \max\{\|f\|_\infty, CM \operatorname{Lip} f\}$ controls how much f can vary within any M-cube of the unit lattice. The sourced densities ρ_k^w arise from ρ_0^w through the operations of §§ 3.1 and 4.17, and every choice in them is the one made in the run at $w = 0$. These choices are the tests χ, the thresholds

ε_k, δ_k, the radii, the sizes $\check{R}_k$, the backgrounds $U_k(\mathbf{V})$ (which minimise the action with no source), the scaling $\mathcal{A}_k = g_k \mathcal{A}_k^{\mathrm{s}}$, the covariances $C^{(k)}$, the gauge-fixing weight and the reference laws of $\mathbf{R}$.

Proposition 5.65 (Kernels; same histories; exactness)**.** To each history h of level k there corresponds a positive kernel $\mathsf{K}_h(U, dV_k)$, built from the choices of the run and from nothing else. For Φ bounded and gauge invariant, the operations carry $\Phi\rho_0$ to

$$\sum_h \tau_h[\Phi],$$

$$\tau_h[\Phi] = \int \Phi(U)\, \rho_0(U)\, \mathsf{K}_h(U, \cdot)\, dU,$$

$$\sum_h \tau_h[\Phi](\mathcal{V}_k) = \int \Phi\rho_0\, dU.$$

For the choice $\Phi = e^{A(w,U)}$ one obtains $\rho_k^w\, dV_k = \sum_h \tau_h^w$, where the sum runs over the histories of ρ_k. Each τ_h^w is a complex measure which, as a function of (c, z), is entire, and $\tau_h^0 = \tau_h \geq 0$. Moreover $\int \rho_k^w\, dV_k = Z(w)$ for $k \leq K$. ∎

Each step consists of finitely many moves. The first is the insertion of $1 = \int dV\, \delta(\mathcal{M}(U)V^{-1})$. The second is a family of decompositions of unity $1 = \sum \prod \chi \prod \chi^c$. The third is a gauge-fixing factor, of total integral 1 over the gauge group; its insertion is harmless since $\Phi\rho_0$, like each a_p, is gauge invariant. The fourth is a set of changes of variables. The last occurs inside $\mathbf{R}$: a term is replaced by the product of a fixed probability law and its integral over a component. Every move is linear in the density and leaves the total integral unchanged, and no move requires positivity. The tests depend on the fields only, so the regions, labels and histories are the same as at $w = 0$.

Proposition 5.66 (Exact insertion of block observables)**.** Given a history h of level k, denote by $Y(h)$ the union of all components that $\mathbf{R}$ has acted upon at levels not exceeding k. A bond c of $\mathbb{T}^{(k)}$ is said to be far from $Y(h)$ when neither the two blocks $B^k(c_\pm)$ nor the bare bonds that join them meet $Y(h)^\sim$. There exists a probability measure P_k on triples (U, h, V), each made of a bare field, a history and a block field, with four properties. First, its marginal on U is $\mu_{K,\ell,\beta}$. Second, under its marginal on (h, V) each bounded gauge-invariant $\Phi(h, V)$ has integral $Z(0)^{-1} \sum_h \int \Phi(h, V)\, \tau_h(dV)$. Third, it is invariant under $(U, h, V) \mapsto (U^u, h, V^{u|})$, where u is a gauge transformation of the bare lattice and $u|$ is its restriction to the sites of $\mathbb{T}^{(k)}$. Fourth, under it $V = \mathcal{M}^k U$ holds on every bond far from $Y(h)$. Consequently, setting $F^{(h)}(V) := \mathbb{E}_{\mathsf{P}_k}[F(\mathcal{M}^k U) \mid h, V]$, which is the conditional expectation with respect to P_k, taken in a version that is invariant in V, one has for each bounded gauge-invariant F on $\mathcal{V}_k$

$$\langle F(\mathcal{M}^k U)\rangle_{K,\ell,\beta} = \frac{\sum_h \int F^{(h)}\, d\tau_h}{\sum_h \int d\tau_h},$$

$$|F^{(h)}| \leq \sup |F|,$$

$$F^{(h)} = F \text{ if } F \text{ reads only bonds far from } Y(h);$$

in particular, $F^{(h)} = F$ for any history in which $\mathbf{R}$ has acted on none of the components. In precisely this sense ρ_k represents the block averages of the bare field. ∎

The mechanism has three parts. First, $\mathbf{T}$ integrates $F \circ \mathcal{M}$ exactly against the density one level lower. Second, on each component on which it acts, $\mathbf{R}$ puts a reference law in place of the law of the block field; this alters the joint distribution of $\mathcal{M}^k U$ and V_k only in a neighbourhood of that component, and it leaves every mass as it was. Third, the steps are glued in succession, each

being regarded as a transport of the gauge-averaged terms. All the assertions are identities between measures and involve neither a constant nor a smallness condition. The role of the proposition is to fix what ρ_k means; nothing is derived from it. In the two-point law §6.1.1.2, a single scale is not isolated by this coupling, because, inside the region it heals, the law of the block field is replaced by $\mathbf{R}$. It is isolated by the exactness with the sources read at depth s, that is, by $\sum_h \tau_h^w(\mathcal{V}_{K-s}) = Z(w)$ (Prop. 5.65, together with §5.20.1.2).

Why the frozen choices still suffice. Bałaban's theorems concern a single real coupling at each step (from these numbers the profiles build the function $g_k^2(x)$ of 119 (1988), (2.24), which coincides with g_k^2 on the last domain), a background that minimises the action appearing in the exponent, and a single coupling-constant counterterm in each step. For this reason the source is not built into that framework but is carried alongside it. In terms of the running shift $\zeta_k(\cdot; w)$ of §5.20.1.2,

$$-A(x_k(\cdot) - \zeta_k(\cdot; w), U) = -A(x_k(\cdot), U) + \sum_{y \in \mathbb{T}^{(k)}} A(h_y^{(k)} \zeta_k(\cdot; w), U),$$

where $h_y^{(k)}$ denote Bałaban's tent functions (Comm. Math. Phys. 119 (1988), (3.40), (3.65)). On the set where $|\partial \mathbf{U} - \mathbf{1}| < \alpha_{0,k} \eta_k^2$ one has

$$|A(h_y^{(k)} \zeta_k(\cdot; w), \mathbf{U})| \le C \alpha_{0,k}^2 \sup |\zeta_k| \le C \varrho \, g_k^2 (\log x_k)^{2q} \ll 1 :$$

it is merely an additional local term, small and analytic, once g is chosen small after ϱ. When the fields are real, $|e^{-A(x_k(\cdot) - \zeta_k, U)}| \le e^{-(1 - C\varrho g^2) A(x_k(\cdot), U)}$; hence a failed test still costs $e^{-c p_0 (g_k)^2}$.

Proposition 5.67 (Reality and positivity at real sources)**.** All the choices are real, and so are the f_i; consequently each move takes real data to real data. Hence, at real (c, z), each of the following quantities is real: the boundary terms, the increments, the activities $E^{(j),w}, R^{(j),w}$ (at real fields), the vacuum constants, the running shifts, and also the $e_j(X; w)$. The exponents are therefore real at real block fields. Suppose moreover that the source is a real constant with $|\zeta| < 2\varrho$. Then, in the operations $\mathbf{T}_k(Z)$ and $\mathbf{T}'_i(X)$ of §5.20.1.2, every exponential factor is strictly positive at real pairs, and

$$x_k(\cdot) - \zeta_k(\cdot; \zeta) > 0,$$
$$\tau_h^\zeta \ge 0 :$$

thus the stability bounds of the run rely on a positivity that is lost at complex w and is still available at real shifts. Furthermore, ρ_0^ζ coincides with the bare density at $\beta - \zeta$, written with the choices belonging to β: these are two presentations of a single measure § 5.2.3. The choices of a reference run do not depend on β. Take a real β whose run has room 2ϱ in the tube, and a real ζ with $|\zeta| < \varrho$; along such a run one then obtains $x_j^\natural(\beta - \zeta) = x_j^\natural(\beta) - \zeta_j(\zeta)$ and $b_j^\natural(\beta - \zeta) = b_j(\zeta)$. These facts enter the runs with real shifts in § 5.2.3 and in appendix A.8, as well as the monotone map that sends the bare coupling to the renormalised coupling § 5.2 and appendix A.8. The conjugated copies of §§ 5.20 and 5.22 and Props. 5.75 and 5.76 make no use of these facts; holomorphy is all they require (with the common-normalizer condition in eq. 5.cdxli). ∎

Changing the real constant in the same frame. Let the reference run at β have room $2r$ through the prefix and take a real $|v| < r$. At the bare level the sourced input is the unsourced one at $\beta - v$. All choices belong to the fixed frame. Suppose the two descriptions agree through level k. The ordinary formation at level k can then be applied to both. The a priori bound by $r/3$ on each

increment beyond its common one-loop term gives

$$|\theta_{k+1} - x^{\natural}_{k+1}(\beta - v)| \le \mathsf{t}_{k+1} - 2r + |\zeta_k(v)| + 2r/3 < \mathsf{t}_{k+1}.$$

Use $|\zeta_k(v)| \le 4|v|/3$ here. This establishes the next admission before comparing the output coefficients. The source-independent kernels act on the same bare input in the two descriptions. Apply the same-frame comparison of presentations from § 5.2.3. It compares the centered totals on the admitted hereditary windows for the common companion. The pointed quadratic readout of those totals uses the same physical convention. This gives $b^{\natural}_{k+1}(\beta - v) = b_{k+1}(v)$ and, by the update of the action weight, $x^{\natural}_{k+1}(\beta - v) = x^{\natural}_{k+1}(\beta) - \zeta_{k+1}(v)$.

The same comparison identifies the unsourced vacuum constant at $\beta - v$. In the all-small-field term, the two densities are continuous on a common open neighbourhood of the identity in the block field. They agree as measures there. The normalized action and activity totals vanish at the trivial background. Evaluate the two densities at the identity, where all the tests pass, to obtain

$$E_{(k)}[\beta - v] = E_{(k)}[\beta] + \sum_{j \le k} \sum_{X} e_j(X; v).$$

This uses equality of the small-field term on the neighbourhood, with the same Haar normalization. Equality of the total partition functions would not determine the individual vacuum constants. A comparison in a different frame needs its own overlap and coupling estimate.

For the normalization at an intermediate witness k_0, sum the constant-source increment bounds only through that witness. The regular term costs $C\theta_j^{-3/2}(\log \theta_j)^{\mathfrak{p}_0}$ at level j. The marked term costs $Cr^{-1}\theta_j^{-\nu}$ by Cauchy's estimate on the composed source disc. Along the reference flow, $\theta_j \ge \theta_{k_0} + c(k_0 - j) - C_0$ on this prefix. The sums give, with $\nu = (\kappa_0 - 4)/2 > 1$,

$$|\zeta'_{k_0}(0) - 1| \le C\left[\theta_{k_0}^{-1/2}(\log \theta_{k_0})^{\mathfrak{p}_0} + r^{-1}\theta_{k_0}^{1-\nu}\right].$$

This bound uses the actual prefix, at the same normalization as the physical source derivatives.

The scope of the mechanism. Because $\mathcal{O}(f) = A(f, U)$ is a weighted action, its source is a coupling shift. A different bounded gauge-invariant local observable Φ need not be a weighted plaquette action. The exact insertion identity can transport $e^{z\Phi}$, but does not put it in the variable-Wilson format or supply its counterterms, source stability or uniform correlation estimates. Those require a separately closed source construction.

Exact insertion of block observables: a coupling of bare field, history and block field. **(i) The setting and the objects.**

Purpose of the block. The effective density of level k decomposes into a finite sum of positive measures indexed by the histories h of level k: $\rho_k\, dV_k = \sum_h \tau_h$. A single application of $\mathbf{T}$ integrates $F \circ \mathcal{M}$ exactly against the full density one level lower, so after k applications one has $\int F\, d(\mathbf{T}^k \rho_0) = \int F(\mathcal{M}^k U)\, \rho_0(U)\, dU$. In the case where $\mathbf{R}$ acts nowhere, $\rho_k\, dV_k$ is the image of $\rho_0\, dU$ under $\mathcal{M}^k$, and the assertion of this block then amounts to the definition of an image measure. Where $\mathbf{R}$ acts on a component, a reference law takes the place of the law of the block field inside that component. This modifies how $\mathcal{M}^k U$ and V_k are jointly distributed on bonds close to that component, and no mass changes. What this block asserts is that the replacements fit together into a single probability measure, defined on triples (bare field, history, block field), with respect to which, away from the components acted upon, the block field coincides with the block average of the bare field. No small factor and no constant enters Statements 1 to 6.

Spaces. Each $\mathcal{V}_j$ is a finite product of copies of the compact group G, and is therefore a compact metrisable space. Finite sets are given the discrete topology, and all measures are Borel measures.

Terms as the measures produced by moves. In parts §5.20.1.1 and § 5.20 the construction is regarded as a chain of moves, a move being a finite family of positive kernels with one kernel for each outcome. The outcomes of the moves of the steps $1, \dots, j$, listed in order, form a string, and such a string is called a history of level j. We write $\mathsf{K}_h(U, \cdot)$ for the kernel obtained by composing

along the string h; it runs from $\mathcal{V}_0$ to the states reached after step j. The admissible histories of level j, namely the strings for which $\mathsf{K}_h \neq 0$, form the finite set $\mathfrak{H}_j$, and $\mathfrak{H}_0 = \{\varnothing\}$. The term of a history $h \in \mathfrak{H}_j$ is defined as the measure into which its string transforms the bare measure, taken as a measure in the block field. Its mass is $\mathsf{t}(h)$:

$$\tau_h(\mathcal{B}) := \int \rho_0(U)\, \mathsf{K}_h(U, \{V_j \in \mathcal{B}\})\, dU,$$

$$\tau_\varnothing = \rho_0\, dU,$$

$$\mathsf{t}(h) := \tau_h(\mathcal{V}_j).$$

The statements and the proofs below use the terms only through this definition.

The two stages of a history. Given $h' \in \mathfrak{H}_{j+1}$, we write $h'|_j \in \mathfrak{H}_j$ for the sub-list of those entries of h' that come from the steps and from the large-field operations of the levels $\leq j$. The strings at the stage reached once the moves of $\mathbf{T}$ of step $j+1$ have been made form a finite set, $\mathfrak{H}^{\mathbf{T}}_{j+1}$. Its elements are the lists $a = (h; P, Q, \Xi)$. Each is formed by some $h \in \mathfrak{H}_j$ together with the outcomes of those three moves of $\mathbf{T}$ which are built from tests:

- $P = P_{j+1}$, the outcome of the plaquette tests (at step 1, the pair (P_0, P_1));
- $Q = Q_{j+1}$, the outcome of the approximate fluctuation tests;
- $\Xi = \Xi_{j+1}$, the outcome of the fluctuation tests together with the recursion along the boundary, which determines $\Omega_{j+1}, \Lambda_{j+1}, S_{j+1}$.

For such an a we set $a|_j := h$. We let $\tau^{\mathbf{T}}_a$ be the measure produced by the string a, taken in the variable V_{j+1}, so that it is a measure on $\mathcal{V}_{j+1}$. The class of the large-field operation of level $j+1$ is denoted $\mathcal{Q}_{j+1}(a)$; it is a set of components whose argument is a string, and it is distinct from the constraint map $\mathcal{Q}_k(U; A)$ introduced in the Notation. It is determined by the entries of a and consists of every component of Z_{j+1} that satisfies the class conditions (part §4.17.1.3). The union of its members is

$$\mathsf{Z}(a) := \bigcup_{X \in \mathcal{Q}_{j+1}(a)} X \subset T_\ell.$$

The components that leave the class in the course of the operation (the drop-outs) are contained in $\mathsf{Z}(a)$, and so are the second-class components, although both kinds remain live. For $h' \in \mathfrak{H}_{j+1}$, the sub-list $h'^{\mathbf{T}} \in \mathfrak{H}^{\mathbf{T}}_{j+1}$ consists of $h'|_j$ together with the outcomes P, Q, Ξ of step $j+1$. Every other entry of h' is an outcome or a class produced within the large-field operation of level $j+1$.

The region of action. Let $h \in \mathfrak{H}_j$. We denote by $Y(h) \subset T_\ell$ the union of those components of the large-field regions of h which have been acted on by the large-field operation at one of the levels $\leq j$; equivalently, it is the union of the class components of these levels. In particular $Y(\varnothing) = \emptyset$, and

$$Y(h') = Y(h'|_j) \cup \mathsf{Z}(h'^{\mathbf{T}})$$

$$(h' \in \mathfrak{H}_{j+1}).$$

Clear and far bonds. To a bond c of $\mathbb{T}^{(j)}$ we attach the set of bare sites $\mathrm{Bl}_j(c) := B^j(c_-) \cup B^j(c_+)$, where $B^0(y) = \{y\}$. We say that the bond c is clear of a set Y when $\mathrm{Bl}_j(c) \cap Y = \emptyset$. A bond c of $\mathbb{T}^{(k)}$ is called far from $Y(h)$ when neither $\mathrm{Bl}_k(c)$ nor any bare bond joining its two blocks meets $Y(h)^\sim$. This set is the union of the enlarged sets $X^\sim$, one for each of the class components X whose union is $Y(h)$. The enlargement adds to such a component one layer of cubes of $\pi_{j(X)}$, and $j(X)$ is the level at which the operation acted on X.

Gauge averages. The G-valued gauge transformations of $\mathbb{T}^{(j)}$ form the group $\mathcal{G}_j$. It is a finite product of copies of G and carries the Haar probability measure dv. It acts through $V \mapsto V^v$, with $(V^v)^{v_0} = V^{v_0 v}$. Given $u \in \mathcal{G}_i$ with $i \le j$, we write $u|_j \in \mathcal{G}_j$ for the gauge transformation obtained from it by restriction to the sites of $\mathbb{T}^{(j)}$. A function on $\mathcal{V}_j$, or a measure on it, is called invariant if this action leaves it invariant. The same convention applies on the configurations over a set of bonds closed under reversal, and on $\mathfrak{H} \times \mathcal{V}_j$, where the group acts trivially on the finite set $\mathfrak{H}$. Given a bounded Borel function F and a finite positive measure τ on $\mathcal{V}_j$, we define

$$\bar{F}(V) := \int F(V^v)\, dv,$$

$$\int F\, d\bar{\tau} := \int \bar{F}\, d\tau.$$

(ii) The statements. Throughout, β is real, $0 \le s \le K$ and $k = K - s$. The run is assumed to satisfy, through level k, the inductive description of the effective densities (part §4.17.1.45), with $x_0, \dots, x_k \ge \gamma^{-2}$, together with the two-sided flow bound (part § 5.2.1). Of all these data, the proofs use only the fact that the moves of the steps $1, \dots, k$ are formed, that is, they use nothing beyond a reference sequence and a torus that is large compared with the cubes of the levels $\le k$.

1. The step, taken term by term. Let $0 \le j < k$ and $h \in \mathfrak{H}_j$. Then each $\tau_a^{\mathbf{T}}$ with $a|_j = h$ is a finite non-negative measure on $\mathcal{V}_{j+1}$. Moreover, every bounded Borel invariant F on $\mathcal{V}_{j+1}$ satisfies the following identity (the transport identity of the stage $\mathbf{T}$):

$$\sum_{a \in \mathfrak{H}_{j+1}^{\mathbf{T}} :\, a|_j = h} \int F\, d\tau_a^{\mathbf{T}} = \int (F \circ \mathcal{M})\, d\tau_h.$$

Now let a be such that $a|_j = h$, and let F be a bounded Borel invariant function on $\mathcal{V}_{j+1}$ which does not depend on the variables $V(b)$ for which b is a bond of $\mathbb{T}^{(j+1)}$ having at least one end point in $\mathsf{Z}(a)$. Then (the transport identity of the stage $\mathbf{R}$)

$$\sum_{h' \in \mathfrak{H}_{j+1} :\, h'^{\mathbf{T}} = a} \int F\, d\tau_{h'} = \int F\, d\tau_a^{\mathbf{T}}.$$

After summation over $h \in \mathfrak{H}_j$, the first identity expresses the exactness of $\mathbf{T}$ on the whole density. Taking $F \equiv 1$ in both identities gives

$$\sum_{h' \in \mathfrak{H}_{j+1} :\, h'|_j = h} \mathsf{t}(h') = \mathsf{t}(h) :$$

thus the step divides the mass of a history among its continuations. The law of the histories (part §§ 5.20 and 5.22) is based on this identity.

2. Gauge averages. Suppose a compact group $\mathcal{G}$ with Haar probability dv acts continuously on a compact metrisable space $\mathcal{X}$. Define $\bar{F}(x) = \int F(v \cdot x)\, dv$ and $\int F\, d\bar{\tau} = \int \bar{F}\, d\tau$. (a) The function $\bar{F}$ is bounded, Borel and invariant, with $\sup |\bar{F}| \le \sup |F|$, and $\bar{F} = F$ when F is invariant. The measure $\bar{\tau}$ is finite, positive and invariant. It satisfies $\int F\, d\bar{\tau} = \int F\, d\tau$ for invariant F, and $\bar{\tau}(N) = \tau(N)$ for invariant Borel sets N. (b) Every invariant finite positive measure ν satisfies $\int \bar{F}\, d\nu = \int F\, d\nu$. Hence, if two invariant finite positive measures give the same integral to every bounded Borel invariant function, they are equal. (c) Let $r : \mathcal{G} \to \mathcal{G}'$ be surjective, and let $q : \mathcal{X} \to \mathcal{X}'$ be a Borel map with $q(v \cdot x) = r(v) \cdot q(x)$. Then the image under q of any invariant measure is invariant, and $f \circ q$ is invariant whenever f is invariant.

3. Gluing. Let $\mathcal{X}_1, \mathcal{X}_2, \mathcal{Y}$ be compact metrisable spaces and let $p_i : \mathcal{X}_i \to \mathcal{Y}$ be Borel maps. Let λ_i be finite positive measures on $\mathcal{X}_i$ such that $(p_1)_* \lambda_1 = (p_2)_* \lambda_2$. Then $\mathcal{X}_1 \times \mathcal{X}_2$ carries a

finite positive measure λ with the following two properties: its marginals are λ_1 and λ_2, and $p_1(x_1) = p_2(x_2)$ for λ-almost every (x_1, x_2).

4. Clear bonds. (a) One has $c_-, c_+ \in \mathrm{Bl}_j(c)$. (b) Let c be a bond of $\mathbb{T}^{(j+1)}$, and let b be a bond of $\mathbb{T}^{(j)}$ whose two end points lie in the double block $B(c_-) \cup B(c_+)$. Then $\mathrm{Bl}_j(b) \subset \mathrm{Bl}_{j+1}(c)$. (c) Every bond of $\mathbb{T}^{(k)}$ that is far from $Y(h)$ is clear of $Y(h)$.

5. A single step as a transport of the averaged terms. Let $0 \le j < k$. Then $(\mathfrak{H}_j \times \mathcal{V}_j) \times (\mathfrak{H}_{j+1} \times \mathcal{V}_{j+1})$ carries a finite positive measure m_{j+1} with the following properties: (a) the marginal on $\mathfrak{H}_j \times \mathcal{V}_j$ equals $\sum_{h \in \mathfrak{H}_j} \delta_h \otimes \bar{\tau}_h$; (b) the marginal on $\mathfrak{H}_{j+1} \times \mathcal{V}_{j+1}$ equals $\sum_{h' \in \mathfrak{H}_{j+1}} \delta_{h'} \otimes \bar{\tau}_{h'}$; (c) for m_{j+1}-almost every $((h, V), (h', V'))$ both of the following hold: $h'|_j = h$, and $V'(c) = \mathcal{M}(V)(c)$ for every bond c of $\mathbb{T}^{(j+1)}$ that has no end point in $\mathsf{Z}(h'^{\mathbf{T}})$.

6. Exact insertion of block observables. We write $Z = Z_{K,\ell,\beta} = Z(0)$ and $\mu = \mu_{K,\ell,\beta}$, and $\mathbb{E}_{\mathsf{P}}$ denotes the expectation with respect to a probability measure P. (a) On $\mathcal{V}_0 \times \mathfrak{H}_k \times \mathcal{V}_k$ there exists a probability measure P with the following four properties. Its marginal on $\mathcal{V}_0$ equals μ. Its marginal on $\mathfrak{H}_k \times \mathcal{V}_k$ is built from the averaged terms:

$$\bar{\nu}_k := Z^{-1} \sum_{h \in \mathfrak{H}_k} \delta_h \otimes \bar{\tau}_h.$$

Hence this marginal assigns to every bounded Borel function $\Phi(h, V)$ that is invariant in V the integral $Z^{-1} \sum_h \int \Phi(h, V)\, \tau_h(dV)$, and the law of the histories of level k is the marginal of P on $\mathfrak{H}_k$: for $\mathsf{E}_k \subset \mathfrak{H}_k$, $\mathsf{P}(h \in \mathsf{E}_k) = Z^{-1} \sum_{h \in \mathsf{E}_k} \mathsf{t}(h) = \tilde{\nu}(\mathsf{E}_k)$. It is invariant under the maps $(U, h, V) \mapsto (U^u, h, V^{u|_k})$, $u \in \mathcal{G}_0$. Conditionally on (U, h), the third coordinate agrees with $\mathcal{M}^k U$ on every bond that is clear of $Y(h)$, and in particular on every bond far from $Y(h)$. (b) For each bounded Borel invariant function F on $\mathcal{V}_k$, let

$$F^{(h)}(V) := \mathbb{E}_{\mathsf{P}}\big[F(\mathcal{M}^k U) \,\big|\, h, V\big]$$

be chosen among the versions of the conditional expectation as one that is invariant in V. Versions with this invariance exist, and for each h any two of them coincide τ_h-almost everywhere. Each of them satisfies

$$\big\langle F(\mathcal{M}^k U) \big\rangle_{K,\ell,\beta} = \frac{\sum_h \int F^{(h)}\, d\tau_h}{\sum_h \int d\tau_h},$$

$$|F^{(h)}| \le \sup |F| \ \tau_h\text{-almost everywhere},$$

$$\sum_h \int d\tau_h = Z.$$

(c) Suppose F depends only on the variables of bonds that are clear of $Y(h)$; this is the case in particular when its dependence is confined to bonds far from $Y(h)$. Then one has $F^{(h)} = F$ almost everywhere with respect to τ_h. In particular, for every history in which the large-field operation has acted on none of the components, $F^{(h)} = F$. Without this restriction one has, in general, $F^{(h)} \ne F$. Here the depth s is arbitrary, $0 \le s \le K$, and $k = K - s$.

7. Bounded block observables. Take F as in Statement 6 (b) and level-k cells $\mathfrak{c}_1, \dots, \mathfrak{c}_n$, and denote by $\mathsf{E}_k := \bigcup_{i \le n} \mathfrak{lv}_k(\mathfrak{c}_i) \subset \mathfrak{H}_k$ the event that at least one of these cells is contained in the live region of level k. Then

$$\Big| \big\langle F(\mathcal{M}^k U) \big\rangle_{K,\ell,\beta} - \frac{\sum_{h \notin \mathsf{E}_k} \int F^{(h)}\, d\tau_h}{\sum_h \int d\tau_h} \Big| = \Big| \frac{\sum_{h \in \mathsf{E}_k} \int F^{(h)}\, d\tau_h}{\sum_h \int d\tau_h} \Big|$$

$$\leq \sup|F|\;\tilde{\nu}(\mathsf{E}_k) \leq n\,C\,\sup|F|\,e^{-c_{\mathrm{r}}p_0(g_k)}.$$

The final inequality is valid whenever the data satisfy the hypotheses of the one-cell bound of part §§ 5.20 and 5.22, with C and c_{r} the constants of that bound. It holds uniformly in K, β, ℓ and the positions of the cells. In this form the bound applies to an observable of bounded modulus that depends on the block variables of a bounded number of cells at its own scale. In the expectation of such an observable, the part contributed by the histories of E_k is smaller than any power of g_k.

(iii) Proofs.

On 1. All three assertions are contained in part § 5.20. There they are properties of the measures that the moves produce, taken one string at a time. Positivity. All kernels of all moves are non-negative. Hence the measure into which a string turns $\rho_0\,dU \geq 0$ is finite and non-negative. The stage **T**. In its first move, **T** sets $V_{j+1} := \mathcal{M}(V_j)$, and no later move of **T** alters V_{j+1}. The reason is that the Faddeev–Popov draw uses gauge transformations supported on the sites $B(y)\setminus\{y\}$, and these act on V_j without touching V_{j+1}. Hence at this stage the identity is valid for every bounded Borel F. As an equality of measures, $\sum_{a|_j=h}\tau_a^{\mathbf{T}} = \mathcal{M}_*\tau_h$. The stage **R**. Given a class component X, denote by $\Lambda^{\mathrm{pr}}\cap X$ its box. This box lies well inside X, and **R** integrates the block field over it. Denote by $\mathfrak{b}_X$ the set of bonds of $\mathbb{T}^{(j+1)}$ that have an end point in that box. The moves of **R** alter V_{j+1} on the bonds of the sets $\mathfrak{b}_X$ and nowhere else. Hence at this stage the identity is valid for every bounded Borel F that does not depend on these bonds. Each bond of $\mathfrak{b}_X$ has both end points in X. Therefore every function that depends on no bond with an end point in $\mathsf{Z}(a)$ belongs to this class. The property that matters is that F does not depend on $\mathfrak{b}_X$. Gauge invariance by itself is not enough. A counterexample is the plaquette function that appears in two places: in the block of this part that treats the curvature observable as a source (in the proof concerning the steps as stochastic kernels: the two refinements), and in part § 5.20 (stability; in the proof of Statement 2 on exactness: which functions are transported). Why history by history. Each move, once summed over its outcomes, is a probability kernel at each state. For a move consisting of tests, exactly one outcome takes the value 1. For the gauge fixing the property holds because the gauge fixing is carried out as a procedure, namely as a draw. Let $\mathfrak{G}\geq 0$ be normalised by $\int\mathfrak{G}(V^u)\,du = 1$. Then the measure

$$(\tau\kappa_{\mathfrak{G}})(\mathcal{B}) = \int\tau(dV)\int du\,\mathfrak{G}(V^u)\,\mathbf{1}_{\mathcal{B}}(V^u)$$

has the same mass as τ, for every measure τ. The product $\mathfrak{G}\,\tau$, by contrast, has mass $\int\mathfrak{G}\,d\tau$. If τ is invariant under the group with respect to which $\mathfrak{G}$ is normalised, this coincides with the mass of τ, and then $\tau\kappa_{\mathfrak{G}} = \mathfrak{G}\,\tau$. Suppose that the bare weight is invariant under gauge transformations and that the representatives transform covariantly. Then, history by history, the term on which a gauge-fixing move acts has this invariance (see the treatment of the steps as stochastic kernels at the beginning), and for such a term the draw coincides with the product. The invariance meant here is invariance under the group of the move. A term need not be invariant under the full gauge group of the level. For a measure without this invariance, replacing the draw by the product does not conserve the mass in general. The masses considered from here on are the ones produced by the draw, which preserves them for any bounded Borel weight. Likewise, the axial frames are drawn before the tests that use them. The test that follows the gauge fixing along the tree lacks gauge invariance. Therefore the mass a_r that passes it depends on the frame r. Suppose the components that leave kept a single frame e while the survivors were averaged over the frames. Then the mass after the reference laws would be off by $\bar{a}-a_e$, with $\bar{a}$ the average of the a_r, and already for $F\equiv 1$ the identity of the stage **R** would not hold. Part § 5.20 contains both points (exactness: frames first; the product form). Why it does not suffice to have the identities in their form summed over the histories. Let $\mathcal{V}_0=\mathcal{V}_1=\mathcal{V}_2=\{0,1\}^2$, where bond 1 is exterior and bond 2 is the component. Let $\mathcal{M}=\mathrm{id}$, let the gauge group be trivial, and let μ be the uniform law on the diagonal. Level 1

has two outcomes a_1, a_2, with $\tau^{\mathbf{T}}_{a_1} = \tau^{\mathbf{T}}_{a_2} = \mu/2$, and $\mathbf{R}$ acts on bond 2 in a_2 alone. Thus $\tau_{a_1} = \mu/2$, while τ_{a_2} is the product of the marginal of $\mu/2$ on bond 1 with the uniform law on bond 2. At level 2 no $\mathbf{R}$ acts, and every history has a single continuation, a'_i for a_i. We set $\tau_{a'_1} := \tau_{a_2}$, $\tau_{a'_2} := \tau_{a_1}$. Every term is non-negative. The totals are transported exactly, and the masses are conserved history by history, $\mathsf{t}(a'_i) = \mathsf{t}(a_i)$. Moreover, $\mathbf{R}$ replaces a term by the product of its exterior marginal with a probability law. On $\{h = a'_1\}$, however, $Y(a'_1) = \emptyset$ forces $V = U$, which is a point of the diagonal. Yet $\tau_{a'_1}$ assigns mass $\frac{1}{4}$ to the complement of the diagonal. Hence no P exists. The failure lies in the identity of the stage $\mathbf{T}$ applied to the one history a_1 by itself. Below, the two identities are applied to invariant F alone.

Proof of 2. (a) The map $(v, x) \mapsto F(v \cdot x)$ is bounded and Borel, so Fubini's theorem shows that $\bar{F}$ is Borel. Right invariance of Haar measure gives $\bar{F}(v_0 \cdot x) = \int F(vv_0 \cdot x)\, dv = \bar{F}(x)$. The functional $F \mapsto \int \bar{F}\, d\tau$ is positive and continuous under monotone limits. It therefore defines a measure $\bar{\tau}$. With $F_{v_0}(x) = F(v_0 \cdot x)$, left invariance yields $\overline{F_{v_0}} = \bar{F}$, and this is the invariance of $\bar{\tau}$. The remaining assertions follow from $\bar{F} = F$ for invariant F, applied to indicators as well. (b) If ν is invariant, Fubini's theorem gives $\int \bar{F}\, d\nu = \int dv \int F(v \cdot x)\, \nu(dx) = \int F\, d\nu$. Hence ν is determined by the values it assigns to the integrals of the invariant functions $\bar{F}$. (c) This follows by a change of variables. By this device, the proof converts the two identities of Statement 1 into equalities of measures, namely equalities between the gauge averages of the terms. The proof applies these identities to invariant functions only.

Proof of 3. We apply the disintegration theorem in the following form. Let $\mathcal{X}$, $\mathcal{Y}$ be compact metrisable, let $p : \mathcal{X} \to \mathcal{Y}$ be Borel, let λ be a finite positive measure on $\mathcal{X}$, and let $\nu = p_*\lambda$. Then there exist probability measures $\kappa(y; \cdot)$ on $\mathcal{X}$, Borel in y, such that $\lambda(B) = \int \nu(dy)\, \kappa(y; B)$ and $\kappa(y; p^{-1}\{y\}) = 1$ for ν-almost every y. Set $\nu = (p_i)_*\lambda_i$, and let $\kappa_i(y; \cdot)$ be a disintegration of this kind of λ_i along p_i. Define

$$\lambda(B) := \int \nu(dy)\, \big(\kappa_1(y; \cdot) \otimes \kappa_2(y; \cdot)\big)(B).$$

The integrand is Borel in y, because the sets B with this property form a Dynkin class that contains the products of Borel sets. We have $\lambda(B_1 \times \mathcal{X}_2) = \int \nu(dy)\kappa_1(y; B_1) = \lambda_1(B_1)$, and the second marginal is treated in the same way. The set $\{p_1(x_1) = p_2(x_2)\}$ is Borel and contains $p_1^{-1}\{y\} \times p_2^{-1}\{y\}$. For ν-almost every y, the latter set has full measure under $\kappa_1(y; \cdot) \otimes \kappa_2(y; \cdot)$.

Proof of 4. (a) Every site lies in its own block. (b) For a site y of $\mathbb{T}^{(j+1)}$, its cube is the union, over the sites $x \in B(y)$ of $\mathbb{T}^{(j)}$, of their cubes. After intersection with the bare lattice this becomes $B^{j+1}(y) = \bigcup_{x \in B(y)} B^j(x)$. Consequently $B^j(b_\pm) \subset B^{j+1}(c_-) \cup B^{j+1}(c_+)$. (c) This follows from $Y(h) \subset Y(h)^\sim$.

Proof of 5. The measures τ_h, $\tau_{h'}$, $\tau^{\mathbf{T}}_a$ are all finite and non-negative. Consequently the gauge average of each of them is a finite, positive and invariant measure. Step 1 (the stage $\mathbf{T}$: Radon–Nikodym). Let $h \in \mathfrak{H}_j$ be fixed. The map $\mathcal{M}$ is covariant, $\mathcal{M}(V^v) = \mathcal{M}(V)^{v|_{j+1}}$, as well as Borel measurable, and $v \mapsto v|_{j+1}$ is a surjection of $\mathcal{G}_j$ onto $\mathcal{G}_{j+1}$. Therefore $\mathcal{M}_*\bar{\tau}_h$ is invariant, and $F \circ \mathcal{M}$ is invariant whenever F is. Let F be any bounded Borel function on $\mathcal{V}_{j+1}$. Applying the transport identity of the stage $\mathbf{T}$ to the invariant function $\bar{F}$, we obtain

$$\sum_{a|_j = h} \int F\, d\bar{\tau}^{\mathbf{T}}_a = \sum_{a|_j = h} \int \bar{F}\, d\tau^{\mathbf{T}}_a = \int (\bar{F} \circ \mathcal{M})\, d\tau_h = \int (\bar{F} \circ \mathcal{M})\, d\bar{\tau}_h = \int \bar{F}\, d(\mathcal{M}_*\bar{\tau}_h) = \int F\, d(\mathcal{M}_*\bar{\tau}_h).$$

Hence $\sum_{a|_j = h} \bar{\tau}^{\mathbf{T}}_a = \mathcal{M}_*\bar{\tau}_h$. In particular $0 \le \bar{\tau}^{\mathbf{T}}_a \le \mathcal{M}_*\bar{\tau}_h$. Therefore $\bar{\tau}^{\mathbf{T}}_a = f_a \cdot \mathcal{M}_*\bar{\tau}_h$ for Borel functions $0 \le f_a \le 1$ with $\sum_{a|_j = h} f_a = 1$ almost everywhere. Define $\mathsf{m}^{\mathbf{T}}_a$ as the image of $(f_a \circ \mathcal{M})\, \bar{\tau}_h$ under $V \mapsto (V, \mathcal{M}(V))$. It is a measure on pairs $(V, W) \in \mathcal{V}_j \times \mathcal{V}_{j+1}$. The marginals of this measure are $(f_a \circ \mathcal{M})\bar{\tau}_h$ and $\bar{\tau}^{\mathbf{T}}_a$, and almost surely $W = \mathcal{M}(V)$. What the identity transports is the sum

over the outcomes and nothing more. It does not fix which portion of τ_h goes to which a, and the only splitting it supplies is the one by $f_a \circ \mathcal{M}$. Accordingly, under the coupling just constructed, the outcome a is drawn with probability $f_a(\mathcal{M}(V))$, and it is not determined by the tests. Step 2 (the stage **R**: restriction off the class). Fix a. Let $\mathrm{Bd}(a)$ denote the set of bonds of $\mathbb{T}^{(j+1)}$ having no end point in $\mathsf{Z}(a)$. Let $\mathcal{V}(a)$ be the space of configurations on $\mathrm{Bd}(a)$, and let $\mathrm{res}_a : \mathcal{V}_{j+1} \to \mathcal{V}(a)$ be the restriction map, which commutes with the gauge action. Set $\lambda_a := \sum_{h'^{\mathbf{T}}=a} \delta_{h'} \otimes \bar{\tau}_{h'}$. Let f be a bounded Borel invariant function on $\mathcal{V}(a)$. Then $f \circ \mathrm{res}_a$ is invariant and depends on no bond with an end point in $\mathsf{Z}(a)$. The transport identity of the stage **R** therefore gives

$$\sum_{h'^{\mathbf{T}}=a} \int (f \circ \mathrm{res}_a)\, d\bar{\tau}_{h'} = \sum_{h'^{\mathbf{T}}=a} \int (f \circ \mathrm{res}_a)\, d\tau_{h'} = \int (f \circ \mathrm{res}_a)\, d\tau_a^{\mathbf{T}} = \int (f \circ \mathrm{res}_a)\, d\bar{\tau}_a^{\mathbf{T}}.$$

The measure $(\mathrm{res}_a)_* \bar{\tau}_a^{\mathbf{T}}$ and the image of λ_a under $(h', V') \mapsto \mathrm{res}_a(V')$ are both invariant, so Statement 2 (b) shows that they are equal. The identity says nothing about the bonds with an end point in $\mathsf{Z}(a)$. For this reason the two measures are glued over $\mathcal{V}(a)$. The result is a measure $\mathsf{m}_a^{\mathbf{R}}$ on $\mathcal{V}_{j+1} \times (\mathfrak{H}_{j+1} \times \mathcal{V}_{j+1})$ with marginals $\bar{\tau}_a^{\mathbf{T}}$ and λ_a. Almost surely under it, $\mathrm{res}_a(V') = \mathrm{res}_a(W)$ and $h'^{\mathbf{T}} = a$. Step 3 (composition). On the middle field W, the marginal of $\mathsf{m}_a^{\mathbf{T}}$ and the marginal of $\mathsf{m}_a^{\mathbf{R}}$ are both equal to $\bar{\tau}_a^{\mathbf{T}}$. We glue over $\mathcal{V}_{j+1}$ and discard the middle field. We then attach to V the label $h = a|_j$ and sum over a. The resulting measure is m_{j+1}. (a): $\sum_{a|_j=h} (f_a \circ \mathcal{M}) \bar{\tau}_h = \bar{\tau}_h$. (b): $\sum_a \lambda_a = \sum_{h'} \delta_{h'} \otimes \bar{\tau}_{h'}$, because each h' has exactly one first stage. (c): almost surely $W = \mathcal{M}(V)$, $\mathrm{res}_a(V') = \mathrm{res}_a(W)$ and $h'^{\mathbf{T}} = a$, so that $h'|_j = a|_j = h$.

Proof of 6. For $0 \le j \le k$ we define

$$\mathsf{A}_j := \{(U, h, V) \in \mathcal{V}_0 \times \mathfrak{H}_j \times \mathcal{V}_j : V(b) = (\mathcal{M}^j U)(b) \text{ for every bond } b \text{ of } \mathbb{T}^{(j)} \text{ clear of } Y(h)\},$$

which is a Borel set. Step 1 (induction on the level). We show that, for each $0 \le j \le k$, the space $\mathcal{V}_0 \times \mathfrak{H}_j \times \mathcal{V}_j$ carries a probability measure P_j with the following three properties. Its projection to $\mathcal{V}_0$ is μ. Its marginal on $\mathfrak{H}_j \times \mathcal{V}_j$ equals $Z^{-1} \sum_{h \in \mathfrak{H}_j} \delta_h \otimes \bar{\tau}_h$, which is the family of averaged terms. Finally, $\mathsf{P}_j(\mathsf{A}_j) = 1$. To begin the induction: the measure $\rho_0\, dU$ is invariant, so $\bar{\tau}_\varnothing = \tau_\varnothing$, and we take for P_0 the push-forward of μ under $U \mapsto (U, \varnothing, U)$. Passage from j to $j+1$: by the second property together with Statement 5 (a), the measures P_j and $Z^{-1}\mathsf{m}_{j+1}$ induce the same marginal on $\mathfrak{H}_j \times \mathcal{V}_j$. We glue the two over this space and project the result onto (U, h', V'), which gives P_{j+1}. The first two properties carry over, and Statement 5 (b) is what carries over the second. As for the third, with probability one we have $(U, h, V) \in \mathsf{A}_j$, $h'|_j = h$, and $V'(c) = \mathcal{M}(V)(c)$ outside $\mathsf{Z}(h'^{\mathbf{T}})$. Take a bond c clear of $Y(h') = Y(h) \cup \mathsf{Z}(h'^{\mathbf{T}})$. Both of its end points belong to $\mathrm{Bl}_{j+1}(c)$, so neither lies in $\mathsf{Z}(h'^{\mathbf{T}})$, and therefore $V'(c) = \mathcal{M}(V)(c)$. For each bond b of the double block of c we have $\mathrm{Bl}_j(b) \subset \mathrm{Bl}_{j+1}(c)$. Hence such a bond is clear of $Y(h)$, and $V(b) = (\mathcal{M}^j U)(b)$. Because $\mathcal{M}(V)(c)$ depends on V only through these bonds, we obtain

$$V'(c) = \mathcal{M}(V)(c) = \mathcal{M}(\mathcal{M}^j U)(c) = (\mathcal{M}^{j+1} U)(c).$$

Step 2 (symmetrisation). The group $\mathcal{G}_0$ acts on triples by $(U, h, V) \mapsto (U^u, h, V^{u|_k})$. Averaging P_k over this action gives a measure P, and it is invariant under the action. Since μ and $\bar{\nu}_k = Z^{-1} \sum_h \delta_h \otimes \bar{\tau}_h$ are both invariant, P still has the first two properties. Covariance gives $\mathcal{M}^k(U^u) = (\mathcal{M}^k U)^{u|_k}$, so the condition $V(b) = (\mathcal{M}^k U)(b)$ holds exactly when $V^{u|_k}(b) = (\mathcal{M}^k(U^u))(b)$ does. Thus the set A_k is invariant, and $\mathsf{P}(\mathsf{A}_k) = \mathsf{P}_k(\mathsf{A}_k) = 1$. Let Φ be a function and N a Borel set, both invariant in V, with sections N_h. Then Statement 2 (a) gives

$$\int \Phi\, d\bar{\nu}_k = Z^{-1} \sum_h \int \Phi(h, V)\, \tau_h(dV),$$

$$\bar\nu_k(N) = Z^{-1}\sum_h \tau_h(N_h).$$

This is why the statement is restricted to invariant Φ and F. The marginal of P consists of the averaged terms, and the averaged terms coincide with the terms only on invariant functions and invariant sets. Step 3 (assertion (a)). The marginals and the invariance were obtained in Step 2. A function of h alone is invariant in V, and this yields the law of the histories. Because $\mathsf{P}(\mathsf{A}_k) = 1$, for almost every (U, h) a regular conditional distribution of V given (U, h) is concentrated on the configurations that agree with $\mathcal{M}^k U$ on every bond clear of $Y(h)$, and so in particular on every far bond. Step 4 (invariant versions). Suppose that F is invariant, and set $\mathsf{f}(U) := F(\mathcal{M}^k U)$. This function is invariant under $\mathcal{G}_0$. Take an arbitrary version Ψ of $\mathbb{E}_{\mathsf{P}}[\mathsf{f}(U)|h, V]$ with $|\Psi| \le \sup|F|$, so that $\mathbb{E}_{\mathsf{P}}[\mathsf{f}(U)\phi(h, V)] = \int \Psi\phi\, d\bar\nu_k$ for every bounded Borel ϕ. Each $v \in \mathcal{G}_k$ is of the form $u|_k$, where $u \in \mathcal{G}_0$ takes the value $v(y)$ on the block $B^k(y)$. Using the invariance of P, of f and of $\bar\nu_k$, we get

$$\mathbb{E}_{\mathsf{P}}[\mathsf{f}(U)\phi(h,V)] = \mathbb{E}_{\mathsf{P}}[\mathsf{f}(U^u)\phi(h,V^v)] = \mathbb{E}_{\mathsf{P}}[\mathsf{f}(U)\phi(h,V^v)] = \int \Psi(h,V)\phi(h,V^v)\,d\bar\nu_k = \int \Psi(h,V^{v^{-1}})\phi(h,V)\,d\bar\nu_k.$$

The first equality holds because P is jointly invariant. We integrate over v, apply Fubini's theorem and use the fact that Haar measure is invariant under inversion. It follows that $F^{(h)}(V) := \int \Psi(h, V^v)\, dv$ is also a version, and that it is Borel, invariant in V and bounded by $\sup|F|$. Any two invariant versions differ only on an invariant Borel set N with $\bar\nu_k(N) = 0$. Since the terms are non-negative, this gives $\tau_h(N_h) = 0$ for every h. Step 5 (assertion (b)). Take an invariant version. Then

$$\langle F(\mathcal{M}^k U)\rangle = \int \mathsf{f}\, d\mu = \mathbb{E}_{\mathsf{P}}[\mathsf{f}(U)] = \int F^{(h)}(V)\,\bar\nu_k(dh, dV) = Z^{-1}\sum_h \int F^{(h)}\, d\tau_h.$$

The last equality uses the invariance of $F^{(h)}$ in V. Taken by itself, the defining property of a conditional expectation yields the integral against $\bar\nu_k$ and nothing more. No use can be made at this point of a non-invariant version. Consider a term in which some component has been put into the second class by the large-field operation at level k. By part § 5.20, such a term carries, in the level-k block field, a Dirac distribution on the gauge tree of the box. Consequently τ_h and $\bar\tau_h$ are mutually singular for such a term. A version is determined $\bar\tau_h$-almost everywhere, and in this case the integral against τ_h of a version that is not invariant is then arbitrary. Taking $\Phi \equiv 1$ gives $\sum_h \int d\tau_h = Z$. For every h, any invariant version coincides τ_h-almost everywhere with the version constructed in Step 4, and that version is bounded by $\sup|F|$. Step 6 (assertion (c)). Fix h_0, and suppose that F depends only on bonds clear of $Y(h_0)$. On $\mathsf{A}_k \cap \{h = h_0\}$ we then have $F(\mathcal{M}^k U) = F(V)$. It follows that

$$\mathbb{E}_{\mathsf{P}}[\mathsf{f}(U)\mathbf{1}_{\{h=h_0\}}\phi(V)] = \mathbb{E}_{\mathsf{P}}[F(V)\mathbf{1}_{\{h=h_0\}}\phi(V)]$$

for every bounded Borel ϕ. In other words, $F^{(h_0)} = F$ $\bar\nu_k$-almost everywhere on $\{h_0\} \times \mathcal{V}_k$. The set on which this equality fails is invariant in V and is therefore a τ_{h_0}-null set. When the operation has acted on no component of h_0, we have $Y(h_0) = \emptyset$, and then every bond is clear.

Proof of 7. The equality comes from splitting the numerator of Statement 6 (b) into the part over E_k and the part over its complement. We have $\tau_h \ge 0$, and $|F^{(h)}| \le \sup|F|$ holds τ_h-almost everywhere. Hence $|\int F^{(h)}\, d\tau_h| \le \sup|F|\,\mathsf{t}(h)$. Summing this over $h \in \mathsf{E}_k$ and dividing by $Z = \sum_h \mathsf{t}(h)$ gives $\sup|F|\,\tilde\nu(\mathsf{E}_k)$. Subadditivity together with the one-cell bound $\tilde\nu(\mathfrak{w}_k(\mathfrak{c})) \le C\, e^{-c_{\mathrm{r}} p_0(g_k)}$ of part §§ 5.20 and 5.22 yields $\tilde\nu(\mathsf{E}_k) \le n\, C\, e^{-c_{\mathrm{r}} p_0(g_k)}$. For a history outside E_k, no one of the cells lies in a large-field region that is live at level k.

(iv) What is used.

From other parts. The following results of part § 3.1.1 are used. The map $\mathcal{M}$ is Borel. It is gauge covariant: $\mathcal{M}(V^v) = \mathcal{M}(V)^{v'}$, where v' is obtained from v by restriction to the sites of the coarser lattice. It is local: $\mathcal{M}(V)(c)$ depends on V only through the bonds of the double block of c. From part § 5.20 we use three things. The first is the description of the construction as a chain of moves. The second is that every kernel is non-negative, so that every τ_h and every $\tau_a^{\mathbf{T}}$ is a finite non-negative measure. The third is the pair of transport identities of Statement 1. From part §4.17.1.3 we use the class of the large-field operation, which enters through the recursion $Y(h') = Y(h'|_j) \cup \mathsf{Z}(h'^{\mathbf{T}})$. Statements 1 to 6 use nothing further from the construction: no bound, no analyticity, and no form of a term. Only Statement 7 draws in addition on part §§ 5.20 and 5.22, through the one-cell bound there on the law of the histories.

Where it is used. Statement 7 is obtained by combining Statement 6 with the one-cell bound of part §§ 5.20 and 5.22. The other parts draw on the identity $\sum_{h'|_j=h} \mathsf{t}(h') = \mathsf{t}(h)$, which is the last line of Statement 1. This identity makes the laws of the histories of the levels $j \le K$ a consistent family of probability measures. That family is the law $\tilde{\nu}$ of part §§ 5.20 and 5.22. The bounds on large fields rest on it, and through them so do the laws at one scale. For $k \ge 1$ the observables of the theorem do not have the form $F(\mathcal{M}^k U)$. Indeed, $\mathcal{O}(f) = \sum_p f(x(p))\, a_p(U)$ is a function of the bare plaquettes, and Statement 6 cannot be applied to it. These observables are carried as sources (first block of this part). A single scale is isolated by exactness with sources at depth s:

$$\sum_{h \in \mathfrak{H}_{K-s}} \tau_h^w(\mathcal{V}_{K-s}) = Z(w),$$

whence, at $c = 0$,

$$\Big\langle \exp \sum_i z_i\, \mathcal{O}(f_i) \Big\rangle_{K,\ell,\beta} = \frac{Z(w)}{Z(0)}$$

and this quantity is an expectation, taken over the histories, the block fields and the stored variables at level $K - s$, of the exponential of the sourced exponent minus the unsourced one.

Against Bałaban's papers. Bałaban presents the operation $\mathbf{R}$ in a simplified form in the Introduction of Comm. Math. Phys. 122 (1989) 175, formula (0.3) on p. 176:

$$(\mathbf{R}\rho)(V) = \sum_Z \rho(Z'', V) \int dV|_{Z'}\, \rho(Z, V) / \int dV|_{Z'}\, \rho(Z'', V).$$

This form has the normalisation property $\int dV\, (\mathbf{R}\rho)(V) = \int dV\, \rho(V)$ ((0.4)). In the procedure as actually carried out, the denominators integrate parts of the densities that are localised near Z' (p. 177), and the normalisation property takes the form (1.102), p. 201. The objects treated in Bałaban's papers are densities. On observables, he states that his Theorem 1 "is the basis for many other applications, for example, for an analysis of expectation values of physical observables, like loop variables, averaged loop variables", and calls these problems that "deserve detailed analysis and further publication" (Comm. Math. Phys. 122 (1989) 355, p. 356).

5.20.1.2. Analytic dependence on the source; locality and decay in tame and in wild regions. **Objects.** We write $w(p) := w(x(p))$. When the source is a constant shift $w \equiv \zeta$, the increments $b_j(\zeta)$ are computed on the infinite lattice in the same way as b_j. They therefore do not depend on the torus, and they are real for real ζ. The running shift is $\zeta_k = \zeta + \sum_{j \le k} [b_j(\zeta) - b_j(0)]$, and the inverse couplings are $x_k - \zeta_k$. For a general source, let p_y denote a bare plaquette at y. Then

$$\zeta_k(p; w) = w(p) + \sum_{j \le k} \varphi_j(p) \sum_{y \in \mathbb{T}^{(j)}} h_y^{(j)}(p)\, [b_j(w(p_y)) - b_j(0)]\ :$$

in other words, the coupling counterterm of level j is applied locally, and its coefficient is taken at the local value of the source.

Proposition 5.68 (The format with sources)**.** The density ρ_k^w has the form of §4.17.1.45, with $x_k(\cdot)$ replaced by $x_k(\cdot)-\zeta_k(\cdot;w)$. Its activities $E^{(j),w}, R^{(j),w}, B^{(j),w}$ are holomorphic in (c,z), depend on w only through its values on $X^{\sim 1}$, and obey the bounds valid at $w=0$ up to constants. Moreover $|\zeta_k(p;w)| \le \frac{4}{3}\sup|w|$, and for every constant $|\zeta_*| < 2\varrho$

$$|E^{(j),w}(X,\cdot) - E^{(j),\zeta_*}(X,\cdot)| \le E_0\,\varrho^{-1}\mathsf{m}_j\, e^{-\kappa d_j(X)},$$
$$\mathsf{m}_j := \max\Big\{\sup_{X^{\sim 1}}|w-\zeta_*|,\, CML^{j-K}\,\mathrm{Lip}\,w\Big\}. \qquad \blacksquare$$

Strategy: induction on the levels, on a complex polydisc of data. On a complex polydisc of data (activities, boundary terms, shifts) centred at the real run, a single step (the cluster expansion of § 4.5 together with **R** of §4.17.1.3–§ 4.17) is local, analytic and uniformly bounded. This is the step on the complex class of §4.17.1.45. Its small-field core is the holomorphic map on local families required by the ordinary-step hypothesis. The sourced data remain inside the polydisc, so holomorphy follows by composition. Locality follows because an activity at X depends on the data, and on the outcomes of the history, only near X (locality in the history, §4.17.1.1).

The marginal term. When the source is constant, the step preserves the symmetries of § 4.2, so once $b_{k+1}(\zeta)A$ has been extracted only irrelevant terms remain. The estimate is the one of § 5.2.3. There is a remainder $\le C g_k^{\kappa_0-4}$, coming from the $R^{(j)}$, which is holomorphic in ζ on a disc of radius $C\varrho$. Apart from this remainder, the increment b_{k+1} is holomorphic in the shifts ζ_j of the earlier levels on discs of radius cx_j, where $c \le \sigma_\star$. These discs make up the polydisc of § 4.5, taken along the run's own couplings. On these discs its non-constant part, less the remainder, is bounded by $Cg_k(\log x_k)^{\mathfrak{p}_0}$. Cauchy's estimate together with $x_k \ge g^{-2} + c(K-k)$ yields

$$|b_{k+1}(\zeta) - b_{k+1}(0)| \le \mathsf{a}_{[K-k-1]}|\zeta|,$$
$$\sum_{s<K}\mathsf{a}_{[s]} \le C\sum_k [g_k^3(\log x_k)^{\mathfrak{p}_0} + \varrho^{-1}g_k^{\kappa_0-4}] \le \tfrac{1}{6},$$

where the quantity $\mathsf{a}_{[s]}$ carries a depth index and denotes, for the step that arrives at depth s, the Lipschitz number of its increment (the same letter is used in § 5.2.3 and § 5.2 and appendix A.8). Consequently $|\zeta_k'(0)-1| \le \frac{1}{6}$. On the smaller real interval $|\zeta| \le \varrho$ one obtains $\zeta_k'(\zeta) \in [\frac{2}{3}, \frac{4}{3}]$: on $|\zeta| < 2\varrho$ the holomorphic quotient $[b_j(\zeta)-b_j(0)]/\zeta$ is bounded by $\mathsf{a}_{[K-j]}$, and Cauchy's estimate on the half-disc gives $|b_j'(\zeta)| \le 2\mathsf{a}_{[K-j]}$. The sum was chosen at most $1/6$ for this fixed loss. No derivative bound throughout the full open source disc follows from the radial inequality alone. On X a general source is compared with $\zeta_* = w(p_*)$, where p_* is a bare plaquette of $X^{\sim 1}$. When $\mathsf{m}_j < \varrho$, the slope displayed above comes from the Schwarz lemma, applied along a line of data and not along a line of sources. This line is a family, holomorphic in t, of data of the level below; at $t=0$ it coincides with the data of ζ_*, and at $t=1$ it coincides on $X^{\sim 1}$ with those of w. For $|t| < \varrho/\mathsf{m}_j$ the family remains in the convex polydisc. When $\mathsf{m}_j \ge \varrho$, the slope is simply the sup bound. Subtract from each of the two activities the normaliser part $\mathbf{N}^{(j)}$ of § 4.5, which is common to both; what is left of each is at most $\frac{1}{2}E_0e^{-\kappa d_j(X)}$. The slope is thus a clause that the induction carries along. The defect estimate uses the sharper, normalizer-subtracted bound of the local data-line argument. The induction carries similar slopes for $R^{(j)}$ and $B_{\mathcal{S}}^{(j)}$, and also, among the operations treated below, for the excess exponents Ψ_ω. The slope does not follow, up to the boundary of the class of sources $\{\|w\|_\flat < 2\varrho\}$, from holomorphy in w, from the bound, from locality and from the agreement at constant sources. Moreover, $\mathsf{m}_j \le e^{d_j(X)}L^{-(K-j)}\|w\|_\flat$. The part that

no counterterm renormalises therefore carries a factor $L^{-(K-j)}$, which is summable over the scales. In short, a gradient of the source is irrelevant.

Wild regions. Two kinds of operations act on the sourced exponent. The first kind consists of the operations $\mathbf{T}_k(Z)$ on the components on which $\mathbf{R}$ has not acted. These are built from the choices of the run only, namely delta functions, tests, gauge-fixing factors and conditional Gaussian weights, and they contain no Wilson weight §4.17.1.3. On such components the source appears in the function on which these operations act, that is, in the Wilson weight and in the boundary terms. Taken on its own, $\mathbf{T}_k(Z)1$ carries no small factor: for a component produced by failed plaquette tests, this quantity is $\le \exp[C\sum_{t\le k}\#_t(Z_t\cap Z)]$. Three places carry the small factor §4.17.1.40: the Gaussian forms of the fluctuation fields, the kept gauge-fixing exponents and the Wilson weight. Only the last of these depends on the source; the first two are built from the choices of the run. The pointwise majorant of §4.17.1.40 gives one factor $e^{-cp_0(g_t)^2}$ for each creation, at the coupling of that creation's own step, and one constant for each site. It is used in the form it has at $w=0$; the sourced exponent and the unsourced one are compared further below, on the supports of the tests.

The second kind concerns a component X on which $\mathbf{R}$ has acted at some level i (whether it was healed, remained as an island, or dropped out because a test of a stage or of the main step failed). For such a component the source enters inside its operation $\mathbf{T}'_i(X)$. Except for a drop-out, this operation contains the Wilson factor of the core. When the component is healed or is an island, the operation coincides with that of Bałaban, Comm. Math. Phys. 122 (1989) 355, (1.71). Let ω run over the local histories of X, let ξ be the stored variables and $\mu_\omega\ge 0$ their measures, and let $\mathbf{p}=(\mathbf{U},\mathbf{J})$ be the pairs on $X^\sim$ at which a term evaluates the factor of X §4.17.1.40. For each ω, e^{Φ_ω} is replaced by $e^{\Phi_\omega+\Psi_\omega}$, with an *excess exponent*

$$\Psi_\omega = A\big(\hat\zeta\,\phi_{\mathrm{c}}\mathbf{1}_X, U''(\xi)\big) + [V^w - V^0](X^\sim) + \sum_{\omega'}\Psi_{\omega'}\,:$$

its summands are the Wilson term, weighted by the shift $\hat\zeta$ of the prepared regions ($|\hat\zeta|\le C\varrho$, and $0\le\phi_{\mathrm{c}}\le 1$ is the cut-off on the old core, § 4.17) and evaluated at the background $U''(\xi)$ of the inner region, which is constructed from the stored fields; the difference between the interaction terms V of $\mathbf{R}$ with and without the source; and the exponents of the inner histories ω', namely those of the swallowed islands and drop-outs. The operation modified in this way is written $\mathbf{T}'_i(X;w)$. (In the case of a drop-out, the operation §4.17.1.40 is taken to include the Wilson weight that the carrying term has over X. The operations that are kept are built from the choices of the run only, and none of their factors depends on the source, which appears in that Wilson weight and in the boundary terms. The exponent is then $\sum_{\omega'}\Psi_{\omega'}$ alone, the contribution of the swallowed islands, and it obeys the bound given below.) The exponent Ψ_ω is holomorphic in (c,z), is real for real pairs and real sources, vanishes at $w=0$, and depends on w only on $X^{\sim 2}$. The background U'' is real, and its regularity holds only at the scale on which the respective layer was stored. Where $\phi_{\mathrm{c}}\ne 0$, one has $|\partial U''-\mathbf{1}|\le C\check R_n^2\varepsilon_n\eta_n^2\le C\eta_n^2$ on $\Omega_n\setminus\Omega_{n+1}$, so the bound $a_p\le C\eta_n^4$ balances the $C\eta_n^{-4}$ plaquettes per site of level n (on the bare layer $a_p\le 2$, since G is compact). Each layer therefore contributes $O(\varrho)$ for each site of its own scale:

$$|\Psi_\omega|\le \bar f\,\#(\omega),$$
$$\bar f = C(1+\varrho),$$

where, as in §4.17.1.45, $\#(\omega)$ is the number of stored sites, each being counted at its own level. Since $\#(\omega)$ is additive over the inner histories, the constant after a step is the maximum of $\bar f$ and the constant of the new term, not their sum. Hence the constant does not increase with the level. The perturbation is extensive and must therefore be paid in advance. This is done by the

mass bound of §4.17.1.40, §4.17.1.45, which the induction maintains with a reserve weight $e^{c\mathsf{v}(\omega)}$, where $c = c_{\mathrm{r}}$ is the reserve constant of §4.17.1.45, one fixed number (the same constant stands in every weight $e^{c\mathsf{v}}$ of this part except the raised one in the bound of the whole label, whereas in the exponents that contain p_0 the letter c denotes a generic constant). Here the space-time volume $\mathsf{v}(\omega)$ counts $\log x_j$ for each stored site of level j:

$$\mathsf{M}(i,X) := \frac{1}{|W_4|}\sum_{\omega} e^{c\mathsf{v}(\omega)} \int \sup_{\mathbf{p}} \left|e^{\Phi_\omega(\xi;\mathbf{p})}\right| \mu_\omega(d\xi) \le e^{-(2-c')p_0(g_i)},$$

with the supremum over the complex pairs taken inside the integral and the sum. This form applies to a healed component and to an island. In the case of a drop-out, the integral contains the Wilson weight of the carrying term over X. The supremum is then taken over the evaluations at ξ, that is, over the real configurations which are compatible with the stored variables, are regular layer by layer, and give the listed tests their recorded values, with the exception of the test whose failure made X a drop-out; these evaluations form a set fibred over ξ §4.17.1.40. A product set contains a flat $\mathbf{U}$, at which $e^{-A} = 1$, and the supremum over such a set removes the factor. Bałaban's bound is $\mathbf{T}'_k(X)1 \le e^{-(2-c')p_0(g_k)}$ (ibid., (1.89), where Bałaban has $2(1+\beta_0)^{-1}$ in place of $2-c'$), for each fixed pair and without the weight. If g is small enough once ϱ has been fixed, then $c\log x_j \ge 2\bar{f}$. Consequently $|e^{\Psi_\omega}| \le e^{c\mathsf{v}(\omega)}$, and for every bounded function F of the stored variables and every evaluation of this kind

$$\left|\mathbf{T}'_i(X;w)F\right| = \left|\frac{1}{|W_4|}\sum_{\omega}\int F\, e^{\Phi_\omega+\Psi_\omega}\, d\mu_\omega\right| \le \sup|F|\,\mathsf{M}(i,X) \le \sup|F|\, e^{-(2-c')p_0(g_i)}.$$

Under a positive measure, the modulus of a complex factor serves as a bound for it; in the case of the small complex term produced by a complex pair, this is the device of Bałaban, Comm. Math. Phys. 122 (1989) 355, (1.73). The estimate just displayed is made on the moduli, and a sourced component retains the small factor of the unsourced one. That $\mathbf{T}'_i(X;w)F$ is holomorphic follows from dominated convergence together with the theorems of Morera and Fubini. For a healed component, the exponentiation of § 4.17 produces its local terms from this bound, each at most $e^{-p_0(g_i)}e^{-\kappa d_i(Y)}$ on a domain Y, just as at $w = 0$.

The whole label. Bounds on the individual components cannot be multiplied together, even though each is uniform in the variables outside its own component, because the background couples the components. (For $\mathbf{1}\{|\xi_1 - \xi_2| < \epsilon\}$ on $[0,T]^2$, every integral in one variable is $\le 2\epsilon$, while the double integral equals $2\epsilon T - \epsilon^2$.) By Tonelli, what does multiply is a product of functions of disjoint groups of variables; here the factors of this one product are the pieces of the pointwise majorant together with the integrands of the masses. For this reason the supremum is placed inside the integral. The result is the integrated bound of §4.17.1.40, with one factor for each creation and one mass for each outermost component on which $\mathbf{R}$ has acted. If every e^{Φ_ω} in $\mathbf{T}'_k(\mathcal{S})$ is replaced by $e^{c\mathsf{v}(\omega)}|e^{\Phi_\omega}|$, then

$$\int \mathbf{T}'_k(\mathcal{S})\Big[\chi_{k,\mathcal{S}}\, e^{-A(x_k(\cdot),U_k(\mathbf{V}))}\Big]dV_k \le \exp\Big[C\sum_{j\le k}(1+\log x_j)\,\#_j(Z_j)\Big]\prod_{(t,\mathfrak{c})} e^{-cp_0(g_t)}\prod_{(i,X)}\mathsf{M}(i,X),$$

where the first product runs over those creation cells, outside these components, that belong to a single history with the label $\mathcal{S}$, namely the history for which this product is largest (the creation cells are not determined by the label, §4.17.1.40). The second runs only over the outermost ones among the components on which $\mathbf{R}$ has acted and which remain in the label. (In §4.17.1.40 the bound carries $p_0(g_t)^2$ and has no logarithm. The sum over the labels needs only the present weaker form, because $CM^4\check{R}_t^5\log x_t \le \frac{1}{2}c\,p_0(g_t)$ (the logarithmic cost allowed in the rarity hypothesis): half of the exponent pays the cost and the other half is kept. Inside a drop-out the mass has

consumed the factors of the creations, and whatever the label still shows there is paid for by the weight. The masses are required at a weight that exceeds the $e^{c\mathsf{v}(\omega)}$ spent by the sources by a constant; this constant is determined by the per-site cost of the part of the label visible inside the drop-out, and its choice comes after that of ϱ and before the ceiling on g is set (the mass bound is valid at every weight $c \geq 0$, §4.17.1.40, and the masses $\mathsf{M}(i, X)$ in the display are understood at the raised weight). The surplus comes from that part of $\mathsf{M}(i, X)$ which is made up of the local histories admitted by the label; on these, $\mathsf{v}(\omega)$ is bounded below by the volume shown §4.17.1.40, §4.17.1.45.) Once its vacuum constant is set aside, the sourced term of the label is dominated by the left-hand side multiplied by $e^{C(1+\varrho)}$ for each bond of the determining set, for three reasons. The moduli are taken inside the integrals. In the operations, $|e^{\Psi_\omega}| \leq e^{c\mathsf{v}(\omega)}$. On the supports of the tests, what remains of the sourced exponent once the Wilson term $-A(x_k(\cdot), U_k)$ and the vacuum constant are set aside is at most $C(1 + \varrho)$ for each bond of the determining set § 5.20. Consider the weighted sum over the labels of §4.17.1.45, that is, the total masses summed over $\mathcal{S}$, where the credit pays the costs. In it every component C of Z_k keeps a factor

$$\exp\big[-cp_0(g_k) - \kappa_1 |C|_k\big],$$

where $|C|_k$ is measured in cells. In §4.17.1.40 the factor is taken per site, with $\#_k(C) \geq d_k(C)$ in place of $|C|_k$. This contains the form "with the additional term $-\kappa_1 d_k(X)$", which Bałaban gives for a general region (ibid., p. 387).

The cost of the reserve. A factor $e^{c \log x_j}$ for each site amounts to at most $e^{cM^4 \check{R}_j^5}$ for each cell. Its form is that of the volume costs $CM^4 \check{R}_n^5$ per cell and per step, which, until $\mathbf{R}$ acts on a component, are covered by the credit $\kappa_j(Z)$ of that component: an amount $p_0(g_j)^2$ for each cube, deposited at its creation, pays for $\check{R}_j^6$ for each cube (ibid., (1.79)–(1.81), in which both the deposit and the cost are measured by the linear size $d'_j(Z) + 1$, counted in MR_j-cubes, and (1.89); these bound $\mathbf{T}'_k(X)1$ for the operation of (1.71), which contains the Wilson factor). Bałaban already pays $O(1) \log g_j^{-2}$ for each site of Z_j, on account of the normalisation constants (ibid., p. 380). The spare weight, whose constant c is free, holds once g is small enough, the constant c being chosen first §4.17.1.40.

Proposition 5.69 (Sources along a frame)**.** Let $(\theta_j)_{j \leq n}$ be a frame such that $\theta_j \geq \gamma_J^{-2}$ throughout, and let (x_j) be the run along it starting from a real β, as in the frame setting of § 5.2. The cube sizes are $\check{R}_j = R(\min_{i \leq j} \theta_i)$. All the other auxiliary choices of level j (the reference laws, the thresholds, the gauge-fixing weight and the scaling of the fluctuation field) are made with θ_j, whereas the weight of the action obeys its own recursion. The reference line is a frame. A first-passage run, that is, the run starting from a first-passage datum (K, β), is the run taken along its own couplings. The run is said to stay in the tube if $|\theta_j - x_j| \leq \mathsf{t}_j$ for all j, and to have room r if $|\theta_j - x_j| + r \leq \mathsf{t}_j$. Suppose that the run has room $2r$ for $r = \varrho$ or for $r = 2\varrho$. Then, for all sources with $\|w\|_\flat < r$ and for the steps of the frame, Props. 5.63, 5.65, 5.67 and 5.68 are valid, with upper stability and the floor holding with $C(1 + \log \theta_n)$ for each site of $\mathbb{T}^{(n)}$. They hold on every admissible pair (ℓ, K) of a torus and a cutoff, admissibility meaning $K \geq n$ and $\ell L^{K-i} \geq CM\check{R}_i$ for $i \leq n$, as in § 5.2.2; all of this is uniform in ℓ, in K, in n, in β and in the frame. Moreover

$$\mathsf{z}_j(\zeta) := (\theta_j - x_j) + \zeta_j(\zeta) \in D(0, \mathsf{t}_j),$$

$$\sum_{j \leq n} \sup_{0 < |\zeta| < r} \frac{|b_j(\zeta) - b_j(0)|}{|\zeta|} \leq \tfrac{1}{3}. \qquad \blacksquare$$

The mechanism is as follows. A step depends separately on the choice θ_j and on the weight $x_j - \zeta_j$. For given remainders, its small-field part is holomorphic in their difference $\mathsf{z}_j(\zeta)$ on $D(0, \sigma_\star \theta_j) \supset D(0, \mathsf{t}_j)$ (the small-field holomorphy hypothesis in the ordinary step, on the disc

stated here); this difference is the deviation $\mathsf{z}_j = \theta_j - x_j$ displaced by the running shift. The full step is formed at a real deviation in the tube, together with a complex shift about it of modulus at most $\bar{c}$ (the complex class of §4.17.1.45). Consequently, as far as the increments are concerned and with the remainders $R^{(i)}$ held fixed, a deviation from the frame and a constant source are the same variable. The remainders depend on the source as well, and the activities in the two presentations of Prop. 5.67 need not coincide term by term. The room is what keeps the complex excursion $|\zeta_j| \le \frac{4}{3}|\zeta|$ inside the disc. The objects constructed do not depend on the tube. A single statement therefore serves the first-passage runs, the comparison of two cutoffs in appendices A.8 and A.9, and the runs with frozen choices that occur in § 5.2.3 and Prop. A.15; it is used with $r = \varrho$, where the room is 2ϱ, and with $r = 2\varrho$, where the room is 4ϱ (which is the room $C\varrho$ of Prop. 5.63).

Reading at depth s. For the run stopped at level $K-s$, everything above holds with the following changes. The norm $\|f\|_{\sharp,s} = \max\{\|f\|_\infty, CML^{-s}\,\mathrm{Lip}\,f\}$ replaces $\|f\|_\sharp$, the Lipschitz constant being measured at scale L^{-s}. The sums are restricted to $j \le K-s$. Either normaliser may be used, each with exponents of its own: E of the whole run, or the vacuum constant $E_{(K-s)}$ of the first $K-s$ steps. The number of sites of $\mathbb{T}^{(K-s)}$ replaces ℓ^4, and $L^{-(K-s-j)}$ replaces $L^{-(K-j)}$. The exponents of the two global bounds also depend on s §6.1.1.2.

Proposition 5.70 (Large fields are rare uniformly in the volume)**.** (Live regions, creation sets, lineages, marks and ages, as well as the law $\tilde{\nu}$ of the histories, are defined in §4.17.1.1, and the prepaid bounds are stated in §4.17.1.45. They have here the meaning given there.) For g small, for every level $0 \le k \le K$ (with $\mathcal{Z}_0 = \emptyset$) and every level-$k$ cell $\mathfrak{c}$, one has uniformly

$$\tilde{\nu}\big(\mathfrak{lv}_k(\mathfrak{c})\big) = \tilde{\nu}\big(\mathfrak{c} \subset \mathcal{Z}_k\big) \le Ce^{-cp_0(g_k)},$$

$$\tilde{\nu}\big(\mathfrak{c} \text{ is created at step } k\big) = \tilde{\nu}\big(\mathfrak{c} \subset \mathcal{W}_k^{\wedge}\big) \le Ce^{-cp_0(g_k)}. \qquad \blacksquare$$

The second bound ($k \ge 1$) is a consequence of the first, because creation cells are live. Indeed, $\mathbf{R}$ touches a component only if no creation has occurred in it during the last N_k steps. Hence a component of Z_k containing a creation cell of step k is not healed at step k, which gives $\mathcal{W}_k^{\wedge} \subset \mathcal{Z}_k$. The same inclusion also bounds the event that $\mathfrak{c}$ lies in a live component in which a creation occurs at step k. The theorem relies on the following point. A label's determining set starts from a neighbourhood of its first region, $\Omega_0 = \Omega_1^{\sim}$, and not from the whole torus. Its background therefore requires the bare field to be regular only on that neighbourhood (see under Lem. 5.72). If regularity is required at every bare plaquette of T_ℓ, then a single rough plaquette, wherever it lies, leaves the step-1 background undefined. By totality §4.17.1.1, every test formed with that background then fails at every place, so that Z_1 equals the whole torus and is never healed. Moreover, at fixed (β, g), the probability that such a plaquette is present tends to 1 as $\ell \to \infty$. The theorem concerns only the law of the histories of the run; it gives no bound on the law that $\mu_{K,\ell,\beta}$ induces on the bare plaquettes inside a box of fixed physical size.

The proof is a first-moment argument. For a level-n cell $\mathfrak{z}$, let $\mathcal{E}_{\mathfrak{z}}^{(n)}(k)$ be the survival event of §4.17.1.1, namely $\mathfrak{z} \subset \mathcal{Z}_i$ for $n \le i \le k$. The event $\mathcal{E}_{\mathfrak{c}}^{(k)}(K) \subset \mathfrak{lv}_k(\mathfrak{c})$ is the never-healed event. (i) If healing takes place at a level $j > k$, the relative removal comparison of §5.20.2.6 gives the factor $Ce^{-cp_0(g_k)-cp_0(g_j)}$, summable in j. This use requires the outer-operation comparison on the actual completed positive terms. (ii) We use Vol_i, Dep_i, the mark $\Theta^{(i)}$ with its dials w, c_a, and the end mark $\Theta = \Theta^{(K)}$, all as in §4.17.1.1. For a component live at level i, along its lineage Vol_i charges every live cell, at every level $j \le i$, with twice the weight $\mathsf{r}_j = M^4\check{R}_j^5$, while Dep_i contributes $cp_0(g_j)^2(d'_j(W)+1)$ for every component W that is created. By §4.17.1.45, the upper bound of § 5.20 stays valid once the mark is prepaid: for the run stopped at an arbitrary level when the dial is $\mathsf{w} = 0$, and at the last level when the dial is $\mathsf{w} > 0$. In the form needed here, the prepaid bound takes up the large-field bound one lineage at a time and consists of two statements,

independent of each other. *Volumes are paid by events.* Let $\mathsf{r}_j = M^4 \check{R}_j^5$. For every history, and for every component X of Z_k, whether live or healed at step k, one has

$$\sum_{(j,C)} \mathsf{r}_j \, |C|_j \le c_G \sum_{(j,C)} \mathsf{r}_j \check{R}_j \, e(j,C),$$

$$e(j,C) := \big(|\operatorname{pred}| - 1\big)^+ + |\operatorname{pred} \cap \operatorname{Act}_{j-1}| + \sum_W \big(d'_j(W) + 1\big),$$

where both sums run over the components of the lineage of X. Here pred denotes the set of components live at level $j-1$ that are contained in C; Act_{j-1} is the set of those components that are live even though $\mathbf{R}$ of level $j-1$ has acted on them; W ranges over those components of the step-j creation set that lie in C; and the constant c_G is determined by L, M and the remaining constants fixed ahead of the ceiling γ imposed on the couplings, and by nothing else. The coupling $\gamma_\star$ is taken small. The inequality is a statement on the geometry of the regions alone; neither a field nor a measure enters. At levels where nothing happens, a component, measured in cells of its own level, shrinks geometrically until it becomes small; and once a component has stayed small over $\check{R}_j + 1$ levels without any event, it belongs to the class of $\mathbf{R}$. *Masses by lineage traces.* The lineage trace $\mathfrak{l}$ of X is the list of the regions $\mathsf{Z}_j(X)$, of the sets created in them, and of the sets Act_j. We set

$$\operatorname{wt}(\mathfrak{l}) := \exp\big[C_{\mathrm{chg}} \operatorname{Vol}_k - \operatorname{Dep}_k - \sum_{(j,W)} 2p_0(g_j) - \sum_j |\operatorname{Act}_j| \, p_0(g_j)\big]$$

, where C_{chg} is the charge constant; it is defined by the requirement that, for each cell of level j, the sums over the admissible regions, the Gaussian integrals, the normalisation constants and the vacuum-energy counterterms contribute a factor no larger than $e^{2C_{\mathrm{chg}}\mathsf{r}_j}$. This constant is determined by the parameters of the construction alone, and it is fixed before $\gamma_\star$. Let $\mathscr{L}$ be any set of lineage traces of level k, and sum over those histories of level k whose live components carry the lineage traces $\mathscr{L}$. Then

$$e^{-E_{(k)}} \sum_h \mathsf{t}(h) \le e^{C|\mathbb{T}^{(k)}|} \prod_{\mathfrak{l} \in \mathscr{L}} \operatorname{wt}(\mathfrak{l}) :$$

indeed, one unfolds the majorants of §4.17.1.40, passing through the components that $\mathbf{R}$ has acted on, as far as the creations, and one sums the entropy with the lineage trace held fixed. No factor arises from a joining, since a creation at a leaf of the lineage pays for it (Prop. 4.63, move (3)). The relation $\mathsf{r}_j \check{R}_j = M^4 \check{R}_j^6 \ll p_0(g_j)$ is valid for all couplings not exceeding the anchor coupling $\gamma_\star$. This anchor coupling is fixed small only once L, M, the dials w and c_a, and the rate c' introduced below (a constant of this proof only, different from the small c' of the mass bound) have been fixed, and it does not exceed γ_J; g is subsequently taken small depending on $\gamma_\star$. Consequently the volumes, the entropy of the lineage traces and the mark are all paid for by the events. Precisely, take the rate $c' \ge 0$, which is fixed before $\gamma_\star$ and may be given any value; the value $c' = 1$ already outweighs the number $C^{|X|_k}$ of connected sets X of cells containing a given cell, because $\mathsf{r}_k \ge M^4$. Then for every set of lineage traces ending in X,

$$\sum_{\mathfrak{l}} \operatorname{wt}(\mathfrak{l}) \, e^{\mathsf{w} \operatorname{Dep}_k + c_a \operatorname{Vol}_k} \le e^{-c' \mathsf{r}_k |X|_k} \, e^{-2 \min p_0(g_{j_0})}$$

, where j_0 denotes the first level of a lineage. It is this bound that prepays the mark. There are two reasons for carrying out the argument at the last level, with the end mark. The first reason applies already at $\mathsf{w} = 0$. The gain obtained from the first moment in (iii) is the quotient of two

quantities: the exponent in the moment bound, equal to $\mathsf{B}(y) = C(1 + \log y)$ for every site of the lattice on which the run stops, and the product of $c_a \mathsf{r}_k$ with the count of level-k cells. When the run is stopped at level k, the quotient equals $\mathsf{B}(x_k)(M\check{R}_k)^4/(c_a\mathsf{r}_k) \approx \log x_k/(c_a\check{R}_k)$, and this carries no smallness of order $e^{-cp_0(g_k)}$. At the last level there are fewer sites, by the factor $L^{-4(K-k)}$. The second reason concerns the alternative of (iv) in which the torus shrinks. There a single deposit is weighed against the volume, and this requires $\mathsf{w} > 0$. Whatever value the dial takes, the end mark registers only lineages of components still live at the last level, that is, only the never-healed lineages. This is why the healed part is split off in (i). Together with the floor one obtains $\mathbb{E}_{\tilde{\nu}} e^{\Theta} \le e^{C_g \ell^4}$, hence $\mathbb{E}_{\tilde{\nu}}\Theta \le C_g \ell^4$ by Jensen's inequality. On the other hand Vol_K charges every never-healed cell, so $\sum_{\mathfrak{c}} \mathbf{1}_{\mathcal{E}_{\mathfrak{c}}^{(k)}(K)} \le \Theta/(c_a \mathsf{r}_k)$, where $c_a > 0$. (iii) Translations by level-k cells commute with the steps up to level k; therefore $\tilde{\nu}(\mathfrak{w}_k(\mathfrak{c}))$ is the same for every $\mathfrak{c}$. The event $\mathcal{E}_{\mathfrak{c}}^{(k)}(K)$, by contrast, depends on coarser partitions and is not covariant. Averaging gives

$$\tilde{\nu}\big(\mathfrak{w}_k(\mathfrak{c})\big) \le \max_{\mathfrak{c}'} \tilde{\nu}\big(\mathfrak{w}_k(\mathfrak{c}') \setminus \mathcal{E}_{\mathfrak{c}'}^{(k)}(K)\big) + \frac{C_g \ell^4}{c_a \mathsf{r}_k \, \#\{\mathfrak{c}'\}} \le C e^{-cp_0(g_k)} + C_g L^{-4(K-k)}.$$

(iv) To make $K - k$ large, the run is extended on the same bare lattice up to a larger reference coupling, not exceeding the anchor coupling $\gamma_\star$. This is a fixed upper limit for the reference couplings. It is chosen small once the dials w and c_a are fixed, in such a way that (i), (ii) and the comparison made in the first alternative below are valid at each reference coupling up to $\gamma_\star$; only afterwards is g chosen small, depending on $\gamma_\star$. This continuation leaves the law up to level k unchanged § 5.2.4. Steps (i)–(iii) are applied to the continued run, with Θ the mark at its last level.

Let $K^\dagger$ be the largest admissible continued first-passage level, $g^\dagger$ its reference coupling, and $m^\dagger = m - (K^\dagger - K)$ its torus exponent. Write $K_\star$ for first passage at the anchor coupling and $X_\star = \gamma_\star^{-2}$. Choose $c_{\mathrm{gap}} = \lceil c_5/\lambda_\sharp \rceil + 1$. If $K^\dagger + c_{\mathrm{gap}} \le K_\star$, the lower flow bound produces a strict new record low within those c_{gap} steps. A strict record low is a first-passage level for a reference threshold there. For a record q, use $g' = (x_q - B_\sharp)^{-1/2}$, or the anchor if $x_q - B_\sharp < X_\star$, in which case $q = K_\star$. The new reference coupling is at least $g^\dagger$ and at most the anchor. The admissible side floor $m_\star(g')$ is nonincreasing with g'. Thus the record would still be admissible if $m^\dagger \ge m_\star(g^\dagger) + c_{\mathrm{gap}}$. Maximality gives the opposite inequality. The terminal site count is then at most $C(1+\log(g^\dagger)^{-2})^{4r}$ by the side floor. Every surviving lineage carries a birth, with mark at least $\mathsf{w} c_{\mathsf{D}}(p_0(g_k) - 1)^2$ by the drift. For the chosen anchor ceiling this mark pays the terminal moment exponent and leaves $cp_0(g_k)$, since $2\mathfrak{p}_0 > 4r + 1$.

Otherwise $K^\dagger > K_\star - c_{\mathrm{gap}}$. The upper flow bound and the first-passage windows give

$$p^\dagger := K^\dagger - K > \frac{g^{-2} - X_\star - B_\sharp}{B_\sharp} - c_{\mathrm{gap}}.$$

The continued exponent $m^\dagger = m - p^\dagger$ is nonnegative. In particular every side with $m \le m_V(g) = \lfloor (g^{-2} - X_\star - B_\sharp)/B_\sharp \rfloor - c_{\mathrm{gap}}$ falls in the volume case. This gives the horizon $\ell_V(g) = 2L^{m_V(g)}$. Use the averaged never-healed bound at this terminal level. Writing $u = K - k$, division by the number of level-k cells gives

$$\frac{\mathsf{B}(x_{K^\dagger})}{2c_a \check{R}_k} L^{-4(p^\dagger + u)}.$$

Here $x_{K^\dagger} \le g^{-2} + B_\sharp$ while $x_k \le (g^{-2} + B_\sharp)(1 + u)$. Consequently the required extra depth is at most $C(1 + \log g^{-2})^{\mathfrak{p}_0}$, uniformly in u: the u-dependent part of $cp_0(g_k)$ is bounded by $4u \log L + C$ because $(\log(1+u))^{\mathfrak{p}_0} = o(u)$. The available depth $p^\dagger$ exceeds this bound for small g. The average is thus at most $Ce^{-cp_0(g_k)}$. Combine it with the pointwise healed estimate by averaging the live-prefix

events, whose probabilities are translation invariant at level k. This uses the common prefix of the two runs.

Proposition 5.71 (Ages)**.** Denote by C the component live at level k that contains $\mathfrak{c}$, by $\mathsf{Z}_j(C)$ its lineage regions, and by j_0 the least level at which these regions are non-empty. The three ages of C are the age read in the label, the age of the lineage, $\mathrm{age}_k(C;h) = k - j_0$, and the seed age. These ages and the events $\mathfrak{lw}_k^{=a}(\mathfrak{c})$, $\mathfrak{lw}_k^{\geq a}(\mathfrak{c})$, $\mathfrak{sd}_k^{=a}(\mathfrak{c})$ are defined as in §4.17.1.1. The age read in the label refers to the oldest region of the label that appears in C: $\mathrm{age}^{\mathrm{lab}}(C) = k - \min\{t : C \cap Z_t \neq \emptyset\}$, where $Z_t = \Lambda_t^c$ is the large-field region of scale t in the label $\mathcal{S}(h)$ of the history. The proposition states that the three ages are ordered,

$$\mathrm{age}^{\mathrm{lab}}(C) \;\leq\; \mathrm{age}_k(C;h) \;\leq\; \text{seed age},$$

and it gives a decay in the second of them. For the first inequality, note that every point of $C \cap Z_t$ belongs to the live region at each level between t and k. It therefore belongs to $\mathsf{Z}_t(C)$, and so $t \geq j_0$. For the second inequality, no component of $\mathsf{Z}_{j_0}(C)$ contains a component that is live at the preceding level. Each such component is therefore created at step j_0, and it is contained in C. The second inequality may be strict. Suppose a region created at a step $t < j_0$ is contained in C. Then it was healed at some step before k, and the live region of some later level covers its place anew. As a point set it counts for the seed level. It contributes no factor, however, since no cell of it stays live through all the levels from t to k. In terms of events, $\mathfrak{sd}_k^{=a}(\mathfrak{c}) \subset \bigcup_{a' \leq a} \mathfrak{lw}_k^{=a'}(\mathfrak{c})$. Let $b \geq 1$, let $c_a \geq C(b)\log L$ (a floor on c_a, which is chosen after L), and let g be small depending on b and c_a. Then, uniformly, with constants C on the right-hand sides that may depend on b and c_a,

$$\tilde{\nu}\big(\mathfrak{lw}_k^{=a}(\mathfrak{c})\big) \leq C e^{-c p_0(g_k)} (2L^b)^{-a},$$

$$\tilde{\nu}\big(\mathfrak{lw}_k^{=a}(\mathfrak{c}) \cap \mathcal{E}_{\mathfrak{c}}^{(k)}(K)\big) \leq C (2L^b)^{-a-(K-k)}.$$

Both bounds decay in the age of the lineage, and so does the age series given further below. Suppose the seed level is formed only from regions that are healed at no step before k. Then the two bounds and the age series hold in the seed age as well, that is, for $\mathfrak{sd}_k^{=a}(\mathfrak{c})$: the seed level equals j_0, the two ages coincide, and $\mathfrak{sd}_k^{=a}(\mathfrak{c}) = \mathfrak{lw}_k^{=a}(\mathfrak{c})$. When the seed level is formed from all regions, the price of survival yields no decay. Two facts show this. First, a level-k cell contains of the order of $L^{4(k-t)}$ cells of level t, whereas the bound per creation has an exponent that is merely polylogarithmic in $k - t$. Second, take $k = K$ and $a = K - 1$. Applied there, the bound gives at most $C(2L^b)^{-(K-1)}$, at fixed g and for every K, for the probability that a fixed cell at the final level is live and occupies the place of a region created at step 1. Yet the tests of the last step depend only on $x_{K-1}, x_K \approx g^{-2}$. A first-moment argument yields only a density, and never a factor per level for a single cell. For this reason the chessboard estimate is now applied to the large fields, which is a second use of reflection positivity.

■

Lemma 5.72 (Chessboard for the law of the histories)**.** Put $q_a = e^{-c_a}$, where $c_a \geq c^{-1}\log(2L^4)$, equivalently $L^4 q_a^c \leq \frac{1}{2}$. In this condition, in the two bounds below and wherever these bounds are quoted, the constant is $c = \frac{1}{16}$; it is fixed before c_a, and it arises as the root, taken over the tiles, of the bound for a pattern that is repeated in one tile out of 2^4. Fix $\epsilon > 0$ and take g small depending on c_a, ϵ. A *tiling* is a translate, by a multiple of the side length of a level-k cell, of the partition of the torus into cubes of side $\mathsf{l}_\square$, called *tiles*, whose faces lie in mirror planes that pass through midpoints of bonds. In each tile, the cube with the same centre and with side $\frac{3}{4}\mathsf{l}_\square$ is called its *core*. A cell, of whatever level, is called *deep* in a tile when the level-k cell containing it lies in the core of that tile. Then for each level $k \leq K$ there is a tile side $\mathsf{l}_\square \approx M\check{R}_k$, measured in level-$k$ units, which

is a power of L and spans at least sixteen level-k cells, with the following property. Let $n \le k$, let $\mathfrak{z}$ be a level-n cell situated deep inside a tile $\square_0$ of a tiling, and let $\square_1, \dots, \square_r$ be further tiles, all distinct. Then

$$\tilde{\nu}\big(\mathcal{E}^{(n)}_{\mathfrak{z}}(k)\big) \le \mathsf{l}_{\square}^{-4}\, q_a^{c(k-n+1)},$$

$$\sum_{\mathfrak{c}_1,\dots,\mathfrak{c}_r} \tilde{\nu}\Big(\mathcal{E}^{(n)}_{\mathfrak{z}}(k) \cap \bigcap_{i\le r} \mathfrak{fv}_k(\mathfrak{c}_i)\Big) \le \mathsf{l}_{\square}^{-4}\, q_a^{c(k-n+1)}\, \epsilon^r,$$

where the sum runs over the level-k cells $\mathfrak{c}_i$ lying deep inside $\square_i$. The first bound is valid for every level-n cell on every torus, including tori too small to carry a tiling with tiles of this side. The bounds are independent of how many tiles the torus contains. Survival costs a factor q_a^c per level, and every further tile that contains a deep live cell costs a factor ϵ. ■

5.20.2. *Radial windows and bare-field witnesses.* This section supplies the conditional-density and witness steps in Lem. 5.72. It concerns the actual positive law of a finite history prefix, at zero source. The all-history hypotheses $\mathcal{H}_{\mathrm{step}}$ of §3.1.1.1 remain in force. In particular, the background and Gaussian estimates below are required on the actual determining regions and stored-field fibres. A regular homogeneous construction does not supply these hypotheses on its own.

The collar cutoff used here is the component-local cutoff specified in §4.17.1.3. Its exponentially close global variant can support analytic estimates, but it cannot replace this sharp cutoff in a comparison of two runs. With the local cutoff, the comparison needs windows only at the ordinary split of each step. No window at a preparatory completion or collar test will be removed.

5.20.2.1. The finite prefix and its real estimates. Fix k, and measure lengths in level-k units. Write

$$s_j = M\check{R}_j L^{-(k-j)}, \qquad r_j = M^4\check{R}_j^5, \qquad N_j = \check{R}_j, \qquad q_a = e^{-c_a}, \qquad \widehat{q} = q_a^{1/16}.$$

The cells are nested, $s_j/s_{j-1} \in \{1, L\}$, and the flow satisfies

$$c(j-i)-C \le x_i - x_j \le C(j-i)+C, \qquad \check{R}_i = R\Big(\min_{t\le i} x_t\Big), \qquad (\log x)^r \le R(x) \le L\max\{1, (\log x)^r\}.$$

All constants below may depend on the fixed construction parameters, including L, but not on the depth, the torus or the chosen history.

We record the real estimates that enter the argument. These are formation and mass estimates, rather than the class-invariance target. The background response on a regular collar has the form

$$\big|V^{\mathrm{bg}}(b)V^{\mathrm{bg}\prime}(b)^{-1} - 1\big| \le C_{\mathrm{dec}} e^{-\delta_{\mathrm{dec}} d(b,A^c)} \varepsilon_{\mathrm{rim}}. \tag{5.cdxii}$$

The two determining sets and their data must coincide in A, its rim must lie in the common top domain, and the actual data must lie below the regularity ceiling. Near the top domain, $\varepsilon_{\mathrm{rim}} \le C\varepsilon_j$. A factor $C(\log x_j)^r$ on rebuilt distant layers is allowed; two additional cells of collar pay for it. The first determining region is $\Omega_0 = \Omega_1^{\sim}$, rather than the whole torus.

In the literal nonpivot coordinates at a split, the smooth real exponent satisfies

$$|\partial_{a_b}\mathcal{F}| \le C_{\mathrm{sl}} p_0(g_j). \tag{5.cdxiii}$$

This estimate includes the Haar, coarea and nonlinear terms. Its quadratic contribution follows from an exponentially summable precision row applied to fields bounded by Cp_0. For the ordinary remainder, the domains through the tested bond have a bounded rooted sum with the ordinary decay. A complex margin $c/p_0(g_j)$ in the scaled coordinate thus suffices by Cauchy's inequality. For the preparatory remainder, use the coordinate $a_b = (\mathrm{i}g_i)^{-1}\log(V_i(b)C_b^{-1})$ at its actual collar level i, with C_b its test centre. The stage domains meeting the tested bond also meet the preparatory collar. Their relative length does not charge the class box of side $O(n_{\mathrm{sm}}\check{R}_j)$ cells. Rooting at one of its cells and summing the exterior continuations therefore costs $C(n_{\mathrm{sm}}\check{R}_j)^4$. The complex margin in this collar coordinate is $c\delta_i'/g_i$, bounded below by a fixed constant. Cauchy's inequality with

the formed activity bounds on that disc gives a contribution bounded by $C(n_{\mathrm{sm}}\check{R}_j)^4 g_i/\delta_i'$. The waiting-window bounds compare the scales, and $\mathfrak{p}_0 > 4r$ makes this at most $Cp_0(g_j)$. This uses the admitted collar margin with the relative rooted sum, and the ordinary margin with the ordinary rooted sum.

For an isolated box of free coordinates, write its conditional exponent as

$$H_e(y) = \tfrac{1}{2}\langle y, K_{II} y\rangle + \langle y, K_{IO} A_O\rangle - P_e(y). \tag{5.cdxiv}$$

Here I is the set of free bonds touching e^{+1}, whereas the marked bonds touch e. The real Gaussian estimates and nonlinear formation bounds give

$$K \geq a_- I, \qquad |K(b,b')| \leq a_+ e^{-c|b-b'|}, \qquad |A_O| \leq Cp_0(g_{j-1}), \tag{5.cdxv}$$

$$\sup_I |\nabla P_e| \leq \epsilon_{\mathrm{nl}}, \qquad \|\nabla^2 P_e\|_{\ell^2\to\ell^2} \leq a_-/2, \qquad \epsilon_{\mathrm{nl}} \longrightarrow 0. \tag{5.cdxvi}$$

Distances in these formulas are in level-$(j-1)$ lattice units. The bounds are needed after conditioning on the exterior datum. Positivity of Haar measure alone supplies none of them.

Finally, let $t(h) \geq 0$ be the masses of the prefix and $Z = \sum_h t(h)$. The marked upper bound and the all-small lower bound, with the same vacuum constant E_k, are

$$e^{-E_k}\sum_h t(h) e^{\Theta_k(h)} \leq e^{C_+ V_k}, \qquad e^{-E_k} Z \geq e^{-C_-(x_k)V_k}, \qquad C_+ + C_-(x) \leq B(x) := C_B(1+\log x). \tag{5.cdxvii}$$

Here $V_k = |\mathbb{T}^{(k)}|$, and

$$\Theta_k = c_a \sum_{X \text{ live at } k} \mathrm{Vol}(X), \qquad \mathrm{Vol}(X) = 2\sum_{j\leq k} r_j |\mathsf{Z}_j(X)|_j.$$

The charge dial in the mass estimate has been increased before choosing the coupling ceiling. Dividing the two inequalities in eq. 5.cdxvii cancels E_k and gives

$$\mathbb{E}_{\widetilde{\nu}} e^{\Theta_k} \leq e^{B(x_k)V_k}. \tag{5.cdxviii}$$

In particular, if n_j distinct level-j cells are forced into surviving lineages, their probability is at most $e^{B(x_k)V_k} q_a^{2\sum_j r_j n_j}$.

5.20.2.2. The coordinates read after a split. The independent split coordinates are the actual nonpivot link logs

$$a_b = g_i^{-1} i^{-1} \log\left(V_i(b) V_i^{\mathrm{bg}}(b)^{-1}\right), \qquad i = j-1.$$

They are not coordinates obtained by a global nonlinear Landau projection. The next field V_j is held fixed, and the literal averaging constraint determines the pivots.

Lemma 5.73 (The split coordinate and the later decisions)**.** Fix all other stage coordinates. On either one-sided shell about any of the three split thresholds, the conditional likelihood of a joint survival event and of later ordinary-split no-window conditions is constant along each radial ray. Existing sharp factors are constant there as well. This statement uses the actual outward determining rows and the margins of the ordinary step. ■

Proof. We describe the readsets, including the pivots. In a single averaging step the pivot for a coarse bond is the middle fine bond of its central line. Internal block contours contain no pivot. A pivot therefore occurs in its own coarse constraint only. The local implicit-function theorem solves that constraint, with the other nonpivot variables held fixed. Changing a_b changes only b and the face pivots in its one-step contour carrier. All constants in this finite-dimensional calculation are C_L; no lower bound uniform in L is being asserted.

The ordinary split reads cubes in $\Omega_j^{\sim -1}$. Their full one-step carriers stay inside Ω_j, with a cell margin. Consequently none of the changed bonds belongs to the actual row Γ_i, whose bonds have no endpoint in Ω_j. The background datum, the coarser field, and the older retained data are

unchanged. The same is true of data used by joined strip backgrounds: their additional tower data are coarse lifts, and their inherited fine rows are these same outward rows. Thus other stored nonpivot coordinates acquire no hidden change through a background centre.

The plaquette tests depend on V_j. The approximate fluctuation tests exclude pivots, use the threshold $2\delta_i$, and have exponentially small centre offsets. They remain constant throughout the shells about $\delta_i^{\flat}, \theta_c\delta_i, \delta_i$. The soft contour cutoffs are identically one there, since their contours remain within $C_L\delta_i$ and the parameter order gives $C_L\delta_i \ll \varepsilon_i$. For the density tests one needs the quantitative new-test implication in Lem. 4.57, not merely their previously recorded answer. With the new coarse plaquette and approximate fluctuation tests passed, that implication gives, on each old density cube in the one-cell enlargement,

$$|\partial U_{i,\square}(V_i) - 1| \le \left(C_0 L^{-2} + C_L A_1/A_0 + Ce^{-c\delta_0 M_2 \check{R}_j}\right)\varepsilon_i\eta_i^2 \le \tfrac{1}{4}\varepsilon_i\eta_i^2.$$

The construction fixes these quantitative margins in this order: the coarse-curvature floor on L, the sufficiently small ratio A_1/A_0 after the finite-L response and pivot constants, and the coupling ceiling after the collar constants. The shell enlarges the fluctuation threshold by $g_i W_\sigma = o(\delta_i)$, so the same bound remains below $\varepsilon_i\eta_i^2/2$. Here η_i is the bare spacing in level-i units. At $i = 0$, $U_{0,\square}(V_0) = V_0$ on the tested plaquettes, so these are the bare P_0 tests on the full changed carrier. Thus fixing the earlier P_0 entry imposes no additional shell cut. Old tests near a changed pivot are either removed in the strip operation or identically one by its one-cell margin.

Now compare two states on the same ray, on the same side of a threshold. All three interval flags agree. The ordinary recursion uses only these flags, so its geometry, its creation entries and its integration region agree. If the coordinate is integrated, its slot is deleted and no subsequent move reads it.

If it is retained, it is still excluded from all later determining data: the old domains only expand. Preparatory gauge reads lie in the outer collars and use these datum rows and fixed coarser fields. The other preparatory tests use backgrounds localised to the actual component. A completion test can read the stored coordinate itself, but then it tests the same previously stored modulus at the old threshold. At a passed original cube test that answer is already one. A failed original cube test is a creation; while its level remains in the preparatory window, the class condition prohibits acting on its component. Once it leaves that window, its old coordinate is below the levels of those completion tests. No recomputation with a new centre is permitted.

The remaining moves are fixed-law gauge or Haar draws, a frame draw, coarse tree maps, or the coarse reference draw. The reference draw and its normalizer read the exterior coarse datum and the fixed component frame; they read no omitted fine row. Their read coordinates have already been shown to agree. Its final density test therefore has the same answer when the later coins agree. The old retained values can remain as passengers; the assertion is about future decisions, not about a full state kernel that carries such passengers unchanged.

Induction through the finite list of moves proves that the indicated future events have equal indicators for equal later coins. Integration over those independent coins gives the asserted constant likelihood. This argument would fail if a retained fine staple were admitted into a later datum row. Fixed coarse means by themselves would not exclude that possibility. □

5.20.2.3. A deterministic family of split tests. At step j, after Ω_j and its single global fluctuation field have been formed, record the three interval flags for every candidate nonpivot bond read by a cube in $\Omega_j^{\sim -1}$. Then run the original recursion using these flags. It uses exactly the flags it would have queried before. Forgetting the unused flags recovers the original history at every configuration, so this refinement changes neither its law nor the moment eq. 5.cdxviii. It introduces no new coin and no sum over possible recursion branches.

This precomputation is legitimate because the split uses one global field, not a field recentered at a hypothetical explored component. No such refinement is made for the context-dependent

preparatory collar cutoffs. Order the diagnostic split sites by step and then by a fixed bond order. Each potential lattice bond now has one position in this list; an inactive site has $T_\sigma = 0$.

Write $p_i = p_0(g_i)$, and set

$$W_\sigma = (C_{\mathrm{sl}} p_i)^{-1}, \qquad \mathfrak{t}_\sigma = \{\mathfrak{b}_i^\flat, \theta_c \mathfrak{b}_i, \mathfrak{b}_i\}, \qquad \mathfrak{b}_i = \delta_i / g_i.$$

Choose the ceiling so that every distinct threshold is at least $2W_\sigma$ from the other thresholds and is itself at least $2W_\sigma$. Let $W_\sigma(u)$ denote the event that the site is active and its modulus is within u of a threshold.

Lemma 5.74 (The conditional radial estimate)**.** Let E be a joint survival event, intersected with any set of no-window conditions at diagnostic sites later in this deterministic order. Then

$$\mathbb{P}(E \cap W_\sigma(u)) \le C \, \frac{u}{W_\sigma} \, \mathbb{P}(E), \qquad 0 \le u \le W_\sigma. \tag{5.cdxix}$$

■

Proof. Disintegrate the stage law in the independent coordinates (s, a_σ). It has density $e^{\mathcal{F}_s(a)} \chi_s(a) \Pi(ds)\, da$, including all its Haar and coarea factors. By Lem. 5.73, the product of χ_s and the conditional future likelihood is constant on every one-sided shell. Conditions at other diagnostic sites of the same step depend on their fixed nonpivot coordinates, so they have the same property. The stage slope is bounded by

$$C_{\mathrm{sl}} p_0(g_j) \le C_{\mathrm{sl}}(p_i + 1) \le 2/W_\sigma.$$

Along a ray in a shell of length W_σ, the exponential density changes by at most e^2, and the radial Jacobian r^2 by at most four. Its initial portion of length u therefore carries at most $4e^2 u / W_\sigma$ times the full shell mass. Apply this separately above and below each threshold. The shells are disjoint. Integration over the angle and s proves eq. 5.cdxix. □

For a tile of side $\ell_\square$, put

$$\begin{aligned} \bar{p}_j &= \max_{\max\{0, j-N_j-1\} \le i \le j} p_0(g_i), \\ \delta_{\mathrm{dec}} D_j &= C + 4 \log \ell_\square + (4 \log L + \log 2)(k-j) \\ &\quad + 4(N_j + 1) \log L + 2 \log \bar{p}_j, \\ u_\sigma &= 2 C_{\mathrm{dec}} e^{-\delta_{\mathrm{dec}} \lceil D_j \rceil} \varepsilon_j / g_{j-1}. \end{aligned}$$

The diagnostic sites of a tile satisfy the more economical count

$$|\mathfrak{S}_j^{\mathrm{split}}[\square]| \le C \ell_\square^4 L^{4(k-j+1)} \le C \ell_\square^4 L^{4(k-j+N_j+1)}.$$

Thus the original, stronger schedule still pays them. The flow over $N_j + 1$ levels gives $\varepsilon_j / g_i \le C \bar{p}_j$, and hence

$$\sum_{\sigma \in \mathfrak{S}_j^{\mathrm{split}}[\square]} C u_\sigma / W_\sigma \le C e^{-C} 2^{-(k-j)}. \tag{5.cdxx}$$

This displays the cancellation of all four count factors, including the retained allowance $L^{4(N_j+1)}$. Before making any removal, enlarge the additive constant C so that the same calculation gives $u_\sigma / W_\sigma \le C' e^{-C} \le \min\{1, (2C_{\mathrm{shell}})^{-1}\}$ at every carried site, where $C_{\mathrm{shell}} \ge 1$ is the constant in eq. 5.cdxix. Thus both $u_\sigma \le W_\sigma$ and $\eta_\sigma = C_{\mathrm{shell}} u_\sigma / W_\sigma \le 1/2$ hold uniformly, including at large levels.

Let $\mathrm{NoWin}^{\mathrm{split}}(\square)$ exclude these diagnostic windows. Remove them in reverse deterministic order. Applying eq. 5.cdxix at each removal and using $1 - \eta \ge e^{-2\eta}$, with $\eta \le 1/2$, gives

$$\mathbb{P}\left(E \cap \bigcap_{\ell=1}^{r} \mathrm{NoWin}^{\mathrm{split}}(\square_\ell) \right) \ge C_{\mathrm{win}}^{-r} \mathbb{P}(E) \tag{5.cdxxi}$$

for distinct tiles. The constant in D_j can make $C_{\mathrm{win}} > 1$ as close to one as desired. This multiplicative estimate uses the conditional denominator in eq. 5.cdxix; an unconditional estimate on the probability of each window would not give eq. 5.cdxxi.

5.20.2.4. The bare-field witnesses. Realise every normalized measurable draw by an independent uniform coin, in the fixed relative chart of its site or component. There are finitely many potential draw sites in a finite prefix. Unused coins are ignored. The resulting run is Borel, has the prescribed kernel law, and, in the tiling regime, commutes with translations by $s_k\mathbb{Z}^4$. Here is the convention needed for the last assertion. The torus side is at least $2\ell_\square$, and $\ell_\square > \max\{128, 2n_{\mathrm{sm}}\}s_k$. Every component on which $\mathbf{R}$ draws fits in a nonwrapping aligned box of side at most $n_{\mathrm{sm}}s_j$, less than half the torus side. Its bounding box has a unique centre modulo the torus and has no nontrivial translation stabilizer. Number its bonds by their offsets from that centre, in the fixed frame. For each such shape and frame choose one Borel coordinate isomorphism, and use the same isomorphism for every translate. Translated states then have identical coordinate laws; their quantile maps with the translated centre coin agree exactly. Site and bond draws use the corresponding fixed local charts.

Induction over the moves proves covariance with this convention; it does not follow from Borel quantile existence alone. On a smaller torus an arbitrary Borel realization suffices for the direct moment argument below, and no covariant realization for a component with a nontrivial stabilizer is asserted. Integration is the deletion of a slot; no later move may read a deleted slot.

Let $m = dU \otimes d\xi$ be product Haar times Lebesgue, and $\mathbb{P} = \mu_{\mathrm{Wilson}} \otimes d\xi$. Since the Wilson action is nonnegative, $\mathbb{P} \le Z^{-1}m$. The tile coordinates contain only bare bonds whose two endpoints are strictly inside the tile and coins placed strictly inside it. For a Borel event A, define

$$F_\square(A) = \left\{x_\square : m^\square\{y : (x_\square, y) \notin A\} = 0\right\}, \qquad \phi_{\square,A}(U) = \int \mathbf{1}_{F_\square(A)}(U|_\square, \xi_\square)\, d\xi_\square.$$

Tonelli's theorem proves that these are measurable. The cylinder over $F_\square(A)$ is contained in A modulo a $\mathbb{P}$-null set, and $0 \le \phi_{\square,A} \le 1$. For distinct tiles,

$$\mathbb{P}\left(\bigcap_\ell \mathrm{pr}_{\square_\ell}^{-1} F_{\square_\ell}(A_\ell)\right) = \left\langle \prod_\ell \phi_{\square_\ell,A_\ell} \right\rangle_{\mathrm{Wilson}}. \tag{5.cdxxii}$$

The exterior measure in the definition is the product reference measure, not a conditional Wilson measure.

These functions need not be gauge invariant. If the chosen statement of chessboard positivity is restricted to gauge-invariant functions, average each ϕ over the gauge variables at all endpoints it reads. Those site sets are disjoint between tiles. Wilson gauge invariance makes the product expectation in eq. 5.cdxxii unchanged. This averaging preserves the support, the interval $[0, 1]$, and translation covariance. No reflection covariance of an individual coin realization is needed.

5.20.2.5. The isolated-cell payment. Suppose $e^{+2} \cap Z_j = \varnothing$. With I as in eq. 5.cdxiv, varying its free coordinates inside $\prod_{b\in I}\{|y_b| < \mathfrak{b}_{j-1}\}$ changes no explored collar or rim answer: every actual such query touches a stage set $Z^{[n]} \subset Z_j$, at index distance at least three from e, and therefore reads no bond of I. The original split queries that do read I are the full-threshold cube tests, which stay passed. The precomputed lower-threshold diagnostic flags may change, but none of them is read by this recursion. Sum over these unused flags before taking the warm-cell conditional law. The isolation factorization concerns the original geometry and column event, not the artificially refined diagnostic string. The approximate tests stay passed by their factor-two margin and their centre offset. At $j = 1$, the bare density margin used in Lem. 5.73 also applies. Thus isolation factors as the indicator of this product window times an exterior indicator. The region, the later run, and a column event are fixed by that exterior datum. Earlier gauge coins must be integrated out: fixing them would constrain the free coordinates to contour level sets. Later coins are independent of the current stage.

Here is the quantitative conditional estimate. Fix a marked bond b of e, and let $w(b') = e^{-\mu|b'-b|}$. For small fixed $\mu > 0$, eq. 5.cdxv gives

$$\left\| \tfrac{1}{2}(wKw^{-1} + w^{-1}Kw) - K \right\| \le C\mu, \qquad \tfrac{1}{2}(wKw^{-1} + w^{-1}Kw) \ge a_- I/2.$$

The same estimates hold for the full principal compression K_{II}. The exterior force has weighted norm at most $Cp_0(g_{j-1})e^{-cLM\check{R}_j}$; the nonlinear force has norm at most $C\epsilon_{\mathrm{nl}}$. In particular both together are small, even though the unweighted exterior force need not be small.

For completeness, a product of convex windows admits the comparison

$$|\nu_{H_1}(f) - \nu_{H_0}(f)| \le \lambda^{-1}\mathrm{Lip}_w(f)\, \|w \sup |\nabla H_1 - \nabla H_0|\|_2$$

when the conjugated symmetric part of H_0'' is at least λI. To see this, approximate the windows by separate convex walls in each bond coordinate and couple the two overdamped Langevin diffusions with the same Brownian motion. Weighted squared distance contracts under the first drift; the remaining force is bounded by the displayed weighted norm. The convex wall terms have the same sign because the weight is scalar on each bond. Pass to stationary couplings and then to hard walls. No convexity of H_1 is needed for this comparison.

Compare eq. 5.cdxiv with the even centered quadratic form on the same symmetric window. The comparison gives $|\nu_{H_e}(y_b^\alpha)| \le 1$ after reducing the coupling ceiling. By eq. 5.cdxvi, $H_e'' \ge a_- I/2$. Comparing it with $H_e - ty_b^\alpha$ and integrating in t yields

$$\nu_{H_e}\left(|y_b^\alpha - \nu_{H_e} y_b^\alpha| \ge u\right) \le 2e^{-a_- u^2/4}.$$

There are at most $C(LM\check{R}_j)^4$ marked bonds. A union bound therefore gives

$$\nu_{H_e}(e \text{ warm}) \le C(LM\check{R}_j)^4 e^{-c(\mathfrak{b}_{j-1}^{\flat\flat}/\sqrt{3}-1)^2} \le q_a^{r_j}.$$

The last inequality uses $2\mathfrak{p}_0 > 5r$, with c_a chosen before the coupling ceiling. Peeling disjoint boxes gives the product of these factors, relative to any column event. No derivative of the curvature or independent-current matching is used in this concentration argument.

5.20.2.6. From the witnesses to the chessboard bound. The remaining geometric and mass ingredients are the following ones of the actual history construction. First, the boxing/cascade recursion is local in its interval flags. Its pieces belong to the c_{lay}-clusters of hot cells, and each piece stays within $c_h = n_{\mathrm{sm}} + 1$ cells of its cluster. This follows by keeping an original seed in every component: a small bounding box is within $n_{\mathrm{sm}} - 1$ cells of that seed, at every round. No monotonicity of the boxing map is required. The same recursion on two common clusters gives the same labelled piece even if distant clusters require more stages.

Second, retain the following exact representation premise on the actual completed positive terms at a healing step. With the live region and its internal histories fixed, the term for a family $\mathfrak{I}$ of healed components is

$$\mathbf{O}_{\mathfrak{I}}[W(\mathfrak{I})], \qquad W(\mathfrak{I}) = \left[\prod_{X\in\mathfrak{I}} \mathbf{T}'_X\right] \exp \sum_Y V_{\mathfrak{I}}(Y).$$

Each $\mathbf{T}'_X$ integrates against a nonnegative measure. To see heredity in the native history, restrict to the clusters of all roots other than X. At each level the separation of distinct clusters exceeds the reach of the bond tests, pivots and collars. Hulls and enlargements are taken on the separated regions, and boxing acts on one component at a time. The class test also reads births in $(j - \mathsf{N}_j, j]$ and marks in $[j - \mathsf{N}_j, j)$. A birth at s can have been healed at an earlier level $i < j$ only if $s \le i - \mathsf{N}_i$; for a mark the inequality is strict. Since $\mathsf{N}_i \ge \mathsf{N}_j$, $i - \mathsf{N}_i < j - \mathsf{N}_j$, so no item read by the class test can have been erased in this way. These two facts give the same tests and classes on the remaining components. Each old component belongs to the cluster of its last successor. For $l \le j$, healing X fills X into Ω_l and Λ_l and removes it from S_l. Omitting the cluster leaves these same completed

label entries on X. The family parametrization over this label drops just X and its isolated inner history.

Write $\mathbf{O}_{\mathfrak{I}}$ for the positive outer operation on the common real exterior variables. For removal of X we need

$$\mathbf{O}_{\mathfrak{I}} \le \mathbf{O}_{\mathfrak{I}\setminus\{X\}}$$

on nonnegative inputs. In the completed-output representation, fix the final determining sequence $\bar{\mathbf{\Omega}}$, its datum and the standing components, including the second-class components Y_ι. The outer plaquette factor in the native output formula is

$$\chi_j\left(\left(\bar{\Omega}_j \cap \bigcap_{\iota} (Y_\iota^c)^{\sim 2}\right)^{\sim 4}\right).$$

On the healed branch the passed inset test is incorporated in this output factor. The preparation's shell guards occur in the island integral. In both terms of the comparison, evaluate the output at the same retained datum, including the restored top field. The completed determining rows and the live-component operations are then common, as are this outer factor and the action factored outside the bracket. The localized interactions that read the inner history remain inside $W(\mathfrak{I})$. In a representation with a top mask depending on the family, we must compare the preparation and ordinary masks on the same level-j cubes at the actual exterior field. The implication from the preparation mask to the ordinary mask is sufficient, since multiplication by these ordered indicators is composed with the same positive maps. Resetting the coupling profile identifies the scale of the tests. The comparison also needs their actual readouts and supports.

Removal leaves the terms away from X unchanged and independent of its integration variables. For the near terms on the insertion side, follow a chain of cubes from the enlarged strip of X to a cube outside the exempt strips. Such a cube exists for the interaction domains of the bracket. The first exit from the strip touches X's strip and is outside every other strip by their separation. On the removal side, root the term at a cube of X's enlarged strip, now outside the remaining exempt strips. In either case the interaction tree meets its root cube. There are $C\check{R}_j^4$ possible roots. The rooted entropy bound and the strict decay reserve give a finite geometric sum per root. Thus the sum of the absolute near terms in the two brackets is bounded by $C\check{R}_j^4$ on their common admitted core. This uses the actual localized interaction domains and the strip separation.

Consequently, pointwise inside the bracket, inserting X, with its internal history in $\mathscr{H}$, costs at most $t_X(\mathscr{H})e^{C\check{R}_j^4}$, where $t_X(\mathscr{H}) = \sum_{\mathfrak{h}\in\mathscr{H}} \mathbf{T}'_{X,\mathfrak{h}}1$. The trace mass bound of Props. 4.63 and 4.64, with its charge dial increased by c_a, gives

$$K_H \sup_X t_X(\mathscr{H})e^{C\check{R}_j^4} \le \sup_{\mathfrak{h}\in\mathscr{H}} q_a^{2\sum_{i\le j} r_i|\mathsf{Z}_i(\mathfrak{h})|_i}.$$

Here the unspent factor $e^{-cp_0(g_j)}$ pays both the strip exponent and the finite shape count K_H. The condition $\mathfrak{p}_0 \ge 6r+1$ pays every increased lineage charge, whose logarithmic degree is at most $6r$. The supremum in this estimate is on that same actual core and is taken before the outer integration.

More precisely, let $E_{c,\mathscr{H}}$ specify a healed member at the cell c, and let E_c allow any internal history there. If B ignores earlier healed members and is unchanged by deleting the member at c, heredity and the pointwise comparison give

$$\widetilde{\nu}(B \cap E_{c,\mathscr{H}}) \le \left(\sup_{\mathfrak{h}\in\mathscr{H}} q_a^{\mathrm{Vol}(\mathfrak{h})}\right)\widetilde{\nu}(B \setminus E_c).$$

Apply $\mathbf{O}_{\mathfrak{I}}$ to the bracket bound and dominate it on the reduced bracket by $\mathbf{O}_{\mathfrak{I}\setminus\{X\}}$. Then apply the same subsequent positive kernels to the two sides. A joint survival event has the required blindness: healing removes an entire component, so its earlier lineage cannot also be the ancestor of a component still live later. Peeling in increasing healing time therefore proves the product bound

for several exclusive healed members. A bound on total polymer mass alone would not justify this relative comparison.

Use the collar lengths

$$\begin{aligned} f_j &= \left\lceil \frac{M_1(\lceil D_j \rceil + C)}{M\check{R}_j} \right\rceil, \\ c_{\mathrm{adj}}(j) &= 2c_h + c_{\mathbf{R}} + f_j + C, \\ m(j) &= 1 + 2(k-j), \qquad m_0(j) = 6(c_{\mathrm{adj}}(j)+1)m(j). \end{aligned}$$

At each level reserve $m_0(j)+f_j+C(c_h+c_{\mathbf{R}})$ additional cells, beyond the preceding depth expressed in the new units. The additional two buffers have this same order. The total physical width is bounded by

$$CM\check{R}_k + CM_1 \sum_{n\ge 0} (1+n) L^{-n} \big(\log \ell_\square + n + \check{R}_{k-n} + \log \bar{p}_{k-n} \big).$$

Since $\check{R}_{k-n} \le C(1+n)\check{R}_k$, it is at most

$$M\check{R}_k \left[C_1 + C_2 \frac{\log \ell_\square + C + \log\log x_k}{\check{R}_k} \right].$$

Choose a power of L with $\ell_\square \asymp M\check{R}_k$, with its fixed constant large enough for $\ell_\square > \max\{128, 2n_{\mathrm{sm}}\}s_k$ and $\ell_\square \ge 32 \max_j B_j s_j$. The width bound above allows this choice, since $\log \ell_\square / \check{R}_k$ tends to zero. The core of side $3\ell_\square/4$ leaves every ancestor of a deep cell at depth at least $4B_j$ in its level-j partition.

The comparison of runs is an induction on the level. Measure the preceding common depth in level-j cells and round up to T_0. The double blocks for a level-j average stay in the cells of their centres, so the current field is common on this inner lattice. Its inherited live hull Z' is also common from T_0 on. A cell of this hull can be touched by a component of the preceding live region from outside the inner lattice. The induction therefore keeps the entire preceding components that meet the common region, including those that only touch it.

At a step $j \ge 2$, a plaquette test is inserted outside $Z'^{\sim 4}$. Its localized background, including the existence check, uses only the full four-layer cube, which lies in the cell and its neighbours. Thus the failure set P_j is common from T_0+4, its hull from T_0+5, and the gauge region $(P_j'^{\sim 1} \cup Z'^{\sim 5})^c$ from T_0+6. The gauge draws use their own blocks before any draw and the same tile coins at the drawn sites. For a tested bond b both runs therefore obtain $u(b_-)V_{j-1}(b)u(b_+)^{-1}$ with the same factors. The approximate test reads this field and the background of its local test cube. Its one-cell neighbourhood puts Q_j at depth T_0+7 and Q_j' at T_0+8. The union defining Ω_j^c is common from T_0+10.

At $j=1$ the inherited live hull is empty. Start at $T_0 = 1$ with P_0'. A failed bare cube contributing to its hull there has side at most $L^{-1}s_1$. Its density test adds at most $L^{-2}s_1$ to the read, so the bare plaquettes tested there are at distance at least $(1-L^{-1}-L^{-2})s_1 > 0$ from the tile boundary. The bare bonds of the tile determine this first hull. With the first-step masks above, the depths for P_1, P_1', the gauge region, Q_1 and Q_1' are $T_0+1, T_0+2, T_0+3, T_0+4, T_0+5$, respectively. The first Ω_1^c is therefore common from T_0+7. The bare regularity test supplies regularity on the first determining region $\Omega_0 = \Omega_1^{\sim}$.

Call the common depth for Ω_j^c A_0. The components of Ω_j^c truncated to depths at least A_0 that touch its outer layer are common. One with diameter exceeding $m_0(j)$ is a first kind of trigger. Otherwise these components lie at depths at most $A_0 + m_0(j)$. From the region of depth at least A_0 remove their closed $4M_1$-neighbourhoods and take the resulting union of M_1-cubes. Its $4M_1$ inner rim lies in Ω_j: a bad cell meeting the rim would either touch the outer layer or be adjacent to a removed cluster, because $10M_1 < M\check{R}_j$ in level-j units. In either case it would itself have been

removed. The region contains depth $A_1 = A_0 + m_0(j) + 2$. At depth $A_2 = A_1 + f_j$ the tested bonds have distance at least $f_j M \check{R}_j - 5M_1 \geq \lceil D_j \rceil$ from its complement.

Use the common retained determining rows and data in this region and the regular rim in eq. 5.cdxii. At the first step the determining region includes $\Omega_1^{\sim}$. A retained bad cell in the comparison region is in a component disjoint from the outer layer, so its touching cells are still in the common Ω_1 data. In either case the post-draw link is common. Bi-invariance of the SU(2) angle bounds the difference of the two relative angle moduli by $\frac{\pi}{2}|V^{\mathrm{bg}}(V^{\mathrm{bg}\prime})^{-1} - 1|/g_{j-1}$, which is strictly less than the excluded window width u_σ. A relative group element at the cut locus has angle π, and the perturbed angle is still above the largest small-field threshold. Thus the split interval flags and the hot-cell clusters at depth A_2 are common. A long boundary cluster of these hot cells gives the second kind of trigger.

Use the depth t in level-j cells. Set $h_j = m_0(j) + c_{\mathrm{adj}}(j) + c_h + 3$ and reserve the two depths $A_\alpha = A_2 + h_j$ and $B_j = A_\alpha + h_j$. These layers are included in the width estimate above. If there is no trigger, a cluster of hot cells at adjacency range $c_{\mathrm{adj}}(j)$ meeting $t < \theta + c_{\mathrm{adj}}(j) + 1$ has its c_h-hull in $t < \theta + h_j$, for $\theta = A_2, A_\alpha$. This follows by adding its diameter budget $m_0(j)$ and the hull enlargement. In particular any cluster reaching the c_h-neighbourhood of a cell at depth A_α or greater stays beyond $A_2 + c_{\mathrm{adj}}(j)$. Thus exterior hot cells cannot join it at the smaller boxing adjacency c_{lay}.

Let $\mathcal{K}$ be the union of the common hot clusters meeting $t < A_\alpha + c_{\mathrm{adj}}(j) + 1$. Its c_h-hull stays below B_j. If a boxing cluster meets $\mathcal{K}$ or leaves the common region $t \geq A_2$, its output pieces stay in this hull or within c_h of $t < A_2$. The other clusters, their labels and their earlier-stage data are common by the induction. They have separation greater than $c_{\mathrm{adj}}(j)$ from $\mathcal{K}$. An output piece is within c_h of its cluster, so its distance from $\mathcal{K}^{+(c_h+1)}$ is at least $c_{\mathrm{adj}}(j) + 1 - c_h - (c_h + 1) = c_{\mathrm{adj}}(j) - 2c_h$. Choose the fixed additive constant in $c_{\mathrm{adj}}(j)$ so that this is at least $c_{\mathbf{R}} + f_j + 4$. The same depth buffer separates the piece from the unmatched outer data. Thus the $c_{\mathbf{R}}$-enlargement used by a component-local preparatory read is inside the common data. To apply this at a preparatory stage, keep the fields before the operation as well as the fields after each draw. The latter can overwrite a field whose earlier value the next test reads.

For a component X in the common inner region, the initial record consists of those incident fields, the trace of the label, the current failed tests and cascade cells, the earlier live components with their records, and the coins assigned to X. Append the answers in their order of use. Equality of this record compares physical link values and discrete decisions. Relative coordinates formed with different global centres can still differ. Their later uses must be through an original recorded answer or a local read, as in the completion test below. The waiting, quietness and single-component tests are functions of this trace. For an eligible component the boxes $X \cap Z_i$ and $X \cap \Omega_i^c$ in the waiting window give the prepared nest and every collar. Hence their row masks, including the incident rows, are common.

At stage zero and at each successor stage, the regularity test uses the localized background with the joined datum of X and its prescribed lifts of the top field there. Its determining rows and their stored values are common. An undefined local background gives the same failure on both sides. For a stage integrating level ι, the gauge draw reads complete blocks with centres in $((\Omega_{\iota+1}^{\sim 3})^c \setminus Z''_{\iota+1}) \cap X$ that lack the earlier gauge factor. That mask is common. Their contours are computed from the original retained V_ι links of these blocks, which are stored as Haar links. The common contour and the same normalized draw coin give the same gauge transformation at each site. The resulting incident links agree and the coarse centres are fixed. The approximate test then reads these links against the common local background.

Consider an ordinary completion test at a stage with stored level ι, and put $t = \iota + 1$. The completion reads a free bond with an endpoint in Ω_t from a test cube of the ordinary boundary layer inside the quiet component X. Its full bond read lies in $\mathrm{Box}_t(X)^{+2} \subset X \cap Z_t$, where $\mathrm{Box}_t(X) = \Omega_t^c \cap X$. The waiting condition excludes creations there in the stage window. The hull of the failed

cubes at level t is at distance at least five level-t cells from $\mathrm{Box}_t(X)$. Since the read of a failed cube stays in its own cell and the touching cells, the completion bond is outside the failed cube masks. It therefore belongs to a previously passed mask or to the plain strip. On the passed mask the recorded ordinary bound makes the completion indicator one.

In the plain strip the test reads the stored relative logarithm $\frac{1}{i}\log(V_\iota(b)V_\iota^{\mathrm{bg,loc}}(b)^{-1})$. The centre is the axial representative outside the nodes of the joined problem on the corresponding strip component of $\Lambda_t^c \cap \Lambda_\iota$. The strip nodes fix this representative at the test bond. The joined problem uses the pre-operation V_t, V_ι of $X \cap Z_t$. The lineage in the waiting window stayed live through each intermediate operation, so none of those operations wrote its fields. The level-$t+1$ gauge draw writes outside the enlarged live region. Hence the inputs have their stored values and are common in the two runs. This gives the same completion answer even when the test fails. It is used before testing the new sharp ball. The latter uses the localized next-stage centre and the post-draw link, both common. On each outcome keep its actual prefix.

On a passing stage the deleted slots are the stored nonpivot fluctuations on $Z_{\iota+1} \cap \Omega_{\iota+1} \cap X$ and the collar nonpivots on $(\Omega_{\iota+1}^c \setminus Z''_{\iota+1}) \cap X$. The translation to the expansion centre is an integration coordinate for the latter group. The real weight can use the global expansion background here. The physical retained state keeps the prescribed $V_{\iota+1}$ row and the surviving slots in their own charts. On a failure of the local sharp test, the stored prefix uses the untransformed real variables. Thus the two runs agree on the retained physical record through either branch. If a later read uses a translated coordinate that is actually retained, its dependence must be checked separately.

After the last collar, the extendibility test asks for an extension of the common exterior coarse field over the main box in the component problem. If this passes, the Haar refresh uses the box coins and keeps the old variables for the prepared datum. The same component frame fixes both trees and the entrance root, so the tree gauge gives the same shell variables. The main fluctuation test reads the resulting field in the prepared shells, and the last-shell test reads the innermost stored field with its prescribed identity data. The localized reference law has the same exterior, frame, tree slice and cutoff in the two runs. Its denominator is the integral of that same density on the same slice, so the draw and any denominator failure agree. The test of $\chi_j(X^{\sim -6})$ is made on the newly drawn field in the inset test cubes. Its full local test neighbourhoods stay in X. Thus the healing decision agrees. The subsequent deletions or inert declarations, the rotation on the healed branch and the label update are functions of the common record and the component coins. Every bond in the final deep region either meets one of these common components or is unchanged. The common surviving labels specify which lower slots remain. This is the induction output at level j.

For the induction on the actual run we must verify that any alternative stage or main-step chart still reads the same incident rows and masks just identified. This is the local-read input to the witness construction below. A sharp test centred on a global determining set of earlier outcomes requires a separate comparison.

If there is a trigger, take its first level J. The preceding survival record is common. At J use the first kind of trigger if it occurs, since the truncated bad components are already common. Otherwise the split comparison gives the common second kind of trigger. Along a cluster path starting at its outer end, mark the first cells reaching successive index radii $3(c_{\mathrm{adj}}(J)+1)(i-1)$, $1 \le i \le m(J)$. From the initial cell one of the diameter endpoints is farther than $m_0(J)/2$. Take a path to it. Its bounded steps give a pattern of $m(J)$ cells, mutually separated by at least $2c_{\mathrm{adj}}(J)+3$. Choose the cluster and the path by a fixed order of cells in the tile. The choice uses only the common data at the first trigger. For a marked cell e, a bad domain or a failed full test puts a cell of Z_J in e^{+2}. The remaining hot kind is warm whenever e^{+2} contains no Z_J-cell. Thus the pattern event is that each mark has a cell of Z_J within two cells or is isolated and warm. For a deep cell z whose lineage survives from n to k, every completion of the tile data has the same survival decisions below J and the marked alternative at J. The status of an individual mark as isolated may vary with the exterior. With no trigger, the decisions agree through k by the same induction on the verified local

reads. These assertions use the split windows only for the original run. Under the local-read input the two types of events give witness sets in the sense of §5.20.2.4. Apply the no-window bound eq. 5.cdxxi, then take the union over witnesses in each tile and use eq. 5.cdxxii for the resulting products.

Keep the tiles whose indices relative to the original tile are even in all four coordinates. The marks in their e^{+2} neighbourhoods are disjoint. In each replica classify each marked cell as isolated and warm, or choose a cell of Z_J within two cells of it. The warm-cell bound can be peeled relative to the event for these chosen cells and the survival columns. The remaining event reads the regions and lineages unchanged by varying the free variables of an isolated box. Every warm mark thus pays $q_a^{r_J}$.

Classify the chosen cells and the column, if $J > n$, by fate. Group objects absorbed by the same healed component and specify its healing level $h \geq J$. Index the component by the level-h parent of its first object. Its lineage at J must contain exactly the level-J cells of that group among the chosen objects. For a group carrying the column, also require its lineage to contain every prescribed column cell at levels $n, \dots, J-1$. These restrictions allow the relative healing bound to be applied once per distinct component. A group with d chosen marks pays $r_J d$ units of the exponent. It has a cell at each level in $(J, h]$, which pays a further $h - J$ units. A column pays one unit per level in $[n, J-1]$. The column's top is used to specify its fate and is not charged as a second mark if it is already a chosen cell. No healed component serves different replicas. Even after moving to nearby chosen cells and projecting to the healing grid, their separation across an intervening tile is at least $\ell_\square - 6s_k > n_{\mathrm{sm}} s_k$. Apply eq. 5.cdxviii once to the surviving marks and columns in all replicas, so there is a single volume factor. Each replica has at most $m+1$ objects. The choices of nearby cells and the group labels cost at most $(C(m+1))^{m+1}$; summing the healing times costs at most $(1-q_a)^{-(m+1)}$. The cost per replica is at most

$$v_{J,\mathbf{e}} = \big(C(m+1)(1-q_a)^{-1}\big)^{m+1} q_a^{(J-n)_+ + r_J m}.$$

This includes the sum over absorption times. The pure-survival alternative has cost $v_{\mathrm{I}} = q_a^{\sum_{j=n}^{k} r_j}$. The whole replicated probability carries only the single factor $e^{B(x_k)V_k}$.

The tile grid has even side count in each coordinate. In the chessboard dissemination, the copies in the even-index class are pure translates, since two face reflections restore the orientation in that coordinate. Integrate the independent coins before dissemination. On the kept tiles, restore the coin integral using eq. 5.cdxxii. By translation covariance of the run, each witness cylinder forces its translated event almost surely. Dropping the other copies, which lie in $[0,1]$, therefore bounds the product by the probability of the replicated events. The chessboard root is of order $|\mathcal{T}|$ and the even-index class has $|\mathcal{T}|/16$ tiles. Consequently the chessboard norm of a witness is at most

$$e^{B(x_k)\ell_\square^4} v_{\mathfrak{w}}^{1/16}.$$

There are at most $\ell_\square^4 L^{4(k-J)}(m_0(J)+1)^{4(m-1)}$ patterns at level J. To bound the pattern count, put $u = x_k + k - J + C_*$ with C_* fixed. The flow gives $N_J = \check{R}_J \leq C(\log u)^r$. The inverse coupling at any level in the range defining $\bar{p}_J$ is at most $C(u + (\log u)^r) \leq C'u$, so $\log \bar{p}_J \leq C(1 + \log\log u)$. Also $\log \ell_\square \leq C(1+\log\log u)$. The displayed formula for D_J now gives $D_J \leq C(1+k-J+(\log u)^r)$. Using $f_J \leq C(1+D_J)$, we obtain

$$m_0(J) \leq C(1+k-J)(1+k-J+(\log u)^r) \leq Cu^2.$$

The lower flow bound gives $r_J \geq cM^4(\log u)^{5r}$ after choosing the coupling ceiling. Apart from the starting position factor $\ell_\square^4 L^{4(k-J)}$, the logarithms of the enumeration and absorption factors are at most $Cm(J)\log u$. Half the surplus $(r_J - 1)m$ pays this pattern count and its absorption prefactor. The remaining half supplies the factor $\widehat{q}^{(r_k-1)/2}$. Also

$$(J-n)_+ + m \geq (k-n+1) + (k-J).$$

Choose $c_a \geq 16\log(2L^4)$. Then $\sum_{J\leq k}(L^4\widehat{q})^{k-J} \leq 2$. Finally $r \geq 2$ gives $5r > 4r+1$, so, by taking the ceiling small, the single-level surplus pays $\ell_\square^8 e^{B(x_k)\ell_\square^4}$, C_{win}, and any fixed requested extra factor ϵ. The witness sum is therefore at most $\ell_\square^{-4}\widehat{q}^{\,k-n+1}$; the sum for an additional tile's deep level-k live cell is at most ϵ. Their product proves the two bounds of Lem. 5.72.

For the first bound the anchor can be anywhere in the torus. When the tiling exists, shift it by level-k cells so the level-k parent of the anchor lies in the central cell of its tile. The tile side and cell side are powers of the odd integer L, so the central cell is well defined and the shift is allowed. The side of the stopped lattice is twice a power of L, so whenever it is at least $2\ell_\square$, the tile count per side is even. If the torus is too small to contain this tiling, then $V_k \leq (2\ell_\square)^4$. On survival, $\Theta_k \geq 2c_a r_k(k-n+1)$. The same surplus comparison gives

$$B(x_k)V_k \leq c_a r_k, \qquad e^{B(x_k)V_k - 2c_a r_k(k-n+1)} \leq \ell_\square^{-4}\widehat{q}^{\,k-n+1}.$$

Thus the first bound follows directly from the stopped moment on every admitted prefix, without a tiling. Only the joint-tile statement requires actual tiles. The argument concerns survival and hence lineage age; it does not assign an age penalty to a seed that was healed before the lineage began.

Proof of Prop. 5.71. Anchor: choose any cell $\mathfrak{z}$ of the lineage region $\mathsf{Z}_n(C)$ at the first level $n = j_0 = k-a$ of the lineage. The lineage regions are nested and contained in the live regions. Hence $\mathfrak{z}$ stays live throughout the levels $n,\dots,k$, so that the event $\mathcal{E}_{\mathfrak{z}}^{(n)}(k)$ occurs, and the cell belongs to the component of $\mathfrak{c}$. This is the point at which the age of the lineage enters: a region healed in the meantime provides no such cell. *Sum over anchors:* a box of side $\mathsf{l}_\square$ contains at most $\mathsf{l}_\square^4 L^{4(k-n)}$ anchors, and each costs $\mathsf{l}_\square^{-4} q_a^{c(k-n+1)}$. For an anchor at a distance of $i \geq i_0$ boxes, the live component is connected and therefore passes through a chain of at least i distinct adjacent boxes, each of which contains a live cell. These cells need not be deep. Consider the 2^4 tilings whose face planes are displaced by $\pm\lfloor n_\square/4\rfloor$ cells from a corner of the level-k cell containing $\mathfrak{z}$, where $n_\square \geq 16$ is the number of level-k cells along a side of a tile. Every level-k cell is deep in a tile of one of these tilings, and $\mathfrak{z}$ is deep in all of them. By the pigeonhole principle a single tiling serves at least $i/16$ of the cells. These cells lie in at least $i/16^2$ distinct tiles, since a tile meets at most 2^4 boxes, and at least i/i_0 of these tiles, where $i_0 = 2\cdot 16^2$, are different from the tile of $\mathfrak{z}$. For a fixed reference chain, the choice of a shift, the subset of its indices served by that shift and the choice of one of at most 2^4 shifted tiles per index cost at most $16\cdot 2^i\cdot 2^{4i}$ choices. The chain itself has at most 3^{4i} choices. The second bound of the lemma sums the deep-cell positions inside the selected tiles. Hence with $C_{\mathrm{ent}} = 2^5 3^4$ and $\epsilon = (2C_{\mathrm{ent}})^{-i_0}$, the additional factor for such an anchor is at most $16\,C_{\mathrm{ent}}^i \epsilon^{i/i_0} = 16\cdot 2^{-i}$, which is summable against the $(2i+3)^4$ boxes of a shell. Finally $L^4 q_a^c \leq (2L^b)^{-2}$, and this is exactly the floor $c_a \geq C(b)\log L$ imposed in the statement of the proposition. Therefore $\tilde{\nu}(\mathfrak{lb}_k^{=a}(\mathfrak{c})) \leq C(2L^b)^{-2a}$. The geometric mean of this bound and that of Prop. 5.70 is the first bound. Summing it gives the age series

$$\sum_{a\geq 0} L^{ba}\,\tilde{\nu}(\mathfrak{lb}_k^{\geq a}(\mathfrak{c})) \leq Ce^{-cp_0(g_k)},$$

which two later parts use. In §6.1.1.2 the series is needed with rate $b \geq 8$, because for two points a live component of age a carries a weight of up to L^{8a}; there the series underlies the two-point law, Thm. A (c)(3), (c)(4). In Thm. A(c)(5) the series enters with rate $b > 16$, on each torus, since a weight L^{16a} arises for four points; this concerns clause (c)(5). The expansions of §6.1.1.2 and Thm. A(c)(5) require the age up to which a live component stores fields. This is the age read in the label. For a component holding an island, the block of which retains the fields of its core §4.17.1.3, it is at most the age of the lineage. Here, as in Thm. A(c)(5), the series is formulated with the age of the lineage, and this age is never smaller. The series therefore covers what these expansions require:

$$\{\mathfrak{c} \subset C,\, \mathrm{age}^{\mathrm{lab}}(C) \geq a\} \subset \mathfrak{lw}_k^{\geq a}(\mathfrak{c}).$$

For the second bound, each old live cell is carried into its whole parent cell at the next scale until its component is healed. A merge retains its lineage through that level. Thus if $\mathfrak{c}$ has lineage age a at level k and survives through K, its parent at level K has lineage age at least $a + K - k$. Sum the first bound at K over these terminal ages.

The two-term age bound. The prepaid trace bounds give a second age estimate on the tori without using the tiles and windows of Lem. 5.72. Decompose the event according to whether the lineage is healed at some level $j > k$ or stays live to the last level K' of the run or its prolongation. Fix $c_a > 0$ and choose the coupling ceiling small enough for the estimates below.

The event is empty if $k - a < 1$. For $j \geq k$ write $\mathfrak{c}^{(j)}$ for the cell at level j containing $\mathfrak{c}$. For a top X at level j, let $\mathscr{G}_X$ be the traces with first level $1 \leq j_0 \leq k - a$. If the component at $\mathfrak{c}$ is carried into X, its lineage up to level k lies in the lineage of X. On $\mathfrak{lw}_k^{\geq a}(\mathfrak{c})$ this puts the trace of X in $\mathscr{G}_X$. Every level from j_0 through j contributes a cell to the lineage. At $\mathsf{w} = 0$ the mark is therefore at least $\bar{c}_a(a + j - k + 1)$, where $\bar{c}_a = 2c_a M^4$. The prepaid trace bound in the proof of Prop. 5.70 gives

$$\sum_{\mathfrak{l} \in \mathscr{G}_X} \mathrm{wt}(\mathfrak{l}) \leq e^{-\bar{c}_a(a+j-k+1)} e^{-c_{\mathrm{cell}}|X|_j} e^{-2\min_{1 \leq i \leq k-a} p_0(g_i)}.$$

We fix c_{cell} before choosing the coupling ceiling.

Survival to the terminal level. For each history in the survival event distinguish the unique terminal component containing the cell $\mathfrak{c}^{(K')}$. Group the histories by their terminal traces. Positivity and the mass bound at fixed traces bound the unnormalized mass, after removing $e^{E_{(K')}}$, by

$$e^{C|\mathbb{T}^{(K')}|} \left[\sum_{X \ni \mathfrak{c}^{(K')}} \sum_{\mathfrak{l} \in \mathscr{G}_X} \mathrm{wt}(\mathfrak{l}) \right] \left[\sum_{\mathscr{C}} \prod_{Y \in \mathscr{C}} \sum_{\mathfrak{l}:\, \mathrm{top} = Y} \mathrm{wt}(\mathfrak{l}) \right].$$

Here X ranges over connected terminal cell unions, and $\mathscr{C}$ over collections of distinct such unions. For the second bracket we drop the exclusion with the distinguished component. Choose c_{cell} so that the sum of $e^{-c_{\mathrm{cell}}|X|_{K'}}$ over connected sets containing a given cell is at most one. The first bracket is then at most $e^{2 - 2p_0(g_k) - \bar{c}_a(a + K' - k + 1)}$, by the drift $p_0(g_i) \geq p_0(g_k) - 1$ for $i \leq k$. For the second bracket the unrestricted trace bound gives total activity at most one per cell. The product bound $1 + z \leq e^z$ then gives at most $e^{|\mathbb{T}^{(K')}|}$, since the number of cells is bounded by the number of sites. Divide once by the partition function floor at K', using $\mathsf{B}(x_{K'})$ to absorb both volume charges and the floor. For a prolongation by p levels on the same bare lattice, the number of terminal sites is $(\ell L^{-p})^4$, and the history law through level k is the same. A Markov bound on the global end mark alone would lose the factor $e^{-2p_0(g_k)}$.

Healing at level $j > k$. Use the outer-operation domination of the relative healing comparison above. Inside the bracket, deletion of the healed member with top X costs at most its restricted trace sum times $e^{C_I \check{R}_j^4 + C}$. A healed top is an aligned box of at most n_{sm} cells per side, so at most $(\sum_{n=1}^{n_{\mathrm{sm}}} n)^4$ such boxes contain $\mathfrak{c}^{(j)}$. Discard the rooted size charge and split the oldest creation payment using the drift to g_k and g_j. For $0 < c_{\mathrm{r}} < 1/2$ the exponents allocated to these scales are $c_{\mathrm{r}} p_0(g_k)$ and $(2 - c_{\mathrm{r}}) p_0(g_j)$, up to an additive constant. The ceiling gives $C_I \check{R}_j^4 \leq (1/2 - c_{\mathrm{r}}) p_0(g_j)$. The probability at this healing level is bounded by

$$C e^{-\bar{c}_a(a+j-k)} e^{-c_{\mathrm{r}} p_0(g_k)} e^{-3p_0(g_j)/2}.$$

The flow lower bound gives a bounded number of visits to any x_j-interval of length $\lambda_\sharp$, and the ceiling gives $e^{-3p_0(g_j)/2} \leq x_j^{-3}$. Thus the last factor is summable in j, uniformly in the run. This yields the first term of the bound without using the terminal partition function floor.

In particular, for each level-k cell $\mathfrak{c}$ and $a \geq 0$,

$$\tilde{\nu}\big(\mathfrak{lv}_k^{\geq a}(\mathfrak{c})\big) \leq Ce^{-\bar{c}_a a}\, e^{-cp_0(g_k)} + C\exp\Big[\mathsf{B}(x_{K'})\,|\mathbb{T}^{(K')}| - 2p_0(g_k) - \bar{c}_a\,(a + K' - k + 1)\Big],$$

where $K' \geq K$ is the last level of an arbitrary prolongation of the run § 5.2.4, the run itself counting as one of its prolongations. The first term contains no volume, and no condition ties k, a, K and ℓ together. The consequences of the bound follow from its second term, case by case; none of the cases is a hypothesis of any statement. (1) There is a side $\ell_V(g)$, tending to infinity as $g \to 0$, with the following property. For every torus with $\ell \leq \ell_V(g)$, the prolongation of (iv) terminates in the alternative where the torus is reduced to polylogarithmic size. In that case the gain $2p_0(g_k)$, of degree $\mathfrak{p}_0$ in the logarithm, dominates the volume term, of degree $4r + 1$. The bound is then $Ce^{-\bar{c}_a a}e^{-cp_0(g_k)}$ at every level. (2) The last lattice of the reference torus $T_{\ell_\star(g)}$ of Prop. 6.3 has polylogarithmic size even without prolongation. Hence that torus satisfies the same bound at all levels. (3) If $\ell > \ell_V(g)$, the torus satisfies the same bound at those levels at which $\bar{c}_a(K' - k + 1) \geq \mathsf{B}(x_{K'})|\mathbb{T}^{(K')}|$. This is a floor on the depth that depends on the torus; once it is met at one level, it is met at every lower level. At all of these levels, if $\bar{c}_a \geq b\log L + \log 2$, summing the bound gives the age series at the rate b. Thus this road alone supplies what §6.1.1.2 uses, namely the age series at the rate $b \geq 8$, at all levels of the reference torus, as used in Thm. A (c)(3), (c)(4). For Thm. A(c)(5) it supplies the series at $b > 16$ on each torus. On the tori of (1) it holds at all levels; on the tori of (3) it holds at the levels below one that depends on ℓ. The additional four-point hypothesis in the short-distance statement has exactly this quantifier structure: $\mathsf{d} \leq \mathsf{d}_4$ if $\ell \leq \ell_4(g)$, and $\mathsf{d} \leq \mathsf{d}_4(\ell)$ otherwise, where $\ell_4(g) := \max\{\ell_V(g), \ell_\star(g)\}$. The contribution of Lem. 5.72, through its tiles, windows and triggers, is decay at all levels and on all tori, with constants independent of ℓ. This is uniformity as $\ell \to \infty$. The same age threshold therefore works on every torus. Passing a four-point estimate to limits on $\mathbb{R}^4$ additionally requires the actual normalized interacting comparison and its $o(g^8)$ remainder, with the prescribed test factors, uniformly along the volume and cutoff limits; see eq. 6.ccxxxiii. These further inputs are not deduced here. The torus and additional-hypothesis restrictions in Thm. A(c)(5) remain in force.

The age constant. In §4.17.1.45 two roles of c_a are described; the one at work here is role (A), the dial of the mark. Its effect on the law of the histories is $C(2L^b)^{-a-(K-k)}$. Role (B), the age rate q_a of the aged load Σ_k, concerns the pairs of lists belonging to a class of the kind given there §4.17.1.45, and it is used in appendices A.8 and A.9. The floors on c_a listed in §4.17.1.45 concern histories, rotations and age-resolved rarity. Prop. 5.71 imposes the third of these.

Cutoffs beyond first passage. The dilation identity of § 5.2.4 gives

$$S^T_{n;K,\ell,\beta}(f_1,\dots,f_n) = S^T_{n;K-q,\,L^q\ell,\,\beta}(f_1^{[q]},\dots,f_n^{[q]}),$$
$$\|f^{[q]}\|_\sharp \le \|f\|_\sharp,$$

since $\mathcal{O}$ carries no power of the spacing. The same relation holds between $\mathbb{L}_K$ and $\mathbb{L}_{K-q}$. Let $\beta \ge g^{-2} + B_\sharp$ and $q \ge 0$, and suppose that $K - q = K(\beta, g)$. The bounds of Prop. 5.64 are then valid at the cutoff K, with $(L^q\mathsf{d}; L^qR; L^q\ell)$ in place of $(\mathsf{d}; R; \ell)$.

5.21. **Stability from a normalized history sum.** This is a conditional deduction of upper stability and the unsourced partition floor. It isolates the estimates a renormalization construction must supply. In particular it does not replace those estimates by a polymer convergence theorem. A representation with a sum over large-field labels and label-dependent integral operations need not be a single coarse Wilson integral multiplied by one uniformly small polymer gas.

Let the final four-dimensional periodic lattice have N sites and $4N$ independently oriented bonds. Haar measure on each SU(2) bond is normalized. Let w range over a specified ordinary complex source neighborhood containing zero. Assume there are complex measures $\tau_S(w)$, indexed by a finite set of labels, such that

$$Z(w) = \sum_S \tau_S(w)(\mathcal{V}), \qquad \tau_S(0) \ge 0. \tag{5.cdxxiii}$$

Assume all normalizers and scalar quantities below are well defined; E is the real vacuum constant of the unsourced all-small-field label. Put

$$\mathcal{E}(w) = \sum_{j,X} e_j(X; w), \qquad \mathcal{E}(0) = 0, \qquad \widetilde{Z}(w) = e^{-E-\mathcal{E}(w)} Z(w).$$

At each finite cutoff the sums are finite. The argument requires their locality and uniform estimates; it uses no sum over cutoffs.

A total-variation criterion for upper stability. Let $\mathcal{E}_S(w)$ denote the vacuum constant in the exponent of the term carrying label S. Suppose there are nonnegative weights W_S and real label costs $q(S)$ satisfying

$$\|e^{\mathcal{E}_S(w)}\tau_S(w)\|_{\mathrm{TV}} \le W_S, \qquad \mathrm{Re}[-E - \mathcal{E}(w) - \mathcal{E}_S(w)] \le C_0 N + q(S), \tag{5.cdxxiv}$$

and the weighted label bound

$$\sum_S e^{q(S)} W_S \le e^{C_1 N}. \tag{5.cdxxv}$$

The constants are independent of cutoff, volume and w in the specified source neighborhood. Then

$$|\widetilde{Z}(w)| \le e^{(C_0+C_1)N}. \tag{5.cdxxvi}$$

Indeed multiply each measure in eq. 5.cdxxiii by the common normalization, insert and remove its own $e^{\mathcal{E}_S(w)}$, take total variations, apply eq. 5.cdxxiv, and sum by eq. 5.cdxxv. This proof takes no logarithm of $Z(w)$ and asserts no absence of its complex zeros.

In a history construction, W_S is obtained by bounding the unsourced integrations after inserting a reserve factor such as $e^{cv(\omega)}$ on every local history. The ordinary source changes an island exponent by an extensive amount, $|\Psi_\omega(w)| \le C_r\, n(\omega)$, where sites are counted in their own scales. A sufficient reserve is $cv(\omega) \ge C_r n(\omega)$, uniformly in the history and the exterior arguments. This is an ordinary mass reserve for one presentation, and is distinct from estimates comparing marked or weighted derivatives of two different presentations.

It is essential that the majorant used for each component be a function of its own integration variables. A bound on its integral at each fixed exterior field is insufficient in a coupled product. For example, let both component factors be $f_1(x,y) = f_2(x,y) = \mathbf{1}_{|x-y|<\epsilon}$ on $[0,T]^2$. Each integral

of a factor in its own variable is at most 2ϵ, whereas the double integral of their product has order ϵT, not ϵ^2. A bound with the supremum over the admissible exterior pair *inside* the local integral supplies a component majorant; Tonelli then permits multiplication when the integration-variable groups are disjoint. The construction must also specify which Wilson factors are allotted to those majorants. An operation with no Wilson factor of its own has no independent plaquette-tail gain.

Neither eq. 5.cdxxiv nor eq. 5.cdxxv is a consequence of the last triangle inequality. They encode ordinary renormalization estimates: suppression of creation events, control of their genealogies, the allocation of Wilson and Gaussian factors without reuse, and payment of the missing vacuum normalizers of every large-field label. In particular, estimates with the volume of each fine lattice cannot simply be summed to obtain a bound with the volume N of the final lattice.

The floor from one positive label. Let $x \geq 1$ be the last inverse coupling and set $r = cx^{-1/2}$ with a fixed $0 < c \leq 1$. Write

$$\mathcal{B}_r = \{V : \|V_b - 1\|_{\mathrm{op}} < r \text{ for every bond } b\}.$$

Suppose the all-small-field label S_0 has a density with respect to final product Haar measure satisfying

$$d\tau_{S_0}(0)(V) \geq e^E \chi(V) \exp[-x\mathcal{A}(V) - C_I N]\, dV, \quad \chi = 1 \text{ on } \mathcal{B}_r, \quad \mathcal{A}(V) \leq C_A r^2 N \text{ on } \mathcal{B}_r. \tag{5.cdxxvii}$$

Here $\mathcal{A}$ may be the action of a minimizing background. If so, inclusion in the characteristic set and the displayed action bound are background estimates which must be verified; they do not follow merely by renaming $\mathcal{A}$ as a coarse action.

For normalized Haar measure there is an absolute $c_H > 0$ such that the operator-norm ball of radius $r \leq 1$ about the identity has measure at least $c_H r^3$. To prove this write $U = \cos\theta\, 1 + i \sin\theta\, n \cdot \sigma$, so $\|U - 1\|_{\mathrm{op}} = 2\sin(\theta/2)$ and normalized Haar measure is $(2\pi^2)^{-1} \sin^2\theta\, d\theta\, d\Omega(n)$. The interval $0 \leq \theta \leq r$ lies in this ball. Since $\sin\theta \geq 2\theta/\pi$ on $[0, 1]$, its measure is at least $8r^3/(3\pi^3)$. Decreasing the constant handles the irrelevant choice of an open instead of a closed ball.

Positivity in eq. 5.cdxxiii permits restriction to S_0, then to $\mathcal{B}_r$. Independence of the $4N$ Haar bonds gives

$$\begin{aligned} e^{-E} Z(0) &\geq e^{-(C_I + C_A c^2)N} (c_H c^3 x^{-3/2})^{4N} \\ &\geq \exp[-(C + 6\log x)N]. \end{aligned} \tag{5.cdxxviii}$$

Since $\mathcal{E}(0) = 0$, this is the required floor for $\widetilde{Z}(0)$. The constant depends on the fixed choices and on C_I, C_A, but not on cutoff or volume.

5.21.0.1. The terminal tests on a product of link balls. For the all-small-field label at the terminal level $n = K$ the determining sequence on the torus is constant. Its only data are the final bonds V. Its final label has no standing regions. The invariant form of the step gives the density

$$\chi_n(T_\ell)\, \exp\{E - x_n A(U_n(V)) + F_n(U_n(V))\}\, dV.$$

The equality of E with minus the empty label's vacuum sum holds at this depth. The pointed regular estimate on the all-small-field label has only one layer, at level n. For the level-j family let $h = n - j$. The tents at all its base points sum to one. Subtract its vacuum value and the corresponding tent-action term. The constant-strength cancellation gives a bound per base point of $CE_0[(1+h)L^{-\beta_{\mathrm{T}} h} + L^{-h}]L^{-4h}$. There are $L^{4h} N_n$ base points. Thus these terms sum to at most CN_n.

The vacuum-subtracted remainder has bound $C(1+h)L^{-4h}\theta_j^{-\kappa_0/2}$ in the rooted sum at a level-j site. Use the second-order estimate on the near liftable domains, and the norm tail for the other domains, including wrapping ones. The same site conversion gives $C(1+h)\theta_j^{-\kappa_0/2} N_n$. The lower flow bound and comparison of the frame with the flow make the sum over h bounded, since $\kappa_0/2 > 2$. On this empty standing label, the remainders include the terms left by healed histories and there

is no boundary term. The resulting $F_n \geq -C_I N_n$ and positivity justify keeping just this label in the lower bound.

We verify the remaining assertions of eq. 5.cdxxvii for this density. Write $\eta_n = L^{-n}$ in level-n units, and assume that $|V(b)-\mathbf{1}| < r$ on every level-n bond. Telescoping the unitary products gives $|\partial V - \mathbf{1}| < 4r$. For a density cube $\square$ take its actual test tower in $A = \square^{\sim 4}$. Its determining datum is obtained by lifting V from the top. To see the size of this datum, give a finer link the identity within a top block and the corresponding oriented $V(c)$ across a top block face. Each elementary plaquette of this lift is either trivial or has the holonomy of the corresponding top plaquette. The one-step average of this lift is its coarser lift: the two endpoint contours are trivial within each block, and every joining segment has the same single face value, or the identity if it crosses no top face. This verifies the same description on the averaged rows of the tower.

All determining plaquette holonomies, including those made using averages across the interfaces, are holonomies of the face lift. Thus the datum has radius $4r$. Apply the linear regularity theorem on the prescribed separated density tower, with its tied radii and with $4r$ below the datum ceiling. The tested plaquettes lie in $\square^{\sim}$, inside the top member of the tower by the test-cube geometry. Their defects are therefore at most $4B_3 r\eta_n^2$. The constant-sequence background of V satisfies the same bound on the whole torus. Take $r = c_0 x_n^{-1/2}$. On the real tube x_n and θ_n are comparable, and $\varepsilon_n\sqrt{\theta_n}$ is bounded below. A fixed c_0 therefore makes $4B_3 r < \varepsilon_n$ and places the datum and background below their respective regularity ceilings. Every factor of the terminal density test is one on the product of balls. The earlier fluctuation tests are accounted for in the integrated density and its regular and remainder totals.

There are $6N_0$ bare plaquettes and $N_0 = L^{4n}N_n$. Using the Wilson normalization gives

$$A(U_n(V)) \leq \tfrac{1}{2}(4B_3 r)^2 \eta_n^4\, 6N_0 = 48 B_3^2 r^2 N_n.$$

The action cost $x_n A(U_n(V))$ per final site is bounded. The four Haar balls per site give the $6\log x_n$ term in eq. 5.cdxxviii. At $n = 0$ the same conclusion follows directly by telescoping the bare plaquette products, with no background construction.

Local energy bounds and the first-passage window. Suppose the source families are actual functions of the source values in a fixed-layer enlargement X^+, holomorphic in its finite-dimensional source parameters and real at real arguments. Suppose their differences at the trivial field obey

$$|\Delta E_j(X,z;w)| \leq C_E e^{-\kappa d_j(X)}, \qquad |\Delta R_j(X;w)| \leq C_R e^{-\kappa d_j(X)}.$$

These are source-minus-unsourced differences: assume explicitly $\Delta E_j(X,z;0) = \Delta R_j(X;0) = 0$. Let the number of possible base points $z \in X$ be at most $C_M(d_j(X)+1)$, as follows for domains covered by the vertices of a cubical spanning tree. For $e_j(X;w) = \sum_{z\in X} \Delta E_j(X,z;w) + \Delta R_j(X;w)$,

$$|e_j(X;w)| \leq C_\delta e^{-(\kappa-\delta)d_j(X)} \qquad (\delta > 0). \tag{5.cdxxix}$$

This uses only $(m+1)e^{-\delta m} \leq C_\delta$. If the rooted entropy rate is $\kappa_D < \kappa$, choose $0 < \delta < \kappa - \kappa_D$. The resulting energy decay still exceeds entropy. Locality, holomorphy, realness and vanishing at zero pass through the finite base-point sum. Dependence on the source function itself is a structural hypothesis here; it is not implied by a bound on a chosen list of source coefficients.

If a first-passage argument supplies $g^{-2} < x \leq g^{-2} + B_\sharp$, then the constants in eq. 5.cdxxvi and eq. 5.cdxxviii, when bounded by $C(1+\log x)$, are uniform over the cutoff data at this fixed g. This is the final elementary step producing constants of the form C_g. It supplies no proof that an admissible run reaches that window.

What a pointwise polymer criterion would supply. There is a different sufficient mechanism when a construction actually has a pointwise hard-core gas. Let its polymers be nonempty subsets of a

set of N sites. Suppose, uniformly in a real coarse field V, its real activities obey

$$\sum_{\gamma' \not\sim \gamma} |z_{\gamma'}(V)| e^{a(\gamma')} \le a(\gamma), \qquad \sup_x \sum_{\gamma \ni x} |z_\gamma(V)| e^{a(\gamma)} \le b. \tag{5.cdxxx}$$

These inequalities remain valid after multiplication of all activities by any $t \in [0,1]$. The convergent cluster expansion makes its partition function $\Xi_V(t)$ nonzero and continuous on this interval. It is real there and $\Xi_V(0) = 1$; hence $\Xi_V(1) > 0$. The absolute cluster theorem bounds its logarithm by $\sum_\gamma |z_\gamma| e^{a(\gamma)} \le bN$: sum the rooted cluster bound and then use the nonempty site supports in eq. 5.cdxxx. The branch is the one continued from $t = 0$. It is real on the interpolation, and therefore gives

$$e^{-bN} \le \Xi_V(1) \le e^{bN}.$$

For complex source activities, the corresponding uniform criterion gives $|\Xi_V(w)| \le e^{bN}$. With a verified common Haar/Wilson integral representation and its action estimates, these facts could be used to prove stability and a floor.

Applying this lemma to a label/history representation requires an identity with that integral. Local polymer exponentiations on small-field regions, or exponentiation of the sum over healed components, do not by themselves provide a pointwise gas over all final coarse fields. For the history representation the concrete sufficient conditions are eq. 5.cdxxiv, eq. 5.cdxxv, and eq. 5.cdxxvii; verification of those ordinary conditions remains necessary for the construction.

5.22. **Source bounds for periodic Wilson states.** Reflection positivity transfers ordinary complex source bounds from finite Wilson tori to arbitrary periodic states through their DLR equations. The argument requires neither geometric forgetting for derivatives of inherited activities nor moment bounds for Gibbs states of a perturbed interaction.

Fix an odd integer $L \ge 3$, $a = L^{-K} \le 1$, a coupling $\beta > 0$, and $\mathcal{O}_a(f) = \sum_p f(x(p)) a_p$. All sums defining this observable are finite when f has compact support. Although its supremum norm may grow as a^{-4}, it is bounded at each fixed spacing. Every periodic Wilson DLR state is invariant under all lattice isometries and positive at all lattice mirrors by Prop. A.11.

5.22.1. *An exact domination lemma, before source estimates.* Use a cubic board of cells of side $l = L^j \ge a$, whose face planes cut links in half. Take cubic tori of side $\ell = 2L^m$ with $m \ge \max(0, j)$, having an even number of cells in every direction. Every sequence of sides tending to infinity in this family eventually accommodates any fixed coarser board of allowed side. All sequences in this section are taken in this family. When localization cubes have side Ma_s with the construction convention $M = L^{m_0}$, restrict to the cofinal subfamily $m \ge m_0$. Then $\ell/(Ma_s) = 2L^{m-m_0+s}$ is an even integer at every scale, and all localization partitions are retained. The cell algebra consists of bounded functions of bonds lying wholly inside the open cell; crossing bonds belong to no cell algebra. Reflected copies are conjugated when the reflecting isometry has negative determinant. For a cell function F define

$$p_{l,\ell}(F) = \left\langle \prod_C F^{[C]} \right\rangle_{a,\ell,\beta}^{1/N},$$
$$N = (\ell/l)^4. \tag{5.cdxxxi}$$

The number inside the root is nonnegative. To see this, split the board at one face and the opposite face. The disseminated product is $\Theta\Phi\,\Phi$ for the product of its copies in one half. Finite-torus reflection positivity applies.

We will use, including for complex functions,

$$\left|\left\langle \prod_C F_C \right\rangle\right| \le \prod_C p_{l,\ell}(F_C),$$
$$p(F+G) \le p(F) + p(G),$$
$$p(\bar{F}) = p(F),$$
$$p(1) = 1. \tag{5.cdxxxii}$$

Here are details extending the finite chessboard argument to this form. In one direction transport all slab profiles to one reference slab, using conjugated reflection for odd copies. For an arbitrary filling of the n slabs, n even, reflection Cauchy–Schwarz bounds the squared absolute partition integral by the two integrals obtained by retaining either half and reflecting it. Normalize label j by $(Z_j^{\mathrm{dis}} + \epsilon)^{1/n}$, where $Z_j^{\mathrm{dis}} \ge 0$ is its fully disseminated integral and $\epsilon > 0$. The normalized filling of maximum absolute value exists. Both reflected fillings also maximize whenever that maximum is positive: each is nonnegative, bounded by the same maximum, and their product bounds its square. A constant run in a maximizing filling can be doubled by retaining the half starting at that run. Once its length reaches $n/2$, a constant filling results. Every constant normalized filling is at most one. Thus the maximum is at most one; the zero maximum is trivial. Letting $\epsilon \downarrow 0$ proves the one-direction bound even when some disseminated integral is zero. Applying this argument successively to the four coordinate directions proves the first inequality in eq. 5.cdxxxii: coordinate reflections commute, and each successive dissemination copies the preceding pattern literally in the other coordinates. The resulting exponent for each original cell is $1/N$.

Expanding the disseminated product for $F + G$ and applying the first inequality term by term bounds it by $\sum_{k=0}^{N} \binom{N}{k} p(F)^k p(G)^{N-k}$. Its nonnegative Nth root proves the triangle inequality. Exactly half the copies are conjugated, giving $p(zF) = |z|p(F)$; conjugating the entire nonnegative disseminated integral proves $p(\bar{F}) = p(F)$. In particular $p(\mathrm{Re}(e^{-i\phi}F)) \le p(F)$.

5.22.1.1. Domination. For any periodic Wilson DLR state ρ, any finite collection of occupied cells, and any sequence of compatible tori $\ell_k \to \infty$,

$$\left|\rho\left(\prod_C F_C\right)\right| \le \liminf_{k\to\infty} \prod_C p_{l,\ell_k}(F_C). \tag{5.cdxxxiii}$$

Only finitely many factors differ from 1.

First consider a bounded real function H of one cell. Sum its mirror copies over a cube of side ℓ. Reflection invariance of ρ makes the expectation of the sum equal to $N\rho(H)$. Conditional Jensen bounds that expectation by the supremum, over exterior configurations, of the logarithm of its conditional exponential moment. Replace the exterior condition by periodic bonds. After adjoining dummy independent Haar variables if necessary, the two integrals use the same variables. Only boundary plaquettes change; each changes the Hamiltonian by at most 2β, and there are $O((\ell/a)^3)$ such plaquettes. Source factors do not cross the outer cell faces and therefore agree in the two integrals. It follows that

$$\rho(H) \le \log p_{l,\ell}(e^H) + C\beta l^4 a^{-3} \ell^{-1}. \tag{5.cdxxxiv}$$

This estimate is uniform in the exterior configuration, not in a; the volume limit is taken with a fixed.

For bounded real G, put $H = \log(1 + tG)$ with $0 < t < (2\|G\|_\infty)^{-1}$. The elementary inequalities $\log(1 + tG) \ge tG - Ct^2\|G\|_\infty^2$ and $\log p(1 + tG) \le tp(G)$ turn eq. 5.cdxxxiv, first as $\ell_k \to \infty$ and then as $t \downarrow 0$, into $\rho(G) \le \liminf_k p_{l,\ell_k}(G)$. For complex F choose ϕ so that $\rho(\mathrm{Re}(e^{-i\phi}F)) = |\rho(F)|$ and use the last consequence of eq. 5.cdxxxii.

Finally enclose the finitely many occupied fine cells strictly inside one larger coarse board cell. The one-cell inequality applies to $G = \prod_C F_C$. On every compatible torus the fine-board chessboard inequality gives $p_{\mathrm{coarse},\ell}(G) \le \prod_C p_{l,\ell}(F_C)$: disseminate the coarse cell, apply eq. 5.cdxxxii on the fine board, and take the coarse root. Choose the larger cell as a sufficiently large allowed integral power of the fine side, so that both boards are compatible along the chosen torus sequence. This proves eq. 5.cdxxxiii. The lower limit along *any* sequence is essential below.

The perturbation $\log(1 + tG)$ in this proof is arbitrary and need not be an admissible linear source. No renormalization-group bound was applied to it or to its Gibbs states. It enters only an exact Jensen and boundary-comparison argument and the elementary seminorm inequality.

The precise ordinary torus hypothesis. Here is one sufficient source hypothesis, stated to expose the actual input to the transfer. Fix constants $r, A, C_\pm > 0$, an entropy rate κ_0, and $\kappa > \kappa_0$, uniformly over the selected cutoff–coupling data (a, β) and allowed tori. Let $\|f\|_\sharp = \max(\|f\|_\infty, c_M\|\nabla f\|_\infty)$. For every finite family of real Lipschitz torus functions require the ordinary complex source bound, on $\sum_i |z_i|\|f_i\|_\sharp < 2r$,

$$|Z_{a,\ell}(W)| \le \exp\left[V_{a,\ell} + \operatorname{Re} U_{a,\ell}(W) + \operatorname{Re}\sum_{s=0}^{K}\sum_{D} e_{a,\ell,s,D}(W) + C_+\ell^4\right],$$
$$Z_{a,\ell}(0) \ge e^{V_{a,\ell} - C_-\ell^4}. \tag{5.cdxxxv}$$

The terms in this expression have the following meaning and bounds. The source-independent vacuum number $V_{a,\ell}$ is real.

The point counterterm is $U_{a,\ell}(W) = \sum_p u_{a,\ell,p}(W(x(p)))$, where each u is holomorphic and zero at zero. For each fixed cutoff its norm on smaller discs is bounded uniformly in p and ℓ; this bound may depend on the cutoff. The domain term $e_{a,\ell,s,D}$ is holomorphic in the ordinary source variables, is zero at zero, depends only on $W|_{D^+}$, and satisfies $|e_{a,\ell,s,D}| \le Ae^{-\kappa m(D)}$. Put $a_s = L^{-s}$ and require $\operatorname{diam} D^+ \le C_0 a_s(m(D) + 1)$. The domain entropy bound, in a convenient localized form, is

$$\sum_{D:\, D^+\cap B\neq\varnothing,\, m(D)\ge t} e^{-\kappa m(D)} \le C(R+1)^4 a_s^{-4} e^{-ct} \tag{5.cdxxxvi}$$

for every radius-R ball B and $t \ge 0$, uniformly in its position. A count per unit cube bounded by $e^{\kappa_0 m}$ implies this form after absorbing a fixed polynomial in m into a smaller entropy gap. All domain sums used are normally convergent on smaller source balls. Source representation independence is required: local terms depend on the source function on their indicated region, not on the particular list representing that function.

The condition in eq. 5.cdxxxv is an upper bound and a floor, not a claim that $Z(W)$ is zero-free on the whole source ball. Its ordinary source class includes the global real Lipschitz profiles constructed by mirror periodization. This last inclusion matters. For $w = \sum_i z_i f_i$ in one cell, the periodized source is

$$W^{\mathrm{ch}} = \sum_i z_i H_i^+ + \sum_i \bar{z}_i H_i^-. \tag{5.cdxxxvii}$$

The even and odd copy profiles $H_i^\pm$ are real. If each support is at distance $b > a$ from the faces, the profiles vanish near the faces and their Lipschitz norms are at most $\|f_i\|_\sharp$. Hence $\|W^{\mathrm{ch}}\|_\flat \le 2\sum_i |z_i|\|f_i\|_\sharp$, independently of the number of cells. The ordinary radius $2r$ in eq. 5.cdxxxv therefore supplies the radius r used below. One first regards the two lists of coefficients as independent

complex variables; evaluating one at the conjugate of the other does not require a new analytic extension.

Holomorphic cell normalizers from the ordinary hypothesis. Suppose m radius-R support balls lie at distance at least a fixed $b > 1$ from the faces of a cell C of side l. For each torus form a normalizer $E_{C,\ell}(z)$ as follows. Retain in $U + \sum e$ only the point terms and domains whose complete dependence region lies in one open board cell. Evaluate each retained term at the source copy in that cell, conjugate its value if that cell has odd parity, sum over cells, and divide by N. The result is holomorphic in z, even though eq. 5.cdxxxvii itself involves $\bar{z}$: on an odd cell the retained term has the form $g(\bar{z})$, and $\overline{g(\bar{z})}$ is holomorphic. Its real part is unchanged by this outer conjugation. Also $E_{C,\ell}(0) = 0$.

Any discarded source-dependent domain meets a support ball and reaches a cell face. Its diameter is at least b, so $m(D) \geq b/(C_0 a_s) - 1$. Counting once per ball, eq. 5.cdxxxvi shows that the discarded contribution divided by N is at most

$$B_{\mathrm{face}}(m, R, b) \leq CAm(R+1)^4 \sum_{s \geq 0} a_s^{-4} e^{-c'b/a_s} < \infty. \tag{5.cdxxxviii}$$

The sum is independent of the cutoff. Overcounting domains meeting several balls only enlarges this upper bound. Point counterterms whose source is nonzero lie strictly inside cells, so none are discarded. Combining eq. 5.cdxxxi, eq. 5.cdxxxv and eq. 5.cdxxxviii proves

$$\begin{aligned} p_{l,\ell}(e^{\mathcal{O}_a(w)}) &\leq \exp[\mathrm{Re}\, E_{C,\ell}(z) + B_C], \\ B_C &= (C_+ + C_-) l^4 + B_{\mathrm{face}}(m, R, b). \end{aligned} \tag{5.cdxxxix}$$

At a fixed cutoff the normalizers are locally bounded uniformly in ℓ. For the point terms this follows from their fixed-cutoff bound and $O(l^4 a^{-4})$ plaquettes per cell. For the domain terms use eq. 5.cdxxxvi with $t = 0$ and the finite sum of a_s^{-4} over $s \leq K$. These bounds may grow with K, which is harmless for the next step. Montel's theorem therefore gives a subsequence on which $E_{C,\ell}$ converges locally uniformly to a holomorphic E_C. Explicitly, Cauchy estimates on smaller source balls give equicontinuity there; Arzela–Ascoli and a diagonal selection over a countable compact exhaustion give the subsequence, and Cauchy's formula proves holomorphy of its limit. For finitely many occupied cells take one common subsequence. It is chosen from the torus normalizers alone, independently of every infinite-volume state.

There is a second uniform bound that does not grow with K. If $n \geq 2$ sources in the same cell have pairwise support distance at least d, a point counterterm has zero mixed derivative, because a point cannot belong to two of the supports. A contributing domain must meet all supports; in particular its diameter is at least d. Cauchy's formula and eq. 5.cdxxxvi, with the first support ball as anchor, give

$$\begin{aligned} |\partial_1 \cdots \partial_n E_C(0)| &\leq D_n(d; R) \prod_i \|f_i\|_\sharp, \\ D_n(d; R) &= CA(R+1)^4 \max(1, d^{-4})(n/r)^n. \end{aligned} \tag{5.cdxl}$$

Indeed the remaining scale sum is $\sum_s a_s^{-4} e^{-c'd/a_s} \leq C \max(1, d^{-4})$. This follows by splitting at $a_s \approx d$; the tail becomes a convergent sum $Cd^{-4} \sum_{j \geq 0} L^{4j} e^{-c'' L^j}$. The same estimate holds before the Montel limit, and hence after it.

All-state generating functions, cumulants and mean differences.

Proposition 5.75 (Ordinary torus source estimates imply all-state bounds)**.** Assume the ordinary torus source hypothesis in eq. 5.cdxxxv–eq. 5.cdxxxvi, including the stated locality, point counterterms and uniformity, and the proved lattice symmetries. Then eq. 5.cdxliii, eq. 5.cdxliv and eq. 5.cdxlv hold for every periodic Wilson DLR state, uniformly in the cutoff, the states and the support positions. Their constants depend only on the displayed source constants, the number of entries, the support radius and, for higher cumulants, the support separation. ■

Let $f_1, \dots, f_n$ have radius-R supports at arbitrary positions. A zero test makes every corresponding multilinear assertion trivial, so such tests may be omitted. Choose an allowed side $l \le C_L n(R+b+1)$ and a board offset that puts each support ball at distance at least b from every face. For each coordinate, the forbidden offsets occupy at most $2n(R+b+a)$ of a period; choosing l larger than a fixed multiple of this number leaves an allowed link-plane offset. Thus at most n cells are occupied. Put $w_C = \sum_{i:\,\mathrm{supp}\, f_i \subset C} z_i f_i$. Apply eq. 5.cdxxxiii to $F_C = e^{\mathcal{O}_a(w_C)}$ along the common Montel subsequence in eq. 5.cdxxxix. For every periodic DLR state ρ this gives

$$\begin{aligned} \left| e^{-E(z)} \rho\left(e^{\sum_i z_i \mathcal{O}_a(f_i)} \right) \right| &\le Q, \\ E &= \sum_C E_C, \\ Q &= e^{\sum_C B_C}, \\ \sum_i |z_i| \|f_i\|_\sharp &< r. \end{aligned} \tag{5.cdxli}$$

The same E and Q serve all states at the datum. The number Q depends on n, R and the fixed source constants, but not on the cutoff, state, or support positions. This conclusion uses the lower limit along the selected sequence, rather than an upper limit over all tori.

Write the normalized holomorphic function in eq. 5.cdxli as h_ρ. It has $h_\rho(0) = 1$ and $|h_\rho| \le Q$. A logarithm is needed only near zero, where it exists by continuity. For any nonempty subset $B \subset \{1, \dots, n\}$, Cauchy's formula, with all other variables zero, gives

$$|\partial_B h_\rho(0)| \le Q(|B|/r)^{|B|} \prod_{i \in B} \|f_i\|_\sharp.$$

The finite derivative identity for a logarithm now implies

$$|\partial_1 \cdots \partial_n \log h_\rho(0)| \le r^{-n} \sum_\pi (|\pi| - 1)! Q^{|\pi|} \prod_{B \in \pi} |B|^{|B|} \prod_i \|f_i\|_\sharp. \tag{5.cdxlii}$$

The sum is over set partitions. No uniform zero-free polydisc is needed for this identity at zero.

For $n \ge 2$, the cumulant is the sum of the mixed derivatives of E and $\log h_\rho$. If more than one cell is occupied, the mixed derivative of $E = \sum_C E_C$ is zero because every summand lacks a variable. If one cell is occupied, use eq. 5.cdxl. eq. 5.cdxl–eq. 5.cdxlii prove the all-state separated cumulant bound

$$|\kappa_n^\rho(\mathcal{O}_a(f_1), \dots, \mathcal{O}_a(f_n))| \le C_n(d; R) \prod_i \|f_i\|_\sharp. \tag{5.cdxliii}$$

Here C_n is explicit as eq. 5.cdxlii, with the uniform choice of Q above, plus D_n. This proves a finite constant for each n; no factorial growth assertion beyond these formulas is being made.

For one radius-R test choose one cell with $l \le C_L(R+b+1)$ and the corresponding common normalizer E_f. Cauchy directly gives

$$|\rho(\mathcal{O}_a(f)) - E_f'(0)| = |h_\rho'(0)| \le (Q_R/r)\|f\|_\sharp.$$

Consequently, for every pair of periodic states at the same datum,

$$|\rho(\mathcal{O}_a(f)) - \nu(\mathcal{O}_a(f))| \le (2Q_R/r)\|f\|_\sharp. \tag{5.cdxliv}$$

The common derivative $E_f'(0)$ may diverge with the cutoff. The estimate controls its differences from the means and hence differences between the means. The argument also covers separated reflected copies: their support radii, distances and test norms are unchanged.

If desired the one-point constant can be chosen to grow only as $(R+1)^4$. Choose a fixed smooth partition of unity by translates of compactly supported functions of uniformly bounded derivative. Its members have supports in balls of a fixed radius R_0, only $C(R+1)^4$ meet a radius-R support, and $\|f\psi_j\|_\sharp \le C\|f\|_\sharp$. Apply eq. 5.cdxliv to these pieces with the fixed constant Q_{R_0} and add. Such a partition is obtained by dividing a nonnegative compact bump, positive on a unit cube, by the positive periodic sum of its integer translates.

5.22.1.2. Short-distance differences between states. At a common first-passage datum let D_n be the difference of the connected n-point functions of two periodic states. For tests in a common unit ball, eq. 5.cdxli uses the same energy E in both states. The source bound admits overlapping supports. Thus the mixed derivative of E cancels, and eq. 5.cdxlii gives

$$|D_n(F_1,\dots,F_n)| \le C_n \prod_i \|F_i\|_\sharp.$$

Take $a_K \le \mathsf{d}$ and put the supports of the f_i in a ball of radius $5\mathsf{d}$. For small enough d, pack $N \ge c\mathsf{d}^{-4}$ disjoint lattice translates of this ball inside a unit ball, with gaps between the copies. For each argument $i < n$ use an independent family of uniform signs ϵ_v^i on the translates. In the last argument use $\epsilon_v^n = \prod_{i<n}\epsilon_v^i$. Write $F_i = \sum_v \epsilon_v^i \tau_v f_i$. Since every summand vanishes at the boundary of its own ball, both the supremum and Lipschitz bounds of f_i hold for F_i. In the multilinear expansion the sign average keeps exactly the terms whose translations all coincide. Translation invariance makes that average $ND_n(f_1,\dots,f_n)$. For $n=1$ take the ordinary sum with all signs one. Therefore

$$|D_n(f_1,\dots,f_n)| \le C_n\mathsf{d}^4 \prod_i \|f_i\|_\sharp \le C_{n,\vartheta}\mathsf{d}^{4-n}\prod_i \|f_i\|_\infty,$$

where the last bound uses $\mathrm{Lip}\, f_i \le \|f_i\|_\infty/(\vartheta\mathsf{d})$.

Finally put $b_1 = C_1(R)$ and $b_m = C_m(d;R)$ for $m \ge 2$. The moment–cumulant formula and invariance of higher cumulants under constant shifts give

$$\left|\rho\prod_{i=1}^k[\mathcal{O}_a(f_i) - \nu(\mathcal{O}_a(f_i))]\right| \le \left(\sum_{\pi\in\mathrm{Part}\{1,\dots,k\}}\prod_{B\in\pi} b_{|B|}\right)\prod_i \|f_i\|_\sharp. \tag{5.cdxlv}$$

Singleton blocks use eq. 5.cdxliv; the other blocks use eq. 5.cdxliii. This is the centered moment estimate required when a component is centered at a different state or mixture, with the same uniformity in both states.

Thus eq. 5.cdxxxv–eq. 5.cdxxxvi, source locality, the stated ordinary holomorphy, and the proved lattice symmetries imply the bounds used under G4. The transfer requires no weighted analytic extension. The analytic construction must still prove the actual ordinary source bound, entropy gap, upper stability and floor. A torus statement controlling only separated mixed derivatives of a logarithm does not state the cell estimate in eq. 5.cdxxxix or a common one-point normalizer. Those concrete conditions must be retained when replacing an all-state input by the transfer theorem.

5.23. **Factorial moment bounds from rooted domains.** We strengthen the spatial counting used in the ordinary-source transfer. The resulting estimates have factorial growth in the number of entries and a separation singularity of order proportional to that number. Only the ordinary finite-torus source estimates are used. In particular, neither weighted forgetting nor source bounds for perturbed Gibbs states are added.

Retain the ordinary source upper bound and floor in eq. 5.cdxxxv, the point-local counterterms, and the proved chessboard domination in eq. 5.cdxxxiii. Let $L \geq 3$ be odd, $a = L^{-K}$, and $a_s = L^{-s}$, $0 \leq s \leq K$. All constants below are uniform over the chosen cutoff–coupling data and all large allowed tori. The physical source norm is $\|f\|_\sharp = \max(\|f\|_\infty, c_M\|\nabla f\|_\infty)$.

The stronger count and its geometric origin. The localization parameter obeys $M = L^{m_0}$ for a fixed integer $m_0 \geq 0$. Use the cofinal family of original tori $\ell = 2L^m$, $m \geq m_0$. At every actual scale, $\ell/(Ma_s) = 2L^{m-m_0+s}$ is an even integer, so all localization partitions tile the torus; no scale terms are omitted. Every fixed power-of-L board also tiles all sufficiently large tori in this family, with even cell counts. At scale s use a cubical partition of side $\rho_s = Ma_s$. A localization domain X is a union of these cubes. It admits a finite connected cubical edge tree meeting every cube of X, and $m(X)$ is the minimum number of edges of such a tree. The source dependence region is contained in a fixed number of neighbouring cube layers around X. This includes source values at plaquette barycentres whose plaquette has a corner in a one-layer enlargement of X, because a bare plaquette has diameter $O(a) \leq O(a_s)$.

For a fixed cube Q the elementary rooted entropy estimate is

$$\#\{X : Q \subset X,\, m(X) = m\} \leq Ce^{\kappa_D m}. \tag{5.cdxlvi}$$

Here is a proof for the stated domain indexing. A minimizing tree with m edges has at most $m+1$ vertices. Every cube meeting an edge of the cubical grid contains an endpoint of that edge; hence every cube of X contains a tree vertex. Choose a tree vertex at a corner of Q in at most 16 ways. Traversing each tree edge twice gives a closed walk of length $2m$ from that corner. There are at most 8^{2m} such walks. At most $16(m+1)$ cubes contain a vertex of the walk, and a subset of these specifies X. Thus

$$\# \leq 16\, 8^{2m} 2^{16(m+1)}.$$

This covers all admissible domains; restricting the domain class can only reduce the count. The same upper bound applies on the finite cubical quotient.

Suppose $|e_{s,X}(w)| \leq Ae^{-\kappa m(X)}$, where $\kappa > \kappa_D$. Summing eq. 5.cdxlvi gives a positive entropy gap $c > 0$. If the dependence region sees a radius-u ball, some cube of X lies within $C\rho_s$ of that ball. There are at most $C(1 + u/a_s)^4$ choices of this root cube. Consequently

$$\sum_{X:\, X^+ \text{ sees } B,\, m(X)\geq t} e^{-\kappa m(X)} \leq C(1 + u/a_s)^4 e^{-ct}. \tag{5.cdxlvii}$$

There is no factor a_s^{-4} when $u \ll a_s$. This improvement uses the fixed-layer dependence geometry and rooted indexing; a bound only by $C(u+1)^4 a_s^{-4}$ would not suffice for the argument below. Also, summing over every possible root cube, with harmless overcounting, gives the global estimate

$$\sum_X e^{-\kappa m(X)} \leq C\ell^4 a_s^{-4}. \tag{5.cdxlviii}$$

The tree representation implies $\operatorname{diam} X^+ \leq Ca_s(m(X) + 1)$, as needed in these estimates.

A polynomial bound for one finely supported cell source. Let a board have side $\eta \le 1$ and one real test v supported in a ball of radius $\delta \le \eta$, lying at distance at least δ from the faces. Assume $\delta \ge 16a$. Periodize zv by reflected copies, conjugating odd copies. Its ordinary source norm is at most $2|z|\|v\|_\sharp$. Thus ordinary source radius $2r$ allows $|z| < r/\|v\|_\sharp$, independently of the number of cells.

Construct the interior-cell holomorphic normalizer $E_{v,\ell}(z)$ as in eq. 5.cdxxxix. The discarded terms are the source-dependent domains whose full dependence region is not contained in one cell. Such a domain sees a copy of the support ball and reaches a face, so $m(X) \ge c_0\delta/a_s - C_0$. Let $N = (\ell/\eta)^4$. The discarded sum divided by N has three scale ranges.

For $a_s \ge \eta$, discard even the source restriction and use eq. 5.cdxlviii. The cost is at most $CA(\eta/a_s)^4$ per scale and per cell; its sum is bounded by a geometric series. For $\delta \le a_s < \eta$, eq. 5.cdxlvii gives a bounded cost per scale, and there are at most $C(1+\log(\eta/\delta))$ such scales. For $a_s < \delta$, eq. 5.cdxlvii and the forced domain size give a cost at most

$$CA(1+\delta/a_s)^4 e^{-c'\delta/a_s}.$$

The sum is bounded uniformly in δ by comparison with $\sum_{j\ge 0}(1+L^j)^4 e^{-c''L^j}$. All ranges may be truncated at the actual cutoff, which only decreases the bounds. The point-local counterterms are retained: their nonzero sources are strictly inside cells. Their possibly large first derivatives cause no face error.

The stability and floor cost per cell is $(C_+ + C_-)\eta^4 \le C_+ + C_-$. We obtain

$$p_{\eta,\ell}(e^{z\mathcal{O}_a(v)}) \le \exp[\mathrm{Re}\, E_{v,\ell}(z)]\, Q_0(\eta/\delta)^p \tag{5.cdxlix}$$

for constants $Q_0 \ge 1$, $p \ge 0$ independent of a, η, δ and the support position. Fixed-cutoff local boundedness and Montel selection supply a limiting holomorphic univariate normalizer E_v. For any finite list of pieces use one common subsequence. It is chosen from the torus data and is independent of the infinite-volume states.

Board offsets and a genuine product source polydisc. Consider n tests supported in radius-R balls, with distinct supports at distance at least $d > 0$. Put $\bar{d} = \min(d, 1)$, and choose an allowed board side, a power of L, such that

$$\frac{\bar{d}}{16L} < \eta \le \frac{\bar{d}}{16},$$
$$\delta = \frac{\eta}{64n}. \tag{5.cdl}$$

First suppose $a \le \delta/16$; thus this board side is a multiple of the mesh. A fixed smooth partition of unity at mesh proportional to δ splits each f_i into pieces $v_{i,k}$ with

$$\#\{k : v_{i,k} \ne 0\} \le C(1+R/\delta)^4,$$
$$\mathrm{supp}\, v_{i,k} \subset B_{i,k}(\delta),$$
$$\|v_{i,k}\|_\sharp \le C\delta^{-1}\|f_i\|_\sharp. \tag{5.cdli}$$

For example, scale a smooth compact partition obtained from translates of a bump divided by their positive periodic sum. Scaling its bounded first derivatives gives the last inequality. A zero piece contributes nothing and is omitted.

Choose one piece from each original test. In any coordinate the offsets for which one enclosing radius-δ ball comes within δ of a face occupy an interval of length at most 4δ modulo η. On the

discrete link-plane offsets, one such interval contains at most $4\delta/a+2$ candidates. The n intervals exclude at most $n(4\delta/a+2)<\eta/a$ candidates by eq. 5.cdl and the fine-mesh assumption. There is an allowed offset in each coordinate. Hence all chosen balls have the required margin. Different original tests occupy different cells: a four-dimensional side-η cube has Euclidean diameter $2\eta<d$, whereas their nonempty supports are d apart.

Write these chosen pieces as $v_1,\dots,v_n$ and their different cells as $C_1,\dots,C_n$. By eq. 5.cdxlix each has a univariate normalizer E_i and a bound

$$\begin{aligned} Q_i &\le Cn^p, \\ r_i &= r/\|v_i\|_\sharp. \end{aligned} \tag{5.cdlii}$$

Product domination in eq. 5.cdxxxiii is applied to the separate cell factors. It therefore gives, for every subset S and every periodic state ρ,

$$\left|\exp\left[-\sum_{i\in S}E_i(z_i)\right]\rho\left(e^{\sum_{i\in S}z_i\mathcal{O}_a(v_i)}\right)\right|\le\prod_{i\in S}Q_i \qquad (|z_i|<r_i, i\in S). \tag{5.cdliii}$$

This is a product polydisc; no bound on its combined source norm is needed. Each chessboard factor in eq. 5.cdxxxiii uses only one periodized piece and its own ordinary source disc. Inactive factors are exactly 1, with $p(1)=1$, so eq. 5.cdliii contains the product over the active subset, not the product over all n indices. The same normalizers work for every state, including a second centering state ν.

Common centering and factorial order. Define $X_i=\mathcal{O}_a(v_i)-E_i'(0)$. Differentiating the normalized function in eq. 5.cdliii once in each active variable, and using product Cauchy, gives

$$|\rho\prod_{i\in S}X_i|\le\prod_{i\in S}(Q_i/r_i). \tag{5.cdliv}$$

Only the first derivative of each univariate E_i occurs; no bound on its higher derivatives is required. Applying the one-variable case to ν gives $|\nu(\mathcal{O}_a(v_i))-E_i'(0)|\le Q_i/r_i$. Expanding the product centered at ν over its active subsets and using eq. 5.cdliv therefore proves

$$\left|\rho\prod_{i=1}^n[\mathcal{O}_a(v_i)-\nu(\mathcal{O}_a(v_i))]\right|\le 2^n\prod_i(Q_i/r_i). \tag{5.cdlv}$$

This explains both the necessity of subset bounds in eq. 5.cdliii and the role of a common normalizer. Uniform moments for auxiliary perturbed Gibbs states are not being presumed.

The sum in eq. 5.cdlv over the piece choices from eq. 5.cdli, using linearity of centering. The board may depend on that finite piece choice; every constant is independent of that choice. Since $\delta\le 1$,

$$\begin{aligned} \left|\rho\prod_{i=1}^n[\mathcal{O}_a(f_i)-\nu(\mathcal{O}_a(f_i))]\right| &\le [C(R+1)^4n^p\delta^{-5}]^n\prod_i\|f_i\|_\sharp \\ &\le B_0(R)^n n^{(p+5)n}\bar{d}^{-5n}\prod_i\|f_i\|_\sharp. \end{aligned} \tag{5.cdlvi}$$

If $a>\delta/16$, dispense with the partition and the board. A radius-R ball contains at most $C(1+R/a)^4$ plaquette barycentres. Since $0\le a_p\le 2$,

$$\|\mathcal{O}_a(f)-\nu(\mathcal{O}_a(f))\|_\infty\le C(R+1)^4a^{-4}\|f\|_\sharp\le C(R+1)^4\delta^{-4}\|f\|_\sharp.$$

Its product bound is absorbed by eq. 5.cdlvi, because δ is comparable to $\bar{d}/n$, $p \geq 0$ and $\bar{d} \leq 1$. This handles every cutoff, including the case when the prospective side η is smaller than the mesh.

Choose the fixed integer $k = \lceil p \rceil + 5$. The elementary inequality $n! \geq (n/e)^n$, obtained by integrating $\log x$, converts eq. 5.cdlvi to

$$\left| \rho \prod_{i=1}^{n} [\mathcal{O}_a(f_i) - \nu(\mathcal{O}_a(f_i))] \right| \leq B(R)^n (n!)^k \max(1, d^{-5n}) \prod_i \|f_i\|_\sharp. \tag{5.cdlvii}$$

Proposition 5.76 (Graded moments and cumulants)**.** Under the ordinary torus source hypotheses and the rooted domain geometry stated above, eq. 5.cdlvii holds for every pair of periodic Wilson DLR states at the same datum. The same moment and cumulant bounds hold for finite Wilson tori in the compatible family above a fixed minimum side, with the integrating and centering state equal to that torus law. For $n \geq 2$ their cumulants satisfy

$$|\kappa_n^\rho(\mathcal{O}_a(f_1), \ldots, \mathcal{O}_a(f_n))| \leq [2B(R)]^n (n!)^k \max(1, d^{-5n}) \prod_i \|f_i\|_\sharp. \tag{5.cdlviii}$$

For the finite-torus assertion, use the univariate normalizers $E_{i,\ell}$ directly. The finite chessboard inequality gives eq. 5.cdliii with those normalizers and the same Q_i on the product polydisc. The remaining estimates use the torus distance for the support separation. They are finite and use no volume limit. The chosen sub-unit power-of-L board tiles every compatible torus above the fixed minimum side.

The constants are independent of the cutoff, the states and support positions. The exponent k is one fixed finite integer. One may take $B(R) = C(R+1)^4$ with a constant fixed by the source and domain parameters. ∎

To verify the cumulant assertion without an unnecessary extra factorial, center at ρ and apply the moment–cumulant formula. For a partition with block sizes $n_1, \ldots, n_m$, use eq. 5.cdlvii on each block and $\prod_j n_j! \leq n!$. The remaining combinatorial bound is

$$\sum_\pi (|\pi| - 1)! \prod_{D \in \pi} (|D|!)^k \leq (n!)^{k-1} \sum_\pi (|\pi| - 1)! \prod_{D \in \pi} |D|! \leq 2^{n-1} (n!)^k.$$

For exactly m blocks, the middle sum equals $\frac{n!}{m}\binom{n-1}{m-1}$: order the blocks, sum over positive compositions of n, and cancel their factorial denominators. Summing over m proves the inequality and eq. 5.cdlviii.

The separation singularity in this theorem is d^{-5n}, not a single d^{-5} prefactor independent of n. This is a factorial-growth estimate at each fixed separation, with diagonal order proportional to n. It is sufficient for extensions whose derivative order may grow linearly with the number of entries. A stronger separation law requires an additional argument. Likewise the theorem does not prove the ordinary analytic source and stability hypotheses themselves.

One may also recover a presentation with a separation prefactor independent of n, at the cost of one factorial power. Put $x = \max(1, d^{-5})$. The exponential series gives $x^n \le n!e^x$, so eq. 5.cdlvii implies

$$\left|\rho \prod_{i=1}^{n} [\mathcal{O}_a(f_i) - \nu(\mathcal{O}_a(f_i))]\right| \le e^{\max(1,d^{-5})} B(R)^n (n!)^{k+1} \prod_i \|f_i\|_\sharp. \tag{5.cdlix}$$

Thus $A(d;R) = e^{\max(1,d^{-5})}$ is one possible prefactor in that notation. Its separation behavior is much worse than a polynomial. The quantitative extension uses the sharper graded form in eq. 5.cdlvii, rather than attempting to infer diagonal order from this rewritten bound.

5.24. **Compactness of separated correlations.** A sequence of cutoff data with uniform source estimates has a subsequence whose separated correlations converge. We also prove existence of Wilson Gibbs states at each fixed datum. Constructing the cutoff data and their uniform estimates, comparing different cutoffs, tuning, and proving uniqueness of continuum limits require further arguments.

The fixed-cutoff Gibbs measure argument. Fix a spacing $a > 0$, a finite inverse coupling $\beta \ge 0$, and the compact group $G = \mathrm{SU}(2)$. Choose one orientation of each nearest neighbor bond of $a\mathbb{Z}^4$ and denote this countable set by E. A configuration is $U \in \mathcal{X} = G^E$; the opposite bond has matrix U_e^{-1}. The product topology is compact and metrizable. For example, enumerating the bonds gives a metric $\sum_{r\ge1} 2^{-r} \min(1, \|U_{e_r} - V_{e_r}\|)$. Continuous functions depending on finitely many bonds are uniformly dense in $C(\mathcal{X})$. Indeed, replacing all coordinates after the first N by a fixed configuration changes the metric by at most 2^{-N}, and uniform continuity proves this approximation.

Let $A_p(U) = 1 - \frac{1}{2}\operatorname{Re}\operatorname{Tr} U_p \in [0,2]$ be the Wilson plaquette action. For a finite bond window W, let $\mathcal{P}(W)$ be the plaquettes meeting W and let ∂W be their bonds outside W. Define

$$\gamma_W F(U) = \frac{\int_{G^W} F(VU) \exp[-\beta \sum_{p\in\mathcal{P}(W)} A_p(VU)]\, dV}{\int_{G^W} \exp[-\beta \sum_{p\in\mathcal{P}(W)} A_p(VU)]\, dV}. \tag{5.cdlx}$$

Here VU replaces the variables in W, and Haar measure is normalized. The denominator lies between $e^{-2\beta|\mathcal{P}(W)|}$ and one. The normalized density is therefore a continuous function of the finite variables $(V, U_{\partial W})$. If F is continuous local, $\gamma_W F$ is continuous local, depending only on the remaining variables of F and on ∂W. This proves the Feller property directly; integration of a uniformly continuous function over the compact finite product preserves continuity. By uniform approximation it also holds for $F \in C(\mathcal{X})$.

Take Wilson probability measures on periodic tori with integer periods $N \to \infty$, and lift them periodically to $\mathcal{X}$. The finite partition function is finite and strictly positive because the integrand is continuous and positive on a compact space. The lifted measures have a weakly convergent subsequence. One direct compactness argument is as follows. The space $C(\mathcal{X})$ is separable: on a compact metric space finite approximations by distance functions with rational coefficients form a countable uniformly dense family. Successively extract convergence of the integrals on this family. Uniform approximation then defines a bounded positive functional on all of $C(\mathcal{X})$, with value one at the constant one. Its Riesz probability measure is the weak limit. This argument simultaneously gives convergence against every continuous function, not merely the chosen countable family.

The lifted torus measure is not itself an infinite-volume DLR state: remote periodic copies can determine an inside variable. The identity needed for passage to the limit is a local one. Fix W

and a continuous local F. For all sufficiently large periods, the finite bonds involved in W, ∂W and F embed injectively into the torus, and the plaquettes meeting W are precisely their usual torus counterparts. Integrating the variables in that torus window first gives exactly

$$\widetilde{\mu}_N(\gamma_W F) = \widetilde{\mu}_N(F). \tag{5.cdlxi}$$

Both integrands are continuous local. Their weak limits satisfy $\mu(\gamma_W F) = \mu(F)$. Cylinder approximation and $\|\gamma_W F\|_\infty \le \|F\|_\infty$ extend this identity to all continuous F. Equivalently the Borel probability measures $\mu\gamma_W$ and μ agree. The identity consequently holds for every bounded Borel F.

It is also the conditional-law formulation of DLR. For bounded outside-measurable g, the kernel is proper and satisfies $\gamma_W(gF) = g\gamma_W F$. Hence $\mu(gF) = \mu(g\gamma_W F)$, and $\gamma_W F$ is outside measurable. Thus eq. 5.cdlx is the conditional expectation given the bonds outside W. This holds for every finite window.

The torus lifts are invariant under lattice translations and the hypercubic group. These transformations preserve Haar measure and the Wilson action, including when a chosen bond orientation is reversed. Weak convergence preserves their invariance, so μ is an invariant Wilson DLR state. In particular it is periodic, with the whole lattice as a period group. The same Feller argument makes the class of translation-invariant DLR states weakly closed, hence compact, and its defining identities make it convex. No compactness of the union over arbitrary period groups is being asserted.

If a source argument uses a prescribed compatible cofinal family of tori, take that family in this proof. No measure estimate beyond compactness is needed. Reflection positivity and the stronger symmetry theorems for every periodic DLR state are separate results; this measure argument proves neither their all-state estimates nor any uniformity as $a \downarrow 0$.

Recognition of limits on sufficiently large tori. Fix (K, β) and a finite bond window W. Use the same cofinal family of admissible tori for the periodic lifts and for their limit points in $\mathcal{G}^{\mathrm{tor}}_{K,\beta}$. Write h_W for the density in eq. 5.cdlx. If the torus is large enough for $W \cup \partial W$ to embed, the marginal on W of its periodic lift has Haar density

$$m_{\widetilde{\mu}_N}(V) = \int h_W(V \mid U_{\partial W})\, d\widetilde{\mu}_N(U).$$

Indeed, for functions of W the identity eq. 5.cdlxi needs only this embedding. The same formula gives the marginal m_μ of any fixed-cutoff torus limit point. All these densities are bounded below by $e^{-2\beta|\mathcal{P}(W)|}$. The functions $h_W(V \mid \cdot)$ of the exterior variables form a compact family in the uniform norm as V ranges over G^W. A finite uniform net and weak convergence thus show that the densities converge uniformly along any torus subsequence converging weakly to μ. The lower bound on the densities gives uniform convergence of their ratios to one. In particular

$$\inf_{\mu \in \mathcal{G}^{\mathrm{tor}}_{K,\beta}} \left\| m_\mu / m_{\widetilde{\mu}_N} - 1 \right\|_\infty \longrightarrow 0 \qquad (N \to \infty).$$

Otherwise a sequence of violating tori would have a weak limit point in this very class and the ratios along that subsequence would tend to one.

For each datum (K, β) take W to be the bond window of physical radius $K + 1$ in the max norm, centered at the origin, and choose a physical side threshold $\ell_\flat(K, \beta)$. Choose it so that every larger admissible torus contains the window without wrapping and the preceding infimum is at most e^{-1/a_K}. We may also take the threshold at least $2L^K$. Choose a limit state μ with ratio error $\epsilon \le 2e^{-1/a_K}$. This single state serves for all orders and all fixed compact supports eventually contained in the window. In fact, if $Y_i = \mathcal{O}_K(f_i)$ and the test supports lie in a box of radius R

within the window with a margin for their plaquettes, then

$$\left|\widetilde{\mu}_N\left(\prod_{i\in I} Y_i\right) - \mu\left(\prod_{i\in I} Y_i\right)\right| \le \epsilon \prod_{i\in I} C(1+R/a_K)^4 \|f_i\|_\infty.$$

The density ratio gives this inequality directly by integration. Expanding cumulants in moments gives an error at order n bounded by $C_n\epsilon(1+R/a_K)^{4n}\prod_i \|f_i\|_\infty$. Expansion over subsets gives the same estimate for the products centered in their respective states. Along first-passage data at fixed g with $K\to\infty$, these errors tend to zero for every fixed R and n. Thus each separated-correlation limit from tori above these thresholds has the same separated cumulants of order $n\ge 2$ and the same self-centered functional as a volume-first limit. Conversely, for a volume-first sequence, at each cutoff choose an arbitrarily large torus from the subsequence realizing the state at that cutoff so its density ratio satisfies the same tolerance. The same comparison realizes its continuum limit by tori above the thresholds.

5.24.0.1. The diagonal reflection form on these tori. Let $\Theta F = \overline{F\circ\theta^*}$ for a diagonal lattice reflection through the origin. The window W is invariant under θ. For a bounded Borel F of the bonds on its positive side, $\Theta F\cdot F$ is a function of W. The fixed-cutoff state μ is reflection positive by Prop. A.11. The same density ratio as above gives

$$\widetilde{\mu}_N(\Theta F\cdot F) \;\ge\; -\epsilon\,\widetilde{\mu}_N(|\Theta F|\,|F|) \;\ge\; -\epsilon\,\widetilde{\mu}_N(|F|^2).$$

The first expression is real by reflection invariance of the torus law. The second inequality uses Cauchy–Schwarz and that invariance. Thus the same side threshold controls the reflection form simultaneously for every such F. Along growing tori above the threshold, the error tends to zero for $F=F_K$ whenever $\epsilon\widetilde{\mu}_N(|F_K|^2)\to 0$. In particular this holds for fixed polynomials in finitely many compactly supported lattice observables, centered in the torus law if desired, by the plaquette count above.

The continuum hypothesis. Fix an odd $L\ge 3$ and a sequence $a_j = L^{-K_j}\downarrow 0$. At the same datum (a_j,β_j) let ρ_j,ν_j be periodic Wilson DLR states. The all-mirror lattice theorem supplies the following properties used here: both states are invariant under unit lattice translations and the hypercubic group, and ρ_j is reflection positive at coordinate and diagonal lattice mirrors. The centering state ν_j is invariant at those mirrors as well. These properties are lattice inputs to the compactness argument.

The infinite-lattice case includes volume-first limits when the states are torus limit points. Direct growing-torus limits are treated below with the finite-torus bounds. The density comparison above identifies limits from tori with sides at least $\ell_\flat(K,\beta)$ with volume-first limits. For general growing tori and for removal of volume after a fixed-torus continuum limit, diagonal reflection positivity requires a separate argument. Tuned fixed-torus limits, when obtained by matched-cutoff estimates, remain separate from the subsequence theorem.

For $f\in C_c^1(\mathbb{R}^4)$ write

$$\mathcal{O}_j(f) = \sum_p f(x_p)A_p, \qquad X_j(f) = \mathcal{O}_j(f) - \nu_j(\mathcal{O}_j(f)), \qquad m_{j,n}(\mathbf{f}) = \rho_j\prod_{i=1}^n X_j(f_i). \tag{5.cdlxii}$$

There is one plaquette for each coordinate plane and lattice site, so six orientations are counted once each; x_p is its barycenter. There is no power of a_j in the observable. At a fixed cutoff these are bounded continuous random variables with finite bond dependence. Set $m_{j,0}=1$ and complexify multilinearly.

Let $\mathfrak{S}_n$ be the tuples of compact C^1 tests with pairwise disjoint supports. A zero entry gives value zero and can be set aside. The separation is the minimum distance of the supports, not that of balls containing them. Every support may lie in its own radius-R ball; these balls may overlap.

Use the fixed physical norm

$$\|f\|_\sharp = \max(\|f\|_\infty, c_M\|\nabla f\|_\infty), \qquad c_M > 0 \text{ fixed.}$$

The ordinary source hypothesis is used only through the following proved implications of it. At each fixed order there is a bound

$$|m_{j,n}(\mathbf{f})| \le C_n(d;R)\prod_i \|f_i\|_\sharp \tag{5.cdlxiii}$$

when the supports have distance at least $d > 0$ and the stated radius bound. Bounds holding eventually on each fixed support neighborhood suffice for extraction. The one-point version is included: for different centering and integrating states it is a uniform bound on their mean difference, not on either unsubtracted mean.

For the quantitative conclusion assume the graded bound, with one finite integer $k \ge 1$ and $B(R) < \infty$ independent of j and the two states,

$$|m_{j,n}(\mathbf{f})| \le B(R)^n (n!)^k \max(1, d^{-5n}) \prod_i \|f_i\|_\sharp. \tag{5.cdlxiv}$$

For $n = 1$ take $d = 1$. A harmless change of B includes complex entries. These are all-state bounds at the indicated data and also apply to the pair (ρ_j, ρ_j) used for canonical recentering. The ordinary-source theorem obtains this bound from D1 and its actual rooted domain geometry, including the fixed-layer dependence and strict entropy gap. A weaker counting assertion alone does not supply this hypothesis. Those source hypotheses, and existence of the given sequence of cutoff data with common constants, are not proved by this section.

5.24.0.2. Compactness on growing tori. For a direct sequence of compatible tori of sides $\ell_j \to \infty$, use its Wilson law for both integration and centering. For any fixed finite collection of bounded support neighborhoods in $\mathbb{R}^4$, choose a box containing them and a fixed margin. Once its diameter is less than $\ell_j/4$, the quotient map to the torus is injective and preserves distances on the box. Periodize each test by summing its translates by the periods. The copies are disjoint, so the sup and Lipschitz norms are preserved. The torus distances between the projected supports are the original distances. The finite-torus part of Prop. 5.76 thus gives eq. 5.cdlxiii and eq. 5.cdlxiv eventually on each fixed support neighborhood.

The countable extraction below only uses eventual bounds on each such neighborhood, so the same diagonal sequence works for the tori. Each graded inequality passes to the limit on its fixed tuple. The extension to diagonal-flat Schwartz tests is then made on that limit functional. Translations, hypercubic symmetries and coordinate mirror positivity pass with the same fixed margins. For coordinate reflection positivity on a torus, the projected supports lie inside one half of the torus cut by the rounded mirror and its antipodal mirror for all large j. The limiting diagonal mirror is a separate case. The half-period is an integer number of lattice spacings, so the antipodal mirror has the same site or link type as the chosen mirror. The fixed support margin accommodates the full diameter of every sampled plaquette. Diagonal positivity uses the additional recognition or another diagonal reflection-positivity input.

A fixed admissible torus. For Thm. A(a)(ii), fix T_ℓ and take any sequence of first-passage data with $K_j \to \infty$. The first bound in Prop. 5.64 applies directly to the connected functions. On this torus the factor $e^{nc_V\ell^4}$ is fixed at each n. Use a countable basis of rational coordinate boxes on the torus. Every tuple of separated compact supports is contained in finite unions of such boxes whose closures remain separated. In each union choose the smooth cutoff exhaustion used below and multiply by rational real Fourier polynomials on T_ℓ. Smoothing and the C^1 convergence of the Fourier approximants give a countable dense family with support in that union. Diagonal extraction on this list of tuples and all $n \ge 2$ gives one subsequence. Telescoping with the torus bound on each product of the fixed support neighborhoods, as in eq. 5.cdlxvi, extends convergence to every separated C^1 tuple. Each canonically centered moment is the finite sum of products of

connected functions over partitions without singletons. The same subsequence therefore gives all the centered moments. The bound and permutation and hypercubic symmetries pass to the limit. Round a torus translation to the lattice modulo ℓ and use the same telescoping estimate on fixed neighborhoods of its translated supports. This gives all torus translations in the limit.

On a product of separated coordinate neighborhoods take smooth cutoffs χ_i supported there and equal to one near the support projections of a joint test F. For $k_i \in \mathbb{Z}^4$ write $e_{k_i}(x) = e^{2\pi i k_i \cdot x/\ell}$. Define the distribution of the connected limit there by

$$F \longmapsto \sum_{k \in \mathbb{Z}^{4n}} \widehat{F}(k) S_n(\chi_1 e_{k_1}, \dots, \chi_n e_{k_n}).$$

Each entry has $\|\chi_i e_{k_i}\|_\sharp \le C(1+|k_i|)$. The torus bound thus costs at most $C(1+|k|)^n$, and $5n+1$ derivatives of F give summable Fourier decay after this cost in dimension $4n$. The series agrees with S_n on separated smooth tensors. The same Fourier approximation with cutoffs gives density of these tensors on each product neighborhood. Thus the local definitions agree on overlaps and give a unique distribution of locally finite order off the diagonals. The same argument applies to the centered moments, using their finite partition formula. Coordinate mirror positivity passes to these limits by the moving-test argument below, using supports compactly contained in one of the two slabs between the rounded link mirror and its antipodal mirror.

If these fixed-torus limits are taken along $\ell_j \to \infty$ and then converge on separated tuples in $\mathbb{R}^4$, the resulting limit is also realized by one growing-torus cutoff sequence. List the countable separated tuples below, projecting them to T_{ℓ_j} once their support neighborhoods embed. At ℓ_j, choose one index from its defining cutoff sequence with $K \ge j$ and with error at most $1/j$ from the fixed-torus limit on the first j listed tuples whose neighborhoods embed. These are finitely many requirements on a convergent sequence. For any fixed separated tuple, choose an approximant from this list with the same separated neighborhoods. The per-ball bound in Prop. 5.64 holds eventually on this product with one constant for the finite cutoffs and the inner limits. Telescoping at both ends gives twice the approximation error plus $1/j$ between the selected finite-torus values and the inner limits. First let j tend to infinity, then let the approximation error tend to zero. The same indices give the centered moments by the finite partition formula. This argument uses the uniform per-ball bound on the growing family.

A countable support-adapted family. Let $\mathcal{U}$ be the bounded finite unions of open boxes with rational endpoints. This is countable. Every finite collection of separated compact sets K_i has neighborhoods $U_i \in \mathcal{U}$ such that $K_i \subset U_i$ and the closures $\overline{U}_i$ are pairwise separated. Cover each compact set by sufficiently small rational boxes inside a neighborhood of radius less than one fourth of its distance from the other sets, then take a finite subcover. The neighborhoods can also stay in any prescribed open support regions.

For each $U \in \mathcal{U}$ we now exhibit a countable family dense in $C^1_c(U)$ for the C^1 norm, with every approximant supported in U. Fix a nonnegative smooth unit-mass bump ϱ in the unit ball. For integers $m \ge 1$ let

$$\chi_{U,m} = \mathbf{1}_{\{x : \operatorname{dist}(x, U^c) > 1/(2m)\}} * \varrho_{1/(8m)}.$$

It is smooth, compactly supported in U, takes values in $[0,1]$, and equals one where $\operatorname{dist}(x, U^c) \ge 1/m$. Use finite rational linear combinations of

$$\chi_{U,m}(x)\cos(2\pi q \cdot x/N), \qquad \chi_{U,m}(x)\sin(2\pi q \cdot x/N), \quad q \in \mathbb{Z}^4,\ N \in \mathbb{N}, \tag{5.cdlxv}$$

and take their rational span over all indices. Call this real countable family D_U.

To verify density, fix $f \in C^1_c(U)$ and choose m such that $\chi_{U,m} = 1$ on a neighborhood of its support. Extend f by zero, smooth it on a sufficiently small scale, and view the resulting smooth compact function as periodic on a large integer-side box. Smoothing converges to f in C^1 because f and its gradient are uniformly continuous. The Fourier partial sums of this smooth periodic function converge with their first derivatives: repeated integration by parts makes the differentiated Fourier

series absolutely summable. Approximate their finitely many real sine and cosine coefficients by rationals and multiply by $\chi_{U,m}$. The multiplication is bounded in C^1, and $\chi_{U,m}f = f$. This produces the required approximants. Complexification gives the countable dense family $D_U + iD_U$ for complex tests.

List all tuples $(g_1, \dots, g_n)$, for every $n \geq 1$ and every ordered list of pairwise separated $\overline{U}_i$, with $g_i \in D_{U_i} + iD_{U_i}$. This is one countable list. By eq. 5.cdlxiii the values of each coordinate form a bounded sequence; finite initial values do not matter if a bound starts eventually. Extract an infinite subsequence for convergence of the first coordinate, then a subsequence of it for the second, and so on. If I_r is the rth nested infinite index set, choose increasing $j_r \in I_r$. For each fixed coordinate its values along (j_r) eventually lie in the convergent subsequence for that coordinate. All listed values therefore converge along this single subsequence.

Extension to every separated tuple. The following telescoping estimate explains why that one extraction is enough. If $\operatorname{supp} f_i \cup \operatorname{supp} g_i$ lies in $\overline{U}_i$, let $d_U > 0$ be a lower bound for their pairwise distances, taking $d_U = 1$ when $n = 1$, and choose a common finite radius bound R_U. Multilinearity and eq. 5.cdlxiii give, uniformly in the relevant tail,

$$|m_{j,n}(\mathbf{f}) - m_{j,n}(\mathbf{g})| \leq C_n(d_U; R_U) \sum_{i=1}^{n} \|f_i - g_i\|_\sharp \prod_{h \neq i} \max(\|f_h\|_\sharp, \|g_h\|_\sharp). \tag{5.cdlxvi}$$

Every mixed tuple in this telescoping sum is separated because the neighborhoods, rather than just the two endpoint tuples, carry the separation.

Given any $\mathbf{f} \in \mathfrak{S}_n$, choose such rational neighborhoods and approximate each entry in C^1 by the corresponding countable family. The right side of eq. 5.cdlxvi tends to zero uniformly in j. For a fixed approximating tuple its values converge along (j_r). Comparing two indices through this tuple shows that $m_{j_r,n}(\mathbf{f})$ is Cauchy. Define its limit by $\sigma_n(\mathbf{f})$. This definition is independent of all neighborhoods and approximants because it is the limit of the original sequence at that tuple. The subsequence has not been chosen again for $\mathbf{f}$; convergence now holds on every separated compact C^1 tuple simultaneously. Set $\sigma_0 = 1$.

Passing in eq. 5.cdlxvi proves the same local continuity estimate for σ_n. On compact subsets of each product of the fixed support spaces, convergence is uniform: choose a finite C^1 net in that compact set, apply pointwise convergence at its centers, and use eq. 5.cdlxvi at both ends. Its constants are uniform on a norm-bounded set.

Multilinearity passes whenever all tuples in the proposed identity are admissible. In particular the union of the alternative supports in a single entry must stay separated from the other entries. Permutation symmetry, conjugation and reality pass directly from their exact lattice identities. The bounds in eq. 5.cdlxiii and eq. 5.cdlxiv pass with their original constants at every fixed tuple, since absolute-value inequalities are closed under convergence. Their radius and separation parameters concern the original supports, not the larger neighborhoods used to construct the limit.

Moving arguments, real symmetries, and rounded mirrors. Suppose $f_{i,j} \to f_i$ in C^1 and all their supports lie in fixed pairwise separated compact neighborhoods. eq. 5.cdlxvi and fixed-tuple convergence give

$$m_{j_r,n}(f_{1,j_r}, \dots, f_{n,j_r}) \longrightarrow \sigma_n(f_1, \dots, f_n). \tag{5.cdlxvii}$$

This is also valid for every finite combination of tuples. For translations the needed C^1 convergence has the explicit modulus

$$\|\tau_v f - \tau_w f\|_\sharp \leq \max\{\|\nabla f\|_\infty |v - w|, c_M \omega_{\nabla f}(|v - w|)\} \longrightarrow 0, \tag{5.cdlxviii}$$

where $\omega_{\nabla f}$ is the uniform modulus of continuity of the compactly supported continuous gradient. No second derivative is required. The chain rule gives the corresponding convergence for a convergent sequence of Euclidean isometries.

For any fixed real translation v choose lattice vectors $v_j \in a_j\mathbb{Z}^4$ with $v_j \to v$. Both states are invariant under v_j, so $m_{j,n}(\tau_{v_j}\mathbf{f}) = m_{j,n}(\mathbf{f})$. The moving-test statement proves real translation invariance of σ_n. Hypercubic invariance is exact at every spacing and passes without rounding.

For clarity the reflection limit also works at arbitrary affine positions. A coordinate mirror $x_\mu = c$ can be approximated by link mirrors with $c_j \in a_j(\mathbb{Z} + 1/2)$ and $c_j \to c$. A diagonal mirror $x_\mu \pm x_\lambda = c$ is approximated by $c_j \in a_j\mathbb{Z}$. In unit-normal coordinates its offset grid is $a_j\mathbb{Z}/\sqrt{2}$. These are different conventions for the same diagonal mirror; no factor $\sqrt{2}$ is discarded. The reflected tests converge in C^1: the linear reflection part is fixed and the translation parts differ by $2(c_j - c)e_\mu$ in the coordinate case, and by $(c_j - c)(e_\mu \pm e_\lambda)$ in the diagonal case.

Let F be a finite complex combination of separated tuples whose entries lie strictly on one open side of the limiting mirror. Its compact support has positive distance from the mirror. For all fine cutoffs the barycenters sampled by those entries and all the bonds of their plaquettes lie on the same side of the rounded mirror: a plaquette has diameter $O(a_j)$, while that distance is fixed. Form its centered lattice polynomial F_j. Reflection invariance of ν_j makes centering commute with the lattice reflection. The moment identity gives

$$\rho_j(\overline{\theta_j F_j} F_j) = m_j(\theta_j \overline{F} \cdot F).$$

The left side is nonnegative where the lattice mirror has exact positivity. For a diagonal mirror through the origin on tori above the local density threshold, its negative part tends to zero by the diagonal reflection-form bound above. Every tuple on the right has a common positive support separation for large j: internal separations are fixed and the two sides are separated by the mirror gap. Apply eq. 5.cdlxvii to get

$$\sigma(\theta \overline{F} \cdot F) \geq 0. \tag{5.cdlxix}$$

Reflection invariance itself passes in the same way. The argument applies at coordinate mirrors for both lattice constructions and at diagonal mirrors for periodic states. For tori above the local density threshold, the diagonal conclusion at the origin extends to affine translates by translation invariance of the limit. A diagonal torus limit outside this threshold requires its own mirror estimate.

The one-point limit and the connected functions. The first moment is determined more precisely by the plaquette means. Unit translations and hypercubic symmetry make

$$\delta_j = \rho_j(A_p) - \nu_j(A_p)$$

independent of the plaquette position and its coordinate plane. Define

$$R_j(f) = \frac{a_j^4}{6} \sum_p f(x_p), \qquad c_j = 6a_j^{-4}\delta_j. \tag{5.cdlxx}$$

Then $m_{j,1}(f) = c_j R_j(f)$. Each orientation samples a translate of $a_j\mathbb{Z}^4$. Comparing its sample value to the integral on the corresponding side-a_j cube gives

$$R_j(f) \longrightarrow \int f, \qquad |R_j(f) - \textstyle\int f| \leq C_R a_j \|\nabla f\|_\infty \tag{5.cdlxxi}$$

for a test supported in a fixed radius-R ball. Indeed the sample and any point of its cube are at distance at most $2a_j$; only $C(R+1)^4 a_j^{-4}$ cubes can contribute. Average the six such estimates. The same bound is independent of the center of that ball.

Choose one real smooth compact bump f_* with $\int f_* = 1$. Its limiting moment exists by the preceding extraction, and $R_j(f_*) \to 1$. Consequently along the extracted subsequence

$$c_j = \frac{m_{j,1}(f_*)}{R_j(f_*)} \longrightarrow c \in \mathbb{R}, \qquad \sigma_1(f) = c \int f. \tag{5.cdlxxii}$$

In particular the coefficients are bounded on that tail. This bound is uniform over both states: $|c_j| \le 2B(R_*)\|f_*\|_\sharp$ once $R_j(f_*) \ge 1/2$, and $|c| \le B(R_*)\|f_*\|_\sharp$. This argument uses the uniform one-point comparison and the common plaquette means; it is not a deduction from formal continuum translation invariance alone. If $\rho_j = \nu_j$, then $c = 0$.

All connected limits are finite algebraic consequences, requiring no common disc of analyticity for an exponential. For a fixed n work in the algebra with $z_i^2 = 0$ and let

$$M_j(z) = 1 + \sum_{\varnothing \ne A \subset [n]} m_{j,|A|}(\mathbf{f}_A) \prod_{i \in A} z_i.$$

The logarithm is the finite series for $\log(1 + (M_j - 1))$. Its coefficient of $\prod_{i=1}^n z_i$ is

$$K_{j,n}(\mathbf{f}) = \sum_{\pi \in \mathrm{Part}[n]} (-1)^{|\pi|-1}(|\pi| - 1)! \prod_{A \in \pi} m_{j,|A|}(\mathbf{f}_A). \tag{5.cdlxxiii}$$

In fact the term of degree r in the logarithm counts the $r!$ orders of each partition with r blocks and divides by r, giving $(r-1)!$. Every subtuple is separated. Hence these finite polynomials converge to the same formula with σ in place of m_j; denote it by S_n. Conversely the finite exponential series gives $\sigma_n = \sum_\pi \prod_{A \in \pi} S_{|A|}$, since its factor $1/r!$ cancels the orders of the blocks.

Adding constants to the $X_j(f_i)$ multiplies M_j by $\exp(\sum_i b_i z_i)$, so changes its logarithm only in degree one. Thus for $n \ge 2$, $K_{j,n}$ is the usual cumulant of the raw variables $\mathcal{O}_j(f_i)$ under ρ_j, independent of the centering state. The graded cumulant bound passes as well:

$$|S_n(\mathbf{f})| \le [2B(R)]^n (n!)^k \max(1, d^{-5n}) \prod_i \|f_i\|_\sharp, \qquad n \ge 2. \tag{5.cdlxxiv}$$

One may obtain it either from the uniform lattice cumulant bound, or directly from eq. 5.cdlxxiii: if block sizes are n_h, $\prod_h n_h! \le n!$, and

$$\sum_\pi (|\pi| - 1)! \prod_{A \in \pi} (|A|!)^k \le (n!)^{k-1} \sum_\pi (|\pi| - 1)! \prod_{A \in \pi} |A|! \le 2^{n-1}(n!)^k.$$

For exactly r blocks the last unweighted sum is $\frac{n!}{r}\binom{n-1}{r-1}$, by ordered compositions of n.

There is a canonical centered functional distinct from the original two-state moments. Center at ρ_j instead of ν_j; expanding each product and using convergence of $m_{j,1}$ gives along the same subsequence

$$\sigma_n^0(\mathbf{f}) = \sum_{A \subset [n]} \sigma_{|A|}(\mathbf{f}_A) \prod_{i \notin A} (-c \textstyle\int f_i) = \sum_{\pi \in \mathrm{Part}_{\ge 2}[n]} \prod_{A \in \pi} S_{|A|}. \tag{5.cdlxxv}$$

It has empty value one, singleton zero, the same higher connected functions, and all the bounds and positivity properties above. For the bounds use the lattice pair (ρ_j, ρ_j) before passing to the limit, so no uncontrolled constant from recentering is introduced.

The quantitative flat extension and closed inequalities. For completeness the topology retained by this compactness statement is explicit. Let $\Delta_n = \bigcup_{i<h}\{x_i = x_h\}$ and let $\mathcal{S}_{\Delta,n}$ be the Schwartz functions whose every jet vanishes on Δ_n. Use

$$p_s(\Phi) = \max_{|\alpha| \le s} \sup_x (1 + |x|^2)^{s/2} |\partial^\alpha \Phi(x)|.$$

On unit-radius supports, eq. 5.cdlxiv and $\|f\|_\sharp \le \max(1, c_M)\max(\|f\|_\infty, \|\nabla f\|_\infty)$ give the quantitative flat-extension hypothesis with separation exponent five. For either limit the theorem gives a unique extension Σ_n with

$$|\Sigma_n(\Phi)| \le C^n (n!)^{k+32} p_{20n}(\Phi). \tag{5.cdlxxvi}$$

The same bound holds for the centered and connected extensions, after changing C. Separated smooth tensors are dense in $\mathcal{S}_{\Delta,n}$.

For the infinite-lattice inputs the same uniform estimate holds for the extensions $\Sigma_{j,n}$. In the direct torus construction the graded bound is eventual on each fixed compact tuple. Distant

supports can project onto the same part of a torus at finite j, so that argument gives the flat estimate on the limit through the fixed compact tuples.

For the infinite-lattice sequence, given Φ approximate it by a finite sum T of smooth separated tensors in p_{20n}. The common estimate controls both $\Sigma_{j,n}(\Phi - T)$ and $\Sigma_n(\Phi - T)$ uniformly in j. On T convergence has already been proved. Taking first the cutoff limit and then the approximation error to zero proves

$$\Sigma_{j_r,n}(\Phi) \longrightarrow \Sigma_n(\Phi). \tag{5.cdlxxvii}$$

The positive-time version of the density theorem uses tensors strictly above the mirror. Their reflected products are flat on all cross diagonals, including the time boundary. Thus the limit's symmetry, reality, real translations and reflection positivity extend by this density and the continuity of tensor products in Schwartz seminorms. For each mirror with the lattice positivity input, use its corresponding positive half-space. It follows by an orthogonal change of variables in the flat-space density construction; it does not assume invariance of the functional under that change of variables.

More generally any closed inequality involving finitely many fixed tuple evaluations and continuous functions of them passes to the limit. The statement allows moving tests as in eq. 5.cdlxvii, convergent scalar coefficients and eventual uniform bounds. Its proof is simply convergence of those finitely many numbers followed by continuity and closedness of the inequality. It does not turn strict inequalities into uniform strict bounds or provide estimates whose constants diverge with the cutoff.

The proposition records the infinite-lattice construction, including its diagonal mirror input. Direct growing tori give the compact-tuple convergence, limiting flat extension and coordinate reflection positivity by the preceding argument.

Proposition 5.77 (Subsequential continuum compactness)**.** For every input sequence satisfying the lattice properties and two-state uniform bounds above, there is one subsequence whose moments converge on every separated compact C^1 tuple. The limiting functional is admissibly multilinear, symmetric and real, invariant under real translations and the hypercubic group, and reflection positive at every coordinate and diagonal affine mirror. It retains the graded bound and its quantitative diagonal-flat extension. Its first moment is $c\int f$, its higher connected functions are the limits of the integrating states' cumulants, and eq. 5.cdlxxv is their canonical centered functional. Moving arguments and fixed-tuple closed inequalities pass as stated above. ■

The preceding sections prove every assertion. The fixed-cutoff measure argument ensures that invariant Wilson DLR states are available at each given datum. It supplies neither a sequence of admissible renormalization data with $K_j \to \infty$ nor D1's uniform source bounds. It also supplies no identification of two different extracted limits, no matched-cutoff convergence rate, and no claim of a full Yang–Mills observable theory.

5.25. **Extension to diagonal-flat Schwartz tests.** A graded bound for compact separated moments implies an extension to diagonal-flat Schwartz tests, with explicit dependence on the number of arguments. The graded bound for the gauge theory remains a hypothesis. We construct the local distributions by Fourier series on product boxes and track the constants as the number of variables grows.

5.25.1. *The hypothesis and the quantitative conclusion.* Use $\|f\|_1 = \max(\|f\|_\infty, \|\nabla f\|_\infty)$ on $C^1_c(\mathbb{R}^4;\mathbb{C})$. Suppose S_n, $n \geq 1$, are complex multilinear forms on tuples with pairwise separated compact supports. Multilinearity is required whenever the supports in the identity remain separated. Fix numbers $1 \leq b < \infty$, $\kappa \geq 0$ and $\alpha \geq 0$, independent of n, such that

$$|S_n(f_1,\dots,f_n)| \leq b^n (n!)^\kappa \max(1, d^{-\alpha n}) \prod_i \|f_i\|_1 \tag{5.cdlxxviii}$$

if every support lies in a ball of radius one, placed anywhere, and different supports have Euclidean distance at least $d > 0$. For $n = 1$ only the radius and norm conditions are imposed. A centered first moment equal to zero is included. Real forms may first be complexified, changing b by an absolute factor. Neither invariance nor positivity is needed for the extension.

Put $A = \lceil \alpha \rceil$, $k_* = A + 15$, and $\Delta_n = \bigcup_{i<j} \{x_i = x_j\}$. Let $\mathcal{S}_{\Delta,n}$ be the Schwartz functions on $\mathbb{R}^{4n}$ whose derivatives of every order vanish on Δ_n. For $n = 1$ this is the ordinary Schwartz space. Write

$$p_k(\varphi) = \max_{|\beta| \le k} \sup_x (1 + |x|^2)^{k/2} |\partial^\beta \varphi(x)|. \tag{5.cdlxxix}$$

Proposition 5.78 (Quantitative diagonal-flat extension)**.** There is $C < \infty$, depending on b, α and fixed numerical cutoff choices but not on n, and a unique continuous linear functional Σ_n on $\mathcal{S}_{\Delta,n}$ agreeing with S_n on smooth separated tensors. It satisfies

$$|\Sigma_n(\varphi)| \le C^n (n!)^{\kappa+32} p_{k_* n}(\varphi), \qquad n \ge 1. \tag{5.cdlxxx}$$

Finite sums of smooth compact separated tensors are dense in this diagonal-flat space in its Schwartz topology. In its positive-time subspace, the tensors may be chosen with supports compactly contained in $\{x_4 > 0\}$. These are constrained density assertions, not density in the unrestricted Schwartz space across the diagonals. ∎

The conclusion is an H1 growth estimate with linear derivative order and a fixed factorial exponent; its constants are intentionally not optimized. For example, $\alpha = 5$ gives the sufficient order $20n$. The probability-theory normalization $\Sigma_0(1) = 1$ may be added separately.

The radius-one formulation loses no information about a hypothesis stated as

$$B(R)^n (n!)^\kappa \max(1, d^{-\alpha n}),$$

with constants independent of support positions. Choose a fixed smooth unit-scale partition in each slot. A function supported in a radius-R ball is a sum of at most $C(1+R)^4$ supported pieces of radius at most one, each with C^1 norm at most $C\|f\|_1$. Supports of pieces from different slots retain the original separation. Multilinearity and eq. 5.cdlxxviii therefore imply the same bound with

$$B_{\mathrm{eff}}(R) \le Cb(1+R)^4. \tag{5.cdlxxxi}$$

Thus only $B(1)$ is used, even if the supplied large-radius upper bound was exponential. This assertion concerns a separate ball for each entry, not one ball enclosing the entire configuration.

5.25.2. *Cutoffs with controlled derivatives.* We fix all cutoffs once. They can be chosen in Gevrey class two, with

$$0 \le \eta \le 1, \qquad |\partial^\beta \eta| \le C^{|\beta|+1} (|\beta|!)^2. \tag{5.cdlxxxii}$$

Here is an explicit construction and a justification of this bound. Set $a(t) = e^{-1/t}$ for $t > 0$ and $a(t) = 0$ for $t \le 0$. On the complex circle $|z - t| = t/2$ one has $\mathrm{Re}(1/z) \ge 2/(9t)$. Cauchy's formula bounds $|a^{(m)}(t)|$ by $m!(2/t)^m e^{-2/(9t)}$. Maximizing $t^{-m} e^{-c/t}$ and using $(m/e)^m \le m!$ gives $C^{m+1}(m!)^2$. The step

$$s(t) = \frac{a(t)}{a(t) + a(1-t)}$$

is set equal to zero for $t \le 0$ and one for $t \ge 1$. Its denominator on $[0,1]$ is at least e^{-2}. The same derivative bound holds for its reciprocal and hence for s. For precision, differentiate $D(1/D) = 1$ inductively; after division by $(m!)^2$, the convolution coefficients are $1/\binom{m}{r} \le 1$. If $|D^{(r)}| \le C_D^{r+1}(r!)^2$, a derivative constant $C_D(1+2e^2C_D)$ for the reciprocal closes the resulting geometric sum. Products obey the same bound by Leibniz's formula. This proves eq. 5.cdlxxxii, including flatness at the join points.

Choose an even η_0 equal to one on $[-1,1]$, zero outside $[-2,2]$, and nonincreasing on $[0,\infty)$, using $s(2-|t|)$ in the transition. Put $\psi(t) = \eta_0(2t)$ and $D(t) = \sum_{z\in\mathbb{Z}} \psi(t-z)$. At most three summands occur locally and $D \ge 1$. Then $\chi_1 = \psi/D$ is supported in $[-1,1]$, has eq. 5.cdlxxxii, and $\sum_z \chi_1(t-z) = 1$. Define

$$\chi(u) = \prod_{a=1}^{4} \chi_1(u_a),$$

$$\chi^+(u) = \prod_{a=1}^{4} \eta_0(u_a),$$

$$\zeta(u) = 1 - \chi^+(u). \tag{5.cdlxxxiii}$$

Thus $\sum_{z\in\mathbb{Z}^4} \chi(u-z) = 1$, $\chi^+ = 1$ on the support of χ, and ζ is zero for $|u|_\infty \le 1$ and one for $|u|_\infty \ge 2$. All these functions lie between zero and one and have eq. 5.cdlxxxii, with an absolute constant. In particular, undifferentiated factors cost one.

We will repeatedly use the following elementary product estimate. For N factors of this kind at scale h, every derivative of order q is bounded by

$$(CN)^q (q!)^2 h^{-q}. \tag{5.cdlxxxiv}$$

An absolute prefactor is allowed at $q = 0$. Indeed, distribute the multiindex among the factors by Leibniz. The sum of the multinomial coefficients is N^q, and the product of the factorials of the assigned derivative orders is at most $q!$. Derivatives of pair differences $x_i - x_j$ introduce only signs. This argument produces no factor C^N when most factors are undifferentiated.

5.25.3. *An explicit local functional on a product box.* First, tensor relations cause no ambiguity. In a finite relation between separated tensors, let d be smaller than every separation. Localize all entries by one finite smooth partition on sets of diameter less than $d/8$. A product of slot sets with two sets closer than $d/2$ has a zero factor in every original tensor: otherwise two original supports would be closer than d. On every remaining product the slot sets are separated, and the form is an ordinary multilinear map on their supported function spaces. It factors through their algebraic tensor product. The latter embeds injectively in functions on the Cartesian product: for the finite-dimensional function spans in any relation, finitely many point evaluations give invertible matrices in each slot. A zero tensor therefore has zero value. Summing the localization proves independence of the tensor presentation.

Here is a quantitative local extension. Let $0 < h \le 1/16$, $\ell = h/64$, and let slot centers be $c_i \in \mathbb{R}^4$. Suppose the cubes $Q_i^+ = c_i + [-2\ell, 2\ell]^4$ have pairwise distance at least $h/2$. They lie in balls of radius one. For a smooth function u supported in $\prod_i (c_i + [-\ell,\ell]^4)$, take its Fourier series on the product of period-8ℓ cubes centered at these points. Write its coefficients as $c_{k_1,\dots,k_n}$, $k_i \in \mathbb{Z}^4$. Multiplying the series by $\prod_i \chi^+((x_i - c_i)/\ell)$ gives a series of separated tensors equal to u

everywhere. There is no boundary ambiguity: u vanishes near the boundaries of the period cubes. Explicitly, with $Q = \prod_i (c_i + [-4\ell, 4\ell]^4)$, the coefficients are

$$c_{k_1,\dots,k_n} = (8\ell)^{-4n} \int_Q u(x) \exp\left[-\frac{i\pi}{4\ell} \sum_i k_i \cdot (x_i - c_i)\right] dx.$$

In the scaled variables $y_i = (x_i - c_i)/(4\ell)$, apply $(1 - \pi^{-2}\Delta_i)^4$ in every four-dimensional slot. Integration by parts gives

$$|c_{k_1,\dots,k_n}| \le C^n J_{8n,\ell}(u) \prod_i (1 + |k_i|^2)^{-4},$$

$$J_{q,\ell}(u) = \max_{|\beta| \le q} \ell^{|\beta|} \|\partial^\beta u\|_\infty. \tag{5.cdlxxxv}$$

The sum of absolute coefficients in the product differential operator is at most C^n; no factor depending unspecifiedly on dimension is hidden here. The localized exponential in slot i has C^1 norm at most $C\ell^{-1}(1 + |k_i|)$. Since

$$\sum_{k \in \mathbb{Z}^4} (1 + |k|)(1 + |k|^2)^{-4} < \infty,$$

eq. 5.cdlxxviii makes the series of its moment values absolutely convergent and gives a local functional $T(u)$ with

$$|T(u)| \le (Cb)^n (n!)^\kappa h^{-(\alpha+1)n} J_{8n,\ell}(u). \tag{5.cdlxxxvi}$$

For a tensor supported in the inner cubes it agrees with S_n: its Fourier coefficients factor slot by slot, the partial sums converge in C^1, and eq. 5.cdlxxviii permits passage through the multilinear form.

These local extensions glue to a unique functional T on $C_c^\infty(\mathbb{R}^{4n} \setminus \Delta_n)$. To check this explicitly, in an overlap choose smaller product boxes enclosing the compact support under consideration. Fourier expansions with cutoffs inside the overlap converge with every fixed number of derivatives and consist of separated tensors. Two local functionals agreeing on those tensors therefore agree there, by their finite-order bounds in eq. 5.cdlxxxvi. A finite partition of unity on a compact off-diagonal set now defines T independently of the cover. For original separated tensors of larger radius, first partition each entry into sufficiently small pieces and use multilinearity. This also shows that the glued T agrees with the given form on all smooth compact separated tensors, not just on the boxes used in its construction. This gluing argument is used for consistency, not for the numerical bound: the latter uses only the fixed explicit grid and eq. 5.cdlxxxvi.

5.25.4. *Dyadic configuration layers and the factorial accounting.* Set $h_j = 2^{-j}/16$ and define

$$P_h(x) = \prod_{i<l} \zeta((x_i - x_l)/h),$$

$$D_0 = P_{h_0},$$

$$D_j = P_{h_j} - P_{h_{j-1}}$$

$$(j \ge 1). \tag{5.cdlxxxvii}$$

For $n = 1$ the empty product is one and $D_j = 0$ for $j \ge 1$. Off the diagonals $\sum_{j \ge 0} D_j = 1$. A fine layer has support

$$h_j \le \min_{i<l} |x_i - x_l|_\infty \le 4h_j. \tag{5.cdlxxxviii}$$

The statements about support also hold for its derivatives. Monotonicity of the one-variable cutoffs makes $0 \le D_j \le 1$, although positivity of these differences is not needed for the estimates.

At level j set $\ell_j = h_j/64$ and use the product grid partition

$$\chi_{j,z}(x) = \prod_{i=1}^{n} \chi(x_i/\ell_j - z_i),$$
$$z \in (\mathbb{Z}^4)^n. \tag{5.cdlxxxix}$$

Retain a box if $\chi_{j,z} D_j$ is not identically zero. At one point of that box every pair has distance greater than h_j; its centers therefore have pair distance at least $h_j - 2\ell_j$. The expanded cubes $\ell_j z_i + [-2\ell_j, 2\ell_j]^4$ have pair distance at least $h_j - 6\ell_j > h_j/2$. Thus eq. 5.cdlxxxvi applies to $u_{j,z} = \chi_{j,z} D_j \varphi$. A box with only boundary contact can be ignored if its smooth localized function is zero.

There are at most Cn^2 pair and grid factors in such a cutoff. Using eq. 5.cdlxxxiv, Leibniz's formula and $q = 8n$, we obtain the uniform scaled-derivative cost

$$(Cn^2)^{8n}((8n)!)^2 \le C^n (n!)^{32}. \tag{5.cdxc}$$

Here $n^{16n} \le e^{16n}(n!)^{16}$, while $(8n)! \le 8^{8n}(n!)^8$. This accounts for every dependence on the number of pair cutoffs; the $\binom{n}{2}$ undifferentiated factors do not create C^{n^2}.

On a fine layer choose a pair with $|x_i - x_l|_\infty \le 4h_j$ and replace x_i by x_l. The resulting point y lies on a diagonal and $|x - y| \le 8h_j$. Every derivative of φ vanishes there. Taylor's integral remainder along this segment gives, for $|\beta| \le 8n$ and any positive integer m,

$$|\partial^\beta \varphi(x)| \le \frac{(Ch_j)^m}{m!} C^{k_* n} (1 + |x|^2)^{-k_* n/2} p_{k_* n}(\varphi) \tag{5.cdxci}$$

when $8n + m \le k_* n$. The derivative of order m along a vector uses at most four coordinate directions; its multinomial sum is bounded by the m-th power of that vector's ℓ^1 norm. Also $|x - y| \le 1/2$, so the weight at every point of the segment is comparable with that at x, at cost $C^{k_* n}$. These are exponential, not factorial, costs in n.

Take

$$m = (A + 5)n + 2,$$
$$8n + m = (A + 13)n + 2 \le (A + 15)n. \tag{5.cdxcii}$$

In differentiating $u_{j,z}$, a derivative on a cutoff costs h_j^{-1}, which its scaled-derivative factor in J_{8n,ℓ_j} cancels. Derivatives remaining on φ are bounded by eq. 5.cdxci. Since $h_j \le 1$, unused positive powers of h_j can be discarded. eq. 5.cdlxxxiv, eq. 5.cdxc–eq. 5.cdxcii consequently imply

$$J_{8n,\ell_j}(u_{j,z}) \le C_A^n (n!)^{32} h_j^m p_{k_* n}(\varphi) \sup_{x \in \operatorname{supp} \chi_{j,z}} (1 + |x|^2)^{-k_* n/2}. \tag{5.cdxciii}$$

For $j = 0$ use the same estimate without h_j^m, by the ordinary Schwartz derivative bound; all powers of the fixed h_0 are absorbed in C_A^n.

The grid sum has exactly the needed size. Since $1 + \sum_i |x_i|^2 \ge \prod_i (1 + |x_i|^2)^{1/n}$, and grid points in one slot differ from its box by a fixed distance less than one,

$$\sum_z \sup_{x\in\operatorname{supp}\chi_{j,z}} (1+|x|^2)^{-k_*n/2} \le C_A^n \prod_{i=1}^n \sum_{z_i\in\mathbb{Z}^4} (1+|\ell_j z_i|^2)^{-k_*/2} \le C_A^n h_j^{-4n}. \tag{5.cdxciv}$$

The one-slot sum is bounded by $C_A\ell_j^{-4}$ by comparison with the integral of $(1+|u|^2)^{-k_*/2}$ on $\mathbb{R}^4$; it is finite because $k_*>4$. No number of configurations depending exponentially on n^2 is being counted.

Combine eq. 5.cdlxxxvi, eq. 5.cdxciii and eq. 5.cdxciv. Since $\alpha\le A$, the sum of absolute values at every fine level is at most

$$C^n(n!)^{\kappa+32}p_{k_*n}(\varphi)h_j^{m-(\alpha+5)n} \le C^n(n!)^{\kappa+32}p_{k_*n}(\varphi)h_j^2. \tag{5.cdxcv}$$

The coarse level has the same bound without h_j^2. The dyadic sum $\sum_j h_j^2$ is finite. We may therefore define

$$\Sigma_n(\varphi)=\sum_{j\ge0}\sum_z T(\chi_{j,z}D_j\varphi). \tag{5.cdxcvi}$$

This absolutely convergent sum proves eq. 5.cdlxxx and continuity in its stated seminorm. It agrees with T on compact off-diagonal tests, because there only finitely many layers and boxes occur and their partition sums to one. In particular it has the required tensor values.

5.25.5. *Density, the time boundary, and independence of choices.* We give the density details also needed for uniqueness. First cut off infinity by $Q_R(x)=\prod_i\chi^+(x_i/R)$. For a fixed Schwartz seminorm p_k, an extra L powers of decay give

$$p_k((1-Q_R)\varphi)\le C_{n,k,L}R^{-L}p_{k+L}(\varphi), \qquad R\ge1. \tag{5.cdxcvii}$$

This follows by Leibniz, since the nonzero terms have at least one $|x_i|\ge R$ and cutoff derivatives cost R^{-1}. Next cut out all pairwise diagonals with P_h. Where a derivative of $1-P_h$ can contribute, some pair is within $2h$. Applying the same Taylor argument with arbitrarily many extra derivatives gives

$$p_k((1-P_h)\varphi)\le C_{n,k,M}h^{M-k}p_{k+M}(\varphi). \tag{5.cdxcviii}$$

For each fixed n,k choose $M>k$. Thus compact tests supported away from every diagonal are dense in $\mathcal{S}_{\Delta,n}$. For example both displayed density constants may be bounded by $C^{k+M+L+1}(n^2+1)^k(k!)^2$, with unused parameters omitted. This follows from eq. 5.cdlxxxiv and the same directional Taylor remainder; these constants are not used in place of the uniform estimate in eq. 5.cdxcv. On each such compact set, use a finite product-box partition and the Fourier expansions of eq. 5.cdlxxxv. Their partial sums converge with every derivative and have a common compact support. They consequently converge in the Schwartz topology and are finite sums of separated smooth tensors. This proves the asserted constrained tensor density. Choosing the truncations and Fourier partial sums successively for the first $1,2,\dots$ seminorms gives a single convergent sequence.

For positive time define $\mathcal{S}_{\Delta,+,n}$ by requiring all derivatives to vanish outside $\{x_{1,4}>0,\dots,x_{n,4}>0\}\setminus\Delta_n$. In addition to the preceding cutoffs, set

$$B_h(x)=\prod_{i=1}^n s(x_{i,4}/h-1). \tag{5.cdxcix}$$

It is zero if some time is at most h and one if all are at least $2h$. Where $1 - B_h$ contributes to a positive-time test, a time is within $2h$ of zero. Taylor expansion to that forbidden hyperplane, where every jet vanishes, proves the analogue of eq. 5.cdxcviii. Hence $Q_R P_h B_h \varphi$ approximates every positive-time test in the full Schwartz topology. Product boxes for this approximation can be expanded while staying strictly above time zero. Their tensor approximants are therefore strictly positive-time separated tensors.

There is also a uniform quantitative version: replace P_h throughout eq. 5.cdlxxxvii by $P_h B_h$. A fine layer then has all pair distances and all times at least h, while one pair distance or one time is at most $4h$. The expanded grid boxes retain separation and positive time. There are still at most Cn^2 cutoff factors. Taylor to the nearest forbidden pair diagonal or time hyperplane supplies eq. 5.cdxci, so exactly the same k_* and factorial exponent apply. This verifies explicitly that the time-boundary constraint introduces no hidden growth in n.

Uniqueness now follows without a choice of a particular partition. Any two continuous extensions agreeing on separated tensors agree on a dense subspace and hence on all of $\mathcal{S}_{\Delta,n}$. Thus the Fourier boxes, cutoffs, dyadic grids and orders of their absolutely convergent sums do not change the resulting functional. The same statement holds on the positive-time subspace.

If the compact moments are symmetric, real or invariant under a specified Euclidean motion, their extensions inherit that identity by density and continuity of the test-function transformations. If the compact moments are reflection positive, take finite positive-time Schwartz sequences and approximate their components by the tensors just constructed. For a negative-time and positive-time factor, their tensor product is flat on all cross diagonals: away from the time boundary one factor already vanishes, and at time zero its jets vanish. Internal diagonals are handled by their own factors. The tensor product is continuous in Schwartz seminorms. Thus the reflected quadratic forms converge, and their nonnegativity passes to these full sequence domains. This proves the distribution-space implication once eq. 5.cdlxxviii is supplied; it is not a proof of the gauge-theory estimate in eq. 5.cdlxxviii.

5.25.5.1. The graded-source limit and phase consequence. Under Prop. 5.76, every pointwise limit on separated tuples of its two-state centered moments retains eq. 5.cdlvii: for each fixed tuple the inequality is closed and its constants are uniform in cutoff and in both states. The phase-interface theorem realizes each of its component forms by such sequences of pure lattice states, centered at the specified reference states. Therefore the same bound holds for those components, including their possibly nonzero first moments; their one-point shifts are covered by the all-state comparison bound. The fixed equivalence between $\|\cdot\|_\sharp$ and $\|\cdot\|_1$ is absorbed in the radius-one constant b. Thus eq. 5.cdlxxviii applies with $\alpha = 5$ and the graded factorial exponent k. These limits and components consequently have the unique diagonal-flat extensions with order $20n$ and factorial exponent $k+32$, uniformly over the stated phase realization. This uses that realization rather than assuming an unrelated component decomposition has the same bounds. It supplies neither contact terms across the diagonals nor full rotational covariance.

5.25.6. *Why the precise hypotheses matter.* Position uniformity cannot be replaced by unrestricted exponential growth in an enclosing radius. For example, $S_1(f) = \int e^{|x|^2} f(x)\,dx$ has finite bounds for compact tests in every fixed ball about the origin, with an exponential radius cost, but has no continuous extension to Schwartz space agreeing with those values. Translated nonnegative compact bumps, damped by $e^{-|x|^2/2}$, converge to zero in every Schwartz seminorm while their values diverge. Their supports can be chosen in successively more distant unit balls. This example violates the position-uniform radius-one hypothesis. Under that actual hypothesis in eq. 5.cdlxxxi rules out this obstruction.

Nor may a prefactor such as e^{cn^2} be silently called factorial growth of a fixed order. A lognormal random constant field has moments $S_n(f_1, \dots, f_n) = e^{n^2/2} \prod_i \int f_i$. It is translation invariant and reflection positive, but violates every estimate in eq. 5.cdlxxx with fixed linear seminorm order and

fixed factorial exponent. To see this, take unit-integral Gevrey-two bumps in n separated unit balls with centers at distance $O(n)$ from the origin. Their tensor is diagonal-flat. For every fixed k, its seminorm p_{kn} grows at most $C^n(n!)^{C_k}$, by eq. 5.cdlxxxii and the polynomial location weights. Its moment grows like $e^{n^2/2}$, which is larger than $C^n(n!)^q$ for every fixed q. Thus the graded, fixed-exponent assumption in eq. 5.cdlxxviii has mathematical content; reflection positivity alone does not supply it.

6. Short distances and Euclidean covariance

The short-distance calculation identifies the free-curvature coefficient after running the coupling to the separation scale. It requires a conditional interacting law, including the exterior configuration and the differentiated normalizer. We give these requirements explicitly before the comparison and insertion estimates. The angular arguments and the reflection-positive Hilbert-space construction provide the covariance needed in §7.

6.1. **The short-distance coefficient and nontriviality.** The short-distance conclusions below retain the interacting extraction and state-transfer inputs of Group F. In particular the exact conditional identities of § 6.3 require the actual complete-law identification, connected force and boundary estimates, a common reference for conditional means, and marked bad-event moments with rates relative to the g^4 signal. The classical Gaussian cutoff theorem supplies a reference estimate; the interacting inputs require separate proof. The assertions of Props. 6.1 and 6.3 are proof-outline conclusions conditional on that remaining package.

6.1.1. *Two-point short-distance asymptotics, torus four-point positivity, and universality.* At short distances, the connected two-point function behaves to leading order like that of N^2-1 free fields. On the tori the same holds for the connected four-point function, restricted to nearly collinear quadruples, which form the class $\mathfrak{T}^{(4)}$ introduced in Thm. A(c)(5). The strength of these free fields equals the running coupling evaluated at the observation scale; it is non-zero and tends to zero only logarithmically.

Objects. The test pairs are $(f,h)\in\mathfrak{T}_{\mathbb{E}}(\mathsf{d};\vartheta)$, that is, pairs of non-negative Lipschitz bumps, each supported in a ball of radius d, placed at a distance from one another that lies between d and $6\mathsf{d}$ (§6.1.1.1). The free kernel is $\mathcal{K}_{\mathrm{free}}(f,h)=\iint f(x)h(y)|x-y|^{-8}dx\,dy$. We take the coupling $g(\mathsf{d})=g_{K-s(\mathsf{d})}(\beta)$, as well as the normalisation $\zeta_K(\mathsf{d})=\zeta'_{K-s(\mathsf{d})}(\beta)$, at the level whose spacing is smaller than d by a power of $\log g(\mathsf{d})^{-2}$.

Proposition 6.1 (non-triviality; Thm. A(c)(3)-(5))**.** Assume $g\le\gamma_W(\vartheta)$. Then there exist an integer $n_0\ge 1$, a threshold $\mathsf{d}_0=\mathsf{d}_0(g,\vartheta)\in(0,1)$, a function $\bar\eta:(0,\mathsf{d}_0]\to[0,\frac13]$ that is non-decreasing with $\bar\eta(\mathsf{d})\to 0$ as $\mathsf{d}\to 0$, and, for each $\ell\in\mathfrak{L}(g)$, a function $\bar\varepsilon_\ell:(0,\mathsf{d}_0]\to[0,\frac12]$ that is non-decreasing with $\bar\varepsilon_\ell(\mathsf{d})\to 0$ as $\mathsf{d}\to 0$, with the following properties whenever $\mathsf{d}\le\mathsf{d}_0$. (i) Take any first-passage datum with $K-s(\mathsf{d})\ge n_0$, any $\ell\in\mathfrak{L}(g)$ and any pair in $\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$. Then

$$\langle\mathcal{O}(f);\mathcal{O}(h)\rangle_{K,\ell,\beta}=\zeta_K(\mathsf{d})^2\,\tfrac{3(N^2-1)}{\pi^4}\,g(\mathsf{d})^4\,\mathcal{K}_{\mathrm{free}}(f,h)\,(1+\varepsilon),$$
$$|\varepsilon|\le\tfrac12,$$

and moreover $|\varepsilon|\le\bar\varepsilon_\ell(\mathsf{d})$, the inequality $|\frac{11N^2}{12\pi^2}g(\mathsf{d})^2\ln(1/\mathsf{d})-1|\le\bar\eta(\mathsf{d})$ holds, and $|\zeta_K(\mathsf{d})-1|$ is bounded by the same right-hand side; all of these bounds are uniform in (K,β). The same identity is also valid in every periodic Gibbs state at β on $\mathbb{L}_K$, with the same $\zeta_K(\mathsf{d})$ and $g(\mathsf{d})$ and with an error satisfying $|\varepsilon|\le\frac12$ (§6.1.1.1). (ii) On pairs of class C^1, every continuum limit obtained from Thm. A(a)(ii) on the torus T_ℓ, as well as every one obtained from Thm. A(a)(iii) on $\mathbb{R}^4$, obeys

$$S_2(f,h) = \tfrac{3(N^2-1)}{\pi^4}\big(\tfrac{12\pi^2}{11N^2}\big)^2 \frac{\mathcal{K}_{\mathrm{free}}(f,h)}{(\ln(1/\mathsf{d}))^2}\,(1+\eta),$$
$$c \le 1+\eta \le C,$$

and on each T_ℓ one has $\eta \to 0$ as $\mathsf{d} \to 0$. (iii) On the tori, let $\vartheta \le \vartheta_\square$, the number from Thm. A(c)(5), which depends only on the fixed data. For the additional four-point clause, assume there exist an integer $n_4 \ge n_0$, a number $\mathsf{d}_4 \in (0, \mathsf{d}_0]$, a side $\ell_4(g) \in \mathfrak{L}(g)$, together with, for each $\ell \in \mathfrak{L}(g)$ with $\ell > \ell_4(g)$, a threshold $\mathsf{d}_4(\ell) \in (0, \mathsf{d}_4]$, with the following property. Suppose $K - s(\mathsf{d}) \ge n_4$. Suppose $\mathsf{d} \le \mathsf{d}_4$ when $\ell \le \ell_4(g)$, and $\mathsf{d} \le \mathsf{d}_4(\ell)$ otherwise. Then for every quadruple in $\mathfrak{T}^{(4)}_{T_\ell}(\mathsf{d};\vartheta)$ (Thm. A(c)(5)), the following bounds hold uniformly in (K,β). Here c, C are constants that are allowed to depend on g as well as on the shape ϑ.

$$c\,(N^2-1)\,g(\mathsf{d})^8 \prod_i \|f_i\|_\infty \le S^T_{4;K,\ell,\beta}(f_1,\dots,f_4) \le C\,(N^2-1)\,g(\mathsf{d})^8 \prod_i \|f_i\|_\infty.$$

Clause (iii) is a statement about the lattice functions on the tori. For four points, Thm. A asserts nothing about the limits, and nothing about $\mathbb{R}^4$. ■

Additional comparison estimates are required for four-point positivity on arbitrary shapes or continuum components and for rigidity of general short-distance germs.

6.1.1.1. The short-distance two-point law and its asymptotic-freedom normalisation. **Objects.** The class $\mathfrak{T}_{\mathbb{E}}(\mathsf{d};\vartheta)$ consists of the pairs of non-zero Lipschitz functions $f, h \ge 0$ satisfying

$$\operatorname{supp} f \subset B(x,\mathsf{d}),$$
$$\operatorname{supp} h \subset B(y,\mathsf{d}),$$
$$\mathsf{d} \le \operatorname{dist}(\operatorname{supp} f, \operatorname{supp} h) \le 6\mathsf{d},$$
$$\textstyle\int f \ge \vartheta \|f\|_\infty \mathsf{d}^4,$$
$$\operatorname{Lip} f \le \tfrac{\|f\|_\infty}{\vartheta \mathsf{d}},$$

where the final two conditions are required of h as well. It follows that $c\vartheta^2\|f\|_\infty\|h\|_\infty \le \mathcal{K}_{\mathrm{free}}(f,h) \le C\|f\|_\infty\|h\|_\infty$. On $\Lambda^2\mathbb{R}^4$ the free curvature propagator is given by

$$\mathbb{K}(u) = \pi^{-2}|u|^{-4}\,\Lambda^2 \mathsf{I}(\hat u),$$
$$\mathsf{I}(\hat u) = 1 - 2\hat u \hat u^{\mathsf{T}},$$
$$\hat u = u/|u|,$$

which is the two-point tensor of $\mathrm{d}A$ when A is a one-form with covariance $\delta_{\mu\nu}(4\pi^2|x|^2)^{-1}$. Since $\Lambda^2\mathsf{I}$ is an involution, $\operatorname{tr}\mathbb{K}(u)^2 = 6\pi^{-4}|u|^{-8}$. On $\Lambda_{K,\ell}$, let $F = \mathrm{d}A$ be the free lattice curvature, where d denotes the lattice coboundary; this curvature is the standard Gaussian on exact 2-cochains with a single colour and unit coupling, and $\mathbb{E}_{\mathrm{free}}$ denotes the expectation with respect to it. Its background part at depth s is $F_B := \mathbb{E}_{\mathrm{free}}[F \mid Q_{K-s}(\mathbf{1})A$ modulo closed forms], and we set

$$C^{\mathrm{free}}_K = C^{\mathrm{bg}}_{K,s} + C^{\mathrm{fl}}_{K,s},$$
$$C^{\mathrm{bg}}_{K,s}(p,q) = \mathbb{E}_{\mathrm{free}}\,F_B(p)F_B(q),$$

$$\mathcal{K}_{K,s}(f,h)=\sum_{p,q} f(x(p))h(x(q))\,C^{\mathrm{bg}}_{K,s}(p,q)^2.$$

Let Q^{pl}_j be the average of plaquettes, that is, of 2-cochains, over blocks of order j; it is the counterpart of the straight average of 1-forms. Because $\mathrm{d}\,Q_j(\mathbf{1})=\mathrm{d}\,Q^{\mathrm{st}}_j(\mathbf{1})=Q^{\mathrm{pl}}_j\mathrm{d}$, the conditioning above is conditioning on $Q^{\mathrm{pl}}_{K-s}F$. The $C^{\mathrm{bg}}_{K,K-j}$ are therefore kernels of orthogonal projections whose ranges are nested, and the slices

$$\mathcal{C}_{(j)}:=C^{\mathrm{bg}}_{K,K-j+1}-C^{\mathrm{bg}}_{K,K-j}$$
$$(C^{\mathrm{bg}}_{K,K}=C^{\mathrm{free}}_K),$$
$$C^{\mathrm{fl}}_{K,s}=\sum_{1\le j\le K-s}\mathcal{C}_{(j)},$$

are the covariances of mutually independent pieces of F. For the free field, the renormalisation group thus reduces to an orthogonal decomposition.

Statement. The error in Prop. 6.1(i) satisfies $|\varepsilon|\le\frac12$ and, in addition, $|\varepsilon|\le\bar\varepsilon_\ell(\mathsf{d})\to0$ on each torus of the family, uniformly in the cutoff, the bare coupling, the pair and its position. The same identity, with the same $\zeta_K(\mathsf{d})$ and $g(\mathsf{d})$ and with $|\varepsilon|\le\frac12$, holds for $\langle\mathcal{O}(f);\mathcal{O}(h)\rangle_\nu$ in every periodic Gibbs state ν at β on $\mathbb{L}_K$. The constant arises as follows: $a_p\approx\frac12|F(p)|^2$; for each colour, Gaussian half-squares have covariance $\frac12(\text{covariance})^2$; and $\frac{N^2-1}{2}\cdot\frac{6}{\pi^4}=\frac{3(N^2-1)}{\pi^4}$.

Two variants. (i) *The one-loop coupling.* Define $\mathsf{g}(\mathsf{d}):=\big(g^{-2}+\frac{11N^2}{12\pi^2}\ln\frac{1}{\mathsf{d}}\big)^{-1/2}$, which depends on d and g only. Lem. 6.2, together with $0\le s\log L-\ln\frac{1}{\mathsf{d}}-\ln C_W(s)<\log L$, gives

$$\big|g(\mathsf{d})^{-2}-\mathsf{g}(\mathsf{d})^{-2}\big|\le o(s)+C+\tfrac{c_0}{\log L}\ln C_W(s)=o(s(\mathsf{d})),$$

uniformly in (K,β). Hence $g(\mathsf{d})/\mathsf{g}(\mathsf{d})\to1$, and if $\mathsf{g}(\mathsf{d})$ is put in place of the running coupling $g(\mathsf{d})$ of the construction, the law still holds below a different threshold, with error functions that differ but are of the same type. (ii) *Cutoffs off first passage.* Let $K'=K(\beta,g)$, and let $b=K-K'$, which may be positive, zero or negative. The dilation identity of § 5.2.4 then takes the form

$$\langle\mathcal{O}(f);\mathcal{O}(h)\rangle_{K,\ell,\beta}=\langle\mathcal{O}(f^{[b]});\mathcal{O}(h^{[b]})\rangle_{K',L^b\ell,\beta},$$

and the map $f\mapsto f^{[b]}$ sends $\mathfrak{T}_{T_\ell}(\mathsf{d};\vartheta)$ into $\mathfrak{T}_{T_{L^b\ell}}(L^b\mathsf{d};\vartheta)$ and leaves $\mathcal{K}_{\mathrm{free}}$ invariant. Consequently, whenever $L^b\ell\in\mathfrak{L}(g)$, $L^b\mathsf{d}\le\mathsf{d}_0$ and $K'-s(L^b\mathsf{d})\ge n_0$, the law at $(K',L^b\ell,\beta)$ with separation $L^b\mathsf{d}$ determines the behaviour of $\mu_{K,\ell,\beta}$ at separation d. For K large and under the conditions of clause (d)(1), § 5.2.4 and § 5.2 and appendix A.8 give $|b|\le C(\varrho)$ for the tuned data and for each of their offsets $t\in\mathsf{J}$. When $b<0$, the identity is to be read in the direction from $(K',L^b\ell)$ to (K,ℓ).

Proof outline. (1) The local analysis is carried out on a single torus, the smallest one, $T_{\ell_\star(g)}$. On that torus the law holds by Prop. 6.3 of §6.1.1.2, and its kernel there is $\frac{N^2-1}{2}\mathcal{K}_{K,s}$. Reflection positivity then carries it from this torus to the remaining volumes and to the Gibbs states. For the Gibbs states this is the only route available, because the expansion deals with torus measures. The local analysis can also be carried out on the other tori. Thm. A(c)(5) does this, with four sources, on every torus of the family, but only below a threshold on d which may decrease with ℓ. The transfer, by contrast, yields a single threshold d_0 common to all of them. (2) Comparison of lattice and continuum. Let u be the shortest vector representing $x(p)-x(q)$ on T_ℓ, with $|u|\approx\mathsf{d}$, let $\sigma_{pq}:=a_K^4\pi^{-2}|u|^{-4}$, and let $\mathbb{K}(\cdot)_{pq}$ be the entry belonging to the planes of p,q. Then

$$|C_K^{\mathrm{free}}(p,q) - a_K^4\,\mathbb{K}(u)_{pq}| \le C\Big[\Big(\frac{a_K}{|u|}\Big)^{1/2} + \Big(\frac{|u|}{\ell}\Big)^4\Big]\sigma_{pq}$$

$$(10a_K \le |u| \le \tfrac{1}{10}\ell),$$

$$|C_{K,s}^{\mathrm{fl}}(p,q)| \le o(1)\,\sigma_{pq}.$$

The first estimate uses the exponent and range proved in appendix A.4. Its range is compatible with the witness requirements. The fixed room floor gives $C_W(s) \ge C_{W,0}A_0^2$ for the admitted small couplings. Increase n_0 so that $10L^{-n_0} \le C_{W,0}A_0^2$. Then

$$\frac{a_K}{\mathsf{d}} = \frac{L^{-(K-s)}}{L^s\mathsf{d}} \le \frac{L^{-n_0}}{C_W(s)} \le \frac{1}{10}.$$

For two supports in the stated shape class, every short difference has $\mathsf{d} \le |u| \le 10\mathsf{d}$: the upper bound is the minimum support distance plus the two support diameters. Decrease d_0 so that $\mathsf{d}_0 \le \ell_\star(g)/100$. Consequently $10a_K \le |u| \le \ell/10$ on every admitted torus. The increase of n_0 uses only constants fixed before g; the decrease of d_0 is made after g, as in the witness construction.

The first inequality compares second differences of the Green's function on the lattice torus with those of the continuum Green's function. Because $a_K/\mathsf{d} \le L^{-n_0}/C_W(s)$, the first term on its right is small by the choice of n_0, and it tends to 0 with d uniformly in K, since the room increases. On the reference torus the second term is at most $C(\mathsf{d}/\ell_\star)^4$, which is small with d_0 and tends to 0 with d. The second inequality is §6.1.1.2(c). Approximating by a Riemann sum, at a relative cost of $Ca_K/(\vartheta^3\mathsf{d})$, yields $\mathcal{K}_{K,s} = 6\pi^{-4}\mathcal{K}_{\mathrm{free}}(1+o(1))$. (3) The common-normalizer transfer estimate in eq. 6.vi, under its stated comparison hypotheses, transfers the law to every T_ℓ and every ν at a cost of $C_2\mathsf{d}^2\|f\|_\infty\|h\|_\infty$, in which $C_2 = C_2(g,\vartheta)$ denotes the constant of that lemma at $n=2$. Take $\mathsf{d} \le \mathsf{d}_0(g,\vartheta)$; there Lem. 6.2 gives $\frac{11N^2}{12\pi^2}g(\mathsf{d})^2\ln(1/\mathsf{d}) \ge \frac{2}{3}$. In this range the cost is a fraction of the main term. That fraction goes to 0 as d does, and so do the $o(1)$ of (2) and the R of Prop. 6.3. The threshold d_0 is chosen at the point where these three errors combined give $|\varepsilon| \le \frac{1}{2}$. The main term is $\ge c\vartheta^2(\ln(1/\mathsf{d}))^{-2}\|f\|_\infty\|h\|_\infty$. Consequently, on every torus, the error goes to 0 as d does.

The same convergence extends to periodic Gibbs states and to $\mathbb{R}^4$ provided the reference-comparison and common-normalizer hypotheses of eq. 6.vi hold uniformly along the defining sequence.

(4) The quantities $\zeta_K(\mathsf{d})$ and $g(\mathsf{d})$ are controlled by Lem. 6.2.

6.1.1.2. Isolating a single scale: coupling and normalisation at the scale d. **Objects.** We set $s = s(\mathsf{d})$ and $k_0 := K - s$, the witness level. We also write $x := x_{k_0}(\beta) = g(\mathsf{d})^{-2}$. The separation measured in level-k_0 units is $\hat{\mathsf{d}} := L^s\mathsf{d} \in [C_W(s), LC_W(s))$. This inclusion holds for $\mathsf{d} < 1$, in which case $s(\mathsf{d}) \ge 1$, and it follows from the minimality of $s(\mathsf{d})$ and the monotonicity of C_W. We write $\underline{x}_s := g^{-2} + \lambda_\sharp s - c_5$ and $\bar{x}_s := g^{-2} + B_\sharp(s+1)$ for the ends of the envelope of § 5.2.1; they do not depend on (K,β), and they are non-decreasing in s. The room $C_W(s) = C_{W,0}\,p_0(\bar{x}_s^{-1/2})^2$ contains a constant $C_{W,0}$, chosen large after L, M and ϑ and before g. The lower bounds placed on this constant are $\vartheta\,C_W(s) \ge CM$, which is used in the Strategy, and those required in (b) and (c) below. The room tends to infinity, but $\log C_W(s) = o(s)$.

Lemma 6.2 (coupling and normalisation)**.** Uniformly in (K,β), and with no torus entering § 5.2.2, one has

$$|x_{K-s} - x_K - c_0 s| \le o(s),$$

$$\zeta_k' = 1 + \sum_{j\le k} b_j'(0) \in [\tfrac{2}{3}, \tfrac{4}{3}],$$

$$|\zeta'_{K-s} - 1| \le C g_{K-s}\, p_0(g_{K-s}).$$ ■

The first bound is the flow statement of § 5.2.1. Because $s(\mathsf{d}) \log L / \ln(1/\mathsf{d}) \to 1$, it yields $g(\mathsf{d})^{-2}/\ln(1/\mathsf{d}) \to c_0/\log L = \frac{11N^2}{12\pi^2}$. For the other two bounds, note that a constant source $w \equiv \zeta$ acts as a shift of β in which every auxiliary choice is held frozen. The increment $b_{j+1}(\zeta)$ is holomorphic for $|\zeta| < 2\varrho$, and its slope is $\le C\big[g_j^3 (\log x_j)^{\mathfrak{p}_0} + \varrho^{-1} g_j^{\kappa_0 - 4}\big]$ §5.20.1.2. Since $x_j \ge x_k + \lambda_\sharp(k-j) - C$, the slopes summed over the levels $j < k$ are at most $C\big[g_k (\log x_k)^{\mathfrak{p}_0} + \varrho^{-1} g_k^{\kappa_0 - 6}\big]$. This bound is a function of x_k alone, it tends to 0, and for $\kappa_0 \ge 7$ it is $\le C g_k\, p_0(g_k)$.

Proposition 6.3 (the law on the reference torus)**.** Let $g \le \gamma_W(\vartheta)$, let (K, β) be a first-passage datum, and suppose that $\mathsf{d} \le \mathsf{d}_0$, $K - s(\mathsf{d}) \ge n_0$ and $(f, h) \in \mathfrak{T}_{T_{\ell_\star(g)}}(\mathsf{d}; \vartheta)$. Then on the torus $T_{\ell_\star(g)}$, uniformly in (K, β) and in the pair, one has

$$\langle \mathcal{O}(f); \mathcal{O}(h) \rangle = \zeta_{k_0}'^{\,2}\, \tfrac{N^2-1}{2}\, x^{-2}\, \mathcal{K}_{K,s}(f, h)\, (1 + R),$$
$$|R| \le \tfrac{1}{4},$$
$$R \to 0 \,(s \to \infty).$$ ■

Strategy. In this proof the pair (f, h) is also written (f_1, f_2). We set $k_1 := k_0 + \mathsf{n}$ with $\mathsf{n} := \lceil 2 \log_L \bar{x}_s \rceil$. Here g is fixed first. Then d_0 is chosen, small enough that $\mathsf{n} \le s(\mathsf{d})$ whenever $\mathsf{d} \le \mathsf{d}_0$. This gives $k_1 \le K$. Reduce d_0 further to fit the complete clean read boxes and their padded neighborhood into a single lattice patch on the reference torus. The geometry of the two localized readouts below bounds the physical diameter by

$$C(\mathsf{r} + L\mathsf{D} + M\check{R}_{k_0}) L^{-(s-\mathsf{n})}.$$

Here $L^{\mathsf{n}} \le L\bar{x}_s^2$, and $\bar{x}_s$ is affine in s at fixed g. The factors $\mathsf{r}, \mathsf{D}, \check{R}_{k_0}$ grow at most polynomially in $1 + s$, uniformly in the first-passage datum. Thus the displayed diameter tends uniformly to zero as $s \to \infty$. Taking it below $\ell_\star(g)/4$ gives the common lift used in the box comparisons. Moreover $x_j \approx x$ for $k_0 \le j \le k_1$, because $B_\sharp \mathsf{n} \ll x$. The sourced construction §5.20.1.1 is carried out up to a level $k \le k_1$, with the sources measured at depth $K - k$. At depth s one has $\|f\|_{\sharp,s} = \|f\|_\infty$, because $\vartheta\, C_W(s) \ge CM$ for $\mathsf{d} \le \mathsf{d}_0(g, \vartheta)$. Above the witness level, $\|f\|_{\sharp, K-k} \le L^{k-k_0} \|f\|_\infty$, so the disc in z becomes smaller. The identities for the derivatives at $z = 0$ remain valid, and in the bounds each derivative in a source costs a factor L^{k-k_0}. One then has $Z(z)/Z(0) = \mathbb{E}_{\mathsf{P}^{(k)}}\, e^{\Phi^{(k)}(z)}$. In what follows we write $\mathsf{P} = \mathsf{P}^{(k_0)}$, which is a law distinct from the law P_{k_0} of Prop. 5.66 that appears further on, and $\Phi = \Phi^{(k_0)}$. Everything stated about these two objects in this paragraph holds at each level $k \le k_1$, with k in place of k_0 and $K - k$ in place of s. P is a probability law on triples. Each triple consists of a class $[h]$ of histories of level k_0 sharing one standing part $h^\flat$ §4.17.1.1, a block field $V \in \mathcal{V}_{k_0}$, and the stored fields. The law is determined by the unsourced sums $\tau_{[h]} := \sum_{h' \in [h]} \tau_{h'}$. Here $\tau_{[h]}(dV)$ is an integral over the stored fields, which are the fields left unintegrated at level k_0. Its integrand is the exponential of the exponent of the sum, integrated against a positive measure composed of the gauge-fixing factors, the characteristic functions, the delta functions of the averaging constraints and Haar measures. Dividing this integrand by $Z(0)$ yields the law on triples. Φ is the difference between the sourced and the unsourced exponent of $\tau_{[h]}$. An exponent is attached to such a sum and not to an individual τ_h §4.17.1.1, §4.17.1.45. The marginal of P in $([h], V)$ equals $Z(0)^{-1} \tau_{[h]}(dV)$. Tested on gauge-invariant functions, this marginal agrees with the push-forward under $h \mapsto [h]$ of the (h, V)-marginal of P_{k_0}, the law on triples (bare field, history, block field) of Prop. 5.66 §5.20.1.1. Projected further onto $[h]$, the marginal is the image of $\tilde{\nu}$. Both laws therefore extend one and the same law of $[h]$ and of the gauge orbit of V, but they add different variables. P_{k_0} adds the bare field and h, the particular member of the class.

P adds the stored fields, on which the part of Φ attached to the live components depends. With the notation $\Phi_B^{(k)} := \prod_{i\in B} \partial_{z_i} \Phi^{(k)}|_{z=0}$ we have

$$\langle \mathcal{O}(f_1); \mathcal{O}(f_2)\rangle = \langle \Phi_1^{(k)}; \Phi_2^{(k)}\rangle_{\mathsf{P}^{(k)}} + \mathbb{E}_{\mathsf{P}^{(k)}} \Phi_{12}^{(k)}$$
$$(1 \le k \le k_1).$$

All of these are expressions for one and the same quantity. At $k = k_0$ the first term on the right is the main term and the second term is an error ((a), (b)). At $k = k_1$ the roles are exchanged, and the law is obtained at that level ((c)). Both $\Phi_i^{(k)}$ and $\Phi_{12}^{(k)}$ are derivatives with respect to the sources. The argument uses conservation of mass alone, $\sum_h \tau_h^w(\mathcal{V}_k) = Z(w)$. In other words, it uses exactness with the sources measured at depth $K-k$, which is the depth s when $k = k_0$ (Prop. 5.65 of §5.20.1.1, together with §5.20.1.2). This is what isolates the scale; no conclusion is drawn from P_{k_0}, the coupling with the bare field. The law of $\mathcal{M}^{k_0}U$ is not identified with the marginal of P in V, because inside what $\mathbf{R}$ heals it replaces the law of the block field. Under P_{k_0}, the equality $V = \mathcal{M}^{k_0}U$ is valid only on bonds at a large distance from the components that $\mathbf{R}$ has acted on. For this reason the sources are carried through the steps, and no observable is conditioned.

(a) *Locality.* The source enters the exponent in three ways. *Pointed pieces:* away from the live components, the coupling term is a sum of functions, each of which depends on a single value of w; this holds because the running shift at a plaquette is a finite sum of functions of this kind. As $f_1 f_2 = 0$, these pieces give no mixed derivative. *Regular local terms:* these are indexed by $j \le k_0$ and $X \in \mathcal{D}_j$, and they depend on w only on $X^\sim$ §5.20.1.2. Every domain that meets both supports satisfies $d_j(X) \ge c\,\hat{\mathsf{d}}/M$. By Cauchy bounds on the source disc, the contribution of these terms to Φ_{12} is bounded pointwise by $C\hat{\mathsf{d}}^4 e^{-c\kappa\hat{\mathsf{d}}/M} \prod \|f_i\|_\infty$. This is of order $e^{-c\,p_0^2}$, where, in this step, p_0 without an argument stands for $p_0(g_{k_0})$. It is therefore smaller than any power of $g(\mathsf{d})$. *The part attached to the live components:* only a cruder bound is used for this part. Consider a class $[h]$ in which a live component is in contact with a support, that is, lies within distance $c\,\mathsf{d}$ of it. The label is common to the class, and hence each live component, its lineage and the age of that lineage are common to the class as well; the members of the class differ only in healed components. Let $a(h)$ denote the largest lineage age among such components (Prop. 5.71 of §§ 5.20 and 5.22). On such a class the mixed derivative of this part is bounded by

$$(\log x)^C \Big[e^{-c\,\hat{\mathsf{d}}} + L^{C\check{R}_{k_0}} L^{8a(h)} \Big] \prod \|f_i\|_\infty .$$

It is the mean that is small: $|\mathbb{E}_{\mathsf{P}} \Phi_{12}| \le o(1)\, x^{-2} \prod \|f_i\|_\infty$. This follows from the age series of §§ 5.20 and 5.22 (Prop. 5.71 and its proof), taken with rate $b \ge 8$. The growth of the amplitude and the decay of the series are governed by one and the same age, namely the age of the lineage. The age recorded in the label restarts wherever $\mathbf{R}$ leaves a component standing as an island, and it is not larger. The number of cells of level k_0 in contact with the supports is $\le C\hat{\mathsf{d}}^4$, so

$$\mathbb{E}_{\mathsf{P}} \Big[\mathbf{1}\{\text{contact}\}\, L^{8a(h)} \Big] \le C\hat{\mathsf{d}}^4 \sum_{a\ge 0} L^{8a} \max_{\mathfrak{c}} \tilde{\nu}\big(\mathfrak{lv}_{k_0}^{\ge a}(\mathfrak{c})\big) \le C\hat{\mathsf{d}}^4 e^{-c\,p_0(g(\mathsf{d}))},$$

and this bound overcomes $(\log x)^C L^{C\check{R}_{k_0}} x^2$ because $\mathfrak{p}_0 \gg r$. *At level k_0 the two-point function draws on this series twice*, both times at a rate $b \ge 8$. At level k_1, in (c), it draws on the series once, at the same rate. The first use is the present one, the bound on the mean of the live part's mixed derivative. The second occurs in (b), in the covariances of the live first derivatives; there the product of two such derivatives carries $L^{4a} \cdot L^{4a}$. At level k_0 these two uses are the source of the weight L^{8a} for two points. At level k_1 its source is (c1). Of the growth, only one property enters: as a function of the age, it is no faster than geometric. A different exponent, or an additional

factor polynomial in a, is handled by a larger rate b, that is, by a larger c_a, a constant that may be increased freely. Hence the theorem on the ages (Prop. 5.71) underlies Thm. A (c)(3) and (c)(4). It also underlies (c)(5), for which Thm. A(c)(5) employs this series with a larger rate.

(b) *The marginal part.* Here the coupling carries the source. Let $h_y^{(j)} \geq 0$, with $\sum_y h_y^{(j)} = 1$, be the tent functions of §5.20.1.1, §5.20.1.2. These weights have range CMa_{K-j} and serve to localise the counterterm of step j. At a fixed level these are the functions h_z in the source profiles. Let p_y be a bare plaquette at y. At points where every profile φ_j, $j \leq k_0$, takes the value 1, that is, off the live components, the running shift satisfies, by its formula in §5.20.1.2,

$$\partial_{z_i} \zeta_{k_0}(p; w)\big|_0 = f_i(x(p)) + \sum_{j \leq k_0} b_j'(0) \sum_{y \in \mathbb{T}^{(j)}} h_y^{(j)}(p)\, f_i(x(p_y)) =: \zeta_{k_0}' \tilde{f}_i(p),$$

$$|\tilde{f}_i - f_i| \leq \tfrac{CM}{\vartheta \hat{\mathsf{d}}} \sum_j |b_j'(0)|\, \|f_i\|_\infty;$$

the factor $\zeta_K(\mathsf{d})^2$ arises here. Differentiating $-A(x_{k_0}(\cdot) - \zeta_{k_0}(\cdot\,; w), U_{k_0})$ gives

$$\Phi_i = \mathrm{const} + \zeta_{k_0}' X_i + (\mathrm{irrelevant}) + (\mathrm{live}) + (\mathrm{far}),$$

$$X_i(V) := \sum_p \tilde{f}_i(p)\, \chi_{k_0}(\square(p))\, a_p(U_{k_0, \square(p)}(V)).$$

In this formula $\square(p)$ is the test cube containing $x(p)$, and $U_{k_0,\square(p)}$ is the background localised about that cube, hence near $\mathrm{supp}\, f_i$. The test $\chi_{k_0}(\square(p))$ acts as a window: a plaquette contributes only if its localised background passes the plaquette test. Consequently X_i depends on V only near the support and is the same for every history. It also satisfies $|X_i| \leq Cx^{-1} p_0^2\, \hat{\mathsf{d}}^4 \|f_i\|_\infty$. Away from the large-field regions, the global background differs from the localised one by an amount exponentially small in the radius of localisation; this difference is the term (far). The size of the irrelevant terms is $\leq C(\log x)^{3\mathfrak{p}_0 - q} x^{-1} \hat{\mathsf{d}}^4 \|f_i\|_\infty$. The physical bound uses the source-family coefficient and a quadratic estimate at the real field size, together with the Wilson subtraction at each grouped core. The factor x^{-1} cannot be dispensed with. The law uses $p_0^2\, \hat{\mathsf{d}}^8 (\log x)^{3\mathfrak{p}_0 - q} \to 0$ as the coefficient of x^{-2}, and Cauchy bounds over the source radius ϱ yield no power of x. The factor comes from the bound on the regular family produced by a step (§4.17.1.45, "Dependence of the regular family on the array"). That bound also holds for shifts of the coupling that vary with position, and it weights a difference of shifts by $CM^4 \alpha_{0,k}^2$. We apply it to the sourced and the unsourced arrays of §5.20.1.2. The remainders enter through the slope towards the constants carried by the induction of §5.20.1.2, and they are small because of their size. The result is a linear recursion for the slopes in the source, and the small factor in front of the bound closes this recursion over the levels. Its solution is $\approx x_j^{-1} (\log x_j)^{\mathfrak{p}_0 + q}$, and this takes the place of ϱ^{-1}. The oscillating part of the source, namely what remains after subtracting from the source its value at a single point of the domain, has a slope bounded in the same way, with an extra factor $L^{-(k_0 - j)}$. The live terms become larger as the age increases. If a live component has $\mathrm{age}^{\mathrm{lab}} = a$, the fields stored in it are those of every level from $k_0 - a$ onwards, together with those of the next finer level. For a component holding an island, the block of which retains the fields of its core §4.17.1.3, the stored fields begin at the first level of its lineage. Over one cube of level k_0, the plaquette sums of these fields contain at most $L^{4(a+1)}$ terms. For the live terms, only their first two moments on the events of contact enter, so that, as in (a), it suffices that their growth be geometric in the age. Let h be a history in which a live component is in contact with $\mathrm{supp}\, f_i$. On such a history the live contribution to Φ_i is bounded by $(\log x)^C \big[e^{-c\hat{\mathsf{d}}} + L^{C\check{R}_{k_0}(a(h)+2)^C L^{4a(h)}}\big] \|f_i\|_\infty$. The power of the age in this bound is the one of Thm. A(c)(5). In the bound $Ce^{-cp_0(g_j)}(2L^b)^{-a}$ of Prop. 5.71, the factor 2^{-a} absorbs this power, so the age series is used at the same rate. This live contribution

has two covariances. One is with the live contribution of the other factor, with weight $L^{8a(h)}$. The other is with those functions of V that occur in the other factor, with weight $L^{4a(h)}$, taken against their supremum bound. By the age series of §§ 5.20 and 5.22, as in (a), both of these covariances are $o(1)\,x^{-2}\prod\|f_i\|_\infty$.

From the two witness insertions to the physical sources. Let λ_{k_0} be the marginal in V of the unsourced law P on the class, V and the stored variables. Write $\mathsf{r} = \mathsf{d}/a_s$, $x = x_{k_0}$, and set $u = x^{-2}\|f_1\|_\infty\|f_2\|_\infty$. The identity obtained by differentiating conservation of mass at the witness is

$$S_2^T(f_1, f_2) = \mathbb{E}_{\mathsf{P}}\Phi_{12} + \mathrm{Cov}_{\mathsf{P}}(\Phi_1, \Phi_2).$$

The law on the right is the full unsourced law. In particular, integrating out the stored variables before differentiating would add the conditional covariance of their two source derivatives to the mixed term.

Use the decomposition $\Phi_i = c_i + \zeta'_{k_0}X_i + I_i + B_i$ at k_0. Here

$$c_i = \partial_{z_i}\sum_{j\le k_0}\sum_{X} e_j(X; w_z)\Big|_{z=0}.$$

This is the same number for every class. The difference between the actual share of source energy in a class and this sum is included in B_i. Count the missing shares of the actual resummed label in the level units of its retained regions. A healed region already removed by resummation enters through its refiled activity and the compensated label. It is not a present lineage contact of the witness. The coefficient of X_i is obtained by summing the single-point source shifts with their own tents. The sourced exponent on this ordinary carrier uses the running weight $x_{k_0} - \zeta_{k_0}$ and the regular members with their tent counterterms already subtracted. The difference of those counterterms is the accumulated shift $\zeta_{k_0} - w$ there. One can first cancel it against the shift in the weight and then extract the Wilson part from the unsubtracted regular derivative. Equivalently keep ζ'_{k_0} in the readout and differentiate the already subtracted regular members. In either form the term $b'_j(0)f_i(x(p_y))A(h_y^{(j)}, \cdot)$ is counted once. There is no analogous mixed derivative of a pointed piece: it is a function of one source value and $f_1 f_2 = 0$.

For I_i, apply the regular formation estimate eq. 4.clxxvii and the window estimates eq. 4.clxxx–eq. 4.clxxxi at $n = k_0$. The real regular windows used here have complete retained data of size $C\delta_n$. In the interior of the padded small-field set, use eq. 3.cdiii and the local gauge estimate on the full stencil of the localized background. The input is the passed tests on the retained datum with its regular background profile, rather than just the curvature cutoff in X_i. The fixed factors in the real gauge estimate fit the regular analytic radius for small g. The independent-coordinate room in the window estimates is the room of the ordinary regular formation rule. Writing $a = n - j$ and $s_n = \delta_n/\alpha_n$, the near part per witness window is bounded by

$$C\|f_i\|_\infty\sum_{j\le n} A_j\left[\left(\frac{\delta_n}{\alpha_j}\right)^2 L^{-\alpha_H a} + \left(\frac{\delta_n}{\alpha_j}\right)^3 L^{-2a}\right] + C\delta_n^2 s_n\,\mathrm{Lip}(f_i)\sum_{j\le n} h_j.$$

Indeed $A_j/\alpha_j^2 \le Cs_n$, $\delta_n/\alpha_j \le Cs_n(1+a)^{1/2}$, and $h_j = h_n L^{-a}$. Both geometric series, including the polynomial in a in the cubic term, give $C\delta_n^2 s_n\|f_i\|_{\sharp,n}$. Here $\|f_i\|_{\sharp,n} \le C\|f_i\|_\infty$ by the lower bound on r. The quadratic term that survives the Wilson subtraction is included in this estimate. Its coefficient is bounded by A_j/α_j^2, so the small factor s_n comes from the formed source derivative. The cubic Cauchy estimate alone would leave this quadratic term at the size of the insertion.

For the ordinary old remainder, the vacuum subtraction gives $(\delta_n/\alpha_j)^2 L^{-4a}$ on the admitted core, which pays for the fine-site count. Its derivative has the additional factor $x_j^{-\kappa_0/2}$ from the

ordinary remainder bound and eq. 4.clxxvi. Its contribution per witness window is at most

$$C\|f_i\|_\infty \delta_n^2 \sum_{j\le n} x_j^{-\kappa_0/2}\alpha_j^{-2} \le C\|f_i\|_\infty x^{-1}p_0^2 x^{2-\kappa_0/2}.$$

The last inequality uses $\alpha_j^{-2} \le Cx_j$ and the lower flow bound. It is useful to retain the small field ratio until after the sum. The common-background, common-base and pointed assumptions of the ordinary formation estimate are needed for these estimates. A bound for a constant source on a larger disc cannot furnish the estimate for the derivative in f_i.

For a long ordinary domain entirely in a homogeneous nonwrapping window, the same vacuum subtraction is performed in its axial chart. At age a the complete field has potential at most $C(2+\operatorname{diam}_j X)\delta_n L^{-2a}$ in that chart, and the normalized current has size at most $C\delta_n L^{-3a}$. The first derivative at the trivial pair is zero by constant colour invariance. The quadratic Cauchy bound therefore contributes at most

$$CA_j\|f_i\|_\infty e^{-\kappa d_j(X)}(2+\operatorname{diam}_j X)^C(\delta_n/\alpha_j)^2 L^{-4a}.$$

The bound is used only on the admitted core. If the polynomially enlarged field ratio is at least the half-disc threshold, the unexpanded ordinary core bound gives the same majorant after enlarging the constant. For domains reaching beyond a fixed enlargement of the witness window around their source anchor, a spare exponential in $d_j(X) \gtrsim L^a$ pays for the polynomial and gives a summable gain after the fine-site count. This argument concerns only the homogeneous domains. For an ordinary domain depending on f_i whose enlargement reaches a live region outside the contact neighborhood, the length is at least $c\mathfrak{r}L^a/M - C$, since the determining set must also reach $\operatorname{supp} f_i$. For a winding domain the bound is $c(\ell/a_s)L^a/M - C$. The ordinary source-disc estimate and the rooted count then put these terms into the exponentially small far error. No small homogeneous field ratio is used on them. Inside the contact neighborhood the live terms are estimated by the age series. The formation and remainder bounds have to hold on the actual regular domains of the unsourced histories.

Let $i(x) = C[(\log x)^{3\mathfrak{p}_0-q} + p_0^2 x^{2-\kappa_0/2}]$, allowing a larger constant for the ordinary far sum. The estimates give $\|I_i\|_\infty \le C\mathfrak{r}^4 x^{-1} i(x)\|f_i\|_\infty$ for the regular part and $\|X_i\|_\infty \le C\mathfrak{r}^4 x^{-1}p_0^2\|f_i\|_\infty$.
A bound using the physical readings. The correlation expansion gives a second way to obtain the irrelevant-source bound. On the same bare torus it starts with the product Haar measure, the same plaquette profiles, and the action $A(\beta - w, U)$. Its extra bare constant is source independent and cancels in $Z(w)/Z(0)$. One can therefore use this calculation for the physical generating function with its own standing histories and running coefficients.

Use the vacuum-subtracted source derivatives of that expansion on their original domains. Write a_j for its tube radius and v_{j-1} for the running maximum of its source weights. The source-disc estimate gives the majorant $Cv_{j-1}a_j^2$ before decay and the norm of the test function. The old remainder has the separate majorant $Cr_{\mathrm{s}}^{-1}g_j^{\kappa_0}$, with r_{s} the source radius. These bounds precede evaluation at a field.

At witness level n, write $t = L^{j-n}$, $\nu = 1+n-j$ and $D_X = \operatorname{diam}_j X + 5M$. First group the constant-source terms contained in the ν-layer enlargement of the anchor cube with their Wilson subtraction at coefficient $b_j'(0)$. For the remaining constant-source terms and for the nonconstant-source differences and old remainders, retain a domain in the short sum when

$$D_X t \le \rho/16, \qquad C_M D_X t^2 \mathfrak{e}/a_j \le 1,$$

where ρ is the witness separation in its lattice units and $\mathfrak{e} = C_{\mathrm{w}}\varepsilon_n$ is a fixed bound for the real-field amplitude. The native grouped core estimate uses its analytic box extension on the field-rescaling disc. At the real endpoint the extension must give the same field and current as the read on the smaller box, up to gauge. The grouped coefficient vanishes to second order at the trivial field.

Schwarz's estimate on this disc gives the factor

$$Cv_{j-1}a_j^2\left(\frac{\mathfrak{e}_Q}{a_n}\right)^2\left(\frac{a_n}{a_j}\right)^2\nu^3(1+a_n/a_j)t^{4+\hat{\alpha}},\qquad \mathfrak{e}_Q\le C\varepsilon_n.$$

Here $\hat{\alpha}>0$ is the exponent in the grouped core estimate. The grouped core uses the same counterterm coefficient as the running source. For the individual terms in the rest of the short sum, the vacuum value and the first field derivative vanish after subtraction and constant colour invariance. In the real box gauge the quadratic estimate costs $(C_MD_Xt^2\mathfrak{e}/a_j)^2$ times the original majorant. Thus the two radii cancel in the grouped core, and a_j^2 cancels in each individual term with majorant $Cv_{j-1}a_j^2$.

The factor t^4 pays for the $O(|S^+|t^{-4})$ older anchors over the enlarged witness support S^+. In the core the remaining factor $\nu^3(1+a_n/a_j)t^{\hat{\alpha}}$ is summable by the running radius comparison. The constant-source tails outside the core have length at least $\nu-1$, and the absolute domain sum pays for D_X^2 and gives exponential decay in ν. The comparison of nonconstant and constant sources uses the Lipschitz profile and has the summable factor $L^{-(n-j)/2}$. For the old remainder the corresponding age sum is bounded by its memory majorant $Cr_{\mathrm{s}}^{-1}\mathfrak{m}_n$.

Outside the grouped cores, if either short-domain inequality fails, then

$$d_j(X)+5\ge c\,t^{-1}\min\{\rho/M,\ a_n/(M\mathfrak{e})\}.$$

This uses $D_X\le 4M(d_j(X)+5)$ and the bound on ta_n/a_j. Use twice the original analytic majorant on these long terms. The length gain pays for the anchor count and the ratio $a_n^2/\mathfrak{e}^2$ at weak coupling. Indeed $a_n/\mathfrak{e}$ grows as $(\log x)^{q-\mathfrak{p}_0}$ and the lower witness bound for ρ grows as $(\log x)^{2\mathfrak{p}_0}$. The exponential decay at the smaller of these two scales pays for the logarithmic ratio. With the kept part this gives

$$|I(f)|\le C|S^+|\,\|f\|_{\sharp,n}\big[\widehat{v}_{n-1}\varepsilon_n^2+r_{\mathrm{s}}^{-1}\mathfrak{m}_n\big],\qquad \widehat{v}_{n-1}=\max_{j\le n}v_{j-1}.$$

At the witness, $\widehat{v}_{n-1}\le C(\log x)^{\mathfrak{p}_0-q}$ and $\mathfrak{m}_n\le Cx^{-(\kappa_0-2)/2}$. This gives the power $3\mathfrak{p}_0-q$ in $i(x)$. The source bound on the analytic tube by itself has the larger power $\mathfrak{p}_0+q$.

The localized function of V evaluates only the shortened sum at the prolonged box field and is zero off its window. For the global field add the whole long sum. On a history with no live label meeting the neighborhood $S_i^{+\rho/4}$ and with the window test passed, evaluate the shortened sum at its real background and add only those long domains in its ordinary region $\Lambda_j(h)$ at each level. The buffer $M\check{R}_jt\le\rho/8$ puts the source anchors inside the ordinary region with their required margin. The source profiles there are one. Use equality with the whole-lattice constituents on the admitted ordinary reads. In the joint law multiply this irrelevant insertion and the corresponding main insertion by the same per-support indicator. The long terms use their own original pair domains. The history bound is uniform before averaging the integrated variables of a history. The discarded indicators, the replacement by the localized main insertion, and any history-dependent share of the vacuum subtraction enter the separate comparison terms.

In a standing history, the geometric part of the reading comparison uses the empty trace. The record sums out a healed component and keeps its contribution in the ordinary activities and constants. The live regions are those still present at the witness level. On the event that none meets $N_i=S_i^{+\rho/4}$, the standing trace is empty throughout N_i. For the grouped core, νL^{j-n} is bounded independently of j, so its enlarged anchor cubes have bounded size in witness units. The box-extension margins of the core also have fixed size in those units, so its reads fit inside $S_i^{+3\rho/16}$. For the other short terms the diameter bound $D_Xt\le\rho/16$ and the source support put their complete reads inside $S_i^{+3\rho/16}$. This leaves $\rho/16$ for rounding to the larger history cells. By

eq. 6.xxxv, this rounding costs at most $C_{\mathrm{tr}} M \check{R}_n$ in witness units. The lower witness bound for ρ dominates this cost at weak coupling. Thus all these short trace windows are inside N_i.

For the history background on that window the preceding levels obey $x_j \geq x_n - C$. It follows that $g_j \leq \sqrt{2}\, g_n$ when $x_n \geq 2C$. The degree function $g(\log g^{-2})^{\mathfrak{p}_0}$ is increasing for $\log g^{-2} > 2\mathfrak{p}_0$. Take x_n large enough that this interval contains $\sqrt{2}\, g_n$. Then $\varepsilon_j \leq \sqrt{2}\, \varepsilon_n$ throughout the older levels. The native data bounds therefore use one real-field amplitude at the witness. On each common admitted pair domain, Lem. 6.5 and Prop. 6.6 identify the short history terms with their own whole-lattice terms, for the same input list and pointing. This identity also holds for their first source derivatives. The long terms keep their history-dependent domains and use the long-domain bound.

In this reading keep the variables integrated inside each history in the unsourced joint probability law. Let $\mathcal{G}$ be the data fixed for its inner integral. In the representation with a fixed reference measure and a source-dependent exponent $A(z)$, the history's first jet is $\mathbb{E}[\partial_i A(0) \mid \mathcal{G}]$. Its mixed jet is the conditional mean of $\partial_{12} A(0)$ plus the conditional covariance of the first derivatives. Total covariance gives

$$\mathbb{E}[\mathrm{Cov}(\partial_1 A(0), \partial_2 A(0) \mid \mathcal{G})] + \mathrm{Cov}(\mathbb{E}[\partial_1 A(0) \mid \mathcal{G}], \mathbb{E}[\partial_2 A(0) \mid \mathcal{G}]) = \mathrm{Cov}(\partial_1 A(0), \partial_2 A(0)).$$

Apply the uniform physical bound to the three irrelevant covariances in this joint law. These pair one irrelevant insertion with the other main insertion, or the two irrelevant insertions with each other. The two sources' factors give the bound $C\mathsf{r}^8 x^{-2} i(x)(p_0^2 + i(x)) \prod_i \|f_i\|_\infty$. This accounts for their shares both inside and between histories, giving the same covariance error in the bound below. The bound on these three covariances holds under any probability law with the stated physical supremum bounds. To use it in the standing-class decomposition here, it suffices to identify the evaluated first-source sum, with its vacuum and Wilson subtractions, on its admitted real pairs. The empty trace comparison gives equality within one whole-lattice recipe. For a transfer between the two expansions one must also match their formed whole-lattice sums and counterterms. The mixed derivative of the exponent itself uses the support and age estimates below.

In the mixed derivative the regular support must reach both original supports, and at age a its length is at least $c\mathsf{r} L^a/M - C$. The Cauchy cost in the slow sources is uniform at the witness. The anchor count times the spare exponential is summable in a and gives $Ce^{-c'\mathsf{r}} \prod_i \|f_i\|_\infty$. For the remaining terms put $\epsilon_{\mathrm{far}} = e^{-c\mathsf{r}} + e^{-c\check{R}_{k_0}}$ and $\Upsilon = L^{C\check{R}_{k_0}}$. The localized-background error uses both radii: the collar at the test cube and the distance to the nearest live region. The bound on B_i has a factor $\epsilon_{\mathrm{far}} + \mathbf{1}_{\mathcal{B}_i} \Upsilon L^{4a_i}$; the live mixed derivative has the corresponding L^{8a} bound. Here the ages are lineage ages. Summing over the $C\mathsf{r}^4$ cells in contact with the supports and applying Prop. 5.71 at rate at least eight gives

$$\mathbb{E}_{\mathsf{P}}\big[\mathbf{1}_{\mathcal{B}} L^{8a}\big] \leq C\mathsf{r}^4 e^{-cp_0}.$$

Since X_i depend only on V, covariance bilinearity and $|\mathrm{Cov}(A,B)| \leq 2\|A\|_\infty \|B\|_\infty$ for the bounded regular terms give

$$\begin{aligned} S_2^T(f_1, f_2) &= (\zeta'_{k_0})^2 \,\mathrm{Cov}_{\lambda_{k_0}}(X_1, X_2) + E, \\ |E|/u &\leq C\mathsf{r}^8 i(x)(p_0^2 + i(x)) + x^2 (\log x)^C \big[\epsilon_{\mathrm{far}} + \Upsilon^2 \mathsf{r}^4 e^{-cp_0}\big]. \end{aligned}$$

The second line uses the first and second contact moments for the covariances involving B_i, and the first moment of the mixed derivative. In the witness window $\mathsf{r} \leq C(\log x)^{2\mathfrak{p}_0}$; the first term tends to zero for $q > 21\mathfrak{p}_0$ and sufficiently large κ_0. The other terms tend to zero by the lower window bound, $r > 1$, and $\mathfrak{p}_0 > \max\{1, r\}$.

The preceding error bound is uniform under the bare lattice translations used in the witness covariance estimate. The physical left-hand side is invariant under these translations, so average the identity over the same translations whenever the witness estimate is an average. The covariance estimate is applied with the widened profiles $\tilde{f}_i$. The widening is at most CMa_s, and these supports

still have separation comparable to d. In the resulting bare kernel sum, replacing $\tilde{f}_i$ by f_i costs at most $C\|\tilde{f}_i - f_i\|_\infty \|f_{3-i}\|_\infty$ times the coupling factor x^{-2}. Indeed the squared curvature kernel on the separated supports has a bounded absolute sum with the two plaquette volume factors. By (b) the cost relative to u is $O(M/(\vartheta \mathsf{r}))$. This replacement precedes the Riemann estimate, which uses the Lipschitz bounds of the original profiles.

(c) *Gaussian extraction: the free template.* The covariance of the bounded witness fields X_i is obtained from the generating function on the fluctuation segment. Denote by $\mathcal{H}_j B$ the exact bare 2-cochain of smallest norm that satisfies $Q_j^{\mathrm{pl}} \mathcal{H}_j B = \mathrm{d}B$. It is the curvature of the linear minimiser, namely $\mathrm{d}\,\mathsf{H}_j$ at $U = \mathbf{1}$, a point at which $J = 0$ and the three Green's functions of § 3.12.1 agree. We then have

$$\mathcal{C}_{(j)} = \mathcal{H}_{j-1} C_j^{\mathrm{tr}} \mathcal{H}_{j-1}^*,$$

$$|\mathcal{H}_j(p,b)| \le C\eta_j^2 e^{-\delta |x(p) - x_b|/a_{K-j}},$$

$$|\mathcal{C}_{(j)}(p,q)| \le C\eta_{j-1}^4 e^{-\delta |x(p)-x(q)|/a_{K-j+1}}.$$

The free propagator framework is developed in [B84a]. For the centered blocks used here, the uniform pointwise estimate follows from Prop. 3.182 and its exponentially weighted version. Rescale a level-j block to side one and put $h = \eta_j$. At the trivial flat reference, let $H_h B$ be the solution with fine force zero and coarse datum B. In the notation of eq. 3.mccxlvii,

$$H_h = K_0^{-1} Q^\sharp (Q K_0^{-1} Q^\sharp)^{-1}.$$

This uses the straight row in eq. 3.mccxxiv. The gauge penalty vanishes: testing the first saddle equation against $D\lambda$ with $q\lambda = 0$ eliminates both its curl term and its multiplier term, while eq. 3.mccxlix makes the remaining gauge test surjective. Thus $RD^* H_h B = 0$. The first equation is consequently the Euler equation for the curl energy on $QA = B$, and its curvature is the least-norm exact curvature with datum $\mathrm{d}B$. Indeed, if $f = \mathrm{d}v$ is exact and $Q^{\mathrm{pl}} f = \mathrm{d}B$, then $Qv - B$ is a closed coarse one-form. On the trivial periodic lattice it is a coarse gradient plus a constant. The identities $qE_q = 1$, $QD = \mathrm{d}_c q$ and $Q(\text{constant}) = \text{constant}$ change v by a closed fine one-form to impose $Qv = B$ without changing f. This identifies the curvature with $\mathcal{H}_j$ modulo closed forms.

Take B to be a unit vector on the coarse bond b. The weighted bound eq. 3.mccxlv gives

$$|\mathrm{d}_h H_h(p,b)| \le C e^{-\delta\, \mathrm{dist}_{\mathrm{phys}}(p,b)}.$$

Its constants are uniform in the fine subdivision and in the volume. A raw bare plaquette curvature is $h^2 \mathrm{d}_h H_h$; restoring the physical block side therefore proves the factor η_j^2 and the distance scale a_{K-j} in the displayed estimate for $\mathcal{H}_j$. This argument uses the scalar equation and projector smoothing in § 3.43, so it does not require a differentiated resolvent to map L^2 into L^8. The symbol C_j^{tr} denotes the single-step covariance in a tree gauge,

$$C_j^{\mathrm{tr}} := \iota \left(\iota^* \mathcal{H}_{j-1}^* \mathcal{H}_{j-1}\, \iota\right)^{-1} \iota^*,$$

$$\mathrm{d}'\, C_j^{\mathrm{tr}}\, \mathrm{d}'^* = C_j^{\mathrm{curv}},$$

where ι embeds into the 1-forms B of level $j-1$ the subspace of those that satisfy $B = 0$ on one contour for each site (a tree in each block, imposed sharply) together with $Q^{\mathrm{st}} B = 0$. In this formula Q^{st} denotes the straight average over the blocks of the single step from level $j-1$ to level j, at $U = \mathbf{1}$. It is thus the covariance, on that subspace, of the quadratic form $\langle B, \mathcal{H}_{j-1}^* \mathcal{H}_{j-1} B\rangle = \|\mathcal{H}_{j-1} B\|^2$, which equals twice the free action of the minimiser with datum B. On the subspace one has $\|\mathcal{H}_{j-1} B\|^2 \ge c\|\mathrm{d}'B\|^2 \ge c_L \|B\|^2$. It differs from the covariance $C^{(j-1)}$ of § 3.12.1-§ 4.5 at the trivial background in two ways. The gauge differs: there it is imposed softly, on the means over all

contours. The constraint also differs: the straight average is not the linearisation of $\mathcal{M}$. The two gauges induce the same curvature law.

Let d' and d'' be the fine and coarse coboundaries, and A the plaquette average for this step. The flat identities $d''Q = d''Q^{\mathrm{st}} = \mathsf{A}d'$ give the common image of the two constraints under d'. To see surjectivity, take $h = d'B$ with $\mathsf{A}h = 0$. The coarse defect QB is closed. Remove its constant harmonic part by a fine constant cochain, since the average on constants is invertible. Remove its exact part by a fine gradient whose scalar has the prescribed values at the coarse centers. For Q^{st} prescribe the corresponding block means of the scalar instead. Both corrections preserve h.

In $QB = 0$ the fibre of h consists of gradients with scalar zero at the coarse centers after subtraction of one constant. A fine harmonic difference would give a nonzero coarse harmonic difference. On that fibre the contour mean to a site changes by the scalar at the site minus its value at the center. Integration of the normalized quadratic contour weight in these coordinates has the same value on each fibre. In $Q^{\mathrm{st}}B = 0$, subtract the tree path sums by a gradient with zero block means to set the tree bonds to zero. A gradient vanishing on the trees has a scalar constant on each block, and $Q^{\mathrm{st}}d'\lambda = 0$ then forces these constants to agree. The tree condition therefore selects one lift of each h. All changes of coordinates are linear, with constant measure factors canceled in normalization.

Consequently both flat quadratic laws induce the Gaussian with curvature precision V_{j-1}^{+} restricted to the exact curvatures satisfying $\mathsf{A}h = 0$, where V_{j-1} is the unconditioned covariance of $Q_{j-1}^{\mathrm{pl}}F$. Its covariance is C_j^{curv}. Transport by $\mathcal{H}_{j-1}$, which depends only on h, gives the same bare slice $\mathcal{C}_{(j)}$ in either gauge. This calculation uses the full quadratic laws. The comparison to the cut windows uses the separate window estimates.

Summing over $j \le k_0$, where the coarsest slice gives the dominant contribution, one obtains the following bound. Here $t = |x(p) - x(q)|/a_s \ge \frac{1}{2}C_W(s)$, and $\delta_{\mathfrak{fl}} > 0$ is a rate formed from the rates δ of the slices, depending only on L and the dimension.

$$|C_{K,s}^{\mathfrak{fl}}(p,q)| \le C\big(\tfrac{a_K}{a_s}\big)^4 e^{-\delta_{\mathfrak{fl}}t} = C\pi^2 t^4 e^{-\delta_{\mathfrak{fl}}t}\,\sigma_{pq}\,:$$

the fluctuation part is small because the curvature of the linear minimiser decays. Because $t^4e^{-\delta_{\mathfrak{fl}}t}$ is decreasing only for $t \ge 4/\delta_{\mathfrak{fl}}$, the room is given the floor $C_{W,0}A_0^2 \ge 8/\delta_{\mathfrak{fl}}$. Over $j > k_0$ the sum of the exponentials reconstitutes the power law: $\sum_i L^{-4i}e^{-tL^{-i}} \le Ct^{-4}$.

The interacting measure. We apply the formula of the Strategy at level k_1, whose backgrounds are flat on the scale of the pair. Only the classes $[h]$ occur, and no finer class of histories: the sourced exponent of level k_1 is the one that §5.20.1.2 provides. (c1) *The first member is an error term.* Consider the terms of the levels $k_0 < j \le k_1$ and a source that is rough on their scale. Their slopes in such a source remain $\le Cx^{-1}(\log x)^{\mathfrak{p}_0+q}$, measured against $\|f_i\|_\infty$ (the norm $\|f_i\|_{\sharp,K-j}$ of the level of the term can exceed it by a factor of up to L^{j-k_0}). The recursion for these slopes closes since $\mathsf{n} \le C\log x$ (its size is set through $\bar{x}_s \le Cx$). Hence $\mathsf{n}^2(\log x)^{\mathfrak{p}_0-q} \le C(\log x)^{2+\mathfrak{p}_0-q}$ is small ($q - \mathfrak{p}_0 > 2$, g small). Suppose first that no large-field region touches the supports at level k_1. Then $|\Phi_i^{(k_1)} - c_i| \le o(1)\,x^{-1}\|f_i\|_\infty$ for a constant c_i, where $o(1)$ refers to $\mathsf{d} \to 0$, uniformly in the datum. In the other case $|\Phi_i^{(k_1)} - c_i|$ grows no faster than $L^{\mathsf{n}}L^{4a}\|f_i\|_\infty$ in the age a of the touching region. The age series of §§ 5.20 and 5.22, taken at the single level k_1 with $b \ge 8$, beats this growth: $L^{\mathsf{n}} \le L\bar{x}_s^2$ is a power of x, while the series lies below every power. Consequently $|\langle\Phi_1^{(k_1)};\Phi_2^{(k_1)}\rangle_{\mathsf{P}^{(k_1)}}| \le o(1)\,x^{-2}\prod\|f_i\|_\infty$. (c2) *The second member becomes one number.* The quantity $\Phi_{12}^{(k_1)}$ is the sum of two parts. The first is the mixed derivative of the sum $\sum_{j\le k_1}\sum_X e_j(X;w)$ of localised source energies from § 5.20. The second is a part that depends on the fields and is bounded only crudely, by a quantity of order $L^{2\mathsf{n}}$ for each bond of the determining set; by § 5.20, the size of that set is counted by the unit volumes of level k_1 together with the sites $\sum_{j\le k_1}\#_j(Z_j)$ of the large-field regions (every source derivative costs a factor L^{n}). We translate the pair by the points v

of a grid $\mathfrak{P} \subset \Lambda_{s,\ell}$ whose spacing is comparable to d, and we write $f_i^v := f_i(\cdot - v)$ for the translated sources. The left-hand side of the formula is unchanged by this. The sum over v of the mixed derivatives equals the mixed derivative evaluated at the two array sources $\sum_v \epsilon_v f_i^v$, whose norms are those of f_i, averaged over the signs $\epsilon \in \{\pm 1\}^{\mathfrak{P}}$ (this is the sign-averaging argument proving eq. 6.vi). The crude bound is therefore divided by the number $\approx L^{4\mathsf{n}}\hat{\mathsf{d}}^{-4}$ of pairs contained in a unit volume of level k_1. What survives is $L^{-2\mathsf{n}}\hat{\mathsf{d}}^4 \le \bar{x}_s^{-4}\hat{\mathsf{d}}^4$, and this requirement fixes n. There remains the number

$$c_{12} := |\mathfrak{P}|^{-1} \sum_{v \in \mathfrak{P}} \partial_{z_1}\partial_{z_2} \sum_{j \le k_1} \sum_{X} e_j(X; z_1 f_1^v + z_2 f_2^v)|_0$$

. (c3) *The number is evaluated.* The number c_{12} is the value at a single point of the mean of $\Phi_{12}^{(k_1)}$ over $\mathfrak{P}$, namely of the term without large-field regions, taken at the trivial block field. Its exact single-step recursion runs from level k to level $k+1$ for $k_0 \le k < k_1$ and is evaluated at the flat terminal field. The measures there are positive, so a remainder is carried through a step by its supremum alone. For each step, two inequalities on a single fluctuation integral § 4.5 are then required. That integral belongs to the small-field step taken over the whole lattice, with a background that is real and regular and with coupling $x_j \approx x$. The first inequality concerns the covariance of two sums of background plaquette energies. The second concerns the mean value of a bilinear form whose arguments are the background plaquette angles. The expansion of § 4.5 gives terms for these comparisons. We must bound the interacting errors on the actual fluctuation integral, using Cauchy estimates on its admitted polydisc, and control the Taylor error in replacing the regular background by the flat one. At the flat background, the object that agrees with the free template is the curvature covariance C_j^{curv} introduced above. It determines $\mathcal{C}_{(j)}$, and a_p depends on nothing else.

6.1.1.3. The clean restriction at fixed exterior data. The clean restriction has to be made on a standing class with its stored variables still present. Fix one such class and keep its exterior datum fixed. At a step of level t above the first level of the clean window, add a small complex constant s to the input boundary term at each level-$(t-1)$ live cube near the clean neighbourhood. The modified boundary system belongs to the complex box of the step identity for s in a neighbourhood of zero. The real moves are independent of the boundary system, so an input term with N marked cubes acquires just e^{sN}. With the stored variables retained, the summed step identity over the fixed output standing part therefore equates two polynomials in $z = e^s$ near $z = 1$. The constant coefficient is the sum of the allowed histories and, on the bracket side, the same bracket with the marked inner histories omitted. The standing input at the first level already enforces its live condition. Take the coefficient algebraically, keeping the original moves, shifts and counterterms. Healings at or below the first level of the window have no such mark. With profiles redrawn after healing and compensated at each plaquette, the coupling lag vanishes on the clean box.

To estimate the quotient, restrict the inner-history measure of each bracket to the histories allowed by the mark. At the same stored datum a restriction decreases its total variation. An inadmissible collar gets weight zero. Keep the empty-family factor, the relative geometry and the interaction potential on each surviving tuple. Thus every absolute activity majorant of the full bracket still applies to the restricted one, including the small factor for each healed collar. The rooted logarithm bound for the relative bracket is $Ce^{-5p_0(g_t)/4}$ per root, with the same field-independent majorant for both sums. Only terms that meet a deleted inner-history choice or collar can differ. Root a changed term in the healed collar $C^{\sim}$ of the deleted choice. In the fixed output label S' we have $C^{\sim 2} \subset \Lambda_t(S')$. Thus $C^{\sim}$ is contained in the one-layer erosion of $\Lambda_t(S')$ and avoids the fixed hulls of the standing components at all levels. The ordinary rooted entropy sum applies there. The marked input cube is in the padded box of the deleted choice. Its possible roots lie within the operation's read radius. The number of such roots per marked cube is polynomial in

$\check{R}_t$, and that polynomial is absorbed by decreasing the reserve in $e^{-5p_0(g_t)/4}$. Summing the two absolute bounds over the resulting roots gives, with $0 < c < 1/4$,

$$0 \le \Delta_t := \log K_t - \log K_t^{\mathrm{cl}} \le C c_t e^{-(1+c)p_0(g_t)}.$$

The lower bound follows by positivity before exponentiation. This comparison uses the standing-class identity with the stored variables, the absolute bracket majorants and the full carrier count. The restriction is of the coefficients of that identity. The covariance and counterterms used to form it are those of the full input.

Positive kernels preserve the resulting domination in later moves. Thus, with $\epsilon_j = C\sum_{t\le j} c_t e^{-(1+c)p_0(g_t)}$ over the marked levels, the clean and full incoming measures have ratio $r_{j-1} \in [e^{-\epsilon_{j-1}}, 1]$ on their common standing variables. At step j, use only the strings of that step that remain clean on both sides. At a fixed output datum, let $\mathbb{E}_j^{\mathrm{ref}}$ be their positive conditional expectation with the full incoming measure. Disintegration and normalization give

$$\mathbb{E}_j^{\mathrm{cl}} F = \frac{\mathbb{E}_j^{\mathrm{ref}}(r_{j-1}F)}{\mathbb{E}_j^{\mathrm{ref}} r_{j-1}}, \qquad e^{-\epsilon_{j-1}} \le \frac{r_{j-1}}{\mathbb{E}_j^{\mathrm{ref}} r_{j-1}} \le e^{\epsilon_{j-1}}.$$

These statements are on the common positive output support, up to null sets of its marginal. The lower bound splits off a reference kernel with weight $e^{-\epsilon_{j-1}}$. Composing the splits leaves a reference branch of weight $P_0 = \exp(-\sum_{k_0<j\le k_1} \epsilon_{j-1})$ on the clean classes. The lost mass is at most the unclean mass plus $1 - P_0 \le \mathsf{n}\epsilon_{k_1}$. This branch uses the original run weights and no counterterms formed from a restricted exponent.

6.1.1.4. The product window on a clean interior box. At step $j = i + 1$ let I be the free bonds of the comparison box. In input-level units let r_{dec} bound the diameter of the enlarged decision cube plus the read of its affected pivot rows. Form I^+ by enlarging I by r_{dec}. Include every decision cube whose enlarged stencil meets I^+ in the clean neighborhood. In the native box geometry a sufficient collar width over I, in these same units, is

$$\max\{100M\check{R}_i + 4r_{\mathrm{dec}},\ 111LM\check{R}_j\}.$$

The first width covers the complete decision stencils. The second places the box on the ordinary side of the live and creation buffers. The native cube count for these enlargements costs a fixed factor bounded by $CL^8 + (2L)^4$, which the rarity exponent absorbs. Use the clean condition before the integrations and healing of step j. The box and its read margin are then inside Λ_j: no creation or live component meets that margin and no operation of $\mathbf{R}$ at this step acts there. In particular no rim, collar or strip test reaches the changed bonds.

Write $\widehat{g}_i = \theta_i^{-1/2}$ for the coordinate scale of the reference frame. Fix $W = V_j$ and the exterior variables of the positive standing class. Take the averaging fibre in the original block axial frame at $V_* = V_i^{\mathrm{bg}}(W)$, with the older prescribed datum held fixed. A change of the literal nonpivot coordinates $\widehat{g}_i y_b$ for $b \in I$ changes only those bonds and the pivots in their one-step read sets. Every such pivot solves its one coarse constraint on the admitted chart. The change leaves the determining rows, the coarse field W, and the older stored variables fixed. In particular the coarse plaquette tests and the test backgrounds computed from W have the same values.

On the box the exact fluctuation test of each retained cube is the full-threshold bound on the supremum of the free bond logarithms that it reads. Its product over the cubes therefore imposes exactly $|\widehat{g}_i y_b| < \delta_i$ for $b \in I$. Every such bond is tested by a cube inside the ordinary split. The tests involving both box and exterior bonds contribute these same bounds and restrictions on the fixed exterior. There is no additional pivot cutoff. The pivot change of variables fixes the nonpivot coordinates and has a single solution on this support, with pivot values of size at most $C_L\delta_i$.

To check the converse, keep the exterior of an admitted clean term and vary the box coordinates throughout these balls. On the affected approximate tests the fixed centers differ from V_* by at most a small fraction of δ_i by the real collar comparison on the test cube. Thus the exact threshold

δ_i implies the approximate threshold $2\delta_i$ at every changed nonpivot read. The contour cutoffs also pass, since their deviation is at most $C_L\delta_i$ and the gauge cutoff uses ε_i. The new-test implication of Lem. 4.57 applies on the full old density carriers and gives the old threshold with its fixed margin. At $i = 0$ check the bare plaquette test on each affected plaquette. The complete ordinary read margin contains all four bonds. At the retained background its plaquette deviation is at most $CL^{-2}\varepsilon_1$ in bare units, by the curvature estimate of that lemma. Relative to this background the nonpivots on the window and its fixed exterior are within δ_0, and the solved pivots are within $C_L\delta_0$. Telescoping the four unitary factors bounds the plaquette deviation by $CL^{-2}\varepsilon_1 + 4C_L\delta_0$. The adjacent frame radii bound $\varepsilon_1/\varepsilon_0$. Choose L for the curvature term, then use the ordinary ceiling on δ_0/ε_0 to put $CL^{-2}(\varepsilon_1/\varepsilon_0) + 4C_L\delta_0/\varepsilon_0$ strictly below one on the whole window. This check uses the estimates on the changed fields and their read sets. The earlier recorded answers alone would not show that they still pass. All other tests or decisions of the step are exterior reads, or use only the fixed W. The local geometry and the clean restriction of this step are therefore constant throughout the box window. The gauge coins inside the box have been integrated in the gauge-fixing factor. Insert the normalized ordinary gauge weight on the summed standing input before conditioning on the exterior. The ordinary gauge group fixes the coarse centres and the older held datum sites. The incoming density and insertion are invariant on these orbits, and the normalized gauge weight has orbit integral one. A draw at an interior site can move an incident exterior fine bond, so disintegrate at the exterior *after* this gauge integration. That exterior contains the retained coarse field, the older stored data used by the standing term, and the fine exterior free coordinates. A fine pivot is reconstructed from its constraint even if its location is outside I. Include its full double block when forming the test read. The exterior may also contain coins of later moves. Integrate out the coins of the collar or ordinary gauge draws made before the tests. At an interior site such a coin fixes the contour and would impose an extra equation on the free box coordinates. Its integrated contribution is the gauge-fixing factor in the coarea integral. The ordinary strip centre, its relative coordinate and Jacobian use the held fine exterior rows and the coarse strip datum. Thus a strip selector in its original coordinate is also fixed in this box integral.

Check the clean margin from the large-field region Z_j before healing, on the whole enlarged family of test reads, including the reconstructed pivots. The collar and rim exploration reads only the successive enlarging large-field sets, all contained in Z_j, the complement of the small-field integration region. The selected exterior decisions therefore stay fixed as the free box varies. When the earlier density indicator reads the changed box, the prospective new-test implication above proves its value on the full product window before it can be kept among the fixed answers. Sum out the unused interior diagnostic flags and apply the coarea formula to the term summed over its standing part.

It follows that on the admitted clean reference class the indicator in eq. 6.xi is the product $\prod_{b\in I} \mathbf{1}\{|y_b| < b_j\}$ with $b_j = \delta_i/\widehat{g}_i$, times an exterior indicator. Use the exact coarea density of eq. 6.x in both numerator and denominator. The retained gauge-fixing exponential and the Haar and pivot densities still occur in the weight. At fixed exterior, freezing the other coordinates of the full quadratic form gives the compression on I and its linear cross force. The exterior mixing measure is proportional to $Z_e m(de)$, since the same box integral Z_e normalizes every inserted observable. This identifies the window and integration variables for the matched conditional comparison.

The positive run also specifies how to impose later tests. On the admitted box fibre let $s_j(e)$ be the state after deleting the integrated slots of step j. The retained field W, the exterior fine slots, the older stored slots and the completed label are fixed by e. The pivots which change with y are integrated slots, since the full read margin separates the box from the retained strip. Hence $s_j(e)$ is constant as y varies on this fibre. The state includes all retained outcomes read by the subsequent moves. The subsequent draws, tests and integrations form a positive kernel $\mathsf{K}_j(s_j(e), d\omega)$ on the later records. Use the kernel conditional on any later coins already included in e, with the remaining coins integrated in the kernel. For a selection A of these records and the fixed exterior,

set $k_A(e) = \mathsf{K}_j(s_j(e), A_e)$. The conditional quotient with a box insertion F is then

$$\frac{\int m(de)\, k_A(e) Z_e\, \nu_e(F)}{\int m(de)\, k_A(e) Z_e}.$$

This follows by performing the box integral before the later kernel integral, for a finite positive denominator. For a standing class sum the positive terms on their shared retained data, keeping each term's own Z_e and $k_A(e)$ in the mixture. A selection on a field already deleted by the run can change the conditional domain and must be kept in the joint integral.

6.1.1.5. Composition on the retained states. Let $\widehat{\tau}_j$ be the measure on the disjoint union of the level-j state spaces, with the block field, all still stored slots and the labels used by the run. Use the original positive operations of step j as a map $\mathcal{T}_j$ on measures, including the sums at merges. For the carried insertion G put $\widehat{\tau}^G_{k_0} = G\widehat{\tau}_{k_0}$ and apply the same operations to $\widehat{\tau}^G_j$ and $\widehat{\tau}_j$. A test splits a measure by its indicator and complement, an integration or change of variables is a positive kernel, and a merge adds the images from its incoming summands. Use gauge fixing as a normalized orbit draw. The gauge-invariant insertions used here can pass through this draw. These operations conserve total mass after summing the labels. At each output, the carried density obeys

$$G^{[j]} = \frac{d\widehat{\tau}^G_j}{d\widehat{\tau}_j} = \mathbb{E}_j G^{[j-1]}, \qquad \mathbb{E}_j F = \frac{d\mathcal{T}_j(F\widehat{\tau}_{j-1})}{d\mathcal{T}_j(\widehat{\tau}_{j-1})}.$$

Positivity gives $|\mathbb{E}_j F| \le \|F\|_\infty$, and $F = 1$ gives $\mathbb{E}_j 1 = 1$. At a merge the two measures in this derivative are the sums of the respective images. In particular the merge uses the conditional weights of the original run.

For a terminal class retained by the run or by an explicit refinement of it, restrict the terminal measures to that class and disintegrate over $V_{k_1} = W$. The conditional expectation of G is the conditional mean of $G^{[k_1]}$ under the uninserted terminal class measure. This follows by testing the carried terminal measure against a bounded function of W. Thus the recursion composes on complete states before projecting to the block field. For a refined class that imposes conditions on earlier deleted slots, retain those conditions in the positive integrals used to form the refinement. We must check the product window and the box estimates for its actual conditional law before applying the Gaussian comparison to that class.

The deletion at an isolated ordinary cell gives a useful case for later prepared and healed exterior observations. At output level j, take I on the bonds of e^{+1} for a cell e, and use the event $e^{+2} \cap Z_j = \varnothing$. The rim width is less than one cell. Each test of the cascade reads within that width of its current stage set. Run the cascade from the data off I, with the output field, the older determining data and the coins fixed, until a stage set first approaches the padded cell. Up to that point the two cascades have the same reads. Thus whether the approach occurs is determined by this exterior. On the event of no approach, the complete post-step state depends only on that exterior. The slots of I are integrated in the ordinary product window.

Write ν_ξ for their conditional law at exterior ξ. Later observations are obtained by positive kernels from the retained state. For a bounded insertion G in I and a later test function F, the integral therefore has the form $\int \nu_\xi(G)\, K_\xi(F)\, d\pi(\xi)$, where π is the pre-deletion exterior measure on this event and K_ξ is the later kernel. Disintegrating over a later prepared or healed observation takes a positive mixture of the ν_ξ. On a class with a common reference law, a uniform normalized density bound against that reference is preserved by this mixture. A finer condition on an earlier deleted coordinate instead enters the original conditional integral.

For two bounded witness reads, also carry $G_1 G_2$ as a single insertion. Put

$$T_j = (G_1 G_2)^{[j]} - G_1^{[j]} G_2^{[j]}.$$

Then $T_{k_0} = 0$ and the recursion on the retained states gives

$$T_j = \mathbb{E}_j T_{j-1} + \mathrm{Cov}_j(G_1^{[j-1]}, G_2^{[j-1]}).$$

Here $\mathrm{Cov}_j(F, H) = \mathbb{E}_j(FH) - \mathbb{E}_j F\, \mathbb{E}_j H$. Indeed, expand the carried product at level $j-1$, apply $\mathbb{E}_j$, and subtract the product of the propagated means. Thus T_{k_1} is the sum of the one-step covariances transported by the later $\mathbb{E}_j$'s, with the weights of the actual merges. In the terminal class, let $\mathbb{E}^W$ denote conditioning on $V_{k_1} = W$ in the uninserted law. The covariance under the terminal conditional law Π of the original reads is

$$\mathrm{Cov}_\Pi(G_1, G_2) = \mathbb{E}^W T_{k_1} + \mathrm{Cov}^W(G_1^{[k_1]}, G_2^{[k_1]}).$$

Only the errors in the propagated means contribute to the last covariance if their leading terms depend on W alone. This is the decomposition used when estimating one-step covariances separately. 6.1.1.6. The two one-step Gaussian comparisons. Fix $k_0 < j \le k_1$. Work in level-$(j-1)$ units on a nonwrapping box containing the read sets in its interior. Write $V_* = V_{j-1}^{\mathrm{bg}}(V_j)$, and take derivatives in the bond logarithms $V = e^{iZ} V_*$. Let y have the centered Gaussian law of the admitted reference box with its covariance $C^{(j-1)}$ on free coordinates. This is the covariance with the reference frame scaling and the actual quadratic weight $\tau_{j-1} = x_{j-1}/\theta_{j-1}$ in the Wilson term. The trace over bond or free coordinates includes the colour slots. The first variation of the physical bond logarithms is $\widehat{g}_{j-1}\mathsf{C}y$, where C is the actual pivot tangent map. Set $C_* = \mathsf{C}C^{(j-1)}\mathsf{C}^{\mathsf{T}}$. Use the uniform covariance bound and the derivative bounds for the actual pivot map, including its higher jets, in the norm of these readouts. The box dimension and the real window radius are bounded by fixed powers of $\log x$. In the following error bounds $g = x^{-1/2} \asymp \widehat{g}_{j-1}$.

Use the normalized conditional law on the clean class, with its actual product window, coarea factors and exterior mixture as in § 6.3. The comparison requires the complete quadratic form and the same chart for the source numerators and the denominator. Assume the exterior comparison on the read set has density ratio between $e^{-\Xi_j}$ and e^{Ξ_j}, with $0 \le \Xi_j \le 1$, the Gaussian window tail is $0 \le t_j \le 1/2$, and the old activity, marked remainder and any coupling lag contributions to the reference box tilt have total bound θ_j^{E}. The intrinsic odd cubic and even higher terms have bounds $Cgn_j\mathsf{b}_j^3$ and $Cg^2 n_j \mathsf{b}_j^4$, respectively, where n_j counts the box coordinates and b_j is the window radius. These parity bounds use an intrinsic reference with every exterior fluctuation argument set to zero. In particular they apply to the homogeneous step with no stored exterior. To use them on a box with stored exterior one must include the change from those arguments in the exterior comparison. The generating-function estimate with the stored values fixed uses the full intrinsic oscillation instead, as in §6.3.0.4. These estimates and the gauge-equivariant means used below require a common admitted analytic tube with room around the symmetric product window and a small bound for the complete tilt.

Use the following formulas at a fixed compatible exterior, with the matched reference having zero exterior fluctuation arguments. For an exterior mixture they apply with a single reference only after the density and readout comparisons to that reference. If the reference varies with the exterior, averaging the conditional covariances also requires the covariance of the conditional means.

Let G, G_1, G_2 be the gauge-invariant readouts, with common tube bound M and Taylor derivative bounds μ_m. Put $\bar{\mu} = \max_m \mu_m$ over the derivatives used here. Write κ for the curvature size of V_* on the read set, κ_i for its curvature size on the box of G_i, and $\kappa_{\mathcal{B}}$ for the curvature size of V_j on the window box. The curvatures are measured in their respective levels, with $\ell'\kappa$ small for a read box of side ℓ' in the common nonwrapping gauge. On the admitted clean boxes use $\kappa, \kappa_1, \kappa_2 \le Cgp_0$. This permits the variation of an old tilt to be bounded by its size times the oscillation of the readout, using the first derivative bound $C\mu_2\kappa$ and the second-order Taylor bound. The mean and covariance comparisons have the form

$$\mathbb{E}_j G = G(V_*) + \tfrac{1}{2}\widehat{g}_{j-1}^2 \,\mathrm{tr}(C_* D^2 G(V_*)) + E_1,$$

$$\operatorname{Cov}_j(G_1,G_2)=\widehat{g}_{j-1}^{2}\langle DG_1(V_*),C_*DG_2(V_*)\rangle + \tfrac{1}{2}\widehat{g}_{j-1}^{4}\operatorname{tr}(D^2G_1(V_*)C_*D^2G_2(V_*)C_*)+E_2.$$

Put $s_i=g\mu_2\kappa_i+g^2\bar{\mu}$. For readouts compared in the same coordinates the required errors are bounded by

$$|E_1|\le C(\log x)^C\{(g^4+\theta_j^{\mathrm{E}}g^2)\bar{\mu}+(g^2+(\log x)^{-2q})\mu_2\kappa\kappa_{\mathcal{B}}\}+C\operatorname{osc}(G)\Xi_j+C(\log x)^C(\mathsf{M}+\bar{\mu})\sqrt{t_j},$$
$$|E_2|\le C(\log x)^C\{(g^2+\theta_j^{\mathrm{E}})(g^4\bar{\mu}^2+g^2\mu_2^2\kappa_1\kappa_2)+(g+\theta_j^{\mathrm{E}})g^3\mu_2\bar{\mu}(\kappa_1+\kappa_2)\}$$
$$+C(\log x)^C\Xi_j s_1s_2+C(\log x)^C(\mathsf{M}+\bar{\mu})^2\sqrt{t_j}.$$

If the exterior changes the readout map, let η_i bound the supremum difference between its i-th pullback and the reference pullback on the same window. It costs η_i in the mean. For the covariance subtract the reference constants $G_i(V_*)$ first. Writing B_i for the resulting reference suprema, the pullback error is bounded by $2\eta_1(B_2+\eta_2)+2\eta_2B_1$. Indeed, under the positive joint law of the exterior and the box, expand the covariance in one pullback difference at a time and use $|\operatorname{Cov}(A,B)|\le 2\|A\|_\infty\|B\|_\infty$. Here $B_i\le C(\log x)^Cs_i$ by the same Taylor bounds as in the density comparison.

Expand the readouts in the actual pivot chart and compare with the centered Gaussian on its symmetric window. The two linear variations give the first covariance term. It vanishes at the flat reference, since a gauge-invariant scalar has zero first variation there. The two quadratic variations give the second term by Wick contraction. For the mean, the trace is the quadratic variation. The extra terms pairing DG with the quadratic pivot increment, the odd cubic tilt, or the linear old-activity increment use the vanishing of their gauge-equivariant means at flat data.

For the covariance, first work on the symmetric product window at the matched reference. Write $\mathbb{E}_0$ for the normalized restricted Gaussian, P for its tilt and $A_i=G_i-\mathbb{E}_0G_i$. Put $\delta=\|P\|_\infty$. Expanding the normalized expectation at $P=0$ gives $\mathbb{E}_PH=\mathbb{E}_0H+\operatorname{Cov}_0(H,P)+O(\delta^2\mathbb{E}_0|H-\mathbb{E}_0H|)$. Indeed $\mathbb{E}_0e^P\ge e^{-\delta}$ for this real tilt, and the linear term of its inverse subtracts $\mathbb{E}_0P$. Apply the identity to A_1A_2 and then to each A_i. Subtracting the shifted means gives

$$\operatorname{Cov}_P(G_1,G_2)=\operatorname{Cov}_0(G_1,G_2)+\operatorname{Cov}_0(A_1A_2,P)-\operatorname{Cov}_0(G_1,P)\operatorname{Cov}_0(G_2,P)+O(\delta^2\|A_1\|_2\|A_2\|_2).$$

The norms here are under the restricted Gaussian. The product of shifted means has the same quadratic size in δ.

The even part of A_i is $O(g^2\bar{\mu})$ and the odd part is $O(g\mu_2\kappa_i+g^3\bar{\mu})$, in the Gaussian norms, with the fixed logarithmic factors from the window. The two same-parity products pair only with P^+, whose size is $O(g^2+\theta_j^{\mathrm{E}})$ with those factors. The mixed products pair only with P^-, of size $O(g+\theta_j^{\mathrm{E}})$. This gives the two displayed orders for the linear tilt correction. Since θ_j^{E} is small, δ^2 is bounded by a fixed logarithmic factor times $g^2+\theta_j^{\mathrm{E}}$. The quadratic normalization terms have the same orders. In the untilted covariance, the extra quadratic pivot term is multiplied by the first derivative of the readout. Its pairing with the main quadratic term, and the pairing of a linear term with a cubic term, cost $O(g^4\mu_2\bar{\mu}(\kappa_1+\kappa_2))$. The remaining higher-order pairings cost $O(g^6\bar{\mu}^2)$, with the logarithmic factors included. In passing to the full Gaussian apply Cauchy–Schwarz on the window complement to the products and to each mean. This gives the $\sqrt{t_j}$ error in the comparison.

For a common pair of pullbacks the exterior density ratio in the normalized comparison is bounded by $e^{\pm\Xi_j}$. The bound for the covariance from the product of two centered suprema therefore costs $C(e^{\Xi_j}-1)\prod_i\|G_i-G_i(V_*)\|_\infty$. The Taylor bounds on the window give $\|G_i-G_i(V_*)\|_\infty\le C(\log x)^Cs_i$. For $\Xi_j\le 1$ this is the stated exterior term. The bound applies to the positive exterior mixture relative to the same reference density. For the first two means the argument uses fixed-radius analytic tubes in the retained datum V_j and the corresponding bounds for the mean on those tubes. For the regular old origins the required joint chart is obtained by combining the retained field and the solved free field before taking the old averages. In the matched local recipe

write $W = e^{iA}$, let $p_k(W)$ be its background link field, and use $e^{i\Phi_{p_k(W)}(u)}p_k(W)$ with the physical free variable u. At $u = 0$ the induced datum agrees with the coarser background by the nested constraint identity. The local contour and pivot charts bound the logarithm of the combined field by $C(\|A\|_\infty + \|u\|_\infty)$. On a joint disc $\|A\|_\infty, \|u\|_\infty < c\alpha_k$ we need the actual maps of the matched recipe to enter the same window section as in eq. 4.cxcv. The constant c must leave room for the sum of the two displacements. Along the retained axial line its actual size is at most $C\,\mathrm{side}(\mathcal{B})\kappa_{\mathcal{B}}$, which must fit on the smaller disc.

The window cancellation uses this combined field: form the old averages, use their common critical window solution, and subtract the vacuum and its own tent counterterm at each old pointing. Sum at that pointing before applying the common bracket. The full pointed quadratic tensor is zero. The excluded domains give a long-domain tail, with length at least $2/\sigma_i - 2$ for a window at relative spacing $\sigma_i = L^{-(k-i)}$. The affine and Hölder estimates then give the pointed remainder with the gain $\rho_i^2\sigma_i^{4+\alpha_H} + \rho_i^3\sigma_i^6$ in units of e_i. Here ρ_i is the ratio of the output radius to the old radius. Counting the σ_i^{-4} pointings in a current-scale window gives the age weight ω_{k+1-i} of eq. 4.cxcviii. Once the actual old readers are identified on this joint chart, the bound there is uniform in (A, u) for the same filed families.

Apply Cauchy's estimate in one retained direction and one physical free coordinate. For a regular term on Y it costs

$$Ce_i\omega_{k+1-i}\alpha_k^{-2}e^{-\lambda d_k(Y)}.$$

If the free coordinate is outside the full carrier its derivative is zero. The first free derivative of the scalar regular sum vanishes at the matched flat recipe with zero exterior free coordinates, by constant colour invariance. Integrating the mixed derivative along the retained axial line and rooting at the free bond gives $CE_0\alpha_k^{-2}\,\mathrm{side}(\mathcal{B})\kappa_{\mathcal{B}}$. The box side costs a fixed power of $\log x$. This estimate concerns the regular part of the old Wick vector. The vacuum-only remainders, Wilson lag and boundary factors use their own origin bounds. A factor with a fixed noncentral spectator need not have zero free derivative at the flat value of the varied datum. The factor $g^2\alpha^{-2}$ gives the inverse logarithmic power in E_1. These bounds, paired with the small first derivative of G, give the product $\kappa\kappa_{\mathcal{B}}$.

The three vectors in the mean correction can be computed from the actual chart coefficients. Write Φ_V for the pivot graph in eq. 4.ccxxx, with $\mathsf{C} = D\Phi_V(0)$, and put $C = C^{(j-1)}$. For two free directions u, v, the quadratic pivot coefficient is given by eq. 4.ccxx. In particular, $D^2\Phi_V(0)[u, v] = -hD^2q_V(0)[\mathsf{C}u, \mathsf{C}v]$ for the pivot right inverse h of $Dq_V(0)$. Write the intrinsic first coefficient of the exponent as $P_1^{\mathrm{int}}(y) = \ell[y] + T[y, y, y]/6$, so that the first-order exponent is $\widehat{g}_{j-1}P_1^{\mathrm{int}}(y)$. Here ℓ is the first log-coarea derivative including Haar density, and T is the signed third derivative of the complete centered action and contour gauge, with the action weight retained as in eq. 4.cccxviii. Write ℓ_{old} for the first derivative of the actual old exponent after substitution in the same graph. Take this derivative in the physical free coordinate $u = \widehat{g}_{j-1}y$, with the sign of the old term in the exponent.

Use first the full Gaussian for these polynomial contractions. Wick's formula gives the three vectors before multiplication by $\widehat{g}_{j-1}^2$:

$$\frac{1}{2}D^2\Phi_V(0) : C, \qquad \mathsf{C}C(\ell + \tfrac{1}{2}T : C), \qquad \mathsf{C}C\ell_{\mathrm{old}}.$$

The contraction in $T : C$ is over two of its three indices. For example, the second identity follows by pairing y with one of the three factors of the cubic. Each pairing leaves the same contraction, hence the coefficient $1/2$. The other two identities use only the second Gaussian moment. For a polynomial vector A, passage back to the real Gaussian conditioned on the window costs at most $C\sqrt{t_j}\|A\|_{L^2(\gamma)}$ when $t_j \le 1/2$. This follows by subtracting the two normalized expectations and using Cauchy–Schwarz on the complement of the window. The polynomial moments are bounded using the same chart jets and covariance. This tail comparison uses the real Gaussian.

For the homogeneous chart near the trivial datum, compose the first two contractions with the actual intermediate field $p_k(W)$ of eq. 4.cccvi. The fixed output radius for that map, the outer chart radius in eq. 4.ccxxx, and the precision bounds in eq. 4.cclxxxv give a common smaller fixed complex radius for the Wick polynomials. The action family in eq. 4.cclxxix and the local coarea factors in eq. 4.ccliv bound the intrinsic coefficients on the read coordinates. At the trivial datum each vector is fixed by all constant colour rotations, so it vanishes. Cauchy's estimate on this homogeneous datum ball bounds each of the first two vectors by $C(\log x)^C \|\log W\|_\infty$ on a fixed smaller ball, with the polylogarithmic factors from the box count. The third vector is bounded by the norm of the summed old first derivative times the pivot and covariance bounds.

For the conditional calculation first compare with its matched local law with zero exterior free coordinates. The density-ratio and readout comparisons account for that change. For the Wick tensors of the local law, keep its box, free and pivot masks, guard and weights fixed. Compare its retained datum to the trivial datum in the same chart. On the nonwrapping box choose the coarse axial logarithm A_W with $\|A_W\|_\infty \le C\,\mathrm{side}(\mathcal{B})\kappa_{\mathcal{B}}$ in coarse units. Use the line e^{itA_W} in the common admitted datum ball. The local pivot jets and coarea derivative use complete fixed stencils. For the cubic action use the rooted coefficient bounds of eq. 4.cclxxix, and for the covariance use the resolvent identity for the same precision and mask. They give a holomorphic comparison on a smaller fixed radius. Expand each contraction by replacing one coefficient at a time along that line. The fixed number of contracted indices costs at most a power of the box count. Both tensors are zero at the trivial datum since all the fixed data there preserve constant colour rotations. The size of $\mathcal{B}$ is bounded by a power of $\log x$, so the first two vectors of the matched local law are bounded by $C(\log x)^C\kappa_{\mathcal{B}}$. For the old regular near terms on this intrinsic box, take the retained angular datum and the physical free coordinate u as separate analytic arguments. Use the jointly admitted old-box substitution of eq. 4.cxlii, with no fixed colour spectators. Keep the box and its filing fixed. In its smaller ball, the amplitude bound eq. 4.cxcviii holds as the retained datum varies through angular size $c\alpha_k$, $k = j-1$, while $\sup|u| \le c\alpha_k$. Here the pointed counterterm is removed before summing the old terms. The bound uses the same age sum $\mathcal{E}_k$ on this joint ball.

To take the first derivative of this sum, put the retained field on the line e^{itA_W}. For a free bond b, let $F_b(t)$ be the u_b-derivative at $u = 0$ of the summed regular exponent in the chart. On the trivial retained datum this derivative vanishes. Indeed the whole intrinsic substitution commutes with constant colour rotations, and their adjoint action has no invariant covector. Cauchy's estimate in u_b, followed by the rooted sum, bounds $|F_b(t)|$ by $C\mathcal{E}_k/\alpha_k$ on $|t|\|A_W\|_\infty < c\alpha_k$. On the half datum ball, the zero at $t = 0$ then gives

$$\|\ell^{\mathrm{reg}}_{\mathrm{old}}\|_\infty \le C\mathcal{E}_k\alpha_k^{-2}\|A_W\|_\infty.$$

This takes the derivative in the physical coordinate. After multiplication by $g_{j-1}^2\mathsf{C}C$, the summable covariance and pivot rows give the bound $C\mathcal{E}_k(\log x)^{-2q}\,\mathrm{side}(\mathcal{B})\kappa_{\mathcal{B}}$ for the regular part of the third vector. Pairing it with the derivative of an invariant readout supplies the second curvature factor, by the same vanishing derivative at the trivial field on the readout's own box. For ordinary spectator-free remainders with the same joint domain, replace $\mathcal{E}_k$ by $\sum_{i\le k}\rho_i^2 r_i$. The vacuum-only bound supplies the amplitude in the same calculation, and eq. 4.ccv sums these weights along the flow.

For ambient coefficients one must also count carriers reaching the exterior. Their exponential tails are the separate boundary error. Terms with fixed colour spectators or without the joint activity-radius domain need their own first derivative bound.

The quadratic part of the first covariance term is a bilinear form in the retained angles. Its mean at later steps requires the first comparison for the carried readout, with its own derivative bounds. On the admitted conditional kernels of the run, let $G_i^{[j]}$ be the propagated conditional mean of the i-th readout. Write it as $M_{i,j} + J_{i,j}$, with $M_{i,j}$ the background read and its accumulated second-order traces. The normalized conditional expectation propagates $J_{i,j}$ in supremum norm.

In the conditional covariance its mixed contribution is bounded by

$$|\mathrm{Cov}_j(M,J)| \le \|J\|_\infty \, \mathbb{E}_j|M - \mathbb{E}_j M|.$$

For the scalar plaquette read on the clean boxes, the mean-value bound for its background part uses the curvature of the retained datum. With a fluctuation of size g times a power of $\log x$, the bound is a power of $\log x$ times $g\kappa_{\mathcal{B}} + g^2$. The trace part already has a factor g^2 and its fixed-radius derivatives cost only powers of $\log x$. The corresponding contribution of two remainders is at most $2\|J_1\|_\infty \|J_2\|_\infty$. This is why the iteration can use bounds on the values of the old remainders.

For the principal half-square term at p, q, put $c_j = x_{j-1}^{-1} \mathcal{C}_{(j)}(p,q)$ at the flat reference. At a fixed retained background with curvature readouts m_p, m_q, the part linear in each fluctuation has the flat-matrix term $c_j \langle m_p, m_q \rangle$. It has a quadratic zero at the flat retained datum. In a later step $l > j$, its leading second-order trace increases its conditional mean by $(N^2-1)c_j c_l$. The two symmetric terms in the Hessian of the bilinear form cancel the $1/2$ of the trace. This produces the cross-level term in the principal covariance, in addition to the single-step term $(N^2-1)c_j^2/2$. The normalized comparison on the actual run must bound the error in these transported traces on the common admitted chart. It also must bound the covariance of the remaining terminal means. The calculation uses the conditional kernels of the run, with the standing data carried by them. Conditioning the initial law on a plain composite block average defines a different law when the operations have redrawn exterior fields.

6.1.1.7. Carrying a scalar plaquette read to a coarser datum. On the nested clean boxes let $T_p(V)$ be the logarithm of a level-k_0 plaquette read from the background at level $k_0 + i$, in the common chart. Use at each finer step the actual datum induced by the coarser background. The finer constraint fibre is contained in the coarser one. Restriction and uniqueness of the regular orbit identify the composed read with the read from the single level-$(k_0 + i)$ background. This applies on the exact common constraints. When using two different localization boxes, keep their collar comparison as a separate error.

The fixed complex Landau tube for that background gives $\|T_p\| \le CL^{-2i}$ on the tube, with a constant independent of i. The real bound at the datum is $CL^{-2i}\kappa(V)$. Cauchy's estimate on a fixed smaller datum ball gives $\|DT_p\| + \|D^2 T_p\| \le CL^{-2i}$. Let G be a scalar read of a fixed family of these holonomies, gauge invariant on the nonwrapping box with the other arguments fixed. At the trivial field constant colour rotations leave the datum fixed. Their adjoint action on each logarithm has no invariant linear functional, so $DG(0) = 0$. Thus in the read chart $\|DG(T)\| \le C\mu_2 \|T\|$. For the traced plaquette weight this cancellation also follows by taking the trace of its traceless linear logarithm. The two chain-rule terms give

$$\|D(G \circ T)(V)\| \le C\mu_2 L^{-4i} \kappa(V),$$
$$\|D^2(G \circ T)(V)\| \le C\mu_2 L^{-4i}.$$

Indeed the first part of the second derivative has two first derivatives of T, while the part with $D^2 T$ gains the other factor through $DG(T)$. With the bounded one-step Gaussian covariance the trace of this latter part at depth i, including the fluctuation scale, costs $C(\log x)^C g_{k_0+i}^2 \mu_2 L^{-4i} \kappa(V)$. It sums geometrically over the clean segment. These are the derivative estimates for this scalar readout family. A matrix plaquette has only the L^{-2i} estimate for its derivatives. The covariance kernels and the differentiated old-origin sums have their own bounds in the iteration.

Prop. 4.22 proves the mean and covariance replacement for the specified plaquette readouts of the flat homogeneous classical-pullback fibre. For the actual clean run, the bounds just stated use the normalized box law, the full tilted exponent, exterior control, and uniform derivative bounds for the carried readouts. These inputs must be verified on the same chart.

6.1.1.8. The transported covariance on the fixed datum chart. At step j use the local reference with its masks and all other chart choices fixed. For a level-k_0 plaquette p let $T_p(u; W)$ be its logarithm in the common frame, after the physical free displacement u and the actual pivot substitution. Write

$m_p = T_p(0;W)$ and $R_{j,p}(W) = D_u T_p(0;W)$. The physical matrix pairing between two plaquette reads is

$$K_{j,pq}(W) = \widehat{g}_{j-1}^2 R_{j,p}(W) C^{(j-1)}(W) R_{j,q}(W)^{\mathsf{T}}.$$

All transposes here use the real coordinate inner products and their bilinear continuations on the complex chart. The first covariance term for the two half-squares is exactly $\langle m_p, K_{j,pq}(W) m_q\rangle$. The second-order term in the mean of one half-square is

$$\tfrac{1}{2}\operatorname{tr} K_{j,pp}(W) + \tfrac{1}{2}\widehat{g}_{j-1}^2 \langle m_p, D_u^2 T_p(0;W) : C^{(j-1)}(W)\rangle.$$

The trace is in the colour space. The second summand uses the composed derivative and therefore includes the quadratic pivot contribution to the trace. When using the separate bond Taylor and pivot terms, count that contribution only once.

The fixed complex datum tube and the nested curvature bound give $\|R_{j,p}\|_{\ell^\infty \to \mathfrak{g}} \le C\eta_{j-1}^2$ on a smaller tube. To see this, apply the plaquette bound to the read with the perturbed datum on the tube and use Cauchy's estimate in a physical free direction, with the bounded pivot tangent included. In particular this bounds the row in ℓ^2. The covariance is uniformly bounded on the same smaller tube by the precision estimates for the fixed masks. Therefore $K_{j,pq}$ and any fixed number of its datum derivatives have bounds of size $C_m x_{j-1}^{-1}\eta_{j-1}^4$, allowing fixed powers of $\log x$ from the box and chart constants. This follows by applying Cauchy's estimate to the bounded product on the larger common tube. The scalar half-square also has the corresponding fixed-order derivative bound without the factor x_{j-1}^{-1}, by applying Cauchy's estimate to the bilinear square of the holomorphic logarithm. These bounds use fixed readsets and the admitted common datum radius.

Along the coarse axial line, subtracting $K_{j,pq}(\mathbf{1})$ costs this same bound times the size of the coarse potential. That size is bounded by the box side times its curvature. The second summand of the mean trace contains m_p itself and so also gains the small curvature of the datum. At the flat datum first compare the box covariance with the full flat covariance restricted to the read set, at the same relative weight τ_{j-1}. The Schur correction in the flat marginal comparison bounds this discrepancy on the read set by $Ce^{-c\mathsf{D}}$, where D is its distance from the edge in the box units. The ℓ^1 bounds on the two readout rows give a cost at most $Cx_{j-1}^{-1}\eta_{j-1}^4 e^{-c\mathsf{D}}$ in the physical pairing. With this marginal the local read gives the local unit-weight slice times x_{j-1}^{-1} in each colour, since $\widehat{g}_{j-1}^2\tau_{j-1}^{-1} = x_{j-1}^{-1}$. Then compare the local and full readout rows by eq. 6.i. Thus the difference from c_j times the identity in colour is bounded by

$$C(\log x)^C x_{j-1}^{-1}\eta_{j-1}^4\big(\operatorname{side}(\mathcal{B})\kappa_{\mathcal{B}} + e^{-c\mathsf{D}} + e^{-cd_{j-1,p}} + e^{-cd_{j-1,q}}\big).$$

In the curved one-line covariance this bound is multiplied by $|m_p||m_q|$. Its part coming from $K_{j,pq}(W) - K_{j,pq}(\mathbf{1})$ has zero quadratic jet at the flat datum, since both m's and the matrix difference vanish there. The flat matrix error can contribute to the quadratic jet, with the exponential losses in the displayed bound and the two factors from the derivatives of the plaquette reads.

In the later mean step the derivative of a trace already carries a factor of x_{j-1}^{-1}. Taking another Gaussian trace at level $l > j$ costs at most $C(\log x)^C x_{l-1}^{-1}x_{j-1}^{-1}\eta_{j-1}^4$ for the mean. The sum of the η_{j-1}^4 factors is geometric. The number of later steps is $O(\log x)$, so these bounds retain a power of x^{-1} beyond the first trace. The principal quadratic term of the transported covariance instead contributes the cross-level contraction described above.

On an admitted clean terminal datum the background half-square contributes at most $C\|\phi_i\|_\infty x^{-1}\mathsf{r}^4 p_0^2 L^{-4\mathsf{n}}$ to the mean after summing its plaquettes. Here n is the number of transported steps and r is the read box size at the initial scale. This follows from the terminal curvature bound $Cp_0 x^{-1/2}$ and the area factor $L^{-2\mathsf{n}}$ in the nested read. Subtract the deterministic reference

mean

$$c_i = \frac{N^2-1}{2} \sum_p \phi_i(x(p)) \sum_j x_{j-1}^{-1} \mathcal{C}_{(j)}(p,p).$$

The trace differences above sum in j by the same geometric bound. The remaining one-step errors propagate by the supremum norm. Thus in using total covariance on the clean terminal law the covariance of the means costs at most $2\|\mathbb{E}(G_1 \mid W) - c_1\|_\infty \|\mathbb{E}(G_2 \mid W) - c_2\|_\infty$, with the conditioning index enlarged to include any class variable. The estimates must be uniform over that index on the clean set. In the generating-function argument the average is taken before the logarithm, and this separate covariance estimate is unnecessary.

6.1.1.9. The linear readout and the free slice. At level $q = j-1$, write $H_q^\square(p)$ for the linearized curvature read of the localized problem for p, with its input row extended by zero to the whole level-q torus. Write $H_q(p)$ for the linearized read of the regular critical problem on that torus with every level-q bond prescribed. Use the same averaging scheme in both problems. Let $d_{q,p}$ be the distance in level-q spacings to the boundary of the localization, with the full read stencil and the fixed collar width included in measuring the margin.

For a real coarse 1-form A, take $V(t)(b) = \exp(itA(b))$ on the whole level-q lattice. Its test holonomies have defect at most $C|t|\|A\|_\infty$, so the homogeneous torus problem and the prescribed joined localization are admitted for small $|t|\|A\|_\infty$. Use the restriction of $V(t)$ on common rows and its prescribed face lift on the new collar rows. By eq. 3.cdi, the discrepancy from the induced datum is bounded by $C|t|\|A\|_\infty$. The weighted response eq. 3.cdii therefore bounds the difference of the plaquette defects in the comparison frame by $C\eta_q^2|t|\|A\|_\infty e^{-cd_{q,p}}$. The comparison can also be made in the ancestor frame of the plaquette. The tower transport and the normalized node factor equal one at the trivial field and are continuous on the datum line. Consequently their conjugations have identity linear part on a defect vanishing at $t=0$. Dividing by $|t|$ and letting t tend to zero gives

$$\left|(H_q^\square(p) - H_q(p))A\right| \le C\eta_q^2 e^{-cd_{q,p}} \|A\|_\infty.$$

For one colour this is the ℓ^1 norm bound of the row, by duality. In the same norm the two rows are bounded by $C\eta_q^2$, from the flat response bound on the torus and the corresponding bound on the localized datum ball. The constants are independent of volume and depth. Here we differentiate a real path with a fixed determining sequence in each problem. The argument uses the actual full read domain and gives no derivative estimate for changing history labels or for an old activity.

Let Γ_j^0 be the covariance of the complete flat fluctuation law on the free bonds of the level-$(j-1)$ torus in the step's contour gauge, at unit relative Wilson weight. In the actual frame coordinates the weight is τ_{j-1}. On curvature reads the actual flat covariance is τ_{j-1}^{-1} times the one at unit weight, by the constrained curvature form and the separate Gaussian integration along gauge directions. Consequently $\widehat{g}_{j-1}^2 \tau_{j-1}^{-1} = x_{j-1}^{-1}$ is the physical covariance factor multiplying the unit-weight slice. Use one colour and put

$$R_{j,p}^\square = H_{j-1}^\square(p)\mathsf{C}^0, \qquad R_{j,p} = H_{j-1}(p)\mathsf{C}^0.$$

Here C^0 is the flat pivot inclusion of the actual step. Its fixed finite range and bounded row sums preserve the bounds for the readout rows. All free bonds used by $R_{j,p}^\square$ lie in $\mathcal{S}_j$. Thus after removing the windows its covariance under the flat read-set marginal is exactly

$$\mathcal{C}_j^\square(p,p') = R_{j,p}^\square \Gamma_j^0 (R_{j,p'}^\square)^{\mathsf{T}}.$$

The whole-torus covariance is $\mathcal{C}_{(j)}(p,p') = R_{j,p}\Gamma_j^0(R_{j,p'})^{\mathsf{T}}$, by the identification of the curvature covariance in the contour and tree gauges. It is this whole-torus operator that is the orthogonal slice. Since $\|\Gamma_j^0\|_{2\to 2} \le C$, subtract the two products one row at a time and use the row bounds to obtain

$$\left|\mathcal{C}_j^\square(p,p') - \mathcal{C}_{(j)}(p,p')\right| \le C\eta_{j-1}^4 \left(e^{-cd_{j-1,p}} + e^{-cd_{j-1,p'}}\right). \tag{6.i}$$

The Gaussian used in both products is the same complete flat Gaussian. The comparison of an interacting box marginal with its flat restriction is the preceding separate estimate.

6.1.1.10. The error on the generating function disc. Let $d_* = \min_{p,j} d_{j-1,p}$ for the plaquettes and levels in the segment. Since $x_{j-1} \asymp x$ and $\eta_{j-1} = L^{-(j-1-k_0)}\eta_{k_0}$, the difference of the covariances of the two sums of independent slices is bounded entrywise by $Cx^{-1}\eta_{k_0}^4 e^{-cd_*}$. To estimate the determinant without a factor counting the total number of covariance entries, use the same standard Gaussian on the direct sum of the flat free bond spaces for both sums. For the localized read its row is

$$u_p = \bigoplus_{j=k_0+1}^{k_1} x_{j-1}^{-1/2} R_{j,p}^{\square} (\Gamma_j^0)^{1/2}.$$

Let v_p denote the row with $R_{j,p}$ in its place. The geometric sum of the squared row bounds gives

$$\|u_p\| + \|v_p\| \le Cx^{-1/2}\eta_{k_0}^2, \qquad \|u_p - v_p\| \le Cx^{-1/2}\eta_{k_0}^2 e^{-cd_*}.$$

For $z_p = \sum_a \lambda_a \phi_a(x(p))$, the matrices of the quadratic Gaussian exponents are $K_\square = \sum_p z_p u_p^{\mathsf{T}} u_p$ and $K = \sum_p z_p v_p^{\mathsf{T}} v_p$, with the factor $1/2$ outside the quadratic form. The trace norm of a rank-one term $u^{\mathsf{T}} v$ is $\|u\|\|v\|$. Hence

$$\|K_\square - K\|_1 \le Cx^{-1}\eta_{k_0}^4 e^{-cd_*} \sum_p |z_p|, \qquad \|K_\square\|_1 + \|K\|_1 \le Cx^{-1}\eta_{k_0}^4 \sum_p |z_p|.$$

For n profiles, the radii

$$|\lambda_a| \le \frac{c_* x}{n\mathsf{r}^4 p_0^2 \|\phi_a\|_\infty}$$

and the count $C\mathsf{r}^4\eta_{k_0}^{-4}$ of plaquettes per profile give $x^{-1}\eta_{k_0}^4 \sum_p |z_p| \le Cc_* p_0^{-2}$. Zero profiles may be omitted. Take c_* small enough that the two operator norms are at most $1/2$. The logarithms of the Gaussian generating functions are then given by the convergent trace series, each with the common colour factor $3/2$ for SU(2). Telescoping each difference of matrix powers and using the trace norm on its changed factor bounds the difference of these logarithms by

$$Cc_* p_0^{-2} e^{-cd_*}.$$

The same bound, with another fixed constant, holds for the generating functions. Cauchy's inequality multiplies this cost by the product of the inverse radii for the insertions being differentiated. For the actual test box and the later boxes, the native margins give $d_* \ge c_0 \min\{\mathsf{D}, \check{R}_{k_0}\} - C_0$. Taking $\mathsf{D} = A \log x$ in this range therefore makes the cost smaller than any prescribed inverse power of $\log x$ after a fixed number of differentiations. The comparison allows the exact whole-torus free projection and its ring sums to be used for the Gaussian target. The normalized nonlinear conditional comparison and the removal of the fluctuation windows enter separately.

For the free calculation, write F_j for the background curvature at level j, with covariance $G_j = C^{\mathrm{bg}}_{K,K-j}$. The nested projections give $F_{j-1} = F_j + Z_j$, where Z_j is independent of F_j and has covariance $\mathcal{C}_{(j)}$. The recursion is evaluated at the zero terminal field $F_{k_1} = 0$. In this Gaussian law F_j is the sum of the slices with $j < l \le k_1$, and its covariance is $G_j^\circ = G_j - G_{k_1}$. For one colour, the covariance of the two half-squares at p, q, conditional on F_j, is

$$\frac{1}{2}\mathcal{C}_{(j)}(p,q)^2 + F_j(p)F_j(q)\mathcal{C}_{(j)}(p,q).$$

The second term comes from the two terms linear in Z_j when the half-squares are expanded at F_j. Averaging over F_j with the terminal field fixed gives

$$\frac{1}{2}\left(G_{j-1}^\circ(p,q)^2 - G_j^\circ(p,q)^2\right).$$

Thus the free contributions telescope to $\frac{1}{2}(G_{k_0}-G_{k_1})(p,q)^2$. For separated supports $|G_{k_0}(p,q)|\le Ca_K^4\mathsf{d}^{-4}$, and the coarse covariance bound gives $|G_{k_1}(p,q)|\le CL^{-4k_1}$. The error in replacing the square by $G_{k_0}(p,q)^2$ includes a product of these two kernels and the square of the coarse kernel. Each support has at most $C(\mathsf{d}/a_K)^4$ plaquettes. After smearing, the error is therefore at most

$$C\left[(\hat{\mathsf{d}}L^{-\mathsf{n}})^4+(\hat{\mathsf{d}}L^{-\mathsf{n}})^8\right]\|f_1\|_\infty\|f_2\|_\infty.$$

This tends to zero at the stated choice of n.

Three facts are used here. The source arrives at level k_0 in the form $\zeta'_{k_0}\tilde{f}_i$ ((b)). Next, $a_p\approx\frac{1}{2}|F(p)|^2$, where $F(p)$, an element of the Lie algebra, is the plaquette angle at p of $U_{k_0,\square}(V)$ (the same angle is written Y_p in Thm. A(c)(5)). Finally, half-squares of Gaussian variables have covariance $\frac{1}{2}$(covariance)^2 for each colour. With the two interacting comparisons and the earlier reductions, we obtain

$$\left|\langle\mathcal{O}(f_1);\mathcal{O}(f_2)\rangle-\zeta_{k_0}'^2\,\tfrac{N^2-1}{2}\,x^{-2}\,\mathcal{K}_{K,s}(\tilde{f}_1,\tilde{f}_2)\right|\le o(1)\,x^{-2}\prod\|f_i\|_\infty.$$

The formula at level k_0, combined with (a) and (b), gives the same estimate for $\zeta_{k_0}'^2\langle X_1;X_2\rangle_{\mathsf{P}}$ in the average over $\mathfrak{P}$. *Positions.* The cumulants are invariant under the translations of $\Lambda_{K,\ell}$. A translation $v\in\Lambda_{s,\ell}$ carries blocks onto blocks, which finer translations do not do. Hence

$$C^{\mathrm{bg}}_{K,s}(p+v,q+v)=C^{\mathrm{bg}}_{K,s}(p,q),$$

$$\mathcal{K}_{K,s}(f,h)=\frac{1}{|\Lambda_{s,\ell}|}\sum_{v\in\Lambda_{s,\ell}}\mathcal{K}_{K,s}(f(\cdot-v),h(\cdot-v))\,:$$

since both sides of the law are invariant, it is enough to make the comparison in the average over such positions.

(d) *Wild histories.* For two points, the relevant event is that a large-field region touches the supports at the single level k_1 ((c1)). The weight of these histories is $\le C(M\check{R}_{k_1})^4e^{-cp_0(g_{k_1})}$, and no union over the levels is taken. In this bound the prefactor counts the sites of level k_1 in one cell of that level, and $O(1)$ cells touch the supports. For four points Thm. A(c)(5), the unclean histories are the ones that matter. At each level j of the segment of Thm. A(c)(5), $k_0-O(L\check{R}_{k_0})\le j\le k_0+r_4$, the number of cells of level j near the supports, that is, within $C\hat{\mathsf{d}}$ cells of level k_0, is $\le C\hat{\mathsf{d}}^4L^{4\max\{k_0-j,0\}}$; altogether there are $\le e^{CL\check{R}_{k_0}}$ such cells. A component on which $\mathbf{R}$ acts at a level j of the segment contains a component that is live at level $j-1$. The one-cell bounds of §§ 5.20 and 5.22 give $\tilde{\nu}(\mathfrak{c}\subset\mathcal{Z}_j)\le Ce^{-cp_0(g_j)}$, and the same bound for creation at step j. Moreover $p_0(g_j)\ge cp_0(g(\mathsf{d}))$, because $x_j\ge x^{1/2}$. Hence $\tilde{\nu}(\text{unclean})\le e^{CL\check{R}_{k_0}-cp_0(g(\mathsf{d}))}$, and this is small because $\mathfrak{p}_0\gg r$. Even here these one-cell bounds must hold uniformly in the volume (prolonged run, translation covariance up to the level, first moment of the prepaid end mark: §§ 5.20 and 5.22). The reason is that $T_{\ell_\star}$ contains $(\ell_\star/Ma_s)^4$ cubes of level k_0, and this number is unbounded.

What must be overcome. The natural size of a factor is $x^{-1}\|f_i\|_\infty$. The bound on X_i is larger by $p_0^2\,\hat{\mathsf{d}}^4=(\log x)^{C\mathfrak{p}_0}$, and the irrelevant terms are controlled by sup bounds only. This is the reason for $q\gg\mathfrak{p}_0$ and for the super-polynomial smallness in (a), (d). The effective action is that of Bałaban (Comm. Math. Phys. 109 (1987) 249; 122 (1989) 175, 355); the sources and the extraction are added to it.

The setting and the objects. In §6.1.1.1 the two-point law is stated for two bumps f,h whose size and mutual separation are both $\approx\mathsf{d}$. They are placed on a lattice of spacing $a_K\ll\mathsf{d}$ inside a volume of side $\ell\gg\mathsf{d}$. Isolating the scale d has two aspects. *From below*: $K-s(\mathsf{d})$ renormalisation steps integrate out the scales between a_K and d. In the leading term these scales survive only through two numbers of the run: $g(\mathsf{d})$, the coupling, and $\zeta_K(\mathsf{d})$, the normalisation. *From above*: comparing T_ℓ

with a single fixed torus eliminates the scales from d up to ℓ, together with the volume itself. This comparison costs $O(\mathsf{d}^2)$, which is of lower order than the main term $\approx (\ln \frac{1}{\mathsf{d}})^{-2}$. This part defines the two numbers, bounds them uniformly in the cutoff and in the bare coupling, and carries out the comparison of volumes. The Gaussian main term is extracted in §6.1.1.1, at the level $K - s(\mathsf{d})$ on the fixed torus. There the cumulant is written in terms of a law on resummed histories and block fields (the history-resummation part of the conditional-law hypothesis). Resummed histories are classes of histories that become equal once their entries inside healed components are deleted. The large-field operation $\mathbf{R}$ sums over the outcomes inside the components it heals, so an exponent is attached to such a class rather than to an individual history; write h for this history class.

6.1.1.11. The law passed to the limits.

Corollary 6.4 (Thm. A(c)(4))**.** Let S be a continuum limit, either on T_ℓ of Thm. A(a)(ii) or on $\mathbb{R}^4$ of Thm. A(a)(iii). Let $\mathsf{d} \le \mathsf{d}_0$, and let $(f,h) \in \mathfrak{T}_{\mathbb{E}}(\mathsf{d};\vartheta)$ be a pair of class C^1. Then the conclusion of Prop. 6.1(ii) is valid with

$$(1-\bar\varepsilon)\big(1-\bar\eta(\mathsf{d})\big)^4 \le 1+\eta \le (1+\bar\varepsilon)\big(1+\bar\eta(\mathsf{d})\big)^4,$$

in which $\bar\eta$ and $\bar\varepsilon_\ell$ denote the functions of Prop. 6.1. Their values lie in $[0,\frac13]$ and $[0,\frac12]$ respectively, and both tend to 0 with d. The quantity $\bar\eta(\mathsf{d})$ is a common bound for $|\zeta_K(\mathsf{d})-1|$ and for $|\frac{11N^2}{12\pi^2} g(\mathsf{d})^2 \ln\frac{1}{\mathsf{d}} - 1|$. On T_ℓ one has $\bar\varepsilon = \bar\varepsilon_\ell(\mathsf{d})$, and on $\mathbb{R}^4$ one has $\bar\varepsilon = \frac12$. Consequently $S_2(f,h) > 0$, and on $\mathbb{R}^4$ we have $\dim \mathcal{H}(S) \ge 2$. ■□

Proof outline. The argument follows from the uniformity in Prop. 6.1(i). In the lattice law, the only quantities that depend on (K,ℓ,β) are $\zeta_K(\mathsf{d}) = 1+\eta''$, $g(\mathsf{d})^2 \propto (1+\eta')/\ln(1/\mathsf{d})$ and ε. Therefore

$$\frac{S^T_{2;K,\ell,\beta}(f,h)\,(\ln(1/\mathsf{d}))^2}{\frac{3(N^2-1)}{\pi^4}(\frac{12\pi^2}{11N^2})^2 \mathcal{K}_{\mathrm{free}}(f,h)} = (1+\eta'')^2(1+\eta')^2(1+\varepsilon)$$

takes its value in the closed interval of the display above, for every datum. For fixed d, we have $K_j - s(\mathsf{d}) \ge n_0$ once j is large. Every pair on $\mathbb{R}^4$ is contained in some ball of radius $5\mathsf{d}$. Hence, once transported to T_{ℓ_j}, it belongs to the class and keeps the same free kernel. A closed condition is preserved in the limit. The bound on ε that clause (c)(4) records on $\mathbb{R}^4$ is the only one in Prop. 6.1(i) that is free of any side, namely $|\varepsilon| \le \frac12$.

The Hilbert space. Let f_{d} be a C^1 bump whose support is the ball $B(2\mathsf{d}e_4, \mathsf{d})$. Then $(\theta f_{\mathsf{d}}, f_{\mathsf{d}})$ is a test pair. Its shape ϑ_1 is fixed and does not depend on d, because the bumps are dilates of a single profile. When $\vartheta > \vartheta_1$, the pair may fail to belong to $\mathfrak{T}_{\mathbb{R}^4}(\mathsf{d};\vartheta)$. We take the ceiling $\gamma_W(\vartheta)$ smaller than the ceiling under which Prop. 6.1(i) is valid at the shape ϑ_1. That theorem is applied there with the witness depth $s_1(\cdot)$, which is built from the room constant $C_{W,0}$ of that shape, and with the thresholds of the same shape. Once d is below its threshold, the three inequalities $\zeta'_{K-s_1(\mathsf{d})} \ge \frac23$, $1+\varepsilon \ge \frac12$ and $g^{-2}_{K-s_1(\mathsf{d})} \le \bar x_{s_1(\mathsf{d})}$ bound $S^T_{2;K_j,\ell_j,\beta_j}(\theta f_{\mathsf{d}}, f_{\mathsf{d}})$ from below by a positive number that does not depend on j. Consequently § 6.16 gives

$$\big\|[(f_{\mathsf{d}})]\big\|^2 = \sigma_S(\theta f_{\mathsf{d}}, f_{\mathsf{d}}) = S_2(\theta f_{\mathsf{d}}, f_{\mathsf{d}}) > 0,$$
$$\langle \Omega, [(f_{\mathsf{d}})]\rangle = \sigma_S((f_{\mathsf{d}})) = 0.$$

Periodic Gibbs states. The expansion concerns torus measures. The law is carried over to these states by eq. 6.vi, under its common-normalizer and reference-comparison hypotheses. That lemma relies on the periodic-state domination in eq. 5.cdxxxiii, according to which the chessboard dominates the periodic Gibbs states. Precisely: let F be a function of a single board cell of side $\mathsf{l} = O(1)$, and on T_ℓ let the chessboard seminorm be denoted $\|\cdot\|_{\mathrm{ch},\ell}$. Then

$$|\nu(F)| \le \limsup_{\ell\to\infty} \|F\|_{\mathrm{ch},\ell}.$$

This is a separate lemma. It is not the chessboard estimate, whose subject is torus measures. Its proof, taken in the form that involves all mirror copies, combines four ingredients: the reflections of the cells leave ν invariant § 5.24 and appendix A.6; Jensen's inequality is used together with the Gibbs property; the boundary condition is changed at the cost of a surface term, which is weighed against a volume; and finally the seminorm enters.

Ergodic components. At (K_j, β_j), each periodic Gibbs state obeys the law. A closed condition that all of these states satisfy is inherited by the components § 7.7. This yields Thm. B(3)(ii): for a volume-first limit, each of its ergodic components satisfies the law in the form it takes on $\mathbb{R}^4$, with d_0 and $\bar\eta$ unchanged.

6.2. **Covariance bounds for interacting extraction.** The finite insertion, renewed pointing, window formation and linear bootstrap sections derive the ordinary regular first response under their explicit unmarked, background, domain and slope inputs. They retain the potential normalization through local integration and give the near bound $C\delta^2 s$ on complete nonwrapping fields of size δ. The remainder below must also include the separately bounded marked, remote and boundary contributions. None of these estimates proves an interacting Gaussian law.

6.2.1. *Covariance bookkeeping with the retained source normalization.* A source-disc bound alone cannot replace the potential estimate. Indeed $H(z,B) = (z/\rho)(B/\alpha)^4$ is bounded by one on its product unit balls, but its source derivative at $B = \epsilon$ is $\rho^{-1}(\epsilon/\alpha)^4$. Relative to $\alpha^2(\epsilon/\alpha)^3$, this is $x(\log x)^{p-3q}/\rho$ when $\alpha = x^{-1/2}(\log x)^q$ and $\epsilon = x^{-1/2}(\log x)^p$; it diverges at fixed ρ. The finite construction supplies its response by differentiating the inserted potential, with the stated unmarked hypotheses.

Here is the corresponding covariance bookkeeping. Let λ be any probability measure, put $n_i = \|f_i\|_\infty$, fix $p > 0$, and suppose, uniformly in $x \to \infty$,

$$\begin{aligned} \|\mathsf{O}_i\|_\infty &\le C x^{-1} r^4 (\log x)^{2p} n_i, \\ \|\mathsf{I}_i\|_\infty &\le C x^{-1} r^4 [(\log x)^{3p-q} + x^{-a}] n_i, \qquad a > 0, \\ r &\le C_r (\log x)^{2p}, \qquad q > 21p. \end{aligned} \tag{6.ii}$$

Then

$$\begin{aligned} &|\operatorname{Cov}_\lambda(\mathsf{O}_1, \mathsf{I}_2)| + |\operatorname{Cov}_\lambda(\mathsf{I}_1, \mathsf{O}_2)| + |\operatorname{Cov}_\lambda(\mathsf{I}_1, \mathsf{I}_2)| \\ &\qquad \le C' x^{-2} \big[(\log x)^{21p-q} + (\log x)^{18p} x^{-a}\big] n_1 n_2 = o(x^{-2} n_1 n_2). \end{aligned} \tag{6.iii}$$

To prove it use $|\operatorname{Cov}(A,B)| \le 2\|A\|_\infty \|B\|_\infty$. The product of the two support counts costs $r^8 \le C(\log x)^{16p}$, and the a priori insertion costs another $(\log x)^{2p}$. For large x the bracket defining I_i is at most one, so its square is absorbed by itself. This gives eq. 6.iii. The numerical inequality $q > 21p$ is a sufficient bound for this particular sup-norm step; it does not claim to be the optimal requirement for the full extraction.

6.2.2. *A scalar transfer estimate with a genuinely common normalizer.* Let ρ, τ be two probability measures with real bounded observables $\mathcal{O}(f)$ linear in a real compactly supported Lipschitz function f. They may be measures on different configuration spaces. Write

$$D(f,h) = \operatorname{Cov}_\rho(\mathcal{O}(f), \mathcal{O}(h)) - \operatorname{Cov}_\tau(\mathcal{O}(f), \mathcal{O}(h)).$$

Assume that D is invariant under joint translations by $a\mathbb{Z}^4$. Set

$$\|f\|_\sharp = \max(\|f\|_\infty, c_M \operatorname{Lip} f)$$

with c_M fixed.

Suppose there are $r_0 > 0$ and $Q \geq 1$ such that, for every pair F_1, F_2 supported in one common unit ball, there is one holomorphic function $E(z_1, z_2)$, $E(0) = 0$, serving both measures, for which

$$\left| e^{-E(z)} \omega\Big(e^{z_1 \mathcal{O}(F_1) + z_2 \mathcal{O}(F_2)} \Big) \right| \leq Q, \qquad \omega \in \{\rho, \tau\}, \tag{6.iv}$$

on $|z_1| \|F_1\|_\sharp + |z_2| \|F_2\|_\sharp < r_0$. The constants are uniform in the ball position and in the measures. The tests in this hypothesis may overlap and have either sign.

Put $c_i = \|F_i\|_\sharp / r_0$ and denote the normalized holomorphic function in eq. 6.iv by h_ω. It has $h_\omega(0) = 1$. Schwarz's lemma along rays gives $|h_\omega(z) - 1| \leq (Q+1) \sum_i c_i |z_i|$. On the bidisc with radii $[4(Q+1)c_i]^{-1}$ its principal logarithm has modulus at most $\log 2$. Thus Cauchy's formula gives

$$|D(F_1, F_2)| \leq 32 \log 2 \, (Q+1)^2 r_0^{-2} \|F_1\|_\sharp \|F_2\|_\sharp. \tag{6.v}$$

The derivatives of E cancel because it is the same function for both measures. If an entry is zero the conclusion is immediate. No global zero-free statement for the original generating functions is needed.

Suppose now $a \leq d$ is sufficiently small, and f, h have supports in one common ball of radius $5d$, with $\mathrm{Lip}\, f \leq \|f\|_\infty / (\vartheta d)$ and the analogous condition for h. Choose a lattice multiple $r' \in [5d, 10d)$ and $N \geq c d^{-4}$ translates of that ball, with centers in $4r'\mathbb{Z}^4$, all lying inside a common unit ball. For each choice of signs $\varepsilon_v = \pm 1$ let $F^\varepsilon = \sum_v \varepsilon_v \tau_v f$ and $H^\varepsilon = \sum_v \varepsilon_v \tau_v h$. The translated balls are disjoint. Since the functions vanish on their ball boundaries, these signed sums do not increase either supremum or Lipschitz norm. Independent signs with the same sign used in the two arguments give

$$\mathbb{E}_\varepsilon D(F^\varepsilon, H^\varepsilon) = N D(f, h).$$

Apply eq. 6.v and divide by N. The result is

$$|D(f, h)| \leq C(Q, r_0, c_M) \vartheta^{-2} d^2 \|f\|_\infty \|h\|_\infty. \tag{6.vi}$$

The exponent is $2 = 4 - 2$: the number of copies gains four powers and the two Lipschitz norms lose one power each.

If a reference law has a main term bounded below by $c_\vartheta x^{-2} \|f\|_\infty \|h\|_\infty$, its relative transfer error is at most $C_\vartheta d^2 x^2$. In the actual short-distance shape class, each nonnegative entry also has the explicit mass lower bound $\int f \geq \vartheta d^4 \|f\|_\infty$, and the two supports are at distance between d and $6d$. These requirements give $\iint f(x) h(y) |x - y|^{-8} \, dx \, dy \geq 10^{-8} \vartheta^2 \|f\|_\infty \|h\|_\infty$. The mass requirement is additional to the Lipschitz bound in eq. 6.vi. At a witness depth with $x \leq \bar{x}_s = O(s)$ and $d < L C_W(s) L^{-s}$, where C_W is polylogarithmic in s, this tends to zero uniformly. For a fixed $\vartheta = 1/12000$ all constants remain finite; their dependence on that fixed shape may affect the permitted coupling and distance thresholds.

The common-normalizer hypothesis in eq. 6.iv is not a consequence of having unrelated bounded normalizers for the two measures. There are two different ways of supplying it. If the scalar stability construction already provides one common cell normalizer for every periodic infinite-volume Gibbs state at the same datum, then eq. 6.iv holds for any two such states. This includes a reference state obtained by taking finite-torus limits, provided its Gaussian estimate was uniform along arbitrarily large tori for each fixed cutoff and pair. The source $z_1 F_1 + z_2 F_2$ is a single cell source, even when the supports overlap. This route needs no comparison with a fixed finite torus.

If, instead, the reference Gaussian estimate has only been proved on one prescribed finite torus, a comparison of that torus with the infinite-volume states remains necessary. Compatible local energies and vacuum-density/floor bounds give one scalar chessboard route to eq. 6.iv. Bounds valid on each torus with a distance threshold tending to zero as its side grows do not by themselves furnish an infinite reference with the needed fixed-pair estimate. In either route, a typed Ward insertion estimate is not needed for eq. 6.v–eq. 6.vi.

A local companion criterion for vacuum-density compatibility. For the finite-reference route just described, the following criterion can remove a separate vacuum-pressure comparison assumption. It is not needed for comparison of infinite-volume states when their common cell normalizer is already available. It does not follow from upper stability and a partition floor alone. Fix $L > 1$ and a cutoff K, and put $a_j = L^{j-K}$ for $0 \le j \le K$. Take compatible side lengths ℓ for which ℓ/a_j is integral. Suppose the vacuum constant has the form

$$E_{\ell,K} = \sum_{j=0}^{K} \left[(\ell/a_j)^4 c_{j,K} + \sum_{z \in (a_j\mathbb{Z}/\ell\mathbb{Z})^4} \sum_{D \in \mathcal{D}_{\ell,j}(z)} v_{\ell,j}(D,z) \right]. \tag{6.vii}$$

The numbers $c_{j,K}$, the coupling sequence and every auxiliary choice are the same for all ℓ. There is a translation-covariant infinite companion $v_{\infty,j}$ and an integer length $m(D) \ge 0$. For both families, assume that the number of domains of length m at a root is at most $C_D e^{\kappa_D m}$ and $|v_{\ell,j}(D,z)|, |v_{\infty,j}(D,z)| \le C_E e^{-\kappa_E m}$ with $\kappa_E > \kappa_D$. The constants are uniform in j, K, ℓ. Assume that rooted domains with $m < \ell/(C_0 a_j)$ correspond bijectively to companion domains, with equal energy values. This is an explicit locality and nonwrapping assumption, with C_0 fixed. It is stronger than simply asserting that the two models use the same formula for their energies.

Absolute convergence defines $\bar{v}_j = \sum_{D \in \mathcal{D}_{\infty,j}(0)} v_{\infty,j}(D,0)$ and $A_K = \sum_{j=0}^{K} a_j^{-4}(c_{j,K} + \bar{v}_j)$. At each root all terms below the stated threshold cancel in the difference from the companion. Sum the two tails of the geometric series with rate $\delta = \kappa_E - \kappa_D > 0$. Division by the physical volume therefore gives

$$\left| \ell^{-4} E_{\ell,K} - A_K \right| \le C \sum_{j=0}^{K} a_j^{-4} e^{-c\ell/a_j}. \tag{6.viii}$$

For every fixed $\ell_0 > 0$ and $\ell \ge \ell_0$, this is bounded by $C_{\ell_0} e^{-c\ell/2}$, independently of K. Indeed, with $b = K - j$,

$$\sum_{b \ge 0} L^{4b} e^{-c\ell L^b} \le e^{-c\ell/2} \sum_{b \ge 0} L^{4b} e^{-(c\ell_0/2)L^b} \le C_{\ell_0} e^{-c\ell/2}. \tag{6.ix}$$

Thus two compatible tori obey the same bound with $2C_{\ell_0} e^{-c \min(\ell,\ell')/2}$. The divergent common number A_K need not converge as the cutoff is removed; it cancels exactly in the comparison.

In a localized source argument, equality of the corresponding nonwrapping source energies also supplies the same interior-cell counterterm for all tori. Bounds on the energies crossing the cell faces and the appropriate partition floor are still needed to obtain eq. 6.iv. This lemma addresses its vacuum-density datum only. If the coupling sequence or local normalizer numbers differ between tori, or if nonwrapping compatibility has not been established for the actual formation rule, eq. 6.viii has not been proved for that construction.

A common source normalizer from the reference torus. Fix one first-passage datum and let $\ell_\star$ be the least compatible side at its reference coupling. Choose board cells of side $\mathsf{l}_c = L^m$ such that a unit ball can be placed in a cell with a fixed positive face clearance, and $\ell_\star/\mathsf{l}_c$ is even. This is possible with a fixed m for the small reference couplings under consideration. Each compatible side is an integer multiple of $\ell_\star$. Translate the two tests jointly by a bare lattice vector to the interior of the designated cell. Write $w_z = z_1 F_1 + z_2 F_2$ and $N_\ell = (\ell/\mathsf{l}_c)^4$. Disseminate this source by reflection and conjugation, using $\bar{z}$ in the odd cells, and call the resulting source W_z. Its supports have the same face clearance in each cell. The dissemination of $e^{\mathcal{O}(w_z)}$ is $e^{\mathcal{O}(W_z)}$; plaquettes in the support have all their corners in a single cell. Joining the reflected profiles across faces leaves the Lipschitz bound unchanged since each profile vanishes near the faces. Treating even and odd copies separately gives $\|W_z\|_\flat \le 2 \sum_i |z_i| \|F_i\|_\sharp$. In particular, $\sum_i |z_i| \|F_i\|_\sharp < \varrho$ is within the source domain of Prop. 5.63.

The numerator of the chessboard seminorm is nonnegative, and

$$\|e^{\mathcal{O}(w_z)}\|_{\mathrm{ch},\ell} = \left(\frac{Z_\ell(W_z)}{Z_\ell(0)}\right)^{1/N_\ell}, \qquad Z_\ell(W_z) \le Z_{\ell_\star}(W_z)^{(\ell/\ell_\star)^4}.$$

To see the second inequality, vary one side at a time, keeping the other three fixed. Across a cell face there is no source. After integrating the crossing bonds the Wilson transfer factor is the product of the gauge-invariant projection and the convolution with the product of the one-bond class functions $c(u) = e^{-\beta(1-\mathrm{tr}(u)/2)}$. These operators commute. The character coefficients of c are nonnegative: expand $e^{(\beta/2)\chi_{1/2}}$ in powers and decompose each tensor power of the defining representation. Hence the crossing factor is positive, with self-adjoint square root S. In a cross-section there are finitely many bonds, and the convolution is trace class since the nonnegative character series at the identity converges. The intervening slab operator with the internal source has the form $A = SBS$, and its mirror copy is A^*. The trace of the shorter period is $\mathrm{Tr}\, T$, where T is the appropriate integer power of AA^*. Repeating the period b times gives $\mathrm{Tr}\, T^b$, at most $(\mathrm{Tr}\, T)^b$ by positivity. Applying this in the four directions proves the comparison. The intermediate rectangular tori here use the same Wilson integrals and need no renormalisation construction.

The actual vacuum constants have the decomposition

$$E_{\ell,K} = \sum_{j<K} \left[\mathsf{E}_{\mathrm{I}}^{(j)} + \sum_y \sum_{X \ni y} E^{(j+1)}(X, \mathbf{1}, y) + \sum_X R^{(j+1)}(X, \mathbf{1}) \right].$$

In $\mathsf{E}_{\mathrm{I}}^{(j)}$, the Haar and Gaussian factor is one scalar per nonpivot bond and the gauge-fixing factor is one scalar per cell of the next lattice. The common coupling sequence and fixed choice of gauges give the same two scalars for the two tori, and both counts per unit volume are the same. Point the remainder term by assigning $R^{(j+1)}(X, \mathbf{1})/|X|$ to each root of X. A domain whose complete window lifts to a cube of side at most $\min(\ell, \ell_\star)/4$ has the same rooted value on either torus by the common companion rule of § 5.2.2. The density of these terms agrees, since the rooted family has the periods of the smaller side. For the other domains the length at depth u is at least $c\ell_\star L^u/M - C$. Summing over their roots and then over u gives

$$\left| E_{\ell,K}/\ell^4 - E_{\ell_\star,K}/\ell_\star^4 \right| \le C \sum_{u \ge 0} L^{4u} e^{-c\ell_\star L^u/M}.$$

Thus the common companion comparison applies to the vacuum constant in the sourced construction. Neither vacuum density need have a cutoff-independent limit.

Use upper stability of Prop. 5.63 on $T_{\ell_\star}$ and the partition floor on T_ℓ. Writing $\mathcal{E}_\star(W_z) = \sum_{j,X} e_j(X; W_z)$ for the reference source energy, the preceding estimates give

$$\|e^{\mathcal{O}(w_z)}\|_{\mathrm{ch},\ell} \le \exp\left[N_{\ell_\star}^{-1} \,\mathrm{Re}\, \mathcal{E}_\star(W_z) + C_g |_c^4 \right].$$

Define $E^\circ(z)$ on the reference torus as the average over its cells of the sums of $e_j(X; \cdot)$ with the complete source footprint contained in one open cell. This includes the barycentres of all bare plaquettes with a corner in $X^{\sim 1}$. Evaluate the energy on that cell's reflected source. In an odd cell, also conjugate the answer. Such terms have the form $\overline{h(\bar{z})}$ with h holomorphic, so E° is holomorphic and $E^\circ(0) = 0$. Its real part is exactly the corresponding interior portion of $N_{\ell_\star}^{-1} \,\mathrm{Re}\, \mathcal{E}_\star(W_z)$. This identification uses locality on the complete source footprint and compares the same energy function in each fixed domain.

A nonzero omitted term must meet a cell face and reach a plaquette whose barycentre is in a source support. The face clearance implies a lower bound $[cL^u/M - C]_+$ on its length at depth u, with the additional bare reach included in C. There are at most $C(1 + L^u/M)^4$ possible roots per cell near a support. The length entropy is smaller than the rate κ_e of Prop. 5.63. Thus the omitted

sum per cell is bounded independently of K by

$$C\sum_{u\geq 0}(1+L^u/M)^4\exp\big(-(\kappa_e-\kappa_{\mathcal{D}})[cL^u/M-C]_+\big).$$

The same E° now gives eq. 6.iv for the reference torus and every compatible torus, by the one-cell chessboard inequality. It gives that estimate for every periodic Gibbs state of the theorem by eq. 5.cdxxxiii. The constants are uniform in the datum. This construction compares with the prescribed finite reference side; it does not require taking a subsequential torus limit.

By Prop. A.11 and the composition of a site reflection with the adjacent link reflection, the states in the theorem are invariant under bare lattice translations. Therefore eq. 6.vi applies to their differences from the reference. Its error is $C_{g,\vartheta}\mathsf{d}^2\|f_1\|_\infty\|f_2\|_\infty$; relative to the nonnegative reference main term in the stated shape class, this is bounded by $C_{g,\vartheta}\mathsf{d}^2x^2$. It tends uniformly to zero by the witness depth bounds. The comparison extends the reference estimate whenever the physical source remainder and the reference witness covariance have the estimates used above.

6.3. **The conditional law and its connected errors.** The short-distance coefficient concerns an interacting probability law. Comparing quadratic kernels does not identify that law. We specify the conditional measure and its complete exponent, then derive the boundary terms, connected errors and exterior average. Applying these identities requires the run-window identification and uniform estimates stated at the end. The primary route is the resummed clean run; an averaging-fibre transfer is a distinct alternative.

Identify the measure before estimating its density. Fix the exterior sigma-field of the run window, with value e. The identity concerns that specified sigma-field and the resummed positive term. A finer history or interpolation resolution need not be positive or carry the same conditional measure.

In an admitted real pivot chart write the raw logarithms as $a=(p,u)$. Suppose the constraints $q(p,u)=0$ have invertible D_pq and solution $p=\varphi(u)$. If the original coordinate density is $w(p,u)$, the constrained density is

$$w(\varphi(u),u)|\det D_pq(\varphi(u),u)|^{-1}\,du. \tag{6.x}$$

This is the change of variables from (p,u) to $(q(p,u),u)$, followed by evaluation of the constraint delta function. Haar factors, delta normalizations and the exact gauge-quotient density belong in w. A gauge-fixing factor requires the separate identity inserting it.

Under a canonical linear rescaling $u=gCy$ retain its determinant. For a variable-dependent map use its full Jacobian. The domain is the image of the actual integration set; a product window is an identity to establish.

On a positive component the complete conditional density may be written

$$d\nu_e(y)=Z_e^{-1}1_{D_e}(y)\exp\{-\tfrac12 y^{\mathsf T}Q_ey-\ell_e^{\mathsf T}y+P_e(y)\}\,dy. \tag{6.xi}$$

The actual extraction rule fixes Q_e. The complete P_e contains old activities, nonlinear action, coarea and Haar logarithms, boundary and stored terms, and healed-origin remainders. A factor outside this exponential is carried in every numerator, or absorbed where its logarithm exists. Zeros require a domain or a decomposition.

An arbitrary choice of Q_e would make eq. 6.xi merely a rewriting. The extraction rule, sigma-field, domain, normalizer and observable numerators all belong to the required identification. For the exact calculus below assume a bounded piecewise smooth domain and a density in $C^1(\overline{D})$, or in $W^{1,1}(D)$ with the indicated integrable traces and integration-by-parts identity. All displayed force integrals are required to be integrable. A rectangle suffices. Unbounded domains require an exhaustion and justified boundary limits.

The complete Hessian and the exterior force. Set

$$V_e = \tfrac{1}{2} y^{\mathsf{T}} Q_e y + \ell_e^{\mathsf{T}} y - P_e(y), \qquad r_e = \nabla P_e - \ell_e. \tag{6.xii}$$

The Hessian is $Q_e - P_e''$. The old-activity theorem controls its stated old-origin sum, not every other term in P_e. Relative to a specified reference the exact force difference is

$$\nabla V_e - \nabla V_0 = (Q_e - Q_0)y + (\ell_e - \ell_0) - \nabla(P_e - P_0). \tag{6.xiii}$$

In the actual run this includes the marginal Schur force $Q^{\mathrm{sc}} y$, the integrated exterior force $Q^e_{IO} y_O$, the newly stored exterior force ℓ^e_E, the interior precision discrepancy and the interaction derivative. Let $\mathcal{B}_{\mathrm{ext}}$ consist of $O \cup E$ and the complement of the padded common region on which the two operator geometries and coordinate masks agree. The force bound required at an interior coordinate is

$$|f_b| \le C b_j e^{-\mu d(b, \mathcal{B}_{\mathrm{ext}})} \tag{6.xiv}$$

uniformly in the admitted exterior data.

6.3.0.1. The determining data and the response to the free box. The restriction identity eq. 3.cccxciii identifies the box background on its induced datum. In the step from j to $j+1$, fixing V_{j+1} and the exterior free fluctuations still leaves the retained lower determining data as arguments of the background and the conditional precision.

One part of those arguments drops out exactly. Let E_1 be the fine bonds incident to Ω_1, E_0 their complement, and $C_0 \subset E_0$ the bonds in a plaquette touching E_1. The bare datum fixes all of E_0. Enlarge C_0 by the portions of E_0 read by the positive-level double blocks or the fixed axial gauge stencil. In the outward-row convention the additional averaging reads come from the level-one outward rows, within the reach of a level-one double block. Fix the nest and axial frame. If two data agree at every positive level and on this enlarged set, the constraints on E_1 and their chosen gauge slice coincide. For any field on that slice, the Wilson actions agree on plaquettes touching E_1, and the other plaquettes give a scalar difference independent of the field on E_1. It follows that the constrained axial solve and the constrained Wilson Hessian on E_1 agree exactly. The scalar cancels in normalized box expectations. In this argument the deep bare fields may be rough. The other terms of the density have their own reads.

On the native small-field support the retained rows below j have the zone and interface plaquette bounds of the determining nest. At the top level, start with the bound $2\varepsilon_j$ on V_j. Replacing an interior bond by the average from the next step changes its factor by at most $c'\varepsilon_j$, with the same bound in the reversed orientation. Telescoping the product of four unitary factors gives at most $4c'\varepsilon_j$ in additional plaquette error. This also covers mixed top plaquettes at the cut. At the action collar in zone zero, the native small-field guard gives the bare bound ε_0. These are bounds on the actual base datum before varying the free box.

For a free box variation about one base datum, extend its coordinate y by zero on the other fluctuation bonds, writing $\widetilde{y}$ for the extension. In the native top rows the variation is multiplication by $e^{iB(y)}$, where

$$B(y) = g_j \mathbf{C}\widetilde{y} - \mathbf{h}\,\widetilde{D}(g_j \mathbf{C}\widetilde{y}).$$

Here $\mathbf{C}$ is the linearized constraint lift, $\mathbf{h}$ the corridor operator and $\widetilde{D}$ the analytic correction vanishing to second order. On a tube of size Y, its quadratic bound gives $\|B\|_\infty \le 2 g_j c_C Y$, below the datum radius by the native ceiling. For a unit direction in one bond b, the derivative of B is $O(g_j)$. The linear lift reaches only b and the pivots of its incident block pairs. The corridor correction reads those block pairs and their adjacent corridors. Consequently this derivative reaches at most $(4d{+}2)^2$ top bonds within $4L{+}2$ input-level units of b. The clean guard puts both endpoints of every bond in that support inside Ω_{j+1}, including the pivots. All lower rows are left fixed.

On the admitted retained datum let $\mathcal{H}(B)$ be the analytic response at its fixed base U_0. For real parameters the solution with the varied top rows is gauge equivalent to $e^{i\eta\mathcal{H}(B(y))}U_0$. The finite support of $D_b B$ makes the chain rule a sum over the bounded list of top rows. The derivative

bound of the retained response applies on these carriers. Use its comparison of distances to the top rows and identify the derivative in its plaquette estimate with the covariant difference at U_0. This gives, for $H(y) = \mathcal{H}(B(y))$,

$$\eta|D_b H(y)(x)| \le C g_j L^{-n(x)} E(x,b), \qquad \eta^2 |D_{U_0} D_b H(y)(p)| \le C g_j L^{-2n(x)} E(x,b).$$

Here $n(x)$ is the retained zone and D_{U_0} is the covariant lattice difference at the plaquette p based at x. The response weight is $E(x,b) = \min\{1, e^{-c\rho(x,b)+c_1}\}$, where ρ is the retained response distance with the top spacing as unit. The constant c_1 allows for the displacement of the rows of $D_b B$ from b. On a smaller tube the two orders of Cauchy differentiation give a mixed bound with the minimum of the two weights and the inverse tube margin. This constructs the read and its derivatives on a fixed labelled nest with admitted restricted data and response. The comparison of masks or of positive-level data across different exteriors still uses their matching estimates.

6.3.0.2. Terms outside the box that read the background. The ordinary pre-strip chart also allows nonzero fixed exterior fluctuations. In that chart let z be the other original free coordinates. The native real windows give $\|z\|_\infty \le C b_j$. The complete level-j datum uses the multiplier $e^{iB_{\mathrm{full}}(y,z)}$, with the same lift and nonlinear correction as above, applied to the full vector (y,z). Keep the outgoing target and all lower data fixed. Apply the bound for the lift and its correction with $R = \max(Y, \|z\|_\infty)$. On the joint chart the smallness of $g_j c_C R$ gives $\|B_{\mathrm{full}}(y,z)\|_\infty \le 2 g_j c_C R$ and $\|D_y B_{\mathrm{full}}\|_{\infty\to\infty} \le 2 g_j c_C$. The individual derivative D_b uses the same rows within $4L+2$ of b. These statements concern the original free slots, including an exterior slot once its identification with an original free coordinate has been made.

Choose y_* on the supported real fibre for the fixed z. Use the actual datum there as the base. The relative logarithm for a variation is

$$A_{z,*}(y) = \frac{1}{i} \log\Big(e^{iB_{\mathrm{full}}(y,z)} e^{-iB_{\mathrm{full}}(y_*,z)} \Big).$$

It is zero at y_*. On the fixed exponential and logarithm balls its norm is at most $C_L g_j Y$ and its derivative in one unit bond direction is at most $C_L g_j$. For the size bound use the segment from y_* to y; for the derivative use the fixed second factor of the relative product. Outside the full top carriers of I the product is the identity. The native response theorem is applied at the real regular datum on the supported fibre, with $A_{z,*}$ in its input ball. Its chain rule gives the same bond and plaquette estimates as above, with a fixed change of constants. If the base is needed at zero, the same segment varies only the clean top carriers. Telescoping the four factors of each affected top plaquette enlarges its real regularity bound by $C_L g_j \|y_*\|_\infty$. The lower interface and bare collar reads are unchanged when the full carriers are in the padded top interior.

For the ordinary strip substitution this identification is exact at fixed retained data. The global centre is determined by the outgoing block field, the level-j field off the incident stock of Ω_{j+1}, and the lower determining rows. The joined local centre uses the outgoing rows and the old retained rows of the output label. The latter are fixed when the free box is varied. In particular the old level-j determining member used here has both endpoints outside the released region. In the join for the strip, the inner cut is in the top layer and its new rows lift the fixed outgoing rows. The outer cut is in the previous layer outside Ω_{j+1} and lifts the fixed old rows there. The averaged local strip field is the one in the axial frame of this joined set. Neither centre reads the shifted strip nonpivots. Thus both centres, and their relative shift on a strip bond, are fixed throughout this variation. The real cutoff in the inverse strip map is a function of that bond's stored coordinate. Consequently a held ordinary post-strip nonpivot yields a fixed pre-strip nonpivot, including on the cutoff's transition shell. The Jacobian of that held slot is also fixed. The box coordinates themselves are unchanged on the ordinary passing mask, and the pivots are recomputed at the same target. The ordinary response calculation therefore applies to these conditional slots.

At the top preparatory stage of this same output level the two centres stay fixed under the original clean-box variation as well. The stage member contains the old Ω_{j+1}. Its old level-j determining

rows are the rows outside that member, with neither endpoint inside it. Thus they miss the box variation and its complete pivot support inside the old Ω_{j+1}. Keep the added coarse targets and exterior data fixed. The joined background uses the lift of the fixed output field on the old inner region and the retained datum on the outer region. The two stage centres and the bondwise cutoff translation are then fixed when the original box varies. At a held real stage slot this again fixes its inverse coordinate. A variation of the released collar variables themselves must use their stage chart.

There is also an exact fixed-coordinate case for the healed reference draw. If its component and full retained exterior rows are outside the padded box read, fix those rows in the exterior disintegration. The basic centre and the normalized reference law depend only on that retained exterior of the component. With its relative reference draw held fixed, the restored physical field $V'V_\Lambda$ is then fixed, and the reference density is scalar in y. The previous box variables have already been integrated in this kernel. The completed global background and the main-step density subsequently read the replacement rows, and their comparison uses those actual reads.

Consider a term f whose remaining dependence on y is its gauge-invariant evaluation on the responding background, restricted to a bond set X_f. Its other arguments are held fixed. The response estimates give size $C_H\Theta(x)$ in each of the bond and plaquette units above, where $\Theta(x) = g_j Y \sum_{b\in I} E(x,b)$ for a tube of size Y. Integrate from the base, where the response vanishes, along the segment to y. For the term's analytic norm divide the two units by their radii $\rho_f^{(1)}$ and $\rho_f^{(2)}$. Write $r_f^{-1} = \max_i(\rho_f^{(i)})^{-1}$. If $C_H r_f^{-1} \max_{x\in V(X_f)} \Theta(x) \le 1/2$, the restricted field lies in the half ball for that norm. A term bounded by M_f on the full ball has the continuation $\varphi_f(y) = f(W^{(y)}|_{X_f})$, where $W^{(y)} = e^{i\eta H(y)}U_0$ in the response gauge at its base. Apply Cauchy's estimate in the direction of the response derivative inside the remaining half ball. With $w_f = g_j M_f/r_f$, this gives

$$|D_b\varphi_f(y)| \le 2C_H w_f \max_{x\in V(X_f)} E(x,b).$$

On a smaller tube with margin r the mixed derivative has the bound $2C_H w_f/r$ times the smaller of the two endpoint weights for b, b'.

Use the scaled path distance in E. A path from zone n to I crosses each intermediate zone. If those zones have width at least M in their own units, each contributes at least M to the scaled distance. Split the response exponential with rates $\mu = \nu$ equal to half the rate in E. The count and the ordinary top-scale distance give, for a level-j cell c,

$$E(x,b) \le Ce^{-\nu d_j(c,b)} e^{-\mu M(j-1-n)_+}, \qquad x \in c \cap Z_n.$$

Select a maximizing endpoint of each X_f. Suppose for any such choice the total w_f in one cell and zone is at most $C(1+j-n)^q L^{4(j-n)}$. The age sum is finite by $L^4 e^{-\mu M} \le 1/2$. If all the reads are at distance at least $D_{\mathrm{far}} \ge 1$ from I in level-j units, the cell sum is at most $CD_{\mathrm{far}}^{d-1} e^{-\nu D_{\mathrm{far}}}$. This bounds the sum of the absolute derivatives at each $b \in I$. It gives an entrywise mixed derivative bound after division by the tube margin.

For the uncompensated frozen coupling, the far term is the sum of Wilson plaquette terms with coefficients $g_j(x_p)^{-2} - g_j^{-2}$, fixed with the exterior and bounded by $C(1+j-n(p))$. Their nonzero plaquette reads are outside the collar of width D_{far}. Plaquettes entirely in the fixed deep bare region have zero derivative. At each responding plaquette the source trace estimate gives $Cg_j(\bar\varepsilon + \theta_*)L^{-4n(p)}E(x_p,b)$ before multiplying by the coefficient. Here $\bar\varepsilon$ bounds the base curvature in its zone units and θ_* is the native trace threshold for Θ on the lag support. On that support the unweighted top distance gives $\Theta(x) \le Cg_j Y D_{\mathrm{far}}^{d-1} e^{-2\nu D_{\mathrm{far}}}$. Use the collar width for which this is at most θ_*. The trace derivative gains one factor of the curvature and one from its variation. There are at most CL^{4j} plaquettes in a level-j cell, so the same cell and age sums give the force bound

$$Cg_j(\bar\varepsilon + \theta_*) D_{\mathrm{far}}^{d-1} e^{-\nu D_{\mathrm{far}}}.$$

For the mixed Hessian of the lag, the plaquette trace bound uses the weight

$$r^{-1}\min\{E(x_p,b),E(x_p,b')\}.$$

In the cell sum the spatial exponential uses the larger of the two distances from c to b,b'. This maximum is at least $D_{\mathrm{far}}-C_d$, at least $(d_j(b,b')-C_d)/2$, and at least $d_j(c,b)$. Use one third of the exponential for each of these three bounds. The last bound sums over cells, and the zone sum is unchanged. Thus the lag Hessian entries are bounded by

$$Cr^{-1}g_j(\bar\varepsilon+\theta_*)e^{-\nu D_{\mathrm{far}}/3}e^{-\nu d_j(b,b')/6}.$$

The weighted row and column sums have the same small prefactor at any smaller fixed rate. At fixed exterior either force estimate bounds the oscillation of the corresponding exponent on the product window by its per-bond bound times $C|I|b_j$, by integrating along a segment in the window. The ratio of the two normalized real densities with and without these terms is between the exponentials of minus and plus this oscillation. Fixed spectator arguments require a compatible gauge comparison of the partial read. A term whose current or stored-colour arguments vary with y must include their derivatives as well.

6.3.0.3. The exterior terms on a matched chart. Fix the retained datum and the coordinate frame of the read set. Write I for the integrated bonds in the box and O for the other integrated bonds of this step. Write E for the newly external fluctuation slots of the same ordinary step, with their fixed values a_E in scaled coordinates. Let Q^e be the complete precision on $I\sqcup O$ before the exterior integrated coordinates are frozen. It includes the ratio correction to the contour gauge quadratic. The conditional mean of the ordinary fluctuation Gaussian gives the additional linear term in the negative log-density

$$\ell^e_E=\tau_k\mathsf{A}^e_{I\sqcup O,E}a_E,\qquad k=j-1.$$

Indeed, for the ordinary Gaussian with precision $\tau_k\mathsf{A}^e_{I\sqcup O,I\sqcup O}$ its mean is $-(\tau_k\mathsf{A}^e_{I\sqcup O,I\sqcup O})^{-1}\ell^e_E$. The gauge compensation has separate integrated and exterior blocks. It changes the precision and leaves this linear term unchanged. Keep the E-slots fixed, including any real near values. At a comparison datum e_0, the centered Gaussian reference on I has covariance $\big((Q^{e_0})^{-1}\big)_{II}$. Its precision is

$$K=Q^{e_0}_{II}-S,\qquad S=Q^{e_0}_{IO}(Q^{e_0}_{OO})^{-1}Q^{e_0}_{OI}.$$

The reference exponent is $-y^{\mathsf T}Ky/2+P_0(y)$. The difference of the gradients of the two negative log-densities gives the terms

$$Sy+Q^e_{IO}y_O+(\ell^e_E)|_I+(Q^e_{II}-Q^{e_0}_{II})y-\nabla(P_e-P_0)(y).$$

All scalar factors depending only on e cancel in this comparison at fixed exterior. This includes the square-completion term of the ordinary move with two external fluctuation arguments: its inverse chain may reach inside the box, but both arguments are fixed. The Schur term S above has two interior arguments and comes from taking the Gaussian marginal of the reference.

The Schur and integrated exterior forces follow from the real precision bounds. Suppose $Q^{e_0}\ge a_-I$, and that Q^{e_0} and Q^e have uniform exponentially weighted row and column sums. Minimizing the quadratic form over the exterior gives $K\ge a_-I$. For a weight e^{φ} with small Lipschitz constant μ, conjugation changes $Q^{e_0}_{OO}$ in operator norm by at most $C\mu$, by its weighted row and column bounds. Its inverse therefore remains bounded in the conjugated norm. Taking $\varphi(b)=\mu d(b,b')$ gives exponential decay of its entries and, at a smaller rate, summable weighted rows and columns. For $b\in I$, the first factor $Q^{e_0}_{IO}(b,\cdot)$ in S already crosses distance $d(b,O)$. Summing the other two factors gives

$$|(Sy)_b|+|(Q^e_{IO}y_O)_b|\le Cb_je^{-cd(b,O)}$$

on the stated windows $|y|_\infty,|y_O|_\infty\le Cb_j$. The old exterior tier bounds give $\sup_E|a_E|\le Cb_j$ on the real support. The weighted row bound for the ordinary cross block therefore gives

$$|(\ell^e_E)_b|\le Cb_je^{-cd(b,E)}.$$

All three bounds hold at the read with the box padding, since both O and E are outside the box. They also hold with the distance to $\mathcal{B}_{\mathrm{ext}}$.

For the change of the interior precision compare the full precision terms before freezing the exterior. Use the local walk expansion with the current, pivot, and relative contour gauge contributions in the same chart.

Fix the outgoing datum V_j and take a collared box W in the common clean top layer of two exteriors. Choose the same collar grid in both settings, with the padding for all the backward blocks and pivots. On W all the layer traces through level j coincide. The only original determining rows with both endpoints in W are top rows, which read the fixed V_j. The joined determining set therefore coincides, and both joined data are formed by the same face lift of those rows. In particular both ambient backgrounds compare to one joined background by eq. 3.cdii. The common comparison box gives one frame for the interior. If a lower read uses the average of either ambient background, apply the local average estimate on its entire double block in that frame. This gives closeness of the two averaged data from the background estimate.

On a union of common standard cells with padding in W, the identical traces give identical indices, row types, outward endpoints, and local carriers of the native walk terms. For the ordinary compressed terms use the part of this union on which both steps have the same integration, pivot, and conditioning masks. A masked term uses this identity only on its full carrier, including the endpoint blocks of its masks.

For the native conditioned covariance, the local resolvent at a cube q uses the sum of the form terms with footprints inside $q^\sharp$, compressed to the kept bonds in $Y' \cap q^*$. Thus one must match the full $q^\sharp$ read and the compression on $Y' \cap q^*$, even for an inverse entry whose endpoints are well inside q^*. On this common support write $R_q^a = (x_a + T_q^a)^{-1}$ for the local resolvent of either datum $a = e, e_0$. Then

$$R_q^e - R_q^{e_0} = -R_q^e\big[(x_e - x_{e_0})I + T_q^e - T_q^{e_0}\big]R_q^{e_0}.$$

Equal data and equal scalar shifts give equality. For nearby data the same identity uses the native inverse bounds on the admitted domain. An extra chain of a labeled reference must have this same form family and compression, or its own comparison.

Write $d_b = d(b, \mathcal{B}_{\mathrm{ext}})$, with the cell and stencil padding included in the common region. Compare the terms in row b whose full carriers stay within distance $d_b/2$ of b. The other terms have charged length at least $cd_b - C$, including the margins of their local read sets. The absolute weighted row bound for each precision gives a cost $Ce^{-c'd_b}$ for this part, with the column variable summed. This distance is bounded by the distance to the actual cut even when the set of exterior integrated bonds has holes.

For the short carriers, compare the terms with common masks in the frame of W. The real pair difference, in local units on a carrier, is bounded by $C\epsilon_0 \sup_{z \text{ on the carrier}} e^{-c_0 d(z, W^c)}$. The real estimates for the potential and its first differences give the field bound, and the full-bond estimates give the current bound. The adjoint rotations preserve the free-bond balls.

The ordinary relative contour square has a direct local comparison. The paths in one block avoid the middle crossing pivots. Thus $G_U h_U = 0$ for those contour rows, and their differential $G_U C_U$ from free coordinates is $G_U \iota$ by eq. 3.cxii. At a common block it has the same path stencils and free mask in both settings. The row coefficients in eq. 3.cclxxiv are analytic in the block links. Their difference is bounded by the field difference on the block, in the common node frame. The weight of the gauge quadratic is one in $Q_\tau = \tau K_A + K_G$. The difference of the two Gram terms is the sum of a bounded row times a row difference and its counterpart with the difference in the first factor. The fixed number of contours per block then gives the short row estimate for this part of the complete precision.

On a common local operator domain with fixed room, Cauchy's estimate on the segment between the two pairs bounds the difference of a short term by its absolute analytic bound times C times the pair difference. Thus the absolute row sum of short differences is at most $C\epsilon_0 e^{-c_0 d_b/2}$. Together

with the long terms this gives

$$\sum_{b'\in I} |(Q^{e}_{II} - Q^{e_0}_{II})(b,b')| \le Ce^{-c'd_b}.$$

Here the row estimate uses the matching for each short summand of the complete precision. For the reference used in the clean mixture, we must file any additional boundary chains under the same local comparison or bound their short contributions separately. Multiplication by $|y|_\infty \le Cb_j$ gives the force bound at depth d_b.

For the interaction force use the full carriers of the evaluated members of P_e. Write $m(Y)$ for their length, with $\operatorname{diam} Y \le C_0(1+m(Y))$ and $\sum_{Y\ni b} e^{-\kappa m(Y)} \le C$. For a carrier reaching $\mathcal{B}_{\mathrm{ext}}$, a derivative bound $v_1 e^{-2\kappa m(Y)}$ on each evaluation gives the needed distance factor by spending half the decay. For a carrier contained in the matched interior, use the matched derivative bound

$$|D_b(P_{e,Y} - P_{0,Y})| \le v_2 e^{-2\kappa m(Y)} \sup_{z\in Y} |U_e(z) - U_{e_0}(z)|$$

in a common gauge. If the last difference is bounded by $\epsilon_0 e^{-c_0 d(z,\mathcal{B}_{\mathrm{ext}})}$ at z, take $c' \le \min(c_0, \kappa/C_0)$ and use $d(b,\mathcal{B}_{\mathrm{ext}}) \le d(z,\mathcal{B}_{\mathrm{ext}}) + C_0(1+m(Y))$. The rooted sum gives $C(v_1+v_2\epsilon_0)e^{-c'd(b,\mathcal{B}_{\mathrm{ext}})}$. The target in eq. 6.xiv follows when $v_1 + v_2\epsilon_0 \le Cb_j$. For the regular old-origin members, the common-domain amplitude estimate in eq. 4.ccvii also suffices, by Cauchy's inequality and the same rooted sum. The force bound must be checked for the other members of the complete exponent.

The readout must be compared in addition to the density. On a common admitted pivot chart eq. 4.ccxxx, the joint Lipschitz bound and finite double-block support of Φ_V bound the difference of the two evaluated link fields by $C\epsilon_0 e^{-c'\mathsf{D}}$ on the read set, when the complete pivot support has distance at least D from $\mathcal{B}_{\mathrm{ext}}$. Apply the background estimate to the coarse field entering the graph, on its whole pivot support. This uses the actual averaging estimate if the comparison was in fine variables, and a fixed margin for the pivot support. Thus a readout with Lipschitz constant L_G costs $\eta_G \le CL_G\epsilon_0 e^{-c'\mathsf{D}}$. For the scalar insertions there is one more small field factor. Put $\Delta = C\epsilon_0 e^{-c'\mathsf{D}}$ for the difference of the two fluctuating link fields at the same free coordinate, after common gauge transport on the full pivot read. Let κ_* bound the curvature of the reference *evaluated* field on the window, including its fluctuation. In the axial gauge of this reference its link logarithms on a box of side ℓ_G are bounded by $C\ell_G\kappa_*$. The short linkwise interpolation to the other field then has logarithms bounded by $C(\ell_G\kappa_* + \Delta)$. The first derivative of the invariant scalar vanishes at the flat field. On the small-field logarithm chart, use the second derivative bound from zero to each point of the path to estimate its first derivative, then integrate along the path. This gives

$$\eta_G \le C\mu_2(\ell_G\kappa_* + \Delta)\Delta.$$

For a Wilson plaquette in SU(2) the same bound follows by subtracting the squared norms of the two plaquette deviations from the identity.

6.3.0.4. Intrinsic terms and the reference Hessian. Work in the canonical free coordinates y, with $g = \widehat{g}_{j-1}$ and a tested radius $b_j \ge 1$. In the intrinsic exponent the action and contour-gauge terms are composed with the actual pivot graph before their quadratic parts are extracted. Their remainders have the form $g^{-2}R_{3,Y}(gy)$, with their signs and fixed coupling weights. Here $R_{3,Y}$ vanishes to order three, and the complete quadratic terms are included in the precision. Any linear term of the full action must be accounted for in the choice of background or in the explicit linear force. At the physical critical background, the tangent of the pivot graph at fixed targets lies in the tangent space of the constraint. The first variation of the Wilson action there vanishes on that tangent space. The contour-gauge action has zero first variation at the axial centre, where its contours are the identity. Extract these Taylor coefficients on the formed carriers before bounding the cubic remainders. Freezing a nonzero exterior argument later still creates the linear terms from the extracted quadratic. The other intrinsic terms are $L_Y(gy) - L_Y(0)$, the local log Haar and pivot coarea factors of eq. 4.ccliv. In particular, the pivot Jacobian is the derivative in the

eliminated coordinates of the actual constraint. The physical cubic and linear coefficients are the ones in eq. 4.cccxviii.

On a fixed unscaled complex radius, let B_Y bound the composed action and the coarea functions on their full carriers. For the homogeneous action these bounds are supplied by eq. 4.cclxxix and its composition with the outer graph in eq. 4.ccxcv. For the local Haar and pivot factors they follow from the joint chart and eq. 4.ccliv. Suppose that gB_* stays within a fixed smaller radius on every complete carrier. Here B_* includes $2b_j$, the real held bound $C_O\widehat{b}_j$ on the other integrated slots, and the full bound on the original exterior slots a_E. Leave a fixed fraction of b_j beyond the real windows for the complex coordinate variations, and use $B_* \le Cb_j$ with the fixed native window constants. Taylor's formula on this joint chart gives the respective bounds $CB_Y gb_j^3$ and $CB_Y gb_j$. For the coarea Hessian subtract its linear Taylor term before estimating derivatives. The subtracted function is bounded by $CB_Y g^2 b_j^2$ there. This subtraction is only for the derivative estimate. The linear term remains in the density. Two coordinate Cauchy estimates at a real point in the tested box give the entry bound

$$|D_b D_{b'} P_{\mathrm{int},Y}| \le CB_Y(gb_j + g^2)\,\mathbf{1}_{\{b,b'\in Y^+\}},$$

where Y^+ is the support of the term in the actual chart. The radii in both derivatives are fixed fractions of b_j. For a repeated coordinate the factorial is 2!. The rooted exponential bound for B_Y absorbs the polynomial number of bonds in Y^+ and the factor $e^{\mu \operatorname{diam} Y^+}$ for a smaller fixed μ. The weighted row and column sums therefore give

$$\|T_\varphi P''_{\mathrm{int}} T_\varphi^{-1}\|_{2\to 2} \le C(gb_j + g^2), \qquad |\varphi(b) - \varphi(b')| \le \mu d(b,b').$$

This bound holds on the proved homogeneous chart, and on a matched reference box with the same rooted intrinsic bounds.

The Haar and coarea part of this calculation applies directly on the labeled clean double blocks at nonzero held coordinates. Each such term is the actual one-step Haar factor or the inverse derivative in the private middle pivot. In a clean complete block the map, the target and the nonpivot and pivot masks are those of the joint local chart. The regularity of the whole double block and the held radii just specified supply its analytic bounds. The number and length of these carriers through one bond are bounded. Two derivatives give Cg^2 in the weighted rows and columns of this part, with the linear coarea term still in the density.

The Wilson action on an ordinary determining set has a direct weighted Hessian estimate on the same chart. Use its exact response package at the physical critical reference. Write F for the bare action in the relative potential, K_0 for the linearized rows, H for their right lift and $\widetilde{\mathsf{G}}$ for the constrained Green operator. The nonlinear row linearization $A = T(A')$ of [B85e] makes the rows linear: $\mathcal{Q}(T(A')) = K_0 A'$. With L_0 the constrained action Hessian, put

$$V_{\mathrm{nl}}(A') = F(T(A')) - F(0) - (F'(0), A') - \tfrac{1}{2}(A', \mathsf{L}_0 A').$$

All pairings and adjoints in this calculation use the native potential, source and row weights. The second variation estimate gives, on a fixed absolute ball,

$$\|V''_{\mathrm{nl}}(A')w\|^\sharp_{\theta,y_0} \le C\|A'\|_\infty \|w\|_{\theta,y_0}.$$

Here θ is a spatial weight rate and can be a fixed positive number below the native response rate. This estimate includes the second derivative of the nonlinear rows contracted with the physical multiplier. The determining-set response package supplies it with constants uniform in the depth and the volume.

For a reduced row datum b, its critical potential is $T(A'(b))$, where

$$A'(b) = \mathsf{H}b + \mathcal{A}_0(b), \qquad K_0\mathcal{A}_0 = 0,$$
$$\mathcal{A}_0 + \widetilde{\mathsf{G}} V'_{\mathrm{nl}}(A') = 0.$$

The gauge condition is $\mathsf{R}D^{\mathsf{T}}\mathcal{A}_0 = 0$. Shrink the fixed absolute datum radius once so that $\|\widetilde{\mathsf{G}}V_{\mathrm{nl}}''(A')\| \le 1/2$ in these weighted spaces. This is possible since $\|A'(b)\|_\infty \le C\|b\|_\infty$. Differentiating the stationary equation gives

$$DA'(b) = (I + \widetilde{\mathsf{G}}V_{\mathrm{nl}}''(A'))^{-1}\mathsf{H}.$$

Set $\mathcal{E}(b) = F(T(A'(b)))$ and $\mathsf{Z} = \mathsf{H}^\sharp\mathsf{L}_0\mathsf{H}$. The identities $\mathsf{L}_0\mathsf{H} = K_0^\sharp\mathsf{Z}$ and $\mathsf{H}^\sharp\mathsf{L}_0 = \mathsf{Z}K_0$ remove the cross terms containing $\mathcal{A}_0$. Stationarity removes its first variations. Thus

$$\mathcal{E}'(b) = \omega_0 + \mathsf{Z}b + \mathsf{H}^\sharp V_{\mathrm{nl}}'(A'),$$
$$\mathcal{E}''(b) - \mathsf{Z} = \mathsf{H}^\sharp V_{\mathrm{nl}}''(A')(I + \widetilde{\mathsf{G}}V_{\mathrm{nl}}''(A'))^{-1}\mathsf{H}.$$

The inverse has norm at most two. The weighted response bounds for H, its adjoint and $\widetilde{\mathsf{G}}$ therefore bound the last operator by $C\|b\|_\infty$. On the highest active rows the scale weights are one. Apply the estimate to a single row and Lie algebra component, with the weight centred at its assigned top site. The output entries then decay exponentially in the distance between the two assigned sites. There are boundedly many active row components per highest-level site. The native exponential sum, at any smaller fixed rate $\mu < \theta$, therefore gives absolute weighted row sums. Symmetry gives the column sums, and the weighted Schur estimate gives the required 2-norm bound.

For this application the varying original datum is confined to the highest level of the ordinary determining set. In the reduced datum, an outward node gauge reads only the interior datum one level below. Those lower increments are zero in the relative chart. There are no inward tying rows. Consequently $\mathsf{b}(B) = B$ on the full varying top stock, as in eq. 3.dxlix; the preceding estimate is already in its physical row coordinates.

Now compose with the actual one-step pivot graph $B = \Phi(v)$, written relative to that reference, so $\Phi(0) = 0$. If $\mathsf{K}(v)$ denotes the Hessian of $\mathcal{E} \circ \Phi$, the full chain rule is

$$\mathsf{K}(v) = D\Phi(v)^\sharp\mathcal{E}''(\Phi(v))D\Phi(v) + \mathcal{E}'(\Phi(v))\, D^2\Phi(v),$$

where the last term denotes contraction in the row coordinate. The joint pivot chart bounds its first three derivatives on the complete double blocks. Their supports have bounded diameter and overlap. The physical multiplier satisfies $\|\omega_0\|_\infty \le C\alpha$. To compare these Hessians, write $J(v) = D\Phi(v)$, $H_2(v) = D^2\Phi(v)$ and $e(v) = \mathcal{E}'(\Phi(v)) - \omega_0$. The difference is the sum of

$$J(v)^\sharp[\mathcal{E}''(\Phi(v)) - \mathsf{Z}]J(v),$$
$$J(v)^\sharp\mathsf{Z}J(v) - J(0)^\sharp\mathsf{Z}J(0),$$
$$e(v)H_2(v), \qquad \omega_0[H_2(v) - H_2(0)].$$

The first term uses the reduced-action estimate. The second and fourth use the local derivatives of the pivot graph. For the third, $\|e(v)\|_\infty \le C\|v\|_\infty$ follows by integrating its bounded Hessian along the segment. Each term therefore gives

$$\|T_\varphi[\mathsf{K}(v) - \mathsf{K}(0)]T_\varphi^{-1}\|_{2\to 2} \le C\|v\|_\infty.$$

In particular, the change in the contraction $\mathcal{E}'(0)D^2\Phi$ is included here. Its value at zero is part of the complete precision eq. 3.dlii; it cannot be discarded before subtracting that precision. For $v = g(y, y_O, a_E)$, the Hessian of the g^{-2}-scaled action remainder is exactly $\mathsf{K}(v) - \mathsf{K}(0)$. This proves its weighted bound Cgb_j throughout the full held window, including nonzero original exterior coordinates. The local contour action has the same bound from its complete block chart after its actual quadratic is subtracted. Together with the Haar and coarea estimate above this proves the intrinsic Hessian estimate on the ordinary labeled clean stock. If lower-level data are also released, keep their reduced-datum derivatives and physical dual weights; the highest-level identification just used does not apply to those directions.

There is a separate parity condition in the value estimate. When all exterior fluctuation arguments of the analytic intrinsic reference are zero, its odd part is bounded by $Cn_j g b_j^3$. The even parts of the action remainder and the coarea increment start at orders four and two, respectively, so their sum is bounded by $Cn_j g^2 b_j^4$. A newly stored a_E held at a nonzero value can enter a cubic with two interior slots, which is even in the interior. With these stored values fixed use the full intrinsic oscillation bound $Cn_j g b_j^3$. On the joint analytic chart the segment (y, ty_O, a_E), $0 \le t \le 1$, still gives the force comparison for setting the other integrated fluctuations to zero. At fixed background and fixed quadratic extraction, the mixed Hessian bound just proved gives

$$|D_b\{P_{\mathrm{int}}(y, y_O) - P_{\mathrm{int}}(y, 0)\}| \le C(gb_j + g^2)b_j e^{-\mu d(b,O)}.$$

Indeed, integrate the mixed derivatives along that segment and sum the weighted row over O. The physical exterior variables remain in the actual law.

To reach the parity reference, this comparison must also set the exterior slots of the current-step analytic intrinsic terms to zero. Use the raw fluctuation coordinates before the strip substitution, at a fixed pair and with the physical tests and Jacobian held in the original density. The larger analytic carrier boxes for these intrinsic terms allow a segment in their current-step stored arguments. On the common admitted box, integrating the mixed Hessian along $(y, 0, ta_E)$ bounds the extra change of interior force by $C(gb_j + g^2)b_j e^{-\mu d(b,E)}$ at bond b. This is a comparison of analytic evaluations. The strip selectors and older stored fields are still present in the physical law and require their own bounds. The frozen quadratic's linear force is accounted for separately.

The regular old-origin contribution is controlled by eq. 4.ccii and its weighted version eq. 4.cciii once the matched reference box reads the same subtracted family on its larger analytic domain. The vacuum-only remainder uses the separate age sum eq. 4.ccv on its admitted class. When the profiles have been redrawn with compensation after healing, the coupling lag vanishes in the clean box. A term whose arguments, including its background read, are fixed by the exterior contributes a scalar to the box. It cancels from the normalized density at that exterior and remains in the exterior mixing weight. A boundary term that still depends on y requires its own weighted Hessian bound in the reference. An estimate for the difference of two exterior forces alone does not supply this. Thus the proved intrinsic bound and the admitted old and vacuum-only bounds can be spent against the Schur gap in eq. 4.cciv. The remaining reference terms must fit in the same gap before applying eq. 6.xv.

6.3.0.5. Freezing only the exterior fluctuations. For a fixed full chart of the small-field step, the pivot graph and the extracted precision are determined before freezing the exterior free coordinates. Keep the retained datum, background, stored spectators and coordinate sets of that chart fixed. In the auxiliary reference put the other integrated fluctuations equal to zero in the smooth exponent and keep the same full precision for forming the centered Gaussian marginal on I. The newly external values a_E remain fixed in the smooth interaction. The force from their cross quadratic is ℓ_E^e in the physical exponent, and dropping it for the centered reference costs the estimate above. In this comparison the interior precision difference vanishes. The Schur correction and the force $Q_{IO}y_O$ are still present. The field readout on its declared read set is also unchanged: each of its pivot values reads only the free bonds of its defining double block, and those bonds are included in the read set. This is the graph in eq. 4.ccxxx. The joint step box for the intrinsic action and coarea factors contains the segment (y, ty_O) with a_E held fixed. The auxiliary zero is an evaluation on that analytic chart. The exterior tests and the recorded exterior values in the physical law are kept.

For any smooth exponent member P admitting the common segment (y, ty_O), its change of interior force is exactly

$$D_b\{P(y, y_O) - P(y, 0)\} = \int_0^1 \sum_{b' \in O} D_b D_{b'} P(y, ty_O)(y_O)_{b'}\, dt.$$

A weighted Hessian row bound A on the full segment therefore gives $CAb_j e^{-\mu d(b,O)}$ on that exterior window. This applies to the intrinsic estimate of §6.3.0.4. It applies to the old terms when their weighted estimate is available on the joint free-coordinate domain, including the exterior slots of these derivatives. A bound on interior derivatives at a fixed exterior does not supply those slots. When the exterior changes the background, stored fields or local masks, freezing its fluctuations first gives an exterior-dependent reference. The direct flat comparison below uses the common rotated read window and the full row estimates to obtain one reference density for the mixture.

6.3.0.6. Charging a boundary tail to the reference Hessian. Let W_I be the union of the current cubes covering the entire free-coordinate box I, with N_W cubes. Suppose the surviving boundary members satisfy the majorants of Prop. 6.9 on one common complex ball of free radius $R_y \geq 2b_j$, after their actual substitutions. Write B_∂ for their sum. Terms independent of the interior coordinates can be set aside as scalars at fixed exterior. Then eq. 6.xlii gives

$$\sup_{\|y\|_\infty < R_y} |B_\partial(y) - B_\partial(0)| \leq CB_* N_W e^{-cs}.$$

Coordinate Cauchy on the real window gives bounds $CB_* N_W R_y^{-1} e^{-cs}$ for each first derivative and $CB_* N_W R_y^{-2} e^{-cs}$ for each Hessian entry. If m_I counts the scalar coordinates and D_I is the diameter of I in the metric of the weights, the weighted Schur bound is

$$\|T_\varphi B''_\partial T_\varphi^{-1}\|_{2\to 2} \leq CB_* N_W m_I R_y^{-2} e^{-cs+\mu D_I}, \qquad \operatorname{Lip}\varphi \leq \mu.$$

This uses the diameter of the box, so it does not require a diameter estimate for an individual carrier in terms of its length after exempt hulls have been removed.

For example, the additional margin $s \geq s_* + (\mu/c)D_I + C_1 \log x$ makes this contribution small when the coordinate counts and B_* grow polylogarithmically. In this inequality the collar and box distances use the units of their respective estimates, with the fixed conversion included in the constants. The enlarged collar still has polylogarithmic size when the original box and the trace padding do. For two exteriors satisfying the same bound, the first derivatives cost at most $CB_* N_W R_y^{-1} e^{-cs}$ in their difference. As a uniform force error it adds at most $Cn_j b_j B_* N_W R_y^{-1} e^{-cs}$ to the logarithmic marginal error. This follows by using the summable weights in eq. 6.xv. The bound requires the larger complex ball for these evaluated boundary members and the stated clean collar around W_I. A real oscillation bound alone gives no Hessian estimate.

For the current ordinary-step boundary and kept potentials one can use the native analytic domain directly. Fix the actual exterior at an admitted support point, with its adapted pair and stored arguments. In the original scaled free coordinates on the clean integrated bonds, the complex step box has radius $3b_j$. Vary y on the ball of radius $2b_j$ and keep every other argument fixed. The support has room in those other coordinates, including the cut strip evaluation. The pinned substitution preserves the stored arguments. Hence the boundary and kept potentials on this ball use their original full carriers and the inherited amplitudes of the ordinary step.

Apply two coordinate Cauchy estimates on each carrier Y. A derivative in the bonds b, b' can occur only if the complete carrier reads both of them. The two root cubes, with the pivot padding included, are nonexempt in the clean box. A hull tree must meet both and spends their separation in ordinary edge length, including the part of the path that passes through any exempt hull. Splitting the rate between this distance and the rooted sum eq. 6.xxxix gives

$$|D_b D_{b'} P_\partial(y)| \leq C_a b_j^{-2} \vartheta_k \big[\mathsf{B}(\log\theta_k)^{-a} + \mathsf{B}_\Phi e^{-L\check{R}_j}\big]\, e^{-\mu_0 d(b,b')}, \qquad k = j-1.$$

Here P_∂ sums these current-step substituted potentials. Their inverse-logarithmic bound uses the native crossing length. The weighted Schur sum is uniform in the number and diameter of the free bonds for $\operatorname{Lip}\varphi < \mu_0$ with a fixed rate reserve. This estimate is for interior derivatives at fixed exterior and can be used in the reference Hessian.

From the exterior force to the normalized marginal. Use the same literal product of free-bond balls $D = \prod_{b\in I} D_b$ for the two real densities. Write

$$H_a(y) = \tfrac{1}{2} y^{\mathsf{T}} A_a y + \ell_a^{\mathsf{T}} y - P_a(y), \qquad d\nu_a = Z_a^{-1} 1_D e^{-H_a}\, dy, \qquad a = 0, 1.$$

Here $a = 0$ denotes the reference with the Gaussian marginal precision K, and $a = 1$ the actual density at fixed exterior. The two real potentials are C^2 in the interior. For $T_\varphi y(b) = e^{\varphi(b)} y(b)$, suppose

$$\langle v, T_\varphi A_0 T_\varphi^{-1} v\rangle \geq \lambda_A |v|^2, \qquad \|T_\varphi P_0'' T_\varphi^{-1}\| \leq \epsilon_0 < \lambda_A.$$

Use $\lambda = \lambda_A - \epsilon_0$, and let δ_b bound the component at b of $\nabla(H_1 - H_0)$. For a real function G with $|G(y) - G(y')| \leq L_G |T_\varphi(y - y')|$, the common-momentum comparison gives

$$|\nu_1(G) - \nu_0(G)| \leq \frac{L_G}{\lambda} \left(\sum_{b\in I} e^{2\varphi(b)} \delta_b^2 \right)^{1/2}. \tag{6.xv}$$

Here is the argument including the hard boundary. First shrink each ball to radius r_b, and choose two intermediate radii between r_b and the original radius. Smooth the metric projection onto the larger intermediate ball by an even mollifier. With support sufficiently small, the resulting map r, taken block by block, is the identity on the smaller intermediate balls, takes values in the original domain and has $\|Dr\| \leq 1$. Replace P_a by $P_a \circ r$ and add the same wall $\epsilon^{-1} \sum_b w(|y_b|^2 - r_b^2)$ to both energies. Take w smooth, convex and nondecreasing, zero on the negative half-line, with positive derivative on the positive half-line and affine for large arguments. The wall is smooth and convex, with bounded Hessian for fixed ϵ. Its Hessian has a positive lower bound off the smaller intermediate ball. In the Hessian of $P_0 \circ r$, the term $Dr^{\mathsf{T}} P_0''(r) Dr$ has weighted norm at most ϵ_0 because Dr commutes with the block weight. The remaining term is block diagonal, vanishes on the smaller intermediate balls in its respective blocks and is dominated by the wall Hessian for small ϵ. Thus the weighted coercivity remains λ on the whole space. The wall cancels from the difference force $f = \nabla(H_1^\epsilon - H_0^\epsilon)$.

For these smooth energies the Hamiltonian flow $\dot q = p,\ \dot p = -\nabla H_a^\epsilon(q)$ preserves the product of its Gibbs law and the standard momentum Gaussian. Energy conservation and zero phase-space divergence give this identity. Start the two flows with the same Gaussian momentum. For $v = T_\varphi(q^1 - q^0)$ their difference satisfies $\ddot v = -M(s) v - T_\varphi f(q^1(s))$ and $\dot v(0) = 0$. The matrix $M(s)$ is the conjugated average of the reference Hessian on the segment between the two positions. It satisfies $\langle z, M(s) z\rangle \geq \lambda |z|^2$, and its norm is at most some Λ at fixed ϵ. The integral equation, for small t, gives

$$|v(t)| \leq (1 - \lambda t^2/2 + C\Lambda^2 t^4)|v(0)| + (1 + C\Lambda t^2) \int_0^t (t - s) |T_\varphi f(q^1(s))|\, ds.$$

Indeed, the average of $M(s)$ with weights $2(t - s)/t^2$ has the same lower bound and norm. Substituting $v(0)$ under the Hessian integral gives the displayed contraction by squaring its norm. One further use of the integral equation bounds the error from $v(s) - v(0)$ by the fourth-order term and the indicated increase of the force bound.

Refresh the common momentum every t units of time. The marginal position laws stay invariant, so the mean of the force integral equals $t^2 \nu_1^\epsilon(|T_\varphi f|)/2$. Iterate the displayed contraction and use the resulting bound on the mean weighted separation for a Lipschitz extension of G. Letting $t \downarrow 0$ at fixed ϵ bounds the observable difference by $L_G \nu_1^\epsilon(|T_\varphi f|)/\lambda$. As $\epsilon \downarrow 0$, the smooth measures converge to the Gibbs laws on the shrunken balls. The walls have quadratic growth outside a bounded set, which gives an integrable majorant for both densities and their first moments for all sufficiently small ϵ. The force has at most linear growth and equals the original force on the shrunken balls. Dominated convergence therefore applies also to its mean. Finally let the shrunken radii increase to the original radii. The bounded reference Hessian and difference force give bounded gradients

on the original bounded convex window, so the last limit is also dominated. This proves eq. 6.xv. Only the reference Hessian was used for coercivity.

Now assume the same coercivity bounds for all weights with Lipschitz constant at most μ. Retain the read coordinates $u = y|_S$ and integrate over $R = I \setminus S$. Write m_a for their marginal densities and ν_a^u for the conditional laws on R. The product window makes the integration domain independent of u, so for $s \in S$

$$\partial_s \log(m_1/m_0) = -\nu_1^u(\partial_s(H_1 - H_0)) - (\nu_1^u - \nu_0^u)(\partial_s H_0).$$

The first term costs δ_s. In the second, apply eq. 6.xv to each unit colour direction of $\partial_s H_0$, with $\varphi(b) = -\mu d(b,s)$ on R. The weighted coercivity survives compression. The Lipschitz constant is at most

$$L_s = \sup_{y \in D} \left(\sum_{b \in R} e^{2\mu d(b,s)} \|H_0''(b,s)\|^2 \right)^{1/2}.$$

Thus the derivative norm is at most $\delta_s + \lambda^{-1} L_s (\sum_{b \in R} e^{-2\mu d(b,s)} \delta_b^2)^{1/2}$. Integrating along segments in the product of read balls bounds the oscillation of $\log(m_1/m_0)$ by the sum of these bounds times the corresponding ball diameters. Since both marginals have mass one, this oscillation bound Ξ_j gives $e^{-\Xi_j} \le m_1/m_0 \le e^{\Xi_j}$.

If the forces obey eq. 6.xiv and the weighted Hessian column bound makes L_s uniform, the triangle inequality $d(b,s) + d(b, \mathcal{B}_{\mathrm{ext}}) \ge d(s, \mathcal{B}_{\mathrm{ext}}) \ge \mathsf{D}$ leaves a summable exponential after extracting a smaller factor $e^{-c'\mathsf{D}}$. The derivative bound at a read coordinate is then $C b_j e^{-c'\mathsf{D}}$. Multiplying by the diameters and the read-coordinate count gives $\Xi_j \le C n_j b_j^2 e^{-c'\mathsf{D}}$. The normalized density bound persists under the positive exterior mixture when each exterior is compared with the same reference density in the common read coordinates. The readout discrepancy η_G remains a separate term.

6.3.0.7. The readout as the retained datum varies. At a fixed output of the step, the exterior mixture keeps the same new field V_j. The following comparison is uniform as that field varies between outputs. Write $\mathsf{r} = L^s \mathsf{d}$ for the separation in witness-level units and $p_0 = p_0(g)$. Put V_j in the axial gauge of the nonwrapping comparison box and extend that gauge constantly on each block to the previous level. Denote the rotated free coordinates by $\widetilde{y}$. If the box has side l in its own units and plaquette bound κ_j, its logarithm satisfies $\|A_j\|_\infty \le C l \kappa_j$. This follows by writing the axial link as a product of at most Cl conjugated plaquettes. For the intermediate background $V_* = \mathcal{M}^{j-1} U_j(V_j)$ in its block axial representative, the contour variables are one, and the actual one-step average is V_j. Apply Lem. 3.8 against the trivial field on the full read double blocks. The intermediate plaquettes are bounded by $C\kappa_j$ by the local averaging estimate. Consequently its logarithm A_* in the same frame obeys $\|A_*\|_\infty \le C(l+1)\kappa_j$.

Let $\mathsf{C}^0 = D\Phi_1(0)$ be the flat pivot inclusion on those blocks. On the joint chart eq. 4.ccxxx, $D\Phi_{V_*}(0) - \mathsf{C}^0$ has norm at most $C\|A_*\|_\infty$ in that frame. The chart remainder is quadratic in the free variables. Multiplying the physical fluctuation by V_* and using the local logarithm of the product gives

$$\|A_{j-1} - A_* - \widehat{g}_{j-1} \mathsf{C}^0 \widetilde{y}\|_\infty \le C(\widehat{g}_{j-1} b_j \|A_*\|_\infty + \widehat{g}_{j-1}^2 b_j^2).$$

Here A_{j-1} is the logarithm of the actual fluctuating field in the chosen frame. The first error includes the commutator from multiplying the two exponentials.

For a bare plaquette p, denote the localized background plaquette minus one at level $j-1$ by $P_p^{(j-1)}(A)$. On its admitted fixed-radius datum ball this readout is bounded by $C\eta_{j-1}^2$. Its derivative at zero is $i\mathcal{H}_{j-1}^{\square}$, the linear curvature response of that same localized background. Cauchy's estimate

on a smaller ball and the preceding inequality give

$$\begin{aligned}\left|P_p^{(j-1)}(A_{j-1}) - P_p^{(j-1)}(A_*) - i\widehat{g}_{j-1}(\mathcal{H}_{j-1}^{\square}\mathsf{C}^0\widetilde{y})(p)\right| \\ \leq C\eta_{j-1}^2(\widehat{g}_{j-1}b_j\|A_*\|_\infty + \widehat{g}_{j-1}^2 b_j^2).\end{aligned}$$

6.3.0.8. The two localized readouts of the retained field. The actual retained datum of step j includes V_j and the lower rows that remain stored. On a fixed output, V_j is common to the exterior mixture. The regular critical field U at an exterior in that mixture still depends on the exterior. In the interior the literal shift in the fluctuation integral uses $V_* = \mathcal{M}^{j-1}U$ in block axial gauge, so $\mathcal{M}(V_*) = V_j$ there. On the remaining level-$(j-1)$ rows keep their stored values, and keep the older rows. This is the datum induced on the truncated sequence by U. Indeed the retained rows already prescribe their averages, and the newly exposed rows read the average V_*. Each equality uses the full double blocks. A variation at fixed induced datum extends by U through the frozen collar. The finer constraints imply the old ones by induction in the averaging, and the fixed inner bonds of crossing plaquettes make their contribution constant. Thus eq. 3.cccxciii identifies the critical field of the truncated problem with U up to residual gauge. This argument keeps the same cut rows and averaging scheme as the step.

For $q = j-1, j$, let $D_{q,p}$ be the domain of the localized minimizer used to read p. At $q = k_0$ this is the prescribed neighborhood of the test cube of p, including its four outer layers. At larger q take the box around the two supports with the prescribed order D margin in level-q units, rounded out on the collar grid. Include all top blocks of the plaquette read and all the double blocks of the retained rows needed by the pivots. The latter cost $O(L)$ extra spacings on the level-$(j-1)$ free-bond grid. The lower box and these double blocks are in the read set of step j, with the extra D margin to its integration-box edge. The upper box reads V_j, which is already fixed in this conditional integral.

Use the clean neighborhood with enough room for these fixed enlargements and their collars. In level-k_1 units the distance from the pair to the farthest of these enlarged boxes is at most $C(\mathsf{r}+\check{R}_{k_0})$: the side at q is at most $C(\mathsf{r}L^{-(q-k_0)} + \mathsf{D} + \check{R}_{k_0})$ in level-q units, and $q \leq k_1$. The same bound allows a further margin of this size in level-k_1 units by increasing the fixed neighborhood constant. The clean-history union bound counts cells meeting the enlarged neighborhood in the same way. On this event there are no standing regions or operations on these boxes and their padded collars, at the two levels of the step. The padded collars lie in the ambient top layer, both for the full problem and for the problem truncated at level $j-1$.

On the padded boxes, the passed coarse tests bound the plaquettes of the stored field V_j by Cgp_0. For V_* use the corresponding bound on the full backward cells of the ambient background and the one-step averaging estimate. This is a bound on the averaged critical field at zero fluctuation. In obtaining the local profile of the ambient background, the fixed ceilings on data outside the clean neighborhood enter with the exponential weight from their distance to the padded boxes. The intervening homogeneous layers have thickness at least $c(\mathsf{r} + \check{R}_{k_0})$ in the native spacings. Their contribution is bounded by $Ce^{-c(\mathsf{r}+\check{R}_{k_0})}$, which is smaller than gp_0 in the short-distance range.

For each $q = j-1, j$, compare the local critical field with the restriction of the same ambient field U. The common rows are those of V_* for the lower comparison and of V_j for the upper comparison. Each localization uses the face lift of its own prescribed datum. The alignment in eq. 3.cdi is performed on each collar using the Cgp_0 local profile. Thus the collar source is of that size. In eq. 3.cdii the source size enters linearly, while the fixed regularity ceilings admit the response path. At a plaquette p the result in the response gauges is bounded by $C\eta_q^2 gp_0 e^{-cd_{q,p}}$, where $d_{q,p}$ is its distance to the localization boundary in level-q units, with the fixed read stencils included in the margin. Write $\Delta_j = \min_{q,p} d_{q,p}$. The native test-cube and box margins give

$$\Delta_j \geq c_0 \min\{\mathsf{D}, \check{R}_{k_0}\} - C_0.$$

For a matrix comparison, express each plaquette defect in the frame at its level-q ancestor. This is conjugation by the product of the averaged contours from the ancestor to its base point. Use the coarse axial frame of the common box as the input frame, extended in the block axial representative for V_*. On the complete order-q double blocks of the ancestor read, write the collar comparison in its aligned real frame as $U' = \exp(i\eta_q A)U$. The potential bound on these blocks gives $a := \|A\|_\infty \le Cgp_0 e^{-c(d_{q,p}-C)}$. The fixed enlargement in the exponent is the size of the full blocks in level-q units. The straightened logarithm and its node factor at level $n \le q$ obey

$$\|a_n^{\mathrm{str}}\|_\infty + \|v_n^{\pm 1} - 1\|_\infty \le CL^n \eta_q a.$$

Apply the one-step contour estimate on the complete block of each contour from the next ancestor to the current one. Covariance with v_n gives a change at most $CL^n\eta_q a$. The product of real unitary transports changes by at most the sum of the changes, which is bounded by Ca. In the collar comparison the gauge at the top is v_q^{-1}, and this differs from one by at most Ca as well. The other gauge factors cancel between successive contours.

For a plaquette defect P, changing the transport to the top therefore changes its conjugation by at most $C|P|a$. The curvature bound gives $|P| \le C\eta_q^2 gp_0$, and the response bound already controls the change of P itself. Together they give $C\eta_q^2 gp_0 e^{-cd_{q,p}}$ for the change in the ancestor-frame defect. On U the two ancestor-frame reads agree. The additional contour from the level-j ancestor to the level-$(j-1)$ ancestor is the identity in the block axial representative used to form V_*.

We use this comparison for the plaquettes in their localized Landau representatives. On each actual local datum ball in the common input frame, its logarithmic datum, including the face lift, is bounded by $Cl\kappa_j$. The normalized potential on the ancestor block has this bound by the weighted response from the flat datum. The same contour length sum makes its transport differ from the identity by $Cl\kappa_j$. Thus each conjugation changes a defect of size $C\eta_q^2\kappa_j$ by at most $C\eta_q^2 l\kappa_j^2$. This argument uses only paths in each localized tower, even when the initial test cubes are disjoint. The common coarse axial gauge has one residual constant rotation, which acts on both plaquettes and leaves the flat Gaussian law invariant. These errors, together with the two collar errors, bound the difference of the lower and upper readouts at zero fluctuation by

$$C\eta_{j-1}^2\big[gp_0 e^{-c\Delta_j} + (l\kappa_j)^2\big].$$

Thus, with $l \le C\mathsf{r}$, $\kappa_j \le Cgp_0$ and $b_j \le Cp_0$, the remainder after retaining the level-j background and the flat linear fluctuation is bounded by

$$C\eta_{j-1}^2\big[(\mathsf{r}gp_0)^2 + gp_0 e^{-c\Delta_j}\big].$$

The term involving Δ_j uses both native margins. For $\mathsf{D} = A\log x$, with A fixed, the initial test margin dominates D when x is sufficiently large.

6.3.0.9. The common part of the two operator recipes. For a fixed step use the same tiling, masks and guard prescription in the two boxes. When comparing a row b, include in a carrier every row of a local coefficient, each endpoint double block of its pivot, its full current stencil, the contour reads, and any held guard used in forming its local Gaussian coefficient. A carrier contained in the common interior uses the same functional recipe on a common admitted pair domain. This follows term by term: its masks are those of the empty standing trace, its pivot is the same local analytic solution, and its walks and guard parameters are specified by the same kit. For the Gaussian root this is the fixed-root restriction of eq. 4.ccxciv. In particular, one must preserve the held guard before restricting a Gaussian functional. Independence of the summed logarithm from auxiliary guard choices is an identity after summation, which does not identify terms filed with different guards.

With d_b the distance to the edge of the common interior, take carriers lying within $d_b/2$ of b including their guards. They are therefore indexed in both recipes. Any term left out of this matching either reaches that edge or has a read or a guard reaching it. Such a term has charged

length at least $cd_b - C$ for the intrinsic and precision families; the fixed guard widths only change C. The absolute rooted majorants give a row sum $Ce^{-c'd_b}$ for the unmatched terms. On each matched carrier use the actual difference of its two potential and current reads before taking Cauchy bounds. For comparison at the same retained datum this difference is the exponential collar error. For comparison with the flat datum it includes the small background displacement $C\mathsf{r}gp_0$. This yields the short-carrier estimates in the real precision and intrinsic force comparisons. Regular old activities also require their own retained-datum radii and origin estimates, and standing old boundary terms use eq. 6.xlii.

6.3.0.10. The Gaussian comparison in the rotated coordinates. The rotation of the actual free variables uses the block-constant extension of the retained axial gauge. In this frame let C_{ax} be the old block-axial centre field. Let u be the Landau gauge of the comparison, restricted to the old sites and with the same residual normalization at the tops. By covariance of the average the old field in the Landau frame is C^u_{ax}. If the Landau potential and its first difference have size ϵ in the upper units, the one-level average estimate bounds each of its old link defects by $CL^{-1}\epsilon$. For a site x and its block centre y, the symmetric mean of the complete block contours satisfies

$$T_{y,x}(C^u_{\mathrm{ax}}) = u(y)u(x)^{-1}.$$

This uses eq. 4.ccxvi and the axial condition $T_{y,x}(C_{\mathrm{ax}}) = 1$. Each path has at most $d(L-1)$ links. Telescoping the path products and applying the local mean estimate bounds their symmetric mean's distance from the identity by $C\epsilon$. At the centre $u(y)$ is the inherited node factor of the localized response, or its inverse according to the direction of the transfer. Its normalized node estimate gives $|u(y) - 1| \le C\epsilon$. Together these estimates give the same bound on $|u(x) - 1|$ at every free site of the complete block. Here $\epsilon \le C\mathsf{r}gp_0$ on the admitted localized core.

On the compression to the admitted blocks, or on short carriers with all their endpoint blocks there, the native precision and covariance transform by the bondwise adjoint matrix R_u at the initial sites. For either matrix A, a spatial weight T_φ commutes with R_u, and

$$\|T_\varphi(R_u^{-1}AR_u - A)T_\varphi^{-1}\| \le 2\|R_u - 1\|\,\|T_\varphi A T_\varphi^{-1}\|.$$

The same estimate holds in the weighted row and column sums, by applying the bound blockwise to the two endpoint rotations. The return to the actual read frame therefore costs $C\epsilon$. It uses the matched masks and full block contours. The other carriers retain the separate exterior decay. In the changed gauge use the relative contour eq. 3.cclxxiv at the transformed centre. It is conjugated at the block node, so its quadratic weight is transported with the same free-bond mask.

On short carriers inside the box, the rotated precision and the flat one use the same local recipe and the same ratio $\tau_{j-1} = x_{j-1}/\theta_{j-1}$ in the Wilson quadratic. The contour-gauge weight retains its original normalization. At the flat pair the actual and straight linearized averages give the same constrained exact curvature space. In that space the action metric at this level is S^+_{j-1}, obtained from the energy of the flat minimizer. The soft contour weight integrates to a constant on each curvature fibre, since its contour values translate along the block gauge orbit. Multiplying the curvature metric by τ_{j-1} gives the covariance factor τ^{-1}_{j-1}. This is the argument of eq. 6.clii with the multiscale action metric. In physical units $\widehat{g}^2_{j-1}\tau^{-1}_{j-1} = x^{-1}_{j-1}$.

6.3.0.11. Matching the precision at a fixed exterior datum. Use the free stock of the step before the strip substitution, with the middle pivot of every eliminated coarse constraint removed. The interior integration stock includes the free bonds with at least one endpoint in the small-field region. In the native prepared chart, the operator at the physical current is $C_V^{\mathsf{T}}\Delta^{(j-1)}C_V$ on that full stock [B87, (2.3)–(2.11)]. The identification uses the perfect action on the actual lower determining set. Its Hessian is the first quadratic term. The change of variables in the plain outer constraint contracts its second derivative with the current of that perfect action. The relative contour square supplies the remaining term.

The same algebra identifies the symmetric-contour form on an admitted compression of a physical graded branch. On the active top datum, keep the finer arguments of the outward node factor fixed. The relative datum then has derivative equal to the active injection and second derivative zero on those directions, as in eq. 3.dxlix. Consequently the reduced action Hessian is the compression of $(QG_1Q^\sharp)^{-1}-a$ by eq. 3.dxlv. At the physical current $j_U = Q^\sharp\omega$, both right lifts have adjoint readout ω, since they are right inverses of the same Q. Convert this Hessian and the coarse current to the raw bond coordinates by eq. 3.dliv. In the actual next pivot graph the constraint contraction uses $h_V^\sharp \widehat{J}$, where $\widehat{J}$ is that raw coarse current. Equation eq. 3.dlii therefore gives the outer action Hessian including its curvature term. Adding the relative contour Gram in the same free coordinates gives eq. 3.dliii. This is the compressed physical value of the completed form when the two constructions use the same branch, active mask and contour gauge.

Freezing exterior fluctuation coordinates in this formed quadratic leaves its principal interior block as the conditional precision. If the full form is K and the fixed coordinates are a_{E}, its mean is $-K_{\mathrm{II}}^{-1}K_{\mathrm{IE}}a_{\mathrm{E}}$. The exterior Schur term is independent of the integrated coordinates. For the comparison Gaussian, form the full precision at the reference datum with the exterior fluctuation set to zero. Its Gaussian marginal on the box has covariance $(K^{-1})_{\mathrm{II}}$ and precision $K_{\mathrm{II}} - K_{\mathrm{IE}}K_{\mathrm{EE}}^{-1}K_{\mathrm{EI}}$, with K now evaluated at that reference datum. This accounts for the Schur correction in the force comparison. The interactions and cutoffs enter the normalized comparison separately.

At unit ratio use the inverse walks of this matched step precision on each of the two nests. The native operator theorem gives this identification and inversion on its admitted prepared geometry. The homogeneous symmetric-contour construction gives it by eq. 5.cclix and eq. 5.cclxxii. For a general labeled nest in the symmetric construction we must admit the same completed carrier families on the actual free stock, including the separation and current bounds of Prop. 3.112. The local algebra above identifies their physical matrix wherever this formation is admitted. The two-level real representation gives, on the padded top territory, a rotated potential of size $C\epsilon/L$, its first difference of size $C\epsilon/L^2$, and scaled current of size $C\epsilon$, all in input units. Let $\epsilon_\star$ bound the complete pair in the local operator norm on this territory. Here $\epsilon_\star \le Crgp_0$.

The clean geometry identifies the top territory, with its free and pivot indices, in both nests. A walk whose enlarged carrier lies there is consequently assigned the same kernel T_ω in both expansions. In the local extension use $\chi_\omega = (1-(2L/M)d_{\mathrm{w}}(\,\cdot\,,[\omega]))_+$, where d_{w} is the operator's weighted distance and $[\omega]$ is the full read. Thus the cutoff is one on the read and its support lies in the padded territory. If (A', J') is the rotated pair based at the identity, the local ball for the operator admits

$$z \longmapsto (e^{iz\eta_{j-1}\chi_\omega A'},\, z\chi_\omega J'), \qquad |z| < R_\star := \frac{\varepsilon_1}{2\epsilon_\star},$$

where ε_1 is the fixed operator radius. The current on this disc is the independent coordinate of the operator. At $z=1$ locality gives the physical kernel on this walk, and at $z=0$ it gives the flat kernel. Weak coupling on the comparison window gives $R_\star \ge 2$. If $\epsilon_\star = 0$ the short-walk values coincide. Otherwise summing the Cauchy bounds for the positive Taylor orders gives $2/R_\star$ times the walk majorant. The sum of the rooted majorants is bounded. For a walk whose enlarged carrier exits the territory, the tree from a read bond to that exit has length at least the distance to the edge minus the fixed padding allowance of the walk domains. The two outer sums are therefore $O(e^{-c\mathsf{D}})$. Returning to the read frame costs $C\epsilon$ by the endpoint estimate above. Thus at unit ratio the entrywise comparison follows directly for the full covariances restricted to the read set.

For the ratio tube use the affine precision $\tau(K_A+K_G)+(1-\tau)K_G$ from eq. 4.ccxxiv. Take a common compression I on complete blocks between the read set and the edge of the matched territory, with a fixed fraction of the margin on each side. Retain part of the exponential walk rate for the distance between the two endpoints in the preceding estimates. The weighted row and column sums then bound the difference of the compressed unit-ratio inverses by $C(\epsilon_\star + e^{-c\mathsf{D}})$. The

local relative gauge forms on I differ by $C\epsilon$ in the same norms. This follows from the fixed-radius contour derivatives on the matched complete blocks.

Write C_a for a unit-ratio inverse on I, G_a for the corresponding compressed gauge form, and $\lambda = (1-\tau)/\tau$. Its inverse at ratio τ is $\tau^{-1}(1+\lambda C_a G_a)^{-1}C_a$. The weighted row and column norms are submultiplicative. The tube makes $|\lambda|\|C_a G_a\|$ small there. After taking out the common τ^{-1}, the difference has a middle factor between $(1+\lambda C_1 G_1)^{-1}$ and $(1+\lambda G_2 C_2)^{-1}$. That middle factor is

$$C_1 - C_2 - \lambda C_1(G_1 - G_2)C_2.$$

This gives the same bound on I at the actual ratio. To return to the full covariance, use the corresponding weighted inverse bound on each full free-bond stock. For a full precision Q_τ and its inverse Γ_τ, block multiplication gives

$$(\Gamma_\tau)_{II} - (Q_\tau)_{II}^{-1} = -(Q_\tau)_{II}^{-1}(Q_\tau)_{II^c}(\Gamma_\tau)_{I^c I}.$$

At a real admitted ratio, symmetry and the other block equation give the factorization

$$(\Gamma_\tau)_{II} - (Q_\tau)_{II}^{-1} = B(\Gamma_\tau)_{I^c I^c}B^*, \qquad B = (Q_\tau)_{II}^{-1}(Q_\tau)_{II^c}.$$

For a linear read supported in I, the weighted inverse and precision bounds give an $\ell^2(I^c)$ tail for its row of B. The tail spends its exponential rate on the distance from the full read to I^c. Include the contour stencil and the plaquette stencil before measuring this distance. If the two reads have depths d_1, d_2 after that padding, the bounded exterior covariance gives an error $Ce^{-c(d_1+d_2)}$ for unit read norms. This implies the required $Ce^{-c\mathsf{D}}$ bound on the buffered read set. The endpoint rotations return this estimate to the read frame. The ratio comparison thus requires the native unit-ratio inverse and the affine gauge correction on the actual free stock, with the same pivot elimination. The resulting bound in scaled free coordinates is

$$\varpi_j \le C\big[\mathsf{r}gp_0 + e^{-c\mathsf{D}}\big]$$

against the flat covariance. This comparison uses the complete precision, including the current and contour gauge terms, in the common local recipe. In a physical plaquette pairing each linear readout also carries its own factor $\widehat{g}_{j-1}\eta_{j-1}^2$.

Let C_0, C_1 be the flat and rotated Gaussian covariances on the read coordinates, and put $C_t = (1-t)C_0 + tC_1$. The uniform precision bounds give common positive lower and upper eigenvalue bounds for C_t, including after compression to the read coordinates. If h_t is its centered Gaussian density and $D = C_1 - C_0$, then

$$\partial_t h_t(y) = \tfrac{1}{2}h_t(y)(y^{\mathsf{T}}C_t^{-1}DC_t^{-1}y - \mathrm{tr}(C_t^{-1}D)).$$

The bracket has mean zero under h_t and second moment $2\|C_t^{-1/2}DC_t^{-1/2}\|_{\mathrm{HS}}^2$, by Wick's formula. Integrating the L^1 bound on the derivative gives a density difference at most $C\|D\|_{\mathrm{HS}}$. On the common product window, normalization costs at most another fixed factor when its Gaussian masses are at least one half. The scalar read dimension is at most $3n_j$, so the entrywise bound above costs $Cn_j\varpi_j$ in the normalized Gaussian comparison. The nonlinear box tilt and exterior comparison enter separately.

6.3.0.12. The change in the intrinsic force. For the intrinsic terms of §6.3.0.4, take the same zero-exterior fluctuation argument for both backgrounds. Put $a_j = C\mathsf{r}gp_0$, which bounds the rotated background potential on the short carriers. On their common fixed-radius chart, the value bounds $CB_Y gb_j^3$ for the action remainder and $CB_Y gb_j$ for the coarea term also hold on the larger fluctuation ball. Cauchy in a fluctuation variable costs C/b_j. Cauchy in the background direction on its fixed radius, followed by integration to the flat datum, costs Ca_j. Thus the short-carrier part of the force difference at bond b is at most $C(gb_j^2 + g)a_j$ after the rooted sum. For the long carriers take

the two force bounds separately and use the distance to the edge as in the precision comparison. The resulting bound is

$$C(gb_j^2+g)\big[a_j+e^{-cd(b,O)}\big].$$

The additional comparison of nonzero exterior fluctuations uses the mixed Hessian estimate on the admitted segment. The old terms need their own retained-datum radii and origin sums. The fixed-radius derivative used here applies to the intrinsic terms.

6.3.0.13. A common flat reference for the exterior mixture. Use the same flat precision on the common free-coordinate box I and the same product window for every admitted exterior. In eq. 6.xv, take the reference potential to be the quadratic of this flat law. The weighted gap of its precision gives the required coercivity and the weighted column bound. The reference has no nonlinear tilt or exterior linear force.

First set aside the old-origin exponent, whose oscillation costs at most $2\epsilon_j^{\mathrm{old}}$ in the normalized log density. Keep the actual intrinsic action and coarea terms in the other potential. Write $a_j = C\mathsf{r}gp_0$ as in the precision comparison. In the rotated coordinates the difference of quadratic forces is bounded on a row by the precision row estimate times b_j. The Schur and exterior forces have the row bounds at O and E. The intrinsic value bounds on the larger ball, followed by one coordinate Cauchy estimate and the rooted sum, give $C(gb_j^2+g)$ for its force on that row at the actual fixed exterior. Thus, with $B=\partial I\cup O\cup E$, the total force difference after the old-origin removal is bounded by

$$\delta_b \le Cb_j\big[a_j+gb_j+e^{-cd(b,B)}\big] \qquad (b_j\ge 1).$$

This uses the complete precision row estimate, including the matching of the gauge weight and the current stencil.

At a read coordinate $s\in S$, the marginal log-density estimate following eq. 6.xv weights the remaining force components by $e^{-\mu d(b,s)}$. The two constant terms sum uniformly, and the last term gains $e^{-c'\mathsf{D}}$ when $d(s,B)\ge \mathsf{D}$. The read-window diameters are at most Cb_j. Restoring the old-origin exponent, the normalized read marginal therefore has density ratio to the same flat box-window marginal in $[e^{-\Xi},e^{\Xi}]$, where

$$\Xi \le 2\epsilon_j^{\mathrm{old}} + C|S|b_j^2\big[a_j+gb_j+e^{-c'\mathsf{D}}\big].$$

The pointwise ratio passes through the positive exterior mixture in these common coordinates. This route only uses the Hessian of the flat reference in the force lemma. The actual chart, rotated readouts and selectors must still be admitted on the common product window. Removing the flat box window and comparing with the full flat marginal use the Gaussian tail and Schur estimates above.

6.3.0.14. Admission on the real conditional window. For each positive string entering the exterior mixture, use the completed label of its averaging step before $\mathbf{R}$ at that level. Keep the actual stored fields, frames and cut rows of that label. In particular a component that will be treated later by an exterior operation is still counted in this exponent, at the stage where the fluctuation integral is performed.

At a point of the support of this step, put $\mathcal{A}_k=\widehat{g}_k(y,y_O)$ on the nonpivot bonds that touch Ω_{k+1}. In the strip use the corresponding cut datum, background and shifted fluctuation. For every carrier Y the step field facts give

$$(U_{k+1},0,J(U_{k+1}))|_Y \in \widetilde{\mathcal{U}}^{1/2}_{\sigma^{\mathbf{T}}}(Y).$$

Here $\sigma^{\mathbf{T}}$ is the graded structure of the completed label. In addition the free variable is in the half-radius version of the step box

$$\begin{cases} \widehat{g}_k^{-1}|\mathcal{A}_k(b)|<3\mathfrak{b}_k & \text{on free bonds touching } \Lambda_{k+1},\\ \widehat{g}_k^{-1}|\mathcal{A}_k(b)|<2C_1^{\mathrm{fl}}\mathfrak{b}_k & \text{on the other free bonds of } \Omega_{k+1}. \end{cases}$$

In this display the bond types are those of the step's integration and exterior sets. The full product of this box with the graded field domain is the domain used by the old-origin substitution. The bound uses $|\mathcal{A}_k| < \delta_k = \widehat{g}_k \mathfrak{b}_k$ on the integration bonds and the bound by $\mathsf{c}_{\mathrm{fl}} \widehat{g}_k \mathfrak{b}_k$ on the other free bonds, or $\mathsf{c}_{\mathrm{fl}}^{\sharp} \widehat{g}_k \mathfrak{b}_k$ after the strip change. Choose C_1^{fl} above these two constants.

The first factor has four requirements. The passed plaquette and fluctuation tests, and the retained tests on the older or prepared rows, give the regularity of the completed datum layer by layer. The bound includes the prescribed shell factor at a prepared component. Near Ω_{k+1} that factor is one, and its far contribution to the profile estimate is bounded by $C(\log x_k)^r e^{-cLM\check{R}_{k+1}}$. The layer estimates of the regular critical field then give the pointwise plaquette and current bounds in their own units. The current is that of the restricted action, including its prescribed boundary stencil. The complex potential slot is zero. On each addressed gauge cube the local transport estimate gives the required real gauge with potential and first difference bounded by the same layer size times the fixed cube-size constant. For each addressed tower, restrict the real critical configuration with its own averages. The restriction identity and the inherited layer bounds give the tower and its plaquette and current estimates in the tower units. On a core the same restriction with the core scale gives its two radius bounds. These are the four clauses of the step membership criterion. Their half-radius form uses $C\varepsilon_i^{[r]} \le c_1 \check{\alpha}_{0,i}$ at each layer. Here $\varepsilon_i^{[r]}$ includes its actual layer factor, $\check{\alpha}_{0,i}$ is the minimum of the two adjacent radii (the single radius at level zero), and $2c_1$ is below the fixed lower bound on the graded coefficients. The test thresholds and radii are chosen in that order. This also admits the structure of any earlier member addressed by the label.

On the clean interior the exact product-window identity supplies these support points when y ranges over $\mathcal{W}_I$. Its old-test implication uses the same coarse datum on full double blocks, the margin from δ_k to the approximate test at $2\delta_k$, and the old plaquette bound from the factors L^{-2} and δ_k/ε_k. Exterior conditions are those of the chosen compatible string. The reference branch retains the same step strings and tests. At a fixed output its normalized weight relative to the clean kernel is bounded above and below by positive constants. Thus the test-support properties carry to that branch under disintegration. For a conditional oscillation estimate the resulting almost-everywhere admission suffices. A bound for a larger complex product of varying exterior data is a different assertion.

Finally the actual step substitution on this domain has the pivots of B' solved before it is used in any older carrier. At complete parameters its averages are $e^{iB'} V_k^{\mathrm{bg}}$, its second member is the actual current of $U^{\sharp}$, and it preserves the prescribed transports at sites with stored data. The older datum induced by the literal fluctuation is regular with half the margin on the support of the tests. This is the regular evaluation clause for the outgoing background of the fluctuation integral. The identity by origins therefore applies on this domain to both the old exponent and its local evaluated potentials.

6.3.0.15. Which remainder is counted in a window. An ordinary $R^{(i)}(X)$ enters the old remainder sum only at its declared action-region place. The remainder of the fluctuation integral is assigned there when the entire domain X is at least one level-i cell inside the corresponding small-field region. The remainder produced by healing is assigned there on the same one-cell interior of its output action region. The term is a function of the core pair at that place. A remainder of the fluctuation integral in the rim, and a term of the healing exponentiation outside its prescribed interior, are filed as boundary terms. Membership in a larger clean box at a later time does not change that filing. An earlier term can enter the ordinary sum after healing only through the actual identity for the refiled term.

For an ordinary $R^{(i)}(X)$, apply the window function of its admitted old core, subtracting its value at $(\mathbf{1}, 0)$. If the window is at scale $m \ge i$, write $\sigma = L^{-(m-i)}$ and ρ for the ratio of output to old radii. Its contribution is bounded by

$$C r_i e^{-\kappa d_i(X)} (2 + d_i(X))^2 \rho^2 \sigma^4.$$

This is the vacuum subtraction estimate. At a fixed old base the rooted domain sum absorbs the polynomial. The $C\sigma^{-4}$ possible old bases in the current window cancel the scale factor. Thus the contribution per window is $C\rho^2 r_i$, with no additional geometric gain in the age. The far members stay at their original places with their rooted decay, as in the step's remainder potential bound. Summation in i is the flow sum eq. 4.ccv. Neither a remainder at the rim nor an exponent with stored colour data enters this sum. Their actual boundary carriers, with the hull family of the label at evaluation, have the separate boundary and stored-factor bounds.

6.3.0.16. Removing the old interaction at fixed exterior data. For the generating function it suffices to bound the oscillation of each old contribution on the real conditional window. Use the reference branches of the clean coupling, after removing the separate restriction weights. Fix a compatible exterior e of the output ω, including its retained datum and all stored spectators. Write $k = j - 1$ and put

$$\vartheta_k = \frac{\delta_k}{\alpha_k} \asymp (\log x)^{\mathfrak{p}_0 - q}, \qquad \delta_k = \widehat{g}_k b_j.$$

Use the complete evaluated potentials of the fluctuation step. The input field in each potential is the pair of the critical background and its current, with the physical free coordinates $\mathcal{A} = \widehat{g}_k(y, y_O(e))$. Its output field $U^\sharp$ is the actual solution at the new datum $e^{iB'} V_k^{\mathrm{bg}}$; B' includes the solved pivot entries. At the complete parameters the levelled substitution reads its actual current and preserves the stored variables. Thus the sum over a given origin equals the difference of that origin evaluated at $U^\sharp$ and at the unfluctuating $U^\flat$.

For the regular family after the pointed counterterm subtraction, the old remainders and the Wilson weight difference, the step estimates on the admitted real box give

$$|V_e^o(Y; y)| \le C\vartheta_k \, S_k^o \, e^{-\lambda d_k(Y)}. \tag{6.xvi}$$

The bound also holds for an ordinary defect if present, with its separate size. In this formula the full fluctuation box has width at most a fixed multiple of b_j, including the fixed exterior coordinates. The normalization constant from $U^\flat$ depends on e alone. The sizes are

$$S_k^E = \sum_{i \le k} \omega_{k+1-i} e_i, \qquad S_k^R = \sum_{i \le k} \rho_i^2 r_i,$$
$$S_k^c \le C M^4 \sup_m \alpha_{0,m}^2 \, \mathsf{b}_\star.$$

Here c denotes the Wilson lag origin, $\mathsf{b}_\star$ bounds the increments of inverse coupling, and the source in the generating function is carried as an insertion in the unsourced step. The regular sum is at most CE_0 by eq. 4.cxcix. For the spectator-free remainders with $r_i \le C x_i^{-s}$ and $s > 2$, eq. 4.ccv bounds S_k^R by Cx^{2-s}. The Wilson size is bounded uniformly. In its formation the level-m Wilson members have size at most $C\alpha_{0,m}^2 (k - m + 1)\mathsf{b}_\star$; the spatial and layer sum is already taken in the displayed bound for S_k^c. Replacing it by the global supremum of $x_k(\cdot) - x_k$ would lose uniformity.

The source of the lag in the unsourced density is $\sum_{i \le k} b_i(1 - \varphi_i)$. When the profiles of healed regions are redrawn, their changes are compensated at each bare plaquette in the Wilson sum. The remaining lag is supported in the standing large-field region. On the clean box, including its trace window, every lag potential whose full carrier stays there vanishes, by the whole-lattice identification of the step potentials. An exterior lag can still vary with y through the background substitution. Use eq. 6.xvi for that localized origin. Its contributing carriers reach the nonempty trace, so the spatial rate can also pay for that distance. The bound without this extra distance gain already suffices here. With profiles kept before compensation, the same bound for the evaluated lag origin is used wherever the step estimate is admitted. A bound on the volume-weighted lag of an individual history cannot be attributed to a resummed class unless the lag is common to that class.

On the positive history resolution before this compensation, the weighted lag can be controlled without excluding all old creations under the box. Let n_j be the size of the level-j window. Count the older level-i cells under it whose profile φ_i is less than one somewhere. Each enters with weight $(j-i+1)L^{-4(j-i)}$. There are at most $Cn_jL^{4(j-i)}$ cells to test, and a profile can differ from one only next to a live or newly created cell at that level. The one-cell probability bound therefore gives the first moment of the weighted count:

$$\mathbb{E}\,\mathfrak{S}_j^{\mathrm{lag}} \le Cn_j \sum_{i<j}(j-i+1)e^{-cp_0(g_i)}.$$

Below the witness level n, the running lower bound $x_i \ge x_n+\lambda_\sharp(n-i)-C$ makes this series summable. Use half the exponent at x_n-C, and use the other half to dominate any fixed inverse power of x_n+n-i. Between n and the end of the fluctuation segment the couplings are comparable to that of the witness. Its logarithmic number of levels contributes only a polynomial prefactor. Markov's inequality at the threshold $\tau_\star n_j$, followed by the sum over the segment, gives an exceptional weight $Ce^{-c'p_0(g_n)}$ for fixed $\tau_\star>0$. This uses the volume of each fine cell in the window units. A count of all older test failures with unit weight would lose that factor. The bound concerns the positive refinement on which the lag is recorded. In the compensated class use the refiled exponent. A restriction by the finer lag event must use its own joint law.

On this finer resolution, fix the exterior and keep the original profiles in the Wilson exponent. On the admitted complex box for both matched fields, the curvature estimate bounds the difference of their plaquette actions per unit volume in window units by $Cg^2b_j^2$. A level-i cell occupies $L^{-4(j-i)}$ of a level-j cell. The running bound on the increments of inverse coupling bounds the missing profile coefficient there by $C(j-i+1)$ in the lag count. Summing the action differences over the lagging cells therefore gives

$$|\mathbb{P}_j^{\mathrm{lag}}| \le Cg^2b_j^2\,\mathfrak{S}_j^{\mathrm{lag}} \le Cg^2b_j^2\,\tau_\star n_j.$$

On a smaller concentric box the Cauchy estimate in two directions spends Cb_j^{-2}, so the Hessian has size $Cg^2\tau_\star n_j$. For its spatially weighted estimate the decay of the two matched minimizer responses is used on the same profile cells. These estimates read the lag at fixed exterior on the finer resolution.

At fixed exterior, subtract the value at one point y_* of the product window. A term independent of the interior coordinates then cancels. Each other full carrier meets a current cube of the box, with its prescribed fixed enlargement for pivot reads. After summing its old origins, root that carrier at such a cube. The strict rate reserve in eq. 6.xvi pays for the rooted count. This gives

$$\sup_{y\in\mathcal{W}_I}|P_{e,\mathrm{reg}}(y)-P_{e,\mathrm{reg}}(y_*)| \le Cn_j\vartheta_k(S_k^E+S_k^R+S_k^c),$$

with the ordinary defect size added if present. This is a current-scale count. The old base points of a regular activity have already been summed in its window, after the counterterm of that base point has been subtracted.

For surviving boundary and kept factors, apply the ordinary step estimates on the admitted support just described. The actual support point, including the cut strip point, lies in the half-radius domain of the substitution. The pinned levelling map uses the actual output datum and its transformed current. It preserves the stored arguments. Thus the identities for the two evaluated exponents use the same spectators, and their estimates apply at the original fixed exterior.

For the filed boundary system, apply the one-term-per-slot bound of Theorem 10(e) by origin, with its stored-field and dial domains. Its amplitude is $C\mathsf{B}\vartheta_k$ at the retained hull rate. Its carrier crosses the small-field boundary, with hull length at least $c_2L\check{R}_j-2$. For any fixed a the bound is $C_a\mathsf{B}\vartheta_k(\log\theta_k)^{-a}$ times the hull decay at a smaller rate. Spend a fixed part of the available rate on the crossing length and retain a rate above the rooted entropy. For a kept factor use its

inherited carrier oscillation bound. The factor form gives $C_a\mathsf{B}_\Phi\vartheta_k e^{-L\check{R}_j}$ times the same decay. In the difference at y and y_*, all terms depending only on the fixed exterior cancel.

The root cubes for the remaining terms lie outside the alive exempt hulls by the clean padding. The rooted sum eq. 6.xxxix thus gives, on labels with these inherited bounds,

$$Cn_j\vartheta_k(S_k^E + S_k^R + S_k^c + C_a\mathsf{B}(\log\theta_k)^{-a} + C_a\mathsf{B}_\Phi e^{-L\check{R}_j})$$

for the old oscillation, including the ordinary defect size when present. This application is to the real support. The induction of the inherited bounds on all surviving histories is a separate input to the exterior mixture.

6.3.0.17. Terms attached to healed regions. In the native positive history representation, a boundary term produced at healing is filed at its healing level. Its localization box can reach the read after the region has disappeared, or when the region lies outside the padded read. To use the estimate on attached terms, test the terms themselves. At read level j, an attached term X filed at $i' < j$ has weight $L^{-4\nu}x_{i'}^{2q+2}M_X d_{i'}(X)^2$, where $\nu = j - i'$, M_X bounds the term and $d_{i'}(X)$ is the size of its localization box in level-i' units. Sum over boxes meeting the read. The native localization estimate bounds the expectation per fine anchor by $CL^{-4\nu}x_{i'}^{2q+2}q_{i'}^{\mathrm{pk}}$, where $q_{i'}^{\mathrm{pk}}$ bounds the mass of the creation or healing event that supplies a term at that position. This uses the sum of $M_X d_{i'}(X)^2$ at a fixed anchor and the spatial decay for terms attached to regions away from the read. The fine-anchor count $Cn_jL^{4\nu}$ cancels the factor $L^{-4\nu}$. The per-bond version uses the same bound with the localization overlap constant.

With $q_{i'}^{\mathrm{pk}} \le Ce^{-c_{\mathrm{cr}}p_0(g_{i'})}$, the resulting level sum is bounded by $Cn_je^{-c_{\mathrm{cr}}p_0/2}$. For $i' < k_0$ use both running bounds on $x_{i'}$. Half the exponent controls the sum of the polynomial weights in $x + B_\sharp(k_0 - i')$, since $\mathfrak{p}_0 > 1$. The levels of the fluctuation segment have comparable coupling and logarithmic cardinality. Their polynomial prefactor is absorbed by the reserve in the exponent since $p_0(g_{i'})/p_0 \to 1$ uniformly in that segment. Markov's inequality at $e^{-c_{\mathrm{cr}}p_0/4}$ gives a failure mass at most $Cn_je^{-c_{\mathrm{cr}}p_0/4}$. This estimates the budget with all filing levels.

For the weighted derivative estimate on the older list $i' < k_0 - \mathfrak{l}$, use its axial-gauge representative and the vanishing first jet at the flat field. The Schwarz bound uses the fine fluctuation $CL^{-2\nu}g_jb_j$ and $(g_j/R_{i'}^{\mathrm{an}})^2 \le C(1 + B_\sharp\nu)$. Thus the term has second-order size at most $CM_X\kappa_{\mathrm{ax}}(X)^2L^{-4\nu}(1 + B_\sharp\nu)b_j^2$, where $\kappa_{\mathrm{ax}}(X)$ bounds the axial representative on the box. Apply Cauchy's estimate on the admitted tube for the Hessian. The weighted row sum contributes $(d_{i'}(X)L^{-\nu}+C)^4e^{c_P(d_{i'}(X)L^{-\nu}+C)}$. For the localized boundary terms the exponential uses part of their decay. For the frozen-coupling term the older-box bound $d_{i'}(X) \le 100M\check{R}_{i'} + C$ and the choice of the lag depth bound $d_{i'}(X)L^{-\nu}$ uniformly. The remaining factors are polynomial in the old inverse coupling, so the same first-moment argument applies with an enlarged fixed power. On this per-bond event the native gradient bound is $C\widehat{b}_j(b_j^2 + p_0(g_j))e^{-cp_0}$, with the pivot-map norm included in C. Integrating on the product window gives an oscillation at most $Cn_jb_j\widehat{b}_j(b_j^2 + p_0(g_j))e^{-cp_0}$, which can be added to the old oscillation when this decomposition is used. We must use the localization and derivative estimates for the actual terms of the conditional density. The all-filings mass estimate alone does not extend the older-list derivative bound to terms filed later.

For the native ordinary remainder, retain the localization of each term in the gradient estimate. The volume factor of a level-i' remainder cancels the number of its fine placements. The remaining old-level sum in the gradient is bounded by $Cg_jx_j^{(3-\kappa_0)/2}\,\mathrm{polylog}(x_j)$, with $x_j = g_j^{-2}$. At fixed background apply Cauchy's estimate in the second coordinate, including a coordinate of the exterior, on the joint window tube. Let m_j be the available complex coordinate margin there. Only terms whose read contains both coordinates can contribute. Their localization reserve pays the weighted

rooted sum, giving

$$\sup_b \sum_{b'} e^{c\,d(b,b')} |\partial_b \partial_{b'} P_j^{\mathbf{R}}| \le \frac{C}{m_j}\, g_j x_j^{(3-\kappa_0)/2} \operatorname{polylog}(x_j).$$

For the bound with b_j^{-1} one needs a complex margin proportional to b_j. An absolute or inverse-logarithmic margin still gives a small Hessian for the chosen $\kappa_0 > 4$. This estimate is for the ordinary remainder with its native localization and chart. The terms from marked or retained spectators need their own mixed bounds.

The per-bond event for this comparison is imposed on the incoming list of formed terms. At a compatible exterior keep its filing levels, localization boxes and recorded coupling lags, along with the spectators and the history data that determine this list. The geometric clean guard fixes the free-bond mask and each dependence shell. The bound M_X is a supremum on the admitted domain of the formed term, including its recorded lag factor where present. It is fixed as the real interior coordinate varies. The axial factor and the distance and count weights also depend only on this exterior record and the fixed geometry. The pass indicator for the budget is therefore an exterior factor on the whole product window. At a merge sum the positive incoming terms with their own list and indicator. A test on a list filed after this step must be imposed at its own incoming comparison or justified by the later-record kernel on the actual retained state.

Let $P_{e,\mathrm{old}}$ denote the sum of the admitted contributions just bounded. Suppose the resulting bound for $\sup_{\mathcal{W}_I} |P_{e,\mathrm{old}}(y) - P_{e,\mathrm{old}}(y_*)|$ is $\epsilon_j^{\mathrm{old}}$, uniformly over the exterior data. For each e, keep the same domain, quadratic extraction, intrinsic terms, stored variables and readout, and remove $P_{e,\mathrm{old}}$ only in an auxiliary normalized interior law E_e^{int}. Then

$$e^{-2\epsilon_j^{\mathrm{old}}} \le \frac{dE_e}{dE_e^{\mathrm{int}}} \le e^{2\epsilon_j^{\mathrm{old}}}.$$

Indeed the ratio is the remaining exponential divided by its mean under E_e^{int}. The estimate costs $(e^{2\epsilon_j^{\mathrm{old}}} - 1) \sup |G|$ for a bounded insertion, uniformly in ω and e. Integrate this bound against the original exterior mixing measure. The intrinsic box comparison and the flat read-set approximation are performed next, for each exterior, and have the same flat target.

The loss $C n_j (\log x)^{\mathfrak{p}_0 - q}$ uses one power of the window-to-radius ratio. It suffices for the generating function route with any fixed number of insertions by choosing q after the polynomial logarithmic losses in the source radii. A Hessian at fixed exterior cannot supply a derivative with respect to a different retained datum or a stored variable. Such a derivative would require one common analytic domain for the two complete evaluations. The normalized removal uses the real admission just proved on the conditional support, with its retained rows and margins. Each surviving boundary or stored factor must also retain its declared carrier and oscillation majorant at that evaluation.

Adding $a^{\mathsf{T}} y$ to P_e leaves its Hessian unchanged and changes its force by a. It translates the Gaussian mean even on all of $\mathbb{R}^n$. Centering a quadratic observable removes its scalar mean, but not the linear term caused by that field translation. We retain this term below.

Integration by parts with its boundary. Suppress e, let Q be real symmetric and invertible, and set $C = Q^{-1}$. Define

$$\mathcal{B}_j(F) = Z^{-1} \int_{\partial D} n_j(y) F(y) e^{-V(y)}\, dS(y). \tag{6.xvii}$$

The divergence theorem, with $\partial_j V = (Qy)_j - r_j$, gives

$$\mathbb{E}[(QY)_j F] = \mathbb{E}[\partial_j F + r_j F] - \mathcal{B}_j(F). \tag{6.xviii}$$

On a rectangle this is the difference of its two face integrals. The boundary has not been discarded.

The auxiliary linear-source function $M(z) = \mathbb{E}e^{z^{\mathsf{T}}Y}$ is entire on a bounded window. Since $M(0) = 1$, its logarithm $W(0) = 0$ is defined near zero. There put

$$\mathbb{E}_z F = \frac{\mathbb{E}(Fe^{z^{\mathsf{T}}Y})}{M(z)}, \qquad \beta_j(z) = \frac{\mathcal{B}_j(e^{z^{\mathsf{T}}Y})}{M(z)}. \tag{6.xix}$$

Apply eq. 6.xviii to the exponential and multiply by C. This proves the exact connected identity

$$\partial_i W(z) = \sum_j C_{ij}\{z_j + \mathbb{E}_z r_j - \beta_j(z)\}. \tag{6.xx}$$

The source here is an auxiliary linear field source. It is not an identification with a physical $\mathrm{tr}(F^2)$ numerator.

Let $m = \mathbb{E}Y$, let Σ be the covariance and let κ denote joint cumulants. Differentiation gives

$$\begin{aligned}
m_i &= \sum_j C_{ij}\{\mathbb{E}r_j - \beta_j(0)\},\\
\Sigma_{ia} - C_{ia} &= \sum_j C_{ij}\{\kappa(r_j, Y_a) - \partial_a\beta_j(0)\},\\
\kappa(Y_i, Y_a, Y_b) &= \sum_j C_{ij}\{\kappa(r_j, Y_a, Y_b) - \partial_a\partial_b\beta_j(0)\},\\
\kappa(Y_i, Y_a, Y_b, Y_c) &= \sum_j C_{ij}\{\kappa(r_j, Y_a, Y_b, Y_c) - \partial_a\partial_b\partial_c\beta_j(0)\}.
\end{aligned} \tag{6.xxi}$$

Each derivative differentiates the numerator and $1/M$. The first produces a covariance; the next remove the disconnected partitions and give the mixed cumulants. Equivalently, compare the convergent local power series in eq. 6.xx.

Put $Z_a = Y_a - m_a$. The boundary derivatives are explicitly

$$\begin{aligned}
\beta_{j,a}(0) &= \mathcal{B}_j(Z_a),\\
\beta_{j,ab}(0) &= \mathcal{B}_j(Z_aZ_b) - \Sigma_{ab}\mathcal{B}_j(1),\\
\beta_{j,abc}(0) &= \mathcal{B}_j(Z_aZ_bZ_c) - \Sigma_{ab}\mathcal{B}_j(Z_c) - \Sigma_{ac}\mathcal{B}_j(Z_b)\\
&\quad - \Sigma_{bc}\mathcal{B}_j(Z_a) - \kappa(Y_a, Y_b, Y_c)\mathcal{B}_j(1).
\end{aligned} \tag{6.xxii}$$

These follow by differentiating the quotient. In particular an unnormalized face-moment estimate is not automatically the same estimate for a connected boundary derivative.

Replacing traces by weighted collar moments. On the product window $D = \prod_b K_b$, let $b(j)$ be the bond carrying the coordinate j. Choose ψ_j depending only on $y_{b(j)}$, vanishing near $\partial K_{b(j)}$, equal to one off its collar of width δ_j, with $|\partial_j\psi_j| \le c/\delta_j$. Integration by parts in the j direction has no boundary contribution for $\psi_j e^{z^{\mathsf{T}}y}$: on the $b(j)$ face the cutoff vanishes, and the other faces have zero j-normal component. Equivalently, use Fubini and integrate only in the factor $K_{b(j)}$. Comparing it with eq. 6.xviii proves

$$\mathcal{B}_j(e^{z^{\mathsf{T}}Y}) = -\mathbb{E}\left[e^{z^{\mathsf{T}}Y}\{\partial_j\psi_j + (1-\psi_j)((QY)_j - r_j(Y) - z_j)\}\right]. \tag{6.xxiii}$$

Indeed the compactly supported formula equates $\mathbb{E}[\psi_j(QY - r)_j e^{zY}]$ with $\mathbb{E}[(\partial_j\psi_j + z_j\psi_j)e^{zY}]$. Adding the part with $1 - \psi_j$ gives the sign and terms above.

Thus moments of the form

$$\mathbb{E}\Big[1_{\mathrm{collar}}(1 + |Y|_{\mathrm{loc}})^q\{\delta_j^{-1} + |(QY)_j| + |r_j(Y)| + 1\}\Big] \tag{6.xxiv}$$

control the corresponding numerator derivatives, with the collar of the single bond $b(j)$. The local norm means the finite tested coordinate list with its specified weights. The derivatives of $1/M$ also use its ordinary moments.

Up to order three there are only finitely many product and quotient terms. If their ordinary moments are uniformly bounded and the weighted collar moments are $O(\epsilon)$, the boundary derivatives in eq. 6.xxi are $O(\epsilon)$ with the resulting finite moment factors. This follows directly by differentiating eq. 6.xxiii and eq. 6.xix. An unweighted collar probability does not prove eq. 6.xxiv; polynomial losses in the force and cutoff remain.

For a bond in the buffered interior the normalized box comparison supplies one way to bound this moment. Suppose the marginal at that bond has density ratio at most e^{Ξ} relative to the reference on the same product window. At the reference, the Gaussian is centered, the coordinate variances are uniformly bounded, the full window has Gaussian probability at least $1/2$, and the tilt has uniformly bounded oscillation. If the bond window contains the Euclidean ball of radius $c_0 b$, a collar of Euclidean width at most $c_0 b/2$ is contained in $\{|y_{b(j)}| \geq c_0 b/2\}$. The Gaussian one-bond tail, divided by the full window probability and multiplied by the exponential of the tilt oscillation, is at most Ce^{-cb^2}. The density ratio transfers this bound at the same bond with factor e^{Ξ}.

On the product window, the uniform precision row bound gives $|(QY)_j| \leq Cb$. The retained linear force and the perturbation gradient use their admitted row bounds. When these, the tested coordinate sizes and the inverse collar width are bounded by fixed powers of $1 + b$, eq. 6.xxiv is bounded by a polynomial in $1+b$ times $e^{\Xi - cb^2}$. This uses the density comparison at the differentiated bond, with its full read margin. The bound is uniform over exterior data for which these constants are uniform, and passes to their positive mixture.

In eq. 6.xxi, the boundary derivatives are contracted against a row of the Gaussian inverse. Take a tested coordinate at depth at least $2D_0$ from the edge after accounting for its read. For bonds at distance at most D_0 from this coordinate, use the collar bound with the common buffer. On the remaining bonds use the deterministic polynomial bound in eq. 6.xxiv from the full window and the uniform force and gradient bounds. The weighted row estimate for the inverse makes the contribution of these remaining bonds at most $C\,\mathrm{poly}(1 + b)e^{-cD_0}$. The quotient derivatives use only moments of the fixed tested list, so they retain the same split.

When e varies, the bound must be uniform in e or integrated in the order specified by the actual exterior law. A fixed-exterior bound is not a uniformly integrable envelope. This matters for a clean restriction formed by healed components.

The covariance of quadratic insertions. Let

$$O_i = Y^{\mathsf{T}} A_i Y + b_i^{\mathsf{T}} Y + c_i, \qquad A_i^{\mathsf{T}} = A_i, \qquad d_i = b_i + 2A_i m. \tag{6.xxv}$$

Only the symmetric part contributes to a scalar quadratic form. Expansion at $Y = m + Z$ gives $O_i - \mathbb{E}O_i = Z^{\mathsf{T}} A_i Z - \operatorname{tr}(A_i\Sigma) + d_i^{\mathsf{T}} Z$. Multiplication yields

$$\begin{aligned}\operatorname{Cov}(O_1, O_2) = {} & 2\operatorname{tr}(A_1\Sigma A_2\Sigma) + d_1^{\mathsf{T}}\Sigma d_2 \\ & + \sum_{a,b,c} (A_1)_{ab}(d_2)_c \kappa(Z_a, Z_b, Z_c) \\ & + \sum_{a,b,c} (A_2)_{ab}(d_1)_c \kappa(Z_a, Z_b, Z_c) \\ & + \sum_{a,b,c,d} (A_1)_{ab}(A_2)_{cd} \kappa(Z_a, Z_b, Z_c, Z_d).\end{aligned} \tag{6.xxvi}$$

The fourth centred moment is its fourth cumulant plus three pairings. Subtracting the quadratic means removes $\Sigma_{ab}\Sigma_{cd}$. The other two pairings contract to the trace above. The mixed terms give the third cumulants, and the two linear terms give $d_1^{\mathsf{T}}\Sigma d_2$. This accounts for every term.

The centred Gaussian reference with covariance C_0 and purely quadratic insertions is $2\operatorname{tr}(A_1 C_0 A_2 C_0)$. Its identification needs the mean term, covariance discrepancy and third and fourth cumulants in the actual tested contractions.

For one sufficient dimension-explicit estimate put

$$a_i = \sum_{a,b} |(A_i)_{ab}|, \quad v_i = \sum_a |(d_i)_a|, \qquad \|\Sigma\|_{\max}, \|C_0\|_{\max} \le K. \tag{6.xxvii}$$

If $\|\Sigma - C_0\|_{\max} \le \epsilon_2$ and the tested third and fourth cumulants are bounded by ϵ_3, ϵ_4, then

$$\begin{aligned} &|\operatorname{Cov}(O_1, O_2) - 2\operatorname{tr}(A_1 C_0 A_2 C_0)| \\ &\quad \le 4K\epsilon_2 a_1 a_2 + K v_1 v_2 + \epsilon_3(a_1 v_2 + a_2 v_1) + \epsilon_4 a_1 a_2. \end{aligned} \tag{6.xxviii}$$

Expand one covariance factor at a time in the trace. Each product differs by at most $2K\epsilon_2$, and the prefactor two gives the first term. Absolute summation gives the others. Weighted localized variants use the same actual contractions.

If $a_i \le Cr^4 \|f_i\|_\infty$ and $r \le C(\log x)^{p_r}$, the first and fourth errors cost $(\log x)^{8p_r}$. They require $\epsilon_2, \epsilon_4 = o((\log x)^{-8p_r})$ to be $o(\|f_1\|_\infty \|f_2\|_\infty)$. A claim that errors merely tend to zero does not suffice. The other rates are read from the mean and third-moment terms.

Each physical quadratic insertion ordinarily carries g^2, so the reference covariance carries g^4. Its errors and the physical remainders must be small relative to that signal. A fixed source-disc estimate cannot replace a missing potential normalization by an inverse power of g.

Exterior averaging adds the conditional means. Suppose the resummed representation has a positive probability realization with exterior variable E, clean event A and correct observable pullbacks. Conditional expectation gives

$$\begin{aligned} \operatorname{Cov}(O_1, O_2 \mid A) = \mathbb{E}\{&\operatorname{Cov}(O_1, O_2 \mid E, A) \mid A\} \\ &+ \operatorname{Cov}(\mathbb{E}(O_1 \mid E, A), \mathbb{E}(O_2 \mid E, A) \mid A). \end{aligned} \tag{6.xxix}$$

Add and subtract the product of conditional means to prove it. It applies to a positive conditional law, not separately to signed expansion terms.

Choose one deterministic number a_i^0 per observable and prove

$$|\mathbb{E}(O_i \mid e, A) - a_i^0| \le \delta_i \quad \text{for every admitted clean } e. \tag{6.xxx}$$

Then the last covariance in eq. 6.xxix has absolute value at most $2\delta_1\delta_2$. Choosing a different reference for every exterior would not prove this. The backgrounds, insertion coefficients and conditional moments need one common reference.

Put $q = \mathbb{P}(A^c)$, assume $0 < q < 1$ and $|O_i| \le B_i$ on the whole positive probability space. Decomposing by A gives

$$\begin{aligned} \operatorname{Cov}(O_1, O_2) = (1-q)\operatorname{Cov}(O_1, O_2 \mid A) + q\operatorname{Cov}(O_1, O_2 \mid A^c) \\ + q(1-q)(m_1^A - m_1^{A^c})(m_2^A - m_2^{A^c}). \end{aligned} \tag{6.xxxi}$$

If every clean conditional covariance differs from a fixed Γ by at most ϵ, this implies

$$|\operatorname{Cov}(O_1, O_2) - \Gamma| \le \epsilon + 2\delta_1\delta_2 + q(6B_1B_2 + |\Gamma|). \tag{6.xxxii}$$

The bad covariance is at most $2B_1B_2$ and the product of mean differences is at most $4B_1B_2$. These prove the constants. At $q = 0$ use the clean formula directly; at $q = 1$ there is no clean estimate to use.

For unbounded insertions retain the marked bad-event moments. For instance Cauchy–Schwarz bounds $|\mathbb{E}[1_{A^c} O_1 O_2]|$ by $(\mathbb{E}[1_{A^c} O_1^2])^{1/2} (\mathbb{E}[1_{A^c} O_2^2])^{1/2}$. Centering also requires the corresponding first moments. A bad-event probability alone does not bound them.

An unnormalized clean sum has to be divided by its actual mass. Source derivatives of that scalar contribute unless the exact numerator and denominator have already been realized as this conditional probability. A normalizer from another volume or resolved class cannot be substituted.

Uniform shapes and the remaining model estimates. The required positive shapes have $f_i \geq 0$, support diameters $O(d)$, separation between d and $6d$, and

$$\int f_i \geq \vartheta d^4 \|f_i\|_\infty, \qquad \|\nabla f_i\|_\infty \leq C_{\mathrm{shape}} d^{-1} \|f_i\|_\infty. \tag{6.xxxiii}$$

The integral against $|x-y|^{-8}$ has a lower bound $c_\vartheta \|f_1\|_\infty \|f_2\|_\infty$. The maximum distance is a fixed multiple of d and the two masses supply d^8. The free comparison theorem gives the lattice version at sufficiently fine compatible mesh, with this uniform derivative bound. In fact, partition into mesh cells of side $a \leq d$. The oscillation on each cell is at most $Ca\|\nabla f_i\|_\infty$. Only cells in the support enlarged by Ca contribute, whose total volume is $O(d^4)$ by the diameter condition. Therefore the Riemann mass error is at most $CC_{\mathrm{shape}}(a/d)d^4\|f_i\|_\infty$, uniformly in the position of the support. Taking $a/d \leq c\vartheta/C_{\mathrm{shape}}$ gives sampled mass at least $(\vartheta/2)d^4\|f_i\|_\infty$. The actual source class permits $C_{\mathrm{shape}} = \vartheta^{-1}$, in which case a sufficient floor is $a/d \leq c\vartheta^2$. Alternatively the sampled mass itself can be imposed. Smoothness and continuum mass without a uniform derivative bound do not supply this floor.

All errors in eq. 6.xxxii, with physical remainders, must be $o(g^4\|f_1\|_\infty\|f_2\|_\infty)$ uniformly on the whole shape class, positions and required reference volumes. The fixed $\vartheta = 1/12000$ allows dependence on that value followed by one uniform cutoff floor.

A reference estimate uniform along arbitrarily large tori at each fixed datum and test pair passes to a reference infinite-volume state and uses common-normalizer transfer. An estimate only on one prescribed finite torus needs the separate finite-reference and vacuum comparison. These are alternatives; neither follows from a local kernel.

The first remaining task is the literal identity in eq. 6.xi, with the actual exterior sigma-field, domain, numerator and normalizer. Next bound the complete Hessian and the individual forces. Then bound the tested connected force and boundary quantities in eq. 6.xxi, using either the trace or collar identity. The common-reference mean comparison in eq. 6.xxx and marked bad moments are separate. Their rates must dominate the window and coefficient losses in eq. 6.xxviii.

These estimates would supply the interacting covariance on the run route. The proved free, remainder and state-transfer implications then have their needed inputs. The alternative averaging-fibre route additionally requires its measure and carried-weight identification; it is not an extra conjunct once the run route closes.

6.3.1. *A clean collar and the actual local filing.* A local term made inside a sufficiently clean collar uses the whole-lattice recipe. The collar must contain the actual trace window, which includes a layer of cells of side $M\check{R}_j\xi_j$. A layer of $M\xi_j$-cubes is smaller and does not suffice for that assertion. We first prove the geometric and finite-filing statements, then sum the boundary tails. The whole-lattice comparison keeps the same input list; in particular it does not erase regular marked members produced by healed histories.

6.3.2. *The two grids and the trace window.* Use physical spacings $\xi_j = L^j\xi_0$. Let π_j be the grid of cubes of side $M\xi_j$, and let $\mathcal{P}_j$ be the cell grid of side $M\check{R}_j\xi_j$. The grids and all labels are the ones fixed by the run. Assume the actual cube ratio

$$\check{R}_j \leq L\check{R}_{j+1}, \qquad j < n. \tag{6.xxxiv}$$

Thus the physical cell size is nondecreasing: $\check{R}_j\xi_j \leq \check{R}_n\xi_n$. Distances in current cube units mean physical distance divided by $M\xi_n$, using the periodic graph metric when needed.

For a region X, its trace window X^{+1} rounds it out, at each level j, to all cells of $\mathcal{P}_j$ that meet it, and adds one cell layer. Include the fixed finite additional layers used by the actual step in this definition. Let c_{tr} be a bound for their number. Rounding and adding these layers gives the explicit inclusion

$$X^{+1} \subset \{x : \mathrm{dist}_\infty(x, X) \leq C_{\mathrm{tr}} M\check{R}_n\xi_n\}, \qquad C_{\mathrm{tr}} = C(1 + c_{\mathrm{tr}}). \tag{6.xxxv}$$

Indeed rounding to one cell moves each coordinate by at most one cell side; every additional layer costs at most one more. Apply eq. 6.xxxiv at each level. No sum over the number of levels occurs, because the window is the union of these levelwise neighborhoods, not their successive enlargement.

Let $\mathcal{S}_n$ be a standing label and put $Z_n = \Lambda_n^c$. For W, a set of current cubes, write W_R for its R-neighborhood in current cube units. The clean-collar condition used below is

$$W_R \subset \Lambda_n, \qquad s := R - C_{\mathrm{tr}}\check{R}_n - C_0 > 0. \tag{6.xxxvi}$$

Here C_0 covers the fixed cube collars of the local potential and integration formulas. They are specified as part of the recipe, rather than changed after a bound is proved.

6.3.3. *What remains in a standing history.* The standing part deletes the entire cluster of each healed component. All data which are still stored belong to an island, a drop-out or an unhealed component. The following conclusion uses this actual deletion rule.

Lemma 6.5 (The standing trace in the clean region)**.** Every retained date, mark and stored variable of a standing history of level n lies in Z_n, with at least one endpoint there for a stored bond. Outside Z_n, the action and determining sequences have no remade layers and coincide with the all-small-field sequences. Consequently the standing trace and its reading on X^{+1} are empty when $X^{+1} \subset \Lambda_n$. ■

Proof. Induct on the moves of a step. A new large-field outcome is recorded only on the component where its test fails; it is contained in that step's large-field region. At a subsequent step, an old component and all its standing contents are either carried, merged into another live component, or treated by the large-field operation. Carrying and merging retain the contents inside the new live component. An island or a drop-out retains its contents inside its own component. Healing deletes that component's entire cluster, including its older dates, fields, frames and marks; none of those deleted variables remains as a free integration variable in the standing part.

The only modifications of an earlier label sequence take place inside a component of a later operation. If the component is healed, its old regions are replaced by the healed output. If it remains standing, it lies in Z_n. The remade levels in eq. 2.xxii are modified inside precisely such a standing drop-out. Hence no remade layer survives outside Z_n. The nested unmodified label regions give the all-small-field sequence there. This proves the induction. An empty trace follows by restriction to X^{+1}. □

An alive island I has a hull $[I]_j$, the union of the closed level-j cubes meeting I. At birth this gives the one-layer enlargement. The nested partitions give

$$([I]_j)^{\uparrow n} = [I]_n, \qquad [I]_n \subset Z_n^{\sim c_I}, \tag{6.xxxvii}$$

where c_I is the fixed hull collar. The first identity uses the same standing island on both sides; an island absorbed before level n is no longer in this family. The second follows from seating of an alive island in the final live region and from the definition of its hull. Thus an alive exempt hull cannot enter the interior of a collar which excludes $Z_n^{\sim c_I}$. The possible failure of a finer place condition inside a remade drop-out occurs in Z_n and does not change this conclusion.

The hulls have a bounded corner multiplicity. Write u_j for the physical side of a level-j localization cube and b_i for the physical side of a level-i history cell. An island born at i has sides between $21b_i$ and $100b_i$. Its level-j hull has sides at most $100b_i + 2u_j$, hence at most $100\check{R}_j + 2$ cubes per direction since $b_i \le b_j$. Use the torus margin $204b_j < \ell$ for the following count. If a hull contains a corner v, its island meets the box of radius u_j about v, and both lift into the box of radius $102b_j$ about the lifted corner. When $21b_i \ge u_j$, the island contains a box of side u_j within the box of radius $2u_j$ about that corner. Distinct such islands are disjoint, so volume comparison gives at most 4^4 of these boxes. When $21b_i < u_j$, each island side is shorter than $5u_j$ and each hull side contains at most six cubes. An aligned interval of length nu_j containing the corner has at most

$n+1$ positions. Summing over $1 \le n \le 6$ gives at most 27^4 distinct small hulls containing the corner. The exempt family consists of geometric hulls, so coincident hulls count once here. Thus $m_H \le 4^4 + 27^4$, also for the surviving subfamily or its hulls on the next grid with the same torus margin.

6.3.4. *Equality of the filed recipes.* We use the actual local representation of the step: walk terms are filed at their full carriers; pivot charts, means and Jacobians carry their defining double blocks; Mayer selections retain the whole selected potential domain; a cutoff selection retains its bond collar; and a connected logarithmic term is filed at the union of its polymer domains. The normalized Gaussian functions are restricted by their prescribed hereditary rules. These are the literal recipes of § 4.5, not merely numerical bounds on a gas. Every dependency must be included in the carrier before this convention is applied.

Proposition 6.6 (The whole-lattice recipe on an empty trace)**.** Fix the reference numbers, the physical action weights and one coherent input list. Let $E_h(X,\cdot)$ be a filed term made from a standing history h, and let $E_\emptyset(X,\cdot)$ be the term made by the whole-lattice step with that same input. If $X^{+1} \subset \Lambda_n(\mathcal{S}_n)$, then

$$E_h(X,P) = E_\emptyset(X,P) \tag{6.xxxviii}$$

on every common admitted local pair domain. The assertion applies to the local Gaussian normalizer, the mismatch leg, the interaction terms, and their declared pointing. It asserts equality of functions at the same pair P, rather than equality of two generally different global backgrounds. ■

Proof. By Lem. 6.5, the trace on the entire set of cells used by a carrier in X is empty. The local regions, tiers and integration-bond types therefore coincide with the whole-lattice ones. In particular a carrier contained in X has no external or crossing integration bond. Its cutoff has the ordinary full threshold. A mean term whose full carrier is contained in X cannot reach an external bond, so that term is zero. The unlocalized global Gaussian mean can still contain walks from the exterior; their full carriers are not contained in X.

Each local differential coefficient and each walk term depends only on the pair, regions and bonds in its full carrier. At the same P, these data agree.

For the axial Gaussian, include the multiplier in this carrier check. The second variation of the pivot row is local to its double block, but it is paired at the pivot with the restricted value of $\mathsf{H}_1^{\mathsf{T}}\mathbf{J}$. Expand this multiplier into its own walk terms before filing the pivot contribution. Attach each such walk at the pivot and retain its padded hull and current read. For the same indexed tree in the two axial kernels, match every layer on that hull and the admitted pair at its inner bonds. The active mask is the same on bonds with both endpoints in the hull. The extensions by zero and restrictions therefore give the same matrix entries in the two settings, term by term. This gives equality for the native axial covariance and precision terms after the free-coordinate and mask identifications. If every term shorter than m_0 matches, the two remaining native tails give a kernel difference bounded by

$$C2^{-m_0} e^{-\delta d_{\mathrm{ax}}(b,b')},$$

where d_{ax} is the distance between bond origins in the axial lattice units. Indeed the absolute sum at length m has the bound $C2^{-m-1}$ times this spatial exponential. Summing the geometric series and then the rows and columns with a smaller spatial weight gives the weighted Schur bound $C2^{-m_0}$. An escaping multiplier branch is part of this same tail.

A local pivot chart and its coarea factor agree by uniqueness of their admitted local solution. The contour gauge and its normalizer use the same contours and reference weight. Coherence of the input identifies each old activity and its own counterterm used in X. Hence all individual potentials agree, including the fields and current produced by their prescribed local substitutions. This last assertion is on their common admitted domain; it does not enlarge either domain.

Now apply the finite expansion. A Mayer term is a product of the same selected potentials and the same cutoff indicators. Its local Gaussian functional has the same coefficients and the same real integration variables. Every coarse-parameter derivative and integral in its polymer activity therefore agrees. A logarithmic term on X uses only polymers contained in X, evaluated by hereditary restriction, so every summand in its connected series agrees. Absolute convergence, or equality at each finite truncation followed by the same absolute majorant, proves eq. 6.xxxviii. Bond roots and midpoint tents are the same in both recipes, so the conclusion also passes to their actual pointing. □

The same proof shows that a nonzero difference between the two recipes must have a carrier whose trace window meets the nonempty standing trace. It cannot be filed on an interior domain and then assigned a different, smaller carrier.

6.3.5. *Hull entropy rooted outside the exempt regions.* We supply the geometric sum needed for old boundary terms. On the grid of level j, let H_j be the hulls of the islands still alive at level n. Their multiplicity at a cube corner is at most m_H. Define $d_j^\natural(X)$ by minimizing the length of a tree of cube edges in X, which meets every cube outside a selected subfamily of complete hulls and meets every selected hull. Edges inside a hull still cost their ordinary length; only the requirement to visit all its cubes is removed.

Lemma 6.7 (A root sum independent of island sizes)**.** If the cube q is contained in no exempt hull, then for $\kappa > c_E$,

$$\sum_{X \ni q} e^{-\kappa d_j^\natural(X)} \le C_H \sum_{m \ge 0} e^{-(\kappa - c_E)m}, \qquad c_E = 2\log 8 + (16 + m_H)\log 2. \tag{6.xxxix}$$

The same count with a given root vertex v holds for domains whose chosen minimizing tree contains v, including vertices in an exempt hull. It also bounds the sum restricted to tree lengths $m \ge D$ by starting the series at D. The constants are independent of the side lengths and volumes of the hulls, the torus size and the number of levels. ■

Proof. Choose, for each X, a minimizing tree and an exempt subfamily by a fixed ordering. Its length in cube-edge units is an integer m. Since q is not exempt, the tree meets one of its at most sixteen corners. Root it at the first such corner. A tree with m edges has a depth-first tour of length $2m$; the lattice graph has degree at most eight. There are at most $16 \cdot 8^{2m}$ choices for this rooted tree, with an overcount which only helps.

Every nonexempt cube of X meets a tree vertex, and at most sixteen cubes meet one vertex. Every selected hull also meets a tree vertex, and at most m_H hulls pass through that vertex. Thus X is recovered by at most $16 + m_H$ binary choices per vertex: choose the incident cubes and the full hulls. There are at most $m+1$ vertices. This gives at most $16 \cdot 8^{2m} 2^{(16+m_H)(m+1)}$ encoded domains. The description is permitted to count disconnected or inadmissible choices as well. Summation with $e^{-\kappa m}$ proves eq. 6.xxxix. For a specified vertex root the same encoding starts at that vertex, and the factor sixteen for its choice is unnecessary. Large hulls add one choice, rather than their volume. □

Lemma 6.8 (A carrier crossing the clean collar)**.** Suppose a level-j carrier meets W, and its trace window meets the standing trace. Under eq. 6.xxxvi and eq. 6.xxxvii, after increasing C_0 by the fixed hull collar,

$$d_j^\natural(X) \ge c_* L^{n-j} s - C_*. \tag{6.xl}$$

The constants depend on the fixed grids and collars only. ■

Proof. The trace padding eq. 6.xxxv shows that the carrier reaches to current distance at least s from W. All current coarsenings of its exempt hulls lie outside the clean interior by eq. 6.xxxvii. An admissible tree must therefore meet a cube near W, which is not exempt, and either a second

nonexempt cube at distance at least s, or a selected hull outside that interior. In the latter case the tree still has to reach that hull. Its connecting path pays ordinary geometric length, including every portion inside a hull.

In physical units that path has length at least $cM\xi_n s - CM\xi_j$, allowing for the two touched cube corners. Division by $M\xi_j$ gives eq. 6.xl. The same argument on the periodic graph uses its intrinsic distance. It does not assume that the whole torus has one nonwrapping chart. □

6.3.6. *Summing old boundary origins in their actual units.* Let the observation region contain N_W current cubes. It meets at most $CL^{4(n-j)}N_W$ cubes at level j. Allow a fixed-origin multiplicity A_j, and suppose the actual local boundary majorants satisfy

$$b_j(X) \le B_j e^{-\kappa d_j^{\natural}(X)}, \qquad A_j B_j \le B_*(1+n-j)^p L^{q(n-j)}. \tag{6.xli}$$

In this hypothesis $d_j^{\natural}$ uses only hulls still alive at the final level n. This is an explicit admission condition on the surviving boundary filing, not a consequence of a creation-time bound using a larger exempt family. A term whose carrier contains no island absorbed between j and n has the same two lengths, so its old bound suffices. At an absorbed hull use the refiling described below.

A simple example shows the necessity. Let H be a large exempt box and q an adjacent cube. The old hull length of $H \cup q$ is zero. After H is removed from the exempt family, a covering tree must visit its many cubes and has length tending to infinity with their number. A bounded nonzero function filed on $H \cup q$ can satisfy the old majorant and fail every proposed bound with the same amplitude and the new length. Deleting the hull from the norm without refiling the function is therefore invalid.

6.3.6.1. Boundary terms at an absorbed island. If a carrier X of origin level j contains the level-j hull of an absorbed island, X meets the treated region and enters the near list of that operation. Consume the old member in the localized interaction. Fill its slot from the reduced history obtained by erasing the past of the treated components, and subtract that same replacement in the localized interaction. The erased history is a genuine history by the reduction rule. The two changes preserve the sum of the exponents.

For an old boundary slot its donor has the same surviving hulls in X as the completed label, so its hull length equals the required output length. At the new level the donor's length is at least the required length. Carry the actual donor member with its birth core and its own level and origin constant. Members made by the current operation use their new bounds, and the pointed regular entries are formed anew with the output base points. Each slot holds one term, and at most four boundary origins occur at one level and carrier. This transfers the decay to the output length without using the old member's discount for the absorbed hull.

At the new level t there is also a change of slot for a domain disjoint from the treated region. Put $A = \Lambda_t(\mathcal{S}^+)$ and write Z_1, Z_2 for the unions of treated components of the first and second classes. The two output action regions are $A \cup Z_1$ and $A \cup Z_1 \cup Z_2$. The one-layer window X_t^{+1} can reach one of the treated components even when X does not. The window criterion tests the new filing by inclusion of X_t^{+1} in these regions. For a former $\mathbf{T}$-boundary member with X disjoint from the treated region, if an outside cell of its window survives, the member remains in the $\mathbf{T}$ slot. If all those cells are treated and at least one is in Z_2, the member goes into the island remainder slot H. If all are in Z_1, it goes into the regular remainder range.

The unpointed contribution supplied by $\mathbf{T}$ occupies exactly one of these places. The change of filing gives no second copy of the function. The rule that treats a carrier as near whenever its window meets the treated region supplies the same member as the rule by slots, from the reduced history. In particular a member in the regular remainder range is supplied on that range in the donor's pre-healing label. On the narrower carry region we must identify the passed function with that member before applying its regular remainder bound. Use the comparison of the unfiled functions under erasure of the deleted histories, with its carrier and stage hypotheses. For the

pointed sum, split the old exterior base points into the still exterior points and the points newly in the one-layer erosion of $A \cup Z_1 \cup Z_2$. The latter points enter the regular sum once.

Apply the carriage bound to the standing list of an output label. At a level and carrier the bound is per origin. A carried or donor member occupies that slot with its own bound. A member formed at this step occupies its new slot. The bounds for the nonpointed origins are therefore returned with amplitude

$$\max\{\mathsf{B}_{\mathrm{in}}, \mathsf{B}^{\mathbf{T}}, \mathsf{B}^{\mathbf{S}}, \mathsf{B}^{\mathbf{R}}\}$$

in the output hull length. For the pointed origin use the new sum of regular members over its output base points. Its bound is the pointed norm summed over those points.

For the ordinary formation the localized Gaussian normalizer has amplitude B_{N}. The other three boundary kinds have amplitude at most one each on the declared step box once the coupling ceiling makes their fluctuation factor times the inherited sizes small. Thus $\mathsf{B}^{\mathbf{T}}$ can be taken as $\mathsf{B}_{\mathsf{N}} + 3$, with the ceiling chosen after the inherited bounds. The stage and main-step origins require their own formation amplitudes on their actual domains.

For the interaction part of the stage amplitude, factor the old term evaluated at the new stage background with zero collar field out of the stage integral. The integrated field's switch is removed at that evaluation. The remaining potential is its difference from that evaluation. Each cluster of the input-dependent logarithm therefore contains such a difference vertex. The stage Cauchy bound gives it a factor $CB_*(\log \theta_t)^{-e}$, where B_* bounds the inherited entries on the admitted collar charts. Sum the rooted cluster bounds over the performed stages. Even when the collars overlap, the count is at most $C(\log \theta_t)^r$, and $e > r$ gives a uniform bound at the stage rate. The input-independent normalizers use their own summability estimate, with the field-independent scalars kept outside the carrier logarithms. For a fixed domain, the sum of the stage terms formed on it occupies one stage slot. The Wilson profile differences enter the action profile sum separately.

The identity on the retained variables of a standing list is the sum over the real histories that return it. The resulting measure uses these filed terms and keeps the stored variables of that list. Hence a uniform bound over the actual output lists gives the bound at the next exterior disintegration, which uses this positive measure. There is no separate exponent of each history in that sum.

The same filing applies at source w and at zero. A carried or refilled member keeps its birth function and field space. The output read uses the cut old list and the restriction of the stored tuple to the carrier. At the fields already integrated by a stage, the old function is read at zero and the associated parameters are omitted. This restriction is independent of w.

On the fixed source class of radius r_{s} at a stop k, the sourced step bounds the difference of a boundary member from its zero-source value by

$$2r_{\mathrm{s}}^{-1}\|w\|_{\flat,K-k}\, B_{j,o} e^{-\kappa d_j^{\natural}(\mathcal{S};X)}.$$

Here $B_{j,o}$ is the boundary majorant's amplitude at origin o and $\mathcal{S}$ is the label at which the member is read. The difference reads w on $X^{\sim}$. At a refilled slot the donor length is at least the required output length, so apply the same bound to the restriction on the output list. Apply the sourced formation estimate to each newly formed boundary member on its own domain. The pointed origin uses its new sum of pointings, with the spare decay paying their count. Repeating the step yields the bound in the surviving hull norm at the witness, on the same source class. This uses the sourced induction with its frozen frame, tube room and constant-source data. The output evaluation must lie in the native restricted field space.

A polynomial in $\log \theta_n$ can be included in B_*. The factor A_j counts the actual origins and slot multiplicities; a fine-origin incidence is not counted as one coarse incidence. Suppose every term being summed has a boundary carrier as in Lem. 6.8.

Proposition 6.9 (The old boundary tail)**.** If $\kappa/2 > c_E$, there are $s_*, c, C > 0$, depending on the displayed rates and powers but not on n, such that for $s \geq s_*$,

$$\sum_{j\leq n} A_j \sum_{\substack{X \text{ meeting } W \\ X \text{ has a boundary carrier}}} b_j(X) \leq CB_* N_W e^{-cs}. \tag{6.xlii}$$

The same estimate bounds the oscillation when $b_j(X)$ is a bound on the complete evaluated local term on one common chart; subtraction of two values costs a factor two. ■

Proof. Root a domain at a fine cube meeting W. None of these cubes is contained in an exempt hull. Spend half its decay on eq. 6.xl, and sum the remaining half with Lem. 6.7. The level-j contribution is bounded by

$$CB_j A_j N_W L^{4(n-j)} e^{-c\kappa L^{n-j} s}.$$

Put $a = n - j$. Equation eq. 6.xli leaves the scalar sum $\sum_{a\geq 0}(1+a)^p L^{(q+4)a} e^{-c\kappa L^a s}$. Factor out $e^{-c\kappa s/2}$. The remaining sum is uniformly bounded for $s \geq s_*$: its ratio at successive integers is bounded by a fixed power times $e^{-c\kappa s(L-1)L^a/2}$, which is at most one half once s_* is fixed sufficiently large. The finite initial factor is harmless. This proves eq. 6.xlii without a cutoff-dependent sum over old levels. □

For example, choose

$$R \geq C_{\mathrm{tr}} \check{R}_n + C_0 + s_* + C_1 \log x. \tag{6.xliii}$$

Then the tail is bounded by $CB_* N_W x^{-cC_1}$. If $\check{R}_n \leq C(\log x)^r$, the enlarged window still has only $O((\operatorname{diam} W + R)^4)$ current cubes, hence a polylogarithmic number of retained coarse link variables. The cell margin and the exponential margin are paid separately in eq. 6.xliii.

6.3.7. *The input which the clean region retains.* Equation eq. 6.xxxviii compares two recipes with the same coherent input list. A regular marked member generated from a previously healed region belongs to that list even when the standing trace is empty. Its small norm is estimated as a marked input, rather than set to zero. Likewise, the complementary Gaussian cutoff factors used in the finite expansion of a passed sharp test can have their roots inside the clean region. Their size is the rooted Gaussian tail of eq. 4.cxxxix; they are not boundary-history carriers.

Thus the local complete exponent separates into the whole-lattice Gaussian and regular interaction recipe, the retained small marked input, the cutoff correction, and the boundary tail eq. 6.xlii. To identify a conditional law in retained coarse coordinates one must in addition identify its physical background substitution and its coarea/domain image on one admitted chart. A bound for $E_h(X, P)$ as a function of P does not prove that two physical substitutions have the same arguments. The geometric proof here establishes the local recipe and its boundary-carrier tail before that separate analytic step.

The generating function on the fluctuation segment. Write $\mathsf{r} = \hat{\mathsf{d}}$, $g = x^{-1/2}$ and $\bar{v} = gp_0$. Let $\phi_a = \widetilde{f}_a$, $a = 1, 2$, and let X_a be the two bounded, windowed observables of the witness field. The tents move a support by at most CMa_s, so the enlarged supports are still separated by at least $\mathsf{d}/2$ when the room constant is large. Use the union of their bare plaquette sets in the readout construction. Each set has at most $C\mathsf{r}^4 \eta_{k_0}^{-4}$ plaquettes.

For step j, let γ_j^0 be the flat Gaussian marginal on the free read bonds and let $\mathcal{W}_j$ be their product of balls. Denote by T_j a uniform bound for the change of expectation of a function of these bonds of supremum at most one, from the rotated actual law at a compatible exterior to $\gamma_j^0(\cdot \mid \mathcal{W}_j)$. The bounds above give, on the admitted reference branches,

$$T_j \leq C\left[e^{2\epsilon_j^{\mathrm{old}}} - 1 + e^{\Xi_j} - 1 + n_j \widehat{g}_{j-1} b_j^3 + t_j + n_j \varpi_j\right], \qquad \Xi_j \leq C n_j b_j^2 e^{-c\mathsf{D}}.$$

Here $t_j \le Cn_j e^{-cb_j^2}$ also bounds the Gaussian error from the box and read-set cutoffs. For this bounded generating function remove both the old-origin tilt and the intrinsic remainder at each original fixed exterior e. If P is a removed part, the density ratio on the product window is $e^P/\langle e^P\rangle$, where the expectation uses the normalized law after that removal. Thus the ratio lies between $e^{-\operatorname{osc} P}$ and $e^{\operatorname{osc} P}$. The old-origin bound gives the first term in T_j. The full intrinsic oscillation at the original stored values is at most $Cn_j\widehat{g}_{j-1}b_j^3$ by §6.3.0.4. This is the third term in T_j at weak coupling. Both removals use the same product window and the same physical background.

The resulting quadratic law on the box has precision Q^e_{II} and linear term $Q^e_{IO}y_O + (\ell^e_E)|_I$. Compare it with the centered law on the same window whose precision is $K_e = (((Q^e)^{-1})_{II})^{-1}$. The force difference is exactly

$$(Q^e_{II} - K_e)y + Q^e_{IO}y_O + (\ell^e_E)|_I.$$

The first term is the Schur correction. All three terms are estimated with the quadratic weighted row bounds at e. The gap and marginal-density argument give the term with Ξ_j, without a comparison of nonlinear evaluations at different exterior values. Remove the centered Gaussian box window, compare its read covariance with the flat one, and impose the flat read window. The two window changes are included in t_j. Use the same Wilson ratio x_{j-1}/θ_{j-1} in the flat precision, as in the Gaussian comparison in the rotated coordinates. The final flat read law is the same for every compatible exterior. Average the normalized estimates against the original positive exterior mixture.

Take the source radii

$$|\lambda_a| \le \Lambda_a = \frac{c_* x}{2\mathsf{r}^4 p_0^2 \|\phi_a\|_\infty} \quad (a = 1, 2), \qquad z_p = \sum_{a=1}^{2} \lambda_a \phi_a(x(p)).$$

Zero profiles may be omitted. The constants below hold with a fixed reserve in these radii. Let ξ_j be independent flat read variables with the windowed law just specified. Put

$$M_{\le j}(p) = i \sum_{k_0 < l \le j} \widehat{g}_{l-1}(H^{\square}_{l-1}\mathsf{C}^0\xi_l)(p), \qquad Z_j(P) = \mathbb{E}\exp\left(\tfrac{1}{2}\sum_p z_p |M_{\le j}(p) + P(p)|^2\right).$$

The norm in the exponent is a norm of real-field matrices. The complex variables are the source coefficients. The row bounds of the linear readout, $b_l \le Cp_0$ and the geometric sum of η_{l-1}^2 give $\sup_p |M_{\le j}(p)| \le C\eta_{k_0}^2\bar{v}$. The same size holds for the remaining plaquette backgrounds and the intermediate arguments in a one-step replacement. Together with the count of plaquettes this gives $|Z_j| \le C$ and

$$|Z_j(P) - Z_j(P')| \le C\frac{\sup_p |P(p) - P'(p)|}{\eta_{k_0}^2\bar{v}}.$$

Indeed one integrates the change of the exponent along the segment between P and P'. Both bounds hold on the same source polydisc.

At each remaining level use one prescribed complete axial tree on the common read box, with its root and all paths included in the read. Gauge-transforming the field before putting it in this fixed gauge changes the resulting field by one constant rotation. Use the tower frames of the localized readouts within this gauge. Simultaneous rotation of all arguments of Z_j leaves it unchanged, since the flat laws and their balls are colour invariant and the flat maps act separately on each colour. Hence

$$Y_j(V_j) = Z_j(P^{(j)}(A^{\mathrm{ax}}(V_j)))$$

is a bounded gauge-invariant function on the admitted readout domain. It can be set to one outside that domain. The read sets contain the entire dependence on the tree and the old readouts. In particular an operation on an exterior component of a clean string leaves that read unchanged. At $j = k_0$ take $Y_{k_0} = e^{\lambda_1 X_1 + \lambda_2 X_2}$ on the whole space. On the clean branch it is Z_{k_0} of the localized

plaquette defects. The equality uses $a_p(U) = \frac{1}{2}|U(\partial p) - 1|^2$ in the normalized SU(2) norm and the fact that the witness tests pass there.

The one-step identity for the localized defects replaces the argument of Z_{j-1}, on the actual fluctuation window, by

$$i\widehat{g}_{j-1}H_{j-1}^{\square}\mathsf{C}^0\widetilde{y} + P^{(j)}(A^{\mathrm{ax}}(V_j)).$$

Its pointwise error is bounded by $C\eta_{j-1}^2[(\mathsf{r}gp_0)^2 + gp_0e^{-c\Delta_j}]$, with Δ_j the native readout margin. This comparison uses one colour convention for both supports. The residual constant rotation drops out of Z_{j-1}. The remaining expression depends only on the free read coordinates and the retained V_j. The T_j bound and one integration with $\gamma_j^0(\cdot \mid \mathcal{W}_j)$ now give

$$\left|\mathbb{E}_j^{\mathrm{ref}}Y_{j-1} - Y_j\right| \le C\left[L^{-2(j-1-k_0)}(\mathsf{r}^2gp_0 + e^{-c\Delta_j}) + T_j\right].$$

For the unconditioned comparison below, $\mathbb{E}_j^{\mathrm{ref}}$ is the box law obtained from the original adjacent-level law after summing the standing parts and holding the compatible exterior fixed. Its good current selector is then constant on the product window. The comparison to γ_j^0 is included in T_j. The last integration is exactly the independent addition of the j-th slice in the definition of Z_j. No change to the exterior weights is made.

For the unconditioned two-point function use the positive joint law of each adjacent pair of levels. Let μ_j be the standing law with its stored variables retained, and let π_j be the joint law of the original step from μ_{j-1} to μ_j, including the string of moves. Its marginals give the exact telescoping identity

$$\mathbb{E}_{\mu_{k_0}}Y_{k_0} - \mathbb{E}_{\mu_{k_1}}Y_{k_1} = \sum_{j=k_0+1}^{k_1}\mathbb{E}_{\pi_j}(Y_{j-1} - Y_j).$$

In the j-th term sum the earlier histories over their standing parts in both the inserted and uninserted measures. Use the good current condition on the complete read box when it selects an exterior weight of the resulting integrals. At fixed exterior, the indicator comparison for that step makes it constant on the product window, with the redundant interior flags summed out. Disintegrate at the ordinary output before the large-field moves. On the good current part those moves are exterior to the read and preserve Y_j, including under the gauge changes. The coarea integral with the insertion has the same weight as the integral without it. After normalization the box expectation is mixed with the exterior law weighted by the uninserted box integral. If d_j denotes the bound for the box comparison and p_j the mass of the excluded current part, the j-th term has modulus at most $d_j + 2Cp_j$, using the bounded extensions of both Y's. For a good condition defined on finer strings, we must verify this factorization after the standing-part sum.

Thus this comparison uses the summed one-step exceptional mass. It allows the earlier healed histories in the incoming standing law. On the prescribed reference torus the age and creation bounds control the probability of a bad current part by summing the cells that meet the padded read, with the lag estimate where the reference requires it. A comparison conditioned on a terminal exterior and on the full string uses the same-exterior estimates separately. If a finer class changes this box law by a normalized Radon–Nikodym factor in $[e^{-\epsilon}, e^{\epsilon}]$, the expectation of the bounded insertion changes by at most $(e^{\epsilon} - 1)\|Y_{j-1}\|_\infty$. In that comparison charge the class restriction at the step where it enters.

6.3.7.1. The top field and the Gaussian windows. To bound the last field, first use the native level-k_1 plaquette tests. The top read of the comparison uses its own localized minimizer. Its box need not be a test cube. Include the complete backward corner blocks for every top plaquette in its determining read. Assign each coarse plaquette to its native test cube by its barycenter. One test layer is wider than the complete corner stencil. At a passed native test the tested background has bare plaquette defect at most $\varepsilon_{k_1}\eta_{k_1}^2$ on those blocks. It has the prescribed level-k_1 averages on the four rows of the plaquette. The local averaging estimate of Prop. 3.4 therefore gives $|\partial V_{k_1} - 1| \le 2\varepsilon_{k_1}$ there.

The localized Landau curvature estimate on the top comparison box then gives

$$\sup_p |P^{(k_1)}(p)| \le C\eta_{k_1}^2 \varepsilon_{k_1} \le C\eta_{k_1}^2 g p_0.$$

The last inequality uses the running coupling bound over the segment. The transfer from the native test uses the scalar curvature of the block datum.

A failed or undefined native test is a creation. A skipped test lies in the margin of an earlier live component. An operation at the top touching a complete read has a live predecessor. Thus the terminal exception is covered by the live and creation cells of levels $k_1 - 1, k_1$ meeting the padded reads and operation margins, before the terminal field is formed. On the clean part, subsequent exterior moves do not read or change the determining data of these plaquettes. The same finite-band rarity estimate bounds p_{top}, with the top stencils included in its cell count. On the part of the original terminal law where this test fails, set the top argument P to zero. The bounded extensions allow this replacement at cost Cp_{top}, where p_{top} is the mass of that part. Keep the original terminal law in the expectation of the resulting $Z_{k_1}(P)$.

Write $Z_\infty(P)$ for the unwindowed flat read expectation and C_{loc} for its one-colour covariance. Set $F_a = \mathrm{diag}(\phi_a(x(p)))$ and $h = \lambda_1 F_1 + \lambda_2 F_2$. Decompose the real matrix field P orthogonally into its component in the three-colour Gaussian subspace, with coordinates p_c, and its perpendicular component $P^\perp$. Then

$$\frac{Z_\infty(P)}{Z_\infty(0)} = \exp E_P(\lambda), \qquad E_P(\lambda) = \frac{1}{2}\sum_{c=1}^{3} p_c^{\mathsf{T}} h(1 - C_{\mathrm{loc}}h)^{-1} p_c + \frac{1}{2}\sum_p h_{pp}|P^\perp(p)|^2.$$

To obtain this identity, write each colour as $C_{\mathrm{loc}}^{1/2}$ times a standard Gaussian and complete the square for real sources. This also works for singular C_{loc}. The weighted trace bound gives convergence and continuation to the source polydisc. More explicitly, put $D_\lambda = \sum_a |\lambda_a||F_a|$ and write $h = D_\lambda^{1/2} S D_\lambda^{1/2}$ with $\|S\| \le 1$. The trace of $D_\lambda^{1/2} C_{\mathrm{loc}} D_\lambda^{1/2}$ is at most a fixed constant less than one, also for the doubled radii after reducing c_*. The Neumann series therefore bounds $|E_P|$ by $C\sum_p (D_\lambda)_{pp}|P(p)|^2$. The plaquette count, the radii and the top-field bound make this $O(L^{-4\mathsf{n}})$.

Let $t = \sum_j C n_j e^{-cb_j^2}$. The joint flat window conditions the full Gaussian on an event of complement probability at most t. At each fixed P the determinant and the bound for E_P, applied to the doubled real sources, bound the second moment of the insertion uniformly. Cauchy–Schwarz bounds the deleted integral by $C\sqrt{t}$, and the conditioning denominator costs Ct. Consequently the terminal mixture

$$Z_{\mathrm{term}} = Z_\infty(0)\, \mathbb{E}_{\mathrm{term}} e^{E_P}$$

approximates the windowed mixture within $C\sqrt{t}$. Here $\mathbb{E}_{\mathrm{term}}$ is expectation under the original terminal law with the argument P defined above. The Gaussian integral is evaluated at each fixed terminal datum before taking that expectation.

Finally, the local/global flat comparison replaces the centered determinant $Z_\infty(0)$ by the whole-torus one, with error $Cp_0^{-2}e^{-cd_*}$. Write Z_{free} for this determinant. Its covariance in one colour is $C^x = \sum_{j=k_0+1}^{k_1} x_{j-1}^{-1} \mathcal{C}_{(j)}$. The trace series gives a fixed positive lower bound for $|Z_{\mathrm{free}}|$ on the polydisc.

For the unconditioned law the generating function differs from Z_{term} by at most

$$C\left[\mathsf{r}^2 g p_0 + \max_j e^{-c\Delta_j} + \sum_j T_j + \sqrt{t} + \sum_j p_j + p_{\mathrm{top}}\right].$$

The joint laws and their conditional kernels can be chosen independently of λ. The bounds may first be imposed on a countable dense subset of the polydisc and then extended by analyticity of the bounded insertions. This compares the generating functions before taking logarithms.

6.3.7.2. Returning to the physical two-point function. The trace series bounds $|Z_\infty(0)|$ below. For large x, the bound on E_P also bounds $|Z_{\mathrm{term}}|$ below. The quotients of the witness generating function by Z_{term} and of $Z_\infty(0)$ by Z_{free} lie in a fixed disc about one when the corresponding errors are small. The two log differences then have supremum $C\delta$, with δ the sum of the generating-function error above and $p_0^{-2}e^{-cd_*}$. Mixed Cauchy differentiation costs $C\delta/(\Lambda_1\Lambda_2)$. Relative to $x^{-2}\|\phi_1\|_\infty\|\phi_2\|_\infty$ this is the factor $C(\mathsf{r}^4p_0^2)^2$ times δ.

The remaining mixed derivative is that of $\log \mathbb{E}_{\mathrm{term}}e^{E_P}$ at zero. Put

$$A_a(P)=\frac{1}{2}\sum_p \phi_a(x(p))|P(p)|^2.$$

Expansion of the resolvent to first order in h gives this derivative as

$$\mathbb{E}_{\mathrm{term}}\sum_{c=1}^{3} p_c^{\mathsf{T}}F_1C_{\mathrm{loc}}F_2p_c+\mathrm{Cov}_{\mathrm{term}}(A_1,A_2).$$

Indeed, $\partial_a E_P(0)=A_a(P)$ and the two orders of $F_1, C_{\mathrm{loc}}, F_2$ give equal quadratic forms by transposition. The covariance comes from differentiating the logarithm after averaging. On the union of the test supports $|C_{\mathrm{loc}}(p,p')|\le Cx^{-1}\eta_{k_0}^4$ by the diagonal bound and positivity. With at most $C\mathsf{r}^4\eta_{k_0}^{-4}$ plaquettes and $|P|\le C\eta_{k_0}^2 gp_0L^{-2\mathsf{n}}$, the two terms are bounded at scale $x^{-2}\|\phi_1\|_\infty\|\phi_2\|_\infty$ by $C\mathsf{r}^8p_0^2L^{-4\mathsf{n}}$ and $C\mathsf{r}^8p_0^4L^{-8\mathsf{n}}$, respectively.

Take $\mathsf{D}=A\log x$, with A fixed sufficiently large. This fits the test margins since $\check{R}_{k_0}/\log x\to\infty$. The witness room gives $\mathsf{r}\asymp p_0^2$, and $\mathsf{D}+\check{R}_{k_0}=O(\mathsf{r})$ in the chosen range of exponents. The box count gives $n_j\le C\mathsf{r}^4$, and $\mathsf{n}=O(\log x)$. Thus the activity contribution to the old oscillations, estimated with one power of the ratio of radii, obeys

$$\sum_j Cn_j(\log x)^{\mathfrak{p}_0-q}\le C(\log x)^{1+9\mathfrak{p}_0-q}.$$

After the mixed Cauchy loss this is $O((\log x)^{1+29\mathfrak{p}_0-q})$, so take $q>1+29\mathfrak{p}_0$ as well as the earlier lower bounds. On the admitted branches the other old terms use the power and collar bounds in the old oscillation estimate. The intrinsic and covariance terms involving $x^{-1/2}$ have only fixed logarithmic prefactors. The two collar distances and the box margin give $e^{-c\mathsf{D}}$ up to fixed constants, which also absorbs any fixed logarithmic power. The window error uses $b_j\asymp p_0$, and $L^{-4\mathsf{n}}\le Cx^{-8}$.

For the geometric exceptions, the union of padded reads fits in the level-k_1 neighborhood counted above. Its cover at level j costs at most $C(\mathsf{r}+\mathsf{D}+\check{R}_{k_0})^4L^{4(k_1-j)}$ cells. Sum the one-cell bounds for live and creation events over these covers, including the predecessors of the top tests. This bounds their contribution to $p_{\mathrm{top}}+\sum_j p_j$ by

$$C(1+\mathsf{n})(\mathsf{r}+\mathsf{D}+\check{R}_{k_0})^4L^{4\mathsf{n}}e^{-cp_0}.$$

Here $L^{4\mathsf{n}}=O(x^8)$ and $\mathfrak{p}_0>1$, so this too is negligible after the Cauchy loss. This count uses the indicated geometric exceptions. When the native decomposition with attached terms supplies the conditional box estimate, charge the failure of its per-bond budget separately. Its bound summed over the segment is $C\sum_j n_je^{-cp_0}$, also negligible after the Cauchy loss. For this native budget split the adjacent law at the incoming list of that step. The indicator is fixed at each exterior, so the box normalization on the passing part is unchanged. On its complement the bounded difference of the two insertions costs at most a constant times the failure mass. Other refinements must still use the factorization of their actual history law.

The flat covariance contribution is $\frac{3}{2}\sum_{p,p'}\phi_1(x(p))\phi_2(x(p'))C^x(p,p')^2$. In the linear combination of the slices, replacing x_{j-1}^{-1} by x^{-1} costs at most $C(B_\sharp\mathsf{n}+c_5)x^{-2}$ times the sum of the absolute slice kernels. On separated supports that sum is bounded by $C\eta_{k_0}^4\mathsf{r}^{-4}$ by the exponential slice bound and its geometric sum. Multiplying by the sum of the two covariances to compare their squares,

and then summing the plaquettes, gives a relative error $C(B_\sharp \mathsf{n} + c_5)/x$ in the smeared covariance at scale x^{-2}. The cross term and square from the omitted coarse covariance give the displayed coarse-complement error of the free calculation.

Apply this conclusion in the physical-source identity at k_0. There the small mixed-source term and the errors in the two first derivatives are the source estimates in (a) and (b). The leading covariance is ${\zeta'_{k_0}}^2\langle X_1; X_2\rangle$. This identifies the covariance of the bounded witness observables in the physical-source identity. The source remainder bounds at k_0, the nondegeneracy for the chosen class of tests, and the reference-law transfer to the stated Gibbs states keep their separate roles.

6.4. **Positive history sums and averaged removal of exceptional runs.** The passage from a run window to a connected correlation has two normalizations. One first sums the positive histories that represent the same physical event, and then normalizes their total mass. Conditioning each history separately generally gives different weights. The purpose of this section is to perform these operations in their correct order and to estimate the error from exceptional runs without a uniform lower bound on the clean probability at every exterior field. The resulting estimates concern the second and fourth connected correlations used in the short-distance argument.

All probability measures in this section are real and positive at zero source. A differentiated polymer resolution can have signed or complex terms; it must first be identified with the positive physical measure before the statements below apply. No estimate on the absolute values of a different resolution makes that identification automatically.

6.4.1. *Pushforward measures and the multiplicity of a history class.* Let H be finite or countable. For $h \in H$ let σ_h be a finite positive measure on a standard Borel space Ω_h, and let $T_h : \Omega_h \longrightarrow \Omega$ be measurable, where Ω is also standard Borel. Assume

$$0 < Z := \sum_{h\in H} \sigma_h(\Omega_h) < \infty. \tag{6.xliv}$$

The common physical measure is

$$\mu = Z^{-1} \sum_{h\in H} (T_h)_* \sigma_h. \tag{6.xlv}$$

For a nonnegative measurable F, its integral is the sum of the integrals of $F \circ T_h$. This follows first for indicators from the definition of a pushforward and then for general F by monotone convergence. For an integrable complex F, apply this result to its absolute value before taking real and imaginary parts.

Suppose H is partitioned into classes $\mathcal{C}$. Define the unnormalized measure of a class by

$$\Sigma_C = \sum_{h\in C} (T_h)_* \sigma_h, \qquad C \in \mathcal{C}. \tag{6.xlvi}$$

Then $Z\mu = \sum_C \Sigma_C$. This is an equality of positive measures, including all Jacobians and all tests present in the individual histories. Rearrangement of the sum is justified by positivity. It does not require a choice of one representative configuration for the class.

Proposition 6.10 (The density of a resummed class)**.** Suppose there is a positive measure ν_C such that $(T_h)_*\sigma_h = a_h \nu_C$ for every $h \in C$, with $a_h \geq 0$ measurable. Then

$$\Sigma_C = a_C \nu_C, \qquad a_C = \sum_{h\in C} a_h. \tag{6.xlvii}$$

In particular the multiplicity can be taken outside an integral over the class variables only if a_C is constant almost everywhere in those variables. ■

Proof. For every nonnegative F, monotone convergence gives $\sum_h \int F a_h \, d\nu_C = \int F \sum_h a_h \, d\nu_C$. This proves the equality of measures. A constant factor outside all such integrals is equivalent to a constant Radon–Nikodym density. □

Equality of the coordinate domains, or independence of the histories from the source parameter, does not imply constancy of a_C. For example the two measures dy and $(1+y^2)\,dy$ on $[-1,1]$ have the same domain and no source dependence, but their sum has density $2+y^2$. Both densities contribute to a source integral $\int e^{zX(y)}\,d\Sigma_C(y)$. If an exact change of variables shows that all the pushforwards are equal to the same measure, the finite cardinality of the class is a constant multiplicity. For an infinite class of identical nonzero positive finite measures, eq. 6.xliv would fail. Thus an infinite counting factor cannot be discarded using a formal cancellation of infinite normalizers.

When $a_C > 0$ is differentiable, $\log a_C$ belongs in the complete conditional exponent of § 6.3. There is an additional derivative contribution even if every summand has a simple Hessian. For a finite class write $a_h = e^{p_h}$ and $w_h = e^{p_h}/\sum_j e^{p_j}$. Direct differentiation yields

$$\nabla \log a_C = \sum_h w_h \nabla p_h, \tag{6.xlviii}$$

$$\nabla^2 \log a_C = \sum_h w_h \nabla^2 p_h + \sum_h w_h (\nabla p_h - \bar{p})(\nabla p_h - \bar{p})^{\mathsf{T}}, \quad \bar{p} = \sum_h w_h \nabla p_h. \tag{6.xlix}$$

For a countable class, a sufficient condition is that, on every compact subset K of the coordinate chart,

$$\sum_{h\in C} \sup_K a_h(1 + |\nabla p_h|^2 + \|\nabla^2 p_h\|) < \infty, \qquad a_h = e^{p_h},$$

and that $a_C = \sum_h a_h$ stays positive. These bounds justify differentiation and each separate weighted sum in eq. 6.xlviii and eq. 6.xlix. Uniform convergence of $\sum_h a_h$ and its first two derivative series alone does not justify separating two divergent terms in the logarithmic Hessian.

The second term in eq. 6.xlix is the covariance of the slopes of the summands. For $p_\pm(y) = \pm My$, each individual Hessian vanishes, whereas $(\log(2\cosh My))''$ equals M^2 at zero. Control of individual Hessians consequently does not control the Hessian after a positive history sum. The slopes or the complete summed density must also be controlled.

6.4.2. *The exterior law at zero source.* Let $\pi : \Omega \longrightarrow E$ be the exterior variable and let $\lambda = \pi_*\mu$. Standard Borel spaces admit a probability kernel $e \mapsto \mu_e$ with

$$\int F\,d\mu = \int_E \left(\int F\,d\mu_e \right) \lambda(de) \tag{6.l}$$

for every integrable F. Equalities involving this kernel are understood outside a λ-null set, on which any fixed version may be used. At a finite lattice cutoff all spaces used here are finite products of compact Lie groups, Euclidean spaces and discrete history sets, so this measurability setting applies.

The weights of separate histories in μ_e involve their conditional masses at e. In particular they are not in general the unconditional weights $\sigma_h(\Omega_h)/Z$. One way to see the distinction is to retain the history label on the disjoint union $\bigsqcup_h \Omega_h$, push it to the pair (h, e), and then disintegrate over e. The resulting conditional law of h supplies exactly the required weights. Summing first and disintegrating first give the same joint integrals when these conditional weights are retained, by eq. 6.l.

For bounded real physical insertions $X_1, \dots, X_n$, put

$$F(z) = \int_E \mu_e\Big(e^{\sum_i z_i X_i}\Big)\,\lambda(de). \tag{6.li}$$

The exterior law λ in this formula is the law at zero source and stays fixed during differentiation. This follows by inserting the exponential in eq. 6.l; no approximation is being made. All changes of exterior weights induced by a source are already present in the conditional generating functions. Factoring out an e-dependent source normalizer from those functions would require retaining that factor inside the outer integral. Dividing it out before exterior averaging would change F and its connected derivatives.

6.4.3. *What an unconditional rarity estimate implies.* Let $C \subset \Omega$ be the clean event, measurable in the joint physical and history space. Enlarge Ω to that joint space when the event is not determined by the physical field alone. Write

$$q(e) = \mu_e(C), \qquad p = \mu(C^c) = \int_E (1 - q(e))\lambda(de). \tag{6.lii}$$

The number p can be exponentially small although $q(e)$ is zero on a set of exterior fields of positive measure. For example take $E = \{0, 1\}$, $\lambda(\{1\}) = p$, and let cleanliness be exactly the event $e = 0$. Then $q(1) = 0$. Thus no positive lower bound on all the conditional clean probabilities follows from small p.

For $0 < \eta < 1$, define

$$G_\eta = \{e : q(e) \geq \eta\}. \tag{6.liii}$$

Since $1 - q > 1 - \eta$ on its complement, integration gives

$$\lambda(G_\eta^c) \leq \frac{p}{1 - \eta}. \tag{6.liv}$$

More generally the set where $q < 1 - \delta$ has mass at most p/δ. The point is that the exceptional exteriors must be retained in the error estimate; they do not disappear from a conditional statement.

For $q(e) > 0$, define the clean conditional kernel

$$\mu_e^C(F) = q(e)^{-1}\mu_e(F1_C). \tag{6.lv}$$

This kernel is a probability measure regardless of how small $q(e)$ is. Its existence says nothing about a uniform bound on its density or its derivatives. Such bounds require the actual complete conditional law. The next comparison uses only positivity.

6.4.4. *A resampling with the same exterior marginal.* It is convenient to compare laws on the feature space $\mathbb{R}^n$. Let $X = (X_1, \ldots, X_n)$ be measurable and let $\rho = X_*\mu$. Let $G \subset \{q > 0\}$ be a measurable admitted set of exteriors. For $e \in G$, use the law of X under μ_e^C; for $e \notin G$, use the point mass at $0 \in \mathbb{R}^n$. Their mixture with respect to the unchanged λ is a probability measure ν_G.

We use the total variation convention $\|\rho - \nu\|_{\mathrm{TV}} = \sup_A |\rho(A) - \nu(A)|$ for probabilities. For bounded f, the corresponding integral bound is $|\rho(f) - \nu(f)| \leq 2\|f\|_\infty \|\rho - \nu\|_{\mathrm{TV}}$. This convention is fixed to avoid an ambiguous factor of two.

Proposition 6.11 (Averaged clean resampling)**.** With the preceding definitions,

$$\|\rho - \nu_G\|_{\mathrm{TV}} \leq p + \lambda(G^c). \tag{6.lvi}$$

If $G = \{q > 0\}$, the right hand side can be improved to p. If $G = G_\eta$, the bound $p + p/(1 - \eta)$ is valid. ∎

Proof. At a fixed e with $q > 0$, the full conditional probability is $q\mu_e^C + (1 - q)\mu_e^{C^c}$, with the second term omitted when $q = 1$. Its distance from μ_e^C is at most $1 - q$. Pushforward under X does not increase total variation, since preimages of feature events are physical events. At $e \notin G$ the distance from the chosen point mass is at most one. Integrate these bounds and take the supremum over feature events to obtain $\int_G (1 - q)\, d\lambda + \lambda(G^c) \leq p + \lambda(G^c)$. When $G = \{q > 0\}$, the integral of $1 - q$ on G^c is exactly $\lambda(G^c)$, so the first displayed expression equals p. The last assertion follows from eq. 6.liv. □

Global conditioning on C gives another probability law, $\mu(\cdot \mid C)$. Its exterior marginal is $q(e)\lambda(de)/(1 - p)$ when $p < 1$. Thus global conditioning and clean conditional resampling have different exterior marginals. The latter is useful when a run-window estimate is to be integrated against the original exterior law. This is the comparison that Prop. 6.11 supplies.

6.4.5. *Weighted exceptional moments.* A small exceptional probability alone does not control an unbounded insertion. The estimate needed for a product of insertions can be stated without global moment bounds on every physical field. For each nonempty $S \subset \{1, \dots, n\}$ let

$$d_S = \mu\left(1_{C^c} \prod_{i\in S} |X_i|\right), \qquad M_S = \prod_{i\in S} M_i, \tag{6.lvii}$$

where $M_i > 0$. Assume these numbers are finite and that $|X_i| \le M_i$ on C, up to a μ-null set. The clean bound can use insertions from which fixed deterministic constants have been subtracted. It cannot subtract an exterior-dependent random constant without changing the global connected correlations.

Proposition 6.12 (Mixed moment error with weighted tails)**.** For every nonempty S,

$$\left|\rho\left(\prod_{i\in S} x_i\right) - \nu_G\left(\prod_{i\in S} x_i\right)\right| \le d_S + [p + \lambda(G^c)]M_S, \tag{6.lviii}$$

$$\left|\rho\left(\prod_{i\in S} x_i\right)\right| \le M_S + d_S, \qquad \left|\nu_G\left(\prod_{i\in S} x_i\right)\right| \le M_S. \tag{6.lix}$$

When $G = \{q > 0\}$ the term $p + \lambda(G^c)$ in the first bound can be replaced by p. ■

Proof. Write $X_S = \prod_{i\in S} X_i$. On G the difference between a full conditional moment and a clean conditional moment is

$$\mu_e(X_S 1_{C^c}) - \frac{1 - q(e)}{q(e)} \mu_e(X_S 1_C). \tag{6.lx}$$

The absolute value of the second term is at most $(1 - q(e))M_S$, because $|\mu_e(X_S 1_C)| \le q(e)M_S$. This cancellation of $q(e)$ is the reason that a uniform clean probability floor is unnecessary here. On G^c the comparison moment is zero. The full moment there has a bad part controlled by the corresponding portion of d_S, and a clean part at most $q(e)M_S$. Integrating these estimates gives

$$d_S + M_S \left[\int_G (1 - q)\, d\lambda + \int_{G^c} q\, d\lambda\right]. \tag{6.lxi}$$

The bracket is at most $p + \lambda(G^c)$. If $G = \{q > 0\}$, its second integral is zero and its first is at most p. Finally, the full absolute moment is at most $M_S + d_S$ and the comparison law is supported in the clean feature box together with the origin. This proves the remaining bounds. □

The defect d_S is precisely where the Gaussian marks and unspent reserves of § 4.18 are used. If the history construction only bounds p, the weighted defects still require proof. If it bounds the marked mass in the actual normalized positive law, Prop. 6.12 converts that estimate directly into a mixed moment error. An unnormalized mass estimate must first be divided by a valid partition floor; the proposition does not provide that floor.

For comparison, suppose $q_0 > 1$ and $\|X_i\|_{L^{nq_0}(\mu)} \le B_i$. Hölder's inequality, first on the exceptional event and then on the product, gives

$$d_S \le p^{1-1/q_0} \prod_{i\in S} B_i, \qquad 1 \le |S| \le n. \tag{6.lxii}$$

Indeed the product has L^{q_0} norm at most the product of the $L^{|S|q_0}$ norms of its entries, and those norms are bounded by the stated L^{nq_0} norms on a probability space. In a fourth connected correlation, uniform eighth moments therefore give a square-root gain from rarity. A direct marked-mass estimate can give a stronger gain and does not need these global eighth moments.

6.4.6. *Connected correlations as polynomials in mixed moments.* For nonempty S, write $m_S = \mathbb{E}\prod_{i\in S} X_i$. The joint cumulant is the polynomial

$$\kappa_S(X) = \sum_{\pi\in\Pi(S)} (-1)^{|\pi|-1}(|\pi|-1)! \prod_{B\in\pi} m_B, \tag{6.lxiii}$$

where $\Pi(S)$ is the set of set partitions of S into nonempty blocks. The indices distinguish repeated insertions as well as different ones. Consequently the same formula gives diagonal moments and derivatives in repeated directions.

For bounded variables, eq. 6.lxiii follows by expanding $\log(1+u) = \sum_{k\ge1}(-1)^{k-1}u^k/k$ in a neighborhood of the origin and extracting the coefficient of $\prod_{i\in S} z_i$ in $\log \mathbb{E}e^{\sum z_i X_i}$. An unordered partition with k blocks arises in $k!$ orders in u^k, and the factor $1/k$ leaves $(k-1)!$. For variables with the finite mixed absolute moments occurring in the formula, use this polynomial as the definition; bounded truncations and dominated convergence recover it from the usual moment definition. No moment generating function is asserted to exist for an unbounded variable.

Let $\left\{ {n \atop k} \right\}$ denote the number of partitions of $\{1,\dots,n\}$ into k blocks. Define

$$b_n(D) = \sum_{k=1}^{n} k! \left\{ {n \atop k} \right\} (1+D)^{k-1}, \qquad D \ge 0. \tag{6.lxiv}$$

In particular

$$b_2(D) = 1 + 2(1+D), \tag{6.lxv}$$

$$b_4(D) = 1 + 14(1+D) + 36(1+D)^2 + 24(1+D)^3. \tag{6.lxvi}$$

Thus $b_2(0) = 3$ and $b_4(0) = 75$.

Proposition 6.13 (Connected error from weighted exceptional runs)**.** Assume $d_S \le DM_S$ for every nonempty $S \subset \{1,\dots,n\}$. Put $r_G = p + \lambda(G^c)$. Then

$$|\kappa_\rho(x_1,\dots,x_n) - \kappa_{\nu_G}(x_1,\dots,x_n)| \le (D + r_G) b_n(D) \prod_{i=1}^{n} M_i. \tag{6.lxvii}$$

For $G = \{q > 0\}$ one may use $r_G = p$. ■

Proof. For each partition with k blocks, subtract the product of its comparison moments from the product of its physical moments. Use the telescoping product identity, replacing one factor at a time. The replaced factor has error at most $(D + r_G)M_B$ by Prop. 6.12; every other factor has modulus at most $(1+D)M_{B'}$. Since the blocks partition all indices, the product of these bounds is $k(D+r_G)(1+D)^{k-1}\prod_i M_i$. Multiply by $(k-1)!$ from eq. 6.lxiii, sum over partitions, and group them by k. This is exactly eq. 6.lxiv. □

The same proof gives a useful comparison for two arbitrary bounded feature laws with coordinate bounds M_i. If their total variation distance is at most δ, each mixed moment differs by at most $2\delta M_S$, and all moments have modulus at most M_S. Therefore

$$|\kappa_\rho - \kappa_\nu| \le 2\delta b_n(0) \prod_i M_i. \tag{6.lxviii}$$

The constants are 6δ for a covariance and 150δ for a fourth cumulant after the product of the bounds is factored out. These constants are explicit sufficient bounds; no optimality is needed in the application.

6.4.7. *Auxiliary source logarithms on the bounded feature box.* Bounded clean insertions also justify the connected interpretation analytically. Let ρ and ν be supported in $\prod_i[-M_i, M_i]$. For

$$t = \sum_i |z_i| M_i \le c < \log 2, \tag{6.lxix}$$

their generating functions satisfy

$$|F_\rho(z) - 1| \le e^t - 1 < 1, \qquad |F_\rho(z)| \ge 2 - e^c, \tag{6.lxx}$$

and the same inequalities hold for ν. The estimate follows from $|e^w - 1| \le e^{|w|} - 1$ and integration. The principal logarithm is therefore single valued throughout this region.

If the total variation distance is at most δ, then $|F_\rho - F_\nu| \le 2\delta e^c$. The straight segment joining their values remains in the disc $|w-1| \le e^c - 1$, where $|1/w| \le (2-e^c)^{-1}$. Integrating the derivative of the logarithm along that segment gives

$$|\log F_\rho(z) - \log F_\nu(z)| \le \frac{2\delta e^c}{2 - e^c}. \tag{6.lxxi}$$

For a polydisc with radii $c/(nM_i)$, Cauchy's formula bounds the derivative with one occurrence of every variable by the right hand side times $n^n c^{-n} \prod_i M_i$. The polynomial bound in eq. 6.lxviii is usually smaller for $n = 2, 4$, but the logarithmic estimate explains the absence of a normalization singularity.

These auxiliary sources are inserted into a fixed positive law at zero source. Their region of holomorphy is not a claim that the renormalization charts extend to sources of size M_i^{-1}. One may use this elementary generating function to estimate a real-law comparison without analytically continuing any history map to that larger source domain. If the physical insertion itself has been replaced by a derivative of a source-dependent effective action, its higher source derivatives must be retained separately.

6.4.8. *The law of total cumulance.* The clean resampling still averages over the exterior field. The conditional means cannot be discarded. The exact expression for all the resulting terms is useful at order four.

Let $e \mapsto \xi_e$ be a probability kernel on $\mathbb{R}^n$, let α be an exterior probability, and let $\xi = \int \xi_e \, \alpha(de)$. For nonempty S set $c_S(e) = \kappa_{\xi_e}(x_i : i \in S)$. Suppose the coordinates are bounded, or that all moments and integrals used below are finite with an integrable dominating function. Then

$$\kappa_\xi(x_1, \dots, x_n) = \sum_{\pi \in \Pi_n} \kappa_\alpha(c_B(e) : B \in \pi). \tag{6.lxxii}$$

An outer cumulant of one entry means its expectation.

To prove the formula for bounded coordinates, write $K_e(z) = \log \xi_e(e^{\sum_i z_i x_i})$ on a common small polydisc. Its coefficient with distinct indices in S is $c_S(e)$. Expand

$$\log \int e^{K_e(z)} \, \alpha(de) = \sum_{k \ge 1} \frac{1}{k!} \kappa_\alpha(K_e(z), \dots, K_e(z)). \tag{6.lxxiii}$$

The series converges near zero because $K_e(0) = 0$ and its supremum tends to zero there. To obtain the coefficient of $z_1 \cdots z_n$, distribute the distinct indices among k nonempty entries. These distributions are ordered partitions. The factor $1/k!$ removes their ordering, giving eq. 6.lxxii. For finite moments one can instead perform the same computation in the algebra of polynomials with $z_i^2 = 0$. Every series truncates at degree n, so the proof uses only the assumed moments and integrability.

At orders two and three the identity reads

$$\kappa_{12} = \mathbb{E}_\alpha c_{12} + \mathrm{Cov}_\alpha(c_1, c_2), \tag{6.lxxiv}$$

$$\kappa_{123} = \mathbb{E}_\alpha c_{123} + \mathrm{Cov}_\alpha(c_1, c_{23}) + \mathrm{Cov}_\alpha(c_2, c_{13}) + \mathrm{Cov}_\alpha(c_3, c_{12}) + \kappa_\alpha(c_1, c_2, c_3). \tag{6.lxxv}$$

At order four it reads

$$\begin{aligned}\kappa_{1234} = \mathbb{E}_\alpha c_{1234} &+ \sum_{i=1}^{4} \mathrm{Cov}_\alpha(c_i, c_{\{1,2,3,4\}\setminus\{i\}}) \\ &+ \sum_{\{ij\}\{k\ell\}} \mathrm{Cov}_\alpha(c_{ij}, c_{k\ell}) \\ &+ \sum_{\{ij\},\{k,\ell\}=\{1,2,3,4\}\setminus\{i,j\}} \kappa_\alpha(c_{ij}, c_k, c_\ell) \\ &+ \kappa_\alpha(c_1, c_2, c_3, c_4). \end{aligned} \tag{6.lxxvi}$$

The second sum has the three pair partitions 12|34, 13|24, 14|23. The third has six choices of the pair $\{i, j\}$. The displayed terms account respectively for partitions with one, two, three and four blocks.

A mixture of centered Gaussian linear laws already illustrates one essential term: its conditional fourth cumulants vanish, but covariance of the conditional covariances can give a nonzero global fourth cumulant. Similarly, for quadratic Gaussian insertions the conditional one-point functions need not be constant when the covariance depends on e. Subtracting the global one-point function does not remove their exterior variation. The mixed outer cumulants in eqs. 6.lxxiv and 6.lxxvi must be estimated.

6.4.9. *A fixed reference and quantitative exterior errors.* For $k \geq 1$ define

$$a_k = \sum_{j=1}^{k} (j-1)! \left\{ {k \atop j} \right\}. \tag{6.lxxvii}$$

The first four values are $1, 2, 6, 26$. The moment polynomial implies $|\kappa(Y_1, \dots, Y_k)| \leq a_k \prod_i \|Y_i\|_\infty$. For $k \geq 2$ this cumulant is unchanged if a deterministic constant is added to any one entry. For bounded variables the invariance follows because such an addition changes the logarithm of the generating function only by a linear term. The polynomial identity extends it to finite moments.

Proposition 6.14 (Exterior averaging around a deterministic reference)**.** In the setting of eq. 6.lxxii, suppose constants γ_S and nonnegative ϵ_S satisfy

$$|c_S(e) - \gamma_S| \leq \epsilon_S \tag{6.lxxviii}$$

for every nonempty S, almost everywhere under α. Then

$$|\kappa_\xi(x_1, \dots, x_n) - \gamma_{\{1,\dots,n\}}| \leq \epsilon_{\{1,\dots,n\}} + \sum_{\substack{\pi \in \Pi_n \\ |\pi| \geq 2}} a_{|\pi|} \prod_{B \in \pi} \epsilon_B. \tag{6.lxxix}$$

■

Proof. The one-block term of eq. 6.lxxii has error at most $\epsilon_{\{1,\dots,n\}}$. In every other term subtract γ_B from each entry, using invariance of an outer cumulant of order at least two under deterministic translations. Its entries now have suprema at most ϵ_B. The bound following eq. 6.lxxvii and the triangle inequality prove the result. □

For two insertions the bound is

$$\epsilon_{12} + 2\epsilon_1\epsilon_2. \tag{6.lxxx}$$

For four insertions it is

$$\begin{aligned}\epsilon_{1234} + 2\sum_i \epsilon_i \epsilon_{\{1,2,3,4\}\setminus\{i\}} + 2 \sum_{\{ij\}\{k\ell\}} \epsilon_{ij}\epsilon_{k\ell} \\ + 6 \sum_{\{ij\},\{k,\ell\}=\{1,2,3,4\}\setminus\{i,j\}} \epsilon_{ij}\epsilon_k\epsilon_\ell + 26\epsilon_1\epsilon_2\epsilon_3\epsilon_4. \end{aligned} \tag{6.lxxxi}$$

This form identifies exactly how accurately the conditional means and the lower connected correlations must be controlled. It is not necessary to demand the same absolute accuracy from quantities that appear only as products. Conversely, a bound on the conditional covariance alone is not enough for the fourth connected correlation.

6.4.10. *Discarding unadmitted exteriors before reference comparison.* Suppose the conditional reference estimates hold only on G, with $\lambda(G) > 0$. Define the probability $\widehat{\nu}_G = \lambda(G)^{-1} \int_G X_* \mu_e^C \, \lambda(de)$. This removes the artificial point mass used in ν_G. Both measures are supported in the clean feature box, and

$$\|\nu_G - \widehat{\nu}_G\|_{\mathrm{TV}} \le \lambda(G^c). \tag{6.lxxxii}$$

Indeed $\nu_G = \lambda(G)\widehat{\nu}_G + \lambda(G^c)\delta_0$. The comparison with a deterministic reference can now be made under the genuine exterior probability $\lambda(\,\cdot \mid G)$.

Combining the preceding propositions gives the complete estimate

$$\begin{aligned} |\kappa_\rho - \gamma_{\{1,\dots,n\}}| \le{}& (D + p + \lambda(G^c)) b_n(D) \prod_i M_i \\ &+ 2\lambda(G^c) b_n(0) \prod_i M_i \\ &+ \epsilon_{\{1,\dots,n\}} + \sum_{\substack{\pi \in \Pi_n \\ |\pi| \ge 2}} a_{|\pi|} \prod_{B \in \pi} \epsilon_B. \end{aligned} \tag{6.lxxxiii}$$

Here $d_S \le D M_S$ and the conditional cumulants on G are within ϵ_S of γ_S. The first line compares the physical law to clean resampling, the second removes the exceptional exterior mass, and the remaining lines average the conditional reference estimates. Every normalization in this comparison is a normalization of a positive probability.

If a chart-admission set $A \subset E$ is also required, take $G = A \cap G_\eta$. Its exceptional mass obeys

$$\lambda(G^c) \le \lambda(A^c) + \frac{p}{1 - \eta}. \tag{6.lxxxiv}$$

A model proof must supply the first term when admission is not automatic. It cannot infer admission solely from the existence of a conditional probability. The useful point is that a lower bound on q is only needed on the retained exteriors, and their discarded mass is explicitly accounted for.

6.4.11. *Short-distance powers and the marked reserve.* We apply eq. 6.lxxxiii with fixed order $n = 2$ or $n = 4$, inverse running coupling $x \longrightarrow \infty$, and $N_i = \|f_i\|_\infty$. Suppose the admitted clean readouts satisfy

$$M_i \le C_i x^{-1} (\log x)^{A_i} N_i, \qquad A_i \ge 0. \tag{6.lxxxv}$$

For the two-point readout bound used in § 6.2, one can take a fixed exponent after substituting the permitted polylogarithmic window radius. No optimization of that exponent is required; it must be recorded before spending the rarity reserve.

Suppose the normalized marked mass gives, for fixed $c, b > 0$,

$$D + p + \lambda(A^c) \le C \exp[-c(\log x)^b]. \tag{6.lxxxvi}$$

Then for fixed $\eta \in (0,1)$ the first two lines of eq. 6.lxxxiii are

$$o\left(x^{-n} \prod_i N_i \right). \tag{6.lxxxvii}$$

To check the assertion, $D \to 0$, so $b_n(D)$ stays bounded. The remaining factor, after $x^{-n} \prod_i N_i$ is removed, is at most a constant times $\exp[-c(\log x)^b + (\sum_i A_i) \log\log x]$. Its exponent tends to $-\infty$ for every $b > 0$. The same conclusion follows from a power bound $O(x^{-a})$ with $a > 0$. A logarithmic rarity bound $O((\log x)^{-P})$ instead requires $P > \sum_i A_i$.

The estimate is a consequence of the normalized, marked hypothesis eq. 6.lxxxvi. The unnormalized unmarked sum of histories does not imply it. The distinction between a stored volume mark and a field moment mark remains exactly the one in § 4.18; the present argument uses the field moments in d_S rather than replacing them by volume moments.

For the conditional errors put $L_S = \prod_{i\in S} x^{-1}N_i$. If the bounds on G have the form $\epsilon_S \le \delta_S(x)L_S$, then the last two lines of eq. 6.lxxxiii, divided by $L_{\{1,\dots,n\}}$, are at most

$$\delta_{\{1,\dots,n\}} + \sum_{\substack{\pi\in\Pi_n \\ |\pi|\ge 2}} a_{|\pi|} \prod_{B\in\pi} \delta_B. \tag{6.lxxxviii}$$

Thus it suffices that the displayed products tend to zero. In particular conditional mean errors that are bounded at their natural scale and actually tend to zero make every partition containing a singleton negligible, provided the other conditional errors remain bounded at their natural scales. A conditional two-point error $o(x^{-2}N_iN_j)$ controls its pair-pair contribution to order four, provided the second pair error stays bounded on its natural scale. The one-block fourth-order error must itself be $o(x^{-4}\prod_i N_i)$; averaging cannot improve it.

6.4.12. *The estimates still required from the interacting model.* The comparison just proved removes an unnecessary demand for a uniform conditional clean-probability floor. It gives a rigorous way to spend the positive history reserve in connected correlations. Its remaining inputs are concrete.

First, the history sum and its class identifications must be equalities for the zero-source physical measure, with their pushforward densities retained as in Prop. 6.10. Second, the complete conditional exponent and the physical readout must be identified on the admitted clean chart. Third, the marked normalized mass must give the d_S bounds. Fourth, the conditional cumulants, including their means and their source contact terms, must obey eq. 6.lxxviii against one fixed reference. These are the places where the background inverse, formation and interacting force estimates enter.

For $n = 2$, the reference coefficient is the free-curvature two-point coefficient, and the errors in eq. 6.lxxx must be small relative to its positive lower bound for the permitted shapes. For $n = 4$, the reference is the connected four-point kernel and the errors are those in eq. 6.lxxxi. The free kernel and its sign calculation do not prove the conditional interacting estimate. The running-coupling conversion is applied only after the comparison at scale x^{-n} has been proved.

These observations also specify a useful order for closing the short-distance argument. The conditional one-block errors and the marked normalization come first. Quantitative shape and grid errors are then compared with the established leading coefficient. Exceptional exteriors and lower conditional cumulants are controlled by the explicit products above. No additional conditional clean-mass theorem is needed for this averaged route.

6.5. **Effective source derivatives and separated contact terms.** Integrating a subset of the fields turns an affine physical source into a nonlinear effective source action. Its mixed derivatives contribute to connected correlations even at zero source. This section derives the exact finite-order formula and estimates the mixed terms from their spatial carriers. It explains when the covariance of the first source responses suffices, and when additional contact insertions must be retained.

6.5.1. *A fixed integration domain and its source derivatives.* Let μ be a probability on a measurable space. Suppose $A(z, y)$ is holomorphic in $z = (z_1, \dots, z_n)$ near zero, measurable in y, and $A(0, y) = 0$. Assume that $e^{A(z,y)}$ and the source derivatives used below have a common integrable bound on a smaller polydisc. These hypotheses justify differentiation under the integral. Put

$$F(z) = \mu(e^{A(z,\cdot)}), \qquad W(z) = \log F(z), \quad W(0) = 0. \tag{6.lxxxix}$$

Since $F(0) = 1$, the logarithm exists on a neighborhood of zero. For nonempty $B \subset \{1, \dots, n\}$ define the distinct-index jet

$$A_B(y) = \prod_{i \in B} \partial_{z_i} A(z, y)\Big|_{z=0}. \tag{6.xc}$$

The integral in eq. 6.lxxxix is on a fixed measurable domain and with a fixed zero-source measure. A source-dependent change of variables must first be written on such a domain with its complete Jacobian. Differentiating a sharp indicator of a moving domain is not justified by holomorphy of the action alone.

Proposition 6.15 (The partition formula for effective source jets)**.** Under these assumptions,

$$\partial_{z_1} \cdots \partial_{z_n} W(0) = \sum_{\pi \in \Pi_n} \kappa_\mu(A_B : B \in \pi). \tag{6.xci}$$

An outer cumulant with one entry is its expectation. ■

Proof. For the coefficient with each index appearing once, work in the commutative algebra in which $z_i^2 = 0$. Every power series then truncates at degree n, so no interchange of infinite series is required. In this algebra $A(z, y) = \sum_{\varnothing \neq B} z_B A_B(y)$, where $z_B = \prod_{i \in B} z_i$. The cumulant expansion of $\log \mu(e^A)$ is $\sum_{k=1}^n \kappa_\mu(A, \dots, A)/k!$. To obtain $z_1 \cdots z_n$, the k entries must use disjoint nonempty index sets whose union is all indices. These are the ordered partitions into k blocks. The factor $1/k!$ removes their ordering, proving the formula. The dominated analytic assumptions identify this finite coefficient with the indicated derivative of W. □

At order two,

$$W_{12}(0) = \mu(A_{12}) + \mathrm{Cov}_\mu(A_1, A_2). \tag{6.xcii}$$

At order three,

$$\begin{aligned} W_{123}(0) = \mu(A_{123}) &+ \mathrm{Cov}_\mu(A_{12}, A_3) + \mathrm{Cov}_\mu(A_{13}, A_2) \\ &+ \mathrm{Cov}_\mu(A_{23}, A_1) + \kappa_\mu(A_1, A_2, A_3). \end{aligned} \tag{6.xciii}$$

At order four,

$$\begin{aligned} W_{1234}(0) = \mu(A_{1234}) &+ \sum_i \mathrm{Cov}_\mu(A_i, A_{\{1,2,3,4\}\setminus\{i\}}) \\ &+ \sum_{\{ij\}\{k\ell\}} \mathrm{Cov}_\mu(A_{ij}, A_{k\ell}) \\ &+ \sum_{\{ij\},\{k,\ell\}=\{1,2,3,4\}\setminus\{i,j\}} \kappa_\mu(A_{ij}, A_k, A_\ell) \\ &+ \kappa_\mu(A_1, A_2, A_3, A_4). \end{aligned} \tag{6.xciv}$$

The sums have respectively four, three and six terms. Thus controlling only A_i does not control either $\mu(A_{12})$ or the additional fourth-order terms.

6.5.2. *A quantitative bound for all contact contributions.* Let $M_i > 0$ and suppose $\|A_i\|_\infty \le M_i$. For each block B of size at least two suppose

$$\|A_B\|_\infty \le \eta_B \prod_{i \in B} M_i. \tag{6.xcv}$$

Using the cumulant constants a_k defined in eq. 6.lxxvii, the partition formula gives

$$|W_{1\cdots n}(0) - \kappa_\mu(A_1, \dots, A_n)| \le \prod_i M_i \sum_{\substack{\pi \in \Pi_n \\ \pi \neq \{\{1\},\dots,\{n\}\}}} a_{|\pi|} \prod_{\substack{B \in \pi \\ |B| \ge 2}} \eta_B. \tag{6.xcvi}$$

Every term in the sum contains at least one mixed block. The proof is to apply $|\kappa(Y_1,\dots,Y_k)| \le a_k \prod \|Y_j\|_\infty$ to each term of eq. 6.xci. It applies also to complex-valued jets, since the moment polynomial and its absolute-value bound are algebraic.

For example, the second-order error is at most $\eta_{12} M_1 M_2$. At order four it is at most

$$\prod_i M_i \Big[\eta_{1234} + 2 \sum_i \eta_{\{1,2,3,4\}\setminus\{i\}} + 2 \sum_{\{ij\}\{k\ell\}} \eta_{ij}\eta_{k\ell} + 6 \sum_{\{i,j\}\subset\{1,2,3,4\}} \eta_{ij} \Big]. \tag{6.xcvii}$$

In this formula the third-order jets occur paired with one first-order jet. There is no isolated third-order coefficient missing from the estimate.

If $A_i = Q_i + I_i$ with $\|Q_i\|_\infty \le M_i$ and $\|I_i\|_\infty \le \rho_i M_i$, multilinearity gives

$$|\kappa(A_1,\dots,A_n) - \kappa(Q_1,\dots,Q_n)| \le a_n \left[\prod_i (1+\rho_i) - 1 \right] \prod_i M_i. \tag{6.xcviii}$$

This follows by expanding in the nonempty subset of entries where I_i is used. Combining eqs. 6.xcvi and 6.xcviii accounts separately for the first response error and for every mixed response. The latter cannot be absorbed into the former without a bound on the mixed jets.

6.5.3. *Local carriers and a common source polydisc.* Let $\mathcal{D}$ be a family of finite domains in a graph with cells v. Let $d(X) \ge 0$ be a connected-size function. For each X suppose $H_X(z,y)$ is holomorphic on the same closed source polydisc $|z_i| \le r_i$, measurable in y on one admitted field domain, and obeys

$$\sup_{z,y} |H_X(z,y)| \le h_X. \tag{6.xcix}$$

Assume H_X is independent of z_i when its physical carrier $X^\sim$ misses $\operatorname{supp} f_i$. All source parameters here are held independently; a single bound for a source in one direction would not justify this mixed Cauchy estimate.

Suppose $H = \sum_X H_X$ converges normally on smaller polydiscs. For a nonempty B its mixed jet is

$$H_B(y) = \sum_{X:\, X^\sim \cap \operatorname{supp} f_i \neq \varnothing \text{ for all } i\in B} \partial_B H_X(0,y). \tag{6.c}$$

Multivariate Cauchy estimates give

$$\|H_B\|_\infty \le \left(\prod_{i\in B} r_i^{-1} \right) \sum_{X \text{ meeting every support in } B} h_X. \tag{6.ci}$$

There is no factorial for distinct variables. Repeated directions can be introduced as distinct source variables; the usual factorial for a single-variable derivative then follows from coefficient multiplicities.

The preceding statement requires a common field domain for all H_X after the actual substitution. If the norm in eq. 6.xcix is established only before substitution, the substituted family must be shown to stay inside that domain. Cauchy's formula does not repair a failure of domain admission.

6.5.4. *Exponential suppression of separated mixed jets.* Assume that a nonnegative local weight has the incidence bound

$$\sup_v \sum_{X\ni v} e^{\kappa d(X)} h_X \le E_0. \tag{6.cii}$$

Suppose every carrier that meets a selected support has an assigned root in a finite set W, and that assigning a root never reduces the estimate on the right: more explicitly, the sum over such domains is bounded by $\sum_{v\in W} \sum_{X\ni v}$ with the same h_X and $d(X)$. Assume also that if $X^\sim$ meets two of the

separated supports, then $d(X) \geq D_*$. These are the geometric root and carrier properties needed in the estimate. Then

$$\sum_{X \text{ meeting every support in } B} h_X \leq |W| E_0 e^{-\kappa D_*}, \qquad |B| \geq 2. \tag{6.ciii}$$

Indeed factor $e^{-\kappa d(X)}$ out of the weighted incidence norm and use $d(X) \geq D_*$ in each contributing term. The root count is paid once by $|W|$. This proves

$$\|H_B\|_\infty \leq |W| E_0 e^{-\kappa D_*} \prod_{i \in B} r_i^{-1}, \qquad |B| \geq 2. \tag{6.civ}$$

A localization carrier may be an enlargement of X. If its enlargement has radius r_{loc} and the supports are separated by d_*, then the size lower bound must use $d_* - 2r_{\mathrm{loc}}$ in the appropriate units. In particular it is not enough to say that the original domains are disjoint. The separation has to survive the actual enlargement used in the source dependence.

If there are several origin levels, apply eq. 6.civ with their own incidence bound, root count, carrier units and source radii, and sum those numerical estimates. A coarse root count cannot be used for fine-origin terms without the filing estimate that transfers their incidence weights. This is the same bookkeeping requirement as in eq. 6.cxcii.

6.5.5. *The short-distance contact budget.* Let $x \to \infty$ and $N_i = \|f_i\|_\infty > 0$. Suppose the first responses satisfy

$$\|A_i\|_\infty \leq C x^{-1} (\log x)^P N_i, \tag{6.cv}$$

and all mixed jets of the regular effective source arise from families satisfying eq. 6.civ. Suppose their summed numerical bound is

$$\|A_B\|_\infty \leq C_n (\log x)^Q e^{-c(\log x)^b} \prod_{i \in B} N_i, \qquad |B| \geq 2, \tag{6.cvi}$$

with $b > 1$. This follows, for example, when $D_* \geq c'(\log x)^b$, the root and norm factors are polylogarithmic, and $r_i^{-1} \leq C_n N_i$.

For a partition in eq. 6.xci with s singleton blocks and at least one mixed block, its cumulant is bounded by

$$C_n x^{-s} (\log x)^{Q'} e^{-c'(\log x)^b} \prod_i N_i. \tag{6.cvii}$$

For fixed $n \leq 4$ this is $o(x^{-n} \prod_i N_i)$. In fact after division by $x^{-n} \prod_i N_i$, the exponent has the form $(n-s)\log x - c'(\log x)^b + Q' \log\log x$, which tends to $-\infty$ because $b > 1$. Summing the finitely many partitions proves

$$W_{1\cdots n}(0) = \kappa_\mu(A_1, \ldots, A_n) + o\left(x^{-n} \prod_i N_i \right), \qquad n = 2, 4. \tag{6.cviii}$$

This establishes the regular separated contact estimate with the actual source radius loss retained.

The condition $b > 1$ in this argument is stronger than the $b > 0$ sufficient for rare-history errors in eq. 6.lxxxvii. Here the mixed jets were not assumed to have the natural factor $x^{-|B|}$, so the exponential must overcome a fixed positive power of x. If the geometric decay is only $e^{-c \log x} = x^{-c}$, its coefficient must exceed the missing source powers. If the mixed jets already carry their full natural powers, $b > 0$ suffices. These are different uses of the same exponential reserve and should not be conflated.

6.5.6. *Normalizer derivatives and the order of averaging.* The formulas also fix the effect of a chosen analytic normalizer. If $\widetilde{F}(z) = e^{-E(z)}F(z)$ with deterministic E, then

$$\partial_B \log F(0) = \partial_B E(0) + \partial_B \log \widetilde{F}(0). \tag{6.cix}$$

A bound on $\widetilde{F}$ alone does not bound the left side until the required mixed derivatives of E are estimated. If E is a sum of terms each depending on only one source variable, its mixed distinct-index derivatives vanish for $|B| \geq 2$. For local terms involving several source variables, the carrier estimate above supplies the needed bound.

For a conditional normalizer $E_e(z)$, the identity must first be kept inside the exterior average:

$$F(z) = \int e^{E_e(z)} \widetilde{F}_e(z)\, \lambda(de). \tag{6.cx}$$

Replacing this by $\int \widetilde{F}_e(z)\lambda(de)$ would remove all the source derivatives of E_e and their exterior fluctuations. Apply the partition formula to the full conditional logarithm, and then the law of total cumulance of § 6.4. This order retains both the source contact terms and the conditional means.

At an actual clean run the remaining verification is therefore precise. The original positive physical source integral must equal the fixed-domain effective integral. Its first responses must have the stated readout and irrelevant remainder. Every mixed response must retain its own carrier and source radius. The marked history reserve controls the exceptional contributions. Under these inputs, eq. 6.cviii and the averaged comparison prove the connected extraction step without discarding any differentiated normalizer.

Physical source derivatives at the four-support witness. Use the sourced density of §5.20.1.1 at level $k_0 = K - s$. The fixed-domain representation in the two-point strategy applies with $w = \sum_{i=1}^{4} z_i f_i$. Use the exponent of each resummed standing class, including the exponents of its integral operations. The measures of the classes carry the unsourced gauge, averaging and test factors on the joint space with stored fields. In particular, an averaging delta can make the conditional measure in the block field singular at fixed stored fields. Exactness of both operations for the sourced total gives $Z(w)/Z(0) = \mathbb{E}e^{\Phi(w)}$, with Φ the complete exponent minus its zero-source value. If the representation extracts deterministic local source energies as a separate factor, include them in Φ in this identity.

The contact of a live component with a support and the age of its lineage are already functions of the standing class. For an event involving an individual history and the block field together, let τ_h^0 and $\tau_{[h]}^0$ denote their zero-source block-field marginals before normalization. Multiply the class joint measure by $d\tau_h^0/d\tau_{[h]}^0(V)$ on the sheet for $h \in [h]$. These weights are independent of w and sum to one almost everywhere. The lift gives the required individual zero-source joint marginal and leaves the sum of the sourced integrals unchanged. The exponent on a sheet is still the exponent of the resummed class.

6.5.6.1. Mixed derivatives on island carriers. Separate the differences of regular activities, remainders and boundary terms with no islands, together with the label's pointed shares of the source energies. The boundary terms involving islands form a second sum. In the remaining sum keep the coupling pieces and the exponents of the integral operations. Each nested operation exponent enters once, in the operation that contains it. The carrying Wilson and boundary terms enter in their own sums when a drop-out has no exponent for them in its integral. All differences use the same stored tuple at source w and at zero.

Write $\rho_{\mathsf{d}} = \mathsf{d}L^s$ for the support separation in witness lattice units. At origin level $j \leq k_0$ the separation in localization cubes is $n_j = (\rho_{\mathsf{d}}/M)L^{k_0-j}$. The rooms ensure $n_j \geq C\check{R}_j$ and give source radii $c\varrho/\|f_i\|_\infty$ at the witness. At fixed fields, the running shift and each counterterm weight are sums of functions of single plaquette source values, with fixed profiles and tents. By linearity of the

Wilson action in the weight, this also applies to the separate carrying Wilson terms of a drop-out. Their mixed derivatives vanish at disjoint supports regardless of the size of the carrying region. An elementary operation whose source dependence lies in its component of diameter $C\check{R}_j$ misses at least one of any two of the separated supports. Apply this criterion to the actual carrier of each sourced operation exponent, including any subtraction from its normalizer.

For a formed operation the fixed-variable source formula uses the twice-enlarged component, with the already formed inner exponents included in the same read. At level j the component at formation is in a cube of side $100M\check{R}_j$ in its own units. Its enlarged physical diameter is at most $CM\check{R}_j a_{K-j}$, below the support separation by the witness room. Thus the mixed derivative of this complete exponent vanishes. This applies also to the executed operations retained inside a drop-out. The Wilson and boundary factors never integrated by those operations keep their own carriers. The calculation uses the exponent at fixed stored fields and zero-source reference measure throughout. In particular it involves no division by a complex sourced operation mass.

For the regular activities and local energies, use the enlarged carrier in eq. 6.ciii. An island boundary term uses the tree length through the cells outside the exempt hulls, with the tree also meeting each selected hull as in Lem. 6.7. By the hull geometry at eq. 6.xxxvii, every source contact in an enlarged carrier is within $C(1+\check{R}_j)$ cells of a vertex of this tree. The two contacts of an active carrier therefore give length at least $n_j - C - C\check{R}_j$. Edges inside a hull still count in the original lattice metric. For the root pin the tree at a vertex near the first contact. There are at most Cn_j^4 such vertices. The rooted walk and hull incidence count in that lemma apply at fixed history, before taking the essential supremum. The exempt hulls are those in the surviving filing at the witness. For a term whose old bound discounted an absorbed hull, apply the donor refiling after eq. 6.xli on the admitted source box before using the incidence estimate.

For the boundary differences use the sourced bound on the retained list at its actual field values. The witness room and monotonicity of the depth norms give $\|w\|_{\flat,K-j} \le \varrho/2$ for every origin on $|z_i| \le \varrho/(8\|f_i\|_\infty)$, after choosing the fixed source radius ϱ within the sourced class. Cauchy differentiation uses this common disc in the sources at fixed stored fields. If an estimate holds almost everywhere for each source separately, impose it first on a countable dense set and extend it there by holomorphy of the actual read. With the exponential incidence bound for the actual island and regular terms, Cauchy's estimate thus costs $Cn_j^4 e^{-cn_j}\prod_{i\in B}(\|f_i\|_\infty/\varrho)$ for a mixed derivative in B, $|B|\ge 2$. Sum the levels using $n_j = n_{k_0}L^{k_0-j}$ to obtain

$$\|\Phi_B\|_\infty \le C(\rho_{\mathsf{d}}/M)^4 e^{-c\rho_{\mathsf{d}}/M}\prod_{i\in B}(\|f_i\|_\infty/\varrho). \tag{6.cxi}$$

This estimate uses the source localization on the resummed standing classes, with the relative incidence bound on the same source polydisc. For an operation that has a larger source carrier one must include it in the mixed-jet estimate with its actual size.

6.5.6.2. First derivatives on live histories. Let B_i be the event that a live component at the witness meets the fixed neighbourhood of support i. On this event let a_i be the greatest lineage age among those components. Each of its a_i+1 stored levels can increase the spatial count in the first source derivative. Write its upper weight as $y_i = \mathbf{1}_{B_i}(a_i+2)^C L^{4(a_i+1)}$. A renewal of the label does not reset the count of retained levels. For a nonempty set of indices I, $\prod_{i\in I} y_i \le \sum_{i\in I} y_i^4$, since every nonzero y_i is at least one. On B_i an attaining component meets one of at most $C\rho_{\mathsf{d}}^4$ witness cells. Summing over these cells and then the ages gives

$$\mathbb{E}\prod_{i\in I} y_i \le C\rho_{\mathsf{d}}^4 Q_{17}(k_0), \qquad Q_b(k_0) = \sum_{a\ge 0} L^{ba}\sup_{\mathfrak{c}}\mathbb{P}(\text{lineage age } a \text{ at } \mathfrak{c}). \tag{6.cxii}$$

The four powers cost L^{16a} and the remaining power pays for the polynomial in the age. This is a normalized history estimate, with the volume and depth range of the age bound in force.

For a first derivative keep the witness, the class, the guards, the background and the stored tuple fixed and restrict the source to $w = tf_i$. This is the one-direction derivative of the very same exponent used in eq. 6.cxi. The source terms on the regular domain depend on the function w, so restricting the other three parameters to zero changes no integration domain or zero-source law. The derivative of the full source energy of the empty standing label is a deterministic constant c_i. For another standing label $\mathcal{S}$ the difference is the first derivative of $\sum_j[\mathsf{T}_j(\mathcal{S};w) - \mathsf{T}_j(\mathcal{S}_\emptyset;w)]$, with T_j the label's share of the source energy. Only a missing term near Z_j^+ whose enlarged carrier meets the source can contribute. This includes a missing one-point gauge term if it is included in the source energy. The small-field regions are nested in the reduced standing label, so the earlier missing sites lie in the witness live footprint. The padding at level j has physical width at most $CM\check{R}_j a_{K-j} \le CM\check{R}_{k_0} a_s$ by eq. 2.xx. Off B_i the padded missing sites are therefore separated from the source by a fixed fraction of d. A contributing carrier has $d_j(X) \ge c\rho_{\mathsf{d}} L^{k_0-j}/M$. Rooting the sum near the source, the weighted carrier count and Cauchy's estimate at the fixed source radius give a bound $C\|f_i\|_\infty \sum_{j\le k_0} m_j^4 e^{-cm_j}$, where $m_j = \rho_{\mathsf{d}} L^{k_0-j}/M$. This is at most $C\|f_i\|_\infty \rho_{\mathsf{d}}^4 e^{-c'\rho_{\mathsf{d}}/M}$. On B_i the missing share belongs to the live error.

Off contact, the counterterm identity for the running weight gives the profile of the insertion,

$$\partial_{z_i}\zeta_{k_0}(p;w)|_0 = f_i(x(p)) + \sum_{j\le k_0} b_j'(0) \sum_y h_y^{(j)}(p) f_i(x(p_y)) = \zeta_{k_0}' \widetilde{f}_i(p).$$

Here $b_j'(0)$ differentiates the whole running increment with respect to the input source. At constant input the tent sum is one and the coefficient is ζ_{k_0}'. The tents have range at most CMa_s, so the bounded sum of $|b_j'(0)|$ and the lower bound on $|\zeta_{k_0}'|$ give $|\widetilde{f}_i - f_i| \le CMa_s \operatorname{Lip}(f_i)$. Use $\widetilde{f}_i$ as the profile of the bounded readout G_i in the four-insertion comparison below. Its window tests all cubes needed to read the enlarged support. Include these padded carriers in the neighborhood defining B_i. On the actual zero-source joint law, the witness characteristic functions imply passage of the individual cube tests off B_i. This is a statement about the same retained fields and the localized witness chart.

On the admitted complete regular carriers the ordinary one-direction bounds eq. 4.clxxvii and eq. 4.clxxx–eq. 4.clxxxi apply to this restriction pointwise. After subtraction of the vacuum value and the tent action, the quadratic coefficient is bounded by $C\delta_{k_0}/\alpha_{k_0}$, using the formed source slope. At age $a = k_0 - j$, the quadratic estimate retains a geometric factor $L^{-\alpha_H a}$ after the fine-site count. The cubic term retains L^{-2a}, up to a polynomial in a. Summing gives $C\delta_{k_0}^3/\alpha_{k_0}$ per witness window and per unit source norm. For an old ordinary remainder the quadratic field factor pays the fine-site count and the extra factor $x_j^{-\kappa_0/2}$ is summed with $\alpha_j^{-2} \le Cx_j$ by the lower flow bound. This gives the additional $x^{2-\kappa_0/2}$ below. For the localized readout off B_i, the four-layer enlargement of the test cube lies in the ordinary region. The history background restricted there is the local minimizer at its own averages by the nested-minimizer property. In a common gauge compare it with the test minimizer on this enlargement. The logarithmic difference of their averages is zero on the common interior rows and is bounded by $C\varepsilon_{k_0}$ on the outer layer. The stored fields enter through these actual boundary averages. The collar response gives a bound $C\varepsilon_{k_0} e^{-c\check{R}_{k_0}}$ for the change of the local potential and its first lattice derivatives in the plaquette read. In bare plaquette units, with $\eta = \eta_{k_0}$, the plaquette change is bounded by $C\varepsilon_{k_0}\eta^2 e^{-c\check{R}_{k_0}}$. Both plaquette defects have size at most $C\varepsilon_{k_0}\eta^2$, so

$$\tfrac{1}{2}\big|\|U_p - \mathbf{1}\|^2 - \|U_p' - \mathbf{1}\|^2\big| \le C\varepsilon_{k_0}^2 \eta^4 e^{-c\check{R}_{k_0}}.$$

The separate contribution from the distant live region uses the same curvature scale with decay $e^{-c\rho_{\mathsf{d}}}$. Summing with the witness profile cancels η^4 against the bare plaquette count and gives the action bound below. These bounds use complete ordinary charts. Long or wrapping carriers that leave those charts use the far bound, or the live estimate if they reach a local contact.

6.5.6.3. Matching the windowed insertions. Let J_i denote the main Wilson response in the physical derivative on B_i^c, before replacing its background by the witness minimizer. Extend $J_i\mathbf{1}_{B_i^c}$ by zero on B_i. On the same joint law put $O_i = \zeta'_{k_0} G_i(V)$, using the witness window even on B_i. Write $F_i = \|f_i\|_\infty$. On the witness window the localized minimizer passes its own plaquette test and there are at most $C\rho_{\mathsf{d}}^4\eta_{k_0}^{-4}$ plaquettes in the enlarged support. Off that window $G_i = 0$. Thus $\|O_i\|_\infty \le Cx^{-1}\rho_{\mathsf{d}}^4 p_0^2 F_i$ on the entire joint law. Then

$$J_i\mathbf{1}_{B_i^c} - O_i = (J_i - O_i)\mathbf{1}_{B_i^c} - O_i\mathbf{1}_{B_i}.$$

The two-radius comparison above bounds the first term on the actual off-contact window, uniformly in the stored fields. It has size at most $Cx^{-1}\rho_{\mathsf{d}}^4 p_0^2 F_i(e^{-c\check{R}_{k_0}} + e^{-c\rho_{\mathsf{d}}})$. The second term is bounded by the windowed supremum times $\mathbf{1}_{B_i}$. This term occurs because the witness window can pass on a live history. Expand the difference of the fourth cumulants by multilinearity. A term containing a deleted slot $O_i\mathbf{1}_{B_i}$ costs its L^1 bound times the suprema of the other slots, by the moment formula for the cumulant. With $B = \bigcup_i B_i$, the total deletion cost is at most

$$C\,\mathbb{P}(B)\,x^{-4}\rho_{\mathsf{d}}^{16}p_0^8\prod_i F_i.$$

Here $\mathbb{P}(B) \le C\rho_{\mathsf{d}}^4 Q_{17}(k_0)$. The off-contact defect costs the same prefactor with $\mathbb{P}(B)$ replaced by $e^{-c\check{R}_{k_0}} + e^{-c\rho_{\mathsf{d}}}$. Thus the replacement of the physical main entries uses the actual joint law and the localized comparison on B_i^c. The age-weighted bound below is still needed for the live parts of the physical derivatives.

The off-contact ordinary error has size at most $Cx^{-1}\rho_{\mathsf{d}}^4 F_i$ times $(\log x)^{3\mathfrak{p}_0-q} + p_0^2 x^{2-\kappa_0/2}$, where the last term pays the ordinary remainders. The far error costs a polynomial in ρ_{d} times $F_i e^{-c\rho_{\mathsf{d}}}$. For the live estimate, differentiate the outermost standing operation exponents. An operation can contribute only when its twice-enlarged carrier meets the source. By the diameter bound above and eq. 2.xx, its entire stored stock is then in the witness source neighborhood of diameter $C\mathsf{d}$. Its lineage is among those counted by a_i, and the oldest retained level is at least $k_0 - a_i$. This also covers the additional waiting-window stock of a drop-out, since its predecessor already belongs to the lineage at the start of the window. At a fixed level j the stocks of distinct outermost operations are disjoint. Nested stocks enter their outer exponent once. A site can occur once more through a drop-out, since intersecting standing marks have disjoint active level windows. The total site count at that level, with this multiplicity, is bounded by $C\rho_{\mathsf{d}}^4 L^{4(k_0-j)}$. Apply the Cauchy bound for the sourced excess exponent, which is bounded by a fixed constant times its stored site count on the fixed source disc. The derivative sum is at most

$$C\varrho^{-1}F_i\rho_{\mathsf{d}}^4\sum_{j=k_0-a_i}^{k_0} L^{4(k_0-j)} \le C\varrho^{-1}F_i\rho_{\mathsf{d}}^4 L^{4a_i}.$$

For the separate carrying Wilson terms, a level-j site has $O(\eta_j^{-4})$ bare plaquettes, each with action bounded by $C\eta_j^4$ on the corresponding regular layer. At the bare layer use boundedness of the plaquette action. This gives a fixed cost per site. The boundary terms use their own rooted carrier estimates. With the source footprint and nested storage bounds above, the live error has bound $s_i y_i$ with $s_i = C(\log x)^C\rho_{\mathsf{d}}^8 L^{C\check{R}_{k_0}}F_i$. This includes the differentiated operation exponent. The joint law throughout is the one used for the mixed jets, and the law of the four G_i is its block-field marginal.

The lineage estimate of Prop. 5.71 with $b = 17$ gives $Q_{17}(k_0) \le Ce^{-cp_0}$ on its admitted runs. The fixed-torus alternative uses the same age exponent at the witness depth where its terminal bound applies. In particular the L^4 error is at most $Cs_i\rho_{\mathsf{d}}e^{-cp_0/4}$. Here $\check{R}_{k_0} = o(p_0)$ and $p_0/\log x \to \infty$, so it is $o(x^{-1})F_i$. In every partition with at least two blocks subtract the deterministic constant from each first jet, using invariance of the cumulant under those translations. The one-block partition

has only the four-source mixed jet. Hölder's inequality in the moment formula bounds each of the fourteen partitions containing a mixed source derivative by the product of its mixed-jet bounds and the L^4 bounds for its first jets. Relative to $x^{-4}\prod_i F_i$ this costs at most $x^4(\log x)^C e^{-c\rho_{\mathsf{d}}/M}$. In the partition into singletons expand the centered entries. Each cumulant term with a live factor costs at most $C\prod_i s_i\,\rho_{\mathsf{d}}^4 Q_{17}(k_0)$ by eq. 6.cxii in the corresponding moments. No independence between contacts is needed. The relative cost is bounded by $Cx^4\rho_{\mathsf{d}}^{36}(\log x)^C L^{C\check{R}_{k_0}}e^{-cp_0}$. In the prescribed room $\check{R}_{k_0}=o(p_0)$ and p_0 is of order $(\log x)^{\mathfrak{p}_0}$ with $\mathfrak{p}_0>1$, so this tends to zero. An ordinary error replacing one of the four main terms costs $C\rho_{\mathsf{d}}^{16}(\log x)^{9\mathfrak{p}_0-q}$ relative to the same scale, apart from the exponentially small pieces. The ordinary remainder has an extra inverse power of x that absorbs every fixed logarithmic power. In the prescribed room $\rho_{\mathsf{d}}\asymp(\log x)^{2\mathfrak{p}_0}$ this requires $q>41\mathfrak{p}_0$. The far error is negligible by the same separation as in eq. 6.cxi. This reduces the physical fourth derivative to the fourth cumulant of the bounded readouts, with the four running source factors retained.

6.5.6.4. Counting the near reads on a finite torus. In the native four-source decomposition there is a second bound for the near first derivatives. It uses the cells actually read by each slot. For source i let T_i count the cells of each stratum $\Omega_n\setminus\Omega_{n+1}$, $n<k_0$, in its enlarged support, measured in units of level n. Include the internal strata of pocket strips with their own units. The action costs a constant on each such cell and on the level-k_0 portion, whose volume is $O(\rho_{\mathsf{d}}^4)$. In counting localization and volume terms, a stratum of level n measured at level $j\ge n$ contributes $L^{-4(j-n)}$ times its own count. The sum over reading levels therefore costs $C(\rho_{\mathsf{d}}^4+T_i)$. For the regular source terms the fine-anchor factor $L^{4(n-j)}$ cancels the same volume power in their decay, leaving the summable $L^{-\alpha(n-j)}$. The boundary, potential and remainder bounds are summed on their admitted native carriers. Together these give, on the i-th live-contact event,

$$|\lambda_i^{\mathrm{near}}|\le C\mathfrak{p}_\flat(\bar{x}_s)(A_i+2)^{c_\flat}(\rho_{\mathsf{d}}^4+T_i)u_i,$$

where A_i is the greatest age read by that source and u_i its source norm in this decomposition.

For a live component Z let $D(Z)$ sum $M^4\check{R}_j^5 d_j'(Z_j\cap Z)$ over its history, including the internal pocket sequences. Here d_j' is the geometric size in the history weight. Pack the cells at level j by a maximal family whose block indices are at least four apart in the max norm. The family has at most d_j' members, and each covers at most 7^4 cells of the original region. The pocket strip costs a further factor at most 2^4, and the level conversion costs at most L^4. Consequently T_i is bounded by a fixed multiple of the sum of the charges of the live components meeting that read. Each surviving level contributes at least one, so $D(Z)\ge\mathrm{age}(Z)+1$.

Sum the charge of the live component of each read block, counting it again if the block occurs in another window. Call this sum D_{read}. It dominates A_i+1 on a live contact and controls every T_i. A product of at most four near derivatives, divided by their deterministic factors, is bounded on its contact event by CD_{read}^p for $p\ge4(1+c_\flat)$. If each window meets at most N_{read} blocks, the power-mean inequality bounds its expectation by $C(4N_{\mathrm{read}})^p$ times the largest rooted p-th charge moment.

On the prolonged finite torus the dialled history bound pays $e^{-c_a n}$ for charge at least n. If the component heals at k, restrict the healed sum to charge at least $n+k-k_0$ and use the healed-block bound. This yields a geometric sum over $k>k_0$. A component still live at $\hat{K}$ has charge at least $n+\hat{K}-k_0$. Use the terminal modulus and partition-function lower bound with the charge dial. The additional decay in $\hat{K}-k_0$ pays the terminal volume under the source's depth condition $c_a(\hat{K}-k_0)\ge\mathsf{B}(g^\dagger)\hat{N}^4$, or the below-horizon bound applies. In this range the resulting rooted moment is bounded by

$$Ce^{-\theta_q p_0(g_{k_0})}\sum_{n\ge1}(n+1)^p e^{-c_a n}.$$

This gives the near-slot moment estimate where the actual four-source members have these own-scale bounds and the dialled mass estimates apply. A volume-uniform terminal charge estimate would be needed for the same argument outside that range.

6.6. **Sharp cutoff boundaries in the classical Gaussian law.** The boundary term in the conditional integration-by-parts identity need not be small merely because a Gaussian precision has a gap. The event that every bond is small can have a probability tending to zero with the volume. Dividing a whole-space tail estimate by that probability would therefore give no uniform conditional estimate. In the centered Gaussian reference law the symmetry and convexity of all the other cutoffs give a different argument: their conditional probability decreases on every radial ray. A radial boundary density can then be compared with an inner radial integral in the same direction. The normalization cancels before directions or bonds are summed.

This section proves that comparison for possibly singular physical bond covariances. It derives the resulting second through eighth moment errors, including the covariance and third and fourth connected errors of localized quadratic curvature observables. The actual flat pivot chart and contour gauge supply a concrete fixed-step application. The measure here is classical Gaussian. Haar and coarea factors, old activities, nonlinear action weights and conditional exterior forces are additional terms in the complete conditional law of § 6.3.

Physical blocks, actual covariance entries and the cutoff law. Let Y be a centered Gaussian vector in a finite-dimensional real space, with covariance $G \geq 0$, and let $Z = CY$. The blocks Z_b belong to $\mathbb{R}^3$; repeated or linearly dependent blocks are allowed. All covariances in this section are the actual matrix entries

$$\Gamma_{bc} = \mathbb{E}[Z_b Z_c^{\mathsf{T}}] = C_b G C_c^{\mathsf{T}}, \qquad \Gamma = CGC^{\mathsf{T}}. \tag{6.cxiii}$$

For every tested block choose $b_b > 0$, put

$$\mathcal{A} = \bigcap_b \{|Z_b| < b_b\}, \qquad \nu(P) = \frac{\mathbb{E}[P(Z)1_{\mathcal{A}}]}{\mathbb{P}(\mathcal{A})}, \qquad d_b = \operatorname{rank}\Gamma_{bb} \leq 3, \tag{6.cxiv}$$

and choose an upper marginal scale $\sigma_b > 0$ with $\Gamma_{bb} \leq \sigma_b^2 I$. There is no lower eigenvalue assumption. A sufficiently small neighborhood of zero in the latent Gaussian support is contained in $\mathcal{A}$, so its probability is positive. If $d_b = 0$, then $Z_b = 0$ almost surely and that constraint has no boundary contribution. All blocks occurring in a polynomial to be bounded below are included among the tested blocks.

For a finite volume no lower bound on $\mathbb{P}(\mathcal{A})$ uniform in the volume is assumed or needed. The estimates concern these normalized finite-volume laws. They do not define conditioning on a simultaneous infinite collection of cutoff events, whose intersection could have probability zero.

Marginal log-concavity from one-dimensional transport. We first prove the integration fact needed for the conditional cutoff probability. Suppose nonnegative integrable functions f, g, h on $\mathbb{R}$, positive in the interiors of their interval supports, satisfy, for $0 < t < 1$,

$$h((1-t)x + ty) \geq f(x)^{1-t} g(y)^t. \tag{6.cxv}$$

It is enough initially that the functions be continuous and the positive densities be smooth on their supports, as will hold in the application. Write $F = \int f$, $G_1 = \int g$, and assume both are positive. The increasing quantile map T from density f/F to g/G_1 is continuously differentiable in the interior and satisfies $T'(x) = f(x)G_1/[Fg(T(x))]$. The increasing map $T_t(x) = (1-t)x + tT(x)$ has derivative at least $T'(x)^t$, by the weighted arithmetic-geometric mean inequality. Changing

variables on its image and using eq. 6.cxv gives

$$\begin{aligned}\int h &\geq \int h(T_t(x))T_t'(x)\,dx \\ &\geq \int f(x)^{1-t}g(T(x))^t T'(x)^t\,dx = F^{1-t}G_1^t.\end{aligned} \tag{6.cxvi}$$

One first integrates over compact subintervals of the positive support and then lets them exhaust it. All terms are nonnegative. This also handles unbounded interval endpoints and densities vanishing at the boundary. If either integral is zero the conclusion is immediate.

The same assertion holds in $\mathbb{R}^q$. Here is the induction needed to avoid importing a stronger integration theorem. Split $x = (x', u)$, $y = (y', v)$. For fixed x', y', apply the one-dimensional assertion in u, v, with the slice of h at $(1-t)x' + ty'$. The resulting marginals obey the same inequality in $q-1$ dimensions. Apply the induction hypothesis to those marginals and use Tonelli. Slices of zero integral cause no problem. In the application all positive supports are open convex sets and all slices have these interval supports. Products of smooth compact cutoffs with a Gaussian have integrable dominating functions for their derivatives on every compact set of external parameters. Their marginals are therefore continuous, smooth where required, and positive on the interiors of their convex supports. These facts justify each induction step directly in this class of functions.

In particular, if $k(z, w)$ is log-concave jointly in z, w, and its w slices are of the preceding integrable class, then

$$w_0(z) = \int k(z, w)\,dw \quad \text{is log-concave in } z. \tag{6.cxvii}$$

Indeed use the joint log-concavity inequality for the slices at z, z' and $(1-t)z+tz'$, then integrate by eq. 6.cxvi and its induction. The function k need not be integrable over the external variable z. If $k(-z, -w) = k(z, w)$, its marginal is even. An even log-concave function satisfies $w_0(rn) \leq w_0(sn)$ whenever $0 \leq s \leq r$: write sn as a convex combination of rn and $-rn$. The argument below uses this ray monotonicity and requires no volume estimate for the complete cutoff probability.

Smooth compact approximants with an inner plateau. For $0 < \delta \leq 1/2$, construct a radial cutoff as follows. Put $\phi(t) = 0$ for $t \leq 0$, and on $0 < t < 1$ set

$$\begin{aligned}\phi''(t) &= \frac{e^{-1/t^2}}{(1-t)^4}, \qquad \phi(t) = \int_0^t (t-s)\frac{e^{-1/s^2}}{(1-s)^4}\,ds, \\ \psi_{b,\delta}(z) &= \begin{cases} e^{-\phi((|z|/b-1+\delta)/\delta)}, & |z| < b, \\ 0, & |z| \geq b. \end{cases}\end{aligned} \tag{6.cxviii}$$

The function ϕ is convex, nondecreasing, flat at zero and diverges at least as a positive multiple of $(1-t)^{-2}$ near one. Consequently $\psi_{b,\delta}$ is smooth on all of $\mathbb{R}^3$, including the sphere $|z| = b$, is even and log-concave, and equals one on $|z| \leq (1-\delta)b$. The smoothness at zero follows because it is constant near zero; the smoothness at the outer sphere follows from the exponential decay of $e^{-\phi}$ faster than every inverse power of $1-t$. Convexity of $z \mapsto \phi((|z|/b - 1 + \delta)/\delta)$ follows from convexity of the norm and monotonic convexity of ϕ, extended by $+\infty$ at and beyond one.

As $\delta \downarrow 0$, these cutoffs increase pointwise to $1_{\{|z|<b\}}$. To check monotonicity inside the ball, the argument of ϕ is $1 - (1 - |z|/b)/\delta$, which decreases as δ decreases. Define

$$\chi_\delta(Z) = \prod_b \psi_{b_b,\delta}(Z_b), \qquad \nu_\delta(P) = \frac{\mathbb{E}[P(Z)\chi_\delta(Z)]}{\mathbb{E}\chi_\delta(Z)}. \tag{6.cxix}$$

Every denominator is positive, and Gaussian dominated convergence gives $\nu_\delta(P) \to \nu(P)$ for every polynomial P. No independence of the tested coordinates enters this statement.

The plateau in eq. 6.cxviii is essential to the estimate: the derivative is supported in the outer annulus $[(1-\delta)b, b]$. A smooth cutoff whose derivative remained nonzero close to zero would not by itself yield the exponential boundary tail established next.

Conditional Gaussian support and monotonicity on rays. Fix a block b of positive rank and let $V_b = \operatorname{ran}\Gamma_{bb}$, of dimension $d = d_b$. Orthogonal decomposition of a whitened latent Gaussian gives, when conditioning on $Z_b = z \in V_b$,

$$Z_a = \Gamma_{ab}\Gamma_{bb}^{+}z + D_aW \quad (a \neq b), \qquad w_{b,\delta}(z) = \mathbb{E}_W \prod_{a\neq b} \psi_{b_a,\delta}(\Gamma_{ab}\Gamma_{bb}^{+}z + D_aW). \tag{6.cxx}$$

Here W is a standard Gaussian independent of Z_b, possibly in dimension zero, and Γ_{bb}^{+} is the inverse on V_b, extended by zero. For completeness, write the latent standard Gaussian as its orthogonal projection onto the row space of the map defining Z_b and the perpendicular projection. The first projection is determined by z, and the second is an independent standard Gaussian on its subspace. This proves eq. 6.cxx, including singular full covariances. There is no bound asserted on $\Gamma_{ab}\Gamma_{bb}^{+}$.

The integrand defining $w_{b,\delta}$, multiplied by the Gaussian density of W, is jointly log-concave in (z, W) and unchanged under $(z, W) \mapsto (-z, -W)$. The formula in eq. 6.cxvii therefore proves that $w_{b,\delta}$ is even and log-concave on V_b. Its value decreases along every ray. It is positive near zero, although it may vanish elsewhere because of exact dependencies between blocks. Such zeros strengthen the inequalities below and require no separate boundary trace convention.

A normalized boundary flux with no lower marginal variance. Define the normalized smooth flux

$$\beta_{b,\delta} = \frac{\mathbb{E}[|\nabla\psi_{b_b,\delta}(Z_b)| \prod_{a\neq b}\psi_{b_a,\delta}(Z_a)]}{\mathbb{E}\chi_\delta(Z)}. \tag{6.cxxi}$$

For $d_b = 0$, set $\beta_b^* = 0$; the flux is zero because the cutoff is constant near zero. In positive rank the radial gradient lies in V_b, so the ambient Euclidean norm in eq. 6.cxxi is also its intrinsic gradient norm. Set $B = b_b$, $a = (1-\delta)B$, and use polar coordinates $z = rn$ in V_b. Up to a constant independent of r, n, the intrinsic Gaussian density is $\exp[-q(n)r^2/2]$, with $q(n) = n^{\mathsf{T}}(\Gamma_{bb}|_{V_b})^{-1}n \geq \sigma_b^{-2}$.

For a fixed direction let

$$\begin{aligned} D_n &= \int_0^B \psi_{B,\delta}(rn)w_{b,\delta}(rn)r^{d-1}e^{-q(n)r^2/2}\,dr, \\ I(q,a) &= \int_0^a s^{d-1}e^{-qs^2/2}\,ds. \end{aligned} \tag{6.cxxii}$$

For every $r \in [a, B]$, ray monotonicity and the inner plateau give $D_n \geq w_{b,\delta}(rn)I(q(n), a)$. The integral of the nonnegative radial derivative $-\partial_r\psi_{B,\delta}$ on $[a, B]$ is one. Thus the corresponding numerator in direction n is at most

$$D_n \frac{B^{d-1}e^{-q(n)a^2/2}}{I(q(n), a)}. \tag{6.cxxiii}$$

This inequality remains true if $D_n = 0$, because the preceding pointwise inequality then forces the numerator to vanish. In particular it does not divide by a possibly zero directional weight.

There are constants depending only on $d \leq 3$ such that $I(q, a) \geq c_d \min(a^d, q^{-d/2})$. If $a \leq q^{-1/2}$, integrate over $[0, a]$, where the exponential is at least $e^{-1/2}$. Otherwise integrate over $[0, q^{-1/2}]$. Since $B/2 \leq a \leq B$, the ratio in eq. 6.cxxiii is bounded by $C_dB^{-1}[1 + (qB^2)^{d/2}]e^{-qB^2/8}$. Taking the supremum over all $q \geq \sigma_b^{-2}$ gives

$$\boxed{\beta_{b,\delta} \leq \beta_b^* := \frac{C_d}{b_b}\left[1 + \left(\frac{b_b}{\sigma_b}\right)^d\right] e^{-c_d b_b^2/\sigma_b^2}} \qquad (1 \leq d = d_b \leq 3). \tag{6.cxxiv}$$

To justify the last maximization explicitly, put $x = qB^2$, $x_0 = B^2/\sigma_b^2$. The product $(1 + x^{d/2})e^{-x/16}$ is bounded on $[0, \infty)$. For $x \geq x_0$, the remaining factor is at most $e^{-x_0/16}$; increasing the constant gives the displayed bound, with for example a smaller fixed c_d. Now integrate eq. 6.cxxiii over directions. The determinant normalization of the intrinsic Gaussian and the total integral of D_n

cancel. This proves eq. 6.cxxiv uniformly in δ, the volume, the number of other constraints, their dependencies and every lower marginal eigenvalue.

For $b_b \geq \sigma_b$ the useful simpler consequence is

$$\beta_b^* \leq C_d b_b^{d-1} \sigma_b^{-d} e^{-c_d b_b^2/\sigma_b^2}. \tag{6.cxxv}$$

The qualification is necessary. At small radius the normalized radial flux can be of order d/b_b; an estimate proportional to $b_b^{d-1}\sigma_b^{-d}$ alone would be false there. The proof has retained that regime in eq. 6.cxxiv.

Gaussian integration by parts before the sharp limit. For a polynomial P in the ambient physical blocks, ordinary Gaussian integration by parts, applied to a whitened latent vector, gives

$$\begin{aligned} \nu_\delta[Z_i P] &= \sum_j \Gamma_{ij} \nu_\delta[\nabla_j P] + \sum_b \Gamma_{ib} \mathcal{L}_{b,\delta}(P), \\ \mathcal{L}_{b,\delta}(P) &= \frac{\mathbb{E}[P(Z) \nabla \psi_{b_b,\delta}(Z_b) \prod_{a \neq b} \psi_{b_a,\delta}(Z_a)]}{\mathbb{E}\chi_\delta(Z)}. \end{aligned} \tag{6.cxxvi}$$

The gradients are column vectors and Γ_{ib} are the corresponding 3 by 3 blocks. Degeneracy of Γ does not alter the formula: in the latent variables its two chain-rule factors are exactly the factors producing CGC^{T}. Gaussian domination justifies integration by parts even when the cutoffs have no compact support in null directions of C.

Let $\|P\|_{\mathcal{B}}$ be the supremum on the product of closed balls $\mathcal{B} = \prod_b \{|z_b| \leq b_b\}$. Then

$$|\mathcal{L}_{b,\delta}(P)| \leq \beta_b^* \|P\|_{\mathcal{B}}. \tag{6.cxxvii}$$

For $L = \sum_i \ell_i^{\mathsf{T}} Z_i$, define its canonical sharp defect by

$$\mathcal{D}_L(P) = \nu[LP] - \sum_{i,j} \ell_i^{\mathsf{T}} \Gamma_{ij} \nu[\nabla_j P]. \quad |\mathcal{D}_L(P)| \leq \left(\sum_{i,b} |\ell_i| \, \|\Gamma_{ib}\| \beta_b^* \right) \|P\|_{\mathcal{B}}. \tag{6.cxxviii}$$

Indeed the left side is the limit of the left side of eq. 6.cxxvi contracted with ℓ, while eq. 6.cxxvii bounds the right side uniformly. This proves the sharp identity and its estimate without assigning a meaning to a product of a boundary delta function with another sharp indicator on the same surface.

For repeated or dependent constraints the separate sharp limits of $\mathcal{L}_{b,\delta}$ need not be intrinsic: the allocation between coincident constraint surfaces can depend on the smoothing schedule. The total defect in eq. 6.cxxviii is intrinsic, because its definition uses only sharp conditional polynomial expectations. This is the only boundary object needed for the moment estimates. When a classical smooth-boundary free-coordinate trace exists, $\nabla 1_D = -n \, dS$, so this defect is the negative of the covariance contracted with the outward boundary functional in the conditional-law identity. This also fixes the sign convention.

The actual weighted Schur accounting. Choose positive amplitude scales a_b and positive cell weights μ_b. Assume the following explicit two-sided bound, using the covariance entries of eq. 6.cxiii:

$$S = \max \left\{ \sup_i \sum_j \mu_j \frac{\|\Gamma_{ij}\|}{a_i a_j}, \sup_j \sum_i \mu_i \frac{\|\Gamma_{ij}\|}{a_i a_j} \right\} < \infty, \qquad \varepsilon_\partial = \sup_b \frac{a_b \beta_b^*}{\mu_b}. \tag{6.cxxix}$$

The two sums give the usual bounded operator on weighted counting ℓ^2 after normalizing by a_b: weighted Cauchy in each row and then the column bound give norm at most S. For the boundary estimate the more specific computation is

$$\frac{1}{a_i} \sum_b \|\Gamma_{ib}\| \beta_b^* = \sum_b \mu_b \frac{\|\Gamma_{ib}\|}{a_i a_b} \frac{a_b \beta_b^*}{\mu_b} \leq S \varepsilon_\partial. \tag{6.cxxx}$$

Thus, with $U(L) = \sum_i a_i|\ell_i|$,

$$|\mathcal{D}_L(P)| \le \delta(L)\|P\|_{\mathcal{B}}, \qquad \delta(L) = S\varepsilon_\partial U(L). \tag{6.cxxxi}$$

There is no number of ambient coordinates in this estimate. The factor μ_b^{-1} in ε_∂ cannot be dropped: coordinate Gaussian integration by parts uses the ordinary partial derivative with respect to that coordinate. A weighted kernel convention does not change that derivative.

For example, if a precision is written as a self-adjoint operator K in pairing $\sum_b \mu_b y_b \cdot z_b$, its coordinate density has Euclidean precision MK, where $M = \mathrm{diag}(\mu_b I_3)$. Its covariance matrix is $(MK)^{-1}$. If $(K^{-1}f)_i = \sum_j \mu_j \mathsf{G}(i,j) f_j$, then $K^{-1} = \mathsf{G}M$, and $(MK)^{-1} = \mathsf{G}$. Accordingly $\mathsf{G}(i,j)$, rather than $\mu_j \mathsf{G}(i,j)$, is the covariance entry used here. This calculation prevents either suppressing or inserting a second cell-volume factor during the physical coordinate conversion.

All linear marginals remain Gaussian dominated. There is a useful moment bound independent of the cutoff radii. For a real linear statistic L, put $v(L)^2 = \mathbb{E}L^2 = \ell^{\mathsf{T}}\Gamma\ell$. If this is zero then $L = 0$ almost surely. Otherwise condition the latent Gaussian on $L = x$. The argument of eq. 6.cxx, now including every cutoff, gives an even log-concave weight $w(x)$ for its one-dimensional Gaussian density, first for ν_δ and then by the sharp limit. Therefore w decreases as $|x|$ increases. For any nonnegative increasing function f of $|x|$,

$$\nu[f(|L|)] \le \mathbb{E}f(|v(L)N|), \qquad N \sim N(0,1). \tag{6.cxxxii}$$

To see the normalization explicitly, let R, R' be independent absolute Gaussian variables of this variance. The product $(f(R) - f(R'))(w(R) - w(R'))$ is nonpositive. Its expectation is twice $\mathbb{E}[fw] - \mathbb{E}[f]\mathbb{E}[w]$. Divide by $\mathbb{E}[w] > 0$, and truncate f followed by monotone convergence if necessary. This proves the assertion without a lower bound on the cutoff probability.

In particular the law is centered and, for $m \ge 1$,

$$\nu[L^{2m}] \le (2m-1)!!\, v(L)^{2m}, \qquad \nu[e^{tL}] \le e^{t^2 v(L)^2/2}, \qquad \mathrm{Cov}_\nu(Z) \le \Gamma. \tag{6.cxxxiii}$$

The exponential assertion follows by symmetry and the increasing function $\cosh(t|L|)$. The covariance ordering follows by testing every linear functional. Every joint moment of an odd number of linear statistics is zero, because the entire law is invariant under $Z \mapsto -Z$.

Covariance and fourth-order defects of linear statistics. For linear statistics L_r, write

$$\begin{aligned} &\Gamma^L_{rs} = \mathbb{E}[L_r L_s], \quad \Sigma_{rs} = \nu[L_r L_s], \quad E_{rs} = \Sigma_{rs} - \Gamma^L_{rs}, \\ &B_r = \sum_i b_i|\ell_{r,i}|, \quad \delta_r = S\varepsilon_\partial U(L_r), \quad v_r = v(L_r). \end{aligned} \tag{6.cxxxiv}$$

Here B_r bounds $|L_r|$ on the entire cutoff box, whereas v_r is its untruncated Gaussian standard deviation. These two quantities have different uses. Taking $P = L_s$ in eq. 6.cxxviii gives

$$|E_{rs}| \le \min(\delta_r B_s, \delta_s B_r), \qquad |\Sigma_{rs}| + |\Gamma^L_{rs}| \le 2v_r v_s. \tag{6.cxxxv}$$

The second estimate follows separately for its two terms from eq. 6.cxxxiii and Gaussian Cauchy–Schwarz. Every third joint moment of linear statistics is exactly zero.

Apply the same identity to $P = L_2L_3L_4$. The derivative of this product produces exactly three pairings, and hence

$$\nu[L_1L_2L_3L_4] = \sum_{j=2}^{4} \Gamma^L_{1j}\Sigma_{\{2,3,4\}\setminus\{j\}} + \mathcal{D}_{L_1}(L_2L_3L_4). \tag{6.cxxxvi}$$

In this formula $\Sigma_{\{r,s\}} = \Sigma_{rs}$. Subtracting the three Gaussian pairings gives an error at most $\delta_1 B_2 B_3 B_4 + \sum_{j=2}^{4} |\Gamma^L_{1j}||E_{\{2,3,4\}\setminus\{j\}}|$. Subtracting instead the three pairings with the actual covariance Σ gives the fourth connected moment and the explicit bound

$$\begin{aligned}|\kappa_\nu(L_1,L_2,L_3,L_4)| \le \delta_1\big(&B_2B_3B_4 + B_2v_3v_4\\ &+ B_3v_2v_4 + B_4v_2v_3\big).\end{aligned} \tag{6.cxxxvii}$$

Indeed replace $\Gamma^L_{1j} - \Sigma_{1j}$ by $-E_{1j}$, use $|E_{1j}| \le \delta_1 B_j$, and bound each remaining Σ by its two Gaussian standard deviations. Any of the four indices can be chosen as the first one. The identity proves the connected estimate without assuming a small fourth cumulant.

For example, $\mathrm{Cov}_\nu(L_1L_2, L_3L_4) = \kappa_\nu(L_1,L_2,L_3,L_4) + \Sigma_{13}\Sigma_{24} + \Sigma_{14}\Sigma_{23}$. Together with eq. 6.cxxxv this gives the covariance error of two factored quadratic observables relative to their untruncated Gaussian Wick contractions. The higher degree estimates below also control three and four such observables.

The moment recursion through degree eight. Let a collection of localized linear statistics obey

$$B_r \le R\,U_r, \qquad v_r \le v\,U_r, \qquad U_r = U(L_r), \qquad D = S\varepsilon_\partial. \tag{6.cxxxviii}$$

One may take $R = \max_{i\in\mathcal{S}} b_i/a_i$ and $v = \max_{i\in\mathcal{S}} \sigma_i/a_i$, where $\mathcal{S}$ is the union of the supports of these statistics. The variance bound follows from $\|\Gamma_{ij}\| \le \sigma_i\sigma_j$, itself a consequence of Gaussian Cauchy–Schwarz. Thus neither R nor v requires a maximum over coordinates outside their supports.

Let $M_\nu(1,\dots,n) = \nu[\prod_{r=1}^n L_r]$ and let M_0 denote the corresponding untruncated Gaussian moment. For even n, integration by parts gives the exact recurrence

$$M_\nu(1,\dots,n) - M_0(1,\dots,n) = \sum_{j=2}^{n} \Gamma^L_{1j}(M_\nu - M_0)(\{2,\dots,n\}\setminus\{j\}) + \mathcal{D}_{L_1}\Big(\prod_{r=2}^{n} L_r\Big). \tag{6.cxxxix}$$

For $n = 2$ the empty moment difference is zero. Odd moments vanish for both laws. Consequently

$$\begin{aligned}|M_\nu(1,\dots,2m) - M_0(1,\dots,2m)| &\le d_{2m}\prod_{r=1}^{2m} U_r,\\ d_2 = DR, \qquad d_n &= DR^{n-1} + (n-1)v^2 d_{n-2}.\end{aligned} \tag{6.cxl}$$

This is a direct induction: each derivative pairing contributes at most $v^2U_1U_j$, and the boundary defect contributes at most $DU_1\prod_{r=2}^n B_r$. In particular

$$\begin{aligned}d_2 &= DR,\\ d_4 &= D(R^3 + 3v^2R),\\ d_6 &= D(R^5 + 5v^2R^3 + 15v^4R),\\ d_8 &= D(R^7 + 7v^2R^5 + 35v^4R^3 + 105v^6R).\end{aligned} \tag{6.cxli}$$

All constants are independent of the total number of Gaussian coordinates. The factors U_r retain the actual localization and scale of each inserted statistic.

Absolute mixed moments have a second bound useful when subtracting products of moments. Hölder with $2m$ equal exponents and eq. 6.cxxxii gives, for either law,

$$\Big|\mathbb{E}_*\prod_{r=1}^{2m} L_r\Big| \le (2m-1)!!\prod_{r=1}^{2m} v_r \le (2m-1)!!v^{2m}\prod_{r=1}^{2m} U_r, \qquad \mathbb{E}_* \in \{\nu,\mathbb{E}\}. \tag{6.cxlii}$$

No independence of the L_r is used. The simultaneous use of the support bound in the defect and the variance bound in unaffected moment factors avoids an unnecessary cutoff radius in every term.

Localized quadratic observables and their connected errors. Write each real quadratic observable in a finite factored form

$$Q_j = \sum_\alpha c_{j,\alpha} L_{j,\alpha,1} L_{j,\alpha,2}, \qquad \mathcal{A}(Q_j) = \sum_\alpha |c_{j,\alpha}| U(L_{j,\alpha,1}) U(L_{j,\alpha,2}). \tag{6.cxliii}$$

Choose common R, v for this finite list as in eq. 6.cxxxviii. Every product of $p \le 4$ quadratic observables is a sum of words of degree $2p$. eq. 6.cxli and eq. 6.cxlii can therefore be summed with absolute coefficients, with factor $\prod_j \mathcal{A}(Q_j)$. This step counts the inserted coefficients, not all coordinates of the integration space.

For clarity define the joint cumulant by the finite partition formula

$$\kappa_*(Q_1, \dots, Q_p) = \sum_{\pi \in \mathfrak{P}_p} (-1)^{|\pi|-1} (|\pi| - 1)! \prod_{A \in \pi} \mathbb{E}_* \prod_{j \in A} Q_j. \tag{6.cxliv}$$

Write $\Delta_p = \kappa_\nu(Q_1, \dots, Q_p) - \kappa_0(Q_1, \dots, Q_p)$. The following explicit estimates result:

$$\begin{aligned} |\nu Q_1 - \mathbb{E} Q_1| &\le d_2 \mathcal{A}(Q_1), \\ |\Delta_2| &\le (d_4 + 2v^2 d_2) \mathcal{A}(Q_1) \mathcal{A}(Q_2), \\ |\Delta_3| &\le (d_6 + 3v^2 d_4 + 15 v^4 d_2) \prod_{j=1}^{3} \mathcal{A}(Q_j), \\ |\Delta_4| &\le (d_8 + 4v^2 d_6 + 30 v^4 d_4 + 156 v^6 d_2) \prod_{j=1}^{4} \mathcal{A}(Q_j). \end{aligned} \tag{6.cxlv}$$

Here the subscript zero always refers to the original untruncated Gaussian with covariance Γ. In particular its quadratic third and fourth cumulants are generally nonzero. They are the Gaussian Wick values, not quantities being set equal to zero.

We give the product bookkeeping in eq. 6.cxlv. By multilinearity it is enough to use one factored monomial for each Q_j. For covariance the full moment difference is bounded by d_4; replacing the product of its two means costs $2v^2 d_2$. At order three the three terms of partition type $(2,1)$ cost at most $3(v^2 d_4 + 3v^4 d_2)$. The product of three means, with coefficient two, costs at most $6v^4 d_2$. Together with d_6, these give the third line. At order four, the four partitions of type $(3,1)$ cost $4(v^2 d_6 + 15 v^6 d_2)$; the three of type $(2,2)$ cost $18 v^4 d_4$. The six of type $(2,1,1)$, with coefficient two, cost $12 v^4 d_4 + 72 v^6 d_2$. The final product of four means, with coefficient six, costs $24 v^6 d_2$. Their sum gives exactly the fourth line. Each estimate uses the telescoping difference of products, bounding unchanged factors by eq. 6.cxlii. This establishes the constants without an assumed connected-moment estimate.

Substituting eq. 6.cxli, the three connected error polynomials multiplying D are respectively

$$\begin{aligned} \mathcal{P}_2(R, v) &= R^3 + 5v^2 R, \\ \mathcal{P}_3(R, v) &= R^5 + 8v^2 R^3 + 39 v^4 R, \\ \mathcal{P}_4(R, v) &= R^7 + 11 v^2 R^5 + 85 v^4 R^3 + 411 v^6 R. \end{aligned} \tag{6.cxlvi}$$

The same method covers mixed linear and quadratic observables: expand each quadratic into its two factors, use eq. 6.cxxxix for each resulting word, and then the finite partition formula. An odd total linear degree vanishes before taking absolute values. No additional probabilistic assumption is required for this mixed case.

The reference Gaussian Wick values themselves can be calculated without a nonsingular physical covariance. If $Q_j = Z^{\mathsf{T}} A_j Z$ with real symmetric A_j, whitening the latent Gaussian gives, for $p \ge 1$,

$$\kappa_0(Q_1, \dots, Q_p) = \frac{2^{p-1}}{p} \sum_{\pi \in S_p} \mathrm{tr}(\Gamma A_{\pi(1)} \cdots \Gamma A_{\pi(p)}). \tag{6.cxlvii}$$

Indeed, near zero the logarithm of the Gaussian exponential generating function is

$$\frac{1}{2}\sum_{k\geq 1}\frac{2^k}{k}\operatorname{tr}\left[\left(\Gamma\sum_j t_j A_j\right)^k\right].$$

Taking the coefficient of $t_1\cdots t_p$ gives eq. 6.cxlvii. This also identifies precisely the contractions to which the sharp-cutoff cumulants are compared.

The fixed-step flat chart realizes the covariance hypothesis. We now specialize to the actual first plain-average fibre at the identity, with fixed odd block size L, in current unit-spacing coordinates. Let P_L be the plain derivative of the nonlinear average, C_0 its pivot inclusion with identity free restriction, and $\mathcal{T}$ the internal contour matrix. The definitions and the actual first-step precision are

$$\begin{aligned}&\operatorname{ran}\mathsf{C}_0=\ker P_L,\qquad (\mathsf{C}_0 u)|_{\text{free}}=u,\\ &\mathsf{K}_L=\mathsf{C}_0^{\mathsf{T}}(\mathrm{d}_1^{\mathsf{T}}\mathrm{d}_1+\mathcal{T}^{\mathsf{T}}\mathcal{T})\mathsf{C}_0,\quad Y\sim N(0,\mathsf{K}_L^{-1}),\quad Z=\mathsf{C}_0 Y.\end{aligned}\tag{6.cxlviii}$$

The quadratic form here is exactly the first-step form, with no previous finer-level minimization. This is the classical reference with action weight $x=\theta$, so $\tau=x/\theta=1$. The canonical normalization $\theta g^2=1$ then gives coefficient one both to the Wilson curl form and to the prescribed actual soft gauge. In particular $\mathcal{T}$ is not a divergence gauge or a freely chosen projector.

The pivot matrix has finite range with at most C_L nonzero blocks in each row and column and entries bounded by C_L. The column assertion follows by the same double-block support as the row assertion: a free bond belongs to only finitely many double blocks when L is fixed. The flat gap and the exponential inverse estimate of § 3.6 give $\|\mathsf{K}_L^{-1}(b,c)\|\leq C_L e^{-\mu_L d(b,c)}$, uniformly on the admitted tori. Consequently

$$\Gamma=\mathsf{C}_0\mathsf{K}_L^{-1}\mathsf{C}_0^{\mathsf{T}},\quad \max\left(\sup_b\sum_c\|\Gamma_{bc}\|,\sup_c\sum_b\|\Gamma_{bc}\|\right)\leq S_L,\quad \Gamma_{bb}\leq\sigma_L^2 I.\tag{6.cxlix}$$

To check volume uniformity, a four-dimensional shell of distance m has at most $C_L(m+1)^3$ bond indices in block coordinates. Summing $e^{-\mu_L m}$ over these shells is finite; periodic image sums have the same upper bound. Multiplying the finite row and column sums of C_0, its transpose and the inverse proves the two sums in eq. 6.cxlix. The diagonal bound follows, for example, from either sum. Increase σ_L to at least one if convenient.

Thus eq. 6.cxxix is realized with $a_b=\mu_b=1$, $S=S_L$, independent of the number of blocks. This is a statement at a fixed current spacing and fixed L. Pointwise decay in a multilevel physical metric would not alone give its unweighted analogue: the number and normalization of coordinates at different scales must still be charged in the actual weighted norm in eq. 6.cxxix.

Why the actual contour gauge leaves no physical quadratic residue. The following range calculation justifies integrating out this soft gauge; it does not apply to an arbitrary soft gauge. Let $\mathcal{S}_0$ be the real site fields vanishing at every block center. For any target internal contour data $s=(s_{y,x})$, $x\in B(y)\setminus\{y\}$, define the site field Rs by $(Rs)(x)=s_{y,x}$ and $(Rs)(y)=0$. Blocks partition the sites, so this is well defined. The plain-gradient identity and the internal path telescoping identity give exactly

$$\begin{aligned}&P_L\mathrm{d}_0 Rs=0,\qquad \mathrm{d}_1\mathrm{d}_0 Rs=0,\qquad (\mathcal{T}\mathrm{d}_0 Rs)_{y,x}=s_{y,x},\\ &\|Rs\|_2=\|s\|_2,\qquad \|\mathrm{d}_0 Rs\|_2\leq 4\|s\|_2.\end{aligned}\tag{6.cl}$$

Here d_0 is the forward site gradient and d_1 the plaquette circulation. The last norm estimate follows by summing $|f(x+e_\mu)-f(x)|^2\leq 2|f(x+e_\mu)|^2+2|f(x)|^2$ over four directions. Internal contour means average paths, but on a gradient every one of those paths gives the same difference $f(x)-f(y)$; hence no path count or L^{-4} factor occurs in the third identity. The constraint identity is $P_L\mathrm{d}_0 f=\mathrm{d}_L(f|_{\text{centers}})$, not a site-average gradient identity.

In particular the image of $\mathcal{T}$ on $\ker \mathrm{d}_1 \cap \ker P_L$ is the entire internal target space. It equals the image of $\mathcal{T}$ on $\ker P_L$. This is the precise range condition that prevents a leftover physical gauge penalty. If an arbitrary alternative gauge failed this condition, integrating its gauge directions could instead leave a positive quadratic form in curvature.

Put $\mathcal{E}_C = \mathrm{d}_1 \ker P_L$. The linear map

$$\ker P_L \longrightarrow \mathcal{E}_C \oplus \mathcal{S}_0, \qquad a \longmapsto (\mathrm{d}_1 a, \mathcal{T} a) \quad \text{is a bijection.} \tag{6.cli}$$

It is onto because any initial lift of a prescribed curvature in $\mathcal{E}_C$ can be adjusted to any prescribed contour data by adding $\mathrm{d}_0 Rs$ from eq. 6.cl. It is one-to-one because the flat gap on $\ker P_L$ says that both $\mathrm{d}_1 a = 0$ and $\mathcal{T} a = 0$ force $a = 0$. Equivalently use the actual internal-bond and crossing-face zero-mode argument of § 3.4.

Apply the constant linear change of variables $Y \mapsto (F, s) = (\mathrm{d}_1 \mathsf{C}_0 Y, \mathcal{T} \mathsf{C}_0 Y)$. Its Jacobian is a nonzero constant between the indicated Euclidean spaces. The quadratic exponent in eq. 6.cxlviii becomes exactly $\frac{1}{2}\|F\|^2 + \frac{1}{2}\|s\|^2$. The constant Jacobian and the integral in s cancel from normalized curvature expectations. Therefore

$$F = \mathrm{d}_1 Z \text{ is a standard Gaussian on } \mathcal{E}_C, \qquad \mathrm{d}_1 \Gamma \mathrm{d}_1^{\mathsf{T}} = P_{\mathcal{E}_C}. \tag{6.clii}$$

The same calculation with any prescribed positive quadratic form in s changes only the gauge normalization. It works here because eq. 6.cl was proved for the actual gauge. It would be invalid merely to call a quadratic penalty a gauge term and omit the range check.

If the action weight is instead $x = \tau\theta$, $\tau > 0$, multiply the curl term in eq. 6.cxlviii by τ, leaving the reference soft gauge coefficient one. The same bijection gives exponent $\frac{\tau}{2}\|F\|^2 + \frac{1}{2}\|s\|^2$, so the corresponding covariance satisfies $\mathrm{d}_1 \Gamma_\tau \mathrm{d}_1^{\mathsf{T}} = \tau^{-1} P_{\mathcal{E}_C}$. All numerical reference applications below use $\tau = 1$. No uniform covariance bound as $\tau \downarrow 0$ is inferred.

The precise relation with the full free-curvature comparison. Let $\mathcal{E} = \mathrm{ran}\, \mathrm{d}_1$ on the same finite periodic lattice, with each positive plaquette counted once. Generally $\mathcal{E}_C$ is a proper subspace of $\mathcal{E}$. With $\mathcal{E}_\perp = \mathcal{E} \cap \mathcal{E}_C^\perp$, ordinary orthogonal decomposition gives

$$P_{\mathrm{ex}} = P_{\mathcal{E}} = P_{\mathcal{E}_C} + P_{\mathcal{E}_\perp}, \qquad F_{\mathrm{full}} = F_C + F_\perp, \tag{6.cliii}$$

where the two summands on the right can be taken independent standard Gaussians on their respective subspaces. This proves that the law of F_{full} is the full exact-cochain Gaussian used in the free-kernel comparison. Conditional on $F_\perp$, its centered fluctuation has exactly the curvature law in eq. 6.clii. Conversely, integrating this complementary Gaussian supplies the missing curvature covariance.

The decomposition is an exact classical Gaussian identity. It does not identify the law of a coarse mean in an interacting run. In particular it does not permit replacing $P_{\mathcal{E}_C}$ by P_{ex} in a single fixed-fibre Gaussian estimate. Fine fluctuations alone omit the second term of eq. 6.cliii; exterior or coarse averaging is a separate covariance contribution in the complete conditional-law identity.

There is a finite-dimensional expression for this distinction which needs no chosen coarse-coordinate parametrization. Let Q_C be any matrix whose kernel inside $\mathcal{E}$ is $\mathcal{E}_C$. Then

$$P_{\mathcal{E}_C} = P_{\mathrm{ex}} - P_{\mathrm{ex}} Q_C^{\mathsf{T}} (Q_C P_{\mathrm{ex}} Q_C^{\mathsf{T}})^{+} Q_C P_{\mathrm{ex}}. \tag{6.cliv}$$

To prove the formula, set $B = Q_C P_{\mathrm{ex}}$ restricted to $\mathcal{E}$. Singular-value decomposition shows that $B^{\mathsf{T}}(BB^{\mathsf{T}})^{+}B$ is the orthogonal projection onto $(\ker B)^\perp$ within $\mathcal{E}$. Subtract it from the identity on $\mathcal{E}$, then extend by zero on $\mathcal{E}^\perp$. This yields exactly eq. 6.cliv. The identity is not an estimate that the correction is small. Such an estimate on a particular physical window would require its own quantitative proof.

At fine physical spacing η, write the canonical raw curvature as F. The physical curvature of this canonically normalized field is $\eta^{-2}F$. The full exact-cochain covariance is therefore $\eta^{-4} P_{\mathrm{ex}}$, precisely the normalization of Requirement F1 in the dependency catalogue. Its plaquette components are

sampled at their own barycentres. For three independent color copies and $Q_f = \frac{1}{2}\sum_p f(x_p)|F_p|^2$, Gaussian Wick contraction gives

$$\mathrm{Cov}_0(Q_f, Q_h) = \frac{1}{2}\sum_{p,q} f(x_p)h(x_q)\|\mathbb{E}[F_p F_q^{\mathsf{T}}]\|_{\mathrm{HS}}^2. \tag{6.clv}$$

The physical Riemann weight η^4 multiplying the squared physical curvature cancels its η^{-4} scaling in Q_f. Thus the $\mathrm{tr}(F^2)$ readout is the unweighted plaquette sum displayed here; inserting an additional η^4 into that sum would change the observable.

For the full law in eq. 6.cliii, eq. 6.clv has the F1 kernel and its proved separated-smearing comparison. For the first-step fluctuation it has $P_{\mathcal{E}_C}$ instead. The cutoff estimates compare each specified Gaussian reference with its own Wick kernel. They transfer F1 only after the reference used in the application has actually been identified with, or compared to, the full exact-cochain reference. The decomposition records how this works in the free Gaussian theory without asserting it for a conditional interacting measure.

Uniform cutoff errors for the curvature polynomial. Take all physical tangent bond radii equal to $B \geq \sigma_L$ in eq. 6.cxlviii, and retain counting weights and unit amplitudes. From eq. 6.cxxv, including all ranks from one through three and enlarging the fixed constants,

$$\sup_b \beta_b^* \leq \beta_L(B) := C_L B^2 e^{-c_L B^2}, \qquad D \leq S_L \beta_L(B), \qquad R = B, \quad v \leq \sigma_L. \tag{6.clvi}$$

The bounds remain valid if some physical blocks are deterministic. For a sampled test function let $W_f = \sum_p |f(x_p)|$, and define the canonical quadratic curvature polynomial

$$Q_f(Z) = \frac{1}{2}\sum_p f(x_p)|\mathrm{d}_1 Z(p)|^2, \qquad \mathcal{A}(Q_f) \leq 24 W_f. \tag{6.clvii}$$

Indeed a fixed color component of a plaquette curl has four coefficients of absolute value one, hence $U \leq 4$. There are three colors and the coefficient is $f/2$, so $(1/2)\cdot 3 \cdot 4^2 = 24$. Each positive direction pair is already counted once in the sum over p. This calculation requires no estimate on the number of unobserved bonds.

For $p = 2, 3, 4$, equations in eq. 6.cxlv– eq. 6.clvi now imply, with $\Delta_p = \kappa_\nu(Q_{f_1}, \dots, Q_{f_p}) - \kappa_0(Q_{f_1}, \dots, Q_{f_p})$,

$$\begin{aligned} |\nu Q_f - \mathbb{E} Q_f| &\leq C_L \beta_L(B) B W_f, \\ |\Delta_p| &\leq C_{L,p}\beta_L(B)(B^{2p-1} + B^{2p-3} + \cdots + B)\prod_{j=1}^{p} W_{f_j}. \end{aligned} \tag{6.clviii}$$

Here all expectations are normalized, including the sharp product cutoff law. The third and fourth lines implicit in the second formula use sixth and eighth underlying linear moments. The vanishing of odd linear moments alone would not establish them. The Gaussian values on the right are the actual cycle contractions in eq. 6.cxlvii, with the constrained curvature projection of eq. 6.clii in this particular application.

If each test is supported in a ball of current radius $r \geq 1$, then $W_f \leq C(r+1)^4\|f\|_\infty$. Thus the covariance error has the expected two localized volume factors $C(r+1)^8\|f\|_\infty\|h\|_\infty$, rather than a factor proportional to the volume of the torus. At orders three and four the factors are $(r+1)^{12}$ and $(r+1)^{16}$. These are insertion-support counts; all constants in their coefficients are uniform over torus size and support positions.

The nonlinear Wilson readout on the actual joint chart. The Gaussian law need not be restricted to a merely formal observable. At the flat background let $a = \Phi_0(u)$ be the actual nonlinear fibre chart of § 3.3, so u is the free raw logarithm and $D\Phi_0(0) = \mathsf{C}_0$. Its domain contains $\|u\|_\infty < r_L$, with $r_L = cL^{-9}$ after reducing the fixed constant. The finite local support and the uniform holomorphic bounds on that joint domain give, on a smaller real ball,

$$\begin{aligned} \|\Phi_0(u) - \mathsf{C}_0 u\|_\infty &\le C_L \|u\|_\infty^2, \\ A_p(e^{i\Phi_0(u)}) &= \frac{1}{2}|\mathrm{d}_1 \mathsf{C}_0 u(p)|^2 + R_p(u), \\ |R_p(u)| &\le C_L \|u\|_\infty^3. \end{aligned} \tag{6.clix}$$

For the second assertion use the holomorphic extension $1 - \frac{1}{2}\operatorname{tr} U_p$, which equals the real Wilson readout on SU(2). Its constant and linear terms at the identity are zero. Differentiating the four bond products shows its quadratic term is $\frac{1}{2}|\mathrm{d}_1 \mathsf{C}_0 u(p)|^2$, with Pauli pairing $|X|^2 = \frac{1}{2}\operatorname{tr} X^2$. To bound the remainder, apply one-variable Cauchy to the actual local function of tu on a disc of radius proportional to $r_L/\|u\|_\infty$. The joint chart bounds this function uniformly; summing the Taylor terms of order at least three on a smaller disc gives the displayed cubic estimate. The first assertion follows by the same argument at order two. A plaquette and its pivot dependencies occupy a fixed number of double blocks, so neither remainder constant contains the lattice volume.

Set $u = gY$. Under the full physical tangent cutoff $|\mathsf{C}_0 Y(b)| < B$, the identity free restriction implies $\|Y\|_\infty \le B$. Thus $gB < r_L/2$ places every field of this cutoff event in the actual chart, and for $O_f = \sum_p f(x_p) A_p$,

$$O_f(e^{i\Phi_0(gY)}) = g^2 Q_f(Z) + \mathcal{R}_f(Y), \quad |\mathcal{R}_f| \le C_L g^3 B^3 W_f, \quad |Q_f| \le C_L B^2 W_f. \tag{6.clx}$$

This uses the literal sampled $\mathrm{tr}(F^2)$ sum, with no extra fine-cell volume factor. The probability measure in this identity is still the Gaussian cutoff measure, rather than the nonlinear action measure.

The finite partition definition of a cumulant gives $|\kappa(X_1, \dots, X_p)| \le C_p \prod_j \|X_j\|_\infty$. Expand each slot of eq. 6.clx. If $gB \le 1$, the terms with at least one remainder have total absolute value at most $C_{L,p} g^{2p+1} B^{2p+1} \prod_j W_{f_j}$. Indeed a remainder replaces one factor $g^2 B^2$ by $g^3 B^3$; the finite sum over subsets costs at most a constant times $gB(1+gB)^{p-1}$. Together with eq. 6.clviii, this proves

$$\begin{aligned} &|\kappa_\nu(O_{f_1}, \dots, O_{f_p}) - g^{2p} \kappa_0(Q_{f_1}, \dots, Q_{f_p})| \\ &\quad \le C_{L,p} g^{2p} \left[\beta_L(B)(B^{2p-1} + B^{2p-3} + \cdots + B) + gB^{2p+1}\right] \prod_{j=1}^{p} W_{f_j}, \qquad p = 2, 3, 4. \end{aligned} \tag{6.clxi}$$

The same reasoning for one insertion gives mean error $C_L g^2 [\beta_L(B) B + gB^3] W_f$. This is a concrete sharp-cutoff Gaussian clean-law estimate with the actual nonlinear observable numerator. It includes neither the nonlinear change in its probability density nor any missing coarse-mean contribution to the chosen free reference.

Literal radius conversions and comparison of cutoff sets. At physical fine spacing η, raw bond logarithm and physical potential obey $a_b = \eta A_b$. In canonical tangent variables $a_b = gZ_b$. Consequently the conversions are

$$|a_b| < \alpha_b \iff |Z_b| < \alpha_b/g, \qquad |\ell_b A_b| < r_b \iff |Z_b| < \frac{\eta r_b}{g \ell_b}. \tag{6.clxii}$$

The second relation refers to the tangent physical potential; ℓ_b is the actual scale multiplying it in that field bound. For the nonlinear chart use eq. 6.clix: its raw field differs from gZ by at most $C_L g^2 B^2$ on the event in question. Any comparison with a cutoff on the nonlinear field must charge this error and preserve a stated radius margin.

A free-coordinate box and the complete physical tangent product box are not equal events. With $c_L = \|\mathsf{C}_0\|_{\infty\to\infty} \geq 1$, one has only

$$\{\|Y\|_\infty < B/c_L\} \subset \{\|\mathsf{C}_0 Y\|_\infty < B\} \subset \{\|Y\|_\infty < B\}. \tag{6.clxiii}$$

The first inclusion uses the actual row norm of the pivot map; the second uses its identity free restriction. Inclusion alone does not compare the corresponding normalized conditional expectations. The proof in this section works directly with the middle, complete physical event, so requires no false identification of these boxes. If an actual integration rule instead cuts only free coordinates, apply the abstract theorem to those actual independent-coordinate blocks and carry the observable coefficient map separately; it is a different cutoff law with its own explicitly specified covariance and radii.

For growing canonical radius $B = c_*(\log\theta)^s$, $s > 0$, and $g = \theta^{-1/2}$, every fixed power of $\log\theta$ times $\beta_L(B)$ tends to zero, as does every fixed such power times g. The chart condition $gB < r_L/2$ holds eventually. Thus for support radii bounded by any fixed power of $\log\theta$, the bracket and localized volume factors in eq. 6.clxi tend to zero, uniformly in torus volume. This is an absolute error relative to the displayed insertion norms. Converting it into a relative error for a specified two-point coefficient also requires a proved lower bound for that coefficient in the same test class.

What passes to limits and what remains an interaction estimate. The finite-volume cutoff laws have Gaussian-dominated moments for every fixed local linear statistic by eq. 6.cxxxii. In the fixed-step setting their marginal variances are bounded uniformly by eq. 6.cxlix. These bounds give tightness on each finite collection of bond coordinates. A diagonal subsequence on the countable lattice therefore has compatible finite-dimensional limits. Uniform higher local moments give uniform integrability, so the local polynomial estimates above pass to every such limit. This conclusion does not assert uniqueness of the limit, and does not identify it with conditioning an infinite-volume Gaussian on a global event of positive probability. It only uses uniform finite-volume bounds.

The boundary proof also identifies its precise obstruction in a general conditional law. If in actual free coordinates, with $G > 0$, its complete density is $Z_e^{-1}\chi(CY)\exp[-\frac{1}{2}Y^{\mathsf{T}}G^{-1}Y + R_e(Y)]$, integration by parts has, in addition to the boundary defect, the force term

$$\mathbb{E}_{\nu_e}[P(CY)\,(CG\nabla R_e(Y))_i]\,. \tag{6.clxiv}$$

This follows by differentiating the entire density, with its actual normalizer; all integrability and chart-domain conditions must hold. An arbitrary exterior linear force can translate the law toward the boundary, and an arbitrary old activity can destroy the even log-concavity used in eq. 6.cxx. Neither its small Hessian alone nor the Gaussian boundary theorem bounds these two effects.

The same radial argument would remain valid for a further even log-concave factor in the physical variables, because that factor can be included in the joint log-concave integrand defining $w_{b,\delta}$. Its new force term still has to be bounded in the complete integration-by-parts identity. No such convexity, symmetry or force bound is asserted for the resummed interacting conditional density.

Accordingly the result established here is the boundary and connected-error estimate for the specified centered classical Gaussian sector, with a realized fixed-step covariance and actual observable coordinates. Applying it to the interacting extraction requires the complete measure, exterior averaging, old-activity and force identifications of the conditional-law section, as well as the correct full free reference for F1. It supplies a proved reference estimate for that boundary task, rather than assuming the interacting extraction law or an all-depth unweighted covariance bound.

6.7. **A normalized Gaussian comparison on a finite run window.** The conditional interacting law can be compared directly with a Gaussian when the complete residual exponent has small oscillation on the tested window. The oscillation is measured after removing a field-independent constant, so an exterior-dependent vacuum energy causes no loss. This route is useful when the number of retained coordinates and their tested ranges grow only as powers of the logarithm

of the inverse coupling. It estimates the normalizer together with the numerator and avoids an integration-by-parts argument at a sharp boundary.

The criterion below concerns the complete exponent. In particular it includes the coarea density, Haar density, history multiplicity and all old activities that vary with the retained coordinates. Bounds on the nonlinear Wilson action alone are insufficient for its application. We first prove the comparison, then give explicit estimates for several of the factors that occur when the criterion is verified.

6.7.1. *The tested domain and the reference Gaussian.* Let γ be a centered Gaussian on $\mathbb{R}^N$ with positive definite covariance C. If the Gaussian is supported on a proper linear subspace, choose orthonormal coordinates on that support throughout; the statements then have exactly the same meaning. Let $v = \max_j C_{jj}$. For $R > 0$, put

$$B_R = \{y : \max_j |y_j| \le R\}. \tag{6.clxv}$$

Let D be measurable and contain B_R, up to a Gaussian null set. The domain may be nonconvex. Define $t = \gamma(D^c)$. The one-dimensional Gaussian bound and the union bound give

$$t \le 2N \exp[-R^2/(2v)]. \tag{6.clxvi}$$

Here and below a zero marginal variance can simply be omitted. For completeness, for a marginal Y with variance σ^2, $\mathbb{E}e^{sY} = e^{s^2\sigma^2/2}$. Markov's inequality at $s = R/\sigma^2$ bounds $\mathbb{P}(Y \ge R)$ by $e^{-R^2/(2\sigma^2)}$; apply the same estimate to $-Y$. This argument uses no independence between coordinates.

Suppose a real measurable function P on D has oscillation at most 2α in the following specified sense: there is a real constant c such that

$$|P(y) - c| \le \alpha \quad \text{for } \gamma\text{-almost every } y \in D. \tag{6.clxvii}$$

The constant may depend on the exterior configuration. Set

$$d\nu = Z^{-1} 1_D e^{P-c}\, d\gamma, \qquad Z = \gamma(1_D e^{P-c}). \tag{6.clxviii}$$

If $t < 1$, this is a probability and

$$e^{-\alpha}(1-t) \le Z \le e^{\alpha}(1-t). \tag{6.clxix}$$

The sharper upper bound displayed here is useful: the clean domain mass belongs to the same reference in both numerator and denominator. Let $\gamma_D = \gamma(\,\cdot \mid D)$. Then

$$e^{-2\alpha} \le \frac{d\nu}{d\gamma_D} \le e^{2\alpha} \quad \text{almost everywhere.} \tag{6.clxx}$$

This proves non-vanishing of the real normalizer with its correct domain factor. It does not assert a complex source-domain estimate.

Proposition 6.16 (Weighted comparison of the normalized laws)**.** For every $H \in L^2(\gamma)$,

$$|\nu(H) - \gamma(H)| \le \frac{e^{2\alpha} - 1 + t}{1-t}\,\gamma(|H|) + \frac{\gamma(|H|1_{D^c})}{1-t} \tag{6.clxxi}$$

$$\le \frac{e^{2\alpha} - 1 + t + \sqrt{t}}{1-t}\,\|H\|_{L^2(\gamma)}. \tag{6.clxxii}$$

Also $\nu(|H|) \le e^{2\alpha}(1-t)^{-1}\gamma(|H|)$ for every nonnegative integrable $|H|$. ■

Proof. The density estimate eq. 6.clxx gives $|\nu(H) - \gamma_D(H)| \le (e^{2\alpha} - 1)\gamma_D(|H|)$. On the other hand,

$$\gamma_D(H) - \gamma(H) = \frac{t\gamma(H) - \gamma(H1_{D^c})}{1-t}.$$

Taking absolute values proves eq. 6.clxxi. Cauchy–Schwarz bounds its last tail integral by $\sqrt{t}\|H\|_2$, and $\gamma(|H|) \le \|H\|_2$. The last assertion follows by direct integration of eq. 6.clxx. □

The estimate only uses a Gaussian measure of the symmetric difference from its full support through t. Thus one can replace a complicated tested domain by a convenient inner box, provided that box actually lies in the image of the physical tests. An upper bound on the size of every retained coordinate does not prove this inner inclusion. The latter is a separate domain-admission calculation.

6.7.2. *Dimension-explicit moments of quadratic readouts.* Let A be real symmetric and set

$$L(A) = \operatorname{tr}|C^{1/2}AC^{1/2}|, \qquad Q_A(y) = y^{\mathsf{T}}Ay - \operatorname{tr}(AC). \tag{6.clxxiii}$$

The absolute value in $L(A)$ is the operator absolute value of a real symmetric matrix. This definition also covers signed profiles. It often gives a better bound than the matrix dimension times a supremum of its entries.

For an integer $k \geq 1$, diagonalize $C^{1/2}AC^{1/2}$ with eigenvalues λ_j. Under γ, its quadratic form has the law $\sum_j \lambda_j G_j^2$, where the G_j are independent standard Gaussians. The triangle inequality in L^k yields

$$\|Q_A\|_{L^k(\gamma)} \leq u_k L(A), \qquad u_k = ((2k-1)!!)^{1/k} + 1. \tag{6.clxxiv}$$

Indeed $\|G_j^2\|_k = ((2k-1)!!)^{1/k}$ and $|\sum_j \lambda_j| \leq \sum_j |\lambda_j|$. The Gaussian even moment formula used here follows by integration by parts: $\mathbb{E}G^{2k} = (2k-1)\mathbb{E}G^{2k-2}$ with $\mathbb{E}G^0 = 1$. The estimate includes singular matrices A; if $L(A) = 0$, the quadratic readout is identically zero on the Gaussian support.

For readouts $Q_i = Q_{A_i}$ and any nonempty set S of s indices, Hölder gives

$$\left\| \prod_{i\in S} Q_i \right\|_{L^2(\gamma)} \leq u_{2s}^s \prod_{i\in S} L_i, \qquad L_i = L(A_i). \tag{6.clxxv}$$

There is no factor depending on N apart from the actual sizes L_i. This matters when an observable sums over a window: its coefficient mass must be counted once in L_i, rather than once for each moment estimate.

Let $0 < g \leq 1$. Suppose the physical readouts on D are

$$X_i(y) = c_i + g^2[Q_i(y) + I_i(y)], \qquad |I_i(y)| \leq \rho_i L_i. \tag{6.clxxvi}$$

The constants c_i are deterministic at the fixed exterior; their variation with the exterior is retained when applying the law of total cumulance. Assume $L_i > 0$ and $0 \leq \rho_i \leq 1$. An entry with $L_i = 0$ may instead be handled separately using a positive bound for its remainder.

Proposition 6.17 (Mixed moments of the physical readouts)**.** Assume $\alpha \leq 1/4$ and $t \leq 1/2$. For every nonempty S of size $s \leq n$,

$$\left| \nu\left(\prod_{i\in S} (X_i - c_i) \right) - g^{2s}\gamma\left(\prod_{i\in S} Q_i \right) \right| \leq K_n \left(\alpha + \sqrt{t} + \max_i \rho_i \right) g^{2s} \prod_{i\in S} L_i, \tag{6.clxxvii}$$

where K_n depends only on n. The absolute mixed moments of both displayed laws are at most $K_n g^{2s} \prod_{i\in S} L_i$. ∎

Proof. First compare the product of Q_i under ν and γ. Apply Prop. 6.16 and eq. 6.clxxv. Since $\alpha \leq 1/4$ and $t \leq 1/2$, the coefficient there is at most a fixed multiple of $\alpha + \sqrt{t}$.

Next expand $\prod(Q_i + I_i) - \prod Q_i$ by the nonempty subset of positions at which the remainder I_i is used. Its absolute expectation is bounded by

$$\sum_{\varnothing \neq J \subset S} \prod_{i\in J} (\rho_i L_i)\, \nu\left(\prod_{i\in S\setminus J} |Q_i| \right).$$

The final assertion of Prop. 6.16 and Hölder bound the last moment by a constant depending only on n times $\prod_{i\in S\setminus J} L_i$. The sum of the remainder coefficients is at most $\prod_{i\in S}(1+\rho_i) - 1 \leq$

$(2^n-1)\max_i \rho_i$. Multiply by g^{2s} to obtain eq. 6.clxxvii. The same estimates without subtraction bound the absolute mixed moments. □

6.7.3. *The conditional connected coefficient.* The cumulant polynomial in § 6.4 now proves, for $2 \le n \le 4$,

$$\left|\kappa_\nu(X_1,\dots,X_n) - g^{2n}\kappa_\gamma(Q_1,\dots,Q_n)\right| \le K_n'\left(\alpha+\sqrt{t}+\max_i \rho_i\right) g^{2n}\prod_i L_i. \tag{6.clxxviii}$$

Here the constants c_i disappear by translation invariance of a cumulant of order at least two. To see the stated dependence, use eq. 6.clxxvii for the one replaced block in the telescoping product proof of Prop. 6.13. The other blocks have the moment bound of Prop. 6.17. There are only finitely many partitions at fixed n, giving K_n'.

For the one-point function the corresponding estimate is

$$|\nu(X_i) - c_i| \le K_1(\alpha+\sqrt{t}+\rho_i)g^2 L_i, \tag{6.clxxix}$$

because $\gamma(Q_i)=0$. When exteriors vary, the error from $c_i(e)$ relative to a fixed reference must be added to this bound; it is not removed by centering the global observable.

The reference quadratic cumulants have a finite trace formula. Put $B(z)=\sum_i z_i C^{1/2}A_iC^{1/2}$. For $2\|B(z)\|<1$, diagonalizing the real Gaussian integral first for real z near zero, and then continuing analytically, gives

$$\begin{aligned}\log\gamma\left(e^{\sum_i z_iQ_i}\right) &= -\operatorname{tr}B(z) - \tfrac{1}{2}\log\det(I-2B(z)) \\ &= \sum_{m\ge 2}\frac{2^{m-1}}{m}\operatorname{tr}B(z)^m.\end{aligned} \tag{6.clxxx}$$

The determinant branch equals one at zero. The logarithmic series converges absolutely in operator norm on every smaller ball, so coefficient extraction is legitimate. In particular

$$\kappa_\gamma(Q_1,Q_2) = 2\operatorname{tr}(A_1CA_2C). \tag{6.clxxxi}$$

At order four it is the sum over closed four-cycles, with the coefficient and colour multiplicity fixed in the free curvature calculation. Thus eq. 6.clxxviii controls an interacting connected coefficient, rather than merely a change in its quadratic precision.

6.7.4. *A finite-dimensional graph for the constraints.* We next estimate one contribution to P. Let $p\in\mathbb{R}^m$ be pivot variables and $y\in\mathbb{R}^N$ retained variables. Suppose a real C^1 constraint is $q(p,y)=0$. On an open neighborhood of the retained domain assume it has one admitted solution $p=\varphi(y)$ and that $D_pq(\varphi(y),y)$ is invertible. These assumptions concern the actual constraint and its chosen branch; the implicit function theorem supplies only a local solution until uniqueness and domain coverage are checked.

Let Q be a fixed invertible pivot matrix and suppose

$$D_pq(\varphi(y),y) = Q[I+E(y)], \qquad \|E(y)\|\le\delta<1. \tag{6.clxxxii}$$

The operator norm here is Euclidean. Then

$$\left|\log\frac{|\det D_pq(\varphi(y),y)|}{|\det Q|}\right| \le \frac{m\delta}{1-\delta}. \tag{6.clxxxiii}$$

Indeed $I+sE$ is invertible for $0\le s\le 1$. Its real determinant remains positive, since it equals one at $s=0$. The logarithmic derivative is $\operatorname{tr}[(I+sE)^{-1}E]$, whose absolute value is at most $m\delta/(1-s\delta)$. Integrating and using the weaker constant $1/(1-\delta)$ proves the estimate. The same proof works for a complex branch of the determinant logarithm on a norm ball, but the probability comparison itself remains real.

Consequently the coarea factor $|\det D_pq|^{-1}$ has a field-dependent logarithm bounded, after subtraction of $-\log|\det Q|$, by the right hand side of eq. 6.clxxxiii. The dimension m is essential. An

operator perturbation $\delta = o(1)$ alone does not imply a small total determinant logarithm if $m\delta$ is large.

The identity underlying this coarea factor is elementary but fixes which determinant occurs. The map $(p, y) \mapsto (q(p, y), y)$ has block-triangular derivative with determinant $\det D_p q$. Changing variables in an integral with the constraint delta function and then evaluating at $q = 0$ gives

$$\int f(p, y)\delta(q(p, y))\, dp\, dy = \int f(\varphi(y), y)|\det D_p q(\varphi(y), y)|^{-1}\, dy \tag{6.clxxxiv}$$

on the admitted branch. With several branches the right side is summed over them. A gauge quotient or a change of the retained coordinates contributes its own factor and must be included as well.

For a useful quantitative graph estimate, suppose

$$q(p, y) = Qp + Ry + g\mathcal{N}(p, y), \tag{6.clxxxv}$$

where $\|Q^{-1}\| \le K$ and $\mathcal{N}$ is Lipschitz in p with constant L_p on a specified closed pivot ball for each y in the retained domain. Put $p_0(y) = -Q^{-1}Ry$. If

$$gKL_p \le \tfrac{1}{2}, \qquad gK \sup_y |\mathcal{N}(p_0(y), y)| \le b/2, \tag{6.clxxxvi}$$

and the radius-b ball about every $p_0(y)$ lies in the declared pivot domain, the map $p \mapsto p_0(y) - gQ^{-1}\mathcal{N}(p, y)$ is a contraction of that ball. Thus

$$|\varphi(y) - p_0(y)| \le 2gK|\mathcal{N}(p_0(y), y)|. \tag{6.clxxxvii}$$

The proof is the contraction inequality with its inhomogeneous term moved to the left. If $\|D_p\mathcal{N}\| \le L_p$, then eq. 6.clxxxii holds with $\delta = gKL_p$. This is a finite-window criterion with a specified radius. For an actual multiscale constraint, a uniform K and coverage of the entire tested set still have to be proved.

6.7.5. *Haar factors and the action remainder.* The analytic Haar density and the bare Wilson remainder are estimated in eq. 4.dlvii and eq. 4.dlviii. In particular, at a fixed coupling, M link logarithms ga with $|a| \le R_1$ and $gR_1 \le 1/4$ contribute at most $CMg^2R_1^2$ to the Haar logarithm after subtraction of its value at zero. The coordinate graph may change a to a function of y; the tested bound must then hold on that whole graph. The fixed Jacobian powers of g cancel in normalized expectations. Source-dependent powers must be retained.

Let S be a real analytic action on a finite coordinate chart and write $a = gTy$, where T is a specified linear map. Assume the quadratic reference has been extracted exactly, so

$$g^{-2}S(gTy) = c_S + \ell^{\mathsf{T}}y + \tfrac{1}{2}y^{\mathsf{T}}Qy + R_S(y). \tag{6.clxxxviii}$$

If $S'(0) = 0$ and the quadratic form is the actual Hessian, then $\ell = 0$. At a noncritical reference the linear term must be retained or removed by a verified Gaussian shift. Suppose that for every $y \in D$ the full segment $a = tgTy$, $0 \le t \le 1$, lies in the declared analytic chart, and that on that segment the third derivative obeys

$$|D^3S(a)[Ty, Ty, Ty]| \le K_S(y). \tag{6.clxxxix}$$

Taylor's integral remainder gives

$$|R_S(y)| \le \tfrac{1}{6}gK_S(y). \tag{6.cxc}$$

The power g comes from the cubic numerator divided by g^2. In a local action with M terms, each depending on a uniformly bounded list of coordinates of size at most R_1, a uniform third-derivative bound gives $K_S(y) \le CMR_1^3$. This assertion requires those local derivative bounds; when S is the Wilson action composed with a constrained background, the derivatives of that background are part of them.

If the graph of the constraints is nonlinear, it is equally valid to take the third derivative of the composed action on the graph. Replacing that derivative by the bare Wilson third derivative

would omit derivatives of φ. The complete constraint-curvature term in the Hessian is also part of the reference Q in eq. 6.clxxxviii. The present comparison does not justify replacing it by a flat Feynman precision.

6.7.6. *Old activities that meet the window.* Let W be a finite set of retained cells. Suppose an old interaction has a convergent representation

$$H(y) = \sum_X H_X(y), \tag{6.cxci}$$

where a term is independent of y if X does not meet W. Let y_* be one admitted reference. If

$$\sup_{v\in W} \sum_{X\ni v} \sup_{y\in D} |H_X(y) - H_X(y_*)| \le \epsilon, \tag{6.cxcii}$$

then

$$\sup_{y\in D} |H(y) - H(y_*)| \le |W|\epsilon. \tag{6.cxciii}$$

To prove this, bound the sum over terms that meet W by the sum over all incidences (v, X) with $v \in W \cap X$. Each contributing term is counted at least once. The positive bound in eq. 6.cxcii gives absolute and uniform convergence of the varying part, so the subtraction and sum are justified.

The norm in eq. 6.cxcii is a norm of the evaluated terms on one common admitted chart. A formal norm at different centers, or a bound after a different levelled substitution, cannot be inserted without an estimate transferring it to this domain. Terms with old fine-scale origins can meet the retained window in many fine cells. Their incidences must be charged in their actual units; the number $|W|$ cannot be taken to be the coarse window volume unless the formation theorem has already performed that accounting.

This qualification is one reason to state the criterion in terms of the complete oscillation rather than claim that every polylogarithmic coarse window automatically has a small residual exponent. The primitive spatial filing and the admitted substitution are what decide whether eq. 6.cxcii is available.

6.7.7. *Putting the complete exponent together.* Suppose the actual constrained positive density, after a fixed change to its retained coordinates, has the form

$$1_D(y) \exp\{-\tfrac{1}{2} y^{\mathsf{T}} Q y + P(y)\}\, dy, \tag{6.cxciv}$$

with the same Q as the covariance $C = Q^{-1}$. A Gaussian shift may be made first if its determinant, domain image and readout change are retained. Suppose there are field-independent constants subtracting the following parts of P, and that their residual absolute bounds are respectively

$$\alpha_{\mathrm{act}}, \quad \alpha_{\mathrm{Haar}}, \quad \alpha_{\mathrm{coarea}}, \quad \alpha_{\mathrm{old}}, \quad \alpha_{\mathrm{hist}}, \quad \alpha_{\mathrm{bdry}}. \tag{6.cxcv}$$

They denote the nonlinear action, Haar density, coarea and coordinate Jacobians, old activities, resummed history density, and the remaining boundary or stored-field terms. Then eq. 6.clxvii holds with their sum

$$\alpha = \alpha_{\mathrm{act}} + \alpha_{\mathrm{Haar}} + \alpha_{\mathrm{coarea}} + \alpha_{\mathrm{old}} + \alpha_{\mathrm{hist}} + \alpha_{\mathrm{bdry}}. \tag{6.cxcvi}$$

This follows from addition of logarithms on the specified positive branch and the triangle inequality. A factor that can vanish must be kept in the domain or resolved into positive components; it cannot simply be assigned a bounded logarithm.

The preceding estimates give, for example,

$$\alpha \le CMgR_1^3 + CMg^2R_1^2 + \frac{m\delta}{1-\delta} + |W|\epsilon + \alpha_{\mathrm{hist}} + \alpha_{\mathrm{bdry}}. \tag{6.cxcvii}$$

Every quantity on the right is measured in the actual retained coordinate chart. This is a sufficient estimate, not a substitute for the actual determinant or old-core calculation.

6.7.8. *A scale criterion for the second and fourth correlations.* Write $g = x^{-1/2}$, $x \to \infty$. Suppose all of the following bounds hold uniformly on the admitted exteriors:

$$\begin{aligned} N + M + m + |W| &\le C(\log x)^A, & R_1 &\le C(\log x)^B, \\ \delta + \epsilon + \alpha_{\mathrm{hist}} + \alpha_{\mathrm{bdry}} &\le Cx^{-a}(\log x)^E, & a &> 0, \\ L_i &\le C(\log x)^{A_i}\|f_i\|_\infty, & \rho_i &\le Cx^{-a}(\log x)^E. \end{aligned} \tag{6.cxcviii}$$

Suppose also that D contains a box for which $R^2/v \ge c(\log x)^b$, $b > 0$. Then eqs. 6.clxvi and 6.cxcvii give

$$\alpha \le Cx^{-\min(a,1/2)}(\log x)^{E'}, \qquad t \le C(\log x)^A e^{-c'(\log x)^b}. \tag{6.cxcix}$$

Thus eq. 6.clxxviii yields

$$\kappa_\nu(X_1, \dots, X_n) = x^{-n}\kappa_\gamma(Q_1, \dots, Q_n) + o\left(x^{-n}\prod_i \|f_i\|_\infty\right), \qquad n = 2, 4. \tag{6.cc}$$

Indeed every power of $\log x$ in $\prod L_i$ is outweighed both by a positive power of x and by $e^{-c'(\log x)^b/2}$. The same estimates give the lower conditional errors relative to their conditional Gaussian values. If C or A_i varies with the exterior, those Gaussian cumulants must also be compared with one fixed reference before applying the exterior averaging formula. If $c_i(e)$ varies, its error relative to a deterministic reference must likewise be added to eq. 6.clxxix.

The finite-window criterion supplies one explicit route to the interacting law. It has no boundary-flux assumption and does not require convexity of the image of the tests. Its cost is the complete total-oscillation bound and an inner Gaussian box on one actual chart. When fine-scale origins, stored terms or class multiplicities prevent a small total oscillation, the force method of § 6.3 is the appropriate alternative; one cannot infer eq. 6.cc from a small old-activity Hessian in that situation.

6.8. **The four-point curvature kernel: quantitative sign and lattice stability.** The free curvature calculation fixes the normalization of every separated Gaussian cycle. Its collinear four-point value is positive, but a continuity argument at one configuration does not give a uniform margin on the class of test functions used in the short-distance assertion. This subsection supplies that margin. It also computes a negative configuration, proves that its sign persists on an explicit open set, and transfers both signs to the free lattice reference at a uniform mesh floor. These are statements about the leading kernel. They do not compare an interacting measure with its Gaussian reference.

We use the covariance convention of the free curvature chapter. Put $\mathcal{V} = \Lambda^2\mathbb{R}^4$, with orthonormal basis $e_P = e_\mu \wedge e_\nu$, $P = (\mu, \nu)$, $\mu < \nu$. Thus $\dim \mathcal{V} = 6$. The position-space curvature kernel is denoted by K. The lattice comparison used below is precisely eq. A.iii; no stronger lattice error exponent is assumed.

6.8.1. *Reflections and the closed-ring formula.* For a unit vector $u \in \mathbb{R}^4$ define

$$s_u = I - 2u \otimes u, \qquad R_u = \Lambda^2 s_u, \qquad P_u\omega = u \wedge \iota_u\omega. \tag{6.cci}$$

Here ι_u denotes contraction. The map P_u is the orthogonal projection onto $u \wedge u^\perp$, so $R_u = I - 2P_u$. In a basis u, v_1, v_2, v_3, R_u is minus the identity on $u \wedge v_i$ and plus the identity on $v_i \wedge v_j$. Consequently it is a symmetric orthogonal involution, with three eigenvalues of each sign. For $x \ne 0$ the normalization is

$$\mathsf{K}(x) = \frac{R_{\widehat{x}}}{\pi^2|x|^4}. \tag{6.ccii}$$

Indeed, differentiating $G(x) = (4\pi^2|x|^2)^{-1}$ gives

$$\partial_\mu\partial_\rho G(x) = -\frac{\delta_{\mu\rho} - 4\widehat{x}_\mu\widehat{x}_\rho}{2\pi^2|x|^4}. \tag{6.cciii}$$

Substitution into the curvature covariance gives the entries of $\pi^{-2}|x|^{-4}(I-2P_{\widehat{x}})$. This verifies that the reflection in eq. 6.cci acts first on $\mathbb{R}^4$; its second exterior power, rather than a rank-one reflection of $\mathcal{V}$, occurs in the curvature kernel.

Let $\mathcal{R}$ be the three unoriented cycles (1234), (1243), (1324). For pairwise distinct points put

$$k_4(x_1,x_2,x_3,x_4)=\sum_{\varsigma\in\mathcal{R}}\operatorname{tr}_{\mathcal{V}}\prod_{j=1}^{4}\mathsf{K}(x_{\varsigma(j)}-x_{\varsigma(j+1)}),\qquad \varsigma(5)=\varsigma(1). \tag{6.cciv}$$

The products are ordered along the cycle. A cyclic change of its starting point preserves the trace. Transposition reverses the order, and $\mathsf{K}(-x)=\mathsf{K}(x)$, so reversing a cycle also preserves its contribution.

Proposition 6.18 (The trace of a closed ring)**.** Let $w_1,\dots,w_4\neq 0$ and $\sum_j w_j=0$. There is an $\alpha\in[0,\pi]$ such that

$$\operatorname{tr}_{\mathcal{V}}[\mathsf{K}(w_1)\mathsf{K}(w_2)\mathsf{K}(w_3)\mathsf{K}(w_4)]=\frac{2+4\cos\alpha}{\pi^8}\prod_{j=1}^{4}|w_j|^{-4}. \tag{6.ccv}$$

In particular the absolute value of the angular factor is at most six. ∎

Proof. Exterior powers preserve products. Thus the angular matrix in the left side is Λ^2T, where $T=s_{\widehat{w_1}}\cdots s_{\widehat{w_4}}\in\mathrm{SO}(4)$. The space $V=\operatorname{span}\{w_1,\dots,w_4\}$ has dimension at most three. Each factor preserves V and fixes $V^\perp$. If $\dim V=3$, then $T|_V\in\mathrm{SO}(3)$ has an eigenvalue one: the other nonreal eigenvalues come in conjugate pairs, while the product of all eigenvalues is one. If $\dim V\le 2$, the fixed space $V^\perp$ already has dimension at least two. In all cases T fixes a plane and rotates its orthogonal complement. Its eigenvalues are $1,1,e^{i\alpha},e^{-i\alpha}$.

For any four-dimensional matrix T, summing its diagonal two-by-two minors gives

$$\operatorname{tr}(\Lambda^2T)=\tfrac12\{(\operatorname{tr}T)^2-\operatorname{tr}(T^2)\}. \tag{6.ccvi}$$

Equivalently this trace is the sum of the six pairwise products of the eigenvalues. For the eigenvalues just found it is $2+4\cos\alpha$. Multiplication by the four radial factors in eq. 6.ccii proves the formula. □

An orientation-preserving or orientation-reversing orthogonal change of coordinates conjugates every angular matrix by the same Λ^2O. Hence k_4 is invariant under isometries and homogeneous of degree -16. Permuting its arguments permutes the three cycles. For separated integrable functions define

$$\mathcal{B}_4(f_1,f_2,f_3,f_4)=\int_{(\mathbb{R}^4)^4}k_4(x_1,x_2,x_3,x_4)\prod_{i=1}^{4}f_i(x_i)\,dx_1\cdots dx_4. \tag{6.ccvii}$$

If every two supports have distance at least $d>0$, absolute convergence and Prop. 6.18 give

$$|\mathcal{B}_4(\underline{f})|\le 18\pi^{-8}d^{-16}\prod_i\|f_i\|_{L^1}. \tag{6.ccviii}$$

The sixteen powers of length in the measure cancel the homogeneity of k_4. Thus simultaneous dilations of the four arguments leave $\mathcal{B}_4$ unchanged.

6.8.2. *A uniform positive cone of configurations.* For $0<\vartheta\le 1$ consider nonzero nonnegative $f_i\in C_c^1(\mathbb{R}^4)$ with $F_i=\|f_i\|_\infty$, each supported in a ball of radius d, and satisfying, for $i\neq j$,

$$\begin{aligned} d\le\operatorname{dist}(\operatorname{supp}f_i,\operatorname{supp}f_j)\le 6d,\\ \int f_i\ge\vartheta d^4F_i,\qquad \operatorname{Lip}f_i\le F_i/(\vartheta d).\end{aligned} \tag{6.ccix}$$

The positive-cone condition is that one unit vector e works for every pair of points in distinct supports:

$$|w-(w\cdot e)e|\le |w|/8, \qquad w=x_i-x_j, \quad i\neq j. \tag{6.ccx}$$

Proposition 6.19 (Quantitative positive margin)**.** Under eq. 6.ccix and eq. 6.ccx, every ring has angular factor greater than four, and

$$\mathcal{B}_4(\underline{f})\ge c_4\vartheta^4\prod_i F_i>0, \qquad c_4=12\pi^{-8}10^{-16}. \tag{6.ccxi}$$

■

Proof. Write $\omega=\arcsin(1/8)$. The unoriented line of every edge is at angle at most ω from $\mathbb{R}e$. The angle between the lines of any two edges is therefore at most 2ω. Two reflections with normal lines at angle ψ give a rotation by 2ψ on their span and the identity on its orthogonal complement.

For an orthogonal map O put $a(O)=\sup_{|v|=1}\angle(v,Ov)$. The spherical triangle inequality gives $a(O_1O_2)\le a(O_1)+a(O_2)$. Grouping the four reflections into two pairs therefore gives $a(T)\le 8\omega$. For the closed ring in Prop. 6.18, $a(T)=\alpha$: in the rotating plane the maximal displacement angle is exactly α, and a fixed-plane component cannot increase it.

Three applications of the double-angle identity yield

$$\cos(2\omega)=\frac{31}{32}, \qquad \cos(4\omega)=\frac{449}{512}, \qquad \cos(8\omega)=\frac{70529}{131072}>\frac12. \tag{6.ccxii}$$

The first two positive cosines, starting from $0<\omega<\pi/2$, put 8ω in $(0,\pi)$. Consequently $8\omega<\pi/3$ and

$$2+4\cos\alpha\ge\frac{136065}{32768}>4. \tag{6.ccxiii}$$

Any two points in distinct supports have distance at most $10d$: choose points realizing their minimum distance at most $6d$ and add the two support diameters, each at most $2d$. Each of the three rings thus contributes at least $4\pi^{-8}(10d)^{-16}$. Integrating the resulting bound and using the four mass floors proves eq. 6.ccxi. □

There is an independent matrix check on the positive margin. If ψ is the angle between the lines of u and v, then $\|R_u-R_v\|_{\mathrm F}=4\sin\psi$. To see this, choose coordinates in which the two normals lie in a coordinate plane. The product s_us_v rotates that plane through 2ψ; on Λ^2 it has two fixed coordinates and two copies of that rotation. Taking the Frobenius norm gives the stated identity. Under eq. 6.ccx, telescoping four orthogonal factors gives $\|R_{u_1}\cdots R_{u_4}-I\|_{\mathrm F}\le 2$. For a real orthogonal six-dimensional matrix A, $\operatorname{tr}A=6-\|A-I\|_{\mathrm F}^2/2$. This gives the weaker but still uniform floor four without the closed-ring eigenvalue calculation.

The nearly collinear class in the four-point assertion satisfies the cone condition. Indeed, if $\operatorname{supp}f_i\subset B(c_i,d/16)$ and the c_i lie on $\mathbb{R}e$, the transverse part of x_i-x_j has length at most $d/8$; the separation in eq. 6.ccix makes this at most $|x_i-x_j|/8$. This class is nonempty with explicit constants. Take $\psi(x)=(1-|x|^2)_+^2$ and $f_i(x)=\psi(16(x-c_i)/d)$, with consecutive centres separated by $5d/4$ along the line. Then

$$\int\psi=\frac{\pi^2}{12}, \qquad \operatorname{Lip}\psi=\frac{8}{3\sqrt3}, \qquad 0<\vartheta\le\min\left\{\frac{\pi^2}{12\cdot16^4},\frac{3\sqrt3}{128}\right\} \tag{6.ccxiv}$$

suffices for all the shape conditions. The first integral follows by radial integration; the gradient norm is the maximum of $4r(1-r^2)$ on $[0,1]$. The smallest support separation is $9d/8$, and the largest minimum separation is less than $6d$.

6.8.3. *A negative transverse configuration.*

Proposition 6.20 (A negative box with an explicit neighbourhood)**.** Let $h>0$ and let e_1',e_2',e_3' be an orthonormal triple. Put

$$x_1^0=y, \quad x_2^0=y+he_1', \quad x_3^0=y+h(e_1'+e_2'), \quad x_4^0=y+h(e_1'+e_2'+e_3'). \tag{6.ccxv}$$

Then

$$k_4(\underline{x}^0) = -\frac{7}{36\pi^8}h^{-16}. \tag{6.ccxvi}$$

If $|x_i - x_i^0| \le h/200000$ for all i, then

$$k_4(\underline{x}) \le -\frac{7}{72\pi^8}h^{-16}. \tag{6.ccxvii}$$

Thus nonzero nonnegative tests supported in these four balls have a strictly negative integrated box. ∎

Proof. Isometries and homogeneity reduce the calculation to $h = 1$, $y = 0$ and the first three coordinate vectors. Every reflection fixes the fourth coordinate. Direct multiplication on the three-dimensional span gives the following exact values:

cycle	$(\lvert w_1\rvert^2, \lvert w_2\rvert^2, \lvert w_3\rvert^2, \lvert w_4\rvert^2)$	$\operatorname{tr}_{\mathcal{V}} \Lambda^2 T$	$\prod_j \lvert w_j\rvert^{-4}$
1234	$(1,1,1,3)$	-2	$1/9$
1243	$(1,2,1,2)$	0	$1/16$
1324	$(2,1,2,3)$	4	$1/144$

(6.ccxviii)

For the first row, the first three reflections multiply to $-I$ on that span, so the product has eigenvalues $1, -1, -1$ there. For the second row the product is a three-cycle of the coordinate axes. For the third row its trace on $\mathbb{R}^4$ is three, giving angular factor four by eq. 6.ccvi. All entries may also be obtained from the rational formula $s_w = I - 2ww^{\mathsf{T}}/(w^{\mathsf{T}}w)$, so no choice of square root affects the calculation. The sum is $-2/9 + 4/144 = -7/36$, proving eq. 6.ccxvi.

For completeness the neighbourhood can be chosen without a compactness argument. The normal-form calculation above also gives $\|R_u - R_v\| \le 2\sin\psi$. Differentiating eq. 6.ccii in a unit direction gives

$$\|\mathsf{K}(z)\| \le \pi^{-2}|z|^{-4}, \qquad \|\partial_v \mathsf{K}(z)\| \le 6\pi^{-2}|z|^{-5}. \tag{6.ccxix}$$

If $|x_i - x_i^0| \le \rho h$ and $\rho \le 1/20$, every edge on the straight path between the configurations has length at least $9h/10$. On this path the kernel norm is less than $2\pi^{-2}h^{-4}$ and its directional derivative norm less than $11\pi^{-2}h^{-5}$. Each edge moves by at most $2\rho h$. Telescoping a product of four factors, using $|\operatorname{tr} A| \le 6\|A\|$, bounds the change in all three rings by

$$|k_4(\underline{x}) - k_4(\underline{x}^0)| \le 12672\rho\pi^{-8}h^{-16}. \tag{6.ccxx}$$

At $\rho = 1/200000$ this is smaller than $7\pi^{-8}h^{-16}/72$. This proves eq. 6.ccxvii. Integration against $\prod_i f_i \ge 0$ proves the last assertion, more precisely with upper bound $-7h^{-16}\prod_i \int f_i/(72\pi^8)$. □

The restriction to a nearly collinear class therefore has mathematical content. Positivity of the covariance operator does not imply positivity of every separated fourth cumulant of its quadratic density.

6.8.4. *The finite Gaussian coefficient and the* SU(2) *normalization.* Let $\xi^1, \dots, \xi^q$ be independent centred Gaussian vectors with the same real positive semidefinite covariance C on a finite coordinate set. For real diagonal matrices D_i set $Q_i = \frac{1}{2}\sum_{a=1}^{q} (\xi^a)^{\mathsf{T}} D_i \xi^a$. On a neighbourhood of the origin,

$$\log \mathbb{E} \exp\Big(\sum_i z_i Q_i\Big) = \frac{q}{2}\sum_{m\ge 1} \frac{1}{m} \operatorname{tr}\left(C\sum_i z_i D_i\right)^m. \tag{6.ccxxi}$$

The formula follows by diagonalizing $C^{1/2}(\sum_i z_i D_i)C^{1/2}$ at real parameters and then using the convergent matrix logarithm near zero; a singular C is handled on its range. Taking the coefficient of $z_1 z_2 z_3 z_4$, and grouping the 24 words by cyclic rotation and reversal, gives

$$\kappa(Q_1, Q_2, Q_3, Q_4) = q \sum_{\varsigma \in \mathcal{R}} \operatorname{tr}(C D_{\varsigma(1)} C D_{\varsigma(2)} C D_{\varsigma(3)} C D_{\varsigma(4)}). \tag{6.ccxxii}$$

Each unoriented ring has eight words, canceling $2m = 8$. Replacing C by $x^{-1}C$ multiplies this by x^{-4}. Centering the Q_i does not change this mixed cumulant.

For the Wilson observable of SU(2) use Pauli matrices normalized by $\frac{1}{2}\operatorname{tr}(\sigma^a\sigma^b) = \delta_{ab}$. For real $v \in \mathbb{R}^3$,

$$1 - \tfrac{1}{2}\operatorname{tr}\exp(iv\cdot\sigma) = 1 - \cos|v| = \tfrac{1}{2}|v|^2 + O(|v|^4). \tag{6.ccxxiii}$$

Hence the Gaussian quadratic reference has $q = 3$, with exactly the factor $1/2$ used above. If each first source derivative has coefficient ζ, its four-point leading term is

$$3\zeta^4 x^{-4}\mathcal{B}_4(\underline{f}). \tag{6.ccxxiv}$$

There is no additional factor $1/2$ in this four-point formula. The quartic Wilson Taylor term in eq. 6.ccxxiii, the interacting density, and the actual source remainders have not been discarded by a claim that their cumulants are small. Estimating their combined contribution is the separate comparison problem.

For comparison with three points, the Hodge star satisfies $*R_u* = -R_u$. Inserting $** = I$ between the factors and taking a trace shows that every odd ring of the continuum kernel vanishes. This is a kernel identity at separated points, rather than an assertion that the quadratic density is a Gaussian random field.

6.8.5. *Uniform comparison with the free lattice box.* Let $a = \ell/N$, $N \in \mathbb{N}$, and let P_{ex} be the orthogonal projection onto the exact lattice two-cochains. Plaquettes $p = (y, P)$ have barycentres $x(p) = y + ac_P$, with $c_P = (e_\mu + e_\nu)/2$ for $P = (\mu, \nu)$. For any real symmetric covariance $C(p, q)$ define the lattice box

$$\mathcal{B}_{4,a}[C](\underline{f}) = \sum_{\varsigma\in\mathcal{R}}\sum_{p_1,\dots,p_4}\prod_i f_i(x(p_i))\prod_{j=1}^{4} C(p_{\varsigma(j)}, p_{\varsigma(j+1)}). \tag{6.ccxxv}$$

In particular the fourth cumulant of the finite free Wilson quadratic reference is $3\mathcal{B}_{4,a}[P_{\mathrm{ex}}]$. There is no missing Riemann factor in this expression: $P_{\mathrm{ex}}(p, q) = a^4\mathsf{K}_{a,\ell}(p, q)$, so a product of four covariance entries contributes a^{16}.

Proposition 6.21 (Uniform lattice error on the full shape class)**.** Suppose eq. 6.ccix holds on a common short lift of the four supports to $\mathbb{R}^4$, with $a \le d/100$ and $\ell \ge 200d$. There is a numerical C, depending only on the constant in eq. A.iii, such that

$$\left|\mathcal{B}_{4,a}[P_{\mathrm{ex}}](\underline{f}) - \mathcal{B}_4(\underline{f})\right| \le C\left\{(a/d)^{1/2} + (d/\ell)^4 + \frac{a}{\vartheta d}\right\}\prod_i F_i. \tag{6.ccxxvi}$$

The bound is uniform in positions, directions, shapes and the number of lattice sites. The cone condition is unnecessary for this error estimate. ∎

Proof. Write the lattice box using four factors $a^4\mathsf{K}_{a,\ell}$. On distinct supports every relevant distance lies between d and $10d$. The comparison in eq. A.iii, entrywise and hence in six-dimensional operator norm, gives

$$\|\mathsf{K}_{a,\ell}(p, q) - \mathsf{K}(x(p) - x(q))\| \le Cd^{-4}\{(a/d)^{1/2} + (d/\ell)^4\}. \tag{6.ccxxvii}$$

All covariance factors are bounded by Cd^{-4}. Replacing the four factors one at a time therefore costs at most $Cd^{-16}\{(a/d)^{1/2} + (d/\ell)^4\}$ per term, before the a^{16} factor and the four test functions. For each orientation, the number of grid points in a ball of radius d is at most $C(d/a)^4$, by placing disjoint mesh cubes about the points. Summation in four variables and in the finitely many orientations gives the first two error terms in eq. 6.ccxxvi.

The remaining sum still samples the test functions and kernels at orientation-dependent barycentres. Move $y_i + ac_{P_i}$ to y_i, one factor at a time. Along these movements a nonzero test function stays in its support enlarged by Ca, and distinct supports remain at distance at least $d/2$. The

test-function cost is at most $CaF_i/(\vartheta d)$, and eq. 6.ccxix bounds each kernel cost by Cad^{-5}. The same grid count bounds the entire change by $Ca\prod_i F_i/(\vartheta d)$.

At the common grid sites y_i, summing the four orientation indices gives precisely the matrix trace in eq. 6.cciv. The result is $a^{16}\sum_{y_1,\dots,y_4} k_4(\underline{y})\prod_i f_i(y_i)$. This last Riemann sum has the same error bound. In fact the integrand has derivative at most $C\vartheta^{-1}d^{-17}\prod_i F_i$ on the product of the enlarged supports, whose sixteen-dimensional volume is Cd^{16}. Its oscillation in a mesh cell is at most the derivative bound times Ca. Summing cell errors gives $Ca\prod_i F_i/(\vartheta d)$. No diagonal singularity is crossed: the product of the four tests vanishes in a neighbourhood of every coincidence diagonal, so the integrand extends by zero there. These three replacements prove the claim. □

Corollary 6.22 (A uniform lattice sign floor)**.** There is a numerical $c>0$ such that the cone class satisfies

$$\mathcal{B}_{4,a}[P_{\mathrm{ex}}](\underline{f}) \ge \tfrac{1}{2}c_4\vartheta^4\prod_i F_i \tag{6.ccxxviii}$$

whenever $a/d \le c\vartheta^8$ and $d/\ell \le c\vartheta$. ■□

Proof. Choose c so that the right side of eq. 6.ccxxvi is at most $c_4\vartheta^4\prod_i F_i/2$. This is possible because the three displayed errors are then bounded by constants times $c^{1/2}\vartheta^4$, $c^4\vartheta^4$ and $c\vartheta^7$, respectively, and $\vartheta\le 1$. The same choice can enforce $a\le d/100$ and $\ell\ge 200d$. Apply Prop. 6.19. □

For any fixed nonzero nonnegative C^1 tests supported in the negative balls of Prop. 6.20, the same comparison gives a strictly negative free lattice box at a sufficiently fine mesh and sufficiently large torus. To apply the uniform version choose $d=9h/10$ and a fixed positive ϑ smaller than all four mass and inverse Lipschitz ratios. The supports then meet the separation and size conditions in eq. 6.ccix. The continuum negative margin is independent of the mesh, whereas the error in eq. 6.ccxxvi tends to zero.

6.8.6. *Background covariances and perturbed profiles.* The continuum sign also applies to the background covariance once its Gaussian complement has the required decay. The exact decomposition and the decay estimate are different facts. Let E be the space of exact two-cochains and let $A: E\to W$ be the actual linear coarse flux map. For a standard Gaussian F on E put $N=E\cap\ker A$ and $Y=E\ominus N$. The restriction of A to Y is injective, so AF determines the Y component of F. Its N component is an independent standard Gaussian. Thus

$$\mathrm{Cov}(\mathbb{E}[F\mid AF]) = P_Y,\quad \mathrm{Cov}(F-\mathbb{E}[F\mid AF]) = P_N,\quad P_{\mathrm{ex}} = P_Y+P_N. \tag{6.ccxxix}$$

This finite-dimensional identity holds for the specified flux map; it contains no spatial decay assertion. To identify P_Y with the background covariance of a connection integral, one must also verify that the actual constraint induces this flux map, that its gauge integration has the prescribed normalization, and that the closed and zero modes have been treated consistently. The Gaussian projection identity does not replace those identifications.

If, in addition, its fluctuation complement satisfies

$$|P_N(p,q)| \le C(a/b)^4 e^{-\delta|x(p)-x(q)|/b},\quad |x(p)-x(q)|\ge b, \tag{6.ccxxx}$$

then the proof of Prop. 6.21 gives

$$\begin{aligned} |\mathcal{B}_{4,a}[P_Y](\underline{f}) - \mathcal{B}_4(\underline{f})| &\le C\left\{(a/d)^{1/2}+(d/\ell)^4+\mathfrak{e}(d/b)+\frac{a}{\vartheta d}\right\}\prod_i F_i,\\ \mathfrak{e}(t) &= \sup_{u\ge t} u^4 e^{-\delta u},\qquad d\ge b. \end{aligned} \tag{6.ccxxxi}$$

Indeed $a^{-4}|P_N(p,q)|$ is bounded by $C|x(p)-x(q)|^{-4}\mathfrak{e}(d/b)$ at the relevant distances. Only the first covariance-replacement step changes. For the actual flat witness, take the flux map Q^{pl}_{K-s} and $b=a_s$. The flat conditioning of §6.1.1.1 identifies P_N with $C^{\mathrm{fl}}_{K,s}$. The sum of the finer slices gives

the tail at the separated distances used here, including the enlarged profiles at distance at least $\mathsf{d}/2$. The flat gauge comparison identifies the curvature law of each slice with the flat step law. Other flux maps require their own complement estimate.

The form is also stable under the small changes of profiles produced by Gaussian extraction. Retain $a \le d/100$. Suppose a covariance satisfies $|C(p,q)| \le C_0 a^4 d^{-4}$ between the distinct enlarged supports. Let $h_i(p)$ depend on the plaquette orientation if necessary, with $|h_i(p)| \le \varepsilon F_i$, supported within distance $d/8$ of $\operatorname{supp} f_i$, and $0 \le \varepsilon \le 1$. The enlarged supports still have separation at least $3d/4$ and volume $O(d^4)$. The absolute grid-mass bound $a^4 \sum_p |h_i(p)| \le C\varepsilon F_i d^4$ uses only that support bound and the point count. Expanding by four-linearity, the fifteen nonempty choices of perturbed arguments give

$$|\mathcal{B}_{4,a}[C](f_1+h_1,\dots,f_4+h_4) - \mathcal{B}_{4,a}[C](\underline{f})| \le C\{(1+\varepsilon)^4 - 1\} \prod_i F_i \le C'\varepsilon \prod_i F_i. \tag{6.ccxxxii}$$

No differentiability or positivity of the h_i is needed. Orientation-dependent perturbations therefore cause no additional covariance or Riemann-sum assumption.

6.8.6.1. The normalized comparison of four bounded insertions. The four-point witness may require the larger room $C_W^{\square}(s) = c_W C_W(s)$, with a fixed $c_W \ge 1$ chosen for the shape class. In this comparison write s for the depth selected with that room and $k_0 = K - s$, $x = x_{k_0}$, $\rho_{\mathsf{d}} = L^s \mathsf{d}$. At the witness take the four windowed plaquette reads G_i, with profiles ϕ_i supported in the allowed enlargements. Use a segment of r steps and a single nonwrapping box for the four supports at each step. At the first input level k_0, include the full datum box of each native localized witness minimizer, with its four test layers. Their size in witness units is included in $C\rho_{\mathsf{d}}$. At a later input level $j-1$, use one localized minimizer cube around the union of the four supports, with a margin of order D in input units. Use one prescribed rooted axial tree for the common frame. Its root and all its paths belong to the read. A simultaneous change of gauge leaves only the rotation at this common root in the four transformed arguments.

For the input fluctuations, define S_j from these full reads by adjoining all free coordinates within the fixed double-block reach of the one-step pivot and reconstruction maps. The exact dependence on S_j follows from the finite range of those maps. For the comparison integral I_j, add a further D-margin to a cube containing these coordinates. With O_j the free exterior, this gives $S_j \subset I_j$, $\#S_j \le n_j = \#I_j$, and $d(S_j, O_j) \ge D$ in input mesh units. In particular

$$n_j \le C(\rho_{\mathsf{d}} L^{-(j-1-k_0)} + D + C_L)^4.$$

These are the read-set conditions of the normalized product-window estimate.

Use a separate neighborhood for the clean clause. It contains I_j, every native test whose reads can reach it, their full row carriers, and the operation margins. If an operation touches one of these reads, its immediately preceding live component is within the diameter of the operation component. The enlargement by $(200\check{R}_j + 1)M$ in the appropriate level units covers this possibility even when the old live cells do not intersect I_j. These layers enter the clean-history count. The conditional comparison integral retains its own box and its distance D to the exterior. The condition is imposed before that step's healing: no creation or operation meets this enlargement, and no live component meets it on either side of the step. Use the compensated standing representation. The coupling lag has no term supported wholly inside the box, since the profiles of healed regions have been redrawn and the standing large regions miss the padded neighborhood. A lag term reaching the box through an exterior trace uses the fixed-exterior old-origin oscillation estimate.

Fix the coarse field, the older stored slots and the exterior free coordinates after the ordinary gauge draw. Use the product-window calculation on the union of all four reads. An affected pivot is reconstructed on its full double block, including when its bond lies outside I_j. For every query that reads the changed coordinates the interior window implication gives the approximate and old density tests throughout the literal product of balls. No operation or its guards meet these carriers.

For an exterior query the complete read is fixed. Induction along the original order of the queries gives the same exterior outcomes throughout the product, including when the next query depends on an earlier answer. Thus at fixed compatible exterior the selector is constant on this product.

On a clean standing part the numerator with insertion and the uninserted denominator therefore use the same interior window and the same exterior measure. Writing Z_{e} for the uninserted interior integral and $m(d\mathsf{e})$ for the exterior measure with its retained weights, the normalized conditional law on the box is averaged with measure proportional to $Z_{\mathsf{e}}\, m(d\mathsf{e})$. A merge or an operation on the exterior takes a further positive mixture of these quotients. The pointwise clean quotient gives the reference branch with the original weights for these allowed strings. On this branch take the same cutoff and the same flat slices for all four profiles. Use the common-frame estimate for the nested localized readouts on their union, including the separate first witness minimizers. The hybrid through the adjacent-level coupling is defined with the bounded function $\exp(\sum_i \lambda_i G_i)$, with independent source coordinates. Setting the top field to zero yields a single Gaussian determinant independent of the terminal datum and of the history. On the polydisc

$$|\lambda_i| \le \Lambda_i = \frac{cx}{4\rho_{\mathsf{d}}^4 p_0^2 \|\phi_i\|_\infty},$$

the diagonal covariance bound and the count of plaquettes give a uniform trace norm bound on the quadratic Gaussian exponent. Zero profiles can be removed. The same bound on a larger disc gives a uniform positive lower bound for the determinant's modulus and the second moment needed to remove the Gaussian windows.

Apply the one-step comparisons on this disc to the positive law of the original segment. The nonlinear readout error sums with the geometric factor $L^{-2(j-1-k_0)}$. The tilt, covariance and exterior errors sum in the bounds for the normalized box law. Charge a finer history restriction at the steps where its density changes that law. The native top tests give the bound on the top field for all four read sets on the common admitted box. This field is present in the Gaussian model as a shift; the linear cross term is odd under simultaneous reflection of the Gaussian variables. Pairing the reflected points makes its effect quadratic in the shift. This and the top square cost $O(L^{-4r})$ on the polydisc. In the cumulant calculation these errors correspond to open chains with two ends at the top curvature. The normalization by the uninserted box mass is made at each step. In particular decay of the free covariance alone does not give the exterior error, which uses the decay of the response of the tilted product-window law to the exterior force.

For an unconditional comparison, average the generating functions before taking logarithms. A clean sum at terminal datum W can also use the normalized mixture of the clean class laws, with weights $d\mathsf{Q}(h,\cdot)/d\mathsf{Q}_{\mathrm{cl}}(W)$ on the clean classes. Here Q_{cl} is the sum of their terminal measures. This gives a common determinant for the clean mixture when the clean comparison is uniform over its constituents. It uses a refinement for which cleanness is a property of each class. The actual exponent and the density of the refined classes have to be kept in the normalized comparison. On an exceptional part the G_i still have their windowed sup bounds. Hence the probability of that part, times the bounded insertion estimate, suffices here. The physical first derivatives instead use eq. 6.cxii.

For the unconditional comparison use the adjacent positive laws π_j of the two-insertion calculation, now on the union of the four complete read sets. Take Y_j to be the bounded generating function with all slices through j replaced by flat ones. The same law π_j supplies the two marginals in $\mathbb{E}_{\pi_j}(Y_{j-1} - Y_j)$. At a compatible good exterior, where the selector equals one throughout the product window, its conditional bound is

$$C\left[L^{-2(j-1-k_0)}(\rho_{\mathsf{d}}^2 x^{-1/2} p_0 + e^{-c\Delta_j}) + T_j\right].$$

Here T_j is the normalized box error of the two-insertion calculation, evaluated with the four-read coordinate count. The bound for changing the plaquette field in the exponent has the same constant

up to a fixed factor because $\sum_i |\lambda_i| \|\phi_i\|_\infty \le cx/(\rho_{\mathsf{d}}^4 p_0^2)$. The sum over steps compares the two original endpoint expectations. On a bad current part of mass p_j, both integrands have a bounded extension, so it contributes at most Cp_j. The top-field error, the deleted Gaussian windows and the localized-to-flat comparison give, in all,

$$\delta \le C\left[\rho_{\mathsf{d}}^2 x^{-1/2} p_0 + \max_j e^{-c\Delta_j} + \sum_j T_j + L^{-4r} + \sqrt{t} + p_0^{-2} e^{-cd_*} + \sum_j p_j + p_{\mathrm{top}}\right].$$

The notation t, d_*, p_{top} has the same meaning as in the two-insertion calculation. This argument uses the uninserted exterior weights at each step. To use this bound on the adjacent laws, we must estimate the box errors on the actual four-read domains and the masses of the incompatible exteriors.

There is also a comparison using the clean restriction of the original run. On a clean class use the reference branch of weight P_0 from the pointwise clean quotient, with the stored variables and exterior datum kept in the disintegration. The box and its padding are the same in the restriction and the comparison. Thus each reference conditional expectation uses the original positive weights for the allowed strings. Compose the carried densities on the complete retained states, and project to the terminal datum at the end. Replace one slice at a time in these conditional expectations. The remaining outer composition has sup norm at most one, so the conditional errors add. For the fixed conditional laws, prove the disc estimate on a countable dense set and extend by analyticity, which gives a common exceptional null set for the source parameters. Write Z^0 for the resulting deterministic flat determinant and δ_{ref} for the uniform relative error of this reference branch. If $\nu(\mathrm{cl})$ is the full law's mass of clean classes, averaging the reference and remaining branches gives

$$\left|\frac{\mathbb{E}\exp(\sum_i \lambda_i G_i)}{Z^0} - 1\right| \le C\big(\delta_{\mathrm{ref}} + 1 - P_0\nu(\mathrm{cl})\big).$$

Indeed the reference weight is $P_0\nu(\mathrm{cl})$, and on the complementary branch the uniform bounds on G_i and the lower bound on $|Z^0|$ control the ratio. This is a bound on the unconditional generating function before taking its logarithm. The hypotheses of this route are the support-adapted clean restriction, the verified bracket majorants, and the conditional comparisons on the common product windows.

For the clean class sum, the excluded histories are counted by live cells meeting the padded box from the lower waiting band through the end of the segment. A union bound over this band and the r steps costs at most $C(1 + r + \check{R}_{k_0})(\rho_{\mathsf{d}} + M\check{R}_{k_0})^4 L^{4r + C\check{R}_{k_0}}$ times the normalized bound per cell. Here the factor in L converts the count from the terminal scale to the finest scale in the band. The healed histories outside this criterion are included in the standing sum. For an estimate by adjacent-step selectors, the corresponding count uses their actual current readsets. In either case spend the rarity reserve after these counts and after the source loss below.

If δ bounds the resulting relative error of the averaged generating function on the polydisc and $\delta < 1/2$, the two logarithms differ by $C\delta$. Mixed Cauchy gives an error at order four of at most

$$C\delta \prod_i \Lambda_i^{-1} \le C\delta\, x^{-4} \rho_{\mathsf{d}}^{16} p_0^8 \prod_i \|\phi_i\|_\infty.$$

The comparison therefore requires that $\delta\rho_{\mathsf{d}}^{16} p_0^8$ tend to zero. For instance, if the old-origin removal has $\epsilon_j^{\mathrm{old}} \le C n_j (\log x)^{\mathfrak{p}_0 - q}$ on the four-read box and $n_j \le C\rho_{\mathsf{d}}^4$, its contribution after Cauchy is bounded by

$$Cr\, \rho_{\mathsf{d}}^{20} p_0^8 (\log x)^{\mathfrak{p}_0 - q}.$$

For $r = O(\log\log x)$ and $\rho_{\mathsf{d}} \asymp (\log x)^{2\mathfrak{p}_0}$, this tends to zero when $q > 49\mathfrak{p}_0$.

On the admitted clean reference windows take $D = A\log x$ and $r = \lceil B \log_L \log x \rceil$. The native margins give $\Delta_j, d_* \ge c_0 \min\{D, \check{R}_{k_0}\} - C_0$, and $\check{R}_{k_0}/\log x$ grows without bound. Hence the geometric errors, including $n_j b_j^2 e^{-cD}$, beat the Cauchy factor. In T_j, the cubic density term costs

$Cn_j x^{-1/2} p_0^3$, the covariance term is bounded by $Cn_j(\rho_{\mathsf{d}} x^{-1/2} p_0 + e^{-cD})$, and $t_j \le Cn_j e^{-cb_j^2}$ with b_j comparable to p_0 on the segment. These and the square root of the total window error also beat every required logarithmic power. Removing the top field costs CL^{-4r}, so take $4B > 40\mathfrak{p}_0$ for $\rho_{\mathsf{d}}^{16} p_0^8 L^{-4r} \to 0$. The loss of the clean reference branch contains both $1 - \nu(\mathrm{cl})$ and $1 - P_0 \le r\epsilon_{k_1}$. The padded cell count and the rooted bound on the deleted histories bound their sum by

$$C(1 + r + \check{R}_{k_0})^C (\rho_{\mathsf{d}} + M\check{R}_{k_0})^4 L^{4r + C\check{R}_{k_0}} e^{-cp_0}.$$

Since $\check{R}_{k_0} = o(p_0)$, this loss tends to zero after multiplication by the Cauchy factor. The terminal test reads are included in the clean neighborhood. With the fixed-torus age estimate, use the unprolonged run and write $h = g^{-2}$. The last lattice has ℓ^4 sites and $x_K \le h + B_\sharp$. In the survival term of Prop. 5.71 a sufficient depth floor is

$$s_{\mathrm{vol}}(g, \ell) = \max\left\{0, \left\lceil \frac{\mathsf{B}(h + B_\sharp)\ell^4}{\bar{c}_a} \right\rceil - 1\right\}.$$

At $k = k_0 + r$, the condition $s - r \ge s_{\mathrm{vol}}(g, \ell)$ gives $\mathsf{B}(x_K)\ell^4 - \bar{c}_a(K - k + 1) \le 0$. The survival term retains $e^{-\bar{c}_a a - 2p_0(g_k)}$, and the healed term already has age decay without a volume factor. For $\bar{c}_a \ge 17 \log L + \log 2$ the weighted age sum is geometric and gives the required bound for $Q_{17}(k)$. All earlier reads meet the same depth floor. At fixed g, the segment has $r \le s/2$ above a fixed depth, so $s \ge 2s_{\mathrm{vol}}(g, \ell)$ there suffices. The witness rule, whose room is at least one, gives this depth for sufficiently small separation. In particular a cutoff of the form $\mathsf{d} \le e^{-C_g(1+\ell^4)}$ suffices for this age condition, in addition to the formation and clean-box conditions. These choices require the stated nonwrapping room and the native density and marginal estimates on the common comparison boxes.

If the comparison is also made at fixed clean class and datum, the same disc controls the proper conditional cumulants relative to the deterministic reference values. On the clean part the multi-block terms in total cumulance contain products of at least two conditional errors. They are absent when comparing the averaged generating functions directly.

The determinant has covariance $\sum_{j=k_0+1}^{k_0+r} x_{j-1}^{-1} \mathcal{C}_{(j)}$. On separated supports the slice bound gives a summable absolute kernel. Replacing x_{j-1}^{-1} by x^{-1} then costs $C(B_\sharp r + c_5)/x$ relative to the natural smeared scale. Each term in the omitted ring sum has at least one coarse edge, which costs L^{-4r} in witness units. Since $L^r \ge \rho_{\mathsf{d}}$ by the segment choice, any further coarse edge costs at most the separated edge bound $C\rho_{\mathsf{d}}^{-4}$. Counting the four supports gives $C\rho_{\mathsf{d}}^4 L^{-4r}$ at the same relative scale. These two comparisons are made on the ring itself and do not pay the source-polydisc loss. To compare the enlarged witness profiles with the original profiles in the ring, use eq. 6.ccxxxii with relative perturbation $CM/(\vartheta\rho_{\mathsf{d}})$. Indeed the enlargement is at most CMa_s, and the slope bound is in units of $\vartheta\mathsf{d}$, with $\mathsf{d}/a_s = \rho_{\mathsf{d}}$. Choose the room so that $CMa_s \le \mathsf{d}/8$ and the relative perturbation is at most one. The witness comparison and the ring use the same enlarged profiles before this replacement. To pay that loss for the $O(L^{-4r})$ term in the generating function, take L^{4r} larger than ρ_{d}^{16} times the prescribed power of $\log x$. A segment of length $O(\log\log x)$ suffices in the stated room, with its constant chosen for the error budget. Fix the reference coupling first. As the witness depth tends to infinity, $x \asymp \bar{x}_s$ and $\bar{x}_s = O_g(1 + s)$, so $r \le s - 1$ and $s - r \to \infty$. The padded neighborhoods have side at most $C(\rho_{\mathsf{d}} + D + M\check{R}_{k_0})L^{-(s-r)}$ in physical units. Their polylogarithmic prefactor is dominated by the mesh decay. Take the separation threshold small enough that this side is below a fixed fraction of the torus side and that the stated volume floor holds at $s - r$. The resulting drift is $o(x)$. The box-law estimates must apply to the actual normalized laws on these boxes.

The coupling in the four-point assertion uses the original two-point depth, say s_2. To compare, the room functions obey

$$\frac{C_W(u+t)}{C_W(u)} \le \bigl(1+\log(1+B_\sharp t)\bigr)^{2\mathfrak{p}_0} \qquad (u,t\ge 0),$$

since $\bar{x}_u \ge e$ and $\bar{x}_{u+t} = \bar{x}_u + B_\sharp t$. Choose a fixed integer t_0 so that L^{t_0} exceeds c_W times the right side. By minimality of the witness depths, $s_2 \le s \le s_2 + t_0$: the original room condition at s_2 implies the enlarged one at $s_2 + t_0$. Thus a required formation floor $K - s \ge n$ follows from $K - s_2 \ge n + t_0$. The two-sided flow bound gives $|x_{K-s} - x_{K-s_2}| \le B_\sharp t_0 + c_5$. Consequently

$$x_{K-s}^{-4} = g(\mathsf{d})^8\bigl(1 + O(x_{K-s_2}^{-1})\bigr).$$

This transfers the leading size and the error scale of the four-point comparison to the coupling used in the assertion. The shape margin is taken from the ring at the chosen witness.

6.8.7. *The exact remaining interacting estimate.* This closes the sign, nondegeneracy and shape stability of the continuum free box, and its comparison with the full free lattice reference. It supplies the corresponding background conclusion whenever the explicitly named complement tail is proved. With the running law $x^{-1} \asymp (\log(1/d))^{-1}$ and $\zeta \to 1$, these bounds give size $(\log(1/d))^{-4}$ for the leading four-point term. They do not prove that running law.

To obtain the interacting four-point assertion, the unresolved comparison must identify the actual normalized history/block-field law and establish, on its whole prescribed class,

$$S_4^T(\underline{f}) = 3\zeta^4 x^{-4}\mathcal{B}_{4,a}[P_Y](\underline{f}) + \mathcal{R}_4(\underline{f}), \qquad |\mathcal{R}_4(\underline{f})| \le \epsilon_4(d)\, x^{-4}\prod_i F_i, \quad \epsilon_4(d) \longrightarrow 0. \tag{6.ccxxxiii}$$

For a relative error at most one half, the actual error must in particular beat the fixed shape margin proportional to ϑ^4. Every contribution belongs in $\mathcal{R}_4$: nonquadratic Wilson terms, the interacting conditional density, source jets, cutoff boundaries, fluctuations not represented by P_Y, and the histories excluded from the clean comparison. The required constants must have the stated cutoff, support-position, reference-volume and state uniformity; the torus-only four-point claim must retain its torus scope.

The bounded windowed insertions use the probability of an unclean history in the normalized comparison. The physical first source derivatives can carry the age weight L^{16a}. They use the weighted sum in eq. 6.cxii or a direct bound for the four physical insertions. Nor does a bound on the interior Hessian determine the exterior force in the fixed conditional density. Those estimates concern the interacting construction, and none is inferred from the sign of the free box.

6.9. **A uniform one-field insertion bound.** An estimate for each correlation function does not automatically give an estimate in the Hilbert norm of an arbitrary polynomial of fields. The argument below obtains that stronger norm by repeated reflections at a fixed lattice spacing, before taking a continuum limit. Its analytic hypotheses are the ordinary scalar source upper bound and floor; no typed insertion estimate is required.

Assume the all-mirror invariance and reflection positivity of every periodic Wilson DLR state, the ordinary torus source hypothesis in eq. 5.cdxxxv, and the fixed-layer rooted domain geometry in eq. 5.cdxlvi–eq. 5.cdxlviii. The localization size is $M = L^{m_0}$, with L odd. Torus limits use the cofinal compatible family $\ell = 2L^m$, $m \ge m_0$. Every scale cube Ma_s then tiles the torus, since $\ell/(Ma_s) = 2L^{m-m_0+s}$. Every fixed board of power-of-L side, and every fixed coarser such board, tiles all sufficiently large tori in this family. No scale terms are omitted.

A raw-algebra reflection argument. Fix the spacing $a = L^{-K}$ and a periodic state ρ. Take a board cell C of side $l = L^j \ge a$, centered at a lattice site, with lower time face t. Its face planes are link

mirrors: l/a is odd. Its middle time plane is a site mirror. Both are supplied by the all-mirror theorem. For bounded finite-support functions above t put

$$\langle B, A\rangle_t = \rho(\overline{\theta_t B}A), \qquad \|A\|_t^2 = \langle A, A\rangle_t. \tag{6.ccxxxiv}$$

For a bounded complex cell function D, let $q_D = \limsup_k p_{l,\ell_k}(D)$ along a compatible cofinal sequence. Define, on the raw half-space algebra,

$$T_D A = D\,\tau_{le_4}A, \qquad D^* = \overline{\theta_{t+l/2}D}.$$

Reflection and translation invariance give the formal adjoint identity $\langle B, T_D A\rangle_t = \langle T_{D^*}B, A\rangle_t$. No bounded operator on a quotient has been assumed.

Put $K_D = T_{D^*}T_D$ and $m_j = \langle A, K_D^j A\rangle_t$. Formal adjoints and reflection Cauchy–Schwarz imply

$$m_{2r} = \|K_D^r A\|_t^2, \qquad m_{2r+1} = \|T_D K_D^r A\|_t^2, \qquad m_j^2 \le m_{j-1}m_{j+1}. \tag{6.ccxxxv}$$

For odd $j = 2r+1$ the last inequality uses the two vectors $K_D^r A, K_D^{r+1}A$; for even $j = 2r \ge 2$ it uses $T_D K_D^{r-1}A, T_D K_D^r A$. Thus it is a raw semidefinite-form calculation, not an application of a spectral theorem to an operator whose boundedness is still to be proved. If $m_0 = 0$, Cauchy–Schwarz already gives $m_1 = 0$. Otherwise log-convexity gives

$$m_1^N \le m_0^{N-1} m_N. \tag{6.ccxxxvi}$$

The expression for m_N consists of the left endpoint $\overline{\theta_t A}$, a column of $2N$ alternating translated copies of D^* and D, and the right endpoint $\tau_{2Nle_4}A$. Choose a coarse side $l' = L^r l$ large enough to contain A in a coarse half-space cell above t, spatially aligned with C. The oddness of L^r makes all coarse faces fine-board faces. Take N through sufficiently large multiples of L^r.

On this coarse grid retain fixed endpoint zones containing both endpoint supports and, if convenient, a fixed number c_A of the adjacent column factors. Bound the complete endpoint functions by their fixed-cutoff supremum norms. This yields a finite constant $C(a, A, D)$ independent of N. In the middle, every occupied coarse cell is a product of fine cell functions only. Fine-to-coarse chessboard refinement and product domination in eq. 5.cdxxxiii apply there. Grid-preserving translates and reflected conjugates of D have the same fine seminorm. Therefore, for every $\epsilon > 0$,

$$m_N \le C(a, A, D)(q_D + \epsilon)^{2N - c_A}. \tag{6.ccxxxvii}$$

The volume limit is taken first for each fixed N. The estimate uses no assertion $p(AD) \le \|A\|_\infty p(D)$: a mixed endpoint block, if present, is bounded in its entirety by its supremum norm. The endpoint zones contain only a bounded number of coarse cells; their factor count and supremum bound stay fixed as the column grows.

Taking the Nth root in eq. 6.ccxxxvi–eq. 6.ccxxxvii, then sending $N \to \infty$ and finally $\epsilon \downarrow 0$, gives

$$\|T_D A\|_t \le q_D \|A\|_t, \qquad |\rho(\overline{\theta_t B} D \tau_{le_4}A)| \le q_D \|B\|_t \|A\|_t. \tag{6.ccxxxviii}$$

The same order of limits proves this when $q_D = 0$; no negative power of q_D is introduced. It also proves that null vectors are mapped to null vectors. Only now may the map be regarded as a bounded operator on the Hilbert quotient. Endpoint constants may depend on the number of fields inside A, but they disappear before eq. 6.ccxxxviii. The final constant does not depend on that number, on B, or on their polynomial coefficients.

One ordinary source and a common centering. Let h be real and supported in the ball of radius $l/4$ around the cell center. For $a \le l/64$, apply the three-range count in eq. 5.cdxlix with board side $\eta = l$, support radius $\delta = l/4$ and the same face margin. Since $\eta/\delta = 4$, its face cost is bounded uniformly as $l \downarrow 0$. Coarse scales use the global count divided by the number of cells, not a separate root count at every source copy. Upper stability and the floor cost only Cl^4. Thus, for $l \le 1$, there are holomorphic univariate normalizers with

$$E_{h,\ell}(0) = 0, \qquad p_{l,\ell}(e^{z\mathcal{O}_a(h) - E_{h,\ell}(z)}) \le Q_0, \qquad |z|\|h\|_\sharp < r_0, \tag{6.ccxxxix}$$

where $Q_0, r_0 > 0$ are independent of the spacing and cell side. They are also independent of the surrounding fields. At a fixed cutoff these normalizers are locally bounded uniformly in the torus, so a compatible Montel subsequence gives one E_h, independently of every infinite-volume state.

The remaining coarse mesh ratios can also be included. If $3 \le l/a < 64$, the ball and its contributing plaquettes lie wholly inside the cell, and $\|\mathcal{O}_a(h)\|_\infty \le C64^4\|h\|_\infty$. Set $E_{h,\ell} = 0$ and enlarge Q_0 in eq. 6.ccxxxix. The only smaller allowed ratio is $l/a = 1$; then a radius-$a/4$ ball about a lattice site contains no plaquette barycentre, so $\mathcal{O}_a(h) = 0$. Consequently eq. 6.ccxxxix covers every allowed ratio, with the same constants.

For fixed z put $D_z = e^{z\mathcal{O}_a(h) - E_h(z)}$. Homogeneity of the seminorm and the locally uniform convergence of the normalizers give $q_{D_z} \le Q_0$. Apply eq. 6.ccxxxviii and then Cauchy's formula on the one source disc. The expectation is holomorphic, because at fixed cutoff all observables are bounded. We obtain

$$\left|\rho\left(\overline{\theta_t B}[\mathcal{O}_a(h) - E_h'(0)]\tau_{le_4}A\right)\right| \le (Q_0/r_0)\|h\|_\sharp\|B\|_t\|A\|_t.$$

Taking $A = B = 1$ in a second arbitrary periodic state ν bounds $|\nu(\mathcal{O}_a(h)) - E_h'(0)|$ by the same expression without the outside norms. The case $D = 1$ of eq. 6.ccxxxviii handles the scalar error. Hence

$$\left|\rho\left(\overline{\theta_t B}[\mathcal{O}_a(h) - \nu(\mathcal{O}_a(h))]\tau_{le_4}A\right)\right| \le C\|h\|_\sharp\|B\|_t\|A\|_t. \tag{6.ccxl}$$

Both states are arbitrary. Large cutoff-dependent normalizer derivatives cancel; uniform moments of perturbed Gibbs states have not been assumed. In using eq. 6.ccxxxviii for D_z, endpoint supremum bounds may depend on z and the cutoff. They disappear pointwise before the uniform bound $q_{D_z} \le Q_0$ is used in Cauchy's formula.

Continuum passage with fixed support gaps. Suppose $a_j \downarrow 0$ and two periodic states ρ_j, ν_j at each datum give convergent centered separated moments

$$\sigma_j(f_1, \dots, f_n) = \rho_j \prod_i [\mathcal{O}_{a_j}(f_i) - \nu_j(\mathcal{O}_{a_j}(f_i))] \longrightarrow \sigma(f_1, \dots, f_n).$$

Use the uniform all-state local moment bounds already proved to obtain equicontinuity under small C^1 test changes. In particular, on each fixed separated tuple, $\sigma_j(\tau_{v_j}X) \to \sigma(X)$ whenever $v_j \to 0$. This follows directly by multilinear telescoping and $\|\tau_v f - f\|_\sharp \to 0$ for compact C^1 functions. The constants in this passage may depend on the fixed tuple and its length; they do not enter the constant in eq. 6.ccxl. Approximating a fixed real translation by lattice translations also proves translation invariance of σ, since both states in each datum are lattice-translation invariant.

Let F, G be arbitrary finite complex linear combinations of field products whose entries have compact supports strictly in positive time and are separated within each product. Define $\|[F]\|_\sigma^2 = \sigma(\theta\overline{F} \cdot F)$. This is nonnegative as a limit of the lattice reflection forms. For a fixed power $l \le 1$ and a real h supported in $\bar{B}((\vec{y}, 0), l/4)$, choose the nearest spatial lattice point $\vec{y}_j$ and put

$$v_j = (\vec{y}_j - \vec{y}, a_j/2), \quad t_j = a_j/2, \quad x_{*,j} = (\vec{y}_j, l/2 + a_j/2).$$

For all sufficiently fine cutoffs l/a_j is odd. Therefore $x_{*,j}$ is a lattice site and the cell faces are link mirrors. The shifted inserted test is explicitly $h_j = \tau_{v_j + (l/2)e_4}h$, supported around that cell center. The identity $\theta_{t_j}\tau_{v_j} = \tau_{v_j}\theta_0$ identifies the lattice reflected norm tuples with the corresponding small translations of the continuum norm tuples. No half-mesh translation invariance of ν_j is being assumed.

Apply eq. 6.ccxl to the centered lattice polynomials associated to $\tau_{v_j}F$ and $\tau_{v_j}G$. All insertion and norm tuples have fixed positive support gaps for large j. Their values converge by the preceding equicontinuity, including when a limiting Hilbert norm is zero. Thus

$$|\sigma(\theta\overline{G} \cdot (\tau_{(l/2)e_4}h) \cdot \tau_{le_4}F)| \le C\|h\|_\sharp\|[G]\|_\sigma\|[F]\|_\sigma. \tag{6.ccxli}$$

The constant is independent of the numbers of entries and the finite polynomial coefficients of F, G. Arbitrary centering states ν_j are retained throughout.

For completeness time translations are contractions in these norms. At fixed spacing, reflection Cauchy–Schwarz gives $\|\tau_u A\|_t^2 \le \|A\|_t \|\tau_{2u} A\|_t$ for every positive lattice time u. Iteration against $\|\tau_{2^N u} A\|_t \le \|A\|_\infty$ proves contraction. The same fixed-gap limit passage proves it for every positive real time in the continuum. It uses no bounded insertion operator in advance.

A C^1 insertion estimate near the time mirror. We record the form needed independently of any plate or angular argument. Let $0 < \eta \le 1$, and let $f \in C_c^1$ be supported in $\{\eta \le x_4 \le 3\eta\}$, with support diameter at most R. Take the largest power l of L below η and partition f at mesh cl into pieces in radius-$l/4$ balls. There are at most $C_L(1+R)^3\eta^{-3}$ pieces: only the three spatial directions have unbounded length in units of η. Since f vanishes on both time faces, $\|f\|_\infty \le \eta \|\partial_4 f\|_\infty$. This cancels the $1/l$ from the partition derivative, because $\eta/l \le L$. Each piece has sharp norm at most $C\|f\|_{C^1}$.

Translate a piece to the middle of its side-l time slab and use eq. 6.ccxli. Its center time lies between $\eta - l/4$ and $3\eta + l/4$; both outside factors therefore retain a positive margin of at least $\eta - 3l/4$. Time contraction absorbs those margins. Summing gives

$$|\sigma(\theta\overline{G} \cdot (f) \cdot \tau_{4\eta e_4} F)| \le C(1+R)^3 \eta^{-3} \|f\|_{C^1} \|[G]\|_\sigma \|[F]\|_\sigma.$$

Taking $G = (f) \cdot \tau_{4\eta e_4} F$ and dividing its norm proves the field-operator estimate; splitting real and imaginary parts handles complex f. We may weaken η^{-3} to η^{-4} and write

$$\|[(f) \cdot \tau_{4\eta e_4} F]\|_\sigma \le C(R)\eta^{-4} \|f\|_{C^1} \|[F]\|_\sigma. \tag{6.ccxlii}$$

Proposition 6.23 (Uniform one-field insertion)**.** Under the stated ordinary source, rooted-domain and lattice-symmetry hypotheses, every limit just described satisfies eq. 6.ccxli–eq. 6.ccxlii. Choose a fixed $0 < c < 1/64$. If a real $k \in C_c^1$ is supported in a ball of radius $c\eta$ centered at y, $|y_4| \le \eta/8$, and all entries of F, G lie strictly above $\eta/4$, then

$$|\sigma(\theta\overline{F} \cdot (k) \cdot G)| \le C\eta^{-4} \|k\|_\sharp \|F\|_{(\eta/4)} \|G\|_{(\eta/4)}, \quad \|F\|_{(\eta/4)} = \|[\tau_{-(\eta/4)e_4} F]\|_\sigma. \tag{6.ccxliii}$$

The constant depends only on the ordinary source and localization constants and the fixed geometric choices. It is independent of the defining sequence, the states, support positions, and the numbers of fields in the surrounding finite complex polynomials. ∎

Indeed put $\xi = c\eta$ and translate the insertion in time by $2\xi - y_4$, placing it in $[\xi, 3\xi]$. After this translation the left and right polynomials are, respectively, $\tau_{y_4 - 2\xi} F$ and $\tau_{-y_4 - 2\xi} G$, the latter followed by $\tau_{4\xi e_4}$. Each is a positive time translate of its $\eta/4$-downshift by at least $\eta/8 - 2\xi > 0$. Apply eq. 6.ccxlii, reflection Cauchy–Schwarz and contraction, absorbing the fixed factor c^{-4}. This proves eq. 6.ccxliii. The theorem asserts the C^1 insertion bound only. It does not assert a plate density theorem, an L^1 insertion bound, or full rotational covariance.

6.10. **Analytic insertion densities and the plate bound.** This section derives an L^1 insertion bound from the proved C^1 Hilbert insertion bound. All operators are regularized by positive Euclidean time. No unsandwiched point field is presumed to exist. The exact spectral cube cone is used, but rotational covariance is not.

Let σ be a symmetric, real, time-reflection-invariant, translation-invariant functional on admissible separated tests, with its reflection-positive Hilbert reconstruction. Write $\mathcal{D}$ for the dense classes of finite complex positive-time polynomials and $\Omega = [1]$. Use the probability normalization $\sigma(1) = 1$, so Ω is a unit vector fixed by all translations. Positive time translation is e^{-tH}, spatial translation is $U(\mathbf{x}) = e^{i\mathbf{x}\cdot\mathbf{P}}$, and these operators strongly commute. Assume their joint spectral measure is carried by

$$E \ge 0, \qquad |q_j| \le E \quad (j = 1, 2, 3). \tag{6.ccxliv}$$

Also assume local C^1 continuity of the separated moments, as proved from the ordinary source bounds. This continuity is used for fixed finite polynomials only; none of its field-number constants enter the uniform bounds below.

The insertion input has fixed constants $C_I, c_I > 0$, $p \geq 0$, with $c_I < 1/64$. For $0 < \eta \leq 1$, let

$$I(F; h; G) = \sigma(\theta\overline{F} \cdot (h) \cdot G), \qquad \|F\|_{(\eta/4)} = \|[\tau_{-(\eta/4)e_4} F]\|.$$

Whenever the entries of F, G lie strictly above $\eta/4$ and h is real with support in a radius-$c_I\eta$ ball centered at y, $|y_4| \leq \eta/8$, the input is

$$|I(F; h; G)| \leq C_I \eta^{-p} \|h\|_\sharp \|F\|_{(\eta/4)} \|G\|_{(\eta/4)}. \tag{6.ccxlv}$$

The norm is $\|h\|_\sharp = \max(\|h\|_\infty, c_M \|\nabla h\|_\infty)$. Splitting real and imaginary parts permits complex h, at a fixed factor two. The preceding insertion theorem supplies $p = 4$; keeping p general makes the additional five powers transparent.

Bounded forms, real translations and holomorphy. For h supported in $\bar{B}(0, c_I\eta)$, define the form on raw positive-time polynomials by

$$b_{h,\eta}(A, B) = I(\tau_{(\eta/4)e_4} A; h; \tau_{(\eta/4)e_4} B).$$

By eq. 6.ccxlv it descends through null vectors and extends to a bounded sesquilinear form on the Hilbert space. Riesz representation gives a unique bounded operator $B_{h,\eta}$, with

$$\|B_{h,\eta}\| \leq C\eta^{-p} \|h\|_\sharp, \quad B_{h,\eta}[A] = [(\tau_{(\eta/4)e_4} h) \cdot \tau_{(\eta/2)e_4} A]. \tag{6.ccxlvi}$$

The vector identity follows by testing against the dense polynomial classes and translating the insertion form. Its displayed product is admissible: the insertion is strictly positive in time and is separated from the shifted entries of A. Thus this identity proves the action of the bounded operator; it does not postulate a point field on the quotient.

For complex $z = (z_4, \mathbf{z})$ put

$$T(z) = \int e^{-z_4 E + i\mathbf{z}\cdot\mathbf{q}} \, d\Pi(E, \mathbf{q}).$$

The spectral cone gives $\|T(z)\| \leq 1$ if $\operatorname{Re} z_4 \geq \sum_j |\operatorname{Im} z_j|$. It is operator-norm holomorphic in the strict interior. Indeed, on a neighbourhood with margin $\epsilon > 0$, derivatives of total order m of the integrand are bounded by $E^m e^{-\epsilon E}$. Its supremum is at most $m!\epsilon^{-m}$. The exponential Taylor series consequently converges uniformly in spectral variables on smaller complex neighbourhoods, so its spectral integrals converge in operator norm and may be differentiated there.

Now take h supported in $\bar{B}(0, c_I\eta/2)$, and set $\Phi = [\tau_{-(\eta/4)e_4} F]$, $\Psi = [\tau_{-(\eta/4)e_4} G]$. For real x, $|x_4| < \eta/8$, direct use of eq. 6.ccxlvi gives

$$I(F; \tau_x h; G) = \langle \Phi, M_{h,\eta}(x) \Psi \rangle, \quad M_{h,\eta}(z) = T(\tfrac{\eta}{8} e_4 + z) B_{h,\eta/2} T(\tfrac{\eta}{8} e_4 - z). \tag{6.ccxlvii}$$

For clarity, the real time factors in this formula are $e^{-(\eta/8 + x_4)H}$ and $e^{-(\eta/8 - x_4)H}$; the spatial factors are $U(\mathbf{x})$ and $U(-\mathbf{x})$. Substitution in eq. 6.ccxlvi, followed by one common time translation, recovers exactly $\theta\overline{F} \cdot (\tau_x h) \cdot G$. Both factors in eq. 6.ccxlvii are contractions on

$$\mathcal{O}_\eta = \{z : |\operatorname{Re} z_4| + \textstyle\sum_{j=1}^3 |\operatorname{Im} z_j| < \eta/8\}. \tag{6.ccxlviii}$$

Hence $M_{h,\eta}$ is operator-norm holomorphic there and $\|M_{h,\eta}(z)\| \leq C\eta^{-p} \|h\|_\sharp$. There is no restriction on real spatial translations.

A scaled compact parametrix, with its distribution proof. In four dimensions let $G(x) = -(64\pi^2)^{-1}|x|^2\log|x|$, with value zero at the origin. It is C^1. For $x \neq 0$,

$$\Delta(|x|^2\log|x|) = 8\log|x| + 6, \quad \Delta\log|x| = 2|x|^{-2}, \quad \Delta|x|^{-2} = 0.$$

The first two identities also hold as distributions: Green's formula on $\{|x| > \epsilon\}$ has boundary errors tending to zero, respectively of orders $O(\epsilon^4|\log\epsilon|)$ and $O(\epsilon^2)$, up to smaller terms. In the third identity the boundary term is nonzero. Against a compact smooth test f, the inner normal is $-\partial_r$, so the term containing $f\partial_n|x|^{-2}$ tends to $-2|S^3|f(0) = -4\pi^2 f(0)$; the other boundary term is $O(\epsilon)$. Thus

$$\Delta|x|^{-2} = -4\pi^2\delta_0, \qquad \Delta^3 G = \delta_0. \tag{6.ccxlix}$$

Choose real $\chi \in C_c^\infty(B(0,1))$ equal to one near zero. Put $n = \chi G$ and $s = \delta_0 - \Delta^3 n$. Since $\chi = 1$ near the singularity, eq. 6.ccxlix cancels its delta term. Every remaining Leibniz term contains a derivative of χ supported away from zero, where G is smooth. Therefore $s \in C_c^\infty(B(0,1))$. For $0 < r \leq 1$ define

$$n_r(x) = r^2 n(x/r) = -\frac{\chi(x/r)|x|^2\log(|x|/r)}{64\pi^2}, \qquad s_r(x) = r^{-4}s(x/r). \tag{6.ccl}$$

Distributional scaling in four dimensions gives $\Delta^3 n_r = \delta_0 - s_r$: six derivatives contribute r^{-6}, the prefactor contributes r^2, and the scaled delta contributes r^4. Ordinary differentiation gives

$$\|n_r\|_\sharp \leq Cr, \qquad \|s_r\|_\sharp \leq Cr^{-5}. \tag{6.ccli}$$

Both supports lie in $\bar{B}(0,r)$. The logarithm is $\log(|x|/r)$, not $\log|x|$ after simply cutting off G. The subtracted quadratic polynomial removes a spurious $|\log r|$ in eq. 6.ccli.

For smooth compact k, convolution with eq. 6.ccxlix–eq. 6.ccl gives

$$k = (\Delta^3 k) * n_r + k * s_r. \tag{6.cclii}$$

This can be verified directly against compact smooth tests by Fubini and distributional integration by parts. Alternatively, the compact C^1 kernel n_r is integrable, so all derivatives of its convolution with k may be placed on k under the integral; the distribution identity then yields eq. 6.cclii. Both sides are C^1, so their distributional equality is also pointwise equality.

A bounded analytic density and the L^1 bound. Fix $\epsilon = \min(c_I/8, 1/1024)$ and put $r = \epsilon\eta$. Both n_r and s_r fit the source ball used in eq. 6.ccxlvii. Define the operator-valued holomorphic function

$$V_\eta(z) = \Delta_z^3 M_{n_r,\eta}(z) + M_{s_r,\eta}(z). \tag{6.ccliii}$$

On the smaller domain $|\operatorname{Re} z_4| + \sum_j |\operatorname{Im} z_j| < \eta/32$, a polydisc of radius $\eta/128$ lies in $\mathcal{O}_\eta$. Cauchy's inequalities bound the six derivatives by $6!(128/\eta)^6$ times the bound in eq. 6.ccxlvii. The sum of coefficients of Δ^3 is 4^3. eq. 6.ccli therefore give

$$\|V_\eta(z)\| \leq C\eta^{-p}(\eta^{-6}r + r^{-5}) \leq C\eta^{-(p+5)}. \tag{6.ccliv}$$

Every constant is independent of the surrounding polynomials and their numbers of entries.

Let $k \in C_c^\infty$ have support in $|x_4| < \eta/64$. For a continuous compact ϕ and a compact C^1 kernel q, Riemann sums for $\phi * q = \int \phi(x)\tau_x q\,dx$ converge in C^1, with common compact support. Local C^1 continuity of the insertion form consequently implies $I(F;\phi * q;G) = \int \phi(x) I(F;\tau_x q;G)\,dx$. All convolution supports here stay strictly between the two outside time supports. Applying this observation to eq. 6.cclii, then integrating six derivatives off k onto the analytic matrix element in eq. 6.ccxlvii, proves

$$I(F;k;G) = \int k(x)\langle\Phi, V_\eta(x)\Psi\rangle\,dx. \tag{6.cclv}$$

The integrations by parts have no boundary terms, since k is compactly supported in the open strip. This proves the density identity, not just an inequality.

Mollification carries eq. 6.cclv to compact C^1 tests in the strip: the left side converges by local C^1 continuity and the right side by L^1 convergence and eq. 6.ccliv. Thus it has the unique continuous extension to every L^1 function on that strip, with

$$|I(F;k;G)| \le C\eta^{-(p+5)}\|k\|_{L^1}\|F\|_{(\eta/4)}\|G\|_{(\eta/4)}. \tag{6.cclvi}$$

For an L^1 test this formula defines the extended insertion; no pointwise field interpretation is required. The operator weak integral $\int k(x)V_\eta(x)\,dx$ is well-defined by its bounded sesquilinear forms and has the same norm bound. For compact C^1 tests it agrees with the regularized field product in eq. 6.ccxlvi. If two choices of η admit the same outside polynomials and test strip, their forms agree first on compact smooth tests and then on the common L^1 domain by density. Thus the extension is compatible on intersecting pulled-down domains.

Different compact parametrices give the same density on the real strip: eq. 6.cclv makes their continuous matrix-element differences zero against every compact smooth test. They therefore agree pointwise. Their holomorphic continuations agree on connected overlapping domains, by uniqueness from an open set of real points, first in scalar matrix elements and then as bounded operators.

Plates and compatible point-plate operators. Split $x = (t, u, x_\perp)$, with $x_\perp \in \mathbb{R}^2$. For $\varphi \in L^1(\mathbb{R}^2)$ set

$$V_{\varphi,\eta}(t,u) = \int \varphi(x_\perp)V_\eta(t,u,x_\perp)\,dx_\perp. \tag{6.cclvii}$$

The weak integral is a bounded operator, of norm at most $C\eta^{-(p+5)}\|\varphi\|_1$. It is operator-norm holomorphic for $|\operatorname{Re} t| + |\operatorname{Im} u| < \eta/32$. Indeed the integrands are uniformly bounded and holomorphic on smaller neighbourhoods, so their locally uniformly convergent Taylor series may be integrated. For an L^1 plane profile α supported in $|t| < \eta/64$, Tonelli and eq. 6.cclv–eq. 6.cclvi give

$$|I(F;\alpha\otimes\varphi;G)| \le C\|\varphi\|_1\eta^{-\wp}\|\alpha\|_1\|F\|_{(\eta/4)}\|G\|_{(\eta/4)}, \qquad \wp = p+5. \tag{6.cclviii}$$

Profiles may extend arbitrarily far in the spatial coordinate u. For smooth compact profiles and labels this is the original insertion; all other stated L^1 cases are its continuous extension.

For $0 < a \le 1/4$ define the regularized point-plate operator

$$B_{\varphi;a} = V_{\varphi,4a}(0,0), \qquad \|B_{\varphi;a}\| \le Ca^{-\wp}\|\varphi\|_1. \tag{6.cclix}$$

It is the operator-norm limit of the regularized insertions of $\alpha_\epsilon \otimes \varphi$, where α_ϵ is a unit-integral smooth plane approximate identity at zero. This follows from the operator-norm continuity in eq. 6.cclvii. For $0 < a < b \le 1/4$,

$$B_{\varphi;b} = e^{-(b-a)H}B_{\varphi;a}e^{-(b-a)H}. \tag{6.cclx}$$

To prove it, first use a compact smooth plate supported in the central strips of both scales. The vector formula in eq. 6.ccxlvi, or its dense matrix-element identity, proves the relation for its two regularized operators. Approximate the plane point by α_ϵ and pass in operator norm. Finally approximate an L^1 label by smooth compact labels using eq. 6.cclviii. This proves compatibility on the stated domains without assuming an unbounded field common to the scales.

The same calculation gives real translation compatibility. For $0 < a' < a \le 1/4$ and $|t| < a/16$,

$$V_{\varphi,4a}(t,u) = e^{-(a-a'+t)H}U(ue_1)B_{\varphi;a'}U(-ue_1)e^{-(a-a'-t)H} \tag{6.cclxi}$$

whenever $a - a' > |t|$. One proves this first for a sufficiently narrow smooth approximate identity, using the real translation calculation of eq. 6.ccxlvii, and then passes in operator norm. The right side also supplies the corresponding holomorphic continuation. These identities are suitable for composing independently regularized plate insertions.

Proposition 6.24 (Analytic density and plate bound)**.** Under eq. 6.ccxliv–eq. 6.ccxlv and the stated local continuity, the insertion has the analytic operator density in eq. 6.ccliii–eq. 6.cclv, the uniform L^1 bound in eq. 6.cclvi, and the plate bound in eq. 6.cclviii with exponent $\wp = p + 5$. The point-plate operators in eq. 6.cclix obey the compatibility and translation identities in eq. 6.cclx–eq. 6.cclxi. All constants are independent of the numbers of outside fields. No additional typed source estimate is used. ∎

As a useful spectral consequence, suppose φ is real. Reflection of real, time-even approximate identities proves $B^*_{\varphi;a} = B_{\varphi;a}$. Let $v_{\varphi,a} = B_{\varphi;a}\Omega$ and let $\mu_{\varphi,a}$ be its finite positive joint spectral measure for (H, P_1). Since $H\Omega = 0$, eq. 6.cclx gives $\mu_{\varphi,b} = e^{-2(b-a)E}\mu_{\varphi,a}$. The positive measure $\mu_\varphi = e^{2aE}\mu_{\varphi,a}$ is thus independent of a, is finite on bounded-energy sets, and may have infinite total mass. For $t > 0$ choose $0 < a < \min(t/2, 1/4)$ and put

$$K_\varphi(t,u) = \langle v_{\varphi,a}, e^{-(t-2a)H+iuP_1} v_{\varphi,a}\rangle = \int e^{-tE+iuq}\, d\mu_\varphi(E,q). \tag{6.cclxii}$$

It is independent of a, finite, and its spectral measure is supported on $E \geq |q|$. The integral is holomorphic on $\mathrm{Re}\, t > |\mathrm{Im}\, u|$: on smaller neighbourhoods all differentiated integrands are bounded by a polynomial in E times a strictly decaying exponential, integrable by eq. 6.cclxii at a smaller positive real time.

To identify this as the original two-plate kernel, take real plane approximate identities even in time, first with a smooth compact real label φ. The vectors $[(\tau_{ae_4}\alpha_\epsilon) \otimes \varphi]$ converge to $v_{\varphi,a}$ by eq. 6.cclix. Their matrix elements in eq. 6.cclxii are precisely the two-plate moments at time separation t, with the narrow profiles, by reflection and common translation. For compact profiles with a fixed positive time gap, choose a smaller than half that gap. Convolution approximates those profiles in C^1, while the approximating matrix elements are uniformly bounded by the two vector norm bounds. Fubini and dominated convergence therefore recover the integral of K_φ against the original profiles. Smooth compact approximation of an L^1 label uses the uniform bounds above. The kernel here is the full two-point moment kernel; for connected moments one applies the construction to the centered functional. This is a statement in the chosen coordinate frame. No diagonal-frame cone, contact value at $t = 0$, or rotational covariance is asserted here.

6.11. **Analytic planar kernels and the finite angular band.** A plate estimate uniform in the numbers of outside fields yields the analytic planar kernels used in the angular argument. Rotation invariance also requires the short-distance two-point estimate. Both estimates are hypotheses in this section.

The assumptions and the meaning of a product. Write a point of Euclidean space as (y, z), where $y = (y^0, y^1) = (x_4, x_1)$ and $z = (x_2, x_3)$. Put $\theta(y^0, y^1) = (-y^0, y^1)$. A tuple is a finite ordered list of compactly supported C^1 test functions with pairwise disjoint supports. Products of tuples mean concatenation, whenever the concatenation still has disjoint supports. We use finite complex linear combinations of such tuples, including the empty tuple 1. The notation (h) denotes the tuple with its single entry h.

Assume the following properties of a centered functional σ. It is multilinear in the entries, symmetric, real on real entries, and invariant under simultaneous real translations and the hypercubic group. Its values satisfy $\sigma(1) = 1$ and $\sigma((h)) = 0$. On every fixed collection of separated compact support sets it is continuous for the C^1 norms, with a bound independent of the positions of those sets when their diameters and mutual distances are fixed. No bound on the growth of these constants with the number of entries is assumed in this section.

Let $\mathscr{A}_+$ denote the combinations of tuples supported strictly in $\{x_4 > 0\}$. Reflection positivity and the reconstruction give a Hilbert space $\mathcal{H}$, a dense class map $F \mapsto [F]$, and $\Omega = [1]$, such that

$$\langle [F], [G]\rangle = \sigma(\theta\overline{F} \cdot G), \qquad \|\Omega\| = 1. \tag{6.cclxiii}$$

The inner product is antilinear in the first entry. The reconstructed time and planar spatial translations have the actions

$$e^{-tH}[F] = [\tau_{te_4}F], \qquad e^{ixP}[F] = [\tau_{xe_1}F], \qquad t \geq 0, \tag{6.cclxiv}$$

where H, P are strongly commuting self-adjoint operators, $H \geq 0$, and

$$H \geq |P|, \qquad H\Omega = P\Omega = 0. \tag{6.cclxv}$$

Only this one planar consequence of the exact cube cone is used.

For F supported above a height a define the pulled-down norm by

$$\|F\|_{(a)} = \|[\tau_{-ae_4}F]\|.$$

A label is a real $\varphi \in C_c^1(\mathbb{R}^2)$ in the transverse variables. A profile is a complex $\alpha \in C_c^1(\mathbb{R}^2)$. The corresponding plate is $\alpha \otimes \varphi$. Fix constants $0 < c < 1/4$ and $\wp > 0$. The plate input is that, for each label, a constant $C_\varphi < \infty$ satisfies

$$\begin{aligned} &|\sigma(\theta\overline{F} \cdot (\alpha \otimes \varphi) \cdot G)| \\ &\qquad \leq C_\varphi \eta^{-\wp} \|\alpha\|_1 \|F\|_{(\eta/4)} \|G\|_{(\eta/4)} \end{aligned} \tag{6.cclxvi}$$

for $0 < \eta \leq 1$, profiles supported in $|y^0| < c\eta$, and F, G supported strictly above $\eta/4$. This includes arbitrary finite combinations and arbitrary numbers of entries in F, G. The constant C_φ is the same for all these choices and all η. Every concatenation in this inequality has separated supports. The plate theorem obtained from the ordinary-source insertion estimate supplies precisely this input, with a fixed small c and $\wp = p + 5$.

Bounded plates and their padding. For a profile in the strip of eq. 6.cclxvi put $u = \eta/4$. On the dense classes define

$$B_\varphi[\alpha; \eta][G] = \big[(\tau_{ue_4}(\alpha \otimes \varphi)) \cdot \tau_{2ue_4}G\big]. \tag{6.cclxvii}$$

The plate on the right lies at times strictly between 0 and $2u$; the other entries lie strictly above $2u$. Thus the right side is an actual class in $\mathcal{H}$.

To prove that it depends only on $[G]$, test it against $[F]$ and translate the full tuple down by u. The resulting expression is

$$\langle [F], B_\varphi[\alpha; \eta][G] \rangle = \sigma\big(\theta\overline{\tau_{ue_4}F} \cdot (\alpha \otimes \varphi) \cdot \tau_{ue_4}G\big).$$

The two pulled-down norms on the right are $\|[F]\|$ and $\|[G]\|$. By eq. 6.cclxvi, density and Hilbert duality,

$$\|B_\varphi[\alpha; \eta][G]\| \leq C_\varphi \eta^{-\wp} \|\alpha\|_1 \|[G]\|. \tag{6.cclxviii}$$

In particular null classes map to null classes. There is a unique bounded extension to the whole Hilbert space, with the same norm bound. No domain containing an unspecified number of fields remains in this statement.

The definitions also fix all padding conventions. If $\eta' \leq \eta$ and the profile is supported in $|y^0| < c\eta'$, then, with $d = (\eta - \eta')/4$,

$$B_\varphi[\alpha; \eta] = e^{-dH} B_\varphi[\alpha; \eta'] e^{-dH}. \tag{6.cclxix}$$

Indeed both sides applied to $[G]$ give the right side of eq. 6.cclxvii; density proves equality of the bounded operators. Realness, symmetry, reflection invariance and a translation by $2u$ similarly give

$$B_\varphi[\alpha; \eta]^* = B_\varphi[\overline{\alpha} \circ \theta; \eta]. \tag{6.cclxx}$$

For example, conjugate the expression defining $\langle [G], B_\varphi[\alpha; \eta][F] \rangle$, reflect all its entries, translate by $2u$, and permute the entries. The resulting expression is $\langle [F], B_\varphi[\overline{\alpha} \circ \theta; \eta][G] \rangle$. The label remains unchanged because it is real and the reflection acts only on the plane.

The tube and its bounded translations. For $v = (v^0, v^1) \in \mathbb{C}^2$ use the linear coordinates $v_\pm = v^0 \pm i v^1$, and set

$$\mathbb{V} = \{\operatorname{Re} v_+ > 0, \operatorname{Re} v_- > 0\} = \{\operatorname{Re} v^0 > |\operatorname{Im} v^1|\}, \qquad v^* = (\overline{v^0}, -\overline{v^1}). \tag{6.cclxxi}$$

This is an open convex cone. Its real points are the half-plane $\mathbb{H} = \{v^0 > 0\}$. Both entrywise conjugation and the star preserve it, and $\operatorname{Re} v \in \mathbb{V}$ whenever $v \in \mathbb{V}$. These assertions follow directly from the two strict real linear inequalities in eq. 6.cclxxi. Notice the distinct identities

$$(v^*)_\pm = \overline{v_\pm}, \qquad v^* = -\theta v \quad \text{for real } v.$$

The joint spectral measure of (H, P) is carried by $\{(E, q) : E \geq |q|\}$. Define on the closed tube

$$T(v) = \int e^{-v^0 E + i v^1 q} \, d\mathsf{P}(E, q). \tag{6.cclxxii}$$

Writing $E_\pm = (E \pm q)/2 \geq 0$ gives $-v^0 E + i v^1 q = -v_+ E_- - v_- E_+$. Consequently

$$\|T(v)\| \leq 1, \quad T(v)T(w) = T(v+w), \quad T(v)^* = T(v^*), \quad T(v)\Omega = \Omega. \tag{6.cclxxiii}$$

The last assertion uses $H\Omega = P\Omega = 0$. Dominated convergence in the spectral measure of each vector proves strong continuity on the closed tube.

The map T is holomorphic in operator norm on $\mathbb{V}$. Here is a direct proof, which also records why the exact cone is appropriate. At a point with $\operatorname{Re} v_\pm \geq r > 0$, expand in increments $\Delta_\pm$. The coefficient of $\Delta_+^j \Delta_-^k$ is the bounded functional-calculus operator with multiplier

$$\frac{(-E_-)^j (-E_+)^k}{j!k!} e^{-v_+ E_- - v_- E_+}.$$

Its norm is at most r^{-j-k}, since $x^j e^{-x}/j! \leq 1$ for $x \geq 0$. The resulting two power series converge in operator norm when $|\Delta_\pm| < r$. For real v with $v^0 \geq 0$, the reconstruction gives

$$T(v)[F] = [\tau_{v^0 e_4 + v^1 e_1} F]. \tag{6.cclxxiv}$$

Let R_z be the complex rotation with the usual matrix

$$R_z = \begin{pmatrix} \cos z & -\sin z \\ \sin z & \cos z \end{pmatrix}.$$

In light coordinates $(R_z v)_\pm = e^{\pm i z} v_\pm$. In particular

$$R_{is}\mathbb{V} = \mathbb{V}, \qquad (R_{is} v)^* = R_{is}(v^*) \quad (s \in \mathbb{R}). \tag{6.cclxxv}$$

These are algebraic identities: the multipliers $e^{\mp s}$ are positive real numbers. For a tuple $v = (v_i)$ in the tube define

$$d_{\mathbb{V}}(v) = \min\{1, \min_{i,\pm} \operatorname{Re}(v_i)_\pm\}.$$

Then its exact transformation and a useful lower bound are

$$d_{\mathbb{V}}(R_{is} v) = \min\{1, \min_i e^{-s} \operatorname{Re}(v_i)_+, \min_i e^{s} \operatorname{Re}(v_i)_-\} \geq e^{-|s|} d_{\mathbb{V}}(v). \tag{6.cclxxvi}$$

The same assertions hold in every quarter-turn frame $\mathbb{V}_\nu = R_\nu \mathbb{V}$, $\nu \in (\pi/2)\mathbb{Z}$. The rotations commute; R_ν is real; and the real points of $\mathbb{V}_\nu$ are $\{e_\nu \cdot v > 0\}$, where $e_\nu = R_\nu(1, 0)$.

We will use the following elementary uniqueness principle. A holomorphic function on a connected open subset of $\mathbb{C}^N$ which vanishes on a nonempty relatively open subset of $\mathbb{R}^N$ vanishes identically. At an interior real point all real derivatives are zero, so its complex Taylor series is zero; overlapping balls then propagate the equality along a path. The same proof applies to Hilbert-valued holomorphic functions after testing against each vector. An antiholomorphic variable is treated by conjugating that variable first. This last use will always take place on domains stable under conjugation.

Products with smooth plane profiles. For $n \geq 1$ define the open convex domain

$$\mathbb{D}^{+}_{n,\eta} = \{\mathbf{x} : x_1 - \eta e_0/4 \in \mathbb{V},\ x_{i+1} - x_i - \eta e_0/2 \in \mathbb{V}\ (i < n)\}. \tag{6.cclxxvii}$$

For profiles α_i in the strip $|y^0| < c\eta$ and labels φ_i, abbreviate $B_i = B_{\varphi_i}[\alpha_i; \eta]$ and put

$$\begin{aligned}\Psi_\eta[\vec{\alpha}](\mathbf{x}) = T(x_1 - \eta e_0/4)B_1 T(x_2 - x_1 - \eta e_0/2)B_2 \cdots \\ T(x_n - x_{n-1} - \eta e_0/2)B_n \Omega.\end{aligned} \tag{6.cclxxviii}$$

It is Hilbert-valued holomorphic, and

$$\|\Psi_\eta[\vec{\alpha}](\mathbf{x})\| \leq \prod_{i=1}^{n} C_{\varphi_i} \eta^{-\wp} \|\alpha_i\|_1. \tag{6.cclxxix}$$

The definitions remain valid on the closed domain using strong continuity of T. When all x_i are real and the displayed non-strict time inequalities hold, direct substitution gives

$$\Psi_\eta[\vec{\alpha}](\mathbf{x}) = \Big[\prod_{i=1}^{n} \tau_{x_i}(\alpha_i \otimes \varphi_i) \Big]. \tag{6.cclxxx}$$

For clarity this is an induction from the right. The rightmost operator first puts its plate at time $\eta/4$. The preceding translation moves it by $x_n - x_{n-1} - \eta e_0/2$; the next plate operator translates that entry forward by $\eta e_0/2$ and inserts its own plate at $\eta e_0/4$. The two centers are therefore $\eta e_0/4$ and $\eta e_0/4 + x_n - x_{n-1}$. Iteration and the leftmost translation put all centers at the x_i. Spatial shifts follow the same addition rule. Time gaps and $c < 1/4$ ensure that every tuple in this induction is separated.

Define the smoothed scalar kernel on $U_\eta = (\eta e_0/2 + \mathbb{V})^{n-1}$ by

$$K_{n,\eta}[\vec{\alpha}](v) = \langle \Omega, B_1 T(v_1 - \eta e_0/2)B_2 \cdots T(v_{n-1} - \eta e_0/2)B_n \Omega \rangle. \tag{6.cclxxxi}$$

It is holomorphic and bounded by the right side of eq. 6.cclxxix. If the y_i are real and $y^0_{i+1} - y^0_i \geq \eta/2$, then

$$\sigma\Big((\tau_{y_i}(\alpha_i \otimes \varphi_i))_{i=1}^{n} \Big) = K_{n,\eta}[\vec{\alpha}](v(\mathbf{y})), \qquad v(\mathbf{y})_i = y_{i+1} - y_i. \tag{6.cclxxxii}$$

Translate the first center to $(\eta/4, 0)$ and use eq. 6.cclxxx; taking its scalar product with Ω proves the formula. Translation invariance then restores the original centers. Set $K_0 = 1$ and $K_1 = 0$, consistently with the empty and centered one-point moments.

Removing the plane smoothing. Fix η and choose real radial nonnegative smooth bumps $\beta_{i,j}$, each of mass one, whose support radii tend to zero. For all large j the bumps fit in $D(0, c\eta)$. The kernels $K_{n,\eta}[\vec{\beta}_j]$ have a common bound

$$C_\eta = \prod_i C_{\varphi_i} \eta^{-\wp n} \quad \text{on } U_\eta. \tag{6.cclxxxiii}$$

The bumps need not have the same radii or shapes in different entries. This freedom is needed later for mixed inner products.

Here are the details identifying every limit. Let b_i be C^1 profiles whose successive support time gaps exceed η. For all large j, their convolutions with $\beta_{i,j}$ have separated supports in fixed compact sets. The identity

$$b_i * \beta_{i,j} = \int b_i(y) \tau_y \beta_{i,j}\, dy$$

holds also as a C^1 Riemann integral. Indeed, for fixed j, the integrand and its first derivatives in the test variable are uniformly continuous on a common compact set. Riemann sums converge there in C^1, with their supports in the same compact neighborhood. Multilinearity, the assumed separated C^1 continuity, and successive Riemann summation in all entries give

$$\sigma\big(((b_i * \beta_{i,j}) \otimes \varphi_i)_i\big) = \int \prod_i b_i(y_i)\, K_{n,\eta}[\vec{\beta}_j](v(\mathbf{y}))\, d\mathbf{y}. \tag{6.cclxxxiv}$$

The integrands are continuous and compactly supported; thus Fubini and the successive integrals are legitimate.

A locally bounded family of holomorphic functions has locally uniformly convergent subsequences here. One can see this directly: Cauchy estimates on smaller polydiscs bound first derivatives, Arzelà–Ascoli gives convergence there, and a diagonal choice on a countable exhaustion gives convergence on all compact subsets. The Cauchy integral formula shows that the limit is holomorphic. Apply this observation to eq. 6.cclxxxiii. Along a subsequence let the limit be K'. Since $b_i * \beta_{i,j} \to b_i$ in C^1, passage in eq. 6.cclxxxiv gives

$$\sigma\big((b_i \otimes \varphi_i)_i\big) = \int \prod_i b_i(y_i) K'(v(\mathbf{y}))\, d\mathbf{y}. \tag{6.cclxxxv}$$

All differences in this integral lie in a compact subset of U_η, so locally uniform convergence suffices.

There is at most one holomorphic K' with this property. If two continuous densities give the displayed integrals, choose the b_i to be unit-mass bumps in successively smaller discs around any fixed real configuration with gaps $> \eta$. Their integrals converge to the values of the two densities there. These values coincide. Such differences contain the nonempty relatively open set $\{v \text{ real} : v_i^0 > \eta\}$ in the connected set U_η; the uniqueness principle proves equality on U_η. Consequently the entire smoothing family converges, not merely a subsequence. Otherwise a sequence staying away on a compact set would have a further subsequence with the same uniquely determined limit. Denote it by $K_{n,\eta}$.

The argument proves a useful strengthened conclusion: for any choice of unit-mass bumps in the different entries with maximal support radius tending to zero, the corresponding kernels converge to the same $K_{n,\eta}$, locally uniformly on U_η. No relation between their rates of shrinkage is required.

If $\eta' < \eta$, the two limiting kernels agree on U_η. Both satisfy eq. 6.cclxxxv for gaps $> \eta$, and the same uniqueness proof applies. Since $\bigcup_{\eta>0} U_\eta$ equals $\mathbb{V}^{n-1}$, this constructs a unique holomorphic function

$$K_n(v; \vec{\varphi}) \quad \text{on } \mathbb{V}^{n-1}.$$

It represents σ on every time-ordered family of profiles:

$$\sigma\big((\alpha_i \otimes \varphi_i)_i\big) = \int \prod_i \alpha_i(y_i) K_n(v(\mathbf{y}); \vec{\varphi})\, d\mathbf{y}. \tag{6.cclxxxvi}$$

Choose η below all the support gaps to obtain this formula. It also proves reality at every real point of the tube, by testing against real profiles concentrated near the chosen points.

The bound retains exactly one factor per field:

$$|K_n(v; \vec{\varphi})| \le \Big(\prod_i C_{\varphi_i}\Big) d_{\mathbb{V}}(v)^{-\wp n}. \tag{6.cclxxxvii}$$

Indeed $v \in U_\eta$ whenever $\eta < \min\{1, 2 \min_{i,\pm} \operatorname{Re}(v_i)_\pm\}$. Take the limit up to this upper endpoint in eq. 6.cclxxxiii; that endpoint is at least $d_{\mathbb{V}}(v)$. Neither a derivative of a shrinking bump nor an estimate whose constant depends on outside field counts has entered this step.

Real overlap identities and the four frames. For a quarter turn ν define $K_n^{(\nu)}(v; \vec{\varphi}) = K_n(R_{-\nu} v; \vec{\varphi})$ on $\mathbb{V}_\nu^{n-1}$. Hypercubic invariance and the change of variables $y_i = R_\nu y_i'$ show that eq. 6.cclxxxvi holds with $K_n^{(\nu)}$ whenever the profiles are ordered in the direction e_ν. The rotations act only on profiles and leave the labels unchanged.

It is useful to account for permutations as well. For a permutation π of $\{1, \dots, n\}$ let

$$\mathcal{U}_{\nu,\pi} = \{\mathbf{z} \in (\mathbb{C}^2)^n : R_{-\nu}(z_{\pi(i+1)} - z_{\pi(i)}) \in \mathbb{V} \ (i < n)\}. \tag{6.cclxxxviii}$$

On it use K_n with these ordered differences and the correspondingly permuted labels. Each domain is open, convex, nonempty, invariant under simultaneous complex translations, and satisfies $\operatorname{Re} \mathbf{z} \in \mathcal{U}_{\nu,\pi}$ whenever $\mathbf{z} \in \mathcal{U}_{\nu,\pi}$.

Two of these kernels agree on every nonempty intersection. The intersection is convex and hence connected. It contains the real part of each of its points, so contains a relatively open real ball. At any real configuration in the intersection take small discs about its points which preserve both strict orders. The moment of profiles supported in these discs has both continuous kernel densities, by eq. 6.cclxxxvi, the frame identity and symmetry of σ. Testing with unit-mass profiles shrinking to the points shows equality of the two densities at that configuration. The holomorphic uniqueness principle now gives equality on the whole complex intersection. This proves the overlap identity without assuming arbitrary-angle rotation symmetry.

Thus the ordered kernels glue to a single holomorphic function $\mathbf{K}_n(\mathbf{z};\vec{\varphi})$ on

$$\mathcal{U}_n^{\mathrm{ord}} = \bigcup_{\nu,\pi} \mathcal{U}_{\nu,\pi}. \tag{6.cclxxxix}$$

This union is connected. To check this explicitly, a real configuration can be assigned any prescribed strict time ordering and, independently, any prescribed strict abscissa ordering. Hence each 0-frame permutation domain intersects every $\pi/2$-frame permutation domain. These convex domains form a connected union. The other two frames give the same domains with their orderings reversed, so add no disconnected components. Its real points form

$$\mathrm{Ord}_n = \{\mathbf{y} : \text{all times are distinct or all abscissae are distinct}\}.$$

The function is translation invariant, symmetric in the labelled points, real on real configurations, and invariant under the quarter turns. These properties follow first from the corresponding moment densities on real open sets and then from uniqueness. Imaginary rotations preserve each domain $\mathcal{U}_{\nu,\pi}$ and hence preserve their union.

Point product vectors and their Gram kernel. Put

$$\mathbb{D}_n^+ = \{\mathbf{x} : x_1 \in \mathbb{V},\ x_{i+1} - x_i \in \mathbb{V}\ (i < n)\}. \tag{6.ccxc}$$

This is the union of the nested $\mathbb{D}_{n,\eta}^+$ as η decreases to zero. For $n = 0$ there is one empty configuration and its vector will be Ω.

We prove strong convergence of the smooth product vectors, rather than analytically continuing an inequality. Fix η, choose a real radial unit-mass bump β_ϵ supported in $D(0,\epsilon)$, and take $\epsilon < c\eta$. Use it in every entry of eq. 6.cclxxviii, writing the result as $\Psi_{\eta,\epsilon}(\mathbf{x};\vec{\varphi})$.

If $\mathbf{x} \in \mathbb{D}_{n,\eta}^+$ and $\mathbf{y} \in \mathbb{D}_{m,\eta}^+$, define the list of differences

$$J(\mathbf{x},\mathbf{y}) = \big((x_n - x_{n-1})^*, \dots, (x_2 - x_1)^*, x_1^* + y_1, y_2 - y_1, \dots, y_m - y_{m-1}\big). \tag{6.ccxci}$$

If $n = 0$, omit the left list and the central entry. If $m = 0$, omit the central entry and the right list. The labels of this list are $(\varphi_n, \dots, \varphi_1, \psi_1, \dots, \psi_m)$. Every entry belongs to $\eta e_0/2 + \mathbb{V}$: for the central one subtracting $\eta e_0/2$ gives the sum $(x_1 - \eta e_0/4)^* + (y_1 - \eta e_0/4)$ in the open cone. The other entries follow from star invariance.

For different radii ϵ, ϵ' the exact identity is

$$\begin{aligned}&\langle \Psi_{\eta,\epsilon}(\mathbf{x};\vec{\varphi}), \Psi_{\eta,\epsilon'}(\mathbf{y};\vec{\psi})\rangle \\ &\quad = K_{n+m,\eta}[\underbrace{\beta_\epsilon, \dots, \beta_\epsilon}_{n}, \underbrace{\beta_{\epsilon'}, \dots, \beta_{\epsilon'}}_{m}](J(\mathbf{x},\mathbf{y})),\end{aligned} \tag{6.ccxcii}$$

with the reversed left labels specified above. For real positions this is just eq. 6.cclxxx, the Hilbert inner product, symmetry, and eq. 6.cclxxxii. Radiality and realness are important: $\theta\overline{\tau_x\beta_\epsilon} = \tau_{\theta x}\beta_\epsilon$ for real x. The left side is antiholomorphic in $\mathbf{x}$ and holomorphic in $\mathbf{y}$; the right side has exactly the same properties because the star is conjugate linear. The product domains are convex, conjugation stable and have real interior points. Mixed holomorphic uniqueness therefore proves eq. 6.ccxcii for complex positions too. For total length one both sides are zero; for length zero both equal one.

The strengthened smoothing conclusion after eq. 6.cclxxxv shows that the right side converges to $K_{n+m}(J(\mathbf{x},\mathbf{y}))$ when both radii tend to zero, with no restriction on their relative rate. Convergence

is uniform on compact sets of the position domains. In particular, for a compact subset of $\mathbb{D}^+_{n,\eta}$, expand

$$\|\Psi_{\eta,\epsilon}(\mathbf{x}) - \Psi_{\eta,\epsilon'}(\mathbf{x})\|^2$$

as its four mixed inner products. All four tend uniformly to the same diagonal kernel value; their signed sum tends to zero. This proves a locally uniform strong Cauchy property. The limit $\Psi_\eta(\mathbf{x};\vec{\varphi})$ is Hilbert-valued holomorphic: uniform convergence on smaller polydiscs passes the vector Cauchy formula and its power series to the limit.

These vectors are independent of η on overlaps. For any fixed sufficiently small physical bump and real positions in an overlap, both definitions give the same class in eq. 6.cclxxx. Vector holomorphic uniqueness on the connected overlap proves equality for complex positions, and then the strong limits agree. We have therefore constructed

$$\Psi(\mathbf{x};\vec{\varphi}) \in \mathcal{H}, \qquad \mathbf{x} \in \mathbb{D}^+_n, \tag{6.ccxciii}$$

holomorphically, with the bound $\|\Psi(\mathbf{x};\vec{\varphi})\| \le \prod_i C_{\varphi_i}\eta^{-\wp}$ for every admissible η. On the domain where $v \in \overline{\mathbb{V}}$, the semigroup rule also gives

$$\Psi(\mathbf{x}+v;\vec{\varphi}) = T(v)\Psi(\mathbf{x};\vec{\varphi}). \tag{6.ccxciv}$$

It holds before removing the bumps and hence afterward.

For labelled positive configurations define

$$\mathrm{Gr}(\mathsf{x},\mathsf{y}) = K_{n+m}(J(\mathbf{x},\mathbf{y});\varphi_n,\dots,\varphi_1,\vec{\psi}).$$

Use the empty and one-point conventions already specified. The construction proves the exact equality

$$\mathrm{Gr}(\mathsf{x},\mathsf{y}) = \langle\Psi(\mathsf{x}),\Psi(\mathsf{y})\rangle. \tag{6.ccxcv}$$

Thus this is a positive semidefinite kernel at complex positions, for arbitrary finite families and different lengths, including the empty configuration.

Finally $R_{is}\mathbb{D}^+_n = \mathbb{D}^+_n$, and eq. 6.cclxxv gives $J(R_{is}\mathbf{x},R_{is}\mathbf{y}) = R_{is}J(\mathbf{x},\mathbf{y})$. Consequently

$$\begin{aligned}\mathrm{Gr}_s(\mathsf{x},\mathsf{y}) &:= K_{n+m}(R_{is}J(\mathbf{x},\mathbf{y});\text{the same labels})\\ &= \langle\Psi(R_{is}\mathsf{x}),\Psi(R_{is}\mathsf{y})\rangle\end{aligned} \tag{6.ccxcvi}$$

is positive semidefinite for every real s. This is an actual Gram representation; positivity has not been inferred from analytic continuation of a positive scalar function.

Narrow pencils and the entire angle function. Fix $0 < \phi_0 < \pi/4$ and let

$$\mathrm{Sec} = \{v \in \mathbb{R}^2 \setminus \{0\} : |\arg(v^0 + iv^1)| < \phi_0\}.$$

For $n \ge 2$, a pencil is a configuration with every consecutive difference in this sector. The cases $n = 0,1$ have their separate constant kernels fixed above. Write $r_{\mathbf{y}} = \min_i |y_{i+1} - y_i| > 0$. For a pencil with differences v_i and labels $\vec{\varphi}$ put, on the vertical strip

$$S_\nu = \{z = t + is : |t - \nu| < \pi/2 - \phi_0\}, \qquad \nu \in (\pi/2)\mathbb{Z},$$

the function

$$a^{(\nu)}_{\mathbf{y}}(z) = K_n(R_{z-\nu}v;\vec{\varphi}). \tag{6.ccxcvii}$$

It is well defined and holomorphic. In fact, writing $(v_i)_\pm = |v_i|e^{\pm i\phi_i}$ with $|\phi_i| < \phi_0$ gives

$$\mathrm{Re}(R_{t+is-\nu}v_i)_\pm = |v_i|e^{\mp s}\cos(\phi_i + t - \nu) > 0. \tag{6.ccxcviii}$$

Only neighboring strips intersect. Their intersection is exactly

$$S_\nu \cap S_{\nu+\pi/2} = \{t + is : \nu + \phi_0 < t < \nu + \pi/2 - \phi_0\},$$

which is nonempty and connected. At a real angle there, all the rotated differences lie strictly in the quadrant between the two frame normals. Small discs about the rotated points are ordered in both frames. The proved real overlap identity therefore identifies the two values in eq. 6.ccxcvii on this

real interval. The one-variable identity theorem identifies the holomorphic functions throughout the intersection.

The strip width $\pi - 2\phi_0$ exceeds their spacing $\pi/2$. They cover the complex plane and glue to a unique entire function $a_{\mathbf{y}}(z;\vec{\varphi})$. The construction immediately gives

$$a_{\mathbf{y}}(z + \pi/2) = a_{\mathbf{y}}(z). \tag{6.ccxcix}$$

Indeed use the patch with center $\nu + \pi/2$ for the shifted argument and the patch with center ν for the original argument; the arguments of K_n are identical. For real t it equals $\mathbf{K}_n(R_t\mathbf{y};\vec{\varphi})$, since a suitable frame orders this rotated pencil. In particular it is real on the real axis, and $a_{\mathbf{y}}(is) = K_n(R_{is}v;\vec{\varphi})$.

For every real t choose a nearest quarter turn ν, so $|t - \nu| \le \pi/4$. Set $c_0 = \cos(\phi_0 + \pi/4) > 0$. By eq. 6.ccxcviii, every light-coordinate margin is at least $c_0 r_{\mathbf{y}} e^{-|s|}$. Hence

$$|a_{\mathbf{y}}(t + is)| \le \Big(\prod_i C_{\varphi_i}\Big) \min\{1, c_0 r_{\mathbf{y}} e^{-|s|}\}^{-\wp n} \le A_{\mathbf{y},\vec{\varphi}} e^{\wp n|s|}, \tag{6.ccc}$$

where $A_{\mathbf{y},\vec{\varphi}} = \prod_i C_{\varphi_i} \max\{1, (c_0 r_{\mathbf{y}})^{-\wp n}\}$. This proves the claimed exponential type with its linear dependence on n; an uncontrolled prefactor depending on n would not change that conclusion.

The finite band and analytic harmonic coefficients. Define

$$c_m(\mathbf{y};\vec{\varphi}) = \frac{2}{\pi} \int_0^{\pi/2} a_{\mathbf{y}}(t;\vec{\varphi}) e^{-4imt}\, dt. \tag{6.ccci}$$

Cauchy's theorem on a rectangle of height s and periodicity give

$$c_m = \frac{2}{\pi} \int_0^{\pi/2} a_{\mathbf{y}}(t + is) e^{-4im(t+is)}\, dt, \qquad |c_m| \le A_{\mathbf{y},\vec{\varphi}} e^{4ms + \wp n|s|}.$$

The vertical sides cancel exactly. Sending s to $-\infty$ for $4m > \wp n$, and to $+\infty$ for $4m < -\wp n$, proves

$$c_m = 0 \quad \text{if } 4|m| > \wp n, \qquad a_{\mathbf{y}}(z) = \sum_{4|m| \le \wp n} c_m e^{4imz}. \tag{6.cccii}$$

For the equality, periodicity makes $a(z)$ a single-valued holomorphic function of $w = e^{4iz}$ on $\mathbb{C}\setminus\{0\}$. Its Laurent coefficients are the displayed c_m, so its Laurent series is the finite sum after the vanishing just proved. Equivalently one may use the uniformly convergent Fourier series of the smooth periodic function on the real axis and then the identity theorem. Reality implies $c_{-m} = \overline{c_m}$.

We next construct coefficients beyond pencils, without assuming continuity across different ordering charts. Put $N = \lfloor \wp n/4 \rfloor$ and on planar configurations introduce the angular vector field

$$\mathcal{L} = \sum_{i=1}^{n} (-y_i^1 \partial_{y_i^0} + y_i^0 \partial_{y_i^1}).$$

On translation-invariant functions it is the same angular vector field in the consecutive-difference coordinates. Let

$$P(X) = \prod_{m=-N}^{N} (X - 4im), \qquad p_m(X) = \prod_{\substack{-N \le j \le N \\ j \ne m}} \frac{X - 4ij}{4im - 4ij}.$$

The finite-band identity gives $P(\mathcal{L})\mathbf{K}_n = 0$ on the relatively open real set of pencils. The left side is holomorphic on the connected domain $\mathcal{U}_n^{\mathrm{ord}}$. Uniqueness therefore proves this equation everywhere on that domain. Define there

$$c_m = p_m(\mathcal{L})\mathbf{K}_n \quad (|m| \le N). \tag{6.ccciii}$$

On pencils these are the Fourier coefficients in eq. 6.ccci: each polynomial p_m selects its corresponding distinct exponential. The polynomial identities $\sum_m p_m = 1$ and $(X - 4im)p_m(X) =$ constant $P(X)$ give

$$\sum_m c_m = \mathbf{K}_n, \qquad \mathcal{L}c_m = 4imc_m. \tag{6.ccciv}$$

The coefficients are holomorphic on the ordered domain, symmetric in labelled points and invariant under simultaneous translations. On its real points $c_{-m} = \overline{c_m}$: the angular vector field has real coefficients, the density is real there, and $\overline{p_m(X)} = p_{-m}(X)$ for real X. For translation invariance, note that applying $\mathcal{L}$ to a translation-invariant function again gives one: the extra term under a common shift is a linear combination of the total derivatives $\sum_i \partial_{y_i^\mu}$, which vanish.

Since imaginary rotations preserve the domain, the last equation in eq. 6.ccciv can be integrated along them:

$$c_m(R_{is}\mathbf{z}) = e^{-4ms}c_m(\mathbf{z}), \qquad \mathbf{K}_n(R_{is}\mathbf{z}) = \sum_m e^{-4ms}c_m(\mathbf{z}). \tag{6.cccv}$$

The same equation gives $c_m(R_t\mathbf{y}) = e^{4imt}c_m(\mathbf{y})$ on any real interval whose rotated configurations remain in Ord_n.

These coefficients extend uniquely and real-analytically to every configuration of distinct real planar points. Here is the full gluing argument. Let $\mathcal{G}_n$ consist of configurations with distinct points and with no two nonzero difference vectors perpendicular. This is open and dense: its complement, in addition to the collisions, is a finite union of zero sets of nonzero real polynomials. Each polynomial is nonzero, as can also be checked by taking distinct points on one line. Every real rotation of a configuration in $\mathcal{G}_n$ is ordinary. Failure of ordinariness would produce a time tie and an abscissa tie, whose nonzero difference vectors are perpendicular. Thus the preceding angular ordinary differential equation proves covariance for every real angle on $\mathcal{G}_n$.

For an arbitrary distinct configuration $\mathbf{y}$, choose an angle t at which all its rotated times are distinct. Only finitely many angles are forbidden. In a neighborhood of $\mathbf{y}$ define the extension by

$$\widetilde{c}_m(\mathbf{u}) = e^{-4imt}c_m(R_t\mathbf{u}). \tag{6.cccvi}$$

This is real-analytic. Two choices agree on the dense set $\mathcal{G}_n$ in every overlap, by the covariance already proved there; continuity makes them agree everywhere in the overlap. They consequently glue, agree with the original coefficients on ordinary configurations, and obey exact covariance

$$\widetilde{c}_m(R_t\mathbf{y}) = e^{4imt}\widetilde{c}_m(\mathbf{y}) \quad (t \in \mathbb{R}). \tag{6.cccvii}$$

This last equality, permutation symmetry, translation invariance and $\widetilde{c}_{-m} = \overline{\widetilde{c}_m}$ follow by continuity from $\mathcal{G}_n$. Uniqueness follows because Ord_n is dense in the collision-free set. The collision-free set itself is connected: one may move its points successively along polygonal paths avoiding the finitely many positions occupied by the other points, with small preliminary moves if necessary. Removing finitely many points does not disconnect the plane.

Set $\overline{m}(0) = \overline{m}(1) = 0$; for $n \geq 2$ let $\overline{m}(n)$ be the largest $|m|$ occurring for any pencil and any real labels, or zero if none occurs. It is a finite integer and

$$\overline{m}(n) \leq \wp n/4. \tag{6.cccviii}$$

Coefficients with $|m| > \overline{m}(n)$ vanish identically on the ordered domain by uniqueness, and then on every collision-free configuration by eq. 6.cccvi. If $\overline{m}(n) > 0$, some pencil and labels have $c_{-\overline{m}(n)} \neq 0$, by reality.

Boost positivity without time ordering and the top kernel. For arbitrary labelled configurations of k and l distinct points in the real open half-plane, define

$$\widetilde{\mathrm{Gr}}_s(\mathsf{x},\mathsf{y}) = \sum_m e^{-4ms}\widetilde{c}_m^{(k+l)}(\theta\mathbf{x},\mathbf{y};\vec{\varphi},\vec{\psi}). \tag{6.cccix}$$

All points in this combined configuration are distinct: the two groups lie on opposite sides of the mirror. Empty tuples have the conventions $\widetilde{c}_m^{(0)} = \delta_{m0}$ and $\widetilde{c}_m^{(1)} = 0$.

This kernel is positive semidefinite for every real s. First suppose each individual configuration has distinct times. Sort those times. The combined reflected configuration is then ordinary, and eq. 6.cccv, symmetry and eq. 6.ccxcvi identify eq. 6.cccix with the proved Gram kernel. For a finite family with time ties, perturb the positions slightly inside the positive half-plane to remove all such ties. The reflected combined configurations remain collision free. Every coefficient is continuous there, so their finite matrices converge entrywise to the desired matrix. Positive semidefiniteness is closed under this limit. This proves the assertion with time ties, arbitrary lengths and the empty configuration included.

In particular, fix $k \geq 1$, put $m_* = \overline{m}(2k)$, and set $c_* = \widetilde{c}_{-m_*}^{(2k)}$. On k-point configurations the kernel

$$(\mathsf{x}, \mathsf{y}) \longmapsto c_*(\theta\mathbf{x}, \mathbf{y}; \vec{\varphi}, \vec{\psi}) \tag{6.cccx}$$

is positive semidefinite: multiply the matrix in eq. 6.cccix by $e^{-4m_* s}$ and send $s \to +\infty$. All the other finitely many terms tend to zero.

There is also an exact formulation across an arbitrary line $\mathfrak{l}$. Let $b \in \mathfrak{l}$ and let $R_\nu e_0$ be its normal pointing into a chosen side $\mathfrak{l}_+$. The map $r(y) = R_{-\nu}(y - b)$ sends that side to $\mathbb{H}$ and intertwines its reflection $\theta_\mathfrak{l}$ with θ. Using translation invariance and eq. 6.cccvii, eq. 6.cccx becomes positivity of

$$(\mathsf{x}, \mathsf{y}) \longmapsto e^{4im_*\nu} c_*(\theta_\mathfrak{l}\mathbf{x}, \mathbf{y}; \vec{\varphi}, \vec{\psi}) \tag{6.cccxi}$$

on k-point configurations in $\mathfrak{l}_+$. The compensating unit phase is essential. This is a statement about the top coefficient, and does not assume reflection positivity of the original functional at every line.

The cut inequality, including the empty tuple. The preceding constructions imply

$$2\overline{m}(k + l) \leq \overline{m}(2k) + \overline{m}(2l) \qquad (k, l \geq 0). \tag{6.cccxii}$$

If the left top frequency is zero there is nothing to prove. Otherwise choose a pencil and labels whose negative top coefficient is nonzero. When $k, l \geq 1$, translate the midpoint between y_k, y_{k+1} to the origin, and put

$$x_i = \theta y_{k+1-i} \ (1 \leq i \leq k), \qquad z_j = y_{k+j} \ (1 \leq j \leq l),$$

with the correspondingly reversed left labels. Both are positive configurations. Their first points and all consecutive differences are in Sec: reflection followed by a minus sign changes the sign of a real spatial coordinate and preserves this sector. The first points are the two reflected halves of the cut difference.

The combined reflected configuration for x, z is exactly the original pencil. The doubled configurations for x, x and z, z are also pencils. Their central differences are respectively $2x_1^0 e_0$ and $2z_1^0 e_0$, and all other differences are sector vectors or their stars. The Gram Cauchy–Schwarz inequality gives

$$|a_{\mathbf{y}}(is)|^2 \leq a_{(\theta x_k, \ldots, \theta x_1, x_1, \ldots, x_k)}(is) a_{(\theta z_l, \ldots, \theta z_1, z_1, \ldots, z_l)}(is).$$

The left side has leading rate $e^{8\overline{m}(k+l)s}$ as $s \to +\infty$, with nonzero coefficient. Each right factor has rate at most $e^{4\overline{m}(2k)s}$ or $e^{4\overline{m}(2l)s}$ by its finite band. Comparison proves eq. 6.cccxii.

If $k = 0$, translate a nonzero pencil so that its first point is in Sec. Apply Gram Cauchy–Schwarz to the empty vector Ω and its product vector. The empty diagonal entry is one, so the same comparison gives $2\overline{m}(l) \leq \overline{m}(2l)$. This is exactly the required inequality with $\overline{m}(0) = 0$. The case $l = 0$ follows by symmetry, and the centered one-point and fully empty cases are immediate. Thus the initial convexity information has not been lost by omitting the vacuum vector.

The two-point interface and its positive plane measure. For two points, Ord_2 is the entire real collision-free set: two distinct points cannot have both coordinates equal. Translation invariance therefore gives a real-analytic function $G_{\varphi,\psi}(r)$ for every real $r \neq 0$, equal to $K_2(r;\varphi,\psi)$ when $r^0 > 0$. Its global coefficients obey

$$\begin{aligned} G_{\varphi,\psi}(r) &= \sum_m \widetilde{c}_m(r;\varphi,\psi), \\ \widetilde{c}_m(R_t r) &= e^{4imt}\widetilde{c}_m(r), \\ K_2(R_{is} r) &= \sum_m e^{-4ms}\widetilde{c}_m(r) \qquad (r^0 > 0). \end{aligned} \tag{6.cccxiii}$$

The last identity holds for every real s, not merely near zero; it is eq. 6.cccv. Every sum is finite, with $|m| \le \overline{m}(2) \le \wp/2$. The kernel in eq. 6.cccix for one-point configurations is positive semidefinite for every s. A quarter-turn frame with projection at least $|r|/\sqrt{2}$ also gives the useful bound

$$|G_{\varphi,\psi}(r)| \le C_\varphi C_\psi \min\{1, |r|/\sqrt{2}\}^{-2\wp}. \tag{6.cccxiv}$$

The density formula for two points also holds for arbitrary disjoint compact plane-profile supports, even if those whole supports have no strict coordinate ordering. Let their distance be $d > 0$ and partition each profile into finitely many pieces supported in sets of diameter less than $d/(10\sqrt{2})$. For each pair of pieces, two chosen points in their supports differ in at least one coordinate by $d/\sqrt{2}$ or more. The small diameters then make the two entire piece supports strictly ordered in that coordinate, in one of the four frames. Apply eq. 6.cclxxxvi and the frame identity to this pair. Summing the finitely many identities by bilinearity proves

$$\sigma(\alpha\otimes\varphi, \beta\otimes\psi) = \iint \alpha(y)\beta(y')G_{\varphi,\psi}(y'-y)\,dy\,dy'$$

for every such pair of profiles. All partitions can be smooth; their products with the original C^1 profiles remain C^1 and have supports in the original separated sets.

There is a positive spectral plane measure for equal real labels. Let $\zeta_s = \Psi((s,0);\varphi)$, $s > 0$. By eq. 6.ccxciv, $\zeta_{s'} = e^{-(s'-s)H}\zeta_s$ if $s' > s$. Let $\mu_s(B) = \langle \zeta_s, \mathsf{P}(B)\zeta_s\rangle$, using the joint spectral measure of (H,P). The locally finite positive measure

$$d\mu_\varphi(E,q) = e^{2sE}\,d\mu_s(E,q) \tag{6.cccxv}$$

is independent of s, because the preceding vector identity multiplies μ_s by $e^{-2(s'-s)E}$. It is carried by $E \ge |q|$ and satisfies

$$K_2(v;\varphi,\varphi) = \int e^{-v^0E + iv^1q}\,d\mu_\varphi(E,q) \qquad (v \in \mathbb{V}). \tag{6.cccxvi}$$

For real $v = (t,x)$ choose $0 < s < t/2$ and use eq. 6.ccxcv and eq. 6.ccxciv: the left side is $\langle \zeta_s, T(t-2s,x)\zeta_s\rangle$, which is the displayed integral. In particular $\int e^{-tE}\,d\mu_\varphi < \infty$ for every $t > 0$. On a compact sub-tube the multiplier is bounded by e^{-dE} for some $d > 0$. Its derivatives are controlled by a smaller exponential weight because $|q| \le E$. The integral is therefore holomorphic there, and uniqueness extends its equality from real points to all of $\mathbb{V}$.

No uniqueness of the vacuum or absence of a zero-energy mass is asserted here. A positive plane measure in a diagonal-reflection picture requires the reflection-positivity input for that picture; it is not inferred from the coordinate cone. Likewise, eliminating the nonzero two-point harmonics requires a separate ultraviolet estimate and the diagonal-frame argument. The polygon and integer arguments can use eq. 6.cccxi and eq. 6.cccxii, but are not contained in this section. Finally, the analytic extension of coefficients to distinct planar points does not identify the original moment functional on every exceptional support arrangement. That distributional passage, in particular arrangements with coincident planar projections and separated transverse supports, requires a separate proof.

6.12. **Eliminating two-point angular harmonics.** This section proves an implication for the two-point function. Its inputs are a continuous planar kernel with its axis-frame spectral realization, the exact cone in that frame, one diagonal reflection, and a uniform short-distance bound on a specified class of positive shapes. It constructs the diagonal spectral measure and proves its cone support; a diagonal cone and continuous rotation covariance are not assumed. The angular continuation needed at order two is proved directly. In particular this argument does not require an all-orders highest-harmonic theorem.

The kernel and Hilbert realization are analytic inputs to this section. They are supplied by the planar-kernel construction when its insertion, continuity and reconstruction hypotheses hold. This section does not prove the interacting source estimates or the short-distance input used below.

6.12.1. *The precise two-point interface.* Write a point of $\mathbb{R}^4$ as (y,z), with $y=(y^0,y^1)=(x_4,x_1)$ and $z=(x_2,x_3)$. On the plane let $e_0=(1,0)$, let R_t be the rotation

$$R_t(u,v)=(u\cos t-v\sin t,u\sin t+v\cos t),\qquad \theta(u,v)=(-u,v).$$

The same notation denotes these maps on $\mathbb{R}^4$, leaving z fixed. Put $n=R_{\pi/4}e_0$. The reflection at $n^\perp$ is $\theta_n=R_{\pi/4}\theta R_{-\pi/4}$.

Let S_2 be a real symmetric bilinear form on pairs of real $C_c^1(\mathbb{R}^4)$ tests with disjoint supports, complexified bilinearly. It is translation- and W_4-invariant and satisfies coordinate reflection positivity. It also satisfies

$$S_2(\theta_n\overline{f},f)\geq 0\quad (f\in C_c^1(\{n\cdot y>0\};\mathbb{C})). \tag{6.cccxvii}$$

Positivity at any one diagonal normal in four dimensions implies this choice by hypercubic conjugation. All scalar products below are conjugate linear in the first variable.

The kernel interface is the following explicit assumption. For real $\varphi,\psi\in C_c^1(\mathbb{R}^2)$ there are continuous functions $G_{\varphi,\psi}$ on $\mathbb{R}^2\setminus\{0\}$ such that

$$S_2(\alpha\otimes\varphi,\beta\otimes\psi)=\iint\alpha(y)\beta(y')G_{\varphi,\psi}(y'-y)\,dy\,dy' \tag{6.cccxviii}$$

whenever the compact supports of $\alpha,\beta\in C_c^1(\mathbb{R}^2;\mathbb{C})$ are disjoint. There is a Hilbert space with strongly commuting self-adjoint operators H,P, with $H\geq|P|$, and vectors $\zeta_{\varphi,a}$, $a>0$, such that

$$\begin{aligned}\zeta_{\varphi,b}&=e^{-(b-a)H}\zeta_{\varphi,a}\quad (b>a),\\ G_{\varphi,\psi}(t,x)&=\langle\zeta_{\varphi,a},e^{-(t-2a)H+ixP}\zeta_{\psi,a}\rangle\quad (t>2a).\end{aligned} \tag{6.cccxix}$$

This is a point-kernel interface, not a bound on unsmeared field operators. At order two every distinct planar pair is in an ordinary coordinate frame. Consequently the two-point specialization of the planar kernel theorem supplies eq. 6.cccxviii–eq. 6.cccxix without an exceptional planar set.

When S_2 is the connected function of a functional with unit vacuum vector Ω, apply the interface to $\zeta_{\varphi,a}-\langle\Omega,\zeta_{\varphi,a}\rangle\Omega$. The vacuum is translation invariant, so its orthogonal projection commutes with H,P; this subtraction preserves eq. 6.cccxix, positivity and the cone. Thus the interface here concerns the connected kernel when the ultraviolet hypothesis concerns the connected function. No uniqueness of the vacuum is required.

Kernel uniqueness follows by testing eq. 6.cccxviii with small nonnegative approximate identities at two distinct points. The same test shows that the kernels inherit reality, bilinearity in the labels, and all symmetries of S_2 that preserve the plane. In particular, for $G_\varphi=G_{\varphi,\varphi}$,

$$G_\varphi(R_{\pi/2}y)=G_\varphi(y),\qquad G_\varphi(y^0,-y^1)=G_\varphi(y),\qquad G_\varphi(y)\in\mathbb{R}. \tag{6.cccxx}$$

6.12.2. *The axis measure and the plate norm.* Fix a real label φ. Let P be the joint spectral measure of (H, P) and set

$$\mu_{\varphi,a}(B) = \langle \zeta_{\varphi,a}, \mathsf{P}(B)\zeta_{\varphi,a}\rangle, \qquad d\mu_{\varphi}(E,q) = e^{2aE}\, d\mu_{\varphi,a}(E,q). \tag{6.cccxxi}$$

The first measure is finite and positive. eq. 6.cccxix gives $d\mu_{\varphi,b} = e^{-2(b-a)E} d\mu_{\varphi,a}$, so the second measure is independent of a. It is positive and carried by

$$\mathcal{C} = \{(E,q) \in \mathbb{R}^2 : E \geq |q|\}.$$

For every $t > 0$ choose $a < t/2$ in eq. 6.cccxix. Then

$$G_\varphi(t,x) = \int e^{-tE+ixq}\, d\mu_\varphi(E,q), \qquad \Gamma_\varphi(t) = \int e^{-tE}\, d\mu_\varphi = G_\varphi(t,0) < \infty. \tag{6.cccxxii}$$

The measure is locally finite and σ-finite: on $\{E \leq N\}$ its mass is at most $e^{tN}\Gamma_\varphi(t)$. The function Γ_φ is nonnegative and nonincreasing. By a quarter turn, any $y \neq 0$ has a positive first coordinate at least $|y|/\sqrt{2}$. Therefore

$$|G_\varphi(y)| \leq \Gamma_\varphi(|y|/\sqrt{2}). \tag{6.cccxxiii}$$

This bound will give the diagonal reconstruction its bounded time orbits. It uses no decay at large distance. Define also

$$\mathbb{V} = \{v \in \mathbb{C}^2 : \operatorname{Re} v^0 > |\operatorname{Im} v^1|\}, \qquad K_\varphi(v) = \int e^{-v^0 E + iv^1 q}\, d\mu_\varphi. \tag{6.cccxxiv}$$

On a compact subset of this tube the absolute multiplier is at most e^{-cE} for some $c > 0$. Its derivatives are dominated by a smaller exponential because $|q| \leq E$. Thus K_φ is holomorphic and agrees with G_φ on the real positive-time half-plane.

On positive-time tests put $\|[f]\|^2 = S_2(\theta\overline{f}, f)$; reflection positivity and polarization give the associated Hilbert seminorm and its triangle inequality. For α supported in $\{y^0 > 0\}$, eq. 6.cccxviii and eq. 6.cccxxii give

$$\|[\alpha \otimes \varphi]\|^2 = \int \left| \iint \alpha(t,x) e^{-tE+ixq}\, dt\, dx \right|^2 d\mu_\varphi(E,q). \tag{6.cccxxv}$$

Indeed, if $t \geq \delta > 0$ on the support, the absolute integral before interchanging the variables is bounded by $\|\alpha\|_{L^1}^2 \Gamma_\varphi(2\delta)$. This proves Fubini's applicability, including when the unweighted measure has infinite mass.

6.12.3. *The ultraviolet class and fixed shapes.* For $d > 0$ and $\vartheta > 0$, let $\mathfrak{T}(d;\vartheta)$ consist of pairs of nonzero, nonnegative $C_c^1(\mathbb{R}^4)$ tests f, h, each supported in a ball of radius d, satisfying

$$\begin{aligned} &d \leq \operatorname{dist}(\operatorname{supp} f, \operatorname{supp} h) \leq 6d, \\ &\int f \geq \vartheta \|f\|_\infty d^4, \qquad \operatorname{Lip} f \leq \|f\|_\infty/(\vartheta d), \end{aligned} \tag{6.cccxxvi}$$

and the analogous conditions for h. The balls may be placed anywhere. The ultraviolet input is a finite nonnegative function u, with $u(d) \to 0$ as $d \downarrow 0$, such that for $0 < d \leq d_0$,

$$0 \leq S_2(f,h) \leq u(d)\|f\|_\infty \|h\|_\infty \quad ((f,h) \in \mathfrak{T}(d;\vartheta_*)). \tag{6.cccxxvii}$$

Here the universal $\vartheta_* > 0$ is chosen below, before S_2. A hypothesis at a smaller shape parameter also suffices. A hypothesis only on a narrower positive-shape class must be checked against this choice; that inclusion is part of the ultraviolet premise.

The following explicit shapes fix all regularity constants. Choose $a \in C_c^1((2,3))$, between zero and one, equal to one on $[9/4, 11/4]$, zero outside $[17/8, 23/8]$, with cubic smoothstep ramps on the two intervening intervals. Then $\int a = 5/8$ and $\|a'\|_\infty \leq 12$. Let $I = [-1/4, 1/4]$, $\tau = \mathbf{1}_I * \mathbf{1}_I$, and

$$b = 12(\tau * \tau), \qquad c_b = \int b = 3/4. \tag{6.cccxxviii}$$

This is a nonnegative C_c^2 spline, supported in $[-1,1]$, with maximum one and $\|b'\|_\infty = 2$. Explicitly it is $1-6s^2+6|s|^3$ for $|s| \le 1/2$, and $2(1-|s|)^3$ for $1/2 \le |s| \le 1$. With the Fourier convention $\widehat{b}(q) = \int b(s)e^{isq}\,ds$, convolution gives

$$\widehat{b}(q) = c_b \left(\frac{\sin(q/4)}{q/4} \right)^4, \tag{6.cccxxix}$$

where the quotient has value one at zero.

For completeness a transverse partition can use $q(s) = b(4s/3)$, $D(s) = \sum_{j\in\mathbb{Z}} q(s-j)$ and $\chi(s) = q(s)/D(s)$. There are at most two nonzero summands, $D \ge 2/27$ by choosing a nearest integer, and $\sum_j \chi(s-j) = 1$. The bounds $\|q'\|_\infty \le 8/3$ and $\|D'\|_\infty \le 16/3$ give $\|\chi'\|_\infty \le 1008$. Put $\psi(z) = \chi(z_1)\chi(z_2)$. It takes values in $[0,1]$, has support in $[-3/4, 3/4]^2$, and its integer translates sum to one. Let $\widetilde{\psi}(z) = \rho(z_1)\rho(z_2)$, where $\rho = 1$ on $[-1,1]$, $\rho = 0$ outside $[-3/2, 3/2]$, and on each intervening interval use the cubic smoothstep. Thus

$$\int \widetilde{\psi} = 25/4, \quad \|\nabla\widetilde{\psi}\|_\infty \le 6, \quad \|\nabla\psi\|_\infty \le 2016, \quad \psi \le \widetilde{\psi}.$$

All these functions are globally C^1. The numerical choice

$$\vartheta_* = 1/12000 \tag{6.cccxxx}$$

will suffice for all tests constructed next.

6.12.4. *Dilation removes the spline zeros.* For $h > 0$ and $1 \le \lambda \le 2$ put

$$\alpha_h^\lambda(t,x) = a(t/h)b(4x/(\lambda h)), \qquad f_h^\lambda = \alpha_h^\lambda \otimes \varphi.$$

The time transform and spatial transform are

$$A_h(E) = \int a(t/h)e^{-tE}\,dt, \qquad B_{h,\lambda}(q) = \frac{c_b \lambda h}{4} \left(\frac{\sin(\lambda h q/16)}{\lambda h q/16} \right)^4.$$

For $E \ge 0$, $A_h(E) \ge (h/2)e^{-3hE}$. There is an absolute $c > 0$ such that

$$\int_1^2 |B_{h,\lambda}(q)|^2\,d\lambda \ge ch^2(1+h^2q^2)^{-4}. \tag{6.cccxxxi}$$

Here is a proof that includes every zero of the transform. Set $x = h|q|/16$. For $0 \le x \le 1$, the sinc in the display is bounded below on $0 \le \lambda x \le 2$, giving a fixed positive lower bound after division by h^2. For $x \ge 1$, that divided integral is at least

$$cx^{-8} \int_1^2 \sin^8(\lambda x)\,d\lambda = cx^{-9} \int_x^{2x} \sin^8 v\,dv.$$

The number $c_1 = \min_{r\in[0,\pi]} \int_r^{r+1} \sin^8 v\,dv$ is positive: the integral is continuous, periodic and never zero. An interval of length $x \ge 1$ contains $\lfloor x \rfloor \ge x/2$ successive intervals of length one. The last display is therefore at least cx^{-8}. Both regimes imply eq. 6.cccxxxi, with constants independent of h, q.

The finite number $C_* = \sup_{r\ge 0} e^{-r}(1+r^2)^4$ and $|q| \le E$ imply

$$e^{-6hE}(1+h^2q^2)^{-4} \ge C_*^{-1}e^{-7hE}.$$

Using eq. 6.cccxxv, eq. 6.cccxxxi and nonnegative Fubini yields

$$\int_1^2 \|[f_h^\lambda]\|^2\,d\lambda \ge ch^4\Gamma_\varphi(7h). \tag{6.cccxxxii}$$

The cone is essential here. Without it, arbitrarily large $|q|$ at small E could make the spline factor too small.

6.12.5. *Positive pedestals put the plate in the ultraviolet class.* Assume $\varphi \neq 0$, choose $r_1 > 0$ with $\operatorname{supp}\varphi \subset B(0, r_1)$, and take $0 < h \leq \min(1, r_1, 2d_0/5)$. Put $A = \|\varphi\|_\infty + h\|\nabla\varphi\|_\infty > 0$ and

$$\psi_i(z) = \psi(z/h - i), \quad \widetilde{\psi}_i(z) = \widetilde{\psi}(z/h - i), \quad \varphi_i^- = 2A\widetilde{\psi}_i, \quad \varphi_i^+ = 2A\widetilde{\psi}_i + \psi_i\varphi.$$

Only indices in $\mathcal{I} = \{i : \psi_i\varphi \not\equiv 0\}$ are needed. Its size is at most $(2r_1/h + 3)^2 \leq 25r_1^2h^{-2}$. For a fixed λ let $h_i^\pm = \alpha_h^\lambda \otimes \varphi_i^\pm$. Then

$$f_h^\lambda = \sum_{i\in\mathcal{I}} (h_i^+ - h_i^-), \qquad A\widetilde{\psi}_i \leq \varphi_i^\pm \leq 3A\widetilde{\psi}_i. \tag{6.cccxxxiii}$$

In particular these tests are nonnegative, nonzero, and have sup norm between A and $3A$.

To check the support geometry in eq. 6.cccxxvi, consider each shape. Each support lies in the ball of radius $d = 5h/2$ about $(5h/2, 0, hi)$: its four coordinate half-widths are at most $h/2, h/2, 3h/2, 3h/2$, whose squared sum is $5h^2 < d^2$. The reflected ball also has radius d. The distance between the support and its time reflection is at least $4h$ and at most $6h$, hence between d and $6d$.

Moreover, direct product integration gives

$$\int h_i^\pm \geq Ah^4 \frac{\int a \int b \int \widetilde{\psi}}{4} = \frac{375}{512} Ah^4.$$

Since $\|h_i^\pm\|_\infty \leq 3A$, the integral condition holds for every $\vartheta \leq 1/160$. The product derivative rule gives

$$\operatorname{Lip} h_i^\pm \leq \frac{A}{h}\,[3(12 + 4\cdot 2) + 2\cdot 6 + 2016 + 1] = 2089A/h.$$

As $\|h_i^\pm\|_\infty \geq A$, eq. 6.cccxxx gives the Lipschitz condition: $2089A/h \leq \|h_i^\pm\|_\infty/(\vartheta_* d)$. The reflected tests have the same integrals and norms. Thus $(\theta h_i^\pm, h_i^\pm) \in \mathfrak{T}(5h/2; \vartheta_*)$.

By eq. 6.cccxxvii, reflection positivity and the Hilbert triangle inequality,

$$\|[f_h^\lambda]\| \leq \sum_i (\|[h_i^+]\| + \|[h_i^-]\|) \leq 6\,\#\mathcal{I}\,A\sqrt{u(5h/2)}.$$

Combining its square with eq. 6.cccxxxii proves

$$\Gamma_\varphi(7h) \leq Cr_1^4(\|\varphi\|_\infty + \|\nabla\varphi\|_\infty)^2 h^{-8} u(5h/2). \tag{6.cccxxxiv}$$

Consequently

$$t^8\Gamma_\varphi(t) \longrightarrow 0 \quad (t \downarrow 0). \tag{6.cccxxxv}$$

The zero label is immediate. The two factors of h^{-4} in eq. 6.cccxxxiv have different origins: normalization in eq. 6.cccxxxii, and the square of the number of transverse cells. No uniform bound on a compact profile's Fourier transform from below was used.

6.12.6. *A short-distance estimate from the ordinary sources.* There is a second way to obtain eq. 6.cccxxxv. It uses reflection positivity at coordinate mirrors, hypercubic and translation invariance, and the joint spectrum $E \geq |p_i|$, $1 \leq i \leq 3$, of the Euclidean Hilbert evolution. For the connected function use the one-field vectors projected off the unit vector Ω. The projection commutes with the evolution and spatial translations. The short-distance input in this argument is an upper bound on a single family of positive mirror pairs.

6.12.6.1. The four-dimensional spectral measure. For a compact C^1 test f in the positive half-space, put

$$\widetilde{f}(E,p)=\int f(x)e^{-x_4E+i\vec{x}\cdot p}\,dx,\qquad m_f(B)=\langle[f],\mathsf{P}(B)[f]\rangle,$$

where P is the joint spectral measure and the brackets refer to the connected form. Integration of positive-time translations against another such test gives $[h*f]=\widetilde{h}(H,P)[f]=\widetilde{f}(H,P)[h]$. The integrals exist in the Hilbert space by strong continuity and compact support. Consequently $|\widetilde{h}|^2m_f=|\widetilde{f}|^2m_h$. Choose a nonnegative $a\in C_c^1((2,3))$ with positive integral and let $v(s)=e^{-s}\mathbf{1}_{[0,1]}(s)$, $v^\vee(s)=v(-s)$, and $b=(v*v^\vee)^{*2}$. This is an even nonnegative C_c^1 function supported in $[-2,2]$. Its Fourier transform satisfies

$$\widehat{b}(q)=\frac{|1-e^{-1+iq}|^4}{(1+q^2)^2}\geq\frac{c}{(1+q^2)^2}.$$

Set $B_1(x)=a(x_4)\prod_{i=1}^3 b(4x_i)$. Its transform is nonzero on the entire spectral carrier, since the time factor is a positive Laplace transform. Thus $d\rho^{(4)}=|\widetilde{B}_1|^{-2}\,dm_{B_1}$ is a positive measure on that carrier, with $dm_f=|\widetilde{f}|^2d\rho^{(4)}$ for each f. Polarization gives the analogous mixed formula. This construction uses only one-field vectors, before taking any pointwise values of a kernel.

For $h>0$ let $B_h(x)=B_1(x/h)$. The time transform is bounded below by che^{-3hE}, and the spatial Fourier bound just proved gives, on the carrier,

$$|\widetilde{B}_h(E,p)|^2\geq ch^8e^{-6hE}(1+h^2E^2)^{-12}\geq c'h^8e^{-ChE}.$$

In particular $s_{\mathrm{a}}(t):=\int e^{-tE}\,d\rho^{(4)}$ is finite for every $t>0$, by taking $Ch\leq t$ and bounding the integral by $(c'h^8)^{-1}\|[B_h]\|^2$. It is nonnegative and nonincreasing. Each support $\operatorname{supp}B_h$ lies in a ball of radius $d=5h/2$ centered at $(5h/2)e_4$, since its four coordinate half-widths are at most $h/2$. Its distance to its reflected support lies between $4h$ and $6h$. Its integral is a positive constant times h^4, its supremum is fixed, and its Lipschitz constant is $O(h^{-1})$. Thus $(\theta B_h,B_h)$ belongs to $\mathfrak{T}(5h/2;\vartheta_1)$ for some fixed $\vartheta_1>0$. If this family of pairs satisfies

$$S_2(\theta B_h,B_h)\leq u(5h/2)\|B_h\|_\infty^2,\qquad d^2u(d)\longrightarrow 0\quad(d\downarrow 0),\tag{6.cccxxxvi}$$

then $s_{\mathrm{a}}(t)\leq Ct^{-8}u(ct)$ for small t, by setting $h=t/C$ in the spectral estimate.

For $t>0$ the continuous kernel on the half-space of displacement vectors is

$$S(te_4+\vec{x})=\int e^{-tE+i\vec{x}\cdot p}\,d\rho^{(4)}(E,p).$$

Indeed the mixed spectral formula gives this kernel when the two test supports lie on opposite sides of the coordinate mirror. By translation it gives the same formula whenever their time projections are strictly ordered. For each nonzero displacement there is a coordinate whose positive or negative projection is nonzero. Use the hypercubic symmetries and small boxes about the two points to obtain the kernel on that neighborhood. Two such formulas agree on overlaps because both represent the same bilinear form there, as can also be seen by testing with approximate identities in the two boxes. Compact separated test pairs are handled by partitions into finitely many such boxes. Choosing a coordinate of largest modulus proves

$$|S(x)|\leq s_{\mathrm{a}}(|x|_\infty)\qquad(x\neq 0).\tag{6.cccxxxvii}$$

6.12.6.2. The transverse integral. For a real transverse label φ, the definition of the plane kernel and uniqueness on separated planar tests give

$$\Gamma_\varphi(t)=\int_{\mathbb{R}^2}S(te_4+\zeta)(\varphi*\varphi^\vee)(\zeta)\,d\zeta,\qquad\varphi^\vee(\zeta)=\varphi(-\zeta).$$

The weight is bounded in modulus by $\|\varphi\|_2^2$, by Cauchy–Schwarz. If φ is supported in a ball of radius R, the weight vanishes for $|\zeta|>2R$. Use eq. 6.cccxxxvii on the square $|\zeta|_\infty\leq t$ and on

the annuli with inner radii $2^i t$ and outer radii $2^{i+1}t$, $i \geq 0$. The area of an annulus is bounded by $C(2^i t)^2$. For $0 < t \leq \delta$ with δ sufficiently small this gives

$$\Gamma_\varphi(t) \leq C\|\varphi\|_2^2 \left[t^{-6} \sum_{2^i t \leq \delta} 2^{-6i} u(c2^i t) + R^2 s_{\mathrm{a}}(\delta) \right]. \tag{6.cccxxxviii}$$

In this sum the central square is included in the $i = 0$ term. Annuli with inner radius greater than δ are bounded using monotonicity of s_{a}. After multiplication by t^8, the sum is bounded by $C \sup_{0<d\leq c\delta} d^2 u(d)$, since its terms are $c^{-2}2^{-8i}(c2^i t)^2 u(c2^i t)$. For fixed δ the remaining term vanishes as $t \downarrow 0$. Then let $\delta \downarrow 0$. This proves eq. 6.cccxxxv.

6.12.6.3. The stopped source bound on the mirror pair. For a periodic-state limit of first-passage data at g, assume the ordinary source systems of Prop. 5.63 at their first-passage stops, including the bounds in eq. 5.cdxxxv–eq. 5.cdxxxvi. The radius, localized energy bound, entropy gap and geometric constants must be uniform over the stops used here. The sum of the upper stability and floor constants at a stop with inverse coupling x must be bounded by $C(1 + \log x)$, as in the stopped estimate of Prop. 5.63. Uniformity at a single reference coupling would give no bound on this sum along the varying stops. The argument also uses the flow inequalities and independence of the self-calibrated prefix from its last level and the reference coupling, together with the exact dilation eq. 5.lxxiv for the periodic Gibbs states.

Call j a record if $x_j < x_i$ for every $i < j$. The indices 0 and K are records at first passage. For $j_0 = 1 + \lceil c_5/\lambda_\sharp \rceil$, the lower bound $x_i - x_{i+j_0} \geq \lambda_\sharp j_0 - c_5 > 0$ gives a new record within j_0 steps of any record $i \leq K - j_0$. When fewer steps remain, K itself supplies the next record. For each fixed small $d > 0$ and all sufficiently large K, choose $b \geq 0$ so that $K - b$ is a record and

$$cL^{-j_0-1} \leq L^b d \leq c, \qquad b \leq \log_L(c/d),$$

where $c \leq 1$ is fixed. In fact, if $n = \lfloor \log_L(c/d) \rfloor$, the depths of successive records have gaps at most j_0, so we may take the greatest record depth $b \leq n$. For $b > 0$, set $g_b^{-2} = x_{K-b} - B_\sharp$. Then $g_b < g$ and $K - b$ is the first-passage cutoff at g_b: every earlier inverse coupling exceeds $g_b^{-2} + B_\sharp = x_{K-b}$. The upper flow bound and the original first-passage window give

$$x_{K-b} \leq g^{-2} + B_\sharp(b + 1).$$

For $b = 0$ use the original reference coupling. At each stop the ordinary source systems are needed on arbitrarily large compatible tori. Their minimum side may depend on g_b, since the all-state transfer takes the side to infinity at fixed cutoff and coupling.

Dilation by L^b takes a periodic state at cutoff K to one at cutoff $K - b$. It preserves the two-source cumulant exactly, with test functions replaced by $f_i^{[b]}(x) = f_i(L^{-b}x)$. For $(f_1, f_2) \in \mathfrak{T}(d; \vartheta_1)$, the rescaled tests have support radii at most one, a fixed positive separation floor, and $\|f_i^{[b]}\|_\sharp \leq C\|f_i\|_\infty$. The latter uses $\mathrm{Lip}\, f_i^{[b]} \leq \|f_i\|_\infty/(\vartheta_1 L^b d)$. In eq. 5.cdxxxix–eq. 5.cdxlii take $n = 2$ and use a cell side bounded by a fixed constant. There are at most two occupied cells. Therefore the normalized generating function has bound $Q \leq \exp[C(1 + \log x_{K-b})]$. The mixed derivative of the holomorphic normalizer is bounded by eq. 5.cdxl at the fixed separation floor. For the remaining factor h_ρ, use $\partial_{12} \log h_\rho(0) = \partial_{12} h_\rho(0) - \partial_1 h_\rho(0)\partial_2 h_\rho(0)$. Cauchy's formula bounds this expression by $C(Q + Q^2) \prod_i \|f_i^{[b]}\|_\sharp$. Returning to the original unit yields the all-state estimate

$$|S_2^T[\rho](f_1, f_2)| \leq C\big[1 + g^{-2} + \log(1/d)\big]^C \|f_1\|_\infty \|f_2\|_\infty. \tag{6.cccxxxix}$$

The constants do not depend on the cutoff, the periodic state, or the support positions. The estimate passes to the continuum limit at each fixed $d > 0$. For the mirror pair this is eq. 6.cccxxxvi, since d^2 times any fixed power of $1 + g^{-2} + \log(1/d)$ tends to zero. Thus the ordinary source estimates at the running stops give eq. 6.cccxxxv for periodic-state limits. The angular argument below

uses this limit and the two mirror representations. The coefficient law has a separate role in the short-distance assertions.

6.12.7. *Entire angular continuation and the critical Fourier modes.* For $r > 0$, define on $|\operatorname{Re} z| < \pi/2$

$$g_r^{[0]}(z) = \int \exp[-rE\cos z + irq\sin z]\, d\mu_\varphi(E,q). \tag{6.cccxl}$$

Writing $z = u + is$, the absolute integrand is $\exp[-r\cos u(E\cosh s + q\sinh s)]$. On the cone this is at most $\exp[-r\cos u\, e^{-|s|}E]$. On a compact substrip its derivatives are dominated by a polynomial in E times a smaller exponential. Thus the integral is holomorphic. Its real restriction is $G_\varphi(r\cos u, r\sin u)$.

On $|\operatorname{Re} z - k\pi/2| < \pi/2$ define $g_r^{[k]}(z) = g_r^{[0]}(z - k\pi/2)$. On real overlaps these functions agree by eq. 6.cccxx. The complex overlaps are connected vertical strips with real intervals, so the identity theorem glues them. Their union is the complex plane. The resulting g_r is entire and $\pi/2$-periodic. For every real u, s choose k with $|u - k\pi/2| \le \pi/4$ to obtain

$$|g_r(u+is)| \le \Gamma_\varphi(re^{-|s|}/\sqrt{2}) = o(e^{8|s|}). \tag{6.cccxli}$$

The last little-o, uniform in u, is exactly eq. 6.cccxxxv.

Define $a_m(r) = \frac{2}{\pi}\int_0^{\pi/2} g_r(u)e^{-4imu}\,du$. Cauchy's theorem on a rectangle of height s, with cancellation of its vertical sides by periodicity, gives

$$|a_m(r)| \le e^{4ms} \sup_{u\in[0,\pi/2]} |g_r(u+is)|.$$

For $m \ge 2$ take $s \to -\infty$; for $m \le -2$ take $s \to +\infty$. eq. 6.cccxli gives $a_m(r) = 0$ in both cases, including $|m| = 2$. A mere $O(t^{-8})$ bound on Γ would not eliminate that boundary mode. Fourier uniqueness for a continuous periodic function now gives, first on the real line and then everywhere by the identity theorem,

$$g_r(z) = a_0(r) + a_1(r)e^{4iz} + a_{-1}(r)e^{-4iz}. \tag{6.cccxlii}$$

One elementary uniqueness proof uses the positive Fejer kernels: their integrals are one, their mass outside every fixed neighbourhood of zero tends to zero, and their convolution converges uniformly on continuous periodic functions. A function with all Fourier coefficients zero has all these convolutions zero. Thus no convergence assumption on a formal Fourier series is implicit in eq. 6.cccxlii.

6.12.8. *The positive measure in the diagonal frame.* Put $D(t,x) = G_\varphi(R_{\pi/4}(t,x))$ for $t > 0$. We construct its positive measure without assuming a diagonal cone. For $y = (t,x)$, $t > 0$, introduce a formal vector e_y with

$$\langle e_y, e_{y'}\rangle_D = D(t+t', x'-x). \tag{6.cccxliii}$$

This kernel is positive semidefinite. To see it, approximate each planar point $R_{\pi/4}y$ by a nonnegative radial smooth mollifier supported strictly on the positive side of the diagonal mirror, tensor it with φ, and apply eq. 6.cccxvii to an arbitrary complex linear combination. The formula in eq. 6.cccxviii and continuity away from zero give eq. 6.cccxliii in the limit. Thus its quotient and completion define a Hilbert space.

On finite sums define $T_s e_{(t,x)} = e_{(t+s,x)}$, $s \ge 0$, and $U_b e_{(t,x)} = e_{(t,x+b)}$, $b \in \mathbb{R}$. The U_b preserve the form and have inverse U_{-b}. The T_s are symmetric and commute with the U_b. If $v = \sum c_i e_{(t_i,x_i)}$ and $\delta = 2\min_i t_i > 0$, eq. 6.cccxxiii gives a finite bound on $\|T_s v\|$ for every $s \ge 0$, namely its square is at most $\Gamma_\varphi(\delta/\sqrt{2})(\sum|c_i|)^2$. On the unquotiented vector space Cauchy–Schwarz gives

$$\|T_s v\|^2 = \langle v, T_{2s}v\rangle_D \le \|v\|\,\|T_{2s}v\|.$$

If $\|v\| = 0$ this preserves the null space. Otherwise iterate and use the bounded orbit to get $\|T_s v\| \le \|v\|$. Therefore T_s descend to self-adjoint contractions. They are positive because $T_s =$

$T_{s/2}^2$. Continuity of D implies strong continuity of both families on point vectors, hence on their completion.

There are strongly commuting generators $H_D \geq 0$ and P_D with $T_s = e^{-sH_D}$ and $U_b = e^{ibP_D}$. For specificity, T_1 is injective: $T_1 v = 0$ implies $T_{1/2} v = 0$, then all its dyadic roots annihilate v, and strong continuity gives $v = 0$. Define $H_D = -\log T_1$ by spectral calculus. Uniqueness of positive roots proves the semigroup identity first at rational times and then at all times. The spectral theorem for the commuting unitary group supplies P_D and the joint spectral measure. This is the elementary Hilbert reconstruction for eq. 6.cccxliii, not an OS analytic-continuation theorem.

Let $v_a = e_{(a,0)}$. If P_D is that joint spectral measure, define

$$\nu_a(B) = \langle v_a, \mathsf{P}_D(B) v_a \rangle_D, \qquad d\nu(E,q) = e^{2aE}\, d\nu_a(E,q). \tag{6.cccxliv}$$

As $v_b = T_{b-a} v_a$ for $b > a$, ν is independent of a. It is a positive, locally finite, σ-finite measure on $[0,\infty) \times \mathbb{R}$, and for $t > 0$

$$D(t,x) = \int e^{-tE+ixq}\, d\nu, \qquad \int e^{-tE}\, d\nu = D(t,0) < \infty. \tag{6.cccxlv}$$

Indeed choose $a < t/2$ and evaluate $\langle v_a, T_{t-2a} U_x v_a \rangle_D$. Nothing so far asserts $|q| \leq E$ for ν.

6.12.9. *Fourier–Laplace uniqueness and the boost identity.* We shall use the following uniqueness fact. Positive Borel measures ρ_1, ρ_2 on $[0,\infty) \times \mathbb{R}$ with finite $\int e^{-tE} d\rho_i$ for every $t > 0$ are equal if their transforms $\int e^{-tE+ixq} d\rho_i$ agree for all $t > 0, x \in \mathbb{R}$. For each real x these transforms are holomorphic in t on $\mathrm{Re}\, t > 0$, by exponential domination, so the identity theorem gives equality at $t = a + i\tau$. These are precisely the joint Fourier transforms of the finite measures $e^{-aE}\rho_i$. Finite Fourier transforms determine their measures: convolve their difference with a Gaussian, apply the Gaussian Fourier integral and Fubini to get zero, and let the Gaussian width tend to zero against compactly supported continuous functions. Hence the weighted measures, and then the original measures, agree. The same proof applies to finite positive linear combinations on either side of an identity.

For $s \in \mathbb{R}$ define the Lorentz map of spectral variables

$$L_s(E,q) = (E\cosh s + q \sinh s,\, E \sinh s + q \cosh s), \qquad \mu_s = (L_s)_* \mu_\varphi. \tag{6.cccxlvi}$$

This preserves $\mathcal{C}$ and $L_s L_t = L_{s+t}$. Its energy satisfies $E_s \geq e^{-|s|} E$, so $\int e^{-tE} d\mu_s \leq \Gamma_\varphi(t e^{-|s|})$. Define $F_s(t,x) = \int e^{-tE+ixq} d\mu_s$ for $t > 0$. Direct multiplication of the rotation matrix by is gives

$$F_s(v) = K_\varphi(R_{is} v) = g_r(\theta + is) \quad (v = R_\theta(re_0),\ v^0 > 0), \tag{6.cccxlvii}$$

where the middle expression uses eq. 6.cccxxiv. In fact $R_{is}(t,x) = (t\cosh s - ix \sinh s, x \cosh s + it \sinh s)$, and its exponential multiplier is exactly $e^{-tE_s + ixq_s}$. Thus eq. 6.cccxlvii holds on the whole real half-plane.

Write $C = \cosh 4 > 1$. The three terms in eq. 6.cccxlii obey the algebraic identity

$$(C-1) g_r(\theta + \pi/4) + g_r(\theta + i) + g_r(\theta - i) = (C+1) g_r(\theta).$$

Using eq. 6.cccxxii, eq. 6.cccxlv and eq. 6.cccxlvii and then transform uniqueness gives the identity of positive measures

$$(C-1)\nu + \mu_1 + \mu_{-1} = (C+1)\mu_\varphi. \tag{6.cccxlviii}$$

Every measure has the required finite Laplace weights. In particular

$$0 \leq \nu \leq \frac{C+1}{C-1} \mu_\varphi. \tag{6.cccxlix}$$

The diagonal measure is therefore carried by the exact cone. This conclusion was deduced from eq. 6.cccxlviii, not inserted as an additional reflection-positivity consequence.

For $r > 0$ and real s the function $w_s(E,q) = \exp[-r(E\cosh s + q\sinh s)]$ is positive on the cone and at most $e^{-re^{-|s|}E}$. It is integrable against all four measures in eq. 6.cccxlviii; for $\mu_{\pm1}$ use $L_sL_{\pm1} = L_{s\pm1}$. Integrating eq. 6.cccxlviii against w_s yields

$$(C-1)\int w_s\,d\nu + F_{s+1}(r,0) + F_{s-1}(r,0) = (C+1)F_s(r,0).$$

Substitute eq. 6.cccxlii into this identity to obtain

$$\begin{aligned} F_s(r,0) &= a_0(r) + a_1(r)e^{-4s} + a_{-1}(r)e^{4s} \geq 0,\\ \int w_s\,d\nu &= a_0(r) - a_1(r)e^{-4s} - a_{-1}(r)e^{4s} \geq 0. \end{aligned} \tag{6.cccl}$$

The first positivity comes from the axis measure, the second from the diagonal measure. Divide by e^{4s} and let $s \to +\infty$: the two limits give $a_{-1}(r) \geq 0$ and $a_{-1}(r) \leq 0$. Then let $s \to -\infty$ to get both signs for $a_1(r)$. The limits also show these coefficients are real, if that was not already deduced from eq. 6.cccxx. Thus $a_1(r) = a_{-1}(r) = 0$ for every $r > 0$.

Proposition 6.25 (Two-point angular elimination)**.** Under eq. 6.cccxvii–eq. 6.cccxix and the short-distance hypothesis in eq. 6.cccxxvi–eq. 6.cccxxx, every equal-real-label plane kernel is radial:

$$G_\varphi(R_t y) = G_\varphi(y) \quad (y \neq 0,\ t \in \mathbb{R}).$$

Every nonzero angular coefficient of the two-point kernel vanishes, including coefficients with mixed transverse labels. ■

Proof. The preceding argument proves the first assertion. It uses the short-distance hypothesis only through eq. 6.cccxxxv, so the same conclusion holds under the other stated hypotheses when that limit is obtained from § 6.12.6. Symmetry of S_2 gives $G_{\varphi,\psi}(y) = G_{\psi,\varphi}(-y)$; invariance under the planar half turn gives $G_{\psi,\varphi}(-y) = G_{\psi,\varphi}(y)$. Therefore the mixed kernel is symmetric in its real labels and

$$G_{\varphi,\psi} = \tfrac{1}{4}(G_{\varphi+\psi} - G_{\varphi-\psi}).$$

Both terms on the right are radial. Complex labels follow by bilinear extension. Uniqueness of Fourier coefficients identifies this conclusion with vanishing of every $m \neq 0$ coefficient in any finite-angular-band construction of the same two-point kernel. In its notation, $\overline{m}(2) = 0$. □

This is the order-two conclusion used by a later all-orders argument. It does not prove that argument, nor does it replace the required extension from planar plate kernels to arbitrary separated four-dimensional tuples. The ultraviolet vanishing and the kernel/spectral interface remain the stated premises. No rotational Ward identity, infrared decay, diagonal cone assumption, or full O(4) covariance was used.

6.13. **Euclidean covariance from the two-point angular law.** The analytic kernels and boost Gram identities of § 6.11, together with the two-point elimination of § 6.12, give Euclidean covariance at every order and on distributions. The argument requires the two-point ultraviolet estimate and uses no rotational Ward identity.

Work first with the centered scalar hierarchy. Compact separated moments are symmetric, translation invariant and hypercubic invariant, have position-uniform local C^1 bounds, and are reflection positive in every coordinate picture. In each such picture the Hilbert generators have the exact cube cone $H_j \geq |P_{j,r}|$ for every tangential coordinate $r \neq j$. Their off-diagonal distributions are those constructed in § 5.25.

6.13.1. *The harmonic data used below.* The chosen plane has coordinates (x_4, x_1), and R_u denotes its rotation through u. A plate label is a real smooth compact function of the two transverse coordinates. The analytic kernel construction supplies, for each n, real-analytic functions

$$c_m(y_1, \dots, y_n; \varphi_1, \dots, \varphi_n), \qquad y_i \in \mathbb{R}^2, \quad y_i \neq y_j, \tag{6.ccli}$$

multilinear in the labels, with only $|m| \leq Cn$ possibly nonzero. Their sum is the moment density on the ordinary ordering charts. They are symmetric under simultaneous permutations of positions and labels, translation invariant in the plane, and satisfy

$$c_m(R_u \mathbf{y}; \vec{\varphi}) = e^{4imu} c_m(\mathbf{y}; \vec{\varphi}), \qquad c_{-m} = \overline{c_m} \quad \text{for real labels.} \tag{6.cclii}$$

An ordinary chart means that all time coordinates are distinct, or that all abscissae are distinct; a fixed ordering is imposed on its connected component. The functions in eq. 6.ccli already extend over the nongeneric planar configurations with distinct points. This is the analytic-projector conclusion of the kernel construction, not a density assertion about distributions on the exceptional set.

Let b_n be the greatest absolute index of a harmonic that is not identically zero for some choice of real labels. Set $b_n = 0$ if there is no nonzero nonconstant harmonic, and put $b_0 = 0$ for the vacuum kernel. Thus b_n are nonnegative integers and $b_n \leq Cn$. The boost Gram identity gives

$$2b_{k+l} \leq b_{2k} + b_{2l}, \qquad k, l \geq 0. \tag{6.ccliii}$$

For clarity, this inequality compares global degrees. A nonzero coefficient can be tested on an open configuration set split into k negative and l positive times: real analyticity and connectedness prevent a nonzero coefficient from vanishing on that whole open set. Cauchy–Schwarz for the boosted product vectors bounds its largest exponential by the two diagonal ones. The empty product is the vacuum vector, so the cases $k = 0$ or $l = 0$ are included.

For $m_* = b_{2k}$ the coefficient c_{-m_*} is a positive kernel on reflected pairs of k-point configurations in a half-plane. This follows by multiplying the positive boost Gram matrix by $e^{-4m_* s}$ and taking $s \to +\infty$. The covariance in eq. 6.ccclii transports it to every oriented line: after multiplication by the fixed unit phase for that line it is again positive. We use the zero-row consequence of this positivity, which is unchanged by that phase. All these statements have explicit product-vector proofs in the preceding kernel section; analytic continuation of a nonnegative real-valued function alone would not prove them.

6.13.2. *The polygon divisibility argument, with label release.* Fix $k \geq 1$, put $m_* = b_{2k}$, and abbreviate $H = c_{-m_*}$. We prove

$$k \mid 2m_*. \tag{6.ccccliv}$$

Suppose otherwise. For any real plate label φ, place that same label at every vertex of a regular $2k$-gon. Rotation through π/k permutes the labelled vertices. Symmetry and eq. 6.ccclii imply

$$H(\mathbf{y}; \varphi, \dots, \varphi) = e^{-4\pi i m_*/k} H(\mathbf{y}; \varphi, \dots, \varphi) = 0. \tag{6.ccclv}$$

The polygon may be translated; rotation about its center has the same effect by translation invariance. The assumption that eq. 6.cccliv fails is exactly what makes the phase in eq. 6.ccclv different from one.

Call a labelled set S a zero set if H vanishes on every $2k$-point extension of it, with all remaining real labels arbitrary. A full polygon with the labels above is a zero set, since no completion is then needed. Here is the reduction of such a set. Suppose it is contained, with its labels, in $\theta Q \cup Q$, where Q has $j \leq k$ points strictly on one side of a line, and the reflected labels agree. Fill Q by any $k - j$ additional points A on that side, with arbitrary real labels. The positive kernel has zero diagonal at $Q \cup A$, because its diagonal configuration contains S. Its entire row is therefore zero. Indeed, positivity of the 2×2 Gram matrix gives $|K(X, Y)|^2 \leq K(X, X)K(Y, Y)$; a zero diagonal makes the right side zero. Consequently

$$H\big(\theta(Q \cup A) \cup B\big) = 0 \tag{6.ccclvi}$$

for every k-point B on the other side of the reflected set, with arbitrary real labels.

For each fixed choice of these free labels, eq. 6.ccclvi holds on a nonempty open set of all the free positions: the reflected points of A and the points of B can be varied independently in their respective half-planes. The configuration space of these $2k - j$ free points, avoiding each other and the fixed reflected Q, is connected. It is the complement of a finite union of affine sets of codimension two. One elementary proof is to join two configurations through a generic intermediate point; the intermediate points whose segment meets one of these affine sets form a finite union of measure-zero cones. Real analyticity and the identity principle extend eq. 6.ccclvi to this whole configuration space. It follows that θQ itself is a zero set, with no restriction on the remaining labels or half-planes. This is the label-release step; it is stronger than just showing that a special diagonal configuration has zero value.

Apply this reduction to the polygon. Reflection in the perpendicular bisector of an edge pairs its vertices and has no vertex on its line. It first reduces the full polygon to a consecutive arc of k vertices. A consecutive arc of $j \leq k$ vertices fits into the reflected union of an arc of $\lceil j/2 \rceil$ vertices. To check the odd case, index the vertices by integers modulo $2k$. For the arc $0, \dots, j - 1$, reflect indices by $i \mapsto j - i$ when j is odd; the union has $j + 1$ vertices, paired without a fixed vertex. For even j, use $i \mapsto j - 1 - i$ and the arc itself is paired. These are edge-bisector reflections, and

the selected half has the stated size. All its labels are still φ. Repeated reduction reaches a single labelled point.

Translation covariance makes every point with label φ a zero set. Since φ was arbitrary, every labelled $2k$-point configuration has zero value: choose any one of its labels as that single fixed label and use the unrestricted completion. Thus H is identically zero, contradicting the definition of m_*. This proves eq. 6.cccliv. When $m_* = 0$ the divisibility is automatic; no contradiction argument is needed.

6.13.3. *Discrete convexity reduces every order to two points.* Set $a_k = b_{2k}$, $a_0 = 0$. eq. 6.ccliii with $k-1, k+1$ gives

$$2a_k \le a_{k-1} + a_{k+1}, \qquad 0 \le a_k \le C'k. \tag{6.ccclvii}$$

The integer increments $d_k = a_k - a_{k-1}$ are nondecreasing, initially nonnegative, and bounded. If one increment exceeded the linear-growth constant, all later increments would do so, contradicting eq. 6.ccclvii. Hence they are eventually a fixed integer d. For large k write

$$a_k = dk - D, \qquad D = \sum_{j\ge 1}(d - d_j) \ge 0, \tag{6.ccclviii}$$

where only finitely many summands are nonzero. By eq. 6.cccliv, $k \mid 2a_k$ for every k, so every sufficiently large k divides $2D$. Thus $D = 0$. Each summand in eq. 6.ccclviii is nonnegative, and every increment equals $d = a_1$. Therefore

$$b_{2k} = kb_2, \qquad 2b_n \le b_{2n} = nb_2. \tag{6.ccclix}$$

The second inequality is the empty-tuple case of eq. 6.ccliii. Convexity and linear growth without eq. 6.cccliv would not suffice: $a_k = \max(0, k-1)$ is a counterexample. The polygon argument is what removes that defect.

The two-point elimination theorem gives $b_2 = 0$ under its ultraviolet shape estimate. eq. 6.ccclix then give $b_n = 0$ for every n. Consequently the ordinary-chart kernel is invariant under rotations in the chosen plane. The following argument upgrades this fact to the actual off-diagonal distributions, including their exceptional supports.

6.13.4. *A local analytic translation lemma.* Fix a coordinate j and divide the entries of a compact separated tuple into two groups whose supports lie on opposite sides of a coordinate hyperplane, with a positive gap. Their scalar moment is a matrix element in the coordinate-j Hilbert picture. After removing some unused gap, it has the form

$$\sigma(F_- \cdot \tau_z F_+) = \langle \Phi, e^{-(d+z_j)H_j} U_{z_\parallel} \Psi \rangle, \qquad d > 0, \tag{6.ccclx}$$

for real z sufficiently small. Here the left group has been reflected to define Φ, both groups have been shifted away from their mirror, and d is the remaining normal separation. This is exactly the compact-tuple translation dictionary, applied to products in each group. Its vectors need not be point fields, and no coincident product is used.

Let v be any real four-vector. The right side with $z = sv$ is holomorphic for small complex s. The precise sufficient domain is

$$d + v_j \operatorname{Re} s > \sum_{r \ne j} |v_r|\,|\operatorname{Im} s|. \tag{6.ccclxi}$$

Indeed the joint spectral cone $|P_{j,r}| \le H_j$ bounds the spectral exponential by one there. On every smaller domain a positive damping margin controls all powers of H_j and $P_{j,r}$, so difference quotients converge in operator norm. This proves holomorphy and local uniform bounds, rather than formally translating test functions to complex arguments. Translating the negative group instead gives the same conclusion with the corresponding signs. In particular, any specified half of a strict coordinate cut may be moved in any real direction, with a holomorphic parameter near zero.

6.13.5. *Removing the exceptional distributional support.* For $n \geq 2$ put

$$\mathcal{C}_n = \{(x_1, \dots, x_n) \in (\mathbb{R}^4)^n : x_i \neq x_l \ (i \neq l)\}.$$

Let T_n be the off-diagonal moment distribution. For a fixed rotation R_u define its difference by

$$D_u(f_1 \otimes \cdots \otimes f_n) = \sigma(f_1, \dots, f_n) - \sigma(R_u f_1, \dots, R_u f_n), \qquad (R_u f)(x) = f(R_u^{-1} x). \tag{6.ccclxii}$$

Both terms are well-defined distributions on $\mathcal{C}_n$.

First suppose all original and all rotated time coordinates at a configuration are distinct. Choose small product support boxes keeping both time orderings strict. In those boxes approximate each smooth test in C^1 by finite sums $\alpha(x_4, x_1)\varphi(x_2, x_3)$, with supports staying inside the boxes. For example, a product Fourier expansion with fixed compact cutoffs gives these approximants with all first derivatives. The rotation-invariant ordinary-chart kernel gives equality for the plate approximants. The local moment bounds pass it to the original tests. Local product-tensor density then says that D_u vanishes on this ordinary open set. This step alone would not rule out a distribution concentrated at the remaining time ties.

Fix any $y \in \mathcal{C}_n$. For each coordinate take all threshold hyperplanes halfway between successive distinct coordinate levels of the finite set $\{y_1, \dots, y_n\}$. Each threshold divides the indices into two nonempty sets, and these finitely many cuts together separate every pair of indices. No point lies on one of their hyperplanes. There exist $u_0 > 0$ and a product neighborhood W of y such that, for every $|u| < u_0$, each cut has the same index partition for x and for $R_u x$, for all $x \in W$, with a uniform positive gap in both configurations. To see this, take the minimum distance of the finitely many original points from these finitely many hyperplanes. Rotation is continuous and the points are bounded, so their displacement and then the size of W can be made smaller than a quarter of that minimum. Also take $u_0 < \pi/3$. The cuts and their partitions are kept fixed throughout the following local argument.

Fix $|u| < u_0$. At $x \in W$ let

$$q(x) = \#\{i < l : x_{i,4} = x_{l,4}\} + \#\{i < l : (R_u x_i)_4 = (R_u x_l)_4\}. \tag{6.ccclxiii}$$

The set $\{q \leq q_0\}$ is open: every inequality between distinct time coordinates remains an inequality in a sufficiently small neighborhood. We prove by induction that D_u vanishes on this open set for each integer q_0. The case $q_0 = 0$ is the ordinary-chart argument above.

Take a point with $q = q_0 > 0$. Choose a tied pair in either frame and one of the fixed common cuts that separates it. Let A be the positive side of that cut. Translate every entry in A by se_4. Original cross-group time differences have slope 1 or -1 in s; rotated cross-group differences have slope $\cos u$ or $-\cos u$. All slopes are nonzero. Internal differences in either group are unchanged. For all sufficiently small nonzero real s, no previously nonzero difference becomes zero, while every previously zero cross difference becomes nonzero. At least the chosen tied pair is of this kind, so the number in eq. 6.ccclxiii decreases.

Choose one such s_* and a small real interval I about it. Shrink the product neighborhood V of the original point so that its translates stay in W, all previously nonzero differences stay nonzero, and all cross-group differences in both frames are nonzero for every $s \in I$. Every configuration in the translated product boxes then has $q \leq q_0 - 1$. By induction, D_u vanishes there as a distribution.

Now take arbitrary smooth slot tests supported in V. Their difference in eq. 6.ccclxii, after the group translation, is a holomorphic function of s near the real segment joining zero to I. For the first moment term this is eq. 6.ccclx–eq. 6.ccclxi with $v = e_4$. For the rotated term the same common cut separates the same two groups, and the direction is $v = R_u e_4$. Strict gaps persist on the chosen segment; shrinking the segment and its complex neighborhood supplies eq. 6.ccclxi in both cases. This is why both configurations were required to share the cut. The holomorphic function is zero on I, hence is zero at zero. Thus D_u vanishes on every such product tensor in V. The explicit local Fourier tensor density for distributions makes it vanish on all tests in V.

This proves the induction step. Since $q \le n(n-1)$, the induction terminates and D_u vanishes on W. It excludes a singular part on any of the exceptional time-tie sets, including configurations with coincident plane projections but separated transverse positions. At no point was the kernel integral assumed analytic across those sets.

6.13.6. *All angles, all planes, and the Schwartz domain.* We have proved local invariance for every sufficiently small u on a neighborhood of each point of $\mathcal{C}_n$. Let a compactly supported off-diagonal test be given, and fix a finite angle v. The union of the supports of its rotations between zero and v is a compact subset of $\mathcal{C}_n$. Cover it by finitely many neighborhoods just constructed, and take the minimum of their positive angle bounds. Partition v into finitely many smaller steps. On each step the difference distribution is zero on the current support, so composition gives invariance through v.

Thus all separated smooth moments are invariant in the chosen plane. Compact C^1 tests follow by smooth approximation on fixed separated neighborhoods. The hypercubic group conjugates this circle of rotations to rotations in every coordinate plane. Givens elimination of the columns of an orthogonal matrix expresses every member of SO(4) as a finite product of such plane rotations: successively zero the entries below each diagonal position, using a rotation in the corresponding two-coordinate plane, and finish with diagonal signs. A coordinate reflection supplies the other component of O(4).

For $n = 1$, the translation-invariant one-point functional is $c \int f$ and is already invariant. Restoring such a mean to a centered field uses a finite subset expansion on each compact tuple; this commutes with every rotation because $\int Rf = \int f$. Finally rotations act continuously on Schwartz space, preserve every pairwise coincidence diagonal, and preserve flatness there. The diagonal-flat extension and its constrained tensor density therefore transfer the same invariance to that full test domain.

Proposition 6.26 (All-orders angular closure)**.** Under the moment, cone, analytic-kernel and two-point hypotheses stated above, the full scalar hierarchy is O(4) invariant, both on separated compact C^1 tuples and on its unique diagonal-flat Schwartz extension. The polygon, discrete-convexity and exceptional-support steps require no further analytic rotation assumption. ■

This closes a conditional Euclidean-covariance implication. It neither proves the ordinary interacting source estimates nor the ultraviolet two-point law, and it does not identify the scalar cyclic theory with a complete family of Yang–Mills observables.

6.14. **Displacement calculus and the exact lattice Ward identity.** This section derives the finite-lattice rotation identity, including its contact terms and its shell flux. The identity is exact before any renormalization or continuum limit. Its separate terms identify which estimates a Ward proof of Euclidean covariance needs. In particular the divergence of the displacement field contributes to the flux and must be retained.

6.14.1. *Plaquette holonomies based at a bond.* Work on the periodic lattice $a\mathbb{Z}^4/\ell\mathbb{Z}^4$ with ℓ/a an integer at least three. Write $\mathcal{B}_+$ for the positive bonds. A configuration is a point of $\mathrm{SU}(2)^{\mathcal{B}_+}$; for a reverse bond $\bar{b}$ set $U_{\bar{b}} = U_b^{-1}$. Let x_b be the midpoint of a bond and x_p the barycentre of a plaquette. A plaquette is counted once in a sum over p, and a sum over its sides uses its four positive bonds, irrespective of the orientation of its boundary.

If b is either orientation of a side of p, let $W_{b,p}$ be the holonomy of the plaquette boundary that starts at b_- and traverses b first. Its first matrix factor is U_b. Reversing the orientation of this boundary inverts the holonomy and leaves its trace unchanged. Set

$$W = w^0(W)I + i\mathbf{w}(W), \qquad w^0(W) = \tfrac{1}{2}\operatorname{tr} W, \qquad \mathbf{w}(W) = \frac{W - W^*}{2i}. \tag{6.ccclxiv}$$

With the Pauli basis t_α and inner product $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr}(XY)$, $\mathbf{w} = \sum_{\alpha=1}^{3} w^\alpha t_\alpha$ and $(w^0)^2+|\mathbf{w}|^2 = 1$. The plaquette density is $a_p = 1 - w^0(W_{b,p})$, independent of the chosen side and base point. Thus

$$0 \le a_p \le 2, \qquad |\mathbf{w}(W_{b,p})|^2 \le 2a_p. \tag{6.ccclxv}$$

For a positive bond, define ∂_b^α by differentiating $U_b \mapsto e^{ist_\alpha}U_b$. The inverse bond changes by the inverse of this matrix. The same definition for an oriented negative bond is understood as a vector field on the positive bond coordinates. Since the four bonds in a plaquette are distinct, a derivative at one of its sides differentiates exactly one factor in the based holonomy.

Lemma 6.27 (The derivatives of the scalar and vector parts)**.** For a side b of p,

$$\partial_b^\alpha w^0_{b,p} = -w^\alpha_{b,p}, \qquad \partial_b^\alpha a_p = w^\alpha_{b,p}, \tag{6.ccclxvi}$$

$$\partial_b^\alpha w^\gamma_{b,p} = \delta_{\alpha\gamma} w^0_{b,p} + \sum_\delta \varepsilon_{\alpha\gamma\delta} w^\delta_{b,p}, \qquad \sum_\alpha \partial_b^\alpha w^\alpha_{b,p} = 3w^0_{b,p}. \tag{6.ccclxvii}$$

The derivatives of a_p vanish on bonds not in its boundary. ■

Proof. Write $W_{b,p} = U_b R$, where R has no dependence on that bond coordinate. Its derivative is $it_\alpha W_{b,p}$. Expand this product with $t_\alpha t_\gamma = \delta_{\alpha\gamma} I + i\sum_\delta \varepsilon_{\alpha\gamma\delta} t_\delta$. Its scalar part is $-w^\alpha$ and its vector part is the right side of eq. 6.ccclxvii. The alternating term vanishes when $\gamma = \alpha$, giving the final sum. □

6.14.2. *The displacement vector field.* For ρ different from the direction of b, let $p(b, \pm\rho)$ be the plaquette spanned by b and $\pm a e_\rho$, and abbreviate its based vector part by $\mathbf{w}_{b,\pm\rho}$. Put

$$\mathbf{u}_{b,\rho} = -\tfrac{1}{2}(\mathbf{w}_{b,+\rho} - \mathbf{w}_{b,-\rho}), \qquad \mathbf{u}_{b,\mu(b)} = 0,$$

$$\mathbf{v}_b[\eta] = a^{-1}\sum_\rho \eta^\rho(x_b)\mathbf{u}_{b,\rho}, \qquad \mathsf{D}_\eta = \sum_{b\in\mathcal{B}_+,\alpha} v_b^\alpha[\eta]\partial_b^\alpha. \tag{6.ccclxviii}$$

Only the finitely many values of η at bond midpoints are used. No regularity in the spatial variable is needed for this definition. Its coefficients are smooth functions of the group variables, even if η is merely a prescribed list of numbers.

For b a side of p define the elementary density

$$d_{b,p,\rho} = \langle \mathbf{u}_{b,\rho}, \mathbf{w}_{b,p}\rangle. \tag{6.ccclxix}$$

Then Lem. 6.27 gives

$$\mathsf{D}_\eta a_p = a^{-1}\sum_{b\subset\partial p,\rho} \eta^\rho(x_b) d_{b,p,\rho}. \tag{6.ccclxx}$$

All densities here are bounded functions on a compact configuration space, so every finite-cutoff derivative and integral used below exists.

6.14.3. *Gauge covariance and bond reversal.* A site gauge u acts by $U_b \mapsto u(b_-)U_b u(b_+)^{-1}$. Its intermediate factors cancel in a based loop, hence

$$W_{b,p}(U^u) = u(b_-)W_{b,p}(U)u(b_-)^{-1}. \tag{6.ccclxxi}$$

The same conjugation acts on $\mathbf{w}$, $\mathbf{u}$ and $\mathbf{v}$. Since the adjoint action preserves the colour inner product, $d_{b,p,\rho}$ is gauge invariant. The transformation rule for the bond derivative is the corresponding adjoint transformation of its direction. The two rules cancel in their contraction, so

$$\mathsf{D}_\eta(F\circ(\,\cdot\,)^u) = (\mathsf{D}_\eta F)\circ(\,\cdot\,)^u. \tag{6.ccclxxii}$$

The exact reversal relations are

$$W_{\bar b,p} = U_b^{-1}W_{b,p}^{-1}U_b, \qquad \mathbf{w}_{\bar b,p} = -\operatorname{Ad}(U_b^{-1})\mathbf{w}_{b,p},$$

$$\mathbf{u}_{\bar b,\rho} = -\operatorname{Ad}(U_b^{-1})\mathbf{u}_{b,\rho}, \qquad \mathbf{v}_{\bar b}[\eta] = -\operatorname{Ad}(U_b^{-1})\mathbf{v}_b[\eta]. \tag{6.ccclxxiii}$$

The first identity follows by following the inverse boundary from the other end of the same bond; the others use inversion and conjugation of the vector part. The midpoint is unchanged. At a fixed configuration, $\partial_{\bar b}^{X} = -\partial_b^{\mathrm{Ad}(U_b)X}$. It follows that $\mathbf{v}_{\bar b}\cdot\partial_{\bar b} = \mathbf{v}_b\cdot\partial_b$. Thus the vector field D_η is independent of which orientation is chosen for each unoriented bond, and $d_{\bar b,p,\rho} = d_{b,p,\rho}$.

For a signed permutation S of the spatial coordinates and a lattice translation, the image of a transverse positive direction can be negative. Extend $\mathbf{u}_b(\xi) = \sum_\rho \xi^\rho \mathbf{u}_{b,\rho}$ linearly to every $\xi\in\mathbb{R}^4$. Interchanging $+\rho$ and $-\rho$ changes its sign. Consequently, for the isometry $\mathsf{r}(x) = Sx + v$,

$$d_{\mathsf{r}b,\mathsf{r}p}(S\xi)(\mathsf{r}U) = d_{b,p}(\xi)(U), \qquad \mathsf{D}_{\mathsf{r}\cdot\eta}(F^{\mathsf{r}}) = (\mathsf{D}_\eta F)^{\mathsf{r}}, \tag{6.ccclxxiv}$$

where $(\mathsf{r}\cdot\eta)(x) = S\eta(\mathsf{r}^{-1}x)$. Bond reversals that arise in this formula are harmless by eq. 6.ccclxxiii. This verifies covariance under coordinate reflections as well as orientation-preserving permutations.

6.14.4. *The exact Haar divergence.* The divergence of a vector field written in left-invariant orthonormal frames is $\operatorname{div}\mathsf{D}_\eta = \sum_{b,\alpha}\partial_b^\alpha v_b^\alpha$. By eq. 6.ccclxvii,

$$\operatorname{div}\mathsf{D}_\eta = \frac{3}{2a}\sum_{b\in\mathcal{B}_+,\rho\neq\mu(b)} \eta^\rho(x_b)[a_{p(b,+\rho)} - a_{p(b,-\rho)}]. \tag{6.ccclxxv}$$

The coefficient three is the dimension of the colour algebra. The sign follows from $w^0 = 1 - a_p$ and the minus sign in the definition of $\mathbf{u}$.

Define the midpoint central difference by

$$\nabla_\rho^c\eta^\rho(x) = a^{-1}[\eta^\rho(x + \tfrac{a}{2}e_\rho) - \eta^\rho(x - \tfrac{a}{2}e_\rho)]. \tag{6.ccclxxvi}$$

A plaquette in the $\mu\nu$ plane occurs twice in eq. 6.ccclxxv for each of its two directions. For the direction ν, its lower μ side has midpoint $x_p - ae_\nu/2$ and sign plus; its upper μ side has midpoint $x_p + ae_\nu/2$ and sign minus. The other direction is identical. Regrouping the finite sum therefore gives

$$\operatorname{div}\mathsf{D}_\eta = -\tfrac{3}{2}\sum_{p\text{ in plane }\mu\nu} a_p[\nabla_\mu^c\eta^\mu(x_p) + \nabla_\nu^c\eta^\nu(x_p)]. \tag{6.ccclxxvii}$$

The notation in this formula sums over all plaquettes once; the two displayed directions are the directions of that plaquette and have no further implicit sum.

Proposition 6.28 (Integration by parts with the sourced Wilson measure)**.** For a smooth vector field D on the finite product of groups and a smooth function F,

$$\int \mathsf{D}F\,dU = -\int F\operatorname{div}\mathsf{D}\,dU. \tag{6.ccclxxviii}$$

For $A(c,U) = \sum_p c(p)a_p(U)$ and $Z(c) = \int e^{-A(c,U)}dU \neq 0$, the normalized, possibly complex, expectation satisfies

$$\langle \mathsf{D}A(c,\cdot)\rangle_c = \langle \operatorname{div}\mathsf{D}\rangle_c. \tag{6.ccclxxix}$$

■

Proof. Left multiplication preserves Haar measure, so $\int \partial_b^\alpha H\,dU = 0$ for smooth H. Apply it to each product $v_b^\alpha F$, expand by the product rule, and sum to prove eq. 6.ccclxxviii. Apply that identity to $F = e^{-A}$ and use $\mathsf{D}e^{-A} = -(\mathsf{D}A)e^{-A}$. Division by the stipulated nonzero integral proves eq. 6.ccclxxix. Compactness justifies differentiation under each integral. The proof works for complex coefficients by linearity; positivity is only needed later for estimates. □

6.14.5. *Orbital, tensor and remainder densities.* The four positive sides of a plaquette have lever arms $x_b - x_p = \pm a e_\mu/2, \pm a e_\nu/2$. Set

$$\begin{aligned} t_\rho(p) &= \sum_{b \subset \partial p} d_{b,p,\rho}, \\ s_{\lambda\rho}(p) &= \sum_{b \subset \partial p} a^{-1}(x_b - x_p)_\lambda d_{b,p,\rho}, \\ r_\rho(p) &= t_\rho(p) - \tfrac{1}{2}[a_{p+ae_\rho} - a_{p-ae_\rho}]. \end{aligned} \tag{6.ccclxxx}$$

These are local gauge-invariant polynomials. Under the signed-permutation isometries, s is a rank-two tensor and r a vector: this follows from eq. 6.ccclxxiv, the transformation of the lever arm by S, and the signed transformation of the central difference.

The sphere relation and $2ab \le a^2 + b^2$ give

$$|d_{b,p,\rho}| \le 1, \qquad |d_{b,p,\rho}| \le \tfrac{1}{2}[a_{p(b,+\rho)} + a_{p(b,-\rho)}] + a_p. \tag{6.ccclxxxi}$$

Indeed $|\mathbf{u}| \le 1$, while the product with $|\mathbf{w}_{b,p}|$ is bounded by $\frac{1}{4}(|\mathbf{w}_+|^2 + |\mathbf{w}_-|^2) + \frac{1}{2}|\mathbf{w}_{b,p}|^2$. If $N(p)$ includes the plaquettes sharing a side with p and the translates $p \pm a e_\rho$, these inequalities imply

$$|s_{\lambda\rho}(p)| + |t_\rho(p)| + |r_\rho(p)| \le C \sum_{p' \in N(p)} a_{p'}. \tag{6.ccclxxxii}$$

The constant is absolute and independent of a, ℓ. Each term depends only on bonds within $2a$ of the plaquette barycentre in the sup norm. These bounds identify the energy weight needed to estimate an insertion; they do not themselves supply decay of a connected correlation.

6.14.6. *A cutoff rotation and the product rule.* Let $\omega \in \mathfrak{so}(4)$. Choose a Lipschitz function χ supported in $(-\ell/2 + 2a, \ell/2 - 2a)^4$ and read it on the torus through representatives in $[-\ell/2, \ell/2)^4$. Define $\eta(x) = \chi(x)\omega x$. The support margin ensures that every shift used in a nonzero term is represented by its ordinary Euclidean shift. The linear field ωx alone is not periodic; this support condition is essential for the next identities.

Put

$$h_{\omega,\chi}(p) = a^{-1} \sum_{b \subset \partial p, \rho} [\chi(x_b) - \chi(x_p)](\omega x_b)_\rho d_{b,p,\rho}. \tag{6.ccclxxxiii}$$

Using $\omega x_b = \omega x_p + \omega(x_b - x_p)$ in eq. 6.ccclxx gives the exact product rule

$$\begin{aligned} \mathsf{D}_{\chi\omega x} a_p = {} & \chi(x_p) a^{-1} \sum_\rho (\omega x_p)_\rho t_\rho(p) \\ & + \chi(x_p) \sum_{\rho,\lambda} \omega_{\rho\lambda} s_{\lambda\rho}(p) + h_{\omega,\chi}(p). \end{aligned} \tag{6.ccclxxxiv}$$

Only the antisymmetric part of s contributes to its contraction with ω. This is an algebraic tensor statement and does not assume rotation symmetry of the quantum measure.

For a plaquette weight f define

$$(\nabla^s_\omega f)(p) = \sum_\rho (\omega x_p)_\rho \frac{f(p + a e_\rho) - f(p - a e_\rho)}{2a}. \tag{6.ccclxxxv}$$

For any weight c, define

$$\begin{aligned} \mathcal{S}_\chi[c] &= \sum_p c(p)\chi(x_p) \sum_{\rho,\lambda} \omega_{\rho\lambda} s_{\lambda\rho}(p), \\ \mathcal{K}_\chi[c] &= \sum_p c(p)\chi(x_p) a^{-1} \sum_\rho (\omega x_p)_\rho r_\rho(p). \end{aligned} \tag{6.ccclxxxvi}$$

Omit the subscript χ when its factor is absent.

The central-difference part of the first line of eq. 6.ccclxxxiv, summed against c, is

$$\sum_{p,\rho}(c\chi)(p)(\omega x_p)_\rho \frac{a_{p+ae_\rho}-a_{p-ae_\rho}}{2a} = -\mathcal{O}(\nabla^s_\omega(c\chi)). \tag{6.ccclxxxvii}$$

To verify it, shift the finite plaquette sum separately in its plus and minus terms. In direction ρ, the ρ component of ωx is unchanged by that shift because $\omega_{\rho\rho}=0$. The compact support margin removes every seam ambiguity. This proves the identity without a discrete product-rule approximation.

6.14.7. *The exact sourced identity.* Let w be a complex plaquette weight and $c=\beta-w$, with $\beta>0$. Assume $\chi=1$ on the Euclidean $2a$-neighborhood of the representatives of all plaquettes where $w\neq 0$. This implies $w\chi=w$ and $w(p)h_{\omega,\chi}(p)=0$. Define the source-independent shell insertion

$$\mathcal{B}_{\omega,\chi}=\beta\mathcal{O}(\nabla^s_\omega\chi)-\beta\sum_p h_{\omega,\chi}(p)+\operatorname{div}\mathsf{D}_{\chi\omega x}. \tag{6.ccclxxxviii}$$

For $Z(w)=\int e^{-A(\beta-w,U)}dU\neq 0$, denote the normalized expectation by $\langle\cdot\rangle_w$.

Proposition 6.29 (Finite-lattice rotational Ward identity)**.** Under the stated support and normalization hypotheses,

$$\langle\mathcal{O}(\nabla^s_\omega w)\rangle_w=-\langle\mathcal{S}_\chi[c]+\mathcal{K}_\chi[c]\rangle_w+\langle\mathcal{B}_{\omega,\chi}\rangle_w. \tag{6.ccclxxxix}$$

The divergence contribution has the explicit shell form

$$\operatorname{div}\mathsf{D}_{\chi\omega x}=-\tfrac{3}{2}\sum_{p\text{ in plane }\mu\nu}a_p\sum_{\rho\in\{\mu,\nu\}}(\omega x_p)_\rho\nabla^c_\rho\chi(x_p). \tag{6.cccxc}$$

■

Proof. Sum eq. 6.ccclxxxiv against c. Insert the definition of r and use eq. 6.ccclxxxvii. Since $c\chi=\beta\chi-w$ and $wh=0$, this gives

$$\mathsf{D}_{\chi\omega x}A(c,U)=\mathcal{S}_\chi[c]+\mathcal{K}_\chi[c]-\beta\mathcal{O}(\nabla^s_\omega\chi)+\mathcal{O}(\nabla^s_\omega w)+\beta\sum_p h_{\omega,\chi}(p).$$

Take normalized expectations and apply eq. 6.ccclxxix. Rearrangement proves eq. 6.ccclxxxix. For the divergence, substitute $\eta=\chi\omega x$ in eq. 6.ccclxxvii. Its component $(\omega x)_\rho$ is constant along the ρ direction, again because $\omega_{\rho\rho}=0$. The remaining central difference acts only on χ, proving eq. 6.cccxc. □

Every occurrence of χ in the flux is a difference of its values at points at distance at most a. In the first term of eq. 6.ccclxxxviii, split $\chi(x_p+ae_\rho)-\chi(x_p-ae_\rho)$ at x_p. The second term already has a bond-midpoint difference, and the last is eq. 6.cccxc. Thus a plaquette contributes nothing when χ is constant on its a-neighborhood. This local assertion does not license setting $\chi=1$ globally on a torus while treating ωx as periodic.

6.14.8. *Connected derivatives and the contact terms.* Choose weights $f_1,\dots,f_n$ and a single cutoff χ equal to one on the prescribed neighborhood of all their supports. Set $w(z)=\sum_i z_i f_i$. Since the group product is compact and $0\le a_p\le 2$, $Z(w(z))$ and every inserted numerator are entire functions of $z\in\mathbb{C}^n$. Also $Z(0)>0$. Continuity gives a common polydisc about zero on which Z is nonzero and all the preceding normalized identities hold holomorphically.

For a bounded insertion Y independent of z,

$$\partial_{z_1}\cdots\partial_{z_n}\langle Y\rangle_{w(z)}\big|_{z=0}=\langle Y;\mathcal{O}(f_1);\dots;\mathcal{O}(f_n)\rangle. \tag{6.cccxci}$$

This follows by adding one more source sY to Z and differentiating $\partial_s\log Z(s,z)$ at $s=0$. Boundedness justifies all derivatives under the integral, and local non-vanishing justifies the logarithm.

Differentiate eq. 6.ccclxxxix once in each of the n source variables. The source on its left gives n terms: differentiating its explicit factor z_i leaves a cumulant with the remaining $n-1$ sources.

The same product rule applies to $\mathcal{S}_\chi[\beta - w] + \mathcal{K}_\chi[\beta - w]$. Since $\chi = 1$ at the source plaquettes, its resulting contact insertions are $\mathcal{S}[f_i] + \mathcal{K}[f_i]$. Consequently

$$\begin{aligned}\sum_{i=1}^{n} & S_n^T(f_1, \dots, \nabla_\omega^s f_i, \dots, f_n) \\ &= -\beta\langle(\mathcal{S}_\chi + \mathcal{K}_\chi)[1]; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle \\ &\quad + \sum_{i=1}^{n}\langle(\mathcal{S} + \mathcal{K})[f_i]; (\mathcal{O}(f_j))_{j\neq i}\rangle \\ &\quad + \langle\mathcal{B}_{\omega,\chi}; \mathcal{O}(f_1); \dots; \mathcal{O}(f_n)\rangle. \end{aligned} \tag{6.cccxcii}$$

For $n = 1$ the contact term is an expectation. No assumption of disjoint supports is used for this exact finite-lattice identity. Such separation enters estimates of its connected terms, rather than the algebraic differentiation.

6.14.9. *The shell estimate and the continuum use.* For a cutoff χ_R equal to one on a ball of radius R and zero outside a ball of radius $2R$, assume $\operatorname{Lip}\chi_R \le C/R$, $a \le R/10$, and $2R+2a < \ell/2$, retaining the earlier torus support margin. The difference form of the flux, together with eq. 6.ccclxxxi, gives

$$|\mathcal{B}_{\omega,\chi_R}| \le C'(\beta+1)|\omega| \sum_{p:\, R-3a\le|x_p|\le 2R+3a} a_p. \tag{6.cccxciii}$$

To check the factors, a difference of χ_R contributes at most its point separation times C/R, cancelling the displayed a^{-1} in each term. In this shell $|x_p| + a \le 3R$, which cancels the remaining R^{-1}. The bounded neighbor multiplicity in eq. 6.ccclxxxi only changes C'. The divergence term has the same estimate with the factor β replaced by one.

This bound supplies integrability at fixed cutoff. It does not make the connected shell insertion tend to zero as $R \to \infty$: the shell contains an increasing number of plaquettes. A Ward proof of continuum rotation invariance must establish vanishing of the last line of eq. 6.cccxcii, as well as the stated ultraviolet estimates for the spin and orbital terms. Clustering without a rate sufficient for the shell sum is not such an estimate.

For $f \in C_c^3$ in a fixed bounded region, Taylor's integral formula yields

$$\|\nabla_\omega^s f - (\omega x)\cdot\nabla f\|_\infty \le Ca^2|\omega| \sup_{x\in\operatorname{supp} f + B_{2a}} (1+|x|)|D^3 f(x)|. \tag{6.cccxciv}$$

For $f \in C_c^4$ supported in a fixed ball of radius R_f, the same calculation after one spatial derivative gives

$$\|\nabla_\omega^s f - (\omega x)\cdot\nabla f\|_{C^1} \le Ca^2|\omega|(1+R_f+a)(\|D^3 f\|_\infty + \|D^4 f\|_\infty).$$

Indeed differentiation also acts on the linear coefficient $(\omega x)_\rho$, producing a central-difference error for f; the other term is the same error for a first derivative of f. Both have the displayed bound. The error is supported in $\operatorname{supp} f + B_a$. On any fixed separated tuple these enlarged supports remain separated for small a. The uniform correlation bound in the declared C^1 test norm, together with multilinearity, therefore passes the left side of the Ward identity to the infinitesimal rotation derivative for smooth tests. If the spin, orbital and shell terms all vanish on that same limiting route, the limiting functionals satisfy the infinitesimal rotation identity. Integrating along one-parameter rotations gives rotation covariance on the smooth test class, and continuity extends it to the declared test space. The exact finite-lattice calculation identifies all terms to be controlled; it does not replace their interacting estimates.

6.15. **Classical** SU(2) **invariants and the smooth Wilson action.** The rotational Ward calculation distinguishes a local density from its integral, and a classical identity from an estimate in the lattice measure. We make those distinctions explicit. First we classify the local gauge-invariant

polynomials through dimension six under the full hypercubic group. We prove the finite enumeration, its independence, and the quotients by divergences and by the field equation. We then derive the smooth-field Wilson expansion with the same SU(2) normalization as the lattice construction. Its pointwise coefficient contains a divergence which disappears only from an unweighted compact-support integral. None of these classical calculations bounds that coefficient under a rough interacting lattice measure.

6.15.1. *Conventions and the local polynomial spaces.* The Lie algebra consists of traceless Hermitian matrices. Write $X = \vec{x} \cdot \sigma$, using Pauli matrices with $\frac{1}{2}\,\mathrm{tr}(\sigma^a\sigma^b) = \delta_{ab}$, and set

$$\langle X, Y\rangle = \tfrac{1}{2}\,\mathrm{tr}(XY), \qquad \langle X, Y, Z\rangle = \tfrac{i}{2}\langle X, [Y, Z]\rangle. \tag{6.cccxcv}$$

Multiplication of the three Pauli matrices gives

$$XY = (\vec{x}\cdot\vec{y})I + i(\vec{x}\times\vec{y})\cdot\sigma, \quad i[X,Y] = -2(\vec{x}\times\vec{y})\cdot\sigma, \quad \langle X,Y,Z\rangle = -\det(\vec{x},\vec{y},\vec{z}). \tag{6.cccxcvi}$$

In particular the triple bracket is real and alternating. This fixes the sign of the cubic invariant below.

Let A_μ be a smooth connection on $\mathbb{R}^4$. Define

$$D_\alpha X = \partial_\alpha X + i[A_\alpha, X], \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + i[A_\mu, A_\nu], \qquad J^{\mathrm{c}}_\nu = \sum_\mu D_\mu F_{\mu\nu}. \tag{6.cccxcvii}$$

The superscript distinguishes this continuum field equation from a current defined as a positive action gradient. For $S_0 = \frac{1}{4}\int \sum_{\mu,\nu} |F_{\mu\nu}|^2$, a compactly supported variation gives $\delta S_0 = -\int \sum_\nu \langle J^{\mathrm{c}}_\nu, \delta A_\nu\rangle$. Thus its sign is the opposite of that positive gradient; the equation $J^{\mathrm{c}} = 0$ is unaffected.

Commuting $\partial_\alpha + iA_\alpha$ as operators, and using the Jacobi identity, proves

$$\begin{aligned} [D_\alpha, D_\beta]X &= i[F_{\alpha\beta}, X], \\ D_\alpha F_{\mu\nu} + D_\mu F_{\nu\alpha} + D_\nu F_{\alpha\mu} &= 0, \\ \partial_\alpha\langle X, Y\rangle &= \langle D_\alpha X, Y\rangle + \langle X, D_\alpha Y\rangle. \end{aligned} \tag{6.cccxcviii}$$

The same product rule holds for the triple bracket. Under a site gauge u, the connection is $A^u_\alpha = uA_\alpha u^{-1} + i(\partial_\alpha u)u^{-1}$. Conjugation of $\partial + iA$ shows that every iterated covariant derivative of F transforms by $X \mapsto uXu^{-1}$.

Give $D_{\alpha_1}\cdots D_{\alpha_k}F_{\mu\nu}$ dimension $k+2$. Let $\mathcal{V}_\delta$ be the real gauge-invariant polynomials in these variables which are homogeneous of dimension δ, identified as functions of (A, x). Constant polynomials form $\mathcal{V}_0$. No naked connection variable is admitted in these spaces. A total derivative of dimension six will mean

$$\partial\mathcal{V}_5 = \left\{ \sum_\rho \partial_\rho v_\rho : v_\rho \in \mathcal{V}_5 \right\}. \tag{6.cccxcix}$$

This is a quotient by local gauge-invariant currents, not a statement that a divergence vanishes pointwise. We keep the current ideal separate: $\mathcal{I}_6$ is spanned by

$$\langle J^{\mathrm{c}}_\nu, D_\alpha F_{\kappa\lambda}\rangle, \qquad \langle D_\alpha J^{\mathrm{c}}_\nu, F_{\kappa\lambda}\rangle. \tag{6.cd}$$

It contains all dimension-six gauge-invariant multiples of the field equation and its covariant derivatives in this polynomial algebra. Indeed a factor $D^k J^{\mathrm{c}}$ leaves dimension $3-k$ for its cofactor. For $k = 0, 1$ this gives precisely the two displayed families; $k = 2$ has no possible cofactor, and $k = 3$ is a linear colour expression, whose gauge average is zero.

An orthogonal map acts on a connection by $(R\cdot A)_\mu(x) = \sum_\nu R_{\mu\nu}A_\nu(R^{-1}x)$. All indices in an iterated derivative of F are tensor indices. The group W_4 consists of every signed coordinate permutation, including orientation-reversing ones. We write $\mathcal{V}_\delta^{W_4}$ for scalar invariants. For tensor channels, $\mathrm{Hom}_{W_4}(M, \mathcal{V}_\delta)$ denotes equivariant linear maps from the indicated real tensor space M.

6.15.2. *The finite invariant enumeration.*

Lemma 6.30 (Colour and index reduction)**.** The spaces below are spanned by the displayed families:

$$\begin{aligned}\mathcal{V}_4 &= \operatorname{span}\{\langle F_{\mu\nu}, F_{\kappa\lambda}\rangle\},\\ \mathcal{V}_5 &= \operatorname{span}\{\langle F_{\mu\nu}, D_\alpha F_{\kappa\lambda}\rangle\},\\ \mathcal{V}_6 &= \operatorname{span}\{\langle F_{\mu\nu}, D_\alpha D_\beta F_{\kappa\lambda}\rangle, \langle D_\alpha F_{\mu\nu}, D_\beta F_{\kappa\lambda}\rangle, \langle F_{\mu\nu}, F_{\kappa\lambda}, F_{\sigma\tau}\rangle\}.\end{aligned} \tag{6.cdi}$$

For $1 \le \delta \le 3$, $\mathcal{V}_\delta = 0$. For a tensor monomial with index tuple I, averaging over all coordinate signs kills it unless its set of odd-multiplicity indices matches the set prescribed by the output channel. A scalar requires the empty set; a vector component ρ requires $\{\rho\}$; an off-diagonal two-tensor component $\rho\sigma$ requires $\{\rho, \sigma\}$. ■

Proof. Each variable has dimension at least two, so through dimension six a monomial has at most three colour vectors. Average its coefficient tensor over constant SU(2) gauges. Conjugation acts as all of SO(3) on the colour vectors. Invariant coefficient tensors of ranks one, two and three are respectively zero, multiples of δ_{ab}, and multiples of ϵ_{abc}. Here is an elementary verification. Rotations through π about coordinate axes kill a rank-one invariant and all off-diagonal rank-two entries; quarter turns equate the remaining diagonal entries. For rank three, the same half turns require each of the three coordinate labels to occur an odd number of times. They therefore occur once each. Proper signed permutations make the remaining coefficients alternating, giving ϵ. These arguments classify coefficient tensors before evaluating any differential identities, so additional relations among the jets cannot enlarge the spanning space. The possible dimensions now give exactly eq. 6.cdi. Linear colour expressions, including a single high derivative of F, average to zero.

A coordinate sign vector ε multiplies a monomial by $\prod_j \varepsilon_j^{n_j}$, where n_j is its index count. Averaging this character against the scalar, vector, or two-tensor character gives the stated parity rule. Averaging subsequently over coordinate permutations gives orbit sums; for a fixed output component use the subgroup fixing its distinguished indices. This works with spanning families and does not assume that all formal monomials are independent. □

The central element $-I$ acts on $\mathcal{V}_\delta$ by $(-1)^\delta$. Hence odd-dimensional scalar invariants vanish, and the vector channel vanishes at even dimension. At dimension four the scalar parity patterns allow only squares of an individual $F_{\mu\nu}$; permutation averaging gives

$$\mathcal{V}_4^{W_4} = \mathbb{R}\sum_{\mu,\nu} |F_{\mu\nu}|^2, \qquad \mathcal{V}_5^{W_4} = 0. \tag{6.cdii}$$

Define the following three symmetric two-tensors and vectors:

$$\begin{aligned} T^1_{\alpha\beta} &= \sum_\kappa \langle F_{\alpha\kappa}, F_{\beta\kappa}\rangle, \quad T^2_{\alpha\beta} = \delta_{\alpha\beta}\sum_{\kappa,\lambda} |F_{\kappa\lambda}|^2,\\ T^3_{\alpha\beta} &= \delta_{\alpha\beta}\sum_\kappa |F_{\alpha\kappa}|^2, \qquad v^1_\rho = \partial_\rho \sum_{\alpha,\beta} |F_{\alpha\beta}|^2,\\ v^2_\rho &= \sum_\nu \langle F_{\rho\nu}, J^c_\nu\rangle, \qquad v^3_\rho = \partial_\rho \sum_\kappa |F_{\rho\kappa}|^2. \end{aligned} \tag{6.cdiii}$$

There is no sum on a repeated free index in T^3 or v^3.

Proposition 6.31 (The low-dimensional tensor channels)**.** The symmetric two-tensor channel of dimension four has basis T^1, T^2, T^3. Its antisymmetric channel is zero. The vector channel of dimension five has basis v^1, v^2, v^3, and

$$v^1_\rho = \sum_\alpha \partial_\alpha T^2_{\alpha\rho}, \quad v^3_\rho = \sum_\alpha \partial_\alpha T^3_{\alpha\rho}, \quad v^2_\rho = \sum_\alpha \partial_\alpha (T^1_{\alpha\rho} - \tfrac{1}{4} T^2_{\alpha\rho}). \tag{6.cdiv}$$

Moreover $\Lambda^2\mathbb{R}^4$ is irreducible for the full W_4, and has no equivariant image in $\mathbb{R}$, $\mathcal{V}_4$, or $\mathcal{V}_5$. ■

Proof. For a diagonal output component $\rho\rho$, sign parity requires even counts. In a product of two F's this leaves $\sum_\kappa |F_{\rho\kappa}|^2$ and the sum of the squares whose indices avoid ρ. These are the same span as the diagonal components of T^2 and T^3. For $\rho \neq \sigma$, the only nonzero parity pattern is $\rho\sigma\kappa\kappa$ with κ different from the two distinguished indices. Its orbit sum is $\sum_\kappa \langle F_{\rho\kappa}, F_{\sigma\kappa}\rangle$. It is symmetric in ρ, σ, so the antisymmetric channel is zero, and the full two-tensor space is the three-dimensional span claimed, subject to the independence verification below.

For the vector component ρ, a dimension-five monomial has five indices. The odd-set rule gives only multiplicities $(3,2)$ or $(1,2,2)$; (5) is zero by antisymmetry of F. Up to signs and orbit multiplicities the resulting list is

$$\begin{aligned} a_\rho &= \sum_\kappa \langle F_{\rho\kappa}, D_\rho F_{\rho\kappa}\rangle, \\ b^1_\rho &= \sum_{\sigma,\tau\neq\rho} \langle F_{\sigma\tau}, D_\rho F_{\sigma\tau}\rangle, \\ b^2_\rho &= \sum_{\sigma,\tau\neq\rho,\,\sigma\neq\tau} \langle F_{\rho\sigma}, D_\tau F_{\tau\sigma}\rangle, \\ b^3_\rho &= \sum_{\sigma,\tau\neq\rho,\,\sigma\neq\tau} \langle F_{\sigma\tau}, D_\sigma F_{\rho\tau}\rangle. \end{aligned} \tag{6.cdv}$$

This list is exhaustive: in the $(3,2)$ pattern the derivative has the distinguished index and both antisymmetric pairs contain it; in the $(1,2,2)$ pattern it either occurs on the derivative, or on one of the two curvature factors, giving the last three entries. The product rule and Bianchi identity give

$$v^3 = 2a, \qquad v^1 = 2b^1 + 4a, \qquad v^2 = a + b^2, \qquad \tfrac{1}{4}v^1 = a + b^3. \tag{6.cdvi}$$

For the last identity contract Bianchi with $F_{\kappa\lambda}$ and sum: the two terms with a derivative on an index of that curvature are equal, while the remaining one is half the derivative of $\sum |F|^2$. Explicitly

$$\sum_{\kappa,\lambda} \langle F_{\kappa\lambda}, D_\kappa F_{\rho\lambda}\rangle = \tfrac{1}{2}\sum_{\kappa,\lambda} \langle F_{\kappa\lambda}, D_\rho F_{\kappa\lambda}\rangle = \tfrac{1}{4} v^1_\rho. \tag{6.cdvii}$$

The left side is $a_\rho + b^3_\rho$. The same contracted identity, applied to the divergence of T^1, proves the third formula in eq. 6.cdiv; the other two are immediate product rules.

Independence can be checked on genuine connections rather than formal jets. Multiply all polynomials in the following tests by a smooth cutoff equal to one near zero, and set $A_\mu = q_\mu \sigma^3$. At zero, only the displayed q_μ are nonzero:

q	(v^1_1, v^2_1, v^3_1)
$q_2 = x_1 + x_1^2/2$	$(4,1,2)$
$q_3 = x_2 + x_1 x_2$	$(4,0,0)$
$q_2 = x_1 + x_3^2/2$	$(0,1,0)$

(6.cdviii)

The determinant is eight. For the tensor channel, the field $q_3 = x_1 + x_2$ has $T^1_{12} = 1$, while $T^2_{12} = T^3_{12} = 0$. The field $q_2 = x_1$ has $(T^1_{33}, T^2_{33}, T^3_{33}) = (0,2,0)$ and $(T^1_{11}, T^2_{11}, T^3_{11}) = (1,2,1)$. These three evaluations prove tensor independence.

Finally the sign subgroup acts on the six coordinate wedges by six distinct characters $\varepsilon_\rho\varepsilon_\sigma$. Averaging against one character projects onto its one-dimensional coordinate wedge. Any invariant subspace containing a nonzero vector therefore contains a coordinate wedge. Permutations act transitively on the six wedges, so it contains them all. This proves irreducibility. Its image in the trivial representation is zero, its dimension-four image is the antisymmetric channel already computed, and its dimension-five image is zero by the action of the central element $-I$. □

6.15.3. *Dimension six as densities.* Define

$$\begin{aligned} O_1 &= \sum_{\mu,\nu} |D_\mu F_{\mu\nu}|^2, & O_2 &= \sum_{\alpha,\mu,\nu} |D_\alpha F_{\mu\nu}|^2, \\ O_3 &= \sum_{\nu} |J^{\mathrm{c}}_\nu|^2, & O_F &= \sum_{\mu,\nu,\rho} \langle F_{\mu\nu}, F_{\nu\rho}, F_{\rho\mu}\rangle, \\ Q &= \sum_{\alpha} \partial_\alpha^2 \sum_{\mu} |F_{\alpha\mu}|^2, & d_i &= \sum_{\rho} \partial_\rho v^i_\rho. \end{aligned} \tag{6.cdix}$$

In O_1 the derivative index is tied to one curvature index before squaring. It is not O_3, which first sums the derivative contributions. In particular $d_3 = Q$.

Lemma 6.32 (An exact off-shell identity)**.** For every smooth connection, pointwise,

$$O_2 = 2O_3 - 4O_F + \tfrac{1}{2} d_1 - 2d_2. \tag{6.cdx}$$

■

Proof. Differentiate Bianchi and commute the two covariant derivatives:

$$\sum_{\rho} D_\rho D_\rho F_{\alpha\beta} = D_\alpha J^{\mathrm{c}}_\beta - D_\beta J^{\mathrm{c}}_\alpha + 2i \sum_{\rho} [F_{\rho\alpha}, F_{\rho\beta}]. \tag{6.cdxi}$$

Pair with $F_{\alpha\beta}$ and sum. Antisymmetry gives twice $T = \sum_{\alpha,\beta} \langle F_{\alpha\beta}, D_\alpha J^{\mathrm{c}}_\beta\rangle$ from the first two terms. The commutator term is $4O_F$ by eq. 6.cccxcv and relabeling the three indices. The product rules give

$$d_1 = 2O_2 + 4T + 8O_F, \qquad d_2 = O_3 + T. \tag{6.cdxii}$$

Eliminate T. No integration or field equation was used. □

Proposition 6.33 (Complete off-shell scalar basis)**.** The space $\mathcal{V}_6^{W_4}$ has basis

$$O_1,\ O_2,\ O_3,\ d_1,\ d_2,\ Q. \tag{6.cdxiii}$$

The cubic-in-F invariant subspace is exactly $\mathbb{R} O_F$. ■

Proof. We first give an exhaustive reduction. In a cubic curvature monomial regard each antisymmetric pair as an edge between its two coordinate labels. The sign rule requires even degree at every label. Three non-loop edges can then form only a triangle; the colour triple bracket gives its oriented contraction O_F. Permutation averaging changes only its multiplicity and sign. Thus the cubic subspace is at most one-dimensional.

For the $DF\,DF$ terms the six index counts are $(4,2)$ or $(2,2,2)$; the all-equal case is zero. The full orbit list, up to signs and factors, is

$$\begin{aligned} &O_1, \qquad S_1 = \sum_{a,b,c}^{\neq} |D_a F_{bc}|^2, \\ &S_2 = \sum_{a,b,c}^{\neq} \langle D_a F_{ac}, D_b F_{bc}\rangle, \qquad S_3 = \sum_{a,b,c}^{\neq} \langle D_a F_{bc}, D_b F_{ac}\rangle. \end{aligned} \tag{6.cdxiv}$$

Here the superscript means that the three labels are distinct. The $(4,2)$ pattern gives O_1. In $(2,2,2)$, either a derivative repeats a curvature index in each factor, giving S_2, or all three labels occur in each factor. Equal derivative labels then give S_1 and different ones give S_3. Expanding the sums and contracting Bianchi yields

$$O_2 = S_1 + 2O_1, \qquad O_3 = O_1 + S_2, \qquad S_1 = 2S_3. \tag{6.cdxv}$$

For the last equality multiply $D_a F_{bc} = D_b F_{ac} - D_c F_{ab}$ by $D_a F_{bc}$ and sum over distinct indices. Interchanging b, c makes the second cross term the negative of the first, giving exactly $2S_3$.

It remains to reduce $FDDF$. The pointwise identity

$$\langle F_{\mu\nu}, D_\alpha D_\beta F_{\kappa\lambda}\rangle = \partial_\alpha \langle F_{\mu\nu}, D_\beta F_{\kappa\lambda}\rangle - \langle D_\alpha F_{\mu\nu}, D_\beta F_{\kappa\lambda}\rangle \tag{6.cdxvi}$$

requires no commutation of D_α, D_β. After averaging over W_4, the first term is the divergence of an equivariant vector of dimension five. By Prop. 6.31 it lies in $\mathrm{span}\{d_1, d_2, Q\}$. The second term was just enumerated. The scalar space is therefore spanned by $O_1, O_2, O_3, O_F, d_1, d_2, Q$, and eq. 6.cdx reduces this to the six members in eq. 6.cdxiii.

We verify all six dimensions on actual compactly supported connections. Use the commuting cutoff connections of the previous proof. The following three choices give the indicated first three columns; since $F(0) = 0$, their last three columns are $(d_1, d_2, Q) = (2O_2, O_3, 2O_1)$:

q	O_1	O_2	O_3
$q_2 = x_1^2/2$	1	2	1
$q_3 = x_1 x_2$	0	4	0
$q_2 = (x_1^2 - x_3^2)/2$	2	4	0

(6.cdxvii)

Their three-by-three determinant is -8. Two further commuting tests, $q_2 = x_1 + x_1^3/6$ and $q_2 = x_1 + x_1 x_3^2/2$, have $DF(0) = 0$ and give respectively $(d_1, d_2, Q) = (4, 1, 2)$ and $(4, 1, 0)$. For the last test choose constant antisymmetric $f_{\mu\nu}$ with $f_{12} = \sigma^1$, $f_{23} = \sigma^2$, $f_{31} = \sigma^3$ and $f_{\mu 4} = 0$, and put $A_\mu = \frac{1}{2}\sum_\nu x_\nu f_{\nu\mu}$ near zero. Then $F(0) = f$, $DF(0) = 0$, and $O_F(0) = -6$. Moreover $D_\mu^2 F_{\mu\nu}(0) = 0$: the ordinary second derivative of $i[A_\mu, A_\nu]$ in the μ direction has a zero factor, and $A(0) = \partial_\mu A_\mu(0) = 0$. Thus $Q(0) = 0$. The curvature Laplacian and eq. 6.cdxii give $(d_1, d_2, Q) = (-12, 9, 0)$. These last three tests have $O_1 = O_2 = O_3 = 0$ and divergence matrix

$$\begin{pmatrix} 4 & 1 & 2 \\ 4 & 1 & 0 \\ -12 & 9 & 0 \end{pmatrix}, \qquad \det = 96. \tag{6.cdxviii}$$

The full six-by-six evaluation matrix is block triangular with determinant -768. This proves independence as functions of connections, and also shows $O_F \neq 0$. The spanning result is therefore a basis result, not merely a list of possible operators. □

6.15.4. *Rotational scalars, divergences and the field equation.*

Proposition 6.34 (The two density obstructions and the single integrated obstruction)**.** One has the direct sum

$$\mathcal{V}_6^{W_4} = \mathrm{span}\{O_2, O_3, d_1, d_2\} \oplus \mathbb{R}O_1 \oplus \mathbb{R}Q, \qquad \mathcal{V}_6^{\mathrm{O}(4)} = \mathrm{span}\{O_2, O_3, d_1, d_2\}. \tag{6.cdxix}$$

Modulo total derivatives the scalar space has basis O_1, O_2, O_3. Modulo rotational scalars and total derivatives its dimension is one, represented by O_1. This last obstruction remains nonzero after also quotienting by $\mathcal{I}_6$. Modulo divergences and $\mathcal{I}_6$ alone, a basis is given by the classes of O_1, O_2, with $O_2 \equiv -4O_F$. For the dimension-four symmetric tensor channel, a combination $c_1T^1 + c_2T^2 + c_3T^3$ is $\mathrm{O}(4)$-equivariant exactly when $c_3 = 0$. ■

Proof. Complete contraction of indices shows that O_2, O_3, O_F are rotational scalars and that T^1, T^2, v^1, v^2 have their stated full orthogonal tensor types. Their divergences d_1, d_2 are therefore scalars. To prove that no further linear combination is a scalar, fix a radial cutoff equal to one near zero and use

$$A_3(x) = \phi(p \cdot x)\sigma^3, \qquad A_\mu = 0 \quad (\mu \neq 3), \qquad |p| = 1,\ p_3 = 0 \tag{6.cdxx}$$

near zero. Rotations fixing e_3 take this family from p to Rp. At zero direct differentiation gives

$$O_1 = \phi''(0)^2 \sum_\alpha p_\alpha^4, \qquad Q = 2\{\phi''(0)^2 + \phi'(0)\phi'''(0)\} \sum_\alpha p_\alpha^4. \tag{6.cdxxi}$$

The factor $\sum p_\alpha^4$ changes from one to one half under a rotation taking e_1 to $(e_1+e_2)/\sqrt{2}$. For $\phi(t)=t^2/2$, invariance of aO_1+bQ would force $a+2b=0$; for $\phi(t)=t+t^3/6$ it would force $b=0$. Hence $a=b=0$. Together with the six-dimensional basis this proves eq. 6.cdxix.

The W_4-scalar part of $\partial\mathcal{V}_5$ equals $\operatorname{span}\{d_1,d_2,Q\}$. Indeed average a current over the finite group with its vector action; the divergence of the average is the scalar average of its divergence. The vector classification then applies. These three divergences are independent by the density basis. Quotienting gives the asserted three-dimensional basis. The off-shell identity also gives $O_2\equiv 2O_3-4O_F$ in this quotient.

The same sign and permutation enumeration in eq. 6.cd gives

$$\mathcal{I}_6^{W_4}=\operatorname{span}\{O_3,d_2\}. \tag{6.cdxxii}$$

For $\langle J_\nu^{\mathrm{c}}, D_\alpha F_{\kappa\lambda}\rangle$ the only nonzero even index pattern contracts ν with one curvature index and α with the other, yielding O_3. For $\langle D_\alpha J_\nu^{\mathrm{c}}, F_{\kappa\lambda}\rangle$ the same pattern yields d_2-O_3 by the product rule. The alternative pairing sets the two curvature indices equal and is zero. Both surviving invariants are rotational scalars. If an invariant belonged to a sum of the rotation, divergence and current subspaces, averaging that membership over W_4 would give precisely these invariant subspaces. Thus the extra quotient cannot remove the class of O_1. This proves the on-shell quotient assertion without setting J^{c} to zero inside an off-shell density identity. The divergence subspace together with the current ideal has basis d_1,d_2,Q,O_3, by the six-density basis. Its quotient is therefore two-dimensional, with basis O_1,O_2; the exact off-shell relation reduces there to $O_2=-4O_F$. In particular O_F is not a divergence, even after the field-equation quotient.

For the tensor assertion, T^1,T^2 already have full orthogonal covariance. In the test family with $\phi'(0)=1$, $T^3_{\alpha\alpha}=p_\alpha^2+\delta_{\alpha 3}$ and every off-diagonal entry is zero. At $p=e_1$, rotating the tensor $\operatorname{diag}(1,0,1,0)$ by $\pi/4$ in the first two coordinates produces a nonzero off-diagonal entry. The tensor T^3 evaluated on the rotated connection still has zero off-diagonal entries. It is therefore not equivariant, and subtracting the equivariant T^1,T^2 proves the claim. □

The conclusion has three distinct meanings. There are two nonrotational directions as local densities, represented by O_1 and Q. In the integral modulo the specified divergences only O_1 remains. In the further algebraic quotient by the field equation it still remains. None of these statements removes a weighted divergence: for a compactly supported test h,

$$\int hQ=\sum_\alpha\int(\partial_\alpha^2 h)\sum_\mu|F_{\alpha\mu}|^2. \tag{6.cdxxiii}$$

This term must be retained when the local Ward calculation uses a nonconstant cutoff.

6.15.5. *Parallel transport and centred plaquettes.* For a piecewise C^1 path $\gamma:[0,T]\to\mathbb{R}^4$, define its transport by the right-ordered differential equation

$$P'(t)=iP(t)\eta(t),\qquad P(0)=I,\qquad \eta(t)=\sum_\mu A_\mu(\gamma(t))\dot\gamma_\mu(t). \tag{6.cdxxiv}$$

Its value at T is denoted $\Pi_\gamma[A]$. Iterating the integral equation gives the absolutely convergent series

$$\Pi_\gamma=I+\sum_{n\ge1}i^n\int_{0<t_1<\cdots<t_n<T}\eta(t_1)\cdots\eta(t_n)\,dt_1\cdots dt_n. \tag{6.cdxxv}$$

The nth term is bounded by $(\int|\eta|)^n/n!$, proving existence and allowing termwise differentiation for smooth finite parameter families on compact parameter sets. Uniqueness follows by iterating the integral equation for a difference, with the same factorial bound. Since η is Hermitian and traceless, $(PP^*)'=0$ and $(\det P)'=i\operatorname{tr}(\eta)\det P=0$. Thus the transport is in SU(2).

Concatenation multiplies transports in path order; reversing a path inverts its transport. These follow either from uniqueness or by reversing the integral equation. Differentiating $u(\gamma(0))P(t)u(\gamma(t))^{-1}$ shows that

$$\Pi_\gamma[A^u] = u(\gamma(0))\Pi_\gamma[A]u(\gamma(T))^{-1}. \tag{6.cdxxvi}$$

The sign in A^u is thereby checked against the transport ordering, rather than chosen independently. The same differentiation gives

$$\frac{d}{dt}\{P(t)X(\gamma(t))P(t)^{-1}\} = P(t)\left(\sum_\mu \dot\gamma_\mu(t)D_\mu X(\gamma(t))\right)P(t)^{-1}. \tag{6.cdxxvii}$$

Fix $\mu < \nu$. Let $c(t)$, $0 \le t \le 4$, traverse the boundary of the unit square centred at zero in the (μ,ν) plane, starting at $-(e_\mu + e_\nu)/2$ and following $+e_\mu, +e_\nu, -e_\mu, -e_\nu$ at unit speed. Extend it periodically. Thus $c(t+2) = -c(t)$. Define

$$W_{\mu\nu}(s,y) = \Pi_{y+sc}[A], \qquad G_{\mu\nu}(s,y) = 1 - \tfrac{1}{2}\operatorname{tr} W_{\mu\nu}(s,y). \tag{6.cdxxviii}$$

These are smooth functions of (s,y) for real s, including $s=0$. Apply the ODE separately on the four sides to see the smooth parameter dependence despite the corners. The trace is real for SU(2). Gauge transformations conjugate W at its base point, so G is gauge invariant.

The centred plaquette has the exact parity

$$G_{\mu\nu}(-s,y) = G_{\mu\nu}(s,y). \tag{6.cdxxix}$$

Indeed changing s to $-s$ starts the same oriented square halfway round its boundary. If its two half-transports are P_1, P_2, the two full transports are P_1P_2 and P_2P_1, whose traces are equal. This is parity of the centred traced plaquette; it does not assert that a matrix-valued transport based at a fixed corner is itself even.

Lemma 6.35 (The pointwise plaquette coefficients)**.** For every smooth connection,

$$G_{\mu\nu}(s,y) = \frac{s^4}{2}|F_{\mu\nu}(y)|^2 + \frac{s^6}{24}\langle F_{\mu\nu}, (D_\mu^2 + D_\nu^2)F_{\mu\nu}\rangle(y) + O(s^8). \tag{6.cdxxx}$$

On a compact set of y the remainder is uniform. For a compactly supported connection it has a common compact support when $|s| \le 1$. ∎

Proof. Use radial gauge about the selected point y. Let $u(y+z)$ be the transport of A along the straight segment from y to $y+z$, and put $B = A^u$. By eq. 6.cdxxvi, every radial segment for B has transport one. Hence

$$z \cdot B(y+z) = 0, \qquad B(y) = 0, \qquad \partial_\lambda B_\kappa(y) = \tfrac{1}{2}F_{\lambda\kappa}(y). \tag{6.cdxxxi}$$

The last two conclusions follow by differentiating the first identity once and twice, and then using the definition of F. This gauge need not be compactly supported; it is used only to identify coefficients at y. The uniform remainder will be estimated using the original plaquette function.

Put $f = F_{\mu\nu}(y)$. In the series eq. 6.cdxxv, the coefficient on the square is

$$\eta_s(t) = s\sum_\kappa B_\kappa(y+sc(t))\dot c_\kappa(t) = s^2\eta_0(t) + O(s^3), \qquad \eta_0(t) = \tfrac{1}{2}(c_\mu\dot c_\nu - c_\nu\dot c_\mu)f. \tag{6.cdxxxii}$$

All leading matrices are multiples of the single f. Let $H = \int_0^4 \eta_s(t)\,dt$ and let W_n denote the ordered integral of n coefficient factors, without i^n. The trace of H is zero. Symmetry of $\mathrm{tr}(XY)$ gives $\operatorname{tr} W_2 = \frac{1}{2}\operatorname{tr} H^2$. The order-$s^6$ part of $\operatorname{tr} W_3$ is zero, because it is proportional to $\operatorname{tr} f^3 = 0$; replacing one leading coefficient by its $O(s^3)$ remainder costs $O(s^7)$. All $n \ge 4$ terms cost $O(s^8)$ by the factorial estimate. Consequently

$$G_{\mu\nu}(s,y) = \tfrac{1}{2}|H|^2 + O(s^7). \tag{6.cdxxxiii}$$

This is where the SU(2) trace algebra is used.

Ordinary Stokes' formula in the square gives $H = \int_{[-s/2,s/2]^2} \psi(y + z_\mu e_\mu + z_\nu e_\nu)\, dz_\mu dz_\nu$, where $\psi = \partial_\mu B_\nu - \partial_\nu B_\mu$. Taylor expansion on the symmetric square yields

$$H = s^2\psi(y) + \frac{s^4}{24}(\partial_\mu^2 + \partial_\nu^2)\psi(y) + O(s^6). \tag{6.cdxxxiv}$$

The odd moments vanish and each quadratic moment equals $s^4/12$. At y one has $\psi = f$. Also, for $i = \mu$ or ν, $\partial_i^2[B_\mu, B_\nu](y) = 0$: both potentials vanish, and in the remaining product one factor is $\partial_i B_i(y) = 0$. For the same reason $D_i^2 F_{\mu\nu}(y) = \partial_i^2 F_{\mu\nu}(y)$. Therefore

$$H = s^2 f + \frac{s^4}{24}(D_\mu^2 + D_\nu^2)F_{\mu\nu}(y) + O(s^6). \tag{6.cdxxxv}$$

Insert this into eq. 6.cdxxxiii. It identifies the coefficients of orders four and six, with an initial $O(s^7)$ remainder. The original function $G(s, y)$ is smooth and even. Thus all odd Taylor coefficients vanish and Taylor's theorem gives an $O(s^8)$ remainder, bounded by $|s|^8 \sup_{|t|\le 1} |\partial_s^8 G(t, y)|/8!$ on $|s| \le 1$. On compact sets this supremum is finite. If A has compact support, any square of side at most one sufficiently far from that support has all its transports equal to one. The plaquette function and all coefficients then vanish. The required uniform support and remainder follow on the fixed enlarged compact set. □

There is a direct noncommuting check on both coefficients. For a connection constant near the square, with $A_1 = r\sigma^1$, $A_2 = r\sigma^2$ and the other components zero, multiplication of the four exponentials gives

$$G_{12}(s, y) = 2\sin^4(sr) = 2r^4 s^4 - \tfrac{4}{3} r^6 s^6 + O(s^8). \tag{6.cdxxxvi}$$

Here $F_{12} = -2r^2\sigma^3$ and $D_1^2 F_{12} = D_2^2 F_{12} = -4r^2 F_{12}$, giving exactly the two coefficients in eq. 6.cdxxx. Ordinary derivatives of F vanish in this test, so replacing the covariant derivatives by ordinary ones would fail the check.

6.15.6. *Summing over the lattice with a controlled remainder.* For a compactly supported connection let its support lie inside $(-\ell/2 + 1, \ell/2 - 1)^4$. Assume $0 < a \le 1$ and $\ell/(2a) \in \mathbb{N}$. Define the torus link variables by transport along each positive bond. Since A vanishes in the boundary collar, a plaquette crossing a torus face agrees with the corresponding square in $\mathbb{R}^4$. The complete unweighted action is therefore exactly

$$A(U) = \sum_{\mu<\nu} \sum_{x \in a\mathbb{Z}^4} G_{\mu\nu}\left(a, x + \tfrac{a}{2}(e_\mu + e_\nu)\right). \tag{6.cdxxxvii}$$

Only finitely many summands are nonzero. The same formula with a weight at the barycentre defines $A(h, U)$.

We record the quadrature estimate so the error in the action is quantitative. For $\varphi \in C_c^\infty(\mathbb{R}^4)$ and any offset $v \in \mathbb{R}^4$,

$$\left| a^4 \sum_{n \in \mathbb{Z}^4} \varphi(a(n + v)) - \int \varphi \right| \le C a^5 \max_i \|\partial_i^5 \varphi\|_{L^1}. \tag{6.cdxxxviii}$$

To prove it, periodize $r \mapsto \varphi(ar)$ on the unit torus. Its Fourier coefficient at $k \in \mathbb{Z}^4$ is $a^{-4}\widehat{\varphi}(2\pi k/a)$, with $\widehat{\varphi}(\xi) = \int e^{-i\xi\cdot x}\varphi(x)\, dx$. The Fourier series converges absolutely after five integrations by parts. Evaluating it at v gives

$$a^4 \sum_n \varphi(a(n + v)) = \sum_{k \in \mathbb{Z}^4} e^{2\pi i k\cdot v}\widehat{\varphi}(2\pi k/a). \tag{6.cdxxxix}$$

For $k \ne 0$ choose a coordinate with $|k_i| = |k|_\infty$ and integrate five times in that coordinate. The term is bounded by $(a/(2\pi|k|_\infty))^5 \|\partial_i^5\varphi\|_1$. The number of points in the shell $|k|_\infty = m$ is $O(m^3)$, so the nonzero-frequency sum converges and proves eq. 6.cdxxxviii. The offset occurs only in phases, giving the asserted uniformity.

Proposition 6.36 (The weighted and unweighted smooth Wilson expansion)**.** For a fixed smooth compactly supported A and a smooth weight h, bounded with its derivatives near that support, one has

$$A(h,U)=\frac{1}{4}\int h\sum_{\mu,\nu}|F_{\mu\nu}|^2+a^2\int h\left(-\frac{O_1}{24}+\frac{Q}{48}\right)+\mathcal{E}_{A,h}(a),\qquad |\mathcal{E}_{A,h}(a)|\le C_{A,h}a^4. \tag{6.cdxl}$$

The constant depends on finitely many smooth norms and a support bound for A,h, and is independent of the torus side, the number of sites and the coupling. In particular,

$$\beta A(U)=\frac{\beta}{4}\int\sum_{\mu,\nu}|F_{\mu\nu}|^2-\frac{a^2\beta}{24}\int O_1+O_A(\beta a^4). \tag{6.cdxli}$$

The divergence is removed only in this unweighted formula. ■

Proof. Denote the coefficients in eq. 6.cdxxx by $c_4^{\mu\nu}$ and $c_6^{\mu\nu}$. They and the remainders have support in one fixed compact set K. There are at most $C(\operatorname{diam}K+1)^4a^{-4}$ relevant mesh points. The sum of the pointwise $O(a^8)$ remainders is consequently $O_{A,h}(a^4)$.

Apply eq. 6.cdxxxviii to $hc_4^{\mu\nu}$ and $hc_6^{\mu\nu}$, with $v=(e_\mu+e_\nu)/2$. The errors in replacing their sums by integrals are $O_{A,h}(a^5)$ and $O_{A,h}(a^7)$ after the additional a^2 factor. Both are bounded by $C_{A,h}a^4$. The leading coefficient satisfies $\sum_{\mu<\nu}c_4^{\mu\nu}=\frac{1}{4}\sum_{\mu,\nu}|F_{\mu\nu}|^2$. For the next coefficient set

$$Y=\sum_{\mu,\nu}\langle F_{\mu\nu},D_\mu^2F_{\mu\nu}\rangle. \tag{6.cdxlii}$$

Pairing the two ordered orientations gives $\sum_{\mu<\nu}c_6^{\mu\nu}=Y/24$ exactly. Applying the product rule twice gives $Q=2O_1+2Y$. Thus the summed pointwise coefficient is $-O_1/24+Q/48$, proving eq. 6.cdxl. For $h=1$, the compact support makes $\int Q=0$. Multiplication by β proves eq. 6.cdxli with its stated dependence on β. □

Equivalently the weighted second coefficient is

$$-\frac{1}{24}\int hO_1+\frac{1}{48}\sum_\mu\int(\partial_\mu^2h)\sum_\nu|F_{\mu\nu}|^2. \tag{6.cdxliii}$$

This displays the source or shell contribution explicitly. It is an integration-by-parts identity for smooth fields, not a reason to omit Q before localizing the action.

The expansion may be differentiated for a specified smooth finite parameter family, but not by differentiating a bare big-O inequality. Suppose $A(t,x),h(t,x)$ are smooth for t in a compact interval, have a common support bound for the connections, and the derivatives used above are uniformly bounded. Apply the Taylor proof directly to $\partial_t^mG(s,y;t)$: this function is still even in s, and its eighth s derivative is uniformly bounded on the compact parameter set. The quadrature proof applies to the differentiated coefficients as well. It follows that for each fixed m,

$$\left|\partial_t^m\mathcal{E}_{A(t),h(t)}(a)\right|\le C_ma^4. \tag{6.cdxliv}$$

The hypotheses are smooth classical family bounds; they are not uniform bounds on fields distributed by the lattice Gibbs law.

6.15.7. *What the classical calculation supplies to the Ward problem.* The classification excludes scalar, antisymmetric and vector partners in the stated dimensions by finite colour and coordinate algebra. It proves that the dimension-six rotational obstruction has two directions as a density, one after the specified divergences are removed, and still one after the field-equation quotient. The Wilson expansion identifies the coefficient of that last direction as $-a^2\beta/24$, while retaining the weighted divergence with coefficient $a^2\beta/48$. These conclusions are off-shell identities or controlled classical asymptotics in their declared test spaces.

To make the bulk term of the interacting rotational Ward identity vanish, one still needs the actual typed insertion estimates, their source formation and normalization, the required treatment of equation-of-motion insertions under the measure, and the shell or volume transfer bounds. A smooth-field expansion is not uniform on the rough lattice fields in that measure. The factors a^2 and the absence of lower-dimensional partners do not, by themselves, prove the quantum estimate. The finite exact Ward identity and the interacting bounds remain distinct parts of that argument.

6.16. **Euclidean evolution from separated moments.** This section proves a conditional reconstruction on compactly supported separated test tuples. The time contraction follows from a bound which is uniform in the positions of finitely many supports. No estimate on arbitrary Schwartz sequences, and no bound uniform in the number of fields, enters that argument. A further lattice approximation hypothesis gives a spectral cone with a cubical spatial section. The round cone and continuous rotations require additional input.

6.16.1. *The precise compact test domain.* Put $\mathcal{D} = C_c^1(\mathbb{R}^4;\mathbb{C})$ and use the norm

$$\|f\|_1 = \max\{\|f\|_\infty, \|\nabla f\|_\infty\}.$$

A separated tuple $X = (f_1, \dots, f_n)$ has compact supports at pairwise positive distance. The empty tuple is denoted by 1. Write $\mathscr{V}$ for the complex vector space of finite formal linear combinations of these tuples. Multilinear and permutation relations may be imposed at the outset; equivalently they will be null relations in the forms below. Concatenation $X \cdot Y$ is used only if the supports of all its entries remain separated. There is no claim that $\mathscr{V}$ is an algebra under unrestricted concatenation.

Our translation convention is $\tau_v f(x) = f(x - v)$, so its support is translated by v. For the time reflection $\theta(x_1, x_2, x_3, x_4) = (x_1, x_2, x_3, -x_4)$ put $\theta f = f \circ \theta$. These maps act entrywise on tuples and linearly on formal combinations. The bar conjugates both the coefficients and the test functions. Let $\mathscr{V}_+$ consist of combinations whose entries have supports compactly contained in $\mathbb{E}_+ = \{x_4 > 0\}$.

Assume normalized symmetric multilinear moments S_n on this separated domain. Write $\sigma(X) = S_n(f_1, \dots, f_n)$, $\sigma(1) = 1$, and extend linearly. They obey the following hypotheses. They preserve conjugation, are translation invariant, and are invariant under the coordinate reflections. Their reflection-positive form satisfies

$$q(F, F) := \sigma(\overline{\theta F} \cdot F) \geq 0 \qquad (F \in \mathscr{V}_+). \tag{6.cdxlv}$$

This is positivity for every finite linear combination, including the empty tuple, rather than just for individual monomials. The concatenations in eq. 6.cdxlv and its polarized form are separated: reflected supports lie strictly below time zero, and the others strictly above it.

For every integer $n \geq 1$, $R < \infty$ and $d > 0$, assume a finite number $M_n(d, R)$ such that

$$|S_n(f_1, \dots, f_n)| \leq M_n(d, R) \prod_{i=1}^{n} \|f_i\|_1, \tag{6.cdxlvi}$$

whenever each support is in a ball of radius R, centered anywhere, and their pairwise distances are at least d. The centers need not lie in a common bounded region. For one entry the separation condition is empty. These are moment bounds, not merely cumulant bounds. Bounds on cumulants give eq. 6.cdxlvi after the finite moment–cumulant sum, if the needed one-point bound is included. There is no restriction here on the growth of M_n as n increases or d decreases.

Proposition 6.37 (Compact-tuple Euclidean reconstruction)**.** Under eq. 6.cdxlv–eq. 6.cdxlvi and the stated invariances, the completion of $\mathscr{V}_+$ by its reflection form is a separable Hilbert space $\mathcal{H}$. The empty tuple has class Ω of norm one. The formulas

$$\begin{gathered} T_t[F] = [\tau_{te_4} F] \\ (t \geq 0), \\ U_x[F] = [\tau_{(x,0)} F] \\ (x \in \mathbb{R}^3) \end{gathered} \tag{6.cdxlvii}$$

define, respectively, a strongly continuous semigroup of positive self-adjoint contractions and a strongly continuous unitary representation. They commute and fix Ω. There is a unique nonnegative self-adjoint operator H with $T_t = e^{-tH}$. Each T_t, $t > 0$, is injective. The spatial generators P_i and H commute strongly. ∎

6.16.2. *The quotient and the contraction proof.* Positivity of the quadratic form gives Hermiticity by polarization and the Cauchy–Schwarz inequality $|q(F,G)|^2 \leq q(F,F)q(G,G)$. Thus $\mathscr{N} = \{F : q(F,F) = 0\}$ is a linear subspace orthogonal to everything. The quotient $\mathscr{V}_+/\mathscr{N}$, with inner product q, is pre-Hilbert; its completion defines $\mathcal{H}$. In particular $\|\Omega\|^2 = \sigma(1) = 1$ and $\langle \Omega, [F] \rangle = \sigma(F)$.

Time and spatial translations in eq. 6.cdxlvii preserve their positive-time domain. Translation invariance and $\theta \tau_{te_4} = \tau_{-te_4} \theta$ give

$$\begin{aligned} q(F, \tau_{te_4} G) &= q(\tau_{te_4} F, G), \\ q(\tau_{(x,0)} F, \tau_{(x,0)} G) &= q(F,G). \end{aligned} \tag{6.cdxlviii}$$

The second identity proves that spatial translations preserve null vectors and induce isometries with inverse U_{-x}. For the first identity, boundedness of the induced time maps remains to be proved; symmetry alone would not suffice.

Fix $F = \sum_\alpha c_\alpha X_\alpha \in \mathscr{V}_+$. There are $\delta > 0$, $d > 0$ and $R < \infty$ such that every entry has time coordinate at least δ, every individual tuple has internal separation at least d, and each entry has support in a ball of radius R. For every $s \geq 0$, every concatenation $\overline{\theta \tau_{se_4} X_\alpha} \cdot \tau_{se_4} X_\beta$ has internal separation at least $d_* = \min\{d, 2\delta\} > 0$. Indeed its two groups lie below $-\delta - s$ and above $\delta + s$. Moving the groups apart changes their centers but not their radii, norms or internal separations. Applying eq. 6.cdxlvi to the finitely many pairs gives

$$\sup_{s \geq 0} \|\tau_{se_4} F\|_q \leq C_F < \infty. \tag{6.cdxlix}$$

The same assertion covers empty entries and constants by normalization. Its essential hypothesis is that the bound is independent of support positions. A bound with a growing enclosing-volume factor would not prove eq. 6.cdxlix.

Write $a(s) = \|\tau_{se_4} F\|_q$. Symmetry and Cauchy–Schwarz give

$$a(t)^2 = q(F, \tau_{2te_4} F) \leq a(0) a(2t).$$

Iteration yields

$$a(t) \leq a(0)^{1-2^{-k}} a(2^k t)^{2^{-k}} \leq a(0)^{1-2^{-k}} C_F^{2^{-k}}. \tag{6.cdl}$$

If $a(0) = 0$, the first inequality already proves $a(t) = 0$. Otherwise let $k \to \infty$ to obtain $a(t) \le a(0)$. Consequently time translations preserve $\mathscr{N}$ and extend to contractions on $\mathcal{H}$. Their symmetry in eq. 6.cdxlviii makes each bounded extension self-adjoint. Moreover

$$q(F, \tau_{te_4} F) = q(\tau_{(t/2)e_4} F, \tau_{(t/2)e_4} F) \ge 0,$$

so every T_t is positive. The semigroup law is inherited from the exact translation law on the dense domain.

6.16.3. *Continuity, generators and elementary consequences.* For $f \in C^1_c$, uniform continuity of ∇f gives

$$\|\tau_h f - f\|_1 \le \max\left\{ |h| \|\nabla f\|_\infty, \sup_{|x-y|\le|h|} |\nabla f(x) - \nabla f(y)| \right\} \longrightarrow 0. \tag{6.cdli}$$

This statement need not hold in the Lipschitz norm for a merely Lipschitz function. To apply it to a moment, enclose the old and slightly moved supports of each entry in one common neighborhood. For sufficiently small shifts these neighborhoods retain half their original positive separation. The multilinear telescoping identity and eq. 6.cdxlvi then bound the change of a moment by a constant times the sum of the C^1 changes of its entries. Differences of functions are evaluated only on those fixed neighborhoods.

Expanding $\|[\tau_h F] - [F]\|^2$ into its finitely many moment terms and using this observation proves convergence to zero for $h = te_4$, $t \downarrow 0$, and for spatial $h \to 0$. The contraction and unitary bounds extend this continuity from the dense quotient domain to all of $\mathcal{H}$. The semigroup and group laws then give strong continuity at every parameter.

Here is a construction of the generator which also rules out a hidden zero-transfer subspace. If $T_1 v = 0$, positivity and $T_1 = T_{1/2}^2$ give $T_{1/2} v = 0$. Repeated halving and strong continuity at zero imply $v = 0$. Thus the spectral measure of T_1 has no atom at zero. Define the densely defined self-adjoint operator $H = -\log T_1$ by functional calculus; its spectrum is nonnegative. Uniqueness of positive roots gives $T_{m/n} = T_1^{m/n}$ for nonnegative rationals, and strong continuity gives $T_t = e^{-tH}$ for all $t \ge 0$. Functional calculus also proves uniqueness of H and injectivity of every T_t, $t > 0$. Since translations fix the empty tuple, $H\Omega = 0$ and $U_x \Omega = \Omega$.

The commuting spatial unitaries have self-adjoint generators $U_x = e^{i \sum_i x_i P_i}$. They commute with T_1 and hence with its spectral projections and with H. The joint spectral theorem, equivalently the spectral theorem for the strongly continuous abelian unitary group $(s, x) \mapsto e^{isH} U_x$, supplies one projection-valued measure E on $[0, \infty) \times \mathbb{R}^3$. Here and below strong commutation means commutation of these spectral projections. For $F, G \in \mathscr{V}_+$, $t \ge 0$ and $x \in \mathbb{R}^3$ the dictionary is

$$\begin{aligned} \sigma(\overline{\theta F}\, \tau_{(x,t)} G) &= \langle [F], e^{-tH} U_x [G] \rangle \\ &= \int e^{-tE + ix\cdot p} \langle [F], \mathsf{E}(dE\, dp)[G] \rangle. \end{aligned} \tag{6.cdlii}$$

For completeness, the quotient is separable. Cover compact positive-time supports by finite unions of rational boxes with closures still in $\mathbb{E}_+$; choose these unions separately for each entry so that they remain separated. There are countably many such finite systems. On each fixed bounded union, the C^1 space is separable: restriction of a function and its four derivatives embeds it isometrically in a finite product of separable continuous-function spaces. Choose countable dense subsets inside the supported C^1 spaces, and rational complex coefficients for formal combinations. Telescoping with eq. 6.cdxlvi, on the fixed separated neighborhoods, shows that the resulting countable set is total in $\mathcal{H}$.

The formal description does not create extra vectors from zero tensors. For a finite homogeneous relation $\sum_\alpha c_\alpha f_{\alpha,1} \otimes \cdots \otimes f_{\alpha,n} = 0$ whose tuples are separated, choose $d > 0$ below all their separations. Localize every entry with one finite smooth partition covering their supports, whose sets have diameter less than $d/8$. In a product of partition sets, either two sets have distance less than $d/2$, in which case every original tensor has a zero factor there, or all the slot sets remain separated. In the latter case the moment is multilinear on the product of their supported function spaces and hence is a linear functional on their algebraic tensor product. The map from this tensor product into functions on the Cartesian product is injective: on the finitely many function spans involved, independent functions can be distinguished by finitely many point evaluations, and the corresponding evaluation matrices invert each slot. Thus the localized zero tensor has zero moment. Summing proves the assertion. Applying it also after concatenation with a reflected test shows that a zero tensor is a reflection-null vector. This is an algebraic compact-support fact, distinct from the Schwartz density input below.

Any realization with a dense image of the same test domain and the same scalar products is therefore canonically unitarily equivalent, by sending each test class to its other realization; the generators intertwine by density. The construction can use any coordinate as time if its RP hypothesis is supplied. For $c \geq 0$ and F supported strictly above time c, the exact identity is $[F] = T_c[\tau_{-ce_4}F]$.

Any further linear symmetry of the moments which preserves positive time and commutes with θ acts isometrically on this quotient. In particular, if the moments also have the spatial signed-permutation symmetry W_3, then $V_w[F] = [wF]$ is unitary, $V_w T_t = T_t V_w$ and $V_w U_x V_w^{-1} = U_{wx}$. Uniqueness of the joint spectral measure gives the corresponding covariance under $(E, p) \mapsto (E, wp)$ and hence W_3 symmetry of the joint spectrum. This optional finite symmetry is not continuous rotation invariance.

Spectral dominated convergence gives $T_t \to P_0 = 1_{\{0\}}(H)$ strongly as $t \to \infty$. Hence the large-time reflected correlations equal the matrix elements of P_0. In particular, $\ker H = \mathbb{C}\Omega$ exactly when their limits, or their Cesàro limits, equal $\overline{\sigma(F)}\sigma(G)$ for every positive-time F, G. If $\sigma(f) = 0$ and $S_2(\theta\bar{f}, f) > 0$ for one positive-time test, then $[(f)]$ is a nonzero vector orthogonal to Ω, so $\dim \mathcal{H} \geq 2$; this requires an actual positive two-point input.

A shift of the fields does not change this representation. More precisely, let m be a complex linear, translation- and reflection-invariant functional on $\mathcal{D}$, satisfying $m(\bar{f}) = \overline{m(f)}$ and bounded on each fixed compact support. Define

$$C_m(f_1, \ldots, f_n) = \sum_{A \subset \{1,\ldots,n\}} \prod_{i \notin A} [-m(f_i)]\, (f_i)_{i \in A},$$

$$\sigma^{\mathrm{c}}(F) = \sigma(C_m F).$$

Then C_{-m} is the inverse of C_m, and on every admissible concatenation $C_m(FG) = C_m(F)C_m(G)$. It commutes with the bar, reflections and translations. Therefore $[F]_{\sigma^{\mathrm{c}}} \mapsto [C_m F]_\sigma$ is a unitary intertwining the two evolutions. Its finite expansion also transfers in eq. 6.cdxlvi, since local boundedness and translation invariance bound m uniformly on balls of each fixed radius. This algebraic statement does not estimate the size of the one-point difference between two cutoff states.

In the particular unweighted Wilson-density normalization, the one-point calibration can be written explicitly. Let two states ρ_j, ν_j on spacing a_j be translation and hypercubic invariant, and put $\mathcal{O}_j(f) = \sum_p f(x(p))a_p$. Their plaquette mean difference $d_j = \rho_j(a_p) - \nu_j(a_p)$ is independent of p. Define

$$R_j(f) = \frac{a_j^4}{6} \sum_p f(x(p)),$$

$$c_j = 6a_j^{-4} d_j.$$

Then $\rho_j(\mathcal{O}_j(f)) - \nu_j(\mathcal{O}_j(f)) = c_j R_j(f)$ exactly, and the six shifted Riemann sums give $R_j(f) \to \int f$. If these one-point differences converge to $m(f)$ on all compact C^1 tests, choose one f_0 with $\int f_0 = 1$. The ratio at f_0 shows $c_j \to m(f_0) = c$, and consequently $m(f) = c \int f$ for every test. A uniform one-point bound at f_0 bounds $|c|$; this bound is an input, not a consequence of translation invariance alone. When the two states coincide, $c_j = c = 0$.

6.16.4. *Coarse lattice inequalities and the continuum cube cone.* The spectral cone needs additional information. Assume infinite-lattice probability measures ν_j at spacings $a_j \downarrow 0$, invariant under lattice translations and reflection positive on the closed side of all coordinate and diagonal lattice mirrors. Their moment forms will be denoted by $S_n^{(j)}$. For periodic Wilson DLR states these hypotheses are proved in Prop. A.11. For other lattice measures they remain explicit assumptions. The normal transfer T_j and tangential translation $V_{j,i}$ in their reflection Hilbert spaces then have the carrier established in the plateau argument:

$$\begin{aligned} \sinh(E/2) &\geq |\sin(p/2)|, \\ T_j &= e^{-E}, \\ V_{j,i} &= e^{ip}. \end{aligned} \tag{6.cdliii}$$

Here E is dimensionless and may be infinity on $\ker T_j$, while p is a circle variable. Only this lattice carrier, rather than a round relativistic cone, is assumed from that argument.

One lattice distinction is useful. For spacing a, the site mirror at zero and the link mirror at $-a/2$ have the same positive-side lattice algebra. Their forms obey $q_{\text{link}}(A, B) = q_{\text{site}}(A, TB)$, by translation invariance. Consequently $[A]_{\text{link}} \mapsto T^{1/2}[A]_{\text{site}}$ is an isometry onto the closure of $\operatorname{Ran} T^{1/2}$, namely $(\ker T)^\perp$. It intertwines the two normal transfers, since this holds on the common local domain. Thus the link transfer is injective, even if the site transfer has a kernel. This proves the relation between the two pictures without discarding site-kernel contributions in eq. 6.cdlv.

For every integer $k \geq 1$, convexity of sinh and the elementary inequality $|\sin(kv)| \leq k|\sin v|$ give

$$\sinh(kE/2) \geq k \sinh(E/2) \geq k|\sin(p/2)| \geq |\sin(kp/2)|. \tag{6.cdliv}$$

The sine inequality follows by induction from the addition formula and $|\cos v| \leq 1$. Thus functional calculus gives the bounded operator inequality

$$(I - T_j^k)^2 \geq T_j^k (2I - V_{j,i}^k - V_{j,i}^{-k}). \tag{6.cdlv}$$

At $T_j = 0$ this inequality holds directly, so no division by a transfer eigenvalue is required. Its quadratic form is a five-point inequality. If $s = ka_j$ and $\Phi_{j,F}(v) = \nu_j(\overline{\theta F_j}\, \tau_v F_j)$, it reads

$$\Phi_{j,F}(0) - 4\Phi_{j,F}(se_4) + \Phi_{j,F}(2se_4) + \Phi_{j,F}(se_4 + se_i) + \Phi_{j,F}(se_4 - se_i) \geq 0. \tag{6.cdlvi}$$

To pass to the limit, use the following approximation between cutoffs. Assume linear local lattice bounded Borel representatives $\iota_j(f)$ for compact C^1 tests. Their supports lie within distance $\epsilon_j \to 0$ of $\operatorname{supp} f$. They commute with coordinate reflection and with translations by $a_j\mathbb{Z}^4$. Let F_j be the product representative of each tuple, extended linearly, and let $S_n^{(j)}$ be its moments. Assume convergence on every fixed separated compact tuple to S_n, and assume eq. 6.cdxlvi for these lattice moments with constants independent of j once j is sufficiently large for the fixed n, d, R. For $\operatorname{tr}(F^2)$ representatives their supremum norms may depend on j; the asserted uniform control is on these moments.

If $v_j \to v$, the supports of $\tau_{v_j} f$ and $\tau_v f$ lie in fixed slightly enlarged sets. Whenever the limiting tuple is separated, these sets retain a positive separation. Telescoping and the uniform moment bound give

$$S_n^{(j)}(f_{1,j}, \dots, f_{n,j}) - S_n^{(j)}(f_1, \dots, f_n) \longrightarrow 0 \tag{6.cdlvii}$$

for such translated tests, by eq. 6.cdli. Together with fixed-tuple convergence, this proves convergence at the moving arguments.

The same argument also explains inheritance of the Hilbert hypotheses from these approximants, when it is needed. A continuum translation is approached by lattice vectors v_j; exact lattice covariance and eq. 6.cdlvii pass its moment identity to the limit. Coordinate reflections through the origin need no rounding. For any fixed $F \in \mathscr{V}_+$ the representatives F_j lie on the positive side for large j, so their finite quadratic RP inequalities pass directly to eq. 6.cdxlv. Normalization, conjugation, multilinearity and the uniform moment bounds pass by their finite defining relations. The lattice mirror hypotheses themselves are supplied by Prop. A.11 in the periodic Wilson application.

There is also a finite-torus scalar-product statement, with a narrower conclusion. If fixed separated moments converge from growing periodic tori and satisfy uniform C^1 bounds on the fixed neighborhoods under consideration, RP at the link mirror $x_4 = a_j/2$ gives

$$q(F, G) = \lim_j \nu_j(\overline{\theta_{a_j/2} F_j}\, G_j).$$

Indeed $\theta_{a_j/2} f = \tau_{a_j e_4} \theta f$, whose difference from θf tends to zero in C^1, and the same telescoping estimate applies. This compares scalar products as numbers; it asserts no maps between the cutoff Hilbert spaces. Finite-torus positivity of this kind does not by itself supply the global lattice semigroup or eq. 6.cdliii.

Fix $t > 0$ and choose integers $k_j \geq 1$ with $k_j a_j \to t$. For positive-time F, its lattice representative is supported on the positive side of the fixed site mirror for large j. All five moment expressions in eq. 6.cdlvi are separated uniformly: the negative reflected group and every translated positive group retain their original positive-time margin. Internal separations are unchanged. Therefore eq. 6.cdlvii passes eq. 6.cdlvi to its continuum version. In $\mathcal{H}$ this is

$$B_{t,i} := (I - T_t)^2 - T_t(2I - U_{te_i} - U_{-te_i}) \geq 0. \tag{6.cdlviii}$$

Initially its form is nonnegative on the dense compact-tuple domain; since $B_{t,i}$ is bounded and self-adjoint, the inequality holds on all of $\mathcal{H}$. No differentiation of a cutoff correlation at $t = 0$ has been used. The fixed-time coarse inequality is passed first.

In the joint functional calculus its multiplier is

$$b_{t,i}(E, p) = 1 - 4e^{-tE} + e^{-2tE} + 2e^{-tE}\cos(tp_i) = 2e^{-tE}[\cosh(tE) + \cos(tp_i) - 2]. \tag{6.cdlix}$$

Take a countable sequence $t_m \downarrow 0$, for example $t_m = 1/m$. Positivity of eq. 6.cdlviii means that the joint projection-valued measure gives zero mass to $\{b_{t_m,i} < 0\}$. Remove the countable union of those sets for the three spatial indices. At every remaining spectral point,

$$0 \leq \lim_{m\to\infty} \frac{\cosh(t_m E) + \cos(t_m p_i) - 2}{t_m^2} = \tfrac{1}{2}(E^2 - p_i^2).$$

Since $E \geq 0$, this proves

$$\mathsf{E}\{(E, p) : E < \max_i |p_i|\} = 0. \tag{6.cdlx}$$

In particular $\operatorname{Dom} H \subset \operatorname{Dom} P_i$ and $\|P_i\psi\| \le \|H\psi\|$ on that domain. The corresponding form-domain statement follows by integrating $|p_i| \le E$ rather than their squares. Zero-energy vectors are spatially invariant. Consequently vectors fixed by all Euclidean time and spatial translations are exactly $\ker H$. This statement concerns invariant Hilbert vectors; identifying them with an ergodic decomposition of correlation states needs the separate phase argument. The cone in eq. 6.cdlx has a cubical spatial section. Nothing here proves $E \ge |p|_2$ or supplies an O(4) action. Finite periodic torus laws cannot simply be substituted for the infinite lattice hypotheses at arbitrarily large separations; any required infinite-volume reflection-positive state must first be obtained.

6.16.5. *What the full Schwartz sequence domain additionally needs.* The compact-tuple construction above does not establish the claimed Schwartz-domain growth property. Here is a precise sufficient additional input for that extension. For $n \ge 2$ put $\Delta_n = \bigcup_{i<j}\{x_i = x_j\}$ and let $\mathcal{S}_{\Delta,n}$ be the Schwartz functions on $\mathbb{R}^{4n}$ whose derivatives of every order vanish on Δ_n. Use $\mathcal{S}_{\Delta,0} = \mathbb{C}$ and the usual Schwartz space for $n = 1$. Define the seminorms

$$p_k(\varphi) = \max_{|\alpha|\le k} \sup_x (1 + |x|^2)^{k/2} |\partial^\alpha \varphi(x)|.$$

The positive-time subspace $\mathcal{S}_{\Delta,+,n}$ consists of functions all of whose derivatives vanish outside $\{x_{1,4} > 0, \dots, x_{n,4} > 0\} \setminus \Delta_n$. This includes flatness at the time boundary, as well as at the diagonals.

The separate continuity/growth input, called H1 here, supplies continuous linear moment functionals Σ_n on these full diagonal-flat spaces, agreeing with the compact smooth separated tensors, and constants $c, C < \infty$, $k \in \mathbb{N}$, $\varkappa < \infty$ such that

$$|\Sigma_n(\varphi)| \le c C^n (n!)^{\varkappa} p_{kn}(\varphi). \tag{6.cdlxi}$$

This is a condition on moments, not just connected functions. It is not a consequence of having a finite $M_n(d, R)$ for every fixed triple of parameters. The Density lemma of § 7.14 proves that finite linear combinations of smooth compact separated tensors are dense in $\mathcal{S}_{\Delta,n}$, and those with strictly positive-time supports are dense in $\mathcal{S}_{\Delta,+,n}$, in their Schwartz subspace topologies. These assertions include the necessary cutoffs at infinity, at the time boundary, and near the diagonals. They cannot be replaced by ordinary tensor density without those constraints.

Using H1 and that density lemma, the finite Schwartz sequence space $\bigoplus_{n\ge 0} \mathcal{S}_{\Delta,+,n}$ has form $\sum_{m,n} \Sigma_{m+n}(\overline{\theta F_m} \otimes G_n)$. The tensor here lies in the diagonal-flat space: an internal coincidence is killed by the corresponding factor, and a coincidence between a negative and a positive time argument forces the common time to zero, where the factors are flat. Tensor multiplication is continuous in the Schwartz topologies. Thus H1 and tensor density transfer the compact reflection positivity and show that its quotient completion is canonically the same $\mathcal{H}$. Compact C^1 tuples themselves are approximated by compact smooth ones inside the same separated positive-time neighborhoods, using eq. 6.cdxlvi. The already proved contraction and unitary groups then extend by this identification. The use of H1 here is its continuity on the enlarged domain; eq. 6.cdlxi also records the additional quantitative growth claim needed when that OS hypothesis is asserted. The compact construction does not supply the factorial constants in H1; the constrained tensor-density proof is given in the cited section.

7. Reconstruction and canonical components

This section passes from the Euclidean functionals to scalar quantum fields. We first construct the spectral and common-domain objects and compare their moments with the reconstructed theory. The compact decomposition then identifies the canonical components and their vacuum spaces. The final reconstruction statement records the precise growth, positivity, covariance and clustering

hypotheses. The intervening observable-algebra results specify the information carried by the scalar field and the additional requirements for extending it to gauge observables.

7.1. **Complex translations and reconstruction domains.** The diagonal-flat growth theorem controls the order of the moment functionals. The Euclidean Hilbert construction gives the Hilbert space and its evolution. We match their domains here, prove the round spectral cone and a joint complex translation theorem for that evolution, and specify the field domain obtained once the Minkowski distributions have been constructed. The translation theorem here has fixed-vector scope. The actual field-product continuation is constructed in § 7.2; the explicit reconstruction norm estimates and the locality theorem are matched in § 7.3.

The exact test-space match. Let $\mathcal{S}_{\Delta,n}$ be the Schwartz functions on $\mathbb{R}^{4n}$ whose every derivative vanishes whenever two four-dimensional arguments coincide. Write

$$p_j(f) = \max_{|\alpha|\le j} \sup_x (1+|x|^2)^{j/2} |\partial^\alpha f(x)|. \tag{7.i}$$

The quantitative flat-extension theorem gives, under its actual source hypotheses, normalized symmetric moments with

$$|S_n(f)| \le A B^n (n!)^{k+32} p_{20n}(f). \tag{7.ii}$$

This is the diagonal-flat version of the growth requirement. If $B_+ = \max(1, B)$, the elementary inequality $n! \ge 2^{n-1}$ gives

$$B^n \le B_+ (n!)^{\log_2 B_+}, \qquad \sigma_n = A B_+ (n!)^{k+32+\lceil \log_2 B_+ \rceil}. \tag{7.iii}$$

Thus $|S_n(f)| \le \sigma_n p_{20n}(f)$, with one linear order coefficient and factorial σ_n. No unquantified change of seminorm order has occurred.

The applicable external result is the corrected scalar theorem $E' \to R'$ with $E0'$, in [OS75, Section IV.1, pp. 287–289]. The exact diagonal-flat seminorm comparison is given in Prop. 7.5. In particular the proved weighted order $20n$ and factorial exponent $k+32$ imply the safe total-monomial order $40n$, with an explicit additional factorial exponent twenty and exponential constant. Thus the actual growth is sufficient without identifying two seminorm conventions by an unspecified dimension-dependent constant. We retain its Euclidean covariance, positivity, symmetry, reality and clustering requirements. The argument here does not replace that result by an assertion about independently continuous correlations or assign an ordinary Euclidean contact prescription.

The positive-time space already used in § 6.16 is denoted by $\mathcal{S}_{\Delta,+,n}$. Its functions are flat outside the region of distinct arguments with all times positive. Define also

$$\begin{aligned} O_n &= \{0 < t_1 < t_2 < \cdots < t_n\}, \\ \mathcal{S}_{\mathrm{ord},n} &= \{f \in \mathcal{S}(\mathbb{R}^{4n}) : \partial^\alpha f = 0 \text{ on } O_n^c \text{ for every } \alpha\}. \end{aligned} \tag{7.iv}$$

The ordered spaces, used in the historical positivity convention [OS73, Sections 2–3], are subspaces of $\mathcal{S}_{\Delta,+,n}$. Set both degree-zero spaces to $\mathbb{C}$, and use finite direct-sum sequences throughout.

Let $\mathcal{K}$ be the completed full positive-time quotient, with its proved form

$$q(F,G) = \sum_{m,n} S_{m+n}(\Theta \overline{F_m} \otimes G_n). \tag{7.v}$$

The historical reflected form also reverses the first group of arguments. Scalar permutation symmetry makes the two forms equal on the ordered subspace. Restriction of eq. 7.v consequently gives an isometry of its ordered quotient into $\mathcal{K}$. Denote its closed range by $\mathcal{K}_{\mathrm{ord}}$.

Proposition 7.1 (The ordered sector and its inherited evolution)**.** The subspace $\mathcal{K}_{\mathrm{ord}}$ contains the vacuum and reduces the positive-time contractions $T_t = e^{-tH}$ and all spatial translations $U_a = e^{ia\cdot P}$. Their restrictions are strongly continuous, have the restricted joint spectral measure, and have the same nonnegative Hamiltonian in functional calculus. Separability, spatial clustering and a one-dimensional kernel of H, when established on $\mathcal{K}$, pass to this sector. ∎

Proof. Simultaneous positive-time translation preserves strict time ordering, as do all simultaneous spatial translations. The corresponding operators therefore preserve the ordered core and its closed range. For a bounded self-adjoint T_t, invariance of a closed subspace M also makes $M^\perp$ invariant: if $x \perp M$ and $y \in M$, then $\langle T_t x, y\rangle = \langle x, T_t y\rangle = 0$. Thus its projection commutes with every T_t, and hence with the spectral projections of H. Both signs of each spatial translation preserve the range, so its projection commutes with the spatial unitaries and their spectral projections as well. This proves the joint spectral restriction and the domain identity

$$\operatorname{Dom} H_{\mathrm{ord}} = \operatorname{Dom} H \cap \mathcal{K}_{\mathrm{ord}}, \qquad H_{\mathrm{ord}} = H|_{\operatorname{Dom} H_{\mathrm{ord}}}. \tag{7.vi}$$

Strong continuity restricts. The degree-zero vector belongs to the ordered sector and still has norm one. A closed subspace of a separable Hilbert space is separable. A clustering matrix-element limit restricts to any pair of its vectors. Finally, a vector annihilated by the restricted Hamiltonian lies in $\ker H$; if the latter is $\mathbb{C}\Omega$, the restricted kernel is exactly that line. □

Equality $\mathcal{K}_{\mathrm{ord}} = \mathcal{K}$ is not needed in this theorem or for these transfers to the reconstructed sector. It is not a Schwartz density statement: evaluation at a configuration with equal positive times and distinct spatial points vanishes on every ordered test, while it need not vanish on a test in $\mathcal{S}_{\Delta,+,n}$. A claim of equality of the completed Hilbert spaces would require an additional argument in the Hilbert seminorm. The imported reconstruction supplies fields on its ordered realization. The theorem above does not extend their field operators or their Lorentz boosts to a larger space.

Implementing spatial rotations on the actual quotient. Suppose now that the established Euclidean functional has spatial SO(3) covariance. This is an explicit input from the full covariance route. Neither the finite hypercubic group nor reflection positivity alone supplies it. For $R \in \mathrm{SO}(3)$ let

$$(r_R f)(\mathbf{x}_1, t_1, \dots, \mathbf{x}_n, t_n) = f(R^{-1}\mathbf{x}_1, t_1, \dots, R^{-1}\mathbf{x}_n, t_n). \tag{7.vii}$$

The pullback preserves all the flat and positive-time spaces, commutes with reflection, and preserves the form in eq. 7.v. It consequently preserves its null space and gives an isometry $V_R[F] = [r_R F]$ with inverse $V_{R^{-1}}$. This defines unitaries on the actual completed quotient.

We check their continuity. For each fixed n, j, the chain rule and the orthogonality of R bound $p_j(r_R f)$ locally uniformly by a constant times $p_j(f)$. Near identity write $R = e^A$, where A is skew and $\|A\| \le C\|R - I\|$. Differentiation along e^{sA} inserts $\sum_i (A\mathbf{x}_i)\cdot\nabla_{\mathbf{x}_i}$, up to the pullback sign. Its coordinate factors consume one weight and its derivative consumes one derivative. Thus

$$p_j(r_R f - f) \le C_{n,j}\|R - I\| p_{j+1}(f) \tag{7.viii}$$

for R in a fixed neighbourhood of identity. Continuity at every rotation follows by composition. Tensor multiplication is continuous in the Schwartz topology, and each S_{m+n} is continuous on its flat domain. Expanding $\|[r_R F] - [F]\|^2 = q(r_R F - F, r_R F - F)$ therefore gives convergence to zero for each finite-sequence class. These classes are dense; unitarity extends the convergence to every vector of $\mathcal{K}$. Hence $R \mapsto V_R$ is a strongly continuous unitary representation. The same proof applies to $\mathcal{K}_{\mathrm{ord}}$, since rotations preserve its core.

On the dense core, the exact coordinate identities are

$$V_R T_t = T_t V_R, \qquad V_R U_a V_R^{-1} = U_{Ra}. \tag{7.ix}$$

They extend to the Hilbert space by boundedness. Uniqueness of the self-adjoint generators gives the unbounded-operator statements

$$\begin{aligned} V_R \operatorname{Dom} H &= \operatorname{Dom} H, & V_R H V_R^{-1} &= H, \\ V_R \operatorname{Dom}(a\cdot P) &= \operatorname{Dom}((Ra)\cdot P), & V_R (a\cdot P) V_R^{-1} &= (Ra)\cdot P. \end{aligned} \tag{7.x}$$

The last form states the convention without an ambiguity about whether a matrix acts on momentum components or on the translating vector. It also identifies the transformed spectral projections.

Proposition 7.2 (The round cone on the Euclidean Hilbert space)**.** Suppose the joint spectral measure E of (H, P) has the established cube carrier $E \geq \max_i |p_i|$. Under the spatial covariance just implemented,

$$\mathsf{E}\{(E,p) : E < |p|_2\} = 0. \tag{7.xi}$$

This holds on the full $\mathcal{K}$ and on its reducing ordered sector. In particular $\mathrm{Dom}\, H \subset \mathrm{Dom}\, |P|$, and $\||P|\psi\| \leq \|H\psi\|$ on that domain. ∎

Proof. Rotating the first coordinate carrier with eq. 7.x gives $\mathsf{E}\{E < |n \cdot p|\} = 0$ for every unit vector $n = Re_1$. Choose a countable dense set n_j on the unit sphere, and remove the countable union of these spectral null sets. At every remaining point, $|n_j \cdot p| \leq E$ for every j. Continuity in n and $\sup_{|n|=1} |n \cdot p| = |p|_2$ prove eq. 7.xi. Integrating $|p|_2^2 \leq E^2$ against a vector's spectral measure gives the stated domain inclusion and norm bound. Integrating $|p|_2 \leq E$ also gives the corresponding form-domain statement. Restriction to a reducing subspace restricts the same measure. □

This argument uses spatial rotations and the already constructed joint spectrum. It does not construct a Lorentz boost.

A joint tube of bounded complex translations. Use the joint measure just proved, and define the open parameter tube

$$\mathcal{T} = \{(z,w) \in \mathbb{C} \times \mathbb{C}^3 : \delta(z,w) := \mathrm{Re}\, z - |\,\mathrm{Im}\, w|_2 > 0\}. \tag{7.xii}$$

Joint functional calculus defines

$$\mathcal{V}(z,w) = \int e^{-zE+iw\cdot p}\, \mathsf{E}(dE\, dp). \tag{7.xiii}$$

On the carrier eq. 7.xi its multiplier has modulus at most $e^{-\delta(z,w)E}$. Thus it is a bounded operator of norm at most one. The notation $e^{-zH+iw\cdot P}$ will mean precisely this bounded operator.

Proposition 7.3 (Complex translations and their boundary)**.** The map $\mathcal{V}$ is jointly holomorphic in operator norm on $\mathcal{T}$. For $a \geq 0$, a three-dimensional multiindex α, and $m = a + |\alpha|$,

$$\|\partial_z^a \partial_w^\alpha \mathcal{V}(z,w)\| \leq m!\, \delta(z,w)^{-m}. \tag{7.xiv}$$

It obeys the addition law and its unitary real boundary is

$$\lim_{\substack{s,y\to 0 \\ s>|y|_2}} \mathcal{V}(s+it, x+iy) = e^{-itH} U_x \quad \text{strongly}. \tag{7.xv}$$

For every pair of Hilbert vectors its matrix coefficients have tempered boundary values in (t,x), with Fourier carrier $E \geq |p|_2$ in the convention $e^{-itE+ix\cdot p}$. Every assertion also holds on $\mathcal{K}_{\mathrm{ord}}$. ∎

Proof. The differentiated multiplier is $(-E)^a (ip)^\alpha e^{-zE+iw\cdot p}$. Its absolute value is at most $E^m e^{-\delta E}$, and

$$E^m e^{-\delta E} \leq m! \delta^{-m}, \tag{7.xvi}$$

as follows from the m-th term of the exponential series. This gives eq. 7.xiv. It also justifies norm differentiation: on a small parameter ball with $\delta \geq \delta_0 > 0$, the scalar exponential Taylor remainder after its linear term is bounded by $C|h|^2 E^2 e^{-\delta_0 E/2}$. Its spectral supremum tends to zero quadratically in $|h|$. The same argument after any fixed number of derivatives proves joint complex differentiability and local norm-convergent Taylor series.

Multipliers multiply, so $\mathcal{V}(z,w)\mathcal{V}(z',w') = \mathcal{V}(z+z', w+w')$; the sum stays in the tube. For the boundary in eq. 7.xv, the multipliers converge pointwise on the spectrum and remain bounded by one. Spectral dominated convergence proves strong convergence for every vector, along every such approach. In particular there is no choice of regulator path in this bounded-operator statement.

For fixed ξ, η, let $\mu_{\xi,\eta}(A) = \langle \xi, \mathsf{E}(A)\eta\rangle$. Its total variation is at most $\|\xi\|\|\eta\|$: for a finite disjoint partition, apply Cauchy–Schwarz to $\sum_j \|\mathsf{E}(A_j)\xi\|\|\mathsf{E}(A_j)\eta\|$. Therefore

$$\langle \xi, e^{-itH} U_x \eta\rangle = \int e^{-itE+ix\cdot p}\, d\mu_{\xi,\eta}(E,p) \tag{7.xvii}$$

is bounded and continuous. For a Schwartz test g on $\mathbb{R}^4$, its distributional pairing has absolute value at most

$$\|\xi\|\|\eta\| \int |g(a)|\, da \le C\|\xi\|\|\eta\| p_5(g). \tag{7.xviii}$$

Dominated convergence with this integrable majorant proves the boundary convergence in $\mathcal{S}'$ as well. The measure in eq. 7.xvii is supported on the forward cone, proving its stated carrier. The reducing-subspace result follows from the common restricted spectral calculus. □

There is also an explicit dense domain for the translation generators:

$$\mathcal{D}_{\text{heat}} = \bigcup_{s>0} \operatorname{Ran} e^{-sH}. \tag{7.xix}$$

The ranges are nested, so this is a linear space; strong continuity at zero makes it dense. The cone and eq. 7.xvi give, for $v = e^{-sH}\eta$,

$$v \in \operatorname{Dom}(H^a P^\alpha), \qquad \|H^a P^\alpha v\| \le m! s^{-m} \|\eta\|. \tag{7.xx}$$

It is invariant under the real translations and spatial rotations. It is also invariant under each generator: reserve half of the heat factor and apply the bound to the other half. Thus generator and domain assertions in this bridge do not depend on a formal exponential with an unspecified domain. Field invariance of this heat domain is not asserted.

The field-product reconstruction beyond this lemma. The coefficients in eq. 7.xvii have a finite measure and hence a simple boundary. The fixed-vector translation theorem alone does not construct arbitrary field products. In particular the strict-gap insertion bound in eq. 6.ccxlii does not permit globally undoing e^{-sH} or putting $s = 0$ in an unbounded field formula.

§ 7.2 supplies the additional construction under the actual UI/PD inputs. It defines the bounded sharp-time Riesz family, removes its heat factors only on finite energy bands, and constructs the spacetime fields on a common energy domain. Its exact sewing, simultaneous forward-tube continuation and full cone tempered boundary apply to arbitrary field products. It also derives Lorentz covariance from actual G2. These results extend the fixed-vector statement in eq. 7.xiv to field products.

§ 7.3 checks the hypotheses of the corrected reconstruction theorem by explicit seminorm estimates. Analytic uniqueness identifies the constructed moments with those of the theorem, giving local commutativity and the stronger linear output-order estimate. The theorem does not assert preservation of the Euclidean factorial exponent. Nor does either construction assign ordinary Euclidean contact values: the diagonal-flat input continues to have its stated domain.

The common field domain after a Minkowski boundary exists. We state the operator hypotheses explicitly. Suppose the continuation step has produced tempered distributions W_n on all of $\mathcal{S}(\mathbb{R}^{4n})$, with $W_0 = 1$, and the positive Wightman form. This is a new assumption in the rest of this section. Let

$$\begin{aligned} \mathcal{B} &= \bigoplus_{n\ge 0}^{\text{finite}} \mathcal{S}(\mathbb{R}^{4n}), \qquad (FG)_n = \sum_{r+s=n} F_r \otimes G_s, \\ F_n^*(x_1,\dots,x_n) &= \overline{F_n(x_n,\dots,x_1)}, \qquad \omega(F) = \sum_n W_n(F_n). \end{aligned} \tag{7.xxi}$$

The empty component is the unit. Positivity means $\omega(F^*F) \ge 0$ for every finite sequence, with no separation restriction. Its polarized form is $\langle F, G\rangle = \omega(F^*G)$, and its null space is $\mathcal{N}_W = \{F : \omega(F^*F) = 0\}$.

Proposition 7.4 (The field-domain construction)**.** The completion of $\mathcal{B}/\mathcal{N}_W$ has a common dense invariant field domain $\mathcal{D} = \mathcal{B}/\mathcal{N}_W$. For a one-variable Schwartz test f, left multiplication by its degree-one sequence defines $\phi(f)$ there. It is linear in f, satisfies $\phi(f) \subset \phi(\bar{f})^*$, and is closable. For each $\eta \in \mathcal{D}$, the map $f \mapsto \phi(f)\eta$ is continuous from Schwartz space to the Hilbert space. The vacuum is cyclic for the field polynomials. Distributional Poincare covariance and spacelike commutativity of W, when supplied, give the corresponding operator identities on $\mathcal{D}$. ■

Proof. Positivity gives Cauchy–Schwarz. A null element A is therefore orthogonal to every element. For arbitrary B,

$$\|BA\|^2 = \omega(A^*B^*BA) = \langle A, B^*BA\rangle = 0. \tag{7.xxii}$$

All expressions are finite algebra products. Thus $\mathcal{N}_W$ is a left ideal, which is exactly what is required for left multiplication to be well-defined on the quotient. Let $\iota(f)$ denote the degree-one sequence. Then

$$\phi(f)[G] = [\iota(f)G], \qquad \langle [F], \phi(f)[G]\rangle = \langle \phi(\bar{f})[F], [G]\rangle. \tag{7.xxiii}$$

The domain is dense by construction and invariant under every such left multiplication. The adjoint domain of $\phi(f)$ contains this same dense domain, by the second identity. Hence $\phi(f)$ is closable. For real f it is symmetric on $\mathcal{D}$; essential self-adjointness has not been inferred.

Fix $\eta = [G]$, with finitely many nonzero degrees. Its image norm squared is a finite sum of terms

$$W_{r+s+2}(G_r^* \otimes \bar{f} \otimes f \otimes G_s). \tag{7.xxiv}$$

Temperedness of each distribution and continuity of tensor multiplication give, for this fixed G, an integer j and constant C_G such that the absolute sum is at most $C_G p_j(f)^2$. Thus $\|\phi(f)\eta\| \le C_G^{1/2} p_j(f)$, proving the asserted Hilbert-valued distribution continuity. The order and constant here may depend on the fixed vector; no degree-independent operator norm has been postulated.

Finite sums of ordinary Schwartz tensors are dense in every $\mathcal{S}(\mathbb{R}^{4n})$. This can be seen directly by cutting off infinity and using Fourier series on a product box, whose partial sums converge in every Schwartz seminorm. The norm of a homogeneous degree-n vector is continuous under this convergence, because its squared norm uses the single tempered distribution W_{2n} and a tensor product. Consequently the polynomial vectors $\phi(f_1)\cdots\phi(f_n)\Omega$ have dense span. Their span is itself a common dense invariant domain if that smaller domain is preferred.

A scalar Poincare pullback α_g is an algebra automorphism of $\mathcal{B}$ commuting with the involution. If $\omega \circ \alpha_g = \omega$, the prescription $\mathcal{U}_g[F] = [\alpha_g F]$ gives a unitary with its exact inverse. The usual Schwartz continuity of affine pullbacks and the same finite form expansion prove strong continuity. On the common domain the algebra identity gives

$$\mathcal{U}_g \phi(f) \mathcal{U}_g^{-1} = \phi(\alpha_g f). \tag{7.xxv}$$

This derives the implementation once distributional covariance has been supplied; it does not derive a Lorentz identity from the spatial rotation argument.

Finally suppose the Minkowski distributions give the adjacent commutator identity for tests f, h with spacelike-separated supports and arbitrary spectator tests. Its pairing with F^* and G says

$$\langle [F], [\phi(f), \phi(h)][G]\rangle = 0 \quad (F, G \in \mathcal{B}). \tag{7.xxvi}$$

The commutator maps the common domain into itself. Density of the test classes therefore makes it zero there. This proves operator locality from that distributional identity. □

7.2. **Tempered fields from analytic insertions.** We construct spacetime-smeared fields from the actual Euclidean insertion estimates. The construction starts with the bounded analytic density of § 6.10, derives a compatible sharp-time Riesz family, and removes its heat factors only on finite energy bands. Temporal smearing then gives a common invariant dense field domain and positive tempered vacuum moments. Exact Euclidean sewing also gives a joint positive-gap continuation

and its tempered real-time boundary. No unrestricted Minkowski moment sequence is assumed in this construction.

The hypotheses are those already declared for UI and PD: the actual ordinary-source and symmetry inputs, the separated-moment continuity, and the reflection-positive Hilbert evolution with its exact cube cone. Write $T_s = e^{-sH}$, $U_x = e^{ix\cdot P}$, and let $\mathcal{K}_{\mathrm{ord}} \subset \mathcal{K}$ be the reducing ordered sector of § 7.1. The round cone is available when the spatial rotation hypothesis of that section is supplied. All estimates below can first be made on $\mathcal{K}$ and then restricted to $\mathcal{K}_{\mathrm{ord}}$. The field's cyclic space will be identified with this ordered sector.

The sharp family is a defined bounded form. The PD input is an actual operator-norm analytic function $V_\eta(t,x)$, uniform in real $x \in \mathbb{R}^3$, with

$$\|V_\eta(t,x)\| \le C\eta^{-N}, \qquad N = p+5 = 9, \tag{7.xxvii}$$

on its retained complex neighbourhood. The value $N = 9$ uses the proved UI exponent $p = 4$. eq. 6.cclv identifies this density with the original insertion form between pulled-down outside polynomials. For $0 < s \le 1/4$ and $h \in L^1(\mathbb{R}^3)$, define

$$\Phi_s(h) = \int h(x) V_{4s}(0,x)\, d^3x, \qquad \|\Phi_s(h)\| \le C s^{-N} \|h\|_1. \tag{7.xxviii}$$

The weak integral is defined by bounded sesquilinear forms and Riesz representation. Norm continuity of the density also permits a Bochner integral. This definition uses no unsandwiched sharp-time operator.

We verify the identification and consistency needed later. Choose a fixed nonnegative even real $\chi \in C_c^\infty(\mathbb{R})$, with integral one, and set $\chi_\epsilon(t) = \epsilon^{-1}\chi(t/\epsilon)$. This temporal approximate identity has $\|\chi_\epsilon\|_1 = 1$ and support shrinking to zero. For compact smooth h, the actual regularized insertion operator for $k_\epsilon(t,x) = \chi_\epsilon(t)h(x)$ has core form

$$\langle [A], \Phi_s(k_\epsilon)[B]\rangle = I(\tau_s A; k_\epsilon; \tau_s B), \qquad \Phi_s(k_\epsilon)[B] = [(\tau_s k_\epsilon)\cdot \tau_{2s} B]. \tag{7.xxix}$$

Here $\tau_s = \tau_{se_4}$, and the second identity is exactly the eq. 6.ccxlvi vector identity, extended to a compact spatial profile by its proved density formula. The two time supports in the displayed product are separated for small ϵ. eq. 6.cclv gives $\Phi_s(k_\epsilon) = \int \chi_\epsilon(t)h(x)V_{4s}(t,x)\, dt, dx$. A temporal Cauchy derivative of eq. 7.xxvii is bounded uniformly in real x, so

$$\|\Phi_s(k_\epsilon) - \Phi_s(h)\| \le C s^{-N-1}\epsilon \|h\|_1. \tag{7.xxx}$$

Thus the sharp limit is an operator-norm limit of actual core insertions. Approximation in spatial L^1 extends this conclusion to the domain in eq. 7.xxviii.

Testing eq. 7.xxix after an additional time shift of both outside polynomials gives

$$\Phi_{s+r}(h) = T_r \Phi_s(h) T_r \quad (0 < s < s+r \le 1/4). \tag{7.xxxi}$$

Indeed $\langle A, T_r\Phi_s(k)T_r B\rangle$ is the insertion form with outside entries shifted by $s+r$. PD's compatibility on the overlapping pulled-down domains identifies it with the $s+r$ form. Pass to the norm limit in eq. 7.xxx, then to spatial L^1 labels. Reflection with the even approximants and spatial translation invariance give, by the same dense-form calculation,

$$\Phi_s(h)^* = \Phi_s(\bar{h}), \qquad U_y \Phi_s(h) U_y^{-1} = \Phi_s(\tau_y h). \tag{7.xxxii}$$

If spatial rotations are supplied, their actual quotient unitaries similarly intertwine $\Phi_s(h)$ and $\Phi_s(r_R h)$.

For larger s, define Φ_s using the right side of eq. 7.xxxi with any smaller fixed heat parameter. The small-parameter identity proves independence of that choice and then eq. 7.xxxi for all $s, r > 0$. Its safe global estimate is $C\min(s,1/4)^{-N}\|h\|_1$. Only the small-s polynomial estimate is used below; no decay as $s \to \infty$ is required.

The ordered sector is preserved. To see the domain explicitly, compact tests supported inside $0 < t_1 < \cdots < t_n$ can be covered by finitely many product boxes with strictly ordered time intervals. A product-box Fourier approximation gives sums of ordered tensors. For a general ordered Schwartz test, first cut off infinity and then cut out the time boundary and adjacent equal-time hyperplanes. Flatness on those boundaries makes the cutoff errors tend to zero in every Schwartz seminorm by Taylor's remainder. Thus such compact ordered tensors are dense in the ordered quotient norm. For each of them, eq. 7.xxix prepends an insertion near time s and shifts all old entries by $2s$, giving another ordered vector. The bounded norm limit therefore preserves $\mathcal{K}_{\mathrm{ord}}$. The adjoint in eq. 7.xxxii does the same, so this subspace reduces every $\Phi_s(h)$.

Removing heat on finite bands. Let $P_m = \mathbf{1}_{[m,m+1)}(H)$, $m = 0, 1, \ldots$. Their orthogonal sum is identity. The notation

$$A_{mn}(h) = (e^{sH} P_m)\Phi_s(h)(P_n e^{sH}) : P_n\mathcal{K} \longrightarrow P_m\mathcal{K} \tag{7.xxxiii}$$

uses bounded outer operators, of norms at most $e^{s(m+1)}$ and $e^{s(n+1)}$. If a larger heat parameter is used, eq. 7.xxxi cancels the extra factors on these same finite bands. Hence eq. 7.xxxiii is independent of $s > 0$. Taking $s = [8(1+m+n)]^{-1}$ in eq. 7.xxviii gives

$$\|A_{mn}(h)\| \le C_A(1+m+n)^N \|h\|_1, \qquad A_{mn}(h)^* = A_{nm}(\bar{h}). \tag{7.xxxiv}$$

All spatial covariance identities restrict to these blocks because the spatial unitaries commute with H. The same construction can be made at a spatial point using $\Phi_s(x) = V_{4s}(0, x)$. Compatibility and the block bound hold pointwise in x, by continuity and eq. 7.xxxi tested against spatial labels, and $A_{mn}(h) = \int h(x) A_{mn}(x)\, dx$.

The unregularized block matrix in eq. 7.xxxiv need not define a Hilbert-space operator at a sharp time. Its polynomially growing columns have not been shown square summable. We next define the spacetime-smeared operator, for which the missing summability is supplied by time integration.

General spacetime tests and the block estimate. For $g \in \mathcal{S}(\mathbb{R} \times \mathbb{R}^3)$, write $g_t(x) = g(t, x)$. The bounded linear map $h \mapsto A_{mn}(h)$ on spatial L^1 makes $t \mapsto A_{mn}(g_t)$ differentiable in operator norm, with derivatives $A_{mn}(\partial_t^j g_t)$. Define directly

$$B_{mn}(g) = \int_{\mathbb{R}} e^{itH} A_{mn}(g_t) e^{-itH}\, dt, \qquad \mathfrak{t}_q(g) = \sum_{j=0}^{q} \|\partial_t^j g\|_{L^1(\mathbb{R}^4)}. \tag{7.xxxv}$$

This is a norm integral between the fixed bands. It is already a definition for arbitrary spacetime Schwartz tests, not just tensors.

Pull out the scalar phase $e^{it(m-n)}$. The remaining generators $(H-m)|_{P_m\mathcal{K}}$ and $(H-n)|_{P_n\mathcal{K}}$ have norm at most one. A derivative of their conjugation costs at most two. Leibniz's rule followed by q integrations by parts therefore gives, when $|m-n| \ge 1$, the bound with factor $3^q|m-n|^{-q}\mathfrak{t}_q(g)$. Boundary terms vanish by the Schwartz decay in temporal L^1. For $m = n$ use the original norm integral. Together these estimates give the explicit safe bound

$$\|B_{mn}(g)\| \le C_A 6^q (1+m+n)^N (1+|m-n|)^{-q} \mathfrak{t}_q(g). \tag{7.xxxvi}$$

The adjoint blocks satisfy $B_{mn}(g)^* = B_{nm}(\bar{g})$. The proof used actual derivatives of a bounded map into fixed energy bands and did not differentiate an unbounded sharp-time field.

Put $W = 1 + H$. On the m-th band, powers of W have norm at most $(m+2)^k$, while $\|W^{-l}P_n\| \le (n+1)^{-l}$. The elementary comparison

$$\frac{(m+2)^k (1+m+n)^N}{(n+1)^{N+k}} \le 2^{N+k}(1+|m-n|)^{N+k} \tag{7.xxxvii}$$

shows exactly how the energy weights are paid. Set $q = N+k+2$, and let $Z_2 = 2\sum_{d\ge 0}(1+d)^{-2} < \infty$. The scalar row and column sums of the weighted blocks are at most $C_A 2^{N+k} 6^{N+k+2} Z_2 \mathrm{t}_{N+k+2}(g)$. Block Schur consequently gives

$$\|W^k \phi(g) W^{-(N+k)}\| \le C_k \mathrm{t}_{N+k+2}(g) \le C'_k p_{N+k+7}(g). \tag{7.xxxviii}$$

For the last inequality use the ordinary four-dimensional Schwartz seminorm: its weight of order at least five is integrable, and it controls all the indicated temporal derivatives. These are actual seminorm estimates for non-tensor spacetime tests.

Here is the operator and domain construction behind eq. 7.xxxviii. For $k = 0$, the block matrix $B_{mn}(g)W^{-N}P_n$ defines a bounded operator by Schur. Multiply it on the right by W^N on $\operatorname{Dom} W^N$ to define $\phi(g)$. For finite-energy vectors its blocks are exactly eq. 7.xxxv. For $\xi \in \operatorname{Dom} W^{N+k}$, approximate by $Q_R\xi$, where $Q_R = \sum_{m=0}^{R} P_m$. The weighted Schur estimate makes both $\phi(g)Q_R\xi$ and $W^k\phi(g)Q_R\xi$ converge. Closedness of W^k proves the claimed domain inclusion and the operator identity in eq. 7.xxxviii.

It follows that

$$\mathcal{D}_\infty = \bigcap_{k\ge 0} \operatorname{Dom} H^k \quad \text{is dense and} \quad \phi(g)\mathcal{D}_\infty \subset \mathcal{D}_\infty. \tag{7.xxxix}$$

Finite-energy vectors make it dense. The adjoint-block identity, first on such vectors and then by their graph-norm approximation, gives

$$\langle \xi, \phi(g)\eta\rangle = \langle \phi(\bar g)\xi, \eta\rangle \quad (\xi,\eta \in \mathcal{D}_\infty). \tag{7.xl}$$

Thus each field restricted to this common invariant domain is closable, and it is symmetric for real tests. eq. 7.xxxviii also proves continuity into every graph seminorm for each fixed domain vector. Essential self-adjointness is not inferred.

All energy projections, sharp insertions and real translations reduce $\mathcal{K}_{\mathrm{ord}}$. The same block limits therefore preserve $\mathcal{D}_\infty \cap \mathcal{K}_{\mathrm{ord}}$. Spatial covariance and time smearing give exact domain identities

$$\begin{aligned} \mathscr{U}(t,y) = e^{itH}U_y, \qquad & \mathscr{U}(t,y)\mathcal{D}_\infty = \mathcal{D}_\infty, \\ \mathscr{U}(t,y)\phi(g)\mathscr{U}(t,y)^{-1} &= \phi(\tau_{(t,y)}g). \end{aligned} \tag{7.xli}$$

Spatial rotations have the analogous covariance when supplied. Our time convention is fixed by the positive sign in e^{itH}. With the standard Minkowski pairing $Et - p\cdot x$, its spatial momentum is $-P$; the round cone is unchanged by this sign.

Positive tempered moments with an explicit extension. The vacuum belongs to $\mathcal{D}_\infty$, and eq. 7.xxxviii can be iterated. Normalize $W_0 = \|\Omega\|^2 = 1$. For each fixed n,

$$\|\phi(g_1)\cdots\phi(g_n)\Omega\| \le D_n \prod_{j=1}^{n} p_{M_n}(g_j), \qquad M_n = nN + 7. \tag{7.xlii}$$

Indeed the energy exponent needed at the next factor increases by N at each step; the largest test order is $nN+7$. Define the vacuum moments on tensors by $W_n(g_1\otimes\cdots\otimes g_n) = \langle\Omega, \phi(g_1)\cdots\phi(g_n)\Omega\rangle$. Multilinearity factors through the algebraic tensor product. Its map into functions is injective on every finite collection: choose point evaluations distinguishing the finite-dimensional function span in each slot and invert those evaluation matrices. Tensor relations therefore introduce no ambiguity.

We prove a fixed-order tempered estimate explicitly. Take a smooth unit product-grid partition and expand each localized test on period-eight boxes in every four-dimensional slot. Multiply each Fourier wave by a fixed larger compact cutoff. Its one-slot p_M norm is at most $C_M(1+|z|)^M(1+|l|)^M$, where $z \in \mathbb{Z}^4$ is the box center and $l \in \mathbb{Z}^4$ its Fourier index. Choose $2r = 2\lceil (M_n+6)/2\rceil$, and integrate $(1-\Delta_i)^r$ in each slot when estimating the coefficients. This gives the factor $\prod_i (1+|l_i|^2)^{-r}$. Derivatives of the fixed cutoffs cost a finite constant depending on n, r. With

$$K_n = n(M_n + 8), \tag{7.xliii}$$

the seminorm $p_{K_n}(F)$ pays both their total derivative order $2rn \le n(M_n+7)$ and the per-slot location weight $(1+|z_i|)^{-(M_n+8)}$. The latter follows from $1+\sum_i |x_i|^2 \ge \prod_i (1+|x_i|^2)^{1/n}$, with fixed-box weight comparisons. After the tensor bound in eq. 7.xlii, the absolute series is controlled by the products of

$$\sum_{z\in\mathbb{Z}^4} (1+|z|)^{-8} < \infty, \qquad \sum_{l\in\mathbb{Z}^4} (1+|l|)^{M_n}(1+|l|^2)^{-r} < \infty. \tag{7.xliv}$$

The second sum converges since $2r - M_n \ge 6 > 4$. The Fourier series of the localized smooth tests converges in every required seminorm. For a tensor its coefficients factor, so the absolutely summed value agrees with its original multilinear value. This constructs the unique continuous extension, with

$$|W_n(F)| \le D'_n p_{K_n}(F) \quad (F \in \mathcal{S}(\mathbb{R}^{4n})). \tag{7.xlv}$$

The same absolute series, using the vector bound in eq. 7.xlii, constructs the Hilbert-valued vacuum-product distribution. Higher graph seminorms are treated by starting the iteration of eq. 7.xxxviii at the corresponding energy exponent.

Thus W_n is tempered at every fixed n. The displayed order $K_n = O(n^2)$ is sufficient for that assertion; no linear order bound or preserved factorial moment exponent is claimed here. For finite tensor sequences, eq. 7.xl gives

$$\sum_{r,s} W_{r+s}(F_r^* \otimes F_s) = \Big\| \sum_s \phi(F_s)\Omega \Big\|^2 \ge 0. \tag{7.xlvi}$$

Here F_r^* reverses and conjugates its arguments, and for a tensor $\phi(F_s)\Omega$ means its field product. The Hilbert-valued extension just constructed defines the same vector for a general Schwartz F_s. Ordinary tensor density and eq. 7.xlv then prove eq. 7.xlvi on all finite unrestricted Schwartz sequences. This proves the required positive Minkowski moment functional from the actual insertion input.

Exact Euclidean words and joint positive-gap analyticity. Let $z = (z_1, \dots, z_{n-1})$ have positive real components and take spatial labels $h_j \in L^1(\mathbb{R}^3)$. Choose small $a_j > 0$ with $a_j + a_{j+1} < \mathrm{Re}\, z_j$. Define

$$\begin{aligned} \mathcal{E}_n(z;h) &= \langle \Omega, \Phi_{a_1}(h_1)\mathcal{P}(z,a;h)\Omega\rangle, \\ \mathcal{P}(z,a;h) &= \prod_{j=1}^{n-1} \left[T_{z_j - a_j - a_{j+1}} \Phi_{a_{j+1}}(h_{j+1}) \right]. \end{aligned} \tag{7.xlvii}$$

The product is ordered by increasing j from left to right and is identity for $n = 1$. Here $T_z = e^{-zH}$ is the actual bounded analytic semigroup. Changing a_j to a smaller b_j inserts the two factors $T_{a_j - b_j}$ of eq. 7.xxxi; they join their adjacent heat factors and give precisely the formula with b_j. The endpoint heat factors fix Ω. Two arbitrary allowed choices are compared through a common smaller choice. Thus eq. 7.xlvii is independent of every auxiliary parameter.

On a neighbourhood of any positive-real-gap point, keep the a_j fixed. The formula is a finite product of bounded norm-holomorphic operators. The matching identities just proved glue these formulas on overlaps, giving joint analyticity on $\mathrm{Re}\, z_j > 0$ for every gap simultaneously. With

$$r = \min\{1/8, \tfrac{1}{4} \min_j \mathrm{Re}\, z_j\}, \qquad |\mathcal{E}_n(z;h)| \le C^n r^{-Nn} \prod_j \|h_j\|_1, \tag{7.xlviii}$$

the bound follows by taking every $a_j = r$. When $n = 1$, fix $a_1 = 1/8$; its vacuum value is independent of that choice.

We verify that this is the actual Euclidean continuation. For real ordered times $\tau_1 < \cdots < \tau_n$, replace every sharp label by $\chi_\epsilon(t) h_j(x)$ in the allowed core formula. Starting at the rightmost factor, repeated use of eq. 7.xxix puts the first insertion at a_1 and the later ones at $a_1 + (\tau_j - \tau_1)$, with the corresponding narrow profiles. A common translation by $\tau_1 - a_1$ identifies its vacuum expectation

with the original Euclidean moment. Operator-norm convergence in eq. 7.xxx therefore identifies the point-time limit with $\mathcal{E}_n(\tau_{j+1}-\tau_j;h)$.

For compact temporal profiles in disjoint ordered intervals there is a fixed positive gap. Choose the auxiliary heat parameters uniformly below that gap. Approximate the profiles by their convolutions with the narrow kernels. The original moments converge by local C^1 continuity, while eq. 7.xlviii permits Fubini and dominated convergence on the operator side. This recovers their exact original moment values. Product-box Fourier approximation gives the same identification for general compact ordered tests, not only time-space tensors. No diagonal Euclidean contact prescription is involved.

The corresponding vector formula, when all absolute times are positive, is

$$T_{\tau_1-a_1}\Phi_{a_1}(h_1)\mathcal{P}((\tau_{j+1}-\tau_j)_j,a;h)\Omega, \tag{7.xlix}$$

with $a_1<\tau_1$. The same core argument and norm limits identify its ordered-profile integrals with the actual ordered Euclidean vectors. It is a bounded formula throughout the retained positive gaps.

A joint tempered boundary, not separate limits. For positive gap regulators ϵ_j, define on tensor tests

$$W_n^\epsilon(g_1\otimes\cdots\otimes g_n)=\langle\Omega,\phi(g_1)T_{\epsilon_1}\phi(g_2)\cdots T_{\epsilon_{n-1}}\phi(g_n)\Omega\rangle. \tag{7.l}$$

Every heat contraction has norm at most one in every energy graph norm. The proof of eq. 7.xlii–eq. 7.xlv therefore gives, uniformly in all $\epsilon_j\ge 0$,

$$|W_n^\epsilon(F)|\le D_n' p_{K_n}(F),\qquad W_n^\epsilon(F)\longrightarrow W_n(F)\quad\text{as all }\epsilon_j\downarrow 0. \tag{7.li}$$

For tensors the convergence follows by replacing the finitely many heat factors one at a time. Strong convergence in every graph norm, combined with eq. 7.xxxviii, controls each term. For an arbitrary Schwartz test, first approximate it by finite tensors in p_{K_n}. The uniform bound controls the two approximation errors, and tensor convergence controls the fixed approximation. This proves joint, regulator-path-independent convergence in $\mathcal{S}'$.

To identify it as the boundary of eq. 7.xlvii, first insert the finite total-energy cutoff Q_R at every field. All generators and field blocks are then bounded, so time integration gives exactly

$$z_j=\epsilon_j-i(t_{j+1}-t_j) \tag{7.lii}$$

in that formula. For fixed positive regulators, its intermediate energy-band sums are absolutely convergent: eq. 7.xxxiv contributes only fixed polynomial factors, while every intermediate energy is damped by its positive gap. The convergence of the cutoff expressions is uniform in the real times and bounded by an integrable product of the spatial L^1 norms of the time slices of the tests. Fubini and dominated convergence therefore identify their integrals with eq. 7.l. On the field side, the cutoff products converge by the graph bounds and $Q_R\to I$ in every graph norm. eq. 7.li–eq. 7.lii consequently prove a jointly tempered boundary of the actual sewn Euclidean functions. Independent one-variable boundary statements have not been substituted for it.

The full spectral carrier and the cyclic sector. Assume now the supplied round cone. The finite-energy cutoff point field $A_R(x)=\sum_{m,n\le R}A_{mn}(x)$ is bounded, spatially covariant, and has bounded translation generators on its cutoff range. Its vacuum words have the usual difference-variable form with one joint translation between each adjacent pair of fields. If a test in one difference variable has Fourier transform vanishing on the forward carrier, integrating that translation gives zero by the joint spectral theorem. Other differences and spectator fields can be held fixed during this integration. On a compact Fourier set outside the product carrier, choose a finite cover by product boxes, in each of which one difference momentum is outside its carrier. A subordinate partition and product Fourier approximation reduce each localized test to the just-proved zero identities. This proves the full product support condition.

The cutoff vacuum distributions converge to W_n by eq. 7.xxxviii and the same uniform extension estimate as in eq. 7.li. Their difference-variable limits are legitimate distributions: fix a Schwartz test ρ of integral one in the common center coordinate, and pair the full distributions with $\rho(x_1)\psi(x_2 - x_1, \dots, x_n - x_{n-1})$. This is a continuous map from difference tests to full tests. The cutoff identities show that the limiting difference functional is independent of ρ and that its pairing with a full test uses the center integral of that test. Thus it is the actual difference-variable distribution of W_n. Support in a fixed closed set passes to this tempered limit. Hence

$$\operatorname{supp} \widehat{W_n^{\mathrm{diff}}} \subset (\overline{V}_+)^{n-1}, \qquad \overline{V}_+ = \{(E,p) : E \geq |p|_2\}, \tag{7.liii}$$

with the Fourier convention of eq. 7.xli and eq. 7.lii. In unreduced momenta this is the corresponding condition on all partial sums, not merely a bound on total translations of one fixed vector. Without the spatial rotation input the same proof retains the cube carrier provided by eq. 6.ccxliv.

Let $\mathcal{H}_\phi$ be the closed span of field-polynomial vacuum vectors. They lie in $\mathcal{K}_{\mathrm{ord}}$, and eq. 7.xli makes their span invariant under both signs of real translations. Thus $\mathcal{H}_\phi$ reduces H, its heat factors and its energy projections. The reverse inclusion, and hence cyclicity, follows as below.

First, polynomial vectors are graph-dense in $\mathcal{H}_\phi \cap \operatorname{Dom} W^l$ for every l. For a smooth compact spectral cutoff $c_R(H)$, its inverse Fourier transform is integrable. Therefore $c_R(H)$ is an integral of real time translations. Applied to a polynomial vector, that integral is approximated in the W^l graph norm by finite sums of translated polynomial vectors: the orbit is graph-norm continuous and its graph norm is constant. Given an arbitrary vector in the stated domain, first cut its spectrum off and then approximate the cutoff vector by these smoothed polynomial approximants. Finally let the cutoff tend to identity. This proves graph density. The energy estimate in eq. 7.xxxviii now shows that every $\phi(g)$ maps $\mathcal{H}_\phi \cap \mathcal{D}_\infty$ into $\mathcal{H}_\phi$.

For compact smooth h, the bounded operators $T_s\phi(\chi_\epsilon \otimes h)T_s$ converge in operator norm to $\Phi_s(h)$. On finitely many bands this follows from the temporal approximate identity and the bounded band generators. The tails converge uniformly: their block norms are bounded by

$$C(1+m+n)^N e^{-s(m+n)}\|h\|_1, \tag{7.liv}$$

independently of ϵ, since $\|\chi_\epsilon\|_1 = 1$. The row and column sums outside a growing finite square tend to zero. The limiting blocks are exactly those of $\Phi_s(h)$ by eq. 7.xxxiii. Each approximating operator preserves $\mathcal{H}_\phi$: the right heat factor sends its vectors into all energy domains, the preceding graph-density argument handles the field, and the left factor preserves the subspace. The norm limit does so as well. Spatial L^1 approximation extends this to all labels in eq. 7.xxviii.

Every vector in the exact Euclidean sewing formula in eq. 7.xlix therefore belongs to $\mathcal{H}_\phi$. Its ordered-profile integrals are the dense ordered Euclidean vectors established above. Consequently

$$\mathcal{H}_\phi = \mathcal{K}_{\mathrm{ord}}. \tag{7.lv}$$

This proves the actual cyclic realization and its inherited separability, evolution and phase vacuum conclusions. It makes no assertion that $\mathcal{K}_{\mathrm{ord}} = \mathcal{K}$.

The joint forward tube and its full cone boundary. The preceding argument also supplies the simultaneous complex spatial variables under the round-cone hypothesis; the following bounded formula proves joint continuation. Work on $\mathcal{K}_{\mathrm{ord}}$ and put

$$S(z,w) = e^{-zH + iw\cdot P}, \qquad \operatorname{Re} z > |\operatorname{Im} w|_2, \qquad \Phi_a^0 = V_{4a}(0,0). \tag{7.lvi}$$

The joint spectral proof in § 7.1 makes S a norm-holomorphic contraction on this tube. It also shows directly that derivatives of total order l on a smaller tube with margin δ have norm at most $l!\delta^{-l}$, because $|P|_2 \leq H$. The point operator Φ_a^0 is the norm limit of the spatial approximate identities in eq. 7.xxviii, by continuity of V_{4a}. It therefore preserves the ordered sector, has norm at most Ca^{-N}, and satisfies exactly the same heat consistency.

For $\delta_j = \operatorname{Re} z_j - |\operatorname{Im} w_j|_2 > 0$, choose $a_j + a_{j+1} < \delta_j$ and define

$$\mathscr{E}_n(z,w) = \left\langle \Omega, \Phi^0_{a_1} \prod_{j=1}^{n-1} \left[S(z_j - a_j - a_{j+1}, w_j) \Phi^0_{a_{j+1}} \right] \Omega \right\rangle, \qquad r = \min\{1/8, \tfrac{1}{4} \min_j \delta_j\}, \qquad |\mathscr{E}_n(z,w)| \le C^n r^{-Nn}. \tag{7.lvii}$$

The product has the same order as eq. 7.xlvii. Decreasing the auxiliary a_j and joining their extra heat factors to the adjacent joint semigroups proves independence, exactly as before: the heat factors commute with S, and every resulting joint semigroup remains inside its contraction tube. Fixed auxiliary parameters on a neighbourhood give joint norm holomorphy, and their independence glues the neighbourhoods. Taking every $a_j = r$ proves the bound. For $n = 1$ use the constant vacuum value with $a_1 = 1/8$.

For real spatial gaps, spatial covariance of the point operators identifies this formula with the spatial point version of eq. 7.xlvii. Spatial approximate identities converge in norm, so the exact Euclidean sewing already proved identifies it with the original ordered Euclidean density and with its spatially smeared values. Thus eq. 7.lvii is a jointly defined function of all $4(n-1)$ complex difference variables, not merely a family indexed by real spatial labels.

Let $\epsilon_j > |y_j|_2$, and insert $R_j = S(\epsilon_j, iy_j) = e^{-\epsilon_j H - y_j \cdot P}$ between the fields in eq. 7.l, obtaining $W_n^{\epsilon,y}$. The round cone gives $\|R_j\| \le 1$. These operators commute with W, are contractions in every graph norm, and converge strongly there to identity as $(\epsilon_j, y_j) \to 0$ within that cone. The last assertion is dominated convergence in the joint spectral measure: the multipliers are bounded by one and converge pointwise to one. The proof of eq. 7.li therefore gives

$$|W_n^{\epsilon,y}(F)| \le D'_n p_{K_n}(F), \qquad W_n^{\epsilon,y}(F) \longrightarrow W_n(F) \quad \text{for every } F \in \mathcal{S}(\mathbb{R}^{4n}). \tag{7.lviii}$$

This is a simultaneous limit along every cone-admissible approach.

Finite energy cutoffs identify the integrated function in eq. 7.lvii with $W_n^{\epsilon,y}$ at

$$z_j = \epsilon_j - i(t_{j+1} - t_j), \qquad w_j = x_{j+1} - x_j + iy_j. \tag{7.lix}$$

Indeed the inserted spectral multipliers are now bounded by $e^{-(\epsilon_j - |y_j|_2)H}$. Hence each intermediate energy sum has the same absolute polynomial-times-exponential bound as in eq. 7.lii, uniformly in all real coordinates. Finite-cutoff Fubini, then dominated convergence, proves the identity on tensor tests. The bound in eq. 7.lvii makes its fixed-regulator integral continuous on Schwartz tests, while eq. 7.lviii does the same on the field side; tensor density proves equality on every such test. eq. 7.lviii is therefore the actual joint tempered boundary of eq. 7.lvii, with all complex spatial and time regulators allowed by the round cone. In the convention of eq. 7.xli, these are difference points $(t + i\epsilon, x + iy)$ with imaginary part in the forward cone.

Lorentz covariance from the actual Euclidean rotations. For this step assume the full scalar Euclidean covariance G2 of the actual diagonal-flat moments. The finite hypercubic group alone does not supply this hypothesis. In particular G2 includes every real rotation in a time–space coordinate plane. No Lorentz covariance of the field has been used in constructing the tube.

Put $\zeta_j = (iz_j, w_j)$. The domain of eq. 7.lvii is precisely $\operatorname{Im} \zeta_j \in V_+$ for every gap. Denote the same holomorphic function by $\mathcal{F}_n(\zeta)$. Its Euclidean slice is $\zeta_j = (i\tau_j, r_j)$, with $\tau_j > 0$ and $r_j \in \mathbb{R}^3$. We first justify pointwise use of G2 on this slice. A compact neighbourhood of any such Euclidean configuration has strictly ordered time coordinates. The sewing formula represents its actual moment functional by the continuous point density in eq. 7.lvii. For a sufficiently small fixed Euclidean rotation, its rotated neighbourhood still has strictly ordered times. Apply G2 to arbitrary smooth tests supported in this neighbourhood. The scalar change-of-variables identity, whose Jacobian is one, makes the two continuous densities equal there. Their difference is zero against every compact smooth test and hence zero pointwise. Thus the local rotation identity is an identity for the actual sewn function, not an imposed extension across the diagonals. Common

Euclidean translations allow the first point to be set at zero, so rotations act linearly on all the difference variables.

Choose a time–space rotation with infinitesimal action $\dot\tau_j = r_j^k$, $\dot r_j^k = -\tau_j$, simultaneously for every gap. Differentiating the just-proved pointwise identity and using $\zeta_j^0 = i\tau_j$ gives

$$i\mathcal{B}_k\mathcal{F}_n = 0 \quad \text{on the Euclidean slice,} \qquad \mathcal{B}_k = \sum_{j=1}^{n-1}\left(\zeta_j^k\partial_{\zeta_j^0} + \zeta_j^0\partial_{\zeta_j^k}\right). \tag{7.lx}$$

The derivative exists because the point density is holomorphic on a complex neighbourhood of that slice. The slice is maximal totally real: the complex linear coordinate change $(\tau_j, r_j) \mapsto (i\tau_j, r_j)$ sends real open boxes to its open pieces. A holomorphic function vanishing on such a box vanishes on a complex neighbourhood, by applying the one-variable identity theorem successively in each coordinate. The product tube is connected, indeed convex. Apply this observation to the holomorphic function $\mathcal{B}_k\mathcal{F}_n$; it follows that it vanishes on the entire tube.

Every real proper orthochronous boost preserves V_+, so its whole real orbit stays inside the tube. Differentiating $s \mapsto \mathcal{F}_n(\Lambda_k(s)\zeta)$ now gives zero at every s. The function is constant along that orbit. Spatial rotations have the same invariance, either by G2 on the slice and the same uniqueness argument, or by their already implemented quotient unitaries. These transformations generate $\mathrm{SO}_+(1,3)$. Explicitly, for any Λ in this group write $\Lambda e_0 = (\gamma, \gamma v)$, where $|v| < 1$. The boost in direction v sends e_0 to this vector. Its inverse times Λ fixes e_0 and is a spatial proper rotation. A directional boost is a spatial rotation conjugate of a coordinate boost. Therefore

$$\mathcal{F}_n(\Lambda\zeta) = \mathcal{F}_n(\zeta) \quad (\Lambda \in \mathrm{SO}_+(1,3)). \tag{7.lxi}$$

The boundary passage uses the full cone limit proved above. If $\eta_j \in V_+$, then also $\Lambda^{-1}\eta_j \in V_+$. Changing real variables $\xi = \Lambda\chi$ in the integral of $\mathcal{F}_n(\xi + i\eta)$ against $f(\Lambda^{-1}\xi)$ changes its regulator to $\Lambda^{-1}\eta$, by eq. 7.lxi, and leaves the Jacobian one. Both regulator families have the same joint tempered limit by eq. 7.lviii. Consequently

$$W_n^{\mathrm{diff}}(f\circ\Lambda^{-1}) = W_n^{\mathrm{diff}}(f), \qquad W_n(\alpha_\Lambda F) = W_n(F), \tag{7.lxii}$$

where α_Λ pulls every argument back by Λ^{-1}. The second identity follows from the center integration defining the difference distribution, which commutes with a determinant-one simultaneous linear change of coordinates. The Schwartz pullback is continuous for every fixed Λ. This proves distributional Lorentz covariance without a separate boundary-exchange assertion.

The covariant field representation on the ordered sector. Let $X(F)$ denote a finite sum of the vacuum-product vectors constructed in eq. 7.xlii–eq. 7.xlvi. Their span is dense in $\mathcal{K}_{\mathrm{ord}}$ by eq. 7.lv. Define there

$$U(\Lambda)X(F) = X(\alpha_\Lambda F), \qquad U(\Lambda)\Omega = \Omega. \tag{7.lxiii}$$

eq. 7.xlvi and eq. 7.lxii, applied by polarization to two finite sequences, preserve their inner product. In particular a zero vector is sent to zero. The map is thus well-defined and isometric on the actual quotient, and its inverse is the same construction for Λ^{-1}. It extends to a unitary on the ordered sector. The pullback composition law gives a representation of $\mathrm{SO}_+(1,3)$ on this sector. On the spatial rotation subgroup it equals the quotient unitary already constructed: both maps have the same action on every field-polynomial vacuum vector, whose span is dense by eq. 7.lv.

If $\Lambda \to I$, its pullbacks converge in every Schwartz seminorm. This follows by the mean-value formula for the derivatives and polynomial weights on a bounded matrix neighbourhood of identity; one additional derivative and weight suffice for each fixed seminorm. The Hilbert-valued extension after eq. 7.xlv then makes $X(\alpha_\Lambda F) \to X(F)$. Unitarity and density extend this convergence to all vectors. Hence U is strongly continuous. The translation implementation already satisfies

$\mathscr{U}(a)X(F) = X(\tau_a F)$. The exact pullback identity therefore gives

$$\begin{aligned} U(\Lambda)\mathscr{U}(a)U(\Lambda)^{-1} &= \mathscr{U}(\Lambda a), \\ \mathscr{U}(a)U(\Lambda)\mathscr{U}(b)U(M) &= \mathscr{U}(a+\Lambda b)U(\Lambda M). \end{aligned} \tag{7.lxiv}$$

The second line exhibits the strongly continuous proper orthochronous Poincare representation, with the positive spectral cone already established. The notation M in this formula denotes another Lorentz matrix.

Covariance also holds on the energy domain. Stone's theorem applied to the first identity in eq. 7.lxiv gives the exact self-adjoint identity $U(\Lambda)HU(\Lambda)^{-1} = \gamma H + b \cdot P$, where $(\gamma, b) = \Lambda e_0$, $\gamma > |b|$. The right side is defined by the joint spectral calculus. The round cone gives

$$(\gamma - |b|)H \le \gamma H + b \cdot P \le (\gamma + |b|)H. \tag{7.lxv}$$

These are pointwise inequalities in that calculus. Their positive lower and upper constants show that the domains of every positive integer power of the two operators agree, with equivalent graph norms. Thus $U(\Lambda)\mathcal{D}_\infty = \mathcal{D}_\infty$ on the ordered sector. The polynomial intertwining identity follows directly from eq. 7.lxiii and multiplication of the field polynomials. Polynomial vectors are graph-dense in every ordered-sector energy domain by the proof preceding eq. 7.liv, now using eq. 7.lv. The graph bound in eq. 7.xxxviii, and the equivalent graph norms just proved, extend that identity to the common domain:

$$U(\Lambda)\phi(g)U(\Lambda)^{-1} = \phi(\alpha_\Lambda g) \quad \text{on } \mathcal{D}_\infty \cap \mathcal{K}_{\text{ord}}. \tag{7.lxvi}$$

This is a common-domain covariant scalar field construction on the actual cyclic space. No action of these Lorentz unitaries on $\mathcal{K} \ominus \mathcal{K}_{\text{ord}}$ is needed or asserted.

The construction has supplied common-domain Hermitian fields, positive tempered moments, the full translation spectral carrier, ordered Euclidean matching, a joint forward-tube tempered boundary, and Lorentz covariance under G2. Spacelike locality still requires the permutation and complex-continuation argument. It does not follow from the block estimate or the real boost identities alone and has not been asserted here. All underlying ordinary-source and ultraviolet gauge-theory hypotheses retain their declared conditional scope.

7.3. **Quantitative reconstruction and local commutativity.** The fields of § 7.2 were constructed before any local-commutativity conclusion was used. Their common domain, positive moments, ordered Euclidean values and joint tempered boundary are therefore available for a comparison with the corrected reconstruction theorem. We make that comparison explicitly. It supplies locality on the actual ordered Hilbert sector, together with the stronger distribution-order conclusion of the imported theorem. The proof does not infer locality from a spectral cone alone.

Throughout this section the ordinary-source, ultraviolet and symmetry hypotheses retain the scope declared in the preceding sections. In particular, full scalar Euclidean covariance means G2 on the actual diagonal-flat moments. It is a condition on the chosen state or component, not a conclusion drawn merely from the symmetry of a mixture. Clustering is used only for the conclusions that explicitly invoke a clustering component.

The distribution space and its numerical seminorms. For $d \ge 1$, use the global Euclidean seminorm

$$p_j^{(d)}(f) = \max_{|\alpha| \le j} \sup_{x \in \mathbb{R}^d} (1 + |x|^2)^{j/2} |\partial^\alpha f(x)|. \tag{7.lxvii}$$

The derivative order is the total multiindex order. The weight is one global weight in d variables. Neither a separate weight in every argument nor a derivative order separately bounded in every coordinate is being substituted for it. These choices matter when $d = 4n$ and j itself grows with n.

Let $\Delta_n = \bigcup_{i<j}\{x_i = x_j\}$, and let $\mathcal{S}_{\Delta,n}$ consist of the functions in $\mathcal{S}(\mathbb{R}^{4n})$ whose every partial derivative vanishes on Δ_n. It has the induced Schwartz topology. It is closed: evaluation of a fixed derivative at a fixed point of Δ_n is a continuous functional, and the space is the intersection of their

kernels. No quotient by contact distributions has been chosen in defining the test space. Rather, such distributions are invisible when a functional is restricted to this space.

For comparison with derivative-sum conventions, put

$$s_j^{(d)}(f) = \sum_{|\alpha|\le j} \sup_x (1+|x|^2)^{j/2} |\partial^\alpha f(x)|.$$

The exact count of multiindices is $\binom{d+j}{j}$. One proof appends a nonnegative slack coordinate so that $|\alpha| \le j$ becomes a composition of j into $d+1$ parts; the usual placement of separators then counts the compositions. Consequently

$$p_j^{(d)}(f) \le s_j^{(d)}(f) \le \binom{d+j}{j} p_j^{(d)}(f) \le 2^{d+j} p_j^{(d)}(f). \tag{7.lxviii}$$

At $d = 4n$, $j = 20n$, the cost is at most 2^{24n}. Thus a switch between a maximum and a sum costs one exponential in n, not an unspecified equivalence constant depending on the dimension. The total derivative order remains $20n$.

The global weight can also be compared with a coordinate maximum weight, but that comparison has a different numerical cost. Since $|x|_\infty \le |x|_2 \le \sqrt{d}\,|x|_\infty$, the reverse weight comparison costs $d^{j/2}$. For the displayed values this is $(4n)^{10n}$, which is still factorial growth. Indeed

$$\log(n!) = \sum_{r=1}^{n} \log r \ge \int_1^n \log t\, dt = n\log n - n + 1$$

implies $n^n \le e^n n!$, and hence

$$(4n)^{10n} \le (4e)^{10n} (n!)^{10}. \tag{7.lxix}$$

This estimate explains why a change of seminorm convention must record its dimension cost. We keep eq. 7.lxvii for the proved extension and give the following explicit comparison for the theorem interface.

Proposition 7.5 (The quantitative seminorm ledger)**.** For $f \in \mathcal{S}(\mathbb{R}^d)$ and an integer $r \ge 0$, put

$$\begin{aligned} q_{\mathrm{tot},r}(f) &= \max_{|\alpha|+|\beta|\le r} \sup_x |x^\alpha \partial^\beta f(x)|, \\ q_{\mathrm{sep},r}(f) &= \max_{|\alpha|\le r,\ |\beta|\le r} \sup_x |x^\alpha \partial^\beta f(x)|. \end{aligned} \tag{7.lxx}$$

Then

$$\begin{gathered} q_{\mathrm{tot},r} \le q_{\mathrm{sep},r} \le p_r, \qquad p_r \le (1+d)^r q_{\mathrm{sep},r}, \\ p_r \le (1+d)^r q_{\mathrm{tot},2r}. \end{gathered} \tag{7.lxxi}$$

In particular, in dimension $4n$,

$$p_{20n}(f) \le [(5e)^{20}]^n (n!)^{20} q_{\mathrm{tot},40n}(f). \tag{7.lxxii}$$

■

Proof. A monomial of degree at most r is bounded by $(1+|x|^2)^{r/2}$, which proves the first two comparisons. For the reverse comparison use $\sqrt{1+|x|^2} \le 1 + \sum_{i=1}^d |x_i|$, and expand its r-th power. All monomials have total degree at most r, and the sum of their nonnegative coefficients is exactly $(1+d)^r$. This proves the separate-order bound. The constraints $|\alpha|, |\beta| \le r$ imply $|\alpha|+|\beta| \le 2r$, giving the last line. Finally $1 + 4n \le 5n$, and the elementary factorial estimate preceding eq. 7.lxix gives $(5n)^{20n} \le [(5e)^{20}]^n (n!)^{20}$. □

The comparison distinguishes total combined order from separate monomial and derivative orders. It also proves a sufficient safe match with a doubled linear order, independently of which of these explicit seminorm conventions is used to state the standard theorem. No appeal to a dimension-dependent topological equivalence is needed.

The exact factorial input. For the actual scalar moments the quantitative extension has already proved, with one fixed finite k,

$$S_0 = 1, \qquad |S_n(f)| \le AB^n (n!)^{k+32} p_{20n}^{(4n)}(f), \quad f \in \mathcal{S}_{\Delta,n}. \tag{7.lxxiii}$$

The coefficient 20 controls every order at once. This is stronger information than continuity at each fixed n. The constant A can be replaced by $\max(1, A)$, and similarly put $B_+ = \max(1, B)$. For $n \ge 1$, the factors $2, 3, \dots, n$ give $n! \ge 2^{n-1}$. Therefore

$$B^n \le B_+ (n!)^{\log_2 B_+}, \qquad \sigma_n = AB_+ (n!)^{k+32+\lceil \log_2 B_+ \rceil} \tag{7.lxxiv}$$

is one factorial-growth majorant, and $|S_n(f)| \le \sigma_n p_{20n}(f)$. If the derivative-sum convention is used, its harmless exponential cost in eq. 7.lxviii is absorbed by replacing B by $2^{24}B$ before the same argument. Moreover the seminorm comparison gives

$$|S_n(f)| \le A\widetilde{B}^n (n!)^{k+52} q_{\mathrm{tot},40n}(f), \qquad \widetilde{B} = B(5e)^{20}. \tag{7.lxxv}$$

Absorbing $\widetilde{B}^n$ as in eq. 7.lxxiv gives a factorial majorant at the one fixed order $40n$. Since $q_{\mathrm{tot},40n} \le p_{40n}$, this also gives the weighted version at that order. Thus both the direct weighted route and the safe total-monomial route meet a fixed linear-order hypothesis. The increase by twenty in the factorial exponent is explicit, and no order proportional to n^2 has entered this Euclidean input.

The extension in eq. 7.lxxiii acts on precisely the coincidence-flat Schwartz space. A stronger assertion on unrestricted Euclidean tests is not needed for the corrected linear-growth hypothesis. In particular there is no step that assigns a distribution supported on Δ_n, then invokes the theorem as if that assignment had been fixed by the separated data.

It is useful to distinguish three estimates at this point. The formula in eq. 7.lxxiii is a proved implication of the declared source assumptions. The energy-band construction gives a possibly larger, quadratic in n, order for unrestricted Minkowski moments. The corrected reconstruction theorem below has its own stronger linear-order Minkowski conclusion. We will obtain that conclusion for the constructed moments only after identifying the two constructions. The identification does not use that stronger bound in advance.

The reflected products really belong to the flat domain. Let $\mathcal{S}_{\Delta,+,n}$ be the flat functions whose jets vanish whenever some time coordinate is nonpositive. Set $\theta(\mathbf{x}, t) = (\mathbf{x}, -t)$. For $F_m \in \mathcal{S}_{\Delta,+,m}$ and $G_n \in \mathcal{S}_{\Delta,+,n}$, consider

$$(\Theta\overline{F_m} \otimes G_n)(x_1, \dots, x_{m+n}) = \overline{F_m(\theta x_1, \dots, \theta x_m)} G_n(x_{m+1}, \dots, x_{m+n}). \tag{7.lxxvi}$$

Every derivative of this product vanishes on a coincidence within the first group, because the corresponding derivative of F_m does; the same applies within the second group. At a cross-group coincidence, simultaneous support of the two factors requires its time coordinate to be both nonpositive and nonnegative. It is therefore zero. At that time boundary every jet of at least one factor vanishes. Leibniz's rule proves vanishing of all derivatives of the product there as well. Thus eq. 7.lxxvi belongs to $\mathcal{S}_{\Delta,m+n}$, including the case in which one degree is zero.

Tensor products and reflection are continuous for the stated seminorms. For example, for each fixed j, m, n,

$$p_j^{(4(m+n))}(\Theta\overline{F} \otimes G) \le 2^{j/2} p_j^{(4m)}(F) p_j^{(4n)}(G). \tag{7.lxxvii}$$

The estimate follows by distributing a derivative between the two groups and using $1 + |x|^2 + |y|^2 \le 2(1 + |x|^2)(1 + |y|^2)$. Reflection changes no derivative magnitude or Euclidean length. This proves joint continuity on any fixed pair of degrees. Consequently the full finite-sequence reflected form

$$q(F, G) = \sum_{m,n} S_{m+n}(\Theta\overline{F_m} \otimes G_n) \tag{7.lxxviii}$$

is well-defined. Its positivity is the actual full-sequence positivity proved by the constrained density argument. The sum is finite; no completion in the number of arguments is being assumed.

The historical ordered space consists of the Schwartz functions whose jets vanish outside $0 < t_1 < \cdots < t_n$. It is a subspace of $\mathcal{S}_{\Delta,+,n}$: a coincidence forces equal times and hence lies outside the strict ordered region. The historical reflected product reverses the first argument group. Scalar permutation symmetry makes its value under S_{m+n} equal to eq. 7.lxxviii, by the permutation reversing just those m slots. This establishes the positivity convention required by the imported theorem on its actual test space.

The isometry between ordered quotients follows directly. Two ordered sequences with zero difference in eq. 7.lxxviii have the same image in the full quotient. Conversely a zero image has zero ordered norm because the two forms agree. Complete this isometry and denote its closed range by $\mathcal{K}_{\mathrm{ord}}$. The resulting Hilbert space is the same one on which the band fields were shown cyclic. An equality with the larger full positive-time space is not part of this argument.

Symmetry, reality and the clustering clause. Scalar permutation symmetry and reality are retained properties of the actual Euclidean functional. On separated tensors these are the identities for products of the same real scalar observable; the quantitative flat extension and constrained tensor density transfer them to the entire flat test space. G2 transfers in the same way. Thus the theorem is applied to the chosen functional itself. No invariance of a component is inferred solely from invariance of its barycenter.

For a clustering component, the compact-tuple cluster estimate gives on its ordered compact core

$$\langle [F], U_a[G]\rangle \longrightarrow \langle [F], \Omega\rangle\langle \Omega, [G]\rangle \quad (|a| \longrightarrow \infty), \tag{7.lxxix}$$

where a is spatial. To extend this to any two ordered Hilbert vectors ξ, η, choose compact-core approximants ξ_l, η_l. Unitarity bounds the matrix-element error, uniformly in a, by

$$\|\xi - \xi_l\|\,\|\eta\| + \|\xi_l\|\,\|\eta - \eta_l\|. \tag{7.lxxx}$$

The corresponding rank-one vacuum expression has the same bound, since $\|\Omega\| = 1$. First take the far-translation limit at fixed l, and then let l grow. This proves eq. 7.lxxix on the whole sector, hence in particular on the historical ordered Schwartz sequences. It proves the exact cluster clause without assuming translation-uniform Schwartz bounds for approximate tests.

Several components need not satisfy eq. 7.lxxix after mixing. The cluster-free part of reconstruction still applies to their positive mixture, whereas a unique-vacuum conclusion must retain the component or rank-one cluster assumption. A polynomial decay estimate on compact tuples does not become the same numerical rate on all Hilbert vectors by the preceding density proof.

One verified reconstruction import. We use the corrected scalar theorem $E' \to R'$ of [OS75, Section IV.1, pp. 287–289]. In the notation fixed above, with the exact input match in Prop. 7.5 and eq. 7.lxxv, its import is the following. Normalized coincidence-flat moments with a factorial majorant at one linear order, scalar Euclidean covariance, ordered reflection positivity, reality and permutation symmetry reconstruct positive local tempered scalar moments with the prescribed ordered Euclidean restrictions, the forward spectrum and relativistic covariance. They obey

$$|\widetilde{W}_n(F)| \le a b^{n^2} p_{wn}^{(4n)}(F) \quad (F \in \mathcal{S}(\mathbb{R}^{4n})) \tag{7.lxxxi}$$

for some fixed a, b, w. Adding the cluster clause gives the clustering conclusion. The axiom-subset form without clustering is the clause specified in [OS75, Introduction, Remark 6], with [OS73, Section 3, Remark 3(b)].

This is one standard theorem import. Its locality conclusion includes the permutation and complex-continuation step. That step is not an additional gauge-model estimate, nor is its proof asserted to follow from the band estimate. The previous sections have matched every model-facing input and its domain.

We specify the time convention before comparing moments. If the imported formulation uses a negative-imaginary-time tube, let $D(t,x) = (-t,x)$. Transport its moments by this real linear involution. The transported field has time action e^{+itH} when the original one uses e^{-itH}, and its Euclidean slice is $(i\tau, r)$. Positivity is unchanged, because pullback by D commutes with reversal and complex conjugation. Locality is unchanged, because D preserves the spacelike relation. Lorentz covariance is transported by $\Lambda \mapsto D\Lambda D$, which again belongs to $\mathrm{SO}_+(1,3)$. Finally $p_j(F \circ D) = p_j(F)$, so eq. 7.lxxxi is unchanged. Equivalently, if $\widetilde{E}_n(z;r)$ is the ordered-gap function, the two conventions have boundaries

$$\widetilde{W}_n^{(-)} : z_j = \epsilon_j + it_j, \qquad \widetilde{W}_n^{(+)} : z_j = \epsilon_j - it_j. \tag{7.lxxxii}$$

In the rest of this section $\widetilde{W}$ denotes the second, which is the convention of the constructed band fields. This is a coordinate convention, not a change of the Euclidean data.

Identifying the local moments without assuming locality. Let $E_n(z;r)$ be the point-insertion ordered-gap function from the band construction, with the first Euclidean point fixed at zero. Let $\widetilde{E}_n(z;r)$ be the ordered-gap function of the import. Both are jointly holomorphic for all $\operatorname{Re} z_j > 0$, and continuous in real spatial gaps r. Their Euclidean restrictions represent the same actual moment functional on compact strictly ordered configurations. The sewing proof gives that assertion for E_n; the prescribed Euclidean restriction is part of the reconstruction import for $\widetilde{E}_n$.

We verify equality of the densities. On any compact ordered neighbourhood the actual moment is represented by the continuous band density. The imported function represents the same functional there. Their difference integrates to zero against every compact smooth test. If it were nonzero at a point, continuity would give a smaller ball and a fixed complex phase on which its real part had a strict sign; a nonnegative test in that ball would contradict the zero integral. Thus the two continuous densities agree pointwise. After common center integration this gives

$$E_n(\tau;r) = \widetilde{E}_n(\tau;r) \quad \text{for every } \tau_j > 0 \text{ and real } r. \tag{7.lxxxiii}$$

Fix r. For fixed positive real values of every variable except z_1, the difference is a holomorphic function of z_1 in the right half-plane and vanishes on its positive real interval. The one-variable identity theorem makes it zero for every such z_1. Keep this first variable at an arbitrary complex value and apply the same argument to z_2. Continue through all the gap variables. It follows that

$$E_n(z;r) = \widetilde{E}_n(z;r) \quad \text{whenever every } \operatorname{Re} z_j > 0. \tag{7.lxxxiv}$$

This induction uses functions already jointly holomorphic on one product domain. It does not replace a joint boundary statement by separate tempered limits.

At every strictly positive regulator, eq. 7.lxxxiv identifies the two integrals against arbitrary spacetime Schwartz tests. The band integrals are well-defined and have their regulator-uniform seminorm bound; the import has its actual tempered boundary in the convention in eq. 7.lxxxii. Let all regulators tend to zero jointly. The independently proved band boundary therefore gives

$$W_n(F) = \widetilde{W}_n(F) \quad \text{for every } F \in \mathcal{S}(\mathbb{R}^{4n}). \tag{7.lxxxv}$$

For $n = 1$ this is the same equality of constant vacuum means; for $n = 0$ it is normalization. The argument has never used locality of W. Uniqueness among already local theories alone would not have justified eq. 7.lxxxv; equality of the actual analytic functions and of their unrestricted boundaries does justify it.

In particular eq. 7.lxxxi now holds for the constructed moments. This is an imported improvement of their growth bound, obtained after identification. Their original energy-band proof required only its own explicit fixed-order estimates and was not circularly based on this improvement.

Distributional locality and the full common domain. For adjacent slots $j, j+1$, let π_j swap those slots. The imported locality and eq. 7.lxxxv say exactly

$$(W_n - W_n \circ \pi_j)|_{\{(x_j - x_{j+1})^2 < 0\}} = 0. \tag{7.lxxxvi}$$

This is a distributional identity on that open subset of all $4n$ real coordinates; spectator arguments are unrestricted. It is not merely a commutation identity for a two-point function. It applies first to compact smooth tests in the open subset. Infinity cutoffs then extend it to a Schwartz test supported there: the cutoffs converge in every Schwartz seminorm, and each cutoff has compact support contained in the same open subset.

Take compact smooth g, h with spacelike separated supports. Let ξ, η be any two finite field-polynomial vacuum vectors. The adjoint identity on the common domain writes $\langle \xi, [\phi(g), \phi(h)]\eta \rangle$ as a finite sum of the expressions in eq. 7.lxxxvi, with the reversed conjugate bra labels before the adjacent g, h pair and arbitrary ket labels after it. Each is zero. The polynomial bra vectors are dense in $\mathcal{K}_{\mathrm{ord}}$, and the commutator ket is an actual Hilbert vector by invariance of the common domain. Hence the commutator is zero on every polynomial ket.

There is a further domain issue before asserting the same identity on $\mathcal{D}_\infty$. Let $W = 1 + H$ and retain the proved energy exponent $N = 9$. Applying the band bound first with $k = 0$, then with $k = N$, gives

$$\begin{aligned} \|\phi(g)\phi(h)\eta\| &\le C_0 C_N p_{2N+7}(g) p_{2N+7}(h) \|W^{2N}\eta\|, \\ \|[\phi(g), \phi(h)]\eta\| &\le 2C_0 C_N p_{2N+7}(g) p_{2N+7}(h) \|W^{2N}\eta\|. \end{aligned} \tag{7.lxxxvii}$$

Here $\eta \in \mathrm{Dom}\, W^{2N}$: the first field maps that domain into $\mathrm{Dom}\, W^N$, where the second is defined. Thus both products are actual continuous maps from this graph domain to the Hilbert space. Polynomial vectors are dense in that graph domain on the ordered sector, by the spectral-smoothing argument and cyclic identification already proved. Approximate η there and use eq. 7.lxxxvii. We obtain

$$[\phi(g), \phi(h)]\eta = 0 \quad \left(\eta \in \mathrm{Dom}\, W^{2N} \cap \mathcal{K}_{\mathrm{ord}}\right), \tag{7.lxxxviii}$$

in particular on the invariant common domain.

The same statement holds for Schwartz labels with spacelike separated supports. Multiply each label by smooth infinity cutoffs; their supports remain spacelike separated, and they converge in every seminorm appearing in eq. 7.lxxxvii. The two product maps therefore converge in the stated graph-to-Hilbert norm. This proves the extension without imposing a uniform spacelike distance at infinity.

eq. 7.lxxxviii is the local commutativity of the constructed unbounded Wightman field. It does not say that arbitrarily selected self-adjoint extensions have commuting spectral projections. Real test fields have been proved symmetric and closable, while essential self-adjointness and strong spectral commutation require their own operator criteria. No exponentials of such unspecified extensions are needed for eq. 7.lxxxviii.

The stronger vector and energy seminorm conclusions. The scalar bound in eq. 7.lxxxi also improves the vacuum-product test-space estimate. For a general $F \in \mathcal{S}(\mathbb{R}^{4n})$, let $X_n(F)$ be its actual Hilbert-valued vacuum product. On finite tensors positivity and the adjoint identity give $\|X_n(F)\|^2 = W_{2n}(F^* \otimes F)$. Both sides are continuous under the already proved unrestricted extensions, so this equality holds for every F. Tensor seminorm comparison then gives

$$\|X_n(F)\| \le A_n p^{(4n)}_{2wn}(F), \qquad A_n = \sqrt{a}\, b^{2n^2} 2^{wn/2}. \tag{7.lxxxix}$$

One may increase a, b, w to make them at least one. The harmless power of two comes from eq. 7.lxxvii; the specific larger constant displayed is sufficient. Reversal and conjugation do not change the seminorm. This estimate is compatible with the actual band-vector extension, because their values on the dense tensor space already agree.

There is an explicit energy version of eq. 7.lxxxix. Set $D_t = \sum_{j=1}^{n} \partial_{t_j}$. Translation covariance and continuity of the vector-valued test map imply

$$H^l X_n(F) = i^l X_n(D_t^l F), \qquad p_j(D_t^l F) \le n^l p_{j+l}(F). \tag{7.xc}$$

For the first identity, the simultaneous test translation is $F(t_1 - s, \dots, t_n - s)$; its derivative is $-D_t F$, while the vector translation is e^{isH}. Difference quotients converge in the Schwartz topology and therefore in the Hilbert space. Stone's closed generator criterion proves the domain identity for $l = 1$; induction proves every l. The second estimate is the multinomial expansion, whose absolute coefficients sum to n^l, followed by the monotonicity of the weights. Consequently

$$\|W^l X_n(F)\| \le A_n (1+n)^l p_{2wn+l}(F). \tag{7.xci}$$

This proves continuity of the vacuum-product distribution into every energy graph domain with a linear test order in n at fixed l. It does not give an essential-self-adjointness conclusion: such a conclusion would need control of repeated products of the same unbounded field on a suitable set of analytic vectors, which is a different estimate.

The exact conditional conclusion.

Proposition 7.6 (The scalar reconstruction on its actual cyclic sector)**.** Suppose the actual scalar moments satisfy eq. 7.lxxiii, the full positive form in eq. 7.lxxviii, reality, permutation symmetry, G2 and the UI/PD insertion hypotheses. The energy-band field on $\mathcal{D}_\infty \cap \mathcal{K}_{\mathrm{ord}}$ is Hermitian, tempered, Poincare covariant, local and cyclic, with the full forward spectrum. Its Euclidean correlations are exactly the original diagonal-flat moments in their declared sense. The conclusion uses one verified corrected reconstruction theorem import for the permutation and locality step and its stronger output-order bound. For a component satisfying eq. 7.lxxix, the corresponding cluster conclusion holds. No rank-one cluster assertion is required for the cluster-free local conclusion. ■

Proof. The field, domain, sewing, spectral and joint-boundary arguments were proved before the import. eq. 7.lxxxiii–eq. 7.lxxxv identify its moments; eq. 7.lxxxvi–eq. 7.lxxxviii prove locality on the actual energy domain. The remaining assertions are the already established covariance, positivity and cyclicity. □

The assertion concerns this scalar cyclic sector. Gauge-theory existence, full observable identification and a positive mass gap require separate arguments.

7.4. **Unrestricted test functions and joint boundary estimates.** This section proves the test-space statements used in constructing the band fields. Its estimates apply to general spacetime tests and to Hilbert-valued vacuum products. They also justify simultaneous regulator limits, the passage to difference variables and the spectral support in every cut. The arguments use explicit partitions and Fourier coefficients. No unrestricted Euclidean contact extension and no unquantified tensor-product assertion enter the proof.

Write $\mathcal{S}_d = \mathcal{S}(\mathbb{R}^d)$ and use $p_j^{(d)}$ from eq. 7.lxvii. Constants indexed by fixed integers n, j, r below are finite constants for those integers. A bound at every fixed field order is sufficient for the unrestricted temperedness constructed here. The stronger linear-order conclusion of the corrected reconstruction import has already been distinguished from these constructive estimates.

A product partition with controlled support. Choose $b \in C_c^\infty(\mathbb{R}^4)$, nonnegative, supported in $(-1,1)^4$ and positive on $[-3/4, 3/4]^4$. The function $d(x) = \sum_{z \in \mathbb{Z}^4} b(x - z)$ is positive, smooth and periodic. Only finitely many terms meet any point; the number is uniformly bounded. Positivity on one compact fundamental cell gives a strictly positive lower bound for d. Therefore

$$\chi_z(x) = \frac{b(x-z)}{d(x)}, \qquad \sum_z \chi_z(x) = 1, \qquad \sup_z \|\partial^\alpha \chi_z\|_\infty < \infty. \tag{7.xcii}$$

The derivative bound follows by repeatedly differentiating the reciprocal of the smooth periodic function with its positive lower bound. Every χ_z is supported in $z+[-1,1]^4$. For a fixed degree n, use the product partition $\chi_Z(X)=\prod_{i=1}^n \chi_{z_i}(x_i)$, where $Z=(z_1,\dots,z_n)$. It still sums to one and is locally finite. A derivative of prescribed total order has a bound depending only on that order and n, with no dependence on the locations.

Fix also $\eta\in C_c^\infty((-3,3)^4)$ equal to one on $[-1,1]^4$. Put $\kappa=2\pi/8$. On the period-eight box in each slot, expand $\chi_Z(X)F(X)$ in its ordinary smooth Fourier series. Its coefficients are

$$c_{Z,L}(F)=8^{-4n}\int_{[-4,4]^{4n}} \chi_Z(Z+Y)F(Z+Y)e^{-i\kappa\sum_i l_i\cdot y_i}\,dY, \tag{7.xciii}$$

where $L=(l_1,\dots,l_n)\in(\mathbb{Z}^4)^n$. The localized function vanishes in a neighbourhood of the periodic boundary. Its periodic extension is consequently smooth with all derivatives matching. There is no boundary term when derivatives are integrated off it.

Define one-slot waves

$$v_{z,l}(x)=\eta(x-z)e^{i\kappa l\cdot(x-z)}. \tag{7.xciv}$$

Multiplying the periodic Fourier series by the larger cutoffs gives the global identity $\chi_Z F=\sum_L c_{Z,L}\bigotimes_i v_{z_i,l_i}$. Indeed on the support of $\chi_Z F$ the cutoffs are one; off that support inside the periodic box the represented periodic function is zero; outside the larger cutoffs both sides are zero. The reason for using two nested cutoffs is precisely to obtain this global identity without leaving noncompact plane waves.

The coefficient estimate and the exact order budget. On the support of χ_Z, the distances between z_i and x_i are bounded by two in Euclidean norm. Thus their polynomial weights of any fixed order are comparable, uniformly in the centers. Also

$$\prod_{i=1}^n(1+|x_i|^2)^{1/n}\le 1+\sum_{i=1}^n|x_i|^2. \tag{7.xcv}$$

Each factor on the left is at most the common right-hand side before taking the product and the n-th root. Apply $\prod_i(1-\kappa^{-2}\Delta_{y_i})^r$ in eq. 7.xciii, where r is a positive integer. Its Fourier multiplier is $\prod_i(1+|l_i|^2)^r$. The derivative order used in all slots together is at most $2rn$. Leibniz's rule, eq. 7.xcii and eq. 7.xcv therefore give, whenever $J\ge 2rn$,

$$|c_{Z,L}(F)|\le C_{n,r,J}p_J^{(4n)}(F)\prod_{i=1}^n(1+|z_i|)^{-J/n}(1+|l_i|^2)^{-r}. \tag{7.xcvi}$$

The integration boxes have fixed volume, so no location-dependent volume factor is present. Derivatives of the fixed partition cost only $C_{n,r,J}$. The single global weight gives decay J/n in each slot.

For every nonnegative integer M, differentiating eq. 7.xciv gives

$$p_M^{(4)}(v_{z,l})\le C_M(1+|z|)^M(1+|l|)^M. \tag{7.xcvii}$$

Each derivative differentiates either the fixed cutoff or its exponential. The latter contributes a power of κl, and the former has a fixed derivative bound. On its support $1+|x|\le C(1+|z|)$, proving the weight factor. Since there are only finitely many derivative splittings up to order M, their sum is absorbed in C_M.

Choose

$$2r=2\left\lceil\frac{M+6}{2}\right\rceil,\qquad K=n(M+8). \tag{7.xcviii}$$

Then $M+6\le 2r\le M+7$, so $2rn\le K$. Combining eq. 7.xcvi–eq. 7.xcviii yields

$$\sum_{Z,L}|c_{Z,L}(F)|\prod_{i=1}^n p_M^{(4)}(v_{z_i,l_i})\le C_{n,M}p_K^{(4n)}(F). \tag{7.xcix}$$

Here the sums to be checked are exactly $\sum_z (1+|z|)^{-8}$ and $\sum_l (1+|l|)^M (1+|l|^2)^{-r}$, in four dimensions. For completeness, the shell $|z|_\infty = k$, $k \geq 1$, has $(2k+1)^4 - (2k-1)^4 \leq C(k+1)^3$ points. Euclidean and maximum norms in this fixed dimension differ by at most two. The first shell sum is dominated by $\sum (k+1)^{-5}$. The second is dominated by $\sum (k+1)^{3+M-2r}$, whose exponent is at most -3. Both converge. Taking their n-fold products proves eq. 7.xcix with a finite constant at the specified field order.

Actual Schwartz convergence of the tensor series. The identity used in eq. 7.xcix also converges in the full Schwartz topology, not merely in the one displayed seminorm. Fix an arbitrary target order j. A tensor in independent slots satisfies

$$p_j^{(4n)}\left(\bigotimes_i f_i\right) \leq \prod_i p_j^{(4)}(f_i), \tag{7.c}$$

because a total derivative splits between the slots and $1+\sum |x_i|^2 \leq \prod_i (1+|x_i|^2)$. Repeat eq. 7.xcvi with $2r \geq j+6$, and choose $J \geq \max\{2rn, n(j+8)\}$. eq. 7.xcvii and eq. 7.c now make the tensor series absolutely summable in $p_j^{(4n)}$. Since F is Schwartz, its p_J in this argument is finite for every such choice. This proves absolute summability in every target seminorm.

The Schwartz space is complete for this family: a Cauchy sequence converges with each derivative uniformly on compact sets; the limits are successive derivatives by the fundamental theorem of calculus, and the weighted Cauchy bounds put the resulting function in every seminorm. Therefore the tensor series has a Schwartz sum. Its pointwise sum is F, by the locally finite partition and the smooth periodic Fourier identities. Its Schwartz sum is hence also F. Finite sums of tensors are consequently dense in the ordinary unrestricted Schwartz space. Absolute convergence permits any enumeration of the indices, with no conditional rearrangement.

There is also no algebraic tensor ambiguity. For a finite-dimensional space E of functions, the evaluation functionals span E^*: otherwise a nonzero vector of their annihilator would be a function vanishing at every point. Select a basis of evaluations and invert its matrix on a basis of E. Applying these evaluation matrices in each slot shows that the natural map from a finite tensor product of such function spaces to functions on the product is injective. Every finite tensor relation lies in one such product. A multilinear functional therefore has a well-defined linear value on the actual finite tensor functions used above.

Scalar, vector and energy-graph extensions. Let $\mathcal{E}$ be a Banach space and suppose that an actual n-linear map satisfies

$$\|\mathcal{L}(f_1, \dots, f_n)\|_{\mathcal{E}} \leq D \prod_i p_M^{(4)}(f_i). \tag{7.ci}$$

Define its value on a general $F \in \mathcal{S}_{4n}$ by

$$\mathcal{L}(F) = \sum_{Z,L} c_{Z,L}(F)\, \mathcal{L}(v_{z_1,l_1}, \dots, v_{z_n,l_n}). \tag{7.cii}$$

The series is absolutely convergent in $\mathcal{E}$ by eq. 7.xcix, and $\|\mathcal{L}(F)\| \leq D C_{n,M} p_{n(M+8)}(F)$. Linearity follows from linearity of every coefficient and absolute summation. On a tensor $F = \bigotimes_i f_i$, the localized coefficients factor into their one-slot coefficients. The one-slot series converges in p_M; expand the finite product and pass to its absolutely convergent series using eq. 7.ci. This proves that eq. 7.cii agrees with the original multilinear value. Ordinary tensor density proves uniqueness of its continuous extension. In particular a different choice of cutoffs gives the same answer.

For the actual band fields put $\mathcal{L}(f_1,\dots,f_n)=\phi(f_1)\cdots\phi(f_n)\Omega$. Iteration of the weighted field bound gives, at every integer $l\ge 0$,

$$\begin{aligned}\|W^l\mathcal{L}(f_1,\dots,f_n)\| &\le D_{n,l}\prod_i p_{M_{n,l}}(f_i),\\ M_{n,l}=nN+l+7,\qquad & K_{n,l}=n(nN+l+15).\end{aligned}\tag{7.ciii}$$

At the first field the required right energy exponent is $l+N$; at the next it is $l+2N$, and so on. The largest temporal test order in the band estimate is exactly $nN+l+7$. Apply eq. 7.cii first in the Hilbert space, then in each energy graph space with its $\|W^l\cdot\|$ norm. The partial tensor sums are the same vectors. They converge both in the Hilbert space and after applying W^l. Closedness of W^l identifies the two limits. Hence the extension $X_n(F)$ satisfies

$$X_n(F)\in\mathcal{D}_\infty,\qquad \|W^lX_n(F)\|\le C_{n,l}p_{K_{n,l}}(F).\tag{7.civ}$$

This constructs the vector-valued distribution in all its graph seminorms. Coherence between graph orders is a consequence of closedness, not an assumed nuclear extension into an intersection of unbounded-operator domains.

If a fixed field $\phi(g)$ is applied to these vectors, the energy bound makes it continuous from the graph norm of W^{l+N} to that of W^l. Passing finite tensors to the limit proves

$$\phi(g)X_n(F)=X_{n+1}(g\otimes F)\quad\text{in every energy graph norm.}\tag{7.cv}$$

This is the precise multiplication identity needed for the unrestricted polynomial representation. It remains valid for non-tensor F; no product of two Euclidean separated-tuple symbols has been used to define it.

Uniform families and the topology of the boundary limit. Suppose $\mathcal{L}_\alpha$ obey eq. 7.ci with common D,M, and converge to $\mathcal{L}_0$ on every finite tensor function. Their extensions have a common p_K bound. For fixed F, choose a finite tensor approximation in that seminorm. The approximation error is bounded uniformly in α; convergence handles the one fixed finite approximation. This proves $\mathcal{L}_\alpha(F)\to\mathcal{L}_0(F)$ for every unrestricted test. It is valid for simultaneous parameters and therefore does not impose an order on their limits.

In fact the convergence is uniform on bounded Schwartz sets. Here is an elementary compactness argument for exactly the seminorm needed. For a bounded set $\mathcal{B}\subset\mathcal{S}_d$, smooth infinity cutoffs satisfy

$$\sup_{F\in\mathcal{B}}p_K((1-\chi_R)F)\le C_KR^{-1}\sup_{F\in\mathcal{B}}p_{K+1}(F)\longrightarrow 0.\tag{7.cvi}$$

The derivative bounds on χ_R and the extra power of decay prove this by Leibniz's rule. At fixed R, the cut-off set has common compact support and uniformly bounded derivatives through order $K+1$. The derivatives through order K are uniformly equicontinuous by the mean-value formula. Successive Arzela–Ascoli extraction for their finitely many derivatives proves precompactness in p_K. Combining this with eq. 7.cvi gives a finite p_K net for $\mathcal{B}$, with centers in the Schwartz space.

Apply the common bound to the distance from a net center. Convergence at its finitely many centers then gives

$$\sup_{F\in\mathcal{B}}\|\mathcal{L}_\alpha(F)-\mathcal{L}_0(F)\|_{\mathcal{E}}\longrightarrow 0.\tag{7.cvii}$$

Thus scalar convergence is in the strong tempered-distribution topology of uniform convergence on bounded Schwartz sets. The Banach-valued statement and the energy-graph version follow from the same finite-net proof. This strengthens the joint pointwise-in-test boundary passage and records exactly which uniform estimate makes the strengthening legitimate.

For the regulated field moments the intermediate operators are $e^{-\epsilon_jH-y_j\cdot P}$, with $\epsilon_j>|y_j|_2$. The cone makes them contractions in every energy graph norm, and the joint spectral theorem makes them converge strongly there to identity. The product field estimates therefore have the same constants as without regulators. Replacing the finitely many intermediate contractions one at a

time proves convergence on tensors in every needed graph norm. eq. 7.cvi–eq. 7.cvii then prove simultaneous strong distributional convergence on unrestricted tests. The construction does not rely on a separately tempered Laplace transform in each gap.

Center integration with an explicit continuous right inverse. For $n \geq 2$ let

$$C_n(a,\xi) = (a, a+\xi_1, \dots, a+\xi_1+\cdots+\xi_{n-1}), \qquad I_nF(\xi) = \int_{\mathbb{R}^4} F(C_n(a,\xi))\,da. \tag{7.cviii}$$

The linear coordinate change has determinant one. Its inverse is $a = x_1$, $\xi_j = x_{j+1} - x_j$, and

$$|a|^2 + \sum_j |\xi_j|^2 \leq 5 \sum_i |x_i|^2, \qquad |C_n(a,\xi)| \leq n|(a,\xi)|.$$

The first estimate follows from $|x_{j+1} - x_j|^2 \leq 2|x_{j+1}|^2 + 2|x_j|^2$. For the second, each cumulative sum is bounded by Cauchy–Schwarz and then the n slot bounds are added.

A derivative in a difference coordinate differentiates at most n original slots, with coefficients of magnitude one. Separate the center weight of order five from the remaining weight of order j. The first preceding comparison shows that

$$p_j^{(4(n-1))}(I_nF) \leq C_4\, 5^{(j+5)/2} n^j p_{j+5}^{(4n)}(F), \quad C_4 = \int_{\mathbb{R}^4} (1+|a|^2)^{-5/2}\,da < \infty. \tag{7.cix}$$

Derivatives may be taken under the integral by the same integrable bound. Thus center integration is an actual continuous map of Schwartz spaces.

Choose $\rho \in \mathcal{S}_4$ with integral one and put

$$J_\rho\psi(X) = \rho(x_1)\psi(x_2 - x_1, \dots, x_n - x_{n-1}). \tag{7.cx}$$

Then $I_nJ_\rho\psi = \psi$. The coordinate comparison above, the chain rule and the separate-variable weight inequality give

$$p_j^{(4n)}(J_\rho\psi) \leq (3n)^j p_j^{(4)}(\rho) p_j^{(4(n-1))}(\psi). \tag{7.cxi}$$

There are at most three summands when an original coordinate derivative hits the center factor or the adjacent difference factors; iterating gives 3^j. The factor n^j pays the global weight under C_n. These safe constants suffice without a dimension-independent assertion about the coordinate change. For $n = 1$, center integration is simply the integral of a Schwartz function and the right inverse is $c \mapsto c\rho$.

Factoring every translation-invariant distribution. We prove the factorization, including independence of ρ. In center coordinates simultaneous spacetime translation changes only a. A translation-invariant distribution annihilates each $\partial_{a_j}\Psi$, because translation difference quotients converge in the Schwartz topology to that derivative. It remains to identify the kernel of center integration.

Choose a one-dimensional Schwartz function ρ_1 of integral one. For a Schwartz function $f(a,\xi)$, define $P_jf = \rho_1(a_j)\int f\,da_j$. Each P_j is continuous on the full Schwartz space: an extra weight of order two gives an integrable majorant in its one integration variable, and the remaining weights are paid from the global weight. More explicitly $p_j(P_if) \leq C_{j,\rho_1} p_{j+2}(f)$. The P_j commute.

If a Schwartz function $g(t,z)$ has integral zero in t for every parameter z, set $Qg(t,z) = \int_{-\infty}^t g(s,z)\,ds$. For $t \geq 0$ it also equals $-\int_t^\infty g(s,z)\,ds$. In either tail $|s| \geq |t|$. The extra weight of order two therefore proves $p_j(Qg) \leq (1+\pi)p_{j+2}(g)$. Derivatives in z pass through the integral and still have zero integral; positive t derivatives are derivatives of g. Thus Q is a continuous Schwartz primitive and $\partial_t Qg = g$.

The telescoping identity

$$f - P_1P_2P_3P_4f = \sum_{j=1}^4 P_1\cdots P_{j-1}(1-P_j)f = \sum_{j=1}^4 \partial_{a_j}\Psi_j \tag{7.cxii}$$

uses this primitive in the j-th variable of each summand. The summand has zero integral in that variable, and all preceding P_i commute with its derivative. Every Ψ_j is Schwartz. If $\int f(a,\xi)\,da = 0$, the product $P_1P_2P_3P_4f$ vanishes; this proves that the kernel of center integration is exactly the sum of these derivative ranges.

It follows that every translation-invariant tempered distribution T has the unique difference distribution

$$t(\psi) = T(J_\rho\psi), \qquad T(F) = t(I_nF). \tag{7.cxiii}$$

Both maps are continuous by eq. 7.cix–eq. 7.cxi. Independence of ρ follows because two right inverses differ by an element of the just-identified kernel. Uniqueness follows from $I_nJ_\rho = 1$. This proof applies to the unrestricted Minkowski distributions. It does not require assigning a value to an arbitrary Euclidean contact distribution.

Finite spectral cutoffs and their actual graph limits. Let $Q_R = \sum_{m=0}^{R} P_m$ for the energy bands used earlier, and set $\phi_R(g) = Q_R\phi(g)Q_R$. These are bounded operators with finite energy range, even though the individual band subspaces can be infinite dimensional. Their weighted field bounds have the same constants as the original fields, since Q_R commutes with W and has norm one. For $\xi \in \mathcal{D}_\infty$,

$$\begin{aligned}\|W^l(\phi_R(g) - \phi(g))\xi\| \le{}& \|(Q_R - 1)W^l\phi(g)\xi\| \\ &+ C_l\mathbf{t}_{N+l+2}(g)\|W^{N+l}(Q_R - 1)\xi\| \longrightarrow 0.\end{aligned} \tag{7.cxiv}$$

Both terms tend to zero by the spectral theorem. Iterating the graph estimate for products and replacing one factor at a time proves convergence of every finite cutoff vacuum product. A uniform graph estimate is retained at each replacement, so the vectors on which the remaining field factors act are in the exact domains required. Applying the extension and uniform-family lemma gives convergence of the scalar and vector distributions on all unrestricted tests, including strong convergence on bounded Schwartz sets.

At a spatial point the bounded cutoff field is $A_R(x) = \sum_{m,n\le R} A_{mn}(x)$. The generators of its translations on $Q_R\mathcal{K}_{\mathrm{ord}}$ are bounded by the spectral cone and the bounded energy range. Its spacetime translate is therefore a norm-continuous bounded point field. Its vacuum products have an ordinary difference-variable expression, with one joint translation between every adjacent pair of fields. No finite-dimensionality of a band is needed for this assertion.

The carrier in every difference variable. Write a difference variable as $\xi = (t,x)$, and use physical momentum $p = -P$. The joint translation is $\mathscr{U}(\xi) = e^{i(Et - p\cdot x)}$ in joint spectral variables. If the test transform $f^+(E,p) = \int f(t,x)e^{i(Et-p\cdot x)}\,dt,dx$ vanishes on the forward cone, then

$$\int f(\xi)\mathscr{U}(\xi)\,d\xi = f^+(H,-P) = 0. \tag{7.cxv}$$

This follows by Fubini against scalar spectral measures; the integrable norm bound of f permits the operator weak integral. Insert this identity at any one of the intermediate translations of a cutoff vacuum word. The other difference tests and the bounded spectator operators can be held fixed during that integration. Thus its Fourier distribution has its carrier in the cone in each cut.

To obtain product support rigorously, take a compact Fourier set outside the product of the closed cones. Around each point choose a product box in which one designated momentum is outside its cone. A finite subcover and a smooth subordinate partition reduce an arbitrary test on that compact set to finitely many tests on such boxes. The product Fourier construction, with cutoffs inside each box, approximates each by finite tensors in the test topology. Every tensor gives zero by eq. 7.cxv. Hence the complete test gives zero. This proves the joint support statement in all momentum variables.

The difference distributions of the cutoffs converge to the actual difference distribution, because eq. 7.cxiii uses the continuous fixed right inverse J_ρ. Fourier transformation is continuous on

Schwartz space: differentiating its integral introduces powers of the original coordinates, while multiplication by a Fourier coordinate is integration by parts. An extra integrable weight gives, for instance, $p_j^{(d)}(\widehat{f}) \le C_{d,j} p_{2j+d+2}^{(d)}(f)$. Thus convergence passes through Fourier transformation. A test outside a fixed closed carrier was zero at every cutoff and remains zero in the limit. Consequently

$$\operatorname{supp} \widehat{W}_n^{\mathrm{diff}} \subset (\overline{V}_+)^{n-1}. \tag{7.cxvi}$$

The unreduced momentum statement follows by the determinant-one center-and-difference coordinate change. It is the corresponding condition on every partial sum. The center Fourier variable is zero by translation invariance. Nothing here replaces these multiple carrier conditions by the spectrum of a single vector.

An explicit absolute bound before removing the cutoffs. There is a useful stronger justification of Fubini at a strictly interior tube point. In a vacuum word set the endpoint band indices to zero, since the vacuum has energy zero. Every field block contributes at most $C(1+m_j+m_{j+1})^N$, and each intermediate cone regulator with margin $\delta_j > 0$ contributes at most $e^{-\delta_j m_j}$. Since $1+m+n \le (1+m)(1+n)$, the absolute sum is bounded by

$$C^n \prod_{j=1}^{n-1} \sum_{m\ge 0} (1+m)^{2N} e^{-\delta_j m} < \infty. \tag{7.cxvii}$$

There is no band-dimension factor: each path is estimated in operator norm before its band indices are summed. For fixed integer q, comparison with the integral of $(1+x)^q e^{-\delta x}$ gives $\sum_{m\ge 0}(1+m)^q e^{-\delta m} \le C_q(1+\delta^{-q-1})$. For $0<\delta\le 1$, expand $(1+x)^q$ and integrate its monomials; for $\delta \ge 1$, dominate by the fixed sum with $\delta = 1$. This proves the asserted bound in both ranges.

The series in eq. 7.cxvii is uniform in all real coordinate translations. Its tails outside a growing finite band cube tend to zero by absolute summability. Integrating against the absolute value of a Schwartz spacetime test is therefore legitimate, and dominated convergence removes the cutoffs. This proves the fixed-regulator identity with the actual holomorphic point function. The bound blows up at the boundary and is used only in the interior. The regulator-uniform Schwartz estimate, which is required for the joint boundary, comes separately from the field energy bound and graph contractions. The joint boundary therefore follows from the regulator-uniform estimate without interchanging the energy cutoff and a singular boundary limit.

Ordered cutoffs in every Schwartz seminorm. We give the constrained density proof also needed for exact sewing. Let $O_n = \{0 < t_1 < \cdots < t_n\}$, and suppose that every jet of F vanishes outside O_n. Put $\ell_0 = t_1$, $\ell_j = t_{j+1} - t_j$ for $1 \le j < n$. Choose a smooth one-variable cutoff c equal to zero on $(-\infty, 1]$ and one on $[2,\infty)$, and set $C_\delta(t) = \prod_{j=0}^{n-1} c(\ell_j/\delta)$. Derivatives of total order r cost at most $C_{n,r}\delta^{-r}$.

On the support of $(1-C_\delta)F$, a nonzero value of a derivative of F can occur only with all $\ell_j > 0$ and some $\ell_j \le 2\delta$. Move to that hyperplane along its normal. The distance is at most 2δ; all jets of F on the hyperplane are zero. Taylor's formula with integral remainder to order M, applied also to every derivative of F, gives an extra factor $C_M\delta^M$. The normal segment has bounded length for $\delta \le 1$, so the global weights are uniformly comparable along it. Leibniz's rule consequently proves, for $M > j$,

$$p_j((1-C_\delta)F) \le C_{n,j,M}\delta^{M-j} p_{j+M}(F). \tag{7.cxviii}$$

The same estimate covers points where a cutoff derivative is nonzero, because they lie in one of these boundary strips. A finite sum over the possible hyperplanes costs only a constant at fixed n, j, M. Thus $C_\delta F \to F$ in every Schwartz seminorm.

After cutting off infinity, these tests have compact support a positive distance from the ordered boundary. Choose a sufficiently fine product grid in the full configuration space. A product box meeting that compact support has its one-slot time intervals strictly ordered and strictly positive,

after slightly enlarging them while retaining their gaps. Only finitely many boxes occur. Localize and apply the product Fourier expansion with its larger cutoffs still inside those enlarged intervals. Every tensor term then has precisely the required support order. Its finite partial sums converge in every Schwartz seminorm. This proves density by compact ordered tensors, including the time boundary and all adjacent equal-time hyperplanes, rather than only a pointwise ordering of generic configurations.

Exact sewing beyond tensor profiles. For a compact ordered time configuration let $\rho = \min\{1, t_1, t_2 - t_1, \dots, t_n - t_{n-1}\}$. The point vector sewing formula, with all auxiliary heat parameters less than $\rho/4$, gives a vector $\mathcal{V}_n(X)$ and the scalar Euclidean density $\mathcal{E}_n(X)$, with

$$\|\mathcal{V}_n(X)\| + |\mathcal{E}_n(X)| \le C_n \rho^{-Nn}. \tag{7.cxix}$$

For the vector formula the first positive absolute time is needed to make its outer heat operator bounded. For the scalar formula it is harmless to include that additional minimum. The constants are uniform in the real spatial coordinates. On a compact set with a fixed positive ordered gap these are bounded continuous functions, by the actual bounded semigroup and insertion formulas.

For compact ordered tensors the narrow-time core insertion formula proves their exact original Euclidean scalar and vector values. The Fourier approximations just proved converge in the original test-space seminorms, and also in L^1 on this common compact set. The first convergence controls the original moments and quotient vectors; the second, with eq. 7.cxix, controls their sewn integrals. Passing the finite tensor identity to the limit proves exact sewing for every compact ordered test.

There is also an absolute-integral formulation for a general ordered Schwartz test. If $\rho < 1$, select its smallest boundary distance and use Taylor's flatness along that hyperplane to order $M = Nn+2$. If $\rho = 1$, the ordinary Schwartz bound already suffices. With $J = 4n + 2$, in either case

$$|F(X)| \le C_n \rho^{Nn+2} (1 + |X|^2)^{-J/2} p_{J+Nn+2}(F). \tag{7.cxx}$$

The finite normal displacement in the Taylor argument preserves the weight comparison. The indicated seminorm pays both its derivative order and its weight. Multiplying by eq. 7.cxix leaves $C_n \rho^2 (1 + |X|^2)^{-J/2}$, which is integrable on $\mathbb{R}^{4n}$. It follows that

$$\int_{O_n} (\|\mathcal{V}_n(X)\| + |\mathcal{E}_n(X)|) |F(X)| \, dX \le C_n p_{(N+4)n+4}(F). \tag{7.cxxi}$$

For $N = 9$, this is at most a p_{17n} bound for $n \ge 1$. In particular it is compatible with the available p_{20n} Euclidean continuity. No factorial-in-n assertion for the constants of this auxiliary ordered estimate is needed or inferred.

The integral in eq. 7.cxxi is a Bochner integral for the vector value: the vector sewing function is norm continuous on compact ordered sets, and the displayed majorant is integrable. The ordered cutoffs converge in Schwartz topology by eq. 7.cxviii. The formula in eq. 7.cxxi also controls the difference of their integrals. Thus the compact sewing identity extends to every ordered Schwartz test, both as a scalar identity and as the actual ordered quotient vector. This proves the precise unrestricted and constrained test-space bridges used in the field construction while leaving the ordinary Euclidean contact domain unassigned.

7.5. **The scalar polynomial representation and its domains.** The unrestricted moments now define a positive noncommutative polynomial state. We identify its quotient representation with the actual band fields, including their generators, graph cores and bounded heat-regularized insertions. The construction is on $\mathcal{K}_{\mathrm{ord}}$. The larger positive-time quotient need not be identified with it.

The algebra in this section is an algebra of Minkowski test sequences. Its concatenations are available on all Schwartz functions because the unrestricted joint boundary has been proved. The earlier separated Euclidean tuple domain, on which arbitrary concatenations need not be admissible, is not being declared an algebra by analogy.

The full test-sequence algebra. Let

$$\mathfrak{B} = \bigoplus_{n\ge 0}^{\text{finite}} \mathcal{S}(\mathbb{R}^{4n}), \qquad \mathcal{S}(\mathbb{R}^0) = \mathbb{C}. \tag{7.cxxii}$$

Every element has finitely many nonzero components. Define

$$\begin{aligned} (FG)_n &= \sum_{r+s=n} F_r \otimes G_s, \qquad 1 = (1, 0, \ldots), \\ F_n^*(x_1, \ldots, x_n) &= \overline{F_n(x_n, \ldots, x_1)}. \end{aligned} \tag{7.cxxiii}$$

Associativity follows by grouping the same finite triple sum over degrees; the unit is immediate. Reversal interchanges the two blocks of a tensor, so $(FG)^* = G^*F^*$, and a second reversal and conjugation gives $(F^*)^* = F$. These are identities of actual Schwartz functions, including components that are not finite tensor sums.

The topology is the locally convex direct-sum topology of the Schwartz spaces. A linear map from it to a locally convex space is continuous precisely when its restriction to each degree is continuous. On any prescribed finite set of degrees, multiplication is jointly continuous: the tensor seminorm estimate in eq. 7.c bounds each summand, and the degree sum is finite. No completion over infinitely many field degrees is needed for the state or the representation below.

Set

$$\omega(F) = \sum_n W_n(F_n), \qquad X(F) = \sum_n X_n(F_n). \tag{7.cxxiv}$$

The scalar and vector sums are finite. The proved unrestricted positivity and tensor-density passage give

$$\omega(F^*G) = \langle X(F), X(G)\rangle, \qquad \omega(1) = 1. \tag{7.cxxv}$$

Thus the polynomial state has been obtained from the actual moments, not imposed on an undefined Euclidean product. It is continuous on every degree. Its topology and positivity do not assert a C^*-norm bound for arbitrary polynomial multiplications.

Cauchy–Schwarz and the null left ideal. For completeness the quotient construction can be made directly from positivity. The form $q(F, G) = \omega(F^*G)$ is Hermitian and positive semidefinite. Hermiticity follows either from eq. 7.cxxv, or by expanding the nonnegative real number $\omega((F+zG)^*(F+zG))$ for real and imaginary values of z. For $q(G, G) > 0$, choose $z = -q(G, F)/q(G, G)$ in the quadratic form. If $q(G, G) = 0$, its linear coefficient must vanish for all complex z. These two cases prove

$$|q(F, G)|^2 \le q(F, F)q(G, G). \tag{7.cxxvi}$$

Consequently a null vector is orthogonal to every element.

Let $\mathfrak{N} = \{F : q(F, F) = 0\}$. It is a linear subspace by eq. 7.cxxvi. It is also a left ideal. If $F \in \mathfrak{N}$ and $A \in \mathfrak{B}$, put $B = A^*AF$, which is another finite algebra element. Orthogonality gives

$$\|AF\|_q^2 = \omega(F^*A^*AF) = q(F, A^*AF) = 0. \tag{7.cxxvii}$$

All terms are finite algebra products; no norm bound on left multiplication has been used. The ideal need not be a right ideal, and no tracial property has been assumed.

Complete $\mathfrak{B}/\mathfrak{N}$ to a Hilbert space $\mathcal{H}_\omega$, and let $\mathcal{D}_{\text{alg}} = \mathfrak{B}/\mathfrak{N}$ be its dense quotient domain. Define

$$\pi(A)[F] = [AF], \qquad \Omega_\omega = [1]. \tag{7.cxxviii}$$

The left ideal makes this well-defined. The domain is invariant under every $\pi(A)$, the algebra law is exact there, and $\langle [F], \pi(A)[G]\rangle = \langle \pi(A^*)[F], [G]\rangle$. The adjoint's domain therefore contains the dense subspace $\mathcal{D}_{\text{alg}}$, with $\pi(A^*) \subset \pi(A)^*$. Every $\pi(A)$ is closable. For $A = A^*$ it is symmetric on this domain; its closure is not thereby proved self-adjoint.

For a one-field label $g \in \mathcal{S}_4$, let $\iota(g) = (0, g, 0, \ldots)$. The corresponding GNS field is $\phi_\omega(g) = \pi(\iota(g))$. It depends linearly on g, and $\phi_\omega(g)^*$ contains $\phi_\omega(\bar{g})$ on the common quotient domain. All of these assertions are about defined Hilbert-space operators.

The exact unitary identification with the ordered sector. eq. 7.cxxv defines an isometry

$$J : \mathcal{D}_{\mathrm{alg}} \longrightarrow \mathcal{K}_{\mathrm{ord}}, \qquad J[F] = X(F). \tag{7.cxxix}$$

It is well-defined because $\mathfrak{N}$ is exactly the kernel of the actual vector map. Conversely every vector in that kernel is null, so no additional quotient has entered. The range contains every finite field-polynomial vacuum vector. Their span is dense in the ordered sector by the bounded sewing and cyclicity proof. Therefore the continuous extension of J is onto. It is a unitary from $\mathcal{H}_\omega$ to $\mathcal{K}_{\mathrm{ord}}$, sending Ω_ω to the actual vacuum Ω.

For a general non-tensor component the multiplication identity in eq. 7.cv says

$$J\phi_\omega(g)[F] = \phi(g)J[F]. \tag{7.cxxx}$$

The right side is defined because $X(F) \in \mathcal{D}_\infty$. For finite tensors it is the ordinary product identity. Passing to a general component in the energy graph norm of order N proves exactly the same equality; passing only in Hilbert norm would not have sufficed for the unbounded field on the right.

This identifies the field representation, not merely its numerical vacuum moments. In particular the cyclic Hilbert space in the reconstruction theorem is the actual closed ordered range. The unitary eq. 7.cxxix supplies no fields on $\mathcal{K} \ominus \mathcal{K}_{\mathrm{ord}}$, and it does not require that complement to vanish.

Translations, generators and spectral projections. Let τ_a shift every spacetime argument by the same real vector in the active test convention. Translation invariance of W preserves eq. 7.cxxv, so $U_\omega(a)[F] = [\tau_a F]$ extends to a unitary on the GNS space. The vector-valued Schwartz continuity makes it strongly continuous on the dense quotient domain; unitarity extends that continuity to all vectors. The actual band-field covariance gives

$$JU_\omega(t, y)J^{-1} = e^{itH}U_y \quad \text{on } \mathcal{K}_{\mathrm{ord}}. \tag{7.cxxxi}$$

This equality of bounded groups is first verified on the dense polynomial vectors and then extended. Stone's uniqueness theorem therefore identifies their self-adjoint generators with their exact domains. In particular

$$\begin{aligned} J\,\mathrm{Dom}\,H_\omega &= \mathrm{Dom}\,H \cap \mathcal{K}_{\mathrm{ord}}, \qquad & JH_\omega J^{-1} &= H_{\mathrm{ord}}, \\ Je^{-sH_\omega}J^{-1} &= T_s|_{\mathcal{K}_{\mathrm{ord}}}, \qquad & J\mathsf{E}_\omega(A)J^{-1} &= \mathsf{E}(A)|_{\mathcal{K}_{\mathrm{ord}}}. \end{aligned} \tag{7.cxxxii}$$

The last assertion holds for every Borel set of joint energy and momentum. Indeed the separate strongly commuting groups in eq. 7.cxxxi have the same joint spectral representation after conjugation. One can also obtain it first for their bounded Fourier transforms, then for the generated spectral projections by the uniqueness part of joint functional calculus. Thus there is no new choice of Hamiltonian or of momentum in the quotient representation.

For any integer l, the same functional calculus gives $J\,\mathrm{Dom}(1 + H_\omega)^l = \mathrm{Dom}\,W^l \cap \mathcal{K}_{\mathrm{ord}}$. The common energy domain is transported exactly. The signs in eq. 7.cxxxi are those of the band construction; physical spatial momentum in $Et - p \cdot x$ is $-P$. Its joint cone and all-cut moment spectrum are consequently the already proved ones.

Integrated polynomial operators on the energy domain. The vector extension can be strengthened to an operator extension. For a finite product of m fields and $l \geq 0$, iteration of the energy bound gives

$$\left\| W^l \phi(g_1) \cdots \phi(g_m) W^{-(mN+l)} \right\| \leq C_{m,l} \prod_i p_{mN+l+7}(g_i). \tag{7.cxxxiii}$$

The product is initially defined on the invariant energy domain. The bound gives its unique bounded extension between the indicated weighted spaces. Regard this as an m-linear map into the Banach space of bounded operators from $\operatorname{Dom} W^{mN+l}$, with norm $\|W^{mN+l}\xi\|$, to $\operatorname{Dom} W^{l}$, with norm $\|W^{l}\xi\|$. The constructive Fourier lemma applies in that actual Banach space. For $F \in \mathcal{S}(\mathbb{R}^{4m})$ it defines $\Phi_m(F)$ with

$$\|W^{l}\Phi_m(F)W^{-(mN+l)}\| \le C'_{m,l}p_{m(mN+l+15)}(F). \tag{7.cxxxiv}$$

The partial Fourier sums at different l are the same polynomial operators. On a vector in a higher energy domain their Hilbert limits agree; closedness of W^l identifies their graph limits. Hence these definitions give one coherent operator on $\mathcal{D}_\infty$, mapping it into itself.

For finite tensors, concatenation and reversal are the ordinary field-product rules. For general F, G, approximate each by finite tensors in every seminorm needed by eq. 7.cxxxiv. Before composing, use the estimate with the larger graph exponent required by the left factor. Both compositions then converge on the same domain. This proves

$$\begin{aligned} \Phi_m(F)\Phi_n(G) &= \Phi_{m+n}(F\otimes G) \quad \text{on } \mathcal{D}_\infty, \\ \langle \xi, \Phi_m(F)\eta\rangle &= \langle \Phi_m(F^*)\xi, \eta\rangle \quad (\xi,\eta\in\mathcal{D}_\infty). \end{aligned} \tag{7.cxxxv}$$

The tensor approximations converge in graph norms, so this passage does not apply a merely closable operator to an uncontrolled Hilbert-space limit. Each $\Phi_m(F)$ restricted to the common domain is closable, because its adjoint contains the densely defined conjugate operator in eq. 7.cxxxv.

The finite sum $\Phi(A) = \sum_m \Phi_m(A_m)$ is now an actual representation of $\mathfrak{B}$ on the invariant common domain. It extends its GNS polynomial realization: on $J\mathcal{D}_{\mathrm{alg}}$, eq. 7.cxxxv and vacuum cyclicity give $\Phi(A)J[F] = J[AF]$. Thus the unrestricted algebra product agrees with the field product on the energy domain.

Polynomial vectors form an energy graph core. Let $\mathcal{P}$ be the span of finite tensor field-polynomial vacuum vectors. It is dense in the ordered sector and is contained in every energy domain. It is invariant under real time translations by field covariance. We prove, for each fixed $l \ge 0$,

$$\overline{\mathcal{P}}^{\,\|W^l\cdot\|} = \operatorname{Dom} W^l \cap \mathcal{K}_{\mathrm{ord}}. \tag{7.cxxxvi}$$

Choose $c_R \in C_c^\infty(\mathbb{R})$, equal to one on $[0,R]$, with $0 \le c_R \le 1$ on the positive axis and support there below $2R+2$. The inverse Fourier transform is integrable, and the spectral theorem gives $c_R(H) = \int \check{c}_R(t)e^{itH}\,dt$ in the strong sense, with a consistent Fourier normalization.

For $\zeta \in \mathcal{P}$, this is also a Bochner integral in the W^l graph norm. The translated vector has constant graph norm $\|W^l\zeta\|$; its orbit is graph continuous, because e^{itH} commutes with W^l and is strongly continuous on the vector $W^l\zeta$. First truncate the integrable Fourier tail. On the remaining compact time interval, uniform continuity gives finite simple-function or Riemann-sum approximations in the graph norm. Every sample is a translated polynomial vector, hence remains in $\mathcal{P}$. Therefore $c_R(H)\zeta$ belongs to its graph closure.

Now let $\xi \in \operatorname{Dom} W^l \cap \mathcal{K}_{\mathrm{ord}}$. At fixed R, choose $\zeta_j \in \mathcal{P}$ converging to ξ in Hilbert norm. Boundedness of $W^l c_R(H)$ makes $c_R(H)\zeta_j \to c_R(H)\xi$ in graph norm. The previous paragraph puts all these approximants in the graph closure of $\mathcal{P}$. Finally $c_R(H)\xi \to \xi$ in that graph norm by dominated spectral convergence. This proves eq. 7.cxxxvi, including the order of the two approximations and the bounded operator which justifies the first.

The dense GNS domain $J\mathcal{D}_{\mathrm{alg}}$ contains $\mathcal{P}$, so it is also an energy graph core. Consequently the extension from the polynomial quotient to $\mathcal{D}_\infty$ does not create a different closure of a fixed field. For example,

$$\begin{aligned} \overline{\phi(g)|_{\mathcal{P}}} &= \overline{\phi(g)|_{J\mathcal{D}_{\mathrm{alg}}}} \\ &= \overline{\phi(g)|_{\mathcal{D}_\infty\cap\mathcal{K}_{\mathrm{ord}}}} \\ &= \overline{\phi(g)|_{\operatorname{Dom} W^N\cap\mathcal{K}_{\mathrm{ord}}}}. \end{aligned} \tag{7.cxxxvii}$$

To prove the last equality, approximate a vector in $\operatorname{Dom} W^N$ by polynomial vectors in that graph norm. Their field images converge by the $k = 0$ energy bound. Thus its graph lies in the closure of the polynomial graph. The reverse graph inclusions follow from the domain inclusions. All operators in eq. 7.cxxxvii are closable, by the conjugate-field adjoint identity. The same proof applies to a fixed integrated polynomial of degree at most m, using W^{mN}.

Poincare implementation on both natural domains. The reconstructed moments are Poincare covariant. Pullback of tests therefore gives a null-preserving isometry $[F] \mapsto [\alpha_{a,\Lambda} F]$ on the quotient. Its inverse is the inverse group transformation. It extends to a unitary, with exact group law. Schwartz continuity on every fixed degree makes it strongly continuous on the quotient domain, and unitarity extends this to the completion. On polynomial vectors it has exactly the band representation's action, so eq. 7.cxxix intertwines the two representations on the entire Hilbert spaces.

The quotient domain is invariant under this action. The energy domain is invariant as well, for a different reason. If $(\gamma, b) = \Lambda e_0$, the exact translation group law and Stone's theorem give $U(\Lambda) H U(\Lambda)^{-1} = \gamma H + b \cdot P$. The round cone yields the positive comparisons

$$(\gamma - |b|) H \le \gamma H + b \cdot P \le (\gamma + |b|) H. \tag{7.cxxxviii}$$

These are inequalities in one joint spectral representation. They imply equality of every integer-power domain and equivalence of its graph norms, since $\gamma > |b|$. For a fixed group element, the action is consequently continuous in each energy graph norm. The field intertwining law on the polynomial domain extends by eq. 7.cxxxvi and eq. 7.cxxxiv to the full common energy domain, for both single fields and integrated words.

This verifies two distinct useful domains: the algebraically invariant quotient domain and the complete intersection of energy domains. They need not be equal as sets. They give the same canonical field closures by eq. 7.cxxxvii, and the exact same Hilbert representation by eq. 7.cxxix. No choice of an unspecified larger extension is hidden in the comparison.

Local polynomial algebras and the locality ideal. For an open spacetime set O, let $\mathfrak{B}_{\mathrm{pol}}(O)$ be the unital algebra generated by the degree-one labels in $C_c^\infty(O)$, and use the actual representation Φ on $\mathcal{D}_\infty$. Isotony is immediate from inclusion of supports. Poincare transformations carry this algebra to that of the transformed region, by the already proved common-domain covariance.

If O_1 and O_2 are spacelike separated, every generator from the first commutes with every generator from the second on the invariant domain. Repeatedly move adjacent generators past one another, applying the proved field commutator at each step. Every intermediate vector remains in the common domain. Hence the represented polynomial algebras commute there:

$$\begin{gathered} [\Phi(A), \Phi(B)]\xi = 0, \\ A \in \mathfrak{B}_{\mathrm{pol}}(O_1), \quad B \in \mathfrak{B}_{\mathrm{pol}}(O_2), \quad \xi \in \mathcal{D}_\infty. \end{gathered} \tag{7.cxxxix}$$

One may also use compact jointly smeared words supported in O_i^m. Cover that compact support by product boxes whose larger cutoffs still lie in O_i, and apply the support-preserving Fourier approximation. The operator graph bound in eq. 7.cxxxiv then passes the same commutation identity to those general smearing functions. This is a topological extension of the local polynomial algebra, not an assertion that every such smearing is literally a finite tensor sum.

Let $\mathfrak{I}_{\mathrm{loc}}$ be the two-sided algebra ideal generated by $\iota(g)\iota(h) - \iota(h)\iota(g)$ with spacelike separated supports. Each generator acts as zero on the GNS domain by locality. Left and right multiplication preserve that domain, so every element of the ideal acts as zero. In particular

$$\mathfrak{I}_{\mathrm{loc}} \subset \mathfrak{N}, \qquad \pi(\mathfrak{I}_{\mathrm{loc}}) = 0. \tag{7.cxl}$$

The positive scalar state therefore factors through the local polynomial quotient. The ideal is invariant under the scalar Poincare pullbacks because those preserve spacelike separation. It is also invariant under star: the star of a generating commutator is the negative commutator of

the conjugate labels, with their supports unchanged. This provides an actual local polynomial observable structure. It does not assert strong commutation of spectral projections of closures of real fields, or select self-adjoint extensions of them. A bounded local operator net needs an additional operator criterion if it is to be built through such spectral calculi.

Recovering the actual bounded insertion operators. The preceding equivalence can be checked against the original bounded Euclidean insertions. On the ordered sector the energy-band proof established the operator-norm limit

$$T_s\phi(\chi_\epsilon \otimes h)T_s \longrightarrow \Phi_s(h) \quad (\epsilon \longrightarrow 0). \tag{7.cxli}$$

Here χ_ϵ is the fixed nonnegative scaled temporal approximate identity chosen in the band construction, with $\|\chi_\epsilon\|_1 = 1$, and h is a compact smooth spatial label. On finitely many bands the limit follows from the time approximate identity and bounded generators. The remaining block tails are bounded by $C(1+m+n)^N e^{-s(m+n)}\|h\|_1$, with vanishing absolute row and column tails uniformly in ϵ. Thus eq. 7.cxli is a bounded-operator limit, not a formal sharp-time substitution.

The same construction on the GNS representation is conjugated by J at every ϵ, by eq. 7.cxxx–eq. 7.cxxxii. The bounded limit is consequently the actual original Riesz operator $\Phi_s(h)$. Spatial approximate identities give the point operator $\Phi_s(x) = V_{4s}(0,x)$ in operator norm, because the analytic density is norm continuous in real space. This identifies more than the vacuum state of the insertion: it identifies the operator between every pair of Hilbert vectors.

There is an exact formula for its real Euclidean time translate. For $0 < s \le 1/4$ and $|t| < s$, choose $0 < a < s - |t|$. Then

$$V_{4s}(t,x) = T_{s+t-a}\Phi_a(x)T_{s-t-a}. \tag{7.cxlii}$$

To verify it, first replace the point label by a narrow compact spacetime approximation and test on ordered compact core vectors. The right heat factor shifts the old entries by $s-t-a$. The insertion at scale a puts the new entry at time a and shifts the old entries by another $2a$. The left heat factor finally puts the new entry at $s+t$ and all old entries at their original times plus $2s$. This is precisely the actual core formula with outside entries shifted by s and the central insertion at t. Both sides are bounded. Dense core equality and the norm point limits prove eq. 7.cxlii wherever the original density is defined. The displayed bounded formula also extends it to the larger interval $|t| < s$.

For the jointly complex translate, write $S(z,w) = e^{-zH+iw\cdot P}$. On the common domain where its two factors are contractions, the formula is

$$V_{4s}(z_4, z_{\mathrm{sp}}) = S(s+z_4-a, z_{\mathrm{sp}})\Phi_a(0)S(s-z_4-a, -z_{\mathrm{sp}}), \tag{7.cxliii}$$

with $|\operatorname{Re} z_4| + |\operatorname{Im} z_{\mathrm{sp}}|_2 < s-a$. At real arguments it agrees with eq. 7.cxlii and spatial covariance. Both sides are operator-norm holomorphic on the intersection with the original PD strip. Testing matrix elements and applying the several-variable identity theorem from a real open box proves their equality on connected common neighbourhoods. In this way the reconstructed generators and field identify the actual PD analytic density, with its branch fixed by the original insertion form. An arbitrary bounded coefficient function having the same vacuum expectation would not have sufficed for this conclusion.

7.6. **Conditional measures on compact metric spaces.** The conditional laws used in the phase construction take values in the Borel probability measures on a compact metric configuration space. The conditioning field may be completed and need not be countably generated as a literal collection of sets. It is the Borel field of the *target* space that is countably generated.

Proposition 7.7 (Compact-metric conditional law)**.** Let X be a nonempty compact metric space, let μ be a Borel probability measure on X, and let $\overline{\mathcal{B}}_\mu$ be the μ-completion of its Borel field $\mathcal{B}(X)$. Suppose

$$\mathcal{N}_\mu \subset \mathcal{H} \subset \overline{\mathcal{B}}_\mu, \tag{7.cxliv}$$

where $\mathcal{H}$ is a σ-field and $\mathcal{N}_\mu$ denotes all subsets of Borel μ-null sets. There exists a map $K : X \to \mathcal{P}(X)$, where $\mathcal{P}(X)$ carries its weak topology, with the following properties.

The map K is both Borel measurable and $\mathcal{H}$-measurable. For every bounded Borel $f : X \to \mathbb{C}$,

$$Kf(x) := \int_X f(y)\, K(x, dy)$$
$$\text{is a version of } E_\mu[f \mid \mathcal{H}]. \tag{7.cxlv}$$

In particular $x \mapsto K(x, B)$ is $\mathcal{H}$-measurable for every $B \in \mathcal{B}(X)$, and

$$\int_H K(x, B)\, \mu(dx) = \mu(H \cap B)$$
$$(H \in \mathcal{H}, B \in \mathcal{B}(X)). \tag{7.cxlvi}$$

The measure on the right is understood on the completion. If K' is another probability-valued kernel satisfying eq. 7.cxlv, then there is a single μ-null set N such that

$$K(x, \cdot) = K'(x, \cdot)$$
$$(x \notin N) \tag{7.cxlvii}$$

as Borel probability measures. One may take N to be a Borel null set. Consequently the law $K_*\mu$ on $\mathcal{P}(X)$ is independent of the choice of conditional kernel. ∎

Borel representatives and countably many tests. We first record why completion causes no obstacle to the Borel version in this statement. Every bounded $\overline{\mathcal{B}}_\mu$-measurable real function u has a Borel representative. Indeed, for each rational r replace $\{u > r\}$ by a Borel set differing from it inside a Borel null set. Outside the union of those countably many null sets, the rational level cuts reconstruct u. Their supremum, clipped to a bound for u, is a Borel function equal to u there. If u was $\mathcal{H}$-measurable, this representative remains $\mathcal{H}$-measurable: the change is confined to a Borel null set, all of whose subsets belong to $\mathcal{H}$ by eq. 7.cxliv. This observation will be applied only countably many times.

Choose a countable dense set $\{x_j\}$ in X. Let $D \subset C(X; \mathbb{R})$ be the collection generated by the constant 1 and the functions $y \mapsto d(y, x_j)$ under rational linear combinations, multiplication, pointwise minimum and pointwise maximum. It is countable, is a rational vector lattice and algebra, and separates points: if $x \neq y$, a dense point sufficiently near x gives different distances from x and y. Its uniform closure is a real algebra containing the constants and separating points, so the Stone–Weierstrass theorem makes D uniformly dense in $C(X; \mathbb{R})$.

For each $f \in D$ take a conditional expectation u_f of f given $\mathcal{H}$. These conditional expectations can be obtained as the orthogonal projections of f onto the closed subspace $L^2(\mathcal{H}, \mu)$ of $L^2(\mu)$. The projection identity against 1_H, $H \in \mathcal{H}$, gives

$$\int_H u_f\, d\mu = \int_H f\, d\mu. \tag{7.cxlviii}$$

Testing the sets on which $u_f < 0$, or u_f exceeds an upper bound for f, proves positivity and the usual L^∞ bound. Linearity and uniqueness follow from the same identity. Choose for each u_f a bounded representative that is both Borel and $\mathcal{H}$-measurable, using the preceding observation.

There are only countably many assertions

$$u_1 = 1,$$
$$u_{rf+sg} = ru_f + su_g$$
$$(f, g \in D, r, s \in \mathbb{Q}),$$
$$u_f \geq 0$$
$$(f \in D, f \geq 0 \text{ on } X). \tag{7.cxlix}$$

Each holds almost surely. Their exceptional sets therefore have a single Borel, $\mathcal{H}$-measurable null union N_0. This is the simultaneous exceptional set needed for the construction. No intersection over all continuous or all Borel functions is taken.

Pointwise Riesz representation. For $x \notin N_0$ define $L_x(f) = u_f(x)$ on D. It is rational linear, positive on nonnegative elements of D, and $L_x(1) = 1$. If rational $r > \|f\|_\infty$, then $r \pm f \in D$ are nonnegative, so

$$|L_x(f)| \leq r.$$

Letting rational r decrease to $\|f\|_\infty$ proves $|L_x(f)| \leq \|f\|_\infty$. Uniform density consequently gives a unique continuous extension to $C(X; \mathbb{R})$. The extension is real linear: approximate a real scalar by rationals and use the norm bound. It is positive: if $f \geq 0$ is continuous, choose $f_j \in D$ with $\|f_j - f\|_\infty < 1/j$; then $f_j + 1/j \geq 0$ and $L_x(f_j) \geq -1/j$. It still takes the value one at 1.

The Riesz representation theorem supplies a unique regular Borel probability measure $K(x, \cdot)$ representing L_x. For $x \in N_0$ set $K(x, \cdot) = \delta_{x_*}$ at one fixed $x_* \in X$. Thus $K(x, \cdot)$ is a probability measure at *every* point. Its values on the null set are a choice, with no conditional interpretation required.

We verify measurability of the probability-valued map. Enumerate D as $\{d_j\}$ and define

$$I : \mathcal{P}(X) \longrightarrow \mathbb{R}^{\mathbb{N}},$$
$$I(\rho) = (\rho(d_j))_j. \tag{7.cl}$$

The map is continuous and injective, by uniform density and uniqueness in Riesz representation. The weak probability space $\mathcal{P}(X)$ is metrized by $\sum_j 2^{-j}|\rho(d_j) - \rho'(d_j)|/(1 + 2\|d_j\|_\infty)$: uniform density identifies convergence in this metric with weak convergence. It is compact. Indeed, compactness follows by taking a diagonal subsequence of integrals on D, extending the resulting positive normalized functional as above, and applying Riesz again. Hence I is a homeomorphism onto its compact image in the Hausdorff space $\mathbb{R}^{\mathbb{N}}$.

Every coordinate of $I \circ K$ equals u_{d_j} outside N_0 and the constant $d_j(x_*)$ on N_0. It is both Borel and $\mathcal{H}$-measurable. The countable product map is therefore measurable for both fields; composing with the inverse of I proves the asserted measurability of K. This also supplies an explicit countable determining family for uniqueness.

All bounded Borel tests and uniqueness. For $f \in D$, the equality $Kf = u_f$ holds outside N_0; thus eq. 7.cxlviii holds with Kf in place of u_f. Uniform approximation extends it to every real continuous f, and Kf is measurable as a uniform limit of the corresponding measurable integrals.

Here are the remaining monotone-class details. The collection of bounded real Borel functions f for which Kf is measurable for both fields and

$$\int_H Kf\,d\mu = \int_H f\,d\mu$$
$$\text{for every } H \in \mathcal{H} \tag{7.cli}$$

is a vector space containing $C(X;\mathbb{R})$. It is closed under uniformly bounded increasing limits, by monotone convergence for each probability $K(x,\cdot)$ and for μ; subtracting a constant reduces to nonnegative functions. The bounded functional monotone-class theorem therefore makes it all bounded Borel functions, since the continuous algebra generates $\mathcal{B}(X)$. One can see the latter directly from a countable dense family of distance functions and the countable base of metric balls. Complex functions follow by their real and imaginary parts. This proves eq. 7.cxlv, and indicators give eq. 7.cxlvi. In particular it also verifies directly that K is a probability kernel, including measurability of every Borel-set evaluation.

For uniqueness, apply uniqueness of conditional expectation to Kd_j and $K'd_j$ for the countably many $d_j \in D$. Outside a single null set all these integrals agree. Uniform density then gives equality on every continuous function, and Riesz uniqueness gives eq. 7.cxlvii. The null set belongs to the completed Borel field and hence is contained in a Borel null set, as asserted. Changing a map on a null set does not change its probability-valued law. The law has barycenter μ, because eq. 7.cli with $H = X$ gives

$$\mu(f) = \int_{\mathcal{P}(X)} \rho(f)\,(K_*\mu)(d\rho)$$
$$(f \text{ bounded Borel}). \tag{7.clii}$$

The integrand is Borel on $\mathcal{P}(X)$: this is immediate for continuous f, and the same bounded monotone-class argument proves it for every bounded Borel f. □

The completed invariant field and the exact null-set scope. For a countable group of Borel, μ-preserving transformations, the field of events in $\overline{\mathcal{B}}_\mu$ invariant modulo μ under every group element is a σ-field containing $\mathcal{N}_\mu$. Thus it satisfies eq. 7.cxliv. In particular the theorem applies to the completed translation-invariant field of a compact metric lattice configuration space. The conditional measures are Borel probabilities on that configuration space; no countable generation of the completed invariant field is needed.

The uniqueness assertion is simultaneous for the resulting probability measures. The assertion that Kf is a conditional-expectation version holds separately for each bounded Borel f and, of course, for any preselected countable collection on one common conull set. It does not assert equality outside one universal exceptional set to an arbitrary previously selected version for *every* Borel function.

Nor does the theorem assert pointwise properness $K(x,H) = 1_H(x)$ simultaneously for all conditioning events. Even when those events are Borel, such a demand is false for completed fields. For example let μ be nonatomic and let $\mathcal{H} = \{H \in \overline{\mathcal{B}}_\mu : \mu(H) \in \{0,1\}\}$ be the null/co-null field. The conditional kernel is $K(x,\cdot) = \mu$ almost surely. If simultaneous properness held outside one null set, then at any point x of that conull set the conditioning event $\{x\} \in \mathcal{H}$ would require $K(x,\{x\}) = 1$, whereas $\mu(\{x\}) = 0$. This does not contradict eq. 7.cxlv: for a fixed Borel $H \in \mathcal{H}$, it gives $K(x,H) = 1_H(x)$ almost surely, with an exceptional set that may depend on H. eq. 7.cxlvi is the corresponding integral identity. The theorem concerns the countably generated Borel target $\mathcal{B}(X)$; no regular-conditional-law assertion for an arbitrary non-countably-generated target field is being made.

7.7. **Canonical phases from lattice estimates.** This section gives a conditional implication with completely specified probabilistic objects. The finite-spacing law G3 is proved in appendix A.7; it

is used before any continuum component is constructed. The all-state moment and one-point comparison estimate G4 is derived from the ordinary torus source hypothesis D1 in § 5.22. Conditional probability measures are constructed in § 7.6. The argument here deduces the canonical phase measure from those precise ingredients.

In this section a pure state means an extreme point of the translation-invariant DLR class $\mathcal{G}_j$, equivalently an ergodic state for lattice translations. Extremality in the larger class of all, possibly noninvariant, DLR states is not used and is not identified with this property without an additional argument.

Put $a_j = L^{-K_j} \downarrow 0$, with L an integer, and let $\Omega_j = \mathrm{SU}(2)^{\mathcal{B}_j}$ be the space of bond configurations on $a_j\mathbb{Z}^4$. The set $\mathcal{B}_j$ contains one orientation of each bond. Its countability makes Ω_j compact and metrizable: enumerate the bonds and use the sum of 2^{-m} times bounded metrics on the group factors. The Wilson specification γ_W integrates finitely many bonds W with density proportional to $\exp[-\beta_j \sum_{p:p\cap W\neq\varnothing} a_p]$. The denominator is positive, and compactness of the integration space shows that $\gamma_W F$ is continuous and local whenever F is. Define $\mathcal{G}_j$ to consist of the translation-invariant probability measures with $\rho(\gamma_W F) = \rho(F)$ for every finite W and continuous local F. The equations are weakly closed, so $\mathcal{G}_j$ is compact and convex. Its nonemptiness follows from the fixed-spacing torus-limit construction, and a centering state $\nu_j \in \mathcal{G}_j$ is fixed. G1 supplies the lattice symmetries and mirror positivity used in G3.

Let $\mathfrak{T}$ be the set of finite tuples of real C^1 compactly supported functions with pairwise disjoint supports, including the empty tuple. Concatenation is used only when its resulting supports are disjoint. For $X = (f_1, \dots, f_k)$ define the bounded continuous local function

$$X_j = \prod_{i=1}^{k} [\mathcal{O}_j(f_i) - \nu_j(\mathcal{O}_j(f_i))],$$

$$\varnothing_j = 1.$$

Assume $\nu_j(X_j) \to \sigma(X)$ on $\mathfrak{T}$. For support radii R and separation d, put $b_1 = C_1(R)$ and $b_n = C_n(d; R)$ for $n \geq 2$, and define

$$M_k(d; R) = \sum_{\pi \in \mathrm{Part}(\{1,\dots,k\})} \prod_{J\in\pi} b_{|J|},$$

$$M_0 = 1.$$

The ordinary moment–cumulant identity, applied under any $\rho \in \mathcal{G}_j$, gives

$$|\rho(X_j)| \leq M_k(d; R) \prod_i \|f_i\|_\sharp.$$

Indeed, a singleton block contributes the difference between the means in ρ and ν_j, and a larger block contributes the corresponding cumulant under ρ. This is why a bound on one-point differences is needed. Telescoping a difference of two products gives equicontinuity with the same constants whenever all supports remain in fixed separated containers.

7.7.1. *Decomposition at one spacing.* Let $\mathcal{I}$ denote the completed invariant σ-field for a fixed $\nu \in \mathcal{G}_j$. Prop. 7.7 gives a probability kernel $\omega \mapsto \nu^\omega$ satisfying $\nu^\omega(F) = E_\nu[F \mid \mathcal{I}](\omega)$ for bounded Borel F. We verify that the components are states of the required class.

First, a bounded invariant density preserves the DLR equations. To prove it, approximate such a density h in $L^1(\nu)$ by bounded continuous local functions h_n. This approximation is available because cylinder functions are uniformly dense in continuous functions on the compact product space, and continuous functions are dense in L^1 of a Borel probability measure. Translate h_n so

its support misses W. Invariance of h and ν leaves the approximation error unchanged, while $\gamma_W(h_nF) = h_n\gamma_W F$ for the translated function. Passing to the limit proves $\nu(h\gamma_W F) = \nu(hF)$.

Consequently $\nu^\omega(\gamma_W F) = \nu^\omega(F)$ almost surely. Likewise $\nu^\omega(\tau_v F) = \nu^\omega(F)$ almost surely for every lattice translation v. These statements can be imposed on one common full-measure set: finite bond sets and translations form countable families, and continuous functions on the compact configuration space have a countable dense family of continuous cylinder functions. Supremum-norm continuity then extends the equations to all continuous local functions. A monotone-class argument extends the integral identities to bounded Borel functions. Thus $\nu^\omega \in \mathcal{G}_j$ almost surely.

The law G3 proves that almost every such state is ergodic. Fix a continuous local A. Applied to ν^ω and then integrated, its far-translation limit gives

$$\int \mathrm{Pl}_{\nu^\omega}(A, A)\, d\nu(\omega) = \mathrm{Pl}_\nu(A, A) = \int |\nu^\omega(A)|^2\, d\nu(\omega).$$

The first passage to the limit uses the bound $\|A\|_\infty^2$ and dominated convergence. The difference of the integrands on the two sides is the nonnegative variance of the invariant conditional expectation of A in ν^ω. It is therefore zero almost surely. Impose this on a countable dense family of continuous local A. Density in $L^2(\nu^\omega)$ then says that every invariant L^2 function is constant, which is ergodicity.

Write $\mathcal{E}_j$ for these ergodic states. They are exactly the extreme points of $\mathcal{G}_j$. A nontrivial invariant event decomposes a state into two DLR states by the invariant-density argument. Conversely, the Radon–Nikodym density in a convex decomposition of an ergodic invariant state is invariant, and hence constant. Moreover $\mathcal{E}_j$ is closed. If $\rho_n \in \mathcal{E}_j$ tends weakly to ρ, G3 gives a decay estimate uniform in n. Passing first to this weak limit and then to far translations identifies $\mathrm{Pl}_\rho(A, A)$ with $|\rho(A)|^2$. The same dense-set argument proves that ρ is ergodic. Thus $\mathcal{E}_j$ is compact.

Let D_j be the law of ν_j^ω under ν_j. It is supported on $\mathcal{E}_j$, has barycenter ν_j, and does not depend on versions of the kernel. For this last assertion choose one countable determining family of continuous cylinder functions. Conditional expectations are unique almost surely for that entire family simultaneously; equality on the family implies equality of the probability measures. This also establishes measurability of the probability-valued map and of every evaluation used below.

Uniqueness among all decompositions supported on $\mathcal{E}_j$ requires one more observation. For any such law with barycenter ν_j, the uniform law G3 identifies the integral of a product of finitely many state evaluations with successive far-translation limits of ν_j. These moments are therefore the same as those of D_j. Evaluations of continuous local functions are continuous and separate measures in the compact space $\mathcal{E}_j$. Their polynomial algebra is dense by Stone–Weierstrass, so equal mixed moments imply equal laws.

7.7.2. *Continuum evaluation coordinates.* Choose countable dense test sets as follows. Let $\mathcal{Q}$ contain all finite unions of closed boxes with rational corners. The space of C^1 functions vanishing off $Q \in \mathcal{Q}$, with its C^1 norm, is separable: embed it by its value and four first derivatives in a finite product of separable continuous-function spaces. Choose a countable dense set in each space and enlarge their union by the countable group of L-adic translations and by W_4. Let $\mathfrak{T}^\circ$ consist of the separated tuples from this union. It is countable. Every separated tuple can be approximated by such tuples with supports in fixed separated containers: cover each original support by a finite union of sufficiently small rational boxes inside its chosen open container and use the dense set for that union.

For each $X \in \mathfrak{T}^\circ$ choose once and for all a moment bound M_X from the preceding estimate. Define the compact metrizable product

$$\mathsf{K} = \prod_{X \in \mathfrak{T}^\circ} [-M_X, M_X],$$

$$e_j(\rho) = (\rho(X_j))_{X\in\mathfrak{T}^\circ},$$
$$\lambda_j = (e_j)_* D_j.$$

The evaluation map is continuous for each fixed j. In particular λ_j is a well-defined probability measure on K.

Proposition 7.8 (Conditional canonical decomposition)**.** Under G1, the finite-spacing law G3, the estimates G4, and the stated convergence of $\nu_j(X_j)$, the measures λ_j converge weakly to one probability measure λ. Put $\mathsf{X} = \operatorname{supp}\lambda$. Every $\xi \in \mathsf{X}$ defines a unique functional σ_ξ on $\mathfrak{T}$, obtained as a limit along pure states $\rho_j \in \mathcal{E}_j$ centered at ν_j. The maps $\xi \mapsto \sigma_\xi(X)$ are continuous and separate points. Furthermore

$$\sigma(X) = \int_{\mathsf{X}} \sigma_\xi(X)\, d\lambda(\xi).$$

Their joint moments are the successive far-translation limits of σ. More precisely, for $q \geq 2$,

$$\int_{\mathsf{X}} \prod_{i=1}^{q} \sigma_\xi(X^i)\, d\lambda(\xi) = \lim_{|v_q|\to\infty} \cdots \lim_{|v_2|\to\infty} \sigma(X^1 \cdot \tau_{v_2} X^2 \cdots \tau_{v_q} X^q).$$

The innermost limit is taken first. At each stage the remaining groups must have disjoint supports; every sufficiently distant next translation is then admitted. Each of these intermediate limits exists. These properties determine the compact space, measure and functionals uniquely among compact metrizable full-support representations with continuous point-separating evaluations and the displayed mixed-moment property, up to the homeomorphism preserving all evaluations. Each component clusters with a bound $C(X,Y)|v|^{-2}$ for sufficiently distant separated products, where $C(X,Y)$ and the distance threshold are uniform in ξ. ∎

Proof. The crucial uniformity comes from reflection sizes rather than supremum norms of X_j. Choose an integer physical half-side ϖ so that all supports under consideration lie in $(-\varpi+1, \varpi-1)^4$. This also leaves room for the corners of plaquettes whose barycenters are in those supports. Then $r_j = \varpi/a_j$ is an integer. Each expression defining $\mathcal{N}_{\rho,r_j}(X_j)$ is the moment of two separated reflected copies of X. G4 bounds it uniformly in j and ρ by a constant determined by $M_{2|X|}$ and the test norms. For a displacement v common to the lattices,

$$\frac{r_j}{|v|_\infty/a_j - 2r_j} = \frac{\varpi}{|v|_\infty - 2\varpi}.$$

Thus G3 gives, in every pure state, factorization of the two separated products with error at most $C(X,Y)|v|^{-2}$, uniformly in the cutoff. The norms of the individual representatives, which can diverge, do not enter this bound.

For finitely many coordinates $X^1, \ldots, X^q$, integrate this factorization over D_j. Choose v_q first, then v_{q-1}, down to v_2, with each new vector sufficiently far from the already fixed union. In the factorization at that stage the enclosing support box and its constant may depend on the fixed outer translations, but not on j. Repeating the estimate and multiplying errors only by the uniform individual moment bounds makes the total error smaller than any prescribed ϵ, uniformly in j. At that one fixed finite configuration the remaining single moment converges by the hypothesis on ν_j. The mixed coordinate moments of λ_j are therefore Cauchy. Their precise iterated-limit representation will follow after the components and their uniform clustering have been constructed. Polynomials in finitely many coordinates separate points of K and contain constants. By Stone–Weierstrass they are uniformly dense in its continuous functions. Compactness therefore supplies a weak limit, and the convergent mixed moments make that limit unique along the full sequence.

If ξ belongs to its support, every neighborhood of ξ has positive λ measure. Portmanteau implies that its preimage has positive D_j measure for all sufficiently large j. Use a nested countable neighborhood base and a diagonal choice to obtain $\rho_j \in \mathcal{E}_j$ with $e_j(\rho_j) \to \xi$. The uniform telescoping estimate extends convergence from $\mathfrak{T}^\circ$ to every tuple, giving σ_ξ. It also proves independence from the choice of ρ_j and continuity in ξ, by uniformly approximating a given evaluation with dense-set evaluations. Multilinearity, symmetry and the moment bounds pass to this limit.

Barycenters and higher mixed moments pass to the limit first on the countable set and then on all tuples by the same uniform approximation. The evaluations separate points because their countable subfamily consists of the coordinates of K. If another compact full-support representation has the same mixed moments, embed both spaces by *all* their tuple evaluations into $\mathbb{R}^{\mathfrak{T}}$. Their images have compact union. Polynomials in finitely many of these coordinates are dense in the continuous functions on that union, again by Stone–Weierstrass. Equality of mixed moments therefore gives equality of the two image measures. Each compact image equals the support of its image measure. The images consequently coincide, and injectivity and compactness identify the two spaces by a homeomorphism preserving the measure and every evaluation. This argument does not presume that a countable subfamily separates an arbitrary competing representation.

Finally pass the pure-state factorization estimate to the limit for L-adic translations. Continuity of C^1 tests in the norm $\|\cdot\|_\sharp$ extends it to real translations, preserving the uniform constants after a fixed enlargement of the support margin. This proves the asserted clustering. Finally hold $v_3, \dots, v_q$ fixed and let $|v_2|$ tend to infinity in the barycenter identity. Uniform component clustering separates X^2 from the fixed union of all the other groups. The uniform moment bounds justify passing this limit through the integral. Then let $|v_3|$ tend to infinity, and continue in that order. This proves existence of every intermediate limit and the displayed mixed-moment formula, with the innermost limit taken first. □

7.7.2.1. Plaquette long-range order and the phase means. Let $\mathcal{I}_j$ be the invariant sigma-field of the lattice state ν_j. For a pure state ρ define its rescaled plaquette mean difference by

$$C_j(\rho) = 6a_j^{-4}\big[\rho(A_p) - \nu_j(A_p)\big].$$

Choose a real singleton f_* from the determining family with $\int f_* \neq 0$. By eq. 5.cdlxx, its coordinate in $e_j(\rho)$ is $R_j(f_*)C_j(\rho)$, with $R_j(f_*) \to \int f_*$. The coordinate is uniformly bounded. Weak convergence of λ_j therefore gives convergence of $\int C_j^2\, dD_j$ to $\int \mathfrak{c}_\xi^2\, d\lambda(\xi)$, where $\sigma_\xi((f)) = \mathfrak{c}_\xi \int f$. The barycenter has zero first moment, so $\int \mathfrak{c}_\xi\, d\lambda = 0$.

The finite-spacing conditional law in G3 gives $\rho(A_p) = E_{\nu_j}[A_p \mid \mathcal{I}_j]$ under the ergodic decomposition. Hence

$$\kappa_S := \int \mathfrak{c}_\xi^2\, d\lambda(\xi) = \lim_j 36a_j^{-8}\,\mathrm{Var}_{\nu_j}\big(E_{\nu_j}[A_p \mid \mathcal{I}_j]\big).$$

The lattice plateau estimate identifies the unscaled variance with the far-translation limit of $\nu_j(A_p\,\tau_v A_p) - \nu_j(A_p)^2$ at that fixed spacing. For separated f, h, take the barycenter of the two-point moment and subtract the product of the means within each phase. This gives

$$S_2(f,h) = \int S_2^\xi(f,h)\, d\lambda(\xi) + \kappa_S \int f \int h.$$

In the zero-energy representation, the projection of a singleton with nonzero integral has a nonconstant phase coordinate when $\kappa_S > 0$. Its variance tests the singleton sector of the vacuum criterion.

For separated C^1 tests in one ball of radius $5\mathsf{d}$ with $\mathrm{Lip}\, f_i \le \|f_i\|_\infty/(\vartheta\mathsf{d})$, take the realizing states of two phases at the same cutoffs and apply the short-distance state comparison. Higher cumulants are invariant under the centering shifts. At order one use the same centering state for both means.

On taking limits the oscillation across phases of the cumulant of a subtuple B is at most

$$C_{|B|,\vartheta}\mathsf{d}^{4-|B|}\prod_{i\in B}\|f_i\|_\infty.$$

The same comparison includes the mixture by using its defining states. Apply eq. 6.lxxvi with the phase law λ. In each outer cumulant with at least two blocks subtract a fixed reference phase value from every argument. The bounded-variable cumulant estimate gives $O(\mathsf{d}^{4|\pi|-4}\prod_i\|f_i\|_\infty)$ for a partition π. Hence

$$\left|S_4(f_1,\dots,f_4)-\int S_4^\xi(f_1,\dots,f_4)\,d\lambda(\xi)\right|\le C_\vartheta\mathsf{d}^4\prod_i\|f_i\|_\infty.$$

Where the interacting lower bound of order $\bar{x}_s^{-4}\prod_i\|f_i\|_\infty$ holds for the mixture, $\mathsf{d}^4\bar{x}_s^4\to 0$ makes the phase average positive for sufficiently small d. By continuity the phases with positive fourth cumulant on that tuple then form an open set of positive λ-measure.

7.7.2.2. Isometries of the decomposition. Let $r(x)=Rx+c$ be an isometry preserving σ. The family $\sigma_\xi\circ r^{-1}$ has the same mixed moments as the original family. Indeed, $r^{-1}\tau_v=\tau_{R^{-1}v}r^{-1}$ on tuples, and the far-translation limits above allow all directions of approach to infinity. Apply the mixed-moment formula to the tuples $r^{-1}X^i$ with translations $R^{-1}v_i$, then use the invariance of σ. For one evaluation use the barycenter formula. The transformed evaluations are continuous and separate points since r is a bijection of the test tuples. Uniqueness gives a homeomorphism h_r with $(h_r)_*\lambda=\lambda$ and $\sigma_{h_r(\xi)}=\sigma_\xi\circ r^{-1}$. Uniqueness of the point with those evaluations also gives $h_{rs}=h_rh_s$.

Suppose now that σ is O(4)-invariant. Each realized component is fixed by the coordinate reflection θ_1 by the lattice symmetry of its realizing states and the continuum passage in G1. The centering state has that symmetry as well. For $R\in\mathrm{O}(4)$, $\sigma_\xi\circ R=\sigma_{h_{R^{-1}}(\xi)}$ is another such component. It follows that

$$\sigma_\xi(R\theta_1R^{-1}X)=\sigma_{h_{R^{-1}}(\xi)}(\theta_1R^{-1}X)=\sigma_{h_{R^{-1}}(\xi)}(R^{-1}X)=\sigma_\xi(X).$$

These hyperplane reflections generate O(4). Thus every component is invariant under this group. Invariance under translations and rotations also transfers coordinate reflection positivity to each affine mirror by conjugation.

7.8. **The invariant Hilbert space of scalar phases.** The canonical decomposition is a statement about correlation forms. A density argument below identifies its zero-energy Hilbert space without assuming that the linear span of phase evaluations is an algebra.

Let X be a compact metrizable phase space, let λ be a Borel probability measure of full support on X, and let σ_ξ be its component correlation forms. They act on finite separated tuples of compactly supported test functions and their complex linear span, including the empty tuple 1. Assume that

$$f_F(\xi):=\sigma_\xi(F)\in C(X),$$
$$\sigma(F)=\int_X f_F\,d\lambda,$$
$$f_1=1. \tag{7.cliii}$$

Assume that the functions f_F separate points of X, and that the components preserve conjugation, time reflection, and translations. Concatenation is used only when all its supports are separated.

The quantitative phase construction supplies uniform clustering: for any fixed separated tuples F,G, once their translated supports are disjoint,

$$\sup_{\xi\in X}|\sigma_\xi(F\,\tau_{te_4}G)-\sigma_\xi(F)\sigma_\xi(G)|\longrightarrow 0$$
$$(t\longrightarrow\infty). \tag{7.cliv}$$

Finite linear combinations inherit this property by summation. Uniform separated moment estimates ensure that all functions displayed in eq. 7.cliii–eq. 7.cliv are bounded, for each fixed tuple. The phases need not have the same one-point mean.

Assume that the reflection-positive quotient has been constructed and completed to a Hilbert space $\mathcal{H}$, with dense classes $[F]$ of separated tuples whose compact supports lie strictly in $x_4>0$, and their linear combinations. Suppose positive time translations induce a strongly continuous contraction semigroup e^{-tH} with a nonnegative self-adjoint generator H and

$$\langle[F],e^{-tH}[G]\rangle=\sigma(\overline{\theta F}\,\tau_{te_4}G),$$
$$t>0. \tag{7.clv}$$

These are the explicit Hilbert-space hypotheses; this section does not derive them from qualitative convergence alone. Write $P_0=1_{\{0\}}(H)$ for the zero-energy projection.

Proposition 7.9 (The zero-energy representation)**.** Under eq. 7.cliii–eq. 7.clv, there is a unique unitary

$$U:\ker H\longrightarrow L^2(X,\lambda)$$
such that
$$UP_0[F]=f_F. \tag{7.clvi}$$

The vacuum vector [1] maps to the constant function 1. If X has exactly $m<\infty$ points, then $\dim\ker H=m$. If X is infinite, then $\ker H$ has countably infinite Hilbert dimension. ■

Proof. The spectral theorem and dominated convergence give $e^{-tH}\to P_0$ strongly as $t\to\infty$. For fixed F,G, use eq. 7.cliii, reflection invariance, conjugation, and eq. 7.cliv in eq. 7.clv to obtain

$$\begin{aligned}\langle P_0[F],P_0[G]\rangle&=\lim_{t\to\infty}\int_X\sigma_\xi(\overline{\theta F}\,\tau_{te_4}G)\,\lambda(d\xi)\\&=\int_X\overline{f_F(\xi)}f_G(\xi)\,\lambda(d\xi).\end{aligned} \tag{7.clvii}$$

The limit is justified by uniform clustering, applied to the finite linear combinations as well as to individual tuples. In particular, if $P_0[F]=0$, then $f_F=0$ in L^2. Thus the rule in eq. 7.clvi is well defined and isometric on the span of the displayed vectors. That span is dense in $\ker H$, since the classes $[F]$ are dense in $\mathcal{H}$ and P_0 is a bounded projection. Its extension is an isometry with closed range. It remains to prove surjectivity.

Let $\mathcal{V}$ be the complex linear span in $C(X)$ of all the phase evaluations f_F with F supported at positive times. Any separated tuple can be moved into this half-space by a positive time translation, which leaves its phase evaluations unchanged. Consequently $\mathcal{V}$ still separates points of X. It contains 1 and is closed under complex conjugation.

For positive-time tuples F,G, the concatenation $F\,\tau_{te_4}G$ is a positive-time separated tuple for all sufficiently large t. By eq. 7.cliv,

$$f_{F\,\tau_{te_4}G}\longrightarrow f_Ff_G$$

uniformly on X.

It follows first for generators, and then for their finite linear combinations, that products of elements of $\mathcal{V}$ belong to its uniform closure. The uniform closure is itself closed under products: approximate each of two factors uniformly by elements of $\mathcal{V}$ and use boundedness of the approximants. It is therefore a unital conjugation-closed subalgebra separating points. The complex Stone–Weierstrass theorem makes that closure equal to $C(X)$. Since continuous functions are dense in $L^2(X,\lambda)$, the image in eq. 7.clvi is dense and closed, hence is the whole space.

The stated vector correspondence determines the unitary on a dense set, proving uniqueness. It sends $[1] = P_0[1]$ to 1, since translation fixes the empty tuple.

For a finite phase space, full support gives positive mass to each of its m points, so L^2 has dimension m. For an infinite compact metric space, L^2 is separable. One way to see this is to take a countable uniformly dense subset of $C(X)$ and then its rational complex linear span, which is dense in L^2. Its dimension cannot be finite: for every integer m choose m distinct points and disjoint open neighbourhoods of them. Full support gives each neighbourhood positive measure, so their indicator functions are m nonzero mutually orthogonal vectors. The dimension is therefore countably infinite. This proves all assertions. □

In particular, a single phase gives a one-dimensional zero-energy space under these hypotheses. Several phases can produce several vacua, even when each component has a unique vacuum. An infinite phase set is recorded by the extended natural count ∞ in dimension statements; its set-theoretic cardinality is not equated with Hilbert dimension. Nothing in this identification gives a strictly positive lower spectral bound on $(\ker H)^{\perp}$.

7.9. **The Hilbert realization of scalar phases.** The canonical phase decomposition first specifies scalar correlation forms. We now identify its ordered Hilbert realization, including its common-domain fields and zero-energy subspace. The proof constructs the measurable Hilbert field and shows that the mixed cyclic space fills its direct integral by proving density of the polynomial sections.

The phase space used here is the canonical *scalar* phase space. Its evaluations are those of the specified scalar observable. They separate this space by its construction. This statement does not say that a single scalar observable distinguishes every state of a larger gauge-observable algebra.

The actual phase hypotheses and their uniform extension. Let X be compact metrizable, let λ be a Borel probability measure of full support, and write S^{ξ} for the component scalar moments. The canonical decomposition gives, on separated compact tuples F,

$$f_F(\xi) = S^{\xi}(F) \in C(X), \qquad S(F) = \int_X f_F(\xi)\, d\lambda(\xi), \qquad f_1 = 1. \tag{7.clviii}$$

These evaluations separate points. Their reality, translations and permutation identities hold componentwise. We use the actual component G2 covariance and the uniform component source bounds; they are not inferred from symmetry of the mixture. The scalar phase construction supplies uniform clustering of each fixed pair of separated compact tuples after sufficiently far translations.

The quantitative scalar route has constants uniform on this state class. In particular, on the coincidence-flat Schwartz domain,

$$\sup_{\xi\in X} |S^{\xi}_n(F)| \le AB^n (n!)^{k+32} p_{20n}(F). \tag{7.clix}$$

The UI/PD insertion constants and the energy exponent $N = 9$ are uniform as well. These are the state-uniform conclusions of the declared ordinary-source hypotheses. They retain those hypotheses; no new uniform bound is extracted from compactness alone.

Constrained tensor density extends eq. 7.clviii to every flat Schwartz test: approximate in p_{20n}, use eq. 7.clix to make the approximation error uniform in ξ, and take the same limit in the barycenter

identity. It also makes $\xi \mapsto S_n^{\xi}(F)$ continuous, as a uniform limit of the compact-tuple evaluations. In particular the reflected forms $q_\xi(F,G)$, for any fixed finite ordered Schwartz sequences, have continuous Gram entries and satisfy

$$q(F,G) = \int_X q_\xi(F,G)\, d\lambda(\xi). \tag{7.clx}$$

The products in these forms are in the flat domain by their proved time-boundary vanishing. Their degree sums are finite.

If the original determining family was defined with compact C^1 labels, smooth labels still suffice. Choose fixed disjoint open containers around the original supports and mollify inside them in the C^1 norm. Telescoping the product and using the uniform separated moment bound gives

$$\sup_{\xi} |f_{(h_1,\dots,h_n)}(\xi) - f_{(h_1',\dots,h_n')}(\xi)| \le C_{h,\mathcal{U}} \sum_i \|h_i - h_i'\|_{C^1}. \tag{7.clxi}$$

The approximating labels have a common norm bound and stay in the fixed separated containers, so the constant is uniform along the approximation. This proves that the continuous scalar coordinates remain determined by smooth compact labels.

Ordered evaluations still separate the phase space. Let $\mathcal{V}_{\mathrm{ord}}$ be the complex span of evaluations of compact smooth tuples with strictly ordered positive-time supports, including the empty tuple. We prove that its uniform closure contains every evaluation in eq. 7.clviii.

Take a compact configuration set away from all coincidence diagonals. At a fixed configuration $(x_1,\dots,x_n)$, choose a unit vector v for which the numbers $v\cdot x_i$ are all distinct. Such a vector exists: the finite product $\prod_{i<j} v\cdot(x_i-x_j)$ is a nonzero homogeneous polynomial. If it vanished on the whole sphere, homogeneity would make it zero everywhere, a contradiction. Select a proper Euclidean rotation whose new time projection is this v, permute the slots into increasing projected-time order, and translate all times to be strictly positive. These strict inequalities persist in an open neighbourhood of the configuration.

A finite subcover of the compact set and a smooth subordinate partition split any supported test into finitely many such pieces. Componentwise G2, permutation symmetry and translation invariance move each piece into a compact ordered positive-time test. The support remains off the coincidence diagonals under all these transformations. Ordered support-preserving Fourier approximation expresses that piece as a Schwartz limit of finite ordered tensors. The formula in eq. 7.clix makes their evaluation errors uniform on X. Applying this to a separated tensor, and then using eq. 7.clxi for the original C^1 family, proves

$$\{f_F : F \text{ in the determining separated family}\} \subset \overline{\mathcal{V}_{\mathrm{ord}}}^{\|\cdot\|_\infty}. \tag{7.clxii}$$

Thus ordered evaluations retain point separation. This is a statement about scalar phase evaluations and their uniform norm, not a rotation of the reflection-positive Hilbert plane or a claim that the full positive-time quotient equals its ordered sector.

Uniform time clustering gives the multiplicative part. For two ordered compact tuples F,G, the concatenation $F\cdot\tau_{te_4}G$ is again ordered for all sufficiently large t, and

$$f_{F\cdot\tau_{te_4}G} \longrightarrow f_F f_G \quad \text{uniformly on } X. \tag{7.clxiii}$$

Finite linear combinations obey the same assertion by summation. The uniform closure of $\mathcal{V}_{\mathrm{ord}}$ is therefore closed under products: first multiply finite linear combinations using eq. 7.clxiii, then approximate each of two general closure elements uniformly and use boundedness of the approximants. Reality gives closure under conjugation, and the empty tuple gives the unit. It separates points by eq. 7.clxii. The complex Stone–Weierstrass theorem consequently yields

$$\overline{\mathcal{V}_{\mathrm{ord}}}^{\|\cdot\|_\infty} = C(X). \tag{7.clxiv}$$

The proof supplies the algebra property; it has not assumed that the original linear span of phase evaluations was already an algebra.

One countable core for all component Hilbert spaces. Write $\mathcal{K}_\xi$ for the component ordered Hilbert sector. Choose a countable family $(F_j)_{j\ge1}$ of finite ordered compact tensor sequences, with $F_1 = 1$, whose images are dense in every $\mathcal{K}_\xi$. Here is an explicit way to make the choice common. Use rational boxes with strictly ordered positive-time intervals, fixed nested compact cutoffs, and the countable Fourier waves on rational grids. Finite sums with rational complex coefficients form a countable family. For any ordered compact test the support-preserving Fourier construction approximates it in every required Schwartz seminorm using such boxes and waves; rational coefficients approximate the finitely many retained coefficients. Infinity and ordered-boundary cutoffs handle general ordered Schwartz tests. The uniform bound in eq. 7.clix, applied to the reflected form, makes these approximations uniform in the component Hilbert norms.

Put

$$v_j(\xi) = [F_j]_\xi, \qquad g_{ij}(\xi) = \langle v_i(\xi), v_j(\xi)\rangle = q_\xi(F_i, F_j). \tag{7.clxv}$$

The Gram entries are continuous by eq. 7.clx. In particular $\|v_j(\xi)\|$ is uniformly bounded on the compact phase space for each fixed j. Zero vectors at some phases cause no difficulty; no constant rank is assumed.

Construct a measurable orthonormal family by Gram–Schmidt with zeros retained. Recursively set

$$u_j = v_j - \sum_{i<j} e_i\langle e_i, v_j\rangle, \qquad e_j = \begin{cases} u_j/\|u_j\|, & \|u_j\| > 0,\\ 0, & \|u_j\| = 0. \end{cases} \tag{7.clxvi}$$

Every coefficient is a Borel function of the finite Gram data already constructed. The set $E_j = \{\|u_j\| > 0\}$ is Borel. At each phase the nonzero e_j form an orthonormal basis, because their finite spans successively recover those of the total fundamental family v_j. This explicit procedure handles phase-dependent ranks and possible vanishing Gram determinants.

A concrete direct-integral construction. A section ψ is called measurable when its coordinates $\psi_j(\xi) = \langle e_j(\xi), \psi(\xi)\rangle$ are Borel after null modifications. Its fiber norm is $\sum_j |\psi_j(\xi)|^2$. Define

$$\begin{aligned} \mathcal{D} &= \int_X^{\oplus} \mathcal{K}_\xi \, d\lambda(\xi),\\ \|\psi\|_{\mathcal{D}}^2 &= \int_X \sum_j |\psi_j(\xi)|^2 d\lambda(\xi) < \infty. \end{aligned} \tag{7.clxvii}$$

Its elements are measurable sections with finite displayed norm, modulo equality almost everywhere. Coordinates vanish off E_j, where the corresponding basis vector is zero. This is the Hilbert direct sum $\bigoplus_j L^2(E_j, \lambda)$, so completeness is immediate. Its scalar product is the integral of the fiber scalar products. The construction is independent up to its natural fiberwise unitary of the Gram–Schmidt presentation: another common total measurable family has measurable transition coefficients, and finite orthogonal projections followed by their strong limits identify the same sections and norms.

The elementary sections $\mathbf{1}_A v_j$, with A Borel, span a dense subspace. If a section were orthogonal to all of them, then $\int_A \langle v_j(\xi), \psi(\xi)\rangle d\lambda = 0$ for every A, j. The integrands are integrable, since the fundamental section has bounded norm and ψ is square integrable. They vanish almost everywhere. Countability gives one full-measure set for all j, and totality of the fibers then makes the section zero.

Continuous coefficients already suffice:

$$\operatorname{span}\{c(\xi)v_j(\xi) : c \in C(X),\ j \ge 1\} \text{ is dense in } \mathcal{D}. \tag{7.clxviii}$$

Indeed a Borel probability on a compact metric space is regular. Approximate the indicator of a Borel set in $L^2(\lambda)$ by continuous functions bounded by one: choose a compact subset and an

open superset with arbitrarily small measure difference, then use a continuous function equal to one on the compact set and zero outside the open set. The uniform bound on $\|v_j\|$ transfers that approximation to the section norm. The preceding elementary-section density proves eq. 7.clxviii.

The mixed quotient embeds isometrically. Let $\mathcal{K}_{\mathrm{mix}}$ be the ordered sector of the barycenter functional S. On its common ordered core set

$$J_0[F](\xi) = [F]_\xi. \tag{7.clxix}$$

For finite linear combinations of the F_j the section is measurable by construction. Any other ordered test is approximated uniformly in the component Hilbert norms by the common family. The section obtained in this way is measurable and agrees with the actual quotient vector in every fiber. Its norm is integrable by the same uniform estimate. The formula in eq. 7.clx gives

$$\langle J_0[F], J_0[G]\rangle_{\mathcal{D}} = q(F, G). \tag{7.clxx}$$

Thus null vectors are sent to zero and the rule extends to an isometry $J : \mathcal{K}_{\mathrm{mix}} \to \mathcal{D}$. Denote its closed range by M. At this stage no onto statement has been used.

The section $\Omega(\xi) = \Omega_\xi = [1]_\xi$ is the image of the mixed vacuum and has norm one. Each component Hilbert space is the actual cyclic scalar space established earlier. Neither $\mathcal{K}_{\mathrm{mix}}$ nor a fiber has been replaced by the full positive-time quotient.

Measurable evolution and exact range invariance. For a fixed $s \geq 0$, the component heat operator sends $[F]_\xi$ to $[\tau_{se_4}F]_\xi$. Its images of the common core are measurable, by uniform test approximation. Its norm is at most one. It therefore defines a bounded decomposable operator $T_s^{\oplus}\psi(\xi) = T_s^{\xi}\psi(\xi)$ on $\mathcal{D}$. The same reasoning applies to both signs of spatial translations. Fiberwise strong continuity and the bound $2\|\psi(\xi)\|$ give strong continuity on $\mathcal{D}$ by dominated convergence. They obey their group and semigroup laws pointwise and hence on this space.

The exact common-core translation identity gives

$$JT_s^{\mathrm{mix}} = T_s^{\oplus}J, \qquad JU_y^{\mathrm{mix}} = U_y^{\oplus}J. \tag{7.clxxi}$$

Thus M is invariant under the heat operators and both spatial translation signs. Since each heat operator is self-adjoint, invariance also makes $M^\perp$ invariant. Thus M reduces them and their spectral projections, including the strong limit of the heat semigroup as $s \to \infty$.

The insertion family is measurable as well. For a narrow smooth label k, its actual core vector identity is

$$\Phi_s^{\xi}(k)[F]_\xi = [(\tau_{se_4}k)\cdot\tau_{2se_4}F]_\xi. \tag{7.clxxii}$$

The right side is a common ordered test when the central temporal support is sufficiently narrow. Its Gram entries are continuous in ξ. The operator-norm approximation to the sharp spatial insertion has error $Cs^{-N-1}\epsilon\|h\|_1$, uniformly in the phase. Hence the sharp operators $\Phi_s^{\xi}(h)$ form a measurable bounded family with common norm $Cs^{-N}\|h\|_1$ for small s, and the safe extended bound for larger s.

eq. 7.clxxii proves range invariance at the vector level. Before taking a sharp limit, its right-hand test is independent of the phase and is exactly the corresponding mixed test. Thus the decomposable insertion sends $J[F]$ to the image under J of the actual mixed inserted vector. Pass to the uniform norm sharp limit, and then extend from the dense core. We obtain

$$J\Phi_s^{\mathrm{mix}}(h) = (\Phi_s(h))^{\oplus}J. \tag{7.clxxiii}$$

The conjugate insertion has the same property, so M reduces these bounded operators. A calculation of matrix elements only between vectors in M would not by itself have proved this invariance; the actual common test in eq. 7.clxxii is the needed step. Spatial approximate identities give the same assertion for the point operator, with uniform norm convergence from the analytic PD derivative bound.

Component zero energy is exactly the vacuum line. Uniform compact-tuple component clustering first gives the rank-one spatial matrix-element limit on the ordered compact core. The unitarity approximation argument extends it to all vectors of $\mathcal{K}_\xi$ for each fixed phase. If $\psi \in \ker H_\xi$, the cube cone already implies that its spatial momentum is zero. Indeed the joint spectral carrier on $E = 0$ has $p = 0$, so $U_y^\xi \psi = \psi$ for every spatial y. Apply the rank-one cluster limit to its own matrix element. It gives $\|\psi\|^2 = |\langle \Omega_\xi, \psi\rangle|^2$, which forces ψ onto the vacuum line. Conversely that line has energy zero. Therefore

$$\ker H_\xi = \mathbb{C}\Omega_\xi, \qquad P_0^\xi [F]_\xi = f_F(\xi)\Omega_\xi. \tag{7.clxxiv}$$

This proof uses component clustering. It does not apply that rank-one assertion to a nontrivial mixture.

Strong spectral convergence gives $T_s^\xi \to P_0^\xi$ as $s \to \infty$. Dominated convergence of the squared section norms makes $T_s^\oplus \to (P_0)^\oplus$ strongly on $\mathcal{D}$. Since M reduces the heat family, it contains the projection of every one of its vectors. For an ordered compact F,

$$J P_0^{\mathrm{mix}}[F] = f_F(\xi)\Omega_\xi \in M. \tag{7.clxxv}$$

eq. 7.clxiv and $\|\Omega_\xi\| = 1$ now give $c(\xi)\Omega_\xi \in M$ for every $c \in C(X)$. This is the supply of phase coefficients needed for surjectivity.

Bounded sewn words make the range onto. Fix a compact ordered test. Its times and adjacent gaps have a strictly positive common margin. Choose the auxiliary heat parameters below a quarter of that margin. Write the configuration as $X = ((\mathbf{x}_i, t_i))_{i=1}^n$. At every point of its compact support its exact vector is a bounded sewn word

$$\begin{aligned} B_\xi(X)\Omega_\xi &= T^\xi_{t_1 - a_1} \Phi^\xi_{a_1}(\mathbf{x}_1)\, \mathcal{Q}_\xi(X)\Omega_\xi, \\ \mathcal{Q}_\xi(X) &= \prod_{j=1}^{n-1} \left[T^\xi_{t_{j+1} - t_j - a_j - a_{j+1}} \Phi^\xi_{a_{j+1}}(\mathbf{x}_{j+1}) \right]. \end{aligned} \tag{7.clxxvi}$$

The product is ordered from left to right with increasing j, and is identity when $n = 1$. The operator norms are bounded uniformly in the phase and in the support variables. The positive heat margins also give norm continuity of these words uniformly in the phase: for heat use the bound on $He^{-\delta H}$; for spatial insertions use the uniform PD Cauchy derivative. Thus their decomposable operators have a norm-continuous compactly parameterized family.

Every factor preserves M by eq. 7.clxxi–eq. 7.clxxiii. Applying a word to $c\Omega \in M$ consequently gives the section $c(\xi)B_\xi(X)\Omega_\xi \in M$. Integrate against the fixed compact test. The uniform norm bound permits a Bochner integral, approximated in norm by finite sums. The range is closed, so the result is in M. Exact ordered sewing in each fiber identifies that result with $c(\xi)[F]_\xi$. For the fundamental core this proves

$$c(\xi) v_j(\xi) \in M \quad (c \in C(X),\ j \geq 1). \tag{7.clxxvii}$$

One may use the ordered Schwartz Bochner estimate to make the same assertion for all ordered tests, but the compact fundamental family already suffices.

These sections are dense in $\mathcal{D}$ by eq. 7.clxviii. Since M is closed, we have proved the onto identification

$$J : \mathcal{K}_{\mathrm{mix}} \xrightarrow{\cong} \int_X^\oplus \mathcal{K}_\xi \, d\lambda(\xi). \tag{7.clxxviii}$$

No density of constant sections was assumed. The zero-energy projection first produced nonconstant phase coefficients; actual bounded insertions and sewing then propagated them to a fiberwise total family. This proves precisely why the complete direct integral is obtained for this scalar phase construction.

Exact generator domains in the direct integral. The heat family on $\mathcal{D}$ is a strongly continuous self-adjoint contraction semigroup, hence has a nonnegative self-adjoint generator $H^{\oplus}$. Its resolvent is

$$(1+H^{\oplus})^{-1}=\int_0^\infty e^{-s}T_s^{\oplus}\,ds,\qquad ((1+H^{\oplus})^{-1}\psi)(\xi)=(1+H_\xi)^{-1}\psi(\xi). \tag{7.clxxix}$$

The integral is strongly convergent, and fiberwise integration follows from the integrable bound $e^{-s}\|\psi(\xi)\|$. The resolvent identifies the generator domain as follows.

If ψ is in that domain, write it as the resolvent applied to $f\in\mathcal{D}$. In each fiber it then lies in $\operatorname{Dom} H_\xi$, and $H_\xi\psi(\xi)=f(\xi)-\psi(\xi)$, a square-integrable section. Conversely, if the fiber domain condition holds and that section is measurable and square integrable, put $f(\xi)=(1+H_\xi)\psi(\xi)$. The pointwise resolvent identity recovers ψ, placing it in the global generator domain. Iterating the same argument with resolvent powers gives

$$\begin{aligned}\operatorname{Dom}(1+H^{\oplus})^l=\{\psi:&\psi(\xi)\in\operatorname{Dom}(1+H_\xi)^l \text{ a.e.},\\ &\int_X\|(1+H_\xi)^l\psi(\xi)\|^2d\lambda<\infty\}.\end{aligned} \tag{7.clxxx}$$

The domain includes the measurability of the displayed section; equivalently it is the range of the measurable resolvent power. The mixed generator is unitarily identified with this operator by eq. 7.clxxi and eq. 7.clxxviii, including its exact domain.

The common global energy domain requires eq. 7.clxxx for *every* integer l. Membership in the fiberwise intersection alone does not ensure these integrability conditions. Since the intersection is countable, a vector in the global domain is in the fiber common domain on one common full-measure set. This distinction is retained in every field formula below.

Spatial unitary groups are also decomposable by eq. 7.clxxi. Stone's theorem identifies their generator domains in the same way. Their groups commute with the heat semigroup, so their joint spectral calculus is the componentwise one. One can see this first for products of bounded resolvents and Fourier–Laplace transforms, then extend to bounded Borel functions by bounded pointwise spectral approximation and strong limits. In particular all band projections and bounded inverse-heat factors on a band are measurable decomposable operators.

Measurable band fields and the global common domain. For fixed $m,n,s>0$, the component block

$$A^\xi_{mn}(h)=(e^{sH_\xi}P^\xi_m)\Phi^\xi_s(h)(P^\xi_n e^{sH_\xi}) \tag{7.clxxxi}$$

is a measurable bounded operator field: its outer factors are bounded Borel functions of the measurable energy generator and its middle factor was constructed above. Its norm has the uniform bound $C(1+m+n)^N\|h\|_1$, by the same small-heat optimization. Time smearing is a norm integral between the fixed bands. Approximation by simple temporal sums, or scalar dominated integration of its matrix entries, makes the resulting blocks measurable. The weighted block-Schur estimate has common constants over all phases.

It follows that

$$L^\xi_g=\phi_\xi(g)(1+H_\xi)^{-N}\quad\text{is measurable and}\quad\sup_\xi\|L^\xi_g\|\le Cp_{N+7}(g). \tag{7.clxxxii}$$

To justify measurability of the full matrix, first retain finitely many bands. Those operators are measurable by the preceding construction. Their strong limit is the weighted field matrix: the bounded Schur operator has those blocks, and the finite orthogonal projections tend strongly to identity. The uniform operator bound passes to the limit. This proves eq. 7.clxxxii without postulating a globally unsandwiched sharp-time field.

Every step defining the blocks commutes with the unitary eq. 7.clxxviii: the energy functional calculus, the actual bounded insertion, the fixed-band inverse heat, the temporal integral and finally

the uniformly bounded strong matrix limit. Therefore the actual mixed field has the identity

$$(J\phi_{\mathrm{mix}}(g)\psi)(\xi) = \phi_\xi(g)(J\psi)(\xi) \quad \left(\psi \in \mathcal{D}_\infty^{\mathrm{mix}}\right). \tag{7.clxxxiii}$$

For its construction on $\mathrm{Dom}(1+H^{\mathrm{mix}})^N$, apply the bounded decomposable field $(L_g)^\oplus$ to the global vector $(1+H^\oplus)^N J\psi$. The integrability condition in eq. 7.clxxx makes this defined. The stronger uniform weighted bounds show that it preserves the global common energy domain. Thus eq. 7.clxxxiii is an exact common-domain formula, not a claim about a maximal direct-integral domain for arbitrarily chosen self-adjoint field extensions.

The same operator-valued Fourier construction gives this identity for every integrated scalar polynomial. Its graph bounds are uniform because the input field energy estimates are uniform. Consequently the full actual scalar polynomial representation, with its common invariant domain, is decomposed by eq. 7.clxxviii.

Measurability of the reconstructed Poincare action. We prove measurability for each component directly. Fixed Minkowski polynomial vacuum vectors in a component form measurable sections. To verify it, start with the measurable vacuum section and use finite band approximations to each smeared field. All are measurable. Their products converge in every required energy graph norm by the uniform field bounds. Pointwise limits of their measurable coordinates are measurable. The general joint-smearing statement follows from the uniform vector Fourier extension.

Choose a countable dense family of Minkowski Schwartz labels and their finite polynomial words. These vectors are total in every component by its cyclic scalar reconstruction. Their Gram entries are measurable by the preceding paragraph. They can therefore be used as another measurable fundamental family, with a Gram–Schmidt construction exactly as in eq. 7.clxvi.

For a fixed Poincare element (a,Λ), the component unitary sends every member of this total family to the same word with transformed tests. The result is a measurable section, by the just-proved test continuity and field measurability. Finite orthogonal projections in that family then prove measurability on the original fundamental sections: expand each original section in its measurable orthonormal coordinates, apply the unitary to a finite expansion, and take the fiber norm limit. The operator norm is one throughout. This constructs the decomposable unitary

$$(U^\oplus(a,\Lambda)\psi)(\xi) = U_\xi(a,\Lambda)\psi(\xi). \tag{7.clxxxiv}$$

The component group laws give its group law. Fiberwise strong continuity and the bound by $2\|\psi(\xi)\|$ give strong continuity after integration. No continuous choice of a basis in the fibers is necessary.

On a mixed Minkowski polynomial vacuum vector the intertwining identity follows from eq. 7.clxxxiii, the invariant component vacua and the common transformed tests. Those vectors are dense in the mixed ordered sector, and both maps are bounded unitaries. Thus

$$JU_{\mathrm{mix}}(a,\Lambda)J^{-1} = U^\oplus(a,\Lambda). \tag{7.clxxxv}$$

All covariance statements here use actual component G2. A symmetry which only permuted components would instead require an accompanying action on the phase variable; that different case is not silently substituted for the componentwise invariance assumed in this scalar route.

Phase multipliers and zero-energy multiplicity. For $c \in L^\infty(X,\lambda)$, multiplication of sections defines a bounded operator M_c of norm at most $\|c\|_\infty$. It commutes with the decomposable Poincare operators and with energy functional calculus. It preserves every global graph domain by eq. 7.clxxx, and the pointwise field identity gives

$$M_c\phi(g)\psi = \phi(g)M_c\psi \quad (\psi \in \mathcal{D}_\infty). \tag{7.clxxxvi}$$

The same is true for integrated polynomial operators on that domain. This is a precise commuting multiplier structure for the scalar representation. It is not an assertion about the center of an unspecified bounded local gauge-observable net.

By eq. 7.clxxiv, the global zero-energy sections are exactly $c(\xi)\Omega_\xi$ with $c \in L^2(X,\lambda)$. The unitary identification therefore gives

$$\ker H^{\mathrm{mix}} \cong L^2(X,\lambda), \qquad P_0^{\mathrm{mix}}[F] \longmapsto f_F, \qquad \Omega^{\mathrm{mix}} \longmapsto 1. \tag{7.clxxxvii}$$

This recovers the phase evaluation rule on the actual ordered sector. Its onto property follows from eq. 7.clxiv, or directly from the full direct-integral identification, and does not assume equality with the larger positive-time Hilbert space. The same sections are invariant under the full reconstructed Poincare group, because every component vacuum is invariant. Conversely any Poincare-invariant vector has energy zero.

If X has exactly m points, full support gives positive mass at every one and $\dim\ker H^{\mathrm{mix}} = m$. If X is infinite, $L^2(X,\lambda)$ is separable: continuous functions have a countable uniformly dense family and are dense in L^2. Its dimension is not finite, because for any m one can choose m distinct points with disjoint open neighbourhoods; full support gives positive measure to each neighbourhood, whose normalized indicators are orthogonal. The Hilbert dimension is therefore countably infinite. It is not the set-theoretic cardinality of the phase space.

The precise uniform spectral-gap criterion. The direct integral also identifies exactly the uniform spectral statement that a further estimate would have to prove. Put $\Pi_m^\xi = \mathbf{1}_{(0,m)}(H_\xi)$ for $m > 0$. These projections are measurable and decompose under J. One explicit verification uses the bounded resolvent $R_\xi = (1 + H_\xi)^{-1}$. Continuous functions of R_ξ decompose by polynomial approximation and eq. 7.clxxix. Continuous functions on $[0,1]$, bounded by one and increasing to the indicator of $((1+m)^{-1}, 1)$, then converge strongly to Π_m^ξ. The same approximation holds for the global resolvent, and dominated section convergence identifies the two limits. Hence

$$\begin{aligned} J\mathbf{1}_{(0,m)}(H^{\mathrm{mix}})J^{-1} &= \int_X^\oplus \Pi_m^\xi \, d\lambda(\xi), \\ \mathbf{1}_{(0,m)}(H^{\mathrm{mix}}) = 0 \iff \Pi_m^\xi &= 0 \text{ a.e.} \end{aligned} \tag{7.clxxxviii}$$

For the reverse implication in the last equivalence integrate the squared norm of a projected section. For the forward implication apply the zero global projection to every fundamental section v_j. The integrals of their nonnegative squared projected norms vanish. Countability and fiber totality make the component projection zero almost everywhere.

Define the extended component gap by $g(\xi) = \sup\{q \in \mathbb{Q}_{>0} : \Pi_q^\xi = 0\}$, with supremum zero for the empty set and infinity when there are no positive energies. It is measurable, since each zero-projection event is the countable intersection of $\{\|\Pi_q^\xi v_j(\xi)\| = 0\}$. Using rational thresholds in eq. 7.clxxxviii gives

$$g_{\mathrm{mix}} = \operatorname*{ess\,inf}_{\xi\in X} g(\xi). \tag{7.clxxxix}$$

Thus positive component gaps almost everywhere are insufficient unless their essential lower bound is positive. No continuity or positive lower bound of this measurable function follows merely from the continuous finite Gram entries.

An elementary Hilbert-family example verifies that distinction. Take $X = [0,1]$ with Lebesgue measure, a vacuum vector in every fiber, and one extra orthogonal unit vector e for $x > 0$. There is no extra vector at $x = 0$. Use fundamental sections $v_1 = \Omega$ and $v_2(x) = \sqrt{x}\, e$, with $v_2(0) = 0$, and let $H_x e = xe$. Their nonconstant Gram entry is x; its heat-translated entry is xe^{-tx}. Both are continuous at zero. Every fiber has a one-dimensional zero-energy space. Its gap is x for $x > 0$, and infinity at zero, while the direct integral has gap zero. This example establishes the spectral point about Gram data; it is not asserted to be a gauge-theory phase model.

The equivalent heat estimate is

$$\begin{gathered}\mathbf{1}_{(0,m)}(H^{\mathrm{mix}}) = 0 \quad \Longleftrightarrow \\ \|e^{-tH^{\mathrm{mix}}} - P_0^{\mathrm{mix}}\| \le e^{-mt} \quad (t \ge 0).\end{gathered} \tag{7.cxc}$$

The forward implication is the spectral supremum on positive energies. Conversely a nonzero spectral projection in $(0, m)$ has a nonzero subprojection in some compact interval strictly below m. A unit vector in that subspace has a slower exponential lower bound, contradicting the displayed norm estimate. Here P_0^{mix} is the whole phase vacuum space in eq. 7.clxxxvii, not just the projection onto the one vector Ω^{mix}.

A dense-core version can be useful without an operator-norm estimate as input. Suppose one common $m > 0$ satisfies, for every vector ζ in a dense core, an estimate

$$\begin{gathered}0 \le \langle \zeta, (e^{-tH^{\mathrm{mix}}} - P_0^{\mathrm{mix}})\zeta \rangle \le C_\zeta e^{-mt}, \\ t \text{ sufficiently large.}\end{gathered} \tag{7.cxci}$$

Each C_ζ may depend on the vector. Its positive spectral measure cannot charge $(0, c]$ for any $c < m$: such a charge would give a lower bound proportional to e^{-ct}, contradicting eq. 7.cxci. Thus the projection onto $(0, m)$ kills the dense core and hence vanishes. Conversely eq. 7.cxc implies eq. 7.cxci with $C_\zeta = \|\zeta\|^2$. These are exact conditional criteria, not an estimate of a positive m. The available compact-tuple clustering does not itself supply eq. 7.cxci.

The identification concerns the scalar correlation phase space and its scalar fields. Extending it to a larger gauge-observable family requires the additional observable and state-consistency data specified separately.

7.10. **The gauge-invariant cutoff algebra.** The scalar field constructed above has a precise polynomial observable sector. To compare it with a gauge theory one first needs the larger gauge-invariant algebra whose observables would have to be transported and identified. We construct that algebra at a cutoff, prove concrete dense families, and state the exact measure-preservation consequence of testing them. These are algebraic statements about the actual bond variables. A continuum limit for the additional observable family is a further requirement.

Bonds, orientations and the full local gauge action. Let $\Gamma = (V, E)$ be a finite or countable locally finite graph, and select one orientation E_+ of its unoriented bonds. The hypercubic lattice and its admissible periodic quotients are the intended cases. Write

$$K = \mathrm{SU}(2), \qquad \Omega_\Gamma = K^{E_+}, \qquad U_{\bar{e}} = U_e^{-1}. \tag{7.cxcii}$$

For a finite composable oriented path $\gamma = e_1 \cdots e_r$, use the ordered product $U_\gamma = U_{e_1} \cdots U_{e_r}$; the empty path has holonomy identity. With $s(e), t(e)$ its endpoints, a site gauge $g = (g_v)_{v \in V}$ acts by

$$(g \cdot U)_e = g_{s(e)} U_e g_{t(e)}^{-1}, \qquad (g \cdot U)_\gamma = g_{s(\gamma)} U_\gamma g_{t(\gamma)}^{-1}. \tag{7.cxciii}$$

The path identity follows by cancelling every adjacent $g_v^{-1} g_v$. It is compatible with reversed paths and with insertion or deletion of immediate retracings.

For a closed path the normalized trace

$$w_\gamma(U) = \tfrac{1}{2} \operatorname{Tr} U_\gamma \tag{7.cxciv}$$

is real, lies in $[-1, 1]$, and is gauge invariant. Reality follows from the conjugate eigenvalues of an SU(2) matrix. Cyclic changes of a loop's basepoint and reversal do not change this trace. The matrix holonomy of an open path is instead gauge covariant at its two endpoints; it is not a scalar gauge-invariant observable by itself.

The gauge group $\mathcal{G} = K^V$ has its compact product topology. Its action on Ω_Γ is continuous: each output bond depends continuously on only that bond and its two site variables. For countable sets,

both product spaces are compact metrizable. An action of all site gauges is meant here. Invariance under gauges fixing selected centers is a different, generally weaker condition.

Continuous local functions and their uniform completion. For finite $F \subset E_+$, let $\mathcal{A}_F = C(K^F)$, embedded into $C(\Omega_\Gamma)$ by ignoring the remaining bonds. If $F \subset F'$, this is an injective unital star map and preserves the supremum norm. Any two local functions belong to the algebra for their finite union of bonds. The union is therefore a unital star algebra.

In the infinite case enumerate the bonds and choose the bounded bi-invariant matrix metric $d_K \le 1$. The product metric $d_\Omega(U,U') = \sum_j 2^{-j} d_K(U_{e_j}, U'_{e_j})$ is compatible with the topology. Fix a reference configuration. Replacing all coordinates after the first N by their reference values changes this metric by at most 2^{-N}. For $f \in C(\Omega_\Gamma)$, uniform continuity shows that the corresponding cylinder functions converge uniformly to f. Hence

$$\mathcal{A} = \overline{\bigcup_{F \Subset E_+} \mathcal{A}_F}^{\|\cdot\|_\infty} = C(\Omega_\Gamma). \tag{7.cxcv}$$

This argument gives the concrete meaning of the quasilocal completion in the compact bond model. It does not include an additional charged field at infinity or a choice of boundary frame for an open path.

Let $S(F)$ be the finite set of endpoints of the bonds in F. A local function is invariant under every full site gauge exactly when it is invariant under $K^{S(F)}$, since no other gauge coordinate enters its value. Define

$$\mathcal{A}_F^{\mathrm{inv}} = C(K^F)^{K^{S(F)}}, \qquad \mathcal{A}^{\mathrm{inv}} = \{f \in C(\Omega_\Gamma) : f(g \cdot U) = f(U)\}. \tag{7.cxcvi}$$

These are unital closed star subalgebras. The local embeddings preserve invariance. Finite products of Wilson traces belong to the appropriate local invariant algebra.

Finite Haar averaging and the actual projection. For $f \in \mathcal{A}_F$, set

$$(\mathcal{E} f)(U) = \int_{K^{S(F)}} f(g \cdot U)\, dg, \tag{7.cxcvii}$$

with normalized product Haar measure and gauges outside $S(F)$ equal to identity. The result is continuous and depends only on F. If a larger endpoint set is used the additional integrations do not change the integrand. If the same cylinder function is represented in a larger bond algebra, the answer is again the same function. Thus eq. 7.cxcvii is one well-defined linear map on the local union.

Haar invariance makes its range gauge invariant. It is positive, preserves the unit and star, and satisfies

$$\|\mathcal{E} f\|_\infty \le \|f\|_\infty, \qquad \mathcal{E}^2 = \mathcal{E}, \qquad \mathcal{E} b = b \quad (b \text{ invariant}). \tag{7.cxcviii}$$

For idempotence, average twice and combine the two site gauges; the normalized product Haar integral is unchanged. All the integrals are on finite compact groups at this stage. The norm bound extends $\mathcal{E}$ uniquely to $C(\Omega_\Gamma)$, still positive, unital and contractive.

The extended image is precisely $\mathcal{A}^{\mathrm{inv}}$. Images of local approximants are invariant, so their uniform limits are invariant. Conversely, if b is invariant and $f_N \to b$ are cylinder approximants, then $\|\mathcal{E} f_N - b\|_\infty \le \|f_N - b\|_\infty$: inside eq. 7.cxcvii, replace $f_N(g \cdot U)$ by $b(g \cdot U) = b(U)$. Thus $\mathcal{E} b = b$. In particular

$$\mathcal{A}^{\mathrm{inv}} = \overline{\bigcup_F \mathcal{A}_F^{\mathrm{inv}}}^{\|\cdot\|_\infty}. \tag{7.cxcix}$$

No assertion that arbitrary cylinder approximants are themselves invariant was needed; averaging makes them so without increasing their approximation error.

For invariant b, c, the finite integral also gives $\mathcal{E}(bfc) = b(\mathcal{E} f)c$, and uniform approximation extends this to the full algebra. The same pointwise argument proves complete positivity: an

average of positive matrices of continuous functions remains positive when tested against every constant coefficient vector. Thus $\mathcal{E}$ is an actual conditional expectation with the stated range and norm.

The compact orbit space and invariant states. The product metric can be chosen gauge invariant, since left and right multiplication preserve the matrix metric on each bond. On gauge orbits define

$$d_{\mathrm{orb}}([U],[V]) = \min_{g\in\mathcal{G}} d_\Omega(U, g\cdot V). \tag{7.cc}$$

The minimum exists by compactness of $\mathcal{G}$ and continuity. Symmetry follows by inversion of the gauge, and the triangle inequality follows by composing two gauges and using that they act by isometries. If the distance is zero, a minimizing gauge gives $U = g\cdot V$, so the orbits coincide. Thus it is a metric. The orbit map is continuous because $d_{\mathrm{orb}}([U],[V]) \le d_\Omega(U,V)$. The compact domain maps onto a Hausdorff metric space, so this is its quotient topology. In particular the orbit space is compact metrizable and all orbits are closed.

Invariant continuous functions descend and lift through this quotient. Hence

$$\mathcal{A}^{\mathrm{inv}} \cong C(\Omega_\Gamma/\mathcal{G}) \tag{7.cci}$$

isometrically as unital star algebras. States of the invariant algebra are therefore probability measures on this compact orbit space. There is no choice of a possibly discontinuous gauge-fixed representative in this assertion.

Every state ω of $\mathcal{A}^{\mathrm{inv}}$ has the gauge-invariant extension $\widehat{\omega}(f) = \omega(\mathcal{E}f)$ to $\mathcal{A}$. It is positive and normalized by eq. 7.cxcviii. Conversely a gauge-invariant state τ on $\mathcal{A}$ satisfies $\tau(\mathcal{E}f) = \tau(f)$. For a cylinder function this follows by integrating its state value over the finite gauge group; the map into continuous functions is norm continuous, so the bounded functional passes through the integral. Uniform approximation handles all f. Thus

$$\begin{aligned} \mathrm{St}(\mathcal{A}^{\mathrm{inv}}) &\longleftrightarrow \mathrm{St}_{\mathcal{G}}(\mathcal{A}), \\ \omega \longmapsto \omega\circ\mathcal{E}, &\qquad \tau \longmapsto \tau|_{\mathcal{A}^{\mathrm{inv}}}. \end{aligned} \tag{7.ccii}$$

is an affine bijection; St denotes normalized positive states and its subscript denotes gauge invariance. By the Riesz representation theorem these are also the corresponding gauge-invariant probability measures on bond configurations. In particular the *full* invariant state determines its unique gauge-invariant lift. A state on a proper scalar subalgebra has a different information content, addressed below.

Gauge invariance of the actual cutoff Gibbs states. For a plaquette p, let $A_p = 1 - \frac{1}{2}\operatorname{Re}\operatorname{Tr} U_p$. It is gauge invariant by eq. 7.cxciii. On a finite periodic lattice, the Wilson density with respect to product bond Haar measure is therefore gauge invariant, and the measure itself is invariant by the left and right invariance of every bond integral. Its positive normalization does not affect the identity.

The same fact holds for a Wilson DLR state on the infinite locally finite graph with its local plaquette specification. Let g be supported on finitely many sites and let W contain every bond incident to those sites. This is finite by local finiteness. All outside bonds are fixed under this gauge. In the conditional kernel γ_W, change the inside variables by that gauge. Product Haar measure is preserved, and each plaquette term meeting W has unchanged trace, including the plaquettes containing fixed outside bonds. Consequently

$$\gamma_W(f\circ g) = \gamma_W f, \qquad \mu(f\circ g) = \mu(\gamma_W(f\circ g)) = \mu(f). \tag{7.cciii}$$

The second identity is the DLR equation. It applies to a local continuous f; its additional outside variables are simply held fixed in the conditional integral.

For an arbitrary site gauge and a local f, only the endpoint values of the finitely many bonds in its support matter. Copy those values to a finitely supported gauge and set the rest to identity.

It has exactly the same action on f, so eq. 7.cciii proves invariance under the arbitrary gauge on all local tests. eq. 7.cxcv then gives invariance on the whole continuous algebra. Thus the cutoff Gibbs and DLR states supply actual states of $\mathcal{A}^{\mathrm{inv}}$, not merely states of its $\mathrm{tr}(F^2)$ part.

This proof is at a fixed cutoff and does not establish joint continuum limits for all these observables. Gauge invariance of a cutoff state and existence of a particular scalar scaling limit have distinct analytical contents.

A concrete dense family of invariant polynomials. On a finite bond set, let $\mathcal{P}_F$ be the unital star algebra generated by the matrix entries of each U_e and their complex conjugates. It separates points of K^F because the defining matrix representation is faithful. It is closed under conjugation. Stone–Weierstrass gives uniform density in $C(K^F)$. Apply the contractive expectation:

$$\overline{\mathcal{E}\mathcal{P}_F}^{\|\cdot\|_\infty} = \mathcal{A}_F^{\mathrm{inv}}. \tag{7.cciv}$$

The averaged polynomial is still a matrix-entry polynomial. Indeed substitute eq. 7.cxciii in each monomial, expand its finitely many indices, and integrate the gauge-dependent coefficients. Only constant coefficients change. It follows that $\mathcal{E}\mathcal{P}_F = \mathcal{P}_F \cap \mathcal{A}_F^{\mathrm{inv}}$, which is itself a unital star algebra. Its density is therefore an algebraic density statement with a concrete averaging map, not a formal span assumption.

The invariant tensors in this family can also be read directly. At each site collect the fundamental, dual and conjugate representation factors carried by the incident matrix entries of a monomial. They form one finite-dimensional unitary representation $R_v(g)$. Haar averaging at that site is

$$P_v = \int_K R_v(g)\, dg. \tag{7.ccv}$$

It is an orthogonal projection onto the invariant tensors: inversion of g proves $P_v^* = P_v$, two Haar integrals prove $P_v^2 = P_v$, and left invariance gives $R_v(h)P_v = P_v$. Conversely it fixes every invariant vector. Choosing an orthonormal basis of that finite invariant space writes P_v as a finite sum of rank-one tensor contractions. Taking the product over vertices shows exactly how an averaged matrix monomial becomes a finite sum of gauge-invariant tensor networks.

This argument uses tensor powers of the defining representation and their invariant projections. No classification of all irreducible representations is required for the density proof. In the SU(2) case the next section gives a second complete family using only closed Wilson trace words and an explicit orbit-separation argument.

Based gauges and physical invariance. The gauge group fixing one root in a connected graph is a proper subgroup of the full site gauge group. A rooted holonomy can be invariant under the former and still transform by conjugation under the remaining root value. Likewise a coordinate chart which fixes block centers may retain coarse bond matrices with their endpoint covariance.

For example, under a gauge fixing all internal path vertices but changing the two endpoints, an open holonomy still obeys the two factors in eq. 7.cxciii. Its individual matrix entries therefore do not belong to $\mathcal{A}^{\mathrm{inv}}$. A gauge-fixed background or a free coordinate in a constrained chart is useful geometric data, but its covariance must be included when an observable is read out. Full gauge-invariant traces or invariant index contractions remove the remaining endpoint action. The distinction prevents a uniqueness theorem modulo center-fixing gauges from being mistaken for a complete specification of physical gauge observables.

Boundary frames can be kept deliberately, in which case the relevant algebra is covariant under the boundary group rather than fixed by it. Its endpoints, transformation law and any later invariant contraction must then be part of the observable type. The invariant algebra in eq. 7.cxcvi uses no unspecified boundary frame.

What a dense insertion test really proves. Suppose a finite-cutoff operation gives a bounded linear readout T from the invariant continuous algebra to another continuous algebra, and probability states μ, ν on the two sides. If $\mu(P) = \nu(TP)$ for every member of a uniformly dense invariant polynomial family, then for any invariant continuous f, choose $P_l \to f$ uniformly and estimate

$$|\mu(f) - \nu(Tf)| \le (1 + \|T\|)\|f - P_l\|_\infty + |\mu(P_l) - \nu(TP_l)| \longrightarrow 0. \tag{7.ccvi}$$

For a positive unital Markov readout, $\|T\| = 1$; if a different bounded norm is used it is the displayed norm which must be checked. Thus exactness on a concrete complete invariant family can establish exactness on the entire continuous gauge-observable algebra.

If the induced full measures are gauge invariant, equality of their invariant states identifies their gauge-invariant lifts by eq. 7.ccii, hence identifies their Borel probability measures. Their integrals then agree on all bounded Borel insertions. For a change of variables or a probability kernel this is an actual measure identity once its induced measure and gauge covariance have been specified. Without that covariance, eq. 7.ccvi still states exactly the invariant continuous identity that was proved and does not silently add a stronger lift.

Preserving only the normalization tests the single element 1, not the dense family in eq. 7.cciv. Preserving scalar tr(F^2) moments tests another subalgebra whose size must be determined. The next two sections make the finite SU(2) complete family and the scalar restriction explicit, so that a proposed larger gauge-theory identification can be checked against defined observables rather than inferred from one sector's existence.

7.11. **Complete Wilson coordinates for a finite** SU(2) **graph.** We prove a concrete completeness result for the cutoff observable algebra. On each finite graph, finitely many Wilson traces of words of length at most three in a fundamental loop family separate every gauge orbit. Their polynomials are uniformly dense in all continuous gauge-invariant functions. We also describe their exact compact coordinate image. The result is kinematic; it does not assume a Gibbs measure or a continuum limit of these coordinates.

Tree frames with a specified root. First take a connected finite graph with vertex set V and one orientation for each bond in E_+. Choose a root o and a spanning tree T. For $v \in V$, let p_v be the tree path from o to v, using inverse matrices whenever a stored orientation is traversed backwards. Set

$$H_v(U) = U_{p_v}, \qquad H_o = I, \qquad (\mathcal{T}U)_e = H_{s(e)} U_e H_{t(e)}^{-1}. \tag{7.ccvii}$$

This is the actual site gauge with values $g_v = H_v$. Every tree bond becomes identity. For a tree edge traversed from parent v to child w, the path identity is $H_w = H_v U_{vw}$, so the transformed bond is $H_v U_{vw} H_w^{-1} = I$. The reversed stored orientation gives the inverse identity and hence also identity.

Enumerate the non-tree bonds as $e_1, \dots, e_r$, where $r = |E_+| - |V| + 1$, and put

$$h_i(U) = H_{s(e_i)} U_{e_i} H_{t(e_i)}^{-1} = U_{p_{s(e_i)} e_i p_{t(e_i)}^{-1}}. \tag{7.ccviii}$$

Thus h_i is the holonomy of a specified closed path based at the root. All these functions are continuous finite products and inverses of actual bond variables. Every tuple in K^r is obtained: set every tree bond to identity and assign the desired tuple to the non-tree bonds.

For an arbitrary site gauge k, the path transformation law gives

$$H_v(k \cdot U) = k_o H_v(U) k_v^{-1}, \qquad h_i(k \cdot U) = k_o h_i(U) k_o^{-1}. \tag{7.ccix}$$

In particular the tuple is invariant under gauges fixing the root. If two configurations have the same tuple, their tree gauges send them to the same tree-identity configuration. The composition of the inverse of one tree gauge and the other is a gauge fixing the root and mapping one configuration to the other. This proves that based gauge orbits are exactly the fibers of the tuple map.

Under the full site group the only residual action on this tuple is simultaneous conjugation. A constant site gauge keeps all tree bonds identity and conjugates every non-tree bond, so every such conjugation is realized. Conversely eq. 7.ccix shows that a full gauge transformation can do no more. We have therefore proved a continuous bijection of compact orbit spaces

$$K^{E_+}/K^V \;\cong\; K^r/\operatorname{Ad} K. \tag{7.ccx}$$

Both quotient spaces are Hausdorff by the compact orbit metric construction. The continuous bijection is consequently a homeomorphism. A tree gauge is not a discarded formal coordinate choice; its exact root covariance is what produces the remaining conjugation quotient.

Several components and changing the tree. For a graph with finitely many connected components choose a root and spanning tree in each. Its orbit space is the product of the corresponding $K^{r_c}/\operatorname{Ad} K$, with an independent root conjugation in every component. In particular a component that is a tree has one orbit and contributes only constant invariant functions. Products of invariant functions from the different components separate this product space. No closed path joining distinct components is introduced.

The construction is independent of the chosen forest at the level of observables. In tree-identity gauge, any closed path is a word in the non-tree bond matrices and their inverses; tree traversals contribute identity. New fundamental loops from another tree are therefore continuous word functions of the old tuple. The converse is true by interchanging the trees. After quotienting by root conjugation these are the same intrinsic orbit coordinates related by homeomorphisms. We will prove completeness using one convenient forest, without attributing physical meaning to its choice.

If a larger finite graph is used, the old closed paths still define exactly the old functions on its bond configurations. The embedding of local invariant algebras is the isometric one proved earlier, not a prescription to identify two different root frames. This will make the final dense Wilson family compatible with nested local windows.

Real quaternion coordinates and the product signs. Let $\sigma_1,\sigma_2,\sigma_3$ be the Pauli matrices, with $\sigma_i\sigma_j = \delta_{ij}I + i\varepsilon_{ijk}\sigma_k$. Every $H \in \mathrm{SU}(2)$ has the unique form

$$H = aI + i\,b\cdot\sigma, \qquad a\in\mathbb{R}, \quad b\in\mathbb{R}^3, \qquad a^2+|b|^2 = 1. \tag{7.ccxi}$$

For example the usual matrix form $\left(\begin{smallmatrix}\alpha & \beta\\ -\bar\beta & \bar\alpha\end{smallmatrix}\right)$, $|\alpha|^2+|\beta|^2=1$, gives $a=\operatorname{Re}\alpha$, $b_3=\operatorname{Im}\alpha$, $b_1=\operatorname{Im}\beta$, $b_2=\operatorname{Re}\beta$. Conversely eq. 7.ccxi is unitary with determinant one. Its inverse is $aI - ib\cdot\sigma$.

In these conventions direct multiplication gives

$$\begin{aligned}(aI+ib\cdot\sigma)(cI+id\cdot\sigma) = {}& (ac - b\cdot d)I\\ &+ i\,(ad+cb-b\times d)\cdot\sigma.\end{aligned} \tag{7.ccxii}$$

The minus sign in the cross product is fixed by the two factors of i. It is useful to retain that sign explicitly when extracting the triple trace.

Write $H_i = a_iI + ib_i\cdot\sigma$. The normalized traces of one, two and three matrices are

$$\begin{aligned} a_i &= \tfrac12\operatorname{Tr}H_i,\\ c_{ij} &= \tfrac12\operatorname{Tr}(H_iH_j) = a_ia_j - b_i\cdot b_j,\\ c_{ijk} &= \tfrac12\operatorname{Tr}(H_iH_jH_k)\\ &= a_ia_ja_k - a_kb_i\cdot b_j - a_ib_j\cdot b_k - a_jb_i\cdot b_k + (b_i\times b_j)\cdot b_k. \end{aligned} \tag{7.ccxiii}$$

This follows by taking the scalar part of eq. 7.ccxii, then multiplying by the third factor. All the displayed traces are real. Thus the selected Wilson traces recover

$$\begin{gathered} G_{ii} = 1-a_i^2, \qquad G_{ij} = a_ia_j - c_{ij} = b_i\cdot b_j,\\ D_{ijk} = c_{ijk} - a_ia_ja_k + a_kG_{ij} + a_iG_{jk} + a_jG_{ik} = \det(b_i,b_j,b_k). \end{gathered} \tag{7.ccxiv}$$

Pairs with $i < j$ and triples with $i < j < k$ suffice; the diagonal Gram entries are already fixed by eq. 7.ccxi.

Conjugation is exactly proper spatial rotation. Conjugation of the traceless Hermitian matrices $b\cdot\sigma$ by an SU(2) matrix is a real orthogonal map of $\mathbb{R}^3$. Indeed it preserves that real three-dimensional space and its inner product $\frac{1}{2}\operatorname{Tr}((b\cdot\sigma)(d\cdot\sigma)) = b\cdot d$. It has determinant one: the map depends continuously on the connected group SU(2), while its determinant takes only the values ±1 and is one at identity.

Every proper rotation occurs. For a unit axis n and angle θ, choose $q = \cos(\theta/2)I - i\sin(\theta/2)n\cdot\sigma$. Using the Pauli identity twice gives

$$\begin{aligned} q(b\cdot\sigma)q^{-1} = \big[&\cos\theta\, b + \sin\theta\,(n\times b) \\ &+ (1-\cos\theta)(n\cdot b)n\big]\cdot\sigma. \end{aligned} \tag{7.ccxv}$$

This is the rotation about n. Any proper real orthogonal matrix in dimension three fixes some axis and acts as a two-dimensional rotation on its orthogonal plane. To see the fixed axis, use its real eigenvalue and determinant one: the remaining eigenvalues are either a conjugate unit pair or real signs, which forces at least one eigenvalue one. Its plane action has determinant one and hence is a rotation. Thus eq. 7.ccxv realizes every element of SO(3).

It follows that simultaneous conjugation of the H_i leaves the a_i fixed and rotates all the b_i by one common proper rotation. The completeness problem is now a finite real linear-algebra problem, including degenerate spans.

Gram data and orientation separate every orbit. Suppose two tuples have equal selected data in eq. 7.ccxiii. Their scalar parts agree, and eq. 7.ccxiv gives the same Gram matrix and the same oriented triple determinants. Define on the span of the first family $T(\sum_i t_i b_i) = \sum_i t_i b'_i$. If the first sum is zero, its squared norm is $t^T G t = 0$; the same Gram matrix makes the second sum zero. Thus the map is well-defined. It preserves inner products and is an isometry onto the second span. Extend it by orthonormal bases of the complements to an element of O(3).

If the span has rank three, some triple determinant is nonzero. Its equality in the two families forces the determinant of this isometry to be $+1$. If the rank is at most two, the complement has a nonzero dimension. Reverse one vector in the chosen complementary image basis if necessary; this changes the full determinant while leaving all the prescribed images $b_i \mapsto b'_i$ untouched. In every rank there is therefore a proper rotation taking the first family to the second. Lift it by eq. 7.ccxv. We have proved

$$\begin{aligned} &(a_i, c_{ij}, c_{ijk}) = (a'_i, c'_{ij}, c'_{ijk}) \\ \iff\quad &H'_i = qH_iq^{-1} \text{ for one } q\in \mathrm{SU}(2). \end{aligned} \tag{7.ccxvi}$$

The reverse implication is trace invariance. The proof includes central matrices, collinear families and coplanar families; it has not divided by a possibly vanishing triple determinant in these cases.

For rank three the orientation data cannot simply be omitted. For example $i\sigma_1, i\sigma_2, i\sigma_3$ and $i\sigma_1, i\sigma_2, -i\sigma_3$ have the same scalar parts and pair Gram entries, but their triple traces have opposite sign. No common proper rotation maps the two triples. The triple coordinate is the part of the trace family which records this distinction.

The exact compact coordinate image. Let Z_r be the image in $\mathbb{R}^{M_r}$, $M_r = r + \binom{r}{2} + \binom{r}{3}$, of the selected data. It is compact as the continuous image of K^r. The data map factors through the simultaneous-conjugation quotient and is injective there by eq. 7.ccxvi. Hence

$$K^r/\operatorname{Ad} K \;\cong\; Z_r. \tag{7.ccxvii}$$

This is a homeomorphism of compact spaces, not just a one-to-one generic-coordinate parametrization.

There is also an explicit test for membership in the image. Given real a_i, c_{ij}, c_{ijk}, form the symmetric matrix G and numbers D_I from eq. 7.ccxiv, with increasing triples I. They are realizable precisely when

$$\begin{gathered} G \geq 0, \qquad \operatorname{rank} G \leq 3, \qquad G_{ii} = 1 - a_i^2, \\ D_I D_J = \det G_{I,J} \quad \text{for all increasing triples } I, J. \end{gathered} \tag{7.ccxviii}$$

Here $G_{I,J}$ is the indicated three-by-three cross Gram matrix. When there are fewer than three indices the second line has no conditions.

Necessity is immediate from a family of real vectors: $G = B^T B$ is positive with rank at most three, and $\det(B_I^T B_J) = \det B_I \det B_J$. For sufficiency, the real spectral theorem supplies vectors $b_i \in \mathbb{R}^3$ with the given positive Gram matrix. Their lengths and the diagonal condition make $a_i I + i b_i \cdot \sigma$ elements of K. If the rank is below three, every three-by-three principal Gram determinant is zero, so the conditions $D_I^2 = 0$ force all the desired triple determinants to vanish, as they do for the reconstructed vectors.

If the rank is three, choose a triple I_0 with positive principal Gram determinant. Its desired D_{I_0} has the same nonzero absolute value as the determinant of those three reconstructed vectors. Apply one common orthogonal reflection if necessary to arrange equality of their signs. For every other triple J, the cross-determinant condition gives $D_{I_0} D_J = \det G_{I_0,J}$. The reconstructed vectors obey the same equality with their actual determinants. Division by this one nonzero D_{I_0} forces all remaining triple determinants to agree. The formula in eq. 7.ccxiii now gives every desired trace. This proves sufficiency, including all degenerate cases.

These are finite polynomial equalities and inequalities: positive semidefiniteness is expressed by principal minors, and rank at most three by the vanishing higher minors. For a positive matrix, vanishing of all four-by-four principal minors already suffices for that rank bound, because four independent Gram vectors would have a positive principal determinant. The cross-determinant conditions link the orientations of different triples. Requiring only their squared determinants would not impose one common orientation across all indices.

Finite Wilson trace words are a complete family. Return to the fundamental based loops γ_i in eq. 7.ccviii. The coordinates in eq. 7.ccxiii are exactly the Wilson traces of

$$\gamma_i, \qquad \gamma_i \gamma_j \ (i < j), \qquad \gamma_i \gamma_j \gamma_k \ (i < j < k). \tag{7.ccxix}$$

These are closed paths based at the same root; repetitions, intersections and retracings are allowed. Each is contained in the original finite graph. The coordinate family for separate connected components is used componentwise and then multiplied as scalar functions.

The unital complex polynomial algebra in these real-valued coordinates is closed under conjugation and separates the compact orbit space. Stone–Weierstrass and eq. 7.ccx, eq. 7.ccxvii therefore give

$$\overline{\mathbb{C}[w_{\gamma_i}, w_{\gamma_i\gamma_j}, w_{\gamma_i\gamma_j\gamma_k}]}^{\|\cdot\|_\infty} = C(K^{E_+})^{K^V}. \tag{7.ccxx}$$

This proves uniform completeness of an actual finite gauge-invariant observable family. It requires neither a generic full-rank assumption nor a separate trace-word invariant-theory assumption. The real linear algebra and compactness above supplied separation in every rank.

The number of coordinates is finite, with $M_r \leq (r+1)^3$. Each fundamental loop has length at most $2(|V| - 1) + 1$; the longest selected word has at most three times that length. Thus the construction supplies explicit finite paths within the original support window. This is a statement about a complete test family, not a uniform low-degree approximation rate for every continuous observable.

Exact trace relations at the cutoff. Some consistency relations can be written without the tree coordinates. For $B \in \mathrm{SU}(2)$, its characteristic equation gives $B + B^{-1} = (\operatorname{Tr} B)I$. Multiply by A and take the trace to obtain

$$\begin{aligned} \operatorname{Tr} A \operatorname{Tr} B &= \operatorname{Tr}(AB) + \operatorname{Tr}(AB^{-1}), \\ w_A w_B &= \tfrac{1}{2}(w_{AB} + w_{AB^{-1}}). \end{aligned} \tag{7.ccxxi}$$

For based closed paths this is an exact identity of continuous bond functions. Empty paths have trace one, retracings can be deleted, reversal leaves the trace fixed, and conjugating a loop by a connecting path leaves its trace fixed. These relations, together with the coordinate-image conditions in eq. 7.ccxviii, give concrete finite compatibility tests for a proposed bare loop readout.

They are identities of the bounded cutoff variables. If a different renormalized observable is defined by cutoff-dependent normalizations or mixing, its transformed relations must be derived from that definition. The unchanged bare identities are not automatically identities between unspecified renormalized composite fields at coincident points.

From a finite complete family to the quasilocal algebra. Every finite local gauge-invariant function is uniformly approximated by the Wilson polynomials just constructed on its own finite graph. Combining this with eq. 7.cxcix gives

$$\mathcal{A}^{\mathrm{inv}} = \overline{\mathbb{C}[w_\gamma : \gamma \text{ a finite closed edge path}]}^{\|\cdot\|_\infty}. \tag{7.ccxxii}$$

A polynomial in finitely many traces is still local; its support is contained in the finite union of the paths' bonds. Conversely no enlargement of a local graph was needed to approximate its invariant functions. The statement is therefore compatible with local support and with the isometric inclusions of finite windows.

On a countable graph the finite closed paths form a countable family. Polynomials with rational complex coefficients in them are a countable uniformly dense family of the invariant algebra. A proposed bounded observable transport can consequently be tested on an explicit determining family, subject to the boundedness and gauge-covariance conditions of eq. 7.ccvi.

At one finite cutoff a state of the invariant algebra is determined by all joint polynomial moments of the selected coordinates. Indeed these moments give its values on the dense polynomial algebra in eq. 7.ccxx, and a state is norm continuous. Equivalently they determine a probability measure on the compact image Z_r. It is the joint moments which are required; separate one-variable laws alone do not specify the correlations of the coordinates.

No continuum path limit has been taken in this argument. The countability of lattice paths, exact finite graph relations and compact bounded moment problem are the cutoff kinematics. Geometric rounding independence, renormalized loop limits, their joint correlations with the scalar field and their physical reconstruction remain separate observable readout questions.

7.12. **The scalar observable and its information.** The complete cutoff invariant algebra has now been described without reference to a continuum limit. We compare it with the actual scalar observable used in the construction. This comparison has two different outcomes. At a fixed cutoff the scalar algebra is a proper subalgebra, and we exhibit the missing information explicitly. In the continuum, the scalar distributions do determine their own cyclic representation and canonical field closures. Neither statement supplies the joint continuum law of other gauge observables.

The actual cutoff normalization. Write $a > 0$ for a fixed lattice spacing. Count each elementary positive coordinate-plane plaquette once, write x_p for its barycenter in physical coordinates, and let U_p be its oriented holonomy. The observable is

$$\begin{aligned} A_p(U) &= 1 - \tfrac{1}{2}\operatorname{ReTr} U_p \in [0,2], \\ O_a(f) &= \sum_p f(x_p) A_p, \qquad X_a(f) = O_a(f) - \rho_a(O_a(f)). \end{aligned} \tag{7.ccxxiii}$$

There is no power of a multiplying this displayed sum. Any proposed different normalization or mixing is a different readout and requires its own estimates. For $f \in C_c^\infty(\mathbb{R}^4)$ the sum is finite; for Schwartz f it converges uniformly as a function of U, since the plaquette barycenters form a finite union of translates of $a\mathbb{Z}^4$. In particular it is a bounded continuous invariant function at this fixed a.

Let $\mathfrak{A}_a^{\mathrm{inv}}$ be the invariant algebra constructed above, with the finite periodic version used on a torus. Define

$$\mathfrak{A}_{a,d} = C^*(1, A_p : \ p \text{ an elementary plaquette}) = C^*(1, O_a(f) : f \in C_c^\infty(\mathbb{R}^4)) \ \subset \ \mathfrak{A}_a^{\mathrm{inv}}. \quad \text{(7.ccxxiv)}$$

For the second equality, the inclusion from right to left follows from the finite sum in eq. 7.ccxxiii. Conversely the barycenters of distinct positive coordinate-plane plaquettes are distinct: their fractional coordinate pattern records the plane, and their integer parts record the translate. Around any one barycenter choose a smooth bump which is one there and zero at every other barycenter. The corresponding $O_a(f)$ is exactly A_p. The same argument uses smooth periodic functions on a finite torus. Adding or subtracting the constants $\rho_a(O_a(f))$ leaves the unital algebra unchanged.

This is an equality of algebras at a fixed cutoff. The bump isolating a single plaquette may depend on a. For example $f_{a,p}(x) = \chi((x - x_p)/(ca))$, with a fixed sufficiently small c, has derivatives of order r of size $O(a^{-r})$. Consequently the algebraic equality does not give a uniform estimate for the isolation of a plaquette as $a \downarrow 0$. It cannot be substituted for the fixed-test estimates used in a continuum limit.

The compact scalar quotient and its moments. Put $Z_a = \Omega_a/\mathcal{G}_a$ and let Y_a be the image of the continuous map

$$\pi_a : Z_a \longrightarrow [0,2]^{\mathcal{P}_a}, \qquad [U] \longmapsto (A_p(U))_p. \quad \text{(7.ccxxv)}$$

The countable product on the right is compact metrizable. Its image Y_a is compact, and π_a is a surjection onto that image. The unital polynomial algebra in the coordinate functions on Y_a is conjugation closed and separates its points. Stone–Weierstrass and the isometric pullback therefore identify

$$\mathfrak{A}_{a,d} = \pi_a^* C(Y_a), \qquad \mathfrak{A}_a^{\mathrm{inv}} = C(Z_a). \quad \text{(7.ccxxvi)}$$

For an infinite graph this includes uniform limits of local scalar functions. It does not require that a scalar function have finite support after taking that limit.

All joint polynomial moments of the A_p determine a state on $\mathfrak{A}_{a,d}$. Indeed they determine its values on the coordinate polynomials, and for any continuous b and polynomial P the difference between two states is bounded by $2\|b - P\|_\infty$. Taking a uniformly approximating sequence proves uniqueness on $C(Y_a)$. The same statement applies to all joint moments of the $O_a(f)$ or the centered variables, provided the centering constants are specified. It is a statement about complete bounded cutoff moment data. A sequence of continuum separated-point limits is not, by itself, this complete fixed-cutoff data set.

What is lost by π_a can be seen without a dimension count or an invariant-theory assumption.

An explicit small-curvature loss of relative color data. Use the unit lattice $\mathbb{Z}^4$ and the Pauli matrices of eq. 7.ccxi. Let $0 < \alpha < \pi/2$ and set

$$A = e^{i\alpha\sigma_3}, \qquad B_\parallel = e^{i\alpha\sigma_3}, \qquad B_\perp = e^{i\alpha\sigma_1}. \quad \text{(7.ccxxvii)}$$

Consider two configurations. In both, all positively oriented bonds are identity except the bond $c_1 = (0, e_1)$, which is A, and the bond $c_2 = (4e_4, 4e_4 + e_1)$. The latter is $B_\parallel$ in the first configuration and $B_\perp$ in the second. Reversed bonds have inverse values as always.

No elementary plaquette contains both special bonds. Each elementary plaquette holonomy is consequently identity, $A^{\pm 1}$, or $B^{\pm 1}$. The plaquettes meeting a special bond are the same in the two configurations, and

$$\tfrac{1}{2}\,\mathrm{Tr}\, A^{\pm 1} = \tfrac{1}{2}\,\mathrm{Tr}\, B_\parallel^{\pm 1} = \tfrac{1}{2}\,\mathrm{Tr}\, B_\perp^{\pm 1} = \cos\alpha. \quad \text{(7.ccxxviii)}$$

Every plaquette action is therefore equal in the two configurations. Each nonidentity plaquette is within $O(\alpha)$ of identity in matrix norm. Thus the example has arbitrarily small plaquette curvature and requires no large-field exceptional set.

Let γ_1 be the positive 12-plaquette based at zero with first edge c_1, and let γ_2 be its translate by $4e_4$, based at $4e_4$. Their holonomies are A and B, respectively. Let P be the straight path from zero to $4e_4$ along coordinate four. All its bonds are identity. The closed path $\gamma = \gamma_1 P \gamma_2 P^{-1}$ has holonomy AB, so that

$$\begin{gathered} w_\gamma(U_\parallel) = \cos(2\alpha), \qquad w_\gamma(U_\perp) = \cos^2\alpha, \\ w_\gamma(U_\parallel) - w_\gamma(U_\perp) = -\sin^2\alpha \neq 0. \end{gathered} \tag{7.ccxxix}$$

The first equality uses the same axis twice. For the second, the scalar part of the product in eq. 7.ccxii is $\cos^2\alpha - \sin^2\alpha\, e_3 \cdot e_1 = \cos^2\alpha$. These configurations are not gauge equivalent, because the normalized trace of every closed path is invariant under the full site gauge group.

Let $\mathbb{E}_a$ be the Haar expectation of eq. 7.cxcvii. Evaluation at either configuration followed by $\mathbb{E}_a$ is a normalized positive, fully gauge-invariant state on $C(\Omega_a)$. Its restriction to $C(Z_a)$ is simply evaluation at that orbit. The two states agree on $\mathfrak{A}_{a,d}$, hence on every product of all the scalar observables in eq. 7.ccxxiii, but disagree on w_γ. This proves that $\mathfrak{A}_{a,d}$ is a proper subalgebra, including in an arbitrarily small-plaquette region.

These two states are algebraic cutoff examples. No assertion that they are Wilson Gibbs states, DLR states, reflection-positive states, or scaling limits is involved. Their role is exact: positivity, gauge invariance, and the complete scalar marginal do not logically determine the full invariant state. Additional dynamical hypotheses could select a unique extension within a smaller class; such a selection would be a further theorem.

A separate loss of periodic holonomy data. On a periodic four-dimensional lattice with side length $N \geq 3$, put $h_\theta = e^{i\theta\sigma_3/N}$ on every positive coordinate-one bond and identity on every other positive bond. All elementary plaquettes are identity, whereas the closed path winding once around coordinate one has

$$A_p = 0 \quad \text{for every } p, \qquad w_{\text{wind}} = \tfrac{1}{2}\operatorname{Tr} h_\theta^N = \cos\theta. \tag{7.ccxxx}$$

For different $\theta \in [0, \pi]$ these are different full gauge orbits with exactly the same scalar plaquette data. A periodic gauge transformation changes the winding holonomy only by conjugation. A nonperiodic change of frame which removes h_θ on the covering lattice is not a periodic gauge transformation and cannot identify these torus orbits.

The complete graph coordinates of eq. 7.ccxix retain such winding loops. Discarding them would change the observable algebra. This example concerns different holonomy data; it does not contradict uniqueness on a fixed nonlinear coarse fibre in the first-step background theorem. Together, eq. 7.ccxxix and eq. 7.ccxxx distinguish a local loss of relative nonabelian orientation from a global loss of periodic holonomy.

Exactly what abstract state extension gives. For any state ν on $C(Y_a)$ consider the set

$$\mathcal{E}_a(\nu) = \{\mu \in \operatorname{Prob}(Z_a) : (\pi_a)_*\mu = \nu\}. \tag{7.ccxxxi}$$

We use the Riesz correspondence between states and probability measures on compact spaces. The set in eq. 7.ccxxxi is convex and weakly closed in the compact state space of $C(Z_a)$. It is nonempty for every ν. Here is a proof which does not choose conditional measures on individual fibres.

Regard ν as a norm-one unital functional on the closed subalgebra $\pi_a^* C(Y_a)$. The complex Hahn–Banach theorem extends it to a functional ℓ on $C(Z_a)$ with

$$\ell|_{\pi_a^* C(Y_a)} = \nu, \qquad \ell(1) = 1, \qquad \|\ell\| = 1. \tag{7.ccxxxii}$$

Such a functional is positive. To check this directly, let b be real-valued with $\|b\|_\infty \le 1$, and write $\ell(b) = u + iv$. For every real t,

$$|1 + it\ell(b)|^2 \le \|1 + itb\|_\infty^2 \le 1 + t^2. \tag{7.ccxxxiii}$$

After expanding the left side and taking $t \to 0$ with both signs we obtain $v = 0$. Thus ℓ is real on real functions. If $0 \le b \le 1$, then $|\ell(b)| \le 1$ and $|1 - \ell(b)| \le 1$, since also $\|1 - b\|_\infty \le 1$. These two real inequalities give $0 \le \ell(b) \le 1$. Scaling proves positivity for every nonnegative b. Riesz now supplies the measure μ, and equality of the restrictions is precisely the pushforward condition in eq. 7.ccxxxi.

Composing this extension with $\mathbb{E}_a$ gives a full gauge-invariant probability law on the bond configuration space. This last lift is unique once μ on the *full* orbit space has been specified, by eq. 7.ccii. The Hahn–Banach step generally has no such uniqueness. Its conclusion concerns a positive bounded cutoff state; it supplies no action, DLR specification, ultraviolet estimates, reflection positivity, or continuum observable normalization.

The nonuniqueness can be located exactly for a point scalar state. For $y \in Y_a$,

$$\mathcal{E}_a(\delta_y) = \mathrm{Prob}(\pi_a^{-1}\{y\}), \qquad \mathcal{E}_a(\delta_y) \text{ is a singleton } \iff \pi_a^{-1}\{y\} \text{ is a singleton.} \tag{7.ccxxxiv}$$

Indeed a measure with pushforward δ_y gives mass one to the closed fibre, and conversely any measure on that fibre has that pushforward. Distinct points of the fibre give distinct Dirac measures. The two orbit points in eq. 7.ccxxix lie in one such nonsingleton fibre. We do not infer that every possible scalar state has multiple extensions.

More globally, equality $\mathfrak{A}_{a,d} = C(Z_a)$ is equivalent to injectivity of π_a. If it is injective, compactness makes it a homeomorphism onto Y_a, so its pullback is onto $C(Z_a)$. Conversely, if the pullback is all of $C(Z_a)$, equal images would force all continuous functions to agree at two points; continuous functions on a compact metric space separate points. This criterion isolates the precise algebraic reason that the scalar marginal does not determine the full cutoff law.

Ordinary positivity does not give reflection positivity. The distinction is already visible on a two-site classical system. Let $s_-, s_+ \in \{-1, 1\}$ and give mass $1/2$ to each of $(s_-, s_+) = (-1, 1), (1, -1)$. This is a positive normalized probability measure invariant under the reflection exchanging the sites. Take the positive-time function $F = s_+$ and let Θ act by exchange and complex conjugation. Then

$$\mu((\Theta F)F) = \mu(s_- s_+) = -1. \tag{7.ccxxxv}$$

Thus even reflection symmetry and ordinary probability positivity do not imply OS positivity. The latter is a distinct family of Gram inequalities pairing reflected positive-time words. An abstract extension from eq. 7.ccxxxii is not an extension satisfying those inequalities unless they too are proved. This example is a logical distinction between two positivity requirements, not a purported counterexample to the Wilson construction.

The scalar continuum distributions determine a cyclic theory. There is also a positive uniqueness statement, now for the actual continuum scalar output. Let two vacuum representations have the same full Minkowski distributions W_n on every $\mathcal{S}((\mathbb{R}^4)^n)$, with their common invariant polynomial domains generated from their vacua. For finite scalar words define

$$U(\varphi_1(f_1) \cdots \varphi_1(f_n)\Omega_1) = \varphi_2(f_1) \cdots \varphi_2(f_n)\Omega_2. \tag{7.ccxxxvi}$$

The formula is extended linearly over finite sums, including the empty word. The squared norm of any such sum is a finite sum of values of W_{m+n} on the reversed, complex-conjugate first word and the second word. Equality of all distributions proves equality of these Gram forms. In particular a null sum on one side is null on the other, and eq. 7.ccxxxvi is well defined and isometric. Its range is the other polynomial core, so it extends to a unitary between the two scalar cyclic Hilbert spaces.

Concatenating one further test function proves the field intertwining on these polynomial cores. Taking closures of the graphs proves equality of the corresponding canonical closed operators whenever these closures exist. Vacuum covariance gives intertwining of translations and proper orthochronous Lorentz transformations first on the core and then on the Hilbert spaces. For translations this also intertwines their joint spectral measures, the energy operator, all its powers and their domains. The common energy domains and canonical closures constructed in RD are therefore part of this determined scalar representation. This argument does not add essential self-adjointness or a bounded local observable net to the established conclusions.

Under their stated hypotheses, EB and QRM supply the full W_n and identify their cyclic space with $\mathcal{K}_{\mathrm{ord}}$. The formula in eq. 7.ccxxxvi thus gives uniqueness in the precise cyclic sense just proved. It makes no assertion that $\mathcal{K}_{\mathrm{ord}}$ exhausts the entire Euclidean OS space, or that adding an independent gauge observable leaves this cyclic space unchanged. A larger joint observable theory contains this scalar representation as the closure of its scalar vacuum words; surjectivity onto the larger space is an additional condition.

Bounded cutoff moments and unbounded continuum fields. At a fixed cutoff one has the elementary bound

$$\|O_a(f)\|_\infty \le 2\sum_p |f(x_p)|, \qquad \|X_a(f)\|_\infty \le 4\sum_p |f(x_p)|. \tag{7.ccxxxvii}$$

For a fixed nonzero physical test function the right side can grow as $a \downarrow 0$. Thus the compact moment argument for a fixed a does not pass to a common compact spectral interval in the continuum. Under those hypotheses, the continuum field has the graph bound, with $W = 1 + H$,

$$\|W^k \varphi(g) W^{-(N+k)}\| \le C_k\, \mathfrak{t}_{N+k+2}(g), \qquad N = 9. \tag{7.ccxxxviii}$$

Here $\mathfrak{t}_q$ is the finite spacetime test seminorm defined in EB. This gives a common invariant energy-smooth domain, products, closability, and the canonical graph identifications of RD. Repeated use increases the energy order and the required test seminorm. It does not give a bound of the form $\|\varphi(g)^n\Omega\| \le C^n n!$ for every Schwartz g with a fixed constant. Such an analytic-vector assertion has not been used to assign a spectral calculus to the field.

A concrete scalar moment calculation shows why positivity and a bound exponential in n^2 alone cannot replace compact moment determinacy. On $(0,\infty)$ let

$$d\mu_\epsilon(x) = \frac{e^{-(\log x)^2/2}}{x\sqrt{2\pi}}\big(1 + \epsilon \sin(2\pi \log x)\big)\, dx, \qquad -1 \le \epsilon \le 1. \tag{7.ccxxxix}$$

These are nonnegative measures. With $z = \log x$, their sine contribution to every nonnegative integer moment is

$$\begin{aligned} &\frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-z^2/2 + nz} \sin(2\pi z)\, dz \\ &\qquad = e^{(n^2 - 4\pi^2)/2} \sin(2\pi n) = 0, \qquad n = 0, 1, 2, \dots . \end{aligned} \tag{7.ccxl}$$

For completeness the complex Gaussian identity used here follows first for real u by completing the square in $\int e^{-z^2/2+uz}\, dz = \sqrt{2\pi} e^{u^2/2}$. Both sides are entire in u by dominated convergence on compact complex sets, so the one-variable identity theorem permits $u = n + 2\pi i$ and taking imaginary parts. The formula in eq. 7.ccxl at $n = 0$ proves normalization. For every ϵ the moments are

$$\int x^n\, d\mu_\epsilon(x) = e^{n^2/2}, \qquad n \ge 0, \tag{7.ccxli}$$

while the measures differ when ϵ differs, since their densities differ on an interval. There are therefore bounded continuous functions whose integrals differ, although all these polynomial moments agree.

This calculation is not an alternative continuum model satisfying the other hypotheses. It proves the narrower implication needed here: polynomial positivity and $e^{O(n^2)}$ moment growth alone

do not determine all bounded functions of an unbounded variable. The imported OS output has an ab^{n^2} type distribution bound; no compact moment or analytic-vector conclusion has been inferred from that form of bound. The actual scalar Wightman functional and its canonical cyclic graph representation remain determined by eq. 7.ccxxxvi. The distinction concerns further spectral completions or additional observables, not an ambiguity in those already constructed distributions.

Scalar phase labels and full gauge-state labels. The compact phase space used in SPH is separated by scalar evaluation functions $f_F(\xi)$ of ordered Euclidean words. Its points distinguish those scalar functionals. They are not, by this definition, points of Z_a, nor are they automatically all extremal states of a continuum gauge-observable algebra. If two proposed joint observable states have the same scalar restriction, all f_F agree for those restrictions even when a mixed or nonscalar correlation differs.

Conversely a theorem specifying a physical class of extensions could prove that its restriction map to scalar states is injective. The finite-cutoff examples show that this would require the additional class restrictions, not just gauge invariance and positivity. In a continuum construction the required conditions include actual joint readouts, their compatible limits, their positivity and covariance, and their relation to the scalar normalization in eq. 7.ccxxiii. The next section formulates these conditions as observable statements and proves the algebraic and representation consequences once they hold.

7.13. **Joint gauge-observable extension and identification.** We give precise conditions under which further observables extend the constructed scalar theory. They concern joint moments, not separate one-observable laws. We prove the resulting positivity, consistency, domain and identification statements, and distinguish them from the model estimates needed for a continuum readout. The scalar reconstruction already established in EB and QRM is retained throughout; none of its locality conclusions is inferred from an unconstructed larger observable family.

Typed words and the scalar restriction. Let I be a set of observable types with a distinguished scalar type 0. Each type has a complex test space E_α, an antilinear involution $f \mapsto f^*$ into E_{α^*}, and a specified localization region for each local test. For point fields take $E_\alpha = \mathcal{S}(\mathbb{R}^4)$, with the appropriate finite tensor components if necessary. A loop type instead needs an explicit space of geometric labels or smearing tests and a support convention; it is not silently a point field. Types are required to have compatible involutions, so $(\alpha^*)^* = \alpha$ and $(f^*)^* = f$.

First form the unital free algebra of finite typed words, with complex linearity in each test. The star is

$$(q_{\alpha_1}(f_1)\cdots q_{\alpha_n}(f_n))^* = q_{\alpha_n^*}(f_n^*)\cdots q_{\alpha_1^*}(f_1^*). \tag{7.ccxlii}$$

Let $\mathcal{I}$ be a declared two-sided star ideal of further relations and let $\mathcal{A}$ be the quotient. The relations may encode exact trace identities, conjugate observables, or other proved identifications. A normalization-dependent continuum identity is not included merely because the corresponding bare cutoff variables satisfy it. This distinction applies in particular to eq. 7.ccxxi.

There is a specified unital star map ι from the scalar word algebra into $\mathcal{A}$, sending its field generator to q_0. If an algebraic embedding is required, its injectivity is an additional algebraic property to check. For the Hilbert-space restriction below the necessary condition is equality of the scalar functional:

$$\omega(\iota(F)) = \omega_d(F) \quad \text{for every scalar word sum } F, \tag{7.ccxliii}$$

where ω_d is the full scalar Minkowski functional already constructed. Any scalar relation killed by ι must consequently be null for that functional. The readout identifying q_0 is the actual centered normalization in eq. 7.ccxxiii, not an independently chosen scalar spectator.

The exact mixed positivity and relation tests. A candidate joint moment family gives a complex linear functional ω on finite word sums. Its elementary requirements are normalization, hermiticity and all finite Gram inequalities:

$$\begin{aligned} &\omega(1) = 1, \qquad \omega(F^*) = \overline{\omega(F)}, \\ &\sum_{i,j=1}^{r} \overline{c_i} c_j \, \omega(F_i^* F_j) \geq 0 \quad (r < \infty, c_i \in \mathbb{C}). \end{aligned} \tag{7.ccxliv}$$

Here the F_i may involve different types and different word lengths. Separate positivity for each type is insufficient. The inequality for all finite sums is equivalent to $\omega(F^*F) \geq 0$ for every element F of the free algebra.

The ideal must also be annihilated. For a star-closed set of generators R of $\mathcal{I}$, an exact equivalent test, under eq. 7.ccxliv, is

$$\begin{aligned} &\omega(ARB) = 0 \quad \text{for every } A, B \text{ and every } R; \\ &\iff \quad \omega((RB)^* RB) = 0 \quad \text{for every } B \text{ and every } R. \end{aligned} \tag{7.ccxlv}$$

To prove the reverse implication, apply Cauchy–Schwarz to the positive form $\langle C, D\rangle = \omega(C^*D)$ with $C = A^*$ and $D = RB$. Its second diagonal entry is zero, so $\omega(ARB) = 0$. The forward implication uses $A = B^*R^*$. Linearity then annihilates the entire generated ideal. Both R and R^* are included in the declared generating set. Merely requiring $\omega(R) = 0$ would not suffice: a centered real variable can have zero mean and positive variance.

Even pairwise consistency of positive laws misses a genuine mixed constraint. Consider three proposed signs $A, B, C \in \{-1, 1\}$ with uniform individual marginals. Give (A, B) and (B, C) the law of two identical uniform signs, and (A, C) the law of two opposite uniform signs. Every pair law is positive and all their one-sign restrictions agree. A common joint positive state would nevertheless have

$$\omega((A - B + C)^2) = 3 - 2\omega(AB) + 2\omega(AC) - 2\omega(BC) = -3. \tag{7.ccxlvi}$$

This contradicts eq. 7.ccxliv. Thus a family of compatible two-observable laws is not a substitute for positive matrices of all finite words.

Test continuity has a concrete meaning. For a fixed word of point-field types, require a bound

$$|\omega(q_{\alpha_1}(f_1) \cdots q_{\alpha_n}(f_n))| \leq C_{\boldsymbol{\alpha}} \prod_{j=1}^{n} p_M(f_j) \tag{7.ccxlvii}$$

for some finite M, which may depend on the word. Different orders in the factors can be replaced by their maximum. This is equivalent to continuity of that multilinear form on the product of the Schwartz spaces: continuity at zero gives a product neighborhood where its absolute value is at most one, and rescaling each nonzero factor gives eq. 7.ccxlvii. The seminorms contain the supremum norm, so a zero factor seminorm means the test is zero.

The explicit tensor expansion of RTE, applied with this M, gives a unique full Schwartz distribution for the word, with

$$|W_{\boldsymbol{\alpha}}(F)| \leq C'_{\boldsymbol{\alpha}} p_{n(M+8)}(F), \qquad F \in \mathcal{S}((\mathbb{R}^4)^n). \tag{7.ccxlviii}$$

This does not postulate a tensor theorem without a norm: the absolutely summable grid/Fourier expansion proved in RTE bounds $\sum |c_\nu| \prod_j p_M(f_{\nu,j})$ by that full-test seminorm. The same construction works for vector-valued forms whenever the right side of eq. 7.ccxlvii bounds a norm in the stated complete target. Conversely a full distribution bound implies a simple-tensor bound, since $p_r(f_1 \otimes \cdots \otimes f_n) \leq \prod_j p_r(f_j)$.

Thus one may complete each fixed typed-word space to its full Schwartz space. Star reversal and concatenation are continuous. Positivity on this completed finite-degree algebra follows by approximating each of the finitely many full tests by tensor sums: every entry of its Gram matrix

converges, and finite positive matrices are closed under entrywise limits. The ideal test is imposed on this completed algebra as well, or extended there by a proved continuity bound for the specified relations. Other test types need their own explicitly stated continuity/completion criterion; eq. 7.ccxlviii does not assign them point-field test spaces.

These degreewise bounds do not by themselves verify the linear growth hypothesis of a Euclidean reconstruction theorem. Such a theorem has a further quantitative hypothesis across all word lengths. QRM checked that hypothesis for the scalar family; it has not checked it for arbitrary additional types.

The joint GNS construction and the scalar subspace. Assume eq. 7.ccxliii–eq. 7.ccxlv and the specified test continuity. On $\mathcal{A}$ use the positive form $\langle A, B\rangle = \omega(A^*B)$. Its Cauchy–Schwarz inequality follows by applying positivity to $A + zB$ for every complex z. The null space $\mathcal{N}$ is a left ideal. Indeed if $\omega(B^*B) = 0$, then

$$\omega((AB)^*AB) = \langle B, A^*AB\rangle = 0 \tag{7.ccxlix}$$

by Cauchy–Schwarz. All the words in the second argument are elements of the algebra, so their norms are finite. There is no implicit bounded-operator estimate in this proof.

Complete $\mathcal{D} = \mathcal{A}/\mathcal{N}$ to $\mathcal{H}$, put $\Omega = [1]$, and act by left multiplication. It is well defined by eq. 7.ccxlix and gives an invariant dense polynomial domain. On that domain the star identity gives

$$\langle \pi(A^*)v, w\rangle = \langle v, \pi(A)w\rangle, \qquad v, w \in \mathcal{D}. \tag{7.ccl}$$

In particular each field is closable, because its adjoint contains its declared star partner on a dense domain. Positivity alone gives neither boundedness nor essential self-adjointness of these fields.

The equality in eq. 7.ccxliii gives an isometry from the scalar polynomial space to $\mathcal{H}$, with

$$J[F]_d = [\iota(F)], \qquad \mathcal{H}_d = \overline{\pi(\iota(\mathcal{A}_d))\Omega}, \qquad J : \mathcal{K}_{\mathrm{ord}} \longrightarrow \mathcal{H}_d \text{ unitary.} \tag{7.ccli}$$

The norm identity is exactly the scalar restriction applied to F^*G; it proves that null vectors are respected. Surjectivity is onto the displayed closed subspace by its definition. It is onto the entire joint Hilbert space precisely when the scalar vacuum words are dense there. That density is an additional condition, not a consequence of matching the scalar moments.

Already for bounded commutative algebras one can see the distinction. The state $\nu \otimes \frac{1}{2}(\delta_{-1} + \delta_1)$ on $C(Y \times \{-1, 1\})$ restricts to ν on $C(Y)$. Its GNS scalar subspace is $L^2(\nu) \otimes \mathbb{C}1$, whereas the sign observable supplies the orthogonal nonzero vector $1 \otimes s$. This is an algebraic example of additional degrees of freedom with exactly the same scalar state; it makes no claim about a continuum gauge extension of the constructed model.

There is a useful exact compression criterion in this bounded case. For a representation of a unital star algebra by bounded operators, let P project onto a closed scalar subspace. Compression is multiplicative on the full algebra if and only if that subspace is reducing. One direction is immediate. For the other, a multiplicative compression would give

$$P\pi(A^*)(1 - P)\pi(A)P = P\pi(A^*A)P - P\pi(A^*)P\pi(A)P = 0. \tag{7.cclii}$$

The left side is the square of the norm of $(1 - P)\pi(A)P$ in quadratic-form language. Hence this operator vanishes for every A, and also for A^*, proving reduction. If Ω is in the subspace and is cyclic for the full algebra, reduction then forces the subspace to be all of $\mathcal{H}$. For unbounded fields an arbitrary projection need not even preserve their domains, so no unqualified compression representation is asserted.

Consistency across finite observable families. Suppose a directed system of unital star algebras $\mathcal{A}_J$ has compatible injective star maps for $J \subset K$. Suppose normalized positive functionals ω_J satisfy

$$\omega_K|_{\mathcal{A}_J} = \omega_J. \tag{7.ccliii}$$

There is then a unique normalized positive functional on their algebraic inductive limit. Define it on an element by using any stage containing that element. A common larger stage and eq. 7.ccliii make the value independent of the choice. Any finite word sum belongs to a common stage, so its square has a nonnegative expectation there; this proves positivity on the union. The same argument proves uniqueness. For a system of isometric unital C^* embeddings the state has norm one on the union and extends to its norm completion. An unbounded field algebra has no such automatic C^* completion; its degreewise test continuity is checked as in eq. 7.ccxlvii–eq. 7.ccxlviii.

If global relations are first specified in one free quotient, the maps from smaller quotients need not be injective automatically. One can instead prescribe all finite mixed moments in that global quotient and apply eq. 7.ccxliv–eq. 7.ccxlv directly. Adding a new type must not silently impose a previously untested relation on an old type.

A practical sufficient route to eq. 7.ccliii is uniqueness of a matched finite-family reconstruction. Suppose the same Euclidean joint data on every overlap have reconstructions unique in the exact analytic/boundary class being used. Restrict a larger reconstruction to the smaller family. It has the smaller Euclidean data and the same required class properties. Uniqueness makes it the smaller reconstruction. Thus the extensions at successive stages are consistent. Both the class membership and the uniqueness theorem must be checked for the actual new types; scalar uniqueness alone is not that theorem.

Covariance and its continuity on the joint space. Assume a declared spacetime symmetry group acts by star automorphisms α_g of $\mathcal{A}$, respecting the relations, and that $\omega \circ \alpha_g = \omega$. Then

$$U_g[A] = [\alpha_g A] \tag{7.ccliv}$$

is well defined and isometric; $\alpha_{g^{-1}}$ gives its inverse. It implements the action on the common polynomial domain, fixes Ω, and satisfies the group law there and hence everywhere. Strong continuity follows if, for every polynomial A, $g \mapsto \omega(A^*\alpha_g A)$ is continuous at the identity, because

$$\|U_g[A] - [A]\|^2 = 2\omega(A^*A) - 2\,\mathrm{Re}\,\omega(A^*\alpha_g A). \tag{7.cclv}$$

Density and the unitary norm then extend continuity to all vectors. For point-field types this condition follows from the actual continuous test action and the joint moment bounds. For geometric loop labels it is a condition on that declared label topology and their joint moments, not a consequence of a name assigned to a loop.

When the scalar inclusion intertwines these actions, its closed cyclic subspace is invariant under U_g and $U_{g^{-1}}$, hence reducing. The unitary in eq. 7.ccli intertwines the established scalar symmetries with their restrictions. It still says nothing about additional joint states orthogonal to that subspace.

Use the physical translation convention $U(a) = \exp(i(a^0 H - \mathbf{a}\cdot\mathbf{P}))$; in the notation of RHI the physical spatial momentum here is minus its spatial generator. Let E be the joint spectral measure of the strongly continuous translation group. Positivity of energy-momentum has the following exact dense test:

$$E(\mathbb{R}^4 \setminus \overline{V}_+) = 0 \quad\Longleftrightarrow\quad \operatorname{supp}\mu_A \subset \overline{V}_+ \ \text{ for every } A \in \mathcal{A}, \qquad \mu_A(B) = \|E(B)[A]\|^2. \tag{7.cclvi}$$

Indeed the forward implication is immediate. Conversely the spectral projection of the open complement kills every vector of the dense polynomial domain, and hence is zero. The measures μ_A are the positive Fourier spectral measures of $\omega(A^*\alpha_a A)$. Checking only the vacuum coefficient checks just the atom at zero, and checking only single-field vectors checks only their span. The criterion is for all words, unless an independent density theorem reduces that test family.

Locality as an all-spectator moment identity. For spacelike separated declared localization regions, put $R = [q_\alpha(f), q_\beta(g)]$. The exact joint locality criterion is

$$\omega(ARB) = 0 \quad \text{for every polynomial } A, B. \tag{7.cclvii}$$

It is equivalent to $\pi(R) = 0$ on the common polynomial domain. For if eq. 7.cclvii holds, the vector $\pi(R)[B]$ has zero inner product with every polynomial vector $[C]$, by taking $A = C^*$. Density forces the vector to vanish. Conversely vanishing of those vectors gives eq. 7.cclvii. Equivalently the norm-square test in eq. 7.ccxlv can be used for these commutator relations.

Equality of only the two-point functions with exchanged entries does not test eq. 7.cclvii, since it leaves out the spectators. Once the elementary commutators vanish on an invariant common domain, the Leibniz rule for commutators proves locality of their localized polynomial algebras on that domain. For extended geometric observables the regions are their actual supports, not nominal base points. No spacelike commutativity is inferred from two disjoint curves unless their complete declared localization regions satisfy the required condition.

This is a criterion for the larger moment family. For the scalar family the necessary identity is already a conclusion of QRM's matched corrected OS theorem. For a new family it must follow from that family's actual joint analytic/permutation argument or a reconstruction theorem whose hypotheses have been checked for it. It is not used here as an assumption purporting to prove that same joint continuation.

A precise common-domain criterion for added point fields. We next prove how verified energy bounds upgrade the polynomial construction. Restrict this paragraph to point-field Schwartz types, or to other test types with the same smooth translation action and moment continuity. Let the joint representation above contain only types meeting that assumption in this domain argument; otherwise the conclusion applies to the cyclic subspace of the qualified subfamily, without a graph-core assertion for the larger space. Let that representation have the forward spectral property. Its polynomial test domain $\mathcal{D}$ lies in $\bigcap_k \mathrm{Dom}\, H^k$. For a full word test F translation in time translates all its arguments; continuity of the two-word norm form permits every Schwartz difference quotient in Hilbert norm. Stone's generator identity gives

$$H^k[F] = i^k[D_0^k F], \qquad D_0 = \sum_j \partial_{x_j^0}. \tag{7.cclviii}$$

Iterating proves the assertion for all k. The sign agrees with $f_a(x) = f(x-a)$ and $U(t) = e^{itH}$.

The same domain is a graph core for every $(1+H)^k$. To see this, choose a smooth compactly supported scalar cutoff c_R equal to one on $[0,R]$ and tending pointwise to one on $[0,\infty)$, with $|c_R| \le 1$. For fixed R, $(1+H)^k c_R(H)$ is bounded, so dense polynomial vectors approximate $c_R(H)\psi$ in graph norm after applying $c_R(H)$. For a polynomial vector v, Fourier inversion expresses $c_R(H)v$ as an integral of $U(t)v$ with a Schwartz coefficient. This integral is approximated by finite Riemann sums in graph norm: on compact time intervals the translated full test is graph continuous by eq. 7.cclviii, and the tail is bounded by the L^1 tail times $\|(1+H)^k v\|$. The Riemann sums are polynomial vectors. Finally $c_R(H)\psi \to \psi$ in graph norm for $\psi \in \mathrm{Dom}(1+H)^k$ by spectral dominated convergence. This proves the core claim with both approximations in their required order.

Set $\mathsf{E} = 1 + H$. Suppose for each added type there is a nonnegative integer N_α such that, for every integer $k \ge 0$, the *actual* polynomial operator satisfies

$$\|\mathsf{E}^k \pi(q_\alpha(f))v\| \le C_{\alpha,k}\, p_{M_{\alpha,k}}(f)\, \|\mathsf{E}^{k+N_\alpha} v\|, \qquad v \in \mathcal{D}. \tag{7.cclix}$$

These are bounds to prove on the stated joint space. The scalar case with $N_0 = 9$ is already available on its own scalar cyclic space. In a larger representation even the action of the scalar field on vectors created by new observables requires the corresponding mixed bound. The scalar result proves neither that larger-space bound nor eq. 7.cclix for a Wilson-loop or composite type.

The core claim and closedness of E^k extend the operator coherently to $\mathcal{D}_\infty = \bigcap_k \mathrm{Dom}\, H^k$, mapping it into itself. More explicitly choose polynomial v_j approaching $v \in \mathcal{D}_\infty$ in graph norms through order j. For each fixed k, eq. 7.cclix makes the field images Cauchy in the E^k graph norm. Their Hilbert limits coincide for all k, and closedness identifies the higher energy powers of that

same limit. The extension is independent of the approximating sequence. The star identity extends by the same graph approximations for both vectors, so the extended operators remain closable.

For $k = 0$ the operator also extends from $\mathrm{Dom}\,\mathsf{E}^{N_\alpha}$ to the Hilbert space with its stated graph bound. Approximating in that graph norm shows that this extension lies in the closure of the original polynomial operator. Conversely all three operators, on $\mathcal{D}$, on $\mathcal{D}_\infty$, and on $\mathrm{Dom}\,\mathsf{E}^{N_\alpha}$, have the same closure: their graphs are nested, and every graph point of the largest restriction is approximated by the smallest. This is a canonical closure statement, not essential self-adjointness.

For a word let $S_j = \sum_{i<j} N_{\alpha_i}$ and $S = \sum_i N_{\alpha_i}$. Inserting the appropriate energy powers between factors gives, first on $\mathcal{D}_\infty$ and then as a bounded weighted map,

$$\|\mathsf{E}^l \pi(q_{\alpha_1}(f_1)) \cdots \pi(q_{\alpha_n}(f_n)) \mathsf{E}^{-(l+S)}\| \le \prod_{j=1}^{n} C_{\alpha_j, l+S_j} p_{M_{\alpha_j, l+S_j}}(f_j). \tag{7.cclx}$$

Indeed the jth weighted factor is $\mathsf{E}^{l+S_j} \pi(q_{\alpha_j}(f_j)) \mathsf{E}^{-(l+S_j+N_{\alpha_j})}$, and the adjacent powers cancel on the energy-smooth domain. Their bounded extensions give the asserted norm inequality. Applying the vector-valued RTE expansion with the maximum of the displayed test orders then gives a full non-tensor word operator with an explicit finite Schwartz seminorm bound. The vacuum vector has $\mathsf{E}^r \Omega = \Omega$, so this also bounds every fixed mixed vacuum distribution.

If eq. 7.cclvii holds, the product estimate with energy order $N_\alpha + N_\beta$ extends its commutator identity from the polynomial domain to $\mathrm{Dom}\,\mathsf{E}^{N_\alpha + N_\beta}$, using the graph core just proved. In particular it holds on $\mathcal{D}_\infty$. This extension is operator locality on a declared common domain; strong commutation of self-adjoint spectral calculi would require further hypotheses.

If the full joint Poincare representation is also established, its forward cone implies $c_\Lambda H \le U_\Lambda H U_\Lambda^{-1} \le C_\Lambda H$ for each fixed proper orthochronous Lorentz transformation, in the joint spectral sense. The time component of a transformed forward momentum is bounded above and below by positive constants times its energy. It follows that $U_\Lambda \mathcal{D}_\infty = \mathcal{D}_\infty$, with equivalent energy graph norms. Covariance on the polynomial domain then extends by eq. 7.cclix to this common domain, with the specified finite-component test transformations. No unrestricted inverse heat operator or all-energy band sum is needed for this domain argument.

Actual cutoff readouts and renormalization. A continuum label requires a concrete cutoff function on the same bond configuration space and the same real measure as the scalar construction. Denote it by $Q_{a,\alpha}(f)$. It must be fully site-gauge invariant. For compactly localized tests it also has a declared finite support, including every connecting path, smearing region and auxiliary loop. A typical linear mixing prescription is

$$Q_{a,\alpha}(f) = \sum_{\beta \in J_\alpha} Z_{a,\alpha\beta} B_{a,\beta}(D_{a,\alpha\beta} f) + c_{a,\alpha}(f) 1, \tag{7.cclxi}$$

where J_α is finite for this prescription, the bare functions $B_{a,\beta}$ and lattice smearing maps $D_{a,\alpha\beta}$ are specified, and the subtraction $c_{a,\alpha}$ is specified and linear. The formula in eq. 7.cclxi is a template for a definition, not an assertion that suitable coefficients or bounds exist. A proposed readout has to supply its actual coefficients, their normalization, and the moment bounds after mixing. Diverging Z factors cannot be ignored because each bare Wilson trace is bounded.

The star and Euclidean reflection operations must be matched by that definition. In particular the real cutoff identities are

$$Q_{a,\alpha^*}(f^*) = \overline{Q_{a,\alpha}(f)}, \qquad \Theta_a Q_{a,\alpha}(f) = Q_{a,\vartheta\alpha}(\vartheta f), \tag{7.cclxii}$$

with the antilinear test/reflection action on the right specified for the type. For a tensor observable the reflection can act on its components. For a loop it acts on its entire oriented path. The complex coefficients and counterterms in eq. 7.cclxi must respect these identities or their explicitly proved limiting versions. The constant counterterm has empty localization support; the nonconstant part has its actual support.

For a point-field readout a useful localization requirement is

$$\operatorname{supp} Q_{a,\alpha}(f) \subset (\operatorname{supp} f)^{\epsilon_a}, \qquad \epsilon_a \downarrow 0, \tag{7.cclxiii}$$

where the left side denotes the physical locations of all bonds on which the function depends. For an extended type replace supp f by its declared geometric region. It is this condition, not the name of the observable, that allows a fixed positive-time gap to survive at small cutoff. It does not by itself give a moment bound or independence of the lattice rounding of a curve.

The Euclidean readout map takes products into the commutative algebra of bond functions. It is not a homomorphism from the noncommutative Minkowski field algebra. A Euclidean family must first be reconstructed, with its ordered analytic continuation and boundary conventions, before it can supply the joint Minkowski functional used in eq. 7.ccxliii–eq. 7.cclx. Ordinary positive probability expectations of the commuting cutoff functions do not alone prove Minkowski positivity and spectrum.

Passing the actual reflected Gram matrices to a limit. Let ρ_a be real probability measures with a proved lattice OS reflection inequality. Write Θ_a for its antilinear reflection, including complex conjugation. For positive-time cutoff functions $F_{a,1}, \dots, F_{a,r}$ the actual inequality is

$$\sum_{i,j} \overline{c_i} c_j \, \rho_a((\Theta_a F_{a,i}) F_{a,j}) \geq 0. \tag{7.cclxiv}$$

Suppose the word tests defining these functions have a fixed positive-time gap $\delta > 0$ and their readouts satisfy eq. 7.cclxiii. If the discrete reflection plane is displaced by $b_a \to 0$ from the continuum plane, then for all sufficiently small a, $\epsilon_a + |b_a| < \delta$. Every bond support is on the required positive side. Thus eq. 7.cclxiv applies to the actual readouts rather than a formally localized replacement.

Suppose also that each entry in eq. 7.cclxiv has a joint limit given by the proposed Euclidean data. Passing a finite sum to the limit gives a positive continuum reflected Gram matrix. This is a complete proof of that implication: for each fixed vector of coefficients the quadratic form is a limit of nonnegative real numbers. One must nevertheless verify that the reflected word test for each entry belongs to the space on which the limit is known. A separated or diagonal-flat distribution cannot be evaluated on an arbitrary coincident-point test without this check.

For the scalar family QRM performs the reflected-product diagonal-flat check and the enlargement of the positive-time quotient. For a proposed new family one can start with compact strictly ordered positive-time words away from all internal diagonals, verify that their reflected products are in the declared joint limit space, and then prove the density/insertion estimate needed to enlarge that space. This explicitly identifies the further test-space task. It does not invent Euclidean contact terms to make an undefined Gram entry exist.

Likewise lattice symmetries pass to any joint limit when their actual readouts intertwine the corresponding test actions and the limiting distributions are continuous in the needed topology. Discrete lattice rotations alone do not prove continuous Euclidean rotational covariance. The scalar G2 argument supplies it for its own family. An extension needs the matching assertion for its mixed data if that assertion enters its reconstruction theorem.

Real OS inequalities are used at real physical measures. An analytic continuation of coarse backgrounds or of external currents is a separate analytic operation. A complex source functional is not a probability state merely because its restriction at zero real source is positive.

A useful bounded loop compactness statement. There is a limited joint compactness result available without any renormalized insertion estimate. Choose countably many labelled closed paths $\gamma_{a,j}$ at each cutoff, and set $Y_{a,j} = w_{\gamma_{a,j}} \in [-1, 1]$. The labels include their chosen lattice rounding. For any sequence $a_k \downarrow 0$ there is a subsequence on which all finite joint polynomial moments of these bounded real coordinates converge.

Indeed the finite coordinate monomials form a countable set, and each expectation is bounded by one. Repeated subsequences and a diagonal selection give a common subsequence. Define L on coordinate polynomials by these limits. For $K = [-1, 1]^{\mathbb{N}}$,

$$|L(P)| \le \|P\|_{C(K)}, \qquad L(1) = 1, \qquad L(P) \ge 0 \quad \text{if } P \ge 0 \text{ on } K. \tag{7.cclxv}$$

The inequalities follow by applying them before taking the limit, since every vector of readout values belongs to K. Coordinate polynomials are conjugation closed and separate points of the compact product. Their uniform closure is $C(K)$, so eq. 7.cclxv extends L uniquely to a positive state. Riesz gives a probability measure on K with exactly the selected joint moments. This proves full finite-coordinate consistency, not just convergence of separate one-loop laws.

If a fixed finite-coordinate polynomial relation R holds for the readouts at every cutoff, then

$$L(|R|^2) = 0, \tag{7.cclxvi}$$

so $R = 0$ almost surely in the limit. More generally the same conclusion holds if the coefficients of relations R_a converge to those of R, their finite coordinate set is fixed, and $R_a(Y_a) = 0$. In fact $R_a \to R$ uniformly on K, whence $\rho_a(|R(Y_a)|^2) \to 0$. This is a direct way to retain a specified finite collection of trace relations, provided the rounded labels preserve their concatenations and inverses. If those label identifications are not exact, a quantitative replacement estimate is needed.

This compactness statement need not give a nontrivial limit: the state concentrated at $Y_j = 1$ is permitted. It gives no rounding independence, continuous spacetime covariance, renormalized local composite limit, or mixed law with the unbounded scalar distributions. It also gives no uniform control after multiplying these coordinates by diverging renormalization constants. Those are different assertions from the bounded probability statement just proved.

What mixed bounds a bounded loop actually needs. Let Y_a be any cutoff loop polynomial with $|Y_a| \le 1$ and let $Q_a = \prod_{j=1}^{n} X_a(f_j)$. At the same real cutoff measure one has

$$|\rho_a(Y_a Q_a)| \le \rho_a\Big(\prod_j |X_a(f_j)|\Big), \qquad |\rho_a(Y_a Q_a)|^2 \le \rho_a(|Q_a|^2). \tag{7.cclxvii}$$

The first is the triangle inequality and the second is Cauchy–Schwarz with $\rho_a(|Y_a|^2) \le 1$. For example a sufficient further estimate is

$$\sup_a \rho_a\Big(\prod_j |X_a(f_j)|^2\Big) \le C_n \prod_j p_M(f_j)^2. \tag{7.cclxviii}$$

It gives uniform boundedness and test continuity for the mixed word in eq. 7.cclxvii. A complete mixed subsequence argument would then also require consistency of every such word, the actual relation tests, and the reflected Gram limits where they are used. It would not yet infer its joint Minkowski continuation without a matched reconstruction.

The separated signed scalar distribution bound does not itself evaluate the right side of eq. 7.cclxviii. That expression is an absolute repeated product at one real cutoff measure. Its test is not automatically in a diagonal-flat Euclidean domain. If a stronger insertion estimate is available, its exact observables, repeated products and exterior variables must be matched before using it here. In particular a formal arbitrary-current coefficient is not identified with the needed physical mixed product merely by naming it $E^{(2)}$. The coefficient/readout identification must be proved on its stated domain wherever it is used.

For a fixed family, uniformly bounded absolute moments of every required degree permit diagonal moment subsequences on a countable dense test set. Uniform multilinear bounds of the form in eq. 7.ccxlvii then extend the limits to all tests by its explicit tensor argument. Positivity and relation identities pass through finite matrices. The uniform bounds are essential in this extension from a countable test set; convergence on a dense set alone does not establish distributional convergence.

A quantitative path-sliding comparison. There is a concrete geometric estimate useful in checking rounding consistency. Suppose two fine paths with the same endpoints are connected by an explicit finite sequence of elementary plaquette moves and retracings. An elementary move replaces two sides of a plaquette by the other two sides with the same endpoints. If A and B are the two path products, AB^{-1} is the plaquette holonomy with the appropriate orientation and base point. All other prefix and suffix products are unitary. Hence the norm of the change in the whole path product is exactly $\|I - U_p\|$ for that move, up to conjugation or inversion, both of which preserve this norm. Telescoping along the supplied sequence gives

$$\|U_\gamma - U_{\gamma'}\|_{\mathrm{op}} \le \sum_{\ell=1}^{M} \|I - U_{p_\ell}\|_{\mathrm{op}}. \tag{7.cclxix}$$

Retracings give zero change. Repeated use of a plaquette is counted with its multiplicity. For closed paths the same bound applies to normalized traces, because $|\operatorname{Tr} C|/2 \le \|C\|_{\mathrm{op}}$ for two-by-two matrices.

For $U \in \mathrm{SU}(2)$ its eigenvalues are $e^{\pm i\theta}$, so the actual plaquette action satisfies

$$\|I - U_p\|_{\mathrm{op}}^2 = 2(1 - \tfrac{1}{2}\operatorname{ReTr} U_p) = 2A_p. \tag{7.cclxx}$$

Taking $L^q(\rho_a)$ norms in eq. 7.cclxix, with $q \ge 2$, and using the triangle inequality therefore gives the explicit sufficient comparison

$$\|w_\gamma - w_{\gamma'}\|_{L^q(\rho_a)} \le \sqrt{2} \sum_{\ell=1}^{M} \|A_{p_\ell}\|_{L^{q/2}(\rho_a)}^{1/2}. \tag{7.cclxxi}$$

This is a true estimate for the real unitary cutoff variables. If the right side tends to zero for the specified rounding moves, it proves convergence of their trace difference in that norm. Mere proximity of two physical curves does not bound the number of moves or this sum. A proof must count the actual filling and provide the needed plaquette norm rate. A uniform pointwise small-curvature estimate $\|I - U_p\| \le \varepsilon_a$ only yields $M\varepsilon_a$, which need not tend to zero.

No filling is asserted between paths of different winding class on a torus. Indeed eq. 7.ccxxx rules out any estimate which would remove nontrivial flat winding holonomies by such a claim. The supplied move sequence, endpoints and periodic identifications are part of the hypotheses of eq. 7.cclxix.

A controlled change of readout scheme. Suppose two readout prescriptions $Q_{a,j}$ and $Q'_{a,j}$ for a fixed word satisfy, for that word length n,

$$\|Q_{a,j} - Q'_{a,j}\|_{L^n(\rho_a)} \longrightarrow 0, \qquad \sup_a (\|Q_{a,j}\|_{L^n} + \|Q'_{a,j}\|_{L^n}) < \infty. \tag{7.cclxxii}$$

Then their joint word expectations have the same limit whenever one exists. Expand the difference of products by replacing one factor at a time. Holder's inequality with n factors bounds the jth term by

$$\|Q_{a,j} - Q'_{a,j}\|_{L^n} \prod_{i<j} \|Q_{a,i}\|_{L^n} \prod_{i>j} \|Q'_{a,i}\|_{L^n}, \tag{7.cclxxiii}$$

which tends to zero. The case $n = 1$ is just its L^1 bound. Versions with different Holder exponents or proved exterior insertion norms give the corresponding same conclusion, but their actual norms must be stated. For bounded loops, eq. 7.cclxxi can supply the difference norm in eq. 7.cclxxii once its right side is controlled. For renormalized fields the differences include their coefficients and counterterms as well.

There is also a useful two-limit density criterion. Suppose for each intended readout $Q_{a,j}$ there are polynomials $P_{a,j}^{(r)}$ in a fixed listed family of loop readouts, with coefficients and finite labels

independent of a for each r. Suppose all fixed-polynomial joint moments have limits, the required L^n norms are uniformly bounded, and

$$\lim_{r\to\infty} \limsup_{a\downarrow 0} \|Q_{a,j} - P_{a,j}^{(r)}\|_{L^n(\rho_a)} = 0 \tag{7.cclxxiv}$$

for every fixed word length involved. The product telescoping bound gives a uniform limiting error $C_n\varepsilon_r$. For each r take the known polynomial-word limit. Applying the same error bound for r and s shows these limits are Cauchy as $r, s \to \infty$. The limiting value is then the unique limit for the original word: its limsup distance from the rth polynomial limit is at most $C_n\varepsilon_r$. This proves the criterion without interchanging an uncontrolled pair of limits.

The fixed-cutoff density theorem in eq. 7.ccxx alone gives no such criterion. Its approximating degrees, loop labels and coefficients may depend arbitrarily on a. Convergence of each fixed loop polynomial does not control a moving polynomial with diverging coefficients. The condition in eq. 7.cclxxiv, or an actual alternative uniform insertion norm, is the missing quantitative link in such a use of cutoff completeness.

Exact laws and dense-observable transport. At a fixed real cutoff, a genuine integration or change of variables may induce a bounded observable map T_a and an identity for a determining class. The dense-class norm estimate in eq. 7.ccvi, applied to the complete Wilson polynomial family in eq. 7.ccxxii, extends that identity to every continuous invariant observable. If the actual induced full bond measure is gauge invariant, equality there determines the full gauge-invariant probability law by eq. 7.ccii, and hence all bounded Borel readouts with that law.

The hypotheses include the map's actual observable type, its boundedness at that cutoff, and the identity on a complete class. An identity only for the scalar partition function, its scalar derivatives, or one special insertion is not the determining-class identity for all Wilson polynomials. A based gauge-fixed calculation must also be transported to the full-site invariant functions before the orbit-space state statement applies. None of these requirements is replaced by preservation of the constant function alone.

Across different cutoffs the norms and readouts in this argument may change. A finite-cutoff equality is exact, but any passage to a renormalized joint continuum family still needs the continuity and limit estimates just described. This separates an exact integral identity from the independent estimates needed to read a continuum observable through it.

Conditional joint reconstruction and physical identification. The previous statements give a precise conditional extension theorem. Suppose a declared family of joint Minkowski word distributions satisfies the scalar restriction eq. 7.ccxliii, mixed positivity eq. 7.ccxliv, the relation test eq. 7.ccxlv, and the specified test continuity (for point-field words, eq. 7.ccxlvii–eq. 7.ccxlviii). Suppose its symmetry action obeys the invariance and continuity used in eq. 7.ccliv–eq. 7.cclv, its full polynomial spectral measures pass eq. 7.cclvi, and its spacelike moment identities pass eq. 7.cclvii. Then it has a normalized positive cyclic representation, closable fields on a common invariant polynomial domain, the stated strongly continuous covariance and forward spectrum, and operator locality on that domain. It contains the constructed scalar representation as exactly the closed subspace in eq. 7.ccli. If the additional whole-space smoothness and energy bounds in eq. 7.cclix are verified, it also has the common energy-smooth domain and canonical closure statements proved above. All assertions follow from the explicit arguments in eq. 7.ccxlix–eq. 7.cclx; no boundedness or domain property of an added type is imported from its scalar restriction.

If the data are first supplied for finite families, eq. 7.ccliii gives the unique global functional once their actual overlap consistency is proved. A matched finite-family Euclidean reconstruction with uniqueness can prove that consistency as explained there. Its theorem must match the new types, the joint positive-time test domain, the growth seminorms and the unrestricted boundary values.

For the scalar family those checks are the established EB/QRM arguments. For a larger family they are explicit new checks, not a generic consequence of citing an OS theorem.

For a gauge-theory identification this conditional theorem must be attached to actual readouts of the same lattice theory. In addition to the compatible joint limits it must identify their renormalizations, exact limiting relations, localization, and relation to the scalar $\mathrm{tr}(F^2)$. One must state why the chosen observable family is sufficient for the intended continuum gauge-invariant theory. The finite-graph theorem supplies that sufficiency at a fixed cutoff. Passing it to the continuum needs a proved uniform criterion such as eq. 7.cclxxiv, or another stated reconstruction of the target observable algebra. Bare cutoff completeness does not silently perform that passage.

Nor do scalar cluster or spectral assertions automatically become assertions on the enlarged Hilbert space. The scalar subspace is reducing for the joint spacetime symmetries, so its spectral measure is a restriction of the full one. Restriction does not exclude additional spectrum or translation-invariant vectors in its orthogonal complement. For example let any constructed scalar theory be tensored with $\mathbb{C}^2$, choose the unit vector $\eta = (1,1)/\sqrt{2}$, and put

$$\widehat{\Omega} = \Omega_d \otimes \eta, \qquad \widehat{U}(g) = U_d(g) \otimes I, \qquad \widehat{\varphi}(f) = \varphi_d(f) \otimes I,$$
$$\chi(f) = \Big(\int f(x)\,dx\Big) I \otimes \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{7.cclxxv}$$

The added field is bounded for each test and central. Its moments are tempered, it is a scalar under proper orthochronous Poincare transformations, and it is local. All mixed moments are those of this tensor representation, so they are positive. The scalar restriction is unchanged. The vectors η and $\mathrm{diag}(1,-1)\eta$ span $\mathbb{C}^2$, so the full polynomial vacuum space is the full tensor space. Both copies of any scalar zero-energy vector have zero energy. In particular a unique scalar vacuum need not become a unique joint vacuum. This is an explicit extension illustrating the logical distinction; it is not claimed to be an actual readout from the lattice gauge measure.

A full joint cluster assertion would resolve this particular vacuum issue. If for every pair of polynomial vectors the spatial translation coefficients cluster to their vacuum product, density and unitarity extend that convergence to the weak operator limit

$$U(0,\mathbf{a}) \;\rightharpoonup\; |\Omega\rangle\langle\Omega| \qquad (|\mathbf{a}| \to \infty). \tag{7.cclxxvi}$$

The extension uses fixed polynomial approximants for both vectors and the uniform unitary norm, so no uniformity over all polynomials is being assumed. By the forward cone, $H = 0$ implies $\mathbf{P} = 0$. A zero-energy vector is therefore invariant under every spatial translation. Applying eq. 7.cclxxvi to it proves $\ker H = \mathbb{C}\Omega$. This is a proof from a *full joint* cluster assertion; scalar clustering only supplies the statement on the scalar cyclic subspace. The phase mixture discussed in SPH retains its own explicit invariant subspace instead of assuming such a joint purity statement.

Bounded Euclidean multiplication is not yet an OS field. The ordinary bounded loop result in eq. 7.cclxv and the measure inequalities in eq. 7.cclxvii do not give an OS operator norm. This failure can be demonstrated at the level of a finite reflection-positive system. Let s_- and s_+ be independent uniform signs, with reflection exchanging them and conjugating coefficients. For positive-time functions F, G of s_+ the OS form is

$$\langle [F], [G] \rangle_{\mathrm{OS}} = \overline{\mathbb{E} F(s_+)}\, \mathbb{E} G(s_+). \tag{7.cclxxvii}$$

It is positive semidefinite, so this is genuinely a reflection-positive probability system. But $F = s_+$ is null. The positive-time function $Y = s_+$ has ordinary supremum norm one, whereas $YF = 1$ has OS norm one. Thus multiplication by Y does not even preserve the OS null space. There can be no bound $\|[YF]\|_{\mathrm{OS}} \le C\|[F]\|_{\mathrm{OS}}$ for this insertion. This example asserts only the insufficiency of ordinary boundedness and reflection positivity; it is not assigned the continuous-time smoothness hypotheses of the scalar reconstruction.

An added sharp-time point type, with spatial test $h \in \mathcal{S}(\mathbb{R}^3)$, instead needs its precise time-shifted two-exterior form $B_{s,\alpha,h}(F,G)$, defined from its actual Euclidean readout, with a bound

$$|B_{s,\alpha,h}(F,G)| \le C_{\alpha,s} p_M(h) \|[F]\|_{\mathrm{OS}} \|[G]\|_{\mathrm{OS}}. \tag{7.cclxxviii}$$

Here p_M is the weighted Schwartz seminorm on $\mathbb{R}^3$. The form is antilinear in F and linear in G and h. A different label space needs its own specified continuous seminorm. This bound makes it vanish if either exterior vector is null, so it descends to the actual quotient. Density and Riesz then give a unique bounded operator $\Phi_{s,\alpha}(h)$ on that specified exterior Hilbert space, with norm at most $C_{\alpha,s} p_M(h)$. The reflection adjoint, spatial covariance and heat consistency must be verified from the same placements. When that space has an established self-adjoint contraction heat semigroup T_r, an exact form identity $B_{s+r}(F,G) = B_s(T_r F, T_r G)$ gives $\Phi_{s+r} = T_r \Phi_s T_r$ by density and boundedness.

If the exterior space is an ordered sector inside a larger OS space, matrix elements alone specify a compression. The actual vector-sewing identity is needed to show that the insertion maps ordered words to the ordered sector; the star partner's sewing and adjoint identities then give reduction. A sharp-time definition also needs its actual narrow-time limit or density estimate. If a proved bound in eq. 7.cclxxviii has a polynomial small-s loss, the heat-consistent energy-band construction of EB supplies the subsequent field method, with its explicitly checked test continuity and domain orders. Point tubes and Lorentz continuation still need their actual density and covariance checks. For a new type none of these model inputs follows from $|w_\gamma| \le 1$ or from the scalar insertion estimate. This separates the bounded cutoff observable, the OS Riesz operator, and the eventual Minkowski field with their exact intervening requirements.

The resulting scope of the tr(F^2) *construction.* The established scalar theory has its unrestricted tempered distributions, positive cyclic representation, forward spectrum, Lorentz covariance, local field relations and the exact phase/Hilbert decomposition within the stated scalar scope. The finite-cutoff gauge analysis also gives an actual complete Wilson observable family and the precise information lost by taking only plaquette actions. These are substantial compatible conclusions.

A full Yang–Mills identification additionally needs the sufficient joint gauge-observable family and its actual continuum readouts, including mixed limits, relations, positivity, covariance, locality and the stated relation to the scalar density. The conditional extension theorem organizes those requirements and proves what follows from them; it does not verify their model estimates. Nontriviality of the scalar field guarantees a nontrivial scalar subspace in any extension satisfying eq. 7.ccxliii, but does not specify all its added degrees of freedom or their dynamics.

7.14. **Conditional scalar reconstruction.**

7.14.1. *Reconstruction: hypotheses, consequences, and scope.* All growth constants below are positive, factorial exponents are nonnegative, and derivative-order coefficients are positive integers. D2 and H1 denote the conditional graded-moment and diagonal-flat growth conclusions. G2 denotes scalar rotation covariance derived under the ordinary-source, mirror, cone and ultraviolet hypotheses stated below.

The passage from Euclidean correlation functions to a quantum field requires several logically separate inputs. Compactness of separated correlations gives an off-diagonal distribution. Reflection positivity gives a Hilbert space. Uniform growth in the number of arguments is needed for the corrected reconstruction theorem. Full Euclidean covariance and clustering have to be established in their own right. Keeping these steps separate prevents a qualitative convergence statement from carrying quantitative information it does not contain.

The Euclidean test domain. Write

$$D_n = \{(x_1, \dots, x_n) \in (\mathbb{R}^4)^n : x_i \ne x_j \text{ whenever } i \ne j\}.$$

Let $\mathcal{S}_{\neq,n}$ be the closed subspace of the Schwartz space $\mathcal{S}(\mathbb{R}^{4n})$ consisting of functions whose every derivative vanishes on the complement of D_n. These functions are flat on all coincidence diagonals. Let $\mathcal{S}_{+,n}$ consist of those elements that are also flat outside the region in which all time coordinates are positive. Set $\mathcal{S}_{\neq,0} = \mathcal{S}_{+,0} = \mathbb{C}$. The time reflection is $\theta(x_1, x_2, x_3, x_4) = (x_1, x_2, x_3, -x_4)$, and

$$(\Theta\overline{\varphi})(x_1, \dots, x_n) = \overline{\varphi(\theta x_1, \dots, \theta x_n)}.$$

Symmetry of the functions makes reversal of the order of the arguments immaterial in this scalar setting.

These are valid test domains for the historical Osterwalder–Schrader reconstruction problem. They do not assert that the Euclidean functions have a preferred extension to every Schwartz test. A distribution supported on a coincidence diagonal vanishes on $\mathcal{S}_{\neq,n}$. Such a distribution therefore cannot be recovered from separated correlations. Any later use of contact terms, composite operator identities at coincident points, or local Ward identities has to specify the additional extension that it uses. This observation does not obstruct reconstruction on the stated domain.

Lemma 7.10 (Density of separated tensors)**.** Finite sums of tensors $f_1 \otimes \cdots \otimes f_n$, with smooth compactly supported f_i having pairwise disjoint supports, are dense in $\mathcal{S}_{\neq,n}$. The corresponding assertion holds in $\mathcal{S}_{+,n}$ when all supports lie at positive times. ■

The proof, including the simultaneous infinity, diagonal and positive-time cutoffs and the product Fourier approximation, is given in § 5.25.

The lemma is a statement about the flat subspace. It gives no density claim in the full Schwartz space, where evaluation on the diagonal is a continuous functional that vanishes on every such tensor.

What convergence preserves. Let σ_j be centred Euclidean moment functionals at finite cutoffs. Suppose that for every separated smooth tuple X the numbers $\sigma_j(X)$ converge to $\sigma(X)$. Suppose also that each σ_j is reflection positive for finite families of internally separated positive-time tuples. Different tuples in a family may overlap. This last qualification is essential: reflection positivity is a matrix condition, not a condition on one tuple at a time.

Lemma 7.11 (Closed reflection positivity)**.** For positive-time tuples $X_1, \dots, X_r$ and complex coefficients $c_1, \dots, c_r$,

$$\sum_{i,k=1}^{r} \overline{c}_i c_k \, \sigma(\theta\overline{X}_i \cdot X_k) \geq 0. \qquad ■$$

Proof. Each X_i is internally separated. Its reflected entries have negative times, so $\theta\overline{X}_i \cdot X_k$ is a separated tuple for every pair (i, k). The displayed expression is the limit of the corresponding finite sum for σ_j, all of which are nonnegative. □

Suppose, in addition, that the limiting functionals extend continuously to $\mathcal{S}_{\neq,n}$. Reflection positivity then holds on finite Schwartz sequences $\Phi = (\varphi_0, \dots, \varphi_N)$ with $\varphi_n \in \mathcal{S}_{+,n}$:

$$\sum_{n,m=0}^{N} S_{n+m}(\Theta\overline{\varphi}_n \otimes \varphi_m) \geq 0.$$

Indeed, the mixed tensor belongs to $\mathcal{S}_{\neq,n+m}$. Coincidences within one factor are killed by its flatness; a coincidence across the two factors has time coordinate zero or lies outside the allowed half-space of one factor, so is killed there. Approximate each φ_n by the tensors in Lem. 7.10, using a seminorm strong enough for all the finitely many mixed terms, and pass to the limit. Permutation symmetry and invariance under any fixed Euclidean isometry pass in exactly the same way once they hold on separated tensors.

There is one further step when translations are initially known only for vectors in a dense subgroup, for example the union of nested lattice translation groups. Given an arbitrary vector v, choose subgroup vectors $v_k \to v$. Translation is continuous on the smooth tests, and the local bounds permit passage from $S_n(\tau_{v_k}\varphi) = S_n(\varphi)$ to $S_n(\tau_v\varphi) = S_n(\varphi)$. This argument does not generate rotations from the finite hypercubic group.
The quantitative reconstruction hypothesis. Fix the seminorms

$$|\varphi|_k = \max_{|\alpha|\le k} \sup_x (1+|x|^2)^{k/2} |\partial^\alpha \varphi(x)|.$$

Theorem C (Scalar reconstruction)**.** For reconstruction, assume a sequence of symmetric functionals $S_n \in \mathcal{S}'_{\neq,n}$, with $S_0 = 1$, and constants $A, B, k_0, p < \infty$, independent of n, such that

$$|S_n(\varphi)| \le A B^n (n!)^p |\varphi|_{k_0 n}. \tag{7.cclxxix}$$

Assume full Euclidean invariance, reflection positivity on the domains just described, and the cluster property for separated positive-time groups of arguments. Under these hypotheses the corrected Osterwalder–Schrader reconstruction theorem gives a Hermitian scalar Wightman field with positive energy, locality, Poincare covariance, and a unique vacuum in its cyclic representation. ■□

Reality in the reconstruction import follows here from reflection positivity. The sesquilinear form on the finite positive-time sequences is Hermitian by polarization. Its degree-zero and degree-n entries give $S_n(\Theta\bar{\varphi}) = \overline{S_n(\varphi)}$ on positive-time tests. A compact separated tensor can be translated into the positive half-space. Translation invariance of S_n on the tensor and on its reflected conjugate then gives the identity before the translation. Separated tensor density and continuity give reality on the entire diagonal-flat space.

We apply the corrected reconstruction theorem of Osterwalder and Schrader [OS75, Sect. IV.1, Thm. $E' \to R'$, p. 287], with the test space and growth hypothesis just stated.

The exact all-order seminorm comparison is proved in Prop. 7.5. § 7.2 gives the direct field, tube-boundary and covariance construction. § 7.3 identifies its moments with those supplied by the corrected reconstruction theorem, which gives locality. The representation is the ordered sector; its common domains and scalar phase decomposition are developed in the reconstruction sections. The cited theorem applies by the hypothesis checks given there.

The condition in eq. 7.cclxxix must not be replaced by the weaker assertion that every S_n is separately continuous with unspecified constants. The growth of the distribution order and the constants in n matters to the existence of tempered boundary values in the analytic continuation. Likewise, a local off-diagonal kernel theorem alone does not establish eq. 7.cclxxix.

7.14.1.1. The explicit rooted-source and dyadic extension route. The rooted-source and dyadic extension estimates give a quantitative proof of eq. 7.cclxxix, with controlled dependence on n. Prop. 5.76 derives moments directly from the ordinary source hypothesis D1 and the proved G1 chessboard domination, using the actual domain geometry in eq. 5.cdxlvi–eq. 5.cdxlviii. At scale $a_s = L^{-s}$, domains are unions of Ma_s cubes, indexed by those unions, and their source dependence occupies only a fixed number of neighboring cube layers. Their size is the minimum cubical edge-tree length meeting all cubes, with source-energy decay above its rooted entropy rate. Here $M = L^{m_0}$, and the cofinal tori $\ell = 2L^m$, $m \ge m_0$, tile every actual scale with even cube counts. No scale is discarded for a failure to tile. The weak abstract eq. 5.cdxxxvi count alone does not furnish the sharper rooted and global counts needed for this proof.

For real compact C^1 tests, the theorem gives one fixed finite integer k, $B(R) = C(R+1)^4$, and the two-state centered bound

$$\left|\rho\prod_{i=1}^{n}[\mathcal{O}_a(f_i)-\nu(\mathcal{O}_a(f_i))]\right|\le B(R)^n(n!)^k\max\{1,d^{-5n}\}\prod_i\|f_i\|_\sharp. \tag{7.cclxxx}$$

Each support lies in its own radius-R ball, placed anywhere; the separation is d. The same factorial exponent bounds higher cumulants. Uniformity lets eq. 7.cclxxx pass to the separated continuum limits and their component limits. This is requirement D2 conditional on the actual D1 source representation and G1, not a new independent analytic estimate.

Prop. 5.78 uses only the unit-radius version, with the fixed equivalence of the sharp and C^1 norms absorbed in $B(1)$. Its product Fourier construction uses $8n$ derivatives; Gevrey-two pair/grid cutoffs cost at most $C^n(n!)^{32}$. Dyadic Taylor flatness to order $10n+2$ pays the local h^{-6n} cost and the h^{-4n} configuration grid count, leaving a summable dyadic square. It yields eq. 7.cclxxix with

$$k_0=20,$$
$$p=k+32.$$

The proof includes independence of tensor presentations, overlap gluing, cutoffs at infinity and every pairwise diagonal, and the positive-time boundary. Its constrained tensor density supplies the full-sequence reflection-positive form. Thus H1 is a proved implication of D2 here; the ordinary analytic RG source estimates in D1 are still assumptions. Neither separate continuity with arbitrary order constants nor weak counting in eq. 5.cdxxxvi alone may be substituted for this GS/QF route. Hilbert space and clustering. § 6.16 proves the quotient construction on compact positive-time tuples from their invariant reflection-positive moments and position-uniform per-ball bounds. Fixed separation of the reflected supports bounds the translated orbits before iterated Cauchy–Schwarz proves contraction. The resulting time semigroup is strongly continuous, positive and self-adjoint, and spatial translations are strongly continuous unitaries. The stated lattice approximation also gives the cube spectral cone. The GS/QF growth and density argument then identifies its completion with the full positive-time Schwartz sequence quotient. These deductions retain their explicit hypotheses; positivity alone is not being substituted for the continuity and growth checks.

For completeness, clustering on a dense class of vectors extends to the whole Hilbert space. Let $U(v)$ denote the spatial unitary translations, and suppose that for dense vectors ξ,η,

$$\langle\xi,U(v)\eta\rangle\longrightarrow\langle\xi,\Omega\rangle\langle\Omega,\eta\rangle$$
$$(|v|\to\infty).$$

Choose dense approximants $\xi_k\to\xi$ and $\eta_k\to\eta$. Unitarity bounds the difference of the matrix elements, uniformly in v, by

$$\|\xi-\xi_k\|\,\|\eta\|+\|\xi_k\|\,\|\eta-\eta_k\|.$$

The vacuum matrix elements obey the same estimate. First let $|v|\to\infty$ at fixed k, then let $k\to\infty$. The argument proves the qualitative cluster limit. It does not preserve a numerical decay rate on arbitrary vectors without further quantitative control of the approximations.

Phases and full rotational covariance. The phase-interface construction realizes the components with the G1 mirror properties and the D1/GS/QF moment bounds. Once the angular argument gives covariance of the mixture, the action on its canonical decomposition and coordinate reflection invariance give covariance of every component. The state-uniform two-point extraction is used to obtain the two-point law for the realizing states of each component. The convergent cutoff one-point differences give the real phase mean $c_\xi\int f$, with $\sup_\xi|c_\xi|$ bounded by the one-point transfer. To restore it on the Schwartz domains, let $I_A\varphi$ be the integral of φ over the variables with indices

in A. Differentiation under the integral preserves flatness on every remaining coincidence diagonal. If the original test is supported at positive ordered times, then so is the marginal on the retained arguments, with flatness at its boundary. Indeed any violation of a retained order or positivity constraint already violates the original one.

Take $k_0 > 4$, increasing the seminorm order if needed. Writing $x = (y, z)$ with $z = (z_i)_{i\in A}$, use

$$(1 + |x|^2)^{k_0 n/2} \geq (1 + |y|^2)^{k_0(n-|A|)/2} \prod_{i\in A} (1 + |z_i|^2)^{k_0/2}.$$

The derivatives of order at most $k_0(n - |A|)$ in y therefore give $|I_A\varphi|_{k_0(n-|A|)} \leq C_{\mathrm{int}}^{|A|} |\varphi|_{k_0 n}$, where $C_{\mathrm{int}} = \int_{\mathbb{R}^4} (1+|z|^2)^{-k_0/2}\, dz$. For A the full set of indices this also bounds the vacuum integral. The restored moment is

$$S_{\xi,n}(\varphi) = \sum_{A\subset\{1,\dots,n\}} c_\xi^{|A|} S^{\mathrm{c}}_{\xi,n-|A|}(I_A\varphi).$$

It agrees with the finite subset expansion on separated tensors, hence with the continuous extension by density. For a centered bound with exponential base B and nonnegative factorial exponent p, the binomial sum gives the same exponent and seminorm order with base $B + C_{\mathrm{int}} \sup_\xi |c_\xi|$, after including the vacuum in the prefactor. On the ordered sequences the same marginal maps define inverse shifts with c_ξ and $-c_\xi$. The reflected pairing identity on separated tensors extends by continuity and gives the same intertwining unitary as on compact tuples. On the cyclic field domain the centered insertion subtracts $c_\xi \int f$ times the identity.

Similarly, invariance under the hypercubic group, coordinate reflection positivity, and translation invariance do not by themselves imply invariance under O(4). The finite group is not dense in the rotation group. Any route through diagonal mirrors, a sharp energy–momentum cone, bounded insertions, or the exclusion of nonconstant angular harmonics needs an additional theorem. The following chain now supplies that theorem conditionally, before the external reconstruction theorem is invoked.

Prop. 6.26 completes the scalar covariance implication. Its preceding sections prove, in order, the uniform Hilbert insertion bound, the analytic L^1 density, the product kernels and continued Gram identities, and the ultraviolet elimination of the two-point harmonics. Polygon divisibility and common coordinate cuts then remove all higher harmonics and exceptional distributional support. The proof works with finite support boxes; density of generic point configurations alone would not exclude a singular part.

The angular continuation uses eq. 6.cccxxxv. The dependency catalogue obtains it through the positive-shape bound eq. 6.cccxxvii, deduced from F2–F3 on the stated all-state test class. The second route in § 6.12.6 uses the bound for a fixed family of reflected bumps from the ordinary source estimates at the running stops. Both deductions apply to periodic-state limits with the coordinate and diagonal mirrors and the exact cone. The finite-window density comparison includes growing tori above its side threshold. General growing-torus limits require their own mirror and cone argument.

The application uses the interacting D1 estimates and the F2–F3 extraction and normalization lemmas in their full forms. These are the model statements invoked to obtain the nontrivial scalar theory. The OS analytic-continuation theorem is still imported, and the identification with full Yang–Mills observables remains separate.

What the reconstructed field establishes. When the limiting observable is the centred scalar tr(F^2), the theorem above constructs its scalar observable sector. A nonzero two-point function on separated tests excludes a field that is a multiple of the identity. A nonzero higher cumulant excludes Gaussianity of that scalar field. Neither conclusion alone identifies a full Yang–Mills observable algebra. In particular, a Wick square of a free Gaussian field already has nonzero higher cumulants. Establishing a Yang–Mills theory in a stronger sense requires the stated additional observables, identities, and compatibility relations that define that sense.

Nor does polynomial clustering establish a mass gap. The positive measure $Ee^{-E}\,dE$ on $[0,\infty)$ has Laplace transform

$$\int_0^\infty Ee^{-E}e^{-tE}\,dE = (1+t)^{-2}.$$

Its support reaches zero, although its transform decays quadratically. A positive spectral gap requires an estimate that excludes spectral support in an entire interval above the vacuum. That question remains separate from the existence and reconstruction of the scalar sector described here.

Appendix A. Auxiliary lattice and comparison estimates

These calculations supply the finite-dimensional normalization, kernel decay, polymer estimates and reflection-positive limits used in the main argument. Each result retains its own parameter range and boundary conditions. The comparison calculations at the end specify the domains needed when the coupling or cutoff changes.

A.1. **Block normalization and a uniform scalar gap.**

A.1.1. *Block means, adjoints, and a uniform scalar Poincaré inequality.* An adjoint depends on the inner products of both its domain and its range. This matters when a block mean maps a fine lattice to a coarse lattice. The following finite-dimensional calculation fixes the convention and gives the scalar estimate used in a gauge-fixed Gaussian comparison. It applies to real or complex scalar functions; a finite number of color components are handled by summing the same estimate component by component.

Fix an integer $m \geq 1$ and put $\eta = m^{-1}$. A block consists of the sites

$$B = \{0, \eta, \dots, (m-1)\eta\}^4.$$

Its associated physical cell has side $m\eta = 1$ and volume one. Translation of the block, including translation by half a fine spacing to use cell centers, does not change the calculation. Let $\mathcal{B}$ be a finite disjoint family of such blocks, for example the blocks of a periodic torus whose side is an integer number of coarse cells. Write Λ_f for their union and Λ_c for the set indexing the blocks. A coarse function has one value per block. We count every fine bond once, with positive orientation. In the block Neumann form only bonds whose two endpoints belong to the same block are counted.

The physical inner products are

$$\langle f, g\rangle_f = \eta^4 \sum_{x\in\Lambda_f} \overline{f(x)}g(x),$$

$$\langle u, v\rangle_c = \sum_{B\in\mathcal{B}} \overline{u(B)}v(B),$$

$$\langle a, b\rangle_e = \eta^4 \sum_{e} \overline{a(e)}b(e).$$

The last sum can mean all fine bonds of the periodic lattice or just the internal bonds of the blocks; the operator being considered determines which set is used. The same weight is used on fine sites and fine bonds. The physical difference is $\mathrm{d}_\eta f(x, x+\eta e_\mu) = \eta^{-1}[f(x+\eta e_\mu) - f(x)]$. The unscaled combinatorial difference is $\mathrm{d}_0 = \eta \mathrm{d}_\eta$. These two operators must not be interchanged in an estimate with a mesh-independent constant.

Proposition A.1 (The mean, its adjoint, and its projection)**.** Define the block mean $Q : H_f \to H_c$ by

$$(Qf)(B) = \eta^4 \sum_{x\in B} f(x) = m^{-4} \sum_{x\in B} f(x).$$

For the physical inner products just specified, its adjoint is $(Q^*u)(x) = u(B)$ for $x \in B$. Consequently $QQ^* = 1$, and $P = Q^*Q$ is the orthogonal projection onto the functions constant in each block. In particular,

$$\|f\|_f^2 = \|(1-P)f\|_f^2 + \|Qf\|_c^2. \qquad \blacksquare$$

Proof. Directly from the definitions,

$$\langle Qf, u\rangle_c = \sum_B \eta^4 \sum_{x\in B} \overline{f(x)} u(B) = \langle f, Q^*u\rangle_f.$$

Since a block contains m^4 sites and $\eta^4 m^4 = 1$, its mean applied to a constant is that constant. Thus $QQ^* = 1$. It follows that $P^* = P$ and $P^2 = Q^*(QQ^*)Q = P$. Its range is precisely the block-constant functions, and the displayed norm identity is the orthogonal decomposition by that projection. No limiting argument is involved. □

Proposition A.2 (A block Neumann estimate)**.** Let $\mathrm{d}_{\eta,N}$ denote the physical difference restricted to internal bonds of each block. If $m \geq 2$, then

$$\|(1-P)f\|_f^2 \leq \frac{\eta^2}{4\sin^2(\pi\eta/2)} \|\mathrm{d}_{\eta,N} f\|_e^2 \leq \tfrac{1}{4}\|\mathrm{d}_{\eta,N} f\|_e^2.$$

For $m = 1$ the left side is zero. If d_η includes bonds between blocks, the same bound holds with d_η on the right. With the combinatorial difference the exact first bound is instead

$$\|(1-P)f\|_f^2 \leq \frac{1}{4\sin^2(\pi/(2m))} \|\mathrm{d}_{0,N} f\|_e^2. \qquad \blacksquare$$

Proof. First consider a one-dimensional path with m vertices indexed by $j = 0, \ldots, m-1$. Its Neumann graph Laplacian is the matrix associated with the quadratic form $\sum_{j=0}^{m-2} |f_{j+1} - f_j|^2$. For an interior vertex it acts as $2f_j - f_{j-1} - f_{j+1}$; at the two endpoints it acts as $f_0 - f_1$ and $f_{m-1} - f_{m-2}$. For $k = 0, \ldots, m-1$, set

$$v_k(j) = \cos\bigl(\pi k(j + \tfrac{1}{2})/m\bigr).$$

The cosine addition formula gives, in the interior,

$$2v_k(j) - v_k(j-1) - v_k(j+1) = 4\sin^2(\pi k/(2m)) v_k(j).$$

At the left endpoint,

$$\cos(\theta/2) - \cos(3\theta/2) = 4\sin^2(\theta/2)\cos(\theta/2),$$
$$\theta = \pi k/m.$$

The right endpoint follows by reflection. Thus these are eigenvectors with eigenvalues $4\sin^2(\pi k/(2m))$. The eigenvalues are distinct because sine is strictly increasing on $[0, \pi/2)$; the nonzero eigenvectors therefore form an orthogonal basis. The $k = 0$ vector is constant.

On the four-dimensional block the graph Laplacian is the sum of the four one-dimensional matrices acting on separate tensor factors. Tensor products of the v_k therefore form an eigenbasis. The eigenvalues of the physical Neumann Laplacian $\mathrm{d}_{\eta,N}^{*}\mathrm{d}_{\eta,N}$ are

$$4\eta^{-2}\sum_{\mu=1}^{4}\sin^2(\pi k_\mu/(2m)),$$
$$0\le k_\mu<m.$$

Its only zero mode is the constant vector, and the smallest positive eigenvalue is $\lambda_N = 4\eta^{-2}\sin^2(\pi/(2m))$. Multiplication of the site and bond inner products by the common constant η^4 does not change this matrix or these eigenvalues. Expanding $f-Pf$ in the eigenbasis gives $\lambda_N\|f-Pf\|_f^2\le\|\mathrm{d}_{\eta,N}f\|_e^2$ on one block and hence on their disjoint union.

Finally, concavity of sine on $[0,\pi/2]$ gives $\sin t\ge 2t/\pi$ there. At $t=\pi/(2m)$ this yields $\sin(\pi/(2m))\ge 1/m=\eta$, so $\lambda_N\ge 4$. Adding the nonnegative contributions of bonds between blocks can only increase the Dirichlet form. Substitution of $\mathrm{d}_{0,N}=\eta\mathrm{d}_{\eta,N}$ gives the combinatorial version. □

These estimates immediately give a uniform scalar gap on a periodic fine lattice. If $\mathcal{L}_\eta=\mathrm{d}_\eta^{*}\mathrm{d}_\eta$ and $\mathcal{A}=\mathcal{L}_\eta+Q^{*}Q$, then

$$\langle f,\mathcal{A}f\rangle_f=\|\mathrm{d}_\eta f\|_e^2+\|Qf\|_c^2\ge\|(1-P)f\|_f^2+\|Qf\|_c^2=\|f\|_f^2.$$

Thus $\mathcal{A}$ is invertible and $\|\mathcal{A}^{-1}\|\le 1$, independently of the mesh and number of blocks. This proves a scalar block-constrained gap. It supplies neither a vector-field coercivity theorem nor a mass gap for a physical Hamiltonian; those are different assertions.

A uniform inverse for a coarse Gram matrix. The same convention also makes a useful projection formula precise. On the periodic lattice put $G'=\mathcal{A}^{-1}$, with $\mathcal{A}$ as above. There is a numerical constant $c>0$, independent of m and the number of blocks, such that

$$QG'^2Q^{*}\ge c1.$$

Here is a finite-dimensional proof, including a choice of the test functions. For $m\ge 2$ define on a block indexed by $j=(j_1,\dots,j_4)$

$$\phi_m(j)=16\prod_{\mu=1}^{4}\sin^2(\pi(j_\mu+\tfrac12)/m).$$

The sum of the m roots $e^{2\pi i(j+1/2)/m}$ is zero. Taking real parts in $\sin^2 t=(1-\cos 2t)/2$ shows that the one-dimensional arithmetic mean of each sine-square factor is $1/2$. Thus the mean of ϕ_m on a block is one. For a coarse function u, define $Tu(x)=u(B)\phi_m(x)$ on the block B. Then $QT=1$ and $\|T\|\le 16$.

There is also a bound $\|\mathcal{L}_\eta Tu\|_f\le C\|u\|_c$ uniform in m. To see it, the second physical difference of $\sin^2(\pi t)$ at interior sites is bounded by the supremum $2\pi^2$ of its second derivative. At an end site, its value is $\sin^2(\pi\eta/2)\le\pi^2\eta^2/4$. A second difference that crosses a block boundary therefore contributes at most a constant times the absolute coarse values in the two neighboring blocks: the factor η^{-2} is canceled by this endpoint bound. The other three sine-square factors are bounded by one. After summing the four coordinate differences, a pointwise bound is $C\sum_{B'\sim B}|u(B')|$, where the sum includes B and its face neighbors. Squaring, summing with weight η^4, and using the bounded number of neighbors proves the norm estimate. For $m=1$, set $T=1$; the unit-spacing

Laplacian has norm at most 16, so the same conclusion holds. Consequently $\|\mathcal{A}Tu\|_f \le C\|u\|_c$ for all m.

Self-adjointness and $G'\mathcal{A} = 1$ now give

$$\|u\|_c^2 = \langle Q^*u, Tu\rangle_f = \langle G'Q^*u, \mathcal{A}Tu\rangle_f \le C\|G'Q^*u\|_f\|u\|_c.$$

This proves the claimed positive lower bound for QG'^2Q^*.

The map $V = G'Q^*$ is injective, its Gram matrix is $V^*V = QG'^2Q^*$, and elementary linear algebra shows that

$$R = 1 - G'Q^*(QG'^2Q^*)^{-1}QG'$$

is the orthogonal projection onto $\ker(QG')$. This kernel equals $\mathcal{L}_\eta \ker Q$. Indeed, if $Qv = 0$ then $\mathcal{A}v = \mathcal{L}_\eta v$, so $QG'\mathcal{L}_\eta v = 0$. Conversely, if $QG'h = 0$, put $v = G'h$. Then $Qv = 0$ and $h = \mathcal{A}v = \mathcal{L}_\eta v$. The expression for R is therefore well defined with a uniformly invertible coarse Gram matrix. Every adjoint in this argument uses the physical inner products fixed above.

Translation to plain Euclidean coordinates. There is an equivalent convention with unweighted coordinate inner products, but the averaging matrix has to change. Let $S_f f = \eta^2 f$ on fine sites, $S_e a = \eta^2 a$ on fine bonds, and $S_c u = u$ on coarse sites. These maps are unitaries from the physical spaces just defined to the spaces with plain coordinate sums. The correctly conjugated averaging matrix is

$$\widetilde{Q} = S_c Q S_f^{-1},$$
$$(\widetilde{Q}v)(B) = \eta^2 \sum_{x\in B} v(x),$$
$$(\widetilde{Q}^*u)(x) = \eta^2 u(B).$$

Each row has m^4 entries of size $\eta^2 = m^{-2}$, so its squared Euclidean length is one. Rows have disjoint supports, proving $\widetilde{Q}\widetilde{Q}^* = 1$. The conjugated projection is $\widetilde{P} = \widetilde{Q}^*\widetilde{Q} = S_f P S_f^{-1}$. Because sites and bonds have the same physical weight, $S_e \mathrm{d}_\eta S_f^{-1}$ is the same physical difference matrix $\eta^{-1}(\mathrm{shift} - 1)$. Therefore the entire Poincaré estimate survives unitary conjugation with its exact constant.

In contrast, suppose that one retains the arithmetic mean matrix $\overline{Q}v = \eta^4 \sum_{x\in B} v(x)$ while using plain inner products on both lattices. Its plain adjoint is $(\overline{Q}^*u)(x) = \eta^4 u(B)$, and

$$\overline{Q}\overline{Q}^* = \eta^4 1,$$
$$P = \eta^{-4}\overline{Q}^*\overline{Q}.$$

Thus $\overline{Q}^*\overline{Q}$ itself is not the projection. For a nonzero constant fine function and $m > 1$, the expression $\|f - \overline{Q}^*\overline{Q}f\|^2$ is positive whereas $\mathrm{d}_\eta f = 0$. This gives an immediate check against accidentally combining the arithmetic mean with the wrong adjoint. Either the physical weighted convention or the conjugated unweighted convention is consistent; mixing them is not.

Consequences for cochains and Gaussian formulas. When a Gaussian curvature is first defined using the plain cochain inner product, its covariance is an orthogonal projection for that inner product. After changing to physical coordinates, transform the random vector and the covariance together. For a linear random vector F with covariance C in plain coordinates and a linear coordinate change T, the covariance of TF is TCT^*. An orthogonal projection is carried to an orthogonal projection when the coordinate change is interpreted as the unitary between the two specified Hilbert spaces;

an arbitrary rescaling of its coordinate entries without the corresponding inner product change has a different covariance.

For a Hilbert-space operator $B : H_1 \to H_2$, unitary changes of coordinates S_1, S_2 give

$$\widetilde{B} = S_2 B S_1^{-1},$$
$$\widetilde{B}^* = S_1 B^* S_2^{-1}.$$

In particular products, orthogonal projections, positive inverses and Moore–Penrose inverses commute with this conjugation. A conditional Gaussian covariance written as $C - CB^*(BCB^*)^+BC$ remains the same Hilbert-space operator after every factor is conjugated. This observation permits a proof to begin in plain finite cochain coordinates and then use physical block coordinates. The change must be announced before the first adjoint, constraint, or Poincaré estimate that uses the latter convention. Every discrete derivative must also say whether it is d_0 or d_η. These declarations prevent a missing power of the mesh from being hidden inside an adjoint symbol.

A.1.2. *A scalar block resolvent with uniform off-diagonal decay.* This finite-dimensional lemma supplies a scalar part of the propagator input. All distances are physical distances. The argument uses a relative form bound and a Neumann series; the exponentially conjugated operator is generally not self-adjoint.

Let a four-dimensional periodic lattice of spacing $\eta = 1/m$, $m \geq 1$, be partitioned into blocks of m^4 sites and physical side one. Its torus has an integer number of blocks in each direction. On site functions use $\langle u, v\rangle = \eta^4 \sum_x \overline{u_x} v_x$. On positively oriented fine bonds use the same weight. Put

$$(\mathrm{d}_\eta u)(x, x + \eta e_\mu) = \eta^{-1}(u_{x+\eta e_\mu} - u_x),$$
$$\mathcal{L}_\eta = \mathrm{d}_\eta^* \mathrm{d}_\eta.$$

The block mean Q takes the arithmetic mean of the m^4 fine values. Its coarse inner product is the unweighted sum, so Q^* embeds a coarse value as a constant on its block. Thus $P = Q^*Q$ is an orthogonal projection and its matrix entries, as an operator acting on coordinate vectors, are $P_{xy} = \eta^4$ when x, y are in the same block and zero otherwise.

Write dist_η for shortest-path distance on the periodic fine graph, with length η per bond. The distance includes wrapping paths. The diameter of a single block in this metric is at most 4. The block Poincaré inequality proves

$$A_\eta := \mathcal{L}_\eta + P \geq 1,$$
$$\|u\| \leq \|A_\eta^{1/2} u\|,$$
$$\|\mathrm{d}_\eta u\| \leq \|A_\eta^{1/2} u\|.$$

These statements also hold when $m = 1$, since then $P = 1$.

Proposition A.3 (A mesh-independent Combes–Thomas bound)**.** Let w be a real function with $|w_x - w_y| \leq \eta$ on every neighboring pair. For real t let M_t multiply by e^{tw_x} and set $A_t = M_t A_\eta M_t^{-1} = A_\eta + E_t$. There is a numerical constant C, independent of the mesh, volume and function w, such that, for $|t| \leq 1$,

$$\|A_\eta^{-1/2} E_t A_\eta^{-1/2}\| \leq C(|t| + t^2).$$

Consequently one may choose a fixed $t_0 > 0$, independent of those parameters, so that $\|A_t^{-1}\| \leq 2$ for $|t| \leq t_0$. For arbitrary nonempty sets of sites X, Y this gives

$$\|1_X A_\eta^{-1} 1_Y\| \le 2\exp[-t_0 \operatorname{dist}_\eta(X,Y)]. \qquad \blacksquare$$

Proof. Set $A = A_\eta$. First consider the contribution $E_t^{\mathcal{L}}$ from the Laplacian. On a positively oriented bond with endpoints x, y, put $\delta = w_x - w_y$, $a = e^{t\delta} - 1$ and $b = e^{-t\delta} - 1$. The diagonal entries of the conjugated edge Laplacian do not change. Its off-diagonal contribution to $\langle u, E_t^{\mathcal{L}} v\rangle$ is therefore

$$-\eta^4 \eta^{-2}(a\overline{u_x}v_y + b\overline{u_y}v_x).$$

The exact identity

$$a\overline{u_x}v_y + b\overline{u_y}v_x = a(\overline{u_x} - \overline{u_y})v_y + b\overline{u_y}(v_x - v_y) + (a+b)\overline{u_y}v_y$$

exposes a difference of u or v in the terms linear in t. Since $|\delta| \le \eta \le 1$ and $|t| \le 1$,

$$|a|, |b| \le e|t|\eta,$$
$$|a+b| = 2|\cosh(t\delta) - 1| \le et^2\eta^2.$$

Each vertex occurs as the head of at most four positively oriented bonds. Cauchy–Schwarz applied to the first term, summed over the bonds, consequently gives $2e|t|\|\mathrm{d}_\eta u\|\|v\|$. The second term is bounded by $2e|t|\|u\|\|\mathrm{d}_\eta v\|$. The last term is bounded by $4et^2\|u\|\|v\|$. In particular,

$$|\langle u, E_t^{\mathcal{L}} v\rangle| \le 2e|t|(\|\mathrm{d}_\eta u\|\|v\| + \|u\|\|\mathrm{d}_\eta v\|) + 4et^2\|u\|\|v\|.$$

The two powers of η^{-1} in the Laplacian have been canceled by the Lipschitz difference and the difference of a test vector. Estimating the off-diagonal matrix entries directly would miss this cancellation.

For the block term, the neighbor condition implies $|w_x - w_y| \le 4$ for sites in one block. Its conjugation error has entries

$$(E_t^P)_{xy} = \eta^4 1_{\{x,y \text{ in one block}\}}(e^{t(w_x - w_y)} - 1).$$

Their absolute row sums and column sums are at most $4e^4|t|$, because a block has m^4 sites and $m^4\eta^4 = 1$. The Schur test therefore yields $\|E_t^P\| \le 4e^4|t|$. The common site weight η^4 does not alter this Schur estimate. Using the two energy bounds preceding the proposition gives

$$|\langle u, E_t v\rangle| \le C(|t| + t^2)\|A^{1/2}u\|\|A^{1/2}v\|,$$

where, for example, $C = 8e + 4e^4$ suffices. Substitute $u = A^{-1/2}f$ and $v = A^{-1/2}g$ and take the supremum over unit vectors f, g. This proves the asserted relative operator bound.

Choose $t_0 = \min\{1, (4C)^{-1}\}$. Then $B_t = A^{-1/2}E_t A^{-1/2}$ has norm at most $1/2$ when $|t| \le t_0$. The exact factorization and its inverse are

$$A_t = A^{1/2}(1 + B_t)A^{1/2},$$
$$A_t^{-1} = A^{-1/2}(1 + B_t)^{-1}A^{-1/2}.$$

The Neumann series gives $\|(1+B_t)^{-1}\| \le 2$ and hence $\|A_t^{-1}\| \le 2$. This step requires no assertion that A_t is self-adjoint or ordered below by a positive scalar.

Finally take $w_x = \operatorname{dist}_\eta(x, Y)$ and $t = t_0$. The triangle inequality makes w admissible, and $w = 0$ on Y. If $D = \operatorname{dist}_\eta(X, Y)$, then $w \ge D$ on X. Similarity of the two inverses gives

$$1_X A^{-1} 1_Y = 1_X M_t^{-1} A_t^{-1} M_t 1_Y.$$

The rightmost multiplication is the identity on Y, while the norm of the leftmost multiplication is at most $e^{-t_0 D}$. Combining these statements with $\|A_t^{-1}\| \le 2$ proves the final estimate. □

The estimate concerns operator norms between sets in the weighted lattice Hilbert space. It does not assert a mesh-independent bound for a Green function kernel normalized relative to the physical counting measure $\eta^4 \sum_x$. Converting a coordinate matrix to that kernel introduces its own power of η. The scalar lemma likewise does not establish the non-Abelian vector-field or multiscale background estimates. Those require their additional gauge constraints and response bounds.
Dirichlet restrictions and the distance being estimated. The same estimate holds for a Dirichlet compression of this particular operator, with constants independent of the shape of the region. To state it without ambiguity, let D be any set of sites of the periodic fine lattice. Let H_D consist of the functions supported in D, with the inherited weighted inner product, and let $J_D : H_D \to H$ be extension by zero. Define

$$A_D = J_D^* A_\eta J_D.$$

In particular the block term is the compression of the *original* projection P. On a block intersected by the boundary it is not a new mean taken over just the surviving sites. Replacing it by that new mean would define a different operator and is not justified by this corollary.

For $u \in H_D$, the quadratic form is

$$\langle u, A_D u\rangle = \|\mathrm{d}_\eta J_D u\|^2 + \|Q J_D u\|_c^2 \ge \|u\|^2.$$

It controls the physical gradient of the zero extension, including boundary bonds. Restrict an admissible ambient weight w to D and conjugate by its exponential on H_D. Because multiplication by this exponential commutes with the site indicator 1_D, the new error form is exactly the restriction of E_t to zero-extended vectors. The estimates in the proof therefore give

$$\|A_D^{-1/2}(M_t A_D M_t^{-1} - A_D) A_D^{-1/2}\| \le C(|t| + t^2).$$

The same Neumann argument implies, for $X, Y \subset D$,

$$\|1_X A_D^{-1} 1_Y\| \le 2e^{-t_0 \,\mathrm{dist}_\eta(X,Y)}.$$

Here the distance is computed in the ambient periodic lattice. It may use a path outside D. No estimate in the possibly much larger intrinsic distance of a region with holes has been asserted.

For comparison with continuum notation, the shortest-path distance is the periodic physical Manhattan distance, and hence is no smaller than the periodic Euclidean distance. The displayed bound therefore implies the same bound with Euclidean distance in its exponent. One can alternatively take w to be Euclidean distance to Y directly, since its change across a fine bond is at most η. Both formulations concern physical distances; using the number of bonds without the factor η would incorrectly strengthen the assertion as the mesh is refined.

The calculation explains the uniform mechanism: the scalar energy controls one physical difference, the weight varies by only $O(t\eta)$ across an edge, and its genuinely quadratic error is $O(t^2\eta^2)$. The block projection has range at most one physical cell and row mass one. Consequently the conjugation cost depends on the dimension and physical block size, with no factor proportional to the number of fine sites in a block or the torus.

A.2. **Convergence of the polymer logarithm.** The small-field integration must produce activities satisfying a summability criterion. Once that estimate is available, the passage from activities to a local logarithm is a combinatorial theorem. We prove that theorem here, so its convergence mechanism is independent of the particular gauge-field coordinates. The estimate on the activities remains a separate analytic task.

A.2.1. *Finite polymer systems.* A finite polymer system consists of a finite set $\mathcal{P}$ and a symmetric incompatibility relation $\not\sim$ on it. Every polymer is incompatible with itself. Give each $X \in \mathcal{P}$ a complex activity z_X. A compatible family contains no two incompatible polymers. Its partition function is

$$Z(z) = \sum_{\substack{\mathcal{F}\subset\mathcal{P}\\ \mathcal{F}\text{ compatible}}} \prod_{X\in\mathcal{F}} z_X.$$

The empty family contributes one. There are no questions of convergence in this polynomial. The problem is to construct its logarithm and bound it in terms of local activities without a loss exponential in $|\mathcal{P}|$.

For a list $\mathbf{X} = (X_1, \dots, X_n)$, let $H(\mathbf{X})$ be the simple graph on the labeled vertices $1, \dots, n$ whose edges are the pairs with $X_i \not\sim X_j$. Repeated polymers are permitted in lists; their distinct vertices are adjacent by self-incompatibility. Define

$$\phi(\mathbf{X}) = \sum_{\substack{F\subset E(H(\mathbf{X}))\\ ([n],F)\text{ connected}}} (-1)^{|F|},$$

$$n \geq 2,$$

$$\phi(X_1) = 1.$$

If $H(\mathbf{X})$ is disconnected the sum is zero.

Lemma A.4 (The connected graph identity)**.** As a formal power series at $z = 0$,

$$\log Z(z) = \sum_{n\geq 1} \frac{1}{n!} \sum_{X_1,\dots,X_n\in\mathcal{P}} \phi(\mathbf{X}) \prod_{i=1}^{n} z_{X_i}. \qquad \blacksquare$$

Proof. The ordered-list expression for Z is

$$Z(z) = \sum_{n\geq 0} \frac{1}{n!} \sum_{\mathbf{X}\in\mathcal{P}^n} \prod_i z_{X_i} \prod_{i<j} (1 - \mathbf{1}_{\{X_i \not\sim X_j\}}).$$

The product is one on pairwise compatible lists and zero otherwise. Such lists contain no repetition, and their $n!$ orderings give exactly the original polynomial. Expand the product as a sum over edge sets. Each graph decomposes uniquely into its connected components. The weight factors over the components, including the activity at each vertex. Expanding the exponential of the proposed connected-graph series enumerates a partition of the labeled vertices into these components. The factor for ordering the components is canceled by the exponential's factorial, and the factorials within the components cancel their vertex orderings. This reconstructs the ordered-list expression coefficient by coefficient. Its constant term is one, so its formal logarithm is the connected series. $\square$

A.2.2. *The tree bound, including the cancellation.*

Lemma A.5 (A hard-core tree inequality)**.** For a finite simple graph H on $n \geq 1$ labeled vertices,

$$\left| \sum_{\substack{F \subset E(H) \\ ([n],F)\ \text{connected}}} (-1)^{|F|} \right| \leq \#\{\text{spanning trees of } H\}.$$

For $n = 1$ the empty tree is counted once. ■

Proof. Fix a total order of the edges of H. Call an edge e eligible for an edge set F if the endpoints of e can be joined by a path in F consisting only of edges smaller than e. This definition does not require $e \in F$. If at least one edge is eligible, choose the smallest eligible edge e and toggle its membership in F.

This is an involution. Eligibility of e depends only on edges smaller than e, which were not changed; the same holds for every smaller edge, so the smallest eligible edge stays the same. Toggling e preserves the connected components, because its endpoints already have a path using smaller edges. It reverses the parity of $|F|$. Thus all connected edge sets with an eligible edge cancel in pairs in the alternating sum.

A surviving edge set has no cycle: if it contained one, the largest edge of that cycle would be eligible. Consequently every surviving connected edge set is a spanning tree. Each contributes the same sign $(-1)^{n-1}$, and there can be no more survivors than spanning trees. This proves the bound. The same argument also proves that the alternating sum has sign $(-1)^{n-1}$ when it is nonzero. □

In particular,

$$|\phi(X_1, \ldots, X_n)| \leq \sum_{T\ \text{tree on}\ [n]} \prod_{\{i,j\} \in E(T)} \mathbf{1}_{\{X_i \not\sim X_j\}}.$$

The cancellation in this inequality is essential. Taking absolute values before summing over connected graphs would count all connected subgraphs, which need not admit the rooted-tree bound used next.

A.2.3. *The rooted-tree estimate.* Let $a_X \geq 0$ be real numbers and suppose

$$\sum_{Y : Y \not\sim X} |z_Y| e^{a_Y} \leq a_X \quad \text{for every } X \in \mathcal{P}. \tag{A.i}$$

This is the polymer criterion used here [KP86]. For a fixed root polymer X, let $\mathcal{T}_X$ be the total nonnegative weight of all rooted labeled trees, with the root fixed to have polymer X, every nonroot vertex of polymer Y carrying weight $|z_Y|$, and every edge requiring incompatibility of its endpoint polymers. A tree on m vertices has the factor $1/(m-1)!$ for the nonroot vertex labels.

One can define this sum without assuming its convergence. Start with $T_X^{(0)} = 1$ and set

$$T_X^{(h+1)} = \exp\left(\sum_{Y : Y \not\sim X} |z_Y| T_Y^{(h)} \right).$$

The quantity $T_X^{(h)}$ sums trees of height at most h. To see this, delete the root. Each child becomes the root of a tree of height at most $h-1$, and an unordered collection of r child trees contributes the factor $1/r!$ in the exponential. Expanding those exponentials and assigning the nonroot labels gives exactly $1/(m-1)!$: the multinomial coefficient for assigning labels to the child

subtrees cancels their factorials and the ordering of the r children. This also verifies the formula for every fixed degree before any infinite sum is taken.

The iterates increase with h. Moreover $T_X^{(h)} \le e^{a_X}$ for every h: this is true at $h = 0$, and the induction step is precisely eq. A.i. Monotone convergence proves

$$\mathcal{T}_X = \lim_{h\to\infty} T_X^{(h)} \le e^{a_X}. \tag{A.ii}$$

Proposition A.6 (Absolute convergence and a nonzero partition function)**.** Under eq. A.i the connected series converges absolutely and defines a number $L(z)$ with $e^{L(z)} = Z(z)$. In particular $Z(z) \neq 0$. For each root polymer X,

$$\sum_{n\ge1} \frac{1}{(n-1)!} \sum_{X_2,\dots,X_n} |\phi(X, X_2, \dots, X_n)| \prod_{i=2}^{n} |z_{X_i}| \le e^{a_X}.$$

Also $|L(z)| \le \sum_X |z_X| e^{a_X}$. If the criterion holds throughout a complex activity domain with a uniform majorant, the series converges locally normally there and gives the branch of $\log Z$ that equals zero at zero. ■

Proof. The tree inequality bounds the root series by $\mathcal{T}_X$, proving its estimate by eq. A.ii. Rooting each nonempty connected list at its first vertex bounds the absolute unrooted series by $\sum_X |z_X| e^{a_X}$: replacing $1/n!$ by $1/(n-1)!$ only enlarges it.

For $|t| < 1$ the same majorant applies to the activities tz_X, and the connected series is analytic in t there. Near zero its exponential is $Z(tz)$ by the formal identity and absolute convergence. The analytic identity theorem gives $e^{L(tz)} = Z(tz)$ on the entire unit disc. The absolute majorant is summable also at $|t| = 1$, so the connected series is continuous as t tends radially to one. Taking that limit proves $Z(z) = e^{L(z)} \neq 0$.

On a domain with a uniform summable majorant, the Weierstrass criterion gives normal convergence on smaller domains and hence holomorphy. The branch statement follows by agreement near zero and continuation within the connected component of that domain containing zero. This statement does not assert that an arbitrary disconnected activity domain has a single preferred logarithm. □

For a countable polymer set the same statements hold in a finite region whenever $\sum_X |z_X| e^{a_X} < \infty$ there, by exhaustion and the displayed absolute majorant. In an infinite lattice the total unrooted sum is generally extensive and infinite. The rooted bound remains meaningful if the incompatibility sums satisfy the criterion, and it is the form used to define local connected expansions. No finite infinite-volume partition function is asserted.

A.2.4. *Locality, weighted bounds, and analytic dependence.* If a polymer activity depends only on field variables in an enlargement of that polymer, each connected term depends only on the union of those enlargements. This is a direct consequence of its product formula. Extra exponential weights can be incorporated in the activity majorant before the criterion is checked. The root estimate then retains those weights. Holomorphic dependence on external fields or sources passes to the connected expansion on every domain where the same absolute majorant is valid.

The theorem does not prove that gauge-field fluctuation activities satisfy eq. A.i. In a multiscale application their majorant must already include the fluctuation covariance, the number of localization domains, source radii, boundary costs, and any required derivatives. Likewise, a tree bound for a small-field expansion does not by itself sum large-field histories. Stating the activity estimate separately keeps the proved combinatorial step from assuming its analytic conclusion.

A.3. **Correlations of the free curvature.** This section computes the noncoincident correlations of $\mathrm{tr}(F^2)$ for a free curvature in four Euclidean dimensions. The calculation fixes a normalization and proves the cycle identities used in the short-distance comparison. It does not identify an interacting lattice measure with this model. That comparison, including its error, remains a separate estimate.

A.3.1. *A Gaussian field defined without a pointwise potential.* Fix a positive number g and an integer $d_G \geq 1$; for the adjoint color space of SU(2), $d_G = 3$. Coordinates in $\mathbb{R}^4$ are denoted by $x_1, \ldots, x_4$. Our Fourier transform is

$$\widehat{f}(k) = \int_{\mathbb{R}^4} e^{-ik\cdot x} f(x)\, dx,$$

$$f(x) = (2\pi)^{-4} \int e^{ik\cdot x} \widehat{f}(k)\, dk.$$

Let G be the tempered distribution with Fourier transform $|k|^{-2}$. In four dimensions this multiplier is locally integrable at zero. Its inverse transform, away from zero, is

$$G(x) = \frac{1}{4\pi^2 |x|^2}.$$

One direct verification uses $|k|^{-2} = \int_0^\infty e^{-t|k|^2}\, dt$ and the product Gaussian integral. The inverse transform of $e^{-t|k|^2}$ is $(4\pi t)^{-2} e^{-|x|^2/(4t)}$. Integrating in t, with $u = |x|^2/(4t)$, gives $1/(4\pi^2|x|^2)$. The same identity holds as tempered distributions by Fubini: the integral $\int |\varphi(x)|\, |x|^{-2} dx$ is finite for every Schwartz function φ, controlling the absolute integral of the heat-kernel formula.

For real Schwartz functions define

$$(f, h)_G = (2\pi)^{-4} \int_{\mathbb{R}^4} \frac{\overline{\widehat{f}(k)} \widehat{h}(k)}{|k|^2}\, dk.$$

This is a positive real bilinear form. The integral is finite near zero because $\int_0^1 r^3 r^{-2} dr < \infty$, and is finite at infinity by Schwartz decay. Quotient by its null space and complete to a real Hilbert space $\mathcal{H}_G$. It is separable: the map $f \mapsto |k|^{-1} \widehat{f}(k)$ embeds it into a separable real form of $L^2(\mathbb{R}^4)$.

Choose an orthonormal basis (e_j) and independent real standard normal variables (N_j) on their countable product probability space. For $f \in \mathcal{H}_G$, the series

$$W(f) = \sum_j (f, e_j)_G N_j$$

converges in L^2 and defines a centered jointly Gaussian family with $\mathbb{E}[W(f)W(h)] = (f, h)_G$. Finite partial sums prove the covariance formula and joint Gaussianity; convergence of their characteristic functions proves it for the limit. Independent copies give families $A_\mu^a(f) = g W_\mu^a(f)$ for $1 \leq \mu \leq 4$, $1 \leq a \leq d_G$. Their covariance is

$$\mathbb{E}[A_\mu^a(f) A_\nu^b(h)] = g^2 \delta_{ab} \delta_{\mu\nu} (f, h)_G.$$

No value $A_\mu^a(x)$ at a point is used in this definition.

Define the curvature as the Gaussian distribution

$$F_{\mu\nu}^a(f) = -A_\nu^a(\partial_\mu f) + A_\mu^a(\partial_\nu f),$$

$$\mu < \nu.$$

There are six independent coordinate labels $P = (\mu, \nu)$, although the six curvature components need not be independent random variables. Differentiating the covariance as a distribution gives

$$\mathbb{E}[F^a_{\mu\nu}(x) F^b_{\rho\sigma}(y)] = g^2 \delta_{ab} K_{\mu\nu,\rho\sigma}(x - y),$$

where the notation on the left denotes a distributional kernel and

$$K_{\mu\nu,\rho\sigma}(x) = -\delta_{\nu\sigma}\partial_\mu\partial_\rho G(x) + \delta_{\nu\rho}\partial_\mu\partial_\sigma G(x) + \delta_{\mu\sigma}\partial_\nu\partial_\rho G(x) - \delta_{\mu\rho}\partial_\nu\partial_\sigma G(x).$$

The signs follow from differentiating one argument at x and the other at y: $\partial_{y_\rho} G(x - y) = -\partial_{x_\rho} G(x - y)$. At nonzero separation this is an ordinary smooth matrix-valued function.

A.3.2. *The six-dimensional curvature matrix.* Give $\Lambda^2\mathbb{R}^4$ the Euclidean inner product for which $(e_\mu \wedge e_\nu)_{\mu<\nu}$ is an orthonormal basis. Put $r = |x|$, $u = x/r$, and $R_u = I - 2u \otimes u$. This R_u is the orthogonal reflection normal to u. Write $\Lambda^2 R_u$ for its induced operator on two-forms.

Proposition A.7 (The curvature coefficient)**.** For $x \neq 0$,

$$K(x) = \frac{1}{\pi^2 r^4} \Lambda^2 R_u,$$
$$K(x)^{\mathsf{T}} K(x) = \frac{1}{\pi^4 r^8} I_6,$$
$$\operatorname{tr}(K(x)^{\mathsf{T}} K(x)) = \frac{6}{\pi^4 r^8}.$$ ■

Proof. Twice differentiating G gives

$$\partial_\mu\partial_\rho G(x) = \frac{1}{2\pi^2 r^4}(4u_\mu u_\rho - \delta_{\mu\rho}).$$

At $u = e_4$ this Hessian is diagonal with entries $(-1, -1, -1, 3)/(2\pi^2 r^4)$. Inserting it in the component formula for K shows that K is diagonal: it has eigenvalue $1/(\pi^2 r^4)$ on $e_1 \wedge e_2, e_1 \wedge e_3, e_2 \wedge e_3$, and eigenvalue $-1/(\pi^2 r^4)$ on $e_1 \wedge e_4, e_2 \wedge e_4, e_3 \wedge e_4$. This is exactly the action of $\Lambda^2 R_{e_4}$ times the stated scalar. Both the differentiated radial kernel and the claimed expression transform by conjugation under orthogonal changes of coordinates, proving the first identity for every u. Since $\Lambda^2 R_u$ is an orthogonal involution on a six-dimensional space, its squared norm matrix is I_6, giving the other two identities. □

Fix the standard orientation and let $J = *$ be the Hodge operator on two-forms. It satisfies $J^2 = I$. For an orthogonal map R,

$$J(\Lambda^2 R) = (\det R)(\Lambda^2 R)J.$$

To check the formula, use the defining identity $\alpha \wedge *\beta = \langle \alpha, \beta \rangle \, dx_1 \wedge \cdots \wedge dx_4$ and the fact that the top exterior power of R multiplies the oriented volume by $\det R$. Since $\det R_u = -1$, $JK(x) = -K(x)J$. Every product of an odd number of matrices $K(x_i)$ therefore anticommutes with J and has trace zero: conjugation by J preserves its trace while changing its sign.

A.3.3. *Regularized densities and connected cycle contractions.* Take a real smooth even function ρ supported in the unit ball with $\int \rho = 1$ and put $\rho_\varepsilon(x) = \varepsilon^{-4}\rho(x/\varepsilon)$. Smear every curvature component with its translate to define $F^a_{\varepsilon,P}(x)$. This is a Gaussian variable with a smooth covariance kernel. Its centered quadratic density is

$$q_\varepsilon(x) = \frac{1}{2}\sum_{a=1}^{d_G}\sum_{P=1}^{6}\big(F^a_{\varepsilon,P}(x)^2 - \mathbb{E}[F^a_{\varepsilon,P}(x)^2]\big).$$

On any compact set its covariance is continuous and bounded. The integral $q_\varepsilon(f) = \int f(x)q_\varepsilon(x)dx$ for compactly supported smooth f is defined by L^2 limits of Riemann sums. The continuous covariance makes those sums Cauchy. The Gaussian moment formula gives convergence of their moments of each fixed order as well: all the integrands are finite sums of products of bounded continuous covariances on a fixed compact set.

For clarity, first fix distinct points $x_1, \dots, x_n$ at a fixed ε. For one color let C_{ij} be the 6×6 covariance block of $F_\varepsilon(x_i)$ and $F_\varepsilon(x_j)$, including the factor g^2. Put $Z = \mathrm{diag}(z_1 I_6, \dots, z_n I_6)$ and let C be the full block covariance matrix. The Gaussian integral yields, near $z = 0$,

$$\log \mathbb{E}\exp\big(\textstyle\sum_i z_i q_\varepsilon(x_i)\big) = -\frac{d_G}{2}\log\det(I - CZ) - \frac{d_G}{2}\sum_i z_i \,\mathrm{tr}\, C_{ii}.$$

Here C is positive semidefinite; write the Gaussian vector as $C^{1/2}N$ and diagonalize the real symmetric quadratic form for small real z. The one-dimensional Gaussian integral proves the determinant identity there. Holomorphic continuation proves it in a complex neighborhood of zero. Sylvester's determinant identity identifies $\det(I - C^{1/2}ZC^{1/2})$ with $\det(I - CZ)$, including singular C.

The last term only subtracts the first cumulants. Expand the logarithm of the determinant by its convergent matrix power series. For $n \geq 2$ the joint cumulant is consequently

$$\kappa_n(q_\varepsilon(x_1), \dots, q_\varepsilon(x_n)) = \frac{d_G}{2n}\sum_{\sigma \in S_n}\mathrm{tr}\,\big(C_{\sigma(1)\sigma(2)}\cdots C_{\sigma(n)\sigma(1)}\big).$$

Indeed only the term of degree n in $\frac{d_G}{2}\sum_{m\geq 1}\mathrm{tr}((CZ)^m)/m$ contributes to the mixed derivative. Each permutation records the order of the n distinct vertices of the cycle. The factor $1/n$ removes its cyclic repetitions; the factor $1/2$ remains even when $n = 2$.

A.3.4. *The noncoincident limit and its consequences.* Let the test functions $f_1, \dots, f_n$ have pairwise disjoint compact supports, and let $d > 0$ be their minimum separation. The regularized off-diagonal kernel is $C_{ij} = g^2(K * \rho_\varepsilon * \rho_\varepsilon)(x_i - x_j)$. For $\varepsilon < d/4$ the convolution samples K only away from zero. It converges uniformly, with every fixed derivative, on the compact set of relevant differences. The cycle formula and ordinary dominated convergence therefore give a finite limit of the smeared connected correlations. Its kernel is

$$C_n(x_1, \dots, x_n) = \frac{d_G g^{2n}}{2n}\sum_{\sigma\in S_n}\mathrm{tr}\,\big(K(x_{\sigma(1)} - x_{\sigma(2)})\cdots K(x_{\sigma(n)} - x_{\sigma(1)})\big),$$
$$x_i \neq x_j.$$

This defines separated correlations. It does not assert convergence of $q_\varepsilon(f)$ in L^2 when the same test is used twice; coincidence singularities require a different distributional treatment.

Corollary A.8 (Two points, odd cycles, and a positive four-point kernel)**.** The limiting connected two-point kernel is

$$C_2(x,y) = \frac{3d_G g^4}{\pi^4 |x-y|^8}.$$

Every odd connected kernel C_n with $n \geq 3$ vanishes at noncoincident arguments. If four distinct points are collinear, then

$$C_4(x_1,x_2,x_3,x_4) = \frac{6d_G g^8}{\pi^8} \sum_{\text{three unoriented cycles } (i,j) \text{ in the cycle}} \prod |x_i - x_j|^{-4} > 0.$$

For four sufficiently small disjoint neighborhoods of such points the four-point kernel remains strictly positive throughout their product. ■□

Proof. For two points, the permutation formula is $d_G g^4 \operatorname{tr}(K(x-y)K(y-x))/2$. The kernel is even and symmetric, so the curvature coefficient proposition proves the stated value. For odd n, every trace in the cycle sum vanishes by the Hodge anticommutation identity. For collinear points all difference directions give the same reflection $R_u = R_{-u}$. The product of four curvature matrices is the positive scalar product of their radial factors times $(\Lambda^2 R_u)^4 = I$. Every trace is therefore six times that scalar. The 24 permutations divide into three unoriented cycles, each represented eight times, canceling the factor $2n = 8$. Positivity on small neighborhoods follows from continuity and the strictly positive value at the chosen configuration. □

For nonnegative nonzero separated tests this gives

$$C_2(f,h) = \frac{3d_G g^4}{\pi^4} \iint \frac{f(x)h(y)}{|x-y|^8}\,dx\,dy > 0.$$

Thus the coefficient $3(N^2-1)/\pi^4$ used for SU(N) is the exact free coefficient in this normalization. For SU(2) it is $9/\pi^4$. Recovering this coefficient from the interacting lattice theory still requires its ultraviolet comparison and uniform remainder estimate. The four-point calculation also shows why non-Gaussianity of the scalar density alone cannot certify an interacting underlying gauge theory: this density already has a positive connected four-point kernel in an explicitly Gaussian model.

A.4. **The free lattice curvature comparison.** This section proves a free-field input of the construction. The covariance of a Gaussian exact lattice two-form is an orthogonal projection. After multiplication by the physical factor a^{-4}, its entries converge away from coincidence to the continuum curvature kernel. We prove an error estimate uniform in the lattice spacing and the torus side, and compute the resulting separated quadratic-density coefficient. No interacting measure is compared with this Gaussian model here.

A.4.1. *Cochains, Fourier conventions, and the statement.* Let $a > 0$, $\ell > 0$, and $N = \ell/a \in \mathbb{N}$. The lattice is $\Lambda_{a,\ell} = a\mathbb{Z}^4/\ell\mathbb{Z}^4$. A one-form has coordinates $A_\mu(x)$, one per positive bond based at x. A two-form has coordinates $F_{\mu\nu}(x)$ for $\mu < \nu$, based at the lower vertex of its positively oriented plaquette. Both cochain spaces use the plain coordinate inner products. Define

$$(\mathrm{d}_1 A)_{\mu\nu}(x) = a^{-1}\{A_\nu(x+ae_\mu) - A_\nu(x) - A_\mu(x+ae_\nu) + A_\mu(x)\}.$$

The factor a^{-1} does not affect the range. Let P_{ex} be the orthogonal projection onto $\operatorname{Ran}\mathrm{d}_1$. It is the covariance of a centered Gaussian vector which is standard on that subspace. Potential gauge directions need not be inverted to define this curvature covariance. The constant two-forms are excluded because they are not in $\operatorname{Ran}\mathrm{d}_1$.

Write $P = (\mu,\nu)$, $\mu < \nu$, and put $c_P = (e_\mu + e_\nu)/2$. The centre of the plaquette (x,P) is $x + ac_P$. The physical lattice kernel is

$$\mathsf{K}_{a,\ell}^{PQ}(x,y) = a^{-4} P_{\mathrm{ex}}((x,P),(y,Q)).$$

For a displacement on the torus, its length means distance to $\ell\mathbb{Z}^4$. Representatives are chosen in $[-\ell/2,\ell/2]^4$ when needed.

The continuum Fourier transform on $\mathbb{R}^4$ is

$$\widehat{f}(k) = \int e^{-ik\cdot x} f(x)\,dx,$$

$$f(x) = (2\pi)^{-4} \int e^{ik\cdot x} \widehat{f}(k)\,dk.$$

Let $G(x) = 1/(4\pi^2|x|^2)$ and define the continuum curvature distribution by

$$\begin{aligned}\mathsf{K}_{\mu\nu,\rho\sigma}(x) = &-\delta_{\nu\sigma}\partial_\mu\partial_\rho G(x) + \delta_{\nu\rho}\partial_\mu\partial_\sigma G(x)\\ &+ \delta_{\mu\sigma}\partial_\nu\partial_\rho G(x) - \delta_{\mu\rho}\partial_\nu\partial_\sigma G(x).\end{aligned}$$

At $x \neq 0$ this is an ordinary smooth matrix. Its contact terms at zero are retained in its Fourier definition and are not squared in the statements below.

Proposition A.9 (Free lattice curvature comparison)**.** There is a numerical constant C such that, for every a, ℓ as above and every pair of plaquettes with centre displacement

$$u = x - y + a(c_P - c_Q),$$

$$10a \le r = \mathrm{dist}(u, \ell\mathbb{Z}^4) \le \ell/10,$$

one has

$$\left|\mathsf{K}_{a,\ell}^{PQ}(x,y) - \mathsf{K}_{PQ}(u)\right| \le C r^{-4}\left\{(a/r)^{1/2} + (r/\ell)^4\right\}. \tag{A.iii}$$

The continuum kernel on the right is evaluated at the short representative of u. The constants do not depend on the orientation labels, the lattice spacing, the number of sites, or the torus side. ■

The exponent $1/2$ is sufficient for convergence and is not claimed optimal. Keeping the barycentre phases in a sharper symbol expansion can improve the lattice power. The proof here deliberately keeps a periodic raw symbol and moves to barycentres only after the Fourier estimate.

A.4.2. *The exact projection symbol.* The torus frequencies are $k \in \delta\mathbb{Z}^4$, where $\delta = 2\pi/\ell$. Choose one Brillouin representative of each frequency modulo $(2\pi/a)\mathbb{Z}^4$ and denote this set by $\mathcal{Q}_a$. Its coordinates lie between $-\pi/a$ and π/a, with one endpoint omitted when necessary. Define

$$t_{a,\mu}(k) = a^{-1}(e^{iak_\mu} - 1),$$

$$\omega_a(k) = \sum_\mu |t_{a,\mu}(k)|^2 = 4a^{-2}\sum_\mu \sin^2(ak_\mu/2).$$

Let $B_a(k) : \mathbb{C}^4 \to \Lambda^2\mathbb{C}^4$ be

$$(B_a(k)v)_{\mu\nu} = t_{a,\mu}(k)v_\nu - t_{a,\nu}(k)v_\mu.$$

This is exactly the Fourier symbol of d_1. Direct multiplication gives

$$B_a^* B_a = \omega_a I - t_a t_a^*,$$

$$B_a t_a = 0.$$

Consequently $(B_aB_a^*)^2 = \omega_a B_aB_a^*$. For $k \neq 0$ in the Brillouin zone,

$$P_a(k) = B_a(k)B_a(k)^*/\omega_a(k)$$

is the orthogonal projection onto $\operatorname{Ran} B_a(k)$. Set $P_a(0) = 0$. Thus $\|P_a(k)\| \leq 1$ and its rank at nonzero frequency is three. At frequency zero the exact-form subspace is zero.

The unitary finite Fourier transform and its inverse are

$$\widehat{F}(k) = N^{-2} \sum_{x \in \Lambda_{a,\ell}} e^{-ik\cdot x} F(x),$$

$$F(x) = N^{-2} \sum_{k \in \mathcal{Q}_a} e^{ik\cdot x} \widehat{F}(k).$$

Therefore

$$a^{-4} P_{\mathrm{ex}}((x,P),(y,Q)) = \ell^{-4} \sum_{k \in \mathcal{Q}_a} e^{ik\cdot(x-y)} [P_a(k)]_{PQ}. \tag{A.iv}$$

Indeed the projection entry before multiplication by a^{-4} has prefactor N^{-4}, and $a^{-4}N^{-4} = \ell^{-4}$.

For the continuum symbol replace t_a by $t_0(k) = ik$. Put

$$P_0(k) = B_0(k)B_0(k)^*/|k|^2$$

$$(k \neq 0),$$

$$P_0(0) = 0.$$

It is again a norm-one rank-three projection. With the specified Fourier convention its inverse transform is precisely the distribution K defined above. On the torus define the distribution

$$\mathsf{K}_\ell(z) = \ell^{-4} \sum_{k \in \delta\mathbb{Z}^4} e^{ik\cdot z} P_0(k). \tag{A.v}$$

The bounded Fourier coefficients define a periodic distribution. Below we show that it is smooth away from the origin and identify its finite-volume correction.

A.4.3. *Uniform derivative and counting estimates.* For each multi-index α there is a constant C_α such that, in the Brillouin zone away from zero,

$$\|\partial^\alpha P_a(k)\| \leq C_\alpha |k|^{-|\alpha|}. \tag{A.vi}$$

The same estimate holds for P_0. To prove it, use $2|\sin(t/2)| \geq 2|t|/\pi$ for $|t| \leq \pi$. Hence $c|k|^2 \leq \omega_a(k) \leq |k|^2$ throughout the Brillouin zone. Differentiating the exponential gives $|\partial^\alpha t_a(k)| \leq C_\alpha |k|^{1-|\alpha|}$; for positive derivative order this uses $a|k| \leq 2\pi$. The product rule gives the analogous bounds of order $2 - |\alpha|$ for ω_a and $B_aB_a^*$. Induction in the identity $\omega_a\omega_a^{-1} = 1$ bounds derivatives of the reciprocal. The quotient rule proves eq. A.vi. All functions involved are periodic; near a Brillouin face the same bounds hold in periodic coordinates, where $|k|$ is comparable to a^{-1}.

For $a|k| \leq c_0$, with a fixed small $c_0 > 0$, Taylor's formula gives $t_a(k) = ik + O(a|k|^2)$. The preceding denominator bounds imply

$$\|P_a(k) - P_0(k)\| \leq Ca|k|. \tag{A.vii}$$

For example, the numerator difference is $O(a|k|^3)$ and the denominator difference is $O(a|k|^3)$, while both denominators are comparable to $|k|^2$. Both symbols have been assigned zero at $k = 0$, so that frequency contributes no error.

We will use two elementary grid bounds. If $R \geq C\delta$ and $M > 4$, then

$$\ell^{-4} \#\{k \in \delta\mathbb{Z}^4 : |k| \leq 2R\} \leq CR^4, \tag{A.viii}$$

$$\ell^{-4} \sum_{k \in \delta \mathbb{Z}^4} (R + |k|)^{-M} \leq C_M R^{4-M}. \tag{A.ix}$$

For a proof, place a cube of side δ about each grid point. The cubes are disjoint up to boundaries. If $R \geq C\delta$, the values of $R + |k|$ and $R + |q|$ on such a cube differ by bounded factors. The second sum is bounded by a constant times $(2\pi)^{-4} \int (R + |q|)^{-M} dq$, which is $C_M R^{4-M}$ after scaling. The first bound follows by enclosing the cubes in a ball of radius $3R$.

A.4.4. *Summation by parts on the frequency grid.* Fix a smooth cutoff χ equal to one on $|k| \leq 1$ and zero on $|k| \geq 2$. Suppose

$$R \geq 2M\delta,$$
$$2R < \pi/(2a).$$

Extend $\chi(k/R)$ periodically on the frequency torus of side $2\pi/a$. It is smooth because its support stays away from the Brillouin faces. The high-frequency lattice symbol

$$H_a(k) = (1 - \chi(k/R)) P_a(k)$$

is smooth on that frequency torus, including at zero. Write

$$\rho_a(k) = \mathrm{dist}(k, (2\pi/a)\mathbb{Z}^4).$$

The derivative estimates and the cutoff product rule give

$$\|\partial_\mu^M H_a(k)\| \leq C_M (R + \rho_a(k))^{-M}.$$

Let $\Delta_\mu H(k) = H(k + \delta e_\mu) - H(k)$. The iterated fundamental theorem of calculus yields

$$\Delta_\mu^M H(k) = \delta^M \int_{[0,1]^M} \partial_\mu^M H(k + \delta(t_1 + \cdots + t_M) e_\mu)\, dt.$$

The torus distance ρ_a changes by at most $M\delta$ along this segment. Since $R \geq 2M\delta$, $R + \rho_a$ remains comparable to its value at k. Thus

$$\|\Delta_\mu^M H_a(k)\| \leq C_M \delta^M (R + \rho_a(k))^{-M}. \tag{A.x}$$

Now let $z \in a\mathbb{Z}^4/\ell\mathbb{Z}^4$, $z \neq 0$, and choose a short representative. Its length r_z is at most ℓ, and some coordinate satisfies $|z_\mu| \geq r_z/2$. On $|z_\mu| \leq \ell/2$,

$$|e^{-i\delta z_\mu} - 1| = 2|\sin(\pi z_\mu/\ell)| \geq 4|z_\mu|/\ell \geq 2r_z/\ell.$$

The phase $e^{ik\cdot z}$ is periodic on the frequency grid because z/a is integral. Finite summation by parts therefore gives the exact identity

$$(e^{-i\delta z_\mu} - 1)^M \sum_{k \in \mathcal{Q}_a} e^{ik\cdot z} H_a(k) = \sum_{k \in \mathcal{Q}_a} e^{ik\cdot z} \Delta_\mu^M H_a(k).$$

There is no Brillouin boundary term. Apply eq. A.x and eq. A.ix, using $\rho_a(k) = |k|$ on the chosen representatives. The result is

$$\left\| \ell^{-4} \sum_{k \in \mathcal{Q}_a} e^{ik\cdot z} H_a(k) \right\| \leq C_M r_z^{-M} R^{4-M}. \tag{A.xi}$$

The powers of the torus side cancel: the inverse difference factor contributes $(\ell/r_z)^M$ and the frequency difference contributes $\delta^M = (2\pi/\ell)^M$.

The continuum torus tail obeys the same estimate. Here $H_0(k) = (1 - \chi(k/R))P_0(k)$ on the infinite reciprocal grid. Its derivatives satisfy the same bounds with $|k|$ in place of ρ_a. The Mth differences are absolutely summable for $M > 4$. Multiplication of a Fourier distribution by $(e^{-i\delta z_\mu} - 1)^M$ shifts its Fourier coefficients and hence gives the same identity distributionally. The right side is an absolutely convergent continuous Fourier series. On the open set where the multiplying factor is nonzero, division by that smooth factor identifies the tail with an ordinary continuous function and proves

$$\|\mathsf{K}_\ell^{\mathrm{high}}(z)\| \le C_M r_z^{-M} R^{4-M}. \tag{A.xii}$$

This distributional argument avoids assuming convergence of an undifferentiated order-zero Fourier series.

A.4.5. *Comparison with the continuum torus kernel.* On the common low-frequency support, the finite and infinite reciprocal grids agree. eq. A.vii and eq. A.viii give

$$\left\| \ell^{-4} \sum_k e^{ik\cdot z} \chi(k/R)(P_a(k) - P_0(k)) \right\| \le CaR^5.$$

Combining this with the two high-frequency bounds yields, for $M > 4$,

$$\|\mathsf{K}_{a,\ell}^{\mathrm{raw}}(z) - \mathsf{K}_\ell(z)\| \le C_M\{aR^5 + r_z^{-M} R^{4-M}\}. \tag{A.xiii}$$

Here $\mathsf{K}_{a,\ell}^{\mathrm{raw}}$ is the matrix Fourier sum in eq. A.iv at the common plaquette-origin displacement z.

Take $M = 9$ and choose

$$R = \frac{D}{r_z}(r_z/a)^{1/10},$$

where D is a fixed sufficiently large numerical constant. Because $r_z \le \ell$, this ensures $R \ge 18\delta$. For r_z/a larger than a fixed numerical threshold C_0, it also ensures $2aR < c_0$ and $2R < \pi/(2a)$. Both terms in eq. A.xiii are then at most a constant times

$$r_z^{-4}(a/r_z)^{1/2}.$$

Indeed $aR^5 = D^5 r_z^{-4}(a/r_z)^{1/2}$ and $r_z^{-9}R^{-5} = D^{-5}r_z^{-4}(a/r_z)^{1/2}$.

For $1 \le r_z/a \le C_0$, the lattice Fourier sum has norm at most a^{-4} because $\|P_a\| \le 1$. The continuum bound $\|\mathsf{K}_\ell(z)\| \le Cr_z^{-4}$, established in the next section, shows that their difference is at most $C'a^{-4}$. Increasing the numerical constant by a factor depending only on C_0 bounds this by $Cr_z^{-4}(a/r_z)^{1/2}$. Consequently, for every nonzero raw lattice displacement,

$$\|\mathsf{K}_{a,\ell}^{\mathrm{raw}}(z) - \mathsf{K}_\ell(z)\| \le Cr_z^{-4}(a/r_z)^{1/2}. \tag{A.xiv}$$

All thresholds used here are fixed numbers. They impose no relation between the physical volume and the ultraviolet cutoff beyond $\ell/a \in \mathbb{N}$.

A.4.6. *The finite-volume correction and contact terms.* The continuum torus heat kernel is

$$p_t^\ell(x) = \sum_{m\in\mathbb{Z}^4} (4\pi t)^{-2} e^{-|x+m\ell|^2/(4t)} = \ell^{-4} \sum_{k\in\delta\mathbb{Z}^4} e^{-t|k|^2} e^{ik\cdot x}.$$

For completeness, the first expression is an absolutely convergent periodic sum for $t > 0$. Its Fourier coefficient at k is the ordinary Gaussian Fourier integral $\ell^{-4}e^{-t|k|^2}$, which proves the second expression. Define the mean-zero torus Green distribution by

$$G_\ell(x) = \int_0^\infty (p_t^\ell(x) - \ell^{-4})\, dt.$$

Its nonzero Fourier coefficient is $\ell^{-4}|k|^{-2}$ and its zero coefficient is zero. Applying the same second-derivative combination as in the definition of K gives eq. A.v.

We record the uniform local correction. At $\ell = 1$ and $|x| \le 1/4$,

$$\begin{aligned} G_1(x) - G(x) = &\int_0^1 \sum_{m\neq 0} (4\pi t)^{-2} e^{-|x+m|^2/(4t)}\, dt - 1 \\ &+ \int_1^\infty (p_t^1(x) - 1)\, dt - \int_1^\infty (4\pi t)^{-2} e^{-|x|^2/(4t)}\, dt. \end{aligned}$$

Every fixed derivative of the first integral is uniformly bounded: differentiating a Gaussian gives a polynomial times a Gaussian, while $|x+m| \ge c|m|$ and $\sum_{m\neq 0} e^{-c|m|^2/t} \le Ce^{-c'/t}$ for $0 < t \le 1$. The resulting power of t^{-1} is integrable against this last exponential. For the second integral, its differentiated Fourier series is bounded by a constant times e^{-ct} for $t \ge 1$. Derivatives of the last integral are bounded by $C_\alpha t^{-2-|\alpha|/2}$ on this ball and are integrable. Thus $G_1 - G$ extends smoothly across zero with bounded derivatives of every fixed order.

Scaling gives $G_\ell(x) = \ell^{-2} G_1(x/\ell)$ and the corresponding homogeneity for G. In particular,

$$\|\mathsf{K}_\ell(z) - \mathsf{K}(z)\| \le C\ell^{-4} \quad (0 < |z| \le \ell/4). \tag{A.xv}$$

The same argument gives an extra factor ℓ^{-1} for each additional derivative of the difference. On the remainder of the unit torus, away from zero, the heat-kernel formula and its derivatives converge uniformly on compact sets. Scaling and compactness therefore also give the global bounds

$$\|\partial^\alpha \mathsf{K}_\ell(z)\| \le C_\alpha r_z^{-4-|\alpha|},$$
$$z \neq 0.$$

These justify the estimate used in the bounded r_z/a regime above.

All these statements concern the kernels away from coincidence. The symbols have not discarded their order-zero distributional terms. For example, the trace of $P_0(k)$ is three for $k \neq 0$. Accordingly the trace of the torus kernel as a distribution is $3(\delta_{\mathrm{per}} - \ell^{-4})$; away from zero it is $-3\ell^{-4}$. The trace of the free noncoincident kernel is zero. This is consistent with eq. A.xv; it would be inconsistent to replace the full distribution by its noncoincident formula before taking Fourier coefficients.

A.4.7. *Moving to barycentres and proving the theorem.* Let u be the centre displacement in eq. A.iii and let $z = u - a(c_P - c_Q)$ be its raw-origin displacement. The shift has length at most a. Choose representatives so $|u| = r \le \ell/10$ and $|z - u| \le a$. Then

$$0.9r \le |z| \le 1.1r < \ell/4.$$

Use eq. A.xiv for the raw kernel and eq. A.xv for the continuum torus kernel at z. The free kernel is homogeneous of degree -4 and smooth away from zero, so $\|\nabla \mathsf{K}(v)\| \le C|v|^{-5}$. The segment from z to u stays at distance at least $0.9r$ from zero. Hence

$$|\mathsf{K}_{PQ}(z) - \mathsf{K}_{PQ}(u)| \le Car^{-5}.$$

Combining the three differences proves

$$|\mathsf{K}_{a,\ell}^{PQ}(x,y) - \mathsf{K}_{PQ}(u)| \le C\{r^{-4}(a/r)^{1/2} + \ell^{-4} + ar^{-5}\}.$$

Since $a/r \le 1$, the last term is absorbed by the first, and $\ell^{-4} = r^{-4}(r/\ell)^4$. This proves eq. A.iii.

The argument also proves convergence to K_ℓ uniformly on any fixed separated subset of the torus, when plaquette centres are used in both arguments. In this statement use the global derivative bound for K_ℓ to move from origins to centres. No assumption that a chosen separation is small relative to ℓ is needed for this fixed-torus convergence.

A.4.8. *The separated quadratic-density coefficient.* Let ξ_p^b, $1 \le b \le d_G$, be independent color copies of the Gaussian exact two-form, each with covariance $g^2 P_{\mathrm{ex}}$. For a test f on the torus define the finite-dimensional centered density

$$Q_a(f) = \frac{1}{2}\sum_{b=1}^{d_G}\sum_p f(x_p)\{(\xi_p^b)^2 - \mathbb{E}(\xi_p^b)^2\}.$$

Here x_p is the plaquette barycentre. Equivalently, $a^{-2}\xi_p^b$ is the physical curvature and the displayed sum includes its a^4 Riemann weight. For two tests of positive mutual separation, the Gaussian identity $\mathrm{Cov}(X^2, Y^2) = 2\,\mathrm{Cov}(X,Y)^2$ gives

$$\mathrm{Cov}(Q_a(f), Q_a(h)) = \frac{d_G g^4}{2}\, a^8 \sum_{p,q} f(x_p)h(x_q)\,|\mathsf{K}_{a,\ell}(p,q)|^2.$$

For real kernels the square here is the ordinary real square of the scalar entry. The uniform separated convergence just proved, followed by ordinary shifted-grid Riemann sums, gives

$$\lim_{a\downarrow 0}\mathrm{Cov}(Q_a(f), Q_a(h)) = \frac{d_G g^4}{2}\iint f(x)h(y)\sum_{P,Q}\mathsf{K}_{\ell,PQ}(x-y)^2\,dx\,dy.$$

For continuous compactly supported separated tests the Riemann-sum integrand is continuous after setting it to zero in a neighbourhood of the diagonal; for C^1 tests it is C^1 there. There is therefore no use of an integral of $|x-y|^{-8}$ through coincidence.

Now keep the tests in a fixed compact subset of $\mathbb{R}^4$ and let $\ell \to \infty$. The correction estimate makes $\mathsf{K}_\ell \to \mathsf{K}$ uniformly on the relevant separated differences. Direct differentiation of G shows

$$\mathsf{K}(x) = \frac{1}{\pi^2|x|^4}\Lambda^2(I - 2\widehat{x}\otimes\widehat{x}).$$

One can check the coefficient at $\widehat{x} = e_4$: the six eigenvalues are three copies of $1/(\pi^2|x|^4)$ and three copies of its negative. Orthogonal covariance proves the identity in every direction. The induced two-form reflection is orthogonal, and therefore

$$\sum_{P,Q}\mathsf{K}_{PQ}(x)^2 = \frac{6}{\pi^4|x|^8}.$$

It follows that the infinite-volume separated covariance is exactly

$$\frac{3d_G g^4}{\pi^4}\iint f(x)h(y)|x-y|^{-8}\,dx\,dy. \tag{A.xvi}$$

For SU(2), $d_G = 3$, so the coefficient is $9g^4/\pi^4$ in this convention. Higher separated Gaussian cumulants can likewise be passed through their finite cycle formulas using the same kernel convergence.

There is also a relative finite-volume statement for nonnegative pairs. Suppose every cross-distance between the supports lies between d and $C_{\mathrm{sh}}d$, and $C_{\mathrm{sh}}d \le \ell/10$. The correction estimate and the Frobenius norm of the free kernel give pointwise

$$\left|\sum_{P,Q}\mathsf{K}_{\ell,PQ}(v)^2-\frac{6}{\pi^4|v|^8}\right|\le C\left(\frac{|v|}{\ell}\right)^4|v|^{-8}.$$

Indeed the difference of the squares is bounded by $C|v|^{-4}\ell^{-4}+C\ell^{-8}$. Multiplication by $f(x)h(y)\ge 0$ and integration show that the limiting torus covariance differs from eq. A.xvi by a relative error at most $C'_{\mathrm{sh}}(d/\ell)^4$, provided both tests are nonzero. The mesh error tends to zero for each such fixed test pair by the uniform kernel estimate and the Riemann-sum argument above. A rate relative to a varying family of tests would require their norms and nondegeneracy to be specified.

This proves the free covariance comparison and the free density coefficient. It asserts neither an L^2 limit of $Q_a(f)$ with the same test repeated nor an interacting short-distance asymptotic. Establishing the latter requires a comparison of the interacting measure with this Gaussian model and a remainder smaller than the leading term.

A.5. **Link reflections of periodic Gibbs states.** This section concerns one fixed lattice and one fixed positive coupling. No continuum limit or renormalization estimate is used. The conclusion quantifies over every periodic DLR state. In particular, the state is not assumed to arise as a limit of torus measures. The finite-dimensional transfer calculation below, together with a local elimination identity, is what permits that quantifier.

Set the lattice spacing equal to one, let $G=\mathrm{SU}(2)$, and use normalized Haar measure. On one orientation of the bonds of $\mathbb{Z}^4$ assign variables $U_b\in G$; reverse orientation means $U_{\bar b}=U_b^{-1}$. For an elementary plaquette set

$$a_p=1-\tfrac12\,\mathrm{tr}\,U_p,$$
$$0\le a_p\le 2.$$

The trace is real for $G=\mathrm{SU}(2)$. Fix $\beta>0$. A DLR state is a probability measure whose conditional law on any finite bond set D, given all other bonds, has density proportional to $\exp[-\beta\sum_{p:p\cap D\neq\varnothing}a_p]$. A state is periodic when it is invariant under a subgroup of finite index of $\mathbb{Z}^4$. A link reflection is reflection in a plane $\{x_\mu=i+\frac12\}$, with $i\in\mathbb{Z}$.

Proposition A.10 (Link symmetry of every periodic Wilson state)**.** Every periodic DLR state of the four-dimensional SU(2) Wilson interaction at $\beta>0$ is invariant under every link reflection. Consequently every such state is invariant under $2\mathbb{Z}^4$. ∎

The proof uses positivity and strict positivity of different objects. The one-bond convolution operator is positive as an operator, with strictly positive Peter–Weyl eigenvalues. Its original kernel is strictly positive pointwise. An auxiliary operator inserted between two half-kernels will be a contraction, and need not be self-adjoint or positive. Confusing these conditions would invalidate the variational step.

A lower tangent valid for an arbitrary periodic state. Let ϕ be a real continuous local observable and let $m\mathbb{Z}^4$ be a subgroup of periods of a DLR state ν. For torus sides s divisible by m, write

$$W_s=\sum_{v\in m\mathbb{Z}^4/s\mathbb{Z}^4}\tau_v\phi,$$
$$Z_s(\phi)=\int e^{W_s-\beta\sum_p a_p}\,dU.$$

Here local observables are evaluated on the periodic extension of the torus configuration. Suppose

$$Z_s(\phi) \le Z_s(0)$$
$$\text{for all sufficiently large multiples } s \text{ of } m. \tag{A.xvii}$$

Then

$$\nu(\phi) \le 0. \tag{A.xviii}$$

To prove this implication, take a cube D_s of sites of side s, and let B_s be the positively oriented bonds whose initial sites belong to the cube. Choose the representatives of the source anchors inside D_s, and denote their sum, evaluated with an arbitrary outside configuration, by $\widetilde{W}_s$. This is a finite local function; terms near the boundary can involve outside bonds. The number of anchors is exactly $M_s = (s/m)^4$. The DLR equation and Jensen's inequality give

$$\log \gamma_{B_s}(e^{\widetilde{W}_s})(U') \ge \gamma_{B_s}(\widetilde{W}_s)(U').$$

The conditional partition-function ratio on the left can be compared with $Z_s(\phi)/Z_s(0)$ uniformly in U'. Interior terms are identical under the identification of B_s with the torus bond coordinates. Only plaquettes and translates of ϕ in a fixed-width boundary layer can differ. Their number is at most $C_{\phi,m}s^3$, and each difference is bounded by $C(\beta + \|\phi\|_\infty)$. Thus

$$\log \frac{Z_s(\phi)}{Z_s(0)} \ge \nu(\widetilde{W}_s) - C_{\phi,m,\beta}s^3 = M_s\nu(\phi) - C_{\phi,m,\beta}s^3.$$

In the first inequality we integrated the conditional inequality against ν and used the DLR equation. The equality uses only $m\mathbb{Z}^4$ invariance. Divide by M_s and let $s \to \infty$ through multiples of m. The assumption in eq. A.xvii proves eq. A.xviii.

This argument needs neither differentiability of an infinite-volume pressure nor representation of ν by finite torus measures. The local source is fixed during the limit $s \to \infty$. If the source later depends on a parameter tending to zero, that parameter is kept fixed until the surface error has disappeared. Constant parts of ϕ cause no difficulty with the anchored definition of $\widetilde{W}_s$: they occur with exactly the same multiplicity in the box and torus source sums.

The one-bond convolution and its smooth conjugations. Define

$$c(g) = e^{-\beta(1-\frac{1}{2}\operatorname{tr} g)},$$
$$(CF)(w) = \int_G c(wv^{-1})F(v)\,dv.$$

The kernel is real, symmetric, and bounded below by $e^{-2\beta}$. Let Π_l be the Peter–Weyl projection corresponding to the irreducible representation of spin $l \in \frac{1}{2}\mathbb{N}_0$ and dimension $d_l = 2l + 1$. Schur's lemma and the Weyl integration formula give

$$C = \sum_l c_l \Pi_l,$$
$$c_l = \frac{2e^{-\beta}}{\beta} I_{2l+1}(\beta) > 0. \tag{A.xix}$$

For completeness, write g with eigenvalues e^{it}, e^{-it} and use $\chi_l(g) = \sin((2l+1)t)/\sin t$. The character coefficient is

$$d_l c_l = \frac{2e^{-\beta}}{\pi}\int_0^\pi e^{\beta\cos t}\sin t\sin(d_l t)\,dt = e^{-\beta}(I_{2l}-I_{2l+2})(\beta).$$

The Bessel recurrence gives eq. A.xix. Its series

$$I_n(\beta)=\sum_{k\ge 0}\frac{(\beta/2)^{n+2k}}{k!(n+k)!}$$

has positive terms and implies factorial decay in n. Moreover,

$$\frac{I_{n+1}}{I_n}\le\frac{\beta}{2(n+1)},$$
$$\frac{I_n}{I_{n+1}}=\frac{2(n+1)}{\beta}+\frac{I_{n+2}}{I_{n+1}}\le C_\beta(1+n). \tag{A.xx}$$

Consequently both ratios between neighboring c_l grow at most polynomially in l. Factorial decay also shows that C is trace class and that $C^{1/2}$ has a real symmetric smooth kernel.

For a finite block $\mathcal{B}$ of spatial bonds put

$$\mathcal{C}=C_{\mathcal{B}}=\bigotimes_{b\in\mathcal{B}}C,$$
$$c_{\boldsymbol{l}}=\prod_{b\in\mathcal{B}}c_{l_b},$$
$$\Pi_{\boldsymbol{l}}=\bigotimes_{b\in\mathcal{B}}\Pi_{l_b}.$$

Its kernel is $\mathcal{C}(u,v)=\prod_{b\in\mathcal{B}}c(u_b v_b^{-1})$. In particular it is strictly positive on the compact product space. It commutes with the simultaneous vertex gauge action on its two arguments.

Let f be a polynomial in matrix coordinates and their conjugates on $G^{\mathcal{B}}$. Define on $C^\infty(G^{\mathcal{B}})$

$$\widehat{f}F=\sum_{\boldsymbol{l},\boldsymbol{l}'}\frac{c_{\boldsymbol{l}}}{c_{\boldsymbol{l}'}}\Pi_{\boldsymbol{l}'}M_f\Pi_{\boldsymbol{l}}F, \tag{A.xxi}$$

where M_f denotes multiplication by f. This is a continuous map of the smooth Frechet space into itself. Indeed, if the degree in coordinate b is d_b, the Clebsch–Gordan rule makes the summand zero unless $|l_b-l'_b|\le d_b/2$. By eq. A.xx, the remaining ratios are bounded by a fixed polynomial in the indices. Smooth functions have Peter–Weyl coefficients decreasing faster than every polynomial, as follows by applying powers of the Casimir. The finite-dimensional reproducing-kernel estimate bounds the supremum norm on a block by a polynomial times its L^2 norm. The same holds after any fixed number of derivatives. These facts give convergence of eq. A.xxi with all derivatives and the asserted continuity.

The notation $\widehat{f}=\mathcal{C}^{-1}M_f\mathcal{C}$ is therefore legitimate on this smooth domain when interpreted by eq. A.xxi. It does not assert that $\mathcal{C}^{-1}$ acts on every smooth function or that $\widehat{f}$ is bounded on L^2. The formal L^2 adjoint on smooth functions is given by the same finite-band series with the coefficient ratios reversed and f replaced by $\overline{f}$. Those ratios also have polynomial growth, so this formal adjoint maps smooth functions continuously to smooth functions. In particular no unbounded inverse is applied outside its justified spectral domain in the following identity:

$$\mathcal{C}\widehat{f}F=M_f\mathcal{C}F. \tag{A.xxii}$$

It follows by multiplying eq. A.xxi by $\mathcal{C}$ and summing the projections. In kernel form, for every w,

$$f(w)\int \mathcal{C}(v,w)F(v)\,dv = \int \mathcal{C}(v,w)(\widehat{f}F)(v)\,dv. \tag{A.xxiii}$$

The equality is pointwise, since the series and both integrals define smooth functions. All these assertions also hold for smooth families depending on extra variables on a finite compact product: differentiation in those variables commutes with the series. The operator $\widehat{f}$ commutes with multiplication by functions of variables on which it does not act, and operators acting on disjoint variable sets commute.

A transfer contraction inequality. Fix a direction, called time, and take a cubic torus of side s. Let $\mathcal{E}$ be its spatial bonds in one layer. On $L^2(G^{\mathcal{E}})$, let P be vertex gauge averaging and let V be β times the spatial plaquette action in that layer. The operator

$$L = PC_{\mathcal{E}}^{1/2}e^{-V}C_{\mathcal{E}}^{1/2}P$$

is positive and trace class. The convolution, its square root, and e^{-V} all commute with P. Integrating the time-like bonds in the torus integral inserts P between successive layers. Using cyclicity of the trace gives

$$Z_s(0) = \operatorname{Tr} L^s. \tag{A.xxiv}$$

Here normalized Haar measure makes the gauge average an orthogonal projection without an additional volume factor.

Let Ξ be a real bounded operator on $L^2(G^{\mathcal{B}})$ commuting with the block vertex gauge action, and suppose $\|\Xi\| \le 1$. Suppose the kernel

$$k_\Xi = \text{kernel of } \mathcal{C}^{1/2}\Xi\mathcal{C}^{1/2}$$

is continuous and strictly positive pointwise. In two adjacent layers, use transported spatial variables u, v and set

$$\phi_\Xi(u,v) = \log\frac{k_\Xi(u,v)}{\mathcal{C}(u,v)}. \tag{A.xxv}$$

This is a real continuous gauge-invariant local observable.

Place translates of the block so far apart that blocks in a single time step have disjoint bond sets. Multiplication by e^{ϕ_Ξ} replaces the corresponding factor $\mathcal{C}$ by $\mathcal{C}^{1/2}\Xi\mathcal{C}^{1/2}$. Thus the full kernel in step j is $C_{\mathcal{E}}^{1/2}\Xi_j C_{\mathcal{E}}^{1/2}$, where Ξ_j is a tensor product of copies of Ξ and identities and has norm at most one. Gauge averaging the time-like bonds gives the trace of the product of $e^{-V}C_{\mathcal{E}}^{1/2}\Xi_j C_{\mathcal{E}}^{1/2}P$. Moving the first half-kernel and layer weight cyclically to the end, and using commutation with P, rewrites this trace as

$$\operatorname{Tr}(\Xi_1 L\Xi_2 L\cdots\Xi_s L).$$

The unperturbed steps have $\Xi_j = 1$. Schatten Holder gives

$$|\operatorname{Tr}(\Xi_1 L\cdots\Xi_s L)| \le \prod_{j=1}^{s}\|\Xi_j L\|_s \le \|L\|_s^s = \operatorname{Tr} L^s. \tag{A.xxvi}$$

The last equality uses $L \geq 0$. The modified integral is positive because every replaced kernel is positive pointwise. Therefore eq. A.xxvi is precisely the partition-function inequality in eq. A.xvii for a sufficiently sparse periodic array of ϕ_Ξ.

Given an arbitrary periodic state ν, choose m large enough that $m\mathbb{Z}^4$ is a subgroup of its periods and the copies are disjoint. eq. A.xvii–eq. A.xviii imply

$$\nu(\phi_\Xi) \leq 0. \tag{A.xxvii}$$

The norm contraction and the lower tangent are the only ingredients of this conclusion. No self-adjointness of Ξ and no differentiability of a limiting pressure has been used.

Two adjacent layers cannot carry an antisymmetric mean. Let $Y(u, v)$ be a real continuous function on $G^{\mathcal{B}} \times G^{\mathcal{B}}$ that is invariant under simultaneous vertex gauge transformations and antisymmetric: $Y(v, u) = -Y(u, v)$. Then every periodic DLR state satisfies

$$\nu(Y(U_i, U_{i+1})) = 0. \tag{A.xxviii}$$

To prove this, first suppose $K(u, v) = Y(u, v)\mathcal{C}(u, v)$ is a polynomial. Its integral operator has finite Peter–Weyl support. Antisymmetry makes K a real skew-adjoint operator. On that finite-dimensional support define

$$D = \mathcal{C}^{-1/2} K \mathcal{C}^{-1/2}.$$

Every eigenvalue c_l is strictly positive, so the definition is valid. Extend D by zero on the orthogonal complement. It is a finite-rank real skew-adjoint operator, commutes with gauge transformations, and $\Xi_\varepsilon = e^{\varepsilon D}$ is a real orthogonal operator for real ε of either sign. In particular $\|\Xi_\varepsilon\| = 1$.

The kernel difference from $\mathcal{C}$ has finite Peter–Weyl support and admits, uniformly in (u, v),

$$k_{\Xi_\varepsilon}(u, v) = \mathcal{C}(u, v) + \varepsilon K(u, v) + O(\varepsilon^2).$$

Since $\mathcal{C} \geq e^{-2\beta|\mathcal{B}|}$, this kernel is strictly positive for sufficiently small $|\varepsilon|$. Consequently $\phi_{\Xi_\varepsilon} = \varepsilon Y + O(\varepsilon^2)$ uniformly. Apply eq. A.xxvii, first at each fixed nonzero ε after taking the thermodynamic limit in the lower-tangent argument. For both signs it gives $\varepsilon\nu(Y) \leq C\varepsilon^2$. Letting $\varepsilon \to 0$ proves eq. A.xxviii in this case.

For general Y, approximate $Y\mathcal{C}$ uniformly by polynomials on the compact product group. Antisymmetrize each approximant and average it over the simultaneous gauge group. These operations preserve polynomiality, because a finite matrix-coordinate degree spans a finite-dimensional representation, and preserve the uniform approximation. Divide by the strictly positive kernel $\mathcal{C}$. The resulting functions Y_n converge uniformly to Y and have the already proved property $\nu(Y_n) = 0$. This proves eq. A.xxviii. The argument also specifies why injectivity of $\mathcal{C}$, rather than merely its operator positivity, is needed.

The exact one-layer DLR elimination identity. Fix a finite stack of layers $[a, b]$. Write $T_{j,y}$ for the time-like bond from (j, y) to $(j + 1, y)$ and $\mathring{U}_{j,e}$ for a spatial bond in layer j. Define

$$g_{j,y} = T_{a,y} \cdots T_{j-1,y},$$
$$U_{j,\langle y,y'\rangle} = g_{j,y} \mathring{U}_{j,\langle y,y'\rangle} g_{j,y'}^{-1}. \tag{A.xxix}$$

These are temporal-gauge coordinates on the stack, with the empty product at $j = a$. A gauge-invariant local function of the stack is exactly a function of the U_j invariant under one common vertex gauge action in all layers. This follows by setting the time-like bonds to the

identity successively. For any fixed finite observable only finitely many vertices are involved, so only finite-support gauge transformations are needed.

Let $a < j < b$ and resample the spatial bonds $\{j\} \times \mathcal{B}$, leaving every other original bond conditioned. The matrices $g_{j,y}$ use only time-like bonds and hence are fixed during this resampling. The substitution $\mathring{U}_j \mapsto v = U_j|_{\mathcal{B}}$ preserves Haar measure. The two time-like plaquettes attached to each resampled spatial bond produce the factors below; all spatial plaquettes that meet the block produce the finite potential $V_j(v;\eta)$, where η denotes the other spatial variables of layer j. Thus the conditional density is exactly

$$\begin{aligned} &\frac{1}{z(w,w',\eta)}\mathcal{C}(w,v)e^{-V_j(v;\eta)}\mathcal{C}(v,w')\,dv, \\ &w = U_{j-1}|_{\mathcal{B}}, \\ &w' = U_{j+1}|_{\mathcal{B}}. \end{aligned} \tag{A.xxx}$$

No plaquette meeting the resampled bonds is omitted: it is either spatial in layer j or one of the two time-like plaquettes of that bond. The potential can involve spatial neighbors outside $\mathcal{B}$, but they lie in the same layer and in a finite enlargement of the block.

Let f be a real polynomial on $G^{\mathcal{B}}$ and let Φ be smooth in finitely many transported variables from layers j and above, and independent of the spatial variables of layer $j-1$. Define

$$\Phi^\flat = \frac{e^{V_j(v;\eta)}}{\mathcal{C}(v,w')}\widehat{f}_v\left[e^{-V_j(v;\eta)}\mathcal{C}(v,w')\Phi\right]. \tag{A.xxxi}$$

All factors are smooth on a finite compact product, and the denominator is strictly positive. eq. A.xxi therefore proves that $\Phi^\flat$ is smooth. Its new dependence is confined to layer j and the already present higher layers; it acquires no spatial variable in layer $j-1$.

Every DLR state, periodic or not, satisfies the exact identity

$$\nu\big(f(U_{j-1}|_{\mathcal{B}})\Phi\big) = \nu(\Phi^\flat). \tag{A.xxxii}$$

Indeed, condition on all bonds except the specified block in layer j. The conditional numerator on the left is

$$f(w)\int \mathcal{C}(v,w)e^{-V_j(v;\eta)}\mathcal{C}(v,w')\Phi\,dv.$$

Apply eq. A.xxiii to the smooth bracketed function. Substitution of eq. A.xxxi identifies the result with the numerator of the conditional expectation on the right. The same normalizer $z(w,w',\eta)$ occurs on both sides; it is not differentiated or canceled through an operator. Integration of this conditional identity gives eq. A.xxxii. Reflection of the construction yields the corresponding identity that eliminates an upper outer layer.

Removing two outer layers and preserving reflection. Let θ be reflection about the middle of a stack with $2k+2$ layers, numbered $i-k,\dots,i+k+1$. Let X be a real gauge-invariant polynomial observable in that stack. In temporal coordinates, a finite sum of products writes it as

$$X = \sum_\alpha f_\alpha(w)h_\alpha(z)\Phi_\alpha,$$

where $w = U_{i-k}|_{\mathcal{B}}$ and $z = U_{i+k+1}|_{\mathcal{B}}$ are the outer layers and Φ_α depends on the inner layers. Take one finite block containing all the variables that occur in this decomposition. The potentials used during elimination can enlarge this block spatially, but only finitely and only in the inner layers.

Use eq. A.xxxii to remove $f_\alpha(w)$ through the first inner layer $u = U_{i-k+1}|_{\mathcal{B}}$, and then its reflected version to remove $h_\alpha(z)$ through the last inner layer $v = U_{i+k}|_{\mathcal{B}}$. The result is

$$\nu(X) = \nu(EX),$$
$$EX = \sum_\alpha \Upsilon^{-1} \widehat{f}_{\alpha,u} \widehat{h}_{\alpha,v} (\Upsilon \Phi_\alpha). \tag{A.xxxiii}$$

When the two inner layers are not adjacent, Υ is the product of their two factors e^{-V} and their bond kernels toward the remaining interior. When $k = 1$ they are adjacent and

$$\Upsilon = e^{-V(u;\eta)} e^{-V(v;\eta')} \mathcal{C}(u, v). \tag{A.xxxiv}$$

The common bond kernel occurs once, not twice. To see the cancellation directly, the first elimination introduces $e^{V(u)}/\mathcal{C}(u, v)$ outside its conjugated operator. In the upper elimination the conditional numerator contains $e^{-V(v)}\mathcal{C}(u, v)$, canceling that denominator before the second operator acts. The operators act on distinct variables and commute. Moving the remaining independent factors through them gives exactly eq. A.xxxiv. This calculation is also a check that normalizing factors from different conditional laws have not been interchanged.

Let R denote literal reversal of the array of transported layer coordinates. An actual reflection of the original bond configuration induces R followed by a common vertex gauge transformation arising from the change of temporal-gauge base. The spatial potential is the same function of its own layer variables at both ends, and $\mathcal{C}(u, v)$ is symmetric. Reflection interchanges u and v, including all their spatial neighbors, and preserves Υ. It also sends $\widehat{f}_u$ to $\widehat{f}_v$. Consequently coordinate elimination commutes with literal reversal:

$$E(RX) = R(EX). \tag{A.xxxv}$$

The output need not be a polynomial, but it is a smooth function of finitely many inner-layer variables.

Every DLR state is invariant under finite-support vertex gauge transformations. This follows by applying its conditional law to the finite set of bonds incident to the transformed vertices and changing variables using Haar invariance and plaquette conjugacy. We may therefore replace EX in expectation by its average under the common temporal-gauge action; denote it by $\mathcal{A}$. This averaging commutes with R. Once averaged, the choice of the lowest layer in eq. A.xxix disappears: changing the base from $i-k$ to $i-k+1$ conjugates all remaining transported variables by one common vertex gauge transformation. The averaged function is invariant under that transformation, even when its values are the actual time-like transport matrices. This last statement is a pointwise identity of invariant functions, and does not invoke invariance of the state under a field-dependent change of variables. Thus the averaged output is a continuous gauge-invariant observable of the smaller stack of $2k$ layers. Since X is gauge invariant, its actual reflection is represented by RX in temporal coordinates. After averaging the common gauge factor on the output also disappears. Thus the identity needed on the original bond space is $\mathcal{A}E(\theta X) = \theta \mathcal{A} EX$.

Proof of the link-reflection theorem. Fix the reflection θ in $x_\mu = i + \frac{1}{2}$. We prove by induction on k that every real gauge-invariant polynomial in the stack $[i - k + 1, i + k]$ has equal expectation to its reflection.

For $k = 1$, temporal gauge writes the observable as $\Psi(U_i, U_{i+1})$. Reflection interchanges the two arguments, up to a common gauge transformation, which does not affect Ψ. The difference from its reflection is the function $Y(u, v) = \Psi(u, v) - \Psi(v, u)$, to which eq. A.xxviii applies. Hence the expectations agree.

Assume the assertion for k. For a polynomial X in the stack with $2k + 2$ layers, eq. A.xxxiii–eq. A.xxxv and gauge averaging reduce the comparison of $\nu(X)$ and $\nu(\theta X)$ to the same comparison

for a continuous gauge-invariant function of the smaller stack. Invariant polynomials are uniformly dense in the invariant continuous functions on a finite product of compact matrix groups: first apply Stone–Weierstrass to matrix coordinates and their conjugates, then average the approximants over the finite vertex gauge group. The induction hypothesis therefore applies to this continuous output. It gives $\nu(\mathcal{A}EX) = \nu(\theta\mathcal{A}EX)$ and proves the next step.

Every gauge-invariant continuous local function fits in a finite symmetric stack and is uniformly approximated as above. We have proved equality of its expectation and that of its reflection. For a general continuous local F, gauge average over the finite set of vertices relevant to its support. Gauge invariance of both ν and its reflected state shows that both expectations of F equal the expectations of that gauge average. Thus $\nu(F) = \nu(\theta F)$ for all continuous local F, which determine the probability measure on the compact product configuration space. The state is reflection invariant. The direction μ and integer i were arbitrary. Composing reflections in two successive link planes gives translation by $2e_\mu$, proving the stated $2\mathbb{Z}^4$ invariance. □

Passage to all mirrors. The common period is the input to the independent fold-and-separation proof in appendix A.6. That section supplies adapted diagonal split tori, finite crossing kernels, the full chessboard inequality, and the arbitrary-state separation argument. The link theorem above does not require those later ingredients.

A.6. **Reflection positivity of periodic Wilson states.** The result in this section concerns the complete algebra of bonds on a closed side of a mirror. Its proof does not identify an arbitrary Gibbs state with a limit of unperturbed torus measures. Instead, a pressure inequality produces, for each separating observable, a suitable limit of *perturbed* torus measures. The common period supplied by link reflection invariance is what makes the separation argument work.

We use the following result proved in the preceding link-symmetry argument: every periodic DLR state of the four-dimensional SU(2) Wilson specification, at $\beta > 0$, is invariant under all coordinate link reflections and consequently under $P = 2\mathbb{Z}^4$. No uniqueness or extremality is assumed here.

Let $\Omega = \mathrm{SU}(2)^{\mathcal{B}}$, with one group variable for each positively oriented nearest-neighbour bond; reversing a bond inverts its variable. Put

$$a_p(U) = 1 - \tfrac{1}{2}\operatorname{tr} U_p,$$

$$c(g) = \exp[-\beta(1 - \tfrac{1}{2}\operatorname{tr} g)].$$

The trace is real for SU(2), and $0 \le a_p \le 2$. The product topology makes Ω compact and metrizable. A state is periodic if it is invariant under some finite-index subgroup of $\mathbb{Z}^4$. Denote by $\mathcal{G}_\beta^P$ the compact convex set of Wilson DLR states invariant under P.

The normals considered below are

$$\mathcal{N} = \{e_\mu : 1 \le \mu \le 4\} \cup \{e_\mu + e_\lambda, e_\mu - e_\lambda : 1 \le \mu < \lambda \le 4\}.$$

For $n \in \mathcal{N}$ write $q = |n|^2$ and

$$\theta_c x = x - 2(n \cdot x - c)n/q.$$

The admissible positions are $c \in \frac{1}{2}\mathbb{Z}$ when $q = 1$ and $c \in \mathbb{Z}$ when $q = 2$. These and only these positions in the stated families give lattice reflections. Let $\mathcal{A}_c^+$ be the bounded Borel functions of bonds both of whose endpoints satisfy $n \cdot x \ge c$, and define $\mathcal{A}_c^-$ similarly. The antilinear reflection is $\Theta_c F = \overline{F \circ \theta_c^*}$.

Proposition A.11 (All-mirror positivity of periodic states)**.** Every periodic Wilson DLR state at $\beta > 0$ is invariant under every admissible reflection θ_c in the sixteen families above. For every such mirror,

$$\nu(\Theta_c F\, F) \geq 0$$
$$(F \in \mathcal{A}_c^+ \text{ or } F \in \mathcal{A}_c^-). \tag{A.xxxvi}$$

In particular, the conclusion applies to every bounded observable on any finite stack contained in either closed half-space, regardless of the number of layers in that stack. ■

Composing the reflections in $x_\mu = 0$ and $x_\mu = 1/2$ gives translation by one lattice unit. Coordinate reflections give sign changes, and reflection in $x_\mu - x_\lambda = 0$ exchanges two coordinates. Thus every such state is invariant under all lattice translations and the full signed-permutation group. At $\beta = 0$, the DLR equations directly force every finite-bond marginal to be product Haar; that unique product state has the same properties.

Positive crossing kernels on split tori. Write an element of SU(2) as a unit quaternion $s(g) \in S^3 \subset \mathbb{R}^4$. Then

$$\tfrac{1}{2}\,\mathrm{tr}(gh^{-1}) = s(g) \cdot s(h).$$

Consequently

$$c(gh^{-1}) = e^{-\beta} \sum_{m=0}^{\infty} \frac{\beta^m}{m!} \sum_{i_1,\dots,i_m=1}^{4} \prod_{j=1}^{m} s_{i_j}(g) \prod_{j=1}^{m} s_{i_j}(h). \tag{A.xxxvii}$$

Every coefficient is nonnegative. Each feature has absolute value at most one, and the sum of the coefficients is $e^{3\beta}$, so the series converges absolutely and uniformly. Tensor products of finitely many such expansions are again uniformly convergent positive Gram expansions. The coarse total coefficient bound is immaterial; each torus used below is finite.

A torus is called split for the mirror normal n if its period lattice has the form

$$\Gamma = \Gamma_{\parallel} \oplus \mathbb{Z}(2zn/q),$$
$$\Gamma_{\parallel} \subset n^{\perp}, \tag{A.xxxviii}$$

where $\Gamma_{\parallel}$ has rank three, $2zn/q$ is a lattice vector, and z is a sufficiently large integer. The reflection has two fixed cuts, at heights c and $c + z$, on this torus. The two intervening closed halves are exchanged. All tori below have arbitrarily large shortest nonzero period, so that no small-quotient ambiguity affects plaquettes.

We verify reflection positivity on these tori for all bounded half-space functions. Partition the bond variables into the interiors of the two halves, the bonds lying on the cuts, and, when present, the bonds crossing a cut. A coordinate site cut has no plaquette with vertices strictly on both sides. Conditional on its boundary bonds, the action is a sum of a positive-side action, its reflected action, and an action on the boundary. Its reflection form is thus an integral of absolute squares.

At a diagonal cut, signed permutations reduce the geometry to $n = e_1 + e_2$, $c \in \mathbb{Z}$. A bond changes the height $x_1 + x_2$ by 0 or 1 in absolute value, so no bond crosses from a strictly negative height to a strictly positive one. The only plaquettes meeting both strict sides are in the $(1, 2)$ plane, with heights $c-1, c, c, c+1$. Each is cut into two two-bond paths between the two vertices at height c. Orienting both paths in the same direction expresses its Wilson factor as $c(P_+P_-^{-1})$. Reflection interchanges the paths, and eq. A.xxxvii supplies the required Gram expansion. Plaquettes in all other planes lie in a closed half. The bonds on the mirror, in directions 3 and 4, are fixed pointwise. Conditioning on them and multiplying the crossing expansions again gives an integral of squares. The same argument applies at the second cut.

For a coordinate link cut there are crossing bonds. Integrate them before evaluating a half-space function, which cannot depend on them. Let X and Y be the collections of bonds parallel to the cut in its two adjacent layers. The product crossing kernel before integration is

$$C(X,Y) = \prod_b c(X_b Y_b^{-1}).$$

It is a positive kernel by eq. A.xxxvii. If $T = (T_x)$ is a collection of group elements at the vertices of one boundary layer, set $(T \cdot X)_{xy} = T_x X_{xy} T_y^{-1}$. Let P_0 be Haar averaging under this action. Simultaneous gauge covariance gives $C(T \cdot X, T \cdot Y) = C(X,Y)$, since each plaquette trace changes by conjugation. Hence the associated integral operator commutes with the orthogonal projection P_0. Integration of the crossing bonds gives the kernel of

$$P_0 C = C P_0 = P_0 C P_0 \geq 0. \tag{A.xxxix}$$

Equivalently, in each Gram feature of the product expansion one may Haar-average both its X argument and its Y argument. Absolute uniform convergence justifies this operation. The crossing bonds at the two cuts are disjoint, so the two constructions multiply. This argument places no gauge-invariance condition on the test function.

In all three geometries the resulting reflection form has the form

$$\int \Theta_c F' F \, e^{-\beta \sum_p a_p} \, dU = \sum_\gamma b_\gamma \int \overline{I_\gamma(F';Y)} I_\gamma(F;Y) \, d\lambda(Y),$$
$$b_\gamma \geq 0, \tag{A.xl}$$

where Y denotes the fixed boundary variables; in the link case Y is absent. The functions I_γ integrate over one half, with its Boltzmann factor and the corresponding Gram feature. The boundary Boltzmann factor is included in the positive measure λ. Fubini's theorem follows from the coefficient summability above. This proves the Cauchy–Schwarz inequality for the reflection form and reflection positivity for all bounded half-space functions. Haar invariance, orientation reversal, and invariance of the plaquette trace also prove reflection invariance of the finite measure.

If E is a bounded real function on one closed half, multiplying the density by $e^{E+\theta_c E}$ preserves both properties: insert Fe^E in eq. A.xl. This observation will apply to folded perturbations.

A finite one-direction chessboard proof. Fix parallel slab faces at heights $c + kH$, where $H \geq 2$ is an integer. Use a split torus with normal height period $2MH$, $M \geq 1$. Its $2M$ slabs are indexed by $\mathbb{Z}/(2M)$. The group generated by reflection in the slab faces acts simply transitively on the slabs: its translations change the index by an even integer and its reflections send k to $2j - 1 - k$. Thus a function on any one slab has a well-defined copy on every other slab obtained using this group. Boundary bonds may belong to two closed slabs; this causes no problem, since the products are functions of the same boundary variables and eq. A.xl permits those variables in both half-algebras.

For positive bounded slab functions $f_0, \ldots, f_{2M-1}$ with strictly positive infima, denote by $Z^{\mathrm{dis}}(f_j)$ the partition function obtained by copying f_j into every slab. We claim

$$\int \prod_{j=0}^{2M-1} f_j \, e^{-\beta \sum_p a_p} \, dU \leq \prod_{j=0}^{2M-1} Z^{\mathrm{dis}}(f_j)^{1/(2M)}. \tag{A.xli}$$

Here is a proof valid for every M, without a power-of-two restriction. Transport the $2M$ functions to one reference slab and retain their labels. For any filling $\sigma : \mathbb{Z}/(2M) \to \{0, \ldots, 2M-1\}$, let $Z(\sigma)$ be the partition function obtained by placing profile $\sigma(k)$ in slab k. Put

$$w_j = Z^{\mathrm{dis}}(f_j)^{1/(2M)},$$
$$T(\sigma) = Z(\sigma)/\prod_k w_{\sigma(k)}.$$

All these numbers are positive, and every constant filling has $T = 1$. Reflection Cauchy–Schwarz at any pair of opposite slab faces gives

$$T(\sigma)^2 \le T(\sigma_+)T(\sigma_-), \tag{A.xlii}$$

where σ_+ retains one semicircle of labels and reflects it into the other, and σ_- does the reverse. Indeed, the unnormalized inequality is eq. A.xl; the two denominators together contain each original denominator factor twice.

There are finitely many fillings. If σ maximizes T, eq. A.xlii forces both reflected fillings to maximize it as well. Take a cyclic constant run of length l in a maximizing filling and reflect at the face at the start of that run. If $l \le M$, retaining the half beginning there gives a maximizing filling with a run of length $2l$. If $l \ge M$, retain a semicircle inside the run and reflect it; the result is constant. Iterating therefore produces a constant maximizing filling. The maximum is one. Applying this to the original filling proves eq. A.xli.

This argument uses reflection positivity only for finite tori. It does not infer the positivity of an arbitrary DLR state from the positivity of one adjacent-layer kernel.

A lower tangent for an arbitrary periodic DLR state. Let Q be a fixed finite-index period subgroup, let A be a real continuous local observable, and form the locally finite interaction array $\mathbb{A} = (\tau_v A)_{v\in Q}$. Write $\kappa = [\mathbb{Z}^4 : Q]^{-1}$. Let $\Gamma_i \subset Q$ be rectangular adapted period lattices with every side tending to infinity. On the corresponding tori put

$$Z_i(t\mathbb{A}) = \int \exp\left[t\sum_{v\in Q/\Gamma_i}(\tau_v A)^{\Gamma_i} - \beta\sum_p a_p\right] dU.$$

For every Q-invariant Wilson DLR state ν and every fixed real t,

$$\frac{1}{|D_i|}\log\frac{Z_i(t\mathbb{A})}{Z_i(0)} \ge t\kappa\nu(A) - o(1). \tag{A.xliii}$$

The error may depend on A, Q, β, t, but is uniform in ν.

For completeness, choose a fundamental parallelepiped D_i and integrate the positively oriented bonds anchored in it, using arbitrary external bonds as boundary conditions. Let W_i sum the source copies whose entire support is interior to this bond set. Conditional Jensen and the DLR equation give

$$\log\frac{Z^t_{D_i}(\eta)}{Z^0_{D_i}(\eta)} \ge t\gamma_{D_i}W_i(\eta),$$
$$\int \gamma_{D_i}W_i\, d\nu = \nu(W_i).$$

The open and torus Hamiltonians differ only in a boundary layer of fixed thickness, determined by the range of A and the plaquettes. Each energy term is bounded; each bond is met by only a fixed finite number of array terms. Thus their logarithmic partition ratios differ by at most $C_A(\beta + |t|\|A\|)|\partial_R D_i|$, uniformly in η. The number of omitted array anchors is at most $C_A|\partial_R D_i|$. Since the remaining Q-translates all have expectation $\nu(A)$, integrating the Jensen bound proves

eq. A.xliii, with an error bounded by a constant times $|\partial_R D_i|/|D_i|$. This ratio tends to zero for the adapted rectangular tori, including the diagonal ones. No differentiability of infinite-volume pressure is used.

Fold domination with a fixed common period. Fix one admissible mirror (n, c). Choose an even positive integer h and set

$$\begin{aligned} w &= hn \in P, \\ H &= n \cdot w = hq, \\ P_{\parallel} &= P \cap n^{\perp}, \\ Q &= P_{\parallel} \oplus \mathbb{Z}w, \\ Q_f &= P_{\parallel} \oplus \mathbb{Z}(2w). \end{aligned} \tag{A.xliv}$$

Both Q and Q_f are finite-index subgroups of P and $[Q : Q_f] = 2$. The tangent lattice $P_{\parallel}$ has an orthogonal integer basis: for a coordinate normal use the three other vectors $2e_j$; for $e_1 \pm e_2$ use $2(e_1 \mp e_2), 2e_3, 2e_4$, with the corresponding signed permutations for the other cases. Thus

$$\Gamma_{M,J} = JP_{\parallel} \oplus \mathbb{Z}(2Mw) \tag{A.xlv}$$

is an adapted rectangular split torus. Its shortest period tends to infinity as $M, J \to \infty$.

Assume for the moment that A is supported in the slab $c \le n \cdot x \le c + H$. The unfolded array contains all its translates by Q. The folded array contains translates of A in the even slabs and the reflected copies $\theta_{c+H}A$ in the odd slabs, with period Q_f. It is invariant under reflection at every face $c + kH$. In eq. A.xli take

$$f_k = \exp\left[t \sum_{v \in P_{\parallel}/JP_{\parallel}} (\tau_{v+kw}A)^{\Gamma_{M,J}}\right],$$
$$t > 0.$$

Their product is the unfolded perturbation. Their disseminated partition functions are identical: normal translation by kw carries one dissemination to another, permutes the slab faces, and preserves the unperturbed Wilson measure. Each is the folded partition function. Hence

$$Z_{M,J}(t\mathbb{A}^{\mathrm{u}}) \le Z_{M,J}(t\mathbb{A}^{\mathrm{f}}). \tag{A.xlvi}$$

Let $\mu_{M,J,t}$ be the normalized folded measure, and define

$$p_{M,J}(t) = |D_{M,J}|^{-1} \log[Z_{M,J}(t\mathbb{A}^{\mathrm{f}})/Z_{M,J}(0)].$$

It is convex, $p_{M,J}(0) = 0$, and therefore $p_{M,J}(t) \le t p'_{M,J}(t)$ for $t > 0$. The folded measure is invariant under Q_f and under θ_{c+H}. All its source terms consequently have the same expectation as A. There are $\kappa|D_{M,J}|$ terms, where $\kappa = [\mathbb{Z}^4 : Q]^{-1}$, so

$$p'_{M,J}(t) = \kappa \mu_{M,J,t}(A).$$

Combining this identity, eq. A.xliii, and eq. A.xlvi gives, for every $\nu \in \mathcal{G}^P_\beta$,

$$t\kappa\nu(A) - o(1) \le p_{M,J}(t) \le t\kappa\mu_{M,J,t}(A). \tag{A.xlvii}$$

The finite folded measure is invariant and reflection positive at every face $c + kH$. Indeed, its perturbation is $E + \theta_{c+kH}E$, where E is the sum of its copies lying in one closed half of the torus. Copies on a boundary can be assigned half weight to each side. The observation following eq. A.xl applies.

First fix $t > 0$ and let $M, J \to \infty$. Compactness gives a limit ρ_t. It is a DLR state for the folded, finite-range perturbed interaction, is Q_f-periodic, is invariant and reflection positive at all its slab faces, and satisfies $\nu(A) \le \rho_t(A)$. Here DLR passage is local and exact: any fixed collection of bonds, its interaction neighbourhood, and the support of a cylinder test embed injectively in all sufficiently large tori. The finite torus conditional equation is then exactly the required local DLR equation on that test. Its kernel is continuous, so the equation passes to a weak limit. A finite test on one half-space eventually lies in a single torus half; hence its reflection inequality passes to the limit as well.

Next let $t \downarrow 0$ along a subsequence for which ρ_t converges to ρ_0. On any fixed finite set, only finitely many source copies meet that set. Their bounded total contribution is $O(t)$ uniformly in the exterior configuration. The perturbed conditional densities thus converge uniformly to the Wilson conditional density. It follows that ρ_0 is a Wilson DLR state. It retains Q_f periodicity, reflection invariance and positivity at the slab faces, and $\nu(A) \le \rho_0(A)$.

Now apply the previously proved link-symmetry theorem to ρ_0. Its finite-index period Q_f is enough for that theorem; the result upgrades its period to P. We have proved the following domination:

$$\begin{aligned} &A \text{ supported in } [c, c+H], \quad \nu \in \mathcal{G}_\beta^P \quad \Longrightarrow \quad \exists \rho_0 \in \mathcal{G}_\beta^P : \\ &\rho_0 \text{ is invariant and positive at } (n, c), \qquad \nu(A) \le \rho_0(A). \end{aligned} \tag{A.xlviii}$$

The order of limits is essential: the surface error in eq. A.xlvii is divided by $t\kappa$, so volume tends to infinity before t tends to zero.

Separation closes the full half-space statement. Let $K = \mathcal{G}_\beta^P$ and let $K_c \subset K$ consist of the states invariant under θ_c and positive on every continuous cylinder function in the positive half-space. The set K is compact and convex: the DLR kernels preserve continuous cylinder functions, and all its defining equations and period equations are closed affine conditions. The set K_c is also closed and convex. It is nonempty, for example by the folded construction with $A = 0$.

For any real continuous local A, choose $v \in P$ so that $\tau_v A$ is supported strictly above the fixed mirror c. Then choose the even h large enough that this support lies below $c + H$. Such a v exists because $n \cdot (2ke_j) = 2k$ for a coordinate j with $n_j = 1$. Apply eq. A.xlviii to $\tau_v A$. Both ν and the resulting ρ_0 are P-invariant, so

$$\begin{aligned} \nu(A) = \nu(\tau_v A) \le \rho_0(\tau_v A) = \rho_0(A), \\ \rho_0 \in K_c. \end{aligned} \tag{A.xlix}$$

If some $\nu \in K$ were outside K_c, separation of a point from a compact convex set of measures would supply a real continuous function A with $\nu(A) > \sup_{\rho \in K_c} \rho(A)$. Continuous cylinder functions are uniformly dense in $C(\Omega)$, so A could be chosen local while retaining a strict inequality. This contradicts eq. A.xlix. Therefore $K = K_c$.

This step explains why the common period cannot be omitted. A state produced by folding initially has only the normal period $2w$. Such a large period need not move a test straddling the mirror into one slab. The link-symmetry theorem supplies $2\mathbb{Z}^4$ for both states in eq. A.xlix, independently of the observable and slab width. No claim that periodic DLR states are torus-selected is needed.

The mirror was arbitrary, so every periodic state is invariant and positive at all the stated mirrors on continuous local half-space functions. Finally, on the compact metrizable product of the positive half-space bonds, continuous cylinder functions are dense in $L^2(\nu)$. For bounded Borel F choose such F_j with $\|F_j - F\|_2 \to 0$. Reflection invariance makes Θ_c an isometry of $L^2(\nu)$, and Cauchy–Schwarz gives

$$|\nu(\Theta_c F_j F_j) - \nu(\Theta_c F F)| \le \|F_j - F\|_2(\|F_j\|_2 + \|F\|_2) \longrightarrow 0.$$

This proves eq. A.xxxvi on the complete positive-half algebra. Replacing a negative-half function by its reflected conjugate proves the other half. In particular, arbitrary finite stacks and their bounded Borel tests are covered. □

As a further consequence, every periodic state is invariant under all unit lattice translations: in direction e_μ the product of the site reflection at 0 and the link reflection at $1/2$ is translation by e_μ. It is also invariant under the hypercubic group. The coordinate reflections at zero generate coordinate sign changes, and the diagonal reflections with normal $e_\mu - e_\lambda$ interchange coordinates. These transformations generate all signed permutations.

Scope of the implication. Positive semidefinite contractions of an adjacent-layer kernel can produce a one-layer positivity test, just as skew contractions produce an antisymmetric invariance test. That observation alone supplies no identity reducing arbitrary odd or even stacks to a single layer. The proof above uses a different route: exact finite-torus positivity on the full half-algebra, a finite chessboard inequality, a lower tangent for each arbitrary periodic DLR state, and separation after the common period is established. Thus the all-stack and diagonal conclusions do not rest on a separate layer-elimination positivity rule. The result is at a fixed lattice spacing and coupling; it makes no continuum-limit or interacting renormalization-group assertion.

A.7. **A uniform lattice plateau estimate.** The estimate in this section is a consequence of translation invariance and reflection positivity in coordinate and diagonal directions. No infrared bound, mass gap, or differentiability of a pressure is assumed. The Gibbs equations are not needed in the proof. Their role in the Wilson application is to place every periodic Wilson Gibbs state in the reflection-positive class by Prop. A.11, proved independently in appendix A.6. The state here is a measure on the infinite lattice, rather than a finite-torus law lifted periodically.

By dilation it suffices initially to take lattice spacing one. Let ν be a translation-invariant probability measure on lattice configurations. A local function means a bounded Borel function of finitely many bonds. The algebra $\mathfrak{A}(S)$ contains the functions of bonds with both endpoints in S. Assume reflection positivity on the closed side of every coordinate mirror $x_\mu = c$, $c \in \frac{1}{2}\mathbb{Z}$, and every diagonal mirror $x_\mu \pm x_\lambda = c$, $c \in \mathbb{Z}$. These are sixteen families. Reflections act linearly on functions by pullback, without complex conjugation. Write θ_μ for reflection in $x_\mu = 0$.

For $r \ge 1$ an integer put $Q_r = [-r, r]^4$. For $A \in \mathfrak{A}(Q_r)$ define

$$\mathcal{N}_{\nu,r}(A) = \max_{\mu,\gamma} \nu\big(\overline{\theta_\mu \gamma A}\, \tau_{2re_\mu} \gamma A\big), \tag{A.l}$$

where γ ranges over the sixteen coordinate sign changes. The terms in this maximum are nonnegative squared reflection norms. Let $\mathcal{I}$ be the completed invariant σ-field for all lattice translations and set

$$\mathrm{Pl}_\nu(A, B) = \nu\big(\overline{E_\nu[A \mid \mathcal{I}]} E_\nu[B \mid \mathcal{I}]\big). \tag{A.li}$$

Proposition A.12 (Uniform plateau bound)**.** There is a numerical constant C such that, for the measures just specified, $A, B \in \mathfrak{A}(Q_r)$, and $y \in \mathbb{Z}^4$ with $n = |y|_\infty - 2r \ge 1$,

$$\left|\nu(\overline{A}\tau_y B) - \mathrm{Pl}_\nu(A,B)\right| \le C\left[(r/n)^2 + (r/n)^3\right]\mathcal{N}_{\nu,r}(A)^{1/2}\mathcal{N}_{\nu,r}(B)^{1/2}. \tag{A.lii}$$

The constant is independent of the measure, the support size, and any coupling. At lattice spacing a, the same statement holds with $Q_r = a[-r,r]^4$ and $n = |y|_\infty/a - 2r$. ■

We use the spectral theorem for commuting bounded normal operators and the mean ergodic theorem for a unitary operator. All estimates supplying the power in eq. A.lii, including the momentum-density estimate, are proved below.

A.7.1. *Reflection Hilbert spaces and their spectral carriers.* Reflection positivity and translation invariance first imply invariance under each of the indicated reflections. On one closed side, positivity makes $\langle F, G\rangle_\theta = \nu(\overline{\theta F}G)$ Hermitian. Testing it against 1 shows $\nu(F) = \nu(\theta F)$ for functions supported on that side. Any local function can be translated into the side by a lattice vector normal to the mirror. Reflecting reverses that translation; translation invariance therefore proves the equality for the original function. The reflections consequently act unitarily on $L^2(\nu)$.

In the coordinate picture μ, complete the quotient of local functions on $x_\mu \ge 0$ for the positive semidefinite form

$$\langle F, G\rangle_\mu = \nu(\overline{\theta_\mu F}G).$$

Denote the class of F by $[F]_\mu$. Its norm is at most $\|F\|_\infty$. Translation by e_μ induces a positive self-adjoint contraction T_μ, and tangential coordinate translations induce commuting unitaries $U_{\mu,\lambda}$, $\lambda \ne \mu$.

Here are the details of this elementary reconstruction. The normal translation is symmetric on the pre-Hilbert space by translation invariance and $\theta_\mu\tau_{e_\mu} = \tau_{-e_\mu}\theta_\mu$. Its quadratic form is nonnegative by reflection positivity at the mirror $x_\mu = -1/2$. Repeated Cauchy–Schwarz gives

$$\|T_\mu[F]\| \le \|[F]\|^{1-2^{-k}}\|T_\mu^{2^k}[F]\|^{2^{-k}}.$$

The last norm is at most $\|F\|_\infty$. Letting $k \to \infty$ proves contraction and preservation of null vectors. The bounded symmetric extension is self-adjoint and positive. Tangential translations preserve the form and have inverses of the same kind, so are unitaries. Their commutations on local functions pass to the completion. In particular, for integers $s \ge 0$ and tangential $w \in \mathbb{Z}^4$,

$$\langle [F]_\mu, T_\mu^s U_\mu^w [G]_\mu\rangle = \nu(\overline{\theta_\mu F}\tau_{se_\mu + w}G). \tag{A.liii}$$

There is a joint spectral measure with variables $(E,\eta) \in [0,\infty] \times (-\pi,\pi]^3$ for which $T_\mu = e^{-E}$ and $U_{\mu,\lambda} = e^{i\eta_\lambda}$. The value $E = \infty$ represents a possible kernel of T_μ. The following carrier bound is essential:

$$\begin{gathered}\cosh E + \cos\eta_\lambda \ge 2\\ (\lambda \ne \mu),\\ E \ge \kappa|\eta|_\infty,\\ \kappa = \frac{2}{\pi}\log(1+\sqrt{2}).\end{gathered} \tag{A.liv}$$

We now derive it from the diagonal mirrors.

A.7.1.1. A polynomial-measure lemma. Suppose a finite positive measure m on the closed unit disc satisfies

$$\int |p(z)|^2\, dm(z) \le C_0 \sup_{[0,1]} |p| \sup_{|z|=1} |p| \tag{A.lv}$$

for every polynomial. Then it gives zero mass to $\{|1+z|^2 < 4|z|\}$.

To prove this without invoking an extremal-function theorem, set

$$L(\zeta) = \operatorname{Log} \frac{1+\zeta}{1-\zeta},$$
$$f_s(z) = \frac{\cos(sL(\sqrt{z}))}{\cosh(\pi s/2)},$$
$$s > 0.$$

The Cayley transform has positive real part in the disc. Thus L is holomorphic, odd, and satisfies $|\operatorname{Im} L| < \pi/2$. The cosine is even in $\sqrt{z}$, so f_s is a single-valued holomorphic function of z in the disc. Put $\varphi(z) = |\operatorname{Im} L(\sqrt{z})|$. The identity $|\cos(u+iv)|^2 = \cos^2 u + \sinh^2 v$ gives

$$|f_s| \le 1,$$
$$\sup_{[0,1)} |f_s| \le \cosh(\pi s/2)^{-1},$$
$$|f_s(z)| \ge \frac{\sinh(s\varphi(z))}{\cosh(\pi s/2)}.$$

Apply eq. A.lv to Taylor approximants of $f_s(qz)$, $0 < q < 1$, which converge uniformly on the closed disc. The resulting inequality, followed by Fatou as $q \uparrow 1$, is

$$\int \sinh^2(s\widetilde{\varphi}(z))\, dm(z) \le C_0 \cosh(\pi s/2).$$

Here $\widetilde{\varphi} = \varphi$ inside the disc, equals $\pi/2$ on the boundary except at 1, and may be set to zero at 1. Indeed the radial limits of φ at the other boundary points equal $\pi/2$. For each $\epsilon > 0$ this inequality forces $m(\{\widetilde{\varphi} \ge \pi/4 + \epsilon\}) = 0$ by letting $s \to \infty$. Finally, if $z = \zeta^2$ inside the disc,

$$\tan \varphi(z) = \frac{2|\operatorname{Im} \zeta|}{1 - |\zeta|^2},$$
$$|1+\zeta^2|^2 - 4|\zeta|^2 = (1-|\zeta|^2)^2 - 4(\operatorname{Im}\zeta)^2.$$

Thus $\varphi > \pi/4$ is precisely the excluded set. The same assertion holds on the boundary, where only $z = 1$ is not excluded. This proves the lemma.

Fix $\lambda \ne \mu$ and a local A on $x_\mu \ge 0$. Choose an integer $R \ge 1$ enclosing its support in $0 \le x_\mu \le R$, $|x_\lambda| \le R$. Let θ' be reflection in $x_\mu + x_\lambda = R$, put $B = \theta'\theta_\mu A$, $A_0 = \tau_{2Re_\mu} A$, and $d_\pm = e_\mu \pm e_\lambda$. Both B and A_0 lie on the positive side of this diagonal mirror. Reconstruction at that mirror gives a positive contraction T_+ for d_+ and a unitary V_- for d_-. The preceding reconstruction proof applies: positivity of T_+ uses the admissible diagonal mirror one height unit behind, and its contraction uses the bounded local representative. For integers $k, l \ge 0$,

$$\langle [A]_\mu, T_\mu^{2R+k+l} U_{\mu,\lambda}^{k-l} [A]_\mu \rangle = \langle [B]', T_+^k V_-^l [A_0]' \rangle. \tag{A.lvi}$$

Both sides are the same lattice correlation: their displacement is $2Re_\mu + kd_+ + ld_-$. No analytic continuation is used here.

Push the spectral measure of $[A]_\mu$, multiplied by t^{2R}, forward by $z = t\zeta$, where $t = e^{-E}$ and $\zeta = e^{i\eta_\lambda}$. Expanding a polynomial and its conjugate in eq. A.lvi gives

$$\int |p|^2 \, dm = \langle [B]', p(T_+)\overline{p}(V_-)[A_0]' \rangle,$$

$$\overline{p}(z) = \sum_k \overline{p_k} z^k.$$

The spectral theorem bounds this by the right side of eq. A.lv, with $C_0 = \|[B]'\|\|[A_0]'\|$. The lemma excludes $|1 + t\zeta|^2 < 4t$. Since $t^{2R} > 0$ for $t > 0$, the joint spectral measure of $[A]_\mu$ itself vanishes there. The local classes are dense, so the operator spectral measure vanishes there. At $t = 0$ the equivalent operator inequality is automatically satisfied. We have proved

$$(1 - T_\mu)^2 \geq T_\mu(2 - U_{\mu,\lambda} - U_{\mu,\lambda}^*).$$

For $0 < t \leq 1$ this is equivalent to the first assertion of eq. A.liv. The curve is $E \geq 2 \operatorname{arcsinh} |\sin(\eta_\lambda/2)|$. Concavity of $x \mapsto \operatorname{arcsinh} \sin x$ on $[0, \pi/2]$ puts it above its endpoint chord, giving the constant κ. Intersect the three carriers to obtain eq. A.liv.

A.7.2. *The zero-energy plateau is the invariant conditional field.* Write $\Pi_\mu = 1_{\{E=0\}}$; spectral dominated convergence gives $T_\mu^s \to \Pi_\mu$ strongly as $s \to \infty$. By eq. A.liv, every tangential unitary equals the identity on this zero-energy space. Let $\mathcal{I}_\mu$ denote the invariant field for the single translation τ_{e_μ}. The mean ergodic theorem on $L^2(\nu)$ and eq. A.liii give

$$\langle [F]_\mu, \Pi_\mu [G]_\mu \rangle = \nu(\overline{\theta_\mu F} E_\nu[G \mid \mathcal{I}_\mu]). \tag{A.lvii}$$

Reflection acts as a unitary involution on $L^2(\mathcal{I}_\mu, \nu)$. That involution is positive. Indeed, approximate an invariant function J in L^2 by bounded local G_k and translate each approximant into $x_\mu \geq 0$. The error is unchanged because J is invariant. Reflection positivity then passes to the L^2 limit and gives $\langle \theta_\mu J, J \rangle \geq 0$. A positive self-adjoint involution has only the eigenvalue $+1$, so reflection fixes J. Consequently eq. A.lvii equals

$$\nu(\overline{E_\nu[F \mid \mathcal{I}_\mu]} E_\nu[G \mid \mathcal{I}_\mu]).$$

The map $\Pi_\mu[F]_\mu \mapsto E_\nu[F \mid \mathcal{I}_\mu]$ is therefore an isometry onto a dense subspace of $L^2(\mathcal{I}_\mu, \nu)$. Density follows by the same shifted local approximations. It intertwines the tangential translations. Since those translations are the identity on Π_μ, every $\mathcal{I}_\mu$-measurable function is invariant under all four coordinate translations. Hence $\mathcal{I}_\mu = \mathcal{I}$ modulo ν, and

$$\langle [F]_\mu, \Pi_\mu [G]_\mu \rangle = \nu(\overline{E_\nu[F \mid \mathcal{I}]} E_\nu[G \mid \mathcal{I}]). \tag{A.lviii}$$

This also proves that $E_\nu[\theta_\mu F \mid \mathcal{I}] = E_\nu[F \mid \mathcal{I}]$. The possible T_μ kernel has energy infinity and does not affect any of these zero-energy identities.

A.7.3. *A positive time filter and a bounded momentum density.* For $A \in \mathfrak{A}(Q_r)$ define its cube vector $\Phi_A^\mu = [\tau_{re_\mu} A]_\mu$. Translation invariance shows $\|\Phi_A^\mu\|^2 = \nu(\overline{\theta_\mu A} \tau_{2re_\mu} A)$. Thus eq. A.l is the maximum of the corresponding squared norms of γA. In particular $\mathcal{N}^{1/2}$ is a seminorm, is invariant under the sign-change group, and is at most the supremum norm.

Let b be the nonnegative periodic triangle function supported on $[-\pi/4, \pi/4]$, normalized to have mean one. Its Fourier coefficients are

$$b_m = \left(\frac{\sin(m\pi/8)}{m\pi/8}\right)^2,$$
$$b_0 = 1.$$

This follows by writing the triangle as the convolution of two interval indicators and normalizing their zeroth coefficients. Let z_m be the even period-four sequence $(0, 1, -2, 1)$ and set

$$\widehat{q}(u) = \sum_{m\in\mathbb{Z}} z_m b_m e^{imu},$$
$$\Psi(q) = 2\sum_{m\geq 1} z_m b_m q^{m-1},$$
$$0 \leq q \leq 1. \tag{A.lix}$$

Both series converge absolutely. The identity $z_m = \frac{1}{2}i^m - (-1)^m + \frac{1}{2}(-i)^m$ gives

$$\widehat{q}(u) = \tfrac{1}{2}b(u+\pi/2) - b(u+\pi) + \tfrac{1}{2}b(u-\pi/2).$$

In particular $\widehat{q}$ vanishes whenever $\operatorname{dist}(u, 2\pi\mathbb{Z}) \leq \pi/4$. The crucial energy property is

$$0 \leq \Psi \leq C,$$
$$\Psi(1) = 0,$$
$$\Psi(e^{-x}) \geq c\tanh x$$
$$(x > 0). \tag{A.lx}$$

To verify its nontrivial lower bound, use the Poisson kernel

$$P_x(u) = \sum_m e^{-|m|x}e^{imu} = \frac{\sinh x}{\cosh x - \cos u}.$$

Since $z_0 = 0$, the Poisson integral yields

$$\Psi(e^{-x}) = \frac{e^x}{2\pi}\int_{|v|\leq\pi/4} b(v)[\tfrac{1}{2}P_x(v-\pi/2) + \tfrac{1}{2}P_x(v+\pi/2) - P_x(v-\pi)]\,dv.$$

Writing $C_x = \cosh x$, the bracket is

$$\sinh x\left[\frac{C_x}{C_x^2 - \sin^2 v} - \frac{1}{C_x + \cos v}\right] \geq \frac{\tanh x}{\sqrt{2}(C_x + 2^{-1/2})}.$$

For this inequality first replace the first fraction by $1/C_x$, then use $\cos v \geq 2^{-1/2}$ and monotonicity of $q/(C_x + q)$. The factor $e^x/(C_x + 2^{-1/2})$ has a positive lower bound, and b has mean one. This proves the lower bound. The upper bound follows from absolute convergence of eq. A.lix, and $\Psi(1) = \widehat{q}(0) = 0$. These statements extend continuously to $q = 0$, where $\Psi(0) = 2b_1 > 0$.

First assume $\theta_\mu A = \epsilon A$, $\epsilon \in \{1, -1\}$, and put $\Phi = \Phi^\mu_A$. For $s \in \mathbb{Z}$ and tangential $w \in \mathbb{Z}^3$, define

$$g_A(s, w) = \epsilon\nu(\overline{A}\tau_{se_\mu + w}A).$$

For $s \geq 2r$, the cube-crossing identity from eq. A.liii gives

$$g_A(s,w) = \langle \Phi, T_\mu^{s-2r} U_\mu^w \Phi \rangle,$$
$$|g_A(s,w)| \le \mathcal{N}(A). \tag{A.lxi}$$

Reflection in $x_\mu = 0$ and translation invariance give $g_A(-s,w) = g_A(s,w)$ and $g_A(s,-w) = \overline{g_A(s,w)}$. Define

$$K_A(w) = \sum_{m\in\mathbb{Z}} z_m b_m g_A(2rm, w).$$

The term at $m = 0$ vanishes. Every other time separation has absolute value at least $2r$, so eq. A.lxi bounds this sum by $C\mathcal{N}(A)$, uniformly in w. By the same identities,

$$K_A(w) = \langle \Phi, \Psi(T_\mu^{2r}) U_\mu^w \Phi \rangle = \int e^{i\eta\cdot w}\, dM_A(\eta), \tag{A.lxii}$$

where M_A is the momentum pushforward of the positive measure

$$\Psi(e^{-2rE})\, d\langle \Phi, \mathsf{E}_\mu(E,\eta)\Phi\rangle.$$

Thus M_A is positive; the signs in z_m do not compromise positivity because eq. A.lx controls their spectral sum.

For $|w|_\infty \ge 2r$, choose $\lambda \ne \mu$ at a maximal coordinate. Replace w by $-w$ if necessary, which only conjugates $K_A(w)$, and suppose $w_\lambda \ge 2r$. In the coordinate picture λ, for every integer s the same correlation is

$$g_A(s,w) = \epsilon \left\langle \Phi^\lambda_{\theta_\lambda A}, T_\lambda^{w_\lambda - 2r} U^s_{\lambda,\mu} \prod_{\lambda' \ne \lambda,\mu} U^{w_{\lambda'}}_{\lambda,\lambda'} \Phi^\lambda_A \right\rangle.$$

Sum this identity at $s = 2rm$ against $z_m b_m$. In this picture the filter becomes the bounded spectral function $\widehat{q}(2r\eta'_\mu)$ of a unitary translation. Whenever it is nonzero, $|\eta'_\mu| > \pi/(8r)$, since the gap around zero modulo 2π includes the ordinary interval about zero. Aliases at other multiples of 2π do not invalidate this implication. The carrier eq. A.liv consequently gives $E' \ge \alpha = \kappa\pi/(8r)$ on the filter support. Both cube vectors have norm at most $\mathcal{N}(A)^{1/2}$. Hence

$$|K_A(w)| \le C\mathcal{N}(A) e^{-\alpha(|w|_\infty - 2r)}$$
$$(|w|_\infty \ge 2r). \tag{A.lxiii}$$

There are $O(r^3)$ points with $|w|_\infty < 2r$ and $O(k^2)$ points on the shell $|w|_\infty = k$. Since α is a fixed constant times r^{-1}, eq. A.lxiii and the previous bound give

$$\sum_w |K_A(w)| \le C\mathcal{N}(A) \left[r^3 + \sum_{j\ge 0} (r^2 + j^2) e^{-\alpha j} \right] \le C r^3 \mathcal{N}(A).$$

Therefore $h_A(\eta) = (2\pi)^{-3} \sum_w K_A(w) e^{-i\eta\cdot w}$ is continuous and bounded by $Cr^3\mathcal{N}(A)$. It has the same Fourier coefficients as M_A. Uniqueness of a finite measure from its Fourier coefficients follows by density of trigonometric polynomials in the continuous functions on the torus. Consequently

$$M_A = h_A(\eta)\, d\eta,$$

$$0 \le h_A \le Cr^3\mathcal{N}(A). \tag{A.lxiv}$$

This is a derived bound for an energy-weighted momentum measure. It is not an assumed bound for the unweighted spectral measure.

A.7.4. *Removing the filter and proving the plateau estimate.* Let $m_\Phi = \langle \Phi, \mathsf{E}_\mu(\cdot)\Phi\rangle$. Since M_A has a density, it has no atom at $\eta = 0$. By eq. A.lx, its weight is strictly positive at every finite $E > 0$. Thus m_Φ gives zero mass to $\{0 < E < \infty, \eta = 0\}$. The energy-zero contribution is precisely the plateau and was not removed by this inference: its weight is zero. The point $E = \infty$ contributes zero to T_μ^n for $n \ge 1$.

For $E > 0$ the function $e^{-nE}/\tanh(2rE)$ decreases in E. Using eq. A.lx, then eq. A.liv, and then eq. A.lxiv, gives

$$\begin{aligned}\langle \Phi, T_\mu^n(1-\Pi_\mu)\Phi\rangle &\le C\int_{\mathbb{T}^3} \frac{e^{-n\kappa|\eta|_\infty}}{\tanh(2r\kappa|\eta|_\infty)} h_A(\eta)\, d\eta \\ &\le Cr^3\mathcal{N}(A)\int_0^\infty u^2 e^{-n\kappa u}\left(1+\frac{1}{2r\kappa u}\right) du \\ &\le C\mathcal{N}(A)\big[(r/n)^3 + (r/n)^2\big]. \end{aligned} \tag{A.lxv}$$

In the second line we used $1/\tanh v \le 1 + 1/v$ and the three-dimensional cube shell volume $24u^2du$. The apparent singularity at $u = 0$ is integrable. The pushforward step can include $E = \infty$ and other unused spectral mass, since the integrand is nonnegative; it does not add an atom at $\eta = 0$ by eq. A.lxiv.

For an arbitrary complex A, write $A = A_+ + A_-$ with $A_\pm = (A \pm \theta_\mu A)/2$. The seminorm property and sign-change invariance imply $\mathcal{N}(A_\pm) \le \mathcal{N}(A)$. The operator $D_n = T_\mu^n(1-\Pi_\mu)$ is positive, so its quadratic form at a sum is at most twice the sum of its two quadratic forms. Applying eq. A.lxv to A_+ and A_- proves eq. A.lxv, with an enlarged numerical constant, for every A.

Finally suppose $y_\mu = |y|_\infty > 0$. The cube-crossing identity is

$$\nu(\overline{A}\tau_y B) = \langle \Phi^\mu_{\theta_\mu A}, T_\mu^n U_\mu^{y_\perp}\Phi^\mu_B\rangle.$$

The Π_μ term is eq. A.li by eq. A.lviii, conditional invariance under translations and reflection, and $U_\mu^{y_\perp}\Pi_\mu = \Pi_\mu$. For the remaining term, Cauchy–Schwarz in the positive form of D_n bounds the modulus by the geometric mean of the two quantities in eq. A.lxv. The tangential unitary commutes with D_n, so does not change that form. Also $\mathcal{N}(\theta_\mu A) = \mathcal{N}(A)$. This proves eq. A.lii. For a negative maximal coordinate use $\nu(\overline{A}\tau_{-y}B) = \overline{\nu(\overline{B}\tau_y A)}$ and conjugate symmetry of the plateau. Dilation restores spacing a and replaces distances by their numbers of lattice steps, proving the stated scaling. □

For an ergodic state the invariant conditional expectations are constants, so eq. A.lii is clustering. For a mixture its generally nonzero invariant plateau is retained. When local continuum representatives have large supremum norms, the reflection sizes in eq. A.l can still be bounded by separated reflected moments. This is the reason eq. A.lii, rather than a supremum-norm substitute, is the useful estimate for a continuum phase construction. No spectral gap above the invariant subspace has been proved: the filter weight vanishes linearly near zero energy, and removing it is exactly what produces the inverse-square term.

Corollary A.13 (Closedness of the ergodic reflection-positive class)**.** At fixed spacing, the translation-invariant measures satisfying these mirror-positivity hypotheses form a weakly closed set, and so does its subset of ergodic measures. ■□

Proof. Translation invariance and each positivity inequality on continuous local functions pass to a weak limit. Reflection invariance passes as well, and approximation in L^2 extends positivity to bounded Borel local functions. Thus the full reflection-positive class is closed.

Suppose now that ν_j are ergodic in this class and converge weakly to ν. For fixed continuous local A, B in a common cube, their plateaux under ν_j are $\overline{\nu_j(A)}\nu_j(B)$. Passing to the limit in eq. A.lii gives

$$|\nu(\overline{A}\tau_y B) - \overline{\nu(A)}\nu(B)| \le C[(r/n)^2 + (r/n)^3]\|A\|_\infty \|B\|_\infty.$$

In particular the correlation converges to the product along $y = ke_\mu$, $k \to \infty$. Its Cesaro means, with $A = B$, identify the squared L^2 norm of $E_\nu[A \mid \mathcal{I}_\mu]$ by the mean ergodic theorem. That norm is $|\nu(A)|^2$, so the conditional expectation is constant. Continuous local functions are dense in $L^2(\nu)$; hence $\mathcal{I}_\mu$, and therefore $\mathcal{I}$, is trivial. This is ergodicity of ν. □

A.8. **Coupling coordinates and conditional comparisons.** The coupling, matched-run and Ward estimates below imply their stated conclusions by elementary analytic arguments. The interacting estimates themselves remain hypotheses. The pointed families, companions, physical Hessian extractor, midpoint tents, list action and radius weights are defined in § 2.2. The pointed cancellation, nonlinear window estimate and finite formation sections have their stated background and domain hypotheses. None computes the one-loop trace or proves a matched-cutoff or typed Ward estimate.

Vacuum accounting and the stronger coupling inputs. Write $P_\circ = (\mathbf{1}, 0)$. With $E_{(0)} = 0$, the accumulated vacuum normalization is defined by

$$\begin{aligned} E_{(k+1)} &= E_{(k)} + \mathsf{e}_k, \qquad E = E_{(K)}, \\ \mathsf{e}_k &= \mathfrak{n}_k + \sum_X \left[\sum_{z \in X} E^{(k+1)}(X, P_\circ, z) + R^{(k+1)}(X, P_\circ) \right]. \end{aligned} \tag{A.lxvi}$$

Here $\mathfrak{n}_k$ is the assigned field-independent step number. For $\tau = x/\theta$, its Gaussian scaling part is $-3|\mathfrak{X}_k| \log \tau/2$; $\mathfrak{X}_k$ is the set of non-pivot bonds, with three real color coordinates per bond. Haar and gauge-fixing counts are included in that assigned number. The family in eq. A.lxvi contains the unnormalized local Gaussian numbers: $\mathbf{N}^\circ = \mathbf{N} + \mathbf{N}^\circ(P_\circ)$. Those numbers must not also be added to $\mathfrak{n}_k$. Equivalently, using $\mathbf{N}$ requires adding the removed local numbers once separately. This convention counts every constant once. At fixed calibration, $\mathfrak{n}_k, \mathbf{N}^\circ, \mathbf{G}_\tau$ have zero source derivative. Source energies are the differences of the displayed E- and R-values between sourced and unsourced runs. More explicitly, the localized source energy of level j is

$$e_j(X; w) = \sum_{z \in X} \{E^{(j),w}(X, P_\circ, z) - E^{(j),0}(X, P_\circ, z)\} + R^{(j),w}(X, P_\circ) - R^{(j),0}(X, P_\circ).$$

The source vacuum exponent is the sum of these values over levels and local domains. This definition introduces no additional arbitrary normalizer. The zero source derivative of the Gaussian pieces refers to the frozen potential presentation. Moving the source into the Gaussian action weight changes the determinant and is another presentation, whose numbers must be compared by its own sum rule.

Requirement B3 retains an actual one-loop input:

$$b_i = b_i^0 + f_i + q_i, \qquad c_0 = \frac{11N^2}{12\pi^2} \log L, \qquad \left| \sum_{i=1}^{r} (b_i^0 - c_0) \right| \le C_0. \tag{A.lxvii}$$

Here $N = 2$, and b_i^0 is independent of the bare coupling and admissible torus. A summable estimate $|b_i^0 - c_0| \le CL^{-\sigma i}$, $\sigma > 0$, suffices. The asserted normalization requires the actual lattice determinant/trace calculation and its zero-mode convention. Gaussian integration alone does not establish it.

Some later slope conclusions require more than eq. A.lxvii. Fix a calibration $\theta_0, \dots, \theta_K$, independent of the varied bare coupling. Put $\xi_j = \theta_j - x_j$. Require the actual unmarked f_i and marked q_i, for $1 \le i \le K$, to be jointly holomorphic on the independent earlier-variable polydiscs with radii

$$\begin{aligned} r_j^{\mathrm{u}} &= c\theta_j, & \sup|f_i| &\le C\theta_i^{-1/2}(\log\theta_i)^p, \\ r_j^{\mathrm{m}} &= c\theta_j(\log\theta_j)^{-C_*}, & \sup|q_i| &\le C\theta_i^{-p_R}, \qquad 0 \le j < i. \end{aligned} \tag{A.lxviii}$$

The constants $p, C_* \ge 0$, $p_R > 0$, and $c > 0$ are fixed. The actual runs must remain in the half polydiscs, with common localization, counterterms and admissible domains. These are stronger actual coupling inputs, including the marked tube. They are not inferred from holomorphy of the bare partition function, nor from a running-radius activity bound followed by two field derivatives. The marked exponent is often $p_R = (\kappa_0 - 4)/2$, but any later use must retain the actual proved loss and tube radius. All norms and source Taylor expansions here use fixed declared domains; shrinking the field domains while differentiating would be a different claim. Compatibility of the companion and earlier cutoff prefix also remains part of the construction.

A dimension-free polydisc lemma and its precise consequence.

Lemma A.14 (The bounded-polydisc estimate, code 1*e*.3.7)**.** If F is holomorphic on $\prod_{j=1}^d D(0, r_j)$ and $|F| \le B$, then

$$\sum_j r_j \left(1 - \frac{|\xi_j|^2}{r_j^2}\right) |\partial_j F(\xi)| \le B. \tag{A.lxix}$$

On the half polydisc this implies

$$\begin{aligned} \left|\sum_j v_j \partial_j F(\xi)\right| &\le \tfrac{4}{3} B \max_j |v_j|/r_j, \\ |F(\xi) - F(\xi')| &\le \tfrac{4}{3} B \max_j |\xi_j - \xi_j'|/r_j. \end{aligned} \tag{A.lxx}$$

No constant depends on d. ■

Proof. The case $B = 0$ is immediate. Set $a_j = \xi_j/r_j$ and $\phi_a(z) = (a + z)/(1 + \bar{a}z)$. Choose unit complex numbers ϵ_j so the terms in the derivative at zero of

$$H(z) = F\big(r_j \phi_{a_j}(\epsilon_j z)\big)_{j=1}^d$$

have the same argument. Since $\phi_a'(0) = 1 - |a|^2$, the modulus of that derivative is the left side of eq. A.lxix. The one-variable Schwarz lemma, applied after a disc automorphism sending $H(0)/B$ to zero, gives $|H'(0)| \le B(1 - |H(0)/B|^2) \le B$. The Schwarz lemma itself follows by applying the maximum principle to a holomorphic function vanishing at zero divided by z, first on smaller circles and then letting their radius increase to one. An attained boundary value makes H constant. On the half polydisc, eq. A.lxix gives $\sum_j r_j |\partial_j F| \le 4B/3$. This proves the first estimate of eq. A.lxx; integration along the segment from ξ to ξ', which stays in that half polydisc, proves the second. □

Assume $\theta_j \ge \theta_i$ for $j < i$, with all centers sufficiently large that $t/(\log t)^{C_*}$ is increasing. Applying the lemma separately in eq. A.lxviii bounds the absolute row sum by

$$a_i := \sum_{j<i} |\partial_{\xi_j}(f_i + q_i)| \le C\left[\theta_i^{-3/2}(\log\theta_i)^p + \theta_i^{-p_R-1}(\log\theta_i)^{C_*}\right]. \tag{A.lxxi}$$

Suppose also that the calibration has backward linear drift $\theta_i \ge h + c_1(K - i)$, $c_1 > 0$. Comparison with integrals gives

$$A := \sum_{i=1}^K a_i \le C\left[h^{-1/2}(\log h)^p + h^{-p_R}(\log h)^{C_*}\right] \longrightarrow 0.$$

Indeed substitute $t = hu$ in each integral; the integrals of $u^{-1-\delta}(1+|\log u|)^r$ are finite for $\delta > 0$. A drift with a fixed additive error is equivalent after increasing h.

For the actual differentiable recursion $x_0(\beta) = \beta$, $x_i(\beta) = \beta - \sum_{r\le i} b_r(\beta)$, put $M_i = \max_{j\le i} |x_j'|$. Calibration is held fixed, so $|b_i'| \le a_i M_{i-1}$ and $M_i \le (1+a_i)M_{i-1}$. Consequently

$$M_K \le \prod_i (1+a_i) \le e^A, \qquad \sum_i |b_i'| \le \prod_i (1+a_i) - 1 \le e^A - 1. \tag{A.lxxii}$$

The middle estimate telescopes the successive products. In particular

$$|x_K'(\beta) - 1| \le \sum_i |b_i'| \le e^A - 1 = O\Big(h^{-1/2}(\log h)^p + h^{-p_R}(\log h)^{C_*}\Big).$$

On witness prefixes with terminal $h \to \infty$, under the same uniformly admitted hypotheses, this makes the constant-source normalization tend to one with the displayed error. Identifying that prefix and derivative with the actual F3 witness remains an input; the estimate does not establish that identification. At weak coupling $A \le \log(4/3)$, hence the last sum is at most $1/3$. Constant sources varying β to $\beta - \zeta$ have the same absolute slopes. This supplies the slope premise in eq. 4.clxxv under eq. A.lxviii and the admitted drift/domain conditions. It does not prove those conditions. Reality gives $2/3 \le x_K'(\beta) \le 4/3$ on an admitted real interval. Integrating the derivative proves the uniform terminal bi-Lipschitz bound. If the same row estimates hold on a whole complex offset disc, write $\Xi_K(t) = h - t + R_K(t)$; then $R_K(0) = 0$ and $\sup|R_K'| \le 1/3$. One can construct the matching inverse without any global uniqueness assertion. For $|t| < \varrho$, solve

$$s = h - \Xi_K(t) + R_{K+1}(s)$$

on the closed disc $|s| \le 2\varrho$, assuming the functions are defined on a neighborhood of that disc. Its right side maps the disc strictly inside itself and has Lipschitz constant at most $1/3$. Iteration converges geometrically to its unique solution, uniformly on smaller t-discs. The iterates are holomorphic, so the solution is holomorphic; reality gives a real solution at real t. The bounds $|\Xi_K(t) - h| \le 4|t|/3$ and $|\Xi_{K+1}(s) - h| \ge 2|s|/3$ give $|\mathfrak{n}_K(t)| \le 2|t|$. This supplies the coordinate part of E2 when its actual full-disc premises are available, rather than proving the existence of those premises. Nor does a polydisc supremum give forgetting in $i - j$: $F_i(\xi) = B\xi_0/r_0$ is bounded by B for every i, with unchanged first derivative.

The fixed-torus coordinate and the matched limit. Fix a torus T_ℓ, $h = g^{-2}$, and a separated tuple of $n \ge 2$ compact Lipschitz tests, with the fixed norm $\|\cdot\|_\sharp$ of the source convention. Every bound below includes their product norm; its constant may depend on order and positive separation. Let $\beta_K^\natural$ be its torus-independent tuned bare center and write depth as $[s] = K - k$. Define

$$\begin{aligned} \mathsf{J} &= [-\varrho/4, \varrho/4], \qquad \Xi_K(t) = x_K^\natural(\beta_K^\natural - t), \\ F_{K,n}(t;\mathbf{f}) &= S^T_{n;K,\ell,\beta_K^\natural - t}(\mathbf{f}), \qquad \mathfrak{n}_K = \Xi_{K+1}^{-1} \circ \Xi_K. \end{aligned} \tag{A.lxxiii}$$

The inverse is the specified holomorphic matching branch. Inputs E2–E3 require its existence on a common neighborhood, reality on the real axis, uniform upper and lower real Lipschitz bounds for Ξ_K, $\Xi_K(0) = h$, and $|\Xi_K(t) - h + t| \le |t|/3$. On a fixed smaller complex neighborhood of J, require

$$\begin{aligned} &|\mathfrak{n}_K - t| + |\mathfrak{n}_K' - 1| + |\Xi_{K+1} - \Xi_K| \le C\eta_K, \\ &|F_{K,n}(t) - \mathfrak{n}_K'(t)^n F_{K+1,n}(\mathfrak{n}_K(t))| \le C_{n,\ell,\mathbf{f}}\epsilon_K, \qquad \sum_K (\eta_K + \epsilon_K) < \infty. \end{aligned} \tag{A.lxxiv}$$

The one-run estimates below supply the uniform holomorphic bound for $F_{K,n}$ on a slightly wider strip, for example $\mathcal{N}_\ell = \{|\operatorname{Re} t| < \varrho/2,\ |\operatorname{Im} t| < \upsilon_\ell\}$, where $\upsilon_\ell = c\varrho e^{-C(g)\ell^4} > 0$. Matching paths stay in that strip after shrinking it. Constants and cutoff floors may depend on this fixed ℓ. Often $\eta_K = \Upsilon_K \le CK^{-3/2}(\log(2+K))^{p+1}$; a separate remainder error Π_K is part of ϵ_K, not a polarization tensor.

The scalar coordinate and the local energies. The two estimates in eq. A.lxxiv use different comparisons. For the coordinate, let $b^A_{[u]}(t)$ and $b^B_{[u]}(t)$ be the scalar increments at the same depth for the two tuned runs. Compare them at the matched constant offsets t and $\mathfrak{n}_K(t)$, using the inverse branch already constructed. They include their marked remainders. We must show on a common offset disc, with fixed room about $|t| \le \varrho$, that

$$\sup_t |b^B_{[u]}(\mathfrak{n}_K(t)) - b^A_{[u]}(t)| \le C(\rho^{K-u} + \Pi_K), \qquad 0 \le u < K. \tag{A.lxxv}$$

Here $0 < \rho < 1$ and Π_K is the error from the unmatched large-field terms at the bare end. The derivative of each increment, including the extra increment $b^B_{[K]}$, has the one-run bound $Ca_{[u]}$, with $a_{[u]} \le C\theta^{-3/2}_{[u]}(\log \theta_{[u]})^p$. This uses the sensitivity estimate on the actual full disc as in eq. A.lxxii. The derivative loses one power of $\theta_{[u]}$ from the holomorphy radius in the bound for the part beyond one loop.

A.8.0.1. Scalar increments of partnered runs. Use the decomposition eq. 5.xxxii into the one-loop increment, the unmarked analytic coefficient f, and the marked correction q. The one-loop difference has a geometric bound at equal depth by eq. 5.cccxi at equal isotropic weights. The unmarked coefficient at equal deviations uses the transfer of the regular chain, with the fixed physical extraction radius. Changing deviations within one cutoff uses the weighted forgetting estimate on the admitted product of offset discs. At the step where a changed deviation enters, Cauchy's estimate divides the one-step bound by a radius of order the inverse coupling. Subsequent steps propagate the resulting difference by the regular age weights. This gives the single-run forcing coefficient of order $\theta^{-3/2}_{[s+1]}(\log \theta_{[s+1]})^p$ at depth s, times a geometrically decreasing kernel in the depth lag.

For the marked correction, first change the deviations of the longer run to those of the shorter run on the shared depths, keeping its extra deviation and its own remainders. This first difference uses the one-cutoff marked estimate with fixed remainder data. Next compare the two cutoffs at equal deviations. The latter difference is the difference of the two marked coefficients minus the difference of the two unmarked coefficients, with all four readouts on their common running radii. The gate for inserting an old remainder is the one of eq. 5.xxxi. It admits the remainder to the regular exponent at the age with the prescribed lattice symmetries. Its coefficient estimate thereafter uses the squared radius ratio. If χ_t is the distance of the two remainder families of depth t, the contribution on reading the coefficient at depth $s < t$ is at most $C\theta_{[t+1]}\chi_t$. Indeed, the extraction costs the inverse square of the running radius at the output, which cancels that radius squared in the ratio. There is no geometric age factor in this bound.

The regular comparison of the four chains bounds the equal-deviation coefficient difference by

$$C\left[\theta_{[s+1]}(1+K-s)\lambda^{K-s} + \Pi_K + \sum_{s<t<K} \theta_{[t+1]}\chi_t\right] \tag{A.lxxvi}$$

for some $\lambda < 1$. The first term pays for the running extraction radius and the geometric mesh defects of the partners. The bare-end remainder terms in Π_K are estimated by their sizes. The one-cutoff bound for each marked correction also bounds this difference by $C\theta^{-\nu}_{[s+1]}$, with $\nu = (\kappa_0 - 4)/2$. For $b = (1+K-s)\lambda^{K-s}$, $\min\{b\theta, \theta^{-\nu}\} \le b^{\nu/(\nu+1)}$. Retain the last two terms of eq. A.lxxvi and apply this inequality to the first. A slightly slower geometric rate absorbs the polynomial in $K-s$.

We must bound the weighted sum of remainder distances for the actual matched histories. Where $\theta_{[K]} \le 2\theta_{[s+1]}$, the two individual size bounds already give $C\Pi_K$ for this sum. Away from that range the comparison uses the two parts of the new remainder. The ordinary step puts in one part from clusters that contain a recent remainder. On the common admitted carriers let D_j be the norm of the difference of the recent remainder inputs of origin j before forming the localized vertices, and A_j the larger of their two norms. Let η be the size of the change of the other inputs in their admitted

units. For the ordinary contribution at level t, the difference of the cluster sums has bound

$$C\mathsf{s}_{t-1}\sum_{j:\,M\check{R}_j>L^{t-j}} w_{t-1,j}(D_j+\eta A_j).$$

Indeed, interpolate the recent inputs on the common domain. Differentiating the logarithm expansion marks one varied recent vertex. Its bound is $C\mathsf{s}_{t-1}w_{t-1,j}D_j$, and the rooted sum over the other vertices is uniform in that interpolation. At fixed recent inputs, every cluster in the selected sum has a recent vertex, including at complex values of the other inputs. Apply the bound with that vertex marked on the complex neighborhood of the path changing the other inputs. Cauchy's estimate on discs of radius proportional to η^{-1} along this path gives the factor η without losing the recent vertex size. The supports and age gate must be common, with the recent-vertex formation bound valid along the enlarged complex path for the other inputs, including the operators. This path uses a fixed reserve in the admitted norms. The other part comes from integrating histories in the marked operation. On a single lattice, the weighted extension estimates that part by the rare mass times the sum of the loads at level n, each with weight $(k-n+1)q_a^{(k-n-1)_+}$ at level k. The exponent next to a live component can read its retained layer repeatedly. Before integration the factor $(k-n+1)$ counts these uses. A bounded price per use is paid by strengthening the survival reserve under the eventual integral. Thus the geometric age weight for that part belongs to the integrated history. The direct recent-input coefficient has no rare-event factor. Its window has length $m_s=O(\log_L(M\check{R}_s))$. In the units of the remainder size at depth s, the ratios of the sizes over this window are bounded. On the profile S of eq. A.lxxix, the window can cost $Cm_s(1+m_s)\rho^{-m_s}$. This is a power of $\log\theta_{[s]}$ and is absorbed by the logarithmically small coefficient. The integrated part requires the full comparison of weighted histories, including the shell and availability estimates below.

Here is the summation required of the local comparison. Put $a_t=\theta_{[t+1]}\chi_t$ and $q_t=C\theta_{[t]}^{-\nu-1}(\log\theta_{[t]})^c$. A sufficient estimate after separating the direct recent-input contribution is

$$a_t\le q_t\left\{\lambda^{K-t}+(1+\varrho^{-1})\left[\sum_{u=t+1}^{K}\lambda^{u-t-1}y_u+\mathsf{h}+\sum_{j=t+1}^{K-1}a_j\right]\right\}+\sum_{j=t+1}^{K-1}P_{tj}a_j+P_{tK}\mathsf{h},$$

where $\mathsf{h}\le C\theta_{[K]}^{-\nu-1}$ is the extra bare-end input. The P_{tj} are the coefficients of the ordinary recent contribution in the weighted units. For fixed j, they vanish unless $j>t$ and the origin j is still recent at t. There are $O(1+\log\log\theta_{[j]})$ such depths. In this window $\theta_{[t+1]}/\theta_{[j+1]}$ is bounded, and the radius loss is at most $C(1+j-t)$. The column sum is therefore bounded by $C\mathsf{s}^{\mathrm{run}}(\theta_{[j]})(1+\log\log\theta_{[j]})^2$. The coupling ceiling makes its supremum small. This charge has no rare-mass factor.

Sum over $r<t<K$. The triangular term with q_ta_j has total coefficient at most $(1+\varrho^{-1})\sum_t q_t$, which is also small at the coupling ceiling. Absorb these two column bounds on the left. The extra input contributes $C\mathsf{h}$. For the coefficient of y_u, split the sum in t at $(u+r)/2$. On the earlier portion the geometric factor gives $C\theta_{[r+1]}^{-\nu}\lambda_H^{u-r-1}$ for a fixed $\lambda_H<1$ larger than $\sqrt{\lambda}$. On the later portion $\theta_{[t]}\asymp\theta_{[u]}$ and the geometric series gives $C\theta_{[u]}^{-\nu-1}(\log\theta_{[u]})^c$. The term $\sum_t q_t\lambda^{K-t}$ has the same split with $u=K$. These bounds give the weighted tail estimate used below, with the bare-end powers included in Π_K. The local estimate still requires the compared history masses on their actual test domains.

A.8.0.2. Feedback from the terminal pin. Let $y_u=\sup|\Delta_u|$ be the deviation difference on the matched offset disc and let D_s be the supremum of the increment difference in eq. A.lxxv. The pin gives $y_u\le\sum_{s<u}D_s$. On this disc the required remainder-distance estimate bounds the last sum in

eq. A.lxxvi by

$$C(\rho_0^{K-s}+\Pi_K)+C\sum_{u>s}\left[(1+\varrho^{-1})\theta_{[s+1]}^{-\nu}\rho_0^{u-s-1}+p_u\right]y_u,\qquad p_u=C\theta_{[u]}^{-\nu}(\log\theta_{[u]})^b.$$

With this estimate and the one-cutoff variation at fixed remainders, the marked difference obeys the same bound, with a rate $\rho_0<1$ large enough to include the interpolation loss. Add the one-loop and the two unmarked differences. For $r_s=\rho^{K-s}+\Pi_K$, with $\rho_0<\rho<1$, the result is

$$D_s\le Cr_s+\sum_{u>s}(k_s\rho_0^{u-s-1}+p_u)y_u, \tag{A.lxxvii}$$

where k_s is a sum of the one-cutoff sensitivity and $C(1+\varrho^{-1})\theta_{[s+1]}^{-\nu}$. The feedback acts in the opposite direction to the pin. Use the finite maximum $Y=\max_s D_s/r_s$. Then $y_u\le Y\mathsf{S}_u$. Against this profile, the geometric kernel in eq. A.lxxvii costs $Ck_s[\rho^{K-s}+(s+1)\Pi_K]$. The factor $(s+1)k_s$ is small at weak coupling since $s+1\le\theta_{[s+1]}/c_0$. The polynomial kernel costs at most $\Pi_K\sum_u up_u+CK^{-\nu}(\log(2+K))^b$, by the convolution estimate for eq. A.lxxix. Use weak coupling for the first term and the independent prefactor of Π_K for the second. Thus the kernel acting on S costs at most $r_s/2$, and $Y\le C+Y/2$. This proves eq. A.lxxv once the marked comparison has the stated bound.

A.8.0.3. The comparison families on the common scaffold. The coupled comparison supplies the matched increment estimate when its response and comparison rows apply to the actual pair of runs. Write r for depth from the terminal lattice. At the unmatched bare end $r=K$, place the longer run's family on the shorter run's scaffold by the transfer that preserves its total. On the common depths, proceed in decreasing r, and in the order of the production blocks at fixed r. For each new block use the shorter run's production with the comparison families already constructed in the input slots. This list uses the longer run's totals. It belongs to the domain of the response estimate because every inserted family obeys the bound in the units of its own component.

There are two representatives of the difference in each component c. The first adds the change M_r^c caused by the changed inputs of the shorter run's production and the change N_r^c from that production to the longer run's. The middle total cancels. The second subtracts the shorter family directly from the longer family transferred to its scaffold. Its norm is at most $(C_V+1)u_r^c$, where C_V bounds the transfer and u_r^c is the one-run unit in this component. Keep the first representative if its norm is within this bound and otherwise use the second. The norm is a supremum on the entire admitted parameter, source and torus domains. The choice is therefore fixed across those domains. The relations between the components that involve only their totals survive the choice. The source-defect components vanish at constant sources in both representatives, and the local and frozen parts of a remainder component are chosen together. The resulting family has the required local reads and is holomorphic on the admitted domain. Its norm g_r^c satisfies

$$g_r^c\le(C_V+1)u_r^c,\qquad g_r^c\le\|M_r^c\|+\|N_r^c\|.$$

In the second case the second inequality follows because the first representative exceeded the threshold. This closes the construction of the next input list.

Use the blocks $P=(\mathsf{P},Q)$, $A=(P^\flat,D)$ and $R=R'+R''$ in their prescribed reading units. Here D is the difference between the local source read and its value with the source frozen at the base point. The pointed block A is scaled by an inverse power of $\log\theta_{[r]}$, and Q,R carry the inverse squared coupling in their units. At the matched depth r_0 the running-coupling difference is zero. The increment functional on the P block then bounds the difference d_{s+1} of the two running couplings by

$$d_{s+1}\le A_0\sum_{r_0\le u\le s}g_u^P,\qquad r_0\le s<K.$$

Insert this sum into the response rows, including their weighted coupling tails. The resulting inequalities have nonnegative coefficients in the block norms. They may involve both earlier and later depths.

For the envelope put $e_r = \rho^{K-r} + \Pi_K$. The three block bounds to check are $C_E e_r$, $4C_E e_r$ and $4C_E \widehat{r}_r e_r$, respectively. Here $\widehat{r}_r$ is the one-run remainder coefficient of order $\theta_{[r]}^{1-\kappa_0/2}$, enlarged by the window factors and one factor of $1+r$ from the coupling sum. Its bare-end value sets the size of the floor Π_K. For the geometric response kernel q^{s-r}, use $q \le \rho^2$ and $e_{r+j} \le \rho^{-j} e_r$. The coupling sum up to r is at most $\rho^{K-r}/(1-\rho) + (r - r_0 + 1)\Pi_K$; convolution with the geometric kernel pays the linear factor in the lag. For the unweighted remainder sum use $\sum_s \widehat{r}_s \le 1$ and the one-run comparison $\widehat{r}_s \le C_T \rho^{-(K-s)/2} \widehat{r}_K$. This bounds $\sum_{s>r} \widehat{r}_s e_s$ by $2\Pi_K$ with the prescribed prefactor in the floor. In the short remainder window of length ℓ_r, the relative one-run bound $\widehat{r}_{r+j} \le 2(1+j)\widehat{r}_r$ leaves at most $C\ell_r^2 \rho^{-\ell_r}$ times the coefficient of that window. The small-coupling bounds for the three rows make the sum of the weighted envelope terms at most half the envelope in each row, and the direct forcing at most the other half. The transferred bare-end family satisfies the envelope there.

To apply these tests simultaneously, take the maximum of the ratios of the actual block norms to their proposed bounds. If the maximum were $t > 1$, it would occur in a response row. At that row the two half-envelope bounds give $t \le (t+1)/2$. Thus every block satisfies its envelope. In particular the coupling increment difference is $O(e_r)$ on the common offset disc, and the running-coupling difference has the preceding summed bound. This application requires the matched productions and the response rows in the same running radii as the increment functional.

For a comparison of the pointed remainder shares, make the volume allocation in each actual run before comparing them. The allocation depends on the original filing domains. For example, file a fixed analytic function first on a cube wholly inside one filing cell, and then on a larger domain containing that cube and meeting a second cell. The total function is unchanged. The second cell receives zero in the first case and the positive volume fraction of the function in the second. At a shared filing domain the two actual allocations have the same geometric weights, so their difference is the weighted sum of the differences of the individual terminal reads. That sum requires the paired terminal-read estimate on its actual carriers.

Let i'' be the filing level of a remainder from level i, and $t_i = L^{i-i''}$ the mesh ratio. The share $\mathrm{sh}(X, y)$ at y is the fraction of the level-i cubes of X in that filing cell. Evaluate $R^{(i)}(X)$ on the box background formed from the actual level-i averages in the padded box of X, and subtract $R^{(i)}(X, 1)$. The assigned domain $Y(X, y)$ contains that box and Y^+ contains the old source read. The box-background and restriction estimates on the filing radii give holomorphy. On admitted backgrounds restriction of the critical orbit identifies the box evaluation with the original one. Summing the shares over y gives one for each X, so the sum is the vacuum-subtracted total. At constant sources the identity before summing y, with boxes and shares transported under the lattice symmetries, gives covariance of the pointed total.

For the one-run compact estimate, take the box background along the complex radial line in the coarse datum. Its inverse analytic radius is bounded by $C(d_i(X)+1)t_i^2 w_{i''-1,i}^{1/2}$. The first derivative of the remainder at the identity vanishes. The second-order Schwarz estimate then gives the square of this inverse radius for the vacuum-subtracted value. This is the t_i^4 gain in the one-run filing norm.

Write $S_i^+[R]$ for the difference norm of the level-i remainder and $\widehat{\vartheta}_i$ for the decay in the datum comparison. At the shared filing slice the paired read bound gives $S_i^+[R] + \widehat{\vartheta}_i g_i^{\kappa_0}$ times the size of the individual read. For compact X of diameter D_X, that size has the factor $D_X^2 t_i^4$ times $w_{i''-1,i} e^{-\kappa_T d_i(X)}$, with $w_{i''-1,i}$ the crossing weight. There are $O(\check{R}_i^4)$ fine anchors in a filing cell. The diameter cost is bounded by $CM^2(d_i(X)+5)^2$ and $\check{R}_i^4 M^2 t_i^4 \le CM^{-2}$. The target domains of compact reads have bounded length. For long X, the length is at least $t_i^{-1} - 2$. The output weight

spends a fixed fraction of the input length decay, and another fraction leaves $Ce^{-cM\check{R}_i}$ to pay the anchor count. The crossing weight at this filing level is bounded below. The rooted sums give

$$S_{i''}^{+}[\mathsf{RF}R] \le Cw_{i''-1,i}(S_i^{+}[R] + \widehat{\vartheta}_i g_i^{\kappa_0}).$$

This estimate uses the compact and long terminal read hypotheses at the actual shared slice.

Localize the difference at each y by the alternating box sum on the shorter run's scaffold. Summing the boxes cancels the proper restrictions and leaves the full-background difference at that y. This gives a pointed comparison family on the admitted domains. Subtract its frozen value at $W(p_y)$ to obtain the source-defect component, which vanishes at constant sources. The factor of two for that subtraction is paid in changing from the Q unit to the pointed A unit under $E_0C_tg^{-2} \ge 2$, with E_0, C_t the unit constants. Add these two filed differences to the comparisons of the ordinary Q and D outputs. The hybrid production uses the filed components already in its input. The filing operation is carried out on the two actual runs.

An older remainder is read as such only in the finite band before filing. After that band its share is part of the pointed regular family. The P and A inequalities gain a sum of remainder distances over the filing depths and a term from the relative datum comparison. At most two depths file at r, each within the window of length ℓ_r. The relative datum term costs $C\rho^{-\ell_r}\mathfrak{r}_r$ times the geometric envelope, where $\mathfrak{r}_r$ is the one-run remainder size in its reading unit. The sum of the remainder distances in that window costs $C\ell_r^2\rho^{-\ell_r}\widehat{r}_r$ times the same envelope. These are the additional terms in the three row smallness test. When the finite-band response and source rows hold, the maximum argument then gives the geometric envelope with the pin at the terminal coupling of the run with filed shares.

A.8.0.4. The change of offset. The two terminal couplings agree at the matched offsets. Subtract this equality at $t = 0$ from the equality at t. The bare terms leave $t - \mathfrak{n}_K(t)$, so the matching defect is, up to sign, the sum of the centered increment differences and the centered extra increment. The one-run derivative bounds and $|\mathfrak{n}_K(t)| \le 2|t|$ bound each centered difference by $C|t|a_{[u]}$. By the Schwarz lemma and eq. A.lxxv it is also bounded by $C|t|(\rho^{K-u} + \Pi_K)/\varrho$. The extra increment costs $C|t|a_{[K]}$. Thus

$$|\mathfrak{n}_K(t) - t| \le |t|\Upsilon_K, \qquad \Upsilon_K = Ca_{[K]} + C\sum_{u<K} \min\{a_{[u]}, (\rho^{K-u} + \Pi_K)/\varrho\}. \tag{A.lxxviii}$$

For the $O(\log(2+K))$ depths nearest K, use $a_{[u]}$ and $\theta_{[u]} \asymp K$ as $K \to \infty$. Farther away, sum the geometric bound, choosing the multiple of $\log(2+K)$ so its sum is $O(K^{-2})$. The other cost is $CK\Pi_K/\varrho$. For $\Pi_K = O(K^{-5/2})$ this gives $\Upsilon_K = O_\varrho(K^{-3/2}(\log(2+K))^{p+1})$ for $p \ge 0$. The bound on Ξ'_{K+1} gives the same rate for $|\Xi_{K+1}(t) - \Xi_K(t)|$ by matching. Cauchy's estimate on a smaller disc bounds the derivative of $\mathfrak{n}_K - \mathrm{id}$ by $C\Upsilon_K$. This gives the first line of eq. A.lxxiv. Writing $N_K = -\Xi'_K(0)$, differentiation of $\Xi_{K+1} \circ \mathfrak{n}_K = \Xi_K$ gives $N_{K+1}\mathfrak{n}'_K(0) = N_K$. Hence N_K converges to a number in $[2/3, 4/3]$, with error bounded by $C\sum_{k\ge K}\Upsilon_k$.

A.8.0.5. The comparison through the sourced run. For the full comparison, index the reference choices by the depth from unit scale. In the two runs the cubes, thresholds, waiting depths and radii then agree at common depths. Cut the histories at a depth T proportional to K, with $K - T$ greater than the waiting depth there by the fixed geometric margin. On the subcollection with no creation at $u \ge T$, the bare-end rules lie beyond the retained births. After matching the real decisions, pair the labels with the same regions and retained rows at common depths. Comparison of the weights still uses their actual stored variables and factors. Consider a record formed with root at common depth $s < T$, containing a birth at depth $u \ge T$. The path to that birth occupies at least $u - s$ levels of the record. Each node on the path contributes at least M^4 sites with its own logarithmic weight, as in eq. 4.cdlxxvii and eq. 4.dxxii. On the common increasing depth frame these logarithmic weights are at least a fixed multiple of $\log\theta_{[s]}$. Thus $v^\natural \ge c_* M^4 \log\theta_{[s]}(u - s)$.

Use the mass bounds at $c' > c$ on the same raised floor and write

$$e^{cv^\natural} \le \theta_{[s]}^{-c_*(c'-c)M^4(T-s)} e^{c'v^\natural}$$

on this subcollection. The inequality holds before the upper integral, with the same localized root sum. Thus for $s \le \alpha' K < T = \alpha K + O(1)$, with $0 < \alpha' < \alpha < 1$, this unpaired part has an exponential cutoff gain. There is no extra factor for choosing a position or the first such birth because the subcollection is counted once in the mass of the whole rooted record. A birth that has been healed and erased from that record is accounted for through the terms it leaves in the recursion. The duration for which a formed mark is later carried belongs to the live-component reserve and the global label comparison. Its stored genealogy is fixed.

For a visible creation on the outer tree one can also retain its own reserve in eq. 4.dx. In the proof of eq. 4.dxi, choose a creation at $u \ge T$ on this subcollection in place of the arbitrary leaf. The other entropy and volume allocations are unchanged. The factor $\exp[-2p_0(g_{[u]})]$ gives a cutoff gain faster than any power of K for $T \asymp K$, even if this visible outer record is short. This estimate uses the actual creation reserve. A failure whose witness has been left for a later operation, or an earlier birth internal to an inner island, uses its own mass.

For the remainders formed at $u \ge T$, use the individual size $C\theta_{[u]}^{-\kappa_0/2}$ and the readout weight $C\theta_{[u]}$, with the logarithmic factors of the norm. On the weak-coupling reference line $\theta_{[u]} \asymp K$ for $T \le u \le K$. The sum over these $O(K)$ depths, including the unpaired last remainder, is at most Π_K of order $K^{-\nu}$ times a power of $\log(2+K)$, where $\nu = (\kappa_0 - 4)/2$. This part enters the comparison at lower depths without an age factor. A first disagreement in a short history at fixed common depth s can occur with no ancestral path reaching T. Its birth-scale bound depends on $\theta_{[s]}$, so one still needs the local near-threshold comparison.

Write $\Delta_u(t)$ for the difference of the two deviations at depth u and at the matched offsets $t, \mathfrak{n}_K(t)$. The matching identity sets $\Delta_0 = 0$. The deviation recursion, with the same reference increment in both runs, expresses Δ_u as a signed sum of the scalar increment differences through depth $u-1$. With

$$\mathsf{S}_u = \frac{\rho^{K+1-u}}{1-\rho} + u\Pi_K, \tag{A.lxxix}$$

the scalar estimate eq. A.lxxv gives $\sup|\Delta_u| \le C\mathsf{S}_u$. If the field v takes values in a smaller offset disc, Cauchy's estimate there bounds the derivative of Δ_u by $C\mathsf{S}_u$ with a fixed radius cost. Thus the mismatch of the pointwise deviation fields is at most $C\mathsf{S}_u$ in the maximum of the supremum norm and the depth-scaled Lipschitz seminorm of the admitted source field.

We must show on the two sourced trajectories that the distance X_s of their outputs at depth s satisfies

$$X_s \le C\mathsf{S}_{s+1} + C\sum_{u \ge s} \alpha_{s,u}\mathsf{S}_u + \sum_{u>s} \varkappa_{s,u} X_u.$$

Use a distance that compares values on common fields, with exponential localization and the units of the defect and remainder families. The small multiple of the individual sizes needed for information lost by the transfer goes in the one-step defect. The unmatched family at depth K also goes in that defect. Its scalar deviation can still enter the input sum.

For the sourced trajectory the size of its defect can be read off at formation. Let $i \le m$ and use the source norm in the units of level m. In the defect at a base point z, compare with the constant $w(p_z)$. A tree meeting all cubes of X has $d_i(X)$ edges of length ML^{i-K}. The prescribed enlargement adds a bounded number of cube lengths to the diameter. Thus

$$\sup_{p \in X^\sim} |w(p) - w(p_z)| \le CML^{i-K}(1 + d_i(X))\,\mathrm{Lip}\, w.$$

Use the constant in the scaled source norm so that the local norm of this difference is at most $e^{d_i(X)}\mathsf{l}_i$, where l_i is its Lipschitz part. This uses the boundedness of $(1+d)e^{-d}$. For a source ball

of radius r, the unit-slope estimate on the admitted line bounds the defect by $r^{-1}e^{d_i(X)}\mathsf{l}_i$ times the rate-$\kappa+1$ decay. Since $\mathsf{l}_i \le L^{-(m-i)}\|w\|_{\flat,K-m}$, this gives

$$|D^{(i),w}(X,\cdot,z)| \le r^{-1}\|w\|_{\flat,K-m}L^{-(m-i)}e^{-\kappa d_i(X)}.$$

For a drive in the unit-scale norm take $m = K$, giving size L^{-u} at depth $u = K - i$, from the moment the entry is made. The old defect is passed on unchanged. The stronger admission bound required by the next box comes from this original estimate on the actual source trajectory. A one-step estimate on free input arrays must assume that stronger bound on the inherited entries.

For a newly formed paired defect, put the two step maps on the same slope line in the source. At depth s, take a common base point y of the enlarged read and set

$$w_\lambda = w(y) + \lambda(w - w(y)).$$

The partnered source uses the pointwise matching map applied to w_λ. In each run subtract the formation map at its own frozen value at the base point of the family. Both defects vanish at $\lambda = 0$, even if the two frozen points differ. The Lipschitz bound on the full read admits the line up to radius $R \asymp L^s/(1 + d(X))$. Suppose the unscaled two-run array comparison on this enlarged disc bounds the difference of the two formation defects by $\mathcal{B}_X e^{-(\kappa+\delta)d(X)}$ with $\delta > 0$. For $R \ge 2$ Schwarz's lemma gives at $\lambda = 1$

$$CL^{-s}(1 + d(X))\mathcal{B}_X e^{-(\kappa+\delta)d(X)}.$$

For $R < 2$, the same bound follows from the unscaled comparison at the actual source, since $L^{-s}(1 + d(X))$ is then bounded below. The spare decay pays for $1 + d(X)$. The output unit L^{-s} is therefore recovered for the paired defect. To separate the displacement of the frozen points, its bound on the line is $C|\lambda| \operatorname{Lip}(w)$ times their physical separation, using the derivative of the matching map. The slope estimate gives that same displacement cost at $\lambda = 1$. The step still needs the unscaled array comparison on the admitted line.

On the common all-small label the carried Wilson weight can be compared separately. At matched levels use the same physical tent grid. Write $w_A = w$ and $w_B = \mathfrak{n}_K \circ w$, and let T^a_u denote tent interpolation at depth u, sampling at the barycentres of the plaquettes p^a_y based at the tent vertices. Set

$$f^a_u(x) = b^a_{[u]}(w_a(x)) - b^a_{[u]}(0).$$

Here the increments are written as functions of the constant source in their respective runs. Denote the constant-source shift at depth s by $\zeta^a(t)$ and the actual shift on plaquettes by c^a. The formula for the carried shift on this label gives, at a common physical position,

$$c^a(x) - \zeta^a(w_a(x)) = \sum_u (T^a_u - I)f^a_u(x),$$

where the sum is over the increments already made. The difference of the two errors is

$$\sum_{u \text{ common}} (T^A_u - I)(f^B_u - f^A_u) + \sum_{u \text{ common}} (T^B_u - T^A_u)f^B_u + (T^B_* - I)f^B_*.$$

The star is the extra first increment in run B. For a common depth, the constant-source recursion expresses the difference of the increments as the difference of two successive matched coupling mismatches. Subtracting its value at zero and using Cauchy's estimate on a smaller source disc gives

$$\operatorname{Lip}(f^B_u - f^A_u) \le C\varrho^{-1} \operatorname{Lip}(w)(\mathsf{m}_u + \mathsf{m}_{u+1}).$$

If $\xi_u \asymp L^{-u}$ is the tent width, $T^A_u - I$ costs at most $C\xi_u$ times that Lipschitz bound. The ratio ξ_u/ξ_s gives the geometric share in the source units at depth s. In the second sum only the sampling plaquettes move, by $O(a_K)$. The unmatched tent has width $O(a_K)$. These two terms together cost $Ca_K \operatorname{Lip}(w)$, using the summable derivative bounds for the increments. For later quadrature, first

differences of the multilinear tents give $\mathrm{Lip}(c^a) \le C\,\mathrm{Lip}(w)$. Indeed the samples at neighboring vertices differ by at most $C\xi_u\,\mathrm{Lip}(f_u^a)$, which pays for the ξ_u^{-1} in a tent derivative.

In the action the scalar coefficient is $x^a - c^a$. The difference of its frozen parts is the paired constant-source coupling mismatch, and the two tent errors give the additional difference just estimated. This uses the scalar weight as a potential in the frozen step. The Gaussian structure and the independent analytic current retain their own arguments.

For an inherited gauge-invariant defect, separate the family difference on the transferred fine argument from the change between the two evaluation arguments of the flat family. After subtraction of the vacuum values, the first part uses the vacuum-only window estimate of Prop. 4.6, with its squared ratio of field radii. The size already present in the inherited defect supplies $L^{-(u-s)}$ in the output units. The second part uses the actual paired evaluation fields on the complete reads. It is a separate term in the unscaled array comparison.

The regular frozen families are compared one source value at a time, with their own counterterm. A displaced barycenter costs the Lipschitz norm of v times its displacement. The source defect at depth u already has the size factor L^{-u}. Relative to the output unit L^{-s}, this contributes $L^{-(u-s)}$ times the squared ratio of the field radii. Marked remainders have the squared ratio and their own power bound. In the recent window the term passed on by the ordinary operation has a logarithmically small coefficient multiplying the input difference. It must be included when closing the full recursion. In the marked operation the dependence before the mass estimate can instead cost a power of $\log\theta_{[s]}$. There the weighted mass and the age sum must supply the smallness.

The required kernels have a geometric part c_{u-s} and a remainder part p_u of size $C\theta_{[u]}^{-\nu}(\log\theta_{[u]})^b$ with a fixed constant. For $q \ge 0$ compare S_{s+q+1} with S_{s+1} using the factors ρ^{-q} and $1+q$ on the two terms of eq. A.lxxix. Thus the geometric part preserves the profile if both $\sum_q c_q\rho^{-q}$ and $\sum_q(1+q)c_q$ are finite. For the remainder kernel, $u(1+K-u) \ge K$ for $1 \le u \le K$. The polynomial size bound for p_u and this inequality give $\sum_{u\le K} p_u\rho^{K-u} \le CK^{-\nu}(\log(2+K))^b$. The other part costs $\Pi_K\sum_u(u+1)p_u$. This sum can be made small at weak coupling by $\nu > 2$ and $\theta_{[u]} = g^{-2} + c_0 u$. In the geometric convolution at the bare end, smallness relative to Π_K instead comes from choosing the prefactor of Π_K large after the kernel constants. The same bounds apply to the input kernel on S_u, where boundedness suffices. With small weighted moments of c_{u-s} in the feedback kernel, its full action on S_{u+1} is at most $\mathsf{S}_{s+1}/2$. Backward induction gives $X_s \le C'\mathsf{S}_{s+1}$ uniformly in K.

The local energy comparison at $(\mathbf{1}, 0)$ uses this profile. A carrier of size $d(X)$ has at most $CM^4(1+d(X))$ base points. A fixed reserve in the exponential decay pays for this count, giving the local estimate with $C\mathsf{S}_{s+1}e^{-\kappa_e d(X)}$. Since $\mathsf{S}_{s+1} \le C(1+s)(\rho^{K-s}+\Pi_K)$, this is the rate used for the mixed derivatives. For the terminal integral we must also compare the weights on the common terminal fields, with the same scalar at the source and at its real center. The corresponding terminal estimate uses the kernels at depth zero and gives $C(g,\ell)\mathsf{S}_1$ on the admitted source ball. Moving the stop by a fixed number of depths restores the corresponding local energies. The same profile bounds their differences at this fixed stop.

A.8.0.6. The source-independent part of the terminal comparison. For a native healed component, form the two remainders on the same prepared determining sets. Keep the real frozen data, masks, Haar coordinates and weights of the common geometry. With the cutoff ζ_0 of the component, the last term is

$$K_5 = -\log\int e^{-S(B')}\,d\mu(B'), \qquad S(B') = x_m\big[A(\zeta_0, U^{B'}) - A(\zeta_0, U_0)\big].$$

The common measure $d\mu$ includes the Haar density, the gauge constraint and the sharp integration domain. The remaining terms of the remainder come from the exact Wilson and quadratic split. In its notation the remainder is $\mathbf{C}_m = K_5 + K_3 - (E_{12} + E_3 + E_4 + K_1 + K_2 + K_4)$. The Wilson terms use the same weights in both runs. For K_5, divide the two integrals before taking the logarithm. On the real data the ratio is the expectation of $e^{-(S_B - S_A)}$ under the normalized A integral. Thus

its log difference costs at most $\sup|S_B - S_A|$ on the same cut. This comparison uses the whole action difference in S with its own centre.

At real common data the layerwise bound on the paired minimizers is obtained by telescoping their data defects. The paired data at each occurrence have bounded depth. The weighted response bound controls the change from the defect at each step, and the common local index fixes the rows. For a Wilson cell this first bounds the curvature difference, after transporting to the same frame. The action estimate multiplies it by the bound on the two curvatures and sums the plaquettes of the cell. Before multiplying by x_n, the result is $CM^4\mathsf{e}_m^2(\vartheta_A^n + C_\chi L^{-n_\chi})$, where n_χ is the cutoff's scale exponent on the cell. The native cutoffs have $n_\chi \geq n-1$, so the cutoff term has the same decay. Two-constants interpolation on the fixed comparison discs gives the square root of the rate on the common complex data. The geometric mean with the one-run bound decaying into the core then leaves $\vartheta_A^{n/4}$ for these Wilson cell terms. The normalization and the cutoff block use their whole-term estimates.

In localizing E_{12} and E_3, assign the switch-independent subtraction to the smallest-box piece of its own cell. For the other pieces it disappears under the switch difference. The two-run estimate applies at the natural binary vertices, where the boxes contain the component and the required cell padding. The complex polydisc bound uses the sum of the separate one-run majorants. The interpolation estimate combines the two bounds, with a rate reserve for the tuple count. This gives the native class estimate

$$\sum_{Y\subset C^\sharp} \sup\bigl|\delta\mathfrak{v}^{\mathbf{C},(n)}(Y)\bigr| \leq C\vartheta_w^{n/8}, \qquad 0<\vartheta_w<1.$$

The class occurs only at the prepared levels in the component and at the two endpoint levels of the split. There is no coupling-difference term from these common weights. This supplies the source-independent load for the matched native remainder. Its use in a terminal read requires the same measure, masks and normalization above. A remainder depending on the variable source uses the separate excess comparison. A source-independent factor that varies with the label still occurs in the weights of both stop integrals.

A.8.0.7. The exterior sum of the Wilson remainder. For the outer part of E_3, the unit on a cell Δ is the weighted Wilson value at the box field minus the value at the root, with the same cutoff and plaquette base points. Its core K_Δ is the level-m cube of Δ with four layers of padding. For a treated component C let H_C be its protected block with one layer of padding and put $\epsilon_\Delta = \sum_C e^{-\delta_P d^\wedge(\Delta,H_C)}$. At the switch vertices the paired Wilson estimate is bounded by $CM^4 x_l \mathsf{e}_m^2 \epsilon_\Delta^{1/2}\langle l\rangle^{1/4}$, where l is the level of Δ and $\langle l\rangle$ is the two-cutoff rate at that level. On the contour the four Wilson values are bounded by $CM^4 x_l \mathsf{e}_m^2$, using their admitted plaquette bounds on the complete core. The switch estimate gives the square root of the product and an exponential in the number of switched cells. The switch sum at an exterior cell Q spends this last exponential to pay for the tree weight and leaves $Ce^{-\operatorname{dist}_m(Q,K_\Delta)}$.

Use subadditivity to distribute $\epsilon_\Delta^{1/4}$ over C. For each C, split its decay into four equal parts. The first pays to replace $x_l\langle l\rangle^{1/8}$ by $Cx_m\vartheta_\diamond^m$, using the flow and the crossed zones, where $\vartheta_\diamond$ is the exterior two-cutoff rate. The second leaves the gain $e^{-5\check{R}_m}$ from the separation of the outer cells from H_C. The third joins the distance from Q to K_Δ to that from Δ to H_C. The fourth sums the cells relative to the boundary of H_C. The separated-block count sums the resulting weights at Q over C. Thus the weighted norm is at most $Cx_m\mathsf{e}_m^2 e^{-5\check{R}_m}\vartheta_\diamond^m$. The factor $x_m\mathsf{e}_m^2$ is polynomial in $\log x_m$, and for a class $n\leq m$ one has $\vartheta_\diamond^m \leq \vartheta_\diamond^n \leq C\Delta_n^c$. The analogous one-run vertex estimate starts with ϵ_Δ and no two-cutoff rate, leaving $\epsilon_\Delta^{1/2}$ after the switch estimate. Three of its eight decay parts pay the lift, anchor and cell count, and the remaining five give $e^{-25\check{R}_m}$.

For E_{12}, first compare the field in a box G with the restriction from a box containing both G and the treated component. The determining data differ only on the new outer rows near the

cut. Then compare the restriction of the larger field to the field of the component's own box. In this comparison only the component's boundary shells supply the mismatch. Both mismatches are outside the core collar and are $O(\mathsf{e}_m)$. The two localized responses give the one-run depth bound on the cell, which combines with the paired vertex bound and the contour majorant. In the exterior sum the switch family must reach from a core inside the treated component to an exterior cell. Its minimum length leaves $Ce^{-c\check{R}_m}$ after paying for the tree weight and the family count. The depth-weighted cell count costs a polynomial in the logarithm of x_m. This gives a small coefficient times the class rate. Retain the class of the original cell or endpoint row. The outer E_3 terms at older levels use the classes in the untreated live regions. The full pieces confined to a treated component use the value estimate at its own box, rather than this exterior gain.

For the source-free remainder on the aligned scaffold, the exterior estimate is used on the filed domains Y_4. At an exterior root Q, terms filed as a value on one treated component do not occur. The label lab_4 of each remaining piece is formed from the four successive switch families. Filing adds whole touched components of Z, so $Q \in Y_4$ implies $Q \in \mathrm{lab}_4$ and $d_{m,Z}(Y_4) \le d_m(\mathrm{lab}_4)$. For the admitted outgoing pieces the label also reaches the source cell of a nonempty switch family. The protected margin between such a cell and Z^c gives $d_m(\mathrm{lab}_4) \ge \check{R}_m - 2$.

With $a = 31\kappa/20$, spend one additional unit of ordinary length weight to extract $e^{-(\check{R}_m-2)}$. The four successive rooted switch sums each spend at most two more units of weight. The rate condition $\kappa_1^\natural \ge c_l(a+9)+\lambda_*$ pays these losses, where c_l is the length allowance per switch cell and λ_* pays the connected count. What remains is the rooted sum of the original units of class n, whose supports have length zero. Its weights are $(2\beta_v\mathfrak{B}_v)^{1/2}$, with β_v the paired binary bound and $\mathfrak{B}_v$ the one-run contour majorant. At an interior cube only units of its treated component contribute. At an exterior cube there is just the outer unit there. The classwise cell sum gives $C\vartheta_{\mathbf{C}}^n \mathsf{N}_{\mathbf{C}}(m)$, where $\mathsf{N}_{\mathbf{C}}(m)$ is polynomial in $\check{R}_m$. Thus the filed load satisfies

$$\sup_{Q\not\subset Z} \sum_{Y_4 \ni Q} e^{a d_{m,Z}(Y_4)} \sup |\delta \mathfrak{v}^{\mathbf{C},(n)}(Y_4)| \le Ce^{-(\check{R}_m-2)}\vartheta_{\mathbf{C}}^n \mathsf{N}_{\mathbf{C}}(m).$$

This sum uses the common switch scaffold and the native piece bounds at its admitted reads.

A.8.0.8. The reads on a live strip. Use two sets for a stored term on a live strip. Its true domain X specifies its fields and dependencies. For counting, choose an optimal tree for its relative length and use the union X^θ of the cells met by the tree. Take the fat core of X^θ for the ordinary expansion. The thickening bound here uses the ordinary tree length of X^θ. A strip block whose enlargement reads X^+ outside this core carries a separate parameter for each response increment that reads it. The initial core contains the tree and its prescribed padding. For a selected set D of these parameters the new core adds only their four-layer blocks and the corridors, with the fixed final enlargement. This includes the complete carrier of the coefficient selected by D.

In the native kept term of a component Y, subtract the value at its declared reference. The true domain is $Y^\sharp$ with its assigned adjacent cell $\Delta_1(Y)$. Both lie in the one-layer enlargement of the strip, and the tree domain is $\Delta_1(Y)$ alone. At a fixed level the strips born there are separated, so at most one strip per face can choose the same cell. This gives at most $2d$ choices at that cell. The reference value stays in the original identity for that label. The nonempty switch difference and the plain derivative both kill that constant. On the admitted tube the resulting coefficient is therefore bounded using the oscillation of the kept term. At a fixed anchor the summed oscillation budget is at most $2dc_H^{\mathrm{kp}} C_\Sigma^U \check{R}_m^{p_\Sigma}$. On the admitted strip discs of radii $r_a = r_0 e^{E_a}$, a nonempty difference in the parameters $a \in D$ gains $\prod_{a\in D}(2/r_0)e^{-E_a}$. Indeed its alternating vertex sum is the integral of the mixed derivative over the unit cube. The remaining radius at each real point is at least $r_a/2$. The domain and locality rules for the strip expansion then treat this coefficient as an ordinary stored term with its enlarged core. For a plain-block derivative the source bound carries the separate decay in the plain block and in the domain's distance to the shift.

In the kept family the source separation supplies $E \geq \varpi(\frac{5}{2}\check{R}_m - 8)$, with $\varpi = \delta_1 M/8$, in the factor reserved for the strip count. All read blocks lie within eight level-m units of $Y^\sharp$, and the source is outside the collar of width $5\check{R}_m/2$. For a plain-block derivative split the domain factor $e^{-E/2}$ into two factors $e^{-E/4}$. The first pays the original level sum under $\varpi/8 \geq 2\log(2L^d)$. For a nonempty switch family split each e^{-E_a} in halves. One half per block pays the original block sum under $\varpi/2 \geq (\bar{a}+2)a_d + 1$ and $\varpi/4 \geq 4\log(2L^d)$, with $\bar{a}, a_d$ the exponents of the block count. A spare half from any selected block pays the kept count. The plain block's own decay is retained. In either case a factor at least as small as $e^{-\varpi(\frac{5}{2}\check{R}_m - 8)/4}$ remains for the polynomial anchor budget. In particular their effective contribution is bounded by a fixed constant times

$$C_\Sigma^U \sup_{R \geq 1} R^{p_\Sigma} e^{-\varpi(\frac{5}{2}R - 8)/4}.$$

This uses the source separation on the actual response paths as well as the native strip-domain rule.

A.8.0.9. Room for the enlarged strip parameters. In one run, let Ω^{s} be the source region and $D(x)$ the distance to it in the scaled path metric. For a datum b supported there, write P_b for the exponential smearing of $|b|$ at rate δ_1. The geometric summation bound gives $\mathsf{P}_b(x) \leq c_1 \|b\|_\infty e^{-\delta_1 D(x)/2}$. The strip letters grow with distance at rate $\delta_1/8$. On the enlarged blocks their envelope is $\max\{c_0^\chi, r_0^\chi C_\Delta e^{\delta_1 D(x)/8}\}$, with c_0^χ the bound for ordinary letters and C_Δ the fixed enlargement cost. In the partition of unity, at most 2^d cutoffs overlap, and the ratio of the local scale to a cutoff's scale is at most L. The first and second differences cost L and L^2, respectively. Use $C_\chi = 2^d C_\zeta L^2$ for both derivative bounds, with C_ζ the partition bound, and take the radius floor $8C_\chi$ before rescaling the strip radii.

Multiplying the letter envelope by $C_\chi \mathsf{P}_b$ bounds the resulting majorant by

$$C_\chi c_1 \|b\|_\infty \max\{c_0^\chi e^{-\delta_1 D(x)/2}, r_0^\chi C_\Delta e^{-3\delta_1 D(x)/8}\}.$$

Dividing the strip radii by C_χ cancels that factor in the strip term. The ordinary letter bound still carries it. For the pointwise room, if $|b(y)| \leq \varepsilon u(y)$ on the source region and $\sum_y e^{-\delta_1 d(x,y)/2} u(y) \leq C_T u(x)$ in the same units, split the smearing exponent in half. The identical profile estimate then holds with $c_1\|b\|_\infty$ replaced by $C_T \varepsilon u(x)$. To use the enlarged discs in the actual box we must fit the first bound under the absolute response ceiling and the second under $\ell_\iota(x)a(x)/4$, where ℓ_ι is the label scale and a is the local response majorant. This requires comparing the native datum budget with the local room on the whole box, including its covered and inactive sites.

For the two-run reading difference at a foreign site, hold the leaf function of the second run fixed and subtract its evaluation at the first run's read with the second run's common datum slots inserted. Write $h(t)$ for this difference in the strip parameters. The native reading comparison uses the full one-run discs, the relative row of the additional strip shift, and the net datum comparison on the true hull. It bounds h by $C\vartheta^\sharp \|F\|$ on those discs, with $\vartheta^\sharp$ taken at the smaller of the leaf level and the lowest zone of its true hull. Cauchy's estimate as above bounds the mixed strip piece by $C\vartheta^\sharp \|F\| \prod_{a \in D} 2/r_a$. In the subsequent affine piece estimate use the smaller radii $r_a/4$. Both bounds in the geometric mean carry the full product on these radii, so the square root falls on $\vartheta^\sharp$. A change of the leaf function at a shared read uses the ordinary one-run strip filing. At the component's own treatment the terms touching its Gaussian core use the comparison about the moved real centre.

A.8.0.10. Counting stored domains away from live strips. At level ℓ let W_ℓ be the union of the four-layer enlargements of the level-ℓ covers of the live strips. Its touching components are the islands of this counting convention. File a domain by including every island it meets. This changes its index and preserves the true dependency set. The relative length $d_\ell^\circ(Y)$ is the least length of a tree in the filed Y that meets every cube outside W_ℓ. The length is measured along the tree, including any path through an island.

For the blind filing used in this count, healing a component absorbs the terms filed on it, including terms indexed by a domain containing one of its islands. If such a term survives until $m \geq \ell$, its excused islands at ℓ survive with it. Their births belong to one component after the youngest birth, and healing that component would absorb the stored term with the rest of its local terms. The coarser cover of its excused set lies in W_m. Indeed, for $m > \ell$ the four finer layers have width $4L^{-(m-\ell)} < 1$ in level-m cells. The cover thus adds at most one layer about the coarser cover of the still-live strips. If a level-m cube of the filed domain is outside W_m, it contains a level-ℓ cube of Y outside its excused set. The old relative tree meets that cube, so it serves as a tree for the filed domain at level m with length at most $L^{-(m-\ell)} d_\ell^\circ(Y)$.

At an exterior level-ℓ root, the blind relative domain count gives a bounded sum with rate $\lambda_0 \geq \mathsf{k}_d + 1$. In that count each cube meets at most 3^d other cubes, and a neighbor belongs to at most one island. Now take an exterior level-m root Q at distance at least one from W_m. Set $y = L^{m-\ell}$. Its y^d subcubes can serve as level-ℓ roots. Islands whose births have healed can be dropped for this count since no surviving domain includes them. If a surviving domain counted at Q contains an island I, the distance from a root subcube in $Y \cap Q$ to I is at least y. A chain in Y from I to that subcube contains a first cube off I. It touches I and lies outside all the islands. The relative tree meets it and the root subcube, so its length is at least $y - 3$. At rate $\lambda \geq \mathsf{k}_d + 2$, spend one unit of the decay on this distance. The remaining rooted sum over each subcube costs at most C_{fib}. The total is at most

$$C_{\mathrm{fib}} y^d e^{-(y-3)},$$

uniformly bounded for $y \geq 1$. If the root is an island I itself at level ℓ, the pure domain $Y = I$ contributes one. Every larger domain containing I has an exterior cube touching I. Rooting at these cubes bounds the sum by $1 + 3^d C_{\mathrm{fib}} N_\ell(I)$, with $N_\ell(I)$ the number of cells of I in this partition.

An output of $\mathbf{T}_\ell$ that meets an old island also meets Ω_ℓ. The old-strip buffer gives distance at least $3\check{R}_\ell - 6$ from that island to Ω_ℓ. The same first exterior cube argument gives $d_\ell^\circ(Y) \geq 3\check{R}_\ell - 8$. One unit of the rate therefore pays the crossing weight of this store. These are counts on the filed domains. Analytic estimates use the true dependency sets and the cores of their actual trees.

A.8.0.11. The one-run sum at a strip tree cell. For a term F formed at $j < m$, write $A = X^\theta$ for its tree domain and $\mathfrak{D}_F$ for the tube on its true domain. Its size after the reference subtraction is bounded on $\mathfrak{D}_F$ by $2c_1^\flat e^{-(1+\beta/2)\kappa d_j(A)}$ for $\mathbf{R}$ and by $x^{\mathbf{T}} e^{-\kappa_{\mathrm{out}} d_j(A)}$ for $\mathbf{T}$. The formation estimate for a core at the reading level gives

$$d_m(K(A)) \leq c_H\big(1 + L^{-(m-j)} d_j(A)\big).$$

For $r \leq 8\kappa/5$, the floor on L gives $r c_H / L \leq a_\flat/4$. Thus the core weight $e^{r d_m(K(A))}$ spends at most $a_\flat/4$ of the old length decay, besides $e^{r c_H}$. The density norm spends a further $a_\flat/4$. The remaining rate for the healing term is $(1 + \beta/2)\kappa - a_\flat/2 \geq c_d + 1 + \lambda_\theta$, and the fluctuation term has at least the same reserve. First count all true domains with the same A, at cost at most $e^{\lambda_\theta(d_j(A)+1)}$. Then sum the trees through the specified root with the remaining rate $c_d + 1$. The two families together cost

$$e^{r c_H + \lambda_\theta} K(d)(2c_1^\flat + x^{\mathbf{T}}).$$

Preparatory chain terms add their stage bounds in the tree norm. At $j = m$ use the stronger creation rate for the terms that carry it, since the core length has no scale reduction there. This proves the stated rooted sum for the admitted one-run terms. For the paired estimates the coefficient and domain comparisons are needed at each reading site.

For the strip class, take an actual reading cell $\Delta \subset C \cap \Omega_n$ of level n in the native grading of the treated component C. Every term in this class carries a pocket. Let $\mathsf{b}(m-j)$ be the bound for the weighted sum at a tree root of level j, including the reserve $\varsigma_0 d_j(A)$, $\varsigma_0 = a_\flat/4$, for $j < m$. A true read reaches Δ either through the one-layer enlargement of A or through a pocket strip enlarged by three layers. Every attached pocket has a tree root within two cells of its strip.

If $j = n - u < n$, the pocket was born no later than j, and its strip satisfies $\mathrm{dist}_n(Y^\sharp, \Omega_n) \geq 5/2$. Thus its three-layer enlargement at level j cannot reach $\Delta \subset \Omega_n$. A direct read needs a tree cube near Δ as well as the cube near the pocket. The tree length is at least $(\frac{5}{2}L^u - 6)_+$. There are at most $(L^u + 2)^d$ choices for the root near Δ. The resulting count over $u \geq 1$ is bounded by $C^{<,\theta}\mathsf{b}(m-n)$, since

$$C^{<,\theta} = C_{\mathsf{b}} \sum_{u \geq 1} (1+u)^{p_{\mathsf{b}}} (L^u + 2)^d e^{-\varsigma_0 (\frac{5}{2}L^u - 6)_+} < \infty,$$

using the polynomial lag bound for b. For $j \geq n$, a direct read or a pocket of side at most one has a root in a fixed neighborhood of Δ. A larger pocket has all sides at least $1/101$ in its level-j units. Its strip must meet the seven-cell box about Δ, and the disjoint strips then fit at most 909^d such pockets against that box. A pocket of largest side s_j supplies at most $(s_j + 7)^d$ roots. If three level-j layers reach a quarter of the pocket's birth radius, $s_j \leq 1212$. Otherwise use the full neighborhood of the birth strip in the screened core. Its width at birth level m' is $5\check{R}_{m'}/2$ for $m' < h$, and $(\check{R}_h + 1)/2$ for $m' = h$, where h is the healing level of the core. Its cells have level at most m', so a contour leaving it from Δ spends at least w in contour length per traversed birth-level unit, with w the depth unit of the screened grading. Thus $\varrho(\Delta) \geq 1 + \check{R}_{m'}/4$, and the root count is at most $C(1 + \varrho(\Delta))^d$, with $\varrho(\Delta)$ its contour-depth variable. Summing these levels costs $C(m - n + 1)(1 + \varrho(\Delta))^d \mathsf{b}(m-n)$.

On an outer rim cell $\Delta \subset (C^\sharp \setminus C) \cap \Omega_m$ of level m, consider terms whose cores lie in $C^\sharp$. A pocket-only read at this cell is impossible. An own pocket is at least $9\check{R}_m$ from C^c. A foreign pocket is far enough from $C^\sharp$ that its root within two cells would put the core outside $C^\sharp$. The strip sum on this rim consequently costs just $(3^d + C^{<,\theta})\mathsf{b}(0)$. The estimates use the native reading sets. They supply the local count with no factor depending on the age of a pocket.

A.8.0.12. The strip sum in a healing component. Use the strip terms whose full admitting cores lie in C, so that their true reads X_F^{r} are contained in C. Write C for the never-integrated core of C, $h = m - N$ for its healing level, and $W(F)$ for the nonnegative weight in the cell estimate. If $X_F^{\mathsf{r}} \subset \mathsf{C}$, choose a cell of the true read closest in contour distance to C^c. The contour factor $e^{-\delta_P \hat{D}_F/4}$, with $\hat{D}_F = d^\wedge(X_F^{\mathsf{r}}, \mathsf{C}^c)$, is at most the corresponding factor for this cell. The rooted sum therefore reduces to

$$\sum_{\Delta \subset \mathsf{C}} e^{-\delta_P d^\wedge(\Delta, \mathsf{C}^c)/4} \sum_{F:\, X_F^{\mathsf{r}} \cap \Delta \neq \varnothing} W(F).$$

At a cell of depth ϱ and level n, crossing the intervening zones gives $h - n \leq \varrho$. Consequently $m - n + 1 \leq (N+2)\varrho$, and the polynomial lag bound for b gives $\mathsf{b}(m-n) \leq C(1+\varrho)^{p_{\mathsf{b}}} \mathsf{b}(N+1)$. The strip cell estimate is then at most $C(N+2)\mathsf{b}(N+1)e^{\varrho}$. Use half the screening exponent to pay e^{ϱ}: $d^\wedge(\Delta, \mathsf{C}^c) \geq (\varrho - 1)w$ and $\delta_P w/8 \geq 2$. The remaining exponential sum is bounded by the contour cell-sum estimate times the number $\mathsf{n}_\partial(m)$ of boundary cells of the closed core complement. Thus the screened strip sum is bounded by $C\mathsf{n}_\partial(m)(N+2)\mathsf{b}(N+1)$.

If the true read leaves C, root it at a cell in the closure of $C \setminus \mathsf{C}$. There $\varrho = 1$ and $h \leq n \leq m$. The strip cell estimate, summed over these cells, costs $C\mathsf{n}_{\mathrm{sh}}(m)(N+1)\mathsf{b}(N)$, with $\mathsf{n}_{\mathrm{sh}}(m)$ the cell count of this region. This case includes a term whose tree stays in C while its true read leaves it. These are one-run estimates for the strip part of the healing sum.

A.8.0.13. Oscillation of a stored term with framed slots. Consider the read of a stored term v with retained variables that rotate at sites inside the blocks. Use two admitted seeds with the same frozen component datum. In the screened core the two reads can be brought to one real centre U^s using the real gauges of the frozen-centre construction. Write their fields as $q_i = e^{i\eta X_i} U^s$ in that gauge, for $i = 0, 1$. If g_i is the change to this gauge and Q_i is the product of the frame factors of the original presentation, the physical field is q_i acted on by $(g_i Q_i)^{-1}$. Hence both physical reads can use the same value of the retained variables.

Set $X_\lambda = (1-\lambda)X_0 + \lambda X_1$ and interpolate the logarithms of the factors $c_i = g_i Q_i$ in the same way. With $q_\lambda = e^{i\eta X_\lambda} U^s$ and c_λ the exponential of the interpolated logarithm, compute the slice datum $B[q_\lambda]$ and its critical representative $V(B[q_\lambda])$. Let $u[U]$ be the site map to the block frame for a field U about U^s. The chart argument of the term is

$$u[V(B[q_\lambda])]\, u[q_\lambda]^{-1} c_\lambda.$$

The two site maps convert the frame of the configuration to that of the slice. They are evaluated at the actual rotation sites, including sites inside a block. At the endpoints the physical-frame identity gives the two original values with the retained variables held fixed.

At a rotation site of depth d_s, the screened field and frame estimates bound the log of this product by the local rotation allowance times $C(1+2|\lambda|)e^{-\delta_1 d_s/4}$, with the available margin in the constant. For a deep term let D_v be the depth of its read, so $d_s \geq D_v - c$ on each determining stencil. The core response bound keeps the potential and current on the admitted slice, and the site bounds keep the frame in its rotation domain on the disc $\rho_v = e^{\delta_1 D_v/8}$. If N_v bounds the holomorphic term minus its fixed constant on that domain, subtract its value at zero and use Schwarz's inequality at one. The oscillation is at most $2N_v/\rho_v \leq 2N_v e^{-\delta_P \widehat{D}_v/2}$ in the scaled depth $\widehat{D}_v$. When $\rho_v = 1$ or the term is outside the deep class, use the two endpoint majorants, giving $2N_v$. These are the two bounds for the oscillation sum at the component's own healing.

For the deep oscillation sum, assign a read to one of its nearest core-boundary cells. The depth rate is measured from the full read. One part of the rate pays for bringing the unit weight to that boundary, including the finer cells in its layers. A second part pays for the sum over read locations relative to the boundary. This bounds the sum of the screened oscillations by the boundary count times the per-cell unit bound. The absolute values of pieces filed inside the component are charged at their own interior stage cells. For those pieces the switch sum costs at most a constant times $e^{4d_m(K_v)}$, with K_v the core of the stored term. The original carrier tree length bounds $d_m(K_v)$, so four units of the original carrier decay cover this cost. An arrested term whose label leaves the component is counted in the exterior sum at its collar.

A.8.0.14. The exact datum in the frame of a kept factor. For the kept factors whose frame sites are top block nodes, let Σ be those sites, $V(B)$ the critical slice about the real centre, and c the frame argument of the kept factor. In the frame of the real inner variables the datum is

$$[M(V(B))^{[c^{-1}]}] = [(e^{iB} D_c)^{[c^{-1}\tau_B]}].$$

Here D_c is the real centre datum and τ_B is the node map of the slice. Brackets identify data related by a node gauge equal to the identity on Σ. A further frame d sends c to cd, since $d^{-1}c^{-1} = (cd)^{-1}$. The same slice is used. The logarithmic polar norms add as an upper bound, so composition spends the corresponding part of the frame radius.

At a leaf admitting the frame extension, write $c = e^Y w$ with w unitary. The regauging map $\rho[Y]$ takes the value e^{-Y} at Σ. If $\widehat{w}$ is a real extension of w, the framed value equals the unframed value at $(V(B)^{\rho[Y]})^{\widehat{w}^{-1}}$ with the original real inner variables. This follows by cancelling the real rotation of the variables against the real gauge. The resulting field is critical, with the exact datum displayed above. The polar extension spends $\|Y\|/(8\varepsilon_{\mathrm{fr}})$ of the slice room, where $\varepsilon_{\mathrm{fr}}$ is the frame radius. The real gauge spends none. Thus half of the frame radius costs at most $1/16$ of the slice room. For admitted charts, equality on overlapping parameter families follows first at the real values from uniqueness of the real orbit and invariance of the kept factor under the datum stabilizer. Holomorphy of the frame extension on the product of the slice and frame domains then extends that equality. The polar representative supplies a pointwise tube witness. This argument concerns frames at top nodes. The interior rotation sites of the preceding estimate need the full site map there.

For paired centres with common datum on the rows at Σ, use the same B on those rows. Denote the additional collar datum of the finer run by b_{col}. Matching the exact datum in the two birth

frames uses the finer frame $h(B)c$, where

$$h(B) = \tau^{\mathsf{B}}_{B\oplus b_{\mathrm{col}}}(\tau^{\mathsf{A}}_{B})^{-1}\Big|_{\Sigma}.$$

Indeed $(h(B)c)^{-1}\tau^{\mathsf{B}}$ and $c^{-1}\tau^{\mathsf{A}}$ agree on Σ. The far ends of the rows are compared modulo the same trivial-on-Σ gauges. A common chart then acts by right multiplication on both frames and leaves the collar variables alone. To apply this formula at the mixed complex reads we must construct the paired bases on the actual joined rows, choose the finer lift, and fit both c and $h(B)c$ in their frame budgets on the contour. The matching identity uses the admitted charts. The extension on each older factor must also be returned through the nesting step before that factor is read again.

On complete stencils the curvature and the Wilson current give part of the room estimate for a polar regauge. Each plaquette is conjugated at its base point, and the six plaquettes in a current stencil are transported to the base of the bond. Consequently the current is conjugated there in the same level units. If z bounds the polar log of the regauge, the two operator norms of the regauge and its inverse are at most e^{z}. Each of these weighted entries therefore grows by at most e^{2z}, with a relative increase bounded by $2ze^{2z}$. This controls the terms that transform by conjugation. The chart potential also reads the variation of the lifted gauge between sites. Its first differences and the changes of the joined boundary rows still have to fit in the room for the mixed contour.

On a matched closed set of determining rows one can reconstruct a critical field for the twisted datum locally. Take its physical row logarithm b about the admitted real datum. If the frame is near a nonidentity unitary section, use the real reference transported by that section. Convert b to the reduced row logarithm $b_{\natural}$ using the inverse of the outward-row correction. Solve the response equations with input $b_{\natural}$ and all switches at one. Its exact average is the specified physical datum acted on by the response's node factor. Removing this factor by a fixed analytic extension of the inverse node map gives the required datum. The response and the local row maps are holomorphic for small rows, and the stationary equation extends from the real data by holomorphy. This reconstruction gives an exact local representative whenever the physical row disc lies in the response radius. The actual paired contour must still give that disc on the closed rows and leave the required margins for the potential and the frame.

A.8.0.15. The relative frame at an already admitted later read. Let Y be a live component born at step m, with padded strip $(Y^{\sharp})^{+}$. The strip remains in the unreblocked large-field neighbourhood. Its separation from the younger determining sets leaves the read indexed by the birth level m. Use the actual paired data at this site. Write D_0 for the first run's datum with the finer run's collar entries inserted, d_{net} for the net phase, and g_{cen} for the centre map in the site's exact representation. Let u^{A} and u^{B} be the site frame corrections in the two reads and c^{A} the first frame. The path for the second run's kept factor uses the arguments

$$D_s = g_{\mathrm{cen}}^{s} e^{is d_{\mathrm{net}}} D_0 g_{\mathrm{cen}}^{-s}, \qquad c_s = e^{sZ}c^{\mathsf{A}}, \qquad e^{Z} = u^{\mathsf{B}}(u^{\mathsf{A}})^{-1}.$$

In the datum formula g_{cen}^{s} and its inverse act at the respective endpoints. At $s = 0$ the arguments are the mixed read, and at $s = 1$ they are the second run's actual read. Hold the real inner variables of Y fixed.

The bound for Z first compares the fine site map to the one constructed on the matched coarse inputs. It bounds the logarithm of their quotient by the local radius times the relative decay. A local Lipschitz estimate then replaces the coarse inputs by those of the actual first run. This gives the smallness of Z in units of the read margin. Use the centre-map and net-phase bounds in the same units. At a site with these room estimates the path stays admitted for $|s| < (8C_{\mathrm{read}}\vartheta_{Y}^{\sharp})^{-1}$, where $\vartheta_{Y}^{\sharp}$ is the relative decay of the read at level m. Schwarz applied after subtracting the value at zero bounds the reading change by $8C_{\mathrm{read}}\vartheta_{Y}^{\sharp}$ times the two-point oscillation of the kept factor at fixed inner variables.

For a nonempty switch coefficient subtract the reading change at the empty switch point. The contour bound of the resulting difference is twice the oscillation: each of the two read maps stays in the admitted datum domain. On the vertices use the reading estimate for both values. The geometric mean leaves $C(\vartheta_Y^\sharp)^{1/2}$ times the oscillation, in addition to the switch coefficient decay. Thus the two kept factors together have a reading bound $C\vartheta_f^{m/2}\check{R}_m^{p_\Sigma}$ in their class, by the oscillation sum at their birth. For the kept quadratic form the actual datum read is the block average on the exterior part of the healing core. It has no independent frame argument. This later-read calculation uses the admitted old contours and their comparison maps.

A.8.0.16. The kept Gaussian at a component's own read. For a read contained in the Gaussian core, use the supported real field U^s of the prepared determining set. The moved centre on the padded read has the form $(e^{i\eta\mathfrak{X}}U^s)^v$, with $\mathfrak{X}$ real and small in the local response norm. To test its admission in the older domain, put a real cutoff on $\mathfrak{X}$ which is one on the full clause read and zero outside the next padded set, still within the core. If the cutoff width is $u/8$, then on any older clause cube meeting it, $\xi_l|\nabla\chi| \le 8L/M$ since $\xi_l \le L\xi_j$ and $u = M\xi_j$ in the level-j read. The product rule preserves the bound on the potential and its first difference in the clause units. In the real gauge of U^s this is within the available radius of the older clause. Extend the site gauge v off the padded read and apply it after the cutoff. The resulting ambient field agrees with the moved centre on the required reads. The argument uses the margin from the read to the artificial strata of the core.

On an aligned label at healing, consider a live interrupted attempt W nested in the operated component. Use the common frozen data and the common collar variable in the two block-centre frames. The stored quadratic q_W is evaluated at that variable. Its native expansion is a sum of units T_e on complete carriers. At an admitted real vertex its sum is the retained quadratic on the physical pair. The comparison uses the determining set refined by the attempt at both vertices. Add and subtract the second unit at the first read expressed in the common datum. This separates the change of the stored coefficient from the change of its read. The reading difference vanishes if the two data agree. For a positive displacement d_e, the line in the native unit tube has radius $r_G/(4d_e)$. With unit majorant G_e, Schwarz's inequality bounds the reading part at the vertex by $4d_eG_e/r_G$. The extension in the switch parameters uses the geometric mean of this bound and the one-run majorant. Together with the stored mark, the resulting size is at most $C\varpi(e)\Delta_{n_e}^c$, where Δ_n^c is the classical load of zone n and $\varpi(e)$ is the native unit weight.

The overlap bound sums these weights by the visited collar cells. For all attempts nested in C, at a fixed zone n, the absolute vertex differences thus sum to at most $CN_n^{\mathrm{vis}}(C)\Delta_n^c$, with $N_n^{\mathrm{vis}}(C)$ the number of visited cells of that zone. Multiplying by a zone parameter of radius E_1/Δ_n, with $\Delta_n^c \le \Delta_n$, costs at most CE_1 per such cell. The rooted sum for the analytic units also retains the spatial weights. Their two tree lengths are bounded by a fixed multiple of the number of constituent cells. Spend the matching part of the native unit decay on these weights and sum the remaining overlap weights at the chosen root. This gives a bound $C\Delta_n^c$ at that root.

A.8.0.17. Composed reads through the preparatory stages. In one run let a chain enter at stage n_e and descend through stages to a stored read at $\nu \le n_e$. Use the fixed real centre U_c of the entry. On the padded leaf hull X^+ write $\mathbf{U} = e^{i\eta A'}U_c$. Extract the datum $B[\mathbf{U}]$ by the chart's nonlinear averaging map on the induced box set $\mathfrak{S}$. For this datum solve in the Landau chart at U_c with its own response $H^\dagger J(U_c)$, obtaining $V(B[\mathbf{U}])$. The stored term $F(X;\cdot)$ is evaluated at

$$S(\mathbf{U}) = (V(B[\mathbf{U}]), J(V(B[\mathbf{U}])))|_X.$$

The current in this pair is recomputed on the slice. At real points the chart represents the native leaf datum up to gauge. On the complex disc the slice is its holomorphic continuation. With no moves, the entry U_e is already critical on the induced box set by restriction from the entry chart. Uniqueness in that chart gives $S(U_e) = (U_e, J(U_e))|_X$.

For the radius check let T_i^a be the stage i bound on member E_a of the field and current table. Put $\mathrm{rm}_i^a = \mathsf{T}_i^a - \mathsf{T}_{i+1}^a$ and $G^a = \mathsf{T}_\nu^a - \mathsf{T}_{n_e}^a$. The stage-room bounds give $\mathrm{rm}_i^a \ge 0$. Let a_i^0 be the size share of the move at stage i, with the local stencil suprema included by the slowness bounds. The classical entry condition at the stage bounds $\sup(|h_i| + \ell_i|\nabla h_i|) \le a_i^0/K_o$ on its local read. The composite logarithm $\mathcal{Y}$ of the moves over U_e then has local size at most $2\ell\sum_i a_i^0/K_o$ by the small-product estimate, with ℓ the slice scale. On the admitted chart, the change of the box datum on each cube is bounded by four times the supremum of $|\mathcal{Y}|$ on that cube. This follows from the linear average bound and the remainder at U_e, using the classical ceiling $C_w|A_e| \le 2$ there, where $U_e = e^{i\eta A_e}U_c$. The local Landau response bounds and slowness therefore control the change of the member with scale power e_a by

$$\frac{16B_\natural C_{\mathrm{sl}}}{K_o}\,\ell^{1-e_a}\sum_{\nu\le i<n_e} a_i^0.$$

Here $e_a = 1, 2, 3$ for the potential, its first difference and the current respectively. For the plaquette member use the gradient bound with its extra η^2 factor. The real-centre member and the cube condition are inherited from the entry.

At a point in the units of the read, let i_1 be the first stage that raises its level, going from ν toward n_e. Before i_1 the converted share at stage i is at most $80\mathsf{r}_1\,\mathrm{rm}_i^a$, where $\mathsf{r}_1 = C_1/C_0$ is the ratio in the stage table. The later shares cannot be charged to their own stage rooms in the finer read units. Instead the share sum from i_1 onward is at most $C_\Sigma\bar{u}_{i_1}^0$, with $\bar{u}_i^a$ the local stage radii. The units at the first raising stage are still the read units, and $\bar{u}_{i_1}^a \le 40\,\mathrm{rm}_{i_1}^a$. Charge this whole part to the first raising room. The two parts spend disjoint rooms, so the total converted share is at most $400\mathsf{r}_1 G^a$. Without a raising stage only the first bound is needed. The fixed choice $K_o \ge 2^{18}B_\natural C_{\mathrm{sl}}\mathsf{r}_1$ then makes the displacement at most $G^a/8$. Adding the entry size gives

$$E_a(S(\mathbf{U})) \le \mathsf{T}_{n_e}^a + G^a/8 = \mathsf{T}_\nu^a - 7G^a/8.$$

Where the point has been raised beyond the leaf's own read level $n_F(x)$, convert the read table to that scale. If the level $n_0(x)$ before the chain exceeds $n_F = n_F(x)$, the factor in this comparison is bounded by $K_u L^{-u}3^{-v}\sqrt{u+v} \le K_u/L$, where $u = n_0 - n_F \ge 1$ and $v \ge 0$ is the excess of the current label level over n_0. At $n_0 = n_F$ the stage-table bound $(9/4)3^{-v}\sqrt{v} \le 3/4$ for $v \ge 1$ gives the same allowance. In the leaf's units the entry spends at most $3/4$ of its radius. For the displacement, the conversion gains at least one power ℓ_{n_F}/ℓ, and the share sum at the read stage is at most $C_\Sigma\bar{u}_\nu^0$. The unit comparison and the choice of K_o then bound the displacement by $1/200$ of the leaf's radius. Thus the total is at most $3/4 + 1/200 < 4/5 - 1/40$. On the label-equal part, use the scheduled table containment $\mathsf{T}_\nu^a \le (c_F^+ - \varrho_F)\mathsf{u}_F^a$, where $c_F^+\mathsf{u}_F^a$ is the leaf's full radius and $\varrho_F\mathsf{u}_F^a$ is its reserve. Fresh terms of the top layer of the healing step and terms left from a preparatory chain use their separate containment clauses, with the latter's own reserve. For the composite plaquette and current members with index at most the birth index, restriction of the slice chart identifies them with its own plaquette and current members. These checks give the one-run stored read on the admitted preparatory chain. The paired Gaussian hull comparison has its own admission.

A.8.0.18. The common datum at a preparatory vertex. Take two aligned native vertex reads over a preparatory seed. In run B let v_0 be the common datum with its prescribed B completion. Write $v_s = e^{isd_{\mathrm{net}}}v_0$ for the interpolation to the actual datum v_1, after removing the real block-centre frame factors from the two ends of the read. The logarithm d_{net} is of the resulting relative group product. In the fixed chart the response bound controls the change from v_1 in each member by $|1-s|D^a$, where $D^a = 2B_\natural\ell^{-e_a}(|d_{\mathrm{net}}|)^{\approx}$ uses the exponentially smeared modulus of the mismatch. The class of the read is governed by $q = \vartheta^{(j\wedge n_X)/2}$, where j is the formation level and n_X is a lower bound for the own read levels on X.

At a point where the label level equals the leaf's own read level, split the mismatch into cells where a move acts and cells of the entry absent from the common data. For a move cell the level l is the label level and satisfies $l \le j$. The noncommon cells have $l = j$ below the label or l below

both j and the label. On a move cell of level l the share bound is $C\vartheta^{l/2}$ times its size share. Here $l \ge j \wedge n_X$, so the preceding first-raising sum bounds this contribution to D^a by CqG^a. At a noncommon cell the mismatch is at most $C\vartheta^{l/2}$ times its local datum unit. For the smeared sum, split $e^{-2\delta_1 d}$ at distance d into rates $\delta_1, \delta_1/4, \delta_1/4, \delta_1/2$. The first converts the zone factor $\vartheta^{l/2}$ to Cq by screening. The second pays the change of local radii, the third yields the depth factor $C2^{-(j-n_F)}$ at read level n_F, and the fourth sums the rooted kernel. The depth factor is proportional to the leaf's tube fraction ϱ_F, so these cells cost $Cq\varrho_F \mathsf{u}_F^a$, with u_F^a the leaf's member unit. At a raised point convert to the earlier leaf units as above, using the screened and smeared bound with the depth factor omitted. With the constant of the comparison enlarged once, the two estimates are

$$D^a \le C_{TL} q\,(G^a + \varrho_F \mathsf{u}_F^a) \quad \text{at a label-equal point,} \qquad D^a \le (C_{TL}/10) q\,\mathsf{u}_F^a \quad \text{at a raised point.}$$

Use the datum comparison in the range $8C_{TL}q \le 1$. This follows from the coupling ceiling $8C_{TL}\vartheta^{n_c/2} \le 1$ when both the formation level and the chosen lower read level are at least n_c. Take $R = (8C_{TL}q)^{-1} \ge 1$. For $|s| \le R$, $|1-s| \le 2R$, so the first estimate spends at most a quarter of $G^a + \varrho_F \mathsf{u}_F^a$ and the second spends at most $\mathsf{u}_F^a/40$. The one-run slice margins admit these displacements in the quarter-room tube. At a raised point the largest coefficient used is $3/4 + 1/200 + 1/40$, below the quarter-room allowance $4/5 - 3/(4 \cdot 40)$. At v_0 the factor $|1-s|$ is one, which gives the smaller eighth-room tube. The datum radius itself is checked before solving for the slice. On the noncommon entries the local datum unit carries the scale factor $L^{l-\iota}$ relative to the current level ι. The comparison floor gives $\vartheta^{l/2}L^{l-\iota} \le \vartheta^{\iota/2} \le q$; on the move cells use the share bound. Thus $R \sup|d_{\mathrm{net}}|$ is controlled by the input datum ceiling and the fixed choice of C_{TL}. Together with the baseline datum budget this puts the full path in the chart's datum ball, as required for the box response estimate.

On this disc the difference $\widehat{F}^{\mathsf{B}}(v_s) - \widehat{F}^{\mathsf{B}}(v_0)$ is holomorphic, vanishes at zero, and is bounded by $2\sup|\widehat{F}^{\mathsf{B}}|$ on the admitted tube. Schwarz's lemma at $s = 1$ gives $16C_{TL}q \sup|\widehat{F}^{\mathsf{B}}|$. This supplies the reading comparison on the native preparatory vertices, using the native share and noncommon datum estimates. For a family with extra strip letters the same argument requires these datum bounds on that family.

A.8.0.19. The kept-term sum in a preparatory integration. At preparatory stage $n+1$, take a native kept term v of a live region Y born at $m < k$. Its pointed and window slots are empty. Its field read uses the padded strip hull $(Y^\sharp)^+$. The active response blocks are those whose enlargements meet this hull. At zero block parameters the move is the identity, so put this value of v in the base. Expand the remaining difference first in the nonempty active block sets A, then in cell switches for the box at fixed A.

For the cell expansion, leave the cells of the enlarged core off the mandatory core at switch value one. Partition the other cells into grains, which are unions of cells. Give all cells of a grain one switch parameter. The restricted family is holomorphic on the grain polydisc. On a zero face, deleting the grain one cell at a time uses the one-machine zero-face identity. At each grain vertex the value is the sum of cell coefficients whose selected cells lie in the enabled grains and the fixed core cells. Sorting by the set of grains met, then inverting, shows that each grain coefficient is the sum of all cell coefficients meeting exactly those grains. The subset of fixed core cells in this sum is unrestricted.

Use Cauchy's estimate on this grain coefficient before taking absolute values of the cell pieces. If $\mathcal{Y}$ is the selected grain set, its bound is the oscillation of v on its admitted read, times

$$e^{-\kappa_1^{(1)}|\mathcal{Y}|} \prod_{\square \in A} (r_\square - 1)^{-1}.$$

Here $r_\square = c_R/\vartheta(\square) \ge 2$, so each inverse radius costs at most $2\vartheta(\square)/c_R$. Piece locality requires every wall component of the selected grains to touch the enlarged core. This uses the preparatory

locality of the full padded read of v. The coefficient also reads the moved base on the enlarged core and selected grains and the stored variable Φ_Y.

Let R_n be the moved region at the stage, $w = M/M_1$, and S_v the native one-cell anchor of the strip. Use level $h+1$ for its corridors in a treated component and level k in an untreated one. The native corridor from a block $\square$ to S_v has fewer than $d(x_\square+1)$ cubes, where $x_\square$ is the distance in its corridor units. The treated and untreated corridor bounds give $x_\square + 1 \le 110 d^\wedge(\square, R_n)/w$. Thus the length weight spends at most $\delta_P d^\wedge(\square, R_n)/4$ when $\delta_P w/4 \ge 110 d(a_k+1)$. The inverse block radius decays at rate $\delta_P/2$ in this distance. After paying the corridor and the fixed flat-format costs, the block weight $y_{v,\square}$ is bounded by $K_2^\flat \vartheta e^{-\delta_P d^\wedge(\square,R_n)/4}$. The weighted grain activity is bounded by $x^U = \exp[-\kappa_1^{(1)} + (a_k+1) + 3^d c_\flat]$. The flat grain floor $2e^2 \widehat{b}_t^2 N^\flat x^U \le 1$, with $\widehat{b}_t$ the adjacency bound and $N^\flat$ the core size bound per block, gives the shape sum at most $e^{|A|}$. The factor e^{-1} per block reserved by the flat-format bound pays this cost. Put $\Sigma_v = \sum_\square y_{v,\square}$ over the blocks of the full read. The confined or two-box block count gives $\Sigma_v \le 2c_{\mathrm{p}} K_2^\flat \vartheta V'$, where V' bounds the contour block sum for a single component of R_n. The coupling ceiling gives $\Sigma_v \le 1/(3N) \le 1/2$, where N is the number of young levels of the step. The remaining block subset sum at a specified anchor $\square_0$ costs at most $e^{\Sigma_v} y_{v,\square_0}$.

At a target filing cell, the anchor can reach the cell through its own core or corridor, or through an attached wall component of grains. In the first case the cell is within $103\check{R}_k$ of the block in level-k units. In the second case the positional grain bound supplies the factor $e^{-(\mathrm{dist}_k - 103\check{R}_k)_+}$. This gives the same position factor for both cases. For the other root sum, a cell meeting S_v is within that distance of every eligible block of v, and the nonempty subset sum is bounded by $\Sigma_v e^{\Sigma_v}$.

The near-strip geometry gives $d^\wedge(\square, R_n) \ge w(5\check{R}_m/2 - 6) \ge w(\check{R}_k - 5)$ for every eligible block. In an own chain it uses the screened collar of $Y^\sharp$; at $m = h$ use the width in level-$(h+1)$ units and $L\check{R}_{h+1} \ge \check{R}_h$. In a foreign chain the collar is inside the untreated component. The hull count gives at most 2^d regions at the same birth level per block. Their polynomial oscillation bound is absorbed at rate $\delta_P/16$ in $d^\wedge$, leaving rate $3\delta_P/16$ in the block sum. Use the uniform bound for $y_{v,\square}$ before summing over the leaves at the block, since their corridors can differ.

This last sum must use the ordinary old blocks together with all blocks meeting the padded kept hulls. The old-block summation proof extends to this enlarged class. Every added block lies in the same cover and has the same lower bound $w(\check{R}_k - 5)$ on its distance to R_n, hence to each component of R_n. In the component sum at rate $\delta_P/8$, half the exponent pays V'. The near component shells then gain $e^{-(\check{R}_k - 5)}$, and the shells at distance $l\check{R}_k$ gain $e^{-l\check{R}_k}$. The outer shells around the target cell gain the position factor from the anchor bound. The original polynomial shell counts and tail floor therefore bound the enlarged block sum by the same V'. The kept contribution to each of the two rooted one-run sums is at most $CK_2^\flat \vartheta V'$ times the kept oscillation constant. This uses the native one-run tube and preparatory piece locality.

At a preparatory stage use the same unit identity after admitting the stage's endpoint maps in the local Gaussian tube. On each complete hull align the parameter-independent axial centres $U_c^{\mathsf{A}}, U_c^{\mathsf{B}}$ by a real gauge Γ, so that their coarse comparison is $\mathcal{M}((U_c^{\mathsf{B}})^\Gamma) = e^{i\eta a_c} U_c^{\mathsf{A}}$. The paired chart gives the corresponding bounds for the centre displacement, the gauge at the datum points and the displacement of the two chart fields on the parameter disc. In the local length units each is bounded by the local radius times the chart decay. When comparing the actual bond datum, take these bounds on both endpoint top blocks and on the smeared averaging contour. In addition to the chart decay there is the error L^{-m} of the averaging contour at separation m in levels. At the preparatory site both errors are bounded by $C\vartheta_2^{n_e/2}$ for the lowest zone n_e of the unit hull. The affine estimate keeps the radius factor, since it occurs in both bounds before the geometric mean, and gives decay $\vartheta_2^{n_e/4}$. The weighted pile over a finite set of cells $\mathcal{A}$ is at most $C|\mathcal{A}|\Pi_G^+(x_{k'})\Delta_n^c$

for the class (n, k'). Apply this to the cells meeting the actual preparatory block and use its block count. This gives the stage load $C\Pi_G^{\mathrm{blk}}(x_{k'})\Delta_n^c$, with Π_G^{blk} including that count.

The small factor for a block uses the count and the unit's local radius together. For each unit meeting the padded response block, choose a common site in its carrier and a current cell at that site. The cell count of $\mathcal{A}$ is bounded by the square of the window bound at this current level, by the spread condition of the response block. The current level and the level of the largest chart radius of the unit differ by at most $N_e + 3$, where N_e is two plus its number of collar cells. The flow then bounds both the window size and the old collar polylogarithm by functions of the coupling at the largest-radius cell, with an increment $O(N_e + 1)$ in inverse squared coupling. If Θ_e is that radius and ϱ_e^2 the old polylogarithm, the reserve e^{-N_e} absorbs the extra dependence on N_e and gives

$$|\mathcal{A}|\varrho_e^2\Theta_e e^{-N_e} \le C\Pi_G^{\mathrm{blk}}(x_{k'}).$$

Use the factor $1/|\mathcal{A}|$ from this bound against the positive unit pile over the cells of $\mathcal{A}$. For a nonempty response coefficient, subtract the value of the unit at zero before applying the anchored Cauchy estimate. The mixed derivative kills that constant. Its rim-anchor sets are empty because the retained collar variable is held fixed in this integration.

The layer bound in the first comparison uses the actual chain vertices at unit dilation and the cut vertices with dilation parameter between zero and one. In the product rule for these reads the dilation member can be evaluated at unit radius. It costs at most $16c_H \sup_p m_p$, where m_p is the amplitude of the source at the row p and c_H is the product-rule constant. The source bound gives $\sup_p m_p \le C\alpha_{0,m}$. This estimates the member directly from the amplitude. On the outer layer the available row room can include the large window weight. That weight remains in the room for the other members of the layer estimate, with the coupling of the same window.

These estimates concern the stored quadratic, in its own collar variables. The new Gaussian integral and its determinant have their separate comparison. At healing, for each unit, machine, filing domain and real hull centre, we must give holomorphic bond configurations on the full padded hull for all subsets of its switches. The configurations must agree bond by bond when a switch is set to zero and its membership in the subset is changed. At the binary vertices they must recover the physical axial reads in the fixed chart. If the far read sees no operated component, the unit must have its root value in the corresponding source frame. The family must stay in the hull tube and satisfy the deep spatial bound. For far units one can use the weaker contour estimate below. The vertex comparison and the rooted estimates apply to the unit values in this assembly.

For the all-one vertex take a fixed wall-connected box G containing the complete read and the full enlargements of the operated components it meets. Use the restriction rule for the determining sets and the matched frames at the glued seam. Put the four stages of the construction on this same box. The starting field is the restriction of the root. On the outer rows the shell restriction gives identical rows and levels in the successive determining sets. Each increment and the intervening gluing are supported away from these block pairs. Consequently the next stage inherits the root average on these rows. On the inner rows use the prescribed prepared data and the frozen data on their respective parts. At the glued stage the cut data are the identity on one side and averages of the matched top data on the other. Composition of the averages gives the prescribed coarse rows there. Thus the last stage has exactly the native datum of the single box. At an admitted datum, uniqueness of the critical orbit identifies the last field in the fixed axial frame. If there are no operated components in the box, there are only outer rows, and the last field and the restricted root are in the same orbit. For the Gaussian value this comparison uses the retained variables transported in the same gauge.

When the one-run and comparison presentations are used at a natural vertex in the same cutoff, their physical boundary functions must agree at a fixed collar variable. Consider the two admitted presentations as continuous families on an open set of real core data, with the same core datum and absolutely axial endpoints. On a dense set of those data the endpoints belong to one orbit at real

parameters. Remove the two accumulated frame maps to obtain the relating gauge of the physical reads. Axial transport makes this gauge constant on each core block. On the graph of core data rows the gauge at one end of a row is transported by that row to the other. At a reference node the gauge must commute with every loop holonomy. If two core-loop holonomies are free coordinates of the datum, their noncommuting values are dense. For such a pair in SU(2) the common centralizer is $\{1, -1\}$. The gauge is then the same central element on the blocks containing the hull base points, so it fixes the hull datum and the collar variable. Continuity in the real core data gives equality also at reducible data. Holomorphy in the presentation parameters then gives equality on the common parameter disc at this vertex. This use of continuity requires two admitted families over an open set. At an isolated reducible datum a stabilizer can rotate the collar variable.

The chart source of a link is fixed by the handed increment in the physical frame. If $\mathfrak{Q}_{s-1}$ is the accumulated frame map before the link, and the chart updates the average at c by $e^{ib_s(c)}$, frame covariance conjugates that factor by $\mathfrak{Q}_{s-1}(c_-)^{-1}$ in the physical average. To realize the prescribed increment $\beta_s(c)$ one therefore uses $b_s(c) = \mathrm{Ad}_{\mathfrak{Q}_{s-1}(c_-)}\,\beta_s(c)$. This is a recursive holomorphic assignment on the admitted tube, since the frame is known before the link. It preserves the size at real data, and the complex frame bound multiplies the size by at most $e^{2\cdot 1.07 K_u \varrho_\Pi}$. A row set to the previous object's own average has zero increment in both frames. For a kept Gaussian the increment may also enter its quadratic in the carrier frame, so its literal value must be kept in that quadratic.

For the zero-face check, the local fixed-point comparison uses switches on enlarged walk carriers. In top-level units give a switch cell Δ a halo $\mathcal{N}(\Delta)$ of width $3LM_1$. Multiply a walk and its transpose by the product of the switches whose halos meet its enlarged localization set, including switches of cells outside the enabled box. A shell row of level n depends on the box within $3s_n$ of the row, and $3s_n \le 3LM_1$. Thus it is the halo condition that removes walks whose rows could change when a cell is deleted.

If the walk has contour tree length $\widehat{d}$, the geometry floor places all its switch cells within $M/4$ of the tree. There are at most $C_d(1 + \widehat{d}/M)$ such cells by the covering count. For $|s(\Delta)| \le e^{\kappa_1}$ the product costs at most $e^{C_d\kappa_1} e^{\delta_0 \widehat{d}/2}$ under $\delta_0 M \ge 2C_d\kappa_1$. This leaves half of the original walk decay.

For one native response link with a fixed base, set a switch to zero and compare the two boxes. The surviving walk words have the same intermediate local stencils, levels and base values off the halo. The first kernel and its local output stencil kill the output on the halo. The last switched kernel reads only input stencils away from the halo. Where the prescribed data agree on those stencils, the two systems therefore have the same equations on the common part and the same weighted uniqueness ball for the response extended by zero. For the ordinary slot system this applies with the slot source on core rows and the prescribed current slot of the fixed base. For the sourced slot system the two extra terms are

$$-\mathcal{G}_0(1-\theta)Q^*\theta_1 H^\dagger_{\mathbf{\Omega}_1(W)} J_1, \qquad -\mathcal{G}_0(\theta J_{1,2}),$$

where W is the box of the link. Switch the cut-geometry operator $H^\dagger$ and the outer kernel $\mathcal{G}_0$ by the same rule. The two J sources are fixed in the core, and the operators between the switched kernels read only local stencils. This makes the sourced terms obey the same zero-face restriction. Rim data read from a fixed configuration agree on the common stencils as well. Uniqueness gives equality of the two outputs on the common bonds and the base on the halo. To apply this check to the Gaussian family, we must verify those walk and row identities for the boxes generated by the unit's switch subsets, including the junction data, and identify the reconstructed bonds in the fixed physical frame on the full padded read.

A.8.0.20. The exterior sum of the Gaussian units. At healing, form the core K_e of a unit by padding its hull by one layer of level-m cells and adjoining the whole protected blocks it touches. For a set of enabled exterior cells J, read in the component $G_e(J)$ of this core after adding J and the protected

blocks joined to it. A unit is far if its padded hull avoids the enlarged operated region. For a far unit subtract the value at the root before taking the alternating sum. This makes the vertex value zero when the box sees no operated component. For a deep unit use the full value, with its hull inside the collar of one treated component. The other units are shallow. The coefficient of a set B of cells is the alternating sum of the values for $J \subset B$. The sum over B recovers the full vertex value, with the root subtraction for a far unit. If a cell of B is outside $G_e(B)$, pairing subsets which differ at that cell gives zero. Thus each connected part of B must attach to the core, counting cells sharing a protected block as adjacent.

At an admitted far read let b_e bound the displacement of the hull datum from its root value at every vertex and let r_e be the local chart radius on the hull. To estimate b_e, compare the exact average of the last field with that of the root, restoring both endpoint frames. The frame corrections are products of the factors for the intervening layers. Their logarithms are bounded by the sum of the layer increments. The two averages can be compared on the disc of the same real centre, where Cauchy's estimate gives the corresponding Lipschitz bound for their logarithms. If the increments at the switch vertices decay away from the enlarged operated blocks $C^\sharp$, uniformly on the admitted tube of hull data, this gives

$$b_e \le C\alpha_{0,m} \sum_C e^{-\delta_P d^\wedge(X_e^+, C^\sharp)}.$$

Here $\alpha_{0,m}$ is the step amplitude and $d^\wedge$ is distance measured along paths in the local scale units. On the complex polydisc it suffices that the total size of the increments is bounded by the chart radius. Both estimates are on the same switch family.

For a function of that family bounded by V on the vertices and by H on the polydisc of radius ρ_1, the alternating coefficient for B is bounded by

$$\Big(\frac{4}{\rho_1 - 1}\Big)^{|B|/2} (VH)^{1/2}.$$

Indeed the vertex sum gives $2^{|B|}V$. Face agreement lets one replace the vertex value in the alternating sum by its integrated mixed derivative in all the switches of B: the terms on a zero face cancel in pairs. Cauchy's estimate gives $(2/(\rho_1 - 1))^{|B|}H$. Switches outside B stay at one. Take the geometric mean of these two bounds.

Write ν_e for the majorant of the unit and $\nu_e^\times$ for that of its common-datum difference. With the fixed tube constants absorbed, the one-run difference from the root has vertex bound $C\nu_e b_e$ and contour bound $C\nu_e r_e$. The change of the common-datum part has the same pair of bounds with ν_e replaced by $\nu_e^\times L^{-\theta j_e/4}$, where j_e is the level of the arrest. For the reading part of the two-run difference from the root, use the better of the separate one-run vertex estimates and the direct vertex comparison. With n_{X_e} the lowest zone on the hull this gives a vertex bound

$$C\nu_e \min\{b_e,\ r_e \vartheta_2^{n_{X_e}/2}\} \le C\nu_e (b_e r_e)^{1/2} \vartheta_2^{n_{X_e}/4}.$$

Its contour bound is $C\nu_e r_e$. Thus in addition to the common switch factor the reading coefficient has size at most $C\nu_e r_e^{3/4} b_e^{1/4} \vartheta_2^{n_{X_e}/8}$. The other two coefficients have the factor $(b_e r_e)^{1/2}$. The quarter-power loss in the reading part therefore affects the spatial and coupling gain as well as the zone rate. The comparison uses the zone rate bound $\vartheta_2^{n/8} \le \vartheta_f^{n/2} \le C\Delta_n^c$ for the reading class $n \le n_{X_e}$, with ϑ_f the decay rate at the later reading sites. Keep the common-datum part in its own class, at j_e or at the level of its filing cell.

For the sum at an exterior cell Q, file the piece for (e, B) on the union of the core, the cells of B, the protected blocks they touch and their attached treated components. Weight it by $e^{a d_{m,Z}}$, where $a = 31\kappa/20$ and $d_{m,Z}$ is its tree length outside the large-field region. If the coefficient is nonzero, this label connects K_e to Q. The length is at most $C(N_e + |B|)$, where N_e is two plus the number of collar cells of the unit. Its ordinary length is at least $\mathrm{dist}_m(Q, K_e) - 1$. Bound the relative length

by the ordinary one and use an extra unit of ordinary length weight to factor out $e^{1-\operatorname{dist}_m(Q,K_e)}$, with distance in level-m units. If c_l is the length allowance per switch cell, take the coefficient rate $\kappa_1^\natural$ so that $\kappa_1^\natural-(a+1)c_l$ is at least the rate required to sum connected families at the boundary of the core. This sum costs at most e^{CN_e}, since the number of core cells is $O(N_e)$ and the adjacency has bounded degree. The unit weights have a corresponding reserve in N_e. What remains to sum is the unit weight times its spatial profile and the factor $e^{-\operatorname{dist}_m(Q,K_e)}$.

For each C in the far profile, use one fraction of its spatial decay to continue that last distance to $C^\sharp$. Use a second fraction to move the decay from the hull to the filing cell of e. The hull has $O(N_e)$ touching cells, so this uses another fixed exponential in N_e. The remaining weight at each filing cell is summable over all live arrests at once. The distance sum over filing cells is bounded by a constant times the number of exterior boundary cells of $C^\sharp$. For this step the distance rate times the width of a cell in its own units must be large enough to pay for subdivision across levels. Finally the separated blocks satisfy $\sum_C e^{-\operatorname{dist}_m(Q,C^\sharp)}\le C$, by counting them in shells around Q. The boundary count costs at most a fixed power of the separation scale.

The unit weight contains a polynomial in the logarithm of the inverse squared coupling at j_e. For the common-datum comparison it is absorbed in the classical load. For the one-run estimate and the weaker reading estimate use a further fraction of the spatial decay: the path from the hull to the boundary crosses the intermediate zones, giving exponential decay in $(m-j_e-2)_+$ after the hull-size cost. The flow bounds the logarithm at j_e by its value at m plus $C\log(2+m-j_e)$. Thus the cost after summation is polynomial in the logarithm at m. Denote that cost by P_m and the common upper bound on the boundary-cell counts by $N_{\partial,m}$. In the exterior norm (the sum of the weighted suprema over the labels containing Q, followed by a supremum over exterior Q), the far one-run bound is $CN_{\partial,m}P_m\alpha_{0,m}^{1/2}$. The two-run bound for class n is

$$CN_{\partial,m}(\alpha_{0,m}^{1/2}+P_m\alpha_{0,m}^{1/4})\Delta_n^c.$$

There are enough spatial fractions in the weakest rate $\delta_P/4$: the cell-summation rate may be chosen at most $\delta_P/16$ in the scaled distance.

For the deep units, use a fixed vertex as the reference. The reference has zero alternating coefficient for nonempty B. Only those B enter the exterior norm because the core is inside the operated region. On a deep hull use the admitted contour bound on the datum increment from this vertex. The preceding comparison then uses the full contour gain $C\alpha_{0,m}e^{-\delta_P d^\wedge(X_e^+,\mathsf{C}_C^c)}$ in place of b_e, where C_C is the collar of C. Cauchy's bound for one run and the vertex/contour bound for the reading difference retain, respectively, $\alpha_{0,m}$ and $\alpha_{0,m}^{1/2}$. Sum at the exterior boundary of the collar in its own level units.

The shallow units use the tube majorant even when their hulls meet the boundary. If the arrest is inside a treated collar but its hull reaches outside, the collar-cell carrier reaches across the collar margin. If the arrest is outside all treated components but its hull meets an enlarged one, the carrier spans the separation from that arrest to the enlargement. Either way, $N_e\ge c\check R_m-C$, where $\check R_m$ is the separation scale. Spend the spare exponential in N_e to get $Ce^{-c'\check R_m}$; in the one-run sum reserve part of it for the polylogarithm at the arrest level, since the carrier has a cell for each intervening level. The same boundary sum now applies with this carrier gain in place of the small spatial amplitude.

All three estimates are small after the boundary and logarithmic costs. For the two-run direction, the dial radius cancels the classical load against the full load. These are the exterior Gaussian contributions to the terminal comparison.

A.8.0.21. Box pieces for a term at its own healing. Consider an interior unit v at the healing site **L**. By the pocket-extinction rule, each of its pockets dies inside the integration region at this step. Use its full fat core K_v in every box of the four links, along with the mandatory hubs $H_C^{(t)}$ of link t. The eligible cells at link t are the cells of its partition that meet neither the interior of any

$H_C^{(t)}$ nor K_v°. Thus a switch never cuts the core. At zero switch on Δ, the one-machine locality identity identifies the restricted response and averaging operators on K_v° for boxes differing by Δ. The moved point reads the operators through a function supported in K_v°, so it agrees for the two boxes. This applies to a shallow interior term as well as a deep one. A kept term reading a foreign live strip still uses the strip parameters above.

For a tuple of finite eligible switch sets $\vec{B} = (B_1, \dots, B_4)$, fix nested regions Y_t before varying the vertices. Take Y_t to be the union of the level-m cube cover of K_v and of the cells of $B_1, \dots, B_t$, enlarged by the components of Z it meets. Thus Y_t depends on the full sets B_s, $s \le t$. If $J_t \subset B_t$ is the enabled set, the box in Y_t consists of the fixed core and hubs there together with the cells of J_t. Hubs outside Y_t are separate wall components and are unread. Build the chain from link 4 to link 1, using at each new chart the parent readings on its layer and the prescribed data at the original sites.

For the analytic handoff to the next link, let c be a young site in that chart, with padded read $P^+(c)$. In a common frame write $b_t^{\mathrm{off}}(c)$ for the parent layer increment before the move and $b_t^{\mathrm{nat}}(c)$ for its increment after the move. Use the handed increment

$$b_t^{\mathrm{off}}(c) + \prod_{\Delta:\ \mathcal{N}(\Delta)\cap P^+(c)\neq\varnothing} s_t(\Delta)\big(b_t^{\mathrm{nat}}(c) - b_t^{\mathrm{off}}(c)\big),$$

where the product uses the link's switch cells and $\mathcal{N}(\Delta)$ is the switch halo. Compose its exponential, after frame transport, with the baseline group datum. At original sites the prescribed increment is zero. If a zero switch meets the padded read, the layer datum is the parent datum on both faces. If its halo misses the read, the natural increment is unchanged by the local response identity. At all ones the handoff is the native layer datum. In particular, the row 2 continuation uses the switched datum even off its box. Keeping the old exterior field without that switch would give two different layer data on a zero face when the toggled cube contains that read. On the admitted reference tube the number of halo factors per young site is bounded and the layer estimate keeps the new datum in its chart.

On the admitted switch discs the responses, layer averages and chart maps are holomorphic. Their composition at a fixed $\vec{B}$ has the same zero-face agreement: the regions are fixed, the parent of the toggled link is fixed, and the remaining links read its output only. For the invariant unit evaluation $F_{\vec{B}}$ on this family let $T_v(\vec{B})$ be its alternating sum over $J_t \subset B_t$ at unit switch values. For $\vec{B} \neq \vec{\varnothing}$ the mixed Cauchy estimate at radius $\rho > 1$ gives

$$|T_v(\vec{B})| \le \left(\frac{2}{\rho - 1}\right)^{\sum_t |B_t|} \inf_{c_0} \sup |F_{\vec{B}} - c_0|.$$

This estimate needs only the family for the given full tuple.

To recover the endpoint, take $D = C$ or $C^\sharp$, and sum over switch sets contained in D. Each Y_t is contained in D because D contains K_v and contains every component of Z it meets. In the sum over B_4, the regions Y_1, Y_2, Y_3 are fixed. The link 4 value for J_4 depends only on J_4 and the hubs reached by the reads from the fixed smaller regions or connected to their wall components. Extra hubs included only by $B_4 \setminus J_4$ are separate summands. At the full enabled set every cube of D outside the fixed core and hubs is enabled. Any hub in D belongs to a component of Z that meets the core or those enabled cells, so the enlargement includes it. Since D is wall connected and the other hubs are separated from it, Möbius inversion on its eligible cells gives the all-enabled link 4 on D.

The chart at Y_3 is now the restriction of that natural vertex. Indeed its block set is the induced one, its layer data are the readings of that vertex, and original sites of Y_3 are original sites of D with the same datum. The restriction rule and uniqueness of the regular critical orbit identify the two charts, up to the complex gauge in the chart. The same identification at complex data follows on the admitted connected seed discs by holomorphy. At fixed J_3, the next links use this restriction

only on the enabled box and on the reads passed to Y_2. The native vertex depends on the parent on its box, and its output off the box is the parent. These reads are independent of any extra part of Y_3 included through $B_3 \setminus J_3$; the box determining set is the label restricted and eroded at the enabled box itself. Thus the sum over B_3 gives the all-enabled link 3 on D. For link 2, use the natural vertex with the prescribed data on the two sides. Its tilted naturality gives the critical chart with the original datum passed to link 1. The same restriction and inversion first at link 2 and then at link 1 give

$$\sum_{\vec{B}:\bigcup B_t\subset D\ \forall t} T_v(\vec{B}) = F_v(X_v;(q^D,J(q^D))|_{X_v}).$$

Here q^D is the field of the four natural links on the full box D. Restriction to the hull followed by its invariant read gives the last expression. This proves the endpoint identity with the tuple-dependent regions, using the native chart restriction and locality on the admitted one-run tubes.

A.8.0.22. Counting the four successive box reads. For a kept term whose construction uses four successive links on boxes with a fixed core K, count the switched cells in their order of dependence, starting with the link read by the term. Let B_t be the switched cells of link t. Start with $\mathcal{R}_0 = K^+$, the core and its touching cells. At link t adjoin B_t and the protected blocks to K. Keep the wall components that meet $\mathcal{R}_{t-1}$, and add those components and their touching cells to obtain $\mathcal{R}_t$. These sets contain the possible reads of the first t links at any vertex with smaller switch sets.

The alternating coefficient is zero if a switch cell lies outside the retained wall components of its link. Removing it leaves every read unchanged at each vertex paired with that removal. For a nonempty tuple the coefficient is also zero if none of the retained components meets a protected block. In that case every link is at its base on the required reads. A connected part of B_t may attach to the already reached region without carrying a source of its own. Count such parts by their first cell touching the reached region, directly or through a protected block. Following a wall path to $\mathcal{R}_{t-1}$ gives such an attachment: as long as the path uses switched cells and their shared protected blocks it stays in one connected part, and the first passage into the core already meets K^+.

On this geometry the cells at the core wall are level-m cubes. Together with the bounded size of a protected block and the bounded adjacency between cells, this gives $C(1 + d_m(K) + \sum_{s<t} |B_s|)$ choices of an attachment cell. Here d_m is ordinary tree length in level-m units. The filing label is built by joining each part at its attachment and including the parents of its cells and its touched protected blocks. This costs at most $c_l|B_t|$ additional tree length at link t. This construction uses the same filing rule for membership in each box. The coefficient bound is used on this larger set of allowable families.

At the chosen connected-family decay rate, the sum for a single attachment cell is bounded by the convergent geometric series for connected cell sets through that cell. With the rate margin for the cell geometry, the product over possible attachments is at most $\exp\{(d_m(K)+1)/4+1+\sum_{s<t}|B_s|\}$. Sum the four links in reverse order, starting with link four. Each summation spends at most one unit of switch decay at each earlier link. After allowing for the tree weight e^{rd_m}, a margin of three units above the connected-family rate suffices. The four sums together cost $e^5 e^{(r+1)d_m(K)}$.

In the exterior estimate the label of a nonzero piece reaches a protected block inside the operated region and also the exterior anchor Q. Its ordinary length is at least $9\check{R}_m + m_\sharp - 2$ and at least the distance from Q to the enlarged component containing K. Here $m_\sharp$ is the width of the exterior enlargement in level-m units. Spend one unit of length decay on each lower bound. The preceding count with $r = a + 2$ leaves the weight $e^{(a+3)d_m(K)}$ for the ordinary unit size sum. The remaining distance to Q is summable over separated components. The factor $e^{-(9\check{R}_m+m_\sharp-2)}$ pays the strip and shallow-cell counts. This uses the four-link filing and its stated locality at the vertices.

A.8.0.23. Old kept terms at a later healing step. For a kept term of Y outside the region Z being treated at step k, take the four-link expansion on its admitted native family. The empty tuple reads

no response: its boxes are the hubs of Z, separated from the padded kept hull. The remaining pieces are split by the first link with a nonempty switch set. The oscillation on the admitted tube bounds the piece for $\vec{B}$ by $\mathfrak{O}(v)e^{-\kappa_1^{(1)}\sum_t |B_t|}$. The wall path from the padded hull to a source leaves the strip's collar. The depth and the number of selected cells are therefore at least $5\check{R}_m/2-1$ for a strip born at m, as needed for the kept-anchor pile. Its label connects $Q_Y(k)$ outside Z to a source cell in Z, so the native separation bound gives length at least $\check{R}_k-2$. Spend one unit of the length decay on this separation, starting the remaining pile at rate $a+1$.

For the rest of the sum use the native weighted piles $\mathsf{Q}_r[\mathfrak{L}^t]$, with t the number of links and r the rate of the label weight. The reading sets give the distance condition at each link. Summing the last link bounds the pile by $(1+C_\Sigma^{\mathrm{sum}}C_e)\mathsf{Q}_{r+2}[\mathfrak{L}^{t-1}]$ for $t\geq 2$. At the first link the kept anchor contributes $4C_\Sigma^{\mathrm{sum}}\mathsf{T}^*_{r+2}$, where T^*_r is the kept-anchor pile and the four counts come from the first nonempty link. The rate floor $\kappa_1^{(1)}\geq c_l(a+9)+\lambda_*$ permits the four sums and the one unit already spent. Since the anchor pile is at most $\mathfrak{n}^{\mathrm{kept,U}}$, the exterior norm at rate a costs

$$e^{-(\check{R}_k-2)}(1+C_\Sigma^{\mathrm{sum}}C_e)^3\,4C_\Sigma^{\mathrm{sum}}\mathfrak{n}^{\mathrm{kept,U}}.$$

If Y lies inside a treated component C, its padded hull lies in the never-integrated core C of C. On the admitted switch polydisc, compare with a fixed shape point over the same seed and retained datum in the same frame. The framed dilation radius is at least $e^{\delta_P D'/2}$, with $D'=d^\wedge((Y^\sharp)^+,\mathsf{C}^c)$. Schwarz's lemma bounds the difference by $\mathfrak{O}(v)e^{-\delta_P D'/2}$ throughout the switch polydisc. Subtracting the fixed value leaves every nonempty mixed coefficient unchanged. The switch Cauchy estimate then retains the same switch-count decay, with the improved oscillation weight. For any exterior piece at least one selected family leaves C: if all families stay inside, so do their corridors and hubs, which lie in the smallest boxes containing their endpoints. Assign Y a cell of C containing a hull point nearest to C^c. The distance of that cell to C^c is at most D', and at most 2^d regions use a given cell. Since $D'\geq w(\check{R}_m-1)/2$ for a region born at m, one quarter of the exponent absorbs the polynomial oscillation at that birth level. The other quarter sums by the boundary cell estimate for C. Thus the deep kept contribution is bounded by the boundary cell count times $C\mathfrak{n}^{\mathrm{kept,U}}$. Both bounds use the native kept tube, the locality of its pieces and the one-run geometry at healing.

A.8.0.24. The exponent of a nested history. At an admitted root read, separate the exponent of each node into its own list and quadratic form, the kept members of its immediate children, and the Gaussians of its live arrests. A child's kept member is read as its empty-switch value plus its nonempty pieces at the parent. For each such member write the two-run difference as the family difference evaluated at a common datum plus the change of read in one family. At the evaluation of a treated root the two runs use the same seed, so the choice of seed introduces no reading mark. Retain the reading marks of the individual stored terms, including the read of a child at its parent's vertex. In the child's family difference, recurse only on the empty-switch value, at the child's datum furnished by the parent's empty-switch evaluation. Keep the nonempty pieces at the parent. The levels decrease along branches and the recursion is an identity at every node. In the resulting sum each own unit is charged once, with the original class for each family mark, reading mark, form and Gaussian.

Write $N(Y)$ for the count of determining sites by stratum in the padded region of Y at its own step, and $K_{\mathrm{cell}}(m_Y)$ for the one-run per-cell bound at that node's step m_Y. The tree totals are

$$N_{\mathrm{tree}}=\sum_Y N(Y),\qquad V_{\mathrm{tree}}=\sum_Y K_{\mathrm{cell}}(m_Y)N(Y).$$

The children inject into the parent's stage cells. The live arrest collars at a node also charge those cells, with disjoint arrest regions. Hence the sum of the child count and the arrest collar-cell count is at most $2N(Y)$. At a real root the retained Gaussian value is nonpositive, and the one-run oscillation bounds its increase on the admitted tube. The node's own quadratic form has bounded

real part and costs a fixed amount at its node. The screened nonempty pieces of a child cost a fixed amount per child: subtract its empty-switch value and apply the depth estimate at fixed collar variable before the switch sum. Summing by generation gives the real-part bound $V_{\mathrm{tree}} + CN_{\mathrm{tree}}$ in either run.

For the comparison on a class dial, the per-cell family bounds give a contribution proportional to N_{tree} after cancelling the load. In the child-family contribution of a nonempty switch set, subtract the family mark at the empty-switch datum before taking the alternating sum. The comparison at fixed collar variable gives the depth factor for this difference on the contour. It pays the polynomial size of the child's family pile. For a child the change-of-read estimate at its parent's vertex has factor ϑ_f^n. For the nonempty pieces take the geometric mean of the vertex and screened contour bounds, leaving $\vartheta_f^{n/2}$ and enough depth decay to pay the child sum. The dial radius is at most $C\vartheta_f^{-n/4}$, by the lower bound on the full load. The absolute contributions from reads are therefore bounded by fixed multiples of V_{tree} and N_{tree}. The family part of the node's own form is charged at its determining rows. To bound the reading part, use the mixed-datum line from the common datum at zero to the second run's datum at one. At the node's step m_Y it lies in the tube for $|s| < R$, with $R = (8C_{\mathrm{read}}\vartheta_Y^\sharp)^{-1} \geq 2$. Here $\vartheta_Y^\sharp \leq \vartheta_f^{m_Y}$ is the relative reading rate and C_{read} includes the frame allowance. Along the line the imaginary part of the form is bounded by its small one-run majorant Σ_Y. For $\Sigma_Y > 0$, mapping the strip $|\operatorname{Im} w| < \Sigma_Y$ to the disc by $w \mapsto \tanh(\pi w/(4\Sigma_Y))$ and using Schwarz–Pick bounds the change by

$$\frac{2\Sigma_Y}{\pi} \log \frac{R+1}{R-1} \leq \frac{2\Sigma_Y}{R}.$$

Use a slightly larger strip and then its limit if needed. This also gives zero when $\Sigma_Y = 0$. The dial bound $C\vartheta_f^{-m_Y/4}$ makes the cost at most one per node for the class m_Y, by the smallness of Σ_Y. For the Gaussians use the dial bound per arrest collar cell. The preceding count charges these costs to N_{tree}. Thus the real-part estimate on the interpolated exponent has the same two tree totals, with the prescribed enlargements of the coefficients.

For a first-class component C reached from a larger residual domain, take the average of the larger root over the determining set of $(C^\sharp)^+$, including its boundary rows. Use the same real inner variables Φ_C and construct the slice on $C^\sharp$ from this averaged datum. On a finer separated box the real restriction is critical. Indeed, extend a competitor on the smaller fibre by the original root outside the box. The finer constraints preserve the larger averages. Any plaquette meeting the cut has its inside bonds on the frozen fine boundary rows, so its action is unchanged. The action difference is exactly the difference of the two smaller-box actions. This also identifies the restricted current on the bonds whose six plaquettes are retained. The inherited layer bounds and regularity of the induced datum give the real orbit identification when their bounds lie in the box radii.

On complete carriers protected by the padding, the restriction of the shell hierarchy preserves the levels of the rows. Indeed, the shell at a given level is obtained by intersecting its ambient member with the box eroded at that level, so it is unchanged on blocks away from both cut collars. The cut can still create new boundary rows. On each complete double block the exact average is holomorphic about an admitted real reference, with both endpoint node factors and the bounded averaged logarithm. The datum on the collar must be computed by this read. Even on a matched real collar the aligned datum can be $e^{iB_1}V^W$, where V^W is the native box datum and B_1 is the small collar correction.

To use this restriction in the residual production, we must match the finer separated sequence with the native padded determining set of C, including the boundary rows where the averaging contours change. The slice with the assigned current must land in the half tube of the kept-factor bound. For the paired and mixed reads we must also fit the quarter-tube and eighth-tube reserves

and the allowance for the frame change. Real restriction on the matched rows gives the real endpoint of this test. The second-class kept-member read uses its own admitted box.

A.8.0.25. Arguments of the production formula. For an aligned label use the recorded direct units, the read terms from earlier productions, and the operator and normalizing entries. The direct entries include the coupling and source coefficients and the constants in the fifth load. At a fluctuation step the operator entries are the covariance, the walk kernel and its composed quadratic form. The remaining entries include the localized potentials, Jacobians and the normalizer with its stored boundary share. At a healing step use the corresponding chain and kernel entries.

For the large-field integral, on its native history tree the exponent consists of the raw Wilson term, the inner integrals' quadratic forms and source weights, and the normalization constants at each node, together with the kept exponent of the output. The latter is the fresh form and kept potential, with the child terms and arrested Gaussians inserted as in the nested sum. Haar factors, tree constraints, the tests and the inner integral kernels are part of the production too. When a source is held fixed, some of these factors still depend on the cutoff through the coupling, density or chart. In particular the old inner quadratics have their own cutoff differences even when independent of the later background. The histories of an inner integral also act on the outer potential terms whose domains contain that integral's component.

Use the localized remainder after the Wilson and quadratic cancellation. In the resulting array its terms take the place of the raw brackets they represent. The kept factors of a pocket are separate entries until they are read into a later exponent. For a live arrest use the Gaussian piece, its shifted value minus the base, its grain piece, or its retained weight, according to the reading site. Each is taken at the site and in the variables in which it occurs. The new integral's determinant is a separate normalizing entry.

On a common chart for the integrated and retained variables, the measures, constraints, tests and reference factors must either be identified by the alignment or entered with their actual changes. With these identifications, substitution of the two arrays into the same integral formula gives the two productions. The componentwise family and reading identities then give $x^{\mathsf{B}} - x^{\mathsf{A}} = \sum_z W^z$ in the entries of that formula. This check uses the full list of native factors at the production, including the terms carried by its inner histories.

A.8.0.26. Bounds for the production on the class discs. At a fluctuation half-step the class derivative splits into an operator direction and a marked potential direction. The Gaussian units enter the latter through their weighted pile. The marked expansion at the convex base is linear in the marked potential, and bounds its derivative by the pile times the native output decay. The one-cell pile and the fluctuation weight therefore add at most $C\theta_i \Pi_G^+(x_i)\Delta_n^{\mathsf{c}}$ to the class-n input. Here θ_i bounds the inverse fluctuation weight. Every filed Gaussian unit holds a retained collar cube, so it enters the large-field activity. The list of outputs wholly in the clean region does not read it. The operator direction keeps its own bound.

For a live arrest W outside the components being treated, evaluate the retained quadratic at the new root on the top storey. Write D_W^z for the sum of its marks of class $z = (n, k')$ and $N(W)$ for the number of filing cells holding its collar. The vertex bound gives $|D_W^z| \le K_G N(W) \Delta_n^{\mathsf{c}}$. For $0 \le \lambda \le 1$ and $|\varepsilon| \le \Lambda_z$, convexity of the real part gives

$$\left|\exp\left((1-\lambda) q_W^{\mathsf{A}} + \lambda q_W^{\mathsf{B}} + \varepsilon D_W^z\right)\right| \le \max_{\mathsf{m}} |e^{q_W^{\mathsf{m}}}|\, e^{K_G E_1^z N(W)}.$$

The collar cells of distinct live arrests are charged on their disjoint regions. Keep the maximum separately for each arrest. The exterior difference from the root to the healed field is filed in the potential and uses the exterior sum above. At the last lattice the weight is read at the root without intervening links. For an arrest nested in a treated component C, use the collar-cell bound inside the exponent of the healer. Its retained variable is integrated there. For a class n present in that arrest the birth index satisfies $j(C) \le n + 1$, giving age at least $m - n - 1$ in the estimate for the healed term.

Before applying the strip estimate, take the convex bound on the one-run potentials and add the class direction. The exterior remainder and the exterior Gaussian each contribute a one-run norm and a two-run norm multiplied by the dial radius. The class direction is bounded by using $\Lambda_z \Delta_n^c \le E_1^z$. The four coefficients tend to zero under the same coupling ceiling, so each may be bounded by one before multiplying by the native strip factor. In the rim direction the two exterior marks enter the linear potential with their original classical load Δ_n^c. At a preparatory stage let ν_ζ be the array of positive unit bounds for the response direction ζ, and μ the moment bound which controls the stage holomorphy. The stage disc has radius $1/(4\mu[\nu_\zeta])$ for a nonzero moment bound. Convexity gives the one-run majorant on the base, and the block estimate gives the additive majorant on the disc. The staged leaf estimate reads this direction bound on the common chart. The Gaussian collar coordinate stays fixed there.

The exterior bracket of the fifth comparison row uses the one-run outer cell estimate in the coefficient of its own unknown. It therefore multiplies the fifth load in that row.

These modulus bounds are for the unnormalized activities. In the activity and age estimates the new costs are the nested exponent, the collar cells of the untreated arrests, and the bounded remainder per treated component. Charge them in the entropy constant and use the age bound at each generation. For the residual output with filing domain X', the same production map $\mathcal{P}_{X'}$ is used at the two arrays of slot values. Apply the modulus bounds to its values on the disc $x^\lambda + \varepsilon W^z$, where x^λ is the convex array and W^z is the class direction. Cauchy's estimate can then bound the corresponding first variation by the supremum on that disc divided by Λ_z. The own classes use their healed exponent bound. This is the passage from the component estimates to the class directions used in the terminal sum.

A.8.0.27. Accumulation of the stored boundary loads. Fix the birth zone n. Let u_j be the partial sum of the class-n loads of stored boundary terms after the first j productions that create such a term. The next production reads at most $a_n + u_j$, where a_n is the class input before this feedback. A fluctuation step adds its store with the crossing weight $w_{\ell,n}$ of its output. A healing step adds its residual store with its residual comparison coefficient. Thus the two estimates take the form

$$u_{j+1} \le (1+\mu_j)u_j + \mu_j a_n + b_j, \qquad u_0 = 0.$$

The b_j are the weighted classical inhomogeneous bounds at the fluctuation steps. The first two crossing weights are one. For $\ell \ge n+2$, let t be the depth of level $\ell - 1$ from the terminal level. The weight is bounded by a multiple of $\exp[-(2/L)(\log(x_* - 2c_2 + \lambda_{\text{flow}} t))^{r_{\text{pr}}}]$, where x_* is the lower bound on the inverse squared coupling and $\lambda_{\text{flow}} > 0$ is the flow lower bound. The depths for different ℓ are distinct. Since $r_{\text{pr}} > 1$, the terms are bounded by a summable tail uniformly for x_* above a fixed lower bound. The tail tends to zero as x_* increases. This gives $\sum_\ell w_{\ell,n} \le 3$ at small coupling.

The fluctuation coefficients are bounded by their uniform bound times $w_{\ell,n}$, and the residual coefficients already have a summable bound. With the enlarged smallness budget of the class estimates these give $\sum_j \mu_j \le 1/40$. Iterating the inequality and bounding the products of $1+\mu_j$ gives

$$u_j \le e^{1/40}\left(a_n \sum_{i<j} \mu_i + \sum_{i<j} b_i\right) \le a_n/38 + e^{1/40} \sum_{i<j} b_i.$$

The new classical marks are bounded by the existing class load and so fit in its right member. The fifth-row term stays in its own coefficient. Only these crossing-weighted stores and the residual stores enter the accumulated load.

The zone of a slot is assigned before its load is formed. A resolved slot inherits an anchor in the zone of its formation. An ordinary zone-b anchor is in Ω_{b+1}^c, and a prepared zone uses the quiet window to give the same birth bound. Trace the anchor through the later reads of that slot. As long as the containing component survives, its successors retain that birth and meet the read domain.

A second-class or arrested successor retains its record. If the containing component is healed, the interactions reading its inner history are summed in the healing bracket. A slot whose zone occurs only in the summed sequence is no longer a resolved read of that zone. Its old age factor is spent in the own-output estimate at production. File the resulting whole slot, with its norm, at the later of its stored level and the block zone. At subsequent reads use the live-incidence condition of the storage rule for that new class. For $b \le k-2$, the terminal comparison requires each sensitive label in class b to have a live component born by $b+1$. This is checked using the resolved interpolation on the actual reads and the block incidence for the own slots. The birth of an erased component gives no terminal age.

A.8.0.28. The envelope for the full zone load. The preceding recurrence sums the stores resolved in one birth zone. The boundary terms produced in healing and filed at their own level also enter the load. Let ℓ_j be the full load in zone j, including the regular and remainder reads, both boundary loads, the shift and increment reads, and the classical index terms. Put $s = K-j$, $e_s = \rho^{K-s} + \Pi_K$, and $\mathsf{l}_s = (\log \theta_{[s]})^{c_1}$, with the exponent covering the logarithmic weights of the comparison. For a lag m use

$$e_{s+m} \le \rho^{-m} e_s, \qquad \mathsf{l}_{s+m} \le \widehat{\mathsf{l}}_m \mathsf{l}_s,$$

where $\widehat{\mathsf{l}}_m$ grows at most as a power of $1+\log(1+m)$. Use the constituent response envelopes on the actual common slot reads. On a live second-class strip we must verify the relative-format version of those bounds on its stored reads, as well as the ordinary version on the other slots. In the weighted term norms S_i^+, at $i = j-m$ they give

$$S_i^+[E] \le C \mathsf{l}_{s+m} e_{s+m}, \qquad S_i^+[R] \le C \theta_{[s+m]}^{-1} \widehat{r}_{s+m} e_{s+m}.$$

The first bound adds the anchored, quadratic and source-defect entries in their respective reading units. The weight for the E terms is bounded by $C(m+1)^3 q^m$. Since $q/\rho < 1$, its sum is $O(\mathsf{l}_s e_s)$. The R weight instead contains $\theta_{[s+m]}/\theta_{[s]}$. It cancels the inverse coupling in the R reading unit. The unweighted remainder tail $\sum_{r>s} \widehat{r}_r e_r \le 2\Pi_K$ then gives $O(e_s)$ for the R load.

The gauge load convolves the shifts $O((1+s+m)e_{s+m})$ with L^{-m}, and the increments $O(e_{s+m})$ with $L^{-m/2}$. The prefactor is bounded by $C\theta_{[s]}^{-1}(\log \theta_{[s]})^{2q_\flat}$. Both convolutions are summable relative to e_s. The ratio $(1+s)/\theta_{[s]}$ is bounded by the flow. Thus this load is $O(\mathsf{l}_s^2 e_s)$. For the classical E and boundary entries the kernels are $C(m+1)^3 q^m$ and $L^{-m/2}$, respectively, times the class rate at i, bounded by $C\rho^i$. The classical R entry with $g_i^{\kappa_0}$ has instead the coupling weight of the R row. After its cancellation the sum contains $\theta_{[r]}^{1-\kappa_0/2} \rho^{K-r}$ at depth r. The one-run comparison of remainder coefficients bounds its sum by $C\Pi_K$. The last classical entry costs $C\mathsf{l}_s e_s$ by the index rate and the logarithmic weight. Together these give $O(\mathsf{l}_s^2 e_s)$ before the two boundary loads are added.

At healing level r, the own output bound uses the load in each zone $i \le r$ with its terminal age weight. Reading that output in zone j costs $L^{-(j-r)/2}$. Interchanging the sums over $r \le j$ and $i \le r$, and adding the resolved boundary estimate above, gives

$$\ell_j \le a_j^0 + \sum_{i \le j} \mathcal{K}(j,i) \ell_i, \qquad \mathcal{K}(j,i) = \frac{4(1+\epsilon_0)}{E_0} \sum_{i \le r \le j} L^{-(j-r)/2} c_1^\flat(g_r)(r-i+1) q_a^{(r-i-1)_+}.$$

Here ϵ_0 is the resolved-load coefficient of the own contribution. The direct and classical bounds just proved enter a_j^0 with fixed constants. The additional own inhomogeneity at r is bounded by $C c_1^\flat(g_r)(q_a^{r-n_c} + \vartheta^r)$ before the $L^{-(j-r)/2}$ convolution, with $\vartheta \le \rho$. The factor $q_a^{-n_c}$ is bounded by a power of $\log(X + CK)$, where $X = \theta_{[0]}$. Split that logarithm at depth $K-r$: the factor that grows with $\theta_{[K-r]}$ is absorbed by $c_1^\flat(g_r) \le C\theta_{[K-r]}^{-\kappa_0/2}$, and the factor growing with r by $(q_a/\rho)^r$. The resulting inhomogeneity after convolution is $O(e_s)$. Hence $a_j^0 \le C\omega_j$, where $\omega_j = \mathsf{l}_{K-j}^2 e_{K-j}$.

For the weighted supremum norm with weight ω_j, write $\xi = j - r$ and $\eta = r - i$. The ratio ω_i/ω_j is at most $\widehat{\mathrm{l}}^2_{\xi+\eta}\rho^{-(\xi+\eta)}$. The two geometric factors $L^{-\xi/2}$ and $q_a^{(\eta-1)_+}$ make the resulting double sum finite, since $L^{-1/2} < \rho$ and $q_a < \rho$. The norm of $\mathcal{K}$ is therefore bounded by $C \sup_r c_1^\flat(g_r) < 1$ at weak coupling. The loads are finite by the a priori dial bound. Taking the weighted norm in the displayed inequality gives $\ell_j \le C\omega_j$. For the stop at depth $s_\circ = K - k$, the same lag comparison yields

$$\sum_{j\le k}(k-j+1)q_a^{(k-j-1)_+}\ell_j \le C\mathrm{l}^2_{s_\circ}e_{s_\circ}\sum_{m\ge 0}(m+1)q_a^{(m-1)_+}\widehat{\mathrm{l}}^2_m\rho^{-m} \le C'\mathrm{l}^2_{s_\circ}e_{s_\circ}.$$

This uses the two-run fluctuation and healing bounds on the admitted slots and the actual crossing weights.

A.8.0.29. Removing the old histories from a remainder. For a history-created second-group remainder $R'^{(k)}(X)$, retain in each operated component only the sequences whose first index exceeds $n < k$. Write $0 < q_{\mathrm{op}} \le 1/2$ for their geometric forgetting rate. Denote the resulting formula by $R'^{(k)}_{>n}(X)$. Make this restriction on the full component operation, including its last exponential, before forming the receiving cluster. The following estimate uses its admitted holomorphic formation chart, with absolute subfamily majorants and the doubled whole-component input allowed by the age-dial construction.

Put $a_0 = k - n$, $p_k = p_0(g_k)$, $P = p_0^\sharp(n) = \min_{j\le n} p_0(g_j)$, and $E_k = \exp[-2p_k/(1+\beta_0)]$. Here $0 \le \beta_0 < 1$ is the degree-comparison constant, and the age comparison gives $P \ge p_k/(1+\beta_0)$. An age-a sub-sum in the removed part, $a \ge a_0$, has absolute majorant at most $\mathrm{e}\, e^{-2P}q_{\mathrm{op}}^{a+1}\sup|F|$. Multiply all these old sub-sums, in every operated component, by the same variable μ. Use the disc

$$|\mu| \le R_\mu := q_{\mathrm{op}}^{-a_0}\exp\bigl[2P - 2p_k/(1+\beta_0)\bigr].$$

The whole old input on this disc is bounded by

$$\mathrm{e}E_k\sup|F|\sum_{a\ge a_0}q_{\mathrm{op}}^{a+1-a_0} \le \mathrm{e}E_k\sup|F|, \qquad q_{\mathrm{op}} \le \tfrac{1}{2}.$$

The younger input has the same bound without the dial. Thus their sum stays within the allowed $2\mathrm{e}E_k\sup|F|$ input. This sum is made at the level of the history-dependent absolute majorants, with the last exponential split separately in each history. No unchanged per-age bound is needed after the old part is enlarged. The receiving formation therefore remains holomorphic and has its bound $\mathfrak{B}_X = e^{-p_k}e^{-\kappa d_k(X)}$ on the disc. At $\mu = 1$ it is the original remainder and at $\mu = 0$ its younger part. Since $R_\mu \ge 2$, the one-variable Schwarz estimate gives

$$\begin{aligned}|R'^{(k)}(X) - R'^{(k)}_{>n}(X)| &\le 2\mathfrak{B}_X/R_\mu\\ &\le 2e^{-\kappa d_k(X)}e^{-(1-\beta_0)P}q_{\mathrm{op}}^{k-n}.\end{aligned}$$

The second inequality follows from $p_k \le (1+\beta_0)P$. Multiplication by the domain weight gives the same bound in the flat remainder norm.

The restriction also identifies which old uses can remain. A direct use of zone $n' < n$ requires an operated sequence with first index at most $n'+1 \le n$, so it has been removed. If every live witness of a leaf on this reader belongs to one of its operated components, a live witness of level $e \le n$ likewise requires first index at most e, and has been removed. Old zone copies can still occur inside an owned leaf formed at $e > n$. The carry bound gives them the factor

$$C_* q_{\mathrm{op}}^{e-n'-1} \le C_* q_{\mathrm{op}}^{n-n'}.$$

Such a copy can occur only when the containing operated sequence starts by e. Thus the younger remainder retains these controlled old-zone reads. In using the live-witness exclusion we must check the stated witness containment on the actual receiving carriers.

A.8.0.30. Aligning the nested stored terms. An aligned outer sequence can still read a stored term whose own healed histories start below n_c. Restrict those histories before using the comparison on zones at or above $n_c - 1$. For a first-group stored leaf formula ψ formed at level e, define ψ^{AL} recursively in its formation level. Restrict the sequence sum of each healed component in its cluster to first index at least n_c. Whenever a nested occurrence loses its witness under this restriction, replace it by its own aligned formula. The recursion terminates since nested formation levels decrease. This changes the formula on its original variables and preserves the full carriers used in its majorant.

First consider the branch with per-cell leaf piles $\epsilon_e^{\mathrm{all}}$ satisfying

$$\epsilon_{\mathrm{tot}}^{\mathrm{all}} := \sum_e \epsilon_e^{\mathrm{all}} \le \tfrac{1}{16}, \qquad C_{\mathrm{AL}} := (1 - 4\epsilon_{\mathrm{tot}}^{\mathrm{all}})^{-1} \le \tfrac{4}{3}.$$

Use the same admitted per-use domains and the two independent age and leaf dials as in the carry estimate, with the operation rate q_{op}. If $\mathfrak{B}_\psi$ is the original full-carrier majorant, then

$$|\psi - \psi^{\mathrm{AL}}| \le C_{\mathrm{AL}} \mathfrak{B}_\psi q_{\mathrm{op}}^{(e-n_c)_+}.$$

Indeed, removing its own unaligned histories costs at most $\mathfrak{B}_\psi q_{\mathrm{op}}^{(e-n_c)_+}$ by geometric forgetting. For nested occurrences formed at $r < e$, the induction hypothesis gives the per-cell dial size $C_{\mathrm{AL}} \epsilon_r^{\mathrm{all}} q_{\mathrm{op}}^{(r-n_c)_+}$. The age dial has radius $q_{\mathrm{op}}^{-(e-r)}$, and the leaf dial has the reciprocal of that size as its radius. The mixed difference estimate therefore bounds this level's contribution by

$$4 C_{\mathrm{AL}} \epsilon_r^{\mathrm{all}} \mathfrak{B}_\psi q_{\mathrm{op}}^{e-r+(r-n_c)_+} \le 4 C_{\mathrm{AL}} \epsilon_r^{\mathrm{all}} \mathfrak{B}_\psi q_{\mathrm{op}}^{(e-n_c)_+}.$$

Summing over r closes the induction because $1 + 4 C_{\mathrm{AL}} \epsilon_{\mathrm{tot}}^{\mathrm{all}} = C_{\mathrm{AL}}$. The constant is independent of the formation level. If only a uniform per-level pile bound is available, the strengthened carry estimate above remains available; the present constant specifically uses the summed small pile.

Run the affine comparison on these fully aligned formulas, including all their nested occurrences. Every retained use is then in the zones where the comparison input applies. The discarded nested differences enter the unmatched term of an operated component. A reader $\mathbf{T}'_k(X)F$ gives a level-e difference the additional factor q_{op}^{k-e}. Since

$$(k - e) + (e - n_c)_+ \ge k - n_c,$$

the transported difference has the same age power as the own unaligned histories. The reader production factors and full carriers are the same in these two estimates. Sum $C_{\mathrm{AL}} \epsilon_e^{\mathrm{all}}$ over the formation levels. The own unmatched bound is multiplied by

$$C_{\mathrm{align}} := 1 + C_{\mathrm{AL}} \epsilon_{\mathrm{tot}}^{\mathrm{all}} \le \tfrac{13}{12}.$$

Keep the terminal rate q_a fixed and choose the operation age reserve so that $q_{\mathrm{op}} \le \min\{q_a, 1/2\}$. Take the admitted dial and history bounds at this reserve before fixing the coupling ceiling. For $k \ge n_c$, the operation error is then bounded by a constant times $q_a^{k-n_c}$. This construction uses the common scaffold, admitted dial domains and zone comparison of the two runs; it does not supply those inputs on an unadmitted read. For a first-group term still stored at the terminal step, the slot argument is more direct. Let Ψ_u be the term of level e occupying such a slot, and set $H_u^{(e)} = \Psi_u - \Psi_u^{\mathrm{AL}}$. Count each outer stored slot once; its aligned formula already includes the nested replacements. On its admitted full reading domain the preceding bound gives the per-cell size

$$\rho_e = C_{\mathrm{AL}} \epsilon_e^{\mathrm{all}} q_{\mathrm{op}}^{(e-n_c)_+}.$$

For nonzero ρ_e, use the slot path $x_u(\nu_e) = \Psi_u^{\mathrm{AL}} + \nu_e H_u^{(e)}$, with the other level slots held at their current endpoints. Its values at zero and one are the aligned and original slots. Omit levels with zero pile. The stored-family value and activity budgets are linear in this pile. Thus a single level-e parameter multiplying $H_u^{(e)}$ has a disc of radius a fixed multiple of ρ_e^{-1} in the terminal slot. Reserve

the corresponding one-unit value and activity allowance, with the coupling ceiling making the radius at least two. Keep the ramps, tests and real geometry fixed. The terminal modulus proof uses these value and activity bounds and the same raw Wilson factors, so its bound on this disc is $CA^\sharp$.

A still-stored boundary term has an anchor in the enlargement of the level-e large-field region in the current label. Its sensitive terminal labels therefore have a live component of age at least $(k-e-1)_+$. Subtract the insensitive labels and use their age-weighted modulus bound on the leaf disc. The Schwarz estimate, with arbitrary labelwise Borel phases, bounds the sum of the resulting total variations by

$$CA^\sharp C_{\mathrm{AL}} \epsilon_e^{\mathrm{all}} q_{\mathrm{op}}^{(e-n_c)_+} q_a^{(k-e-1)_+}.$$

For $k \geq n_c$, summation over e gives $C(q_a) A^\sharp C_{\mathrm{AL}} \epsilon_{\mathrm{tot}}^{\mathrm{all}} q_a^{k-n_c}$. This argument uses the full leaf reading domain, including its configuration tube. We must check that the actual terminal read lies in that tube and that its slot has the reserved value and activity allowance.

When a leaf has been absorbed by a healing operation, its change can enter both a younger boundary term and a second-group regular remainder. For the former use the operation-reader estimate and then the surviving younger boundary slot. For the latter we must identify the recomputed list and terminal slots and verify their comparison bounds. Such absorption does not by itself supply an old live witness at the terminal step. After these terminal transport checks, write C_{term} for the fixed coefficient of the total unmatched bound.

A.8.0.31. Summing the changes of the aligned terminal terms. On the common terminal variables first compare the fully aligned labels with the ramps and the stop-level production slots of the ramp move fixed to those of run A. On a standing second-class component the common variable is V', with $V = V' V_\Lambda^{\mathsf{c}}$ in run c. On a removed component with stored shell variables use $V'' = V' M_{\mathbf{B}''}(U_0^{\mathsf{c}})$. Elsewhere keep the raw variables. In the order of the reference construction each multiplier is determined by the already fixed exterior. Conditional right translation thus preserves the free Haar factors, and the recorded tree constraints act on the common relative variables. Both physical charts push forward from the same reference space for this label and frame. The two physical tree slices can have different images. All total variations of aligned differences below are computed on the common chart. Write $\tau_h^{B|A}$ for the B term with these A ramps and production slots, and τ_h for a term along the interpolation to it. Test the family against arbitrary Borel φ_h with $|\varphi_h| \leq 1$. The first variation of $\sum_h \int \varphi_h \, d\tau_h$ is the sum of the directions indexed by a charged birth zone b and a read level $t \in [b, k]$, where k is the stop level. Use the slot comparison at the common endpoints, so the sum of these directions is the actual interpolated derivative.

Write ℓ_b for the load of a zone. Its direction has an analytic disc of radius E_1/ℓ_b in the aligned comparison, with zero loads omitted. Terms that read no slot of that direction are constant on the disc. Subtract the sum over those insensitive labels before applying Cauchy's inequality. Put $a = (k-b-1)_+$ and let $m_a(h)$ count the live components of age at least a. If $a > 0$, the incidence check gives $m_a(h) \geq 1$ on each remaining label. The age modulus bound on the disc allows multiplication of each term by $\mu^{m_a(h)}$, with $|\mu| = q_a^{-a}$. The age reserve pays this multiplier in the sum of the absolute masses, bounded by $A^\sharp$. The sum over the sensitive labels without the multiplier is therefore at most $A^\sharp q_a^a$. When $a = 0$ the ordinary modulus bound suffices. The common integrable majorant permits the holomorphic integration against the tests. Cauchy's inequality, integration along the interpolation and summation over t give

$$\left| \sum_h \int \varphi_h \, d(\tau_h^{B|A} - \tau_h^A) \right| \leq \frac{A^\sharp}{E_1} \sum_{b \leq k} (k-b+1) q_a^{(k-b-1)_+} \ell_b$$

at the fixed ramps. For each label take the conjugate polar phase of its difference measure as the test. The left side becomes the sum of its total variations, so the comparison holds before summing the label masses.

For the joint estimate of the ramp change and the recent unaligned labels, keep the objects of run B at its real center. On the aligned labels use the move that interpolates the ramps $t^{\mathsf{A}} + s(t^{\mathsf{B}} - t^{\mathsf{A}})$ and the small production slots that read only V_k. On the common slot chart, each such interpolated slot is bounded by $1 + 20C_\chi \vartheta_\chi^k |s|$ times the enlarged support indicator in run B. The indicator is fixed on the real integration space. There are at most N_w^4 slots on the last lattice. Their product is bounded by e times the enlarged support on the disc

$$|s| \le (20C_\chi N_w^4 \vartheta_\chi^k)^{-1}.$$

In using this disc, the admitted fillings and the labelwise modulus estimate must hold with doubled component majorants. The weight bound pays this doubling from the component exponent and sums the fillings. It permits the insertion of the same arbitrary bounded Borel phases as above. On the disc the integral with these phases is bounded by a fixed multiple of $A^\sharp$. Cauchy's inequality bounds its change from $s = 0$ to $s = 1$ by $CN_w^4 \vartheta_\chi^k A^\sharp$. Here the disc contains $|s| \le 2$ for sufficiently large k. The two endpoints have the B objects and, respectively, the A and B ramps and stop-level slots.

For each run separately, multiply the common unaligned labels born above n_c by μ. The admitted multiplier disc has radius $(4C_\chi^\sharp \vartheta_\chi^k)^{-1}$, with the same modulus bound. The sum for this class is linear in μ, so its coefficient has size at most $CC_\chi^\sharp \vartheta_\chi^k A^\sharp$. Taking the supremum over the labelwise phases gives the variation sum for these terms. This gives the joint bound $C_{\mathrm{test}} \vartheta_\chi^k A^\sharp$, using the common-slot, support and modulus bounds of the admitted ramp moves. Add it once to the fixed-ramp comparison. The labels born at or below the common-level threshold n_c which are left unpaired contribute at most $C_{\mathrm{term}} q_a^{k-n_c} A^\sharp$, including the nested remainders when the terminal marked-leaf check just described holds on the summed-pile branch. With no nested terms take $C_{\mathrm{term}} = 4$. Thus the total of the aligned differences and the unaligned masses is bounded by

$$A^\sharp \left[E_1^{-1} \sum_{b \le k} (k - b + 1) q_a^{(k-b-1)_+} \ell_b + C_{\mathrm{term}} q_a^{k-n_c} + C_{\mathrm{test}} \vartheta_\chi^k \right].$$

At a fixed stop depth $s_\circ = K - k$, the alignment level satisfies $n_c \le C + C \log\log(x_0 + C)$ on the admitted runs. Since $x_0 \le C_g(1 + K)$, the factor $q_a^{-n_c}$ grows at most as a fixed power of $\log(2+K)$. For $q_a < \rho < 1$ this power is absorbed by $(q_a/\rho)^k$. Thus the old-label term is $O(\rho^{K-s_\circ})$, as is the joint ramp term if $\vartheta_\chi \le \rho$. With the full load envelope, the three terms therefore cost $C_{s_\circ} A^\sharp (\rho^{K-s_\circ} + \Pi_K)$ at a fixed stop, at both the source and its real centre. This is the error for the scalar comparison below once the actual terminal labels have the stated slot interpolation, age incidence, marked-leaf transport and jointly controlled ramp.

A.8.0.32. The normalized integral at the stop. For the second line of eq. A.lxxiv, put $v = t + \sum_i z_i f_i$. Use v in run A and $\mathfrak{n}_K \circ v$ in run B. Take the real centers at $t_1 = \mathrm{Re}\, t$ and $\mathfrak{n}_K(t_1)$. The sources relative to the centers are $v - t_1$ and $\mathfrak{n}_K \circ v - \mathfrak{n}_K(t_1)$, respectively.

For the source-class comparison, let r_J be the parameter-disc radius and take $|t_1| \le r_J/4$. Choose the residual source $W = v - t_1$ in the class of radius $3r_J/8$. Its amplitude is at most $1.6(3r_J/8) = 0.6r_J$, and its Lipschitz bound is $3/8$ of the bound in the radius-r_J class. Thus the values of $t_1 + W$ lie in $\bar{D}(0, 0.85r_J)$. Write $\mathfrak{n}_K(z) = z + e(z)$, with $|e(z)| \le \Upsilon_K |z|$ on $D(0, r_J)$. Cauchy's estimate on the smaller disc gives $\sup |\mathfrak{n}_K'| \le 1 + (20/3)\Upsilon_K \le 8/3$ when $\Upsilon_K \le 1/4$. Integration along the segment joining $t_1 + W(x)$ and $t_1 + W(y)$ gives the Lipschitz bound of the radius-r_J class for the residual source of run B. Its amplitude is at most

$$\|W\|_\infty + \sup_x |e(t_1 + W(x))| + |e(t_1)| \le (0.6 + 0.85/4 + 0.25/4) r_J = 7r_J/8.$$

The matched real center satisfies $|\mathfrak{n}_K(t_1)| \le (5/4)|t_1|$. Choose the real center interval with room for this change and the source radius in both runs. These bounds admit the matched residual source on the local pair, including its spatial variation.

For one run, take w relative to its real center and let $Z(w)$ be the corresponding bare sourced partition function. At level j write $\mathsf{V}_j(X;w)$ for the sum of the regular anchored terms and the remainder produced at that level, evaluated at the trivial field, together with the one-point gauge terms extracted by the completed gauge steps. Group the latter on their single M-cubes and include each at the step at which it is extracted. The gauge term at a pin is $\log[\mathfrak{z}(a_j(y;w))/\mathfrak{z}(x_j)]$, with the reference coefficient x_j of that run at its real center. Set

$$e_j(X;w) = \mathsf{V}_j(X;w) - \mathsf{V}_j(X;0), \qquad E_u^{\mathrm{src}}(w) = \sum_X e_{K-u}(X;w).$$

The values of these energies are independent of where the run is stopped. For a stop at depth s put $k = K - s$ and define

$$I_k(w) = \exp\left(-E_{(k)} - \sum_{j=1}^{k}\sum_X e_j(X;w)\right)\int \rho_k^w.$$

Here $E_{(k)}$ is the source-independent vacuum constant and the integral includes the stored variables. Exact preservation of mass under the first k operations gives $\int \rho_k^w = Z(w)$.

In the native history representation, the boundary energy is extracted globally. A label $\mathcal{S}$ produces its regular share only on the carriers contained in its small-field regions $\Lambda_j(\mathcal{S})$. Thus its term in $I_k(w)$ contains the compensating exponent

$$-\sum_{j\le k}\sum_{X\not\subset\Lambda_j(\mathcal{S})} e_j(X;w).$$

This includes the negative copy of the centered gauge term on each omitted cube. The compensating exponent is the unit-field evaluation of stored boundary families at their respective levels, and their real-center vacuum constants are stored separately. A common multiplicative normalization in a run is chosen once at its real center and applies to every label and source value in the comparison.

Consequently the normalized stop factor is

$$\widetilde{Z}^{[s]}(w) = \frac{I_k(w)}{I_k(0)} = \exp\left(-\sum_{u=s}^{K-1} E_u^{\mathrm{src}}(w)\right)\frac{Z(w)}{Z(0)}.$$

Increasing the stop depth from s to $s+1$ therefore multiplies the stop factor by $e^{E_s^{\mathrm{src}}(w)}$. In the logarithms normalized at $w=0$ this adds exactly $E_s^{\mathrm{src}}(w)$. For a fixed stop depth, its cost in the bound for $|\widetilde{Z}|$ is at most $\exp(C\ell^4\sum_{u<s}L^{4u})$, using the connected-domain sum for each transferred energy. This cost is independent of the cutoff at a fixed s.

At a fixed stop depth $s_\circ$, the one-run positive lower bound and stability upper bound give $\widetilde{Z}(0)=1$ and $|\widetilde{Z}|\le Q_\ell$ on the source ball, with Q_ℓ fixed at fixed torus. If the radius is r, then on the ball of radius $r/(2(Q_\ell+1))$ the Schwarz lemma on each complex line gives $|\widetilde{Z}-1|\le 1/2$. The logarithm there is single-valued and Lipschitz on the range.

To apply the history estimate to the stop integral, choose one positive scalar a_{c} for each run and real center, in the one-run vacuum normalization. Let P, Q be the resulting stop integrals $a_{\mathsf{B}} I_k^{\mathsf{B}}$ and $a_{\mathsf{A}} I_k^{\mathsf{A}}$, at the matched residual sources. Use these same scalars at zero residual source. Let τ_h^{c} denote the normalized history measure in the chart of its label. Its total mass is unchanged by the chart substitution. With H_{al} the aligned labels, the difference of the scalar readouts is

$$P - Q = \sum_{h\in H_{\mathrm{al}}}(\tau_h^{\mathsf{B}} - \tau_h^{\mathsf{A}})(1) + \sum_{h\in H^{\mathsf{B}}\setminus H_{\mathrm{al}}}\tau_h^{\mathsf{B}}(1) - \sum_{h\in H^{\mathsf{A}}\setminus H_{\mathrm{al}}}\tau_h^{\mathsf{A}}(1).$$

Pair the first sum in the common label charts and bound each evaluation at 1 by the corresponding variation. The sum estimate above therefore gives both $|P-Q|\le A_\ell^\sharp\varepsilon$ and $|P_0-Q_0|\le A_\ell^\sharp\varepsilon$, where subscript 0 denotes the zero residual source at those same real centers. In terms of the original

two integrals the common comparison scalar is $a_{\mathsf{A}}/a_{\mathsf{B}}$. The one-run lower bound gives $P_0 \geq B_\ell^{-1}$ in this normalization. Subtracting the two ratios gives

$$\left|\frac{P}{P_0} - \frac{Q}{Q_0}\right| \leq B_\ell(|P-Q| + |Q/Q_0|\,|P_0 - Q_0|) \leq B_\ell A_\ell^\sharp(1+Q_\ell)\varepsilon.$$

The logarithms on the smaller ball have the same error up to a fixed factor. This uses the history estimate for the actual tests and stored variables at both arguments.

A.8.0.33. A uniform bound on the wider strip. The admitted interval I_K in eq. 5.lxv leaves the stated room for the source. Its terminal image contains $h \pm \varrho$, and the upper slope bound for x_K there is at most $5/3$. The preimages of these two values are each at least $3\varrho/5$ from $\beta_K^\natural$. Thus the one-run estimates apply at the real centers $|\operatorname{Re} t| < \varrho/2$. Let $r_\ell > 0$ be the radius of the smaller source ball just obtained from the Schwarz bound. For $|\operatorname{Im} t| < r_\ell/2$, take a product disc in the z_i on which $\sum_i |z_i|\|f_i\|_\sharp < r_\ell/2$. The residual source $w = \mathrm{i}\operatorname{Im} t + \sum_i z_i f_i$ lies in the smaller ball, with real center $\operatorname{Re} t$. There the stop factor is nonzero and its logarithm has a uniform bound. The local energy factors in the exact partition-function identity are exponentials. Thus the original partition function is nonzero on this disc and the mixed derivative of its logarithm includes a uniformly bounded stop term by Cauchy's inequality.

In each individual local energy the single-point frozen share depends on at most one z_i at a plaquette. For $n \geq 2$ it disappears from the mixed derivative. The remaining terms contributing at depth s have enlarged carriers meeting every support. The individual energy bound and the separation of the supports then give the summable majorant $C(\ell L^s)^4 e^{-c\mathsf{d}L^s/M}$ after the connected-domain count. Cauchy's inequality on the same fixed product disc bounds their mixed derivatives uniformly in K. Together with the stop term this bounds $F_{K,n}$ on the wider strip. At each finite cutoff the bare integral is jointly entire in the bare coefficient and the finitely many source parameters. At nonzero values its mixed logarithmic derivative is independent of the local logarithm branch and is holomorphic in t. This gives holomorphy across the real-center slices used to obtain the bound.

A.8.0.34. Mixed derivatives and the separation of supports. At a plaquette barycenter at most one f_i is nonzero. Every mixed derivative in distinct z_i of the pointwise change $\mathfrak{n}_K \circ v$ thus vanishes from order two on. The mixed derivative of the full logarithm for run B is $\mathfrak{n}_K'(t)^n F_{K+1,n}(\mathfrak{n}_K(t))$. The same argument kills the sitewise frozen pieces of the local energies for $n \geq 2$. In the other local terms the derivative selects only enlarged carriers $X^\sim$ meeting every support. If the separation is at least $\mathsf{d} > 0$, this forces $d(X) \geq \mathsf{d}L^s/M - C$ at depth s, where $d(X)$ is the size used in the connected-domain sum.

On a shared carrier at depth s, decompose the trivial-field evaluation $\mathsf{V}_{K-s}(X;w)$ into the sum of the frozen regular terms at their separate base points and the evaluation of the defect, remainder and gauge terms. For the frozen term at a base point y the source enters only through $w(p_y)$. The full mixed derivative kills this share in each run separately. Represent the difference of the remaining sums at the matched sources by a local family $G_s(X;v)$ on the common scaffold. Its sum gives the difference by the total identities of the comparison families. The centered difference uses $G_s(X;v) - G_s(X;t_1)$, with the same real centers as above.

The one-point part of G_s on a single M-cube is the sum over its pins y of the paired log normalizer differences. On the common scaffold these are

$$\log \mathfrak{z}(\theta_{[s]} - \zeta_{(s)}^{\mathrm{B}}(p_y)) - \log \mathfrak{z}(\theta_{[s]} - \zeta_{(s)}^{\mathrm{A}}(p_y)),$$

up to constants independent of the source. Here $\zeta_{(s)}$ is the running shift at the gauge step being compared. For the common gauge cutoff write $\mathfrak{z}(a) = \int \chi e^{-aQ}\, dH$, where dH is Haar measure and $0 \leq Q \leq \epsilon_s^2$ on the cutoff support. On the strip $|\operatorname{Im} a|\epsilon_s^2 \leq 1$ its real part is at least $\cos(1)\mathfrak{z}(\operatorname{Re} a)$, and differentiating under the integral gives $|(\log \mathfrak{z})'(a)| \leq \epsilon_s^2/\cos(1)$. Apply this on the segment joining the admitted weights. The pointwise shift bound gives $C(1+s)(\rho^{K-s} + \Pi_K)$ for the difference

at a pin, and the fixed number of pins per cube preserves this rate. On the plateau this is the pointwise bound for the local pair. On a common transition layer the two shift fields interpolate their consecutive-depth values with the same coefficient $\lambda(p_y) \in [0,1]$. Their difference is thus the interpolation of the two paired differences. If the two adjacent depths are $s, s+1$, write $e_s = \rho^{K-s} + \Pi_K$. Then $e_{s+1} \le \rho^{-1} e_s$ and $s+2 \le 2(1+s)$, so the layer has the same $C(1+s)e_s$ bound. The consecutive reads in this argument are those of the actual paired residual sources. At each pin we must check the common profile and admission of both reads. For different interpolation coefficients the additional term is their difference times the change of one run's shift between the two depths.

For the compensating gauge copy in the terminal integral, the standing aligned label supplies the same regions and cell partition in the two runs. The fixed profile is a product over rounded complement cells. At a pin only cells within one quarter of their side length contribute a factor different from one. Thus the common pin and the matching cells of the standing label give identical profile values. In a restoration step the sequence in force can differ from the retained label. A gauge read there requires this check for the sequence that actually supplies it.

The argument of the normalizer on an omitted cube is the stored gauge-weight entry at its own level j, read at the terminal step. The mark of this entry is charged in the running-weight load. Write $\mathfrak{g}_j = g_j^2 (\log x_j)^{2q_\flat}$ for the load weight and E_1 for the bound on the product of this weight and the moved slot along the dial. The change of the gauge exponent on a cube is at most $CE_1 \epsilon_j^2 / \mathfrak{g}_j$, by the strong logarithmic derivative bound above. Since $\epsilon_j^2 \le C g_j^2 (\log x_j)^{2\mathfrak{p}_0}$ and $q_\flat \ge \mathfrak{p}_0$, this is CE_1 per cube on the admitted dial. The negative sign of this copy is retained in the exponent. The global gauge term uses the local energy comparison.

An older-layer read also has a summable bound when the pointing measures match, even if the active depths at the pin are far from the nominal gauge depth. At each shared depth u, the pointing measure is the sum of the tents at the pin times the profile value, with atoms at their source sampling points. Its mass is at most one. Use the two real centers $t_1, \mathfrak{n}_K(t_1)$ and put $W_A = v - t_1$, $W_B = \mathfrak{n}_K(v) - \mathfrak{n}_K(t_1)$. By the exact shift formula, the difference of the residual shifts is $W_B - W_A$ plus the sum of the paired centered increments integrated against the common pointing measures, and the extra finest increment of the longer run. At an atom with source value v_*, the shared increment difference obeys

$$|\delta b_u(v_*) - \delta b_u(t_1)| \le C \min\{a_{[u]}, e_u/\varrho\}\, |v_* - t_1|.$$

The two bounds follow from the one-run slopes and Cauchy's estimate on the smaller disc, respectively. The extra increment costs $C a_{[K]} \|W_A\|_\infty$. Since $\mathfrak{n}_K' - 1 = O(\Upsilon_K)$, integration along the source segment bounds $W_B - W_A$ by $C\Upsilon_K \|W_A\|_\infty$. The same sum as in eq. A.lxxviii then bounds the residual shift difference by $C\Upsilon_K \|W_A\|_\infty$. If the pointing measures differ, their difference acts on a one-run centered increment and requires a separate comparison.

For the centered logarithm, the full gauge arguments also contain the real-center inverse couplings. Denote their difference at the pin by d_s. The two logarithm differences, at the source and at its center, are bounded together by $C\epsilon_s^2 (2d_s + \Upsilon_K \|W_A\|_\infty)$ on the same admitted gauge strip. When the centers are the scalar centers, their difference uses the paired scalar estimate.

The shift reads the source within two depth-s spacings of the pin. The gauge term therefore uses the enlarged carrier test for mixed derivatives. Since its carrier is a single cube, once $CML^{-s} < \mathsf{d}$ it cannot meet two of the separated supports and its mixed derivative vanishes. The older-layer estimate thus adds only $C_{n,\ell,\mathbf{f}} \Upsilon_K$ to the mixed-derivative comparison when used on gauge pins with the indicated locality and compared real centers.

For a sharper comparison, consider a pin p in a common native zone of actual depth $m < K$. Its profiles are zero for $u < m$, possibly partial at m, and one for $u > m$ among the available inputs, including the extra finest input of the longer run. Use this calculation where that pattern and the common tents apply to the full gauge read, and the re-centered shift formula uses the

same increment functions as the frozen matching, as in eq. 5.lxxiii. Let $D_u(z) = \delta b_u(z) - \delta b_u(t_1)$ for the shared increments, and let D_K denote the centered extra increment at the matched offset. Terminal matching at $v(p)$ and at t_1 gives

$$W_B(p) - W_A(p) + \sum_{u=0}^{K} D_u(v(p)) = 0.$$

Replace each tent sample by its value at the pin in the residual shift difference. The identity cancels the full sum, leaving the omitted increments for $u < m$ and the omitted fraction at m. On the smaller source disc their total is at most $C(1+m)e_m\|W_A\|_\infty$, since $e_u \le e_m$ for $u \le m$. For the error of this replacement, the sample distance is $O(L^{-u})$ and the increment difference has derivative bounded by Ce_u/ϱ on the shared depths. Hence the error is at most

$$C\,\mathrm{Lip}(v)\left(\sum_{u=m}^{K-1} e_u L^{-u} + a_{[K]}L^{-K}\right) \le C\,\mathrm{Lip}(v)e_m L^{-m},$$

using $L^{-1} < \rho$ and bounded $a_{[K]}$. The real-center coefficient is compared separately as above. If the extra input is not fully present, the omitted fraction of $D_K(v(p))$ must also be included.

The logarithm on a cube is a sum over individual pins. At one pin the source halo has radius $O(L^{-m})$. Once this is below a fixed fraction of d, the halo misses at least one support and the mixed derivative at that pin vanishes. On the remaining pins with $s \le m$, both depths are bounded in terms of the separation. The bound $(1+m)e_m = O_{\mathsf{d}}(\rho^K + \Pi_K)$ then gives the sharper mixed-derivative rate after the finite cube count and Cauchy's estimate. The identification of the actual zone and the profile pattern is needed for this use of the rate.

For the remaining two parts use the total identities at the unit field, where the regular background is the unit field and the current is zero. The source defect D contributes the sum over base points of its comparison representative with weight $E_0\mathsf{l}_s$. The remainder contributes one representative per carrier with weight $(C_t\theta_{[s]})^{-1}$. These are the weights of the native comparison units, with $\mathsf{l}_s \le C(1+s)$ and $\theta_{[s]} \ge h$. On the admitted source polydisc the A and R envelopes therefore bound these two parts together by $C(1+s)(1+d(X))(\rho^{K-s}+\Pi_K)e^{-\kappa^c d(X)}$. Here κ^c is the decay rate in the comparison norm, and the factor $1+d(X)$ counts the base points in the defect. Absorb that factor by taking $\kappa_e < \kappa^c$. This gives the following bound for the regular parts, and for the complete family at depths where the gauge pins satisfy the nominal-depth estimate. Use it on the shared depths through $K/2$:

$$|G_s(X;v)| \le C(1+s)(\rho^{K-s}+\Pi_K)e^{-\kappa_e d(X)}.$$

We must verify the comparison envelopes and the pin estimates for the actual paired trajectories on this polydisc. Summing by an anchor cube and reserving exponential decay after the connected-domain count costs at most $C(\ell L^s)^4 e^{-c\mathsf{d}L^s/M}$. This cost times $(1+s)\rho^{-s}$ is summable. Thus Cauchy's estimate gives $C_{n,\ell,\mathbf{f}}(\rho^K + \Pi_K)$ for the mixed derivative from these depths. For deeper shared depths, and for the one extra depth of run B, use the two individual local energy bounds. The same carrier count gives $C_{n,\ell,\mathbf{f}}e^{-c'\mathsf{d}L^{K/2}/M}$. For the stop logarithms use Cauchy's estimate on the smaller polydisc, with the fixed-torus radius above. In particular the second line of eq. A.lxxiv follows if the stop error ε has a summable bound and the shared-depth estimate holds. At $t=0$ the identity for N_K cancels the derivative factor in the comparison of $N_K^{-n}F_{K,n}(0)$, with no additional error from the matching derivative.

For summability one can stop the shared-depth comparison earlier. For any $q > 1$ take s_K with $c\mathsf{d}L^{s_K}/M \ge 2q\log K$ and $s_K = O(\log_L \log K)$. The individual local bounds give a summable tail above s_K, including the carrier count. This leaves the comparison of the terminal normalized integral at the fixed stop and the shared-depth estimate through s_K. One can also transfer the shallower totals into the stop factor at depth s_K using the exact identity above. In that form the stop factor must be compared at the moving depth. For sources near their real centers, the Schwarz

estimate for each local energy and the connected-domain sum give $|\sum_{u<s} E_u^{\mathrm{src}}(w)| \le C_\ell L^{4s}\|w\|_\sharp$. The stop factor at depth zero is close to one on a fixed source ball. The exponential of the transferred energies is also close to one on the ball of radius $c_\ell L^{-4s}$. At $s = s_K$ the resulting mixed Cauchy estimate costs only a fixed power of $\log K$, provided the comparison of the two moving-stop factors is established on that ball.

Proposition A.15 (The conditional terminal-coupling limit)**.** These inputs imply uniform convergence $F_{K,n} \to S^{[t]}_{\ell,n}$ on the smaller strip and $\Xi_K \to \Xi_\infty$. The latter is a real bi-Lipschitz homeomorphism from J onto $I \supset [h - \varrho/6, h + \varrho/6]$. Set

$$S^{(\ell,x)} = S^{[\Xi_\infty^{-1}(x)]}_\ell, \qquad x \in I.$$

If $t_K \in \mathsf{J}$ and $\Xi_K(t_K) \to x$, then $F_{K,n}(t_K) \to S_n^{(\ell,x)}$, without a subsequence. ■

Proof. Cauchy's estimate on the wider strip bounds $F'_{K,n}$ uniformly. Thus replacing $\mathfrak{n}_K(t)$ by t in eq. A.lxxiv, and its derivative factor by one, costs $C\eta_K$. The resulting increments are summable, uniformly on the smaller strip. Their limit is holomorphic and continuous on J. The coordinate increments are summable too. Its real Lipschitz bounds pass to the limit; the endpoint bounds give the asserted interval. Uniform coordinate convergence gives $\Xi_\infty(t_K) \to x$; the lower Lipschitz bound forces $t_K \to \Xi_\infty^{-1}(x)$. Uniform convergence of F_K and continuity of its limit finish the proof. □

This gives the terminal-coupling limit in part (d) of Thm. A under the stated comparison premises; infinite-volume uniqueness requires a separate argument. The depth coordinate $\Xi_{\infty,s}(t) = \lim_K x^\natural_{K-s}(\beta^\natural_K - t)$ requires the corresponding fixed-depth comparison. If its envelope is

$$\lambda s - C \le \Xi_{\infty,s}(t) - \Xi_{\infty,0}(t) \le \Lambda s + C,$$

then $\Xi_{\infty,s}(t)^{-1/2}$ decreases in order as $s^{-1/2}$; at length L^{-s} this is logarithmic running. The reference value $g_{[s]} = \theta_{[s]}^{-1/2}$ is a calibration, not this arbitrary member's running coupling. Common-prefix matching must also control $\hat{t}_{K,s} = \beta^\natural_{K-s}(g_{[s]}) - \beta^\natural_K(g)$. If it converges to $\hat{t}_s$, exact finite dilation passes to $S^{[t]}_\ell[g] = S^{[t+\hat{t}_s]}_{L^s\ell}[g_{[s]}](\mathbf{f}^{[s]})$, wherever both offsets are in their windows. For the first-passage application let K' be the stopping index of the run whose choices follow the coupling, at bare inverse coupling β. The scaffold comparison in E4 must give a live frozen run of length K' from the same bare value, in the admitted real tuning class. Write y for its terminal inverse coupling. Write $\omega(g) \le 1$ for the terminal cost in eq. 5.lv. The first-passage window then gives

$$h - \omega(g) \le y \le h + B_\sharp + \omega(g).$$

Use the increasing inverse of the frozen terminal coupling as a function of the bare value in this class. Its value at h is $\beta^\natural_{K'}$. The inverse tuning estimate has Lipschitz constant at most $3/2$, so

$$-\tfrac{3}{2}\omega(g) \le \beta - \beta^\natural_{K'} \le \tfrac{3}{2}(B_\sharp + \omega(g)).$$

For $\varrho \ge 6(B_\sharp + 1)$ this puts the offset $\beta^\natural_{K'} - \beta$ in J. The identification uses uniqueness of the admitted real frozen run at terminal value y.

Conversely, for sufficiently large K and $\beta = \beta^\natural_K - t$ with $t \in \mathsf{J}$, the forward comparison at its first-passage index K' gives $|\beta^\natural_{K'} - \beta^\natural_K| \le \varrho/2$. The matched-depth estimates below give a lower spacing bound between the tuned bare centres. To see this, compare their couplings at depth $\min(K, K')$ by telescoping in the run length. The geometric error and the summed floor give a fixed bound there. The remaining depth interval of the longer run then contributes at least $\lambda|K - K'| - C$ by the flow bound. This gives $|K - K'| \le C(\varrho)$. The comparison between the ordinary and the frozen bookkeeping, including admission of the frozen run, is the input to this argument. Exact lattice dilation uses $f^{[b]}(y) = f(L^{-b}y)$, without an observable prefactor, and $T_\ell \mapsto T_{L^b\ell}$. Conditional

offset limits and depth shifts pass through that identity only when the offsets converge and both torus floors hold. The change-of-reference claims retain these matching premises. The tuned bare increments must also satisfy:

$$\beta^{\natural}_{K+1}(g)-\beta^{\natural}_{K}(g)\longrightarrow c_0,\qquad \beta^{\natural}_{K+b}(g)-\beta^{\natural}_{K}(g)\longrightarrow bc_0\quad (b\in\mathbb{Z}\text{ fixed}). \tag{A.lxxx}$$

The second assertion follows by finite telescoping from the first. The following proof uses the stronger comparison hypotheses. Write $x^{[i]}_{[s]}$ for the tuned depth-s coupling in the run of length i, with $x^{[i]}_{[0]}=h$ and $x^{[i]}_{[i]}=\beta^{\natural}_i$. Require $x^{[i]}_{[v]}-x^{[i]}_{[u]}$, for $u<v$, to lie in $[\lambda(v-u)-C,\Lambda(v-u)+C]$, and

$$|x^{[i+1]}_{[s]}-x^{[i]}_{[s]}|\le C(L^{-c(i-s)}+s\Pi_i),\qquad \sup_{s\ge1}s\sum_{i\ge s}\Pi_i<\infty.$$

These constants and the last bound are fixed before ϱ. Comparing two bare centers at the same depth K', then telescoping in the run length, bounds their extra discrepancy by $C\sum_{i\ge K'}L^{-c(i-K')}+CK'\sum_{i\ge K'}\Pi_i\le C'$. The spacing bounds therefore give a uniform bound A on $\alpha_K=\beta^{\natural}_{K+1}-\beta^{\natural}_{K}$. Choose $\varrho>20A$.

Each cluster value α of α_K gives $S^{[t]}_{\ell}=S^{[t-\alpha]}_{L\ell}(\mathbf{f}^{[1]})$ by exact finite dilation and uniform moving-offset convergence. Two cluster values $\alpha^-<\alpha^+$ give an offset period $\omega=\alpha^+-\alpha^-$ on an open interval. Holomorphy in the offset extends that identity throughout the overlapping strips. Repeated blocking can then restore every offset modulo ω into $[0,\omega)$; all intermediate offsets stay in J by the choice of ϱ. Require, additionally, one fixed positive shape Ψ, a floor ℓ_0, and $r_0>0$, such that

$$0<\mathfrak{m}(r)\le S^{[t]}_{\ell,2}(\Psi_r)\le\mathfrak{M}(r),\quad \mathfrak{M}(r)\longrightarrow0,\qquad t\in\mathsf{J},\ \ell\ge\ell_0,\ 0<r\le r_0.$$

Normalization by the two fixed supremum norms is understood. The required uniformity in offset, shape placement and arbitrarily large tori is part of the stated tuned two-point comparison. For $r_m=L^{-m}r_0$, blocking gives $\mathfrak{m}(r_0)\le\mathfrak{M}(r_m)\to0$, a contradiction. Thus α_K has one cluster value and converges. Finally eq. A.lxvii–eq. A.lxviii, backward drift and $x^{\natural}_K=h$ make the average nonlinear increment error tend to zero, so $\beta^{\natural}_K/K\to c_0$. Cesaro averaging identifies the unique cluster value with c_0. Without these two-run and uniform-envelope premises, eq. A.lxxx requires a separate comparison estimate. With eq. A.lxxx, exact finite dilation and the moving-offset limit give

$$S^{[t]}_{\ell}(\mathbf{f})=S^{[t-bc_0]}_{L^b\ell}(\mathbf{f}^{[b]})\qquad (b\ge1),$$

when both offsets lie in their windows and the torus floors hold. Indeed the same bare coupling has the second offset $t-[\beta^{\natural}_{K+b}-\beta^{\natural}_{K}]$, which converges to $t-bc_0$. Thus the original blocking target is preserved without claiming it follows from one-step potential estimates.

Comparison of distinct offsets requires the following additional two-point assumption. Fix a nonempty class of nonnegative C^2 pairs: each test is supported in a radius-r ball, the two supports have distance in $[r,6r]$, and each test satisfies $\int f\ge\vartheta r^4\|f\|_\infty$ and $\mathrm{Lip}\,f\le\|f\|_\infty/(\vartheta r)$. Define $\sigma\approx_2\sigma'$ by an error $Cr^2\|f\|_\infty\|h\|_\infty$, uniformly on this class at all sufficiently small r. Set $(D_\mu\sigma)_2(f,h)=\sigma_2(f(\cdot/\mu),h(\cdot/\mu))$. Require $\sigma_t\approx_2 D_\mu\sigma_{t'}$, with $\mu=\exp((t-t')\log L/c_0)$, and uniform two-point upper and lower envelopes of order $(\log(1/r))^{-2}$. A germ cannot be fixed by D_μ, $\mu\ne1$. For $\mu>1$, apply its error bound to one fixed rescaled shape, with normalized value $Q(r)$, at radii $r\mu^{-a}$. The upper envelope gives $Q(r\mu^{-a})\to0$, so

$$Q(r)\le C\sum_{a\ge1}(r\mu^{-a})^2=\frac{Cr^2}{\mu^2-1}.$$

This contradicts the logarithmic lower envelope for arbitrarily small r. For $\mu<1$, invert the dilation. Thus equality of two different-offset limits would contradict rigidity. The comparison retains its

shape, placement, offset and torus-floor quantifiers; it is an additional premise, not a consequence of matching alone.

Why a geometric relabelling needs another construction. Take three test tuples of the same fixed correlation order, for example three pairs, with observable values

$$F_K(t) = (1,\, 1+t,\, 1+\Pi_K) = \alpha + \beta t + \gamma \Pi_K, \qquad \Pi_K = K^{-q}, \quad q > 1.$$

Here $\alpha = (1,1,1)$, $\beta = (0,1,0)$, $\gamma = (0,0,1)$, with determinant one. On $|t| \le 1/4$ the values are positive and uniformly bounded. With $\Xi_K(t) = h - t$ and $\mathfrak{n}_K = \mathrm{id}$, all summable matching inequalities hold; their increments are $O(K^{-q-1})$. Suppose a reparametrization $t \mapsto \tau_K(t)$ and one common normalization $a_K(t)$, bounded above and away from zero, made their increments geometrically small. At a fixed t, geometric summability forces a_K, the first component, to converge geometrically to a nonzero limit. The third component minus the first is $a_K \Pi_K$, tends to zero, and would also be geometrically small. This contradicts its polynomial lower bound. The same argument allows arbitrary reparametrizations: their value does not enter either the first component or the third-minus-first. Only a common, nonvanishing field normalization is required. This is a counterexample to the proposed implication from numerical inequalities, not an assertion that these three numbers already form a physical Yang–Mills state. Thus those inequalities alone do not supply the stronger second-coordinate claim. The construction with filed remainders uses its own running coordinate. At the filing step add the vacuum-subtracted pointed shares to E. Extract their Wilson coefficient on the filing lattice and subtract the corresponding counterterm. The vacuum values go into the scalar normalization and the shares still in the filing band remain in the unfiled part. The shifted Wilson coefficient and the gauge-fixing coefficient both use the new running shift. The filing identity is applied within the same admitted production. If x_K is the coupling before this transfer, the two zero-source coordinates at the same bare datum satisfy

$$\hat{x}_K = x_K - \sum_{i:\, i'' \le K} \mathrm{coeff}_{A,i''}\big(\mathsf{RF}[R^{(i)}]\big).$$

Here i'' is the actual filing level and $\mathrm{coeff}_{A,i''}$ denotes the coefficient of the vacuum-subtracted total at that level. On a common admitted complex offset disc, denote this sum by $T_K(t)$. The filing norm and coefficient estimate bound each summand by

$$C\theta_{i''} w_{i''-1,i} \|R^{(i)}\| \le C\theta_i^{1-\kappa_0/2}.$$

The inequality uses the comparison of radii at the filing depth and the one-run remainder bound. Summing over the depths of the input remainders, where the inverse coupling grows at least linearly, gives

$$\sup_t |T_K(t)| \le C h^{2-\kappa_0/2} \qquad (\kappa_0 > 4).$$

Cauchy's estimate gives the same small bound for the derivative on a fixed smaller disc. Thus the coordinate $\Xi_K(t) - T_K(t)$ of this presentation keeps a lower real Lipschitz bound when Ξ_K has one and the coupling is small. The ordinary pin at h is then at $h - T_K(0)$ in the filed coordinate at that bare datum.

For the two-cutoff estimate align the remainders by birth depth s from the terminal lattice. If the two runs use the same scaffold, the filing delay a_s at that depth is fixed by the scaffold. The remainder is included precisely if $a_s \le s$, so the inclusion test is common on shared depths. Terms at an extra bare depth can be bounded by their one-run size. If the schedules differ, their inclusion tests must be compared as well.

On the admitted paired paths write χ_s for the termwise difference norm of the two remainders at birth depth s, and ι_s for the relative datum error of their filed read. The termwise filing estimate

and the Wilson coefficient bound give

$$|\delta T| \le C \sum_{\substack{s \text{ shared} \\ a_s \le s}} (\theta_{[s]}\chi_s + \iota_s \theta_{[s]}^{1-\kappa_0/2}) + C\theta_{[K+1]}^{1-\kappa_0/2}.$$

Here the coefficient estimate at the filing level uses the crossing weight to return to the unit at birth. Each filed remainder occurs once in the sum. The termwise norm uses the actual filed slices of the runs.

In the comparison pinned in the filed coordinate, the filed response rows bound $\theta_{[s]}\chi_s$ by $C\widehat{r}_s\rho^{K-s}$. The datum term has the same bound on the admitted paths. Using $\widehat{r}_s \le C\rho^{-(K-s)/2}\widehat{r}_K$ and summing the geometric factor gives an $O(\widehat{r}_K)$ bound for δT, including the bare end. Thus a geometric comparison at each fixed depth may still yield a polynomial error for the sum of the transferred coefficients. If the paired reads are available only through depth $K/2$, their contribution is bounded by a geometric factor $\rho^{K/2}$ times the summable one-run weights. The other depths are bounded separately by the one-run coefficient sum, which on that depth interval is $O_g(K^{2-\kappa_0/2})$. With $\widehat{r}_K$ of order $\theta_{[K]}^{2-\kappa_0/2}$ times a power of its logarithm, the bound is summable for $\kappa_0 > 6$.

On a common admitted terminal interval for the filed runs, let u be the filed terminal coupling and write $\mathcal{T}_K(u)$ for the sum T_K at its corresponding bare datum. The pre-transfer coordinate of this same presentation is $X_K(u) = u + \mathcal{T}_K(u)$. On a smaller interval it has a uniform positive lower Lipschitz bound by the one-run derivative estimate and the admitted inverse for u. For a pre-transfer target x, put $u_K = X_K^{-1}(x)$. Comparing $X_{K+1}(u_K)$ with x gives

$$|u_{K+1} - u_K| \le C \sup_u |\mathcal{T}_{K+1}(u) - \mathcal{T}_K(u)|.$$

This transfers the summable bound to the change of pin within the same presentations. Its use for the separately constructed ordinary run requires the matching of its production and source normalization to that presentation.

The optional Ward target on arbitrary defining sequences. In this section $S_{j,n}$ and S_n, $n \ge 2$, are *connected* forms. Require convergence on every smooth compact separated tuple on $\mathbb{R}^4$, along the specified defining sequence. On every fixed collection of separated compact support neighborhoods require uniform local multilinear C^1 bounds. Mollification inside slightly enlarged separated neighborhoods then extends the forms and their convergence to compact C^1 tuples: first approximate each entry, use the uniform telescoping bound, and then pass to the limit for the smooth approximants. Assume full signed-permutation symmetry W_4 of the limit. The specified lattice probability laws realize these connected forms; the mixed cumulants below use those laws. For torus approximants a fixed R must fit in one common chart for all sufficiently large j. For $\omega \in \mathfrak{so}(4)$, set $\delta_\omega f = -(\omega x)\cdot\nabla f$ and

$$D_{j,\omega}^s f(x) = \sum_\rho (\omega x)_\rho \frac{f(x + a_j e_\rho) - f(x - a_j e_\rho)}{2a_j}.$$

Here are actual additional Ward inputs, independent of the angular route. Choose smooth $\chi_R = 1$ on B_R, zero outside B_{2R}, with derivatives bounded by $C_r R^{-r}$. For every j, ω, R, supply fixed annular local functions

$$\mathfrak{B}_{j,\omega,R} = \sum_{\Delta \in \mathcal{A}_{j,R}} \mathfrak{B}_{j,\omega,R}^{\Delta}.$$

There are at most CR^4 physical unit boxes; each summand is a specified gauge-invariant Borel function of links in a fixed enlargement of its box, linear in ω, supported near $R - Ca_j \le |x| \le 2R + Ca_j$. The functions may contain the divergent Wilson weight. They are supplied by the actual

change of variables, independently of the tests, not defined as their desired rotation defect. Require the exact finite identity

$$\sum_i S_{j,n}(f_1,\dots,D^s_{j,\omega}f_i,\dots,f_n) = \mathfrak{U}_{j,\omega,R}(\mathbf{f}) + \langle \mathfrak{B}_{j,\omega,R}; \mathcal{O}_j(f_1);\dots;\mathcal{O}_j(f_n)\rangle^T. \tag{A.lxxxi}$$

In eq. A.lxxxi, $\mathcal{O}_j(\mathbf{f})$ denotes the separate list of cumulant arguments, not their coincident product. The bulk functional $\mathfrak{U}$ must also be the specified correlation from that same Ward construction; it is not an arbitrary error chosen after the boundary functions. For every fixed sufficiently large R, require $\mathfrak{U}_{j,\omega,R} \to 0$ and the analogous sum with $D^s_{j,\omega}f_i + \delta_\omega f_i$ to tend to zero. These include the actual typed anisotropy and volume-transfer estimates where needed, and are not proved by scalar source bounds.

Multilinear telescoping and local C^1 continuity differentiate the limit's rotation orbit on smooth tuples. Applying the assumed convergence at the fixed derivative tests in eq. A.lxxxi, without differentiating a cutoff limit, gives

$$\lim_j \langle \mathfrak{B}_{j,\omega,R}; \mathcal{O}_j(\mathbf{f})\rangle^T = -\delta_\omega S_n(\mathbf{f}). \tag{A.lxxxii}$$

It is independent of R. Consequently the condition in part (e) of Thm. A

$$\liminf_{R\to\infty} \limsup_{j\to\infty} \left| \langle \mathfrak{B}_{j,\omega,R}; \mathcal{O}_j(\mathbf{f})\rangle^T \right| = 0 \tag{A.lxxxiii}$$

is equivalent to vanishing of this derivative. The inner limit always has R fixed.

For completeness, one nonzero generator suffices. Assume also the invariant one-point convention $c\int f$ and zeroth moment one, or use the centered functional. Let G be the closed subgroup of orthogonal transformations preserving the connected forms. Closedness follows from C^1 continuity on separated tuples. Equivalently, define G from the centered moment forms obtained by the finite sum over partitions of these cumulants; the inverse partition formula gives the same group. The invariant one-point function $c\int f$ makes the uncentered moment group the same. Let $\mathcal{A}$ consist of skew matrices whose entire one-parameter group lies in G. It is closed under real scalar multiplication and addition: the finite-matrix product formula

$$e^{t(A+B)} = \lim_{r\to\infty} (e^{tA/r}e^{tB/r})^r$$

follows by expanding each factor to second order and telescoping the product error, which is $O(r^{-1})$. Closedness of G then applies. It is also stable under conjugation by W_4. For $a < b$, sign matrices $D_\epsilon = \mathrm{diag}(\epsilon)$ give the projection

$$2^{-4} \sum_{\epsilon\in\{\pm1\}^4} \epsilon_a\epsilon_b D_\epsilon A D_\epsilon = A_{ab}E_{ab}.$$

Thus any nonzero element of $\mathcal{A}$ supplies a coordinate-plane generator; permutations supply all of them. Successive plane rotations reduce any determinant-one orthogonal matrix to the identity column by column. Hence G contains SO(4), and its coordinate reflections give all O(4). The condition in eq. A.lxxxiii for a nonzero ω, on every smooth tuple, gives its whole one-parameter group by applying the zero derivative also at each rotated tuple. Conversely full invariance makes eq. A.lxxxii zero for every ω. No Lie-subgroup theorem is needed. Indeed an infinite closed G has nonidentity $g_r \to I$; the normalized skew logarithms have a nonzero subsequential limit A. Integer powers $g_r^{\lfloor t/\|\log g_r\|\rfloor} \to e^{tA}$ supply the required one-parameter group. Therefore such a W_4-invariant group is finite or all of O(4).

Infrared meaning and the Hilbert-space consequence. A uniform mixed-insertion estimate $|\langle \mathfrak{B}^{\Delta}; \mathcal{O}_j(\mathbf{f})\rangle^T| \le CD^{-4-\delta}$, $\delta > 0$, at distance D from the fixed supports, implies eq. A.lxxxiii: its shell sum is at most $CR^4R^{-4-\delta}$. Decay D^{-4} alone yields only a bounded sum. It also suffices that the sum of absolute mixed insertions tend to zero along one sequence $R_l \to \infty$, after $\limsup_j$. Ordinary per-ball density bounds do not give either estimate for these different, weight-bearing insertions. If an older survival argument is used, its event $\mathcal{L}_{\mathfrak{c},k,a,K}$ means that cell $\mathfrak{c}$ is live at level k, its component lineage starts at $k-a$, and it remains live at every level $k,\dots,K$. Its weighted law is an additional history input.

Assume the relevant centered or recentered moment functional also satisfies the HB quotient hypotheses. Once full scalar rotation invariance is supplied, spatial $R \in \mathrm{O}(3)$ acts on the HB quotient by $V_R[F] = [(R \oplus 1)F]$. It preserves the positive form and null space, has inverse $V_{R^{-1}}$, fixes Ω, commutes with time evolution and conjugates a spatial translation by x to one by Rx. Local C^1 continuity proves strong continuity on dense test vectors, and unitarity extends it to the completion. The joint spectral measure is consequently covariant under $(E,p) \mapsto (E,Rp)$. For limits for which HB's cube inequalities have been established, they give $E \ge |p|$: use a countable dense set of rotations, remove their common null set, and approximate a rotation carrying any given p to $|p|e_1$. Centering is the map C_m proved in § 6.16 and changes none of these conclusions. In the alternative notation, $\mathsf{sh}_\xi = C_{-m_\xi}$, $m_\xi(f) = c_\xi \int f$, so its inverse is C_{m_ξ}. No operator for a general time-mixing Euclidean rotation is asserted. External OS reconstruction, F4 identification of the full theory and any mass-gap estimate remain separate. The periodic-state angular proof retains its own mirror, cone and ultraviolet interface; the optional Ward premises above do not silently enlarge that state class.

A.9. **Weighted derivatives and matched presentations.** This section formulates weighted derivative estimates and comparisons of matched presentations on their specified domains. These stronger estimates are separate from the ordinary scalar source argument and its angular continuation. We prove the analytic, convolution and normalization steps. The remaining input is the corresponding extension of the actual formation and history maps, with the weights and reserves specified here. No survival probability is used as an enlarged input radius.

A direction is a list, including its own coefficient. Fix a real frame and all its reference operations. A direction at old level j contains a pointed regular family $\dot{E}^{(j)}$, any awaiting remainder $\dot{R}^{(j)}$, and the specified weight or vertex amplitude. Its regular family has the actual gauge covariance and the covariance of its pointed total required by the chosen channel. Its own coefficient is differentiated with that family.

For the scalar channel let T be the fixed physical Hessian readout. If E^λ is an admitted holomorphic family, put $b^\lambda = T[E^\lambda]$. With the frame fixed the map T is linear and independent of λ, so

$$\dot{b} = T[\dot{E}], \qquad \frac{d}{d\lambda}[E^\lambda - b^\lambda h_z A]\big|_0 = \dot{E} - \dot{b}\, h_z A. \tag{A.lxxxiv}$$

The tent h_z and background convention are the assigned frozen ones. If the readout itself changes between two frames, its difference is an additional term; it is not covered by eq. A.lxxxiv. For a tensor channel the scalar coefficient is replaced by the full assigned vector of counter-insertions. Its variation is taken at the same time as that of its own family.

This rule has analytic content at a single old base. Marginal cancellation is applied to the whole covariant member, including its own counterterm, before the bracket interpolation. A noncovariant defect is a separate input with its own scale-small norm. Cauchy differentiation of a partition function cannot give that defect the cancellation of another member.

The norms below are norms at the birth level, on the actual hereditary cores. Write $\mathsf{e}_j, \mathsf{r}_j$ for the regular and remainder norms. Let τ_j be the amplitude and k its fixed generating test. The norms also cover the associated frozen families at every admitted constant source ζ and every constant

generator h_0 in the channel with $|h_0| \le \|\mathsf{k}\|_\infty$. The amplitude norms below are the corresponding suprema over ζ. This frozen coverage is part of the weighted input class; it is not inferred from a norm of only the varying test. Put

$$\mathsf{z}_j = \|\tau_j\|_\infty \|\mathsf{k}\|_\infty, \quad \mathsf{b}_j = \|\tau_j - \tau_{j-1}\|_\infty \|\mathsf{k}\|_\infty, \quad \mathsf{z}_n^\sharp = \sum_{j \le n} L^{-(n-j)}(\mathsf{z}_j + \mathsf{z}_{j-1}). \tag{A.lxxxv}$$

Use $\mathsf{z}_{-1} = 0$ and the actual first-entry convention. The symbols here measure this direction, not the deviations of the running coupling.

Let ς_ν be the actual forgetting weights and let $w_{n,j} = (\alpha_n/\alpha_j)^2$ be the squared radius ratios. If a remainder born at j is refiled at j'', define

$$\begin{aligned} \mathcal{L}_n = &\sum_{j \le n} \varsigma_{n+1-j} \mathsf{e}_j + \sum_{j \le n \le j''} w_{n,j} \mathsf{r}_j \\ &+ \mathfrak{g}_n \left[\mathsf{z}_n^\sharp + \sum_{j \le n} L^{-c(n-j)} \mathsf{b}_j \right]. \end{aligned} \tag{A.lxxxvi}$$

The second sum includes precisely the remainders still awaiting refiling. A bare stored insertion contributes its stated additional size $C\mathfrak{g}_0 N_{\mathrm{bare}} \|\mathsf{k}\|_\infty$ at level zero. The size N_{bare} is counted from that insertion and is not silently set to one.

The vertex response factor is of order $\mathfrak{g}_n = x_n^{-1}(\log x_n)^C$ on the coupling-independent analytic route. For a member analytic only on a running field radius the corresponding gain is generally the square of the fluctuation-to-radius ratio. These two factors are different. The parity calculation below explains the distinction.

Weighted extension and the derivative rows. Attach one dial $\lambda_{n,k'}$ to the entry when it is used in zone n at step k'. A useful actual class requires a joint extension with axis radii

$$|\lambda_{n,k'}| < c/\mathcal{L}_n. \tag{A.lxxxvii}$$

When a load is zero its direction has no contribution and that axis can be omitted. The class also specifies the joint multi-axis budget; individual axis discs alone do not give their Cartesian product.

The required analytic assertion has two parts. In the small-field map the differentiated potential norm must retain $C\varkappa_k \mathcal{L}_k$, where $\varkappa_k$ is the actual fluctuation-to-radius ratio. For a large-field output, setting one dial to zero must change its assigned term by a bound proportional to the share of histories retaining that zone, on the same radius in eq. A.lxxxvii. The small factor and the age bound are bounds of the extension, rather than consequences of its holomorphy.

Here are the exact implications once that class has been proved. The inserted finite-map theorem applied to the potential derivative gives its regular potential row. In addition the actual refile operation must carry a remainder born at j with norm at most $Cw_{k,j}\mathsf{r}_j$, and the new birth has norm σ_{k+1}. These primitive refile bounds give the complete regular row below. For the amplitude row one must also use the actual channel update and its normalization:

$$\mathcal{T}_{k+1}[\dot{E}^{(k+1),\mathrm{fr}}_{\zeta,h_0}] = [\tau_{k+1}(\zeta) - \tau_k(\zeta)]h_0. \tag{A.lxxxviii}$$

Here $\mathcal{T}_{k+1}$ is the own scalar or tensor coefficient readout in the fixed chosen channel. Its field-radius bound is $|\mathcal{T}_{k+1}F| \le C\alpha_{k+1}^{-2}\|F\|$. Apply it to every frozen constant generator; taking $|h_0| = \|\mathsf{k}\|_\infty$, or the supremum over its unit directions for a linear amplitude map, proves the displayed amplitude-increment bound. If $\|\mathsf{k}\|_\infty = 0$ that bound is immediate. Without this actual update and frozen coverage the argument bounds only the own coefficient $\mathcal{T}_{k+1}[\dot{E}]$, not the separately defined amplitude increment. A coefficient readout costs the inverse square of its physical field radius:

$$\mathsf{e}_{k+1} \le C\varkappa_k \mathcal{L}_k + C \sum_{j \text{ refiled at } k} w_{k,j}\mathsf{r}_j + \sigma_{k+1}, \qquad \mathsf{b}_{k+1} \le C\alpha_{k+1}^{-2}\mathsf{e}_{k+1}. \tag{A.lxxxix}$$

If a different normalized coefficient convention is used, its explicit conversion replaces the last factor. In Gaussian coordinates that conversion includes the square of the coupling. It cannot be charged once in $w_{k,j}$ and a second time as another old-radius loss:

$$\alpha_n^{-2} w_{n,j} = \alpha_j^{-2}. \tag{A.xc}$$

Suppose a single zone dial has holomorphic output difference $F(\lambda) - F(0)$ bounded on eq. A.lxxxvii by $C\epsilon_k q_a^{(k-n-1)_+}$, with $0 < q_a < 1$. Cauchy's formula gives

$$|F'(0)| \le C'\epsilon_k q_a^{(k-n-1)_+} \mathcal{L}_n. \tag{A.xci}$$

There are $k - n + 1$ possible uses of this zone by step k. Summing its actual dial derivatives gives the aged load

$$\mathcal{A}_k = \sum_{n \le k} (k - n + 1) q_a^{(k-n-1)_+} \mathcal{L}_n. \tag{A.xcii}$$

This derivation uses an already contracted direction at each use, with its own radius. It does not apply Cauchy again on a radius chosen at the original birth.

The own part of a new boundary term can have this age-weighted bound. Dependence on an inner stored layer in a live component is different: its boundary exponent can carry order-one coefficients counted at each later use. Its small factor belongs to the component's mass integral. That inner part must be paid under that mass where the component enters. Placing the same age factor in an exponent norm would claim a stronger estimate than eq. A.xci.

The exact convolution requirements. Assume the actual forgetting kernel satisfies $\varsigma_\nu \le C\nu^4 L^{-c(\nu-1)}$. For a fixed θ with $L^{-c} < \theta < 1$ put

$$S_\theta = \sum_{\nu \ge 1} \varsigma_\nu \theta^{-\nu} < \infty. \tag{A.xciii}$$

The convergence is geometric after a polynomial factor. If the shifts agree, the unmarked row $u_{k+1} \le \varkappa_k \sum_{j \le k} \varsigma_{k+1-j} u_j$ propagates $u_j \le A\theta^j$ whenever $\sup_k \varkappa_k S_\theta \le 1$. Substitute the induction bound, factor $A\theta^{k+1}$, and the remaining sum is at most S_θ. Thus θ is chosen first and the weak-coupling floor after it. The weight itself is not generated by this induction.

For the age kernel define $a_\nu = (\nu + 1) q_a^{(\nu-1)_+}$. On sequences decaying as θ^n its norm is

$$A_\theta = \sum_{\nu \ge 0} a_\nu \theta^{-\nu} < \infty \quad \Longleftrightarrow \quad \theta > q_a. \tag{A.xciv}$$

For $\nu \ge 1$ the ratio in the geometric series is q_a/θ, with the extra factor $\nu + 1$. At equality the terms do not even tend to zero. Consequently $\mathcal{L}_n \le B\theta^n$ implies $\mathcal{A}_k \le BA_\theta \theta^k$ under this strict inequality. The first finitely many ages are included in the same finite constant. They are not discarded by an asymptotic argument.

Squared radius weights often grow polynomially in age under the admitted frame. If $w_{n,j} \le C(n - j + 2)^A$ and a refiled remainder has $\mathsf{r}_j \le B\theta^j \eta^{n-j}$ for its actual allowed range, then

$$\sum_{j \le n} w_{n,j} \mathsf{r}_j \le CB\theta^n \sum_{a \ge 0} (a + 2)^A (\eta/\theta)^a. \tag{A.xcv}$$

This is finite for $\eta < \theta$. A remainder merely bounded by $B\theta^j$ has no such conclusion for growing weights. The recent-window remainders must instead be treated by their own recursive small factor and finite window length. These are distinct origin classes.

More generally if a nonnegative sequence satisfies $e_{n+1} \le \epsilon \sum_{\nu \ge 1} K_\nu e_{n+1-\nu} + s_{n+1}$, with $e_j = 0$ at negative indices, the weighted supremum norm gives

$$\sup_{n \ge 0} \theta^{-n} e_n \le \frac{e_0 + \sup_{n \ge 1} \theta^{-n} s_n}{1 - \epsilon \sum_{\nu \ge 1} K_\nu \theta^{-\nu}} \tag{A.xcvi}$$

when the denominator is positive. Prove it first up to a finite horizon, take the maximum, and move its coefficient to the left. The same bound is independent of that horizon. This is the discrete contraction used after the actual weighted rows are obtained; it does not establish those rows.

A single-zone incidence budget on its original mass. A zone parameter can occur in several copied potentials, counterterms and scalar anchors. To exhibit its true domain, list these occurrences before estimating a complex exponential. Let $B_p(Y) \geq 0$ be the specified reserve quantities of a component, including any size reserve, and let $a_{pt} \geq 0$ be the charge of occurrence t against reserve p. Suppose the actual exponent increment satisfies

$$|\Psi(\lambda, Y, e) - \Psi(0, Y, e)| \leq \sum_p B_p(Y) \sum_t a_{pt} |\lambda_t| \tag{A.xcvii}$$

on its admitted parameter domain. All occurrences and their scalar constants are included. An oscillation norm alone does not bound a scalar anchor.

The needed unsourced reserve is the weighted mass

$$\int^* \sup_{e \in \mathcal{E}_\omega(Y)} |w^0_\omega(Y, e)| \exp\left\{ \sum_p r_p B_p(Y) \right\} dY \leq M_\omega. \tag{A.xcviii}$$

The set $\mathcal{E}_\omega(Y)$ is the actual admissible external fibre at the stored variable Y. The upper integral is the infimum of integrals of measurable majorants; one may equivalently assume the displayed common integrable envelope as part of the premise. The parameter-dependent integrands are measurable in Y and pointwise holomorphic on their joint source domain. Fixed integration sets and this common envelope justify the dominated Cauchy argument below. The original inner Wilson factor is part of w^0_ω with its prescribed original weight. This assertion does not replace it by a freely reduced coupling. The positive reserve must have been established in the original mass estimate with its exact allocation.

On the joint domain

$$\sum_t a_{pt} |\lambda_t| < r_p \quad \text{for every } p, \tag{A.xcix}$$

equations in eq. A.xcvii–eq. A.xcviii bound the perturbed component mass by M_ω. The proof multiplies the absolute value of the original integrand by the exponential majorant and integrates. The exterior supremum stays inside the stored-variable integral. For fixed integration sets, this same domination on smaller parameter domains justifies holomorphy of the integral by the Cauchy formula.

For polydisc radii R_t it suffices to prove $\sum_t a_{pt} R_t \leq (1 - \epsilon) r_p$ simultaneously. Single-axis bounds $a_{pt} R_t \leq r_p$ do not imply this sum-budget condition. This is the no-reuse incidence test: every copy is charged once, and all copies sharing a reserve are summed.

For one dial with load $\mathcal{L}_n$, an incidence estimate $\sum_t a_{pt} \leq C_p \mathcal{L}_n$ gives radius $c/\mathcal{L}_n$, with c chosen from the unused r_p/C_p. Its radius does not increase as the zone ages. Age rarity is a bound on M_ω or its sum, and yields the factor in eq. A.xci. The zone-reserve test section proves why placing its reciprocal into the input radius fails. The application uses eq. A.xcvii–eq. A.xcviii for the complete one-zone list and its assigned mass, including the original inner Wilson factor and dropout fibres.

For m source directions sharing one potential allowance, radii proportional to $c/(m\mathcal{L}_i)$ give a valid sum-budget disc. Its mixed derivative costs at most $M(m/c)^m \prod_i \mathcal{L}_i$. Since $m^m \leq e^m m!$, this has a factorial majorant. It is incorrect to use the two-direction spare-slot radii for arbitrary m without summing the shared charge.

The even-coupling vertex gain. There is an exact analytic gain for a vertex admitted on coupling-independent field radii. Let q be an auxiliary coupling, distinct from any constraint derivative. Suppose the actual finite integral and localized formation are holomorphic for $|q| < R_q$ and $|u| < u_0$. The parameter u multiplies the specified vertex. Assume the Gaussian and cutoffs are fixed and

even, the potentials are unchanged by $(q, Y) \mapsto (-q, -Y)$, and at $q = 0$ the output is independent of u. Write its normed output as $\Phi(q, u)$ and suppose $\|\Phi\| \le M$.

Cauchy in u bounds $F(q) = \partial_u \Phi(q, 0)$ by $2M/u_0$. Changing the integration variable $Y \mapsto -Y$ proves $F(-q) = F(q)$, term by term only when the actual localization rule respects that change of variables. Independence at zero gives $F(0) = 0$. Therefore

$$\|F(g)\| \le \frac{2M}{u_0}|g/R_q|^2, \qquad \|F(g) - g^2 L\| \le C\frac{M}{u_0}|g/R_q|^4, \quad L = \tfrac{1}{2}F''(0), \quad |g| \le R_q/2. \tag{A.c}$$

For a proof expand $F(q) = \sum_{j\ge 1} a_{2j} q^{2j}$. Cauchy bounds $\|a_{2j}\| \le (2M/u_0)R_q^{-2j}$. Alternatively the even function defines a holomorphic function of q^2 and Schwarz proves the first bound. The geometric tail starting with $j = 2$ proves the second.

The radius R_q is determined by the actual potential and field cutoff. If the cut field has size $b_k \asymp (\log x_k)^{p_0}$, a cubic intrinsic remainder $q^{-2}V_{\ge 3}(qY)$ has removable value zero and size $O(|q|b_k^3)$. With the other potential and core estimates admitting this whole disc, one may take $R_q = cb_k^{-3}$. Further proved polynomial logarithmic losses replace the exponent three by their total exponent. Then $(g/R_q)^2$ has the form $x_k^{-1}(\log x_k)^C$. A mere upper bound on a possible radius would not prove this gain. If the admitted disc is instead only $R_q = c\alpha_k/b_k$ on a running field radius, the ratio squared is $c^{-2}(\delta_k/\alpha_k)^2$, where $\delta_k = gb_k$. Parity alone does not replace that smaller radius by a coupling-independent one.

The one-loop coefficient has a direct normalized formula. Suppose the inserted vertex and other potentials have $S = qS_1 + q^2 S_2 + \cdots$ and $V = qV_1 + \cdots$. Differentiate $\log \mathbb{E}_0 e^{V+uS}$ in u and expand at zero. Parity makes $\mathbb{E}_0 S_1 = \mathbb{E}_0 V_1 = 0$. The coefficient of q^2 is consequently

$$\mathbb{E}_0 S_2 + \mathbb{E}_0(S_1 V_1) - \mathbb{E}_0 S_1\, \mathbb{E}_0 V_1 = \mathbb{E}_0 S_2 + \mathrm{Cov}_0(S_1, V_1). \tag{A.ci}$$

The expectation includes the actual cutoff and its normalizer. Replacing it by an uncut Gaussian requires the separate cutoff-tail estimate. The formula in eq. A.ci does not compute its tensor eigenvalue or its sign.

Matched reference measures on a common domain. Two local coefficient lists can be compared by differentiating a common finite integral only after their variables, sharp sets and numerators have been pulled to the same reference domain. Let

$$Z_t(z) = \int_D e^{V_t(y,z)}\, d\mu_{A_t}(y), \qquad d\mu_{A_t} = (2\pi)^{-N/2} \det(A_t)^{1/2} e^{-y^{\mathsf{T}} A_t y/2}\, dy. \tag{A.cii}$$

Assume the determinant branch, matrices and integrands are holomorphic on a common neighbourhood of the interpolation $0 \le t \le 1$, with fixed D and uniform domination there. Assume the generating function has a common nonzero neighbourhood of $z = 0$. Differentiation gives exactly

$$\frac{d}{dt}\log Z_t(z) = \tfrac{1}{2}\,\mathrm{Tr}(A_t^{-1}\dot{A}_t) + \left\langle \dot{V}_t - \tfrac{1}{2} y^{\mathsf{T}} \dot{A}_t y \right\rangle_{t,z}. \tag{A.ciii}$$

This follows by differentiating the determinant and the quadratic exponent. A change-of-variables Jacobian belongs in V_t with its own derivative. If the cutoff moves, eq. A.ciii is not the complete derivative unless the induced boundary term or an exact fixed-domain pullback has been supplied.

For the normalized moment generator $\log Z_t(z) - \log Z_t(0)$, the trace cancels if A_t has no source dependence. With $L_t = \dot{V}_t - \frac{1}{2} y^{\mathsf{T}} \dot{A}_t y$ the derivative is

$$\langle L_t(z)\rangle_{t,z} - \langle L_t(0)\rangle_{t,0}. \tag{A.civ}$$

A subtracted vacuum energy has its derivative as well; it cancels between two presentations only when it is the genuinely common energy. If V_t includes a transported observable, $\dot{V}_t$ includes that observable's coordinate derivative. It is not just the difference of old interaction coefficients.

Suppose the right side of eq. A.civ, including every required scalar energy term, has holomorphic supremum at most ϵ_t on a common polydisc of source radii r_i. Integrating in t and using coordinate Cauchy gives for a mixed cumulant derivative

$$|\Delta\kappa_{i_1,\dots,i_m}| \le \gamma! \prod_i r_i^{-\gamma_i} \int_0^1 \epsilon_t \, dt, \tag{A.cv}$$

where γ_i is the number of repetitions of coordinate i. For distinct coordinates there is no $m!$. A common source radius with repeated derivatives has the indicated factorial. This is a cumulant bound because it differentiates a logarithm. For raw moments use the actual normalized generator $M_t(z) = Z_t(z)/Z_t(0)$. If $|M_t(z)| \le Q$ on the same polydisc and $G_t = \partial_t \log M_t$ is bounded by ϵ_t, then $M_1 - M_0 = \int_0^1 M_t G_t \, dt$. Coordinate Cauchy gives the raw-moment version of eq. A.cv with the additional factor Q. Scalar energy subtractions must be restored or included in the identified target generator before it is called a physical moment. Thus a coefficient comparison gives a matched moment estimate only after the actual common domain, scalar normalization, source generators and expectation bound have been proved.

For adjacent cutoff estimates the resulting error sequence has to be summable in the telescoping theorem. A direct two-cutoff bound by $\epsilon_{\min(K,K')}$ with $\epsilon_K \to 0$ is another sufficient route. Bounded derivatives at each cutoff, with no comparison of the generators across cutoffs, supply neither. The comparison-capsule section gives the resulting fixed-torus Cauchy limit in its exact scope.

A.9.0.1. Transfer between the two bare lattices. At a common physical depth transfer a regional pair (U, J) by blocking U one step and taking the cell mean of the current J. On common rows the composition of the straightened constraints identifies the retained block field. The local Jacobian and the block gauge factors agree when formed from this same field by the same rule. The other quantities to compare are the regional propagators and kernels, the localized Wilson response, the localized Gaussian normalizer, and the fields used to evaluate earlier activities. Define these on the two cutoffs by the same regional rule before taking differences. A walk term indexed by a finite set S of cubes is obtained from the alternating sum of the regional fluctuation operators of $Y \subset S$, each compressed to the interior of Y. The switched precision is linear in those walk terms. The normalizer uses the logarithm of an operator built on the recursive tower of each region. On the full torus the sequences are constant and the two regional rules use the same uncut fluctuation operator for their respective totals.

A.9.0.2. The analytic line on ordinary carriers. Take the sharp ordinary data $U, A', \mathbf{J}$ with radii a_0, a_1, c_0, where $a_1^2 \le C a_0$. Let $h = \eta/L$ be the sharp spacing in the common units. Use the line

$$A_z = A_{\mathrm{R}} + z A_{\mathrm{I}}, \qquad \mathbf{U}_z = e^{ihA_z} U, \qquad \mathbf{J}_z = J_{\mathrm{R}} + z J_{\mathrm{I}}, \qquad |z| \le R,$$

where the real and imaginary parts are in the real Lie algebra and $R > 1 + \sqrt{2}$ is fixed. The target is at $z = i$. The line has a fixed real base U.

For a plaquette p, let $d_U A'(p)$ be the sum of its four signed increments transported to the base of p in U. The first-order expansion of $\partial(e^{ihA'}U)(p)(\partial U(p))^{-1}$ is $1 + ih d_U A'(p)$, with error $Ch^2 a_1^2$. Both endpoint curvatures are bounded by $a_0 h^2$, so $|d_U A'| \le Ch(a_0 + a_1^2)$. The transports in U preserve the real Lie algebra. Applying the same expansion to A_z therefore bounds the curvature along the line by $C_R h^2 (a_0 + a_1^2)$.

The complete double blocks of a flat gauge cube and their neighboring translates lie in its 4η halo. Use the one-layer sharp collar when this halo is contained there. A cube whose halo fits within the sharp chart size can use the given gauge on the halo. Otherwise start from the sharp gauge on the cube, rounding inward by at most a fixed number of fine layers. Work on the lattice lift of the cube, using the covering charts on a short torus.

For the axial continuation add one whole face layer at a time. In the signed outward direction d, let $W(x)$ be the last gauged normal link from $x - hd$ to the face point x. With g the gauge already

defined, set

$$g(x+hd) = W(x)^{-1} g(x) U_d(x).$$

The new gauged normal link is then $W(x)$. For a tangential direction μ, the plaquette identity for the gauged links V gives

$$V_\mu(x+hd) = W(x)^{-1} (\partial U^g)_{d\mu}(x) V_\mu(x) W(x+he_\mu).$$

The plaquette error and the tangential change of W are both $O(h^2 a_0)$. Commuting W past V_μ costs $O(h^2 a_0^2)$, since both links differ from one by $O(ha_0)$. Thus each new tangential link differs from its old face value by $O(h^2 a_0)$. For tangential first differences compare the two corresponding old links and add these two errors. Normal links inherit their tangential differences and agree on successive links of each extended ray. The small logarithm divided by h therefore preserves the field and first-difference bounds, with a fixed constant at each added layer. Extend successive faces of the enlarged rectangle to include the corners. The halo takes $O(L)$ fine layers in each direction, so the total constant is independent of the cutoff.

In the resulting sharp chart write $U = e^{iha}$. The fixed one-step logarithm map has bounded first and second derivatives on a small chart and is the same function on translated stencils. Applying it to the real base gives a coarse gauge potential and first difference bounded by Ca_0. For the complex factor use

$$F(a,B) = \frac{1}{i\eta} \log\Big(\mathcal{M}(e^{ihB} e^{iha}) \mathcal{M}(e^{iha})^{-1} \Big).$$

It vanishes when $B = 0$. The derivative in B is bounded by C_L, and the mixed derivative bound gives $|D_a F| \le C_L h \sup |B|$. Comparing the complete neighboring stencils therefore bounds $F(a, A_z)$ and its first covariant difference by $C_R a_1$. The coarse plaquette estimate uses exchanges of adjacent steps in the averaging paths: the $O(L^2)$ resulting sharp plaquette errors are transported by products of bounded length. This bounds the two coarse curvatures by $C_{L,R}\eta^2(a_0 + a_1^2)$. The finite transport in the cell mean bounds the independent current by $C_{L,R} c_0$.

These are the single-scale clauses on any ordinary domain covered by these charts. In either tower rule, at an address of depth p below level j, put $\lambda = L^{-(j-p)}$. If a_0', a_1' bound the resulting curvature and complex factor, the tower theorem gives the bound $K_{\mathrm{red}}(\lambda a_0' + \lambda^2 (a_1')^2)$ at that address. In its tower subcall the independent current is set to zero. The current slot of the evaluation pair keeps its own radius. Taking the sharp radii smaller than the paired running radii by one fixed factor gives, on both lines, $a_i' \le b_i$ for the corresponding field radii, $a_0' + a_1' \le a^\sharp$, and $K_{\mathrm{red}}(a_0' + (a_1')^2) \le b_0$. Thus the line and its transfer lie in their cores at every tower address. On the real diameter the line gives real pairs in a fixed small radius, with real chart bound $C_{L,R}(a_0 + a_1)$. This is enough for the real regional estimates stated at a fixed small radius. For a shifted or graded read the same argument requires its own complete collar and bounds.

A.9.0.3. The regional row form. Write $\sharp$ for the finer bare lattice and $\flat$ for the other lattice. For the switched kernel use scale sequences on a fixed region Y. At physical spacing ξ_n the direct rule keeps precisely the cubes of side $LM_1\xi_n$ in Y whose distance from Y^c is at least $\mathsf{r}_t \xi_n$. The tests defining a member at a common physical scale therefore agree cube by cube: $\Omega_{n+1}^\sharp(Y) = \Omega_n^\flat(Y)$ for $n \ge 1$. At the lower cut include the full double blocks and the neighboring stencils of the first rows retained. Use real pairs of small radius whose transfers are also in the small-radius class. Extend flat row data by zero off the common interior and outward rows. Let F be the fluctuation form before the row penalty, Q the row map, and set

$$\mathsf{G} = (\mathsf{F} + Q^{\mathsf{T}} \xi_c^{-2} Q)^{-1}, \qquad \mathsf{M} = Q \mathsf{G} Q^{\mathsf{T}}, \qquad \mathsf{H} = \mathsf{G} Q^{\mathsf{T}} \mathsf{M}^{-1}.$$

Here ξ_c denotes the local row scale in the diagonal penalty. Let $\mathbb{P}$ be the cell mean and q_1 the straightened linearized bond average for one step. For $f = \mathsf{H}^\flat B_1$ and $h = \mathsf{H}^\sharp B_2^\uparrow$, write $\lambda_i = (\mathsf{M}^{-1} - a_{\mathrm{row}}) B_i$ for the row multiplier, with $a_{\mathrm{row}} = \mathrm{diag}(\xi_c^{-2})$ in that run. Then $\mathsf{F}\mathsf{H} B_i = Q^{\mathsf{T}} \lambda_i$. Set

$\mathsf{d} = Q^\sharp \mathbb{P}^\mathsf{T} f - B_1^\uparrow$. Use $\mathbb{P}^\mathsf{T} f$ in the sharp equation and $q_1 h$ in the flat one. The row penalties agree on the shared rows. If $Q^\flat q_1 h = B_2$, the identity is

$$\begin{aligned}(B_1^\uparrow, (\mathsf{M}^\sharp)^{-1} B_2^\uparrow)_\sharp - (B_1, (\mathsf{M}^\flat)^{-1} B_2)_\flat \\ = \mathfrak{D}(f,h) - (\mathsf{d}, \lambda_2), \qquad \mathfrak{D}(f,a) = (\mathbb{P}^\mathsf{T} f, \mathsf{F}^\sharp a)_\sharp - (f, \mathsf{F}^\flat q_1 a)_\flat.\end{aligned} \tag{A.cvi}$$

On common rows $Q^\flat q_1 h = B_2$ by composition of the straightened averages. For an unmatched bottom row use the complete double block in the sharp bare rim. Its fine links are rows of the sharp determining set with zero prescribed values, so its average is zero. If a regional cut changes this relation, keep the extra term $(Q^\flat q_1 h - B_2, \lambda_1^\flat)$ on the right of eq. A.cvi. The prolongation defect pairs with the sharp row multiplier. Its size is $O(\eta \nabla f)$ on the common interior, where η is the coarser of the two bare meshes.

The consistency estimate for $\mathfrak{D}$ uses the fields cut off at the unmatched rim. In scaled local first-order norms $n_1(y), n_2(y)$, the second of which includes the Hölder quotient of the gradient, its bound is

$$C(1+u_0) \sum_y (\eta/\xi_y)^\beta n_1(y) n_2(y), \qquad 0 < \beta < 1.$$

The row currents must satisfy $\xi_c^3 |\mathsf{j}^\sharp(c)| \le 1$ and $\xi_c^3 |\mathsf{j}^\sharp(c) - \mathsf{j}^\flat(c)| \le u_0 (\eta/\xi_c)^\beta$ on the common rows. The term linear in the prescribed current uses the exact transpose pairing $(\mathbb{P}^\mathsf{T} f, J)_\sharp = (f, \mathbb{P} J)_\flat$. In the Landau projection use the straightened site and field transfers q_0, q_1, for which $q_1 D_U = D_W q_0$ on the complete carriers, with $W = \mathcal{M} U$. Write $\mathbb{P}^\mathrm{f}$ for the covariant face mean. On the full stencils the site Laplacian commutator is

$$(\Delta^{\circ\flat} q_0 - q_0 \Delta^{\circ\sharp}) \lambda = D_W^\mathsf{T} (q_1 - \mathbb{P}^\mathrm{f}) D_U \lambda - (q_0 D_U^\mathsf{T} - D_W^\mathsf{T} \mathbb{P}^\mathrm{f}) D_U \lambda.$$

This adds and subtracts $D_W^\mathsf{T} \mathbb{P}^\mathrm{f} D_U \lambda$ after intertwining. The last term is bounded by $C \eta^{-1} \varpi \sup |D_U \lambda|$ on its box, with ϖ the plaquette defect there. The unit-scale curvature bound gives $\varpi \le C\eta^2$. The first term is the divergence of the small face-mean difference. After summation by parts the divergence term is paired with a controlled first difference. The resulting site resolvent term is controlled in cell L^2 and is used in the final pairing. A second application of a propagator to that term would need a different bound.

In screening the rim, crossing one layer pays for the growth of the number of tops by at most L^4 and for the growth of the consistency weight by L^β. The resulting sum is

$$\sum_y (\eta/\xi_y)^\beta e^{-\delta_0 d(y,y')} \le C (\eta/\xi_{y'})^\beta.$$

First work with the forms at prescribed row current zero. For the bond force $J^\sharp$, put

$$a_* = (1 - \mathsf{H}^\sharp Q^\sharp) \mathsf{G}^\sharp J^\sharp, \qquad \mathsf{j}_*^\sharp = (\mathsf{H}^\sharp)^\mathsf{T} J^\sharp.$$

Then $Q^\sharp a_* = 0$, and multiplying out the inverse equations gives $\mathsf{F}^\sharp a_* = J^\sharp - Q^{\sharp\mathsf{T}} \mathsf{j}_*^\sharp$. The exact transpose pairing for $J^\flat = \mathbb{P} J^\sharp$ therefore gives

$$(B_1, \mathsf{j}_*^\flat) - (B_1^\uparrow, \mathsf{j}_*^\sharp) = \mathfrak{D}(f, a_*) + (\mathsf{d}, \mathsf{j}_*^\sharp) + (Q^\flat q_1 a_*, \lambda_1^\flat).$$

The last term vanishes on the matched constraint carriers by the same common-row and bare-rim calculation. The weak form estimate with the weighted fields f, a_* and the screening sum then give the induced current difference. Insert these currents in eq. A.cvi. At unit rows the entrywise rate is $CL^{-\beta k}$ when the step has k finer levels. The consistency estimate supplies the power of the mesh and the propagation bound screens the bottom layers.

For the switched kernel use the common regional alternating sums with switches held at one on the prescribed collar. The alternating sums have a real smallness bound with a cost exponential in the number of cubes and an independent analytic bound with exponential decay. To combine the two, write the complex pair on the line through its real and imaginary parts whose real arguments

give real pairs. This line and its transfer must be admitted on a disc of radius $R > 1 + \sqrt{2}$, with the target at i. The two-constants estimate uses the bound on the real diameter and the analytic bound on the disc. Retain a sufficiently small positive power of the real smallness so the remaining localization absorbs the growth of the alternating sums. With the summable spatial decay this gives a bound $CL^{-\alpha_K k}$ on the differences of the inverse covariance and the auxiliary Gaussian kernel, for some $\alpha_K > 0$.

A.9.0.4. The normalization on the regional towers. The tower for the determinant is defined by the recursive regional rule. To compare its operators, let $\mathbb{E}^\sharp, \mathbb{E}^\flat$ be the inverses of the two first-order systems with bilinear forms $\mathfrak{K}^\sharp$ and $\mathfrak{K}^\flat$. For right-hand sides $G^\sharp, G^\flat$, put $\Phi^\sharp = \mathbb{E}^{\sharp\mathsf{T}} G^\sharp$ and $\Psi^\flat = \mathbb{E}^\flat G^\flat$. If $\mathfrak{i}$ prolongs the flat variables and $\mathfrak{a}$ averages the sharp variables, the inverse identities give

$$(G^\sharp, \mathfrak{i}\Psi^\flat)_\sharp - (\mathfrak{a}\Phi^\sharp, G^\flat)_\flat = \mathfrak{K}^\sharp(\Phi^\sharp, \mathfrak{i}\Psi^\flat) - \mathfrak{K}^\flat(\mathfrak{a}\Phi^\sharp, \Psi^\flat).$$

Use multilinear prolongation $\mathfrak{i}_0$ and straightened averaging $\mathfrak{a}_0$. Compare real pairs of small radius with transfers of small radius. The flat test field Ψ vanishes below a fixed margin above the first common layer. Require $\xi_c^3 |\mathfrak{j}^\sharp(c)| \le 1$ and $\xi_c^3 |\mathfrak{j}^\flat(c)| \le 1$ on every row of the two systems, respectively.

For a common bond row the chain rule gives $Q^\sharp = Q^\flat q_1$, and for a common top it gives $Q'^\sharp = Q'^\flat q_0$. This cancels the terms with the averaged fine field. For the reversed pairing with a prolonged coarse field B, the bond-row error is $Q^\flat(q_1\mathfrak{i}_0 - 1)B$. On a complete carrier the flat means reproduce constants. The local comb and the first difference of B bound the L^2 norm of the expression before the row map by $C\eta\|\nabla B\|_2 + C_L\varpi\|B\|_2$. The raw bond-row coefficients give $|\xi_c Q^\flat(c)F| \le C\xi_c^{-1}\|F\|_2$ on its full cells. This gives the mesh factor in the pairing with the raw multiplier. The reversed top pairing uses the estimate for $q_0\mathfrak{i}_0 - 1$ in its site scale.

For the top mass use $a'(y) = \kappa_n \xi_y^{-2}$, where $\kappa_0 = \kappa_1 = 1$ and $\kappa_{n+1}^{-1} = 1 + L^{-2}\kappa_n^{-1}$ for $n \ge 1$. The insertion of a fine level changes n to $n+1$ on a common top, so

$$|a'^\sharp(y) - a'^\flat(y)| \le \xi_y^{-2}(\eta/\xi_y)^2.$$

The local site row bound makes this a weak error of the same order in the scaled cell norms.

For the site slot p the site equation gives a first-difference bound. With $g_0, D^\mathsf{T} x$ the two parts of its source, test on a complete collar against $\chi^2 p$, where $\chi = 1$ on the cell and $|D\chi| \le C/\xi_y$. Covariant summation by parts and the local bound on Q' give

$$\xi_y \|Dp\|_{2,y} \le C(\|p\|_2 + \xi_y^2 \|g_0 - Q'^\mathsf{T} r\|_2 + \xi_y \|x\|_2),$$

with the norms on the right taken on the collar. The current in x is used in norm. The row-current term in x is $-\Delta^{(2)}[\mathfrak{j}](A - Dv)$. Its one-cell bound uses the raw amplitude $\xi_c^3|\mathfrak{j}(c)|$, with no difference of the multiplier. For the Laplacian pairing with p, apply the difference of the two gradient means to the other slot. In flat coordinates the derivative of the multilinear prolongation is the prolongation of each directional derivative, constant along its coarse segment and multilinear transversely. This map and its volume transpose reproduce constants on complete incidences. Together with $D_W q_0 = q_1 D_U$ this moves the difference of means onto the gradient of the other slot, costing one mesh times its extra L^2 difference. In the local comb the covariant corrections cost $C_L\varpi/\eta$ on site values and $C_L\varpi$ on gradient means. Thus p requires only its first-difference bound. The site equations for v, w, q have L^2 sources controlled by the cell norms and first differences of the bond field. In their admitted regular gauge, localizing to a cell and using the flat identity $\sum_{\mu,\nu} \|\partial_\mu \partial_\nu f\|_2^2 = \|\Delta_0 f\|_2^2$ gives their interior second-difference bounds. The cutoff terms and the difference from the covariant site Laplacian cost a bounded first difference and value in cell units. In the scaled cell norms of the first-order systems this gives

$$|\mathfrak{K}^\sharp[\mathfrak{j}^\sharp](\Phi, \mathfrak{i}_0\Psi) - \mathfrak{K}^\flat[\mathfrak{j}^\flat](\mathfrak{a}_0\Phi, \Psi)| \le C \sum_y \left(\frac{\eta}{\xi_y} + \mathfrak{u}(y)\right) [\![\Phi]\!]^\vee_{(y)} [\![\Psi]\!]_{(y)},$$

where $\mathsf{u}(y) = \max_{c \text{ at } y} \xi_c^3 |\mathsf{j}^\sharp(c) - \mathsf{j}^\flat(c)|$ on the common rows. The unmatched layers are screened by the cutoff and the exponential propagation bound. First use the estimate with zero row currents. The screening sum with exponent one then gives $\xi_c^3 |\mathsf{j}_*^\sharp(c) - \mathsf{j}_*^\flat(c)| \le Ca\,\eta/\xi_c$ on common rows, where a is the pair radius and $\mathsf{j}_* = \mathsf{H}^\mathsf{T} J$. Using these induced currents in the consistency estimate gives the real entrywise rate CL^{-k} on the common interior bonds. The exponential kernel estimate on the larger complex domain and the two-constants estimate give a Schur norm bound $CL^{-k/2}$ on the smaller domain.

A.9.0.5. Common layers of a recursive tower. Suppose the two calls use the same physical first member Y, with aligned grids, and the same recursive rule. At the next grid of side s, assume that Y is a union of its cells. If the previous member was obtained by removing k layers of side s/L from Y, a new cell at distance ts from Y^c has clearance $ts - ks/L$ when it is contained in that member. The distance index t is an integer. This follows by thickening the union of excluded closed cells in the sup metric, so the same rule applies at faces and corners.

The rule that requires clearance at least s therefore removes $1 + \lceil k/L \rceil$ new layers. Starting from $k = 0$ its counts are $1, 2, 2, \ldots$. Since the finer call has one more erosion before the first common scale, the two members agree at the second common erosion. This is the rule of the minimal determining tower with M_1-cubes.

For the rule with $\mathfrak{M}$-cubes and strict clearance $> R_1 s/L$, the new count is $1 + \lfloor (R_1 + k)/L \rfloor$. If $R_1 = L^a$, the count after n erosions for $1 \le n \le a$ is $1 + \sum_{b=1}^{n} L^{a-b}$. The count at $n = a$ is a fixed point of the recursion. Hence the members agree after a common erosions for this rule. The full torus and empty regions agree directly.

These calculations require alignment of Y through the indicated first common grids. Once a member agrees, the same recursive rule at the same physical grids gives agreement thereafter. The common determining rows and tops are obtained from the coincident members above this fixed bottom margin. A row at level n uses its predecessor member for the complete double block and its successor member for the row type. In the M_1 tower take $n \ge 3$ before allowing for the fixed stencil margin. Use the same rule one level above synchronization in the strict-clearance tower. Cut below the complete carriers and include the full stencils in the unit-error halo before the form comparison.

The fine tower is one erosion ahead at the first common grid, so its member there is contained in Y. Monotonicity preserves this inclusion at each common physical scale. Consequently the local fine spacing is at most the local coarse spacing. The one-run kernel bounds can use the coarser weighted distance, with a fixed reduction in the exponential rate. Indeed, away from layer boundaries the length weights are ordered pointwise. On a boundary, use the smaller of the two adjacent spacings as the weight convention, which is equivalent up to a factor of two to the length counting both adjacent layers: a boundary segment of length t contributes $t \max(\xi_1^{-1}, \xi_2^{-1})$ under the first convention and $t(\xi_1^{-1} + \xi_2^{-1})$ under the second. A coarse cell in the omitted layers contains only a bounded number of finer cells, since these layers span a fixed number of meshes. Thus the same shell count applies to the one-run terms in the unit-error halo.

For the inner window $\square_0 = \square^{\sim 5}$, the raw determining tower uses the strict-clearance rule. In units of the output level the original π_k-cells have side M/L. The inequalities $M \ge L^4 M_1$ and $M_1 \ge R_1 \mathfrak{M}$ give alignment throughout this tower when the two calls use the same physical window. The inner and removed-window reads lie in $\square^{\sim 2}$, and $\square^{\sim 3}$ lies in the protected interior of $\square_0$. The native pillar bound puts its $4L^n$-halo in the free level-n pillars in bare units. This includes the full plaquette and bond-current stencils for the inner reads. The inverse solves use the whole tower, with its unmatched lower data screened as above. If the depth is shorter than the fixed margin needed for synchronization, the individual bounds suffice with a larger constant at those bounded depths.

For the inner transfer at old level $1 \le j \le k$, use $X^\dagger$, the enlargement of X by one layer of π_{j-1}, as the read collar. In level-k units its width is $ML^{-(k-j+1)}$, at most M/L and at least ML^{-k}. It contains the one-step stencils and keeps $X^\dagger$ strictly inside $\square^{\sim 3}$ for $X \subset \square^{\sim 2}$. This places the transfer in the ordinary field and current bounds on the protected collar. For the inner input of Prop. 3.135, the same proof gives the old analytic core on the collar itself. Its complete bare stencils reach at most two bare steps from $X^\dagger$ and remain in $\square^{\sim 3}$, where the flat representative has the bounds eq. 3.mcx. For a tower address (p, Z) of $X^\dagger$, $Z \subset \square^{\sim 3} \subset \square_{(j)}$, so the exact change of gauge eq. 3.mcxvi applies to the restriction of the native response to Z. Its conjugation norm is less than $6/5$. For a truncation at depth $n' \le p$, the curvature and current bounds after this change are

$$\frac{6}{5}\frac{a_0}{8}L^{-2j}, \qquad \frac{6}{5}\frac{a_0}{16}L^{-3(j-n')}.$$

These are below $(a_0/4)L^{-(j-p)}L^{-2p}$ and $(a_0/4)L^{-(j-p)}L^{2(n'-p)}$, respectively, since $n' \le p \le j$. This gives both hereditary rules for the triple on $X^\dagger$, with core radii $(a_0/4, a_1/2, a_0/4)$, for the admitted native substitution. The pair-orbit evaluation is still on the activity domain X. Use the distance with the thinner read collar and a single sharp-to-flat radius loss for the one bare transfer. The frame and shifted inputs in the paired comparison must still be identified on that collar.

On a common row the plain relative datum has an exact transfer. First transfer the shifted sharp field and its sharp base separately. Their relative logarithm on a flat bond is $\mathsf{Q}_1^\sharp(\mathbf{U}; \mathsf{a})$. On the common small logarithm branches the semigroup identity gives

$$\mathsf{Q}_{q+1}^\sharp(\mathbf{U}; \mathsf{a}) = \mathsf{Q}_q^\flat(\mathcal{M}\mathbf{U}; \mathsf{Q}_1^\sharp(\mathbf{U}; \mathsf{a})).$$

Compare this input to the actual flat shifted input in the common frame on the complete carrier. At fixed base the row estimate eq. 3.mcxv bounds the datum difference by CL^q times the supremum difference of the dimensionless input logs. For the conjugation of the input potential, only the frame value at the initial site enters this bound. The change of base and the node factors of the straightened chart must also be matched. The baseline inner datum uses the output radius $a_{0,k+1}$. Its ratio to the old radius is at most $C(n+1)^{1/2}$, without a scale gain for that baseline.

The Gaussian compression uses the calculation for its specified tower. For any other response at a vertex, the same physical subregion and cutoff must be used at both depths, and the sequence of its core must be identified before applying the tower calculation. If the cutoff ζ is applied after the solve, the extra term in a common averaging row is

$$[q_1, \zeta]A(c) = \sum_b q_1(c, b)(\zeta(b_-) - \zeta(c_-))A(b).$$

Here the cutoff is evaluated at the initial endpoints of the bonds. This term vanishes on a plateau containing the whole carrier. For a smooth cutoff it costs at most $C\eta\|\nabla\zeta\|_\infty \sup_{\mathrm{supp}\, q_1(c,\cdot)} |A|$ for the complete one-step stencil of mesh η, by its bounded row sum. One must check the cutoff on the actual vertex read, including any derivative of this commutator needed in its transfer estimate. A characteristic cutoff on a straddling carrier gives only the individual row bound.

For the raw switched response a protected case is available. The field on an active wall component is its own restricted solve, with support in $\mathbb{Y}^1$, and its reduced current has support in $\mathbb{Y}^2$. The margins of $\mathbb{Y}^3$ keep the full row, plaquette and first-difference stencils that meet either support inside that component. Its indicator therefore adds no commutator on these native reads, by Prop. 3.175. In the specified raw sequence, if every partition cube meeting a cell hull has parameter one, the field, projected field and reduced current equal the all-one fields on that sequence. Indeed, every term of their operator series uses only these cubes, and the local structures agree throughout the sequence. This is the all-one identification in Prop. 3.175. When it holds on both actual regional sequences, use their undecoupled comparison with their own determining data.

For other switch assignments, the raw operator at a vertex keeps the walk terms of the ambient sequence supported on the active cubes. The Möbius sum defining a named vertex selects the

undecoupled response of its designated region. To identify the two, one must first identify their linear lifts on the specified row data. At the all-one vertex $QH_1 = 1$ on the rows. For the proper mask the linearized constraint is $QH_1(s)B'$, by Prop. 3.175. The support property alone therefore supplies no lift identity for a different designated regional solve.

For a named vertex using the direct region rule one can test the required top inset on its full read carrier K. If the required inset has width $r_{\mathrm{in}}\xi_{\mathrm{top}}$, it suffices that

$$\mathrm{dist}_\infty(K, Y^c) \geq (\mathsf{r}_{\mathrm{t}} + LM_1 + r_{\mathrm{in}})\xi_{\mathrm{top}}.$$

Every cube of the top grid meeting this neighborhood of K then passes the direct clearance condition. For an outer vertex the region and the reads used in this check must be those of its own supply.

Write A_Y^- for the interior compression of the tower operator and $T_Y = 1 - c_N^{-1}\mathsf{A}_Y^-$, with $\|T_Y\| \leq q < 1$ on the matched domain. At a free bond b the root of the logarithm is

$$f_b(Y) = 3\log c_N - \sum_{m\geq 1} m^{-1}\,\mathrm{tr}(T_Y^m)_{bb}.$$

Use the same alternating sum over regions $Y \subset X$ containing the determining cubes of b. Subtract the value of each sum at the trivial pair, weight by the tents of the base points, and multiply by $-1/2$. The alternating sum and partition of unity restore the normalized log determinant. The local pivot terms agree on the common retained rows because their coefficients are functions of the same retained field.

The difference of the roots is bounded by the operator norm difference times $C\sum_{m\geq 1} q^{m-1}$, since telescoping T_Y^m produces m terms and the logarithm divides by m. The sum over regions thus costs at most $C2^{|X|}L^{-k/2}$, before localization. Independently, a chain missing a cube cancels by switching that cube in the alternating sum. Short chains meeting all cubes pay the distance of the carrier. Long chains are summed using q^m. The split at a small multiple of the carrier length in units of the regional mesh retains exponential decay of the localized sum. Interpolating the two summed bounds with a small power of the norm difference absorbs the factor $2^{|X|}$ and retains the required exponential decay together with a positive power of L^{-k}. This also applies to the vacuum subtraction.

A.9.0.6. The regional Wilson increment. For each action carrier X and each vertex subregion Z containing X, let H_Z be the response on the core of Z times the prescribed cutoff. The localized interaction uses the action increment $\mathrm{Br}_{X,Z} = a_X(e^{i\eta H_Z}U) - a_X(U)$. For $Y \supset X$ sum these increments over $Z = X \cup S$, $S \subset Y \setminus X$, with sign $(-1)^{|Y\setminus X|-|S|}$, and take the third-order Taylor remainder in the fluctuation. Inversion at the full vertex restores the Wilson increment with the same subtraction. The one-run bound with the decoupling parameters in their larger polydisc provides decay in the number of cubes of $Y \setminus X$.

For a matched real vertex write U' for the perturbed fine field, W' for the perturbed coarse field, and $W = \mathcal{M}U$. Put $D = A^\flat \circ \mathcal{M} - A^\sharp$. Subtract the increments in the two actions to get

$$\begin{aligned}(A^\sharp(U') - A^\sharp(U)) - (A^\flat(W') - A^\flat(W))\\ = -(D(U') - D(U)) + (A^\flat(\mathcal{M}U') - A^\flat(W')).\end{aligned} \tag{A.cvii}$$

The local version includes the plaquette shell of the carrier. The second term compares configurations on the coarse lattice. Use the response estimate on the matched real cores and the continuity of the action there. The first term uses consistency of the action under a regular perturbation of a possibly rough background.

For this consistency, the exterior derivative commutes with the linearized averaging of bond fields. In unscaled plaquette logarithms the induced average on two-forms is L^2 times a probability average of fine plaquettes. The difference of the flat quadratic actions is therefore $-L^4/2$ times the sum of the corresponding flux variances, plus the shell term. If the perturbation changes the curvature from F to $F + \gamma$, the change of variance on a block is $2\,\mathrm{Cov}(F, \gamma) + \mathrm{Var}(\gamma)$, where the

covariance uses the Lie algebra inner product. Subtract the mean of γ and use Cauchy–Schwarz. The rough F enters through its size, and the oscillation of γ supplies a positive power of the mesh from the Hölder bound on the covariant gradient of the perturbation. In the localized consistency estimate the shell and the nonlinear errors from multiplication of plaquettes are kept with this same carrier. This is an estimate of the variation $D(U') - D(U)$, since the variance of F itself need not be small.

The first variation of this quadratic defect in the chosen cell frame is

$$-L^4 \sum_P \mathrm{Cov}_P(F, \delta_A F) + d(\mathrm{shell})[A].$$

Subtract the block mean of $\delta_A F$ in the covariance. The Hölder bound on the covariant gradient of the test variation then gives the mesh gain while the rough curvature is used only in norm. Keep the differentiated nonlinear plaquette errors with the same full carrier.

For the nonlinear force itself, keep the transports in the plaquette chart fixed and separate the explicit background holonomy. Use h for the bare spacing in one run, and write $\Pi^0 = c_0\mathbf{1} + iH_0$ for the background plaquette. The perturbed plaquette has trace $\mathrm{tr}(\Pi^0 e^{i\theta})$, where $\theta = d\gamma + n$ and the four γ's are the increments transported to the plaquette base. Put $\sigma = \max_i |\gamma_i|$ and $w = |\Pi^0 - 1|$. The chart bounds n by $C\sigma^2$. In the differentiated expansion the terms of order zero and one are the background force and the Hessian. The remaining explicit dependence on Π^0 is in the two oriented remainders of the plaquette force. Replacing (c_0, H_0) by $(1, 0)$ in those remainders changes them by $C\sigma^2 w$. Indeed $H - H_0$ and $H' - H_0$ change by at most $Cw|\theta|$ and each is multiplied by an $O(\sigma)$ factor. The other changed terms have a factor n, a second-order exponential remainder, or $(\cos\sqrt{\langle\theta,\theta\rangle} - 1)H_0$. Each has the same $C\sigma^2 w$ bound, since $|c_0 - 1| = w^2/2$ and $|H_0| \le w$. The transport of the output to its bond base is kept in both terms. At the local scale ξ use $w \le a_0(h/\xi)^2$, $\sigma \le h \sup|A|$, and the factor h^{-3} in the bond force. The resulting scaled force error is at most

$$Ca_0(h/\xi)\,(\xi \sup|A|)^2.$$

Thus the two cutoffs can bound their explicit rough holonomy remainders separately in the weak pairing.

To compare the remaining force, put $N(A)_{\mu\nu} = i[A_\mu, A_\nu]$. The second-order term of $\log(e^{-i\gamma_3}e^{-i\gamma_4}e^{i\gamma_1}e^{i\gamma_2})/i$ at $\gamma_3 = \gamma_1,\ \gamma_2 = \gamma_4$ is $i[\gamma_1, \gamma_4]$. On the full stencil the opposite increments differ by $h^2\nabla_U A$. Consequently

$$h^{-2}\theta = D_U A + N(A) + O\big((h/\xi)\xi^{-2}(a^2 + a^3)\big), \qquad a = \xi \sup|A| + \xi^2 \sup|\nabla_U A|.$$

Differentiating the same finite expansion in a test $\widehat{A}$ replaces $a^2 + a^3$ in the error by $C(a + a^2)b$, where $b = \xi \sup|\widehat{A}| + \xi^2 \sup|\nabla_U \widehat{A}|$. For the plaquette with the explicit holonomy removed the action is $\frac{1}{2}\langle\theta,\theta\rangle + O(|\theta|^4)$. Subtract the derivative of its quadratic action. Dividing by the cell measure h^4, the leading nonlinear weak density is

$$\langle N(A), D_U\widehat{A}\rangle + \langle D_U A + N(A), N'_A\widehat{A}\rangle.$$

The omitted density is bounded by $C(h/\xi)\xi^{-4}a^2 b$ on a small first-order ball.

On complete common cells compare this density at $(U, A, \widehat{A})$ and $(W, q_1 A, q_1\widehat{A})$, with $W = \mathcal{M}U$. Now take h and $\eta = Lh$ to be the two meshes. For a coarse $\mu\nu$-plaquette the flat per-area mean Q_2^0 assigns weight L^{-6} to the L^2 fine plaquettes of each translate based at a site in its initial block of L^4 sites. Stokes on each translated square gives $D_\eta Q_1^0 = Q_2^0 D_h$. Each fine plaquette occurs L^2 times over the incident coarse plaquettes. The lattice volume ratio is L^4, so the volume transpose of Q_2^0 reproduces constants on complete incidences.

Use the transports of a local comb to carry the fine plaquette values to the coarse base and call this mean Q_2. In a common comb on the enlarged stencil the one-step row coefficients and these transports differ from their flat values by $C_L\varpi$, with ϖ the maximum fine plaquette defect there.

The coarse links in this frame are also within $C_L\varpi$ of one. Taking the difference of the two local operators therefore gives

$$|D_W q_1 B - Q_2 D_U B| \le C_L \eta^{-1} \varpi \sup |B|$$

on that stencil. The estimate compares the row coefficients at each of the neighboring rows using their sup bound. It requires no difference estimate on the plaquette defect.

Use this identity for $B = A, \widehat{A}$. In the first pairing move Q_2 onto the coarse $N(q_1 A)$ by the volume transpose. In the pairing with $D_U A$, move it onto $N'_{q_1 A}(q_1 \widehat{A})$. Constant reproduction and the first-difference bounds compare these adjoint means with the corresponding fine products at cost $C\eta$ times a product difference and $C_L\varpi$ times its size. The purely algebraic pairing uses the same volume quadrature. Since $\varpi \le C(\eta/\xi)^2$, all three errors are bounded by $C(\eta/\xi)\xi^{-4}a^2 b$ in density, hence by $C(\eta/\xi)a^2 b$ per cell. Only first differences of the test are used. If a common smooth cutoff multiplies the product, its difference enters in that same quadrature bound. At a cubical face with an incomplete incidence strip of width $O(\eta)$, the density bound $C\xi^{-4}a^2 b$ and the strip volume $O(\eta\xi^3)$ give the same cell estimate. Keep any wider transition or unmatched row in its native cutoff estimate. This comparison is for the physical Wilson remainder on its admitted common carriers.

For the scalar weight carried by a sourced step, let c_h, c_η be the two plaquette weights on an admitted common carrier. Use S_h for the fine Wilson increment density, with its background subtracted, or its first variation. Include the lattice volumes in the sums. On complete incidences one has the exact identity

$$\sum_p h^4 c_h(p) S_h(p) = \sum_P \eta^4 c_\eta(P)(Q_2^0 S_h)(P) + \sum_p h^4 \big[c_h(p) - ((Q_2^0)^* c_\eta)(p)\big] S_h(p).$$

The transpose is taken for these lattice volumes. Extend the coarse weight to the common physical stencil, using its tent formula for the actual source. Every coarse barycentre contributing to the transpose at p is within $C\eta$ of the barycentre of p. Constant reproduction therefore bounds the coefficient in brackets by the paired weight difference at that barycentre plus $C\eta\,\mathrm{Lip}(c_\eta)$. Compare the first sum to the coarse action increment with the same weight c_η. In the variance calculation each block difference is multiplied by $c_\eta(P)$. The oscillation of the curvature increment still gives the gain against the rough background. In the curl pairings one includes c_η in the coarse test product before transposition. Its first difference adds the same $C\eta\,\mathrm{Lip}(c_\eta)$ error, times the bound on the corresponding pairing density.

For a plaquette filed to a single cube, incomplete incidences lie in the strip along its faces. A plaquette filed to two or four cubes already lies in that strip, since it crosses a face. On a fixed admitted carrier its volume is $O_M(\eta\xi^3)$. The bound for the normalized increment or variation density then gives the shell cost $O_M(\eta/\xi)$ in cell units, multiplied by the size of the weight. The physical variation of the weight member is the sum of the plaquette variations with their individual $c(p)$. It is used inside this weak pairing. The current argument of the response remains independent.

In particular let U_* be an actual fine critical field on the uncut core, $W_* = \mathcal{M}U_*$, and $P_* = d\mathcal{M}$ in compatible tangent coordinates. Take A with its complete action and row stencils where the vertex cutoff equals one and away from all unmatched row readouts. The physical row differentials then obey $K^\sharp A = K^\flat P_* A$. Write J^a for the inner-action gradient at cutoff a, which is the Wilson current on full stencils. With the fine multiplier ω_*, stationarity gives

$$\big\langle J^\flat(W_*) - K^{\flat\mathsf{T}}\omega_*, P_* A\big\rangle_\flat = dD_{U_*}[A].$$

If the carrier reaches an unmatched row or the action shell, keep its term in the comparison. This identity tests the physical coarse residual on transferred variations. The transition shell of a cutoff response retains its own force. With a general fine field the fine residual $\langle J^\sharp - K^{\sharp\mathsf{T}}\omega, A\rangle_\sharp$ also enters. The full equation with an independent current still requires the weak source estimate.

Use the common real-to-complex interpolation disc for the vertex bounds and then three circles in the fluctuation amplitude. A sufficiently small positive power of the real bound retains the required carrier decay after the alternating sum. On the resulting admitted disc the cubic remainder is bounded in powers of the unscaled fluctuation $\mathsf{t}^{-1/2}B$. Its contribution to the interaction is multiplied by t, so the cubic bound is $C\mathsf{t}^{-1/2}\|B\|^3$ times the mesh gain and the carrier decay. On $\mathsf{t}^{-1/2}\|B\| \le C\delta(\mathsf{t})$ this is $C\delta(\mathsf{t})\|B\|^2$ times those same factors. Compare the two vertex constructions with the same reference weight t here. A difference of actual scalar action weights is paid on the coupling disc in the step comparison.

A.9.0.7. Transport of earlier activities. Use the same regional rule to construct the response and its reduced current at the two spacings. The row data are related by composition of the relative averages. At vertices of the decoupling parameters the response is the undecoupled response of the selected subregion, multiplied by its cutoff. Thus the comparison uses responses with specified determining sets. Insert the averaged fine response into the coarser response equation. Its error in the field equation is measured in a weak pairing with one controlled difference on the test field. The nonlinear constraint on a common row composes with the actual one-step transfer. Write η for the coarser bare spacing, $h = \eta/L$, and $\breve{q}(A) = \eta^{-1}\mathcal{Q}^1(U^\sharp, hA)$ for that transfer in the straightened chart. If the comparison uses its linearization $q_1 A$, the Taylor bound on the full carrier P_c gives

$$|\breve{q}(A)(c) - q_1 A(c)| \le C_L(h^2/\eta) \sup_{\mathsf{P}_c} |A|^2.$$

Thus in the row scale ξ_y the discrepancy is at most $C(\eta/\xi_y)(\xi_y \sup_{\mathsf{P}_c}|A|)^2$. Use either the exact transfer or this small value defect in the constraint comparison. The same row estimate in a variation $\widehat{A}$ gives

$$|(D\breve{q}(A) - q_1)\widehat{A}(c)| \le C_L(h^2/\eta) \sup_{\mathsf{P}_c} |A| \sup_{\mathsf{P}_c} |\widehat{A}|.$$

Indeed the derivative of the dimensionless one-step remainder is bounded by $C_L h \sup_{\mathsf{P}_c}|A|$, and the input and output changes of units give h/η. On common rows the actual constraint derivatives obey $K^\sharp(A) = K^\flat(\breve{q}(A))D\breve{q}(A)$. When a weak equation uses $q_1\widehat{A}$ in place of the exact tangent transfer, the constraint-multiplier error is

$$\langle \omega,\, K^\flat(\breve{q}(A))(D\breve{q}(A) - q_1)\widehat{A}\rangle.$$

This error uses the row estimate on the full carriers and the actual dual weights of the multiplier. The Wilson and prescribed-current terms of the field equation require their own comparisons. The noncommon rows are in the lowest layers. For the weighted contraction put $e(y) = \eta/\xi_y$ above the fixed bottom cut and $e(y) = 1$ below it. Define $\mathsf{E}(x) = \sup_y e^{-\mu' d(x,y)} e(y)$, where d is the regional distance and μ' is smaller than the propagation rate. The triangle inequality gives $\mathsf{E}(y)/\mathsf{E}(x) \le e^{\mu' d(x,y)}$. Thus a kernel decaying faster than this weight varies retains a summable bound in the norm with weight $\mathsf{E}^{-\gamma}$. A layer r spacings below the top can enlarge the bare-to-local mesh ratio by at most CL^r, and each intervening layer has the fixed distance floor. Choosing the floor so the exponential decay absorbs L^r bounds E at the top by CL^{-m} in a tower with m levels. The reduced current is estimated from the row multiplier, without applying two differences to a response known only in a first-order norm. The comparison of the reduced current is used well inside the top layer.

Here is the passage from the cell estimate to the field norm used at a vertex. In units of one common cell let f be the difference of the transferred fine field and the coarse field, written on the common mesh of spacing b. Let E bound its lattice L^1 norm on the cell and a fixed collar. An L^2 estimate there gives this input by Cauchy–Schwarz. Use the first-difference bound and the common β_H-Hölder bound for the gradients. Both bounds are on the same chart and cover the full neighboring stencils. At an interior site x, average f on a cube of side $r \ge 2b$ near x. The number of sites is comparable to $(r/b)^4$. Thus the value at x has norm at most $C(Er^{-4} + r)$. For a gradient, average the difference of f between this cube and its translate by a mesh multiple of

size comparable to r in the gradient direction, divided by that translation length. Telescoping the difference and using the Hölder bound on every intervening stencil compares this average to $\nabla f(x)$ with error Cr^{β_H}. The two cube sums are bounded by CEr^{-4}, so

$$|\nabla f(x)| \le C(Er^{-5} + r^{\beta_H}).$$

Take r comparable to $\max\{b, E^{1/(5+\beta_H)}\}$ for the gradient, and to $\max\{b, E^{1/5}\}$ for the value. When either choice exceeds the collar radius the original bounded field norm gives the result with a larger constant. The resulting bounds are $C(E^{1/5} + b)$ and $C(E^{\beta_H/(5+\beta_H)} + b^{\beta_H})$, respectively. The same bounds hold at the edge of an aligned cell when the cell supplies the weak input and all the one-sided stencils. Choose the averaging cube inward in every coordinate and translate it in the inward sign of the derivative direction. Telescoping then uses only the given Hölder bound in the cell. For a backward difference the base of the stencil is one mesh from the forward base, which is covered by the same error. The small radii can be chosen below a fixed fraction of the cell side, and larger radii are covered by the original first-order bound. Consequently a weak estimate with a power of the local mesh ratio gives a positive power in the first-order field norm on the common interior. At the unmatched bottom use its unit bound in the screened weight. The multiplier comparison used for the current is separate from this interpolation.

To obtain the weak input for that interpolation, write $\lambda_y = \eta/\xi_y$ and let $f^\sharp, f^\flat$ be the actual bond sources in the direct regional comparison. In the scaled cell norms the source condition is

$$\xi_y \|f^\flat\|_{L^2(\Delta_y)} \le N(y), \qquad |(f^\sharp, \widehat{A})_\sharp - (f^\flat, q_1 \widehat{A})_\flat| \le \sum_y \lambda_y^\gamma N(y) \|\widehat{A}\|_y.$$

Inject $f^\sharp, f^\flat$ into the bond slots. Take $\Phi^\sharp = \mathbb{E}^\sharp f^\sharp$ and $\Psi^\flat = (\mathbb{E}^\flat)^{\mathsf{T}} g$, where g tests the wanted flat output in its own slot. Write the bilinear form with its operator acting in the second argument. Then the inverse identity is

$$\begin{aligned}(g, \mathfrak{a}_0 \Phi^\sharp - \mathbb{E}^\flat f^\flat)_\flat = {} & \mathfrak{K}^\flat(\Psi^\flat, \mathfrak{a}_0 \Phi^\sharp) - \mathfrak{K}^\sharp(\mathfrak{i}_0 \Psi^\flat, \Phi^\sharp) \\ & + (\mathfrak{i}_0 \Psi^\flat, f^\sharp)_\sharp - (\Psi^\flat, f^\flat)_\flat.\end{aligned}$$

To use the form estimate in the first line, write S for the constant self-transpose map on the first-order slots given by

$$S(A; v, w, \sigma; p, q, r) = (A; p + w, q + v, r - \sigma; v, w, \sigma).$$

For the full system add the multiplier slot $\omega \mapsto -\omega$. The reduced system obeys $\mathbb{K}^{\mathsf{T}} S = S\mathbb{K}$. In the full system the extra pairings in $(S\Psi', \mathbb{K}\Psi)$ are $-(QA', \omega) - (\omega', QA)$, so the same identity holds. The common transfer and cut use the same site map on v, w, p, q and the same top map on σ, r. They commute with this slot change. Thus $\mathfrak{K}(\Psi, \Phi) = \mathfrak{K}(S\Phi, S^{-1}\Psi)$ puts the first line in the order of the form estimate, with the same support at the cut. The source condition in the second line is tested at the prolonged bond component $\mathfrak{i}_0 \Psi^\flat$. It leaves the quadrature error pairing $f^\flat$ with the difference of $q_1 \mathfrak{i}_0 \Psi^\flat$ and $\Psi^\flat$ on their bond slots. On complete common cells the flat averaging and prolongation reproduce constants, so the difference of the means costs one mesh times one difference. The common comb frame adds a term proportional to the plaquette radius. Use the local L^2 bound on the prolonged test and its differences in this pairing. At the fixed-margin cut use the weighted kernel decay for the lower unmatched contributions. This gives the cell-mean comparison for the fields and the separate row comparison for the multipliers under the stated source condition.

In the middle part of the projected inverse $G_1 D R D^{\mathsf{T}} G_1$, use the first-order system with the multiplier and its equation removed. Its inverse contains the block G_1. Put $u^a = R^a D^{a\mathsf{T}} G_1^a f^a$ for $a = \sharp, \flat$. For the actual force with its weighted supremum bound, the first-difference Hölder bound for $G_1 f$ and the smoothing part of R give a scaled site Hölder bound of some exponent $\beta > 0$ for u. On a unit cell the site mean error E then gives a supremum error at most $C(Er^{-4} + r^\beta)$ by

averaging on a cube of side r around the test site. Use r of order $E^{1/(4+\beta)}$ above the mesh floor. This retains a positive mesh power in the site supremum norm. Split the second solve as

$$q_1 G_1^\sharp D_U u^\sharp - G_1^\flat D_W u^\flat = q_1 G_1^\sharp D_U u^\sharp - G_1^\flat D_W q_0 u^\sharp + G_1^\flat D_W (q_0 u^\sharp - u^\flat).$$

The last term uses the single-run differentiated-source kernel on the improved site norm. The first line uses the weak comparison again, now with the actual gradient source. On complete common stencils $D_W q_0 u^\sharp = q_1 D_U u^\sharp$. For such sources the weak source defect is the pairing of $D_U u^\sharp$ with $(1 - q_1^\mathsf{T} q_1)\widehat{A}$.

At the flat pair the per-length bond average has row mass one. In the weighted transpose the mass is also one, since a fine link of a given direction occurs in L paths counted across the incident coarse rows and the measure ratio is L^4. Both means only read a fixed number of bare blocks. Thus $(1-q_1^\mathsf{T} q_1)\widehat{A}$ costs $C\eta \sup|\nabla \widehat{A}|$. In a common comb on the full union of incident double blocks the straightened row coefficients differ from the flat ones by $C_L \varpi$, where ϖ is the plaquette bound there. This adds at most $C_L \varpi \sup|\widehat{A}|$. Since $\varpi \le C(\eta/\xi_y)^2$, the cell pairing gains a mesh ratio using the scaled L^2 bound of the gradient source.

For that bound, use the bond-force entry of the reduced first-order inverse on the admitted undecoupled regional frame, including its independent current. For a force $f_{y'}$ supported at one source cell, put $A = G_1 f_{y'}$. The cell estimate for this entry is

$$\xi_y^{-1}\|A\|_{L^2(\Delta_y)} + \|\nabla A\|_{L^2(\Delta_y)} + \xi_y \|\nabla\nabla A\|_{L^2(\Delta_y)} \le C e^{-\delta d(y,y')} \xi_{y'} \|f_{y'}\|_{L^2(\Delta_{y'})}.$$

The stencils in this estimate carry the covariant differences of that frame. In $Du = DD^\mathsf{T} A - D(1-R)D^\mathsf{T} A$, the first term on full interior stencils is bounded by the covariant second differences on the neighboring cells, with the curvature times A if their order is exchanged. For the second term, with η_a the bare mesh in the run, the kernel from z to y is bounded by $C(\eta_a/\xi_z)^4 \xi_y^{-2} e^{-\delta d(y,z)}$. Cauchy–Schwarz on the source cell and the volume of the output cell give the L^2 contribution $Ce^{-\delta d(y,z)}\xi_z^{-2}\|A\|_{L^2(\Delta_z)}$. After multiplying by ξ_y, absorb ξ_y/ξ_z in the exponential by the scale-conversion bound. Summing the cell estimates yields the same scaled L^2 force bound for Du with a reduced decay rate. Thus the second gradient-source comparison uses an off-shell one-run estimate on complete common cells. To return its output to the first-order field norm, use the differentiated-source Hölder bound as well. On the complete stencils $D_\mu u = -\nabla_\mu^\mathsf{T} X_{\mu\mu}$, where $X_{\mu\mu}(x) = \mathrm{Ad}(U_\mu(x))u(x + he_\mu)$ on the lattice of mesh h. The bound for $\nabla G_1 \nabla^\mathsf{T}$ applies with an output Hölder exponent strictly below the Hölder exponent of this transported site input. Together with the mean comparison this gives the required first-order interpolation for the second solve. At the cut the cutoff-gradient and incomplete-row terms must be kept in the lower-layer estimate. This second solve uses the site estimate just interpolated. The terminal L^2 correction for the site Laplacian commutator is only paired in the first comparison.

The lift, inverse row map and induced current used by the nonlinear equation are compared at zero prescribed row current, with the actual independent bond current still present. Use $m(c) = \xi_c$ to convert from per-length rows to raw rows. If H, S are the per-length lift and inverse row map, the raw quantities are Hm^{-1} and $m^3 S m^{-1}$. The row mass converts by the same rule as S; on a common row it agrees at the two cutoffs. Hence the comparison of P uses the difference of the converted S's.

At a common source row c', inject $m^{-1}e_{c'}$ in the constraint slot of the full first-order system. Its bond component is the raw lift applied to $e_{c'}$. Its multiplier component, multiplied by m^3, is the raw inverse row map applied to $e_{c'}$. For an output row c, use $m^{-1}e_c$ in the dual row slot, so the pairing with its multiplier has exactly the factor $m(c)^3$. For the output field, test in one cell at y by a bond force g with $\xi_y\|g\|_2 \le 1$. The source-row terms agree in the two inverse identities because the common source row and its normalized injection are the same on the complete carrier.

The form difference then controls the inverse row entry and the scaled cell L^2 difference of the lift. This comparison uses both directions of the form pairing, with the quadrature errors above.

Before comparison the raw row and lift kernels are bounded by $C(\xi_y/\xi_{c'})e^{-\delta d(y,y(c'))}$ in their respective scaled norms. The scale conversion absorbs the displayed ratio in a smaller exponential. In the form estimate use the two localized solutions centered at $y(c')$ and y. Summing their product against the mesh weight gives $C\mathsf{E}(y)^{\beta}e^{-\delta' d(y,y(c'))}$ for some $\beta > 0$ and a smaller rate δ'. This follows by the triangle inequality for d and $\mathsf{E}(z)/\mathsf{E}(y) \le e^{\mu' d(y,z)}$, leaving a summable exponential in the intermediate cell z. For the lift apply the cell interpolation to the difference divided by its one-run exponential envelope. The one-run gradient Hölder bound and the scaled cell error give a positive power of $\mathsf{E}(y)$ in the first-order norm with a remaining exponential in $d(y, y(c'))$. Thus, after reducing the exponent and the rate,

$$\|(\Delta H)e_{c'}\|_{X_y} + |\Delta P(c, c')| \le C\mathsf{E}(y)^{\alpha}e^{-\delta' d(y,y(c'))}, \qquad y = y(c)$$

for the row entry, and for any output cell y in the lift term. Here X_y is the scaled first-order field norm. A uniformly bounded number of rows per top and the summable distance bound give the weighted operator comparisons. For a row input bounded by E^{γ} times its norm, the output is bounded by $\mathsf{E}^{\alpha+\gamma}$ times that norm. Take $\gamma\mu'$ below the remaining rate.

For the induced current use a bond-force input $\mathbf{J}$. Elimination gives the multiplier component $-H^{\mathsf{T}}\mathbf{J}$, so its negative, multiplied by m^3, is the raw ϕ. If the transferred current is $\mathbf{J}^{\flat} = \mathbb{P}\mathbf{J}^{\sharp}$, the source discrepancy in the same row test pairs $\mathbf{J}^{\sharp}$ with the difference between the prolongation and $\mathbb{P}^{\mathsf{T}}$ of the bond test. The two means reproduce constants in the flat frame. On their common carrier the error costs η times one difference and the small comb correction. This gives the mesh gain in ϕ without differentiating the independent current.

Enlarge the lower cut by the full stencils of the rows and their neighboring cells. For the admitted local-box cut the first-order commutator is supported on this enlarged transition and has a uniform local weak L^2 bound. The extra constraint and multiplier terms are bounded there by their own complete row coefficients times differences of the cutoff. Use $e = 1$ in the transition, and screen its terms by the localized kernels. In particular a source row c' in the cut gives only the one-run exponential there, which is paid since $\mathsf{E}(y) \ge e^{-\mu' d(y,y(c'))}$. This treats the admitted direct sequences on common complete upper carriers. A complex comparison uses a common admitted line and the two-constants bound with a reserve in the localization rate.

For the nonlinear field equation, compare the force on the undecoupled solves at $s = 1$ before multiplying by the vertex cutoff. The form of the force is eq. 3.cdxlv. For a test $\widehat{A}$ put $x = dX(T)[\widehat{A}]$, and use raw row pairings. Moving the transposes to the test gives the identity

$$\begin{aligned}(\mathsf{W}_s(T), \widehat{A}) = &-(x - d\mathcal{C}_2(T)[\widehat{A}], \phi) - (\widehat{Q}\widehat{A}, P_s X)\\ &-(x, P_s\widehat{Q}T) + (x, P_s X) + (\mathcal{V}(Y), \widehat{A} - H_s x).\end{aligned}$$

The first four terms are row pairings. The last uses the weak Wilson estimate on the same carriers, with the adjusted test $\widehat{A} - H_s x$.

On the common rows the exact nonlinear composition identifies $\mathcal{C}^{\sharp}(B) - \mathcal{C}^{\flat}(q_1 B)$ with the difference of the flat full row map at $\breve{q}(B)$ and $q_1 B$. The common linear part cancels. Use the value and derivative estimates for $\breve{q} - q_1$ above, together with the local Lipschitz estimate for the row derivative on the admitted ball. This gives the small weak row error and its derivative, in the full row norms. The quadratic part has the same estimate by extracting the degree-two term on a fluctuation disc.

In $X = \mathcal{C}(T - H_s X)$, transfer T by q_1. On any common row stock where the weighted lift comparison holds, the difference of the right sides is the row error, plus the effect of the lift difference on $X = O(\|T\|^2)$, plus a term bounded by $C\|T\|\|H_s\|$ times the difference of X. Absorb the last term by the contraction margin. The same comparison for dX follows from the differentiated equation, or from Cauchy's estimate in the fluctuation at the fixed admitted pair. In the first pairing

retain the cancellation $X(T) = \mathcal{C}_2(T) + O(T^3)$ at each cutoff. On a smaller complex fluctuation disc the difference of $x - d\mathcal{C}_2(T)[\widehat{A}]$ then has quadratic order in T and the same mesh gain. The other row terms have quadratic order by $X = O(\|T\|^2)$ and $x = O(\|T\|\|\widehat{A}\|)$. These estimates use the row comparison for P_s and for $\phi = H_s^{\mathsf{T}}\mathbf{J}$ at the actual rows. The latter is the induced current comparison, with the transferred independent current kept in the pair.

The lift comparison also controls the difference between the two Y's and between the two adjusted tests in the first-order field norm. Thus on the admitted direct regional charts the Wilson pairing and the three row-operator comparisons give the force defect in the norm of the one-difference test. The estimate for the vertex field uses the common smooth cutoff after the solves. For each native vertex mask we must check its row and stencil carriers against these estimates, or apply the corresponding comparison on the actual recursive tower. Keep the cutoff commutators and incomplete stencils in the estimate at the lower margin.

For the inner evaluation, anchor both frames at the same base point. The mixed transport of a quaternion from y to x is $P_{xy}v = E(x,y)v\mathbf{U}(x,y)^{-1}$. At real data it is an isometry. In the frame equation write $D_b = \nabla_b + C_bS_b$ at spacing b, with $C_b = (P-1)/b$ on positive edges and S_b the shift to their terminal sites. Thus $\mathcal{K}_b = D_b^*D_b$, and the forcing is $F_b = (\theta + \Delta_b\theta)w_{0,b}$, where $\Delta_b = \nabla_b^*\nabla_b$.

Here is the comparison on a fixed box whose Dirichlet faces are common nodal planes, with zero values at their nodes on both meshes. Use regular gauges for $E, \mathbf{U}$ on the box and all double blocks of the coarse path and energy reads. Work in these gauges with $\|C_b\|_\infty + \|\nabla_b C_b\|_\infty \le C\varepsilon_{\mathrm{g}}$. Let a_b be the unit anchor and take the cutoff to vanish near the boundary. The comb has bounded physical length, so $\|w_{0,b} - a_b\|_\infty \le C\varepsilon_{\mathrm{g}}$. For the one-run normalization use

$$(\mathcal{K}_b + 1)(w_b - \theta a_b) = (\theta + \Delta_b\theta)(w_{0,b} - a_b) - (\mathcal{K}_b - \Delta_b)(\theta a_b).$$

Pairing forward and backward edges bounds the last term by $C\varepsilon_{\mathrm{g}}$. In fact the sum of the two transport increments in a coordinate direction is a difference of the forward coefficients, times b, plus $O(b^2\varepsilon_{\mathrm{g}}^2)$. The remaining terms contain one difference of θ. The resolvent has supremum norm at most one by the equation at a site of maximal quaternion norm. It follows that $|w_b|$ is bounded below on each fixed positive plateau of θ when ε_{g} is small.

To compare the forcing, use the same comb order on the box lift. Put $\epsilon_E = \eta^{-1}\sup|E^\flat - \mathcal{M}E^\sharp|$ in the aligned gauges. Use the restriction of the fine gauge for the coarse $\mathbf{U}$, so $\mathbf{U}^\flat = \mathcal{M}\mathbf{U}^\sharp$ in these frames. Each coarse transport differs from its straight fine product by $O_L(h^2) + O(\eta\epsilon_E)$, where $h = \eta/L$, by the one-step path comparison. Telescope the products from the base to a coarse site. For a nearby fine site, append the short cell path and put it in the native comb order by interchanging neighboring steps. There are at most C_L/h such exchanges, each with curvature error $O(h^2)$. The path terms therefore differ by $C(\eta + \epsilon_E + |a_h - a_\eta|)$ on paired cells. Away from the cells at the breaks of the piecewise polynomial θ, the two second differences of the cutoff agree up to $C\eta$. They are bounded in the break cells, of total volume $O(\eta)$. The forcing discrepancy thus costs $C(\eta^{1/2} + \epsilon_E + |a_h - a_\eta|)$ in physical L^2.

For the mixed energy let R denote restriction to coarse sites and I multilinear prolongation of a coarse test g. The two forms are $\mathcal{A}_b(v,g) = (v,g)_b + (D_bv, D_bg)_b$ in the physical-volume pairings. The one-step path comparison and the first differences in the good gauges bound the discrepancy of the per-length coefficients on corresponding stencils by $C(\eta + \epsilon_E)$. The flat gradient of Ig is an interpolation of the parallel coarse gradients. Its volume transpose preserves constants on each complete incident stencil. In the flat gradient terms replace this mean of $\nabla_h w_h$ by the mean along the straight segment giving $\nabla_\eta R w_h$. Their difference costs $C\eta\|\nabla_h^2 w_h\|_2$ by telescoping over the fixed number of fine edges in the stencil. In terms with C_bS_b, move coefficient factors onto w_h or $\nabla_h w_h$ before this quadrature. Differences then cost one difference of the coefficient and one of the field, or the coefficient discrepancy above. The mass terms use the constant-preserving site transpose. On a common grid-aligned Dirichlet box, boundary sites are zero and the coarse edges

paired with a nonzero test gradient have their full transverse incidence. This yields

$$|\mathcal{A}_\eta(Rw_h, g) - \mathcal{A}_h(w_h, Ig)| \le C(\eta + \epsilon_E)\|w_h\|_{H^2_h}\|g\|_{H^1_\eta}.$$

The one-run H^2_h bound follows by writing $(\Delta_h + 1)w_h = F_h - (\mathcal{K}_h - \Delta_h)w_h$. The last term is bounded in L^2 by $C\varepsilon_{\mathrm{g}}(\|w_h\|_2 + \|\nabla_h w_h\|_2)$. Energy controls these first-order norms. Odd reflection across the Dirichlet faces and commutation of the flat differences bound the second differences by the flat Laplacian norm. Pair the frame equations with $g = Rw_h - w_\eta$, using Ig on the fine lattice. The source bound and the same site quadrature give

$$\|Rw_h - w_\eta\|_{H^1_\eta} \le C(\eta^{1/2} + \epsilon_E + |a_h - a_\eta|).$$

For the Dirichlet realization one can also remove a thin boundary strip before comparing coefficients. Taking the norm in the mixed equation gives $(\Delta_b + 1)|w_b| \le |F_b|$. In a coordinate of width D, a multiple of $s(D - s)$, with s the distance to one face in that coordinate, dominates the bounded forcing and is nonnegative on the boundary. Comparison in each coordinate bounds $|w_b|$ by C times the distance to the boundary. Setting zero boundary data on a common inset at distance at most $r + \eta$ changes the solution there by at most $C(r + \eta)$ in supremum norm, by the homogeneous mixed equation. The coefficient and forcing comparison can then be made on that inset.

On an interior read the first differences are uniformly bounded by the local bootstrap eq. 3.dcccxci. Its proof applies componentwise to these orthogonal transports: the flat principal part is the same, and in the regular gauge the first-order coefficient is $O(\varepsilon_{\mathrm{g}})$, while the zeroth coefficient and the forcing have bounded finite L^p norms. The energy estimate supplies its outer H^1 hypothesis. All cutoffs fit in the fixed read collar. A maximum of size s for the difference therefore gives an L^2 contribution at least cs^6 in dimension four. Below mesh radius use the single site with weight η^4 and $s \lesssim \eta$. The resulting supremum error has the power $1/3$ of the L^2 error. The plateau lower bound permits normalization in SU(2) with the same error. This supplies the frame value for conjugating the inner datum on the common carrier.

The native box has faces between sites and is a union of complete paired cells. For the realization with only inner edges, use its inner-edge scalar Laplacian for $\Delta_b\theta$ and take θ constant in the normal directions near the faces. In particular a cutoff vanishing in a boundary collar has the same forcing comparison as above. Use the mixed cell mean P_0 from the fine sites to a coarse centre. Its transports are those of the two averaged contours. Let Q_P average fine bond fields along the straight L-step paths from one cell to the neighboring cell, transported to the initial centre. The weight of each fine edge occurrence in the path sum is L^{-5}. Telescoping the path differences gives

$$D_\eta P_0 = Q_P D_h + \mathcal{R}, \qquad \|\mathcal{R}\|_{2\to 2} \le C(\eta + \epsilon_E).$$

Indeed the mismatch of the two end transports is $O_L(h^2) + O(\eta\epsilon_E)$ on each path, and the path difference is divided by η.

For coarse v, let $I_P v$ be the multilinear prolongation, transporting the coarse corner values by short paths to the centre of the cell containing the fine site and from there to that site. The mass defect $I_P v - P_0^* v$ costs $C\eta\|D_\eta v\|_2$. In a local comb gauge on a complete stencil the gradients of $I_P v$ and $Q_P^* D_\eta v$ average the neighboring parallel slopes, with total weight one in each case. In the latter average this follows by counting the paths through the fine bond in its own direction. Their difference is bounded by $C\eta$ times the second differences of v. Restoring the transports costs $C(\eta + \epsilon_E)\|v\|_2$, by the complete stencil curvature bound and the same link comparison. At a cell face reflect the field and the tangential links, with identity transport on the link to the mirror site. The straddling plaquettes are then flat and the outward coarse slope is zero. The reflected stencil therefore gives the same estimate at the inner boundary.

For the second-difference bound put $v = w_\eta$ and $\psi = D_\eta v$. If T_μ denotes an orthogonal transport followed by one translation, the difference $T_\mu T_\nu - T_\nu T_\mu$ is a two-step transport times the plaquette

holonomy minus the identity. Consequently $[D_\mu, D_\nu]$, and the corresponding forward-backward commutator, have bounded L^2 operator norm under the normalized curvature bound. Commuting one difference through $D^\sharp D$ gives, with the shifted carriers retained,

$$D_\mu D^\sharp_\nu D_\nu v = D^\sharp_\nu D_\nu D_\mu v + [D_\mu, D^\sharp_\nu] D_\nu v + D^\sharp_\nu [D_\mu, D_\nu] v.$$

Thus the equation for ψ has source $D_\eta F_\eta$, a divergence of a field with norm at most $C\|v\|_2$, and a term of norm at most $C\|\psi\|_2$. Pair with ψ. Summing the source pairing over components gives $(D_\eta F_\eta, D_\eta v)_\eta = (F_\eta, F_\eta - v)_\eta$. Move the divergence onto the differences of ψ and use Young's inequality to absorb half their square norm. The site energy estimate then yields $\|D_\eta \psi\|_2 \le C\|F_\eta\|_2$. For the inner-edge boundary convention, perform the calculation after the preceding even reflection of the site field and links. The added normal difference is zero and the straddling plaquette is flat, so restriction gives the same estimate on the original box without a boundary term.

The exact form difference is

$$\mathcal{A}_h(I_P v, f) - \mathcal{A}_\eta(v, P_0 f) = (I_P v - P_0^* v, f)_h \\ + (D_h I_P v - Q_P^* D_\eta v, D_h f)_h - (D_\eta v, \mathcal{R} f)_\eta.$$

With $f = w_h - I_P v$, the forcing difference and these defects give the same L^2 error as in the nodal-plane case. Use the regular gauges on the read collar for the first-difference bound and the supremum interpolation. In those gauges the covariant prolongation differs from the ordinary one by $O(\eta)$ there.

For the native real box, use the common good gauge of the input $\mathbf{U}$, normalized at y_0 and restricted from fine to coarse. Here a_0 bounds the actual real potential in that gauge, including any real increment. The box and its one-step carrier fit the ordinary gauge cube by the size hierarchy. The actual box datum in this gauge is formed by the full iterated averages of the gauged input. It is gauge equivalent to the native comb datum by a small node transformation. Indeed the difference of the two gauges, normalized at y_0, is the inverse comb transport in the good gauge, hence close to one by the potential bound. The small normalized-gauge and real-base change argument in the proof of Prop. 3.135, on these complete box reads and the common smaller logarithm ball, identifies the two representatives of the box orbit.

The ordinary datum in the common good gauge agrees on common rows by composition and covariance of the averages. On a row whose full preceding outward read is also common, the reduced data agree by Prop. 3.176 and eq. 3.cccxi. At the trivial base, Cor. 3.178 identifies the Landau potential of the box minimizer with $\mathbf{H}(\mathbf{1}; B_\natural)$ on the closed noninward row stock, for its actual ordinary logarithmic datum and reduced datum, on the fixed small ball of that result. On a row of spacing ξ_c, the averaging bound at the trivial base gives $|B(c)| \le C\xi_c a_0$ in the good gauge. The same profile for $B_\natural$ follows from the local change of data, since its row lengths are comparable. Integrating the screened derivative of Cor. 3.178 from zero bounds the potential by Ca_0 and its first difference at a cell y by Ca_0/ξ_y. In the sum one absorbs the ratio of the source and output row lengths in the exponential kernel. Thus on the protected upper reads the regional response estimate compares the two Landau potentials at these data.

The actual coarse average of the fine minimizer uses the one-step node factors as well as the straightened potential. In the trivial-base chart the one-step factor differs from one by $C_L h \sup|A|$. Comparing the same holomorphic map on two translated blocks bounds its change by $C_L h\eta \sup|\nabla_h A|$. In the normalized coarse link the two endpoint factors therefore cost at most $C_L h(\sup|\nabla_h A| + \sup|A|^2)$. Here the quadratic term comes from commuting the near-identity factor past the coarse link. Include also the nonlinear straightened transfer error. This supplies the actual E-coefficient estimate on the matched protected reads.

The anchor in a Landau representative uses its gauge at y_0. By Prop. 3.176, the inverse of that gauge factor there is $v_B(y_0)$. The native comb node factor is one at this site. For the accumulated factor, separate the fine one-step factor from the recursion. The remaining product is the flat factor

at the exactly straightened fine input. The one-step factor is $1 + O(h \sup |A|)$. On the complete common top pillar the flat factor has a bounded holomorphic chart by Cor. 3.178, and Cauchy's inequality bounds its change by the supremum input error there. Hence the protected response comparison also compares the Landau anchor factors, when the original anchors coincide in the common input frame.

For the inner-edge realization the matched interior comparison suffices, with a shrinking boundary strip. In fixed box units let $C\eta < r < 1$. The accumulated erosion of the native tower through scale ξ is at most $C\xi$, by summing the successive mesh widths. Thus stencils at distance at least r from the faces have local scale $\xi_y \geq cr$. The fixed unmatched lower layers lie in the strip. By eq. 3.cdliii, the ratio of two row spacings is at most a fixed factor times $\exp((\log L)d/\rho)$, where d is the regional distance. Taking a fixed power $\sigma \in (0, 1]$ small enough for the screening rate gives $\mathsf{E}(y) \leq C(\eta/\xi_y)^\sigma$. On the complete common rows outside the strip the response bound and the one-step node correction give the per-length box coefficient error

$$\epsilon_E(r) \leq C\big(\eta^\alpha r^{-1-\alpha} + \eta/r\big), \qquad \alpha > 0.$$

The one-run mixed coefficient is bounded throughout the box by the potential bound, and has first difference at most C/r outside the strip.

For this application one need not bound the normalized curvature throughout the strip. Write the mixed equation as

$$\begin{aligned}(\Delta_b + 1)w_b = F_b - \nabla_b^*(C_b S_b w_b) \\ - S_b^* C_b^* \nabla_b w_b - S_b^* C_b^* C_b S_b w_b.\end{aligned}$$

Reflect the scalar site functions evenly at each cell face. In the divergence term reflect an edge component oddly across the faces normal to its direction, with zero flux across those faces, and evenly across the transverse faces. Then the extended flat equation uses the usual lattice differences. Integrating the heat kernel bound eq. 3.dcxlv with the mass factor e^{-t} gives a uniform L^∞ bound for the first derivative of the flat resolvent. Its derivative followed by a divergence costs at most $C(1 + |\log b|)$, since the second derivative of the heat kernel has summed bound $C \max(t, b^2)^{-1}$. The mixed maximum principle bounds w_b. The term linear in $\nabla_b w_b$ is absorbed by smallness of the coefficient in the common gauge. Hence

$$\|\nabla_b w_b\|_\infty \leq C(1 + |\log b|).$$

The same logarithmic bound holds after adding the bounded transport term.

Apply the nodal restriction and multilinear test prolongation to the weak forms in these gauges, using even extension for the scalar quadrature on the doubled box. Split the complete interpolation stencils by whether they meet the strip, including the $O(\eta)$ stencil halo. The forms on the strip of width Cr cost at most $Cr^{1/2}(1 + |\log \eta|)$ times the test energy norm, by the volume of the strip and the preceding bounds. In the interior the flat equation with a cutoff changing on length r gives an L^2 bound Cr^{-2} on the second differences. The interior quadrature therefore costs $C(\eta r^{-2} + \epsilon_E(r))$. For the forcing the coordinate combs from y_0 to the points of the smaller rectangle stay within that rectangle. Use the per-length comparison on their full carriers. On the rim use the uniform bound on F_b, which requires the stated face property of the cutoff. This yields the energy error

$$C(r^{1/2}(1 + |\log \eta|) + \eta r^{-2} + \epsilon_E(r) + \eta^{1/2} + |a_h - a_\eta|).$$

Take $r = \eta^\gamma$ with $0 < \gamma < \min\{1/2, \alpha/(1 + \alpha)\}$. The mesh terms are $O(\eta^\nu)$ for any $\nu < \min\{\gamma/2, 1 - 2\gamma, \alpha - (1 + \alpha)\gamma\}$. On the read collar the one-run interior Lipschitz estimate and plateau normalization give the frame comparison with the cube root of the energy error. The plateau lower bound here also follows by comparing the forms for w_b and θa_b in energy, which uses only the small supremum of the mixed coefficient; interpolate the result on the interior collar. To apply this argument to the named smoothing rule use its actual edge stock, cutoff at the faces and anchored real input.

To compare regular activities use their own counterterms before estimating the old evaluations. Write $G^\sharp, G^\flat$ for partnered families, $e^\sharp, e^\flat$ for their scalar coefficients, and $A^\sharp, A^\flat$ for the actions. With $\widetilde{G} = G - eA$ in each run the exact decomposition is

$$\begin{aligned}\widetilde{G}^\sharp = (G^\sharp - \mathbb{M}^* G^\flat - (e^\sharp - e^\flat)A^\sharp) \\ + e^\flat(\mathbb{M}^* A^\flat - A^\sharp) + \mathbb{M}^* \widetilde{G}^\flat.\end{aligned} \tag{A.cviii}$$

The second-order test on a linear potential of constant curvature extracts e. The linearized averaging reproduces that potential in the coarser units. Thus the first bracket obeys the same cancellation as a regular counterterm-subtracted family. The action defect in the second term obeys the same quadratic cancellation. In a fluctuation insertion its value at the background is removed by the fluctuation derivative. Its variation is bounded by the regular perturbation estimate for the plaquette defect. The two powers of the radius in that estimate pay for extracting $e^\flat$ by Cauchy's inequality.

In the last term compare the two evaluations of the same counterterm-subtracted function. Integrating one derivative in each of N decoupling parameters gives an alternating sum of its 2^N vertex values. Use the regional response and frame comparison on the reduced polydiscs where the full vertex carrier admits it. For an outer evaluation, use the domain field at $B(t)$ plus w times its window direction. Write $\rho_j = a_{0,k+1}/a_{0,j}$ and $\sigma_j = L^{-(k-j)}$ in the partnered units. The one-run field bound on the full $X^\sim$ gives a direction of size at most $C\sigma_j\delta(\mathsf{t}) \le C\mathsf{s}\rho_j\sigma_j a_{0,j}$. The field at $B(t)$ has size at most $C\mathsf{s}a_{0,j}$, since $\sigma_j\rho_j \le C$. After allowing for the fixed transfer loss in the background radius, the disc $|w| \le c/(\mathsf{s}\rho_j\sigma_j)$ therefore fits the old radius on each cutoff. Choose c smaller than the available radius fraction divided by the one-run bound. The same disc gives the Cauchy cost $C\mathsf{s}\rho_j\sigma_j$ for the derivative.

For the raw switched outer field, restrict to the wall component supplying its read of X. If all determining rows of that component have complete averaging carriers disjoint from the window direction, its row datum is independent of the window parameter. Indeed the parameter enters the literal averages only through $(t + t_\square\zeta_\square)H_k$ on those carriers. The potential and assigned current of the restricted response therefore both have zero window derivative. When this holds at both cutoffs the two activity derivatives vanish at this vertex, even when their baseline fields differ.

For the named supply the vertex reduction is instead given by its alternating sum. Write F_a for the cutoff times the undecoupled field and current of the region Y_a, using that region's datum. At a vertex with active set v, only subsets $S \subset v$ survive the switch monomials. The coefficient of F_a is $\sum_{a\subset S\subset v}(-1)^{|S\setminus a|}$. It is zero for $a \subsetneq v$ and one for $a = v$. Thus the exact datum for the nonzero comparison is the one on Y_v's determining set. If the complete read lies inside the cutoff plateau of a common region Y_v, and its own sequence satisfies the direct top-inset criterion above, the full-switch response estimate applies on those reads even when v is a proper subset of the ambient cubes. The assignment of the region's own core and datum is needed before using that criterion. In particular the raw masked lift applied to a datum supported on a missing row can vanish on the read while the named all-one lift has that row as its linear constraint.

The outer and removed tails can be bounded before comparing the vertex fields. For an outer X away from the window, the window derivative has the distance gain of the one-run estimate. For X meeting $\square^\sim$ and leaving $\square^{\sim 2}$, $d_j(X) \ge \sigma_j^{-1} - C$, so use part of the old-domain decay instead. Each of the two removed directions already has this gain on its full carrier in eq. 4.cli. After the window count, the factor e^{-c/σ_j} in these estimates leaves the localization reserve. With age $n = k + 1 - j$, $\rho_j \le C(n+1)^{1/2}$ and $\sigma_j^{-1} = L^{n-1}$. Thus the two one-run bounds themselves give a factor $L^{-\alpha j}$ together with the fixed exponential age weight when $L^{n-1} \ge A(j+1)$, with A fixed sufficiently large. Use the actual outer vertex and removed segment comparisons at the remaining ages.

The first removed segment is affine in the two shift coordinates of its common response chart. On a matched complete carrier write it as $A_t = A_0 + t\Delta A,\ J_t = J_0 + t\Delta J$, where the current slot includes the fixed base current. To compare the segment at the two cutoffs, put the two endpoints

in the same frames before interpolating. Let h be the fine spacing in these units and let T be the relative potential transfer. Its second derivative in the potential slots is $O(h)$: the holomorphic map on the dimensionless stencil is evaluated at hA_t and its logarithm is divided by Lh. Consequently the difference between $T(A_t)$ and the affine interpolation of the transferred endpoints is bounded by

$$Ch\,t(1-t)\|\Delta A\|^2.$$

For the neighboring-stencil estimate take local units in which the reference scale is one. Compare the stencils by parallel transport. The base link arguments of the stencil differ by $O(h^2\kappa)$, where κ is the bound on the base curvature on the intervening plaquette strips. At an interpolated potential the difference of the angular arguments is bounded by $Ch^2 \sup_t \|\nabla_U A_t\|$. The second derivatives differ by Ch times the sum of these two argument differences. The two direction slots also vary, each by at most $Ch\|\nabla_U \Delta A\|$. After division by the mesh, the first difference of the affine defect is bounded by $Ct(1-t)$ times

$$h\|\Delta A\|\|\nabla_U \Delta A\| + h^2\|\Delta A\|^2(\kappa + \sup_t \|\nabla_U A_t\|).$$

All these norms use the complete neighboring stencils in the same local units. For the current transfer $\mathbb{P}_{\mathbf{U}_t} J_t$, a derivative of each path transport costs $Ch\|\Delta A\|$. Its second derivative and the product rule give the affine defect bounded by

$$Ct(1-t)(h\|\Delta A\|\|\Delta J\| + h^2\|\Delta A\|^2 \sup_t \|J_t\|).$$

Together with the transferred endpoint estimates, these defects give the comparison on this first segment wherever the paths are admitted in the common radius. The current used here is the assigned affine current.

For the second removed segment use the old box datum of [B87, (3.22)–(3.25)]. Write a for the upper fluctuation in angular units, u for the window parameter, and χ for the cutoff equal to one on $\square^{\sim 3}$ and zero outside $\square^{\sim 4}$. Denote the relative log of the full average at a row c by Q_c. In a common box frame the two relative data are

$$D_{\mathrm{mid}}(c) = \chi(c) Q_c((t + u\zeta_\square)a), \qquad D_{\mathrm{in}}(c) = Q_c((t\chi + u\zeta_\square)a).$$

Set $R = D_{\mathrm{mid}} - D_{\mathrm{in}}$. The row path runs from the inner term to the preceding term with $D_s = D_{\mathrm{in}} + sR$. For this comparison use complete old-row carriers, with every carrier reading $\zeta_\square$ in the plateau of χ and with $\chi(c) = 1$ on its row. On that carrier $R(c) = 0$. On the other carriers both data are independent of u. Thus R is independent of u, and its support is in the transition enlarged by the row reads. At $u = 0$ the fixed-base estimate for the full row logarithm bounds $R(c)$ by $C|t|\delta_k L^{q-k}$ on a level-q row of the old box.

The relative path multiplies the actual unperturbed row field V_0 on the left. Use the same native node gauge for both factors. The datum of the box solve is the absolute log

$$b_s(c) = \frac{1}{i}\log(e^{iD_s(c)} V_0(c)).$$

On the native second-removed path with the common rebuilt box base, the inner datum b_0 is the full average of its top-masked perturbation. Use the extra row datum $B'_s = b_s - b_0$ in the old-box response with its original truncated determining set. The product log has $B'_s = 0$ off the support of R and gives $|B'_s(c)| \le C|s|\,|R(c)|$. It is independent of u, since D_{in} is independent of u wherever R is nonzero. The admitted fluctuation margin keeps $\sup|B'_s| < \mathsf{c}_3 a_0$ in Prop. 3.135. Apply the local naturalizer to this absolute row difference before using the reduced response estimate. Its fixed reach enlarges the source support and leaves the separation from the inner read. In level-j units the increment of the potential, its first difference and the assigned current on the inner read is bounded by

$$C|st|\delta_k L^{j-k}\exp(-cML^{k-j}).$$

Here the response norm is the one on the native top read, and the distance is from the enlarged transition to that read. At $j = k$ use the thinner inner collar where this separation persists after adding the complete transfer and first-difference stencils. At full old switches the potential and its assigned current are those of the native critical pair in the box Landau representative.

Compare the row data before applying the paired native box estimates. The difference of the R's is the difference of the two full-average discrepancies with the common cutoff on the first. For either mask $m = t + u\zeta_{\square}$ or $m = t\chi + u\zeta_{\square}$, first compare the shifted fields at the one-step carrier. Replacing the mask there by its value at the coarse initial site costs at most $C_L\eta \operatorname{Lip}(m) \sup|a^{\sharp}|$ in the relative log. Moving this constant through the log costs at most $C_L(1 + \|m\|_{\infty})^2 \sup|a^{\sharp}|^2$ by Taylor's estimate at zero. The remaining input difference is that of the transferred unmasked angular shift and the flat shift, multiplied by m. The exact composition for the plain relative log and its fixed-base bound CL^q for q remaining stages transfer these errors to the old row. A difference between the rebuilt base fields or their node frames is accounted for in the same row comparison. Taking differences in the analytic product log gives

$$|\delta B'_s| \le C|s|(|\delta R| + R_*(|\delta D_{\mathrm{in}}| + |\delta V_0|)), \qquad R_* = \max_{a=\sharp,\flat} |R^a|.$$

This follows by writing B'_s as the integral of the derivative of the product log in the R direction. The common node frames and paired full-row comparisons supply the terms on the right. For paired unperturbed row fields the last term vanishes by nonlinear composition. With the native old-box response estimate this compares the whole segment on the admitted common top reads. Apply the scalar real-to-complex comparison on the admitted common lines before using Cauchy's estimate in the window parameter. For another continuation of this segment one must first identify its assigned base, row representative and current.

To use a paired response estimate at an activity, transfer the actual evaluation pair. On an admitted common read let $W = \mathcal{M}U$, let $A^{\sharp}, A^{\flat}$ be the two response potentials, and write $J^a_{\mathrm{tot}} = J^a_{\mathrm{base}} + i^a$ for their assigned currents. Use the paired base currents $J^{\flat}_{\mathrm{base}} = q_0 J^{\sharp}_{\mathrm{base}}$. The transfer of the fine pair averages $U' = e^{ihA^{\sharp}}U$ and uses $\mathbb{P}_{U'} J^{\sharp}_{\mathrm{tot}}$. Apply the inverse of the node gauge v of the straightened average to both components. The resulting potential relative to W is exactly $\breve{q}(A^{\sharp})$. At the start of a coarse bond c, the resulting current is

$$\operatorname{Ad}(v(c_-)^{-1})(\mathbb{P}_{U'} J^{\sharp}_{\mathrm{tot}})(c).$$

On the full double-block stencils each path transport and the node gauge change by at most $C_L h \sup|A^{\sharp}|$. This follows by telescoping the bounded products and using the regular contour mean on the admitted ball. The current in this display therefore differs from $q_0 J^{\sharp}_{\mathrm{tot}}$ by at most $C_L h \sup|A^{\sharp}| \sup|J^{\sharp}_{\mathrm{tot}}|$. No derivative of the current is used.

For the potential the local Taylor error of $\breve{q} - q_1$ is quadratic. Compare the remainders on neighboring stencils after parallel transport. The potential inputs then differ by $C_L h \sup|\nabla_U A^{\sharp}|$. The transported base links differ by $C_L(h/\xi_y)^2$ when the normalized base curvature is bounded in the local units. This is the product estimate on the small plaquette strips joining the translated paths. The Taylor bounds with one more derivative give the same mesh gain in the scaled first-difference norm. Writing a_y for the scaled first-order norm of the fine potential on the full stencil and j_y for the scaled supremum norm of the total fine current, the errors beyond the paired response comparison are bounded by

$$C(\eta/\xi_y)a_y^2 \quad \text{and} \quad C(\eta/\xi_y)a_y j_y.$$

The remaining current difference is $q_0 i^{\sharp} - i^{\flat}$ because the base parts cancel. The remaining potential difference is $q_1 A^{\sharp} - A^{\flat}$. The relative log between the two perturbed fields obeys the same first-order bound, by the product and logarithm estimates on the small chart. The response comparison thus supplies the evaluation mismatch on these complete reads once its two native data and currents have been identified. For the reduced currents this use is on the common interior top reads.

To pass from a real evaluation mismatch to the value of a gauge-invariant activity, fix a real vertex and use a relative gauge for which the potential, its first covariant difference and the current mismatch have size at most $2b_0\vartheta$, where ϑ bounds the infimum in the mismatch norm. A zero infimum is handled by taking the bound to zero. Write the flat evaluation as $(\mathbf{U},\mathbf{J})$ and the two mismatches as A, I. Use the scalar activity on the disc of pairs

$$(e^{i\eta_j wA}\mathbf{U},\ \mathbf{J}+wI), \qquad |w|\le c/\vartheta.$$

The flat evaluation must have the half field radii and a fixed margin below b_0 in the current slot. Use $0<b_0\le b_1$, $b_0+b_1\le a^\sharp$ and $b_1^2\le \mathsf{c}_\natural b_0$. The outer substitution theorem admits the disc when the potential and its covariant difference stay below $b_0/[128K_{\mathrm{red}}(1+\mathsf{c}_\natural)]$, and $|\mathbf{J}+wI|<b_0$. The fixed choice of c ensures both bounds. When the mismatch is small enough that the disc radius is at least two, Cauchy's inequality bounds the change from 0 to 1 by $C\vartheta$ times the activity norm. Larger mismatches use the individual bounds. The value at 1 is the fine evaluation after transfer and gauge change, so gauge invariance identifies its value. The relative gauge is chosen only on real data. For complex interpolation use the scalar difference of the original two evaluations, which are holomorphic on their admitted lines.

The single-lattice bound on the larger polydisc retains the decay in the size of the carrier. Combining the smallness at the vertices with this decay costs part of the localization exponent, which pays for the 2^N terms. In the inner term subtract the value at the flat pair and use the vanishing of the first derivative there. The quadratic bound on the two nearby linear-response evaluations then retains the small radius factor in the difference. To retain an age weight as well, the one-run estimate for this difference uses the second renormalization at a Hölder exponent strictly larger than the exponent in the age weight. The reserve in the age exponent pays for interpolation.

The resulting regular commutation defect of an input formed with j fine levels has a factor $\omega_{k+1-j}\min\{1, CL^{-\alpha j}\}$ in the step with $k+1$ fine levels, with some $\alpha>0$. The unmatched input at the finest scale has the age weight ω_{k+1}. A marked remainder uses its squared radius ratio in place of the age weight. To obtain the regular two-run bound, sum these defects and join the two inverse covariances and the two auxiliary Gaussian kernels by line segments in their admitted complex balls. Cauchy's estimate on the available radius bounds the change in the small-field expansion by the size of the Gaussian and potential differences. The change in a scalar weight costs $C|x^\sharp-x^\flat|/\mathsf{t}$ on a disc of radius $c\mathsf{t}$. The Gaussian normalizer is compared separately.

In the covariant regular chain this gives, at equal deviations, an inequality of the form

$$D_s \le C\lambda^{K-s} + \varepsilon \sum_{s<t<K} \omega_{t-s} D_t,$$

where D_s is the distance between the families at depth s. Choose $\lambda<1$ larger than the geometric rates of the mesh defects and age weights, absorbing polynomial factors in $K-s$. Choose ε with $\varepsilon\sum_{m\ge1}\omega_m\lambda^{-m}<1$. Downward induction then gives $D_s\le C'\lambda^{K-s}$. The constant-source coefficient is read from this distance by the same quadratic test. The full step still needs the sourced estimate on position-dependent arrays and the matching of the marked histories. Those terms have no common covariant quadratic cancellation to which the regular age estimate can be applied.

A.9.0.8. Sharp background tests on the matched histories. The approximate fluctuation test of §4.17.1.1 compares the nonpivot values of $V_k(\mathcal{M}^k U_{t,\square}(V_t))^{-1}$ with the identity. The threshold is $2\delta_k$. Even when the two retained fields agree, the regional background $U_{t,\square}$ is computed with a different bare depth. Work on a common partial history up to the first differing decision. One must first compare the availability of the reads, since the test convention assigns failure to an undefined read.

Consider a newly inserted ordinary plaquette test on a cube $\square'$ tested in both runs. On the common prefix the incoming positive density tests cover the entire footprint of the one-step average for the lifted datum on $\square'^{\sim 4}$. To see this on either lattice, measure lengths in units of the incoming $k=t-1$ lattice. Put $s=LM_2\check{R}_t$ and $W=LM\check{R}_t$. The four enlargements cost at most $4s$, and

completing the one-step averaging blocks adds at most L. A level-k density cube covering a corner of a plaquette in this footprint has side at most Ls, by the neighboring-scale bound for $\check{R}$. The total $4s + L + Ls < W$, since $W \geq L^2 s$. The ordinary plaquette test region has four coarse layers to the old large region. Even in the first step there is one coarse layer to the hull of the failed bare tests. Thus all the incoming density cubes just selected have their positive test, and none is in an omitted second-class inset.

The full bare averaging blocks of a level-k plaquette with a corner in such a density cube lie in its one-layer enlargement. The positive density test gives bare curvature less than $\varepsilon_k \eta_k^2$ there. The local small-curvature average estimate gives $|\partial V_k - 1| \leq 2\varepsilon_k$ on the one-step footprint, since the localized orbit has the prescribed V_k there. At $k = 0$ use the bare test itself. The ordinary averaging row prescribes $V_t = \mathcal{M} V_k$ on the top bonds inside $\square'^{\sim 4}$. Thus the one-step bound gives $|\partial V_t - 1| < 3L^2 \varepsilon_k$ there. At weak coupling $\varepsilon_1 = 3L^2 \varepsilon_k \leq a_1$ and $B_3 \varepsilon_1 \leq \varepsilon_{\mathrm{v}} \leq a_0$. The real existence and orbit comparison theorem therefore admits the lifted datum at the test radius ε_{v}, independently at both cutoffs.

A newly inserted approximate fluctuation test is reached only on a cube whose own plaquette test passed. It uses that same orbit on the fourfold box. The third enlargement lies in the top member with a margin containing the full averaging blocks of its top sites. Earlier determining rows have no endpoint there, so the only nodes in the axial-gauge carrier are its top sites. On each of their bare blocks the orbit curvature is bounded by $B_3 \varepsilon_1 \eta_t^2$ by the admission just proved. Local averaging bounds the level-k plaquettes in the contour blocks by $2L^{-2} B_3 \varepsilon_1$. At the same weak-coupling floor, $L^{-2} B_3 \varepsilon_1 \leq c_2^{\mathrm{elem}}$, so the contour means exist. For $k = 0$ use the bare curvature bound. The regular orbit on this box therefore has an axial representative with a unique center on the tested bonds.

These admission checks apply to ordinary tests at their first insertion on the common prefix. A carried test reproduced on its original reads belongs to its original decision.

For a failed ordinary plaquette test, the fourfold read can be exposed in the raw incoming stock. Take the complete read with every fine double block of its top rows inside the raw footprint of the failed component. In the first step the incoming field there is raw Haar, without earlier averaging constraints. Keep the nonpivots fixed and use each top row's private middle link to expose its generated top output. The literal row Haar kernel has mass at most $2L^9$ on a generated target. Distinct rows read none of the other's pivots, so the joint upper bound is the product of these row factors times Haar on the top read. The remaining factors use the true inverse graphs and their reference majorants.

The admitted top curvature bound on this read puts its axial logs in the ball of radius $CD\varepsilon_k$, for a box of side D in the test units. The radial argument at the ordinary threshold ε_t then gives the factor $Cn\vartheta$ in the top Haar bound. The row costs grow at most exponentially in the number of top rows of the read. This calculation concerns the first split and a later reference stock in which these generated values remain free. At a later ordinary step the selected incoming pivots must remain free after the earlier eliminations. A subsequent prescribed outgoing row can constrain the exposed top values and must be included in that reference chart.

The marked tests have their own admission arguments.

At stage zero the first regularity guard uses the untouched incoming datum, with the incoming profile. Its positive branch enters the first prospective guard. For that guard and the later prospective guards, use the promoted datum bounds on the entering support. These bounds, Ce_q in each layer q, use the actual promoted averages before their delta functions are integrated. Localizing the stage to its class component gives the joined determining set with its prescribed collar. A tested holonomy in the joined set either agrees with a tested holonomy on the global datum, is one in the lifted collar, or equals a tested top holonomy of the datum being lifted. Hence the joined datum is $\bar{\varepsilon}$-regular for $\bar{\varepsilon} > C \max_q e_q$. At weak coupling take $\bar{\varepsilon} \leq a_1$ and $B_3 \bar{\varepsilon} \leq \varepsilon_{\mathrm{v}} \leq a_0$. The regular orbit for the localized set in force therefore exists before the prospective guard is applied. The inset

and crossing tests that use this same orbit also have defined reads. Passing their layer thresholds requires their separate curvature estimates.

For the last-shell test take $h = m_\flat$ and the full fourfold test box in $Z''_{h+1} \cap \Omega_h$. The entering prepared datum bounds its original level-h plaquette defects by Ce_h. Keep the specified links meeting the inner cutoff. Every link set to one belongs to $\mathbb{B}_0$, so the preceding positive main-shell test and the basic reference bound give distance at most $\delta'_m + CL^{h-m}\varepsilon_m$ from one at that link. Telescoping at most four replaced factors bounds the plaquette defects of the artificial datum by

$$Ce_h + 4\delta'_m + CL^{h-m}\varepsilon_m.$$

This lies in the fixed lifted-datum existence radius at weak coupling, independently of the waiting depth. This proves admission before the last-shell decision, whose half-threshold is a smaller requirement.

On a passing last-shell cube the local test transfers to the glued background U^0 at the same plaquettes. In the tested curvature units its bound is

$$\tfrac{1}{2}\varepsilon_h + C\varepsilon\big(L^{-\mathsf{N}} + e^{-\delta_0 D_2}\big).$$

Here N is the waiting length and D_2 is the clearance of the tested enlargement inside its full fourfold box, in level-h units. This is a one-run comparison of the two lifted fields from the same inner datum. The band for the last-shell decision is a function of that retained level-h datum. In the framed integral it stays fixed through the V' integration at fixed datum, so its thickness must be estimated in the retained-field law. On the identity side of the mask, a tested plaquette with a complete vacuum top box around its carrier has normalized curvature at most $Ce\,e^{-cD}$, where e is the admitted radius and D the distance to that box's complement in level units. This is the real collar comparison with the vacuum at the same retained data. Such plaquettes cannot attain a maximum in the threshold band once this bound is below $\varepsilon_h/4$. The remaining test depends on the actual determining links of the retained h-datum. The ordinary radial change holding those links fixed gives no shell gain for this test.

The effect of a change of site gauge before masking can be assigned to the erased rows. Write J for the map setting the erased links to one. With gauge action $V_b^g = g(b_-)V_b g(b_+)^{-1}$, pull $J(V^g)$ back by g^{-1}. Its kept links are those of $J(V)$, and its erased links are $g(b_-)^{-1}g(b_+)$. The plaquette norms of the constrained background are invariant under this gauge pullback. In the lift to the complete test tower an induced link is either unchanged or copies one changed link between level-h blocks. If the erased-link jumps have logarithms of size at most b and their logarithmic interpolation lies in the common regular domain, apply the curvature response bound along it. In the units of the test the change at a plaquette is at most Cbe^{-cD}, where D is its tower distance to the affected determining rows. Use this comparison in each fixed tower before comparing the two cutoffs. It isolates the gauge contribution to the band on the actual read.

There is a shell estimate on a different chart, when all kept links in the complete test read are free and can be scaled together from the vacuum. In a wholly kept box with a full product Haar stock, fix a rooted axial tree. The tree change of variables leaves independent Haar on the n non-tree links. The bound in Lem. 3.22 puts their logarithms in a ball of radius $r \le CDe_h$ in the maximum bond norm for a box of side D. The test is gauge invariant on this whole read, so the tree gauge factors can be held fixed. In a box meeting the cut use this calculation only on a product log-Haar chart in which scaling all kept logs preserves the identity mask.

Along a ray write the log datum as tB with $\sup|B| = 1$, $0 \le t \le r$. The kept logs give the corresponding lifted row logs, and the masked row logs are zero. Thus the vacuum continuation of Prop. 3.117 admits a fixed complex disc in the radial variable, on the separated tower with complete top test stencils. For a tested plaquette p let $F_p(t)$ be its matrix curvature increment divided by the squared bare mesh in level-h units, in the holomorphic slice. The uniform bound eq. 3.mlxiv bounds this function on the disc, and $F_p(0) = 0$. Use r at most half the disc radius. The real norm

is the norm on the actual orbit. Cauchy's estimate gives

$$|tF_p'(t) - F_p(t)| \le Cr^2.$$

Put $\lambda = \varepsilon_h/2$. For real fields F_p is a quaternion, so the operator norm is Euclidean in its four real coordinates. Where $|F_p|$ lies between $\lambda/2$ and $3\lambda/2$, its derivative in $\log t$ is at least $\lambda/2 - Cr^2$. If $Cr^2 \le \lambda/4$, every active maximizer has slope at least $\lambda/4$ in this range. The band of relative width $\vartheta \le 1/4$ for the maximum can be crossed only upward. It therefore occupies an interval of length at most 8ϑ in $\log t$ on any ray.

Along each ray the Haar radial density is $t^{3n-1}w(t)$, where w is the product of the sinc-squared Jacobians and is decreasing on the small ball. Its cumulative mass H satisfies $tH'(t) \le 3nH(t)$. An interval of logarithmic length 8ϑ therefore contributes at most $24n\vartheta$ times the mass of the whole ray. Integrating the ray directions gives the same relative bound for the product Haar ball. For $n = 0$ the band is empty. To use it in the weighted history estimate, these retained coordinates must survive the outgoing inverse as a product stock. The band is taken in the stored frame before the exterior supremum. On a wholly kept box the exterior compensator acts by a gauge on the full datum, so the band is the same for every admitted exterior. For a cut box use the actual mask-preserving pullback to check this property. The remaining reference integral must have its bound uniformly in these coordinates.

These conditions hold for the reference integral of an unmarked late last-shell drop-out before a later constraint, when the full level-h box read, including its endpoints and lift halo, is kept and lies in Ω_h. In (1.145) those links are in the dV_h Haar stock, separate from the relative field V'. The ordinary strip pivots at fixed V_h are already eliminated. Unresolved raw targets have both endpoints in Ω_h^c in the old core and are outside this stock. With the test stock fixed, eliminate those remaining targets against their own averaging constraints and keep the old strip variables in their original charts. The unmarked late product envelope after eq. 4.diii then has the same conditional integral bound for every value of the test stock. The radial restriction gives the stated gain in that reference integral, with any already justified event payments. Subsequent constraints that prescribe some of these links require the bound on the corresponding conditional chart.

The same reference calculation applies to a test box crossing the identity cut if its full level-h read lies in Ω_h and all its kept links are in the retained Haar stock. In the class geometry the inner cut is one coarse layer from the hull of the old core and two layers from the outer prepared face. In level-h units write $A = LM\check{R}_{h+1}$ for the coarse layer and $d = LM_2\check{R}_h$ for the test cube side. A fourfold box crossing the cut has width $9d$. Even after the endpoint padding, $9d + 2 < A$, by $\check{R}_h \le L\check{R}_{h+1}$, $M \ge L^2M_2$, and $L \ge 13$. Thus the full padded read clears the old core and stays in the prepared member. Its kept links, including those crossing the cut, are in the second Haar stock of (1.143). The gauge-nest convention excludes the crossing links from $\mathbb{B}_0$, so (1.145) retains them in dV_h. Use this stock after the compensating change of (4c). At fixed exterior and frame this change is a Haar translation at the endpoints of every kept link. With those translated coordinates fixed, the last-shell test sees their masked field directly. Read the other factors at the inverse translation. The conditional reference bound is uniform in those values, so the band in these coordinates can be restricted before the exterior supremum.

To obtain the free logarithmic chart, form the finite graph of the full read and contract each connected component of its identity edges. On the quotient choose a rooted spanning tree from the kept edges. The tree change of variables is triangular: after fixing the tree links it multiplies each non-tree link on the left and right by fixed group elements. It leaves independent Haar on the non-tree links. The site gauge is constant on each contracted component, so every identity link stays at one.

If the artificial plaquette defects on the read are at most e, first apply the ordinary axial estimate on the full box. All its transformed links differ from one by at most CDe. Along any original identity edge the two axial site gauges therefore differ by at most CDe. Replace the gauge on each

contracted component by its value at a chosen vertex. Paths in each component have length at most CD^4, so this changes the gauged kept links by at most CD^5e. Choose the quotient tree by shortest paths from the root. Its depth is at most CD, since contraction cannot increase distance. Putting the field in that gauge costs at most another CD, by telescoping the tree paths. Thus the non-tree logarithms lie in the product ball of radius CD^6e when that radius is small. Apply the radial estimate when its square meets the half-threshold condition. The whole kept graph must be free in this calculation, including the kept crossing links.

The same last-shell restriction applies on a marked late branch whose triangular lower reference chart leaves this full kept Haar stock. Hold its recorded post-gauge coordinates and the inner variables fixed while integrating the lower Lebesgue blocks. The uniform inverse Jacobian and Gaussian bounds give the same conditional majorant e^{CV° as for the triangular reference integral. The bounds for closed inner leaves remain functions of their own stored variables. The last-shell band is Borel in the fixed kept stock and can be inserted before the compatible-evaluation supremum. Integrate it with the Haar and quotient calculation above. This gives the factor $Cn\vartheta$ in the weighted outer integral, with the same inner masses and charged events. The argument requires the stock in the recorded coordinates to remain free after the lower eliminations. For the comparison of runs, the paired reads must place their disagreements in this band uniformly over the admitted evaluations.

The final density test that classifies a restored component uses the reference draw. On that support the restored top field has plaquette defects bounded by $C(M)\varepsilon_m$. The estimate follows by changing the at most four plaquette factors from the reference field, including on the face where the reference agrees with the exterior. Each fourfold density box based in the six-cell inset is contained in the component. Apply the same lifted-datum existence estimate on those full boxes before either classification outcome.

On a complete fourfold test read wholly in the fresh box, first use the normalized reference law at fixed exterior, before prescribing new outgoing rows or splitting the two classes. In a fixed frame the denominator is the centre action factor times $(\theta^{-3/2}\sigma_0)^{n_{\mathsf{G}}}$ times the admitted fluctuation normalizer. The latter has bounded log. The centre action costs at most $Cp_0(g_m)^2(M\check{R}_m)^4$ in the exponent, and n_{G} is at most $C(M\check{R}_m)^4$. Hence the inverse denominator is bounded by $\exp[C(\log\theta)^c]$ on this slice.

The numerator density is at most one relative to the off-tree Haar product. In physical coordinates the non-tree translations preserve Haar. For the gauge-invariant test and its top curvature bound the choice of tree values is immaterial, by site gauges fixing the root. Thus one may integrate the tree values against Haar as well when bounding that event, and discard fresh variables off its read. The top curvature bound places the read in the small axial logarithm ball. The radial estimate at threshold $\lambda = \varepsilon_m$ gives Haar band mass at most $Cn\vartheta$, where n counts the free links of the read. The fixed-exterior probability of this band is at most $C\vartheta\exp[C(\log\theta)^c]$. For the weighted history envelope after lower eliminations, the same radial factor applies where the complete product stock is still free, the remaining reference bound is uniform with that stock fixed, and the band is Borel in the recorded coordinates.

The six-cell inset places the test box inside the component. A test box entirely outside the restored stock reads only the retained field, and a box meeting the restoration boundary can read fixed nonzero links as well as the new draw.

There is a crossing case with the fixed links separated from the tested stencils. Fix the physical tree values first. On a chart whose remaining free links have product Haar reference, use the tree frame to write the retained logs as B_{fix} and the free logs along a ray as tB, $0 \le t \le r$. The fixed logs are held at every t. If all the fixed edges, including the physical reference-tree edges, extend to a common tree with compatible prescribed root gauges, their values can be gauged to the identity. Otherwise retain their actual B_{fix}. The fixed-coordinate part of the lifted datum is formed by the same block copies. Let $F_p(t)$ be the normalized matrix plaquette read of this datum. On the fixed

small response disc the previous Cauchy calculation now gives

$$|tF_p'(t) - (F_p(t) - F_p(0))| \le Cr^2.$$

It is enough that $\sup_p |F_p(0)| + Cr^2 \le \lambda/4$. The active norm slopes in the band are then at least $\lambda/4$, giving the same $Cn\vartheta$ radial bound.

To check the centre condition, interpolate from the vacuum to B_{fix} with the free logs zero. When the complete lifted carriers of all nonzero fixed logs are at distance at least D_0 from the tested stencils, the weighted response eq. 3.mlxvii gives $\sup_p |F_p(0)| \le C\|B_{\mathrm{fix}}\| e^{-cD_0}$ in the normalized curvature units. Thus the radial comparison also applies when the full test box meets the old field but its tested stencils and their padding are separated from the nonzero fixed data by the prescribed collar. This uses the actual product stock in the fixed tree frame and the admitted interpolated data.

For a crossing chart with unscreened fixed rows we must control the centre curvature on its own constraints. The two class weights are compared before removing any implied reference cutoff or frame factor on the first-class side. In the quiet nest, the final test bases in the old enlarged hull lie in the region excluded from the earlier ordinary plaquette-test insertion. The inherited top plaquette bound there admits the background, while the band of its derived bare curvature still needs control in the retained-field integral.

In each case compare like reads on the part of the history where the earlier decisions agree. For a plaquette test the numerical quantity is the curvature at the finest scale of the local minimizer divided by η_t^2 on each respective lattice. A bound on the response at the common top alone does not control that read.

There is a mesh comparison on a matched input to these plaquette tests. For the ordinary and final density tests, take an aligned fourfold box and the same level-i field on the entire lifted read in a common frame. The third enlargement lies in the top member of the separated tower. The tested first enlargement and its complete difference stencils are therefore inside that member. If the top plaquette defect is at most e_i, the common comb gauge bounds the top logarithms by $N = CM_2\check{R}_i e_i$, and the block lift copies them to the induced rows. The comparison of common trial fields below now applies to this read: the complete common row data coincide, the lower unmatched rows are screened, and on the top stencils the first differences have the same two-mesh and Hölder bounds. Pair the literal plaquettes of each orientation by inward rounding. For a bare mesh η in level-i units this gives $|q_\sharp - q_\flat| \le CN\eta^{\min(\alpha,\beta_H)}$ for the maxima. It also applies to a last-shell box when the complete masked top datum is common. Use the ordinary incoming or final reference curvature bound for e_i, or the admitted masked-field bound for the last shell. With the running threshold bounds their ratios to the common test threshold cost at most powers of the scale parameters in this comparison. This gives the numerical part of the test comparison on matched input.

On the region where the two reads are defined and the lesser value is at most twice the threshold, we must show a relative comparison with a positive power of the ratio of the bare mesh to the tested scale, allowing the powers of $\log\theta$ in the history estimates. If the two tested numbers q_A, q_B differ by at most $\vartheta\lambda$ at threshold λ, then

$$|\mathbf{1}_{q_A<\lambda} - \mathbf{1}_{q_B<\lambda}| \le \mathbf{1}_{|q_A-\lambda|\le\vartheta\lambda}.$$

For the approximate test q_A can be the maximum over its nonpivot bonds. The shell is then determined by at least one bond attaining that maximum. The same argument applies to each subsequent decision on the part of the history where the preceding decisions agree.

On a disagreement with $\vartheta \le 1/2$, both numerical reads are at least $\lambda/2$. For the tests with an energy witness obtained by forcing the read below threshold at small energy, run that calculation with threshold $\lambda/2$. In the ordinary plaquette and approximate tests this reduces the witness constant. In the main and last-shell tests it also requires a smaller bound on the reference-field

errors at weak coupling. The witness can therefore be taken on either history, including the passing one. Using it in the mass requires the unspent share of the action or form on that same history. A stopped root whose payment was reserved for a later read uses that reserve only there. This gives the corresponding ordinary-creation or marked-event energy on the full compatible carrier. Its use depends on which action or Gaussian share is available there. The thickness of the near-threshold set still has to enter its integral.

At fixed coarse datum the approximate test has a fixed center on each tested bond. In the strip where the cutoff equals one, the substitution gives $V_k(b)D_b^{-1} = W'X_b$ with X_b fixed by the retained exterior datum and D_b the cube center at the bond. Thus on this strip the read is a radius of a translated stored coordinate. Where the cutoff is less than one, the strip's center reserve keeps the approximate test below $2\delta_k$. In the admitted deep ordinary box the global radius bound and the local-to-global offset also keep it away from $2\delta_k$. Both observations use the margins in the ordinary step. To estimate the first approximate failure, the runs must be compared before the test has fixed Q_t and hence Ω_t. The ordinary fluctuation chart on $\Omega_t^{\sim -1}$ is then unavailable on the failing side.

Before the Q_t split one can use the gauge draw itself on a tested footprint in the gauged blocks. The approximate-test cubes are inset one coarse layer in the gauge region. Their two test layers and one bond reach stay in that region by the cube-size floor. Fix the input before this draw and the preceding plaquette decisions. For a fine test bond choose an endpoint x off the top lattice in a block where the draw is made, and hold the other draws fixed. If y is the block centre, the coordinate $W_x = V_k(y,x)u(x)^{-1}$ has the normalized one-site density of the gauge-fixing factor. Its Haar density is at most $C\mathfrak{t}_k^{3/2}$, by the lower bound on the gauge normalizer. Here $\mathfrak{t}_k = g_k^{-2}$ is the coefficient of the auxiliary gauge weight: it is x_k in the own run and θ_k in the reference run. The transformation is the identity at the top sites, so the coarse average and the local approximate test centre at the bond stay fixed. The relative tested link is a fixed left and right translate of W_x or W_x^{-1}.

The Haar volume of the band about $2\delta_k$ is $C\vartheta\delta_k^3$, for either side of the threshold. Thus the conditional probability for this endpoint is at most $C\vartheta\mathfrak{t}_k^{3/2}\delta_k^3 \le C\vartheta p_0(g_k)^3$. Sum over the bonds that can attain the test maximum. Earlier approximate decisions may be retained as indicators bounded by one in this estimate. This gives a band bound relative to the positive mass before the gauge draw, for both outcomes of the test. The residual half of the gauge exponent in the ordinary-event majorant has the same pointwise bound with the original normalizer. Its mass is bounded uniformly above and is at least one, since the positive exponent cost has been halved. The same band factor therefore applies in that residual reference integral. Any test gain uses the assigned Wilson and gauge share before this integral. To apply it inside a later history mass, the chosen endpoint coordinate and fixed centre must still be available in its reference integral.

For a bond from a block centre to an adjacent site x in that block, the contour mean is just the link and the local axial centre is the identity on it. The test on this bond is therefore radial in the actual W_x. Its threshold in eigenangle is $r_0 = 2\arcsin\delta_k$, inside the gauge draw cutoff by the threshold ratio. In the angular coordinate r the density is proportional to $\sin^2 r\, e^{-\mathfrak{t}_k(1-\cos r)}$ on this chart. On both shells of width $w = g_k/(Cp_0(g_k))$ at r_0, its log derivative has absolute value bounded by $C(\delta_k^{-1} + \mathfrak{t}_k\delta_k) \le C'/w$. The same bound holds for the residual half weight. A one-sided band of width $u \le w$ has at most Cu/w times the mass of that side's shell of width w. Thus a relative band about this single-bond decision has cost $C\vartheta p_0(g_k)^2$ in its own pass or failure mass. Other tests that read this same gauge variable can restrict the comparison shells; they retain their indicators in the actual prefix law.

On the failing side of a first ordinary approximate decision, the enlarged failed cube is outside Ω_t. A complete endpoint block of the test in that enlargement uses the old raw Haar field and the averaging delta in the retained one-step operation. Use the rooted tree coordinates of that block, with the top site fixed and all incident links of the moving sites included. They separate the block

quotient and the site gauges with Jacobian one. At fixed quotient the change from the site gauge to W_x is a Haar translation and inversion. The outgoing average is independent of these site gauges. The centre is thus fixed in the reference coordinates, both if the outgoing datum is retained and if it is eliminated against its own averaging delta. The band is a function of the recorded stock before the exterior-pair supremum.

On an untouched piece with this complete block free of incoming constraints on the moving sites, keep the quotient fixed while applying the residual product majorant. Extract the assigned failure gain first. The remaining one-site gauge factor is the half-weight above, and restricting it to the band gives $C\vartheta p_0(g_k)^3$ with the other factors in the same majorant. The first step has no earlier averaging constraints. At a later step the same argument requires that the earlier constraints and reads leave these gauge coordinates available. A passing coordinate integrated in Λ_t uses its own conditional window.

There is a local estimate before the Q_t split. Fix V_t, the recorded plaquette decisions, and the other free variables in a complete row chart for $\mathcal{M}V_k = V_t$. On a branch where the actual middle pivots can be eliminated and b is a free bond, its reference factor is Haar. Reconstruct the pivots and evaluate the density and gauge factors at those reconstructed values. The centre $D_b(V_t)$ is fixed. A band of relative width ϑ about the group-norm threshold $2\delta_k$ has Haar measure at most $C\vartheta\delta_k^3$. In $SU(2)$ the operator-norm distance from the identity is $2\sin(\phi/2)$, which has derivative bounded away from zero at these small radii. Thus the restricted integral in this slot is at most $C\vartheta\delta_k^3$ times a pointwise upper bound for the reconstructed density in the slot, including its pivot Jacobians. A union over the free bonds that can attain the maximum gives the bound on these charts. In the full pre-Q_t history one must still control those densities uniformly in the retained coordinates in the order of the weighted mass integral.

The main-shell fluctuation test has a different shell estimate. On the preceding positive exterior and extendibility support, the box critical-orbit construction gives V_Λ and the mixed reference center V_0 from the fixed exterior. These are available before the main-shell decision at each cutoff. In one retained frame use the relative links $V'(b) = V''(b)V_0(b)^{-1}$. First eliminate the central delta chain from the top down. The relative Haar translation and the normalized tree constraints then leave the free indices $(\mathbb{B}_0 \cap X) \setminus rT_0$. The test reads the largest $SU(2)$ eigenangle of these links, with $\phi(-1) = \pi$. Thus the logarithm cut gives a numerical failure at the threshold $\delta'_m < \pi$. On each free Haar factor the band $|\phi - \delta'_m| \le \vartheta\delta'_m$ has measure at most $C\vartheta(\delta'_m)^3$, by integrating $\sin^2\phi\, d\phi$. The condition is independent of the exterior in the relative coordinates of that run and frame. At a matched stored link and frame, the difference of the two relative angles is at most the group distance of the two centers by the bi-invariant triangle inequality. The maximum obeys the same bound. Comparing the centers between the two cutoffs still uses the response comparison. In the admitted passing-side product envelope eq. 4.diii without earlier marks, use the product majorant in the free B-coordinates after bounding the exterior dependence. Replace a scaled B-ball of radius δ'_m/g_m by its inner annulus of relative width ϑ. Its volume is at most 3ϑ times the ball volume. Sum over the free coordinates that can realize the maximum, with the same other factors in the envelope. This gives a relative $Cn_B\vartheta$ bound for that product estimate, where n_B is the free-coordinate count.

On the failing side, consider a late drop-out with no earlier marks in the core. The completed frame operation leaves normalized Haar in each non-tree shell slot. In its recorded frame it is the relative link V' that is held fixed in the exterior supremum in eq. 4.cdlxxi. Changing the exterior reconstructs $V'' = V'V_0$ at that same slot. The shell indicator is therefore a function of the stored variable. It can multiply the pointwise majorant before the exterior supremum in eq. 4.dviii. The unmarked late reference bound after eq. 4.diii is independent of the shell Haar variables. Restricting that product envelope to the band gains $Cn_B\vartheta(\delta'_m)^3$ relative to its full Haar integral, with the same strip factors. Any event payments used here are the ones already justified on that failing history. To use the bound for disagreement, the paired centres must place the disagreeing records in this band uniformly over the admitted evaluations.

The failing-side gain also holds for marked late branches with the triangular reference charts of eq. 4.dii, provided the selected shell slots remain free Haar factors. Write the outer reference measure as $dv_{\mathrm{sh}}\, d\mu^{\circ}_{v_{\mathrm{sh}}}$, where v_{sh} comprises the relative shell slots and dv_{sh} is product Haar. Keep v_{sh} and the inner variables fixed during the successive integrations of the lower Lebesgue blocks. Their inverse Jacobian bounds and Gaussian reserves are uniform in these group arguments. The strong majorants of the closed leaves in eq. 4.dviii depend on their own stored variables. Thus

$$\int g_{\omega}(v_{\mathrm{sh}}, \cdot; \xi_{\mathrm{in}})\, d\mu^{\circ}_{v_{\mathrm{sh}}} \le e^{C_5 V^{\circ}}$$

uniformly in v_{sh} and ξ_{in}. Integrate only the shell band in dv_{sh}. The right side then acquires the factor $Cn_B\vartheta(\delta'_m)^3$ before the same sum and integration of the inner histories. The event payments and volume weights are those of the original history.

This applies with closed inner leaves and older regular strip charts whose pivot eliminations are complete. An older raw row with a free Haar target can also be eliminated by substituting its average for the target in the remaining reads, starting from the highest such row. Here the substitutions must leave the same lower reference charts, with their bounds uniform in the fixed shell. Fixing a shell link can change the coarse datum of an ordinary pivot chart. The unresolved lower raw targets of the frozen core lie inside the inner cut. They are separate from the native shell slots.

For prescribed targets in the first outgoing rows of the late stop, fix the reconstructed exterior and the old shell. Each row that reads the fresh box has a free middle Haar link, including a row crossing its face. The old frame tree acts on the old field. Use the joint inverse of the fresh rows at the prescribed targets, with their native branch version, and evaluate the retained weights at the reconstructed links. The factor per row is bounded by $2L^9 + 1$, uniformly in the old shell. These factors use the existing first-outgoing volume count.

An older restored box also gives its joint first-outgoing density at prescribed targets in its protected row stock. There is a two-level case in which one such target is the fixed middle link of the next raw row. Suppose the complete two-level backward blocks of this row lie in that restored box and the row is still carried in the ordinary raw graph. Its alternate face link belongs to the same first-outgoing stock and is left free. The restored cutoff bounds the plaquettes on the backward blocks. The local averaging estimate in Prop. 3.6, applied to the complete two-level carriers, gives small curvature on the next double block. The regular alternate pivot then gives the joint bound at the fixed middle link and fixed next output, with the same other stored variables. Both inverse densities enter the old core integral. They are uniform in the relative shell and use its slots only through the fixed parameters of their charts. A prescribed output in a different graph still requires the joint bound for that graph with the shell fixed.

For a sharp preparatory collar test there is a conditional shell bound in its native free exterior stock. Fix the other stage variables, including the top datum determining the centre. In the scaled coordinate the threshold is $t = \delta'_i/g_i$. Use the width $W = (C_{\mathrm{sl}} p_0(g_i))^{-1}$. The surviving ordinary completion factors use the separate strip stock. The other test on this coordinate is the approximate collar bound at $2\delta'_i$, with the regularity read fixed in this conditional integration. In the local convention the two centres coincide. In the global convention their scaled separation is at most $W/4$ by the collar response. Since $W = o(t)$, the approximate test passes on both one-sided shells of width W. The density slope in eq. 5.cdxiii is at most $2/W$ there, using the waiting-window comparison of p_0. The radial Jacobian varies by a fixed factor. Consequently restricting one of these shells to distance $u \le W$ from the threshold gains Cu/W in the conditional stage mass. In angular units a relative width ϑ corresponds to $u = \vartheta\delta'_i/g_i$.

In the part of the outer collar receiving a fresh normalized gauge draw, the same one-site argument applies to the local sharp test before its decision. Take a tested bond with a noncoarse endpoint in a complete new block and keep the other variables fixed. The site gauge must fix the determining

nodes of the prospective centre. The relative link then has a translated one-site density bounded by $C\mathsf{t}_i^{3/2}$. A band at angle δ_i' has conditional mass at most $C\vartheta(\delta_i'\sqrt{\mathsf{t}_i})^3 \le C\vartheta p_1(g_i)^3$. On this band the same-centre approximate test on the bond passes. This calculation uses the untransformed local prefix, on either side of the sharp threshold. If the bond joins the block centre to its neighbour and the stage uses the same axial gauge, the test is radial in that draw. The preceding one-sided shell argument with δ_i' gives the bound $C\vartheta p_1(g_i)^2$ relative to the single-bond outcome. For the full maximum use the predecision union bound over the bonds with such a draw. Carried strip links and the other selectors still use their own reference factors.

For a stopped component, the top datum determining this centre can be reconstructed from the carrying field in the compatible-evaluation supremum. The conditional collar bound fixes that datum first. To use it in the marked integral we need a relative shell coordinate in the actual stored stock, or a uniform centre-spread bound in a fixed chart.

On a global sharp failure with the full sharp mask in the transformed free exterior stock, this band is fixed in the stored coordinates W_b'. For relative width $\vartheta \le 1/8$, the band about δ_i' lies below the onset $4\delta_i'/3$ of the cutoff substitution. Indeed, above the onset the original global angle is at least $4\delta_i'/3 - \delta_i'/8$, outside the band. Below the onset it is exactly the angle of W_b'. The substitution there is a right translation with Haar Jacobian one at fixed exterior. The maximum band is therefore Borel in the stored stock. On the part of the weighted envelope with these slots still free, restrict one slot that attains the maximum and use its Haar volume $C\vartheta(\delta_i')^3$. Summing over the free slots gives the band bound with the same conditional majorant for the other variables. A tested bond outside the transformed mask has to keep its own read. In particular the local sharp failure uses the preceding conditional calculation on its untransformed prefix.

For the other marked histories we must show that restricting the weighted mass of the operation to this shell gains a factor $C\vartheta^a \exp[C(\log\theta)^c]$ over its large-field bound, for some $a > 0$. Here θ is the frame at the finest depth held in the histories counted by that operation. The comparison must also sum these restricted masses over the possible first decisions with the localization weights of the recursion. The window estimate eq. 5.cdxix uses Lem. 5.73, which fixes the other stage coordinates and varies a literal nonpivot logarithm. To use it on a derived background read would require a change of variables controlling the conditional mass and preserving the readsets of the surviving history. Even for a literal translated coordinate the remaining tests can have distinct centers. The full shell comparison uses the stage law after summing the earlier gauge draws. A bound on the normalized positive survival probability at a fixed exterior pair does not bound the mass in eq. 4.cdlxxiv, where the exterior-pair supremum and the mark weights enter before the integration and summation. For the extendibility test the infimum is a function of the prescribed top exterior alone. Its shell indicator is constant on the central fibre at fixed exterior.

On the complete exterior carriers of the extendibility witness there is a uniform separation at low energy. Let E be the assigned exterior Wilson share at level i. The axial extension with the old exterior and the lifted-datum estimate give a candidate whose tested curvature, in level-i units, is at most $C(M)\sqrt{E/\theta_i}$. The infimum is no larger. Hence $E \le c(M)\theta_i\varepsilon_i^2$, with $c(M)$ chosen for the half-threshold, puts the read at most ε_i. This leaves a margin of at least ε_i to the test value $2\varepsilon_i$. Apply this on both runs when the complete carriers are available in both. There is no disagreement on that low-energy part. On the rest, a shell gain requires a bound using the exterior read. The central fibre estimate supplies no such bound.

The two infima can be compared on regular trials near their common threshold. For a trial V let $q_b(V)$ be the normalized tested supremum at cutoff b, and put $q_b^* = \inf_V q_b(V)$. The box clearance puts the full averaging blocks of the top plaquettes changed by the trial, including its outer layer, in the tested top inset. If $q_b(V) \le 5\varepsilon_i$, the local averaging bound gives defect at most $2q_b(V)$ on each of those coarse plaquettes. With the prepared exterior regularity on the unchanged carriers, the same trial bonds are therefore admitted at the other cutoff, at weak coupling. Use the actual extension class over that exterior.

On aligned class boxes the near-threshold trial supplies a small datum on the entire lift. A top plaquette of the box whose full averaging carrier is inside the tested inset has defect at most $10\varepsilon_i$ by the local average estimate. The remaining top plaquettes are within a fixed top stencil width of the outside of that inset. Choose a density cube containing one of their corners. It lies in the carried positive region $\Omega_i^{\sim 4}$. Its one-layer enlargement contains the full read of the plaquette, in the top part of its own fourfold tower. The clearance between the trial box and this read leaves that density datum unchanged. The positive test and the same averaging bound give defect at most $2\varepsilon_i$ there. The lift on the class box uses only its top links and complete backward blocks. In a rectangular lift the top field in a common rooted comb gauge has logarithms bounded by $N = CM\check{R}_i\varepsilon_i$, and the induced lower row logs have the same bound.

The test lift is on the minimal recursive tower of the component. Use the same physical comb on the top field in both runs. The block lift is the identity on internal bonds and copies a top link on each bond crossing its corresponding block face. Each one-step averaging path is either internal, with transport one, or crosses that face exactly once, with transport equal to the copied top link. Its contour transports are one. Thus all paths in the mean give the same value, and the one-step average of the finer lift is exactly the coarser lift. Hence the literal prescribed datum agrees on complete common rows, and the reduced datum agrees where its outward read also agrees. The physical recursive tower has the two-cell inset at every level above its first lower layer. The two cutoffs therefore have the same upper regions on this aligned box. The reduced outward read uses only the adjacent lower block, so the unmatched stock stays in a fixed number of lower layers after enlarging it by that read. At the trivial base, Cor. 3.178 identifies the Landau potential of the test lift with the response to these actual reduced data. Use the two-depth response estimate on the common top layer, with the error weight screening the unmatched lower stock. On the full common top stencils it gives a difference bounded by $CN\eta^{\alpha}$ between the transferred fine potential and the coarse potential, including one difference. The individual first differences have a Hölder bound of size CN there.

To compare the literal plaquettes, in this chart expand the four links in the mesh. The normalized curvature has the linear curl and the quadratic commutator of the potential, with an error $C\eta(N + N^2 + N^3)$. At the trivial base the per-length linear transfer is the straight path mean: the coarse component is the average of the transported fine component along the L-step straight paths, with total weight one. A coarse difference of this mean is an average of fine differences, by telescoping the L fine translations. The difference from any of the nearby fine first differences costs $CN\eta^{\beta_H}$ on the common stencil. The same pairing compares the potential values in the commutator. Each tested plaquette has a same-orientation partner within $C\eta$ by rounding inward in the common tested box. The clearance to the tower boundary puts these partners on complete top stencils. Taking the maxima in both directions therefore gives

$$|q_\sharp(V) - q_\flat(V)| \le CM\check{R}_i\varepsilon_i\,\eta^{\min(\alpha,\beta_H)}.$$

Here the box, the exterior data and the top trial are the matched ones. The unmatched bottom rows use their one-run bounds and screening. In the class range $\check{R}_i \le i$, so in the flat level units $\eta = L^{-i}$ this gives a relative error CL^{-ci} for some $c > 0$. If the exterior links instead differ by at most b_{e} in a common top frame, use the same free trial links and substitute the other exterior. The lift copies these links or the identity. Its induced row logs differ by at most Cb_{e}, provided both data stay in the small chart. At a single cutoff the screened derivative of Cor. 3.178 gives an additional Cb_{e} in the normalized curvature on the tested top stencils. This term uses the actual exterior matching estimate.

Suppose the smaller infimum is at most $4\varepsilon_i$, and the comparison of the two bare lifts on these common trials gives $|q_\sharp(V) - q_\flat(V)| \le d$, with $d < \varepsilon_i/2$. Use a trial arbitrarily close to the smaller infimum. Its value bounds the other infimum above by the first plus d. Trials approaching that

other infimum also lie below $5\varepsilon_i$, so reversing the argument gives $|q_\sharp^* - q_\flat^*| \le d$. This reduces the pointwise transfer to the uniform local comparison on the near-threshold trial class.

The infimum also has a radial band estimate on a free exterior logarithmic chart. To see this on the aligned class box, fix a rooted tree of the exterior graph. The exterior of the inner rectangle has paths of length $O(D)$ to the root along the outer faces, where $D = CM\check{R}_i$ is the box side. Choose the tree with those path lengths. Include the outer crossing links of the test read as leaves. At an aligned rectangular face each outside site on such a link has just one neighbor inside the box. Extend it into the inner rectangle by axial columns from a face. The added tree links are free trial links. In the extended tree gauge all paths to the root still have length $O(D)$, and this gauge keeps the prescribed exterior in its original tree chart.

A near-minimizing trial in the threshold band has top plaquette defects at most $C\varepsilon_i$, by the preceding trial and density bounds. Its ordinary full-box axial gauge has links within $CD\varepsilon_i$ of one. Each link in the fixed extended tree gauge is a loop of length $O(D)$ in that field. Thus the exterior and free trial logs in the fixed chart are bounded by $r = CD^2\varepsilon_i$.

Write $\lambda = 2\varepsilon_i$. For a combined vector B of exterior and trial logs with $\sup|B| \le r$, the vacuum continuation of the test lift is holomorphic on the fixed datum ball. For each normalized matrix plaquette read F_p, its Taylor series at the vacuum gives, uniformly for $0 \le u \le 1$,

$$|F_p(uB) - uF_p(B)| \le Cu(1-u)r^2.$$

Indeed the constant term is zero and the linear term cancels. Each term of degree $m \ge 2$ has factor $|u^m - u| \le (m-1)u(1-u)$, and Cauchy's bound sums the resulting series.

On an exterior ray let $t_1 < t_2$ both lie in the band of relative width $\vartheta \le 1/4$. Take a trial arbitrarily close to the infimum at t_2 and multiply every log, exterior and interior, by $u = t_1/t_2$. The tree links remain fixed and the scaled trial lies in the admitted class at t_1. The previous inequality and passage to the infimum give

$$q^*(t_1) \le uq^*(t_2) + Cu(1-u)r^2.$$

The condition $Cr^2 \le \lambda/4$ holds at weak coupling: D is bounded by a fixed power of $\log g_i^{-2}$, while $\varepsilon_i = g_i p_0(g_i)$ and p_0 is a fixed power of that logarithm. Hence $r^2/\lambda \le CD^4\varepsilon_i \to 0$ as $g_i \to 0$. The band bounds then give $1 - u \le C\vartheta$ and $u \ge c > 0$. The band's logarithmic radial diameter is therefore $O(\vartheta)$. The Haar radial estimate gives a relative bound $Cn_{\mathrm{e}}\vartheta$ on a product ball of n_{e} exterior non-tree links.

At the native extendibility stop the top exterior is fixed in the infimum defining q^*. The product bound enters a history integral only where those exterior links form a free stock in a containing operation and the uniform reference bound is taken with that stock fixed. A prescribed outgoing row can have its whole double block in the exterior. In that case its constraint belongs to the same stock, and the weighted shell requires the bound on the constrained coordinates.

There is an integrated exterior stock when this extendibility stop is immediately below the floor $h = i + 1$ of a containing operation that reaches its shell or last-shell stop. Take an old component whose tested box and stencil pad are disjoint from the other marked charts. Include both its tested exterior and its previously generated targets in Z_i'' when listing the ordinary h-rows that read it. With the full row carriers in the eight-cell raw buffer, every target on this list belongs to the integrated inner field in (1.145). The old prepared targets in Z_i'' and the old exterior links retain their separate factors in the joint integral. Integrate the listed h-targets against their own Haar deltas, substituting their row values in every remaining factor. The other h-rows do not read the listed old stock. In the absence of a later prescribed target on this list, this leaves product Haar on the old exterior, with the old inner graph evaluated on its own targets.

Apply the pointwise reference majorant of the containing operation before taking the exterior supremum. On its admitted chart the majorant is uniform in the free old group links after the other reference factors have been bounded. The exterior tree change preserves their Haar factors. The

band depends only on these stored exterior coordinates at level i. It therefore costs $Cn_{\mathrm{e}}\vartheta$ relative to the product-ball majorant for this chart. With identical top exteriors the relative tolerance above is $O(D\eta^{\min(\alpha,\beta_H)})$. Since $n_{\mathrm{e}} \le CD^4$ and $\check{R}_i \le i$, the resulting $CD^5\eta^{\min(\alpha,\beta_H)}$ is still geometric in i.

In the same immediate-floor geometry, take an old second-class component of level i with a final density test. Use the complete fourfold read and its row carriers in the raw buffer and include its old exterior and restored links in the list of reads. If all affected h-targets are integrated, the same Haar delta elimination leaves product Haar on the old exterior of the test. A read wholly outside the old proposal box uses only this stock. The negative test of the old second-class component belongs to the containing core's payment. Keep its actual failed cube for that payment before bounding the remaining reference factors.

For a read crossing the old proposal boundary, use the physical off-tree links in the recorded frame. The old reference centre is determined by the old exterior stock. At fixed old exterior the relative non-tree Haar factors give the physical ones by right translation. After bounding the reference density, restore the interior site gauges for this invariant band event, fixing the exterior sites. This replaces the interior tree constraint by normalized Haar integration. Together with the old exterior factors, Fubini gives a product Haar majorant on the complete physical test read. The complete top curvature bound and the radial estimate at $\lambda = \varepsilon_i$ give a factor $Cn\vartheta$ relative to its product majorant, with n the free-link count. The reference normalization is bounded uniformly on the admitted slice; use that bound once if it has not already been included in the pointwise envelope. The band here is a function of the recorded physical links and can be imposed before the compatible-exterior supremum.

Some of these estimates also hold with prescribed h-targets. Consider rows crossing a face of the old trial box Λ^{prop}. Its faces are aligned with the h-blocks. The middle link of such a row crosses the face and is among the free trial links in (1.142). It is thus outside the exterior read of the infimum. The trial box includes a full M-layer beyond Z_i'', so this link is also separate from the old prepared targets. It is an original Haar link in the ordinary raw average. Solve the prescribed row in this link, keeping its two endpoint blocks and all weights that read the solved link in the joint integral. Distinct rows have private middle links. The row kernel has mass at most $2L^9 + 1$ per prescribed target, uniformly with the exterior stock fixed. Each such row still reads the old exterior in its outside endpoint block.

There is another choice of link for rows that cross the outer face of the old rectangular component. The middle link can now be in the test read, since the lift uses links meeting that component. Orient the row outward and choose an axial link on the part of the straight path strictly in the outside endpoint block. Take it beyond the test pad and before or at the outside coarse centre. The choice is available for large L and for the complete face carriers in the raw buffer, disjoint from other marked charts. This link also has original Haar measure.

The carried positive density cubes in the outer collar supply the regularity for this second choice. At the face their tested enlargements cover the two endpoint blocks, including the full averaging reads of the top plaquettes there. Thus the top defects on the double block are at most $C\varepsilon_i$, also on a failing extendibility branch. Choose a rooted tree of the double block avoiding the chosen link. The small-plaquette estimate puts every admitted field into a log ball of radius $C_L\varepsilon_i$ in this gauge, which depends only on the other links. At the vacuum the derivative of the logarithm of the row in the chosen link is $m\mathbf{1}$, with $m \ge L^{-3}$. Indeed, the L starts on the straight axis in the contour formula each contribute the link once in the positive direction. The off-axis starts can read it only on the reversed contour in the outside block, also in the positive direction. Regularity of (1.28) gives derivative distance at most $m/2$ from $m\mathbf{1}$ on a small convex log ball. There is at most one inverse there, with Haar density bounded by C_L.

An outside endpoint block of a face-crossing row has only one neighboring block in the aligned rectangle. Distinct such rows therefore use different outside blocks, and none reads the chosen link of another. First use the private middle-link kernels for the inner-face rows, conditional on all

the outer-face links. The outer-face rows do not read those middle links. Then use the outer-face inverses, with all retained factors evaluated at the solved links. The resulting bound has the same $Cn_e\vartheta$ from the exterior stock, with the product of the row constants in the reference bound. Those constants are charged to the number of prescribed rows. Other affected targets must be integrated or have their own inverse estimate with the test stock held fixed. In particular a row wholly in the exterior can constrain the quotient links. Its site-gauge variables give another case in which the band estimate applies.

For the final density test there is also an outside-pivot construction relative to its complete fourfold read S. Take a rectangle S aligned with the next averaging blocks. A prescribed row crossing its face has a straight-axis link strictly in the outside endpoint block, beyond S and before the outer centre. Use that link as its pivot when it remains in the raw Haar stock. The six-cell inset leaves the complete double block inside the component. The physical field on the final reference support has top plaquette defects at most $C(M)\varepsilon_i$ there. The tree avoiding this link therefore gives the same small regular inverse ball as in the outer-face argument, with constants depending on L, M. Different face-crossing rows use different outside blocks. Solving them consumes no link of the test read. Other rows reading these pivots must have their targets integrated or be solved in a compatible inverse order. A row whose whole double block lies inside S has no outside pivot of this kind.

In the rooted exterior tree coordinates write the actual exterior field as $V(b) = g_{b_-}^{-1}\widehat{V}(b)g_{b_+}$, where $\widehat{V}$ is the tree-gauged field. The change separates product Haar on the non-tree links and on the nonroot site gauges. The extendibility infimum depends only on $\widehat{V}$, since a site gauge extends to the trial fields over the inner box. Suppose the remaining prescribed rows have their full double blocks in the exterior stock and form a forest of coarse bonds. Choose a root for each tree, using the fixed root of the exterior gauge if the tree contains it. Take the nonroot coarse sites to have their full fine link stars in the free stock. By covariance each prescribed value is

$$W(c) = g_{c_-}^{-1}\mathcal{M}(\widehat{V})(c)g_{c_+}.$$

For fixed quotient links and fixed root gauges solve the rows in the child gauges, starting at the roots. The conditional Haar factor is one for each solved row. Substitute the gauges in all remaining reads before applying the reference bound. This keeps the full quotient Haar integral for the band. The previous face-row inverses, if needed, can be taken first with these gauges fixed, since the forest rows read none of their pivots. This forest calculation also applies to the final density read on a physical stock with these free site gauges. A reference tree delta still present before classification must be included in the joint gauge calculation with the prescribed rows. Its removal after a successful class test does not supply the gauge variables for the preceding decision. On an already ungauged stock the plaquette norms are gauge invariant, so at fixed quotient links the values prescribed on a forest change the solved site gauges and leave the test value unchanged. The children must have their complete link stars in the free stock. If the prescribed graph has cycles, solving a spanning forest leaves a constraint on the ordered product of the quotient row values around each fundamental cycle. That product must equal the product of the prescribed values conjugated by the root gauge. The remaining estimate is on these constrained quotient coordinates.

Some of the remaining cycle constraints can be treated together. Use the same forest and give each nonforest row its private fine middle link. The exterior tree can avoid all these links. A transverse plaquette detour replaces each forbidden link in an exterior path by a nonmiddle crossing and two within-block links. The root paths still have length $O(D)$. Thus the near-minimizing trial estimate in this tree again gives the log radius $r = CD^2\varepsilon_i$.

Let x denote the non-tree exterior logs apart from the selected pivots. For a coarse vertex v, let $A_v(x)$ be the ordered product of the quotient row values from its forest root to v, and T_v the product of the prescribed values on that path. If the root gauge is g_0, the solved gauge at v is

$A_v(x)^{-1}g_0T_v$. The remaining equation for a row c from v to w is therefore

$$\mathcal{M}(\widehat{V})(c) = A_v(x)^{-1}Z_cA_w(x), \qquad Z_c = g_0T_vW(c)T_w^{-1}g_0^{-1}.$$

The forest paths read none of the selected pivots. Each remaining row reads only its own pivot from this stock. Thus the inverse in all these links is diagonal even when the forest paths overlap. At the vacuum the single-row pivot derivative is $L^{-3}\mathbf{1}$ in the oriented log coordinates by (1.31). The regular one-row chart gives a unique inverse $p_c = P_c(x)$ with conditional Haar density bounded by C_L per row on the near-band chart. In taking the radial integral keep the prescribed values and root gauges fixed.

Write ℓ_* for one plus the greatest sum of the two forest path lengths used by any such row. The geometry gives $\ell_* \le CD^4$. On the near-band chart $|Z_c - \mathbf{1}| \le C_L\ell_*r$. On a complex radial disc of radius $c_L/(\ell_*r)$, the forest products stay in a fixed small ball by the product estimate. The target of each one-row inverse stays in its regular ball at weak coupling. Cauchy's estimate on this disc gives, for $0 \le u \le 1$,

$$|P_c(ux) - uP_c(x)| \le C_L(1-u)(|Z_c - \mathbf{1}| + \ell_*^2r^2).$$

Indeed $P_c(0)$ has size $C_L|Z_c - \mathbf{1}|$. The linear part cancels, and the terms of degree at least two cost $C_L(1-u)\ell_*^2r^2$.

To compare two radii on this graph, scale a near-minimizing trial at the larger radius, then correct all its selected exterior pivots to their values at the smaller radius. The corrections are to the non-tree links in the fixed chart. Their lifted data and their reduced data use the full local carriers of the corrected links. For pivots whose target products satisfy $|Z_c - \mathbf{1}| \le c_L\varepsilon_i$ use the unweighted normalized response bound. For the others suppose the complete carriers are in the middle of the two-cell outer density collar, a distance $d_0 \ge cM\check{R}_i$ from all tested stencils. In eq. 3.mlxvii use the weight given by distance to the union of their carriers. This gives a normalized curvature response bounded by Ce^{-cd_0} times the supremum of their corrections. Including the curvature scaling error of the trial itself, the extra term in the radial inequality is bounded by

$$C_L(1-u)\big[c_L\varepsilon_i + \ell_*^2r^2 + e^{-cd_0}(\ell_*r + \ell_*^2r^2)\big].$$

The terms containing r^2 are $o(\varepsilon_i)$ by the logarithmic class size. The screened term linear in r is also $o(\varepsilon_i)$, since $\ell_*D^2e^{-cd_0}$ tends to zero. Choose c_L small so the total coefficient of $1-u$ is below $\lambda/4$. The earlier logarithmic radial diameter bound applies to x. The product of the one-row inverse bounds is charged to the number of prescribed rows, and the band still contributes $Cn_e\vartheta$. For a nonscreened pivot at a general target product one must control the curvature produced by its offset on the constrained exterior chart.

For a final test it suffices to screen the pivots from the actual tested stencils. With $r = CD^2\varepsilon_i$, the required bound on the linear offset is

$$C_Le^{-cd_0}\ell_*D^2 \le c_0.$$

Since $\ell_* \le CD^4$, a sufficiently large multiple of $\log(2+D)$ for d_0 suffices. The outer part of the fourfold test box can provide this clearance to its tested first enlargement. Use the complete lifted pivot carriers in measuring that distance.

There is a sufficient rule for choosing the forest in the screened case. Declare a prescribed edge near when the complete lifted carrier of its available private pivot is within d_0 of the tested stencils. If those edges form a forest, extend it to a spanning forest of the prescribed graph. Every remaining chord then has a far pivot, so its cycle-product offset enters only through the screened carriers above. The forest rows are solved in the site gauges. The criterion uses the whole pivot carrier, including its double block and induced outward rows. A cycle among the near edges leaves a near chord under every choice of forest, and its invariant target product remains in the centre comparison.

To compare the two cutoffs on this chart, take the same prescribed graph, forest and exterior tree at the common top level. Use the same remaining quotient links in the two reference integrals.

The row maps at levels i and $h = i + 1$ have the same double blocks at matched depths. The forest products $A_v(x)$ therefore agree. Different prescribed forest values change only the solved site gauges. The test infimum is invariant under those gauges. For the remaining rows the only prescribed values entering the quotient graph are the root-conjugated cycle products Z_c. If these agree, so do the solved quotient links. This also treats prescribed forests, for which there are no cycle products. If the two prescribed fields are related by coarse site gauges u_v, their path products satisfy $T_v^\flat = u_0 T_v^\sharp u_v^{-1}$. Choose the root of the exterior tree at a fine site outside the prescribed graph. Pair the free forest root gauges by $g_0^\flat = g_0^\sharp u_0^{-1}$. This preserves their Haar measures and gives equal Z_c on all remaining rows. For the test read alone the two prescribed forests may even differ, provided their site gauges are free on the same exterior stock. Each forest removes only its own gauge coordinates, so the test uses the same quotient field in the two integrals. The same observation applies to additional branches attached to a common cyclic part at one vertex each. Such branches leave its cycle products unchanged. Use the reference bound of each graph for its other factors.

For unequal products compare the two one-row inverses at fixed x. The targets differ by $A_v(x)^{-1}(Z_c^\sharp - Z_c^\flat)A_w(x)$. On the common regular ball the difference of the solved log links is at most $C_L|Z_c^\sharp - Z_c^\flat|$. At fixed root gauges the difference of the products is bounded by the sum of the prescribed row differences along the two forest paths and the remaining row. This follows by telescoping products of real unitary matrices in a common root frame. The regular Haar inverse density for such a row is the same function of x and Z_c at both cutoffs. Its logarithm is Lipschitz in the solved log link on the fixed regular ball, by the determinant formula and the local inverse bound. Thus the absolute log ratio of these row factors is at most $C_L \sum_c |Z_c^\sharp - Z_c^\flat|$. In particular these factors agree for equal products.

Write b_{near} for the maximum product difference on the unscreened rows and b_{far} for the maximum on the screened rows. The other exterior links agree on the quotient graph. Transfer a near-minimizing trial from the passing side by keeping its free interior links and replacing the exterior pivots by those of the other graph. First compare the two lifts at the original trial, then replace the pivots in the other lift. The pointwise comparison at the original trial and the unweighted and weighted response bounds on the two sets of pivot carriers give an increase of at most

$$d = CD\varepsilon_i\,\eta^{\min(\alpha,\beta_H)} + C_L(b_{\mathrm{near}} + e^{-cd_0}b_{\mathrm{far}}).$$

Here the changed data must stay in the common regular ball. If one test passes and the other fails, the failing infimum is at most the passing infimum plus d. Consequently both infima are within d of λ. It suffices to use the band of the reference graph on the remaining quotient stock. When $d/\lambda \le 1/4$ and the radial bound applies to this graph, the factor is $Cn_{\mathrm{e}}d/\lambda$ in the common reference majorant. The inverse row densities remain in that majorant. A geometric bound in the bare mesh uses a mesh-small product discrepancy in the normalized width d/λ. The reference graph's prescribed cycle products must be fixed at this stage, as in the fixed-output convention for the compatible evaluation supremum. The other graph can vary there if the comparison by d is uniform. The reference band is then independent of that evaluation. Integrate it at fixed reference root gauges, and then integrate those gauges against Haar.

Replacing the sharp tests by Lipschitz functions allows pointwise telescoping, with a separate estimate for the modified one-run operation. The marked remainders and hence the tuned coordinates in that argument belong to the modified recursion.

Calibration maps and their admitted cocycle. For the same bare integral let $C_a(\beta)$ and $C_b(\beta)$ be two actual terminal-coupling coordinates, on domains where the branches are holomorphic and invertible. Define

$$H_{ab} = C_b \circ C_a^{-1}. \tag{A.cix}$$

On an admitted common domain with the same bare branch, one has

$$H_{bc} \circ H_{ab} = H_{ac}, \qquad H'_{ab}(x) = \frac{C'_b(\beta)}{C'_a(\beta)}, \qquad \beta = C_a^{-1}(x). \tag{A.cx}$$

These follow by composition and the chain rule. Every intermediate inverse must be the admitted branch. Geometric overlap of two complex discs does not establish that fact.

Moreover $|H_{ab}(x) - x| = |C_b(\beta) - C_a(\beta)|$. This is the actual calibration discrepancy to estimate. Equality after a change of physical units does not bound it. If the two presentations are known to represent the same bare integral on an admitted real open overlap, their holomorphic identities extend from that overlap by the identity theorem. The real admission is an input, as in the flow section.

Typed closure is a family of finite-grade estimates. A nonscalar insertion must come with its own gauge-covariant source core, counter-insertions, and tensor readout. At finite grade, label every source monomial and all cross terms created by the actual finite map. If several directions share one potential or zone reserve, their joint radii obey eq. A.xcix, with the multi-direction cost following it. A scalar α^2 gain is not copied into another channel without its own field-degree and counter-insertion calculation.

Parity proves eq. A.c only for the actually admitted unmarked family. Its one-loop coefficient in eq. A.ci, the marked corrections, and the finite-grade closure still have to be computed for that channel. For the optional anisotropy Ward route an estimate of size $K^{-c_*}(\log K)^A$ multiplied by $g_0^{-1} \asymp K^{1/2}$ tends to zero from this bound if $c_* > 1/2$. At equality the logarithmic factor must itself tend to zero; a mere bounded-power estimate does not settle it. A fixed-torus estimate does not give a growing-volume Ward limit unless its volume factors are controlled at that rate.

The outstanding weighted task is therefore the actual finite-map extension with the weights in eq. A.lxxxvi, not just ordinary source holomorphy. The outstanding zone task is the complete incidence and original-weight reserve in eq. A.xcvii–eq. A.xcix. The matched task is the actual common-domain generator estimate in eq. A.civ–eq. A.cv, with its summable or direct two-cutoff error. Typed closure requires these assertions at the stated finite grade. These are separate optional routes; the estimates they require are not additional hypotheses of the ordinary scalar angular argument.

Appendix B. Index of notation and conventions

The table gives the principal notation and links to its definitions.

Notation	Meaning	Definition
$\mu_e, q(e), \nu_G$	Zero-source conditional law, clean conditional probability, and exterior-preserving clean resampling.	eq. 6.l; eq. 6.lii; Prop. 6.11
$d_S, b_n(D)$	Weighted exceptional mixed moment and its connected-error polynomial.	eq. 6.lvii; eq. 6.lxiv
A_B	Mixed source derivative of the complete effective action.	eq. 6.xc
$\alpha, L(A)$	Complete residual oscillation and Gaussian quadratic coefficient size.	eq. 6.clxvii; eq. 6.clxxiii
$\mathsf{D}_\eta, \mathcal{B}_{\omega,\chi}$	Lattice displacement vector field and its exact shell flux.	eq. 6.ccclxviii; eq. 6.ccclxxxviii
Lattices, fields, and action		
$G, \mathfrak{g}, N, d_G$	$G = \mathrm{SU}(2)$, $N = 2$, and $d_G = N^2 - 1 = 3$; Hermitian traceless matrices with $\langle X, Y\rangle = \frac{1}{2}\operatorname{tr}(XY)$.	gauge convention; appendix A.3

Notation	Meaning	Definition
L, M	Odd blocking integer $L > 1$ and localization cube side M. The rooted-source construction takes $M = L^{m_0}$ and compatible tori $\ell = 2L^m$.	localization cubes; eq. 5.cdxlvi
K, j, s, a_s, η_j	Cutoff depth K, level j, remaining depth $s = K - j$, physical spacing $a_s = L^{-s}$, and bare spacing $\eta_j = L^{-j}$ in level-j units.	level and depth; gauge convention
$\Lambda_{K,\ell}, \mathbb{T}^{(j)}$	Finite torus lattice $(L^{-K}\mathbb{Z}/\ell\mathbb{Z})^4$ and the lattice retained after j blockings.	lattice units; gauge convention
$U_b, U_p, \partial U$	Bond variable, ordered plaquette transport, and its alternate notation. Reversal gives $U_{\bar{b}} = U_b^{-1}$.	bond fields; gauge convention
$a_p, A(w,U), \beta$	Plaquette action $1 - \frac{1}{2}\,\mathrm{Re}\,\mathrm{tr}\,U_p$, weighted action $\sum_p w(x(p))a_p$, and bare inverse coupling $\beta = g_0^{-2}$.	action and sources; first passage
$\mathcal{O}(f), S_n^T$	Smeared action density $\sum_p f(x(p))a_p$ and its joint cumulants. No spacing prefactor occurs in $\mathcal{O}$.	observables; action and sources
Averages and propagators		
$\mathcal{M}, Q_k, Q_k^\sharp$	Contour block average, linearization of its plain iterate, and straightened linearization after removal of the coarse gauge factor.	eq. 2.iv; iterated averages
Q_k^{st}, Q'_k	Straight covariant bond average and site-function average.	iterated averages
$\mathrm{d}_0, \mathrm{d}_1, \mathrm{d}_\eta$	Unscaled scalar difference, plaquette coboundary, and physical difference; in the scalar block convention $\mathrm{d}_0 = \eta\mathrm{d}_\eta$.	block inner products; curvature kernels
$Q, P, \mathcal{A}, G'$	Scalar arithmetic mean, block-constant projection $P = Q^*Q$, operator $\mathcal{A} = \mathrm{d}_\eta^*\mathrm{d}_\eta + P$, and inverse $G' = \mathcal{A}^{-1}$.	block mean; scalar operator; scalar projection
$R, \mathbf{R}, R^{(j)}$	Scalar orthogonal projection onto $\ker(QG')$, large-field operation, and local remainder family, respectively.	scalar projection; § 4.17; activity families
$P_{\mathrm{ex}}, K, \mathsf{K}$	Projection onto exact lattice two-forms and continuum curvature covariance at unit coupling. The physical lattice kernel is $a^{-4}P_{\mathrm{ex}}(p,q)$.	curvature kernels; appendix A.3
$U_j(V), \mathsf{G}, \mathsf{H}$	Constrained regular background, constrained inverse, and response operator, with the gauges and domains specified in the background hypotheses.	background orbit; background operators
$\Phi_\Lambda(B), r_B$ Z_y	Potential/current background on the datum ball $\sup_c \lvert B(c)\rvert < r_B$ and local output $Z_y = (\xi_y H, \xi_y^2 \nabla H, \xi_y^3 J)$.	eq. 4.xxv; eq. 4.xxvi
Couplings and sources		
x_j, g_j, b_j	Running inverse coupling $x_j = x_{j-1} - b_j$, $x_0 = \beta$, and coupling $g_j = x_j^{-1/2}$.	flow recursion; frames and tests
c_0, b_j^0	Leading drift c_0 and designated one-loop increment b_j^0. The numerical flow theorem assumes an interval-sum law for these increments.	flow recursion; eq. A.lxvii
$\theta_j, \bar{g}_j, x_j^\natural$	Auxiliary frame, its coupling $\bar{g}_j = \theta_j^{-1/2}$, and inverse coupling computed with auxiliary choices frozen at $\theta_j = g^{-2} + c_0(K-j)$.	frames and tests; frozen coupling
$K(\beta, g)$ $\mathcal{W}_K(g)$	First entry into $g^{-2} < x_K \le g^{-2} + B_\sharp$ and the set of bare couplings with this stopping index.	first passage

Notation	Meaning	Definition
w, c, z, f_i	Source triple with function $w = c + \sum_i z_i f_i$ on each fixed finite real test family.	source triple
$\|\cdot\|_{\sharp,s}, \|\cdot\|_{\flat,s}$ $\mathcal{W}_r^{(s)}$	Test norm $\max(\|f\|_\infty, c_M L^{-s} \operatorname{Lip} f)$, source-triple norm $\|c\| + \sum_i \|z_i\| \|f_i\|_{\sharp,s}$, and its open radius-$r$ ball.	source norms
$h_y^{(j)}, \varphi_j, \zeta_j$	Anchor partition, fixed coupling profile, and running source shift defined by the exact counterterm identity.	source profiles
ϱ, ρ $\alpha_{0,j}, \alpha_{1,j}$	Source or coupling-disc radii in the stated estimate and complex field radii. Their meanings are local to that estimate.	source triple; frames and tests; eq. 2.v
$p_0, \check{R}_j$ ε_j, δ_j	Deposit scale, localization/waiting scale, and curvature and fluctuation thresholds formed from the fixed auxiliary frame.	frames and tests; eq. 2.xx
Local families and histories		
$E^{(j)}, R^{(j)}$ $B^{(j)}$	Regular activities, remainder activities, and boundary activities; pointed families also carry a base point.	activity families; eq. 2.xiv
$\mathsf{b}(E), e_j, \mathcal{E}$	Scalar marginal extractor, local source energy, and its sum. The vacuum constant E has a different role.	eq. 4.xix; local source energies
$d_j(X), \kappa, \kappa_{\mathcal{D}}$	Localization tree length, decay exponent, and domain-counting exponent. Summability requires decay greater than the counting exponent.	tree length; eq. 5.cdxlvi
Ω_j, Λ_j, Z_j	Regular region, buffered inner region, and complement $Z_j = \Lambda_j^c$ in the specified history geometry.	history regions
h, ω	Full history and local history inside a stored operation.	histories; stored operations
Ξ_ω, μ_ω Φ_ω, Ψ_ω	Stored integration space, positive reference measure, exponent, and excess source exponent. The integration domains and evaluation fibers are part of the construction.	eq. 2.xxv; eq. 5.xiv; excess source exponent
$\Theta^{(k)}, q_a, \Sigma_k$	History deposit/volume mark, age-survival rate $q_a = e^{-c_a}$, and integrated aged load.	eq. 2.xxx; survival rate; aged load
$\mathfrak{C}_k, \mathcal{U}_j^c$	Class of complete presentations and complex pair domain. The current in a pair is an independent coordinate.	presentations; eq. 2.v
$\mathcal{G}_z, \mathcal{K}_z, A_z$	Complete pointed total, its response-composed Hessian, and midpoint tent action for the same member.	eq. 4.ii; eq. 4.iv; eq. 4.xxi
$\rho, \sigma, a, a_{\text{out}}$ (window)	Field-radius ratio $\rho = a_{\text{out}}/a \geq 1$, scale ratio σ, member radius, and output radius.	eq. 4.xxix
$\tau_S(w), N$ (history sum)	Complex label measures, positive at zero, and the site count in the ordinary stability criterion.	eq. 5.cdxxiii
$E(z), E_{\ell,K}, A_K$ (transfer)	Common cell normalizer, finite-torus vacuum constant, and companion vacuum density used in the finite-reference route.	eq. 6.iv; eq. 6.vii; eq. 6.viii
Limits and reconstruction		

Notation	Meaning	Definition
$\mathfrak{T}_{\mathbb{E}}(\mathsf{d};\vartheta)$ $\mathfrak{T}^{(4)}_{\mathbb{E}}(\mathsf{d};\vartheta)$	Positive test pairs with quantitative mass and Lipschitz bounds, and quadruples whose pairs belong to this class, with small supports around collinear centres.	positive test pairs; four-point tests
$\mathfrak{S},\sigma$	Separated compact test tuples, including the empty tuple, and their moment functional, extended to finite linear combinations.	separated tests; moment functional
$\mathsf{d}(X)$	Minimum distance between distinct test supports; support-containing balls do not define this distance.	separated tests
σ_S,σ^0	Canonical centered functional: the partition sum over blocks of size at least two, with empty value one and singleton value zero.	centered functional; eq. 5.cdlxxv
$\mathcal{S}_{\neq},\lvert\varphi\rvert_k$	Schwartz tests flat with all derivatives on every coincidence diagonal, and the weighted derivative seminorm, written $p_k(\varphi)$ in the extension proof.	diagonal-flat space; eq. 5.cdlxxix
$m(X),X^+$	Minimum cubical tree-edge count and the fixed-layer source-dependence enlargement in the rooted geometry.	eq. 5.cdxlvi; eq. 5.cdxlvii
$k,B(R)$	Factorial exponent and support-radius factor in the graded all-state estimate; one may take $B(R)=C(R+1)^4$.	Prop. 5.76
$k_{\mathrm{E}},\kappa_{\mathrm{E}}$	Diagonal-flat extension exponents; the quantitative theorem permits $k_{\mathrm{E}}=20$ and $\kappa_{\mathrm{E}}=k+32$.	Prop. 5.78
$\mathsf{p},\wp$	Uniform insertion exponent and analytic plate-density exponent $\wp=\mathsf{p}+5$.	Prop. 6.23; Prop. 6.24
$G_{\varphi,\psi},\mu_\varphi$ Γ_φ,L_s	Transversely smeared planar kernel, positive spectral measure, Laplace norm, and spectral boost.	eq. 6.cccxviii; eq. 6.cccxxi; eq. 6.cccxxii; eq. 6.cccxlvi
$\vartheta_*,u(d),c_m,b_n$	Fixed positive-shape threshold, ultraviolet error function, angular coefficients, and largest nonconstant angular frequency.	eq. 6.cccxxvi; eq. 6.cccxxvii; eq. 6.ccli; harmonic degree
$\mathcal{G}^{\mathrm{tor}}$	Limits of periodically lifted torus measures at fixed spacing and coupling.	torus-limit states; eq. 5.cdlx
$\mathcal{G}_j,\mathcal{E}_j,D_j$	Translation-invariant DLR class, its ergodic states, and the decomposition law of the centering state on those states.	invariant DLR states; ergodic decomposition
$\mathcal{I},\mathrm{Pl}_\nu$ $\mathcal{N}_{\nu,r}$	Invariant sigma-field, conditional-expectation plateau pairing, and reflected size in the lattice plateau estimate.	plateau quantities; Prop. A.12
$\mathsf{X},\lambda,\sigma_\xi$	Compact support of the phase-coordinate law, that probability law, and its continuous component evaluations.	phase coordinates
$\theta,\mathcal{H},H,P$ Ω,P_0	Time reflection, Hilbert completion of its positive form, time and momentum generators, empty-tuple vector, and projection onto $\ker H$.	eq. 6.cdxlv; eq. 6.cdxlvii; § 7.8
$\mathfrak{U},\mathfrak{B},W_4$	Ward defect, shell insertion, and signed coordinate permutation group, with the specified covariance and flux conventions.	Ward insertion; eq. 4.xvii

Conventions. Lengths are physical unless a level or local lattice unit is stated. In particular, a_{K-j} and η_j express different unit conventions. An adjoint always uses the stated inner products on both spaces.

Constants may depend on fixed geometric and analytic parameters. The stated parameter dependence and uniformity are part of each estimate. A radius used for coefficient extraction is the radius in that estimate.

Concatenation of test tuples is used only when the resulting supports remain separated. Centering changes the first cumulant and leaves higher cumulants unchanged. Two-state moments retain the stated centering state.

The letters K, R, P, ρ, N have several local uses, distinguished above by their arguments and spaces. The window exponent is a local Hölder exponent; it is independent of the field analyticity radii.

In the usual Wilson coupling convention, $\beta = 2N/g_{\mathrm{std}}^2$. All logarithms are natural. Near-identity group logarithms use the branches specified in the defining chart. A pair current is independent of its field coordinate unless the pair is explicitly restricted to a background graph. Each use of a construction retains its stated existence, domain, and uniformity hypotheses.

Analytic operators and domains. The following symbols refer to the constructions in the analytic estimates. Their transposes use the physical measures unless a counting normalization is displayed.

Notation	Meaning	Definition
$Y, Y^{\flat}, Z$	Balanced first-order site space, its source space under pairing with $\ell^2 u$, and the site second-difference energy space.	eq. 3.cmxxix.
$\mathcal{A}_\alpha, \mathcal{F}_p$	Strong field space and normalized finite-p force space. A cell force has norm $\xi^{3-4/p}\|f\|_p$.	eq. 3.dccclxxxvii; eq. 3.mclxxxiii.
$\widehat{F}, \widehat{M}$	Site columns and their positive Gram matrix in counting top coordinates; the column injection is $q^{\sharp}\Xi^{-4}$.	eq. 3.cmxcv.
Λ, S, Π_U	Linear gauge restoration, its complement $S = 1 - D\Lambda$, and the selected finite gauge chain with derivative S.	eq. 3.mcxxxix; eq. 3.mcxlii.
G_F, G_0, G, G_1	Positive field primitive, projected zero-current inverse, first current inverse, and second current inverse.	eq. 3.cmxcix; eq. 3.miii; eq. 3.mxiii.
G_0^a, G^a, H^a; ω^a, G_1^a, H_1^a	Weighted canonical inverses and lifts. The multiplier is $(H^a)^{\sharp}J$, formed before the second inverse. Their auxiliary field output has two finite-p differences.	eq. 3.mcccclxxxiii; eq. 3.mcccclxxxiv; eq. 3.mcccclxxxv; eq. 3.mcccclxxxvi.
H, H_1, ω_*	First and second row lifts, and the multiplier readout $H^{\sharp}J$. The readout uses the first lift.	eq. 3.mvii; eq. 3.mxiv; eq. 3.mix.
K_4, G_4, M_4	Fourth-order site precision, its inverse, and the counting coarse matrix qG_4q^*. They give a new strong family for G_sR.	eq. 3.mclxv; eq. 3.mclxxvii; eq. 3.mclxxx.
$\mathcal{S}_\alpha, Y^{2,\alpha}$	Fourth-order source class $\Delta C_2^\alpha + D^{\sharp}L_3^\infty + L_4^\infty$ and its strong site output space.	eq. 3.mclxix; eq. 3.mxxiv.
$\mathcal{L}_{U,J}, A(b), C(B)$	Selected auxiliary action, its constrained potential, and its finite gauge completion. The potential has $D_bA(0) = H_1$; the literal completion has $D_BC(0) = SH_1Db_U(0)$.	eq. 3.mcxlviii; eq. 3.mclii; eq. 3.mcliv.

Notation	Meaning	Definition
$\Gamma_P, K_\tau(P)$	Mass-subtracted coarse form and its outer plain-constraint precision, including the multiplier times the second chart derivative.	eq. 5.cclix.
$\mathcal{W}_p, \mathcal{L}_{0,a}$	Field norm with two physical L^p differences, and the full weighted trial saddle. The principal perturbation acts from $\mathcal{W}_p$ to $\mathcal{F}_p$.	eq. 3.mcccli; eq. 3.mcccliv; eq. 3.mccclvii.
$M_{\mathrm{c}}, M_{\mathrm{G}}, M_{\mathrm{h}}$	Carrier cube size, Gaussian parameter cube size, and a prescribed history cube size. Their separation requirements are stated at their respective uses.	eq. 3.cmlxxxvii; eq. 5.cclxix.
$D_W(s), R_W(s)$	Local Gaussian covariance and its root. At complete parameters they give the actual matrix marginal.	eq. 5.cclxxi; eq. 5.cclxxii.
$p_j^{\mathrm{lit}}, \mathsf{p}_j$	Literal intermediate mean of the physical critical orbit. Its raw bond amplitude is bounded by C/L times the output amplitude.	eq. 3.mcccvi; eq. 4.cccxcii.
$e_{s,z}, T_{i,k}$	Original fine-plaquette action pointing and complete physical transport with exact hat refinement.	eq. 4.ccclviii; eq. 4.cdxvii.
$\mathfrak{H}_{r,\kappa,\gamma_{\mathrm{per}}}$	Compatible pointed interaction space: fixed analytic radius, carrier decay and finite-period tail bound.	eq. 4.ccclxxxviii.
$\mathbf{F}_j(x,h), N_j$	Vacuum-subtracted formation map and physical Gaussian logarithm in the literal regular step. N_j is independent of x, h.	eq. 4.cdxiv; eq. 4.cdv.
$\mathcal{B}_{j,e}$	Complete rooted sphere family: the moving-centre logarithmic weight jet minus the fixed-kernel jet. Its six curvature readouts vanish.	eq. 4.cdxxxix; Cor. 4.55.
$\widetilde{\beta}_a^{\mu\nu}(z)$	Weighted massive Wilson coefficient in one plane, with vector regulator $z\mathsf{A}$ before congruence.	eq. 5.cli; eq. 5.clii.
$x_j, h_j, \|\cdot\|_{j,*}$	Tuned coupling, zero-readout regular profile, and horizon-adapted profile norm.	eq. 5.cccxxi; eq. 5.cccxvi.
$P_{j\ell}, U_{ji}, B^+, K^+$	Backward scalar and forward profile propagators of the mixed-end linear response, with positive majorizing kernels.	eq. 5.cccxliii; eq. 5.cccxlv.
$T_N, O_N, \nu_{N,t}$	Bare tuned scalar insertion, its nonconstant conditional coarse image, and the capped regular terminal law.	eq. 5.ccclxxxi; eq. 5.ccclxxviii.
$\delta\alpha_N, \psi_N$	Bare tuning with its value at the reference endpoint subtracted, and its local inverse in the bare scalar source. Neither subtracts a random field.	Lem. 5.60 and Cor. 5.62.

B.1. **Proof references.** The table locates the component arguments. Each result is used with the hypotheses stated at its cited location.

Proof task	Arguments and precise references
Projected Landau inverse	Local gauge regularity in § 3.16; the all-mesh projected gap and actual-slice saddle inverse in § 3.17; curvature-only periodic localization in § 3.18.

Proof task	Arguments and precise references
Normalized interacting extraction	Positive history aggregation, weighted exceptional moments and exterior cumulants in § 6.4; mixed source derivatives in § 6.5; the complete-oscillation Gaussian criterion in § 6.7.
Four-point and Ward calculations	Quantitative free-kernel signs and shape comparison in § 6.8; exact Haar divergence and contact terms in § 6.14; off-shell invariant classification and the weighted smooth Wilson expansion in § 6.15.
Averaging and the flat reference	Pivot coordinates, the contour-gauge gap, bounded lifts, and exponential kernels: §§ 3.3 to 3.6.
First constrained background	Nonflat first-step precision, the regular critical orbit, and the relative-current extension: §§ 3.7, 3.8 and 3.10.
Multiscale backgrounds	Pairings, core domains, hierarchy identities, nonlinear responses, and local Hölder estimates: §§ 3.9, 3.11, 3.14 and 3.15. Layered precision is treated in § 3.13.
Local interactions	The member's quadratic counterterm is given by eq. 4.xix and eq. 4.xxiv. The nonlinear window estimate is Prop. 4.5; insertion and the source bootstrap are Props. 4.8 and 4.10.
Histories and stability	History counting and mass estimates: § 4.18. The ordinary stability upper bound and positive floor are eq. 5.cdxxvi and eq. 5.cdxxviii, under their respective hypotheses.
Flow and coupling comparison	First-passage arithmetic: § 5.1. Joint-source comparison and the terminal coupling limit: Lem. A.14 and Prop. A.15. Weighted comparison domains: appendix A.9.
States and continuum limits	Periodic-state mirror positivity: Prop. A.11. Common state normalizers and factorial moment bounds: Props. 5.75 and 5.76. Subsequence compactness and flat-test extension: Props. 5.77 and 5.78.
Free and interacting two-point functions	The free-kernel estimate is eq. A.iii. Conditional covariance and Gaussian cutoff calculations are in §§ 6.3 and 6.6; the extraction and common-normalizer transfer estimates are in § 6.2.
Insertions and rotations	Uniform insertion, plate density, the planar kernel, and harmonic elimination: Props. 6.23 to 6.26 and § 6.11.
Fields and operator domains	Energy-band fields: § 7.2. The matched scalar reconstruction is Prop. 7.6; its representation domains are in § 7.5.
Phases and spectra	The conditional probability kernel is Prop. 7.7. Phase coordinates, the vacuum space, and the scalar direct integral are treated in §§ 7.7 to 7.9.
Gauge observables	The cutoff orbit coordinates and continuum extension criteria are in §§ 7.11 and 7.13. The scalar observable information is characterized in § 7.12.

B.2. **References for the analytic constructions.** These references distinguish the completed operators from their local coefficient families, and the regular step from its complementary history operations. Each construction retains its stated reference, geometry and parameter domain.

Construction	Proof and scope
Strong response at an independent current	The actual Wilson stencil, finite-p bootstrap and current adjoints are in §§ 3.25 and 3.26. The completed strong estimates do not require DJ.
Local positive inverses	Prop. 3.102 uses genuine Dirichlet compressions, slow cutoffs and full geometry carriers. The positive coarse inverse is Prop. 3.103.
Named current families	Prop. 3.112 constructs $G_0, G, H, \omega_*, G_1, H_1$ in that order, with normalized physical row transposes. It retains the bare current readout separately.
Strong restoration coefficients	Props. 3.154, 3.159 and 3.160 prove the fourth-order realization, its exact G_sR identity and its new $C^{2,\alpha}$ coefficient family.

Construction	Proof and scope
Strong saddle coefficients	Prop. 3.163 uses the finite-p force classes and the strong restoration. Its current ceiling may shrink with the fixed Hölder exponent and carrier rate.
Nonlinear current germ	Props. 3.149 to 3.151 give the selected auxiliary action, its physical matching and its current update. The H_1, SH_1 and literal datum jets are distinct.
All-degree nonlinear coefficients	Props. 3.167 to 3.169 give the convergent connected tree series for that selected germ, including joint base and datum variables. Local-data suprema precede all carrier sums.
Weighted connected germ	Props. 3.205, 3.207 and 3.208 construct the weighted principal and current inversions before the actual nonlinear tree series. Prop. 3.209 identifies its admitted physical curvature-atlas branch.
Actual hierarchical row density	Prop. 3.187 computes the one-step second jet and the full chain sum. Uniformity in depth does not make the nonlinear radius independent of L.
Homogeneous critical fields	Prop. 3.183 constructs the physical orbit. Flat reference descent is Prop. 3.192; Prop. 3.120 also allows a current-regular trial.
Curvature-only homogeneous data	Props. 3.193 and 3.194 construct the trial from a smooth quaternionic bundle, then correct its actual mean and current. The atlas allows large coarse holonomies and nontrivial bundle topology.
Weighted physical atlas	Lem. 3.195 and Props. 3.199, 3.201 and 3.202 retain two L^p differences before perturbing the principal part. The physical gap includes both constraint curvatures and the row-mass subtraction.
Outer nonflat Gaussian	Props. 5.41 and 5.42 prove the real physical gap and the canonical local Gaussian extension near that reference, retaining the mass subtraction and chart curvature.
Cutoff covariance	Prop. 4.25 uses the middle crossing pivots and real colour-ball cutoff. It proves covariance of the actual regular integral before pointing or filing.
Physical irrelevance	Prop. 4.36 uses the literal intermediate field, complete Ward cancellation, block interpolation and explicit finite-period tails. Individual local pieces need not be invariant.
Fixed-radius formation	Prop. 4.46 and Cors. 4.47 and 4.48 admit the old interaction on its own ball, prove the coupling and profile slopes, and close one regular step on the common radius.
Fixed-kernel stress response	Prop. 4.54 and Cor. 4.55 compare the moving weight jet with the fixed kernel by a normalized rooted sphere family. Prop. 5.54 uses the corrected profile forcing in its coupling-weighted memory sum.
Actual scalar one-loop limit	The bare Wilson calculation is Prop. 5.9; physical gauge change is eq. 5.civ. The regulator and Taylor-vertex comparisons are §§ 5.7, 5.9 and 5.10.
Diagonal one-loop response	Props. 5.14, 5.47, 5.48 and 5.50 retain the actual weight parameter, its principal-part perturbation, the weighted regulator and its differentiated Gaussian polarization.
Regular tuning	Prop. 5.51 solves the mixed end conditions. eqs. 5.cccxxxi and 5.cccxxxii differentiate one frozen presentation, with constants uniform in the horizon.
Bounded initial irrelevant profiles	Props. 5.52 and 5.53 retune the same finite regular problem. eqs. 5.cccxlvii and 5.ccccliv give the mixed-end inverse and all fixed-order initial-profile jets.

Construction	Proof and scope
Scalar normalization and nonvanishing	Props. 5.55 and 5.59 retain the numerical vacuum derivative before centering and prove a positive variance in the capped regular law. Prop. 5.61 extracts a nondegenerate coarse subsequence at fixed volume and cutoff.

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